
HPX Documentation

2.0.0

The STE || AR Group

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USER DOCUMENTATION

Welcome to the *HPX* documentation!

If you're new to *HPX* you can get started with the *Quick start* guide. Don't forget to read the *Terminology* section to learn about the most important concepts in *HPX*. The *Examples* give you a feel for how it is to write real *HPX* applications and the *Manual* contains detailed information about everything from building *HPX* to debugging it. There are links to blog posts and videos about *HPX* in *Additional material*.

You can find a comprehensive list of contact options on [Support for deploying and using HPX¹](#). Do not hesitate to contact us if you can't find what you are looking for in the documentation!

See *Citing HPX* for details on how to cite *HPX* in publications. See *HPX users* for a list of institutions and projects using *HPX*.

There are also available a [PDF](#) version of this documentation as well as a [Single HTML Page](#).

¹ <https://github.com/TheHPXProject/hpx/blob/master/.github/SUPPORT.md>

WHAT IS *HPX*?

HPX is a C++ Standard Library for Concurrency and Parallelism. It implements all of the corresponding facilities as defined by the C++ Standard. Additionally, in *HPX* we implement functionalities proposed as part of the ongoing C++ standardization process. We also extend the C++ Standard APIs to the distributed case. *HPX* is developed by the STE||AR group (see *People*).

The goal of *HPX* is to create a high quality, freely available, open source implementation of a new programming model for conventional systems, such as classic Linux based Beowulf clusters or multi-socket highly parallel SMP nodes. At the same time, we want to have a very modular and well designed runtime system architecture which would allow us to port our implementation onto new computer system architectures. We want to use real-world applications to drive the development of the runtime system, coining out required functionalities and converging onto a stable API which will provide a smooth migration path for developers.

The API exposed by *HPX* is not only modeled after the interfaces defined by the C++11/14/17/20 ISO standard. It also adheres to the programming guidelines used by the Boost collection of C++ libraries. We aim to improve the scalability of today's applications and to expose new levels of parallelism which are necessary to take advantage of the exascale systems of the future.

HPX is a member of the High-Performance Software Foundation ([HPSF](https://hpsf.io)²) network, which signals its commitment to the open-source, high-performance computing (HPC) community. By joining [HPSF](https://hpsf.io)³ - and, concurrently, the Linux Foundation - *HPX* gains access to a broader collaborative ecosystem that promotes shared standards, interoperable tools, and joint research initiatives, all of which accelerates the development of scalable parallel applications. This dual affiliation enhances *HPX*'s credibility, attracting contributors and industry partners who value the rigorous governance and transparent development practices upheld by both organizations. Consequently, *HPX* can more effectively influence the direction of future HPC frameworks while benefiting from the collective expertise, resources, and industry backing provided by [HPSF](https://hpsf.io)⁴ and the Linux Foundation.

² <https://hpsf.io/projects/hpx/>

³ <https://hpsf.io/projects/hpx/>

⁴ <https://hpsf.io/projects/hpx/>

WHAT'S SO SPECIAL ABOUT *HPX*?

- *HPX* exposes a uniform, standards-oriented API for ease of programming parallel and distributed applications.
- It enables programmers to write fully asynchronous code using hundreds of millions of threads.
- *HPX* provides unified syntax and semantics for local and remote operations.
- *HPX* makes concurrency manageable with dataflow and future based synchronization.
- It implements a rich set of runtime services supporting a broad range of use cases.
- *HPX* exposes a uniform, flexible, and extendable performance counter framework which can enable runtime adaptivity
- It is designed to solve problems conventionally considered to be scaling-impaired.
- *HPX* has been designed and developed for systems of any scale, from hand-held devices to very large scale systems.
- It is the first fully functional implementation of the ParalleX execution model.
- *HPX* is published under a liberal open-source license and has an open, active, and thriving developer community.

2.1 Quick start

The following steps will help you get started with *HPX*. Before getting started, make sure you have all the necessary prerequisites, which are listed in `_prerequisites`. After *Installing HPX*, you can check how to run a simple example *Hello, World!*. *Writing task-based applications* explains how you can get started with *HPX*. You can refer to our *Migration guide* if you use other APIs for parallelism (like OpenMP, MPI or Intel Threading Building Blocks (TBB)) and you would like to convert your code to *HPX* code.

2.1.1 Installing *HPX*

The easiest way to install *HPX* on your system is by choosing one of the steps below:

1. **vcpkg**

You can download and install *HPX* using the `vcpkg`⁵ dependency manager:

```
$ vcpkg install hpx
```

2. **Spack**

Another way to install *HPX* is using `Spack`⁶:

⁵ <https://github.com/Microsoft/vcpkg>

⁶ <https://spack.readthedocs.io/en/latest/>

```
$ spack install hpx
```

3. Conan

You can also use the [Conan](https://conan.io/)⁷ package manager to build and use *HPX* in your projects. First, export the *HPX* recipe from the *HPX* repository to your local Conan cache:

```
$ git clone https://github.com/TheHPXProject/hpx.git
$ cd hpx
$ conan create . --build=missing
```

This builds *HPX* and makes it available in your local Conan cache. You can then use it in your Conan-based projects by adding it to your `conanfile.txt`:

```
[requires]
hpx/2.0.0

[generators]
CMakeDeps
CMakeToolchain
```

Then use `find_package(HPX REQUIRED)` in your `CMakeLists.txt` as usual. Conan will automatically handle building *HPX* with your chosen configuration options and provide the necessary CMake integration files.

4. Fedora

Installation can be done with [Fedora](https://fedoraproject.org/wiki/DNF)⁸ as well:

```
$ dnf install hpx*
```

5. Arch Linux

HPX is available in the [Arch User Repository \(AUR\)](https://wiki.archlinux.org/title/Arch_User_Repository)⁹ as `hpx` too.

More information or alternatives regarding the installation can be found in the *Building HPX*, a detailed guide with thorough explanation of ways to build and use *HPX*.

2.1.2 Hello, World!

To get started with this minimal example you need to create a new project directory and a file `CMakeLists.txt` with the contents below in order to build an executable using [CMake](https://www.cmake.org)¹⁰ and *HPX*:

```
cmake_minimum_required(VERSION 3.19)
project(my_hpx_project CXX)
find_package(HPX REQUIRED)
add_executable(my_hpx_program main.cpp)
target_link_libraries(my_hpx_program HPX::hpx HPX::wrap_main hpx::iostreams_component)
```

The next step is to create a `main.cpp` with the contents below:

⁷ <https://conan.io/>

⁸ <https://fedoraproject.org/wiki/DNF>

⁹ https://wiki.archlinux.org/title/Arch_User_Repository

¹⁰ <https://www.cmake.org>

```
// Including 'hpx/hpx_main.hpp' instead of the usual 'hpx/hpx_init.hpp' enables
// to use the plain C-main below as the direct main HPX entry point.
#include <hpx/hpx_main.hpp>
#include <hpx/iostream.hpp>

int main()
{
    // Say hello to the world!
    hpx::cout << "Hello World!\n" << std::flush;
    return 0;
}
```

Then, in your project directory run the following:

```
$ mkdir build && cd build
$ cmake -DHPX_DIR=</path/to/hpx/installation> ..
$ make all
$ ./my_hpx_program
```

```
$ ./my_hpx_program
Hello World!
```

The program looks almost like a regular C++ hello world with the exception of the two includes and `hpx::cout`.

- When you include `hpx_main.hpp` *HPX* makes sure that `main` actually gets launched on the *HPX* runtime. So while it looks almost the same you can now use futures, `async`, parallel algorithms and more which make use of the *HPX* runtime with lightweight threads.
- `hpx::cout` is a replacement for `std::cout` to make sure printing never blocks a lightweight thread. You can read more about `hpx::cout` in *The HPX I/O-streams component*.

Note:

- You will most likely have more than one `main.cpp` file in your project. See the section on *Using HPX with CMake-based projects* for more details on how to use `add_hpx_executable`.
- `HPX::wrap_main` is required if you are implicitly using `main()` as the runtime entry point. See *Re-use the `main()` function as the main HPX entry point* for more information.
- `hpx::iostreams_component` is optional for a minimal project but lets us use the *HPX* equivalent of `std::cout`, i.e., the *HPX The HPX I/O-streams component* functionality in our application.
- You do not have to let *HPX* take over your main function like in the example. See *Starting the HPX runtime* for more details on how to initialize and run the *HPX* runtime.

Caution: Ensure that *HPX* is installed with `HPX_WITH_DISTRIBUTED_RUNTIME=ON` to prevent encountering an error indicating that the `HPX::iostreams_component` target is not found.

When including `hpx_main.hpp` the user-defined `main` gets renamed and the real `main` function is defined by *HPX*. This means that the user-defined `main` must include a return statement, unlike the real `main`. If you do not include the return statement, you may end up with confusing compile time errors mentioning `user_main` or even runtime errors.

2.1.3 Writing task-based applications

So far we haven't done anything that can't be done using the C++ standard library. In this section we will give a short overview of what you can do with *HPX* on a single node. The essence is to avoid global synchronization and break up your application into small, composable tasks whose dependencies control the flow of your application. Remember, however, that *HPX* allows you to write distributed applications similarly to how you would write applications for a single node (see *Why HPX?* and *Writing distributed applications*).

If you are already familiar with `async` and `future` from the C++ standard library, the same functionality is available in *HPX*.

The following terminology is essential when talking about task-based C++ programs:

- **lightweight thread**: Essential for good performance with task-based programs. Lightweight refers to smaller stacks and faster context switching compared to OS threads. Smaller overheads allow the program to be broken up into smaller tasks, which in turns helps the runtime fully utilize all processing units.
- **async**: The most basic way of launching tasks asynchronously. Returns a `future<T>`.
- **future<T>**: Represents a value of type `T` that will be ready in the future. The value can be retrieved with `get` (blocking) and one can check if the value is ready with `is_ready` (non-blocking).
- **shared_future<T>**: Same as `future<T>` but can be copied (similar to `std::unique_ptr` vs `std::shared_ptr`).
- **continuation**: A function that is to be run after a previous task has run (represented by a future). `then` is a method of `future<T>` that takes a function to run next. Used to build up dataflow DAGs (directed acyclic graphs). `shared_futures` help you split up nodes in the DAG and functions like `when_all` help you join nodes in the DAG.

The following example is a collection of the most commonly used functionality in *HPX*:

```
#include <hpx/algorithm.hpp>
#include <hpx/future.hpp>
#include <hpx/init.hpp>

#include <iostream>
#include <random>
#include <vector>

void final_task(hpx::future<hpx::tuple<hpx::future<double>, hpx::future<void>>>>)
{
    std::cout << "in final_task" << std::endl;
}

int hpx_main()
{
    // A function can be launched asynchronously. The program will not block
    // here until the result is available.
    hpx::future<int> f = hpx::async([]() { return 42; });
    std::cout << "Just launched a task!" << std::endl;

    // Use get to retrieve the value from the future. This will block this task
    // until the future is ready, but the HPX runtime will schedule other tasks
    // if there are tasks available.
    std::cout << "f contains " << f.get() << std::endl;
}
```

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```

// Let's launch another task.
hpx::future<double> g = hpx::async([]() { return 3.14; });

// Tasks can be chained using the then method. The continuation takes the
// future as an argument.
hpx::future<double> result = g.then([](hpx::future<double>&& gg) {
    // This function will be called once g is ready. gg is g moved
    // into the continuation.
    return gg.get() * 42.0 * 42.0;
});

// You can check if a future is ready with the is_ready method.
std::cout << "Result is ready? " << result.is_ready() << std::endl;

// You can launch other work in the meantime. Let's sort a vector.
std::vector<int> v(1000000);

// We fill the vector synchronously and sequentially.
hpx::generate(hpx::execution::seq, std::begin(v), std::end(v), &std::rand);

// We can launch the sort in parallel and asynchronously.
hpx::future<void> done_sorting =
    hpx::sort(hpx::execution::par(           // In parallel.
              hpx::execution::task),        // Asynchronously.
              std::begin(v), std::end(v));

// We launch the final task when the vector has been sorted and result is
// ready using when_all.
auto all = hpx::when_all(result, done_sorting).then(&final_task);

// We can wait for all to be ready.
all.wait();

// all must be ready at this point because we waited for it to be ready.
std::cout << (all.is_ready() ? "all is ready!" : "all is not ready...")
          << std::endl;

return hpx::local::finalize();
}

int main(int argc, char* argv[])
{
    return hpx::local::init(hpx_main, argc, argv);
}

```

Try copying the contents to your `main.cpp` file and look at the output. It can be a good idea to go through the program step by step with a debugger. You can also try changing the types or adding new arguments to functions to make sure you can get the types to match. The type of the `then` method can be especially tricky to get right (the continuation needs to take the future as an argument).

Note: HPX programs accept command line arguments. The most important one is `--hpx:threads=N` to set the number of OS threads used by HPX. HPX uses one thread per core by default. Play around with the example above

and see what difference the number of threads makes on the `sort` function. See *Launching and configuring HPX applications* for more details on how and what options you can pass to *HPX*.

Tip: The example above used the construction `hpx::when_all(...).then(...)`. For convenience and performance it is a good idea to replace uses of `hpx::when_all(...).then(...)` with `dataflow`. See *Dataflow* for more details on `dataflow`.

Tip: If possible, try to use the provided parallel algorithms instead of writing your own implementation. This can save you time and the resulting program is often faster.

2.1.4 Next steps

If you haven't done so already, reading the *Terminology* section will help you get familiar with the terms used in *HPX*.

The *Examples* section contains small, self-contained walkthroughs of example *HPX* programs. The *Local to remote* example is a thorough, realistic example starting from a single node implementation and going stepwise to a distributed implementation.

The *Manual* contains detailed information on writing, building and running *HPX* applications.

2.2 Tutorials

This section provides structured, video-based learning resources for *HPX*. The tutorials complement the reference documentation with guided, example-driven explanations of core concepts and advanced features.

Each tutorial introduces the relevant concepts and then applies them step by step through practical code examples. The focus is on demonstrating how to design, implement, and reason about parallel and distributed applications using *HPX*. While the main documentation emphasizes API descriptions and detailed reference material, the tutorial series adopts a practical, code-oriented approach.

The tutorials are intended for a broad audience: from beginners setting up *HPX* for the first time to advanced users exploring performance tuning, custom executors, and SIMD vectorization. Together, they form a structured learning path that lowers the entry barrier while progressively introducing more advanced features.

The complete video series is available on YouTube:

[HPX Tutorial Playlist](#)¹¹

The code is available on Github:

HPX Tutorial Code <https://github.com/STELLAR-GROUP/HPX_Tutorials_Code>

¹¹ <https://www.youtube.com/watch?v=OZWTmrIWUF0&list=PL7vEgTL3FalZkvNqssEP9-tXQ5rEwZy2l&index=1>

2.2.1 Available Modules

- **Introduction to |hpx|**

Foundational overview of *HPX*, including its architecture, design principles, and key capabilities in the context of high-performance computing.

- **Building |hpx| on Windows/Unix**

Step-by-step guidance on installing *HPX* and setting up the environment based on your operating system.

- **Running |hpx| applications**

Demonstration on how to build and run a simple Hello World example using *HPX*.

- **Algorithms in |hpx|**

Overview of the parallel algorithms provided by *HPX* and explanation on how execution policies unlock high-performance C++ computing.

- **Vectorization and SIMD in |hpx|**

Introduction to data-parallel programming, SIMD execution policies, and vectorization techniques relevant to modern C++ and HPC workloads.

- **Futures and Asynchronous Programming**

Explanation of the asynchronous programming model in *HPX*, including futures, *async*, and dataflow-based execution patterns.

- **Performance Analysis and Optimization**

Guidance on profiling *HPX* applications, analyzing performance counters, identifying bottlenecks, and applying optimization strategies.

- **Task Scheduling and Custom Executors**

Detailed discussion of the execution model in *HPX*, built-in executors, and mechanisms for defining custom execution strategies.

- **Migration Guides**

Best practices and recommendations for transitioning from other HPC frameworks to *HPX*.

- **HPX in Practice**

Examples of open-source libraries and applications using *HPX*, illustrating real-world usage patterns and ecosystem integration.

2.3 Examples

The following sections analyze some examples to help you get familiar with the *HPX* style of programming. We start off with simple examples that utilize basic *HPX* elements and then begin to expose the reader to the more complex and powerful *HPX* concepts. Section *Building tests and examples* shows how you can build the examples.

2.3.1 Asynchronous execution

The Fibonacci sequence is a sequence of numbers starting with 0 and 1 where every subsequent number is the sum of the previous two numbers. In this example, we will use *HPX* to calculate the value of the *n*-th element of the Fibonacci sequence. In order to compute this problem in parallel, we will use a facility known as a future.

As shown in the Fig. ?? below, a future encapsulates a delayed computation. It acts as a proxy for a result initially not known, most of the time because the computation of the result has not completed yet. The future synchronizes the access of this value by optionally suspending any *HPX*-threads requesting the result until the value is available. When a future is created, it spawns a new *HPX*-thread (either remotely with a *parcel* or locally by placing it into the thread

queue) which, when run, will execute the function associated with the future. The arguments of the function are bound when the future is created.

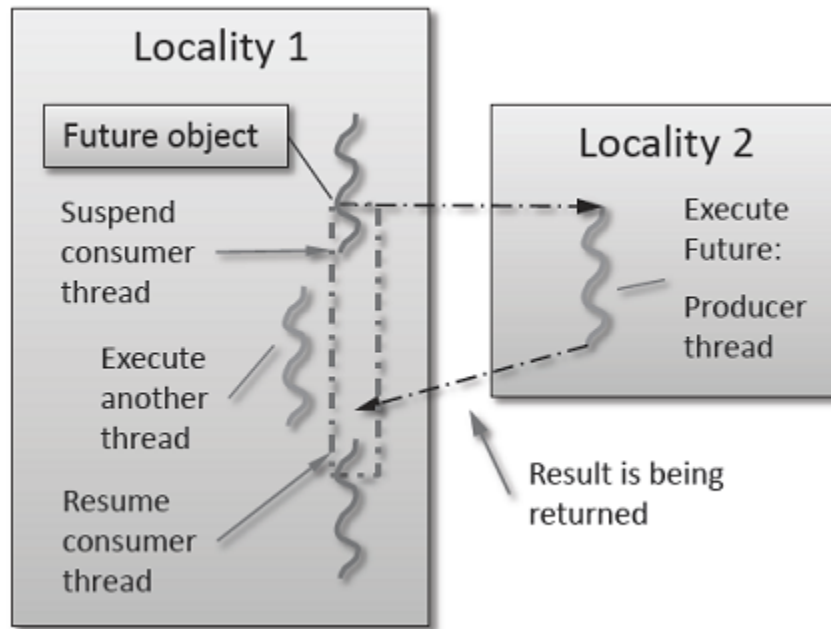


Fig. 2.1: Schematic of a future execution.

Once the function has finished executing, a write operation is performed on the future. The write operation marks the future as completed, and optionally stores data returned by the function. When the result of the delayed computation is needed, a read operation is performed on the future. If the future's function hasn't completed when a read operation is performed on it, the reader *HPX*-thread is suspended until the future is ready. The future facility allows *HPX* to schedule work early in a program so that when the function value is needed it will already be calculated and available. We use this property in our Fibonacci example below to enable its parallel execution.

Setup

The source code for this example can be found here: `fibonacci_local.cpp`.

To compile this program, go to your *HPX* build directory (see *Building HPX* for information on configuring and building *HPX*) and enter:

```
$ make examples.quickstart.fibonacci_local
```

To run the program type:

```
$ ./bin/fibonacci_local
```

This should print (time should be approximate):

```
fibonacci(10) == 55
elapsed time: 0.002430 [s]
```

This run used the default settings, which calculate the tenth element of the Fibonacci sequence. To declare which Fibonacci value you want to calculate, use the `--n-value` option. Additionally you can use the `--hpx:threads` option to declare how many OS-threads you wish to use when running the program. For instance, running:


```
$ ./bin/fibonacci --n-value 20 --hpx:threads 4
```

Will yield:

```
fibonacci(20) == 6765
elapsed time: 0.062854 [s]
```

Walkthrough

Now that you have compiled and run the code, let's look at how the code works. Since this code is written in C++, we will begin with the `main()` function. Here you can see that in *HPX*, `main()` is only used to initialize the runtime system. It is important to note that application-specific command line options are defined here. *HPX* uses [Boost.Program_options](https://www.boost.org/doc/html/program_options.html)¹² for command line processing. You can see that our programs `--n-value` option is set by calling the `add_options()` method on an instance of `hpx::program_options::options_description`. The default value of the variable is set to 10. This is why when we ran the program for the first time without using the `--n-value` option the program returned the 10th value of the Fibonacci sequence. The constructor argument of the description is the text that appears when a user uses the `--hpx:help` option to see what command line options are available. `HPX_APPLICATION_STRING` is a macro that expands to a string constant containing the name of the *HPX* application currently being compiled.

In *HPX* `main()` is used to initialize the runtime system and pass the command line arguments to the program. If you wish to add command line options to your program you would add them here using the instance of the Boost class `options_description`, and invoking the public member function `.add_options()` (see [Boost Documentation](https://www.boost.org/doc/)¹³ for more details). `hpx::init` calls `hpx_main()` after setting up *HPX*, which is where the logic of our program is encoded.

```
int main(int argc, char* argv[])
{
    // Configure application-specific options
    hpx::program_options::options_description desc_commandline(
        "Usage: " HPX_APPLICATION_STRING " [options]");

    // clang-format off
    desc_commandline.add_options()
        ("n-value",
         hpx::program_options::value<std::uint64_t>()->default_value(10),
         "n value for the Fibonacci function")
        ;
    // clang-format on

    // Initialize and run HPX
    hpx::local::init_params init_args;
    init_args.desc_cmdline = desc_commandline;

    return hpx::local::init(hpx_main, argc, argv, init_args);
}
```

The `hpx::init` function in `main()` starts the runtime system, and invokes `hpx_main()` as the first *HPX*-thread. Below we can see that the basic program is simple. The command line option `--n-value` is read in, a timer (`hpx::chrono::high_resolution_timer`) is set up to record the time it takes to do the computation, the `fibonacci` function is invoked synchronously, and the answer is printed out.

¹² https://www.boost.org/doc/html/program_options.html

¹³ <https://www.boost.org/doc/>

```
int hpx_main(hpx::program_options::variables_map& vm)
{
    hpx::threads::add_scheduler_mode(
        hpx::threads::policies::scheduler_mode::fast_idle_mode);

    // extract command line argument, i.e. fib(N)
    std::uint64_t n = vm["n-value"].as<std::uint64_t>();

    {
        // Keep track of the time required to execute.
        hpx::chrono::high_resolution_timer t;

        std::uint64_t r = fibonacci(n);

        char const* fmt = "fibonacci({1}) == {2}\\nelapsed time: {3} [s]\\n";
        hpx::util::format_to(std::cout, fmt, n, r, t.elapsed());
    }

    return hpx::local::finalize();    // Handles HPX shutdown
}
```

The `fibonacci` function itself is synchronous as the work done inside is asynchronous. To understand what is happening we have to look inside the `fibonacci` function:

```
std::uint64_t fibonacci(std::uint64_t n)
{
    if (n < 2)
        return n;

    hpx::future<std::uint64_t> n1 = hpx::async(fibonacci, n - 1);
    std::uint64_t n2 = fibonacci(n - 2);

    return n1.get() + n2;    // wait for the Future to return their values
}
```

This block of code looks similar to regular C++ code. First, if `(n < 2)`, meaning `n` is 0 or 1, then we return 0 or 1 (recall the first element of the Fibonacci sequence is 0 and the second is 1). If `n` is larger than 1 we spawn two new tasks whose results are contained in `n1` and `n2`. This is done using `hpx::async` which takes as arguments a function (function pointer, object or lambda) and the arguments to the function. Instead of returning a `std::uint64_t` like `fibonacci` does, `hpx::async` returns a future of a `std::uint64_t`, i.e. `hpx::future<std::uint64_t>`. Each of these futures represents an asynchronous, recursive call to `fibonacci`. After we've created the futures, we wait for both of them to finish computing, we add them together, and return that value as our result. We get the values from the futures using the `get` method. The recursive call tree will continue until `n` is equal to 0 or 1, at which point the value can be returned because it is implicitly known. When this termination condition is reached, the futures can then be added up, producing the `n`-th value of the Fibonacci sequence.

Note that calling `get` potentially blocks the calling `HPX`-thread, and lets other `HPX`-threads run in the meantime. There are, however, more efficient ways of doing this. `examples/quickstart/fibonacci_futures.cpp` contains many more variations of locally computing the Fibonacci numbers, where each method makes different tradeoffs in where asynchrony and parallelism is applied. To get started, however, the method above is sufficient and optimizations can be applied once you are more familiar with `HPX`. The example *Dataflow* presents dataflow, which is a way to more efficiently chain together multiple tasks.

2.3.2 Parallel algorithms

This program will perform a matrix multiplication in parallel. The output will look something like this:

```
Matrix A is :
4 9 6
1 9 8

Matrix B is :
4 9
6 1
9 8

Resultant Matrix is :
124 93
130 82
```

Setup

The source code for this example can be found here: `matrix_multiplication.cpp`.

To compile this program, go to your *HPX* build directory (see *Building HPX* for information on configuring and building *HPX*) and enter:

```
$ make examples.quickstart.matrix_multiplication
```

To run the program type:

```
$ ./bin/matrix_multiplication
```

or:

```
$ ./bin/matrix_multiplication --n 2 --m 3 --k 2 --s 100 --l 0 --u 10
```

where the first matrix is $n \times m$ and the second $m \times k$, s is the seed for creating the random values of the matrices and the range of these values is $[l, u]$

This should print:

```
Matrix A is :
4 9 6
1 9 8

Matrix B is :
4 9
6 1
9 8

Resultant Matrix is :
124 93
130 82
```

Notice that the numbers may be different because of the random initialization of the matrices.

Walkthrough

Now that you have compiled and run the code, let's look at how the code works.

First, `main()` is used to initialize the runtime system and pass the command line arguments to the program. `hpx::init` calls `hpx_main()` after setting up HPX, which is where our program is implemented.

```
int main(int argc, char* argv[])
{
    using namespace hpx::program_options;
    options_description cmdline("usage: " HPX_APPLICATION_STRING " [options]");
    // clang-format off
    cmdline.add_options()
        ("n",
         hpx::program_options::value<std::size_t>()->default_value(2),
         "Number of rows of first matrix")
        ("m",
         hpx::program_options::value<std::size_t>()->default_value(3),
         "Number of columns of first matrix (equal to the number of rows of "
         "second matrix)")
        ("k",
         hpx::program_options::value<std::size_t>()->default_value(2),
         "Number of columns of second matrix")
        ("seed,s",
         hpx::program_options::value<unsigned int>(),
         "The random number generator seed to use for this run")
        ("l",
         hpx::program_options::value<int>()->default_value(0),
         "Lower limit of range of values")
        ("u",
         hpx::program_options::value<int>()->default_value(10),
         "Upper limit of range of values");
    // clang-format on
    hpx::local::init_params init_args;
    init_args.desc_cmdline = cmdline;

    return hpx::local::init(hpx_main, argc, argv, init_args);
}
```

Proceeding to the `hpx_main()` function, we can see that matrix multiplication can be done very easily.

```
int hpx_main(hpx::program_options::variables_map& vm)
{
    using element_type = int;

    // Define matrix sizes
    std::size_t const rowsA = vm["n"].as<std::size_t>();
    std::size_t const colsA = vm["m"].as<std::size_t>();
    std::size_t const rowsB = colsA;
    std::size_t const colsB = vm["k"].as<std::size_t>();
    std::size_t const rowsR = rowsA;
    std::size_t const colsR = colsB;

    // Initialize matrices A and B
```

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```

std::vector<int> A(rowsA * colsA);
std::vector<int> B(rowsB * colsB);
std::vector<int> R(rowsR * colsR);

// Define seed
unsigned int seed = std::random_device{}();
if (vm.count("seed"))
    seed = vm["seed"].as<unsigned int>();

gen.seed(seed);
std::cout << "using seed: " << seed << std::endl;

// Define range of values
int const lower = vm["l"].as<int>();
int const upper = vm["u"].as<int>();

// Matrices have random values in the range [lower, upper]
std::uniform_int_distribution<element_type> dis(lower, upper);
auto generator = std::bind(dis, gen);
hpx::ranges::generate(A, generator);
hpx::ranges::generate(B, generator);

// Perform matrix multiplication
hpx::experimental::for_loop(hpx::execution::par, 0, rowsA, [&](auto i) {
    hpx::experimental::for_loop(0, colsB, [&](auto j) {
        R[i * colsR + j] = 0;
        hpx::experimental::for_loop(0, rowsB, [&](auto k) {
            R[i * colsR + j] += A[i * colsA + k] * B[k * colsB + j];
        });
    });
});

// Print all 3 matrices
print_matrix(A, rowsA, colsA, "A");
print_matrix(B, rowsB, colsB, "B");
print_matrix(R, rowsR, colsR, "R");

return hpx::local::finalize();
}

```

First, the dimensions of the matrices are defined. If they were not given as command-line arguments, their default values are 2 x 3 for the first matrix and 3 x 2 for the second. We use standard vectors to define the matrices to be multiplied as well as the resultant matrix.

To give some random initial values to our matrices, we use `std::uniform_int_distribution`¹⁴. Then, `std::bind()` is used along with `hpx::ranges::generate()` to yield two matrices A and B, which contain values in the range of [0, 10] or in the range defined by the user at the command-line arguments. The seed to generate the values can also be defined by the user.

The next step is to perform the matrix multiplication in parallel. This can be done by just using an `hpx::experimental::for_loop` combined with a parallel execution policy `hpx::execution::par` as the outer loop of the multiplication. Note that the execution of `hpx::experimental::for_loop` without specifying an execu-

¹⁴ https://en.cppreference.com/w/cpp/numeric/random/uniform_int_distribution

tion policy is equivalent to specifying `hpx::execution::seq` as the execution policy.

Finally, the matrices A, B that are multiplied as well as the resultant matrix R are printed using the following function.

```
void print_matrix(std::vector<int> const& M, std::size_t rows, std::size_t cols,
    char const* message)
{
    std::cout << "\nMatrix " << message << " is:" << std::endl;
    for (std::size_t i = 0; i < rows; i++)
    {
        for (std::size_t j = 0; j < cols; j++)
            std::cout << M[i * cols + j] << " ";
        std::cout << "\n";
    }
}
```

2.3.3 Asynchronous execution with actions

This example extends the *previous example* by introducing *actions*: functions that can be run remotely. In this example, however, we will still only run the action locally. The mechanism to execute *actions* stays the same: `hpx::async`. Later examples will demonstrate running actions on remote *localities* (e.g. *Remote execution with actions*).

Setup

The source code for this example can be found here: `fibonacci.cpp`.

To compile this program, go to your *HPX* build directory (see *Building HPX* for information on configuring and building *HPX*) and enter:

```
$ make examples.quickstart.fibonacci
```

To run the program type:

```
$ ./bin/fibonacci
```

This should print (time should be approximate):

```
fibonacci(10) == 55
elapsed time: 0.00186288 [s]
```

This run used the default settings, which calculate the tenth element of the Fibonacci sequence. To declare which Fibonacci value you want to calculate, use the `--n-value` option. Additionally you can use the `--hpx:threads` option to declare how many OS-threads you wish to use when running the program. For instance, running:

```
$ ./bin/fibonacci --n-value 20 --hpx:threads 4
```

Will yield:

```
fibonacci(20) == 6765
elapsed time: 0.233827 [s]
```

Walkthrough

The code needed to initialize the *HPX* runtime is the same as in the *previous example*:

```
int main(int argc, char* argv[])
{
    // Configure application-specific options
    hpx::program_options::options_description desc_commandline(
        "Usage: " HPX_APPLICATION_STRING " [options]");

    desc_commandline.add_options()("n-value",
        hpx::program_options::value<std::uint64_t>()->default_value(10),
        "n value for the Fibonacci function");

    // Initialize and run HPX
    hpx::init_params init_args;
    init_args.desc_cmdline = desc_commandline;

    return hpx::init(argc, argv, init_args);
}
```

The `hpx::init` function in `main()` starts the runtime system, and invokes `hpx_main()` as the first *HPX*-thread. The command line option `--n-value` is read in, a timer (`hpx::chrono::high_resolution_timer`) is set up to record the time it takes to do the computation, the *fibonacci action* is invoked synchronously, and the answer is printed out.

```
int hpx_main(hpx::program_options::variables_map& vm)
{
    // extract command line argument, i.e. fib(N)
    std::uint64_t n = vm["n-value"].as<std::uint64_t>();

    {
        // Keep track of the time required to execute.
        hpx::chrono::high_resolution_timer t;

        // Wait for fib() to return the value
        fibonacci_action fib;
        std::uint64_t r = fib(hpx::find_here(), n);

        char const* fmt = "fibonacci({1}) == {2}\\nelapsed time: {3} [s]\\n";
        hpx::util::format_to(std::cout, fmt, n, r, t.elapsed());
    }

    return hpx::finalize();    // Handles HPX shutdown
}
```

Upon a closer look we see that we've created a `std::uint64_t` to store the result of invoking our *fibonacci_action* `fib`. This *action* will launch synchronously (as the work done inside of the *action* will be asynchronous itself) and return the result of the Fibonacci sequence. But wait, what is an *action*? And what is this *fibonacci_action*? For starters, an *action* is a wrapper for a function. By wrapping functions, *HPX* can send packets of work to different processing units. These vehicles allow users to calculate work now, later, or on certain nodes. The first argument to our *action* is the location where the *action* should be run. In this case, we just want to run the *action* on the machine that we are currently on, so we use `hpx::find_here`. To further understand this we turn to the code to find where *fibonacci_action* was defined:

```
// forward declaration of the Fibonacci function
std::uint64_t fibonacci(std::uint64_t n);

// This is to generate the required boilerplate we need for the remote
// invocation to work.
HPX_PLAIN_ACTION(fibonacci, fibonacci_action)
```

A plain *action* is the most basic form of *action*. Plain *actions* wrap simple global functions which are not associated with any particular object (we will discuss other types of *actions* in *Components and actions*). In this block of code the function `fibonacci()` is declared. After the declaration, the function is wrapped in an *action* in the declaration `HPX_PLAIN_ACTION`. This function takes two arguments: the name of the function that is to be wrapped and the name of the *action* that you are creating.

This picture should now start making sense. The function `fibonacci()` is wrapped in an *action* `fibonacci_action`, which was run synchronously but created asynchronous work, then returns a `std::uint64_t` representing the result of the function `fibonacci()`. Now, let's look at the function `fibonacci()`:

```
std::uint64_t fibonacci(std::uint64_t n)
{
    if (n < 2)
        return n;

    // We restrict ourselves to execute the Fibonacci function locally.
    hpx::id_type const locality_id = hpx::find_here();

    // Invoking the Fibonacci algorithm twice is inefficient.
    // However, we intentionally demonstrate it this way to create some
    // heavy workload.

    fibonacci_action fib;
    hpx::future<std::uint64_t> n1 = hpx::async(fib, locality_id, n - 1);
    hpx::future<std::uint64_t> n2 = hpx::async(fib, locality_id, n - 2);

    return n1.get() +
        n2.get();    // wait for the Futures to return their values
}
```

This block of code is much more straightforward and should look familiar from the *previous example*. First, if `(n < 2)`, meaning `n` is 0 or 1, then we return 0 or 1 (recall the first element of the Fibonacci sequence is 0 and the second is 1). If `n` is larger than 1 we spawn two tasks using `hpx::async`. Each of these futures represents an asynchronous, recursive call to `fibonacci`. As previously we wait for both futures to finish computing, get the results, add them together, and return that value as our result. The recursive call tree will continue until `n` is equal to 0 or 1, at which point the value can be returned because it is implicitly known. When this termination condition is reached, the futures can then be added up, producing the `n`-th value of the Fibonacci sequence.

2.3.4 Remote execution with actions

This program will print out a hello world message on every OS-thread on every *locality*. The output will look something like this:

```
hello world from OS-thread 1 on locality 0
hello world from OS-thread 1 on locality 1
hello world from OS-thread 0 on locality 0
hello world from OS-thread 0 on locality 1
```

Setup

The source code for this example can be found here: `hello_world_distributed.cpp`.

To compile this program, go to your *HPX* build directory (see *Building HPX* for information on configuring and building *HPX*) and enter:

```
$ make examples.quickstart.hello_world_distributed
```

To run the program type:

```
$ ./bin/hello_world_distributed
```

This should print:

```
hello world from OS-thread 0 on locality 0
```

To use more OS-threads use the command line option `--hpx:threads` and type the number of threads that you wish to use. For example, typing:

```
$ ./bin/hello_world_distributed --hpx:threads 2
```

will yield:

```
hello world from OS-thread 1 on locality 0
hello world from OS-thread 0 on locality 0
```

Notice how the ordering of the two print statements will change with subsequent runs. To run this program on multiple localities please see the section *How to use HPX applications with PBS*.

Walkthrough

Now that you have compiled and run the code, let's look at how the code works, beginning with `main()`:

```
// Here is the main entry point. By using the include 'hpx/hpx_main.hpp' HPX
// will invoke the plain old C-main() as its first HPX thread.
int main()
{
    // Get a list of all available localities.
    std::vector<hpx::id_type> localities = hpx::find_all_localities();

    // Reserve storage space for futures, one for each locality.
    std::vector<hpx::future<void>> futures;
```

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```

futures.reserve(localities.size());

for (hpx::id_type const& node : localities)
{
    // Asynchronously start a new task. The task is encapsulated in a
    // future, which we can query to determine if the task has
    // completed.
    typedef hello_world_foreman_action action_type;
    futures.push_back(hpx::async<action_type>(node));
}

// The non-callback version of hpx::wait_all takes a single parameter,
// a vector of futures to wait on. hpx::wait_all only returns when
// all of the futures have finished.
hpx::wait_all(futures);
return 0;
}

```

In this excerpt of the code we again see the use of futures. This time the futures are stored in a vector so that they can easily be accessed. `hpx::wait_all` is a family of functions that wait on for an `std::vector<>` of futures to become ready. In this piece of code, we are using the synchronous version of `hpx::wait_all`, which takes one argument (the `std::vector<>` of futures to wait on). This function will not return until all the futures in the vector have been executed.

In *Asynchronous execution with actions* we used `hpx::find_here` to specify the target of our actions. Here, we instead use `hpx::find_all_localities`, which returns an `std::vector<>` containing the identifiers of all the machines in the system, including the one that we are on.

As in *Asynchronous execution with actions* our futures are set using `hpx::async<>`. The `hello_world_foreman_action` is declared here:

```

// Define the boilerplate code necessary for the function 'hello_world_foreman'
// to be invoked as an HPX action.
HPX_PLAIN_ACTION(hello_world_foreman, hello_world_foreman_action)

```

Another way of thinking about this wrapping technique is as follows: functions (the work to be done) are wrapped in actions, and actions can be executed locally or remotely (e.g. on another machine participating in the computation).

Now it is time to look at the `hello_world_foreman()` function which was wrapped in the action above:

```

void hello_world_foreman()
{
    // Get the number of worker OS-threads in use by this locality.
    std::size_t const os_threads = hpx::get_os_thread_count();

    // Populate a set with the OS-thread numbers of all OS-threads on this
    // locality. When the hello world message has been printed on a particular
    // OS-thread, we will remove it from the set.
    std::set<std::size_t> attendance;
    for (std::size_t os_thread = 0; os_thread < os_threads; ++os_thread)
        attendance.insert(os_thread);

    // As long as there are still elements in the set, we must keep scheduling
    // HPX-threads. Because HPX features work-stealing task schedulers, we have

```

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```

// no way of enforcing which worker OS-thread will actually execute
// each HPX-thread.
while (!attendance.empty())
{
    // Each iteration, we create a task for each element in the set of
    // OS-threads that have not said "Hello world". Each of these tasks
    // is encapsulated in a future.
    std::vector<hpx::future<std::size_t>> futures;
    futures.reserve(attendance.size());

    for (std::size_t worker : attendance)
    {
        // Asynchronously start a new task. The task is encapsulated in a
        // future that we can query to determine if the task has completed.
        //
        // We give the task a hint to run on a particular worker thread
        // (core) and suggest binding the scheduled thread to the given
        // core, but no guarantees are given by the scheduler that the task
        // will actually run on that worker thread. It will however try as
        // hard as possible to place the new task on the given worker
        // thread.
        hpx::execution::parallel_executor exec(
            hpx::threads::thread_priority::bound);

        hpx::threads::thread_schedule_hint hint(
            hpx::threads::thread_schedule_hint_mode::thread,
            static_cast<std::int16_t>(worker));

        // Annotate the created threads with the given thread name
        hpx::execution::experimental::annotating_executor annotating_exec(
            hpx::execution::experimental::with_hint(exec, hint),
            std::string("hello_world_worker#") + std::to_string(worker));

        futures.push_back(
            hpx::async(annotating_exec, hello_world_worker, worker));
    }

    // Wait for all of the futures to finish. The callback version of the
    // hpx::wait_each function takes two arguments: a vector of futures,
    // and a binary callback. The callback takes two arguments; the first
    // is the index of the future in the vector, and the second is the
    // return value of the future. hpx::wait_each doesn't return until
    // all the futures in the vector have returned.
    hpx::spinlock mtx;
    hpx::wait_each(hpx::unwrapping([&](std::size_t t) {
        if (std::size_t(-1) != t)
        {
            std::lock_guard<hpx::spinlock> lk(mtx);
            attendance.erase(t);
        }
    })),
        futures);

```

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```

    }
}

```

Now, before we discuss `hello_world_foreman()`, let's talk about the `hpx::wait_each` function. The version of `hpx::wait_each` invokes a callback function provided by the user, supplying the callback function with the result of the future.

In `hello_world_foreman()`, an `std::set<>` called `attendance` keeps track of which OS-threads have printed out the hello world message. When the OS-thread prints out the statement, the future is marked as ready, and `hpx::wait_each` in `hello_world_foreman()`. If it is not executing on the correct OS-thread, it returns a value of -1, which causes `hello_world_foreman()` to leave the OS-thread id in `attendance`.

```

std::size_t hello_world_worker(std::size_t desired)
{
    // Returns the OS-thread number of the worker that is running this
    // HPX-thread.
    std::size_t current = hpx::get_worker_thread_num();
    if (current == desired)
    {
        // The HPX-thread has been run on the desired OS-thread.
        char const* msg = "hello world from OS-thread {1} on locality {2}\n";

        hpx::util::format_to(hpx::cout, msg, desired, hpx::get_locality_id())
            << std::flush;

        return desired;
    }

    // This HPX-thread has been run by the wrong OS-thread, make the foreman
    // try again by rescheduling it.
    return std::size_t(-1);
}

```

Because HPX features work stealing task schedulers, there is no way to guarantee that an action will be scheduled on a particular OS-thread. This is why we must use a guess-and-check approach.

2.3.5 Components and actions

The accumulator examples demonstrate the use of components. Components are C++ classes that expose methods as a type of HPX action. These actions are called component actions. There are three examples: - accumulator - template accumulator - template function accumulator

Components are globally named, meaning that a component action can be called remotely (e.g., from another machine). There are two accumulator examples in HPX.

In the *Asynchronous execution with actions* and the *Remote execution with actions*, we introduced plain actions, which wrapped global functions. The target of a plain action is an identifier which refers to a particular machine involved in the computation. For plain actions, the target is the machine where the action will be executed.

Component actions, however, do not target machines. Instead, they target component instances. The instance may live on the machine that we've invoked the component action from, or it may live on another machine.

The components in these examples expose three different functions:

- `reset()` - Resets the accumulator value to 0.

- `add(arg)` - Adds `arg` to the accumulators value.
- `query()` - Queries the value of the accumulator.

These examples create an instance of the (template or template function) accumulator, and then allow the user to enter commands at a prompt, which subsequently invoke actions on the accumulator instance.

Accumulator

Setup

The source code for this example can be found here: `accumulator_client.cpp`.

To compile this program, go to your *HPX* build directory (see *Building HPX* for information on configuring and building *HPX*) and enter:

```
$ make examples.accumulators.accumulator
```

To run the program type:

```
$ ./bin/accumulator_client
```

Once the program starts running, it will print the following prompt and then wait for input. An example session is given below:

```
commands: reset, add [amount], query, help, quit
> add 5
> add 10
> query
15
> add 2
> query
17
> reset
> add 1
> query
1
> quit
```

Walkthrough

Now, let's take a look at the source code of the accumulator example. This example consists of two parts: an *HPX* component library (a library that exposes an *HPX* component) and a client application which uses the library. This walkthrough will cover the *HPX* component library. The code for the client application can be found here: `accumulator_client.cpp`.

An *HPX* component is represented by two C++ classes:

- **A server class** - The implementation of the component's functionality.
- **A client class** - A high-level interface that acts as a proxy for an instance of the component.

Typically, these two classes both have the same name, but the server class usually lives in different sub-namespaces (server). For example, the full names of the two classes in accumulator are:

- `examples::server::accumulator` (server class)

- `examples::accumulator` (client class)

The server class

The following code is from `server/accumulator.hpp`¹⁵.

All *HPX* component server classes must inherit publicly from the *HPX* component base class: `hpx::components::component_base`

The accumulator component inherits from `hpx::components::locking_hook`. This allows the runtime system to ensure that all action invocations are serialized. That means that the system ensures that no two actions are invoked at the same time on a given component instance. This makes the component thread safe and no additional locking has to be implemented by the user. Moreover, an accumulator component is a component because it also inherits from `hpx::components::component_base` (the template argument passed to `locking_hook` is used as its base class). The following snippet shows the corresponding code:

```
class accumulator
: public hpx::components::locking_hook<
    hpx::components::component_base<accumulator>>
```

Our accumulator class will need a data member to store its value in, so let's declare a data member:

```
argument_type value_;
```

The constructor for this class simply initializes `value_` to 0:

```
accumulator()
: value_(0)
{
}
```

Next, let's look at the three methods of this component that we will be exposing as component actions:

Here are the action types. These types wrap the methods we're exposing. The wrapping technique is very similar to the one used in the *Asynchronous execution with actions* and the *Remote execution with actions*:

```
HPX_DEFINE_COMPONENT_ACTION(accumulator, reset)
HPX_DEFINE_COMPONENT_ACTION(accumulator, add)
HPX_DEFINE_COMPONENT_ACTION(accumulator, query)
```

The last piece of code in the server class header is the declaration of the action type registration code:

```
HPX_REGISTER_ACTION_DECLARATION(
    examples::server::accumulator::reset_action, accumulator_reset_action)

HPX_REGISTER_ACTION_DECLARATION(
    examples::server::accumulator::add_action, accumulator_add_action)

HPX_REGISTER_ACTION_DECLARATION(
    examples::server::accumulator::query_action, accumulator_query_action)
```

Note: The code above must be placed in the global namespace.

¹⁵ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fdbc0c8044e7/examples/accumulators/server/accumulator.hpp>

The rest of the registration code is in `accumulator.cpp`¹⁶

```

////////////////////////////////////
// Add factory registration functionality.
HPX_REGISTER_COMPONENT_MODULE()

////////////////////////////////////
typedef hpx::components::component<examples::server::accumulator>
    accumulator_type;

HPX_REGISTER_COMPONENT(accumulator_type, accumulator)

////////////////////////////////////
// Serialization support for accumulator actions.
HPX_REGISTER_ACTION(
    accumulator_type::wrapped_type::reset_action, accumulator_reset_action)
HPX_REGISTER_ACTION(
    accumulator_type::wrapped_type::add_action, accumulator_add_action)
HPX_REGISTER_ACTION(
    accumulator_type::wrapped_type::query_action, accumulator_query_action)

```

Note: The code above must be placed in the global namespace.

The client class

The following code is from `accumulator.hpp`¹⁷

The client class is the primary interface to a component instance. Client classes are used to create components:

```

// Create a component on this locality.
examples::accumulator c = hpx::new<examples::accumulator>(hpx::find_here());

```

and to invoke component actions:

```
c.add(hpx::launch::apply, 4);
```

Clients, like servers, need to inherit from a base class, this time, `hpx::components::client_base`:

```

class accumulator
: public hpx::components::client_base<accumulator, server::accumulator>

```

For readability, we typedef the base class like so:

```

typedef hpx::components::client_base<accumulator, server::accumulator>
    base_type;

```

Here are examples of how to expose actions through a client class:

There are a few different ways of invoking actions:

¹⁶ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/examples/accumulators/accumulator.cpp>

¹⁷ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/examples/accumulators/accumulator.hpp>

- **Non-blocking:** For actions that don't have return types, or when we do not care about the result of an action, we can invoke the action using fire-and-forget semantics. This means that once we have asked *HPX* to compute the action, we forget about it completely and continue with our computation. We use `hpx::post` to invoke an action in a non-blocking fashion.

```
void reset(hpx::launch::apply_policy)
{
    HPX_ASSERT(this->get_id());

    typedef server::accumulator::reset_action action_type;
    hpx::post(action_type(), this->get_id());
}
```

- **Asynchronous:** Futures, as demonstrated in *Asynchronous execution*, *Asynchronous execution with actions*, and the *Remote execution with actions*, enable asynchronous action invocation. Here's an example from the accumulator client class:

```
hpx::future<argument_type> query(hpx::launch::async_policy)
{
    HPX_ASSERT(this->get_id());

    typedef server::accumulator::query_action action_type;
    return hpx::async(action_type(), this->get_id());
}
```

- **Synchronous:** To invoke an action in a fully synchronous manner, we can simply call `hpx::sync` which is semantically equivalent to `hpx::async().get()` (i.e., create a future and immediately wait on it to be ready). Here's an example from the accumulator client class:

```
void add(argument_type arg)
{
    HPX_ASSERT(this->get_id());

    typedef server::accumulator::add_action action_type;
    action_type()(this->get_id(), arg);
}
```

Note that `this->get_id()` references a data member of the `hpx::components::client_base` base class which identifies the server accumulator instance.

`hpx::id_type` is a type which represents a global identifier in *HPX*. This type specifies the target of an action. This is the type that is returned by `hpx::find_here` in which case it represents the *locality* the code is running on.

Template accumulator

Walkthrough

The server class

The following code is from `server/template_accumulator.hpp`¹⁸.

¹⁸ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/examples/accumulators/server/template_accumulator.hpp

Similarly to the accumulator example, the component server class inherits publicly from `hpx::components::component_base` and from `hpx::components::locking_hook` ensuring thread-safe method invocations.

```
template <typename T>
class template_accumulator
: public hpx::components::locking_hook<
    hpx::components::component_base<template_accumulator<T>>>
```

The body of the template accumulator class remains mainly the same as the accumulator with the difference that it uses templates in the data types.

```
typedef T argument_type;

template_accumulator()
: value_(0)
{
}

////////////////////////////////////
// Exposed functionality of this component.

/// Reset the components value to 0.
void reset()
{
    // set value_ to 0.
    value_ = 0;
}

/// Add the given number to the accumulator.
void add(argument_type arg)
{
    // add value_ to arg, and store the result in value_.
    value_ += arg;
}

/// Return the current value to the caller.
argument_type query() const
{
    // Get the value of value_.
    return value_;
}

////////////////////////////////////
// Each of the exposed functions needs to be encapsulated into an
// action type, generating all required boilerplate code for threads,
// serialization, etc.
HPX_DEFINE_COMPONENT_ACTION(template_accumulator, reset)
HPX_DEFINE_COMPONENT_ACTION(template_accumulator, add)
HPX_DEFINE_COMPONENT_ACTION(template_accumulator, query)
```

The last piece of code in the server class header is the declaration of the action type registration code. `REGISTER_TEMPLATE_ACCUMULATOR_DECLARATION(type)` declares actions for the specified type, while `REGISTER_TEMPLATE_ACCUMULATOR(type)` registers the actions and the component for the specified type, using macros

to handle boilerplate code.

```
#define REGISTER_TEMPLATE_ACCUMULATOR_DECLARATION(type)      \
    HPX_REGISTER_ACTION_DECLARATION(                          \
        examples::server::template_accumulator<type>::reset_action, \
        HPX_PP_CAT(__template_accumulator_reset_action_, type)) \
                                                                    \
    HPX_REGISTER_ACTION_DECLARATION(                          \
        examples::server::template_accumulator<type>::add_action, \
        HPX_PP_CAT(__template_accumulator_add_action_, type)) \
                                                                    \
    HPX_REGISTER_ACTION_DECLARATION(                          \
        examples::server::template_accumulator<type>::query_action, \
        HPX_PP_CAT(__template_accumulator_query_action_, type)) \
                                                                    \
    /**/

#define REGISTER_TEMPLATE_ACCUMULATOR(type)                  \
    HPX_REGISTER_ACTION(                                       \
        examples::server::template_accumulator<type>::reset_action, \
        HPX_PP_CAT(__template_accumulator_reset_action_, type)) \
                                                                    \
    HPX_REGISTER_ACTION(                                       \
        examples::server::template_accumulator<type>::add_action, \
        HPX_PP_CAT(__template_accumulator_add_action_, type)) \
                                                                    \
    HPX_REGISTER_ACTION(                                       \
        examples::server::template_accumulator<type>::query_action, \
        HPX_PP_CAT(__template_accumulator_query_action_, type)) \
                                                                    \
    typedef ::hpx::components::component<                    \
        examples::server::template_accumulator<type>>         \
        HPX_PP_CAT(__template_accumulator_, type);           \
    HPX_REGISTER_COMPONENT(HPX_PP_CAT(__template_accumulator_, type)) \
                                                                    \
    /**/
```

Note: The code above must be placed in the global namespace.

Finally, `HPX_REGISTER_COMPONENT_MODULE()` in file `server/template_accumulator.cpp`¹⁹ adds the factory registration functionality.

¹⁹ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/examples/accumulators/server/template_accumulator.cpp

The client class

The client class of the template accumulator can be found in `template_accumulator.hpp`²⁰ and is very similar to the client class of the accumulator with the only difference that it uses templates and hence can work with different types.

Template function accumulator

Walkthrough

The server class

The following code is from `server/template_function_accumulator.hpp`²¹.

The component server class inherits publicly from `hpx::components::component_base`.

```
class template_function_accumulator
: public hpx::components::component_base<template_function_accumulator>
```

`typedef hpx::spinlock mutex_type` defines a `mutex_type` as `hpx::spinlock` for thread safety, while the code that follows exposes the functionality of this component.

```
////////////////////////////////////
// Exposed functionality of this component.

/// Reset the value to 0.
void reset()
{
    // Atomically set value_ to 0.
    std::lock_guard<mutex_type> l(mtx_);
    value_ = 0;
}

/// Add the given number to the accumulator.
template <typename T>
void add(T arg)
{
    // Atomically add value_ to arg, and store the result in value_.
    std::lock_guard<mutex_type> l(mtx_);
    value_ += static_cast<double>(arg);
}

/// Return the current value to the caller.
double query() const
{
    // Get the value of value_.
    std::lock_guard<mutex_type> l(mtx_);
    return value_;
}
```

²⁰ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/examples/accumulators/template_accumulator.hpp

²¹ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/examples/accumulators/server/template_function_accumulator.hpp

- `reset()`: Resets the accumulator value to 0 in a thread-safe manner using `std::lock_guard`.
- `add()`: Adds a value to the accumulator, allowing any type `T` that can be cast to double.
- `query()`: Returns the current value of the accumulator in a thread-safe manner.

To define the actions for `reset()` and `query()` we can use the macro `HPX_DEFINE_COMPONENT_ACTION`. However, actions with template arguments require special type definitions. Therefore, we use `make_action()` to define `add()`.

```

////////////////////////////////////
// Each of the exposed functions needs to be encapsulated into an
// action type, generating all required boilerplate code for threads,
// serialization, etc.

HPX_DEFINE_COMPONENT_ACTION(template_function_accumulator, reset)
HPX_DEFINE_COMPONENT_ACTION(template_function_accumulator, query)

// Actions with template arguments (see add<>() above) require special
// type definitions. The simplest way to define such an action type is
// by deriving from the HPX facility make_action.
template <typename T>
struct add_action
    : hpx::actions::make_action<void> (template_function_accumulator::*)(
        T),
        &template_function_accumulator::template add<T>,
        add_action<T>::type
{
};

```

The last piece of code in the server class header is the action registration:

```

HPX_REGISTER_ACTION_DECLARATION(
    examples::server::template_function_accumulator::reset_action,
    managed_accumulator_reset_action)

HPX_REGISTER_ACTION_DECLARATION(
    examples::server::template_function_accumulator::query_action,
    managed_accumulator_query_action)

```

Note: The code above must be placed in the global namespace.

The rest of the registration code is in `accumulator.cpp`²²

```

////////////////////////////////////
// Add factory registration functionality.
HPX_REGISTER_COMPONENT_MODULE()

////////////////////////////////////
typedef hpx::components::component<
    examples::server::template_function_accumulator>
    accumulator_type;

```

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²² <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/examples/accumulators/accumulator.cpp>

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```

HPX_REGISTER_COMPONENT(accumulator_type, template_function_accumulator)

////////////////////////////////////
// Serialization support for managed_accumulator actions.
HPX_REGISTER_ACTION(accumulator_type::wrapped_type::reset_action,
    managed_accumulator_reset_action)
HPX_REGISTER_ACTION(accumulator_type::wrapped_type::query_action,
    managed_accumulator_query_action)

```

Note: The code above must be placed in the global namespace.

The client class

The client class of the template accumulator can be found in `template_function_accumulator.hpp`²³ and is very similar to the client class of the accumulator with the only difference that it uses templates and hence can work with different types.

2.3.6 Dataflow

HPX provides its users with several different tools to simply express parallel concepts. One of these tools is a *local control object (LCO)* called dataflow. An LCO is a type of component that can spawn a new thread when triggered. They are also distinguished from other components by a standard interface that allow users to understand and use them easily. A Dataflow, being an LCO, is triggered when the values it depends on become available. For instance, if you have a calculation X that depends on the results of three other calculations, you could set up a dataflow that would begin the calculation X as soon as the other three calculations have returned their values. Dataflows are set up to depend on other dataflows. It is this property that makes dataflow a powerful parallelization tool. If you understand the dependencies of your calculation, you can devise a simple algorithm that sets up a dependency tree to be executed. In this example, we calculate compound interest. To calculate compound interest, one must calculate the interest made in each compound period, and then add that interest back to the principal before calculating the interest made in the next period. A practical person would, of course, use the formula for compound interest:

$$F = P(1 + i)^n$$

where F is the future value, P is the principal value, i is the interest rate, and n is the number of compound periods.

However, for the sake of this example, we have chosen to manually calculate the future value by iterating:

$$I = Pi$$

and

$$P = P + I$$

²³ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/examples/accumulators/template_function_accumulator.hpp

Setup

The source code for this example can be found here: `interest_calculator.cpp`.

To compile this program, go to your *HPX* build directory (see *Building HPX* for information on configuring and building *HPX*) and enter:

```
$ make examples.quickstart.interest_calculator
```

To run the program type:

```
$ ./bin/interest_calculator --principal 100 --rate 5 --cp 6 --time 36
Final amount: 134.01
Amount made: 34.0096
```

Walkthrough

Let us begin with `main`. Here we can see that we again are using `Boost.Program_options` to set our command line variables (see *Asynchronous execution with actions* for more details). These options set the principal, rate, compound period, and time. It is important to note that the units of time for `cp` and `time` must be the same.

```
int main(int argc, char** argv)
{
    options_description cmdline("Usage: " HPX_APPLICATION_STRING " [options]");

    cmdline.add_options()("principal", value<double>()->default_value(1000),
        "The principal [$]")("rate", value<double>()->default_value(7),
        "The interest rate [%]")("cp", value<int>()->default_value(12),
        "The compound period [months]")("time",
        value<int>()->default_value(12 * 30),
        "The time money is invested [months]");

    hpx::init_params init_args;
    init_args.desc_cmdline = cmdline;

    return hpx::init(argc, argv, init_args);
}
```

Next we look at `hpx_main`.

```
int hpx_main(variables_map& vm)
{
    {
        using hpx::dataflow;
        using hpx::make_ready_future;
        using hpx::shared_future;
        using hpx::unwrapping;
        hpx::id_type here = hpx::find_here();

        double init_principal =
            vm["principal"].as<double>();           //Initial principal
        double init_rate = vm["rate"].as<double>(); //Interest rate
        int cp = vm["cp"].as<int>();                 //Length of a compound period
```

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```

int t = vm["time"].as<int>();    //Length of time money is invested

init_rate /= 100;    //Rate is a % and must be converted
t /= cp;    //Determine how many times to iterate interest calculation:
//How many full compound periods can fit in the time invested

// In non-dataflow terms the implemented algorithm would look like:
//
// int t = 5;    // number of time periods to use
// double principal = init_principal;
// double rate = init_rate;
//
// for (int i = 0; i < t; ++i)
// {
//     double interest = calc(principal, rate);
//     principal = add(principal, interest);
// }
//
// Please note the similarity with the code below!

shared_future<double> principal = make_ready_future(init_principal);
shared_future<double> rate = make_ready_future(init_rate);

for (int i = 0; i < t; ++i)
{
    shared_future<double> interest =
        dataflow(unwrapping(calc), principal, rate);
    principal = dataflow(unwrapping(add), principal, interest);
}

// wait for the dataflow execution graph to be finished calculating our
// overall interest
double result = principal.get();

std::cout << "Final amount: " << result << std::endl;
std::cout << "Amount made: " << result - init_principal << std::endl;
}

return hpx::finalize();
}

```

Here we find our command line variables read in, the rate is converted from a percent to a decimal, the number of calculation iterations is determined, and then our shared_futures are set up. Notice that we first place our principal and rate into shares futures by passing the variables `init_principal` and `init_rate` using `hpx::make_ready_future`.

In this way `hpx::shared_future<double> principal` and `rate` will be initialized to `init_principal` and `init_rate` when `hpx::make_ready_future<double>` returns a future containing those initial values. These shared futures then enter the for loop and are passed to `interest`. Next `principal` and `interest` are passed to the reassignment of `principal` using a `hpx::dataflow`. A dataflow will first wait for its arguments to be ready before launching any callbacks, so `add` in this case will not begin until both `principal` and `interest` are ready. This loop continues for each compound period that must be calculated. To see how `interest` and `principal` are calculated in the loop, let us look at `calc_action` and `add_action`:

```
// Calculate interest for one period
double calc(double principal, double rate)
{
    return principal * rate;
}

////////////////////////////////////
// Add the amount made to the principal
double add(double principal, double interest)
{
    return principal + interest;
}
```

After the shared future dependencies have been defined in `hpx_main`, we see the following statement:

```
double result = principal.get();
```

This statement calls `hpx::future::get` on the shared future `principal` which had its value calculated by our for loop. The program will wait here until the entire dataflow tree has been calculated and the value assigned to `result`. The program then prints out the final value of the investment and the amount of interest made by subtracting the final value of the investment from the initial value of the investment.

2.3.7 Local to remote

When developers write code they typically begin with a simple serial code and build upon it until all of the required functionality is present. The following set of examples were developed to demonstrate this iterative process of evolving a simple serial program to an efficient, fully-distributed *HPX* application. For this demonstration, we implemented a 1D heat distribution problem. This calculation simulates the diffusion of heat across a ring from an initialized state to some user-defined point in the future. It does this by breaking each portion of the ring into discrete segments and using the current segment's temperature and the temperature of the surrounding segments to calculate the temperature of the current segment in the next timestep as shown by Fig. ?? below.

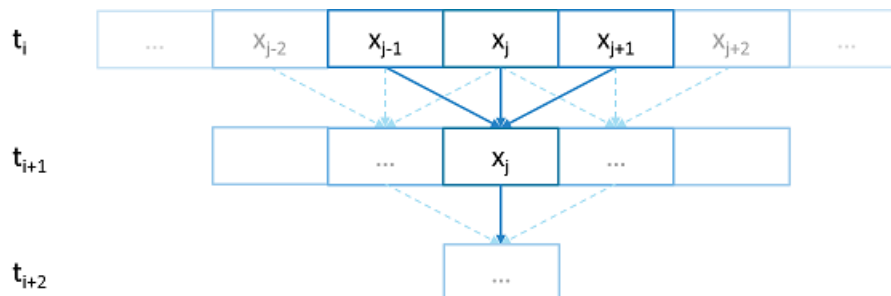


Fig. 2.2: Heat diffusion example program flow.

We parallelize this code over the following eight examples:

- Example 1
- Example 2
- Example 3
- Example 4
- Example 5

- Example 6
- Example 7
- Example 8

The first example is straight serial code. In this code we instantiate a vector `U` that contains two vectors of doubles as seen in the structure `stepper`.

```
struct stepper
{
    // Our partition type
    typedef double partition;

    // Our data for one time step
    typedef std::vector<partition> space;

    // Our operator
    static double heat(double left, double middle, double right)
    {
        return middle + (k * dt / (dx * dx)) * (left - 2 * middle + right);
    }

    // do all the work on 'nx' data points for 'nt' time steps
    space do_work(std::size_t nx, std::size_t nt)
    {
        // U[t][i] is the state of position i at time t.
        std::vector<space> U(2);
        for (space& s : U)
            s.resize(nx);

        // Initial conditions: f(0, i) = i
        for (std::size_t i = 0; i != nx; ++i)
            U[0][i] = double(i);

        // Actual time step loop
        for (std::size_t t = 0; t != nt; ++t)
        {
            space const& current = U[t % 2];
            space& next = U[(t + 1) % 2];

            next[0] = heat(current[nx - 1], current[0], current[1]);

            for (std::size_t i = 1; i != nx - 1; ++i)
                next[i] = heat(current[i - 1], current[i], current[i + 1]);

            next[nx - 1] = heat(current[nx - 2], current[nx - 1], current[0]);
        }

        // Return the solution at time-step 'nt'.
        return U[nt % 2];
    }
};
```

Each element in the vector of doubles represents a single grid point. To calculate the change in heat distribution, the temperature of each grid point, along with its neighbors, is passed to the function `heat`. In order to improve readability,

references named `current` and `next` are created which, depending on the time step, point to the first and second vector of doubles. The first vector of doubles is initialized with a simple heat ramp. After calling the heat function with the data in the `current` vector, the results are placed into the `next` vector.

In example 2 we employ a technique called futurization. Futurization is a method by which we can easily transform a code that is serially executed into a code that creates asynchronous threads. In the simplest case this involves replacing a variable with a future to a variable, a function with a future to a function, and adding a `.get()` at the point where a value is actually needed. The code below shows how this technique was applied to the `struct stepper`.

```
struct stepper
{
    // Our partition type
    typedef hpx::shared_future<double> partition;

    // Our data for one time step
    typedef std::vector<partition> space;

    // Our operator
    static double heat(double left, double middle, double right)
    {
        return middle + (k * dt / (dx * dx)) * (left - 2 * middle + right);
    }

    // do all the work on 'nx' data points for 'nt' time steps
    hpx::future<space> do_work(std::size_t nx, std::size_t nt)
    {
        using hpx::dataflow;
        using hpx::unwrapping;

        // U[t][i] is the state of position i at time t.
        std::vector<space> U(2);
        for (space& s : U)
            s.resize(nx);

        // Initial conditions: f(0, i) = i
        for (std::size_t i = 0; i != nx; ++i)
            U[0][i] = hpx::make_ready_future(double(i));

        auto Op = unwrapping(&stepper::heat);

        // Actual time step loop
        for (std::size_t t = 0; t != nt; ++t)
        {
            space const& current = U[t % 2];
            space& next = U[(t + 1) % 2];

            // WHEN U[t][i-1], U[t][i], and U[t][i+1] have been computed, THEN we
            // can compute U[t+1][i]
            for (std::size_t i = 0; i != nx; ++i)
            {
                next[i] =
                    dataflow(hpx::launch::async, Op, current[idx(i, -1, nx)],
                        current[i], current[idx(i, +1, nx)]);
            }
        }
    }
}
```

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```

    }

    // Now the asynchronous computation is running; the above for-loop does not
    // wait on anything. There is no implicit waiting at the end of each timestep;
    // the computation of each U[t][i] will begin as soon as its dependencies
    // are ready and hardware is available.

    // Return the solution at time-step 'nt'.
    return hpx::when_all(U[nt % 2]);
}
};

```

In example 2, we redefine our partition type as a `shared_future` and, in main, create the object `result`, which is a future to a vector of partitions. We use `result` to represent the last vector in a string of vectors created for each timestep. In order to move to the next timestep, the values of a partition and its neighbors must be passed to `heat` once the futures that contain them are ready. In *HPX*, we have an LCO (Local Control Object) named `Dataflow` that assists the programmer in expressing this dependency. `Dataflow` allows us to pass the results of a set of futures to a specified function when the futures are ready. `Dataflow` takes three types of arguments, one which instructs the dataflow on how to perform the function call (async or sync), the function to call (in this case `Op`), and futures to the arguments that will be passed to the function. When called, dataflow immediately returns a future to the result of the specified function. This allows users to string dataflows together and construct an execution tree.

After the values of the futures in dataflow are ready, the values must be pulled out of the future container to be passed to the function `heat`. In order to do this, we use the *HPX* facility `unwrapping`, which underneath calls `.get()` on each of the futures so that the function `heat` will be passed doubles and not futures to doubles.

By setting up the algorithm this way, the program will be able to execute as quickly as the dependencies of each future are met. Unfortunately, this example runs terribly slow. This increase in execution time is caused by the overheads needed to create a future for each data point. Because the work done within each call to `heat` is very small, the overhead of creating and scheduling each of the three futures is greater than that of the actual useful work! In order to amortize the overheads of our synchronization techniques, we need to be able to control the amount of work that will be done with each future. We call this amount of work per overhead grain size.

In example 3, we return to our serial code to figure out how to control the grain size of our program. The strategy that we employ is to create “partitions” of data points. The user can define how many partitions are created and how many data points are contained in each partition. This is accomplished by creating the struct `partition`, which contains a member object `data_`, a vector of doubles that holds the data points assigned to a particular instance of `partition`.

In example 4, we take advantage of the partition setup by redefining `space` to be a vector of `shared_futures` with each future representing a partition. In this manner, each future represents several data points. Because the user can define how many data points are in each partition, and, therefore, how many data points are represented by one future, a user can control the grainsize of the simulation. The rest of the code is then futurized in the same manner as example 2. It should be noted how strikingly similar example 4 is to example 2.

Example 4 finally shows good results. This code scales equivalently to the OpenMP version. While these results are promising, there are more opportunities to improve the application’s scalability. Currently, this code only runs on one *locality*, but to get the full benefit of *HPX*, we need to be able to distribute the work to other machines in a cluster. We begin to add this functionality in example 5.

In order to run on a distributed system, a large amount of boilerplate code must be added. Fortunately, *HPX* provides us with the concept of a *component*, which saves us from having to write quite as much code. A component is an object that can be remotely accessed using its global address. Components are made of two parts: a server and a client class. While the client class is not required, abstracting the server behind a client allows us to ensure type safety instead of having to pass around pointers to global objects. Example 5 renames example 4’s struct `partition` to `partition_data` and adds serialization support. Next, we add the server side representation of the data in the structure `partition_server`. `Partition_server` inherits from `hpx::components::component_base`, which contains a server-side component

boilerplate. The boilerplate code allows a component's public members to be accessible anywhere on the machine via its Global Identifier (GID). To encapsulate the component, we create a client side helper class. This object allows us to create new instances of our component and access its members without having to know its GID. In addition, we are using the client class to assist us with managing our asynchrony. For example, our client class `partition`'s member function `get_data()` returns a future to `partition_data` `get_data()`. This struct inherits its boilerplate code from `hpx::components::client_base`.

In the structure `stepper`, we have also had to make some changes to accommodate a distributed environment. In order to get the data from a particular neighboring partition, which could be remote, we must retrieve the data from all of the neighboring partitions. These retrievals are asynchronous and the function `heat_part_data`, which, amongst other things, calls `heat`, should not be called unless the data from the neighboring partitions have arrived. Therefore, it should come as no surprise that we synchronize this operation with another instance of `dataflow` (found in `heat_part`). This `dataflow` receives futures to the data in the current and surrounding partitions by calling `get_data()` on each respective partition. When these futures are ready, `dataflow` passes them to the `unwrapping` function, which extracts the `shared_array` of doubles and passes them to the `lambda`. The `lambda` calls `heat_part_data` on the `locality`, which the middle partition is on.

Although this example could run distributed, it only runs on one *locality*, as it always uses `hpx::find_here()` as the target for the functions to run on.

In example 6, we begin to distribute the partition data on different nodes. This is accomplished in `stepper::do_work()` by passing the GID of the *locality* where we wish to create the partition to the partition constructor.

```
for (std::size_t i = 0; i != np; ++i)
    U[0][i] = partition(localities[locidx(i, np, nl)], nx, double(i));
```

We distribute the partitions evenly based on the number of localities used, which is described in the function `locidx`. Because some of the data needed to update the partition in `heat_part` could now be on a new *locality*, we must devise a way of moving data to the *locality* of the middle partition. We accomplished this by adding a switch in the function `get_data()` that returns the end element of the buffer `data_` if it is from the left partition or the first element of the buffer if the data is from the right partition. In this way only the necessary elements, not the whole buffer, are exchanged between nodes. The reader should be reminded that this exchange of end elements occurs in the function `get_data()` and, therefore, is executed asynchronously.

Now that we have the code running in distributed, it is time to make some optimizations. The function `heat_part` spends most of its time on two tasks: retrieving remote data and working on the data in the middle partition. Because we know that the data for the middle partition is local, we can overlap the work on the middle partition with that of the possibly remote call of `get_data()`. This algorithmic change, which was implemented in example 7, can be seen below:

```
// The partitioned operator, it invokes the heat operator above on all elements
// of a partition.
static partition heat_part(
    partition const& left, partition const& middle, partition const& right)
{
    using hpx::dataflow;
    using hpx::unwrapping;

    hpx::shared_future<partition_data> middle_data =
        middle.get_data(partition_server::middle_partition);

    hpx::future<partition_data> next_middle = middle_data.then(
        unwrapping([middle](partition_data const& m) -> partition_data {
            HPX_UNUSED(middle);
```

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```

        // All local operations are performed once the middle data of
        // the previous time step becomes available.
        std::size_t size = m.size();
        partition_data next(size);
        for (std::size_t i = 1; i != size - 1; ++i)
            next[i] = heat(m[i - 1], m[i], m[i + 1]);
        return next;
    }));

    return dataflow(hpx::launch::async,
        unwrapping([left, middle, right](partition_data next,
            partition_data const& l, partition_data const& m,
            partition_data const& r) -> partition {
                HPX_UNUSED(left);
                HPX_UNUSED(right);

                // Calculate the missing boundary elements once the
                // corresponding data has become available.
                std::size_t size = m.size();
                next[0] = heat(l[size - 1], m[0], m[1]);
                next[size - 1] = heat(m[size - 2], m[size - 1], r[0]);

                // The new partition_data will be allocated on the same locality
                // as 'middle'.
                return partition(middle.get_id(), std::move(next));
            }),
        std::move(next_middle),
        left.get_data(partition_server::left_partition), middle_data,
        right.get_data(partition_server::right_partition));
}

```

Example 8 completes the futurization process and utilizes the full potential of *HPX* by distributing the program flow to multiple localities, usually defined as nodes in a cluster. It accomplishes this task by running an instance of *HPX* main on each *locality*. In order to coordinate the execution of the program, the `struct stepper` is wrapped into a component. In this way, each *locality* contains an instance of `stepper` that executes its own instance of the function `do_work()`. This scheme does create an interesting synchronization problem that must be solved. When the program flow was being coordinated on the head node, the GID of each component was known. However, when we distribute the program flow, each partition has no notion of the GID of its neighbor if the next partition is on another *locality*. In order to make the GIDs of neighboring partitions visible to each other, we created two buffers to store the GIDs of the remote neighboring partitions on the left and right respectively. These buffers are filled by sending the GID of newly created edge partitions to the right and left buffers of the neighboring localities.

In order to finish the simulation, the solution vectors named `result` are then gathered together on *locality* 0 and added into a vector of spaces `overall_result` using the *HPX* functions `gather_id` and `gather_here`.

Example 8 completes this example series, which takes the serial code of example 1 and incrementally morphs it into a fully distributed parallel code. This evolution was guided by the simple principles of futurization, the knowledge of grainsize, and utilization of components. Applying these techniques easily facilitates the scalable parallelization of most applications.

2.3.8 Serializing user-defined types

In order to facilitate the sending and receiving of complex datatypes HPX provides a serialization abstraction.

Just like boost, hpx allows users to serialize user-defined types by either providing the serializer as a member function or defining the serialization as a free function.

Unlike Boost HPX doesn't acknowledge second unsigned int parameter, it is solely there to preserve API compatibility with Boost Serialization

This tutorial was heavily inspired by [Boost's serialization concepts](#)²⁴.

Setup

The source code for this example can be found here: `custom_serialization.cpp`.

To compile this program, go to your *HPX* build directory (see *Building HPX* for information on configuring and building *HPX*) and enter:

```
$ make examples.quickstart.custom_serialization
```

To run the program type:

```
$ ./bin/custom_serialization
```

This should print:

```
Rectangle(Point(x=0,y=0),Point(x=0,y=5))  
gravity.g = 9.81%
```

Serialization Requirements

In order to serialize objects in HPX, at least one of the following criteria must be met:

In the case of default constructible objects:

- The object is an empty type.
- Has a serialization function as shown in this tutorial.
- All members are accessible publicly and they can be used in structured binding contexts.

Otherwise:

- They need to have special serialization support.

²⁴ https://www.boost.org/doc/libs/1_79_0/libs/serialization/doc/serialization.html

Member function serialization

```

struct point_member_serialization
{
    int x{0};
    int y{0};

    // Required when defining the serialization function as private
    // In this case it isn't
    // Provides serialization access to HPX
    friend class hpx::serialization::access;

    // Second argument exists solely for compatibility with boost serialize
    // it is NOT processed by HPX in any way.
    template <typename Archive>
    void serialize(Archive& ar, unsigned int const)
    {
        // clang-format off
        ar & x & y;
        // clang-format on
    }
};

// Allow bitwise serialization
HPX_IS_BITWISE_SERIALIZABLE(point_member_serialization)

```

Notice that `point_member_serialization` is defined as bitwise serializable (see *Bitwise serialization for bitwise copyable data* for more details). HPX is also able to recursively serialize composite classes and structs given that its members are serializable.

```

struct rectangle_member_serialization
{
    point_member_serialization top_left;
    point_member_serialization lower_right;

    template <typename Archive>
    void serialize(Archive& ar, unsigned int const)
    {
        // clang-format off
        ar & top_left & lower_right;
        // clang-format on
    }
};

```

Free function serialization

In order to decouple your models from HPX, HPX also allows for the definition of free function serializers.

```
struct rectangle_free
{
    point_member_serialization top_left;
    point_member_serialization lower_right;
};

template <typename Archive>
void serialize(Archive& ar, rectangle_free& pt, unsigned int const)
{
    // clang-format off
    ar & pt.lower_right & pt.top_left;
    // clang-format on
}
```

Even if you can't modify a class to befriend it, you can still be able to serialize your class provided that your class is default constructable and you are able to reconstruct it yourself.

```
class point_class
{
public:
    point_class(int x, int y)
        : x(x)
        , y(y)
    {
    }

    point_class() = default;

    [[nodiscard]] int get_x() const noexcept
    {
        return x;
    }

    [[nodiscard]] int get_y() const noexcept
    {
        return y;
    }

private:
    int x;
    int y;
};

template <typename Archive>
void load(Archive& ar, point_class& pt, unsigned int const)
{
    int x, y;
    ar >> x >> y;
    pt = point_class(x, y);
}
```

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```

}

template <typename Archive>
void save(Archive& ar, point_class const& pt, unsigned int const)
{
    ar << pt.get_x() << pt.get_y();
}

// This tells HPX that you have spilt your serialize function into
// load and save
HPX_SERIALIZATION_SPLIT_FREE(point_class)

```

Serializing non default constructable classes

Some classes don't provide any default constructor.

```

class planet_weight_calculator
{
public:
    explicit planet_weight_calculator(double g)
        : g(g)
    {
    }

    template <class Archive>
    friend void save_construct_data(
        Archive&, planet_weight_calculator const*, unsigned int);

    [[nodiscard]] double get_g() const
    {
        return g;
    }

private:
    // Provides serialization access to HPX
    friend class hpx::serialization::access;
    template <class Archive>
    void serialize(Archive&, unsigned int const)
    {
        // Serialization will be done in the save_construct_data
        // Still needs to be defined
    }

    double g;
};

```

In this case you have to define a `save_construct_data` and `load_construct_data` in which you do the serialization yourself.

```

template <class Archive>
inline void save_construct_data(Archive& ar,

```

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```

    planet_weight_calculator const* weight_calc, unsigned int const)
{
    ar << weight_calc->g;    // Do all of your serialization here
}

template <class Archive>
inline void load_construct_data(
    Archive& ar, planet_weight_calculator* weight_calc, unsigned int const)
{
    double g;
    ar >> g;

    // ::new(ptr) construct new object at given address
    hpx::construct_at(weight_calc, g);
}

```

Bitwise serialization for bitwise copyable data

When sending non arithmetic types not defined by `std::is_arithmetic`²⁵, HPX has to (de)serialize each object separately. However, if the class you are trying to send consists only of bitwise copyable datatypes, you may mark your class as such. Then HPX will serialize your object bitwise instead of element wise. This has enormous benefits, especially when sending a vector/array of your class. To define your class as such you need to call `HPX_IS_BITWISE_SERIALIZABLE(T)` with your desired custom class.

```

struct point_member_serialization
{
    int x{0};
    int y{0};

    // Required when defining the serialization function as private
    // In this case it isn't
    // Provides serialization access to HPX
    friend class hpx::serialization::access;

    // Second argument exists solely for compatibility with boost serialize
    // it is NOT processed by HPX in any way.
    template <typename Archive>
    void serialize(Archive& ar, unsigned int const)
    {
        // clang-format off
        ar & x & y;
        // clang-format on
    }
};

// Allow bitwise serialization
HPX_IS_BITWISE_SERIALIZABLE(point_member_serialization)

```

²⁵ https://en.cppreference.com/w/cpp/types/is_arithmetic

2.3.9 Termination detection

This example demonstrates how to use *HPX*'s termination detection API to ensure all asynchronous work has completed before shutting down the runtime. This is particularly useful when you need to guarantee that all posted tasks have finished execution before the application exits.

Overview

The termination detection API provides a way to wait for all local *HPX*-threads to complete their work. This is essential in scenarios where you spawn asynchronous tasks using *hpx::post* or *hpx::async* and need to ensure they all finish before the runtime shuts down.

The API supports several use cases:

- **Basic usage:** Wait indefinitely for all tasks to complete
- **Timeout:** Wait for a specified duration
- **Deadline:** Wait until a specific time point
- **Cancellation:** Support for cancellation tokens to interrupt the wait

Setup

The source code for this example can be found here: `termination_detection.cpp`.

To compile this program, go to your *HPX* build directory (see *Building HPX* for information on configuring and building *HPX*) and enter:

```
$ make examples.quickstart.termination_detection
```

To run the program type:

```
$ ./bin/termination_detection
```

This will execute all four examples demonstrating different usage patterns of the termination detection API.

Walkthrough

The example demonstrates four different usage patterns of the termination detection API. Let's examine each one:

Basic usage

The simplest form waits indefinitely for all local threads to complete:

This is useful when you want to ensure all work is done before shutting down, and you don't have time constraints.

Timeout usage

You can specify a maximum duration to wait:

The function returns `true` if all threads completed within the timeout, or `false` if the timeout elapsed. This is useful when you want to give tasks a reasonable amount of time to complete but don't want to wait indefinitely.

Deadline usage

Similar to timeout, but you specify an absolute time point:

This is useful when you have a specific deadline by which all work must be completed.

Cancellation support

The most flexible form supports cancellation tokens:

This allows external control over the wait operation. You can request cancellation from another thread, which is useful in scenarios like graceful shutdown with user interruption support.

API Reference

The termination detection API is available in the `hpx::local` namespace:

`void hpx::local::termination_detection()`

Wait indefinitely for all local *HPX*-threads to complete.

`bool hpx::local::termination_detection(hpx::chrono::steady_duration const &timeout)`

Wait for all local *HPX*-threads to complete, with a timeout.

Parameters

timeout – Maximum duration to wait

Returns

`true` if all threads completed, `false` if timeout elapsed

`bool hpx::local::termination_detection(hpx::chrono::steady_time_point const &deadline)`

Wait for all local *HPX*-threads to complete, until a deadline.

Parameters

deadline – Absolute time point to wait until

Returns

`true` if all threads completed, `false` if deadline passed

`bool hpx::local::termination_detection(hpx::stop_token stop_token, hpx::chrono::steady_duration const &timeout)`

Wait for all local *HPX*-threads to complete, with cancellation support.

Parameters

- **stop_token** – Token that can be used to request cancellation
- **timeout** – Maximum duration to wait (defaults to maximum duration)

Returns

`true` if all threads completed, `false` if timeout elapsed or stop was requested

Use cases

The termination detection API is particularly useful in the following scenarios:

- **Graceful shutdown:** Ensure all background tasks complete before exiting
- **Testing:** Verify that all spawned tasks have finished in unit tests
- **Batch processing:** Wait for all work items in a batch to complete
- **Resource cleanup:** Ensure all tasks using shared resources have finished before cleanup

See also

- `hpx::post` - Fire-and-forget asynchronous execution
- `hpx::async` - Asynchronous execution with a future
- `hpx::finalize` - Shut down the *HPX* runtime

2.4 Manual

The manual is your comprehensive guide to *HPX*. It contains detailed information on how to build and use *HPX* in different scenarios.

2.4.1 Prerequisites

Supported platforms

At this time, *HPX* supports the following platforms. Other platforms may work, but we do not test *HPX* with other platforms, so please be warned.

Table 2.1: Supported Platforms for *HPX*

Name	Minimum Version	Architectures
Linux	2.6	x86-32, x86-64, k1om
BlueGeneQ	V1R2M0	PowerPC A2
Windows	Any Windows system	x86-32, x86-64
Mac OSX	Any OSX system	x86-64
ARM	Any ARM system	Any architecture
RISC-V	Any RISC-V system	Any architecture

Supported compilers

The table below shows the supported compilers for *HPX*.

Table 2.2: Supported Compilers for *HPX*

Name	Minimum Version	Latest tested
GNU Compiler Collection (g++) ²⁶	12.0	15.0
clang: a C language family frontend for LLVM ²⁷	16.0	20.0
Visual C++ ²⁸ (x64)	2022	2022

Software and libraries

The table below presents all the necessary prerequisites for building *HPX*.

Table 2.3: Software prerequisites for *HPX*

	Name	Minimum Version	Latest tested
Build System	CMake ²⁹	3.20	4.1
Required Libraries	Boost ³⁰	1.71.0	1.88.0
	Portable Hardware Locality (HWLOC) ³¹	1.5	2.4

The most important dependencies are [Boost](#)³² and [Portable Hardware Locality \(HWLOC\)](#)³³. The installation of Boost is described in detail in Boost's [Getting Started](#)³⁴ document. A recent version of hwloc is required in order to support thread pinning and NUMA awareness and can be found in [Hwloc Downloads](#)³⁵.

HPX is written in 99.99% Standard C++ (the remaining 0.01% is platform specific assembly code). As such, *HPX* is compilable with almost any standards compliant C++ compiler. The code base takes advantage of C++ language and standard library features when available.

Note: When building Boost using gcc, please note that it is required to specify a `cxxflags=-std=c++20` command line argument to b2 (bjam).

Note: In most configurations, *HPX* depends only on header-only Boost. Boost.Filesystem is required if the standard library does not support filesystem. The following are not needed by default, but are required in certain configurations: Boost.Chrono, Boost.DateTime, Boost.Log, Boost.LogSetup, Boost.Regex, and Boost.Thread.

Depending on the options you chose while building and installing *HPX*, you will find that *HPX* may depend on several other libraries such as those listed below.

Note: In order to use a high speed parcellport, we currently recommend configuring *HPX* to use MPI so that MPI can be used for communication between different localities. Please set the CMake variable `MPI_CXX_COMPILER` to your MPI C++ compiler wrapper if not detected automatically.

Table 2.4: Optional software prerequisites for *HPX*

Name	Minimum version
google-perftools ³⁶	1.7.1
jemalloc ³⁷	2.1.0
mi-malloc ³⁸	1.0.0
Performance Application Programming Interface (PAPI)	

²⁶ <https://gcc.gnu.org>

²⁷ <https://clang.llvm.org/>

²⁸ <https://msdn.microsoft.com/en-us/visualc/default.aspx>

²⁹ <https://www.cmake.org>

³⁰ <https://www.boost.org/>

³¹ <https://www.open-mpi.org/projects/hwloc/>

³² <https://www.boost.org/>

³³ <https://www.open-mpi.org/projects/hwloc/>

³⁴ https://www.boost.org/more/getting_started/index.html

³⁵ <https://www.open-mpi.org/software/hwloc/v1.11>

2.4.2 Getting HPX

Download a tarball of the latest release from [HPX Downloads](#)³⁹ and unpack it or clone the repository directly using git:

```
$ git clone https://github.com/TheHPXProject/hpx.git
```

It is also recommended that you check out the latest stable tag:

```
$ cd hpx
```

```
$ git checkout v2.0.0
```

2.4.3 Building HPX

Basic information

The build system for *HPX* is based on [CMake](#)⁴⁰, a cross-platform build-generator tool which is not responsible for building the project but rather generates the files needed by your build tool (GNU make, Visual Studio, etc.) for building *HPX*. If CMake is not already installed in your system, you can download it and install it here: [CMake Downloads](#)⁴¹.

Once [CMake](#)⁴² has been run, the build process can be started. The build process consists of the following parts:

- The *HPX* core libraries (target `core`): This forms the basic set of *HPX* libraries.
- *HPX* Examples (target `examples`): This target is enabled by default and builds all *HPX* examples (disable by setting `HPX_WITH_EXAMPLES:BOOL=Off`). *HPX* examples are part of the `all` target and are included in the installation if enabled.
- *HPX* Tests (target `tests`): This target builds the *HPX* test suite and is enabled by default (disable by setting `HPX_WITH_TESTS:BOOL=Off`). They are not built by the `all` target and have to be built separately.
- *HPX* Documentation (target `docs`): This target builds the documentation, and is not enabled by default (enable by setting `HPX_WITH_DOCUMENTATION:BOOL=On`. For more information see *Documentation*.

The *HPX* build process is highly configurable through [CMake](#)⁴³, and various [CMake](#)⁴⁴ variables influence the build process. A list with the most important [CMake](#)⁴⁵ variables can be found in the section that follows, while the complete list of available [CMake](#)⁴⁶ variables is in *CMake options*. These variables can be used to refine the recipes that can be found at *Platform specific build recipes*, a section that shows some basic steps on how to build *HPX* for a specific platform.

In order to use *HPX*, only the core libraries are required. In order to use the optional libraries, you need to specify them as link dependencies in your build (See *Creating HPX projects*).

³⁶ <https://code.google.com/p/gperftools>

³⁷ <http://jemalloc.net>

³⁸ <http://microsoft.github.io/mimalloc/>

³⁹ <https://hpx.stellar-group.org/downloads/>

⁴⁰ <https://www.cmake.org>

⁴¹ <https://www.cmake.org/cmake/resources/software.html>

⁴² <https://www.cmake.org>

⁴³ <https://www.cmake.org>

⁴⁴ <https://www.cmake.org>

⁴⁵ <https://www.cmake.org>

⁴⁶ <https://www.cmake.org>

Most important CMake options

While building *HPX*, you are provided with multiple CMake options which correspond to different configurations. Below, there is a set of the most important and frequently used CMake options.

HPX_WITH_MALLOC

Use a custom allocator. Using a custom allocator tuned for multithreaded applications is very important for the performance of *HPX* applications. When debugging applications, it's useful to set this to `system`, as custom allocators can hide some memory-related bugs. Note that setting this to something other than `system` requires an external dependency.

HPX_WITH_CUDA

Enable support for CUDA. Use `CMAKE_CUDA_COMPILER` to set the CUDA compiler. This is a standard CMake⁴⁷ variable, like `CMAKE_CXX_COMPILER`.

HPX_WITH_PARCELPORTR_MPI

Enable the MPI parcellport. This enables the use of MPI for the networking operations in the *HPX* runtime. The default value is `OFF` because it's not available on all systems and/or requires another dependency. However, it is the recommended parcellport.

HPX_WITH_PARCELPORTR_TCP

Enable the TCP parcellport. Enables the use of TCP for networking in the runtime. The default value is `ON`. However, it's only recommended for debugging purposes, as it is slower than the MPI parcellport.

HPX_WITH_PARCELPORTR_LCI

Enable the LCI parcellport. This enables the use of LCI for the networking operations in the *HPX* runtime. The default value is `OFF` because it's not available on all systems and/or requires another dependency. However, this experimental parcellport may provide better performance than the MPI parcellport. Please refer to *Using the LCI parcellport* for more information about the LCI parcellport.

HPX_WITH_APEX

Enable APEX integration. [APEX](https://uo-oaciss.github.io/apex/quickstarthpx/)⁴⁸ can be used to profile *HPX* applications. In particular, it provides information about individual tasks in the *HPX* runtime.

HPX_WITH_GENERIC_CONTEXT_COROUTINES

Enable Boost. Context for task context switching. It must be enabled for non-x86 architectures such as ARM and Power.

HPX_WITH_MAX_CPU_COUNT

Set the maximum CPU count supported by *HPX*. The default value is 64, and should be set to a number at least as high as the number of cores on a system including virtual cores such as hyperthreads.

HPX_WITH_CXX_STANDARD

Set a specific C++ standard version e.g. `HPX_WITH_CXX_STANDARD=23`. The default and minimum value is 20. Possible values are 20, 23, or 26.

HPX_WITH_EXAMPLES

Build examples.

HPX_WITH_TESTS

Build tests.

⁴⁷ <https://www.cmake.org>

⁴⁸ <https://uo-oaciss.github.io/apex/quickstarthpx/>

HPX_WITH_DEBUG_POSTFIX

Set the postfix for debug libraries. The default is `d`. This variable is used to set `CMAKE_DEBUG_POSTFIX` and is only relevant for Debug builds or multi-configuration generators. The default rarely needs to be changed. It ensures that generated debug binaries have a different name than release binaries, which is important to avoid ABI problems when both debug and release binaries are installed on the same system.

For a complete list of available [CMake](#)⁴⁹ variables that influence the build of *HPX*, see *CMake options*.

Build types

[CMake](#)⁵⁰ can be configured to generate project files suitable for builds that have enabled debugging support or for an optimized build (without debugging support). The [CMake](#)⁵¹ variable used to set the build type is `CMAKE_BUILD_TYPE` (for more information see the [CMake Documentation](#)⁵²). Available build types are:

- **Debug:** Full debug symbols are available as well as additional assertions to help debugging. To enable the debug build type for the *HPX* API, the C++ Macro `HPX_DEBUG` is defined.
- **RelWithDebInfo:** Release build with debugging symbols. This is most useful for profiling applications
- **Release:** Release build. This disables assertions and enables default compiler optimizations.
- **RelMinSize:** Release build with optimizations for small binary sizes.

Important: We currently don't guarantee ABI compatibility between Debug and Release builds. Please make sure that applications built against *HPX* use the same build type as you used to build *HPX*. For CMake builds, this means that the `CMAKE_BUILD_TYPE` variables have to match and for projects not using [CMake](#)⁵³, the `HPX_DEBUG` macro has to be set in debug mode.

Using CMake Presets

HPX provides a `CMakePresets.json` file which includes a variety of pre-defined build configurations. These presets allow you to easily configure the build for common scenarios without needing to manually specify multiple CMake variables.

To use a preset, you can use the `--preset` option with CMake:

```
$ cmake --preset <preset-name>
$ cmake --build --preset <preset-name>
```

Some of the available presets include:

- **default:** Standard release build with tests and examples enabled.
- **minimal:** Minimal build with only core features (no tests, examples, or tools).
- **full:** Full build with all standard features enabled.
- **debug:** Debug build with symbols and debug-optimized settings.
- **performance:** Build optimized for performance analysis with APEX profiling.
- **cuda:** Build with CUDA support (requires CUDA toolkit).

⁴⁹ <https://www.cmake.org>

⁵⁰ <https://www.cmake.org>

⁵¹ <https://www.cmake.org>

⁵² https://cmake.org/cmake/help/latest/variable/CMAKE_BUILD_TYPE.html

⁵³ <https://www.cmake.org>

- `sycl`: Build with SYCL support (requires compatible compiler).
- `sanitizer-address`: Build with AddressSanitizer enabled.

For a full list of available presets, you can run:

```
$ cmake --list-presets
```

Platform specific build recipes

Unix variants

Once you have the source code and the dependencies and assuming all your dependencies are in paths known to `CMake`⁵⁴, the following gets you started:

1. First, set up a separate build directory to configure the project:

```
$ mkdir build && cd build
```

2. To configure the project you have the following options:

- To build the core *HPX* libraries and examples, and install them to your chosen location (recommended):

```
$ cmake -DCMAKE_INSTALL_PREFIX=/install/path ..
```

Tip: If you want to change `CMake`⁵⁵ variables for your build, it is usually a good idea to start with a clean build directory to avoid configuration problems. It is especially important that you use a clean build directory when changing between Release and Debug modes.

- To install *HPX* to the default system folders, simply leave out the `CMAKE_INSTALL_PREFIX` option:

```
$ cmake ..
```

- If your dependencies are in custom locations, you may need to tell `CMake`⁵⁶ where to find them by passing one or more options to `CMake`⁵⁷ as shown below:

```
$ cmake -DBoost_ROOT=/path/to/boost
      -DHWloc_ROOT=/path/to/hwloc
      -DTcmalloc_ROOT=/path/to/tcmalloc
      -DJemalloc_ROOT=/path/to/jemalloc
      [other CMake variable definitions]
      /path/to/source/tree
```

For instance:

```
$ cmake -DBoost_ROOT=/packages/boost -DHWloc_ROOT=/packages/hwloc -DCMAKE_
INSTALL_PREFIX=/packages/hpx ~/downloads/hpx_1.5.1
```

- If you want to try *HPX* without using a custom allocator pass `-DHPX_WITH_MALLOC=system` to `CMake`⁵⁸:

⁵⁴ <https://www.cmake.org>

⁵⁵ <https://www.cmake.org>

⁵⁶ <https://www.cmake.org>

⁵⁷ <https://www.cmake.org>

⁵⁸ <https://www.cmake.org>

```
$ cmake -DCMAKE_INSTALL_PREFIX=/install/path -DHPX_WITH_MALLOC=system ..
```

Note: Please pay special attention to the section about *HPX_WITH_MALLOC:STRING* as this is crucial for getting decent performance.

Important: If you are building *HPX* for a system with more than 64 processing units, you must change the CMake⁵⁹ variable *HPX_WITH_MAX_CPU_COUNT* (to a value at least as big as the number of (virtual) cores on your system). Note that the default value is 64.

Caution: Compiling and linking *HPX* needs a considerable amount of memory. It is advisable that at least 2 GB of memory per parallel process is available.

3. Once the configuration is complete, to build the project you run:

```
$ cmake --build . --target install
```

Windows

Note: The following build recipes are mostly user-contributed and may be outdated. We always welcome updated and new build recipes.

To build *HPX* under Windows 10 x64 with Visual Studio 2015:

- Download the CMake V3.19 installer (or latest version) from [here](#)⁶⁰
- Download the hwloc V1.11.0 (or the latest version) from [here](#)⁶¹ and unpack it.
- Download the latest Boost libraries from [here](#)⁶² and unpack them.
- Build the Boost DLLs and LIBs by using these commands from Command Line (or PowerShell). Open CMD/PowerShell inside the Boost dir and type in:

```
.\bootstrap.bat
```

This batch file will set up everything needed to create a successful build. Now execute:

```
.\b2.exe link=shared variant=release,debug architecture=x86 address-model=64  
↪threading=multi --build-type=complete install
```

This command will start a (very long) build of all available Boost libraries. Please, be patient.

- Open CMake-GUI.exe and set up your source directory (input field 'Where is the source code') to the *base directory* of the source code you downloaded from *HPX*'s GitHub pages. Here's an example of CMake path settings, which point to the Documents/GitHub/hpx folder:

⁵⁹ <https://www.cmake.org>

⁶⁰ <https://blog.kitware.com/cmake-3-19-0-available-for-download/>

⁶¹ <https://www.open-mpi.org/software/hwloc/v2.11/>

⁶² <https://www.boost.org/users/download/>

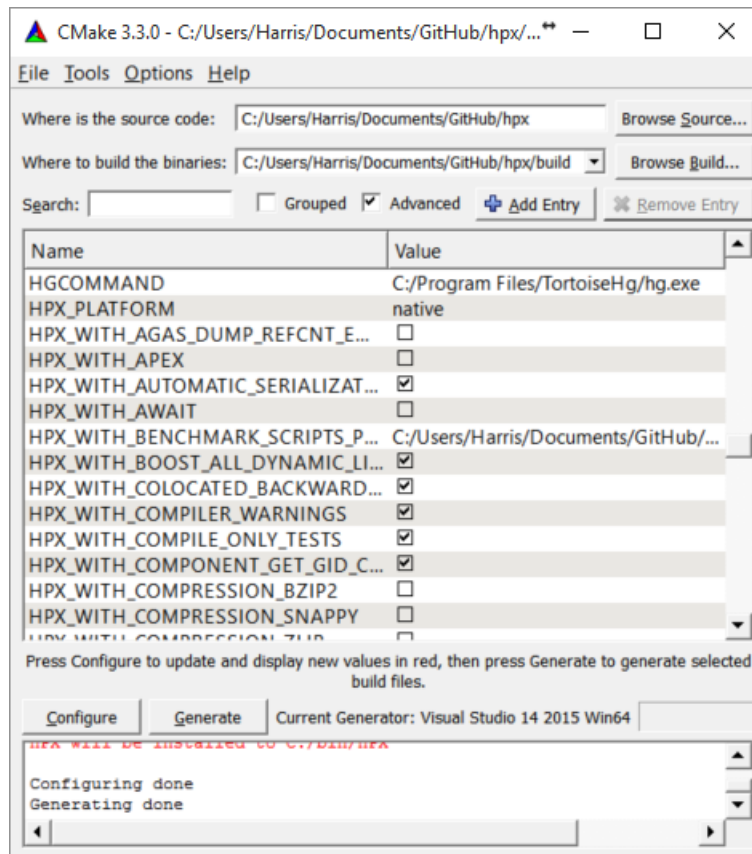


Fig. 2.3: Example CMake path settings.

Inside ‘Where is the source-code’ enter the base directory of your *HPX* source directory (do not enter the “src” sub-directory!). Inside ‘Where to build the binaries’ you should put in the path where all the building processes will happen. This is important because the building machinery will do an “out-of-tree” build. CMake will not touch or change the original source files in any way. Instead, it will generate Visual Studio Solution Files, which will build *HPX* packages out of the *HPX* source tree.

- Set new configuration variables (in CMake, not in Windows environment): Boost_ROOT, Hwloc_ROOT, Asio_ROOT, CMAKE_INSTALL_PREFIX. The meaning of these variables is as follows:
 - Boost_ROOT the *HPX* root directory of the unpacked Boost headers/cpp files.
 - Hwloc_ROOT the *HPX* root directory of the unpacked Portable Hardware Locality files.
 - Asio_ROOT the *HPX* root directory of the unpacked ASIO files. Alternatively use HPX_WITH_FETCH_ASIO with value True.
 - CMAKE_INSTALL_PREFIX the *HPX* root directory where the future builds of *HPX* should be installed.

Note: *HPX* is a very large software collection, so it is not recommended to use the default C:\Program Files\hpx. Many users may prefer to use simpler paths *without* whitespace, like C:\bin\hpx or D:\bin\hpx etc.

To insert new env-vars click on “Add Entry” and then insert the name inside “Name”, select PATH as Type and put the path-name in the “Path” text field. Repeat this for the first three variables.

This is how variable insertion will look:

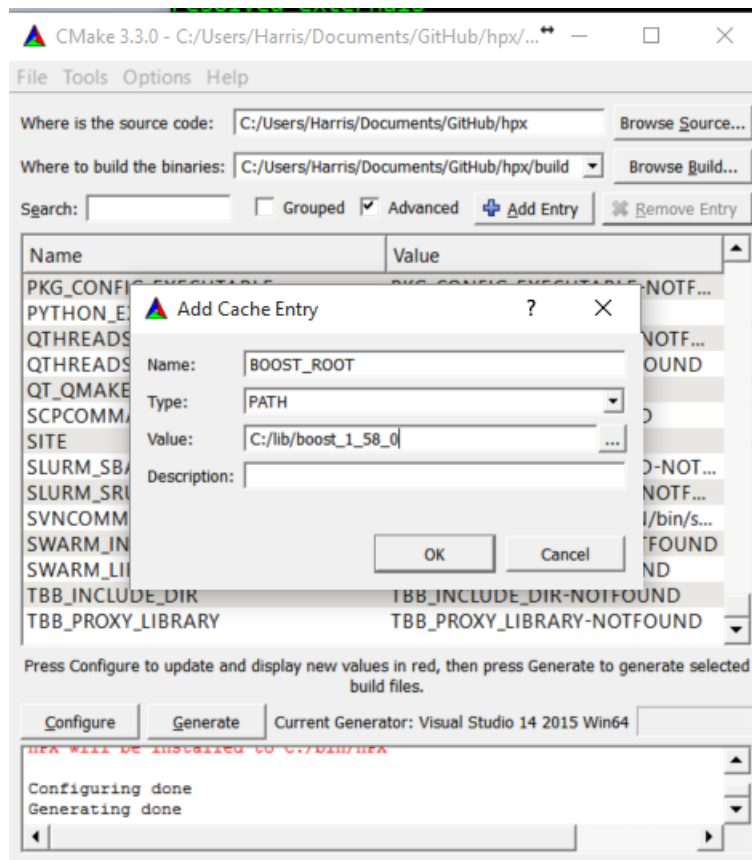


Fig. 2.4: Example CMake adding entry.

Alternatively, users could provide `Boost_LIBRARYDIR` instead of `Boost_ROOT`; the difference is that `Boost_LIBRARYDIR` should point to the subdirectory inside Boost root where all the compiled DLLs/LIBs are. For example, `Boost_LIBRARYDIR` may point to the `bin.v2` subdirectory under the Boost rootdir. It is important to keep the meanings of these two variables separated from each other: `Boost_DIR` points to the ROOT folder of the Boost library. `Boost_LIBRARYDIR` points to the subdir inside the Boost root folder where the compiled binaries are.

- Click the ‘Configure’ button of CMake-GUI. You will be immediately presented with a small window where you can select the C++ compiler to be used within Visual Studio. This has been tested using the latest v14 (a.k.a C++ 2015) but older versions should be sufficient too. Make sure to select the 64Bit compiler.
- After the generate process has finished successfully, click the ‘Generate’ button. Now, CMake will put new VS Solution files into the BUILD folder you selected at the beginning.
- Open Visual Studio and load the `HPX.sln` from your build folder.
- Go to `CMakePredefinedTargets` and build the `INSTALL` project:

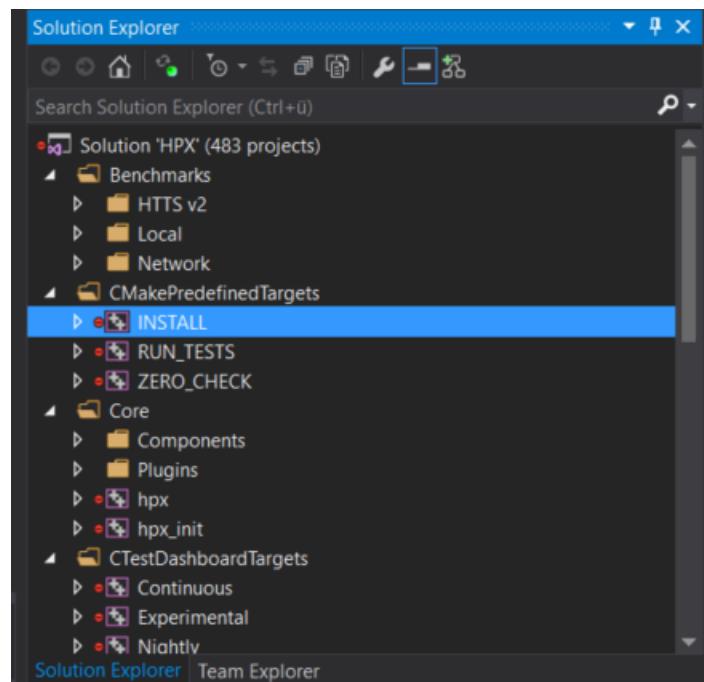
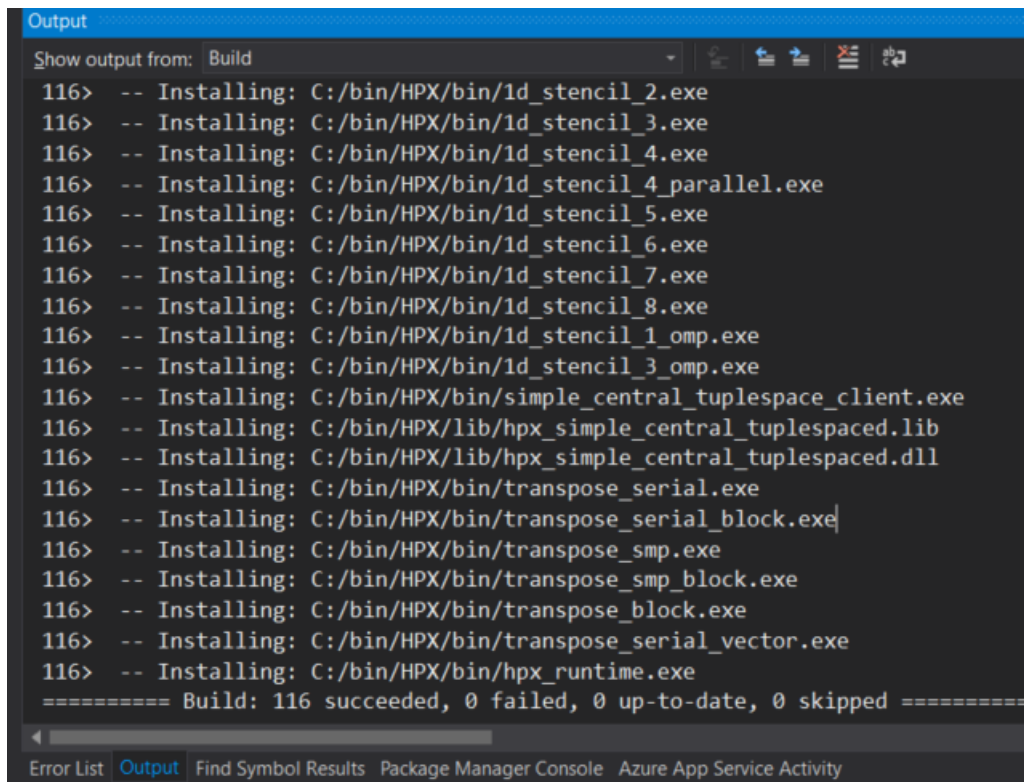


Fig. 2.5: Visual Studio `INSTALL` target.

It will take some time to compile everything, and in the end you should see an output similar to this one:

2.4.4 CMake options

In order to configure *HPX*, you can set a variety of options to allow CMake to generate your specific makefiles/project files. A list of the most important CMake options can be found in *Most important CMake options*, while this section includes the comprehensive list.



```
Output
Show output from: Build
116> -- Installing: C:/bin/HPX/bin/1d_stencil_2.exe
116> -- Installing: C:/bin/HPX/bin/1d_stencil_3.exe
116> -- Installing: C:/bin/HPX/bin/1d_stencil_4.exe
116> -- Installing: C:/bin/HPX/bin/1d_stencil_4_parallel.exe
116> -- Installing: C:/bin/HPX/bin/1d_stencil_5.exe
116> -- Installing: C:/bin/HPX/bin/1d_stencil_6.exe
116> -- Installing: C:/bin/HPX/bin/1d_stencil_7.exe
116> -- Installing: C:/bin/HPX/bin/1d_stencil_8.exe
116> -- Installing: C:/bin/HPX/bin/1d_stencil_1_omp.exe
116> -- Installing: C:/bin/HPX/bin/1d_stencil_3_omp.exe
116> -- Installing: C:/bin/HPX/bin/simple_central_tuplespace_client.exe
116> -- Installing: C:/bin/HPX/lib/hpx_simple_central_tuplespaced.lib
116> -- Installing: C:/bin/HPX/lib/hpx_simple_central_tuplespaced.dll
116> -- Installing: C:/bin/HPX/bin/transpose_serial.exe
116> -- Installing: C:/bin/HPX/bin/transpose_serial_block.exe
116> -- Installing: C:/bin/HPX/bin/transpose_smp.exe
116> -- Installing: C:/bin/HPX/bin/transpose_smp_block.exe
116> -- Installing: C:/bin/HPX/bin/transpose_block.exe
116> -- Installing: C:/bin/HPX/bin/transpose_serial_vector.exe
116> -- Installing: C:/bin/HPX/bin/hpx_runtime.exe
===== Build: 116 succeeded, 0 failed, 0 up-to-date, 0 skipped =====
Error List Output Find Symbol Results Package Manager Console Azure App Service Activity
```

Fig. 2.6: Visual Studio build output.

Variables that influence how *HPX* is built

The options are split into these categories:

- *Generic options*
- *Build Targets options*
- *Thread Manager options*
- *AGAS options*
- *Parcelport options*
- *Profiling options*
- *Debugging options*
- *Modules options*

Generic options

- `HPX_WITH_AUTOMATIC_SERIALIZATION_REGISTRATION:BOOL`
- `HPX_WITH_BUILD_BINARY_PACKAGE:BOOL`
- `HPX_WITH_CHECK_MODULE_DEPENDENCIES:BOOL`
- `HPX_WITH_COMPILER_WARNINGS:BOOL`
- `HPX_WITH_COMPILER_WARNINGS_AS_ERRORS:BOOL`
- `HPX_WITH_COMPRESSION_BZIP2:BOOL`
- `HPX_WITH_COMPRESSION_SNAPPY:BOOL`
- `HPX_WITH_COMPRESSION_ZLIB:BOOL`
- `HPX_WITH_CONTRACTS_MODE:STRING`
- `HPX_WITH_CUDA:BOOL`
- `HPX_WITH_CXX20_STD_EXECUTION_POLICES:INTERNAL`
- `HPX_WITH_CXX_MODULES:BOOL`
- `HPX_WITH_CXX_STANDARD:STRING`
- `HPX_WITH_DATAPAR_BACKEND:STRING`
- `HPX_WITH_DEPRECATION_WARNINGS:BOOL`
- `HPX_WITH_DISABLED_SIGNAL_EXCEPTION_HANDLERS:BOOL`
- `HPX_WITH_DYNAMIC_HPX_MAIN:BOOL`
- `HPX_WITH_FAULT_TOLERANCE:BOOL`
- `HPX_WITH_FULL_RPATH:BOOL`
- `HPX_WITH_GCC_VERSION_CHECK:BOOL`
- `HPX_WITH_GENERIC_CONTEXT_COROUTINES:BOOL`
- `HPX_WITH_GIT_HOOKS:BOOL`
- `HPX_WITH_HIDDEN_VISIBILITY:BOOL`
- `HPX_WITH_HIP:BOOL`
- `HPX_WITH_HIPSYCL:BOOL`
- `HPX_WITH_IGNORE_COMPILER_COMPATIBILITY:BOOL`
- `HPX_WITH_LOGGING:BOOL`
- `HPX_WITH_MALLOC:STRING`
- `HPX_WITH_MODULES_AS_STATIC_LIBRARIES:BOOL`
- `HPX_WITH_MODULE_COMPATIBILITY_HEADERS:BOOL`
- `HPX_WITH_NICE_THREADLEVEL:BOOL`
- `HPX_WITH_PARCEL_COALESCING:BOOL`
- `HPX_WITH_PKGCONFIG:BOOL`
- `HPX_WITH_PRECOMPILED_HEADERS:BOOL`
- `HPX_WITH_PSEUDO_DEPENDENCIES:BOOL`

- `HPX_WITH_RUN_MAIN_EVERYWHERE:BOOL`
- `HPX_WITH_STACKOVERFLOW_DETECTION:BOOL`
- `HPX_WITH_STATIC_LINKING:BOOL`
- `HPX_WITH_SUPPORT_NO_UNIQUE_ADDRESS_ATTRIBUTE:BOOL`
- `HPX_WITH_SYCL:BOOL`
- `HPX_WITH_SYCL_FLAGS:STRING`
- `HPX_WITH_UNITY_BUILD:BOOL`
- `HPX_WITH_VIM_YCM:BOOL`
- `HPX_WITH_ZERO_COPY_SERIALIZATION_THRESHOLD:STRING`

HPX_WITH_AUTOMATIC_SERIALIZATION_REGISTRATION:BOOL

Use automatic serialization registration for actions and functions. This affects compatibility between HPX applications compiled with different compilers (default ON)

HPX_WITH_BUILD_BINARY_PACKAGE:BOOL

Build HPX on the build infrastructure on any LINUX distribution (default: OFF).

HPX_WITH_CHECK_MODULE_DEPENDENCIES:BOOL

Verify that no modules are cross-referenced from a different module category (default: OFF)

HPX_WITH_COMPILER_WARNINGS:BOOL

Enable compiler warnings (default: ON)

HPX_WITH_COMPILER_WARNINGS_AS_ERRORS:BOOL

Turn compiler warnings into errors (default: OFF)

HPX_WITH_COMPRESSION_BZIP2:BOOL

Enable bzip2 compression for parcel data (default: OFF).

HPX_WITH_COMPRESSION_SNAPPY:BOOL

Enable snappy compression for parcel data (default: OFF).

HPX_WITH_COMPRESSION_ZLIB:BOOL

Enable zlib compression for parcel data (default: OFF).

HPX_WITH_CONTRACTS_MODE:STRING

Contract evaluation mode for pre-C++26 compilers, options are: ENFORCE, QUICK_ENFORCE, OBSERVE, IGNORE

HPX_WITH_CUDA:BOOL

Enable support for CUDA (default: OFF)

HPX_WITH_CXX20_STD_EXECUTION_POLICES:INTERNAL

HPX_WITH_CXX_MODULES:BOOL

Enable exposing C++20 modules (default: OFF).

HPX_WITH_CXX_STANDARD:STRING

Set the C++ standard to use when compiling HPX itself. (default: 20)

HPX_WITH_DATAPAR_BACKEND:STRING

Define which vectorization library should be used. Options are: VC, EVE, STD_EXPERIMENTAL_SIMD, SVE, EMULATE, NONE

HPX_WITH_DEPRECATED_WARNINGS:BOOL

Enable warnings for deprecated facilities (default: ON).

HPX_WITH_DISABLED_SIGNAL_EXCEPTION_HANDLERS:BOOL

Disables the mechanism that produces debug output for caught signals and unhandled exceptions (default: OFF)

HPX_WITH_DYNAMIC_HPX_MAIN:BOOL

Enable dynamic overload of system main() (Linux and Apple only, default: ON)

HPX_WITH_FAULT_TOLERANCE:BOOL

Build HPX to tolerate failures of nodes, i.e. ignore errors in active communication channels (default: OFF)

HPX_WITH_FULL_RPATH:BOOL

Build and link HPX libraries and executables with full RPATHs (default: ON)

HPX_WITH_GCC_VERSION_CHECK:BOOL

Don't ignore version reported by gcc (default: ON)

HPX_WITH_GENERIC_CONTEXT_COROUTINES:BOOL

Use Boost.Context as the underlying coroutines context switch implementation.

HPX_WITH_GIT_HOOKS:BOOL

Enable git commit hooks for formatting checks on commit (default: OFF).

HPX_WITH_HIDDEN_VISIBILITY:BOOL

Use -fvisibility=hidden for builds on platforms which support it (default OFF)

HPX_WITH_HIP:BOOL

Enable compilation with HIPCC (default: OFF)

HPX_WITH_HIPSYCL:BOOL

Use hipsycl cmake integration (default: OFF)

HPX_WITH_IGNORE_COMPILER_COMPATIBILITY:BOOL

Ignore compiler incompatibility in dependent projects (default: ON).

HPX_WITH_LOGGING:BOOL

Build HPX with logging enabled (default: ON).

HPX_WITH_MALLOC:STRING

Define which allocator should be linked in. Options are: system, tcmalloc, jemalloc, mimalloc, tbbmalloc, and custom (default is: tcmalloc)

HPX_WITH_MODULES_AS_STATIC_LIBRARIES:BOOL

Compile HPX modules as STATIC (whole-archive) libraries instead of OBJECT libraries (Default: ON)

HPX_WITH_MODULE_COMPATIBILITY_HEADERS:BOOL

Generate backwards-compatibility headers for HPX Modules (default: OFF)

HPX_WITH_NICE_THREADLEVEL:BOOL

Set HPX worker threads to have high NICE level (may impact performance) (default: OFF)

HPX_WITH_PARCEL_COALESCING:BOOL

Enable the parcel coalescing plugin (default: ON).

HPX_WITH_PKGCONFIG:BOOL

Enable generation of pkgconfig files (default: ON on Linux without CUDA/HIP, otherwise OFF)

HPX_WITH_PRECOMPILED_HEADERS:BOOL

Enable precompiled headers for certain build targets (experimental) (default OFF)

HPX_WITH_PSEUDO_DEPENDENCIES:BOOL

Turn compiler warnings into errors (default: ON)

HPX_WITH_RUN_MAIN_EVERYWHERE:BOOL

Run hpx_main by default on all localities (default: OFF, deprecated, will be removed).

HPX_WITH_STACKOVERFLOW_DETECTION:BOOL

Enable stackoverflow detection for HPX threads/coroutines (default: OFF, debug: ON).

HPX_WITH_STATIC_LINKING:BOOL

Compile HPX statically linked libraries (Default: OFF)

HPX_WITH_SUPPORT_NO_UNIQUE_ADDRESS_ATTRIBUTE:BOOL

Enable the use of the `[[no_unique_address]]` attribute (default: ON)

HPX_WITH_SYCL:BOOL

Enable support for Sycl (default: OFF)

HPX_WITH_SYCL_FLAGS:STRING

Sycl compile flags for selecting specific targets (default: empty)

HPX_WITH_UNITY_BUILD:BOOL

Enable unity build for certain build targets (default OFF)

HPX_WITH_VIM_YCM:BOOL

Generate HPX completion file for VIM YouCompleteMe plugin

HPX_WITH_ZERO_COPY_SERIALIZATION_THRESHOLD:STRING

The threshold in bytes to when perform zero copy optimizations (default: 8192)

Build Targets options

- *HPX_WITH_ASIO_TAG:STRING*
- *HPX_WITH_COMPILE_ONLY_TESTS:BOOL*
- *HPX_WITH_DISTRIBUTED_RUNTIME:BOOL*
- *HPX_WITH_DOCUMENTATION:BOOL*
- *HPX_WITH_DOCUMENTATION_OUTPUT_FORMATS:STRING*
- *HPX_WITH_EXAMPLES:BOOL*
- *HPX_WITH_EXAMPLES_HDF5:BOOL*
- *HPX_WITH_EXAMPLES_OPENMP:BOOL*
- *HPX_WITH_EXAMPLES_QT4:BOOL*
- *HPX_WITH_EXAMPLES_QTHREADS:BOOL*
- *HPX_WITH_EXAMPLES_TASKBENCH:BOOL*
- *HPX_WITH_EXAMPLES_TBB:BOOL*
- *HPX_WITH_EXECUTABLE_PREFIX:STRING*
- *HPX_WITH_FAIL_COMPILE_TESTS:BOOL*

- *HPX_WITH_FETCH_APEX:BOOL*
- *HPX_WITH_FETCH_ASIO:BOOL*
- *HPX_WITH_FETCH_BOOST:BOOL*
- *HPX_WITH_FETCH_GASNET:BOOL*
- *HPX_WITH_FETCH_HWLOC:BOOL*
- *HPX_WITH_FETCH_LCI:BOOL*
- *HPX_WITH_FETCH_LCW:BOOL*
- *HPX_WITH_IO_COUNTERS:BOOL*
- *HPX_WITH_LCI_BOOTSTRAP_MPI:BOOL*
- *HPX_WITH_LCI_TAG:STRING*
- *HPX_WITH_LCW_TAG:STRING*
- *HPX_WITH_NANOBENCH:BOOL*
- *HPX_WITH_PARALLEL_LINK_JOBS:STRING*
- *HPX_WITH_TESTS:BOOL*
- *HPX_WITH_TESTS_BENCHMARKS:BOOL*
- *HPX_WITH_TESTS_EXAMPLES:BOOL*
- *HPX_WITH_TESTS_EXTERNAL_BUILD:BOOL*
- *HPX_WITH_TESTS_HEADERS:BOOL*
- *HPX_WITH_TESTS_REGRESSIONS:BOOL*
- *HPX_WITH_TESTS_UNIT:BOOL*
- *HPX_WITH_THRUST:BOOL*
- *HPX_WITH_TOOLS:BOOL*

HPX_WITH_ASIO_TAG:STRING

Asio repository tag or branch

HPX_WITH_COMPILE_ONLY_TESTS:BOOL

Create build system support for compile time only HPX tests (default ON)

HPX_WITH_DISTRIBUTED_RUNTIME:BOOL

Enable the distributed runtime (default: ON). Turning off the distributed runtime completely disallows the creation and use of components and actions. Turning this option off is experimental!

HPX_WITH_DOCUMENTATION:BOOL

Build the HPX documentation (default OFF).

HPX_WITH_DOCUMENTATION_OUTPUT_FORMATS:STRING

List of documentation output formats to generate. Valid options are `html;singlehtml;latexpdf;man`. Multiple values can be separated with semicolons. (default `html`).

HPX_WITH_EXAMPLES:BOOL

Build the HPX examples (default ON)

HPX_WITH_EXAMPLES_HDF5:BOOL

Enable examples requiring HDF5 support (default: OFF).

HPX_WITH_EXAMPLES_OPENMP:BOOL

Enable examples requiring OpenMP support (default: OFF).

HPX_WITH_EXAMPLES_QT4:BOOL

Enable examples requiring Qt4 support (default: OFF).

HPX_WITH_EXAMPLES_QTHREADS:BOOL

Enable examples requiring QThreads support (default: OFF).

HPX_WITH_EXAMPLES_TASKBENCH:BOOL

Enable the task-bench example (fetched from upstream StanfordLegion/task-bench, default: OFF).

HPX_WITH_EXAMPLES_TBB:BOOL

Enable examples requiring TBB support (default: OFF).

HPX_WITH_EXECUTABLE_PREFIX:STRING

Executable prefix (default none), **‘hpx_’** useful for system install.

HPX_WITH_FAIL_COMPILE_TESTS:BOOL

Create build system support for fail compile HPX tests (default ON)

HPX_WITH_FETCH_APEX:BOOL

Use FetchContent to fetch APEX. By default an installed APEX will be used. (default: OFF)

HPX_WITH_FETCH_ASIO:BOOL

Use FetchContent to fetch Asio. By default an installed Asio will be used. (default: OFF)

HPX_WITH_FETCH_BOOST:BOOL

Use FetchContent to fetch Boost. By default an installed Boost will be used. (default: OFF)

HPX_WITH_FETCH_GASNET:BOOL

Use FetchContent to fetch GASNET. By default an installed GASNET will be used. (default: OFF).

HPX_WITH_FETCH_HWLOC:BOOL

Use FetchContent to fetch Hwloc. By default an installed Hwloc will be used. (default: OFF)

HPX_WITH_FETCH_LCI:BOOL

Use FetchContent to fetch LCI. By default an installed LCI will be used. (default: OFF)

HPX_WITH_FETCH_LCW:BOOL

Use FetchContent to fetch LCW. By default an installed LCW will be used. (default: OFF)

HPX_WITH_IO_COUNTERS:BOOL

Enable IO counters (default: ON)

HPX_WITH_LCI_BOOTSTRAP_MPI:BOOL

Configure the autofetched LCI with mpi bootstrap support (default: OFF)

HPX_WITH_LCI_TAG:STRING

LCI repository tag or branch

HPX_WITH_LCW_TAG:STRING

LCW repository tag or branch

HPX_WITH_NANOBENCH:BOOL

Use Nanobench for performance tests. Nanobench will be fetched using FetchContent (default: OFF)

HPX_WITH_PARALLEL_LINK_JOBS:STRING

Number of Parallel link jobs while building hpx (only for Ninja as generator) (default 2)

HPX_WITH_TESTS:BOOL

Build the HPX tests (default ON)

HPX_WITH_TESTS_BENCHMARKS:BOOL

Build HPX benchmark tests (default: ON)

HPX_WITH_TESTS_EXAMPLES:BOOL

Add HPX examples as tests (default: ON)

HPX_WITH_TESTS_EXTERNAL_BUILD:BOOL

Build external cmake build tests (default: ON)

HPX_WITH_TESTS_HEADERS:BOOL

Build HPX header tests (default: OFF)

HPX_WITH_TESTS_REGRESSIONS:BOOL

Build HPX regression tests (default: ON)

HPX_WITH_TESTS_UNIT:BOOL

Build HPX unit tests (default: ON)

HPX_WITH_THRUST:BOOL

Enable support for NVIDIA Thrust integration (default: ON when CUDA is enabled, OFF otherwise)

HPX_WITH_TOOLS:BOOL

Build HPX tools (default: OFF)

Thread Manager options

- *HPX_COROUTINES_WITH_SWAP_CONTEXT_EMULATION:BOOL*
- *HPX_COROUTINES_WITH_THREAD_SCHEDULE_HINT_RUNS_AS_CHILD:BOOL*
- *HPX_WITH_COROUTINE_COUNTERS:BOOL*
- *HPX_WITH_IO_POOL:BOOL*
- *HPX_WITH_MAX_CPU_COUNT:STRING*
- *HPX_WITH_MAX_NUMA_DOMAIN_COUNT:STRING*
- *HPX_WITH_SCHEDULER_LOCAL_STORAGE:BOOL*
- *HPX_WITH_SPINLOCK_DEADLOCK_DETECTION:BOOL*
- *HPX_WITH_SPINLOCK_POOL_NUM:STRING*
- *HPX_WITH_STACKTRACES:BOOL*
- *HPX_WITH_STACKTRACES_DEMANGLE_SYMBOLS:BOOL*
- *HPX_WITH_STACKTRACES_STATIC_SYMBOLS:BOOL*
- *HPX_WITH_THREAD_BACKTRACE_DEPTH:STRING*
- *HPX_WITH_THREAD_BACKTRACE_ON_SUSPENSION:BOOL*
- *HPX_WITH_THREAD_CREATION_AND_CLEANUP_RATES:BOOL*
- *HPX_WITH_THREAD_CUMULATIVE_COUNTS:BOOL*

- *HPX_WITH_THREAD_IDLE_RATES:BOOL*
- *HPX_WITH_THREAD_LOCAL_STORAGE:BOOL*
- *HPX_WITH_THREAD_MANAGER_IDLE_BACKOFF:BOOL*
- *HPX_WITH_THREAD_QUEUE_WAITTIME:BOOL*
- *HPX_WITH_THREAD_STACK_MMAP:BOOL*
- *HPX_WITH_THREAD_STEALING_COUNTS:BOOL*
- *HPX_WITH_THREAD_TARGET_ADDRESS:BOOL*
- *HPX_WITH_TIMER_POOL:BOOL*
- *HPX_WITH_WORK_REQUESTING_SCHEDULERS:BOOL*

HPX_COROUTINES_WITH_SWAP_CONTEXT_EMULATION:BOOL

Emulate SwapContext API for coroutines (Windows only, default: OFF)

HPX_COROUTINES_WITH_THREAD_SCHEDULE_HINT_RUNS_AS_CHILD:BOOL

Futures attempt to run associated threads directly if those have not been started (default: OFF)

HPX_WITH_COROUTINE_COUNTERS:BOOL

Enable keeping track of coroutine creation and rebind counts (default: OFF)

HPX_WITH_IO_POOL:BOOL

Disable internal IO thread pool, do not change if not absolutely necessary (default: ON)

HPX_WITH_MAX_CPU_COUNT:STRING

HPX applications will not use more that this number of OS-Threads (empty string means dynamic) (default: "")

HPX_WITH_MAX_NUMA_DOMAIN_COUNT:STRING

HPX applications will not run on machines with more NUMA domains (default: 8)

HPX_WITH_SCHEDULER_LOCAL_STORAGE:BOOL

Enable scheduler local storage for all HPX schedulers (default: OFF)

HPX_WITH_SPINLOCK_DEADLOCK_DETECTION:BOOL

Enable spinlock deadlock detection (default: OFF)

HPX_WITH_SPINLOCK_POOL_NUM:STRING

Number of elements a spinlock pool manages (default: 128)

HPX_WITH_STACKTRACES:BOOL

Attach backtraces to HPX exceptions (default: ON)

HPX_WITH_STACKTRACES_DEMANGLE_SYMBOLS:BOOL

Thread stack back trace symbols will be demangled (default: ON)

HPX_WITH_STACKTRACES_STATIC_SYMBOLS:BOOL

Thread stack back trace will resolve static symbols (default: OFF)

HPX_WITH_THREAD_BACKTRACE_DEPTH:STRING

Thread stack back trace depth being captured (default: 20)

HPX_WITH_THREAD_BACKTRACE_ON_SUSPENSION:BOOL

Enable thread stack back trace being captured on suspension (default: OFF)

HPX_WITH_THREAD_CREATION_AND_CLEANUP_RATES:BOOL

Enable measuring thread creation and cleanup times (default: OFF)

HPX_WITH_THREAD_CUMULATIVE_COUNTS:BOOL

Enable keeping track of cumulative thread counts in the schedulers (default: ON)

HPX_WITH_THREAD_IDLE_RATES:BOOL

Enable measuring the percentage of overhead times spent in the scheduler (default: OFF)

HPX_WITH_THREAD_LOCAL_STORAGE:BOOL

Enable thread local storage for all HPX threads (default: OFF)

HPX_WITH_THREAD_MANAGER_IDLE_BACKOFF:BOOL

HPX scheduler threads do exponential backoff on idle queues (default: ON)

HPX_WITH_THREAD_QUEUE_WAITTIME:BOOL

Enable collecting queue wait times for threads (default: OFF)

HPX_WITH_THREAD_STACK_MMAP:BOOL

Use mmap for stack allocation on appropriate platforms

HPX_WITH_THREAD_STEALING_COUNTS:BOOL

Enable keeping track of counts of thread stealing incidents in the schedulers (default: OFF)

HPX_WITH_THREAD_TARGET_ADDRESS:BOOL

Enable storing target address in thread for NUMA awareness (default: OFF)

HPX_WITH_TIMER_POOL:BOOL

Disable internal timer thread pool, do not change if not absolutely necessary (default: ON)

HPX_WITH_WORK_REQUESTING_SCHEDULERS:BOOL

Enable work requesting scheduler (default: ON)

AGAS options

- *HPX_WITH_AGAS_DUMP_REFCNT_ENTRIES:BOOL*

HPX_WITH_AGAS_DUMP_REFCNT_ENTRIES:BOOL

Enable dumps of the AGAS refcnt tables to logs (default: OFF)

Parcelport options

- *HPX_WITH_NETWORKING:BOOL*
- *HPX_WITH_PARCELPORTR_ACTION_COUNTERS:BOOL*
- *HPX_WITH_PARCELPORTR_COUNTERS:BOOL*
- *HPX_WITH_PARCELPORTR_GASNET:BOOL*
- *HPX_WITH_PARCELPORTR_LCI:BOOL*
- *HPX_WITH_PARCELPORTR_LCI_LOG:STRING*
- *HPX_WITH_PARCELPORTR_LCI_PCOUNTER:STRING*
- *HPX_WITH_PARCELPORTR_LCW:BOOL*
- *HPX_WITH_PARCELPORTR_LCW_LOG:STRING*

- *HPX_WITH_PARCELPOR_T_LCW_PCOUNTER:STRING*
- *HPX_WITH_PARCELPOR_T_LIBFABRIC:BOOL*
- *HPX_WITH_PARCELPOR_T_MPI:BOOL*
- *HPX_WITH_PARCELPOR_T_TCP:BOOL*
- *HPX_WITH_PARCEL_PROFILING:BOOL*

HPX_WITH_NETWORKING:BOOL

Enable support for networking and multi-node runs (default: ON)

HPX_WITH_PARCELPOR_T_ACTION_COUNTERS:BOOL

Enable performance counters reporting parcellport statistics on a per-action basis.

HPX_WITH_PARCELPOR_T_COUNTERS:BOOL

Enable performance counters reporting parcellport statistics.

HPX_WITH_PARCELPOR_T_GASNET:BOOL

Enable the GASNET based parcellport.

HPX_WITH_PARCELPOR_T_LCI:BOOL

Enable the LCI based parcellport.

HPX_WITH_PARCELPOR_T_LCI_LOG:STRING

Enable the LCI-parcellport-specific logger

HPX_WITH_PARCELPOR_T_LCI_PCOUNTER:STRING

Enable the LCI-parcellport-specific performance counter

HPX_WITH_PARCELPOR_T_LCW:BOOL

Enable the LCW based parcellport.

HPX_WITH_PARCELPOR_T_LCW_LOG:STRING

Enable the LCW-parcellport-specific logger

HPX_WITH_PARCELPOR_T_LCW_PCOUNTER:STRING

Enable the LCW-parcellport-specific performance counter

HPX_WITH_PARCELPOR_T_LIBFABRIC:BOOL

Enable the libfabric based parcellport. This is currently an experimental feature

HPX_WITH_PARCELPOR_T_MPI:BOOL

Enable the MPI based parcellport.

HPX_WITH_PARCELPOR_T_TCP:BOOL

Enable the TCP based parcellport.

HPX_WITH_PARCEL_PROFILING:BOOL

Enable profiling data for parcels

Profiling options

- *HPX_LIKWID_WITH_LIKWID:BOOL*
- *HPX_WITH_APEX:BOOL*
- *HPX_WITH_ITTNOTIFY:BOOL*
- *HPX_WITH_PAPI:BOOL*
- *HPX_WITH_TRACY:BOOL*

HPX_LIKWID_WITH_LIKWID:BOOL

Enable Likwid instrumentation support.

HPX_WITH_APEX:BOOL

Enable APEX instrumentation support.

HPX_WITH_ITTNOTIFY:BOOL

Enable Amplifier (ITT) instrumentation support.

HPX_WITH_PAPI:BOOL

Enable the PAPI based performance counter.

HPX_WITH_TRACY:BOOL

Enable Tracy instrumentation support.

Debugging options

- *HPX_WITH_ASSERTS_AS_CONTRACT_ASSERTS:BOOL*
- *HPX_WITH_ATTACH_DEBUGGER_ON_TEST_FAILURE:BOOL*
- *HPX_WITH_CONTRACTS:BOOL*
- *HPX_WITH_PARALLEL_TESTS_BIND_NONE:BOOL*
- *HPX_WITH_SANITIZERS:BOOL*
- *HPX_WITH_TESTS_COMMAND_LINE:STRING*
- *HPX_WITH_TESTS_DEBUG_LOG:BOOL*
- *HPX_WITH_TESTS_DEBUG_LOG_DESTINATION:STRING*
- *HPX_WITH_TESTS_MAX_THREADS_PER_LOCALITY:STRING*
- *HPX_WITH_THREAD_DEBUG_INFO:BOOL*
- *HPX_WITH_THREAD_DESCRIPTION_FULL:BOOL*
- *HPX_WITH_THREAD_GUARD_PAGE:BOOL*
- *HPX_WITH_VALGRIND:BOOL*
- *HPX_WITH_VERIFY_LOCKS:BOOL*
- *HPX_WITH_VERIFY_LOCKS_BACKTRACE:BOOL*

HPX_WITH_ASSERTS_AS_CONTRACT_ASSERTS:BOOL

Swap `hpx_assert` with `hpx_contract_assert`

HPX_WITH_ATTACH_DEBUGGER_ON_TEST_FAILURE:BOOL

Break the debugger if a test has failed (default: OFF)

HPX_WITH_CONTRACTS:BOOL

Enable C++ contracts support in HPX

HPX_WITH_PARALLEL_TESTS_BIND_NONE:BOOL

Pass `-hpx:bind=none` to tests that may run in parallel (cmake `-j` flag) (default: OFF)

HPX_WITH_SANITIZERS:BOOL

Configure with sanitizer instrumentation support.

HPX_WITH_TESTS_COMMAND_LINE:STRING

Add given command line options to all tests run

HPX_WITH_TESTS_DEBUG_LOG:BOOL

Turn on debug logs (`-hpx:debug-hpx-log`) for tests (default: OFF)

HPX_WITH_TESTS_DEBUG_LOG_DESTINATION:STRING

Destination for test debug logs (default: `cout`)

HPX_WITH_TESTS_MAX_THREADS_PER_LOCALITY:STRING

Maximum number of threads to use for tests (default: 0, use the number of threads specified by the test)

HPX_WITH_THREAD_DEBUG_INFO:BOOL

Enable thread debugging information (default: OFF, implicitly enabled in debug builds)

HPX_WITH_THREAD_DESCRIPTION_FULL:BOOL

Use function address for thread description (default: OFF)

HPX_WITH_THREAD_GUARD_PAGE:BOOL

Enable thread guard page (default: ON)

HPX_WITH_VALGRIND:BOOL

Enable Valgrind instrumentation support.

HPX_WITH_VERIFY_LOCKS:BOOL

Enable lock verification code (default: OFF, enabled in debug builds)

HPX_WITH_VERIFY_LOCKS_BACKTRACE:BOOL

Enable thread stack back trace being captured on lock registration (to be used in combination with `HPX_WITH_VERIFY_LOCKS=ON`, default: OFF)

Modules options

- *HPX_ALLOCATOR_SUPPORT_WITH_CACHING:BOOL*
- *HPX_COMMAND_LINE_HANDLING_LOCAL_WITH_JSON_CONFIGURATION_FILES:BOOL*
- *HPX_DATASTRUCTURES_WITH_ADAPT_STD_TUPLE:BOOL*
- *HPX_DATASTRUCTURES_WITH_ADAPT_STD_VARIANT:BOOL*
- *HPX_FILESYSTEM_WITH_BOOST_FILESYSTEM_COMPATIBILITY:BOOL*
- *HPX_FUNCTIONAL_WITH_BOOST_PLACEHOLDERS:BOOL*
- *HPX_IOSTREAM_WITH_WIDE_STREAMS:BOOL*
- *HPX_ITERATOR_SUPPORT_WITH_BOOST_ITERATOR_TRAVERSAL_TAG_COMPATIBILITY:BOOL*

- *HPX_LOGGING_WITH_SEPARATE_DESTINATIONS:BOOL*
- *HPX_SERIALIZATION_WITH_ALLOW_CONST_TUPLE_MEMBERS:BOOL*
- *HPX_SERIALIZATION_WITH_ALLOW_RAW_POINTER_SERIALIZATION:BOOL*
- *HPX_SERIALIZATION_WITH_ALL_TYPES_ARE_BITWISE_SERIALIZABLE:BOOL*
- *HPX_SERIALIZATION_WITH_BOOST_TYPES:BOOL*
- *HPX_SERIALIZATION_WITH_SUPPORTS_ENDIANESS:BOOL*
- *HPX_TOPOLOGY_WITH_ADDITIONAL_HWLOC_TESTING:BOOL*
- *HPX_WITH_POWER_COUNTER:BOOL*
- *HPX_WITH_SUPERVISION:BOOL*

HPX_ALLOCATOR_SUPPORT_WITH_CACHING:BOOL

Enable caching allocator. (default: ON)

HPX_COMMAND_LINE_HANDLING_LOCAL_WITH_JSON_CONFIGURATION_FILES:BOOL

Enable reading JSON formatted configuration files on the command line.

(default: OFF)

HPX_DATASTRUCTURES_WITH_ADAPT_STD_TUPLE:BOOL

Enable compatibility of `hpx::get` with `std::tuple`. (default: ON)

HPX_DATASTRUCTURES_WITH_ADAPT_STD_VARIANT:BOOL

Enable compatibility of `hpx::get` with `std::variant`.

(default: OFF)

HPX_FILESYSTEM_WITH_BOOST_FILESYSTEM_COMPATIBILITY:BOOL

Enable `Boost.FileSystem` compatibility. (default: OFF)

HPX_FUNCTIONAL_WITH_BOOST_PLACEHOLDERS:BOOL

Enable support for Boost placeholder types. (default: OFF)

HPX_IOSTREAM_WITH_WIDE_STREAMS:BOOL

Enable wide IO streams. (default: OFF)

HPX_ITERATOR_SUPPORT_WITH_BOOST_ITERATOR_TRAVERSAL_TAG_COMPATIBILITY:BOOL

Enable `Boost.Iterator` traversal tag compatibility. (default: OFF)

HPX_LOGGING_WITH_SEPARATE_DESTINATIONS:BOOL

Enable separate logging channels for AGAS, timing, and parcel transport. (default: ON)

HPX_SERIALIZATION_WITH_ALLOW_CONST_TUPLE_MEMBERS:BOOL

Enable serializing `std::tuple` with `const` members. (default: OFF)

HPX_SERIALIZATION_WITH_ALLOW_RAW_POINTER_SERIALIZATION:BOOL

Enable serializing raw pointers. (default: OFF)

HPX_SERIALIZATION_WITH_ALL_TYPES_ARE_BITWISE_SERIALIZABLE:BOOL

Assume all types are bitwise serializable. (default: OFF)

HPX_SERIALIZATION_WITH_BOOST_TYPES:BOOL

Enable serialization of certain Boost types. (default: OFF)

HPX_SERIALIZATION_WITH_SUPPORTS_ENDIANESS:BOOL

Support endian conversion on inout and output archives. (default: OFF)

HPX_TOPOLOGY_WITH_ADDITIONAL_HWLOC_TESTING:BOOL

Enable HWLOC filtering that makes it report no cores, this is purely an option supporting better testing - do not enable under normal circumstances. (default: OFF)

HPX_WITH_POWER_COUNTER:BOOL

Enable use of performance counters based on pwr library (default: OFF)

HPX_WITH_SUPERVISION:BOOL

Enable supervision functionalities. (default: OFF)

Additional tools and libraries used by HPX

Here is a list of additional libraries and tools that are either optionally supported by the build system or are optionally required for certain examples or tests. These libraries and tools can be detected by the *HPX* build system.

Each of the tools or libraries listed here will be automatically detected if they are installed in some standard location. If a tool or library is installed in a different location, you can specify its base directory by appending `_ROOT` to the variable name as listed below. For instance, to configure a custom directory for Boost, specify `Boost_ROOT=/custom/boost/root`.

Boost_ROOT:PATH

Specifies where to look for the Boost installation to be used for compiling *HPX*. Set this if CMake is not able to locate a suitable version of Boost. The directory specified here can be either the root of an installed Boost distribution or the directory where you unpacked and built Boost without installing it (with staged libraries).

Hwloc_ROOT:PATH

Specifies where to look for the hwloc library. Set this if CMake is not able to locate a suitable version of hwloc. Hwloc provides platform- independent support for extracting information about the used hardware architecture (number of cores, number of NUMA domains, hyperthreading, etc.). *HPX* utilizes this information if available.

Papi_ROOT:PATH

Specifies where to look for the PAPI library. The PAPI library is needed to compile a special component exposing PAPI hardware events and counters as *HPX* performance counters. This is not available on the Windows platform.

Amplifier_ROOT:PATH

Specifies where to look for one of the tools of the Intel Parallel Studio product, either Intel Amplifier or Intel Inspector. This should be set if the CMake variable `HPX_USE_ITT_NOTIFY` is set to ON. Enabling ITT support in *HPX* will integrate any application with the mentioned Intel tools, which customizes the generated information for your application and improves the generated diagnostics.

In addition, some of the examples may need the following variables:

Hdf5_ROOT:PATH

Specifies where to look for the Hierarchical Data Format V5 (HDF5) include files and libraries.

2.4.5 Using *HPX* on Compiler Explorer

Compiler Explorer⁶³ (CE) is a browser-based tool that compiles C++ code in real time and shows assembly, program output, and optimization results side by side. It is widely used for quick experiments, conference demos, and sharing reproducible snippets. This page explains how to build *HPX* for CE, how to link against its static libraries from a raw compiler command, and how to use the *sandbox header* that *HPX* ships specifically for constrained environments.

Sandbox environment constraints

CE executes compiled programs inside a Linux container (nsjail) with no network access and limited resources. The constraints that matter most for *HPX* are:

- No network namespace — sockets cannot be opened. Any attempt to initialise the distributed runtime will fail with `EACCES` at startup.
- A small PID budget (typically 32 PIDs) — *HPX* starts cleanly within this limit when the worker thread count is kept low.
- A short execution timeout — CE's public instance allows roughly 10 seconds. An *HPX* fork or self-hosted instance can relax this to 30 seconds, which is enough for benchmarks on problem sizes of one million elements or more.
- Static linking is required — the sandbox does not augment `LD_LIBRARY_PATH`, so shared *HPX* libraries are not found at runtime.

Building *HPX* for Compiler Explorer

HPX ships a `godbolt-minimal` configure preset in `CMakePresets.json` that encodes all the flags needed for a CE-compatible build:

```
$ cmake --preset godbolt-minimal
$ cmake --build --preset godbolt-minimal
```

This preset enables:

- `HPX_WITH_STATIC_LINKING=ON` — bundles everything into the `.a` archives that CE links against.
- `HPX_WITH_NETWORKING=OFF` — disables the `parcels` so *HPX* never attempts to open a network socket.
- `HPX_WITH_FETCH_ASIO=ON` — downloads Asio via `FetchContent`, removing the need for a system-level Asio installation.
- `HPX_WITH_MALLOC=system` — uses the system allocator; avoids a `jemalloc` or `tcmalloc` dependency.
- `HPX_WITH_TESTS=OFF`, `HPX_WITH_EXAMPLES=OFF`, `HPX_WITH_DOCUMENTATION=OFF` — skips everything that CE does not need.

The preset produces four static libraries under `build/godbolt-minimal/lib/`:

```
libhpx_wrap.a
libhpx_init.a
libhpx.a
libhpx_core.a
```

⁶³ <https://godbolt.org>

These are the only artifacts that CE consumes.

Note: The `HPX_WITH_CXX_STANDARD` variable pins the C++ standard for the build (e.g. `-DHPX_WITH_CXX_STANDARD=20`). This is *HPX*'s own cache variable and is distinct from `CMAKE_CXX_STANDARD`, which *HPX* rejects unless `HPX_USE_CMAKE_CXX_STANDARD` is also set.

Linking without CMake

CE's backend compiles user code with a raw `g++` or `clang++` invocation. The complete set of flags needed is:

```
$ g++ -std=c++20 -O2 -o my_program my_program.cpp \
    -isystem /path/to/hpx/include \
    -isystem /path/to/boost/include \
    -L/path/to/hpx/lib \
    -DHPX_APPLICATION_EXPORTS \
    -Wl,-wrap=main \
    -lhpx_wrap -lhpx_init -lhpx -lhpx_core \
    -lpthread -ldl -lrt
```

Two details here are easy to get wrong:

Library link order. The order `hpx_wrap` → `hpx_init` → `hpx` → `hpx_core` must be preserved. Reversing it produces undefined-reference errors because `hpx_wrap` depends on symbols in `hpx_init`, which depends on the full runtime in `hpx`, which in turn depends on the core library.

The `-Wl,-wrap=main` flag. Including `hpx/hpx_main.hpp` (see *Re-use the `main()` function as the main HPX entry point*) works by re-routing control through *HPX*'s own entry point before the user's `main` is called. On Linux this is implemented via the linker's `--wrap` option; the flag must therefore appear on the *linker* command line, not merely in the compile flags. Without it, the *HPX* runtime is never initialised and all API calls crash at startup. See *Linux implementation* for a detailed explanation of the mechanism.

Important: `-DHPX_APPLICATION_EXPORTS` must be passed as a preprocessor definition when compiling application code against the static libraries. Omitting it causes link failures related to *HPX*'s symbol visibility macros.

Writing *HPX* code for Compiler Explorer

The simplest way to write an *HPX* program for CE is to include `hpx/hpx_main.hpp`. This header arranges for the *HPX* runtime to be initialised before `main` runs, so the body of `main` can call any *HPX* API function directly:

```
#include <hpx/hpx_main.hpp>
#include <hpx/algorithm.hpp>
#include <hpx/execution.hpp>

#include <iostream>
#include <numeric>
#include <vector>

int main()
{
    std::vector<int> v(1'000'000);
```

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```

std::iota(v.begin(), v.end(), 0);

long long sum = hpx::transform_reduce(
    hpx::execution::par, v.begin(), v.end(), 0LL,
    std::plus<>{}, [](int x) { return static_cast<long long>(x); });

std::cout << "sum = " << sum << "\n";
}

```

Caution: Include `hpx/hpx_main.hpp` in exactly one translation unit — the file that contains `main`. Including it in more than one file causes a multiple definition error for the `include_libhpx_wrap` variable that controls runtime initialisation.

The `hpx/experimental/sandbox.hpp` header

HPX ships a header-only toolkit at `hpx/experimental/sandbox.hpp` designed specifically for code running in constrained environments. It provides:

- **Environment introspection** — `hpx::experimental::sandbox::detect_environment()` returns an `environment_info` struct describing the number of physical cores, processing units, NUMA domains, and active HPX worker threads. The `is_sandbox` flag is set when the `COMPILER_EXPLORER` or `HPX_SANDBOX` environment variable is present, letting code adapt its behaviour automatically.
- **Timing** — `hpx::experimental::sandbox::measure(fn, iterations)` runs a callable `fn` for `iterations` repetitions (with one warmup pass) and returns the mean execution time in milliseconds.
- **Comparative benchmarking** — `hpx::experimental::sandbox::benchmark(label, seq_fn, par_fn, iterations)` measures both a sequential and a parallel version of the same computation, computes speedup and parallel efficiency, and returns a `benchmark_report` struct. Calling `report.print(std::cout)` produces a fixed-width table that renders cleanly in CE's output pane.

A typical usage pattern looks like this:

```

#include <hpx/hpx_main.hpp>
#include <hpx/algorithm.hpp>
#include <hpx/execution.hpp>
#include <hpx/experimental/sandbox.hpp>

#include <algorithm>
#include <iostream>
#include <numeric>
#include <vector>

int main()
{
    namespace sb = hpx::experimental::sandbox;

    sb::describe_environment(std::cout);

    std::vector<int> data(500'000);
    std::iota(data.begin(), data.end(), 0);
}

```

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```

auto report = sb::benchmark(
    "transform_reduce",
    [&]() {
        hpx::transform_reduce(
            hpx::execution::seq,
            data.begin(), data.end(), 0LL,
            std::plus<{}>, [](int x) { return (long long)x; });
    },
    [&]() {
        hpx::transform_reduce(
            hpx::execution::par,
            data.begin(), data.end(), 0LL,
            std::plus<{}>, [](int x) { return (long long)x; });
    });

report.print(std::cout);
}

```

The output includes sequential and parallel mean times, speedup, parallel efficiency, and a verdict (Excellent scaling, Good scaling, Moderate scaling, Limited scaling, or No speedup).

Note: All functions in `hpx/experimental/sandbox.hpp` must be called from within a running *HPX* runtime, i.e., from an *HPX* thread. Calling them before `hpx::init` or after `hpx::finalize` is undefined behaviour. When using `hpx/hpx_main.hpp`, the body of `main` satisfies this requirement automatically.

Known limitations in sandboxed environments

- **Single locality only.** The distributed runtime can be compiled in (`HPX_WITH_DISTRIBUTED_RUNTIME=ON`) and actions on locality 0 work normally, but there is no way to launch a second locality from within CE's sandbox. Code that calls `hpx::find_all_localities()` or `hpx::get_num_localities()` will always see exactly one locality.
- **Networking is disabled.** `HPX_WITH_NETWORKING=OFF` means all parcelport-dependent functionality (remote actions, distributed data structures) is unavailable regardless of what the code requests.
- **Thread count is limited by the sandbox's CPU allocation.** CE's public instance typically exposes two cores. Use `--hpx:threads=N` on the command line or `hpx::init_params::cfg` in code to set the worker count explicitly rather than relying on hardware detection. See *Launching and configuring HPX applications* for details.
- **macOS link flag differs.** The `-Wl,-wrap=main` flag is Linux-specific. On macOS the linker uses `-Wl,-e,_initialize_main` instead. CE runs Linux containers, so this only matters when building the CE integration locally on macOS for testing.

2.4.6 Migration guide

The Migration Guide serves as a valuable resource for developers seeking to transition their parallel computing applications from different APIs (i.e. OpenMP, Intel Threading Building Blocks (TBB), MPI) to *HPX*. *HPX*, an advanced C++ library, offers a versatile and high-performance platform for parallel and distributed computing, providing a wide range of features and capabilities. This guide aims to assist developers in understanding the key differences between different APIs and *HPX*, and it provides step-by-step instructions for converting code to *HPX* code effectively.

Some general steps that can be used to migrate code to *HPX* code are the following:

1. Install *HPX* using the *Quick start* guide.
2. Include the *HPX* header files:

Add the necessary header files for *HPX* at the beginning of your code, such as:

```
#include <hpx/init.hpp>
```

3. Replace your code with *HPX* code using the guide that follows.
4. Use *HPX*-specific features and APIs:

HPX provides additional features and APIs that can be used to take advantage of the library's capabilities. For example, you can use the *HPX* asynchronous execution to express fine-grained tasks and dependencies, or utilize *HPX*'s distributed computing features for distributed memory systems.

5. Compile and run the *HPX* code:

Compile the converted code with the *HPX* library and run it using the appropriate *HPX* runtime environment.

OpenMP

The OpenMP API supports multi-platform shared-memory parallel programming in C/C++. Typically it is used for loop-level parallelism, but it also supports function-level parallelism. Below are some examples on how to convert OpenMP to *HPX* code:

OpenMP parallel for loop

Parallel for loop

OpenMP code:

```
#pragma omp parallel for
for (int i = 0; i < n; ++i) {
    // loop body
}
```

HPX equivalent:

```
#include <hpx/algorithm.hpp>

hpx::experimental::for_loop(hpx::execution::par, 0, n, [&](int i) {
    // loop body
});
```

In the above code, the OpenMP `#pragma omp parallel for` directive is replaced with `hpx::experimental::for_loop` from the *HPX* library. The loop body within the lambda function will be executed in parallel for each iteration.

Private variables

OpenMP code:

```
int x = 0;

#pragma omp parallel for private(x)
for (int i = 0; i < n; ++i) {
    // loop body
}
```

HPX equivalent:

```
#include <hpx/algorithm.hpp>

hpx::experimental::for_loop(hpx::execution::par, 0, n, [&](int i) {
    int x = 0; // Declare 'x' as a local variable inside the loop body
    // loop body
});
```

The variable *x* is declared as a local variable inside the loop body, ensuring that it is private to each thread.

Shared variables

OpenMP code:

```
int x = 0;

#pragma omp parallel for shared(x)
for (int i = 0; i < n; ++i) {
    // loop body
}
```

HPX equivalent:

```
#include <hpx/algorithm.hpp>

std::atomic<int> x = 0; // Declare 'x' as a shared variable outside the loop

hpx::experimental::for_loop(hpx::execution::par, 0, n, [&](int i) {
    // loop body
});
```

To ensure variable *x* is shared among all threads, you simply have to declare it as an atomic variable outside the *for_loop*.

Number of threads

OpenMP code:

```
#pragma omp parallel for num_threads(2)
for (int i = 0; i < n; ++i) {
    // loop body
}
```

HPX equivalent:

```
#include <hpx/algorithm.hpp>
#include <hpx/execution.hpp>

hpx::execution::experimental::num_cores nc(2);

hpx::experimental::for_loop(hpx::execution::par.with(nc), 0, n, [&](int i) {
    // loop body
});
```

To declare the number of threads to be used for the parallel region, you can use `hpx::execution::experimental::num_cores` and pass the number of cores (*nc*) to `hpx::experimental::for_loop` using `hpx::execution::par.with(nc)`. This example uses 2 threads for the parallel loop.

Reduction

OpenMP code:

```
int s = 0;

#pragma omp parallel for reduction(+: s)
for (int i = 0; i < n; ++i) {
    s += i;
    // loop body
}
```

HPX equivalent:

```
#include <hpx/algorithm.hpp>
#include <hpx/execution.hpp>

int s = 0;

hpx::experimental::for_loop(hpx::execution::par, 0, n, reduction(s, 0, plus<>()), [&
↪](int i, int& accum) {
    accum += i;
    // loop body
});
```

The reduction clause specifies that the variable *s* should be reduced across iterations using the *plus<>* operation. It initializes *s* to 0 at the beginning of the loop and accumulates the values of *s* from each iteration using the + operator. The lambda function representing the loop body takes two parameters: *i*, which represents the loop index, and *accum*, which is the reduction variable *s*. The lambda function is executed for each iteration of the loop. The reduction ensures that the *accum* value is correctly accumulated across different iterations and threads.

Schedule

OpenMP code:

```
int s = 0;

// static scheduling with chunk size 1000
#pragma omp parallel for schedule(static, 1000)
for (int i = 0; i < n; ++i) {
    // loop body
}
```

HPX equivalent:

```
#include <hpx/algorithm.hpp>
#include <hpx/execution.hpp>

hpx::execution::experimental::static_chunk_size cs(1000);

hpx::experimental::for_loop(hpx::execution::par.with(cs), 0, n, [&](int i) {
    // loop body
});
```

To define the scheduling type, you can use the corresponding execution policy from `hpx::execution::experimental`, define the chunk size (cs, here declared as 1000) and pass it to the to `hpx::experimental::for_loop` using `hpx::execution::par.with(cs)`.

Accordingly, other types of scheduling are available and can be used in a similar manner:

```
#include <hpx/execution.hpp>
hpx::execution::experimental::dynamic_chunk_size cs(1000);
```

```
#include <hpx/execution.hpp>
hpx::execution::experimental::guided_chunk_size cs(1000);
```

```
#include <hpx/execution.hpp>
hpx::execution::experimental::auto_chunk_size cs(1000);
```

OpenMP single thread

OpenMP code:

```
{ // parallel code
    #pragma omp single
    {
        // single-threaded code
    }
    // more parallel code
}
```

HPX equivalent:

```
#include <hpx/mutex.hpp>

hpx::mutex mtx;

{ // parallel code
  { // single-threaded code
    std::scoped_lock l(mtx);
  }
  // more parallel code
}
```

To make sure that only one thread accesses a specific code within a parallel section you can use `hpx::mutex` and `std::scoped_lock` to take ownership of the given mutex `mtx`. For more information about mutexes please refer to *Mutex*.

OpenMP tasks

Simple tasks

OpenMP code:

```
// executed asynchronously by any available thread
#pragma omp task
{
  // task code
}
```

HPX equivalent:

```
#include <hpx/future.hpp>

auto future = hpx::async([](){
  // task code
});
```

or

```
#include <hpx/future.hpp>

hpx::post([](){
  // task code
}); // fire and forget
```

The tasks in *HPX* can be defined simply by using the `async` function and passing as argument the code you wish to run asynchronously. Another alternative is to use `post` which is a fire-and-forget method.

Tip: If you think you will like to synchronize your tasks later on, we suggest you use `hpx::async` which provides synchronization options, while `hpx::post` explicitly states that there is no return value or way to synchronize with the function execution. Synchronization options are listed below.

Task wait

OpenMP code:

```
#pragma omp task
{
    // task code
}

#pragma omp taskwait
// code after completion of task
```

HPX equivalent:

```
#include <hpx/future.hpp>

hpx::async([](){
    // task code
}).get(); // wait for the task to complete

// code after completion of task
```

The `get()` function can be used to ensure that the task created with `hpx::async` is completed before the code continues executing beyond that point.

Multiple tasks synchronization

OpenMP code:

```
#pragma omp task
{
    // task 1 code
}

#pragma omp task
{
    // task 2 code
}

#pragma omp taskwait
// code after completion of both tasks 1 and 2
```

HPX equivalent:

```
#include <hpx/future.hpp>

auto future1 = hpx::async([](){
    // task 1 code
});

auto future2 = hpx::async([](){
    // task 2 code
});
```

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```
auto future = hpx::when_all(future1, future2).then([](auto&&){
    // code after completion of both tasks 1 and 2
});
```

If you would like to synchronize multiple tasks, you can use the `hpx::when_all` function to define which futures have to be ready and the `then()` function to declare what should be executed once these futures are ready.

Dependencies

OpenMP code:

```
int a = 10;
int b = 20;
int c = 0;

#pragma omp task depend(in: a, b) depend(out: c)
{
    // task code
    c = 100;
}
```

HPX equivalent:

```
#include <hpx/future.hpp>

int a = 10;
int b = 20;
int c = 0;

// Create a future representing 'a'
auto future_a = hpx::make_ready_future(a);

// Create a future representing 'b'
auto future_b = hpx::make_ready_future(b);

// Create a task that depends on 'a' and 'b' and executes 'task_code'
auto future_c = hpx::dataflow(
    []() {
        // task code
        return 100;
    },
    future_a, future_b);

c = future_c.get();
```

If one of the arguments of `hpx::dataflow` is a future, then it will wait for the future to be ready to launch the thread. Hence, to define the dependencies of tasks you have to create futures representing the variables that create dependencies and pass them as arguments to `hpx::dataflow`. `get()` is used to save the result of the future to the desired variable.

Nested tasks

OpenMP code:

```
#pragma omp task
{
    // Outer task code
    #pragma omp task
    {
        // Inner task code
    }
}
```

HPX equivalent:

```
#include <hpx/future.hpp>

auto future_outer = hpx::async([]() {
    // Outer task code

    hpx::async([]() {
        // Inner task code
    });
});
```

or

```
#include <hpx/future.hpp>

auto future_outer = hpx::post([]() { // fire and forget
    // Outer task code

    hpx::post([]() { // fire and forget
        // Inner task code
    });
});
```

If you have nested tasks, you can simply use nested `hpx::async` or `hpx::post` calls. The implementation is similar if you want to take care of synchronization:

OpenMP code:

```
#pragma omp taskwait
{
    // Outer task code
    #pragma omp taskwait
    {
        // Inner task code
    }
}
```

HPX equivalent:

```
#include <hpx/future.hpp>

auto future_outer = hpx::async([]() {
    // Outer task code

    hpx::async([]() {
        // Inner task code
    }).get();    // Wait for the inner task to complete
});

future_outer.get();    // Wait for the outer task to complete
```

Task yield

OpenMP code:

```
#pragma omp task
{
    // code before yielding
    #pragma omp taskyield
    // code after yielding
}
```

HPX equivalent:

```
#include <hpx/future.hpp>
#include <hpx/thread.hpp>

auto future = hpx::async([](){
    // code before yielding
});

// yield execution to potentially allow other tasks to run
hpx::this_thread::yield();

// code after yielding
```

After creating a task using `hpx::async`, `hpx::this_thread::yield` can be used to reschedule the execution of threads, allowing other threads to run.

Task group

OpenMP code:

```
#pragma omp taskgroup
{
    #pragma omp task
    {
        // task 1 code
    }
}
```

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```

#pragma omp task
{
    // task 2 code
}

```

HPX equivalent:

```

#include <hpx/task_group.hpp>

// Declare a task group
hpx::experimental::task_group tg;

// Run the tasks
tg.run([](){
    // task 1 code
});
tg.run(
    // task 2 code
);

// Wait for the task group
tg.wait();

```

To create task groups, you can use `hpx::experimental::task_group`. The function `run()` can be used to run each task within the task group, while `wait()` can be used to achieve synchronization. If you do not care about waiting for the task group to complete its execution, you can simply remove the `wait()` function.

OpenMP sections

OpenMP code:

```

#pragma omp sections
{
    #pragma omp section
    // section 1 code
    #pragma omp section
    // section 2 code
} // implicit synchronization

```

HPX equivalent:

```

#include <hpx/future.hpp>

auto future_section1 = hpx::async([](){
    // section 1 code
});
auto future_section2 = hpx::async([](){
    // section 2 code
});

```

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```
// synchronization: wait for both sections to complete
hpx::wait_all(future_section1, future_section2);
```

Unlike tasks, there is an implicit synchronization barrier at the end of each *sections* directive in OpenMP. This synchronization is achieved using `hpx::wait_all` function.

Note: If the *nowait* clause is used in the *sections* directive, then you can just remove the `hpx::wait_all` function while keeping the rest of the code as it is.

Intel Threading Building Blocks (TBB)

Intel Threading Building Blocks (TBB) provides a high-level interface for parallelism and concurrent programming using standard ISO C++ code. Below are some examples on how to convert Intel Threading Building Blocks (TBB) to HPX code:

parallel_for

Intel Threading Building Blocks (TBB) code:

```
auto values = std::vector<double>(10000);

tbb::parallel_for( tbb::blocked_range<int>(0, values.size()),
                  [&](tbb::blocked_range<int> r)
{
    for (int i=r.begin(); i<r.end(); ++i)
    {
        // loop body
    }
});
```

HPX equivalent:

```
#include <hpx/algorithm.hpp>

auto values = std::vector<double>(10000);

hpx::experimental::for_loop(hpx::execution::par, 0, values.size(), [&](int i) {
    // loop body
});
```

In the above code, `tbb::parallel_for` is replaced with `hpx::experimental::for_loop` from the HPX library. The loop body within the lambda function will be executed in parallel for each iteration.

parallel_for_each

Intel Threading Building Blocks (TBB) code:

```

auto values = std::vector<double>(10000);

tbb::parallel_for_each(values.begin(), values.end(), [&]() {
    // loop body
});

```

HPX equivalent:

```

#include <hpx/algorithm.hpp>

auto values = std::vector<double>(10000);

hpx::for_each(hpx::execution::par, values.begin(), values.end(), [&]() {
    // loop body
});

```

By utilizing `hpx::for_each` and specifying a parallel execution policy with `hpx::execution::par`, it is possible to transform `tbb::parallel_for_each` into its equivalent counterpart in HPX.

parallel_invoke

Intel Threading Building Blocks (TBB) code:

```

tbb::parallel_invoke(task1, task2, task3);

```

HPX equivalent:

```

#include <hpx/future.hpp>

hpx::wait_all(hpx::async(task1), hpx::async(task2), hpx::async(task3));

```

To convert `tbb::parallel_invoke` to HPX, we use `hpx::async` to asynchronously execute each task, which returns a future representing the result of each task. We then pass these futures to `hpx::when_all`, which waits for all the futures to complete before returning.

parallel_pipeline

Intel Threading Building Blocks (TBB) code:

```

tbb::parallel_pipeline(4,
    tbb::make_filter<void, int>(tbb::filter::serial_in_order,
        [](tbb::flow_control& fc) -> int {
            // Generate numbers from 1 to 10
            static int i = 1;
            if (i <= 10) {
                return i++;
            }
            else {

```

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```

        fc.stop();
        return 0;
    }
}) &
tbb::make_filter<int, int>(tbb::filter::parallel,
    [](int num) -> int {
        // Multiply each number by 2
        return num * 2;
    }) &
tbb::make_filter<int, void>(tbb::filter::serial_in_order,
    [](int num) {
        // Print the results
        std::cout << num << " ";
    })
);

```

HPX equivalent:

```

#include <iostream>
#include <vector>
#include <ranges>
#include <hpx/algorithm.hpp>

// generate the values
auto range = std::views::iota(1) | std::views::take(10);

// materialize the output vector
std::vector<int> results(10);

// in parallel execution of pipeline and transformation
hpx::ranges::transform(
    hpx::execution::par, range, result.begin(), [](int i) { return 2 * i; });

// print the modified vector
for (int i : result)
{
    std::cout << i << " ";
}
std::cout << std::endl;

```

The line `auto range = std::views::iota(1) | std::views::take(10);` generates a range of values using the `std::views::iota` function. It starts from the value 1 and generates an infinite sequence of incrementing values. The `std::views::take(10)` function is then applied to limit the sequence to the first 10 values. The result is stored in the `range` variable.

Hint: A view is a lightweight object that represents a particular view of a sequence or range. It acts as a read-only interface to the original data, providing a way to query and traverse the elements without making any copies or modifications.

Views can be composed and chained together to form complex pipelines of operations. These operations are evaluated lazily, meaning that the actual computation is performed only when the result is needed or consumed.

Since views perform lazy evaluation, we use `std::vector<int> results(10);` to materialize the vector that will store the

transformed values. The `hpx::ranges::transform` function is then used to perform a parallel transformation on the range. The transformed values will be written to the `results` vector.

Hint: Ranges enable loop fusion by combining multiple operations into a single parallel loop, eliminating waiting time and reducing overhead. Using ranges, you can express these operations as a pipeline of transformations on a sequence of elements. This pipeline is evaluated in a single pass, performing all the desired operations in parallel without the need to wait between them.

In addition, *HPX* enhances the benefits of range fusion by offering parallel execution policies, which can be used to optimize the execution of the fused loop across multiple threads.

parallel_reduce

Reduction

Intel Threading Building Blocks (TBB) code:

```
auto values = std::vector<double>{1,2,3,4,5,6,7,8,9};

auto total = tbb::parallel_reduce(
    tbb::blocked_range<int>(0, values.size()),
    0.0,
    [&](tbb::blocked_range<int> r, double running_total)
    {
        for (int i=r.begin(); i<r.end(); ++i)
        {
            running_total += values[i];
        }

        return running_total;
    },
    std::plus<double>());
```

HPX equivalent:

```
#include <hpx/numeric.hpp>

auto values = std::vector<double>{1,2,3,4,5,6,7,8,9};

auto total = hpx::reduce(
    hpx::execution::par, values.begin(), values.end(), 0, std::plus{});
```

By utilizing `hpx::reduce` and specifying a parallel execution policy with `hpx::execution::par`, it is possible to transform `tbb::parallel_reduce` into its equivalent counterpart in *HPX*. As demonstrated in the previous example, the management of intermediate results is seamlessly handled internally by *HPX*, eliminating the need for explicit consideration.

Transformation & Reduction

Intel Threading Building Blocks (TBB) code:

```
auto values = std::vector<double>{1,2,3,4,5,6,7,8,9};

auto transform_function(double current_value){
    // transformation code
}

auto total = tbb::parallel_reduce(
    tbb::blocked_range<int>(0, values.size()),
    0.0,
    [&](tbb::blocked_range<int> r, double transformed_val)
    {
        for (int i=r.begin(); i<r.end(); ++i)
        {
            transformed_val += transform_function(values[i]);
        }
        return transformed_val;
    },
    std::plus<double>());
```

HPX equivalent:

```
#include <hpx/numeric.hpp>

auto values = std::vector<double>{1,2,3,4,5,6,7,8,9};

auto transform_function(double current_value)
{
    // transformation code
}

auto total = hpx::transform_reduce(hpx::execution::par, values.begin(),
    values.end(), 0, std::plus{}),
    [&](double current_value) { return transform_function(current_value); });
```

In situations where certain values require transformation before the reduction process, *HPX* provides a straightforward solution through `hpx::transform_reduce`. The `transform_function()` allows for the application of the desired transformation to each value.

parallel_scan

Intel Threading Building Blocks (TBB) code:

```
tbb::parallel_scan(tbb::blocked_range<size_t>(0, input.size()),
    0,
    [&input, &output](const tbb::blocked_range<size_t>& range, int& partial_sum, bool is_
    ↪final_scan) {
        for (size_t i = range.begin(); i != range.end(); ++i) {
            partial_sum += input[i];
        }
    },
    output);
```

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```

        if (is_final_scan) {
            output[i] = partial_sum;
        }
    }
    return partial_sum;
},
[](int left_sum, int right_sum) {
    return left_sum + right_sum;
}
);

```

HPX equivalent:

```

#include <hpx/numeric.hpp>

hpx::inclusive_scan(hpx::execution::par, input.begin(), input.end(),
    output.begin(),
    [](const int& left, const int& right) { return left + right; });

```

`hpx::inclusive_scan` with `hpx::execution::par` as execution policy can be used to perform a prefix scan in parallel. The management of intermediate results is seamlessly handled internally by *HPX*, eliminating the need for explicit consideration. `input.begin()` and `input.end()` refer to the beginning and end of the sequence of elements the algorithm will be applied to respectively. `output.begin()` refers to the beginning of the destination, while the last argument specifies the function which will be invoked for each of the values of the input sequence.

Apart from `hpx::inclusive_scan`, *HPX* provides its users with `hpx::exclusive_scan`. The key difference between inclusive scan and exclusive scan lies in the treatment of the current element during the scan operation. In an inclusive scan, each element in the output sequence includes the contribution of the corresponding element in the input sequence, while in an exclusive scan, the current element in the input sequence does not contribute to the corresponding element in the output sequence.

parallel_sort

Intel Threading Building Blocks (TBB) code:

```

std::vector<int> numbers = {9, 2, 7, 1, 5, 3};

tbb::parallel_sort(numbers.begin(), numbers.end());

```

HPX equivalent:

```

#include <hpx/algorithm.hpp>

std::vector<int> numbers = {9, 2, 7, 1, 5, 3};

hpx::sort(hpx::execution::par, numbers.begin(), numbers.end());

```

`hpx::sort` provides an equivalent functionality to `tbb::parallel_sort`. When given a parallel execution policy with `hpx::execution::par`, the algorithm employs parallel execution, allowing for efficient sorting across available threads.

task_group

Intel Threading Building Blocks (TBB) code:

```
// Declare a task group
tbb::task_group tg;

// Run the tasks
tg.run(task1);
tg.run(task2);

// Wait for the task group
tg.wait();
```

HPX equivalent:

```
#include <hpx/task_group.hpp>

// Declare a task group
hpx::experimental::task_group tg;

// Run the tasks
tg.run(task1);
tg.run(task2);

// Wait for the task group
tg.wait();
```

HPX drew inspiration from Intel Threading Building Blocks (TBB) to introduce the `hpx::experimental::task_group` feature. Therefore, utilizing `hpx::experimental::task_group` provides an equivalent functionality to `tbb::task_group`.

MPI

MPI is a standardized communication protocol and library that allows multiple processes or nodes in a parallel computing system to exchange data and coordinate their execution.

List of MPI-HPX functions

MPI function	HPX equivalent
<i>MPI_Allgather</i>	<i>hpx::collectives::all_gather</i>
<i>MPI_Allreduce</i>	<i>hpx::collectives::all_reduce</i>
<i>MPI_Alltoall</i>	<i>hpx::collectives::all_to_all</i>
<i>MPI_Barrier</i>	<i>hpx::distributed::barrier</i>
<i>MPI_Bcast</i>	<i>hpx::collectives::broadcast_to()</i> and <i>hpx::collectives::broadcast_from()</i> used with <i>get()</i>
<i>MPI_Comm_size</i>	<i>hpx::get_num_localities</i>
<i>MPI_Comm_rank</i>	<i>hpx::get_locality_id()</i>
<i>MPI_Exscan</i>	<i>hpx::collectives::exclusive_scan()</i> used with <i>get()</i>
<i>MPI_Gather</i>	<i>hpx::collectives::gather_here()</i> and <i>hpx::collectives::gather_there()</i> used with <i>get()</i>
<i>MPI_Irecv</i>	<i>hpx::collectives::get()</i>
<i>MPI_Isend</i>	<i>hpx::collectives::set()</i>
<i>MPI_Reduce</i>	<i>hpx::collectives::reduce_here</i> and <i>hpx::collectives::reduce_there</i> used with <i>get()</i>
<i>MPI_Scan</i>	<i>hpx::collectives::inclusive_scan()</i> used with <i>get()</i>
<i>MPI_Scatter</i>	<i>hpx::collectives::scatter_to()</i> and <i>hpx::collectives::scatter_from()</i>
<i>MPI_Wait</i>	<i>hpx::collectives::get()</i> used with a future i.e. <i>setf.get()</i>

MPI_Send & MPI_Recv

Let's assume we have the following simple message passing code where each process sends a message to the next process in a circular manner. The exchanged message is modified and printed to the console.

MPI code:

```
#include <cstdint>
#include <cstdint>
#include <iostream>
#include <mpi.h>
#include <vector>

constexpr int times = 2;

int main(int argc, char *argv[]) {
    MPI_Init(&argc, &argv);

    int num_localities;
    MPI_Comm_size(MPI_COMM_WORLD, &num_localities);

    int this_locality;
    MPI_Comm_rank(MPI_COMM_WORLD, &this_locality);

    int next_locality = (this_locality + 1) % num_localities;
    std::vector<int> msg_vec = {0, 1};

    int cnt = 0;
    int msg = msg_vec[this_locality];
```

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```

int recv_msg;
MPI_Request request_send, request_recv;
MPI_Status status;

while (cnt < times) {
    cnt += 1;

    MPI_Isend(&msg, 1, MPI_INT, next_locality, cnt, MPI_COMM_WORLD,
              &request_send);
    MPI_Irecv(&recv_msg, 1, MPI_INT, next_locality, cnt, MPI_COMM_WORLD,
              &request_recv);

    MPI_Wait(&request_send, &status);
    MPI_Wait(&request_recv, &status);

    std::cout << "Time: " << cnt << ", Locality " << this_locality
               << " received msg: " << recv_msg << "\n";

    recv_msg += 10;
    msg = recv_msg;
}

MPI_Finalize();
return 0;
}

```

HPX equivalent:

```

#include <hpx/config.hpp>

#if !defined(HPX_COMPUTE_DEVICE_CODE)
#include <hpx/algorithm.hpp>
#include <hpx/hpx_init.hpp>
#include <hpx/modules/collectives.hpp>

#include <cstdint>
#include <cstdint>
#include <iostream>
#include <utility>
#include <vector>

using namespace hpx::collectives;

constexpr char const* channel_communicator_name =
    "/example/channel_communicator/";

// the number of times
constexpr int times = 2;

int hpx_main()
{

```

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```

std::uint32_t num_localities = hpx::get_num_localities(hpx::launch::sync);
std::uint32_t this_locality = hpx::get_locality_id();

// allocate channel communicator
// NOTE: The communicator object must be kept alive until all
// participating sites have successfully connected. Destroying it
// prematurely may cause other sites to hang. Consider using a barrier
// after creation if the communicator would otherwise go out of scope.
auto comm = create_channel_communicator(hpx::launch::sync,
    channel_communicator_name, num_sites_arg(num_localities),
    this_site_arg(this_locality));

std::uint32_t next_locality = (this_locality + 1) % num_localities;
std::vector<int> msg_vec = {0, 1};

int cnt = 0;
int msg = msg_vec[this_locality];

// send values to another locality
auto setf = set(comm, that_site_arg(next_locality), msg, tag_arg(cnt));
auto got_msg = get<int>(comm, that_site_arg(next_locality), tag_arg(cnt));

setf.get();

while (cnt < times)
{
    cnt += 1;

    auto done_msg = got_msg.then([&](auto&& f) {
        int rec_msg = f.get();
        std::cout << "Time: " << cnt << ", Locality " << this_locality
            << " received msg: " << rec_msg << "\n";

        // change msg by adding 10
        rec_msg += 10;

        // start next round
        setf =
            set(comm, that_site_arg(next_locality), rec_msg, tag_arg(cnt));
        got_msg =
            get<int>(comm, that_site_arg(next_locality), tag_arg(cnt));
        setf.get();
    });

    done_msg.get();
}

return hpx::finalize();
}
#endif

int main(int argc, char* argv[])

```

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```

{
  #if !defined(HPX_COMPUTE_DEVICE_CODE)
    hpx::init_params params;
    params.cfg = {"--hpx:run-hpx-main"};
    return hpx::init(argc, argv, params);
  #else
    (void) argc;
    (void) argv;
    return 0;
  #endif
}

```

To perform message passing between different processes in *HPX* we can use a channel communicator. To understand this example, let's focus on the *hpx_main()* function:

- *hpx::get_num_localities(hpx::launch::sync)* retrieves the number of localities, while *hpx::get_locality_id()* returns the ID of the current locality.
- *create_channel_communicator* function is used to create a channel to serve the communication. This function takes several arguments, including the launch policy (*hpx::launch::sync*), the name of the communicator (*channel_communicator_name*), the number of localities, and the ID of the current locality.
- The communication follows a ring pattern, where each process (or locality) sends a message to its neighbor in a circular manner. This means that the messages circulate around the localities, ensuring that the communication wraps around when reaching the end of the locality sequence. To achieve this, the *next_locality* variable is calculated as the ID of the next locality in the ring.
- The initial values for the communication are set (*msg_vec*, *cnt*, *msg*).
- The *set()* function is called to send the message to the next locality in the ring. The message is sent asynchronously and is associated with a tag (*cnt*).
- The *get()* function is called to receive a message from the next locality. It is also associated with the same tag as the *set()* operation.
- The *setf.get()* call blocks until the message sending operation is complete.
- A continuation is set up using the function *then()* to handle the received message. Inside the continuation:
 - The received message value (*rec_msg*) is retrieved using *f.get()*.
 - The received message is printed to the console and then modified by adding 10.
 - The *set()* and *get()* operations are repeated to send and receive the modified message to the next locality.
 - The *setf.get()* call blocks until the new message sending operation is complete.
- The *done_msg.get()* call blocks until the continuation is complete for the current loop iteration.

Having said that, we conclude to the following table:

MPI_Gather

The following code gathers data from all processes to the root process and verifies the gathered data in the root process.

MPI code:

```
#include <iostream>
#include <mpi.h>
#include <numeric>
#include <vector>

int main(int argc, char *argv[]) {
    MPI_Init(&argc, &argv);

    int num_localities, this_locality;
    MPI_Comm_size(MPI_COMM_WORLD, &num_localities);
    MPI_Comm_rank(MPI_COMM_WORLD, &this_locality);

    std::vector<int> local_data; // Data to be gathered

    if (this_locality == 0) {
        local_data.resize(num_localities); // Resize the vector on the root process
    }

    // Each process calculates its local data value
    int my_data = 42 + this_locality;

    for (std::uint32_t i = 0; i != 10; ++i) {

        // Gather data from all processes to the root process (process 0)
        MPI_Gather(&my_data, 1, MPI_INT, local_data.data(), 1, MPI_INT, 0,
                  MPI_COMM_WORLD);

        // Only the root process (process 0) will print the gathered data
        if (this_locality == 0) {
            std::cout << "Gathered data on the root: ";
            for (int i = 0; i < num_localities; ++i) {
                std::cout << local_data[i] << " ";
            }
            std::cout << std::endl;
        }
    }
    std::cout << std::endl;

    MPI_Finalize();
    return 0;
}
```

HPX equivalent:

```
std::uint32_t num_localities = hpx::get_num_localities(hpx::launch::sync);
std::uint32_t this_locality = hpx::get_locality_id();

// test functionality based on immediate local result value
```

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```

auto gather_direct_client = create_communicator(gather_direct_basename,
    num_sites_arg(num_localities), this_site_arg(this_locality));

for (std::uint32_t i = 0; i != 10; ++i)
{
    if (this_locality == 0)
    {
        hpx::future<std::vector<std::uint32_t>> overall_result =
            gather_here(gather_direct_client, std::uint32_t(42));

        std::vector<std::uint32_t> sol = overall_result.get();
        std::cout << "Gathered data on the root:";

        for (std::size_t j = 0; j != sol.size(); ++j)
        {
            HPX_TEST(j + 42 == sol[j]);
            std::cout << " " << sol[j];
        }
        std::cout << std::endl;
    }
    else
    {
        hpx::future<void> overall_result =
            gather_there(gather_direct_client, this_locality + 42);
        overall_result.get();
    }
}

```

This code will print 10 times the following message:

```
Gathered data on the root: 42 43
```

HPX uses two functions to implement the functionality of *MPI_Gather*: *gather_here* and *gather_there*. *gather_here* is gathering data from all localities to the locality with ID 0 (root locality). *gather_there* allows non-root localities to participate in the gather operation by sending data to the root locality. In more detail:

- *hpx::get_num_localities(hpx::launch::sync)* retrieves the number of localities, while *hpx::get_locality_id()* returns the ID of the current locality.
- The function *create_communicator()* is used to create a communicator called *gather_direct_client*.
- If the current locality is the root (its ID is equal to 0):
 - The *gather_here* function is used to perform the gather operation. It collects data from all other localities into the *overall_result* future object. The function arguments provide the necessary information, such as the base name for the gather operation (*gather_direct_basename*), the value to be gathered (*value*), the number of localities (*num_localities*), the current locality ID (*this_locality*), and the generation number (related to the gather operation).
 - The *get()* member function of the *overall_result* future is used to retrieve the gathered data.
 - The next *for* loop is used to verify the correctness of the gathered data (*sol*). *HPX_TEST* is a macro provided by the HPX testing utilities to perform similar testing with the Standard C++ macro *assert*.
- If the current locality is not the root:

- The *gather_there* function is used to participate in the gather operation initiated by the root locality. It sends the data (in this case, the value *this_locality* + 42) to the root locality, indicating that it should be included in the gathering.
- The *get()* member function of the *overall_result* future is used to wait for the gather operation to complete for this locality.

MPI_Scatter

The following code gathers data from all processes to the root process and verifies the gathered data in the root process.

MPI code:

```
#include <iostream>
#include <mpi.h>
#include <vector>

int main(int argc, char *argv[]) {
    MPI_Init(&argc, &argv);

    int num_localities, this_locality;
    MPI_Comm_size(MPI_COMM_WORLD, &num_localities);
    MPI_Comm_rank(MPI_COMM_WORLD, &this_locality);

    int num_localities = num_localities;
    std::vector<int> data(num_localities);

    if (this_locality == 0) {
        // Fill the data vector on the root locality (locality 0)
        for (int i = 0; i < num_localities; ++i) {
            data[i] = 42 + i;
        }
    }

    int local_data; // Variable to store the received data

    // Scatter data from the root locality to all other localities
    MPI_Scatter(&data[0], 1, MPI_INT, &local_data, 1, MPI_INT, 0, MPI_COMM_WORLD);

    // Now, each locality has its own local_data

    // Print the local_data on each locality
    std::cout << "Locality " << this_locality << " received " << local_data
                << std::endl;

    MPI_Finalize();
    return 0;
}
```

HPX equivalent:

```
std::uint32_t num_localities = hpx::get_num_localities(hpx::launch::sync);
HPX_TEST_LTE(std::uint32_t(2), num_localities);
```

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```

std::uint32_t this_locality = hpx::get_locality_id();

auto scatter_direct_client =
    hpx::collectives::create_communicator(scatter_direct_basename,
        num_sites_arg(num_localities), this_site_arg(this_locality));

// test functionality based on immediate local result value
for (std::uint32_t i = 0; i != 10; ++i)
{
    if (this_locality == 0)
    {
        std::vector<std::uint32_t> data(num_localities);
        std::iota(data.begin(), data.end(), 42 + i);

        hpx::future<std::uint32_t> result =
            scatter_to(scatter_direct_client, std::move(data));

        HPX_TEST_EQ(i + 42 + this_locality, result.get());
    }
    else
    {
        hpx::future<std::uint32_t> result =
            scatter_from<std::uint32_t>(scatter_direct_client);

        HPX_TEST_EQ(i + 42 + this_locality, result.get());

        std::cout << "Locality " << this_locality << " received "
            << i + 42 + this_locality << std::endl;
    }
}

```

For `num_localities = 2` and since we run for 10 iterations this code will print the following message:

```

Locality 1 received 43
Locality 1 received 44
Locality 1 received 45
Locality 1 received 46
Locality 1 received 47
Locality 1 received 48
Locality 1 received 49
Locality 1 received 50
Locality 1 received 51
Locality 1 received 52

```

HPX uses two functions to implement the functionality of *MPI_Scatter*: `hpx::scatter_to` and `hpx::scatter_from`. `hpx::scatter_to` is distributing the data from the locality with ID 0 (root locality) to all other localities. `hpx::scatter_from` allows non-root localities to receive the data from the root locality. In more detail:

- `hpx::get_num_localities(hpx::launch::sync)` retrieves the number of localities, while `hpx::get_locality_id()` returns the ID of the current locality.
- The function `hpx::collectives::create_communicator()` is used to create a communicator called `scatter_direct_client`.

- If the current locality is the root (its ID is equal to 0):
 - The data vector is filled with values ranging from $42 + i$ to $42 + i + \text{num_localities} - 1$.
 - The `hpx::scatter_to` function is used to perform the scatter operation using the communicator `scatter_direct_client`. This scatters the data vector to other localities and returns a future representing the result.
 - `HPX_TEST_EQ` is a macro provided by the `HPX` testing utilities to test the distributed values.
- If the current locality is not the root:
 - The `hpx::scatter_from` function is used to collect the data by the root locality.
 - `HPX_TEST_EQ` is a macro provided by the `HPX` testing utilities to test the collected values.

MPI_Allgather

The following code gathers data from all processes and sends the data to all processes.

MPI code:

```
#include <cstdint>
#include <iostream>
#include <mpi.h>
#include <vector>

int main(int argc, char **argv) {
    MPI_Init(&argc, &argv);

    int rank, size;
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    // Get the number of MPI processes
    int num_localities = size;

    // Get the MPI process rank
    int here = rank;

    std::uint32_t value = here;

    std::vector<std::uint32_t> r(num_localities);

    // Perform an all-gather operation to gather values from all processes.
    MPI_Allgather(&value, 1, MPI_UINT32_T, r.data(), 1, MPI_UINT32_T,
                 MPI_COMM_WORLD);

    // Print the result.
    std::cout << "Locality " << here << " has values:";
    for (size_t j = 0; j < r.size(); ++j) {
        std::cout << " " << r[j];
    }
    std::cout << std::endl;

    MPI_Finalize();
}
```

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```

    return 0;
}

```

HPX equivalent:

```

std::uint32_t num_localities = hpx::get_num_localities(hpx::launch::sync);
std::uint32_t here = hpx::get_locality_id();

// test functionality based on immediate local result value
auto all_gather_direct_client =
    create_communicator(all_gather_direct_basename,
        num_sites_arg(num_localities), this_site_arg(here));

std::uint32_t value = here;

hpx::future<std::vector<std::uint32_t>> overall_result =
    all_gather(all_gather_direct_client, value);

std::vector<std::uint32_t> r = overall_result.get();

std::cout << "Locality " << here << " has values:";
for (std::size_t j = 0; j != r.size(); ++j)
{
    std::cout << " " << j;
}
std::cout << std::endl;

```

For num_localities = 2 this code will print the following message:

```

Locality 0 has values: 0 1
Locality 1 has values: 0 1

```

HPX uses the function *all_gather* to implement the functionality of *MPI_Allgather*. In more detail:

- *hpx::get_num_localities(hpx::launch::sync)* retrieves the number of localities, while *hpx::get_locality_id()* returns the ID of the current locality.
- The function *hpx::collectives::create_communicator()* is used to create a communicator called *all_gather_direct_client*.
- The values that the localities exchange with each other are equal to each locality's ID.
- The gather operation is performed using *all_gather*. The result is stored in an *hpx::future* object called *overall_result*, which represents a future result that can be retrieved later when needed.
- The *get()* function waits until the result is available and then stores it in the vector called *r*.

MPI_Allreduce

The following code combines values from all processes and distributes the result back to all processes.

MPI code:

```
#include <stdint>
#include <iostream>
#include <mpi.h>

int main(int argc, char **argv) {
    MPI_Init(&argc, &argv);

    int rank, size;
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    // Get the number of MPI processes
    int num_localities = size;

    // Get the MPI process rank
    int here = rank;

    // Create a communicator for the all reduce operation.
    MPI_Comm all_reduce_direct_client;
    MPI_Comm_split(MPI_COMM_WORLD, 0, rank, &all_reduce_direct_client);

    // Perform the all reduce operation to calculate the sum of 'here' values.
    std::uint32_t value = here;
    std::uint32_t res = 0;
    MPI_Allreduce(&value, &res, 1, MPI_UINT32_T, MPI_SUM,
                  all_reduce_direct_client);

    std::cout << "Locality " << rank << " has value: " << res << std::endl;

    MPI_Finalize();
    return 0;
}
```

HPX equivalent:

```
std::uint32_t const num_localities =
    hpx::get_num_localities(hpx::launch::sync);
std::uint32_t const here = hpx::get_locality_id();

auto const all_reduce_direct_client =
    create_communicator(all_reduce_direct_basename,
                        num_sites_arg(num_localities), this_site_arg(here));

std::uint32_t value = here;

hpx::future<std::uint32_t> overall_result =
    all_reduce(all_reduce_direct_client, value, std::plus<std::uint32_t>{});
```

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```
std::uint32_t res = overall_result.get();
std::cout << "Locality " << here << " has value: " << res << std::endl;
```

For `num_localities = 2` this code will print the following message:

```
Locality 0 has value: 1
Locality 1 has value: 1
```

HPX uses the function *all_reduce* to implement the functionality of *MPI_Allreduce*. In more detail:

- *hpx::get_num_localities(hpx::launch::sync)* retrieves the number of localities, while *hpx::get_locality_id()* returns the ID of the current locality.
- The function *hpx::collectives::create_communicator()* is used to create a communicator called *all_reduce_direct_client*.
- The value of each locality is equal to its ID.
- The reduce operation is performed using *all_reduce*. The result is stored in an *hpx::future* object called *overall_result*, which represents a future result that can be retrieved later when needed.
- The *get()* function waits until the result is available and then stores it in the variable *res*.

MPI_Alltoall

The following code gathers data from and scatters data to all processes.

MPI code:

```
#include <algorithm>
#include <cstdint>
#include <iostream>
#include <mpi.h>
#include <vector>

int main(int argc, char **argv) {
    MPI_Init(&argc, &argv);

    int rank, size;
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    // Get the number of MPI processes
    int num_localities = size;

    // Get the MPI process rank
    int this_locality = rank;

    // Create a communicator for all-to-all operation.
    MPI_Comm all_to_all_direct_client;
    MPI_Comm_split(MPI_COMM_WORLD, 0, rank, &all_to_all_direct_client);

    std::vector<std::uint32_t> values(num_localities);
    std::fill(values.begin(), values.end(), this_locality);
```

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```

// Create vectors to store received values.
std::vector<std::uint32_t> r(num_localities);

// Perform an all-to-all operation to exchange values with other localities.
MPI_Alltoall(values.data(), 1, MPI_UINT32_T, r.data(), 1, MPI_UINT32_T,
             all_to_all_direct_client);

// Print the results.
std::cout << "Locality " << this_locality << " has values:";
for (std::size_t j = 0; j != r.size(); ++j) {
    std::cout << " " << r[j];
}
std::cout << std::endl;

MPI_Finalize();
return 0;
}

```

HPX equivalent:

```

std::uint32_t num_localities = hpx::get_num_localities(hpx::launch::sync);
std::uint32_t this_locality = hpx::get_locality_id();

auto all_to_all_direct_client =
    create_communicator(all_to_all_direct_basename,
                       num_sites_arg(num_localities), this_site_arg(this_locality));

std::vector<std::uint32_t> values(num_localities);
std::fill(values.begin(), values.end(), this_locality);

hpx::future<std::vector<std::uint32_t>> overall_result =
    all_to_all(all_to_all_direct_client, std::move(values));

std::vector<std::uint32_t> r = overall_result.get();
std::cout << "Locality " << this_locality << " has values:";

for (std::size_t j = 0; j != r.size(); ++j)
{
    std::cout << " " << r[j];
}
std::cout << std::endl;

```

For num_localities = 2 this code will print the following message:

```

Locality 0 has values: 0 1
Locality 1 has values: 0 1

```

HPX uses the function *all_to_all* to implement the functionality of *MPI_Alltoall*. In more detail:

- *hpx::get_num_localities(hpx::launch::sync)* retrieves the number of localities, while *hpx::get_locality_id()* returns the ID of the current locality.
- The function *hpx::collectives::create_communicator()* is used to create a communicator called

all_to_all_direct_client.

- The value each locality sends is equal to its ID.
- The all-to-all operation is performed using *all_to_all*. The result is stored in an *hpx::future* object called *overall_result*, which represents a future result that can be retrieved later when needed.
- The *get()* function waits until the result is available and then stores it in the variable *r*.

MPI_Barrier

The following code shows how barrier is used to synchronize multiple processes.

MPI code:

```
#include <cstdlib>
#include <iostream>
#include <mpi.h>

int main(int argc, char **argv) {
    MPI_Init(&argc, &argv);

    std::size_t iterations = 5;

    int rank, size;
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    for (std::size_t i = 0; i != iterations; ++i) {
        MPI_Barrier(MPI_COMM_WORLD);
        if (rank == 0) {
            std::cout << "Iteration " << i << " completed." << std::endl;
        }
    }

    MPI_Finalize();
    return 0;
}
```

HPX equivalent:

```
std::size_t iterations = 5;
std::uint32_t this_locality = hpx::get_locality_id();

char const* const barrier_test_name = "/test/barrier/multiple";

hpx::distributed::barrier b(barrier_test_name);
for (std::size_t i = 0; i != iterations; ++i)
{
    b.wait();
    if (this_locality == 0)
    {
        std::cout << "Iteration " << i << " completed." << std::endl;
    }
}
```


This code will print the following message:

```
Iteration 0 completed.
Iteration 1 completed.
Iteration 2 completed.
Iteration 3 completed.
Iteration 4 completed.
```

HPX uses the function *barrier* to implement the functionality of *MPI_Barrier*. In more detail:

- After defining the number of iterations, we use *hpx::get_locality_id()* to get the ID of the current locality.
- *char const* const barrier_test_name = "/test/barrier/multiple"*: This line defines a constant character array as the name of the barrier. This name is used to identify the barrier across different localities. All participating threads that use this name will synchronize at this barrier.
- Using *hpx::distributed::barrier b(barrier_test_name)*, we create an instance of the distributed barrier with the previously defined name. This barrier will be used to synchronize the execution of threads across different localities.
- Running for all the desired iterations, we use *b.wait()* to synchronize the threads. Each thread waits until all other threads also reach this point before any of them can proceed further.

MPI_Bcast

The following code broadcasts data from one process to all other processes.

MPI code:

```
#include <iostream>
#include <mpi.h>

int main(int argc, char *argv[]) {
    MPI_Init(&argc, &argv);

    int num_localities;
    MPI_Comm_size(MPI_COMM_WORLD, &num_localities);

    int here;
    MPI_Comm_rank(MPI_COMM_WORLD, &here);

    int value;

    for (int i = 0; i < 5; ++i) {
        if (here == 0) {
            value = i + 42;
        }

        // Broadcast the value from process 0 to all other processes
        MPI_Bcast(&value, 1, MPI_INT, 0, MPI_COMM_WORLD);

        if (here != 0) {
            std::cout << "Locality " << here << " received " << value << std::endl;
        }
    }
}
```

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```

    }

    MPI_Finalize();
    return 0;
}

```

HPX equivalent:

```

std::uint32_t num_localities = hpx::get_num_localities(hpx::launch::sync);

std::uint32_t here = hpx::get_locality_id();

auto broadcast_direct_client =
    create_communicator(broadcast_direct_basename,
        num_sites_arg(num_localities), this_site_arg(here));

// test functionality based on immediate local result value
for (std::uint32_t i = 0; i != 5; ++i)
{
    if (here == 0)
    {
        hpx::future<std::uint32_t> result =
            broadcast_to(broadcast_direct_client, i + 42);

        result.get();
    }
    else
    {
        hpx::future<std::uint32_t> result =
            hpx::collectives::broadcast_from<std::uint32_t>(
                broadcast_direct_client);

        uint32_t r = result.get();

        std::cout << "Locality " << here << " received " << r << std::endl;
    }
}

```

For num_localities = 2 this code will print the following message:

```

Locality 1 received 42
Locality 1 received 43
Locality 1 received 44
Locality 1 received 45
Locality 1 received 46

```

HPX uses two functions to implement the functionality of *MPI_Bcast*: *broadcast_to* and *broadcast_from*. *broadcast_to* is broadcasting the data from the root locality to all other localities. *broadcast_from* allows non-root localities to collect the data sent by the root locality. In more detail:

- *hpx::get_num_localities(hpx::launch::sync)* retrieves the number of localities, while *hpx::get_locality_id()* returns the ID of the current locality.
- The function *create_communicator()* is used to create a communicator called *broadcast_direct_client*.

- If the current locality is the root (its ID is equal to 0):
 - The *broadcast_to* function is used to perform the broadcast operation using the communicator *broadcast_direct_client*. This sends the data to other localities and returns a future representing the result.
 - The *get()* member function of the *result* future is used to wait for and retrieve the result.
- If the current locality is not the root:
 - The *broadcast_from* function is used to collect the data by the root locality.
 - The *get()* member function of the *result* future is used to wait for the result.

MPI_Exscan

The following code computes the exclusive scan (partial reductions) of data on a collection of processes.

MPI code:

```
#include <iostream>
#include <mpi.h>
#include <numeric>
#include <vector>

int main(int argc, char *argv[]) {
    MPI_Init(&argc, &argv);

    int num_localities;
    MPI_Comm_size(MPI_COMM_WORLD, &num_localities);

    int here;
    MPI_Comm_rank(MPI_COMM_WORLD, &here);

    // Calculate the value for this locality (here)
    int value = here;

    // Perform an exclusive scan
    std::vector<int> result(num_localities);
    MPI_Exscan(&value, &result[0], 1, MPI_INT, MPI_SUM, MPI_COMM_WORLD);

    if (here != 0) {
        int r = result[here - 1]; // Result is in the previous rank's slot

        std::cout << "Locality " << here << " has value " << r << std::endl;
    }

    MPI_Finalize();
    return 0;
}
```

HPX equivalent:

```
std::uint32_t num_localities = hpx::get_num_localities(hpx::launch::sync);
std::uint32_t here = hpx::get_locality_id();

auto exclusive_scan_client = create_communicator(exclusive_scan_basename,
```

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```

    num_sites_arg(num_localities), this_site_arg(here));

// test functionality based on immediate local result value
std::uint32_t value = here;

hpx::future<std::uint32_t> overall_result = exclusive_scan(
    exclusive_scan_client, value, std::plus<std::uint32_t>{});

uint32_t r = overall_result.get();

if (here != 0)
{
    std::cout << "Locality " << here << " has value " << r << std::endl;
}

```

For `num_localities = 2` this code will print the following message:

```
Locality 1 has value 0
```

HPX uses the function *exclusive_scan* to implement *MPI_Exscan*. In more detail:

- `hpx::get_num_localities(hpx::launch::sync)` retrieves the number of localities, while `hpx::get_locality_id()` returns the ID of the current locality.
- The function `create_communicator()` is used to create a communicator called *exclusive_scan_client*.
- The *exclusive_scan* function is used to perform the exclusive scan operation using the communicator *exclusive_scan_client*. `std::plus<std::uint32_t>{}` specifies the binary associative operator to use for the scan. In this case, it's addition for summing values.
- The `get()` member function of the *overall_result* future is used to wait for the result.

MPI_Scan

The following code Computes the inclusive scan (partial reductions) of data on a collection of processes.

MPI code:

```

#include <iostream>
#include <mpi.h>
#include <numeric>
#include <vector>

int main(int argc, char *argv[]) {
    MPI_Init(&argc, &argv);

    int num_localities;
    MPI_Comm_size(MPI_COMM_WORLD, &num_localities);

    int here;
    MPI_Comm_rank(MPI_COMM_WORLD, &here);

    // Calculate the value for this locality (here)
    int value = here;

```

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```

std::vector<int> result(num_localities);

MPI_Scan(&value, &result[0], 1, MPI_INT, MPI_SUM, MPI_COMM_WORLD);

std::cout << "Locality " << here << " has value " << result[0] << std::endl;

MPI_Finalize();
return 0;
}

```

HPX equivalent:

```

std::uint32_t num_localities = hpx::get_num_localities(hpx::launch::sync);
std::uint32_t here = hpx::get_locality_id();

auto inclusive_scan_client = create_communicator(inclusive_scan_basename,
    num_sites_arg(num_localities), this_site_arg(here));

std::uint32_t value = here;

hpx::future<std::uint32_t> overall_result = inclusive_scan(
    inclusive_scan_client, value, std::plus<std::uint32_t>{});

uint32_t r = overall_result.get();

std::cout << "Locality " << here << " has value " << r << std::endl;

```

For num_localities = 2 this code will print the following message:

```

Locality 0 has value 0
Locality 1 has value 1

```

HPX uses the function *inclusive_scan* to implement *MPI_Scan*. In more detail:

- *hpx::get_num_localities(hpx::launch::sync)* retrieves the number of localities, while *hpx::get_locality_id()* returns the ID of the current locality.
- The function *create_communicator()* is used to create a communicator called *inclusive_scan_client*.
- The *inclusive_scan* function is used to perform the exclusive scan operation using the communicator *inclusive_scan_client*. *std::plus<std::uint32_t>{}* specifies the binary associative operator to use for the scan. In this case, it's addition for summing values.
- The *get()* member function of the *overall_result* future is used to wait for the result.

MPI_Reduce

The following code performs a global reduce operation across all processes.

MPI code:

```
#include <iostream>
#include <mpi.h>

int main(int argc, char *argv[]) {
    MPI_Init(&argc, &argv);

    int num_processes;
    MPI_Comm_size(MPI_COMM_WORLD, &num_processes);

    int this_rank;
    MPI_Comm_rank(MPI_COMM_WORLD, &this_rank);

    int value = this_rank;

    int result = 0;

    // Perform the reduction operation
    MPI_Reduce(&value, &result, 1, MPI_INT, MPI_SUM, 0, MPI_COMM_WORLD);

    // Print the result for the root process (process 0)
    if (this_rank == 0) {
        std::cout << "Locality " << this_rank << " has value " << result
                  << std::endl;
    }

    MPI_Finalize();
    return 0;
}
```

HPX equivalent:

```
std::uint32_t num_localities = hpx::get_num_localities(hpx::launch::sync);
std::uint32_t this_locality = hpx::get_locality_id();

auto reduce_direct_client = create_communicator(reduce_direct_basename,
    num_sites_arg(num_localities), this_site_arg(this_locality));

std::uint32_t value = hpx::get_locality_id();

if (this_locality == 0)
{
    hpx::future<std::uint32_t> overall_result = reduce_here(
        reduce_direct_client, value, std::plus<std::uint32_t>{});

    uint32_t r = overall_result.get();

    std::cout << "Locality " << this_locality << " has value " << r
              << std::endl;
}
```

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```

}
else
{
    hpx::future<void> overall_result =
        reduce_there(reduce_direct_client, std::move(value));
    overall_result.get();
}

```

This code will print the following message:

```
Locality 0 has value 1
```

HPX uses two functions to implement the functionality of *MPI_Reduce*: *reduce_here* and *reduce_there*. *reduce_here* is gathering data from all localities to the locality with ID 0 (root locality) and then performs the defined reduction operation. *reduce_there* allows non-root localities to participate in the reduction operation by sending data to the root locality. In more detail:

- *hpx::get_num_localities(hpx::launch::sync)* retrieves the number of localities, while *hpx::get_locality_id()* returns the ID of the current locality.
- The function *create_communicator()* is used to create a communicator called *reduce_direct_client*.
- If the current locality is the root (its ID is equal to 0):
 - The *reduce_here* function initiates a reduction operation with addition (*std::plus*) as the reduction operator. The result is stored in *overall_result*.
 - The *get()* member function of the *overall_result* future is used to wait for the result.
- If the current locality is not the root:
 - The *reduce_there* initiates a remote reduction operation.
 - The *get()* member function of the *overall_result* future is used to wait for the remote reduction operation to complete. This is done to ensure synchronization among localities.

Migrating to C++26 reflection-based actions

Starting with HPX 2.0, C++26 static reflection (P2996) can be used to define distributed actions without the verbose macro boilerplate previously required. This section describes how to migrate existing action definitions to the new reflection-based API when building with `HPX_WITH_CXX26_REFLECTION=ON`.

Build requirements

Reflection-based actions require a C++26-capable compiler with reflection support enabled:

- GCC 16.0.1 trunk or later (`stellargroup/gcc_trunk_build_env:1`) with `-freflection -std=c++26`
- Clang 22 with P2996 support (`stellargroup/clang_p2996_build_env:1`) with `-freflection-latest -stdlib=libc++`

Reflection support is auto-detected by HPX's CMake build system via `hpx_check_for_cxx26_reflection()` in `cmake/HPX_PerformCxxFeatureTests.cmake`. No manual flag is required when using a supported compiler — `HPX_HAVE_CXX26_REFLECTION` is set automatically.

Existing code requires no changes

The most important point: **existing code using** `HPX_PLAIN_ACTION`, `HPX_DEFINE_COMPONENT_ACTION`, and related macros requires zero source changes. When `HPX_WITH_CXX26_REFLECTION=ON`, these macros automatically use the reflection-based implementation internally. No `HPX_REGISTER_ACTION` calls are needed — registration is automatic.

```
// This works unchanged with or without reflection enabled.
// When HPX_WITH_CXX26_REFLECTION=ON, it automatically uses
// reflect_action<^^some_function> internally.
HPX_PLAIN_ACTION(some_function, some_action)
```

Explicit migration: plain actions

For new code or explicit migration, use `HPX_ACTION` or `reflect_action` directly:

```
// Before: three-step macro sequence
HPX_PLAIN_ACTION(app::compute, compute_action)
HPX_REGISTER_ACTION_DECLARATION(compute_action)
HPX_REGISTER_ACTION(compute_action)

// After: single line, no registration required
HPX_ACTION(app::compute, compute_action)

// Or equivalently, using the template directly:
using compute_action = hpx::actions::reflect_action<^^app::compute>;
```

Explicit migration: component actions

```
// Before: three-step macro sequence
HPX_DEFINE_COMPONENT_ACTION(server, compute, compute_action)
HPX_REGISTER_ACTION_DECLARATION(compute_action)
HPX_REGISTER_ACTION(compute_action)

// After: single line
HPX_COMPONENT_ACTION(server, compute, compute_action)

// Or equivalently:
using compute_action =
    hpx::actions::reflect_component_action<^^server::compute>;
```

Explicit migration: direct plain actions

```
// Before: three-step macro sequence
HPX_PLAIN_DIRECT_ACTION(app::compute, compute_action)
HPX_REGISTER_ACTION_DECLARATION(compute_action)
HPX_REGISTER_ACTION(compute_action)

// After: single line
```

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```
HPX_DIRECT_ACTION(app::compute, compute_action)

// Or equivalently, using the template directly:
using compute_action =
    hpx::actions::reflect_direct_action<^^app::compute>;
```

Explicit migration: direct component actions

Direct actions execute without spawning a new HPX thread — locally they run inline on the calling thread, and remotely on the target locality's existing HPX thread. The reflection-based API preserves this behavior:

```
// Before: three-step macro sequence
HPX_DEFINE_COMPONENT_DIRECT_ACTION(server, compute, compute_action)
HPX_REGISTER_ACTION_DECLARATION(compute_action)
HPX_REGISTER_ACTION(compute_action)

// After: single line
HPX_COMPONENT_DIRECT_ACTION(server, compute, compute_action)

// Or equivalently, using the template directly:
using compute_action =
    hpx::actions::reflect_component_direct_action<^^server::compute>;
```

Overloaded functions

Reflection cannot disambiguate overloaded functions. If your action function is overloaded, keep the existing HPX_PLAIN_ACTION macro:

```
// Not supported with reflection (ambiguous):
// using my_action = hpx::actions::reflect_action<^^overloaded_func>;

// Use the existing macro instead:
HPX_PLAIN_ACTION(static_cast<int>(*)>(int)>(overloaded_func), my_action)
```

Properties available at compile time

Reflection-based action types expose additional compile-time properties:

```
using my_action = hpx::actions::reflect_action<^^app::compute>;

// Number of parameters
static_assert(my_action::arity == 2);

// Function pointer type and value
using fptr = my_action::func_ptr_type;
constexpr auto fp = my_action::func_ptr;
```

2.4.7 Building tests and examples

Tests

To build the tests:

```
$ cmake --build . --target tests
```

To control which tests to run use `ctest`:

- To run single tests, for example a test for `for_loop`:

```
$ ctest --output-on-failure -R tests.unit.modules.algorithms.algorithms.for_loop
```

- To run a whole group of tests:

```
$ ctest --output-on-failure -R tests.unit
```

Examples

- To build (and install) all examples invoke:

```
$ cmake -DHPX_WITH_EXAMPLES=On .  
$ make examples  
$ make install
```

- To build the `hello_world_1` example run:

```
$ make hello_world_1
```

HPX executables end up in the `bin` directory in your build directory. You can now run `hello_world_1` and should see the following output:

```
$ ./bin/hello_world_1  
Hello World!
```

You've just run an example which prints `Hello World!` from the *HPX* runtime. The source for the example is in `examples/quickstart/hello_world_1.cpp`. The `hello_world_distributed` example (also available in the `examples/quickstart` directory) is a distributed hello world program, which is described in *Remote execution with actions*. It provides a gentle introduction to the distributed aspects of *HPX*.

Tip: Most build targets in *HPX* have two names: a simple name and a hierarchical name corresponding to what type of example or test the target is. If you are developing *HPX* it is often helpful to run `make help` to get a list of available targets. For example, `make help | grep hello_world` outputs the following:

```
... examples.quickstart.hello_world_2  
... hello_world_2  
... examples.quickstart.hello_world_1  
... hello_world_1  
... examples.quickstart.hello_world_distributed  
... hello_world_distributed
```

It is also possible to build, for instance, all quickstart examples using `make examples.quickstart`.

2.4.8 Creating HPX projects

Using HPX with pkg-config

How to build HPX applications with pkg-config

After you are done installing *HPX*, you should be able to build the following program. It prints Hello World! on the *locality* you run it on.

```
// Including 'hpx/hpx_main.hpp' instead of the usual 'hpx/hpx_init.hpp' enables
// to use the plain C-main below as the direct main HPX entry point.
#include <hpx/hpx_main.hpp>
#include <hpx/iostream.hpp>

int main()
{
    // Say hello to the world!
    hpx::cout << "Hello World!\n" << std::flush;
    return 0;
}
```

Copy the text of this program into a file called `hello_world.cpp`.

Now, in the directory where you put `hello_world.cpp`, issue the following commands (where `$HPX_LOCATION` is the build directory or `CMAKE_INSTALL_PREFIX` you used while building *HPX*):

```
$ export PKG_CONFIG_PATH=$PKG_CONFIG_PATH:$HPX_LOCATION/lib/pkgconfig
$ c++ -o hello_world hello_world.cpp \
    `pkg-config --cflags --libs hpx_application` \
    -lhpx_iostreams -DHPX_APPLICATION_NAME=hello_world
```

Important: When using `pkg-config` with *HPX*, the `pkg-config` flags must go after the `-o` flag.

Note: *HPX* libraries have different names in debug and release mode. If you want to link against a debug *HPX* library, you need to use the `_debug` suffix for the `pkg-config` name. That means instead of `hpx_application` or `hpx_component`, you will have to use `hpx_application_debug` or `hpx_component_debug`. Moreover, all referenced *HPX* components need to have an appended `d` suffix. For example, instead of `-lhpx_iostreams` you will need to specify `-lhpx_iostreamsd`.

Important: If the *HPX* libraries are in a path that is not found by the dynamic linker, you will need to add the path `$HPX_LOCATION/lib` to your linker search path (for example `LD_LIBRARY_PATH` on Linux).

To test the program, type:

```
$ ./hello_world
```

which should print Hello World! and exit.

How to build *HPX* components with pkg-config

Let's try a more complex example involving an *HPX* component. An *HPX* component is a class that exposes *HPX* actions. *HPX* components are compiled into dynamically loaded modules called component libraries. Here's the source code:

hello_world_component.cpp

```
#include <hpx/config.hpp>
#if !defined(HPX_COMPUTE_DEVICE_CODE)
#include <hpx/iostream.hpp>
#include "hello_world_component.hpp"

#include <iostream>

namespace examples { namespace server {
    void hello_world::invoke()
    {
        hpx::cout << "Hello HPX World!" << std::endl;
    }
}} // namespace examples::server

HPX_REGISTER_COMPONENT_MODULE()

typedef hpx::components::component<examples::server::hello_world>
    hello_world_type;

HPX_REGISTER_COMPONENT(hello_world_type, hello_world)

HPX_REGISTER_ACTION(
    examples::server::hello_world::invoke_action, hello_world_invoke_action)
#endif
```

hello_world_component.hpp

```
#pragma once

#include <hpx/config.hpp>
#if !defined(HPX_COMPUTE_DEVICE_CODE)
#include <hpx/hpx.hpp>
#include <hpx/include/actions.hpp>
#include <hpx/include/components.hpp>
#include <hpx/include/lcos.hpp>
#include <hpx/serialization.hpp>

#include <utility>

namespace examples { namespace server {
    struct HPX_COMPONENT_EXPORT hello_world
        : hpx::components::component_base<hello_world>
    {
        void invoke();
        HPX_DEFINE_COMPONENT_ACTION(hello_world, invoke)
    };
}
```

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```

}} // namespace examples::server

HPX_REGISTER_ACTION_DECLARATION(
    examples::server::hello_world::invoke_action, hello_world_invoke_action)

namespace examples {
    struct hello_world
    : hpx::components::client_base<hello_world, server::hello_world>
    {
        typedef hpx::components::client_base<hello_world, server::hello_world>
            base_type;

        hello_world(hpx::future<hpx::id_type>&& f)
            : base_type(std::move(f))
        {
        }

        hello_world(hpx::id_type&& f)
            : base_type(std::move(f))
        {
        }

        void invoke()
        {
            hpx::async<server::hello_world::invoke_action>(this->get_id())
                .get();
        }
    };
} // namespace examples

#endif

```

hello_world_client.cpp

```

#include <hpx/config.hpp>
#if defined(HPX_COMPUTE_HOST_CODE)
#include <hpx/wrap_main.hpp>

#include "hello_world_component.hpp"

int main()
{
    {
        // Create a single instance of the component on this locality.
        examples::hello_world client =
            hpx::new_<examples::hello_world>(hpx::find_here());

        // Invoke the component's action, which will print "Hello World!".
        client.invoke();
    }

    return 0;
}

```

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```
}  
#endif
```

Copy the three source files above into three files (called `hello_world_component.cpp`, `hello_world_component.hpp` and `hello_world_client.cpp`, respectively).

Now, in the directory where you put the files, run the following command to build the component library. (where `$HPX_LOCATION` is the build directory or `CMAKE_INSTALL_PREFIX` you used while building *HPX*):

```
$ export PKG_CONFIG_PATH=$PKG_CONFIG_PATH:$HPX_LOCATION/lib/pkgconfig  
$ c++ -o libhpx_hello_world.so hello_world_component.cpp \  
  `pkg-config --cflags --libs hpx_component` \  
  -lhpx_iostreams -DHPX_COMPONENT_NAME=hpx_hello_world
```

Now pick a directory in which to install your *HPX* component libraries. For this example, we'll choose a directory named `my_hpx_libs`:

```
$ mkdir ~/my_hpx_libs  
$ mv libhpx_hello_world.so ~/my_hpx_libs
```

Note: *HPX* libraries have different names in debug and release mode. If you want to link against a debug *HPX* library, you need to use the `_debug` suffix for the pkg-config name. That means instead of `hpx_application` or `hpx_component` you will have to use `hpx_application_debug` or `hpx_component_debug`. Moreover, all referenced *HPX* components need to have a appended `d` suffix, e.g. instead of `-lhpx_iostreams` you will need to specify `-lhpx_iostreamsd`.

Important: If the *HPX* libraries are in a path that is not found by the dynamic linker. You need to add the path `$HPX_LOCATION/lib` to your linker search path (for example `LD_LIBRARY_PATH` on Linux).

Now, to build the application that uses this component (`hello_world_client.cpp`), we do:

```
$ export PKG_CONFIG_PATH=$PKG_CONFIG_PATH:$HPX_LOCATION/lib/pkgconfig  
$ c++ -o hello_world_client hello_world_client.cpp \  
  `pkg-config --cflags --libs hpx_application` \  
  -L${HOME}/my_hpx_libs -lhpx_hello_world -lhpx_iostreams
```

Important: When using `pkg-config` with *HPX*, the `pkg-config` flags must go after the `-o` flag.

Finally, you'll need to set your `LD_LIBRARY_PATH` before you can run the program. To run the program, type:

```
$ export LD_LIBRARY_PATH="$LD_LIBRARY_PATH:$HOME/my_hpx_libs"  
$ ./hello_world_client
```

which should print `Hello HPX World!` and exit.

Using HPX with CMake-based projects

In addition to the pkg-config support discussed on the previous pages, *HPX* comes with full CMake support. In order to integrate *HPX* into existing or new CMakeLists.txt, you can leverage the `find_package`⁶⁴ command integrated into CMake. Following, is a Hello World component example using CMake.

Let's revisit what we have. We have three files that compose our example application:

- `hello_world_component.hpp`
- `hello_world_component.cpp`
- `hello_world_client.hpp`

The basic structure to include *HPX* into your CMakeLists.txt is shown here:

```
# Require a recent version of cmake
cmake_minimum_required(VERSION 3.18 FATAL_ERROR)

# This project is C++ based.
project(your_app CXX)

# Instruct cmake to find the HPX settings
find_package(HPX)
```

In order to have CMake find *HPX*, it needs to be told where to look for the `HPXConfig.cmake` file that is generated when *HPX* is built or installed. It is used by `find_package(HPX)` to set up all the necessary macros needed to use *HPX* in your project. The ways to achieve this are:

- Set the `HPX_DIR` CMake variable to point to the directory containing the `HPXConfig.cmake` script on the command line when you invoke CMake:

```
$ cmake -DHPX_DIR=$HPX_LOCATION/lib/cmake/HPX ...
```

where `$HPX_LOCATION` is the build directory or `CMAKE_INSTALL_PREFIX` you used when building/configuring *HPX*.

- Set the `CMAKE_PREFIX_PATH` variable to the root directory of your *HPX* build or install location on the command line when you invoke CMake:

```
$ cmake -DCMAKE_PREFIX_PATH=$HPX_LOCATION ...
```

The difference between `CMAKE_PREFIX_PATH` and `HPX_DIR` is that CMake will add common postfixes, such as `lib/cmake/<project`, to the `CMAKE_PREFIX_PATH` and search in these locations too. Note that if your project uses *HPX* as well as other CMake-managed projects, the paths to the locations of these multiple projects may be concatenated in the `CMAKE_PREFIX_PATH`.

- The variables above may be set in the CMake GUI or curses `ccmake` interface instead of the command line.

Additionally, if you wish to require *HPX* for your project, replace the `find_package(HPX)` line with `find_package(HPX REQUIRED)`.

You can check if *HPX* was successfully found with the `HPX_FOUND` CMake variable.

⁶⁴ https://www.cmake.org/cmake/help/latest/command/find_package.html

Using CMake targets

The recommended way of setting up your targets to use *HPX* is to link to the `HPX::hpx` CMake⁶⁵ target:

```
target_link_libraries(hello_world_component PUBLIC HPX::hpx)
```

This requires that you have already created the target like this:

```
add_library(hello_world_component SHARED hello_world_component.cpp)
target_include_directories(hello_world_component PUBLIC ${CMAKE_CURRENT_SOURCE_DIR})
```

When you link your library to the `HPX::hpx` CMake⁶⁶ target, you will be able use *HPX* functionality in your library. To use `main()` as the implicit entry point in your application you must additionally link your application to the CMake target `HPX::wrap_main`. This target is automatically linked to executables if you are using the macros described below (*Using macros to create new targets*). See *Re-use the `main()` function as the main *HPX* entry point* for more information on implicitly using `main()` as the entry point. If you want the same wrapping behavior without including `hpx/hpx_main.hpp`⁶⁷, link to the `HPX::auto_wrap_main` target instead. This enables the runtime initialization around `main()` unconditionally and is useful for codebases where adding the header to `main.cpp` is impractical

Note: The use of `HPX::auto_wrap_main` is not supported when using the native Windows MSVC toolchain.

If you want to use the facilities exposed by `hpx::runtime_manager` in binaries that were not linked as executables (e.g., in shared libraries), you will need make your cmake target explicitly depend on the `HPX::init` target:

```
add_library(hello_world_component SHARED hello_world_component.cpp)
target_link_libraries(hello_world_component PRIVATE HPX::init)
```

Otherwise you may see compilation errors complaining about the header file `hpx/runtime_manager.hpp` not being found.

Creating a component requires setting two additional compile definitions:

```
target_compile_options(hello_world_component
    HPX_COMPONENT_NAME=hello_world
    HPX_COMPONENT_EXPORTS)
```

Instead of setting these definitions manually you may link to the `HPX::component` target, which sets `HPX_COMPONENT_NAME` to `hpx_<target_name>`, where `<target_name>` is the target name of your library. Note that these definitions should be `PRIVATE` to make sure these definitions are not propagated transitively to dependent targets.

In addition to making your library a component you can make it a plugin. To do so link to the `HPX::plugin` target. Similarly to `HPX::component` this will set `HPX_PLUGIN_NAME` to `hpx_<target_name>`. This definition should also be `PRIVATE`. Unlike regular shared libraries, plugins are loaded at runtime from certain directories and will not be found without additional configuration. Plugins should be installed into a directory containing only plugins. For example, the plugins created by *HPX* itself are installed into the `hpx` subdirectory in the library install directory (typically `lib` or `lib64`). When using the `HPX::plugin` target you need to install your plugins into an appropriate directory. You may also want to set the location of your plugin in the build directory with the `*_OUTPUT_DIRECTORY*` CMake target properties to be able to load the plugins in the build directory. Once you've set the install or output directory of your plugin you need to tell your executable where to find it at runtime. You can do this either by setting the environment variable `HPX_COMPONENT_PATHS` or the ini setting `hpx.component_paths` (see `--hpx:ini`) to the directory containing your plugin.

⁶⁵ <https://www.cmake.org>

⁶⁶ <https://www.cmake.org>

⁶⁷ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/wrap/include/hpx/hpx_main.hpp

Using macros to create new targets

In addition to the targets described above, *HPX* provides convenience macros to hide optional boilerplate code that may be useful for your project. The link to the targets described above. We recommend that you use the targets directly whenever possible as they tend to compose better with other targets.

The macro for adding an *HPX* component is `add_hpx_component`. It can be used in your `CMakeLists.txt` file like this:

```
# build your application using HPX
add_hpx_component(hello_world
  SOURCES hello_world_component.cpp
  HEADERS hello_world_component.hpp
  COMPONENT_DEPENDENCIES iostreams)
```

Note: `add_hpx_component` adds a `_component` suffix to the target name. In the example above, a `hello_world_component` target will be created.

The available options to `add_hpx_component` are:

- `SOURCES`: The source files for that component
- `HEADERS`: The header files for that component
- `DEPENDENCIES`: Other libraries or targets this component depends on
- `COMPONENT_DEPENDENCIES`: The components this component depends on
- `PLUGIN`: Treats this component as a plugin-able library
- `COMPILE_FLAGS`: Additional compiler flags
- `LINK_FLAGS`: Additional linker flags
- `FOLDER`: Adds the headers and source files to this Source Group folder
- `EXCLUDE_FROM_ALL`: Do not build this component as part of the `all` target

After adding the component, the way you add the executable is as follows:

```
# build your application using HPX
add_hpx_executable(hello_world
  SOURCES hello_world_client.cpp
  COMPONENT_DEPENDENCIES hello_world)
```

Note: `add_hpx_executable` automatically adds a `_component` suffix to dependencies specified in `COMPONENT_DEPENDENCIES`, meaning you can directly use the name given when adding a component using `add_hpx_component`.

When you configure your application, all you need to do is set the `HPX_DIR` variable to point to the installation of *HPX*.

Note: All library targets built with *HPX* are exported and readily available to be used as arguments to `target_link_libraries`⁶⁸ in your targets. The *HPX* include directories are available with the `HPX_INCLUDE_DIRS` CMake variable.

⁶⁸ https://www.cmake.org/cmake/help/latest/command/target_link_libraries.html

Using the *HPX* compiler wrapper `hpxcxx`

The `hpxcxx` compiler wrapper helps to compile a *HPX* component, application, or object file, based on the arguments passed to it.

```
$ hpxcxx [--exe=<APPLICATION_NAME> | --comp=<COMPONENT_NAME> | -c] FLAGS FILES
```

The `hpxcxx` command **requires** that either an application or a component is built or `-c` flag is specified. If the build is against a debug build, the `-g` is to be specified while building.

Optional FLAGS

- `-l <LIBRARY> | -l<LIBRARY>`: Links `<LIBRARY>` to the build
- `-g`: Specifies that the application or component build is against a debug build
- `-rd`: Sets `release-with-debug-info` option
- `-mr`: Sets `minsize-release` option

All other flags (like `-o OUTPUT_FILE`) are directly passed to the underlying C++ compiler.

Using macros to set up existing targets to use *HPX*

In addition to the `add_hpx_component` and `add_hpx_executable`, you can use the `hpx_setup_target` macro to have an already existing target to be used with the *HPX* libraries:

```
hpx_setup_target(target)
```

Optional parameters are:

- `EXPORT`: Adds it to the CMake export list `HPXTargets`
- `INSTALL`: Generates an install rule for the target
- `PLUGIN`: Treats this component as a plugin-able library
- `TYPE`: The type can be: `EXECUTABLE`, `LIBRARY` or `COMPONENT`
- `DEPENDENCIES`: Other libraries or targets this component depends on
- `COMPONENT_DEPENDENCIES`: The components this component depends on
- `COMPILE_FLAGS`: Additional compiler flags
- `LINK_FLAGS`: Additional linker flags

If you do not use CMake, you can still build against *HPX*, but you should refer to the section on *How to build HPX components with pkg-config*.

Note: Since *HPX* relies on dynamic libraries, the dynamic linker needs to know where to look for them. If *HPX* isn't installed into a path that is configured as a linker search path, external projects need to either set `RPATH` or adapt `LD_LIBRARY_PATH` to point to where the *HPX* libraries reside. In order to set `RPATHs`, you can include `HPX_SetFullRPATH` in your project after all libraries you want to link against have been added. Please also consult the CMake documentation [here](https://gitlab.kitware.com/cmake/community/wikis/doc/cmake/RPATH-handling)⁶⁹.

⁶⁹ <https://gitlab.kitware.com/cmake/community/wikis/doc/cmake/RPATH-handling>

Using HPX with Makefile

A basic project building with *HPX* is through creating makefiles. The process of creating one can get complex depending upon the use of cmake parameter `HPX_WITH_HPX_MAIN` (which defaults to ON).

How to build *HPX* applications with makefile

If *HPX* is installed correctly, you should be able to build and run a simple Hello World program. It prints Hello World! on the *locality* you run it on.

```
// Including 'hpx/hpx_main.hpp' instead of the usual 'hpx/hpx_init.hpp' enables
// to use the plain C-main below as the direct main HPX entry point.
#include <hpx/hpx_main.hpp>
#include <hpx/iostream.hpp>

int main()
{
    // Say hello to the world!
    hpx::cout << "Hello World!\n" << std::flush;
    return 0;
}
```

Copy the content of this program into a file called `hello_world.cpp`.

Now, in the directory where you put `hello_world.cpp`, create a Makefile. Add the following code:

```
CXX=(CXX) # Add your favourite compiler here or let makefile choose default.

CXXFLAGS=-O3 -std=c++17

Boost_ROOT=/path/to/boost
Hwloc_ROOT=/path/to/hwloc
Tcmalloc_ROOT=/path/to/tcmalloc
HPX_ROOT=/path/to/hpx

INCLUDE_DIRECTIVES=$(HPX_ROOT)/include $(Boost_ROOT)/include $(Hwloc_ROOT)/include

LIBRARY_DIRECTIVES=-L$(HPX_ROOT)/lib $(HPX_ROOT)/lib/libhpx_init.a $(HPX_ROOT)/lib/
↳ libhpx.so $(Boost_ROOT)/lib/libboost_atomic-mt.so $(Boost_ROOT)/lib/libboost_
↳ filesystem-mt.so $(Boost_ROOT)/lib/libboost_program_options-mt.so $(Boost_ROOT)/lib/
↳ libboost_regex-mt.so $(Boost_ROOT)/lib/libboost_system-mt.so -lpthread $(Tcmalloc_
↳ ROOT)/libtcmalloc_minimal.so $(Hwloc_ROOT)/libhwloc.so -ldl -lrt

LINK_FLAGS=$(HPX_ROOT)/lib/libhpx_wrap.a -Wl,-wrap=main # should be left empty for HPX_
↳ WITH_HPX_MAIN=OFF

hello_world: hello_world.o
    $(CXX) $(CXXFLAGS) -o hello_world hello_world.o $(LIBRARY_DIRECTIVES) $(LINK_FLAGS)

hello_world.o:
    $(CXX) $(CXXFLAGS) -c -o hello_world.o hello_world.cpp $(INCLUDE_DIRECTIVES)
```

Important: `LINK_FLAGS` should be left empty if `HPX_WITH_HPX_MAIN` is set to OFF. Boost in the above example

is build with `--layout=tagged`. Actual Boost flags may vary on your build of Boost.

To build the program, type:

```
$ make
```

A successful build should result in `hello_world` binary. To test, type:

```
$ ./hello_world
```

How to build *HPX* components with makefile

Let's try a more complex example involving an *HPX* component. An *HPX* component is a class that exposes *HPX* actions. *HPX* components are compiled into dynamically-loaded modules called component libraries. Here's the source code:

hello_world_component.cpp

```
#include <hpx/config.hpp>
#if !defined(HPX_COMPUTE_DEVICE_CODE)
#include <hpx/iostream.hpp>
#include "hello_world_component.hpp"

#include <iostream>

namespace examples { namespace server {
    void hello_world::invoke()
    {
        hpx::cout << "Hello HPX World!" << std::endl;
    }
}} // namespace examples::server

HPX_REGISTER_COMPONENT_MODULE()

typedef hpx::components::component<examples::server::hello_world>
    hello_world_type;

HPX_REGISTER_COMPONENT(hello_world_type, hello_world)

HPX_REGISTER_ACTION(
    examples::server::hello_world::invoke_action, hello_world_invoke_action)
#endif
```

hello_world_component.hpp

```
#pragma once

#include <hpx/config.hpp>
#if !defined(HPX_COMPUTE_DEVICE_CODE)
#include <hpx/hpx.hpp>
#include <hpx/include/actions.hpp>
#include <hpx/include/components.hpp>
```

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```

#include <hpx/include/lcos.hpp>
#include <hpx/serialization.hpp>

#include <utility>

namespace examples { namespace server {
    struct HPX_COMPONENT_EXPORT hello_world
        : hpx::components::component_base<hello_world>
    {
        void invoke();
        HPX_DEFINE_COMPONENT_ACTION(hello_world, invoke)
    };
}} // namespace examples::server

HPX_REGISTER_ACTION_DECLARATION(
    examples::server::hello_world::invoke_action, hello_world_invoke_action)

namespace examples {
    struct hello_world
        : hpx::components::client_base<hello_world, server::hello_world>
    {
        typedef hpx::components::client_base<hello_world, server::hello_world>
            base_type;

        hello_world(hpx::future<hpx::id_type>&& f)
            : base_type(std::move(f))
        {
        }

        hello_world(hpx::id_type&& f)
            : base_type(std::move(f))
        {
        }

        void invoke()
        {
            hpx::async<server::hello_world::invoke_action>(this->get_id())
                .get();
        }
    };
} // namespace examples

#endif

```

hello_world_client.cpp

```

#include <hpx/config.hpp>
#if defined(HPX_COMPUTE_HOST_CODE)
#include <hpx/wrap_main.hpp>

#include "hello_world_component.hpp"

```

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```

int main()
{
    {
        // Create a single instance of the component on this locality.
        examples::hello_world client =
            hpx::new_<examples::hello_world>(hpx::find_here());

        // Invoke the component's action, which will print "Hello World!".
        client.invoke();
    }

    return 0;
}
#endif

```

Now, in the directory, create a Makefile. Add the following code:

```

CXX=(CXX) # Add your favourite compiler here or let makefile choose default.

CXXFLAGS=-O3 -std=c++17

Boost_ROOT=/path/to/boost
Hwloc_ROOT=/path/to/hwloc
Tcmalloc_ROOT=/path/to/tcmalloc
HPX_ROOT=/path/to/hpx

INCLUDE_DIRECTIVES=$(HPX_ROOT)/include $(Boost_ROOT)/include $(Hwloc_ROOT)/include

LIBRARY_DIRECTIVES=-L$(HPX_ROOT)/lib $(HPX_ROOT)/lib/libhpx_init.a $(HPX_ROOT)/lib/
↳ libhpx.so $(Boost_ROOT)/lib/libboost_atomic-mt.so $(Boost_ROOT)/lib/libboost_
↳ filesystem-mt.so $(Boost_ROOT)/lib/libboost_program_options-mt.so $(Boost_ROOT)/lib/
↳ libboost_regex-mt.so $(Boost_ROOT)/lib/libboost_system-mt.so -lpthread $(Tcmalloc_
↳ ROOT)/libtcmalloc_minimal.so $(Hwloc_ROOT)/libhwloc.so -ldl -lrt

LINK_FLAGS=$(HPX_ROOT)/lib/libhpx_wrap.a -Wl,-wrap=main # should be left empty for HPX_
↳ WITH_HPX_MAIN=OFF

hello_world_client: libhpx_hello_world hello_world_client.o
    $(CXX) $(CXXFLAGS) -o hello_world_client $(LIBRARY_DIRECTIVES) libhpx_hello_world
↳ $(LINK_FLAGS)

hello_world_client.o: hello_world_client.cpp
    $(CXX) $(CXXFLAGS) -o hello_world_client.o hello_world_client.cpp $(INCLUDE_DIRECTIVES)

libhpx_hello_world: hello_world_component.o
    $(CXX) $(CXXFLAGS) -o libhpx_hello_world hello_world_component.o $(LIBRARY_DIRECTIVES)

hello_world_component.o: hello_world_component.cpp
    $(CXX) $(CXXFLAGS) -c -o hello_world_component.o hello_world_component.cpp $(INCLUDE_
↳ DIRECTIVES)

```

To build the program, type:

```
$ make
```

A successful build should result in `hello_world` binary. To test, type:

```
$ ./hello_world
```

Note: Due to high variations in CMake flags and library dependencies, it is recommended to build *HPX* applications and components with `pkg-config` or `CMakeLists.txt`. Writing Makefile may result in broken builds if due care is not taken. `pkg-config` files and CMake systems are configured with CMake build of *HPX*. Hence, they are stable when used together and provide better support overall.

2.4.9 Starting the *HPX* runtime

In order to write an application that uses services from the *HPX* runtime system, you need to initialize the *HPX* library by inserting certain calls into the code of your application. Depending on your use case, this can be done in 3 different ways:

- *Minimally invasive:* Re-use the `main()` function as the main *HPX* entry point.
- *Balanced use case:* Supply your own main *HPX* entry point while blocking the main thread.
- *Most flexibility:* Supply your own main *HPX* entry point while avoiding blocking the main thread.
- *Suspend and resume:* As above but suspend and resume the *HPX* runtime to allow for other runtimes to be used.

Re-use the `main()` function as the main *HPX* entry point

This method is the least intrusive to your code. However, it provides you with the smallest flexibility in terms of initializing the *HPX* runtime system. The following code snippet shows what a minimal *HPX* application using this technique looks like:

```
#include <hpx/hpx_main.hpp>

int main(int argc, char* argv[])
{
    return 0;
}
```

The only change to your code you have to make is to include the file `hpx/hpx_main.hpp`. In this case the function `main()` will be invoked as the first *HPX* thread of the application. The runtime system will be initialized behind the scenes before the function `main()` is executed and will automatically stop after `main()` has returned. For this method to work you must link your application to the CMake⁷⁰ target `HPX::wrap_main`. This is done automatically if you are using the provided macros (*Using macros to create new targets*) to set up your application, but must be done explicitly if you are using targets directly (*Using CMake targets*). All *HPX* API functions can be used from within the `main()` function now. If you cannot or do not want to include `hpx/hpx_main.hpp` in `main.cpp`, you can instead link against `HPX::auto_wrap_main`. That target enables the same runtime startup path without needing the header-triggered option.

Note: The use of `HPX::auto_wrap_main` is not supported when using the native Windows MSVC toolchain.

⁷⁰ <https://www.cmake.org>

Note: The function `main()` does not need to expect receiving `argc` and `argv` as shown above, but could expose the signature `int main()`. This is consistent with the usually allowed prototypes for the function `main()` in C++ applications.

All command line arguments specific to *HPX* will still be processed by the *HPX* runtime system as usual. However, those command line options will be removed from the list of values passed to `argc/argv` of the function `main()`. The list of values passed to `main()` will hold only the commandline options that are not recognized by the *HPX* runtime system (see the section *HPX Command Line Options* for more details on what options are recognized by *HPX*).

Note: In this mode all one-letter shortcuts that are normally available on the *HPX* command line are disabled (such as `-t` or `-l` see *HPX Command Line Options*). This is done to minimize any possible interaction between the command line options recognized by the *HPX* runtime system and any command line options defined by the application.

The value returned from the function `main()` as shown above will be returned to the operating system as usual.

Important: To achieve this seamless integration, the header file `hpx/hpx_main.hpp` defines a macro:

```
#define main hpx_startup::user_main
```

which could result in unexpected behavior.

Important: To achieve this seamless integration, we use different implementations for different operating systems. In case of Linux or macOS, the code present in `hpx_wrap.cpp` is put into action. We hook into the system function in case of Linux and provide alternate entry point in case of macOS. For other operating systems we rely on a macro:

```
#define main hpx_startup::user_main
```

provided in the header file `hpx/hpx_main.hpp`. This implementation can result in unexpected behavior.

Caution: We make use of an *override* variable `include_libhpx_wrap` in the header file `hpx/hpx_main.hpp` to swiftly choose the function call stack at runtime. Therefore, the header file should *only* be included in the main executable. Including it in the components will result in multiple definition of the variable.

Supply your own main *HPX* entry point while blocking the main thread

With this method you need to provide an explicit main-thread function named `hpx_main` at global scope. This function will be invoked as the main entry point of your *HPX* application on the console *locality* only (this function will be invoked as the first *HPX* thread of your application). All *HPX* API functions can be used from within this function.

The thread executing the function `hpx::init` will block waiting for the runtime system to exit. The value returned from `hpx_main` will be returned from `hpx::init` after the runtime system has stopped.

The function `hpx::finalize` has to be called on one of the *HPX* localities in order to signal that all work has been scheduled and the runtime system should be stopped after the scheduled work has been executed.

This method of invoking *HPX* has the advantage of the user being able to decide which version of `hpx::init` to call. This allows to pass additional configuration parameters while initializing the *HPX* runtime system.


```
#include <hpx/hpx_init.hpp>

int hpx_main(int argc, char* argv[])
{
    // Any HPX application logic goes here...
    return hpx::finalize();
}

int main(int argc, char* argv[])
{
    // Initialize HPX, run hpx_main as the first HPX thread, and
    // wait for hpx::finalize being called.
    return hpx::init(argc, argv);
}
```

Note: The function `hpx_main` does not need to expect receiving `argc/argv` as shown above, but could expose one of the following signatures:

```
int hpx_main();
int hpx_main(int argc, char* argv[]);
int hpx_main(hpx::program_options::variables_map& vm);
```

This is consistent with (and extends) the usually allowed prototypes for the function `main()` in C++ applications.

The header file to include for this method of using *HPX* is `hpx/hpx_init.hpp`.

There are many additional overloads of `hpx::init` available, such as the ability to provide your own entry-point function instead of `hpx_main`. Please refer to the function documentation for more details (see: `hpx/hpx_init.hpp`).

Supply your own main *HPX* entry point while avoiding blocking the main thread

With this method you need to provide an explicit main thread function named `hpx_main` at global scope. This function will be invoked as the main entry point of your *HPX* application on the console *locality* only (this function will be invoked as the first *HPX* thread of your application). All *HPX* API functions can be used from within this function.

The thread executing the function `hpx::start` will *not* block waiting for the runtime system to exit, but will return immediately. The function `hpx::finalize` has to be called on one of the *HPX* localities in order to signal that all work has been scheduled and the runtime system should be stopped after the scheduled work has been executed.

This method of invoking *HPX* is useful for applications where the main thread is used for special operations, such as GUIs. The function `hpx::stop` can be used to wait for the *HPX* runtime system to exit and should at least be used as the last function called in `main()`. The value returned from `hpx_main` will be returned from `hpx::stop` after the runtime system has stopped.

```
#include <hpx/hpx_start.hpp>

int hpx_main(int argc, char* argv[])
{
    // Any HPX application logic goes here...
    return hpx::finalize();
}
```

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```
int main(int argc, char* argv[])
{
    // Initialize HPX, run hpx_main.
    hpx::start(argc, argv);

    // ...Execute other code here...

    // Wait for hpx::finalize being called.
    return hpx::stop();
}
```

Note: The function `hpx_main` does not need to expect receiving `argc/argv` as shown above, but could expose one of the following signatures:

```
int hpx_main();
int hpx_main(int argc, char* argv[]);
int hpx_main(hpx::program_options::variables_map& vm);
```

This is consistent with (and extends) the usually allowed prototypes for the function `main()` in C++ applications.

The header file to include for this method of using *HPX* is `hpx/hpx_start.hpp`.

There are many additional overloads of `hpx::start` available, such as the option for users to provide their own entry point function instead of `hpx_main`. Please refer to the function documentation for more details (see: `hpx/hpx_start.hpp`).

Supply your own explicit startup function as the main *HPX* entry point

There is also a way to specify any function (besides `hpx_main`) to be used as the main entry point for your *HPX* application:

```
#include <hpx/hpx_init.hpp>

int application_entry_point(int argc, char* argv[])
{
    // Any HPX application logic goes here...
    return hpx::finalize();
}

int main(int argc, char* argv[])
{
    // Initialize HPX, run application_entry_point as the first HPX thread,
    // and wait for hpx::finalize being called.
    return hpx::init(&application_entry_point, argc, argv);
}
```

Note: The function supplied to `hpx::init` must have one of the following prototypes:

```
int application_entry_point(int argc, char* argv[]);
int application_entry_point(hpx::program_options::variables_map&
vm);
```

Note: If `nullptr` is used as the function argument, *HPX* will not run any startup function on this locality.

Suspending and resuming the *HPX* runtime

In some applications it is required to combine *HPX* with other runtimes. To support this use case, *HPX* provides two functions: `hpx::suspend` and `hpx::resume`. `hpx::suspend` is a blocking call which will wait for all scheduled tasks to finish executing and then put the thread pool OS threads to sleep. `hpx::resume` simply wakes up the sleeping threads so that they are ready to accept new work. `hpx::suspend` and `hpx::resume` can be found in the header `hpx/hpx_suspend.hpp`.

```
#include <hpx/hpx_start.hpp>
#include <hpx/hpx_suspend.hpp>

int main(int argc, char* argv[])
{
    // Initialize HPX, don't run hpx_main
    hpx::start(nullptr, argc, argv);

    // Schedule a function on the HPX runtime
    hpx::post(&my_function, ...);

    // Wait for all tasks to finish, and suspend the HPX runtime
    hpx::suspend();

    // Execute non-HPX code here

    // Resume the HPX runtime
    hpx::resume();

    // Schedule more work on the HPX runtime

    // hpx::finalize has to be called from the HPX runtime before hpx::stop
    hpx::post([]() { hpx::finalize(); });
    return hpx::stop();
}
```

Note: `hpx::suspend` does not wait for `hpx::finalize` to be called. Only call `hpx::finalize` when you wish to fully stop the *HPX* runtime.

Warning:

`hpx::suspend` only waits for local tasks, i.e. tasks on the
current locality, to finish executing. When using `hpx::suspend` in a multi-locality scenario the user is responsible for ensuring that any work required from other localities has also finished.

HPX also supports suspending individual thread pools and threads. For details on how to do that, see the documentation

for `hpx::threads::thread_pool_base`.

Automatically suspending worker threads

The previous method guarantees that the worker threads are suspended when you ask for it and that they stay suspended. An alternative way to achieve the same effect is to tweak how quickly *HPX* suspends its worker threads when they run out of work. The following configuration values make sure that *HPX* idles very quickly:

```
hpx.max_idle_backoff_time = 1000
hpx.max_idle_loop_count = 0
```

They can be set on the command line using `--hpx:ini=hpx.max_idle_backoff_time=1000` and `--hpx:ini=hpx.max_idle_loop_count=0`. See *Launching and configuring HPX applications* for more details on how to set configuration parameters.

After setting idling parameters the previous example could now be written like this instead:

```
#include <hpx/hpx_start.hpp>

int main(int argc, char* argv[])
{
    // Initialize HPX, don't run hpx_main
    hpx::start(nullptr, argc, argv);

    // Schedule some functions on the HPX runtime
    // NOTE: run_as_hpx_thread blocks until completion.
    hpx::run_as_hpx_thread(&my_function, ...);
    hpx::run_as_hpx_thread(&my_other_function, ...);

    // hpx::finalize has to be called from the HPX runtime before hpx::stop
    hpx::post([]() { hpx::finalize(); });
    return hpx::stop();
}
```

In this example each call to `hpx::run_as_hpx_thread` acts as a “parallel region”.

Working of `hpx_main.hpp`

In order to initialize *HPX* from `main()`, we make use of linker tricks.

It is implemented differently for different operating systems. The method of implementation is as follows:

- *Linux*: Using linker `--wrap` option.
- *Mac OSX*: Using the linker `-e` option.
- *Windows*: Using `#define main hpx_startup::user_main`

Linux implementation

We make use of the Linux linker `ld`'s `--wrap` option to wrap the `main()` function. This way any calls to `main()` are redirected to our own implementation of `main`. It is here that we check for the existence of `hpx_main.hpp` by making use of a shadow variable `include_libhpx_wrap`. The value of this variable determines the function stack at runtime.

The implementation can be found in `libhpx_wrap.a`.

Important: It is necessary that `hpx_main.hpp` be not included more than once. Multiple inclusions can result in multiple definition of `include_libhpx_wrap`.

Mac OSX implementation

Here we make use of yet another linker option `-e` to change the entry point to our custom entry function `initialize_main`. We initialize the *HPX* runtime system from this function and call `main` from the initialized system. We determine the function stack at runtime by making use of the shadow variable `include_libhpx_wrap`.

The implementation can be found in `libhpx_wrap.a`.

Important: It is necessary that `hpx_main.hpp` be not included more than once. Multiple inclusions can result in multiple definition of `include_libhpx_wrap`.

Windows implementation

We make use of a macro `#define main hpx_startup::user_main` to take care of the initializations.

This implementation could result in unexpected behaviors.

2.4.10 Launching and configuring *HPX* applications

Configuring *HPX* applications

All *HPX* applications can be configured using special command line options and/or using special configuration files. This section describes the available options, the configuration file format, and the algorithm used to locate possible predefined configuration files. Additionally, this section describes the defaults assumed if no external configuration information is supplied.

During startup any *HPX* application applies a predefined search pattern to locate one or more configuration files. All found files will be read and merged in the sequence they are found into one single internal database holding all configuration properties. This database is used during the execution of the application to configure different aspects of the runtime system.

In addition to the ini files, any application can supply its own configuration files, which will be merged with the configuration database as well. Moreover, the user can specify additional configuration parameters on the command line when executing an application. The *HPX* runtime system will merge all command line configuration options (see the description of the `--hpx:ini`, `--hpx:config`, and `--hpx:app-config` command line options).

The HPX ini file format

All HPX applications can be configured using a special file format that is similar to the well-known [Windows INI file format](#)⁷¹. This is a structured text format that allows users to group key/value pairs (properties) into sections. The basic element contained in an ini file is the property. Every property has a name and a value, delimited by an equal sign '='. The name appears to the left of the equal sign:

```
name=value
```

The value may contain equal signs as only the first '=' character is interpreted as the delimiter between name and value. Whitespace before the name, after the value and immediately before and after the delimiting equal sign is ignored. Whitespace inside the value is retained.

Properties may be grouped into arbitrarily named sections. The section name appears on a line by itself, in square brackets. All properties after the section declaration are associated with that section. There is no explicit “end of section” delimiter; sections end at the next section declaration or the end of the file:

```
[section]
```

In HPX sections can be nested. A nested section has a name composed of all section names it is embedded in. The section names are concatenated using a dot '.':

```
[outer_section.inner_section]
```

Here, `inner_section` is logically nested within `outer_section`.

It is possible to use the full section name concatenated with the property name to refer to a particular property. For example, in:

```
[a.b.c]  
d = e
```

the property value of `d` can be referred to as `a.b.c.d=e`.

In HPX ini files can contain comments. Hash signs '#' at the beginning of a line indicate a comment. All characters starting with '#' until the end of the line are ignored.

If a property with the same name is reused inside a section, the second occurrence of this property name will override the first occurrence (discard the first value). Duplicate sections simply merge their properties together, as if they occurred contiguously.

In HPX ini files a property value `${FOO:default}` will use the environmental variable `FOO` to extract the actual value if it is set and `default` otherwise. No default has to be specified. Therefore, `${FOO}` refers to the environmental variable `FOO`. If `FOO` is not set or empty, the overall expression will evaluate to an empty string. A property value `[$section.key:default]` refers to the value held by the property `section.key` if it exists and `default` otherwise. No default has to be specified. Therefore `[$section.key]` refers to the property `section.key`. If the property `section.key` is not set or empty, the overall expression will evaluate to an empty string.

Note: Any property `[$section.key:default]` is evaluated whenever it is queried and not when the configuration data is initialized. This allows for lazy evaluation and relaxes initialization order of different sections. The only exception are recursive property values, e.g., values referring to the very key they are associated with. Those property values are evaluated at initialization time to avoid infinite recursion.

⁷¹ https://en.wikipedia.org/wiki/INI_file

Built-in default configuration settings

During startup any *HPX* application applies a predefined search pattern to locate one or more configuration files. All found files will be read and merged in the sequence they are found into one single internal data structure holding all configuration properties.

As a first step the internal configuration database is filled with a set of default configuration properties. Those settings are described on a section by section basis below.

Note: You can print the default configuration settings used for an executable by specifying the command line option `--hpx:dump-config`.

The system configuration section

```
[system]
pid = <process-id>
prefix = <current prefix path of core HPX library>
executable = <current prefix path of executable>
```

Property	Description
system.pid	This is initialized to store the current OS-process id of the application instance.
system.prefix	This is initialized to the base directory <i>HPX</i> has been loaded from.
system.executable_prefix	This is initialized to the base directory the current executable has been loaded from.

The *HPX* configuration section

```
[hpx]
location = ${HPX_LOCATION:${system.prefix}}
component_path = ${hpx.location}/lib/hpx:${system.executable_prefix}/lib/hpx:${system.
↪executable_prefix}/../lib/hpx
master_ini_path = ${hpx.location}/share/hpx-<version>:${system.executable_prefix}/share/
↪hpx-<version>:${system.executable_prefix}/../share/hpx-<version>
ini_path = ${hpx.master_ini_path}/ini
os_threads = 1
cores = all
localities = 1
program_name =
cmd_line =
lock_detection = ${HPX_LOCK_DETECTION:0}
throw_on_held_lock = ${HPX_THROW_ON_HELD_LOCK:1}
minimal_deadlock_detection = <debug>
spinlock_deadlock_detection = <debug>
spinlock_deadlock_detection_limit = ${HPX_SPINLOCK_DEADLOCK_DETECTION_LIMIT:1000000}
max_background_threads = ${HPX_MAX_BACKGROUND_THREADS:${hpx.os_threads}}
max_idle_loop_count = ${HPX_MAX_IDLE_LOOP_COUNT:<hpx_idle_loop_count_max>}
max_busy_loop_count = ${HPX_MAX_BUSY_LOOP_COUNT:<hpx_busy_loop_count_max>}
max_idle_backoff_time = ${HPX_MAX_IDLE_BACKOFF_TIME:<hpx_idle_backoff_time_max>}
```

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```
exception_verbosity = ${HPX_EXCEPTION_VERBOSITY:2}
trace_depth = ${HPX_TRACE_DEPTH:20}
handle_signals = ${HPX_HANDLE_SIGNALS:1}
handle_failed_new = ${HPX_HANDLE_FAILED_NEW:1}



### [hpx.stacks]


small_size = ${HPX_SMALL_STACK_SIZE:<hpx_small_stack_size>}
medium_size = ${HPX_MEDIUM_STACK_SIZE:<hpx_medium_stack_size>}
large_size = ${HPX_LARGE_STACK_SIZE:<hpx_large_stack_size>}
huge_size = ${HPX_HUGE_STACK_SIZE:<hpx_huge_stack_size>}
use_guard_pages = ${HPX_THREAD_GUARD_PAGE:1}
```


Property	Description
<code>hpx.location</code>	This is initialized to the id of the <i>locality</i> this application instance is running on.
<code>hpx.component_path</code>	Duplicates are discarded. This property can refer to a list of directories separated by ':' (Linux, Android, and MacOS) or by ';' (Windows).
<code>hpx.master_ini_path</code>	This is initialized to the list of default paths of the main <code>hpx.ini</code> configuration files. This property can refer to a list of directories separated by ':' (Linux, Android, and MacOS) or using ';' (Windows).
<code>hpx.ini_path</code>	This is initialized to the default path where <i>HPX</i> will look for more ini configuration files. This property can refer to a list of directories separated by ':' (Linux, Android, and MacOS) or using ';' (Windows).
<code>hpx.os_threads</code>	This setting reflects the number of OS threads used for running <i>HPX</i> threads. Defaults to number of detected cores (not hyperthreads/PUs).
<code>hpx.cores</code>	This setting reflects the number of cores used for running <i>HPX</i> threads. Defaults to number of detected cores (not hyperthreads/PUs).
<code>hpx.localities</code>	This setting reflects the number of localities the application is running on. Defaults to 1.
<code>hpx.program_name</code>	This setting reflects the program name of the application instance. Initialized from the command line <code>argv[0]</code> .
<code>hpx.cmd_line</code>	This setting reflects the actual command line used to launch this application instance.
<code>hpx.lock_detection</code>	This setting verifies that no locks are being held while a <i>HPX</i> thread is suspended. This setting is applicable only if <code>HPX_WITH_VERIFY_LOCKS</code> is set during configuration in CMake.
<code>hpx.throw_on_hold</code>	This setting causes an exception if during lock detection at least one lock is being held while a <i>HPX</i> thread is suspended. This setting is applicable only if <code>HPX_WITH_VERIFY_LOCKS</code> is set during configuration in CMake. This setting has no effect if <code>hpx.lock_detection=0</code> .
<code>hpx.minimal_deadlock_detection</code>	This setting enables support for minimal deadlock detection for <i>HPX</i> threads. By default this is set to 1 (for Debug builds) or to 0 (for Release, RelWithDebInfo, RelMinSize builds). This setting is effective only if <code>HPX_WITH_THREAD_DEADLOCK_DETECTION</code> is set during configuration in CMake.
<code>hpx.spinlock_deadlock_detection</code>	This setting verifies that spinlocks don't spin longer than specified using the <code>spinlock_deadlock_detection_limit</code> . This setting is applicable only if <code>HPX_WITH_SPINLOCK_DEADLOCK_DETECTION</code> is set during configuration in CMake. By default this is set to 1 (for Debug builds) or to 0 (for Release, RelWithDebInfo, RelMinSize builds).
<code>hpx.spinlock_deadlock_detection_limit</code>	This setting specifies the upper limit of the allowed number of spins that spinlocks are allowed to perform. This setting is applicable only if <code>HPX_WITH_SPINLOCK_DEADLOCK_DETECTION</code> is set during configuration in CMake. By default this is set to 1000000.
<code>hpx.max_background_threads</code>	This setting defines the number of threads in the scheduler, which are used to execute background work. By default this is the same as the number of cores used for the scheduler.
<code>hpx.max_idle_loop_count</code>	By default this is defined by the preprocessor constant <code>HPX_IDLE_LOOP_COUNT_MAX</code> . This is an internal setting that you should change only if you know exactly what you are doing.
<code>hpx.max_busy_loop_count</code>	This setting defines the maximum value of the busy-loop counter in the scheduler. By default this is defined by the preprocessor constant <code>HPX_BUSY_LOOP_COUNT_MAX</code> . This is an internal setting that you should change only if you know exactly what you are doing.
<code>hpx.max_idle_backoff_time</code>	This setting defines the maximum time (in milliseconds) for <code>hpx.max_idle_loop_count</code> iterations. This setting is applicable only if <code>HPX_WITH_THREAD_MANAGER_IDLE_BACKOFF</code> is set during configuration in CMake ⁷² . By default this is defined by the preprocessor constant <code>HPX_IDLE_BACKOFF_TIME_MAX</code> . This is an internal setting that you should change only if you know exactly what you are doing.
<code>hpx.exception_verbosity</code>	This setting defines the verbosity of exceptions. Valid values are integers. A setting of 2 or higher prints all available information. A setting of 1 leaves out the build configuration and environment variables. A setting of 0 or lower prints only the description of the thrown exception and the file name, function, and line number where the exception was thrown. The default value is 2 or the value of the environment variable <code>HPX_EXCEPTION_VERBOSITY</code> .
<code>hpx.trace_depth</code>	This setting defines the number of stack-levels printed in generated stack backtraces. This defaults to 20, but can be changed using the cmake <code>HPX_WITH_THREAD_BACKTRACE_DEPTH</code> configuration setting.
<code>hpx.handle_signals</code>	This setting defines whether <i>HPX</i> will register signal handlers that will print the configuration information (stack backtrace, system information, etc.) whenever a signal is raised. The default is 1. Setting this value to 0 can be useful in cases when generating a core-dump on segmentation faults or similar signals is desired.

The `hpx.threadpools` configuration section

```
[hpx.threadpools]
io_pool_size = ${HPX_NUM_IO_POOL_SIZE:2}
parcel_pool_size = ${HPX_NUM_PARCEL_POOL_SIZE:2}
timer_pool_size = ${HPX_NUM_TIMER_POOL_SIZE:2}
```

Property	Description
<code>hpx.threadpools.io_pool_size</code>	The value of this property defines the number of OS threads created for the internal I/O thread pool.
<code>hpx.threadpools.parcel_pool_size</code>	The value of this property defines the number of OS threads created for the internal parcel thread pool.
<code>hpx.threadpools.timer_pool_size</code>	The value of this property defines the number of OS threads created for the internal timer thread pool.

The `hpx.thread_queue` configuration section

Important: These are the setting control internal values used by the thread scheduling queues in the *HPX* scheduler. You should not modify these settings unless you know exactly what you are doing.

```
[hpx.thread_queue]
min_tasks_to_steal_pending = ${HPX_THREAD_QUEUE_MIN_TASKS_TO_STEAL_PENDING:0}
min_tasks_to_steal_staged = ${HPX_THREAD_QUEUE_MIN_TASKS_TO_STEAL_STAGED:0}
min_add_new_count = ${HPX_THREAD_QUEUE_MIN_ADD_NEW_COUNT:10}
max_add_new_count = ${HPX_THREAD_QUEUE_MAX_ADD_NEW_COUNT:10}
max_delete_count = ${HPX_THREAD_QUEUE_MAX_DELETE_COUNT:1000}
```

Property	Description
<code>hpx.thread_queue.min_tasks_to_steal_pending</code>	The value of this property defines the number of pending <i>HPX</i> threads that have to be available before neighboring cores are allowed to steal work. The default is to allow stealing always.
<code>hpx.thread_queue.min_tasks_to_steal_staged</code>	The value of this property defines the number of staged <i>HPX</i> tasks that need to be available before neighboring cores are allowed to steal work. The default is to allow stealing always.
<code>hpx.thread_queue.min_add_new_count</code>	The value of this property defines the minimal number of tasks to be converted into <i>HPX</i> threads whenever the thread queues for a core have run empty.
<code>hpx.thread_queue.max_add_new_count</code>	The value of this property defines the maximal number of tasks to be converted into <i>HPX</i> threads whenever the thread queues for a core have run empty.
<code>hpx.thread_queue.max_delete_count</code>	The value of this property defines the number of terminated <i>HPX</i> threads to discard during each invocation of the corresponding function.

⁷² <https://www.cmake.org>

The `hpx.components` configuration section

```
[hpx.components]
load_external = ${HPX_LOAD_EXTERNAL_COMPONENTS:1}
```

Property	Description
<code>hpx.components.load_external</code>	This entry defines whether external components will be loaded on this <i>locality</i> . This entry is normally set to 1, and usually there is no need to directly change this value. It is automatically set to 0 for a dedicated AGAS server <i>locality</i> .

Additionally, the section `hpx.components` will be populated with the information gathered from all found components. The information loaded for each of the components will contain at least the following properties:

```
[hpx.components.<component_instance_name>]
name = <component_name>
path = <full_path_of_the_component_module>
enabled = ${hpx.components.load_external}
```

Property	Description
<code>hpx.components.<component_instance_name>.name</code>	This is the name of a component, usually the same as the second argument to the macro used while registering the component with <code>HPX_REGISTER_COMPONENT</code> . Set by the component factory.
<code>hpx.components.<component_instance_name>.path</code>	This is either the full path file name of the component module or the directory the component module is located in. In this case, the component module name will be derived from the property <code>hpx.components.<component_instance_name>.name</code> . Set by the component factory.
<code>hpx.components.<component_instance_name>.enabled</code>	This setting explicitly enables or disables the component. This is an optional property. <code>HPX</code> assumes that the component is enabled if it is not defined.

The value for `<component_instance_name>` is usually the same as for the corresponding `name` property. However, generally it can be defined to any arbitrary instance name. It is used to distinguish between different ini sections, one for each component.

The `hpx.parcel` configuration section

```
[hpx.parcel]
address = ${HPX_PARCEL_SERVER_ADDRESS:<hpx_initial_ip_address>}
port = ${HPX_PARCEL_SERVER_PORT:<hpx_initial_ip_port>}
bootstrap = ${HPX_PARCEL_BOOTSTRAP:<hpx_parcel_bootstrap>}
max_connections = ${HPX_PARCEL_MAX_CONNECTIONS:<hpx_parcel_max_connections>}
max_connections_per_locality = ${HPX_PARCEL_MAX_CONNECTIONS_PER_LOCALITY:<hpx_parcel_max_
↳connections_per_locality>}
max_message_size = ${HPX_PARCEL_MAX_MESSAGE_SIZE:<hpx_parcel_max_message_size>}
max_outbound_message_size = ${HPX_PARCEL_MAX_OUTBOUND_MESSAGE_SIZE:<hpx_parcel_max_
↳outbound_message_size>}
array_optimization = ${HPX_PARCEL_ARRAY_OPTIMIZATION:1}
zero_copy_optimization = ${HPX_PARCEL_ZERO_COPY_OPTIMIZATION:${hpx.parcel.array_
↳optimization}}
```

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```

zero_copy_receive_optimization = ${HPX_PARCEL_ZERO_COPY_RECEIVE_OPTIMIZATION:${hpx.
↪parcel.array_optimization}}
async_serialization = ${HPX_PARCEL_ASYNC_SERIALIZATION:1}
message_handlers = ${HPX_PARCEL_MESSAGE_HANDLERS:0}

```

Property	Description
<code>hpx.parcel.address</code>	This property defines the default IP address to be used for the <i>parcel</i> layer to listen to. This IP address will be used as long as no other values are specified (for instance, using the <code>--hpx:hpx</code> command line option). The expected format is any valid IP address or domain name format that can be resolved into an IP address. The default depends on the compile time preprocessor constant <code>HPX_INITIAL_IP_ADDRESS</code> ("127.0.0.1").
<code>hpx.parcel.port</code>	This property defines the default IP port to be used for the <i>parcel</i> layer to listen to. This IP port will be used as long as no other values are specified (for instance using the <code>--hpx:hpx</code> command line option). The default depends on the compile time preprocessor constant <code>HPX_INITIAL_IP_PORT</code> (7910).
<code>hpx.parcel.bootstrap</code>	This property defines which <i>parcelpport</i> type should be used during application bootstrap. The default depends on the compile time preprocessor constant <code>HPX_PARCEL_BOOTSTRAP</code> ("tcp").
<code>hpx.parcel.max_connections</code>	This property defines how many network connections between different localities are overall kept alive by each <i>locality</i> . The default depends on the compile time preprocessor constant <code>HPX_PARCEL_MAX_CONNECTIONS</code> (512).
<code>hpx.parcel.max_connections_per_locality</code>	This property defines the maximum number of network connections that one <i>locality</i> will open to another <i>locality</i> . The default depends on the compile time preprocessor constant <code>HPX_PARCEL_MAX_CONNECTIONS_PER_LOCALITY</code> (4).
<code>hpx.parcel.max_message_size</code>	This property defines the maximum allowed message size that will be transferrable through the <i>parcel</i> layer. The default depends on the compile time preprocessor constant <code>HPX_PARCEL_MAX_MESSAGE_SIZE</code> (1000000000 bytes).
<code>hpx.parcel.max_outbound_message_size</code>	This property defines the maximum allowed outbound coalesced message size that will be transferrable through the <i>parcel</i> layer. The default depends on the compile time preprocessor constant <code>HPX_PARCEL_MAX_OUTBOUND_MESSAGE_SIZE</code> (1000000 bytes).
<code>hpx.parcel.array_optimization</code>	This property defines whether this <i>locality</i> is allowed to utilize array optimizations during serialization of <i>parcel</i> data. The default is 1.
<code>hpx.parcel.zero_copy_optimization</code>	This property defines whether this <i>locality</i> is allowed to utilize zero copy optimizations during serialization of <i>parcel</i> data. The default is the same value as set for <code>hpx.parcel.array_optimization</code> .
<code>hpx.parcel.zero_copy_receive_optimization</code>	This property defines whether this <i>locality</i> is allowed to utilize zero copy optimizations on the receiving end during de-serialization of <i>parcel</i> data. The default is the same value as set for <code>hpx.parcel.array_optimization</code> .
<code>hpx.parcel.zero_copy_serialization_threshold</code>	This property defines the threshold value (in bytes) starting at which the serialization layer will apply zero-copy optimizations for serialized entities. The default value is defined by the preprocessor constant <code>HPX_ZERO_COPY_SERIALIZATION_THRESHOLD</code> .
<code>hpx.parcel.async_serialization</code>	This property defines whether this <i>locality</i> is allowed to spawn a new thread for serialization (this is both for encoding and decoding parcels). The default is 1.
<code>hpx.parcel.message_handlers</code>	This property defines whether message handlers are loaded. The default is 0.
<code>hpx.parcel.max_background_threads</code>	This property defines how many cores should be used to perform background operations. The default is -1 (all cores).

The following settings relate to the TCP/IP parcelport.

```
[hpx.parcel.tcp]
enable = ${HPX_HAVE_PARCELPORT_TCP:${hpx.parcel.enabled}}
array_optimization = ${HPX_PARCEL_TCP_ARRAY_OPTIMIZATION:${hpx.parcel.array_
↳optimization}}
zero_copy_optimization = ${HPX_PARCEL_TCP_ZERO_COPY_OPTIMIZATION:${hpx.parcel.zero_copy_
↳optimization}}
zero_copy_receive_optimization = ${HPX_PARCEL_TCP_ZERO_COPY_RECEIVE_OPTIMIZATION:${hpx.
↳parcel.zero_copy_receive_optimization}}
zero_copy_serialization_threshold = ${HPX_PARCEL_TCP_ZERO_COPY_SERIALIZATION_THRESHOLD:
↳${hpx.parcel.zero_copy_serialization_threshold}}
async_serialization = ${HPX_PARCEL_TCP_ASYNC_SERIALIZATION:${hpx.parcel.async_
↳serialization}}
parcel_pool_size = ${HPX_PARCEL_TCP_PARCEL_POOL_SIZE:${hpx.threadpools.parcel_pool_size}}
max_connections = ${HPX_PARCEL_TCP_MAX_CONNECTIONS:${hpx.parcel.max_connections}}
max_connections_per_locality = ${HPX_PARCEL_TCP_MAX_CONNECTIONS_PER_LOCALITY:${hpx.
↳parcel.max_connections_per_locality}}
max_message_size = ${HPX_PARCEL_TCP_MAX_MESSAGE_SIZE:${hpx.parcel.max_message_size}}
max_outbound_message_size = ${HPX_PARCEL_TCP_MAX_OUTBOUND_MESSAGE_SIZE:${hpx.parcel.max_
↳outbound_message_size}}
max_background_threads = ${HPX_PARCEL_TCP_MAX_BACKGROUND_THREADS:${hpx.parcel.max_
↳background_threads}}
```

Property	Description
<code>hpx.parcel.tcp.enable</code>	Enables the use of the default TCP parcelport. Note that the initial bootstrap of the overall <i>HPX</i> application will be performed using the default TCP connections. This parcelport is enabled by default. This will be disabled only if MPI is enabled (see below).
<code>hpx.parcel.tcp.array_optimization</code>	This property defines whether this <i>locality</i> is allowed to utilize array optimizations in the TCP/IP parcelport during serialization of parcel data. The default is the same value as set for <code>hpx.parcel.array_optimization</code> .
<code>hpx.parcel.tcp.zero_copy_optimization</code>	This property defines whether this <i>locality</i> is allowed to utilize zero copy optimizations during serialization of parcel data. The default is the same value as set for <code>hpx.parcel.zero_copy_optimization</code> .
<code>hpx.parcel.tcp.zero_copy_receive_optimization</code>	This property defines whether this <i>locality</i> is allowed to utilize zero copy optimizations on the receiving end in the TCP/IP parcelport during de-serialization of <i>parcel</i> data. The default is the same value as set for <code>hpx.parcel.zero_copy_optimization</code> .
<code>hpx.parcel.tcp.zero_copy_serialization_threshold</code>	This property defines the threshold value (in bytes) starting at which the serialization layer will apply zero copy optimizations for serialized entities. The default is the same value as set for <code>hpx.parcel.zero_copy_serialization_threshold</code> .
<code>hpx.parcel.tcp.async_serialization</code>	This property defines whether this <i>locality</i> is allowed to spawn a new thread for serialization in the TCP/IP parcelport (this is both for encoding and decoding parcels). The default is the same value as set for <code>hpx.parcel.async_serialization</code> .
<code>hpx.parcel.tcp.parcel_pool_size</code>	The value of this property defines the number of OS threads created for the internal parcel thread pool of the TCP <i>parcel</i> port. The default is taken from <code>hpx.threadpools.parcel_pool_size</code> .
<code>hpx.parcel.tcp.max_connections</code>	This property defines how many network connections between different localities are overall kept alive by each <i>locality</i> . The default is taken from <code>hpx.parcel.max_connections</code> .
<code>hpx.parcel.tcp.max_connections_per_locality</code>	This property defines the maximum number of network connections that one <i>locality</i> will open to another <i>locality</i> . The default is taken from <code>hpx.parcel.max_connections_per_locality</code> .
<code>hpx.parcel.tcp.max_message_size</code>	This property defines the maximum allowed message size that will be transferrable through the <i>parcel</i> layer. The default is taken from <code>hpx.parcel.max_message_size</code> .
<code>hpx.parcel.tcp.max_outbound_message_size</code>	This property defines the maximum allowed outbound coalesced message size that will be transferrable through the <i>parcel</i> layer. The default is taken from <code>hpx.parcel.max_outbound_connections</code> .
<code>hpx.parcel.tcp.max_background_threads</code>	This property defines how many cores should be used to perform background operations. The default is taken from <code>hpx.parcel.max_background_threads</code> .

The following settings relate to the MPI parcelport. These settings take effect only if the compile time constant `HPX_HAVE_PARCELPOR_T_MPI` is set (the equivalent CMake variable is `HPX_WITH_PARCELPOR_T_MPI` and has to be set to ON).

```
[hpx.parcel.mpi]
enable = ${HPX_HAVE_PARCELPOR_T_MPI:[hpx.parcel.enabled]}
env = ${HPX_HAVE_PARCELPOR_T_MPI_ENV:MV2_COMM_WORLD_RANK,PMI_RANK,OMPI_COMM_WORLD_SIZE,
↪ALPS_APP_PE,PALS_NODEID}
multithreaded = ${HPX_HAVE_PARCELPOR_T_MPI_MULTITHREADED:1}
rank = <MPI_rank>
processor_name = <MPI_processor_name>
array_optimization = ${HPX_HAVE_PARCEL_MPI_ARRAY_OPTIMIZATION:[hpx.parcel.array_
↪optimization]}
zero_copy_optimization = ${HPX_HAVE_PARCEL_MPI_ZERO_COPY_OPTIMIZATION:[hpx.parcel.zero_
↪copy_optimization]}
zero_copy_receive_optimization = ${HPX_HAVE_PARCEL_MPI_ZERO_COPY_RECEIVE_OPTIMIZATION:
↪[hpx.parcel.zero_copy_receive_optimization]}
```

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```
zero_copy_serialization_threshold = ${HPX_PARCEL_MPI_ZERO_COPY_SERIALIZATION_THRESHOLD:
↪$[hpx.parcels.zero_copy_serialization_threshold]}
use_io_pool = ${HPX_HAVE_PARCEL_MPI_USE_IO_POOL:$1}
async_serialization = ${HPX_HAVE_PARCEL_MPI_ASYNC_SERIALIZATION:$[hpx.parcels.async_
↪serialization]}
parcel_pool_size = ${HPX_HAVE_PARCEL_MPI_PARCEL_POOL_SIZE:$[hpx.threadpools.parcel_pool_
↪size]}
max_connections = ${HPX_HAVE_PARCEL_MPI_MAX_CONNECTIONS:$[hpx.parcels.max_connections]}
max_connections_per_locality = ${HPX_HAVE_PARCEL_MPI_MAX_CONNECTIONS_PER_LOCALITY:$[hpx.
↪parcels.max_connections_per_locality]}
max_message_size = ${HPX_HAVE_PARCEL_MPI_MAX_MESSAGE_SIZE:$[hpx.parcels.max_message_
↪size]}
max_outbound_message_size = ${HPX_HAVE_PARCEL_MPI_MAX_OUTBOUND_MESSAGE_SIZE:$[hpx.
↪parcels.max_outbound_message_size]}
max_background_threads = ${HPX_PARCEL_MPI_MAX_BACKGROUND_THREADS:$[hpx.parcels.max_
↪background_threads]}
```


Property	Description
<code>hpx.parcel.mpi.enable</code>	Enables the use of the MPI parcelport. <i>HPX</i> tries to detect if the application was started within a parallel MPI environment. If the detection was successful, the MPI parcelport is enabled by default. To explicitly disable the MPI parcelport, set to 0. Note that the initial bootstrap of the overall <i>HPX</i> application will be performed using MPI as well.
<code>hpx.parcel.mpi.env</code>	This property influences which environment variables (separated by commas) will be analyzed to find out whether the application was invoked by MPI.
<code>hpx.parcel.mpi.multithreaded</code>	This property is used to determine what threading mode to use when initializing MPI. If this setting is 0, <i>HPX</i> will initialize MPI with <code>MPI_THREAD_SINGLE</code> . If the value is not equal to 0, <i>HPX</i> will initialize MPI with <code>MPI_THREAD_MULTII</code> .
<code>hpx.parcel.mpi.rank</code>	This property will be initialized to the MPI rank of the <i>locality</i> .
<code>hpx.parcel.mpi.processor_name</code>	This property will be initialized to the MPI processor name of the <i>locality</i> .
<code>hpx.parcel.mpi.array_optimization</code>	This property defines whether this <i>locality</i> is allowed to utilize array optimizations in the MPI parcelport during serialization of <i>parcel</i> data. The default is the same value as set for <code>hpx.parcel.array_optimization</code> .
<code>hpx.parcel.mpi.zero_copy_optimization</code>	This property defines whether this <i>locality</i> is allowed to utilize zero copy optimizations in the MPI parcelport during serialization of <i>parcel</i> data. The default is the same value as set for <code>hpx.parcel.zero_copy_optimization</code> .
<code>hpx.parcel.mpi.zero_copy_receive_optimization</code>	This property defines whether this <i>locality</i> is allowed to utilize zero copy optimizations on the receiving end in the MPI parcelport during de-serialization of <i>parcel</i> data. The default is the same value as set for <code>hpx.parcel.zero_copy_optimization</code> .
<code>hpx.parcel.mpi.zero_copy_serialization_threshold</code>	This property defines the threshold value (in bytes) starting at which the serialization layer will apply zero-copy optimizations for serialized entities. The default is the same value as set for <code>hpx.parcel.zero_copy_serialization_threshold</code> .
<code>hpx.parcel.mpi.use_io_pool</code>	This property can be set to run the progress thread inside of <i>HPX</i> threads instead of a separate thread pool. The default is 1.
<code>hpx.parcel.mpi.async_serialization</code>	This property defines whether this <i>locality</i> is allowed to spawn a new thread for serialization in the MPI parcelport (this is both for encoding and decoding parcels). The default is the same value as set for <code>hpx.parcel.async_serialization</code> .
<code>hpx.parcel.mpi.parcel_pool_size</code>	The value of this property defines the number of OS threads created for the internal parcel thread pool of the MPI <i>parcel</i> port. The default is taken from <code>hpx.threadpools.parcel_pool_size</code> .
<code>hpx.parcel.mpi.max_connections</code>	This property defines how many network connections between different localities are overall kept alive by each <i>locality</i> . The default is taken from <code>hpx.parcel.max_connections</code> .
<code>hpx.parcel.mpi.max_connections_per_locality</code>	This property defines the maximum number of network connections that one <i>locality</i> will open to another <i>locality</i> . The default is taken from <code>hpx.parcel.max_connections_per_locality</code> .
<code>hpx.parcel.mpi.max_message_size</code>	This property defines the maximum allowed message size that will be transferrable through the <i>parcel</i> layer. The default is taken from <code>hpx.parcel.max_message_size</code> .
<code>hpx.parcel.mpi.max_outbound_message_size</code>	This property defines the maximum allowed outbound coalesced message size that will be transferrable through the <i>parcel</i> layer. The default is taken from <code>hpx.parcel.max_outbound_message_size</code> .
<code>hpx.parcel.mpi.max_background_threads</code>	This property defines how many cores should be used to perform background operations. The default is taken from <code>hpx.parcel.max_background_threads</code> .

The `hpx.agas` configuration section

```
[hpx.agas]
address = ${HPX_AGAS_SERVER_ADDRESS:<hpx_initial_ip_address>}
port = ${HPX_AGAS_SERVER_PORT:<hpx_initial_ip_port>}
service_mode = hosted
dedicated_server = 0
max_pending_refcnt_requests = ${HPX_AGAS_MAX_PENDING_REFCNT_REQUESTS:<hpx_initial_agas_
↪max_pending_refcnt_requests>}
use_caching = ${HPX_AGAS_USE_CACHING:1}
use_range_caching = ${HPX_AGAS_USE_RANGE_CACHING:1}
local_cache_size = ${HPX_AGAS_LOCAL_CACHE_SIZE:<hpx_agas_local_cache_size>}
```

Property	Description
<code>hpx.agas.address</code>	This property defines the default IP address to be used for the AGAS root server. This IP address will be used as long as no other values are specified (for instance, using the <code>--hpx:agas</code> command line option). The expected format is any valid IP address or domain name format that can be resolved into an IP address. The default depends on the compile time preprocessor constant <code>HPX_INITIAL_IP_ADDRESS</code> ("127.0.0.1").
<code>hpx.agas.port</code>	This property defines the default IP port to be used for the AGAS root server. This IP port will be used as long as no other values are specified (for instance, using the <code>--hpx:agas</code> command line option). The default depends on the compile time preprocessor constant <code>HPX_INITIAL_IP_PORT</code> (7009).
<code>hpx.agas.service_mode</code>	This property specifies what type of AGAS service is running on this <i>locality</i> . Currently, two modes exist. The <i>locality</i> that acts as the AGAS server runs in <code>bootstrap</code> mode. All other localities are in <code>hosted</code> mode.
<code>hpx.agas.dedicated_server</code>	This property specifies whether the AGAS server is exclusively running AGAS services and not hosting any application components. It is a boolean value. Set to 1 if <code>--hpx:run-agas-server-only</code> is present.
<code>hpx.agas.max_pending_refcnt_requests</code>	This property defines the number of reference counting requests (increments or decrements) to buffer. The default depends on the compile time preprocessor constant <code>HPX_AGAS_MAX_PENDING_REFCNT_REQUESTS</code> (4096).
<code>hpx.agas.use_caching</code>	This property specifies whether a software address translation cache is used. It is a boolean value. Defaults to 1.
<code>hpx.agas.use_range_caching</code>	This property specifies whether range-based caching is used by the software address translation cache. This property is ignored if <code>hpx.agas.use_caching</code> is false. It is a boolean value. Defaults to 1.
<code>hpx.agas.local_cache_size</code>	This property defines the size of the software address translation cache for AGAS services. This property is ignored if <code>hpx.agas.use_caching</code> is false. Note that if <code>hpx.agas.use_range_caching</code> is true, this size will refer to the maximum number of ranges stored in the cache, not the number of entries spanned by the cache. The default depends on the compile time preprocessor constant <code>HPX_AGAS_LOCAL_CACHE_SIZE</code> (4096).

The `hpx.commandline` configuration section

The following table lists the definition of all pre-defined command line option shortcuts. For more information about commandline options, see the section *HPX Command Line Options*.

```
[hpx.commandline]
aliasing = ${HPX_COMMANDLINE_ALIASING:1}
allow_unknown = ${HPX_COMMANDLINE_ALLOW_UNKNOWN:0}

[hpx.commandline.aliases]
-a = --hpx:agas
-c = --hpx:console
-h = --hpx:help
-I = --hpx:ini
-l = --hpx:localities
-p = --hpx:app-config
-q = --hpx:queuing
-r = --hpx:run-agas-server
-t = --hpx:threads
-v = --hpx:version
-w = --hpx:worker
-x = --hpx:hpx
-0 = --hpx:node=0
-1 = --hpx:node=1
-2 = --hpx:node=2
-3 = --hpx:node=3
-4 = --hpx:node=4
-5 = --hpx:node=5
-6 = --hpx:node=6
-7 = --hpx:node=7
-8 = --hpx:node=8
-9 = --hpx:node=9
```

Note: The short options listed above are disabled by default if the application is built using `#include <hpx/hpx_main.hpp>`. See *Re-use the `main()` function as the main HPX entry point* for more information. The rationale behind this is that in this case the user's application may handle its own command line options, since *HPX* passes all unknown options to `main()`. Short options like `-t` are prone to create ambiguities regarding what the application will support. Hence, the user should instead rely on the corresponding long options like `--hpx:threads` in such a case.

Property	Description
<code>hpx.commandline.aliases</code>	Enable command line aliases as defined in the section <code>hpx.commandline.aliases</code> (see below). Defaults to 1.
<code>hpx.commandline.allow_unknown</code>	Allow for unknown command line options to be passed through to <code>hpx_main()</code> Defaults to 0.
<code>hpx.commandline.aliases.-a</code>	On the commandline <code>-a</code> expands to: <code>--hpx:agas</code> .
<code>hpx.commandline.aliases.-c</code>	On the commandline <code>-c</code> expands to: <code>--hpx:console</code> .
<code>hpx.commandline.aliases.-h</code>	On the commandline <code>-h</code> expands to: <code>--hpx:help</code> .
<code>hpx.commandline.aliases.--help</code>	On the commandline <code>--help</code> expands to: <code>--hpx:help</code> .
<code>hpx.commandline.aliases.-I</code>	On the commandline <code>-I</code> expands to: <code>--hpx:ini</code> .
<code>hpx.commandline.aliases.-l</code>	On the commandline <code>-l</code> expands to: <code>--hpx:localities</code> .
<code>hpx.commandline.aliases.-p</code>	On the commandline <code>-p</code> expands to: <code>--hpx:app-config</code> .
<code>hpx.commandline.aliases.-q</code>	On the commandline <code>-q</code> expands to: <code>--hpx:queuing</code> .
<code>hpx.commandline.aliases.-r</code>	On the commandline <code>-r</code> expands to: <code>--hpx:run-agas-server</code> .
<code>hpx.commandline.aliases.-t</code>	On the commandline <code>-t</code> expands to: <code>--hpx:threads</code> .
<code>hpx.commandline.aliases.-v</code>	On the commandline <code>-v</code> expands to: <code>--hpx:version</code> .
<code>hpx.commandline.aliases.--version</code>	On the commandline <code>--version</code> expands to: <code>--hpx:version</code> .
<code>hpx.commandline.aliases.-w</code>	On the commandline <code>-w</code> expands to: <code>--hpx:worker</code> .
<code>hpx.commandline.aliases.-x</code>	On the commandline <code>-x</code> expands to: <code>--hpx:hpx</code> .
<code>hpx.commandline.aliases.-0</code>	On the commandline <code>-0</code> expands to: <code>--hpx:node=0</code> .
<code>hpx.commandline.aliases.-1</code>	On the commandline <code>-1</code> expands to: <code>--hpx:node=1</code> .
<code>hpx.commandline.aliases.-2</code>	On the commandline <code>-2</code> expands to: <code>--hpx:node=2</code> .
<code>hpx.commandline.aliases.-3</code>	On the commandline <code>-3</code> expands to: <code>--hpx:node=3</code> .
<code>hpx.commandline.aliases.-4</code>	On the commandline <code>-4</code> expands to: <code>--hpx:node=4</code> .
<code>hpx.commandline.aliases.-5</code>	On the commandline <code>-5</code> expands to: <code>--hpx:node=5</code> .
<code>hpx.commandline.aliases.-6</code>	On the commandline <code>-6</code> expands to: <code>--hpx:node=6</code> .
<code>hpx.commandline.aliases.-7</code>	On the commandline <code>-7</code> expands to: <code>--hpx:node=7</code> .
<code>hpx.commandline.aliases.-8</code>	On the commandline <code>-8</code> expands to: <code>--hpx:node=8</code> .
<code>hpx.commandline.aliases.-9</code>	On the commandline <code>-9</code> expands to: <code>--hpx:node=9</code> .

Loading INI files

During startup and after the internal database has been initialized as described in the section *Built-in default configuration settings*, HPX will try to locate and load additional ini files to be used as a source for configuration properties. This allows for a wide spectrum of additional customization possibilities by the user and system administrators. The sequence of locations where HPX will try loading the ini files is well defined and documented in this section. All ini files found are merged into the internal configuration database. The merge operation itself conforms to the rules as described in the section *The HPX ini file format*.

1. Load all component shared libraries found in the directories specified by the property `hpx.component_path` and retrieve their default configuration information (see section *Loading components* for more details). This property can refer to a list of directories separated by ':' (Linux, Android, and MacOS) or by ';' (Windows).
2. Load all files named `hpx.ini` in the directories referenced by the property `hpx.master_ini_path`. This property can refer to a list of directories separated by ':' (Linux, Android, and MacOS) or by ';' (Windows).
3. Load a file named `.hpx.ini` in the current working directory, e.g., the directory the application was invoked from.
4. Load a file referenced by the environment variable `HPX_INI`. This variable is expected to provide the full path name of the ini configuration file (if any).
5. Load a file named `/etc/hpx.ini`. This lookup is done on non-Windows systems only.
6. Load a file named `.hpx.ini` in the home directory of the current user, e.g., the directory referenced by the environment variable `HOME`.
7. Load a file named `.hpx.ini` in the directory referenced by the environment variable `PWD`.
8. Load the file specified on the command line using the option `--hpx:config`.
9. Load all properties specified on the command line using the option `--hpx:ini`. The properties will be added to the database in the same sequence as they are specified on the command line. The format for those options is, for instance, `--hpx:ini=hpx.default_stack_size=0x4000`. In addition to the explicit command line options, this will set the following properties as implied from other settings:
 - `hpx.parcel.address` and `hpx.parcel.port` as set by `--hpx:hpx`
 - `hpx.agas.address`, `hpx.agas.port` and `hpx.agas.service_mode` as set by `--hpx:agas`
 - `hpx.program_name` and `hpx.cmd_line` will be derived from the actual command line
 - **hpx.os_threads and hpx.localities as set by**
 `--hpx:threads` and `--hpx:localities`
 - `hpx.runtime_mode` will be derived from any explicit `--hpx:console`, `--hpx:worker`, or `--hpx:connect`, or it will be derived from other settings, such as `--hpx:node=0`, which implies `--hpx:console`.
10. Load files based on the pattern `*.ini` in all directories listed by the property `hpx.ini_path`. All files found during this search will be merged. The property `hpx.ini_path` can hold a list of directories separated by ':' (on Linux or Mac) or ';' (on Windows).
11. Load the file specified on the command line using the option `--hpx:app-config`. Note that this file will be merged as the content for a top level section `[application]`.

Note: Any changes made to the configuration database caused by one of the steps will influence the loading process for all subsequent steps. For instance, if one of the ini files loaded changes the property `hpx.ini_path`, this will influence the directories searched in step 9 as described above.

Important: The *HPX* core library will verify that all configuration settings specified on the command line (using the `--hpx:ini` option) will be checked for validity. That means that the library will accept only *known* configuration settings. This is to protect the user from unintentional typos while specifying those settings. This behavior can be overwritten by appending a '!' to the configuration key, thus forcing the setting to be entered into the configuration database. For instance: `--hpx:ini=hpx.foo! = 1`

If any of the environment variables or files listed above are not found, the corresponding loading step will be silently skipped.

Loading components

HPX relies on loading application specific components during the runtime of an application. Moreover, *HPX* comes with a set of preinstalled components supporting basic functionalities useful for almost every application. Any component in *HPX* is loaded from a shared library, where any of the shared libraries can contain more than one component type. During startup, *HPX* tries to locate all available components (e.g., their corresponding shared libraries) and creates an internal component registry for later use. This section describes the algorithm used by *HPX* to locate all relevant shared libraries on a system. As described, this algorithm is customizable by the configuration properties loaded from the ini files (see section *Loading INI files*).

Loading components is a two-stage process. First *HPX* tries to locate all component shared libraries, loads those, and generates a default configuration section in the internal configuration database for each component found. For each found component the following information is generated:

```
[hpx.components.<component_instance_name>]
name = <name_of_shared_library>
path = ${component_path}
enabled = ${hpx.components.load_external}
default = 1
```

The values in this section correspond to the expected configuration information for a component as described in the section *Built-in default configuration settings*.

In order to locate component shared libraries, *HPX* will try loading all shared libraries (files with the platform specific extension of a shared library, Linux: *.so, Windows: *.dll, MacOS: *.dylib found in the directory referenced by the ini property `hpx.component_path`).

This first step corresponds to step 1) during the process of filling the internal configuration database with default information as described in section *Loading INI files*.

After all of the configuration information has been loaded, *HPX* performs the second step in terms of loading components. During this step, *HPX* scans all existing configuration sections `[hpx.component.<some_component_instance_name>]` and instantiates a special factory object for each of the successfully located and loaded components. During the application's life time, these factory objects are responsible for creating new and discarding old instances of the component they are associated with. This step is performed after step 1) of the process of filling the internal configuration database with default information as described in section *Loading INI files*.

Application specific component example

This section assumes there is a simple application component that exposes one member function as a component action. The header file `app_server.hpp` declares the C++ type to be exposed as a component. This type has a member function `print_greeting()`, which is exposed as an action `print_greeting_action`. We assume the source files for this example are located in a directory referenced by `$APP_ROOT`:

```
// file: $APP_ROOT/app_server.hpp
#include <hpx/hpx.hpp>
#include <hpx/include/iostreams.hpp>

namespace app
{
    // Define a simple component exposing one action 'print_greeting'
    class HPX_COMPONENT_EXPORT server
    : public hpx::components::component_base<server>
    {
        void print_greeting ()
        {
            hpx::cout << "Hey, how are you?\n" << std::flush;
        }

        // Component actions need to be declared, this also defines the
        // type 'print_greeting_action' representing the action.
        HPX_DEFINE_COMPONENT_ACTION(server, print_greeting, print_greeting_action);
    };
}

// Declare boilerplate code required for each of the component actions.
HPX_REGISTER_ACTION_DECLARATION(app::server::print_greeting_action);
```

The corresponding source file contains mainly macro invocations that define the boilerplate code needed for *HPX* to function properly:

```
// file: $APP_ROOT/app_server.cpp
#include "app_server.hpp"

// Define boilerplate required once per component module.
HPX_REGISTER_COMPONENT_MODULE();

// Define factory object associated with our component of type 'app::server'.
HPX_REGISTER_COMPONENT(app::server, app_server);

// Define boilerplate code required for each of the component actions. Use the
// same argument as used for HPX_REGISTER_ACTION_DECLARATION above.
HPX_REGISTER_ACTION(app::server::print_greeting_action);
```

The following gives an example of how the component can be used. Here, one instance of the `app::server` component is created on the current *locality* and the exposed action `print_greeting_action` is invoked using the global id of the newly created instance. Note that no special code is required to delete the component instance after it is not needed anymore. It will be deleted automatically when its last reference goes out of scope (shown in the example below at the closing brace of the block surrounding the code):


```
// file: $APP_ROOT/use_app_server_example.cpp
#include <hpx/hpx_init.hpp>
#include "app_server.hpp"

int hpx_main()
{
    {
        // Create an instance of the app_server component on the current locality.
        hpx::naming::id_type app_server_instance =
            hpx::create_component<app::server>(hpx::find_here());

        // Create an instance of the action 'print_greeting_action'.
        app::server::print_greeting_action print_greeting;

        // Invoke the action 'print_greeting' on the newly created component.
        print_greeting(app_server_instance);
    }
    return hpx::finalize();
}

int main(int argc, char* argv[])
{
    return hpx::init(argc, argv);
}
```

In order to make sure that the application will be able to use the component `app::server`, special configuration information must be passed to *HPX*. The simplest way to allow *HPX* to ‘find’ the component is to provide special ini configuration files that add the necessary information to the internal configuration database. The component should have a special ini file containing the information specific to the component `app_server`.

```
# file: $APP_ROOT/app_server.ini
[hpx.components.app_server]
name = app_server
path = $APP_LOCATION/
```

Here, `$APP_LOCATION` is the directory where the (binary) component shared library is located. *HPX* will attempt to load the shared library from there. The section name `hpx.components.app_server` reflects the instance name of the component (`app_server` is an arbitrary, but unique name). The property value for `hpx.components.app_server.name` should be the same as used for the second argument to the macro `HPX_REGISTER_COMPONENT` above.

Additionally, a file `.hpx.ini`, which could be located in the current working directory (see step 3 as described in the section *Loading INI files*), can be used to add to the ini search path for components:

```
# file: $PWD/.hpx.ini
[hpx]
ini_path = ${hpx.ini_path}:$APP_ROOT/
```

This assumes that the above ini file specific to the component is located in the directory `$APP_ROOT`.

Note: It is possible to reference the defined property from inside its value. *HPX* will gracefully use the previous value of `hpx.ini_path` for the reference on the right hand side and assign the overall (now expanded) value to the property.

Logging

HPX uses a sophisticated logging framework, allowing users to follow in detail what operations have been performed inside the *HPX* library in what sequence. This information proves to be very useful for diagnosing problems or just for improving the understanding of what is happening in *HPX* as a consequence of invoking *HPX* API functionality.

Default logging

Enabling default logging is a simple process. The detailed description in the remainder of this section explains different ways to customize the defaults. Default logging can be enabled by using one of the following:

- A command line switch `--hpx:debug-hpx-log`, which will enable logging to the console terminal.
- The command line switch `--hpx:debug-hpx-log=<filename>`, which enables logging to a given file `<filename>`.
- Setting an environment variable `HPX_LOGLEVEL=<loglevel>` while running the *HPX* application. In this case `<loglevel>` should be a number between (or equal to) 1 and 5 where 1 means minimal logging and 5 causes all available messages to be logged. When setting the environment variable, the logs will be written to a file named `hpx.<PID>.lo` in the current working directory, where `<PID>` is the process id of the console instance of the application.

Customizing logging

Generally, logging can be customized either using environment variable settings or using by an ini configuration file. Logging is generated in several categories, each of which can be customized independently. All customizable configuration parameters have reasonable defaults, allowing for the use of logging without any additional configuration effort. The following table lists the available categories.

Table 2.5: Logging categories

Category	Category shortcut	Information to be generated	Environment variable
General	None	Logging information generated by different subsystems of <i>HPX</i> , such as thread-manager, parcel layer, LCOs, etc.	<code>HPX_LOGLEVEL</code>
<i>AGAS</i>	<code>AGAS</code>	Logging output generated by the <i>AGAS</i> subsystem	<code>HPX_AGAS_LOGLEVEL</code>
Application	<code>APP</code>	Logging generated by applications.	<code>HPX_APP_LOGLEVEL</code>

By default, all logging output is redirected to the console instance of an application, where it is collected and written to a file, one file for each logging category.

Each logging category can be customized at two levels. The parameters for each are stored in the ini configuration sections `hpx.logging.CATEGORY` and `hpx.logging.console.CATEGORY` (where `CATEGORY` is the category shortcut as listed in the table above). The former influences logging at the source *locality* and the latter modifies the logging behaviour for each of the categories at the console instance of an application.

Levels

All *HPX* logging output has seven different logging levels. These levels can be set explicitly or through environment variables in the main *HPX* ini file as shown below. The logging levels and their associated integral values are shown in the table below, ordered from most verbose to least verbose. By default, all *HPX* logs are set to 0, e.g., all logging output is disabled by default.

Table 2.6: Logging levels

Logging level	Integral value
<debug>	5
<info>	4
<warning>	3
<error>	2
<fatal>	1
No logging	0

Tip: The easiest way to enable logging output is to set the environment variable corresponding to the logging category to an integral value as described in the table above. For instance, setting `HPX_LOGLEVEL=5` will enable full logging output for the general category. Please note that the syntax and means of setting environment variables varies between operating systems.

Configuration

Logs will be saved to destinations as configured by the user. By default, logging output is saved on the console instance of an application to `hpx.<CATEGORY>.<PID>.lo` (where `CATEGORY` and `PID` are placeholders for the category shortcut and the OS process id). The output for the general logging category is saved to `hpx.<PID>.log`. The default settings for the general logging category are shown here (the syntax is described in the section *The HPX ini file format*):

```
[hpx.logging]
level = ${HPX_LOGLEVEL:0}
destination = ${HPX_LOGDESTINATION:console}
format = ${HPX_LOGFORMAT:(T%locality%/%hpxthread%.%hpxphase%/%hpxcomponent%) P%parentloc
↳ %/%hpxparent%.%hpxparentphase% %time%($hh:$mm.$ss.$mili) [%idx%]|\\n}
```

The logging level is taken from the environment variable `HPX_LOGLEVEL` and defaults to zero, e.g., no logging. The default logging destination is read from the environment variable `HPX_LOGDESTINATION`. On any of the localities it defaults to `console`, which redirects all generated logging output to the console instance of an application. The following table lists the possible destinations for any logging output. It is possible to specify more than one destination separated by whitespace.

Table 2.7: Logging destinations

Logging destination	Description
<code>file(<filename>)</code>	Directs all output to a file with the given <code><filename></code> .
<code>cout</code>	Directs all output to the local standard output of the application instance on this <i>locality</i> .
<code>cerr</code>	Directs all output to the local standard error output of the application instance on this <i>locality</i> .
<code>console</code>	Directs all output to the console instance of the application. The console instance has its logging destinations configured separately.
<code>android_log</code>	Directs all output to the (Android) system log (available on Android systems only).

The logging format is read from the environment variable `HPX_LOGFORMAT`, and it defaults to a complex format description. This format consists of several placeholder fields (for instance `%locality%`), which will be replaced by concrete values when the logging output is generated. All other information is transferred verbatim to the output. The table below describes the available field placeholders. The separator character `|` separates the logging message prefix formatted as shown and the actual log message which will replace the separator.

Table 2.8: Available field placeholders

Name	Description
<i>locality</i>	The id of the <i>locality</i> on which the logging message was generated.
<i>hpxthread</i>	The id of the <i>HPX</i> thread generating this logging output.
<i>hpxphase</i>	The phase ⁷⁴ of the <i>HPX</i> thread generating this logging output.
<i>hpxcomponent</i>	The local virtual address of the component which the current <i>HPX</i> thread is accessing.
<i>parentloc</i>	The id of the <i>locality</i> where the <i>HPX</i> thread was running that initiated the current <i>HPX</i> thread. The current <i>HPX</i> thread is generating this logging output.
<i>hpxparent</i>	The id of the <i>HPX</i> thread that initiated the current <i>HPX</i> thread. The current <i>HPX</i> thread is generating this logging output.
<i>hpxparentphase</i>	The phase of the <i>HPX</i> thread when it initiated the current <i>HPX</i> thread. The current <i>HPX</i> thread is generating this logging output.
<i>time</i>	The time stamp for this logging outputline as generated by the source <i>locality</i> .
<i>idx</i>	The sequence number of the logging output line as generated on the source <i>locality</i> .
<i>osthread</i>	The sequence number of the OS thread that executes the current <i>HPX</i> thread.

Note: Not all of the field placeholder may be expanded for all generated logging output. If no value is available for a particular field, it is replaced with a sequence of `'-'` characters.

Here is an example line from a logging output generated by one of the *HPX* examples (please note that this is generated on a single line, without a line break):

```
(T000000000/0000000002d46f90.01/00000000009ebc10) P-----/0000000002d46f80.02 17:49.37.
↪320 [0000000000000004d]
    <info> [RT] successfully created component {0000000100ff0001, 0000000000030002} of
↪type: component_barrier[7(3)]
```

The default settings for the general logging category on the console is shown here:

```
[hpx.logging.console]
level = ${HPX_LOGLEVEL:[hpx.logging.level]}
destination = ${HPX_CONSOLE_LOGDESTINATION:file(hpx.${system.pid}.log)}
format = ${HPX_CONSOLE_LOGFORMAT:|}
```

These settings define how the logging is customized once the logging output is received by the console instance of an application. The logging level is read from the environment variable `HPX_LOGLEVEL` (as set for the console instance of the application). The level defaults to the same values as the corresponding settings in the general logging configuration shown before. The destination on the console instance is set to be a file that's name is generated based on its OS process id. Setting the environment variable `HPX_CONSOLE_LOGDESTINATION` allows customization of the naming scheme for the output file. The logging format is set to leave the original logging output unchanged, as received from one of the localities the application runs on.

⁷⁴ The phase of a *HPX*-thread counts how often this thread has been activated.

HPX Command Line Options

The predefined command line options for any application using `hpx::init` are described in the following subsections.

HPX options (allowed on command line only)

--hpx:help

Print out program usage (default: this message). Possible values: `full` (additionally prints options from components).

--hpx:version

Print out *HPX* version and copyright information.

--hpx:info

Print out *HPX* configuration information.

--hpx:options-file arg

Specify a file containing command line options (alternatively: `@filepath`).

HPX options (additionally allowed in an options file)

--hpx:worker

Run this instance in worker mode.

--hpx:console

Run this instance in console mode.

--hpx:connect

Run this instance in worker mode, but connecting late.

--hpx:run-agas-server

Run *AGAS* server as part of this runtime instance.

--hpx:run-hpx-main

Run the `hpx_main` function, regardless of *locality* mode.

--hpx:hpx arg

The IP address the *HPX* parcellport is listening on, expected format: `address:port` (default: `127.0.0.1:7910`).

--hpx:agas arg

The IP address the *AGAS* root server is running on, expected format: `address:port` (default: `127.0.0.1:7910`).

--hpx:run-agas-server-only

Run only the *AGAS* server.

--hpx:nodefile arg

The file name of a node file to use (list of nodes, one node name per line and core).

--hpx:nodes arg

The (space separated) list of the nodes to use (usually this is extracted from a node file).

--hpx:endnodes

This can be used to end the list of nodes specified using the option `--hpx:nodes`.

--hpx:ifsuffix arg

Suffix to append to host names in order to resolve them to the proper network interconnect.

--hpx:ifprefix arg

Prefix to prepend to host names in order to resolve them to the proper network interconnect.

--hpx:iftransform arg

Sed-style search and replace (s/search/replace/) used to transform host names to the proper network interconnect.

--hpx:force_ipv4

Network hostnames will be resolved to ipv4 addresses instead of using the first resolved endpoint. This is especially useful on Windows where the local hostname will resolve to an ipv6 address while remote network hostnames are commonly resolved to ipv4 addresses.

--hpx:localities arg

The number of localities to wait for at application startup (default: 1).

--hpx:node arg

Number of the node this *locality* is run on (must be unique).

--hpx:ignore-batch-env

Ignore batch environment variables.

--hpx:expect-connecting-localities

This *locality* expects other localities to dynamically connect (this is implied if the number of initial localities is larger than 1).

--hpx:pu-offset

The first processing unit this instance of *HPX* should be run on (default: 0).

--hpx:pu-step

The step between used processing unit numbers for this instance of *HPX* (default: 1).

--hpx:threads arg

The number of operating system threads to spawn for this *HPX locality*. Possible values are: numeric values 1, 2, 3 and so on, *all* (which spawns one thread per processing unit, includes hyperthreads), or *cores* (which spawns one thread per core) (default: *cores*).

--hpx:cores arg

The number of cores to utilize for this *HPX locality* (default: *all*, i.e., the number of cores is based on the number of threads *--hpx:threads* assuming *--hpx:bind=compact*).

--hpx:affinity arg

The affinity domain the OS threads will be confined to, possible values: *pu*, *core*, *numa*, *machine* (default: *pu*).

--hpx:bind arg

he detailed affinity description for the OS threads, see *More details about HPX command line options* for a detailed description of possible values. Do not use with *--hpx:pu-step*, *--hpx:pu-offset* or *--hpx:affinity* options. Implies *--hpx:numa-sensitive* (*--hpx:bind=none*) disables defining thread affinities).

--hpx:use-process-mask

Use the process mask to restrict available hardware resources (implies *--hpx:ignore-batch-env*).

--hpx:print-bind

Print to the console the bit masks calculated from the arguments specified to all *--hpx:bind* options.

--hpx:queuing arg

The queue scheduling policy to use. Options are local, local-priority-fifo, local-priority-lifo, static, static-priority, abp-priority-fifo, local-workrequesting-fifo, local-workrequesting-lifo, local-workrequesting-mc, and abp-priority-lifo (default: local-priority-fifo).

--hpx:high-priority-threads arg

The number of operating system threads maintaining a high priority queue (default: number of OS threads), valid for `--hpx:queuing=abp-priority`, `--hpx:queuingstatic-priority` and `--hpx:queuinglocal-priority` only.

--hpx:numa-sensitive

Makes the scheduler NUMA sensitive.

HPX configuration options**--hpx:app-config arg**

Load the specified application configuration (ini) file.

--hpx:config arg

Load the specified *HPX* configuration (ini) file.

--hpx:ini arg

Add a configuration definition to the default runtime configuration.

--hpx:exit

Exit after configuring the runtime.

HPX debugging options**--hpx:list-symbolic-names**

List all registered symbolic names after startup.

--hpx:list-component-types

List all dynamic component types after startup.

--hpx:dump-config-initial

Print the initial runtime configuration.

--hpx:dump-config

Print the final runtime configuration.

--hpx:debug-hpx-log [arg]

Enable all messages on the *HPX* log channel and send all *HPX* logs to the target destination (default: cout).

--hpx:debug-agas-log [arg]

Enable all messages on the *AGAS* log channel and send all *AGAS* logs to the target destination (default: cout).

--hpx:debug-parcel-log [arg]

Enable all messages on the parcel transport log channel and send all parcel transport logs to the target destination (default: cout).

--hpx:debug-timing-log [arg]

Enable all messages on the timing log channel and send all timing logs to the target destination (default: cout).

--hpx:debug-app-log [arg]

Enable all messages on the application log channel and send all application logs to the target destination (default: cout).

--hpx:debug-clp

Debug command line processing.

--hpx:attach-debugger arg

Wait for a debugger to be attached, possible arg values: `startup` or `exception` (default: `startup`)

HPX options related to performance counters**--hpx:print-counter**

Print the specified performance counter either repeatedly and/or at the times specified by `--hpx:print-counter-at` (see also option `--hpx:print-counter-interval`).

--hpx:print-counter-reset

Print the specified performance counter either repeatedly and/or at the times specified by `--hpx:print-counter-at`. Reset the counter after the value is queried (see also option `--hpx:print-counter-interval`).

--hpx:print-counter-interval

Print the performance counter(s) specified with `--hpx:print-counter` repeatedly after the time interval (specified in milliseconds), (default: 0, which means print once at shutdown).

--hpx:print-counter-destination

Print the performance counter(s) specified with `--hpx:print-counter` to the given file (default: console).

--hpx:list-counters

List the names of all registered performance counters, possible values: `minimal` (prints counter name skeletons), `full` (prints all available counter names).

--hpx:list-counter-infos

List the description of all registered performance counters, possible values: `minimal` (prints info for counter name skeletons), `full` (prints all available counter infos).

--hpx:print-counter-format

Print the performance counter(s) specified with `--hpx:print-counter`. Possible formats in CSV include a format with a header or without any header (see option `--hpx:no-csv-header`). Possible values: `csv` (prints counter values in CSV format with full names as header), `csv-short` (prints counter values in CSV format with short names provided with `--hpx:print-counter` as `--hpx:print-counter shortname`, `full-countername`

--hpx:no-csv-header

Print the performance counter(s) specified with `--hpx:print-counter` and `csv` or `csv-short` format specified with `--hpx:print-counter-format` without header.

--hpx:print-counter-at arg

Print the performance counter(s) specified with `--hpx:print-counter` (or `--hpx:print-counter-reset`) at the given point in time, possible argument values: `startup`, `shutdown` (default), `noshutdown`.

--hpx:reset-counters

Reset all performance counter(s) specified with `--hpx:print-counter` after they have been evaluated.

--hpx:print-counters-locally

Each *locality* prints only its own local counters. If this is used with `--hpx:print-counter-destination=<file>`, the code will append a `".<locality_id>"` to the file name in order to avoid clashes between localities.

Command line argument shortcuts

Additionally, the following shortcuts are available from every *HPX* application.

Table 2.9: Predefined command line option shortcuts

Shortcut option	Equivalent long option
-a	--hpx:agas
-c	--hpx:console
-h	--hpx:help
-I	--hpx:ini
-l	--hpx:localities
-p	--hpx:app-config
-q	--hpx:queuing
-r	--hpx:run-agas-server
-t	--hpx:threads
-v	--hpx:version
-w	--hpx:worker
-x	--hpx:hpx
-0	--hpx:node=0
-1	--hpx:node=1
-2	--hpx:node=2
-3	--hpx:node=3
-4	--hpx:node=4
-5	--hpx:node=5
-6	--hpx:node=6
-7	--hpx:node=7
-8	--hpx:node=8
-9	--hpx:node=9

Note: The short options listed above are disabled by default if the application is built using `#include <hpx/hpx_main.hpp>`. See *Re-use the main() function as the main HPX entry point* for more information. The rationale behind this is that in this case the user's application may handle its own command line options, since *HPX* passes all unknown options to `main()`. Short options like `-t` are prone to create ambiguities regarding what the application will support. Hence, the user should instead rely on the corresponding long options like `--hpx:threads` in such a case.

It is possible to define your own shortcut options. In fact, all of the shortcuts listed above are pre-defined using the technique described here. Also, it is possible to redefine any of the pre-defined shortcuts to expand differently as well.

Shortcut options are obtained from the internal configuration database. They are stored as key-value properties in a special properties section named `hpx.commandline`. You can define your own shortcuts by adding the corresponding definitions to one of the ini configuration files as described in the section *Configuring HPX applications*. For instance, in order to define a command line shortcut `--p`, which should expand to `-hpx:print-counter`, the following configuration information needs to be added to one of the ini configuration files:

```
[hpx.commandline.aliases]
--pc = --hpx:print-counter
```

Note: Any arguments for shortcut options passed on the command line are retained and passed as arguments to the corresponding expanded option. For instance, given the definition above, the command line option:

```
--pc=/threads{locality#0/total}/count/cumulative
```

would be expanded to:

```
--hpx:print-counter=/threads{locality#0/total}/count/cumulative
```

Important: Any shortcut option should either start with a single '-' or with two '--' characters. Shortcuts starting with a single '-' are interpreted as short options (i.e., everything after the first character following the '-' is treated as the argument). Shortcuts starting with '--' are interpreted as long options. No other shortcut formats are supported.

Specifying options for single localities only

For runs involving more than one *locality*, it is sometimes desirable to supply specific command line options to single localities only. When the *HPX* application is launched using a scheduler (like PBS; for more details see section *How to use HPX applications with PBS*), specifying dedicated command line options for single localities may be desirable. For this reason all of the command line options that have the general format `--hpx:<some_key>` can be used in a more general form: `--hpx:<N>:<some_key>`, where `<N>` is the number of the *locality* this command line option will be applied to; all other localities will simply ignore the option. For instance, the following PBS script passes the option `--hpx:pu-offset=4` to the *locality* '1' only.

```
#!/bin/bash
#
#PBS -l nodes=2:ppn=4

APP_PATH=~/.packages/hpx/bin/hello_world_distributed
APP_OPTIONS=

pbsdsh -u $APP_PATH $APP_OPTIONS --hpx:1:pu-offset=4 --hpx:nodes=`cat $PBS_NODEFILE`
```

Caution: If the first application specific argument (inside `$APP_OPTIONS`) is a non-option (i.e., does not start with a - or a --), then it must be placed before the option `--hpx:nodes`, which, in this case, should be the last option on the command line.

Alternatively, use the option `--hpx:endnodes` to explicitly mark the end of the list of node names:

```
$ pbsdsh -u $APP_PATH --hpx:1:pu-offset=4 --hpx:nodes=`cat $PBS_NODEFILE` --
↪hpx:endnodes $APP_OPTIONS
```

More details about HPX command line options

This section documents the following list of the command line options in more detail:

- The command line option `--hpx:bind`

The command line option `--hpx:bind`

This command line option allows one to specify the required affinity of the HPX worker threads to the underlying processing units. As a result the worker threads will run only on the processing units identified by the corresponding bind specification. The affinity settings are to be specified using `--hpx:bind=<BINDINGS>`, where <BINDINGS> have to be formatted as described below.

In addition to the syntax described below, one can use `--hpx:bind=none` to disable all binding of any threads to a particular core. This is mostly supported for debugging purposes.

The specified affinities refer to specific regions within a machine hardware topology. In order to understand the hardware topology of a particular machine, it may be useful to run the `lstopo` tool, which is part of Portable Hardware Locality (HWLOC), to see the reported topology tree. Seeing and understanding a topology tree will definitely help in understanding the concepts that are discussed below.

Affinities can be specified using hwloc tuples. Tuples of hwloc *objects* and associated *indexes* can be specified in the form `object:index`, `object:index-index` or `object:index,...,index`. Hwloc objects represent types of mapped items in a topology tree. Possible values for objects are `socket`, `numanode`, `core` and `pu` (processing unit). Indexes are non-negative integers that specify a unique physical object in a topology tree using its logical sequence number.

Chaining multiple tuples together in the more general form `object1:index1[.object2:index2[...]]` is permissible. While the first tuple's object may appear anywhere in the topology, the Nth tuple's object must have a shallower topology depth than the (N+1)th tuple's object. Put simply: as you move right in a tuple chain, objects must go deeper in the topology tree. Indexes specified in chained tuples are relative to the scope of the parent object. For example, `socket:0.core:1` refers to the second core in the first socket (all indices are zero based).

Multiple affinities can be specified using several `--hpx:bind` command line options or by appending several affinities separated by a `' ; '`. By default, if multiple affinities are specified, they are added.

"all" is a special affinity consisting in the entire current topology.

Note: All "names" in an affinity specification, such as `thread`, `socket`, `numanode`, `pu` or `all`, can be abbreviated. Thus, the affinity specification `threads:0-3=socket:0.core:1.pu:1` is fully equivalent to its shortened form `t:0-3=s:0.c:1.p:1`.

Here is a full grammar describing the possible format of mappings:

```

mappings      ::=  distribution | mapping (";" mapping)*
distribution  ::=  "compact" | "scatter" | "balanced" | "numa-balanced"
mapping       ::=  thread_spec "=" pu_specs
thread_spec   ::=  "thread:" range_specs
pu_specs      ::=  pu_spec ( "." pu_spec ) *
pu_spec       ::=  type ":" range_specs | "~" pu_spec
range_specs   ::=  range_spec ( "," range_spec ) *
range_spec    ::=  int | int "-" int | "all"
type          ::=  "socket" | "numanode" | "core" | "pu"

```

The following example assumes a system with at least 4 cores, where each core has more than 1 processing unit (hardware threads). Running `hello_world_distributed` with 4 OS threads (on 4 processing units), where each of those threads is bound to the first processing unit of each of the cores, can be achieved by invoking:

```
$ hello_world_distributed -t4 --hpx:bind=thread:0-3=core:0-3.pu:0
```

Here, `thread:0-3` specifies the OS threads used to define affinity bindings, and `core:0-3.pu:0` defines that for each of the cores (`core:0-3`) only their first processing unit `pu:0` should be used.

Note: The command line option `--hpx:print-bind` can be used to print the bitmasks generated from the affinity mappings as specified with `--hpx:bind`. For instance, on a system with hyperthreading enabled (i.e. 2 processing units per core), the command line:

```
$ hello_world_distributed -t4 --hpx:bind=thread:0-3=core:0-3.pu:0 --hpx:print-bind
```

will cause this output to be printed:

```
0: PU L#0(P#0), Core L#0, Socket L#0, Node L#0(P#0)
1: PU L#2(P#2), Core L#1, Socket L#0, Node L#0(P#0)
2: PU L#4(P#4), Core L#2, Socket L#0, Node L#0(P#0)
3: PU L#6(P#6), Core L#3, Socket L#0, Node L#0(P#0)
```

where each bit in the bitmasks corresponds to a processing unit the listed worker thread will be bound to run on.

The difference between the four possible predefined distribution schemes (`compact`, `scatter`, `balanced` and `numa-balanced`) is best explained with an example. Imagine that we have a system with 4 cores and 4 hardware threads per core on 2 sockets. If we place 8 threads the assignments produced by the `compact`, `scatter`, `balanced` and `numa-balanced` types are shown in the figure below. Notice that `compact` does not fully utilize all the cores in the system. For this reason it is recommended that applications are run using the `scatter` or `balanced/numa-balanced` options in most cases.

In addition to the predefined distributions it is possible to restrict the resources used by *HPX* to the process CPU mask. The CPU mask is typically set by e.g. `MPI`⁷³ and batch environments. Using the command line option `--hpx:use-process-mask` makes *HPX* act as if only the processing units in the CPU mask are available for use by *HPX*. The number of threads is automatically determined from the CPU mask. The number of threads can still be changed manually using this option, but only to a number less than or equal to the number of processing units in the CPU mask. The option `--hpx:print-bind` is useful in conjunction with `--hpx:use-process-mask` to make sure threads are placed as expected.

2.4.11 Writing single-node applications

Being a C++ Standard Library for Concurrency and Parallelism, *HPX* implements all of the corresponding facilities as defined by the C++ Standard but also those which are proposed as part of the ongoing C++ standardization process. This section focuses on the features available in *HPX* for parallel and concurrent computation on a single node, although many of the features presented here are also implemented to work in the distributed case.

⁷³ https://en.wikipedia.org/wiki/Message_Passing_Interface

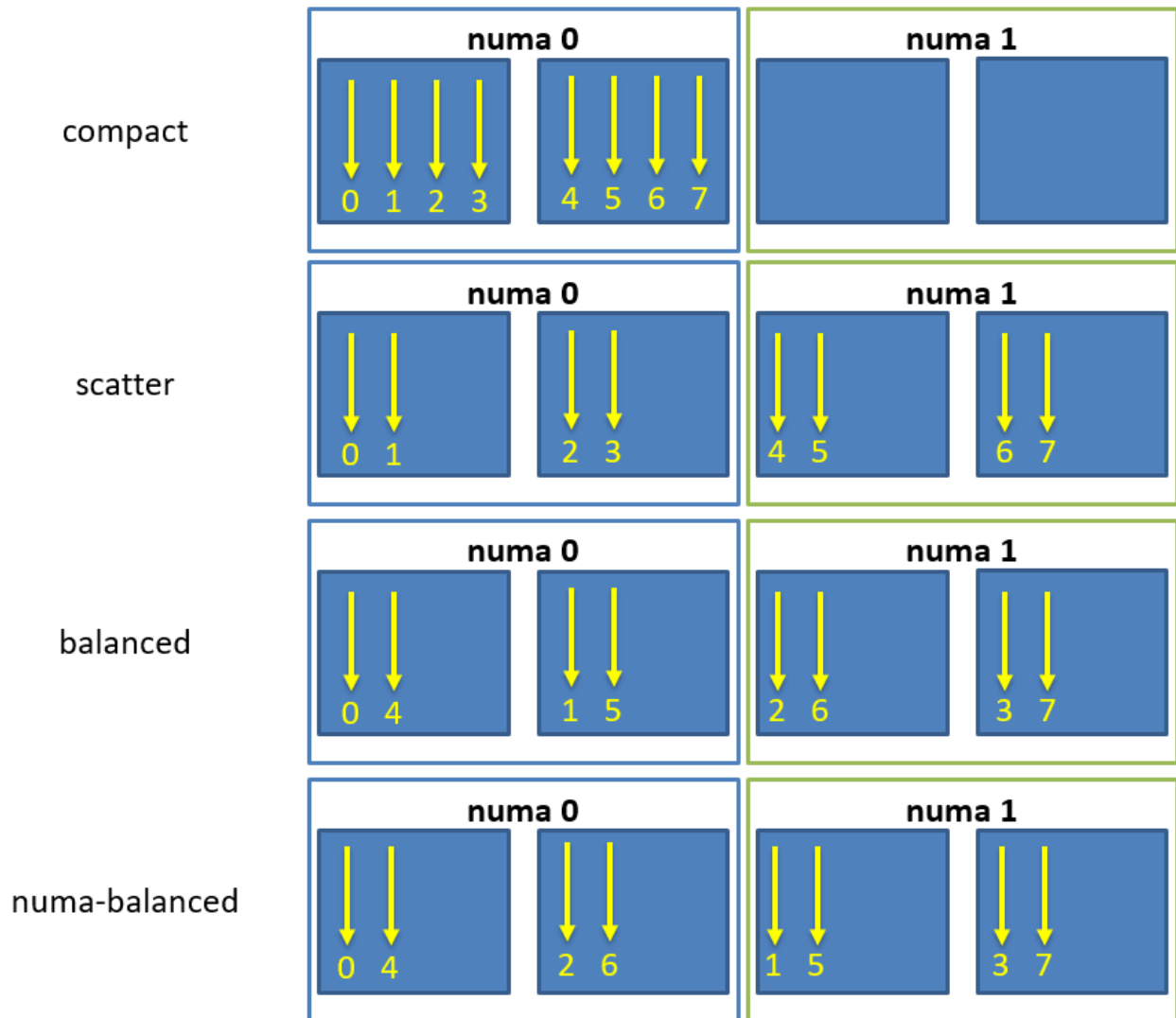


Fig. 2.7: Schematic of thread affinity type distributions.

Synchronization objects

The following objects are providing synchronization for *HPX* applications:

1. *Barrier*
2. *Condition variable*
3. *Latch*
4. *Mutex*
5. *Shared mutex*
6. *Semaphore*
7. *Composable guards*

Barrier

Barriers are used for synchronizing multiple threads. They provide a synchronization point, where all threads must wait until they have all reached the barrier, before they can continue execution. This allows multiple threads to work together to solve a common task, and ensures that no thread starts working on the next task until all threads have completed the current task. This ensures that all threads are in the same state before performing any further operations, leading to a more consistent and accurate computation.

Unlike latches, barriers are reusable: once the participating threads are released from a barrier's synchronization point, they can re-use the same barrier. It is thus useful for managing repeated tasks, or phases of a larger task, that are handled by multiple threads. The code below shows how barriers can be used to synchronize two threads:

```
#include <hpx/barrier.hpp>
#include <hpx/future.hpp>
#include <hpx/init.hpp>

#include <iostream>

int hpx_main()
{
    hpx::barrier b(2);

    hpx::future<void> f1 = hpx::async([&b]() {
        std::cout << "Thread 1 started." << std::endl;
        // Do some computation
        b.arrive_and_wait();
        // Continue with next task
        std::cout << "Thread 1 finished." << std::endl;
    });

    hpx::future<void> f2 = hpx::async([&b]() {
        std::cout << "Thread 2 started." << std::endl;
        // Do some computation
        b.arrive_and_wait();
        // Continue with next task
        std::cout << "Thread 2 finished." << std::endl;
    });
}
```

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```

    f1.get();
    f2.get();

    return hpx::local::finalize();
}

int main(int argc, char* argv[])
{
    return hpx::local::init(hpx_main, argc, argv);
}

```

In this example, two `hpx::future` objects are created, each representing a separate thread of execution. The `wait` function of the `hpx::barrier` object is called by each thread. The threads will wait at the barrier until both have reached it. Once both threads have reached the barrier, they can continue with their next task.

Condition variable

A *condition variable* is a synchronization primitive in *HPX* that allows a thread to wait for a specific condition to be satisfied before continuing execution. It is typically used in conjunction with a mutex or a lock to protect shared data that is being modified by multiple threads. Hence, it blocks one or more threads until another thread both modifies a shared variable (the condition) and notifies the `condition_variable`. The code below shows how two threads modifying the shared variable data can be synchronized using the `condition_variable`:

```

#include <hpx/condition_variable.hpp>
#include <hpx/init.hpp>
#include <hpx/mutex.hpp>
#include <hpx/thread.hpp>

#include <iostream>
#include <string>

hpx::condition_variable cv;
hpx::mutex m;
std::string data;
bool ready = false;
bool processed = false;

void worker_thread()
{
    // Wait until the main thread signals that data is ready
    std::unique_lock<hpx::mutex> lk(m);
    cv.wait(lk, [] { return ready; });

    // Access the shared resource
    std::cout << "Worker thread: Processing data...\n";
    data = "Test data after";

    // Send data back to the main thread
    processed = true;
    std::cout << "Worker thread: data processing is complete\n";
}

```

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```

    // Manual unlocking is done before notifying, to avoid waking up
    // the waiting thread only to block again
    lk.unlock();
    cv.notify_one();
}

int hpx_main()
{
    hpx::thread worker(worker_thread);

    // Do some work
    std::cout << "Main thread: Preparing data...\n";
    data = "Test data before";
    hpx::this_thread::sleep_for(std::chrono::seconds(1));
    std::cout << "Main thread: Data before processing = " << data << '\n';

    // Signal that data is ready and send data to worker thread
    {
        std::lock_guard<hpx::mutex> lk(m);
        ready = true;
        std::cout << "Main thread: Data is ready...\n";
    }
    cv.notify_one();

    // Wait for the worker thread to finish
    {
        std::unique_lock<hpx::mutex> lk(m);
        cv.wait(lk, [] { return processed; });
    }
    std::cout << "Main thread: Data after processing = " << data << '\n';
    worker.join();

    return hpx::local::finalize();
}

int main(int argc, char* argv[])
{
    return hpx::local::init(hpx_main, argc, argv);
}

```

The main thread of the code above starts by creating a worker thread and preparing the shared variable data. Once the data is ready, the main thread acquires a lock on the mutex `m` using `std::lock_guard<hpx::mutex> lk(m)` and sets the ready flag to true, then signals the worker thread to start processing by calling `cv.notify_one()`. The `cv.wait()` call in the main thread then blocks until the worker thread signals that processing is complete by setting the processed flag.

The worker thread starts by acquiring a lock on the mutex `m` to ensure exclusive access to the shared data. The `cv.wait()` call blocks the thread until the ready flag is set by the main thread. Once this is true, the worker thread accesses the shared data resource, processes it, and sets the processed flag to indicate completion. The mutex is then unlocked using `lk.unlock()` and the `cv.notify_one()` call signals the main thread to resume execution. Finally, the new data is printed by the main thread to the console.

Latch

A *latch* is a downward counter which can be used to synchronize threads. The value of the counter is initialized on creation. Threads may block on the latch until the counter is decremented to zero. There is no possibility to increase or reset the counter, which makes the latch a single-use barrier.

In *HPX*, a latch is implemented as a counting semaphore, which can be initialized with a specific count value and decremented each time a thread reaches the latch. When the count value reaches zero, all waiting threads are unblocked and allowed to continue execution. The code below shows how latch can be used to synchronize 16 threads:

```
std::ptrdiff_t num_threads = 16;

/////////////////////////////////////////////////////////////////
void wait_for_latch(hpx::latch& l)
{
    l.arrive_and_wait();
}

/////////////////////////////////////////////////////////////////
int hpx_main(hpx::program_options::variables_map& vm)
{
    num_threads = vm["num-threads"].as<std::ptrdiff_t>();

    hpx::latch l(num_threads + 1);

    std::vector<hpx::future<void>> results;
    for (std::ptrdiff_t i = 0; i != num_threads; ++i)
        results.push_back(hpx::async(&wait_for_latch, std::ref(l)));

    // Wait for all threads to reach this point.
    l.arrive_and_wait();

    hpx::wait_all(results);

    return hpx::local::finalize();
}
```

In the above code, the `hpx_main` function creates a latch object `l` with a count of `num_threads + 1` and `num_threads` number of threads using `hpx::async`. These threads call the `wait_for_latch` function and pass the reference to the latch object. In the `wait_for_latch` function, the thread calls the `arrive_and_wait` method on the latch, which decrements the count of the latch and causes the thread to wait until the count reaches zero. Finally, the main thread waits for all the threads to arrive at the latch by calling the `arrive_and_wait` method and then waits for all the threads to finish by calling the `hpx::wait_all` method.

Mutex

A *mutex* (short for “mutual exclusion”) is a synchronization primitive in *HPX* used to control access to a shared resource, ensuring that only one thread can access it at a time. A mutex is used to protect data structures from race conditions and other synchronization-related issues. When a thread acquires a mutex, other threads that try to access the same resource will be blocked until the mutex is released. The code below shows the basic use of mutexes:

```
#include <hpx/future.hpp>
#include <hpx/init.hpp>
#include <hpx/mutex.hpp>

#include <iostream>

int hpx_main()
{
    hpx::mutex m;

    hpx::future<void> f1 = hpx::async([&m]() {
        std::scoped_lock sl(m);
        std::cout << "Thread 1 acquired the mutex" << std::endl;
    });

    hpx::future<void> f2 = hpx::async([&m]() {
        std::scoped_lock sl(m);
        std::cout << "Thread 2 acquired the mutex" << std::endl;
    });

    hpx::wait_all(f1, f2);

    return hpx::local::finalize();
}

int main(int argc, char* argv[])
{
    return hpx::local::init(hpx_main, argc, argv);
}
```

In this example, two *HPX* threads created using `hpx::async` are acquiring a `hpx::mutex m`. `std::scoped_lock sl(m)` is used to take ownership of the given mutex `m`. When control leaves the scope in which the `scoped_lock` object was created, the `scoped_lock` is destructed and the mutex is released.

Attention: A common way to acquire and release mutexes is by using the function `m.lock()` before accessing the shared resource, and `m.unlock()` called after the access is complete. However, these functions may lead to deadlocks in case of exception(s). That is, if an exception happens when the mutex is locked then the code that unlocks the mutex will never be executed, the lock will remain held by the thread that acquired it, and other threads will be unable to access the shared resource. This can cause a deadlock if the other threads are also waiting to acquire the same lock. For this reason, we suggest you use `std::scoped_lock`, which prevents this issue by releasing the lock when control leaves the scope in which the `scoped_lock` object was created.

Shared mutex

A *shared mutex* is a synchronization primitive that can be used to protect shared data from being simultaneously accessed by multiple threads. In contrast to other mutex types which facilitate exclusive access, a `shared_mutex` has two levels of access:

- *Exclusive access* prevents any other thread from acquiring the mutex, just as with the normal mutex. It does not matter if the other thread tries to acquire shared or exclusive access.
- *Shared access* allows multiple threads to acquire the mutex, but all of them only in shared mode. Exclusive access is not granted until all of the previous shared holders have returned the mutex (typically, as long as an exclusive request is waiting, new shared ones are queued to be granted after the exclusive access).

Shared mutexes are especially useful when shared data can be safely read by any number of threads simultaneously, but a thread may only write the same data when no other thread is reading or writing at the same time. A typical scenario is a database: The data can be read simultaneously by different threads with no problem. However, modification of the database is critical: if some threads read data while another one is writing, the threads reading may receive inconsistent data. Hence, while a thread is writing, reading should not be allowed. After writing is complete, reads can occur simultaneously again. The code below shows how `shared_mutex` can be used to synchronize reads and writes:

```
int const writers = 3;
int const readers = 3;
int const cycles = 10;

using std::chrono::milliseconds;

int hpx_main()
{
    std::vector<hpx::thread> threads;
    std::atomic<bool> ready(false);
    hpx::shared_mutex stm;

    for (int i = 0; i < writers; ++i)
    {
        threads.emplace_back([&ready, &stm, i] {
            std::mt19937 urng(static_cast<std::uint32_t>(std::time(nullptr)));
            std::uniform_int_distribution<int> dist(1, 1000);

            while (!ready)
            { /** wait... ***/
            }

            for (int j = 0; j < cycles; ++j)
            {
                // scope of unique_lock
                {
                    std::unique_lock<hpx::shared_mutex> ul(stm);

                    std::cout << "^^^ Writer " << i << " starting..."
                                << std::endl;
                    hpx::this_thread::sleep_for(milliseconds(dist(urng)));
                    std::cout << "vvv Writer " << i << " finished."
                                << std::endl;
                }
            }
        });
    }
}
```

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```

        hpx::this_thread::sleep_for(milliseconds(dist(urng)));
    }
    });
}

for (int i = 0; i < readers; ++i)
{
    int k = writers + i;
    threads.emplace_back([&ready, &stm, k, i] {
        HPX_UNUSED(k);
        std::mt19937 urng(static_cast<std::uint32_t>(std::time(nullptr)));
        std::uniform_int_distribution<int> dist(1, 1000);

        while (!ready)
        { /*** wait... ***/
        }

        for (int j = 0; j < cycles; ++j)
        {
            // scope of shared_lock
            {
                std::shared_lock<hpx::shared_mutex> sl(stm);

                std::cout << "Reader " << i << " starting..." << std::endl;
                hpx::this_thread::sleep_for(milliseconds(dist(urng)));
                std::cout << "Reader " << i << " finished." << std::endl;
            }
            hpx::this_thread::sleep_for(milliseconds(dist(urng)));
        }
    });
}

ready = true;
for (auto& t : threads)
    t.join();

return hpx::local::finalize();
}

```

The above code creates writers and readers threads, each of which will perform cycles of operations. Both the writer and reader threads use the `hpx::shared_mutex` object `stm` to synchronize access to a shared resource.

- For the writer threads, a `unique_lock` on the shared mutex is acquired before each write operation and is released after control leaves the scope in which the `unique_lock` object was created.
- For the reader threads, a `shared_lock` on the shared mutex is acquired before each read operation and is released after control leaves the scope in which the `shared_lock` object was created.

Before each operation, both the reader and writer threads sleep for a random time period, which is generated using a random number generator. The random time period simulates the processing time of the operation.

Semaphore

Semaphores are a synchronization mechanism used to control concurrent access to a shared resource. The two types of semaphores are:

- counting semaphore: it has a counter that is bigger than zero. The counter is initialized in the constructor. Acquiring the semaphore decreases the counter and releasing the semaphore increases the counter. If a thread tries to acquire the semaphore when the counter is zero, the thread will block until another thread increments the counter by releasing the semaphore. Unlike `hpx::mutex`, an `hpx::counting_semaphore` is not bound to a thread, which means that the acquire and release call of a semaphore can happen on different threads.
- binary semaphore: it is an alias for a `hpx::counting_semaphore<1>`. In this case, the least maximal value is 1. `hpx::binary_semaphore` can be used to implement locks.

```
#include <hpx/init.hpp>
#include <hpx/semaphore.hpp>
#include <hpx/thread.hpp>

#include <iostream>

// initialize the semaphore with a count of 3
hpx::counting_semaphore<> semaphore(3);

void worker()
{
    semaphore.acquire();    // decrement the semaphore's count
    std::cout << "Entering critical section" << std::endl;
    hpx::this_thread::sleep_for(std::chrono::seconds(1));
    semaphore.release();    // increment the semaphore's count
    std::cout << "Exiting critical section" << std::endl;
}

int hpx_main()
{
    hpx::thread t1(worker);
    hpx::thread t2(worker);
    hpx::thread t3(worker);
    hpx::thread t4(worker);
    hpx::thread t5(worker);

    t1.join();
    t2.join();
    t3.join();
    t4.join();
    t5.join();

    return hpx::local::finalize();
}

int main(int argc, char* argv[])
{
    return hpx::local::init(hpx_main, argc, argv);
}
```

In this example, the counting semaphore is initialized to the value of 3. This means that up to 3 threads can access the

critical section (the section of code inside the `worker()` function) at the same time. When a thread enters the critical section, it acquires the semaphore, which decrements the count, while when it exits the critical section, it releases the semaphore, incrementing thus the count. The `worker()` function simulates a critical section by acquiring the semaphore, sleeping for 1 second and then releasing the semaphore.

In the main function, 5 worker threads are created and started, each trying to enter the critical section. If the count of the semaphore is already 0, a worker will wait until another worker releases the semaphore (increasing its value).

Composable guards

Composable guards operate in a manner similar to locks, but are applied only to asynchronous functions. The guard (or guards) is automatically locked at the beginning of a specified task and automatically unlocked at the end. Because guards are never added to an existing task's execution context, the calling of guards is freely composable and can never deadlock.

To call an application with a single guard, simply declare the guard and call `run_guarded()` with a function (`task`):

```
hpx::lcos::local::guard gu;  
run_guarded(gu, task);
```

If a single method needs to run with multiple guards, use a guard set:

```
std::shared_ptr<hpx::lcos::local::guard> gu1(new hpx::lcos::local::guard());  
std::shared_ptr<hpx::lcos::local::guard> gu2(new hpx::lcos::local::guard());  
gs.add(*gu1);  
gs.add(*gu2);  
run_guarded(gs, task);
```

Guards use two atomic operations (which are not called repeatedly) to manage what they do, so overhead should be extremely low.

Execution control

The following objects are providing control of the execution in *HPX* applications:

1. *Futures*
2. *Channels*
3. *Task blocks*
4. *Task groups*
5. *Threads*

Futures

Futures are a mechanism to represent the result of a potentially asynchronous operation. A future is a type that represents a value that will become available at some point in the future, and it can be used to write asynchronous and parallel code. Futures can be returned from functions that perform time-consuming operations, allowing the calling code to continue executing while the function performs its work. The value of the future is set when the operation completes and can be accessed later. Futures are used in *HPX* to write asynchronous and parallel code. Below is an example demonstrating different features of futures:

```

#include <hpx/assert.hpp>
#include <hpx/future.hpp>
#include <hpx/hpx_main.hpp>
#include <hpx/tuple.hpp>

#include <iostream>
#include <utility>

int main()
{
    // Asynchronous execution with futures
    hpx::future<void> f1 = hpx::async(hpx::launch::async, []() {});
    hpx::shared_future<int> f2 =
        hpx::async(hpx::launch::async, []() { return 42; });
    hpx::future<int> f3 =
        f2.then([](hpx::shared_future<int>&& f) { return f.get() * 3; });

    hpx::promise<double> p;
    auto f4 = p.get_future();
    HPX_ASSERT(!f4.is_ready());
    p.set_value(123.45);
    HPX_ASSERT(f4.is_ready());

    hpx::packaged_task<int()> t([]() { return 43; });
    hpx::future<int> f5 = t.get_future();
    HPX_ASSERT(!f5.is_ready());
    t();
    HPX_ASSERT(f5.is_ready());

    // Fire-and-forget
    hpx::post([]() {
        std::cout << "This will be printed later\n" << std::flush;
    });

    // Synchronous execution
    hpx::sync([]() {
        std::cout << "This will be printed immediately\n" << std::flush;
    });

    // Combinators
    hpx::future<double> f6 = hpx::async([]() { return 3.14; });
    hpx::future<double> f7 = hpx::async([]() { return 42.0; });
    std::cout
        << hpx::when_all(f6, f7)
            .then([](hpx::future<
                hpx::tuple<hpx::future<double>, hpx::future<double>>>
                f) {
                hpx::tuple<hpx::future<double>, hpx::future<double>> t =
                    f.get();
                double pi = hpx::get<0>(t).get();
                double r = hpx::get<1>(t).get();
                return pi * r * r;
            })
    );
}

```

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```

        .get()
    << std::endl;

    // Easier continuations with dataflow; it waits for all future or
    // shared_future arguments before executing the continuation, and also
    // accepts non-future arguments
    hpx::future<double> f8 = hpx::async([]() { return 3.14; });
    hpx::future<double> f9 = hpx::make_ready_future(42.0);
    hpx::shared_future<double> f10 = hpx::async([]() { return 123.45; });
    hpx::future<hpx::tuple<double, double>> f11 = hpx::dataflow(
        [](hpx::future<double> a, hpx::future<double> b,
            hpx::shared_future<double> c, double d) {
            return hpx::make_tuple<>(a.get() + b.get(), c.get() / d);
        },
        f8, f9, f10, -3.9);

    // split_future gives a tuple of futures from a future of tuple
    hpx::tuple<hpx::future<double>, hpx::future<double>> f12 =
        hpx::split_future(std::move(f11));
    std::cout << hpx::get<1>(f12).get() << std::endl;

    return 0;
}

```

The first section of the main function demonstrates how to use futures for asynchronous execution. The first two lines create two futures, one for void and another for an integer, using the `hpx::async()` function. These futures are executed *asynchronously* in separate threads using the `hpx::launch::async` launch policy. The third future is created by *chaining* the second future using the `then()` member function. This future multiplies the result of the second future by 3.

The next part of the code demonstrates how to use *promises* and *packaged tasks*, which are constructs used for communicating data between threads. The `promise` class is used to store a value that can be retrieved *later* using a future. The `packaged_task` class represents a task that can be executed *asynchronously*, and its result can be obtained using a future. The last three lines create a packaged task that returns an integer, obtain its future, execute the task, and check whether the future is ready or not.

The code then demonstrates how to use the `hpx::post()` and `hpx::sync()` functions for *fire-and-forget* and *synchronous* execution, respectively. The `hpx::post()` function executes a given function *asynchronously* and *returns immediately* without waiting for the result. The `hpx::sync()` function executes a given function *synchronously* and *waits* for the result before returning.

Next the code demonstrates the use of *combinators*, which are higher-order functions that combine two or more futures into a single future. The `hpx::when_all()` function is used to combine two futures, which return double values, into a tuple of futures. The `then()` member function is then used to compute the area of a circle using the values of the two futures. The `get()` member function is used to retrieve the result of the computation.

The last section demonstrates the use of `hpx::dataflow()`, which is a higher-order function that waits for all the future or `shared_future` arguments to be ready before executing the continuation. The `hpx::make_ready_future()` function is used to create a future with a given value. The `hpx::split_future()` function is used to split a future of a tuple into a tuple of futures. The last line retrieves the value of the second future in the tuple using `hpx::get()` and prints it to the console.

Extended facilities for futures

Concurrency is about both decomposing and composing the program from the parts that work well individually and together. It is in the composition of connected and multicore components where today's C++ libraries are still lacking.

The functionality of `std::future`⁷⁵ offers a partial solution. It allows for the separation of the initiation of an operation and the act of waiting for its result; however, the act of waiting is synchronous. In communication-intensive code this act of waiting can be unpredictable, inefficient and simply frustrating. The example below illustrates a possible synchronous wait using futures:

```
#include <future>
using namespace std;
int main()
{
    future<int> f = async([]() { return 123; });
    int result = f.get(); // might block
}
```

For this reason, *HPX* implements a set of extensions to `std::future`⁷⁶ (as proposed by N4313⁷⁷). This proposal introduces the following key asynchronous operations to `hpx::future`, `hpx::shared_future` and `hpx::async`, which enhance and enrich these facilities.

Table 2.11: Facilities extending `std::future`

Facility	Description
<code>hpx::future::then</code>	In asynchronous programming, it is very common for one asynchronous operation, on completion, to invoke a second operation and pass data to it. The current C++ standard does not allow one to register a continuation to a future. With <code>then</code> , instead of waiting for the result, a continuation is “attached” to the asynchronous operation, which is invoked when the result is ready. Continuations registered using <code>then</code> function will help to avoid blocking waits or wasting threads on polling, greatly improving the responsiveness and scalability of an application.
unwrapping constructor for <code>hpx::future</code>	In some scenarios, you might want to create a future that returns another future, resulting in nested futures. Although it is possible to write code to unwrap the outer future and retrieve the nested future and its result, such code is not easy to write because users must handle exceptions and it may cause a blocking call. Unwrapping can allow users to mitigate this problem by doing an asynchronous call to unwrap the outermost future.
<code>hpx::future::is_ready</code>	There are often situations where a <code>get()</code> call on a future may not be a blocking call, or is only a blocking call under certain circumstances. This function gives the ability to test for early completion and allows us to avoid associating a continuation, which needs to be scheduled with some non-trivial overhead and near-certain loss of cache efficiency.
<code>hpx::make_ready_future</code>	Some functions may know the value at the point of construction. In these cases the value is immediately available, but needs to be returned as a future. By using <code>hpx::make_ready_future</code> a future can be created that holds a pre-computed result in its shared state. In the current standard it is non-trivial to create a future directly from a value. First a promise must be created, then the promise is set, and lastly the future is retrieved from the promise. This can now be done with one operation.

The standard also omits the ability to compose multiple futures. This is a common pattern that is ubiquitous in other asynchronous frameworks and is absolutely necessary in order to make C++ a powerful asynchronous programming language. Not including these functions is synonymous to Boolean algebra without AND/OR.

⁷⁵ <http://en.cppreference.com/w/cpp/thread/future>

⁷⁶ <http://en.cppreference.com/w/cpp/thread/future>

⁷⁷ <http://wg21.link/n4313>

In addition to the extensions proposed by N4313⁷⁸, HPX adds functions allowing users to compose several futures in a more flexible way.

Table 2.12: Facilities for composing `hpx::futures`

Facility	Description
<code>hpx::when_any</code> , <code>hpx::when_any_n</code>	Asynchronously wait for at least one of multiple future or shared_future objects to finish.
<code>hpx::wait_any</code> , <code>hpx::wait_any_n</code>	Synchronously wait for at least one of multiple future or shared_future objects to finish.
<code>hpx::when_all</code> , <code>hpx::when_all_n</code>	Asynchronously wait for all future and shared_future objects to finish.
<code>hpx::wait_all</code> , <code>hpx::wait_all_n</code>	Synchronously wait for all future and shared_future objects to finish.
<code>hpx::when_some</code> , <code>hpx::when_some_n</code>	Asynchronously wait for multiple future and shared_future objects to finish.
<code>hpx::wait_some</code> , <code>hpx::wait_some_n</code>	Synchronously wait for multiple future and shared_future objects to finish.
<code>hpx::when_each</code>	Asynchronously wait for multiple future and shared_future objects to finish and call a function for each of the future objects as soon as it becomes ready.
<code>hpx::wait_each</code> , <code>hpx::wait_each_n</code>	Synchronously wait for multiple future and shared_future objects to finish and call a function for each of the future objects as soon as it becomes ready.

Channels

Channels combine communication (the exchange of a value) with synchronization (guaranteeing that two calculations (tasks) are in a known state). A channel can transport any number of values of a given type from a sender to a receiver:

```
hpx::lcos::local::channel<int> c;
hpx::future<int> f = c.get();
HPX_ASSERT(!f.is_ready());
c.set(42);
HPX_ASSERT(f.is_ready());
std::cout << f.get() << std::endl;
```

Channels can be handed to another thread (or in case of channel components, to other localities), thus establishing a communication channel between two independent places in the program:

```
void do_something(hpx::lcos::local::receive_channel<int> c,
                 hpx::lcos::local::send_channel<> done)
{
    // prints 43
    std::cout << c.get(hpx::launch::sync) << std::endl;
    // signal back
    done.set();
}

void send_receive_channel()
{
    hpx::lcos::local::channel<int> c;
    hpx::lcos::local::channel<> done;
```

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⁷⁸ <http://wg21.link/n4313>

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```

hpx::post(&do_something, c, done);

// send some value
c.set(43);
// wait for thread to be done
done.get().wait();
}

```

Note how `hpx::lcos::local::channel::get` without any arguments returns a future which is ready when a value has been set on the channel. The launch policy `hpx::launch::sync` can be used to make `hpx::lcos::local::channel::get` block until a value is set and return the value directly.

A channel component is created on one *locality* and can be sent to another *locality* using an action. This example also demonstrates how a channel can be used as a range of values:

```

// channel components need to be registered for each used type (not needed
// for hpx::lcos::local::channel)
HPX_REGISTER_CHANNEL(double)

void channel_sender(hpx::lcos::channel<double> c)
{
    for (double d : c)
        hpx::cout << d << std::endl;
}
HPX_PLAIN_ACTION(channel_sender)

void channel()
{
    // create the channel on this locality
    hpx::lcos::channel<double> c(hpx::find_here());

    // pass the channel to a (possibly remote invoked) action
    hpx::post(channel_sender_action(), hpx::find_here(), c);

    // send some values to the receiver
    std::vector<double> v = {1.2, 3.4, 5.0};
    for (double d : v)
        c.set(d);

    // explicitly close the communication channel (implicit at destruction)
    c.close();
}

```

Task blocks

Task blocks in *HPX* provide a way to structure and organize the execution of tasks in a parallel program, making it easier to manage dependencies between tasks. A task block actually is a group of tasks that can be executed in parallel. Tasks in a task block can depend on other tasks in the same task block. The task block allows the runtime to optimize the execution of tasks, by scheduling them in an optimal order based on the dependencies between them.

The `define_task_block`, `run` and the `wait` functions implemented based on N4755⁷⁹ are based on the `task_block` concept that is a part of the common subset of the [Microsoft Parallel Patterns Library \(PPL\)](https://msdn.microsoft.com/en-us/library/dd492418.aspx)⁸⁰ and the [Intel Threading Building Blocks \(TBB\)](https://www.threadingbuildingblocks.org/)⁸¹ libraries.

These implementations adopt a simpler syntax than exposed by those libraries— one that is influenced by language-based concepts, such as `spawn` and `sync` from [Cilk++](https://ericniebler.com/2014/05/19/cilk-plus/)⁸² and `async` and `finish` from [X10](https://ericniebler.com/2014/05/19/cilk-plus/)⁸³. They improve on existing practice in the following ways:

- The exception handling model is simplified and more consistent with normal C++ exceptions.
- Most violations of strict fork-join parallelism can be enforced at compile time (with compiler assistance, in some cases).
- The syntax allows scheduling approaches other than child stealing.

Consider an example of a parallel traversal of a tree, where a user-provided function `compute` is applied to each node of the tree, returning the sum of the results:

```
template <typename Func>
int traverse(node& n, Func && compute)
{
    int left = 0, right = 0;
    define_task_block(
        [&](task_block<>& tr) {
            if (n.left)
                tr.run([&] { left = traverse(*n.left, compute); });
            if (n.right)
                tr.run([&] { right = traverse(*n.right, compute); });
        });

    return compute(n) + left + right;
}
```

The example above demonstrates the use of two of the functions, `hpx::experimental::define_task_block` and the `hpx::experimental::task_block::run` member function of a `hpx::experimental::task_block`.

The `task_block` function delineates a region in a program code potentially containing invocations of threads spawned by the `run` member function of the `task_block` class. The `run` function spawns an *HPX* thread, a unit of work that is allowed to execute in parallel with respect to the caller. Any parallel tasks spawned by `run` within the task block are joined back to a single thread of execution at the end of the `define_task_block`. `run` takes a user-provided function object `f` and starts it asynchronously—i.e., it may return before the execution of `f` completes. The *HPX* scheduler may choose to run `f` immediately or delay running `f` until compute resources become available.

A `task_block` can be constructed only by `define_task_block` because it has no public constructors. Thus, `run` can be invoked directly or indirectly only from a user-provided function passed to `define_task_block`:

⁷⁹ <http://wg21.link/n4755>

⁸⁰ <https://msdn.microsoft.com/en-us/library/dd492418.aspx>

⁸¹ <https://www.threadingbuildingblocks.org/>

⁸² <https://software.intel.com/en-us/articles/intel-cilk-plus/>

⁸³ <https://x10-lang.org/>

```

void g();

void f(task_block<>& tr)
{
    tr.run(g);          // OK, invoked from within task_block in h
}

void h()
{
    define_task_block(f);
}

int main()
{
    task_block<> tr;      // Error: no public constructor
    tr.run(g);           // No way to call run outside of a define_task_block
    return 0;
}

```

Extensions for task blocks

Using execution policies with task blocks

HPX implements some extensions for `task_block` beyond the actual standards proposal N4755⁸⁴. The main addition is that a `task_block` can be invoked with an execution policy as its first argument, very similar to the parallel algorithms.

An execution policy is an object that expresses the requirements on the ordering of functions invoked as a consequence of the invocation of a task block. Enabling passing an execution policy to `define_task_block` gives the user control over the amount of parallelism employed by the created `task_block`. In the following example the use of an explicit `par` execution policy makes the user's intent explicit:

```

template <typename Func>
int traverse(node *n, Func&& compute)
{
    int left = 0, right = 0;

    define_task_block(
        execution::par,          // execution::parallel_policy
        [&](task_block<>& tb) {
            if (n->left)
                tb.run([&] { left = traverse(n->left, compute); });
            if (n->right)
                tb.run([&] { right = traverse(n->right, compute); });
        });

    return compute(n) + left + right;
}

```

This also causes the `hpx::experimental::task_block` object to be a template in our implementation. The template argument is the type of the execution policy used to create the task block. The template argument defaults to `hpx::execution::parallel_policy`.

⁸⁴ <http://wg21.link/n4755>

HPX still supports calling `hpx::experimental::define_task_block` without an explicit execution policy. In this case the task block will run using the `hpx::execution::parallel_policy`.

HPX also adds the ability to access the execution policy that was used to create a given `task_block`.

Using executors to run tasks

Often, users want to be able to not only define an execution policy to use by default for all spawned tasks inside the task block, but also to customize the execution context for one of the tasks executed by `task_block::run`. Adding an optionally passed executor instance to that function enables this use case:

```
template <typename Func>
int traverse(node *n, Func&& compute)
{
    int left = 0, right = 0;

    define_task_block(
        execution::par,                // execution::parallel_policy
        [&](auto& tb) {
            if (n->left)
            {
                // use explicitly specified executor to run this task
                tb.run(my_executor(), [&] { left = traverse(n->left, compute); });
            }
            if (n->right)
            {
                // use the executor associated with the par execution policy
                tb.run([&] { right = traverse(n->right, compute); });
            }
        });

    return compute(n) + left + right;
}
```

HPX still supports calling `hpx::experimental::task_block::run` without an explicit executor object. In this case the task will be run using the executor associated with the execution policy that was used to call `hpx::experimental::define_task_block`.

Task groups

A *task group* in HPX is a synchronization primitive that allows you to execute a group of tasks concurrently and wait for their completion before continuing. The tasks in an `hpx::experimental::task_group` can be added dynamically. This is the HPX implementation of `tbb::task_group` of the Intel Threading Building Blocks (TBB)⁸⁵ library.

The example below shows that to use a task group, you simply create an `hpx::task_group` object and add tasks to it using the `run()` method. Once all the tasks have been added, you can call the `wait()` method to synchronize the tasks and wait for them to complete.

```
#include <hpx/experimental/task_group.hpp>
#include <hpx/init.hpp>
```

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⁸⁵ <https://www.threadingbuildingblocks.org/>

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```
#include <iostream>

void task1()
{
    std::cout << "Task 1 executed." << std::endl;
}

void task2()
{
    std::cout << "Task 2 executed." << std::endl;
}

int hpx_main()
{
    hpx::experimental::task_group tg;

    tg.run(task1);
    tg.run(task2);

    tg.wait();

    std::cout << "All tasks finished!" << std::endl;

    return hpx::local::finalize();
}

int main(int argc, char* argv[])
{
    return hpx::local::init(hpx_main, argc, argv);
}
```

Note: *task groups* and *task blocks* are both ways to group and synchronize parallel tasks, but *task groups* are used to group multiple tasks together as a single unit, while *task blocks* are used to execute a loop in parallel, with each iteration of the loop executing in a separate task. If the difference is not clear yet, continue reading.

A *task group* is a construct that allows multiple parallel tasks to be grouped together as a single unit. The task group provides a way to synchronize all the tasks in the group before continuing with the rest of the program.

A *task block*, on the other hand, is a parallel loop construct that allows you to execute a loop in parallel, with each iteration of the loop executing in a separate task. The loop iterations are executed in a block, meaning that the loop body is executed as a single task.

Threads

A thread in *HPX* refers to a sequence of instructions that can be executed concurrently with other such sequences in multithreading environments, while sharing a same address space. These threads can communicate with each other through various means, such as futures or shared data structures.

The example below demonstrates how to launch multiple threads and synchronize them using a `hpx::latch` object. It also shows how to query the state of threads and wait for futures to complete.

```
#include <hpx/future.hpp>
#include <hpx/init.hpp>
#include <hpx/thread.hpp>

#include <functional>
#include <iostream>
#include <vector>

int const num_threads = 10;

/////////////////////////////////////////////////////////////////
void wait_for_latch(hpx::latch& l)
{
    l.arrive_and_wait();
}

int hpx_main()
{
    // Spawn a couple of threads
    hpx::latch l(num_threads + 1);

    std::vector<hpx::future<void>> results;
    results.reserve(num_threads);

    for (int i = 0; i != num_threads; ++i)
        results.push_back(hpx::async(&wait_for_latch, std::ref(l)));

    // Allow spawned threads to reach latch
    hpx::this_thread::yield();

    // Enumerate all suspended threads
    hpx::threads::enumerate_threads(
        [](hpx::threads::thread_id_type id) -> bool {
            std::cout << "thread " << hpx::thread::id(id) << " is "
                << hpx::threads::get_thread_state_name(
                    hpx::threads::get_thread_state(id))
                << std::endl;
            return true;    // always continue enumeration
        },
        hpx::threads::thread_schedule_state::suspended);

    // Wait for all threads to reach this point.
    l.arrive_and_wait();
}
```

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```

    hpx::wait_all(results);

    return hpx::local::finalize();
}

int main(int argc, char* argv[])
{
    return hpx::local::init(hpx_main, argc, argv);
}

```

In more detail, the `wait_for_latch()` function is a simple helper function that waits for a `hpx::latch` object to be released. At this point we remind that `hpx::latch` is a synchronization primitive that allows multiple threads to wait for a common event to occur.

In the `hpx_main()` function, an `hpx::latch` object is created with a count of `num_threads + 1`, indicating that `num_threads` threads need to arrive at the latch before the latch is released. The loop that follows launches `num_threads` asynchronous operations, each of which calls the `wait_for_latch` function. The resulting futures are added to the vector.

After the threads have been launched, `hpx::this_thread::yield()` is called to give them a chance to reach the latch before the program proceeds. Then, the `hpx::threads::enumerate_threads` function prints the state of each suspended thread, while the next call of `l.arrive_and_wait()` waits for all the threads to reach the latch. Finally, `hpx::wait_all` is called to wait for all the futures to complete.

Hint: An advantage of using `hpx::thread` over other threading libraries is that it is optimized for high-performance parallelism, with support for lightweight threads and task scheduling to minimize thread overhead and maximize parallelism. Additionally, `hpx::thread` integrates seamlessly with other features of *HPX* such as futures, promises, and task groups, making it a powerful tool for parallel programming.

Checkout the examples of *Shared mutex*, *Condition variable*, *Semaphore* to see how *HPX* threads are used in combination with other features.

High level parallel facilities

In preparation for the upcoming C++ Standards, there are currently several proposals targeting different facilities supporting parallel programming. *HPX* implements (and extends) some of those proposals. This is well aligned with our strategy to align the APIs exposed from *HPX* with current and future C++ Standards.

At this point, *HPX* implements several of the C++ Standardization working papers, most notably [N4409](http://wg21.link/n4409)⁸⁶ (Working Draft, Technical Specification for C++ Extensions for Parallelism), [N4755](http://wg21.link/n4755)⁸⁷ (Task Blocks), and [N4406](http://wg21.link/n4406)⁸⁸ (Parallel Algorithms Need Executors).

⁸⁶ <http://wg21.link/n4409>

⁸⁷ <http://wg21.link/n4755>

⁸⁸ <http://wg21.link/n4406>

Using parallel algorithms

A parallel algorithm is a function template declared in the namespace `hpx::parallel`.

All parallel algorithms are very similar in semantics to their sequential counterparts (as defined in the namespace `std`) with an additional formal template parameter named `ExecutionPolicy`. The execution policy is generally passed as the first argument to any of the parallel algorithms and describes the manner in which the execution of these algorithms may be parallelized and the manner in which they apply user-provided function objects.

The applications of function objects in parallel algorithms invoked with an execution policy object of type `hpx::execution::sequenced_policy` or `hpx::execution::sequenced_task_policy` execute in sequential order. For `hpx::execution::sequenced_policy` the execution happens in the calling thread.

The applications of function objects in parallel algorithms invoked with an execution policy object of type `hpx::execution::parallel_policy` or `hpx::execution::parallel_task_policy` are permitted to execute in an unordered fashion in unspecified threads, and are indeterminately sequenced within each thread.

Important: It is the caller's responsibility to ensure correctness, such as making sure that the invocation does not introduce data races or deadlocks.

The example below demonstrates how to perform a sequential and parallel `hpx::for_each` loop on a vector of integers.

```
#include <hpx/algorithm.hpp>
#include <hpx/execution.hpp>
#include <hpx/init.hpp>

#include <iostream>
#include <vector>

int hpx_main()
{
    std::vector<int> v{1, 2, 3, 4, 5};

    auto print = [](int const& n) { std::cout << n << ' '; };

    std::cout << "Print sequential: ";
    hpx::for_each(v.begin(), v.end(), print);
    std::cout << '\n';

    std::cout << "Print parallel: ";
    hpx::for_each(hpx::execution::par, v.begin(), v.end(), print);
    std::cout << '\n';

    return hpx::local::finalize();
}

int main(int argc, char* argv[])
{
    return hpx::local::init(hpx_main, argc, argv);
}
```

The above code uses `hpx::for_each` to print the elements of the vector `v{1, 2, 3, 4, 5}`. At first, `hpx::for_each()` is called without an execution policy, which means that it applies the lambda function `print` to each element in the vector sequentially. Hence, the elements are printed in order.

Next, `hpx::for_each()` is called with the `hpx::execution::par` execution policy, which applies the lambda function `print` to each element in the vector in parallel. Therefore, the output order of the elements in the vector is not deterministic and may vary from run to run.

Parallel exceptions

During the execution of a standard parallel algorithm, if temporary memory resources are required by any of the algorithms and no memory is available, the algorithm throws a `std::bad_alloc` exception.

During the execution of any of the parallel algorithms, if the application of a function object terminates with an uncaught exception, the behavior of the program is determined by the type of execution policy used to invoke the algorithm:

- If the execution policy object is of type `hpx::execution::parallel_unsequenced_policy`, `hpx::terminate` shall be called.
- If the execution policy object is of type `hpx::execution::sequenced_policy`, `hpx::execution::sequenced_task_policy`, `hpx::execution::parallel_policy`, or `hpx::execution::parallel_task_policy`, the execution of the algorithm terminates with an `hpx::exception_list` exception. All uncaught exceptions thrown during the application of user-provided function objects shall be contained in the `hpx::exception_list`.

For example, the number of invocations of the user-provided function object in `for_each` is unspecified. When `hpx::for_each` is executed sequentially, only one exception will be contained in the `hpx::exception_list` object.

These guarantees imply that, unless the algorithm has failed to allocate memory and terminated with `std::bad_alloc`, all exceptions thrown during the execution of the algorithm are communicated to the caller. It is unspecified whether an algorithm implementation will “forge ahead” after encountering and capturing a user exception.

The algorithm may terminate with the `std::bad_alloc` exception even if one or more user-provided function objects have terminated with an exception. For example, this can happen when an algorithm fails to allocate memory while creating or adding elements to the `hpx::exception_list` object.

Parallel algorithms

HPX provides implementations of the following parallel algorithms:

Table 2.13: Non-modifying parallel algorithms of header `hpx/algorithm.hpp`

Name	Description	C++ standard
<code>hpx::adjacent_find</code>	Computes the differences between adjacent elements in a range.	<code>adjacent_find</code> ⁸⁹
<code>hpx::all_of</code>	Checks if a predicate is <code>true</code> for all of the elements in a range.	<code>all_any_none_of</code> ⁹⁰
<code>hpx::any_of</code>	Checks if a predicate is <code>true</code> for any of the elements in a range.	<code>all_any_none_of</code> ⁹¹
<code>hpx::count</code>	Returns the number of elements equal to a given value.	<code>count</code> ⁹²
<code>hpx::count_if</code>	Returns the number of elements satisfying a specific criteria.	<code>count_if</code> ⁹³
<code>hpx::equal</code>	Determines if two sets of elements are the same.	<code>equal</code> ⁹⁴
<code>hpx::find</code>	Finds the first element equal to a given value.	<code>find</code> ⁹⁵
<code>hpx::find_end</code>	Finds the last sequence of elements in a certain range.	<code>find_end</code> ⁹⁶
<code>hpx::find_first_of</code>	Searches for any one of a set of elements.	<code>find_first_of</code> ⁹⁷
<code>hpx::find_if</code>	Finds the first element satisfying a specific criteria.	<code>find_if</code> ⁹⁸
<code>hpx::find_if_not</code>	Finds the first element not satisfying a specific criteria.	<code>find_if_not</code> ⁹⁹
<code>hpx::for_each</code>	Applies a function to a range of elements.	<code>for_each</code> ¹⁰⁰
<code>hpx::for_each_n</code>	Applies a function to a number of elements.	<code>for_each_n</code> ¹⁰¹
<code>hpx::lexicographical_compare</code>	Checks if a range of values is lexicographically less than another range of values.	<code>lexicographical_compare</code> ¹⁰²
<code>hpx::mismatch</code>	Finds the first position where two ranges differ.	<code>mismatch</code> ¹⁰³
<code>hpx::none_of</code>	Checks if a predicate is <code>true</code> for none of the elements in a range.	<code>all_any_none_of</code> ¹⁰⁴
<code>hpx::search</code>	Searches for a range of elements.	<code>search</code> ¹⁰⁵
<code>hpx::search_n</code>	Searches for a number consecutive copies of an element in a range.	<code>search_n</code> ¹⁰⁶

⁸⁹ http://en.cppreference.com/w/cpp/algorithm/adjacent_find⁹⁰ http://en.cppreference.com/w/cpp/algorithm/all_any_none_of⁹¹ http://en.cppreference.com/w/cpp/algorithm/all_any_none_of⁹² <http://en.cppreference.com/w/cpp/algorithm/count>⁹³ http://en.cppreference.com/w/cpp/algorithm/count_if⁹⁴ <http://en.cppreference.com/w/cpp/algorithm/equal>⁹⁵ <http://en.cppreference.com/w/cpp/algorithm/find>⁹⁶ http://en.cppreference.com/w/cpp/algorithm/find_end⁹⁷ http://en.cppreference.com/w/cpp/algorithm/find_first_of⁹⁸ http://en.cppreference.com/w/cpp/algorithm/find_if⁹⁹ http://en.cppreference.com/w/cpp/algorithm/find_if_not¹⁰⁰ http://en.cppreference.com/w/cpp/algorithm/for_each¹⁰¹ http://en.cppreference.com/w/cpp/algorithm/for_each_n¹⁰² http://en.cppreference.com/w/cpp/algorithm/lexicographical_compare¹⁰³ <http://en.cppreference.com/w/cpp/algorithm/mismatch>¹⁰⁴ http://en.cppreference.com/w/cpp/algorithm/all_any_none_of¹⁰⁵ <http://en.cppreference.com/w/cpp/algorithm/search>¹⁰⁶ http://en.cppreference.com/w/cpp/algorithm/search_n

Table 2.14: Modifying parallel algorithms of header `hpx/algorithm.hpp`

Name	Description	C++ standard
<code>hpx::copy</code>	Copies a range of elements to a new location.	<code>exclusive_scan</code> ¹⁰⁷
<code>hpx::copy_n</code>	Copies a number of elements to a new location.	<code>copy_n</code> ¹⁰⁸
<code>hpx::copy_if</code>	Copies the elements from a range to a new location for which the given predicate is <code>true</code>	<code>copy</code> ¹⁰⁹
<code>hpx::move</code>	Moves a range of elements to a new location.	<code>move</code> ¹¹⁰
<code>hpx::fill</code>	Assigns a range of elements a certain value.	<code>fill</code> ¹¹¹
<code>hpx::fill_n</code>	Assigns a value to a number of elements.	<code>fill_n</code> ¹¹²
<code>hpx::generate</code>	Saves the result of a function in a range.	<code>generate</code> ¹¹³
<code>hpx::generate_n</code>	Saves the result of N applications of a function.	<code>generate_n</code> ¹¹⁴
<code>hpx::experimental::reduce_by_key</code>	Performs an inclusive scan on consecutive elements with matching keys, with a reduction to output only the final sum for each key. The key sequence <code>{1,1,1,2,3,3,3,3,1}</code> and value sequence <code>{2,3,4,5,6,7,8,9,10}</code> would be reduced to <code>keys={1,2,3,1}</code> , <code>values={9,5,30,10}</code> .	
<code>hpx::remove</code>	Removes the elements from a range that are equal to the given value.	<code>remove</code> ¹¹⁵
<code>hpx::remove_if</code>	Removes the elements from a range that are equal to the given predicate is <code>false</code>	<code>remove</code> ¹¹⁶
<code>hpx::remove_copy</code>	Copies the elements from a range to a new location that are not equal to the given value.	<code>remove_copy</code> ¹¹⁷
<code>hpx::remove_copy_if</code>	Copies the elements from a range to a new location for which the given predicate is <code>false</code>	<code>remove_copy</code> ¹¹⁸
<code>hpx::replace</code>	Replaces all values satisfying specific criteria with another value.	<code>replace</code> ¹¹⁹
<code>hpx::replace_if</code>	Replaces all values satisfying specific criteria with another value.	<code>replace</code> ¹²⁰
<code>hpx::replace_copy</code>	Copies a range, replacing elements satisfying specific criteria with another value.	<code>replace_copy</code> ¹²¹
<code>hpx::replace_copy_if</code>	Copies a range, replacing elements satisfying specific criteria with another value.	<code>replace_copy</code> ¹²²
<code>hpx::reverse</code>	Reverses the order elements in a range.	<code>reverse</code> ¹²³
<code>hpx::reverse_copy</code>	Creates a copy of a range that is reversed.	<code>reverse_copy</code> ¹²⁴
<code>hpx::rotate</code>	Rotates the order of elements in a range.	<code>rotate</code> ¹²⁵
<code>hpx::rotate_copy</code>	Copies and rotates a range of elements.	<code>rotate_copy</code> ¹²⁶
<code>hpx::shift_left</code>	Shifts the elements in the range left by n positions.	<code>shift_left</code> ¹²⁷
<code>hpx::shift_right</code>	Shifts the elements in the range right by n positions.	<code>shift_right</code> ¹²⁸
<code>hpx::swap_ranges</code>	Swaps two ranges of elements.	<code>swap_ranges</code> ¹²⁹
<code>hpx::transform</code>	Applies a function to a range of elements.	<code>transform</code> ¹³⁰
<code>hpx::unique</code>	Eliminates all but the first element from every consecutive group of equivalent elements from a range.	<code>unique</code> ¹³¹
<code>hpx::unique_copy</code>	Copies the elements from one range to another in such a way that there are no consecutive equal elements.	<code>unique_copy</code> ¹³²

Table 2.15: Set operations on sorted sequences of header `hpx/algorithm.hpp`

Name	Description	C++ standard
<code>hpx::merge</code>	Merges two sorted ranges.	<code>merge</code> ¹³³
<code>hpx::inplace_merge</code>	Merges two ordered ranges in-place.	<code>inplace_merge</code> ¹³⁴
<code>hpx::includes</code>	Returns true if one set is a subset of another.	<code>includes</code> ¹³⁵
<code>hpx::set_difference</code>	Computes the difference between two sets.	<code>set_difference</code> ¹³⁶
<code>hpx::set_intersection</code>	Computes the intersection of two sets.	<code>set_intersection</code> ¹³⁷
<code>hpx::set_symmetric_difference</code>	Computes the symmetric difference between two sets.	<code>set_symmetric_difference</code> ¹³⁸
<code>hpx::set_union</code>	Computes the union of two sets.	<code>set_union</code> ¹³⁹

Table 2.16: Heap operations of header `hpx/algorithm.hpp`

Name	Description	C++ standard
<code>hpx::is_heap</code>	Returns true if the range is max heap.	<code>is_heap</code> ¹⁴⁰
<code>hpx::is_heap_until</code>	Returns the first element that breaks a max heap.	<code>is_heap_until</code> ¹⁴¹
<code>hpx::make_heap</code>	Constructs a max heap in the range [first, last).	<code>make_heap</code> ¹⁴²

¹⁰⁷ http://en.cppreference.com/w/cpp/algorithm/exclusive_scan¹⁰⁸ http://en.cppreference.com/w/cpp/algorithm/copy_n¹⁰⁹ <http://en.cppreference.com/w/cpp/algorithm/copy>¹¹⁰ <http://en.cppreference.com/w/cpp/algorithm/move>¹¹¹ <http://en.cppreference.com/w/cpp/algorithm/fill>¹¹² http://en.cppreference.com/w/cpp/algorithm/fill_n¹¹³ <http://en.cppreference.com/w/cpp/algorithm/generate>¹¹⁴ http://en.cppreference.com/w/cpp/algorithm/generate_n¹¹⁵ <http://en.cppreference.com/w/cpp/algorithm/remove>¹¹⁶ <http://en.cppreference.com/w/cpp/algorithm/remove>¹¹⁷ http://en.cppreference.com/w/cpp/algorithm/remove_copy¹¹⁸ http://en.cppreference.com/w/cpp/algorithm/remove_copy¹¹⁹ <http://en.cppreference.com/w/cpp/algorithm/replace>¹²⁰ <http://en.cppreference.com/w/cpp/algorithm/replace>¹²¹ http://en.cppreference.com/w/cpp/algorithm/replace_copy¹²² http://en.cppreference.com/w/cpp/algorithm/replace_copy¹²³ <http://en.cppreference.com/w/cpp/algorithm/reverse>¹²⁴ http://en.cppreference.com/w/cpp/algorithm/reverse_copy¹²⁵ <http://en.cppreference.com/w/cpp/algorithm/rotate>¹²⁶ http://en.cppreference.com/w/cpp/algorithm/rotate_copy¹²⁷ http://en.cppreference.com/w/cpp/algorithm/shift_left¹²⁸ http://en.cppreference.com/w/cpp/algorithm/shift_right¹²⁹ http://en.cppreference.com/w/cpp/algorithm/swap_ranges¹³⁰ <http://en.cppreference.com/w/cpp/algorithm/transform>¹³¹ <http://en.cppreference.com/w/cpp/algorithm/unique>¹³² http://en.cppreference.com/w/cpp/algorithm/unique_copy¹³³ <http://en.cppreference.com/w/cpp/algorithm/merge>¹³⁴ http://en.cppreference.com/w/cpp/algorithm/inplace_merge¹³⁵ <http://en.cppreference.com/w/cpp/algorithm/includes>¹³⁶ http://en.cppreference.com/w/cpp/algorithm/set_difference¹³⁷ http://en.cppreference.com/w/cpp/algorithm/set_intersection¹³⁸ http://en.cppreference.com/w/cpp/algorithm/set_symmetric_difference¹³⁹ http://en.cppreference.com/w/cpp/algorithm/set_union

Table 2.17: Minimum/maximum operations of header `hpx/algorithm.hpp`

Name	Description	C++ standard
<code>hpx::max_element</code>	Returns the largest element in a range.	<code>max_element</code> ¹⁴³
<code>hpx::min_element</code>	Returns the smallest element in a range.	<code>min_element</code> ¹⁴⁴
<code>hpx::minmax_element</code>	Returns the smallest and the largest element in a range.	<code>minmax_element</code> ¹⁴⁵

Table 2.18: Partitioning Operations of header `hpx/algorithm.hpp`

Name	Description	C++ standard
<code>hpx::nth_element</code>	Partially sorts the given range making sure that it is partitioned by the given element	<code>nth_element</code> ¹⁴⁶
<code>hpx::is_partitioned</code>	Returns <code>true</code> if each true element for a predicate precedes the false elements in a range.	<code>is_partitioned</code> ¹⁴⁷
<code>hpx::partition</code>	Divides elements into two groups without preserving their relative order.	<code>partition</code> ¹⁴⁸
<code>hpx::partition_copy</code>	Copies a range dividing the elements into two groups.	<code>partition_copy</code> ¹⁴⁹
<code>hpx::stable_partition</code>	Divides elements into two groups while preserving their relative order.	<code>stable_partition</code> ¹⁵⁰

Table 2.19: Sorting Operations of header `hpx/algorithm.hpp`

Name	Description	C++ standard
<code>hpx::is_sorted</code>	Returns <code>true</code> if each element in a range is sorted.	<code>is_sorted</code> ¹⁵¹
<code>hpx::is_sorted_until</code>	Returns the first unsorted element.	<code>is_sorted_until</code> ¹⁵²
<code>hpx::sort</code>	Sorts the elements in a range.	<code>sort</code> ¹⁵³
<code>hpx::stable_sort</code>	Sorts the elements in a range, maintain sequence of equal elements.	<code>stable_sort</code> ¹⁵⁴
<code>hpx::partial_sort</code>	Sorts the first elements in a range.	<code>partial_sort</code> ¹⁵⁵
<code>hpx::partial_sort_copy</code>	Sorts the first elements in a range, storing the result in another range.	<code>partial_sort_copy</code> ¹⁵⁶
<code>hpx::experimental::sort_by_key</code>	Sorts a range of data using keys supplied in another range.	

¹⁴⁰ http://en.cppreference.com/w/cpp/algorithm/is_heap¹⁴¹ http://en.cppreference.com/w/cpp/algorithm/is_heap_until¹⁴² http://en.cppreference.com/w/cpp/algorithm/make_heap¹⁴³ http://en.cppreference.com/w/cpp/algorithm/max_element¹⁴⁴ http://en.cppreference.com/w/cpp/algorithm/min_element¹⁴⁵ http://en.cppreference.com/w/cpp/algorithm/minmax_element¹⁴⁶ http://en.cppreference.com/w/cpp/algorithm/nth_element¹⁴⁷ http://en.cppreference.com/w/cpp/algorithm/is_partitioned¹⁴⁸ <http://en.cppreference.com/w/cpp/algorithm/partition>¹⁴⁹ http://en.cppreference.com/w/cpp/algorithm/partition_copy¹⁵⁰ http://en.cppreference.com/w/cpp/algorithm/stable_partition

Table 2.20: Numeric Parallel Algorithms of header `hpx/numeric.hpp`

Name	Description	C++ standard
<code>hpx::adjacent_difference</code>	Calculates the difference between each element in an input range and the preceding element.	<code>adjacent_difference</code> ¹⁵⁷
<code>hpx::exclusive_scan</code>	Does an exclusive parallel scan over a range of elements.	<code>exclusive_scan</code> ¹⁵⁸
<code>hpx::inclusive_scan</code>	Does an inclusive parallel scan over a range of elements.	<code>inclusive_scan</code> ¹⁵⁹
<code>hpx::reduce</code>	Sums up a range of elements.	<code>reduce</code> ¹⁶⁰
<code>hpx::transform_exclusive_scan</code>	Does an exclusive parallel scan over a range of elements after applying a function.	<code>transform_exclusive_scan</code> ¹⁶¹
<code>hpx::transform_inclusive_scan</code>	Does an inclusive parallel scan over a range of elements after applying a function.	<code>transform_inclusive_scan</code> ¹⁶²
<code>hpx::transform_reduce</code>	Sums up a range of elements after applying a function. Also, accumulates the inner products of two input ranges.	<code>transform_reduce</code> ¹⁶³

¹⁵¹ http://en.cppreference.com/w/cpp/algorithm/is_sorted¹⁵² http://en.cppreference.com/w/cpp/algorithm/is_sorted_until¹⁵³ <http://en.cppreference.com/w/cpp/algorithm/sort>¹⁵⁴ http://en.cppreference.com/w/cpp/algorithm/stable_sort¹⁵⁵ http://en.cppreference.com/w/cpp/algorithm/partial_sort¹⁵⁶ http://en.cppreference.com/w/cpp/algorithm/partial_sort_copy¹⁵⁷ http://en.cppreference.com/w/cpp/algorithm/adjacent_difference¹⁵⁸ http://en.cppreference.com/w/cpp/algorithm/exclusive_scan¹⁵⁹ http://en.cppreference.com/w/cpp/algorithm/inclusive_scan¹⁶⁰ <http://en.cppreference.com/w/cpp/algorithm/reduce>¹⁶¹ http://en.cppreference.com/w/cpp/algorithm/transform_exclusive_scan¹⁶² http://en.cppreference.com/w/cpp/algorithm/transform_inclusive_scan¹⁶³ http://en.cppreference.com/w/cpp/algorithm/transform_reduce

Table 2.21: Dynamic Memory Management of header `hpx/memory.hpp`

Name	Description	C++ standard
<code>hpx::destroy</code>	Destroys a range of objects.	<code>destroy</code> ¹⁶⁴
<code>hpx::destroy_n</code>	Destroys a range of objects.	<code>destroy_n</code> ¹⁶⁵
<code>hpx::uninitialized_copy</code>	Copies a range of objects to an uninitialized area of memory.	<code>uninitialized_copy</code> ¹⁶⁶
<code>hpx::uninitialized_copy_n</code>	Copies a number of objects to an uninitialized area of memory.	<code>uninitialized_copy_n</code> ¹⁶⁷
<code>hpx::uninitialized_default_construct</code>	Constructs a range of objects to an uninitialized area of memory.	<code>uninitialized_default_construct</code> ¹⁶⁸
<code>hpx::uninitialized_default_construct_n</code>	Constructs a number of objects to an uninitialized area of memory.	<code>uninitialized_default_construct_n</code> ¹⁶⁹
<code>hpx::uninitialized_fill</code>	Copies an object to an uninitialized area of memory.	<code>uninitialized_fill</code> ¹⁷⁰
<code>hpx::uninitialized_fill_n</code>	Copies an object to an uninitialized area of memory.	<code>uninitialized_fill_n</code> ¹⁷¹
<code>hpx::uninitialized_move</code>	Moves a range of objects to an uninitialized area of memory.	<code>uninitialized_move</code> ¹⁷²
<code>hpx::uninitialized_move_n</code>	Moves a number of objects to an uninitialized area of memory.	<code>uninitialized_move_n</code> ¹⁷³
<code>hpx::uninitialized_value_construct</code>	Constructs objects in an uninitialized area of memory.	<code>uninitialized_value_construct</code> ¹⁷⁴
<code>hpx::uninitialized_value_construct_n</code>	Constructs objects in an uninitialized area of memory.	<code>uninitialized_value_construct_n</code> ¹⁷⁵

Table 2.22: Index-based for-loops of header `hpx/algorithm.hpp`

Name	Description
<code>hpx::experimental::for_loop</code>	Implements loop functionality over a range specified by integral or iterator bounds.
<code>hpx::experimental::for_loop_strided</code>	Implements loop functionality over a range specified by integral or iterator bounds.
<code>hpx::experimental::for_loop_n</code>	Implements loop functionality over a range specified by integral or iterator bounds.
<code>hpx::experimental::for_loop_n_strided</code>	Implements loop functionality over a range specified by integral or iterator bounds.

¹⁶⁴ <http://en.cppreference.com/w/cpp/memory/destroy>¹⁶⁵ http://en.cppreference.com/w/cpp/memory/destroy_n¹⁶⁶ http://en.cppreference.com/w/cpp/memory/uninitialized_copy¹⁶⁷ http://en.cppreference.com/w/cpp/memory/uninitialized_copy_n¹⁶⁸ http://en.cppreference.com/w/cpp/memory/uninitialized_default_construct¹⁶⁹ http://en.cppreference.com/w/cpp/memory/uninitialized_default_construct_n¹⁷⁰ http://en.cppreference.com/w/cpp/memory/uninitialized_fill¹⁷¹ http://en.cppreference.com/w/cpp/memory/uninitialized_fill_n¹⁷² http://en.cppreference.com/w/cpp/memory/uninitialized_move¹⁷³ http://en.cppreference.com/w/cpp/memory/uninitialized_move_n¹⁷⁴ http://en.cppreference.com/w/cpp/memory/uninitialized_value_construct¹⁷⁵ http://en.cppreference.com/w/cpp/memory/uninitialized_value_construct_n

Executor parameters and executor parameter traits

HPX introduces the notion of execution parameters and execution parameter traits. At this point, the only parameter that can be customized is the size of the chunks of work executed on a single *HPX* thread (such as the number of loop iterations combined to run as a single task).

An executor parameter object is responsible for exposing the calculation of the size of the chunks scheduled. It abstracts the (potentially platform-specific) algorithms of determining those chunk sizes.

The way executor parameters are implemented is aligned with the way executors are implemented. All functionalities of concrete executor parameter types are exposed and accessible through a corresponding customization point, e.g. `get_chunk_size()`.

With `executor_parameter_traits`, clients access all types of executor parameters uniformly, e.g.:

```
std::size_t chunk_size =  
    hpx::execution::experimental::get_chunk_size(my_parameter, my_executor,  
        num_cores, num_tasks);
```

This call synchronously retrieves the size of a single chunk of loop iterations (or similar) to combine for execution on a single *HPX* thread if the overall number of cores `num_cores` and tasks to schedule is given by `num_tasks`. The lambda function exposes a means of test-probing the execution of a single iteration for performance measurement purposes. The execution parameter type might dynamically determine the execution time of one or more tasks in order to calculate the chunk size; see `hpx::execution::experimental::auto_chunk_size` for an example of this executor parameter type.

Other functions in the interface exist to discover whether an executor parameter type should be invoked once (i.e., it returns a static chunk size; see `hpx::execution::experimental::static_chunk_size`) or whether it should be invoked for each scheduled chunk of work (i.e., it returns a variable chunk size; for an example, see `hpx::execution::experimental::guided_chunk_size`).

Although this interface appears to require executor parameter type authors to implement all different basic operations, none are required. In practice, all operations have sensible defaults. However, some executor parameter types will naturally specialize all operations for maximum efficiency.

HPX implements the following executor parameter types:

- `hpx::execution::experimental::auto_chunk_size`: Loop iterations are divided into pieces and then assigned to threads. The number of loop iterations combined is determined based on measurements of how long the execution of 1% of the overall number of iterations takes. This executor parameter type makes sure that as many loop iterations are combined as necessary to run for the amount of time specified.
- `hpx::execution::experimental::static_chunk_size`: Loop iterations are divided into pieces of a given size and then assigned to threads. If the size is not specified, the iterations are, if possible, evenly divided contiguously among the threads. This executor parameters type is equivalent to OpenMP's STATIC scheduling directive.
- `hpx::execution::experimental::dynamic_chunk_size`: Loop iterations are divided into pieces of a given size and then dynamically scheduled among the cores; when a core finishes one chunk, it is dynamically assigned another. If the size is not specified, the default chunk size is 1. This executor parameter type is equivalent to OpenMP's DYNAMIC scheduling directive.
- `hpx::execution::experimental::guided_chunk_size`: Iterations are dynamically assigned to cores in blocks as cores request them until no blocks remain to be assigned. This is similar to `dynamic_chunk_size` except that the block size decreases each time a number of loop iterations is given to a thread. The size of the initial block is proportional to `number_of_iterations / number_of_cores`. Subsequent blocks are proportional to `number_of_iterations_remaining / number_of_cores`. The optional chunk size parameter defines the minimum block size. The default minimal chunk size is 1. This executor parameter type is equivalent to OpenMP's GUIDED scheduling directive.

2.4.12 Writing distributed applications

This section focuses on the features of *HPX* needed to write distributed applications, namely the *Active Global Address Space* (AGAS), remotely executable functions (i.e., *actions*), and distributed objects (i.e., *components*).

Global names

HPX implements an *Active Global Address Space* (AGAS) which exposes a single uniform address space spanning all localities an application runs on. AGAS is a fundamental component of the ParalleX execution model. Conceptually, there is no rigid demarcation of local or global memory in AGAS; all available memory is a part of the same address space. AGAS enables named objects to be moved (migrated) across localities without having to change the object's name; i.e., no references to migrated objects have to be ever updated. This feature has significance for dynamic load balancing and in applications where the workflow is highly dynamic, allowing work to be migrated from heavily loaded nodes to less loaded nodes. In addition, immutability of names ensures that AGAS does not have to keep extra indirections ("bread crumbs") when objects move, hence, minimizing complexity of code management for system developers as well as minimizing overheads in maintaining and managing aliases.

The AGAS implementation in *HPX* does not automatically expose every local address to the global address space. It is the responsibility of the programmer to explicitly define which of the objects have to be globally visible and which of the objects are purely local.

In *HPX* global addresses (global names) are represented using the `hpx::id_type` data type. This data type is conceptually very similar to `void*` pointers as it does not expose any type information of the object it is referring to.

The only predefined global addresses are assigned to all localities. The following *HPX* API functions allow one to retrieve the global addresses of localities:

- `hpx::find_here`: retrieves the global address of the *locality* this function is called on.
- `hpx::find_all_localities`: retrieves the global addresses of all localities available to this application (including the *locality* the function is being called on).
- `hpx::find_remote_localities`: retrieves the global addresses of all remote localities available to this application (not including the *locality* the function is being called on).
- `hpx::get_num_localities`: retrieves the number of localities available to this application.
- `hpx::find_locality`: retrieves the global address of any *locality* supporting the given component type.
- `hpx::get_colocation_id`: retrieves the global address of the *locality* currently hosting the object with the given global address.

Additionally, the global addresses of localities can be used to create new instances of components using the following *HPX* API function:

- `hpx::components::new_`: Creates a new instance of the given *Component* type on the specified *locality*.

Note: *HPX* does not expose any functionality to delete component instances. All global addresses (as represented using `hpx::id_type`) are automatically garbage collected. When the last (global) reference to a particular component instance goes out of scope, the corresponding component instance is automatically deleted.

Posting actions

Action type definition

Actions are special types used to describe possibly remote operations. For every global function and every member function which has to be invoked distantly, a special type must be defined. For any global function the special macro `HPX_PLAIN_ACTION` can be used to define the action type. Here is an example demonstrating this:

```
namespace app
{
    void some_global_function(double d)
    {
        cout << d;
    }
}

// This will define the action type 'some_global_action' which represents
// the function 'app::some_global_function'.
HPX_PLAIN_ACTION(app::some_global_function, some_global_action);
```

Important: The macro `HPX_PLAIN_ACTION` has to be placed in global namespace, even if the wrapped function is located in some other namespace. The newly defined action type is placed in the global namespace as well.

If the action type should be defined somewhere not in global namespace, the action type definition has to be split into two macro invocations (`HPX_DEFINE_PLAIN_ACTION` and `HPX_REGISTER_ACTION`) as shown in the next example:

```
namespace app
{
    void some_global_function(double d)
    {
        cout << d;
    }

    // On conforming compilers the following macro expands to:
    //
    //     typedef hpx::actions::make_action<
    //         decltype(&some_global_function), &some_global_function
    //     >::type some_global_action;
    //
    // This will define the action type 'some_global_action' which represents
    // the function 'some_global_function'.
    HPX_DEFINE_PLAIN_ACTION(some_global_function, some_global_action);
}

// The following macro expands to a series of definitions of global objects
// which are needed for proper serialization and initialization support
// enabling the remote invocation of the function`some_global_function`
HPX_REGISTER_ACTION(app::some_global_action, app_some_global_action);
```

The shown code defines an action type `some_global_action` inside the namespace `app`.

Important: If the action type definition is split between two macros as shown above, the name of the action type to

create has to be the same for both macro invocations (here `some_global_action`).

Important: The second argument passed to `HPX_REGISTER_ACTION` (`app_some_global_action`) has to comprise a globally unique C++ identifier representing the action. This is used for serialization purposes.

For member functions of objects which have been registered with AGAS (e.g., ‘components’), a different registration macro `HPX_DEFINE_COMPONENT_ACTION` has to be utilized. Any component needs to be declared in a header file and have some special support macros defined in a source file. Here is an example demonstrating this. The first snippet has to go into the header file:

```
namespace app
{
    struct some_component
    : hpx::components::component_base<some_component>
    {
        int some_member_function(std::string s)
        {
            return std::stoi(s);
        }

        // This will define the action type 'some_member_action' which
        // represents the member function 'some_member_function' of the
        // object type 'some_component'.
        HPX_DEFINE_COMPONENT_ACTION(some_component, some_member_function,
            some_member_action);
    };
}

// Note: The second argument to the macro below has to be systemwide-unique
// C++ identifiers
HPX_REGISTER_ACTION_DECLARATION(app::some_component::some_member_action, some_component_
↪some_action);
```

The next snippet belongs in a source file (e.g., the main application source file) in the simplest case:

```
typedef hpx::components::component<app::some_component> component_type;
typedef app::some_component some_component;

HPX_REGISTER_COMPONENT(component_type, some_component);

// The parameters for this macro have to be the same as used in the corresponding
// HPX_REGISTER_ACTION_DECLARATION() macro invocation above
typedef some_component::some_member_action some_component_some_action;
HPX_REGISTER_ACTION(some_component_some_action);
```

While these macro invocations are a bit more complex than those for simple global functions, they should still be manageable.

The most important macro invocation is the `HPX_DEFINE_COMPONENT_ACTION` in the header file as this defines the action type we need to invoke the member function. For a complete example of a simple component action see `component_in_executable.cpp`.

Reflection-based component actions (experimental)

Note: This feature requires a compiler with C++26 reflection support (`HPX_HAVE_CXX26_REFLECTION`) and is currently experimental.

When reflection support is available, component actions can be defined with a single macro instead of the three-step process described above:

```
namespace app
{
    struct some_component
    : hpx::components::component_base<some_component>
    {
        int some_member_function(std::string s)
        {
            return std::stoi(s);
        }
        // This single line replaces HPX_DEFINE_COMPONENT_ACTION,
        // HPX_REGISTER_ACTION_DECLARATION, and HPX_REGISTER_ACTION.
        HPX_COMPONENT_ACTION(some_component, some_member_function,
            some_member_action);
    };
}
```

The resulting `some_member_action` type can be used exactly like an action type created with `HPX_DEFINE_COMPONENT_ACTION`, with `hpx::post`, `hpx::async`, and `hpx::sync`, or by invoking it directly. No separate registration step is required in the source file.

Action invocation

The process of invoking a global function (or a member function of an object) with the help of the associated action is called ‘posting the action’. Actions can have arguments, which will be supplied while the action is applied. At the minimum, one parameter is required to post any action - the id of the *locality* the associated function should be invoked on (for global functions), or the id of the component instance (for member functions). Generally, *HPX* provides several ways to post an action, all of which are described in the following sections.

Generally, *HPX* actions are very similar to ‘normal’ C++ functions except that actions can be invoked remotely. Fig. ?? below shows an overview of the main API exposed by *HPX*. This shows the function invocation syntax as defined by the C++ language (dark gray), the additional invocation syntax as provided through C++ Standard Library features (medium gray), and the extensions added by *HPX* (light gray) where:

- `f` function to invoke,
- `p...`: (optional) arguments,
- `R`: return type of `f`,
- `action`: action type defined by, `HPX_DEFINE_PLAIN_ACTION` or `HPX_DEFINE_COMPONENT_ACTION` encapsulating `f`,
- `a`: an instance of the type `action`,
- `id`: the global address the action is applied to.

R f(p...)	Synchronous (return R)	Asynchronous (return future<R>)	Fire & Forget (return void)
Functions (direct)	f(p...) C++	async(f, p...)	post(f, p...)
Functions (lazy)	bind(f, p...)(...)	async(bind(f, p...), ...) C++ Library	post(bind(f, p...), ...)
Actions (direct)	HPX_ACTION(f, a) a(id, p...)	HPX_ACTION(f, a) async(a, id, p...)	HPX_ACTION(f, a) post(a, id, p...)
Actions (lazy)	HPX_ACTION(f, a) bind(a, id, p...)(...)	HPX_ACTION(f, a) async(bind(a, id, p...), ...)	HPX_ACTION(f, a) post(bind(a, id, p...), ...) HPX

Fig. 2.8: Overview of the main API exposed by *HPX*.

This figure shows that *HPX* allows the user to **post** actions with a syntax similar to the C++ standard. In fact, all action types have an overloaded function operator allowing to synchronously **post** the action. Further, *HPX* implements `hpx::async` which semantically works similar to the way `std::async` works for plain C++ function.

Note: The similarity of posting an action to conventional function invocations extends even further. *HPX* implements `hpx::bind` and `hpx::function` two facilities which are semantically equivalent to the `std::bind` and `std::function` types as defined by the C++11 Standard. While `hpx::async` extends beyond the conventional semantics by supporting actions and conventional C++ functions, the *HPX* facilities `hpx::bind` and `hpx::function` extend beyond the conventional standard facilities too. The *HPX* facilities not only support conventional functions, but can be used for actions as well.

Additionally, *HPX* exposes `hpx::post` and `hpx::async_continue` both of which refine and extend the standard C++ facilities.

The different ways to invoke a function in *HPX* will be explained in more detail in the following sections.

Posting an action asynchronously without any synchronization

This method ('fire and forget') will make sure the function associated with the action is scheduled to run on the target *locality*. Posting the action does not wait for the function to start running, instead it is a fully asynchronous operation. The following example shows how to **post** the action as defined in the previous section on the local *locality* (the *locality* this code runs on):

```
some_global_action act;    // define an instance of some_global_action
hpx::post(act, hpx::find_here(), 2.0);
```

(the function `hpx::find_here()` returns the id of the local *locality*, i.e. the *locality* this code executes on).

Any component member function can be invoked using the same syntactic construct. Given that `id` is the global address for a component instance created earlier, this invocation looks like:

```
some_component_action act;    // define an instance of some_component_action
hpx::post(act, id, "42");
```

In this case any value returned from this action (e.g. in this case the integer 42 is ignored. Please look at *Action type definition* for the code defining the component action `some_component_action` used.

Posting an action asynchronously with synchronization

This method will make sure the action is scheduled to run on the target *locality*. Posting the action itself does not wait for the function to start running or to complete, instead this is a fully asynchronous operation similar to using `hpx::post` as described above. The difference is that this method will return an instance of a `hpx::future<>` encapsulating the result of the (possibly remote) execution. The future can be used to synchronize with the asynchronous operation. The following example shows how to post the action from above on the local *locality*:

```
some_global_action act;    // define an instance of some_global_action
hpx::future<void> f = hpx::async(act, hpx::find_here(), 2.0);
//
// ... other code can be executed here
//
f.get();    // this will possibly wait for the asynchronous operation to 'return'
```

(as before, the function `hpx::find_here()` returns the id of the local *locality* (the *locality* this code is executed on).

Note: The use of a `hpx::future<void>` allows the current thread to synchronize with any remote operation not returning any value.

Note: Any `std::future<>` returned from `std::async()` is required to block in its destructor if the value has not been set for this future yet. This is not true for `hpx::future<>` which will never block in its destructor, even if the value has not been returned to the future yet. We believe that consistency in the behavior of futures is more important than standards conformance in this case.

Any component member function can be invoked using the same syntactic construct. Given that `id` is the global address for a component instance created earlier, this invocation looks like:

```
some_component_action act;    // define an instance of some_component_action
hpx::future<int> f = hpx::async(act, id, "42");
//
// ... other code can be executed here
//
cout << f.get();    // this will possibly wait for the asynchronous operation to 'return'
↪ 42
```

Note: The invocation of `f.get()` will return the result immediately (without suspending the calling thread) if the result from the asynchronous operation has already been returned. Otherwise, the invocation of `f.get()` will suspend the execution of the calling thread until the asynchronous operation returns its result.

Posting an action synchronously

This method will schedule the function wrapped in the specified action on the target *locality*. While the invocation appears to be synchronous (as we will see), the calling thread will be suspended while waiting for the function to return. Invoking a plain action (e.g. a global function) synchronously is straightforward:

```
some_global_action act;    // define an instance of some_global_action
act(hpx::find_here(), 2.0);
```

While this call looks just like a normal synchronous function invocation, the function wrapped by the action will be scheduled to run on a new thread and the calling thread will be suspended. After the new thread has executed the wrapped global function, the waiting thread will resume and return from the synchronous call.

Equivalently, any action wrapping a component member function can be invoked synchronously as follows:

```
some_component_action act;    // define an instance of some_component_action
int result = act(id, "42");
```

The action invocation will either schedule a new thread locally to execute the wrapped member function (as before, `id` is the global address of the component instance the member function should be invoked on), or it will send a parcel to the remote *locality* of the component causing a new thread to be scheduled there. The calling thread will be suspended until the function returns its result. This result will be returned from the synchronous action invocation.

It is very important to understand that this ‘synchronous’ invocation syntax in fact conceals an asynchronous function call. This is beneficial as the calling thread is suspended while waiting for the outcome of a potentially remote operation. The *HPX* thread scheduler will schedule other work in the meantime, allowing the application to make further progress while the remote result is computed. This helps overlapping computation with communication and hiding communication latencies.

Note: The syntax of posting an action is always the same, regardless whether the target *locality* is remote to the invocation *locality* or not. This is a very important feature of *HPX* as it frees the user from the task of keeping track what actions have to be applied locally and which actions are remote. If the target for posting an action is local, a new thread is automatically created and scheduled. Once this thread is scheduled and run, it will execute the function encapsulated by that action. If the target is remote, *HPX* will send a parcel to the remote *locality* which encapsulates the action and its parameters. Once the parcel is received on the remote *locality* *HPX* will create and schedule a new thread there. Once this thread runs on the remote *locality*, it will execute the function encapsulated by the action.

Posting an action with a continuation but without any synchronization

This method is very similar to the method described in section *Posting an action asynchronously without any synchronization*. The difference is that it allows the user to chain a sequence of asynchronous operations, while handing the (intermediate) results from one step to the next step in the chain. Where `hpx::post` invokes a single function using ‘fire and forget’ semantics, `hpx::post_continue` asynchronously triggers a chain of functions without the need for the execution flow ‘to come back’ to the invocation site. Each of the asynchronous functions can be executed on a different *locality*.

Posting an action with a continuation and with synchronization

This method is very similar to the method described in section *Posting an action asynchronously with synchronization*. In addition to what `hpx::async` can do, the functions `hpx::async_continue` takes an additional function argument. This function will be called as the continuation of the executed action. It is expected to perform additional operations and to make sure that a result is returned to the original invocation site. This method chains operations asynchronously by providing a continuation operation which is automatically executed once the first action has finished executing.

As an example we chain two actions, where the result of the first action is forwarded to the second action and the result of the second action is sent back to the original invocation site:

```
// first action
std::int32_t action1(std::int32_t i)
{
    return i+1;
}
HPX_PLAIN_ACTION(action1);    // defines action1_type

// second action
std::int32_t action2(std::int32_t i)
{
    return i*2;
}
HPX_PLAIN_ACTION(action2);    // defines action2_type

// this code invokes 'action1' above and passes along a continuation
// function which will forward the result returned from 'action1' to
// 'action2'.
action1_type act1;           // define an instance of 'action1_type'
action2_type act2;           // define an instance of 'action2_type'
hpx::future<int> f =
    hpx::async_continue(act1, hpx::make_continuation(act2),
        hpx::find_here(), 42);
hpx::cout << f.get() << "\n";    // will print: 86 ((42 + 1) * 2)
```

By default, the continuation is executed on the same *locality* as `hpx::async_continue` is invoked from. If you want to specify the *locality* where the continuation should be executed, the code above has to be written as:

```
// this code invokes 'action1' above and passes along a continuation
// function which will forward the result returned from 'action1' to
// 'action2'.
action1_type act1;           // define an instance of 'action1_type'
action2_type act2;           // define an instance of 'action2_type'
hpx::future<int> f =
    hpx::async_continue(act1, hpx::make_continuation(act2, hpx::find_here()),
        hpx::find_here(), 42);
hpx::cout << f.get() << "\n";    // will print: 86 ((42 + 1) * 2)
```

Similarly, it is possible to chain more than 2 operations:

```
action1_type act1;           // define an instance of 'action1_type'
action2_type act2;           // define an instance of 'action2_type'
hpx::future<int> f =
    hpx::async_continue(act1,
```

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```

    hpx::make_continuation(act2, hpx::make_continuation(act1)),
    hpx::find_here(), 42);
hpx::cout << f.get() << "\n";    // will print: 87 ((42 + 1) * 2 + 1)

```

The function `hpx::make_continuation` creates a special function object which exposes the following prototype:

```

struct continuation
{
    template <typename Result>
    void operator()(hpx::id_type id, Result&& result) const
    {
        ...
    }
};

```

where the parameters passed to the overloaded function operator `operator()()` are:

- the `id` is the global id where the final result of the asynchronous chain of operations should be sent to (in most cases this is the id of the `hpx::future` returned from the initial call to `hpx::async_continue`. Any custom continuation function should make sure this `id` is forwarded to the last operation in the chain.
- the `result` is the result value of the current operation in the asynchronous execution chain. This value needs to be forwarded to the next operation.

Note: All of those operations are implemented by the predefined continuation function object which is returned from `hpx::make_continuation`. Any (custom) function object used as a continuation should conform to the same interface.

Action error handling

Like in any other asynchronous invocation scheme it is important to be able to handle error conditions occurring while the asynchronous (and possibly remote) operation is executed. In *HPX* all error handling is based on standard C++ exception handling. Any exception thrown during the execution of an asynchronous operation will be transferred back to the original invocation *locality*, where it is rethrown during synchronization with the calling thread.

Important: Exceptions thrown during asynchronous execution can be transferred back to the invoking thread only for the synchronous and the asynchronous case with synchronization. Like with any other unhandled exception, any exception thrown during the execution of an asynchronous action *without* synchronization will result in calling `hpx::terminate` causing the running application to exit immediately.

Note: Even if error handling internally relies on exceptions, most of the API functions exposed by *HPX* can be used without throwing an exception. Please see *Working with exceptions* for more information.

As an example, we will assume that the following remote function will be executed:

```

namespace app
{
    void some_function_with_error(int arg)

```

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```

{
    if (arg < 0) {
        HPX_THROW_EXCEPTION(hpx::error::bad_parameter,
            "some_function_with_error",
            "some really bad error happened");
    }
    // do something else...
}
}

// This will define the action type 'some_error_action' which represents
// the function 'app::some_function_with_error'.
HPX_PLAIN_ACTION(app::some_function_with_error, some_error_action);

```

The use of `HPX_THROW_EXCEPTION` to report the error encapsulates the creation of a `hpx::exception` which is initialized with the error code `hpx::error::bad_parameter`. Additionally it carries the passed strings, the information about the file name, line number, and call stack of the point the exception was thrown from.

We invoke this action using the synchronous syntax as described before:

```

// note: wrapped function will throw hpx::exception
some_error_action act;           // define an instance of some_error_action
try {
    act(hpx::find_here(), -3);    // exception will be rethrown from here
}
catch (hpx::exception const& e) {
    // prints: 'some really bad error happened: HPX(bad parameter)'
    cout << e.what();
}

```

If this action is invoked asynchronously with synchronization, the exception is propagated to the waiting thread as well and is re-thrown from the future's function `get()`:

```

// note: wrapped function will throw hpx::exception
some_error_action act;           // define an instance of some_error_action
hpx::future<void> f = hpx::async(act, hpx::find_here(), -3);
try {
    f.get();                     // exception will be rethrown from here
}
catch (hpx::exception const& e) {
    // prints: 'some really bad error happened: HPX(bad parameter)'
    cout << e.what();
}

```

For more information about error handling please refer to the section *Working with exceptions*. There we also explain how to handle error conditions without having to rely on exception.

Writing components

A component in *HPX* is a C++ class which can be created remotely and for which its member functions can be invoked remotely as well. The following sections highlight how components can be defined, created, and used.

Defining components

In order for a C++ class type to be managed remotely in *HPX*, the type must be derived from the `hpx::components::component_base` template type. We call such C++ class types ‘components’.

Note that the component type itself is passed as a template argument to the base class:

```
// header file some_component.hpp

#include <hpx/include/components.hpp>

namespace app
{
    // Define a new component type 'some_component'
    struct some_component
        : hpx::components::component_base<some_component>
    {
        // This member function is has to be invoked remotely
        int some_member_function(std::string const& s)
        {
            return std::stoi(s);
        }

        // This will define the action type 'some_member_action' which
        // represents the member function 'some_member_function' of the
        // object type 'some_component'.
        HPX_DEFINE_COMPONENT_ACTION(some_component, some_member_function, some_member_
        ↪action);
    };
}

// This will generate the necessary boiler-plate code for the action allowing
// it to be invoked remotely. This declaration macro has to be placed in the
// header file defining the component itself.
//
// Note: The second argument to the macro below has to be systemwide-unique
//       C++ identifiers
//
HPX_REGISTER_ACTION_DECLARATION(app::some_component::some_member_action, some_component_
↪some_action);
```

There is more boiler plate code which has to be placed into a source file in order for the component to be usable. Every component type is required to have macros placed into its source file, one for each component type and one macro for each of the actions defined by the component type.

For instance:

```
// source file some_component.cpp
```

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```

#include "some_component.hpp"

// The following code generates all necessary boiler plate to enable the
// remote creation of 'app::some_component' instances with 'hpx::new_<>()'
//
using some_component = app::some_component;
using some_component_type = hpx::components::component<some_component>;

// Please note that the second argument to this macro must be a
// (system-wide) unique C++-style identifier (without any namespaces)
//
HPX_REGISTER_COMPONENT(some_component_type, some_component);

// The parameters for this macro have to be the same as used in the corresponding
// HPX_REGISTER_ACTION_DECLARATION() macro invocation in the corresponding
// header file.
//
// Please note that the second argument to this macro must be a
// (system-wide) unique C++-style identifier (without any namespaces)
//
HPX_REGISTER_ACTION(app::some_component::some_member_action, some_component_some_action);

```

Defining client side representation classes

Often it is very convenient to define a separate type for a component which can be used on the client side (from where the component is instantiated and used). This step might seem as unnecessary duplicating code, however it significantly increases the type safety of the code.

A possible implementation of such a client side representation for the component described in the previous section could look like:

```

#include <hpx/include/components.hpp>

namespace app
{
    // Define a client side representation type for the component type
    // 'some_component' defined in the previous section.
    //
    struct some_component_client
    : hpx::components::client_base<some_component_client, some_component>
    {
        using base_type = hpx::components::client_base<
            some_component_client, some_component>;

        some_component_client(hpx::future<hpx::id_type> && id)
            : base_type(std::move(id))
        {}

        hpx::future<int> some_member_function(std::string const& s)
        {
            some_component::some_member_action act;

```

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```

        return hpx::async(act, get_id(), s);
    }
};
}

```

A client side object stores the global id of the component instance it represents. This global id is accessible by calling the function `client_base<>::get_id()`. The special constructor which is provided in the example allows to create this client side object directly using the API function `hpx::new_`.

Creating component instances

Instances of defined component types can be created in two different ways. If the component to create has a defined client side representation type, then this can be used, otherwise use the server type.

The following examples assume that `some_component_type` is the type of the server side implementation of the component to create. All additional arguments (see , ... notation below) are passed through to the corresponding constructor calls of those objects:

```

// create one instance on the given locality
hpx::id_type here = hpx::find_here();
hpx::future<hpx::id_type> f =
    hpx::new_<some_component_type>(here, ...);

// create one instance using the given distribution
// policy (here: hpx::colocating_distribution_policy)
hpx::id_type here = hpx::find_here();
hpx::future<hpx::id_type> f =
    hpx::new_<some_component_type>(hpx::colocated(here), ...);

// create multiple instances on the given locality
hpx::id_type here = find_here();
hpx::future<std::vector<hpx::id_type>> f =
    hpx::new_<some_component_type[]>(here, num, ...);

// create multiple instances using the given distribution
// policy (here: hpx::binpacking_distribution_policy)
hpx::future<std::vector<hpx::id_type>> f = hpx::new_<some_component_type[]>(
    hpx::binpacking(hpx::find_all_localities()), num, ...);

```

The examples below demonstrate the use of the same API functions for creating client side representation objects (instead of just plain ids). These examples assume that `client_type` is the type of the client side representation of the component type to create. As above, all additional arguments (see , ... notation below) are passed through to the corresponding constructor calls of the server side implementation objects corresponding to the `client_type`:

```

// create one instance on the given locality
hpx::id_type here = hpx::find_here();
client_type c = hpx::new_<client_type>(here, ...);

// create one instance using the given distribution
// policy (here: hpx::colocating_distribution_policy)
hpx::id_type here = hpx::find_here();
client_type c = hpx::new_<client_type>(hpx::colocated(here), ...);

```

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```

// create multiple instances on the given locality
hpx::id_type here = hpx::find_here();
hpx::future<std::vector<client_type>>> f =
    hpx::new_<client_type[]>(here, num, ...);

// create multiple instances using the given distribution
// policy (here: hpx::binpacking_distribution_policy)
hpx::future<std::vector<client_type>>> f = hpx::new_<client_type[]>(
    hpx::binpacking(hpx::find_all_localities()), num, ...);

```

Using component instances

After having created the component instances as described above, we can simply use them as indicated below:

```

#include <hpx/include/components.hpp>
#include <iostream>
#include <vector>

// Define a simple component
struct some_component : hpx::components::component_base<some_component>
{
    void print() const
    {
        std::cout << "Hello from component instance!" << std::endl;
    }
    HPX_DEFINE_COMPONENT_ACTION(some_component, print, print_action);
};

typedef some_component::print_action print_action;

// Create one instance on the given locality
hpx::id_type here = hpx::find_here();
hpx::future<hpx::id_type> f1 =
    hpx::new_<some_component>(here);

// Get the future value
hpx::id_type instance_id = f1.get();

// Invoke action on the instance
hpx::async<print_action>(instance_id).get();

// Create multiple instances on the given locality
int num = 3;
hpx::future<std::vector<hpx::id_type>>> f2 =
    hpx::new_<some_component[]>(here, num);

// Get the future value
std::vector<hpx::id_type> instance_ids = f2.get();

// Invoke action on each instance

```

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```
for (const auto& id : instance_ids)
{
    hpx::async<print_action>(id).get();
}
```

We can use the component instances with distribution policies the same way.

Segmented containers

In parallel programming, there is now a plethora of solutions aimed at implementing “partially contiguous” or segmented data structures, whether on shared memory systems or distributed memory systems. *HPX* implements such structures by drawing inspiration from Standard C++ containers.

Using segmented containers

A segmented container is a template class that is described in the namespace `hpx`. All segmented containers are very similar semantically to their sequential counterpart (defined in namespace `std` but with an additional template parameter named `DistPolicy`). The distribution policy is an optional parameter that is passed last to the segmented container constructor (after the container size when no default value is given, after the default value if not). The distribution policy describes the manner in which a container is segmented and the placement of each segment among the available runtime localities.

However, only a part of the `std` container member functions were reimplemented:

- (constructor), (destructor), operator=
- operator[]
- begin, cbegin, end, cend
- size

An example of how to use the `partitioned_vector` container would be:

```
#include <hpx/include/partitioned_vector.hpp>

// The following code generates all necessary boiler plate to enable the
// remote creation of 'partitioned_vector' segments
//
HPX_REGISTER_PARTITIONED_VECTOR(double);

// By default, the number of segments is equal to the current number of
// localities
//
hpx::partitioned_vector<double> va(50);
hpx::partitioned_vector<double> vb(50, 0.0);
```

An example of how to use the `partitioned_vector` container with distribution policies would be:

```
#include <hpx/include/partitioned_vector.hpp>
#include <hpx/modules/runtime_distributed.hpp>

// The following code generates all necessary boiler plate to enable the
// remote creation of 'partitioned_vector' segments
```

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```
//
HPX_REGISTER_PARTITIONED_VECTOR(double);

std::size_t num_segments = 10;
std::vector<hpx::id_type> locs = hpx::find_all_localities();

auto layout =
    hpx::container_layout( num_segments, locs );

// The number of segments is 10 and those segments are spread across the
// localities collected in the variable locs in a Round-Robin manner
//
hpx::partitioned_vector<double> va(50, layout);
hpx::partitioned_vector<double> vb(50, 0.0, layout);
```

By definition, a segmented container must be accessible from any thread although its construction is synchronous only for the thread who has called its constructor. To overcome this problem, it is possible to assign a symbolic name to the segmented container:

```
#include <hpx/include/partitioned_vector.hpp>

// The following code generates all necessary boiler plate to enable the
// remote creation of 'partitioned_vector' segments
//
HPX_REGISTER_PARTITIONED_VECTOR(double);

hpx::future<void> fserver = hpx::async(
    []() {
        hpx::partitioned_vector<double> v(50);

        // Register the 'partitioned_vector' with the name "some_name"
        //
        v.register_as("some_name");

        /* Do some code */
    });

hpx::future<void> fclient =
    hpx::async(
        []() {
            // Naked 'partitioned_vector'
            //
            hpx::partitioned_vector<double> v;

            // Now the variable v points to the same 'partitioned_vector' that has
            // been registered with the name "some_name"
            //
            v.connect_to("some_name");

            /* Do some code */
        });
```

Segmented containers

HPX provides the following segmented containers:

Table 2.23: Sequence containers

Name	Description	In header	C++ standard
<code>hpx::partitioned_vector</code>	Dynamic segmented contiguous array.	<code><hpx/include/partitioned_vector.hpp></code>	<code>vector</code> ¹⁷⁶

Table 2.24: Unordered associative containers

Name	Description	In header	C++ standard
<code>hpx::unordered_map</code>	Segmented collection of key-value pairs, hashed by keys, keys are unique.	<code><hpx/include/unordered_map.hpp></code>	<code>unordered_map</code> ¹⁷⁷

Segmented iterators and segmented iterator traits

The basic iterator used in the STL library is only suitable for one-dimensional structures. The iterators we use in HPX must adapt to the segmented format of our containers. Our iterators are then able to know when incrementing themselves if the next element of type T is in the same data segment or in another segment. In this second case, the iterator will automatically point to the beginning of the next segment.

Note: Note that the dereference operation `operator *` does not directly return a reference of type T& but an intermediate object wrapping this reference. When this object is used as an l-value, a remote write operation is performed; When this object is used as an r-value, implicit conversion to T type will take care of performing remote read operation.

It is sometimes useful not only to iterate element by element, but also segment by segment, or simply get a local iterator in order to avoid additional construction costs at each dereferencing operations. To mitigate this need, the `hpx::traits::segmented_iterator_traits` are used.

With `segmented_iterator_traits` users can uniformly get the iterators which specifically iterates over segments (by providing a segmented iterator as a parameter), or get the local begin/end iterators of the nearest local segment (by providing a per-segment iterator as a parameter):

```
#include <hpx/include/partitioned_vector.hpp>

// The following code generates all necessary boiler plate to enable the
// remote creation of 'partitioned_vector' segments
//
HPX_REGISTER_PARTITIONED_VECTOR(double);

using iterator = hpx::partitioned_vector<T>::iterator;
using traits    = hpx::traits::segmented_iterator_traits<iterator>;

hpx::partitioned_vector<T> v;
std::size_t count = 0;
```

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¹⁷⁶ <http://en.cppreference.com/w/cpp/container/vector>

¹⁷⁷ http://en.cppreference.com/w/cpp/container/unordered_map

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```
auto seg_begin = traits::segment(v.begin());
auto seg_end   = traits::segment(v.end());

// Iterate over segments
for (auto seg_it = seg_begin; seg_it != seg_end; ++seg_it)
{
    auto loc_begin = traits::begin(seg_it);
    auto loc_end   = traits::end(seg_it);

    // Iterate over elements inside segments
    for (auto lit = loc_begin; lit != loc_end; ++lit, ++count)
    {
        *lit = count;
    }
}
```

Which is equivalent to:

```
hpx::partitioned_vector<T> v;
std::size_t count = 0;

auto begin = v.begin();
auto end   = v.end();

for (auto it = begin; it != end; ++it, ++count)
{
    *it = count;
}
```

Using views

The use of multidimensional arrays is quite common in the numerical field whether to perform dense matrix operations or to process images. It exist many libraries which implement such object classes overloading their basic operators (e.g. +, -, *, (), etc.). However, such operation becomes more delicate when the underlying data layout is segmented or when it is mandatory to use optimized linear algebra subroutines (i.e. BLAS subroutines).

Our solution is thus to relax the level of abstraction by allowing the user to work not directly on n-dimensionnal data, but on “n-dimensionnal collections of 1-D arrays”. The use of well-accepted techniques on contiguous data is thus preserved at the segment level, and the composability of the segments is made possible thanks to multidimensional array-inspired access mode.

Preface: Why SPMD?

Although *HPX* refutes by design this programming model, the *locality* plays a dominant role when it comes to implement vectorized code. To maximize local computations and avoid unneeded data transfers, a parallel section (or Single Programming Multiple Data section) is required. Because the use of global variables is prohibited, this parallel section is created via the RAII idiom.

To define a parallel section, simply write an action taking a `spmd_block` variable as a first parameter:

```
#include <hpx/modules/collectives.hpp>

void bulk_function(hpx::lcos::spmd_block block /* , arg0, arg1, ... */)
{
    // Parallel section

    /* Do some code */
}
HPX_PLAIN_ACTION(bulk_function, bulk_action);
```

Note: In the following paragraphs, we will use the term “image” several times. An image is defined as a lightweight process whose entry point is a function provided by the user. It’s an “image of the function”.

The `spmd_block` class contains the following methods:

- Team information: `get_num_images`, `this_image`, `images_per_locality`
- Control statements: `sync_all`, `sync_images`

Here is a sample code summarizing the features offered by the `spmd_block` class:

```
#include <hpx/modules/collectives.hpp>

void bulk_function(hpx::lcos::spmd_block block /* , arg0, arg1, ... */)
{
    std::size_t num_images = block.get_num_images();
    std::size_t this_image = block.this_image();
    std::size_t images_per_locality = block.images_per_locality();

    /* Do some code */

    // Synchronize all images in the team
    block.sync_all();

    /* Do some code */

    // Synchronize image 0 and image 1
    block.sync_images(0,1);

    /* Do some code */

    std::vector<std::size_t> vec_images = {2,3,4};

    // Synchronize images 2, 3 and 4
    block.sync_images(vec_images);
}
```

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```

// Alternative call to synchronize images 2, 3 and 4
block.sync_images(vec_images.begin(), vec_images.end());

/* Do some code */

// Non-blocking version of sync_all()
hpx::future<void> event =
    block.sync_all(hpx::launch::async);

// Callback waiting for 'event' to be ready before being scheduled
hpx::future<void> cb =
    event.then(
        [] (hpx::future<void>)
        {

            /* Do some code */

        });

// Finally wait for the execution tree to be finished
cb.get();
}
HPX_PLAIN_ACTION(bulk_test_function, bulk_test_action);

```

Then, in order to invoke the parallel section, call the function `define_spmc_block` specifying an arbitrary symbolic name and indicating the number of images per *locality* to create:

```

void bulk_function(hpx::lcos::spmd_block block, /* , arg0, arg1, ... */)
{
}
HPX_PLAIN_ACTION(bulk_test_function, bulk_test_action);

int main()
{
    /* std::size_t arg0, arg1, ...; */

    bulk_action act;
    std::size_t images_per_locality = 4;

    // Instantiate the parallel section
    hpx::lcos::define_spmc_block(
        "some_name", images_per_locality, std::move(act) /*, arg0, arg1, ... */);

    return 0;
}

```

Note: In principle, the user should never call the `spmd_block` constructor. The `define_spmc_block` function is responsible of instantiating `spmd_block` objects and broadcasting them to each created image.

SPMD multidimensional views

Some classes are defined as “container views” when the purpose is to observe and/or modify the values of a container using another perspective than the one that characterizes the container. For example, the values of an `std::vector` object can be accessed via the expression `[i]`. Container views can be used, for example, when it is desired for those values to be “viewed” as a 2D matrix that would have been flattened in a `std::vector`. The values would be possibly accessible via the expression `vv(i, j)` which would call internally the expression `v[k]`.

By default, the `partitioned_vector` class integrates 1-D views of its segments:

```
#include <hpx/include/partitioned_vector.hpp>

// The following code generates all necessary boiler plate to enable the
// remote creation of 'partitioned_vector' segments
//
HPX_REGISTER_PARTITIONED_VECTOR(double);

using iterator = hpx::partitioned_vector<double>::iterator;
using traits   = hpx::traits::segmented_iterator_traits<iterator>;

hpx::partitioned_vector<double> v;

// Create a 1-D view of the vector of segments
auto vv = traits::segment(v.begin());

// Access segment i
std::vector<double> v = vv[i];
```

Our views are called “multidimensional” in the sense that they generalize to N dimensions the purpose of `segmented_iterator_traits::segment()` in the 1-D case. Note that in a parallel section, the 2-D expression `a(i, j) = b(i, j)` is quite confusing because without convention, each of the images invoked will race to execute the statement. For this reason, our views are not only multidimensional but also “spmd-aware”.

Note: SPMD-awareness: The convention is simple. If an assignment statement contains a view subscript as an l-value, it is only and only the image holding the r-value who is evaluating the statement. (In MPI sense, it is called a Put operation).

Subscript-based operations

Here are some examples of using subscripts in the 2-D view case:

```
#include <hpx/components/containers/partitioned_vector/partitioned_vector_view.hpp>
#include <hpx/include/partitioned_vector.hpp>

// The following code generates all necessary boiler plate to enable the
// remote creation of 'partitioned_vector' segments
//
HPX_REGISTER_PARTITIONED_VECTOR(double);

using Vec = hpx::partitioned_vector<double>;
using View_2D = hpx::partitioned_vector_view<double, 2>;
```

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```

/* Do some code */

Vec v;

// Parallel section (suppose 'block' an spmd_block instance)
{
    std::size_t height, width;

    // Instantiate the view
    View_2D vv(block, v.begin(), v.end(), {height,width});

    // The l-value is a view subscript, the image that owns vv(1,0)
    // evaluates the assignment.
    vv(0,1) = vv(1,0);

    // The l-value is a view subscript, the image that owns the r-value
    // (result of expression 'std::vector<double>(4,1.0)') evaluates the
    // assignment : oops! race between all participating images.
    vv(2,3) = std::vector<double>(4,1.0);
}

```

Iterator-based operations

Here are some examples of using iterators in the 3-D view case:

```

#include <hpx/components/containers/partitioned_vector/partitioned_vector_view.hpp>
#include <hpx/include/partitioned_vector.hpp>

// The following code generates all necessary boiler plate to enable the
// remote creation of 'partitioned_vector' segments
//
HPX_REGISTER_PARTITIONED_VECTOR(int);

using Vec = hpx::partitioned_vector<int>;
using View_3D = hpx::partitioned_vector_view<int,3>;

/* Do some code */

Vec v1, v2;

// Parallel section (suppose 'block' an spmd_block instance)
{
    std::size_t size_x, size_y, size_z;

    // Instantiate the views
    View_3D vv1(block, v1.begin(), v1.end(), {size_x,size_y,size_z});
    View_3D vv2(block, v2.begin(), v2.end(), {size_x,size_y,size_z});

    // Save previous segments covered by vv1 into segments covered by vv2
    auto vv2_it = vv2.begin();

```

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```

auto vv1_it = vv1.cbegin();

for(; vv2_it != vv2.end(); vv2_it++, vv1_it++)
{
    // It's a Put operation
    *vv2_it = *vv1_it;
}

// Ensure that all images have performed their Put operations
block.sync_all();

// Ensure that only one image is putting updated data into the different
// segments covered by vv1
if(block.this_image() == 0)
{
    int idx = 0;

    // Update all the segments covered by vv1
    for(auto i = vv1.begin(); i != vv1.end(); i++)
    {
        // It's a Put operation
        *i = std::vector<float>(elt_size,idx++);
    }
}
}

```

Here is an example that shows how to iterate only over segments owned by the current image:

```

#include <hpx/components/containers/partitioned_vector/partitioned_vector_view.hpp>
#include <hpx/components/containers/partitioned_vector/partitioned_vector_local_view.hpp>
#include <hpx/include/partitioned_vector.hpp>

// The following code generates all necessary boiler plate to enable the
// remote creation of 'partitioned_vector' segments
//
HPX_REGISTER_PARTITIONED_VECTOR(float);

using Vec = hpx::partitioned_vector<float>;
using View_1D = hpx::partitioned_vector_view<float,1>;

/* Do some code */

Vec v;

// Parallel section (suppose 'block' an spmd_block instance)
{
    std::size_t num_segments;

    // Instantiate the view
    View_1D vv(block, v.begin(), v.end(), {num_segments});

    // Instantiate the local view from the view

```

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```

    auto local_vv = hpx::local_view(vv);

    for ( auto i = local_vv.begin(); i != local_vv.end(); i++ )
    {
        std::vector<float> & segment = *i;

        /* Do some code */
    }
}

```

Instantiating sub-views

It is possible to construct views from other views: we call it sub-views. The constraint nevertheless for the subviews is to retain the dimension and the value type of the input view. Here is an example showing how to create a sub-view:

```

#include <hpx/components/containers/partitioned_vector/partitioned_vector_view.hpp>
#include <hpx/include/partitioned_vector.hpp>

// The following code generates all necessary boiler plate to enable the
// remote creation of 'partitioned_vector' segments
//
HPX_REGISTER_PARTITIONED_VECTOR(float);

using Vec = hpx::partitioned_vector<float>;
using View_2D = hpx::partitioned_vector_view<float,2>;

/* Do some code */

Vec v;

// Parallel section (suppose 'block' an spmd_block instance)
{
    std::size_t N = 20;
    std::size_t tilesz = 5;

    // Instantiate the view
    View_2D vv(block, v.begin(), v.end(), {N,N});

    // Instantiate the subview
    View_2D svv(
        block,&vv(tilesz,0),&vv(2*tilesz-1,tilesz-1),{tilesz,tilesz},{N,N});

    if(block.this_image() == 0)
    {
        // Equivalent to 'vv(tilesz,0) = 2.0f'
        svv(0,0) = 2.0f;

        // Equivalent to 'vv(2*tilesz-1,tilesz-1) = 3.0f'
        svv(tilesz-1,tilesz-1) = 3.0f;
    }
}

```

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}

Note: The last parameter of the subview constructor is the size of the original view. If one would like to create a subview of the subview and so on, this parameter should stay unchanged. {N,N} for the above example).

C++ co-arrays

Fortran has extended its scalar element indexing approach to reference each segment of a distributed array. In this extension, a segment is attributed a ?co-index? and lives in a specific *locality*. A co-index provides the application with enough information to retrieve the corresponding data reference. In C++, containers present themselves as a ?smarter? alternative of Fortran arrays but there are still no corresponding standardized features similar to the Fortran co-indexing approach. We present here an implementation of such features in *HPX*.

Preface: co-array, a segmented container tied to a SPMD multidimensional views

As mentioned before, a co-array is a distributed array whose segments are accessible through an array-inspired access mode. We have previously seen that it is possible to reproduce such access mode using the concept of views. Nevertheless, the user must pre-create a segmented container to instantiate this view. We illustrate below how a single constructor call can perform those two operations:

```
#include <hpx/components/containers/coarray/coarray.hpp>
#include <hpx/modules/collectives.hpp>

// The following code generates all necessary boiler plate to enable the
// co-creation of 'coarray'
//
HPX_REGISTER_COARRAY(double);

// Parallel section (suppose 'block' an spmd_block instance)
{
    using hpx::container::placeholders::_;

    std::size_t height=32, width=4, segment_size=10;

    hpx::coarray<double,3> a(block, "a", {height,width,_}, segment_size);

    /* Do some code */
}
```

Unlike segmented containers, a co-array object can only be instantiated within a parallel section. Here is the description of the parameters to provide to the coarray constructor:

Table 2.25: Parameters of coarray constructor

Parameter	Description
<code>block</code>	Reference to a <code>spmd_block</code> object
<code>"a"</code>	Symbolic name of type <code>std::string</code>
<code>{height,width, _}</code>	Dimensions of the coarray object
<code>segment_size</code>	Size of a co-indexed element (i.e. size of the object referenced by the expression <code>a(i,j,k)</code>)

Note that the “last dimension size” cannot be set by the user. It only accepts the constexpr variable `hpx::container::placeholders::_`. This size, which is considered private, is equal to the number of current images (value returned by `block.get_num_images()`).

Note: An important constraint to remember about coarray objects is that all segments sharing the same “last dimension index” are located in the same image.

Using co-arrays

The member functions owned by the coarray objects are exactly the same as those of `spmd` multidimensional views. These are:

- * Subscript-based operations
- * Iterator-based operations

However, one additional functionality is provided. Knowing that the element `a(i,j,k)` is in the memory of the `k`th image, the use of local subscripts is possible.

Note: For `spmd` multidimensional views, subscripts are only global as it still involves potential remote data transfers.

Here is an example of using local subscripts:

```
#include <hpx/components/containers/coarray/coarray.hpp>
#include <hpx/modules/collectives.hpp>

// The following code generates all necessary boiler plate to enable the
// co-creation of 'coarray'
//
HPX_REGISTER_COARRAY(double);

// Parallel section (suppose 'block' an spmd_block instance)
{
    using hpx::container::placeholders::_;

    std::size_t height=32, width=4, segment_size=10;

    hpx::coarray<double,3> a(block, "a", {height,width,_}, segment_size);

    double idx = block.this_image()*height*width;

    for (std::size_t j = 0; j<width; j++)
```

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```

for (std::size_t i = 0; i<height; i++)
{
    // Local write operation performed via the use of local subscript
    a(i,j,_) = std::vector<double>(elt_size,idx);
    idx++;
}

block.sync_all();
}

```

Note: When the “last dimension index” of a subscript is equal to `hpx::container::placeholders::_`, local subscript (and not global subscript) is used. It is equivalent to a global subscript used with a “last dimension index” equal to the value returned by `block.this_image()`.

2.4.13 Running on batch systems

This section walks you through launching *HPX* applications on various batch systems.

How to use *HPX* applications with PBS

Most *HPX* applications are executed on parallel computers. These platforms typically provide integrated job management services that facilitate the allocation of computing resources for each parallel program. *HPX* includes support for one of the most common job management systems, the Portable Batch System (PBS).

All PBS jobs require a script to specify the resource requirements and other parameters associated with a parallel job. The PBS script is basically a shell script with PBS directives placed within commented sections at the beginning of the file. The remaining (not commented-out) portions of the file executes just like any other regular shell script. While the description of all available PBS options is outside the scope of this tutorial (the interested reader may refer to in-depth [documentation](#)¹⁷⁸ for more information), below is a minimal example to illustrate the approach. The following test application will use the multithreaded `hello_world_distributed` program, explained in the section *Remote execution with actions*.

```

#!/bin/bash
#
#PBS -l nodes=2:ppn=4

APP_PATH=~/.packages/hpx/bin/hello_world_distributed
APP_OPTIONS=

pbsdsh -u $APP_PATH $APP_OPTIONS --hpx:nodes=`cat $PBS_NODEFILE`

```

Caution: If the first application specific argument (inside `$APP_OPTIONS`) is a non-option (i.e., does not start with a `-` or a `--`), then the argument has to be placed before the option `--hpx:nodes`, which, in this case, should be the last option on the command line.

Alternatively, use the option `--hpx:endnodes` to explicitly mark the end of the list of node names:

```
$ pbsdsh -u $APP_PATH --hpx:nodes`cat $PBS_NODEFILE` --hpx:endnodes $APP_OPTIONS
```

¹⁷⁸ <http://www.clusterresources.com/torquedocs21/>

The `#PBS -l nodes=2:ppn=4` directive will cause two compute nodes to be allocated for the application, as specified in the option `nodes`. Each of the nodes will dedicate four cores to the program, as per the option `ppn`, short for “processors per node” (PBS does not distinguish between processors and cores). Note that requesting more cores per node than physically available is pointless and may prevent PBS from accepting the script.

On newer PBS versions the PBS command syntax might be different. For instance, the PBS script above would look like:

```
#!/bin/bash
#
#PBS -l select=2:ncpus=4

APP_PATH=~/packages/hpx/bin/hello_world_distributed
APP_OPTIONS=

pbsdsh -u $APP_PATH $APP_OPTIONS --hpx:nodes=`cat $PBS_NODEFILE`
```

`APP_PATH` and `APP_OPTIONS` are shell variables that respectively specify the correct path to the executable (`hello_world_distributed` in this case) and the command line options. Since the `hello_world_distributed` application doesn’t need any command line options, `APP_OPTIONS` has been left empty. Unlike in other execution environments, there is no need to use the `--hpx:threads` option to indicate the required number of OS threads per node; the *HPX* library will derive this parameter automatically from PBS.

Finally, `pbsdsh` is a PBS command that starts tasks to the resources allocated to the current job. It is recommended to leave this line as shown and modify only the PBS options and shell variables as needed for a specific application.

Important: A script invoked by `pbsdsh` starts in a very basic environment: the user’s `$HOME` directory is defined and is the current directory, the `LANG` variable is set to C and the `PATH` is set to the basic `/usr/local/bin:/usr/bin:/bin` as defined in a system-wide file `pbs_environment`. Nothing that would normally be set up by a system shell profile or user shell profile is defined, unlike the environment for the main job script.

Another choice is for the `pbsdsh` command in your main job script to invoke your program via a shell, like `sh` or `bash`, so that it gives an initialized environment for each instance. Users can create a small script `runme.sh`, which is used to invoke the program:

```
#!/bin/bash
# Small script which invokes the program based on what was passed on its
# command line.
#
# This script is executed by the bash shell which will initialize all
# environment variables as usual.
$@
```

Now, the script is invoked using the `pbsdsh` tool:

```
#!/bin/bash
#
#PBS -l nodes=2:ppn=4

APP_PATH=~/packages/hpx/bin/hello_world_distributed
APP_OPTIONS=
```

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```
pbsdsh -u runme.sh $APP_PATH $APP_OPTIONS --hpx:nodes=`cat $PBS_NODEFILE`
```

All that remains now is submitting the job to the queuing system. Assuming that the contents of the PBS script were saved in the file `pbs_hello_world.sh` in the current directory, this is accomplished by typing:

```
$ qsub ./pbs_hello_world_pbs.sh
```

If the job is accepted, `qsub` will print out the assigned job ID, which may look like:

```
$ 42.supercomputer.some.university.edu
```

To check the status of your job, issue the following command:

```
$ qstat 42.supercomputer.some.university.edu
```

and look for a single-letter job status symbol. The common cases include:

- *Q* - signifies that the job is queued and awaiting its turn to be executed.
- *R* - indicates that the job is currently running.
- *C* - means that the job has completed.

The example `qstat` output below shows a job waiting for execution resources to become available:

Job id	Name	User	Time Use	S	Queue
-----	-----	-----	-----	-----	-----
42.supercomputer	...ello_world.sh	joe_user		0 Q	batch

After the job completes, PBS will place two files, `pbs_hello_world.sh.o42` and `pbs_hello_world.sh.e42`, in the directory where the job was submitted. The first contains the standard output and the second contains the standard error from all the nodes on which the application executed. In our example, the error output file should be empty and the standard output file should contain something similar to:

```
hello world from OS-thread 3 on locality 0
hello world from OS-thread 2 on locality 0
hello world from OS-thread 1 on locality 1
hello world from OS-thread 0 on locality 0
hello world from OS-thread 3 on locality 1
hello world from OS-thread 2 on locality 1
hello world from OS-thread 1 on locality 0
hello world from OS-thread 0 on locality 1
```

Congratulations! You have just run your first distributed *HPX* application!

How to use HPX applications with SLURM

Just like PBS (described in section *How to use HPX applications with PBS*), SLURM is a job management system which is widely used on large supercomputing systems. Any HPX application can easily be run using SLURM. This section describes how this can be done.

The easiest way to run an HPX application using SLURM is to utilize the command line tool `srun`, which interacts with the SLURM batch scheduling system:

```
$ srun -p <partition> -N <number-of-nodes> hpx-application <application-arguments>
```

Here, `<partition>` is one of the node partitions existing on the target machine (consult the machine's documentation to get a list of existing partitions) and `<number-of-nodes>` is the number of compute nodes that should be used. By default, the HPX application is started with one *locality* per node and uses all available cores on a node. You can change the number of localities started per node (for example, to account for NUMA effects) by specifying the `-n` option of `srun`. The number of cores per *locality* can be set by `-c`. The `<application-arguments>` are any application specific arguments that need to be passed on to the application.

Note: There is no need to use any of the HPX command line options related to the number of localities, number of threads, or related to networking ports. All of this information is automatically extracted from the SLURM environment by the HPX startup code.

Important: The `srun` documentation explicitly states: “If `-c` is specified without `-n`, as many tasks will be allocated per node as possible while satisfying the `-c` restriction. For instance on a cluster with 8 CPUs per node, a job request for 4 nodes and 3 CPUs per task may be allocated 3 or 6 CPUs per node (1 or 2 tasks per node) depending upon resource consumption by other jobs.” For this reason, it's recommended to always specify `-n <number-of-instances>`, even if `<number-of-instances>` is equal to one (1).

Interactive shells

To get an interactive development shell on one of the nodes, users can issue the following command:

```
$ srun -p <node-type> -N <number-of-nodes> --pty /bin/bash -l
```

After the shell has been opened, users can run their HPX application. By default, it uses all available cores. Note that if you requested one node, you don't need to do `srun` again. However, if you requested more than one node, and want to run your distributed application, you can use `srun` again to start up the distributed HPX application. It will use the resources that have been requested for the interactive shell.

Scheduling batch jobs

The above mentioned method of running HPX applications is fine for development purposes. The disadvantage that comes with `srun` is that it only returns once the application is finished. This might not be appropriate for longer-running applications (for example, benchmarks or larger scale simulations). In order to cope with that limitation, users can use the `sbatch` command.

The `sbatch` command expects a script that it can run once the requested resources are available. In order to request resources, users need to add `#SBATCH` comments in their script or provide the necessary parameters to `sbatch` directly. The parameters are the same as with `run`. The commands you need to execute are the same you would need to start your application as if you were in an interactive shell.

2.4.14 Debugging HPX applications

Using a debugger with HPX applications

Using a debugger such as `gdb` with *HPX* applications is no problem. However, there are some things to keep in mind to make the experience somewhat more productive.

Call stacks in *HPX* can often be quite unwieldy as the library is heavily templated and the call stacks can be very deep. For this reason it is sometimes a good idea compile *HPX* in `RelWithDebInfo` mode, which applies some optimizations but keeps debugging symbols. This can often compress call stacks significantly. On the other hand, stepping through the code can also be more difficult because of statements being reordered and variables being optimized away. Also, note that because *HPX* implements user-space threads and context switching, call stacks may not always be complete in a debugger.

HPX launches not only worker threads but also a few helper threads. The first thread is the main thread, which typically does no work in an *HPX* application, except at startup and shutdown. If using the default settings, *HPX* will spawn six additional threads (used for service thread pools). The first worker thread is usually the eighth thread, and most user codes will be run on these worker threads. The last thread is a helper thread used for *HPX* shutdown.

Finally, since *HPX* is a multi-threaded runtime, the following `gdb` options can be helpful:

```
set pagination off
set non-stop on
```

Non-stop mode allows users to have a single thread stop on a breakpoint without stopping all other threads as well.

Using sanitizers with HPX applications

Warning: Not all parts of *HPX* are sanitizer clean. This means that users may end up with false positives from *HPX* itself when using sanitizers for their applications.

To use sanitizers with *HPX*, turn on `HPX_WITH_SANITIZERS` and turn off `HPX_WITH_STACKOVERFLOW_DETECTION` during `CMake`¹⁷⁹ configuration. It's recommended to also build Boost with the same sanitizers that will be used for *HPX*. The appropriate sanitizers can then be enabled using `CMake` by appending `-fsanitize=address` `-fno-omit-frame-pointer` to `CMAKE_CXX_FLAGS` and `-fsanitize=address` to `CMAKE_EXE_LINKER_FLAGS`. Replace `address` with the sanitizer that you want to use.

Debugging applications using core files

For *HPX* to generate useful core files, *HPX* has to be compiled without signal and exception handlers `HPX_WITH_DISABLED_SIGNAL_EXCEPTION_HANDLERS:BOOL`. If this option is not specified, the signal handlers change the application state. For example, after a segmentation fault the stack trace will show the signal handler. Similarly, unhandled exceptions are also caught by these handlers and the stack trace will not point to the location where the unhandled exception was thrown.

In general, core files are a helpful tool to inspect the state of the application at the moment of the crash (post-mortem debugging), without the need of attaching a debugger beforehand. This approach to debugging is especially useful if the error cannot be reliably reproduced, as only a single crashed application run is required to gain potentially helpful information like a stacktrace.

To debug with core files, the operating system first has to be told to actually write them. On most Unix systems this can be done by calling:

¹⁷⁹ <https://www.cmake.org>

```
$ ulimit -c unlimited
```

in the shell. Now the debugger can be started up with:

```
$ gdb <application> <core file name>
```

The debugger should now display the last state of the application. The default file name for core files is `core`.

2.4.15 Optimizing *HPX* applications

Performance counters

Performance counters in *HPX* are used to provide information as to how well the runtime system or an application is performing. The counter data can help determine system bottlenecks, and fine-tune system and application performance. The *HPX* runtime system, its networking, and other layers provide counter data that an application can consume to provide users with information about how well the application is performing.

Applications can also use counter data to determine how much system resources to consume. For example, an application that transfers data over the network could consume counter data from a network switch to determine how much data to transfer without competing for network bandwidth with other network traffic. The application could use the counter data to adjust its transfer rate as the bandwidth usage from other network traffic increases or decreases.

Performance counters are *HPX* parallel processes that expose a predefined interface. *HPX* exposes special API functions that allow one to create, manage, and read the counter data, and release instances of performance counters. Performance Counter instances are accessed by name, and these names have a predefined structure which is described in the section *Performance counter names*. The advantage of this is that any Performance Counter can be accessed remotely (from a different *locality*) or locally (from the same *locality*). Moreover, since all counters expose their data using the same API, any code consuming counter data can be utilized to access arbitrary system information with minimal effort.

Counter data may be accessed in real time. More information about how to consume counter data can be found in the section *Consuming performance counter data*.

All *HPX* applications provide command line options related to performance counters, such as the ability to list available counter types, or periodically query specific counters to be printed to the screen or save them in a file. For more information, please refer to the section *HPX Command Line Options*.

Performance counter names

All Performance Counter instances have a name uniquely identifying each instance. This name can be used to access the counter, retrieve all related meta data, and to query the counter data (as described in the section *Consuming performance counter data*). Counter names are strings with a predefined structure. The general form of a countername is:

```
/objectname{full_instancename}/countername@parameters
```

where `full_instancename` could be either another (full) counter name or a string formatted as:

```
parentinstancename#parentindex/instancename#instanceindex
```

Each separate part of a countername (e.g., `objectname`, `countername` `parentinstancename`, `instancename`, and `parameters`) should start with a letter ('a'... 'z', 'A'... 'Z') or an underscore character ('_'), optionally followed by letters, digits ('0'... '9'), hyphen ('-'), or underscore characters. Whitespace is not allowed inside a counter name. The characters '/', '{', '}', '#', and '@' have a special meaning and are used to delimit the different parts of the counter name.

The parts `parentinstanceindex` and `instanceindex` are integers. If an index is not specified, *HPX* will assume a default of `-1`.

Two counter name examples

This section gives examples of both simple counter names and aggregate counter names. For more information on simple and aggregate counter names, please see *Performance counter instances*.

An example of a well-formed (and meaningful) simple counter name would be:

```
/threads{locality#0/total}/count/cumulative
```

This counter returns the current cumulative number of executed (retired) *HPX* threads for the *locality* `0`. The counter type of this counter is `/threads/count/cumulative` and the full instance name is `locality#0/total`. This counter type does not require an `instanceindex` or `parameters` to be specified.

In this case, the `parentindex` (the `'0'`) designates the *locality* for which the counter instance is created. The counter will return the number of *HPX* threads retired on that particular *locality*.

Another example for a well formed (aggregate) counter name is:

```
/statistics{/threads{locality#0/total}/count/cumulative}/average@500
```

This counter takes the simple counter from the first example, samples its values every `500` milliseconds, and returns the average of the value samples whenever it is queried. The counter type of this counter is `/statistics/average` and the instance name is the full name of the counter for which the values have to be averaged. In this case, the `parameters` (the `'500'`) specify the sampling interval for the averaging to take place (in milliseconds).

Performance counter types

Every performance counter belongs to a specific performance counter type which classifies the counters into groups of common semantics. The type of a counter is identified by the `objectname` and the `countername` parts of the name.

```
/objectname/countername
```

When an application starts *HPX* will register all available counter types on each of the localities. These counter types are held in a special performance counter registration database, which can be used to retrieve the meta data related to a counter type and to create counter instances based on a given counter instance name.

Performance counter instances

The `full_instancename` distinguishes different counter instances of the same counter type. The formatting of the `full_instancename` depends on the counter type. There are two types of counters: simple counters, which usually generate the counter values based on direct measurements, and aggregate counters, which take another counter and transform its values before generating their own counter values. An example for a simple counter is given *above*: counting retired *HPX* threads. An aggregate counter is shown as an example *above* as well: calculating the average of the underlying counter values sampled at constant time intervals.

While simple counters use instance names formatted as `parentinstancename#parentindex/instancename#instanceindex`, most aggregate counters have the full counter name of the embedded counter as their instance name.

Not all simple counter types require specifying all four elements of a full counter instance name; some of the parts (`parentinstancename`, `parentindex`, `instancename`, and `instanceindex`) are optional for specific counters.

Please refer to the documentation of a particular counter for more information about the formatting requirements for the name of this counter (see *Existing HPX performance counters*).

The `parameters` are used to pass additional information to a counter at creation time. They are optional, and they fully depend on the concrete counter. Even if a specific counter type allows additional parameters to be given, those usually are not required as sensible defaults will be chosen. Please refer to the documentation of a particular counter for more information about what parameters are supported, how to specify them, and what default values are assumed (see also *Existing HPX performance counters*).

Every *locality* of an application exposes its own set of performance counter types and performance counter instances. The set of exposed counters is determined dynamically at application start based on the execution environment of the application. For instance, this set is influenced by the current hardware environment for the *locality* (such as whether the *locality* has access to accelerators), and the software environment of the application (such as the number of OS threads used to execute *HPX* threads).

Using wildcards in performance counter names

It is possible to use wildcard characters when specifying performance counter names. Performance counter names can contain two types of wildcard characters:

- Wildcard characters in the performance counter type
- Wildcard characters in the performance counter instance name

A wildcard character has a meaning which is very close to usual file name wildcard matching rules implemented by common shells (like `bash`).

Table 2.26: Wildcard characters in the performance counter type

Wild-card	Description
*	This wildcard character matches any number (zero or more) of arbitrary characters.
?	This wildcard character matches any single arbitrary character.
[...]	This wildcard character matches any single character from the list of specified within the square brackets.

Table 2.27: Wildcard characters in the performance counter instance name

Wild-card	Description
*	This wildcard character matches any <i>locality</i> or any thread, depending on whether it is used for <code>locality#*</code> or <code>worker-thread#*</code> . No other wildcards are allowed in counter instance names.

Consuming performance counter data

You can consume performance data using either the command line interface, the *HPX* application or the *HPX* API. The command line interface is easier to use, but it is less flexible and does not allow one to adjust the behaviour of your application at runtime. The command line interface provides a convenience abstraction but simplified abstraction for querying and logging performance counter data for a set of performance counters.

Consuming performance counter data from the command line

HPX provides a set of predefined command line options for every application that uses `hpx::init` for its initialization. While there are many more command line options available (see *HPX Command Line Options*), the set of options related to performance counters allows one to list existing counters, and query existing counters once at application termination or repeatedly after a constant time interval.

The following table summarizes the available command line options:

Table 2.28: HPX Command Line Options Related to Performance Counters

Command line option	Description
<code>--hpx:print-counter</code>	Prints the specified performance counter either repeatedly and/or at the times specified by <code>--hpx:print-counter-at</code> (see also option <code>--hpx:print-counter-interval</code>).
<code>--hpx:print-counter-reset</code>	Prints the specified performance counter either repeatedly and/or at the times specified by <code>--hpx:print-counter-at</code> . Reset the counter after the value is queried (see also option <code>--hpx:print-counter-interval</code>).
<code>--hpx:print-counter-interval</code>	Prints the performance counter(s) specified with <code>--hpx:print-counter</code> repeatedly after the time interval (specified in milliseconds) (default:0 which means print once at shutdown).
<code>--hpx:print-counter-destination</code>	Prints the performance counter(s) specified with <code>--hpx:print-counter</code> to the given file (default: console).
<code>--hpx:list-counters</code>	List names of all registered performance counters.
<code>--hpx:list-counter-infos</code>	List detailed description of all registered performance counters.
<code>--hpx:print-counter-format</code>	Prints the performance counter(s) specified with <code>--hpx:print-counter</code> . Possible formats in CVS format with header or without any header (see option <code>--hpx:no-csv-header</code>), possible values: <code>csv</code> (prints counter values in CSV format with full names as header) <code>csv-short</code> (prints counter values in CSV format with shortnames provided with <code>--hpx:print-counter</code> as <code>--hpx:print-counter-shortname,full-countername</code>).
<code>--hpx:no-csv-header</code>	Prints the performance counter(s) specified with <code>--hpx:print-counter</code> and <code>csv</code> or <code>csv-short</code> format specified with <code>--hpx:print-counter-format</code> without header.
<code>--hpx:print-counter-at arg</code>	Prints the performance counter(s) specified with <code>--hpx:print-counter</code> (or <code>--hpx:print-counter-reset</code>) at the given point in time. Possible argument values: <code>startup</code> , <code>shutdown</code> (default), <code>noshutdown</code> .
<code>--hpx:reset-counters</code>	Reset the performance counter(s) specified with <code>--hpx:print-counter</code> after they have been evaluated.
<code>--hpx:print-counter-types</code>	Append counter types description to generated output.
<code>--hpx:print-counter-locally</code>	Prints locally its own local counters.

While the options `--hpx:list-counters` and `--hpx:list-counter-infos` give a short list of all available counters, the full documentation for those can be found in the section *Existing HPX performance counters*.

HPX-Top

HPX-Top is a real-time, terminal-based dashboard designed for monitoring the health and performance of distributed *HPX* applications. Inspired by tools like *htop* and *bt*, it provides a high-level overview of system utilization, network flow, and internal runtime metrics.



Fig. 2.9: HPX-Top Real-time Performance Dashboard

Features

- **Locality-Aware Monitoring:** Automatically detects and displays metrics for all localities in a running distributed application.
- **Thread Pool Utilization:** Visualizes real-time usage of worker threads with instantaneous counts and historical sparkline graphs.
- **Parcel Throughput:** Tracks incoming and outgoing parcels, helping to identify network-bound bottlenecks or communication hotspots.
- **AGAS Health Monitor:** Displays cache hit and miss rates for the Application Global Address Space (AGAS), crucial for diagnosing naming service bottlenecks.

Usage

HPX-Top is implemented as a Python script that acts as a wrapper around your *HPX* application. It automatically configures the necessary performance counters and output formats.

The tool requires the rich Python library for rendering the terminal UI:

```
$ pip install rich
```

To monitor an *HPX* application, launch it through `hpx-top.py`:

```
$ python3 tools/hpx-top.py ./your_application --hpx:threads=4 [other HPX flags]
```

You can also run HPX-Top in mock mode to explore the interface without a running *HPX* application:

```
$ python3 tools/hpx-top.py --mock
```

Technical Details

The tool leverages *HPX*'s built-in performance counter framework. It launches the target application with specific flags (`--hpx:print-counter-interval` and `--hpx:print-counter-format=csv-short`) and parses the standard output in a non-blocking background thread to update the TUI.

HPX Smart Telemetry (`hpx_stat_viewer`)

HPX Smart Telemetry (`hpx_stat_viewer.py`) is an intelligent, lightweight performance dashboard designed to identify performance anomalies in real-time. While `hpx-top` provides a broad system overview, `hpx_stat_viewer` focuses on deep-dive analysis of specific metrics with built-in heuristic alerting.



Fig. 2.10: HPX Smart Telemetry Dashboard with Anomaly Detection

Features

- **Heuristic Anomaly Detection:** Uses an Exponential Moving Average (EMA) to monitor counter trends. It automatically flags sudden spikes (+150%) or drops (-60%) with prominent visual alerts like [SPIKE] and [DROP].
- **Live History Sparklines:** Displays 10-tick historical trend graphs next to every metric, helping developers distinguish between momentary noise and persistent bottlenecks.
- **Smart Filtering and Grouping:** Supports regex-based counter filtering and automatically groups metrics by locality for large-scale distributed runs.
- **Zero-Dependency Core:** Requires no external TUI libraries, making it highly portable for various terminal environments.

Usage

`hpx_stat_viewer.py` reads HPX counter data from standard input in `csv-short` transient format. This allows it to be used for both live monitoring and post-mortem analysis of log files.

To monitor a live application with smart filtering:

```
$ ./your_hpx_app --hpx:print-counter-interval=500 --hpx:print-counter-format=csv-short ..
↩ | \
python3 tools/hpx_stat_viewer.py --filter "threads|utilization"
```

To replay an offline performance log for analysis:


```
$ python3 tools/hpx_stat_viewer.py --replay your_hpx_counters.csv --speed 0.5
```

Technical Details

The dashboard implements a thread-safe parser that handles HPX's CSV-short format. The anomaly detection engine uses a smoothing factor of 0.4 for its EMA calculations, balancing responsiveness with stability. It is officially integrated into the HPX installation rules under the `tools` component.

A simple example

All of the commandline options mentioned above can be tested using the `hello_world_distributed` example.

Listing all available counters `hello_world_distributed --hpx:list-counters` yields:

```
List of available counter instances (replace * below with the appropriate
sequence number)
```

```
-----
/agas/count/allocate /agas/count/bind /agas/count/bind_gid
/agas/count/bind_name ... /threads{locality#*/allocator#*/count/objects
/threads{locality#*/total}/count/stack-recycles
/threads{locality#*/total}/idle-rate
/threads{locality#*/worker-thread#*/idle-rate
```

Providing more information about all available counters, `hello_world_distributed --hpx:list-counter-infos` yields:

```
Information about available counter instances (replace * below with the
appropriate sequence number)
```

```
-----
fullname: /agas/count/allocate helptext: returns the number of invocations of
the AGAS service 'allocate' type: counter_type::raw version: 1.0.0
-----
```

```
-----
fullname: /agas/count/bind helptext: returns the number of invocations of the
AGAS service 'bind' type: counter_type::raw version: 1.0.0
-----
```

```
-----
fullname: /agas/count/bind_gid helptext: returns the number of invocations of
the AGAS service 'bind_gid' type: counter_type::raw version: 1.0.0
-----
```

```
...
```

This command will not only list the counter names but also a short description of the data exposed by this counter.

Note: The list of available counters may differ depending on the concrete execution environment (hardware or software) of your application.

Requesting the counter data for one or more performance counters can be achieved by invoking `hello_world_distributed` with a list of counter names:

```
$ hello_world_distributed \  
  --hpx:print-counter=/threads{locality#0/total}/count/cumulative \  
  --hpx:print-counter=/agas{locality#0/total}/count/bind
```

which yields for instance:

```
hello world from OS-thread 0 on locality 0  
/threads{locality#0/total}/count/cumulative,1,0.212527,[s],33  
/agas{locality#0/total}/count/bind,1,0.212790,[s],11
```

The first line is the normal output generated by `hello_world_distributed` and has no relation to the counter data listed. The last two lines contain the counter data as gathered at application shutdown. These lines have six fields, the counter name, the sequence number of the counter invocation, the time stamp at which this information has been sampled, the unit of measure for the time stamp, the actual counter value and an optional unit of measure for the counter value.

Note: The command line option `--hpx:print-counter-types` will append a seventh field to the generated output. This field will hold an abbreviated counter type.

The actual counter value can be represented by a single number (for counters returning singular values) or a list of numbers separated by `' '` (for counters returning an array of values, like for instance a histogram).

Note: The name of the performance counter will be enclosed in double quotes `''` if it contains one or more commas `,`.

Requesting to query the counter data once after a constant time interval with this command line:

```
$ hello_world_distributed \  
  --hpx:print-counter=/threads{locality#0/total}/count/cumulative \  
  --hpx:print-counter=/agas{locality#0/total}/count/bind \  
  --hpx:print-counter-interval=20
```

yields for instance (leaving off the actual console output of the `hello_world_distributed` example for brevity):

```
threads{locality#0/total}/count/cumulative,1,0.002409,[s],22  
agas{locality#0/total}/count/bind,1,0.002542,[s],9  
threads{locality#0/total}/count/cumulative,2,0.023002,[s],41  
agas{locality#0/total}/count/bind,2,0.023557,[s],10  
threads{locality#0/total}/count/cumulative,3,0.037514,[s],46  
agas{locality#0/total}/count/bind,3,0.038679,[s],10
```

The command `--hpx:print-counter-destination=<file>` will redirect all counter data gathered to the specified file name, which avoids cluttering the console output of your application.

The command line option `--hpx:print-counter` supports using a limited set of wildcards for a (very limited) set of use cases. In particular, all occurrences of `#*` as in `locality#*` and in `worker-thread#*` will be automatically expanded to the proper set of performance counter names representing the actual environment for the executed program. For instance, if your program is utilizing four worker threads for the execution of *HPX* threads (see command line option `--hpx:threads`) the following command line

```
$ hello_world_distributed \
  --hpx:threads=4 \
  --hpx:print-counter=/threads{locality#0/worker-thread#*}/count/cumulative
```

will print the value of the performance counters monitoring each of the worker threads:

```
hello world from OS-thread 1 on locality 0
hello world from OS-thread 0 on locality 0
hello world from OS-thread 3 on locality 0
hello world from OS-thread 2 on locality 0
/threads{locality#0/worker-thread#0}/count/cumulative,1,0.0025214,[s],27
/threads{locality#0/worker-thread#1}/count/cumulative,1,0.0025453,[s],33
/threads{locality#0/worker-thread#2}/count/cumulative,1,0.0025683,[s],29
/threads{locality#0/worker-thread#3}/count/cumulative,1,0.0025904,[s],33
```

The command `--hpx:print-counter-format` takes values `csv` and `csv-short` to generate CSV formatted counter values with a header.

With format as `csv`:

```
$ hello_world_distributed \
  --hpx:threads=2 \
  --hpx:print-counter-format csv \
  --hpx:print-counter /threads{locality#*/total}/count/cumulative \
  --hpx:print-counter /threads{locality#*/total}/count/cumulative-phases
```

will print the values of performance counters in CSV format with the full countername as a header:

```
hello world from OS-thread 1 on locality 0
hello world from OS-thread 0 on locality 0
/threads{locality#*/total}/count/cumulative,/threads{locality#*/total}/count/cumulative-
↪phases
39,93
```

With format `csv-short`:

```
$ hello_world_distributed \
  --hpx:threads 2 \
  --hpx:print-counter-format csv-short \
  --hpx:print-counter cumulative,/threads{locality#*/total}/count/cumulative \
  --hpx:print-counter phases,/threads{locality#*/total}/count/cumulative-phases
```

will print the values of performance counters in CSV format with the short countername as a header:

```
hello world from OS-thread 1 on locality 0
hello world from OS-thread 0 on locality 0
cumulative,phases
39,93
```

With format `csv` and `csv-short` when used with `--hpx:print-counter-interval`:

```
$ hello_world_distributed \
  --hpx:threads 2 \
  --hpx:print-counter-format csv-short \
```

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```
--hpx:print-counter cumulative,/threads{locality#*/total}/count/cumulative \
--hpx:print-counter phases,/threads{locality#*/total}/count/cumulative-phases \
--hpx:print-counter-interval 5
```

will print the header only once repeating the performance counter value(s) repeatedly:

```
cum,phases
25,42
hello world from OS-thread 1 on locality 0
hello world from OS-thread 0 on locality 0
44,95
```

The command `--hpx:no-csv-header` can be used with `--hpx:print-counter-format` to print performance counter values in CSV format without any header:

```
$ hello_world_distributed \
--hpx:threads 2 \
--hpx:print-counter-format csv-short \
--hpx:print-counter cumulative,/threads{locality#*/total}/count/cumulative \
--hpx:print-counter phases,/threads{locality#*/total}/count/cumulative-phases \
--hpx:no-csv-header
```

will print:

```
hello world from OS-thread 1 on locality 0
hello world from OS-thread 0 on locality 0
37,91
```

Consuming performance counter data using the *HPX* API

HPX provides an API that allows users to discover performance counters and to retrieve the current value of any existing performance counter from any application.

Discover existing performance counters

Retrieve the current value of any performance counter

Performance counters are specialized *HPX* components. In order to retrieve a counter value, the performance counter needs to be instantiated. *HPX* exposes a client component object for this purpose:

```
hpx::performance_counters::performance_counter counter(std::string const& name);
```

Instantiating an instance of this type will create the performance counter identified by the given name. Only the first invocation for any given counter name will create a new instance of that counter. All following invocations for a given counter name will reference the initially created instance. This ensures that at any point in time there is never more than one active instance of any of the existing performance counters.

In order to access the counter value (or to invoke any of the other functionality related to a performance counter, like `start`, `stop` or `reset`) member functions of the created client component instance should be called:

```
// print the current number of threads created on locality 0
hpx::performance_counters::performance_counter count(
    "/threads{locality#0/total}/count/cumulative");
hpx::cout << count.get_value<int>().get() << std::endl;
```

For more information about the client component type, see `hpx::performance_counters::performance_counter`

Note: In the above example `count.get_value()` returns a future. In order to print the result we must append `.get()` to retrieve the value. You could write the above example like this for more clarity:

```
// print the current number of threads created on locality 0
hpx::performance_counters::performance_counter count(
    "/threads{locality#0/total}/count/cumulative");
hpx::future<int> result = count.get_value<int>();
hpx::cout << result.get() << std::endl;
```

Providing performance counter data

HPX offers several ways by which you may provide your own data as a performance counter. This has the benefit of exposing additional, possibly application-specific information using the existing Performance Counter framework, unifying the process of gathering data about your application.

An application that wants to provide counter data can implement a performance counter to provide the data. When a consumer queries performance data, the HPX runtime system calls the provider to collect the data. The runtime system uses an internal registry to determine which provider to call.

Generally, there are two ways of exposing your own performance counter data: a simple, function-based way and a more complex, but more powerful way of implementing a full performance counter. Both alternatives are described in the following sections.

Exposing performance counter data using a simple function

The simplest way to expose arbitrary numeric data is to write a function which will then be called whenever a consumer queries this counter. Currently, this type of performance counter can only be used to expose integer values. The expected signature of this function is:

```
std::int64_t some_performance_data(bool reset);
```

The argument `bool reset` (which is supplied by the runtime system when the function is invoked) specifies whether the counter value should be reset after evaluating the current value (if applicable).

For instance, here is such a function returning how often it was invoked:

```
// The atomic variable 'counter' ensures the thread safety of the counter.
boost::atomic<std::int64_t> counter(0);

std::int64_t some_performance_data(bool reset)
{
    std::int64_t result = ++counter;
    if (reset)
        counter = 0;
```

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```

    return result;
}

```

This example function exposes a linearly-increasing value as our performance data. The value is incremented on each invocation, i.e., each time a consumer requests the counter data of this performance counter.

The next step in exposing this counter to the runtime system is to register the function as a new raw counter type using the *HPX* API function `hpx::performance_counters::install_counter_type`. A counter type represents certain common characteristics of counters, like their counter type name and any associated description information. The following snippet shows an example of how to register the function `some_performance_data`, which is shown above, for a counter type named `"/test/data"`. This registration has to be executed before any consumer instantiates, and queries an instance of this counter type:

```

#include <hpx/include/performance_counters.hpp>

void register_counter_type()
{
    // Call the HPX API function to register the counter type.
    hpx::performance_counters::install_counter_type(
        "/test/data",                // counter type name
        &some_performance_data,     // function providing counter_
↪data
        "returns a linearly increasing counter value" // description text (optional)
        ""                          // unit of measure (optional)
    );
}

```

Now it is possible to instantiate a new counter instance based on the naming scheme `"/test{locality#*/total}/data"` where `*` is a zero-based integer index identifying the *locality* for which the counter instance should be accessed. The function `hpx::performance_counters::install_counter_type` enables users to instantiate exactly one counter instance for each *locality*. Repeated requests to instantiate such a counter will return the same instance, i.e., the instance created for the first request.

If this counter needs to be accessed using the standard *HPX* command line options, the registration has to be performed during application startup, before `hpx_main` is executed. The best way to achieve this is to register an *HPX* startup function using the API function `hpx::register_startup_function` before calling `hpx::init` to initialize the runtime system:

```

int main(int argc, char* argv[])
{
    // By registering the counter type we make it available to any consumer
    // who creates and queries an instance of the type "/test/data".
    //
    // This registration should be performed during startup. The
    // function 'register_counter_type' should be executed as an HPX thread right
    // before hpx_main is executed.
    hpx::register_startup_function(&register_counter_type);

    // Initialize and run HPX.
    return hpx::init(argc, argv);
}

```

Please see the code in `simplest_performance_counter.cpp` for a full example demonstrating this functionality.

Implementing a full performance counter

Sometimes, the simple way of exposing a single value as a performance counter is not sufficient. For that reason, *HPX* provides a means of implementing full performance counters which support:

- Retrieving the descriptive information about the performance counter
- Retrieving the current counter value
- Resetting the performance counter (value)
- Starting the performance counter
- Stopping the performance counter
- Setting the (initial) value of the performance counter

Every full performance counter will implement a predefined interface:

```
// Copyright (c) 2007-2026 Hartmut Kaiser
//
// SPDX-License-Identifier: BSL-1.0
// Distributed under the Boost Software License, Version 1.0. (See accompanying
// file LICENSE_1_0.txt or copy at http://www.boost.org/LICENSE_1_0.txt)

#pragma once

#include <hpx/config.hpp>
#include <hpx/modules/async_base.hpp>
#include <hpx/modules/components.hpp>
#include <hpx/modules/execution.hpp>
#include <hpx/modules/functional.hpp>
#include <hpx/modules/futures.hpp>

#include <hpx/performance_counters/counters_fwd.hpp>
#include <hpx/performance_counters/server/base_performance_counter.hpp>

#include <string>
#include <utility>
#include <vector>

#include <hpx/config/warnings_prefix.hpp>

////////////////////////////////////
namespace hpx::performance_counters {
    //////////////////////////////////////
    HPX_CXX_EXPORT struct HPX_EXPORT performance_counter
        : components::client_base<performance_counter,
            server::base_performance_counter>
    {
        using base_type = components::client_base<performance_counter,
            server::base_performance_counter>;

        performance_counter() = default;
    };
}
```

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```

explicit performance_counter(std::string const& name);

performance_counter(
    std::string const& name, hpx::id_type const& locality);

performance_counter(id_type const& id)
    : base_type(id)
{
}

performance_counter(future<id_type>&& id)
    : base_type(HPX_MOVE(id))
{
}

performance_counter(hpx::future<performance_counter>&& c)
    : base_type(HPX_MOVE(c))
{
}

////////////////////////////////////
future<counter_info> get_info() const;
counter_info get_info(
    launch::sync_policy, error_code& ec = throws) const;

future<counter_value> get_counter_value(bool reset) const;
counter_value get_counter_value(
    launch::sync_policy, bool reset, error_code& ec = throws) const;

future<counter_value> get_counter_value() const;
counter_value get_counter_value(
    launch::sync_policy, error_code& ec = throws) const;

future<counter_values_array> get_counter_values_array(bool reset) const;
counter_values_array get_counter_values_array(
    launch::sync_policy, bool reset, error_code& ec = throws) const;

future<counter_values_array> get_counter_values_array() const;
counter_values_array get_counter_values_array(
    launch::sync_policy, error_code& ec = throws) const;

////////////////////////////////////
future<bool> start() const;
bool start(launch::sync_policy, error_code& ec = throws) const;

future<bool> stop() const;
bool stop(launch::sync_policy, error_code& ec = throws) const;

future<void> reset() const;
void reset(launch::sync_policy, error_code& ec = throws) const;

future<void> reinit(bool reset = true) const;

```

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```

void reinit(launch::sync_policy, bool reset = true,
            error_code& ec = throws) const;

////////////////////////////////////
future<std::string> get_name() const;
std::string get_name(
    launch::sync_policy, error_code& ec = throws) const;

private:
    template <typename T>
    static T extract_value(future<counter_value>&& value)
    {
        return value.get().get_value<T>();
    }

public:
    template <typename T>
    future<T> get_value(bool reset = false)
    {
        return get_counter_value(reset).then(hpx::launch::sync,
            hpx::bind_front(&performance_counter::extract_value<T>));
    }
    template <typename T>
    T get_value(
        launch::sync_policy, bool reset = false, error_code& ec = throws)
    {
        return get_counter_value(launch::sync, reset).get_value<T>(ec);
    }

    template <typename T>
    future<T> get_value() const
    {
        return get_counter_value(false).then(hpx::launch::sync,
            hpx::bind_front(&performance_counter::extract_value<T>));
    }
    template <typename T>
    T get_value(launch::sync_policy, error_code& ec = throws) const
    {
        return get_counter_value(launch::sync, false).get_value<T>(ec);
    }
};

// Return all counters matching the given name (with optional wild cards).
HPX_CXX_EXPORT HPX_EXPORT std::vector<performance_counter>
discover_counters(std::string const& name, error_code& ec = throws);
} // namespace hpx::performance_counters

#include <hpx/config/warnings_suffix.hpp>

```

In order to implement a full performance counter, you have to create an *HPX* component exposing this interface. To simplify this task, *HPX* provides a ready-made base class which handles all the boiler plate of creating a component for you. The remainder of this section will explain the process of creating a full performance counter based on the Sine example, which you can find in the directory `examples/performance_counters/sine/`.

The base class is defined in the header file `base_performance_counter.cpp` as:

```
// Copyright (c) 2007-2026 Hartmut Kaiser
//
// SPDX-License-Identifier: BSL-1.0
// Distributed under the Boost Software License, Version 1.0. (See accompanying
// file LICENSE_1_0.txt or copy at http://www.boost.org/LICENSE_1_0.txt)

#pragma once

#include <hpx/config.hpp>
#include <hpx/modules/actions_base.hpp>
#include <hpx/modules/components_base.hpp>
#include <hpx/modules/runtime_local.hpp>

#include <hpx/performance_counters/counters.hpp>
#include <hpx/performance_counters/server/base_performance_counter.hpp>

#ifdef DOXYGEN
//[performance_counter_base_class
namespace hpx::performance_counters {

    template <typename Derived>
    class base_performance_counter;
} // namespace hpx::performance_counters
//]
#endif

//[performance_counter_base_class
namespace hpx::performance_counters {

    HPX_CXX_EXPORT template <typename Derived>
    class base_performance_counter
        : public hpx::performance_counters::server::base_performance_counter
        , public hpx::components::component_base<Derived>
    {
    private:
        using base_type = hpx::components::component_base<Derived>;

    public:
        using type_holder = Derived;
        using base_type_holder =
            hpx::performance_counters::server::base_performance_counter;

        // NOLINTBEGIN(bugprone-crt-p-constructor-accessibility)
        base_performance_counter() = default;

        explicit base_performance_counter(
            hpx::performance_counters::counter_info const& info)
            : base_type_holder(info)
        {
        }
    }
}
//]

```

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```

// NOLINTEND(bugprone-crtip-constructor-accessibility)

// Disambiguate finalize() which is implemented in both base classes
void finalize()
{
    base_type_holder::finalize();
    base_type::finalize();
}

hpx::naming::address get_current_address() const
{
    return hpx::naming::address(
        hpx::naming::get_gid_from_locality_id(hpx::get_locality_id()),
        hpx::components::get_component_type<Derived>(),
        const_cast<Derived*>(static_cast<Derived const*>(this)));
}
};
} // namespace hpx::performance_counters

```

The single template parameter is expected to receive the type of the derived class implementing the performance counter. In the Sine example this looks like:

```

// Copyright (c) 2007-2012 Hartmut Kaiser
//
// SPDX-License-Identifier: BSL-1.0
// Distributed under the Boost Software License, Version 1.0. (See accompanying
// file LICENSE_1_0.txt or copy at http://www.boost.org/LICENSE_1_0.txt)

#pragma once

#include <hpx/config.hpp>
#if !defined(HPX_COMPUTE_DEVICE_CODE)
#include <hpx/hpx.hpp>
#include <hpx/include/lcos_local.hpp>
#include <hpx/include/performance_counters.hpp>
#include <hpx/include/util.hpp>

#include <cstdint>

namespace performance_counters { namespace sine { namespace server {
    //////////////////////////////////////
    //[sine_counter_definition
    class sine_counter
    : public hpx::performance_counters::base_performance_counter<sine_counter>
    {
    public:
        sine_counter()
            : current_value_(0)
            , evaluated_at_(0)
        {
        }
    }
}
}
}

```

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(continued from previous page)

```

explicit sine_counter(
    hpx::performance_counters::counter_info const& info);

    /// This function will be called in order to query the current value of
    /// this performance counter
    hpx::performance_counters::counter_value get_counter_value(bool reset);

    /// The functions below will be called to start and stop collecting
    /// counter values from this counter.
    bool start();
    bool stop();

    /// finalize() will be called just before the instance gets destructed
    void finalize();

protected:
    bool evaluate();

private:
    typedef hpx::spinlock mutex_type;

    mutable mutex_type mtx_;
    double current_value_;
    std::uint64_t evaluated_at_;

    hpx::util::interval_timer timer_;
};
}}} /// namespace performance_counters::sine::server
#endif

```

i.e., the type `sine_counter` is derived from the base class passing the type as a template argument (please see `simplest_performance_counter.cpp` for the full source code of the counter definition). For more information about this technique (called Curiously Recurring Template Pattern - CRTP), please see for instance the corresponding [Wikipedia article](http://en.wikipedia.org/wiki/Curiously_recurring_template_pattern)¹⁸⁰. This base class itself is derived from the `performance_counter` interface described above.

Additionally, a full performance counter implementation not only exposes the actual value but also provides information about:

- The point in time a particular value was retrieved.
- A (sequential) invocation count.
- The actual counter value.
- An optional scaling coefficient.
- Information about the counter status.

¹⁸⁰ http://en.wikipedia.org/wiki/Curiously_recurring_template_pattern

Existing HPX performance counters

The HPX runtime system exposes a wide variety of predefined performance counters. These counters expose critical information about different modules of the runtime system. They can help determine system bottlenecks and fine-tune system and application performance.

Table 2.29: AGAS performance counter `/agas/count/<agas_service>`

Counter type	<code>/agas/count/<agas_service></code> where <code><agas_service></code> is one of the following: <i>primary namespace services:</i> <code>route</code> , <code>bind_gid</code> , <code>resolve_gid</code> , <code>unbind_gid</code> , <code>increment_credit</code> , <code>decrement_credit</code> , <code>allocate</code> , <code>begin_migration</code> , <code>end_migration</code> <i>component namespace services:</i> <code>bind_prefix</code> , <code>bind_name</code> , <code>resolve_id</code> , <code>unbind_name</code> , <code>iterate_types</code> , <code>get_component_typename</code> , <code>num_localities_type</code> <i>locality namespace services:</i> <code>free</code> , <code>localities</code> , <code>num_localities</code> , <code>num_threads</code> , <code>resolve_locality</code> , <code>resolved_localities</code> <i>symbol namespace services:</i> <code>bind</code> , <code>resolve</code> , <code>unbind</code> , <code>iterate_names</code> , <code>on_symbol_namespace_event</code>
Counter instance formatting	<code><agas_instance>/total</code> where <code><agas_instance></code> is the name of the AGAS service to query. Currently, this value will be <code>locality#0</code> where <code>0</code> is the root <i>locality</i> (the id of the <i>locality</i> hosting the AGAS service). The value for <code>*</code> can be any <i>locality</i> id for the following <code><agas_service></code> : <code>route</code> , <code>bind_gid</code> , <code>resolve_gid</code> , <code>unbind_gid</code> , <code>increment_credit</code> , <code>decrement_credit</code> , <code>bin</code> , <code>resolve</code> , <code>unbind</code> , and <code>iterate_names</code> (only the primary and symbol AGAS service components live on all localities, whereas all other AGAS services are available on <code>locality#0</code> only).
Description	Returns the total number of invocations of the specified AGAS service since its creation.

Table 2.30: AGAS performance counter `/agas/<agas_service_category>/count`

Counter type	<code>/agas/<agas_service_category>/count</code> where <code><agas_service_category></code> is one of the following: <code>primary</code> , <code>locality</code> , <code>component</code> or <code>symbol</code>
Counter instance formatting	<code><agas_instance>/total</code> where <code><agas_instance></code> is the name of the AGAS service to query. Currently, this value will be <code>locality#0</code> where <code>0</code> is the root <i>locality</i> (the id of the <i>locality</i> hosting the AGAS service). Except for <code><agas_service_category></code> , <code>primary</code> or <code>symbol</code> for which the value for <code>*</code> can be any <i>locality</i> id (only the primary and symbol AGAS service components live on all localities, whereas all other AGAS services are available on <code>locality#0</code> only).
Description	Returns the overall total number of invocations of all AGAS services provided by the given AGAS service category since its creation.

Table 2.31: AGAS performance counter /agas/<agas_service_category>/count

Counter type	/agas/<agas_service_category>/count where <agas_service_category> is one of the following: <code>primary</code> , <code>locality</code> , <code>component</code> or <code>symbol</code>
Counter instance formatting	<agas_instance>/total where <agas_instance> is the name of the AGAS service to query. Currently, this value will be <code>locality#0</code> where <code>0</code> is the root <i>locality</i> (the id of the <i>locality</i> hosting the AGAS service). Except for <agas_service_category>, <code>primary</code> or <code>symbol</code> for which the value for <code>*</code> can be any <i>locality</i> id (only the primary and symbol AGAS service components live on all localities, whereas all other AGAS services are available on <code>locality#0</code> only).
Description	Returns the overall total number of invocations of all AGAS services provided by the given AGAS service category since its creation.

Table 2.32: AGAS performance counter agas/time/<agas_service>

Counter type	agas/time/<agas_service> where <agas_service> is one of the following: <i>primary namespace services:</i> <code>route</code> , <code>bind_gid</code> , <code>resolve_gid</code> , <code>unbind_gid</code> , <code>increment_credit</code> , <code>decrement_credit</code> , <code>allocate</code> , <code>begin_migration</code> , <code>end_migration</code> <i>component namespace services:</i> <code>bind_prefix</code> , <code>bind_name</code> , <code>resolve_id</code> , <code>unbind_name</code> , <code>iterate_types</code> , <code>get_component_typename</code> , <code>num_localities_type</code> <i>locality namespace services:</i> <code>free</code> , <code>localities</code> , <code>num_localities</code> , <code>num_threads</code> , <code>resolve_locality</code> , <code>resolved_localities</code> <i>symbol namespace services:</i> <code>bind</code> , <code>resolve</code> , <code>unbind</code> , <code>iterate_names</code> , <code>on_symbol_namespace_event</code>
Counter instance formatting	<agas_instance>/total where <agas_instance> is the name of the AGAS service to query. Currently, this value will be <code>locality#0</code> where <code>0</code> is the root <i>locality</i> (the id of the <i>locality</i> hosting the AGAS service). The value for <code>*</code> can be any <i>locality</i> id for the following <agas_service>: <code>route</code> , <code>bind_gid</code> , <code>resolve_gid</code> , <code>unbind_gid</code> , <code>increment_credit</code> , <code>decrement_credit</code> , <code>bin</code> , <code>resolve</code> , <code>unbind</code> , and <code>iterate_names</code> (only the primary and symbol AGAS service components live on all localities, whereas all other AGAS services are available on <code>locality#0</code> only).
Description	Returns the overall execution time of the specified AGAS service since its creation (in nanoseconds).

Table 2.33: AGAS performance counter /agas/<agas_service_category>/time`

Counter type	/agas/<agas_service_category>/time where <agas_service_category> is one of the following: <code>primary</code> , <code>locality</code> , <code>component</code> or <code>symbol</code>
Counter instance formatting	<agas_instance>/total where <agas_instance> is the name of the AGAS service to query. Currently, this value will be <code>locality#0</code> where <code>0</code> is the root <i>locality</i> (the id of the <i>locality</i> hosting the AGAS service). Except for <agas_service_category> <code>primary</code> or <code>symbol</code> for which the value for <code>*</code> can be any <i>locality</i> id (only the primary and symbol AGAS service components live on all localities, whereas all other AGAS services are available on <code>locality#0</code> only).
Description	Returns the overall execution time of all AGAS services provided by the given AGAS service category since its creation (in nanoseconds).

Table 2.34: AGAS performance counter /agas/count/entries

Counter type	/agas/count/entries
Counter instance formatting	locality#*/total where * is the <i>locality</i> id of the <i>locality</i> the AGAS cache should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the number of cache entries resident in the AGAS cache of the specified <i>locality</i> (see <cache_statistics>).

Table 2.35: AGAS performance counter /agas/count/<cache_statistics>

Counter type	/agas/count/<cache_statistics> where <cache_statistics> is one of the following: cache/evictions, cache/hits, cache/insertions, cache/misses
Counter instance formatting	locality#*/total where * is the <i>locality</i> id of the <i>locality</i> the AGAS cache should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i>
Description	Returns the number of cache events (evictions, hits, inserts, and misses) in the AGAS cache of the specified <i>locality</i> (see <cache_statistics>).

Table 2.36: AGAS performance counter /agas/count/<full_cache_statistics>

Counter type	/agas/count/<full_cache_statistics> where <full_cache_statistics> is one of the following: cache/get_entry, cache/insert_entry, cache/update_entry, cache/erase_entry
Counter instance formatting	locality#*/total where * is the <i>locality</i> id of the <i>locality</i> the AGAS cache should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the number of invocations of the specified cache API function of the AGAS cache.

Table 2.37: AGAS performance counter /agas/time/<full_cache_statistics>

Counter type	/agas/time/<full_cache_statistics> where <full_cache_statistics> is one of the following: cache/get_entry, cache/insert_entry, cache/update_entry, cache/erase_entry
Counter instance formatting	locality#*/total where * is the <i>locality</i> id of the <i>locality</i> the AGAS cache should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the overall time spent executing of the specified API function of the AGAS cache.

Table 2.38: Parcel layer performance counter `/data/count/<connection_type>/<operation>`

Counter type	<code>/data/count/<connection_type>/<operation></code> where: <code><operation></code> is one of the following: <code>sent</code> , <code>received</code> <code><connection_type></code> is one of the following: <code>tcp</code> , <code>mpi</code>
Counter instance formatting	<code>locality#*/total</code> where <code>*</code> is the <i>locality</i> id of the <i>locality</i> the overall number of transmitted bytes should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the overall number of raw (uncompressed) bytes sent or received (see <code><operation></code> , e.g. <code>sent</code> or <code>received</code>) for the specified <code><connection_type></code> . The performance counters are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_COUNTERS</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_COUNTERS</code> . The performance counters for the connection type <code>mpi</code> are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_MPI</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_MPI</code> . Please see <i>CMake options</i> for more details.

Table 2.39: Parcel layer performance counter `/data/time/<connection_type>/<operation>`

Counter type	<code>/data/time/<connection_type>/<operation></code> where: <code><operation></code> is one of the following: <code>sent</code> , <code>received</code> <code><connection_type></code> is one of the following: <code>tcp</code> , <code>mpi</code>
Counter instance formatting	<code>locality#*/total</code> where <code>*</code> is the <i>locality</i> id of the <i>locality</i> the total transmission time should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the total time (in nanoseconds) between the start of each asynchronous transmission operation and the end of the corresponding operation for the specified <code><connection_type></code> the given <i>locality</i> (see <code><operation></code> , e.g. <code>sent</code> or <code>received</code>). The performance counters are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_COUNTERS</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_COUNTERS</code> . The performance counters for the connection type <code>mpi</code> are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_MPI</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_MPI</code> . Please see <i>CMake options</i> for more details.

Table 2.40: Parcel layer performance counter `/serialize/count/<connection_type>/<operation>`

Counter type	<code>/serialize/count/<connection_type>/<operation></code> where: <operation> is one of the following: <code>sent</code> , <code>received</code> <connection_type> is one of the following: <code>tcp</code> , <code>mpi</code>
Counter instance formatting	<code>locality#*/total</code> where * is the <i>locality</i> id of the <i>locality</i> the overall number of transmitted bytes should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the overall number of bytes transferred (see <operation>, e.g. <code>sent</code> or <code>received</code> possibly compressed) for the specified <connection_type> by the given <i>locality</i> . The performance counters are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_COUNTERS</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_COUNTERS</code> . The performance counters for the connection type <code>mpi</code> are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_MPI</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_MPI</code> . Please see <i>CMake options</i> for more details.
Description	If the configure-time option <code>-DHPX_WITH_PARCELPORNT_ACTION_COUNTERS=On</code> was specified, this counter allows one to specify an optional action name as its parameter. In this case the counter will report the number of bytes transmitted for the given action only.

Table 2.41: Parcel layer performance counter `/serialize/time/<connection_type>/<operation>`

Counter type	<code>/serialize/time/<connection_type>/<operation></code> where: <operation> is one of the following: <code>sent</code> , <code>received</code> <connection_type> is one of the following: <code>tcp</code> , <code>mpi</code>
Counter instance formatting	<code>locality#*/total</code> where * is the <i>locality</i> id of the <i>locality</i> the serialization time should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the overall time spent performing outgoing data serialization for the specified <connection_type> on the given <i>locality</i> (see <operation>, e.g. <code>sent</code> or <code>received</code>). The performance counters are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_COUNTERS</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_COUNTERS</code> . The performance counters for the connection type <code>mpi</code> are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_MPI</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_MPI</code> . Please see <i>CMake options</i> for more details.
Parameters	If the configure-time option <code>-DHPX_WITH_PARCELPORNT_ACTION_COUNTERS=On</code> was specified, this counter allows one to specify an optional action name as its parameter. In this case the counter will report the serialization time for the given action only.

Table 2.42: Parcel layer performance counter `/parcels/count/routed`

Counter type	<code>/parcels/count/routed</code>
Counter instance formatting	<code>locality#*/total</code> where <code>*</code> is the <i>locality</i> id of the <i>locality</i> the number of routed parcels should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the overall number of routed (outbound) parcels transferred by the given <i>locality</i> . Routed parcels are those which cannot directly be delivered to its destination as the local AGAS is not able to resolve the destination address. In this case a parcel is sent to the AGAS service component which is responsible for creating the destination GID (and is responsible for resolving the destination address). This AGAS service component will deliver the parcel to its final target.
Parameters	If the configure-time option <code>-DHPX_WITH_PARCELPOR_ACTION_COUNTERS=On</code> was specified, this counter allows one to specify an optional action name as its parameter. In this case the counter will report the number of parcels for the given action only.

Table 2.43: Parcel layer performance counter `/parcels/count/<connection_type>/<operation>`

Counter type	<code>/parcels/count/<connection_type>/<operation></code> where: <code><operation></code> is one of the following: <code>sent</code> , <code>received</code> <code><connection_type></code> is one of the following: <code>tcp</code> , <code>mpi</code>
Counter instance formatting	<code>locality#*/total</code> where <code>*</code> is the <i>locality</i> id of the <i>locality</i> the number of parcels should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the overall number of parcels transferred using the specified <code><connection_type></code> by the given <i>locality</i> (see <code><operation></code> , e.g. <code>sent</code> or <code>received</code>). The performance counters are available only if the compile time constant <code>HPX_HAVE_PARCELPOR_COUNTERS</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPOR_COUNTERS</code> . The performance counters for the connection type <code>mpi</code> are available only if the compile time constant <code>HPX_HAVE_PARCELPOR_MPI</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPOR_MPI</code> . Please see <i>CMake options</i> for more details.

Table 2.44: Parcel layer performance counter `/messages/count/<connection_type>/<operation>`

Counter type	<code>/messages/count/<connection_type>/<operation></code> where: <code><operation></code> is one of the following: <code>sent</code> , <code>received</code> <code><connection_type></code> is one of the following: <code>tcp</code> , <code>mpi</code>
Counter instance formatting	<code>locality#*/total</code> where <code>*</code> is the <i>locality</i> id of the <i>locality</i> the number of messages should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the overall number of messages ¹⁸¹ transferred using the specified <code><connection_type></code> by the given <i>locality</i> (see <code><operation></code> , e.g. <code>sent</code> or <code>received</code>) The performance counters are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_COUNTERS</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_COUNTERS</code> . The performance counters for the connection type <code>mpi</code> are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_MPI</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_MPI</code> . Please see <i>CMake options</i> for more details.

Table 2.45: Parcel layer performance counter `/parcelport/count/<connection_type>/zero_copy_chunks/<operation>`

Counter type	<code>/parcelport/count/<connection_type>/zero_copy_chunks/<operation></code> where: <code><operation></code> is one of the following: <code>sent</code> , <code>received</code> <code><connection_type></code> is one of the following: <code>tcp</code> , <code>mpi</code>
Counter instance formatting	<code>locality#*/total</code> where <code>*</code> is the <i>locality</i> id of the <i>locality</i> the overall number of transmitted bytes should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the overall number of zero-copy chunks sent or received (see <code><operation></code> , e.g. <code>sent</code> or <code>received</code>) for the specified <code><connection_type></code> . The performance counters are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_COUNTERS</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_COUNTERS</code> . The performance counters for the connection type <code>mpi</code> are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_MPI</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_MPI</code> . Please see <i>CMake options</i> for more details.

¹⁸¹ A message can potentially consist of more than one *parcel*.

Table 2.46: Parcel layer performance counter `/parcelport/count-max/<connection_type>/zero_copy_chunks/<operation>`

Counter type	<code>/parcelport/count-max/<connection_type>/zero_copy_chunks/<operation></code> where: <code><operation></code> is one of the following: <code>sent</code> , <code>received</code> <code><connection_type></code> is one of the following: <code>tcp</code> , <code>mpi</code>
Counter instance formatting	<code>locality#*/total</code> where <code>*</code> is the <i>locality</i> id of the <i>locality</i> the overall number of transmitted bytes should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the maximum number of zero-copy chunks sent or received per message (see <code><operation></code> , e.g. <code>sent</code> or <code>received</code>) for the specified <code><connection_type></code> . The performance counters are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_COUNTERS</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_COUNTERS</code> . The performance counters for the connection type <code>mpi</code> are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_MPI</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_MPI</code> . Please see <i>CMake options</i> for more details.

Table 2.47: Parcel layer performance counter `/parcelport/size/<connection_type>/zero_copy_chunks/<operation>`

Counter type	<code>/parcelport/size/<connection_type>/zero_copy_chunks/<operation></code> where: <code><operation></code> is one of the following: <code>sent</code> , <code>received</code> <code><connection_type></code> is one of the following: <code>tcp</code> , <code>mpi</code>
Counter instance formatting	<code>locality#*/total</code> where <code>*</code> is the <i>locality</i> id of the <i>locality</i> the overall number of transmitted bytes should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the overall size of zero-copy chunks sent or received (see <code><operation></code> , e.g. <code>sent</code> or <code>received</code>) for the specified <code><connection_type></code> . The performance counters are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_COUNTERS</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_COUNTERS</code> . The performance counters for the connection type <code>mpi</code> are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_MPI</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_MPI</code> . Please see <i>CMake options</i> for more details.

Table 2.48: Parcel layer performance counter `/parcelport/size-max/<connection_type>/zero_copy_chunks/<operation>`

Counter type	<code>/parcelport/size-max/<connection_type>/zero_copy_chunks/<operation></code> where: <operation> is one of the following: <code>sent</code> , <code>received</code> <connection_type> is one of the following: <code>tcp</code> , <code>mpi</code>
Counter instance formatting	<code>locality#*/total</code> where * is the <i>locality</i> id of the <i>locality</i> the overall number of transmitted bytes should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the maximum size of zero-copy chunks sent or received (see <operation>, e.g. <code>sent</code> or <code>received</code>) for the specified <connection_type>. The performance counters are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_COUNTERS</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_COUNTERS</code> . The performance counters for the connection type <code>mpi</code> are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_MPI</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_MPI</code> . Please see <i>CMake options</i> for more details.

Table 2.49: Parcel layer performance counter `/parcelport/count/<connection_type>/<cache_statistics>`

Counter type	<code>/parcelport/count/<connection_type>/<cache_statistics></code> where: <cache_statistics> is one of the following: <code>cache/insertions</code> , <code>cache/evictions</code> , <code>cache/hits</code> , <code>cache/misses</code> , <code>cache/reclaims</code> <connection_type> is one of the following: <code>tcp</code> , <code>mpi</code>
Counter instance formatting	<code>locality#*/total</code> where * is the <i>locality</i> id of the <i>locality</i> the number of messages should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the overall number cache events (evictions, hits, inserts, misses, and reclaims) for the connection cache of the given connection type on the given <i>locality</i> (see <cache_statistics>, e.g. <code>ache/insertions</code> , <code>cache/evictions</code> , <code>cache/hits</code> , <code>cache/misses</code> or <code>cache/reclaims</code>). The performance counters for the connection type <code>mpi</code> are available only if the compile time constant <code>HPX_HAVE_PARCELPORNT_MPI</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPORNT_MPI</code> . Please see <i>CMake options</i> for more details.

Table 2.50: Parcel layer performance counter `/parcelport/count/<connection_type>/cache-reservation-failures`

Counter type	<code>/parcelport/count/<connection_type>/cache-reservation-failures</code> where <code><connection_type></code> is one of the following: <code>tcp</code> , <code>mpi</code>
Counter instance formatting	<code>locality#*/total</code> where <code>*</code> is the <i>locality</i> id of the <i>locality</i> the counter should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the cumulative number of times <code>get_or_reserve()</code> returned <code>false</code> because every connection slot in the connection cache was already checked out (pool saturation). This is distinct from ordinary cache misses, which occur whenever a new connection to a <i>locality</i> is first created and do not cause parcel deferral. A non-zero and growing value of this counter means that outgoing parcels are being silently queued until a connection is reclaimed; raising <code>hpx.max_connections</code> or <code>hpx.max_connections_per_locality</code> in the ini configuration will relieve the saturation. When pool exhaustion is detected, <i>HPX</i> also emits a warning-level log message (throttled to at most once every five seconds) of the form: <code>parcelport connection cache exhausted: N/M connections in use, K deferred reservation(s) since last reset</code>. The performance counters for the connection type <code>mpi</code> are available only if the compile time constant <code>HPX_HAVE_PARCELPOR_MPI</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPOR_MPI</code>. Please see <i>CMake options</i> for more details.

Table 2.51: Parcel layer performance counters `/parcelport/count/<connection_type>/cache-connections` and `/parcelport/count/<connection_type>/cache-max-connections`

Counter type	<code>/parcelport/count/<connection_type>/cache-connections</code> <code>/parcelport/count/<connection_type>/cache-max-connections</code> where <code><connection_type></code> is one of the following: <code>tcp</code> , <code>mpi</code>
Counter instance formatting	<code>locality#*/total</code> where <code>*</code> is the <i>locality</i> id of the <i>locality</i> the counter should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	<code>cache-connections</code> returns the current number of connections tracked by the connection cache for the given connection type on the given <i>locality</i>. This includes both connections that are currently checked out and connections that are idle and available for reuse. <code>cache-max-connections</code> returns the configured upper limit on the total number of connections the cache is allowed to hold. This value is fixed at startup and corresponds to the <code>hpx.max_connections</code> ini key. Together these two counters give the pool utilisation ratio. If <code>cache-connections</code> is consistently equal to <code>cache-max-connections</code> and <code>cache-reservation-failures</code> is increasing, the connection pool is saturated and the limit should be raised. The performance counters for the connection type <code>mpi</code> are available only if the compile time constant <code>HPX_HAVE_PARCELPOR_MPI</code> was defined while compiling the <i>HPX</i> core library (which is not defined by default). The corresponding cmake configuration constant is <code>HPX_WITH_PARCELPOR_MPI</code>. Please see <i>CMake options</i> for more details.

Table 2.52: Parcel layer performance counter `/parcelqueue/length/<operation>`

Counter type	<code>/parcelqueue/length/<operation></code> where <code><operation></code> is one of the following: <code>sent</code> , <code>receive</code>
Counter instance formatting	<code>locality#*/total</code> where <code>*</code> is the <i>locality</i> id of the <i>locality</i> the <i>parcel</i> queue should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the current number of parcels stored in the <i>parcel</i> queue (see <code><operation></code> for which queue to query, e.g. <code>sent</code> or <code>received</code>).

Table 2.53: Thread manager performance counter `/threads/count/cumulative`

Counter type	<code>/threads/count/cumulative</code>
Counter instance formatting	<code>locality#*/total</code> or <code>locality#*/worker-thread#*</code> or <code>locality#*/pool#*/worker-thread#*</code> where: <code>locality#*</code> is defining the <i>locality</i> for which the overall number of retired <i>HPX</i> -threads should be queried for. The <i>locality</i> id (given by the <code>*</code>) is a (zero based) number identifying the <i>locality</i> . pool#* is defining the pool for which the current value of the idle-loop counter should be queried for. worker-thread#* is defining the worker thread for which the overall number of retired <i>HPX</i> -threads should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the 'default' pool.
Description	Returns the overall number of executed (retired) <i>HPX</i> -threads on the given <i>locality</i> since application start. If the instance name is <code>total</code> the counter returns the accumulated number of retired <i>HPX</i> -threads for all worker threads (cores) on that <i>locality</i> . If the instance name is <code>worker-thread#*</code> the counter will return the overall number of retired <i>HPX</i> -threads for all worker threads separately. This counter is available only if the configuration time constant <code>HPX_WITH_THREAD_CUMULATIVE_COUNTS</code> is set to <code>ON</code> (default: <code>ON</code>).

Table 2.54: Thread manager performance counter `/threads/time/average`

Counter type	<code>/threads/time/average</code>
Counter instance formatting	<code>locality#*/total</code> or <code>locality#*/worker-thread#*</code> or <code>locality#*/pool#*/worker-thread#*</code> where: <code>locality#*</code> is defining the <i>locality</i> for which the average time spent executing one <i>HPX</i> -thread should be queried for. The <i>locality</i> id (given by the <code>*</code>) is a (zero based) number identifying the <i>locality</i> . <code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for. <code>worker-thread#*</code> is defining the worker thread for which the average time spent executing one <i>HPX</i> -thread should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the 'default' pool.
Description	Returns the average time spent executing one <i>HPX</i> -thread on the given <i>locality</i> since application start. If the instance name is <code>total</code> the counter returns the average time spent executing one <i>HPX</i> -thread for all worker threads (cores) on that <i>locality</i> . If the instance name is <code>worker-thread#*</code> the counter will return the average time spent executing one <i>HPX</i> -thread for all worker threads separately. This counter is available only if the configuration time constants <code>HPX_WITH_THREAD_CUMULATIVE_COUNTS</code> (default: ON) and <code>HPX_WITH_THREAD_IDLE_RATES</code> are set to ON (default: OFF). The unit of measure for this counter is nanosecond [ns].

Table 2.55: Thread manager performance counter `/threads/time/average-overhead`

Counter type	<code>/threads/time/average-overhead</code>
Counter instance formatting	<code>locality#*/total</code> or <code>locality#*/worker-thread#*</code> or <code>locality#*/pool#*/worker-thread#*</code> where: <code>locality#*</code> is defining the <i>locality</i> for which the average overhead spent executing one <i>HPX</i> -thread should be queried for. The <i>locality</i> id (given by the <code>*</code>) is a (zero based) number identifying the <i>locality</i> . <code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for. <code>worker-thread#*</code> is defining the worker thread for which the average overhead spent executing one <i>HPX</i> -thread should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the 'default' pool.
Description	Returns the average time spent on overhead while executing one <i>HPX</i> -thread on the given <i>locality</i> since application start. If the instance name is <code>total</code> the counter returns the average time spent on overhead while executing one <i>HPX</i> -thread for all worker threads (cores) on that <i>locality</i> . If the instance name is <code>worker-thread#*</code> the counter will return the average time spent on overhead executing one <i>HPX</i> -thread for all worker threads separately. This counter is available only if the configuration time constants <code>HPX_WITH_THREAD_CUMULATIVE_COUNTS</code> (default: ON) and <code>HPX_WITH_THREAD_IDLE_RATES</code> are set to ON (default: OFF). The unit of measure for this counter is nanosecond [ns].

Table 2.56: Thread manager performance counter `/threads/count/cumulative-phases`

Counter type	<code>/threads/count/cumulative-phases</code>
Counter instance formatting	<code>locality#*/total</code> or <code>locality#*/worker-thread#*</code> or <code>locality#*/pool#*/worker-thread#*</code> where: <code>locality#*</code> is defining the <i>locality</i> for which the overall number of executed <i>HPX</i> -thread phases (invocations) should be queried for. The <i>locality</i> id (given by the <code>*</code>) is a (zero based) number identifying the <i>locality</i> . <code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for. <code>worker-thread#*</code> is defining the worker thread for which the overall number of executed <i>HPX</i> -thread phases (invocations) should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the ‘default’ pool.
Description	Returns the overall number of executed <i>HPX</i> -thread phases (invocations) on the given <i>locality</i> since application start. If the instance name is <code>total</code> the counter returns the accumulated number of executed <i>HPX</i> -thread phases (invocations) for all worker threads (cores) on that <i>locality</i> . If the instance name is <code>worker-thread#*</code> the counter will return the overall number of executed <i>HPX</i> -thread phases for all worker threads separately. This counter is available only if the configuration time constant <code>HPX_WITH_THREAD_CUMULATIVE_COUNTS</code> is set to ON (default: ON). The unit of measure for this counter is nanosecond [ns].

Table 2.57: Thread manager performance counter `/threads/time/average-phase`

Counter type	<code>/threads/time/average-phase</code>
Counter instance formatting	<code>locality#*/total</code> or <code>locality#*/worker-thread#*</code> or <code>locality#*/pool#*/worker-thread#*</code> where: <code>locality#*</code> is defining the <i>locality</i> for which the average time spent executing one <i>HPX</i> -thread phase (invocation) should be queried for. The <i>locality</i> id (given by the <code>*</code>) is a (zero based) number identifying the <i>locality</i> . <code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for. <code>worker-thread#*</code> is defining the worker thread for which the average time executing one <i>HPX</i> -thread phase (invocation) should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the ‘default’ pool.
Description	Returns the average time spent executing one <i>HPX</i> -thread phase (invocation) on the given <i>locality</i> since application start. If the instance name is <code>total</code> the counter returns the average time spent executing one <i>HPX</i> -thread phase (invocation) for all worker threads (cores) on that <i>locality</i> . If the instance name is <code>worker-thread#*</code> the counter will return the average time spent executing one <i>HPX</i> -thread phase for all worker threads separately. This counter is available only if the configuration time constants <code>HPX_WITH_THREAD_CUMULATIVE_COUNTS</code> (default: ON) and <code>HPX_WITH_THREAD_IDLE_RATES</code> are set to ON (default: OFF). The unit of measure for this counter is nanosecond [ns].

Table 2.58: Thread manager performance counter `/threads/time/average-phase-overhead`

Counter type	<code>/threads/time/average-phase-overhead</code>
Counter instance formatting	<code>locality#*/total</code> or <code>locality#*/worker-thread#*</code> or <code>locality#*/pool#*/worker-thread#*</code> where: <code>locality#*</code> is defining the <i>locality</i> for which the average time overhead executing one <i>HPX</i> -thread phase (invocation) should be queried for. The <i>locality</i> id (given by the <code>*</code>) is a (zero based) number identifying the <i>locality</i> . <code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for. <code>worker-thread#*</code> is defining the worker thread for which the average overhead executing one <i>HPX</i> -thread phase (invocation) should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the ‘default’ pool.
Description	Returns the average time spent on overhead executing one <i>HPX</i> -thread phase (invocation) on the given <i>locality</i> since application start. If the instance name is <code>total</code> the counter returns the average time spent on overhead while executing one <i>HPX</i> -thread phase (invocation) for all worker threads (cores) on that <i>locality</i> . If the instance name is <code>worker-thread#*</code> the counter will return the average time spent on overhead executing one <i>HPX</i> -thread phase for all worker threads separately. This counter is available only if the configuration time constants <code>HPX_WITH_THREAD_CUMULATIVE_COUNTS</code> (default: ON) and <code>HPX_WITH_THREAD_IDLE_RATES</code> are set to ON (default: OFF). The unit of measure for this counter is nanosecond [ns].

Table 2.59: Thread manager performance counter `/threads/time/overall`

Counter type	<code>/threads/time/overall</code>
Counter instance formatting	<code>locality#*/total</code> or <code>locality#*/worker-thread#*</code> or <code>locality#*/pool#*/worker-thread#*</code> where: <code>locality#*</code> is defining the <i>locality</i> for which the overall time spent running the scheduler should be queried for. The <i>locality</i> id (given by the <code>*</code>) is a (zero based) number identifying the <i>locality</i> . <code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for. <code>worker-thread#*</code> is defining the worker thread for which the overall time spent running the scheduler should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the ‘default’ pool.
Description	Returns the overall time spent running the scheduler on the given <i>locality</i> since application start. If the instance name is <code>total</code> the counter returns the overall time spent running the scheduler for all worker threads (cores) on that <i>locality</i> . If the instance name is <code>worker-thread#*</code> the counter will return the overall time spent running the scheduler for all worker threads separately. This counter is available only if the configuration time constant <code>HPX_WITH_THREAD_IDLE_RATES</code> is set to ON (default: OFF). The unit of measure for this counter is nanosecond [ns].

Table 2.60: Thread manager performance counter `/threads/time/cumulative`

Counter type	<code>/threads/time/cumulative</code>
Counter instance formatting	<code>locality#*/total</code> or <code>locality#*/worker-thread#*</code> or <code>locality#*/pool#*/worker-thread#*</code> where: <code>locality#*</code> is defining the <i>locality</i> for which the overall time spent executing all <i>HPX</i> -threads should be queried for. The <i>locality</i> id (given by the <code>*</code>) is a (zero based) number identifying the <i>locality</i> . <code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for. <code>worker-thread#*</code> is defining the worker thread for which the overall time spent executing all <i>HPX</i> -threads should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the 'default' pool.
Description	Returns the overall time spent executing all <i>HPX</i> -threads on the given <i>locality</i> since application start. If the instance name is <code>total</code> the counter returns the overall time spent executing all <i>HPX</i> -threads for all worker threads (cores) on that <i>locality</i> . If the instance name is <code>worker-thread#*</code> the counter will return the overall time spent executing all <i>HPX</i> -threads for all worker threads separately. This counter is available only if the configuration time constants <code>HPX_THREAD_MAINTAIN_CUMULATIVE_COUNTS</code> (default: ON) and <code>HPX_THREAD_MAINTAIN_IDLE_RATES</code> are set to ON (default: OFF).

Table 2.61: Thread manager performance counter `/threads/time/cumulative-overheads`

Counter type	<code>/threads/time/cumulative-overheads</code>
Counter instance formatting	<code>locality#*/total</code> or <code>locality#*/worker-thread#*</code> or <code>locality#*/pool#*/worker-thread#*</code> where: <code>locality#*</code> is defining the <i>locality</i> for which the overall overhead time incurred by executing all <i>HPX</i> -threads should be queried for. The <i>locality</i> id (given by the <code>*</code>) is a (zero based) number identifying the <i>locality</i> . <code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for. <code>worker-thread#*</code> is defining the worker thread for which the the overall overhead time incurred by executing all <i>HPX</i> -threads should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the 'default' pool.
Description	Returns the overall overhead time incurred executing all <i>HPX</i> -threads on the given <i>locality</i> since application start. If the instance name is <code>total</code> the counter returns the overall overhead time incurred executing all <i>HPX</i> -threads for all worker threads (cores) on that <i>locality</i> . If the instance name is <code>worker-thread#*</code> the counter will return the overall overhead time incurred executing all <i>HPX</i> -threads for all worker threads separately. This counter is available only if the configuration time constants <code>HPX_THREAD_MAINTAIN_CUMULATIVE_COUNTS</code> (default: ON) and <code>HPX_THREAD_MAINTAIN_IDLE_RATES</code> are set to ON (default: OFF). The unit of measure for this counter is nanosecond [ns].

Table 2.62: Thread manager performance counter `threads/count/instantaneous/<thread-state>`

Counter type	<code>threads/count/instantaneous/<thread-state></code> where: <code><thread-state></code> is one of the following: <code>all</code> , <code>active</code> , <code>pending</code> , <code>suspended</code> , <code>terminated</code> , <code>staged</code>
Counter instance formatting	<code>locality#*/total</code> or <code>locality#*/worker-thread#*</code> or <code>locality#*/pool#*/worker-thread#*</code> where: <code>locality#*</code> is defining the <i>locality</i> for which the current number of threads with the given state should be queried for. The <i>locality</i> id (given by the <code>*</code>) is a (zero based) number identifying the <i>locality</i> . <code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for. <code>worker-thread#*</code> is defining the worker thread for which the current number of threads with the given state should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the ‘default’ pool. The <code>staged</code> thread state refers to registered tasks before they are converted to thread objects.
Description	Returns the current number of <i>HPX</i> -threads having the given thread state on the given <i>locality</i> . If the instance name is <code>total</code> the counter returns the current number of <i>HPX</i> -threads of the given state for all worker threads (cores) on that <i>locality</i> . If the instance name is <code>worker-thread#*</code> the counter will return the current number of <i>HPX</i> -threads in the given state for all worker threads separately.

Table 2.63: Thread manager performance counter threads/
wait-time/<thread-state>

Counter type	threads/wait-time/<thread-state> where: <thread-state> is one of the following: pending staged
Counter instance formatting	locality#*/total or locality#*/worker-thread#* or locality#*/pool#*/worker-thread#* where: locality#* is defining the <i>locality</i> for which the average wait time of <i>HPX</i> -threads (pending) or thread descriptions (staged) with the given state should be queried for. The <i>locality</i> id (given by *) is a (zero based) number identifying the <i>locality</i> . pool#* is defining the pool for which the current value of the idle-loop counter should be queried for. worker-thread#* is defining the worker thread for which the average wait time for the given state should be queried for. The worker thread number (given by the *) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the 'default' pool. The staged thread state refers to the wait time of registered tasks before they are converted into thread objects, while the pending thread state refers to the wait time of threads in any of the scheduling queues.
Description	Returns the average wait time of <i>HPX</i> -threads (if the thread state is pending or of task descriptions (if the thread state is staged on the given <i>locality</i> since application start. If the instance name is <code>total</code> the counter returns the wait time of <i>HPX</i> -threads of the given state for all worker threads (cores) on that <i>locality</i> . If the instance name is <code>worker-thread#*</code> the counter will return the wait time of <i>HPX</i> -threads in the given state for all worker threads separately. These counters are available only if the compile time constant <code>HPX_WITH_THREAD_QUEUE_WAITTIME</code> was defined while compiling the <i>HPX</i> core library (default: OFF). The unit of measure for this counter is nanosecond [ns].

Table 2.64: Thread manager performance counter `/threads/idle-rate`

Counter type	<code>/threads/idle-rate</code>
Counter instance formatting	<code>locality#*/total</code> or <code>locality#*/worker-thread#*</code> or <code>locality#*/pool#*/worker-thread#*</code> where: <code>locality#*</code> is defining the <i>locality</i> for which the average idle rate of all (or one) worker threads should be queried for. The <i>locality</i> id (given by the <code>*</code>) is a (zero based) number identifying the <i>locality</i> <code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for. <code>worker-thread#*</code> is defining the worker thread for which the averaged idle rate should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the ‘default’ pool.
Description	Returns the average idle rate for the given worker thread(s) on the given <i>locality</i> . The idle rate is defined as the ratio of the time spent on scheduling and management tasks and the overall time spent executing work since the application started. This counter is available only if the configuration time constant <code>HPX_WITH_THREAD_IDLE_RATES</code> is set to ON (default: OFF).

Table 2.65: Thread manager performance counter `/threads/creation-idle-rate`

Counter type	<code>/threads/creation-idle-rate</code>
Counter instance formatting	<code>locality#*/total</code> or <code>locality#*/worker-thread#*</code> or <code>locality#*/pool#*/worker-thread#*</code> where: <code>locality#*</code> is defining the <i>locality</i> for which the average creation idle rate of all (or one) worker threads should be queried for. The <i>locality</i> id (given by the <code>*</code>) is a (zero based) number identifying the <i>locality</i> . <code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for. <code>worker-thread#*</code> is defining the worker thread for which the averaged idle rate should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the ‘default’ pool.
Description	Returns the average idle rate for the given worker thread(s) on the given <i>locality</i> which is caused by creating new threads. The creation idle rate is defined as the ratio of the time spent on creating new threads and the overall time spent executing work since the application started. This counter is available only if the configuration time constants <code>HPX_WITH_THREAD_IDLE_RATES</code> (default: OFF) and <code>HPX_WITH_THREAD_CREATION_AND_CLEANUP_RATES</code> are set to ON.

Table 2.66: Thread manager performance counter `/threads/cleanup-idle-rate`

Counter type	<code>/threads/cleanup-idle-rate</code>
Counter instance formatting	<code>locality#*/total</code> or <code>locality#*/worker-thread#*</code> or <code>locality#*/pool#*/worker-thread#*</code> where: <code>locality#*</code> is defining the <i>locality</i> for which the average cleanup idle rate of all (or one) worker threads should be queried for. The <i>locality</i> id (given by the <code>*</code>) is a (zero based) number identifying the <i>locality</i> . <code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for. <code>worker-thread#*</code> is defining the worker thread for which the averaged cleanup idle rate should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the 'default' pool.
Description	Returns the average idle rate for the given worker thread(s) on the given <i>locality</i> which is caused by cleaning up terminated threads. The cleanup idle rate is defined as the ratio of the time spent on cleaning up terminated thread objects and the overall time spent executing work since the application started. This counter is available only if the configuration time constants <code>HPX_WITH_THREAD_IDLE_RATES</code> (default: OFF) and <code>HPX_WITH_THREAD_CREATION_AND_CLEANUP_RATES</code> are set to ON.

Table 2.67: Thread manager performance counter `/threadqueue/length`

Counter type	<code>/threadqueue/length</code>
Counter instance formatting	<code>locality#*/total</code> or <code>locality#*/worker-thread#*</code> or <code>locality#*/pool#*/worker-thread#*</code> where: <code>locality#*</code> is defining the <i>locality</i> for which the current length of all thread queues in the scheduler for all (or one) worker threads should be queried for. The <i>locality</i> id (given by the <code>*</code>) is a (zero based) number identifying the <i>locality</i> . <code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for. <code>worker-thread#*</code> is defining the worker thread for which the current length of all thread queues in the scheduler should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the 'default' pool.
Description	Returns the overall length of all queues for the given worker thread(s) on the given <i>locality</i> .

Table 2.68: Thread manager performance counter `/threads/count/stack-unbinds`

Counter type	<code>/threads/count/stack-unbinds</code>
Counter instance formatting	<code>locality#*/total</code> where: * is the <i>locality</i> id of the <i>locality</i> the unbind (madvise) operations should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the total number of <i>HPX</i> -thread unbind (madvise) operations performed for the referenced <i>locality</i> . Note that this counter is not available on Windows based platforms.

Table 2.69: Thread manager performance counter `/threads/count/stack-recycles`

Counter type	<code>/threads/count/stack-recycles</code>
Counter instance formatting	<code>locality#*/total</code> where: * is the <i>locality</i> id of the <i>locality</i> the recycling operations should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the total number of <i>HPX</i> -thread recycling operations performed.

Table 2.70: Thread manager performance counter `/threads/count/stolen-from-pending`

Counter type	<code>/threads/count/stolen-from-pending</code>
Counter instance formatting	<code>locality#*/total</code> where: * is the <i>locality</i> id of the <i>locality</i> the number of ‘stole’ threads should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the total number of <i>HPX</i> -threads ‘stolen’ from the pending thread queue by a neighboring thread worker thread (these threads are executed by a different worker thread than they were initially scheduled on). This counter is available only if the configuration time constant <code>HPX_WITH_THREAD_STEALING_COUNTS</code> is set to <code>ON</code> (default: <code>ON</code>).

Table 2.71: Thread manager performance counter `/threads/count/pending-misses`

Counter type	<code>/threads/count/pending-misses</code>
Counter instance formatting	<code>locality#*/total</code> or <code>locality#*/worker-thread#*</code> or <code>locality#*/pool#*/worker-thread#*</code> where: <code>locality#*</code> is defining the <i>locality</i> for which the number of pending queue misses of all (or one) worker threads should be queried for. The <i>locality</i> id (given by the <code>*</code>) is a (zero based) number identifying the <i>locality</i> <code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for. <code>worker-thread#*</code> is defining the worker thread for which the number of pending queue misses should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the ‘default’ pool.
Description	Returns the total number of times that the referenced worker-thread on the referenced <i>locality</i> failed to find pending <i>HPX</i> -threads in its associated queue. This counter is available only if the configuration time constant <code>HPX_WITH_THREAD_STEALING_COUNTS</code> is set to ON (default: ON).

Table 2.72: Thread manager performance counter `/threads/count/pending-accesses`

Counter type	<code>/threads/count/pending-accesses</code>
Counter instance formatting	<code>locality#*/total</code> or <code>locality#*/worker-thread#*</code> or <code>locality#*/pool#*/worker-thread#*</code> where: <code>locality#*</code> is defining the <i>locality</i> for which the number of pending queue accesses of all (or one) worker threads should be queried for. The <i>locality</i> id (given by the <code>*</code>) is a (zero based) number identifying the <i>locality</i> <code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for. <code>worker-thread#*</code> is defining the worker thread for which the number of pending queue accesses should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the ‘default’ pool.
Description	Returns the total number of times that the referenced worker-thread on the referenced <i>locality</i> looked for pending <i>HPX</i> -threads in its associated queue. This counter is available only if the configuration time constant <code>HPX_WITH_THREAD_STEALING_COUNTS</code> is set to ON (default: ON).

Table 2.73: Thread manager performance counter `/threads/count/stolen-from-staged`

Counter type	<code>/threads/count/stolen-from-staged</code>
Counter instance formatting	<code>locality#*/total</code> or <code>locality#*/worker-thread#*</code> or <code>locality#*/pool#*/worker-thread#*</code> where: <code>locality#*</code> is defining the <i>locality</i> for which the number of <i>HPX</i> -threads stolen from the staged queue of all (or one) worker threads should be queried for. The <i>locality</i> id (given by the <code>*</code>) is a (zero based) number identifying the <i>locality</i> . <code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for. <code>worker-thread#*</code> is defining the worker thread for which the number of <i>HPX</i> -threads stolen from the staged queue should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the ‘default’ pool.
Description	Returns the total number of <i>HPX</i> -threads ‘stolen’ from the staged thread queue by a neighboring worker thread (these threads are executed by a different worker thread than they were initially scheduled on). This counter is available only if the configuration time constant <code>HPX_WITH_THREAD_STEALING_COUNTS</code> is set to ON (default: ON).

Table 2.74: Thread manager performance counter `/threads/count/stolen-to-pending`

Counter type	<code>/threads/count/stolen-to-pending</code>
Counter instance formatting	<code>locality#*/total</code> or <code>locality#*/worker-thread#*</code> or <code>locality#*/pool#*/worker-thread#*</code> where: <code>locality#*</code> is defining the <i>locality</i> for which the number of <i>HPX</i> -threads stolen to the pending queue of all (or one) worker threads should be queried for. The <i>locality</i> id (given by the <code>*</code>) is a (zero based) number identifying the <i>locality</i> . <code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for. <code>worker-thread#*</code> is defining the worker thread for which the number of <i>HPX</i> -threads stolen to the pending queue should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the ‘default’ pool.
Description	Returns the total number of <i>HPX</i> -threads ‘stolen’ to the pending thread queue of the worker thread (these threads are executed by a different worker thread than they were initially scheduled on). This counter is available only if the configuration time constant <code>HPX_WITH_THREAD_STEALING_COUNTS</code> is set to ON (default: ON).

Table 2.75: Thread manager performance counter `/threads/count/stolen-to-staged`

Counter type	<code>/threads/count/stolen-to-staged</code>
Counter instance formatting	<code>locality#*/total</code> or <code>locality#*/worker-thread#*</code> or <code>locality#*/pool#*/worker-thread#*</code> where: <code>locality#*</code> is defining the <i>locality</i> for which the number of <i>HPX</i> -threads stolen to the staged queue of all (or one) worker threads should be queried for. The <i>locality</i> id (given by <code>*</code>) is a (zero based) number identifying the <i>locality</i> . <code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for. <code>worker-thread#*</code> is defining the worker thread for which the number of <i>HPX</i> -threads stolen to the staged queue should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the ‘default’ pool.
Description	Returns the total number of <i>HPX</i> -threads ‘stolen’ to the staged thread queue of a neighboring worker thread (these threads are executed by a different worker thread than they were initially scheduled on). This counter is available only if the configuration time constant <code>HPX_WITH_THREAD_STEALING_COUNTS</code> is set to ON (default: ON).

Table 2.76: Thread manager performance counter `/threads/count/objects`

Counter type	<code>/threads/count/objects</code>
Counter instance formatting	<code>locality#*/total</code> or <code>locality#*/allocator#*</code> where: <code>locality#*</code> is defining the <i>locality</i> for which the current (cumulative) number of all created <i>HPX</i> -thread objects should be queried for. The <i>locality</i> id (given by <code>*</code>) is a (zero based) number identifying the <i>locality</i> . <code>allocator#*</code> is defining the number of the allocator instance using which the threads have been created. <i>HPX</i> uses a varying number of allocators to create (and recycle) <i>HPX</i> -thread objects, most likely these counters are of use for debugging purposes only. The allocator id (given by <code>*</code>) is a (zero based) number identifying the allocator to query.
Description	Returns the total number of <i>HPX</i> -thread objects created. Note that thread objects are reused to improve system performance, thus this number does not reflect the number of actually executed (retired) <i>HPX</i> -threads.

Table 2.77: Thread manager performance counter `/scheduler/`
`utilization/instantaneous`

Counter type	<code>/scheduler/utilization/instantaneous</code>
Counter instance formatting	<code>locality#*/total</code> where: <code>locality#*</code> is defining the <i>locality</i> for which the current (instantaneous) scheduler utilization queried for. The <i>locality</i> id (given by <code>*</code>) is a (zero based) number identifying the <i>locality</i> .
Description	Returns the total (instantaneous) scheduler utilization. This is the current percentage of scheduler threads executing <i>HPX</i> threads.
Parameters	Percent

Table 2.78: Thread manager performance counter `/threads/`
`idle-loop-count/instantaneous`

Counter type	<code>/threads/idle-loop-count/instantaneous</code>
Counter instance formatting	<code>locality#*/worker-thread#*</code> or <code>locality#*/pool#*/worker-thread#*</code> where: <code>locality#*</code> is defining the <i>locality</i> for which the current current accumulated value of all idle-loop counters of all worker threads should be queried. The <i>locality</i> id (given by the <code>*</code>) is a (zero based) number identifying the <i>locality</i> . <code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for. <code>worker-thread#*</code> is defining the worker thread for which the current value of the idle-loop counter should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the 'default' pool.
Description	Returns the current (instantaneous) idle-loop count for the given <i>HPX</i> - worker thread or the accumulated value for all worker threads.

Table 2.79: Thread manager performance counter `/threads/
busy-loop-count/instantaneous`

Counter type	<code>/threads/busy-loop-count/instantaneous</code>
Counter instance formatting	<code>locality#*/worker-thread#*</code> or <code>locality#*/pool#*/worker-thread#*</code> where: <code>locality#*</code> is defining the <i>locality</i> for which the current current accumulated value of all busy-loop counters of all worker threads should be queried. The <i>locality</i> id (given by the <code>*</code>) is a (zero based) number identifying the <i>locality</i> . <code>pool#*</code> is defining the pool for which the current value of the idle-loop counter should be queried for. <code>worker-thread#*</code> is defining the worker thread for which the current value of the busy-loop counter should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> . If no pool-name is specified the counter refers to the 'default' pool.
Description	Returns the current (instantaneous) busy-loop count for the given <i>HPX</i> - worker thread or the accumulated value for all worker threads.

Table 2.80: Thread manager performance counter `/threads/time/
background-work-duration`

Counter type	<code>/threads/time/background-work-duration</code>
Counter instance formatting	<code>locality#*/total</code> or <code>locality#*/worker-thread#*</code> where: <code>locality#*</code> is defining the locality for which the overall time spent performing background work should be queried for. The locality id (given by <code>*</code>) is a (zero based) number identifying the locality. <code>worker-thread#*</code> is defining the worker thread for which the overall time spent performing background work should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> .
Description	Returns the overall time spent performing background work on the given locality since application start. If the instance name is <code>total</code> the counter returns the overall time spent performing background work for all worker threads (cores) on that locality. If the instance name is <code>worker-thread#*</code> the counter will return the overall time spent performing background work for all worker threads separately. This counter is available only if the configuration time constants <code>HPX_WITH_BACKGROUND_THREAD_COUNTERS</code> (default: OFF) and <code>HPX_WITH_THREAD_IDLE_RATES</code> are set to ON (default: OFF). The unit of measure for this counter is nanosecond [ns].

Table 2.81: Thread manager performance counter `/threads/background-overhead`

Counter type	<code>/threads/background-overhead</code>
Counter instance formatting	<code>locality#*/total</code> or <code>locality#*/worker-thread#*</code> where: <code>locality#*</code> is defining the locality for which the background overhead should be queried for. The locality id (given by <code>*</code>) is a (zero based) number identifying the locality. <code>worker-thread#*</code> is defining the worker thread for which the background overhead should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> .
Description	Returns the background overhead on the given locality since application start. If the instance name is <code>total</code> the counter returns the background overhead for all worker threads (cores) on that locality. If the instance name is <code>worker-thread#*</code> the counter will return background overhead for all worker threads separately. This counter is available only if the configuration time constants <code>HPX_WITH_BACKGROUND_THREAD_COUNTERS</code> (default: OFF) and <code>HPX_WITH_THREAD_IDLE_RATES</code> are set to ON (default: OFF). The unit of measure displayed for this counter is 0.1%.

Table 2.82: Thread manager performance counter `/threads/time/background-send-duration`

Counter type	<code>/threads/time/background-send-duration</code>
Counter instance formatting	<code>locality#*/total</code> or <code>locality#*/worker-thread#*</code> where: <code>locality#*</code> is defining the locality for which the overall time spent performing background work related to sending parcels should be queried for. The locality id (given by <code>*</code>) is a (zero based) number identifying the locality. <code>worker-thread#*</code> is defining the worker thread for which the overall time spent performing background work related to sending parcels should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> .
Description	Returns the overall time spent performing background work related to sending parcels on the given locality since application start. If the instance name is <code>total</code> the counter returns the overall time spent performing background work for all worker threads (cores) on that locality. If the instance name is <code>worker-thread#*</code> the counter will return the overall time spent performing background work for all worker threads separately. This counter is available only if the configuration time constants <code>HPX_WITH_BACKGROUND_THREAD_COUNTERS</code> (default: OFF) and <code>HPX_WITH_THREAD_IDLE_RATES</code> are set to ON (default: OFF). The unit of measure for this counter is nanosecond [ns]. This counter will currently return meaningful values for the MPI parcellport only.

Table 2.83: Thread manager performance counter `/threads/background-send-overhead`

Counter type	<code>/threads/background-send-overhead</code>
Counter instance formatting	<code>locality#*/total</code> or <code>locality#*/worker-thread#*</code> where: <code>locality#*</code> is defining the locality for which the background overhead related to sending parcels should be queried for. The locality id (given by <code>*</code>) is a (zero based) number identifying the locality. <code>worker-thread#*</code> is defining the worker thread for which the background overhead related to sending parcels should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> .
Description	Returns the background overhead related to sending parcels on the given locality since application start. If the instance name is <code>total</code> the counter returns the background overhead for all worker threads (cores) on that locality. If the instance name is <code>worker-thread#*</code> the counter will return background overhead for all worker threads separately. This counter is available only if the configuration time constants <code>HPX_WITH_BACKGROUND_THREAD_COUNTERS</code> (default: OFF) and <code>HPX_WITH_THREAD_IDLE_RATES</code> are set to ON (default: OFF). The unit of measure displayed for this counter is 0.1%. This counter will currently return meaningful values for the MPI parcelport only.

Table 2.84: Thread manager performance counter `/threads/time/background-receive-duration`

Counter type	<code>/threads/time/background-receive-duration</code>
Counter instance formatting	<code>locality#*/total</code> or <code>locality#*/worker-thread#*</code> where: <code>locality#*</code> is defining the locality for which the overall time spent performing background work related to receiving parcels should be queried for. The locality id (given by <code>*</code>) is a (zero based) number identifying the locality. <code>worker-thread#*</code> is defining the worker thread for which the overall time spent performing background work related to receiving parcels should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> .
Description	Returns the overall time spent performing background work related to receiving parcels on the given locality since application start. If the instance name is <code>total</code> the counter returns the overall time spent performing background work for all worker threads (cores) on that locality. If the instance name is <code>worker-thread#*</code> the counter will return the overall time spent performing background work for all worker threads separately. This counter is available only if the configuration time constants <code>HPX_WITH_BACKGROUND_THREAD_COUNTERS</code> (default: OFF) and <code>HPX_WITH_THREAD_IDLE_RATES</code> are set to ON (default: OFF). The unit of measure for this counter is nanosecond [ns]. This counter will currently return meaningful values for the MPI parcelport only.

Table 2.85: Thread manager performance counter `/threads/background-receive-overhead`

Counter type	<code>/threads/background-receive-overhead</code>
Counter instance formatting	<code>locality#*/total</code> or <code>locality#*/worker-thread#*</code> where: <code>locality#*</code> is defining the locality for which the background overhead related to receiving should be queried for. The locality id (given by <code>*</code>) is a (zero based) number identifying the locality. <code>worker-thread#*</code> is defining the worker thread for which the background overhead related to receiving parcels should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code> .
Description	Returns the background overhead related to receiving parcels on the given locality since application start. If the instance name is <code>total</code> the counter returns the background overhead for all worker threads (cores) on that locality. If the instance name is <code>worker-thread#*</code> the counter will return background overhead for all worker threads separately. This counter is available only if the configuration time constants <code>HPX_WITH_BACKGROUND_THREAD_COUNTERS</code> (default: <code>OFF</code>) and <code>HPX_WITH_THREAD_IDLE_RATES</code> are set to <code>ON</code> (default: <code>OFF</code>). The unit of measure displayed for this counter is 0.1%. This counter will currently return meaningful values for the MPI parcelport only.

Table 2.86: General performance counter `/runtime/count/component`

Counter type	<code>/runtime/count/component</code>
Counter instance formatting	<code>locality#*/total</code> where: <code>*</code> is the <i>locality</i> id of the <i>locality</i> the number of components should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the overall number of currently active components of the specified type on the given <i>locality</i> .
Parameters	The type of the component. This is the string which has been used while registering the component with <i>HPX</i> , e.g. which has been passed as the second parameter to the macro <code>HPX_REGISTER_COMPONENT</code> .

Table 2.87: General performance counter `/runtime/count/action-invocation`

Counter type	<code>/runtime/count/action-invocation</code>
Counter instance formatting	<code>locality#*/total</code> where: <code>*</code> is the <i>locality</i> id of the <i>locality</i> the number of action invocations should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the overall (local) invocation count of the specified action type on the given <i>locality</i> .
Parameters	The action type. This is the string which has been used while registering the action with <i>HPX</i> , e.g. which has been passed as the second parameter to the macro <code>HPX_REGISTER_ACTION</code> or <code>HPX_REGISTER_ACTION_ID</code> .

Table 2.88: General performance counter `/runtime/count/remote-action-invocation`

Counter type	<code>/runtime/count/remote-action-invocation</code>
Counter instance formatting	<code>locality#*/total</code> where: * is the <i>locality</i> id of the <i>locality</i> the number of action invocations should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the overall (remote) invocation count of the specified action type on the given <i>locality</i> .
Parameters	The action type. This is the string which has been used while registering the action with <i>HPX</i> , e.g. which has been passed as the second parameter to the macro <i>HPX_REGISTER_ACTION</i> or <i>HPX_REGISTER_ACTION_ID</i> .

Table 2.89: General performance counter `/runtime/uptime`

Counter type	<code>/runtime/uptime</code>
Counter instance formatting	<code>locality#*/total</code> where: * is the <i>locality</i> id of the <i>locality</i> the system uptime should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the overall time since application start on the given <i>locality</i> in nanoseconds.

Table 2.90: General performance counter `/runtime/memory/virtual`

Counter type	<code>/runtime/memory/virtual</code>
Counter instance formatting	<code>locality#*/total</code> where: * is the <i>locality</i> id of the <i>locality</i> the allocated virtual memory should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the amount of virtual memory currently allocated by the referenced <i>locality</i> (in bytes).

Table 2.91: General performance counter `/runtime/memory/resident`

Counter type	<code>/runtime/memory/resident</code>
Counter instance formatting	<code>locality#*/total</code> where: * is the <i>locality</i> id of the <i>locality</i> the allocated resident memory should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the amount of resident memory currently allocated by the referenced <i>locality</i> (in bytes).

Table 2.92: General performance counter `/runtime/memory/total`

Counter type	<code>/runtime/memory/total</code>
Counter instance formatting	<code>locality#*/total</code> where: * is the <i>locality</i> id of the <i>locality</i> the total available memory should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> . Note: only supported in Linux.
Description	Returns the total available memory for use by the referenced <i>locality</i> (in bytes). This counter is available on Linux and Windows systems only.

Table 2.93: General performance counter `/runtime/io/read_bytes_issued`

Counter type	<code>/runtime/io/read_bytes_issued</code>
Counter instance formatting	<code>locality#*/total</code> where: * is the <i>locality</i> id of the <i>locality</i> the number of bytes read should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the number of bytes read by the process (aggregate of count arguments passed to <code>read()</code> call or its analogues). This performance counter is available only on systems which expose the related data through the <code>/proc</code> file system.

Table 2.94: General performance counter `/runtime/io/write_bytes_issued`

Counter type	<code>/runtime/io/write_bytes_issued</code>
Counter instance formatting	<code>locality#*/total</code> where: * is the <i>locality</i> id of the <i>locality</i> the number of bytes written should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the number of bytes written by the process (aggregate of count arguments passed to <code>write()</code> call or its analogues). This performance counter is available only on systems which expose the related data through the <code>/proc</code> file system.

Table 2.95: General performance counter `/runtime/io/read_syscalls`

Counter type	<code>/runtime/io/read_syscalls</code>
Counter instance formatting	<code>locality#*/total</code> where: * is the <i>locality</i> id of the <i>locality</i> the number of system calls should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the number of system calls that perform I/O reads. This performance counter is available only on systems which expose the related data through the <code>/proc</code> file system.

Table 2.96: General performance counter `/runtime/io/write_syscalls`

Counter type	<code>/runtime/io/write_syscalls</code>
Counter instance formatting	<code>locality#*/total</code> where: * is the <i>locality</i> id of the <i>locality</i> the number of system calls should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the number of system calls that perform I/O writes. This performance counter is available only on systems which expose the related data through the <code>/proc</code> file system.

Table 2.97: General performance counter `/runtime/io/read_bytes_transferred`

Counter type	<code>/runtime/io/read_bytes_transferred</code>
Counter instance formatting	<code>locality#*/total</code> where: * is the <i>locality</i> id of the <i>locality</i> the number of bytes transferred should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the number of bytes retrieved from storage by I/O operations. This performance counter is available only on systems which expose the related data through the <code>/proc</code> file system.

Table 2.98: General performance counter `/runtime/io/write_bytes_transferred`

Counter type	<code>/runtime/io/write_bytes_transferred</code>
Counter instance formatting	<code>locality#*/total</code> where: * is the <i>locality</i> id of the <i>locality</i> the number of bytes transferred should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the number of bytes retrieved from storage by I/O operations. This performance counter is available only on systems which expose the related data through the <code>/proc</code> file system.

Table 2.99: General performance counter `/runtime/io/write_bytes_cancelled`

Counter type	<code>/runtime/io/write_bytes_cancelled</code>
Counter instance formatting	<code>locality#*/total</code> where: * is the <i>locality</i> id of the <i>locality</i> the number of bytes not being transferred should be queried. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the number of bytes accounted by <code>write_bytes_transferred</code> that has not been ultimately stored due to truncation or deletion. This performance counter is available only on systems which expose the related data through the <code>/proc</code> file system.

Table 2.100: Performance counter `/papi/<papi_event>`

Counter type	<code>/papi/<papi_event></code> where: <papi_event> is the name of the PAPI event to expose as a performance counter (such as <code>PAPI_SR_INS</code>). Note that the list of available PAPI events changes depending on the used architecture. For a full list of available PAPI events and their (short) description use the <code>--hpx:list-counters</code> and <code>--hpx:papi-event-info=all</code> command line options.
Counter instance formatting	<code>locality#*/total</code> or <code>locality#*/worker-thread#*</code> where: <code>locality#*</code> is defining the <i>locality</i> for which the current current accumulated value of all busy-loop counters of all worker threads should be queried. The <i>locality</i> id (given by <code>*</code>) is a (zero based) number identifying the <i>locality</i>. <code>worker-thread#*</code> is defining the worker thread for which the current value of the busy-loop counter should be queried for. The worker thread number (given by the <code>*</code>) is a (zero based) worker thread number (given by the <code>*</code>) is a (zero based) number identifying the worker thread. The number of available worker threads is usually specified on the command line for the application using the option <code>--hpx:threads</code>.
Description	Returns the current count of occurrences of the specified PAPI event. This counter is available only if the configuration time constant <code>HPX_WITH_PAPI</code> is set to <code>ON</code> (default: <code>OFF</code>).

Table 2.101: Performance counter `/statistics/average`

Counter type	<code>/statistics/average</code>
Counter instance formatting	Any full performance counter name. The referenced performance counter is queried at fixed time intervals as specified by the first parameter.
Description	Returns the current average (mean) value calculated based on the values queried from the underlying counter (the one specified as the instance name).
Parameters	Any parameter will be interpreted as a list of up to two comma separated (integer) values, where the first is the time interval (in milliseconds) at which the underlying counter should be queried. If no value is specified, the counter will assume <code>1000</code> [ms] as the default. The second value can be either <code>0</code> or <code>1</code> and specifies whether the underlying counter should be reset during evaluation <code>1</code> or not <code>0</code> . The default value is <code>0</code> .

Table 2.102: Performance counter `/statistics/rolling_average`

Counter type	<code>/statistics/rolling_average</code>
Counter instance formatting	Any full performance counter name. The referenced performance counter is queried at fixed time intervals as specified by the first parameter.
Description	Returns the current rolling average (mean) value calculated based on the values queried from the underlying counter (the one specified as the instance name).
Parameters	Any parameter will be interpreted as a list of up to three comma separated (integer) values, where the first is the time interval (in milliseconds) at which the underlying counter should be queried. If no value is specified, the counter will assume <code>1000</code> [ms] as the default. The second value will be interpreted as the size of the rolling window (the number of latest values to use to calculate the rolling average). The default value for this is <code>10</code> . The third value can be either <code>0</code> or <code>1</code> and specifies whether the underlying counter should be reset during evaluation <code>1</code> or not <code>0</code> . The default value is <code>0</code> .

Table 2.103: Performance counter `/statistics/stddev`

Counter type	<code>/statistics/stddev</code>
Counter instance formatting	Any full performance counter name. The referenced performance counter is queried at fixed time intervals as specified by the first parameter.
Description	Returns the current standard deviation (stddev) value calculated based on the values queried from the underlying counter (the one specified as the instance name).
Parameters	Any parameter will be interpreted as a list of up to two comma separated (integer) values, where the first is the time interval (in milliseconds) at which the underlying counter should be queried. If no value is specified, the counter will assume 1000 [ms] as the default. The second value can be either 0 or 1 and specifies whether the underlying counter should be reset during evaluation 1 or not 0. The default value is 0.

Table 2.104: Performance counter `/statistics/rolling_stddev`

Counter type	<code>/statistics/rolling_stddev</code>
Counter instance formatting	Any full performance counter name. The referenced performance counter is queried at fixed time intervals as specified by the first parameter.
Description	Returns the current rolling variance (stddev) value calculated based on the values queried from the underlying counter (the one specified as the instance name).
Parameters	Any parameter will be interpreted as a list of up to three comma separated (integer) values, where the first is the time interval (in milliseconds) at which the underlying counter should be queried. If no value is specified, the counter will assume 1000 [ms] as the default. The second value will be interpreted as the size of the rolling window (the number of latest values to use to calculate the rolling average). The default value for this is 10. The third value can be either 0 or 1 and specifies whether the underlying counter should be reset during evaluation 1 or not 0. The default value is 0.

Table 2.105: Performance counter `/statistics/median`

Counter type	<code>/statistics/median</code>
Counter instance formatting	Any full performance counter name. The referenced performance counter is queried at fixed time intervals as specified by the first parameter.
Description	Returns the current (statistically estimated) median value calculated based on the values queried from the underlying counter (the one specified as the instance name).
Parameters	Any parameter will be interpreted as a list of up to two comma separated (integer) values, where the first is the time interval (in milliseconds) at which the underlying counter should be queried. If no value is specified, the counter will assume 1000 [ms] as the default. The second value can be either 0 or 1 and specifies whether the underlying counter should be reset during evaluation 1 or not 0. The default value is 0.

Table 2.106: Performance counter `/statistics/max`

Counter type	<code>/statistics/max</code>
Counter instance formatting	Any full performance counter name. The referenced performance counter is queried at fixed time intervals as specified by the first parameter.
Description	Returns the current maximum value calculated based on the values queried from the underlying counter (the one specified as the instance name).
Parameters	Any parameter will be interpreted as a list of up to two comma separated (integer) values, where the first is the time interval (in milliseconds) at which the underlying counter should be queried. If no value is specified, the counter will assume 1000 [ms] as the default. The second value can be either 0 or 1 and specifies whether the underlying counter should be reset during evaluation 1 or not 0. The default value is 0.

Table 2.107: Performance counter `/statistics/rolling_max`

Counter type	<code>/statistics/rolling_max</code>
Counter instance formatting	Any full performance counter name. The referenced performance counter is queried at fixed time intervals as specified by the first parameter.
Description	Returns the current rolling maximum value calculated based on the values queried from the underlying counter (the one specified as the instance name).
Parameters	Any parameter will be interpreted as a list of up to three comma separated (integer) values, where the first is the time interval (in milliseconds) at which the underlying counter should be queried. If no value is specified, the counter will assume 1000 [ms] as the default. The second value will be interpreted as the size of the rolling window (the number of latest values to use to calculate the rolling average). The default value for this is 10. The third value can be either 0 or 1 and specifies whether the underlying counter should be reset during evaluation 1 or not 0. The default value is 0.

Table 2.108: Performance counter `/statistics/min`

Counter type	<code>/statistics/min</code>
Counter instance formatting	Any full performance counter name. The referenced performance counter is queried at fixed time intervals as specified by the first parameter.
Description	Returns the current minimum value calculated based on the values queried from the underlying counter (the one specified as the instance name).
Parameters	Any parameter will be interpreted as a list of up to two comma separated (integer) values, where the first is the time interval (in milliseconds) at which the underlying counter should be queried. If no value is specified, the counter will assume 1000 [ms] as the default. The second value can be either 0 or 1 and specifies whether the underlying counter should be reset during evaluation 1 or not 0. The default value is 0.

Table 2.109: Performance counter `/statistics/rolling_min`

Counter type	<code>/statistics/rolling_min</code>
Counter instance formatting	Any full performance counter name. The referenced performance counter is queried at fixed time intervals as specified by the first parameter.
Description	Returns the current rolling minimum value calculated based on the values queried from the underlying counter (the one specified as the instance name).
Parameters	Any parameter will be interpreted as a list of up to three comma separated (integer) values, where the first is the time interval (in milliseconds) at which the underlying counter should be queried. If no value is specified, the counter will assume 1000 [ms] as the default. The second value will be interpreted as the size of the rolling window (the number of latest values to use to calculate the rolling average). The default value for this is 10. The third value can be either 0 or 1 and specifies whether the underlying counter should be reset during evaluation 1 or not 0. The default value is 0.

Table 2.110: Performance counter `/arithmetics/add`

Counter type	<code>/arithmetics/add</code>
Description	Returns the sum calculated based on the values queried from the underlying counters (the ones specified as the parameters).
Parameters	The parameter will be interpreted as a comma separated list of full performance counter names which are queried whenever this counter is accessed. Any wildcards in the counter names will be expanded.

Table 2.111: Performance counter `/arithmetics/subtract`

Counter type	<code>/arithmetics/subtract</code>
Description	Returns the difference calculated based on the values queried from the underlying counters (the ones specified as the parameters).
Parameters	The parameter will be interpreted as a comma separated list of full performance counter names which are queried whenever this counter is accessed. Any wildcards in the counter names will be expanded.

Table 2.112: Performance counter `/arithmetics/multiply`

Counter type	<code>/arithmetics/multiply</code>
Description	Returns the product calculated based on the values queried from the underlying counters (the ones specified as the parameters).
Parameters	The parameter will be interpreted as a comma separated list of full performance counter names which are queried whenever this counter is accessed. Any wildcards in the counter names will be expanded.

Table 2.113: Performance counter `/arithmetics/divide`

Counter type	<code>/arithmetics/divide</code>
Description	Returns the result of division of the values queried from the underlying counters (the ones specified as the parameters).
Parameters	The parameter will be interpreted as a comma separated list of full performance counter names which are queried whenever this counter is accessed. Any wildcards in the counter names will be expanded.

Table 2.114: Performance counter `/arithmetics/mean`

Counter type	<code>/arithmetics/mean</code>
Description	Returns the average value of all values queried from the underlying counters (the ones specified as the parameters).
Parameters	The parameter will be interpreted as a comma separated list of full performance counter names which are queried whenever this counter is accessed. Any wildcards in the counter names will be expanded.

Table 2.115: Performance counter `/arithmetics/variance`

Counter type	<code>/arithmetics/variance</code>
Description	Returns the standard deviation of all values queried from the underlying counters (the ones specified as the parameters).
Parameters	The parameter will be interpreted as a comma separated list of full performance counter names which are queried whenever this counter is accessed. Any wildcards in the counter names will be expanded.

Table 2.116: Performance counter `/arithmetics/median`

Counter type	<code>/arithmetics/median</code>
Description	Returns the median value of all values queried from the underlying counters (the ones specified as the parameters).
Parameters	The parameter will be interpreted as a comma separated list of full performance counter names which are queried whenever this counter is accessed. Any wildcards in the counter names will be expanded.

Table 2.117: Performance counter `/arithmetics/min`

Counter type	<code>/arithmetics/min</code>
Description	Returns the minimum value of all values queried from the underlying counters (the ones specified as the parameters).
Parameters	The parameter will be interpreted as a comma separated list of full performance counter names which are queried whenever this counter is accessed. Any wildcards in the counter names will be expanded.

Table 2.118: Performance counter `/arithmetics/max`

Counter type	<code>/arithmetics/max</code>
Description	Returns the maximum value of all values queried from the underlying counters (the ones specified as the parameters).
Parameters	The parameter will be interpreted as a comma separated list of full performance counter names which are queried whenever this counter is accessed. Any wildcards in the counter names will be expanded.

Table 2.119: Performance counter `/arithmetics/count`

Counter type	<code>/arithmetics/count</code>
Description	Returns the count value of all values queried from the underlying counters (the ones specified as the parameters).
Parameters	The parameter will be interpreted as a comma separated list of full performance counter names which are queried whenever this counter is accessed. Any wildcards in the counter names will be expanded.

Note: The `/arithmetics` counters can consume an arbitrary number of other counters. For this reason those have to be specified as parameters (a comma separated list of counters appended after a '@'). For instance:

```
$ ./bin/hello_world_distributed -t2 \
  --hpx:print-counter=/threads{locality#0/worker-thread#*}/count/cumulative \
  --hpx:print-counter=/arithmetics/add@/threads{locality#0/worker-thread#*}/count/
↪cumulative
hello world from OS-thread 0 on locality 0
hello world from OS-thread 1 on locality 0
/threads{locality#0/worker-thread#0}/count/cumulative,1,0.515640,[s],25
/threads{locality#0/worker-thread#1}/count/cumulative,1,0.515520,[s],36
/arithmetics/add@/threads{locality#0/worker-thread#*}/count/cumulative,1,0.516445,[s],64
```

Since all wildcards in the parameters are expanded, this example is fully equivalent to specifying both counters separately to `/arithmetics/add`:

```
$ ./bin/hello_world_distributed -t2 \
  --hpx:print-counter=/threads{locality#0/worker-thread#*}/count/cumulative \
  --hpx:print-counter=/arithmetics/add@
  /threads{locality#0/worker-thread#0}/count/cumulative,\
  /threads{locality#0/worker-thread#1}/count/cumulative
```


Table 2.120: Performance counter /coalescing/count/parcels

Counter type	/coalescing/count/parcels
Counter instance formatting	locality#*/total where: * is the <i>locality</i> id of the <i>locality</i> the number of parcels for the given action should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the number of parcels handled by the message handler associated with the action which is given by the counter parameter.
Parameters	The action type. This is the string which has been used while registering the action with <i>HPX</i> , e.g. which has been passed as the second parameter to the macro <i>HPX_REGISTER_ACTION</i> or <i>HPX_REGISTER_ACTION_ID</i> .

Table 2.121: Performance counter /coalescing/count/messages

Counter type	/coalescing/count/messages
Counter instance formatting	locality#*/total where: * is the <i>locality</i> id of the <i>locality</i> the number of messages for the given action should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the number of messages generated by the message handler associated with the action which is given by the counter parameter.
Parameters	The action type. This is the string which has been used while registering the action with <i>HPX</i> , e.g. which has been passed as the second parameter to the macro <i>HPX_REGISTER_ACTION</i> or <i>HPX_REGISTER_ACTION_ID</i> .

Table 2.122: Performance counter /coalescing/count/average-parcels-per-message

Counter type	/coalescing/count/average-parcels-per-message
Counter instance formatting	locality#*/total where: * is the <i>locality</i> id of the <i>locality</i> the number of messages for the given action should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the average number of parcels sent in a message generated by the message handler associated with the action which is given by the counter parameter.
Parameters	The action type. This is the string which has been used while registering the action with <i>HPX</i> , e.g. which has been passed as the second parameter to the macro <i>HPX_REGISTER_ACTION</i> or <i>HPX_REGISTER_ACTION_ID</i> .

Table 2.123: Performance counter /coalescing/time/average-parcel-arrival

Counter type	/coalescing/time/average-parcel-arrival
Counter instance formatting	locality#*/total where: * is the <i>locality</i> id of the <i>locality</i> the average time between parcels for the given action should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns the average time between arriving parcels for the action which is given by the counter parameter.
Parameters	The action type. This is the string which has been used while registering the action with <i>HPX</i> , e.g. which has been passed as the second parameter to the macro <i>HPX_REGISTER_ACTION</i> or <i>HPX_REGISTER_ACTION_ID</i> .

Table 2.124: Performance counter /coalescing/time/parcel-arrival-histogram

Counter type	/coalescing/time/parcel-arrival-histogram
Counter instance formatting	locality#*/total where: * is the <i>locality</i> id of the <i>locality</i> the average time between parcels for the given action should be queried for. The <i>locality</i> id is a (zero based) number identifying the <i>locality</i> .
Description	Returns a histogram representing the times between arriving parcels for the action which is given by the counter parameter. This counter returns an array of values, where the first three values represent the three parameters used for the histogram followed by one value for each of the histogram buckets. The first unit of measure displayed for this counter [ns] refers to the lower and upper boundary values in the returned histogram data only. The second unit of measure displayed [0.1%] refers to the actual histogram data. For each bucket the counter shows a value between 0 and 1000 which corresponds to a percentage value between 0% and 100%.
Parameters	The action type and optional histogram parameters. The action type is the string which has been used while registering the action with <i>HPX</i> , e.g. which has been passed as the second parameter to the macro <i>HPX_REGISTER_ACTION</i> or <i>HPX_REGISTER_ACTION_ID</i> . The action type may be followed by a comma separated list of up-to three numbers: the lower and upper boundaries for the collected histogram, and the number of buckets for the histogram to generate. By default these three numbers will be assumed to be 0 ([ns], lower bound), 1000000 ([ns], upper bound), and 20 (number of buckets to generate).

Note: The performance counters related to *parcel* coalescing are available only if the configuration time constant *HPX_WITH_PARCEL_COALESCING* is set to ON (default: ON). However, even in this case it will be available only for actions that are enabled for parcel coalescing (see the macros *HPX_ACTION_USES_MESSAGE_COALESCING* and *HPX_ACTION_USES_MESSAGE_COALESCING_NOTHROW*).

APEX integration

HPX provides integration with *APEX*¹⁸², which is a framework for application profiling using task timers and various performance counters Huck *et al.*¹⁹⁰. It can be added as a git submodule by turning on the option *HPX_WITH_APEX:BOOL* during *CMake*¹⁸³ configuration. *TAU*¹⁸⁴ is an optional dependency when using *APEX*.

To build *HPX* with *APEX*¹⁸⁵, add *HPX_WITH_APEX=ON*, and, optionally, *Tau_ROOT=\$PATH_TO_TAU* to your *CMake*¹⁸⁶ configuration. In addition, you can override the tag used for *APEX*¹⁸⁷ with the *HPX_WITH_APEX_TAG* option. Please see the *APEX HPX documentation*¹⁸⁸ for detailed instructions on using *APEX*¹⁸⁹ with *HPX*.

¹⁸² <http://uo-oaciss.github.io/apex>

¹⁹⁰ K. A. Huck, A. Porterfield, N Chaimov, H. Kaiser, A. D. Malony, T. Sterling, and R. Fowler. *An autonomic performance environment for exascale*. Supercomputing Frontiers and Innovations, 2015.

¹⁸³ <https://www.cmake.org>

¹⁸⁴ <https://www.cs.uoregon.edu/research/tau/home.php>

¹⁸⁵ <http://uo-oaciss.github.io/apex>

¹⁸⁶ <https://www.cmake.org>

¹⁸⁷ <http://uo-oaciss.github.io/apex>

¹⁸⁸ <https://uo-oaciss.github.io/apex/usage/#hpx-louisiana-state-university>

¹⁸⁹ <http://uo-oaciss.github.io/apex>

References

2.4.16 Using the LCI parcelport

Basic information

The [Lightweight Communication Interface](#)¹⁹¹ (LCI) is an ongoing research project aiming to provide efficient support for applications with irregular and asynchronous communication patterns such as graph analysis, sparse linear algebra, and task-based runtime on modern parallel architectures. Its features include (a) support for more communication primitives such as two-sided send/rcv and one-sided (dynamic or direct) remote put/get (b) better multi-threaded performance (c) explicit user control of communication resource (d) flexible signaling mechanisms such as synchronizer, completion queue, and active message handler. It is designed to be a low-level communication library used by high-level libraries and frameworks.

The LCI parcelport is an experimental parcelport. It aims to provide the best possible communication performance on high-performance computation platforms. Compared to the MPI parcelport, it uses much fewer messages and memory copies to transfer an *HPX* parcel over the network. Its message transmission path involves minimum synchronization points and is almost lock-free. It is expected to be much faster than the MPI parcelport.

Build *HPX* with the LCI parcelport

While building *HPX*, you can specify a set of [CMake](#)¹⁹² variables to enable and configure the LCI parcelport. Below, there is a set of the most important and frequently used CMake variables.

HPX_WITH_PARCELPORT_LCI

Enable the LCI parcelport. This enables the use of LCI for networking operations in the *HPX* runtime. The default value is OFF because it's not available on all systems and/or requires another dependency. However, this experimental parcelport may provide better performance than the MPI parcelport. You must set this variable to ON in order to use the LCI parcelport. All the following variables only make sense when this variable is set to ON.

HPX_WITH_FETCH_LCI

Use FetchContent to fetch LCI. The default value is OFF. If this option is set to OFF. You need to install your own LCI library and *HPX* will try to find it using [CMake](#)¹⁹³ `find_package`. You can specify the location of the LCI installation by the environmental variable `LCI_ROOT`. Refer to the [LCI README](#)¹⁹⁴ for how to install LCI. If this option is set to ON, *HPX* will fetch and build LCI for you. You can use the following [CMake](#)¹⁹⁵ variables to configure this behavior for your platform.

HPX_WITH_LCI_TAG

This variable only takes effect when `HPX_WITH_FETCH_LCI` is set to ON and `FETCHCONTENT_SOURCE_DIR_LCI` is not set. *HPX* will fetch LCI from its github repository. This variable controls the branch/tag LCI will be fetched.

FETCHCONTENT_SOURCE_DIR_LCI

This variable only takes effect when `HPX_WITH_FETCH_LCI` is set to ON. When it is defined, `HPX_WITH_LCI_TAG` will be ignored. It accepts a path to a local version of LCI source code and *HPX* will fetch and build LCI from there. The default value is set conservatively for the stability of *HPX*, but users are welcome to set this variable to `master` for potentially better performance.

¹⁹¹ <https://github.com/uiuc-hpc/lci>

¹⁹² <https://www.cmake.org>

¹⁹³ <https://www.cmake.org>

¹⁹⁴ <https://github.com/uiuc-hpc/lci#readme>

¹⁹⁵ <https://www.cmake.org>

Run HPX with the LCI parcelport

We use the same mechanisms as MPI to launch LCI, so you can use the same way you run MPI parcelport to run LCI parcelport. Typically, it would be `hpxrun.py`, `mpirun`, or `srun`.

`hpxrun.py` serves as a wrapper for `mpirun` and `srun`. If you are using `hpxrun.py`, pass `-p lci` to the scripts. You also need to pass either `-r mpi` or `-r srun` to select the correct run wrapper according to the platform.

If you are using `mpirun` or `srun`, you can just pass `--hpx:ini=hpx.parcels.lci.priority=1000`, `--hpx:ini=hpx.parcels.lci.enable=1`, and `--hpx:ini=hpx.parcels.bootstrap=lci` to the HPX applications.

The `hpxrun.py` argument `-r none` (the default option for the run wrapper) and its corresponding HPX arguments `--hpx:hpx` and `--hpx:agas` do not work for the MPI or the LCI parcelport.

Performance tuning of the LCI parcelport

We encourage users to set the following environmental variables when using the LCI parcelport to get better performance.

```
$ export LCI_SERVER_MAX_SENDS=1024
$ export LCI_SERVER_MAX_RECVS=4096
$ export LCI_SERVER_NUM_PKTS=65536
$ export LCI_SERVER_MAX_CQES=65536
$ export LCI_PACKET_SIZE=12288
```

This setting needs roughly 800MB memory per process. The memory consumption mainly comes from the packets, which can be calculated using $LCI_SERVER_NUM_PKTS \times LCI_PACKET_SIZE$.

In addition, users can tune the following command-line options when using the LCI parcelport to get better performance.

--hpx:ini=hpx.parcels.lci.ndevices=<int>

The number of LCI devices to use. The default value is 2. An LCI device represents a collection of network resources. More devices lead to lower thread contention, but too many devices may lead to load imbalance or hardware overhead.

--hpx:ini=hpx.parcels.lci.progress_type=<worker|rp>

The way to progress the LCI device. The default value is `worker`. The `worker` option uses all worker threads to progress the LCI devices. The `rp` option uses dedicated pinned threads to progress the LCI devices. Normally, the `worker` option gives better performance, but the `rp` option has been observed with better performance on some clusters with prior generation of InfiniBand hardware.

Reference

Yan, Jiakun, Hartmut Kaiser, and Marc Snir. *Understanding the Communication Needs of Asynchronous Many-Task Systems—A Case Study of HPX+ LCI*. arXiv preprint, 2025.

Yan, Jiakun, and Marc Snir. *LCI: a Lightweight Communication Interface for Efficient Asynchronous Multithreaded Communication*. The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC), 2025.

2.4.17 Using the LCW parcelport

Basic information

The LCW parcelport is an advanced MPI parcelport that can utilize the MPICH VCI and Continuation extensions. The exact MPI implementation is wrapped in a thin wrapper layer (the Lightweight Communication Wrapper, or LCW). This wrapper provides an active message, send/recv, completion queue, and device abstraction on top of MPI (with/without extensions), GASNet-EX, and LCI. The GASNet-EX backend of LCW cannot be used in *HPX* for now, as it lacks send/recv support.

Build *HPX* with the LCW parcelport

While building *HPX*, you can specify a set of [CMake](https://www.cmake.org)¹⁹⁶ variables to enable and configure the LCW parcelport. Below, there is a set of the most important and frequently used CMake variables.

HPX_WITH_PARCELPORT_LCW

Enable the LCW parcelport. This enables the use of LCW for networking operations in the *HPX* runtime. The default value is OFF because it's not available on all systems and/or requires another dependency. You must set this variable to ON in order to use the LCW parcelport. All the following variables only make sense when this variable is set to ON.

HPX_WITH_FETCH_LCW

Use FetchContent to fetch LCW. The default value is OFF. If this option is set to OFF. You need to install your own LCW library and *HPX* will try to find it using [CMake](https://www.cmake.org)¹⁹⁷ `find_package`. You can specify the location of the LCW installation by the environmental variable `LCW_ROOT`. Refer to the [LCW README](https://github.com/JiakunYan/lcw#readme)¹⁹⁸ for how to install LCW. If this option is set to ON. *HPX* will fetch and build LCW for you. You can use the following [CMake](https://www.cmake.org)¹⁹⁹ variables to configure this behavior for your platform.

HPX_WITH_LCW_TAG

This variable only takes effect when `HPX_WITH_FETCH_LCW` is set to ON and `FETCHCONTENT_SOURCE_DIR_LCW` is not set. *HPX* will fetch LCW from its github repository. This variable controls the branch/tag LCW will be fetched.

FETCHCONTENT_SOURCE_DIR_LCW

This variable only takes effect when `HPX_WITH_FETCH_LCW` is set to ON. When it is defined, `HPX_WITH_LCW_TAG` will be ignored. It accepts a path to a local version of LCW source code and *HPX* will fetch and build LCW from there.

Run *HPX* with the LCW parcelport

You can configure runtime behavior with the following options.

--hpx:ini=hpx.parcel.lcw.ndevices=<n>

The number of LCW devices to use (default: 2). Each LCW device maps to an MPI communicator or an LCI device. When the MPICH VCI extension is enabled, each MPI communicator will be mapped to a dedicated collection of network resources. More devices lead to lower thread contention, but too many devices may lead to load imbalance or hardware overhead.

¹⁹⁶ <https://www.cmake.org>

¹⁹⁷ <https://www.cmake.org>

¹⁹⁸ <https://github.com/JiakunYan/lcw#readme>

¹⁹⁹ <https://www.cmake.org>

Reference

Yan, Jiakun, Marc Snir, and Yanfei Guo. *Examining MPI and its Extensions for Asynchronous Multithreaded Communication*. In *European MPI Users' Group Meeting*, pp. 122-142. Cham: Springer Nature Switzerland, 2025.

2.4.18 HPX runtime and resources

HPX thread scheduling policies

The HPX runtime has six thread scheduling policies: local-priority, static-priority, local, static, local-workrequesting-fifo, and abp-priority. These policies can be specified from the command line using the command line option `--hpx:queuing`. In order to use a particular scheduling policy, the runtime system must be built with the appropriate scheduler flag turned on (e.g. `cmake -DHPX_THREAD_SCHEDULERS=local`, see *CMake options* for more information).

Priority local scheduling policy (default policy)

The priority local scheduling policy maintains one queue per operating system (OS) thread. The OS thread pulls its work from this queue. By default the number of high priority queues is equal to the number of OS threads; the number of high priority queues can be specified on the command line using `--hpx:high-priority-threads`. High priority threads are executed by any of the OS threads before any other work is executed. When a queue is empty, work will be taken from high priority queues first. There is one low priority queue from which threads will be scheduled only when there is no other work.

For this scheduling policy there is an option to turn on NUMA sensitivity using the command line option `--hpx:numa-sensitive`. When NUMA sensitivity is turned on, work stealing is done from queues associated with the same NUMA domain first, only after that work is stolen from other NUMA domains.

This scheduler is enabled at build time by default using the FIFO (first-in-first-out) queueing policy. This policy can be invoked using `--hpx:queuinglocal-priority-fifo`. The scheduler can also be enabled using the LIFO (last-in-first-out) policy. This is not the default policy and must be invoked using the command line option `--hpx:queuinglocal-priority-lifo`.

Static priority scheduling policy

- invoke using: `--hpx:queuingstatic-priority` (or `-qs`)

The static scheduling policy maintains one queue per OS thread from which each OS thread pulls its tasks (user threads). Threads are distributed in a round robin fashion. There is no thread stealing in this policy.

Local scheduling policy

- invoke using: `--hpx:queuinglocal` (or `-ql`)
- flag to turn on for build: `HPX_THREAD_SCHEDULERS=all` or `HPX_THREAD_SCHEDULERS=local`

The local scheduling policy maintains one queue per OS thread from which each OS thread pulls its tasks (user threads).

Static scheduling policy

- invoke using: `--hpx:queuingstatic`
- flag to turn on for build: `HPX_THREAD_SCHEDULERS=all` or `HPX_THREAD_SCHEDULERS=static`

The static scheduling policy maintains one queue per OS thread from which each OS thread pulls its tasks (user threads). Threads are distributed in a round robin fashion. There is no thread stealing in this policy.

Priority ABP scheduling policy

- invoke using: `--hpx:queuingabp-priority-fifo`
- flag to turn on for build: `HPX_THREAD_SCHEDULERS=all` or `HPX_THREAD_SCHEDULERS=abp-priority`

Priority ABP policy maintains a double ended lock free queue for each OS thread. By default the number of high priority queues is equal to the number of OS threads; the number of high priority queues can be specified on the command line using `--hpx:high-priority-threads`. High priority threads are executed by the first OS threads before any other work is executed. When a queue is empty work will be taken from high priority queues first. There is one low priority queue from which threads will be scheduled only when there is no other work. For this scheduling policy there is an option to turn on NUMA sensitivity using the command line option `--hpx:numa-sensitive`. When NUMA sensitivity is turned on work stealing is done from queues associated with the same NUMA domain first, only after that work is stolen from other NUMA domains.

This scheduler can be used with two underlying queuing policies (FIFO: first-in-first-out, and LIFO: last-in-first-out). In order to use the LIFO policy use the command line option `--hpx:queuingabp-priority-lifo`.

Work requesting scheduling policies

- invoke using: `--hpx:queuinglocal-workrequesting-fifo`, using `--hpx:queuinglocal-workrequesting-lifo`, or using `--hpx:queuinglocal-workrequesting-mc`

The work-requesting policies rely on a different mechanism of balancing work between cores (compared to the other policies listed above). Instead of actively trying to steal work from other cores, requesting work relies on a less disruptive mechanism. If a core runs out of work, instead of actively looking at the queues of neighboring cores, in this case a request is posted to another core. This core now (whenever it is not busy with other work) either responds to the original core by sending back work or passes the request on to the next possible core in the system. In general, this scheme avoids contention on the work queues as those are always accessed by their own cores only.

The HPX resource partitioner

The HPX resource partitioner lets you take the execution resources available on a system—processing units, cores, and numa domains—and assign them to thread pools. By default HPX creates a single thread pool name `default`. While this is good for most use cases, the resource partitioner lets you create multiple thread pools with custom resources and options.

Creating custom thread pools is useful for cases where you have tasks which absolutely need to run without interference from other tasks. An example of this is when using MPI²⁰⁰ for distribution instead of the built-in mechanisms in HPX (useful in legacy applications). In this case one can create a thread pool containing a single thread for MPI communication. MPI tasks will then always run on the same thread, instead of potentially being stuck in a queue behind other threads.

²⁰⁰ https://en.wikipedia.org/wiki/Message_Passing_Interface

Note that *HPX* thread pools are completely independent from each other in the sense that task stealing will never happen between different thread pools. However, tasks running on a particular thread pool can schedule tasks on another thread pool.

Note: It is simpler in some situations to schedule important tasks with high priority instead of using a separate thread pool.

Using the resource partitioner

The `hpx::resource::partitioner` is now created during *HPX* runtime initialization without explicit action needed from the user. To specify some of the initialization parameters you can use the `hpx::init_params`.

The resource partitioner callback is the interface to add thread pools to the *HPX* runtime and to assign resources to the thread pools. In order to create custom thread pools you can specify the resource partitioner callback `hpx::init_params::rp_callback` which will be called once the resource partitioner will be created, see the example below. You can also specify other parameters, see `hpx::init_params`.

To add a thread pool use the `hpx::resource::partitioner::create_thread_pool` method. If you simply want to use the default scheduler and scheduler options, it is enough to call `rp.create_thread_pool("my-thread-pool")`.

Then, to add resources to the thread pool you can use the `hpx::resource::partitioner::add_resource` method. The resource partitioner exposes the hardware topology retrieved using [Portable Hardware Locality \(HWLOC\)](#)²⁰¹ and lets you iterate through the topology to add the wanted processing units to the thread pool. Below is an example of adding all processing units from the first NUMA domain to a custom thread pool, unless there is only one NUMA domain in which case we leave the first processing unit for the default thread pool:

Note: Whatever processing units are not assigned to a thread pool by the time `hpx::init` is called will be added to the default thread pool. It is also possible to explicitly add processing units to the default thread pool, and to create the default thread pool manually (in order to e.g. set the scheduler type).

Tip: The command line option `--hpx:print-bind` is useful for checking that the thread pools have been set up the way you expect.

Difference between the old and new version

In the old version, you had to create an instance of the `resource_partitioner` with `argc` and `argv`.

```
int main(int argc, char** argv)
{
    hpx::resource::partitioner rp(argc, argv);
    hpx::init();
}
```

From *HPX* 1.5.0 onwards, you just pass `argc` and `argv` to `hpx::init()` or `hpx::start()` for the binding options to be parsed by the resource partitioner.

²⁰¹ <https://www.open-mpi.org/projects/hwloc/>


```
int main(int argc, char** argv)
{
    hpx::init_params init_args;
    hpx::init(argc, argv, init_args);
}
```

In the old version, when creating a custom thread pool, you just called the utilities on the resource partitioner instantiated previously.

```
int main(int argc, char** argv)
{
    hpx::resource::partitioner rp(argc, argv);

    rp.create_thread_pool("my-thread-pool");

    bool one_numa_domain = rp.numa_domains().size() == 1;
    bool skipped_first_pu = false;

    hpx::resource::numa_domain const& d = rp.numa_domains()[0];

    for (const hpx::resource::core& c : d.cores())
    {
        for (const hpx::resource::pu& p : c.pus())
        {
            if (one_numa_domain && !skipped_first_pu)
            {
                skipped_first_pu = true;
                continue;
            }

            rp.add_resource(p, "my-thread-pool");
        }
    }

    hpx::init();
}
```

You now specify the resource partitioner callback which will tie the resources to the resource partitioner created during runtime initialization.

```
void init_resource_partitioner_handler(hpx::resource::partitioner& rp)
{
    rp.create_thread_pool("my-thread-pool");

    bool one_numa_domain = rp.numa_domains().size() == 1;
    bool skipped_first_pu = false;

    hpx::resource::numa_domain const& d = rp.numa_domains()[0];

    for (const hpx::resource::core& c : d.cores())
    {
        for (const hpx::resource::pu& p : c.pus())
        {
```

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```
        if (one_numa_domain && !skipped_first_pu)
        {
            skipped_first_pu = true;
            continue;
        }

        rp.add_resource(p, "my-thread-pool");
    }
}

int main(int argc, char* argv[])
{
    hpx::init_params init_args;
    init_args.rp_callback = &init_resource_partitioner_handler;

    hpx::init(argc, argv, init_args);
}
```

Advanced usage

It is possible to customize the built in schedulers by passing scheduler options to `hpx::resource::partitioner::create_thread_pool`. It is also possible to create and use custom schedulers.

Note: It is not recommended to create your own scheduler. The *HPX* developers use this to experiment with new scheduler designs before making them available to users via the standard mechanisms of choosing a scheduler (command line options). If you would like to experiment with a custom scheduler the resource partitioner example `shared_priority_queue_scheduler.cpp` contains a fully implemented scheduler with logging, etc. to make exploration easier.

To choose a scheduler and custom mode for a thread pool, pass additional options when creating the thread pool like this:

```
rp.create_thread_pool("my-thread-pool",
    hpx::resource::policies::local_priority_lifo,
    hpx::policies::scheduler_mode(
        hpx::policies::scheduler_mode::default |
        hpx::policies::scheduler_mode::enable_elasticity));
```

The available schedulers are documented here: `hpx::resource::scheduling_policy`, and the available scheduler modes here: `hpx::threads::policies::scheduler_mode`. Also see the examples folder for examples of advanced resource partitioner usage: `simple_resource_partitioner.cpp` and `oversubscribing_resource_partitioner.cpp`.

2.4.19 Executors

Executors in *HPX* provide a flexible way to control **how, when, and where** tasks are executed. Instead of manually creating threads or managing thread pools, you can hand your tasks to an executor, and it takes care of the details of running them.

This page introduces the concept of executors, the main types available in *HPX*, and how to create custom executors.

What is an executor?

An **executor** is an abstraction that separates the *what* from the *how* of task execution:

- **What:** the work to be performed (e.g. a function or task).
- **How:** whether the task runs synchronously, asynchronously, or in parallel across multiple cores.

By using executors, you can switch between execution strategies without rewriting your algorithms. *HPX* provides a rich set of executors with a unified API.

Main Executors in *HPX*

HPX provides multiple executors. Below are three of the most commonly used:

Parallel Executor

- Default in *HPX*.
- Creates a new HPX thread for every scheduled task.
- Works well for large tasks, but frequent small tasks incur overhead due to thread creation/destruction.

```
#include <hpx/execution.hpp>
#include <hpx/algorithm.hpp>
#include <vector>

std::vector<int> data(100, 1);

hpx::for_each(
    hpx::execution::par,
    data.begin(), data.end(),
    [](int &x) {
        x += 1;
    }
);
```

Fork-Join Executor

- Spawns one thread per CPU core when created.
- Threads are reused across tasks, avoiding repeated creation costs.
- Efficient for workloads with many sequential parallel regions.
- May waste resources if parallel work is limited.

```
#include <hpx/execution.hpp>
#include <hpx/algorithm.hpp>
#include <vector>

std::vector<int> data(100, 1);

hpx::execution::experimental::fork_join_executor exec;

hpx::for_each(
    hpx::execution::par.on(exec),
    data.begin(), data.end(),
    [](int &x) {
        x += 1;
    }
);
```

Sequential Executor

- Executes all tasks synchronously on the calling thread.
- Useful for debugging (deterministic execution) or when parallelism brings no benefit.
- Maintains the same executor-based API while running tasks sequentially.

```
#include <hpx/execution.hpp>
#include <hpx/algorithm.hpp>
#include <vector>

std::vector<int> data(100, 1);

hpx::execution::sequenced_executor seq_exec;

hpx::for_each(
    hpx::execution::seq.on(seq_exec),
    data.begin(), data.end(),
    [](int &x) { x += 1; }
);
```

Executors in real-world applications

A practical example is the **LULESH** mini-application (a hydrodynamics benchmark).

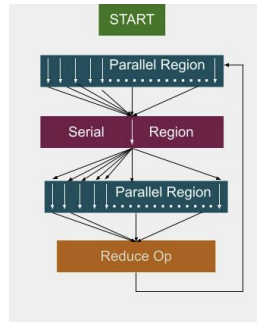


Fig. 2.11: Parallel executor: creates and destroys threads for each loop.

- With the **parallel executor**, each parallel loop creates and destroys new threads, introducing overhead.
- With the **fork-join executor**, threads are created once and reused across loops, reducing overhead and improving performance.

In studies, the fork-join executor achieved significant speedups, in some cases more than twice as fast as traditional OpenMP implementations.

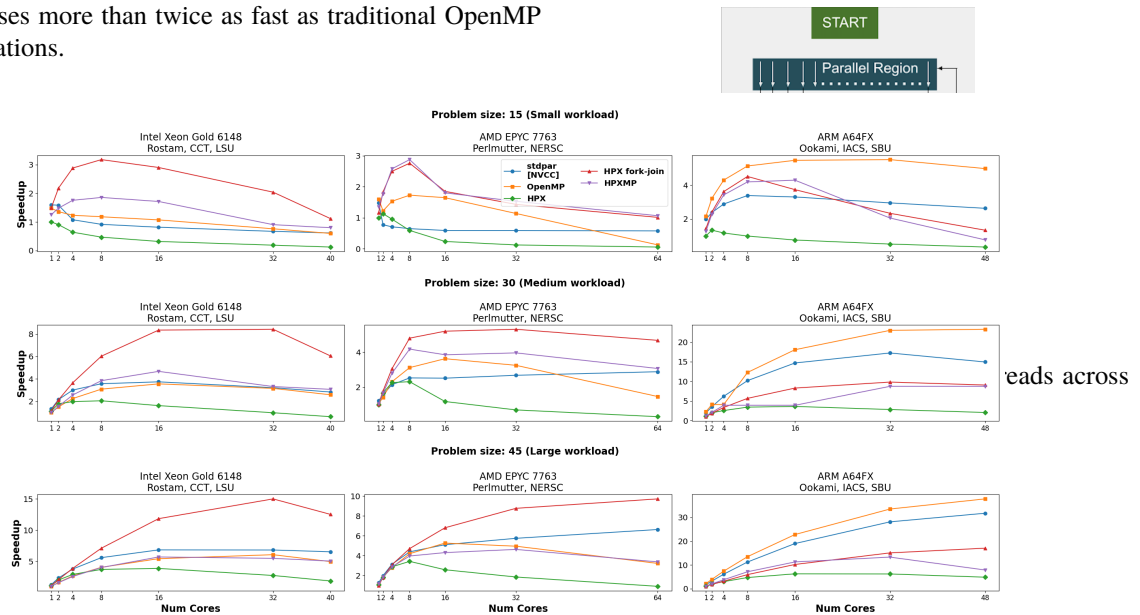


Fig. 2.13: Performance comparison between executors in the LULESH benchmark.

Custom executors

While *HPX* provides a variety of built-in executors, you may sometimes need to adapt task execution to your own requirements. This is where **custom executors** come in. By writing a small wrapper around an existing executor, you can extend its behavior—for example, to add logging, profiling information, or special scheduling rules—while still taking advantage of the *HPX* executor API.

Custom annotating executor

The following example shows how to implement a simple executor that annotates tasks with a string label for easier debugging and profiling.

Note: Annotations do not affect how tasks run or what results they produce. Their main purpose is to give human-readable names to tasks so that they can be identified in profilers, and debuggers.

Full example code

```
#include <hpx/hpx_main.hpp>
#include <hpx/include/parallel_executors.hpp>
#include <hpx/include/async.hpp>
#include <hpx/execution.hpp>
#include <hpx/modules/async_base.hpp>
#include <hpx/modules/threading_base.hpp>

#include <iostream>
#include <string>
#include <utility>

template <typename BaseExecutor>
struct simple_annotating_executor
{
    BaseExecutor base_;
    const char* annotation_;

    simple_annotating_executor(BaseExecutor exec, const char* ann)
        : base_(std::move(exec)), annotation_(ann)
    {}

    // Non-blocking one-way executor
    template <typename F, typename... Ts>
    friend void tag_invoke(hpx::parallel::execution::post_t,
                          simple_annotating_executor const& exec,
                          F&& f, Ts&&... ts)
    {
        hpx::post(
```

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```

        hpx::annotated_function(std::forward<F>(f), exec.annotation_),
        std::forward<Ts>(ts)...);
    }

    // Synchronous execution
    template <typename F, typename... Ts>
    friend auto tag_invoke(hpx::parallel::execution::sync_execute_t,
        simple_annotating_executor const& exec,
        F&& f, Ts&&... ts)
    {
        return hpx::parallel::execution::sync_execute(
            exec.base_,
            hpx::annotated_function(std::forward<F>(f), exec.annotation_),
            std::forward<Ts>(ts)...);
    }
};

// Example functions
int compute_square(int x)
{
    std::cout << "[sync_execute] Running task with annotation\n";
    return x * x;
}

void say_hello()
{
    std::cout << "[post] Running task with annotation\n";
}

int main()
{
    simple_annotating_executor exec(hpx::execution::parallel_executor{}, "my_custom_task
↪");

    // Synchronous execution
    int result = hpx::parallel::execution::sync_execute(exec, &compute_square, 7);
    std::cout << "Result from sync_execute: " << result << "\n";

    // Post a task
    hpx::parallel::execution::post(exec, &say_hello);

    return 0;
}

```

Explanation

The first lines pull in the necessary HPX headers for executors, asynchronous execution, and annotated functions. The key one here is `hpx/threading_base/annotated_function.hpp`, which provides the facility to tag tasks with a string label. We then define a `simple_annotating_executor` that wraps another executor and associates an annotation string with every task:

```
template <typename BaseExecutor>
struct simple_annotating_executor
{
    BaseExecutor base_;
    const char* annotation_;

    simple_annotating_executor(BaseExecutor exec, const char* ann)
        : base_(std::move(exec)), annotation_(ann)
    {}
};
```

The `post` customization schedules a task to run asynchronously. We wrap the task in `hpx::annotated_function` so that it carries the annotation. Executors in HPX customize these operations through `tag_invoke` overloads, which are selected by special tag objects like `post_t` and `sync_execute_t`. This is why the executor interface may look different from a normal member function API.

```
template <typename F, typename... Ts>
friend void tag_invoke(hpx::parallel::execution::post_t,
                      simple_annotating_executor const& exec,
                      F&& f, Ts&&... ts)
{
    hpx::post(
        hpx::annotated_function(std::forward<F>(f), exec.annotation_),
        std::forward<Ts>(ts)...);
}
```

The `sync_execute` customization runs a task immediately and returns the result. Again, we wrap the function with an annotation before executing. The key difference is that `post` schedules a task in a fire-and-forget style (no result is returned), while `sync_execute` blocks until the task finishes and gives you the result back.

```
template <typename F, typename... Ts>
friend auto tag_invoke(hpx::parallel::execution::sync_execute_t,
                      simple_annotating_executor const& exec,
                      F&& f, Ts&&... ts)
{
    return hpx::parallel::execution::sync_execute(
        exec.base_,
        hpx::annotated_function(std::forward<F>(f), exec.annotation_),
        std::forward<Ts>(ts)...);
}
```

Note how we delegate the actual execution to `exec.base_`, the underlying executor. This makes the custom executor lightweight: it only adds annotations, while leaving the scheduling strategy to the base executor (here, a *parallel_executor*).

We define two simple functions to demonstrate both synchronous and asynchronous execution: These functions also print to `std::cout`, but this output is not the actual annotation. Annotations are stored internally by HPX and become visible when you use debugging or profiling tools.


```

int compute_square(int x)
{
    std::cout << "[sync_execute] Running task with annotation\n";
    return x * x;
}

void say_hello()
{
    std::cout << "[post] Running task with annotation\n";
}

```

Finally, in main we create the executor with a base executor and annotation string. We then run one task with `sync_execute` (blocking, returns result) and one with `post` (asynchronous, fire-and-forget). You can also create multiple annotating executors with different strings, so each task gets its own label. This is especially useful in larger applications with many different kinds of tasks, where annotations make it much easier to trace what is happening.

```

int main()
{
    simple_annotating_executor exec(hpx::execution::parallel_executor{}, "my_custom_task
↪");

    // Synchronous execution
    int result = hpx::parallel::execution::sync_execute(exec, &compute_square, 7);
    std::cout << "Result from sync_execute: " << result << "\n";

    // Post a task
    hpx::parallel::execution::post(exec, &say_hello);

    return 0;
}

```

Custom annotating executor with parallel algorithms

The following example demonstrates how to use a custom annotating executor with *HPX* parallel algorithms, such as `for_each`. This allows you to attach annotations to tasks while executing them in parallel.

Full example code

```

#include <hpx/execution.hpp>
#include <hpx/hpx_main.hpp>
#include <hpx/include/parallel_algorithm.hpp>
#include <hpx/include/parallel_executors.hpp>
#include <hpx/modules/threading_base.hpp>

#include <iostream>
#include <utility>
#include <vector>

template <typename BaseExecutor>
struct simple_annotating_executor

```

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```

{
    BaseExecutor base_;
    char const* annotation_;

    using execution_category =
        hpx::traits::executor_execution_category_t<BaseExecutor>;

    simple_annotating_executor(BaseExecutor exec, char const* ann)
    : base_(std::move(exec))
    , annotation_(ann)
    {
    }

    // Bulk async_execute (used by parallel algorithms)
    template <typename F, typename Shape, typename... Ts>
    friend auto tag_invoke(hpx::parallel::execution::bulk_async_execute_t,
        simple_annotating_executor const& exec, F&& f, Shape const& shape,
        Ts&&... ts)
    {
        return hpx::parallel::execution::bulk_async_execute(
            exec.base_,
            hpx::annotated_function(std::forward<F>(f), exec.annotation_),
            shape, std::forward<Ts>(ts)...);
    }
};

namespace hpx::execution::experimental {

    // The annotating executor exposes the same executor categories as its
    // underlying (wrapped) executor.

    template <typename BaseExecutor>
    struct is_two_way_executor<simple_annotating_executor<BaseExecutor>>
    : is_two_way_executor<BaseExecutor>
    {
    };
} // namespace hpx::execution::experimental

int main()
{
    using base_executor = hpx::execution::parallel_executor;
    simple_annotating_executor exec(base_executor{}, "for_each_task");

    std::vector<int> data = {1, 2, 3, 4, 5};

    // Use the custom executor with a parallel algorithm
    hpx::for_each(hpx::execution::par.on(exec), // attach executor
        data.begin(), data.end(), [](int& x) {
            std::cout << "Processing " << x << " on thread "
                << hpx::get_worker_thread_num() << "\n";
            x *= x;
        });
}

```

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```

std::cout << "Squared values: ";
for (int v : data)
    std::cout << v << " ";
std::cout << "\n";

return 0;
}

```

Explanation

Similar as before, the first lines pull in the necessary HPX headers for executors, asynchronous execution, and annotated functions. The key one here is *hpx/threading_base/annotated_function.hpp*, which provides the facility to tag tasks with a string label. We then define a *simple_annotating_executor* that wraps another executor and associates an annotation string with every task:

```

template <typename BaseExecutor>
struct simple_annotating_executor
{
    BaseExecutor base_;
    const char* annotation_;

    using execution_category =
        hpx::traits::executor_execution_category_t<BaseExecutor>;

    simple_annotating_executor(BaseExecutor exec, const char* ann)
        : base_(std::move(exec)), annotation_(ann)
    {}
};

```

Note that we expose the execution category of the custom executor with *using execution_category = hpx::traits::executor_execution_category_t<BaseExecutor>*. We inherit the execution category from the underlying executor (*BaseExecutor*), which ensures that our *simple_annotating_executor* behaves like the base executor in terms of parallelism and task execution capabilities.

The *bulk_async_execute* customization schedules a set of tasks to run asynchronously. We wrap each task in *hpx::annotated_function* so that it carries the annotation. Executors in *HPX* customize these operations through *tag_invoke* overloads, which are selected by special tag objects like *bulk_async_execute_t*. This is why the executor interface may look different from a normal member function API - it uses tag dispatch to integrate seamlessly with the *HPX* parallel algorithms infrastructure.

```

template <typename F, typename Shape, typename... Ts>
friend auto tag_invoke(hpx::parallel::execution::bulk_async_execute_t,
    simple_annotating_executor const& exec, F&& f, Shape const& shape,
    Ts&&... ts)
{
    return hpx::parallel::execution::bulk_async_execute(
        exec.base_,
        hpx::annotated_function(std::forward<F>(f), exec.annotation_),
        shape, std::forward<Ts>(ts)...);
}

```

Note how we delegate the actual execution to *exec.base_*, the underlying executor. This makes the custom executor lightweight: it only adds annotations, while leaving the scheduling strategy to the base executor (here, a *parallel_executor*).

The *hpx::execution::experimental* namespace contains traits that describe executor capabilities, such as whether an executor can run tasks one-way, two-way, or never-blocking. These traits are used internally by *HPX* to verify that an executor is compatible with a given parallel algorithm or execution policy.

```
namespace hpx::execution::experimental {

    // The annotating executor exposes the same executor categories as its
    // underlying (wrapped) executor.

    template <typename BaseExecutor>
    struct is_never_blocking_one_way_executor<
        simple_annotating_executor<BaseExecutor>>
    : is_never_blocking_one_way_executor<BaseExecutor>
    {
    };

    template <typename BaseExecutor>
    struct is_one_way_executor<simple_annotating_executor<BaseExecutor>>
    : is_one_way_executor<BaseExecutor>
    {
    };

    template <typename BaseExecutor>
    struct is_two_way_executor<simple_annotating_executor<BaseExecutor>>
    : is_two_way_executor<BaseExecutor>
    {
    };

}
```

- *is_never_blocking_one_way_executor* indicates whether the executor can schedule tasks in a fire-and-forget style without blocking.
- *is_one_way_executor* indicates support for one-way execution (tasks can be scheduled but no result is returned).
- *is_two_way_executor* indicates support for two-way execution (tasks return a result or a future).

In all cases, the custom executor inherits the capabilities of the base executor, so it integrates seamlessly with *HPX* algorithms.

This design ensures that *simple_annotating_executor* can be used anywhere its underlying executor could be used, while still adding the annotation functionality. It keeps the custom executor lightweight and fully compatible with the parallel algorithms infrastructure.

In main, we create the executor with a base executor and an annotation string, and then use it with a parallel algorithm:

```
int main()
{
    using base_executor = hpx::execution::parallel_executor;
    simple_annotating_executor exec(base_executor{}, "for_each_task");

    std::vector<int> data = {1, 2, 3, 4, 5};

    // Use the custom executor with a parallel algorithm
```

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```

hpx::for_each(hpx::execution::par.on(exec),    // attach executor
    data.begin(), data.end(), [](int& x) {
        std::cout << "Processing " << x << " on thread "
            << hpx::get_worker_thread_num() << "\n";
        x *= x;
    });

std::cout << "Squared values: ";
for (int v : data)
    std::cout << v << " ";
std::cout << "\n";

return 0;
}

```

First, we create a base executor (*parallel_executor*) and wrap it in our *simple_annotating_executor*, providing an annotation string “for_each_task”. This custom executor will attach the annotation to every task it schedules, while delegating actual execution to the base executor.

We then use `hpx::for_each` with a parallel execution policy and attach our custom executor using `par.on(exec)`: `hpx::execution::par.on(exec)` attaches our custom executor to the algorithm. `for_each` internally partitions the work across threads and schedules each task using *bulk_async_execute*. Each task is annotated with “for_each_task”, visible in debuggers and profilers. The results of the parallel computation are stored in the data vector, demonstrating that the algorithm

executed successfully in parallel.

This pattern is especially useful in larger applications with many tasks, as annotations make it much easier to trace and debug the execution of parallel algorithms.

2.4.20 Error handling

Like in any other asynchronous invocation scheme, it is important to be able to handle error conditions occurring while the asynchronous (and possibly remote) operation is executed. In *HPX* all error handling is based on standard C++ exception handling. Any exception thrown during the execution of an asynchronous operation will be transferred back to the original invocation *locality*, where it will be rethrown during synchronization with the calling thread.

The source code for this example can be found here: `error_handling.cpp`.

Working with exceptions

For the following description assume that the function `raise_exception()` is executed by invoking the plain action `raise_exception_type`.

```

void raise_exception()
{
    HPX_THROW_EXCEPTION(
        hpx::error::no_success, "raise_exception", "simulated error");
}
HPX_PLAIN_ACTION(raise_exception, raise_exception_action)

```

The exception is thrown using the macro `HPX_THROW_EXCEPTION`. The type of the thrown exception is `hpx::exception`. This associates additional diagnostic information with the exception, such as file name and line

number, *locality* id and thread id, and stack backtrace from the point where the exception was thrown.

Any exception thrown during the execution of an action is transferred back to the (asynchronous) invocation site. It will be rethrown in this context when the calling thread tries to wait for the result of the action by invoking either `future<>::get()` or the synchronous action invocation wrapper as shown here:

Note: The exception is transferred back to the invocation site even if it is executed on a different *locality*.

Additionally, this example demonstrates how an exception thrown by an (possibly remote) action can be handled. It shows the use of `hpx::diagnostic_information`, which retrieves all available diagnostic information from the exception as a formatted string. This includes, for instance, the name of the source file and line number, the sequence number of the OS thread and the *HPX* thread id, the *locality* id and the stack backtrace of the point where the original exception was thrown.

Under certain circumstances it is desirable to output only some of the diagnostics, or to output those using different formatting. For this case, *HPX* exposes a set of lower-level functions as demonstrated in the following code snippet:

Working with error codes

Most of the API functions exposed by *HPX* can be invoked in two different modes. By default those will throw an exception on error as described above. However, sometimes it is desirable not to throw an exception in case of an error condition. In this case an object instance of the `hpx::error_code` type can be passed as the last argument to the API function. In case of an error, the error condition will be returned in that `hpx::error_code` instance. The following example demonstrates extracting the full diagnostic information without exception handling:

```
hpx::cout << "Error reporting using error code\n";

// Create a new error_code instance.
hpx::error_code ec;

// If an instance of an error_code is passed as the last argument while
// invoking the action, the function will not throw in case of an error
// but store the error information in this error_code instance instead.
raise_exception_action do_it;
do_it(hpx::find_here(), ec);

if (ec)
{
    // Print just the essential error information.
    hpx::cout << "returned error: " << ec.get_message() << "\n";

    // Print all of the available diagnostic information as stored with
    // the exception.
    hpx::cout << "diagnostic information:"
               << hpx::diagnostic_information(ec) << "\n";
}

hpx::cout << std::flush;
```

Note: The error information is transferred back to the invocation site even if it is executed on a different *locality*.

This example shows how an error can be handled without having to resolve to exceptions and that the returned `hpx::error_code` instance can be used in a very similar way as the `hpx::exception` type above. Simply pass it to the `hpx::diagnostic_information`, which retrieves all available diagnostic information from the error code instance as a formatted string.

As for handling exceptions, when working with error codes, under certain circumstances it is desirable to output only some of the diagnostics, or to output those using different formatting. For this case, *HPX* exposes a set of lower-level functions usable with error codes as demonstrated in the following code snippet:

```
hpx::cout << "Detailed error reporting using error code\n";

// Create a new error_code instance.
hpx::error_code ec;

// If an instance of an error_code is passed as the last argument while
// invoking the action, the function will not throw in case of an error
// but store the error information in this error_code instance instead.
raise_exception_action do_it;
do_it(hpx::find_here(), ec);

if (ec)
{
    // Print the elements of the diagnostic information separately.
    hpx::cout << "{what}: " << hpx::get_error_what(ec) << "\n";
    hpx::cout << "{locality-id}: " << hpx::get_error_locality_id(ec)
        << "\n";
    hpx::cout << "{hostname}: " << hpx::get_error_host_name(ec)
        << "\n";
    hpx::cout << "{pid}: " << hpx::get_error_process_id(ec) << "\n";
    hpx::cout << "{function}: " << hpx::get_error_function_name(ec)
        << "\n";
    hpx::cout << "{file}: " << hpx::get_error_file_name(ec) << "\n";
    hpx::cout << "{line}: " << hpx::get_error_line_number(ec)
        << "\n";
    hpx::cout << "{os-thread}: " << hpx::get_error_os_thread(ec)
        << "\n";
    hpx::cout << "{thread-id}: " << std::hex
        << hpx::get_error_thread_id(ec) << "\n";
    hpx::cout << "{thread-description}: "
        << hpx::get_error_thread_description(ec) << "\n\n";
    hpx::cout << "{state}: " << std::hex << hpx::get_error_state(ec)
        << "\n";
    hpx::cout << "{stack-trace}: " << hpx::get_error_backtrace(ec)
        << "\n";
    hpx::cout << "{env}: " << hpx::get_error_env(ec) << "\n";
}

hpx::cout << std::flush;
```

For more information please refer to the documentation of `hpx::get_error_what`, `hpx::get_error_locality_id`, `hpx::get_error_host_name`, `hpx::get_error_process_id`, `hpx::get_error_function_name`, `hpx::get_error_file_name`, `hpx::get_error_line_number`, `hpx::get_error_os_thread`, `hpx::get_error_thread_id`, `hpx::get_error_thread_description`, `hpx::get_error_backtrace`, `hpx::get_error_env`, and `hpx::get_error_state`.

Lightweight error codes

Sometimes it is not desirable to collect all the ambient information about the error at the point where it happened as this might impose too much overhead for simple scenarios. In this case, *HPX* provides a lightweight error code facility that will hold the error code only. The following snippet demonstrates its use:

```
hpx::cout << "Error reporting using an lightweight error code\n";

// Create a new error_code instance.
hpx::error_code ec(hpx::throwmode::lightweight);

// If an instance of an error_code is passed as the last argument while
// invoking the action, the function will not throw in case of an error
// but store the error information in this error_code instance instead.
raise_exception_action do_it;
do_it(hpx::find_here(), ec);

if (ec)
{
    // Print just the essential error information.
    hpx::cout << "returned error: " << ec.get_message() << "\n";

    // Print all of the available diagnostic information as stored with
    // the exception.
    hpx::cout << "error code:" << ec.value() << "\n";
}

hpx::cout << std::flush;
```

All functions that retrieve other diagnostic elements from the `hpx::error_code` will fail if called with a lightweight `error_code` instance.

2.4.21 The *HPX* I/O-streams component

The *HPX* I/O-streams subsystem extends the standard C++ output streams `std::cout` and `std::cerr` to work in the distributed setting of an *HPX* application. All of the output streamed to `hpx::cout` will be dispatched to `std::cout` on the console *locality*. Likewise, all output generated from `hpx::cerr` will be dispatched to `std::cerr` on the console *locality*.

Note: All existing standard manipulators can be used in conjunction with `hpx::cout` and `hpx::cerr`.

In order to use either `hpx::cout` or `hpx::cerr`, application codes need to `#include <hpx/include/iostreams.hpp>`. For an example, please see the following ‘Hello world’ program:

```
// Including 'hpx/hpx_main.hpp' instead of the usual 'hpx/hpx_init.hpp' enables
// to use the plain C-main below as the direct main HPX entry point.
#include <hpx/hpx_main.hpp>
#include <hpx/iostream.hpp>

int main()
{
    // Say hello to the world!
```

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```

    hpx::cout << "Hello World!\n" << std::flush;
    return 0;
}

```

Additionally, those applications need to link with the `iostreams` component. When using CMake this can be achieved by using the `COMPONENT_DEPENDENCIES` parameter; for instance:

```

include(HPX_AddExecutable)

add_hpx_executable(
    hello_world
    SOURCES hello_world.cpp
    COMPONENT_DEPENDENCIES iostreams
)

```

Note: The `hpx::cout` and `hpx::cerr` streams buffer all output locally until a `std::endl` or `std::flush` is encountered. That means that no output will appear on the console as long as either of these is explicitly used.

2.4.22 Supervision dispatch

The supervision dispatch component (`hpx::supervision`, built on top of the *Supervision module* module) lets a locality discover its peers, detect when one of them has gone silent, and safely stop dispatching work to a peer that has latched a terminal lifecycle event - all without every caller needing to make an authoritative, synchronous round-trip for every dispatch.

This page describes the initialization and shutdown ordering applications must follow, the one-shot nature of peer discovery, the distinction between the locally cached peer state and the registry's authoritative state, and how that distinction shows up in the two-tier admission check used by fenced dispatch. For the detailed semantics of individual lifecycle events, `check_admission`, and `dispatch_outcome` values, see *Supervision module*; this page does not repeat that reference material.

Initialization and shutdown ordering

Supervision dispatch is brought up with a single call to `init()` and torn down with a single call to `finalize()`:

```

#include <hpx/supervision_dispatch.hpp>

// Blocking, one-shot startup for this locality.
hpx::supervision::init(hpx::launch::sync);

// ... application runs, dispatching fenced work to peers ...

// Cooperative, one-shot shutdown for this locality.
hpx::supervision::finalize();

```

`init()` performs, in order:

1. Creation of a local `sentinel` and `registry` component for this locality.
2. Publication of `event::started` on the `sentinel` - *before* either component's symbol name is registered with AGAS, so that no peer can discover and join a not-yet-started `sentinel`.

3. Registration of both symbol names.
4. A single `discover_and_join()` pass that joins every peer registry reachable within `discovery_timeout`.

`init()` is idempotent: calling it while already active is a no-op, and concurrent callers attach to a single in-flight initialization rather than racing to create duplicate sentinels/registries. `finalize()` mirrors this ordering in reverse - it publishes `event::completed` for the local sentinel, unregisters both symbol names, and releases the components - and is always safe to call speculatively (e.g. during shutdown) even if `init()` was never called, since it is a no-op unless the runtime is currently active.

Applications should treat `init()/finalize()` as a matched pair around the portion of the program that performs supervised, fenced dispatch, and should not assume any lifecycle event is visible to peers before `init()`'s future (or synchronous call) has completed.

Calling `init()` again after it has already succeeded is a documented no-op: it returns an already-ready future without repeating any of the four steps above, and in particular does **not** re-run `discover_and_join()`. Peers that started, or otherwise became discoverable, after the first `init()` call completed are *not* retroactively added to the registry by a second call - the one-shot discovery behavior described below applies equally to repeated `init()` calls, not just the first one. To pick up such a peer, join it explicitly via `registry::join()`. Calling `init()` while a previous call is still in progress attaches to that same in-flight initialization rather than starting a second one; calling it while `finalize()` is concurrently in progress fails immediately with `hpx::error::invalid_status` rather than queuing behind it.

What is a target, and how is it introduced

A “target” is nothing more than an `hpx::id_type` - the id of any locality or component whose lifecycle should be tracked. It is not a distinct kind of object, and there is no separate handle or wrapper type: the same id you'd use to invoke an action on a component is what you pass as `target` to `publish_event()`, `check_admission()`, `dispatch_work()`, and the rest of the supervision API.

Because `dispatch_work()` resolves `target` via `hpx::colocated()` to find the locality to dispatch to, a target must also be independently resolvable by AGAS to some locality - this is a separate requirement from having recorded supervision state, and it is not something supervision dispatch itself grants. Any id obtained the ordinary way (creating a component, `hpx::find_here()`, a symbolic-name lookup, a peer's joined sentinel/registry id) already satisfies this. If a target id can no longer be resolved to a locality - for instance, because the underlying component was already destroyed and unregistered - `dispatch_work()` fails when `hpx::colocated()` cannot resolve it, *before* either admission check runs; this is a distinct failure mode from `hpx::error::target_fenced` and is not something publishing `event::started` can paper over.

In practice, two kinds of ids show up as targets in this component:

- The sentinel/registry ids of peer localities, once joined via `registry::join()` - these are the targets whose shadow state feeds `registry::snapshot_peers()` and local admission filtering.
- The ids of individual localities/components that fenced work is dispatched against via `dispatch_work(target, epoch, ...)` - these need not be peers' sentinels at all; any component id that has had lifecycle events published for it can be fenced.

Unlike joining a peer registry, there is no explicit “register this target” call for the underlying supervision manager. A target is introduced implicitly, purely as a side effect of publishing its first event: the supervision manager tracks targets in a map keyed by id, and a lookup miss is treated as “this target has never been seen,” with an implicit current epoch of 0 and a default event of `unknown`. The lifecycle-transition validation then requires that the *first* event ever published for a target open with `started` - publishing anything else for a target with no prior recorded state is rejected as an invalid transition. Put simply:

```
// Introduces `some_actor` as a tracked target under epoch 0. This is the
// only step required - no prior registration call exists.
hpx::supervision::publish_event(some_actor, hpx::supervision::event::started, 0);
```

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```
// From this point on, `some_actor` has recorded state: query_state(),
// check_admission(), and dispatch_work() against it now reflect that
// state rather than "unknown."
```

Before a target's first `started` publication, it has no recorded state at all: `query_state()` reports it as `unknown` rather than as any specific lifecycle stage, and `check_admission()` for an epoch-0, never-published target behaves as admitting (there is nothing latched to fence against yet). Once introduced, a target's tracked state persists on the supervision manager that owns it until either it reaches a terminal event that is superseded by a higher epoch (see "Epochs and when to change them" below), or it is explicitly forgotten via `remove_target()` - typically done by callers that know a given id will never be queried or observed again locally, e.g. after a failed registration attempt or once a peer has been evicted from the registry, to avoid accumulating unbounded per-target state.

Removing a target

`hpx::supervision::remove_target()` is the counterpart to introducing a target: it unconditionally erases whatever supervision state is locally recorded for a target id, on a single locality.

```
// Synchronous version, target on a possibly remote locality:
hpx::supervision::remove_target(hpx::launch::sync, locality, target);
```

See *Supervision module* for the full semantics of `remove_target`.

A few points worth calling out:

- **It is a forget operation, not a lifecycle event.** It erases the target's tracked state (epoch, current event, sequence number) and any observers still registered for it, without publishing an event of any kind. This is different from a target reaching a terminal event – removal discards history rather than recording an outcome.
- **It is local only.** Removal only clears state held on the given `locality` (or the calling locality, for the local-only overload). It does not touch `target`'s state on any other locality, and it does not affect AGAS registration or symbolic names for that id.
- **It is safe to call on a target with no recorded state.** This is a documented no-op rather than an error, so callers do not need to first check whether a target was ever introduced before removing it.
- **Republishing afterward reintroduces the target from scratch.** After removal, publishing `event::started` for the same id behaves exactly as if the id had never been seen before - epoch resets to whatever the new `started` publication specifies.
- **Safe to call from within the target's own observer callback.** Calling `remove_target()` on `target` from inside one of `target`'s own observer callbacks is supported and does not deadlock.

Removing a target is the natural cleanup step once it is known to be permanently done with - for example, once a peer has been evicted from the registry, or once a fenced target has reached a terminal event and will not be reincarnated under a new epoch - so that per-target state does not accumulate indefinitely over the lifetime of a long-running application.

Application-visible state machines

Applications observe two distinct, overlapping state machines.

Per-locality dispatcher lifecycle

Tracked once per locality (`dispatch_api.cpp`), driven by `init()/finalize()`:

```
uninitialized -> initializing -> active -> finalizing -> uninitialized
```

- **uninitialized**: no local sentinel/registry exist.
- **initializing**: sentinel/registry creation, `event::started` publication, symbol registration, and the one-shot `discover_and_join()` pass are in progress. Concurrent callers attach to the same in-flight initialization rather than racing.
- **active**: heartbeat and failure-detection background tasks are running.
- **finalizing**: background tasks are being stopped, `event::completed` is published, and both symbol names are being unregistered before component release.

`init()/finalize()` are idempotent no-ops when already in the target state. A failure during `initializing` unregisters any names already registered and returns to `uninitialized` rather than leaving the locality stuck mid-transition.

Per-target lifecycle event

Reported by an actor about itself (`hpx::supervision::event`), tracked per epoch and mirrored into peer shadows:

```
unknown -> started -> running -> suspending -> completed
                \-> failed
                \-> losing_locality
```

`completed` and `failed` are terminal: once published for a target/epoch, later terminal publications for that same target/epoch are rejected as already-terminal, and publications carrying a lower epoch are rejected as stale. `losing_locality` is a non-terminal warning that the hosting locality may become unavailable.

How they interact

The dispatcher-lifecycle state machine governs whether *this locality* can participate at all. The per-target event state machine governs whether *a specific peer* remains eligible for fenced dispatch. Failure detection moves a peer's locally cached shadow toward `failed` only when a timeout on `await_terminal` coincides with an unchanged event sequence number - a timeout alone is not sufficient, since the peer may simply still be `running`. This local `failed` marking never touches the peer's own authoritative state; it only affects what this locality's client-side filtering sees, per the shadow/authoritative distinction above.

Heartbeats: distinguishing idle from dead

While a locality is `active` (see “Application-visible state machines” above), a background heartbeat task runs alongside the failure-detection loop. Its job is narrow but essential: periodically republish `event::running` against the locality’s own local sentinel, purely to keep that sentinel’s event sequence number and timestamp advancing even when the application itself has nothing new to report.

Why this is needed

Failure detection works by calling `await_terminal()` with a timeout against each joined peer’s shadow. On its own, a bare timeout on that call is ambiguous: it fires identically whether the peer has crashed, or the peer is simply alive and quietly idle – `running` is a stable state that can legitimately persist for a long time with no events published at all. Without some independent signal of liveness, failure detection would have no way to tell these two situations apart, and would either fence peers that are perfectly healthy but quiet, or fail to fence peers that have genuinely died.

The heartbeat resolves this by giving every alive peer a steady, low-cost source of forward progress – its event sequence number – independent of whatever the application is actually doing. This lets `peer_is_alive_but_silent()` compare a peer’s sequence number immediately before and after an `await_terminal()` timeout:

- If the sequence number **advanced** during the timeout window, the peer’s heartbeat kept running, so the peer is alive but simply not reporting a terminal event - no action is taken.
- If the sequence number **did not change**, the peer’s own heartbeat task has itself gone silent, which is treated as genuine evidence of a stall, and the failure-detection loop fences the peer locally by publishing `event::failed` against its shadow (never against the peer’s own, remote, authoritative state - see “Shadow state versus the authoritative registry” above).

Operational properties

- **Best-effort.** A failed heartbeat publication attempt does not stop or crash the loop; it is simply retried on the next interval.
- **Fixed interval, tied to detection timing.** The heartbeat republishes at roughly a third of the failure-detection poll timeout, so that under normal conditions several heartbeats land within any single detection window – making a missed heartbeat a meaningful signal rather than noise from timing jitter.
- **Started and stopped in lockstep with failure detection.** Both background tasks are started together, immediately after the one-shot `discover_and_join()` pass inside `init()`, and both are signaled to stop and joined together during `finalize()`, before `event::completed` is published and either symbol name is unregistered.
- **Entirely internal.** Applications do not start, stop, or configure the heartbeat directly, and do not need to publish any events themselves to keep it running – it is a side effect of calling `init()`, and requires no ongoing application action beyond staying inside the `init()/finalize()` bracket.

In short, the heartbeat is what makes failure detection’s timeout-based stall check meaningful at all: without it, a timeout by itself would not be able to distinguish “no news” from “no pulse.”

This is also why fenced dispatch depends on the heartbeat working correctly, not just on failure detection running. `check_admission()` - whether called as the caller’s local, non-authoritative early-out or as the target’s own authoritative re-check inside `invoke_fenced_action()` - ultimately answers “is this target still available” by consulting whatever lifecycle state has been latched for it, which for peers is only ever updated to `failed` once failure detection has genuine evidence of a stall. Without the heartbeat, that evidence would not exist: a silent-but-alive peer and a genuinely dead one would look identical to `await_terminal()`, and admission would have no reliable basis for deciding whether it is safe to keep dispatching work to that locality. In short, the heartbeat is not just an internal detail of failure detection - it is what makes the fencing decisions applications rely on throughout this page trustworthy in the first place.

One-shot discovery and staleness

The `discover_and_join()` pass performed by `init()` runs exactly once. It finds whichever peer sentinels/registries are reachable *at that moment* and joins them; it does not re-run periodically, and peers that join the system later are not automatically discovered by localities that have already initialized. This has two practical implications:

- A locality's set of known peers is fixed at `init()` time (plus any peers joined explicitly afterwards via `registry::join()`). It does not grow on its own as new peers appear.
- Because failure detection (see below) only tracks peers that have actually been joined, a peer that starts too late to be discovered is invisible to both admission filtering and fencing until it is joined explicitly.

Applications that need a stable, complete peer set at some later point should join peers explicitly (`registry::join()`) rather than relying on rediscovery after `init()`.

Shadow state versus the authoritative registry

Once a peer is joined, its lifecycle state is mirrored locally into a *shadow* target owned by the joining registry - a lightweight, locally readable copy of that peer's most recently observed state, updated as lifecycle and heartbeat events arrive. This shadow state is what `registry::snapshot_peers()` and the local admission check consult.

It is important to keep two things distinct:

- **Shadow state** (local): a locally cached view of a peer's lifecycle, updated asynchronously as events are received. It can lag reality by however long it takes an event to propagate, and can go stale entirely if the peer's heartbeat has stopped but failure detection has not yet timed out.
- **Registry state** (remote, authoritative): the actual lifecycle/fencing state tracked by the peer's own supervision manager, on the peer's own locality. This is the only state that is authoritative for admission decisions.

Failure detection narrows - but does not eliminate - the gap between the two: a background loop periodically checks each joined peer's shadow for silence and marks it accordingly, but there is always a window between a peer actually going silent and that silence being locally observed.

How sentinels and registries stay synchronized

The shadow-mirroring described above is not maintained by polling. It is driven entirely by event-based observer callbacks, registered once at `registry::join()` time and then firing automatically for the lifetime of the joined peer.

Registration, at join time

When `registry::join(peer_sentinel, peer_locality)` is called, the local registry does two things before returning a shadow id to the caller:

1. **Seeds the shadow with the peer's current state.** It performs a single, synchronous `query_state()` call against `peer_sentinel` to read whatever epoch the peer is currently in, then publishes `event::started` for that epoch onto a freshly minted shadow id. This seed step exists so the shadow reflects the peer's state as of joining, rather than starting from `unknown` and waiting for the first subsequent event to arrive.
2. **Registers a lifecycle-event observer** on `peer_sentinel` (via `hpx::supervision::register_observer()`), along with an activity-transition observer (via `register_activity_observer()`, currently a no-op placeholder reserved for future use). These registrations live on the peer's own locality and are what make the subsequent synchronization automatic: no further polling or explicit refresh call from the local registry is needed.

Both registrations are made atomically with respect to concurrent `join()` calls for the same peer (a reservation is taken in `peers_` before either remote call is issued), and either observer failing to register cleanly rolls back the other, so a peer is never left half-joined.

Ongoing synchronization

From then on, every event the peer publishes for itself - including the heartbeat's periodic `event::running` republication (see "Heartbeats: distinguishing idle from dead" above) - invokes the registered lifecycle observer callback on the *peer's* locality, which in turn calls back into the joining registry to mirror that event onto the local shadow, at the same epoch it actually occurred in. No round trip is initiated by the joining side to "check in" on a peer; synchronization is entirely push-driven by whichever locality actually observes the peer's own state changing.

If the mirrored event is terminal (`completed` or `failed`), the callback additionally schedules - via `hpx::post()`, deferred rather than inline, since the callback itself does not hold the registry's lock - the peer's eviction from `peers_`. Once evicted, the peer no longer appears in `registry::snapshot_peers()`, and its observer registrations are unregistered on the peer's locality so nothing is left listening for further events from a peer that has already reached a terminal state.

What this does and does not cover

This automatic synchronization keeps the shadow up to date with whatever the peer *chooses to publish* - it is not a substitute for failure detection. A peer that stops publishing anything at all (e.g. because it crashed) generates no event, so no observer callback fires, and the shadow simply stops advancing. Detecting that silence, and deciding whether it means the peer is dead rather than merely idle, is exactly the job of the heartbeat/failure-detection pairing described earlier in this page - the observer-based synchronization described here and the polling-with-timeout failure detection described earlier are complementary, not overlapping, mechanisms.

Because `observers_` is keyed per target and holds a list rather than a single callback, this fan-out is genuinely one-to-many: if several localities have each independently joined the same peer, a single event published by that peer notifies every one of their registries in the same pass, each mirroring the event onto its own local shadow. This reach is scoped to the publishing target, however - only localities that have actually called `join()` against that specific peer are registered as observers and receive its events. A locality that has not joined a given peer is not notified when that peer publishes, regardless of whether it participates in supervision dispatch generally.

Client-side filtering versus fenced-dispatch admission

This shadow/authoritative distinction is exactly why fenced dispatch performs *two* admission checks rather than one:

- **Client-side filtering** (`check_admission()` called directly by the caller, or peer filtering based on `registry::snapshot_peers()`) consults only the locally cached shadow state. It is fast and involves no remote round-trip, but its answer can be stale: it may say a peer is admitted when the peer has, in fact, already latched a terminal event that just hasn't propagated back yet. This check is an optimization - a cheap way to avoid dispatching work that is *already known locally* to be pointless - and must never be treated as authoritative.
- **Fenced-dispatch admission** (the re-check performed inside `invoke_fenced_action()`, once the dispatched action has actually reached the target's own locality via `hpx::colocated`) runs on the same locality/thread that owns the target's authoritative supervision state, with no suspension point between the check and the action invocation that follows it. This is the check that actually decides whether the wrapped action runs, and it is what closes the race between a stale client-side "admitted" result and a peer that has since gone terminal.

In short: use client-side filtering to avoid *obviously* wasted dispatches; rely only on the fenced-dispatch outcome to know whether work actually ran.

Epochs and when to change them

Every lifecycle event and every fenced dispatch is scoped to an epoch (`std::uint64_t`, defaulting to 0). The epoch is what lets a target be “un-fenced” and reused after it has latched a terminal event, instead of being permanently unusable once it completes or fails once.

A target’s epoch only ever increases, and publications are compared against whatever epoch is currently in effect for that target:

- A publication for the **same** epoch as the target’s current one is normal progression (`started` -> `running` -> ... -> a terminal event), subject to the terminal/already-terminal rule described above.
- A publication for a **lower** epoch than the target’s current one is rejected as stale (`publish_result::stale_epoch`) and has no effect - this guards against a delayed or reordered event from an earlier generation of the target clobbering newer state.
- A publication for a **higher** epoch than the target’s current one resets the target: it becomes the new current epoch, the event sequence number resets, and any previously latched terminal event (`completed/ failed`) under the old epoch no longer fences new dispatches. This is the only way to make a target eligible for fenced dispatch again after it has gone terminal.

The same epoch value is consulted end-to-end: `check_admission(target, epoch)` compares the *epoch you pass in* against the target’s current epoch, not just whether the target is terminal in some absolute sense. This is why both `dispatch_work()`’s local check and `invoke_fenced_action()`’s remote re-check take an explicit epoch argument rather than operating on “the” state of the target.

When to keep the epoch the same

Use the same epoch value across all fenced dispatches that belong to the same logical run/incarnation of the target - e.g. repeated or retried calls against a target that is still expected to be `started/running` under the epoch it was initialized with. Fencing is only meaningful when related dispatches share an epoch: if a target has already latched `failed` for epoch N, every further dispatch under epoch N is correctly rejected, which is the intended behavior - it stops work from continuing to target something that has already terminated in this generation.

When to bump the epoch

Bump to a new, higher epoch when you are intentionally starting a new generation of the target and want prior terminal state to stop fencing new work - typically when a target is restarted/reincarnated after failing or completing, and you want dispatch to resume against the restarted instance rather than remain permanently fenced by the old generation’s terminal event. Concretely: publish the target’s own `event::started` under the new epoch as part of bringing the restarted instance up, then use that same, higher epoch value for subsequent dispatches and admission checks against it.

Do not reuse an old epoch value after bumping, and do not bump the epoch merely to “clear” a fence for the *same* running instance - if the target has not actually restarted, admission is correctly rejecting dispatch because the target is genuinely terminal, and forcing an epoch bump only defeats that protection rather than fixing the underlying condition.

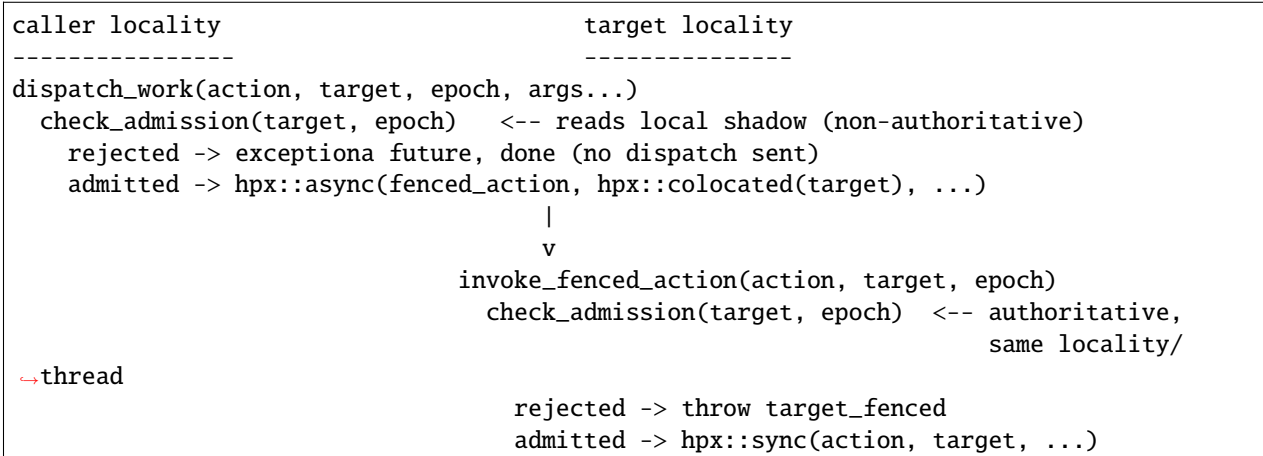
How fenced work is invoked

`hpx::supervision::dispatch_work<action_type>(target, epoch, action, args...)` is the entry point callers use. It performs two distinct steps, on two different localities:

1. **Local, non-authoritative early-out (caller's locality).** `dispatch_work()` first calls `check_admission(target, epoch)` against the caller's own locally cached shadow state. If this reports `dispatch_outcome::rejected_fenced`, `dispatch_work()` returns an already-exceptional future carrying `hpx::error::target_fenced` immediately - no parcel is ever sent, and the wrapped action is never invoked. This is purely an optimization to avoid a wasted round trip; it is not authoritative, since the shadow state it consults can be stale (see the sections above).
2. **Colocated dispatch and authoritative re-check (target's locality).** If the local check does not reject, `dispatch_work()` wraps the caller's action in a generated `fenced_action` and dispatches it via `hpx::async` to `hpx::colocated(target)` - i.e. it runs on the same locality that hosts `target` and owns its authoritative supervision state. The action body that actually runs there is `invoke_fenced_action()`, which performs a *second* `check_admission(target, epoch)` call. Because this call executes on the same locality/thread that owns the target's supervision state, with no suspension point before the wrapped action runs immediately afterward (via `hpx::sync`, invoked directly/locally with no further parcel hop), this second check is authoritative. If it also reports `rejected_fenced` - meaning the target latched a terminal event for this epoch sometime between the caller's stale check and this point - `invoke_fenced_action()` throws `hpx::error::target_fenced` instead of invoking the action.

Note that “dispatched to the target's locality” does not imply a network round-trip: `hpx::colocated(target)` resolves to whichever locality actually hosts `target`, including the invoking locality itself when `target` happens to be local. In that case no parcel is sent, but `invoke_fenced_action()` - and, if admitted, the wrapped action inside it - still each run on a newly spawned HPX thread rather than inline on the calling thread, exactly as an unfenced `hpx::async<Action>(target, args...)` call behaves when `target` is local: it schedules a new thread rather than invoking the action synchronously in place. The one exception, on both sides, is an action explicitly declared as a `direct_action`, which HPX is permitted to invoke inline when the target is local; `fenced_action` itself is an ordinary action, and the wrapped user `Action` follows whatever declaration the caller gave it. Either way, fenced dispatch adds only the two-tier admission check described above - it does not change whether or where the action is scheduled to run relative to unfenced dispatch.

Both the local early-out and the remote re-check surface failure as the same exception type/message shape (`hpx::error::target_fenced`), so callers handle a single error regardless of which side detected the fence:



Only `target` (not `epoch`) is forwarded into the wrapped action's own argument list; the `epoch` is fencing metadata consumed entirely by the dispatch machinery. A convenience overload of `dispatch_work()` accepts an action instance directly rather than only an action type, but follows the identical two-check flow described above.

Relationship to unfenced dispatch

From the caller's point of view, converting an unfenced dispatch into a fenced one requires exactly two changes to the call site:

```
// Unfenced:
hpx::future<result_type> f = hpx::async<Action>(target, args...);

// Fenced:
hpx::future<result_type> f =
    hpx::supervision::dispatch_work<Action>(target, epoch, args...);
```

1. Replace the call to `hpx::async<Action>` with `hpx::supervision::dispatch_work<Action>`.
2. Insert the fencing `epoch` as the second argument, ahead of the action's own arguments (`args...` is otherwise unchanged and is forwarded to `Action` verbatim).

Both a template-argument form and an instance-argument form are available for `dispatch_work()`, mirroring the two ways `hpx::async` can be called:

```
// Action specified as an explicit template argument (as above):
hpx::future<result_type> f =
    hpx::supervision::dispatch_work<Action>(target, epoch, args...);

// Convenience overload: action passed as an instance, deduced instead of
// spelled out explicitly. Note the action instance comes *first*, ahead
// of `target` and `epoch`.
hpx::future<result_type> f =
    hpx::supervision::dispatch_work(Action(), target, epoch, args...);
```

The instance-argument overload is a thin wrapper that simply forwards to the template-argument overload above; the two are otherwise identical in behavior (same two-check admission flow, same exception on fencing, same argument forwarding to `Action`). It exists purely so that call sites already holding an action instance - rather than naming the action type directly - don't need an extra step to obtain one, the same convenience `hpx::async` itself offers by accepting either an action type or an action instance.

Everything else - `target`, the returned `hpx::future<result_type>`, and how the caller consumes it (`.get()`, continuations, etc.) - stays the same. The only additional caller-side responsibility is handling `hpx::error::target_fenced` on that future (see below), since a fenced dispatch never invokes `Action` at all, whereas an unfenced `hpx::async<Action>` call has no equivalent rejection path.

Observing whether an action was fenced

From the caller's point of view, `dispatch_work()` always returns an `hpx::future`. Whether the fence was detected locally (early-out, no dispatch sent) or remotely (authoritative re-check on the target's own locality), the future becomes exceptional in exactly the same way, so callers do not need to special-case which side caught it:

```
hpx::future<result_type> f =
    hpx::supervision::dispatch_work(action, target, epoch, args...);

try
{
    result_type result = f.get();
    // Admitted on both checks; action ran and produced `result`.
```

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```

}
catch (hpx::exception const& e)
{
    if (e.get_error() == hpx::error::target_fenced)
    {
        // Fenced - either the caller's local admission check rejected
        // it before any dispatch was sent, or the target's own
        // authoritative re-check rejected it just before invocation.
        // The wrapped action was NOT invoked in either case.
    }
    else
    {
        throw;    // some other failure (e.g. transport, target action
                  // itself threw)
    }
}
}

```

Two points worth calling out explicitly:

- `e.get_error() == hpx::error::target_fenced` is the single, reliable signal that an action was fenced. Matching on the exception's `what()` text is unnecessary and discouraged, since the message wording is not part of the stable contract; only the error code is.
- A fenced result and a “the action itself threw” result are both surfaced through the same future/exception mechanism, so callers must check `get_error()` to distinguish “not invoked because fenced” from “invoked, but failed.” There is no separate return value, flag, or callback that reports fencing independently of the future's exceptional state.

Because `f.get()` throws synchronously the moment either check rejects, a caller waiting with a timeout (see below) sees ready-with-exception rather than ready-with-result if either side fenced the dispatch before the timeout elapsed.

Worked examples

Fencing a silent peer

```

#include <hpx/supervision_dispatch.hpp>

hpx::supervision::registry local_registry(hpx::find_here());
hpx::supervision::sentinel peer_sentinel(peer_symbolic_name);

hpx::id_type const shadow = local_registry.join(
    hpx::launch::sync, peer_sentinel, peer_locality);

// ... time passes; the peer's heartbeat stops ...

// Client-side filtering: cheap, but only as fresh as the last observed
// heartbeat/lifecycle event mirrored onto `shadow`.
for (auto const& peer : local_registry.snapshot_peers(hpx::launch::sync))
{
    if (peer.state == hpx::supervision::lifecycle_state::failed)
    {
        // Skip obviously-dead peers before even attempting dispatch.
    }
}

```

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```
        continue;
    }
    // Attempt fenced dispatch anyway; the target-side re-check inside
    // invoke_fenced_action() is authoritative and will reject the action
    // with hpx::error::target_fenced if the peer has latched a terminal
    // event we haven't observed locally yet.
}
```

Waiting on a dispatch outcome with a timeout

```
#include <hpx/async_combinators/wait_all_for.hpp>
#include <hpx/supervision_dispatch.hpp>

hpx::future<result_type> f =
    hpx::supervision::dispatch_work(target, epoch, action, args...);

if (hpx::wait_all_for(std::chrono::seconds(5), f) ==
    hpx::future_status::ready)
{
    try
    {
        result_type result = f.get();
    }
    catch (hpx::exception const& e)
    {
        if (e.get_error() == hpx::error::target_fenced)
        {
            // Authoritative rejection: the target latched a terminal
            // event for this epoch before the action could run.
        }
    }
}
else
{
    // Dispatch did not complete within the timeout; the target may be
    // slow, unreachable, or the shadow state used for client-side
    // filtering may be stale.
}
```

See *Supervision module* for the full semantics of `check_admission`, `dispatch_outcome`, and the individual lifecycle events referenced above.

2.4.23 Troubleshooting

Common issues

This section contains commonly encountered problems when compiling or using HPX.

See also the closed issues on [GitHub](#)²⁰² to find out how other people resolved a similar problem. If nothing of that works, you can also open a new issue on [GitHub](#)²⁰³ or contact us using one the options found in [Support for deploying and using HPX](#)²⁰⁴.

`hpx::iostreams_component` target not found

You may see a [CMake](#)²⁰⁵ error message that looks a bit like this:

```
error: `hpx::iostreams_component` target not found
```

Simply ensure that *HPX* is installed with `HPX_WITH_DISTRIBUTED_RUNTIME=ON` to prevent encountering such error(s). This is required if you want to use `hpx::cout`.

Undefined reference to `hpx::cout`

You may see a linker error message that looks a bit like this:

```
hello_world.cpp:(.text+0x5aa): undefined reference to `hpx::cout'
```

This usually happens if you are trying to use *HPX* iostreams functionality such as `hpx::cout` but are not linking against it. The iostreams functionality is not part of the core *HPX* library, and must be linked to explicitly. Typically this can be solved by adding `COMPONENT_DEPENDENCIES iostreams` to a call to `add_hpx_library/add_hpx_executable/hpx_setup_target` if using [CMake](#)²⁰⁶. See *Creating HPX projects* for more details.

Build fails with ASIO error

You may see an error message that looks a bit like this:

```
Cannot open include file asio/io_context.hpp
```

This can be resolved by using `-DHPX_WITH_FETCH_ASIO=ON` to the cmake command line.

See also the corresponding closed [Issue #5404](#)²⁰⁷ for more information.

²⁰² <https://github.com/TheHPXProject/hpx/issues?q=is%3Aissue+is%3Aclosed>

²⁰³ <https://github.com/TheHPXProject/hpx/issues>

²⁰⁴ <https://github.com/TheHPXProject/hpx/blob/master/.github/SUPPORT.md>

²⁰⁵ <https://www.cmake.org>

²⁰⁶ <https://www.cmake.org>

²⁰⁷ <https://github.com/TheHPXProject/hpx/issues/5404>

Build fails with TCMalloc error

You may see an error message that looks a bit like this:

```
Could NOT find TCMalloc (missing: Tcmalloc_LIBRARY Tcmalloc_INCLUDE_DIR)
ERROR: HPX_WITH_MALLOC was set to tcmalloc, but tcmalloc could not be
found.  Valid options for HPX_WITH_MALLOC are: system, tcmalloc, jemalloc,
mimalloc, tbbmalloc, and custom
```

This can be resolved either by defining `HPX_WITH_MALLOC=system` or by installing TCMalloc. This error occurs when users don't specify an option for `HPX_WITH_MALLOC`; in that case, *HPX* will be looking `tcmalloc`, which is the default value.

Useful suggestions

Reducing compilation time

If you want to significantly reduce compilation time, you can just use the local part of *HPX* for parallelism by disabling the distributed functionality. Moreover, you can avoid compiling examples. These can be done with the following flags:

```
-DHPX_WITH_NETWORKING=OFF
-DHPX_WITH_DISTRIBUTED_RUNTIME=OFF
-DHPX_WITH_EXAMPLES=OFF
-DHPX_WITH_TESTS=OFF
```

Linking *HPX* to your application

If you want to avoid installing and linking *HPX*, you can just build *HPX* and then use the following flag on your *HPX* application CMake configuration:

```
-DHPX_DIR=<build_dir>/lib/cmake/HPX
```

Note: For this to work you need not to specify `-DCMAKE_INSTALL_PREFIX` when building *HPX*.

HPX-application build type conformance

Your application's build type should align with the *HPX* build type. For example, if you specified `-DCMAKE_BUILD_TYPE=Debug` during the *HPX* compilation, then your application needs to be compiled with the same flag. We recommend keeping a separate build folder for different build types and just point accordingly to the type you want by using `-DHPX_DIR=<build_dir>/lib/cmake/HPX`.

2.5 Terminology

This section gives definitions for some of the terms used throughout the *HPX* documentation and source code.

Locality

A locality in *HPX* describes a synchronous domain of execution, or the domain of bounded upper response time. This normally is just a single node in a cluster or a NUMA domain in a SMP machine.

Active Global Address Space

AGAS

HPX incorporates a global address space. Any executing thread can access any object within the domain of the parallel application with the caveat that it must have appropriate access privileges. The model does not assume that global addresses are cache coherent; all loads and stores will deal directly with the site of the target object. All global addresses within a Synchronous Domain are assumed to be cache coherent for those processor cores that incorporate transparent caches. The Active Global Address Space used by *HPX* differs from research PGAS²⁰⁸ models. Partitioned Global Address Space is passive in their means of address translation. Copy semantics, distributed compound operations, and affinity relationships are some of the global functionality supported by AGAS.

Process

The concept of the “process” in *HPX* is extended beyond that of either sequential execution or communicating sequential processes. While the notion of process suggests action (as do “function” or “subroutine”) it has a further responsibility of context, that is, the logical container of program state. It is this aspect of operation that process is employed in *HPX*. Furthermore, referring to “parallel processes” in *HPX* designates the presence of parallelism within the context of a given process, as well as the coarse grained parallelism achieved through concurrency of multiple processes of an executing user job. *HPX* processes provide a hierarchical name space within the framework of the active global address space and support multiple means of internal state access from external sources.

Parcel

The Parcel is a component in *HPX* that communicates data, invokes an action at a distance, and distributes flow-control through the migration of continuations. Parcels bridge the gap of asynchrony between synchronous domains while maintaining symmetry of semantics between local and global execution. Parcels enable message-driven computation and may be seen as a form of “active messages”. Other important forms of message-driven computation predating active messages include *dataflow tokens*²⁰⁹, the *J-machine*’s²¹⁰ support for remote method instantiation, and at the coarse grained variations of Unix remote procedure calls, among others. This enables work to be moved to the data as well as performing the more common action of bringing data to the work. A parcel can cause actions to occur remotely and asynchronously, among which are the creation of threads at different system nodes or synchronous domains.

Local Control Object

Lightweight Control Object

LCO

A local control object (sometimes called a lightweight control object) is a general term for the synchronization mechanisms used in *HPX*. Any object implementing a certain concept can be seen as an LCO. This concepts encapsulates the ability to be triggered by one or more events which when taking the object into a predefined state will cause a thread to be executed. This could either create a new thread or resume an existing thread.

The LCO is a family of synchronization functions potentially representing many classes of synchronization constructs, each with many possible variations and multiple instances. The LCO is sufficiently general that it can subsume the functionality of conventional synchronization primitives such as spinlocks, mutexes, semaphores, and global barriers. However due to the rich concept an LCO can represent powerful synchronization and control functionality not widely employed, such as dataflow and futures (among others), which open up enormous

²⁰⁸ <https://www.pgas.org/>

²⁰⁹ http://en.wikipedia.org/wiki/Dataflow_architecture

²¹⁰ <http://en.wikipedia.org/wiki/J%E2%80%93Machine>

opportunities for rich diversity of distributed control and operation.

See `lcops` for more details on how to use LCOs in *HPX*.

Action

An action is a function that can be invoked remotely. In *HPX* a plain function can be made into an action using a macro. See `applying_actions` for details on how to use actions in *HPX*.

Component

A component is a C++ object which can be accessed remotely. A component can also contain member functions which can be invoked remotely. These are referred to as component actions. See *Writing components* for details on how to use components in *HPX*.

2.6 Why *HPX*?

Current advances in high performance computing (HPC) continue to suffer from the issues plaguing parallel computation. These issues include, but are not limited to, ease of programming, inability to handle dynamically changing workloads, scalability, and efficient utilization of system resources. Emerging technological trends such as multi-core processors further highlight limitations of existing parallel computation models. To mitigate the aforementioned problems, it is necessary to rethink the approach to parallelization models. ParalleX contains mechanisms such as multi-threading, *parcels*, *global name space* support, percolation and *local control objects (LCO)*. By design, ParalleX overcomes limitations of current models of parallelism by alleviating contention, latency, overhead and starvation. With ParalleX, it is further possible to increase performance by at least an order of magnitude on challenging parallel algorithms, e.g., dynamic directed graph algorithms and adaptive mesh refinement methods for astrophysics. An additional benefit of ParalleX is fine-grained control of power usage, enabling reductions in power consumption.

2.6.1 ParalleX—a new execution model for future architectures

ParalleX is a new parallel execution model that offers an alternative to the conventional computation models, such as message passing. ParalleX distinguishes itself by:

- Split-phase transaction model
- Message-driven
- Distributed shared memory (not cache coherent)
- Multi-threaded
- Futures synchronization
- *Local Control Objects (LCOs)*
- Synchronization for anonymous producer-consumer scenarios
- Percolation (pre-staging of task data)

The ParalleX model is intrinsically latency hiding, delivering an abundance of variable-grained parallelism within a hierarchical namespace environment. The goal of this innovative strategy is to enable future systems delivering very high efficiency, increased scalability and ease of programming. ParalleX can contribute to significant improvements in the design of all levels of computing systems and their usage from application algorithms and their programming languages to system architecture and hardware design together with their supporting compilers and operating system software.

2.6.2 What is HPX?

High Performance ParalleX (*HPX*) is the first runtime system implementation of the ParalleX execution model. The *HPX* runtime software package is a modular, feature-complete, and performance-oriented representation of the ParalleX execution model targeted at conventional parallel computing architectures, such as SMP nodes and commodity clusters. It is academically developed and freely available under an open source license. We provide *HPX* to the community for experimentation and application to achieve high efficiency and scalability for dynamic adaptive and irregular computational problems. *HPX* is a C++ library that supports a set of critical mechanisms for dynamic adaptive resource management and lightweight task scheduling within the context of a global address space. It is solidly based on many years of experience in writing highly parallel applications for HPC systems.

The two-decade success of the communicating sequential processes (CSP) execution model and its message passing interface (MPI) programming model have been seriously eroded by challenges of power, processor core complexity, multi-core sockets, and heterogeneous structures of GPUs. Both efficiency and scalability for some current (strong scaled) applications and future Exascale applications demand new techniques to expose new sources of algorithm parallelism and exploit unused resources through adaptive use of runtime information.

The ParalleX execution model replaces CSP to provide a new computing paradigm embodying the governing principles for organizing and conducting highly efficient scalable computations greatly exceeding the capabilities of today's problems. *HPX* is the first practical, reliable, and performance-oriented runtime system incorporating the principal concepts of the ParalleX model publicly provided in open source release form.

HPX is designed by the STE||AR²¹¹ Group (Systems Technology, Emergent Parallelism, and Algorithm Research) at Louisiana State University (LSU)²¹²'s Center for Computation and Technology (CCT)²¹³ to enable developers to exploit the full processing power of many-core systems with an unprecedented degree of parallelism. STE||AR²¹⁴ is a research group focusing on system software solutions and scientific application development for hybrid and many-core hardware architectures.

For more information about the STE||AR²¹⁵ Group, see *People*.

2.6.3 What makes our systems slow?

Estimates say that we currently run our computers at well below 100% efficiency. The theoretical peak performance (usually measured in FLOPS²¹⁶—floating point operations per second) is much higher than any practical peak performance reached by any application. This is particularly true for highly parallel hardware. The more hardware parallelism we provide to an application, the better the application must scale in order to efficiently use all the resources of the machine. Roughly speaking, we distinguish two forms of scalability: strong scaling (see Amdahl's Law²¹⁷) and weak scaling (see Gustafson's Law²¹⁸). Strong scaling is defined as how the solution time varies with the number of processors for a fixed **total** problem size. It gives an estimate of how much faster we can solve a particular problem by throwing more resources at it. Weak scaling is defined as how the solution time varies with the number of processors for a fixed problem size **per processor**. In other words, it defines how much more data can we process by using more hardware resources.

In order to utilize as much hardware parallelism as possible an application must exhibit excellent strong and weak scaling characteristics, which requires a high percentage of work executed in parallel, i.e., using multiple threads of execution. Optimally, if you execute an application on a hardware resource with *N* processors it either runs *N* times faster or it can handle *N* times more data. Both cases imply 100% of the work is executed on all available processors in parallel. However, this is just a theoretical limit. Unfortunately, there are more things that limit scalability, mostly

²¹¹ <https://stellar-group.org>

²¹² <https://www.lsu.edu>

²¹³ <https://www.cct.lsu.edu>

²¹⁴ <https://stellar-group.org>

²¹⁵ <https://stellar-group.org>

²¹⁶ <http://en.wikipedia.org/wiki/FLOPS>

²¹⁷ http://en.wikipedia.org/wiki/Amdahl%27s_law

²¹⁸ http://en.wikipedia.org/wiki/Gustafson%27s_law

inherent to the hardware architectures and the programming models we use. We break these limitations into four fundamental factors that make our systems *SLOW*:

- **Starvation** occurs when there is insufficient concurrent work available to maintain high utilization of all resources.
- **Latencies** are imposed by the time-distance delay intrinsic to accessing remote resources and services.
- **Overhead** is work required for the management of parallel actions and resources on the critical execution path, which is not necessary in a sequential variant.
- **Waiting for contention resolution** is the delay due to the lack of availability of oversubscribed shared resources.

Each of those four factors manifests itself in multiple and different ways; each of the hardware architectures and programming models expose specific forms. However, the interesting part is that all of them are limiting the scalability of applications no matter what part of the hardware jungle we look at. Hand-helds, PCs, supercomputers, or the cloud, all suffer from the reign of the 4 horsemen: **Starvation**, **Latency**, **Overhead**, and **Contention**. This realization is very important as it allows us to derive the criteria for solutions to the scalability problem from first principles, and it allows us to focus our analysis on very concrete patterns and measurable metrics. Moreover, any derived results will be applicable to a wide variety of targets.

2.6.4 Technology demands new response

Today's computer systems are designed based on the initial ideas of [John von Neumann](http://qss.stanford.edu/~godfrey/vonNeumann/vnedvac.pdf)²¹⁹, as published back in 1945, and later extended by the [Harvard architecture](http://en.wikipedia.org/wiki/Harvard_architecture)²²⁰. These ideas form the foundation, the execution model, of computer systems we use currently. However, a new response is required in the light of the demands created by today's technology.

So, what are the overarching objectives for designing systems allowing for applications to scale as they should? In our opinion, the main objectives are:

- **Performance**: as previously mentioned, scalability and efficiency are the main criteria people are interested in.
- **Fault tolerance**: the low expected mean time between failures ([MTBF](http://en.wikipedia.org/wiki/Mean_time_between_failures)²²¹) of future systems requires embracing faults, not trying to avoid them.
- **Power**: minimizing energy consumption is a must as it is one of the major cost factors today, and will continue to rise in the future.
- **Generality**: any system should be usable for a broad set of use cases.
- **Programmability**: for programmer this is a very important objective, ensuring long term platform stability and portability.

What needs to be done to meet those objectives, to make applications scale better on tomorrow's architectures? Well, the answer is almost obvious: we need to devise a new execution model—a set of governing principles for the holistic design of future systems—targeted at minimizing the effect of the outlined **SLOW** factors. Everything we create for future systems, every design decision we make, every criteria we apply, have to be validated against this single, uniform metric. This includes changes in the hardware architecture we prevalently use today, and it certainly involves new ways of writing software, starting from the operating system, runtime system, compilers, and at the application level. However, the key point is that all those layers have to be co-designed; they are interdependent and cannot be seen as separate facets. The systems we have today have been evolving for over 50 years now. All layers function in a certain way, relying on the other layers to do so. But we do not have the time to wait another 50 years for a new coherent system to evolve. The new paradigms are needed now—therefore, co-design is the key.

²¹⁹ <http://qss.stanford.edu/~godfrey/vonNeumann/vnedvac.pdf>

²²⁰ http://en.wikipedia.org/wiki/Harvard_architecture

²²¹ http://en.wikipedia.org/wiki/Mean_time_between_failures

2.6.5 Governing principles applied while developing HPX

As it turns out, we do not have to start from scratch. Not everything has to be invented and designed anew. Many of the ideas needed to combat the 4 horsemen already exist, many for more than 30 years. All it takes is to gather them into a coherent approach. We'll highlight some of the derived principles we think to be crucial for defeating **SLOW**. Some of those are focused on high-performance computing, others are more general.

Focus on latency hiding instead of latency avoidance

It is impossible to design a system exposing zero latencies. In an effort to come as close as possible to this goal many optimizations are mainly targeted towards minimizing latencies. Examples for this can be seen everywhere, such as low latency network technologies like [InfiniBand](#)²²², caching memory hierarchies in all modern processors, the constant optimization of existing [MPI](#)²²³ implementations to reduce related latencies, or the data transfer latencies intrinsic to the way we use [GPGPUs](#)²²⁴ today. It is important to note that existing latencies are often tightly related to some resource having to wait for the operation to be completed. At the same time it would be perfectly fine to do some other, unrelated work in the meantime, allowing the system to hide the latencies by filling the idle-time with useful work. Modern systems already employ similar techniques (pipelined instruction execution in the processor cores, asynchronous input/output operations, and many more). What we propose is to go beyond anything we know today and to make latency hiding an intrinsic concept of the operation of the whole system stack.

Embrace fine-grained parallelism instead of heavyweight threads

If we plan to hide latencies even for very short operations, such as fetching the contents of a memory cell from main memory (if it is not already cached), we need to have very lightweight threads with extremely short context switching times, optimally executable within one cycle. Granted, for mainstream architectures, this is not possible today (even if we already have special machines supporting this mode of operation, such as the [Cray XMT](#)²²⁵). For conventional systems, however, the smaller the overhead of a context switch and the finer the granularity of the threading system, the better will be the overall system utilization and its efficiency. For today's architectures we already see a flurry of libraries providing exactly this type of functionality: non-pre-emptive, task-queue based parallelization solutions, such as [Intel Threading Building Blocks \(TBB\)](#)²²⁶, [Microsoft Parallel Patterns Library \(PPL\)](#)²²⁷, [Cilk++](#)²²⁸, and many others. The possibility to suspend a current task if some preconditions for its execution are not met (such as waiting for I/O or the result of a different task), seamlessly switching to any other task which can continue, and to reschedule the initial task after the required result has been calculated, which makes the implementation of latency hiding almost trivial.

Rediscover constraint-based synchronization to replace global barriers

The code we write today is riddled with implicit (and explicit) global barriers. By “global barriers,” we mean the synchronization of the control flow between several (very often all) threads (when using [OpenMP](#)²²⁹) or processes ([MPI](#)²³⁰). For instance, an implicit global barrier is inserted after each loop parallelized using [OpenMP](#)²³¹ as the system synchronizes the threads used to execute the different iterations in parallel. In [MPI](#)²³² each of the communication steps imposes an explicit barrier onto the execution flow as (often all) nodes have to be synchronized. Each of those barriers

²²² <http://en.wikipedia.org/wiki/InfiniBand>

²²³ https://en.wikipedia.org/wiki/Message_Passing_Interface

²²⁴ <http://en.wikipedia.org/wiki/GPGPU>

²²⁵ http://en.wikipedia.org/wiki/Cray_XMT

²²⁶ <https://www.threadingbuildingblocks.org/>

²²⁷ <https://msdn.microsoft.com/en-us/library/dd492418.aspx>

²²⁸ <https://software.intel.com/en-us/articles/intel-cilk-plus/>

²²⁹ <https://openmp.org/wp/>

²³⁰ https://en.wikipedia.org/wiki/Message_Passing_Interface

²³¹ <https://openmp.org/wp/>

²³² https://en.wikipedia.org/wiki/Message_Passing_Interface

is like the eye of a needle the overall execution is forced to be squeezed through. Even minimal fluctuations in the execution times of the parallel threads (jobs) causes them to wait. Additionally, it is often only one of the executing threads that performs the actual reduce operation, which further impedes parallelism. A closer analysis of a couple of key algorithms used in science applications reveals that these global barriers are not always necessary. In many cases it is sufficient to synchronize a small subset of the threads. Any operation should proceed whenever the preconditions for its execution are met, and only those. Usually there is no need to wait for iterations of a loop to finish before you can continue calculating other things; all you need is to complete the iterations that produce the required results for the next operation. Good bye global barriers, hello constraint based synchronization! People have been trying to build this type of computing (and even computers) since the 1970s. The theory behind what they did is based on ideas around static and dynamic dataflow. There are certain attempts today to get back to those ideas and to incorporate them with modern architectures. For instance, a lot of work is being done in the area of constructing dataflow-oriented execution trees. Our results show that employing dataflow techniques in combination with the other ideas, as outlined herein, considerably improves scalability for many problems.

Adaptive locality control instead of static data distribution

While this principle seems to be a given for single desktop or laptop computers (the operating system is your friend), it is everything but ubiquitous on modern supercomputers, which are usually built from a large number of separate nodes (i.e., Beowulf clusters), tightly interconnected by a high-bandwidth, low-latency network. Today's prevalent programming model for those is MPI, which does not directly help with proper data distribution, leaving it to the programmer to decompose the data to all of the nodes the application is running on. There are a couple of specialized languages and programming environments based on PGAS²³³ (Partitioned Global Address Space) designed to overcome this limitation, such as Chapel²³⁴, X10²³⁵, UPC²³⁶, or Fortress²³⁷. However, all systems based on PGAS rely on static data distribution. This works fine as long as this static data distribution does not result in heterogeneous workload distributions or other resource utilization imbalances. In a distributed system these imbalances can be mitigated by migrating part of the application data to different localities (nodes). The only framework supporting (limited) migration today is Charm++²³⁸. The first attempts towards solving related problem go back decades as well, a good example is the Linda coordination language²³⁹. Nevertheless, none of the other mentioned systems support data migration today, which forces the users to either rely on static data distribution and live with the related performance hits or to implement everything themselves, which is very tedious and difficult. We believe that the only viable way to flexibly support dynamic and adaptive *locality* control is to provide a global, uniform address space to the applications, even on distributed systems.

Prefer moving work to the data over moving data to the work

For the best performance it seems obvious to minimize the amount of bytes transferred from one part of the system to another. This is true on all levels. At the lowest level we try to take advantage of processor memory caches, thus, minimizing memory latencies. Similarly, we try to amortize the data transfer time to and from GPGPUs²⁴⁰ as much as possible. At high levels we try to minimize data transfer between different nodes of a cluster or between different virtual machines on the cloud. Our experience (well, it's almost common wisdom) shows that the amount of bytes necessary to encode a certain operation is very often much smaller than the amount of bytes encoding the data the operation is performed upon. Nevertheless, we still often transfer the data to a particular place where we execute the operation just to bring the data back to where it came from afterwards. As an example let's look at the way we usually write our applications for clusters using MPI. This programming model is all about data transfer between nodes. MPI is the prevalent programming model for clusters, and it is fairly straightforward to understand and to use. Therefore,

²³³ <https://www.pgas.org/>

²³⁴ <https://chapel.cray.com/>

²³⁵ <https://x10-lang.org/>

²³⁶ <https://upc.lbl.gov/>

²³⁷ <https://labs.oracle.com/projects/plrg/Publications/index.html>

²³⁸ <https://charm.cs.uiuc.edu/>

²³⁹ [http://en.wikipedia.org/wiki/Linda_\(coordination_language\)](http://en.wikipedia.org/wiki/Linda_(coordination_language))

²⁴⁰ <http://en.wikipedia.org/wiki/GPGPU>

we often write applications in a way that accommodates this model, centered around data transfer. These applications usually work well for smaller problem sizes and for regular data structures. The larger the amount of data we have to churn and the more irregular the problem domain becomes, the worse the overall machine utilization and the (strong) scaling characteristics become. While it is not impossible to implement more dynamic, data driven, and asynchronous applications using MPI, it is somewhat difficult to do so. At the same time, if we look at applications that prefer to execute the code close to the *locality* where the data was placed, i.e., utilizing active messages (for instance based on Charm++²⁴¹), we see better asynchrony, simpler application codes, and improved scaling.

Favor message driven computation over message passing

Today's prevalently used programming model on parallel (multi-node) systems is MPI. It is based on message passing, as the name implies, which means that the receiver has to be aware of a message about to come in. Both codes, the sender and the receiver, have to synchronize in order to perform the communication step. Even the newer, asynchronous interfaces require explicitly coding the algorithms around the required communication scheme. As a result, everything but the most trivial MPI applications spends a considerable amount of time waiting for incoming messages, thus, causing starvation and latencies to impede full resource utilization. The more complex and more dynamic the data structures and algorithms become, the larger the adverse effects. The community discovered message-driven and data-driven methods of implementing algorithms a long time ago, and systems such as Charm++²⁴² have already integrated active messages demonstrating the validity of the concept. Message-driven computation allows for sending messages without requiring the receiver to actively wait for them. Any incoming message is handled asynchronously and triggers the encoded action by passing along arguments and—possibly—continuations. HPX combines this scheme with work-queue based scheduling as described above, which allows the system to almost completely overlap any communication with useful work, thereby minimizing latencies.

2.7 Additional material

- 2-day workshop held at CSCS in 2016
 - Recorded lectures²⁴³
 - Slides²⁴⁴
- Tutorials repository²⁴⁵
- C++ Lectures <https://www.youtube.com/playlist?list=PL7vEgTL3FalY2eBxud1wsfz8OKvE9sd_z>
- Parallel C++ for Scientific Applications Lectures <<https://www.youtube.com/watch?v=RdmXiIilArM&list=PL7vEgTL3Falab59uJ>>
- STEllAR Group blog posts²⁴⁶
- Basic HPX recipes
 - Exporting a free function from a shared library which lives in a namespace, to use as Action²⁴⁷
 - Turning a struct or class into a component and use it's methods²⁴⁸
 - Creating and referencing components in hpx²⁴⁹

²⁴¹ <https://charm.cs.uiuc.edu/>

²⁴² <https://charm.cs.uiuc.edu/>

²⁴³ <https://www.youtube.com/playlist?list=PL1tk5lGm7zvSXfS-sqOOmIJ0lFNjKze18>

²⁴⁴ <https://github.com/STELLAR-GROUP/tutorials/tree/master/cscs2016>

²⁴⁵ <https://github.com/STELLAR-GROUP/tutorials>

²⁴⁶ <http://stellar-group.org/blog/>

²⁴⁷ <https://gitlab.com/-/snippets/1821389>

²⁴⁸ <https://gitlab.com/-/snippets/1822983>

²⁴⁹ <https://gitlab.com/-/snippets/1828131>

2.8 Overview

HPX is organized into different sub-libraries and those in turn into modules. The libraries and modules are independent, with clear dependencies and no cycles. As an end-user, the use of these libraries is completely transparent. If you use e.g. `add_hpx_executable` to create a target in your project you will automatically get all modules as dependencies. See below for a list of the available libraries and modules. Currently these are nothing more than an internal grouping and do not affect usage. They cannot be consumed individually at the moment.

Note: There is a dependency report that displays useful information about the structure of the code. It is available for each commit at [HPX Dependency report](#).

2.8.1 Core modules

affinity

The affinity module contains helper functionality for mapping worker threads to hardware resources.

See the *API reference* of the module for more details.

algorithms

The algorithms module exposes the full set of algorithms defined by the C++ standard. There is also partial support for C++ ranges.

See the *API reference* of the module for more details.

allocator_support

This module provides utilities for allocators. It contains `hpx::util::internal_allocator` which directly forwards allocation calls to `jemalloc`. This utility is mainly useful on Windows.

See the *API reference* of the module for more details.

asio

The asio module is a thin wrapper around the [Boost.Asio](#)²⁵⁰ library, providing a few additional helper functions.

See the *API reference* of the module for more details.

²⁵⁰ https://www.boost.org/doc/libs/release/doc/html/boost_asio.html

assertion

The assertion library implements the macros `HPX_ASSERT` and `HPX_ASSERT_MSG`. Those two macros can be used to implement assertions which are turned off during a release build.

By default, the location and function where the assert has been called from are displayed when the assertion fires. This behavior can be modified by using `hpx::assertion::set_assertion_handler`. When HPX initializes, it uses this function to specify a more elaborate assertion handler. If your application needs to customize this, it needs to do so before calling `hpx::init`, `hpx_main` or using the C-main wrappers.

See the *API reference* of the module for more details.

async_base

The `async_base` module defines the basic functionality for spawning tasks on thread pools. This module does not implement any functionality on its own, but is extended by `async_local` and `async_distributed` with implementations for the local and distributed cases.

See the *API reference* of this module for more details.

async_combinators

This module contains combinators for futures. The `when_*` functions allow you to turn multiple futures into a single future which is ready when all, any, some, or each of the given futures are ready. The `wait_*` combinators are equivalent to the `when_*` functions except that they do not return a future. Those wait for all futures to become ready before returning to the user. Note that the `wait_*` functions will rethrow one of the exceptions from exceptional futures. The `wait_*_nothrow` combinators are equivalent to the `wait_*` functions exception that they do not throw if one of the futures has become exceptional.

The `split_future` combinator takes a single future of a container (e.g. `tuple`) and turns it into a container of futures.

See `lcos_local`, `synchronization`, and `async_distributed` for other synchronization facilities.

See the *API reference* of this module for more details.

async_cuda

This library adds a simple API that enables the user to retrieve a future from a [CUDA](https://www.nvidia.com/object/cuda_home_new.html)²⁵¹ stream. Typically, a user may launch one or more kernels and then get a future from the stream that will become ready when those kernels have completed. It is important to note that multiple kernels may be launched without fetching a future, and multiple futures may be obtained from the helper. Please refer to the unit tests and examples for further examples.

See the *API reference* of this module for more details.

²⁵¹ https://www.nvidia.com/object/cuda_home_new.html

async_local

This module extends *async_base* to provide local implementations of `hpx::async`, `hpx::post`, `hpx::sync`, and `hpx::dataflow`. The *async_distributed* module extends the functionality in this module to work with *actions*.

See the API reference of this module for more details.

async_mpi

The MPI library is intended to simplify the process of integrating [MPI²⁵²](#) based codes with the *HPX* runtime. Any MPI function that is asynchronous and uses an `MPI_Request` may be converted into an `hpx::future`. The syntax is designed to allow a simple replacement of the MPI call with a futurized async version that accepts an executor instead of a communicator, and returns a future instead of assigning a request. Typically, an MPI call of the form

```
int MPI_Isend(buf, count, datatype, rank, tag, comm, request);
```

becomes

```
hpx::future<int> f = hpx::async(executor, MPI_Isend, buf, count, datatype, rank, tag);
```

When the MPI operation is complete, the future will become ready. This allows communication to be integrated cleanly with the rest of *HPX*, in particular the continuation style of programming may be used to build up more complex code. Consider the following example, that chains user processing, sends and receives using continuations...

```
// create an executor for MPI dispatch
hpx::mpi::experimental::executor exec(MPI_COMM_WORLD);

// post an asynchronous receive using MPI_Irecv
hpx::future<int> f_recv = hpx::async(
    exec, MPI_Irecv, &data, rank, MPI_INT, rank_from, i);

// attach a continuation to run when the recv completes,
f_recv.then([=, &tokens, &counter](auto&&)
{
    // call an application specific function
    msg_recv(rank, size, rank_to, rank_from, tokens[i], i);

    // send a new message
    hpx::future<int> f_send = hpx::async(
        exec, MPI_Isend, &tokens[i], 1, MPI_INT, rank_to, i);

    // when that send completes
    f_send.then([=, &tokens, &counter](auto&&)
    {
        // call an application specific function
        msg_send(rank, size, rank_to, rank_from, tokens[i], i);
    });
});
```

The example above makes use of `MPI_Isend` and `MPI_Irecv`, but *any* MPI function that uses requests may be futurized in this manner. The following is a (non exhaustive) list of MPI functions that *should* be supported, though not all have been tested at the time of writing (please report any problems to the issue tracker).

²⁵² https://en.wikipedia.org/wiki/Message_Passing_Interface


```

int MPI_Isend(...);
int MPI_Ibsend(...);
int MPI_Issend(...);
int MPI_Irsend(...);
int MPI_Irecv(...);
int MPI_Imrecv(...);
int MPI_Ibarrier(...);
int MPI_Ibcast(...);
int MPI_Igather(...);
int MPI_Igatherv(...);
int MPI_Iscatter(...);
int MPI_Iscatterv(...);
int MPI_Iallgather(...);
int MPI_Iallgatherv(...);
int MPI_Ialltoall(...);
int MPI_Ialltoallv(...);
int MPI_Ialltoallw(...);
int MPI_Ireduce(...);
int MPI_Iallreduce(...);
int MPI_Ireduce_scatter(...);
int MPI_Ireduce_scatter_block(...);
int MPI_Iscan(...);
int MPI_Iexscan(...);
int MPI_Ineighbor_allgather(...);
int MPI_Ineighbor_allgatherv(...);
int MPI_Ineighbor_alltoall(...);
int MPI_Ineighbor_alltoallv(...);
int MPI_Ineighbor_alltoallw(...);

```

Note that the *HPX* mpi futurization wrapper should work with *any* asynchronous MPI call, as long as the function signature has the last two arguments `MPI_xxx(..., MPI_Comm comm, MPI_Request *request)` - internally these two parameters will be substituted by the executor and future data parameters that are supplied by template instantiations inside the `hpx::mpi` code.

See the API reference of this module for more details.

async_sycl

This module allows creating HPX futures using SYCL²⁵³ events, effectively integrating asynchronous SYCL kernels and memory transfers with HPX. Building on this integration, this module also contains a SYCL executor. This executor encapsulates a SYCL queue. When SYCL queue member functions are launched with this executor, the user can automatically obtain the HPX futures associated with them.

The creation of the HPX futures using SYCL events is based on the same event polling mechanism that the CUDA HPX integration uses. Each registered event gets an associated callback and gets inserted into a callback vector to be polled by the scheduler in between tasks. Once the polling reveals the event is complete, the callback will be called, which in turn sets the future to ready (see `sycl_event_callback.cpp`). There are multiple adaptations for HipSYCL for this: To keep the runtime alive (avoiding the repeated on-the-fly creation of the runtime during the polling), we keep a default queue. Furthermore, we flush the internal SYCL DAG to ensure that the launched SYCL function is actually being executed.

The SYCL executor offers the usual `post` and `async_execute` functions. Additionally, it contains two `get_future` functions. One expects a pre-existing SYCL event to return a future, the other one does not but will launch an empty SYCL kernel instead, to obtain an event (causing higher overhead for the sake of being more convenient). The `post` and

²⁵³ <https://en.wikipedia.org/wiki/SYCL>

async_execute implementations here are actually different for HipSYCL and OneAPI, since the sycl::queue in OneAPI uses a different interface (using a code_location parameter) which requires some adaptations here.

To make this module compile, we use the -fno-sycl and -fsycl compiler parameters for the OneAPI use-case (requiring HPX to be compiled with dpcpp). For HipSYCL we use its cmake integration instead (requiring HPX to be compiled with clang++ and including HipSYCL as a library).

To build with OneAPI, use the CMake Variable HPX_WITH_SYCL=ON. To build with HipSYCL, use HPX_WITH_SYCL=ON and HPX_WITH_HIPSYCL=ON (and make sure find_package will find HipSYCL).

Lastly, the module contains three tests/examples. All three implement a simple vector add example. The first one obtains a future using the free method get_future, the second one uses a single SYCL executor and the last one is using multiple executors called from multiple host threads.

To build the tests, use "make tests.unit.modules.async_sycl" To run the tests, use "ctest -R sycl".

NOTE: Theoretically, this module could work with other SYCL implementations, but was only tested using OneAPI and HipSYCL so far.

See the *API reference* of this module for more details.

batch_environments

This module allows for the detection of execution as batch jobs, a series of programs executed without user intervention. All data is preselected and will be executed according to preset parameters, such as date or completion of another task. Batch environments are especially useful for executing repetitive tasks.

HPX supports the creation of batch jobs through the Portable Batch System (PBS) and SLURM.

For more information on batch environments, see *Running on batch systems* and the API reference for the module.

cache

This module provides two cache data structures:

- `hpx::util::cache::local_cache`
- `hpx::util::cache::lru_cache`

See the *API reference* of the module for more details.

checkpoint_base

The checkpoint_base module contains lower level facilities that wrap simple check-pointing capabilities. This module does not implement special handling for futures or components, but simply serializes all arguments to or from a given container.

This module exposes the `hpx::util::save_checkpoint_data`, `hpx::util::restore_checkpoint_data`, and `hpx::util::prepare_checkpoint_data` APIs. These functions encapsulate the basic serialization functionalities necessary to save/restore a variadic list of arguments to/from a given data container.

See the *API reference* of this module for more details.

command_line_handling_local

TODO: High-level description of the module.

See the API reference of this module for more details.

compute_local

TODO: High-level description of the module.

See the *API reference* of this module for more details.

concepts

This module provides helpers for emulating concepts. It provides the following macros:

- `HPX_CONCEPT_REQUIRES`
- `HPX_HAS_MEMBER_XXX_TRAIT_DEF`
- `HPX_HAS_XXX_TRAIT_DEF`

See the API reference of the module for more details.

concurrency

This module provides concurrency primitives useful for multi-threaded programming such as:

- `hpx::barrier`
- `hpx::util::cache_line_data` and `hpx::util::cache_aligned_data`: wrappers for aligning and padding data to cache lines.
- various lockfree queue data structures

See the API reference of the module for more details.

config

The config module contains various configuration options, typically hidden behind macros that choose the correct implementation based on the compiler and other available options. It also contains platform independent macros to control inlining, export sets and more.

See the *API reference* of the module for more details.

config_registry

The config_registry module is a low level module providing helper functionality for registering configuration entries to a global registry from other modules. The `hpx::config_registry::add_module_config` function is used to add configuration options, and `hpx::config_registry::get_module_configs` can be used to retrieve configuration entries registered so far. `add_module_config_helper` can be used to register configuration entries through static global options.

See the API reference of this module for more details.

contracts

The contracts module provides C++ contracts support for HPX with intelligent fallback to assertions when native contracts are not available. This module implements a forward-compatible API that works across different C++ standards and compiler capabilities.

The module provides three primary macros: `HPX_PRE`, `HPX_POST`, and `HPX_CONTRACT_ASSERT`. These macros automatically adapt their behavior based on compiler capabilities:

- When C++26 native contracts are available (`__cpp_contracts` defined), they map to standard contract syntax
- When `HPX_WITH_CXX26_CONTRACTS=OFF`, preconditions and postconditions become no-ops while contract assertions remain available as enhanced assertions

Configuration

Enable contracts in CMake:

```
cmake -DHPX_WITH_CXX26_CONTRACTS=ON -DCMAKE_CXX_STANDARD=26
```

Enable contract-enhanced assertions (optional):

```
cmake -DHPX_WITH_CXX26_CONTRACTS=ON -DHPX_CONTRACTS_WITH_ASSERTS_AS_CONTRACT_ASSERTS=ON
```

Contract assertions work even when contracts are disabled:

```
cmake -DHPX_WITH_CXX26_CONTRACTS=OFF # HPX_CONTRACT_ASSERT still maps to HPX_ASSERT
```

Advanced Features

Contract-Enhanced Assertions

When `HPX_CONTRACTS_WITH_ASSERTS_AS_CONTRACT_ASSERTS=ON` is enabled, regular `HPX_ASSERT` calls are automatically upgraded to use contract assertions in C++26 mode:

```
void example_function(int value)
{
    HPX_ASSERT(value > 0); // Becomes contract_assert(value > 0) in C++26 mode
    // ... rest of function
}
```

Warning: Enhanced assertions only provide benefits when native C++26 contracts are supported by the compiler. Without native contract support, `HPX_ASSERT` -> `HPX_CONTRACT_ASSERT` -> `HPX_ASSERT` (no enhancement). CMake will issue a warning if you enable this option without native contract support.

This provides enhanced contract semantics throughout your existing codebase without requiring changes to assertion code. The transformation occurs in the contracts module (`contracts.hpp`) where the `HPX_ASSERT` macro is overridden to use `HPX_CONTRACT_ASSERT` when `HPX_WITH_ASSERTS_AS_CONTRACT_ASSERTS=ON`:

- `HPX_WITH_CXX26_CONTRACTS=ON` - Contracts module is enabled
- `HPX_CONTRACTS_WITH_ASSERTS_AS_CONTRACT_ASSERTS=ON` - Assertion enhancement is enabled

The implementation works by redefining `HPX_ASSERT` in `contracts.hpp` to use `HPX_CONTRACT_ASSERT`, which automatically adapts to the current contract mode (native C++26 contracts or assertion fallback). This ensures all existing `HPX_ASSERT` calls throughout the HPX codebase automatically gain contract semantics when available.

Usage Examples

Preconditions and postconditions using declaration syntax (C++26):

```
int divide(int a, int b) HPX_PRE(b != 0)
{
    return a / b;
}

int factorial(int n) HPX_PRE(n >= 0) HPX_POST(r; r > 0)
{
    return n <= 1 ? 1 : n * factorial(n - 1);
}
```

Contract assertions (available in all modes):

```
void process_array(std::vector<int>& arr, size_t index)
{
    HPX_CONTRACT_ASSERT(index < arr.size());
    arr[index] *= 2;
}
```

Design Philosophy

HPX_CONTRACT_ASSERT: Enhanced assertion mechanism

Available even when `HPX_WITH_CONTRACTS=OFF` because it provides value as an enhanced assertion. Maps to `HPX_ASSERT` in all configurations.

HPX_PRE/HPX_POST: True contract syntax

Represent language-level contract semantics. When contracts are enabled but native C++26 contracts are not available, these become no-ops to maintain forward compatibility. When `HPX_WITH_CONTRACTS=OFF`, they are also no-ops. This prepares for C++26 migration where they will be attached to function declarations rather than used in function bodies.

Migration Strategy

The module is designed for smooth migration to C++26 native contracts:

Current (transition mode):

```
int func(int x)
{
    HPX_PRE(x > 0);    // No-op in fallback mode, active in native mode
    return x;
}
```

Target (C++26 native):

```
int func(int x) HPX_PRE(x > 0)
{
    return x;
}
```

Note: In fallback mode, HPX_PRE and HPX_POST become no-ops to maintain forward compatibility and avoid performance overhead. Use HPX_CONTRACT_ASSERT when you need runtime validation in all modes.

Testing

The module includes comprehensive testing with automatic compiler capability detection. Tests are organized into three categories:

- **Declaration tests:** Test C++26 native contract syntax when `__cpp_contracts` is available
- **Fallback tests:** Test assertion fallback behavior when contracts are not natively supported
- **Disabled tests:** Test no-op behavior when contracts are disabled

The test suite automatically detects compiler capabilities at configure time and builds only the appropriate tests for the current configuration.

See the *API reference* of the module for more details.

coroutines

The coroutines module provides coroutine (user-space thread) implementations for different platforms.

See the *API reference* of the module for more details.

datastructures

The datastructures module provides basic data structures (typically provided for compatibility with older C++ standards):

- `hpx::detail::small_vector`
- `hpx::util::basic_any`
- `hpx::util::member_pack`
- `hpx::optional`
- `hpx::tuple`
- `hpx::variant`

See the *API reference* of the module for more details.

debugging

This module provides helpers for demangling symbol names.

See the *API reference* of the module for more details.

errors

This module provides support for exceptions and error codes:

- `hpx::exception`
- `hpx::error_code`
- `hpx::error`

See the *API reference* of the module for more details.

execution

This library implements executors and execution policies for use with parallel algorithms and other facilities related to managing the execution of tasks.

See the *API reference* of the module for more details.

execution_base

The basic execution module is the main entry point to implement parallel and concurrent operations. It is modeled after P0443²⁵⁴ with some additions and implementations for the described concepts. Most notably, it provides an abstraction for execution resources, execution contexts and execution agents in such a way, that it provides customization points that those aforementioned concepts can be replaced and combined with ease.

For that purpose, three virtual base classes are provided to be able to provide implementations with different properties:

- **resource_base:** This is the abstraction for execution resources, that is for example CPU cores or an accelerator.
- **context_base:** An execution context uses execution resources and is able to spawn new execution agents, as new threads of executions on the available resources.
- **agent_base:** The execution agent represents the thread of execution, and can be used to yield, suspend, resume or abort a thread of execution.

executors

The executors module exposes executors and execution policies. Most importantly, it exposes the following classes and constants:

- `hpx::execution::sequenced_executor`
- `hpx::execution::parallel_executor`
- `hpx::execution::sequenced_policy`
- `hpx::execution::parallel_policy`
- `hpx::execution::parallel_unsequenced_policy`

²⁵⁴ <http://wg21.link/p0443>

- `hpx::execution::sequenced_task_policy`
- `hpx::execution::parallel_task_policy`
- `hpx::execution::seq`
- `hpx::execution::par`
- `hpx::execution::par_unseq`
- `hpx::execution::task`

See the *API reference* of this module for more details.

filesystem

This module provides a compatibility layer for the C++17 filesystem library. If the filesystem library is available this module will simply forward its contents into the `hpx::filesystem` namespace. If the library is not available it will fall back to Boost.Filesystem instead.

See the *API reference* of the module for more details.

format

The format module exposes the `format` and `format_to` functions for formatting strings.

See the *API reference* of the module for more details.

functional

This module provides function wrappers and helpers for managing functions and their arguments.

- `hpx::function`
- `hpx::function_ref`
- `hpx::move_only_function`
- `hpx::bind`
- `hpx::bind_back`
- `hpx::bind_front`
- `hpx::util::deferred_call`
- `hpx::invoke`
- `hpx::invoke_r`
- `hpx::invoke_fused`
- `hpx::invoke_fused_r`
- `hpx::mem_fn`
- `hpx::util::one_shot`
- `hpx::util::protect`
- `hpx::util::result_of`
- `hpx::placeholders::_1`

- `hpx::placeholders::_2`
- ...
- `hpx::placeholders::_9`

See the *API reference* of the module for more details.

futures

This module defines the `hpx::future` and `hpx::shared_future` classes corresponding to the C++ standard library classes `std::future`²⁵⁵ and `std::shared_future`²⁵⁶. Note that the specializations of `hpx::future::then` for executors and execution policies are defined in the *execution* module.

See the *API reference* of this module for more details.

hardware

The hardware module abstracts away hardware specific details of timestamps and CPU features.

See the *API reference* of the module for more details.

hashing

The hashing module provides two hashing implementations:

- `hpx::util::fibhash`
- `hpx::util::jenkins_hash`

See the *API reference* of the module for more details.

include_local

This module provides no functionality in itself. Instead it provides headers that group together other headers that often appear together. This module provides local-only headers.

See the *API reference* of this module for more details.

ini

TODO: High-level description of the module.

See the *API reference* of this module for more details.

²⁵⁵ <http://en.cppreference.com/w/cpp/thread/future>

²⁵⁶ http://en.cppreference.com/w/cpp/thread/shared_future

init_runtime_local

TODO: High-level description of the module.

See the API reference of this module for more details.

io_service

This module provides an abstraction over Boost.ASIO, combining multiple `asio::io_contexts` into a single pool. `hpx::util::io_service_pool` provides a simple pool of `asio::io_contexts` with an API similar to `asio::io_context`. `hpx::threads::detail::io_service_thread_pool` wraps `hpx::util::io_service_pool` into an interface derived from `hpx::threads::detail::thread_pool_base`.

See the *API reference* of this module for more details.

iterator_support

This module provides helpers for iterators. It provides `hpx::util::iterator_facade` and `hpx::util::iterator_adaptor` for creating new iterators, and the trait `hpx::util::is_iterator` along with more specific iterator traits.

See the API reference of the module for more details.

itt_notify

This module provides support for profiling with Intel VTune²⁵⁷.

See the API reference of this module for more details.

lci_base

This module provides helper functionality for detecting LCI environments.

See the API reference of this module for more details.

lcos_local

This module provides the following local *LCOs*:

- `hpx::lcos::local::and_gate`
- `hpx::lcos::local::channel`
- `hpx::lcos::local::one_element_channel`
- `hpx::lcos::local::receive_channel`
- `hpx::lcos::local::send_channel`
- `hpx::lcos::local::guard`
- `hpx::lcos::local::guard_set`
- `hpx::lcos::local::run_guarded`
- `hpx::lcos::local::conditional_trigger`

²⁵⁷ <https://software.intel.com/content/www/us/en/develop/tools/vtune-profiler.html>

- `hpx::packaged_task`
- `hpx::promise`
- `hpx::lcos::local::receive_buffer`
- `hpx::lcos::local::trigger`

See *lcos_distributed* for distributed LCOs. Basic synchronization primitives for use in *HPX* threads can be found in *synchronization*. *async_combinators* contains useful utility functions for combining futures.

See the *API reference* of this module for more details.

likwid

TODO: High-level description of the module.

See the *API reference* of this module for more details.

lock_registration

This module contains functionality for registering locks to detect when they are locked and unlocked on different threads.

See the *API reference* of this module for more details.

logging

This module provides useful macros for logging information.

See the *API reference* of the module for more details.

memory

Part of this module is a forked version of `boost::intrusive_ptr` from `Boost.SmartPtr`²⁵⁸.

See the *API reference* of the module for more details.

mpi_base

This module provides helper functionality for detecting `MPI`²⁵⁹ environments.

See the *API reference* of this module for more details.

²⁵⁸ https://www.boost.org/doc/libs/release/libs/smart_ptr/doc/html/smart_ptr.html

²⁵⁹ https://en.wikipedia.org/wiki/Message_Passing_Interface

pack_traversal

This module exposes the basic functionality for traversing various packs, both synchronously and asynchronously: `hpx::util::traverse_pack` and `hpx::util::traverse_pack_async`. It also exposes the higher level functionality of unwrapping nested futures: `hpx::util::unwrap` and its function object form `hpx::util::functional::unwrap`.

See the *API reference* of this module for more details.

plugin

This module provides base utilities for creating plugins.

See the *API reference* of the module for more details.

prefix

This module provides utilities for handling the prefix of an *HPX* application, i.e. the paths used for searching components and plugins.

See the *API reference* of this module for more details.

preprocessor

This library contains useful preprocessor macros:

- `HPX_PP_CAT`: Concatenate two tokens
- `HPX_PP_EXPAND`: Expands a preprocessor token
- `HPX_PP_NARGS`: Determines the number of arguments passed to a variadic macro
- `HPX_PP_STRINGIZE`: Turns a token into a string
- `HPX_PP_STRIP_PARENS`: Strips parenthesis from a token

See the *API reference* of the module for more details.

program_options

The module `program_options` is a direct fork of the `Boost.Program_options`²⁶⁰ library (Boost V1.70.0²⁶¹). In order to be included as an *HPX* module, the `Boost.Program_options` library has been moved to the namespace `hpx::program_options`. We have also replaced all Boost facilities the library depends on with either the equivalent facilities from the standard library or from *HPX*. As a result, the *HPX* `program_options` module is fully interface compatible with `Boost.Program_options` (sans the `hpx` namespace and the `#include <hpx/modules/program_options.hpp>` changes that need to be applied to all code relying on this library).

All credit goes to Vladimir Prus, the author of the excellent `Boost.Program_options` library. All bugs have been introduced by us.

See the *API reference* of the module for more details.

²⁶⁰ https://www.boost.org/doc/html/program_options.html

²⁶¹ https://www.boost.org/doc/libs/1_70_0/doc/html/program_options.html

properties

This module implements the `prefer` customization point for properties in terms of [P2220](https://wg21.link/p2220)²⁶². This differs from [P1393](http://wg21.link/p1393)²⁶³ in that it relies fully on `tag_invoke` overloads and fewer base customization points. Actual properties are defined in modules. All functionality is experimental and can be accessed through the `hpx::experimental` namespace.

See the API reference of this module for more details.

resiliency

In *HPX*, a program failure is a manifestation of a failing task. This module exposes several APIs that allow users to manage failing tasks in a convenient way by either replaying a failed task or by replicating a specific task.

Task replay is analogous to the Checkpoint/Restart mechanism found in conventional execution models. The key difference being localized fault detection. When the runtime detects an error, it replays the failing task as opposed to completely rolling back the entire program to the previous checkpoint.

Task replication is designed to provide reliability enhancements by replicating a set of tasks and evaluating their results to determine a consensus among them. This technique is most effective in situations where there are few tasks in the critical path of the DAG which leaves the system underutilized or where hardware or software failures may result in an incorrect result instead of an error. However, the drawback of this method is the additional computational cost incurred by repeating a task multiple times.

The following API functions are exposed:

- `hpx::resiliency::experimental::async_replay`: This version of task replay will catch user-defined exceptions and automatically reschedule the task *N* times before throwing an `hpx::resiliency::experimental::abort_replay_exception` if no task is able to complete execution without an exception.
- `hpx::resiliency::experimental::async_replay_validate`: This version of replay adds an argument to `async_replay` which receives a user-provided validation function to test the result of the task against. If the task's output is validated, the result is returned. If the output fails the check or an exception is thrown, the task is replayed until no errors are encountered or the number of specified retries has been exceeded.
- `hpx::resiliency::experimental::async_replicate`: This is the most basic implementation of the task replication. The API returns the first result that runs without detecting any errors.
- `hpx::resiliency::experimental::async_replicate_validate`: This API additionally takes a validation function which evaluates the return values produced by the threads. The first task to compute a valid result is returned.
- `hpx::resiliency::experimental::async_replicate_vote`: This API adds a vote function to the basic replicate function. Many hardware or software failures are silent errors which do not interrupt program flow. In order to detect errors of this kind, it is necessary to run the task several times and compare the values returned by every version of the task. In order to determine which return value is “correct”, the API allows the user to provide a custom consensus function to properly form a consensus. This voting function then returns the “correct” answer.
- `hpx::resiliency::experimental::async_replicate_vote_validate`: This combines the features of the previously discussed replicate set. Replicate vote validate allows a user to provide a validation function to filter results. Additionally, as described in replicate vote, the user can provide a “voting function” which returns the consensus formed by the voting logic.

²⁶² <https://wg21.link/p2220>

²⁶³ <http://wg21.link/p1393>

- `hpx::resiliency::experimental::dataflow_replay`: This version of dataflow replay will catch user-defined exceptions and automatically reschedules the task N times before throwing an `hpx::resiliency::experimental::abort_replay_exception` if no task is able to complete execution without an exception. Any arguments for the executed task that are futures will cause the task invocation to be delayed until all of those futures have become ready.
- `hpx::resiliency::experimental::dataflow_replay_validate`: This version of replay adds an argument to dataflow replay which receives a user-provided validation function to test the result of the task against. If the task's output is validated, the result is returned. If the output fails the check or an exception is thrown, the task is replayed until no errors are encountered or the number of specified retries have been exceeded. Any arguments for the executed task that are futures will cause the task invocation to be delayed until all of those futures have become ready.
- `hpx::resiliency::experimental::dataflow_replicate`: This is the most basic implementation of the task replication. The API returns the first result that runs without detecting any errors. Any arguments for the executed task that are futures will cause the task invocation to be delayed until all of those futures have become ready.
- `hpx::resiliency::experimental::dataflow_replicate_validate`: This API additionally takes a validation function which evaluates the return values produced by the threads. The first task to compute a valid result is returned. Any arguments for the executed task that are futures will cause the task invocation to be delayed until all of those futures have become ready.
- `hpx::resiliency::experimental::dataflow_replicate_vote`: This API adds a vote function to the basic replicate function. Many hardware or software failures are silent errors which do not interrupt program flow. In order to detect errors of this kind, it is necessary to run the task several times and compare the values returned by every version of the task. In order to determine which return value is "correct", the API allows the user to provide a custom consensus function to properly form a consensus. This voting function then returns the "correct" answer. Any arguments for the executed task that are futures will cause the task invocation to be delayed until all of those futures have become ready.
- `hpx::resiliency::experimental::dataflow_replicate_vote_validate`: This combines the features of the previously discussed replicate set. Replicate vote validate allows a user to provide a validation function to filter results. Additionally, as described in replicate vote, the user can provide a "voting function" which returns the consensus formed by the voting logic. Any arguments for the executed task that are futures will cause the task invocation to be delayed until all of those futures have become ready.

See the *API reference* of the module for more details.

resource_partitioner

The `resource_partitioner` module defines `hpx::resource::partitioner`, the class used by the runtime and users to partition available hardware resources into thread pools. See *Using the resource partitioner* for more details on using the resource partitioner in applications.

See the API reference of this module for more details.

runtime_configuration

This module handles the configuration options required by the runtime.

See the *API reference* of this module for more details.

runtime_local

TODO: High-level description of the library.

See the *API reference* of this module for more details.

schedulers

This module provides schedulers used by thread pools in the *thread_pools* module. There are currently three main schedulers:

- `hpx::threads::policies::local_priority_queue_scheduler`
- `hpx::threads::policies::static_priority_queue_scheduler`
- `hpx::threads::policies::shared_priority_queue_scheduler`

Other schedulers are specializations or variations of the above schedulers. See the examples of the *resource_partitioner* module for examples of specifying a custom scheduler for a thread pool.

See the *API reference* of this module for more details.

serialization

This module provides serialization primitives and support for all built-in types as well as all C++ Standard Library collection and utility types. This list is extended by *HPX* vocabulary types with proper support for global reference counting. *HPX*'s mode of serialization is derived from Boost's [serialization model](https://www.boost.org/doc/libs/1_72_0/libs/serialization/doc/index.html)²⁶⁴ and, as such, is mostly interface compatible with its Boost counterpart.

The purest form of serializing data is to copy the content of the payload bit by bit; however, this method is impractical for generic C++ types, which might be composed of more than just regular built-in types. Instead, *HPX*'s approach to serialization is derived from the Boost Serialization library, and is geared towards allowing the programmer of a given class explicit control and syntax of what to serialize. It is based on operator overloading of two special archive types that hold a buffer or stream to store the serialized data and is responsible for dispatching the serialization mechanism to the intrusive or non-intrusive version. The serialization process is recursive. Each member that needs to be serialized must be specified explicitly. The advantage of this approach is that the serialization code is written in C++ and leverages all necessary programming techniques. The generic, user-facing interface allows for effective application of the serialization process without obstructing the algorithms that need special code for packing and unpacking. It also allows for optimizations in the implementation of the archives.

See the *API reference* of the module for more details.

²⁶⁴ https://www.boost.org/doc/libs/1_72_0/libs/serialization/doc/index.html

static_reinit

This module provides a simple wrapper around static variables that can be reinitialized.

See the *API reference* of this module for more details.

string_util

This module contains string utilities inspired by the Boost String Algorithms Library.

See the *API reference* of this module for more details.

synchronization

This module provides synchronization primitives that should be used rather than the C++ standard ones in *HPX* threads:

- `hpx::barrier`
- `hpx::binary_semaphore`
- `hpx::call_once`
- `hpx::condition_variable`
- `hpx::condition_variable_any`
- `hpx::counting_semaphore`
- `hpx::lcos::local::event`
- `hpx::latch`
- `hpx::mutex`
- `hpx::no_mutex`
- `hpx::once_flag`
- `hpx::recursive_mutex`
- `hpx::shared_mutex`
- `hpx::sliding_semaphore`
- `hpx::spinlock` (*std::mutex* compatible spinlock)
- `hpx::spinlock_no_backoff` (*boost::mutex* compatible spinlock)
- `hpx::spinlock_pool`
- `hpx::stop_callback`
- `hpx::stop_source`
- `hpx::stop_token`
- `hpx::in_place_stop_token`
- `hpx::timed_mutex`
- `hpx::upgrade_to_unique_lock`
- `hpx::upgrade_lock`

See *lcos_local*, *async_combinators*, and *async_distributed* for higher level synchronization facilities.

See the *API reference* of this module for more details.

tag_invoke

TODO: High-level description of the module.

See the *API reference* of this module for more details.

testing

The testing module contains useful macros for testing. The results of tests can be printed with `hpx::util::report_errors`. The following macros are provided:

- `HPX_TEST`
- `HPX_TEST_MSG`
- `HPX_TEST_EQ`
- `HPX_TEST_NEQ`
- `HPX_TEST_LT`
- `HPX_TEST_LTE`
- `HPX_TEST_RANGE`
- `HPX_TEST_EQ_MSG`
- `HPX_TEST_NEQ_MSG`
- `HPX_SANITY`
- `HPX_SANITY_MSG`
- `HPX_SANITY_EQ`
- `HPX_SANITY_NEQ`
- `HPX_SANITY_LT`
- `HPX_SANITY_LTE`
- `HPX_SANITY_RANGE`
- `HPX_SANITY_EQ_MSG`

See the API reference of the module for more details.

thread_pool_util

This module contains helper functions for asynchronously suspending and resuming thread pools and their worker threads.

See the *API reference* of this module for more details.

thread_pools

This module defines the thread pools and utilities used by the *HPX* runtime. The only thread pool implementation provided by this module is `hpx::threads::detail::scheduled_thread_pool`, which is derived from `hpx::threads::detail::thread_pool_base` defined in the *threading_base* module.

See the API reference of this module for more details.

thread_support

This module provides miscellaneous utilities for threading and concurrency.

See the *API reference* of the module for more details.

threading

This module provides the equivalents of `std::thread` and `std::jthread` for lightweight *HPX* threads:

- `hpx::thread`
- `hpx::jthread`

See the *API reference* of this module for more details.

threading_base

This module contains the base class definition required for threads. The base class `hpx::threads::thread_data` is inherited by two specializations for stackful and stackless threads: `hpx::threads::thread_data_stackful` and `hpx::threads::thread_data_stackless`. In addition, the module defines the base classes for schedulers and thread pools: `hpx::threads::policies::scheduler_base` and `hpx::threads::thread_pool_base`.

See the API reference of this module for more details.

thread_manager

This module defines the `hpx::threads::threadmanager` class. This is used by the runtime to manage the creation and destruction of thread pools. The *resource_partitioner* module handles the partitioning of resources into thread pools, but not the creation of thread pools.

See the API reference of this module for more details.

thrust

The thrust module integrates *HPX* parallel algorithms with NVIDIA Thrust, enabling GPU acceleration using familiar *HPX* algorithm syntax.

Execution Policies

- hpx::thrust::thrust_host_policy**
CPU execution using optimized parallel implementations
- hpx::thrust::thrust_device_policy**
Synchronous GPU execution
- hpx::thrust::thrust_task_policy**
Asynchronous GPU execution returning *HPX* futures

Policy Mappings

The *HPX* thrust policies map directly to Thrust execution policies:

```
// Policy mappings
hpx::thrust::thrust_host_policy    -> thrust::host
hpx::thrust::thrust_device_policy -> thrust::device
hpx::thrust::thrust_task_policy   -> thrust::par_nosync
```

Usage

Synchronous Device Execution

```
#include <hpx/modules/thrust.hpp>
#include <thrust/device_vector.h>

hpx::thrust::thrust_device_policy device{};
thrust::device_vector<int> d_vec(1000, 0);

hpx::fill(device, d_vec.begin(), d_vec.end(), 42);
int sum = hpx::reduce(device, d_vec.begin(), d_vec.end(), 0);
```

Asynchronous Execution with Futures

```
#include <hpx/modules/async_cuda.hpp>
#include <hpx/modules/thrust.hpp>

int hpx_main() {
    // Required for async operations
    hpx::cuda::experimental::enable_user_polling polling_guard("default");

    hpx::thrust::thrust_task_policy task{};
    thrust::device_vector<int> d_vec(1000, 1);

    auto fill_future = hpx::fill(task, d_vec.begin(), d_vec.end(), 42);
    fill_future.get(); // Standard HPX future operations
}
```

Build Requirements

- CUDA Toolkit 12.4.0+
- *HPX* with `HPX_WITH_CUDA=ON`
- Enable with: `cmake -DHPX_WITH_CUDA=ON ...`

See Also

- *async_cuda* - CUDA integration
- *execution* - *HPX* execution policies

timed_execution

This module provides extensions to the executor interfaces defined in the *execution* module that allow timed submission of tasks on thread pools (at or after a specified time).

See the *API reference* of this module for more details.

timing

This module provides the timing utilities (clocks and timers).

See the *API reference* of the module for more details.

topology

This module provides the class `hpx::threads::topology` which represents the hardware resources available on a node. The class is a light wrapper around the [Portable Hardware Locality \(HWLOC\)](https://www.open-mpi.org/projects/hwloc/)²⁶⁵ library. The `hpx::threads::cpu_mask` is a small companion class that represents a set of resources on a node.

See the *API reference* of the module for more details.

tracy

This modules enables the integration of *HPX* applications with the [|tracy|](https://github.com/jacobperron/tracy) profiler.

type_support

This module provides helper facilities related to types.

See the *API reference* of the module for more details.

²⁶⁵ <https://www.open-mpi.org/projects/hwloc/>

util

The util module provides miscellaneous standalone utilities.

See the *API reference* of the module for more details.

version

This module macros and functions for accessing version information about *HPX* and its dependencies.

See the *API reference* of this module for more details.

2.8.2 Main HPX modules

actions

TODO: High-level description of the library.

See the *API reference* of this module for more details.

actions_base

TODO: High-level description of the library.

See the *API reference* of this module for more details.

agas

TODO: High-level description of the module.

See the *API reference* of this module for more details.

agas_base

This module holds the implementation of the four AGAS services: primary namespace, locality namespace, component namespace, and symbol namespace.

See the *API reference* of this module for more details.

async_colocated

TODO: High-level description of the module.

See the *API reference* of this module for more details.

async_distributed

This module contains functionality for asynchronously launching work on remote localities: `hpx::async`, `hpx::post`. This module extends the local-only functions in `libs_async_local`.

See the API reference of this module for more details.

checkpoint

A common need of users is to periodically backup an application. This practice provides resiliency and potential restart points in code. *HPX* utilizes the concept of a **checkpoint** to support this use case.

Found in `hpx/util/checkpoint.hpp`, checkpoints are defined as objects that hold a serialized version of an object or set of objects at a particular moment in time. This representation can be stored in memory for later use or it can be written to disk for storage and/or recovery at a later point. In order to create and fill this object with data, users must use a function called `save_checkpoint`. In code the function looks like this:

```
hpx::future<hpx::util::checkpoint> hpx::util::save_checkpoint(a, b, c, ...);
```

`save_checkpoint` takes arbitrary data containers, such as `int`, `double`, `float`, `vector`, and `future`, and serializes them into a newly created checkpoint object. This function returns a `future` to a `checkpoint` containing the data. Here's an example of a simple use case:

```
using hpx::util::checkpoint;
using hpx::util::save_checkpoint;

std::vector<int> vec{1,2,3,4,5};
hpx::future<checkpoint> save_checkpoint(vec);
```

Once the future is ready, the checkpoint object will contain the `vector` `vec` and its five elements.

`prepare_checkpoint` takes arbitrary data containers (same as for `save_checkpoint`), , such as `int`, `double`, `float`, `vector`, and `future`, and calculates the necessary buffer space for the checkpoint that would be created if `save_checkpoint` was called with the same arguments. This function returns a `future` to a `checkpoint` that is appropriately initialized. Here's an example of a simple use case:

```
using hpx::util::checkpoint;
using hpx::util::prepare_checkpoint;

std::vector<int> vec{1,2,3,4,5};
hpx::future<checkpoint> prepare_checkpoint(vec);
```

Once the future is ready, the checkpoint object will be initialized with an appropriately sized internal buffer.

It is also possible to modify the launch policy used by `save_checkpoint`. This is accomplished by passing a launch policy as the first argument. It is important to note that passing `hpx::launch::sync` will cause `save_checkpoint` to return a `checkpoint` instead of a `future` to a `checkpoint`. All other policies passed to `save_checkpoint` will return a `future` to a `checkpoint`.

Sometimes `checkpoint` s must be declared before they are used. `save_checkpoint` allows users to move pre-created `checkpoint` s into the function as long as they are the first container passing into the function (In the case where a launch policy is used, the `checkpoint` will immediately follow the launch policy). An example of these features can be found below:

```

char character = 'd';
int integer = 10;
float flt = 10.01f;
bool boolean = true;
std::string str = "I am a string of characters";
std::vector<char> vec(str.begin(), str.end());
checkpoint archive;

// Test 1
// test basic functionality
hpx::shared_future<checkpoint> f_archive = save_checkpoint(
    std::move(archive), character, integer, flt, boolean, str, vec);

```

Once users can create checkpoints they must now be able to restore the objects they contain into memory. This is accomplished by the function `restore_checkpoint`. This function takes a `checkpoint` and fills its data into the containers it is provided. It is important to remember that the containers must be ordered in the same way they were placed into the checkpoint. For clarity see the example below:

```

char character2;
int integer2;
float flt2;
bool boolean2;
std::string str2;
std::vector<char> vec2;

restore_checkpoint(data, character2, integer2, flt2, boolean2, str2, vec2);

```

The core utility of `checkpoint` is in its ability to make certain data persistent. Often, this means that the data needs to be stored in an object, such as a file, for later use. *HPX* has two solutions for these issues: stream operator overloads and access iterators.

HPX contains two stream overloads, `operator<<` and `operator>>`, to stream data out of and into `checkpoint`. Here is an example of the overloads in use below:

```

double a9 = 1.0, b9 = 1.1, c9 = 1.2;
std::ofstream test_file_9("test_file_9.txt");
hpx::future<checkpoint> f_9 = save_checkpoint(a9, b9, c9);
test_file_9 << f_9.get();
test_file_9.close();

double a9_1, b9_1, c9_1;
std::ifstream test_file_9_1("test_file_9.txt");
checkpoint archive9;
test_file_9_1 >> archive9;
restore_checkpoint(archive9, a9_1, b9_1, c9_1);

```

This is the primary way to move data into and out of a `checkpoint`. It is important to note, however, that users should be cautious when using a stream operator to load data and another function to remove it (or vice versa). Both `operator<<` and `operator>>` rely on a `.write()` and a `.read()` function respectively. In order to know how much data to read from the `std::istream`, the `operator<<` will write the size of the checkpoint before writing the checkpoint data. Correspondingly, the `operator>>` will read the size of the stored data before reading the data into a new instance of `checkpoint`. As long as the user employs the `operator<<` and `operator>>` to stream the data, this detail can be ignored.

Important: Be careful when mixing `operator<<` and `operator>>` with other facilities to read and write to a checkpoint. `operator<<` writes an extra variable, and `operator>>` reads this variable back separately. Used together the user will not encounter any issues and can safely ignore this detail.

Users may also move the data into and out of a checkpoint using the exposed `.begin()` and `.end()` iterators. An example of this use case is illustrated below.

```
std::ofstream test_file_7("checkpoint_test_file.txt");
std::vector<float> vec7{1.02f, 1.03f, 1.04f, 1.05f};
hpx::future<checkpoint> fut_7 = save_checkpoint(vec7);
checkpoint archive7 = fut_7.get();
std::copy(archive7.begin(),      // Write data to ofstream
          archive7.end(),        // ie. the file
          std::ostream_iterator<char>(test_file_7));
test_file_7.close();

std::vector<float> vec7_1;
std::vector<char> char_vec;
std::ifstream test_file_7_1("checkpoint_test_file.txt");
if (test_file_7_1)
{
    test_file_7_1.seekg(0, test_file_7_1.end);
    auto length = test_file_7_1.tellg();
    test_file_7_1.seekg(0, test_file_7_1.beg);
    char_vec.resize(length);
    test_file_7_1.read(char_vec.data(), length);
}
checkpoint archive7_1(std::move(char_vec));    // Write data to checkpoint
restore_checkpoint(archive7_1, vec7_1);
```

Checkpointing components

`save_checkpoint` and `restore_checkpoint` are also able to store components inside checkpoints. This can be done in one of two ways. First a client of the component can be passed to `save_checkpoint`. When the user wishes to resurrect the component she can pass a client instance to `restore_checkpoint`.

This technique is demonstrated below:

```
// Try to checkpoint and restore a component with a client
std::vector<int> vec3{10, 10, 10, 10, 10};

// Create a component instance through client constructor
data_client D(hpx::find_here(), std::move(vec3));
hpx::future<checkpoint> f3 = save_checkpoint(D);

// Create a new client
data_client E;

// Restore server inside client instance
restore_checkpoint(f3.get(), E);
```

The second way a user can save a component is by passing a `shared_ptr` to the component to `save_checkpoint`.

This component can be resurrected by creating a new instance of the component type and passing a `shared_ptr` to the new instance to `restore_checkpoint`.

This technique is demonstrated below:

```
// test checkpoint a component using a shared_ptr
std::vector<int> vec{1, 2, 3, 4, 5};
data_client A(hpx::find_here(), std::move(vec));

// Checkpoint Server
hpx::id_type old_id = A.get_id();

hpx::future<std::shared_ptr<data_server>> f_a_ptr =
    hpx::get_ptr<data_server>(A.get_id());
std::shared_ptr<data_server> a_ptr = f_a_ptr.get();
hpx::future<checkpoint> f = save_checkpoint(a_ptr);
auto&& data = f.get();

// test prepare_checkpoint API
checkpoint c = prepare_checkpoint(hpx::launch::sync, a_ptr);
HPX_TEST(c.size() == data.size());

// Restore Server
// Create a new server instance
std::shared_ptr<data_server> b_server;
restore_checkpoint(data, b_server);
```

collectives

The `collectives` module exposes a set of distributed collective operations. Those can be used to exchange data between participating sites in a coordinated way. At this point the module exposes the following collective primitives:

- `hpx::collectives::all_gather`: receives a set of values from all participating sites.
- `hpx::collectives::all_reduce`: performs a reduction on data from each participating site to each participating site.
- `hpx::collectives::all_to_all`: each participating site provides its element of the data to collect while all participating sites receive the data from every other site.
- `hpx::collectives::broadcast_to` and `hpx::collectives::broadcast_from`: performs a broadcast operation from a root site to all participating sites.
- `hpx::collectives::exclusive_scan`: performs an exclusive scan operation on a set of values received from all call sites operating on the given base name.
- `hpx::collectives::gather_here` and `hpx::collectives::gather_there`: gathers values from all participating sites.
- `hpx::collectives::inclusive_scan`: performs an inclusive scan operation on a set of values received from all call sites operating on the given base name.
- `hpx::collectives::reduce_here` and `hpx::collectives::reduce_there`: performs a reduction on data from each participating site to a root site.
- `hpx::collectives::scatter_to` and `hpx::collectives::scatter_from`: receives an element of a set of values operating on the given base name.

- `hpx::lcos::broadcast`: performs a given action on all given global identifiers.
- `hpx::distributed::barrier`: distributed barrier.
- `hpx::lcos::fold`: performs a fold with a given action on all given global identifiers.
- `hpx::distributed::latch`: distributed latch.
- `hpx::lcos::reduce`: performs a reduction on data from each given global identifiers.
- `hpx::lcos::spmd_block`: performs the same operation on a local image while providing handles to the other images.

See the *API reference* of the module for more details.

Hierarchical collectives

Hierarchical communicators split a collective into sub-communicators arranged as a tree. This reduces the number of participants in each individual operation, but it also means that one user-visible hierarchical call can touch several internal communicators.

Generation model

Hierarchical collectives use two internal generations for each user generation. For a user generation k , the first internal phase uses $2k - 1$ and the second uses $2k$. Single-pass hierarchical operations and the inter-group exchange of `hpx::collectives::all_to_all` advance the communicator by two generations in one step so that shared hierarchical communicators stay aligned with two-phase collectives.

When a hierarchical communicator instance is shared between collective operations, callers must pass explicit, strictly consecutive positive generation numbers. Two-phase hierarchical collectives, including `hpx::collectives::all_gather`, `hpx::collectives::all_reduce`, `hpx::collectives::all_to_all`, `hpx::collectives::inclusive_scan`, `hpx::collectives::exclusive_scan`, and `hpx::distributed::barrier`, reject the default generation because their phase generations are derived from the explicit user generation.

command_line_handling

The `command_line_handling` module defines and handles the command-line options required by the *HPX* runtime, combining them with configuration options defined by the *runtime_configuration* module. The actual parsing of command line options is handled by the *program_options* module.

See the *API reference* of the module for more details.

components

TODO: High-level description of the module.

See the *API reference* of this module for more details.

components_base

TODO: High-level description of the library.

See the *API reference* of this module for more details.

compute

The compute module provides utilities for handling task and memory affinity on host systems.

See the *API reference* of the module for more details.

distribution_policies

TODO: High-level description of the module.

See the *API reference* of this module for more details.

execution_distributed

Provides distributed sender and receiver facilities for HPX execution.

See the *API reference* of this module for more details.

executors_distributed

This module provides the executor `hpx::parallel::execution::distribution_policy_executor`. It allows one to create work that is implicitly distributed over multiple localities.

See the *API reference* of this module for more details.

include

This module provides no functionality in itself. Instead it provides headers that group together other headers that often appear together.

See the *API reference* of this module for more details.

init_runtime

TODO: High-level description of the library.

See the *API reference* of this module for more details.

lcos_distributed

This module contains distributed *LCOs*. Currently the only LCO provided is `:cpp:class::hpx::lcos::channel`, a construct for sending values from one *locality* to another. See `libs_lcos_local` for local LCOs.

See the API reference of this module for more details.

naming

TODO: High-level description of the module.

See the API reference of this module for more details.

naming_base

This module provides a forward declaration of *address_type*, *component_type* and *invalid_locality_id*.

See the *API reference* of this module for more details.

parcelport_lci

TODO: High-level description of the module.

See the API reference of this module for more details.

parcelport_mpi

TODO: High-level description of the module.

See the API reference of this module for more details.

parcelport_tcp

TODO: High-level description of the module.

See the API reference of this module for more details.

parcelset

TODO: High-level description of the module.

See the *API reference* of this module for more details.

parcelset_base

TODO: High-level description of the module.

See the *API reference* of this module for more details.

performance_counters

This module provides the basic functionality required for defining performance counters. See *Performance counters* for more information about performance counters.

See the API reference of this module for more details.

plugin_factories

TODO: High-level description of the module.

See the *API reference* of this module for more details.

resiliency_distributed

Software resiliency features of *HPX* were introduced in the *resiliency module*. This module extends the APIs to run on distributed-memory systems allowing the user to invoke the failing task on other localities at runtime. This is useful in cases where a node is identified to fail more often (e.g., for certain ALU computes) as the task can now be replayed or replicated among different localities. The API exposed allows for an easy integration with the local only resiliency APIs as well.

Distributed software resilience APIs have a similar function signature and lives under the same namespace of *hpx::resiliency::experimental*. The difference arises in the formal parameters where distributed APIs takes the localities as the first argument, and an action as opposed to a function or a function object. The localities signify the order in which the API will either schedule (in case of Task Replay) tasks in a round robin fashion or replicate the tasks onto the list of localities.

The list of APIs exposed by distributed resiliency modules is the same as those defined in *local resiliency module*.

See the *API reference* of this module for more details.

runtime_components

TODO: High-level description of the module.

See the *API reference* of this module for more details.

runtime_distributed

TODO: High-level description of the module.

See the *API reference* of this module for more details.

segmented_algorithms

Segmented algorithms extend the usual parallel *algorithms* by providing overloads that work with distributed containers, such as partitioned vectors.

See the *API reference* of the module for more details.

statistics

This module provide some statistics utilities like rolling min/max and histogram.

See the API reference of the module for more details.

Supervision module

The supervision module provides lifecycle event publication, state querying, and observer registration for actors/components running on local or remote localities.

Overview

Several core operations are exposed, each with a local (synchronous) call and a remote (locality-qualified, future-returning or `launch::sync_policy`) call:

Publishing events

```
hpx::future<hpx::supervision::publish_result> hpx::supervision::publish_event(hpx::id_type const
&locality, hpx::id_type
const &target,
hpx::supervision::event ev,
std::uint64_t epoch = 0)
```

```
hpx::supervision::publish_result hpx::supervision::publish_event(hpx::launch::sync_policy, hpx::id_type
const &locality, hpx::id_type const
&target, hpx::supervision::event ev,
std::uint64_t epoch = 0, hpx::error_code
&ec = hpx::throws)
```

```
hpx::supervision::publish_result hpx::supervision::publish_event(hpx::id_type const &target,
hpx::supervision::event ev, std::uint64_t
epoch = 0, hpx::error_code &ec =
hpx::throws)
```

Publish a lifecycle event for `target`. Events are visible immediately to local observers; remote observers are notified within roughly one to two parcel round-trips. Publishing non-terminal events is not idempotent - each call creates a distinct, timestamped record. `event::completed` and `event::failed` are latched: the first terminal publication for a target returns `publish_result::applied`; every later terminal publication for that target is a no-op that returns `publish_result::already_terminal`.

Querying lifecycle state

```
hpx::future<hpx::supervision::lifecycle_state> hpx::supervision::query_state(hpx::id_type const &locality,
                                                                           hpx::id_type const &target)
```

```
hpx::supervision::lifecycle_state hpx::supervision::query_state(hpx::launch::sync_policy, hpx::id_type const
                                                                &locality, hpx::id_type const &target,
                                                                hpx::error_code &ec = hpx::throws)
```

```
hpx::supervision::lifecycle_state hpx::supervision::query_state(hpx::id_type const &target, hpx::error_code
                                                                &ec = hpx::throws)
```

Query the most recently observed lifecycle state for `target`. Includes a sequence number for gap detection and a staleness error code for remote queries whose result may lag the latest event.

Registering observers

```
hpx::future<hpx::id_type> hpx::supervision::register_observer(hpx::id_type const &locality, hpx::id_type
                                                            const &target,
                                                            hpx::supervision::lifecycle_callback const
                                                            &callback, std::optional<std::uint64_t>
                                                            epoch_filter = std::nullopt)
```

```
hpx::id_type hpx::supervision::register_observer(hpx::launch::sync_policy, hpx::id_type const &locality,
                                                hpx::id_type const &target,
                                                hpx::supervision::lifecycle_callback const &callback,
                                                std::optional<std::uint64_t> epoch_filter = std::nullopt,
                                                hpx::error_code &ec = hpx::throws)
```

```
hpx::id_type hpx::supervision::register_observer(hpx::id_type const &target,
                                                hpx::supervision::lifecycle_callback const &callback,
                                                std::optional<std::uint64_t> epoch_filter = std::nullopt,
                                                hpx::error_code &ec = hpx::throws)
```

Register callback to be invoked on lifecycle events of `target`, returning an observer handle usable with `unregister_observer`. Local callbacks fire synchronously within the publish call; remote callbacks fire via a retried parcel. If `epoch_filter` is set, the observer only receives notifications (including the initial state snapshot delivered at registration time) whose epoch matches; by default the observer receives notifications for every epoch.

Unregistering observers

```
hpx::future<void> hpx::supervision::unregister_observer(hpx::id_type const &locality, hpx::id_type const
                                                       &observer_handle)
```

```
void hpx::supervision::unregister_observer(hpx::launch::sync_policy, hpx::id_type const &locality,
                                           hpx::id_type const &observer_handle, hpx::error_code &ec =
                                           hpx::throws)
```

```
void hpx::supervision::unregister_observer(hpx::id_type const &observer_handle, hpx::error_code &ec =
                                           hpx::throws)
```

Unregister a previously registered observer. `observer_handle` must have been obtained from `register_observer`; a handle obtained from `register_activity_observer`, or any handle never returned by either registration function, is rejected.

Removing target state

```
hpx::future<void> hpx::supervision::remove_target(hpx::id_type const &locality, hpx::id_type const &target)
```

```
void hpx::supervision::remove_target(hpx::launch::sync_policy, hpx::id_type const &locality, hpx::id_type const &target, hpx::error_code &ec = hpx::throws)
```

```
void hpx::supervision::remove_target(hpx::id_type const &target, hpx::error_code &ec = hpx::throws)
```

Clear all locally tracked state for `target`. Unlike `unregister_observer`, which removes a single previously registered observer handle, `remove_target` unconditionally forgets every piece of local state held for `target` - its recorded lifecycle state and current epoch, and any observers still registered for it - regardless of any specific observer handle. Intended for callers that know `target` will never be queried or observed again locally (e.g. after a failed registration that seeded some state for it, or once a peer has been evicted), so that local state does not accumulate indefinitely.

Waiting for terminal events

```
hpx::future<hpx::supervision::lifecycle_state> hpx::supervision::await_terminal(hpx::id_type const &locality, hpx::id_type const &target, std::uint64_t epoch = 0, std::chrono::steady_clock::duration timeout = std::chrono::steady_clock::duration::max())
```

```
hpx::supervision::lifecycle_state hpx::supervision::await_terminal(hpx::launch::sync_policy, hpx::id_type const &locality, hpx::id_type const &target, std::uint64_t epoch = 0, std::chrono::steady_clock::duration timeout = std::chrono::steady_clock::duration::max(), hpx::error_code &ec = hpx::throws)
```

```
hpx::future<hpx::supervision::lifecycle_state> hpx::supervision::await_terminal(hpx::id_type const &target, std::uint64_t epoch = 0, std::chrono::steady_clock::duration timeout = std::chrono::steady_clock::duration::max())
```

Asynchronously wait, one-shot and without blocking a worker thread, and the `launch::sync_policy` overload waits synchronously until `target` reaches a terminal event (`event::completed` or `event::failed`) within epoch. If a terminal event has already been recorded for `target` in epoch, the returned future is immediately satisfied with that state. Waits are scoped to a single epoch: if `target`'s epoch advances past epoch before a terminal event occurs, any outstanding waiter for the stale epoch is invalidated. A waiter that reaches neither of these outcomes is bounded by `timeout` (or a built-in default if `timeout` is left at its sentinel value `std::chrono::steady_clock::duration::max()`), after which it is swept and invalidated. Dispatch is routed locally when `target`'s supervision manager lives on the calling locality, and remotely (via a registered distributed action) otherwise.

Checking dispatch admission

hpx::supervision::dispatch_outcome *hpx::supervision::check_admission*(*hpx::id_type* const &target, *std::uint64_t* epoch = 0)

Check whether *target* currently admits new dispatch under *epoch*, i.e. whether it is safe to schedule or route work to it right now. This is a pure, `noexcept` local read of the same terminal latch state consulted by `await_terminal`: it returns `dispatch_outcome::rejected_fenced` if *target* has already latched a terminal event (`event::completed` or `event::failed`) under *epoch*, and `dispatch_outcome::admitted` otherwise. Unlike `publish_result`, which reports how a publication was resolved, `dispatch_outcome` answers the separate, consumer-side question of whether dispatch should proceed; it must only be called on the locality *target* lives on.

This admission fence is scoped to a single locality's latch state: it reflects only terminal events published to the supervision manager hosting *target*, and it fails open (returns `admitted`) when *target* is unknown locally, including when its terminal event was only ever published elsewhere. There is no built-in mechanism that propagates terminal latch state across localities - observer registration delivers notifications, not admission state. Any additional locality that needs to fence its own dispatch decisions on *target* must independently mirror or route to the owning locality's state; this module does not provide that for you.

Registering activity observers

hpx::future<hpx::id_type> *hpx::supervision::register_activity_observer*(*hpx::id_type* const &locality, *hpx::supervision::activity_callback* const &callback, *std::optional<std::uint64_t>* epoch_filter = *std::nullopt*)

hpx::id_type *hpx::supervision::register_activity_observer*(*hpx::launch::sync_policy*, *hpx::id_type* const &locality, *hpx::supervision::activity_callback* const &callback, *std::optional<std::uint64_t>* epoch_filter = *std::nullopt*, *hpx::error_code* &ec = *hpx::throws*)

hpx::id_type *hpx::supervision::register_activity_observer*(*hpx::supervision::activity_callback* const &callback, *std::optional<std::uint64_t>* epoch_filter = *std::nullopt*, *hpx::error_code* &ec = *hpx::throws*)

Register callback to observe activity-state transitions (between `hpx::supervision::activity_state::inactive` and `active`) of *all* targets tracked by a locality's supervision manager. Unlike `register_observer`, none of these overloads take a *target* parameter: the feature reports on every target the addressed manager tracks rather than on a single one.

Registering an activity observer for an already-active target triggers a replay of its current state. The snapshot of tracked state and the insertion of the observer are performed atomically under the manager's internal lock, which guarantees exactly-once delivery: the observer receives either the replay or a live transition that raced with registration, but never both and never neither.

Note that the notification itself (replay or live) is delivered after the lock is released. Its delivery order relative to any other concurrent live notification for the same target is *not* guaranteed, only the exactly-once property is guaranteed, not relative ordering.

Activity notifications are a pure discovery/notification signal: unlike `publish_event`, registering or unregistering an activity observer never feeds the terminal latch consulted by `check_admission`.

If `epoch_filter` is set, `callback` only receives notifications - including the registration-time replay - whose epoch matches; by default it receives notifications regardless of epoch.

Unregistering activity observers

```
hpx::future<void> hpx::supervision::unregister_activity_observer(hpx::id_type const &locality,
                                                                hpx::id_type const
                                                                &observer_handle)

void hpx::supervision::unregister_activity_observer(hpx::launch::sync_policy, hpx::id_type const
                                                    &locality, hpx::id_type const &observer_handle,
                                                    hpx::error_code &ec = hpx::throws)

void hpx::supervision::unregister_activity_observer(hpx::id_type const &observer_handle,
                                                    hpx::error_code &ec = hpx::throws)
```

Unregister a previously registered activity observer. `observer_handle` must have been obtained from `register_activity_observer`; a handle obtained from `register_observer`, or any handle never returned by either registration function, is rejected.

See the *API reference* of this module for more details.

2.9 API reference

HPX follows a versioning scheme with three numbers: `major.minor.patch`. We guarantee no breaking changes in the API for patch releases. Minor releases may remove or break existing APIs, but only after a deprecation period of at least two minor releases. In rare cases do we outright remove old and unused functionality without a deprecation period.

We do not provide any ABI compatibility guarantees between any versions, debug and release builds, and builds with different C++ standards.

The public API of HPX is presented below. Clicking on a name brings you to the full documentation for the class or function. Including the header specified in a heading brings in the features listed under that heading.

Note: Names listed here are guaranteed stable with respect to semantic versioning. However, at the moment the list is incomplete and certain unlisted features are intended to be in the public API. While we work on completing the list, if you're unsure about whether a particular unlisted name is part of the public API you can get into contact with us or open an issue and we'll clarify the situation.

2.9.1 Public API

Our API is semantically conforming; hence, the reader is highly encouraged to refer to the corresponding facility in the [C++ Standard](#)²⁶⁶ if needed. All names below are also available in the top-level `hpx` namespace unless otherwise noted. The names in `hpx` should be preferred. The names in sub-namespaces will eventually be removed.

²⁶⁶ <https://en.cppreference.com/w/cpp/header>

hpx/algorithm.hpp

The header `hpx/algorithm.hpp`²⁶⁷ corresponds to the C++ standard library header `algorithm`²⁶⁸. See *Using parallel algorithms* for more information about the parallel algorithms.

ClassesTable 2.125: Classes of header `hpx/algorithm.hpp`

Class	C++ standard
<code>hpx::experimental::reduction</code>	N4808 ²⁶⁹
<code>hpx::experimental::induction</code>	N4808 ²⁷⁰

FunctionsTable 2.126: *hpx* functions of header `hpx/algorithm.hpp`

<i>hpx</i> function	C++ standard
<code>hpx::adjacent_find</code>	<code>std::adjacent_find</code> ²⁷¹
<code>hpx::all_of</code>	<code>std::all_of</code> ²⁷²
<code>hpx::any_of</code>	<code>std::any_of</code> ²⁷³
<code>hpx::copy</code>	<code>std::copy</code> ²⁷⁴
<code>hpx::copy_if</code>	<code>std::copy_if</code> ²⁷⁵
<code>hpx::copy_n</code>	<code>std::copy_n</code> ²⁷⁶
<code>hpx::count</code>	<code>std::count</code> ²⁷⁷
<code>hpx::count_if</code>	<code>std::count_if</code> ²⁷⁸
<code>hpx::ends_with</code>	<code>std::ends_with</code> ²⁷⁹
<code>hpx::equal</code>	<code>std::equal</code> ²⁸⁰
<code>hpx::fill</code>	<code>std::fill</code> ²⁸¹
<code>hpx::fill_n</code>	<code>std::fill_n</code> ²⁸²
<code>hpx::find</code>	<code>std::find</code> ²⁸³
<code>hpx::find_end</code>	<code>std::find_end</code> ²⁸⁴
<code>hpx::find_first_of</code>	<code>std::find_first_of</code> ²⁸⁵
<code>hpx::find_if</code>	<code>std::find_if</code> ²⁸⁶
<code>hpx::find_if_not</code>	<code>std::find_if_not</code> ²⁸⁷
<code>hpx::for_each</code>	<code>std::for_each</code> ²⁸⁸
<code>hpx::for_each_n</code>	<code>std::for_each_n</code> ²⁸⁹
<code>hpx::generate</code>	<code>std::generate</code> ²⁹⁰
<code>hpx::generate_n</code>	<code>std::generate_n</code> ²⁹¹
<code>hpx::includes</code>	<code>std::includes</code> ²⁹²
<code>hpx::inplace_merge</code>	<code>std::inplace_merge</code> ²⁹³
<code>hpx::is_heap</code>	<code>std::is_heap</code> ²⁹⁴
<code>hpx::is_heap_until</code>	<code>std::is_heap_until</code> ²⁹⁵
<code>hpx::is_partitioned</code>	<code>std::is_partitioned</code> ²⁹⁶
<code>hpx::is_sorted</code>	<code>std::is_sorted</code> ²⁹⁷

continues on next page

²⁶⁷ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

²⁶⁸ <http://en.cppreference.com/w/cpp/header/algorithm>

²⁶⁹ <http://wg21.link/n4808>

²⁷⁰ <http://wg21.link/n4808>

Table 2.126 – continued from previous page

<i>hpx</i> function	C++ standard
<i>hpx::is_sorted_until</i>	<code>std::is_sorted_until</code> ²⁹⁸
<i>hpx::lexicographical_compare</i>	<code>std::lexicographical_compare</code> ²⁹⁹
<i>hpx::make_heap</i>	<code>std::make_heap</code> ³⁰⁰
<i>hpx::max_element</i>	<code>std::max_element</code> ³⁰¹
<i>hpx::merge</i>	<code>std::merge</code> ³⁰²
<i>hpx::min_element</i>	<code>std::min_element</code> ³⁰³
<i>hpx::minmax_element</i>	<code>std::minmax_element</code> ³⁰⁴
<i>hpx::mismatch</i>	<code>std::mismatch</code> ³⁰⁵
<i>hpx::move</i>	<code>std::move</code> ³⁰⁶
<i>hpx::none_of</i>	<code>std::none_of</code> ³⁰⁷
<i>hpx::nth_element</i>	<code>std::nth_element</code> ³⁰⁸
<i>hpx::partial_sort</i>	<code>std::partial_sort</code> ³⁰⁹
<i>hpx::partial_sort_copy</i>	<code>std::partial_sort_copy</code> ³¹⁰
<i>hpx::partition</i>	<code>std::partition</code> ³¹¹
<i>hpx::partition_copy</i>	<code>std::partition_copy</code> ³¹²
<i>hpx::experimental::reduce_by_key</i>	<code>reduce_by_key</code> ³¹³
<i>hpx::remove</i>	<code>std::remove</code> ³¹⁴
<i>hpx::remove_copy</i>	<code>std::remove_copy</code> ³¹⁵
<i>hpx::remove_copy_if</i>	<code>std::remove_copy_if</code> ³¹⁶
<i>hpx::remove_if</i>	<code>std::remove_if</code> ³¹⁷
<i>hpx::replace</i>	<code>std::replace</code> ³¹⁸
<i>hpx::replace_copy</i>	<code>std::replace_copy</code> ³¹⁹
<i>hpx::replace_copy_if</i>	<code>std::replace_copy_if</code> ³²⁰
<i>hpx::replace_if</i>	<code>std::replace_if</code> ³²¹
<i>hpx::reverse</i>	<code>std::reverse</code> ³²²
<i>hpx::reverse_copy</i>	<code>std::reverse_copy</code> ³²³
<i>hpx::rotate</i>	<code>std::rotate</code> ³²⁴
<i>hpx::rotate_copy</i>	<code>std::rotate_copy</code> ³²⁵
<i>hpx::search</i>	<code>std::search</code> ³²⁶
<i>hpx::search_n</i>	<code>std::search_n</code> ³²⁷
<i>hpx::set_difference</i>	<code>std::set_difference</code> ³²⁸
<i>hpx::set_intersection</i>	<code>std::set_intersection</code> ³²⁹
<i>hpx::set_symmetric_difference</i>	<code>std::set_symmetric_difference</code> ³³⁰
<i>hpx::set_union</i>	<code>std::set_union</code> ³³¹
<i>hpx::shift_left</i>	<code>std::shift_left</code> ³³²
<i>hpx::shift_right</i>	<code>std::shift_right</code> ³³³
<i>hpx::sort</i>	<code>std::sort</code> ³³⁴
<i>hpx::experimental::sort_by_key</i>	<code>sort_by_key</code> ³³⁵
<i>hpx::stable_partition</i>	<code>std::stable_partition</code> ³³⁶
<i>hpx::stable_sort</i>	<code>std::stable_sort</code> ³³⁷
<i>hpx::starts_with</i>	<code>std::starts_with</code> ³³⁸
<i>hpx::swap_ranges</i>	<code>std::swap_ranges</code> ³³⁹
<i>hpx::transform</i>	<code>std::transform</code> ³⁴⁰
<i>hpx::unique</i>	<code>std::unique</code> ³⁴¹
<i>hpx::unique_copy</i>	<code>std::unique_copy</code> ³⁴²
<i>hpx::experimental::for_loop</i>	N4808 ³⁴³
<i>hpx::experimental::for_loop_strided</i>	N4808 ³⁴⁴
<i>hpx::experimental::for_loop_n</i>	N4808 ³⁴⁵
<i>hpx::experimental::for_loop_n_strided</i>	N4808 ³⁴⁶

271 http://en.cppreference.com/w/cpp/algorithm/adjacent_find
 272 http://en.cppreference.com/w/cpp/algorithm/all_any_none_of
 273 http://en.cppreference.com/w/cpp/algorithm/all_any_none_of
 274 <http://en.cppreference.com/w/cpp/algorithm/copy>
 275 <http://en.cppreference.com/w/cpp/algorithm/copy>
 276 http://en.cppreference.com/w/cpp/algorithm/copy_n
 277 <http://en.cppreference.com/w/cpp/algorithm/count>
 278 <http://en.cppreference.com/w/cpp/algorithm/count>
 279 http://en.cppreference.com/w/cpp/algorithm/ranges/ends_with
 280 <http://en.cppreference.com/w/cpp/algorithm/equal>
 281 <http://en.cppreference.com/w/cpp/algorithm/fill>
 282 http://en.cppreference.com/w/cpp/algorithm/fill_n
 283 <http://en.cppreference.com/w/cpp/algorithm/find>
 284 http://en.cppreference.com/w/cpp/algorithm/find_end
 285 http://en.cppreference.com/w/cpp/algorithm/find_first_of
 286 <http://en.cppreference.com/w/cpp/algorithm/find>
 287 <http://en.cppreference.com/w/cpp/algorithm/find>
 288 http://en.cppreference.com/w/cpp/algorithm/for_each
 289 http://en.cppreference.com/w/cpp/algorithm/for_each_n
 290 <http://en.cppreference.com/w/cpp/algorithm/generate>
 291 http://en.cppreference.com/w/cpp/algorithm/generate_n
 292 <http://en.cppreference.com/w/cpp/algorithm/includes>
 293 http://en.cppreference.com/w/cpp/algorithm/inplace_merge
 294 http://en.cppreference.com/w/cpp/algorithm/is_heap
 295 http://en.cppreference.com/w/cpp/algorithm/is_heap_until
 296 http://en.cppreference.com/w/cpp/algorithm/is_partitioned
 297 http://en.cppreference.com/w/cpp/algorithm/is_sorted
 298 http://en.cppreference.com/w/cpp/algorithm/is_sorted_until
 299 http://en.cppreference.com/w/cpp/algorithm/lexicographical_compare
 300 http://en.cppreference.com/w/cpp/algorithm/make_heap
 301 http://en.cppreference.com/w/cpp/algorithm/max_element
 302 <http://en.cppreference.com/w/cpp/algorithm/merge>
 303 http://en.cppreference.com/w/cpp/algorithm/min_element
 304 http://en.cppreference.com/w/cpp/algorithm/minmax_element
 305 <http://en.cppreference.com/w/cpp/algorithm/mismatch>
 306 <http://en.cppreference.com/w/cpp/algorithm/move>
 307 http://en.cppreference.com/w/cpp/algorithm/all_any_none_of
 308 http://en.cppreference.com/w/cpp/algorithm/nth_element
 309 http://en.cppreference.com/w/cpp/algorithm/partial_sort
 310 http://en.cppreference.com/w/cpp/algorithm/partial_sort_copy
 311 <http://en.cppreference.com/w/cpp/algorithm/partition>
 312 http://en.cppreference.com/w/cpp/algorithm/partition_copy
 313 https://thrust.github.io/doc/group__reductions_gad5623f203f9b3fdcab72481c3913f0e0.html
 314 <http://en.cppreference.com/w/cpp/algorithm/remove>
 315 http://en.cppreference.com/w/cpp/algorithm/remove_copy
 316 http://en.cppreference.com/w/cpp/algorithm/remove_copy
 317 <http://en.cppreference.com/w/cpp/algorithm/remove>
 318 <http://en.cppreference.com/w/cpp/algorithm/replace>
 319 http://en.cppreference.com/w/cpp/algorithm/replace_copy
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 322 <http://en.cppreference.com/w/cpp/algorithm/reverse>
 323 http://en.cppreference.com/w/cpp/algorithm/reverse_copy
 324 <http://en.cppreference.com/w/cpp/algorithm/rotate>
 325 http://en.cppreference.com/w/cpp/algorithm/rotate_copy
 326 <http://en.cppreference.com/w/cpp/algorithm/search>
 327 http://en.cppreference.com/w/cpp/algorithm/search_n
 328 http://en.cppreference.com/w/cpp/algorithm/set_difference
 329 http://en.cppreference.com/w/cpp/algorithm/set_intersection
 330 http://en.cppreference.com/w/cpp/algorithm/set_symmetric_difference
 331 http://en.cppreference.com/w/cpp/algorithm/set_union
 332 <http://en.cppreference.com/w/cpp/algorithm/shift>
 333 <http://en.cppreference.com/w/cpp/algorithm/shift>
 334 <http://en.cppreference.com/w/cpp/algorithm/sort>
 335 https://thrust.github.io/doc/group__sorting_gabe038d6107f7c824cf74120500ef45ea.html
 336 http://en.cppreference.com/w/cpp/algorithm/stable_partition
 337 http://en.cppreference.com/w/cpp/algorithm/stable_sort
 338 http://en.cppreference.com/w/cpp/algorithm/ranges/starts_with
 339 http://en.cppreference.com/w/cpp/algorithm/swap_ranges
 340 <http://en.cppreference.com/w/cpp/algorithm/transform>
 341 <http://en.cppreference.com/w/cpp/algorithm/unique>
 342 http://en.cppreference.com/w/cpp/algorithm/unique_copy
 343 <http://wg21.link/n4808>
 344 <http://wg21.link/n4808>

Table 2.127: *hpx::ranges* functions of header *hpx/algorithm.hpp*

<i>hpx::ranges</i> function	C++ standard
<i>hpx::ranges::adjacent_find</i>	<code>std::adjacent_find</code> ³⁴⁷
<i>hpx::ranges::all_of</i>	<code>std::all_of</code> ³⁴⁸
<i>hpx::ranges::any_of</i>	<code>std::any_of</code> ³⁴⁹
<i>hpx::ranges::copy</i>	<code>std::copy</code> ³⁵⁰
<i>hpx::ranges::copy_if</i>	<code>std::copy_if</code> ³⁵¹
<i>hpx::ranges::copy_n</i>	<code>std::copy_n</code> ³⁵²
<i>hpx::ranges::count</i>	<code>std::count</code> ³⁵³
<i>hpx::ranges::count_if</i>	<code>std::count_if</code> ³⁵⁴
<i>hpx::ranges::ends_with</i>	<code>std::ends_with</code> ³⁵⁵
<i>hpx::ranges::equal</i>	<code>std::equal</code> ³⁵⁶
<i>hpx::ranges::fill</i>	<code>std::fill</code> ³⁵⁷
<i>hpx::ranges::fill_n</i>	<code>std::fill_n</code> ³⁵⁸
<i>hpx::ranges::find</i>	<code>std::find</code> ³⁵⁹
<i>hpx::ranges::find_end</i>	<code>std::find_end</code> ³⁶⁰
<i>hpx::ranges::find_first_of</i>	<code>std::find_first_of</code> ³⁶¹
<i>hpx::ranges::find_if</i>	<code>std::find_if</code> ³⁶²
<i>hpx::ranges::find_if_not</i>	<code>std::find_if_not</code> ³⁶³
<i>hpx::ranges::for_each</i>	<code>std::for_each</code> ³⁶⁴
<i>hpx::ranges::for_each_n</i>	<code>std::for_each_n</code> ³⁶⁵
<i>hpx::ranges::generate</i>	<code>std::generate</code> ³⁶⁶
<i>hpx::ranges::generate_n</i>	<code>std::generate_n</code> ³⁶⁷
<i>hpx::ranges::includes</i>	<code>std::includes</code> ³⁶⁸
<i>hpx::ranges::inplace_merge</i>	<code>std::inplace_merge</code> ³⁶⁹
<i>hpx::ranges::is_heap</i>	<code>std::is_heap</code> ³⁷⁰
<i>hpx::ranges::is_heap_until</i>	<code>std::is_heap_until</code> ³⁷¹
<i>hpx::ranges::is_partitioned</i>	<code>std::is_partitioned</code> ³⁷²
<i>hpx::ranges::is_sorted</i>	<code>std::is_sorted</code> ³⁷³
<i>hpx::ranges::is_sorted_until</i>	<code>std::is_sorted_until</code> ³⁷⁴
<i>hpx::ranges::make_heap</i>	<code>std::make_heap</code> ³⁷⁵
<i>hpx::ranges::max_element</i>	<code>std::max_element</code> ³⁷⁶
<i>hpx::ranges::merge</i>	<code>std::merge</code> ³⁷⁷
<i>hpx::ranges::min_element</i>	<code>std::min_element</code> ³⁷⁸
<i>hpx::ranges::minmax_element</i>	<code>std::minmax_element</code> ³⁷⁹
<i>hpx::ranges::mismatch</i>	<code>std::mismatch</code> ³⁸⁰
<i>hpx::ranges::move</i>	<code>std::move</code> ³⁸¹
<i>hpx::ranges::none_of</i>	<code>std::none_of</code> ³⁸²
<i>hpx::ranges::nth_element</i>	<code>std::nth_element</code> ³⁸³
<i>hpx::ranges::partial_sort</i>	<code>std::partial_sort</code> ³⁸⁴
<i>hpx::ranges::partial_sort_copy</i>	<code>std::partial_sort_copy</code> ³⁸⁵
<i>hpx::ranges::partition</i>	<code>std::partition</code> ³⁸⁶
<i>hpx::ranges::partition_copy</i>	<code>std::partition_copy</code> ³⁸⁷
<i>hpx::ranges::set_difference</i>	<code>std::set_difference</code> ³⁸⁸
<i>hpx::ranges::set_intersection</i>	<code>std::set_intersection</code> ³⁸⁹
<i>hpx::ranges::set_symmetric_difference</i>	<code>std::set_symmetric_difference</code> ³⁹⁰
<i>hpx::ranges::set_union</i>	<code>std::set_union</code> ³⁹¹
<i>hpx::ranges::shift_left</i>	P2440 ³⁹²
<i>hpx::ranges::shift_right</i>	P2440 ³⁹³
<i>hpx::ranges::sort</i>	<code>std::sort</code> ³⁹⁴

continues on next page

Table 2.127 – continued from previous page

<i>hpx::ranges</i> function	C++ standard
<i>hpx::ranges::stable_partition</i>	std::stable_partition ³⁹⁵
<i>hpx::ranges::stable_sort</i>	std::stable_sort ³⁹⁶
<i>hpx::ranges::starts_with</i>	std::starts_with ³⁹⁷
<i>hpx::ranges::swap_ranges</i>	std::swap_ranges ³⁹⁸
<i>hpx::ranges::transform</i>	std::transform ³⁹⁹
<i>hpx::ranges::unique</i>	std::unique ⁴⁰⁰
<i>hpx::ranges::unique_copy</i>	std::unique_copy ⁴⁰¹
<i>hpx::ranges::experimental::for_loop</i>	N4808 ⁴⁰²
<i>hpx::ranges::experimental::for_loop_strided</i>	N4808 ⁴⁰³

`hpx/any.hpp`

The header `hpx/any.hpp`⁴⁰⁴ corresponds to the C++ standard library header `any`⁴⁰⁵.

`hpx::any` is compatible with `std::any`.

```
347 http://en.cppreference.com/w/cpp/algorithm/ranges/adjacent\_find
348 http://en.cppreference.com/w/cpp/algorithm/ranges/all\_any\_none\_of
349 http://en.cppreference.com/w/cpp/algorithm/ranges/all\_any\_none\_of
350 http://en.cppreference.com/w/cpp/algorithm/ranges/copy
351 http://en.cppreference.com/w/cpp/algorithm/ranges/copy
352 http://en.cppreference.com/w/cpp/algorithm/ranges/copy\_n
353 http://en.cppreference.com/w/cpp/algorithm/ranges/count
354 http://en.cppreference.com/w/cpp/algorithm/ranges/count
355 http://en.cppreference.com/w/cpp/algorithm/ranges/ends\_with
356 http://en.cppreference.com/w/cpp/algorithm/ranges/equal
357 http://en.cppreference.com/w/cpp/algorithm/ranges/fill
358 http://en.cppreference.com/w/cpp/algorithm/ranges/fill\_n
359 http://en.cppreference.com/w/cpp/algorithm/ranges/find
360 http://en.cppreference.com/w/cpp/algorithm/ranges/find\_end
361 http://en.cppreference.com/w/cpp/algorithm/ranges/find\_first\_of
362 http://en.cppreference.com/w/cpp/algorithm/ranges/find
363 http://en.cppreference.com/w/cpp/algorithm/ranges/find
364 http://en.cppreference.com/w/cpp/algorithm/ranges/for\_each
365 http://en.cppreference.com/w/cpp/algorithm/ranges/for\_each\_n
366 http://en.cppreference.com/w/cpp/algorithm/ranges/generate
367 http://en.cppreference.com/w/cpp/algorithm/ranges/generate\_n
368 http://en.cppreference.com/w/cpp/algorithm/ranges/includes
369 http://en.cppreference.com/w/cpp/algorithm/ranges/inplace\_merge
370 http://en.cppreference.com/w/cpp/algorithm/ranges/is\_heap
371 http://en.cppreference.com/w/cpp/algorithm/ranges/is\_heap\_until
372 http://en.cppreference.com/w/cpp/algorithm/ranges/is\_partitioned
373 http://en.cppreference.com/w/cpp/algorithm/ranges/is\_sorted
374 http://en.cppreference.com/w/cpp/algorithm/ranges/is\_sorted\_until
375 http://en.cppreference.com/w/cpp/algorithm/ranges/make\_heap
376 http://en.cppreference.com/w/cpp/algorithm/ranges/max\_element
377 http://en.cppreference.com/w/cpp/algorithm/ranges/merge
378 http://en.cppreference.com/w/cpp/algorithm/ranges/min\_element
379 http://en.cppreference.com/w/cpp/algorithm/ranges/minmax\_element
380 http://en.cppreference.com/w/cpp/algorithm/ranges/mismatch
381 http://en.cppreference.com/w/cpp/algorithm/ranges/move
382 http://en.cppreference.com/w/cpp/algorithm/ranges/all\_any\_none\_of
383 http://en.cppreference.com/w/cpp/algorithm/ranges/nth\_element
384 http://en.cppreference.com/w/cpp/algorithm/ranges/partial\_sort
385 http://en.cppreference.com/w/cpp/algorithm/ranges/partial\_sort\_copy
386 http://en.cppreference.com/w/cpp/algorithm/ranges/partition
387 http://en.cppreference.com/w/cpp/algorithm/ranges/partition\_copy
388 http://en.cppreference.com/w/cpp/algorithm/ranges/set\_difference
389 http://en.cppreference.com/w/cpp/algorithm/ranges/set\_intersection
390 http://en.cppreference.com/w/cpp/algorithm/ranges/set\_symmetric\_difference
391 http://en.cppreference.com/w/cpp/algorithm/ranges/set\_union
392 https://wg21.link/p2440
393 https://wg21.link/p2440
394 http://en.cppreference.com/w/cpp/algorithm/ranges/sort
395 http://en.cppreference.com/w/cpp/algorithm/ranges/stable\_partition
396 http://en.cppreference.com/w/cpp/algorithm/ranges/stable\_sort
397 http://en.cppreference.com/w/cpp/algorithm/ranges/starts\_with
398 http://en.cppreference.com/w/cpp/algorithm/ranges/swap\_ranges
399 http://en.cppreference.com/w/cpp/algorithm/ranges/transform
400 http://en.cppreference.com/w/cpp/algorithm/ranges/unique
401 http://en.cppreference.com/w/cpp/algorithm/ranges/unique\_copy
402 http://wg21.link/n4808
403 http://wg21.link/n4808
404 http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include\_local/include/hpx/any.hpp
405 http://en.cppreference.com/w/cpp/header/any
```


Classes

Table 2.128: Classes of header `hpx/any.hpp`

Class	C++ standard
<code>hpx::any</code>	<code>std::any</code> ⁴⁰⁶
<code>hpx::any_nonser</code>	
<code>hpx::bad_any_cast</code>	<code>std::bad_any_cast</code> ⁴⁰⁷
<code>hpx::unique_any_nonser</code>	

Functions

Table 2.129: Functions of header `hpx/any.hpp`

Function	C++ standard
<code>hpx::any_cast</code>	<code>std::any_cast</code> ⁴⁰⁸
<code>hpx::make_any</code>	<code>std::make_any</code> ⁴⁰⁹
<code>hpx::make_any_nonser</code>	
<code>hpx::make_unique_any_nonser</code>	

`hpx/assert.hpp`

The header `hpx/assert.hpp`⁴¹⁰ corresponds to the C++ standard library header `cassert`⁴¹¹.

`HPX_ASSERT` is the *HPX* equivalent to `assert` in `cassert`. `HPX_ASSERT` can also be used in CUDA device code.

Macros

Table 2.130: Macros of header `hpx/assert.hpp`

Macro
<code>HPX_ASSERT</code>
<code>HPX_ASSERT_MSG</code>

⁴⁰⁶ <http://en.cppreference.com/w/cpp/utility/any>

⁴⁰⁷ http://en.cppreference.com/w/cpp/utility/any/bad_any_cast

⁴⁰⁸ http://en.cppreference.com/w/cpp/utility/any/any_cast

⁴⁰⁹ http://en.cppreference.com/w/cpp/utility/any/make_any

⁴¹⁰ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/core/assertion/include/hpx/assert.hpp>

⁴¹¹ <http://en.cppreference.com/w/cpp/header/cassert>

`hpx/barrier.hpp`

The header `hpx/barrier.hpp`⁴¹² corresponds to the C++ standard library header `barrier`⁴¹³ and contains a distributed barrier implementation. This functionality is also exposed through the `hpx::distributed` namespace. The name in `hpx::distributed` should be preferred.

Classes

Table 2.131: Classes of header `hpx/barrier.hpp`

Class	C++ standard
<code>hpx::barrier</code>	<code>std::barrier</code> ⁴¹⁴

Table 2.132: Distributed implementation of classes of header `hpx/barrier.hpp`

Class
<code>hpx::distributed::barrier</code>

`hpx/channel.hpp`

The header `hpx/channel.hpp`⁴¹⁵ contains a local and a distributed channel implementation. This functionality is also exposed through the `hpx::distributed` namespace. The name in `hpx::distributed` should be preferred.

Classes

Table 2.133: Classes of header `hpx/channel.hpp`

Class
<code>hpx::channel</code>

Table 2.134: Distributed implementation of classes of header `hpx/channel.hpp`

Class
<code>hpx::distributed::channel</code>

⁴¹² <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/barrier.hpp>

⁴¹³ <http://en.cppreference.com/w/cpp/header/barrier>

⁴¹⁴ <http://en.cppreference.com/w/cpp/thread/barrier>

⁴¹⁵ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/channel.hpp>

hpx/chrono.hpp

The header `hpx/chrono.hpp`⁴¹⁶ corresponds to the C++ standard library header `chrono`⁴¹⁷. The following replacements and extensions are provided compared to `chrono`⁴¹⁸.

Classes

Table 2.135: Classes of header `hpx/chrono.hpp`

Class	C++ standard
<code>hpx::chrono::high_resolution_clock</code>	<code>std::high_resolution_clock</code> ⁴¹⁹
<code>hpx::chrono::high_resolution_timer</code>	
<code>hpx::chrono::steady_time_point</code>	<code>std::time_point</code> ⁴²⁰

hpx/condition_variable.hpp

The header `hpx/condition_variable.hpp`⁴²¹ corresponds to the C++ standard library header `condition_variable`⁴²².

Classes

Table 2.136: Classes of header `hpx/condition_variable.hpp`

Class	C++ standard
<code>hpx::condition_variable</code>	<code>std::condition_variable</code> ⁴²³
<code>hpx::condition_variable_any</code>	<code>std::condition_variable_any</code> ⁴²⁴
<code>hpx::cv_status</code>	<code>std::cv_status</code> ⁴²⁵

hpx/exception.hpp

The header `hpx/exception.hpp`⁴²⁶ corresponds to the C++ standard library header `exception`⁴²⁷. `hpx::exception` extends `std::exception` and is the base class for all exceptions thrown in *HPX*. `HPX_THROW_EXCEPTION` can be used to throw *HPX* exceptions with file and line information attached to the exception.

⁴¹⁶ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/chrono.hpp

⁴¹⁷ <http://en.cppreference.com/w/cpp/header/chrono>

⁴¹⁸ <http://en.cppreference.com/w/cpp/header/chrono>

⁴¹⁹ http://en.cppreference.com/w/cpp/chrono/high_resolution_clock

⁴²⁰ http://en.cppreference.com/w/cpp/chrono/time_point

⁴²¹ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/condition_variable.hpp

⁴²² http://en.cppreference.com/w/cpp/header/condition_variable

⁴²³ http://en.cppreference.com/w/cpp/thread/condition_variable

⁴²⁴ http://en.cppreference.com/w/cpp/thread/condition_variable_any

⁴²⁵ http://en.cppreference.com/w/cpp/thread/cv_status

⁴²⁶ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/exception.hpp

⁴²⁷ <http://en.cppreference.com/w/cpp/header/exception>

Macros

- `HPX_THROW_EXCEPTION`

Classes

Table 2.137: Classes of header `hpx/exception.hpp`

Class	C++ standard
<code>hpx::exception</code>	<code>std::exception</code> ⁴²⁸

`hpx/execution.hpp`

The header `hpx/execution.hpp`⁴²⁹ corresponds to the C++ standard library header `execution`⁴³⁰. See *High level parallel facilities*, *Using parallel algorithms* and *Executor parameters and executor parameter traits* for more information about execution policies and executor parameters.

Note: These names are only available in the `hpx::execution` namespace, not in the top-level `hpx` namespace.

Constants

Table 2.138: Constants of header `hpx/execution.hpp`

Constant	C++ standard
<code>hpx::execution::seq</code>	<code>std::execution_policy_tag</code> ⁴³¹
<code>hpx::execution::par</code>	<code>std::execution_policy_tag</code> ⁴³²
<code>hpx::execution::par_unseq</code>	<code>std::execution_policy_tag</code> ⁴³³
<code>hpx::execution::task</code>	

⁴²⁸ <http://en.cppreference.com/w/cpp/error/exception>

⁴²⁹ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fdbc0c8044e7/libs/core/include_local/include/hpx/execution.hpp

⁴³⁰ <http://en.cppreference.com/w/cpp/header/execution>

⁴³¹ http://en.cppreference.com/w/cpp/algorithm/execution_policy_tag

⁴³² http://en.cppreference.com/w/cpp/algorithm/execution_policy_tag

⁴³³ http://en.cppreference.com/w/cpp/algorithm/execution_policy_tag

Classes

Table 2.139: Classes of header `hpx/execution.hpp`

Class	C++ standard
<code>hpx::execution::sequenced_policy</code>	<code>std::execution_policy_tag_t</code> ⁴³⁴
<code>hpx::execution::parallel_policy</code>	<code>std::execution_policy_tag_t</code> ⁴³⁵
<code>hpx::execution::parallel_unsequenced_policy</code>	<code>std::execution_policy_tag_t</code> ⁴³⁶
<code>hpx::execution::sequenced_task_policy</code>	
<code>hpx::execution::parallel_task_policy</code>	
<code>hpx::execution::experimental::auto_chunk_size</code>	
<code>hpx::execution::experimental::dynamic_chunk_size</code>	
<code>hpx::execution::experimental::guided_chunk_size</code>	
<code>hpx::execution::experimental::persistent_auto_chunk_size</code>	
<code>hpx::execution::experimental::static_chunk_size</code>	
<code>hpx::execution::experimental::num_cores</code>	

`hpx/functional.hpp`

The header `hpx/functional.hpp`⁴³⁷ corresponds to the C++ standard library header `functional`⁴³⁸. `hpx::function` is a more efficient and serializable replacement for `std::function`.

Constants

The following constants correspond to the C++ standard `std::placeholders`⁴³⁹

Table 2.140: Constants of header `hpx/functional.hpp`

Constant
<code>hpx::placeholders::_1</code>
<code>hpx::placeholders::_2</code>
...
<code>hpx::placeholders::_9</code>

⁴³⁴ http://en.cppreference.com/w/cpp/algorithm/execution_policy_tag_t

⁴³⁵ http://en.cppreference.com/w/cpp/algorithm/execution_policy_tag_t

⁴³⁶ http://en.cppreference.com/w/cpp/algorithm/execution_policy_tag_t

⁴³⁷ http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fdbc0c8044e7/libs/core/include_local/include/hpx/functional.hpp

⁴³⁸ `hpp`

⁴³⁸ <http://en.cppreference.com/w/cpp/header/functional>

⁴³⁹ <http://en.cppreference.com/w/cpp/utility/functional/placeholders>

Classes

Table 2.141: Classes of header `hpx/functional.hpp`

Class	C++ standard
<code>hpx::function</code>	<code>std::function</code> ⁴⁴⁰
<code>hpx::function_ref</code>	P0792 ⁴⁴¹
<code>hpx::move_only_function</code>	<code>std::move_only_function</code> ⁴⁴²
<code>hpx::is_bind_expression</code>	<code>std::is_bind_expression</code> ⁴⁴³
<code>hpx::is_placeholder</code>	<code>std::is_placeholder</code> ⁴⁴⁴
<code>hpx::scoped_annotation</code>	

Functions

`hpx/future.hpp`

The header `hpx/future.hpp`⁴⁴⁵ corresponds to the C++ standard library header `future`⁴⁴⁶. See *Extended facilities for futures* for more information about extensions to futures compared to the C++ standard library.

This header file also contains overloads of `hpx::async`, `hpx::post`, `hpx::sync`, and `hpx::dataflow` that can be used with actions. See *Action invocation* for more information about invoking actions.

Classes

Table 2.142: Classes of header `hpx/future.hpp`

Class	C++ standard
<code>hpx::future</code>	<code>std::future</code> ⁴⁴⁷
<code>hpx::shared_future</code>	<code>std::shared_future</code> ⁴⁴⁸
<code>hpx::promise</code>	<code>std::promise</code> ⁴⁴⁹
<code>hpx::launch</code>	<code>std::launch</code> ⁴⁵⁰
<code>hpx::packaged_task</code>	<code>std::packaged_task</code> ⁴⁵¹

Note: All names except `hpx::promise` are also available in the top-level `hpx` namespace. `hpx::promise` refers to `hpx::distributed::promise`, a distributed variant of `hpx::promise`, but will eventually refer to `hpx::promise` after a deprecation period.

⁴⁴⁰ <http://en.cppreference.com/w/cpp/utility/functional/function>

⁴⁴¹ <http://wg21.link/p0792>

⁴⁴² http://en.cppreference.com/w/cpp/utility/functional/move_only_function

⁴⁴³ http://en.cppreference.com/w/cpp/utility/functional/is_bind_expression

⁴⁴⁴ http://en.cppreference.com/w/cpp/utility/functional/is_placeholder

⁴⁴⁵ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

⁴⁴⁶ <http://en.cppreference.com/w/cpp/header/future>

⁴⁴⁷ <http://en.cppreference.com/w/cpp/thread/future>

⁴⁴⁸ http://en.cppreference.com/w/cpp/thread/shared_future

⁴⁴⁹ <http://en.cppreference.com/w/cpp/thread/promise>

⁴⁵⁰ <http://en.cppreference.com/w/cpp/thread/launch>

⁴⁵¹ http://en.cppreference.com/w/cpp/thread/packaged_task

Table 2.143: Distributed implementation of classes of header `hpx/future.hpp`

Class
<code>hpx::distributed::promise</code>

Functions

Table 2.144: Functions of header `hpx/future.hpp`

Function	C++ standard
<code>hpx::async</code>	<code>std::async</code> ⁴⁵²
<code>hpx::post</code>	
<code>hpx::sync</code>	
<code>hpx::dataflow</code>	
<code>hpx::make_future</code>	
<code>hpx::make_shared_future</code>	
<code>hpx::make_ready_future</code>	P0159 ⁴⁵³
<code>hpx::make_ready_future_alloc</code>	
<code>hpx::make_ready_future_at</code>	
<code>hpx::make_ready_future_after</code>	
<code>hpx::make_exceptional_future</code>	P0159 ⁴⁵⁴
<code>hpx::when_all</code>	P0159 ⁴⁵⁵
<code>hpx::when_any</code>	P0159 ⁴⁵⁶
<code>hpx::when_some</code>	
<code>hpx::when_each</code>	
<code>hpx::wait_all</code>	
<code>hpx::wait_all_nothrow</code>	
<code>hpx::wait_all_n</code>	
<code>hpx::wait_all_n_nothrow</code>	
<code>hpx::wait_all_for</code>	
<code>hpx::wait_all_for_nothrow</code>	
<code>hpx::wait_all_for_n</code>	
<code>hpx::wait_all_for_n_nothrow</code>	
<code>hpx::wait_any</code>	
<code>hpx::wait_any_n</code>	
<code>hpx::wait_any_nothrow</code>	
<code>hpx::wait_any_n_nothrow</code>	
<code>hpx::wait_some</code>	
<code>hpx::wait_some_n</code>	
<code>hpx::wait_some_nothrow</code>	
<code>hpx::wait_some_n_nothrow</code>	
<code>hpx::wait_each</code>	
<code>hpx::wait_each_n</code>	
<code>hpx::wait_each_nothrow</code>	
<code>hpx::wait_each_n_nothrow</code>	

⁴⁵² <http://en.cppreference.com/w/cpp/thread/async>

⁴⁵³ <http://wg21.link/p0159>

⁴⁵⁴ <http://wg21.link/p0159>

⁴⁵⁵ <http://wg21.link/p0159>

⁴⁵⁶ <http://wg21.link/p0159>

hpx/init.hpp

The header `hpx/init.hpp`⁴⁵⁷ contains functionality for starting, stopping, suspending, and resuming the *HPX* runtime. This is the main way to explicitly start the *HPX* runtime. See *Starting the HPX runtime* for more details on starting the *HPX* runtime.

Classes

Table 2.145: Classes of header `hpx/init.hpp`

Class
<code>hpx::init_params</code>
<code>hpx::runtime_mode</code>

Functions

Table 2.146: Functions of header `hpx/init.hpp`

Function
<code>hpx::init</code>
<code>hpx::start</code>
<code>hpx::finalize</code>
<code>hpx::disconnect</code>
<code>hpx::suspend</code>
<code>hpx::resume</code>

hpx/latch.hpp

The header `hpx/latch.hpp`⁴⁵⁸ corresponds to the C++ standard library header `latch`⁴⁵⁹. It contains a local and a distributed latch implementation. This functionality is also exposed through the `hpx::distributed` namespace. The name in `hpx::distributed` should be preferred.

Classes

Table 2.147: Classes of header `hpx/latch.hpp`

Class	C++ standard
<code>hpx::latch</code>	<code>std::latch</code> ⁴⁶⁰

Table 2.148: Distributed implementation of classes of header `hpx/latch.hpp`

Class
<code>hpx::distributed::latch</code>

⁴⁵⁷ http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/init_runtime/include/hpx/init.hpp

⁴⁵⁸ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/latch.hpp>

⁴⁵⁹ <http://en.cppreference.com/w/cpp/header/latch>

⁴⁶⁰ <http://en.cppreference.com/w/cpp/thread/latch>

hpx/mutex.hpp

The header `hpx/mutex.hpp`⁴⁶¹ corresponds to the C++ standard library header `mutex`⁴⁶².

Classes

Table 2.149: Classes of header `hpx/mutex.hpp`

Class	C++ standard
<code>hpx::mutex</code>	<code>std::mutex</code> ⁴⁶³
<code>hpx::no_mutex</code>	
<code>hpx::once_flag</code>	<code>std::once_flag</code> ⁴⁶⁴
<code>hpx::recursive_mutex</code>	<code>std::recursive_mutex</code> ⁴⁶⁵
<code>hpx::spinlock</code>	
<code>hpx::timed_mutex</code>	<code>std::timed_mutex</code> ⁴⁶⁶
<code>hpx::unlock_guard</code>	

Functions

Table 2.150: Functions of header `hpx/mutex.hpp`

Class	C++ standard
<code>hpx::call_once</code>	<code>std::call_once</code> ⁴⁶⁷

hpx/memory.hpp

The header `hpx/memory.hpp`⁴⁶⁸ corresponds to the C++ standard library header `memory`⁴⁶⁹. It contains parallel versions of the copy, fill, move, and construct helper functions in `memory`⁴⁷⁰. See *Using parallel algorithms* for more information about the parallel algorithms.

⁴⁶¹ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/mutex.hpp

⁴⁶² <http://en.cppreference.com/w/cpp/header/mutex>

⁴⁶³ <http://en.cppreference.com/w/cpp/thread/mutex>

⁴⁶⁴ http://en.cppreference.com/w/cpp/thread/once_flag

⁴⁶⁵ http://en.cppreference.com/w/cpp/thread/recursive_mutex

⁴⁶⁶ http://en.cppreference.com/w/cpp/thread/timed_mutex

⁴⁶⁷ http://en.cppreference.com/w/cpp/thread/call_once

⁴⁶⁸ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/memory.hpp

hpp

⁴⁶⁹ <http://en.cppreference.com/w/cpp/header/memory>

⁴⁷⁰ <http://en.cppreference.com/w/cpp/header/memory>

Functions

Table 2.151: *hpx::ranges* functions of header *hpx/memory.hpp*

<i>hpx::ranges</i> function	C++ standard
<i>hpx::ranges::uninitialized_copy</i>	<code>std::uninitialized_copy</code> ⁴⁷¹
<i>hpx::ranges::uninitialized_copy_n</i>	<code>std::uninitialized_copy_n</code> ⁴⁷²
<i>hpx::ranges::uninitialized_default_construct</i>	<code>std::uninitialized_default_construct</code> ⁴⁷³
<i>hpx::ranges::uninitialized_default_construct_n</i>	<code>std::uninitialized_default_construct_n</code> ⁴⁷⁴
<i>hpx::ranges::uninitialized_fill</i>	<code>std::uninitialized_fill</code> ⁴⁷⁵
<i>hpx::ranges::uninitialized_fill_n</i>	<code>std::uninitialized_fill_n</code> ⁴⁷⁶
<i>hpx::ranges::uninitialized_move</i>	<code>std::uninitialized_move</code> ⁴⁷⁷
<i>hpx::ranges::uninitialized_move_n</i>	<code>std::uninitialized_move_n</code> ⁴⁷⁸
<i>hpx::ranges::uninitialized_value_construct</i>	<code>std::uninitialized_value_construct</code> ⁴⁷⁹
<i>hpx::ranges::uninitialized_value_construct_n</i>	<code>std::uninitialized_value_construct_n</code> ⁴⁸⁰

hpx/numeric.hpp

The header *hpx/numeric.hpp*⁴⁸¹ corresponds to the C++ standard library header *numeric*⁴⁸². See *Using parallel algorithms* for more information about the parallel algorithms.

Functions

Table 2.152: *hpx* functions of header *hpx/numeric.hpp*

<i>hpx</i> function	C++ standard
<i>hpx::adjacent_difference</i>	<code>std::adjacent_difference</code> ⁴⁸³
<i>hpx::exclusive_scan</i>	<code>std::exclusive_scan</code> ⁴⁸⁴
<i>hpx::inclusive_scan</i>	<code>std::inclusive_scan</code> ⁴⁸⁵
<i>hpx::reduce</i>	<code>std::reduce</code> ⁴⁸⁶
<i>hpx::transform_exclusive_scan</i>	<code>std::transform_exclusive_scan</code> ⁴⁸⁷
<i>hpx::transform_inclusive_scan</i>	<code>std::transform_inclusive_scan</code> ⁴⁸⁸
<i>hpx::transform_reduce</i>	<code>std::transform_reduce</code> ⁴⁸⁹

⁴⁷¹ http://en.cppreference.com/w/cpp/memory/ranges/uninitialized_copy

⁴⁷² http://en.cppreference.com/w/cpp/memory/ranges/uninitialized_copy_n

⁴⁷³ http://en.cppreference.com/w/cpp/memory/ranges/uninitialized_default_construct

⁴⁷⁴ http://en.cppreference.com/w/cpp/memory/ranges/uninitialized_default_construct_n

⁴⁷⁵ http://en.cppreference.com/w/cpp/memory/ranges/uninitialized_fill

⁴⁷⁶ http://en.cppreference.com/w/cpp/memory/ranges/uninitialized_fill_n

⁴⁷⁷ http://en.cppreference.com/w/cpp/memory/ranges/uninitialized_move

⁴⁷⁸ http://en.cppreference.com/w/cpp/memory/ranges/uninitialized_move_n

⁴⁷⁹ http://en.cppreference.com/w/cpp/memory/ranges/uninitialized_value_construct

⁴⁸⁰ http://en.cppreference.com/w/cpp/memory/ranges/uninitialized_value_construct_n

⁴⁸¹ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/numeric.hpp

⁴⁸² <http://en.cppreference.com/w/cpp/header/numeric>

⁴⁸³ http://en.cppreference.com/w/cpp/algorithm/adjacent_difference

⁴⁸⁴ http://en.cppreference.com/w/cpp/algorithm/exclusive_scan

⁴⁸⁵ http://en.cppreference.com/w/cpp/algorithm/inclusive_scan

⁴⁸⁶ <http://en.cppreference.com/w/cpp/algorithm/reduce>

⁴⁸⁷ http://en.cppreference.com/w/cpp/algorithm/transform_exclusive_scan

⁴⁸⁸ http://en.cppreference.com/w/cpp/algorithm/transform_inclusive_scan

⁴⁸⁹ http://en.cppreference.com/w/cpp/algorithm/transform_reduce

Table 2.153: *hpx::ranges* functions of header `hpx/numeric.hpp`

<i>hpx::ranges</i> function
<i>hpx::ranges::adjacent_difference</i>
<i>hpx::ranges::exclusive_scan</i>
<i>hpx::ranges::inclusive_scan</i>
<i>hpx::ranges::reduce</i>
<i>hpx::ranges::transform_exclusive_scan</i>
<i>hpx::ranges::transform_inclusive_scan</i>
<i>hpx::ranges::transform_reduce</i>

`hpx/optional.hpp`

The header `hpx/optional.hpp`⁴⁹⁰ corresponds to the C++ standard library header `optional`⁴⁹¹. `hpx::optional` is compatible with `std::optional`.

Constants

- `hpx::nullopt`

Classes

Table 2.154: Classes of header `hpx/optional.hpp`

Class	C++ standard
<code>hpx::optional</code>	<code>std::optional</code> ⁴⁹²
<code>hpx::nullopt_t</code>	<code>std::nullopt_t</code> ⁴⁹³
<code>hpx::bad_optional_access</code>	<code>std::bad_optional_access</code> ⁴⁹⁴

`hpx/runtime.hpp`

The header `hpx/runtime.hpp`⁴⁹⁵ contains functions for accessing local and distributed runtime information.

⁴⁹⁰ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/optional.hpp

⁴⁹¹ <http://en.cppreference.com/w/cpp/header/optional>

⁴⁹² <http://en.cppreference.com/w/cpp/utility/optional>

⁴⁹³ http://en.cppreference.com/w/cpp/utility/nullopt_t

⁴⁹⁴ http://en.cppreference.com/w/cpp/utility/optional/bad_optional_access

⁴⁹⁵ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

Typedefs

Table 2.155: Typedefs of header `hpx/runtime.hpp`

Typedef
<code>hpx::startup_function_type</code>
<code>hpx::shutdown_function_type</code>

Functions

Table 2.156: Functions of header `hpx/runtime.hpp`

Function
<code>hpx::find_root_locality</code>
<code>hpx::find_all_localities</code>
<code>hpx::find_remote_localities</code>
<code>hpx::find_locality</code>
<code>hpx::get_colocation_id</code>
<code>hpx::get_locality_id</code>
<code>hpx::get_num_worker_threads</code>
<code>hpx::get_worker_thread_num</code>
<code>hpx::get_thread_name</code>
<code>hpx::register_pre_startup_function</code>
<code>hpx::register_startup_function</code>
<code>hpx::register_pre_shutdown_function</code>
<code>hpx::register_shutdown_function</code>
<code>hpx::get_num_localities</code>
<code>hpx::get_locality_name</code>
<code>hpx::local::termination_detection</code>

`hpx/experimental/scope.hpp`

The header `hpx/experimental/scope.hpp`⁴⁹⁶ corresponds to the C++ standard library header `experimental/scope`⁴⁹⁷.

Classes

Table 2.157: Classes of header `hpx/scope.hpp`

Class	C++ standard
<code>hpx::experimental::scope_exit</code>	<code>std::scope_exit</code> ⁴⁹⁸
<code>hpx::experimental::scope_fail</code>	<code>std::scope_fail</code> ⁴⁹⁹
<code>hpx::experimental::scope_success</code>	<code>std::scope_success</code> ⁵⁰⁰

⁴⁹⁶ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/experimental/scope.hpp

⁴⁹⁷ <http://en.cppreference.com/w/cpp/header/experimental/scope>

⁴⁹⁸ http://en.cppreference.com/w/cpp/experimental/scope_exit

⁴⁹⁹ http://en.cppreference.com/w/cpp/experimental/scope_fail

⁵⁰⁰ http://en.cppreference.com/w/cpp/experimental/scope_success

hpx/semaphore.hpp

The header `hpx/semaphore.hpp`⁵⁰¹ corresponds to the C++ standard library header `semaphore`⁵⁰².

ClassesTable 2.158: Classes of header `hpx/semaphore.hpp`

Class	C++ standard
<code>hpx::binary_semaphore</code>	<code>std::counting_semaphore</code> ⁵⁰³
<code>hpx::counting_semaphore</code>	<code>std::counting_semaphore</code> ⁵⁰⁴

hpx/shared_mutex.hpp

The header `hpx/shared_mutex.hpp`⁵⁰⁵ corresponds to the C++ standard library header `shared_mutex`⁵⁰⁶.

ClassesTable 2.159: Classes of header `hpx/shared_mutex.hpp`

Class	C++ standard
<code>hpx::shared_mutex</code>	<code>std::shared_mutex</code> ⁵⁰⁷

hpx/source_location.hpp

The header `hpx/source_location.hpp`⁵⁰⁸ corresponds to the C++ standard library header `source_location`⁵⁰⁹.

ClassesTable 2.160: Classes of header `hpx/system_error.hpp`

Class	C++ standard
<code>hpx::source_location</code>	<code>std::source_location</code> ⁵¹⁰

⁵⁰¹ http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/semaphore.hpp

⁵⁰² <http://en.cppreference.com/w/cpp/header/semaphore>

⁵⁰³ http://en.cppreference.com/w/cpp/thread/counting_semaphore

⁵⁰⁴ http://en.cppreference.com/w/cpp/thread/counting_semaphore

⁵⁰⁵ http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/shared_mutex.hpp

⁵⁰⁶ http://en.cppreference.com/w/cpp/header/shared_mutex

⁵⁰⁷ http://en.cppreference.com/w/cpp/thread/shared_mutex

⁵⁰⁸ http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/source_location.hpp

⁵⁰⁹ http://en.cppreference.com/w/cpp/header/source_location

⁵¹⁰ http://en.cppreference.com/w/cpp/utility/source_location

hpx/stop_token.hpp

The header `hpx/stop_token.hpp`⁵¹¹ corresponds to the C++ standard library header `stop_token`⁵¹².

Constants

Table 2.161: Constants of header `hpx/stop_token.hpp`

Constant	C++ standard
<code>hpx::nstopstate</code>	<code>std::nstopstate</code> ⁵¹³

Classes

Table 2.162: Classes of header `hpx/stop_token.hpp`

Class	C++ standard
<code>hpx::stop_callback</code>	<code>std::stop_callback</code> ⁵¹⁴
<code>hpx::stop_source</code>	<code>std::stop_source</code> ⁵¹⁵
<code>hpx::stop_token</code>	<code>std::stop_token</code> ⁵¹⁶
<code>hpx::nstopstate_t</code>	<code>std::nstopstate_t</code> ⁵¹⁷

hpx/system_error.hpp

The header `hpx/system_error.hpp`⁵¹⁸ corresponds to the C++ standard library header `system_error`⁵¹⁹.

Classes

Table 2.163: Classes of header `hpx/system_error.hpp`

Class	C++ standard
<code>hpx::error_code</code>	<code>std::error_code</code> ⁵²⁰

⁵¹¹ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/stop_token.hpp

⁵¹² http://en.cppreference.com/w/cpp/header/stop_token

⁵¹³ http://en.cppreference.com/w/cpp/thread/stop_source/nstopstate

⁵¹⁴ http://en.cppreference.com/w/cpp/thread/stop_callback

⁵¹⁵ http://en.cppreference.com/w/cpp/thread/stop_source

⁵¹⁶ http://en.cppreference.com/w/cpp/thread/stop_token

⁵¹⁷ http://en.cppreference.com/w/cpp/thread/stop_source/nstopstate_t

⁵¹⁸ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/system_error.hpp

⁵¹⁹ http://en.cppreference.com/w/cpp/header/system_error

⁵²⁰ http://en.cppreference.com/w/cpp/error/error_code

hpx/task_block.hpp

The header `hpx/task_block.hpp`⁵²¹ corresponds to the `task_block` feature in N4755⁵²². See `using_task_block` for more details on using task blocks.

Classes

Table 2.164: Classes of header `hpx/task_block.hpp`

Class
<code>hpx::experimental::task_canceled_exception</code>
<code>hpx::experimental::task_block</code>

Functions

Table 2.165: Functions of header `hpx/task_block.hpp`

Function
<code>hpx::experimental::define_task_block</code>
<code>hpx::experimental::define_task_block_restore_thread</code>

hpx/experimental/task_group.hpp

The header `hpx/experimental/task_group.hpp`⁵²³ corresponds to the `task_group` feature in oneAPI Threading Building Blocks (oneTBB)⁵²⁴.

Classes

Table 2.166: Classes of header `hpx/experimental/task_group.hpp`

Class
<code>hpx::experimental::task_group</code>

hpx/thread.hpp

The header `hpx/thread.hpp`⁵²⁵ corresponds to the C++ standard library header `thread`⁵²⁶. The functionality in this header is equivalent to the standard library thread functionality, with the exception that the *HPX* equivalents are implemented on top of lightweight threads and the *HPX* runtime.

⁵²¹ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/task_block.hpp

⁵²² <http://wg21.link/n4755>

⁵²³ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/experimental/task_group.hpp

⁵²⁴ https://spec.oneapi.io/versions/1.0-rev-3/elements/oneTBB/source/task_scheduler/task_group/task_group_cls.html

⁵²⁵ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/thread.hpp

⁵²⁶ <http://en.cppreference.com/w/cpp/header/thread>

Classes

Table 2.167: Classes of header `hpx/thread.hpp`

Class	C++ standard
<code>hpx::thread</code>	<code>std::thread</code> ⁵²⁷
<code>hpx::jthread</code>	<code>std::jthread</code> ⁵²⁸

Functions

Table 2.168: Functions of header `hpx/thread.hpp`

Function	C++ standard
<code>hpx::this_thread::yield</code>	<code>std::yield</code> ⁵²⁹
<code>hpx::this_thread::get_id</code>	<code>std::get_id</code> ⁵³⁰
<code>hpx::this_thread::sleep_for</code>	<code>std::sleep_for</code> ⁵³¹
<code>hpx::this_thread::sleep_until</code>	<code>std::sleep_until</code> ⁵³²

`hpx/tuple.hpp`

The header `hpx/tuple.hpp`⁵³³ corresponds to the C++ standard library header `tuple`⁵³⁴. `hpx::tuple` can be used in CUDA device code, unlike `std::tuple`.

Constants

Table 2.169: Constants of header `hpx/tuple.hpp`

Constant	C++ standard
<code>hpx::ignore</code>	<code>std::ignore</code> ⁵³⁵

Classes

Table 2.170: Classes of header `hpx/tuple.hpp`

Class	C++ standard
<code>hpx::tuple</code>	<code>std::tuple</code> ⁵³⁶
<code>hpx::tuple_size</code>	<code>std::tuple_size</code> ⁵³⁷
<code>hpx::tuple_element</code>	<code>std::tuple_element</code> ⁵³⁸

⁵²⁷ <http://en.cppreference.com/w/cpp/thread/thread>

⁵²⁸ <http://en.cppreference.com/w/cpp/thread/jthread>

⁵²⁹ <http://en.cppreference.com/w/cpp/thread/yield>

⁵³⁰ http://en.cppreference.com/w/cpp/thread/get_id

⁵³¹ http://en.cppreference.com/w/cpp/thread/sleep_for

⁵³² http://en.cppreference.com/w/cpp/thread/sleep_until

⁵³³ http://github.com/TheHPXProject/hpx/blob/d402eada7405eedff56d454c9380fbdc0c8044e7/libs/core/include_local/include/hpx/tuple.hpp

⁵³⁴ <http://en.cppreference.com/w/cpp/header/tuple>

⁵³⁵ <http://en.cppreference.com/w/cpp/utility/tuple/ignore>

Functions

Table 2.171: Functions of header `hpx/tuple.hpp`

Function	C++ standard
<code>hpx::make_tuple</code>	<code>std::tuple_element</code> ⁵³⁹
<code>hpx::tie</code>	<code>std::tie</code> ⁵⁴⁰
<code>hpx::forward_as_tuple</code>	<code>std::forward_as_tuple</code> ⁵⁴¹
<code>hpx::tuple_cat</code>	<code>std::tuple_cat</code> ⁵⁴²
<code>hpx::get</code>	<code>std::get</code> ⁵⁴³

`hpx/type_traits.hpp`

The header `hpx/type_traits.hpp`⁵⁴⁴ corresponds to the C++ standard library header `type_traits`⁵⁴⁵.

Classes

Table 2.172: Classes of header `hpx/type_traits.hpp`

Class	C++ standard
<code>hpx::is_invocable</code>	<code>std::is_invocable</code> ⁵⁴⁶
<code>hpx::is_invocable_r</code>	<code>std::is_invocable</code> ⁵⁴⁷

`hpx/unwrap.hpp`

The header `hpx/unwrap.hpp`⁵⁴⁸ contains utilities for unwrapping futures.

Classes

Table 2.173: Classes of header `hpx/unwrap.hpp`

Class
<code>hpx::functional::unwrap</code>
<code>hpx::functional::unwrap_n</code>
<code>hpx::functional::unwrap_all</code>

⁵³⁶ <http://en.cppreference.com/w/cpp/utility/tuple>

⁵³⁷ http://en.cppreference.com/w/cpp/utility/tuple_size

⁵³⁸ http://en.cppreference.com/w/cpp/utility/tuple_element

⁵³⁹ http://en.cppreference.com/w/cpp/utility/tuple/tuple_element

⁵⁴⁰ <http://en.cppreference.com/w/cpp/utility/tuple/tuple/tie>

⁵⁴¹ http://en.cppreference.com/w/cpp/utility/tuple/forward_as_tuple

⁵⁴² http://en.cppreference.com/w/cpp/utility/tuple/tuple_cat

⁵⁴³ <http://en.cppreference.com/w/cpp/utility/tuple/get>

⁵⁴⁴ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/type_traits.hpp

⁵⁴⁵ `hpx`

⁵⁴⁶ http://en.cppreference.com/w/cpp/header/type_traits

⁵⁴⁷ http://en.cppreference.com/w/cpp/types/is_invocable

⁵⁴⁸ http://en.cppreference.com/w/cpp/types/is_invocable

⁵⁴⁹ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/unwrap.hpp

Functions

Table 2.174: Functions of header `hpx/unwrap.hpp`

Function
<code>hpx::unwrap</code>
<code>hpx::unwrap_n</code>
<code>hpx::unwrap_all</code>
<code>hpx::unwrapping</code>
<code>hpx::unwrapping_n</code>
<code>hpx::unwrapping_all</code>

`hpx/version.hpp`

The header `hpx/version.hpp`⁵⁴⁹ provides version information about *HPX*.

Macros

Table 2.175: Macros of header `hpx/version.hpp`

Macro
<code>HPX_VERSION_MAJOR</code>
<code>HPX_VERSION_MINOR</code>
<code>HPX_VERSION_SUBMINOR</code>
<code>HPX_VERSION_FULL</code>
<code>HPX_VERSION_DATE</code>
<code>HPX_VERSION_TAG</code>
<code>HPX_AGAS_VERSION</code>

Functions

Table 2.176: Functions of header `hpx/version.hpp`

Function
<code>hpx::major_version</code>
<code>hpx::minor_version</code>
<code>hpx::subminor_version</code>
<code>hpx::full_version</code>
<code>hpx::full_version_as_string</code>
<code>hpx::tag</code>
<code>hpx::agas_version</code>
<code>hpx::build_type</code>
<code>hpx::build_date_time</code>

⁵⁴⁹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e77/libs/core/version/include/hpx/version.hpp>

hpx/wrap_main.hpp

The header `hpx/wrap_main.hpp`⁵⁵⁰ does not provide any direct functionality but is used for implicitly using `main` as the runtime entry point. See *Re-use the `main()` function as the main HPX entry point* for more details on implicitly starting the HPX runtime.

2.9.2 Public distributed API

Our Public Distributed API offers a rich set of tools and functions that enable developers to harness the full potential of distributed computing. Here, you'll find a comprehensive list of header files, classes and functions for various distributed computing features provided by HPX.

hpx/barrier.hpp

The header `hpx/barrier.hpp`⁵⁵¹ includes a distributed barrier implementation. For information regarding the C++ standard library header `barrier`⁵⁵², see *Public API*.

Classes

Table 2.177: Distributed implementation of classes of header `hpx/barrier.hpp`

Class
<code>hpx::distributed::barrier</code>

Functions

Table 2.178: `hpx` functions of header `hpx/barrier.hpp`

Function
<code>hpx::distributed::wait</code>
<code>hpx::distributed::synchronize</code>

hpx/collectives.hpp

The header `hpx/collectives.hpp`⁵⁵³ contains definitions and implementations related to the collectives operations.

⁵⁵⁰ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/wrap/include/hpx/wrap_main.hpp

⁵⁵¹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/barrier.hpp>

⁵⁵² <http://en.cppreference.com/w/cpp/header/barrier>

⁵⁵³ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/collectives.hpp>

Classes

Table 2.179: *hpx* classes of header *hpx/collectives.hpp*

Class
<i>hpx::collectives::num_sites_arg</i>
<i>hpx::collectives::this_site_arg</i>
<i>hpx::collectives::that_site_arg</i>
<i>hpx::collectives::generation_arg</i>
<i>hpx::collectives::root_site_arg</i>
<i>hpx::collectives::tag_arg</i>
<i>hpx::collectives::arity_arg</i>
<i>hpx::collectives::communicator</i>
<i>hpx::collectives::channel_communicator</i>

Functions

Table 2.180: *hpx* functions of header *hpx/collectives.hpp*

Function
<i>hpx::collectives::all_gather</i>
<i>hpx::collectives::all_reduce</i>
<i>hpx::collectives::all_to_all</i>
<i>hpx::collectives::broadcast_to</i>
<i>hpx::collectives::broadcast_from</i>
<i>hpx::collectives::create_channel_communicator</i>
<i>hpx::collectives::set</i>
<i>hpx::collectives::get</i>
<i>hpx::collectives::create_communicator</i>
<i>hpx::collectives::create_hierarchical_communicator</i>
<i>hpx::collectives::create_local_communicator</i>
<i>hpx::collectives::communicator::set_info</i>
<i>hpx::collectives::communicator::get_info</i>
<i>hpx::collectives::communicator::is_root</i>
<i>hpx::collectives::exclusive_scan</i>
<i>hpx::collectives::gather_here</i>
<i>hpx::collectives::gather_there</i>
<i>hpx::collectives::inclusive_scan</i>
<i>hpx::collectives::reduce_here</i>
<i>hpx::collectives::reduce_there</i>
<i>hpx::collectives::scatter_from</i>
<i>hpx::collectives::scatter_to</i>

hpx/latch.hpp

The header `hpx/latch.hpp`⁵⁵⁴ includes a distributed latch implementation. For information regarding the C++ standard library header `latch`⁵⁵⁵, see *Public API*.

Classes

Table 2.181: Distributed implementation of classes of header `hpx/latch.hpp`

Class
<code>hpx::distributed::latch</code>

Member functions

Table 2.182: *hpx* functions of class `hpx::distributed::latch` from header `hpx/latch.hpp`

Function
<code>hpx::distributed::latch::count_down_and_wait</code>
<code>hpx::distributed::latch::arrive_and_wait</code>
<code>hpx::distributed::latch::count_down</code>
<code>hpx::distributed::latch::is_ready</code>
<code>hpx::distributed::latch::try_wait</code>
<code>hpx::distributed::latch::wait</code>

hpx/async.hpp

The header `hpx/async.hpp`⁵⁵⁶ includes distributed implementations of *hpx::async*, *hpx::post*, *hpx::sync*, and *hpx::dataflow*. For information regarding the C++ standard library header, see *Public API*.

Functions

Table 2.183: Distributed implementation of functions of header `hpx/async.hpp`

Functions
<code>modules_hpx/async_distributed/async.hpp_api</code>
<code>modules_hpx/async_distributed/sync.hpp_api</code>
<code>modules_hpx/async_distributed/post.hpp_api</code>
<code>modules_hpx/async_distributed/dataflow.hpp_api</code>

⁵⁵⁴ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/latch.hpp>

⁵⁵⁵ <http://en.cppreference.com/w/cpp/header/latch>

⁵⁵⁶ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/async_distributed/include/hpx/async.hpp

hpx/components.hpp

The header `hpx/include/components.hpp`⁵⁵⁷ includes the components implementation. A component in *hpx* is a C++ class which can be created remotely and for which its member functions can be invoked remotely as well. More information about how components can be defined, created, and used can be found in *Writing components*. *Components and actions* includes examples on the accumulator, template accumulator and template function accumulator.

Macros

Table 2.184: *hpx* macros of header `hpx/components.hpp`

Macro
<code>HPX_DEFINE_COMPONENT_ACTION</code>
<code>HPX_REGISTER_ACTION_DECLARATION</code>
<code>HPX_REGISTER_ACTION</code>
<code>HPX_REGISTER_COMMANDLINE_MODULE</code>
<code>HPX_REGISTER_COMPONENT</code>
<code>HPX_REGISTER_COMPONENT_MODULE</code>
<code>HPX_REGISTER_STARTUP_MODULE</code>

Classes

Table 2.185: *hpx* classes of header `hpx/components.hpp`

Class
<code>hpx::components::client</code>
<code>hpx::components::client_base</code>
<code>hpx::components::component</code>
<code>hpx::components::component_base</code>
<code>hpx::components::component_commandline_base</code>

Functions

Table 2.186: *hpx* functions of header `hpx/components.hpp`

Function
<code>hpx::new_</code>

⁵⁵⁷ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/include/components.hpp>

2.9.3 Supervision dispatch module

The supervision dispatch module provides peer discovery, cooperative lifecycle management, failure detection, and fenced action dispatch across localities, built on top of the *supervision* module's lifecycle events and admission checks. See that module's reference for the semantics of `hpx::supervision::check_admission` and `hpx::supervision::dispatch_outcome`; this page documents only the dispatch-specific surface layered on top of them. For the narrative on initialization ordering, one-shot discovery, shadow-state semantics, and client-side filtering versus fenced-dispatch admission, see *Supervision dispatch*.

Overview

Lifecycle initialization and shutdown

```
hpx::shared_future<void> hpx::supervision::init(hpx::chrono::steady_duration const &discovery_timeout =
hpx::supervision::default_discovery_timeout)
```

```
void hpx::supervision::init(hpx::launch::sync_policy, hpx::chrono::steady_duration const
&discovery_timeout = hpx::supervision::default_discovery_timeout)
```

Performs one-shot, idempotent initialization of the supervision-dispatch runtime for this locality: creates a local sentinel and registry, publishes `event::started` before either symbol name is registered, registers both names, and performs a single `discover_and_join()` pass. Calling `init()` while already active is a no-op; concurrent callers during initializing attach to the in-flight operation instead of racing it.

```
void hpx::supervision::finalize()
```

Performs one-shot, idempotent teardown of the runtime previously started by `init()`: publishes `event::completed` at a new epoch, unregisters both symbol names, and releases both components. A no-op unless the runtime is currently active.

```
bool hpx::supervision::is_initialized() noexcept
```

Returns whether the runtime is currently active. Never blocks; safe to call from any thread, including while `init()/finalize()` is in flight.

Peer discovery

```
struct hpx::supervision::discovered_peer
```

A peer whose sentinel and registry symbol names both resolved during a `discover_peers()` call. Holds `locality`, `sentinel_client`, and `registry_client`.

```
std::vector<hpx::supervision::discovered_peer> hpx::supervision::discover_peers(hpx::chrono::steady_duration
const &timeout =
hpx::supervision::default_discovery_timeout)
```

Performs a one-time discovery pull: concurrently resolves the pinned sentinel and registry names of every remote locality, bounded by a single wait for `timeout`. Localities that have not yet called `init()` are silently excluded rather than causing a hang or failure.

```
std::vector<hpx::id_type> hpx::supervision::fan_out_join(hpx::supervision::registry const &local_registry,
std::vector<hpx::supervision::discovered_peer>
const &peers)
```

Fans out `join()` calls from `local_registry` to every entry in `peers`. Reuses `registry::join()`'s reservation/idempotency machinery, so repeated or overlapping calls never create more than one shadow per peer sentinel.

```
std::vector<hpx::id_type> hpx::supervision::discover_and_join(hpx::supervision::registry const  
                                                             &local_registry,  
                                                             hpx::chrono::steady_duration const  
                                                             &timeout =  
                                                             hpx::supervision::default_discovery_timeout)
```

Composes a single reactive discovery-and-join pass: one `discover_peers()` pull followed by one `fan_out_join()` call. Introduces no polling, timer, or repeated broadcast of its own.

Sentinel client

class `hpx::supervision::sentinel`

A lightweight, self-supervising client handle. Constructing it with a target locality creates the underlying server-side component; calling `start()` publishes the `started` lifecycle event for it.

```
hpx::future<hpx::supervision::publish_result> start(std::uint64_t epoch = 0) const
```

```
hpx::supervision::publish_result start(hpx::launch::sync_policy, std::uint64_t epoch = 0, hpx::error_code  
                                       &ec = hpx::throws) const
```

```
hpx::future<bool> register_name()
```

```
bool register_name(hpx::launch::sync_policy, hpx::error_code &ec = hpx::throws)
```

```
hpx::future<hpx::id_type> unregister_name() const
```

```
hpx::id_type unregister_name(hpx::launch::sync_policy, hpx::error_code &ec = hpx::throws) const
```

Registry client

class `hpx::supervision::registry`

A lightweight, self-supervising client handle for pairing with peer sentinels; mirrors a peer's lifecycle state locally via `join()` and can itself be discovered by name in AGAS.

```
hpx::future<hpx::id_type> join(hpx::supervision::sentinel const &peer_sentinel, hpx::id_type const  
                              &peer_locality) const
```

```
hpx::id_type join(hpx::launch::sync_policy, hpx::supervision::sentinel const &peer_sentinel, hpx::id_type  
                  const &peer_locality, hpx::error_code &ec = hpx::throws) const
```

Joins a peer sentinel: creates (or reuses) a local shadow target mirroring its lifecycle state, and registers this registry as an observer of the peer's lifecycle/activity events.

```
hpx::future<std::vector<hpx::supervision::server::peer_snapshot>> snapshot_peers() const
```

```
std::vector<hpx::supervision::server::peer_snapshot> snapshot_peers(hpx::launch::sync_policy,  
                                                                    hpx::error_code &ec = hpx::throws)  
                                                                    const
```

Returns a point-in-time snapshot of all fully joined, non-evicting peers.

```
hpx::future<bool> register_name()
```

```
bool register_name(hpx::launch::sync_policy, hpx::error_code &ec = hpx::throws)
```

```
hpx::future<hpx::id_type> unregister_name() const
```



```
hpx::id_type unregister_name(hpx::launch::sync_policy, hpx::error_code &ec = hpx::throws) const
```

```
struct hpx::supervision::server::peer_snapshot
```

Plain-data view of a single joined peer: `peer_sentinel`, `peer_locality`, `shadow`, and `join_epoch`.

Fenced dispatch

```
template<typename Action, typename ...Ts>
decltype(auto) hpx::supervision::dispatch_work(hpx::id_type const &target, std::uint64_t epoch, Ts&&... ts)
```

Dispatches `Action` to `target` under supervision fencing. Performs a cheap, non-authoritative admission check on the caller's locality to short-circuit an already-known-fenced target, then dispatches to the target's own locality (via `hpx::colocated`) where an authoritative re-check against the same latch consulted by `hpx::supervision::check_admission` runs immediately before the wrapped action executes, on the same thread, closing the admission/invocation race. See *Supervision dispatch* for the full client-side-filtering versus fenced-admission narrative and worked examples.

Throws

`hpx::exception` with `hpx::error::target_fenced` if the target has latched a terminal event for epoch since the client-side check; the wrapped action is not invoked in that case.

Testing support

The following are declared in `hpx/supervision_dispatch/testing.hpp` and are **not** part of the stable public dispatch API; they exist solely to make unit tests for this module deterministic:

```
std::vector<hpx::supervision::server::peer_snapshot> hpx::supervision::testing::local_snapshot_peers()

void hpx::supervision::testing::set_failure_detection_poll_timeout_for_testing(hpx::chrono::steady_duration
                                                                              const
                                                                              &timeout)

hpx::id_type hpx::supervision::testing::last_join_shadow()

void hpx::supervision::testing::suspend_heartbeat_for_testing()

bool hpx::supervision::testing::failure_detection_sweep_in_flight_for_testing()
```

See the *API reference* of this module for more details, and *Supervision module* for the underlying lifecycle event, admission, and dispatch-outcome semantics this module builds on.

2.9.4 Full API

The full API of *HPX* is presented below. The listings for the public API above refer to the full documentation below.

Note: Most names listed in the full API reference are implementation details or considered unstable. They are listed mostly for completeness. If there is a particular feature you think deserves being in the public API we may consider promoting it. In general we prioritize making sure features corresponding to C++ standard library features are stable and complete.

actions

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/actions/action_support.hpp

Defined in header `hpx/actions/action_support.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/actions/base_action.hpp

Defined in header `hpx/actions/base_action.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/actions/actions_fwd.hpp

Defined in header `hpx/actions/actions_fwd.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/actions/transfer_base_action.hpp

Defined in header `hpx/actions/transfer_base_action.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/actions/transfer_action.hpp

Defined in header `hpx/actions/transfer_action.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

actions_base

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/actions_base/basic_action_fwd.hpp

Defined in header `hpx/actions_base/basic_action_fwd.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **actions**

template<typename **Component**, typename **Signature**, typename **Derived**>

```
struct basic_action
```

```
    #include <basic_action_fwd.hpp>
```

Template Parameters

- **Component** – component type
- **Signature** – return type and arguments
- **Derived** – derived action class

hpx/actions_base/actions_base_support.hpp

Defined in header `hpx/actions_base/actions_base_support.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace actions
```

hpx/actions_base/reflect_action.hpp

Defined in header `hpx/actions_base/reflect_action.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Reflection-based action definition for HPX remote operations.

HPX_DEFINE_COMPONENT_ACTION

Defined in header `hpx/components.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Warning: doxygenfunction: Unable to resolve function “HPX_DEFINE_COMPONENT_ACTION” with arguments “None”. Candidate function could not be parsed. Parsing error is Error when parsing function declaration. If the function has no return type: Invalid C++ declaration: Expected end of definition or ;. [error at 87] HPX_DEFINE_COMPONENT_ACTION (action_move_semantics, test_movable, test_movable_action) HPX_DEFINE_COMPONENT_ACTION(action_move_semantics _____^ If the function has a return type: Error in declarator If declarator-id with parameters-and-qualifiers: Invalid C++ declaration: Expected identifier in nested name. [error at 28] HPX_DEFINE_COMPONENT_ACTION (action_move_semantics, test_movable, test_movable_action) HPX_DEFINE_COMPONENT_ACTION(action_move_semantics _____^ If parenthesis in noptr-declarator: Error in declarator or parameters-and-qualifiers If pointer to member declarator: Invalid C++ declaration: Expected ‘::’ in pointer to member (function). [error at 50] HPX_DEFINE_COMPONENT_ACTION (action_move_semantics, test_movable, test_movable_action) HPX_DEFINE_COMPONENT_ACTION(action_move_semantics _____^ If declarator-id: Invalid C++ declaration: Expecting “(” in parameters-and-qualifiers. [error at 50] HPX_DEFINE_COMPONENT_ACTION (action_move_semantics, test_movable, test_movable_action) HPX_DEFINE_COMPONENT_ACTION(action_move_semantics _____^

hpx/actions_base/actions_base_fwd.hpp

Defined in header `hpx/actions_base/actions_base_fwd.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace actions
```

hpx/actions_base/plain_action.hpp

Defined in header `hpx/actions_base/plain_action.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace actions
```

```
        namespace traits
```

hpx/actions_base/preassigned_action_id.hpp

Defined in header `hpx/actions_base/preassigned_action_id.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace actions
```

hpx/actions_base/lambda_to_action.hpp

Defined in header `hpx/actions_base/lambda_to_action.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

HPX_REGISTER_ACTION_DECLARATION

Defined in header `hpx/components.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *HPX_REGISTER_ACTION*

Warning: doxygenfunction: Cannot find function “HPX_REGISTER_ACTION_DECLARATION” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

HPX_REGISTER_ACTION

Defined in header `hpx/components.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `HPX_REGISTER_ACTION_DECLARATION`

Warning: doxygenfunction: Cannot find function “HPX_REGISTER_ACTION” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

hpx/actions_base/macros.hpp

Defined in header `hpx/actions_base/macros.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Defines

HPX_REGISTER_ACTION_INVOCATION_COUNT(Action)

HPX_REGISTER_PER_ACTION_DATA_COUNTER_TYPES(Action)

HPX_MAKE_ACTION(func)

HPX_MAKE_DIRECT_ACTION(func)

HPX_DECLARE_ACTION(...)

Declares an action type.

HPX_REGISTER_ACTION_DECLARATION(...)

Declare the necessary component action boilerplate code.

The macro `HPX_REGISTER_ACTION_DECLARATION` can be used to declare all the boilerplate code which is required for proper functioning of component actions in the context of HPX.

The parameter *action* is the type of the action to declare the boilerplate for.

This macro can be invoked with an optional second parameter. This parameter specifies a unique name of the action to be used for serialization purposes. The second parameter has to be specified if the first parameter is not usable as a plain (nonqualified) C++ identifier, i.e. the first parameter contains special characters which cannot be part of a C++ identifier, such as ‘<’, ‘>’, or ‘.’.

```
namespace app
{
    // Define a simple component exposing one action 'print_greeting'
    class HPX_COMPONENT_EXPORT server
    : public hpx::components::component_base<server>
    {
        void print_greeting ()
        {
            hpx::cout << "Hey, how are you?\n" << std::flush;
        }

        // Component actions need to be declared, this also defines the
        // type 'print_greeting_action' representing the action.
        HPX_DEFINE_COMPONENT_ACTION(server,
            print_greeting, print_greeting_action);
    };
}

// Declare boilerplate code required for each of the component actions.
HPX_REGISTER_ACTION_DECLARATION(app::server::print_greeting_action)
```

Example:

Note: This macro has to be used once for each of the component actions defined using one of the *HPX_DEFINE_COMPONENT_ACTION* macros. It has to be visible in all translation units using the action, thus it is recommended to place it into the header file defining the component.

HPX_REGISTER_ACTION_DECLARATION(...)**HPX_REGISTER_ACTION_DECLARATION_1(action)****HPX_REGISTER_ACTION(...)**

Define the necessary component action boilerplate code.

The macro *HPX_REGISTER_ACTION* can be used to define all the boilerplate code which is required for proper functioning of component actions in the context of HPX.

The parameter *action* is the type of the action to define the boilerplate for.

This macro can be invoked with an optional second parameter. This parameter specifies a unique name of the action to be used for serialization purposes. The second parameter has to be specified if the first parameter is not usable as a plain (nonqualified) C++ identifier, i.e. the first parameter contains special characters which cannot be part of a C++ identifier, such as '<', '>', or ':'.

Note: This macro has to be used once for each of the component actions defined using one of the *HPX_DEFINE_COMPONENT_ACTION* or *HPX_DEFINE_PLAIN_ACTION* macros. It has to occur exactly once for each of the actions, thus it is recommended to place it into the source file defining the component.

Note: Only one of the forms of this macro *HPX_REGISTER_ACTION* or *HPX_REGISTER_ACTION_ID* should be used for a particular action, never both.

HPX_REGISTER_ACTION_ID(action, actionname, actionid)

Define the necessary component action boilerplate code and assign a predefined unique id to the action.

The macro *HPX_REGISTER_ACTION* can be used to define all the boilerplate code which is required for proper functioning of component actions in the context of HPX.

The parameter *action* is the type of the action to define the boilerplate for.

The parameter *actionname* specifies a unique name of the action to be used for serialization purposes. The second parameter has to be usable as a plain (nonqualified) C++ identifier, it should not contain special characters which cannot be part of a C++ identifier, such as '<', '>', or ':'.

The parameter *actionid* specifies a unique integer value which will be used to represent the action during serialization.

Note: This macro has to be used once for each of the component actions defined using one of the *HPX_DEFINE_COMPONENT_ACTION* or global actions *HPX_DEFINE_PLAIN_ACTION* macros. It has to occur exactly once for each of the actions, thus it is recommended to place it into the source file defining the component.

Note: Only one of the forms of this macro *HPX_REGISTER_ACTION* or *HPX_REGISTER_ACTION_ID* should be used for a particular action, never both.

HPX_DEFINE_COMPONENT_ACTION(...)

Registers a member function of a component as an action type with HPX.

The macro *HPX_DEFINE_COMPONENT_ACTION* can be used to register a member function of a component as an action type named *action_type*.

The parameter *component* is the type of the component exposing the member function *func* which should be associated with the newly defined action type. The parameter *action_type* is the name of the action type to register with HPX.

```
namespace app
{
    // Define a simple component exposing one action 'print_greeting'
    class HPX_COMPONENT_EXPORT server
    : public hpx::components::component_base<server>
    {
    public:
        void print_greeting() const
        {
            hpx::cout << "Hey, how are you?\n" << std::flush;
        }

        // Component actions need to be declared, this also defines the
        // type 'print_greeting_action' representing the action.
        HPX_DEFINE_COMPONENT_ACTION(server, print_greeting,
            print_greeting_action);
    };
}
```

Example:

The first argument must provide the type name of the component the action is defined for.

The second argument must provide the member function name the action should wrap.

The default value for the third argument (the typename of the defined action) is derived from the name of the function (as passed as the second argument) by appending ‘_action’. The third argument can be omitted only if the second argument with an appended suffix ‘_action’ resolves to a valid, unqualified C++ type name.

Note: The macro *HPX_DEFINE_COMPONENT_ACTION* can be used with 2 or 3 arguments. The third argument is optional.

HPX_DEFINE_PLAIN_ACTION(...)

Defines a plain action type.

```
namespace app
{
    void some_global_function(double d)
    {
        cout << d;
    }

    // This will define the action type 'app::some_global_action' which
    // represents the function 'app::some_global_function'.
    HPX_DEFINE_PLAIN_ACTION(some_global_function, some_global_action);
}
```

Example:

Note: Usually this macro will not be used in user code unless the intent is to avoid defining the `action_type` in global namespace. Normally, the use of the macro *HPX_PLAIN_ACTION* is recommended.

Note: The macro *HPX_DEFINE_PLAIN_ACTION* can be used with 1 or 2 arguments. The second argument is optional. The default value for the second argument (the typename of the defined action) is derived from the name of the function (as passed as the first argument) by appending ‘_action’. The second argument can be omitted only if the first argument with an appended suffix ‘_action’ resolves to a valid, unqualified C++ type name.

HPX_DECLARE_PLAIN_ACTION(...)

Declares a plain action type.

HPX_PLAIN_ACTION(...)

Defines a plain action type based on the given function *func* and registers it with HPX.

The macro *HPX_PLAIN_ACTION* can be used to define a plain action (e.g. an action encapsulating a global or free function) based on the given function *func*. It defines the action type *name* representing the given function. This macro additionally registers the newly define action type with HPX.

The parameter *func* is a global or free (non-member) function which should be encapsulated into a plain action. The parameter *name* is the name of the action type defined by this macro.


```

namespace app {
    void some_global_function(double d) {
        cout << d;
    }
}

// This will define the action type 'some_global_action' which
// represents the function 'app::some_global_function'.
HPX_PLAIN_ACTION(app::some_global_function, some_global_action)

```

Example:

Note: The macro *HPX_PLAIN_ACTION* has to be used at global namespace even if the wrapped function is located in some other namespace. The newly defined action type is placed into the global namespace as well.

Note: The macro *HPX_PLAIN_ACTION_ID* can be used with 1, 2, or 3 arguments. The second and third arguments are optional. The default value for the second argument (the typename of the defined action) is derived from the name of the function (as passed as the first argument) by appending ‘_action’. The second argument can be omitted only if the first argument with an appended suffix ‘_action’ resolves to a valid, unqualified C++ type name. The default value for the third argument is *hpx::components::factory_state::check*.

Note: Only one of the forms of this macro *HPX_PLAIN_ACTION* or *HPX_PLAIN_ACTION_ID* should be used for a particular action, never both.

HPX_PLAIN_ACTION_ID(func, name, id)

Defines a plain action type based on the given function *func* and registers it with HPX.

The macro *HPX_PLAIN_ACTION_ID* can be used to define a plain action (e.g. an action encapsulating a global or free function) based on the given function *func*. It defines the action type *actionname* representing the given function.

The parameter *actionid* specifies a unique integer value which will be used to represent the action during serialization.

The parameter *func* is a global or free (non-member) function which should be encapsulated into a plain action. The parameter *name* is the name of the action type defined by this macro.

The second parameter has to be usable as a plain (nonqualified) C++ identifier, it should not contain special characters which cannot be part of a C++ identifier, such as ‘<’, ‘>’, or ‘.’.

```

namespace app {
    void some_global_function(double d) {
        cout << d;
    }
}

// This will define the action type 'some_global_action' which

```

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```
// represents the function 'app::some_global_function'.
HPX_PLAIN_ACTION_ID(app::some_global_function, some_global_action,
    some_unique_id);
```

Example:

Note: The macro `HPX_PLAIN_ACTION_ID` has to be used at global namespace even if the wrapped function is located in some other namespace. The newly defined action type is placed into the global namespace as well.

Note: Only one of the forms of this macro `HPX_PLAIN_ACTION` or `HPX_PLAIN_ACTION_ID` should be used for a particular action, never both.

`HPX_REGISTER_ACTION_FACTORY_ID`(Name, Id)

`hpx/actions_base/traits/action_remote_result.hpp`

Defined in header `hpx/actions_base/traits/action_remote_result.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **traits**

Typedefs

```
template<typename Result>
using action_remote_result_t = typename action_remote_result<Result>::type

template<typename Result>
struct action_remote_result : public detail::action_remote_result_customization_point<Result>
```

agas

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/agas/addressing_service.hpp

Defined in header `hpx/agas/addressing_service.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace agas
```

```
        struct addressing_service
```

Public Types

```
        using component_id_type = components::component_type
```

```
        using iterate_names_return_type = std::map<std::string, hpx::id_type>
```

```
        using iterate_types_function_type = hpx::function<void(std::string const&,
        components::component_type), true>
```

```
        using mutex_type = hpx::spinlock
```

```
        using gva_cache_type = hpx::util::cache::lru_cache<gva_cache_key, gva,
        hpx::util::cache::statistics::local_full_statistics>
```

```
        using migrated_objects_table_type = std::set<naming::gid_type>
```

```
        using refcnt_requests_type = std::map<naming::gid_type, std::int64_t>
```

```
        using resolved_localities_type = std::map<naming::gid_type, parcelset::endpoints_type>
```

Public Functions

```
HPX_NON_COPYABLE(addressing_service)
```

```
explicit addressing_service(util::runtime_configuration const &ini_)
```

```
~addressing_service() = default
```

```
void bootstrap(parcelset::endpoints_type const &endpoints, util::runtime_configuration &rtcfg)
```

```
void initialize(std::uint64_t rts_lva)
```

```
void adjust_local_cache_size(std::size_t) const
```

Adjust the size of the local AGAS Address resolution cache.

```
inline state get_status() const

inline void set_status(state new_state)

inline naming::gid_type const &get_local_locality(error_code& = throws) const

void set_local_locality(naming::gid_type const &g)

void register_console(parcelset::endpoints_type const &eps)

inline bool is_bootstrap() const

inline bool is_console() const
    Returns whether this addressing_service represents the console locality.

inline bool is_connecting() const
    Returns whether this addressing_service is connecting to a running application.

bool resolve_locally_known_addresses(naming::gid_type const &id, naming::address &addr)
    const

void register_server_instances()

void unregister_server_instances(hpx::error_code &ec = hpx::throws)

void garbage_collect_non_blocking(error_code &ec = throws)

void garbage_collect(error_code &ec = throws)

inline server::primary_namespace &get_local_primary_namespace_service()

inline naming::address::address_type get_primary_ns_lva() const

inline naming::address::address_type get_symbol_ns_lva() const

inline server::component_namespace *get_local_component_namespace_service() const

inline server::locality_namespace *get_local_locality_namespace_service() const

inline server::symbol_namespace &get_local_symbol_namespace_service() const

inline naming::address::address_type get_runtime_support_lva() const

std::uint64_t get_cache_entries(bool) const

std::uint64_t get_cache_hits(bool) const

std::uint64_t get_cache_misses(bool) const

std::uint64_t get_cache_evictions(bool) const

std::uint64_t get_cache_insertions(bool) const

std::uint64_t get_cache_get_entry_count(bool reset) const

std::uint64_t get_cache_insertion_entry_count(bool reset) const

std::uint64_t get_cache_update_entry_count(bool reset) const

std::uint64_t get_cache_erase_entry_count(bool reset) const
```

`std::uint64_t get_cache_get_entry_time(bool reset) const`

`std::uint64_t get_cache_insertion_entry_time(bool reset) const`

`std::uint64_t get_cache_update_entry_time(bool reset) const`

`std::uint64_t get_cache_erase_entry_time(bool reset) const`

`bool register_locality(parcelset::endpoints_type const &endpoints, naming::gid_type &prefix, std::uint32_t num_threads, error_code &ec = throws)`

Add a locality to the runtime.

`parcelset::endpoints_type const &resolve_locality(naming::gid_type const &gid, error_code &ec = throws)`

Resolve a locality to its prefix.

Returns

Returns an empty vector if the locality is not registered.

`bool has_resolved_locality(naming::gid_type const &gid)`

`bool unregister_locality(naming::gid_type const &gid, error_code &ec = throws)`

Remove a locality from the runtime.

`void remove_resolved_locality(naming::gid_type const &gid)`

remove given locality from locality cache

`bool get_console_locality(naming::gid_type &locality, error_code &ec = throws)`

Get locality locality_id of the console locality.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Parameters

- **locality** – [out] The locality_id value uniquely identifying the console locality. This is valid only, if the return value of this function is true.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

This function returns *true* if a console locality_id exists and returns *false* otherwise.

`bool get_localities(std::vector<naming::gid_type> &locality_ids, components::component_type type, error_code &ec = throws) const`

Query for the locality_ids of all known localities.

This function returns the locality_ids of all localities known to the AGAS server or all localities having a registered factory for a given component type.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

Parameters

- **locality_ids** – [out] The vector will contain the prefixes of all localities registered with the AGAS server. The returned vector holds the prefixes representing the runtime_support components of these localities.

- **type** – [in] The component type will be used to determine the set of prefixes having a registered factory for this component. The default value for this parameter is *components::component_enum_type::invalid*, which will return prefixes of all localities.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

```
inline bool get_localities(std::vector<naming::gid_type> &locality_ids, error_code &ec =  
                           throws) const
```

```
hpx::future<std::uint32_t> get_num_localities_async(components::component_type type =  
                                                    to_int(hpx::components::component_enum_type::invalid))  
                                                    const
```

Query for the number of all known localities.

This function returns the number of localities known to the AGAS server or the number of localities having a registered factory for a given component type.

Parameters

type – [in] The component type will be used to determine the set of prefixes having a registered factory for this component. The default value for this parameter is *components::component_type::invalid*, which will return prefixes of all localities.

Returns

A future that becomes ready with the number of matching localities.

```
std::uint32_t get_num_localities(components::component_type type, error_code &ec = throws)  
                                const
```

Query for the number of all known localities.

This function returns the number of localities known to the AGAS server or the number of localities having a registered factory for a given component type.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

Parameters

- **type** – [in] The component type will be used to determine the set of prefixes having a registered factory for this component. The default value for this parameter is *components::component_type::invalid*, which will return prefixes of all localities.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

```
inline std::uint32_t get_num_localities(error_code &ec = throws) const
```

Query for the number of all known localities.

This function returns the number of localities known to the AGAS server or the number of localities having a registered factory for a given component type.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

Parameters

ec – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

```
hpx::future<std::uint32_t> get_num_overall_threads_async() const
```

```

std::uint32_t get_num_overall_threads(error_code &ec = throws) const

hpx::future<std::vector<std::uint32_t>> get_num_threads_async() const

std::vector<std::uint32_t> get_num_threads(error_code &ec = throws) const

components::component_type get_component_id(std::string const &name, error_code &ec =
                                             throws) const

```

Return a unique id usable as a component type.

This function returns the component type id associated with the given component name. If this is the first request for this component name a new unique id will be created.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Parameters

- **name** – [in] The component name (string) to get the component type for.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

The function returns the currently associated component type. Any error results in an exception thrown from this function.

```

void iterate_types(iterate_types_function_type const &f, error_code &ec = throws) const

std::string get_component_type_name(components::component_type id, error_code &ec =
                                     throws) const

inline components::component_type register_factory(naming::gid_type const &locality_id,
                                                    std::string const &name, error_code &ec
                                                    = throws) const

```

Register a factory for a specific component type.

This function allows to register a component factory for a given locality and component type.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Parameters

- **locality_id** – [in] The locality value uniquely identifying the given locality the factory needs to be registered for.
- **name** – [in] The component name (string) to register a factory for the given component type for.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

The function returns the currently associated component type. Any error results in an exception thrown from this function. The returned component type is the same as if the function *get_component_id* was called using the same component name.

```

components::component_type register_factory(std::uint32_t locality_id, std::string const
                                             &name, error_code &ec = throws) const

```

```
bool get_id_range(std::uint64_t count, naming::gid_type &lower_bound, naming::gid_type  
                  &upper_bound, error_code &ec = throws)
```

Get unique range of freely assignable global ids.

Every locality needs to be able to assign global ids to different components without having to consult the AGAS server for every id to generate. This function can be called to preallocate a range of ids usable for this purpose.

Note: This function assigns a range of global ids usable by the given locality for newly created components. Any of the returned global ids still has to be bound to a local address, either by calling *bind* or *bind_range*.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Parameters

- **count** – [in] The number of global ids to be generated.
- **lower_bound** – [out] The lower bound of the assigned id range. The returned value can be used as the first id to assign. This is valid only, if the return value of this function is true.
- **upper_bound** – [out] The upper bound of the assigned id range. The returned value can be used as the last id to assign. This is valid only, if the return value of this function is true.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

This function returns *true* if a new range has been generated (it has been called for the first time for the given locality) and returns *false* if this locality already got a range assigned in an earlier call. Any error results in an exception thrown from this function.

```
inline bool bind_local(naming::gid_type const &id, naming::address const &addr, error_code  
                       &ec = throws)
```

Bind a global address to a local address.

Every element in the HPX namespace has a unique global address (global id). This global address has to be associated with a concrete local address to be able to address an instance of a component using its global address.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Note: Binding a gid to a local address sets its global reference count to one.

Parameters

- **id** – [in] The global address which has to be bound to the local address.
- **addr** – [in] The local address to be bound to the global address.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

This function returns *true*, if this global id got associated with an local address. It returns

false otherwise.

```
inline hpx::future<bool> bind_async(naming::gid_type const &id, naming::address const &addr,
                                   std::uint32_t locality_id)
```

```
inline hpx::future<bool> bind_async(naming::gid_type const &id, naming::address const &addr,
                                   naming::gid_type const &locality)
```

```
bool bind_range_local(naming::gid_type const &lower_id, std::uint64_t count, naming::address
                      const &baseaddr, std::uint64_t offset, error_code &ec = throws)
```

Bind unique range of global ids to given base address.

Every locality needs to be able to bind global ids to different components without having to consult the AGAS server for every id to bind. This function can be called to bind a range of consecutive global ids to a range of consecutive local addresses (separated by a given *offset*).

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Note: Binding a gid to a local address sets its global reference count to one.

Parameters

- **lower_id** – [in] The lower bound of the assigned id range. The value can be used as the first id to assign.
- **count** – [in] The number of consecutive global ids to bind starting at *lower_id*.
- **baseaddr** – [in] The local address to bind to the global id given by *lower_id*. This is the base address for all additional local addresses to bind to the remaining global ids.
- **offset** – [in] The offset to use to calculate the local addresses to be bound to the range of global ids.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

This function returns *true*, if the given range was successfully bound. It returns *false* otherwise.

```
hpx::future<bool> bind_range_async(naming::gid_type const &lower_id, std::uint64_t count,
                                   naming::address const &baseaddr, std::uint64_t offset,
                                   naming::gid_type const &locality)
```

```
inline hpx::future<bool> bind_range_async(naming::gid_type const &lower_id, std::uint64_t
                                         count, naming::address const &baseaddr,
                                         std::uint64_t offset, std::uint32_t locality_id)
```

```
inline bool unbind_local(naming::gid_type const &id, error_code &ec = throws)
```

Unbind a global address.

Remove the association of the given global address with any local address, which was bound to this global address. Additionally it returns the local address which was bound at the time of this call.

Note: You can unbind only global ids bound using the function *bind*. Do not use this function to unbind any of the global ids bound using *bind_range*.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Note: This function will raise an error if the global reference count of the given gid is not zero!

Parameters

- **id** – [in] The global address (id) for which the association has to be removed.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

The function returns *true* if the association has been removed, and it returns *false* if no association existed. Any error results in an exception thrown from this function.

```
inline bool unbind_local(naming::gid_type const &id, naming::address &addr, error_code &ec =  
                           throws)
```

Unbind a global address.

Remove the association of the given global address with any local address, which was bound to this global address. Additionally it returns the local address which was bound at the time of this call.

Note: You can unbind only global ids bound using the function *bind*. Do not use this function to unbind any of the global ids bound using *bind_range*.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Note: This function will raise an error if the global reference count of the given gid is not zero!

Parameters

- **id** – [in] The global address (id) for which the association has to be removed.
- **addr** – [out] The local address which was associated with the given global address (id). This is valid only if the return value of this function is true.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

The function returns *true* if the association has been removed, and it returns *false* if no association existed. Any error results in an exception thrown from this function.

```
inline bool unbind_range_local(naming::gid_type const &lower_id, std::uint64_t count,  
                                error_code &ec = throws)
```

Unbind the given range of global ids.

Note: You can unbind only global ids bound using the function *bind_range*. Do not use this function to unbind any of the global ids bound using *bind*.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Note: This function will raise an error if the global reference count of the given gid is not zero!

Parameters

- **lower_id** – [in] The lower bound of the assigned id range. The value must be the first id of the range as specified to the corresponding call to *bind_range*.
- **count** – [in] The number of consecutive global ids to unbind starting at *lower_id*. This number must be identical to the number of global ids bound by the corresponding call to *bind_range*.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

This function returns *true* if a new range has been generated (it has been called for the first time for the given locality) and returns *false* if this locality already got a range assigned in an earlier call. Any error results in an exception thrown from this function.

```
bool unbind_range_local(naming::gid_type const &lower_id, std::uint64_t count,
                        naming::address &addr, error_code &ec = throws)
```

Unbind the given range of global ids.

Note: You can unbind only global ids bound using the function *bind_range*. Do not use this function to unbind any of the global ids bound using *bind*.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Note: This function will raise an error if the global reference count of the given gid is not zero!

Parameters

- **lower_id** – [in] The lower bound of the assigned id range. The value must be the first id of the range as specified to the corresponding call to *bind_range*.
- **count** – [in] The number of consecutive global ids to unbind starting at *lower_id*. This number must be identical to the number of global ids bound by the corresponding call to *bind_range*.
- **addr** – [out] The local address which was associated with the given global address (id). This is valid only if the return value of this function is true.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

This function returns *true* if a new range has been generated (it has been called for the first time for the given locality) and returns *false* if this locality already got a range assigned in an earlier call.

```
hpx::future<naming::address> unbind_range_async(naming::gid_type const &lower_id,
                                                std::uint64_t count = 1)
```

```
inline bool is_local_address_cached(naming::gid_type const &id, error_code &ec = throws)
```

Test whether the given address refers to a local object.

This function will test whether the given address refers to an object living on the locality of the caller.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Parameters

- **id** – [in] The address to test.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

This function returns *true* if the passed address refers to an object which lives on the locality of the caller.

```
bool is_local_address_cached(naming::gid_type const &id, naming::address &addr,  
                             error_code &ec = throws)
```

```
bool is_local_address_cached(naming::gid_type const &id, naming::address &addr,  
                             std::pair<bool, components::pinned_ptr> &r,  
                             hpx::move_only_function<std::pair<bool,  
                             components::pinned_ptr>(naming::address const&)> &&f,  
                             error_code &ec = throws)
```

```
bool is_local_lva_encoded_address(std::uint64_t msb) const
```

```
inline bool resolve_local(naming::gid_type const &id, naming::address &addr, error_code &ec  
                          = throws)
```

Resolve a given global address (*id*) to its associated local address.

This function returns the local address which is currently associated with the given global address (*id*).

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Parameters

- **id** – [in] The global address (*id*) for which the associated local address should be returned.
- **addr** – [out] The local address which currently is associated with the given global address (*id*), this is valid only if the return value of this function is true.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

This function returns *true* if the global address has been resolved successfully (there exists an association to a local address) and the associated local address has been returned. The function returns *false* if no association exists for the given global address. Any error results in an exception thrown from this function.

```
inline bool resolve_local(hpx::id_type const &id, naming::address &addr, error_code &ec =  
                          throws)
```

```

inline naming::address resolve_local(naming::gid_type const &id, error_code &ec = throws)

inline naming::address resolve_local(hpx::id_type const &id, error_code &ec = throws)

hpx::future_or_value<naming::address> resolve_async(naming::gid_type const &id)

inline hpx::future_or_value<naming::address> resolve_async(hpx::id_type const &id)

hpx::future_or_value<id_type> get_colocation_id_async(hpx::id_type const &id)

bool resolve_full_local(naming::gid_type const &id, naming::address &addr, error_code &ec
                        = throws)

inline bool resolve_full_local(hpx::id_type const &id, naming::address &addr, error_code &ec
                                = throws)

inline naming::address resolve_full_local(naming::gid_type const &id, error_code &ec =
                                           throws)

inline naming::address resolve_full_local(hpx::id_type const &id, error_code &ec = throws)

hpx::future_or_value<naming::address> resolve_full_async(naming::gid_type const &id)

inline hpx::future_or_value<naming::address> resolve_full_async(hpx::id_type const &id)

bool resolve_cached(naming::gid_type const &id, naming::address &addr, error_code &ec =
                    throws)

inline bool resolve_cached(hpx::id_type const &id, naming::address &addr, error_code &ec =
                            throws)

inline bool resolve_local(naming::gid_type const *gids, naming::address *addrs, std::size_t size,
                           hpx::detail::dynamic_bitset<> &locals, error_code &ec = throws)

bool resolve_full_local(naming::gid_type const *gids, naming::address *addrs, std::size_t size,
                        hpx::detail::dynamic_bitset<> &locals, error_code &ec = throws)

bool resolve_cached(naming::gid_type const *gids, naming::address *addrs, std::size_t size,
                    hpx::detail::dynamic_bitset<> &locals, error_code &ec = throws)

hpx::future_or_value<std::int64_t> incref_async(naming::gid_type const &gid, std::int64_t
                                              credits = 1, hpx::id_type const &keep_alive =
                                              hpx::invalid_id)

```

Increment the global reference count for the given id.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Parameters

- **gid** – [in] The global address (id) for which the global reference count has to be incremented.
- **credits** – [in] The number of reference counts to add for the given id.
- **keep_alive** – [in] Id to keep alive (if valid)

Returns

Whether the operation was successful.

```
inline std::int64_t incref(naming::gid_type const &gid, std::int64_t credits = 1, error_code &ec =  
    throws)
```

```
void decref(naming::gid_type const &id, std::int64_t credits = 1, error_code &ec = throws)
```

Decrement the global reference count for the given id.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Parameters

- **id** – [in] The global address (id) for which the global reference count has to be decremented.
- **credits** – [in] The number of reference counts to add for the given id.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

The global reference count after the decrement.

```
hpx::future<iterate_names_return_type> iterate_ids(std::string const &pattern) const
```

Invoke the supplied *hpx::function* for every registered global name.

This function iterates over all registered global ids and returns every found entry matching the given name pattern. Any error results in an exception thrown (or reported) from this function.

Parameters

pattern – [in] pattern (possibly using wildcards) to match all existing entries against

```
bool register_name(std::string const &name, naming::gid_type const &id, error_code &ec =  
    throws) const
```

Register a global name with a global address (id)

This function registers an association between a global name (string) and a global address (id) usable with one of the functions above (bind, unbind, and resolve).

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Parameters

- **name** – [in] The global name (string) to be associated with the global address.
- **id** – [in] The global address (id) to be associated with the global address.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

The function returns *true* if the global name was registered. It returns false if the global name is not registered.

```
hpx::future<bool> register_name_async(std::string const &name, hpx::id_type const &id) const
```

```
bool register_name(std::string const &name, hpx::id_type const &id, error_code &ec = throws)  
    const
```

```
hpx::future<hpx::id_type> unregister_name_async(std::string const &name) const
```

Unregister a global name (release any existing association).

This function releases any existing association of the given global name with a global address (id).

Parameters

name – [in] The global name (string) for which any association with a global address (id) has to be released.

Returns

A future that becomes ready with the global id that was associated with the given name.

`hpx::id_type unregister_name(std::string const &name, error_code &ec = throws) const`

Unregister a global name (release any existing association)

This function releases any existing association of the given global name with a global address (id).

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Parameters

- **name** – [in] The global name (string) for which any association with a global address (id) has to be released.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

The global id that was associated with the given name. Any error results in an exception thrown from this function.

`hpx::future<hpx::id_type> resolve_name_async(std::string const &name) const`

Query for the global address associated with a given global name.

This function returns the global address associated with the given global name.

Parameters

name – [in] The global name (string) for which the currently associated global address has to be retrieved.

Returns

A future holding the global id currently associated with the given global name.

`hpx::id_type resolve_name(std::string const &name, error_code &ec = throws) const`

Query for the global address associated with a given global name.

This function returns the global address associated with the given global name.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Parameters

- **name** – [in] The global name (string) for which the currently associated global address has to be retrieved.
- **ec** – [in,out] This represents the error status on exit. If this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

The global id currently associated with the given global name.

`future<hpx::id_type> on_symbol_namespace_event(std::string const &name, bool
call_for_past_events = false) const`

Install a listener for a given symbol namespace event.

This function installs a listener for a given symbol namespace event. It returns a future which becomes ready as a result of the listener being triggered.

Note: The only event type which is currently supported is *symbol_ns_bind*, i.e. the listener is triggered whenever a global id is registered with the given name.

Parameters

- **name** – [in] The global name (string) for which the given event should be triggered.
- **call_for_past_events** – [in, optional] Trigger the listener even if the given event has already happened in the past. The default for this parameter is *false*.

Returns

A future instance encapsulating the global id which is causing the registered listener to be triggered.

```
void update_cache_entry(naming::gid_type const &gid, gva const &gva, error_code &ec =  
                        throws)
```

Warning: This function is for internal use only. It is dangerous and may break your code if you use it.

```
inline void update_cache_entry(naming::gid_type const &gid, naming::address const &addr,  
                               std::uint64_t count = 0, std::uint64_t offset = 0, error_code &ec  
                               = throws)
```

Warning: This function is for internal use only. It is dangerous and may break your code if you use it.

```
bool get_cache_entry(naming::gid_type const &gid, gva &gva, naming::gid_type &idbase,  
                    error_code &ec = throws) const
```

Warning: This function is for internal use only. It is dangerous and may break your code if you use it.

```
void remove_cache_entry(naming::gid_type const &id, error_code &ec = throws) const
```

Warning: This function is for internal use only. It is dangerous and may break your code if you use it.

```
void clear_cache(error_code &ec = throws) const
```

Warning: This function is for internal use only. It is dangerous and may break your code if you use it.

```
void start_shutdown(error_code &ec = throws)
```

```
hpx::future<std::pair<hpx::id_type, naming::address>> begin_migration(hpx::id_type const &id)  
start/stop migration of an object
```


Returns

Current locality and address of the object to migrate

bool **end_migration**(*hpx::id_type* const &id)

std::pair<bool, *components::pinned_ptr*> **was_object_migrated**(*naming::gid_type* const &gid,
hpx::move_only_function<*components::pinned_ptr*
&&f>)

Maintain list of migrated objects.

hpx::future<void> **mark_as_migrated**(*naming::gid_type* const &gid,
hpx::move_only_function<*std::pair*<bool,
hpx::future<void>>()> &&f, bool
expect_to_be_marked_as_migrating)

Mark the given object as being migrated (if the object is unpinned). Delay migration until the object is unpinned otherwise.

void **unmark_as_migrated**(*naming::gid_type* const &gid_, *hpx::move_only_function*<void()>
&&f)

Remove the given object from the table of migrated objects.

void **pre_cache_endpoints**(*std::vector*<*parcelset::endpoints_type*> const&)

Public Members

mutable *hpx::shared_mutex* **gva_cache_mtx_**

std::shared_ptr<*gva_cache_type*> **gva_cache_**

mutable *mutex_type* **migrated_objects_mtx_** =
mutex_type("addressing_service::migrated_objects_mtx")

migrated_objects_table_type **migrated_objects_table_**

mutable *mutex_type* **console_cache_mtx_** =
mutex_type("addressing_service::console_cache_mtx")

std::uint32_t **console_cache_**

const *std::size_t* **max_refcnt_requests_**

mutex_type **refcnt_requests_mtx_** = *mutex_type*("addressing_service::refcnt_requests_mtx")

std::size_t **refcnt_requests_count_**

bool **enable_refcnt_caching_**

std::shared_ptr<*refcnt_requests_type*> **refcnt_requests_**

```
const service_mode service_type

const runtime_mode runtime_type

const bool caching_

const bool range_caching_

const threads::thread_priority action_priority_

std::uint64_t rts_lva_

std::unique_ptr<component_namespace> component_ns_

std::unique_ptr<locality_namespace> locality_ns_

symbol_namespace symbol_ns_

primary_namespace primary_ns_

std::atomic<hpx::state> state_

naming::gid_type locality_

mutable hpx::shared_mutex resolved_localities_mtx_

resolved_localities_type resolved_localities_
```

Public Static Functions

```
static std::int64_t synchronize_with_async_incref(std::int64_t old_credit, hpx::id_type const &id, std::int64_t compensated_credit)
```

Protected Functions

```
void launch_bootstrap(parcelset::endpoints_type const &endpoints, util::runtime_configuration &rtcfg)

naming::address resolve_full_postproc(naming::gid_type const &id,
                                         primary_namespace::resolved_type const&)

bool bind_postproc(naming::gid_type const &id, gva const &g, future<bool> f)

bool was_object_migrated_locked(naming::gid_type const &id)
    Maintain list of migrated objects.
```

Private Functions

void **send_refcnt_requests**(std::unique_lock<mutex_type> &l, error_code &ec = throws)

Assumes that *refcnt_requests_mtx_* is locked.

void **send_refcnt_requests_non_blocking**(std::unique_lock<mutex_type> &l, error_code &ec)

Assumes that *refcnt_requests_mtx_* is locked.

std::vector<hpx::future<std::vector<std::int64_t>>> **send_refcnt_requests_async**(std::unique_lock<mutex_type> &l)

Assumes that *refcnt_requests_mtx_* is locked.

void **send_refcnt_requests_sync**(std::unique_lock<mutex_type> &l, error_code &ec)

Assumes that *refcnt_requests_mtx_* is locked.

agas_base

See *Public API* for a list of names and headers that are part of the public HPX API.

hpx/agas_base/server/primary_namespace.hpp

Defined in header `hpx/agas_base/server/primary_namespace.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

Variables

HPX_ACTION_USES_MEDIUM_STACK(hpx::agas::server::primary_namespace::allocate_action) **HPX_REGISTER_ACTION**
hpx::naming::address > std_pair_address_id_type

namespace **hpx**

namespace **agas**

Functions

naming::gid_type **bootstrap_primary_namespace_gid**()

hpx::id_type **bootstrap_primary_namespace_id**()

namespace **server**

AGAS's primary namespace maps 128-bit global identifiers (GIDs) to resolved addresses.

The following is the canonical description of the partitioning of AGAS's primary namespace.

-----MSB----- -----LSB-----	
BBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBB	
prefix RC ----identifier----	
MSB	- Most significant bits (bit 64 to bit 127)
LSB	- Least significant bits (bit 0 to bit 63)
prefix	- Highest 32 bits (bit 96 to bit 127) of the MSB. Each locality is assigned a prefix. This creates a 96-bit address space for each locality.
RC	- Bit 88 to bit 92 of the MSB. This is the log2 of the number of reference counting credits on the GID. Bit 93 is used by the locking scheme for gid_types. Bit 94 is a flag which is set if the credit value is valid. Bit 95 is a flag that is set if a GID's credit count is ever split (e.g. if the GID is ever passed to another locality).
identifier	- Bit 64 to bit 86 of the MSB, and the entire LSB. The content of these bits depends on the component type of the underlying object. For all user-defined components, these bits contain a unique 88-bit number which is assigned sequentially for each locality. For <code>\a hpx#components#component_enum_type#runtime_support</code> the high 24 bits are zeroed and the low 64 bits hold the LVA of the component.

The following address ranges are reserved. Some are either explicitly or implicitly protected by AGAS. The letter x represents a single-byte wild card.

```

00000000xxxxxxxxxxxxxxxxxxxxxxxxxxxx
    Historically unused address space reserved for future use.
xxxxxxxxxxxx0000xxxxxxxxxxxxxxxxxxxx
    Address space for LVA-encoded GIDs.
00000001xxxxxxxxxxxxxxxxxxxxxxxxxxxx
    Prefix of the bootstrap AGAS locality.
000000010000000010000000000000001
    Address of the primary_namespace component on the bootstrap AGAS
    locality.
000000010000000010000000000000002
    Address of the component_namespace component on the bootstrap AGAS
    locality.
000000010000000010000000000000003
    Address of the symbol_namespace component on the bootstrap AGAS
    locality.
000000010000000010000000000000004
    Address of the locality_namespace component on the bootstrap AGAS
    locality.

```

Note: The layout of the address space is implementation defined, and subject to change. Never write application code that relies on the internal layout of GIDs. AGAS only guarantees that all assigned

GIDs will be unique.

Variables

```
constexpr char const *const primary_namespace_service_name = "primary/"
```

```
struct primary_namespace : public components::fixed_component_base<primary_namespace>
```

Public Types

```
using mutex_type = hpx::spinlock
```

```
using base_type = components::fixed_component_base<primary_namespace>
```

```
using component_type = std::int32_t
```

```
using gva_table_data_type = std::pair<gva, naming::gid_type>
```

```
using gva_table_type = std::map<naming::gid_type, gva_table_data_type>
```

```
using refcnt_table_type = std::map<naming::gid_type, std::int64_t>
```

```
using resolved_type = hpx::tuple<naming::gid_type, gva, naming::gid_type>
```

Public Functions

```
inline mutex_type &mutex()
```

```
void wait_for_migration_locked(std::unique_lock<mutex_type> &l, naming::gid_type  
                             const &id, error_code &ec)
```

```
inline primary_namespace()
```

```
void finalize() const
```

```
inline void set_local_locality(naming::gid_type const &g)
```

```
void register_server_instance(char const *servicename, std::uint32_t locality_id =  
                             naming::invalid_locality_id, error_code &ec = throws)
```

```
void unregister_server_instance(error_code &ec = throws) const
```

```
bool bind_gid(gva const &g, naming::gid_type id, naming::gid_type const &locality)
```

```
std::pair<hpx::id_type, naming::address> begin_migration(naming::gid_type id)
```

```
bool end_migration(naming::gid_type const &id)

resolved_type resolve_gid(naming::gid_type const &id)

hpx::id_type colocate(naming::gid_type const &id)

naming::address unbind_gid(std::uint64_t count, naming::gid_type id)

std::int64_t increment_credit(std::int64_t credits, naming::gid_type lower,
                               naming::gid_type upper)

std::vector<std::int64_t> decrement_credit(std::vector<hpx::tuple<std::int64_t,
                               naming::gid_type, naming::gid_type>> const
                               &requests)

std::pair<naming::gid_type, naming::gid_type> allocate(std::uint64_t count)

resolved_type resolve_gid_locked(std::unique_lock<mutex_type> &l, naming::gid_type
                                const &gid, error_code &ec)
```

Public Members

```
counter_data counter_data_
```

Private Types

```
using migration_table_type = std::map<naming::gid_type, hpx::tuple<bool, std::size_t,
lcos::local::detail::condition_variable>>
```

```
using free_entry_allocator_type = util::internal_allocator<free_entry>
```

```
using free_entry_list_type = std::list<free_entry, free_entry_allocator_type>
```

Private Functions

```
resolved_type resolve_gid_locked_non_local(std::unique_lock<mutex_type> &l,
                                           naming::gid_type const &gid, error_code
                                           &ec)
```

```
void increment(naming::gid_type const &lower, naming::gid_type const &upper, std::int64_t
               const &credits, error_code &ec)
```

```
void resolve_free_list(std::unique_lock<mutex_type> &l,
                       std::list<refcnt_table_type::iterator> const &free_list,
                       free_entry_list_type &free_entry_list, naming::gid_type const
                       &lower, naming::gid_type const &upper, error_code &ec)
```

```
void decrement_sweep(free_entry_list_type &free_list, naming::gid_type const &lower,
                      naming::gid_type const &upper, std::int64_t credits, error_code &ec)
```

```
void free_components_sync(free_entry_list_type const &free_list, naming::gid_type const
                        &lower, naming::gid_type const &upper, error_code &ec)
    const
```

Private Members

```
mutex_type mutex_ = mutex_type("primary_namespace")
```

```
gva_table_type gvas_
```

```
refcnt_table_type refcnts_
```

```
std::string instance_name_
```

```
naming::gid_type next_id_
```

```
naming::gid_type locality_
```

```
migration_table_type migrating_objects_
```

```
struct counter_data
```

Public Functions

```
counter_data(counter_data const&) = delete
```

```
counter_data(counter_data&&) = delete
```

```
counter_data &operator=(counter_data const&) = delete
```

```
counter_data &operator=(counter_data&&) = delete
```

```
counter_data() = default
```

```
std::int64_t get_bind_gid_count(bool)
```

```
std::int64_t get_resolve_gid_count(bool)
```

```
std::int64_t get_unbind_gid_count(bool)
```

```
std::int64_t get_increment_credit_count(bool)
```

```
std::int64_t get_decrement_credit_count(bool)
```

```
std::int64_t get_allocate_count(bool)
```

```
std::int64_t get_begin_migration_count(bool)
```

```
std::int64_t get_end_migration_count(bool)
```

```
std::int64_t get_overall_count(bool)
std::int64_t get_bind_gid_time(bool)
std::int64_t get_resolve_gid_time(bool)
std::int64_t get_unbind_gid_time(bool)
std::int64_t get_increment_credit_time(bool)
std::int64_t get_decrement_credit_time(bool)
std::int64_t get_allocate_time(bool)
std::int64_t get_begin_migration_time(bool)
std::int64_t get_end_migration_time(bool)
std::int64_t get_overall_time(bool)
void increment_bind_gid_count()
void increment_resolve_gid_count()
void increment_unbind_gid_count()
void increment_increment_credit_count()
void increment_decrement_credit_count()
void increment_allocate_count()
void increment_begin_migration_count()
void increment_end_migration_count()
void enable_all()
```

Public Members

```
api_counter_data bind_gid_
api_counter_data resolve_gid_
api_counter_data unbind_gid_
api_counter_data increment_credit_
api_counter_data decrement_credit_
api_counter_data allocate_
api_counter_data begin_migration_
```



```
api_counter_data end_migration_
```

```
struct api_counter_data
```

Public Members

```
std::atomic<std::int64_t> count_ = 0
```

```
std::atomic<std::int64_t> time_ = 0
```

```
bool enabled_ = false
```

```
struct free_entry
```

Public Functions

```
inline free_entry(agas::gva const &gva, naming::gid_type const &gid, naming::gid_type
                  const &loc)
```

Public Members

```
agas::gva gva_
```

```
naming::gid_type gid_
```

```
naming::gid_type locality_
```

async_colocated

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx::get_colocation_id

Defined in header `hpx/runtime.hpp`⁵⁵⁸.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

⁵⁵⁸ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

hpx::id_type **hpx::get_colocation_id**(*launch::sync_policy*, *hpx::id_type* const &id, *error_code* &ec = *throws*)

Return the id of the locality where the object referenced by the given id is currently located on.

The function *hpx::get_colocation_id()* returns the id of the locality where the given object is currently located.

See also:

hpx::get_colocation_id()

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Parameters

- **id** – [in] The id of the object to locate.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

hpx::future<hpx::id_type> **hpx::get_colocation_id**(*hpx::id_type* const &id)

Asynchronously return the id of the locality where the object referenced by the given id is currently located on.

See also:

hpx::get_colocation_id(launch::sync_policy)

Parameters

- **id** – [in] The id of the object to locate.

async_distributed

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/async_distributed/promise.hpp

Defined in header *hpx/async_distributed/promise.hpp*.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

template<>

```
class promise<void, hpx::util::unused_type> : public lcos::detail::promise_base<void, hpx::util::unused_type,  
lcos::detail::promise_data<void>>
```

Public Functions

promise() = default

constructs a promise object and a shared state.

template<typename **Allocator**>

inline **promise**(*std::allocator_arg_t*, *Allocator* const &a)

constructs a promise object and a shared state. The constructor uses the allocator *to* allocate the memory for the shared state.

promise(*promise* &&other) noexcept = default

constructs a new promise object and transfers ownership of the shared state of other (if any) to the newly-constructed object.

Post

other has no shared state.

~promise() = default

Abandons any shared state.

promise &**operator**=(*promise* &&other) noexcept = default

Abandons any shared state (30.6.4) and then as if `promise(HPX_MOVE(other)).swap(*this)`.

Returns

*this.

inline void **swap**(*promise* &other) noexcept

Exchanges the shared state of *this and other.

Post

*this has the shared state (if any) that other had prior to the call to swap. other has the shared state (if any) that *this had prior to the call to swap.

inline void **set_value**()

atomically stores the value *r* in the shared state and makes that state ready (30.6.4).

Throws

future_error – if its shared state already has a stored value. if shared state has no stored value exception is raised. **promise_already_satisfied** if its shared state already has a stored value or exception. **no_state** if *this has no shared state.

Private Types

using **base_type** = *lcos::detail::promise_base*<void, *hpx::util::unused_type*,
lcos::detail::promise_data<void>>

template<typename **R**, typename **Allocator**>

struct **uses_allocator**<*hpx::distributed::promise*<*R*>, *Allocator*> : public **true_type**

#include <*promise.hpp*> Requires: Allocator shall be an allocator (17.6.3.5)

namespace **hpx**

namespace **distributed**

Functions

```
template<typename Result, typename RemoteResult>
void swap(promise<Result, RemoteResult> &x, promise<Result, RemoteResult> &y) noexcept
```

```
template<typename Result, typename RemoteResult>
```

```
class promise : public lcos::detail::promise_base<Result, RemoteResult,
lcos::detail::promise_data<Result>>
```

#include <promise.hpp> A promise can be used by a single *thread* to invoke a (remote) action and wait for the result. The result is expected to be sent back to the promise using the LCO's `set_event` action

A promise is one of the simplest synchronization primitives provided by HPX. It allows to synchronize on a eager evaluated remote operation returning a result of the type *Result*. The *promise* allows to synchronize exactly one *thread* (the one passed during construction time).

```
// Create the promise (the expected result is a id_type)
hpx::distributed::promise<hpx::id_type> p;

// Get the associated future
future<hpx::id_type> f = p.get_future();

// initiate the action supplying the promise as a
// continuation
apply<some_action>(new continuation(p.get_id()), ...);

// Wait for the result to be returned, yielding control
// in the meantime.
hpx::id_type result = f.get();
// ...
```

Note: The action executed by the promise must return a value of a type convertible to the type as specified by the template parameter *RemoteResult*

Template Parameters

- **Result** – The template parameter *Result* defines the type this promise is expected to return from *promise::get*.
- **RemoteResult** – The template parameter *RemoteResult* defines the type this promise is expected to receive from the remote action.

Public Functions

promise() = default

constructs a promise object and a shared state.

```
template<typename Allocator>
```

```
inline promise(std::allocator_arg_t, Allocator const &a)
```

constructs a promise object and a shared state. The constructor uses the allocator *to* allocate the memory for the shared state.

promise(*promise* &&other) noexcept = default

constructs a new promise object and transfers ownership of the shared state of other (if any) to the newly- constructed object.

Post

other has no shared state.

~promise() = default

Abandons any shared state.

promise &**operator**=(*promise* &&other) noexcept = default

Abandons any shared state (30.6.4) and then as if promise(HPX_MOVE(other)).swap(*this).

Returns

*this.

inline void **swap**(*promise* &other) noexcept

Exchanges the shared state of *this and other.

Post

*this has the shared state (if any) that other had prior to the call to swap. other has the shared state (if any) that *this had prior to the call to swap.

Private Types

```
using base_type = lcos::detail::promise_base<Result, RemoteResult,
lcos::detail::promise_data<Result>>
```

```
template<> unused_type > : public lcos::detail::promise_base< void,
hpx::util::unused_type, lcos::detail::promise_data< void > >
```

Public Functions

promise() = default

constructs a promise object and a shared state.

template<typename **Allocator**>

inline **promise**(std::allocator_arg_t, *Allocator* const &a)

constructs a promise object and a shared state. The constructor uses the allocator *to* allocate the memory for the shared state.

promise(*promise* &&other) noexcept = default

constructs a new promise object and transfers ownership of the shared state of other (if any) to the newly- constructed object.

Post

other has no shared state.

~promise() = default

Abandons any shared state.

promise &**operator**=(*promise* &&other) noexcept = default

Abandons any shared state (30.6.4) and then as if promise(HPX_MOVE(other)).swap(*this).

Returns

*this.

inline void **swap**(*promise* &other) noexcept

Exchanges the shared state of *this and other.

Post

*this has the shared state (if any) that other had prior to the call to swap. other has the shared state (if any) that *this had prior to the call to swap.

inline void **set_value**()

atomically stores the value r in the shared state and makes that state ready (30.6.4).

Throws

future_error – if its shared state already has a stored value. if shared state has no stored value exception is raised. **promise_already_satisfied** if its shared state already has a stored value or exception. **no_state** if *this has no shared state.

Private Types

```
using base_type = lcos::detail::promise_base<void, hpx::util::unused_type,  
lcos::detail::promise_data<void>>
```

namespace **std**

```
template<typename R, typename Allocator> promise< R >,  
Allocator > : public true_type  
#include <promise.hpp> Requires: Allocator shall be an allocator (17.6.3.5)
```

hpx/async_distributed/trigger_lco_fwd.hpp

Defined in header hpx/async_distributed/trigger_lco_fwd.hpp.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

Functions

void **trigger_lco_event**(*hpx::id_type* id, *naming::address* &&addr, bool move_credits = true)

Trigger the LCO referenced by the given id.

Parameters

- **id** – [in] This represents the id of the LCO which should be triggered.
- **addr** – [in] This represents the addr of the LCO which should be triggered.
- **move_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

inline void **trigger_lco_event**(*hpx::id_type* const &id, bool move_credits = true)

Trigger the LCO referenced by the given id.

Parameters

- **id** – [in] This represents the id of the LCO which should be triggered.

- **move_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

void **trigger_lco_event**(*hpx::id_type* id, *naming::address* &&addr, *hpx::id_type* const &cont, bool move_credits = true)

Trigger the LCO referenced by the given id.

Parameters

- **id** – [in] This represents the id of the LCO which should be triggered.
- **addr** – [in] This represents the addr of the LCO which should be triggered.
- **cont** – [in] This represents the LCO to trigger after completion.
- **move_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

inline void **trigger_lco_event**(*hpx::id_type* const &id, *hpx::id_type* const &cont, bool move_credits = true)

Trigger the LCO referenced by the given id.

Parameters

- **id** – [in] This represents the id of the LCO which should be triggered.
- **cont** – [in] This represents the LCO to trigger after completion.
- **move_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

template<typename **Result**>

void **set_lco_value**(*hpx::id_type* id, *naming::address* &&addr, *Result* &&t, bool move_credits = true)

Set the result value for the LCO referenced by the given id.

Parameters

- **id** – [in] This represents the id of the LCO which should receive the given value.
- **addr** – [in] This represents the addr of the LCO which should be triggered.
- **t** – [in] This is the value which should be sent to the LCO.
- **move_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

template<typename **Result**>

void **set_lco_value**(*hpx::id_type* const &id, *Result* &&t, bool move_credits = true)

Set the result value for the (managed) LCO referenced by the given id.

Parameters

- **id** – [in] This represents the id of the LCO which should receive the given value.
- **t** – [in] This is the value which should be sent to the LCO.
- **move_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

template<typename **Result**>

void **set_lco_value_unmanaged**(*hpx::id_type* const &id, *Result* &&t, bool move_credits = true)

Set the result value for the (unmanaged) LCO referenced by the given id.

Parameters

- **id** – [in] This represents the id of the LCO which should receive the given value.
- **t** – [in] This is the value which should be sent to the LCO.
- **move_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

```
template<typename Result>  
void set_lco_value(hpx::id_type id, naming::address &&addr, Result &&t, hpx::id_type const &cont, bool  
    move_credits = true)
```

Set the result value for the LCO referenced by the given id.

Parameters

- **id** – [in] This represents the id of the LCO which should receive the given value.
- **addr** – [in] This represents the addr of the LCO which should be triggered.
- **t** – [in] This is the value which should be sent to the LCO.
- **cont** – [in] This represents the LCO to trigger after completion.
- **move_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

```
template<typename Result>  
void set_lco_value(hpx::id_type const &id, Result &&t, hpx::id_type const &cont, bool move_credits =  
    true)
```

Set the result value for the (managed) LCO referenced by the given id.

Parameters

- **id** – [in] This represents the id of the LCO which should receive the given value.
- **t** – [in] This is the value which should be sent to the LCO.
- **cont** – [in] This represents the LCO to trigger after completion.
- **move_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

```
template<typename Result>  
void set_lco_value_unmanaged(hpx::id_type const &id, Result &&t, hpx::id_type const &cont, bool  
    move_credits = true)
```

Set the result value for the (unmanaged) LCO referenced by the given id.

Parameters

- **id** – [in] This represents the id of the LCO which should receive the given value.
- **t** – [in] This is the value which should be sent to the LCO.
- **cont** – [in] This represents the LCO to trigger after completion.
- **move_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

```
void set_lco_error(hpx::id_type id, naming::address &&addr, std::exception_ptr const &e, bool  
    move_credits = true)
```

Set the error state for the LCO referenced by the given id.

Parameters

- **id** – [in] This represents the id of the LCO which should receive the error value.

- **addr** – [in] This represents the addr of the LCO which should be triggered.
- **e** – [in] This is the error value which should be sent to the LCO.
- **move_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

void **set_lco_error**(*hpx::id_type* id, *naming::address* &&addr, *std::exception_ptr* &&e, bool move_credits = true)

Set the error state for the LCO referenced by the given id.

Parameters

- **id** – [in] This represents the id of the LCO which should receive the error value.
- **addr** – [in] This represents the addr of the LCO which should be triggered.
- **e** – [in] This is the error value which should be sent to the LCO.
- **move_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

inline void **set_lco_error**(*hpx::id_type* const &id, *std::exception_ptr* const &e, bool move_credits = true)

Set the error state for the LCO referenced by the given id.

Parameters

- **id** – [in] This represents the id of the LCO which should receive the error value.
- **e** – [in] This is the error value which should be sent to the LCO.
- **move_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

inline void **set_lco_error**(*hpx::id_type* const &id, *std::exception_ptr* &&e, bool move_credits = true)

Set the error state for the LCO referenced by the given id.

Parameters

- **id** – [in] This represents the id of the LCO which should receive the error value.
- **e** – [in] This is the error value which should be sent to the LCO.
- **move_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

void **set_lco_error**(*hpx::id_type* id, *naming::address* &&addr, *std::exception_ptr* const &e, *hpx::id_type* const &cont, bool move_credits = true)

Set the error state for the LCO referenced by the given id.

Parameters

- **id** – [in] This represents the id of the LCO which should receive the error value.
- **addr** – [in] This represents the addr of the LCO which should be triggered.
- **e** – [in] This is the error value which should be sent to the LCO.
- **cont** – [in] This represents the LCO to trigger after completion.
- **move_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

```
void set_lco_error(hpx::id_type id, naming::address &&addr, std::exception_ptr &&e, hpx::id_type const &cont, bool move_credits = true)
```

Set the error state for the LCO referenced by the given id.

Parameters

- **id** – [in] This represents the id of the LCO which should receive the error value.
- **addr** – [in] This represents the addr of the LCO which should be triggered.
- **e** – [in] This is the error value which should be sent to the LCO.
- **cont** – [in] This represents the LCO to trigger after completion.
- **move_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

```
inline void set_lco_error(hpx::id_type const &id, std::exception_ptr const &e, hpx::id_type const &cont, bool move_credits = true)
```

Set the error state for the LCO referenced by the given id.

Parameters

- **id** – [in] This represents the id of the LCO which should receive the error value.
- **e** – [in] This is the error value which should be sent to the LCO.
- **cont** – [in] This represents the LCO to trigger after completion.
- **move_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

```
inline void set_lco_error(hpx::id_type const &id, std::exception_ptr &&e, hpx::id_type const &cont, bool move_credits = true)
```

Set the error state for the LCO referenced by the given id.

Parameters

- **id** – [in] This represents the id of the LCO which should receive the error value.
- **e** – [in] This is the error value which should be sent to the LCO.
- **cont** – [in] This represents the LCO to trigger after completion.
- **move_credits** – [in] If this is set to *true* then it is ok to send all credits in *id* along with the generated message. The default value is *true*.

hpx::post (distributed)

Defined in header `hpx/async.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Warning: doxygenfunction: Unable to resolve function “hpx::post” with arguments (distributed) in doxygen xml output for project “hpx” from directory: `../hpx_autodoc`. Potential matches:

```
- template<typename Action, typename ...Ts> bool post(hpx::id_type const &id, Ts&&...  
→vs)  
- template<typename Action, typename Client, typename Stub, typename Data, typename ..  
→.Ts> bool post(components::client_base<Client, Stub, Data> const &c, Ts&&... vs)  
- template<typename Action, typename Continuation, typename ...Ts> bool_
```

```

↪ post(Continuation &&c, hpx::id_type const &gid, Ts&&... vs)
- template<typename Action, typename Continuation, typename Client, typename Stub,
↪ typename Data, typename ...Ts> bool post(Continuation &&cont, components::client_
↪ base<Client, Stub, Data> const &c, Ts&&... vs)
- template<typename Action, typename Continuation, typename DistPolicy, typename ...
↪ Ts> bool post(Continuation &&c, DistPolicy const &policy, Ts&&... vs)
- template<typename Action, typename DistPolicy, typename ...Ts> bool post(DistPolicy
↪ const &policy, Ts&&... vs)
- template<typename Action, typename Target, typename ...Ts> bool post(Action &&
↪ action, Target &&target, Ts&&... ts)
- template<typename F, typename ...Ts> bool post(F &&f, Ts&&... ts)

```

hpx/async_distributed/trigger_lco.hpp

Defined in header hpx/async_distributed/trigger_lco.hpp.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

hpx::sync (distributed)

Defined in header hpx/async.hpp.

See *Public API* for a list of names and headers that are part of the public HPX API.

Warning: doxygenfunction: Unable to resolve function “hpx::sync” with arguments (distributed) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:

```

- template<typename Action, typename F, typename ...Ts> auto sync(F &&f, Ts&&... ts) -
↪ decltype(detail::sync_action_dispatch<Action, std::decay_t<F>>::call(HPX_
↪ FORWARD(F, f), HPX_FORWARD(Ts, ts)...))
- template<typename Action, typename Target, typename ...Ts> decltype(auto)
↪ sync(Action &&action, Target &&target, Ts&&... ts)
- template<typename F, typename ...Ts> decltype(auto) sync(F &&f, Ts&&... ts)

```

hpx/async_distributed/base_lco.hpp

Defined in header hpx/async_distributed/base_lco.hpp.

See *Public API* for a list of names and headers that are part of the public HPX API.

template<>

struct **get_lva**<hpx::lcos::base_lco>

Public Static Functions

```
static inline constexpr hpx::lcos::base_lco *call(hpx::naming::address_type lva) noexcept  
template<>  
struct get_lva<hpx::lcos::base_lco const>
```

Public Static Functions

```
static inline constexpr hpx::lcos::base_lco const *call(hpx::naming::address_type lva) noexcept  
  
namespace hpx
```

```
template<> base_lco >
```

Public Static Functions

```
static inline constexpr hpx::lcos::base_lco *call(hpx::naming::address_type lva) noexcept  
  
template<> base_lco const >
```

Public Static Functions

```
static inline constexpr hpx::lcos::base_lco const *call(hpx::naming::address_type lva) noexcept  
  
namespace lcos
```

```
class base_lco
```

#include <base_lco.hpp> The *base_lco* class is the common base class for all LCO's implementing a simple *set_event* action

Subclassed by *hpx::lcos::base_lco_with_value< Result, RemoteResult, ComponentTag >*,
hpx::lcos::base_lco_with_value< void, void, ComponentTag >

Public Types

```
typedef components::managed_component<base_lco> wrapping_type
```

```
typedef base_lco base_type_holder
```

Public Functions

virtual void **set_event**() = 0

virtual void **set_exception**(std::exception_ptr const &e)

virtual void **connect**(hpx::id_type const&)

virtual void **disconnect**(hpx::id_type const&)

virtual **~base_lco**()

Destructor, needs to be virtual to allow for clean destruction of derived objects

void **set_event_nonvirt**()

The *function* `set_event_nonvirt` is called whenever a *set_event_action* is applied on a instance of a LCO. This function just forwards to the virtual function *set_event*, which is overloaded by the derived concrete LCO.

void **set_exception_nonvirt**(std::exception_ptr const &e)

The *function* `set_exception` is called whenever a *set_exception_action* is applied on a instance of a LCO. This function just forwards to the virtual function *set_exception*, which is overloaded by the derived concrete LCO.

Parameters

e – [in] The exception encapsulating the error to report to this LCO instance.

void **connect_nonvirt**(hpx::id_type const &id)

The *function* `connect_nonvirt` is called whenever a *connect_action* is applied on a instance of a LCO. This function just forwards to the virtual function *connect*, which is overloaded by the derived concrete LCO.

Parameters

id – [in] target id

void **disconnect_nonvirt**(hpx::id_type const &id)

The *function* `disconnect_nonvirt` is called whenever a *disconnect_action* is applied on a instance of a LCO. This function just forwards to the virtual function *disconnect*, which is overloaded by the derived concrete LCO.

Parameters

id – [in] target id

HPX_DEFINE_COMPONENT_DIRECT_ACTION (base_lco, set_event_nonvirt, set_event_action) HPX_DEFINE_COMPONENT_DIRECT_ACTION(base_lco

Each of the exposed functions needs to be encapsulated into an action type, allowing to generate all required boilerplate code for threads, serialization, etc.

The *set_event_action* may be used to unconditionally trigger any LCO instances, it carries no additional parameters. The *set_exception_action* may be used to transfer arbitrary error information from the remote site to the LCO instance specified as a continuation. This action carries 2 parameters:

Parameters

std::exception_ptr – [in] The exception encapsulating the error to report to this LCO instance.

set_exception_action HPX_DEFINE_COMPONENT_DIRECT_ACTION (base_lco, connect_nonvirt, connect_action) HPX_DEFINE_COMPONENT_DIRECT_ACTION(base_lco

The *connect_action* may be used to.

The `set_exception_action` may be used to

Public Members

`set_exception_nonvirt`

`set_exception_action` `disconnect_nonvirt`

Public Static Functions

static `components::component_type` `get_component_type`() noexcept

static void `set_component_type`(`components::component_type` type)

hpx::dataflow (distributed)

Defined in header `hpx/async.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Warning: doxygenfunction: Unable to resolve function “hpx::dataflow” with arguments (distributed) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:

```
- template<typename Action, typename Target, typename ...Ts> decltype(auto) ↵  
↵dataflow(Action &&action, Target &&target, Ts&&... ts)  
- template<typename F, typename ...Ts> decltype(auto) dataflow(F &&f, Ts&&... ts)
```

hpx/async_distributed/packaged_action.hpp

Defined in header `hpx/async_distributed/packaged_action.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **lcos**

template<typename **Action**, typename **Result**, bool **DirectExecute**>

class **packaged_action**

#include <packaged_action.hpp> A *packaged_action* can be used by a single *thread* to invoke a (re-
mote) action and wait for the result. The result is expected to be sent back to the *packaged_action*
using the LCO's `set_event` action

A *packaged_action* is one of the simplest synchronization primitives provided by HPX. It allows to
synchronize on an eager evaluated remote operation returning a result of the type *Result*.

Note: The action executed using the *packaged_action* as a continuation must return a value of a type convertible to the type as specified by the template parameter *Result*.

Template Parameters

- **Action** – The template parameter *Action* defines the action to be executed by this *packaged_action* instance. The arguments *arg0*,... *argN* are used as parameters for this action.
- **Result** – The template parameter *Result* defines the type this *packaged_action* is expected to return from its associated future *packaged_action::get_future*.
- **DirectExecute** – The template parameter *DirectExecute* is an optimization aid allowing to execute the action directly if the target is local (without spawning a new thread for this). This template does not have to be supplied explicitly as it is derived from the template parameter *Action*.

```
template<typename Action, typename Result>
```

```
class packaged_action<Action, Result, false> : public hpx::distributed::promise<Result,  
hpx::traits::extract_action<Action>::remote_result_type>
```

```
    Subclassed by hpx::lcos::packaged_action< Action, Result, true >
```

Public Functions

```
inline packaged_action()
```

```
template<typename Allocator>
```

```
inline packaged_action(std::allocator_arg_t, Allocator const &alloc)
```

```
template<typename ...Ts>
```

```
inline void post(hpx::id_type const &id, Ts&&... vs)
```

```
template<typename ...Ts>
```

```
inline void post(naming::address &&addr, hpx::id_type const &id, Ts&&... vs)
```

```
template<typename Callback, typename ...Ts>
```

```
inline void post_cb(hpx::id_type const &id, Callback &&cb, Ts&&... vs)
```

```
template<typename Callback, typename ...Ts>
```

```
inline void post_cb(naming::address &&addr, hpx::id_type const &id, Callback &&cb, Ts&&...  
vs)
```

```
template<typename ...Ts>
```

```
inline void post_p(hpx::id_type const &id, hpx::launch policy, Ts&&... vs)
```

```
template<typename ...Ts>
```

```
inline void post_p(naming::address &&addr, hpx::id_type const &id, hpx::launch policy, Ts&&...  
vs)
```

```
template<typename Callback, typename ...Ts>
```

```
inline void post_p_cb(hpx::id_type const &id, hpx::launch policy, Callback &&cb, Ts&&... vs)
```

```
template<typename Callback, typename ...Ts>
```

```
inline void post_p_cb(naming::address &&addr, hpx::id_type const &id, hpx::launch policy,  
Callback &&cb, Ts&&... vs)
```

```
template<typename ...Ts>
```

```
inline void post_deferred(naming::address &&addr, hpx::id_type const &id, hpx::launch policy,
                          Ts&&... vs)
```

```
template<typename Callback, typename ...Ts>
inline void post_deferred_cb(naming::address &&addr, hpx::id_type const &id, hpx::launch
                             policy, Callback &&cb, Ts&&... vs)
```

Protected Types

```
using action_type = hpx::traits::extract_action<Action>::type
```

```
using remote_result_type = action_type::remote_result_type
```

```
using base_type = hpx::distributed::promise<Result, remote_result_type>
```

Protected Functions

```
template<typename ...Ts>
inline void do_post(naming::address &&addr, hpx::id_type const &id, hpx::launch policy, Ts&&...
                   vs)
```

```
template<typename ...Ts>
inline void do_post(hpx::id_type const &id, hpx::launch policy, Ts&&... vs)
```

```
template<typename Callback, typename ...Ts>
inline void do_post_cb(naming::address &&addr, hpx::id_type const &id, hpx::launch policy,
                       Callback &&cb, Ts&&... vs)
```

```
template<typename Callback, typename ...Ts>
inline void do_post_cb(hpx::id_type const &id, hpx::launch policy, Callback &&cb, Ts&&... vs)
```

```
template<typename Action, typename Result>
```

```
class packaged_action<Action, Result, true> : public hpx::lcos::packaged_action<Action, Result,
false>
```

Public Functions

```
inline packaged_action()
```

Construct a (non-functional) instance of a `packaged_action`. To use this instance its member function `post` needs to be directly called.

```
template<typename Allocator>
inline packaged_action(std::allocator_arg_t, Allocator const &alloc)
```

```
template<typename ...Ts>
inline void post(hpx::id_type const &id, Ts&&... vs)
```

```
template<typename ...Ts>
inline void post(naming::address &&addr, hpx::id_type const &id, Ts&&... vs)
```

```
template<typename Callback, typename ...Ts>
```



```

inline void post_cb(hpx::id_type const &id, Callback &&cb, Ts&&... vs)

template<typename Callback, typename ...Ts>
inline void post_cb(naming::address &&addr, hpx::id_type const &id, Callback &&cb, Ts&&...
vs)

```

Private Types

```
using action_type = packaged_action<Action, Result, false>::action_type
```

hpx/async_distributed/base_lco_with_value.hpp

Defined in header `hpx/async_distributed/base_lco_with_value.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
namespace components
```

```
namespace lcos
```

```
template<typename Result, typename RemoteResult, typename ComponentTag>
```

```
class base_lco_with_value : public hpx::lcos::base_lco, public ComponentTag
```

#include <*base_lco_with_value.hpp*> The `base_lco_with_value` class is the common base class for all LCO's synchronizing on a value. The *RemoteResult* template argument should be set to the type of the argument expected for the `set_value` action.

Template Parameters

- **RemoteResult** – The type of the result value to be carried back to the LCO instance.
- **ComponentTag** – The tag type representing the type of the component (either `component_tag` or `managed_component_tag`).

Public Types

```
using wrapping_type = detail::base_lco_wrapping_type<ComponentTag,
base_lco_with_value>::type
```

```
using base_type_holder = base_lco_with_value
```

Public Functions

inline void **set_value_nonvirt**(*RemoteResult* &&result)

The *function* `set_value_nonvirt` is called whenever a *set_value_action* is applied on this LCO instance. This function just forwards to the virtual function *set_value*, which is overloaded by the derived concrete LCO.

Parameters

result – [in] The result value to be transferred from the remote operation back to this LCO instance.

inline *Result* **get_value_nonvirt**()

The *function* `get_result_nonvirt` is called whenever a *get_result_action* is applied on this LCO instance. This function just forwards to the virtual function *get_result*, which is overloaded by the derived concrete LCO.

HPX_DEFINE_COMPONENT_DIRECT_ACTION (*base_lco_with_value*, *set_value_nonvirt*, *set_value_action*) **HPX_DEFINE_COMPONENT_DIRECT_ACTION**(*base_lco_with_value*

The *set_value_action* may be used to trigger any LCO instances while carrying an additional parameter of any type.

RemoteResult is taken by rvalue ref. This allows for perfect forwarding. When the action thread function is created, the values are moved into the called function. If we took it by const lvalue reference, we would disable the possibility to further move the result to the designated destination.

Parameters

RemoteResult – [in] The type of the result to be transferred back to this LCO instance.

The *get_value_action* may be used to query the value this LCO instance exposes as its ‘result’ value.

Public Members

get_value_nonvirt

Public Static Functions

static inline *components::component_type* **get_component_type**() noexcept

static inline void **set_component_type**(*components::component_type* type)

Protected Types

using **result_type** = *std::conditional_t*<*std::is_void_v*<*Result*>, *util::unused_type*, *Result*>

Protected Functions

~base_lco_with_value() override = default

Destructor, needs to be virtual to allow for clean destruction of derived objects

inline virtual void **set_event()** override

virtual void **set_value**(*RemoteResult* &&result) = 0

virtual *result_type* **get_value()** = 0

inline virtual *result_type* **get_value**(*error_code*&)

template<typename **ComponentTag**>

class **base_lco_with_value**<void, void, *ComponentTag*> : public *hpx::lcos::base_lco*, public *ComponentTag*

#include <base_lco_with_value.hpp> The *base_lco*<void> specialization is used whenever the *set_event* action for a particular LCO doesn't carry any argument.

Template Parameters

void – This specialization expects no result value and is almost completely equivalent to the plain *base_lco*.

Public Types

using **wrapping_type** = detail::base_lco_wrapping_type<*ComponentTag*, *base_lco_with_value*>::type

using **base_type_holder** = *base_lco_with_value*

using **set_value_action** = *base_lco*::set_event_action

Public Functions

inline void **get_value()**

Protected Functions

~base_lco_with_value() override = default

Destructor, needs to be virtual to allow for clean destruction of derived objects

namespace **traits**

hpx/async_distributed/transfer_continuation_action.hpp

Defined in header `hpx/async_distributed/transfer_continuation_action.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx::distributed::promise

Defined in header `hpx/future.hpp`⁵⁵⁹.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

template<typename **Result**, typename **RemoteResult**>

class **promise**

A promise can be used by a single *thread* to invoke a (remote) action and wait for the result. The result is expected to be sent back to the promise using the LCO's `set_event` action

A promise is one of the simplest synchronization primitives provided by HPX. It allows to synchronize on a eager evaluated remote operation returning a result of the type *Result*. The *promise* allows to synchronize exactly one *thread* (the one passed during construction time).

```
// Create the promise (the expected result is a id_type)
hpx::distributed::promise<hpx::id_type> p;

// Get the associated future
future<hpx::id_type> f = p.get_future();

// initiate the action supplying the promise as a
// continuation
apply<some_action>(new continuation(p.get_id()), ...);

// Wait for the result to be returned, yielding control
// in the meantime.
hpx::id_type result = f.get();
// ...
```

Note: The action executed by the promise must return a value of a type convertible to the type as specified by the template parameter *RemoteResult*

Template Parameters

- **Result** – The template parameter *Result* defines the type this promise is expected to return from *promise::get*.
- **RemoteResult** – The template parameter *RemoteResult* defines the type this promise is expected to receive from the remote action.

⁵⁵⁹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

hpx::async (distributed)

Defined in header `hpx/async.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Warning: doxygenfunction: Unable to resolve function “hpx::async” with arguments (distributed) in doxygen xml output for project “hpx” from directory: `../hpx_autodoc`. Potential matches:

```
- template<typename Action, typename F, typename ...Ts> auto async(F &&f, Ts&&... ts)
  ↳-> decltype(detail::async_action_dispatch<Action, std::decay_t<F>>::call(HPX_
  ↳FORWARD(F, f), HPX_FORWARD(Ts, ts)...))
- template<typename Action, typename Target, typename ...Ts> auto async(Action &&
  ↳action, Target &&target, Ts&&... ts)
- template<typename Executor, typename ...Ts> decltype(auto) async(Executor &&exec,
  ↳hpx::sycl::experimental::sycl_executor::queue_function_ptr_t<Ts...> &&f, Ts&&... ts)
- template<typename F, typename ...Ts> decltype(auto) async(F &&f, Ts&&... ts)
```

checkpoint

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/checkpoint/checkpoint.hpp

Defined in header `hpx/checkpoint/checkpoint.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

This header defines the `save_checkpoint` and `restore_checkpoint` functions. These functions are designed to help HPX application developer’s checkpoint their applications. `Save_checkpoint` serializes one or more objects and saves them as a byte stream. `Restore_checkpoint` converts the byte stream back into instances of the objects.

namespace **hpx**

namespace **util**

Functions

`std::ostream &operator<<(std::ostream &ost, checkpoint const &ckp)`

Operator<< Overload

This overload is the main way to write data from a checkpoint to an object such as a file. Inside the function, the size of the checkpoint will be written to the stream before the checkpoint’s data. The `operator>>` overload uses this to read the correct number of bytes. Be mindful of this additional write and read when you use different facilities to write out or read in data to a checkpoint!

Parameters

- **ost** – Output stream to write to.
- **ckp** – Checkpoint to copy from.

Returns

Operator<< returns the ostream object.

```
std::istream &operator>>(std::istream &ist, checkpoint &ckp)
```

Operator>> Overload

This overload is the main way to read in data from an object such as a file to a checkpoint. It is important to note that inside the function, the first variable to be read is the size of the checkpoint. This size variable is written to the stream before the checkpoint's data in the operator<< overload. Be mindful of this additional read and write when you use different facilities to read in or write out data from a checkpoint!

Parameters

- **ist** – Input stream to write from.
- **ckp** – Checkpoint to write to.

Returns

Operator>> returns the ostream object.

```
template<typename T, typename ...Ts>  
hpx::future<checkpoint> save_checkpoint(T &&t, Ts&&... ts)
```

Save_checkpoint

Save_checkpoint takes any number of objects which a user may wish to store and returns a future to a checkpoint object. This function can also store a component either by passing a shared_ptr to the component or by passing a component's client instance to save_checkpoint. Additionally, the function can take a policy as a first object which changes its behavior depending on the policy passed to it. Most notably, if a sync policy is used save_checkpoint will simply return a checkpoint object.

Template Parameters

- **T** – Containers passed to save_checkpoint to be serialized and placed into a checkpoint object.
- **Ts** – More containers passed to save_checkpoint to be serialized and placed into a checkpoint object.
- **U** – This parameter is used to make sure that T is not a launch policy or a checkpoint. This forces the compiler to choose the correct overload.

Parameters

- **t** – A container to restore.
- **ts** – Other containers to restore Containers must be in the same order that they were inserted into the checkpoint.

Returns

Save_checkpoint returns a future to a checkpoint with one exception: if you pass hpx::launch::sync as the first argument. In this case save_checkpoint will simply return a checkpoint.

```
template<typename T, typename ...Ts>  
hpx::future<checkpoint> save_checkpoint(checkpoint &&c, T &&t, Ts&&... ts)
```

Save_checkpoint - Take a pre-initialized checkpoint

Save_checkpoint takes any number of objects which a user may wish to store and returns a future to a checkpoint object. This function can also store a component either by passing a shared_ptr to the component or by passing a component's client instance to save_checkpoint. Additionally, the function can take a policy as a first object which changes its behavior depending on the policy passed to it. Most notably, if a sync policy is used save_checkpoint will simply return a checkpoint object.

Template Parameters

- **T** – Containers passed to `save_checkpoint` to be serialized and placed into a checkpoint object.
- **Ts** – More containers passed to `save_checkpoint` to be serialized and placed into a checkpoint object.

Parameters

- **c** – Takes a pre-initialized checkpoint to copy data into.
- **t** – A container to restore.
- **ts** – Other containers to restore Containers must be in the same order that they were inserted into the checkpoint.

Returns

`Save_checkpoint` returns a future to a checkpoint with one exception: if you pass `hpx::launch::sync` as the first argument. In this case `save_checkpoint` will simply return a checkpoint.

```
template<typename T, typename ...Ts>
hpx::future<checkpoint> save_checkpoint(hpx::launch p, T &&t, Ts&&... ts)
Save_checkpoint - Policy overload
```

`Save_checkpoint` takes any number of objects which a user may wish to store and returns a future to a checkpoint object. This function can also store a component either by passing a `shared_ptr` to the component or by passing a component's client instance to `save_checkpoint`. Additionally, the function can take a policy as a first object which changes its behavior depending on the policy passed to it. Most notably, if a sync policy is used `save_checkpoint` will simply return a checkpoint object.

Template Parameters

- **T** – Containers passed to `save_checkpoint` to be serialized and placed into a checkpoint object.
- **Ts** – More containers passed to `save_checkpoint` to be serialized and placed into a checkpoint object.

Parameters

- **p** – Takes an HPX launch policy. Allows the user to change the way the function is launched i.e. `async`, `sync`, etc.
- **t** – A container to restore.
- **ts** – Other containers to restore Containers must be in the same order that they were inserted into the checkpoint.

Returns

`Save_checkpoint` returns a future to a checkpoint with one exception: if you pass `hpx::launch::sync` as the first argument. In this case `save_checkpoint` will simply return a checkpoint.

```
template<typename T, typename ...Ts>
hpx::future<checkpoint> save_checkpoint(hpx::launch p, checkpoint &&c, T &&t, Ts&&... ts)
Save_checkpoint - Policy overload & pre-initialized checkpoint
```

`Save_checkpoint` takes any number of objects which a user may wish to store and returns a future to a checkpoint object. This function can also store a component either by passing a `shared_ptr` to the component or by passing a component's client instance to `save_checkpoint`. Additionally, the function can take a policy as a first object which changes its behavior depending on the policy passed to it. Most notably, if a sync policy is used `save_checkpoint` will simply return a checkpoint object.

Template Parameters

- **T** – Containers passed to `save_checkpoint` to be serialized and placed into a checkpoint object.
- **Ts** – More containers passed to `save_checkpoint` to be serialized and placed into a checkpoint object.

Parameters

- **p** – Takes an HPX launch policy. Allows the user to change the way the function is launched i.e. async, sync, etc.
- **c** – Takes a pre-initialized checkpoint to copy data into.
- **t** – A container to restore.
- **ts** – Other containers to restore Containers must be in the same order that they were inserted into the checkpoint.

Returns

Save_checkpoint returns a future to a checkpoint with one exception: if you pass `hpx::launch::sync` as the first argument. In this case `save_checkpoint` will simply return a checkpoint.

```
template<typename T, typename ...Ts>
```

```
checkpoint save_checkpoint(hpx::launch::sync_policy sync_p, T &&t, Ts&&... ts)
```

Save_checkpoint - Sync_policy overload

Save_checkpoint takes any number of objects which a user may wish to store and returns a future to a checkpoint object. This function can also store a component either by passing a `shared_ptr` to the component or by passing a component's client instance to `save_checkpoint`. Additionally, the function can take a policy as a first object which changes its behavior depending on the policy passed to it. Most notably, if a sync policy is used `save_checkpoint` will simply return a checkpoint object.

Template Parameters

- **T** – Containers passed to `save_checkpoint` to be serialized and placed into a checkpoint object.
- **Ts** – More containers passed to `save_checkpoint` to be serialized and placed into a checkpoint object.
- **U** – This parameter is used to make sure that **T** is not a checkpoint. This forces the compiler to choose the correct overload.

Parameters

- **sync_p** – `hpx::launch::sync_policy`
- **t** – A container to restore.
- **ts** – Other containers to restore Containers must be in the same order that they were inserted into the checkpoint.

Returns

Save_checkpoint which is passed `hpx::launch::sync_policy` will return a checkpoint which contains the serialized values checkpoint.

```
template<typename T, typename ...Ts>
```

```
checkpoint save_checkpoint(hpx::launch::sync_policy sync_p, checkpoint &&c, T &&t, Ts&&... ts)
```

Save_checkpoint - Sync_policy overload & pre-init. checkpoint

Save_checkpoint takes any number of objects which a user may wish to store and returns a future to a checkpoint object. This function can also store a component either by passing a `shared_ptr` to the component or by passing a component's client instance to `save_checkpoint`. Additionally, the function can take a policy as a first object which changes its behavior depending on the policy passed to it. Most notably, if a sync policy is used `save_checkpoint` will simply return a checkpoint object.

Template Parameters

- **T** – Containers passed to `save_checkpoint` to be serialized and placed into a checkpoint object.
- **Ts** – More containers passed to `save_checkpoint` to be serialized and placed into a checkpoint object.

Parameters

- **sync_p** – `hpx::launch::sync_policy`
- **c** – Takes a pre-initialized checkpoint to copy data into.
- **t** – A container to restore.
- **ts** – Other containers to restore Containers must be in the same order that they were inserted into the checkpoint.

Returns

`Save_checkpoint` which is passed `hpx::launch::sync_policy` will return a checkpoint which contains the serialized values checkpoint.

```
template<typename T, typename ...Ts>
hpx::future<checkpoint> prepare_checkpoint(T const &t, Ts const&... ts)
prepare_checkpoint
```

`prepare_checkpoint` takes the containers which have to be filled from the byte stream by a subsequent `restore_checkpoint` invocation. `prepare_checkpoint` will calculate the necessary buffer size and will return an appropriately sized checkpoint object.

Template Parameters

- **T** – A container to restore.
- **ts** – Other containers to restore. Containers must be in the same order that they were inserted into the checkpoint.

Parameters

- **t** – A container to restore.
- **ts** – Other containers to restore Containers must be in the same order that they were inserted into the checkpoint.

Returns

`prepare_checkpoint` returns a properly resized checkpoint object that can be used for a subsequent `restore_checkpoint` operation.

```
template<typename T, typename ...Ts>
hpx::future<checkpoint> prepare_checkpoint(checkpoint &&c, T const &t, Ts const&... ts)
prepare_checkpoint
```

`prepare_checkpoint` takes the containers which have to be filled from the byte stream by a subsequent `restore_checkpoint` invocation. `prepare_checkpoint` will calculate the necessary buffer size and will return an appropriately sized checkpoint object.

Template Parameters

- **T** – A container to restore.
- **ts** – Other containers to restore. Containers must be in the same order that they were inserted into the checkpoint.

Parameters

- **c** – Takes a pre-initialized checkpoint to prepare
- **t** – A container to restore.
- **ts** – Other containers to restore Containers must be in the same order that they were inserted into the checkpoint.

Returns

`prepare_checkpoint` returns a properly resized checkpoint object that can be used for a subsequent `restore_checkpoint` operation.

```
template<typename T, typename ...Ts>
hpx::future<checkpoint> prepare_checkpoint(hpx::launch p, T const &t, Ts const&... ts)
prepare_checkpoint
```

`prepare_checkpoint` takes the containers which have to be filled from the byte stream by a subsequent `restore_checkpoint` invocation. `prepare_checkpoint` will calculate the necessary buffer size and will return an appropriately sized checkpoint object.

Template Parameters

- **T** – A container to restore.
- **Ts** – Other containers to restore. Containers must be in the same order that they were inserted into the checkpoint.

Parameters

- **p** – Takes an HPX launch policy. Allows the user to change the way the function is launched i.e. async, sync, etc.
- **t** – A container to restore.
- **ts** – Other containers to restore Containers must be in the same order that they were inserted into the checkpoint.

Returns

prepare_checkpoint returns a properly resized checkpoint object that can be used for a subsequent restore_checkpoint operation.

```
template<typename T, typename ...Ts>
hpx::future<checkpoint> prepare_checkpoint(hpx::launch p, checkpoint &&c, T const &t, Ts
                                         const&... ts)
```

prepare_checkpoint

prepare_checkpoint takes the containers which have to be filled from the byte stream by a subsequent restore_checkpoint invocation. prepare_checkpoint will calculate the necessary buffer size and will return an appropriately sized checkpoint object.

Template Parameters

- **T** – A container to restore.
- **Ts** – Other containers to restore. Containers must be in the same order that they were inserted into the checkpoint.

Parameters

- **p** – Takes an HPX launch policy. Allows the user to change the way the function is launched i.e. async, sync, etc.
- **c** – Takes a pre-initialized checkpoint to prepare
- **t** – A container to restore.
- **ts** – Other containers to restore Containers must be in the same order that they were inserted into the checkpoint.

Returns

prepare_checkpoint returns a properly resized checkpoint object that can be used for a subsequent restore_checkpoint operation.

```
template<typename T, typename ...Ts>
void restore_checkpoint(checkpoint const &c, T &t, Ts&... ts)
```

Restore_checkpoint

Restore_checkpoint takes a checkpoint object as a first argument and the containers which will be filled from the byte stream (in the same order as they were placed in save_checkpoint). Restore_checkpoint can resurrect a stored component in two ways: by passing in an instance of a component's shared_ptr or by passing in an instance of the component's client.

Template Parameters

- **T** – A container to restore.
- **Ts** – Other containers to restore. Containers must be in the same order that they were inserted into the checkpoint.

Parameters

- **c** – The checkpoint to restore.
- **t** – A container to restore.
- **ts** – Other containers to restore Containers must be in the same order that they were inserted into the checkpoint.

Returns

Restore_checkpoint returns void.

class **checkpoint**

#include <checkpoint.hpp> Checkpoint Object

Checkpoint is the container object which is produced by `save_checkpoint` and is consumed by a `restore_checkpoint`. A checkpoint may be moved into the `save_checkpoint` object to write the byte stream to the pre-created checkpoint object.

Checkpoints are able to store all containers which are serializable including components.

Public Types

using **const_iterator** = *std::vector<char>::const_iterator*

Public Functions

checkpoint() = default

~checkpoint() = default

checkpoint(*checkpoint* const &c) = default

checkpoint(*checkpoint* &&c) noexcept = default

inline explicit **checkpoint**(*std::vector<char>* const &vec)

inline explicit **checkpoint**(*std::vector<char>* &&vec) noexcept

checkpoint &**operator**=(*checkpoint* const &c) = default

checkpoint &**operator**=(*checkpoint* &&c) noexcept = default

inline *const_iterator* **begin**() const noexcept

inline *const_iterator* **end**() const noexcept

inline *std::size_t* **size**() const noexcept

inline char ***data**() noexcept

inline char const ***data**() const noexcept

Private Functions

template<typename **Archive**>

inline void **serialize**(*Archive* &arch, unsigned int const)

Private Members

`std::vector<char> data_`

Friends

friend class `hpx::serialization::access`

`friend std::ostream &operator<<(std::ostream &ost, checkpoint const &ckp)`

Operator<< Overload

This overload is the main way to write data from a checkpoint to an object such as a file. Inside the function, the size of the checkpoint will be written to the stream before the checkpoint's data. The `operator>>` overload uses this to read the correct number of bytes. Be mindful of this additional write and read when you use different facilities to write out or read in data to a checkpoint!

Parameters

- **ost** – Output stream to write to.
- **ckp** – Checkpoint to copy from.

Returns

Operator<< returns the ostream object.

`friend std::istream &operator>>(std::istream &ist, checkpoint &ckp)`

Operator>> Overload

This overload is the main way to read in data from an object such as a file to a checkpoint. It is important to note that inside the function, the first variable to be read is the size of the checkpoint. This size variable is written to the stream before the checkpoint's data in the `operator<<` overload. Be mindful of this additional read and write when you use different facilities to read in or write out data from a checkpoint!

Parameters

- **ist** – Input stream to write from.
- **ckp** – Checkpoint to write to.

Returns

Operator>> returns the ostream object.

`template<typename T, typename ...Ts>`

`friend void restore_checkpoint(checkpoint const &c, T &t, Ts&... ts)`

`Restore_checkpoint`

`Restore_checkpoint` takes a checkpoint object as a first argument and the containers which will be filled from the byte stream (in the same order as they were placed in `save_checkpoint`). `Restore_checkpoint` can resurrect a stored component in two ways: by passing in an instance of a component's `shared_ptr` or by passing in an instance of the component's client.

Template Parameters

- **T** – A container to restore.
- **Ts** – Other containers to restore. Containers must be in the same order that they were inserted into the checkpoint.

Parameters

- **c** – The checkpoint to restore.
- **t** – A container to restore.

- **ts** – Other containers to restore Containers must be in the same order that they were inserted into the checkpoint.

Returns

Restore_checkpoint returns void.

inline friend bool **operator==(***checkpoint* const &lhs, *checkpoint* const &rhs) noexcept

inline friend bool **operator!=(***checkpoint* const &lhs, *checkpoint* const &rhs) noexcept

collectives

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx::collectives::communicator

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::collectives::create_communicator*
- *hpx::collectives::create_local_communicator*
- *hpx::collectives::create_hierarchical_communicator*
- *hpx::collectives::communicator::set_info*
- *hpx::collectives::communicator::get_info*
- *hpx::collectives::communicator::is_root*

struct **communicator**

A communicator instance represents the list of sites that participate in a particular collective operation.

hpx::collectives::create_communicator

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::collectives::communicator*
- *hpx::collectives::create_local_communicator*
- *hpx::collectives::create_hierarchical_communicator*
- *hpx::collectives::communicator::set_info*
- *hpx::collectives::communicator::get_info*
- *hpx::collectives::communicator::is_root*

```
communicator hpx::collectives::create_communicator(char const *basename, num_sites_arg num_sites =
                                                    num_sites_arg(), this_site_arg this_site =
                                                    this_site_arg(), generation_arg generation =
                                                    generation_arg(), root_site_arg root_site =
                                                    root_site_arg())
```

Create a new communicator object usable with any collective operation

This functions creates a new communicator object that can be called in order to pre-allocate a communicator object usable with multiple invocations of the collective operations (such as *all_gather*, *all_reduce*, *all_to_all*, *broadcast*, etc.).

Parameters

- **basename** – The base name identifying the collective operation
- **num_sites** – The number of participating sites (default: all localities).
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.
- **generation** – The generational counter identifying the sequence number of the collective operation performed on the given base name. This is optional and needs to be supplied only if the collective operation on the given base name has to be performed more than once.
- **root_site** – The site that is responsible for creating the collective support object. This value is optional and defaults to '0' (zero).

Throws

hpx::exception – with *hpx::error::bad_parameter* if *basename* is null or empty, if the (resolved) *num_sites* is zero, if the (resolved) *this_site* is not smaller than *num_sites*, or if the (resolved) *root_site* does not designate a participating site.

Returns

This function returns a new communicator object usable with the collective operation.

```
communicator hpx::collectives::create_communicator(hpx::launch::sync_policy policy, char const
                                                    *basename, num_sites_arg num_sites, this_site_arg
                                                    this_site, generation_arg generation, root_site_arg
                                                    root_site)
```

hpx::collectives::create_local_communicator

Defined in header *hpx/collectives.hpp*.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::collectives::communicator*
- *hpx::collectives::create_communicator*
- *hpx::collectives::create_hierarchical_communicator*
- *hpx::collectives::communicator::set_info*
- *hpx::collectives::communicator::get_info*
- *hpx::collectives::communicator::is_root*

```
communicator hpx::collectives::create_local_communicator(char const *basename, num_sites_arg
                                                         num_sites, this_site_arg this_site,
                                                         generation_arg generation =
                                                         generation_arg(), root_site_arg root_site =
                                                         root_site_arg())
```

Create a new communicator object usable with any local collective operation

This functions creates a new communicator object that can be called in order to pre-allocate a communicator object usable with multiple invocations of the collective operations (such as *all_gather*, *all_reduce*, *all_to_all*, *broadcast*, etc.).

Parameters

- **basename** – The base name identifying the collective operation
- **num_sites** – The number of participating sites
- **this_site** – The sequence number of this invocation (usually the sequence number of the object participating in the collective operation). This value must be in the range [0, num_sites).
- **generation** – The generational counter identifying the sequence number of the collective operation performed on the given base name. This is optional and needs to be supplied only if the collective operation on the given base name has to be performed more than once.
- **root_site** – The site that is responsible for creating the collective support object. This value is optional and defaults to '0' (zero).

Throws

hpx::exception – with *hpx::error::bad_parameter* if *basename* is null or empty, if *num_sites* is unspecified or zero, if *this_site* is unspecified or not smaller than *num_sites*, or if the (resolved) *root_site* does not designate a participating site.

Returns

This function returns a new communicator object usable for all local collective operations.

```
communicator hpx::collectives::create_local_communicator(hpx::launch::sync_policy policy, char const
                                                         *basename, num_sites_arg num_sites,
                                                         this_site_arg this_site, generation_arg
                                                         generation, root_site_arg root_site)
```

hpx::collectives::create_hierarchical_communicator

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::collectives::communicator*
- *hpx::collectives::create_communicator*
- *hpx::collectives::create_local_communicator*
- *hpx::collectives::communicator::set_info*
- *hpx::collectives::communicator::get_info*
- *hpx::collectives::communicator::is_root*

```
hierarchical_communicator hpx::collectives::create_hierarchical_communicator(char const
                                                                    *basename,
                                                                    num_sites_arg
                                                                    num_sites =
                                                                    num_sites_arg(),
                                                                    this_site_arg this_site
                                                                    = this_site_arg(),
                                                                    arity_arg arity =
                                                                    arity_arg(),
                                                                    generation_arg
                                                                    generation =
                                                                    generation_arg(),
                                                                    root_site_arg
                                                                    root_site =
                                                                    root_site_arg(),
                                                                    flat_fallback_threshold_arg
                                                                    threshold =
                                                                    flat_fallback_threshold_arg())
```

Create a new hierarchical communicator object usable with collective operations

This functions creates a new `hierarchical_communicator` object that can be called in order to pre-allocate a communicator object usable with multiple invocations of a collective operation (such as *all_gather*, *all_reduce*, *all_to_all*, *broadcast*, etc.).

Note: The sub-communicators of a `hierarchical_communicator` share one generation sequence per registered name. Every hierarchical collective advances each sub-communicator it touches by exactly two internal generations per call (a single-pass collective such as *broadcast*, *gather*, *scatter* or *reduce*, and the inter-group exchange of *all_to_all*, skip the second generation in one step rather than performing a second round-trip). A single hierarchical communicator instance may therefore be shared freely across collective operations, provided every call uses an explicit, strictly consecutive generation number so the shared sequence stays gap-free. Because the two-phase hierarchical collectives (*all_gather*, *all_reduce*, *all_to_all*, *inclusive_scan*, *exclusive_scan* and the hierarchical *barrier*) derive their phase generations from that explicit number, they require it and reject an auto (default) generation. The non-hierarchical (flat) collectives keep supporting the default generation, falling back to the internal per-communicator counter maintained in the *and_gate*. Once any operation on a communicator has used that default, later operations on the same instance may no longer pass an explicit generation number: the counter's position is no longer reliably known to the caller. The reverse transition, from explicit numbering back to the default, remains valid. All participants of a single operation must use the same generation mode.

Note: Hierarchical collective overloads that return *hpx::future* may still perform internal waits while walking the communicator tree. They should not be treated as fully non-blocking: validation failures are reported through the returned exceptional future, but internal phase hand-offs can throw or block before that future is delivered.

Parameters

- **basename** – The base name identifying the collective operation
- **num_sites** – The number of participating sites (default: all localities).
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.
- **arity** – The arity of the hierarchical tree of communicators to create for the given endpoint. The default arity is 2 and the value must be at least 2.

- **generation** – The generational counter identifying the sequence number of the collective operation performed on the given base name. This is optional and needs to be supplied only if the collective operation on the given base name has to be performed more than once.
- **root_site** – The site that is responsible for creating the collective support object. Hierarchical communicators currently support only site 0 as the root, which is also the default.
- **threshold** – The site-count threshold below which the communicator collapses to a single flat communicator spanning all sites (strict comparison: `num_sites < threshold`). The default is 16; pass 0 to disable the fallback and always build a tree.

Throws

`hpx::exception` – with `hpx::error::bad_parameter` if *basename* is null or empty, if the (resolved) *num_sites* is zero, if the (resolved) *this_site* is not smaller than *num_sites*, if the (resolved) *root_site* is not zero, or if *arity* is smaller than 2.

Returns

This function returns a new communicator object usable with the collective operation.

hpx::collectives::communicator::set_info

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::collectives::communicator`
- `hpx::collectives::create_communicator`
- `hpx::collectives::create_local_communicator`
- `hpx::collectives::create_hierarchical_communicator`
- `hpx::collectives::communicator::get_info`
- `hpx::collectives::communicator::is_root`

void `hpx::collectives::communicator::set_info`(*num_sites_arg* num_sites, *this_site_arg* this_site)
noexcept

Store the number of used sites and the index of the current site for this communicator instance.

Parameters

- **num_sites** – The number of participating sites (default: all localities).
- **this_site** – The sequence number of this site (usually the locality id).

hpx::collectives::communicator::get_info

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::collectives::communicator`
- `hpx::collectives::create_communicator`
- `hpx::collectives::create_local_communicator`

- `hpx::collectives::create_hierarchical_communicator`
- `hpx::collectives::communicator::set_info`
- `hpx::collectives::communicator::is_root`

`std::pair<num_sites_arg, this_site_arg> hpx::collectives::communicator::get_info() const noexcept`
Retrieve the number of used sites and the index of the current site for this communicator instance.

`hpx::collectives::communicator::is_root`

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::collectives::communicator`
- `hpx::collectives::create_communicator`
- `hpx::collectives::create_local_communicator`
- `hpx::collectives::create_hierarchical_communicator`
- `hpx::collectives::communicator::set_info`
- `hpx::collectives::communicator::get_info`

`bool hpx::collectives::communicator::is_root() const`

Return whether this communicator instance represents the root site of the communication operation.

`hpx::collectives::scatter_from`

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::collectives::scatter_to`

`template<typename T>`

`hpx::future<T> hpx::collectives::scatter_from(char const *basename, this_site_arg this_site =
this_site_arg(), generation_arg generation =
generation_arg(), root_site_arg root_site = root_site_arg())`

Scatter (receive) a set of values to different call sites

This function receives an element of a set of values operating on the given base name.

Parameters

- **basename** – The base name identifying the scatter operation
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the scatter operation performed on the given base name. This is optional and needs to be supplied only if the scatter operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.

- **root_site** – The sequence number of the central scatter point (usually the locality id). This value is optional and defaults to 0.

Returns

This function returns a future holding the scattered value. It will become ready once the scatter operation has been completed.

```
template<typename T>
hpx::future<T> hpx::collectives::scatter_from(communicator comm, this_site_arg this_site =
                                             this_site_arg(), generation_arg generation =
                                             generation_arg())
```

Scatter (receive) a set of values to different call sites

This function receives an element of a set of values operating on the given base name.

Scatter (receive) a set of values to different call sites

This function receives an element of a set of values operating on the given base name.

Note: The generation values from corresponding *scatter_to* and *scatter_from* have to match.

Note: The generation values from corresponding *scatter_to* and *scatter_from* have to match.

Parameters

- **comm** – A communicator object returned from *create_communicator*
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.
- **generation** – The generational counter identifying the sequence number of the scatter operation performed on the given base name. This is optional and needs to be supplied only if the scatter operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **comm** – A communicator object returned from *create_communicator*
- **generation** – The generational counter identifying the sequence number of the scatter operation performed on the given base name. This is optional and needs to be supplied only if the scatter operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.

Returns

This function returns a future holding the scattered value. It will become ready once the scatter operation has been completed.

Returns

This function returns a future holding the scattered value. It will become ready once the scatter operation has been completed.

```
template<typename T>
```

```
T hpx::collectives::scatter_from(hpx::launch::sync_policy, char const *basename, this_site_arg this_site =
    this_site_arg(), generation_arg generation = generation_arg(),
    root_site_arg root_site = root_site_arg())
```

Scatter (receive) a set of values to different call sites

This function receives an element of a set of values operating on the given base name.

Parameters

- **basename** – The base name identifying the scatter operation
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the scatter operation performed on the given base name. This is optional and needs to be supplied only if the scatter operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **root_site** – The sequence number of the central scatter point (usually the locality id). This value is optional and defaults to 0.

Returns

This function returns the scattered value. It executes synchronously and directly returns the result.

```
template<typename T>
```

```
T hpx::collectives::scatter_from(hpx::launch::sync_policy, communicator comm, this_site_arg this_site =
    this_site_arg(), generation_arg generation = generation_arg())
```

Scatter (receive) a set of values to different call sites

This function receives an element of a set of values operating on the given base name.

Scatter (receive) a set of values to different call sites

This function receives an element of a set of values operating on the given base name.

Note: The generation values from corresponding *scatter_to* and *scatter_from* have to match.

Note: The generation values from corresponding *scatter_to* and *scatter_from* have to match.

Parameters

- **comm** – A communicator object returned from *create_communicator*
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the scatter operation performed on the given base name. This is optional and needs to be supplied only if the scatter operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **comm** – A communicator object returned from *create_communicator*
- **generation** – The generational counter identifying the sequence number of the scatter operation performed on the given base name. This is optional and needs to be supplied only

if the scatter operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.

- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

Returns

This function returns the scattered value. It executes synchronously and directly returns the result.

Returns

This function returns the scattered value. It executes synchronously and directly returns the result.

hpx::collectives::scatter_to

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::collectives::scatter_from`

```
template<typename T>
hpx::future<T> hpx::collectives::scatter_to(char const *basename, std::vector<T> &&result,
                                           num_sites_arg num_sites = num_sites_arg(), this_site_arg
                                           this_site = this_site_arg(), generation_arg generation =
                                           generation_arg())
```

Scatter (send) a part of the value set at the given call site

This function transmits the value given by *result* to a central scatter site (where the corresponding *scatter_from* is executed)

Parameters

- **basename** – The base name identifying the scatter operation
- **result** – The value to transmit to the central scatter point from this call site.
- **num_sites** – The number of participating sites (default: all localities).
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the scatter operation performed on the given base name. This is optional and needs to be supplied only if the scatter operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.

Returns

This function returns a future holding the scattered value. It will become ready once the scatter operation has been completed.

```
template<typename T>
hpx::future<T> hpx::collectives::scatter_to(communicator comm, std::vector<T> &&result, this_site_arg
                                           this_site = this_site_arg(), generation_arg generation =
                                           generation_arg())
```

Scatter (send) a part of the value set at the given call site

This function transmits the value given by *result* to a central scatter site (where the corresponding *scatter_from* is executed)

Scatter (send) a part of the value set at the given call site

This function transmits the value given by *result* to a central scatter site (where the corresponding *scatter_from* is executed)

Note: The generation values from corresponding *scatter_to* and *scatter_from* have to match.

Note: The generation values from corresponding *scatter_to* and *scatter_from* have to match.

Parameters

- **comm** – A communicator object returned from *create_communicator*
- **result** – The value to transmit to the central scatter point from this call site.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.
- **generation** – The generational counter identifying the sequence number of the scatter operation performed on the given base name. This is optional and needs to be supplied only if the scatter operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **comm** – A communicator object returned from *create_communicator*
- **result** – The value to transmit to the central scatter point from this call site.
- **generation** – The generational counter identifying the sequence number of the scatter operation performed on the given base name. This is optional and needs to be supplied only if the scatter operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.

Returns

This function returns a future holding the scattered value. It will become ready once the scatter operation has been completed.

Returns

This function returns a future holding the scattered value. It will become ready once the scatter operation has been completed.

```
template<typename T>
T hpx::collectives::scatter_to(hpx::launch::sync_policy, char const *basename, std::vector<T> &&result,
                               num_sites_arg num_sites = num_sites_arg(), this_site_arg this_site =
                               this_site_arg(), generation_arg generation = generation_arg())
```

Scatter (send) a part of the value set at the given call site

This function transmits the value given by *result* to a central scatter site (where the corresponding *scatter_from* is executed)

Parameters

- **basename** – The base name identifying the scatter operation
- **result** – The value to transmit to the central scatter point from this call site.

- **num_sites** – The number of participating sites (default: all localities).
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the scatter operation performed on the given base name. This is optional and needs to be supplied only if the scatter operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.

Returns

This function returns the scattered value. It executes synchronously and directly returns the result.

```
template<typename T>
T hpx::collectives::scatter_to(hpx::launch::sync_policy, communicator comm, std::vector<T> &&result,
                               this_site_arg this_site = this_site_arg(), generation_arg generation =
                               generation_arg())
```

Scatter (send) a part of the value set at the given call site

This function transmits the value given by *result* to a central scatter site (where the corresponding *scatter_from* is executed)

Scatter (send) a part of the value set at the given call site

This function transmits the value given by *result* to a central scatter site (where the corresponding *scatter_from* is executed)

Note: The generation values from corresponding *scatter_to* and *scatter_from* have to match.

Note: The generation values from corresponding *scatter_to* and *scatter_from* have to match.

Parameters

- **comm** – A communicator object returned from *create_communicator*
- **result** – The value to transmit to the central scatter point from this call site.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the scatter operation performed on the given base name. This is optional and needs to be supplied only if the scatter operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **comm** – A communicator object returned from *create_communicator*
- **result** – The value to transmit to the central scatter point from this call site.
- **generation** – The generational counter identifying the sequence number of the scatter operation performed on the given base name. This is optional and needs to be supplied only if the scatter operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

Returns

This function returns the scattered value. It executes synchronously and directly returns the result.

Returns

This function returns the scattered value. It executes synchronously and directly returns the result.

hpx::collectives::all_reduce

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

template<typename **T**, typename **F**>

```
hpx::future<std::decay_t<T>> hpx::collectives::all_reduce(char const *basename, T &&result, F &&op,
                                                         num_sites_arg num_sites = num_sites_arg(),
                                                         this_site_arg this_site = this_site_arg(),
                                                         generation_arg generation = generation_arg(),
                                                         root_site_arg root_site = root_site_arg())
```

AllReduce a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

Parameters

- **basename** – The base name identifying the all_reduce operation
- **result** – The value to transmit to all participating sites from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **num_sites** – The number of participating sites (default: all localities).
- **generation** – The generational counter identifying the sequence number of the all_reduce operation performed on the given base name. This is optional and needs to be supplied only if the all_reduce operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **root_site** – The site that is responsible for creating the all_reduce support object. This value is optional and defaults to '0' (zero).

Returns

This function returns a future holding a vector with all values send by all participating sites. It will become ready once the all_reduce operation has been completed.

template<typename **T**, typename **F**>

```
hpx::future<std::decay_t<T>> hpx::collectives::all_reduce(communicator comm, T &&result, F &&op,
                                                         this_site_arg this_site = this_site_arg(),
                                                         generation_arg generation = generation_arg())
```

AllReduce a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

AllReduce a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

Parameters

- **comm** – A communicator object returned from *create_communicator*
- **result** – The value to transmit to all participating sites from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.
- **generation** – The generational counter identifying the sequence number of the all_reduce operation performed on the given base name. This is optional and needs to be supplied only if the all_reduce operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **comm** – A communicator object returned from *create_communicator*
- **result** – The value to transmit to all participating sites from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **generation** – The generational counter identifying the sequence number of the all_reduce operation performed on the given base name. This is optional and needs to be supplied only if the all_reduce operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.

Returns

This function returns a future holding a vector with all values send by all participating sites. It will become ready once the all_reduce operation has been completed.

Returns

This function returns a future holding a vector with all values send by all participating sites. It will become ready once the all_reduce operation has been completed.

Warning: doxygenfunction: Unable to resolve function “hpx::collectives::all_reduce” with arguments (hpx::launch::sync_policy, char const*, T&&, F&&, num_sites_arg, this_site_arg, generation_arg, root_site_arg) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:

```
- template<typename T, typename F> decltype(auto) all_reduce(hpx::launch::sync_policy_
↳ policy, char const *basename, T &&result, F &&op, num_sites_arg num_sites = num_
↳ sites_arg(), this_site_arg this_site = this_site_arg(), generation_arg generation =
↳ generation_arg(), root_site_arg root_site = root_site_arg())
- template<typename T, typename F> decltype(auto) all_reduce(hpx::launch::sync_policy_
↳ policy, communicator comm, T &&result, F &&op, this_site_arg this_site = this_site_
↳ arg(), generation_arg generation = generation_arg())
- template<typename T, typename F> hpx::future<std::decay_t<T>> all_reduce(char const_
↳ *basename, T &&result, F &&op, num_sites_arg num_sites = num_sites_arg(), this_site_
↳ arg this_site = this_site_arg(), generation_arg generation = generation_arg(), root_
↳ site_arg root_site = root_site_arg())
- template<typename T, typename F> hpx::future<std::decay_t<T>> all_
↳ reduce(communicator comm, T &&result, F &&op, this_site_arg this_site = this_site_
↳ arg(), generation_arg generation = generation_arg())
```

Warning: doxygenfunction: Unable to resolve function “hpx::collectives::all_reduce” with arguments (hpx::launch::sync_policy, communicator, T&&, F&&, this_site_arg, generation_arg) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:

```
- template<typename T, typename F> decltype(auto) all_reduce(hpx::launch::sync_policy_
↳policy, char const *basename, T &&result, F &&op, num_sites_arg num_sites = num_
↳sites_arg(), this_site_arg this_site = this_site_arg(), generation_arg generation =_
↳generation_arg(), root_site_arg root_site = root_site_arg())
- template<typename T, typename F> decltype(auto) all_reduce(hpx::launch::sync_policy_
↳policy, communicator comm, T &&result, F &&op, this_site_arg this_site = this_site_
↳arg(), generation_arg generation = generation_arg())
- template<typename T, typename F> hpx::future<std::decay_t<T>> all_reduce(char const_
↳*basename, T &&result, F &&op, num_sites_arg num_sites = num_sites_arg(), this_site_
↳arg this_site = this_site_arg(), generation_arg generation = generation_arg(), root_
↳site_arg root_site = root_site_arg())
- template<typename T, typename F> hpx::future<std::decay_t<T>> all_
↳reduce(communicator comm, T &&result, F &&op, this_site_arg this_site = this_site_
↳arg(), generation_arg generation = generation_arg())
```

hpx::lcos::reduce

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::lcos::reduce_with_index`

Warning: doxygenfunction: Unable to resolve function “hpx::lcos::reduce” with arguments “None”. Candidate function could not be parsed. Parsing error is Error in template parameter list. If parameter: Error when parsing template parameter. If unconstrained type parameter or template type parameter: Invalid C++ declaration: Expected ‘typename’ or ‘class’ in the beginning of template type parameter. [error at 60] template<typename Action, typename ReduceOp, typename ArgN, ...> hpx::future< decltype(Action(hpx::id_type, ArgN,...))> reduce (std::vector< hpx::id_type > const &ids, ReduceOp &&reduce_op, ArgN argN,...) _____^ If constrained type parameter or non-type parameter: Invalid C++ declaration: Expected identifier in nested name. [error at 60] template<typename Action, typename ReduceOp, typename ArgN, ...> hpx::future< decltype(Action(hpx::id_type, ArgN,...))> reduce (std::vector< hpx::id_type > const &ids, ReduceOp &&reduce_op, ArgN argN,...) _____^ If no parameter: Invalid C++ declaration: Expected “,” or “>”. [error at 60] template<typename Action, typename ReduceOp, typename ArgN, ...> hpx::future< decltype(Action(hpx::id_type, ArgN,...))> reduce (std::vector< hpx::id_type > const &ids, ReduceOp &&reduce_op, ArgN argN,...) _____^

hpx::lcos::reduce_with_index

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::lcos::reduce`

Warning: doxygenfunction: Unable to resolve function “hpx::lcos::reduce_with_index” with arguments “None”. Candidate function could not be parsed. Parsing error is Error in template parameter list. If parameter: Error when parsing template parameter. If unconstrained type parameter or template type parameter: Invalid C++ declaration: Expected ‘typename’ or ‘class’ in the beginning of template type parameter. [error at 60] template<typename Action, typename ReduceOp, typename ArgN, ...> hpx::future< decltype(Action(hpx::id_type, ArgN,..., std::size_t))> reduce_with_index (std::vector< hpx::id_type > const &ids, ReduceOp &&reduce_op, ArgN argN,...) _____^ If constrained type parameter or non-type parameter: Invalid C++ declaration: Expected identifier in nested name. [error at 60] template<typename Action, typename ReduceOp, typename ArgN, ...> hpx::future< decltype(Action(hpx::id_type, ArgN,..., std::size_t))> reduce_with_index (std::vector< hpx::id_type > const &ids, ReduceOp &&reduce_op, ArgN argN,...) _____^ If no parameter: Invalid C++ declaration: Expected “,” or “>”. [error at 60] template<typename Action, typename ReduceOp, typename ArgN, ...> hpx::future< decltype(Action(hpx::id_type, ArgN,..., std::size_t))> reduce_with_index (std::vector< hpx::id_type > const &ids, ReduceOp &&reduce_op, ArgN argN,...) _____^

hpx::lcos::broadcast

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::lcos::broadcast_post`
- `hpx::lcos::broadcast_with_index`
- `hpx::lcos::broadcast_post_with_index`

template<typename **Action**>

struct **broadcast**

hpx::lcos::broadcast_post

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::lcos::broadcast`
- `hpx::lcos::broadcast_with_index`
- `hpx::lcos::broadcast_post_with_index`

Warning: doxygenfunction: Unable to resolve function “hpx::lcos::broadcast_post” with arguments “None”. Candidate function could not be parsed. Parsing error is Error in template parameter list. If parameter: Error when parsing template parameter. If unconstrained type parameter or template type parameter: Invalid C++ declaration: Expected ‘typename’ or ‘class’ in the beginning of template type parameter. [error at 41] template<typename Action, typename ArgN, ...> void broadcast_post (std::vector< hpx::id_type > const &ids, ArgN argN,...) _____^ If constrained type parameter or non-type parameter: Invalid C++ declaration: Expected identifier in nested name. [error at 41] template<typename Action, typename ArgN, ...> void broadcast_post (std::vector< hpx::id_type > const &ids, ArgN argN,...) _____^ If no parameter: Invalid C++ declaration: Expected “,” or “>”. [error at 41] template<typename Action, typename ArgN, ...> void broadcast_post (std::vector< hpx::id_type > const &ids, ArgN argN,...) _____^

hpx::lcos::broadcast_with_index

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::lcos::broadcast`
- `hpx::lcos::broadcast_post`
- `hpx::lcos::broadcast_post_with_index`

Warning: doxygenfunction: Unable to resolve function “hpx::lcos::broadcast_with_index” with arguments “None”. Candidate function could not be parsed. Parsing error is Error in template parameter list. If parameter: Error when parsing template parameter. If unconstrained type parameter or template type parameter: Invalid C++ declaration: Expected ‘typename’ or ‘class’ in the beginning of template type parameter. [error at 41] template<typename Action, typename ArgN, ...> hpx::future< std::vector< decltype(Action(hpx::id_type, ArgN,..., std::size_t))> > broadcast_with_index (std::vector< hpx::id_type > const &ids, ArgN argN,...) _____^ If constrained type parameter or non-type parameter: Invalid C++ declaration: Expected identifier in nested name. [error at 41] template<typename Action, typename ArgN, ...> hpx::future< std::vector< decltype(Action(hpx::id_type, ArgN,..., std::size_t))> > broadcast_with_index (std::vector< hpx::id_type > const &ids, ArgN argN,...) _____^ If no parameter: Invalid C++ declaration: Expected “,” or “>”. [error at 41] template<typename Action, typename ArgN, ...> hpx::future< std::vector< decltype(Action(hpx::id_type, ArgN,..., std::size_t))> > broadcast_with_index (std::vector< hpx::id_type > const &ids, ArgN argN,...) _____^

hpx::lcos::broadcast_post_with_index

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::lcos::broadcast`
- `hpx::lcos::broadcast_post`
- `hpx::lcos::broadcast_with_index`

Warning: doxygenfunction: Unable to resolve function “hpx::lcos::broadcast_post_with_index” with arguments “None”. Candidate function could not be parsed. Parsing error is Error in template parameter list. If parameter: Error when parsing template parameter. If unconstrained type parameter or template type parameter: Invalid C++ declaration: Expected ‘typename’ or ‘class’ in the beginning of template type parameter. [error at 41] template<typename Action, typename ArgN, ...> void broadcast_post_with_index (std::vector< hpx::id_type > const &ids, ArgN argN,...) _____^ If constrained type parameter or non-type parameter: Invalid C++ declaration: Expected identifier in nested name. [error at 41] template<typename Action, typename ArgN, ...> void broadcast_post_with_index (std::vector< hpx::id_type > const &ids, ArgN argN,...) _____^ If no parameter: Invalid C++ declaration: Expected “,” or “>”. [error at 41] template<typename Action, typename ArgN, ...> void broadcast_post_with_index (std::vector< hpx::id_type > const &ids, ArgN argN,...) _____^

hpx::collectives::exclusive_scan

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename T, typename F>
hpx::future<std::decay_t<T>> hpx::collectives::exclusive_scan(char const *basename, T &&result, F
                                                                &&op, num_sites_arg num_sites =
                                                                num_sites_arg(), this_site_arg this_site =
                                                                this_site_arg(), generation_arg generation
                                                                = generation_arg(), root_site_arg root_site
                                                                = root_site_arg())
```

Exclusive scan a set of values from different call sites

This function performs an exclusive scan operation on a set of values received from all call sites operating on the given base name.

Note: This no-init overload has MPI_Exscan-like semantics, with a deterministic HPX first result: participating site 0 returns a value-initialized result (`std::decay_t<T>{}`, or `false` for `bool`). Use the overload taking an explicit `init` argument for C++ exclusive-scan semantics where site 0 returns `init`.

Parameters

- **basename** – The base name identifying the `exclusive_scan` operation
- **result** – The value to transmit to all participating sites from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **num_sites** – The number of participating sites (default: all localities).
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the `exclusive_scan` operation performed on the given base name. This is optional and needs to be supplied only if the `exclusive_scan` operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **root_site** – The site that is responsible for creating the `exclusive_scan` support object. This value is optional and defaults to ‘0’ (zero).

Returns

For the participating site *i* this function returns the reduction (calculated according to the function *op*) of the values passed in by the participating sites 0, ..., *i*-1. Participating site 0 returns a value-initialized result. The returned future will become ready once the `exclusive_scan` operation has been completed.

```
template<typename T, typename F>
hpx::future<std::decay_t<T>> hpx::collectives::exclusive_scan(communicator comm, T &&result, F
                                                                &&op, this_site_arg this_site =
                                                                this_site_arg(), generation_arg generation
                                                                = generation_arg())
```

Exclusive scan a set of values from different call sites

This function performs an exclusive scan operation on a set of values received from all call sites operating on the given base name.

Exclusive scan a set of values from different call sites

This function performs an exclusive scan operation on a set of values received from all call sites operating on the given base name.

Note: This no-init overload has MPI_Exscan-like semantics, with a deterministic HPX first result: participating site 0 returns a value-initialized result (`std::decay_t<T>{}`, or `false` for `bool`). Use the overload taking an explicit `init` argument for C++ exclusive-scan semantics where site 0 returns `init`.

Note: This no-init overload has MPI_Exscan-like semantics, with a deterministic HPX first result: participating site 0 returns a value-initialized result (`std::decay_t<T>{}`, or `false` for `bool`). Use the overload taking an explicit `init` argument for C++ exclusive-scan semantics where site 0 returns `init`.

Parameters

- **comm** – A communicator object returned from `create_communicator`
- **result** – The value to transmit to all participating sites from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the `exclusive_scan` operation performed on the given base name. This is optional and needs to be supplied only if the `exclusive_scan` operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **comm** – A communicator object returned from `create_communicator`
- **result** – The value to transmit to all participating sites from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **generation** – The generational counter identifying the sequence number of the `exclusive_scan` operation performed on the given base name. This is optional and needs to be supplied only if the `exclusive_scan` operation on the given base name has to be performed

more than once. The generation number (if given) must be a positive number greater than zero.

- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

Returns

For the participating site *i* this function returns the reduction (calculated according to the function *op*) of the values passed in by the participating sites 0, ..., *i*-1. Participating site 0 returns a value-initialized result. The returned future will become ready once the `exclusive_scan` operation has been completed.

Returns

For the participating site *i* this function returns a future the reduction (calculated according to the function *op*) of the values passed in by the participating sites 0, ..., *i*-1. Participating site 0 returns a value-initialized result. The returned future will become ready once the `exclusive_scan` operation has been completed.

Warning: doxygenfunction: Unable to resolve function “hpx::collectives::exclusive_scan” with arguments (hpx::launch::sync_policy, char const*, T&&, F&&, num_sites_arg, this_site_arg, generation_arg, root_site_arg) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:

```
- template<typename T, typename F> decltype(auto) exclusive_scan(hpx::launch::sync_
  ↳policy, char const *basename, T &&result, F &&op, num_sites_arg num_sites = num_
  ↳sites_arg(), this_site_arg this_site = this_site_arg(), generation_arg generation =
  ↳generation_arg(), root_site_arg root_site = root_site_arg())
- template<typename T, typename F> decltype(auto) exclusive_scan(hpx::launch::sync_
  ↳policy, communicator comm, T &&result, F &&op, this_site_arg this_site = this_site_
  ↳arg(), generation_arg generation = generation_arg())
- template<typename T, typename F> hpx::future<std::decay_t<T>> exclusive_scan(char_
  ↳const *basename, T &&result, F &&op, num_sites_arg num_sites = num_sites_arg(),
  ↳this_site_arg this_site = this_site_arg(), generation_arg generation = generation_
  ↳arg(), root_site_arg root_site = root_site_arg())
- template<typename T, typename F> hpx::future<std::decay_t<T>> exclusive_
  ↳scan(communicator comm, T &&result, F &&op, this_site_arg this_site = this_site_
  ↳arg(), generation_arg generation = generation_arg())
```

Warning: doxygenfunction: Unable to resolve function “hpx::collectives::exclusive_scan” with arguments (hpx::launch::sync_policy, communicator, T&&, F&&, this_site_arg, generation_arg) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:

```
- template<typename T, typename F> decltype(auto) exclusive_scan(hpx::launch::sync_
  ↳policy, char const *basename, T &&result, F &&op, num_sites_arg num_sites = num_
  ↳sites_arg(), this_site_arg this_site = this_site_arg(), generation_arg generation =
  ↳generation_arg(), root_site_arg root_site = root_site_arg())
- template<typename T, typename F> decltype(auto) exclusive_scan(hpx::launch::sync_
  ↳policy, communicator comm, T &&result, F &&op, this_site_arg this_site = this_site_
  ↳arg(), generation_arg generation = generation_arg())
- template<typename T, typename F> hpx::future<std::decay_t<T>> exclusive_scan(char_
  ↳const *basename, T &&result, F &&op, num_sites_arg num_sites = num_sites_arg(),
  ↳this_site_arg this_site = this_site_arg(), generation_arg generation = generation_
  ↳arg(), root_site_arg root_site = root_site_arg())
- template<typename T, typename F> hpx::future<std::decay_t<T>> exclusive_
```

```
↪ scan(communicator comm, T &&result, F &&op, this_site_arg this_site = this_site_
↪ arg(), generation_arg generation = generation_arg())
```

hpx::collectives::inclusive_scan

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<typename T, typename F>
```

```
hpx::future<std::decay_t<T>> hpx::collectives::inclusive_scan(char const *basename, T &&result, F
&&op, num_sites_arg num_sites =
num_sites_arg(), this_site_arg this_site =
this_site_arg(), generation_arg generation
= generation_arg(), root_site_arg root_site
= root_site_arg())
```

Inclusive `inclusive_scan` a set of values from different call sites

This function performs an inclusive scan operation on a set of values received from all call sites operating on the given base name.

Parameters

- **basename** – The base name identifying the `inclusive_scan` operation
- **result** – The value to transmit to all participating sites from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **num_sites** – The number of participating sites (default: all localities).
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the `inclusive_scan` operation performed on the given base name. This is optional and needs to be supplied only if the `inclusive_scan` operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **root_site** – The site that is responsible for creating the `inclusive_scan` support object. This value is optional and defaults to '0' (zero).

Returns

This function returns a future holding a vector with all values send by all participating sites. It will become ready once the `inclusive_scan` operation has been completed.

```
template<typename T, typename F>
```

```
hpx::future<std::decay_t<T>> hpx::collectives::inclusive_scan(communicator comm, T &&result, F
&&op, this_site_arg this_site =
this_site_arg(), generation_arg generation
= generation_arg())
```

Inclusive `inclusive_scan` a set of values from different call sites

This function performs an inclusive scan operation on a set of values received from all call sites operating on the given base name.

Inclusive inclusive_scan a set of values from different call sites

This function performs an inclusive scan operation on a set of values received from all call sites operating on the given base name.

Parameters

- **comm** – A communicator object returned from *create_communicator*
- **result** – The value to transmit to all participating sites from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.
- **generation** – The generational counter identifying the sequence number of the inclusive_scan operation performed on the given base name. This is optional and needs to be supplied only if the inclusive_scan operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **comm** – A communicator object returned from *create_communicator*
- **result** – The value to transmit to all participating sites from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **generation** – The generational counter identifying the sequence number of the inclusive_scan operation performed on the given base name. This is optional and needs to be supplied only if the inclusive_scan operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.

Returns

This function returns a future holding a vector with all values send by all participating sites. It will become ready once the inclusive_scan operation has been completed.

Returns

This function returns a future holding a vector with all values send by all participating sites. It will become ready once the inclusive_scan operation has been completed.

Warning: doxygenfunction: Unable to resolve function “hpx::collectives::inclusive_scan” with arguments (hpx::launch::sync_policy, char const*, T&&, F&&, num_sites_arg, this_site_arg, generation_arg, root_site_arg) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:

```
- template<typename T, typename F> decltype(auto) inclusive_scan(hpx::launch::sync_
  ↳policy, char const *basename, T &&result, F &&op, num_sites_arg num_sites = num_
  ↳sites_arg(), this_site_arg this_site = this_site_arg(), generation_arg generation =
  ↳generation_arg(), root_site_arg root_site = root_site_arg())
- template<typename T, typename F> decltype(auto) inclusive_scan(hpx::launch::sync_
  ↳policy, communicator comm, T &&result, F &&op, this_site_arg this_site = this_site_
  ↳arg(), generation_arg generation = generation_arg())
- template<typename T, typename F> hpx::future<std::decay_t<T>> inclusive_scan(char_
  ↳const *basename, T &&result, F &&op, num_sites_arg num_sites = num_sites_arg(),
  ↳this_site_arg this_site = this_site_arg(), generation_arg generation = generation_
  ↳arg(), root_site_arg root_site = root_site_arg())
- template<typename T, typename F> hpx::future<std::decay_t<T>> inclusive_
```

```

↪scan(communicator comm, T &&result, F &&op, this_site_arg this_site = this_site_
↪arg(), generation_arg generation = generation_arg())

```

Warning: doxygenfunction: Unable to resolve function “hpx::collectives::inclusive_scan” with arguments (hpx::launch::sync_policy, communicator, T&&, F&&, this_site_arg, generation_arg) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:

```

- template<typename T, typename F> decltype(auto) inclusive_scan(hpx::launch::sync_
↪policy, char const *basename, T &&result, F &&op, num_sites_arg num_sites = num_
↪sites_arg(), this_site_arg this_site = this_site_arg(), generation_arg generation =
↪generation_arg(), root_site_arg root_site = root_site_arg())
- template<typename T, typename F> decltype(auto) inclusive_scan(hpx::launch::sync_
↪policy, communicator comm, T &&result, F &&op, this_site_arg this_site = this_site_
↪arg(), generation_arg generation = generation_arg())
- template<typename T, typename F> hpx::future<std::decay_t<T>> inclusive_scan(char_
↪const *basename, T &&result, F &&op, num_sites_arg num_sites = num_sites_arg(),
↪this_site_arg this_site = this_site_arg(), generation_arg generation = generation_
↪arg(), root_site_arg root_site = root_site_arg())
- template<typename T, typename F> hpx::future<std::decay_t<T>> inclusive_
↪scan(communicator comm, T &&result, F &&op, this_site_arg this_site = this_site_
↪arg(), generation_arg generation = generation_arg())

```

hpx::collectives::all_to_all

Defined in header hpx/collectives.hpp.

See *Public API* for a list of names and headers that are part of the public HPX API.

```

template<typename T>
hpx::future<std::vector<T>> hpx::collectives::all_to_all(char const *basename, std::vector<T>
&&local_result, num_sites_arg num_sites =
num_sites_arg(), this_site_arg this_site =
this_site_arg(), generation_arg generation =
generation_arg(), root_site_arg root_site =
root_site_arg())

```

AllToAll a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

Parameters

- **basename** – The base name identifying the all_to_all operation
- **local_result** – A vector of values to transmit to all participating sites from this call site.
- **num_sites** – The number of participating sites (default: all localities).
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the all_to_all operation performed on the given base name. This is optional and needs to be supplied only if the all_to_all operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.

- **root_site** – The site that is responsible for creating the all_to_all support object. This value is optional and defaults to '0' (zero).

Returns

This function returns a future holding a vector with all values send by all participating sites. It will become ready once the all_to_all operation has been completed.

```
template<typename T>
hpx::future<std::vector<T>> hpx::collectives::all_to_all(communicator fid, std::vector<T>
&&local_result, this_site_arg this_site =
this_site_arg(), generation_arg generation =
generation_arg())
```

AllToAll a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

AllToAll a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

Parameters

- **fid** – A communicator object returned from *create_communicator*
- **local_result** – A vector of values to transmit to all participating sites from this call site.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.
- **generation** – The generational counter identifying the sequence number of the all_to_all operation performed on the given base name. This is optional and needs to be supplied only if the all_to_all operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **fid** – A communicator object returned from *create_communicator*
- **local_result** – A vector of values to transmit to all participating sites from this call site.
- **generation** – The generational counter identifying the sequence number of the all_to_all operation performed on the given base name. This is optional and needs to be supplied only if the all_to_all operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.

Returns

This function returns a future holding a vector with all values send by all participating sites. It will become ready once the all_to_all operation has been completed.

Returns

This function returns a future holding a vector with all values send by all participating sites. It will become ready once the all_to_all operation has been completed.

```
template<typename T>
std::vector<T> hpx::collectives::all_to_all(hpx::launch::sync_policy policy, char const *basename,
std::vector<T> &&local_result, num_sites_arg num_sites =
num_sites_arg(), this_site_arg this_site = this_site_arg(),
generation_arg generation = generation_arg(), root_site_arg
root_site = root_site_arg())
```

AllToAll a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

Parameters

- **policy** – The execution policy specifying synchronous execution.
- **basename** – The base name identifying the all_to_all operation
- **local_result** – A vector of values to transmit to all participating sites from this call site.
- **num_sites** – The number of participating sites (default: all localities).
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the all_to_all operation performed on the given base name. This is optional and needs to be supplied only if the all_to_all operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **root_site** – The site that is responsible for creating the all_to_all support object. This value is optional and defaults to '0' (zero).

Returns

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

```
template<typename T>
std::vector<T> hpx::collectives::all_to_all(hpx::launch::sync_policy policy, communicator fid,
                                           std::vector<T> &&local_result, this_site_arg this_site =
                                           this_site_arg(), generation_arg generation = generation_arg())
```

AllToAll a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

AllToAll a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

Parameters

- **policy** – The execution policy specifying synchronous execution.
- **fid** – A communicator object returned from `create_communicator`
- **local_result** – A vector of values to transmit to all participating sites from this call site.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the all_to_all operation performed on the given base name. This is optional and needs to be supplied only if the all_to_all operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **policy** – The execution policy specifying synchronous execution.
- **fid** – A communicator object returned from `create_communicator`
- **local_result** – A vector of values to transmit to all participating sites from this call site.

- **generation** – The generational counter identifying the sequence number of the all_to_all operation performed on the given base name. This is optional and needs to be supplied only if the all_to_all operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

Returns

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

Returns

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

hpx::collectives::broadcast_to

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::collectives::broadcast_from`

```
template<typename T>
hpx::future<T> hpx::collectives::broadcast_to(char const *basename, T &&local_result, num_sites_arg
num_sites = num_sites_arg(), this_site_arg this_site =
this_site_arg(), generation_arg generation =
generation_arg())
```

Broadcast a value to different call sites

This function sends a set of values to all call sites operating on the given base name.

Parameters

- **basename** – The base name identifying the broadcast operation
- **local_result** – A value to transmit to all participating sites from this call site.
- **num_sites** – The number of participating sites (default: all localities).
- **generation** – The generational counter identifying the sequence number of the broadcast operation performed on the given base name. This is optional and needs to be supplied only if the broadcast operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

Returns

This function returns a future holding the value that was sent to all participating sites. It will become ready once the broadcast operation has been completed.

```
template<typename T>
hpx::future<T> hpx::collectives::broadcast_to(communicator comm, T &&local_result, this_site_arg
this_site = this_site_arg(), generation_arg generation =
generation_arg())
```

Broadcast a value to different call sites

This function sends a set of values to all call sites operating on the given base name.

Note: The generation values from corresponding *broadcast_to* and *broadcast_from* have to match.

Parameters

- **comm** – A communicator object returned from *create_communicator*
- **local_result** – A value to transmit to all participating sites from this call site.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.
- **generation** – The generational counter identifying the sequence number of the broadcast operation performed on the given base name. This is optional and needs to be supplied only if the broadcast operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.

Returns

This function returns a future holding the value that was sent to all participating sites. It will become ready once the broadcast operation has been completed.

```
template<typename T>
hpx::future<T> hpx::collectives::broadcast_to(communicator comm, generation_arg generation, T
&&local_result, this_site_arg this_site = this_site_arg())
```

Broadcast a value to different call sites

This function sends a set of values to all call sites operating on the given base name.

Note: The generation values from corresponding *broadcast_to* and *broadcast_from* have to match.

Parameters

- **comm** – A communicator object returned from *create_communicator*
- **local_result** – A value to transmit to all participating sites from this call site.
- **generation** – The generational counter identifying the sequence number of the broadcast operation performed on the given base name. This is optional and needs to be supplied only if the broadcast operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.

Returns

This function returns a future holding the value that was sent to all participating sites. It will become ready once the broadcast operation has been completed.

Warning: doxygenfunction: Unable to resolve function “hpx::collectives::broadcast_to” with arguments (hpx::launch::sync_policy, char const*, T&&, num_sites_arg, this_site_arg, generation_arg) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:

```

- template<typename T> decltype(auto) broadcast_to(hpx::launch::sync_policy policy,
↳char const *basename, T &&local_result, num_sites_arg num_sites = num_sites_arg(),
↳this_site_arg this_site = this_site_arg(), generation_arg generation = generation_
↳arg())
- template<typename T> decltype(auto) broadcast_to(hpx::launch::sync_policy policy,
↳communicator comm, T &&local_result, this_site_arg this_site = this_site_arg(),
↳generation_arg generation = generation_arg())
- template<typename T> hpx::future<T> broadcast_to(char const *basename, T &&local_
↳result, num_sites_arg num_sites = num_sites_arg(), this_site_arg this_site = this_
↳site_arg(), generation_arg generation = generation_arg())
- template<typename T> hpx::future<T> broadcast_to(communicator comm, T &&local_
↳result, this_site_arg this_site = this_site_arg(), generation_arg generation =
↳generation_arg())
- template<typename T> hpx::future<T> broadcast_to(communicator comm, generation_arg
↳generation, T &&local_result, this_site_arg this_site = this_site_arg())

```

Warning: doxygenfunction: Unable to resolve function “hpx::collectives::broadcast_to” with arguments (hpx::launch::sync_policy, communicator, T&&, this_site_arg, generation_arg) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:

```

- template<typename T> decltype(auto) broadcast_to(hpx::launch::sync_policy policy,
↳char const *basename, T &&local_result, num_sites_arg num_sites = num_sites_arg(),
↳this_site_arg this_site = this_site_arg(), generation_arg generation = generation_
↳arg())
- template<typename T> decltype(auto) broadcast_to(hpx::launch::sync_policy policy,
↳communicator comm, T &&local_result, this_site_arg this_site = this_site_arg(),
↳generation_arg generation = generation_arg())
- template<typename T> hpx::future<T> broadcast_to(char const *basename, T &&local_
↳result, num_sites_arg num_sites = num_sites_arg(), this_site_arg this_site = this_
↳site_arg(), generation_arg generation = generation_arg())
- template<typename T> hpx::future<T> broadcast_to(communicator comm, T &&local_
↳result, this_site_arg this_site = this_site_arg(), generation_arg generation =
↳generation_arg())
- template<typename T> hpx::future<T> broadcast_to(communicator comm, generation_arg
↳generation, T &&local_result, this_site_arg this_site = this_site_arg())

```

hpx::collectives::broadcast_from

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::collectives::broadcast_to`

template<typename T>

```

hpx::future<T> hpx::collectives::broadcast_from(char const *basename, this_site_arg this_site =
this_site_arg(), generation_arg generation =
generation_arg(), root_site_arg root_site =
root_site_arg())

```

Receive a value that was broadcast to different call sites

This function sends a set of values to all call sites operating on the given base name.

Parameters

- **basename** – The base name identifying the broadcast operation
- **generation** – The generational counter identifying the sequence number of the broadcast operation performed on the given base name. This is optional and needs to be supplied only if the broadcast operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **root_site** – The site responsible for broadcasting the value. This is optional and defaults to site 0 (zero).

Returns

This function returns a future holding the value that was sent to all participating sites. It will become ready once the broadcast operation has been completed.

```
template<typename T>
hpx::future<T> hpx::collectives::broadcast_from(communicator comm, this_site_arg this_site =
                                                this_site_arg(), generation_arg generation =
                                                generation_arg())
```

Receive a value that was broadcast to different call sites

This function sends a set of values to all call sites operating on the given base name.

Receive a value that was broadcast to different call sites

This function sends a set of values to all call sites operating on the given base name.

Note: The generation values from corresponding *broadcast_to* and *broadcast_from* have to match.

Note: The generation values from corresponding *broadcast_to* and *broadcast_from* have to match.

Parameters

- **comm** – A communicator object returned from *create_communicator*
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the broadcast operation performed on the given base name. This is optional and needs to be supplied only if the broadcast operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **comm** – A communicator object returned from *create_communicator*
- **generation** – The generational counter identifying the sequence number of the broadcast operation performed on the given base name. This is optional and needs to be supplied only

if the broadcast operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.

- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

Returns

This function returns a future holding the value that was sent to all participating sites. It will become ready once the broadcast operation has been completed.

Returns

This function returns a future holding the value that was sent to all participating sites. It will become ready once the broadcast operation has been completed.

```
template<typename T>
T hpx::collectives::broadcast_from(hpx::launch::sync_policy policy, char const *basename, this_site_arg
    this_site = this_site_arg(), generation_arg generation =
    generation_arg(), root_site_arg root_site = root_site_arg())
```

Receive a value that was broadcast to different call sites

This function sends a set of values to all call sites operating on the given base name.

Parameters

- **policy** – The execution policy specifying synchronous execution.
- **basename** – The base name identifying the broadcast operation
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the broadcast operation performed on the given base name. This is optional and needs to be supplied only if the broadcast operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **root_site** – The site responsible for broadcasting the value. This is optional and defaults to site 0 (zero).

Returns

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

```
template<typename T>
T hpx::collectives::broadcast_from(hpx::launch::sync_policy policy, communicator comm, this_site_arg
    this_site = this_site_arg(), generation_arg generation =
    generation_arg())
```

Receive a value that was broadcast to different call sites

This function sends a set of values to all call sites operating on the given base name.

Receive a value that was broadcast to different call sites

This function sends a set of values to all call sites operating on the given base name.

Note: The generation values from corresponding *broadcast_to* and *broadcast_from* have to match.

Note: The generation values from corresponding *broadcast_to* and *broadcast_from* have to match.

Parameters

- **policy** – The execution policy specifying synchronous execution.
- **comm** – A communicator object returned from *create_communicator*
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.
- **generation** – The generational counter identifying the sequence number of the broadcast operation performed on the given base name. This is optional and needs to be supplied only if the broadcast operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **policy** – The execution policy specifying synchronous execution.
- **comm** – A communicator object returned from *create_communicator*
- **generation** – The generational counter identifying the sequence number of the broadcast operation performed on the given base name. This is optional and needs to be supplied only if the broadcast operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.

Returns

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

Returns

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

hpx::distributed::latch

Defined in header [hpx/latch.hpp](#)⁵⁶⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
class latch : public hpx::components::client_base<latch, hpx::lcos::server::latch>
```

Latch is an implementation of a synchronization primitive that allows multiple threads to wait for a shared event to occur before proceeding. This latch can be invoked in a distributed application.

For a local only latch

See also:

hpx::latch.

⁵⁶⁰ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/latch.hpp>

hpx/collectives/macros.hpp

Defined in header `hpx/collectives/macros.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Defines

```

HPX_REGISTER_BROADCAST_POST_ACTION_DECLARATION(...)
HPX_REGISTER_BROADCAST_POST_ACTION_DECLARATION_(...)
HPX_REGISTER_BROADCAST_POST_ACTION_DECLARATION_1(Action)
HPX_REGISTER_BROADCAST_POST_ACTION_DECLARATION_2(Action, Name)
HPX_REGISTER_BROADCAST_POST_ACTION(...)
HPX_REGISTER_BROADCAST_POST_ACTION_(...)
HPX_REGISTER_BROADCAST_POST_ACTION_1(Action)
HPX_REGISTER_BROADCAST_POST_ACTION_2(Action, Name)
HPX_REGISTER_BROADCAST_POST_WITH_INDEX_ACTION_DECLARATION(...)
HPX_REGISTER_BROADCAST_POST_WITH_INDEX_ACTION_DECLARATION_(...)
HPX_REGISTER_BROADCAST_POST_WITH_INDEX_ACTION_DECLARATION_1(Action)
HPX_REGISTER_BROADCAST_POST_WITH_INDEX_ACTION_DECLARATION_2(Action, Name)
HPX_REGISTER_BROADCAST_POST_WITH_INDEX_ACTION(...)
HPX_REGISTER_BROADCAST_POST_WITH_INDEX_ACTION_(...)
HPX_REGISTER_BROADCAST_POST_WITH_INDEX_ACTION_1(Action)
HPX_REGISTER_BROADCAST_POST_WITH_INDEX_ACTION_2(Action, Name)
HPX_REGISTER_BROADCAST_ACTION_DECLARATION(...)
HPX_REGISTER_BROADCAST_ACTION_DECLARATION_(...)
HPX_REGISTER_BROADCAST_ACTION_DECLARATION_1(Action)
HPX_REGISTER_BROADCAST_ACTION_DECLARATION_2(Action, Name)
HPX_REGISTER_BROADCAST_ACTION(...)
HPX_REGISTER_BROADCAST_ACTION_(...)
HPX_REGISTER_BROADCAST_ACTION_1(Action)
HPX_REGISTER_BROADCAST_ACTION_2(Action, Name)
HPX_REGISTER_BROADCAST_ACTION_ID(Action, Name, Id)
HPX_REGISTER_BROADCAST_WITH_INDEX_ACTION_DECLARATION(...)

```

HPX_REGISTER_BROADCAST_WITH_INDEX_ACTION_DECLARATION_(...)
HPX_REGISTER_BROADCAST_WITH_INDEX_ACTION_DECLARATION_1(Action)
HPX_REGISTER_BROADCAST_WITH_INDEX_ACTION_DECLARATION_2(Action, Name)
HPX_REGISTER_BROADCAST_WITH_INDEX_ACTION(...)
HPX_REGISTER_BROADCAST_WITH_INDEX_ACTION_(...)
HPX_REGISTER_BROADCAST_WITH_INDEX_ACTION_1(Action)
HPX_REGISTER_BROADCAST_WITH_INDEX_ACTION_2(Action, Name)
HPX_REGISTER_BROADCAST_WITH_INDEX_ACTION_ID(Action, Name, Id)
HPX_REGISTER_FOLD_ACTION_DECLARATION(...)
HPX_REGISTER_FOLD_ACTION_DECLARATION_(...)
HPX_REGISTER_FOLD_ACTION_DECLARATION_2(Action, FoldOp)
HPX_REGISTER_FOLD_ACTION_DECLARATION_3(Action, FoldOp, Name)
HPX_REGISTER_FOLD_ACTION(...)
HPX_REGISTER_FOLD_ACTION_(...)
HPX_REGISTER_FOLD_ACTION_2(Action, FoldOp)
HPX_REGISTER_FOLD_ACTION_3(Action, FoldOp, Name)
HPX_REGISTER_FOLD_WITH_INDEX_ACTION_DECLARATION(...)
HPX_REGISTER_FOLD_WITH_INDEX_ACTION_DECLARATION_(...)
HPX_REGISTER_FOLD_WITH_INDEX_ACTION_DECLARATION_2(Action, FoldOp)
HPX_REGISTER_FOLD_WITH_INDEX_ACTION_DECLARATION_3(Action, FoldOp, Name)
HPX_REGISTER_FOLD_WITH_INDEX_ACTION(...)
HPX_REGISTER_FOLD_WITH_INDEX_ACTION_(...)
HPX_REGISTER_FOLD_WITH_INDEX_ACTION_2(Action, FoldOp)
HPX_REGISTER_FOLD_WITH_INDEX_ACTION_3(Action, FoldOp, Name)
HPX_REGISTER_REDUCE_ACTION_DECLARATION(...)
HPX_REGISTER_REDUCE_ACTION_DECLARATION_(...)
HPX_REGISTER_REDUCE_ACTION_DECLARATION_2(Action, ReduceOp)
HPX_REGISTER_REDUCE_ACTION_DECLARATION_3(Action, ReduceOp, Name)
HPX_REGISTER_REDUCE_ACTION(...)
HPX_REGISTER_REDUCE_ACTION_(...)
HPX_REGISTER_REDUCE_ACTION_2(Action, ReduceOp)

```

HPX_REGISTER_REDUCE_ACTION_3(Action, ReduceOp, Name)
HPX_REGISTER_REDUCE_WITH_INDEX_ACTION_DECLARATION(...)
HPX_REGISTER_REDUCE_WITH_INDEX_ACTION_DECLARATION_(...)
HPX_REGISTER_REDUCE_WITH_INDEX_ACTION_DECLARATION_2(Action, ReduceOp)
HPX_REGISTER_REDUCE_WITH_INDEX_ACTION_DECLARATION_3(Action, ReduceOp, Name)
HPX_REGISTER_REDUCE_WITH_INDEX_ACTION(...)
HPX_REGISTER_REDUCE_WITH_INDEX_ACTION_(...)
HPX_REGISTER_REDUCE_WITH_INDEX_ACTION_2(Action, ReduceOp)
HPX_REGISTER_REDUCE_WITH_INDEX_ACTION_3(Action, ReduceOp, Name)

```

hpx::collectives::channel_communicator

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::collectives::create_channel_communicator`
- `hpx::collectives::get`
- `hpx::collectives::set`

class **channel_communicator**

A handle identifying the communication channel to use for get/set operations

hpx::collectives::create_channel_communicator

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::collectives::channel_communicator`
- `hpx::collectives::get`
- `hpx::collectives::set`

```

hpx::future<channel_communicator> hpx::collectives::create_channel_communicator(char const
                                                                    *basename,
                                                                    num_sites_arg
                                                                    num_sites =
                                                                    num_sites_arg(),
                                                                    this_site_arg
                                                                    this_site =
                                                                    this_site_arg())

```

Create a new communicator object usable with peer-to-peer channel-based operations

This functions creates a new communicator object that can be called in order to pre-allocate a communicator object usable with multiple invocations of channel-based peer-to-peer operations.

Note: The caller must ensure that the communicator object returned by this function is kept alive (i.e., does not go out of scope) until all participating sites have successfully connected. Destroying the communicator before all sites have reached the creation point may cause the remaining sites to hang indefinitely. Consider using a barrier or similar synchronization mechanism after creating the communicator to ensure all sites are connected before allowing it to be destroyed.

Parameters

- **basename** – The base name identifying the collective operation
- **num_sites** – The number of participating sites (default: all localities).
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

Returns

This function returns a future to a new communicator object usable with the collective operation.

```
channel_communicator hpx::collectives::create_channel_communicator(hpx::launch::sync_policy, char
                                                                    const *basename,
                                                                    num_sites_arg num_sites =
                                                                    num_sites_arg(), this_site_arg
                                                                    this_site = this_site_arg())
```

Create a new communicator object usable with peer-to-peer channel-based operations

This functions creates a new communicator object that can be called in order to pre-allocate a communicator object usable with multiple invocations of channel-based peer-to-peer operations.

Note: The caller must ensure that the communicator object returned by this function is kept alive (i.e., does not go out of scope) until all participating sites have successfully connected. Destroying the communicator before all sites have reached the creation point may cause the remaining sites to hang indefinitely. Consider using a barrier or similar synchronization mechanism after creating the communicator to ensure all sites are connected before allowing it to be destroyed.

Parameters

- **basename** – The base name identifying the collective operation
- **num_sites** – The number of participating sites (default: all localities).
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

Returns

This function returns a new communicator object usable with the collective operation.

hpx::collectives::get

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::collectives::channel_communicator`
- `hpx::collectives::create_channel_communicator`
- `hpx::collectives::set`

```
template<typename T>
hpx::future<T> hpx::collectives::get(channel_communicator comm, that_site_arg site, tag_arg tag =
                                     tag_arg())
```

Send a value to the given site

This function receives a value from the given site based on the given communicator.

Parameters

- **comm** – The channel communicator object to use for the data transfer
- **site** – The source site

Returns

This function returns a `future<T>` that becomes ready once the data transfer operation has finished. The future will hold the received value.

hpx::collectives::set

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::collectives::channel_communicator`
- `hpx::collectives::create_channel_communicator`
- `hpx::collectives::get`

```
template<typename T>
hpx::future<void> hpx::collectives::set(channel_communicator comm, that_site_arg site, T &&value,
                                       tag_arg tag = tag_arg())
```

Send a value to the given site

This function sends a value to the given site based on the given communicator.

Parameters

- **comm** – The channel communicator object to use for the data transfer
- **site** – The destination site
- **value** – The value to send
- **tag** – The (optional) tag identifying the concrete operation

Returns

This function returns a *future<void>* that becomes ready once the data transfer operation has finished.

hpx::collectives::reduce_here

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::collectives::reduce_there*

template<typename **T**, typename **F**>

```
hpx::future<std::decay_t<T>> hpx::collectives::reduce_here(char const *basename, T &&result, F &&op,  
                                                           num_sites_arg num_sites = num_sites_arg(),  
                                                           this_site_arg this_site = this_site_arg(),  
                                                           generation_arg generation =  
                                                           generation_arg())
```

Reduce a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

Parameters

- **basename** – The base name identifying the reduce operation
- **result** – A value to reduce on the central reduction point from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **num_sites** – The number of participating sites (default: all localities).
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.
- **generation** – The generational counter identifying the sequence number of the reduce operation performed on the given base name. This is optional and needs to be supplied only if the reduce operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.

Returns

This function returns a future holding a vector with all values send by all participating sites. It will become ready once the reduce operation has been completed.

template<typename **T**, typename **F**>

```
hpx::future<decay_t<T>> hpx::collectives::reduce_here(communicator comm, T &&result, F &&op,  
                                                       this_site_arg this_site = this_site_arg(),  
                                                       generation_arg generation = generation_arg())
```

Reduce a set of values from different call sites

This function receives a set of values that are the result of applying a given operator on values supplied from all call sites operating on the given base name.

Reduce a set of values from different call sites

This function receives a set of values that are the result of applying a given operator on values supplied from all call sites operating on the given base name.

Note: The generation values from corresponding *reduce_here* and *reduce_there* have to match.

Note: The generation values from corresponding *reduce_here* and *reduce_there* have to match.

Parameters

- **comm** – A communicator object returned from *create_communicator*
- **result** – A value to reduce on the root_site from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.
- **generation** – The generational counter identifying the sequence number of the reduce operation performed on the given base name. This is optional and needs to be supplied only if the reduce operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **comm** – A communicator object returned from *create_communicator*
- **result** – A value to reduce on the root_site from this call site.
- **op** – Reduction operation to apply to all values supplied from all participating sites
- **generation** – The generational counter identifying the sequence number of the reduce operation performed on the given base name. This is optional and needs to be supplied only if the reduce operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.

Returns

This function returns a future holding a value calculated based on the values send by all participating sites. It will become ready once the reduce operation has been completed.

Returns

This function returns a future holding a value calculated based on the values send by all participating sites. It will become ready once the reduce operation has been completed.

Warning: doxygenfunction: Unable to resolve function “hpx::collectives::reduce_here” with arguments (hpx::launch::sync_policy, char const*, T&&, F&&, num_sites_arg, this_site_arg, generation_arg) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:

```
- template<typename T, typename F> decltype(auto) reduce_here(hpx::launch::sync_
  ↳policy, char const *basename, T &&result, F &&op, num_sites_arg num_sites = num_
  ↳sites_arg(), this_site_arg this_site = this_site_arg(), generation_arg generation =
  ↳generation_arg())
- template<typename T, typename F> decltype(auto) reduce_here(hpx::launch::sync_
  ↳policy, communicator comm, T &&result, F &&op, this_site_arg this_site = this_site_
  ↳arg(), generation_arg generation = generation_arg())
- template<typename T, typename F> hpx::future<decay_t<T>> reduce_here(communicator_
  ↳comm, T &&result, F &&op, this_site_arg this_site = this_site_arg(), generation_arg_
```

```

↪ generation = generation_arg())
- template<typename T, typename F> hpx::future<std::decay_t<T>> reduce_here(char_
↪ const *basename, T &&result, F &&op, num_sites_arg num_sites = num_sites_arg(),
↪ this_site_arg this_site = this_site_arg(), generation_arg generation = generation_
↪ arg())

```

Warning: doxygenfunction: Unable to resolve function “hpx::collectives::reduce_here” with arguments (hpx::launch::sync_policy, communicator, T&&, F&&, this_site_arg, generation_arg) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:

```

- template<typename T, typename F> decltype(auto) reduce_here(hpx::launch::sync_
↪ policy, char const *basename, T &&result, F &&op, num_sites_arg num_sites = num_
↪ sites_arg(), this_site_arg this_site = this_site_arg(), generation_arg generation =
↪ generation_arg())
- template<typename T, typename F> decltype(auto) reduce_here(hpx::launch::sync_
↪ policy, communicator comm, T &&result, F &&op, this_site_arg this_site = this_site_
↪ arg(), generation_arg generation = generation_arg())
- template<typename T, typename F> hpx::future<decay_t<T>> reduce_here(communicator_
↪ comm, T &&result, F &&op, this_site_arg this_site = this_site_arg(), generation_arg_
↪ generation = generation_arg())
- template<typename T, typename F> hpx::future<std::decay_t<T>> reduce_here(char_
↪ const *basename, T &&result, F &&op, num_sites_arg num_sites = num_sites_arg(),
↪ this_site_arg this_site = this_site_arg(), generation_arg generation = generation_
↪ arg())

```

hpx::collectives::reduce_there

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::collectives::reduce_here`

template<typename T, typename F>

```

hpx::future<void> hpx::collectives::reduce_there(char const *basename, T &&result, this_site_arg this_site
= this_site_arg(), generation_arg generation =
generation_arg(), root_site_arg root_site =
root_site_arg())

```

Reduce a given value at the given call site

This function transmits the value given by *result* to a central reduce site (where the corresponding *reduce_here* is executed)

Parameters

- **basename** – The base name identifying the reduction operation
- **result** – A future referring to the value to transmit to the central reduction point from this call site.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

- **generation** – The generational counter identifying the sequence number of the reduce operation performed on the given base name. This is optional and needs to be supplied only if the reduce operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **root_site** – The sequence number of the central reduction point (usually the locality id). This value is optional and defaults to 0.

Returns

This function returns a *future<void>*. It will become ready once the reduction operation has been completed.

```
template<typename T>
hpx::future<void> hpx::collectives::reduce_there(communicator comm, T &&result, this_site_arg this_site
                                              = this_site_arg(), generation_arg generation =
                                              generation_arg())
```

Reduce a given value at the given call site

This function transmits the value given by *result* to a central reduce site (where the corresponding *reduce_here* is executed)

Reduce a given value at the given call site

This function transmits the value given by *result* to a central reduce site (where the corresponding *reduce_here* is executed)

Note: The generation values from corresponding *reduce_here* and *reduce_there* have to match.

Note: The generation values from corresponding *reduce_here* and *reduce_there* have to match.

Parameters

- **comm** – A communicator object returned from *create_communicator*
- **result** – A value to reduce on the central reduction point from this call site.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.
- **generation** – The generational counter identifying the sequence number of the reduce operation performed on the given base name. This is optional and needs to be supplied only if the reduce operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **comm** – A communicator object returned from *create_communicator*
- **result** – A value to reduce on the central reduction point from this call site.
- **generation** – The generational counter identifying the sequence number of the reduce operation performed on the given base name. This is optional and needs to be supplied only if the reduce operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.

Returns

This function returns a future holding a value calculated based on the values send by all participating sites. It will become ready once the reduce operation has been completed.

Returns

This function returns a future holding a value calculated based on the values send by all participating sites. It will become ready once the reduce operation has been completed.

```
template<typename T>
void hpx::collectives::reduce_there(hpx::launch::sync_policy, char const *basename, T &&result,
                                   this_site_arg this_site = this_site_arg(), generation_arg generation =
                                   generation_arg(), root_site_arg root_site = root_site_arg())
```

Reduce a given value at the given call site

This function transmits the value given by *result* to a central reduce site (where the corresponding *reduce_here* is executed)

Parameters

- **basename** – The base name identifying the reduction operation
- **result** – A future referring to the value to transmit to the central reduction point from this call site.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.
- **generation** – The generational counter identifying the sequence number of the reduce operation performed on the given base name. This is optional and needs to be supplied only if the reduce operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **root_site** – The sequence number of the central reduction point (usually the locality id). This value is optional and defaults to 0.

```
template<typename T>
void hpx::collectives::reduce_there(hpx::launch::sync_policy, communicator comm, T &&result,
                                   this_site_arg this_site = this_site_arg(), generation_arg generation =
                                   generation_arg())
```

Reduce a given value at the given call site

This function transmits the value given by *result* to a central reduce site (where the corresponding *reduce_here* is executed)

Reduce a given value at the given call site

This function transmits the value given by *result* to a central reduce site (where the corresponding *reduce_here* is executed)

Note: The generation values from corresponding *reduce_here* and *reduce_there* have to match.

Note: The generation values from corresponding *reduce_here* and *reduce_there* have to match.

Parameters

- **comm** – A communicator object returned from *create_communicator*
- **result** – A value to reduce on the central reduction point from this call site.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.
- **generation** – The generational counter identifying the sequence number of the reduce operation performed on the given base name. This is optional and needs to be supplied only if the reduce operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **comm** – A communicator object returned from *create_communicator*
- **result** – A value to reduce on the central reduction point from this call site.
- **generation** – The generational counter identifying the sequence number of the reduce operation performed on the given base name. This is optional and needs to be supplied only if the reduce operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.

hpx::lcos::spmd_block

Defined in header *hpx/collectives.hpp*.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::lcos::define_spmd_block*

struct spmd_block

The class *spmd_block* defines an interface for launching multiple images while giving handles to each image to interact with the remaining images. The *define_spmd_block* function templates create multiple images of a user-defined action and launches them in a possibly separate thread. A temporary *spmd_block* object is created and diffused to each image. The constraint for the action given to the *define_spmd_block* function is to accept a *spmd_block* as first parameter.

hpx::lcos::define_spmd_block

Defined in header *hpx/collectives.hpp*.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::lcos::spmd_block*

template<typename **F**, typename ...**Args**>

hpx::future<void> *hpx::lcos::define_spmd_block*(*std::string* &&name, *std::size_t* images_per_locality, *F*&&, *Args*&&... args)

hpx::distributed::barrier

Defined in header [hpx/barrier.hpp](#)⁵⁶¹.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

explicit **hpx::distributed::barrier**(std::string const &base_name)

Creates a barrier, rank is locality id, size is number of localities

A barrier *base_name* is created. It expects that *hpx::get_num_localities()* participate and the local rank is *hpx::get_locality_id()*.

Parameters

base_name – The name of the barrier

hpx::distributed::barrier(std::string const &base_name, std::size_t num)

Creates a barrier with a given size, rank is locality id

A barrier *base_name* is created. It expects that *num* participate and the local rank is *hpx::get_locality_id()*.

Parameters

- **base_name** – The name of the barrier
- **num** – The number of participating threads

hpx::distributed::barrier(std::string const &base_name, std::size_t num, std::size_t rank)

Creates a barrier with a given size and rank

A barrier *base_name* is created. It expects that *num* participate and the local rank is *rank*.

Parameters

- **base_name** – The name of the barrier
- **num** – The number of participating threads
- **rank** – The rank of the calling site for this invocation

hpx::distributed::barrier(std::string const &base_name, std::vector<std::size_t> const &ranks, std::size_t rank)

Creates a barrier with a vector of ranks

A barrier *base_name* is created. It expects that *ranks.size()* and the local rank is *rank* (must be contained in *ranks*).

Parameters

- **base_name** – The name of the barrier
- **ranks** – Gives a list of participating ranks (this could be derived from a list of locality ids)
- **rank** – The rank of the calling site for this invocation

⁵⁶¹ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/barrier.hpp>

hpx::collectives::gather_here

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::collectives::gather_there`

```
template<typename T>
hpx::future<std::vector<decay_t<T>>> hpx::collectives::gather_here(char const *basename, T &&result,
                                                                    num_sites_arg num_sites =
                                                                    num_sites_arg(), this_site_arg
                                                                    this_site = this_site_arg(),
                                                                    generation_arg generation =
                                                                    generation_arg())
```

Gather a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

Parameters

- **basename** – The base name identifying the gather operation
- **result** – The value to transmit to the central gather point from this call site.
- **num_sites** – The number of participating sites (default: all localities).
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the gather operation performed on the given base name. This is optional and needs to be supplied only if the gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.

Returns

This function returns a future holding a vector with all gathered values. It will become ready once the gather operation has been completed.

```
template<typename T>
hpx::future<std::vector<decay_t<T>>> hpx::collectives::gather_here(communicator comm, T &&result,
                                                                    this_site_arg this_site =
                                                                    this_site_arg(), generation_arg
                                                                    generation = generation_arg())
```

Gather a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

Gather a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

Note: The generation values from corresponding *gather_here* and *gather_there* have to match.

Note: The generation values from corresponding *gather_here* and *gather_there* have to match.

Parameters

- **comm** – A communicator object returned from *create_communicator*
- **result** – The value to transmit to the central gather point from this call site.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.
- **generation** – The generational counter identifying the sequence number of the gather operation performed on the given base name. This is optional and needs to be supplied only if the gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **comm** – A communicator object returned from *create_communicator*
- **result** – The value to transmit to the central gather point from this call site.
- **generation** – The generational counter identifying the sequence number of the gather operation performed on the given base name. This is optional and needs to be supplied only if the gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.

Returns

This function returns a future holding a vector with all gathered values. It will become ready once the gather operation has been completed.

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This function returns a future holding a vector with all gathered values. It will become ready once the gather operation has been completed.

Warning: doxygenfunction: Unable to resolve function “hpx::collectives::gather_here” with arguments (hpx::launch::sync_policy, char const*, T&&, num_sites_arg, this_site_arg, generation_arg) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:

```
- template<typename T> decltype(auto) gather_here(hpx::launch::sync_policy, char_
↳const *basename, T &&result, num_sites_arg num_sites = num_sites_arg(), this_site_
↳arg this_site = this_site_arg(), generation_arg generation = generation_arg())
- template<typename T> decltype(auto) gather_here(hpx::launch::sync_policy,
↳communicator comm, T &&result, this_site_arg this_site = this_site_arg(),
↳generation_arg generation = generation_arg())
- template<typename T> hpx::future<std::vector<decay_t<T>>> gather_here(char const_
↳*basename, T &&result, num_sites_arg num_sites = num_sites_arg(), this_site_arg_
↳this_site = this_site_arg(), generation_arg generation = generation_arg())
- template<typename T> hpx::future<std::vector<decay_t<T>>> gather_here(communicator_
↳comm, T &&result, this_site_arg this_site = this_site_arg(), generation_arg_
↳generation = generation_arg())
```

Warning: doxygenfunction: Unable to resolve function “hpx::collectives::gather_here” with arguments (hpx::launch::sync_policy, communicator, T&&, this_site_arg, generation_arg) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:


```

- template<typename T> decltype(auto) gather_here(hpx::launch::sync_policy, char_
↪const *basename, T &&result, num_sites_arg num_sites = num_sites_arg(), this_site_
↪arg this_site = this_site_arg(), generation_arg generation = generation_arg())
- template<typename T> decltype(auto) gather_here(hpx::launch::sync_policy,
↪communicator comm, T &&result, this_site_arg this_site = this_site_arg(),
↪generation_arg generation = generation_arg())
- template<typename T> hpx::future<std::vector<decay_t<T>>> gather_here(char const_
↪*basename, T &&result, num_sites_arg num_sites = num_sites_arg(), this_site_arg_
↪this_site = this_site_arg(), generation_arg generation = generation_arg())
- template<typename T> hpx::future<std::vector<decay_t<T>>> gather_here(communicator_
↪comm, T &&result, this_site_arg this_site = this_site_arg(), generation_arg_
↪generation = generation_arg())

```

hpx::collectives::gather_there

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::collectives::gather_here`

```

template<typename T>
hpx::future<void> hpx::collectives::gather_there(char const *basename, T &&result, this_site_arg this_site
= this_site_arg(), generation_arg generation =
generation_arg(), root_site_arg root_site =
root_site_arg())

```

Gather a given value at the given call site

This function transmits the value given by *result* to a central gather site (where the corresponding *gather_here* is executed)

Parameters

- **basename** – The base name identifying the gather operation
- **result** – The value to transmit to the central gather point from this call site.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the gather operation performed on the given base name. This is optional and needs to be supplied only if the gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **root_site** – The sequence number of the central gather point (usually the locality id). This value is optional and defaults to 0.

Returns

This function returns a future holding a vector with all gathered values. It will become ready once the gather operation has been completed.

```

template<typename T>
hpx::future<void> hpx::collectives::gather_there(communicator comm, T &&result, this_site_arg this_site
= this_site_arg(), generation_arg generation =
generation_arg())

```

Gather a given value at the given call site

This function transmits the value given by *result* to a central gather site (where the corresponding *gather_here* is executed)

Gather a given value at the given call site

This function transmits the value given by *result* to a central gather site (where the corresponding *gather_here* is executed)

Note: The generation values from corresponding *gather_here* and *gather_there* have to match.

Note: The generation values from corresponding *gather_here* and *gather_there* have to match.

Parameters

- **comm** – A communicator object returned from *create_communicator*
- **result** – The value to transmit to the central gather point from this call site.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.
- **generation** – The generational counter identifying the sequence number of the gather operation performed on the given base name. This is optional and needs to be supplied only if the gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **comm** – A communicator object returned from *create_communicator*
- **result** – The value to transmit to the central gather point from this call site.
- **generation** – The generational counter identifying the sequence number of the gather operation performed on the given base name. This is optional and needs to be supplied only if the gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.

Returns

This function returns a future holding a vector with all gathered values. It will become ready once the gather operation has been completed.

Returns

This function returns a future holding a vector with all gathered values. It will become ready once the gather operation has been completed.

```
template<typename T>
void hpx::collectives::gather_there(hpx::launch::sync_policy, char const *basename, T &&result,
                                   this_site_arg this_site = this_site_arg(), generation_arg generation =
                                   generation_arg(), root_site_arg root_site = root_site_arg())
```

Gather a given value at the given call site

This function transmits the value given by *result* to a central gather site (where the corresponding *gather_here* is executed)

Parameters

- **basename** – The base name identifying the gather operation
- **result** – The value to transmit to the central gather point from this call site.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.
- **generation** – The generational counter identifying the sequence number of the gather operation performed on the given base name. This is optional and needs to be supplied only if the gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **root_site** – The sequence number of the central gather point (usually the locality id). This value is optional and defaults to 0.

Returns

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

```
template<typename T>
void hpx::collectives::gather_there(hpx::launch::sync_policy, communicator comm, T &&result,
                                   this_site_arg this_site = this_site_arg(), generation_arg generation =
                                   generation_arg())
```

Gather a given value at the given call site

This function transmits the value given by *result* to a central gather site (where the corresponding *gather_here* is executed)

Gather a given value at the given call site

This function transmits the value given by *result* to a central gather site (where the corresponding *gather_here* is executed)

Note: The generation values from corresponding *gather_here* and *gather_there* have to match.

Note: The generation values from corresponding *gather_here* and *gather_there* have to match.

Parameters

- **comm** – A communicator object returned from *create_communicator*
- **result** – The value to transmit to the central gather point from this call site.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever *hpx::get_locality_id()* returns.
- **generation** – The generational counter identifying the sequence number of the gather operation performed on the given base name. This is optional and needs to be supplied only if the gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **comm** – A communicator object returned from *create_communicator*

- **result** – The value to transmit to the central gather point from this call site.
- **generation** – The generational counter identifying the sequence number of the gather operation performed on the given base name. This is optional and needs to be supplied only if the gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

Returns

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

Returns

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

hpx::collectives::num_sites_arg

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::collectives::this_site_arg`
- `hpx::collectives::that_site_arg`
- `hpx::collectives::generation_arg`
- `hpx::collectives::root_site_arg`
- `hpx::collectives::tag_arg`
- `hpx::collectives::arity_arg`
- `hpx::collectives::flat_fallback_threshold_arg`

using `hpx::collectives::num_sites_arg = detail::argument_type<detail::num_sites_tag>`

The number of participating sites (default: all localities)

hpx::collectives::this_site_arg

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::collectives::num_sites_arg`
- `hpx::collectives::that_site_arg`
- `hpx::collectives::generation_arg`
- `hpx::collectives::root_site_arg`
- `hpx::collectives::tag_arg`
- `hpx::collectives::arity_arg`

- `hpx::collectives::flat_fallback_threshold_arg`

using `hpx::collectives::this_site_arg` = detail::argument_type<detail::this_site_tag>

The local end of the communication channel.

`hpx::collectives::that_site_arg`

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::collectives::num_sites_arg`
- `hpx::collectives::this_site_arg`
- `hpx::collectives::generation_arg`
- `hpx::collectives::root_site_arg`
- `hpx::collectives::tag_arg`
- `hpx::collectives::arity_arg`
- `hpx::collectives::flat_fallback_threshold_arg`

using `hpx::collectives::that_site_arg` = detail::argument_type<detail::that_site_tag>

The opposite end of the communication channel.

`hpx::collectives::generation_arg`

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::collectives::num_sites_arg`
- `hpx::collectives::this_site_arg`
- `hpx::collectives::that_site_arg`
- `hpx::collectives::root_site_arg`
- `hpx::collectives::tag_arg`
- `hpx::collectives::arity_arg`
- `hpx::collectives::flat_fallback_threshold_arg`

using `hpx::collectives::generation_arg` = detail::argument_type<detail::generation_tag>

The generational counter identifying the sequence number of the operation performed on the given base name. It needs to be supplied only if the operation on the given base name has to be performed more than once. It must be a positive number greater than zero. On any given communicator instance, operations that pass an explicit generation may not follow operations that used the default: the internal counter's position is no longer reliably known to the caller. The reverse transition, from explicit numbering back to the default, remains valid.

hpx::collectives::root_site_arg

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::collectives::num_sites_arg`
- `hpx::collectives::this_site_arg`
- `hpx::collectives::that_site_arg`
- `hpx::collectives::generation_arg`
- `hpx::collectives::tag_arg`
- `hpx::collectives::arity_arg`
- `hpx::collectives::flat_fallback_threshold_arg`

using `hpx::collectives::root_site_arg = detail::argument_type<detail::root_site_tag, 0>`

The site that is responsible for creating the support object of the operation. It defaults to '0' (zero).

hpx::collectives::tag_arg

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::collectives::num_sites_arg`
- `hpx::collectives::this_site_arg`
- `hpx::collectives::that_site_arg`
- `hpx::collectives::generation_arg`
- `hpx::collectives::root_site_arg`
- `hpx::collectives::arity_arg`
- `hpx::collectives::flat_fallback_threshold_arg`

using `hpx::collectives::tag_arg = detail::argument_type<detail::tag_tag, 0>`

The tag identifying the concrete operation.

hpx::collectives::arity_arg

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::collectives::num_sites_arg`
- `hpx::collectives::this_site_arg`

- `hpx::collectives::that_site_arg`
- `hpx::collectives::generation_arg`
- `hpx::collectives::root_site_arg`
- `hpx::collectives::tag_arg`
- `hpx::collectives::flat_fallback_threshold_arg`

using `hpx::collectives::arity_arg` = `detail::argument_type<detail::arity_tag, 2>`

The number of children each of the communication nodes is connected to (default: 2).

`hpx::collectives::flat_fallback_threshold_arg`

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::collectives::num_sites_arg`
- `hpx::collectives::this_site_arg`
- `hpx::collectives::that_site_arg`
- `hpx::collectives::generation_arg`
- `hpx::collectives::root_site_arg`
- `hpx::collectives::tag_arg`
- `hpx::collectives::arity_arg`

using `hpx::collectives::flat_fallback_threshold_arg` =
`detail::argument_type<detail::flat_fallback_threshold_tag, 16>`

The site-count threshold below which a hierarchical_communicator collapses to a single flat communicator spanning all sites. At small site counts, flat collectives outperform hierarchical composition because tree-walking overhead exceeds the synchronization-depth benefit. Pass 0 to disable the fallback and always build a tree.

`hpx::collectives::all_gather`

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

template<typename **T**>

```
hpx::future<std::vector<std::decay_t<T>>>> hpx::collectives::all_gather(char const *basename, T
&&result, num_sites_arg
num_sites = num_sites_arg(),
this_site_arg this_site =
this_site_arg(), generation_arg
generation = generation_arg(),
root_site_arg root_site =
root_site_arg())
```

AllGather a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

Parameters

- **basename** – The base name identifying the all_gather operation
- **result** – The value to transmit to all participating sites from this call site.
- **num_sites** – The number of participating sites (default: all localities).
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the all_gather operation performed on the given base name. This is optional and needs to be supplied only if the all_gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **root_site** – The site that is responsible for creating the all_gather support object. This value is optional and defaults to '0' (zero).

Returns

This function returns a future holding a vector with all values send by all participating sites. It will become ready once the all_gather operation has been completed.

```
template<typename T>
hpx::future<std::vector<std::decay_t<T>>> hpx::collectives::all_gather(comm, T
                                                                    &&result, this_site_arg this_site =
                                                                    this_site_arg(), generation_arg
                                                                    generation = generation_arg())
```

AllGather a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

AllGather a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

Parameters

- **comm** – A communicator object returned from `create_communicator`
- **result** – The value to transmit to all participating sites from this call site.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the all_gather operation performed on the given base name. This is optional and needs to be supplied only if the all_gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **comm** – A communicator object returned from `create_communicator`
- **result** – The value to transmit to all participating sites from this call site.
- **generation** – The generational counter identifying the sequence number of the all_gather operation performed on the given base name. This is optional and needs to be supplied only if the all_gather operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

Returns

This function returns a future holding a vector with all values send by all participating sites. It will become ready once the `all_gather` operation has been completed.

Returns

This function returns a future holding a vector with all values send by all participating sites. It will become ready once the `all_gather` operation has been completed.

```
template<typename T>
std::vector<std::decay_t<T>> hpx::collectives::all_gather(hpx::launch::sync_policy, char const
                                                         *basename, T &&result, num_sites_arg
                                                         num_sites = num_sites_arg(), this_site_arg
                                                         this_site = this_site_arg(), generation_arg
                                                         generation = generation_arg(), root_site_arg
                                                         root_site = root_site_arg())
```

AllGather a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

Parameters

- **basename** – The base name identifying the `all_gather` operation
- **result** – The value to transmit to all participating sites from this call site.
- **num_sites** – The number of participating sites (default: all localities).
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the `all_gather` operation performed on the given base name. This is optional and needs to be supplied only if the `all_gather` operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **root_site** – The site that is responsible for creating the `all_gather` support object. This value is optional and defaults to '0' (zero).

Returns

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

```
template<typename T>
std::vector<std::decay_t<T>> hpx::collectives::all_gather(hpx::launch::sync_policy, communicator comm,
                                                         T &&result, this_site_arg this_site =
                                                         this_site_arg(), generation_arg generation =
                                                         generation_arg())
```

AllGather a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

AllGather a set of values from different call sites

This function receives a set of values from all call sites operating on the given base name.

Parameters

- **comm** – A communicator object returned from `create_communicator`
- **result** – The value to transmit to all participating sites from this call site.

- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.
- **generation** – The generational counter identifying the sequence number of the `all_gather` operation performed on the given base name. This is optional and needs to be supplied only if the `all_gather` operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **comm** – A communicator object returned from `create_communicator`
- **result** – The value to transmit to all participating sites from this call site.
- **generation** – The generational counter identifying the sequence number of the `all_gather` operation performed on the given base name. This is optional and needs to be supplied only if the `all_gather` operation on the given base name has to be performed more than once. The generation number (if given) must be a positive number greater than zero.
- **this_site** – The sequence number of this invocation (usually the locality id). This value is optional and defaults to whatever `hpx::get_locality_id()` returns.

Returns

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

Returns

This function returns a vector with all values send by all participating sites. This function executes synchronously and directly returns the result.

hpx::lcos::fold

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::lcos::fold_with_index`
- `hpx::lcos::inverse_fold`
- `hpx::lcos::inverse_fold_with_index`

Warning: doxygenfunction: Unable to resolve function “hpx::lcos::fold” with arguments “None”. Candidate function could not be parsed. Parsing error is Error in template parameter list. If parameter: Error when parsing template parameter. If unconstrained type parameter or template type parameter: Invalid C++ declaration: Expected ‘typename’ or ‘class’ in the beginning of template type parameter. [error at 73] template<typename Action, typename FoldOp, typename Init, typename ArgN, ...> hpx::future< decltype(Action(hpx::id_type, ArgN,...))> fold (std::vector< hpx::id_type > const &ids, FoldOp &&fold_op, Init &&init, ArgN argN,...) _____^ If constrained type parameter or non-type parameter: Invalid C++ declaration: Expected identifier in nested name. [error at 73] template<typename Action, typename FoldOp, typename Init, typename ArgN, ...> hpx::future< decltype(Action(hpx::id_type, ArgN,...))> fold (std::vector< hpx::id_type > const &ids, FoldOp &&fold_op, Init &&init, ArgN argN,...) _____^ If no parameter: Invalid C++ declaration: Expected “,” or “>”. [error at 73] template<typename Action, typename FoldOp, typename Init, typename ArgN, ...> hpx::future< decltype(Action(hpx::id_type, ArgN,...))> fold (std::vector< hpx::id_type > const &ids, FoldOp &&fold_op, Init &&init, ArgN argN,...) _____^

hpx::lcos::fold_with_index

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::lcos::fold`
- `hpx::lcos::inverse_fold`
- `hpx::lcos::inverse_fold_with_index`

Warning: doxygenfunction: Unable to resolve function “hpx::lcos::fold_with_index” with arguments “None”. Candidate function could not be parsed. Parsing error is Error in template parameter list. If parameter: Error when parsing template parameter. If unconstrained type parameter or template type parameter: Invalid C++ declaration: Expected ‘typename’ or ‘class’ in the beginning of template type parameter. [error at 73] template<typename Action, typename FoldOp, typename Init, typename ArgN, ...> hpx::future< decltype(Action(hpx::id_type, ArgN,..., std::size_t))> fold_with_index (std::vector< hpx::id_type > const &ids, FoldOp &&fold_op, Init &&init, ArgN argN,...) _____^

If constrained type parameter or non-type parameter: Invalid C++ declaration: Expected identifier in nested name. [error at 73] template<typename Action, typename FoldOp, typename Init, typename ArgN, ...> hpx::future< decltype(Action(hpx::id_type, ArgN,..., std::size_t))> fold_with_index (std::vector< hpx::id_type > const &ids, FoldOp &&fold_op, Init &&init, ArgN argN,...) _____

^ If no parameter: Invalid C++ declaration: Expected “,” or “>”. [error at 73] template<typename Action, typename FoldOp, typename Init, typename ArgN, ...> hpx::future< decltype(Action(hpx::id_type, ArgN,..., std::size_t))> fold_with_index (std::vector< hpx::id_type > const &ids, FoldOp &&fold_op, Init &&init, ArgN argN,...) _____^

hpx::lcos::inverse_fold

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::lcos::fold`
- `hpx::lcos::fold_with_index`
- `hpx::lcos::inverse_fold_with_index`

Warning: doxygenfunction: Unable to resolve function “hpx::lcos::inverse_fold” with arguments “None”. Candidate function could not be parsed. Parsing error is Error in template parameter list. If parameter: Error when parsing template parameter. If unconstrained type parameter or template type parameter: Invalid C++ declaration: Expected ‘typename’ or ‘class’ in the beginning of template type parameter. [error at 73] template<typename Action, typename FoldOp, typename Init, typename ArgN, ...> hpx::future< decltype(Action(hpx::id_type, ArgN,...))> inverse_fold (std::vector< hpx::id_type > const &ids, FoldOp &&fold_op, Init &&init, ArgN argN,...) _____^

If constrained type parameter or non-type parameter: Invalid C++ declaration: Expected identifier in nested name. [error at 73] template<typename Action, typename FoldOp, typename Init, typename ArgN, ...> hpx::future< decltype(Action(hpx::id_type, ArgN,...))> inverse_fold (std::vector< hpx::id_type > const &ids, FoldOp &&fold_op, Init &&init, ArgN argN,...) _____

^ If no parameter: Invalid C++ declaration: Expected

“,” or “>”. [error at 73] template<typename Action, typename FoldOp, typename Init, typename ArgN, ...> hpx::future< decltype(Action(hpx::id_type, ArgN,...))> inverse_fold (std::vector< hpx::id_type > const &ids, FoldOp &&fold_op, Init &&init, ArgN argN,...) _____^

hpx::lcos::inverse_fold_with_index

Defined in header `hpx/collectives.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::lcos::fold`
- `hpx::lcos::fold_with_index`
- `hpx::lcos::inverse_fold`

Warning: doxygenfunction: Unable to resolve function “hpx::lcos::inverse_fold_with_index” with arguments “None”. Candidate function could not be parsed. Parsing error is Error in template parameter list. If parameter: Error when parsing template parameter. If unconstrained type parameter or template type parameter: Invalid C++ declaration: Expected ‘typename’ or ‘class’ in the beginning of template type parameter. [error at 73] template<typename Action, typename FoldOp, typename Init, typename ArgN, ...> hpx::future< decltype(Action(hpx::id_type, ArgN,..., std::size_t))> inverse_fold_with_index (std::vector< hpx::id_type > const &ids, FoldOp &&fold_op, Init &&init, ArgN argN,...) _____^ If constrained type parameter or non-type parameter: Invalid C++ declaration: Expected identifier in nested name. [error at 73] template<typename Action, typename FoldOp, typename Init, typename ArgN, ...> hpx::future< decltype(Action(hpx::id_type, ArgN,..., std::size_t))> inverse_fold_with_index (std::vector< hpx::id_type > const &ids, FoldOp &&fold_op, Init &&init, ArgN argN,...) _____^ If no parameter: Invalid C++ declaration: Expected “,” or “>”. [error at 73] template<typename Action, typename FoldOp, typename Init, typename ArgN, ...> hpx::future< decltype(Action(hpx::id_type, ArgN,..., std::size_t))> inverse_fold_with_index (std::vector< hpx::id_type > const &ids, FoldOp &&fold_op, Init &&init, ArgN argN,...) _____^

components

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx::components::client

Defined in header `hpx/components.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

template<typename **Component**, typename **Data** = void>

class **client** : public `hpx::components::client_base<client<Component, void>, Component, void>`

The client class is a wrapper that manages a distributed component. It extends `client_base` with specific Component and Data types.

Template Parameters

- **Component** – The type of the component.
- **Data** – The type of the data associated with the client (default is void).

hpx::components::client_base

Defined in header `hpx/components.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename Derived, typename Stub, typename Data>
```

```
class client_base : public hpx::components::stub_base<ServerComponent>
```

This class template serves as a base class for client components, providing common functionality such as managing shared state, ID retrieval, and asynchronous operations.

Template Parameters

- **Derived** – The derived client component type.
- **Stub** – The stub type used for communication.
- **Data** – The extra data type used for additional information.

hpx/components/basename_registration_fwd.hpp

Defined in header `hpx/components/basename_registration_fwd.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
namespace hpx
```

Functions

```
hpx::future<bool> register_with_basename(std::string base_name, hpx::id_type const &id, std::size_t  
sequence_nr = ~static_cast<std::size_t>(0))
```

Register the given id using the given base name.

The function registers the given ids using the provided base name.

Note: The operation will fail if the given sequence number is not unique.

Parameters

- **base_name** – [in] The base name for which to retrieve the registered ids.
- **id** – [in] The id to register using the given base name.
- **sequence_nr** – [in, optional] The sequential number to use for the registration of the id. This number has to be unique system-wide for each registration using the same base name. The default is the current locality identifier. Also, the sequence numbers have to be consecutive starting from zero.

Returns

A future representing the result of the registration operation itself.

```
bool register_with_basename(hpx::launch::sync_policy, std::string base_name, hpx::id_type const &id,  
                           std::size_t sequence_nr = ~static_cast<std::size_t>(0), error_code &ec =  
                           throws)
```

```
hpx::future<bool> register_with_basename(std::string base_name, hpx::future<hpx::id_type> f,  
                                         std::size_t sequence_nr = ~static_cast<std::size_t>(0))
```

Register the id wrapped in the given future using the given base name.

The function registers the object the given future refers to using the provided base name.

Note: The operation will fail if the given sequence number is not unique.

Parameters

- **base_name** – [in] The base name for which to retrieve the registered ids.
- **f** – [in] The future which should be registered using the given base name.
- **sequence_nr** – [in, optional] The sequential number to use for the registration of the id. This number has to be unique system-wide for each registration using the same base name. The default is the current locality identifier. Also, the sequence numbers have to be consecutive starting from zero.

Returns

A future representing the result of the registration operation itself.

hpx/components/get_ptr.hpp

Defined in header `hpx/components/get_ptr.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Functions

```
template<typename Component>  
hpx::future<std::shared_ptr<Component>> get_ptr(hpx::id_type const &id)
```

Returns a future referring to the pointer to the underlying memory of a component.

The function `hpx::get_ptr` can be used to extract a future referring to the pointer to the underlying memory of a given component.

Note: This function will successfully return the requested result only if the given component is currently located on the calling locality. Otherwise, the function will raise an error.

Note: The component instance the returned pointer refers to can not be migrated as long as there is at least one copy of the returned `shared_ptr` alive.

Parameters

id – [in] The global id of the component for which the pointer to the underlying memory should be retrieved.

Template Parameters

Component – The type of the server side component.

Returns

This function returns a future representing the pointer to the underlying memory for the component instance with the given *id*.

```
template<typename Derived, typename Stub, typename Data>
hpx::future<std::shared_ptr<typename components::client_base<Derived, Stub, Data>::server_component_type>> get_ptr(con
Stu
Da
com
&c
```

Returns a future referring to the pointer to the underlying memory of a component.

The function `hpx::get_ptr` can be used to extract a future referring to the pointer to the underlying memory of a given component.

Note: This function will successfully return the requested result only if the given component is currently located on the calling locality. Otherwise, the function will raise an error.

Note: The component instance the returned pointer refers to can not be migrated as long as there is at least one copy of the returned `shared_ptr` alive.

Parameters

c – [in] A client side representation of the component for which the pointer to the underlying memory should be retrieved.

Returns

This function returns a future representing the pointer to the underlying memory for the component instance with the given *id*.

```
template<typename Component>
std::shared_ptr<Component> get_ptr(launch::sync_policy p, hpx::id_type const &id, error_code &ec =
throws)
```

Returns the pointer to the underlying memory of a component.

The function `hpx::get_ptr_sync` can be used to extract the pointer to the underlying memory of a given component.

Note: This function will successfully return the requested result only if the given component is currently located on the requesting locality. Otherwise, the function will raise an error.

Note: The component instance the returned pointer refers to can not be migrated as long as there is at least one copy of the returned `shared_ptr` alive.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

Parameters

- **p** – [in] The parameter *p* represents a placeholder type to turn make the call synchronous.
- **id** – [in] The global id of the component for which the pointer to the underlying memory should be retrieved.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Template Parameters

Component – The only template parameter has to be the type of the server side component.

Returns

This function returns the pointer to the underlying memory for the component instance with the given *id*.

```
template<typename Derived, typename Stub, typename Data>
std::shared_ptr<typename components::client_base<Derived, Stub, Data>::server_component_type> get_ptr(launch::sync_policy p,
com-
po-
nents::client_base
Stub,
Data>
const
&c,
er-
ror_code
&ec
=
throws)
```

Returns the pointer to the underlying memory of a component.

The function *hpx::get_ptr_sync* can be used to extract the pointer to the underlying memory of a given component.

Note: This function will successfully return the requested result only if the given component is currently located on the requesting locality. Otherwise, the function will raise an error.

Note: The component instance the returned pointer refers to can not be migrated as long as there is at least one copy of the returned *shared_ptr* alive.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

Parameters

- **p** – [in] The parameter *p* represents a placeholder type to turn make the call synchronous.
- **c** – [in] A client side representation of the component for which the pointer to the underlying memory should be retrieved.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

This function returns the pointer to the underlying memory for the component instance with the given *id*.

hpx/components/basename_registration.hpp

Defined in header `hpx/components/basename_registration.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Functions

`std::vector<hpx::future<hpx::id_type>> find_all_from_basename(std::string base_name, std::size_t num_ids)`

Return all registered ids from all localities from the given base name.

This function locates all ids which were registered with the given base name. It returns a list of futures representing those ids.

Return all registered clients from all localities from the given base name.

This function locates all ids which were registered with the given base name. It returns a list of futures representing those ids.

Note: The futures will become ready even if the event (for instance, binding the name to an id) has already happened in the past. This is important in order to reliably retrieve ids from a name, even if the name was already registered.

Note: The futures embedded in the returned client objects will become ready even if the event (for instance, binding the name to an id) has already happened in the past. This is important in order to reliably retrieve ids from a name, even if the name was already registered.

Parameters

- **base_name** – [in] The base name for which to retrieve the registered ids.
- **num_ids** – [in] The number of registered ids to expect.
- **base_name** – [in] The base name for which to retrieve the registered ids.
- **num_ids** – [in] The number of registered ids to expect.

Returns

A list of futures representing the ids which were registered using the given base name.

Template Parameters

Client – The client type to return

Returns

A list of futures representing the ids which were registered using the given base name.

```
std::vector<hpx::future<hpx::id_type>> find_all_from_basename(hpx::launch::sync_policy, std::string  
                                                             base_name, std::size_t num_ids)
```

```
std::vector<hpx::future<hpx::id_type>> find_from_basename(std::string base_name,  
                                                         std::vector<std::size_t> const &ids)
```

Return registered ids from the given base name and sequence numbers.

This function locates the ids which were registered with the given base name and the given sequence numbers. It returns a list of futures representing those ids.

Return registered clients from the given base name and sequence numbers.

This function locates the ids which were registered with the given base name and the given sequence numbers. It returns a list of futures representing those ids.

Note: The futures will become ready even if the event (for instance, binding the name to an id) has already happened in the past. This is important in order to reliably retrieve ids from a name, even if the name was already registered.

Note: The futures embedded in the returned client objects will become ready even if the event (for instance, binding the name to an id) has already happened in the past. This is important in order to reliably retrieve ids from a name, even if the name was already registered.

Parameters

- **base_name** – [in] The base name for which to retrieve the registered ids.
- **ids** – [in] The sequence numbers of the registered ids.
- **base_name** – [in] The base name for which to retrieve the registered ids.
- **ids** – [in] The sequence numbers of the registered ids.

Returns

A list of futures representing the ids which were registered using the given base name and sequence numbers.

Template Parameters

Client – The client type to return

Returns

A list of futures representing the ids which were registered using the given base name and sequence numbers.

```
std::vector<hpx::future<hpx::id_type>> find_from_basename(hpx::launch::sync_policy, std::string
                                                         base_name, std::vector<std::size_t> const
                                                         &ids)
```

```
hpx::future<hpx::id_type> find_from_basename(std::string base_name, std::size_t sequence_nr =
                                             ~static_cast<std::size_t>(0))
```

Return registered id from the given base name and sequence number.

This function locates the id which was registered with the given base name and the given sequence number. It returns a future representing those id.

This function locates the id which was registered with the given base name and the given sequence number. It returns a future representing those id.

Note: The future will become ready even if the event (for instance, binding the name to an id) has already happened in the past. This is important in order to reliably retrieve ids from a name, even if the name was already registered.

Note: The future embedded in the returned client object will become ready even if the event (for instance, binding the name to an id) has already happened in the past. This is important in order to reliably retrieve ids from a name, even if the name was already registered.

Parameters

- **base_name** – [in] The base name for which to retrieve the registered ids.
- **sequence_nr** – [in] The sequence number of the registered id.
- **base_name** – [in] The base name for which to retrieve the registered ids.
- **sequence_nr** – [in] The sequence number of the registered id.

Returns

A representing the id which was registered using the given base name and sequence numbers.

Template Parameters

Client – The client type to return

Returns

A representing the id which was registered using the given base name and sequence numbers.

```
hpx::id_type find_from_basename(hpx::launch::sync_policy, std::string base_name, std::size_t sequence_nr
                                = ~static_cast<std::size_t>(0))
```

```
hpx::future<hpx::id_type> unregister_with_basename(std::string base_name, std::size_t sequence_nr =
                                                    ~static_cast<std::size_t>(0))
```

Unregister the given id using the given base name.

The function unregisters the given ids using the provided base name.

Unregister the given base name.

The function unregisters the given ids using the provided base name.

Parameters

- **base_name** – [in] The base name for which to retrieve the registered ids.
- **sequence_nr** – [in, optional] The sequential number to use for the un-registration. This number has to be the same as has been used with *register_with_basename* before.
- **base_name** – [in] The base name for which to retrieve the registered ids.
- **sequence_nr** – [in, optional] The sequential number to use for the un-registration. This number has to be the same as has been used with *register_with_basename* before.

Returns

A future representing the result of the un-registration operation itself.

Template Parameters

Client – The client type to return

Returns

A future representing the result of the un-registration operation itself.

```
template<typename Client, typename Stub, typename Data>  
hpx::future<bool> register_with_basename(std::string base_name, components::client_base<Client, Stub,  
                                         Data> &client, std::size_t sequence_nr =  
                                         ~static_cast<std::size_t>(0))
```

Register the id wrapped in the given client using the given base name.

The function registers the object the given client refers to using the provided base name.

Note: The operation will fail if the given sequence number is not unique.

Template Parameters

Client – The client type to register

Parameters

- **base_name** – [in] The base name for which to retrieve the registered ids.
- **client** – [in] The client which should be registered using the given base name.
- **sequence_nr** – [in, optional] The sequential number to use for the registration of the id. This number has to be unique system-wide for each registration using the same base name. The default is the current locality identifier. Also, the sequence numbers have to be consecutive starting from zero.

Returns

A future representing the result of the registration operation itself.

components_base

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/components_base/agas_interface.hpp

Defined in header `hpx/components_base/agas_interface.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **agas**

Functions

bool **is_console**()

bool **register_name**(*launch::sync_policy*, *std::string* const &name, *naming::gid_type* const &gid, *error_code* &ec = *throws*)

bool **register_name**(*launch::sync_policy*, *std::string* const &name, *hpx::id_type* const &id, *error_code* &ec = *throws*)

hpx::future<bool> **register_name**(*std::string* const &name, *hpx::id_type* const &id)

hpx::id_type **unregister_name**(*launch::sync_policy*, *std::string* const &name, *error_code* &ec = *throws*)

hpx::future<*hpx::id_type*> **unregister_name**(*std::string* const &name)

hpx::id_type **resolve_name**(*launch::sync_policy*, *std::string* const &name, *error_code* &ec = *throws*)

hpx::future<*hpx::id_type*> **resolve_name**(*std::string* const &name)

hpx::future<*std::uint32_t*> **get_num_localities**(*naming::component_type* type = *naming::component_invalid*)

std::uint32_t **get_num_localities**(*launch::sync_policy*, *naming::component_type* type, *error_code* &ec = *throws*)

inline *std::uint32_t* **get_num_localities**(*launch::sync_policy*, *error_code* &ec = *throws*)

std::string **get_component_type_name**(*naming::component_type* type, *error_code* &ec = *throws*)

hpx::future<*std::vector*<*std::uint32_t*>> **get_num_threads**()

std::vector<*std::uint32_t*> **get_num_threads**(*launch::sync_policy*, *error_code* &ec = *throws*)

hpx::future<*std::uint32_t*> **get_num_overall_threads**()

std::uint32_t **get_num_overall_threads**(*launch::sync_policy*, *error_code* &ec = *throws*)

std::uint32_t **get_locality_id**(*error_code* &ec = *throws*)

inline *hpx::naming::gid_type* **get_locality**()

std::vector<*std::uint32_t*> **get_all_locality_ids**(*naming::component_type* type, *error_code* &ec = *throws*)

```
inline std::vector<std::uint32_t> get_all_locality_ids(error_code &ec = throws)

bool is_local_address_cached(naming::gid_type const &gid, error_code &ec = throws)

bool is_local_address_cached(naming::gid_type const &gid, naming::address &addr, error_code
                             &ec = throws)

bool is_local_address_cached(naming::gid_type const &gid, naming::address &addr,
                             std::pair<bool, components::pinned_ptr> &r,
                             hpx::move_only_function<std::pair<bool,
                             components::pinned_ptr>(naming::address const&)> &&f,
                             error_code &ec = throws)

inline bool is_local_address_cached(hpx::id_type const &id, error_code &ec = throws)

inline bool is_local_address_cached(hpx::id_type const &id, naming::address &addr, error_code
                                     &ec = throws)

inline bool is_local_address_cached(hpx::id_type const &id, naming::address &addr,
                                     std::pair<bool, components::pinned_ptr> &r,
                                     hpx::move_only_function<std::pair<bool,
                                     components::pinned_ptr>(naming::address const&)> &&f,
                                     error_code &ec = throws)

void update_cache_entry(naming::gid_type const &gid, naming::address const &addr, std::uint64_t
                        count = 0, std::uint64_t offset = 0, error_code &ec = throws)

bool is_local_lva_encoded_address(naming::gid_type const &gid)

inline bool is_local_lva_encoded_address(hpx::id_type const &id)

hpx::future_or_value<naming::address> resolve_async(hpx::id_type const &id)

hpx::future<naming::address> resolve(hpx::id_type const &id)

naming::address resolve(launch::sync_policy, hpx::id_type const &id, error_code &ec = throws)

bool resolve_local(naming::gid_type const &gid, naming::address &addr, error_code &ec = throws)

bool resolve_cached(naming::gid_type const &gid, naming::address &addr)

hpx::future<bool> bind(naming::gid_type const &gid, naming::address const &addr, std::uint32_t
                      locality_id)

bool bind(launch::sync_policy, naming::gid_type const &gid, naming::address const &addr,
          std::uint32_t locality_id, error_code &ec = throws)

hpx::future<bool> bind(naming::gid_type const &gid, naming::address const &addr, naming::gid_type
                      const &locality_)

bool bind(launch::sync_policy, naming::gid_type const &gid, naming::address const &addr,
          naming::gid_type const &locality_, error_code &ec = throws)

hpx::future<naming::address> unbind(naming::gid_type const &gid, std::uint64_t count = 1)

naming::address unbind(launch::sync_policy, naming::gid_type const &gid, std::uint64_t count = 1,
                      error_code &ec = throws)
```

```

bool bind_gid_local(naming::gid_type const &gid, naming::address const &addr, error_code &ec =
    throws)

void unbind_gid_local(naming::gid_type const &gid, error_code &ec = throws)

bool bind_range_local(naming::gid_type const &gid, std::size_t count, naming::address const &addr,
    std::size_t offset, error_code &ec = throws)

void unbind_range_local(naming::gid_type const &gid, std::size_t count, error_code &ec = throws)

void garbage_collect_non_blocking(error_code &ec = throws)

void garbage_collect(error_code &ec = throws)

void garbage_collect_non_blocking(hpx::id_type const &id, error_code &ec = throws)
    Invoke an asynchronous garbage collection step on the given target locality.

void garbage_collect(hpx::id_type const &id, error_code &ec = throws)
    Invoke a synchronous garbage collection step on the given target locality.

hpx::id_type get_console_locality(error_code &ec = throws)
    Return an id_type referring to the console locality.

naming::gid_type get_next_id(std::size_t count, error_code &ec = throws)

void decref(naming::gid_type const &id, std::int64_t credits, error_code &ec = throws)

hpx::future_or_value<std::int64_t> incref(naming::gid_type const &gid, std::int64_t credits,
    hpx::id_type const &keep_alive = hpx::invalid_id)

std::int64_t incref(launch::sync_policy, naming::gid_type const &gid, std::int64_t credits = 1,
    hpx::id_type const &keep_alive = hpx::invalid_id, error_code &ec = throws)

std::int64_t replenish_credits(naming::gid_type &gid)

hpx::future_or_value<id_type> get_colocation_id(hpx::id_type const &id)

hpx::id_type get_colocation_id(launch::sync_policy, hpx::id_type const &id, error_code &ec =
    throws)

hpx::future<hpx::id_type> on_symbol_namespace_event(std::string const &name, bool
    call_for_past_events)

hpx::future<std::pair<hpx::id_type, naming::address>> begin_migration(hpx::id_type const &id)

bool end_migration(hpx::id_type const &id)

hpx::future<void> mark_as_migrated(naming::gid_type const &gid,
    hpx::move_only_function<std::pair<bool,
    hpx::future<void>>()> &&f, bool
    expect_to_be_marked_as_migrating)

std::pair<bool, components::pinned_ptr> was_object_migrated(naming::gid_type const &gid,
    hpx::move_only_function<components::pinned_ptr()>
    &&f)

void unmark_as_migrated(naming::gid_type const &gid, hpx::move_only_function<void()> &&f)

```

```
hpx::future<std::map<std::string, hpx::id_type>> find_symbols(std::string const &pattern = "")
std::map<std::string, hpx::id_type> find_symbols(hpx::launch::sync_policy, std::string const &pattern
= "")

naming::component_type register_factory(std::uint32_t prefix, std::string const &name, error_code
&ec = throws)

naming::component_type get_component_id(std::string const &name, error_code &ec = throws)

void destroy_component(naming::gid_type const &gid, naming::address const &addr)

naming::address_type get_primary_ns_lva()

naming::address_type get_symbol_ns_lva()

naming::address_type get_runtime_support_lva()

struct agas_interface_functions &agas_init()
```

hpx/components_base/get_lva.hpp

Defined in header `hpx/components_base/get_lva.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

```
template<typename Component, typename Enable = void>
```

```
struct get_lva
```

#include <get_lva.hpp> The `get_lva` template is a helper structure allowing to convert a local virtual address as stored in a local address (returned from the function `agas::addressing_service::resolve`) to the address of the component implementing the action.

The default implementation uses the template argument *Component* to deduce the type wrapping the component implementing the action. This is used to get the needed address.

Template Parameters

Component – This is the type of the component implementing the action to execute.

Public Static Functions

```
static inline constexpr Component *call(naming::address_type lva) noexcept
```

hpx/components_base/component_type.hpp

Defined in header `hpx/components_base/component_type.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Defines

HPX_COMPONENT_ENUM_TYPE_ENUM_DEPRECATION_MSG

HPX_FACTORY_STATE_ENUM_DEPRECATION_MSG

namespace **hpx**

namespace **components**

Typedefs

using **component_deleter_type** = void (*)(hpx::naming::gid_type const&, hpx::naming::address const&)

Enums

enum class **component_enum_type** : *naming::component_type*

Values:

enumerator **invalid**

enumerator **runtime_support**

enumerator **plain_function**

enumerator **base_lco**

enumerator **base_lco_with_value_unmanaged**

enumerator **base_lco_with_value**

enumerator **latch**

enumerator **barrier**

enumerator **promise**

enumerator **agas_locality_namespace**

enumerator **agas_primary_namespace**

enumerator **agas_component_namespace**

enumerator **agas_symbol_namespace**

enumerator **supervision_manager**

enumerator **last**

enumerator **first_dynamic**

enum class **factory_state** : *std::uint8_t*

Values:

enumerator **enabled**

enumerator **disabled**

enumerator **check**

Functions

constexpr *naming::component_type* **to_int**(*component_enum_type* t) noexcept

constexpr int **to_int**(*factory_state* t) noexcept

bool &**enabled**(*component_type* type)

util::atomic_count &**instance_count**(*component_type* type)

component_deleter_type &**deleter**(*component_type* type)

bool **enumerate_instance_counts**(*hpx::move_only_function*<bool(*component_type*)> const &f)

std::string **get_component_type_name**(*component_type* type)

Return the string representation for a given component type id.

constexpr *component_type* **get_base_type**(*component_type* t) noexcept

The lower short word of the component type is the type of the component exposing the actions.

constexpr *component_type* **get_derived_type**(*component_type* t) noexcept

The upper short word of the component is the actual component type.

constexpr *component_type* **derived_component_type**(*component_type* derived, *component_type* base) noexcept

A component derived from a base component exposing the actions needs to have a specially formatted component type.

constexpr bool **types_are_compatible**(*component_type* lhs, *component_type* rhs) noexcept

Verify the two given component types are matching (compatible)

template<typename **Component**, typename **Enable** = void>

```

char const *get_component_name() noexcept

template<typename Component, typename Enable = void>
char const *get_component_base_name() noexcept

template<typename Component>
component_type get_component_type() noexcept

template<typename Component>
void set_component_type(component_type type)

```

Variables

```

constexpr component_enum_type component_invalid = component_enum_type::invalid

constexpr component_enum_type component_runtime_support =
component_enum_type::runtime_support

constexpr component_enum_type component_plain_function =
component_enum_type::plain_function

constexpr component_enum_type component_base_lco = component_enum_type::base_lco

constexpr component_enum_type component_base_lco_with_value_unmanaged =
component_enum_type::base_lco_with_value_unmanaged

constexpr component_enum_type component_base_lco_with_value =
component_enum_type::base_lco_with_value

constexpr component_enum_type component_latch = component_enum_type::latch

constexpr component_enum_type component_barrier = component_enum_type::barrier

constexpr component_enum_type component_promise = component_enum_type::promise

constexpr component_enum_type component_agas_locality_namespace =
component_enum_type::agas_locality_namespace

constexpr component_enum_type component_agas_primary_namespace =
component_enum_type::agas_primary_namespace

constexpr component_enum_type component_agas_component_namespace =
component_enum_type::agas_component_namespace

constexpr component_enum_type component_agas_symbol_namespace =
component_enum_type::agas_symbol_namespace

```

```
constexpr component_enum_type component_last = component_enum_type::last
```

```
constexpr component_enum_type component_first_dynamic = component_enum_type::first_dynamic
```

```
constexpr factory_state factory_enabled = factory_state::enabled
```

```
constexpr factory_state factory_disabled = factory_state::disabled
```

```
constexpr factory_state factory_check = factory_state::check
```

```
constexpr int component_type_mask = 0x3FF
```

```
constexpr int component_type_shift = 10
```

HPX_REGISTER_STARTUP_MODULE

Defined in header `hpx/components.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Warning: doxygenfunction: Cannot find function “HPX_REGISTER_STARTUP_MODULE” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

hpx::components::component

Defined in header `hpx/components.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::components::component_base`

template<typename **Component**>

class **component**

The *component* class wraps around a given component type, adding additional type aliases and constructors. It inherits from the specified component type.

hpx::components::component_base

Defined in header `hpx/components.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::components::component`

template<typename **Component**>

class **component_base**

`component_base` serves as a base class for components. It provides common functionality needed by components, such as address and ID retrieval. The template parameter *Component* specifies the derived component type.

Subclassed by `Component`

HPX_REGISTER_COMMANDLINE_MODULE

Defined in header `hpx/components.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Warning: doxygenfunction: Cannot find function “HPX_REGISTER_COMMANDLINE_MODULE” in doxygen xml output for project “hpx” from directory: `../hpx_autodoc`

hpx/components_base/server/managed_component_base.hpp

Defined in header `hpx/components_base/server/managed_component_base.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

template<>

struct **init**<*traits::construct_with_back_ptr*>

Public Static Functions

template<typename **Component**, typename **Managed**>
static inline constexpr void **call**(*Component**, *Managed**) noexcept

template<typename **Component**, typename **Managed**, typename ...**Ts**>
static inline void **call_new**(*Component* *&component, *Managed* *this_, *Ts*&&... vs)

template<>

struct **init**<*traits::construct_without_back_ptr*>

Public Static Functions

```
template<typename Component, typename Managed>
static inline void call(Component *component, Managed *this_)

template<typename Component, typename Managed, typename ...Ts>
static inline void call_new(Component *&component, Managed *this_, Ts&&... vs)
```

```
template<>
```

```
struct destroy_backptr<traits::managed_object_is_lifetime_controlled>
```

Public Static Functions

```
template<typename BackPtr>
static inline void call(BackPtr *back_ptr)
```

```
template<>
```

```
struct destroy_backptr<traits::managed_object_controls_lifetime>
```

Public Static Functions

```
template<typename BackPtr>
static inline constexpr void call(BackPtr*) noexcept
```

```
template<>
```

```
struct manage_lifetime<traits::managed_object_is_lifetime_controlled>
```

Public Static Functions

```
template<typename Component>
static inline constexpr void call(Component*) noexcept
```

```
template<typename Component>
static inline void addref(Component *component) noexcept
```

```
template<typename Component>
static inline void release(Component *component) noexcept
```

```
template<>
```

```
struct manage_lifetime<traits::managed_object_controls_lifetime>
```

Public Static Functions

```
template<typename Component>
static inline void call(Component *component) noexcept(noexcept(component->finalize()))
```

```
template<typename Component>
static inline constexpr void addref(Component*) noexcept
```

```
template<typename Component>
static inline constexpr void release(Component*) noexcept
```

```
namespace hpx
```

```
namespace components
```

Functions

```
template<typename Component, typename Derived>
void intrusive_ptr_add_ref(managed_component<Component, Derived> *p) noexcept
```

```
template<typename Component, typename Derived>
void intrusive_ptr_release(managed_component<Component, Derived> *p) noexcept
```

```
template<typename Component, typename Derived>
```

```
class managed_component
```

```
    #include <managed_component_base.hpp>
```

```
template<typename Component, typename Wrapper, typename CtorPolicy, typename DtorPolicy>
```

```
class managed_component_base
```

```
namespace detail_adl_barrier
```

```
    template<typename DtorTag>
```

```
    struct destroy_backptr
```

```
    template<> managed_object_controls_lifetime >
```

Public Static Functions

```
template<typename BackPtr>
static inline constexpr void call(BackPtr*) noexcept
```

```
template<> managed_object_is_lifetime_controlled >
```

Public Static Functions

```
template<typename BackPtr>
static inline void call(BackPtr *back_ptr)

template<typename BackPtrTag>
struct init

template<> construct_with_back_ptr >
```

Public Static Functions

```
template<typename Component, typename Managed>
static inline constexpr void call(Component*, Managed*) noexcept

template<typename Component, typename Managed, typename ...Ts>
static inline void call_new(Component *&component, Managed *this_, Ts&&... vs)

template<> construct_without_back_ptr >
```

Public Static Functions

```
template<typename Component, typename Managed>
static inline void call(Component *component, Managed *this_)

template<typename Component, typename Managed, typename ...Ts>
static inline void call_new(Component *&component, Managed *this_, Ts&&... vs)

template<typename DtorTag>
struct manage_lifetime

template<> managed_object_controls_lifetime >
```

Public Static Functions

```
template<typename Component>
static inline void call(Component *component) noexcept(noexcept(component->finalize()))

template<typename Component>
static inline constexpr void addref(Component*) noexcept

template<typename Component>
static inline constexpr void release(Component*) noexcept

template<> managed_object_is_lifetime_controlled >
```


Public Static Functions

```
template<typename Component>
static inline constexpr void call(Component*) noexcept

template<typename Component>
static inline void addref(Component *component) noexcept

template<typename Component>
static inline void release(Component *component) noexcept
```

hpx/components_base/server/fixed_component_base.hpp

Defined in header `hpx/components_base/server/fixed_component_base.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **components**

```
template<typename Component>
class fixed_component

template<typename Component>
class fixed_component_base
```

hpx/components_base/server/migration_support.hpp

Defined in header `hpx/components_base/server/migration_support.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **components**

```
template<typename BaseComponent, typename Mutex = hpx::spinlock>
struct migration_support : public BaseComponent
    #include <migration_support.hpp> This hook has to be inserted into the derivation chain of any component for it to support migration.
```

Public Types

using **decorates_action** = void

Public Functions

inline **migration_support**()

template<typename **T**, typename ...**Ts**, typename =
std::enable_if_t<!std::is_same_v<std::decay_t<T>, migration_support>>>
inline explicit **migration_support**(*T &&t, Ts&&... ts*)

migration_support(*migration_support const&*) = default

migration_support(*migration_support&&*) = default

migration_support &operator=(*migration_support const&*) = default

migration_support &operator=(*migration_support&&*) = default

~migration_support() = default

inline *naming::gid_type* **get_base_gid**(*naming::gid_type const &assign_gid =*
naming::invalid_gid) const

inline bool **pin**() noexcept

inline bool **unpin**()

inline *std::uint32_t* **pin_count**() const noexcept

inline void **mark_as_migrated**()

inline *hpx::future<void>* **mark_as_migrated**(*hpx::id_type const &to_migrate*)

inline void **unmark_as_migrated**(*hpx::id_type const &to_migrate*)

Public Static Functions

static inline constexpr bool **supports_migration**() noexcept

static inline constexpr void **on_migrated**() noexcept

template<typename **F**>

static inline *threads::thread_function_type* **decorate_action**(*naming::address_type lva, F &&f*)

static inline *std::pair<bool, components::pinned_ptr>* **was_object_migrated**(*hpx::naming::gid_type*
const &*id*, *naming::address_type*
lva)

Protected Functions

```
inline threads::thread_result_type thread_function(threads::thread_function_type &&,
                                                    components::pinned_ptr,
                                                    threads::thread_restart_state state)
```

Private Types

```
using base_type = BaseComponent
```

```
using this_component_type = typename base_type::this_component_type
```

Private Members

```
hpx::intrusive_ptr<detail::migration_support_data<Mutex>> data_
```

```
hpx::promise<void> trigger_migration_
```

```
bool started_migration_ = false
```

```
bool was_marked_for_migration_ = false
```

compute

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/compute/host/target_distribution_policy.hpp

Defined in header `hpx/compute/host/target_distribution_policy.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

distribution_policies

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/distribution_policies/colocating_distribution_policy.hpp

Defined in header `hpx/distribution_policies/colocating_distribution_policy.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace components
```

Variables

const *colocating_distribution_policy* **colocated**

A predefined instance of the co-locating *distribution_policy*. It will represent the local locality and will place all items to create here.

struct **colocating_distribution_policy**

#include <colocating_distribution_policy.hpp> This class specifies the parameters for a distribution policy to use for creating a given number of items on the locality where a given object is currently placed.

Public Functions

constexpr **colocating_distribution_policy**() = default

Default-construct a new instance of a *colocating_distribution_policy*. This policy will represent the local locality.

inline *colocating_distribution_policy* **operator**()(id_type const &id) const

Create a new *colocating_distribution_policy* representing the locality where the given object is current located

Parameters

id – [in] The global address of the object with which the new instances should be colocated on

template<typename **Client**, typename **Stub**, typename **Data**>

inline *colocating_distribution_policy* **operator**()(*client_base*<*Client*, *Stub*, *Data*> const &client) const

Create a new *colocating_distribution_policy* representing the locality where the given object is current located

Parameters

client – [in] The client side representation of the object with which the new instances should be colocated on

template<typename **Component**, typename ...**Ts**>

inline *hpx::future*<*hpx::id_type*> **create**(*Ts*&&... vs) const

Create one object on the locality of the object this distribution policy instance is associated with

Note: This function is part of the placement policy implemented by this class

Parameters

vs – [in] The arguments which will be forwarded to the constructor of the new object.

Returns

A future holding the global address which represents the newly created object

template<bool **WithCount**, typename **Component**, typename ...**Ts**>

inline *hpx::future*<*std::vector*<*bulk_locality_result*>> **bulk_create**(*std::size_t* count, *Ts*&&... vs) const

Create multiple objects colocated with the object represented by this policy instance

Note: This function is part of the placement policy implemented by this class

Parameters

- **count** – [in] The number of objects to create
- **vs** – [in] The arguments which will be forwarded to the constructors of the new objects.

Returns

A future holding the list of global addresses which represent the newly created objects

```
template<typename Action, typename ...Ts>
inline async_result<Action>::type async(launch policy, Ts&&... vs) const

template<typename Action, typename Callback, typename ...Ts>
inline async_result<Action>::type async_cb(launch policy, Callback &&cb, Ts&&... vs) const
```

Note: This function is part of the invocation policy implemented by this class

```
template<typename Action, typename Continuation, typename ...Ts>
inline bool apply(Continuation &&c, launch policy, Ts&&... vs) const
```

Note: This function is part of the invocation policy implemented by this class

```
template<typename Action, typename ...Ts>
inline bool apply(launch policy, Ts&&... vs) const

template<typename Action, typename Continuation, typename Callback, typename ...Ts>
inline bool apply_cb(Continuation &&c, launch policy, Callback &&cb, Ts&&... vs) const
```

Note: This function is part of the invocation policy implemented by this class

```
template<typename Action, typename Callback, typename ...Ts>
inline bool apply_cb(launch policy, Callback &&cb, Ts&&... vs) const
```

```
inline hpx::id_type get_next_target() const
```

Returns the locality which is anticipated to be used for the next async operation

Public Static Functions

```
static inline std::size_t get_num_localities()
```

Returns the number of associated localities for this distribution policy

Note: This function is part of the creation policy implemented by this class

Private Functions

```
inline hpx::id_type resolve_direct_target() const
```

```
template<typename Action>
```

```
struct async_result
```

```
    #include <colocating_distribution_policy.hpp>
```

Note: This function is part of the invocation policy implemented by this class

Public Types

```
using type = hpx::future<typename traits::promise_local_result<typename  
hpx::traits::extract_action<Action>::remote_result_type>::type>
```

hpx/distribution_policies/default_distribution_policy.hpp

Defined in header `hpx/distribution_policies/default_distribution_policy.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace components
```

Variables

```
const default_distribution_policy default_layout
```

A predefined instance of the default *distribution_policy*. It will represent the local locality and will place all items to create here.

```
struct default_distribution_policy
```

#include <default_distribution_policy.hpp> This class specifies the parameters for a simple distribution policy to use for creating (and evenly distributing) a given number of items on a given set of localities.

Public Functions

```
constexpr default_distribution_policy() = default
```

Default-construct a new instance of a *default_distribution_policy*. This policy will represent one locality (the local locality).

```
inline default_distribution_policy operator()(std::vector<id_type> locs) const
```

Create a new *default_distribution* policy representing the given set of localities.

Parameters

locs – [in] The list of localities the new instance should represent

```
inline default_distribution_policy operator()(id_type loc) const
```

Create a new *default_distribution* policy representing the given locality

Parameters

loc – [in] The locality the new instance should represent

```
template<typename Component, typename ...Ts>
```

```
inline hpx::future<hpx::id_type> create(Ts&&... vs) const
```

Create one object on one of the localities associated by this policy instance

Note: This function is part of the placement policy implemented by this class

Parameters

vs – [in] The arguments which will be forwarded to the constructor of the new object.

Returns

A future holding the global address which represents the newly created object

```
template<bool WithCount, typename Component, typename ...Ts>
```

```
inline hpx::future<std::vector<bulk_locality_result>> bulk_create(std::size_t count, Ts&&... vs)
                                                    const
```

Create multiple objects on the localities associated by this policy instance

Note: This function is part of the placement policy implemented by this class

Parameters

- **count** – [in] The number of objects to create
- **vs** – [in] The arguments which will be forwarded to the constructors of the new objects.

Returns

A future holding the list of global addresses that represent the newly created objects

```
template<typename Action, typename ...Ts>
```

```
inline async_result<Action>::type async(launch policy, Ts&&... vs) const
```

```
template<typename Action, typename Callback, typename ...Ts>
```

```
inline async_result<Action>::type async_cb(launch policy, Callback &&cb, Ts&&... vs) const
```

Note: This function is part of the invocation policy implemented by this class

```
template<typename Action, typename Continuation, typename ...Ts>
```

```
inline bool apply(Continuation &&c, launch policy, Ts&&... vs) const
```

Note: This function is part of the invocation policy implemented by this class

```
template<typename Action, typename ...Ts>
```

```
inline bool apply(threads::thread_priority priority, Ts&&... vs) const
```

```
template<typename Action, typename Continuation, typename Callback, typename ...Ts>
```

```
inline bool apply_cb(Continuation &&c, launch policy, Callback &&cb, Ts&&... vs) const
```

Note: This function is part of the invocation policy implemented by this class

```
template<typename Action, typename Callback, typename ...Ts>
inline bool apply_cb(launch policy, Callback &&cb, Ts&&... vs) const
```

```
inline std::size_t get_num_localities() const
```

Returns the number of associated localities for this distribution policy

Note: This function is part of the creation policy implemented by this class

```
inline hpx::id_type get_next_target() const
```

Returns the locality which is anticipated to be used for the next async operation

```
template<typename Action>
```

```
struct async_result
```

```
#include <default_distribution_policy.hpp>
```

Note: This function is part of the invocation policy implemented by this class

Public Types

```
using type = hpx::future<typename traits::promise_local_result<typename
hpx::traits::extract_action<Action>::remote_result_type>::type>
```

hpx/distribution_policies/binpacking_distribution_policy.hpp

Defined in header `hpx/distribution_policies/binpacking_distribution_policy.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
namespace components
```

Variables

```
constexpr char const *const default_binpacking_counter_name =
"/runtime{locality/total}/count/component@"
```

```
const binpacking_distribution_policy binpacked
```

A predefined instance of the binpacking *distribution_policy*. It will represent the local locality and will place all items to create here.

```
struct binpacking_distribution_policy
```

```
#include <binpacking_distribution_policy.hpp> This class specifies the parameters for a binpacking
distribution policy to use for creating a given number of items on a given set of localities. The bin-
packing policy will distribute the new objects in a way such that each of the localities will equalize the
```


number of overall objects of this type based on a given criteria (by default this criteria is the overall number of objects of this type).

Public Functions

inline **binpacking_distribution_policy**()

Default-construct a new instance of a `binpacking_distribution_policy`. This policy will represent one locality (the local locality).

inline *binpacking_distribution_policy* **operator**() (*std::vector<id_type>* const &locs, char const *perf_counter_name = *default_binpacking_counter_name*) const

Create a new *default_distribution* policy representing the given set of localities.

Parameters

- **locs** – [in] The list of localities the new instance should represent
- **perf_counter_name** – [in] The name of the performance counter which should be used as the distribution criteria (by default the overall number of existing instances of the given component type will be used).

inline *binpacking_distribution_policy* **operator**() (*std::vector<id_type>* &&locs, char const *perf_counter_name = *default_binpacking_counter_name*) const

Create a new *default_distribution* policy representing the given set of localities.

Parameters

- **locs** – [in] The list of localities the new instance should represent
- **perf_counter_name** – [in] The name of the performance counter which should be used as the distribution criteria (by default the overall number of existing instances of the given component type will be used).

inline *binpacking_distribution_policy* **operator**() (*id_type* const &loc, char const *perf_counter_name = *default_binpacking_counter_name*) const

Create a new *default_distribution* policy representing the given locality

Parameters

- **loc** – [in] The locality the new instance should represent
- **perf_counter_name** – [in] The name of the performance counter that should be used as the distribution criteria (by default the overall number of existing instances of the given component type will be used).

template<typename **Component**, typename ...**Ts**>

inline *hpx::future<hpx::id_type>* **create**(*Ts*&&... vs) const

Create one object on one of the localities associated by this policy instance

Parameters

- **vs** – [in] The arguments which will be forwarded to the constructor of the new object.

Returns

A future holding the global address which represents the newly created object

template<bool **WithCount**, typename **Component**, typename ...**Ts**>

inline *hpx::future<std::vector<bulk_locality_result>>* **bulk_create**(*std::size_t* count, *Ts*&&... vs) const

Create multiple objects on the localities associated by this policy instance

Parameters

- **count** – [in] The number of objects to create
- **vs** – [in] The arguments which will be forwarded to the constructors of the new objects.

Returns

A future holding the list of global addresses which represent the newly created objects

inline *std::string* const &**get_counter_name**() const

Returns the name of the performance counter associated with this policy instance.

inline *std::size_t* **get_num_localities**() const

Returns the number of associated localities for this distribution policy

Note: This function is part of the creation policy implemented by this class

hpx/distribution_policies/unwrapping_result_policy.hpp

Defined in header `hpx/distribution_policies/unwrapping_result_policy.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **components**

struct **unwrapping_result_policy**

#include <unwrapping_result_policy.hpp> This class is a distribution policy that can be used with actions that return futures. For those actions it is possible to apply certain optimizations if the action is invoked synchronously.

Public Functions

inline explicit **unwrapping_result_policy**(id_type const &id)

template<typename **Client**, typename **Stub**, typename **Data**>

inline explicit **unwrapping_result_policy**(client_base<*Client*, *Stub*, *Data*> const &client)

template<typename **Action**, typename ...**Ts**>

inline *async_result*<*Action*>::type **async**(launch policy, *Ts*&&... vs) const

template<typename **Action**, typename ...**Ts**>

inline *async_result*<*Action*>::type **async**(launch::sync_policy, *Ts*&&... vs) const

template<typename **Action**, typename **Callback**, typename ...**Ts**>

inline *async_result*<*Action*>::type **async_cb**(launch policy, *Callback* &&cb, *Ts*&&... vs) const

template<typename **Action**, typename **Continuation**, typename ...**Ts**>

inline bool **apply**(*Continuation* &&c, launch policy, *Ts*&&... vs) const

Note: This function is part of the invocation policy implemented by this class

template<typename **Action**, typename ...**Ts**>

inline bool **apply**(launch policy, *Ts*&&... vs) const

template<typename **Action**, typename **Continuation**, typename **Callback**, typename ...**Ts**>

```
inline bool apply_cb(Continuation &&c, launch policy, Callback &&cb, Ts&&... vs) const
```

Note: This function is part of the invocation policy implemented by this class

```
template<typename Action, typename Callback, typename ...Ts>
inline bool apply_cb(launch policy, Callback &&cb, Ts&&... vs) const

inline hpx::id_type const &get_next_target() const

template<typename Action>

struct async_result
```

Public Types

```
using type = typename traits::promise_local_result<typename
hpx::traits::extract_action<Action>::remote_result_type>::type
```

hpx/distribution_policies/target_distribution_policy.hpp

Defined in header `hpx/distribution_policies/target_distribution_policy.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
namespace components
```

Variables

```
const target_distribution_policy target
```

A predefined instance of the `target_distribution_policy`. It will represent the local locality and will place all items to create here.

```
struct target_distribution_policy
```

#include <target_distribution_policy.hpp> This class specifies the parameters for a simple distribution policy to use for creating (and evenly distributing) a given number of items on a given set of localities.

Public Functions

target_distribution_policy() = default

Default-construct a new instance of a `target_distribution_policy`. This policy will represent one locality (the local locality).

inline *target_distribution_policy* **operator()** (id_type const &id) const

Create a new `target_distribution_policy` representing the given locality

Parameters

loc – [in] The locality the new instance should represent

template<typename **Component**, typename ...**Ts**>

inline *hpx::future*<*hpx::id_type*> **create**(*Ts*&&... vs) const

Create one object on one of the localities associated by this policy instance

Note: This function is part of the placement policy implemented by this class

Parameters

vs – [in] The arguments which will be forwarded to the constructor of the new object.

Returns

A future holding the global address which represents the newly created object

template<bool **WithCount**, typename **Component**, typename ...**Ts**>

inline *hpx::future*<*std::vector*<bulk_locality_result>> **bulk_create**(*std::size_t* count, *Ts*&&... vs)
const

Create multiple objects on the localities associated by this policy instance

Note: This function is part of the placement policy implemented by this class

Parameters

- **count** – [in] The number of objects to create

- **vs** – [in] The arguments which will be forwarded to the constructors of the new objects.

Returns

A future holding the list of global addresses which represent the newly created objects

template<typename **Action**, typename ...**Ts**>

inline *async_result*<*Action*>::type **async**(*launch* policy, *Ts*&&... vs) const

template<typename **Action**, typename **Callback**, typename ...**Ts**>

inline *async_result*<*Action*>::type **async_cb**(*launch* policy, *Callback* &&cb, *Ts*&&... vs) const

Note: This function is part of the invocation policy implemented by this class

template<typename **Action**, typename **Continuation**, typename ...**Ts**>

inline bool **apply**(*Continuation* &&c, *launch* policy, *Ts*&&... vs) const

Note: This function is part of the invocation policy implemented by this class

template<typename **Action**, typename ...**Ts**>

```
inline bool apply(launch policy, Ts&&... vs) const

template<typename Action, typename Continuation, typename Callback, typename ...Ts>
inline bool apply_cb(Continuation &&c, launch policy, Callback &&cb, Ts&&... vs) const
```

Note: This function is part of the invocation policy implemented by this class

```
template<typename Action, typename Callback, typename ...Ts>
inline bool apply_cb(launch policy, Callback &&cb, Ts&&... vs) const
```

```
inline std::size_t get_num_localities() const
```

Returns the number of associated localities for this distribution policy

Note: This function is part of the creation policy implemented by this class

```
inline hpx::id_type get_next_target() const
```

Returns the locality which is anticipated to be used for the next async operation

```
template<typename Action>
```

```
struct async_result
```

```
    #include <target_distribution_policy.hpp>
```

Note: This function is part of the invocation policy implemented by this class

Public Types

```
using type = hpx::future<typename traits::promise_local_result<typename
hpx::traits::extract_action<Action>::remote_result_type>::type>
```

execution_distributed

See *Public API* for a list of names and headers that are part of the public HPX API.

hpx/execution_distributed/distributed_then_sender.hpp

Defined in header `hpx/execution_distributed/distributed_then_sender.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

Placeholder for the distributed then sender adaptor.

A proper component-based receiver marshalling system (nvexec-style) will be implemented here in a future phase.

hpx/execution_distributed/distributed_continues_on_sender.hpp

Defined in header `hpx/execution_distributed/distributed_continues_on_sender.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Placeholder for the distributed `continues_on` sender adaptor.

A proper component-based receiver marshalling system (nvexec-style) will be implemented here in a future phase.

hpx/execution_distributed/distributed_scheduler.hpp

Defined in header `hpx/execution_distributed/distributed_scheduler.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

P2300-compliant scheduler for cross-locality execution.

Provides a scheduler that dispatches work to a remote HPX locality via the parcelport, bridging the `hpx::future` completion back into the P2300 `set_value` / `set_error` receiver protocol.

Usage:

```
hpx::id_type remote = hpx::find_all_localities()[1];
auto sched = hpx::distributed::distributed_scheduler{remote};
auto result = ex::just(42)
    | ex::continues_on(sched)
    | ex::then([](int x){ return x * 2; })
    | tt::sync_wait();
```

executors_distributed

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/executors_distributed/distribution_policy_executor.hpp

Defined in header `hpx/executors_distributed/distribution_policy_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **execution**

namespace **experimental**

Functions

```
template<typename DistPolicy>
distribution_policy_executor(DistPolicy&&) ->
    distribution_policy_executor<std::decay_t<DistPolicy>>
```

```
template<typename DistPolicy> HPX_DEPRECATED_V(1, 9,
    "hpx::parallel::execution::make_distribution_policy_executor is " "deprecated,
    use " "hpx::parallel::execution::distribution_policy_executor instead") distribution_policy_executor
```

Create a new *distribution_policy_executor* from the given *distribution_policy*.

Parameters

policy – The *distribution_policy* to create an executor from

```
template<typename DistPolicy>
```

```
class distribution_policy_executor
```

#include <distribution_policy_executor.hpp> A *distribution_policy_executor* creates groups of parallel execution agents that execute in threads implicitly created by the executor and placed on any of the associated localities.

Template Parameters

DistPolicy – The distribution policy type for which an executor should be created. The expression *hpx::traits::is_distribution_policy_v<DistPolicy>* must evaluate to true.

include

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/experimental/sandbox.hpp

Defined in header *hpx/experimental/sandbox.hpp*.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace experimental
```

```
        namespace sandbox
```

Functions

```
environment_info detect_environment()
```

Detect and return information about the current environment.

```
inline void describe_environment(std::ostream &os)
```

Print a formatted environment report.

```
template<typename F>
```

double **measure**(*F* &&fn, *std::size_t* iterations = 5)

Measure the mean execution time of a callable.

template<typename **SeqFn**, typename **ParFn**>

benchmark_report **benchmark**(*std::string* label, *SeqFn* &&seq_fn, *ParFn* &&par_fn, *std::size_t* iterations = 5)

Compare sequential and parallel execution of the same work.

struct **benchmark_report**

#include <sandbox.hpp> Holds the results of a comparative benchmark.

Public Functions

void **print**(*std::ostream* &os) const

Print a formatted benchmark report.

Parameters

os – Output stream.

Public Members

std::string **label**

User-provided description.

std::size_t **iterations** = 0

Measurement iterations.

double **seq_mean_ms** = 0.0

Sequential mean time (ms)

double **par_mean_ms** = 0.0

Parallel mean time (ms)

double **speedup** = 0.0

seq_time / par_time

double **efficiency_pct** = 0.0

(speedup / workers) * 100

std::size_t **num_workers** = 0

HPX workers during measurement.

struct **environment_info**

#include <sandbox.hpp> Describes the hardware and runtime environment.

Public Functions

void **print**(*std::ostream* &os) const
Print a formatted environment report.

Parameters

os – Output stream.

Public Members

std::size_t **cores** = 0
Physical CPU cores.

std::size_t **pus** = 0
Processing units (hardware threads)

std::size_t **numa_nodes** = 0
NUMA domains.

std::size_t **hpx_workers** = 0
Active HPX worker threads.

bool **is_sandbox** = false
Likely constrained environment.

init_runtime

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/hpx_start.hpp

Defined in header `hpx/hpx_start.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

hpx::init_params

Defined in header `hpx/init.hpp`⁵⁶².

See *Public API* for a list of names and headers that are part of the public *HPX* API.

struct **init_params**

Parameters used to initialize the HPX runtime through `hpx::init` and `hpx::start`.

⁵⁶² http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/init_runtime/include/hpx/init.hpp

hpx/hpx_init.hpp

Defined in header `hpx/hpx_init.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

hpx::finalize

Defined in header `hpx/init.hpp`⁵⁶³.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::disconnect*

int **hpx::finalize**(double shutdown_timeout, double localwait = -1.0, *hpx::error_code* &ec = throws)

Main function to gracefully terminate the HPX runtime system.

The function `hpx::finalize` is the main way to (gracefully) exit any HPX application. It must be called at least once, but can be called multiple times as well. However, only the first invocation will have effect. It will notify all connected localities to finish execution. Only after all other localities have exited this function will return, allowing to exit the console locality as well.

During the execution of this function the runtime system will invoke all registered shutdown functions (see `hpx::init`) on all localities.

The default value (-1.0) will try to find a globally set timeout value (can be set as the configuration parameter `hpx.shutdown_timeout`), and if that is not set or -1.0 as well, it will disable any timeout, each connected locality will wait for all existing HPX-threads to terminate.

The default value (-1.0) will try to find a globally set wait time value (can be set as the configuration parameter “`hpx.finalize_wait_time`”), and if this is not set or -1.0 as well, it will disable any addition local wait time before proceeding.

This function will block and wait for all connected localities to exit before returning to the caller. It should be the last HPX-function called by any application.

Using this function is an alternative to `hpx::disconnect`, these functions do not need to be called both.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

Parameters

- **shutdown_timeout** – This parameter allows to specify a timeout (in microseconds), specifying how long any of the connected localities should wait for pending tasks to be executed. After this timeout, all suspended HPX-threads will be aborted. Note, that this function will

⁵⁶³ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fdbc0c8044e7/libs/full/init_runtime/include/hpx/init.hpp

not abort any running HPX-threads. In any case the shutdown will not proceed as long as there is at least one pending/running HPX-thread.

- **localwait** – This parameter allows to specify a local wait time (in microseconds) before the connected localities will be notified and the overall shutdown process starts.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

This function will always return zero.

```
inline int hpx::finalize(hpx::error_code &ec = throws)
```

Main function to gracefully terminate the HPX runtime system.

The function `hpx::finalize` is the main way to (gracefully) exit any HPX application. It must be called at least once, but can be called multiple times as well. However, only the first invocation will have effect. It will notify all connected localities to finish execution. Only after all other localities have exited this function will return, allowing to exit the console locality as well.

During the execution of this function the runtime system will invoke all registered shutdown functions (see `hpx::init`) on all localities.

This function will block and wait for all connected localities to exit before returning to the caller. It should be the last HPX-function called by any application.

Using this function is an alternative to `hpx::disconnect`, these functions do not need to be called both.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

Parameters

ec – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

This function will always return zero.

hpx::disconnect

Defined in header `hpx/init.hpp`⁵⁶⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::finalize*

```
int hpx::disconnect(double shutdown_timeout, double localwait = -1.0, hpx::error_code &ec = throws)
```

Disconnect this locality from the application.

The function `hpx::disconnect` can be used to disconnect a locality from a running HPX application.

During the execution of this function the runtime system will invoke all registered shutdown functions (see `hpx::init`) on this locality.

⁵⁶⁴ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/init_runtime/include/hpx/init.hpp

The default value (`-1.0`) will try to find a globally set timeout value (can be set as the configuration parameter “`hpx.shutdown_timeout`”), and if that is not set or `-1.0` as well, it will disable any timeout, each connected locality will wait for all existing HPX-threads to terminate.

The default value (`-1.0`) will try to find a globally set wait time value (can be set as the configuration parameter `hpx.finalize_wait_time`), and if this is not set or `-1.0` as well, it will disable any addition local wait time before proceeding.

This function will block and wait for this locality to finish executing before returning to the caller. It should be the last HPX-function called by any locality being disconnected.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn’t throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

Parameters

- **shutdown_timeout** – This parameter allows to specify a timeout (in microseconds), specifying how long this locality should wait for pending tasks to be executed. After this timeout, all suspended HPX-threads will be aborted. Note, that this function will not abort any running HPX-threads. In any case the shutdown will not proceed as long as there is at least one pending/running HPX-thread.
- **localwait** – This parameter allows to specify a local wait time (in microseconds) before the connected localities will be notified and the overall shutdown process starts.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

This function will always return zero.

```
inline int hpx::disconnect(hpx::error_code &ec = throws)
```

Disconnect this locality from the application.

The function `hpx::disconnect` can be used to disconnect a locality from a running HPX application.

During the execution of this function the runtime system will invoke all registered shutdown functions (see `hpx::init`) on this locality.

This function will block and wait for this locality to finish executing before returning to the caller. It should be the last HPX-function called by any locality being disconnected.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn’t throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

Parameters

ec – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

This function will always return zero.

hpx::init

Defined in header `hpx/init.hpp`⁵⁶⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
inline int hpx::init(std::function<int(hpx::program_options::variables_map&)> f, int argc, char **argv,
                    init_params const &params = init_params())
```

Main entry point for launching the HPX runtime system.

This is the main entry point for any HPX application. This function (or one of its overloads) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as a HPX thread. This overload will not call `hpx_main`.

This is the main entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as a HPX thread.

Note: If the parameter `mode` is not given (defaulted), the created runtime system instance will be executed in console or worker mode depending on the command line arguments passed in `argc/argv`. Otherwise, it will be executed as specified by the parameter `mode`.

Parameters

- **f** – [in] The function to be scheduled as an HPX thread. Usually this function represents the main entry point of any HPX application. If `f` is `nullptr` the HPX runtime environment will be started without invoking `f`.
- **argc** – [in] The number of command line arguments passed in `argv`. This is usually the unchanged value as passed by the operating system (to `main()`).
- **argv** – [in] The command line arguments for this application, usually that is the value as passed by the operating system (to `main()`).
- **params** – [in] The parameters to the `hpx::init` function (See documentation of `hpx::init_params`)

Returns

The function returns the value, which has been returned from the user supplied `f`.

```
inline int hpx::init(std::function<int(int, char**)> f, int argc, char **argv, init_params const &params =
                    init_params())
```

Main entry point for launching the HPX runtime system.

This is the main entry point for any HPX application. This function (or one of its overloads) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as a HPX thread. This overload will not call `hpx_main`.

This is the main entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as a HPX thread.

⁵⁶⁵ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbdc0c8044e7/libs/full/init_runtime/include/hpx/init.hpp

Note: If the parameter `mode` is not given (defaulted), the created runtime system instance will be executed in console or worker mode depending on the command line arguments passed in `argc/argv`. Otherwise, it will be executed as specified by the parameter `mode`.

Parameters

- **f** – [in] The function to be scheduled as an HPX thread. Usually this function represents the main entry point of any HPX application. If `f` is `nullptr` the HPX runtime environment will be started without invoking `f`.
- **argc** – [in] The number of command line arguments passed in `argv`. This is usually the unchanged value as passed by the operating system (to `main()`).
- **argv** – [in] The command line arguments for this application, usually that is the value as passed by the operating system (to `main()`).
- **params** – [in] The parameters to the `hpx::init` function (See documentation of `hpx::init_params`)

Returns

The function returns the value, which has been returned from the user supplied `f`.

inline int `hpx::init`(int argc, char **argv, *init_params* const ¶ms = *init_params*())

Main entry point for launching the HPX runtime system.

This is the main entry point for any HPX application. This function (or one of its overloads) should be called from the users `main()` function. This overload expects a user-defined function named `hpx_main` at global scope, which will be used as the entry point for the HPX application.

This is the main entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as a HPX thread.

Note: If the parameter `mode` is not given (defaulted), the created runtime system instance will be executed in console or worker mode depending on the command line arguments passed in `argc/argv`. Otherwise, it will be executed as specified by the parameter `mode`.

Parameters

- **argc** – [in] The number of command line arguments passed in `argv`. This is usually the unchanged value as passed by the operating system (to `main()`).
- **argv** – [in] The command line arguments for this application, usually that is the value as passed by the operating system (to `main()`).
- **params** – [in] The parameters to the `hpx::init` function (See documentation of `hpx::init_params`)

Returns

The function returns the value, which has been returned from `hpx_main` (or 0 when executed in worker mode).

```
inline int hpx::init(std::nullptr_t f, int argc, char **argv, init_params const &params = init_params())
```

Main entry point for launching the HPX runtime system.

This is the main entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as a HPX thread. This overload will not call `hpx_main`.

This is the main entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as a HPX thread.

Note: If the parameter `mode` is not given (defaulted), the created runtime system instance will be executed in console or worker mode depending on the command line arguments passed in `argc/argv`. Otherwise, it will be executed as specified by the parameter `mode`.

Parameters

- **f** – [in] The function to be scheduled as an HPX thread. Usually this function represents the main entry point of any HPX application. If `f` is `nullptr` the HPX runtime environment will be started without invoking `f`.
- **argc** – [in] The number of command line arguments passed in `argv`. This is usually the unchanged value as passed by the operating system (to `main()`).
- **argv** – [in] The command line arguments for this application, usually that is the value as passed by the operating system (to `main()`).
- **params** – [in] The parameters to the `hpx::init` function (See documentation of `hpx::init_params`)

Returns

The function returns the value, which has been returned from the user supplied `f`.

```
inline int hpx::init(init_params const &params = init_params())
```

Main entry point for launching the HPX runtime system.

This is a simplified main entry point, which can be used to set up the runtime for an HPX application (the runtime system will be set up in console mode or worker mode depending on the command line settings). This overload expects a user-defined function named `hpx_main` at global scope, which will be used as the entry point for the HPX application.

This is a simplified main entry point, which can be used to set up the runtime for an HPX application (the runtime system will be set up in console mode or worker mode depending on the command line settings).

Note: The created runtime system instance will be executed in console or worker mode depending on the command line arguments passed in `argc/argv`. If not command line arguments are passed, console mode is assumed.

Note: If no command line arguments are passed the HPX runtime system will not support any of the default command line options as described in the section ‘HPX Command Line Options’.

Parameters

params – [in] The parameters to the `hpx::init` function (See documentation of `hpx::init_params`)

Returns

The function returns the value, which has been returned from `hpx_main` (or 0 when executed in worker mode).

hpx::start

Defined in header [hpx/init.hpp](http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fdbc0c8044e7/libs/full/init_runtime/include/hpx/init.hpp)⁵⁶⁶.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
inline bool hpx::start(std::function<int(hpx::program_options::variables_map&)> f, int argc, char **argv,
                      init_params const &params = init_params())
```

Main non-blocking entry point for launching the HPX runtime system.

This is the main, non-blocking entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as a HPX thread. It will return immediately after that. Use `hpx::wait` and `hpx::stop` to synchronize with the runtime system's execution. This overload will not call `hpx_main`.

This is the main, non-blocking entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as an HPX thread. It will return immediately after that. Use `hpx::wait` and `hpx::stop` to synchronize with the runtime system's execution.

Note: If the parameter `mode` is not given (defaulted), the created runtime system instance will be executed in console or worker mode depending on the command line arguments passed in `argc/argv`. Otherwise, it will be executed as specified by the parameter `mode`.

Parameters

- **f** – [in] The function to be scheduled as an HPX thread. Usually this function represents the main entry point of any HPX application. If `f` is `nullptr` the HPX runtime environment will be started without invoking `f`.
- **argc** – [in] The number of command line arguments passed in `argv`. This is usually the unchanged value as passed by the operating system (to `main()`).
- **argv** – [in] The command line arguments for this application, usually that is the value as passed by the operating system (to `main()`).
- **params** – [in] The parameters to the `hpx::start` function (See documentation of `hpx::init_params`)

Returns

The function returns *true* if command line processing succeeded and the runtime system was started successfully. It will return *false* otherwise.

⁵⁶⁶ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fdbc0c8044e7/libs/full/init_runtime/include/hpx/init.hpp

```
inline bool hpx::start(std::function<int(int, char**)> f, int argc, char **argv, init_params const &params =
    init_params())
```

Main non-blocking entry point for launching the HPX runtime system.

This is the main, non-blocking entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as a HPX thread. It will return immediately after that. Use `hpx::wait` and `hpx::stop` to synchronize with the runtime system's execution. This overload will not call `hpx_main`.

This is the main, non-blocking entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as an HPX thread. It will return immediately after that. Use `hpx::wait` and `hpx::stop` to synchronize with the runtime system's execution.

Note: If the parameter `mode` is not given (defaulted), the created runtime system instance will be executed in console or worker mode depending on the command line arguments passed in `argc/argv`. Otherwise, it will be executed as specified by the parameter `mode`.

Parameters

- **f** – [in] The function to be scheduled as an HPX thread. Usually this function represents the main entry point of any HPX application. If `f` is `nullptr` the HPX runtime environment will be started without invoking `f`.
- **argc** – [in] The number of command line arguments passed in `argv`. This is usually the unchanged value as passed by the operating system (to `main()`).
- **argv** – [in] The command line arguments for this application, usually that is the value as passed by the operating system (to `main()`).
- **params** – [in] The parameters to the `hpx::start` function (See documentation of `hpx::init_params`)

Returns

The function returns *true* if command line processing succeeded and the runtime system was started successfully. It will return *false* otherwise.

```
inline bool hpx::start(int argc, char **argv, init_params const &params = init_params())
```

Main non-blocking entry point for launching the HPX runtime system.

This is the main, non-blocking entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as a HPX thread. It will return immediately after that. Use `hpx::wait` and `hpx::stop` to synchronize with the runtime system's execution. This overload will not call `hpx_main`.

This is the main, non-blocking entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as an HPX thread. It will return immediately after that. Use `hpx::wait` and `hpx::stop` to synchronize with the runtime system's execution.

Note: If the parameter `mode` is not given (defaulted), the created runtime system instance will be executed in console or worker mode depending on the command line arguments passed in `argc/argv`. Otherwise, it will be

executed as specified by the `parametermode`.

Parameters

- **argc** – [in] The number of command line arguments passed in `argv`. This is usually the unchanged value as passed by the operating system (to `main()`).
- **argv** – [in] The command line arguments for this application, usually that is the value as passed by the operating system (to `main()`).
- **params** – [in] The parameters to the `hpx::start` function (See documentation of `hpx::init_params`)

Returns

The function returns *true* if command line processing succeeded and the runtime system was started successfully. It will return *false* otherwise.

```
inline bool hpx::start(std::nullptr_t f, int argc, char **argv, init_params const &params = init_params())
```

Main non-blocking entry point for launching the HPX runtime system.

This is the main, non-blocking entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as a HPX thread. It will return immediately after that. Use `hpx::wait` and `hpx::stop` to synchronize with the runtime system's execution. This overload will not call `hpx_main`.

This is the main, non-blocking entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as an HPX thread. It will return immediately after that. Use `hpx::wait` and `hpx::stop` to synchronize with the runtime system's execution.

Note: If the parameter `mode` is not given (defaulted), the created runtime system instance will be executed in console or worker mode depending on the command line arguments passed in `argc/argv`. Otherwise, it will be executed as specified by the `parametermode`.

Parameters

- **f** – [in] The function to be scheduled as an HPX thread. Usually this function represents the main entry point of any HPX application. If `f` is `nullptr` the HPX runtime environment will be started without invoking `f`.
- **argc** – [in] The number of command line arguments passed in `argv`. This is usually the unchanged value as passed by the operating system (to `main()`).
- **argv** – [in] The command line arguments for this application, usually that is the value as passed by the operating system (to `main()`).
- **params** – [in] The parameters to the `hpx::start` function (See documentation of `hpx::init_params`)

Returns

The function returns *true* if command line processing succeeded and the runtime system was started successfully. It will return *false* otherwise.

```
inline bool hpx::start(init_params const &params = init_params())
```

Main non-blocking entry point for launching the HPX runtime system.

This is a simplified main, non-blocking entry point, which can be used to set up the runtime for an HPX application (the runtime system will be set up in console mode or worker mode depending on the command line settings). It will return immediately after that. Use `hpx::wait` and `hpx::stop` to synchronize with the runtime system's execution.

This is the main, non-blocking entry point for any HPX application. This function (or one of its overloads below) should be called from the users `main()` function. It will set up the HPX runtime environment and schedule the function given by `f` as an HPX thread. It will return immediately after that. Use `hpx::wait` and `hpx::stop` to synchronize with the runtime system's execution.

Note: The created runtime system instance will be executed in console or worker mode depending on the command line arguments passed in `argc/argv`. If not command line arguments are passed, console mode is assumed.

Note: If no command line arguments are passed the HPX runtime system will not support any of the default command line options as described in the section 'HPX Command Line Options'.

Parameters

params – [in] The parameters to the `hpx::start` function (See documentation of `hpx::init_params`)

Returns

The function returns *true* if command line processing succeeded and the runtime system was started successfully. It will return *false* otherwise.

hpx::suspend

Defined in header `hpx/init.hpp`⁵⁶⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::resume`

```
int hpx::suspend(error_code &ec = throws)
```

Suspend the runtime system.

The function `hpx::suspend` is used to suspend the HPX runtime system. It can only be used when running HPX on a single locality. It will block waiting for all thread pools to be empty. This function only be called when the runtime is running, or already suspended in which case this function will do nothing.

Note: As long as `ec` is not pre-initialized to `hpx::throws` this function doesn't throw but returns the result code using the parameter `ec`. Otherwise it throws an instance of `hpx::exception`.

⁵⁶⁷ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/init_runtime/include/hpx/init.hpp

Parameters

ec – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

This function will always return zero.

hpx::resume

Defined in header [hpx/init.hpp](#)⁵⁶⁸.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::suspend*

int **hpx::resume**(*error_code* &ec = *throws*)

Resume the HPX runtime system.

The function *hpx::resume* is used to resume the HPX runtime system. It can only be used when running HPX on a single locality. It will block waiting for all thread pools to be resumed. This function only be called when the runtime suspended, or already running in which case this function will do nothing.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Parameters

ec – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

This function will always return zero.

naming_base

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/naming_base/unmanaged.hpp

Defined in header *hpx/naming_base/unmanaged.hpp*.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

⁵⁶⁸ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/init_runtime/include/hpx/init.hpp

Functions

`hpx::id_type unmanaged(hpx::id_type const &id)`

The helper function `hpx::unmanaged` can be used to generate a global identifier which does not participate in the automatic garbage collection.

Note: This function allows to apply certain optimizations to the process of memory management in HPX. It however requires the user to take full responsibility for keeping the referenced objects alive long enough.

Parameters

id – [in] The id to generated the unmanaged global id from This parameter can be itself a managed or a unmanaged global id.

Returns

This function returns a new global id referencing the same object as the parameter *id*. The only difference is that the returned global identifier does not participate in the automatic garbage collection.

namespace **naming**

parcelset

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/parcelset/message_handler_fwd.hpp

Defined in header `hpx/parcelset/message_handler_fwd.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/parcelset/connection_cache.hpp

Defined in header `hpx/parcelset/connection_cache.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/parcelset/parcelhandler.hpp

Defined in header `hpx/parcelset/parcelhandler.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/parcelset/parcelset_fwd.hpp

Defined in header hpx/parcelset/parcelset_fwd.hpp.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

parcelset_base

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/parcelset_base/parcelset_base_fwd.hpp

Defined in header hpx/parcelset_base/parcelset_base_fwd.hpp.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Defines

HPX_PARCELPORNT_BACKGROUND_MODE_ENUM_DEPRECATION_MSG

namespace **hpx**

namespace **parcelset**

Typedefs

using **parcel_write_handler_type** = *hpx::function<void(std::error_code const&, parcelset::parcel const&)>*

The type of the function that can be registered as a parcel write handler using the function *hpx::set_parcel_write_handler*.

Note: A parcel write handler is a function which is called by the parcel layer whenever a parcel has been sent by the underlying networking library and if no explicit parcel handler function was specified for the parcel.

Enums

enum class **parcelport_background_mode** : *std::uint8_t*

Type of background work to perform.

Values:

enumerator **flush_buffers**

perform buffer flush operations

enumerator **send**
 perform send operations (includes buffer flush)

enumerator **receive**
 perform receive operations

enumerator **all**
 perform all operations

Functions

inline bool **operator&**(*parcelport_background_mode* lhs, *parcelport_background_mode* rhs)
 char const ***get_parcelport_background_mode_name**(*parcelport_background_mode* mode)

Variables

parcel **empty_parcel**

constexpr *parcelport_background_mode* **parcelport_background_mode_flush_buffers** =
parcelport_background_mode::flush_buffers

constexpr *parcelport_background_mode* **parcelport_background_mode_send** =
parcelport_background_mode::send

constexpr *parcelport_background_mode* **parcelport_background_mode_receive** =
parcelport_background_mode::receive

constexpr *parcelport_background_mode* **parcelport_background_mode_all** =
parcelport_background_mode::all

hpx/parcelset_base/parcelport.hpp

Defined in header `hpx/parcelset_base/parcelport.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/parcelset_base/set_parcel_write_handler.hpp

Defined in header `hpx/parcelset_base/set_parcel_write_handler.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

plugin_factories

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/plugin_factories/parcelport_factory.hpp

Defined in header `hpx/plugin_factories/parcelport_factory.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/plugin_factories/binary_filter_factory.hpp

Defined in header `hpx/plugin_factories/binary_filter_factory.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **plugins**

template<typename **BinaryFilter**>

struct **binary_filter_factory** : public `binary_filter_factory_base`

#include <binary_filter_factory.hpp> The *message_handler_factory* provides a minimal implementation of a message handler's factory. If no additional functionality is required this type can be used to implement the full set of minimally required functions to be exposed by a message handler's factory instance.

Template Parameters

BinaryFilter – The message handler type this factory should be responsible for.

Public Functions

inline **binary_filter_factory**(*util::section* const *global, *util::section* const *local, bool isenabled)

Construct a new factory instance.

Note: The contents of both sections has to be cloned in order to save the configuration setting for later use.

Parameters

- **global** – [in] The pointer to a *hpx::util::section* instance referencing the settings read from the [settings] section of the global configuration file (`hpx.ini`) This pointer may be nullptr if no such section has been found.
- **local** – [in] The pointer to a *hpx::util::section* instance referencing the settings read from the section describing this component type: `[hpx.components.<name>]`, where `<name>` is the instance name of the component as given in the configuration files.
- **isenabled** –

~binary_filter_factory() override = default


```
inline serialization::binary_filter *create(bool compress, serialization::binary_filter *next_filter =
                                         nullptr) override
```

Create a new instance of a message handler

return Returns the newly created instance of the message handler supported by this factory

Protected Attributes

```
util::section global_settings_
```

```
util::section local_settings_
```

```
bool isEnabled_
```

hpx/plugin_factories/message_handler_factory.hpp

Defined in header `hpx/plugin_factories/message_handler_factory.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX API*.

hpx/plugin_factories/plugin_registry.hpp

Defined in header `hpx/plugin_factories/plugin_registry.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX API*.

namespace **hpx**

namespace **plugins**

```
template<typename Plugin, char const *const Name, char const *const Section, char const *const
Suffix>
```

```
struct plugin_registry : public plugin_registry_base
```

#include <plugin_registry.hpp> The `plugin_registry` provides a minimal implementation of a plugin's registry. If no additional functionality is required this type can be used to implement the full set of minimally required functions to be exposed by a plugin's registry instance.

Template Parameters

Plugin – The plugin type this registry should be responsible for.

Public Functions

```
inline bool get_plugin_info(std::vector<std::string> &fillini) override
```

Return the ini-information for all contained components.

Parameters

fillini – [in] The module is expected to fill this vector with the ini-information (one line per vector element) for all components implemented in this module.

Returns

Returns *true* if the parameter *fillini* has been successfully initialized with the registry data of all implemented in this module.

resiliency_distributed

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/resiliency_distributed/async_replay_distributed.hpp

Defined in header `hpx/resiliency_distributed/async_replay_distributed.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **resiliency**

namespace **experimental**

Functions

```
template<typename Action, typename ...Ts>
hpx::future<hpx::util::detail::invoke_deferred_result_t<Action, hpx::id_type, Ts...>> hpx_invoke(async_replay_t,
                                                                                               const
                                                                                               &ids,
                                                                                               Ac-
                                                                                               tion
                                                                                               &&ac-
                                                                                               tion,
                                                                                               Ts&&...
                                                                                               ts)
```

ADL hook for distributed `async_replay` CPO.

Asynchronously launches *action* on localities in *ids*, replaying on the next locality on error (aborts on `abort_replay_exception`).

Parameters

- **tag** – CPO tag (`async_replay_t`).
- **ids** – Target localities to try in order.
- **action** – Action to invoke on each locality.
- **ts** – Arguments forwarded to *action*.

Returns

future with the first successful result.

```
template<typename Pred, typename Action, typename ...Ts>
```

```

hpx::future<hpx::util::detail::invoke_deferred_result_t<Action, hpx::id_type, Ts...>> hpx_invoke(async_replay_validate
                                                                    std::vector<hpx::id_
                                                                    const
                                                                    &ids,
                                                                    Pred
                                                                    &&pred,
                                                                    Ac-
                                                                    tion
                                                                    &&ac-
                                                                    tion,
                                                                    Ts&&...
                                                                    ts)

```

ADL hook for distributed async_replay_validate CPO.

Asynchronously launches *action* on localities in *ids*, validating results with *pred*. Replays on the next locality on error (aborts on abort_replay_exception).

Parameters

- **tag** – CPO tag (async_replay_validate_t).
- **ids** – Target localities to try in order.
- **pred** – Predicate validating each invocation result.
- **action** – Action to invoke on each locality.
- **ts** – Arguments forwarded to *action*.

Returns

future with the first valid result.

hpx/resiliency_distributed/async_replicate_distributed.hpp

Defined in header hpx/resiliency_distributed/async_replicate_distributed.hpp.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **resiliency**

namespace **experimental**

Functions

```

template<typename Vote, typename Pred, typename Action, typename ...Ts>
decltype(auto) hpx_invoke(async_replicate_vote_validate_t, std::vector<hpx::id_type> const &ids,
                          Vote &&vote, Pred &&pred, Action &&action, Ts&&... ts)

```

ADL hook for distributed async_replicate_vote_validate CPO.

Asynchronously launches *action* on each locality in *ids*, validates results with *pred*, then runs valid results through *vote*.

Parameters

- **tag** – CPO tag (async_replicate_vote_validate_t).
- **ids** – Target localities (one invocation per locality).
- **vote** – Voting function selecting among valid results.
- **pred** – Predicate validating each invocation result.
- **action** – Action to invoke on each locality.

- **ts** – Arguments forwarded to *action*.

Returns

future with the voted-upon result.

```
template<typename Vote, typename Action, typename ...Ts>
decltype(auto) hpx_invoke(async_replicate_vote_t, std::vector<hpx::id_type> const &ids, Vote
&&vote, Action &&action, Ts&&... ts)
```

ADL hook for distributed *async_replicate_vote* CPO.

Asynchronously launches *action* on each locality in *ids*, runs valid results through *vote*. Validation uses the default exception check.

Parameters

- **tag** – CPO tag (*async_replicate_vote_t*).
- **ids** – Target localities (one invocation per locality).
- **vote** – Voting function selecting among valid results.
- **action** – Action to invoke on each locality.
- **ts** – Arguments forwarded to *action*.

Returns

future with the voted-upon result.

```
template<typename Pred, typename Action, typename ...Ts>
decltype(auto) hpx_invoke(async_replicate_validate_t, std::vector<hpx::id_type> const &ids, Pred
&&pred, Action &&action, Ts&&... ts)
```

ADL hook for distributed *async_replicate_validate* CPO.

Asynchronously launches *action* on each locality in *ids*, validates results with *pred*. Returns the first valid result.

Parameters

- **tag** – CPO tag (*async_replicate_validate_t*).
- **ids** – Target localities (one invocation per locality).
- **pred** – Predicate validating each invocation result.
- **action** – Action to invoke on each locality.
- **ts** – Arguments forwarded to *action*.

Returns

future with the first valid result.

```
template<typename Action, typename ...Ts>
decltype(auto) hpx_invoke(async_replicate_t, std::vector<hpx::id_type> const &ids, Action
&&action, Ts&&... ts)
```

ADL hook for distributed *async_replicate* CPO.

Asynchronously launches *action* on each locality in *ids*. Validates by checking for exceptions. Returns the first valid result.

Parameters

- **tag** – CPO tag (*async_replicate_t*).
- **ids** – Target localities (one invocation per locality).
- **action** – Action to invoke on each locality.
- **ts** – Arguments forwarded to *action*.

Returns

future with the first non-throwing result.

runtime_components

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/runtime_components/reflect_client.hpp

Defined in header `hpx/runtime_components/reflect_client.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/runtime_components/components_fwd.hpp

Defined in header `hpx/runtime_components/components_fwd.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Functions

`components::server::runtime_support *get_runtime_support_ptr()`

namespace **components**

template<typename **Component**>

struct **component_factory**

namespace **server**

namespace **stubs**

namespace **components**

hpx/runtime_components/component_registry.hpp

Defined in header `hpx/runtime_components/component_registry.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **components**

template<typename **Component**, factory_state **state**>

```
struct component_registry : public component_registry_base
```

#include <component_registry.hpp> The `component_registry` provides a minimal implementation of a component's registry. If no additional functionality is required this type can be used to implement the full set of minimally required functions to be exposed by a component's registry instance.

Template Parameters

Component – The component type this registry should be responsible for.

Public Functions

```
inline bool get_component_info(std::vector<std::string> &fillini, std::string const &filepath, bool  
                                is_static = false) override
```

Return the ini-information for all contained components.

Parameters

- **fillini** – [in] The module is expected to fill this vector with the ini-information (one line per vector element) for all components implemented in this module.
- **filepath** –
- **is_static** –

Returns

Returns *true* if the parameter *fillini* has been successfully initialized with the registry data of all implemented in this module.

```
inline void register_component_type() override
```

Enables this type of registry and sets its destroy mechanism.

hpx/runtime_components/macros.hpp

Defined in header `hpx/runtime_components/macros.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Defines

HPX_REGISTER_MINIMAL_COMPONENT_FACTORY(...)

HPX_REGISTER_COMPONENT(...)

HPX_REGISTER_ENABLED_COMPONENT_FACTORY(ComponentType, componentname)

HPX_REGISTER_DISABLED_COMPONENT_FACTORY(ComponentType, componentname)

HPX_REGISTER_MINIMAL_COMPONENT_FACTORY_(...)

HPX_REGISTER_MINIMAL_COMPONENT_FACTORY_1(ComponentType)

HPX_REGISTER_MINIMAL_COMPONENT_FACTORY_2(ComponentType, componentname)

HPX_REGISTER_MINIMAL_COMPONENT_FACTORY_3(ComponentType, componentname, state)

HPX_REGISTER_MINIMAL_COMPONENT_FACTORY_DYNAMIC(...)

HPX_REGISTER_COMPONENT_DYNAMIC(...)

This macro is used create and to register a minimal component factory for a component type which allows it to be remotely created using the `hpx::new_<>` function. This macro can be invoked with one, two or three arguments

HPX_REGISTER_ENABLED_COMPONENT_FACTORY_DYNAMIC(ComponentType, componentname)

HPX_REGISTER_DISABLED_COMPONENT_FACTORY_DYNAMIC(ComponentType, componentname)

HPX_REGISTER_MINIMAL_COMPONENT_FACTORY_DYNAMIC(...)

HPX_REGISTER_MINIMAL_COMPONENT_FACTORY_DYNAMIC_1(ComponentType)

HPX_REGISTER_MINIMAL_COMPONENT_FACTORY_DYNAMIC_2(ComponentType, componentname)

HPX_REGISTER_MINIMAL_COMPONENT_FACTORY_DYNAMIC_3(ComponentType, componentname, state)

HPX_REGISTER_MINIMAL_COMPONENT_REGISTRY(...)

This macro is used create and to register a minimal component registry with Hpx.Plugin.

HPX_REGISTER_MINIMAL_COMPONENT_REGISTRY(...)

HPX_REGISTER_MINIMAL_COMPONENT_REGISTRY_2(ComponentType, componentname)

HPX_REGISTER_MINIMAL_COMPONENT_REGISTRY_3(ComponentType, componentname, state)

HPX_REGISTER_MINIMAL_COMPONENT_REGISTRY_DYNAMIC(...)

HPX_REGISTER_MINIMAL_COMPONENT_REGISTRY_DYNAMIC(...)

HPX_REGISTER_MINIMAL_COMPONENT_REGISTRY_DYNAMIC_2(ComponentType, componentname)

HPX_REGISTER_MINIMAL_COMPONENT_REGISTRY_DYNAMIC_3(ComponentType, componentname, state)

HPX_REGISTER_DERIVED_COMPONENT_FACTORY(...)

This macro is used create and to register a minimal component factory with Hpx.Plugin. This macro may be used if the registered component factory is the only factory to be exposed from a particular module. If more than one factory needs to be exposed the *HPX_REGISTER_COMPONENT_FACTORY* and *HPX_REGISTER_COMPONENT_MODULE* macros should be used instead.

HPX_REGISTER_DERIVED_COMPONENT_FACTORY(...)

HPX_REGISTER_DERIVED_COMPONENT_FACTORY_3(ComponentType, componentname, basecomponentname)

HPX_REGISTER_DERIVED_COMPONENT_FACTORY_4(ComponentType, componentname, basecomponentname, state)

HPX_REGISTER_DERIVED_COMPONENT_FACTORY_DYNAMIC(...)

HPX_REGISTER_DERIVED_COMPONENT_FACTORY_DYNAMIC(...)

HPX_REGISTER_DERIVED_COMPONENT_FACTORY_DYNAMIC_3(ComponentType, componentname,
basecomponentname)

HPX_REGISTER_DERIVED_COMPONENT_FACTORY_DYNAMIC_4(ComponentType, componentname,
basecomponentname, state)

HPX_DISTRIBUTED_METADATA_DECLARATION(...)

HPX_DISTRIBUTED_METADATA_DECLARATION(...)

HPX_DISTRIBUTED_METADATA_DECLARATION_1(config)

HPX_DISTRIBUTED_METADATA_DECLARATION_2(config, name)

HPX_DISTRIBUTED_METADATA(...)

HPX_DISTRIBUTED_METADATA_...

HPX_DISTRIBUTED_METADATA_1(config)

HPX_DISTRIBUTED_METADATA_2(config, name)

HPX_REGISTER_COMPONENT

Defined in header `hpx/components.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Warning: doxygenfunction: Cannot find function “HPX_REGISTER_COMPONENT” in doxygen xml output for project “hpx” from directory: `../hpx_autodoc`

hpx::new_

Defined in header `hpx/components.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<typename Component, typename ...Ts>  
auto hpx::new_(id_type const &locality, Ts&&... vs)
```

Create one or more new instances of the given *Component* type on the specified locality.

This function creates one or more new instances of the given *Component* type on the specified locality and returns a future object for the global address which can be used to reference the new component instance.

Note: This function requires to specify an explicit template argument which will define what type of component(s) to create, for instance:

```
hpx::future<hpx::id_type> f =  
    hpx::new_<some_component>(hpx::find_here(), ...);  
hpx::id_type id = f.get();
```

Parameters

- **locality** – [in] The global address of the locality where the new instance should be created on.
- **vs** – [in] Any number of arbitrary arguments (passed by value, by const reference or by rvalue reference) which will be forwarded to the constructor of the created component instance.

Returns

The function returns different types depending on its use:

- If the explicit template argument *Component* represents a component type (`traits::is_component<Component>::value` evaluates to true), the function will return an `hpx::future` object instance which can be used to retrieve the global address of the newly created component.

- If the explicit template argument *Component* represents a client side object (`traits::is_client<Component>::value` evaluates to true), the function will return a new instance of that type which can be used to refer to the newly created component instance.

```
template<typename Component, typename ...Ts>
auto hpx::new_(id_type const &locality, std::size_t count, Ts&&... vs)
```

Create multiple new instances of the given *Component* type on the specified locality.

This function creates multiple new instances of the given *Component* type on the specified locality and returns a future object for the global address which can be used to reference the new component instance.

Note: This function requires to specify an explicit template argument which will define what type of component(s) to create, for instance:

```
hpx::future<std::vector<hpx::id_type> > f =
    hpx::new_<some_component[]>(hpx::find_here(), 10, ...);
hpx::id_type id = f.get();
```

Parameters

- **locality** – [in] The global address of the locality where the new instance should be created on.
- **count** – [in] The number of component instances to create
- **vs** – [in] Any number of arbitrary arguments (passed by value, by const reference or by rvalue reference) which will be forwarded to the constructor of the created component instance.

Returns

The function returns different types depending on its use:

- If the explicit template argument *Component* represents an array of a component type (i.e. *Component*[], where `traits::is_component<Component>::value` evaluates to true), the function will return an `hpx::future` object instance which holds a `std::vector<hpx::id_type>`, where each of the items in this vector is a global address of one of the newly created components.
- If the explicit template argument *Component* represents an array of a client side object type (i.e. *Component*[], where `traits::is_client<Component>::value` evaluates to true), the function will return an `hpx::future` object instance which holds a `std::vector<hpx::id_type>`, where each of the items in this vector is a client side instance of the given type, each representing one of the newly created components.

```
template<typename Component, typename DistPolicy, typename ...Ts>
auto hpx::new_(DistPolicy const &policy, Ts&&... vs)
```

Create one or more new instances of the given *Component* type based on the given distribution policy.

This function creates one or more new instances of the given *Component* type on the localities defined by the given distribution policy and returns a future object for global address which can be used to reference the new component instance(s).

Note: This function requires to specify an explicit template argument which will define what type of component(s) to create, for instance:

```
hpx::future<hpx::id_type> f =  
    hpx::new_<some_component>(hpx::default_layout, ...);  
hpx::id_type id = f.get();
```

Parameters

- **policy** – [in] The distribution policy used to decide where to place the newly created.
- **vs** – [in] Any number of arbitrary arguments (passed by value, by const reference or by rvalue reference) which will be forwarded to the constructor of the created component instance.

Returns

The function returns different types depending on its use:

- If the explicit template argument *Component* represents a component type (`traits::is_component<Component>::value` evaluates to true), the function will return an `hpx::future` object instance which can be used to retrieve the global address of the newly created component.
- If the explicit template argument *Component* represents a client side object (`traits::is_client<Component>::value` evaluates to true), the function will return a new instance of that type which can be used to refer to the newly created component instance.

```
template<typename Component, typename DistPolicy, typename ...Ts>  
auto hpx::new_(DistPolicy const &policy, std::size_t count, Ts&&... vs)
```

Create multiple new instances of the given *Component* type on the localities as defined by the given distribution policy.

This function creates multiple new instances of the given *Component* type on the localities defined by the given distribution policy and returns a future object for the global address which can be used to reference the new component instance.

Note: This function requires to specify an explicit template argument which will define what type of component(s) to create, for instance:

```
hpx::future<std::vector<hpx::id_type> > f =  
    hpx::new_<some_component[]>(hpx::default_layout, 10, ...);  
hpx::id_type id = f.get();
```

Parameters

- **policy** – [in] The distribution policy used to decide where to place the newly created.
- **count** – [in] The number of component instances to create
- **vs** – [in] Any number of arbitrary arguments (passed by value, by const reference or by rvalue reference) which will be forwarded to the constructor of the created component instance.

Returns

The function returns different types depending on its use:

- If the explicit template argument *Component* represents an array of a component type (i.e. *Component*[], where `traits::is_component<Component>::value` evaluates to true), the function will return an `hpx::future` object instance which holds a

`std::vector<hpx::id_type>`, where each of the items in this vector is a global address of one of the newly created components.

- If the explicit template argument *Component* represents an array of a client side object type (i.e. *Component*[], where `traits::is_client<Component>::value` evaluates to true), the function will return an `hpx::future` object instance which holds a `std::vector<hpx::id_type>`, where each of the items in this vector is a client side instance of the given type, each representing one of the newly created components.

`hpx/runtime_components/reflect_client_macros.hpp`

Defined in header `hpx/runtime_components/reflect_client_macros.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

`runtime_distributed`

See *Public API* for a list of names and headers that are part of the public *HPX* API.

`hpx/runtime_distributed.hpp`

Defined in header `hpx/runtime_distributed.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

class **runtime_distributed** : public *runtime*

#include <runtime_distributed.hpp> The *runtime* class encapsulates the HPX runtime system in a simple to use way. It makes sure all required parts of the HPX runtime system are properly initialized.

Public Functions

explicit **runtime_distributed**(*util::runtime_configuration* &rtcfg, int (*pre_main)(*runtime_mode*) = nullptr, void (*post_main)() = nullptr)

Construct a new HPX runtime instance

Parameters

- **rtcfg** – Runtime configuration for this instance
- **pre_main** – Function to be called before running the main action of this instance
- **post_main** – Function to be called after running the main action of this instance

~runtime_distributed()

The destructor makes sure all HPX runtime services are properly shut down before exiting.

int **start**(*hpx::function<hpx_main_function_type>* const &func, bool blocking = false) override

Start the runtime system.

Parameters

- **func** – [in] This is the main function of an HPX application. It will be scheduled for execution by the thread manager as soon as the runtime has been initialized. This function is expected to expose an interface as defined by the typedef *hpx_main_function_type*.

- **blocking** – [in] This allows to control whether this call blocks until the runtime system has been stopped. If this parameter is *true* the function *runtime::start* will call *runtime::wait* internally.

Returns

If a blocking is a true, this function will return the value as returned as the result of the invocation of the function object given by the parameter *func*. Otherwise it will return zero.

int **start**(bool blocking = false) override

Start the runtime system.

Parameters

blocking – [in] This allows to control whether this call blocks until the runtime system has been stopped. If this parameter is *true* the function *runtime::start* will call *runtime::wait* internally .

Returns

If a blocking is a true, this function will return the value as returned as the result of the invocation of the function object given by the parameter *func*. Otherwise it will return zero.

int **wait**() override

Wait for the shutdown action to be executed.

Returns

This function will return the value as returned as the result of the invocation of the function object given by the parameter *func*.

void **stop**(bool blocking = true) override

Initiate termination of the runtime system.

Parameters

blocking – [in] This allows to control whether this call blocks until the runtime system has been fully stopped. If this parameter is *false* then this call will initiate the stop action but will return immediately. Use a second call to stop with this parameter set to *true* to wait for all internal work to be completed.

int **finalize**(double shutdown_timeout) override

void **stop_helper**(bool blocking, *std::condition_variable* &cond, *std::mutex* &mtx)

Stop the runtime system, wait for termination.

Parameters

- **blocking** – [in] This allows to control whether this call blocks until the runtime system has been fully stopped. If this parameter is *false* then this call will initiate the stop action but will return immediately. Use a second call to stop with this parameter set to *true* to wait for all internal work to be completed.
- **cond** – Condition used to update all thread when done
- **mtx** – Mutex used by this function to sync all threads

int **suspend**() override

Suspend the runtime system.

int **resume**() override

Resume the runtime system.

bool **report_error**(*std::size_t* num_thread, *std::exception_ptr* const &e, bool terminate_all = true) override

Report a non-recoverable error to the runtime system.

Parameters

- **num_thread** – [in] The number of the operating system thread the error has been detected in.
- **e** – [in] This is an instance encapsulating an exception which lead to this function call.
- **terminate_all** – [in] Kill all localities attached to the currently running application (default: true)

bool **report_error**(*std::exception_ptr* const &*e*, bool *terminate_all* = true) override

Report a non-recoverable error to the runtime system.

Note: This function will retrieve the number of the current shepherd thread and forward to the `report_error` function above.

Parameters

- **e** – [in] This is an instance encapsulating an exception which lead to this function call.
- **terminate_all** – [in] Kill all localities attached to the currently running application (default: true)

int **run**(*hpx::function<hpx_main_function_type>* const &*func*) override

Run the HPX runtime system, use the given function for the main *thread* and block waiting for all threads to finish.

Note: The parameter *func* is optional. If no function is supplied, the runtime system will simply wait for the shutdown action without explicitly executing any main thread.

Parameters

func – [in] This is the main function of an HPX application. It will be scheduled for execution by the thread manager as soon as the runtime has been initialized. This function is expected to expose an interface as defined by the typedef *hpx_main_function_type*. This parameter is optional and defaults to none main thread function, in which case all threads have to be scheduled explicitly.

Returns

This function will return the value as returned as the result of the invocation of the function object given by the parameter **func**.

int **run**() override

Run the HPX runtime system, initially use the given number of (OS) threads in the thread-manager and block waiting for all threads to finish.

Returns

This function will always return 0 (zero).

bool **is_networking_enabled**() override

template<typename **F**>

inline *components::server::console_error_dispatcher::sink_type* **set_error_sink**(*F* &&*sink*)

performance_counters::registry &**get_counter_registry**()

Allow access to the registry counter registry instance used by the HPX runtime.

performance_counters::registry const &**get_counter_registry**() const

Allow access to the registry counter registry instance used by the HPX runtime.

void **register_query_counters**(*std::shared_ptr<util::query_counters>* const &*active_counters*)

void **start_active_counters**(*error_code* &*ec* = *throws*) const

`void stop_active_counters(error_code &ec = throws) const`

`void reset_active_counters(error_code &ec = throws) const`

`void reinit_active_counters(bool reset = true, error_code &ec = throws) const`

`void evaluate_active_counters(bool reset = false, char const *description = nullptr, error_code &ec = throws) const`

`void stop_evaluating_counters(bool terminate = false) const`

`agas::addressing_service &get_agas_client()`
Allow access to the AGAS client instance used by the HPX runtime.

`hpx::threads::threadmanager &get_thread_manager() override`
Allow access to the thread manager instance used by the HPX runtime.

`applier::applier &get_applier()`
Allow access to the applier instance used by the HPX runtime.

`std::string here() const override`
Returns a string of the locality endpoints (usable in debug output)

`naming::address_type get_runtime_support_lva() const`

`naming::gid_type get_next_id(std::size_t count = 1)`

`void init_id_pool_range()`

`util::unique_id_ranges &get_id_pool()`

`void initialize_agas()`
Initialize AGAS operation.

`void add_pre_startup_function(startup_function_type f) override`
Add a function to be executed inside a HPX thread before `hpx_main` but guaranteed to be executed before any startup function registered with `add_startup_function`.

Note: The difference to a startup function is that all pre-startup functions will be (system-wide) executed before any startup function.

Parameters

f – The function ‘f’ will be called from inside a HPX thread before `hpx_main` is executed. This is very useful to setup the runtime environment of the application (install performance counters, etc.)

`void add_startup_function(startup_function_type f) override`

Add a function to be executed inside a HPX thread before `hpx_main`

Parameters

f – The function ‘f’ will be called from inside a HPX thread before `hpx_main` is executed. This is very useful to setup the runtime environment of the application (install performance counters, etc.)

void **add_pre_shutdown_function**(*shutdown_function_type* f) override

Add a function to be executed inside a HPX thread during `hpx::finalize`, but guaranteed before any of the shutdown functions is executed.

Note: The difference to a shutdown function is that all pre-shutdown functions will be (system-wide) executed before any shutdown function.

Parameters

f – The function ‘f’ will be called from inside a HPX thread while `hpx::finalize` is executed. This is very useful to tear down the runtime environment of the application (uninstall performance counters, etc.)

void **add_shutdown_function**(*shutdown_function_type* f) override

Add a function to be executed inside a HPX thread during `hpx::finalize`

Parameters

f – The function ‘f’ will be called from inside a HPX thread while `hpx::finalize` is executed. This is very useful to tear down the runtime environment of the application (uninstall performance counters, etc.)

hpx::util::io_service_pool ***get_thread_pool**(char const *name) override

Access one of the internal thread pools (io_service instances) HPX is using to perform specific tasks. The three possible values for the argument name are “main_pool”, “io_pool”, “parcel_pool”, and “timer_pool”. For any other argument value the function will return zero.

bool **register_thread**(char const *name, *std::size_t* num = 0, bool service_thread = true, *error_code* &ec = *throws*) override

Register an external OS-thread with HPX.

notification_policy_type **get_notification_policy**(char const *prefix, *runtime_local::os_thread_type* type) override

Generate a new notification policy instance for the given thread name prefix

std::uint32_t **get_locality_id**(*error_code* &ec) const override

std::size_t **get_num_worker_threads**() const override

std::uint32_t **get_num_localities**(*hpx::launch::sync_policy*, *error_code* &ec) const override

std::uint32_t **get_initial_num_localities**() const override

hpx::future<std::uint32_t> **get_num_localities**() const override

std::string **get_locality_name**() const override

std::uint32_t **get_num_localities**(*hpx::launch::sync_policy*, *components::component_type* type, *error_code* &ec) const

hpx::future<std::uint32_t> **get_num_localities**(*components::component_type* type) const

std::uint32_t **assign_cores**(*std::string* const &locality_basename, *std::uint32_t* num_threads) override

std::uint32_t **assign_cores**() override

Public Static Functions

static void **register_counter_types**()

Install all performance counters related to this runtime instance.

Private Types

using **used_cores_map_type** = *std::map<std::string, std::uint32_t>*

Private Functions

threads::thread_result_type **run_helper**(*hpx::function<runtime::hpx_main_function_type>* const
&func, int &result)

void **init_global_data**()

void **deinit_global_data**()

void **wait_helper**(*std::mutex* &mtx, *std::condition_variable* &cond, bool &running)

void **init_tss_helper**(char const *context, *runtime_local::os_thread_type* type, *std::size_t*
local_thread_num, *std::size_t* global_thread_num, char const *pool_name, char
const *postfix, bool service_thread)

void **deinit_tss_helper**(char const *context, *std::size_t* num) const

void **init_tss_ex**(*std::string* const &locality, char const *context, *runtime_local::os_thread_type* type,
std::size_t local_thread_num, *std::size_t* global_thread_num, char const
*pool_name, char const *postfix, bool service_thread, *error_code* &ec) const

Private Members

runtime_mode **mode_**

util::unique_id_ranges **id_pool_**

agas::addressing_service **agas_client_**

applier::applier **applier_**

used_cores_map_type **used_cores_map_**

std::unique_ptr<components::server::runtime_support> **runtime_support_**

std::shared_ptr<util::query_counters> **active_counters_**


```
int (*pre_main_)(runtime_mode)
```

```
void (*post_main_())
```

Private Static Functions

```
static void default_errorsink(std::string const&)
```

hpx::find_locality

Defined in header [hpx/runtime.hpp](#)⁵⁶⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

hpx::id_type **hpx::find_locality**(*components::component_type* type, *error_code* &ec = *throws*)

Return the global id representing an arbitrary locality which supports the given component type.

The function `find_locality()` can be used to retrieve the global id of an arbitrary locality currently available to this application which supports the creation of instances of the given component type.

See also:

hpx::find_here(), *hpx::find_all_localities()*

Note: Generally, the id of a locality can be used for instance to create new instances of components and to invoke plain actions (global functions).

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Note: This function will return meaningful results only if called from an HPX-thread. It will return *hpx::invalid_id* otherwise.

Parameters

- **type** – [in] The type of the components for which the function should return any available locality.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

The global id representing an arbitrary locality currently available to this application which supports the creation of instances of the given component type. If no locality supporting the given component type is currently available, this function will return *hpx::invalid_id*.

⁵⁶⁹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/runtime.hpp>

hpx::find_root_locality

Defined in header [hpx/runtime.hpp](#)⁵⁷⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- [hpx::find_all_localities](#)
- [hpx::find_remote_localities](#)

hpx::id_type **hpx::find_root_locality**(*error_code* &ec = *throws*)

Return the global id representing the root locality.

The function `find_root_locality()` can be used to retrieve the global id usable to refer to the root locality. The root locality is the locality where the main AGAS service is hosted.

See also:

`hpx::find_all_localities()`, `hpx::find_locality()`

Note: Generally, the id of a locality can be used for instance to create new instances of components and to invoke plain actions (global functions).

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Note: This function will return meaningful results only if called from an HPX-thread. It will return *hpx::invalid_id* otherwise.

Parameters

ec – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

The global id representing the root locality for this application.

hpx::find_all_localities

Defined in header [hpx/runtime.hpp](#)⁵⁷¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- [hpx::find_root_locality](#)
- [hpx::find_remote_localities](#)

⁵⁷⁰ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

⁵⁷¹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

```
std::vector<hpx::id_type> hpx::find_all_localities(error_code &ec = throws)
```

Return the list of global ids representing all localities available to this application.

The function `find_all_localities()` can be used to retrieve the global ids of all localities currently available to this application.

See also:

`hpx::find_here()`, `hpx::find_locality()`

Note: Generally, the id of a locality can be used for instance to create new instances of components and to invoke plain actions (global functions).

Note: As long as *ec* is not pre-initialized to `hpx::throws` this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of `hpx::exception`.

Note: This function will return meaningful results only if called from an HPX-thread. It will return an empty vector otherwise.

Parameters

ec – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

Returns

The global ids representing the localities currently available to this application.

```
std::vector<hpx::id_type> hpx::find_all_localities(components::component_type type, error_code &ec = throws)
```

Return the list of global ids representing all localities available to this application which support the given component type.

The function `find_all_localities()` can be used to retrieve the global ids of all localities currently available to this application which support the creation of instances of the given component type.

See also:

`hpx::find_here()`, `hpx::find_locality()`

Note: Generally, the id of a locality can be used for instance to create new instances of components and to invoke plain actions (global functions).

Note: As long as *ec* is not pre-initialized to `hpx::throws` this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of `hpx::exception`.

Note: This function will return meaningful results only if called from an HPX-thread. It will return an empty vector otherwise.

Parameters

- **type** – [in] The type of the components for which the function should return the available localities.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

The global ids representing the localities currently available to this application which support the creation of instances of the given component type. If no localities supporting the given component type are currently available, this function will return an empty vector.

hpx::find_remote_localities

Defined in header [hpx/runtime.hpp](#)⁵⁷².

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::find_root_locality*
- *hpx::find_all_localities*

std::vector<hpx::id_type> hpx::find_remote_localities(error_code &ec = throws)

Return the list of locality ids of remote localities supporting the given component type. By default this function will return the list of all remote localities (all but the current locality).

The function *find_remote_localities()* can be used to retrieve the global ids of all remote localities currently available to this application (i.e. all localities except the current one).

See also:

hpx::find_here(), *hpx::find_locality()*

Note: Generally, the id of a locality can be used for instance to create new instances of components and to invoke plain actions (global functions).

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Note: This function will return meaningful results only if called from an HPX-thread. It will return an empty vector otherwise.

⁵⁷² <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e77/libs/full/include/include/hpx/runtime.hpp>

Parameters

ec – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

The global ids representing the remote localities currently available to this application.

```
std::vector<hpx::id_type> hpx::find_remote_localities(components::component_type type, error_code &ec
= throws)
```

Return the list of locality ids of remote localities supporting the given component type. By default this function will return the list of all remote localities (all but the current locality).

The function `find_remote_localities()` can be used to retrieve the global ids of all remote localities currently available to this application (i.e. all localities except the current one) which support the creation of instances of the given component type.

See also:

hpx::find_here(), *hpx::find_locality()*

Note: Generally, the id of a locality can be used for instance to create new instances of components and to invoke plain actions (global functions).

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Note: This function will return meaningful results only if called from an HPX-thread. It will return an empty vector otherwise.

Parameters

- **type** – [in] The type of the components for which the function should return the available remote localities.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

The global ids representing the remote localities currently available to this application.

hpx/runtime_distributed/runtime_support.hpp

Defined in header `hpx/runtime_distributed/runtime_support.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **agas**

Functions

```
struct runtime_components_init_interface_functions &runtime_components_init()
```

namespace **components**

Functions

```
struct counter_interface_functions &counter_init()
```

```
class runtime_support : public hpx::components::stubs::runtime_support  
    #include <runtime_support.hpp> The runtime_support class is the client side representation of a  
    server::runtime_support component
```

Public Functions

```
inline runtime_support(hpx::id_type const &gid = hpx::invalid_id)
```

Create a client side representation for the existing server::runtime_support instance with the given global id *gid*.

```
template<typename Component, typename ...Ts>  
inline hpx::id_type create_component(Ts&&... vs)
```

Create a new component type using the *runtime_support*.

```
template<typename Component, typename ...Ts>  
inline hpx::future<hpx::id_type> create_component_async(Ts&&... vs)
```

Asynchronously create a new component using the *runtime_support*.

```
template<bool WithCount, typename Component, typename ...Ts>  
inline std::vector<hpx::id_type> bulk_create_component(std::size_t count, Ts&&... vs)
```

Asynchronously create N new default constructed components using the *runtime_support*

```
template<bool WithCount, typename Component, typename ...Ts>  
inline hpx::future<std::vector<hpx::id_type>> bulk_create_components_async(std::size_t  
                                                                            count, Ts&&...  
                                                                            vs)
```

Asynchronously create a new component using the *runtime_support*.

```
inline hpx::future<int> load_components_async() const
```

```
inline int load_components() const
```

```
inline hpx::future<void> call_startup_functions_async(bool pre_startup) const
```

```
inline void call_startup_functions(bool pre_startup) const
```

```
inline hpx::future<void> shutdown_async(double timeout = -1) const
```

Shutdown the given runtime system.

```
inline void shutdown(double timeout = -1) const
```

```
inline void shutdown_all(double timeout = -1) const
```

Shutdown the runtime systems of all localities.

```

inline hpx::future<void> terminate_async() const
    Terminate the given runtime system.

inline void terminate() const

inline void terminate_all() const
    Terminate the runtime systems of all localities.

inline void get_config(util::section &ini) const
    Retrieve configuration information.

inline hpx::id_type const &get_id() const

inline naming::gid_type const &get_raw_gid() const

```

Private Types

```
using base_type = stubs::runtime_support
```

Private Members

```
hpx::id_type gid_
```

hpx/runtime_distributed/applier_fwd.hpp

Defined in header `hpx/runtime_distributed/applier_fwd.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
namespace hpx
```

```
    namespace applier
```

Functions

```
applier &get_applier()
```

The function *get_applier* returns a reference to the (thread specific) *applier* instance.

```
applier *get_applier_ptr()
```

The function *get_applier* returns a pointer to the (thread specific) *applier* instance. The returned pointer is NULL if the current thread is not known to HPX or if the runtime system is not active.

```
namespace applier
```

The namespace *applier* contains all definitions needed for the class `hpx::applier::applier` and its related functionality. This namespace is part of the HPX core module.

hpx/runtime_distributed/get_locality_name.hpp

Defined in header `hpx/runtime_distributed/get_locality_name.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Functions

future<std::string> **get_locality_name**(*hpx::id_type* const &id)

Return the name of the referenced locality.

This function returns a future referring to the name for the locality of the given id.

See also:

std::string *get_locality_name*()

Parameters

id – [in] The global id of the locality for which the name should be retrieved

Returns

This function returns the name for the locality of the given id. The name is retrieved from the underlying networking layer and may be different for different parcel ports.

hpx/runtime_distributed/find_here.hpp

Defined in header `hpx/runtime_distributed/find_here.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Functions

hpx::id_type **find_here**(*error_code* &ec = *throws*)

Return the global id representing this locality.

The function `find_here()` can be used to retrieve the global id usable to refer to the current locality.

See also:

hpx::find_all_localities(), *hpx::find_locality*()

Note: Generally, the id of a locality can be used for instance to create new instances of components and to invoke plain actions (global functions).

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

Note: This function will return meaningful results only if called from an HPX-thread. It will return *hpx::invalid_id* otherwise.

Parameters

ec – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

The global id representing the locality this function has been called on.

hpx/runtime_distributed/applier.hpp

Defined in header `hpx/runtime_distributed/applier.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **applier**

class **applier**

#include <applier.hpp> The *applier* class is used to decide whether a particular action has to be issued on a local or a remote resource. If the target component is local a new *thread* will be created, if the target is remote a parcel will be sent.

Public Functions

applier(*applier* const&) = delete

applier(*applier*&&) = delete

applier &**operator**=(*applier* const&) = delete

applier &**operator**=(*applier*&&) = delete

applier()

void **init**(*threads::threadmanager* &tm)

~applier() = default

void **initialize**(*std::uint64_t* rts)

threads::threadmanager &**get_thread_manager**()

Access the *thread-manager* instance associated with this *applier*.

This function returns a reference to the thread manager this *applier* instance has been created with.

```
naming::gid_type const &get_raw_locality(error_code &ec = throws) const
```

Allow access to the locality of the locality this applier instance is associated with.

This function returns a reference to the locality this applier instance is associated with.

```
std::uint32_t get_locality_id(error_code &ec = throws) const
```

Allow access to the id of the locality this applier instance is associated with.

This function returns a reference to the id of the locality this applier instance is associated with.

```
bool get_raw_remote_localities(std::vector<naming::gid_type> &locality_ids,  
                               components::component_type type =  
                               to_int(hpx::components::component_enum_type::invalid),  
                               error_code &ec = throws) const
```

Return list of localities of all remote localities registered with the AGAS service for a specific component type.

This function returns a list of all remote localities (all localities known to AGAS except the local one) supporting the given component type.

Parameters

- **locality_ids** – [out] The reference to a vector of *id_types* filled by the function.
- **type** – [in] The type of the component which needs to exist on the returned localities.

Returns

The function returns *true* if there is at least one remote locality known to the AGAS service (!*prefixes.empty()*).

```
bool get_remote_localities(std::vector<hpx::id_type> &locality_ids,  
                           components::component_type type =  
                           to_int(hpx::components::component_enum_type::invalid),  
                           error_code &ec = throws) const
```

```
bool get_raw_localities(std::vector<naming::gid_type> &locality_ids,  
                        components::component_type type =  
                        to_int(hpx::components::component_enum_type::invalid)) const
```

Return list of *locality_ids* of all localities registered with the AGAS service for a specific component type.

This function returns a list of all localities (all localities known to AGAS except the local one) supporting the given component type.

Parameters

- **locality_ids** – [out] The reference to a vector of *id_types* filled by the function.
- **type** – [in] The type of the component which needs to exist on the returned localities.

Returns

The function returns *true* if there is at least one remote locality known to the AGAS service (!*prefixes.empty()*).

```
bool get_localities(std::vector<hpx::id_type> &locality_ids, error_code &ec = throws) const
```

```
bool get_localities(std::vector<hpx::id_type> &locality_ids, components::component_type  
                  type, error_code &ec = throws) const
```

```
inline naming::gid_type const &get_runtime_support_raw_gid() const
```

By convention the *runtime_support* has a *gid* identical to the prefix of the locality the *runtime_support* is responsible for

```
inline hpx::id_type const &get_runtime_support_gid() const
```

By convention the *runtime_support* has a *gid* identical to the prefix of the locality the *runtime_support* is responsible for

Private Members

threads::threadmanager ***thread_manager_**

hpx::id_type **runtime_support_id_**

hpx/runtime_distributed/runtime_fwd.hpp

Defined in header `hpx/runtime_distributed/runtime_fwd.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Functions

runtime_distributed &**get_runtime_distributed**()

runtime_distributed *&**get_runtime_distributed_ptr**()

naming::gid_type const &**get_locality**()

The function *get_locality* returns a reference to the locality prefix.

void **start_active_counters**(*error_code* &*ec* = *throws*)

Start all active performance counters, optionally naming the section of code.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Note: The active counters are those which have been specified on the command line while executing the application (see command line option `-hpx:print-counter`)

Parameters

ec – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

void **reset_active_counters**(*error_code* &*ec* = *throws*)

Resets all active performance counters.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Note: The active counters are those which have been specified on the command line while executing the application (see command line option `-hpx:print-counter`)

Parameters

ec – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

void **reinit_active_counters**(bool reset = true, *error_code* &ec = *throws*)

Re-initialize all active performance counters.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Note: The active counters are those which have been specified on the command line while executing the application (see command line option `-hpx:print-counter`)

Parameters

- **reset** – [in] Reset the current values before re-initializing counters (default: true)
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

void **stop_active_counters**(*error_code* &ec = *throws*)

Stop all active performance counters.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Note: The active counters are those which have been specified on the command line while executing the application (see command line option `-hpx:print-counter`)

Parameters

ec – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

void **evaluate_active_counters**(bool reset = false, char const *description = nullptr, *error_code* &ec = *throws*)

Evaluate and output all active performance counters, optionally naming the point in code marked by this function.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Note: The output generated by this function is redirected to the destination specified by the corresponding command line options (see `-hpx:print-counter-destination`).

Note: The active counters are those which have been specified on the command line while executing the application (see command line option `-hpx:print-counter`)

Parameters

- **reset** – [in] this is an optional flag allowing to reset the counter value after it has been evaluated.
- **description** – [in] this is an optional value naming the point in the code marked by the call to this function.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

```
serialization::binary_filter *create_binary_filter(char const *binary_filter_type, bool compress,
                                                    serialization::binary_filter *next_filter = nullptr,
                                                    error_code &ec = throws)
```

Create an instance of a binary filter plugin.

Note: As long as `ec` is not pre-initialized to `hpx::throws` this function doesn't throw but returns the result code using the parameter `ec`. Otherwise it throws an instance of `hpx::exception`.

Parameters

- **binary_filter_type** – [in] The type of the binary filter to create
- **compress** – [in] The created filter should support compression
- **next_filter** – [in] Use this as the filter to dispatch the invocation into.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

hpx/runtime_distributed/copy_component.hpp

Defined in header `hpx/runtime_distributed/copy_component.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **components**

Functions

```
template<typename Component>  
future<hpx::id_type> copy(hpx::id_type const &to_copy)
```

Copy given component to the specified target locality.

The function `copy<Component>` will create a copy of the component referenced by `to_copy` on the locality specified with `target_locality`. It returns a future referring to the newly created component instance.

Note: The new component instance is created on the locality of the component instance which is to be copied.

Parameters

to_copy – [in] The global id of the component to copy

Template Parameters

The – only template argument specifies the component type to create.

Returns

A future representing the global id of the newly (copied) component instance.

```
template<typename Component>  
future<hpx::id_type> copy(hpx::id_type const &to_copy, hpx::id_type const &target_locality)
```

Copy given component to the specified target locality.

The function `copy<Component>` will create a copy of the component referenced by `to_copy` on the locality specified with `target_locality`. It returns a future referring to the newly created component instance.

Parameters

- **to_copy** – [in] The global id of the component to copy
- **target_locality** – [in] The locality where the copy should be created.

Template Parameters

The – only template argument specifies the component type to create.

Returns

A future representing the global id of the newly (copied) component instance.

```
template<typename Derived, typename Stub, typename Data>  
Derived copy(client_base<Derived, Stub, Data> const &to_copy, hpx::id_type const &target_locality =  
             hpx::invalid_id)
```

Copy given component to the specified target locality.

The function `copy` will create a copy of the component referenced by the client side object `to_copy` on the locality specified with `target_locality`. It returns a new client side object future referring to the newly created component instance.

Note: If the second argument is omitted (or is `invalid_id`) the new component instance is created on the locality of the component instance which is to be copied.

Parameters

- **to_copy** – [in] The client side object representing the component to copy
- **target_locality** – [in, optional] The locality where the copy should be created (default is same locality as source).

Template Parameters

The – only template argument specifies the component type to create.

Returns

A future representing the global id of the newly (copied) component instance.

hpx/runtime_distributed/get_num_localities.hpp

Defined in header `hpx/runtime_distributed/get_num_localities.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Functions

`hpx::future<std::uint32_t> get_num_localities(components::component_type t)`

Asynchronously return the number of localities which are currently registered for the running application.

The function `get_num_localities` asynchronously returns the number of localities currently connected to the console which support the creation of the given component type. The returned future represents the actual result.

See also:

`hpx::find_all_localities`, `hpx::get_num_localities`

Note: This function will return meaningful results only if called from an HPX-thread. It will return 0 otherwise.

Parameters

t – The component type for which the number of connected localities should be retrieved.

`std::uint32_t get_num_localities(launch::sync_policy, components::component_type t, error_code &ec = throws)`

Synchronously return the number of localities which are currently registered for the running application.

The function `get_num_localities` returns the number of localities currently connected to the console which support the creation of the given component type. The returned future represents the actual result.

See also:

`hpx::find_all_localities`, `hpx::get_num_localities`

Note: This function will return meaningful results only if called from an HPX-thread. It will return 0 otherwise.

Parameters

- **t** – The component type for which the number of connected localities should be retrieved.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

hpx/runtime_distributed/migrate_component.hpp

Defined in header `hpx/runtime_distributed/migrate_component.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace components
```

Functions

```
template<typename Component, typename DistPolicy>
future<hpx::id_type> migrate(hpx::id_type const &to_migrate, [[maybe_unused]] DistPolicy const
                             &policy)
```

Migrate the given component to the specified target locality

The function *migrate*<*Component*> will migrate the component referenced by *to_migrate* to the locality specified with *target_locality*. It returns a future referring to the migrated component instance.

Parameters

- **to_migrate** – [in] The client side representation of the component to migrate.
- **policy** – [in] A distribution policy which will be used to determine the locality to migrate this object to.

Template Parameters

- **Component** – Specifies the component type of the component to migrate.
- **DistPolicy** – Specifies the distribution policy to use to determine the destination locality.

Returns

A future representing the global id of the migrated component instance. This should be the same as *migrate_to*.

```
template<typename Derived, typename Stub, typename Data, typename DistPolicy>
Derived migrate(client_base<Derived, Stub, Data> const &to_migrate, DistPolicy const &policy)
```

Migrate the given component to the specified target locality

The function *migrate*<*Component*> will migrate the component referenced by *to_migrate* to the locality specified with *target_locality*. It returns a future referring to the migrated component instance.

Parameters

- **to_migrate** – [in] The client side representation of the component to migrate.
- **policy** – [in] A distribution policy which will be used to determine the locality to migrate this object to.

Template Parameters

- **Derived** – Specifies the component type of the component to migrate.
- **DistPolicy** – Specifies the distribution policy to use to determine the destination locality.

Returns

A future representing the global id of the migrated component instance. This should be the same as *migrate_to*.

```
template<typename Component>
future<hpx::id_type> migrate(hpx::id_type const &to_migrate, hpx::id_type const &target_locality)
```

Migrate the component with the given id to the specified target locality

The function *migrate*<*Component*> will migrate the component referenced by *to_migrate* to the locality specified with *target_locality*. It returns a future referring to the migrated component instance.

Parameters

- **to_migrate** – [in] The global id of the component to migrate.
- **target_locality** – [in] The locality where the component should be migrated to.

Template Parameters

Component – Specifies the component type of the component to migrate.

Returns

A future representing the global id of the migrated component instance. This should be the same as *migrate_to*.

```
template<typename Derived, typename Stub, typename Data>
Derived migrate(client_base<Derived, Stub, Data> const &to_migrate, hpx::id_type const
               &target_locality)
```

Migrate the given component to the specified target locality

The function *migrate*<*Component*> will migrate the component referenced by *to_migrate* to the locality specified with *target_locality*. It returns a future referring to the migrated component instance.

Parameters

- **to_migrate** – [in] The client side representation of the component to migrate.
- **target_locality** – [in] The id of the locality to migrate this object to.

Template Parameters

Derived – Specifies the component type of the component to migrate.

Returns

A client side representation of representing of the migrated component instance. This should be the same as *migrate_to*.

hpx/runtime_distributed/server/runtime_support.hpp

Defined in header `hpx/runtime_distributed/server/runtime_support.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace components
```

```
        namespace server
```

```
            class runtime_support
```

Public Types

```
            using type_holder = runtime_support
```

Public Functions

explicit **runtime_support**(*hpx::util::runtime_configuration* &cfg)

inline **~runtime_support**()

void **delete_function_lists**()

void **tidy**()

template<typename **Component**>
naming::gid_type **create_component**()

Actions to create new objects.

template<typename **Component**, typename **T**, typename ...**Ts**>
naming::gid_type **create_component**(*T* v, *Ts...* vs)

template<typename **Component**, typename ...**Ts**>
std::vector<naming::gid_type> **bulk_create_component**(*std::size_t* count, *Ts...* vs)

template<typename **Component**, typename ...**Ts**>
std::vector<naming::gid_type> **bulk_create_component_with_count**(*std::size_t* count, *Ts...*
vs)

template<typename **Component**>
naming::gid_type **copy_create_component**(*std::shared_ptr<Component>* const &p, bool)

template<typename **Component**>
naming::gid_type **migrate_component_to_here**(*std::shared_ptr<Component>* const &p,
hpx::id_type)

void **shutdown**(double timeout, *hpx::id_type* const &respond_to)
Gracefully shutdown this runtime system instance.

void **shutdown_all**(double timeout)
Gracefully shutdown runtime system instances on all localities.

void **terminate**(*hpx::id_type* const &respond_to)
Shutdown this runtime system instance.

inline void **terminate_act**(*hpx::id_type* const &id)

void **terminate_all**()
Shutdown runtime system instances on all localities.

inline void **terminate_all_act**()

util::section **get_config**()
Retrieve configuration information.

int **load_components**()
Load all components on this locality.

void **call_startup_functions**(bool pre_startup)

void **call_shutdown_functions**(bool pre_shutdown)

void **garbage_collect()**

Force a garbage collection operation in the AGAS layer.

naming::gid_type **create_performance_counter**(*performance_counters::counter_info* const &*info*)

Create the given performance counter instance.

void **remove_from_connection_cache**(*naming::gid_type* const &*gid*,
parcelset::endpoints_type const &*eps*)

Remove the given locality from our connection cache.

HPX_DEFINE_COMPONENT_ACTION (*runtime_support*, *terminate_act*,
terminate_action) **HPX_DEFINE_COMPONENT_ACTION**(*runtime_support*
termination detection

terminate_all_action **HPX_DEFINE_COMPONENT_ACTION** (*runtime_support*,
remove_from_connection_cache) void **run()**

Start the *runtime_support* component.

void **wait()**

Wait for the *runtime_support* component to notify the calling thread.

This function will be called from the main thread, causing it to block while the HPX functionality is executed. The main thread will block until the *shutdown_action* is executed, which in turn notifies all waiting threads.

void **stop**(double *timeout*, *hpx::id_type* const &*respond_to*, bool *remove_from_remote_caches*)

Notify all waiting (blocking) threads allowing the system to be properly stopped.

Note: This function can be called from any thread.

void **stopped()**

called locally only

void **notify_waiting_main()**

inline bool **was_stopped()** const

void **add_pre_startup_function**(*startup_function_type* *f*)

void **add_startup_function**(*startup_function_type* *f*)

void **add_pre_shutdown_function**(*shutdown_function_type* *f*)

void **add_shutdown_function**(*shutdown_function_type* *f*)

void **remove_here_from_connection_cache()**

void **remove_here_from_console_connection_cache()**

Public Members

terminate_all_act

Public Static Functions

static inline component_type **get_component_type**()

static inline void **set_component_type**(component_type t)

static inline constexpr void **finalize**()

finalize() will be called just before the instance gets destructed

static inline bool **is_target_valid**(hpx::id_type const &id)

Protected Functions

int **load_components**(util::section &ini, naming::gid_type const &prefix,
 agas::addressing_service &agas_client,
 hpx::program_options::options_description &options,
 std::set<std::string> &startup_handled)

bool **load_component**(hpx::util::plugin::dll &d, util::section &ini, std::string const &instance,
 std::string const &component, filesystem::path const &lib,
 naming::gid_type const &prefix, agas::addressing_service
 &agas_client, bool isdefault, bool isenabled,
 hpx::program_options::options_description &options,
 std::set<std::string> &startup_handled)

bool **load_component_dynamic**(util::section &ini, std::string const &instance, std::string
 const &component, filesystem::path lib, naming::gid_type
 const &prefix, agas::addressing_service &agas_client, bool
 isdefault, bool isenabled,
 hpx::program_options::options_description &options,
 std::set<std::string> &startup_handled)

bool **load_startup_shutdown_functions**(hpx::util::plugin::dll &d, error_code &ec)

bool **load_commandline_options**(hpx::util::plugin::dll &d,
 hpx::program_options::options_description &options,
 error_code &ec)

bool **load_component_static**(util::section &ini, std::string const &instance, std::string const
 &component, filesystem::path const &lib, naming::gid_type
 const &prefix, agas::addressing_service &agas_client, bool
 isdefault, bool isenabled,
 hpx::program_options::options_description &options,
 std::set<std::string> &startup_handled)

bool **load_startup_shutdown_functions_static**(std::string const &mod, error_code
 &ec)

```

bool load_commandline_options_static(std::string const &mod,
                                     hpx::program_options::options_description
                                     &options, error_code &ec)

bool load_plugins(util::section &ini, hpx::program_options::options_description &options,
                 std::set<std::string> &startup_handled)

bool load_plugin(hpx::util::plugin::dll &d, util::section &ini, std::string const &instance,
                 std::string const &component, filesystem::path const &lib, bool isenabled,
                 hpx::program_options::options_description &options, std::set<std::string>
                 &startup_handled)

bool load_plugin_dynamic(util::section &ini, std::string const &instance, std::string const
                        &component, filesystem::path lib, bool isenabled,
                        hpx::program_options::options_description &options,
                        std::set<std::string> &startup_handled)

bool load_plugin_static(util::section &ini, std::string const &instance, std::string const
                        &plugin, bool isenabled,
                        hpx::program_options::options_description &options,
                        std::set<std::string> &startup_handled)

std::size_t dijkstra_termination_detection(std::vector<hpx::id_type> const
                                           &locality_ids)

```

Private Types

```

using plugin_map_mutex_type = hpx::spinlock

using plugin_factory_type = plugin_factory

using plugin_map_type = std::map<std::string, plugin_factory_type>

using modules_map_type = std::map<std::string, hpx::util::plugin::dll>

using static_modules_type = std::vector<static_factory_load_data_type>

```

Private Members

```

std::mutex mtx_

std::condition_variable wait_condition_

std::condition_variable stop_condition_

bool stop_called_

```

```
bool stop_done_

bool terminated_

std::thread::id main_thread_id_

std::atomic<bool> shutdown_all_invoked_

plugin_map_mutex_type p_mtx_

plugin_map_type plugins_

modules_map_type &modules_

static_modules_type static_modules_

hpx::spinlock globals_mtx_

std::list<startup_function_type> pre_startup_functions_

std::list<startup_function_type> startup_functions_

std::list<shutdown_function_type> pre_shutdown_functions_

std::list<shutdown_function_type> shutdown_functions_

struct plugin_factory
```

Public Functions

```
inline plugin_factory(std::shared_ptr<plugins::plugin_factory_base> const &f, bool
                      enabled)
```

Public Members

```
std::shared_ptr<plugins::plugin_factory_base> first

bool isEnabled
```

hpx/runtime_distributed/server/copy_component.hpp

Defined in header `hpx/runtime_distributed/server/copy_component.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace components
```

```
        namespace server
```

Functions

```
template<typename Component>
hpx::id_type copy_component_here(hpx::id_type const &to_copy)
```

```
template<typename Component>
future<hpx::id_type> copy_component(hpx::id_type const &to_copy, hpx::id_type const
                                     &target_locality)
```

```
template<typename Component>
```

```
struct copy_component_action : public hpx::actions::action<future<hpx::id_type>
    (*)(hpx::id_type const&, hpx::id_type const&), &copy_component<Component>,
    copy_component_action<Component>>
```

```
template<typename Component>
```

```
struct copy_component_action_here : public hpx::actions::action<hpx::id_type (*)
    (hpx::id_type const&), &copy_component_here<Component>, copy_component_action_here<Component>>
```

hpx/runtime_distributed/stubs/runtime_support.hpp

Defined in header `hpx/runtime_distributed/stubs/runtime_support.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace components
```

```
        namespace stubs
```

```
            struct runtime_support
```

```
                Subclassed by hpx::components::runtime_support
```

Public Static Functions

```
template<typename Component, typename ...Ts>
static inline hpx::future<hpx::id_type> create_component_async(hpx::id_type const &gid,
                                                            Ts&&... vs)
```

Create a new component *type* using the *runtime_support* with the given *targetgid*. This is a non-blocking call. The caller needs to call *future::get* on the result of this function to obtain the global id of the newly created object.

```
template<typename Component, typename ...Ts>
static inline hpx::id_type create_component(hpx::id_type const &gid, Ts&&... vs)
```

Create a new component *type* using the *runtime_support* with the given *targetgid*. Block for the creation to finish.

```
template<bool WithCount, typename Component, typename ...Ts>
static inline hpx::future<std::vector<hpx::id_type>> bulk_create_component_colocated_async(hpx::id_type
                                                                                          const
                                                                                          &gid,
                                                                                          std::size_t
                                                                                          count,
                                                                                          Ts&&...
                                                                                          vs)
```

Create multiple new components *type* using the *runtime_support* colocated with the given *targetgid*. This is a non-blocking call.

```
template<bool WithCount, typename Component, typename ...Ts>
static inline std::vector<hpx::id_type> bulk_create_component_colocated(hpx::id_type
                                                                                          const &gid,
                                                                                          std::size_t
                                                                                          count, Ts&&...
                                                                                          vs)
```

Create multiple new components *type* using the *runtime_support* colocated with the given *targetgid*. Block for the creation to finish.

```
template<bool WithCount, typename Component, typename ...Ts>
static inline hpx::future<std::vector<hpx::id_type>> bulk_create_component_async(hpx::id_type
                                                                                          const
                                                                                          &gid,
                                                                                          std::size_t
                                                                                          count,
                                                                                          Ts&&...
                                                                                          vs)
```

Create multiple new components *type* using the *runtime_support* on the given locality. This is a non-blocking call.

```
template<bool WithCount, typename Component, typename ...Ts>
static inline std::vector<hpx::id_type> bulk_create_component(hpx::id_type const &gid,
                                                            std::size_t count, Ts&&...
                                                            vs)
```

Create multiple new components *type* using the *runtime_support* on the given locality. Block for the creation to finish.

```
template<typename Component, typename ...Ts>
static inline hpx::future<hpx::id_type> create_component_colocated_async(hpx::id_type
                                                                                          const &gid,
                                                                                          Ts&&... vs)
```


Create a new component *type* using the *runtime_support* with the given *targetgid*. This is a non-blocking call. The caller needs to call *future::get* on the result of this function to obtain the global id of the newly created object.

```
template<typename Component, typename ...Ts>
static inline hpx::id_type create_component_colocated(hpx::id_type const &gid, Ts&&...
                                                    vs)
```

Create a new component *type* using the *runtime_support* with the given *targetgid*. Block for the creation to finish.

```
template<typename Component>
static inline hpx::future<hpx::id_type> copy_create_component_async(hpx::id_type const
                                                                    &gid,
                                                                    std::shared_ptr<Component>
                                                                    const &p, bool
                                                                    local_op)
```

```
template<typename Component>
static inline hpx::id_type copy_create_component(hpx::id_type const &gid,
                                                    std::shared_ptr<Component> const &p,
                                                    bool local_op)
```

```
template<typename Component>
static inline hpx::future<hpx::id_type> migrate_component_async(hpx::id_type const
                                                                &target_locality,
                                                                std::shared_ptr<Component>
                                                                const &p, hpx::id_type
                                                                const &to_migrate)
```

```
template<typename Component, typename DistPolicy>
static inline hpx::future<hpx::id_type> migrate_component_async(DistPolicy const &policy,
                                                                std::shared_ptr<Component>
                                                                const &p, hpx::id_type
                                                                const &to_migrate)
```

```
template<typename Component, typename Target>
static inline hpx::id_type migrate_component(Target const &target, hpx::id_type const
                                                                &to_migrate, std::shared_ptr<Component>
                                                                const &p)
```

```
static hpx::future<int> load_components_async(hpx::id_type const &gid)
```

```
static int load_components(hpx::id_type const &gid)
```

```
static hpx::future<void> call_startup_functions_async(hpx::id_type const &gid, bool
                                                    pre_startup)
```

```
static void call_startup_functions(hpx::id_type const &gid, bool pre_startup)
```

```
static hpx::future<void> shutdown_async(hpx::id_type const &targetgid, double timeout = -1)
    Shutdown the given runtime system.
```

```
static void shutdown(hpx::id_type const &targetgid, double timeout = -1)
```

```
static void shutdown_all(hpx::id_type const &targetgid, double timeout = -1)
    Shutdown the runtime systems of all localities.
```

```
static void shutdown_all(double timeout = -1)

static hpx::future<void> terminate_async(hpx::id_type const &targetgid)
    Retrieve configuration information.
    Terminate the given runtime system
static void terminate(hpx::id_type const &targetgid)

static void terminate_all(hpx::id_type const &targetgid)
    Terminate the runtime systems of all localities.
static void terminate_all()

static void garbage_collect_non_blocking(hpx::id_type const &targetgid)

static hpx::future<void> garbage_collect_async(hpx::id_type const &targetgid)

static void garbage_collect(hpx::id_type const &targetgid)

static hpx::future<hpx::id_type> create_performance_counter_async(hpx::id_type const
                                                                &targetgid, perfor-
                                                                mance_counters::counter_info
                                                                const &info)

static hpx::id_type create_performance_counter(hpx::id_type const &targetgid,
                                                performance_counters::counter_info
                                                const &info, error_code &ec = throws)

static hpx::future<util::section> get_config_async(hpx::id_type const &targetgid)
    Retrieve configuration information.
static void get_config(hpx::id_type const &targetgid, util::section &ini)

static void remove_from_connection_cache_async(hpx::id_type const &target,
                                                naming::gid_type const &gid,
                                                parcelset::endpoints_type const
                                                &endpoints)
```

segmented_algorithms

See *Public API* for a list of names and headers that are part of the public HPX API.

hpx/parallel/segmented_algorithms/transform.hpp

Defined in header `hpx/parallel/segmented_algorithms/transform.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

Functions

```
template<typename SegIter, typename OutIter, typename F>
hpx::parallel::util::in_out_result<SegIter, OutIter> hpx_invoke(hpx::transform_t, SegIter first, SegIter
last, OutIter dest, F &&f)
```

```
template<typename ExPolicy, typename SegIter, typename OutIter, typename F>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, hpx::parallel::util::in_out_result<SegIter, OutIter>> hpx_invoke(
```

```
template<typename InIter1, typename InIter2, typename OutIter, typename F>
hpx::parallel::util::in_in_out_result<InIter1, InIter2, OutIter> hpx_invoke(hpx::transform_t, InIter1
first1, InIter1 last1, InIter2
first2, OutIter dest, F &&f)
```

```
template<typename ExPolicy, typename InIter1, typename InIter2, typename OutIter, typename
F>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, hpx::parallel::util::in_in_out_result<InIter1, InIter2, OutIter>> hpx_invoke(
```

```
template<typename InIter1, typename InIter2, typename OutIter, typename F>
hpx::parallel::util::in_in_out_result<InIter1, InIter2, OutIter> hpx_invoke(hpx::transform_t, InIter1
first1, InIter1 last1, InIter2
first2, InIter2 last2, OutIter
dest, F &&f)
```

```
template<typename ExPolicy, typename InIter1, typename InIter2, typename OutIter, typename  
F>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, hpx::parallel::util::in_in_out_result<InIter1, InIter2, OutIter>> hpx
```

hpx/parallel/segmented_algorithms/adjacent_difference.hpp

Defined in header `hpx/parallel/segmented_algorithms/adjacent_difference.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

Functions

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Op>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx_invoke(hpx::adjacent_difference_t,  
                                     ExPolicy &&policy,  
                                     FwdIter1 first,  
                                     FwdIter1 last, FwdIter2  
                                     dest, Op &&op)  
  
template<typename InIter1, typename InIter2, typename Op>  
InIter2 hpx_invoke(hpx::adjacent_difference_t, InIter1 first, InIter1 last, InIter2 dest, Op &&op)
```

hpx/parallel/segmented_algorithms/for_each.hpp

Defined in header `hpx/parallel/segmented_algorithms/for_each.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

Functions

```
template<typename InIter, typename F>
InIter hpx_invoke(hpx::for_each_t, InIter first, InIter last, F &&f)
```

```
template<typename ExPolicy, typename SegIter, typename F>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, SegIter> hpx_invoke(hpx::for_each_t, ExPolicy
&&policy, SegIter first,
SegIter last, F &&f)
```

```
template<typename InIter, typename Size, typename F>
InIter hpx_invoke(hpx::for_each_n_t, InIter first, Size count, F &&f)
```

```
template<typename ExPolicy, typename SegIter, typename Size, typename F>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, SegIter> hpx_invoke(hpx::for_each_n_t,
ExPolicy &&policy,
SegIter first, Size count,
F &&f)
```

hpx/parallel/segmented_algorithms/mismatch.hpp

Defined in header `hpx/parallel/segmented_algorithms/mismatch.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

Functions

```
template<typename SegIter1, typename SegIter2, typename Pred = hpx::parallel::detail::equal_to>
std::pair<SegIter1, SegIter2> hpx_invoke(hpx::mismatch_t, SegIter1 first1, SegIter1 last1, SegIter2
                                         first2, Pred &&pred = Pred())
```

```
template<typename ExPolicy, typename SegIter1, typename SegIter2, typename Pred =
hpx::parallel::detail::equal_to>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, std::pair<SegIter1, SegIter2>> hpx_invoke(hpx::mismatch_t,
                                                                                                     Ex-
                                                                                                     Pol-
                                                                                                     icy
                                                                                                     &&pol-
                                                                                                     icy,
                                                                                                     Se-
                                                                                                     gIter1
                                                                                                     first1,
                                                                                                     Se-
                                                                                                     gIter1
                                                                                                     last1,
                                                                                                     Se-
                                                                                                     gIter2
                                                                                                     first2,
                                                                                                     Pred
                                                                                                     &&pred
                                                                                                     =
                                                                                                     Pred())
```

```
template<typename SegIter1, typename SegIter2, typename Pred = hpx::parallel::detail::equal_to>
std::pair<SegIter1, SegIter2> hpx_invoke(hpx::mismatch_t, SegIter1 first1, SegIter1 last1, SegIter2
                                         first2, SegIter2 last2, Pred &&pred = Pred())
```

```
template<typename ExPolicy, typename SegIter1, typename SegIter2, typename Pred =
hpx::parallel::detail::equal_to>
```

```

hpx::parallel::util::detail::algorithm_result_t<ExPolicy, std::pair<SegIter1, SegIter2>> hpx_invoke(hpx::mismatch_t,
                                                    Ex-
                                                    Pol-
                                                    icy
                                                    &&pol-
                                                    icy,
                                                    Se-
                                                    gIter1
                                                    first1,
                                                    Se-
                                                    gIter1
                                                    last1,
                                                    Se-
                                                    gIter2
                                                    first2,
                                                    Se-
                                                    gIter2
                                                    last2,
                                                    Pred
                                                    &&pred
                                                    =
                                                    Pred())

```

hpx/parallel/segmented_algorithms/adjacent_find.hpp

Defined in header `hpx/parallel/segmented_algorithms/adjacent_find.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

Functions

```

template<typename InIter, typename Pred>
InIter hpx_invoke(hpx::adjacent_find_t, InIter first, InIter last, Pred &&pred = Pred())

template<typename ExPolicy, typename SegIter, typename Pred>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, SegIter> hpx_invoke(hpx::adjacent_find_t,
                                                    ExPolicy &&policy,
                                                    SegIter first, SegIter last,
                                                    Pred &&pred)

```

hpx/parallel/segmented_algorithms/transform_inclusive_scan.hpp

Defined in header `hpx/parallel/segmented_algorithms/transform_inclusive_scan.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **segmented**

Functions

```
template<typename InIter, typename OutIter, typename Op, typename Conv>
OutIter hpx_invoke(hpx::transform_inclusive_scan_t, InIter first, InIter last, OutIter dest, Op &&op,
                    Conv &&conv)

template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Op, typename Conv>
parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx_invoke(hpx::transform_inclusive_scan_t,
                                                                    ExPolicy &&policy,
                                                                    FwdIter1 first, FwdIter1 last, FwdIter2 dest, Op &&op,
                                                                    Conv &&conv, T init)

template<typename InIter, typename OutIter, typename T, typename Op, typename Conv>
OutIter hpx_invoke(hpx::transform_inclusive_scan_t, InIter first, InIter last, OutIter dest, Op &&op,
                    Conv &&conv, T init)

template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename T, typename Op,
typename Conv>
parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx_invoke(hpx::transform_inclusive_scan_t,
                                                                    ExPolicy &&policy,
                                                                    FwdIter1 first, FwdIter1 last,
                                                                    FwdIter2 dest, Op &&op,
                                                                    Conv &&conv, T init)
```

hpx/parallel/segmented_algorithms/exclusive_scan.hpp

Defined in header `hpx/parallel/segmented_algorithms/exclusive_scan.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

Functions

```
template<typename InIter, typename OutIter, typename T, typename Op = std::plus<T>>
OutIter hpx_invoke(hpx::exclusive_scan_t, InIter first, InIter last, OutIter dest, T init, Op &&op =
    Op())
```

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename T, typename Op =
    std::plus<T>>
parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx_invoke(hpx::exclusive_scan_t,
    ExPolicy &&policy,
    FwdIter1 first, FwdIter1
    last, FwdIter2 dest, T init,
    Op &&op = Op())
```

hpx/parallel/segmented_algorithms/inclusive_scan.hpp

Defined in header `hpx/parallel/segmented_algorithms/inclusive_scan.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

Functions

```
template<typename InIter, typename OutIter, typename Op = std::plus<>>
OutIter hpx_invoke(hpx::inclusive_scan_t, InIter first, InIter last, OutIter dest, Op &&op = Op())
```

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Op = std::plus<>>
parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx_invoke(hpx::inclusive_scan_t,
    ExPolicy &&policy,
    FwdIter1 first, FwdIter1
    last, FwdIter2 dest, Op
    &&op = Op())
```

```
template<typename InIter, typename OutIter, typename Op, typename T>
OutIter hpx_invoke(hpx::inclusive_scan_t, InIter first, InIter last, OutIter dest, Op &&op, T &&init)
```

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Op, typename T>
parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx_invoke(hpx::inclusive_scan_t,
    ExPolicy &&policy,
    FwdIter1 first, FwdIter1
    last, FwdIter2 dest, Op
    &&op, T &&init)
```

hpx/parallel/segmented_algorithms/transform_exclusive_scan.hpp

Defined in header `hpx/parallel/segmented_algorithms/transform_exclusive_scan.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **segmented**

Functions

```
template<typename InIter, typename OutIter, typename T, typename Op, typename Conv>  
OutIter hpx_invoke(hpx::transform_exclusive_scan_t, InIter first, InIter last, OutIter dest, T init, Op  
                    &&op, Conv &&conv)
```

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename T, typename Op,  
typename Conv>  
parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx_invoke(hpx::transform_exclusive_scan_t,  
                                ExPolicy &&policy,  
                                FwdIter1 first, FwdIter1  
                                last, FwdIter2 dest, T init,  
                                Op &&op, Conv &&conv)
```

hpx/parallel/segmented_algorithms/equal.hpp

Defined in header `hpx/parallel/segmented_algorithms/equal.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

Functions

```
template<typename SegIter1, typename SegIter2, typename Pred = hpx::parallel::detail::equal_to>  
bool hpx_invoke(hpx::equal_t, SegIter1 first1, SegIter1 last1, SegIter2 first2, Pred &&pred = Pred())
```

```
template<typename ExPolicy, typename SegIter1, typename SegIter2, typename Pred =  
hpx::parallel::detail::equal_to>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx_invoke(hpx::equal_t, ExPolicy  
                                &&policy, SegIter1 first1,  
                                SegIter1 last1, SegIter2  
                                first2, Pred &&pred =  
                                Pred())
```

hpx/parallel/segmented_algorithms/copy.hpp

Defined in header `hpx/parallel/segmented_algorithms/copy.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

Functions

```
template<typename SegIter, typename OutIter>
OutIter hpx_invoke(hpx::copy_t, SegIter first, SegIter last, OutIter dest)
```

```
template<typename ExPolicy, typename SegIter, typename OutIter>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, OutIter> hpx_invoke(hpx::copy_t, ExPolicy
&&policy, SegIter first,
SegIter last, OutIter
dest)
```

hpx/parallel/segmented_algorithms/generate.hpp

Defined in header `hpx/parallel/segmented_algorithms/generate.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

Functions

```
template<typename SegIter, typename F>
SegIter hpx_invoke(hpx::generate_t, SegIter first, SegIter last, F &&f)
```

```
template<typename SegIter, typename Size, typename F>
SegIter hpx_invoke(hpx::generate_n_t, SegIter first, Size count, F &&f)
```

```
template<typename ExPolicy, typename SegIter, typename F>
parallel::util::detail::algorithm_result_t<ExPolicy, SegIter> hpx_invoke(hpx::generate_t, ExPolicy
&&policy, SegIter first,
SegIter last, F &&f)
```

```
template<typename ExPolicy, typename SegIter, typename Size, typename F>
```

```
parallel::util::detail::algorithm_result_t<ExPolicy, SegIter> hpx_invoke(hpx::generate_n_t, ExPolicy
&&policy, SegIter first, Size
count, F &&f)
```

hpx/parallel/segmented_algorithms/minmax.hpp

Defined in header `hpx/parallel/segmented_algorithms/minmax.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

Typedefs

template<typename **T**>

using **minmax_element_result** = *hpx::parallel::util::min_max_result*<**T**>

namespace **segmented**

Typedefs

template<typename **T**>

using **minmax_element_result** = *hpx::parallel::util::min_max_result*<**T**>

Functions

template<typename **SegIter**, typename **F**>

SegIter **hpx_invoke**(*hpx::min_element_t*, *SegIter* first, *SegIter* last, *F* &&f)

template<typename **ExPolicy**, typename **SegIter**, typename **F**>

hpx::parallel::util::detail::algorithm_result_t<*ExPolicy*, *SegIter*> **hpx_invoke**(*hpx::min_element_t*,
ExPolicy &&policy,
SegIter first, *SegIter* last,
F &&f)

template<typename **SegIter**, typename **F**>

SegIter **hpx_invoke**(*hpx::max_element_t*, *SegIter* first, *SegIter* last, *F* &&f)

template<typename **ExPolicy**, typename **SegIter**, typename **F**>

hpx::parallel::util::detail::algorithm_result_t<*ExPolicy*, *SegIter*> **hpx_invoke**(*hpx::max_element_t*,
ExPolicy &&policy,
SegIter first, *SegIter* last,
F &&f)

template<typename **SegIter**, typename **F**>

```
minmax_element_result<SegIter> hpx_invoke(hpx::minmax_element_t, SegIter first, SegIter last, F
&&f)
```

```
template<typename ExPolicy, typename SegIter, typename F>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, minmax_element_result<SegIter>> hpx_invoke(hpx::minmax_ele
Ex-
Pol-
icy
&&pol-
icy,
Se-
gIter
first,
Se-
gIter
last,
F
&&f)
```

hpx/parallel/segmented_algorithms/count.hpp

Defined in header `hpx/parallel/segmented_algorithms/count.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace parallel
```

```
        namespace segmented
```

Functions

```
template<typename InIter, typename T>
std::iterator_traits<InIter>::difference_type hpx_invoke(hpx::count_t, InIter first, InIter last, T const
&value)
```

```
template<typename ExPolicy, typename SegIter, typename T>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, typename std::iterator_traits<SegIter>::difference_type> hpx_invo
```

```
template<typename InIter, typename F>
std::iterator_traits<InIter>::difference_type hpx_invoke(hpx::count_if_t, InIter first, InIter last, F
&&f)
```

```
template<typename ExPolicy, typename SegIter, typename F>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, typename std::iterator_traits<SegIter>::difference_type> hpx_invo
```

hpx/parallel/segmented_algorithms/is_partitioned.hpp

Defined in header `hpx/parallel/segmented_algorithms/is_partitioned.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace parallel
```

namespace **segmented**

Functions

```
template<typename InIter, typename Pred>
bool hpx_invoke(hpx::is_partitioned_t, InIter first, InIter last, Pred &&pred)

template<typename ExPolicy, typename SegIter, typename Pred>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx_invoke(hpx::is_partitioned_t,
                                                                    ExPolicy &&policy, SegIter
                                                                    first, SegIter last, Pred
                                                                    &&pred)
```

hpx/parallel/segmented_algorithms/fill.hpp

Defined in header `hpx/parallel/segmented_algorithms/fill.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

Functions

```
template<typename SegIter, typename T>
SegIter hpx_invoke(hpx::fill_t, SegIter first, SegIter last, T const &value)

template<typename SegIter, typename Size, typename T>
SegIter hpx_invoke(hpx::fill_n_t, SegIter first, Size count, T const &value)

template<typename ExPolicy, typename SegIter, typename T>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, SegIter> hpx_invoke(hpx::fill_t, ExPolicy
                                                                    &&policy, SegIter first,
                                                                    SegIter last, T const
                                                                    &value)

template<typename ExPolicy, typename SegIter, typename Size, typename T>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, SegIter> hpx_invoke(hpx::fill_n_t, ExPolicy
                                                                    &&policy, SegIter first,
                                                                    Size count, T const
                                                                    &value)
```

hpx/parallel/segmented_algorithms/reduce.hpp

Defined in header `hpx/parallel/segmented_algorithms/reduce.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace parallel
```

```
namespace hpx
```

```
    namespace segmented
```

Functions

```
template<typename InIterB, typename InIterE, typename T, typename F>  
T hpx_invoke(hpx::reduce_t, InIterB first, InIterE last, T init, F &&f)
```

```
template<typename ExPolicy, typename InIterB, typename InIterE, typename T, typename F>  
parallel::util::detail::algorithm_result_t<ExPolicy, T> hpx_invoke(hpx::reduce_t, ExPolicy &&policy,  
                                                                    InIterB first, InIterE last, T init, F  
                                                                    &&f)
```

hpx/parallel/segmented_algorithms/replace_copy.hpp

Defined in header `hpx/parallel/segmented_algorithms/replace_copy.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace parallel
```

```
    namespace segmented
```

Functions

```
template<typename SegIter, typename OutIter, typename T>  
OutIter hpx_invoke(hpx::replace_copy_t, SegIter first, SegIter last, OutIter dest, T const &old_value, T  
                                                             const &new_value)
```

```
template<typename ExPolicy, typename SegIter, typename OutIter, typename T>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, OutIter> hpx_invoke(hpx::replace_copy_t,  
                                                                    ExPolicy &&policy,  
                                                                    SegIter first, SegIter last,  
                                                                    OutIter dest, T const  
                                                                    &old_value, T const  
                                                                    &new_value)
```



```

template<typename SegIter, typename OutIter, typename Pred, typename T>
OutIter hpx_invoke(hpx::replace_copy_if_t, SegIter first, SegIter last, OutIter dest, Pred &&pred, T
                    const &new_value)

template<typename ExPolicy, typename SegIter, typename OutIter, typename Pred, typename T>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, OutIter> hpx_invoke(hpx::replace_copy_if_t,
                                                                              ExPolicy &&policy,
                                                                              SegIter first, SegIter last,
                                                                              OutIter dest, Pred
                                                                              &&pred, T const
                                                                              &new_value)

```

hpx/parallel/segmented_algorithms/replace.hpp

Defined in header `hpx/parallel/segmented_algorithms/replace.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

Functions

```

template<typename SegIter, typename T>
SegIter hpx_invoke(hpx::replace_t, SegIter first, SegIter last, T const &old_value, T const
                    &new_value)

template<typename ExPolicy, typename SegIter, typename T>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, SegIter> hpx_invoke(hpx::replace_t, ExPolicy
                                                                              &&policy, SegIter first,
                                                                              SegIter last, T const
                                                                              &old_value, T const
                                                                              &new_value)

template<typename SegIter, typename Pred, typename T>
SegIter hpx_invoke(hpx::replace_if_t, SegIter first, SegIter last, Pred &&pred, T const &new_value)

template<typename ExPolicy, typename SegIter, typename Pred, typename T>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, SegIter> hpx_invoke(hpx::replace_if_t,
                                                                              ExPolicy &&policy,
                                                                              SegIter first, SegIter last,
                                                                              Pred &&pred, T const
                                                                              &new_value)

```

hpx/parallel/segmented_algorithms/all_any_none.hpp

Defined in header `hpx/parallel/segmented_algorithms/all_any_none.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

Functions

```
template<typename InIter, typename F>  
bool hpx_invoke(hpx::none_of_t, InIter first, InIter last, F &&f)
```

```
template<typename ExPolicy, typename SegIter, typename F>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx_invoke(hpx::none_of_t, ExPolicy  
    &&policy, SegIter first,  
    SegIter last, F &&f)
```

```
template<typename InIter, typename F>  
bool hpx_invoke(hpx::any_of_t, InIter first, InIter last, F &&f)
```

```
template<typename ExPolicy, typename SegIter, typename F>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx_invoke(hpx::any_of_t, ExPolicy  
    &&policy, SegIter first,  
    SegIter last, F &&f)
```

```
template<typename InIter, typename F>  
bool hpx_invoke(hpx::all_of_t, InIter first, InIter last, F &&f)
```

```
template<typename ExPolicy, typename SegIter, typename F>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx_invoke(hpx::all_of_t, ExPolicy  
    &&policy, SegIter first,  
    SegIter last, F &&f)
```

hpx/parallel/segmented_algorithms/is_sorted.hpp

Defined in header `hpx/parallel/segmented_algorithms/is_sorted.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

Functions

```
template<typename InIter, typename Pred = hpx::parallel::detail::less>
InIter hpx_invoke(hpx::is_sorted_until_t, InIter first, InIter last, Pred &&pred = Pred())

template<typename ExPolicy, typename SegIter, typename Pred = hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, SegIter> hpx_invoke(hpx::is_sorted_until_t,
                                                                 ExPolicy &&policy,
                                                                 SegIter first, SegIter last,
                                                                 Pred &&pred = Pred())

template<typename InIter, typename Pred = hpx::parallel::detail::less>
bool hpx_invoke(hpx::is_sorted_t, InIter first, InIter last, Pred &&pred = Pred())

template<typename ExPolicy, typename SegIter, typename Pred = hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx_invoke(hpx::is_sorted_t, ExPolicy
                                                                 &&policy, SegIter first,
                                                                 SegIter last, Pred &&pred =
                                                                 Pred())
```

hpx/parallel/segmented_algorithms/transform_reduce.hpp

Defined in header `hpx/parallel/segmented_algorithms/transform_reduce.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **segmented**

Functions

```
template<typename SegIter, typename T, typename Reduce, typename Convert>
std::decay_t<T> hpx_invoke(hpx::transform_reduce_t, SegIter first, SegIter last, T &&init, Reduce
                                                                 &&red_op, Convert &&conv_op)

template<typename ExPolicy, typename SegIter, typename T, typename Reduce, typename
Convert>
parallel::util::detail::algorithm_result_t<ExPolicy, std::decay_t<T>> hpx_invoke(hpx::transform_reduce_t,
                                                                 ExPolicy &&policy,
                                                                 SegIter first, SegIter
                                                                 last, T &&init,
                                                                 Reduce &&red_op,
                                                                 Convert
                                                                 &&conv_op)

template<typename FwdIter1, typename FwdIter2, typename T, typename Reduce, typename
Convert>
```

```
T hpx_invoke(hpx::transform_reduce_t, FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, T init, Reduce
&&red_op, Convert &&conv_op)
```

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename T, typename
Reduce, typename Convert>
parallel::util::detail::algorithm_result_t<ExPolicy, T> hpx_invoke(hpx::transform_reduce_t, ExPolicy
&&policy, FwdIter1 first1,
FwdIter1 last1, FwdIter2 first2, T
init, Reduce &&red_op, Convert
&&conv_op)
```

supervision

See *Public API* for a list of names and headers that are part of the public HPX API.

hpx::supervision::publish_event

Defined in header `hpx/supervision.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::supervision::query_state`
- `hpx::supervision::register_observer`
- `hpx::supervision::unregister_observer`
- `hpx::supervision::remove_target`
- `hpx::supervision::register_activity_observer`
- `hpx::supervision::unregister_activity_observer`
- `hpx::supervision::check_admission`
- `hpx::supervision::await_terminal`
- `hpx::supervision::is_valid_transition`

```
hpx::future<publish_result> hpx::supervision::publish_event(hpx::id_type const &locality, hpx::id_type
const &target, hpx::supervision::event ev,
std::uint64_t epoch = 0)
```

Publish a lifecycle event for a target actor on a possibly remote locality.

The function `publish_event()` asynchronously notifies the supervision manager running on *locality* that *target* has reached the lifecycle state *ev*. Registered observers of *target* are notified as a result.

Note: Publishing is not idempotent: publishing the same event twice creates two distinct records with different timestamps.

Note: Events are immediately visible to observers on the same locality as *target*; remote observers become aware of the event within about 1-2 parcel round-trips to the AGAS authority managing the actor's registration.

Parameters

- **locality** – [in] The locality on which the supervision manager responsible for *target* is running.
- **target** – [in] The actor (or component) for which the event is published.
- **ev** – [in] The lifecycle event to publish for *target*.
- **epoch** – [in] The epoch this publication belongs to. Publications are compared against the target's current epoch: publications for a lower epoch are rejected as stale no-ops, publications for a higher epoch reset the target's sequence number and become the new current epoch, and publications for the current epoch are applied normally (subject to terminal-state latching).

Throws

hpx::exception – if *locality* does not represent a locality, or if *target* does not represent a valid target. The returned future becomes exceptional if the operation fails.

Returns

A future that becomes ready once the event has been recorded by the supervision manager on *locality*, holding *publish_result::applied*, *publish_result::already_terminal* if *target* had already reached a terminal event (*completed* or *failed*) within *epoch*, or *publish_result::stale_epoch* if *epoch* is lower than the target's current epoch.

```
publish_result hpx::supervision::publish_event(hpx::launch::sync_policy, hpx::id_type const &locality,
                                              hpx::id_type const &target, hpx::supervision::event ev,
                                              std::uint64_t epoch = 0, hpx::error_code &ec =
                                              hpx::throws)
```

Publish a lifecycle event for a target actor on a possibly remote locality, blocking until the operation has completed.

This is the synchronous equivalent of *publish_event(hpx::id_type const&, hpx::id_type const&, event)*.

Note: Publishing non-terminal events is not idempotent: publishing the same event twice creates two distinct records with different timestamps. *event::completed* and *event::failed* are latched: the first terminal publication for a target wins, and every later terminal publication for that target is a no-op that returns *publish_result::already_terminal*.

Parameters

- **locality** – [in] The locality on which the supervision manager responsible for *target* is running.
- **target** – [in] The actor (or component) for which the event is published.
- **ev** – [in] The lifecycle event to publish for *target*.
- **epoch** – [in] The epoch this publication belongs to. See the asynchronous remote overload for the epoch-scoped semantics.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Throws

hpx::exception – if *locality* does not represent a locality, or if *target* does not represent a valid target, unless *ec* was not pre-initialized to *hpx::throws*. As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*.

Returns

`publish_result::applied`, `publish_result::already_terminal` if `target` had already reached a terminal event (`completed` or `failed`) within `epoch`, or `publish_result::stale_epoch` if `epoch` is lower than the target's current epoch.

`publish_result hpx::supervision::publish_event(hpx::id_type const &target, hpx::supervision::event ev, std::uint64_t epoch = 0, hpx::error_code &ec = throws)`

Publish a lifecycle event for a target actor on the local locality.

This overload publishes `ev` directly through the local supervision manager without going through AGAS or a remote parcel. It is intended to be called from within an actor or action running on the same locality as `target`.

Note: Publishing non-terminal events is not idempotent: publishing the same event twice creates two distinct records with different timestamps. `event::completed` and `event::failed` are latched: the first terminal publication for a target wins, and every later terminal publication for that target is a no-op that returns `publish_result::already_terminal`.

Note: Local observers of `target` are notified synchronously as part of this call.

Parameters

- **target** – [in] The actor (or component) for which the event is published. Must be local to the calling locality.
- **ev** – [in] The lifecycle event to publish for `target`.
- **epoch** – [in] The epoch this publication belongs to. See the asynchronous remote overload for the epoch-scoped semantics.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

Throws

`hpx::exception` – if `target` does not represent a valid target, unless `ec` was not pre-initialized to `hpx::throws`.

Returns

`publish_result::applied`, `publish_result::already_terminal` if `target` had already reached a terminal event (`completed` or `failed`) within `epoch`, or `publish_result::stale_epoch` if `epoch` is lower than the target's current epoch.

`hpx::supervision::query_state`

Defined in header `hpx/supervision.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::supervision::publish_event`
- `hpx::supervision::register_observer`
- `hpx::supervision::unregister_observer`
- `hpx::supervision::remove_target`

- `hpx::supervision::register_activity_observer`
- `hpx::supervision::unregister_activity_observer`
- `hpx::supervision::check_admission`
- `hpx::supervision::await_terminal`
- `hpx::supervision::is_valid_transition`

`hpx::future<lifecycle_state> hpx::supervision::query_state(hpx::id_type const &locality, hpx::id_type const &target)`

Query the last known lifecycle state of a target actor on a possibly remote locality.

The function `query_state()` asynchronously retrieves the most recently published lifecycle event for *target*, as observed by the supervision manager running on *locality*.

Note: If *target* is not local to *locality*, the query is forwarded to the actor's home locality, adding about one parcel round-trip of latency.

Note: The returned `lifecycle_state::ec` may carry a staleness error code, indicating that the returned state may be stale if a preceding observer parcel has not yet been delivered.

Note: The `lifecycle_state::event_sequence_number` allows clients to detect gaps, e.g. if an observer callback was skipped due to a network failure.

Parameters

- **locality** – [in] The locality on which the supervision manager responsible for *target* is running.
- **target** – [in] The actor (or component) whose state is queried.

Throws

`hpx::exception` – if *locality* does not represent a locality, or if *target* does not represent a valid target. The returned future becomes exceptional if the operation fails.

Returns

A future holding the last event recorded for *target* (i.e., the most recent value in the runtime's event log), together with its timestamp and sequence number.

`lifecycle_state hpx::supervision::query_state(hpx::launch::sync_policy, hpx::id_type const &locality, hpx::id_type const &target, hpx::error_code &ec = hpx::throws)`

Query the last known lifecycle state of a target actor on a possibly remote locality, blocking until the result is available.

This is the synchronous equivalent of `query_state(hpx::id_type const&, hpx::id_type const&)`.

Note: The returned `lifecycle_state::ec` may carry a staleness error code, indicating that the returned state may be stale if a preceding observer parcel has not yet been delivered.

Parameters

- **locality** – [in] The locality on which the supervision manager responsible for *target* is running.
- **target** – [in] The actor (or component) whose state is queried.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Throws

hpx::exception – if *locality* does not represent a locality, or if *target* does not represent a valid target, unless *ec* was not pre-initialized to *hpx::throws*.

Returns

The last event recorded for *target*, together with its timestamp and sequence number.

```
lifecycle_state hpx::supervision::query_state(hpx::id_type const &target, hpx::error_code &ec =  
                                              hpx::throws)
```

Query the last known lifecycle state of a target actor on the local locality.

Parameters

- **target** – [in] The actor (or component) whose state is queried. Must be local to the calling locality.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Throws

hpx::exception – if *target* does not represent a valid target, unless *ec* was not pre-initialized to *hpx::throws*.

Returns

The last event recorded locally for *target*, together with its timestamp and sequence number.

hpx::supervision::register_observer

Defined in header `hpx/supervision.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::supervision::publish_event*
- *hpx::supervision::query_state*
- *hpx::supervision::unregister_observer*
- *hpx::supervision::remove_target*
- *hpx::supervision::register_activity_observer*
- *hpx::supervision::unregister_activity_observer*
- *hpx::supervision::check_admission*
- *hpx::supervision::await_terminal*
- *hpx::supervision::is_valid_transition*

```
hpx::future<hpx::id_type> hpx::supervision::register_observer(hpx::id_type const &locality, hpx::id_type
                                                             const &target, lifecycle_callback const
                                                             &callback, std::optional<std::uint64_t>
                                                             epoch_filter = std::nullopt)
```

Register a callback to observe lifecycle events published by a target actor running on a possibly remote locality.

Note: If *locality* is the locality *target* is running on, the callback is invoked synchronously from within the publish call (best effort; the callback must not block indefinitely).

Note: If *locality* differs from the locality *target* is running on, the callback is invoked asynchronously via a dedicated parcel, with retry semantics (e.g., 3 attempts over 500 ms before logging failure).

Note: The callback must not throw; exceptions thrown from it are logged and do not affect the observer registration.

Parameters

- **locality** – [in] The locality on which the callback should be registered.
- **target** – [in] The actor (or component) to observe.
- **callback** – [in] The callback invoked whenever *target* publishes a lifecycle event.
- **epoch_filter** – [in] If set, restricts notifications to events published under this epoch; events published under any other epoch are silently skipped for this observer. If `std::nullopt` (the default), the observer is notified regardless of epoch.

Throws

`hpx::exception` – if *locality* does not represent a locality, or if *target* does not represent a valid target. The returned future becomes exceptional if the operation fails.

Returns

A future holding the global id of the observer handle. This handle must be passed to `unregister_observer()` in order to stop receiving notifications.

```
hpx::id_type hpx::supervision::register_observer(hpx::launch::sync_policy, hpx::id_type const &locality,
                                                hpx::id_type const &target, lifecycle_callback const
                                                &callback, std::optional<std::uint64_t> epoch_filter =
                                                std::nullopt, hpx::error_code &ec = hpx::throws)
```

Register a callback to observe lifecycle events published by a target actor running on a possibly remote locality, blocking until the registration has completed.

This is the synchronous equivalent of `register_observer(hpx::id_type const&, hpx::id_type const&, lifecycle_callback const&)`.

Parameters

- **locality** – [in] The locality on which the callback should be registered.
- **target** – [in] The actor (or component) to observe.
- **callback** – [in] The callback invoked whenever *target* publishes a lifecycle event.
- **epoch_filter** – [in] If set, restricts notifications to events published under this epoch; see the asynchronous overload for details.

- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Throws

hpx::exception – if *locality* does not represent a locality, or if *target* does not represent a valid target, unless *ec* was not pre-initialized to *hpx::throws*.

Returns

The global id of the observer handle. This handle must be passed to `unregister_observer()` in order to stop receiving notifications.

```
hpx::id_type hpx::supervision::register_observer(hpx::id_type const &target, lifecycle_callback const  
                                                &callback, std::optional<std::uint64_t> epoch_filter =  
                                                std::nullopt, hpx::error_code &ec = hpx::throws)
```

Register a callback to observe lifecycle events published by a target actor on the local locality.

Note: The callback is invoked synchronously from within the publish call (best effort; the callback must not block indefinitely) and must not throw; exceptions thrown from it are logged and do not affect the observer registration.

Parameters

- **target** – [in] The actor (or component) to observe. Must be local to the calling locality.
- **callback** – [in] The callback invoked whenever *target* publishes a lifecycle event.
- **epoch_filter** – [in] If set, restricts notifications to events published under this epoch; events published under any other epoch are silently skipped for this observer. If `std::nullopt` (the default), the observer is notified regardless of epoch.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Throws

hpx::exception – if *target* does not represent a valid target, unless *ec* was initialized to *hpx::throws*.

Returns

The global id of the observer handle. This handle must be passed to `unregister_observer()` in order to stop receiving notifications.

`hpx::supervision::unregister_observer`

Defined in header `hpx/supervision.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::supervision::publish_event*
- *hpx::supervision::query_state*
- *hpx::supervision::register_observer*
- *hpx::supervision::remove_target*
- *hpx::supervision::register_activity_observer*
- *hpx::supervision::unregister_activity_observer*

- *hpx::supervision::check_admission*
- *hpx::supervision::await_terminal*
- *hpx::supervision::is_valid_transition*

hpx::future<void> **hpx::supervision::unregister_observer**(*hpx::id_type* const &locality, *hpx::id_type* const &observer_handle)

Unregister a previously registered observer on a possibly remote locality.

Note: Once the returned future has become ready, no orphaned callbacks fire after unregistration completes.

Parameters

- **locality** – [in] The locality on which the observer was registered.
- **observer_handle** – [in] The handle returned by a prior call to `register_observer()`. Only handles returned from that function are accepted; passing a handle obtained from `register_activity_observer()` (or any foreign handle) is rejected.

Throws

hpx::exception – if *locality* does not represent a locality, or if *observer_handle* does not represent a valid observer handle. The returned future becomes exceptional if the operation fails.

Returns

A future that becomes ready once the observer has been unregistered.

void *hpx::supervision::unregister_observer*(*hpx::launch::sync_policy*, *hpx::id_type* const &locality, *hpx::id_type* const &observer_handle, *hpx::error_code* &ec = *hpx::throws*)

Unregister a previously registered observer on a possibly remote locality, blocking until the operation has completed.

This is the synchronous equivalent of `unregister_observer(hpx::id_type const&, hpx::id_type const&)`.

Note: No orphaned callbacks fire after unregistration completes.

Parameters

- **locality** – [in] The locality on which the observer was registered.
- **observer_handle** – [in] The handle returned by a prior call to `register_observer()`. See the asynchronous overload regarding foreign handles.
- **ec** – [in,out] this represents the error status on exit, if this pre-initialized to *hpx::throws* the function will throw on error instead.

Throws

hpx::exception – if *locality* does not represent a locality, or if *observer_handle* does not represent a valid observer handle, unless *ec* was initialized not to *hpx::throws*.

void *hpx::supervision::unregister_observer*(*hpx::id_type* const &observer_handle, *hpx::error_code* &ec = *hpx::throws*)

Unregister a previously registered observer on the local locality.

Note: No orphaned callbacks fire after unregistration completes.

Parameters

- **observer_handle** – [in] The handle returned by a prior call to `register_observer()`. Must be local to the calling locality. See the remote overload regarding foreign handles.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

Throws

`hpx::exception` – if `observer_handle` does not represent a valid observer handle, unless `ec` was not pre-initialized to `hpx::throws`.

`hpx::supervision::remove_target`

Defined in header `hpx/supervision.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::supervision::publish_event`
- `hpx::supervision::query_state`
- `hpx::supervision::register_observer`
- `hpx::supervision::unregister_observer`
- `hpx::supervision::register_activity_observer`
- `hpx::supervision::unregister_activity_observer`
- `hpx::supervision::check_admission`
- `hpx::supervision::await_terminal`
- `hpx::supervision::is_valid_transition`

`hpx::future<void> hpx::supervision::remove_target(hpx::id_type const &locality, hpx::id_type const &target)`

Clear all locally tracked state for a target on a possibly remote locality.

Unlike `unregister_observer()`, which removes a single previously registered observer handle, `remove_target()` unconditionally forgets every piece of local state held for `target` - its recorded lifecycle state (see `publish_event()` / `query_state()`) and any observers still registered for it (see `register_observer()`) - regardless of any specific observer handle.

Note: This is intended for callers that know `target` will never be queried or observed again locally (e.g. after a failed registration that seeded some state for it, or once a peer has been evicted), so that local state does not accumulate indefinitely. It is not itself a lifecycle event and does not affect `target's` state on any other locality.

Parameters

- **locality** – [in] The locality on which `target's` state is tracked.
- **target** – [in] The actor (or component) whose locally tracked state should be removed.

Throws

hpx::exception – if *locality* does not represent a locality, or if *target* does not represent a valid target. The returned future becomes exceptional if the operation fails.

Returns

A future that becomes ready once *target's* state has been removed.

void *hpx::supervision::remove_target*(*hpx::launch::sync_policy*, *hpx::id_type* const &*locality*, *hpx::id_type* const &*target*, *hpx::error_code* &*ec* = *hpx::throws*)

Clear all locally tracked state for a target on a possibly remote locality, blocking until the operation has completed.

This is the synchronous equivalent of *remove_target*(*hpx::id_type* const&, *hpx::id_type* const&).

Parameters

- **locality** – [in] The locality on which *target's* state is tracked.
- **target** – [in] The actor (or component) whose locally tracked state should be removed.
- **ec** – [in,out] this represents the error status on exit, if this pre-initialized to *hpx::throws* the function will throw on error instead.

Throws

hpx::exception – if *locality* does not represent a locality, or if *target* does not represent a valid target, unless *ec* was initialized not to *hpx::throws*.

void *hpx::supervision::remove_target*(*hpx::id_type* const &*target*, *hpx::error_code* &*ec* = *hpx::throws*)

Clear all locally tracked state for a target on the local locality.

Parameters

- **target** – [in] The actor (or component) whose locally tracked state should be removed. Must be local to the calling locality.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Throws

hpx::exception – if *target* does not represent a valid target, unless *ec* was not pre-initialized to *hpx::throws*.

hpx::supervision::register_activity_observer

Defined in header *hpx/supervision.hpp*.

See *Public API* for a list of names and headers that are part of the public *HPX API*.

See also:

- *hpx::supervision::publish_event*
- *hpx::supervision::query_state*
- *hpx::supervision::register_observer*
- *hpx::supervision::unregister_observer*
- *hpx::supervision::remove_target*
- *hpx::supervision::unregister_activity_observer*
- *hpx::supervision::check_admission*

- *hpx::supervision::await_terminal*
- *hpx::supervision::is_valid_transition*

hpx::future<hpx::id_type> *hpx::supervision::register_activity_observer*(*hpx::id_type* const &*locality*,
activity_callback const
&*callback*,
std::optional<std::uint64_t>
epoch_filter = *std::nullopt*)

Register a callback to observe activity-state transitions of all targets tracked on a possibly remote locality.

Note: For every target tracked on *locality* that is already *activity_state::active* at registration time, the new observer synchronously receives one replay notification (carrying *activity_transition::already_active*) as part of registration, taken under the same lock that serializes activity-state transitions. This guarantees the observer sees exactly one notification for each such target's current activity state: either the replay, or a live transition that raced with registration, but never both and never neither.

Note: If *locality* is the local locality, callbacks (including the registration-time replay) are invoked synchronously from within the call that triggered the transition (best effort; the callback must not block indefinitely).

Note: If *locality* is remote, callbacks are invoked asynchronously via a dedicated parcel, with retry semantics (e.g., 3 attempts over 500 ms before logging failure).

Note: The callback must not throw; exceptions thrown from it are logged and do not affect the observer registration.

Parameters

- **locality** – [in] The locality on which the callback should be registered.
- **callback** – [in] The callback invoked whenever any target tracked on *locality* transitions between *activity_state::inactive* and *activity_state::active*.
- **epoch_filter** – [in] If set, restricts notifications to transitions recorded under this epoch; transitions recorded under any other epoch are silently skipped for this observer. This also applies to the registration-time replay described below. If *std::nullopt* (the default), the observer is notified regardless of epoch.

Throws

hpx::exception – if *locality* does not represent a locality. The returned future becomes exceptional if the operation fails.

Returns

A future holding the global id of the observer handle. This handle must be passed to *unregister_activity_observer()* in order to stop receiving notifications.

hpx::id_type *hpx::supervision::register_activity_observer*(*hpx::launch::sync_policy*, *hpx::id_type* const
&*locality*, activity_callback const &*callback*,
std::optional<std::uint64_t> *epoch_filter* =
std::nullopt, *hpx::error_code* &*ec* =
hpx::throws)

Register a callback to observe activity-state transitions of all targets tracked on a possibly remote locality, blocking until the registration has completed.

This is the synchronous equivalent of `register_activity_observer(hpx::id_type const&, activity_callback const&, std::optional<std::uint64_t>)`.

Parameters

- **locality** – [in] The locality on which the callback should be registered.
- **callback** – [in] The callback invoked whenever any target tracked on *locality* transitions between *activity_state::inactive* and *activity_state::active*.
- **epoch_filter** – [in] If set, restricts notifications to transitions recorded under this epoch; see the asynchronous overload for details, including how this applies to the registration-time replay.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Throws

hpx::exception – if *locality* does not represent a locality, unless *ec* was not pre-initialized to *hpx::throws*.

Returns

The global id of the observer handle. This handle must be passed to `unregister_activity_observer()` in order to stop receiving notifications.

```
hpx::id_type hpx::supervision::register_activity_observer(activity_callback const &callback,
                                                         std::optional<std::uint64_t> epoch_filter =
                                                         std::nullopt, hpx::error_code &ec =
                                                         hpx::throws)
```

Register a callback to observe activity-state transitions of all targets tracked on the local locality.

Note: As for the remote overloads, for every target on the local locality that is already *activity_state::active* at registration time, one *activity_transition::already_active* replay notification is delivered synchronously as part of this call, taken under the same lock that serializes activity-state transitions, so the observer can neither miss nor double-receive any active target's current activity state.

Parameters

- **callback** – [in] The callback invoked whenever any target tracked on the local locality transitions between *activity_state::inactive* and *activity_state::active*.
- **epoch_filter** – [in] If set, restricts notifications to transitions recorded under this epoch, including the registration-time replay; see the remote overload for details. If `std::nullopt` (the default), the observer is notified regardless of epoch.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Throws

hpx::exception – if the operation fails, unless *ec* was initialized to *hpx::throws*.

Returns

The global id of the observer handle. This handle must be passed to `unregister_activity_observer()` in order to stop receiving notifications.

hpx::supervision::unregister_activity_observer

Defined in header `hpx/supervision.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::supervision::publish_event`
- `hpx::supervision::query_state`
- `hpx::supervision::register_observer`
- `hpx::supervision::unregister_observer`
- `hpx::supervision::remove_target`
- `hpx::supervision::register_activity_observer`
- `hpx::supervision::check_admission`
- `hpx::supervision::await_terminal`
- `hpx::supervision::is_valid_transition`

```
hpx::future<void> hpx::supervision::unregister_activity_observer(hpx::id_type const &locality,  
                                                                hpx::id_type const  
                                                                &observer_handle)
```

Unregister a previously registered activity observer on a possibly remote locality.

Note: Once the returned future has become ready, no orphaned callbacks fire after unregistration completes.

Parameters

- **locality** – [in] The locality on which the observer was registered.
- **observer_handle** – [in] The handle returned by a prior call to `register_activity_observer()`. Only handles returned from that function are accepted; passing a handle obtained from `register_observer()` (or any foreign handle) is rejected.

Throws

`hpx::exception` – if *locality* does not represent a locality, or if *observer_handle* does not represent a valid activity-observer handle. The returned future becomes exceptional if the operation fails.

Returns

A future that becomes ready once the observer has been unregistered.

```
void hpx::supervision::unregister_activity_observer(hpx::launch::sync_policy, hpx::id_type const  
                                                  &locality, hpx::id_type const &observer_handle,  
                                                  hpx::error_code &ec = hpx::throws)
```

Unregister a previously registered activity observer on a possibly remote locality, blocking until the operation has completed.

This is the synchronous equivalent of `unregister_activity_observer(hpx::id_type const&, hpx::id_type const&)`.

Note: No orphaned callbacks fire after unregistration completes.

Parameters

- **locality** – [in] The locality on which the observer was registered.
- **observer_handle** – [in] The handle returned by a prior call to `register_activity_observer()`. See the asynchronous overload regarding foreign handles.
- **ec** – [in,out] this represents the error status on exit, if this pre-initialized to `hpx::throws` the function will throw on error instead.

Throws

`hpx::exception` – if *locality* does not represent a locality, or if *observer_handle* does not represent a valid activity-observer handle, unless *ec* was initialized not to `hpx::throws`.

```
void hpx::supervision::unregister_activity_observer(hpx::id_type const &observer_handle,
                                                  hpx::error_code &ec = hpx::throws)
```

Unregister a previously registered activity observer on the local locality.

Note: No orphaned callbacks fire after unregistration completes.

Parameters

- **observer_handle** – [in] The handle returned by a prior call to `register_activity_observer()`. Must be local to the calling locality. See the remote overload regarding foreign handles.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

Throws

`hpx::exception` – if *observer_handle* does not represent a valid activity-observer handle, unless *ec* was not pre-initialized to `hpx::throws`.

hpx::supervision::check_admission

Defined in header `hpx/supervision.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::supervision::publish_event`
- `hpx::supervision::query_state`
- `hpx::supervision::register_observer`
- `hpx::supervision::unregister_observer`
- `hpx::supervision::remove_target`
- `hpx::supervision::register_activity_observer`
- `hpx::supervision::unregister_activity_observer`
- `hpx::supervision::await_terminal`
- `hpx::supervision::is_valid_transition`

dispatch_outcome *hpx::supervision::check_admission*(*hpx::id_type* const &target, *std::uint64_t* epoch = 0)

Check whether *target* currently admits new dispatch under *epoch*.

Queries the local supervision manager's terminal-latch state for *target* within *epoch*, i.e. the same state maintained by *publish_event()* and consulted by *await_terminal()*. Does not publish, mutate state, or notify observers.

Note: This is a pure local read of already-resident latch state, not a query that can fail the way a remote *publish_event()* call can; it never throws and carries no *hpx::error_code* out-parameter.

Note: The terminal latch consulted here is populated only by *publish_event()* calls delivered to *this* locality's supervision manager for *target*. It is not synchronized across localities: a terminal event published to a different locality's manager for the same logical target/epoch is invisible here and does not affect this call's outcome. *register_observer* / *unregister_observer* notifications do not feed this state either. Any locality other than the one hosting *target's* supervision manager that needs to gate its own dispatch decisions on *target's* terminal state must build that cross-locality propagation itself (e.g. registering an observer with the owning locality and re-publishing locally, or routing the admission decision to the owning locality); this API provides no such mechanism.

Parameters

- **target** – [in] The actor (or component) to check. Must be local to the calling locality.
- **epoch** – [in] The epoch dispatch would occur under.

Returns

dispatch_outcome::rejected_fenced if *target* has already latched a terminal event (completed or failed) under *epoch*, *dispatch_outcome::admitted* otherwise (including when *target* is unknown locally or invalid).

hpx::supervision::await_terminal

Defined in header *hpx/supervision.hpp*.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::supervision::publish_event*
- *hpx::supervision::query_state*
- *hpx::supervision::register_observer*
- *hpx::supervision::unregister_observer*
- *hpx::supervision::remove_target*
- *hpx::supervision::register_activity_observer*
- *hpx::supervision::unregister_activity_observer*
- *hpx::supervision::check_admission*
- *hpx::supervision::is_valid_transition*

```
hpx::future<lifecycle_state> hpx::supervision::await_terminal(hpx::id_type const &locality, hpx::id_type
                                                             const &target, std::uint64_t epoch = 0,
                                                             std::optional<std::chrono::steady_clock::duration>
                                                             timeout = std::nullopt)
```

Asynchronously wait for a target actor on a possibly remote locality to reach a terminal lifecycle event (completed or failed) within a given epoch.

The function `await_terminal()` is a one-shot, blocking-free counterpart to `register_observer()` for the specific case of waiting on the terminal transition: unlike an observer, it does not need to be explicitly unregistered, but it also only ever resolves once, for the exact (*target*, *epoch*) pair passed in.

Note: below for this call only. If `std::nullopt` (the default), *locality*'s current default timeout is used instead.

Note: below).

Note: *locality* bounds the lifetime of every waiter it registers on behalf of an outstanding call, so dropping the returned future without *target* ever reaching a terminal event under *epoch* does not leak the corresponding waiter entry indefinitely: it is swept and invalidated at or after *timeout* elapses, enforced by a background timer on *locality* independently of any further activity for *target*, so an idle target's waiter is not left to accumulate until some later, unrelated call happens to sweep it. Callers that need a tighter bound than *timeout* should still race the returned future against an external timeout.

Parameters

- **locality** – [in] The locality on which the supervision manager responsible for *target* is running.
- **target** – [in] The actor (or component) to wait on.
- **epoch** – [in] The epoch *target* must reach a terminal event under for the returned future to resolve. If *target*'s epoch is later superseded by a higher epoch (see the epoch semantics of *publish_event*) before it reaches a terminal event under *epoch*, the returned future becomes exceptional (see
- **timeout** – [in] Overrides the server-enforced timeout described in

Throws

- *below*) – instead of ever resolving with a value.
- **hpx::exception** – if *locality* does not represent a locality, or if *target* does not represent a valid target. The returned future becomes exceptional with a *hpx::error::stale_state* exception, naming *target*, *epoch*, and the epoch that superseded it, if *target*'s epoch advances past *epoch* before reaching a terminal event under it. It also becomes exceptional with a *hpx::error::future_cancelled* exception if neither of the above happens before *timeout* elapses (see

Returns

A future that becomes ready, holding the *lifecycle_state* recorded for *target*, as soon as *target* reaches a terminal event under *epoch*. If *target* has already reached a terminal event under *epoch* at the time of the call, the returned future is ready immediately.

```
lifecycle_state hpx::supervision::await_terminal(hpx::launch::sync_policy, hpx::id_type const &locality,
                                                hpx::id_type const &target, std::uint64_t epoch = 0,
                                                std::optional<std::chrono::steady_clock::duration>
                                                timeout = std::nullopt, hpx::error_code &ec =
                                                hpx::throws)
```

Wait for a target actor on a possibly remote locality to reach a terminal lifecycle event, blocking until it does.

This is the synchronous equivalent of `await_terminal(hpx::id_type const&, hpx::id_type const&, std::uint64_t)`.

Note: This call blocks until *target* reaches a terminal event, its epoch is superseded, or *timeout* elapses.

Parameters

- **locality** – [in] The locality on which the supervision manager responsible for *target* is running.
- **target** – [in] The actor (or component) to wait on.
- **epoch** – [in] The epoch *target* must reach a terminal event under. See the asynchronous overload for details.
- **timeout** – [in] Overrides the server-enforced timeout described in the asynchronous overload for this call only. If `std::nullopt` (the default), *locality*'s current default timeout is used instead.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

Throws

`hpx::exception` – if *locality* does not represent a locality, or if *target* does not represent a valid target, unless *ec* was not pre-initialized to `hpx::throws`. Also throws (or sets *ec* to) a `hpx::error::stale_state` exception, naming *target*, *epoch*, and the epoch that superseded it, if *target*'s epoch advances past *epoch* before reaching a terminal event under it. Also throws (or sets *ec* to) a `hpx::error::future_cancelled` exception if neither of the above happens before *timeout* elapses.

Returns

The *lifecycle_state* recorded for *target* once it reaches a terminal event under *epoch*.

```
hpx::future<lifecycle_state> hpx::supervision::await_terminal(hpx::id_type const &target, std::uint64_t
                                                            epoch = 0,
                                                            std::optional<std::chrono::steady_clock::duration>
                                                            timeout = std::nullopt, hpx::error_code &ec
                                                            = hpx::throws)
```

Asynchronously wait for a target actor on the local locality to reach a terminal lifecycle event within a given epoch.

Note: As for the remote overloads, the returned future becomes exceptional with `hpx::error::stale_state` if *epoch* is superseded, or with `hpx::error::future_cancelled` if *timeout* elapses first.

Parameters

- **target** – [in] The actor (or component) to wait on. Must be local to the calling locality.

- **epoch** – [in] The epoch *target* must reach a terminal event under. See the remote overload for details.
- **timeout** – [in] Overrides the server-enforced timeout described in the asynchronous overload for this call only. If `std::nullopt` (the default), the local locality's default timeout is used instead.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

Throws

`hpx::exception` – if *target* does not represent a valid target, unless *ec* was not pre-initialized to `hpx::throws`.

Returns

A future that becomes ready, holding the *lifecycle_state* recorded for *target*, as soon as *target* reaches a terminal event under *epoch*. See the remote overload for how the returned future behaves if *target's* epoch is superseded before reaching a terminal event under *epoch*.

```
lifecycle_state hpx::supervision::await_terminal(hpx::launch::sync_policy, hpx::id_type const &target,
                                                std::uint64_t epoch = 0,
                                                std::optional<std::chrono::steady_clock::duration>
                                                timeout = std::nullopt, hpx::error_code &ec =
                                                hpx::throws)
```

Synchronously wait for a local supervised target to reach a terminal lifecycle state (completed or failed).

This overload is restricted to targets that are guaranteed to be local to the calling locality. Because no dispatch/AGAS round-trip is required, it is cheaper than the remote `hpx::launch::sync_policy` overload and simply blocks the calling thread on the underlying local future.

Parameters

- **target** – The id of the supervised entity to wait on. Must refer to a component that is local to this locality.
- **epoch** – The join/generation epoch the caller expects *target* to still be part of. Passing an epoch that no longer matches the target's current epoch may cause the wait to resolve immediately with a mismatch-related error.
- **timeout** – Optional upper bound on how long to wait before giving up. If not supplied, waits indefinitely for a terminal state.
- **ec** – Used to report errors instead of throwing an exception. When *target* does not represent a valid id, `hpx::error::bad_parameter` is reported. If the wait is cancelled due to timeout elapsing without a terminal state being reached, `hpx::error::future_cancelled` is reported.

Throws

`hpx::exception` – if *ec* is `hpx::throws` (the default) and an error is encountered; otherwise sets *ec* to reflect the error and returns.

Returns

The terminal *lifecycle_state* observed for *target*, or a default-constructed *lifecycle_state* if an error occurred and *ec* was used to report it.

hpx::supervision::is_valid_transition

Defined in header `hpx/supervision.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::supervision::publish_event`
- `hpx::supervision::query_state`
- `hpx::supervision::register_observer`
- `hpx::supervision::unregister_observer`
- `hpx::supervision::remove_target`
- `hpx::supervision::register_activity_observer`
- `hpx::supervision::unregister_activity_observer`
- `hpx::supervision::check_admission`
- `hpx::supervision::await_terminal`

bool `hpx::supervision::is_valid_transition`(event from, event to) noexcept

Determine whether transitioning from lifecycle event *from* to lifecycle event *to* is permitted by the supervision lifecycle state machine.

event::completed and *event::failed* are terminal states and have no outgoing transitions. *event::losing_locality* is only reachable from *event::started*, *event::running*, or *event::suspending*, and may only transition to *failed*. A missing prior event is represented as *event::unknown*, from which the only valid transition is to *event::started*.

hpx::supervision::testing::set_register_observer_snapshot_hook

Defined in header `hpx/supervision.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

void `hpx::supervision::testing::set_register_observer_snapshot_hook`(std::function<void()> hook)

Set the hook invoked after an observer snapshot is captured.

Note: Used for testing purposes only.

Parameters

hook – Hook to invoke. An empty function clears the hook.

hpx::supervision::supervision_manager

Defined in header `hpx/supervision.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
class supervision_manager
```

hpx/supervision/server/supervision_manager.hpp

Defined in header `hpx/supervision/server/supervision_manager.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace supervision
```

```
        namespace server
```

Variables

```
constexpr char const *const supervision_manager_name = "supervision_manager/"
```

```
struct observer_entry
```

Public Members

```
    hpx::id_type agent
```

```
    std::uint64_t epoch_filter
```

```
struct supervision_manager : public  
    hpx::components::fixed_component_base<supervision_manager>
```

Public Types

```
using base_type = components::fixed_component_base<supervision_manager>
```

Public Functions

supervision_manager()

~supervision_manager()

void **register_server_instance**(char const *service_name, std::uint32_t locality_id,
error_code &ec = throws)

void **unregister_server_instance**(error_code &ec = throws) const

publish_result **publish_event**(hpx::id_type const &target, event ev, std::uint64_t epoch)

lifecycle_state **query_state**(hpx::id_type const &target)

hpx::id_type **register_observer**(hpx::id_type const &target, hpx::id_type const &agent,
std::uint64_t epoch_filter = static_cast<std::uint64_t>(-1))

void **unregister_observer**(hpx::id_type const &observer_handle)

void **remove_target**(hpx::id_type const &target)

Clears all locally tracked state for **target**.

Unlike **unregister_observer()**, which removes a single previously registered observer handle (and leaves any recorded lifecycle state for its target(s) intact), **remove_target()** unconditionally forgets every piece of local bookkeeping this supervision manager holds for **target** - its recorded lifecycle state and current epoch (see **publish_event()**), and any per-target observers still registered for it (see **register_observer()**) - regardless of any specific observer handle. Intended for callers that know **target** will never be queried or observed again locally (e.g. after a failed registration that seeded some state for it, or once a peer has been evicted) and want to reclaim that local state instead of letting it accumulate indefinitely.

hpx::id_type **register_activity_observer**(hpx::id_type const &agent, std::uint64_t
epoch_filter = static_cast<std::uint64_t>(-1))

Registers a locality-scoped activity observer.

Replays targets that are already active when registration takes its snapshot.

The snapshot of the target's tracked state and the insertion of the observer into the tracked set happen atomically under **mtx_**, which guarantees the observer receives exactly one notification for the target's current state: either the replay (if already active at registration time) or a live transition that raced with registration, but never both and never neither.

Delivery of that notification (replay or live) happens after **mtx_** is released and is therefore not ordered relative to any other concurrent live notification for the same target; callers must not assume replay is delivered before or after a racing live event.

void **unregister_activity_observer**(hpx::id_type const &observer_handle)

Unregisters an activity observer.

The handle must have been returned by **register_activity_observer()**. As with **unregister_observer()**, no orphaned callbacks fire after this call completes. **observer_handle** must have been returned by **register_activity_observer()**; a handle returned by **register_observer()** instead (i.e. found in **agents_** rather than **activity_observers_**) is rejected.

hpx::future<lifecycle_state> **await_terminal**(hpx::id_type const &target, std::uint64_t epoch
= 0, std::chrono::steady_clock::duration timeout
= (std::chrono::steady_clock::duration::max)())


```
dispatch_outcome check_admission(hpx::id_type const &target, std::uint64_t epoch = 0)
                                const noexcept
```

Protected Types

```
using waiters_t = std::vector<waiter_entry>
```

```
using stale_waiters_t = std::vector<std::pair<std::uint64_t, waiters_t>>
```

```
using expired_waiters_t = std::vector<expired_waiter>
```

Protected Functions

```
hpx::future<void> fire_events(hpx::id_type const &target, lifecycle_event_notification const
                             &notification)
```

```
hpx::future<void> fire_event(hpx::id_type const &target, hpx::id_type const &agent,
                             lifecycle_event_notification notification)
```

```
hpx::future<void> deliver_activity_notification(activity_notification const
                                                &notification,
                                                std::vector<observer_entry> const
                                                &observers)
```

```
hpx::future<void> fire_activity_event(hpx::id_type const &agent, activity_notification
                                       notification)
```

```
void record_error(hpx::id_type const &target, std::uint64_t expected_sequence_number,
                  hpx::error_code const &ec)
```

```
bool unregister_observer_target(hpx::id_type const &target, hpx::id_type const
                                &observer_handle)
```

```
std::uint64_t current_epoch_for(hpx::id_type const &target) const
```

```
waiters_t drain_terminal_waiters_locked(std::unique_lock<hpx::spinlock> &l,
                                         hpx::id_type const &target, std::uint64_t
                                         epoch, event last_event)
```

```
stale_waiters_t drain_stale_waiters_locked(std::unique_lock<hpx::spinlock> &l,
                                             hpx::id_type const &target, std::uint64_t
                                             epoch)
```

```
apply_result apply_new_epoch_locked(std::unique_lock<hpx::spinlock> &l, hpx::id_type
                                     const &target, supervision::event ev, std::uint64_t
                                     epoch)
```

```
std::optional<apply_result> apply_current_epoch_locked(std::unique_lock<hpx::spinlock>
                                                         &l, hpx::id_type const &target,
                                                         supervision::event ev,
                                                         std::uint64_t epoch)
```

```
expired_waiters_t drain_expired_waiters_locked(std::unique_lock<hpx::spinlock> &l,  
                                                std::chrono::steady_clock::time_point  
                                                now)  
  
stale_waiters_t drain_all_waiters_for_target_locked(std::unique_lock<hpx::spinlock>  
                                                    &l, hpx::id_type const &target)  
  
void sweep_expired_waiters()  
void arm_sweep_timer_locked(std::unique_lock<hpx::spinlock> &&l)  
bool sweep_timer_callback()  
void recompute_earliest_deadline_locked(std::unique_lock<hpx::spinlock> &l)  
void remove_target_from_agents_locked(std::unique_lock<hpx::spinlock> &l,  
                                       hpx::id_type const &agent, hpx::id_type const  
                                       &target)
```

Protected Static Functions

```
static void invalidate_stale_waiters(hpx::id_type const &target, std::uint64_t epoch,  
                                     stale_waiters_t &stale)  
  
static void resolve_terminal_waiters(lifecycle_event_notification const &notification,  
                                       waiters_t &to_resolve)  
  
static void invalidate_expired_waiters(expired_waiters_t &expired)
```

Private Members

```
mutable hpx::spinlock mtx_  
  
hpx::spinlock sweep_timer_mtx_  
  
mutable std::string instance_name_  
  
std::map<hpx::id_type, lifecycle_state> states_  
  
std::map<hpx::id_type, std::vector<observer_entry>> observers_  
  
std::map<hpx::id_type, std::vector<hpx::id_type>> agents_  
  
std::vector<observer_entry> activity_observers_  
  
std::map<hpx::id_type, std::chrono::steady_clock::time_point> activated_at_  
  
std::map<waiter_key, waiters_t> waiters_
```

```
std::chrono::steady_clock::time_point earliest_deadline_ =  
(std::chrono::steady_clock::time_point::max())
```

```
hpx::util::pool_timer sweep_timer_
```

```
std::chrono::steady_clock::time_point armed_deadline_ =  
(std::chrono::steady_clock::time_point::max())
```

```
bool shutting_down_ = false
```

```
std::size_t callbacks_in_flight_ = 0
```

```
hpx::lcos::local::detail::condition_variable shutdown_cv_
```

```
struct apply_result
```

Public Members

```
lifecycle_event_notification notification
```

```
waiters_t to_resolve
```

```
struct expired_waiter
```

Public Members

```
hpx::id_type target
```

```
std::uint64_t epoch
```

```
hpx::promise<lifecycle_state> promise
```

```
struct waiter_entry
```

Public Members

```
hpx::promise<lifecycle_state> promise
```

```
std::chrono::steady_clock::time_point deadline
```

```
struct waiter_key
```

Public Members

`hpx::id_type target`

`std::uint64_t epoch`

Friends

inline friend bool **operator<**(*waiter_key* const &lhs, *waiter_key* const &rhs)

inline friend bool **operator==**(*waiter_key* const &lhs, *waiter_key* const &rhs)

algorithms

See *Public API* for a list of names and headers that are part of the public *HPX* API.

`hpx::experimental::run_on_all`

Defined in header `hpx/task_block.hpp`⁵⁷³.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Warning: doxygenfunction: Unable to resolve function “`hpx::experimental::run_on_all`” with arguments (`ExPolicy&&, T&&, Ts&&...`) in doxygen xml output for project “`hpx`” from directory: `../hpx_autodoc`. Potential matches:

```
- template<typename ExPolicy, typename T, typename ...Ts> decltype(auto) run_on_  
  ↳all(ExPolicy &&policy, T &&t, Ts&&... ts)  
- template<typename T, typename ...Ts> decltype(auto) run_on_all(T &&t, Ts&&... ts)
```

Warning: doxygenfunction: Unable to resolve function “`hpx::experimental::run_on_all`” with arguments (`T&&, Ts&&...`) in doxygen xml output for project “`hpx`” from directory: `../hpx_autodoc`. Potential matches:

```
- template<typename ExPolicy, typename T, typename ...Ts> decltype(auto) run_on_  
  ↳all(ExPolicy &&policy, T &&t, Ts&&... ts)  
- template<typename T, typename ...Ts> decltype(auto) run_on_all(T &&t, Ts&&... ts)
```

⁵⁷³ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/task_block.hpp

hpx::experimental::task_canceled_exception

Defined in header [hpx/task_block.hpp](#)⁵⁷⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::experimental::task_block`
- `hpx::experimental::define_task_block`
- `hpx::experimental::define_task_block_restore_thread`

class **task_canceled_exception** : public `hpx::exception`

The class `task_canceled_exception` defines the type of objects thrown by `task_block::run` or `task_block::wait` if they detect that an exception is pending within the current parallel region.

hpx::experimental::task_block

Defined in header [hpx/task_block.hpp](#)⁵⁷⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::experimental::task_canceled_exception`
- `hpx::experimental::define_task_block`
- `hpx::experimental::define_task_block_restore_thread`

template<typename **ExPolicy** = `hpx::execution::parallel_policy`>

class **task_block**

The class `task_block` defines an interface for forking and joining parallel tasks. The `define_task_block` and `define_task_block_restore_thread` function templates create an object of type `task_block` and pass a reference to that object to a user-provided callable object.

An object of class `task_block` cannot be constructed, destroyed, copied, or moved except by the implementation of the task region library. Taking the address of a `task_block` object via `operator&` or `addressof` is ill formed. The result of obtaining its address by any other means is unspecified.

A `task_block` is active if it was created by the nearest enclosing task block, where “task block” refers to an invocation of `define_task_block` or `define_task_block_restore_thread` and “nearest

enclosing” means the most recent invocation that has not yet completed. Code designated for execution in another thread by means other than the facilities in this section (e.g., using `thread` or `async`) are not enclosed in the task region and a `task_block` passed to (or captured by) such code is not active within that code. Performing any operation on a `task_block` that is not active results in undefined behavior.

The `task_block` that is active before a specific call to the `run` member function is not active within the asynchronous function that invoked `run`. (The invoked function should not, therefore, capture the `task_block` from the surrounding block.)

⁵⁷⁴ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/task_block.hpp

⁵⁷⁵ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/task_block.hpp

Example:

```

define_task_block([&](auto& tr) {
    tr.run([&] {
        tr.run([] { f(); });           // Error: tr is not active
        define_task_block([&](auto& tr) { // Nested task block
            tr.run(f);                 // OK: inner tr is active
            /// ...
        });
    }); /// ...
});

```

Template Parameters

ExPolicy – The execution policy an instance of a task_block was created with. This defaults to *parallel_policy*.

hpx::experimental::define_task_block

Defined in header `hpx/task_block.hpp`⁵⁷⁶.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::experimental::task_canceled_exception`
- `hpx::experimental::task_block`
- `hpx::experimental::define_task_block_restore_thread`

Warning: doxygenfunction: Unable to resolve function “hpx::experimental::define_task_block” with arguments (ExPolicy&&, F&&) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:

```

- template<typename ExPolicy, typename F> decltype(auto) define_task_block(ExPolicy &&
  ↳ policy, F &&f)
- template<typename F> void define_task_block(F &&f)

```

template<typename F>

void `hpx::experimental::define_task_block(F &&f)`

Constructs a task_block, tr, and invokes the expression $f(tr)$ on the user-provided object, f . This version uses *parallel_policy* for task scheduling.

Postcondition: All tasks spawned from f have finished execution. A call to `define_task_block` may return on a different thread than that on which it was called.

Note: It is expected (but not mandated) that f will (directly or indirectly) call `tr.run(callable_object)`.

⁵⁷⁶ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fdbc0c8044e7/libs/core/include_local/include/hpx/task_block.hpp

Template Parameters

F – The type of the user defined function to invoke inside the `define_task_block` (deduced). *F* shall be `MoveConstructible`.

Parameters

f – The user defined function to invoke inside the task block. Given a lvalue *tr* of type `task_block`, the expression, `(void)f(tr)`, shall be well-formed.

Throws

exception_list – specified in Exception Handling.

hpx::experimental::define_task_block_restore_thread

Defined in header `hpx/task_block.hpp`⁵⁷⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::experimental::task_canceled_exception`
- `hpx::experimental::task_block`
- `hpx::experimental::define_task_block`

template<typename **ExPolicy**, typename **F**>

`hpx::parallel::util::detail::algorithm_result_t<ExPolicy> hpx::experimental::define_task_block_restore_thread(ExPolicy &&policy, F &&f)`

Constructs a `task_block`, *tr*, and invokes the expression `f(tr)` on the user-provided object, *f*.

Postcondition: All tasks spawned from *f* have finished execution. A call to `define_task_block_restore_thread` always returns on the same thread as that on which it was called.

Note: It is expected (but not mandated) that *f* will (directly or indirectly) call `tr.run(callable_object)`.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the task block may be parallelized.
- **F** – The type of the user defined function to invoke inside the `define_task_block` (deduced). *F* shall be `MoveConstructible`.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **f** – The user defined function to invoke inside the `define_task_block`. Given an lvalue *tr* of type `task_block`, the expression, `(void)f(tr)`, shall be well-formed.

Throws

exception_list – specified in Exception Handling.

⁵⁷⁷ http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/task_block.hpp

```
template<typename F>
void hpx::experimental::define_task_block_restore_thread(F &&f)
```

Constructs a `task_block`, `tr`, and invokes the expression $f(tr)$ on the user-provided object, f . This version uses *parallel_policy* for task scheduling.

Postcondition: All tasks spawned from f have finished execution. A call to *define_task_block_restore_thread* always returns on the same thread as that on which it was called.

Note: It is expected (but not mandated) that f will (directly or indirectly) call `tr.run(callable_object)`.

Template Parameters

F – The type of the user defined function to invoke inside the `define_task_block` (deduced). F shall be `MoveConstructible`.

Parameters

f – The user defined function to invoke inside the `define_task_block`. Given an lvalue tr of type `task_block`, the expression, `(void)f(tr)`, shall be well-formed.

Throws

exception_list – specified in Exception Handling.

hpx::experimental::task_group

Defined in header `hpx/experimental/task_group.hpp`⁵⁷⁸.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

class **task_group**

A *task_group* represents concurrent execution of a group of tasks. Tasks can be dynamically added to the group while it is executing.

hpx/parallel/util/range.hpp

Defined in header `hpx/parallel/util/range.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **util**

⁵⁷⁸ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e77/libs/core/include_local/include/hpx/experimental/task_group.hpp

Typedefs

```
template<typename Iterator, typename Sentinel = Iterator>
using range = hpx::util::iterator_range<Iterator, Sentinel>
```

Functions

```
template<typename Iter, typename Sent>
range<Iter, Sent> concat(range<Iter, Sent> const &it1, range<Iter, Sent> const &it2)

template<typename Iter1, typename Sent1, typename Iter2, typename Sent2>
range<Iter2, Iter2> init_move(range<Iter2, Sent2> const &dest, range<Iter1, Sent1> const &src)
```

Move objects from the range src to dest.

Parameters

- **dest** – [in] : range where move the objects
- **src** – [in] : range from where move the objects

Returns

range with the objects moved and the size adjusted

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2>
range<Iter2, Sent2> uninit_move(range<Iter2, Sent2> const &dest, range<Iter1, Sent1> const
&src)
```

Move objects from the range src creating them in dest.

Parameters

- **dest** – [in] : range where move and create the objects
- **src** – [in] : range from where move the objects

Returns

range with the objects moved and the size adjusted

```
template<typename Iter, typename Sent>
void destroy_range(range<Iter, Sent> r)
```

destroy a range of objects

Parameters

r – [in] : range to destroy

```
template<typename Iter, typename Sent>
range<Iter, Sent> init(range<Iter, Sent> const &r, typename std::iterator_traits<Iter>::value_type
&val)
```

initialize a range of objects with the object val moving across them

Parameters

- **r** – [in] : range of elements not initialized
- **val** – [in] : object used for the initialization

Returns

range initialized

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename
Compare>
bool is_mergeable(range<Iter1, Sent1> const &src1, range<Iter2, Sent2> const &src2, Compare
comp)
```

: indicate if two ranges have a possible merge

Remark

Parameters

- **src1** – [in] : first range
- **src2** – [in] : second range
- **comp** – [in] : object for to compare elements

Returns

true : they can be merged false : they can't be merged

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename Iter3,
typename Sent3, typename Compare>
```

```
range<Iter3, Sent3> full_merge(range<Iter3, Sent3> const &dest, range<Iter1, Sent1> const
&src1, range<Iter2, Sent2> const &src2, Compare comp)
```

Merge two contiguous ranges src1 and src2 , and put the result in the range dest, returning the range merged.

Parameters

- **dest** – [in] : range where locate the elements merged. the size of dest must be greater or equal than the sum of the sizes of src1 and src2
- **src1** – [in] : first range to merge
- **src2** – [in] : second range to merge
- **comp** – [in] : comparison object

Returns

range with the elements merged and the size adjusted

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename Value,
typename Compare>
```

```
range<Value*> uninit_full_merge(range<Value*> const &dest, range<Iter1, Sent1> const
&src1, range<Iter2, Sent2> const &src2, Compare comp)
```

Merge two contiguous ranges src1 and src2 , and create and move the result in the uninitialized range dest, returning the range merged.

Parameters

- **dest** – [in] : range where locate the elements merged. the size of dest must be greater or equal than the sum of the sizes of src1 and src2. Initially is un-initialize memory
- **src1** – [in] : first range to merge
- **src2** – [in] : second range to merge
- **comp** – [in] : comparison object

Returns

range with the elements merged and the size adjusted

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename
Compare>
```

```
range<Iter2, Sent2> half_merge(range<Iter2, Sent2> const &dest, range<Iter1, Sent1> const
&src1, range<Iter2, Sent2> const &src2, Compare comp)
```

: Merge two buffers. The first buffer is in a separate memory

Parameters

- **dest** – [in] : range where finish the two buffers merged
- **src1** – [in] : first range to merge in a separate memory
- **src2** – [in] : second range to merge, in the final part of the range where deposit the final results
- **comp** – [in] : object for compare two elements of the type pointed by the Iter1 and Iter2

Returns

: range with the two buffers merged

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename Iter3,
typename Sent3, typename Compare>
```

```
bool in_place_merge_uncontiguous(range<Iter1, Sent1> const &src1, range<Iter2, Sent2>
                                const &src2, range<Iter3, Sent3> &aux, Compare comp)
```

: merge two non contiguous buffers src1 , src2, using the range aux as auxiliary memory

Remark

Parameters

- **src1** – [**in**] : first range to merge
- **src2** – [**in**] : second range to merge
- **aux** – [**in**] : auxiliary range used in the merge
- **comp** – [**in**] : object for to compare elements

Returns

true : not changes done false : changes in the buffers

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename
Compare>
```

```
range<Iter1, Sent1> in_place_merge(range<Iter1, Sent1> const &src1, range<Iter1, Sent1> const
                                &src2, range<Iter2, Sent2> &buf, Compare comp)
```

: merge two contiguous buffers (src1, src2) using buf as auxiliary memory

Remark

Parameters

- **src1** – [**in**] : first range to merge
- **src2** – [**in**] : second range to merge
- **buf** – [**in**] : auxiliary memory used in the merge
- **comp** – [**in**] : object for to compare elements

Returns

true : not changes done false : changes in the buffers

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename
Compare>
```

```
void merge_flow(range<Iter1, Sent1> rng1, range<Iter2, Sent2> rbuf, range<Iter1, Sent1> rng2,
                Compare cmp)
```

hpx::experimental::induction

Defined in header [hpx/algorithm.hpp](#)⁵⁷⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename T>
constexpr hpx::parallel::detail::induction_stride_helper<T> hpx::experimental::induction(T &&value,
                                                                                          std::size_t stride)
```

The function template returns an induction object of unspecified type having a value type and encapsulating an initial *value* of that type and, optionally, a stride.

For each element in the input range, a looping algorithm over input sequence *S* computes an induction value from an induction variable and ordinal position *p* within *S* by the formula $i + p * \text{stride}$ if a stride was specified or $i + p$ otherwise. This induction value is passed to the element access function.

If the *value* argument to *induction* is a non-const lvalue, then that lvalue becomes the live-out object for the returned induction object. For each induction object that has a live-out object, the looping algorithm assigns the value of $i + n * \text{stride}$ to the live-out object upon return, where *n* is the number of elements in the input range.

Template Parameters

T – The value type to be used by the induction object.

Parameters

- **value** – [in] The initial value to use for the induction object
- **stride** – [in] The (optional) stride to use for the induction object (default: 1)

Returns

This returns an induction object with value type *T*, initial *value*, and (if specified) *stride*. If *T* is a lvalue of non-const type, *value* is used as the live-out object for the induction object; otherwise there is no live-out object.

```
template<typename T>
constexpr hpx::parallel::detail::induction_helper<T> hpx::experimental::induction(T &&value)
```

hpx::experimental::for_each_index

Defined in header [hpx/algorithm.hpp](#)⁵⁸⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename Mapping, typename Fun>
constexpr void hpx::experimental::for_each_index(Mapping const &mapping, Fun fun)
    Calls fun with every multidimensional index in the domain of mapping (i.e., its extents), in sequential order.
```

If *fun* returns a result, the result is ignored.

The layout mapping is a hint; implementations are encouraged to choose an iteration order that performs well for the given mapping.

Note: Complexity: Applies *fun* exactly the product of all extents of *mapping* times.

⁵⁷⁹ <http://github.com/TheHPXProject/hpx/blob/d402eoad7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

⁵⁸⁰ <http://github.com/TheHPXProject/hpx/blob/d402eoad7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Template Parameters

- **Mapping** – A type meeting the layout mapping requirements. Must expose *index_type*, *extents_type*, *static rank()*, and *extents()*.
- **Fun** – The type of the function/function object to use (deduced). *Fun* must meet *CopyConstructible* and be invocable with exactly *Mapping::rank()* arguments of type *Mapping::index_type*.

Parameters

- **mapping** – The layout mapping whose extents define the iteration domain.
- **fun** – Callable invoked as `fun(i0, i1, ...)` for every multidimensional index in the domain.

```
template<typename ExPolicy, typename Mapping, typename Fun>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy> hpx::experimental::for_each_index(ExPolicy
                                                         &&policy,
                                                         Mapping const
                                                         &mapping,
                                                         Fun fun)
```

Calls *fun* with every multidimensional index in the domain of *mapping* (i.e., its extents), executed according to *policy*.

If *fun* returns a result, the result is ignored.

Note: Complexity: Applies *fun* exactly the product of all extents of *mapping* times.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced).
- **Mapping** – A type meeting the layout mapping requirements. Must meet *CopyConstructible*.
- **Fun** – The type of the function/function object to use (deduced). Must meet *CopyConstructible*.

Parameters

- **policy** – The execution policy to use for scheduling.
- **mapping** – The layout mapping whose extents define the iteration domain.
- **fun** – Callable invoked as `fun(i0, i1, ...)` for every multidimensional index in the domain.

Returns

hpx::future<void> if the execution policy is *sequenced_task_policy* or *parallel_task_policy*, *void* otherwise.

hpx::partial_sort_copy

Defined in header [hpx/algorithm.hpp](#)⁵⁸¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename InIter, typename RandIter, typename Comp = hpx::parallel::detail::less>
RandIter hpx::partial_sort_copy(InIter first, InIter last, RandIter d_first, RandIter d_last, Comp &&comp =
    Comp())
```

Sorts some of the elements in the range [first, last) in ascending order, storing the result in the range [d_first, d_last). At most d_last - d_first of the elements are placed sorted to the range [d_first, d_first + n) where n is the number of elements to sort (n = min(last - first, d_last - d_first)).

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: $O(N \log(\min(D, N)))$, where $N = \text{std::distance}(\text{first}, \text{last})$ and $D = \text{std::distance}(\text{d_first}, \text{d_last})$ comparisons.

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **RandIter** – The type of the destination iterators used (deduced). This iterator type must meet the requirements of a random iterator.
- **Comp** – The type of the function/function object to use (deduced). Comp defaults to detail::less.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **d_first** – Refers to the beginning of the destination range.
- **d_last** – Refers to the end of the destination range.
- **comp** – comp is a callable object. The return value of the INVOKE operation applied to an object of type Comp, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that comp will not apply any non-constant function through the dereferenced iterator. This defaults to detail::less.

Returns

The *partial_sort_copy* algorithm returns a *RandomIt*. The algorithm returns an iterator to the element defining the upper boundary of the sorted range i.e. d_first + min(last - first, d_last - d_first)

⁵⁸¹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

```
template<typename ExPolicy, typename FwdIter, typename RandIter, typename Comp =
hpx::parallel::detail::less>
parallel::util::detail::algorithm_result_t<ExPolicy, RandIter> hpx::partial_sort_copy(ExPolicy &&policy,
                                                                                      FwdIter first, FwdIter
                                                                                      last, RandIter d_first,
                                                                                      RandIter d_last, Comp
                                                                                      &&comp = Comp())
```

Sorts some of the elements in the range `[first, last)` in ascending order, storing the result in the range `[d_first, d_last)`. At most `d_last - d_first` of the elements are placed sorted to the range `[d_first, d_first + n)` where `n` is the number of elements to sort (`n = min(last - first, d_last - d_first)`). Executed according to the policy.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: $O(N \log(\min(D, N)))$, where $N = \text{std::distance}(\text{first}, \text{last})$ and $D = \text{std::distance}(\text{d_first}, \text{d_last})$ comparisons.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **RandIter** – The type of the destination iterators used (deduced) This iterator type must meet the requirements of an random iterator.
- **Comp** – The type of the function/function object to use (deduced). `Comp` defaults to `detail::less`.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **d_first** – Refers to the beginning of the destination range.
- **d_last** – Refers to the end of the destination range.
- **comp** – `comp` is a callable object. The return value of the INVOKE operation applied to an object of type `Comp`, when contextually converted to `bool`, yields `true` if the first argument of the call is less than the second, and `false` otherwise. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator. This defaults to `detail::less`.

Returns

The *partial_sort_copy* algorithm returns a *hpx::future<RandomIt>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *RandomIt* otherwise. The

algorithm returns an iterator to the element defining the upper boundary of the sorted range i.e. `d_first + min(last - first, d_last - d_first)`

`hpx::experimental::reduction_bit_or`

Defined in header `hpx/algorithm.hpp`⁵⁸².

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename T>
constexpr hpx::parallel::detail::reduction_helper<T, std::bit_or<T>> hpx::experimental::reduction_bit_or(T
                                                                    &var)
```

The function template `reduction_bit_or` returns a reduction object of unspecified type having a value type and encapsulating an identity value for the reduction, it uses `std::bit_or{}` as its combiner function, and a live-out object from which the initial value is obtained and into which the final value is stored.

A parallel algorithm uses the `reduction_bit_or` object by allocating an unspecified number of instances, called views, of the reduction's value type. Each view is initialized with the reduction object's identity value, except that the live-out object (which was initialized by the caller) comprises one of the views. The algorithm passes a reference to a view to each application of an element-access function, ensuring that no two concurrently-executing invocations share the same view. A view can be shared between two applications that do not execute concurrently, but initialization is performed only once per view.

Modifications to the view by the application of element access functions accumulate as partial results. At some point before the algorithm returns, the partial results are combined, two at a time, using the `std::bit_or{}` operation until a single value remains, which is then assigned back to the live-out object.

T shall meet the requirements of *CopyConstructible* and *MoveAssignable*.

Note: In order to produce useful results, modifications to the view should be limited to commutative operations closely related to the combiner operation.

Template Parameters

- **T** – The value type to be used by the induction object.
- **Op** – The type of the binary function (object) used to perform the reduction operation.

Parameters

- **var** – [in,out] The life-out value to use for the reduction object. This will hold the reduced value after the algorithm is finished executing.
- **identity** – [in] The identity value to use for the reduction operation. This argument is optional and defaults to a copy of *var*.

Returns

This returns a reduction object of unspecified type having a value type of *T*. When the return value is used by an algorithm, the reference to *var* is used as the live-out object, new views are initialized to a copy of *identity*, and views are combined by invoking the copy of combiner, passing it the two views to be combined.

```
template<typename T>
```

⁵⁸² <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>


```
constexpr hpx::parallel::detail::reduction_helper<T, std::bit_or<T>> hpx::experimental::reduction_bit_or(T
                                                                    &var,
                                                                    T
                                                                    const
                                                                    &iden-
                                                                    tity)
```

hpx::transform

Defined in header [hpx/algorithm.hpp](#)⁵⁸³.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename FwdIter1, typename FwdIter2, typename F>
FwdIter2 hpx::transform(FwdIter1 first, FwdIter1 last, FwdIter2 dest, F &&f)
```

Applies the given function *f* to the range [first, last) and stores the result in another range, beginning at dest.

Note: Complexity: Exactly *last - first* applications of *f*

Template Parameters

- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *transform* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type &a);
```

The signature does not need to have const&. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *Ret* must be such that an object of type *FwdIter2* can be dereferenced and assigned a value of type *Ret*.

Returns

The *transform* algorithm returns a *FwdIter2*. The *transform* algorithm returns a tuple holding an iterator referring to the first element after the input sequence and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename F>
```

⁵⁸³ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

```
parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx::transform(ExPolicy &&policy, FwdIter1 first,
                                                                              FwdIter1 last, FwdIter2 dest, F
                                                                              &&f)
```

Applies the given function f to the range $[first, last)$ and stores the result in another range, beginning at $dest$. Executed according to the policy.

The invocations of f in the parallel *transform* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The invocations of f in the parallel *transform* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Exactly *last - first* applications of f

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the invocations of f .
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *transform* requires F to meet the requirements of *CopyConstructible*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by $[first, last)$. This is an unary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type &a);
```

The signature does not need to have `const&`. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *Ret* must be such that an object of type *FwdIter2* can be dereferenced and assigned a value of type *Ret*.

Returns

The *transform* algorithm returns a `hpx::future<FwdIter2>` if the execution policy is of type *parallel_task_policy* and returns *FwdIter2* otherwise. The *transform* algorithm returns a tuple holding an iterator referring to the first element after the input sequence and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename FwdIter1, typename FwdIter2, typename FwdIter3, typename F>
FwdIter3 hpx::transform(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, FwdIter3 dest, F &&f)
```

Applies the given function f to pairs of elements from two ranges: one defined by $[first1, last1)$ and the other beginning at $first2$, and stores the result in another range, beginning at $dest$.

Note: Complexity: Exactly *last - first* applications of f

Template Parameters

- **FwdIter1** – The type of the source iterators for the first range used (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators for the second range used (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *transform* requires F to meet the requirements of *CopyConstructible*.

Parameters

- **first1** – Refers to the beginning of the first sequence of elements the algorithm will be applied to.
- **last1** – Refers to the end of the first sequence of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by $[first, last)$. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively. The type *Ret* must be such that an object of type *FwdIter3* can be dereferenced and assigned a value of type *Ret*.

Returns

The *transform* algorithm returns a *FwdIter3*. The *transform* algorithm returns a tuple holding an iterator referring to the first element after the first input sequence, an iterator referring to the first element after the second input sequence, and the output iterator referring to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename FwdIter3, typename F>
parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter3> hpx::transform(ExPolicy &&policy, FwdIter1
                                                                    first1, FwdIter1 last1, FwdIter2
                                                                    first2, FwdIter3 dest, F &&f)
```

Applies the given function f to pairs of elements from two ranges: one defined by $[first1, last1)$ and the other beginning at $first2$, and stores the result in another range, beginning at $dest$. Executed according to the policy.

The invocations of *f* in the parallel *transform* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The invocations of *f* in the parallel *transform* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Exactly *last - first* applications of *f*

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the invocations of *f*.
- **FwdIter1** – The type of the source iterators for the first range used (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators for the second range used (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *transform* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the first sequence of elements the algorithm will be applied to.
- **last1** – Refers to the end of the first sequence of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively. The type *Ret* must be such that an object of type *FwdIter3* can be dereferenced and assigned a value of type *Ret*.

Returns

The *transform* algorithm returns a *hpx::future<FwdIter3>* if the execution policy is of type *parallel_task_policy* and returns *FwdIter3* otherwise. The *transform* algorithm returns a tuple holding an iterator referring to the first element after the first input sequence, an iterator referring to the first element after the second input sequence, and the output iterator referring to the element in the destination range, one past the last element copied.

hpx::ends_with

Defined in header `hpx/algorithm.hpp`⁵⁸⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename InIter1, typename InIter2, typename Pred = hpx::parallel::detail::equal_to>
bool hpx::ends_with(InIter1 first1, InIter1 last1, InIter2 first2, InIter2 last2, Pred &&pred = Pred())
    Checks whether the second range defined by [first1, last1) matches the suffix of the first range defined by [first2, last2)
```

The assignments in the parallel *ends_with* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: Linear: at most min(N1, N2) applications of the predicate.

Template Parameters

- **InIter1** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **InIter2** – The type of the begin destination iterators used (deduced). This iterator type must meet the requirements of a input iterator.
- **Pred** – The binary predicate that compares the elements. This defaults to `hpx::parallel::detail::equal_to`.

Parameters

- **first1** – Refers to the beginning of the source range.
- **last1** – Refers to the end of the source range.
- **first2** – Refers to the beginning of the destination range.
- **last2** – Refers to the end of the destination range.
- **pred** – Specifies the binary predicate function (or function object) which will be invoked for comparison of the elements in the two ranges.

Returns

The *ends_with* algorithm returns *bool*. The *ends_with* algorithm returns a boolean with the value true if the second range matches the suffix of the first range, false otherwise.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Pred = ranges::equal_to>
hpx::parallel::util::detail::algorithm_result<ExPolicy, bool>::type hpx::ends_with(ExPolicy &&policy, FwdIter1
    first1, FwdIter1 last1,
    FwdIter2 first2, FwdIter2
    last2, Pred &&pred = Pred())
```

Checks whether the second range defined by [first1, last1) matches the suffix of the first range defined by [first2, last2). Executed according to the policy.

The assignments in the parallel *ends_with* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

⁵⁸⁴ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

The assignments in the parallel *ends_with* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Linear: at most $\min(N_1, N_2)$ applications of the predicate.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the begin destination iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The binary predicate that compares the elements. This defaults to *ranges::equal_to*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the source range.
- **last1** – Refers to the end of the source range.
- **first2** – Refers to the beginning of the destination range.
- **last2** – Refers to the end of the destination range.
- **pred** – Specifies the binary predicate function (or function object) which will be invoked for comparison of the elements in the two ranges.

Returns

The *ends_with* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *ends_with* algorithm returns a boolean with the value true if the second range matches the suffix of the first range, false otherwise.

hpx::search

Defined in header [hpx/algorithm.hpp](#)⁵⁸⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::search_n*

```
template<typename FwdIter, typename FwdIter2, typename Pred = parallel::detail::equal_to>
FwdIter hpx::search(FwdIter first, FwdIter last, FwdIter2 s_first, FwdIter2 s_last, Pred &&op = Pred())
```

Searches the range [first, last) for any elements in the range [s_first, s_last). Uses a provided predicate to compare elements.

The comparison operations in the parallel *search* algorithm execute in sequential order in the calling thread.

⁵⁸⁵ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Note: Complexity: at most $(S*N)$ comparisons where $S = \text{distance}(s_first, s_last)$ and $N = \text{distance}(first, last)$.

Template Parameters

- **FwdIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *search* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

Parameters

- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **s_first** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **s_last** – Refers to the end of the sequence of elements of the algorithm will be searching for.
- **op** – Refers to the binary predicate which returns true if the elements should be treated as equal. the signature of the function should be equivalent to

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

Returns

The *search* algorithm returns a `hpx::future<FwdIter>` if the execution policy is of type *task_execution_policy* and returns *FwdIter* otherwise. The *search* algorithm returns an iterator to the beginning of the first subsequence $[s_first, s_last)$ in range $[first, last)$. If the length of the subsequence $[s_first, s_last)$ is greater than the length of the range $[first, last)$, *last* is returned. Additionally if the size of the subsequence is empty *first* is returned. If no subsequence is found, *last* is returned.

```
template<typename ExPolicy, typename FwdIter, typename FwdIter2, typename Pred =
parallel::detail::equal_to>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::search(ExPolicy &&policy, FwdIter first,
FwdIter last, FwdIter2 s_first,
FwdIter2 s_last, Pred &&op =
Pred())
```

Searches the range $[first, last)$ for any elements in the range $[s_first, s_last)$. Uses a provided predicate to compare elements. Executed according to the policy.

The comparison operations in the parallel *search* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *search* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: at most ($S*N$) comparisons where $S = \text{distance}(s_first, s_last)$ and $N = \text{distance}(first, last)$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *search* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **s_first** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **s_last** – Refers to the end of the sequence of elements of the algorithm will be searching for.
- **op** – Refers to the binary predicate which returns true if the elements should be treated as equal. the signature of the function should be equivalent to

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

Returns

The *search* algorithm returns a `hpx::future<FwdIter>` if the execution policy is of type *task_execution_policy* and returns *FwdIter* otherwise. The *search* algorithm returns an iterator to the beginning of the first subsequence $[s_first, s_last)$ in range $[first, last)$. If the length of the subsequence $[s_first, s_last)$ is greater than the length of the range $[first, last)$, *last* is returned. Additionally if the size of the subsequence is empty *first* is returned. If no subsequence is found, *last* is returned.

```
template<typename FwdIter, typename Searcher>
FwdIter hpx::search(FwdIter first, FwdIter last, Searcher &&searcher)
    Searches the range  $[first, last)$  using a searcher object.
```


Template Parameters

- **FwdIter** – The type of the source iterators (deduced). This iterator type must meet the requirements of a forward iterator.
- **Searcher** – A callable searcher object meeting the C++ Searcher requirements (e.g. `std::boyer_moore_searcher`).

Parameters

- **first** – Refers to the beginning of the sequence to search.
- **last** – Refers to the end of the sequence to search.
- **searcher** – A callable object performing the search.

Returns

An iterator to the beginning of the found subsequence, or *last* if no subsequence is found.

```
template<typename ExPolicy, typename FwdIter, typename Searcher>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::search(ExPolicy &&policy, FwdIter first,
                                                                              FwdIter last, Searcher
                                                                              &&searcher)
```

Searches the range [first, last) using a searcher object. Executed according to the policy.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced).
- **FwdIter** – The type of the source iterators (deduced).
- **Searcher** – A callable searcher object meeting the C++ Searcher requirements.

Parameters

- **policy** – The execution policy to use.
- **first** – Refers to the beginning of the sequence to search.
- **last** – Refers to the end of the sequence to search.
- **searcher** – A callable object performing the search.

Returns

The algorithm returns a `hpx::future<FwdIter>` if the execution policy is a task policy, or *FwdIter* otherwise.

hpx::search_n

Defined in header `hpx/algorithm.hpp`⁵⁸⁶.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::search`

```
template<typename FwdIter, typename FwdIter2, typename Pred = parallel::detail::equal_to>
```

⁵⁸⁶ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

FwdIter **hpx::search_n**(*FwdIter* first, *std::size_t* count, *FwdIter2* s_first, *FwdIter2* s_last, *Pred* &&op = *Pred*())

Searches the range [first, last) for any elements in the range [s_first, s_last). Uses a provided predicate to compare elements.

The comparison operations in the parallel *search_n* algorithm execute in sequential order in the calling thread.

Note: Complexity: at most ($S \cdot N$) comparisons where $S = \text{distance}(s_first, s_last)$ and $N = \text{count}$.

Template Parameters

- **FwdIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *search_n* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to *std::equal_to*<>

Parameters

- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **count** – Refers to the range of elements of the first range the algorithm will be applied to.
- **s_first** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **s_last** – Refers to the end of the sequence of elements of the algorithm will be searching for.
- **op** – Refers to the binary predicate which returns true if the elements should be treated as equal. the signature of the function should be equivalent to

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

Returns

The *search_n* algorithm returns *FwdIter*. The *search_n* algorithm returns an iterator to the beginning of the last subsequence [s_first, s_last) in range [first, first+count). If the length of the subsequence [s_first, s_last) is greater than the length of the range [first, first+count), *first* is returned. Additionally if the size of the subsequence is empty or no subsequence is found, *first* is also returned.

```
template<typename ExPolicy, typename FwdIter, typename FwdIter2, typename Pred =  
parallel::detail::equal_to>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::search_n(ExPolicy &&policy, FwdIter  
first, std::size_t count, FwdIter2  
s_first, FwdIter2 s_last, Pred  
&&op = Pred())
```

Searches the range [first, last) for any elements in the range [s_first, s_last). Uses a provided predicate to compare elements. Executed according to the policy.

The comparison operations in the parallel *search_n* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *search_n* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: at most ($S \cdot N$) comparisons where S = distance(s_first, s_last) and N = count.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *search_n* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **count** – Refers to the range of elements of the first range the algorithm will be applied to.
- **s_first** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **s_last** – Refers to the end of the sequence of elements of the algorithm will be searching for.
- **op** – Refers to the binary predicate which returns true if the elements should be treated as equal. the signature of the function should be equivalent to

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have const &, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

Returns

The *search_n* algorithm returns a `hpx::future<FwdIter>` if the execution policy is of type *task_execution_policy* and returns *FwdIter* otherwise. The *search_n* algorithm returns an iterator to the beginning of the last subsequence [s_first, s_last) in range [first, first+count). If the length of the subsequence [s_first, s_last) is greater than the length of the range [first, first+count),

first is returned. Additionally if the size of the subsequence is empty or no subsequence is found, *first* is also returned.

hpx::adjacent_difference

Defined in header `hpx/algorithm.hpp`⁵⁸⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename FwdIter1, typename FwdIter2>
FwdIter2 hpx::adjacent_difference(FwdIter1 first, FwdIter1 last, FwdIter2 dest)
```

Assigns each value in the range given by result its corresponding element in the range [first, last] and the one preceding it except *result, which is assigned *first.

Note: Complexity: Exactly (last - first) - 1 application of the binary operator and (last - first) assignments.

Template Parameters

- **FwdIter1** – The type of the source iterators used for the input range (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used for the output range (deduced). This iterator type must meet the requirements of a forward iterator.

Parameters

- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **dest** – Refers to the beginning of the sequence of elements the results will be assigned to.

Returns

The *adjacent_difference* algorithm returns a *FwdIter2*. The *adjacent_difference* algorithm returns an iterator to the element past the last element written.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx::adjacent_difference(ExPolicy
                                                                                               &&policy,
                                                                                               FwdIter1 first,
                                                                                               FwdIter1 last,
                                                                                               FwdIter2 dest)
```

Assigns each value in the range given by result its corresponding element in the range [first, last] and the one preceding it except *result, which is assigned *first. Executed according to the policy.

The difference operations in the parallel *adjacent_difference* invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The difference operations in the parallel *adjacent_difference* invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

⁵⁸⁷ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e77/libs/full/include/include/hpx/algorithm.hpp>

Note: Complexity: Exactly (last - first) - 1 application of the binary operator and (last - first) assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the input range (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used for the output range (deduced). This iterator type must meet the requirements of a forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **dest** – Refers to the beginning of the sequence of elements the results will be assigned to.

Returns

The *adjacent_difference* algorithm returns a *hpx::future<FwdIter2>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter2* otherwise. The *adjacent_difference* algorithm returns an iterator to the element past the last element written.

```
template<typename FwdIter1, typename FwdIter2, typename Op>
FwdIter2 hpx::adjacent_difference(FwdIter1 first, FwdIter1 last, FwdIter2 dest, Op &&op)
```

Assigns each value in the range given by result its corresponding element in the range [first, last] and the one preceding it except *result, which is assigned *first

Note: Complexity: Exactly (last - first) - 1 application of the binary operator and (last - first) assignments.

Template Parameters

- **FwdIter1** – The type of the source iterators used for the input range (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used for the output range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Op** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *adjacent_difference* requires *Op* to meet the requirements of *Copy-Constructible*.

Parameters

- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.

- **dest** – Refers to the beginning of the sequence of elements the results will be assigned to.
- **op** – The binary operator which returns the difference of elements. The signature should be equivalent to the following:

```
bool op(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* must be such that objects of type *FwdIter1* can be dereferenced and then implicitly converted to the dereferenced type of *dest*.

Returns

The *adjacent_difference* algorithm returns *FwdIter2*. The *adjacent_difference* algorithm returns an iterator to the element past the last element written.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Op>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx::adjacent_difference(ExPolicy
                                                                                               &&policy,
                                                                                               FwdIter1 first,
                                                                                               FwdIter1 last,
                                                                                               FwdIter2 dest,
                                                                                               Op &&op)
```

Assigns each value in the range given by result its corresponding element in the range [first, last] and the one preceding it except *result, which is assigned *first

The difference operations in the parallel *adjacent_difference* invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The difference operations in the parallel *adjacent_difference* invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Exactly (last - first) - 1 application of the binary operator and (last - first) assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the input range (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used for the output range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Op** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *adjacent_difference* requires *Op* to meet the requirements of *CopyConstructible*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.

- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **dest** – Refers to the beginning of the sequence of elements the results will be assigned to.
- **op** – The binary operator which returns the difference of elements. The signature should be equivalent to the following:

```
bool op(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* must be such that objects of type *FwdIter1* can be dereferenced and then implicitly converted to the dereferenced type of *dest*.

Returns

The *adjacent_difference* algorithm returns a *hpx::future<FwdIter2>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter2* otherwise. The *adjacent_difference* algorithm returns an iterator to the element past the last element written.

hpx::uninitialized_relocate

Defined in header `hpx/algorithm.hpp`⁵⁸⁸.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::uninitialized_relocate_n*

```
template<typename InIter1, typename InIter2, typename FwdIter>
FwdIter hpx::uninitialized_relocate(InIter1 first, InIter2 last, FwdIter dest)
```

Relocates the elements in the range, defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the move-construction of an element, all elements left in the input range, as well as all objects already constructed in the destination range are destroyed. After this algorithm completes, the source range should be freed or reused without destroying the objects.

The assignments in the parallel *uninitialized_relocate* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

Note: Complexity: time: $O(n)$, space: $O(1)$ 1) For “trivially relocatable” underlying types (T) and a contiguous iterator range [first, last): $\text{std::distance}(\text{first}, \text{last}) * \text{sizeof}(T)$ bytes are copied. 2) For “trivially relocatable” underlying types (T) and a non-contiguous iterator range [first, last): $\text{std::distance}(\text{first}, \text{last})$ memory copies of $\text{sizeof}(T)$ bytes each are performed. 3) For “non-trivially relocatable” underlying types (T): $\text{std::distance}(\text{first}, \text{last})$ move assignments and destructions are performed.

Note: Declare a type as “trivially relocatable” using the `HPX_DECLARE_TRIVIALY_RELOCATABLE` macros found in `<hpx/type_support/is_trivially_relocatable.hpp>`.

Template Parameters

- **InIter1** – The type of the source iterator first (deduced). This iterator type must meet the requirements of an input iterator.

⁵⁸⁸ <http://github.com/TheHPXProject/hpx/blob/d402eed7405eedff56d454c9380fbd0c8044e77/libs/full/include/include/hpx/algorithm.hpp>

- **InIter2** – The type of the source iterator last (deduced). This iterator type must meet the requirements of an input iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *uninitialized_relocate* algorithm returns *FwdIter*. The *uninitialized_relocate* algorithm returns the output iterator to the element in the destination range, one past the last element relocated.

```
template<typename ExPolicy, typename InIter1, typename InIter2, typename FwdIter>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::uninitialized_relocate(ExPolicy  
                                                                                               &&policy,  
                                                                                               InIter1 first,  
                                                                                               InIter2 last,  
                                                                                               FwdIter dest)
```

Relocates the elements in the range defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the move-construction of an element, all elements left in the input range, as well as all objects already constructed in the destination range are destroyed. After this algorithm completes, the source range should be freed or reused without destroying the objects.

The assignments in the parallel *uninitialized_relocate* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: time: $O(n)$, space: $O(1)$ 1) For “trivially relocatable” underlying types (T) and a contiguous iterator range [first, last): $\text{std::distance}(\text{first}, \text{last}) * \text{sizeof}(T)$ bytes are copied. 2) For “trivially relocatable” underlying types (T) and a non-contiguous iterator range [first, last): $\text{std::distance}(\text{first}, \text{last})$ memory copies of $\text{sizeof}(T)$ bytes each are performed. 3) For “non-trivially relocatable” underlying types (T): $\text{std::distance}(\text{first}, \text{last})$ move assignments and destructions are performed.

Note: Declare a type as “trivially relocatable” using the `HPX_DECLARE_TRIVIALY_RELOCATABLE` macros found in `<hpx/type_support/is_trivially_relocatable.hpp>`.

Note: Parallel execution requires both *first* and *dest* to be contiguous iterators. If either iterator is non-contiguous, the algorithm cannot determine whether the source and destination ranges overlap, and will silently fall back to sequential execution regardless of the execution policy provided.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.

- **InIter1** – The type of the source iterator first (deduced). This iterator type must meet the requirements of an input iterator.
- **InIter2** – The type of the source iterator last (deduced). This iterator type must meet the requirements of an input iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range. The assignments in the parallel *uninitialized_relocate_n* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

Returns

The *uninitialized_relocate* algorithm returns a *hpx::future<FwdIter>*, if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *uninitialized_relocate* algorithm returns the output iterator to the element in the destination range, one past the last element relocated.

hpx::uninitialized_relocate_n

Defined in header [hpx/algorithm.hpp](#)⁵⁸⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::uninitialized_relocate*

```
template<typename InIter, typename Size, typename FwdIter>
FwdIter hpx::uninitialized_relocate_n(InIter first, Size count, FwdIter dest)
```

Relocates the elements in the range, defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the move-construction of an element, all elements left in the input range, as well as all objects already constructed in the destination range are destroyed. After this algorithm completes, the source range should be freed or reused without destroying the objects.

The assignments in the parallel *uninitialized_relocate_n* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

Note: Complexity: time: $O(n)$, space: $O(1)$ 1) For “trivially relocatable” underlying types (T) and a contiguous iterator range [first, first+count): $\text{count} \times \text{sizeof}(T)$ bytes are copied. 2) For “trivially relocatable” underlying types (T) and a non-contiguous iterator range [first, first+count): count memory copies of $\text{sizeof}(T)$ bytes each are performed. 3) For “non-trivially relocatable” underlying types (T): count move assignments and destructions are performed.

⁵⁸⁹ <http://github.com/TheHPXProject/hpx/blob/d402cead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Note: Declare a type as “trivially relocatable” using the `HPX_DECLARE_TRIVIALLY_RELOCATABLE` macros found in `<hpx/type_support/is_trivially_relocatable.hpp>`.

Template Parameters

- **InIter** – The type of the source iterator first (deduced). This iterator type must meet the requirements of an input iterator.
- **Size** – The type of the argument specifying the number of elements to relocate.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *uninitialized_relocate_n* algorithm returns *FwdIter*. The *uninitialized_relocate_n* algorithm returns the output iterator to the element in the destination range, one past the last element relocated.

```
template<typename ExPolicy, typename InIter, typename Size, typename FwdIter>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::uninitialized_relocate_n(ExPolicy
                                                                                               &&policy,
                                                                                               InIter first,
                                                                                               Size count,
                                                                                               FwdIter
                                                                                               dest)
```

Relocates the elements in the range, defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the move-construction of an element, all elements left in the input range, as well as all objects already constructed in the destination range are destroyed. After this algorithm completes, the source range should be freed or reused without destroying the objects.

The assignments in the parallel *uninitialized_relocate_n* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized_relocate_n* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: time: $O(n)$, space: $O(1)$ 1) For “trivially relocatable” underlying types (T) and a contiguous iterator range [first, first+count): `count*sizeof(T)` bytes are copied. 2) For “trivially relocatable” underlying types (T) and a non-contiguous iterator range [first, first+count): `count` memory copies of `sizeof(T)` bytes each are performed. 3) For “non-trivially relocatable” underlying types (T): `count` move assignments and destructions are performed.

Note: Declare a type as “trivially relocatable” using the `HPX_DECLARE_TRIVIALY_RELOCATABLE` macros found in `<hpx/type_support/is_trivially_relocatable.hpp>`.

Note: Parallel execution requires both *first* and *dest* to be contiguous iterators. If either iterator is non-contiguous, the algorithm cannot determine whether the source and destination ranges overlap, and will silently fall back to sequential execution regardless of the execution policy provided.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **InIter** – The type of the source iterator first (deduced). This iterator type must meet the requirements of an input iterator.
- **Size** – The type of the argument specifying the number of elements to relocate.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *uninitialized_relocate_n* algorithm returns a `hpx::future<FwdIter>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *uninitialized_relocate_n* algorithm returns the output iterator to the element in the destination range, one past the last element relocated.

hpx::reduce_deterministic

Defined in header `hpx/algorithm.hpp`⁵⁹⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename ExPolicy, typename FwdIter, typename F, typename T = typename
std::iterator_traits<FwdIter>::value_type>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> hpx::reduce_deterministic(ExPolicy &&policy,
                                                                                      FwdIter first, FwdIter
                                                                                      last, T init, F &&f)
```

Returns GENERALIZED_SUM(*f*, *init*, **first*, ..., *(*first* + (*last* - *first*) - 1)). Executed according to the policy.

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

⁵⁹⁰ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

The reduce operations in the parallel *copy_if* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate f .

Note: GENERALIZED_SUM(op , a1 , ..., aN) is defined as follows:

- a1 when N is 1
 - $\text{op}(\text{GENERALIZED_SUM}(\text{op}, \text{b1}, \dots, \text{bK}), \text{GENERALIZED_SUM}(\text{op}, \text{bM}, \dots, \text{bN}))$, where:
 - $\text{b1}, \dots, \text{bN}$ may be any permutation of $\text{a1}, \dots, \text{aN}$ and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source begin and end iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *reduce* requires F to meet the requirements of *CopyConstructible*.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`. The types *Type1* *Ret* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to any of those types.

Returns

The *reduce* algorithm returns a `hpx::future<T>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns T otherwise. The *reduce* algorithm returns the result of the generalized sum over the elements given by the input range [first, last).

```
template<typename ExPolicy, typename FwdIter, typename T = typename
std::iterator_traits<FwdIter>::value_type>
util::detail::algorithm_result_t<ExPolicy, T> hpx::reduce_deterministic(ExPolicy &&policy, FwdIter first,
                                                                    FwdIter last, T init)
```

Returns GENERALIZED_SUM(+, init, *first, ..., *(first + (last - first) - 1)). Executed according to the policy.

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *copy_if* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the operator+().

Note: GENERALIZED_SUM(+, a1, ..., aN) is defined as follows:

- a1 when N is 1
 - op(GENERALIZED_SUM(+, b1, ..., bK), GENERALIZED_SUM(+, bM, ..., bN)), where:
 - b1, ..., bN may be any permutation of a1, ..., aN and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source begin and end iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.

Returns

The *reduce* algorithm returns a *hpx::future<T>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *T* otherwise. The *reduce* algorithm returns the result of the generalized sum (applying operator+()) over the elements given by the input range [first, last).

```
template<typename ExPolicy, typename FwdIter>
hpx::parallel::util::detail::algorithm_result<ExPolicy, typename std::iterator_traits<FwdIter>::value_type>::type hpx::reduce_deter
```

Returns GENERALIZED_SUM(+, T(), *first, ..., *(first + (last - first) - 1)). Executed according to the policy.

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the operator+().

Note: The type of the initial value (and the result type) *T* is determined from the *value_type* of the used *FwdIter*.

Note: GENERALIZED_SUM(+, a1, ..., aN) is defined as follows:

- a1 when N is 1
 - op(GENERALIZED_SUM(+, b1, ..., bK), GENERALIZED_SUM(+, bM, ..., bN)), where:
 - b1, ..., bN may be any permutation of a1, ..., aN and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source begin and end iterators used (deduced). This iterator type must meet the requirements of a forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

Returns

The *reduce* algorithm returns a `hpx::future<T>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `T` otherwise (where `T` is the `value_type` of *FwdIter*). The *reduce* algorithm returns the result of the generalized sum (applying operator`+`()) over the elements given by the input range `[first, last)`.

```
template<typename FwdIter, typename F, typename T = typename std::iterator_traits<FwdIter>::value_type>
T hpx::reduce_deterministic(FwdIter first, FwdIter last, T init, F &&f)
```

Returns `GENERALIZED_SUM(f, init, *first, ..., *(first + (last - first) - 1))`. Executed according to the policy.

The difference between *reduce* and *accumulate* is that the behavior of *reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate *f*.

Note: `GENERALIZED_SUM(op, a1, ..., aN)` is defined as follows:

- `a1` when `N` is 1
 - `op(GENERALIZED_SUM(op, b1, ..., bK), GENERALIZED_SUM(op, bM, ..., bN))`, where:
 - `b1, ..., bN` may be any permutation of `a1, ..., aN` and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

- **FwdIter** – The type of the source begin and end iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *reduce* requires *F* to meet the requirements of *CopyConstructible*.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by `[first, last)`. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`. The types *Type1 Ret* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to any of those types.

Returns

The *reduce* algorithm returns *T*. The *reduce* algorithm returns the result of the generalized sum over the elements given by the input range `[first, last)`.

```
template<typename FwdIter, typename T = typename std::iterator_traits<FwdIter>::value_type>
T hpx::reduce_deterministic(FwdIter first, FwdIter last, T init)
```

Returns GENERALIZED_SUM(+, init, *first, ..., *(first + (last - first) - 1)). Executed according to the policy.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the operator+().

Note: GENERALIZED_SUM(+, a1, ..., aN) is defined as follows:

- a1 when N is 1
 - op(GENERALIZED_SUM(+, b1, ..., bK), GENERALIZED_SUM(+, bM, ..., bN)), where:
 - b1, ..., bN may be any permutation of a1, ..., aN and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

- **FwdIter** – The type of the source begin and end iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.

Returns

The *reduce* algorithm returns a *T*. The *reduce* algorithm returns the result of the generalized sum (applying operator+()) over the elements given by the input range [first, last).

```
template<typename FwdIter>
std::iterator_traits<FwdIter>::value_type hpx::reduce_deterministic(FwdIter first, FwdIter last)
```

Returns GENERALIZED_SUM(+, T(), *first, ..., *(first + (last - first) - 1)). Executed according to the policy.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the operator+().

Note: The type of the initial value (and the result type) *T* is determined from the value_type of the used *FwdIter*.

Note: GENERALIZED_SUM(+, a1, ..., aN) is defined as follows:

- a_1 when N is 1
- $op(\text{GENERALIZED_SUM}(+, b_1, \dots, b_K), \text{GENERALIZED_SUM}(+, b_M, \dots, b_N))$, where:
 - $b_1, \dots, b_N$ may be any permutation of $a_1, \dots, a_N$ and
 - $1 < K+1 = M \leq N$.

Template Parameters

FwdIter – The type of the source begin and end iterators used (deduced). This iterator type must meet the requirements of an input iterator.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

Returns

The *reduce* algorithm returns T (where T is the `value_type` of *FwdIter*). The *reduce* algorithm returns the result of the generalized sum (applying `operator+()`) over the elements given by the input range `[first, last)`.

hpx::set_union

Defined in header `hpx/algorithm.hpp`⁵⁹¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename FwdIter3, typename Pred =
hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter3> hpx::set_union(ExPolicy &&policy, FwdIter1
first1, FwdIter1 last1,
FwdIter2 first2, FwdIter2
last2, FwdIter3 dest, Pred
&&op = Pred())
```

Constructs a sorted range beginning at `dest` consisting of all elements present in one or both sorted ranges `[first1, last1)` and `[first2, last2)`. This algorithm expects both input ranges to be sorted with the given binary predicate *pred*. Executed according to the policy.

If some element is found m times in `[first1, last1)` and n times in `[first2, last2)`, then all m elements will be copied from `[first1, last1)` to `dest`, preserving order, and then exactly $\max(n-m, 0)$ elements will be copied from `[first2, last2)` to `dest`, also preserving order.

The resulting range cannot overlap with either of the input ranges.

The application of function objects in parallel algorithm invoked with a sequential execution policy object execute in sequential order in the calling thread (*sequenced_policy*) or in a single new thread spawned from the current thread (for *sequenced_task_policy*).

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

⁵⁹¹ <http://github.com/TheHPXProject/hpx/blob/d402eed7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Note: Complexity: At most $2 \cdot (N1 + N2 - 1)$ comparisons, where $N1$ is the length of the first sequence and $N2$ is the length of the second sequence.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **FwdIter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator or output iterator and sequential execution.
- **Op** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set_union* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*

Returns

The *set_union* algorithm returns a `hpx::future<FwdIter3>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter3* otherwise. The *set_union* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename FwdIter1, typename FwdIter2, typename FwdIter3, typename Pred =  
hpx::parallel::detail::less>
```

```
FwdIter3 hpx::set_union(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, FwdIter2 last2, FwdIter3 dest, Pred
                        &&op = Pred())
```

Constructs a sorted range beginning at *dest* consisting of all elements present in one or both sorted ranges [*first1*, *last1*) and [*first2*, *last2*). This algorithm expects both input ranges to be sorted with the given binary predicate *pred*. Executed according to the policy.

If some element is found *m* times in [*first1*, *last1*) and *n* times in [*first2*, *last2*), then all *m* elements will be copied from [*first1*, *last1*) to *dest*, preserving order, and then exactly `std::max(n-m, 0)` elements will be copied from [*first2*, *last2*) to *dest*, also preserving order.

The resulting range cannot overlap with either of the input ranges.

Note: Complexity: At most $2 \cdot (N_1 + N_2 - 1)$ comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

Template Parameters

- **FwdIter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator or output iterator and sequential execution.
- **Op** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set_union* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*

Returns

The *set_union* algorithm returns a *FwdIter3*. The *set_union* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

hpx::rotate

Defined in header [hpx/algorithm.hpp](#)⁵⁹².

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::rotate_copy*

template<typename **FwdIter**>

FwdIter **hpx::rotate**(*FwdIter* first, *FwdIter* new_first, *FwdIter* last)

Performs a left rotation on a range of elements. Specifically, *rotate* swaps the elements in the range [first, last) in such a way that the element new_first becomes the first element of the new range and new_first - 1 becomes the last element.

The assignments in the parallel *rotate* algorithm execute in sequential order in the calling thread.

Note: Complexity: Linear in the distance between *first* and *last*.

Note: The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable* and *MoveConstructible*.

Template Parameters

FwdIter – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **new_first** – Refers to the element that should appear at the beginning of the rotated range.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

Returns

The *rotate* algorithm returns a *FwdIter*. The *rotate* algorithm returns the iterator to the new location of the element pointed by first, equal to first + (last - new_first).

template<typename **ExPolicy**, typename **FwdIter**>

hpx::parallel::util::detail::algorithm_result_t<*ExPolicy*, *FwdIter*> **hpx::rotate**(*ExPolicy* &&policy, *FwdIter* first, *FwdIter* new_first, *FwdIter* last)

Performs a left rotation on a range of elements. Specifically, *rotate* swaps the elements in the range [first, last) in such a way that the element new_first becomes the first element of the new range and new_first - 1 becomes the last element. Executed according to the policy.

⁵⁹² <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

The assignments in the parallel *rotate* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *rotate* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Linear in the distance between *first* and *last*.

Note: The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable* and *MoveConstructible*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **new_first** – Refers to the element that should appear at the beginning of the rotated range.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

Returns

The *rotate* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *rotate* algorithm returns the iterator equal to *first* + (*last* - *new_first*).

hpx::rotate_copy

Defined in header [hpx/algorithm.hpp](#)⁵⁹³.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::rotate*

```
template<typename FwdIter, typename OutIter>
```

```
OutIter hpx::rotate_copy(FwdIter first, FwdIter new_first, FwdIter last, OutIter dest_first)
```

Copies the elements from the range [*first*, *last*), to another range beginning at *dest_first* in such a way, that the element *new_first* becomes the first element of the new range and *new_first* - 1 becomes the last element.

The assignments in the parallel *rotate_copy* algorithm execute in sequential order in the calling thread.

⁵⁹³ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **OutIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a output iterator.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **new_first** – Refers to the element that should appear at the beginning of the rotated range.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest_first** – Refers to the begin of the destination range.

Returns

The *rotate_copy* algorithm returns a output iterator, The *rotate_copy* algorithm returns the output iterator to the element past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx::rotate_copy(ExPolicy &&policy,
                                                                                      FwdIter1 first, FwdIter1
                                                                                      new_first, FwdIter1 last,
                                                                                      FwdIter2 dest_first)
```

Copies the elements from the range [first, last), to another range beginning at *dest_first* in such a way, that the element *new_first* becomes the first element of the new range and *new_first* - 1 becomes the last element. Executed according to the policy.

The assignments in the parallel *rotate_copy* algorithm execute in sequential order in the calling thread.

The assignments in the parallel *rotate_copy* algorithm execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **new_first** – Refers to the element that should appear at the beginning of the rotated range.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest_first** – Refers to the begin of the destination range.

Returns

The *rotate_copy* algorithm returns a *hpx::future<FwdIter2>* if the execution policy is of type *parallel_task_policy* and returns *FwdIter2* otherwise. The *rotate_copy* algorithm returns the output iterator to the element past the last element copied.

hpx::experimental::reduction

Defined in header [hpx/algorithm.hpp](#)⁵⁹⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename T, typename Op>
constexpr hpx::parallel::detail::reduction_helper<T, std::decay_t<Op>> hpx::experimental::reduction(T
                                                                    &var,
                                                                    T
                                                                    const
                                                                    &iden-
                                                                    tity,
                                                                    Op
                                                                    &&com-
                                                                    biner)
```

The function template returns a reduction object of unspecified type having a value type and encapsulating an identity value for the reduction, a combiner function object, and a live-out object from which the initial value is obtained and into which the final value is stored.

A parallel algorithm uses reduction objects by allocating an unspecified number of instances, called views, of the reduction's value type. Each view is initialized with the reduction object's identity value, except that the live-out object (which was initialized by the caller) comprises one of the views. The algorithm passes a reference to a view to each application of an element-access function, ensuring that no two concurrently-executing invocations share the same view. A view can be shared between two applications that do not execute concurrently, but initialization is performed only once per view.

Modifications to the view by the application of element access functions accumulate as partial results. At some point before the algorithm returns, the partial results are combined, two at a time, using the reduction object's combiner operation until a single value remains, which is then assigned back to the live-out object.

T shall meet the requirements of *CopyConstructible* and *MoveAssignable*. The expression

```
var = combiner(var, var)
```

shall be well-formed.

Note: In order to produce useful results, modifications to the view should be limited to commutative operations closely related to the combiner operation. For example if the combiner is `plus<T>`, incrementing the view would

⁵⁹⁴ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

be consistent with the combiner but doubling it or assigning to it would not.

Template Parameters

- **T** – The value type to be used by the induction object.
- **Op** – The type of the binary function (object) used to perform the reduction operation.

Parameters

- **var** – [in,out] The live-out value to use for the reduction object. This will hold the reduced value after the algorithm is finished executing.
- **identity** – [in] The identity value to use for the reduction operation. This argument is optional and defaults to a copy of *var*.
- **combiner** – [in] The binary function (object) used to perform a pairwise reduction on the elements.

Returns

This returns a reduction object of unspecified type having a value type of *T*. When the return value is used by an algorithm, the reference to *var* is used as the live-out object, new views are initialized to a copy of identity, and views are combined by invoking the copy of combiner, passing it the two views to be combined.

```
template<typename T, typename Op>  
constexpr hpx::parallel::detail::reduction_helper<T, std::decay_t<Op>> hpx::experimental::reduction(T  
                                                                 &var,  
                                                                 Op  
                                                                 &&combiner)
```

hpx::for_each

Defined in header [hpx/algorithm.hpp](#)⁵⁹⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::for_each_n*

```
template<typename InIter, typename F>  
F hpx::for_each(InIter first, InIter last, F &&f)
```

Applies *f* to the result of dereferencing every iterator in the range [first, last).

If *f* returns a result, the result is ignored.

If the type of *first* satisfies the requirements of a mutable iterator, *f* may apply non-constant functions through the dereferenced iterator.

Note: Complexity: Applies *f* exactly *last - first* times.

⁵⁹⁵ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Template Parameters

- **InIter** – The type of the source begin and end iterator used (deduced). This iterator type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). F must meet requirements of *MoveConstructible*.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
<ignored> pred(const Type &a);
```

The signature does not need to have const&. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

Returns

f.

```
template<typename ExPolicy, typename FwdIter, typename F>
util::detail::algorithm_result_t<ExPolicy, void> hpx::for_each(ExPolicy &&policy, FwdIter first, FwdIter last, F
&&f)
```

Applies *f* to the result of dereferencing every iterator in the range [first, last). Executed according to the policy.

If *f* returns a result, the result is ignored.

If the type of *first* satisfies the requirements of a mutable iterator, *f* may apply non-constant functions through the dereferenced iterator.

Unlike its sequential form, the parallel overload of *for_each* does not return a copy of its *Function* parameter, since parallelization may not permit efficient state accumulation.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Applies *f* exactly *last - first* times.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.

- **FwdIter** – The type of the source begin and end iterator used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *for_each* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

`<ignored> pred(const Type &a);`

The signature does not need to have `const&`. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

Returns

The *for_each* algorithm returns a *hpx::future<void>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `void` otherwise.

hpx::for_each_n

Defined in header [hpx/algorithm.hpp](#)⁵⁹⁶.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::for_each*

template<typename **InIter**, typename **Size**, typename **F**>
InIter **hpx::for_each_n**(*InIter* first, *Size* count, *F* &&*f*)

Applies *f* to the result of dereferencing every iterator in the range [first, first + count), starting from first and proceeding to first + count - 1.

If *f* returns a result, the result is ignored.

If the type of *first* satisfies the requirements of a mutable iterator, *f* may apply non-constant functions through the dereferenced iterator.

Note: Complexity: Applies *f* exactly *count* times.

Template Parameters

- **InIter** – The type of the source begin and end iterator used (deduced). This iterator type must meet the requirements of an input iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.

⁵⁹⁶ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **F** – The type of the function/function object to use (deduced). F must meet requirements of *MoveConstructible*.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
<ignored> pred(const Type &a);
```

The signature does not need to have const&. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

Returns

first + *count* for non-negative values of *count* and *first* for negative values.

```
template<typename ExPolicy, typename FwdIter, typename Size, typename F>
util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::for_each_n(ExPolicy &&policy, FwdIter first, Size
                                                                    count, F &&f)
```

Applies *f* to the result of dereferencing every iterator in the range [first, first + count), starting from first and proceeding to first + count - 1. Executed according to the policy.

If *f* returns a result, the result is ignored.

If the type of *first* satisfies the requirements of a mutable iterator, *f* may apply non-constant functions through the dereferenced iterator.

Unlike its sequential form, the parallel overload of *for_each_n* does not return a copy of its *Function* parameter, since parallelization may not permit efficient state accumulation.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Applies *f* exactly *count* times.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.

- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *for_each_n* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
<ignored> pred(const Type &a);
```

The signature does not need to have `const&`. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

Returns

The *for_each_n* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. It returns *first* + *count* for non-negative values of *count* and *first* for negative values.

hpx::set_symmetric_difference

Defined in header `hpx/algorithm.hpp`⁵⁹⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename FwdIter3, typename Pred =
hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result<ExPolicy, FwdIter3>::type hpx::set_symmetric_difference(ExPolicy
&&policy,
FwdIter1
first1,
FwdIter1
last1,
FwdIter2
first2,
FwdIter2
last2,
FwdIter3
dest,
Pred
&&op
=
Pred())
```

Constructs a sorted range beginning at *dest* consisting of all elements present in either of the sorted ranges [first1, last1) and [first2, last2), but not in both of them are copied to the range beginning at *dest*. The resulting range is also sorted. This algorithm expects both input ranges to be sorted with the given binary predicate *pred*. Executed according to the policy.

⁵⁹⁷ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

If some element is found m times in $[first1, last1)$ and n times in $[first2, last2)$, it will be copied to *dest* exactly $std::abs(m-n)$ times. If $m > n$, then the last $m-n$ of those elements are copied from $[first1, last1)$, otherwise the last $n-m$ elements are copied from $[first2, last2)$. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

The application of function objects in parallel algorithm invoked with a sequential execution policy object execute in sequential order in the calling thread (*sequenced_policy*) or in a single new thread spawned from the current thread (for *sequenced_task_policy*).

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most $2 \cdot (N1 + N2 - 1)$ comparisons, where $N1$ is the length of the first sequence and $N2$ is the length of the second sequence.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **FwdIter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator or output iterator and sequential execution.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set_symmetric_difference* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

Returns

The *set_symmetric_difference* algorithm returns a *hpx::future<FwdIter3>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter3* otherwise. The *set_symmetric_difference* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename FwdIter1, typename FwdIter2, typename FwdIter3, typename Pred =  
hpx::parallel::detail::less>  
FwdIter3 hpx::set_symmetric_difference(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, FwdIter2 last2,  
FwdIter3 dest, Pred &&op = Pred())
```

Constructs a sorted range beginning at *dest* consisting of all elements present in either of the sorted ranges [*first1*, *last1*) and [*first2*, *last2*), but not in both of them are copied to the range beginning at *dest*. The resulting range is also sorted. This algorithm expects both input ranges to be sorted with the given binary predicate *pred*.

If some element is found *m* times in [*first1*, *last1*) and *n* times in [*first2*, *last2*), it will be copied to *dest* exactly `std::abs(m-n)` times. If *m*>*n*, then the last *m-n* of those elements are copied from [*first1*,*last1*), otherwise the last *n-m* elements are copied from [*first2*,*last2*). The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

Note: Complexity: At most $2 \cdot (N_1 + N_2 - 1)$ comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

Template Parameters

- **FwdIter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator or output iterator and sequential execution.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set_symmetric_difference* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.

- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

Returns

The *set_symmetric_difference* algorithm returns a *FwdIter3*. The *set_symmetric_difference* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

hpx::mismatch

Defined in header `hpx/algorithm.hpp`⁵⁹⁸.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Pred>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, std::pair<FwdIter1, FwdIter2>> hpx::mismatch(ExPolicy
                                                                                               &&pol-
                                                                                               icy,
                                                                                               FwdIter1
                                                                                               first1,
                                                                                               FwdIter1
                                                                                               last1,
                                                                                               FwdIter2
                                                                                               first2,
                                                                                               FwdIter2
                                                                                               last2,
                                                                                               Pred
                                                                                               &&op)
```

Returns the first mismatching pair of elements from two ranges: one defined by `[first1, last1)` and another defined by `[first2, last2)`. If `last2` is not provided, it denotes `first2 + (last1 - first1)`. Executed according to the policy.

The comparison operations in the parallel *mismatch* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *mismatch* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most $\min(\text{last1} - \text{first1}, \text{last2} - \text{first2})$ applications of the predicate *op* or *operator==*. If *FwdIter1* and *FwdIter2* meet the requirements of *RandomAccessIterator* and $(\text{last1} - \text{first1}) \neq (\text{last2} - \text{first2})$ then no applications of the predicate *op* or *operator==* are made.

⁵⁹⁸ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Note: The two ranges are considered mismatch if, for every iterator *i* in the range `[first1,last1)`, **i* mismatches `*(first2 + (i - first1))`. This overload of `mismatch` uses `operator==` to determine if two elements are mismatch.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of `mismatch` requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as mismatch. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

Returns

The `mismatch` algorithm returns a `hpx::future<std::pair<FwdIter1,FwdIter2>>` if the execution policy is of type `sequenced_task_policy` or `parallel_task_policy` and returns `std::pair<FwdIter1,FwdIter2>` otherwise. If no mismatches are found when the comparison reaches `last1` or `last2`, whichever happens first, the pair holds the end iterator and the corresponding iterator from the other range.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2>
```

```

hpx::parallel::util::detail::algorithm_result_t<ExPolicy, std::pair<FwdIter1, FwdIter2>> hpx::mismatch(ExPolicy
                                                                                               &&pol-
                                                                                               icy,
                                                                                               FwdIter1
                                                                                               first1,
                                                                                               FwdIter1
                                                                                               last1,
                                                                                               FwdIter2
                                                                                               first2,
                                                                                               FwdIter2
                                                                                               last2)

```

Returns the first mismatching pair of elements from two ranges: one defined by [first1, last1) and another defined by [first2, last2). If last2 is not provided, it denotes first2 + (last1 - first1). Executed according to the policy.

The comparison operations in the parallel *mismatch* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *mismatch* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most $\min(\text{last1} - \text{first1}, \text{last2} - \text{first2})$ applications of *operator==*. If *FwdIter1* and *FwdIter2* meet the requirements of *RandomAccessIterator* and $(\text{last1} - \text{first1}) \neq (\text{last2} - \text{first2})$ then no applications of *operator==* are made.

Note: The two ranges are considered mismatch if, for every iterator *i* in the range [first1, last1), *i mismatches *(first2 + (i - first1)). This overload of *mismatch* uses *operator==* to determine if two elements are mismatch.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.

Returns

The *mismatch* algorithm returns a `hpx::future<std::pair<FwdIter1,FwdIter2>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `std::pair<FwdIter1,FwdIter2>` otherwise. If no mismatches are found when the comparison reaches `last1` or `last2`, whichever happens first, the pair holds the end iterator and the corresponding iterator from the other range.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Pred>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, std::pair<FwdIter1, FwdIter2>> hpx::mismatch(ExPolicy
                                                                                               &&pol-
                                                                                               icy,
                                                                                               FwdIter1
                                                                                               first1,
                                                                                               FwdIter1
                                                                                               last1,
                                                                                               FwdIter2
                                                                                               first2,
                                                                                               Pred
                                                                                               &&op)
```

Returns the first mismatching pair of elements from two ranges: one defined by `[first1, last1)` and another defined by `[first2, last2)`. If `last2` is not provided, it denotes `first2 + (last1 - first1)`. Executed according to the policy.

The comparison operations in the parallel *mismatch* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *mismatch* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most `last1 - first1` applications of the predicate *op* or *operator==*. If *FwdIter1* and *FwdIter2* meet the requirements of *RandomAccessIterator* and `(last1 - first1) != (last2 - first2)` then no applications of the predicate *op* or *operator==* are made.

Note: The two ranges are considered mismatch if, for every iterator *i* in the range `[first1, last1)`, **i* mismatches `*(first2 + (i - first1))`. This overload of *mismatch* uses *operator==* to determine if two elements are mismatch.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *mismatch* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as mismatch. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

Returns

The *mismatch* algorithm returns a `hpx::future<std::pair<FwdIter1,FwdIter2>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `std::pair<FwdIter1,FwdIter2>` otherwise. If no mismatches are found when the comparison reaches `last1` or `last2`, whichever happens first, the pair holds the end iterator and the corresponding iterator from the other range.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, std::pair<FwdIter1, FwdIter2>> hpx::mismatch(ExPolicy
                                                                                                     &&pol-
                                                                                                     icy,
                                                                                                     FwdIter1
                                                                                                     first1,
                                                                                                     FwdIter1
                                                                                                     last1,
                                                                                                     FwdIter2
                                                                                                     first2)
```

Returns the first mismatching pair of elements from two ranges: one defined by `[first1, last1)` and another defined by `[first2,last2)`. If `last2` is not provided, it denotes `first2 + (last1 - first1)`. Executed according to the policy.

The comparison operations in the parallel *mismatch* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *mismatch* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most `last1 - first1` applications of *operator==*. If *FwdIter1* and *FwdIter2* meet the requirements of *RandomAccessIterator* and `(last1 - first1) != (last2 - first2)` then no applications of *operator==* are made.

Note: The two ranges are considered mismatch if, for every iterator *i* in the range `[first1,last1)`, **i* mismatches **(first2 + (i - first1))*. This overload of *mismatch* uses *operator==* to determine if two elements are mismatch.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *mismatch* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.

Returns

The *mismatch* algorithm returns a `hpx::future<std::pair<FwdIter1,FwdIter2>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `std::pair<FwdIter1,FwdIter2>` otherwise. If no mismatches are found when the comparison reaches `last1` or `last2`, whichever happens first, the pair holds the end iterator and the corresponding iterator from the other range.

```
template<typename FwdIter1, typename FwdIter2, typename Pred>
std::pair<FwdIter1, FwdIter2> hpx::mismatch(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, FwdIter2 last2,
                                           Pred &&op)
```

Returns the first mismatching pair of elements from two ranges: one defined by `[first1, last1)` and another defined by `[first2, last2)`. If `last2` is not provided, it denotes `first2 + (last1 - first1)`.

Note: Complexity: At most $\min(\text{last1} - \text{first1}, \text{last2} - \text{first2})$ applications of the predicate *op* or *operator==*. If *FwdIter1* and *FwdIter2* meet the requirements of *RandomAccessIterator* and $(\text{last1} - \text{first1}) \neq (\text{last2} - \text{first2})$ then no applications of the predicate *op* or *operator==* are made.

Note: The two ranges are considered mismatch if, for every iterator *i* in the range `[first1, last1)`, **i* mismatches **(first2 + (i - first1))*. This overload of *mismatch* uses *operator==* to determine if two elements are mismatch.

Template Parameters

- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *mismatch* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as mismatch. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

Returns

The *mismatch* algorithm returns a `std::pair<FwdIter1,FwdIter2>`. If no mismatches are found when the comparison reaches `last1` or `last2`, whichever happens first, the pair holds the end iterator and the corresponding iterator from the other range.

```
template<typename FwdIter1, typename FwdIter2>
```

```
std::pair<FwdIter1, FwdIter2> hpx::mismatch(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, FwdIter2 last2)
```

Returns the first mismatching pair of elements from two ranges: one defined by `[first1, last1)` and another defined by `[first2,last2)`. If `last2` is not provided, it denotes `first2 + (last1 - first1)`.

Note: Complexity: At most `min(last1 - first1, last2 - first2)` applications of `operator==`. If *FwdIter1* and *FwdIter2* meet the requirements of *RandomAccessIterator* and `(last1 - first1) != (last2 - first2)` then no applications of `operator==` are made.

Note: The two ranges are considered mismatch if, for every iterator *i* in the range `[first1,last1)`, `*i` mismatches `*(first2 + (i - first1))`. This overload of *mismatch* uses `operator==` to determine if two elements are mismatch.

Template Parameters

- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.

- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.

Returns

The *mismatch* algorithm returns a `std::pair<FwdIter1,FwdIter2>`. If no mismatches are found when the comparison reaches `last1` or `last2`, whichever happens first, the pair holds the end iterator and the corresponding iterator from the other range.

```
template<typename FwdIter1, typename FwdIter2, typename Pred>  
std::pair<FwdIter1, FwdIter2> hpx::mismatch(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, Pred &&op)
```

Returns the first mismatching pair of elements from two ranges: one defined by `[first1, last1)` and another defined by `[first2,last2)`. If `last2` is not provided, it denotes `first2 + (last1 - first1)`.

Note: Complexity: At most `last1 - first1` applications of the predicate *op* or *operator==*. If *FwdIter1* and *FwdIter2* meet the requirements of *RandomAccessIterator* and `(last1 - first1) != (last2 - first2)` then no applications of the predicate *op* or *operator==* are made.

Note: The two ranges are considered mismatch if, for every iterator *i* in the range `[first1,last1)`, **i* mismatches `*(first2 + (i - first1))`. This overload of *mismatch* uses *operator==* to determine if two elements are mismatch.

Template Parameters

- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *mismatch* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as mismatch. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

Returns

The *mismatch* algorithm returns a `std::pair<FwdIter1,FwdIter2>`. If no mismatches are found when the comparison reaches `last1` or `last2`, whichever happens first, the pair holds the end iterator and the corresponding iterator from the other range.

```
template<typename FwdIter1, typename FwdIter2>
std::pair<FwdIter1, FwdIter2> hpx::mismatch(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2)
```

Returns the first mismatching pair of elements from two ranges: one defined by `[first1, last1)` and another defined by `[first2,last2)`. If `last2` is not provided, it denotes `first2 + (last1 - first1)`.

Note: Complexity: At most `last1 - first1` applications of `operator==`. If *FwdIter1* and *FwdIter2* meet the requirements of *RandomAccessIterator* and `(last1 - first1) != (last2 - first2)` then no applications of `operator==` are made.

Note: The two ranges are considered mismatch if, for every iterator *i* in the range `[first1,last1)`, `*i` mismatches `*(first2 + (i - first1))`. This overload of *mismatch* uses `operator==` to determine if two elements are mismatch.

Template Parameters

- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *mismatch* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.

Returns

The *mismatch* algorithm returns a `std::pair<FwdIter1,FwdIter2>`. If no mismatches are found when the comparison reaches `last1` or `last2`, whichever happens first, the pair holds the end iterator and the corresponding iterator from the other range.

hpx::experimental::reduction_plus

Defined in header `hpx/algorithm.hpp`⁵⁹⁹.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<typename T>
constexpr hpx::parallel::detail::reduction_helper<T, std::plus<T>> hpx::experimental::reduction_plus(T
                                                                    &var)
```

The function template *reduction_plus* returns a reduction object of unspecified type having a value type and encapsulating an identity value for the reduction, it uses `std::plus{ }` as its combiner function, and a live-out object from which the initial value is obtained and into which the final value is stored.

A parallel algorithm uses the *reduction_plus* object by allocating an unspecified number of instances, called views, of the reduction's value type. Each view is initialized with the reduction object's identity value, except that the live-out object (which was initialized by the caller) comprises one of the views. The algorithm passes a reference to a view to each application of an element-access function, ensuring that no two concurrently-executing invocations share the same view. A view can be shared between two applications that do not execute concurrently, but initialization is performed only once per view.

Modifications to the view by the application of element access functions accumulate as partial results. At some point before the algorithm returns, the partial results are combined, two at a time, using the `std::plus{ }` operation until a single value remains, which is then assigned back to the live-out object.

T shall meet the requirements of *CopyConstructible* and *MoveAssignable*.

Note: In order to produce useful results, modifications to the view should be limited to commutative operations closely related to the combiner operation.

Template Parameters

- **T** – The value type to be used by the induction object.
- **Op** – The type of the binary function (object) used to perform the reduction operation.

Parameters

- **var** – [in,out] The life-out value to use for the reduction object. This will hold the reduced value after the algorithm is finished executing.
- **identity** – [in] The identity value to use for the reduction operation. This argument is optional and defaults to a copy of *var*.

Returns

This returns a reduction object of unspecified type having a value type of *T*. When the return value is used by an algorithm, the reference to *var* is used as the live-out object, new views are initialized to a copy of identity, and views are combined by invoking the copy of combiner, passing it the two views to be combined.

```
template<typename T>
```

⁵⁹⁹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>


```
constexpr hpx::parallel::detail::reduction_helper<T, std::plus<T>> hpx::experimental::reduction_plus(T
                                                                    &var,
                                                                    T
                                                                    const
                                                                    &identity)
```

hpx::reduce_by_key

Defined in header [hpx/algorithm.hpp](#)⁶⁰⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

Warning: doxygenfunction: Cannot find function “hpx::reduce_by_key” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

hpx::includes

Defined in header [hpx/algorithm.hpp](#)⁶⁰¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Pred =
hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool>::type hpx::includes(ExPolicy &&policy, FwdIter1
                                                                    first1, FwdIter1 last1,
                                                                    FwdIter2 first2, FwdIter2
                                                                    last2, Pred &&op = Pred())
```

Returns true if every element from the sorted range $[first2, last2)$ is found within the sorted range $[first1, last1)$. Also returns true if $[first2, last2)$ is empty. The version expects both ranges to be sorted with the user supplied binary predicate f . Executed according to the policy.

The comparison operations in the parallel *includes* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *includes* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: At most $2 \cdot (N1 + N2 - 1)$ comparisons, where $N1 = \text{std::distance}(first1, last1)$ and $N2 = \text{std::distance}(first2, last2)$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.

⁶⁰⁰ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

⁶⁰¹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *includes* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as includes. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

Returns

The *includes* algorithm returns a `hpx::future<bool>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *includes* algorithm returns true every element from the sorted range `[first2, last2)` is found within the sorted range `[first1, last1)`. Also returns true if `[first2, last2)` is empty.

```
template<typename FwdIter1, typename FwdIter2, typename Pred = hpx::parallel::detail::less>  
bool hpx::includes(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, FwdIter2 last2, Pred &&op = Pred())
```

Returns true if every element from the sorted range `[first2, last2)` is found within the sorted range `[first1, last1)`. Also returns true if `[first2, last2)` is empty. The version expects both ranges to be sorted with the user supplied binary predicate *f*.

Note: At most $2*(N1+N2-1)$ comparisons, where $N1 = \text{std::distance}(\text{first1}, \text{last1})$ and $N2 = \text{std::distance}(\text{first2}, \text{last2})$.

Template Parameters

- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *includes* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as includes. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

Returns

The *includes* algorithm returns a *bool*. The *includes* algorithm returns true every element from the sorted range *[first2, last2)* is found within the sorted range *[first1, last1)*. Also returns true if *[first2, last2)* is empty.

hpx::adjacent_find

Defined in header `hpx/algorithm.hpp`⁶⁰².

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename InIter, typename Pred = hpx::parallel::detail::equal_to>
InIter hpx::adjacent_find(InIter first, InIter last, Pred &&pred = Pred())
```

Searches the range *[first, last)* for two consecutive identical elements.

Note: Complexity: Exactly the smaller of $(result - first) + 1$ and $(last - first) - 1$ application of the predicate where *result* is the value returned

Template Parameters

- **InIter** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an input iterator.
- **Pred** – The type of an optional function/function object to use.

Parameters

- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.

⁶⁰² <http://github.com/TheHPXProject/hpx/blob/d402cead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **pred** – The binary predicate which returns *true* if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* must be such that objects of type *Iter* can be dereferenced and then implicitly converted to *Type1*.

Returns

The *adjacent_find* algorithm returns an iterator to the first of the identical elements. If no such elements are found, *last* is returned.

```
template<typename ExPolicy, typename FwdIter, typename Pred = hpx::parallel::detail::equal_to>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::adjacent_find(ExPolicy &&policy,
                                                                                      FwdIter first, FwdIter
                                                                                      last, Pred &&pred =
                                                                                      Pred())
```

Searches the range `[first, last)` for two consecutive identical elements. This version uses the given binary predicate *pred*

The comparison operations in the parallel *adjacent_find* invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *adjacent_find* invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

This overload of *adjacent_find* is available if the user decides to provide their algorithm their own binary predicate *pred*.

Note: Complexity: Exactly the smaller of $(result - first) + 1$ and $(last - first) - 1$ application of the predicate where *result* is the value returned

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *adjacent_find* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **pred** – The binary predicate which returns *true* if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

Returns

The *adjacent_find* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *adjacent_find* algorithm returns an iterator to the first of the identical elements. If no such elements are found, *last* is returned.

hpx::transform_inclusive_scan

Defined in header `hpx/algorithm.hpp`⁶⁰³.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename InIter, typename OutIter, typename BinOp, typename UnOp>
OutIter hpx::transform_inclusive_scan(InIter first, InIter last, OutIter dest, BinOp &&binary_op, UnOp
&&unary_op)
```

Assigns through each iterator *i* in `[result, result + (last - first))` the value of `GENERALIZED_NONCOMMUTATIVE_SUM(op, conv(*first), ..., conv(*(first + (i - result))))`.

The reduce operations in the parallel *transform_inclusive_scan* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Neither *binary_op* nor *unary_op* shall invalidate iterators or sub-ranges, or modify elements in the ranges `[first,last)` or `[result,result + (last - first))`.

The difference between *inclusive_scan* and *transform_inclusive_scan* is that *transform_inclusive_scan* includes the *i*th input element in the *i*th sum.

Note: Complexity: $O(\text{last} - \text{first})$ applications of each of *binary_op* and *unary_op*.

Note: `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- `a1` when `N` is 1
 - `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where $1 < K+1 = M \leq N$.
-

⁶⁰³ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **BinOp** – The type of *binary_op*.
- **UnOp** – The type of *unary_op*.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **binary_op** – Binary *FunctionObject* that will be applied in to the result of *unary_op*, the results of other *binary_op*, and *init* if provided.
- **unary_op** – Unary *FunctionObject* that will be applied to each element of the input range. The return type must be acceptable as input to *binary_op*.

Returns

The *transform_inclusive_scan* algorithm returns a returns *OutIter*. The *transform_inclusive_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename BinOp, typename UnOp>
parallel::util::detail::algorithm_result<ExPolicy, FwdIter2>::type hpx::transform_inclusive_scan(ExPolicy
                                                                                               &&policy,
                                                                                               FwdIter1
                                                                                               first,
                                                                                               FwdIter1
                                                                                               last,
                                                                                               FwdIter2
                                                                                               dest, BinOp
                                                                                               &&bi-
                                                                                               nary_op,
                                                                                               UnOp
                                                                                               &&unary_op)
```

Assigns through each iterator *i* in [result, result + (last - first)) the value of GENERALIZED_NONCOMMUTATIVE_SUM(op, conv(*first), ..., conv(*(first + (i - result)))). Executed according to the policy.

The reduce operations in the parallel *transform_inclusive_scan* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform_inclusive_scan* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Neither *binary_op* nor *unary_op* shall invalidate iterators or sub-ranges, or modify elements in the ranges [first,last) or [result,result + (last - first)).

The difference between *inclusive_scan* and *transform_inclusive_scan* is that *transform_inclusive_scan* includes the *ith* input element in the *ith* sum.

Note: Complexity: $O(\text{last} - \text{first})$ applications of each of *binary_op* and *unary_op*.

Note: GENERALIZED_NONCOMMUTATIVE_SUM(*op*, *a1*, ..., *aN*) is defined as:

- *a1* when *N* is 1
 - *op*(GENERALIZED_NONCOMMUTATIVE_SUM(*op*, *a1*, ..., *aK*), GENERALIZED_NONCOMMUTATIVE_SUM(*op*, *aM*, ..., *aN*)) where $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **BinOp** – The type of *binary_op*.
- **UnOp** – The type of *unary_op*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **binary_op** – Binary *FunctionObject* that will be applied in to the result of *unary_op*, the results of other *binary_op*, and *init* if provided.
- **unary_op** – Unary *FunctionObject* that will be applied to each element of the input range. The return type must be acceptable as input to *binary_op*.

Returns

The *transform_inclusive_scan* algorithm returns a *hpx::future<FwdIter2>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter2* otherwise. The *transform_inclusive_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename InIter, typename OutIter, typename BinOp, typename UnOp, typename T = typename
std::iterator_traits<InIter>::value_type>
OutIter hpx::transform_inclusive_scan(InIter first, InIter last, OutIter dest, BinOp &&binary_op, UnOp
&&unary_op, T init)
```

Assigns through each iterator *i* in $[\text{result}, \text{result} + (\text{last} - \text{first}))$ the value of GENERALIZED_NONCOMMUTATIVE_SUM(*op*, *init*, *conv*(**first*), ..., *conv*(*(*first* + (*i* - *result*))))).

The reduce operations in the parallel *transform_inclusive_scan* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Neither *binary_op* nor *unary_op* shall invalidate iterators or sub-ranges, or modify elements in the ranges [first,last) or [result,result + (last - first)).

The difference between *inclusive_scan* and *transform_inclusive_scan* is that *transform_inclusive_scan* includes the *i*th input element in the *i*th sum. If *binary_op* is not mathematically associative, the behavior of *transform_inclusive_scan* may be non-deterministic.

Note: Complexity: $O(\text{last} - \text{first})$ applications of each of *binary_op* and *unary_op*.

Note: GENERALIZED_NONCOMMUTATIVE_SUM(*op*, *a1*, ..., *aN*) is defined as:

- *a1* when *N* is 1
 - *op*(GENERALIZED_NONCOMMUTATIVE_SUM(*op*, *a1*, ..., *aK*), GENERALIZED_NONCOMMUTATIVE_SUM(*op*, *aM*, ..., *aN*)) where $1 < K+1 = M \leq N$.
-

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **BinOp** – The type of *binary_op*.
- **UnOp** – The type of *unary_op*.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **binary_op** – Binary *FunctionObject* that will be applied in to the result of *unary_op*, the results of other *binary_op*, and *init* if provided.
- **unary_op** – Unary *FunctionObject* that will be applied to each element of the input range. The return type must be acceptable as input to *binary_op*.
- **init** – The initial value for the generalized sum.

Returns

The *transform_inclusive_scan* algorithm returns a *returns OutIter*. The *transform_inclusive_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.


```

template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename BinOp, typename UnOp,
typename T = typename std::iterator_traits<FwdIter1>::value_type>
parallel::util::detail::algorithm_result<ExPolicy, FwdIter2>::type hpx::transform_inclusive_scan(ExPolicy
                                                                                               &&policy,
                                                                                               FwdIter1
                                                                                               first,
                                                                                               FwdIter1
                                                                                               last,
                                                                                               FwdIter2
                                                                                               dest, BinOp
                                                                                               &&bi-
                                                                                               nary_op,
                                                                                               UnOp
                                                                                               &&unary_op,
                                                                                               T init)

```

Assigns through each iterator *i* in [result, result + (last - first)) the value of GENERALIZED_NONCOMMUTATIVE_SUM(op, init, conv(*first), ..., conv(*(first + (i - result)))). Executed according to the policy.

The reduce operations in the parallel *transform_inclusive_scan* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform_inclusive_scan* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Neither *binary_op* nor *unary_op* shall invalidate iterators or sub-ranges, or modify elements in the ranges [first,last) or [result,result + (last - first)).

The difference between *inclusive_scan* and *transform_inclusive_scan* is that *transform_inclusive_scan* includes the *i*th input element in the *i*th sum. If *binary_op* is not mathematically associative, the behavior of *transform_inclusive_scan* may be non-deterministic.

Note: Complexity: O(last - first) applications of each of *binary_op* and *unary_op*.

Note: GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN) is defined as:

- a1 when N is 1
 - op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN)) where 1 < K+1 = M <= N.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.

- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **BinOp** – The type of *binary_op*.
- **UnOp** – The type of *unary_op*.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **binary_op** – Binary *FunctionObject* that will be applied in to the result of *unary_op*, the results of other *binary_op*, and *init* if provided.
- **unary_op** – Unary *FunctionObject* that will be applied to each element of the input range. The return type must be acceptable as input to *binary_op*.
- **init** – The initial value for the generalized sum.

Returns

The *transform_inclusive_scan* algorithm returns a `hpx::future<FwdIter2>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter2* otherwise. The *transform_inclusive_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

hpx::exclusive_scan

Defined in header `hpx/algorithm.hpp`⁶⁰⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

template<typename **InIter**, typename **OutIter**, typename **T**>

OutIter **hpx::exclusive_scan**(*InIter* first, *InIter* last, *OutIter* dest, *T* init)

Assigns through each iterator *i* in [result, result + (last - first)) the value of GENERALIZED_NONCOMMUTATIVE_SUM(+, init, *first, ..., *(first + (i - result) - 1))

The reduce operations in the parallel *exclusive_scan* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

The difference between *exclusive_scan* and *inclusive_scan* is that *inclusive_scan* includes the *i*th input element in the *i*th sum.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate `std::plus<T>`.

Note: GENERALIZED_NONCOMMUTATIVE_SUM(+, a1, ..., aN) is defined as:

⁶⁰⁴ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- a_1 when N is 1
- `GENERALIZED_NONCOMMUTATIVE_SUM(+,  $a_1, \dots, a_K$ )`
 - `GENERALIZED_NONCOMMUTATIVE_SUM(+,  $a_M, \dots, a_N$ )` where $1 < K+1 = M \leq N$.

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.

Returns

The *exclusive_scan* algorithm returns *OutIter*. The *exclusive_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename T>
util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx::exclusive_scan(ExPolicy &&policy, FwdIter1 first,
                                                                    FwdIter1 last, FwdIter2 dest, T init)
```

Assigns through each iterator i in $[\text{result}, \text{result} + (\text{last} - \text{first}))$ the value of `GENERALIZED_NONCOMMUTATIVE_SUM(+, init, *first, ..., *(first + (i - result) - 1))`

The reduce operations in the parallel *exclusive_scan* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *exclusive_scan* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *exclusive_scan* and *inclusive_scan* is that *inclusive_scan* includes the i th input element in the i th sum.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate `std::plus<T>`.

Note: `GENERALIZED_NONCOMMUTATIVE_SUM(+,  $a_1, \dots, a_N$ )` is defined as:

- a_1 when N is 1
- `GENERALIZED_NONCOMMUTATIVE_SUM(+,  $a_1, \dots, a_K$ )`
 - `GENERALIZED_NONCOMMUTATIVE_SUM(+,  $a_M, \dots, a_N$ )` where $1 < K+1 = M \leq N$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.

Returns

The *exclusive_scan* algorithm returns a `hpx::future<FwdIter2>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter2* otherwise. The *exclusive_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename InIter, typename OutIter, typename T, typename Op>
```

```
OutIter hpx::exclusive_scan(InIter first, InIter last, OutIter dest, T init, Op &&op)
```

Assigns through each iterator *i* in `[result, result + (last - first))` the value of `GENERALIZED_NONCOMMUTATIVE_SUM(binary_op, init, *first, ..., *(first + (i - result) - 1))`.

The reduce operations in the parallel *exclusive_scan* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

The difference between *exclusive_scan* and *inclusive_scan* is that *inclusive_scan* includes the *i*th input element in the *i*th sum. If *op* is not mathematically associative, the behavior of *inclusive_scan* may be non-deterministic.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate *op*.

Note: `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- `a1` when *N* is 1
 - `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where $1 < K+1 = M \leq N$.
-

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **Op** – The type of the binary function object used for the reduction operation.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

Returns

The *exclusive_scan* algorithm returns *OutIter*. The *exclusive_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Op, typename T>
util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx::exclusive_scan(ExPolicy &&policy, FwdIter1 first,
                                                                    FwdIter1 last, FwdIter2 dest, T init,
                                                                    Op &&op)
```

Assigns through each iterator *i* in `[result, result + (last - first))` the value of `GENERALIZED_NONCOMMUTATIVE_SUM(binary_op, init, *first, ..., *(first + (i - result) - 1))`.

The reduce operations in the parallel *exclusive_scan* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *exclusive_scan* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *exclusive_scan* and *inclusive_scan* is that *inclusive_scan* includes the *i*th input element in the *i*th sum. If *op* is not mathematically associative, the behavior of *inclusive_scan* may be non-deterministic.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate *op*.

Note: `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- `a1` when `N` is 1
 - `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Op** – The type of the binary function object used for the reduction operation.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

Returns

The *exclusive_scan* algorithm returns a `hpx::future<OutIter>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *OutIter* otherwise. The *exclusive_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

hpx::inclusive_scan

Defined in header `hpx/algorithm.hpp`⁶⁰⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename InIter, typename OutIter>
```

```
OutIter hpx::inclusive_scan(InIter first, InIter last, OutIter dest)
```

Assigns through each iterator i in $[result, result + (last - first))$ the value of `GENERALIZED_NONCOMMUTATIVE_SUM(+, *first, ..., *(first + (i - result)))`.

The reduce operations in the parallel *inclusive_scan* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

The difference between *exclusive_scan* and *inclusive_scan* is that *inclusive_scan* includes the i th input element in the i th sum.

Note: Complexity: $O(last - first)$ applications of the predicate op , here `std::plus<>()`.

Note: `GENERALIZED_NONCOMMUTATIVE_SUM(+, a1, ..., aN)` is defined as:

- $a1$ when N is 1
 - `GENERALIZED_NONCOMMUTATIVE_SUM(+, a1, ..., aK)`
 - `GENERALIZED_NONCOMMUTATIVE_SUM(+, aM, ..., aN)` where $1 < K+1 = M \leq N$.
-

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *inclusive_scan* algorithm returns *OutIter*. The *inclusive_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2>
```

```
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx::inclusive_scan(ExPolicy &&policy,  
                                                                                      FwdIter1 first,  
                                                                                      FwdIter1 last,  
                                                                                      FwdIter2 dest)
```

⁶⁰⁵ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Assigns through each iterator i in $[result, result + (last - first))$ the value of `GENERALIZED_NONCOMMUTATIVE_SUM(+, *first, ..., *(first + (i - result)))`. Executed according to the policy.

The reduce operations in the parallel *inclusive_scan* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *inclusive_scan* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *exclusive_scan* and *inclusive_scan* is that *inclusive_scan* includes the i th input element in the i th sum.

Note: Complexity: $O(last - first)$ applications of the predicate *op*, here `std::plus<>()`.

Note: `GENERALIZED_NONCOMMUTATIVE_SUM(+, a1, ..., aN)` is defined as:

- $a1$ when N is 1
 - `GENERALIZED_NONCOMMUTATIVE_SUM(+, a1, ..., aK)`
 - `GENERALIZED_NONCOMMUTATIVE_SUM(+, aM, ..., aN)` where $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *inclusive_scan* algorithm returns a `hpx::future<FwdIter2>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `FwdIter2` otherwise. The *inclusive_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename InIter, typename OutIter, typename Op>
```


OutIter **hpx::inclusive_scan**(*InIter* first, *InIter* last, *OutIter* dest, *Op* &&op)

Assigns through each iterator *i* in [result, result + (last - first)) the value of GENERALIZED_NONCOMMUTATIVE_SUM(op, *first, ..., *(first + (i - result))).

The reduce operations in the parallel *inclusive_scan* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

The difference between *exclusive_scan* and *inclusive_scan* is that *inclusive_scan* includes the *i*th input element in the *i*th sum.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate *op*.

Note: GENERALIZED_NONCOMMUTATIVE_SUM(+, a1, ..., aN) is defined as:

- a1 when N is 1
 - GENERALIZED_NONCOMMUTATIVE_SUM(+, a1, ..., aK)
 - GENERALIZED_NONCOMMUTATIVE_SUM(+, aM, ..., aN) where $1 < K+1 = M \leq N$.
-

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Op** – The type of the binary function object used for the reduction operation.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have const&, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

Returns

The *inclusive_scan* algorithm returns *OutIter*. The *inclusive_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

template<typename **ExPolicy**, typename **FwdIter1**, typename **FwdIter2**, typename **Op**>

hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> **hpx::inclusive_scan**(*ExPolicy* &&policy,
FwdIter1 first,
FwdIter1 last,
FwdIter2 dest, *Op*
&&op)

Assigns through each iterator *i* in [result, result + (last - first)) the value of GENERALIZED_NONCOMMUTATIVE_SUM(op, *first, ..., *(first + (i - result))). Executed according to the policy.

The reduce operations in the parallel *inclusive_scan* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *inclusive_scan* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *exclusive_scan* and *inclusive_scan* is that *inclusive_scan* includes the *i*th input element in the *i*th sum.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate *op*.

Note: GENERALIZED_NONCOMMUTATIVE_SUM(+, a1, ..., aN) is defined as:

- a1 when N is 1
 - GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK)
 - GENERALIZED_NONCOMMUTATIVE_SUM(+, aM, ..., aN) where $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Op** – The type of the binary function object used for the reduction operation.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

Returns

The *inclusive_scan* algorithm returns a `hpx::future<FwdIter2>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter2* otherwise. The *inclusive_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename InIter, typename OutIter, typename Op, typename T>
OutIter hpx::inclusive_scan(InIter first, InIter last, OutIter dest, Op &&op, T init)
```

Assigns through each iterator *i* in `[result, result + (last - first))` the value of `GENERALIZED_NONCOMMUTATIVE_SUM(op, init, *first, ..., *(first + (i - result)))`.

The reduce operations in the parallel *inclusive_scan* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

The difference between *exclusive_scan* and *inclusive_scan* is that *inclusive_scan* includes the *i*th input element in the *i*th sum. If *op* is not mathematically associative, the behavior of *inclusive_scan* may be non-deterministic.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate *op*.

Note: `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- $a1$ when N is 1
 - `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where $1 < K+1 = M \leq N$.
-

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Op** – The type of the binary function object used for the reduction operation.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

- **dest** – Refers to the beginning of the destination range.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **init** – The initial value for the generalized sum.

Returns

The *inclusive_scan* algorithm returns *OutIter*. The *inclusive_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Op, typename T>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx::inclusive_scan(ExPolicy &&policy,
                                                                                      FwdIter1 first,
                                                                                      FwdIter1 last,
                                                                                      FwdIter2 dest, Op
                                                                                      &&op, T init)
```

Assigns through each iterator *i* in `[result, result + (last - first))` the value of `GENERALIZED_NONCOMMUTATIVE_SUM(op, init, *first, ..., *(first + (i - result)))`. Executed according to the policy.

The reduce operations in the parallel *inclusive_scan* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *inclusive_scan* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *exclusive_scan* and *inclusive_scan* is that *inclusive_scan* includes the *i*th input element in the *i*th sum. If *op* is not mathematically associative, the behavior of *inclusive_scan* may be non-deterministic.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate *op*.

Note: `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- `a1` when *N* is 1
 - `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.

- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Op** – The type of the binary function object used for the reduction operation.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **init** – The initial value for the generalized sum.

Returns

The *inclusive_scan* algorithm returns a `hpx::future<FwdIter2>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter2* otherwise. The *inclusive_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

hpx::experimental::reduction_bit_and

Defined in header `hpx/algorithm.hpp`⁶⁰⁶.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename T>
constexpr hpx::parallel::detail::reduction_helper<T, std::bit_and<T>> hpx::experimental::reduction_bit_and(T
&var)
```

The function template *reduction_bit_and* returns a reduction object of unspecified type having a value type and encapsulating an identity value for the reduction, it uses `std::bit_and{}` as its combiner function, and a live-out object from which the initial value is obtained and into which the final value is stored.

A parallel algorithm uses the *reduction_bit_and* object by allocating an unspecified number of instances, called views, of the reduction's value type. Each view is initialized with the reduction object's identity value, except that the live-out object (which was initialized by the caller) comprises one of the views. The algorithm passes a reference to a view to each application of an element-access function, ensuring that no two concurrently-executing invocations share the same view. A view can be shared between two applications that do not execute concurrently, but initialization is performed only once per view.

⁶⁰⁶ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e77/libs/full/include/include/hpx/algorithm.hpp>

Modifications to the view by the application of element access functions accumulate as partial results. At some point before the algorithm returns, the partial results are combined, two at a time, using the `std::bit_and{ }` operation until a single value remains, which is then assigned back to the live-out object.

T shall meet the requirements of *CopyConstructible* and *MoveAssignable*.

Note: In order to produce useful results, modifications to the view should be limited to commutative operations closely related to the combiner operation.

Template Parameters

- **T** – The value type to be used by the induction object.
- **Op** – The type of the binary function (object) used to perform the reduction operation.

Parameters

- **var** – [in,out] The live-out value to use for the reduction object. This will hold the reduced value after the algorithm is finished executing.
- **identity** – [in] The identity value to use for the reduction operation. This argument is optional and defaults to a copy of *var*.

Returns

This returns a reduction object of unspecified type having a value type of *T*. When the return value is used by an algorithm, the reference to *var* is used as the live-out object, new views are initialized to a copy of identity, and views are combined by invoking the copy of combiner, passing it the two views to be combined.

```
template<typename T>
constexpr hpx::parallel::detail::reduction_helper<T, std::bit_and<T>> hpx::experimental::reduction_bit_and(T
                                                                    &var,
                                                                    T
                                                                    const
                                                                    &iden-
                                                                    tity)
```

hpx::set_intersection

Defined in header `hpx/algorithm.hpp`⁶⁰⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename FwdIter3, typename Pred =
hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter3> hpx::set_intersection(ExPolicy &&policy,
                                                                    FwdIter1 first1,
                                                                    FwdIter1 last1,
                                                                    FwdIter2 first2,
                                                                    FwdIter2 last2,
                                                                    FwdIter3 dest, Pred
                                                                    &&op = Pred())
```

⁶⁰⁷ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Constructs a sorted range beginning at `dest` consisting of all elements present in both sorted ranges `[first1, last1)` and `[first2, last2)`. This algorithm expects both input ranges to be sorted with the given binary predicate *pred*. Executed according to the policy.

If some element is found *m* times in `[first1, last1)` and *n* times in `[first2, last2)`, the first `std::min(m, n)` elements will be copied from the first range to the destination range. The order of equivalent elements is preserved. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

The application of function objects in parallel algorithm invoked with a sequential execution policy object execute in sequential order in the calling thread (*sequenced_policy*) or in a single new thread spawned from the current thread (for *sequenced_task_policy*).

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most $2 \cdot (N_1 + N_2 - 1)$ comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **FwdIter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator or output iterator with sequential execution.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set_intersection* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.

- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*

Returns

The *set_intersection* algorithm returns a *hpx::future<FwdIter3>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter3* otherwise. The *set_intersection* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename FwdIter1, typename FwdIter2, typename FwdIter3, typename Pred =  
hpx::parallel::detail::less>  
FwdIter3 hpx::set_intersection(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, FwdIter2 last2, FwdIter3 dest,  
Pred &&op = Pred())
```

Constructs a sorted range beginning at *dest* consisting of all elements present in both sorted ranges [*first1*, *last1*) and [*first2*, *last2*). This algorithm expects both input ranges to be sorted with the given binary predicate *pred*.

If some element is found *m* times in [*first1*, *last1*) and *n* times in [*first2*, *last2*), the first `std::min(m, n)` elements will be copied from the first range to the destination range. The order of equivalent elements is preserved. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

Note: Complexity: At most $2 \cdot (N1 + N2 - 1)$ comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

Template Parameters

- **FwdIter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator or output iterator with sequential execution.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set_intersection* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.

- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

Returns

The *set_intersection* algorithm returns a *FwdIter3*. The *set_intersection* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

hpx/parallel/algorithms/iota.hpp

Defined in header `hpx/parallel/algorithms/iota.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

Functions

```
template<typename FwdIter, typename Sent, typename T>
void iota(FwdIter first, Sent last, T value)
```

Fills the range [first, last) with sequentially increasing values, starting with *value* and repetitively evaluating ++*value*.

The assignments in the *iota* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

Note: Complexity: Exactly *std::distance(first, last)* assignments and increments.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input or output iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **T** – The type of the value to be assigned (deduced). This type must be weakly incrementable.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – The value to be stored in the first element.

Returns

The *iota* algorithm returns *void*.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename T>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy> iota(ExPolicy &&policy, FwdIter first, Sent last, T  
value)
```

Fills the range [first, last) with sequentially increasing values, starting with *value* and repetitively evaluating ++*value*.

The assignments in the parallel *iota* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *iota* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Exactly *std::distance(first, last)* assignments and increments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **T** – The type of the value to be assigned (deduced). This type must be weakly incrementable.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – The value to be stored in the first element.

Returns

The *iota* algorithm returns a *hpx::future<void>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *void* otherwise.

hpx::shift_left

Defined in header `hpx/algorithm.hpp`⁶⁰⁸.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename FwdIter, typename Size>
FwdIter hpx::shift_left(FwdIter first, FwdIter last, Size n)
```

Shifts the elements in the range [first, last) by n positions towards the beginning of the range. For every integer i in [0, last - first

- n), moves the element originally at position first + n + i to position first + i.

The assignment operations in the parallel *shift_left* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

Note: Complexity: At most (last - first) - n assignments.

Note: The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable*.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of positions to shift by.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **n** – Refers to the number of positions to shift.

Returns

The *shift_left* algorithm returns *FwdIter*. The *shift_left* algorithm returns an iterator to the end of the resulting range. Specifically: If *n* is 0 or *n* >= (last - first), does nothing and returns *last* and *first* respectively. Otherwise returns *first* + ((last - first) - n).

```
template<typename ExPolicy, typename FwdIter, typename Size>
hpx::parallel::util::detail::algorithm_result<ExPolicy, FwdIter> hpx::shift_left(ExPolicy &&policy, FwdIter
                                                                                   first, FwdIter last, Size n)
```

Shifts the elements in the range [first, last) by n positions towards the beginning of the range. For every integer i in [0, last - first

- n), moves the element originally at position first + n + i to position first + i. Executed according to the policy.

The assignment operations in the parallel *shift_left* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

⁶⁰⁸ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

The assignment operations in the parallel *shift_left* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most (last - first) - n assignments.

Note: The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of positions to shift by.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **n** – Refers to the number of positions to shift.

Returns

The *shift_left* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *shift_left* algorithm returns an iterator to the end of the resulting range. Specifically: If *n* is 0, does nothing and returns *last*. If *n* >= (last - first), does nothing and returns *first*. Otherwise returns *first* + ((last - first) - n).

hpx::lexicographical_compare

Defined in header [hpx/algorithm.hpp](#)⁶⁰⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename InIter1, typename InIter2, typename Pred = hpx::parallel::detail::less>
bool hpx::lexicographical_compare(InIter1 first1, InIter1 last1, InIter2 first2, InIter2 last2, Pred &&pred)
```

Checks if the first range [first1, last1) is lexicographically less than the second range [first2, last2). uses a provided predicate to compare elements.

The comparison operations in the parallel *lexicographical_compare* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

⁶⁰⁹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Note: Complexity: At most $2 * \min(N1, N2)$ applications of the comparison operation, where $N1 = \text{std::distance}(\text{first1}, \text{last})$ and $N2 = \text{std::distance}(\text{first2}, \text{last2})$.

Note: Lexicographical comparison is an operation with the following properties

- Two ranges are compared element by element
 - The first mismatching element defines which range is lexicographically *less* or *greater* than the other
 - If one range is a prefix of another, the shorter range is lexicographically *less* than the other
 - If two ranges have equivalent elements and are of the same length, then the ranges are lexicographically *equal*
 - An empty range is lexicographically *less* than any non-empty range
 - Two empty ranges are lexicographically *equal*
-

Template Parameters

- **InIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an input iterator.
- **InIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an input iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *lexicographical_compare* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **pred** – Refers to the comparison function that the first and second ranges will be applied to

Returns

The *lexicographically_compare* algorithm returns a *bool* if the execution policy object is not passed in. The *lexicographically_compare* algorithm returns true if the first range is lexicographically less, otherwise it returns false. range [first2, last2), it returns false.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Pred =
    hpx::parallel::detail::less>
```

```
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx::lexicographical_compare(ExPolicy
                                                                 &&policy,
                                                                 FwdIter1 first1,
                                                                 FwdIter1 last1,
                                                                 FwdIter2 first2,
                                                                 FwdIter2 last2,
                                                                 Pred &&pred)
```

Checks if the first range [first1, last1) is lexicographically less than the second range [first2, last2). uses a provided predicate to compare elements.

The comparison operations in the parallel *lexicographical_compare* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *lexicographical_compare* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most $2 * \min(N1, N2)$ applications of the comparison operation, where $N1 = \text{std::distance}(\text{first1}, \text{last})$ and $N2 = \text{std::distance}(\text{first2}, \text{last2})$.

Note: Lexicographical comparison is an operation with the following properties

- Two ranges are compared element by element
 - The first mismatching element defines which range is lexicographically *less* or *greater* than the other
 - If one range is a prefix of another, the shorter range is lexicographically *less* than the other
 - If two ranges have equivalent elements and are of the same length, then the ranges are lexicographically *equal*
 - An empty range is lexicographically *less* than any non-empty range
 - Two empty ranges are lexicographically *equal*
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *lexicographical_compare* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.

- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **pred** – Refers to the comparison function that the first and second ranges will be applied to

Returns

The *lexicographically_compare* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *lexicographically_compare* algorithm returns true if the first range is lexicographically less, otherwise it returns false. range [first2, last2), it returns false.

hpx::partition

Defined in header [hpx/algorithm.hpp](#)⁶¹⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::stable_partition*
- *hpx::partition_copy*

template<typename **FwdIter**, typename **Pred**, typename **Proj** = *hpx::identity*>

FwdIter **hpx::partition**(*FwdIter* first, *FwdIter* last, *Pred* &&pred, *Proj* &&proj = *Proj*())

Reorders the elements in the range [first, last) in such a way that all elements for which the predicate *pred* returns true precede the elements for which the predicate *pred* returns false. Relative order of the elements is not preserved.

The assignments in the parallel *partition* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: At most $2 * (\text{last} - \text{first})$ swaps. Exactly *last - first* applications of the predicate and projection.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.

⁶¹⁰ <http://github.com/TheHPXProject/hpx/blob/d402cead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate for partitioning the source iterators. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *partition* algorithm returns *FwdIter*. The *partition* algorithm returns the iterator to the first element of the second group.

```
template<typename ExPolicy, typename FwdIter, typename Pred, typename Proj = hpx::identity>
parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::partition(ExPolicy &&policy, FwdIter first,
                                                                    FwdIter last, Pred &&pred, Proj
                                                                    &&proj = Proj())
```

Reorders the elements in the range [first, last) in such a way that all elements for which the predicate *pred* returns true precede the elements for which the predicate *pred* returns false. Relative order of the elements is not preserved.

The assignments in the parallel *partition* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *partition* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most $2 * (\text{last} - \text{first})$ swaps. Exactly *last - first* applications of the predicate and projection.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.

- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate for partitioning the source iterators. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have const&, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *partition* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *parallel_task_policy* and returns *FwdIter* otherwise. The *partition* algorithm returns the iterator to the first element of the second group.

hpx::stable_partition

Defined in header `hpx/algorithm.hpp`⁶¹¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::partition*
- *hpx::partition_copy*

```
template<typename BidirIter, typename F, typename Proj = hpx::identity>
```

```
BidirIter hpx::stable_partition(BidirIter first, BidirIter last, F &&f, Proj &&proj = Proj())
```

Permutes the elements in the range [first, last) such that there exists an iterator *i* such that for every iterator *j* in the range [first, *i*) INVOKE(*f*, INVOKE(*proj*, **j*)) != false, and for every iterator *k* in the range [*i*, last), INVOKE(*f*, INVOKE(*proj*, **k*)) == false

The invocations of *f* in the parallel *stable_partition* algorithm invoked without an execution policy object executes in sequential order in the calling thread.

Note: Complexity: At most (last - first) * log(last - first) swaps, but only linear number of swaps if there is enough extra memory. Exactly *last - first* applications of the predicate and projection.

Template Parameters

- **BidirIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a bidirectional iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *transform* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

⁶¹¹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Unary predicate which returns true if the element should be ordered before other elements. Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool fun(const Type &a);
```

The signature does not need to have `const&`. The type *Type* must be such that an object of type *BidirIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *f* is invoked.

Returns

The *stable_partition* algorithm returns an iterator *i* such that for every iterator *j* in the range [first, *i*), `f(*j) != false INVOKE(f, INVOKE(proj, *j)) != false`, and for every iterator *k* in the range [*i*, last), `f(*k) == false INVOKE(f, INVOKE(proj, *k)) == false`. The relative order of the elements in both groups is preserved.

```
template<typename ExPolicy, typename BidirIter, typename F, typename Proj = hpx::identity>
parallel::util::detail::algorithm_result_t<ExPolicy, BidirIter> hpx::stable_partition(ExPolicy &&policy,
                                                                                      BidirIter first, BidirIter
                                                                                      last, F &&f, Proj &&proj
                                                                                      = Proj())
```

Permutes the elements in the range [first, last) such that there exists an iterator *i* such that for every iterator *j* in the range [first, *i*) `INVOKE(f, INVOKE(proj, *j)) != false`, and for every iterator *k* in the range [*i*, last), `INVOKE(f, INVOKE(proj, *k)) == false`

The invocations of *f* in the parallel *stable_partition* algorithm invoked with an execution policy object of type *sequenced_policy* executes in sequential order in the calling thread.

The invocations of *f* in the parallel *stable_partition* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most $(last - first) * \log(last - first)$ swaps, but only linear number of swaps if there is enough extra memory. Exactly *last - first* applications of the predicate and projection.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the invocations of *f*.
- **BidirIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a bidirectional iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *transform* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Unary predicate which returns true if the element should be ordered before other elements. Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool fun(const Type &a);
```

The signature does not need to have const&. The type *Type* must be such that an object of type *BidIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *f* is invoked.

Returns

The *stable_partition* algorithm returns an iterator *i* such that for every iterator *j* in the range [first, i), $f(*j) \neq \text{false}$ INVOKE(*f*, INVOKE(*proj*, **j*)) $\neq \text{false}$, and for every iterator *k* in the range [i, last), $f(*k) == \text{false}$ INVOKE(*f*, INVOKE(*proj*, **k*)) $== \text{false}$. The relative order of the elements in both groups is preserved. If the execution policy is of type *parallel_task_policy* the algorithm returns a *future<>* referring to this iterator.

hpx::partition_copy

Defined in header `hpx/algorithm.hpp`⁶¹².

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::partition*
- *hpx::stable_partition*

```
template<typename FwdIter1, typename FwdIter2, typename FwdIter3, typename Pred, typename Proj =
hpx::identity>
std::pair<FwdIter2, FwdIter3> hpx::partition_copy(FwdIter1 first, FwdIter1 last, FwdIter2 dest_true, FwdIter3
dest_false, Pred &&pred, Proj &&proj = Proj())
```

Copies the elements in the range, defined by [first, last), to two different ranges depending on the value returned by the predicate *pred*. The elements, that satisfy the predicate *pred* are copied to the range beginning at *dest_true*. The rest of the elements are copied to the range beginning at *dest_false*. The order of the elements is preserved.

The assignments in the parallel *partition_copy* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the predicate *pred*.

⁶¹² <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Template Parameters

- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range for the elements that satisfy the predicate *pred* (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter3** – The type of the iterator representing the destination range for the elements that don't satisfy the predicate *pred* (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition_copy* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest_true** – Refers to the beginning of the destination range for the elements that satisfy the predicate *pred*
- **dest_false** – Refers to the beginning of the destination range for the elements that don't satisfy the predicate *pred*.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate for partitioning the source iterators. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *partition_copy* algorithm returns `std::pair<OutIter1, OutIter2>`. The *partition_copy* algorithm returns the pair of the destination iterator to the end of the *dest_true* range, and the destination iterator to the end of the *dest_false* range.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename FwdIter3, typename Pred,
typename Proj = hpx::identity>
```

```

parallel::util::detail::algorithm_result_t<ExPolicy, std::pair<FwdIter2, FwdIter3>> hpx::partition_copy(ExPolicy
&&pol-
icy,
FwdIter1
first,
FwdIter1
last,
FwdIter2
dest_true,
FwdIter3
dest_false,
Pred
&&pred,
Proj
&&proj
=
Proj())

```

Copies the elements in the range, defined by [first, last), to two different ranges depending on the value returned by the predicate *pred*. The elements, that satisfy the predicate *pred*, are copied to the range beginning at *dest_true*. The rest of the elements are copied to the range beginning at *dest_false*. The order of the elements is preserved.

The assignments in the parallel *partition_copy* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *partition_copy* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the predicate *pred*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range for the elements that satisfy the predicate *pred* (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter3** – The type of the iterator representing the destination range for the elements that don't satisfy the predicate *pred* (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition_copy* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest_true** – Refers to the beginning of the destination range for the elements that satisfy the predicate *pred*.
- **dest_false** – Refers to the beginning of the destination range for the elements that don't satisfy the predicate *pred*.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate for partitioning the source iterators. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *partition_copy* algorithm returns a `hpx::future<std::pair<OutIter1, OutIter2>>` if the execution policy is of type *parallel_task_policy* and returns `std::pair<OutIter1, OutIter2>` otherwise. The *partition_copy* algorithm returns the pair of the destination iterator to the end of the *dest_true* range, and the destination iterator to the end of the *dest_false* range.

hpx::experimental::sort_by_key

Defined in header `hpx/algorithm.hpp`⁶¹³.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename ExPolicy, typename KeyIter, typename ValueIter, typename Compare = detail::less>
util::detail::algorithm_result_t<ExPolicy, sort_by_key_result<KeyIter, ValueIter>> hpx::experimental::sort_by_key(ExPolicy
&&pol-
icy,
KeyIter
key_first,
KeyIter
key_last,
Val-
ueIter
value_first,
Com-
pare
&&comp
=
Com-
pare())
```

⁶¹³ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Sorts one range of data using keys supplied in another range. The key elements in the range `[key_first, key_last)` are sorted in ascending order with the corresponding elements in the value range moved to follow the sorted order. The algorithm is not stable, the order of equal elements is not guaranteed to be preserved. The function uses the given comparison function object `comp` (defaults to using `operator<()`). Executed according to the policy.

A sequence is sorted with respect to a comparator *comp* if for every iterator *i* pointing to the sequence and every non-negative integer *n* such that *i* + *n* is a valid iterator pointing to an element of the sequence, and `INVOKE(comp, *(i + n), *i) == false`.

comp has to induce a strict weak ordering on the values.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: $O(N \log(N))$, where $N = \text{std::distance}(\text{first}, \text{last})$ comparisons.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **KeyIter** – The type of the key iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **ValueIter** – The type of the value iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Compare** – The type of the function/function object to use (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **key_first** – Refers to the beginning of the sequence of key elements the algorithm will be applied to.
- **key_last** – Refers to the end of the sequence of key elements the algorithm will be applied to.
- **value_first** – Refers to the beginning of the sequence of value elements the algorithm will be applied to, the range of elements must match `[key_first, key_last)`
- **comp** – `comp` is a callable object. The return value of the `INVOKE` operation applied to an object of type `Comp`, when contextually converted to `bool`, yields `true` if the first argument of the call is less than the second, and `false` otherwise. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator.

Returns

The *sort_by_key* algorithm returns a `hpx::future<sort_by_key_result<KeyIter, ValueIter>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *otherwise*. The algorithm returns a pair holding an iterator pointing to the first element after the last element in the input key sequence and an iterator pointing to the first element after the last element in the input value sequence.

hpx::transform_exclusive_scan

Defined in header [hpx/algorithm.hpp](#)⁶¹⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename InIter, typename OutIter, typename BinOp, typename UnOp, typename T = typename
std::iterator_traits<InIter>::value_type>
OutIter hpx::transform_exclusive_scan(InIter first, InIter last, OutIter dest, T init, BinOp &&binary_op,
                                     UnOp &&unary_op)
```

Transforms each element in the range $[first, last)$ with *unary_op*, then computes an exclusive prefix sum operation using *binary_op* over the resulting range, with *init* as the initial value, and writes the results to the range beginning at *dest*. “exclusive” means that the *i*-th input element is not included in the *i*-th sum. Formally, assigns through each iterator *i* in $[dest, d_first + (last - first))$ the value of the generalized noncommutative sum of *init*, *unary_op*(*j)... for every *j* in $[first, first + (i - d_first))$ over *binary_op*, where generalized noncommutative sum GNSUM(*op*, *a*1, ..., *a* N) is defined as follows:

- if $N=1$, *a*1
- if $N > 1$, *op*(GNSUM(*op*, *a*1, ..., *a*K), GNSUM(*op*, *a*M, ..., *a*N)) for any *K* where $1 < K+1 = M \leq N$ In other words, the summation operations may be performed in arbitrary order, and the behavior is nondeterministic if *binary_op* is not associative.

The reduce operations in the parallel *transform_exclusive_scan* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Neither *unary_op* nor *binary_op* shall invalidate iterators or sub-ranges, or modify elements in the ranges $[first, last)$ or $[result, result + (last - first))$.

The behavior of *transform_exclusive_scan* may be non-deterministic for a non-associative predicate.

Note: Complexity: $O(last - first)$ applications of each of *binary_op* and *unary_op*.

Note: GENERALIZED_NONCOMMUTATIVE_SUM(*op*, *a*1, ..., *a*N) is defined as:

- *a*1 when *N* is 1
 - *op*(GENERALIZED_NONCOMMUTATIVE_SUM(*op*, *a*1, ..., *a*K), GENERALIZED_NONCOMMUTATIVE_SUM(*op*, *a*M, ..., *a*N) where $1 < K+1 = M \leq N$.
-

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **BinOp** – The type of *binary_op*.
- **UnOp** – The type of *unary_op*.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

⁶¹⁴ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/lib/full/include/include/hpx/algorithm.hpp>

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.
- **binary_op** – Binary *FunctionObject* that will be applied to the result of *unary_op*, the results of other *binary_op*, and *init*.
- **unary_op** – Unary *FunctionObject* that will be applied to each element of the input range. The return type must be acceptable as input to *binary_op*.

Returns

The *transform_exclusive_scan* algorithm returns a *returns OutIter*. The *transform_exclusive_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename BinOp, typename UnOp,
typename T = typename std::iterator_traits<FwdIter1>::value_type>
parallel::util::detail::algorithm_result<ExPolicy, FwdIter2>::type hpx::transform_exclusive_scan(ExPolicy
                                                                                               &&policy,
                                                                                               FwdIter1
                                                                                               first,
                                                                                               FwdIter1
                                                                                               last,
                                                                                               FwdIter2
                                                                                               dest, T init,
                                                                                               BinOp
                                                                                               &&bi-
                                                                                               nary_op,
                                                                                               UnOp
                                                                                               &&unary_op)
```

Assigns through each iterator *i* in [result, result + (last - first)) the value of GENERALIZED_NONCOMMUTATIVE_SUM(binary_op, init, conv(*first), ..., conv(*(first + (i - result) - 1))). Executed according to the policy.

The reduce operations in the parallel *transform_exclusive_scan* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform_exclusive_scan* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Neither *unary_op* nor *binary_op* shall invalidate iterators or sub-ranges, or modify elements in the ranges [first,last) or [result,result + (last - first)).

The behavior of *transform_exclusive_scan* may be non-deterministic for a non-associative predicate.

Note: Complexity: O(last - first) applications of each of *binary_op* and *unary_op*.

Note: `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- `a1` when `N` is 1
 - `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **BinOp** – The type of *binary_op*.
- **UnOp** – The type of *unary_op*.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.
- **binary_op** – Binary *FunctionObject* that will be applied in to the result of *unary_op*, the results of other *binary_op*, and *init*.
- **unary_op** – Unary *FunctionObject* that will be applied to each element of the input range. The return type must be acceptable as input to *binary_op*.

Returns

The *transform_exclusive_scan* algorithm returns a `hpx::future<FwdIter2>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter2* otherwise. The *transform_exclusive_scan* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

hpx::experimental::reduction_min

Defined in header `hpx/algorithm.hpp`⁶¹⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename T>
constexpr hpx::parallel::detail::reduction_helper<T, hpx::parallel::detail::min_of<T>> hpx::experimental::reduction_min(T
&va
```

The function template `reduction_min` returns a reduction object of unspecified type having a value type and encapsulating an identity value for the reduction, it uses `std::min_of{ }` as its combiner function, and a live-out object from which the initial value is obtained and into which the final value is stored.

A parallel algorithm uses the `reduction_min` object by allocating an unspecified number of instances, called views, of the reduction's value type. Each view is initialized with the reduction object's identity value, except that the live-out object (which was initialized by the caller) comprises one of the views. The algorithm passes a reference to a view to each application of an element-access function, ensuring that no two concurrently-executing invocations share the same view. A view can be shared between two applications that do not execute concurrently, but initialization is performed only once per view.

Modifications to the view by the application of element access functions accumulate as partial results. At some point before the algorithm returns, the partial results are combined, two at a time, using the `std::min_of{ }` operation until a single value remains, which is then assigned back to the live-out object.

`T` shall meet the requirements of *CopyConstructible* and *MoveAssignable*.

Note: In order to produce useful results, modifications to the view should be limited to commutative operations closely related to the combiner operation.

Template Parameters

- **T** – The value type to be used by the induction object.
- **Op** – The type of the binary function (object) used to perform the reduction operation.

Parameters

- **var** – [in,out] The life-out value to use for the reduction object. This will hold the reduced value after the algorithm is finished executing.
- **identity** – [in] The identity value to use for the reduction operation. This argument is optional and defaults to a copy of `var`.

Returns

This returns a reduction object of unspecified type having a value type of `T`. When the return value is used by an algorithm, the reference to `var` is used as the live-out object, new views are initialized to a copy of `identity`, and views are combined by invoking the copy of combiner, passing it the two views to be combined.

```
template<typename T>
```

⁶¹⁵ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

```
constexpr hpx::parallel::detail::reduction_helper<T, hpx::parallel::detail::min_of<T>> hpx::experimental::reduction_min(T  
&va  
T  
cons  
&id  
tity)
```

hpx::equal

Defined in header `hpx/algorithm.hpp`⁶¹⁶.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Pred = detail::equal_to>  
util::detail::algorithm_result_t<ExPolicy, bool> hpx::equal(ExPolicy &&policy, FwdIter1 first1, FwdIter1 last1,  
FwdIter2 first2, FwdIter2 last2, Pred &&op = Pred())
```

Returns true if the range [first1, last1) is equal to the range [first2, last2), and false otherwise. Executed according to the policy.

The comparison operations in the parallel *equal* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *equal* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: $O(\min(\text{last1} - \text{first1}, \text{last2} - \text{first2}))$ applications of the predicate *op*.

Note: The two ranges are considered equal if, for every iterator *i* in the range [first1, last1), **i* equals $*(\text{first2} + (i - \text{first1}))$. This overload of *equal* uses `operator==` to determine if two elements are equal.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

⁶¹⁶ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

Returns

The *equal* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *equal* algorithm returns true if the elements in the two ranges are equal, otherwise it returns false. If the length of the range `[first1, last1)` does not equal the length of the range `[first2, last2)`, it returns false.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2>
util::detail::algorithm_result_t<ExPolicy, bool> hpx::equal(ExPolicy &&policy, FwdIter1 first1, FwdIter1 last1,
                                                           FwdIter2 first2, FwdIter2 last2)
```

Returns true if the range `[first1, last1)` is equal to the range `[first2, last2)`, and false otherwise. Executed according to policy.

The comparison operations in the parallel *equal* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *equal* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: $O(\min(\text{last1} - \text{first1}, \text{last2} - \text{first2}))$ applications of the predicate *std::equal_to*.

Note: The two ranges are considered equal if, for every iterator *i* in the range `[first1, last1)`, **i* equals **(first2 + (i - first1))*. This overload of *equal* uses `operator==` to determine if two elements are equal.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.

- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.

Returns

The *equal* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *equal* algorithm returns true if the elements in the two ranges are equal, otherwise it returns false. If the length of the range [first1, last1) does not equal the length of the range [first2, last2), it returns false.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Pred = detail::equal_to>  
util::detail::algorithm_result_t<ExPolicy, bool> hpx::equal(ExPolicy &&policy, FwdIter1 first1, FwdIter1 last1,  
FwdIter2 first2, Pred &&op = Pred())
```

Returns true if the range [first1, last1) is equal to the range starting at first2, and false otherwise. Executed according to policy.

The comparison operations in the parallel *equal* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *equal* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: $O(\min(\text{last1} - \text{first1}, \text{last2} - \text{first2}))$ applications of the predicate *op*.

Note: The two ranges are considered equal if, for every iterator *i* in the range [first1,last1), **i* equals **(first2 + (i - first1))*. This overload of *equal* uses operator== to determine if two elements are equal.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

Returns

The *equal* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *equal* algorithm returns true if the elements in the two ranges are equal, otherwise it returns false.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2>
util::detail::algorithm_result_t<ExPolicy, bool> hpx::equal(ExPolicy &&policy, FwdIter1 first1, FwdIter1 last1,
                                                         FwdIter2 first2)
```

Returns true if the range `[first1, last1)` is equal to the range `[first2, last2)`, and false otherwise. Executed according to policy.

The comparison operations in the parallel *equal* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *equal* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most *last1 - first1* applications of the predicate *op*.

Note: The two ranges are considered equal if, for every iterator *i* in the range `[first1, last1)`, **i* equals **(first2 + (i - first1))*. This overload of *equal* uses `operator==` to determine if two elements are equal.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.

- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.

Returns

The *equal* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *equal* algorithm returns true if the elements in the two ranges are equal, otherwise it returns false. If the length of the range [first1, last1) does not equal the length of the range [first2, last2), it returns false.

```
template<typename FwdIter1, typename FwdIter2, typename Pred = detail::equal_to>  
bool hpx::equal(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, FwdIter2 last2, Pred &&op = Pred())
```

Returns true if the range [first1, last1) is equal to the range [first2, last2), and false otherwise.

Note: Complexity: At most min(last1 - first1, last2 - first2) applications of the predicate *op*.

Note: The two ranges are considered equal if, for every iterator *i* in the range [first1,last1), **i* equals **(first2 + (i - first1))*. This overload of *equal* uses operator== to determine if two elements are equal.

Template Parameters

- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to *std::equal_to<>*

Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.

- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

Returns

The *equal* algorithm returns a *bool*. The *equal* algorithm returns true if the elements in the two ranges are equal, otherwise it returns false. If the length of the range `[first1, last1)` does not equal the length of the range `[first2, last2)`, it returns false.

```
template<typename FwdIter1, typename FwdIter2>
```

```
bool hpx::equal(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, FwdIter2 last2)
```

Returns true if the range `[first1, last1)` is equal to the range `[first2, last2)`, and false otherwise.

Note: Complexity: At most $\min(\text{last1} - \text{first1}, \text{last2} - \text{first2})$ applications of the predicate *std::equal_to*.

Note: The two ranges are considered equal if, for every iterator *i* in the range `[first1, last1)`, **i* equals **(first2 + (i - first1))*. This overload of *equal* uses `operator==` to determine if two elements are equal.

Template Parameters

- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.

Returns

The *equal* algorithm returns a *bool*. The *equal* algorithm returns true if the elements in the two ranges are equal, otherwise it returns false. If the length of the range `[first1, last1)` does not equal the length of the range `[first2, last2)`, it returns false.

```
template<typename FwdIter1, typename FwdIter2, typename Pred = detail::equal_to>
```

bool *hpx::equal*(*FwdIter1* first1, *FwdIter1* last1, *FwdIter2* first2, *Pred* &&op = *Pred*())

Returns true if the range [first1, last1) is equal to the range [first2, first2 + (last1 - first1)), and false otherwise.

Note: Complexity: At most *last1 - first1* applications of the predicate *op*.

Note: The two ranges are considered equal if, for every iterator *i* in the range [first1, last1), **i* equals **(first2 + (i - first1))*. This overload of *equal* uses *operator==* to determine if two elements are equal.

Template Parameters

- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

Returns

The *equal* algorithm returns a *bool*. The *equal* algorithm returns true if the elements in the two ranges are equal, otherwise it returns false. If the length of the range [first1, last1) does not equal the length of the range [first2, last2), it returns false.

hpx::set_difference

Defined in header `hpx/algorithm.hpp`⁶¹⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename FwdIter3, typename Pred =
hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter3> hpx::set_difference(ExPolicy &&policy,
                                                                                      FwdIter1 first1,
                                                                                      FwdIter1 last1,
                                                                                      FwdIter2 first2,
                                                                                      FwdIter2 last2,
                                                                                      FwdIter3 dest, Pred
                                                                                      &&op = Pred())
```

Constructs a sorted range beginning at *dest* consisting of all elements present in the range $[first1, last1)$ and not present in the range $[first2, last2)$. This algorithm expects both input ranges to be sorted with the given binary predicate *pred*. Executed according to the policy.

Equivalent elements are treated individually, that is, if some element is found m times in $[first1, last1)$ and n times in $[first2, last2)$, it will be copied to *dest* exactly $\text{std::max}(m-n, 0)$ times. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

The application of function objects in parallel algorithm invoked with a sequential execution policy object execute in sequential order in the calling thread (*sequenced_policy*) or in a single new thread spawned from the current thread (for *sequenced_task_policy*).

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most $2 \cdot (N1 + N2 - 1)$ comparisons, where $N1$ is the length of the first sequence and $N2$ is the length of the second sequence.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **FwdIter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a output iterator.

⁶¹⁷ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set_difference* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*

Returns

The *set_difference* algorithm returns a `hpx::future<FwdIter3>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter3* otherwise. The *set_difference* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename FwdIter1, typename FwdIter2, typename FwdIter3, typename Pred =  
hpx::parallel::detail::less>  
FwdIter3 hpx::set_difference(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, FwdIter2 last2, FwdIter3 dest,  
                             Pred &&op = Pred())
```

Constructs a sorted range beginning at *dest* consisting of all elements present in the range `[first1, last1)` and not present in the range `[first2, last2)`. This algorithm expects both input ranges to be sorted with the given binary predicate *pred*.

Equivalent elements are treated individually, that is, if some element is found *m* times in `[first1, last1)` and *n* times in `[first2, last2)`, it will be copied to *dest* exactly `std::max(m-n, 0)` times. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

Note: Complexity: At most $2 \cdot (N_1 + N_2 - 1)$ comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

Template Parameters

- **FwdIter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a forward iterator.
- **FwdIter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set_difference* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*

Returns

The *set_difference* algorithm returns a *FwdIter3*. The *set_difference* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

hpx::copy

Defined in header `hpx/algorithm.hpp`⁶¹⁸.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::copy_n`
- `hpx::copy_if`

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx::copy(ExPolicy &&policy, FwdIter1 first,
                                                                              FwdIter1 last, FwdIter2 dest)
```

Copies the elements in the range, defined by [first, last), to another range beginning at *dest*. Executed according to the policy.

⁶¹⁸ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e77/libs/full/include/include/hpx/algorithm.hpp>

The assignments in the parallel *copy* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *copy* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *copy* algorithm returns a *hpx::future<FwdIter2>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter2* otherwise. The *copy* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename FwdIter1, typename FwdIter2>
FwdIter2 hpx::copy(FwdIter1 first, FwdIter1 last, FwdIter2 dest)
```

Copies the elements in the range, defined by [first, last), to another range beginning at *dest*.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.

- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *copy* algorithm returns a *FwdIter2*. The *copy* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

hpx::copy_n

Defined in header `hpx/algorithm.hpp`⁶¹⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::copy*
- *hpx::copy_if*

```
template<typename ExPolicy, typename FwdIter1, typename Size, typename FwdIter2>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx::copy_n(ExPolicy &&policy, FwdIter1
                                                                    first, Size count, FwdIter2 dest)
```

Copies the elements in the range [first, first + count), starting from first and proceeding to first + count - 1., to another range beginning at dest. Executed according to the policy.

The assignments in the parallel *copy_n* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *copy_n* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *count* assignments, if count > 0, no assignments otherwise.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.

⁶¹⁹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e77/libs/full/include/include/hpx/algorithm.hpp>

- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *copy_n* algorithm returns a *hpx::future<FwdIter2>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter2* otherwise. The *copy_n* algorithm returns Iterator in the destination range, pointing past the last element copied if count>0 or result otherwise.

```
template<typename FwdIter1, typename Size, typename FwdIter2>
FwdIter2 hpx::copy_n(FwdIter1 first, Size count, FwdIter2 dest)
```

Copies the elements in the range [first, first + count), starting from first and proceeding to first + count - 1., to another range beginning at dest.

Note: Complexity: Performs exactly *count* assignments, if count > 0, no assignments otherwise.

Template Parameters

- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *copy_n* algorithm returns a *FwdIter2* . The *copy_n* algorithm returns Iterator in the destination range, pointing past the last element copied if count>0 or result otherwise.

hpx::copy_if

Defined in header [hpx/algorithm.hpp](#)⁶²⁰.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::copy*
- *hpx::copy_n*

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Pred>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx::copy_if(ExPolicy &&policy, FwdIter1
                                                                    first, FwdIter1 last, FwdIter2
                                                                    dest, Pred &&pred)
```

⁶²⁰ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fdbc0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Copies the elements in the range, defined by `[first, last)`, to another range beginning at *dest*. Copies only the elements for which the predicate *f* returns true. The order of the elements that are not removed is preserved. Executed according to the policy.

The assignments in the parallel *copy_if* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *copy_if* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the predicate *f*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *copy_if* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by `[first, last)`. This is an unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

Returns

The *copy_if* algorithm returns a `hpx::future<FwdIter2>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter2* otherwise. The *copy_if* algorithm returns output iterator to the element in the destination range, one past the last element copied.

```
template<typename FwdIter1, typename FwdIter2, typename Pred>
```

FwdIter2 **hpx::copy_if**(*FwdIter1* first, *FwdIter1* last, *FwdIter2* dest, *Pred* &&pred)

Copies the elements in the range, defined by [first, last), to another range beginning at *dest*. Copies only the elements for which the predicate *f* returns true. The order of the elements that are not removed is preserved.

Note: Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the predicate *f*.

Template Parameters

- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *copy_if* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

Returns

The *copy_if* algorithm returns a *FwdIter2*. The *copy_if* algorithm returns output iterator to the element in the destination range, one past the last element copied.

hpx::move

Defined in header `hpx/algorithm.hpp`⁶²¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx::move(ExPolicy &&policy, FwdIter1 first,
                                                                              FwdIter1 last, FwdIter2 dest)
```

Moves the elements in the range [first, last), to another range beginning at *dest*. After this operation the elements in the moved-from range will still contain valid values of the appropriate type, but not necessarily the same values as before the move. Executed according to the policy.

⁶²¹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

The move assignments in the parallel *move* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The move assignments in the parallel *move* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* move assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the move assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *move* algorithm returns a `hpx::future<FwdIter2>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter2* otherwise. The *move* algorithm returns the output iterator to the element in the destination range, one past the last element moved.

```
template<typename FwdIter1, typename FwdIter2>
FwdIter2 hpx::move(FwdIter1 first, FwdIter1 last, FwdIter2 dest)
```

Moves the elements in the range `[first, last)`, to another range beginning at *dest*. After this operation the elements in the moved-from range will still contain valid values of the appropriate type, but not necessarily the same values as before the move.

Note: Complexity: Performs exactly *last - first* move assignments.

Template Parameters

- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.

- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *move* algorithm returns a *FwdIter2*. The *move* algorithm returns the output iterator to the element in the destination range, one past the last element moved.

hpx::experimental::reduction_bit_xor

Defined in header `hpx/algorithm.hpp`⁶²².

See *Public API* for a list of names and headers that are part of the public HPX API.

template<typename T>

constexpr *hpx::parallel::detail::reduction_helper*<T, *std::bit_xor*<T>> *hpx::experimental::reduction_bit_xor*(T &var)

The function template *reduction_bit_xor* returns a reduction object of unspecified type having a value type and encapsulating an identity value for the reduction, it uses *std::bit_xor*{ } as its combiner function, and a live-out object from which the initial value is obtained and into which the final value is stored.

A parallel algorithm uses the *reduction_bit_xor* object by allocating an unspecified number of instances, called views, of the reduction's value type. Each view is initialized with the reduction object's identity value, except that the live-out object (which was initialized by the caller) comprises one of the views. The algorithm passes a reference to a view to each application of an element-access function, ensuring that no two concurrently-executing invocations share the same view. A view can be shared between two applications that do not execute concurrently, but initialization is performed only once per view.

Modifications to the view by the application of element access functions accumulate as partial results. At some point before the algorithm returns, the partial results are combined, two at a time, using the *std::bit_xor*{ } operation until a single value remains, which is then assigned back to the live-out object.

T shall meet the requirements of *CopyConstructible* and *MoveAssignable*.

Note: In order to produce useful results, modifications to the view should be limited to commutative operations closely related to the combiner operation.

Template Parameters

- **T** – The value type to be used by the induction object.
- **Op** – The type of the binary function (object) used to perform the reduction operation.

Parameters

- **var** – [in,out] The life-out value to use for the reduction object. This will hold the reduced value after the algorithm is finished executing.
- **identity** – [in] The identity value to use for the reduction operation. This argument is optional and defaults to a copy of *var*.

Returns

This returns a reduction object of unspecified type having a value type of *T*. When the return value is used by an algorithm, the reference to *var* is used as the live-out object, new views

⁶²² <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

are initialized to a copy of identity, and views are combined by invoking the copy of combiner, passing it the two views to be combined.

```
template<typename T>
constexpr hpx::parallel::detail::reduction_helper<T, std::bit_xor<T>> hpx::experimental::reduction_bit_xor(T
                                                                    &var,
                                                                    T
                                                                    const
                                                                    &iden-
                                                                    tity)
```

hpx::remove

Defined in header [hpx/algorithm.hpp](#)⁶²³.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::remove_if`

```
template<typename FwdIter, typename T = typename std::iterator_traits<FwdIter>::value_type>
FwdIter hpx::remove(FwdIter first, FwdIter last, T const &value)
```

Removes all elements satisfying specific criteria from the range `[first, last)` and returns a past-the-end iterator for the new end of the range. This version removes all elements that are equal to *value*.

The assignments in the parallel *remove* algorithm execute in sequential order in the calling thread.

Note: Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the operator`==()`.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to remove (deduced). This value type must meet the requirements of *CopyConstructible*.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – Specifies the value of elements to remove.

Returns

The *remove* algorithm returns a *FwdIter*. The *remove* algorithm returns the iterator to the new end of the range.

```
template<typename ExPolicy, typename FwdIter, typename T = typename
std::iterator_traits<FwdIter>::value_type>
```

⁶²³ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

`hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::remove(ExPolicy &&policy, FwdIter first, FwdIter last, T const &value)`

Removes all elements satisfying specific criteria from the range `[first, last)` and returns a past-the-end iterator for the new end of the range. This version removes all elements that are equal to *value*. Executed according to the policy.

The assignments in the parallel *remove* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *remove* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the `operator==()`.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to remove (deduced). This value type must meet the requirements of *CopyConstructible*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – Specifies the value of elements to remove.

Returns

The *remove* algorithm returns a `hpx::future<FwdIter>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *remove* algorithm returns the iterator to the new end of the range.

hpx::remove_if

Defined in header `hpx/algorithm.hpp`⁶²⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::remove`

`template<typename FwdIter, typename Pred>`

⁶²⁴ <http://github.com/TheHPXProject/hpx/blob/d402eada7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

FwdIter **hpx::remove_if**(*FwdIter* first, *FwdIter* last, *Pred* &&pred)

Removes all elements satisfying specific criteria from the range [first, last) and returns a past-the-end iterator for the new end of the range. This version removes all elements for which predicate *pred* returns true.

The assignments in the parallel *remove_if* algorithm execute in sequential order in the calling thread.

Note: Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the predicate *pred*.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *remove_if* requires *Pred* to meet the requirements of *Copy-Constructible*.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have *const&*, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

Returns

The *remove_if* algorithm returns a *FwdIter*. The *remove_if* algorithm returns the iterator to the new end of the range.

```
template<typename ExPolicy, typename FwdIter, typename Pred>
```

```
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::remove_if(ExPolicy &&policy, FwdIter  
first, FwdIter last, Pred  
&&pred)
```

Removes all elements satisfying specific criteria from the range [first, last) and returns a past-the-end iterator for the new end of the range. This version removes all elements for which predicate *pred* returns true. Executed according to the policy.

The assignments in the parallel *remove_if* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *remove_if* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the predicate *pred*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *remove_if* requires *Pred* to meet the requirements of *Copy-Constructible*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

Returns

The *remove_if* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *remove_if* algorithm returns the iterator to the new end of the range.

hpx::generate

Defined in header [hpx/algorithm.hpp](#)⁶²⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::generate_n*

```
template<typename ExPolicy, typename FwdIter, typename F>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::generate(ExPolicy &&policy, FwdIter
                                                                    first, FwdIter last, F &&f)
```

Assign each element in range [*first*, *last*) a value generated by the given function object *f*. Executed according to the policy.

⁶²⁵ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

The assignments in the parallel *generate* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *generate* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Exactly *distance(first, last)* invocations of *f* and assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – generator function that will be called. signature of function should be equivalent to the following:

Ret fun ();

The type *Ret* must be such that an object of type *FwdIter* can be dereferenced and assigned a value of type *Ret*.

Returns

The *generate* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise.

```
template<typename FwdIter, typename F>
void hpx::generate(FwdIter first, FwdIter last, F &&f)
```

Assign each element in range [*first*, *last*) a value generated by the given function object *f*.

Note: Complexity: Exactly *distance(first, last)* invocations of *f* and assignments.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – generator function that will be called. signature of function should be equivalent to the following:

`Ret fun();`

The type *Ret* must be such that an object of type *FwdIter* can be dereferenced and assigned a value of type *Ret*.

Returns

The *generate* algorithm returns void.

hpx::generate_n

Defined in header `hpx/algorithm.hpp`⁶²⁶.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::generate*

```
template<typename ExPolicy, typename FwdIter, typename Size, typename F>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::generate_n(ExPolicy &&policy, FwdIter  
first, Size count, F &&f)
```

Assigns each element in range $[first, first+)$ a value generated by the given function object *f*. Executed according to the policy.

The assignments in the parallel *generate_n* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *generate_n* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Exactly *count* invocations of *f* and assignments, for *count* > 0.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Size** – The type of a non-negative integral value specifying the number of items to iterate over.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.

⁶²⁶ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements in the sequence the algorithm will be applied to.
- **f** – Refers to the generator function object that will be called. The signature of the function should be equivalent to

```
Ret fun();
```

The type *Ret* must be such that an object of type *OutputIt* can be dereferenced and assigned a value of type *Ret*.

Returns

The *generate_n* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. *generate_n* algorithm returns iterator one past the last element assigned if *count*>0, first otherwise.

```
template<typename FwdIter, typename Size, typename F>
FwdIter hpx::generate_n(FwdIter first, Size count, F &&f)
```

Assigns each element in range *[first, first+)* a value generated by the given function object *f*.

Note: Complexity: Exactly *count* invocations of *f* and assignments, for *count* > 0.

Template Parameters

- **Size** – The type of a non-negative integral value specifying the number of items to iterate over.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements in the sequence the algorithm will be applied to.
- **f** – Refers to the generator function object that will be called. The signature of the function should be equivalent to

```
Ret fun();
```

The type *Ret* must be such that an object of type *OutputIt* can be dereferenced and assigned a value of type *Ret*.

Returns

The *generate_n* algorithm returns a *FwdIter*. *generate_n* algorithm returns iterator one past the last element assigned if *count*>0, first otherwise.

hpx::min_element

Defined in header [hpx/algorithm.hpp](#)⁶²⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::max_element*
- *hpx::minmax_element*

```
template<typename FwdIter, typename F = hpx::parallel::detail::less>  
FwdIter hpx::min_element(FwdIter first, FwdIter last, F &&f)
```

Finds the smallest element in the range [first, last) using the given comparison function *f*.

The comparisons in the parallel *min_element* algorithm execute in sequential order in the calling thread.

Note: Complexity: Exactly $\max(N-1, 0)$ comparisons, where $N = \text{std::distance}(\text{first}, \text{last})$.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the left argument is less than the right element. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

Returns

The *min_element* algorithm returns *FwdIter*. The *min_element* algorithm returns the iterator to the smallest element in the range [first, last). If several elements in the range are equivalent to the smallest element, returns the iterator to the first such element. Returns last if the range is empty.

```
template<typename ExPolicy, typename FwdIter, typename F = hpx::parallel::detail::less>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::min_element(ExPolicy &&policy, FwdIter  
first, FwdIter last, F &&f)
```

Finds the smallest element in the range [first, last) using the given comparison function *f*. Executed according to the policy.

⁶²⁷ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

The comparisons in the parallel *min_element* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *min_element* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Exactly $\max(N-1, 0)$ comparisons, where $N = \text{std::distance}(\text{first}, \text{last})$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *min_element* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the left argument is less than the right element. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

Returns

The *min_element* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *min_element* algorithm returns the iterator to the smallest element in the range $[\text{first}, \text{last})$. If several elements in the range are equivalent to the smallest element, returns the iterator to the first such element. Returns last if the range is empty.

hpx::max_element

Defined in header `hpx/algorithm.hpp`⁶²⁸.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::min_element*
- *hpx::minmax_element*

⁶²⁸ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

```
template<typename FwdIter, typename F = hpx::parallel::detail::less>  
FwdIter hpx::max_element(FwdIter first, FwdIter last, F &&f)
```

Finds the largest element in the range [first, last) using the given comparison function *f*.

The comparisons in the parallel *min_element* algorithm execute in sequential order in the calling thread.

Note: Complexity: Exactly $\max(N-1, 0)$ comparisons, where $N = \text{std::distance}(\text{first}, \text{last})$.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if left argument is less than the right element. This argument is optional and defaults to `std::less`. the left argument is less than the right element. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

Returns

The *max_element* algorithm returns *FwdIter*. The *max_element* algorithm returns the iterator to the smallest element in the range [first, last). If several elements in the range are equivalent to the smallest element, returns the iterator to the first such element. Returns last if the range is empty.

```
template<typename ExPolicy, typename FwdIter, typename F = hpx::parallel::detail::less>  
hpx::parallel::util::detail::algorithm_result<ExPolicy, FwdIter>::type hpx::max_element(ExPolicy &&policy,  
                                                                                   FwdIter first, FwdIter  
                                                                                   last, F &&f)
```

Removes all elements satisfying specific criteria from the range Finds the largest element in the range [first, last) using the given comparison function *f*. Executed according to the policy.

The comparisons in the parallel *max_element* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *max_element* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Exactly $\max(N-1, 0)$ comparisons, where $N = \text{std::distance}(\text{first}, \text{last})$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *max_element* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if left argument is less than the right element. This argument is optional and defaults to `std::less`. the left argument is less than the right element. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

Returns

The *max_element* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *max_element* algorithm returns the iterator to the smallest element in the range [first, last). If several elements in the range are equivalent to the smallest element, returns the iterator to the first such element. Returns last if the range is empty.

hpx::minmax_element

Defined in header `hpx/algorithm.hpp`⁶²⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::min_element*
- *hpx::max_element*

```
template<typename FwdIter, typename F = hpx::parallel::detail::less>
minmax_element_result<FwdIter> hpx::minmax_element(FwdIter first, FwdIter last, F &&f)
```

Finds the largest element in the range [first, last) using the given comparison function *f*.

The comparisons in the parallel *minmax_element* algorithm execute in sequential order in the calling thread.

⁶²⁹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e77/libs/full/include/include/hpx/algorithm.hpp>

Note: Complexity: At most $\max(\text{floor}(3/2*(N-1)), 0)$ applications of the predicate, where $N = \text{std::distance}(\text{first}, \text{last})$.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the left argument is less than the right element. This argument is optional and defaults to `std::less`. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

Returns

The *minmax_element* algorithm returns a *minmax_element_result<FwdIter>*. The *minmax_element* algorithm returns a pair consisting of an iterator to the smallest element as the min element and an iterator to the largest element as the max element. Returns *minmax_element_result<FwdIter>{first,first}* if the range is empty. If several elements are equivalent to the smallest element, the iterator to the first such element is returned. If several elements are equivalent to the largest element, the iterator to the last such element is returned.

```
template<typename ExPolicy, typename FwdIter, typename F = hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, minmax_element_result<FwdIter>> hpx::minmax_element(ExPolicy
&&pol-
icy,
FwdIter
first,
FwdIter
last,
F
&&f)
```

Finds the largest element in the range `[first, last)` using the given comparison function *f*.

The comparisons in the parallel *minmax_element* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *minmax_element* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most $\max(\text{floor}(3/2*(N-1)), 0)$ applications of the predicate, where $N = \text{std::distance}(\text{first}, \text{last})$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *minmax_element* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the left argument is less than the right element. This argument is optional and defaults to `std::less`. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

Returns

The *minmax_element* algorithm returns a `hpx::future<minmax_element_result<FwdIter>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *minmax_element_result<FwdIter>* otherwise. The *minmax_element* algorithm returns a pair consisting of an iterator to the smallest element as the min element and an iterator to the largest element as the max element. Returns *std::make_pair(first,first)* if the range is empty. If several elements are equivalent to the smallest element, the iterator to the first such element is returned. If several elements are equivalent to the largest element, the iterator to the last such element is returned.

hpx::destroy

Defined in header `hpx/algorithm.hpp`⁶³⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::destroy_n`

template<typename **ExPolicy**, typename **FwdIter**>

⁶³⁰ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

`util::detail::algorithm_result_t<ExPolicy> hpx::destroy(ExPolicy &&policy, FwdIter first, FwdIter last)`

Destroys objects of type `typename iterator_traits<ForwardIt>::value_type` in the range `[first, last)`. Executed according to the policy.

The operations in the parallel *destroy* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The operations in the parallel *destroy* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* operations.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

Returns

The *destroy* algorithm returns a `hpx::future<void>`, if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *void* otherwise.

`template<typename FwdIter>`

`void hpx::destroy(FwdIter first, FwdIter last)`

Destroys objects of type `typename iterator_traits<ForwardIt>::value_type` in the range `[first, last)`.

Note: Complexity: Performs exactly *last - first* operations.

Template Parameters

FwdIter – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

Returns

The *destroy* algorithm returns a *void*

hpx::destroy_n

Defined in header `hpx/algorithm.hpp`⁶³¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::destroy`

```
template<typename ExPolicy, typename FwdIter, typename Size>
util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::destroy_n(ExPolicy &&policy, FwdIter first, Size
                                                                    count)
```

Destroys objects of type `typename iterator_traits<FwdIter>::value_type` in the range `[first, first + count)`. Executed according to the policy.

The operations in the parallel `destroy_n` algorithm invoked with an execution policy object of type `sequenced_policy` execute in sequential order in the calling thread.

The operations in the parallel `destroy_n` algorithm invoked with an execution policy object of type `parallel_policy` or `parallel_task_policy` are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *count* operations, if *count* > 0, no assignments otherwise.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply this algorithm to.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.

Returns

The `destroy_n` algorithm returns a `hpx::future<FwdIter>` if the execution policy is of type `sequenced_task_policy` or `parallel_task_policy` and returns `FwdIter` otherwise. The `destroy_n` algorithm returns the iterator to the element in the source range, one past the last element constructed.

```
template<typename FwdIter, typename Size>
```

⁶³¹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

FwdIter **hpx::destroy_n**(*FwdIter* first, *Size* count)

Destroys objects of type `typename iterator_traits<ForwardIt>::value_type` in the range `[first, first + count)`.

Note: Complexity: Performs exactly *count* operations, if *count* > 0, no assignments otherwise.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply this algorithm to.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.

Returns

The *destroy_n* algorithm returns a *FwdIter*. The *destroy_n* algorithm returns the iterator to the element in the source range, one past the last element constructed.

hpx::make_heap

Defined in header `hpx/algorithm.hpp`⁶³².

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename ExPolicy, typename RndIter, typename Comp>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy> hpx::make_heap(ExPolicy &&policy, RndIter first,
                                                                    RndIter last, Comp &&comp)
```

Constructs a *max heap* in the range `[first, last)`. Executed according to the policy.

The predicate operations in the parallel *make_heap* algorithm invoked with an execution policy object of type *sequential_execution_policy* executes in sequential order in the calling thread.

The comparison operations in the parallel *make_heap* algorithm invoked with an execution policy object of type *parallel_execution_policy* or *parallel_task_execution_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: at most $(3*N)$ comparisons where $N = \text{distance}(\text{first}, \text{last})$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **RndIter** – The type of the source iterators used for algorithm. This iterator must meet the requirements for a random access iterator.

⁶³² <http://github.com/TheHPXProject/hpx/blob/d402cead7405eedff56d454c9380fbd0c8044e77/libs/full/include/include/hpx/algorithm.hpp>

- **Comp** – Comparison function object (i.e. an object that satisfies the requirements of Compare) which returns true if the first argument is less than the second. The signature of the comparison function should be equivalent to the following:

```
bool cmp(const Type1 &a, const Type2 &b);
```

While the signature does not need to have `const &`, the function must not modify the objects passed to it and must be able to accept all values of type (possibly *const*) *Type1* and *Type2* regardless of value category (thus, *Type1 &* is not allowed, nor is *Type1* unless for *Type1* a move is equivalent to a copy. The types *Type1* and *Type2* must be such that an object of type *RandomIt* can be dereferenced and then implicitly converted to both of them.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.
- **comp** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second. The signature of the function should be equivalent to

```
bool comp(const Type &a, const Type &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *RndIter* can be dereferenced and then implicitly converted to *Type*.

Returns

The *make_heap* algorithm returns a *hpx::future<void>* if the execution policy is of type *task_execution_policy* and returns *void* otherwise.

```
template<typename ExPolicy, typename RndIter>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy> hpx::make_heap(ExPolicy &&policy, RndIter first,
                                                                    RndIter last)
```

Constructs a *max heap* in the range [first, last). Uses the operator < for comparisons. Executed according to the policy.

The predicate operations in the parallel *make_heap* algorithm invoked with an execution policy object of type *sequential_execution_policy* executes in sequential order in the calling thread.

The comparison operations in the parallel *make_heap* algorithm invoked with an execution policy object of type *parallel_execution_policy* or *parallel_task_execution_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: at most (3*N) comparisons where *N* = distance(first, last).

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.

- **RndIter** – The type of the source iterators used for algorithm. This iterator must meet the requirements for a random access iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.

Returns

The *make_heap* algorithm returns a *hpx::future<void>* if the execution policy is of type *task_execution_policy* and returns *void* otherwise.

```
template<typename RndIter, typename Comp>
void hpx::make_heap(RndIter first, RndIter last, Comp &&comp)
```

Constructs a *max heap* in the range [first, last).

Note: Complexity: at most $(3*N)$ comparisons where $N = \text{distance}(\text{first}, \text{last})$.

Template Parameters

- **RndIter** – The type of the source iterators used for algorithm. This iterator must meet the requirements for a random access iterator.
- **Comp** – Comparison function object (i.e. an object that satisfies the requirements of Compare) which returns true if the first argument is less than the second. The signature of the comparison function should be equivalent to the following:

```
bool cmp(const Type1 &a, const Type2 &b);
```

While the signature does not need to have `const &`, the function must not modify the objects passed to it and must be able to accept all values of type (possibly *const*) *Type1* and *Type2* regardless of value category (thus, *Type1 &* is not allowed, nor is *Type1* unless for *Type1* a move is equivalent to a copy. The types *Type1* and *Type2* must be such that an object of type *RandomIt* can be dereferenced and then implicitly converted to both of them.

Parameters

- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.
- **comp** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second. The signature of the function should be equivalent to

```
bool comp(const Type &a, const Type &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *RndIter* can be dereferenced and then implicitly converted to *Type*.

Returns

The *make_heap* algorithm returns a *void*.

```
template<typename RndIter>
void hpx::make_heap(RndIter first, RndIter last)
    Constructs a max heap in the range [first, last).
```

Note: Complexity: at most $(3*N)$ comparisons where $N = \text{distance}(\text{first}, \text{last})$.

Template Parameters

RndIter – The type of the source iterators used for algorithm. This iterator must meet the requirements for a random access iterator.

Parameters

- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.

Returns

The *make_heap* algorithm returns a *void*.

hpx::experimental::reduction_max

Defined in header `hpx/algorithm.hpp`⁶³³.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename T>
constexpr hpx::parallel::detail::reduction_helper<T, hpx::parallel::detail::max_of<T>> hpx::experimental::reduction_max(T
&va
```

The function template *reduction_max* returns a reduction object of unspecified type having a value type and encapsulating an identity value for the reduction, it uses `std::max_of{ }` as its combiner function, and a live-out object from which the initial value is obtained and into which the final value is stored.

A parallel algorithm uses the *reduction_max* object by allocating an unspecified number of instances, called views, of the reduction's value type. Each view is initialized with the reduction object's identity value, except that the live-out object (which was initialized by the caller) comprises one of the views. The algorithm passes a reference to a view to each application of an element-access function, ensuring that no two concurrently-executing invocations share the same view. A view can be shared between two applications that do not execute concurrently, but initialization is performed only once per view.

Modifications to the view by the application of element access functions accumulate as partial results. At some point before the algorithm returns, the partial results are combined, two at a time, using the `std::max_of{ }` operation until a single value remains, which is then assigned back to the live-out object.

T shall meet the requirements of *CopyConstructible* and *MoveAssignable*.

Note: In order to produce useful results, modifications to the view should be limited to commutative operations closely related to the combiner operation.

Template Parameters

⁶³³ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **T** – The value type to be used by the induction object.
- **Op** – The type of the binary function (object) used to perform the reduction operation.

Parameters

- **var** – [in,out] The life-out value to use for the reduction object. This will hold the reduced value after the algorithm is finished executing.
- **identity** – [in] The identity value to use for the reduction operation. This argument is optional and defaults to a copy of *var*.

Returns

This returns a reduction object of unspecified type having a value type of *T*. When the return value is used by an algorithm, the reference to *var* is used as the live-out object, new views are initialized to a copy of identity, and views are combined by invoking the copy of combiner, passing it the two views to be combined.

```
template<typename T>
constexpr hpx::parallel::detail::reduction_helper<T, hpx::parallel::detail::max_of<T>> hpx::experimental::reduction_max(T
&va
T
con
&id
tity;
```

hpx::uninitialized_fill

Defined in header `hpx/algorithm.hpp`⁶³⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::uninitialized_fill_n`

```
template<typename FwdIter, typename T>
void hpx::uninitialized_fill(FwdIter first, FwdIter last, T const &value)
```

Copies the given *value* to an uninitialized memory area, defined by the range [first, last). If an exception is thrown during the initialization, the function has no effects.

Note: Complexity: Linear in the distance between *first* and *last*

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **T** – The type of the value to be assigned (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

⁶³⁴ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fdbc0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **value** – The value to be assigned.

Returns

The *uninitialized_fill* algorithm returns nothing

```
template<typename ExPolicy, typename FwdIter, typename T>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy> hpx::uninitialized_fill(ExPolicy &&policy, FwdIter
                                                                    first, FwdIter last, T const
                                                                    &value)
```

Copies the given *value* to an uninitialized memory area, defined by the range [first, last). If an exception is thrown during the initialization, the function has no effects. Executed according to the policy.

The initializations in the parallel *uninitialized_fill* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The initializations in the parallel *uninitialized_fill* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Linear in the distance between *first* and *last*

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **T** – The type of the value to be assigned (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – The value to be assigned.

Returns

The *uninitialized_fill* algorithm returns a *hpx::future<void>*, if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns nothing otherwise.

hpx::uninitialized_fill_n

Defined in header [hpx/algorithm.hpp](#)⁶³⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::uninitialized_fill`

```
template<typename FwdIter, typename Size, typename T>
```

```
FwdIter hpx::uninitialized_fill_n(FwdIter first, Size count, T const &value)
```

Copies the given *value* value to the first count elements in an uninitialized memory area beginning at first. If an exception is thrown during the initialization, the function has no effects.

Note: Complexity: Performs exactly *count* assignments, if count > 0, no assignments otherwise.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **T** – The type of the value to be assigned (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **value** – The value to be assigned.

Returns

The *uninitialized_fill_n* algorithm returns a returns *FwdIter*. The *uninitialized_fill_n* algorithm returns the output iterator to the element in the range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter, typename Size, typename T>
```

```
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::uninitialized_fill_n(ExPolicy  
                                                                                      &&policy,  
                                                                                      FwdIter first,  
                                                                                      Size count, T  
                                                                                      const &value)
```

Copies the given *value* value to the first count elements in an uninitialized memory area beginning at first. If an exception is thrown during the initialization, the function has no effects. Executed according to the policy.

The initializations in the parallel *uninitialized_fill_n* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The initializations in the parallel *uninitialized_fill_n* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

⁶³⁵ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Note: Complexity: Performs exactly *count* assignments, if *count* > 0, no assignments otherwise.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **T** – The type of the value to be assigned (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **value** – The value to be assigned.

Returns

The *uninitialized_fill_n* algorithm returns a `hpx::future<FwdIter>`, if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `FwdIter` otherwise. The *uninitialized_fill_n* algorithm returns the output iterator to the element in the range, one past the last element copied.

hpx::experimental::reduction_multiplies

Defined in header `hpx/algorithm.hpp`⁶³⁶.

See *Public API* for a list of names and headers that are part of the public HPX API.

template<typename T>

constexpr `hpx::parallel::detail::reduction_helper<T, std::multiplies<T>>` `hpx::experimental::reduction_multiplies(T &var)`

The function template *reduction_multiplies* returns a reduction object of unspecified type having a value type and encapsulating an identity value for the reduction, it uses `std::multiplies{ }` as its combiner function, and a live-out object from which the initial value is obtained and into which the final value is stored.

A parallel algorithm uses the *reduction_multiplies* object by allocating an unspecified number of instances, called views, of the reduction's value type. Each view is initialized with the reduction object's identity value, except that the live-out object (which was initialized by the caller) comprises one of the views. The algorithm passes a reference to a view to each application of an element-access function, ensuring that no two concurrently-executing invocations share the same view. A view can be shared between two applications that do not execute concurrently, but initialization is performed only once per view.

Modifications to the view by the application of element access functions accumulate as partial results. At some point before the algorithm returns, the partial results are combined, two at a time, using the `std::multiplies{ }` operation until a single value remains, which is then assigned back to the live-out object.

⁶³⁶ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e77/libs/full/include/include/hpx/algorithm.hpp>

T shall meet the requirements of *CopyConstructible* and *MoveAssignable*.

Note: In order to produce useful results, modifications to the view should be limited to commutative operations closely related to the combiner operation.

Template Parameters

- **T** – The value type to be used by the induction object.
- **Op** – The type of the binary function (object) used to perform the reduction operation.

Parameters

- **var** – [in,out] The life-out value to use for the reduction object. This will hold the reduced value after the algorithm is finished executing.
- **identity** – [in] The identity value to use for the reduction operation. This argument is optional and defaults to a copy of *var*.

Returns

This returns a reduction object of unspecified type having a value type of *T*. When the return value is used by an algorithm, the reference to *var* is used as the live-out object, new views are initialized to a copy of identity, and views are combined by invoking the copy of combiner, passing it the two views to be combined.

```
template<typename T>
constexpr hpx::parallel::detail::reduction_helper<T, std::multiplies<T>> hpx::experimental::reduction_multiplies(T
                                                                    &var,
                                                                    T
                                                                    const
                                                                    &iden-
                                                                    tity)
```

hpx::is_heap

Defined in header `hpx/algorithm.hpp`⁶³⁷.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::is_heap_until`

```
template<typename ExPolicy, typename RandIter, typename Comp = hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx::is_heap(ExPolicy &&policy, RandIter first,
                                                                    RandIter last, Comp &&comp =
                                                                    Comp())
```

Returns whether the range is max heap. That is, true if the range is max heap, false otherwise. The function uses the given comparison function object *comp* (defaults to using operator<()). Executed according to the policy.

comp has to induce a strict weak ordering on the values.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

⁶³⁷ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Linear in the distance between *first* and *last*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **RandIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – *comp* is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that *comp* will not apply any non-constant function through the dereferenced iterator.

Returns

The *is_heap* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *is_heap* algorithm returns whether the range is max heap. That is, true if the range is max heap, false otherwise.

```
template<typename RandIter, typename Comp = hpx::parallel::detail::less>
bool hpx::is_heap(RandIter first, RandIter last, Comp &&comp = Comp())
```

Returns whether the range is max heap. That is, true if the range is max heap, false otherwise. The function uses the given comparison function object *comp* (defaults to using operator<()).

comp has to induce a strict weak ordering on the values.

Note: Complexity: Linear in the distance between *first* and *last*.

Template Parameters

- **RandIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

- **comp** – *comp* is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that *comp* will not apply any non-constant function through the dereferenced iterator.

Returns

The *is_heap* a bool. The *is_heap* algorithm returns whether the range is max heap. That is, true if the range is max heap, false otherwise.

hpx::is_heap_until

Defined in header [hpx/algorithm.hpp](#)⁶³⁸.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::is_heap*

```
template<typename ExPolicy, typename RandIter, typename Comp = hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, RandIter> hpx::is_heap_until(ExPolicy &&policy,
                                             RandIter first, RandIter
                                             last, Comp &&comp =
                                             Comp())
```

Returns the upper bound of the largest range beginning at *first* which is a max heap. That is, the last iterator *it* for which range [first, it) is a max heap. The function uses the given comparison function object *comp* (defaults to using operator<()). Executed according to the policy.

comp has to induce a strict weak ordering on the values.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Linear in the distance between *first* and *last*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **RandIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.

⁶³⁸ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – *comp* is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that *comp* will not apply any non-constant function through the dereferenced iterator.

Returns

The *is_heap_until* algorithm returns a *hpx::future<RandIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *RandIter* otherwise. The *is_heap_until* algorithm returns the upper bound of the largest range beginning at first which is a max heap. That is, the last iterator *it* for which range [first, it) is a max heap.

```
template<typename RandIter, typename Comp = hpx::parallel::detail::less>
RandIter hpx::is_heap_until(RandIter first, RandIter last, Comp &&comp = Comp())
```

Returns the upper bound of the largest range beginning at *first* which is a max heap. That is, the last iterator *it* for which range [first, it) is a max heap. The function uses the given comparison function object *comp* (defaults to using operator<()).

comp has to induce a strict weak ordering on the values.

Note: Complexity: Linear in the distance between *first* and *last*.

Template Parameters

- **RandIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – *comp* is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that *comp* will not apply any non-constant function through the dereferenced iterator.

Returns

The *is_heap_until* algorithm returns a *RandIter*. The *is_heap_until* algorithm returns the upper bound of the largest range beginning at first which is a max heap. That is, the last iterator *it* for which range [first, it) is a max heap.

hpx::reverse

Defined in header [hpx/algorithm.hpp](#)⁶³⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::reverse_copy*

template<typename **BidirIter**>

void **hpx::reverse**(*BidirIter* first, *BidirIter* last)

Reverses the order of the elements in the range [first, last). Behaves as if applying *std::iter_swap* to every pair of iterators first+i, (last-i) - 1 for each non-negative i < (last-first)/2.

The assignments in the parallel *reverse* algorithm execute in sequential order in the calling thread.

Note: Complexity: Linear in the distance between *first* and *last*.

Template Parameters

BidirIter – The type of the source iterators used (deduced). This iterator type must meet the requirements of a bidirectional iterator.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

Returns

The *reverse* algorithm returns *void*.

template<typename **ExPolicy**, typename **BidirIter**>

hpx::parallel::util::detail::algorithm_result_t<*ExPolicy*, void> **hpx::reverse**(*ExPolicy* &&policy, *BidirIter* first, *BidirIter* last)

Reverses the order of the elements in the range [first, last). Behaves as if applying *std::iter_swap* to every pair of iterators first+i, (last-i) - 1 for each non-negative i < (last-first)/2. Executed according to the policy.

The assignments in the parallel *reverse* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *reverse* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Linear in the distance between *first* and *last*.

Template Parameters

⁶³⁹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **BidirIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a bidirectional iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

Returns

The *reverse* algorithm returns a `hpx::future<void>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *void* otherwise.

hpx::reverse_copy

Defined in header `hpx/algorithm.hpp`⁶⁴⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::reverse`

template<typename **BidirIter**, typename **OutIter**>

OutIter **hpx::reverse_copy**(*BidirIter* first, *BidirIter* last, *OutIter* dest)

Copies the elements from the range [first, last) to another range beginning at dest in such a way that the elements in the new range are in reverse order. Behaves as if by executing the assignment $*(dest + (last - first) - 1 - i) = *(first + i)$ once for each non-negative $i < (last - first)$. If the source and destination ranges (that is, [first, last) and [dest, dest+(last-first)) respectively) overlap, the behavior is undefined.

The assignments in the parallel *reverse_copy* algorithm execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **BidirIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a bidirectional iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the begin of the destination range.

⁶⁴⁰ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Returns

The *reverse_copy* algorithm returns an *OutIter*. The *reverse_copy* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename BidirIter, typename FwdIter>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::reverse_copy(ExPolicy &&policy,
                                                                                      BidirIter first, BidirIter
                                                                                      last, FwdIter dest)
```

Copies the elements from the range [first, last) to another range beginning at dest in such a way that the elements in the new range are in reverse order. Behaves as if by executing the assignment $*(dest + (last - first) - 1 - i) = *(first + i)$ once for each non-negative $i < (last - first)$. If the source and destination ranges (that is, [first, last) and [dest, dest+(last-first)) respectively) overlap, the behavior is undefined. Executed according to the policy.

The assignments in the parallel *reverse_copy* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *reverse_copy* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **BidirIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a bidirectional iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the begin of the destination range.

Returns

The *reverse_copy* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *reverse_copy* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

hpx::count

Defined in header `hpx/algorithm.hpp`⁶⁴¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::count_if`

```
template<typename ExPolicy, typename FwdIter, typename T>
util::detail::algorithm_result<ExPolicy, typename std::iterator_traits<FwdIter>::difference_type>::type hpx::count(ExPolicy
&&pol-
icy,
FwdIter
first,
FwdIter
last,
T
const
&value)
```

Returns the number of elements in the range `[first, last)` satisfying a specific criteria. This version counts the elements that are equal to the given *value*. Executed according to the policy.

The comparisons in the parallel *count* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *last - first* comparisons.

Note: The comparisons in the parallel *count* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the comparisons.
- **FwdIter** – The type of the source iterator used (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to search for (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – The value to search for.

⁶⁴¹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Returns

The *count* algorithm returns a *hpx::future<difference_type>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *difference_type* otherwise (where *difference_type* is defined by *std::iterator_traits<FwdIterB>::difference_type*). The *count* algorithm returns the number of elements satisfying the given criteria.

```
template<typename InIter, typename T>  
std::iterator_traits<InIter>::difference_type hpx::count(InIter first, InIter last, T const &value)
```

Returns the number of elements in the range [first, last) satisfying a specific criteria. This version counts the elements that are equal to the given *value*.

Note: Complexity: Performs exactly *last - first* comparisons.

Template Parameters

- **InIter** – The type of the source iterator used (deduced). This iterator type must meet the requirements of an input iterator.
- **T** – The type of the value to search for (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – The value to search for.

Returns

The *count* algorithm returns a *difference_type* (where *difference_type* is defined by *std::iterator_traits<InIter>::difference_type*). The *count* algorithm returns the number of elements satisfying the given criteria.

hpx::count_if

Defined in header [hpx/algorithm.hpp](#)⁶⁴².

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::count*

```
template<typename ExPolicy, typename FwdIter, typename F>  
util::detail::algorithm_result<ExPolicy, typename std::iterator_traits<FwdIter>::difference_type>::type hpx::count_if(ExPolicy  
    &&pol-  
    icy,  
    FwdIter  
    first,  
    FwdIter  
    last,  
    F  
    &&f)
```

⁶⁴² <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Returns the number of elements in the range `[first, last)` satisfying a specific criteria. This version counts elements for which predicate `f` returns true. Executed according to the policy.

Note: Complexity: Performs exactly *last - first* applications of the predicate.

Note: The assignments in the parallel *count_if* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

Note: The assignments in the parallel *count_if* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the comparisons.
- **FwdIter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *count_if* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by `[first, last)`. This is an unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

Returns

The *count_if* algorithm returns `hpx::future<difference_type>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *difference_type* otherwise (where *difference_type* is defined by `std::iterator_traits<FwdIter>::difference_type`). The *count* algorithm returns the number of elements satisfying the given criteria.

```
template<typename InIter, typename F>
std::iterator_traits<InIter>::difference_type hpx::count_if(InIter first, InIter last, F &&f)
```

Returns the number of elements in the range `[first, last)` satisfying a specific criteria. This version counts elements for which predicate `f` returns true.

Note: Complexity: Performs exactly *last - first* applications of the predicate.

Template Parameters

- **InIter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *count_if* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

Returns

The *count_if* algorithm returns *difference_type* (where a *difference_type* is defined by *std::iterator_traits<InIter>::difference_type*). The *count* algorithm returns the number of elements satisfying the given criteria.

hpx::is_partitioned

Defined in header `hpx/algorithm.hpp`⁶⁴³.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename FwdIter, typename Pred>
```

```
bool hpx::is_partitioned(FwdIter first, FwdIter last, Pred &&pred)
```

Determines if the range [first, last) is partitioned.

Note: Complexity: at most (*N*) predicate evaluations where *N* = distance(first, last).

Template Parameters

- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of the function/function object to use (deduced).

Parameters

⁶⁴³ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.
- **pred** – Refers to the unary predicate which returns true for elements expected to be found in the beginning of the range. The signature of the function should be equivalent to

```
bool pred(const Type &a);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

Returns

The *is_partitioned* algorithm returns *bool*. The *is_partitioned* algorithm returns true if each element in the sequence for which *pred* returns true precedes those for which *pred* returns false. Otherwise *is_partitioned* returns false. If the range *[first, last)* contains less than two elements, the function is always true.

```
template<typename ExPolicy, typename FwdIter, typename Pred>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx::is_partitioned(ExPolicy &&policy,
                                                                                      FwdIter first, FwdIter last,
                                                                                      Pred &&pred)
```

Determines if the range *[first, last)* is partitioned. Executed according to the policy.

The predicate operations in the parallel *is_partitioned* algorithm invoked with an execution policy object of type *sequenced_policy* executes in sequential order in the calling thread.

The comparison operations in the parallel *is_partitioned* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: at most (*N*) predicate evaluations where *N* = *distance*(*first*, *last*).

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of the function/function object to use (deduced). *Pred* must be *CopyConstructible* when using a parallel policy.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.

- **pred** – Refers to the unary predicate which returns true for elements expected to be found in the beginning of the range. The signature of the function should be equivalent to

```
bool pred(const Type &a);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

Returns

The *is_partitioned* algorithm returns a *hpx::future<bool>* if the execution policy is of type *task_execution_policy* and returns *bool* otherwise. The *is_partitioned* algorithm returns true if each element in the sequence for which *pred* returns true precedes those for which *pred* returns false. Otherwise *is_partitioned* returns false. If the range [first, last) contains less than two elements, the function is always true.

hpx::fill

Defined in header `hpx/algorithm.hpp`⁶⁴⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::fill_n*

```
template<typename ExPolicy, typename FwdIter, typename T>
util::detail::algorithm_result_t<ExPolicy> hpx::fill(ExPolicy &&policy, FwdIter first, FwdIter last, T value)
```

Assigns the given value to the elements in the range [first, last). Executed according to the policy.

The comparisons in the parallel *fill* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *fill* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be assigned (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

⁶⁴⁴ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e77/libs/full/include/include/hpx/algorithm.hpp>

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – The value to be assigned.

Returns

The *fill* algorithm returns a `hpx::future<void>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *difference_type* otherwise (where *difference_type* is defined by *void*).

```
template<typename FwdIter, typename T>
void hpx::fill(FwdIter first, FwdIter last, T value)
```

Assigns the given value to the elements in the range [first, last).

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be assigned (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – The value to be assigned.

Returns

The *fill* algorithm returns a *void*.

hpx::fill_n

Defined in header `hpx/algorithm.hpp`⁶⁴⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::fill`

```
template<typename ExPolicy, typename FwdIter, typename Size, typename T>
util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::fill_n(ExPolicy &&policy, FwdIter first, Size count, T value)
```

Assigns the given value value to the first count elements in the range beginning at first if count > 0. Does nothing otherwise. Executed according to the policy.

The comparisons in the parallel *fill_n* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

⁶⁴⁵ <http://github.com/TheHPXProject/hpx/blob/d402cead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

The comparisons in the `parallel_fill_n` algorithm invoked with an execution policy object of type `parallel_policy` or `parallel_task_policy` are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *count* assignments, for *count* > 0.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **T** – The type of the value to be assigned (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **value** – The value to be assigned.

Returns

The `fill_n` algorithm returns a `hpx::future<void>` if the execution policy is of type `sequenced_task_policy` or `parallel_task_policy` and returns `difference_type` otherwise (where `difference_type` is defined by `void`).

```
template<typename FwdIter, typename Size, typename T>
FwdIter hpx::fill_n(FwdIter first, Size count, T value)
```

Assigns the given value *value* to the first *count* elements in the range beginning at *first* if *count* > 0. Does nothing otherwise.

Note: Complexity: Performs exactly *count* assignments, for *count* > 0.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **T** – The type of the value to be assigned (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **value** – The value to be assigned.

Returns

The *fill_n* algorithm returns a *FwdIter*.

hpx::starts_with

Defined in header [hpx/algorithm.hpp](#)⁶⁴⁶.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename InIter1, typename InIter2, typename Pred = hpx::parallel::detail::equal_to>
bool hpx::starts_with(InIter1 first1, InIter1 last1, InIter2 first2, InIter2 last2, Pred &&pred = Pred())
```

Checks whether the second range defined by [first1, last1) matches the prefix of the first range defined by [first2, last2)

The assignments in the parallel *starts_with* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: Linear: at most min(N1, N2) applications of the predicate.

Template Parameters

- **InIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **InIter2** – The type of the destination iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Pred** – The binary predicate that compares the elements. This defaults to *hpx::parallel::detail::equal_to*.

Parameters

- **first1** – Refers to the beginning of the source range.
- **last1** – Sentinel value referring to the end of the source range.
- **first2** – Refers to the beginning of the destination range.
- **last2** – Sentinel value referring to the end of the destination range.
- **pred** – Specifies the binary predicate function (or function object) which will be invoked for comparison of the elements in the two ranges.

Returns

The *starts_with* algorithm returns *bool*. The *starts_with* algorithm returns a boolean with the value true if the second range matches the prefix of the first range, false otherwise.

```
template<typename ExPolicy, typename InIter1, typename InIter2, typename Pred = ranges::equal_to>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx::starts_with(ExPolicy &&policy, InIter1
    first1, InIter1 last1, InIter2
    first2, InIter2 last2, Pred
    &&pred = Pred())
```

Checks whether the second range defined by [first1, last1) matches the prefix of the first range defined by [first2, last2). Executed according to the policy.

⁶⁴⁶ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

The assignments in the parallel *starts_with* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: Linear: at most $\min(N1, N2)$ applications of the predicate.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **InIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **InIter2** – The type of the destination iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Pred** – The binary predicate that compares the elements. This defaults to *ranges::equal_to*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the source range.
- **last1** – Sentinel value referring to the end of the source range.
- **first2** – Refers to the beginning of the destination range.
- **last2** – Sentinel value referring to the end of the destination range.
- **pred** – Specifies the binary predicate function (or function object) which will be invoked for comparison of the elements in the two ranges.

Returns

The *starts_with* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *starts_with* algorithm returns a boolean with the value true if the second range matches the prefix of the first range, false otherwise.

hpx::remove_copy

Defined in header [hpx/algorithm.hpp](#)⁶⁴⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::remove_copy_if*

```
template<typename InIter, typename OutIter, typename T = typename std::iterator_traits<InIter>::value_type>
OutIter hpx::remove_copy(InIter first, InIter last, OutIter dest, T const &value)
```

Copies the elements in the range, defined by [first, last), to another range beginning at *dest*. Copies only the elements for which the comparison operator returns false when compare to value. The order of the elements that are not removed is preserved.

⁶⁴⁷ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Effects: Copies all the elements referred to by the iterator it in the range [first,last) for which the following corresponding conditions do not hold: `*it == value`

The assignments in the parallel *remove_copy* algorithm execute in sequential order in the calling thread.

Note: Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the predicate *pred*, here comparison operator.

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **T** – The type that the result of dereferencing *FwdIter1* is compared to.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **value** – Value to be removed.

Returns

The *remove_copy* algorithm returns an *OutIter*. The *remove_copy* algorithm returns the iterator to the element past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename T = typename
std::iterator_traits<InIter>::value_type>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx::remove_copy(ExPolicy &&policy,
                                                                    FwdIter1 first, FwdIter1
                                                                    last, FwdIter2 dest, T const
                                                                    &value)
```

Copies the elements in the range, defined by [first, last), to another range beginning at *dest*. Copies only the elements for which the comparison operator returns false when compare to value. The order of the elements that are not removed is preserved. Executed according to the policy.

Effects: Copies all the elements referred to by the iterator it in the range [first,last) for which the following corresponding conditions do not hold: `*it == value`

The assignments in the parallel *remove_copy* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *remove_copy* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the predicate *pred*, here comparison operator.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type that the result of dereferencing FwdIter1 is compared to.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **value** – Value to be removed.

Returns

The *remove_copy* algorithm returns a *hpx::future<FwdIter2>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter2* otherwise. The *remove_copy* algorithm returns the iterator to the element past the last element copied.

hpx::remove_copy_if

Defined in header [hpx/algorithm.hpp](#)⁶⁴⁸.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::remove_copy*

```
template<typename InIter, typename OutIter, typename Pred>  
OutIter hpx::remove_copy_if(InIter first, InIter last, OutIter dest, Pred &&pred)
```

Copies the elements in the range, defined by [first, last), to another range beginning at *dest*. Copies only the elements for which the predicate *pred* returns false. The order of the elements that are not removed is preserved.

Effects: Copies all the elements referred to by the iterator it in the range [first,last) for which the following corresponding conditions do not hold: INVOKE(pred, *it) != false.

The assignments in the parallel *remove_copy_if* algorithm execute in sequential order in the calling thread.

⁶⁴⁸ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Note: Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the predicate *pred*.

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of the function/function object to use (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the elements to be removed. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

Returns

The *remove_copy_if* algorithm returns an *OutIter*. The *remove_copy_if* algorithm returns the iterator to the element past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Pred>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx::remove_copy_if(ExPolicy &&policy,
                                                                                      FwdIter1 first,
                                                                                      FwdIter1 last,
                                                                                      FwdIter2 dest, Pred
                                                                                      &&pred)
```

Copies the elements in the range, defined by [first, last), to another range beginning at *dest*. Copies only the elements for which the predicate *pred* returns false. The order of the elements that are not removed is preserved. Executed according to the policy.

Effects: Copies all the elements referred to by the iterator it in the range [first,last) for which the following corresponding conditions do not hold: `INVOKE(pred, *it) != false`.

The assignments in the parallel *remove_copy_if* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *remove_copy_if* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the predicate *pred*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *remove_copy_if* requires *Pred* to meet the requirements of *CopyConstructible*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the elements to be removed. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

Returns

The *remove_copy_if* algorithm returns a *hpx::future<FwdIter2>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter2* otherwise. The *remove_copy_if* algorithm returns the iterator to the element past the last element copied.

hpx::shift_right

Defined in header `hpx/algorithm.hpp`⁶⁴⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename FwdIter, typename Size>
FwdIter hpx::shift_right(FwdIter first, FwdIter last, Size n)
```

Shifts the elements in the range [first, last) by *n* positions towards the end of the range. For every integer *i* in [0, last - first - *n*), moves the element originally at position first + *i* to position first

⁶⁴⁹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- $n + i$.

The assignment operations in the parallel *shift_right* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

Note: Complexity: At most $(\text{last} - \text{first}) - n$ assignments.

Note: The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable*.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of positions to shift by.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **n** – Refers to the number of positions to shift.

Returns

The *shift_right* algorithm returns *FwdIter*. The *shift_right* algorithm returns an iterator to the beginning of the resulting range. Specifically: If n is 0 or $n \geq (\text{last} - \text{first})$, does nothing and returns *first* and *last* respectively. Otherwise returns $\text{first} + n$.

```
template<typename ExPolicy, typename FwdIter, typename Size>
hpx::parallel::util::detail::algorithm_result<ExPolicy, FwdIter> hpx::shift_right(ExPolicy &&policy, FwdIter
                                                                    first, FwdIter last, Size n)
```

Shifts the elements in the range $[\text{first}, \text{last})$ by n positions towards the end of the range. For every integer i in $[0, \text{last} - \text{first} - n)$, moves the element originally at position $\text{first} + i$ to position first

- $n + i$. Executed according to the policy.

The assignment operations in the parallel *shift_right* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignment operations in the parallel *shift_right* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most $(\text{last} - \text{first}) - n$ assignments.

Note: The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of positions to shift by.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **n** – Refers to the number of positions to shift.

Returns

The *shift_right* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *shift_right* algorithm returns an iterator to the beginning of the resulting range. Specifically: If *n* is 0, does nothing and returns *first*. If *n* >= (last - first), does nothing and returns *last*. Otherwise returns *first* + *n*.

hpx::stable_sort

Defined in header [hpx/algorithm.hpp](#)⁶⁵⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename RandomIt, typename Comp = hpx::parallel::detail::less, typename Proj = hpx::identity>  
void hpx::stable_sort(RandomIt first, RandomIt last, Comp &&comp = Comp(), Proj &&proj = Proj())
```

Sorts the elements in the range [first, last) in ascending order. The relative order of equal elements is preserved. The function uses the given comparison function object *comp* (defaults to using operator<()).

A sequence is sorted with respect to a comparator *comp* and a projection *proj* if for every iterator *i* pointing to the sequence and every non-negative integer *n* such that *i* + *n* is a valid iterator pointing to an element of the sequence, and INVOKE(comp, INVOKE(proj, *(i + n)), INVOKE(proj, *i)) == false.

comp has to induce a strict weak ordering on the values.

The assignments in the parallel *stable_sort* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: $O(N \log(N))$, where $N = \text{std::distance}(\text{first}, \text{last})$ comparisons.

Template Parameters

- **RandomIt** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*.

⁶⁵⁰ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e77/libs/full/include/include/hpx/algorithm.hpp>

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – `comp` is a callable object. The return value of the INVOKE operation applied to an object of type `Comp`, when contextually converted to `bool`, yields `true` if the first argument of the call is less than the second, and `false` otherwise. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate `comp` is invoked.

Returns

The `stable_sort` algorithm returns `void`.

```
template<typename ExPolicy, typename RandomIt, typename Comp = hpx::parallel::detail::less, typename Proj =
hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy> hpx::stable_sort(ExPolicy &&policy, RandomIt first,
                                                                    RandomIt last, Comp &&comp =
                                                                    Comp(), Proj &&proj = Proj())
```

Sorts the elements in the range `[first, last)` in ascending order. The relative order of equal elements is preserved. The function uses the given comparison function object `comp` (defaults to using `operator<()`). Executed according to the policy.

A sequence is sorted with respect to a comparator `comp` and a projection `proj` if for every iterator `i` pointing to the sequence and every non-negative integer `n` such that `i + n` is a valid iterator pointing to an element of the sequence, and `INVOKE(comp, INVOKE(proj, *(i + n)), INVOKE(proj, *i)) == false`.

`comp` has to induce a strict weak ordering on the values.

The application of function objects in parallel algorithm invoked with an execution policy object of type `sequenced_policy` execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type `parallel_policy` or `parallel_task_policy` are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: $O(N \log(N))$, where $N = \text{std::distance}(\text{first}, \text{last})$ comparisons.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **RandomIt** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – `comp` is a callable object. The return value of the INVOKE operation applied to an object of type `Comp`, when contextually converted to `bool`, yields `true` if the first argument of the call is less than the second, and `false` otherwise. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate `comp` is invoked.

Returns

The `stable_sort` algorithm returns a `hpx::future<void>` if the execution policy is of type `sequenced_task_policy` or `parallel_task_policy` and returns `void` otherwise.

hpx::reduce

Defined in header `hpx/algorithm.hpp`⁶⁵¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename ExPolicy, typename FwdIter, typename F, typename T = typename
std::iterator_traits<FwdIter>::value_type>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> hpx::reduce(ExPolicy &&policy, FwdIter first,
                                                                    FwdIter last, T init, F &&f)
```

Returns GENERALIZED_SUM(`f`, `init`, `*first`, ..., `*(first + (last - first) - 1)`). Executed according to the policy.

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *copy_if* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate `f`.

Note: GENERALIZED_SUM(`op`, `a1`, ..., `aN`) is defined as follows:

- `a1` when `N` is 1
 - `op(GENERALIZED_SUM(op, b1, ..., bK), GENERALIZED_SUM(op, bM, ..., bN))`, where:
 - `b1`, ..., `bN` may be any permutation of `a1`, ..., `aN` and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

⁶⁵¹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source begin and end iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *reduce* requires *F* to meet the requirements of *CopyConstructible*.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`. The types *Type1* *Ret* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to any of those types.

Returns

The *reduce* algorithm returns a `hpx::future<T>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *T* otherwise. The *reduce* algorithm returns the result of the generalized sum over the elements given by the input range [first, last).

```
template<typename ExPolicy, typename FwdIter, typename T = typename
std::iterator_traits<FwdIter>::value_type>
util::detail::algorithm_result_t<ExPolicy, T> hpx::reduce(ExPolicy &&policy, FwdIter first, FwdIter last, T init)
Returns GENERALIZED_SUM(+, init, *first, ..., *(first + (last - first) - 1)). Executed according to the policy.
```

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *copy_if* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the operator+().

Note: GENERALIZED_SUM(+, a1, ..., aN) is defined as follows:

- a_1 when N is 1
 - $op(\text{GENERALIZED_SUM}(+, b_1, \dots, b_K), \text{GENERALIZED_SUM}(+, b_M, \dots, b_N))$, where:
 - $b_1, \dots, b_N$ may be any permutation of $a_1, \dots, a_N$ and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source begin and end iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.

Returns

The *reduce* algorithm returns a `hpx::future<T>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns T otherwise. The *reduce* algorithm returns the result of the generalized sum (applying operator $+$ ()) over the elements given by the input range $[first, last)$.

```
template<typename ExPolicy, typename FwdIter>
```

```
hpx::parallel::util::detail::algorithm_result<ExPolicy, typename std::iterator_traits<FwdIter>::value_type>::type hpx::reduce(ExPol  
&&po  
icy,  
FwdIt  
first,  
FwdIt  
last)
```

Returns $\text{GENERALIZED_SUM}(+, T(), *first, \dots, *(first + (last - first) - 1))$. Executed according to the policy.

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the operator+().

Note: The type of the initial value (and the result type) T is determined from the `value_type` of the used `FwdIter`.

Note: `GENERALIZED_SUM(+, a1, ..., aN)` is defined as follows:

- a_1 when N is 1
 - `op(GENERALIZED_SUM(+, b1, ..., bK), GENERALIZED_SUM(+, bM, ..., bN))`, where:
 - $b_1, \dots, b_N$ may be any permutation of $a_1, \dots, a_N$ and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source begin and end iterators used (deduced). This iterator type must meet the requirements of a forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

Returns

The *reduce* algorithm returns a `hpx::future<T>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns T otherwise (where T is the `value_type` of `FwdIter`). The *reduce* algorithm returns the result of the generalized sum (applying operator+()) over the elements given by the input range `[first, last)`.

```
template<typename FwdIter, typename F, typename T = typename std::iterator_traits<FwdIter>::value_type>
T hpx::reduce(FwdIter first, FwdIter last, T init, F &&f)
```

Returns `GENERALIZED_SUM(f, init, *first, ..., *(first + (last - first) - 1))`. Executed according to the policy.

The difference between *reduce* and *accumulate* is that the behavior of *reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate f .

Note: `GENERALIZED_SUM(op, a1, ..., aN)` is defined as follows:

- a_1 when N is 1
- `op(GENERALIZED_SUM(op, b1, ..., bK), GENERALIZED_SUM(op, bM, ..., bN))`, where:

- $b_1, \dots, b_N$ may be any permutation of $a_1, \dots, a_N$ and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

- **FwdIter** – The type of the source begin and end iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *reduce* requires *F* to meet the requirements of *CopyConstructible*.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`. The types *Type1* *Ret* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to any of those types.

Returns

The *reduce* algorithm returns *T*. The *reduce* algorithm returns the result of the generalized sum over the elements given by the input range [first, last).

```
template<typename FwdIter, typename T = typename std::iterator_traits<FwdIter>::value_type>
T hpx::reduce(FwdIter first, FwdIter last, T init)
```

Returns GENERALIZED_SUM(+, init, *first, ..., *(first + (last - first) - 1)). Executed according to the policy.

The difference between *reduce* and *accumulate* is that the behavior of *reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the operator+().

Note: GENERALIZED_SUM(+, $a_1, \dots, a_N$) is defined as follows:

- a_1 when N is 1
 - $\text{op}(\text{GENERALIZED_SUM}(+, b_1, \dots, b_K), \text{GENERALIZED_SUM}(+, b_M, \dots, b_N))$, where:
 - $b_1, \dots, b_N$ may be any permutation of $a_1, \dots, a_N$ and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

- **FwdIter** – The type of the source begin and end iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.

Returns

The *reduce* algorithm returns a *T*. The *reduce* algorithm returns the result of the generalized sum (applying operator+()) over the elements given by the input range [first, last).

```
template<typename FwdIter>
```

```
std::iterator_traits<FwdIter>::value_type hpx::reduce(FwdIter first, FwdIter last)
```

Returns GENERALIZED_SUM(+, T(), *first, ..., *(first + (last - first) - 1)). Executed according to the policy.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the operator+().

Note: The type of the initial value (and the result type) *T* is determined from the value_type of the used *FwdIter*.

Note: GENERALIZED_SUM(+, a1, ..., aN) is defined as follows:

- a1 when N is 1
 - op(GENERALIZED_SUM(+, b1, ..., bK), GENERALIZED_SUM(+, bM, ..., bN)), where:
 - b1, ..., bN may be any permutation of a1, ..., aN and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

FwdIter – The type of the source begin and end iterators used (deduced). This iterator type must meet the requirements of an input iterator.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

Returns

The *reduce* algorithm returns *T* (where *T* is the value_type of *FwdIter*). The *reduce* algorithm returns the result of the generalized sum (applying operator+()) over the elements given by the input range [first, last).

hpx::uninitialized_value_construct

Defined in header [hpx/algorithm.hpp](#)⁶⁵².

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::uninitialized_value_construct_n`

template<typename **FwdIter**>

void `hpx::uninitialized_value_construct`(*FwdIter* first, *FwdIter* last)

Constructs objects of type `typename iterator_traits<ForwardIt>::value_type` in the uninitialized storage designated by the range by value-initialization. If an exception is thrown during the initialization, the function has no effects.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

FwdIter – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

Returns

The `uninitialized_value_construct` algorithm returns nothing

template<typename **ExPolicy**, typename **FwdIter**>

`hpx::parallel::util::detail::algorithm_result_t<ExPolicy>` `hpx::uninitialized_value_construct`(*ExPolicy* &&policy, *FwdIter* first, *FwdIter* last)

Constructs objects of type `typename iterator_traits<ForwardIt>::value_type` in the uninitialized storage designated by the range by value-initialization. If an exception is thrown during the initialization, the function has no effects. Executed according to the policy.

The assignments in the parallel `uninitialized_value_construct` algorithm invoked with an execution policy object of type `sequenced_policy` execute in sequential order in the calling thread.

The assignments in the parallel `uninitialized_value_construct` algorithm invoked with an execution policy object of type `parallel_policy` or `parallel_task_policy` are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

⁶⁵² <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

Returns

The *uninitialized_value_construct* algorithm returns a *hpx::future<void>*, if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns nothing otherwise.

hpx::uninitialized_value_construct_n

Defined in header [hpx/algorithm.hpp](#)⁶⁵³.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::uninitialized_value_construct*

template<typename **FwdIter**, typename **Size**>

FwdIter **hpx::uninitialized_value_construct_n**(*FwdIter* first, *Size* count)

Constructs objects of type *typename iterator_traits<ForwardIt> ::value_type* in the uninitialized storage designated by the range [first, first + count) by value-initialization. If an exception is thrown during the initialization, the function has no effects.

Note: Complexity: Performs exactly *count* assignments, if *count* > 0, no assignments otherwise.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.

Returns

The *uninitialized_value_construct_n* algorithm returns a *returns FwdIter*. The *uninitialized_value_construct_n* algorithm returns the iterator to the element in the source range, one past the last element constructed.

template<typename **ExPolicy**, typename **FwdIter**, typename **Size**>

⁶⁵³ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::uninitialized_value_construct_n(ExPolicy &&policy, FwdIter first, Size count)

Constructs objects of type `typename iterator_traits<ForwardIt>::value_type` in the uninitialized storage designated by the range `[first, first + count)` by value-initialization. If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized_value_construct_n* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized_value_construct_n* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *count* assignments, if *count* > 0, no assignments otherwise.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.

Returns

The *uninitialized_value_construct_n* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *uninitialized_value_construct_n* algorithm returns the iterator to the element in the source range, one past the last element constructed.

hpx::uninitialized_copy

Defined in header [hpx/algorithm.hpp](#)⁶⁵⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::uninitialized_copy_n`

template<typename **InIter**, typename **FwdIter**>

FwdIter **hpx::uninitialized_copy**(*InIter* first, *InIter* last, *FwdIter* dest)

Copies the elements in the range, defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the copy operation, the function has no effects.

The assignments in the parallel *uninitialized_copy* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *uninitialized_copy* algorithm returns *FwdIter*. The *uninitialized_copy* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

template<typename **ExPolicy**, typename **FwdIter1**, typename **FwdIter2**>

hpx::parallel::util::detail::algorithm_result_t<*ExPolicy*, *FwdIter2*> **hpx::uninitialized_copy**(*ExPolicy* &&policy, *FwdIter1* first, *FwdIter1* last, *FwdIter2* dest)

Copies the elements in the range, defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the copy operation, the function has no effects. Executed according to the policy.

The assignments in the parallel *uninitialized_copy* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

⁶⁵⁴ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

The assignments in the parallel *uninitialized_copy* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *uninitialized_copy* algorithm returns a *hpx::future<FwdIter2>*, if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter2* otherwise. The *uninitialized_copy* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

hpx::uninitialized_copy_n

Defined in header [hpx/algorithm.hpp](#)⁶⁵⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::uninitialized_copy*

template<typename **InIter**, typename **Size**, typename **FwdIter**>
FwdIter **hpx::uninitialized_copy_n**(*InIter* first, *Size* count, *FwdIter* dest)

Copies the elements in the range [first, first + count), starting from first and proceeding to first + count - 1., to another range beginning at dest. If an exception is thrown during the copy operation, the function has no effects.

The assignments in the parallel *uninitialized_copy_n* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *count* assignments, if count > 0, no assignments otherwise.

⁶⁵⁵ <http://github.com/TheHPXProject/hpx/blob/d402cead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *uninitialized_copy_n* algorithm returns a *FwdIter*. The *uninitialized_copy_n* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Size, typename FwdIter2>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx::uninitialized_copy_n(ExPolicy
                                                                                               &&policy,
                                                                                               FwdIter1 first,
                                                                                               Size count,
                                                                                               FwdIter2 dest)
```

Copies the elements in the range [first, first + count), starting from first and proceeding to first + count - 1., to another range beginning at dest. If an exception is thrown during the copy operation, the function has no effects.

The assignments in the parallel *uninitialized_copy_n* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized_copy_n* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *count* assignments, if count > 0, no assignments otherwise.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *uninitialized_copy_n* algorithm returns a `hpx::future<FwdIter2>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter2* otherwise. The *uninitialized_copy_n* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

hpx::experimental::for_loop

Defined in header `hpx/algorithm.hpp`⁶⁵⁶.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::experimental::for_loop_strided`
- `hpx::experimental::for_loop_n`
- `hpx::experimental::for_loop_n_strided`

template<typename **I**, typename ...**Args**>

void `hpx::experimental::for_loop`(`std::decay_t<I>` first, *I* last, *Args*&&... args)

The `for_loop` implements loop functionality over a range specified by integral or iterator bounds. For the iterator case, these algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator.

The execution of `for_loop` without specifying an execution policy is equivalent to specifying `hpx::execution::seq` as the execution policy.

Requires: *I* shall be an integral type or meet the requirements of an input iterator type. The *args* parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function, *f*. *f* shall meet the requirements of *MoveConstructible*.

Effects: Applies *f* to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the *args* parameter pack. The length of the input sequence is *last - first*.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

⁶⁵⁶ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Remarks: If f returns a result, the result is ignored.

Note: As described in the C++ standard, arithmetic on non-random-access iterators is performed using `advance` and `distance`.

Note: The order of the elements of the input sequence is important for determining ordinal position of an application of f , even though the applications themselves may be unordered.

Template Parameters

- **I** – The type of the iteration variable. This could be an (forward) iterator type or an integral type.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last) should expose a signature equivalent to:

```
<ignored> pred(I const& a, ...);
```

The signature does not need to have `const&`. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

```
template<typename ExPolicy, typename I, typename ...Args>
auto hpx::experimental::for_loop(ExPolicy &&policy, std::decay_t<I> first, I last, Args&&... args)
```

The `for_loop` implements loop functionality over a range specified by integral or iterator bounds. For the iterator case, these algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator. Executed according to the policy.

Requires: I shall be an integral type or meet the requirements of an input iterator type. The *args* parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function, f . f shall meet the requirements of *MoveConstructible*.

Effects: Applies f to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the *args* parameter pack. The length of the input sequence is $last - first$.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding f , an additional argument is passed to each application of f as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of f in the input sequence.

Complexity: Applies f exactly once for each element of the input sequence.

Remarks: If f returns a result, the result is ignored.

Note: As described in the C++ standard, arithmetic on non-random-access iterators is performed using advance and distance.

Note: The order of the elements of the input sequence is important for determining ordinal position of an application of f , even though the applications themselves may be unordered.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **I** – The type of the iteration variable. This could be an (forward) iterator type or an integral type.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last) should expose a signature equivalent to:

```
<ignored> pred(I const& a, ...);
```

The signature does not need to have const&. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

Returns

The *for_loop* algorithm returns a `hpx::future<void>` if the execution policy is of type `hpx::execution::sequenced_task_policy` or `hpx::execution::parallel_task_policy` and returns `void` otherwise.

hpx::experimental::for_loop_strided

Defined in header [hpx/algorithm.hpp](#)⁶⁵⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::experimental::for_loop`
- `hpx::experimental::for_loop_n`
- `hpx::experimental::for_loop_n_strided`

template<typename **I**, typename **S**, typename ...**Args**>

void `hpx::experimental::for_loop_strided`(`std::decay_t<I>` first, `I` last, `S` stride, `Args&&...` args)

The `for_loop_strided` implements loop functionality over a range specified by integral or iterator bounds. For the iterator case, these algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator.

The execution of `for_loop` without specifying an execution policy is equivalent to specifying `hpx::execution::seq` as the execution policy.

Requires: *I* shall be an integral type or meet the requirements of an input iterator type. The *args* parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function, *f*. *f* shall meet the requirements of *MoveConstructible*.

Effects: Applies *f* to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the *args* parameter pack. The length of the input sequence is *last - first*.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

Remarks: If *f* returns a result, the result is ignored.

Note: As described in the C++ standard, arithmetic on non-random-access iterators is performed using advance and distance.

Note: The order of the elements of the input sequence is important for determining ordinal position of an application of *f*, even though the applications themselves may be unordered.

Template Parameters

⁶⁵⁷ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **I** – The type of the iteration variable. This could be an (forward) iterator type or an integral type.
- **S** – The type of the stride variable. This should be an integral type.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **stride** – Refers to the stride of the iteration steps. This shall have non-zero value and shall be negative only if I has integral type or meets the requirements of a bidirectional iterator.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last) should expose a signature equivalent to:

```
<ignored> pred(I const& a, ...);
```

The signature does not need to have `const&`. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

```
template<typename ExPolicy, typename I, typename S, typename ...Args>  
auto hpx::experimental::for_loop_strided(ExPolicy &&policy, std::decay_t<I> first, I last, S stride,  
                                         Args&&... args)
```

The `for_loop_strided` implements loop functionality over a range specified by integral or iterator bounds. For the iterator case, these algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator. Executed according to the policy.

Requires: *I* shall be an integral type or meet the requirements of an input iterator type. The *args* parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function, *f*. *f* shall meet the requirements of *MoveConstructible*.

Effects: Applies *f* to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the *args* parameter pack. The length of the input sequence is *last - first*.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

Remarks: If *f* returns a result, the result is ignored.

Note: As described in the C++ standard, arithmetic on non-random-access iterators is performed using advance and distance.

Note: The order of the elements of the input sequence is important for determining ordinal position of an application of f , even though the applications themselves may be unordered.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **I** – The type of the iteration variable. This could be an (forward) iterator type or an integral type.
- **S** – The type of the stride variable. This should be an integral type.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **stride** – Refers to the stride of the iteration steps. This shall have non-zero value and shall be negative only if I has integral type or meets the requirements of a bidirectional iterator.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last) should expose a signature equivalent to:

```
<ignored> pred(I const& a, ...);
```

The signature does not need to have const&. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

Returns

The *for_loop_strided* algorithm returns a `hpx::future<void>` if the execution policy is of type `hpx::execution::sequenced_task_policy` or `hpx::execution::parallel_task_policy` and returns `void` otherwise.

hpx::experimental::for_loop_n

Defined in header `hpx/algorithm.hpp`⁶⁵⁸.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::experimental::for_loop`
- `hpx::experimental::for_loop_strided`
- `hpx::experimental::for_loop_n_strided`

template<typename **I**, typename **Size**, typename ...**Args**>

void `hpx::experimental::for_loop_n`(*I* first, *Size* size, *Args*&&... args)

The `for_loop_n` implements loop functionality over a range specified by integral or iterator bounds. For the iterator case, these algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator.

The execution of `for_loop_n` without specifying an execution policy is equivalent to specifying `hpx::execution::seq` as the execution policy.

Requires: *I* shall be an integral type or meet the requirements of an input iterator type. The *args* parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function, *f*. *f* shall meet the requirements of *MoveConstructible*.

Effects: Applies *f* to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the *args* parameter pack. The length of the input sequence is *last - first*.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

Remarks: If *f* returns a result, the result is ignored.

Note: As described in the C++ standard, arithmetic on non-random-access iterators is performed using advance and distance.

Note: The order of the elements of the input sequence is important for determining ordinal position of an application of *f*, even though the applications themselves may be unordered.

Template Parameters

⁶⁵⁸ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **I** – The type of the iteration variable. This could be an (forward) iterator type or an integral type.
- **Size** – The type of a non-negative integral value specifying the number of items to iterate over.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **size** – Refers to the number of items the algorithm will be applied to.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last) should expose a signature equivalent to:

```
<ignored> pred(I const& a, ...);
```

The signature does not need to have const&. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

```
template<typename ExPolicy, typename I, typename Size, typename ...Args>
auto hpx::experimental::for_loop_n(ExPolicy &&policy, I first, Size size, Args&&... args)
```

The *for_loop_n* implements loop functionality over a range specified by integral or iterator bounds. For the iterator case, these algorithms resemble *for_each* from the Parallelism TS, but leave to the programmer when and if to dereference the iterator. Executed according to the policy.

Requires: *I* shall be an integral type or meet the requirements of an input iterator type. The *args* parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function, *f*. *f* shall meet the requirements of *MoveConstructible*.

Effects: Applies *f* to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the *args* parameter pack. The length of the input sequence is *last - first*.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

Remarks: If *f* returns a result, the result is ignored.

Note: As described in the C++ standard, arithmetic on non-random-access iterators is performed using advance and distance.

Note: The order of the elements of the input sequence is important for determining ordinal position of an application of f , even though the applications themselves may be unordered.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **I** – The type of the iteration variable. This could be an (forward) iterator type or an integral type.
- **Size** – The type of a non-negative integral value specifying the number of items to iterate over.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **size** – Refers to the number of items the algorithm will be applied to.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last) should expose a signature equivalent to:

```
<ignored> pred(I const& a, ...);
```

The signature does not need to have const&. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

Returns

The *for_loop_n* algorithm returns a *hpx::future<void>* if the execution policy is of type *hpx::execution::sequenced_task_policy* or *hpx::execution::parallel_task_policy* and returns *void* otherwise.

hpx::experimental::for_loop_n_strided

Defined in header `hpx/algorithm.hpp`⁶⁵⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::experimental::for_loop`
- `hpx::experimental::for_loop_strided`
- `hpx::experimental::for_loop_n`

template<typename **I**, typename **Size**, typename **S**, typename ...**Args**>

void `hpx::experimental::for_loop_n_strided`(*I* first, *Size* size, *S* stride, *Args*&&... args)

The `for_loop_n_strided` implements loop functionality over a range specified by integral or iterator bounds. For the iterator case, these algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator.

The execution of `for_loop` without specifying an execution policy is equivalent to specifying `hpx::execution::seq` as the execution policy.

Requires: *I* shall be an integral type or meet the requirements of an input iterator type. The *args* parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function, *f*. *f* shall meet the requirements of *MoveConstructible*.

Effects: Applies *f* to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the *args* parameter pack. The length of the input sequence is *last - first*.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

Remarks: If *f* returns a result, the result is ignored.

Note: As described in the C++ standard, arithmetic on non-random-access iterators is performed using advance and distance.

Note: The order of the elements of the input sequence is important for determining ordinal position of an application of *f*, even though the applications themselves may be unordered.

Template Parameters

⁶⁵⁹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **I** – The type of the iteration variable. This could be an (forward) iterator type or an integral type.
- **Size** – The type of a non-negative integral value specifying the number of items to iterate over.
- **S** – The type of the stride variable. This should be an integral type.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **size** – Refers to the number of items the algorithm will be applied to.
- **stride** – Refers to the stride of the iteration steps. This shall have non-zero value and shall be negative only if I has integral type or meets the requirements of a bidirectional iterator.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last) should expose a signature equivalent to:

`<ignored> pred(I const& a, ...);`

The signature does not need to have const&. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

```
template<typename ExPolicy, typename I, typename Size, typename S, typename ...Args>  
auto hpx::experimental::for_loop_n_strided(ExPolicy &&policy, I first, Size size, S stride, Args&&... args)
```

The `for_loop_n_strided` implements loop functionality over a range specified by integral or iterator bounds. For the iterator case, these algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator. Executed according to the policy.

Requires: *I* shall be an integral type or meet the requirements of an input iterator type. The *args* parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function, *f*. *f* shall meet the requirements of *MoveConstructible*.

Effects: Applies *f* to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the *args* parameter pack. The length of the input sequence is *last - first*.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

Remarks: If f returns a result, the result is ignored.

Note: As described in the C++ standard, arithmetic on non-random-access iterators is performed using advance and distance.

Note: The order of the elements of the input sequence is important for determining ordinal position of an application of f , even though the applications themselves may be unordered.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **I** – The type of the iteration variable. This could be an (forward) iterator type or an integral type.
- **Size** – The type of a non-negative integral value specifying the number of items to iterate over.
- **S** – The type of the stride variable. This should be an integral type.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **size** – Refers to the number of items the algorithm will be applied to.
- **stride** – Refers to the stride of the iteration steps. This shall have non-zero value and shall be negative only if I has integral type or meets the requirements of a bidirectional iterator.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last) should expose a signature equivalent to:

```
<ignored> pred(I const& a, ...);
```

The signature does not need to have const&. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

Returns

The *for_loop_n_strided* algorithm returns a `hpx::future<void>` if the execution policy is of type `hpx::execution::sequenced_task_policy` or `hpx::execution::parallel_task_policy` and returns `void` otherwise.

hpx::uninitialized_move

Defined in header [hpx/algorithm.hpp](#)⁶⁶⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::uninitialized_move_n`

template<typename **InIter**, typename **FwdIter**>

FwdIter **hpx::uninitialized_move**(*InIter* first, *InIter* last, *FwdIter* dest)

Moves the elements in the range, defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the initialization, some objects in [first, last) are left in a valid but unspecified state.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *uninitialized_move* algorithm returns *FwdIter*. The *uninitialized_move* algorithm returns the output iterator to the element in the destination range, one past the last element moved.

template<typename **ExPolicy**, typename **FwdIter1**, typename **FwdIter2**>

hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> **hpx::uninitialized_move**(*ExPolicy* &&policy, *FwdIter1* first, *FwdIter1* last, *FwdIter2* dest)

Moves the elements in the range, defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the initialization, some objects in [first, last) are left in a valid but unspecified state. Executed according to the policy.

The assignments in the parallel *uninitialized_move* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized_move* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

⁶⁶⁰ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *uninitialized_move* algorithm returns a `hpx::future<FwdIter2>`, if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter2* otherwise. The *uninitialized_move* algorithm returns the output iterator to the element in the destination range, one past the last element moved.

hpx::uninitialized_move_n

Defined in header `hpx/algorithm.hpp`⁶⁶¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::uninitialized_move`

```
template<typename InIter, typename Size, typename FwdIter>
std::pair<InIter, FwdIter> hpx::uninitialized_move_n(InIter first, Size count, FwdIter dest)
```

Moves the elements in the range `[first, first + count)`, starting from `first` and proceeding to `first + count - 1`, to another range beginning at `dest`. If an exception is thrown during the initialization, some objects in `[first, first + count)` are left in a valid but unspecified state.

Note: Complexity: Performs exactly *count* movements, if *count* > 0, no move operations otherwise.

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.

⁶⁶¹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *uninitialized_move_n* algorithm returns a `std::pair<InIter,FwdIter>`. The *uninitialized_move_n* algorithm returns A pair whose first element is an iterator to the element past the last element moved in the source range, and whose second element is an iterator to the element past the last element moved in the destination range.

```
template<typename ExPolicy, typename FwdIter1, typename Size, typename FwdIter2>
parallel::util::detail::algorithm_result<ExPolicy, std::pair<FwdIter1, FwdIter2>>::type hpx::uninitialized_move_n(ExPolicy
                                                                                                     &&pol-
                                                                                                     icy,
                                                                                                     FwdIter1
                                                                                                     first,
                                                                                                     Size
                                                                                                     count,
                                                                                                     FwdIter2
                                                                                                     dest)
```

Moves the elements in the range `[first, first + count)`, starting from *first* and proceeding to *first* + *count* - 1., to another range beginning at *dest*. If an exception is thrown during the initialization, some objects in `[first, first + count)` are left in a valid but unspecified state. Executed according to the policy.

The assignments in the parallel *uninitialized_move_n* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized_move_n* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *count* movements, if *count* > 0, no move operations otherwise.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *uninitialized_move_n* algorithm returns a `hpx::future<std::pair<FwdIter1,FwdIter2>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `std::pair<FwdIter1,FwdIter2>` otherwise. The *uninitialized_move_n* algorithm returns A pair whose first element is an iterator to the element past the last element moved in the source range, and whose second element is an iterator to the element past the last element moved in the destination range.

hpx::merge

Defined in header `hpx/algorithm.hpp`⁶⁶².

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::inplace_merge*

```
template<typename ExPolicy, typename RandIter1, typename RandIter2, typename RandIter3, typename
Comp = hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, RandIter3> hpx::merge(ExPolicy &&policy, RandIter1
first1, RandIter1 last1, RandIter2
first2, RandIter2 last2, RandIter3
dest, Comp &&comp = Comp())
```

Merges two sorted ranges [first1, last1) and [first2, last2) into one sorted range beginning at *dest*. The order of equivalent elements in each of the original two ranges is preserved. For equivalent elements in the original two ranges, the elements from the first range precede the elements from the second range. The destination range cannot overlap with either of the input ranges. Executed according to the policy.

The assignments in the parallel *merge* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *merge* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs $O(\text{std::distance}(\text{first1}, \text{last1}) + \text{std::distance}(\text{first2}, \text{last2}))$ applications of the comparison *comp* and each projection.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.

⁶⁶² <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **RandIter1** – The type of the source iterators used (deduced) representing the first sorted range. This iterator type must meet the requirements of a random access iterator.
- **RandIter2** – The type of the source iterators used (deduced) representing the second sorted range. This iterator type must meet the requirements of a random access iterator.
- **RandIter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *merge* requires *Comp* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the first range of elements the algorithm will be applied to.
- **last1** – Refers to the end of the first range of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second range of elements the algorithm will be applied to.
- **last2** – Refers to the end of the second range of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **comp** – *comp* is a callable object which returns true if the first argument is less than the second, and false otherwise. The signature of this comparison should be equivalent to:

```
bool comp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *RandIter1* and *RandIter2* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

Returns

The *merge* algorithm returns a `hpx::future<RandIter3>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *RandIter3* otherwise. The *merge* algorithm returns the destination iterator to the end of the *dest* range.

```
template<typename RandIter1, typename RandIter2, typename RandIter3, typename Comp =  
hpx::parallel::detail::less>  
RandIter3 hpx::merge(RandIter1 first1, RandIter1 last1, RandIter2 first2, RandIter2 last2, RandIter3 dest, Comp  
                    &&comp = Comp())
```

Merges two sorted ranges `[first1, last1)` and `[first2, last2)` into one sorted range beginning at *dest*. The order of equivalent elements in each of the original two ranges is preserved. For equivalent elements in the original two ranges, the elements from the first range precede the elements from the second range. The destination range cannot overlap with either of the input ranges.

Note: Complexity: Performs $O(\text{std::distance}(\text{first1}, \text{last1}) + \text{std::distance}(\text{first2}, \text{last2}))$ applications of the comparison *comp* and each projection.

Template Parameters

- **RandIter1** – The type of the source iterators used (deduced) representing the first sorted range. This iterator type must meet the requirements of a random access iterator.

- **RandIter2** – The type of the source iterators used (deduced) representing the second sorted range. This iterator type must meet the requirements of a random access iterator.
- **RandIter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *merge* requires *Comp* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

Parameters

- **first1** – Refers to the beginning of the first range of elements the algorithm will be applied to.
- **last1** – Refers to the end of the first range of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second range of elements the algorithm will be applied to.
- **last2** – Refers to the end of the second range of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **comp** – *comp* is a callable object which returns true if the first argument is less than the second, and false otherwise. The signature of this comparison should be equivalent to:

```
bool comp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *RandIter1* and *RandIter2* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

Returns

The *merge* algorithm returns a *RandIter3*. The *merge* algorithm returns the destination iterator to the end of the *dest* range.

hpx::inplace_merge

Defined in header `hpx/algorithm.hpp`⁶⁶³.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::merge*

```
template<typename ExPolicy, typename RandIter, typename Comp = hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy> hpx::inplace_merge(ExPolicy &&policy, RandIter first,
                                                                              RandIter middle, RandIter last,
                                                                              Comp &&comp = Comp())
```

Merges two consecutive sorted ranges [first, middle) and [middle, last) into one sorted range [first, last). The order of equivalent elements in each of the original two ranges is preserved. For equivalent elements in the original two ranges, the elements from the first range precede the elements from the second range. Executed according to the policy.

⁶⁶³ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

The assignments in the parallel *inplace_merge* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *inplace_merge* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs $O(\text{std::distance}(\text{first}, \text{last}))$ applications of the comparison *comp* and each projection.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **RandIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *inplace_merge* requires *Comp* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the first sorted range the algorithm will be applied to.
- **middle** – Refers to the end of the first sorted range and the beginning of the second sorted range the algorithm will be applied to.
- **last** – Refers to the end of the second sorted range the algorithm will be applied to.
- **comp** – *comp* is a callable object which returns true if the first argument is less than the second, and false otherwise. The signature of this comparison should be equivalent to:

```
bool comp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *RandIter* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

Returns

The *inplace_merge* algorithm returns a *hpx::future<void>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns void otherwise. The *inplace_merge* algorithm returns the source iterator *last*.

```
template<typename RandIter, typename Comp = hpx::parallel::detail::less>
```

```
void hpx::inplace_merge(RandIter first, RandIter middle, RandIter last, Comp &&comp = Comp())
```

Merges two consecutive sorted ranges [first, middle) and [middle, last) into one sorted range [first, last). The order of equivalent elements in each of the original two ranges is preserved. For equivalent elements in the original two ranges, the elements from the first range precede the elements from the second range.

Note: Complexity: Performs $O(\text{std::distance}(\text{first}, \text{last}))$ applications of the comparison *comp* and each projection.

Template Parameters

- **RandIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *inplace_merge* requires *Comp* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

Parameters

- **first** – Refers to the beginning of the first sorted range the algorithm will be applied to.
- **middle** – Refers to the end of the first sorted range and the beginning of the second sorted range the algorithm will be applied to.
- **last** – Refers to the end of the second sorted range the algorithm will be applied to.
- **comp** – *comp* is a callable object which returns true if the first argument is less than the second, and false otherwise. The signature of this comparison should be equivalent to:

```
bool comp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *RandIter* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

Returns

The *inplace_merge* algorithm returns a *void*. The *inplace_merge* algorithm returns the source iterator *last*.

hpx::replace

Defined in header `hpx/algorithm.hpp`⁶⁶⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::replace_if*
- *hpx::replace_copy*
- *hpx::replace_copy_if*

```
template<typename InIter, typename T = typename std::iterator_traits<InIter>::value_type>
void hpx::replace(InIter first, InIter last, T const &old_value, T const &new_value)
```

Replaces all elements satisfying specific criteria with *new_value* in the range [first, last).

Effects: Substitutes elements referred by the iterator it in the range [first, last) with *new_value*, when the following corresponding conditions hold: **it == old_value*

The assignments in the parallel *replace* algorithm execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *last - first* assignments.

⁶⁶⁴ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **T** – The type of the old and new values to replace (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **old_value** – Refers to the old value of the elements to replace.
- **new_value** – Refers to the new value to use as the replacement.

Returns

The *replace* algorithm returns a *void*.

```
template<typename ExPolicy, typename FwdIter, typename T = typename
std::iterator_traits<FwdIter>::value_type>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, void> hpx::replace(ExPolicy &&policy, FwdIter first,
                                                                    FwdIter last, T const &old_value, T
                                                                    const &new_value)
```

Replaces all elements satisfying specific criteria with *new_value* in the range [first, last). Executed according to the policy.

Effects: Substitutes elements referred by the iterator it in the range [first, last) with *new_value*, when the following corresponding conditions hold: **it == old_value*

The assignments in the parallel *replace* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *replace* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **T** – The type of the old and new values to replace (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.

- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **old_value** – Refers to the old value of the elements to replace.
- **new_value** – Refers to the new value to use as the replacement.

Returns

The *replace* algorithm returns a *hpx::future<void>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *void* otherwise.

hpx::replace_if

Defined in header [hpx/algorithm.hpp](#)⁶⁶⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::replace*
- *hpx::replace_copy*
- *hpx::replace_copy_if*

```
template<typename Iter, typename Pred, typename T = typename std::iterator_traits<Iter>::value_type>
void hpx::replace_if(Iter first, Iter last, Pred &&pred, T const &new_value)
```

Replaces all elements satisfying specific criteria (for which predicate *pred* returns true) with *new_value* in the range [first, last).

Effects: Substitutes elements referred by the iterator it in the range [first, last) with *new_value*, when the following corresponding conditions hold: `INVOKE(f, *it) != false`

The assignments in the parallel *replace_if* algorithm execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *last - first* applications of the predicate.

Template Parameters

- **Iter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. (deduced).
- **T** – The type of the new values to replace (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns

⁶⁶⁵ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

true for the elements which need to be replaced. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

- **new_value** – Refers to the new value to use as the replacement.

Returns

The *replace_if* algorithm returns *void*.

```
template<typename ExPolicy, typename FwdIter, typename Pred, typename T = typename  
std::iterator_traits<FwdIter>::value_type>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, void> hpx::replace_if(ExPolicy &&policy, FwdIter  
first, FwdIter last, Pred &&pred,  
T const &new_value)
```

Replaces all elements satisfying specific criteria (for which predicate *f* returns true) with *new_value* in the range [first, last). Executed according to the policy.

Effects: Substitutes elements referred by the iterator it in the range [first, last) with *new_value*, when the following corresponding conditions hold: `INVOKE(f, *it) != false`

The assignments in the parallel *replace_if* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *replace_if* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* applications of the predicate.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. (deduced).
- **T** – The type of the new values to replace (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the elements which need to be replaced. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **new_value** – Refers to the new value to use as the replacement.

Returns

The *replace_if* algorithm returns a *hpx::future<void>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *void* otherwise.

hpx::replace_copy

Defined in header `hpx/algorithm.hpp`⁶⁶⁶.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::replace*
- *hpx::replace_if*
- *hpx::replace_copy_if*

```
template<typename InIter, typename OutIter, typename T = typename std::iterator_traits<OutIter>::value_type>
OutIter hpx::replace_copy(InIter first, InIter last, OutIter dest, T const &old_value, T const &new_value)
```

Copies the all elements from the range [first, last) to another range beginning at *dest* replacing all elements satisfying a specific criteria with *new_value*.

Effects: Assigns to every iterator *it* in the range [result, result + (last - first)) either *new_value* or **(first + (it - result))* depending on whether the following corresponding condition holds: **(first + (i - result)) == old_value*

The assignments in the parallel *replace_copy* algorithm execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *last - first* applications of the predicate.

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **T** – The type of the old and new values (deduced).

Parameters

⁶⁶⁶ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **old_value** – Refers to the old value of the elements to replace.
- **new_value** – Refers to the new value to use as the replacement.

Returns

The *replace_copy* algorithm returns an *OutIter* The *replace_copy* algorithm returns the Iterator to the element past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename T = typename
std::iterator_traits<FwdIter2>::value_type>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx::replace_copy(ExPolicy &&policy,
                                                                                      FwdIter1 first, FwdIter1
                                                                                      last, FwdIter2 dest, T
                                                                                      const &old_value, T
                                                                                      const &new_value)
```

Copies the all elements from the range [first, last) to another range beginning at *dest* replacing all elements satisfying a specific criteria with *new_value*. Executed according to the policy.

Effects: Assigns to every iterator it in the range [result, result + (last - first)) either *new_value* or $*(first + (it - result))$ depending on whether the following corresponding condition holds: $*(first + (i - result)) == old_value$

The assignments in the parallel *replace_copy* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *replace_copy* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* applications of the predicate.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the old and new values (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.

- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **old_value** – Refers to the old value of the elements to replace.
- **new_value** – Refers to the new value to use as the replacement.

Returns

The *replace_copy* algorithm returns a *hpx::future<FwdIter2>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter2* otherwise. The *replace_copy* algorithm returns the Iterator to the element past the last element copied.

hpx::replace_copy_if

Defined in header [hpx/algorithm.hpp](#)⁶⁶⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::replace*
- *hpx::replace_if*
- *hpx::replace_copy*

template<typename **InIter**, typename **OutIter**, typename **Pred**, typename **T** = typename *std::iterator_traits<OutIter>::value_type*>

OutIter **hpx::replace_copy_if**(*InIter* first, *InIter* last, *OutIter* dest, *Pred* &&pred, *T* const &new_value)

Copies the all elements from the range [first, last) to another range beginning at *dest* replacing all elements satisfying a specific criteria with *new_value*.

Effects: Assigns to every iterator it in the range [result, result + (last - first)) either *new_value* or **(first + (it - result))* depending on whether the following corresponding condition holds: *INVOKE(f, *(first + (i - result))) != false*

The assignments in the parallel *replace_copy_if* algorithm execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *last - first* applications of the predicate.

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. (deduced).
- **T** – The type of the new values to replace (deduced).

Parameters

⁶⁶⁷ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the elements which need to be replaced. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

- **new_value** – Refers to the new value to use as the replacement.

Returns

The *replace_copy_if* algorithm returns an *OutIter*. The *replace_copy_if* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Pred, typename T = typename
std::iterator_traits<FwdIter2>::value_type>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx::replace_copy_if(ExPolicy &&policy,
                                                                                          FwdIter1 first,
                                                                                          FwdIter1 last,
                                                                                          FwdIter2 dest, Pred
                                                                                          &&pred, T const
                                                                                          &new_value)
```

Copies the all elements from the range [first, last) to another range beginning at *dest* replacing all elements satisfying a specific criteria with *new_value*.

Effects: Assigns to every iterator it in the range [result, result + (last - first)) either *new_value* or **(first + (it - result))* depending on whether the following corresponding condition holds: `INVOKE(f, *(first + (i - result))) != false`

The assignments in the parallel *replace_copy_if* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *replace_copy_if* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* applications of the predicate.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.

- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *replace_copy_if* requires *Pred* to meet the requirements of *CopyConstructible*. (deduced).
- **T** – The type of the new values to replace (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the elements which need to be replaced. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

- **new_value** – Refers to the new value to use as the replacement.

Returns

The *replace_copy_if* algorithm returns a `hpx::future<FwdIter2>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter2* otherwise. The *replace_copy_if* algorithm returns the iterator to the element in the destination range, one past the last element copied.

hpx::uninitialized_default_construct

Defined in header `hpx/algorithm.hpp`⁶⁶⁸.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::uninitialized_default_construct_n`

template<typename **FwdIter**>

void **hpx::uninitialized_default_construct**(*FwdIter* first, *FwdIter* last)

Constructs objects of type `typename iterator_traits<ForwardIt>::value_type` in the uninitialized storage designated by the range by default-initialization. If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized_default_construct* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

⁶⁶⁸ <http://github.com/TheHPXProject/hpx/blob/d402cead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

FwdIter – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

Returns

The *uninitialized_default_construct* algorithm returns nothing

```
template<typename ExPolicy, typename FwdIter>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy> hpx::uninitialized_default_construct(ExPolicy
                                                                                               &&policy,
                                                                                               FwdIter
                                                                                               first,
                                                                                               FwdIter
                                                                                               last)
```

Constructs objects of type `typename iterator_traits<ForwardIt>::value_type` in the uninitialized storage designated by the range by default-initialization. If an exception is thrown during the initialization, the function has no effects. Executed according to the policy.

The assignments in the parallel *uninitialized_default_construct* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized_default_construct* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

Returns

The *uninitialized_default_construct* algorithm returns a *hpx::future<void>*, if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns nothing otherwise.

hpx::uninitialized_default_construct_n

Defined in header [hpx/algorithm.hpp](#)⁶⁶⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::uninitialized_default_construct*

template<typename **FwdIter**, typename **Size**>

FwdIter **hpx::uninitialized_default_construct_n**(*FwdIter* first, *Size* count)

Constructs objects of type *typename iterator_traits<ForwardIt> ::value_type* in the uninitialized storage designated by the range [first, first + count) by default-initialization. If an exception is thrown during the initialization, the function has no effects.

Note: Complexity: Performs exactly *count* assignments, if *count* > 0, no assignments otherwise.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.

Returns

The *uninitialized_default_construct_n* algorithm returns a *returns FwdIter*. The *uninitialized_default_construct_n* algorithm returns the iterator to the element in the source range, one past the last element constructed.

template<typename **ExPolicy**, typename **FwdIter**, typename **Size**>

hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> **hpx::uninitialized_default_construct_n**(*ExPolicy* &&policy, *FwdIter* first, *Size* count)

Constructs objects of type *typename iterator_traits<ForwardIt> ::value_type* in the uninitialized storage designated by the range [first, first + count) by default-initialization. If an exception is thrown during the initialization, the function has no effects. Executed according to the policy.

⁶⁶⁹ <http://github.com/TheHPXProject/hpx/blob/d402eoad7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

The assignments in the parallel *uninitialized_default_construct_n* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized_default_construct_n* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *count* assignments, if *count* > 0, no assignments otherwise.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.

Returns

The *uninitialized_default_construct_n* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *uninitialized_default_construct_n* algorithm returns the iterator to the element in the source range, one past the last element constructed.

hpx::find

Defined in header [hpx/algorithm.hpp](#)⁶⁷⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::find_if*
- *hpx::find_if_not*
- *hpx::find_end*
- *hpx::find_first_of*

```
template<typename ExPolicy, typename FwdIter, typename T>
util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::find(ExPolicy &&policy, FwdIter first, FwdIter last, T
                                                             const &val)
```

Returns the first element in the range [first, last) that is equal to value. Executed according to the policy.

⁶⁷⁰ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

The comparison operations in the parallel *find* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *find* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most last - first applications of the operator==().

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to find (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **val** – the value to compare the elements to

Returns

The *find* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *find* algorithm returns the first element in the range [first,last) that is equal to *val*. If no such element in the range of [first,last) is equal to *val*, then the algorithm returns *last*.

```
template<typename InIter, typename T>
InIter hpx::find(InIter first, InIter last, T const &val)
```

Returns the first element in the range [first, last) that is equal to value. Executed according to the policy.

Note: Complexity: At most last - first applications of the operator==().

Template Parameters

- **InIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an input iterator.
- **T** – The type of the value to find (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.

- **val** – the value to compare the elements to

Returns

The *find* algorithm returns a *InIter*. The *find* algorithm returns the first element in the range [first,last) that is equal to *val*. If no such element in the range of [first,last) is equal to *val*, then the algorithm returns *last*.

hpx::find_if

Defined in header `hpx/algorithm.hpp`⁶⁷¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::find`
- `hpx::find_if_not`
- `hpx::find_end`
- `hpx::find_first_of`

```
template<typename ExPolicy, typename FwdIter, typename F>  
util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::find_if(ExPolicy &&policy, FwdIter first, FwdIter last,  
                                                                F &&f)
```

Returns the first element in the range [first, last) for which predicate *f* returns true. Executed according to the policy.

The comparison operations in the parallel *find_if* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *find_if* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most last - first applications of the predicate.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.

⁶⁷¹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **f** – The unary predicate which returns true for the required element. The signature of the predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

Returns

The *find_if* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *find_if* algorithm returns the first element in the range `[first,last)` that satisfies the predicate *f*. If no such element exists that satisfies the predicate *f*, the algorithm returns *last*.

```
template<typename InIter, typename F>
InIter hpx::find_if(InIter first, InIter last, F &&f)
```

Returns the first element in the range `[first, last)` for which predicate *f* returns true.

Note: Complexity: At most last - first applications of the predicate.

Template Parameters

- **InIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **f** – The unary predicate which returns true for the required element. The signature of the predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type*.

Returns

The *find_if* algorithm returns a *InIter*. The *find_if* algorithm returns the first element in the range `[first,last)` that satisfies the predicate *f*. If no such element exists that satisfies the predicate *f*, the algorithm returns *last*.

hpx::find_if_not

Defined in header [hpx/algorithm.hpp](#)⁶⁷².

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::find`
- `hpx::find_if`
- `hpx::find_end`
- `hpx::find_first_of`

```
template<typename ExPolicy, typename FwdIter, typename F>  
util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::find_if_not(ExPolicy &&policy, FwdIter first, FwdIter  
                                                                    last, F &&f)
```

Returns the first element in the range [first, last) for which predicate *f* returns false. Executed according to the policy.

The comparison operations in the parallel *find_if_not* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *find_if_not* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most last - first applications of the predicate.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **f** – The unary predicate which returns false for the required element. The signature of the predicate should be equivalent to:

⁶⁷² <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

```
bool pred(const Type &a);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

Returns

The *find_if_not* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *find_if_not* algorithm returns the first element in the range `[first, last)` that does **not** satisfy the predicate *f*. If no such element exists that does not satisfy the predicate *f*, the algorithm returns *last*.

```
template<typename FwdIter, typename F>
FwdIter hpx::find_if_not(FwdIter first, FwdIter last, F &&f)
```

Returns the first element in the range `[first, last)` for which predicate *f* returns false.

Note: Complexity: At most last - first applications of the predicate.

Template Parameters

- **FwdIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **f** – The unary predicate which returns false for the required element. The signature of the predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

Returns

The *find_if_not* algorithm returns a *FwdIter*. The *find_if_not* algorithm returns the first element in the range `[first, last)` that does **not** satisfy the predicate *f*. If no such element exists that does not satisfy the predicate *f*, the algorithm returns *last*.

hpx::find_end

Defined in header `hpx/algorithm.hpp`⁶⁷³.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::find`
- `hpx::find_if`
- `hpx::find_if_not`
- `hpx::find_first_of`

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Pred = detail::equal_to>
util::detail::algorithm_result_t<ExPolicy, FwdIter1> hpx::find_end(ExPolicy &&policy, FwdIter1 first1,
                                                                    FwdIter1 last1, FwdIter2 first2, FwdIter2
                                                                    last2, Pred &&op = Pred())
```

Returns the last subsequence of elements [first2, last2) found in the range [first, last) using the given predicate *op* to compare elements. Executed according to the policy.

The comparison operations in the parallel *find_end* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *find_end* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

This overload of *find_end* is available if the user decides to provide the algorithm their own predicate *op*.

Note: Complexity: at most $S \cdot (N - S + 1)$ comparisons where S = distance(first2, last2) and N = distance(first1, last1).

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *replace* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

⁶⁷³ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **last2** – Refers to the end of the sequence of elements of the algorithm will be searching for.
- **op** – The binary predicate which returns *true* if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively.

Returns

The *find_end* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *find_end* algorithm returns an iterator to the beginning of the last subsequence `[first2, last2)` in range `[first, last)`. If the length of the subsequence `[first2, last2)` is greater than the length of the range `[first1, last1)`, *last1* is returned. Additionally if the size of the subsequence is empty or no subsequence is found, *last1* is also returned.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2>
util::detail::algorithm_result_t<ExPolicy, FwdIter1> hpx::find_end(ExPolicy &&policy, FwdIter1 first1,
                                                                    FwdIter1 last1, FwdIter2 first2, FwdIter2
                                                                    last2)
```

Returns the last subsequence of elements `[first2, last2)` found in the range `[first, last)`. Elements are compared using *operator==*. Executed according to the policy.

The comparison operations in the parallel *find_end* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *find_end* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: at most $S \cdot (N - S + 1)$ comparisons where $S = \text{distance}(\text{first2}, \text{last2})$ and $N = \text{distance}(\text{first1}, \text{last1})$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **last2** – Refers to the end of the sequence of elements of the algorithm will be searching for.

Returns

The *find_end* algorithm returns a `hpx::future<FwdIter>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *find_end* algorithm returns an iterator to the beginning of the last subsequence [first2, last2) in range [first, last). If the length of the subsequence [first2, last2) is greater than the length of the range [first1, last1), *last1* is returned. Additionally if the size of the subsequence is empty or no subsequence is found, *last1* is also returned.

```
template<typename FwdIter1, typename FwdIter2, typename Pred = detail::equal_to>
FwdIter1 hpx::find_end(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, FwdIter2 last2, Pred &&op = Pred())
```

Returns the last subsequence of elements [first2, last2) found in the range [first, last) using the given predicate *op* to compare elements.

This overload of *find_end* is available if the user decides to provide the algorithm their own predicate *op*.

Note: Complexity: at most $S \cdot (N - S + 1)$ comparisons where $S = \text{distance}(\text{first2}, \text{last2})$ and $N = \text{distance}(\text{first1}, \text{last1})$.

Template Parameters

- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *replace* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **last2** – Refers to the end of the sequence of elements of the algorithm will be searching for.

- **op** – The binary predicate which returns *true* if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively.

Returns

The *find_end* algorithm returns a *FwdIter1*. The *find_end* algorithm returns an iterator to the beginning of the last subsequence `[first2, last2)` in range `[first, last)`. If the length of the subsequence `[first2, last2)` is greater than the length of the range `[first1, last1)`, *last1* is returned. Additionally if the size of the subsequence is empty or no subsequence is found, *last1* is also returned.

```
template<typename FwdIter1, typename FwdIter2>
```

```
FwdIter1 hpx::find_end(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2, FwdIter2 last2)
```

Returns the last subsequence of elements `[first2, last2)` found in the range `[first, last)`. Elements are compared using *operator==*.

Note: Complexity: at most $S \cdot (N - S + 1)$ comparisons where $S = \text{distance}(\text{first2}, \text{last2})$ and $N = \text{distance}(\text{first1}, \text{last1})$.

Template Parameters

- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **last2** – Refers to the end of the sequence of elements of the algorithm will be searching for.

Returns

The *find_end* algorithm returns a *FwdIter1*. The *find_end* algorithm returns an iterator to the beginning of the last subsequence `[first2, last2)` in range `[first, last)`. If the length of the subsequence `[first2, last2)` is greater than the length of the range `[first1, last1)`, *last1* is returned. Additionally if the size of the subsequence is empty or no subsequence is found, *last1* is also returned.

hpx::find_first_of

Defined in header `hpx/algorithm.hpp`⁶⁷⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::find`
- `hpx::find_if`
- `hpx::find_if_not`
- `hpx::find_end`

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Pred = detail::equal_to>
util::detail::algorithm_result_t<ExPolicy, FwdIter1> hpx::find_first_of(ExPolicy &&policy, FwdIter1 first,
                                                                    FwdIter1 last, FwdIter2 s_first,
                                                                    FwdIter2 s_last, Pred &&op = Pred())
```

Searches the range `[first, last)` for any elements in the range `[s_first, s_last)`. Uses binary predicate `op` to compare elements. Executed according to the policy.

The comparison operations in the parallel `find_first_of` algorithm invoked with an execution policy object of type `sequenced_policy` execute in sequential order in the calling thread.

The comparison operations in the parallel `find_first_of` algorithm invoked with an execution policy object of type `parallel_policy` or `parallel_task_policy` are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

This overload of `find_first_of` is available if the user decides to provide the algorithm their own predicate `op`.

Note: Complexity: at most $(S*N)$ comparisons where S = distance(`s_first`, `s_last`) and N = distance(`first`, `last`).

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of `equal` requires `Pred` to meet the requirements of `CopyConstructible`. This defaults to `std::equal_to<>`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.

⁶⁷⁴ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **s_first** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **s_last** – Refers to the end of the sequence of elements of the algorithm will be searching for.
- **op** – The binary predicate which returns *true* if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively.

Returns

The *find_first_of* algorithm returns a *hpx::future<FwdIter1>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter1* otherwise. The *find_first_of* algorithm returns an iterator to the first element in the range `[first, last)` that is equal to an element from the range `[s_first, s_last)`. If the length of the subsequence `[s_first, s_last)` is greater than the length of the range `[first, last)`, *last* is returned. Additionally if the size of the subsequence is empty or no subsequence is found, *last* is also returned.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2>
util::detail::algorithm_result_t<ExPolicy, FwdIter1> hpx::find_first_of(ExPolicy &&policy, FwdIter1 first,
                                                                    FwdIter1 last, FwdIter2 s_first,
                                                                    FwdIter2 s_last)
```

Searches the range `[first, last)` for any elements in the range `[s_first, s_last)`. Elements are compared using *operator==*. Executed according to the policy.

The comparison operations in the parallel *find_first_of* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *find_first_of* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: at most $(S*N)$ comparisons where $S = \text{distance}(s_first, s_last)$ and $N = \text{distance}(first, last)$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **s_first** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **s_last** – Refers to the end of the sequence of elements of the algorithm will be searching for.

Returns

The *find_first_of* algorithm returns a *hpx::future<FwdIter1>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter1* otherwise. The *find_first_of* algorithm returns an iterator to the first element in the range [first, last) that is equal to an element from the range [s_first, s_last). If the length of the subsequence [s_first, s_last) is greater than the length of the range [first, last), *last* is returned. Additionally if the size of the subsequence is empty or no subsequence is found, *last* is also returned.

```
template<typename FwdIter1, typename FwdIter2, typename Pred = detail::equal_to>
FwdIter1 hpx::find_first_of(FwdIter1 first, FwdIter1 last, FwdIter2 s_first, FwdIter2 s_last, Pred &&op =
    Pred())
```

Searches the range [first, last) for any elements in the range [s_first, s_last). Uses binary predicate *op* to compare elements.

This overload of *find_first_of* is available if the user decides to provide the algorithm their own predicate *op*.

Note: Complexity: at most ($S \cdot N$) comparisons where $S = \text{distance}(s_first, s_last)$ and $N = \text{distance}(first, last)$.

Template Parameters

- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

Parameters

- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **s_first** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **s_last** – Refers to the end of the sequence of elements of the algorithm will be searching for.

- **op** – The binary predicate which returns *true* if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively.

Returns

The *find_first_of* algorithm returns a *FwdIter1*. The *find_first_of* algorithm returns an iterator to the first element in the range `[first, last)` that is equal to an element from the range `[s_first, s_last)`. If the length of the subsequence `[s_first, s_last)` is greater than the length of the range `[first, last)`, *last* is returned. Additionally if the size of the subsequence is empty or no subsequence is found, *last* is also returned.

```
template<typename FwdIter1, typename FwdIter2>
```

```
FwdIter1 hpx::find_first_of(FwdIter1 first, FwdIter1 last, FwdIter2 s_first, FwdIter2 s_last)
```

Searches the range `[first, last)` for any elements in the range `[s_first, s_last)`. Elements are compared using *operator==*.

Note: Complexity: at most $(S*N)$ comparisons where $S = \text{distance}(s_first, s_last)$ and $N = \text{distance}(first, last)$.

Template Parameters

- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **s_first** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **s_last** – Refers to the end of the sequence of elements of the algorithm will be searching for.

Returns

The *find_first_of* algorithm returns a *FwdIter1*. The *find_first_of* algorithm returns an iterator to the first element in the range `[first, last)` that is equal to an element from the range `[s_first, s_last)`. If the length of the subsequence `[s_first, s_last)` is greater than the length of the range `[first, last)`, *last* is returned. Additionally if the size of the subsequence is empty or no subsequence is found, *last* is also returned.

hpx::unique

Defined in header `hpx/algorithm.hpp`⁶⁷⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::unique_copy`

```
template<typename FwdIter, typename Pred = hpx::parallel::detail::equal_to, typename Proj = hpx::identity>  
FwdIter hpx::unique(FwdIter first, FwdIter last, Pred &&pred = Pred(), Proj &&proj = Proj())
```

Eliminates all but the first element from every consecutive group of equivalent elements from the range [first, last) and returns a past-the-end iterator for the new logical end of the range.

Note: Complexity: Performs not more than *last - first* assignments, exactly *last - first - 1* applications of the predicate *pred* and no more than twice as many applications of the projection *proj*.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *unique* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *pred* is invoked.

Returns

The *unique* algorithm returns *FwdIter*. The *unique* algorithm returns the iterator to the new end of the range.

```
template<typename ExPolicy, typename FwdIter, typename Pred = hpx::parallel::detail::equal_to, typename Proj = hpx::identity>
```

⁶⁷⁵ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

```
parallel::util::detail::algorithm_result<ExPolicy, FwdIter>::type hpx::unique(ExPolicy &&policy, FwdIter first,
                                                                    FwdIter last, Pred &&pred =
                                                                    Pred(), Proj &&proj = Proj())
```

Eliminates all but the first element from every consecutive group of equivalent elements from the range [first, last) and returns a past-the-end iterator for the new logical end of the range. Executed according to the policy.

The assignments in the parallel *unique* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *unique* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs not more than *last - first* assignments, exactly *last - first - 1* applications of the predicate *pred* and no more than twice as many applications of the projection *proj*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *unique* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *pred* is invoked.

Returns

The *unique* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *unique* algorithm returns the iterator to the new end of the range.

hpx::unique_copy

Defined in header `hpx/algorithm.hpp`⁶⁷⁶.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::unique`

```
template<typename InIter, typename OutIter, typename Pred = hpx::parallel::detail::equal_to, typename Proj = hpx::identity>
```

```
OutIter hpx::unique_copy(InIter first, InIter last, OutIter dest, Pred &&pred = Pred(), Proj &&proj = Proj())
```

Copies the elements from the range `[first, last)`, to another range beginning at `dest` in such a way that there are no consecutive equal elements. Only the first element of each group of equal elements is copied.

Note: Complexity: Performs not more than `last - first` assignments, exactly `last - first - 1` applications of the predicate `pred` and no more than twice as many applications of the projection `proj`

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of `unique_copy` requires `Pred` to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by `[first, last)`. This is a binary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate `pred` is invoked.

Returns

The `unique_copy` algorithm returns a returns *OutIter*. The `unique_copy` algorithm returns the destination iterator to the end of the `dest` range.

⁶⁷⁶ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename Pred =
hpx::parallel::detail::equal_to, typename Proj = hpx::identity>
parallel::util::detail::algorithm_result<ExPolicy, FwdIter2>::type hpx::unique_copy(ExPolicy &&policy,
                                                                                   FwdIter1 first, FwdIter1
                                                                                   last, FwdIter2 dest, Pred
                                                                                   &&pred = Pred(), Proj
                                                                                   &&proj = Proj())
```

Copies the elements from the range `[first, last)`, to another range beginning at `dest` in such a way that there are no consecutive equal elements. Only the first element of each group of equal elements is copied. Executed according to the policy.

The assignments in the parallel *unique_copy* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *unique_copy* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs not more than *last - first* assignments, exactly *last - first - 1* applications of the predicate *pred* and no more than twice as many applications of the projection *proj*

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *unique_copy* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by `[first, last)`. This is a binary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *pred* is invoked.

Returns

The *unique_copy* algorithm returns a *hpx::future<FwdIter2>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter2* otherwise. The *unique_copy* algorithm returns the pair of the source iterator to *last*, and the destination iterator to the end of the *dest* range.

hpx::all_of

Defined in header [hpx/algorithm.hpp](#)⁶⁷⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::any_of*
- *hpx::none_of*

```
template<typename ExPolicy, typename FwdIter, typename F>  
util::detail::algorithm_result_t<ExPolicy, bool> hpx::all_of(ExPolicy &&policy, FwdIter first, FwdIter last, F  
    &&f)
```

Checks if unary predicate *f* returns true for all elements in the range [first, last).

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most *last - first* applications of the predicate *f*

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *all_of* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

⁶⁷⁷ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

Returns

The *all_of* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *all_of* algorithm returns true if the unary predicate *f* returns true for all elements in the range, false otherwise. It returns true if the range is empty.

```
template<typename ExPolicy, typename InIter, typename F>
bool hpx::all_of(InIter first, InIter last, F &&f)
```

Checks if unary predicate *f* returns true for all elements in the range [first, last).

Note: Complexity: At most *last - first* applications of the predicate *f*

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *all_of* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

Returns

The *all_of* algorithm returns a *bool*. The *all_of* algorithm returns true if the unary predicate *f* returns true for all elements in the range, false otherwise. It returns true if the range is empty.

hpx::any_of

Defined in header `hpx/algorithm.hpp`⁶⁷⁸.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::all_of`
- `hpx::none_of`

```
template<typename ExPolicy, typename FwdIter, typename F>  
util::detail::algorithm_result_t<ExPolicy, bool> hpx::any_of(ExPolicy &&policy, FwdIter first, FwdIter last, F  
    &&f)
```

Checks if unary predicate *f* returns true for at least one element in the range [first, last).

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most *last - first* applications of the predicate *f*

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *any_of* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

⁶⁷⁸ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Returns

The *any_of* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *any_of* algorithm returns true if the unary predicate *f* returns true for at least one element in the range, false otherwise. It returns false if the range is empty.

```
template<typename InIter, typename F>
bool hpx::any_of(InIter first, InIter last, F &&f)
```

Checks if unary predicate *f* returns true for at least one element in the range [first, last).

Note: Complexity: At most *last - first* applications of the predicate *f*

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *any_of* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have *const&*, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

Returns

The *any_of* algorithm returns a *bool*. The *any_of* algorithm returns true if the unary predicate *f* returns true for at least one element in the range, false otherwise. It returns false if the range is empty.

hpx::none_of

Defined in header [hpx/algorithm.hpp](#)⁶⁷⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::all_of*
- *hpx::any_of*

```
template<typename ExPolicy, typename FwdIter, typename F>
```

⁶⁷⁹ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

`util::detail::algorithm_result_t<ExPolicy, bool> hpx::none_of(ExPolicy &&policy, FwdIter first, FwdIter last, F &&f)`

Checks if unary predicate *f* returns true for no elements in the range [first, last).

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most *last - first* applications of the predicate *f*

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *none_of* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

Returns

The *none_of* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *none_of* algorithm returns true if the unary predicate *f* returns true for no elements in the range, false otherwise. It returns true if the range is empty.

```
template<typename InIter, typename F>  
bool hpx::none_of(InIter first, InIter last, F &&f)
```

Checks if unary predicate *f* returns true for no elements in the range [first, last).

Note: Complexity: At most *last - first* applications of the predicate *f*

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *none_of* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

Returns

The *none_of* algorithm returns a *bool*. The *none_of* algorithm returns true if the unary predicate *f* returns true for no elements in the range, false otherwise. It returns true if the range is empty.

hpx::is_sorted

Defined in header `hpx/algorithm.hpp`⁶⁸⁰.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::is_sorted_until*

```
template<typename FwdIter, typename Pred = hpx::parallel::detail::less>
bool hpx::is_sorted(FwdIter first, FwdIter last, Pred &&pred = Pred())
```

Determines if the range [first, last) is sorted. Uses *pred* to compare elements.

The comparison operations in the parallel *is_sorted* algorithm executes in sequential order in the calling thread.

Note: Complexity: at most $(N+S-1)$ comparisons where N = distance(first, last). S = number of partitions

Template Parameters

- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of an optional function/function object to use.

Parameters

⁶⁸⁰ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.
- **pred** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second argument. The signature of the function should be equivalent to

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

Returns

The *is_sorted* algorithm returns a *bool*. The *is_sorted* algorithm returns true if each element in the sequence [first, last) satisfies the predicate passed. If the range [first, last) contains less than two elements, the function always returns true.

```
template<typename ExPolicy, typename FwdIter, typename Pred = hpx::parallel::detail::less>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx::is_sorted(ExPolicy &&policy, FwdIter first,  
                                     FwdIter last, Pred &&pred =  
                                     Pred())
```

Determines if the range [first, last) is sorted. Uses *pred* to compare elements. Executed according to the policy.

The comparison operations in the parallel *is_sorted* algorithm invoked with an execution policy object of type *sequenced_policy* executes in sequential order in the calling thread.

The comparison operations in the parallel *is_sorted* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: at most $(N+S-1)$ comparisons where N = distance(first, last). S = number of partitions

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *is_sorted* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.

- **pred** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second argument. The signature of the function should be equivalent to

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

Returns

The *is_sorted* algorithm returns a *hpx::future<bool>* if the execution policy is of type *task_execution_policy* and returns *bool* otherwise. The *is_sorted* algorithm returns a *bool* if each element in the sequence [first, last) satisfies the predicate passed. If the range [first, last) contains less than two elements, the function always returns true.

hpx::is_sorted_until

Defined in header `hpx/algorithm.hpp`⁶⁸¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::is_sorted*

```
template<typename FwdIter, typename Pred = hpx::parallel::detail::less>
FwdIter hpx::is_sorted_until(FwdIter first, FwdIter last, Pred &&pred = Pred())
```

Returns the first element in the range [first, last) that is not sorted. Uses a predicate to compare elements or the less than operator.

The comparison operations in the parallel *is_sorted_until* algorithm execute in sequential order in the calling thread.

Note: Complexity: at most $(N+S-1)$ comparisons where N = distance(first, last). S = number of partitions

Template Parameters

- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of an optional function/function object to use.

Parameters

- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.
- **pred** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second argument. The signature of the function should be equivalent to

⁶⁸¹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

Returns

The *is_sorted_until* algorithm returns a *FwdIter*. The *is_sorted_until* algorithm returns the first unsorted element. If the sequence has less than two elements or the sequence is sorted, last is returned.

```
template<typename ExPolicy, typename FwdIter, typename Pred = hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result<ExPolicy, FwdIter>::type hpx::is_sorted_until(ExPolicy
                                                                                          &&policy, FwdIter
                                                                                          first, FwdIter last,
                                                                                          Pred &&pred =
                                                                                          Pred())
```

Returns the first element in the range [first, last) that is not sorted. Uses a predicate to compare elements or the less than operator. Executed according to the policy.

The comparison operations in the parallel *is_sorted_until* algorithm invoked with an execution policy object of type *sequenced_policy* executes in sequential order in the calling thread.

The comparison operations in the parallel *is_sorted_until* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: at most (N+S-1) comparisons where *N* = distance(first, last). *S* = number of partitions

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *is_sorted_until* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.
- **pred** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second argument. The signature of the function should be equivalent to


```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

Returns

The *is_sorted_until* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *task_execution_policy* and returns *FwdIter* otherwise. The *is_sorted_until* algorithm returns the first unsorted element. If the sequence has less than two elements or the sequence is sorted, last is returned.

hpx/parallel/algorithms/transform_reduce_binary.hpp

Defined in header `hpx/parallel/algorithms/transform_reduce_binary.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

hpx::nth_element

Defined in header `hpx/algorithm.hpp`⁶⁸².

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename RandomIt, typename Pred = hpx::parallel::detail::less>
void hpx::nth_element(RandomIt first, RandomIt nth, RandomIt last, Pred &&pred = Pred())
```

nth_element is a partial sorting algorithm that rearranges elements in `[first, last)` such that the element pointed at by *nth* is changed to whatever element would occur in that position if `[first, last)` were sorted and all of the elements before this new *nth* element are less than or equal to the elements after the new *nth* element. Executed according to the policy.

The comparison operations in the parallel *nth_element* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: Linear in `std::distance(first, last)` on average. $O(N)$ applications of the predicate, and $O(N \log N)$ swaps, where $N = \text{last} - \text{first}$.

Template Parameters

- **RandomIt** – The type of the source begin, *nth*, and end iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Pred** – Comparison function object which returns true if the first argument is less than the second. This defaults to `std::less<>`.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **nth** – Refers to the iterator defining the sort partition point

⁶⁸² <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the comparison function object which returns true if the first argument is less than (i.e. is ordered before) the second. The signature of this comparison function should be equivalent to:

```
bool cmp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type must be such that an object of type *randomIt* can be dereferenced and then implicitly converted to *Type*. This defaults to `std::less<>`.

Returns

The *nth_element* algorithms returns nothing.

```
template<typename ExPolicy, typename RandomIt, typename Pred = hpx::parallel::detail::less>
void hpx::nth_element(ExPolicy &&policy, RandomIt first, RandomIt nth, RandomIt last, Pred &&pred =
    Pred())
```

nth_element is a partial sorting algorithm that rearranges elements in `[first, last)` such that the element pointed at by *nth* is changed to whatever element would occur in that position if `[first, last)` were sorted and all of the elements before this new *nth* element are less than or equal to the elements after the new *nth* element.

The comparison operations in the parallel *nth_element* invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *nth_element* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Linear in `std::distance(first, last)` on average. $O(N)$ applications of the predicate, and $O(N \log N)$ swaps, where $N = last - first$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **RandomIt** – The type of the source begin, *nth*, and end iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Pred** – Comparison function object which returns true if the first argument is less than the second. This defaults to `std::less<>`.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **nth** – Refers to the iterator defining the sort partition point
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

- **pred** – Specifies the comparison function object which returns true if the first argument is less than (i.e. is ordered before) the second. The signature of this comparison function should be equivalent to:

```
bool cmp(const Type1 &a, const Type2 &b);
```

The signature does not need to have const&, but the function must not modify the objects passed to it. The type must be such that an object of type *randomIt* can be dereferenced and then implicitly converted to Type. This defaults to `std::less<>`.

Returns

The *nth_element* algorithms returns nothing.

hpx::transform_reduce

Defined in header `hpx/algorithm.hpp`⁶⁸³.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename ExPolicy, typename FwdIter, typename T, typename Reduce, typename Convert>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> hpx::transform_reduce(ExPolicy &&policy,
                                                                 FwdIter first, FwdIter last,
                                                                 T init, Reduce &&red_op,
                                                                 Convert &&conv_op)
```

Returns GENERALIZED_SUM(red_op, init, conv_op(*first), ..., conv_op(*(first + (last - first) - 1))). Executed according to the policy.

The reduce operations in the parallel *transform_reduce* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform_reduce* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *transform_reduce* and *accumulate* is that the behavior of *transform_reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicates *red_op* and *conv_op*.

Note: GENERALIZED_SUM(op, a1, ..., aN) is defined as follows:

- a1 when N is 1
 - op(GENERALIZED_SUM(op, b1, ..., bK), GENERALIZED_SUM(op, bM, ..., bN)), where:
 - b1, ..., bN may be any permutation of a1, ..., aN and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

⁶⁸³ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **Reduce** – The type of the binary function object used for the reduction operation.
- **Convert** – The type of the unary function object used to transform the elements of the input sequence before invoking the reduce function.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.
- **red_op** – Specifies the function (or function object) which will be invoked for each of the values returned from the invocation of *conv_op*. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1*, *Type2*, and *Ret* must be such that an object of a type as returned from *conv_op* can be implicitly converted to any of those types.

- **conv_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

Returns

The *transform_reduce* algorithm returns a *hpx::future<T>* if the execution policy is of type *parallel_task_policy* and returns *T* otherwise. The *transform_reduce* algorithm returns the result of the generalized sum over the values returned from *conv_op* when applied to the elements given by the input range [first, last).

```
template<typename InIter, typename T, typename Reduce, typename Convert>
```

```
T hpx::transform_reduce(InIter first, InIter last, T init, Reduce &&red_op, Convert &&conv_op)
```

Returns GENERALIZED_SUM(red_op, init, conv_op(*first), ..., conv_op(*(first + (last - first) - 1))).

The difference between *transform_reduce* and *accumulate* is that the behavior of *transform_reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicates *red_op* and *conv_op*.

Note: `GENERALIZED_SUM(op, a1, ..., aN)` is defined as follows:

- `a1` when `N` is 1
 - `op(GENERALIZED_SUM(op, b1, ..., bK), GENERALIZED_SUM(op, bM, ..., bN))`, where:
 - `b1, ..., bN` may be any permutation of `a1, ..., aN` and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **Reduce** – The type of the binary function object used for the reduction operation.
- **Convert** – The type of the unary function object used to transform the elements of the input sequence before invoking the reduce function.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.
- **red_op** – Specifies the function (or function object) which will be invoked for each of the values returned from the invocation of *conv_op*. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1*, *Type2*, and *Ret* must be such that an object of a type as returned from *conv_op* can be implicitly converted to any of those types.

- **conv_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by `[first, last)`. This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

Returns

The *transform_reduce* algorithm returns a *T*. The *transform_reduce* algorithm returns the result of the generalized sum over the values returned from *conv_op* when applied to the elements given by the input range `[first, last)`.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename T>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> hpx::transform_reduce(ExPolicy &&policy,
                                                                 FwdIter1 first1, FwdIter1
                                                                 last1, FwdIter2 first2, T
                                                                 init)
```

Returns the result of accumulating `init` with the inner products of the pairs formed by the elements of two ranges starting at `first1` and `first2`. Executed according to the policy.

The operations in the parallel *transform_reduce* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The operations in the parallel *transform_reduce* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: $O(\text{last} - \text{first})$ applications each of *reduce* and *transform*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the first source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the second source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **T** – The type of the value to be used as return values (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the first sequence of elements the result will be calculated with.
- **last1** – Refers to the end of the first sequence of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second sequence of elements the result will be calculated with.
- **init** – The initial value for the sum.

Returns

The *transform_reduce* algorithm returns a *hpx::future<T>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *T* otherwise.

```
template<typename InIter1, typename InIter2, typename T>
T hpx::transform_reduce(InIter1 first1, InIter1 last1, InIter2 first2, T init)
```

Returns the result of accumulating `init` with the inner products of the pairs formed by the elements of two ranges starting at `first1` and `first2`.

Note: Complexity: $O(\text{last} - \text{first})$ applications each of *reduce* and *transform*.

Template Parameters

- **InIter1** – The type of the first source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **InIter2** – The type of the second source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **T** – The type of the value to be used as return values (deduced).

Parameters

- **first1** – Refers to the beginning of the first sequence of elements the result will be calculated with.
- **last1** – Refers to the end of the first sequence of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second sequence of elements the result will be calculated with.
- **init** – The initial value for the sum.

Returns

The *transform_reduce* algorithm returns a *T*.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2, typename T, typename Reduce,
typename Convert>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> hpx::transform_reduce(ExPolicy &&policy,
                                          FwdIter1 first1, FwdIter1
                                          last1, FwdIter2 first2, T init,
                                          Reduce &&red_op, Convert
                                          &&conv_op)
```

Returns the result of accumulating *init* with the inner products of the pairs formed by the elements of two ranges starting at *first1* and *first2*. Executed according to the policy.

The operations in the parallel *transform_reduce* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The operations in the parallel *transform_reduce* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: $O(\text{last} - \text{first})$ applications each of *reduce* and *transform*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the first source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the second source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **T** – The type of the value to be used as return values (deduced).

- **Reduce** – The type of the binary function object used for the multiplication operation.
- **Convert** – The type of the unary function object used to transform the elements of the input sequence before invoking the reduce function.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the first sequence of elements the result will be calculated with.
- **last1** – Refers to the end of the first sequence of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second sequence of elements the result will be calculated with.
- **init** – The initial value for the sum.
- **red_op** – Specifies the function (or function object) which will be invoked for the initial value and each of the return values of *conv_op*. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Ret* must be such that it can be implicitly converted to a type of *T*.

- **conv_op** – Specifies the function (or function object) which will be invoked for each of the input values of the sequence. This is a binary predicate. The signature of this predicate should be equivalent to

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Ret* must be such that it can be implicitly converted to an object for the second argument type of *red_op*.

Returns

The *transform_reduce* algorithm returns a *hpx::future<T>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *T* otherwise.

```
template<typename InIter1, typename InIter2, typename T, typename Reduce, typename Convert>  
T hpx::transform_reduce(ExPolicy &&policy, InIter1 first1, InIter1 last1, InIter2 first2, T init, Reduce  
                        &&red_op, Convert &&conv_op)
```

Returns the result of accumulating *init* with the inner products of the pairs formed by the elements of two ranges starting at *first1* and *first2*.

Note: Complexity: $O(\text{last} - \text{first})$ applications each of *reduce* and *transform*.

Template Parameters

- **InIter1** – The type of the first source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **InIter2** – The type of the second source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **T** – The type of the value to be used as return values (deduced).

- **Reduce** – The type of the binary function object used for the multiplication operation.
- **Convert** – The type of the unary function object used to transform the elements of the input sequence before invoking the reduce function.

Parameters

- **first1** – Refers to the beginning of the first sequence of elements the result will be calculated with.
- **last1** – Refers to the end of the first sequence of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second sequence of elements the result will be calculated with.
- **init** – The initial value for the sum.
- **red_op** – Specifies the function (or function object) which will be invoked for the initial value and each of the return values of *conv_op*. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Ret* must be such that it can be implicitly converted to a type of *T*.

- **conv_op** – Specifies the function (or function object) which will be invoked for each of the input values of the sequence. This is a binary predicate. The signature of this predicate should be equivalent to

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Ret* must be such that it can be implicitly converted to an object for the second argument type of *red_op*.

Returns

The *transform_reduce* algorithm returns a *T*.

```
template<typename ExPolicy, typename Iter, typename Sent, typename T, typename Reduce, typename
Convert>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> hpx::transform_reduce(ExPolicy &&policy, Iter
first, Sent last, T init,
Reduce &&red_op, Convert
&&conv_op)
```

Returns GENERALIZED_SUM(red_op, init, conv_op(*first), ..., conv_op(*(first + (last - first) - 1))).

The reduce operations in the parallel *transform_reduce* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform_reduce* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *transform_reduce* and *accumulate* is that the behavior of *transform_reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicates *red_op* and *conv_op*.

Note: `GENERALIZED_SUM(op, a1, ..., aN)` is defined as follows:

- `a1` when `N` is 1
 - `op(GENERALIZED_SUM(op, b1, ..., bK), GENERALIZED_SUM(op, bM, ..., bN))`, where:
 - `b1, ..., bN` may be any permutation of `a1, ..., aN` and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of a sentinel for `Iter`.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **Reduce** – The type of the binary function object used for the reduction operation.
- **Convert** – The type of the unary function object used to transform the elements of the input sequence before invoking the reduce function.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the first sorted range the algorithm will be applied to.
- **last** – Refers to the end of the second sorted range the algorithm will be applied to.
- **init** – The initial value for the generalized sum.
- **red_op** – Specifies the function (or function object) which will be invoked for each of the values returned from the invocation of *conv_op*. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1*, *Type2*, and *Ret* must be such that an object of a type as returned from *conv_op* can be implicitly converted to any of those types.

- **conv_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by `[first, last)`. This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *Iter* can be dereferenced and

then implicitly converted to `Type`. The type `R` must be such that an object of this type can be implicitly converted to `T`.

Returns

The `transform_reduce` algorithm returns a `hpx::future<T>` if the execution policy is of type `parallel_task_policy` and returns `T` otherwise. The `transform_reduce` algorithm returns the result of the generalized sum over the values returned from `conv_op` when applied to the elements given by the input range `[first, last)`.

```
template<typename Iter, typename Sent, typename T, typename Reduce, typename Convert>
T hpx::transform_reduce(Iter first, Sent last, T init, Reduce &&red_op, Convert &&conv_op)
```

Returns `GENERALIZED_SUM(red_op, init, conv_op(*first), ..., conv_op(*(first + (last - first) - 1))`.

The difference between `transform_reduce` and `accumulate` is that the behavior of `transform_reduce` may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicates `red_op` and `conv_op`.

Note: `GENERALIZED_SUM(op, a1, ..., aN)` is defined as follows:

- `a1` when `N` is 1
 - `op(GENERALIZED_SUM(op, b1, ..., bK), GENERALIZED_SUM(op, bM, ..., bN))`, where:
 - `b1, ..., bN` may be any permutation of `a1, ..., aN` and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

- **Iter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an random access iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for `Iter`.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **Reduce** – The type of the binary function object used for the reduction operation.
- **Convert** – The type of the unary function object used to transform the elements of the input sequence before invoking the reduce function.

Parameters

- **first** – Refers to the beginning of the first sorted range the algorithm will be applied to.
- **last** – Refers to the end of the second sorted range the algorithm will be applied to.
- **init** – The initial value for the generalized sum.
- **red_op** – Specifies the function (or function object) which will be invoked for each of the values returned from the invocation of `conv_op`. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1*, *Type2*, and *Ret* must be such that an object of a type as returned from *conv_op* can be implicitly converted to any of those types.

- **conv_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by `[first, last)`. This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *Iter* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

Returns

The *transform_reduce* algorithm returns *T*. The *transform_reduce* algorithm returns the result of the generalized sum over the values returned from *conv_op* when applied to the elements given by the input range `[first, last)`.

```
template<typename ExPolicy, typename Iter, typename Sent, typename Iter2, typename T>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> hpx::transform_reduce(ExPolicy &&policy, Iter
                                                                    first, Sent last, Iter2 first2, T
                                                                    init)
```

Returns GENERALIZED_SUM(*red_op*, *init*, *conv_op*(**first*), ..., *conv_op*((**first* + (*last* - *first*) - 1))).

The reduce operations in the parallel *transform_reduce* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform_reduce* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *transform_reduce* and *accumulate* is that the behavior of *transform_reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicates *red_op* and *conv_op*.

Note: GENERALIZED_SUM(*op*, *a1*, ..., *aN*) is defined as follows:

- *a1* when *N* is 1
 - *op*(GENERALIZED_SUM(*op*, *b1*, ..., *bK*), GENERALIZED_SUM(*op*, *bM*, ..., *bN*)), where:
 - *b1*, ..., *bN* may be any permutation of *a1*, ..., *aN* and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.

- **Iter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an random access iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an random access iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the first sorted range the algorithm will be applied to.
- **last** – Refers to the end of the second sorted range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **init** – The initial value for the generalized sum.

Returns

The *transform_reduce* algorithm returns a *hpx::future<T>* if the execution policy is of type *parallel_task_policy* and returns *T* otherwise. The *transform_reduce* algorithm returns the result of the generalized sum over the values returned from *conv_op* when applied to the elements given by the input range [first, last).

```
template<typename Iter, typename Sent, typename Iter2, typename T>
```

```
T hpx::transform_reduce(Iter first, Sent last, Iter2 first2, T init)
```

Returns GENERALIZED_SUM(red_op, init, conv_op(*first), ..., conv_op(*(first + (last - first) - 1))).

The difference between *transform_reduce* and *accumulate* is that the behavior of *transform_reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicates *red_op* and *conv_op*.

Note: GENERALIZED_SUM(op, a1, ..., aN) is defined as follows:

- a1 when N is 1
 - op(GENERALIZED_SUM(op, b1, ..., bK), GENERALIZED_SUM(op, bM, ..., bN)), where:
 - b1, ..., bN may be any permutation of a1, ..., aN and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

- **Iter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an random access iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter.

- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an random access iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **first** – Refers to the beginning of the first sorted range the algorithm will be applied to.
- **last** – Refers to the end of the second sorted range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **init** – The initial value for the generalized sum.

Returns

The *transform_reduce* algorithm returns *T*. The *transform_reduce* algorithm returns the result of the generalized sum over the values returned from *conv_op* when applied to the elements given by the input range [first, last).

```
template<typename ExPolicy, typename Iter, typename Sent, typename Iter2, typename T, typename Reduce,  
typename Convert>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> hpx::transform_reduce(ExPolicy &&policy, Iter  
first, Sent last, Iter2 first2, T  
init, Reduce &&red_op,  
Convert &&conv_op)
```

Returns GENERALIZED_SUM(red_op, init, conv_op(*first), ..., conv_op(*(first + (last - first) - 1))).

The reduce operations in the parallel *transform_reduce* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform_reduce* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *transform_reduce* and *accumulate* is that the behavior of *transform_reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicates *red_op* and *conv_op*.

Note: GENERALIZED_SUM(op, a1, ..., aN) is defined as follows:

- a1 when N is 1
 - op(GENERALIZED_SUM(op, b1, ..., bK), GENERALIZED_SUM(op, bM, ..., bN)), where:
 - b1, ..., bN may be any permutation of a1, ..., aN and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an random access iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for **Iter**.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an random access iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **Reduce** – The type of the binary function object used for the reduction operation.
- **Convert** – The type of the unary function object used to transform the elements of the input sequence before invoking the reduce function.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the first sorted range the algorithm will be applied to.
- **last** – Refers to the end of the second sorted range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **init** – The initial value for the generalized sum.
- **red_op** – Specifies the function (or function object) which will be invoked for each of the values returned from the invocation of *conv_op*. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1*, *Type2*, and *Ret* must be such that an object of a type as returned from *conv_op* can be implicitly converted to any of those types.

- **conv_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *Iter* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

Returns

The *transform_reduce* algorithm returns a *hpx::future<T>* if the execution policy is of type *parallel_task_policy* and returns *T* otherwise. The *transform_reduce* algorithm returns the result of the generalized sum over the values returned from *conv_op* when applied to the elements given by the input range [first, last).

```
template<typename Iter, typename Sent, typename Iter2, typename T, typename Reduce, typename Convert>
```

T `hpx::transform_reduce`(*Iter* first, *Sent* last, *Iter2* first2, *T* init, *Reduce* &&red_op, *Convert* &&conv_op)
Returns GENERALIZED_SUM(red_op, init, conv_op(*first), ..., conv_op(*(first + (last - first) - 1))).

The difference between *transform_reduce* and *accumulate* is that the behavior of *transform_reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicates *red_op* and *conv_op*.

Note: GENERALIZED_SUM(op, a1, ..., aN) is defined as follows:

- a1 when N is 1
 - op(GENERALIZED_SUM(op, b1, ..., bK), GENERALIZED_SUM(op, bM, ..., bN)), where:
 - b1, ..., bN may be any permutation of a1, ..., aN and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

- **Iter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an random access iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an random access iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **Reduce** – The type of the binary function object used for the reduction operation.
- **Convert** – The type of the unary function object used to transform the elements of the input sequence before invoking the reduce function.

Parameters

- **first** – Refers to the beginning of the first sorted range the algorithm will be applied to.
- **last** – Refers to the end of the second sorted range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **init** – The initial value for the generalized sum.
- **red_op** – Specifies the function (or function object) which will be invoked for each of the values returned from the invocation of *conv_op*. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have const&, but the function must not modify the objects passed to it. The types *Type1*, *Type2*, and *Ret* must be such that an object of a type as returned from *conv_op* can be implicitly converted to any of those types.

- **conv_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have const&, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *Iter* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

Returns

The *transform_reduce* algorithm returns *T*. The *transform_reduce* algorithm returns the result of the generalized sum over the values returned from *conv_op* when applied to the elements given by the input range [first, last).

```
template<typename ExPolicy, typename Rng, typename T, typename Reduce, typename Convert>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> hpx::transform_reduce(ExPolicy &&policy, Rng
&&rng, T init, Reduce
&&red_op, Convert
&&conv_op)
```

Returns GENERALIZED_SUM(red_op, init, conv_op(*first), ..., conv_op(*(first + (last - first) - 1))).

The reduce operations in the parallel *transform_reduce* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform_reduce* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *transform_reduce* and *accumulate* is that the behavior of *transform_reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicates *red_op* and *conv_op*.

Note: GENERALIZED_SUM(op, a1, ..., aN) is defined as follows:

- a1 when N is 1
 - op(GENERALIZED_SUM(op, b1, ..., bK), GENERALIZED_SUM(op, bM, ..., bN)), where:
 - b1, ..., bN may be any permutation of a1, ..., aN and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **Reduce** – The type of the binary function object used for the reduction operation.
- **Convert** – The type of the unary function object used to transform the elements of the input sequence before invoking the reduce function.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.
- **red_op** – Specifies the function (or function object) which will be invoked for each of the values returned from the invocation of *conv_op*. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1*, *Type2*, and *Ret* must be such that an object of a type as returned from *conv_op* can be implicitly converted to any of those types.

- **conv_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *Iter* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

Returns

The *transform_reduce* algorithm returns a *hpx::future<T>* if the execution policy is of type *parallel_task_policy* and returns *T* otherwise. The *transform_reduce* algorithm returns the result of the generalized sum over the values returned from *conv_op* when applied to the elements given by the input range [first, last).

```
template<typename Rng, typename T, typename Reduce, typename Convert>
```

```
T hpx::transform_reduce(Rng &&rng, T init, Reduce &&red_op, Convert &&conv_op)
```

Returns GENERALIZED_SUM(red_op, init, conv_op(*first), ..., conv_op(*(first + (last - first) - 1))).

The difference between *transform_reduce* and *accumulate* is that the behavior of *transform_reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicates *red_op* and *conv_op*.

Note: GENERALIZED_SUM(op, a1, ..., aN) is defined as follows:

- a1 when N is 1
- op(GENERALIZED_SUM(op, b1, ..., bK), GENERALIZED_SUM(op, bM, ..., bN)), where:

- $b_1, \dots, b_N$ may be any permutation of $a_1, \dots, a_N$ and
- $1 < K+1 = M \leq N$.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **Reduce** – The type of the binary function object used for the reduction operation.
- **Convert** – The type of the unary function object used to transform the elements of the input sequence before invoking the reduce function.

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.
- **red_op** – Specifies the function (or function object) which will be invoked for each of the values returned from the invocation of *conv_op*. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1*, *Type2*, and *Ret* must be such that an object of a type as returned from *conv_op* can be implicitly converted to any of those types.

- **conv_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *Iter* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

Returns

The *transform_reduce* algorithm returns *T*. The *transform_reduce* algorithm returns the result of the generalized sum over the values returned from *conv_op* when applied to the elements given by the input range [first, last).

```
template<typename ExPolicy, typename Rng, typename Iter2, typename T>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> hpx::transform_reduce(ExPolicy &&policy, Rng
                                                                    &&rng, Iter2 first2, T init)
```

Returns the result of accumulating *init* with the inner products of the pairs formed by the elements of two ranges starting at *first1* and *first2*.

The operations in the parallel *transform_reduce* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The operations in the parallel *transform_reduce* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate *op2*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Iter2** – The type of the second source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be used as return values (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second sequence of elements the result will be calculated with.
- **init** – The initial value for the sum.

Returns

The *transform_reduce* algorithm returns a *hpx::future<T>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *T* otherwise.

```
template<typename Rng, typename Iter2, typename T>
T hpx::transform_reduce(Rng &&rng, Iter2 first2, T init)
```

Returns the result of accumulating *init* with the inner products of the pairs formed by the elements of two ranges starting at *first1* and *first2*.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate *op2*.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Iter2** – The type of the second source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be used as return values (deduced).

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second sequence of elements the result will be calculated with.

- **init** – The initial value for the sum.

Returns

The *transform_reduce* algorithm returns *T*.

```
template<typename ExPolicy, typename Rng, typename Iter2, typename T, typename Reduce, typename
Convert>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> hpx::transform_reduce(ExPolicy &&policy, Rng
&&rng, Iter2 first2, T init,
Reduce &&red_op, Convert
&&conv_op)
```

Returns the result of accumulating *init* with the inner products of the pairs formed by the elements of two ranges starting at *first1* and *first2*.

The operations in the parallel *transform_reduce* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The operations in the parallel *transform_reduce* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate *op2*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Iter2** – The type of the second source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **T** – The type of the value to be used as return values (deduced).
- **Reduce** – The type of the binary function object used for the multiplication operation.
- **Convert** – The type of the unary function object used to transform the elements of the input sequence before invoking the reduce function.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second sequence of elements the result will be calculated with.
- **init** – The initial value for the sum.
- **red_op** – Specifies the function (or function object) which will be invoked for the initial value and each of the return values of *op2*. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Ret* must be such that it can be implicitly converted to a type of *T*.

- **conv_op** – Specifies the function (or function object) which will be invoked for each of the input values of the sequence. This is a binary predicate. The signature of this predicate should be equivalent to

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Ret* must be such that it can be implicitly converted to an object for the second argument type of *op1*.

Returns

The *transform_reduce* algorithm returns a *hpx::future<T>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *T* otherwise.

template<typename **Rng**, typename **Iter2**, typename **T**, typename **Reduce**, typename **Convert**>

T **hpx::transform_reduce**(*Rng* &&rng, *Iter2* first2, *T* init, *Reduce* &&red_op, *Convert* &&conv_op)

Returns the result of accumulating *init* with the inner products of the pairs formed by the elements of two ranges starting at *first1* and *first2*.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate *op2*.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Iter2** – The type of the second source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be used as return) values (deduced).
- **Reduce** – The type of the binary function object used for the multiplication operation.
- **Convert** – The type of the unary function object used to transform the elements of the input sequence before invoking the reduce function.

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second sequence of elements the result will be calculated with.
- **init** – The initial value for the sum.
- **red_op** – Specifies the function (or function object) which will be invoked for the initial value and each of the return values of *op2*. This is a binary predicate. The signature of this predicate should be equivalent to should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Ret* must be such that it can be implicitly converted to a type of *T*.

- **conv_op** – Specifies the function (or function object) which will be invoked for each of the input values of the sequence. This is a binary predicate. The signature of this predicate should be equivalent to

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Ret* must be such that it can be implicitly converted to an object for the second argument type of *op1*.

Returns

The *transform_reduce* algorithm returns *T*.

hpx::partial_sort

Defined in header `hpx/algorithm.hpp`⁶⁸⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename RandIter, typename Comp = hpx::parallel::detail::less>
```

```
RandIter hpx::partial_sort(RandIter first, RandIter middle, RandIter last, Comp &&comp = Comp())
```

Places the first middle - first elements from the range [first, last) as sorted with respect to comp into the range [first, middle). The rest of the elements in the range [middle, last) are placed in an unspecified order.

Note: Complexity: Approximately (last - first) * log(middle - first) comparisons.

Template Parameters

- **RandIter** – The type of the source begin, middle, and end iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced). Comp defaults to `detail::less`.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the middle of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – comp is a callable object. The return value of the INVOKE operation applied to an object of type Comp, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that comp will not apply any non-constant function through the dereferenced iterator. It defaults to `detail::less`.

Returns

The *partial_sort* algorithm returns a *RandIter* that refers to *last*.

```
template<typename ExPolicy, typename RandIter, typename Comp = hpx::parallel::detail::less>
parallel::util::detail::algorithm_result_t<ExPolicy, RandIter> hpx::partial_sort(ExPolicy &&policy, RandIter
                                                                    first, RandIter middle,
                                                                    RandIter last, Comp &&comp
                                                                    = Comp())
```

⁶⁸⁴ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Places the first middle - first elements from the range [first, last) as sorted with respect to comp into the range [first, middle). The rest of the elements in the range [middle, last) are placed in an unspecified order.

Note: Complexity: Approximately (last - first) * log(middle - first) comparisons.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **RandIter** – The type of the source begin, middle, and end iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced). Comp defaults to detail::less.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the middle of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – comp is a callable object. The return value of the INVOKE operation applied to an object of type Comp, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that comp will not apply any non-constant function through the dereferenced iterator. It defaults to detail::less.

Returns

The *partial_sort* algorithm returns a *hpx::future<RandIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *RandIter* otherwise. The iterator returned refers to *last*.

hpx::swap_ranges

Defined in header [hpx/algorithm.hpp](#)⁶⁸⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename FwdIter1, typename FwdIter2>
```

```
FwdIter2 hpx::swap_ranges(FwdIter1 first1, FwdIter1 last1, FwdIter2 first2)
```

Exchanges elements between range [first1, last1) and another range starting at *first2*.

The swap operations in the parallel *swap_ranges* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: Linear in the distance between *first1* and *last1*.

⁶⁸⁵ <http://github.com/TheHPXProject/hpx/blob/d402cead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Template Parameters

- **FwdIter1** – The type of the first range of iterators to swap (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the second range of iterators to swap (deduced). This iterator type must meet the requirements of a forward iterator.

Parameters

- **first1** – Refers to the beginning of the first sequence of elements the algorithm will be applied to.
- **last1** – Refers to the end of the first sequence of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second sequence of elements the algorithm will be applied to.

Returns

The *swap_ranges* algorithm returns *FwdIter2*. The *swap_ranges* algorithm returns iterator to the element past the last element exchanged in the range beginning with *first2*.

```
template<typename ExPolicy, typename FwdIter1, typename FwdIter2>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx::swap_ranges(ExPolicy &&policy,
                                                                                      FwdIter1 first1, FwdIter1
                                                                                      last1, FwdIter2 first2)
```

Exchanges elements between range [first1, last1) and another range starting at *first2*. Executed according to the policy.

The swap operations in the parallel *swap_ranges* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The swap operations in the parallel *swap_ranges* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Linear in the distance between *first1* and *last1*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the swap operations.
- **FwdIter1** – The type of the first range of iterators to swap (deduced). This iterator type must meet the requirements of a forward iterator.
- **FwdIter2** – The type of the second range of iterators to swap (deduced). This iterator type must meet the requirements of a forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the first sequence of elements the algorithm will be applied to.
- **last1** – Refers to the end of the first sequence of elements the algorithm will be applied to.

- **first2** – Refers to the beginning of the second sequence of elements the algorithm will be applied to.

Returns

The *swap_ranges* algorithm returns a *hpx::future<FwdIter2>* if the execution policy is of type *parallel_task_policy* and returns *FwdIter2* otherwise. The *swap_ranges* algorithm returns iterator to the element past the last element exchanged in the range beginning with *first2*.

hpx::sort

Defined in header [hpx/algorithm.hpp](#)⁶⁸⁶.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename RandomIt, typename Comp = hpx::parallel::detail::less, typename Proj = hpx::identity>
void hpx::sort(RandomIt first, RandomIt last, Comp &&comp, Proj &&proj = Proj())
```

Sorts the elements in the range [first, last) in ascending order. The order of equal elements is not guaranteed to be preserved. The function uses the given comparison function object comp (defaults to using operator<()).

A sequence is sorted with respect to a comparator *comp* and a projection *proj* if for every iterator *i* pointing to the sequence and every non-negative integer *n* such that *i + n* is a valid iterator pointing to an element of the sequence, and INVOKE(comp, INVOKE(proj, *(i + n)), INVOKE(proj, *i)) == false.

comp has to induce a strict weak ordering on the values.

The assignments in the parallel *sort* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: $O(N \log(N))$, where $N = \text{std::distance}(\text{first}, \text{last})$ comparisons.

Template Parameters

- **RandomIt** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – comp is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that comp will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

Returns

The *sort* algorithm returns *void*.

⁶⁸⁶ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e77/libs/full/include/include/hpx/algorithm.hpp>

```
template<typename ExPolicy, typename RandomIt, typename Comp = hpx::parallel::detail::less, typename Proj =
hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy> hpx::sort(ExPolicy &&policy, RandomIt first, RandomIt
last, Comp &&comp, Proj &&proj)
```

Sorts the elements in the range `[first, last)` in ascending order. The order of equal elements is not guaranteed to be preserved. The function uses the given comparison function object `comp` (defaults to using `operator<()`). Executed according to the policy.

A sequence is sorted with respect to a comparator `comp` and a projection `proj` if for every iterator `i` pointing to the sequence and every non-negative integer `n` such that `i + n` is a valid iterator pointing to an element of the sequence, and `INVOKE(comp, INVOKE(proj, *(i + n)), INVOKE(proj, *i)) == false`.

`comp` has to induce a strict weak ordering on the values.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: $O(N \log(N))$, where $N = \text{std::distance}(\text{first}, \text{last})$ comparisons.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **RandomIt** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – `comp` is a callable object. The return value of the `INVOKE` operation applied to an object of type `Comp`, when contextually converted to `bool`, yields `true` if the first argument of the call is less than the second, and `false` otherwise. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate `comp` is invoked.

Returns

The `sort` algorithm returns a `hpx::future<void>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `void` otherwise.

hpx::ranges::partial_sort_copy

Defined in header [hpx/algorithm.hpp](#)⁶⁸⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename InIter, typename Sent1, typename RandIter, typename Sent2, typename Comp =  
ranges::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>  
partial_sort_copy_result<InIter, RandIter> hpx::ranges::partial_sort_copy(InIter first, Sent1 last, RandIter  
r_first, Sent2 r_last, Comp  
&&comp = Comp(), Proj1  
&&proj1 = Proj1(), Proj2  
&&proj2 = Proj2())
```

Sorts some of the elements in the range [first, last) in ascending order, storing the result in the range [r_first, r_last). At most r_last - r_first of the elements are placed sorted to the range [r_first, r_first + n) where n is the number of elements to sort (n = min(last - first, r_last - r_first)).

The assignments in the parallel *partial_sort_copy* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: $O(N \log(\min(D, N)))$, where $N = \text{std::distance}(\text{first}, \text{last})$ and $D = \text{std::distance}(\text{r_first}, \text{r_last})$ comparisons.

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent1** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter.
- **RandIter** – The type of the destination iterators used (deduced). This iterator type must meet the requirements of a random iterator.
- **Sent2** – The type of the destination sentinel (deduced). This sentinel type must be a sentinel for RandIter.
- **Comp** – The type of the function/object to use (deduced). Comp defaults to `detail::less`.
- **Proj1** – The type of an optional projection function for the input range. This defaults to `hpx::identity`.
- **Proj2** – The type of an optional projection function for the output range. This defaults to `hpx::identity`.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the sentinel value denoting the end of the sequence of elements the algorithm will be applied to.
- **r_first** – Refers to the beginning of the destination range.

⁶⁸⁷ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **r_last** – Refers to the sentinel denoting the end of the destination range.
- **comp** – `comp` is a callable object. The return value of the INVOKE operation applied to an object of type `Comp`, when contextually converted to `bool`, yields `true` if the first argument of the call is less than the second, and `false` otherwise. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator. This defaults to `detail::less`.
- **proj1** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate `comp` is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation after the actual predicate `comp` is invoked.

Returns

The *partial_sort_copy* algorithm returns a *partial_sort_copy_result*<*InIter*, *RandIter*>. The algorithm returns {last, result_first + N}.

```
template<typename ExPolicy, typename FwdIter, typename Sent1, typename RandIter, typename Sent2,
typename Comp = ranges::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
parallel::util::detail::algorithm_result_t<ExPolicy, partial_sort_copy_result<FwdIter, RandIter>> hpx::ranges::partial_sort_co
```

Sorts some of the elements in the range `[first, last)` in ascending order, storing the result in the range `[r_first, r_last)`. At most `r_last - r_first` of the elements are placed sorted to the range `[r_first, r_first + n)` where `n` is the number of elements to sort (`n = min(last - first, r_last - r_first)`).

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: $O(N \log(\min(D, N)))$, where $N = \text{std::distance}(\text{first}, \text{last})$ and $D = \text{std::distance}(\text{r_first}, \text{r_last})$

comparisons.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for FwdIter.
- **RandIter** – The type of the destination iterators used (deduced). This iterator type must meet the requirements of an random iterator.
- **Sent2** – The type of the destination sentinel (deduced). This sentinel type must be a sentinel for RandIter.
- **Comp** – The type of the function/function object to use (deduced). Comp defaults to `detail::less`.
- **Proj1** – The type of an optional projection function for the input range. This defaults to `hpx::identity`.
- **Proj2** – The type of an optional projection function for the output range. This defaults to `hpx::identity`.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the sentinel value denoting the end of the sequence of elements the algorithm will be applied to.
- **r_first** – Refers to the beginning of the destination range.
- **r_last** – Refers to the sentinel denoting the end of the destination range.
- **comp** – `comp` is a callable object. The return value of the INVOKE operation applied to an object of type `Comp`, when contextually converted to `bool`, yields `true` if the first argument of the call is less than the second, and `false` otherwise. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator. This defaults to `detail::less`.
- **proj1** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate `comp` is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation after the actual predicate `comp` is invoked.

Returns

The `partial_sort_copy` algorithm returns a `hpx::future<partial_sort_copy_result<FwdIter, RandIter>>` if the execution policy is of type `sequenced_task_policy` or `parallel_task_policy` and returns `partial_sort_copy_result<FwdIter, RandIter>` otherwise. The algorithm returns `{last, result_first + N}`.

```
template<typename Rng1, typename Rng2, typename Comp = ranges::less, typename Proj1 = hpx::identity,
typename Proj2 = hpx::identity>
```

```

partial_sort_copy_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng2>>> hpx::ranges::partial_sort_copy(Rng1
&&rng
Rng2
&&rng
Comp
&&con
=
Comp(
Proj1
&&pro
=
Proj1(
Proj2
&&pro
=
Proj2(

```

Sorts some of the elements in the range [first, last) in ascending order, storing the result in the range [r_first, r_last). At most r_last - r_first of the elements are placed sorted to the range [r_first, r_first + n) where n is the number of elements to sort (n = min(last - first, r_last - r_first)).

The assignments in the parallel *partial_sort_copy* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: $O(N \log(\min(D, N)))$, where $N = \text{std::distance}(\text{first}, \text{last})$ and $D = \text{std::distance}(\text{r_first}, \text{r_last})$ comparisons.

Template Parameters

- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a input iterator.
- **Rng2** – The type of the destination range used (deduced). The iterators extracted from this range type must meet the requirements of a random iterator.
- **Comp** – The type of the function/function object to use (deduced). Comp defaults to `detail::less`.
- **Proj1** – The type of an optional projection function for the input range. This defaults to `hpx::identity`.
- **Proj2** – The type of an optional projection function for the output range. This defaults to `hpx::identity`.

Parameters

- **rng1** – Refers to the source range.
- **rng2** – Refers to the destination range.
- **comp** – comp is a callable object. The return value of the INVOKE operation applied to an object of type Comp, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that comp will not apply any non-constant function through the dereferenced iterator. This defaults to `detail::less`.
- **proj1** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

- **proj2** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation after the actual predicate *comp* is invoked.

Returns

The *partial_sort_copy* algorithm returns *partial_sort_copy_result*<*range_iterator_t*<*Rng1*>, *range_iterator_t*<*Rng2*>>. The algorithm returns {last, result_first + N}.

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Comp = ranges::less, typename Proj1 =  
hpx::identity, typename Proj2 = hpx::identity>  
parallel::util::detail::algorithm_result_t<ExPolicy, partial_sort_copy_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng2>
```

Sorts some of the elements in the range [first, last) in ascending order, storing the result in the range [r_first, r_last). At most r_last - r_first of the elements are placed sorted to the range [r_first, r_first + n) where n is the number of elements to sort (n = min(last - first, r_last - r_first)).

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: $O(N \log(\min(D, N)))$, where $N = \text{std::distance}(\text{first}, \text{last})$ and $D = \text{std::distance}(\text{r_first}, \text{r_last})$ comparisons.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a forward iterator.
- **Rng2** – The type of the destination range used (deduced). The iterators extracted from this range type must meet the requirements of a random iterator.

- **Comp** – The type of the function/function object to use (deduced). Comp defaults to `detail::less`.
- **Proj1** – The type of an optional projection function for the input range. This defaults to `hpx::identity`.
- **Proj2** – The type of an optional projection function for the output range. This defaults to `hpx::identity`.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the source range.
- **rng2** – Refers to the destination range.
- **comp** – `comp` is a callable object. The return value of the INVOKE operation applied to an object of type `Comp`, when contextually converted to `bool`, yields `true` if the first argument of the call is less than the second, and `false` otherwise. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator. This defaults to `detail::less`.
- **proj1** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate `comp` is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation after the actual predicate `comp` is invoked.

Returns

The `partial_sort_copy` algorithm returns a `hpx::future<partial_sort_copy_result<range_iterator_t<Rng1>, range_iterator_t<Rng2>>>` if the execution policy is of type `sequenced_task_policy` or `parallel_task_policy` and returns `partial_sort_copy_result<range_iterator_t<Rng1>, range_iterator_t<Rng2>>` otherwise. The algorithm returns `{last, result_first + N}`.

hpx::ranges::transform

Defined in header `hpx/algorithm.hpp`⁶⁸⁸.

See *Public API* for a list of names and headers that are part of the public HPX API.

template<typename **ExPolicy**, typename **FwdIter1**, typename **Sent1**, typename **FwdIter2**, typename **F**, typename **Proj** = `hpx::identity`>

`parallel::util::detail::algorithm_result<ExPolicy, ranges::unary_transform_result<FwdIter1, FwdIter2>>>::type hpx::ranges::transi`

⁶⁸⁸ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e77/libs/full/include/include/hpx/algorithm.hpp>

Applies the given function f to the given range rng and stores the result in another range, beginning at $dest$.

The invocations of f in the parallel *transform* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The invocations of f in the parallel *transform* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Exactly $\text{size}(rng)$ applications of f

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the invocations of f .
- **FwdIter1** – The type of the source iterators for the first range used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of a sentinel for FwdIter1.
- **FwdIter2** – The type of the source iterators for the first range used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *transform* requires F to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the first sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type &a);
```

The signature does not need to have `const&`. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *Ret* must be such that an object of type *FwdIter2* can be dereferenced and assigned a value of type *Ret*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate f is invoked.

Returns

The *transform* algorithm returns a `hpx::future<ranges::unary_transform_result<FwdIter1, FwdIter2>>` if the execution policy is of type *parallel_task_policy* and returns

ranges::unary_transform_result<FwdIter1, FwdIter2> otherwise. The *transform* algorithm returns a tuple holding an iterator referring to the first element after the input sequence and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng, typename FwdIter, typename F, typename Proj = hpx::identity>
parallel::util::detail::algorithm_result<ExPolicy, ranges::unary_transform_result<std::ranges::iterator_t<Rng>, FwdIter>> hpx::rang
```

Applies the given function *f* to the given range *rng* and stores the result in another range, beginning at *dest*.

The invocations of *f* in the parallel *transform* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The invocations of *f* in the parallel *transform* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Exactly $\text{size}(\text{rng})$ applications of *f*

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the invocations of *f*.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *transform* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type &a);
```

The signature does not need to have `const&`. The type *Type* must be such that an object of type `range_iterator<Rng>::type` can be dereferenced and then implicitly converted to *Type*. The type *Ret* must be such that an object of type *OutIter* can be dereferenced and assigned a value of type *Ret*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *f* is invoked.

Returns

The *transform* algorithm returns a `hpx::future<ranges::unary_transform_result<range_iterator<Rng>::type, FwdIter>>` if the execution policy is of type *parallel_task_policy* and returns `ranges::unary_transform_result<range_iterator<Rng>::type, FwdIter>` otherwise. The *transform* algorithm returns a tuple holding an iterator referring to the first element after the input sequence and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Sent1, typename FwdIter2, typename Sent2,
typename FwdIter3, typename F, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
parallel::util::detail::algorithm_result<ExPolicy, ranges::binary_transform_result<FwdIter1, FwdIter2, FwdIter3>>::type hpx::range
```

Applies the given function *f* to pairs of elements from two ranges: one defined by *rng* and the other beginning at *first2*, and stores the result in another range, beginning at *dest*.

The invocations of *f* in the parallel *transform* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The invocations of *f* in the parallel *transform* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Exactly $\text{size}(\text{rng})$ applications of f

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the invocations of f .
- **FwdIter1** – The type of the source iterators for the first range used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of a sentinel for FwdIter1.
- **FwdIter2** – The type of the source iterators for the first range used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent2** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of a sentinel for FwdIter2.
- **FwdIter3** – The type of the source iterators for the first range used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *transform* requires F to meet the requirements of *CopyConstructible*.
- **Proj1** – The type of an optional projection function to be used for elements of the first sequence. This defaults to *hpx::identity*
- **Proj2** – The type of an optional projection function to be used for elements of the second sequence. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the first sequence of elements the algorithm will be applied to.
- **last1** – Refers to the end of the first sequence of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second sequence of elements the algorithm will be applied to.
- **last2** – Refers to the end of the second sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`. The types *Type1* and *Type2* must be such that objects of types FwdIter1 and FwdIter2 can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively. The type *Ret* must be such that an object of type *FwdIter3* can be dereferenced and assigned a value of type *Ret*.

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *f* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *f* is invoked.

Returns

The *transform* algorithm returns `A hpx::future<ranges::binary_transform_result<FwdIter1, FwdIter2, FwdIter3>>` if the execution policy is of type *parallel_task_policy* and returns `ranges::binary_transform_result<FwdIter1, FwdIter2, FwdIter3>` otherwise. The *transform* algorithm returns a tuple holding an iterator referring to the first element after the first input sequence, an iterator referring to the first element after the second input sequence, and the output iterator referring to the element in the destination range, one past the last element copied.

```
template<typename FwdIter1, typename Sent1, typename FwdIter2, typename F, typename Proj = hpx::identity>
ranges::unary_transform_result<FwdIter1, FwdIter2> hpx::ranges::transform(FwdIter1 first, Sent1 last,
                                                                    FwdIter2 dest, F &&f, Proj
                                                                    &&proj = Proj())
```

Applies the given function *f* to the given range *rng* and stores the result in another range, beginning at *dest*.

Note: Complexity: Exactly `size(rng)` applications of *f*

Template Parameters

- **FwdIter1** – The type of the source iterators for the first range used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of a sentinel for `FwdIter1`.
- **FwdIter2** – The type of the source iterators for the first range used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *transform* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **first** – Refers to the beginning of the first sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by `[first, last)`. This is an unary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type &a);
```

The signature does not need to have `const&`. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *Ret* must be such that an object of type *FwdIter2* can be dereferenced and assigned a value of type *Ret*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate f is invoked.

Returns

The *transform* algorithm returns `ranges::unary_transform_result<FwdIter1, FwdIter2>`. The *transform* algorithm returns a tuple holding an iterator referring to the first element after the input sequence and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng, typename FwdIter, typename F, typename Proj = hpx::identity>
ranges::unary_transform_result<std::ranges::iterator_t<Rng>, FwdIter> hpx::ranges::transform(Rng &&rng,
                                                                                          FwdIter dest,
                                                                                          F &&f, Proj
                                                                                          &&proj =
                                                                                          Proj())
```

Applies the given function f to the given range *rng* and stores the result in another range, beginning at *dest*.

Note: Complexity: Exactly `size(rng)` applications of f

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *transform* requires F to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type &a);
```

The signature does not need to have `const&`. The type *Type* must be such that an object of type `range_iterator<Rng>::type` can be dereferenced and then implicitly converted to *Type*. The type *Ret* must be such that an object of type *OutIter* can be dereferenced and assigned a value of type *Ret*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate f is invoked.

Returns

The *transform* algorithm returns `ranges::unary_transform_result<range_iterator<Rng>::type, FwdIter>`. The *transform* algorithm returns a tuple holding an iterator referring to the first element after the input sequence and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename FwdIter1, typename Sent1, typename FwdIter2, typename Sent2, typename FwdIter3,
typename F, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
ranges::binary_transform_result<FwdIter1, FwdIter2, FwdIter3> hpx::ranges::transform(FwdIter1 first1,
                                                                                   Sent1 last1,
                                                                                   FwdIter2 first2,
                                                                                   Sent2 last2,
                                                                                   FwdIter3 dest, F
                                                                                   &&f, Proj1
                                                                                   &&proj1 = Proj1(),
                                                                                   Proj2 &&proj2 =
                                                                                   Proj2())
```

Applies the given function *f* to pairs of elements from two ranges: one defined by *rng* and the other beginning at *first2*, and stores the result in another range, beginning at *dest*.

Note: Complexity: Exactly *size(rng)* applications of *f*

Template Parameters

- **FwdIter1** – The type of the source iterators for the first range used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of a sentinel for **FwdIter1**.
- **FwdIter2** – The type of the source iterators for the first range used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent2** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of a sentinel for **FwdIter2**.
- **FwdIter3** – The type of the source iterators for the first range used (deduced). This iterator type must meet the requirements of a forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *transform* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj1** – The type of an optional projection function to be used for elements of the first sequence. This defaults to *hpx::identity*
- **Proj2** – The type of an optional projection function to be used for elements of the second sequence. This defaults to *hpx::identity*

Parameters

- **first1** – Refers to the beginning of the first sequence of elements the algorithm will be applied to.
- **last1** – Refers to the end of the first sequence of elements the algorithm will be applied to.
- **first2** – Refers to the beginning of the second sequence of elements the algorithm will be applied to.
- **last2** – Refers to the end of the second sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate. The signature of this predicate should be equivalent to:


```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively. The type *Ret* must be such that an object of type *FwdIter3* can be dereferenced and assigned a value of type *Ret*.

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *f* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *f* is invoked.

Returns

The *transform* algorithm returns `ranges::binary_transform_result<FwdIter1, FwdIter2, FwdIter3>`. The *transform* algorithm returns a tuple holding an iterator referring to the first element after the first input sequence, an iterator referring to the first element after the second input sequence, and the output iterator referring to the element in the destination range, one past the last element copied.

```
template<typename Rng1, typename Rng2, typename FwdIter, typename F, typename Proj1 = hpx::identity,
typename Proj2 = hpx::identity>
ranges::binary_transform_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng2>, FwdIter> hpx::ranges::transform(
```

Applies the given function *f* to pairs of elements from two ranges: one defined by [first1, last1) and the other beginning at first2, and stores the result in another range, beginning at dest.

Note: Complexity: Exactly min(last2-first2, last1-first1) applications of *f*

Note: The algorithm will invoke the binary predicate until it reaches the end of the shorter of the two given input sequences

Template Parameters

- **Rng1** – The type of the first source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

- **Rng2** – The type of the second source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *transform* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj1** – The type of an optional projection function to be used for elements of the first sequence. This defaults to *hpx::identity*
- **Proj2** – The type of an optional projection function to be used for elements of the second sequence. This defaults to *hpx::identity*

Parameters

- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`. The types *Type1* and *Type2* must be such that objects of types `range_iterator<Rng1>::type` and `range_iterator<Rng2>::type` can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively. The type *Ret* must be such that an object of type *FwdIter* can be dereferenced and assigned a value of type *Ret*.

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *f* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *f* is invoked.

Returns

The *transform* algorithm returns *ranges::binary_transform_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng2>, FwdIter>*. The *transform* algorithm returns a tuple holding an iterator referring to the first element after the first input sequence, an iterator referring to the first element after the second input sequence, and the output iterator referring to the element in the destination range, one past the last element copied.

`hpx::ranges::ends_with`

Defined in header `hpx/algorithm.hpp`⁶⁸⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename Pred =
ranges::equal_to, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
```

⁶⁸⁹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

```
bool hpx::ranges::ends_with(Iter1 first1, Sent1 last1, Iter2 first2, Sent2 last2, Pred &&pred = Pred(), Proj1
&&proj1 = Proj1(), Proj2 &&proj2 = Proj2())
```

Checks whether the second range defined by [first1, last1) matches the suffix of the first range defined by [first2, last2)

The assignments in the parallel *ends_with* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: Linear: at most min(N1, N2) applications of the predicate and both projections.

Template Parameters

- **Iter1** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Iter2** – The type of the begin destination iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent2** – The type of the end destination iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter2.
- **Pred** – The binary predicate that compares the projected elements.
- **Proj1** – The type of an optional projection function for the source range. This defaults to *hpx::identity*
- **Proj2** – The type of an optional projection function for the destination range. This defaults to *hpx::identity*

Parameters

- **first1** – Refers to the beginning of the source range.
- **last1** – Sentinel value referring to the end of the source range.
- **first2** – Refers to the beginning of the destination range.
- **last2** – Sentinel value referring to the end of the destination range.
- **pred** – Specifies the binary predicate function (or function object) which will be invoked for comparison of the elements in the in two ranges projected by proj1 and proj2 respectively.
- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements in the source range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements in the destination range as a projection operation before the actual predicate *is* invoked.

Returns

The *ends_with* algorithm returns *bool*. The *ends_with* algorithm returns a boolean with the value true if the second range matches the suffix of the first range, false otherwise.

```
template<typename ExPolicy, typename FwdIter1, typename Sent1, typename FwdIter2, typename Sent2,
typename Pred = ranges::equal_to, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
```

```
parallel::util::detail::algorithm_result<ExPolicy, bool>::type hpx::ranges::ends_with(ExPolicy &&policy,
                                                                 FwdIter1 first1, Sent1
                                                                 last1, FwdIter2 first2,
                                                                 Sent2 last2, Pred
                                                                 &&pred = Pred(), Proj1
                                                                 &&proj1 = Proj1(),
                                                                 Proj2 &&proj2 =
                                                                 Proj2())
```

Checks whether the second range defined by [first1, last1) matches the suffix of the first range defined by [first2, last2)

The assignments in the parallel *ends_with* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *ends_with* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Linear: at most min(N1, N2) applications of the predicate and both projections.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used(deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **FwdIter2** – The type of the begin destination iterators used deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent2** – The type of the end destination iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter2.
- **Pred** – The binary predicate that compares the projected elements.
- **Proj1** – The type of an optional projection function for the source range. This defaults to *hpx::identity*
- **Proj2** – The type of an optional projection function for the destination range. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the source range.
- **last1** – Sentinel value referring to the end of the source range.
- **first2** – Refers to the beginning of the destination range.
- **last2** – Sentinel value referring to the end of the destination range.

- **pred** – Specifies the binary predicate function (or function object) which will be invoked for comparison of the elements in the in two ranges projected by proj1 and proj2 respectively.
- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements in the source range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements in the destination range as a projection operation before the actual predicate *is* invoked.

Returns

The *ends_with* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *ends_with* algorithm returns a boolean with the value true if the second range matches the suffix of the first range, false otherwise.

```
template<typename Rng1, typename Rng2, typename Pred = ranges::equal_to, typename Proj1 = hpx::identity,
typename Proj2 = hpx::identity>
bool hpx::ranges::ends_with(Rng1 &&rng1, Rng2 &&rng2, Pred &&pred = Pred(), Proj1 &&proj1 = Proj1(),
                           Proj2 &&proj2 = Proj2())
```

Checks whether the second range *rng2* matches the suffix of the first range *rng1*.

The assignments in the parallel *ends_with* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: Linear: at most min(N1, N2) applications of the predicate and both projections.

Template Parameters

- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Rng2** – The type of the destination range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The binary predicate that compares the projected elements.
- **Proj1** – The type of an optional projection function for the source range. This defaults to *hpx::identity*
- **Proj2** – The type of an optional projection function for the destination range. This defaults to *hpx::identity*

Parameters

- **rng1** – Refers to the source range.
- **rng2** – Refers to the destination range.
- **pred** – Specifies the binary predicate function (or function object) which will be invoked for comparison of the elements in the in two ranges projected by proj1 and proj2 respectively.
- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements in the source range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements in the destination range as a projection operation before the actual predicate *is* invoked.

Returns

The *ends_with* algorithm returns *bool*. The *ends_with* algorithm returns a boolean with the value true if the second range matches the suffix of the first range, false otherwise.

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Pred = ranges::equal_to, typename  
Proj1 = hpx::identity, typename Proj2 = hpx::identity>  
hpx::parallel::util::detail::algorithm_result<ExPolicy, bool>::type hpx::ranges::ends_with(ExPolicy &&policy,  
Rng1 &&rng1,  
Rng2 &&rng2, Pred  
&&pred = Pred(),  
Proj1 &&proj1 =  
Proj1(), Proj2  
&&proj2 = Proj2())
```

Checks whether the second range *rng2* matches the suffix of the first range *rng1*.

The assignments in the parallel *ends_with* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *ends_with* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Linear: at most min(N1, N2) applications of the predicate and both projections.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Rng2** – The type of the destination range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The binary predicate that compares the projected elements.
- **Proj1** – The type of an optional projection function for the source range. This defaults to *hpx::identity*
- **Proj2** – The type of an optional projection function for the destination range. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the source range.
- **rng2** – Refers to the destination range.
- **pred** – Specifies the binary predicate function (or function object) which will be invoked for comparison of the elements in the in two ranges projected by proj1 and proj2 respectively.
- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements in the source range as a projection operation before the actual predicate *is* invoked.

- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements in the destination range as a projection operation before the actual predicate *is* invoked.

Returns

The *ends_with* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *ends_with* algorithm returns a boolean with the value true if the second range matches the suffix of the first range, false otherwise.

hpx::ranges::search

Defined in header [hpx/algorithm.hpp](#)⁶⁹⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::ranges::search_n*

```
template<typename FwdIter, typename Sent, typename FwdIter2, typename Sent2, typename Pred =
hpx::ranges::equal_to, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
FwdIter hpx::ranges::search(FwdIter first, Sent last, FwdIter2 s_first, Sent2 s_last, Pred &&op = Pred(), Proj1
&&proj1 = Proj1(), Proj2 &&proj2 = Proj2())
```

Searches the range [first, last) for any elements in the range [s_first, s_last). Uses a provided predicate to compare elements.

The comparison operations in the parallel *search* algorithm execute in sequential order in the calling thread.

Note: Complexity: at most ($S \cdot N$) comparisons where $S = \text{distance}(s_first, s_last)$ and $N = \text{distance}(first, last)$.

Template Parameters

- **FwdIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel used for the first range (deduced). This iterator type must meet the requirements of an sentinel.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the source sentinel used for the second range (deduced). This iterator type must meet the requirements of an sentinel.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *adjacent_find* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function. This defaults to *hpx::identity* and is applied to the elements of type dereferenced *FwdIter*.
- **Proj2** – The type of an optional projection function. This defaults to *hpx::identity* and is applied to the elements of type dereferenced *FwdIter2*.

⁶⁹⁰ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Parameters

- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **s_first** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **s_last** – Refers to the end of the sequence of elements of the algorithm will be searching for.
- **op** – Refers to the binary predicate which returns true if the elements should be treated as equal. the signature of the function should be equivalent to

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of type dereferenced *FwdIter1* as a projection operation before the actual predicate is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of type dereferenced *FwdIter2* as a projection operation before the actual predicate is invoked.

Returns

The *search* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *task_execution_policy* and returns *FwdIter* otherwise. The *search* algorithm returns an iterator to the beginning of the first subsequence [*s_first*, *s_last*) in range [*first*, *last*). If the length of the subsequence [*s_first*, *s_last*) is greater than the length of the range [*first*, *last*), *last* is returned. Additionally if the size of the subsequence is empty *first* is returned. If no subsequence is found, *last* is returned.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename FwdIter2, typename Sent2,
typename Pred = hpx::ranges::equal_to, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
parallel::util::detail::algorithm_result<ExPolicy, FwdIter>::type hpx::ranges::search(ExPolicy &&policy,
FwdIter first, Sent last,
FwdIter2 s_first, Sent2
s_last, Pred &&op =
Pred(), Proj1 &&proj1 =
Proj1(), Proj2 &&proj2 =
Proj2())
```

Searches the range [*first*, *last*) for any elements in the range [*s_first*, *s_last*). Uses a provided predicate to compare elements.

The comparison operations in the parallel *search* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *search* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: at most $(S*N)$ comparisons where $S = \text{distance}(s_first, s_last)$ and $N = \text{distance}(first, last)$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel used for the first range (deduced). This iterator type must meet the requirements of an sentinel.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the source sentinel used for the second range (deduced). This iterator type must meet the requirements of an sentinel.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *adjacent_find* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function. This defaults to *hpx::identity* and is applied to the elements of type dereferenced *FwdIter*.
- **Proj2** – The type of an optional projection function. This defaults to *hpx::identity* and is applied to the elements of type dereferenced *FwdIter2*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **s_first** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **s_last** – Refers to the end of the sequence of elements of the algorithm will be searching for.
- **op** – Refers to the binary predicate which returns true if the elements should be treated as equal. the signature of the function should be equivalent to

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of type dereferenced *FwdIter1* as a projection operation before the actual predicate *is* invoked.

- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of type dereferenced *FwdIter2* as a projection operation before the actual predicate *is* invoked.

Returns

The *search* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *task_execution_policy* and returns *FwdIter* otherwise. The *search* algorithm returns an iterator to the beginning of the first subsequence *[s_first, s_last)* in range *[first, last)*. If the length of the subsequence *[s_first, s_last)* is greater than the length of the range *[first, last)*, *last* is returned. Additionally if the size of the subsequence is empty *first* is returned. If no subsequence is found, *last* is returned.

```
template<typename Rng1, typename Rng2, typename Pred = hpx::ranges::equal_to, typename Proj1 = hpx::identity,
typename Proj2 = hpx::identity>
std::ranges::iterator_t<Rng1> hpx::ranges::search(Rng1 &&rng1, Rng2 &&rng2, Pred &&op = Pred(), Proj1
&&proj1 = Proj1(), Proj2 &&proj2 = Proj2())
```

Searches the range *[first, last)* for any elements in the range *[s_first, s_last)*. Uses a provided predicate to compare elements.

The comparison operations in the parallel *search* algorithm execute in sequential order in the calling thread.

Note: Complexity: at most (*S***N*) comparisons where *S* = distance(*s_first*, *s_last*) and *N* = distance(*first*, *last*).

Template Parameters

- **Rng1** – The type of the examine range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the search range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *adjacent_find* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to *std::equal_to*<>
- **Proj1** – The type of an optional projection function. This defaults to *hpx::identity* and is applied to the elements of *Rng1*.
- **Proj2** – The type of an optional projection function. This defaults to *hpx::identity* and is applied to the elements of *Rng2*.

Parameters

- **rng1** – Refers to the sequence of elements the algorithm will be examining.
- **rng2** – Refers to the sequence of elements the algorithm will be searching for.
- **op** – Refers to the binary predicate which returns true if the elements should be treated as equal. the signature of the function should be equivalent to

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have *const &*, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of *rng1* as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of *rng2* as a projection operation before the actual predicate *is* invoked.

Returns

The *search* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *task_execution_policy* and returns *FwdIter* otherwise. The *search* algorithm returns an iterator to the beginning of the first subsequence [s_first, s_last) in range [first, last). If the length of the subsequence [s_first, s_last) is greater than the length of the range [first, last), *last* is returned. Additionally if the size of the subsequence is empty *first* is returned. If no subsequence is found, *last* is returned.

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Pred = hpx::ranges::equal_to, typename
Proj1 = hpx::identity, typename Proj2 = hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng1>> hpx::ranges::search(ExPolicy
&&policy,
Rng1
&&rng1,
Rng2
&&rng2,
Pred
&&op
=
Pred(),
Proj1
&&proj1
=
Proj1(),
Proj2
&&proj2
=
Proj2())
```

Searches the range [first, last) for any elements in the range [s_first, s_last). Uses a provided predicate to compare elements.

The comparison operations in the parallel *search* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *search* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: at most ($S \cdot N$) comparisons where $S = \text{distance}(s_first, s_last)$ and $N = \text{distance}(first, last)$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.

- **Rng1** – The type of the examine range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the search range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *adjacent_find* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function. This defaults to *hpx::identity* and is applied to the elements of *Rng1*.
- **Proj2** – The type of an optional projection function. This defaults to *hpx::identity* and is applied to the elements of *Rng2*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the sequence of elements the algorithm will be examining.
- **rng2** – Refers to the sequence of elements the algorithm will be searching for.
- **op** – Refers to the binary predicate which returns true if the elements should be treated as equal. the signature of the function should be equivalent to

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of *rng1* as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of *rng2* as a projection operation before the actual predicate *is* invoked.

Returns

The *search* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *task_execution_policy* and returns *FwdIter* otherwise. The *search* algorithm returns an iterator to the beginning of the first subsequence `[s_first, s_last)` in range `[first, last)`. If the length of the subsequence `[s_first, s_last)` is greater than the length of the range `[first, last)`, *last* is returned. Additionally if the size of the subsequence is empty *first* is returned. If no subsequence is found, *last* is returned.

hpx::ranges::search_n

Defined in header `hpx/algorithm.hpp`⁶⁹¹.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::ranges::search*

```
template<typename FwdIter, typename FwdIter2, typename Sent2, typename Pred = hpx::ranges::equal_to,
        typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
```

⁶⁹¹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

```
FwdIter hpx::ranges::search_n(FwdIter first, std::size_t count, FwdIter2 s_first, Sent s_last, Pred &&op =
    Pred(), Proj1 &&proj1 = Proj1(), Proj2 &&proj2 = Proj2())
```

Searches the range [first, last) for any elements in the range [s_first, s_last). Uses a provided predicate to compare elements.

The comparison operations in the parallel *search_n* algorithm execute in sequential order in the calling thread.

Note: Complexity: at most ($S \cdot N$) comparisons where $S = \text{distance}(s_first, s_last)$ and $N = \text{count}$.

Template Parameters

- **FwdIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the source sentinel used for the second range (deduced). This iterator type must meet the requirements of an sentinel.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *adjacent_find* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function. This defaults to *hpx::identity* and is applied to the elements of type dereferenced *FwdIter*.
- **Proj2** – The type of an optional projection function. This defaults to *hpx::identity* and is applied to the elements of type dereferenced *FwdIter2*.

Parameters

- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **count** – Refers to the range of elements of the first range the algorithm will be applied to.
- **s_first** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **s_last** – Refers to the end of the sequence of elements of the algorithm will be searching for.
- **op** – Refers to the binary predicate which returns true if the elements should be treated as equal. the signature of the function should be equivalent to

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of type dereferenced *FwdIter1* as a projection operation before the actual predicate *is* invoked.

- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of type dereferenced *FwdIter2* as a projection operation before the actual predicate *is* invoked.

Returns

The *search_n* algorithm returns *FwdIter*. The *search_n* algorithm returns an iterator to the beginning of the last subsequence [s_first, s_last) in range [first, first+count). If the length of the subsequence [s_first, s_last) is greater than the length of the range [first, first+count), *first* is returned. Additionally, if the size of the subsequence is empty or no subsequence is found, *first* is also returned.

```
template<typename ExPolicy, typename FwdIter, typename FwdIter2, typename Sent2, typename Pred =  
hpx::ranges::equal_to, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>  
hpx::parallel::util::detail::algorithm_result<ExPolicy, FwdIter>::type hpx::ranges::search_n(ExPolicy  
                                                    &&policy,  
                                                    FwdIter first,  
                                                    std::size_t count,  
                                                    FwdIter2 s_first,  
                                                    Sent2 s_last, Pred  
                                                    &&op = Pred(),  
                                                    Proj1 &&proj1 =  
                                                    Proj1(), Proj2  
                                                    &&proj2 =  
                                                    Proj2())
```

Searches the range [first, last) for any elements in the range [s_first, s_last). Uses a provided predicate to compare elements.

The comparison operations in the parallel *search_n* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *search_n* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: at most ($S \cdot N$) comparisons where S = distance(s_first, s_last) and N = count.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the source sentinel used for the second range (deduced). This iterator type must meet the requirements of an sentinel.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *adjacent_find* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

- **Proj1** – The type of an optional projection function. This defaults to *hpx::identity* and is applied to the elements of type dereferenced *FwdIter*.
- **Proj2** – The type of an optional projection function. This defaults to *hpx::identity* and is applied to the elements of type dereferenced *FwdIter2*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **count** – Refers to the range of elements of the first range the algorithm will be applied to.
- **s_first** – Refers to the beginning of the sequence of elements the algorithm will be searching for.
- **s_last** – Refers to the end of the sequence of elements of the algorithm will be searching for.
- **op** – Refers to the binary predicate which returns true if the elements should be treated as equal. the signature of the function should be equivalent to

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of type dereferenced *FwdIter1* as a projection operation before the actual predicate is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of type dereferenced *FwdIter2* as a projection operation before the actual predicate is invoked.

Returns

The *search_n* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *task_execution_policy* and returns *FwdIter* otherwise. The *search_n* algorithm returns an iterator to the beginning of the last subsequence [s_first, s_last) in range [first, first+count). If the length of the subsequence [s_first, s_last) is greater than the length of the range [first, first+count), *first* is returned. Additionally if the size of the subsequence is empty or no subsequence is found, *first* is also returned.

```
template<typename Rng1, typename Rng2, typename Pred = hpx::ranges::equal_to, typename Proj1 = hpx::identity,
typename Proj2 = hpx::identity>
std::ranges::iterator_t<Rng1> hpx::ranges::search_n(Rng1 &&rng1, std::size_t count, Rng2 &&rng2, Pred
&&op = Pred(), Proj1 &&proj1 = Proj1(), Proj2
&&proj2 = Proj2())
```

Searches the range [first, last) for any elements in the range [s_first, s_last). Uses a provided predicate to compare elements.

The comparison operations in the parallel *search* algorithm execute in sequential order in the calling thread.

Note: Complexity: at most (S*N) comparisons where *S* = distance(s_first, s_last) and *N* = distance(first, last).

Template Parameters

- **Rng1** – The type of the examine range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the search range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *adjacent_find* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function. This defaults to `hpx::identity` and is applied to the elements of *Rng1*.
- **Proj2** – The type of an optional projection function. This defaults to `hpx::identity` and is applied to the elements of *Rng2*.

Parameters

- **rng1** – Refers to the sequence of elements the algorithm will be examining.
- **count** – The number of elements to apply the algorithm on.
- **rng2** – Refers to the sequence of elements the algorithm will be searching for.
- **op** – Refers to the binary predicate which returns true if the elements should be treated as equal. the signature of the function should be equivalent to

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of *rng1* as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of *rng2* as a projection operation before the actual predicate *is* invoked.

Returns

The *search* algorithm returns a `hpx::future<FwdIter>` if the execution policy is of type *task_execution_policy* and returns *FwdIter* otherwise. The *search* algorithm returns an iterator to the beginning of the first subsequence [s_first, s_last) in range [first, last). If the length of the subsequence [s_first, s_last) is greater than the length of the range [first, last), *last* is returned. Additionally if the size of the subsequence is empty *first* is returned. If no subsequence is found, *last* is returned.

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Pred = hpx::ranges::equal_to, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
```

```

hpx::parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng1>> hpx::ranges::search_n(ExPolicy
                                                                    &&pol-
                                                                    icy,
                                                                    Rng1
                                                                    &&rng1,
                                                                    std::size_t
                                                                    count,
                                                                    Rng2
                                                                    &&rng2,
                                                                    Pred
                                                                    &&op
                                                                    =
                                                                    Pred(),
                                                                    Proj1
                                                                    &&proj1
                                                                    =
                                                                    Proj1(),
                                                                    Proj2
                                                                    &&proj2
                                                                    =
                                                                    Proj2())

```

Searches the range [first, last) for any elements in the range [s_first, s_last). Uses a provided predicate to compare elements.

The comparison operations in the parallel *search* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *search* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: at most ($S \cdot N$) comparisons where $S = \text{distance}(s_first, s_last)$ and $N = \text{distance}(first, last)$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng1** – The type of the examine range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the search range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *adjacent_find* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function. This defaults to `hpx::identity` and is applied to the elements of *Rng1*.
- **Proj2** – The type of an optional projection function. This defaults to `hpx::identity` and is applied to the elements of *Rng2*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the sequence of elements the algorithm will be examining.
- **count** – The number of elements to apply the algorithm on.
- **rng2** – Refers to the sequence of elements the algorithm will be searching for.
- **op** – Refers to the binary predicate which returns true if the elements should be treated as equal. the signature of the function should be equivalent to

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of *rng1* as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of *rng2* as a projection operation before the actual predicate *is* invoked.

Returns

The *search* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *task_execution_policy* and returns *FwdIter* otherwise. The *search* algorithm returns an iterator to the beginning of the first subsequence [*s_first*, *s_last*) in range [*first*, *last*). If the length of the subsequence [*s_first*, *s_last*) is greater than the length of the range [*first*, *last*), *last* is returned. Additionally if the size of the subsequence is empty *first* is returned. If no subsequence is found, *last* is returned.

hpx::ranges::adjacent_difference

Defined in header [hpx/algorithm.hpp](#)⁶⁹².

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename FwdIter1, typename FwdIter2, typename Sent>
FwdIter2 hpx::ranges::adjacent_difference(FwdIter1 first, Sent last, FwdIter2 dest)
```

Searches the range [*first*, *last*) for two consecutive identical elements.

Note: Complexity: Exactly the smaller of (*result* - *first*) + 1 and (*last* - *first*) - 1 application of the predicate where *result* is the value returned

Template Parameters

- **FwdIter1** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter*.

⁶⁹² <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Parameters

- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *adjacent_difference* algorithm returns an iterator to the first of the identical elements. If no such elements are found, *last* is returned.

```
template<typename Rng, typename FwdIter2>
FwdIter2 hpx::ranges::adjacent_difference(Rng &&rng, FwdIter2 dest)
```

Searches the *rng* for two consecutive identical elements.

Note: Complexity: Exactly the smaller of (result - first) + 1 and (last - first) - 1 application of the predicate where *result* is the value returned

Template Parameters

- **FwdIter2** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *adjacent_difference* algorithm returns an iterator to the first of the identical elements.

```
template<typename ExPolicy, typename FwdIter1, typename Sent, typename FwdIter2>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx::ranges::adjacent_difference(ExPolicy
                                                                                                     &&pol-
                                                                                                     icy,
                                                                                                     FwdIter1
                                                                                                     first,
                                                                                                     Sent
                                                                                                     last,
                                                                                                     FwdIter2
                                                                                                     dest)
```

Searches the range [first, last) for two consecutive identical elements.

Note: Complexity: Exactly the smaller of (result - first) + 1 and (last - first) - 1 application of the predicate where *result* is the value returned

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *adjacent_difference* algorithm returns an iterator to the first of the identical elements. If no such elements are found, *last* is returned.

```
template<typename ExPolicy, typename Rng, typename FwdIter2>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx::ranges::adjacent_difference(ExPolicy
                                                                                               &&pol-
                                                                                               icy,
                                                                                               Rng
                                                                                               &&rng,
                                                                                               FwdIter2
                                                                                               dest)
```

Searches the *rng* for two consecutive identical elements.

Note: Complexity: Exactly the smaller of (result - first) + 1 and (last - first) - 1 application of the predicate where *result* is the value returned

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter2** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.

- **dest** – Refers to the beginning of the destination range.

Returns

The *adjacent_difference* algorithm returns an iterator to the first of the identical elements.

```
template<typename FwdIter1, typename Sent, typename FwdIter2, typename Op>
FwdIter2 hpx::ranges::adjacent_difference(FwdIter1 first, Sent last, FwdIter2 dest, Op &&op)
```

Searches the range [first, last) for two consecutive identical elements.

Note: Complexity: Exactly the smaller of (result - first) + 1 and (last - first) - 1 application of the predicate where *result* is the value returned

Template Parameters

- **FwdIter1** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter.
- **Op** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *adjacent_difference* requires *Op* to meet the requirements of *CopyConstructible*.

Parameters

- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – Binary operation function object that will be applied. The signature of the function should be equivalent to the following:

Ret fun(const Type1 &a, const Type2 &b);
--

The signature does not need to have const &. The types *Type1* and *Type2* must be such that an object of type *iterator_traits<InputIt>::value_type* can be implicitly converted to both of them. The type *Ret* must be such that an object of type *OutputIt* can be dereferenced and assigned a value of type *Ret*.

Returns

The *adjacent_difference* algorithm returns an iterator to the first of the identical elements. If no such elements are found, *last* is returned.

```
template<typename Rng, typename FwdIter2, typename Op>
FwdIter2 hpx::ranges::adjacent_difference(Rng &&rng, FwdIter2 dest, Op &&op)
```

Searches the *rng* for two consecutive identical elements.

Template Parameters

- **FwdIter2** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Op** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *adjacent_difference* requires *Op* to meet the requirements of *Copy-Constructible*.

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – Binary operation function object that will be applied. The signature of the function should be equivalent to the following:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`. The types *Type1* and *Type2* must be such that an object of type *iterator_traits<InputIt>::value_type* can be implicitly converted to both of them. The type *Ret* must be such that an object of type *OutputIt* can be dereferenced and assigned a value of type *Ret*.

Returns

The *adjacent_difference* algorithm returns an iterator to the first of the identical elements.

```
template<typename ExPolicy, typename FwdIter1, typename Sent, typename FwdIter2, typename Op>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx::ranges::adjacent_difference(ExPolicy  
                                                                                               &&pol-  
                                                                                               icy,  
                                                                                               FwdIter1  
                                                                                               first,  
                                                                                               Sent  
                                                                                               last,  
                                                                                               FwdIter2  
                                                                                               dest,  
                                                                                               Op  
                                                                                               &&op)
```

Searches the range [first, last) for two consecutive identical elements.

Note: Complexity: Exactly the smaller of (result - first) + 1 and (last - first) - 1 application of the predicate where *result* is the value returned

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter*.

- **Op** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *adjacent_difference* requires *Op* to meet the requirements of *CopyConstructible*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – Binary operation function object that will be applied. The signature of the function should be equivalent to the following:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`. The types *Type1* and *Type2* must be such that an object of type *iterator_traits<InputIt>::value_type* can be implicitly converted to both of them. The type *Ret* must be such that an object of type *OutputIt* can be dereferenced and assigned a value of type *Ret*.

Returns

The *adjacent_difference* algorithm returns an iterator to the first of the identical elements. If no such elements are found, *last* is returned.

```
template<typename ExPolicy, typename Rng, typename FwdIter2, typename Op>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter2> hpx::ranges::adjacent_difference(ExPolicy
                                                                                               &&pol-
                                                                                               icy,
                                                                                               Rng
                                                                                               &&rng,
                                                                                               FwdIter2
                                                                                               dest,
                                                                                               Op
                                                                                               &&op)
```

Searches the *rng* for two consecutive identical elements.

Note: Complexity: Exactly the smaller of (result - first) + 1 and (last - first) - 1 application of the predicate where *result* is the value returned

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter2** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

- **Op** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *adjacent_difference* requires *Op* to meet the requirements of *CopyConstructible*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – Binary operation function object that will be applied. The signature of the function should be equivalent to the following:

```
Ret fun(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`. The types *Type1* and *Type2* must be such that an object of type *iterator_traits<InputIt>::value_type* can be implicitly converted to both of them. The type *Ret* must be such that an object of type *OutputIt* can be dereferenced and assigned a value of type *Ret*.

Returns

The *adjacent_difference* algorithm returns an iterator to the first of the identical elements.

hpx::ranges::set_union

Defined in header `hpx/algorithm.hpp`⁶⁹³.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<typename ExPolicy, typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename Iter3, typename Pred = hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
```

⁶⁹³ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

```

hpx::parallel::util::detail::algorithm_result<ExPolicy, set_union_result<Iter1, Iter2, Iter3>>::type hpx::ranges::set_union(ExPolicy,
&&policy,
Iter1,
first1,
Sent1,
last1,
Iter2,
first2,
Sent2,
last2,
Iter3,
dest,
Pred,
&&operator=,
Pred(),
Proj1,
&&project,
=,
Proj1(),
Proj2,
&&project,
=,
Proj2())

```

Constructs a sorted range beginning at `dest` consisting of all elements present in one or both sorted ranges `[first1, last1)` and `[first2, last2)`. This algorithm expects both input ranges to be sorted with the given binary predicate `f`.

If some element is found m times in `[first1, last1)` and n times in `[first2, last2)`, then all m elements will be copied from `[first1, last1)` to `dest`, preserving order, and then exactly `std::max(n-m, 0)` elements will be copied from `[first2, last2)` to `dest`, also preserving order.

The resulting range cannot overlap with either of the input ranges.

The application of function objects in parallel algorithm invoked with a sequential execution policy object execute in sequential order in the calling thread (*sequenced_policy*) or in a single new thread spawned from the current thread (for *sequenced_task_policy*).

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most $2 \cdot (N1 + N2 - 1)$ comparisons, where $N1$ is the length of the first sequence and $N2$ is the length of the second sequence.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.

- **Iter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the end source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an sentinel for Iter2.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set_union* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

Returns

The *set_union* algorithm returns a `hpx::future<ranges::set_union_result<Iter1, Iter2, Iter3>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns

ranges::set_union_result<Iter1, Iter2, Iter3> otherwise. The *set_union* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Iter3, typename Pred =
hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, set_union_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng2>, It
```

Constructs a sorted range beginning at *dest* consisting of all elements present in one or both sorted ranges *[first1, last1)* and *[first2, last2)*. This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

If some element is found *m* times in *[first1, last1)* and *n* times in *[first2, last2)*, then all *m* elements will be copied from *[first1, last1)* to *dest*, preserving order, and then exactly *std::max(n-m, 0)* elements will be copied from *[first2, last2)* to *dest*, also preserving order.

The resulting range cannot overlap with either of the input ranges.

The application of function objects in parallel algorithm invoked with a sequential execution policy object execute in sequential order in the calling thread (*sequenced_policy*) or in a single new thread spawned from the current thread (for *sequenced_task_policy*).

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most $2 \cdot (N1 + N2 - 1)$ comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.

- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set_union* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

Returns

The *set_union* algorithm returns a `hpx::future<ranges::set_union_result<Iter1, Iter2, Iter3>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *ranges::set_union_result<Iter1, Iter2, Iter3>* otherwise. where *Iter1* is *range_iterator_t<Rng1>* and *Iter2* is *range_iterator_t<Rng2>* The *set_union* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng1, typename Rng2, typename Iter3, typename Pred = hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
```

```

set_union_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng2>, Iter3> hpx::ranges::set_union(Rng1
&&rng1,
Rng2
&&rng2,
Iter3
dest,
Pred
&&op
=
Pred(),
Proj1
&&proj1
=
Proj1(),
Proj2
&&proj2
=
Proj2())

```

Constructs a sorted range beginning at *dest* consisting of all elements present in one or both sorted ranges *[first1, last1)* and *[first2, last2)*. This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

If some element is found *m* times in *[first1, last1)* and *n* times in *[first2, last2)*, then all *m* elements will be copied from *[first1, last1)* to *dest*, preserving order, and then exactly `std::max(n-m, 0)` elements will be copied from *[first2, last2)* to *dest*, also preserving order.

The resulting range cannot overlap with either of the input ranges.

Note: Complexity: At most $2 \cdot (N1 + N2 - 1)$ comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

Template Parameters

- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set_union* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

Parameters

- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.

- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *Iter* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

Returns

The *set_union* algorithm returns *ranges::set_union_result<Iter1, Iter2, Iter3>*. where *Iter1* is *range_iterator_t<Rng1>* and *Iter2* is *range_iterator_t<Rng2>* The *set_union* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

hpx::ranges::rotate

Defined in header `hpx/algorithm.hpp`⁶⁹⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::ranges::rotate_copy*

template<typename **FwdIter**, typename **Sent**>

subrange_t<*FwdIter*, *Sent*> **hpx::ranges::rotate**(*FwdIter* first, *FwdIter* middle, *Sent* last)

Performs a left rotation on a range of elements. Specifically, *rotate* swaps the elements in the range [first, last) in such a way that the element middle becomes the first element of the new range and middle - 1 becomes the last element.

The assignments in the parallel *rotate* algorithm execute in sequential order in the calling thread.

Note: Complexity: Linear in the distance between *first* and *last*.

Note: The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable* and *MoveConstructible*.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.

⁶⁹⁴ <http://github.com/TheHPXProject/hpx/blob/d402cead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for `FwdIter`.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the element that should appear at the beginning of the rotated range.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

Returns

The *rotate* algorithm returns a *subrange_t*<*FwdIter*, *Sent*>. The *rotate* algorithm returns the iterator equal to `pair(first + (last - middle), last)`.

```
template<typename ExPolicy, typename FwdIter, typename Sent>
parallel::util::detail::algorithm_result<ExPolicy, subrange_t<FwdIter, Sent>>::type hpx::ranges::rotate(ExPolicy
&&pol-
icy,
FwdIter
first,
FwdIter
mid-
dle,
Sent
last)
```

Performs a left rotation on a range of elements. Specifically, *rotate* swaps the elements in the range `[first, last)` in such a way that the element `middle` becomes the first element of the new range and `middle - 1` becomes the last element.

The assignments in the parallel *rotate* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *rotate* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Linear in the distance between *first* and *last*.

Note: The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable* and *MoveConstructible*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for `FwdIter`.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the element that should appear at the beginning of the rotated range.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

Returns

The *rotate* algorithm returns a `hpx::future<subrange_t<FwdIter, Sent>>` if the execution policy is of type *parallel_task_policy* and returns a *subrange_t<FwdIter, Sent>* otherwise. The *rotate* algorithm returns the iterator equal to `pair(first + (last - middle), last)`.

```
template<typename Rng>
subrange_t<std::ranges::iterator_t<Rng>, std::ranges::iterator_t<Rng>> hpx::ranges::rotate(Rng &&rng,
                                                                                          std::ranges::iterator_t<Rng>
                                                                                          middle)
```

Uses *rng* as the source range, as if using *util::begin(rng)* as *first* and *ranges::end(rng)* as *last*. Performs a left rotation on a range of elements. Specifically, *rotate* swaps the elements in the range `[first, last)` in such a way that the element *middle* becomes the first element of the new range and *middle - 1* becomes the last element.

The assignments in the parallel *rotate* algorithm execute in sequential order in the calling thread.

Note: Complexity: Linear in the distance between *first* and *last*.

Note: The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable* and *MoveConstructible*.

Template Parameters

Rng – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a forward iterator.

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the element that should appear at the beginning of the rotated range.

Returns

The *rotate* algorithm returns a *subrange_t<std::ranges::iterator_t<Rng>, std::ranges::iterator_t<Rng>>*. The *rotate* algorithm returns the iterator equal to `pair(first + (last - middle), last)`.

```
template<typename ExPolicy, typename Rng>
parallel::util::detail::algorithm_result<ExPolicy, subrange_t<std::ranges::iterator_t<Rng>, std::ranges::iterator_t<Rng>>> hpx::range
```

Uses *rng* as the source range, as if using *util::begin(rng)* as *first* and *ranges::end(rng)* as *last*. Performs a left

rotation on a range of elements. Specifically, *rotate* swaps the elements in the range [first, last) in such a way that the element *middle* becomes the first element of the new range and *middle* - 1 becomes the last element.

The assignments in the parallel *rotate* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *rotate* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Linear in the distance between *first* and *last*.

Note: The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable* and *MoveConstructible*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the element that should appear at the beginning of the rotated range.

Returns

The *rotate* algorithm returns a `hpx::future <subrange_t<std::ranges::iterator_t<Rng>, std::ranges::iterator_t<Rng>>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *subrange_t<std::ranges::iterator_t<Rng>, std::ranges::iterator_t<Rng>>*, otherwise. The *rotate* algorithm returns the iterator equal to `pair(first + (last - middle), last)`.

hpx::ranges::rotate_copy

Defined in header `hpx/algorithm.hpp`⁶⁹⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::rotate`

```
template<typename FwdIter, typename Sent, typename OutIter>
rotate_copy_result<FwdIter, OutIter> hpx::ranges::rotate_copy(FwdIter first, FwdIter middle, Sent last,
                                                             OutIter dest_first)
```

Copies the elements from the range [first, last), to another range beginning at *dest_first* in such a way, that the element *middle* becomes the first element of the new range and *middle* - 1 becomes the last element.

⁶⁹⁵ <http://github.com/TheHPXProject/hpx/blob/d402cead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

The assignments in the parallel *rotate_copy* algorithm execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for FwdIter.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the element that should appear at the beginning of the rotated range.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest_first** – Output iterator to the initial position of the range where the reversed range is stored. The pointed type shall support being assigned the value of an element in the range [first,last).

Returns

The *rotate_copy* algorithm returns a *rotate_copy_result<FwdIter, OutIter>*. The *rotate_copy* algorithm returns the output iterator to the element past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Sent, typename FwdIter2>
parallel::util::detail::algorithm_result<ExPolicy, rotate_copy_result<FwdIter1, FwdIter2>>::type hpx::ranges::rotate_copy(ExPo
&&p
icy,
Fwd
first,
Fwd
mid-
dle,
Sent
last,
Fwd
dest
```

Copies the elements from the range [first, last), to another range beginning at *dest_first* in such a way, that the element *middle* becomes the first element of the new range and *middle* - 1 becomes the last element.

The assignments in the parallel *rotate_copy* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *rotate_copy* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for FwdIter.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the element that should appear at the beginning of the rotated range.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest_first** – Output iterator to the initial position of the range where the reversed range is stored. The pointed type shall support being assigned the value of an element in the range [first,last).

Returns

The *rotate_copy* algorithm returns `hpx::future< rotate_copy_result<FwdIter1, FwdIter2>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `rotate_copy_result<FwdIter1, FwdIter2>` otherwise. The *rotate_copy* algorithm returns the output iterator to the element past the last element copied.

```
template<typename Rng, typename OutIter>
rotate_copy_result<std::ranges::iterator_t<Rng>, OutIter> hpx::ranges::rotate_copy(Rng &&rng,
                                                                                    std::ranges::iterator_t<Rng>
                                                                                    middle, OutIter
                                                                                    dest_first)
```

Uses *rng* as the source range, as if using `util::begin(rng)` as *first* and `ranges::end(rng)` as *last*. Copies the elements from the range [first, last), to another range beginning at *dest_first* in such a way, that the element *middle* becomes the first element of the new range and *middle* - 1 becomes the last element.

The assignments in the parallel *rotate_copy* algorithm execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a forward iterator.

- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the element that should appear at the beginning of the rotated range.
- **dest_first** – Output iterator to the initial position of the range where the reversed range is stored. The pointed type shall support being assigned the value of an element in the range [first,last).

Returns

The *rotate* algorithm returns a *rotate_copy_result*<std::ranges::iterator_t<Rng>, OutIter>. The *rotate_copy* algorithm returns the output iterator to the element past the last element copied.

```
template<typename ExPolicy, typename Rng, typename OutIter>  
parallel::util::detail::algorithm_result<ExPolicy, rotate_copy_result<std::ranges::iterator_t<Rng>, OutIter>> hpx::ranges::rotate_
```

Uses *rng* as the source range, as if using *util::begin(rng)* as *first* and *ranges::end(rng)* as *last*. Copies the elements from the range [first, last), to another range beginning at *dest_first* in such a way, that the element *new_first* becomes the first element of the new range and *new_first* - 1 becomes the last element.

The assignments in the parallel *rotate_copy* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *rotate_copy* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last* - *first* assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a forward iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the element that should appear at the beginning of the rotated range.
- **dest_first** – Output iterator to the initial position of the range where the reversed range is stored. The pointed type shall support being assigned the value of an element in the range [first,last).

Returns

The `rotate_copy` algorithm returns a `hpx::future<rotate_copy_result<std::ranges::iterator_t<Rng>, OutIter>>` if the execution policy is of type `parallel_task_policy` and returns `rotate_copy_result<std::ranges::iterator_t<Rng>, OutIter>` otherwise. The `rotate_copy` algorithm returns the output iterator to the element past the last element copied.

hpx::ranges::for_each

Defined in header `hpx/algorithm.hpp`⁶⁹⁶.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::for_each_n`

```
template<typename InIter, typename Sent, typename F, typename Proj = hpx::identity>
for_each_result<InIter, F> hpx::ranges::for_each(InIter first, Sent last, F &&f, Proj &&proj = Proj())
```

Applies *f* to the result of dereferencing every iterator in the range [first, last).

If *f* returns a result, the result is ignored.

If the type of *first* satisfies the requirements of a mutable iterator, *f* may apply non-constant functions through the dereferenced iterator.

Note: Complexity: Applies *f* exactly *last - first* times.

Template Parameters

- **InIter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of `for_each` requires *F* to meet the requirements of `CopyConstructible`.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

⁶⁹⁶ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

`<ignored> pred(const Type &a);`

The signature does not need to have `const&`. The type *Type* must be such that an object of type *Iter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

{last, HPX_MOVE(f)} where last is the iterator corresponding to the input sentinel last.

```
template<typename Rng, typename F, typename Proj = hpx::identity>
for_each_result<std::ranges::iterator_t<Rng>, F> hpx::ranges::for_each(Rng &&rng, F &&f, Proj &&proj =
                                                                    Proj())
```

Applies *f* to the result of dereferencing every iterator in the given range *rng*.

If *f* returns a result, the result is ignored.

If the type of *first* satisfies the requirements of a mutable iterator, *f* may apply non-constant functions through the dereferenced iterator.

Note: Complexity: Applies *f* exactly *size(rng)* times.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *for_each* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

`<ignored> pred(const Type &a);`

The signature does not need to have `const&`. The type *Type* must be such that an object of type *Iter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

{std::end(rng), HPX_MOVE(f)}

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename F, typename Proj = hpx::identity>
```

```
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::ranges::for_each(ExPolicy &&policy,
                                                                                          FwdIter first, Sent
                                                                                          last, F &&f, Proj
                                                                                          &&proj = Proj())
```

Applies *f* to the result of dereferencing every iterator in the range [first, last).

If *f* returns a result, the result is ignored.

If the type of *first* satisfies the requirements of a mutable iterator, *f* may apply non-constant functions through the dereferenced iterator.

Unlike its sequential form, the parallel overload of *for_each* does not return a copy of its *Function* parameter, since parallelization may not permit efficient state accumulation.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Applies *f* exactly *last - first* times.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **FwdIter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter*.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *for_each* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
<ignored> pred(const Type &a);
```

The signature does not need to have `const&`. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *for_each* algorithm returns a `hpx::future<FwdIter>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. It returns *last*.

```
template<typename ExPolicy, typename Rng, typename F, typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng>> hpx::ranges::for_each(ExPolicy
                                                                                                     &&pol-
                                                                                                     icy,
                                                                                                     Rng
                                                                                                     &&rng,
                                                                                                     F
                                                                                                     &&f,
                                                                                                     Proj
                                                                                                     &&proj
                                                                                                     =
                                                                                                     Proj())
```

Applies *f* to the result of dereferencing every iterator in the given range *rng*.

If *f* returns a result, the result is ignored.

If the type of *first* satisfies the requirements of a mutable iterator, *f* may apply non-constant functions through the dereferenced iterator.

Unlike its sequential form, the parallel overload of *for_each* does not return a copy of its *Function* parameter, since parallelization may not permit efficient state accumulation.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Applies *f* exactly *size(rng)* times.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *for_each* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
<ignored> pred(const Type &a);
```

The signature does not need to have `const&`. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *for_each* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. It returns *last*.

hpx::ranges::for_each_n

Defined in header `hpx/algorithm.hpp`⁶⁹⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::ranges::for_each*

```
template<typename InIter, typename Size, typename F, typename Proj = hpx::identity>
for_each_n_result<InIter, F> hpx::ranges::for_each_n(InIter first, Size count, F &&f, Proj &&proj = Proj())
```

Applies *f* to the result of dereferencing every iterator in the range [first, first + count), starting from first and proceeding to first + count - 1.

If *f* returns a result, the result is ignored.

If the type of *first* satisfies the requirements of a mutable iterator, *f* may apply non-constant functions through the dereferenced iterator.

Unlike its sequential form, the parallel overload of *for_each* does not return a copy of its *Function* parameter, since parallelization may not permit efficient state accumulation.

Note: Complexity: Applies *f* exactly *count* times.

Template Parameters

- **InIter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an input iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *for_each* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

⁶⁹⁷ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

`<ignored> pred(const Type &a);`

The signature does not need to have `const&`. The type *Type* must be such that an object of type *Iter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

It returns *last*.

```
template<typename ExPolicy, typename FwdIter, typename Size, typename F, typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, FwdIter>::type hpx::ranges::for_each_n(ExPolicy
                                                                                               &&policy,
                                                                                               FwdIter first,
                                                                                               Size count, F
                                                                                               &&f, Proj
                                                                                               &&proj =
                                                                                               Proj())
```

Applies *f* to the result of dereferencing every iterator in the range [first, first + count), starting from first and proceeding to first + count - 1.

If *f* returns a result, the result is ignored.

If the type of *first* satisfies the requirements of a mutable iterator, *f* may apply non-constant functions through the dereferenced iterator.

Unlike its sequential form, the parallel overload of *for_each* does not return a copy of its *Function* parameter, since parallelization may not permit efficient state accumulation.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Applies *f* exactly *count* times.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.

- **FwdIter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *for_each* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
<ignored> pred(const Type &a);
```

The signature does not need to have const&. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *for_each* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. It returns *last*.

hpx::ranges::set_symmetric_difference

Defined in header [hpx/algorithm.hpp](#)⁶⁹⁸.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<typename ExPolicy, typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename Iter3,
typename Pred = hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
```

⁶⁹⁸ <http://github.com/TheHPXProject/hpx/blob/d402eed7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

hpx::parallel::util::detail::algorithm_result<ExPolicy, set_symmetric_difference_result<Iter1, Iter2, Iter3>>::type hpx::ranges::set

Constructs a sorted range beginning at *dest* consisting of all elements present in either of the sorted ranges *[first1, last1)* and *[first2, last2)*, but not in both of them are copied to the range beginning at *dest*. The resulting range is also sorted. This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

If some element is found *m* times in *[first1, last1)* and *n* times in *[first2, last2)*, it will be copied to *dest* exactly $\text{std::abs}(m-n)$ times. If $m > n$, then the last $m-n$ of those elements are copied from *[first1, last1)*, otherwise the last $n-m$ elements are copied from *[first2, last2)*. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

The application of function objects in parallel algorithm invoked with a sequential execution policy object execute in sequential order in the calling thread (*sequenced_policy*) or in a single new thread spawned from the current thread (for *sequenced_task_policy*).

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most $2 \cdot (N1 + N2 - 1)$ comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.

- **Iter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the end source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an sentinel for Iter2.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set_symmetric_difference* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

Returns

The *set_symmetric_difference* algorithm returns a `hpx::future<ranges::set_symmetric_difference_result<Iter1, Iter2, Iter3>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy*

and returns `ranges::set_symmetric_difference_result<Iter1, Iter2, Iter3>` otherwise. The `set_symmetric_difference` algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Iter3, typename Pred =  
hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>  
hpx::parallel::util::detail::algorithm_result<ExPolicy, set_symmetric_difference_result<std::ranges::iterator_t<Rng1>, std::ranges::itera
```

Constructs a sorted range beginning at `dest` consisting of all elements present in either of the sorted ranges `[first1, last1)` and `[first2, last2)`, but not in both of them are copied to the range beginning at `dest`. The resulting range is also sorted. This algorithm expects both input ranges to be sorted with the given binary predicate `f`.

If some element is found m times in `[first1, last1)` and n times in `[first2, last2)`, it will be copied to `dest` exactly $\text{std::abs}(m-n)$ times. If $m > n$, then the last $m-n$ of those elements are copied from `[first1, last1)`, otherwise the last $n-m$ elements are copied from `[first2, last2)`. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

The application of function objects in parallel algorithm invoked with a sequential execution policy object execute in sequential order in the calling thread (*sequenced_policy*) or in a single new thread spawned from the current thread (for *sequenced_task_policy*).

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most $2 \cdot (N1 + N2 - 1)$ comparisons, where $N1$ is the length of the first sequence and $N2$ is the length of the second sequence.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set_symmetric_difference* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

Returns

The *set_symmetric_difference* algorithm returns a `hpx::future<ranges::set_symmetric_difference_result<Iter1, Iter2, Iter3>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `ranges::set_symmetric_difference_result<Iter1, Iter2, Iter3>` otherwise. where *Iter1* is `range_iterator_t<Rng1>` and *Iter2* is `range_iterator_t<Rng2>` The *set_symmetric_difference* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename Iter3, typename Pred
= hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
```

```
set_symmetric_difference_result<Iter1, Iter2, Iter3> hpx::ranges::set_symmetric_difference(Iter1 first1,
                                                                                          Sent1 last1,
                                                                                          Iter2 first2,
                                                                                          Sent2 last2,
                                                                                          Iter3 dest,
                                                                                          Pred &&op =
                                                                                          Pred(), Proj1
                                                                                          &&proj1 =
                                                                                          Proj1(), Proj2
                                                                                          &&proj2 =
                                                                                          Proj2())
```

Constructs a sorted range beginning at *dest* consisting of all elements present in either of the sorted ranges [*first1*, *last1*) and [*first2*, *last2*), but not in both of them are copied to the range beginning at *dest*. The resulting range is also sorted. This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

If some element is found *m* times in [*first1*, *last1*) and *n* times in [*first2*, *last2*), it will be copied to *dest* exactly `std::abs(m-n)` times. If *m*>*n*, then the last *m-n* of those elements are copied from [*first1*,*last1*), otherwise the last *n-m* elements are copied from [*first2*,*last2*). The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

Note: Complexity: At most $2*(N1 + N2 - 1)$ comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

Template Parameters

- **Iter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the end source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an sentinel for Iter2.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set_symmetric_difference* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.

- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *Iter* can be dereferenced and then implicitly converted to *Type1*.

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

Returns

The *set_symmetric_difference* algorithm returns *ranges::set_symmetric_difference_result<Iter1, Iter2, Iter3>*. The *set_symmetric_difference* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng1, typename Rng2, typename Iter3, typename Pred = hpx::parallel::detail::less, typename  
Proj1 = hpx::identity, typename Proj2 = hpx::identity>
```

```
set_symmetric_difference_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng2>, Iter3> hpx::ranges::set_symmetri
```

Constructs a sorted range beginning at *dest* consisting of all elements present in either of the sorted ranges [*first1*, *last1*) and [*first2*, *last2*), but not in both of them are copied to the range beginning at *dest*. The resulting range is also sorted. This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

If some element is found m times in $[first1, last1)$ and n times in $[first2, last2)$, it will be copied to *dest* exactly $\text{std::abs}(m-n)$ times. If $m > n$, then the last $m-n$ of those elements are copied from $[first1, last1)$, otherwise the last $n-m$ elements are copied from $[first2, last2)$. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

Note: Complexity: At most $2 \cdot (N1 + N2 - 1)$ comparisons, where $N1$ is the length of the first sequence and $N2$ is the length of the second sequence.

Template Parameters

- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set_symmetric_difference* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

Parameters

- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

Returns

The *set_symmetric_difference* algorithm returns `ranges::set_symmetric_difference_result<Iter1, Iter2, Iter3>`, where *Iter1* is `range_iterator_t<Rng1>` and *Iter2* is `range_iterator_t<Rng2>`. The *set_symmetric_difference* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

hpx::ranges::mismatch

Defined in header `hpx/algorithm.hpp`⁶⁹⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename ExPolicy, typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename
Pred = equal_to, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, mismatch_result<Iter1, Iter2>>::type hpx::ranges::mismatch(ExPolicy
&&pol-
icy,
Iter1
first1,
Sent1
last1,
Iter2
first2,
Sent2
last2,
Pred
&&op
=
Pred(),
Proj1
&&proj1
=
Proj1(),
Proj2
&&proj2
=
Proj2())
```

Returns true if the range [first1, last1) is mismatch to the range [first2, last2), and false otherwise.

The comparison operations in the parallel *mismatch* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *mismatch* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most $\min(\text{last1} - \text{first1}, \text{last2} - \text{first2})$ applications of the predicate f . If *FwdIter1* and *FwdIter2* meet the requirements of *RandomAccessIterator* and $(\text{last1} - \text{first1}) \neq (\text{last2} - \text{first2})$ then no applications of the predicate f are made.

Note: The two ranges are considered mismatch if, for every iterator i in the range $[\text{first1}, \text{last1})$, $*i$ mismatches $*(\text{first2} + (i - \text{first1}))$. This overload of *mismatch* uses operator== to determine if two elements are mismatch.

Template Parameters

⁶⁹⁹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the source iterators used for the end of the first range (deduced).
- **Iter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the source iterators used for the end of the second range (deduced).
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *mismatch* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function applied to the first range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second range. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as mismatch. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual predicate *is* invoked.

Returns

The *mismatch* algorithm returns a `hpx::future<bool>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *mismatch* algorithm returns true if the elements in the two ranges are mismatch, otherwise it returns false. If the length of the range [first1, last1) does not mismatch the length of the range [first2, last2), it returns false.

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Pred = equal_to, typename Proj1 =
hpx::identity, typename Proj2 = hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, mismatch_result<typename hpx::traits::range_traits<Rng1>::iterator_type, typen
```

Returns `std::pair` with iterators to the first two non-equivalent elements.

The comparison operations in the parallel *mismatch* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *mismatch* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most *last1 - first1* applications of the predicate *f*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng1** – The type of the first source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Rng2** – The type of the second source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *mismatch* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function applied to the first range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second range. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as mismatch. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual predicate *is* invoked.

Returns

The *mismatch* algorithm returns a `hpx::future<std::pair<FwdIter1, FwdIter2>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `std::pair<FwdIter1, FwdIter2>` otherwise. The *mismatch* algorithm returns the first mismatching pair of elements from two ranges: one defined by `[first1, last1)` and another defined by `[first2, last2)`.

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename Pred = equal_to,
typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
mismatch_result<Iter1, Iter2> hpx::ranges::mismatch(Iter1 first1, Sent1 last1, Iter2 first2, Sent2 last2, Pred
&&op = Pred(), Proj1 &&proj1 = Proj1(), Proj2
&&proj2 = Proj2())
```

Returns true if the range `[first1, last1)` is mismatch to the range `[first2, last2)`, and false otherwise.

Note: Complexity: At most $\min(\text{last1} - \text{first1}, \text{last2} - \text{first2})$ applications of the predicate *f*. If *FwdIter1* and *FwdIter2* meet the requirements of *RandomAccessIterator* and $(\text{last1} - \text{first1}) \neq (\text{last2} - \text{first2})$ then no applications of the predicate *f* are made.

Note: The two ranges are considered mismatch if, for every iterator *i* in the range `[first1, last1)`, **i* mismatches $*(\text{first2} + (i - \text{first1}))$. This overload of *mismatch* uses `operator==` to determine if two elements are mismatch.

Template Parameters

- **Iter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the source iterators used for the end of the first range (deduced).
- **Iter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the source iterators used for the end of the second range (deduced).
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *mismatch* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

- **Proj1** – The type of an optional projection function applied to the first range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second range. This defaults to `hpx::identity`

Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as mismatch. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual predicate *is* invoked.

Returns

The *mismatch* algorithm returns *bool*. The *mismatch* algorithm returns true if the elements in the two ranges are mismatch, otherwise it returns false. If the length of the range [first1, last1) does not mismatch the length of the range [first2, last2), it returns false.

```
template<typename Rng1, typename Rng2, typename Pred = equal_to, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
```

```
mismatch_result<typename hpx::traits::range_traits<Rng1>::iterator_type, typename hpx::traits::range_traits<Rng2>::iterator_type> hp
```

Returns `std::pair` with iterators to the first two non-equivalent elements.

Note: Complexity: At most *last1* - *first1* applications of the predicate *f*.

Template Parameters

- **Rng1** – The type of the first source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Rng2** – The type of the second source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *mismatch* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function applied to the first range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second range. This defaults to `hpx::identity`

Parameters

- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as mismatch. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual predicate *is* invoked.

Returns

The *mismatch* algorithm returns `std::pair<FwdIter1, FwdIter2>`. The *mismatch* algorithm returns the first mismatching pair of elements from two ranges: one defined by [*first1*, *last1*) and another defined by [*first2*, *last2*).

hpx::ranges::includes

Defined in header `hpx/algorithm.hpp`⁷⁰⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename ExPolicy, typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename
Pred = hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
```

⁷⁰⁰ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fdbc0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

```

hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx::ranges::includes(ExPolicy &&policy, Iter1
                                                                    first1, Sent1 last1, Iter2
                                                                    first2, Sent2 last2, Pred
                                                                    &&op = Pred(), Proj1
                                                                    &&proj1 = Proj1(),
                                                                    Proj2 &&proj2 =
                                                                    Proj2())

```

Returns true if every element from the sorted range [first2, last2) is found within the sorted range [first1, last1). Also returns true if [first2, last2) is empty. The version expects both ranges to be sorted with the user supplied binary predicate *f*.

The comparison operations in the parallel *includes* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *includes* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: At most $2 \cdot (N1 + N2 - 1)$ comparisons, where $N1 = \text{std::distance}(\text{first1}, \text{last1})$ and $N2 = \text{std::distance}(\text{first2}, \text{last2})$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the end source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an sentinel for Iter2.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *includes* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.

- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as includes. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

Returns

The *includes* algorithm returns a `hpx::future<bool>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *includes* algorithm returns true every element from the sorted range [first2, last2) is found within the sorted range [first1, last1). Also returns true if [first2, last2) is empty.

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename Pred =  
hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>  
bool hpx::ranges::includes(Iter1 first1, Sent1 last1, Iter2 first2, Sent2 last2, Pred &&op = Pred(), Proj1  
                           &&proj1 = Proj1(), Proj2 &&proj2 = Proj2())
```

Returns true if every element from the sorted range [first2, last2) is found within the sorted range [first1, last1). Also returns true if [first2, last2) is empty. The version expects both ranges to be sorted with the user supplied binary predicate *f*.

Note: At most $2*(N1+N2-1)$ comparisons, where $N1 = \text{std::distance}(\text{first1}, \text{last1})$ and $N2 = \text{std::distance}(\text{first2}, \text{last2})$.

Template Parameters

- **Iter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the end source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an sentinel for Iter2.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *includes* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`

- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as includes. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

Returns

The *includes* algorithm returns true every element from the sorted range [first2, last2) is found within the sorted range [first1, last1). Also returns true if [first2, last2) is empty.

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Pred = hpx::parallel::detail::less,
typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx::ranges::includes(ExPolicy &&policy,
                                                                                      Rng1 &&rng1, Rng2
                                                                                      &&rng2, Pred &&op =
                                                                                      Pred(), Proj1 &&proj1 =
                                                                                      Proj1(), Proj2 &&proj2
                                                                                      = Proj2())
```

Returns true if every element from the sorted range [first2, last2) is found within the sorted range [first1, last1). Also returns true if [first2, last2) is empty. The version expects both ranges to be sorted with the user supplied binary predicate *f*.

The comparison operations in the parallel *includes* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *includes* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads,

and indeterminately sequenced within each thread.

Note: At most $2 \cdot (N1 + N2 - 1)$ comparisons, where $N1 = \text{std::distance}(\text{first1}, \text{last1})$ and $N2 = \text{std::distance}(\text{first2}, \text{last2})$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *includes* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as includes. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

Returns

The *includes* algorithm returns a `hpx::future<bool>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *includes* algorithm returns true every element from the sorted range `[first2, last2)` is found within the sorted range `[first1, last1)`. Also returns true if `[first2, last2)` is empty.

```
template<typename Rng1, typename Rng2, typename Pred = hpx::parallel::detail::less, typename Proj1 =  
hpx::identity, typename Proj2 = hpx::identity>
```

```
bool hpx::ranges::includes(Rng1 &&rng1, Rng2 &&rng2, Pred &&op = Pred(), Proj1 &&proj1 = Proj1(),
                          Proj2 &&proj2 = Proj2())
```

Returns true if every element from the sorted range [first2, last2) is found within the sorted range [first1, last1). Also returns true if [first2, last2) is empty. The version expects both ranges to be sorted with the user supplied binary predicate *f*.

Note: At most $2 \cdot (N1 + N2 - 1)$ comparisons, where $N1 = \text{std::distance}(\text{first1}, \text{last1})$ and $N2 = \text{std::distance}(\text{first2}, \text{last2})$.

Template Parameters

- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *includes* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

Parameters

- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as includes. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

Returns

The *includes* algorithm returns true every element from the sorted range [first2, last2) is found within the sorted range [first1, last1). Also returns true if [first2, last2) is empty.

hpx::ranges::adjacent_find

Defined in header [hpx/algorithm.hpp](#)⁷⁰¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename FwdIter, typename Sent, typename Proj = hpx::identity, typename Pred = detail::equal_to>
FwdIter hpx::ranges::adjacent_find(FwdIter first, Sent last, Pred &&pred = Pred(), Proj &&proj = Proj())
```

Searches the range [first, last) for two consecutive identical elements.

Note: Complexity: Exactly the smaller of (result - first) + 1 and (last - first) - 1 application of the predicate where *result* is the value returned

Template Parameters

- **FwdIter** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*
- **Pred** – The type of an optional function/function object to use.

Parameters

- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **pred** – The binary predicate which returns *true* if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *adjacent_find* algorithm returns an iterator to the first of the identical elements. If no such elements are found, *last* is returned.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename Proj = hpx::identity, typename Pred
= detail::equal_to>
```

⁷⁰¹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbdc0c8044e77/libs/full/include/include/hpx/algorithm.hpp>

```
parallel::util::detail::algorithm_result<ExPolicy, FwdIter>::type hpx::ranges::adjacent_find(ExPolicy
                                                                    &&policy,
                                                                    FwdIter first,
                                                                    Sent last, Pred
                                                                    &&pred =
                                                                    Pred(), Proj
                                                                    &&proj =
                                                                    Proj())
```

Searches the range [first, last) for two consecutive identical elements. This version uses the given binary predicate *pred*

The comparison operations in the parallel *adjacent_find* invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *adjacent_find* invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

This overload of *adjacent_find* is available if the user decides to provide their algorithm their own binary predicate *pred*.

Note: Complexity: Exactly the smaller of (result - first) + 1 and (last - first) - 1 application of the predicate where *result* is the value returned

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *adjacent_find* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **pred** – The binary predicate which returns *true* if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *adjacent_find* algorithm returns a `hpx::future<InIter>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *InIter* otherwise. The *adjacent_find* algorithm returns an iterator to the first of the identical elements. If no such elements are found, *last* is returned.

```
template<typename Rng, typename Proj = hpx::identity, typename Pred = detail::equal_to>
hpx::traits::range_traits<Rng>::iterator_type hpx::ranges::adjacent_find(Rng &&rng, Pred &&pred =
                                                                    Pred(), Proj &&proj = Proj())
```

Searches the range *rng* for two consecutive identical elements.

Note: Complexity: Exactly the smaller of $(\text{result} - \text{std::begin}(\text{rng})) + 1$ and $(\text{std::begin}(\text{rng}) - \text{std::end}(\text{rng})) - 1$ applications of the predicate where *result* is the value returned

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*
- **Pred** – The type of an optional function/function object to use.

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – The binary predicate which returns *true* if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *adjacent_find* algorithm returns an iterator to the first of the identical elements. If no such elements are found, *last* is returned.

```
template<typename ExPolicy, typename Rng, typename Proj = hpx::identity, typename Pred = detail::equal_to>
```


parallel::util::detail::algorithm_result<ExPolicy, typename hpx::traits::range_traits<Rng>::iterator_type>::type hpx::ranges::adjac

Searches the range *rng* for two consecutive identical elements.

The comparison operations in the parallel *adjacent_find* invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *adjacent_find* invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

This overload of *adjacent_find* is available if the user decides to provide their algorithm their own binary predicate *pred*.

Note: Complexity: Exactly the smaller of (result - std::begin(rng)) + 1 and (std::begin(rng) - std::end(rng)) - 1 applications of the predicate where *result* is the value returned

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *adjacent_find* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – The binary predicate which returns *true* if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *adjacent_find* algorithm returns a `hpx::future<InIter>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *InIter* otherwise. The *adjacent_find* algorithm returns an iterator to the first of the identical elements. If no such elements are found, *last* is returned.

hpx::ranges::transform_inclusive_scan

Defined in header `hpx/algorithm.hpp`⁷⁰².

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename InIter, typename Sent, typename OutIter, typename BinOp, typename UnOp>
transform_inclusive_scan_result<InIter, OutIter> hpx::ranges::transform_inclusive_scan(InIter first, Sent
                                                    last, OutIter
                                                    dest, BinOp
                                                    &&binary_op,
                                                    UnOp
                                                    &&unary_op)
```

Assigns through each iterator *i* in `[result, result + (last - first))` the value of GENERALIZED_NONCOMMUTATIVE_SUM(`op, conv(*first), ..., conv(*(first + (i - result)))`).

The reduce operations in the parallel *transform_inclusive_scan* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Neither *conv* nor *op* shall invalidate iterators or sub-ranges, or modify elements in the ranges `[first,last)` or `[result,result + (last - first))`.

The behavior of *transform_inclusive_scan* may be non-deterministic for a non-associative predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicates *op* and *conv*.

Note: GENERALIZED_NONCOMMUTATIVE_SUM(`op, a1, ..., aN`) is defined as:

- `a1` when *N* is 1
 - `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN)` where $1 < K+1 = M \leq N$.
-

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.

⁷⁰² <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter`.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **BinOp** – The type of the binary function object used for the reduction operation.
- **UnOp** – The type of the unary function object used for the conversion operation.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **binary_op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **unary_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

Returns

The *transform_inclusive_scan* algorithm returns *transform_inclusive_scan_result<InIter, OutIter>*. The *transform_inclusive_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Sent, typename FwdIter2, typename BinOp,
typename UnOp>
```

parallel::util::detail::algorithm_result<ExPolicy, transform_inclusive_result<FwdIter1, FwdIter2>>::type *hpx::ranges::transform*

Assigns through each iterator *i* in $[\text{result}, \text{result} + (\text{last} - \text{first}))$ the value of `GENERALIZED_NONCOMMUTATIVE_SUM(op, conv(*first), ..., conv(*(first + (i - result))))`.

The reduce operations in the parallel *transform_inclusive_scan* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform_inclusive_scan* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Neither *conv* nor *op* shall invalidate iterators or sub-ranges, or modify elements in the ranges $[\text{first}, \text{last})$ or $[\text{result}, \text{result} + (\text{last} - \text{first}))$.

The behavior of *transform_inclusive_scan* may be non-deterministic for a non-associative predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicates *op* and *conv*.

Note: `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- *a1* when *N* is 1
 - `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

- **BinOp** – The type of the binary function object used for the reduction operation.
- **UnOp** – The type of the unary function object used for the conversion operation.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **binary_op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **unary_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

Returns

The *transform_inclusive_scan* algorithm returns a `hpx::future<transform_inclusive_result<FwdIter1, FwdIter2>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *transform_inclusive_result<FwdIter1, FwdIter2>* otherwise. The *transform_inclusive_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng, typename O, typename BinOp, typename UnOp>
transform_inclusive_scan_result<std::ranges::iterator_t<Rng>, O> hpx::ranges::transform_inclusive_scan(Rng
                                                                                                     &&rng,
                                                                                                     O
                                                                                                     dest,
                                                                                                     BinOp
                                                                                                     &&bi-
                                                                                                     nary_op,
                                                                                                     UnOp
                                                                                                     &&unary_op)
```

Assigns through each iterator *i* in [result, result + (last - first)) the value of GENERALIZED_NONCOMMUTATIVE_SUM(op, conv(*first), ..., conv(*(first + (i - result)))).

The reduce operations in the parallel *transform_inclusive_scan* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Neither *conv* nor *op* shall invalidate iterators or sub-ranges, or modify elements in the ranges [first,last) or [result,result + (last - first)).

The behavior of *transform_inclusive_scan* may be non-deterministic for a non-associative predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicates *op* and *conv*.

Note: `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- `a1` when `N` is 1
 - `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where $1 < K+1 = M \leq N$.
-

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **O** – The type of the iterator representing the destination range (deduced).
- **BinOp** – The type of the binary function object used for the reduction operation.
- **UnOp** – The type of the unary function object used for the conversion operation.

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **binary_op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **unary_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

Returns

The *transform_inclusive_scan* algorithm returns a *transform_inclusive_scan_result*<std::ranges::iterator_t<Rng>, O>. The *transform_inclusive_scan* algorithm returns an input iterator to one past the end of the range and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng, typename O, typename BinOp, typename UnOp>
parallel::util::detail::algorithm_result<ExPolicy, transform_inclusive_scan_result<std::ranges::iterator_t<Rng>, O>>::type hpx::rang
```

Assigns through each iterator *i* in [result, result + (last - first)) the value of GENERALIZED_NONCOMMUTATIVE_SUM(op, conv(*first), ..., conv(*(first + (i - result)))).

The reduce operations in the parallel *transform_inclusive_scan* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform_inclusive_scan* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Neither *conv* nor *op* shall invalidate iterators or sub-ranges, or modify elements in the ranges [first,last) or [result,result + (last - first)).

The behavior of *transform_inclusive_scan* may be non-deterministic for a non-associative predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicates *op* and *conv*.

Note: GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN) is defined as:

- a1 when N is 1
 - op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN) where $1 < K+1 = M \leq N$).
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.

- **O** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator. This iterator type must meet the requirements of an forward iterator.
- **BinOp** – The type of the binary function object used for the reduction operation.
- **UnOp** – The type of the unary function object used for the conversion operation.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **binary_op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **unary_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

Returns

The *transform_inclusive_scan* algorithm returns a `hpx::future<transform_inclusive_scan_result<std::ranges::iterator_t<Rng>, O>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *transform_inclusive_scan_result<std::ranges::iterator_t<Rng>, O>* otherwise. The *transform_inclusive_scan* algorithm returns an input iterator to one past the end of the range and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename InIter, typename Sent, typename OutIter, typename BinOp, typename UnOp, typename T =
typename std::iterator_traits<InIter>::value_type>
transform_inclusive_scan_result<InIter, OutIter> hpx::ranges::transform_inclusive_scan(InIter first, Sent
                                                                    last, OutIter
                                                                    dest, BinOp
                                                                    &&binary_op,
                                                                    UnOp
                                                                    &&unary_op, T
                                                                    init)
```

Assigns through each iterator *i* in [result, result + (last - first)) the value of `GENERALIZED_NONCOMMUTATIVE_SUM(binary_op, init, conv(*first), ..., conv(*(first + (i - result))))`.

The reduce operations in the parallel *transform_inclusive_scan* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Neither *conv* nor *op* shall invalidate iterators or sub-ranges, or modify elements in the ranges $[first, last)$ or $[result, result + (last - first))$.

The behavior of *transform_inclusive_scan* may be non-deterministic for a non-associative predicate.

Note: Complexity: $O(last - first)$ applications of the predicates *op* and *conv*.

Note: `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- `a1` when `N` is 1
 - `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where $1 < K+1 = M \leq N$.
-

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter`.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **BinOp** – The type of the binary function object used for the reduction operation.
- **UnOp** – The type of the unary function object used for the conversion operation.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **binary_op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **unary_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by $[first, last)$. This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

- **init** – The initial value for the generalized sum.

Returns

The *transform_inclusive_scan* algorithm returns *transform_inclusive_scan_result<InIter, OutIter>*. The *transform_inclusive_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Sent, typename FwdIter2, typename BinOp,
typename UnOp, typename T = typename std::iterator_traits<FwdIter1>::value_type>
parallel::util::detail::algorithm_result<ExPolicy, transform_inclusive_result<FwdIter1, FwdIter2>>::type hpx::ranges::transform
```

Assigns through each iterator *i* in $[result, result + (last - first))$ the value of `GENERALIZED_NONCOMMUTATIVE_SUM(binary_op, init, conv(*first), ..., conv(*(first + (i - result))))`.

The reduce operations in the parallel *transform_inclusive_scan* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform_inclusive_scan* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Neither *conv* nor *op* shall invalidate iterators or sub-ranges, or modify elements in the ranges $[first, last)$ or $[result, result + (last - first))$.

The behavior of *transform_inclusive_scan* may be non-deterministic for a non-associative predicate.

Note: Complexity: $O(last - first)$ applications of the predicates *op* and *conv*.

Note: `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- a_1 when N is 1
- $op(\text{GENERALIZED_NONCOMMUTATIVE_SUM}(op, a_1, \dots, a_K), \text{GENERALIZED_NONCOMMUTATIVE_SUM}(op, a_M, \dots, a_N))$ where $1 < K+1 = M \leq N$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for **FwdIter**.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **BinOp** – The type of the binary function object used for the reduction operation.
- **UnOp** – The type of the unary function object used for the conversion operation.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **binary_op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **unary_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

- **init** – The initial value for the generalized sum.

Returns

The *transform_inclusive_scan* algorithm returns a `hpx::future<transform_inclusive_result<FwdIter1, FwdIter2>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *transform_inclusive_result<FwdIter1, FwdIter2>* otherwise. The *transform_inclusive_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng, typename O, typename BinOp, typename UnOp, typename T = typename
std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>
transform_inclusive_scan_result<std::ranges::iterator_t<Rng>, O> hpx::ranges::transform_inclusive_scan(Rng
                                                                &&rng,
                                                                O
                                                                dest,
                                                                BinOp
                                                                &&binary_op,
                                                                UnOp
                                                                &&unary_op,
                                                                T
                                                                init)
```

Assigns through each iterator *i* in `[result, result + (last - first))` the value of `GENERALIZED_NONCOMMUTATIVE_SUM(binary_op, init, conv(*first), ..., conv(*(first + (i - result))))`.

The reduce operations in the parallel *transform_inclusive_scan* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Neither *conv* nor *op* shall invalidate iterators or sub-ranges, or modify elements in the ranges `[first,last)` or `[result,result + (last - first))`.

The behavior of *transform_inclusive_scan* may be non-deterministic for a non-associative predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicates *op* and *conv*.

Note: `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- *a1* when *N* is 1
 - `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where $1 < K+1 = M \leq N$.
-

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **O** – The type of the iterator representing the destination range (deduced).
- **BinOp** – The type of the binary function object used for the reduction operation.
- **UnOp** – The type of the unary function object used for the conversion operation.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **binary_op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **unary_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

- **init** – The initial value for the generalized sum.

Returns

The *transform_inclusive_scan* algorithm returns a *transform_inclusive_scan_result*<`std::ranges::iterator_t<Rng>`, *O*>. The *transform_inclusive_scan* algorithm returns an input iterator to one past the end of the range and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng, typename O, typename BinOp, typename UnOp, typename T =
typename std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>
parallel::util::detail::algorithm_result<ExPolicy, transform_inclusive_scan_result<std::ranges::iterator_t<Rng>, O>>::type hpx::rang
```

Assigns through each iterator *i* in [result, result + (last - first)) the value of `GENERALIZED_NONCOMMUTATIVE_SUM(binary_op, init, conv(*first), ..., conv(*(first + (i - result))))`.

The reduce operations in the parallel *transform_inclusive_scan* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform_inclusive_scan* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Neither *conv* nor *op* shall invalidate iterators or sub-ranges, or modify elements in the ranges [first,last) or [result,result + (last - first)).

The behavior of *transform_inclusive_scan* may be non-deterministic for a non-associative predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicates *op* and *conv*.

Note: `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- `a1` when `N` is 1
 - `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **O** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator. This iterator type must meet the requirements of an forward iterator.
- **BinOp** – The type of the binary function object used for the reduction operation.
- **UnOp** – The type of the unary function object used for the conversion operation.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **binary_op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **unary_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

- **init** – The initial value for the generalized sum.

Returns

The *transform_inclusive_scan* algorithm returns a `hpx::future<transform_inclusive_scan_result<std::ranges::iterator_t<Rng>, O>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *transform_inclusive_scan_result<std::ranges::iterator_t<Rng>, O>* otherwise. The *transform_inclusive_scan* algorithm returns an input iterator to one past the end of the range and an output iterator to the element in the destination range, one past the last element copied.

hpx::ranges::exclusive_scan

Defined in header `hpx/algorithm.hpp`⁷⁰³.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename InIter, typename Sent, typename OutIter, typename T = typename
std::iterator_traits<InIter>::value_type, typename Op = std::plus<T>>
exclusive_scan_result<InIter, OutIter> hpx::ranges::exclusive_scan(InIter first, Sent last, OutIter dest, T init,
                                                                    Op &&op = Op())
```

Assigns through each iterator *i* in `[result, result + (last - first))` the value of `GENERALIZED_NONCOMMUTATIVE_SUM(binary_op, init, *first, ..., *(first + (i - result) - 1))`.

The difference between *exclusive_scan* and *inclusive_scan* is that *inclusive_scan* includes the *i*th input element in the *i*th sum. If *op* is not mathematically associative, the behavior of *inclusive_scan* may be non-deterministic.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate *op*.

Note: `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- *a1* when *N* is 1
 - `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where $1 < K+1 = M \leq N$.
-

Template Parameters

- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter1*.

⁷⁰³ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **Op** – The type of the binary function object used for the reduction operation.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

Returns

The *exclusive_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Sent, typename FwdIter2, typename T = typename  
std::iterator_traits<FwdIter1>::value_type, typename Op = std::plus<T>>  
parallel::util::detail::algorithm_result<ExPolicy, exclusive_scan_result<FwdIter1, FwdIter2>>::type hpx::ranges::exclusive_sca
```

Assigns through each iterator *i* in `[result, result + (last - first))` the value of `GENERALIZED_NONCOMMUTATIVE_SUM(binary_op, init, *first, ..., *(first + (i - result) - 1))`.

The reduce operations in the parallel *exclusive_scan* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *exclusive_scan* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *exclusive_scan* and *inclusive_scan* is that *inclusive_scan* includes the *i*th input element in the *i*th sum. If *op* is not mathematically associative, the behavior of *inclusive_scan* may be non-deterministic.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate *op*.

Note: GENERALIZED_NONCOMMUTATIVE_SUM(*op*, *a1*, ..., *aN*) is defined as:

- *a1* when *N* is 1
 - *op*(GENERALIZED_NONCOMMUTATIVE_SUM(*op*, *a1*, ..., *aK*), GENERALIZED_NONCOMMUTATIVE_SUM(*op*, *aM*, ..., *aN*)) where $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter1*.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **Op** – The type of the binary function object used for the reduction operation.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

Returns

The *exclusive_scan* algorithm returns a `hpx::future<util::in_out_result<FwdIter1, FwdIter2>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *util::in_out_result<FwdIter1, FwdIter2>* otherwise. The *exclusive_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng, typename O, typename T = typename
std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type, typename Op = std::plus<T>>
exclusive_scan_result<std::ranges::iterator_t<Rng>, O> hpx::ranges::exclusive_scan(Rng &&rng, O dest, T
init, Op &&op = Op())
```

Assigns through each iterator *i* in `[result, result + (last - first))` the value of `GENERALIZED_NONCOMMUTATIVE_SUM(+, init, *first, ..., *(first + (i - result) - 1))`

The difference between *exclusive_scan* and *inclusive_scan* is that *inclusive_scan* includes the *i*th input element in the *i*th sum.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate `std::plus<T>`.

Note: `GENERALIZED_NONCOMMUTATIVE_SUM(+, a1, ..., aN)` is defined as:

- `a1` when `N` is 1
 - `GENERALIZED_NONCOMMUTATIVE_SUM(+, a1, ..., aK)`
 - `GENERALIZED_NONCOMMUTATIVE_SUM(+, aM, ..., aN)` where $1 < K+1 = M \leq N$.
-

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **O** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **Op** – The type of the binary function object used for the reduction operation.

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

Returns

The *exclusive_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng, typename O, typename T = typename
std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type, typename Op = std::plus<T>>
parallel::util::detail::algorithm_result<ExPolicy, exclusive_scan_result<std::ranges::iterator_t<Rng>, O>> hpx::ranges::exclusive
```

Assigns through each iterator *i* in [result, result + (last - first)) the value of GENERALIZED_NONCOMMUTATIVE_SUM(+, init, *first, ..., *(first + (i - result) - 1))

The reduce operations in the parallel *exclusive_scan* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *exclusive_scan* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *exclusive_scan* and *inclusive_scan* is that *inclusive_scan* includes the *i*th input element in the *i*th sum.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate *std::plus<T>*.

Note: GENERALIZED_NONCOMMUTATIVE_SUM(+, a1, ..., aN) is defined as:

- a1 when N is 1
 - GENERALIZED_NONCOMMUTATIVE_SUM(+, a1, ..., aK)
 - GENERALIZED_NONCOMMUTATIVE_SUM(+, aM, ..., aN) where $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.

- **O** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **Op** – The type of the binary function object used for the reduction operation.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

Returns

The *exclusive_scan* algorithm returns a `hpx::future<util::in_out_result<std::ranges::iterator_t<Rng>, O>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *util::in_out_result<std::ranges::iterator_t<Rng>, O>* otherwise. The *exclusive_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

hpx::ranges::inclusive_scan

Defined in header `hpx/algorithm.hpp`⁷⁰⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename InIter, typename Sent, typename OutIter, typename Op = std::plus<typename  
std::iterator_traits<InIter>::value_type>>
```

```
inclusive_scan_result<InIter, OutIter> hpx::ranges::inclusive_scan(InIter first, Sent last, OutIter dest, Op  
                                                                    &&op)
```

Assigns through each iterator *i* in `[result, result + (last - first))` the value of `GENERALIZED_NONCOMMUTATIVE_SUM(op, *first, ..., *(first + (i - result)))`.

The reduce operations in the parallel *inclusive_scan* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

The difference between *exclusive_scan* and *inclusive_scan* is that *inclusive_scan* includes the *i*th input element in the *i*th sum.

⁷⁰⁴ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate *op*.

Note: GENERALIZED_NONCOMMUTATIVE_SUM(+, a1, ..., aN) is defined as:

- a1 when N is 1
 - GENERALIZED_NONCOMMUTATIVE_SUM(+, a1, ..., aK)
 - GENERALIZED_NONCOMMUTATIVE_SUM(+, aM, ..., aN) where $1 < K+1 = M \leq N$.
-

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Op** – The type of the binary function object used for the reduction operation.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have const&, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

Returns

The *inclusive_scan* algorithm returns `util::in_out_result<InIter, OutIter>`. The *inclusive_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Sent, typename FwdIter2, typename Op =
std::plus<typename std::iterator_traits<FwdIter1>::value_type>>
```

`parallel::util::detail::algorithm_result<ExPolicy, inclusive_scan_result<FwdIter1, FwdIter2>>::type hpx::ranges::inclusive_scan`

Assigns through each iterator i in $[result, result + (last - first))$ the value of `GENERALIZED_NONCOMMUTATIVE_SUM(op, *first, ..., *(first + (i - result)))`.

The reduce operations in the parallel *inclusive_scan* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *inclusive_scan* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *exclusive_scan* and *inclusive_scan* is that *inclusive_scan* includes the i th input element in the i th sum.

Note: Complexity: $O(last - first)$ applications of the predicate *op*.

Note: `GENERALIZED_NONCOMMUTATIVE_SUM(+, a1, ..., aN)` is defined as:

- $a1$ when N is 1
 - `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK)`
 - `GENERALIZED_NONCOMMUTATIVE_SUM(+, aM, ..., aN)` where $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter`.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Op** – The type of the binary function object used for the reduction operation.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

Returns

The *inclusive_scan* algorithm returns a `hpx::future<util::in_out_result<FwdIter1, FwdIter2>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *util::in_out_result<FwdIter1, FwdIter2>* otherwise. The *inclusive_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng, typename O, typename Op = std::plus<typename
hpx::traits::range_traits<Rng>::value_type>>
inclusive_scan_result<std::ranges::iterator_t<Rng>, O> hpx::ranges::inclusive_scan(Rng &&rng, O dest, Op
&&op)
```

Assigns through each iterator *i* in `[result, result + (last - first))` the value of `GENERALIZED_NONCOMMUTATIVE_SUM(op, *first, ..., *(first + (i - result)))`.

The reduce operations in the parallel *inclusive_scan* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

The difference between *exclusive_scan* and *inclusive_scan* is that *inclusive_scan* includes the *i*th input element in the *i*th sum.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate *op*.

Note: `GENERALIZED_NONCOMMUTATIVE_SUM(+, a1, ..., aN)` is defined as:

- `a1` when *N* is 1
 - `GENERALIZED_NONCOMMUTATIVE_SUM(+, a1, ..., aK)`
– `GENERALIZED_NONCOMMUTATIVE_SUM(+, aM, ..., aN)` where $1 < K+1 = M \leq N$.
-

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

- **O** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Op** – The type of the binary function object used for the reduction operation.

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

Returns

The *inclusive_scan* algorithm returns `util::in_out_result<std::ranges::iterator_t<Rng>, O>`. The *inclusive_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng, typename O, typename Op = std::plus<typename  
hpx::traits::range_traits<Rng>::value_type>>  
parallel::util::detail::algorithm_result<ExPolicy, inclusive_scan_result<std::ranges::iterator_t<Rng>, O>> hpx::ranges::inclusive
```

Assigns through each iterator *i* in `[result, result + (last - first))` the value of `GENERALIZED_NONCOMMUTATIVE_SUM(op, *first, ..., *(first + (i - result)))`.

The reduce operations in the parallel *inclusive_scan* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *inclusive_scan* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *exclusive_scan* and *inclusive_scan* is that *inclusive_scan* includes the *i*th input element in the *i*th sum.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate *op*.

Note: `GENERALIZED_NONCOMMUTATIVE_SUM(+, a1, ..., aN)` is defined as:

- a_1 when N is 1
- `GENERALIZED_NONCOMMUTATIVE_SUM(op,  $a_1, \dots, a_K$ )`
 - `GENERALIZED_NONCOMMUTATIVE_SUM(+,  $a_M, \dots, a_N$ )` where $1 < K+1 = M \leq N$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **O** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Op** – The type of the binary function object used for the reduction operation.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

Returns

The *inclusive_scan* algorithm returns a `hpx::future<util::in_out_result<std::ranges::iterator_t<Rng>, O>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `util::in_out_result<std::ranges::iterator_t<Rng>, O>` otherwise. The *inclusive_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied

```
template<typename InIter, typename Sent, typename OutIter, typename Op, typename T = typename
std::iterator_traits<InIter>::value_type>
inclusive_scan_result<InIter, OutIter> hpx::ranges::inclusive_scan(InIter first, Sent last, OutIter dest, Op
&&op, T init)
```

Assigns through each iterator *i* in $[result, result + (last - first))$ the value of `GENERALIZED_NONCOMMUTATIVE_SUM(op, init, *first, ..., *(first + (i - result)))`.

The reduce operations in the parallel *inclusive_scan* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

The difference between *exclusive_scan* and *inclusive_scan* is that *inclusive_scan* includes the *i*th input element in the *i*th sum. If *op* is not mathematically associative, the behavior of *inclusive_scan* may be non-deterministic.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate *op*.

Note: GENERALIZED_NONCOMMUTATIVE_SUM(*op*, *a*₁, ..., *a*_N) is defined as:

- *a*₁ when *N* is 1
 - *op*(GENERALIZED_NONCOMMUTATIVE_SUM(*op*, *a*₁, ..., *a*_K), GENERALIZED_NONCOMMUTATIVE_SUM(*op*, *a*_M, ..., *a*_N)) where $1 < K+1 = M \leq N$.
-

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Op** – The type of the binary function object used for the reduction operation.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **init** – The initial value for the generalized sum.

Returns

The *inclusive_scan* algorithm returns `util::in_out_result<InIter, OutIter>`. The *inclusive_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Sent, typename FwdIter2, typename Op, typename T = typename std::iterator_traits<FwdIter1>::value_type>
```

`parallel::util::detail::algorithm_result<ExPolicy, inclusive_scan_result<FwdIter1, FwdIter2>>::type hpx::ranges::inclusive_scan`

Assigns through each iterator i in $[result, result + (last - first))$ the value of `GENERALIZED_NONCOMMUTATIVE_SUM(op, init, *first, ..., *(first + (i - result)))`.

The reduce operations in the parallel *inclusive_scan* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *inclusive_scan* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *exclusive_scan* and *inclusive_scan* is that *inclusive_scan* includes the i th input element in the i th sum. If op is not mathematically associative, the behavior of *inclusive_scan* may be non-deterministic.

Note: Complexity: $O(last - first)$ applications of the predicate op .

Note: `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- $a1$ when N is 1
 - $op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, \dots, aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, \dots, aN))$ where $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter`.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Op** – The type of the binary function object used for the reduction operation.

- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **init** – The initial value for the generalized sum.

Returns

The *inclusive_scan* algorithm returns a `hpx::future<util::in_out_result<InIter, OutIter>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *util::in_out_result<InIter, OutIter>* otherwise. The *inclusive_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng, typename O, typename Op, typename T = typename
std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>
inclusive_scan_result<std::ranges::iterator_t<Rng>, O> hpx::ranges::inclusive_scan(Rng &&rng, O dest, Op
&&op, T init)
```

Assigns through each iterator *i* in `[result, result + (last - first))` the value of `GENERALIZED_NONCOMMUTATIVE_SUM(op, init, *first, ..., *(first + (i - result)))`.

The reduce operations in the parallel *inclusive_scan* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

The difference between *exclusive_scan* and *inclusive_scan* is that *inclusive_scan* includes the *i*th input element in the *i*th sum. If *op* is not mathematically associative, the behavior of *inclusive_scan* may be non-deterministic.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate *op*.

Note: `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- `a1` when `N` is 1
 - `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where $1 < K+1 = M \leq N$.
-

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **O** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Op** – The type of the binary function object used for the reduction operation.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have const&, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **init** – The initial value for the generalized sum.

Returns

The *inclusive_scan* algorithm returns *util::in_out_result<std::ranges::iterator_t<Rng>, O>*. The *inclusive_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng, typename O, typename Op, typename T = typename
std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>
parallel::util::detail::algorithm_result<ExPolicy, inclusive_scan_result<std::ranges::iterator_t<Rng>, O>> hpx::ranges::inclusive
```

Assigns through each iterator *i* in $[\text{result}, \text{result} + (\text{last} - \text{first}))$ the value of GENERALIZED_NONCOMMUTATIVE_SUM(*op*, *init*, **first*, ..., **(first + (i - result))*).

The reduce operations in the parallel *inclusive_scan* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *inclusive_scan* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *exclusive_scan* and *inclusive_scan* is that *inclusive_scan* includes the *i*th input element in the *i*th sum. If *op* is not mathematically associative, the behavior of *inclusive_scan* may be non-deterministic.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate *op*.

Note: GENERALIZED_NONCOMMUTATIVE_SUM(*op*, *a1*, ..., *aN*) is defined as:

- *a1* when *N* is 1
 - *op*(GENERALIZED_NONCOMMUTATIVE_SUM(*op*, *a1*, ..., *aK*), GENERALIZED_NONCOMMUTATIVE_SUM(*op*, *aM*, ..., *aN*)) where $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **O** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Op** – The type of the binary function object used for the reduction operation.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **init** – The initial value for the generalized sum.

Returns

The *inclusive_scan* algorithm returns a `hpx::future<util::in_out_result<std::ranges::iterator_t<Rng>, O>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `util::in_out_result<std::ranges::iterator_t<Rng>, O>` otherwise. The *inclusive_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied

hpx::ranges::set_intersection

Defined in header `hpx/algorithm.hpp`⁷⁰⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

```

template<typename ExPolicy, typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename
Iter3, typename Pred = hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename Proj2 =
hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, set_intersection_result<Iter1, Iter2, Iter3>>::type hpx::ranges::set_interse

```

Constructs a sorted range beginning at `dest` consisting of all elements present in both sorted ranges `[first1, last1)` and `[first2, last2)`. This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

If some element is found *m* times in `[first1, last1)` and *n* times in `[first2, last2)`, the first `std::min(m, n)` elements will be copied from the first range to the destination range. The order of equivalent elements is preserved. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

The application of function objects in parallel algorithm invoked with a sequential execution policy object execute in sequential order in the calling thread (*sequenced_policy*) or in a single new thread spawned from the current thread (for *sequenced_task_policy*).

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

⁷⁰⁵ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Note: Complexity: At most $2*(N1 + N2 - 1)$ comparisons, where $N1$ is the length of the first sequence and $N2$ is the length of the second sequence.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Iter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the end source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an sentinel for Iter2.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set_intersection* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

Returns

The *set_intersection* algorithm returns a `hpx::future<ranges::set_intersection_result<Iter1, Iter2, Iter3>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `ranges::set_intersection_result<Iter1, Iter2, Iter3>` otherwise. The *set_intersection* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Iter3, typename Pred =
hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, set_intersection_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng2>, std::ranges::iterator_t<Rng3>>>
```

Constructs a sorted range beginning at *dest* consisting of all elements present in both sorted ranges *[first1, last1)* and *[first2, last2)*. This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

If some element is found *m* times in *[first1, last1)* and *n* times in *[first2, last2)*, the first `std::min(m, n)` elements will be copied from the first range to the destination range. The order of equivalent elements is preserved. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

The application of function objects in parallel algorithm invoked with a sequential execution policy object execute in sequential order in the calling thread (*sequenced_policy*) or in a single new thread spawned from the current thread (for *sequenced_task_policy*).

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most $2 \cdot (N1 + N2 - 1)$ comparisons, where $N1$ is the length of the first sequence and $N2$ is the length of the second sequence.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set_intersection* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

Returns

The *set_intersection* algorithm returns a `hpx::future<ranges::set_intersection_result<Iter1, Iter2, Iter3>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `ranges::set_intersection_result<Iter1, Iter2, Iter3>` otherwise. where *Iter1* is

range_iterator_t<Rng1> and Iter2 is range_iterator_t<Rng2> The *set_intersection* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename Iter3, typename Pred
= hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
set_intersection_result<Iter1, Iter2, Iter3> hpx::ranges::set_intersection(Iter1 first1, Sent1 last1, Iter2 first2,
                                                                    Sent2 last2, Iter3 dest, Pred &&op
                                                                    = Pred(), Proj1 &&proj1 =
                                                                    Proj1(), Proj2 &&proj2 = Proj2())
```

Constructs a sorted range beginning at *dest* consisting of all elements present in both sorted ranges [*first1*, *last1*) and [*first2*, *last2*). This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

If some element is found *m* times in [*first1*, *last1*) and *n* times in [*first2*, *last2*), the first `std::min(m, n)` elements will be copied from the first range to the destination range. The order of equivalent elements is preserved. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

Note: Complexity: At most $2 \cdot (N_1 + N_2 - 1)$ comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

Template Parameters

- **Iter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the end source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an sentinel for Iter2.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set_intersection* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.

- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *Iter1* can be dereferenced and then implicitly converted to *Type1*.

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

Returns

The *set_intersection* algorithm returns *ranges::set_intersection_result<Iter1, Iter2, Iter3>*. The *set_intersection* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng1, typename Rng2, typename Iter3, typename Pred = hpx::parallel::detail::less, typename  
Proj1 = hpx::identity, typename Proj2 = hpx::identity>
```

```
set_intersection_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng2>, Iter3> hpx::ranges::set_intersection(Rng1  
&&  
Rng2  
&&  
Iter3  
dest  
Pred  
&&  
=  
Pred  
Proj1  
&&  
=  
Proj2  
Proj1  
&&  
=  
Proj1
```

Constructs a sorted range beginning at *dest* consisting of all elements present in both sorted ranges [*first1*, *last1*) and [*first2*, *last2*). This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

If some element is found *m* times in [*first1*, *last1*) and *n* times in [*first2*, *last2*), the first `std::min(m, n)` elements will be copied from the first range to the destination range. The order of equivalent elements is preserved. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

Note: Complexity: At most $2 \cdot (N1 + N2 - 1)$ comparisons, where $N1$ is the length of the first sequence and $N2$ is the length of the second sequence.

Template Parameters

- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set_intersection* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

Parameters

- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

Returns

The *set_intersection* algorithm returns `ranges::set_intersection_result<Iter1, Iter2, Iter3>`, where *Iter1* is `range_iterator_t<Rng1>` and *Iter2* is `range_iterator_t<Rng2>`. The *set_intersection* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

hpx/parallel/container_algorithms/iota.hpp

Defined in header `hpx/parallel/container_algorithms/iota.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **ranges**

Functions

```
template<typename FwdIter, typename Sent, typename T>  
iota_result<FwdIter, T> iota(FwdIter first, Sent last, T value)
```

Fills the range [first, last) with sequentially increasing values, starting with *value* and repetitively evaluating ++*value*.

The assignments in the *iota* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

Note: Complexity: Exactly *std::distance(first, last)* assignments and increments.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input or output iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **T** – The type of the target value. This type must meet the requirements of weakly incrementable.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – The value to be stored in the first element.

Returns

The *iota* algorithm returns a *hpx::ranges::iota_result* containing the final iterator and the final value.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename T>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, iota_result<FwdIter, T>> iota(ExPolicy  
                                                                                      &&policy,  
                                                                                      FwdIter first,  
                                                                                      Sent last, T  
                                                                                      value)
```

Fills the range [first, last) with sequentially increasing values, starting with *value* and repetitively evaluating ++*value*.

The assignments in the parallel *iota* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *iota* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Exactly *std::distance(first, last)* assignments and increments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for FwdIter.
- **T** – The type of the target value. This type must meet the requirements of weakly incrementable.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – The value to be stored in the first element.

Returns

The *iota* algorithm returns a *hpx::future<iota_result>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *iota_result* otherwise.

```
template<typename Range, typename T>
iota_result<typename hpx::traits::range_traits<Range>::iterator_type, T> iota(Range &&r, T value)
    Fills the range r with sequentially increasing values, starting with value and repetitively evaluating ++value.
```

The assignments in the *iota* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

Note: Complexity: Exactly *std::ranges::distance(r)* assignments and increments.

Template Parameters

- **Range** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input or output iterator
- **T** – The type of the target value. This type must meet the requirements of weakly incrementable.

Parameters

- **r** – Refers to the sequence of elements the algorithm will be applied to.
- **value** – The value to be stored in the first element.

Returns

The *iota* algorithm returns a *hpx::ranges::iota_result* containing the final iterator and the final value.

```
template<typename ExPolicy, typename Range, typename T>
```

hpx::parallel::util::detail::algorithm_result_t<ExPolicy, iota_result<typename hpx::traits::range_traits<Range>::iterator_type

Fills the range *r* with sequentially increasing values, starting with *value* and repetitively evaluating *++value*.

The assignments in the parallel *iota* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *iota* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Exactly *std::ranges::distance(r)* assignments and increments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Range** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a forward iterator.
- **T** – The type of the target value. This type must meet the requirements of weakly incrementable.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **r** – Refers to the sequence of elements the algorithm will be applied to.
- **value** – The value to be stored in the first element.

Returns

The *iota* algorithm returns a *hpx::future<iota_result>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *iota_result* otherwise.

hpx::ranges::shift_left

Defined in header [hpx/algorithm.hpp](#)⁷⁰⁶.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

template<typename **FwdIter**, typename **Sent**, typename **Size**>
FwdIter **hpx::ranges::shift_left**(*FwdIter* first, *Sent* last, *Size* n)

Shifts the elements in the range [first, last) by *n* positions towards the beginning of the range. For every integer *i* in [0, last - first

- *n*), moves the element originally at position first + *n* + *i* to position first + *i*.

The assignment operations in the parallel *shift_left* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

⁷⁰⁶ <http://github.com/TheHPXProject/hpx/blob/d402cead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Note: Complexity: At most (last - first) - n assignments.

Note: The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable*.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **Size** – The type of the argument specifying the number of positions to shift by.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **n** – Refers to the number of positions to shift.

Returns

The *shift_left* algorithm returns *FwdIter*. The *shift_left* algorithm returns an iterator to the end of the resulting range.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename Size>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::ranges::shift_left(ExPolicy
                                                                    &&policy, FwdIter
                                                                    first, Sent last, Size
                                                                    n)
```

Shifts the elements in the range [first, last) by n positions towards the beginning of the range. For every integer i in [0, last - first

- n), moves the element originally at position first + n + i to position first + i.

The assignment operations in the parallel *shift_left* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignment operations in the parallel *shift_left* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most (last - first) - n assignments.

Note: The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for FwdIter.
- **Size** – The type of the argument specifying the number of positions to shift by.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **n** – Refers to the number of positions to shift.

Returns

The *shift_left* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *shift_left* algorithm returns an iterator to the end of the resulting range.

```
template<typename Rng, typename Size>
```

```
std::ranges::iterator_t<Rng> hpx::ranges::shift_left(Rng &&rng, Size n)
```

Shifts the elements in the range [first, last) by n positions towards the beginning of the range. For every integer i in [0, last - first

- n), moves the element originally at position first + n + i to position first + i.

The assignment operations in the parallel *shift_left* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

Note: Complexity: At most (last - first) - n assignments.

Note: The type of dereferenced *std::ranges::iterator_t<Rng>* must meet the requirements of *MoveAssignable*.

Template Parameters

- **Rng** – The type of the range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of positions to shift by.

Parameters

- **rng** – Refers to the range in which the elements will be shifted.
- **n** – Refers to the number of positions to shift.

Returns

The *shift_left* algorithm returns `std::ranges::iterator_t<Rng>`. The *shift_left* algorithm returns an iterator to the end of the resulting range.

```
template<typename ExPolicy, typename Rng, typename Size>
parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng>> hpx::ranges::shift_left(ExPolicy
                                                                    &&pol-
                                                                    icy,
                                                                    Rng
                                                                    &&rng,
                                                                    Size
                                                                    n)
```

Shifts the elements in the range [first, last) by *n* positions towards the beginning of the range. For every integer *i* in [0, last - first

- *n*), moves the element originally at position first + *n* + *i* to position first + *i*.

The assignment operations in the parallel *shift_left* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignment operations in the parallel *shift_left* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most (last - first) - *n* assignments.

Note: The type of dereferenced `std::ranges::iterator_t<Rng>` must meet the requirements of *MoveAssignable*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of positions to shift by.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the range in which the elements will be shifted.
- **n** – Refers to the number of positions to shift.

Returns

The *shift_left* algorithm returns a `hpx::future<std::ranges::iterator_t<Rng>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `std::ranges::iterator_t<Rng>` otherwise. The *shift_left* algorithm returns an iterator to the end of the resulting range.

hpx::ranges::lexicographical_compare

Defined in header [hpx/algorithm.hpp](#)⁷⁰⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename InIter1, typename Sent1, typename InIter2, typename Sent2, typename Proj1 =  
hpx::identity, typename Proj2 = hpx::identity, typename Pred = hpx::parallel::detail::less>  
bool hpx::ranges::lexicographical_compare(InIter1 first1, Sent1 last1, InIter2 first2, Sent2 last2, Pred  
                                         &&pred = Pred(), Proj1 &&proj1 = Proj1(), Proj2 &&proj2 =  
                                         Proj2())
```

Checks if the first range [first1, last1) is lexicographically less than the second range [first2, last2). uses a provided predicate to compare elements.

The comparison operations in the parallel *lexicographical_compare* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: At most $2 * \min(N1, N2)$ applications of the comparison operation, where $N1 = \text{std::distance}(\text{first1}, \text{last})$ and $N2 = \text{std::distance}(\text{first2}, \text{last2})$.

Note: Lexicographical comparison is an operation with the following properties

- Two ranges are compared element by element
 - The first mismatching element defines which range is lexicographically *less* or *greater* than the other
 - If one range is a prefix of another, the shorter range is lexicographically *less* than the other
 - If two ranges have equivalent elements and are of the same length, then the ranges are lexicographically *equal*
 - An empty range is lexicographically *less* than any non-empty range
 - Two empty ranges are lexicographically *equal*
-

Template Parameters

- **InIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent1** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter1.
- **InIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent2** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter2.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *lexicographical_compare* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function for FwdIter1. This defaults to `hpx::identity`

⁷⁰⁷ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **Proj2** – The type of an optional projection function for FwdIter2. This defaults to *hpx::identity*

Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **pred** – Refers to the comparison function that the first and second ranges will be applied to
- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual predicate *is* invoked.

Returns

The *lexicographically_compare* algorithm returns *bool*. The *lexicographically_compare* algorithm returns true if the first range is lexicographically less, otherwise it returns false. range [first2, last2), it returns false.

```
template<typename ExPolicy, typename FwdIter1, typename Sent1, typename FwdIter2, typename Sent2,
typename Proj1 = hpx::identity, typename Proj2 = hpx::identity, typename Pred = hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx::ranges::lexicographical_compare(ExPolicy
&&pol-
icy,
FwdIter1
first1,
Sent1
last1,
FwdIter2
first2,
Sent2
last2,
Pred
&&pred
=
Pred(),
Proj1
&&proj1
=
Proj1(),
Proj2
&&proj2
=
Proj2())
```

Checks if the first range [first1, last1) is lexicographically less than the second range [first2, last2). uses a provided predicate to compare elements.

The comparison operations in the parallel *lexicographical_compare* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *lexicographical_compare* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most $2 * \min(N1, N2)$ applications of the comparison operation, where $N1 = \text{std::distance}(\text{first1}, \text{last})$ and $N2 = \text{std::distance}(\text{first2}, \text{last2})$.

Note: Lexicographical comparison is an operation with the following properties

- Two ranges are compared element by element
 - The first mismatching element defines which range is lexicographically *less* or *greater* than the other
 - If one range is a prefix of another, the shorter range is lexicographically *less* than the other
 - If two ranges have equivalent elements and are of the same length, then the ranges are lexicographically *equal*
 - An empty range is lexicographically *less* than any non-empty range
 - Two empty ranges are lexicographically *equal*
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for FwdIter1.
- **FwdIter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for FwdIter2.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *lexicographical_compare* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function for FwdIter1. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function for FwdIter2. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.

- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **pred** – Refers to the comparison function that the first and second ranges will be applied to
- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual predicate *is* invoked.

Returns

The *lexicographically_compare* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *lexicographically_compare* algorithm returns true if the first range is lexicographically less, otherwise it returns false. range [first2, last2), it returns false.

```
template<typename Rng1, typename Rng2, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity,
typename Pred = hpx::parallel::detail::less>
bool hpx::ranges::lexicographical_compare(Rng1 &&rng1, Rng2 &&rng2, Pred &&pred = Pred(), Proj1
&&proj1 = Proj1(), Proj2 &&proj2 = Proj2())
```

Checks if the first range rng1 is lexicographically less than the second range rng2. uses a provided predicate to compare elements.

The comparison operations in the parallel *lexicographical_compare* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: At most $2 * \min(N1, N2)$ applications of the comparison operation, where $N1 = \text{std::distance}(\text{std::begin}(\text{rng1}), \text{std::end}(\text{rng1}))$ and $N2 = \text{std::distance}(\text{std::begin}(\text{rng2}), \text{std::end}(\text{rng2}))$.

Note: Lexicographical comparison is an operation with the following properties

- Two ranges are compared element by element
 - The first mismatching element defines which range is lexicographically *less* or *greater* than the other
 - If one range is a prefix of another, the shorter range is lexicographically *less* than the other
 - If two ranges have equivalent elements and are of the same length, then the ranges are lexicographically *equal*
 - An empty range is lexicographically *less* than any non-empty range
 - Two empty ranges are lexicographically *equal*
-

Template Parameters

- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

- **Rng2** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *lexicographical_compare* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function for elements of the first range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function for elements of the second range. This defaults to `hpx::identity`

Parameters

- **rng1** – Refers to the sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Refers to the comparison function that the first and second ranges will be applied to
- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual predicate *is* invoked.

Returns

The *lexicographically_compare* algorithm returns *bool*. The *lexicographically_compare* algorithm returns true if the first range is lexicographically less, otherwise it returns false. range [first2, last2), it returns false.

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Proj1 = hpx::identity, typename Proj2
= hpx::identity, typename Pred = hpx::parallel::detail::less>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx::ranges::lexicographical_compare(ExPolicy
                                                                                                     &&pol-
                                                                                                     icy,
                                                                                                     Rng1
                                                                                                     &&rng1,
                                                                                                     Rng2
                                                                                                     &&rng2,
                                                                                                     Pred
                                                                                                     &&pred
                                                                                                     =
                                                                                                     Pred(),
                                                                                                     Proj1
                                                                                                     &&proj1
                                                                                                     =
                                                                                                     Proj1(),
                                                                                                     Proj2
                                                                                                     &&proj2
                                                                                                     =
                                                                                                     Proj2())
```

Checks if the first range `rng1` is lexicographically less than the second range `rng2`. uses a provided predicate to compare elements.

The comparison operations in the parallel *lexicographical_compare* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *lexicographical_compare* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most $2 * \min(N1, N2)$ applications of the comparison operation, where $N1 = \text{std::distance}(\text{std::begin}(\text{rng1}), \text{std::end}(\text{rng1}))$ and $N2 = \text{std::distance}(\text{std::begin}(\text{rng2}), \text{std::end}(\text{rng2}))$.

Note: Lexicographical comparison is an operation with the following properties

- Two ranges are compared element by element
 - The first mismatching element defines which range is lexicographically *less* or *greater* than the other
 - If one range is a prefix of another, the shorter range is lexicographically *less* than the other
 - If two ranges have equivalent elements and are of the same length, then the ranges are lexicographically *equal*
 - An empty range is lexicographically *less* than any non-empty range
 - Two empty ranges are lexicographically *equal*
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *lexicographical_compare* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function for elements of the first range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function for elements of the second range. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Refers to the comparison function that the first and second ranges will be applied to
- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual predicate *is* invoked.

Returns

The *lexicographically_compare* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *lexicographically_compare* algorithm returns true if the first range is lexicographically less, otherwise it returns false. range [first2, last2), it returns false.

hpx::ranges::partition

Defined in header [hpx/algorithm.hpp](#)⁷⁰⁸.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::ranges::stable_partition*
- *hpx::ranges::partition_copy*

```
template<typename Rng, typename Pred, typename Proj = hpx::identity>
subrange_t<std::ranges::iterator_t<Rng>> hpx::ranges::partition(Rng &&rng, Pred &&pred, Proj &&proj =
    Proj())
```

Reorders the elements in the range *rng* in such a way that all elements for which the predicate *pred* returns true precede the elements for which the predicate *pred* returns false. Relative order of the elements is not preserved.

The assignments in the parallel *partition* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: Performs at most $2 * N$ swaps, exactly N applications of the predicate and projection, where $N = \text{std::distance}(\text{begin}(\text{rng}), \text{end}(\text{rng}))$.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by the range *rng*. This is an unary predicate for partitioning the source iterators. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

⁷⁰⁸ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *partition* algorithm returns *subrange_t<std::ranges::iterator_t<Rng>>* The *partition* algorithm returns a subrange starting with an iterator to the first element of the second group and finishing with an iterator equal to last.

```
template<typename ExPolicy, typename Rng, typename Pred, typename Proj = hpx::identity>
parallel::util::detail::algorithm_result<ExPolicy, subrange_t<std::ranges::iterator_t<Rng>>> hpx::ranges::partition(ExPolicy
&&pol-
icy,
Rng
&&rng,
Pred
&&pred,
Proj
&&proj
=
Proj())
```

Reorders the elements in the range *rng* in such a way that all elements for which the predicate *pred* returns true precede the elements for which the predicate *pred* returns false. Relative order of the elements is not preserved.

The assignments in the parallel *partition* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *partition* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs at most $2 * N$ swaps, exactly N applications of the predicate and projection, where $N = \text{std::distance}(\text{begin}(\text{rng}), \text{end}(\text{rng}))$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.

- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by the range *rng*. This is an unary predicate for partitioning the source iterators. The signature of this predicate should be equivalent to:

`bool pred(const Type &a);`

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *partition* algorithm returns a `hpx::future<subrange_t<std::ranges::iterator_t<Rng>>>` if the execution policy is of type *parallel_task_policy* and returns *subrange_t<std::ranges::iterator_t<Rng>>* The *partition* algorithm returns a subrange starting with an iterator to the first element of the second group and finishing with an iterator equal to last.

```
template<typename FwdIter, typename Sent, typename Pred, typename Proj = hpx::identity>  
subrange_t<FwdIter> hpx::ranges::partition(FwdIter first, Sent last, Pred &&pred, Proj &&proj = Proj())
```

Reorders the elements in the range [first, last) in such a way that all elements for which the predicate *pred* returns true precede the elements for which the predicate *pred* returns false. Relative order of the elements is not preserved.

The assignments in the parallel *partition* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: At most $2 * (\text{last} - \text{first})$ swaps. Exactly *last - first* applications of the predicate and projection.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate for partitioning the source iterators. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *partition* algorithm returns *subrange_t<FwdIter>*. The *partition* algorithm returns a subrange starting with an iterator to the first element of the second group and finishing with an iterator equal to last.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename Pred, typename Proj =
hpx::identity>
parallel::util::detail::algorithm_result<ExPolicy, subrange_t<FwdIter>>::type hpx::ranges::partition(ExPolicy
&&pol-
icy,
FwdIter
first,
Sent
last,
Pred
&&pred,
Proj
&&proj
=
Proj())
```

Reorders the elements in the range `[first, last)` in such a way that all elements for which the predicate *pred* returns true precede the elements for which the predicate *pred* returns false. Relative order of the elements is not preserved.

The assignments in the parallel *partition* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *partition* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most $2 * (\text{last} - \text{first})$ swaps. Exactly *last - first* applications of the predicate and projection.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.

- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate for partitioning the source iterators. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *InIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *partition* algorithm returns a `hpx::future<subrange_t<FwdIter>>` if the execution policy is of type *parallel_task_policy* and returns *subrange_t<FwdIter>* otherwise. The *partition* algorithm returns a subrange starting with an iterator to the first element of the second group and finishing with an iterator equal to last.

hpx::ranges::stable_partition

Defined in header `hpx/algorithm.hpp`⁷⁰⁹.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::ranges::partition*
- *hpx::ranges::partition_copy*

```
template<typename Rng, typename Pred, typename Proj = hpx::identity>
subrange_t<std::ranges::iterator_t<Rng>> hpx::ranges::stable_partition(Rng &&rng, Pred &&pred, Proj
&&proj = Proj())
```

Permutes the elements in the range [first, last) such that there exists an iterator *i* such that for every iterator *j* in the range [first, *i*) `INVOKE(f, INVOKE(proj, *j)) != false`, and for every iterator *k* in the range [*i*, last), `INVOKE(f, INVOKE(proj, *k)) == false`

The invocations of *f* in the parallel *stable_partition* algorithm invoked without an execution policy object executes in sequential order in the calling thread.

⁷⁰⁹ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Note: Complexity: At most $(last - first) * \log(last - first)$ swaps, but only linear number of swaps if there is enough extra memory Exactly *last - first* applications of the predicate and projection.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an bidirectional iterator
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Unary predicate which returns true if the element should be ordered before other elements. Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool fun(const Type &a);
```

The signature does not need to have const&. The type *Type* must be such that an object of type *BidirIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *f* is invoked.

Returns

The *stable_partition* algorithm returns an iterator *i* such that for every iterator *j* in the range [first, i), $f(*j) \neq \text{false}$ INVOKE(*f*, INVOKE(*proj*, **j*)) $\neq \text{false}$, and for every iterator *k* in the range [i, last), $f(*k) == \text{false}$ INVOKE(*f*, INVOKE(*proj*, **k*)) $== \text{false}$. The relative order of the elements in both groups is preserved.

```
template<typename ExPolicy, typename Rng, typename Pred, typename Proj = hpx::identity>
```

```
parallel::util::detail::algorithm_result<ExPolicy, subrange_t<std::ranges::iterator_t<Rng>>> hpx::ranges::stable_partition(Ex
```

```
&&  
icy  
Rn  
&&  
Pr  
&&  
Pr  
&&  
=  
Pr
```

Permutes the elements in the range [first, last) such that there exists an iterator *i* such that for every iterator *j* in the range [first, i) INVOKE(*f*, INVOKE(*proj*, **j*)) $\neq \text{false}$, and for every iterator *k* in the range [i, last), INVOKE(*f*, INVOKE(*proj*, **k*)) $== \text{false}$

The invocations of *f* in the parallel *stable_partition* algorithm invoked with an execution policy object of type *sequenced_policy* executes in sequential order in the calling thread.

The invocations of f in the parallel *stable_partition* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most $(\text{last} - \text{first}) * \log(\text{last} - \text{first})$ swaps, but only linear number of swaps if there is enough extra memory. Exactly *last - first* applications of the predicate and projection.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the invocations of f .
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an bidirectional iterator
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Unary predicate which returns true if the element should be ordered before other elements. Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool fun(const Type &a);
```

The signature does not need to have `const&`. The type *Type* must be such that an object of type *BidirIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate f is invoked.

Returns

The *stable_partition* algorithm returns an iterator i such that for every iterator j in the range [first, i), $f(*j) \neq \text{false}$ INVOKE(f , INVOKE(proj , $*j$)) $\neq \text{false}$, and for every iterator k in the range [i , last), $f(*k) == \text{false}$ INVOKE(f , INVOKE(proj , $*k$)) $== \text{false}$. The relative order of the elements in both groups is preserved. If the execution policy is of type *parallel_task_policy* the algorithm returns a future<> referring to this iterator.

```
template<typename BidirIter, typename Sent, typename Pred, typename Proj = hpx::identity>
subrange_t<BidirIter> hpx::ranges::stable_partition(BidirIter first, Sent last, Pred &&pred, Proj &&proj =
                                                    Proj())
```

Permutes the elements in the range [first, last) such that there exists an iterator i such that for every iterator j in the range [first, i) INVOKE(f , INVOKE(proj , $*j$)) $\neq \text{false}$, and for every iterator k in the range [i , last), INVOKE(f , INVOKE(proj , $*k$)) $== \text{false}$

The invocations of *f* in the parallel *stable_partition* algorithm invoked without an execution policy object executes in sequential order in the calling thread.

Note: Complexity: At most $(last - first) * \log(last - first)$ swaps, but only linear number of swaps if there is enough extra memory Exactly *last - first* applications of the predicate and projection.

Template Parameters

- **BidirIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for BidirIter.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **pred** – Unary predicate which returns true if the element should be ordered before other elements. Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool fun(const Type &a);
```

The signature does not need to have const&. The type *Type* must be such that an object of type *BidirIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *f* is invoked.

Returns

The *stable_partition* algorithm returns an iterator *i* such that for every iterator *j* in the range [first, i), $f(*j) \neq \text{false}$ $\text{INVOKE}(f, \text{INVOKE}(\text{proj}, *j)) \neq \text{false}$, and for every iterator *k* in the range [i, last), $f(*k) == \text{false}$ $\text{INVOKE}(f, \text{INVOKE}(\text{proj}, *k)) == \text{false}$. The relative order of the elements in both groups is preserved.

```
template<typename ExPolicy, typename BidirIter, typename Sent, typename Pred, typename Proj =
hpx::identity>
```

parallel::util::detail::algorithm_result<ExPolicy, subrange_t<BidirIter>>::type *hpx::ranges::stable_partition*(*ExPolicy* &&*policy*, *BidirIter* *first*, *Sent* *last*, *Pred* &&*pred*, *Proj* &&*proj* = *Proj*())

Permutes the elements in the range `[first, last)` such that there exists an iterator `i` such that for every iterator `j` in the range `[first, i)` `INVOKE(f, INVOKE(proj, *j)) != false`, and for every iterator `k` in the range `[i, last)`, `INVOKE(f, INVOKE(proj, *k)) == false`

The invocations of *f* in the parallel *stable_partition* algorithm invoked with an execution policy object of type *sequenced_policy* executes in sequential order in the calling thread.

The invocations of *f* in the parallel *stable_partition* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most $(last - first) * \log(last - first)$ swaps, but only linear number of swaps if there is enough extra memory Exactly *last - first* applications of the predicate and projection.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the invocations of *f*.
- **BidirIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *BidirIter*.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **pred** – Unary predicate which returns true if the element should be ordered before other elements. Specifies the function (or function object) which will be invoked for each of the

elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool fun(const Type &a);
```

The signature does not need to have const&. The type *Type* must be such that an object of type *BidirIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *f* is invoked.

Returns

The *stable_partition* algorithm returns an iterator *i* such that for every iterator *j* in the range [first, *i*), *f*(*j) != false INVOKE(*f*, INVOKE(*proj*, *j)) != false, and for every iterator *k* in the range [*i*, last), *f*(*k) == false INVOKE(*f*, INVOKE(*proj*, *k)) == false. The relative order of the elements in both groups is preserved. If the execution policy is of type *parallel_task_policy* the algorithm returns a future<> referring to this iterator.

hpx::ranges::partition_copy

Defined in header `hpx/algorithm.hpp`⁷¹⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::ranges::partition*
- *hpx::ranges::stable_partition*

```
template<typename Rng, typename OutIter2, typename OutIter3, typename Pred, typename Proj =
hpx::identity>
partition_copy_result<std::ranges::iterator_t<Rng>, OutIter2, OutIter3> hpx::ranges::partition_copy(Rng
&&rng,
Out-
Iter2
dest_true,
Out-
Iter3
dest_false,
Pred
&&pred,
Proj
&&proj
=
Proj())
```

Copies the elements in the range *rng*, to two different ranges depending on the value returned by the predicate *pred*. The elements, that satisfy the predicate *pred* are copied to the range beginning at *dest_true*. The rest of the elements are copied to the range beginning at *dest_false*. The order of the elements is preserved.

The assignments in the parallel *partition_copy* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

⁷¹⁰ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Note: Complexity: Performs not more than N assignments, exactly N applications of the predicate *pred*, where $N = \text{std::distance}(\text{begin}(\text{rng}), \text{end}(\text{rng}))$.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **OutIter2** – The type of the iterator representing the destination range for the elements that satisfy the predicate *pred* (deduced). This iterator type must meet the requirements of an forward iterator.
- **OutIter3** – The type of the iterator representing the destination range for the elements that don't satisfy the predicate *pred* (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition_copy* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest_true** – Refers to the beginning of the destination range for the elements that satisfy the predicate *pred*
- **dest_false** – Refers to the beginning of the destination range for the elements that don't satisfy the predicate *pred*.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate for partitioning the source iterators. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *partition_copy* algorithm returns a `partition_copy_result<std::ranges::iterator_t<Rng>, FwdIter2, FwdIter3>>`. The *partition_copy* algorithm returns the tuple of the source iterator *last*, the destination iterator to the end of the *dest_true* range, and the destination iterator to the end of the *dest_false* range.

```
template<typename ExPolicy, typename Rng, typename FwdIter2, typename FwdIter3, typename Pred,
typename Proj = hpx::identity>
```

```
parallel::util::detail::algorithm_result<ExPolicy, partition_copy_result<std::ranges::iterator_t<Rng>, FwdIter2, FwdIter3>>>::type hpx::
```

Copies the elements in the range *rng*, to two different ranges depending on the value returned by the predicate *pred*. The elements, that satisfy the predicate *pred* are copied to the range beginning at *dest_true*. The rest of the elements are copied to the range beginning at *dest_false*. The order of the elements is preserved.

The assignments in the parallel *partition_copy* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *partition_copy* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs not more than N assignments, exactly N applications of the predicate *pred*, where $N = \text{std::distance}(\text{begin}(\text{rng}), \text{end}(\text{rng}))$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **FwdIter2** – The type of the iterator representing the destination range for the elements that satisfy the predicate *pred* (deduced). This iterator type must meet the requirements of an forward iterator.
- **FwdIter3** – The type of the iterator representing the destination range for the elements that don't satisfy the predicate *pred* (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition_copy* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest_true** – Refers to the beginning of the destination range for the elements that satisfy the predicate *pred*.
- **dest_false** – Refers to the beginning of the destination range for the elements that don't satisfy the predicate *pred*.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate for partitioning the source iterators. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *partition_copy* algorithm returns a `hpx::future<partition_copy_result<std::ranges::iterator_t<Rng>, FwdIter2, FwdIter3>>` if the execution policy is of type *parallel_task_policy* and returns `partition_copy_result<std::ranges::iterator_t<Rng>, FwdIter2, FwdIter3>` otherwise. The *partition_copy* algorithm returns the tuple of the source iterator *last*, the destination iterator to the end of the *dest_true* range, and the destination iterator to the end of the *dest_false* range.

```
template<typename InIter, typename Sent, typename OutIter2, typename OutIter3, typename Pred, typename Proj = hpx::identity>
```

```
partition_copy_result<InIter, OutIter2, OutIter3> hpx::ranges::partition_copy(InIter first, Sent last, OutIter2  
dest_true, OutIter3 dest_false,  
Pred &&pred, Proj &&proj =  
Proj())
```

Copies the elements in the range, defined by [first, last), to two different ranges depending on the value returned by the predicate *pred*. The elements, that satisfy the predicate *pred* are copied to the range beginning at *dest_true*. The rest of the elements are copied to the range beginning at *dest_false*. The order of the elements is preserved.

The assignments in the parallel *partition_copy* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the predicate *f*.

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.

- **OutIter2** – The type of the iterator representing the destination range for the elements that satisfy the predicate *pred* (deduced). This iterator type must meet the requirements of an forward iterator.
- **OutIter3** – The type of the iterator representing the destination range for the elements that don't satisfy the predicate *pred* (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *partition_copy* requires *Pred* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest_true** – Refers to the beginning of the destination range for the elements that satisfy the predicate *pred*
- **dest_false** – Refers to the beginning of the destination range for the elements that don't satisfy the predicate *pred*.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate for partitioning the source iterators. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have const&, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *partition_copy* algorithm returns a *partition_copy_result*<*FwdIter*, *OutIter2*, *OutIter3*>. The *partition_copy* algorithm returns the tuple of the source iterator *last*, the destination iterator to the end of the *dest_true* range, and the destination iterator to the end of the *dest_false* range.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename OutIter2, typename OutIter3,
typename Pred, typename Proj = hpx::identity>
```

`parallel::util::detail::algorithm_result<ExPolicy, partition_copy_result<FwdIter, OutIter2, OutIter3>>::type hpx::ranges::partiti`

Copies the elements in the range, defined by `[first, last)`, to two different ranges depending on the value returned by the predicate *pred*. The elements, that satisfy the predicate *pred* are copied to the range beginning at *dest_true*. The rest of the elements are copied to the range beginning at *dest_false*. The order of the elements is preserved.

The assignments in the parallel *partition_copy* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *partition_copy* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the predicate *f*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **OutIter2** – The type of the iterator representing the destination range for the elements that satisfy the predicate *pred* (deduced). This iterator type must meet the requirements of an forward iterator.
- **OutIter3** – The type of the iterator representing the destination range for the elements that don't satisfy the predicate *pred* (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form,

the parallel overload of *partition_copy* requires *Pred* to meet the requirements of *CopyConstructible*.

- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest_true** – Refers to the beginning of the destination range for the elements that satisfy the predicate *pred*
- **dest_false** – Refers to the beginning of the destination range for the elements that don't satisfy the predicate *pred*.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate for partitioning the source iterators. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *partition_copy* algorithm returns a *hpx::future<partition_copy_result<FwdIter, OutIter2, OutIter3>>* if the execution policy is of type *parallel_task_policy* and returns *partition_copy_result<FwdIter, OutIter2, OutIter3>* otherwise. The *partition_copy* algorithm returns the tuple of the source iterator *last*, the destination iterator to the end of the *dest_true* range, and the destination iterator to the end of the *dest_false* range.

hpx::ranges::transform_exclusive_scan

Defined in header [hpx/algorithm.hpp](#)⁷¹¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename InIter, typename Sent, typename OutIter, typename BinOp, typename UnOp, typename T =
typename std::iterator_traits<InIter>::value_type>
transform_exclusive_scan_result<InIter, OutIter> hpx::ranges::transform_exclusive_scan(InIter first, Sent
```

```
last, OutIter
dest, T init,
BinOp
&&binary_op,
UnOp
&&unary_op)
```

⁷¹¹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Assigns through each iterator i in $[result, result + (last - first))$ the value of `GENERALIZED_NONCOMMUTATIVE_SUM(binary_op, init, conv(*first), ..., conv(*(first + (i - result) - 1)))`.

The reduce operations in the parallel *transform_exclusive_scan* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Neither *conv* nor *op* shall invalidate iterators or sub-ranges, or modify elements in the ranges $[first, last)$ or $[result, result + (last - first))$.

The behavior of *transform_exclusive_scan* may be non-deterministic for a non-associative predicate.

Note: Complexity: $O(last - first)$ applications of the predicates *op* and *conv*.

Note: `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- $a1$ when N is 1
 - $op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, \dots, aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, \dots, aN))$ where $1 < K+1 = M \leq N$.
-

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter`.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **BinOp** – The type of the binary function object used for the reduction operation.
- **UnOp** – The type of the unary function object used for the conversion operation.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.
- **binary_op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **unary_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by `[first, last)`. This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

Returns

The *transform_exclusive_scan* algorithm returns *transform_exclusive_scan_result<InIter, OutIter>*. The *transform_exclusive_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Sent, typename FwdIter2, typename BinOp,
typename UnOp, typename T = typename std::iterator_traits<FwdIter1>::value_type>
parallel::util::detail::algorithm_result<ExPolicy, transform_exclusive_result<FwdIter1, FwdIter2>>::type hpx::ranges::transform
```

Assigns through each iterator *i* in `[result, result + (last - first))` the value of `GENERALIZED_NONCOMMUTATIVE_SUM(binary_op, init, conv(*first), ..., conv(*(first + (i - result) - 1))`.

The reduce operations in the parallel *transform_exclusive_scan* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform_exclusive_scan* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Neither *conv* nor *op* shall invalidate iterators or sub-ranges, or modify elements in the ranges `[first,last)` or `[result,result + (last - first))`.

The behavior of *transform_exclusive_scan* may be non-deterministic for a non-associative predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicates *op* and *conv*.

Note: `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- `a1` when `N` is 1
 - `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `FwdIter`.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **BinOp** – The type of the binary function object used for the reduction operation.
- **UnOp** – The type of the unary function object used for the conversion operation.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.
- **binary_op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **unary_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by `[first, last)`. This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

Returns

The *transform_exclusive_scan* algorithm returns a `hpx::future<transform_exclusive_result<FwdIter1, FwdIter2>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *transform_exclusive_result<FwdIter1, FwdIter2>* otherwise. The *transform_exclusive_scan* algorithm returns an input iterator to the point denoted by the sentinel and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng, typename O, typename BinOp, typename UnOp, typename T = typename
std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>
transform_exclusive_scan_result<std::ranges::iterator_t<Rng>, O> hpx::ranges::transform_exclusive_scan(Rng
&&rng,
O
dest,
T
init,
BinOp
&&binary_op,
UnOp
&&unary_op)
```

Assigns through each iterator *i* in `[result, result + (last - first))` the value of `GENERALIZED_NONCOMMUTATIVE_SUM(binary_op, init, conv(*first), ..., conv(*(first + (i - result) - 1))`.

The reduce operations in the parallel *transform_exclusive_scan* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Neither *conv* nor *op* shall invalidate iterators or sub-ranges, or modify elements in the ranges `[first,last)` or `[result,result + (last - first))`.

The behavior of *transform_exclusive_scan* may be non-deterministic for a non-associative predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicates *op* and *conv*.

Note: `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- *a1* when *N* is 1
 - `op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, ..., aN))` where $1 < K+1 = M \leq N$.
-

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **O** – The type of the iterator representing the destination range (deduced).
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).
- **BinOp** – The type of the binary function object used for the reduction operation.
- **UnOp** – The type of the unary function object used for the conversion operation.

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.
- **binary_op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **unary_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

Returns

The *transform_exclusive_scan* algorithm returns a *transform_exclusive_scan_result*<std::ranges::iterator_t<Rng>, O>. The *transform_exclusive_scan* algorithm returns an input iterator to one past the end of the range and an output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng, typename O, typename BinOp, typename UnOp, typename T =  
typename std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>
```

parallel::util::detail::algorithm_result<ExPolicy, transform_exclusive_scan_result<std::ranges::iterator_t<Rng>, O>>::type hpx::rang

Assigns through each iterator i in $[result, result + (last - first))$ the value of `GENERALIZED_NONCOMMUTATIVE_SUM(binary_op, init, conv(*first), ..., conv(*(first + (i - result) - 1)))`.

The reduce operations in the parallel *transform_exclusive_scan* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *transform_exclusive_scan* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Neither *conv* nor *op* shall invalidate iterators or sub-ranges, or modify elements in the ranges $[first, last)$ or $[result, result + (last - first))$.

The behavior of *transform_exclusive_scan* may be non-deterministic for a non-associative predicate.

Note: Complexity: $O(last - first)$ applications of the predicates *op* and *conv*.

Note: `GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, ..., aN)` is defined as:

- $a1$ when N is 1
 - $op(GENERALIZED_NONCOMMUTATIVE_SUM(op, a1, \dots, aK), GENERALIZED_NONCOMMUTATIVE_SUM(op, aM, \dots, aN))$ where $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **O** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator. This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

- **BinOp** – The type of the binary function object used for the reduction operation.
- **UnOp** – The type of the unary function object used for the conversion operation.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **init** – The initial value for the generalized sum.
- **binary_op** – Specifies the function (or function object) which will be invoked for each of the values of the input sequence. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Ret* must be such that an object of a type as given by the input sequence can be implicitly converted to any of those types.

- **unary_op** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a unary predicate. The signature of this predicate should be equivalent to:

```
R fun(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*. The type *R* must be such that an object of this type can be implicitly converted to *T*.

Returns

The *transform_exclusive_scan* algorithm returns a `hpx::future<transform_exclusive_scan_result<std::ranges::iterator_t<Rng>, O>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *transform_exclusive_scan_result<std::ranges::iterator_t<Rng>, O>* otherwise. The *transform_exclusive_scan* algorithm returns an input iterator to one past the end of the range and an output iterator to the element in the destination range, one past the last element copied.

hpx::ranges::equal

Defined in header `hpx/algorithm.hpp`⁷¹².

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename ExPolicy, typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename
Pred = equal_to, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx::ranges::equal(ExPolicy &&policy, Iter1
first1, Sent1 last1, Iter2 first2,
Sent2 last2, Pred &&op =
Pred(), Proj1 &&proj1 =
Proj1(), Proj2 &&proj2 =
Proj2())
```

⁷¹² <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Returns true if the range [first1, last1) is equal to the range [first2, last2), and false otherwise.

The comparison operations in the parallel *equal* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *equal* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most $\min(\text{last1} - \text{first1}, \text{last2} - \text{first2})$ applications of the predicate *f*.

Note: The two ranges are considered equal if, for every iterator *i* in the range [first1,last1), **i* equals **(first2 + (i - first1))*. This overload of *equal* uses *operator==* to determine if two elements are equal.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the source iterators used for the end of the first range (deduced).
- **Iter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the source iterators used for the end of the second range (deduced).
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function applied to the first range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second range. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.

- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual predicate *is* invoked.

Returns

The *equal* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *equal* algorithm returns true if the elements in the two ranges are equal, otherwise it returns false. If the length of the range [first1, last1) does not equal the length of the range [first2, last2), it returns false.

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Pred = equal_to, typename Proj1 =  
hpx::identity, typename Proj2 = hpx::identity>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx::ranges::equal(ExPolicy &&policy, Rng1  
                                         &&rng1, Rng2 &&rng2,  
                                         Pred &&op = Pred(), Proj1  
                                         &&proj1 = Proj1(), Proj2  
                                         &&proj2 = Proj2())
```

Returns true if the range [first1, last1) is equal to the range starting at first2, and false otherwise.

The comparison operations in the parallel *equal* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparison operations in the parallel *equal* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most *last1 - first1* applications of the predicate *f*.

Note: The two ranges are considered equal if, for every iterator *i* in the range [first1,last1), **i* equals **(first2 + (i - first1))*. This overload of *equal* uses operator== to determine if two elements are equal.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng1** – The type of the first source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Rng2** – The type of the second source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.

- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function applied to the first range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second range. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual predicate *is* invoked.

Returns

The *equal* algorithm returns a `hpx::future<bool>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *equal* algorithm returns true if the elements in the two ranges are equal, otherwise it returns false.

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename Pred = equal_to,
typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
bool hpx::ranges::equal(Iter1 first1, Sent1 last1, Iter2 first2, Sent2 last2, Pred &&op = Pred(), Proj1 &&proj1
= Proj1(), Proj2 &&proj2 = Proj2())
```

Returns true if the range [first1, last1) is equal to the range [first2, last2), and false otherwise.

Note: Complexity: At most $\min(\text{last1} - \text{first1}, \text{last2} - \text{first2})$ applications of the predicate *f*.

Note: The two ranges are considered equal if, for every iterator *i* in the range [first1,last1), **i* equals **(first2 + (i - first1))*. This overload of *equal* uses operator== to determine if two elements are equal.

Template Parameters

- **Iter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the source iterators used for the end of the first range (deduced).

- **Iter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the source iterators used for the end of the second range (deduced).
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function applied to the first range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second range. This defaults to `hpx::identity`

Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual predicate *is* invoked.

Returns

The *equal* algorithm returns true if the elements in the two ranges are equal, otherwise it returns false. If the length of the range [first1, last1) does not equal the length of the range [first2, last2), it returns false.

```
template<typename Rng1, typename Rng2, typename Pred = equal_to, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
bool hpx::ranges::equal(Rng1 &&rng1, Rng2 &&rng2, Pred &&op = Pred(), Proj1 &&proj1 = Proj1(), Proj2 &&proj2 = Proj2())
```

Returns true if the range [first1, last1) is equal to the range starting at first2, and false otherwise.

Note: Complexity: At most *last1 - first1* applications of the predicate *f*.

Note: The two ranges are considered equal if, for every iterator *i* in the range `[first1,last1)`, **i* equals **(first2 + (i - first1))*. This overload of `equal` uses `operator==` to determine if two elements are equal.

Template Parameters

- **Rng1** – The type of the first source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Rng2** – The type of the second source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj1** – The type of an optional projection function applied to the first range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second range. This defaults to `hpx::identity`

Parameters

- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter1* and *FwdIter2* can be dereferenced and then implicitly converted to *Type1* and *Type2* respectively

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual predicate *is* invoked.

Returns

The *equal* algorithm returns true if the elements in the two ranges are equal, otherwise it returns false.

hpx::ranges::set_difference

Defined in header `hpx/algorithm.hpp`⁷¹³.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename ExPolicy, typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename
Iter3, typename Pred = hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename Proj2 =
hpx::identity>
```

⁷¹³ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

hpx::parallel::util::detail::algorithm_result<ExPolicy, set_difference_result<Iter1, Iter3>>::type *hpx::ranges::set_difference*(*Ex*

Constructs a sorted range beginning at *dest* consisting of all elements present in the range *[first1, last1)* and not present in the range *[first2, last2)*. This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

Equivalent elements are treated individually, that is, if some element is found *m* times in *[first1, last1)* and *n* times in *[first2, last2)*, it will be copied to *dest* exactly *std::max(m-n, 0)* times. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

The application of function objects in parallel algorithm invoked with a sequential execution policy object execute in sequential order in the calling thread (*sequenced_policy*) or in a single new thread spawned from the current thread (for *sequenced_task_policy*).

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most $2 \cdot (N1 + N2 - 1)$ comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.

- **Iter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the end source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an sentinel for Iter2.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set_difference* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

Returns

The *set_difference* algorithm returns a `hpx::future<ranges::set_difference_result<Iter1, Iter3>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns

ranges::set_difference_result<Iter1, Iter3> otherwise. The *set_difference* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Iter3, typename Pred =  
hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>  
hpx::parallel::util::detail::algorithm_result<ExPolicy, set_difference_result<std::ranges::iterator_t<Rng1>, Iter3>> hpx::ranges::se
```

Constructs a sorted range beginning at *dest* consisting of all elements present in the range *[first1, last1)* and not present in the range *[first2, last2)*. This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

Equivalent elements are treated individually, that is, if some element is found *m* times in *[first1, last1)* and *n* times in *[first2, last2)*, it will be copied to *dest* exactly *std::max(m-n, 0)* times. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

The application of function objects in parallel algorithm invoked with a sequential execution policy object execute in sequential order in the calling thread (*sequenced_policy*) or in a single new thread spawned from the current thread (for *sequenced_task_policy*).

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most $2 \cdot (N_1 + N_2 - 1)$ comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.

- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set_difference* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

Returns

The *set_difference* algorithm returns a `hpx::future<ranges::set_difference_result<Iter1, Iter3>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `ranges::set_difference_result<Iter1, Iter3>` otherwise. where *Iter1* is `range_iterator_t<Rng1>` and *Iter2* is `range_iterator_t<Rng2>` The *set_difference* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename Iter3, typename Pred
= hpx::parallel::detail::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
set_difference_result<Iter1, Iter3> hpx::ranges::set_difference(Iter1 first1, Sent1 last1, Iter2 first2, Sent2
                                                             last2, Iter3 dest, Pred &&op = Pred(), Proj1
                                                             &&proj1 = Proj1(), Proj2 &&proj2 =
                                                             Proj2())
```

Constructs a sorted range beginning at *dest* consisting of all elements present in the range `[first1, last1)` and not

present in the range `[first2, last2)`. This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

Equivalent elements are treated individually, that is, if some element is found *m* times in `[first1, last1)` and *n* times in `[first2, last2)`, it will be copied to *dest* exactly `std::max(m-n, 0)` times. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

Note: Complexity: At most $2 \cdot (N1 + N2 - 1)$ comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

Template Parameters

- **Iter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an forward iterator.
- **Sent2** – The type of the end source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an sentinel for Iter2.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set_difference* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *InIter* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

Returns

The *set_difference* algorithm returns *ranges::set_difference_result<Iter1, Iter3>*. The *set_difference* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng1, typename Rng2, typename Iter3, typename Pred = hpx::parallel::detail::less, typename
Proj1 = hpx::identity, typename Proj2 = hpx::identity>
set_difference_result<std::ranges::iterator_t<Rng1>, Iter3> hpx::ranges::set_difference(Rng1 &&rng1,
                                                                                      Rng2 &&rng2, Iter3
                                                                                      dest, Pred &&op =
                                                                                      Pred(), Proj1
                                                                                      &&proj1 = Proj1(),
                                                                                      Proj2 &&proj2 =
                                                                                      Proj2())
```

Constructs a sorted range beginning at *dest* consisting of all elements present in the range [*first1*, *last1*) and not present in the range [*first2*, *last2*). This algorithm expects both input ranges to be sorted with the given binary predicate *f*.

Equivalent elements are treated individually, that is, if some element is found *m* times in [*first1*, *last1*) and *n* times in [*first2*, *last2*), it will be copied to *dest* exactly `std::max(m-n, 0)` times. The resulting range cannot overlap with either of the input ranges.

The resulting range cannot overlap with either of the input ranges.

Note: Complexity: At most $2 \cdot (N_1 + N_2 - 1)$ comparisons, where *N1* is the length of the first sequence and *N2* is the length of the second sequence.

Template Parameters

- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.

- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *set_difference* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function applied to the first sequence. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function applied to the second sequence. This defaults to `hpx::identity`

Parameters

- **rng1** – Refers to the first sequence of elements the algorithm will be applied to.
- **rng2** – Refers to the second sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **op** – The binary predicate which returns true if the elements should be treated as equal. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *Iter* can be dereferenced and then implicitly converted to *Type1*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first sequence as a projection operation before the actual predicate *op* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second sequence as a projection operation before the actual predicate *op* is invoked.

Returns

The *set_difference* algorithm returns `ranges::set_difference_result<Iter1, Iter3>`. where *Iter1* is `range_iterator_t<Rng1>` and *Iter2* is `range_iterator_t<Rng2>` The *set_difference* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

hpx::ranges::copy

Defined in header `hpx/algorithm.hpp`⁷¹⁴.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::ranges::copy_n`
- `hpx::ranges::copy_if`

template<typename **ExPolicy**, typename **FwdIter1**, typename **Sent1**, typename **FwdIter**>

⁷¹⁴ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

parallel::util::detail::algorithm_result<ExPolicy, ranges::copy_result<FwdIter1, FwdIter>>::type *hpx::ranges::copy*(*ExPolicy* &&*policy*, *FwdIter1* *iter*, *Sent1* *sent*, *FwdIter* *dest*)

Copies the elements in the range, defined by [first, last), to another range beginning at *dest*.

The assignments in the parallel *copy* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *copy* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **iter** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *copy* algorithm returns a *hpx::future<ranges::copy_result<FwdIter1, FwdIter> >* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *ranges::copy_result<FwdIter1, FwdIter>* otherwise. The *copy* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

template<typename **ExPolicy**, typename **Rng**, typename **FwdIter**>

parallel::util::detail::algorithm_result<ExPolicy, ranges::copy_result<typename hpx::traits::range_traits<Rng>::iterator_type, FwdIter>

Copies the elements in the range *rng* to another range beginning at *dest*.

The assignments in the parallel *copy* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *copy* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly `std::distance(begin(rng), end(rng))` assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *copy* algorithm returns a `hpx::future<ranges::copy_result<iterator_t<Rng>, FwdIter2>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `ranges::copy_result<iterator_t<Rng>, FwdIter2>` otherwise. The *copy* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename FwdIter1, typename Sent1, typename FwdIter>  
ranges::copy_result<FwdIter1, FwdIter> hpx::ranges::copy(FwdIter1 iter, Sent1 sent, FwdIter dest)
```

Copies the elements in the range, defined by [first, last), to another range beginning at *dest*.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **FwdIter1** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **iter** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *copy* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng, typename FwdIter>
ranges::copy_result<typename hpx::traits::range_traits<Rng>::iterator_type, FwdIter> hpx::ranges::copy(Rng
                                                         &&rng,
                                                         FwdIter
                                                         dest)
```

Copies the elements in the range *rng* to another range beginning at *dest*.

Note: Complexity: Performs exactly `std::distance(begin(rng), end(rng))` assignments.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *copy* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

hpx::ranges::copy_n

Defined in header `hpx/algorithm.hpp`⁷¹⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::copy`

⁷¹⁵ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- `hpx::ranges::copy_if`

template<typename **ExPolicy**, typename **FwdIter1**, typename **Size**, typename **FwdIter2**>

`hpx::parallel::util::detail::algorithm_result<ExPolicy, ranges::copy_n_result<FwdIter1, FwdIter2>>::type` `hpx::ranges::copy_n(Ex`

Copies the elements in the range `[first, first + count)`, starting from `first` and proceeding to `first + count - 1.`, to another range beginning at `dest`.

The assignments in the parallel `copy_n` algorithm invoked with an execution policy object of type `sequenced_policy` execute in sequential order in the calling thread.

The assignments in the parallel `copy_n` algorithm invoked with an execution policy object of type `parallel_policy` or `parallel_task_policy` are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *count* assignments, if *count* > 0, no assignments otherwise.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The `copy_n` algorithm returns a `hpx::future<ranges::copy_n_result<FwdIter1, FwdIter2> >` if the execution policy is of type `sequenced_task_policy` or `parallel_task_policy` and returns `ranges::copy_n_result<FwdIter1, FwdIter2>` otherwise. The `copy` algorithm returns the pair of the input iterator forwarded to the first element after the last in the input sequence and the output iterator to the element in the destination range, one past the last element copied.

template<typename **FwdIter1**, typename **Size**, typename **FwdIter2**>

`ranges::copy_n_result<FwdIter1, FwdIter2> hpx::ranges::copy_n(FwdIter1 first, Size count, FwdIter2 dest)`

Copies the elements in the range `[first, first + count)`, starting from `first` and proceeding to `first + count - 1`, to another range beginning at `dest`.

Note: Complexity: Performs exactly *count* assignments, if *count* > 0, no assignments otherwise.

Template Parameters

- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *copy* algorithm returns the pair of the input iterator forwarded to the first element after the last in the input sequence and the output iterator to the element in the destination range, one past the last element copied.

`hpx::ranges::copy_if`

Defined in header `hpx/algorithm.hpp`⁷¹⁶.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::ranges::copy`
- `hpx::ranges::copy_n`

`template<typename ExPolicy, typename FwdIter1, typename Sent1, typename FwdIter, typename Pred, typename Proj> hpx::identity>`

⁷¹⁶ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

hpx::parallel::util::detail::algorithm_result<ExPolicy, ranges::copy_if_result<FwdIter1, FwdIter>>::type hpx::ranges::copy_if(E

Copies the elements in the range, defined by [first, last) to another range beginning at *dest*. The order of the elements that are not removed is preserved.

The assignments in the parallel *copy_if* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *copy_if* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for FwdIter1.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **iter** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – The binary predicate which returns *true* if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *copy_if* algorithm returns a `hpx::future<ranges::copy_if_result<iterator_t<Rng>, FwdIter2>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `ranges::copy_if_result<iterator_t<Rng>, FwdIter2>` otherwise. The *copy_if* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng, typename FwdIter, typename Pred, typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, ranges::copy_if_result<typename hpx::traits::range_traits<Rng>::iterator_type, FwdIter2>>
```

Copies the elements in the range, defined by *rng* to another range beginning at *dest*. The order of the elements that are not removed is preserved.

The assignments in the parallel *copy_if* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *copy_if* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.

- **dest** – Refers to the beginning of the destination range.
- **pred** – The binary predicate which returns *true* if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *copy_if* algorithm returns a `hpx::future<ranges::copy_if_result<iterator_t<Rng>, FwdIter2>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `ranges::copy_if_result<iterator_t<Rng>, FwdIter2>` otherwise. The *copy_if* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename FwdIter1, typename Sent1, typename FwdIter, typename Pred, typename Proj =  
hpx::identity>  
ranges::copy_if_result<FwdIter1, FwdIter> hpx::ranges::copy_if(FwdIter1 iter, Sent1 sent, FwdIter dest, Pred  
                    &&pred, Proj &&proj = Proj())
```

Copies the elements in the range, defined by [first, last) to another range beginning at *dest*. The order of the elements that are not removed is preserved.

Template Parameters

- **FwdIter1** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for FwdIter1.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **iter** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – The binary predicate which returns *true* if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *copy_if* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng, typename FwdIter, typename Pred, typename Proj = hpx::identity>
ranges::copy_if_result<typename hpx::traits::range_traits<Rng>::iterator_type, FwdIter> hpx::ranges::copy_if(Rng
                                                                                                     &&rng,
                                                                                                     FwdIter
                                                                                                     dest,
                                                                                                     Pred
                                                                                                     &&pred,
                                                                                                     Proj
                                                                                                     &&proj
                                                                                                     =
                                                                                                     Proj())
```

Copies the elements in the range, defined by *rng* to another range beginning at *dest*. The order of the elements that are not removed is preserved.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of an optional function/function object to use.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – The binary predicate which returns *true* if the elements should be treated as equal. The signature should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The types *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *copy_if* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

hpx::ranges::move

Defined in header `hpx/algorithm.hpp`⁷¹⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename ExPolicy, typename Iter1, typename Sent1, typename Iter2>
hpx::parallel::util::detail::algorithm_result<ExPolicy, move_result<Iter1, Iter2>>::type hpx::ranges::move(ExPolicy
                                                                                                     &&pol-
                                                                                                     icy,
                                                                                                     Iter1
                                                                                                     first,
                                                                                                     Sent1
                                                                                                     last,
                                                                                                     Iter2
                                                                                                     dest)
```

Moves the elements in the range *rng* to another range beginning at *dest*. After this operation the elements in the moved-from range will still contain valid values of the appropriate type, but not necessarily the same values as before the move.

The assignments in the parallel *copy* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *copy* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly `std::distance(begin(rng), end(rng))` assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the source iterators used for the end of the first range (deduced).
- **Iter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

⁷¹⁷ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Returns

The *move* algorithm returns a `hpx::future<ranges::move_result<iterator_t<Rng>, FwdIter2>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *ranges::move_result<iterator_t<Rng>, FwdIter2>* otherwise. The *move* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element moved.

```
template<typename ExPolicy, typename Rng, typename Iter2>
hpx::parallel::util::detail::algorithm_result<ExPolicy, move_result<std::ranges::iterator_t<Rng>, Iter2>>::type hpx::ranges::move(
```

Moves the elements in the range *rng* to another range beginning at *dest*. After this operation the elements in the moved-from range will still contain valid values of the appropriate type, but not necessarily the same values as before the move.

The assignments in the parallel *copy* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *copy* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly `std::distance(begin(rng), end(rng))` assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Iter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *move* algorithm returns a `hpx::future<ranges::move_result<iterator_t<Rng>, FwdIter2>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *ranges::move_result<iterator_t<Rng>, FwdIter2>* otherwise. The *move* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element moved.

```
template<typename Iter1, typename Sent1, typename Iter2>
```

`move_result<Iter1, Iter2> hpx::ranges::move(Iter1 first, Iter2 last, Iter2 dest)`

Moves the elements in the range *rng* to another range beginning at *dest*. After this operation the elements in the moved-from range will still contain valid values of the appropriate type, but not necessarily the same values as before the move.

Note: Complexity: Performs exactly `std::distance(begin(rng), end(rng))` assignments.

Template Parameters

- **Iter1** – The type of the source iterators used for the first range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the source iterators used for the end of the first range (deduced).
- **Iter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *move* algorithm returns `ranges::move_result<iterator_t<Rng>, FwdIter2>`. The *move* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element moved.

`template<typename Rng, typename Iter2>`

`move_result<std::ranges::iterator_t<Rng>, Iter2> hpx::ranges::move(Rng &&rng, Iter2 dest)`

Moves the elements in the range *rng* to another range beginning at *dest*. After this operation the elements in the moved-from range will still contain valid values of the appropriate type, but not necessarily the same values as before the move.

Note: Complexity: Performs exactly `std::distance(begin(rng), end(rng))` assignments.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Iter2** – The type of the source iterators used for the second range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.

Returns

The *move* algorithm returns a `ranges::move_result<iterator_t<Rng>, FwdIter2>`. The *move* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element moved.

hpx::ranges::remove

Defined in header `hpx/algorithm.hpp`⁷¹⁸.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::remove_if`

```
template<typename Iter, typename Sent, typename Proj = hpx::identity, typename T = typename
hpx::parallel::traits::projected<Iter, Proj>::value_type>
subrange_t<Iter, Sent> hpx::ranges::remove(Iter first, Sent last, T const &value, Proj &&proj = Proj())
```

Removes all elements that are equal to *value* from the range [first, last) and returns a subrange [ret, last), where ret is a past-the-end iterator for the new end of the range.

The assignments in the parallel *remove* algorithm execute in sequential order in the calling thread.

Note: Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the operator==() and the projection *proj*.

Template Parameters

- **Iter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for FwdIter.
- **T** – The type of the value to remove (deduced). This value type must meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – Specifies the value of elements to remove.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *remove* algorithm returns a `subrange_t<FwdIter, Sent>`. The *remove* algorithm returns an object {ret, last}, where ret is a past-the-end iterator for a new subrange of the values all in valid but unspecified state.

```
template<typename Rng, typename Proj = hpx::identity, typename T = typename
hpx::parallel::traits::projected<std::ranges::iterator_t<Rng>, Proj>::value_type>
```

⁷¹⁸ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

```
subrange_t<std::ranges::iterator_t<Rng>> hpx::ranges::remove(Rng &&rng, T const &value, Proj &&proj =  
Proj())
```

Removes all elements that are equal to *value* from the range *rng* and and returns a subrange [ret, util::end(rng)), where ret is a past-the-end iterator for the new end of the range.

The assignments in the parallel *remove* algorithm execute in sequential order in the calling thread.

Note: Complexity: Performs not more than *util::end(rng)*

- *util::begin(rng)* assignments, exactly *util::end(rng) - util::begin(rng)* applications of the operator==() and the projection *proj*.
-

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **T** – The type of the value to remove (deduced). This value type must meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **value** – Specifies the value of elements to remove.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *remove* algorithm returns a *subrange_t<std::ranges::iterator_t<Rng>>*. The *remove* algorithm returns an object {ret, last}, where ret is a past-the-end iterator for a new subrange of the values all in valid but unspecified state.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename Proj = hpx::identity, typename T =  
typename hpx::parallel::traits::projected<FwdIter, Proj>::value_type>  
parallel::util::detail::algorithm_result<ExPolicy, subrange_t<FwdIter, Sent>>::type hpx::ranges::remove(ExPolicy  
                                                    &&pol-  
                                                    icy,  
                                                    FwdIter  
                                                    first,  
                                                    Sent  
                                                    last, T  
                                                    const  
                                                    &value,  
                                                    Proj  
                                                    &&proj  
                                                    =  
                                                    Proj())
```

Removes all elements that are equal to *value* from the range [first, last) and and returns a subrange [ret, last), where ret is a past-the-end iterator for the new end of the range.

The assignments in the parallel *remove* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *remove* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the operator `==()` and the projection *proj*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for FwdIter.
- **T** – The type of the value to remove (deduced). This value type must meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – Specifies the value of elements to remove.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *remove* algorithm returns a `hpx::future<subrange_t<FwdIter, Sent>>`. The *remove* algorithm returns an object {ret, last}, where ret is a past-the-end iterator for a new subrange of the values all in valid but unspecified state.

```
template<typename ExPolicy, typename Rng, typename Proj = hpx::identity, typename T = typename
hpx::parallel::traits::projected<std::ranges::iterator_t<Rng>, Proj>::value_type>
```

```
parallel::util::detail::algorithm_result<ExPolicy, subrange_t<std::ranges::iterator_t<Rng>>> hpx::ranges::remove(ExPolicy  
&&pol-  
icy,  
Rng  
&&rng,  
T  
const  
&value,  
Proj  
&&proj  
=  
Proj())
```

Removes all elements that are equal to *value* from the range *rng* and and returns a subrange [ret, util::end(rng)), where ret is a past-the-end iterator for the new end of the range.

The assignments in the parallel *remove* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *remove* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs not more than *util::end(rng)*

- *util::begin(rng)* assignments, exactly *util::end(rng) - util::begin(rng)* applications of the operator *==()* and the projection *proj*.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **T** – The type of the value to remove (deduced). This value type must meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **value** – Specifies the value of elements to remove.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *remove* algorithm returns a *hpx::future< subrange_t<std::ranges::iterator_t<Rng>>>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *remove* algorithm returns the iterator to the new end of the range.

hpx::ranges::remove_if

Defined in header [hpx/algorithm.hpp](#)⁷¹⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::ranges::remove*

```
template<typename Iter, typename Sent, typename Pred, typename Proj = hpx::identity>
subrange_t<Iter, Sent> hpx::ranges::remove_if(Iter first, Sent sent, Pred &&pred, Proj &&proj = Proj())
```

Removes all elements for which predicate *pred* returns true from the range [first, last) and returns a subrange [ret, last), where ret is a past-the-end iterator for the new end of the range.

The assignments in the parallel *remove_if* algorithm execute in sequential order in the calling thread.

Note: Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the predicate *pred* and the projection *proj*.

Template Parameters

- **Iter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for FwdIter.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *remove_if* requires *Pred* to meet the requirements of *Copy-Constructible*..
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have const&, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *remove_if* algorithm returns a *subrange_t<FwdIter, Sent>*. The *remove_if* algorithm returns

⁷¹⁹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

an object {ret, last}, where ret is a past-the-end iterator for a new subrange of the values all in valid but unspecified state.

```
template<typename Rng, typename Pred, typename Proj = hpx::identity>
subrange_t<std::ranges::iterator_t<Rng>> hpx::ranges::remove_if(Rng &&rng, Pred &&pred, Proj &&proj =
    Proj())
```

Removes all elements that are equal to *value* from the range *rng* and returns a subrange [ret, util::end(rng)), where ret is a past-the-end iterator for the new end of the range.

The assignments in the parallel *remove_if* algorithm execute in sequential order in the calling thread.

Note: Complexity: Performs not more than *util::end(rng)*

- *util::begin(rng)* assignments, exactly *util::end(rng) - util::begin(rng)* applications of the operator==() and the projection *proj*.
-

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *remove_if* requires *Pred* to meet the requirements of *Copy-Constructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have const&, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *remove_if* algorithm returns a *subrange_t<std::ranges::iterator_t<Rng>>*. The *remove_if* algorithm returns an object {ret, last}, where ret is a past-the-end iterator for a new subrange of the values all in valid but unspecified state.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename Pred, typename Proj =
hpx::identity>
```

```
parallel::util::detail::algorithm_result<ExPolicy, subrange_t<FwdIter, Sent>>::type hpx::ranges::remove_if(ExPolicy
                                                                    &&pol-
                                                                    icy,
                                                                    FwdIter
                                                                    first,
                                                                    Sent
                                                                    sent,
                                                                    Pred
                                                                    &&pred,
                                                                    Proj
                                                                    &&proj
                                                                    =
                                                                    Proj())
```

Removes all elements for which predicate *pred* returns true from the range [first, last) and returns a subrange [ret, last), where ret is a past-the-end iterator for the new end of the range.

The assignments in the parallel *remove_if* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *remove_if* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs not more than *last - first* assignments, exactly *last - first* applications of the predicate *pred* and the projection *proj*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for FwdIter.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *remove_if* requires *Pred* to meet the requirements of *Copy-Constructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *remove_if* algorithm returns a `hpx::future<subrange_t<FwdIter, Sent>>`. The *remove_if* algorithm returns an object {ret, last}, where ret is a past-the-end iterator for a new subrange of the values all in valid but unspecified state.

```
template<typename ExPolicy, typename Rng, typename Pred, typename Proj = hpx::identity>
parallel::util::detail::algorithm_result<ExPolicy, subrange_t<std::ranges::iterator_t<Rng>>> hpx::ranges::remove_if(ExPolicy
                                                                 &&pol-
                                                                 icy,
                                                                 Rng
                                                                 &&rng,
                                                                 Pred
                                                                 &&pred,
                                                                 Proj
                                                                 &&proj
                                                                 =
                                                                 Proj())
```

Removes all elements that are equal to *value* from the range *rng* and and returns a subrange [ret, util::end(rng)), where ret is a past-the-end iterator for the new end of the range.

The assignments in the parallel *remove_if* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *remove_if* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs not more than `util::end(rng)`

- `util::begin(rng)` assignments, exactly `util::end(rng) - util::begin(rng)` applications of the operator`==()` and the projection *proj*.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *remove_if* requires *Pred* to meet the requirements of *Copy-Constructible*.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *remove_if* algorithm returns a `hpx::future<subrange_t< std::ranges::iterator_t<Rng>>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *remove_if* algorithm returns an object {ret, last}, where ret is a past-the-end iterator for a new subrange of the values all in valid but unspecified state.

hpx::ranges::generate

Defined in header `hpx/algorithm.hpp`⁷²⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::generate_n`

```
template<typename ExPolicy, typename Rng, typename F>
```

```
hpx::parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng>>> hpx::ranges::generate(ExPolicy
                                                                                                     &&pol-
                                                                                                     icy,
                                                                                                     Rng
                                                                                                     &&rng,
                                                                                                     F
                                                                                                     &&f)
```

Assign each element in range [first, last) a value generated by the given function object *f*

The assignments in the parallel *generate* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *generate* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Exactly *distance(first, last)* invocations of *f* and assignments.

⁷²⁰ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – generator function that will be called. signature of function should be equivalent to the following:

`Ret fun();`

The type *Ret* must be such that an object of type *FwdIter* can be dereferenced and assigned a value of type *Ret*.

Returns

The *replace_if* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. It returns *last*.

```
template<typename ExPolicy, typename Iter, typename Sent, typename F>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, Iter> hpx::ranges::generate(ExPolicy &&policy, Iter  
first, Sent last, F &&f)
```

Assign each element in range [first, last) a value generated by the given function object *f*

The assignments in the parallel *generate* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *generate* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Exactly *distance(first, last)* invocations of *f* and assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source end iterator used (deduced). This iterator type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – generator function that will be called. signature of function should be equivalent to the following:

```
Ret fun();
```

The type *Ret* must be such that an object of type *FwdIter* can be dereferenced and assigned a value of type *Ret*.

Returns

The *replace_if* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. It returns *last*.

```
template<typename Rng, typename F>
```

```
std::ranges::iterator_t<Rng> hpx::ranges::generate(Rng &&rng, F &&f)
```

Assign each element in range [first, last) a value generated by the given function object *f*

Note: Complexity: Exactly *distance(first, last)* invocations of *f* and assignments.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – generator function that will be called. signature of function should be equivalent to the following:

```
Ret fun();
```

The type *Ret* must be such that an object of type *FwdIter* can be dereferenced and assigned a value of type *Ret*.

Returns

The *replace_if* algorithm returns *last*.

```
template<typename Iter, typename Sent, typename F>
```

```
Iter hpx::ranges::generate(Iter first, Sent last, F &&f)
```

Assign each element in range [first, last) a value generated by the given function object *f*

Note: Complexity: Exactly *distance(first, last)* invocations of *f* and assignments.

Template Parameters

- **Iter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source end iterator used (deduced). This iterator type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – generator function that will be called. signature of function should be equivalent to the following:

`Ret fun();`

The type *Ret* must be such that an object of type *FwdIter* can be dereferenced and assigned a value of type *Ret*.

Returns

The *replace_if* algorithm returns *last*.

hpx::ranges::generate_n

Defined in header `hpx/algorithm.hpp`⁷²¹.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::ranges::generate*

```
template<typename ExPolicy, typename FwdIter, typename Size, typename F>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::ranges::generate_n(ExPolicy  
                                                                                          &&policy, FwdIter  
                                                                                          first, Size count, F  
                                                                                          &&f)
```

Assigns each element in range [first, first+count) a value generated by the given function object *g*.

The assignments in the parallel *generate_n* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *generate_n* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Exactly *count* invocations of *f* and assignments, for *count* > 0.

⁷²¹ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements in the sequence the algorithm will be applied to.
- **f** – Refers to the generator function object that will be called. The signature of the function should be equivalent to

Ret <code>fun()</code> ;

The type *Ret* must be such that an object of type *OutputIt* can be dereferenced and assigned a value of type *Ret*.

Returns

The *replace_if* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. It returns *last*.

```
template<typename FwdIter, typename Size, typename F>
FwdIter hpx::ranges::generate_n(FwdIter first, Size count, F &&f)
```

Assigns each element in range [first, first+count) a value generated by the given function object *g*.

Note: Complexity: Exactly *count* invocations of *f* and assignments, for count > 0.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements in the sequence the algorithm will be applied to.
- **f** – Refers to the generator function object that will be called. The signature of the function should be equivalent to

```
Ret fun();
```

The type *Ret* must be such that an object of type *OutputIt* can be dereferenced and assigned a value of type *Ret*.

Returns

The *replace_if* algorithm returns *last*.

hpx::ranges::min_element

Defined in header [hpx/algorithm.hpp](#)⁷²².

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::ranges::max_element*
- *hpx::ranges::minmax_element*

```
template<typename FwdIter, typename Sent, typename F = hpx::parallel::detail::less, typename Proj = hpx::identity>
```

```
FwdIter hpx::ranges::min_element(FwdIter first, Sent last, F &&f = F(), Proj &&proj = Proj())
```

Finds the smallest element in the range [first, last) using the given comparison function *f*.

The comparisons in the parallel *min_element* algorithm execute in sequential order in the calling thread.

Note: Complexity: Exactly $\max(N-1, 0)$ comparisons, where $N = \text{std::distance}(\text{first}, \text{last})$.

Template Parameters

- **FwdIter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for FwdIter.
- **F** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the the left argument is less than the right element. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

⁷²² <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *min_element* algorithm returns *FwdIter*. The *min_element* algorithm returns the iterator to the smallest element in the range [first, last). If several elements in the range are equivalent to the smallest element, returns the iterator to the first such element. Returns last if the range is empty.

```
template<typename Rng, typename F = hpx::parallel::detail::less, typename Proj = hpx::identity>
std::ranges::iterator_t<Rng> hpx::ranges::min_element(Rng &&rng, F &&f = F(), Proj &&proj = Proj())
```

Finds the smallest element in the range [first, last) using the given comparison function *f*.

The comparisons in the parallel *min_element* algorithm execute in sequential order in the calling thread.

Note: Complexity: Exactly $\max(N-1, 0)$ comparisons, where $N = \text{std::distance}(\text{first}, \text{last})$.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the the left argument is less than the right element. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *min_element* algorithm returns a *hpx::traits::range_iterator<Rng>::type* otherwise. The *min_element* algorithm returns the iterator to the smallest element in the range [first, last). If several elements in the range are equivalent to the smallest element, returns the iterator to the first such element. Returns last if the range is empty.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename F = hpx::parallel::detail::less,
typename Proj = hpx::identity>
```

```
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::ranges::min_element(ExPolicy
                                                                 &&policy,
                                                                 FwdIter first,
                                                                 Sent last, F &&f
                                                                 = F(), Proj
                                                                 &&proj = Proj())
```

Finds the smallest element in the range [first, last) using the given comparison function *f*.

The comparisons in the parallel *min_element* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *min_element* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Exactly $\max(N-1, 0)$ comparisons, where $N = \text{std::distance}(\text{first}, \text{last})$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for FwdIter.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *min_element* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the the left argument is less than the right element. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *min_element* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *min_element*

algorithm returns the iterator to the smallest element in the range [first, last). If several elements in the range are equivalent to the smallest element, returns the iterator to the first such element. Returns last if the range is empty.

```
template<typename ExPolicy, typename Rng, typename F = hpx::parallel::detail::less, typename Proj =  

hpx::identity>  

hpx::parallel::util::detail::algorithm_result_t<ExPolicy, std::ranges::iterator_t<Rng>> hpx::ranges::min_element(ExPolicy  

&&pol-  

icy,  

Rng  

&&rng,  

F  

&&f  

=  

F(),  

Proj  

&&proj  

=  

Proj())
```

Finds the smallest element in the range [first, last) using the given comparison function *f*.

The comparisons in the parallel *min_element* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *min_element* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Exactly $\max(N-1, 0)$ comparisons, where $N = \text{std::distance}(\text{first}, \text{last})$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *min_element* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the the left argument is less than the right element. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *min_element* algorithm returns a `hpx::future<hpx::traits::range_iterator<Rng>::type>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *min_element* algorithm returns the iterator to the smallest element in the range `[first, last)`. If several elements in the range are equivalent to the smallest element, returns the iterator to the first such element. Returns `last` if the range is empty.

hpx::ranges::max_element

Defined in header `hpx/algorithm.hpp`⁷²³.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::min_element`
- `hpx::ranges::minmax_element`

```
template<typename FwdIter, typename Sent, typename F = hpx::parallel::detail::less, typename Proj = hpx::identity>
```

```
FwdIter hpx::ranges::max_element(FwdIter first, Sent last, F &&f = F(), Proj &&proj = Proj())
```

Finds the greatest element in the range `[first, last)` using the given comparison function *f*.

The comparisons in the parallel *max_element* algorithm execute in sequential order in the calling thread.

Note: Complexity: Exactly $\max(N-1, 0)$ comparisons, where $N = \text{std::distance}(\text{first}, \text{last})$.

Template Parameters

- **FwdIter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **F** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

⁷²³ <http://github.com/TheHPXProject/hpx/blob/d402eed7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **f** – The binary predicate which returns true if the This argument is optional and defaults to `std::less`. the left argument is less than the right element. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *max_element* algorithm returns a *FwdIter*. The *max_element* algorithm returns the iterator to the smallest element in the range [first, last). If several elements in the range are equivalent to the smallest element, returns the iterator to the first such element. Returns last if the range is empty.

```
template<typename Rng, typename F = hpx::parallel::detail::less, typename Proj = hpx::identity>
std::ranges::iterator_t<Rng> hpx::ranges::max_element(Rng &&rng, F &&f = F(), Proj &&proj = Proj())
```

Finds the greatest element in the range [first, last) using the given comparison function *f*.

The comparisons in the parallel *max_element* algorithm execute in sequential order in the calling thread.

Note: Complexity: Exactly $\max(N-1, 0)$ comparisons, where $N = \text{std::distance}(\text{first}, \text{last})$.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the This argument is optional and defaults to `std::less`. the left argument is less than the right element. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *max_element* algorithm returns a *hpx::traits::range_iterator<Rng>::type* otherwise. The *max_element* algorithm returns the iterator to the smallest element in the range [first, last). If

several elements in the range are equivalent to the smallest element, returns the iterator to the first such element. Returns last if the range is empty.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename F = hpx::parallel::detail::less,
typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::ranges::max_element(ExPolicy
                                                                                      &&policy,
                                                                                      FwdIter first,
                                                                                      Sent last, F &&f
                                                                                      = F(), Proj
                                                                                      &&proj = Proj())
```

Finds the greatest element in the range [*first*, *last*) using the given comparison function *f*.

The comparisons in the parallel *max_element* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *max_element* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Exactly $\max(N-1, 0)$ comparisons, where $N = \text{std::distance}(\text{first}, \text{last})$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *max_element* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the This argument is optional and defaults to *std::less*. the left argument is less than the right element. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have *const* &, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *max_element* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *max_element* algorithm returns the iterator to the smallest element in the range [first, last). If several elements in the range are equivalent to the smallest element, returns the iterator to the first such element. Returns last if the range is empty.

```
template<typename ExPolicy, typename Rng, typename F = hpx::parallel::detail::less, typename Proj =
hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, std::ranges::iterator_t<Rng>> hpx::ranges::max_element(ExPolicy
&&pol-
icy,
Rng
&&rng,
F
&&f
=
F(),
Proj
&&proj
=
Proj())
```

Finds the greatest element in the range [first, last) using the given comparison function *f*.

The comparisons in the parallel *max_element* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *max_element* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Exactly $\max(N-1, 0)$ comparisons, where $N = \text{std::distance}(\text{first}, \text{last})$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *max_element* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.

- **f** – The binary predicate which returns true if the This argument is optional and defaults to `std::less`. the left argument is less than the right element. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *max_element* algorithm returns a `hpx::future<hpx::traits::range_iterator<Rng>::type>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *max_element* algorithm returns the iterator to the smallest element in the range [first, last). If several elements in the range are equivalent to the smallest element, returns the iterator to the first such element. Returns last if the range is empty.

hpx::ranges::minmax_element

Defined in header `hpx/algorithm.hpp`⁷²⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::min_element`
- `hpx::ranges::max_element`

```
template<typename FwdIter, typename Sent, typename F = hpx::parallel::detail::less, typename Proj =  
hpx::identity>  
minmax_element_result<FwdIter> hpx::ranges::minmax_element(FwdIter first, Sent last, F &&f = F(), Proj  
                    &&proj = Proj())
```

Finds the greatest element in the range [first, last) using the given comparison function *f*.

The assignments in the parallel *minmax_element* algorithm execute in sequential order in the calling thread.

Note: Complexity: At most $\max(\text{floor}(3/2*(N-1)), 0)$ applications of the predicate, where $N = \text{std::distance}(\text{first}, \text{last})$.

Template Parameters

- **FwdIter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **F** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

⁷²⁴ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the the left argument is less than the right element. This argument is optional and defaults to `std::less`. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *minmax_element* algorithm returns a *minmax_element_result*<*FwdIter*, *FwdIter*>. The *minmax_element* algorithm returns a *min_max_result* consisting of an iterator to the smallest element as the min element and an iterator to the greatest element as the max element. Returns *minmax_element_result*{first, first} if the range is empty. If several elements are equivalent to the smallest element, the iterator to the first such element is returned. If several elements are equivalent to the largest element, the iterator to the last such element is returned.

```
template<typename Rng, typename F = hpx::parallel::detail::less, typename Proj = hpx::identity>
minmax_element_result<std::ranges::iterator_t<Rng>> hpx::ranges::minmax_element(Rng &&rng, F &&f =
                                                    F(), Proj &&proj =
                                                    Proj())
```

Finds the greatest element in the range [first, last) using the given comparison function *f*.

The assignments in the parallel *minmax_element* algorithm execute in sequential order in the calling thread.

Note: Complexity: At most $\max(\text{floor}(3/2 * (N-1)), 0)$ applications of the predicate, where $N = \text{std::distance}(\text{first}, \text{last})$.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the the left argument is less than the right element. This argument is optional and defaults to `std::less`. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *minmax_element* algorithm returns a *minmax_element_result*<*hpx::traits::range_iterator*<*Rng*>::type, *hpx::traits::range_iterator*<*Rng*>::type>. The *minmax_element* algorithm returns a *min_max_result* consisting of an range iterator to the smallest element as the min element and an range iterator to the greatest element as the max element. If several elements are equivalent to the smallest element, the iterator to the first such element is returned. If several elements are equivalent to the largest element, the iterator to the last such element is returned.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename F = hpx::parallel::detail::less,  
typename Proj = hpx::identity>
```

```
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, minmax_element_result<FwdIter>> hpx::ranges::minmax_element(ExPo  
&&p  
icy,  
FwdI  
first,  
Sent  
last,  
F  
&&f  
=  
F(),  
Proj  
&&p  
=  
Proj(
```

Finds the greatest element in the range [first, last) using the given comparison function *f*.

The comparisons in the parallel *minmax_element* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *minmax_element* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most $\max(\text{floor}(3/2 * (N-1)), 0)$ applications of the predicate, where $N = \text{std::distance}(\text{first}, \text{last})$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.

- **FwdIter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for FwdIter.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *minmax_element* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the the left argument is less than the right element. This argument is optional and defaults to `std::less`. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *minmax_element* algorithm returns a *minmax_element_result*<*FwdIter*, *FwdIter*>. The *minmax_element* algorithm returns a *min_max_result* consisting of an iterator to the smallest element as the min element and an iterator to the greatest element as the max element. Returns *minmax_element_result*{*first*, *first*} if the range is empty. If several elements are equivalent to the smallest element, the iterator to the first such element is returned. If several elements are equivalent to the largest element, the iterator to the last such element is returned.

```
template<typename ExPolicy, typename Rng, typename F = hpx::parallel::detail::less, typename Proj = hpx::identity>
```

```
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, minmax_element_result<std::ranges::iterator_t<Rng>>> hpx::ranges::minm
```

Finds the greatest element in the range [*first*, *last*) using the given comparison function *f*.

The comparisons in the parallel *minmax_element* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *minmax_element* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most $\max(\text{floor}(3/2*(N-1)), 0)$ applications of the predicate, where $N = \text{std::distance}(\text{first}, \text{last})$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *minmax_element* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – The binary predicate which returns true if the the left argument is less than the right element. This argument is optional and defaults to *std::less*. The signature of the predicate function should be equivalent to the following:

```
bool pred(const Type1 &a, const Type1 &b);
```

The signature does not need to have *const &*, but the function must not modify the objects passed to it. The type *Type1* must be such that objects of type *FwdIter* can be dereferenced and then implicitly converted to *Type1*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *minmax_element* algorithm returns a *minmax_element_result*<*hpx::traits::range_iterator*<*Rng*>::type, *hpx::traits::range_iterator*<*Rng*>::type>. The *minmax_element* algorithm returns a *min_max_result* consisting of an range iterator to the smallest element as the min element and an range iterator to the greatest element as the max element. If several elements are equivalent to the smallest element, the iterator to the first such element is returned. If several elements are equivalent to the largest element, the iterator to the last such element is returned.

hpx::ranges::destroy

Defined in header [hpx/algorithm.hpp](#)⁷²⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::destroy_n`

```
template<typename ExPolicy, typename Rng>
hpx::parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng>> hpx::ranges::destroy(ExPolicy
&&policy,
Rng
&&rng)
```

Destroys objects of type `typename iterator_traits<ForwardIt>::value_type` in the range `[first, last)`.

The operations in the parallel *destroy* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The operations in the parallel *destroy* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* operations.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.

Returns

The *destroy* algorithm returns a `hpx::future<void>`, if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *void* otherwise.

```
template<typename ExPolicy, typename Iter, typename Sent>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, Iter> hpx::ranges::destroy(ExPolicy &&policy, Iter
first, Sent last)
```

Destroys objects of type `typename iterator_traits<ForwardIt>::value_type` in the range `[first, last)`.

The operations in the parallel *destroy* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

⁷²⁵ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

The operations in the parallel *destroy* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* operations.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter`.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

Returns

The *destroy* algorithm returns a `hpx::future<void>`, if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *void* otherwise.

```
template<typename Rng>
hpx::traits::range_iterator<Rng>::type hpx::ranges::destroy(Rng &&rng)
    Destroys objects of type typename iterator_traits<ForwardIt>::value_type in the range [first, last).
```

Note: Complexity: Performs exactly *last - first* operations.

Template Parameters

Rng – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

Parameters

rng – Refers to the sequence of elements the algorithm will be applied to.

Returns

The *destroy* algorithm returns *void*.

```
template<typename Iter, typename Sent>
Iter hpx::ranges::destroy(Iter first, Sent last)
    Destroys objects of type typename iterator_traits<ForwardIt>::value_type in the range [first, last).
```

Note: Complexity: Performs exactly *last - first* operations.

Template Parameters

- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter`.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

Returns

The *destroy* algorithm returns *void*.

`hpx::ranges::destroy_n`

Defined in header `hpx/algorithm.hpp`⁷²⁶.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::destroy`

```
template<typename ExPolicy, typename FwdIter, typename Size>
hpx::parallel::util::detail::algorithm_result<ExPolicy, FwdIter>::type hpx::ranges::destroy_n(ExPolicy
                                                                                               &&policy,
                                                                                               FwdIter first,
                                                                                               Size count)
```

Destroys objects of type `typename iterator_traits<ForwardIt>::value_type` in the range `[first, first + count)`.

The operations in the parallel *destroy_n* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The operations in the parallel *destroy_n* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *count* operations, if *count* > 0, no assignments otherwise.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply this algorithm to.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

⁷²⁶ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.

Returns

The *destroy_n* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *destroy_n* algorithm returns the iterator to the element in the source range, one past the last element constructed.

```
template<typename FwdIter, typename Size>  
FwdIter hpx::ranges::destroy_n(FwdIter first, Size count)
```

Destroys objects of type *typename iterator_traits<ForwardIt>::value_type* in the range *[first, first + count)*.

Note: Complexity: Performs exactly *count* operations, if *count* > 0, no assignments otherwise.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply this algorithm to.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.

Returns

The *destroy_n* algorithm returns the iterator to the element in the source range, one past the last element constructed.

hpx::ranges::make_heap

Defined in header [hpx/algorithm.hpp](#)⁷²⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename ExPolicy, typename Iter, typename Sent, typename Comp, typename Proj = hpx::identity>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, Iter> hpx::ranges::make_heap(ExPolicy &&policy, Iter  
first, Sent last, Comp  
&&comp, Proj &&proj  
= Proj{ })
```

Constructs a *max heap* in the range *[first, last)*.

The predicate operations in the parallel *make_heap* algorithm invoked with an execution policy object of type *sequential_execution_policy* executes in sequential order in the calling thread.

⁷²⁷ <http://github.com/TheHPXProject/hpx/blob/d402cead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

The comparison operations in the parallel *make_heap* algorithm invoked with an execution policy object of type *parallel_execution_policy* or *parallel_task_execution_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: at most $(3*N)$ comparisons where $N = \text{distance}(\text{first}, \text{last})$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second. The signature of the function should be equivalent to

```
bool comp(const Type &a, const Type &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *RndIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

Returns

The *make_heap* algorithm returns a *hpx::future<Iter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *Iter* otherwise. It returns *last*.

```
template<typename ExPolicy, typename Rng, typename Comp, typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng>> hpx::ranges::make_heap(ExPolicy
&&pol-
icy,
Rng
&&rng,
Comp
&&comp,
Proj
&&proj
=
Proj{ })
```

Constructs a *max heap* in the range [first, last).

The predicate operations in the parallel *make_heap* algorithm invoked with an execution policy object of type *sequential_execution_policy* executes in sequential order in the calling thread.

The comparison operations in the parallel *make_heap* algorithm invoked with an execution policy object of type *parallel_execution_policy* or *parallel_task_execution_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: at most (3*N) comparisons where $N = \text{distance}(\text{first}, \text{last})$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **comp** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second. The signature of the function should be equivalent to

```
bool comp(const Type &a, const Type &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *RndIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

Returns

The *make_heap* algorithm returns a *hpx::future<Iter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *Iter* otherwise. It returns *last*.

```
template<typename ExPolicy, typename Iter, typename Sent, typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, Iter> hpx::ranges::make_heap(ExPolicy &&policy, Iter
                                                                                       first, Sent last, Proj
                                                                                       &&proj = Proj{ })
```

Constructs a *max heap* in the range [first, last).

The predicate operations in the parallel *make_heap* algorithm invoked with an execution policy object of type *sequential_execution_policy* executes in sequential order in the calling thread.

The comparison operations in the parallel *make_heap* algorithm invoked with an execution policy object of type *parallel_execution_policy* or *parallel_task_execution_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: at most $(3*N)$ comparisons where $N = \text{distance}(\text{first}, \text{last})$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

Returns

The *make_heap* algorithm returns a *hpx::future<Iter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *Iter* otherwise. It returns *last*.

```
template<typename ExPolicy, typename Rng, typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng>> hpx::ranges::make_heap(ExPolicy
                                                                                                     &&pol-
                                                                                                     icy,
                                                                                                     Rng
                                                                                                     &&rng,
                                                                                                     Proj
                                                                                                     &&proj
                                                                                                     =
                                                                                                     Proj{ })
```

Constructs a *max heap* in the range $[\text{first}, \text{last})$.

The predicate operations in the parallel *make_heap* algorithm invoked with an execution policy object of type *sequential_execution_policy* executes in sequential order in the calling thread.

The comparison operations in the parallel *make_heap* algorithm invoked with an execution policy object of type *parallel_execution_policy* or *parallel_task_execution_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: at most $(3*N)$ comparisons where $N = \text{distance}(\text{first}, \text{last})$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

Returns

The *make_heap* algorithm returns a `hpx::future<Iter>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *Iter* otherwise. It returns *last*.

```
template<typename Iter, typename Sent, typename Comp, typename Proj = hpx::identity>  
Iter hpx::ranges::make_heap(Iter first, Sent last, Comp &&comp, Proj &&proj = Proj{ })
```

Constructs a *max heap* in the range `[first, last)`.

Note: Complexity: at most $(3*N)$ comparisons where $N = \text{distance}(\text{first}, \text{last})$.

Template Parameters

- **Iter** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for `Iter1`.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second. The signature of the function should be equivalent to

```
bool comp(const Type &a, const Type &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *RndIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

Returns

The *make_heap* algorithm returns *Iter*. It returns *last*.

```
template<typename Rng, typename Comp, typename Proj = hpx::identity>
std::ranges::iterator_t<Rng> hpx::ranges::make_heap(Rng &&rng, Comp &&comp, Proj &&proj = Proj{ })
```

Constructs a *max heap* in the range [first, last).

Note: Complexity: at most (3*N) comparisons where $N = \text{distance}(\text{first}, \text{last})$.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **comp** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second. The signature of the function should be equivalent to

```
bool comp(const Type &a, const Type &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *RndIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

Returns

The *make_heap* algorithm returns *Iter*. It returns *last*.

```
template<typename Iter, typename Sent, typename Proj = hpx::identity>
Iter hpx::ranges::make_heap(Iter first, Sent last, Proj &&proj = Proj{ })
```

Constructs a *max heap* in the range [first, last).

Note: Complexity: at most (3*N) comparisons where $N = \text{distance}(\text{first}, \text{last})$.

Template Parameters

- **Iter** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for *Iter1*.

- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

Returns

The *make_heap* algorithm returns *Iter*. It returns *last*.

```
template<typename Rng, typename Proj = hpx::identity>
std::ranges::iterator_t<Rng> hpx::ranges::make_heap(Rng &&rng, Proj &&proj = Proj{ })
```

Constructs a *max heap* in the range [first, last).

Note: Complexity: at most (3*N) comparisons where *N* = distance(first, last).

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

Returns

The *make_heap* algorithm returns *Iter*. It returns *last*.

hpx::ranges::uninitialized_fill

Defined in header [hpx/algorithm.hpp](#)⁷²⁸.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::ranges::uninitialized_fill_n*

```
template<typename FwdIter, typename Sent, typename T>
FwdIter hpx::ranges::uninitialized_fill(FwdIter first, Sent last, T const &value)
```

Copies the given *value* to an uninitialized memory area, defined by the range [first, last). If an exception is thrown during the initialization, the function has no effects.

The assignments in the ranges *uninitialized_fill* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

⁷²⁸ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fdbc0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Note: Complexity: Linear in the distance between *first* and *last*

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for FwdIter.
- **T** – The type of the value to be assigned (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **value** – The value to be assigned.

Returns

The *uninitialized_fill* algorithm returns a *FwdIter*. The *uninitialized_fill* algorithm returns the output iterator to the element in the range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename T>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::ranges::uninitialized_fill(ExPolicy
                                                                                               &&pol-
                                                                                               icy,
                                                                                               FwdIter
                                                                                               first,
                                                                                               Sent
                                                                                               last, T
                                                                                               const
                                                                                               &value)
```

Copies the given *value* to an uninitialized memory area, defined by the range [first, last). If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized_fill* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized_fill* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Linear in the distance between *first* and *last*

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for FwdIter.
- **T** – The type of the value to be assigned (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **value** – The value to be assigned.

Returns

The *uninitialized_fill* algorithm returns a *FwdIter*. The *uninitialized_fill* algorithm returns the output iterator to the element in the range, one past the last element copied.

```
template<typename Rng, typename T>  
hpx::traits::range_traits<Rng>::iterator_type hpx::ranges::uninitialized_fill(Rng &&rng, T const &value)
```

Copies the given *value* to an uninitialized memory area, defined by the range [first, last). If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized_fill* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

Note: Complexity: Linear in the distance between *first* and *last*

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **T** – The type of the value to be assigned (deduced).

Parameters

- **rng** – Refers to the range to which the value will be filled
- **value** – The value to be assigned.

Returns

The *uninitialized_fill* algorithm returns a *hpx::traits::range_traits<Rng>::iterator_type*. The *uninitialized_fill* algorithm returns the output iterator to the element in the range, one past the last element copied.

```
template<typename ExPolicy, typename Rng, typename T>
```

```
parallel::util::detail::algorithm_result<ExPolicy, typename hpx::traits::range_traits<Rng1>::iterator_type>::type hpx::ranges::unin
```

Copies the given *value* to an uninitialized memory area, defined by the range [first, last). If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized_fill* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized_fill* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Linear in the distance between *first* and *last*

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **T** – The type of the value to be assigned (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the range to which the value will be filled
- **value** – The value to be assigned.

Returns

The *uninitialized_fill* algorithm returns a `hpx::future<typename hpx::traits::range_traits<Rng>::iterator_type>`, if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `typename hpx::traits::range_traits<Rng>::iterator_type` otherwise. The *uninitialized_fill* algorithm returns the iterator to one past the last element filled in the range.

hpx::ranges::uninitialized_fill_n

Defined in header [hpx/algorithm.hpp](#)⁷²⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::ranges::uninitialized_fill*

template<typename **FwdIter**, typename **Size**, typename **T**>

FwdIter **hpx::ranges::uninitialized_fill_n**(*FwdIter* first, *Size* count, *T* const &value)

Copies the given *value* value to the first count elements in an uninitialized memory area beginning at first. If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized_fill_n* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *count* assignments, if count > 0, no assignments otherwise.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **T** – The type of the value to be assigned (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **value** – The value to be assigned.

Returns

The *uninitialized_fill_n* algorithm returns a returns *FwdIter*. The *uninitialized_fill_n* algorithm returns the output iterator to the element in the range, one past the last element copied.

template<typename **ExPolicy**, typename **FwdIter**, typename **Size**, typename **T**>

hpx::parallel::util::detail::algorithm_result_t<*ExPolicy*, *FwdIter*> **hpx::ranges::uninitialized_fill_n**(*ExPolicy* &&pol-
icy,
FwdIter first,
Size count,
T const
&value)

Copies the given *value* value to the first count elements in an uninitialized memory area beginning at first. If an exception is thrown during the initialization, the function has no effects.

⁷²⁹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

The assignments in the parallel *uninitialized_fill_n* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized_fill_n* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *count* assignments, if *count* > 0, no assignments otherwise.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **T** – The type of the value to be assigned (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **value** – The value to be assigned.

Returns

The *uninitialized_fill_n* algorithm returns a *hpx::future<FwdIter>*, if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *uninitialized_fill_n* algorithm returns the output iterator to the element in the range, one past the last element copied.

hpx::ranges::is_heap

Defined in header [hpx/algorithm.hpp](#)⁷³⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::ranges::is_heap_until*

```
template<typename ExPolicy, typename Rng, typename Comp = hpx::parallel::detail::less, typename Proj =
hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx::ranges::is_heap(ExPolicy &&policy, Rng
&&rng, Comp &&comp =
Comp(), Proj &&proj =
Proj())
```

⁷³⁰ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Returns whether the range is max heap. That is, true if the range is max heap, false otherwise. The function uses the given comparison function object *comp* (defaults to using operator<()).

comp has to induce a strict weak ordering on the values.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs at most N applications of the comparison *comp*, at most 2 * N applications of the projection *proj*, where N = last - first.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **comp** – *comp* is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that *comp* will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* is invoked.

Returns

The *is_heap* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *is_heap* algorithm returns whether the range is max heap. That is, true if the range is max heap, false otherwise.

```
template<typename ExPolicy, typename Iter, typename Sent, typename Comp = hpx::parallel::detail::less,
        typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx::ranges::is_heap(ExPolicy &&policy, Iter
                                                                                   first, Sent last, Comp
                                                                                   &&comp = Comp(), Proj
                                                                                   &&proj = Proj())
```

Returns whether the range is max heap. That is, true if the range is max heap, false otherwise. The function uses the given comparison function object *comp* (defaults to using operator<()).

comp has to induce a strict weak ordering on the values.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs at most N applications of the comparison *comp*, at most $2 * N$ applications of the projection *proj*, where $N = \text{last} - \text{first}$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – *comp* is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to *bool*, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that *comp* will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *is_heap* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *is_heap* algorithm returns whether the range is max heap. That is, true if the range is max heap, false otherwise.

```
template<typename Rng, typename Comp = hpx::parallel::detail::less, typename Proj = hpx::identity>
bool hpx::ranges::is_heap(Rng &&rng, Comp &&comp = Comp(), Proj &&proj = Proj())
```

Returns whether the range is max heap. That is, true if the range is max heap, false otherwise. The function uses the given comparison function object *comp* (defaults to using operator<()).

comp has to induce a strict weak ordering on the values.

Note: Complexity: Performs at most N applications of the comparison *comp*, at most $2 * N$ applications of the projection *proj*, where $N = \text{last} - \text{first}$.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **comp** – *comp* is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to *bool*, yields *true* if the first argument of the call is less than the second, and *false* otherwise. It is assumed that *comp* will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* is invoked.

Returns

The *is_heap* algorithm returns *bool*. The *is_heap* algorithm returns whether the range is max heap. That is, *true* if the range is max heap, *false* otherwise.

```
template<typename Iter, typename Sent, typename Comp = hpx::parallel::detail::less, typename Proj = hpx::identity>
```

```
bool hpx::ranges::is_heap(Iter first, Sent last, Comp &&comp = Comp(), Proj &&proj = Proj())
```

Returns whether the range is max heap. That is, *true* if the range is max heap, *false* otherwise. The function uses the given comparison function object *comp* (defaults to using *operator<()*).

comp has to induce a strict weak ordering on the values.

Note: Complexity: Performs at most *N* applications of the comparison *comp*, at most *2 * N* applications of the projection *proj*, where *N* = *last* - *first*.

Template Parameters

- **Iter** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of a sentinel for *Iter*1.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – *comp* is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to *bool*, yields *true* if the first argument of the call is less than the second, and *false* otherwise. It is assumed that *comp* will not apply any non-constant function through the dereferenced iterator.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *is_heap* algorithm returns *bool*. The *is_heap* algorithm returns whether the range is max heap. That is, true if the range is max heap, false otherwise.

hpx::ranges::is_heap_until

Defined in header `hpx/algorithm.hpp`⁷³¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::is_heap`

```
template<typename ExPolicy, typename Rng, typename Comp = hpx::parallel::detail::less, typename Proj =
hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng>> hpx::ranges::is_heap_until(ExPolicy
&&pol-
icy,
Rng
&&rng,
Comp
&&comp
=
Comp(),
Proj
&&proj
=
Proj())
```

Returns the upper bound of the largest range beginning at *first* which is a max heap. That is, the last iterator *it* for which range [first, it) is a max heap. The function uses the given comparison function object *comp* (defaults to using operator<()).

comp has to induce a strict weak ordering on the values.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs at most N applications of the comparison *comp*, at most 2 * N applications of the projection *proj*, where N = last - first.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.

⁷³¹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **comp** – `comp` is a callable object. The return value of the INVOKE operation applied to an object of type `Comp`, when contextually converted to `bool`, yields `true` if the first argument of the call is less than the second, and `false` otherwise. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate `is` is invoked.

Returns

The `is_heap_until` algorithm returns a `hpx::future<RandIter>` if the execution policy is of type `sequenced_task_policy` or `parallel_task_policy` and returns `RandIter` otherwise. The `is_heap_until` algorithm returns the upper bound of the largest range beginning at `first` which is a max heap. That is, the last iterator `it` for which range `[first, it)` is a max heap.

```
template<typename ExPolicy, typename Iter, typename Sent, typename Comp = hpx::parallel::detail::less,
        typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, Iter> hpx::ranges::is_heap_until(ExPolicy
                                                                 &&policy, Iter
                                                                 first, Sent last,
                                                                 Comp &&comp =
                                                                 Comp(), Proj
                                                                 &&proj = Proj())
```

Returns the upper bound of the largest range beginning at `first` which is a max heap. That is, the last iterator `it` for which range `[first, it)` is a max heap. The function uses the given comparison function object `comp` (defaults to using `operator<()`).

`comp` has to induce a strict weak ordering on the values.

The application of function objects in parallel algorithm invoked with an execution policy object of type `sequenced_policy` execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type `parallel_policy` or `parallel_task_policy` are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs at most N applications of the comparison `comp`, at most $2 * N$ applications of the projection `proj`, where $N = \text{last} - \text{first}$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.

- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – *comp* is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that comp will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* is invoked.

Returns

The *is_heap_until* algorithm returns a *hpx::future<RandIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *RandIter* otherwise. The *is_heap_until* algorithm returns the upper bound of the largest range beginning at first which is a max heap. That is, the last iterator *it* for which range [first, it) is a max heap.

```
template<typename Rng, typename Comp = hpx::parallel::detail::less, typename Proj = hpx::identity>
std::ranges::iterator_t<Rng> hpx::ranges::is_heap_until(Rng &&rng, Comp &&comp = Comp(), Proj
&&proj = Proj())
```

Returns the upper bound of the largest range beginning at *first* which is a max heap. That is, the last iterator *it* for which range [first, it) is a max heap. The function uses the given comparison function object *comp* (defaults to using operator<()).

comp has to induce a strict weak ordering on the values.

Note: Complexity: Performs at most N applications of the comparison *comp*, at most 2 * N applications of the projection *proj*, where N = last - first.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **comp** – *comp* is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that comp will not apply any non-constant function through the dereferenced iterator.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *is_heap_until* algorithm returns *RandIter*. The *is_heap_until* algorithm returns the upper bound of the largest range beginning at *first* which is a max heap. That is, the last iterator *it* for which range [*first*, *it*) is a max heap.

```
template<typename Iter, typename Sent, typename Comp = hpx::parallel::detail::less, typename Proj = hpx::identity>
```

```
Iter hpx::ranges::is_heap_until(Iter first, Sent last, Comp &&comp = Comp(), Proj &&proj = Proj())
```

Returns the upper bound of the largest range beginning at *first* which is a max heap. That is, the last iterator *it* for which range [*first*, *it*) is a max heap. The function uses the given comparison function object *comp* (defaults to using operator<()).

comp has to induce a strict weak ordering on the values.

Note: Complexity: Performs at most N applications of the comparison *comp*, at most 2 * N applications of the projection *proj*, where N = last - first.

Template Parameters

- **Iter** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **comp** – *comp* is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that *comp* will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *is_heap_until* algorithm returns *RandIter*. The *is_heap_until* algorithm returns the upper bound of the largest range beginning at *first* which is a max heap. That is, the last iterator *it* for which range [*first*, *it*) is a max heap.

hpx::ranges::reverse

Defined in header `hpx/algorithm.hpp`⁷³².

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::reverse_copy`

template<typename **Iter**, typename **Sent**>

Iter **hpx::ranges::reverse**(*Iter* first, *Sent* sent)

Reverses the order of the elements in the range [first, last). Behaves as if applying `std::iter_swap` to every pair of iterators `first+i`, `(last-i) - 1` for each non-negative `i < (last-first)/2`.

The assignments in the parallel *reverse* algorithm execute in sequential order in the calling thread.

Note: Complexity: Linear in the distance between *first* and *last*.

Template Parameters

- **Iter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *Iter*.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.

Returns

The *reverse* algorithm returns a *Iter*. It returns *last*.

template<typename **Rng**>

`std::ranges::iterator_t<Rng>` **hpx::ranges::reverse**(*Rng* &&rng)

Uses *rng* as the source range, as if using `util::begin(rng)` as *first* and `ranges::end(rng)` as *last*. Reverses the order of the elements in the range [first, last). Behaves as if applying `std::iter_swap` to every pair of iterators `first+i`, `(last-i) - 1` for each non-negative `i < (last-first)/2`.

The assignments in the parallel *reverse* algorithm execute in sequential order in the calling thread.

Note: Complexity: Linear in the distance between *first* and *last*.

Template Parameters

Rng – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a bidirectional iterator.

⁷³² <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Parameters

rng – Refers to the sequence of elements the algorithm will be applied to.

Returns

The *reverse* algorithm returns a *hpx::traits::range_iterator<Rng>::type*. It returns *last*.

```
template<typename ExPolicy, typename Iter, typename Sent>
```

```
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, Iter> hpx::ranges::reverse(ExPolicy &&policy, Iter  
first, Sent sent)
```

Reverses the order of the elements in the range [first, last). Behaves as if applying `std::iter_swap` to every pair of iterators `first+i`, `(last-i) - 1` for each non-negative `i < (last-first)/2`.

The assignments in the parallel *reverse* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *reverse* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Linear in the distance between *first* and *last*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *Iter*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.

Returns

The *reverse* algorithm returns a *hpx::future<Iter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *Iter* otherwise. It returns *last*.

```
template<typename ExPolicy, typename Rng>
```

```
parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng>> hpx::ranges::reverse(ExPolicy  
                                                    &&pol-  
                                                    icy, Rng  
                                                    &&rng)
```

Uses *rng* as the source range, as if using *util::begin(rng)* as *first* and *ranges::end(rng)* as *last*. Reverses the order of the elements in the range [first, last). Behaves as if applying `std::iter_swap` to every pair of iterators `first+i`, `(last-i) - 1` for each non-negative `i < (last-first)/2`.

The assignments in the parallel *reverse* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *reverse* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Linear in the distance between *first* and *last*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a bidirectional iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.

Returns

The *reverse* algorithm returns a `hpx::future<std::ranges::iterator_t<Rng>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `hpx::future<std::ranges::iterator_t<Rng>>` otherwise. It returns *last*.

hpx::ranges::reverse_copy

Defined in header `hpx/algorithm.hpp`⁷³³.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::reverse`

```
template<typename Iter, typename Sent, typename OutIter>
```

```
reverse_copy_result<Iter, OutIter> hpx::ranges::reverse_copy(Iter first, Sent last, OutIter result)
```

Copies the elements from the range `[first, last)` to another range beginning at `result` in such a way that the elements in the new range are in reverse order. Behaves as if by executing the assignment `*(result + (last - first) - 1 - i) = *(first + i)` once for each non-negative `i < (last - first)`. If the source and destination ranges (that is, `[first, last)` and `[result, result+(last-first))` respectively) overlap, the behavior is undefined.

The assignments in the parallel *reverse_copy* algorithm execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

⁷³³ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **Iter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for Iter.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **result** – Refers to the begin of the destination range.

Returns

The *reverse_copy* algorithm returns a *reverse_copy_result<Iter, OutIter>*. The *reverse_copy* algorithm returns the pair of the input iterator forwarded to the first element after the last in the input sequence and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng, typename OutIter>
reverse_copy_result<std::ranges::iterator_t<Rng>, OutIter> hpx::ranges::reverse_copy(Rng &&rng, OutIter
                                                                    result)
```

Uses *rng* as the source range, as if using *util::begin(rng)* as *first* and *ranges::end(rng)* as *last*. Copies the elements from the range $[first, last)$ to another range beginning at *result* in such a way that the elements in the new range are in reverse order. Behaves as if by executing the assignment $*(result + (last - first) - 1 - i) = *(first + i)$ once for each non-negative $i < (last - first)$. If the source and destination ranges (that is, $[first, last)$ and $[result, result + (last - first))$ respectively) overlap, the behavior is undefined.

The assignments in the parallel *reverse_copy* algorithm execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a bidirectional iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **result** – Refers to the begin of the destination range.

Returns

The *reverse_copy* algorithm returns a *ranges::reverse_copy_result<std::ranges::iterator_t<Rng>, OutIter>*. The *reverse_copy* algorithm returns an object equal to $\{last, result + N\}$ where $N = last - first$

```
template<typename ExPolicy, typename Iter, typename Sent, typename FwdIter>
```

parallel::util::detail::algorithm_result<ExPolicy, reverse_copy_result<Iter, FwdIter>>::type *hpx::ranges::reverse_copy*(*ExPolicy* &&*policy*, *Iter* *first*, *Sent* *last*, *FwdIter* *re-result*)

Copies the elements from the range [*first*, *last*) to another range beginning at *result* in such a way that the elements in the new range are in reverse order. Behaves as if by executing the assignment $*(result + (last - first) - 1 - i) = *(first + i)$ once for each non-negative $i < (last - first)$. If the source and destination ranges (that is, [*first*, *last*) and [*result*, *result*+(*last*-*first*)) respectively) overlap, the behavior is undefined.

The assignments in the parallel *reverse_copy* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *reverse_copy* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *Iter*.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **result** – Refers to the begin of the destination range.

Returns

The *reverse_copy* algorithm returns a *hpx::future<reverse_copy_result<Iter, FwdIter> >* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *reverse_copy_result<Iter, FwdIter>* otherwise. The *reverse_copy* algorithm returns the pair of the input iterator forwarded to the first element after the last in the input sequence and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng, typename OutIter>  
parallel::util::detail::algorithm_result<ExPolicy, reverse_copy_result<std::ranges::iterator_t<Rng>, OutIter>>::type hpx::ranges::r
```

Uses *rng* as the source range, as if using *util::begin(rng)* as *first* and *ranges::end(rng)* as *last*. Copies the elements from the range [first, last) to another range beginning at *result* in such a way that the elements in the new range are in reverse order. Behaves as if by executing the assignment $*(result + (last - first) - 1 - i) = *(first + i)$ once for each non-negative $i < (last - first)$. If the source and destination ranges (that is, [first, last) and [result, result+(last-first)) respectively) overlap, the behavior is undefined.

The assignments in the parallel *reverse_copy* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *reverse_copy* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a bidirectional iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **result** – Refers to the begin of the destination range.

Returns

The *reverse_copy* algorithm returns a *hpx::future*<*ranges::reverse_copy_result*<*std::ranges::iterator_t*<*Rng*>, *OutIter*>> if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *ranges::reverse_copy_result*<*std::ranges::iterator_t*<*Rng*>, *OutIter*> otherwise. The *reverse_copy* algorithm returns an object equal to {last, result + N} where $N = last - first$

hpx::ranges::count

Defined in header `hpx/algorithm.hpp`⁷³⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::count_if`

```
template<typename ExPolicy, typename Rng, typename Proj = hpx::identity, typename T = typename
hpx::parallel::traits::projected<std::ranges::iterator_t<Rng>, Proj>::value_type>
hpx::parallel::util::detail::algorithm_result<ExPolicy, typename std::iterator_traits<typename hpx::traits::range_traits<Rng>::iterator_t
```

Returns the number of elements in the range `[first, last)` satisfying a specific criteria. This version counts the elements that are equal to the given *value*.

The comparisons in the parallel *count* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *last - first* comparisons.

Note: The comparisons in the parallel *count* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the comparisons.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **T** – The type of the value to search for (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

⁷³⁴ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **value** – The value to search for.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *count* algorithm returns a `hpx::future<difference_type>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *difference_type* otherwise (where *difference_type* is defined by `std::iterator_traits<FwdIter>::difference_type`). The *count* algorithm returns the number of elements satisfying the given criteria.

```
template<typename ExPolicy, typename Iter, typename Sent, typename Proj = hpx::identity, typename T =  
typename hpx::parallel::traits::projected<Iter, Proj>::value_type>  
hpx::parallel::util::detail::algorithm_result<ExPolicy, typename std::iterator_traits<Iter>::difference_type>::type hpx::ranges::count
```

Returns the number of elements in the range `[first, last)` satisfying a specific criteria. This version counts the elements that are equal to the given *value*.

The comparisons in the parallel *count* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *last - first* comparisons.

Note: The comparisons in the parallel *count* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the comparisons.
- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter`.
- **T** – The type of the value to search for (deduced).

- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – The value to search for.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *count* algorithm returns a *hpx::future<difference_type>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *difference_type* otherwise (where *difference_type* is defined by *std::iterator_traits<FwdIter>::difference_type*). The *count* algorithm returns the number of elements satisfying the given criteria.

```
template<typename Rng, typename Proj = hpx::identity, typename T = typename
hpx::parallel::traits::projected<std::ranges::iterator_t<Rng>, Proj>::value_type>
std::iterator_traits<typename hpx::traits::range_traits<Rng>::iterator_type>::difference_type hpx::ranges::count(Rng
&&rng,
T
const
&value,
Proj
&&proj
=
Proj())
```

Returns the number of elements in the range [first, last) satisfying a specific criteria. This version counts the elements that are equal to the given *value*.

Note: Complexity: Performs exactly *last - first* comparisons.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **T** – The type of the value to search for (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **value** – The value to search for.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *count* algorithm returns the number of elements satisfying the given criteria.

```
template<typename Iter, typename Sent, typename Proj = hpx::identity, typename T = typename  
hpx::parallel::traits::projected<Iter, Proj>::value_type>  
std::iterator_traits<Iter>::difference_type hpx::ranges::count(Iter first, Sent last, T const &value, Proj &&proj  
= Proj())
```

Returns the number of elements in the range [first, last) satisfying a specific criteria. This version counts the elements that are equal to the given *value*.

Note: Complexity: Performs exactly *last - first* comparisons.

Template Parameters

- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter`.
- **T** – The type of the value to search for (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **value** – The value to search for.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *count* algorithm returns the number of elements satisfying the given criteria.

hpx::ranges::count_if

Defined in header `hpx/algorithm.hpp`⁷³⁵.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::ranges::count`

```
template<typename ExPolicy, typename Rng, typename F, typename Proj = hpx::identity>  
hpx::parallel::util::detail::algorithm_result<ExPolicy, typename std::iterator_traits<typename hpx::traits::range_traits<Rng>::iterator_t
```

⁷³⁵ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Returns the number of elements in the range `[first, last)` satisfying a specific criteria. This version counts elements for which predicate `f` returns true.

Note: Complexity: Performs exactly *last - first* applications of the predicate.

Note: The assignments in the parallel *count_if* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

Note: The assignments in the parallel *count_if* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the comparisons.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *count_if* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by `[first, last)`. This is a unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *count_if* algorithm returns *hpx::future<difference_type>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *difference_type* otherwise (where *difference_type* is defined by *std::iterator_traits<FwdIter>::difference_type*). The *count* algorithm returns the number of elements satisfying the given criteria.

```
template<typename ExPolicy, typename Iter, typename Sent, typename F, typename Proj = hpx::identity>
```

hpx::parallel::util::detail::algorithm_result<ExPolicy, typename std::iterator_traits<Iter>::difference_type>::type hpx::ranges::count_if_parallel

Returns the number of elements in the range [first, last) satisfying a specific criteria. This version counts elements for which predicate *f* returns true.

Note: Complexity: Performs exactly *last - first* applications of the predicate.

Note: The assignments in the parallel *count_if* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

Note: The assignments in the parallel *count_if* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the comparisons.
- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter*.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *count_if* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *count_if* algorithm returns *hpx::future<difference_type>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *difference_type* otherwise (where *difference_type* is defined by *std::iterator_traits<FwdIter>::difference_type*). The *count* algorithm returns the number of elements satisfying the given criteria.

```
template<typename Rng, typename F, typename Proj = hpx::identity>
std::iterator_traits<typename hpx::traits::range_traits<Rng>::iterator_type>::difference_type hpx::ranges::count_if(Rng
                                                                &&rng,
                                                                F
                                                                &&f,
                                                                Proj
                                                                &&proj
                                                                =
                                                                Proj())
```

Returns the number of elements in the range `[first, last)` satisfying a specific criteria. This version counts elements for which predicate *f* returns true.

Note: Complexity: Performs exactly *last - first* applications of the predicate.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *count_if* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by `[first, last)`. This is an unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *count* algorithm returns the number of elements satisfying the given criteria.

```
template<typename Iter, typename Sent, typename F, typename Proj = hpx::identity>
std::iterator_traits<Iter>::difference_type hpx::ranges::count_if(Iter first, Sent last, F &&f, Proj &&proj =
    Proj())
```

Returns the number of elements in the range [first, last) satisfying a specific criteria. This version counts elements for which predicate *f* returns true.

Note: Complexity: Performs exactly *last - first* applications of the predicate.

Template Parameters

- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter*.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *count_if* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have *const&*, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *count* algorithm returns the number of elements satisfying the given criteria.

hpx::ranges::is_partitioned

Defined in header `hpx/algorithm.hpp`⁷³⁶.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename FwdIter, typename Sent, typename Pred, typename Proj = hpx::identity>
```

⁷³⁶ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

bool *hpx::ranges::is_partitioned*(*FwdIter* first, *Sent* last, *Pred* &&pred, *Proj* &&proj = *Proj*())

Determines if the range [first, last) is partitioned.

Note: Complexity: at most (N) predicate evaluations where $N = \text{distance}(\text{first}, \text{last})$.

Template Parameters

- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **Pred** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*.

Parameters

- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.
- **pred** – Refers to the unary predicate which returns true for elements expected to be found in the beginning of the range. The signature of the function should be equivalent to

```
bool pred(const Type &a);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *is_partitioned* algorithm returns *bool*. The *is_partitioned* algorithm returns true if each element in the sequence for which *pred* returns true precedes those for which *pred* returns false. Otherwise *is_partitioned* returns false. If the range [first, last) contains less than two elements, the function is always true.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename Pred, typename Proj =
hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx::ranges::is_partitioned(ExPolicy
&&policy,
FwdIter first,
Sent last, Pred
&&pred, Proj
&&proj =
Proj())
```

Determines if the range [first, last) is partitioned.

The predicate operations in the parallel *is_partitioned* algorithm invoked with an execution policy object of type *sequenced_policy* executes in sequential order in the calling thread.

The comparison operations in the parallel *is_partitioned* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: at most (N) predicate evaluations where $N = \text{distance}(\text{first}, \text{last})$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for FwdIter.
- **Pred** – The type of the function/function object to use (deduced). *Pred* must be *CopyConstructible* when using a parallel policy.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.
- **pred** – Refers to the unary predicate which returns true for elements expected to be found in the beginning of the range. The signature of the function should be equivalent to

```
bool pred(const Type &a);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *is_partitioned* algorithm returns a *hpx::future<bool>* if the execution policy is of type *task_execution_policy* and returns *bool* otherwise. The *is_partitioned* algorithm returns true if each element in the sequence for which *pred* returns true precedes those for which *pred* returns false. Otherwise *is_partitioned* returns false. If the range $[\text{first}, \text{last})$ contains less than two elements, the function is always true.

```
template<typename Rng, typename Pred, typename Proj = hpx::identity>
bool hpx::ranges::is_partitioned(Rng &&rng, Pred &&pred, Proj &&proj = Proj())
```

Determines if the range *rng* is partitioned.

Note: Complexity: at most (N) predicate evaluations where $N = \text{std::size}(\text{rng})$.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`.

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Refers to the unary predicate which returns true for elements expected to be found in the beginning of the range. The signature of the function should be equivalent to

```
bool pred(const Type &a);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *is_partitioned* algorithm returns *bool*. The *is_partitioned* algorithm returns true if each element in the sequence for which *pred* returns true precedes those for which *pred* returns false. Otherwise *is_partitioned* returns false. If the range *rng* contains less than two elements, the function is always true.

```
template<typename ExPolicy, typename Rng, typename Pred, typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx::ranges::is_partitioned(ExPolicy
                                                                                               &&policy, Rng
                                                                                               &&rng, Pred
                                                                                               &&pred, Proj
                                                                                               &&proj =
                                                                                               Proj())
```

Determines if the range [first, last) is partitioned.

The predicate operations in the parallel *is_partitioned* algorithm invoked with an execution policy object of type *sequenced_policy* executes in sequential order in the calling thread.

The comparison operations in the parallel *is_partitioned* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: at most (*N*) predicate evaluations where *N* = `std::size(rng)`.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.

- **Pred** – The type of the function/function object to use (deduced). *Pred* must be *CopyConstructible* when using a parallel policy.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Refers to the unary predicate which returns true for elements expected to be found in the beginning of the range. The signature of the function should be equivalent to

```
bool pred(const Type &a);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *is_partitioned* algorithm returns a *hpx::future<bool>* if the execution policy is of type *task_execution_policy* and returns *bool* otherwise. The *is_partitioned* algorithm returns true if each element in the sequence for which *pred* returns true precedes those for which *pred* returns false. Otherwise *is_partitioned* returns false. If the range *rng* contains less than two elements, the function is always true.

hpx::ranges::fill

Defined in header [hpx/algorithm.hpp](#)⁷³⁷.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::ranges::fill_n*

```
template<typename ExPolicy, typename Rng, typename T = typename  
std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>
```

```
hpx::parallel::util::detail::algorithm_result<ExPolicy, typename hpx::traits::range_traits<Rng>::iterator_type>::type hpx::ranges::fill
```

Assigns the given value to the elements in the range [first, last).

The comparisons in the parallel *fill* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

⁷³⁷ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

The comparisons in the parallel *fill* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **T** – The type of the value to be assigned (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **value** – The value to be assigned.

Returns

The *fill* algorithm returns a *hpx::future<void>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *difference_type* otherwise (where *difference_type* is defined by *void*).

```
template<typename ExPolicy, typename Iter, typename Sent, typename T = typename
std::iterator_traits<Iter>::value_type>
hpx::parallel::util::detail::algorithm_result<ExPolicy, Iter>::type hpx::ranges::fill(ExPolicy &&policy, Iter
first, Sent last, T const
&value)
```

Assigns the given value to the elements in the range [first, last).

The comparisons in the parallel *fill* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *fill* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter*.

- **T** – The type of the value to be assigned (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **value** – The value to be assigned.

Returns

The *fill* algorithm returns a `hpx::future<void>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *difference_type* otherwise (where *difference_type* is defined by *void*).

```
template<typename Rng, typename T = typename std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>  
std::ranges::iterator_t<Rng> hpx::ranges::fill(Rng &&rng, T const &value)
```

Assigns the given value to the elements in the range [first, last).

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **T** – The type of the value to be assigned (deduced).

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **value** – The value to be assigned.

Returns

The *fill* algorithm returns *void*.

```
template<typename Iter, typename Sent, typename T = typename std::iterator_traits<Iter>::value_type>  
Iter hpx::ranges::fill(Iter first, Sent last, T const &value)
```

Assigns the given value to the elements in the range [first, last).

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter`.
- **T** – The type of the value to be assigned (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **value** – The value to be assigned.

Returns

The *fill* algorithm returns *void*.

hpx::ranges::fill_n

Defined in header `hpx/algorithm.hpp`⁷³⁸.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::fill`

```
template<typename ExPolicy, typename Rng, typename T = typename
std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, std::ranges::iterator_t<Rng>> hpx::ranges::fill_n(ExPolicy
&&pol-
icy,
Rng
&&rng,
T
const
&value)
```

Assigns the given value value to the first count elements in the range beginning at first if count > 0. Does nothing otherwise.

The comparisons in the parallel *fill_n* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *fill_n* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **T** – The type of the value to be assigned (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.

⁷³⁸ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **value** – The value to be assigned.

Returns

The *fill_n* algorithm returns a *hpx::future<void>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *difference_type* otherwise (where *difference_type* is defined by *void*).

```
template<typename ExPolicy, typename FwdIter, typename Size, typename T = typename
std::iterator_traits<FwdIter>::value_type>
hpx::parallel::util::detail::algorithm_result<ExPolicy, FwdIter>::type hpx::ranges::fill_n(ExPolicy &&policy,
                                                                                      FwdIter first, Size
                                                                                      count, T const
                                                                                      &value)
```

Assigns the given value value to the first count elements in the range beginning at first if count > 0. Does nothing otherwise.

The comparisons in the parallel *fill_n* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The comparisons in the parallel *fill_n* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *count* assignments, for count > 0.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **T** – The type of the value to be assigned (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **value** – The value to be assigned.

Returns

The *fill_n* algorithm returns a *hpx::future<void>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *difference_type* otherwise (where *difference_type* is defined by *void*).

```
template<typename Rng, typename T = typename std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>
```

`hpx::traits::range_traits<Rng>::iterator_type hpx::ranges::fill_n(Rng &&rng, T const &value)`

Assigns the given value value to the first count elements in the range beginning at first if count > 0. Does nothing otherwise.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **T** – The type of the value to be assigned (deduced).

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **value** – The value to be assigned.

Returns

The *fill_n* algorithm returns an output iterator that compares equal to last.

```
template<typename FwdIter, typename Size, typename T = typename std::iterator_traits<FwdIter>::value_type>
FwdIter hpx::ranges::fill_n(Iterator first, Size count, T const &value)
```

Assigns the given value value to the first count elements in the range beginning at first if count > 0. Does nothing otherwise.

Note: Complexity: Performs exactly *count* assignments, for count > 0.

Template Parameters

- **Iterator** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **T** – The type of the value to be assigned (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **value** – The value to be assigned.

Returns

The *fill_n* algorithm returns an output iterator that compares equal to last.

hpx::ranges::starts_with

Defined in header `hpx/algorithm.hpp`⁷³⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename Pred =
ranges::equal_to, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
```

⁷³⁹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

```
bool hpx::ranges::starts_with(Iter1 first1, Sent1 last1, Iter2 first2, Sent2 last2, Pred &&pred = Pred(), Proj1  
                             &&proj1 = Proj1(), Proj2 &&proj2 = Proj2())
```

Checks whether the second range defined by [first1, last1) matches the prefix of the first range defined by [first2, last2)

The assignments in the parallel *starts_with* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: Linear: at most min(N1, N2) applications of the predicate and both projections.

Template Parameters

- **Iter1** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **Iter2** – The type of the begin destination iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent2** – The type of the end destination iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter2.
- **Pred** – The binary predicate that compares the projected elements.
- **Proj1** – The type of an optional projection function for the source range. This defaults to *hpx::identity*
- **Proj2** – The type of an optional projection function for the destination range. This defaults to *hpx::identity*

Parameters

- **first1** – Refers to the beginning of the source range.
- **last1** – Sentinel value referring to the end of the source range.
- **first2** – Refers to the beginning of the destination range.
- **last2** – Sentinel value referring to the end of the destination range.
- **pred** – Specifies the binary predicate function (or function object) which will be invoked for comparison of the elements in the in two ranges projected by proj1 and proj2 respectively.
- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements in the source range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements in the destination range as a projection operation before the actual predicate *is* invoked.

Returns

The *starts_with* algorithm returns *bool*. The *starts_with* algorithm returns a boolean with the value true if the second range matches the prefix of the first range, false otherwise.

```
template<typename ExPolicy, typename FwdIter1, typename Sent1, typename FwdIter2, typename Sent2,  
        typename Pred = ranges::equal_to, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
```

```

hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx::ranges::starts_with(ExPolicy &&policy,
                                                                 FwdIter1 first1, Sent1
                                                                 last1, FwdIter2 first2,
                                                                 Sent2 last2, Pred
                                                                 &&pred = Pred(),
                                                                 Proj1 &&proj1 =
                                                                 Proj1(), Proj2
                                                                 &&proj2 = Proj2())

```

Checks whether the second range defined by [first1, last1) matches the prefix of the first range defined by [first2, last2)

The assignments in the parallel *starts_with* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *starts_with* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Linear: at most min(N1, N2) applications of the predicate and both projections.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the begin source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent1** – The type of the end source iterators used(deduced). This iterator type must meet the requirements of an sentinel for Iter1.
- **FwdIter2** – The type of the begin destination iterators used deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent2** – The type of the end destination iterators used (deduced). This iterator type must meet the requirements of an sentinel for Iter2.
- **Pred** – The binary predicate that compares the projected elements.
- **Proj1** – The type of an optional projection function for the source range. This defaults to *hpx::identity*
- **Proj2** – The type of an optional projection function for the destination range. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the source range.
- **last1** – Sentinel value referring to the end of the source range.
- **first2** – Refers to the beginning of the destination range.
- **last2** – Sentinel value referring to the end of the destination range.

- **pred** – Specifies the binary predicate function (or function object) which will be invoked for comparison of the elements in the in two ranges projected by proj1 and proj2 respectively.
- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements in the source range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements in the destination range as a projection operation before the actual predicate *is* invoked.

Returns

The *starts_with* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *starts_with* algorithm returns a boolean with the value true if the second range matches the prefix of the first range, false otherwise.

```
template<typename Rng1, typename Rng2, typename Pred = ranges::equal_to, typename Proj1 = hpx::identity,
typename Proj2 = hpx::identity>
bool hpx::ranges::starts_with(Rng1 &&rng1, Rng2 &&rng2, Pred &&pred = Pred(), Proj1 &&proj1 =
    Proj1(), Proj2 &&proj2 = Proj2())
```

Checks whether the second range *rng2* matches the prefix of the first range *rng1*.

The assignments in the parallel *starts_with* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: Linear: at most min(N1, N2) applications of the predicate and both projections.

Template Parameters

- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Rng2** – The type of the destination range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The binary predicate that compares the projected elements.
- **Proj1** – The type of an optional projection function for the source range. This defaults to *hpx::identity*
- **Proj2** – The type of an optional projection function for the destination range. This defaults to *hpx::identity*

Parameters

- **rng1** – Refers to the source range.
- **rng2** – Refers to the destination range.
- **pred** – Specifies the binary predicate function (or function object) which will be invoked for comparison of the elements in the in two ranges projected by proj1 and proj2 respectively.
- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements in the source range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements in the destination range as a projection operation before the actual predicate *is* invoked.

Returns

The *starts_with* algorithm returns *bool*. The *starts_with* algorithm returns a boolean with the value true if the second range matches the prefix of the first range, false otherwise.

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Pred = ranges::equal_to, typename
Proj1 = hpx::identity, typename Proj2 = hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, bool>::type hpx::ranges::starts_with(ExPolicy
                                                                                               &&policy, Rng1
                                                                                               &&rng1, Rng2
                                                                                               &&rng2, Pred
                                                                                               &&pred =
                                                                                               Pred(), Proj1
                                                                                               &&proj1 =
                                                                                               Proj1(), Proj2
                                                                                               &&proj2 =
                                                                                               Proj2())
```

Checks whether the second range *rng2* matches the prefix of the first range *rng1*.

The assignments in the parallel *starts_with* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *starts_with* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Linear: at most min(N1, N2) applications of the predicate and both projections.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Rng2** – The type of the destination range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The binary predicate that compares the projected elements.
- **Proj1** – The type of an optional projection function for the source range. This defaults to *hpx::identity*
- **Proj2** – The type of an optional projection function for the destination range. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the source range.
- **rng2** – Refers to the destination range.
- **pred** – Specifies the binary predicate function (or function object) which will be invoked for comparison of the elements in the in two ranges projected by proj1 and proj2 respectively.

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements in the source range as a projection operation before the actual predicate *is* invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements in the destination range as a projection operation before the actual predicate *is* invoked.

Returns

The *starts_with* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *starts_with* algorithm returns a boolean with the value true if the second range matches the prefix of the first range, false otherwise.

hpx::ranges::remove_copy

Defined in header [hpx/algorithm.hpp](#)⁷⁴⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::ranges::remove_copy_if*

Warning: doxygenfunction: Cannot find function “hpx::ranges::remove_copy” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

hpx::ranges::remove_copy_if

Defined in header [hpx/algorithm.hpp](#)⁷⁴¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::ranges::remove_copy*

Warning: doxygenfunction: Cannot find function “hpx::ranges::remove_copy_if” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

hpx::ranges::shift_right

Defined in header [hpx/algorithm.hpp](#)⁷⁴².

See *Public API* for a list of names and headers that are part of the public HPX API.

template<typename **FwdIter**, typename **Sent**, typename **Size**>

FwdIter **hpx::ranges::shift_right**(*FwdIter* first, *Sent* last, *Size* n)

Shifts the elements in the range [first, last) by n positions towards the end of the range. For every integer i in [0, last - first - n), moves the element originally at position first + i to position first

⁷⁴⁰ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

⁷⁴¹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

⁷⁴² <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- $n + i$.

The assignment operations in the parallel *shift_right* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

Note: Complexity: At most (last - first) - n assignments.

Note: The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable*.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **Size** – The type of the argument specifying the number of positions to shift by.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **n** – Refers to the number of positions to shift.

Returns

The *shift_right* algorithm returns *FwdIter*. The *shift_right* algorithm returns an iterator to the end of the resulting range.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename Size>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::ranges::shift_right(ExPolicy
                                                                                               &&policy,
                                                                                               FwdIter first,
                                                                                               Sent last, Size n)
```

Shifts the elements in the range [first, last) by n positions towards the end of the range. For every integer i in [0, last - first - n), moves the element originally at position first + i to position first

- $n + i$.

The assignment operations in the parallel *shift_right* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignment operations in the parallel *shift_right* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most (last - first) - n assignments.

Note: The type of dereferenced *FwdIter* must meet the requirements of *MoveAssignable*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **Size** – The type of the argument specifying the number of positions to shift by.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **n** – Refers to the number of positions to shift.

Returns

The *shift_right* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *shift_right* algorithm returns an iterator to the end of the resulting range.

```
template<typename Rng, typename Size>
```

```
std::ranges::iterator_t<Rng> hpx::ranges::shift_right(Rng &&rng, Size n)
```

Shifts the elements in the range [first, last) by *n* positions towards the end of the range. For every integer *i* in [0, last - first - *n*), moves the element originally at position first + *i* to position first

- *n* + *i*.

The assignment operations in the parallel *shift_right* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

Note: Complexity: At most (last - first) - *n* assignments.

Note: The type of dereferenced *std::ranges::iterator_t<Rng>* must meet the requirements of *MoveAssignable*.

Template Parameters

- **Rng** – The type of the range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of positions to shift by.

Parameters

- **rng** – Refers to the range in which the elements will be shifted.
- **n** – Refers to the number of positions to shift.

Returns

The *shift_right* algorithm returns *std::ranges::iterator_t<Rng>*. The *shift_right* algorithm returns an iterator to the end of the resulting range.

```
template<typename ExPolicy, typename Rng, typename Size>
parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng>> hpx::ranges::shift_right(ExPolicy
                                                                    &&pol-
                                                                    icy,
                                                                    Rng
                                                                    &&rng,
                                                                    Size
                                                                    n)
```

Shifts the elements in the range [first, last) by n positions towards the end of the range. For every integer i in [0, last - first - n), moves the element originally at position first + i to position first

- n + i.

The assignment operations in the parallel *shift_right* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignment operations in the parallel *shift_right* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most (last - first) - n assignments.

Note: The type of dereferenced *std::ranges::iterator_t<Rng>* must meet the requirements of *MoveAssignable*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Size** – The type of the argument specifying the number of positions to shift by.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the range in which the elements will be shifted.
- **n** – Refers to the number of positions to shift.

Returns

The *shift_right* algorithm returns a *hpx::future<std::ranges::iterator_t<Rng>>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *std::ranges::iterator_t<Rng>* otherwise. The *shift_right* algorithm returns an iterator to the end of the resulting range.

hpx::ranges::stable_sort

Defined in header [hpx/algorithm.hpp](#)⁷⁴³.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename RandomIt, typename Sent, typename Comp = ranges::less, typename Proj = hpx::identity>  
RandomIt hpx::ranges::stable_sort(RandomIt first, Sent last, Comp &&comp = Comp(), Proj &&proj =  
                                Proj())
```

Sorts the elements in the range [first, last) in ascending order. The relative order of equal elements is preserved. The function uses the given comparison function object comp (defaults to using operator<()).

A sequence is sorted with respect to a comparator *comp* and a projection *proj* if for every iterator *i* pointing to the sequence and every non-negative integer *n* such that *i* + *n* is a valid iterator pointing to an element of the sequence, and INVOKE(comp, INVOKE(proj, *(i + n)), INVOKE(proj, *i)) == false.

comp has to induce a strict weak ordering on the values.

The assignments in the parallel *stable_sort* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: $O(N \log(N))$, where $N = \text{std::distance}(\text{first}, \text{last})$ comparisons.

Template Parameters

- **RandomIt** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an random iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *RandomIt*.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **comp** – comp is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that comp will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

Returns

The *stable_sort* algorithm returns *RandomIt*. The algorithm returns an iterator pointing to the first element after the last element in the input sequence.

```
template<typename ExPolicy, typename RandomIt, typename Sent, typename Comp = ranges::less, typename  
Proj = hpx::identity>
```

⁷⁴³ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

```
parallel::util::detail::algorithm_result<ExPolicy, RandomIt>::type hpx::ranges::stable_sort(ExPolicy
                                                                    &&policy,
                                                                    RandomIt first,
                                                                    Sent last, Comp
                                                                    &&comp =
                                                                    Comp(), Proj
                                                                    &&proj =
                                                                    Proj())
```

Sorts the elements in the range `[first, last)` in ascending order. The relative order of equal elements is preserved. The function uses the given comparison function object `comp` (defaults to using `operator<()`).

A sequence is sorted with respect to a comparator `comp` and a projection `proj` if for every iterator `i` pointing to the sequence and every non-negative integer `n` such that `i + n` is a valid iterator pointing to an element of the sequence, and `INVOKE(comp, INVOKE(proj, *(i + n)), INVOKE(proj, *i)) == false`.

`comp` has to induce a strict weak ordering on the values.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: $O(N \log(N))$, where $N = \text{std::distance}(\text{first}, \text{last})$ comparisons.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **RandomIt** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an random iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `RandomIt`.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **comp** – `comp` is a callable object. The return value of the `INVOKE` operation applied to an object of type `Comp`, when contextually converted to `bool`, yields `true` if the first argument of the call is less than the second, and `false` otherwise. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate `comp` is invoked.

Returns

The *stable_sort* algorithm returns a *hpx::future<RandomIt>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *RandomIt* otherwise. The algorithm returns an iterator pointing to the first element after the last element in the input sequence.

```
template<typename Rng, typename Comp = ranges::less, typename Proj = hpx::identity>
std::ranges::iterator_t<Rng> hpx::ranges::stable_sort(Rng &&rng, Comp &&comp = Comp(), Proj &&proj
                                                    = Proj())
```

Sorts the elements in the range [first, last) in ascending order. The relative order of equal elements is preserved. The function uses the given comparison function object *comp* (defaults to using *operator<()*).

A sequence is sorted with respect to a comparator *comp* and a projection *proj* if for every iterator *i* pointing to the sequence and every non-negative integer *n* such that *i + n* is a valid iterator pointing to an element of the sequence, and *INVOKE(comp, INVOKE(proj, *(i + n)), INVOKE(proj, *i)) == false*.

comp has to induce a strict weak ordering on the values.

The assignments in the parallel *stable_sort* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: $O(N \log(N))$, where $N = \text{std::distance}(\text{first}, \text{last})$ comparisons.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **comp** – *comp* is a callable object. The return value of the *INVOKE* operation applied to an object of type *Comp*, when contextually converted to *bool*, yields *true* if the first argument of the call is less than the second, and *false* otherwise. It is assumed that *comp* will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

Returns

The *stable_sort* algorithm returns *std::ranges::iterator_t<Rng>*. It returns *last*.

```
template<typename ExPolicy, typename Rng, typename Comp = ranges::less, typename Proj = hpx::identity>
```

```
parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng>> hpx::ranges::stable_sort(ExPolicy
                                                                 &&pol-
                                                                 icy,
                                                                 Rng
                                                                 &&rng,
                                                                 Comp
                                                                 &&comp
                                                                 =
                                                                 Comp(),
                                                                 Proj
                                                                 &&proj
                                                                 =
                                                                 Proj())
```

Sorts the elements in the range `[first, last)` in ascending order. The relative order of equal elements is preserved. The function uses the given comparison function object `comp` (defaults to using `operator<()`).

A sequence is sorted with respect to a comparator `comp` and a projection `proj` if for every iterator `i` pointing to the sequence and every non-negative integer `n` such that `i + n` is a valid iterator pointing to an element of the sequence, and `INVOKE(comp, INVOKE(proj, *(i + n)), INVOKE(proj, *i)) == false`.

`comp` has to induce a strict weak ordering on the values.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: $O(N \log(N))$, where $N = \text{std::distance}(\text{first}, \text{last})$ comparisons.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **comp** – `comp` is a callable object. The return value of the `INVOKE` operation applied to an object of type `Comp`, when contextually converted to `bool`, yields `true` if the first argument of the call is less than the second, and `false` otherwise. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate `comp` is invoked.

Returns

The *stable_sort* algorithm returns a `hpx::future<std::ranges::iterator_t<Rng>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `std::ranges::iterator_t<Rng>` otherwise. It returns *last*.

hpx::ranges::reduce

Defined in header `hpx/algorithm.hpp`⁷⁴⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename F, typename T = typename
std::iterator_traits<FwdIter>::value_type>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> hpx::ranges::reduce(ExPolicy &&policy, FwdIter
first, Sent last, T init, F &&f)
```

Returns GENERALIZED_SUM(f, init, *first, ..., *(first + (last - first) - 1)).

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *copy_if* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate *f*.

Note: GENERALIZED_SUM(op, a1, ..., aN) is defined as follows:

- a1 when N is 1
 - op(GENERALIZED_SUM(op, b1, ..., bK), GENERALIZED_SUM(op, bM, ..., bN)), where:
 - b1, ..., bN may be any permutation of a1, ..., aN and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel used (deduced). This iterator type must meet the requirements of an forward iterator.

⁷⁴⁴ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *copy_if* requires *F* to meet the requirements of *CopyConstructible*.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have const&. The types *Type1* *Ret* must be such that an object of type *FwdIterB* can be dereferenced and then implicitly converted to any of those types.

- **init** – The initial value for the generalized sum.

Returns

The *reduce* algorithm returns a *hpx::future<T>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *T* otherwise. The *reduce* algorithm returns the result of the generalized sum over the elements given by the input range [first, last).

```
template<typename ExPolicy, typename Rng, typename F, typename T = typename
std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> hpx::ranges::reduce(ExPolicy &&policy, Rng
&&rng, T init, F &&f)
```

Returns GENERALIZED_SUM(f, init, *first, ..., *(first + (last - first) - 1)).

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *copy_if* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate *f*.

Note: GENERALIZED_SUM(op, a1, ..., aN) is defined as follows:

- a1 when N is 1
- op(GENERALIZED_SUM(op, b1, ..., bK), GENERALIZED_SUM(op, bM, ..., bN)), where:
 - b1, ..., bN may be any permutation of a1, ..., aN and

– $1 < K+1 = M \leq N$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *copy_if* requires *F* to meet the requirements of *CopyConstructible*.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have const&. The types *Type1* *Ret* must be such that an object of type *FwdIterB* can be dereferenced and then implicitly converted to any of those types.

- **init** – The initial value for the generalized sum.

Returns

The *reduce* algorithm returns a *hpx::future<T>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *T* otherwise. The *reduce* algorithm returns the result of the generalized sum over the elements given by the input range [first, last).

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename T = typename
std::iterator_traits<FwdIter>::value_type>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> hpx::ranges::reduce(ExPolicy &&policy, FwdIter
first, Sent last, T init)
```

Returns GENERALIZED_SUM(+, init, *first, ..., *(first + (last - first) - 1)).

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *copy_if* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the operator+().

Note: GENERALIZED_SUM(+, a1, ..., aN) is defined as follows:

- a1 when N is 1
 - op(GENERALIZED_SUM(+, b1, ..., bK), GENERALIZED_SUM(+, bM, ..., bN)), where:
 - b1, ..., bN may be any permutation of a1, ..., aN and
 - 1 < K+1 = M <= N.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel used (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.

Returns

The *reduce* algorithm returns a `hpx::future<T>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *T* otherwise. The *reduce* algorithm returns the result of the generalized sum (applying operator+()) over the elements given by the input range [first, last).

```
template<typename ExPolicy, typename Rng, typename T = typename
std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, T> hpx::ranges::reduce(ExPolicy &&policy, Rng
&&rng, T init)
```

Returns GENERALIZED_SUM(+, init, *first, ..., *(first + (last - first) - 1)).

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *copy_if* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the operator+().

Note: GENERALIZED_SUM(+, a1, ..., aN) is defined as follows:

- a1 when N is 1
 - op(GENERALIZED_SUM(+, b1, ..., bK), GENERALIZED_SUM(+, bM, ..., bN)), where:
 - b1, ..., bN may be any permutation of a1, ..., aN and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.

Returns

The *reduce* algorithm returns a `hpx::future<T>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *T* otherwise. The *reduce* algorithm returns the result of the generalized sum (applying operator+()) over the elements given by the input range [first, last).

```
template<typename ExPolicy, typename FwdIter, typename Sent>
hpx::parallel::util::detail::algorithm_result<ExPolicy, typename std::iterator_traits<FwdIter>::value_type>::type hpx::ranges::redu
```

Returns GENERALIZED_SUM(+, T(), *first, ..., *(first + (last - first) - 1)).

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *copy_if* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the operator+().

Note: The type of the initial value (and the result type) T is determined from the `value_type` of the used `FwdIterB`.

Note: `GENERALIZED_SUM(+, a1, ..., aN)` is defined as follows:

- $a1$ when N is 1
 - $\text{op}(\text{GENERALIZED_SUM}(+, b1, \dots, bK), \text{GENERALIZED_SUM}(+, bM, \dots, bN))$, where:
 - $b1, \dots, bN$ may be any permutation of $a1, \dots, aN$ and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel used (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

Returns

The *reduce* algorithm returns a `hpx::future<T>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns T otherwise (where T is the `value_type` of `FwdIterB`). The *reduce* algorithm returns the result of the generalized sum (applying operator+()) over the elements given by the input range `[first, last)`.

```
template<typename ExPolicy, typename Rng>
hpx::parallel::util::detail::algorithm_result<ExPolicy, typename std::iterator_traits<typename hpx::traits::range_traits<Rng>::iterator_t
```

Returns `GENERALIZED_SUM(+, T(), *first, ..., *(first + (last - first) - 1))`.

The reduce operations in the parallel *reduce* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The reduce operations in the parallel *copy_if* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the operator+().

Note: The type of the initial value (and the result type) *T* is determined from the *value_type* of the used *FwdIterB*.

Note: GENERALIZED_SUM(+, a1, ..., aN) is defined as follows:

- a1 when N is 1
 - op(GENERALIZED_SUM(+, b1, ..., bK), GENERALIZED_SUM(+, bM, ..., bN)), where:
 - b1, ..., bN may be any permutation of a1, ..., aN and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.

Returns

The *reduce* algorithm returns a *hpx::future<T>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *T* otherwise (where *T* is the *value_type* of *FwdIterB*). The *reduce* algorithm returns the result of the generalized sum (applying operator+()) over the elements given by the input range [first, last).

```
template<typename FwdIter, typename Sent, typename F, typename T = typename
std::iterator_traits<FwdIter>::value_type>
```

```
T hpx::ranges::reduce(FwdIter first, Sent last, T init, F &&f)
```

Returns GENERALIZED_SUM(f, init, *first, ..., *(first + (last - first) - 1)).

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate f .

Note: `GENERALIZED_SUM(op, a1, ..., aN)` is defined as follows:

- a_1 when N is 1
 - $\text{op}(\text{GENERALIZED_SUM}(\text{op}, b_1, \dots, b_K), \text{GENERALIZED_SUM}(\text{op}, b_M, \dots, b_N))$, where:
 - $b_1, \dots, b_N$ may be any permutation of $a_1, \dots, a_N$ and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

- **FwdIter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel used (deduced). This iterator type must meet the requirements of an forward iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *copy_if* requires F to meet the requirements of *CopyConstructible*.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by `[first, last)`. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`. The types *Type1* *Ret* must be such that an object of type *FwdIterB* can be dereferenced and then implicitly converted to any of those types.

- **init** – The initial value for the generalized sum.

Returns

The *reduce* algorithm returns T . The *reduce* algorithm returns the result of the generalized sum over the elements given by the input range `[first, last)`.

```
template<typename Rng, typename F, typename T = typename
std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>
T hpx::ranges::reduce(Rng &&rng, T init, F &&f)
```

Returns `GENERALIZED_SUM(f, init, *first, ..., *(first + (last - first) - 1))`.

The difference between *reduce* and *accumulate* is that the behavior of *reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the predicate f .

Note: `GENERALIZED_SUM(op, a1, ..., aN)` is defined as follows:

- $a1$ when N is 1
 - $\text{op}(\text{GENERALIZED_SUM}(\text{op}, b1, \dots, bK), \text{GENERALIZED_SUM}(\text{op}, bM, \dots, bN))$, where:
 - $b1, \dots, bN$ may be any permutation of $a1, \dots, aN$ and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of `copy_if` requires F to meet the requirements of `CopyConstructible`.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by $[\text{first}, \text{last})$. This is a binary predicate. The signature of this predicate should be equivalent to:

```
Ret fun(const Type1 &a, const Type1 &b);
```

The signature does not need to have `const&`. The types `Type1` `Ret` must be such that an object of type `FwdIterB` can be dereferenced and then implicitly converted to any of those types.

- **init** – The initial value for the generalized sum.

Returns

The *reduce* algorithm returns T . The *reduce* algorithm returns the result of the generalized sum over the elements given by the input range $[\text{first}, \text{last})$.

```
template<typename FwdIter, typename Sent, typename T = typename std::iterator_traits<FwdIter>::value_type>
T hpx::ranges::reduce(FwdIter first, Sent last, T init)
```

Returns `GENERALIZED_SUM(+, init, *first, ..., *(first + (last - first) - 1))`.

The difference between *reduce* and *accumulate* is that the behavior of *reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the operator `+`.

Note: `GENERALIZED_SUM(+, a1, ..., aN)` is defined as follows:

- $a1$ when N is 1

- `op(GENERALIZED_SUM(+, b1, ..., bK), GENERALIZED_SUM(+, bM, ..., bN))`, where:
 - `b1, ..., bN` may be any permutation of `a1, ..., aN` and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

- **FwdIter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel used (deduced). This iterator type must meet the requirements of an forward iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the generalized sum.

Returns

The *reduce* algorithm returns *T*. The *reduce* algorithm returns the result of the generalized sum (applying operator`+`()) over the elements given by the input range `[first, last)`.

```
template<typename Rng, typename T = typename std::iterator_traits<std::ranges::iterator_t<Rng>>::value_type>
T hpx::ranges::reduce(Rng &&rng, T init)
```

Returns `GENERALIZED_SUM(+, init, *first, ..., *(first + (last - first) - 1))`.

The difference between *reduce* and *accumulate* is that the behavior of *reduce* may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the operator`+`.

Note: `GENERALIZED_SUM(+, a1, ..., aN)` is defined as follows:

- `a1` when `N` is 1
 - `op(GENERALIZED_SUM(+, b1, ..., bK), GENERALIZED_SUM(+, bM, ..., bN))`, where:
 - `b1, ..., bN` may be any permutation of `a1, ..., aN` and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **T** – The type of the value to be used as initial (and intermediate) values (deduced).

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.

- **init** – The initial value for the generalized sum.

Returns

The *reduce* algorithm returns *T*. The *reduce* algorithm returns the result of the generalized sum (applying operator+()) over the elements given by the input range [first, last).

```
template<typename FwdIter, typename Sent>
```

```
std::iterator_traits<FwdIter>::value_type hpx::ranges::reduce(FwdIter first, Sent last)
```

Returns GENERALIZED_SUM(+, T(), *first, ..., *(first + (last - first) - 1)).

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the operator+().

Note: The type of the initial value (and the result type) *T* is determined from the value_type of the used *FwdIterB*.

Note: GENERALIZED_SUM(+, a1, ..., aN) is defined as follows:

- a1 when N is 1
 - op(GENERALIZED_SUM(+, b1, ..., bK), GENERALIZED_SUM(+, bM, ..., bN)), where:
 - b1, ..., bN may be any permutation of a1, ..., aN and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

- **FwdIter** – The type of the source begin iterator used (deduced). This iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the source sentinel used (deduced). This iterator type must meet the requirements of an forward iterator.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

Returns

The *reduce* algorithm returns *T* (where T is the value_type of *FwdIterB*). The *reduce* algorithm returns the result of the generalized sum (applying operator+()) over the elements given by the input range [first, last).

```
template<typename Rng>
```

```
std::iterator_traits<typename hpx::traits::range_traits<Rng>::iterator_type>::value_type hpx::ranges::reduce(Rng  
                                                         &&rng)
```

Returns GENERALIZED_SUM(+, T(), *first, ..., *(first + (last - first) - 1)).

The difference between *reduce* and *accumulate* is that the behavior of reduce may be non-deterministic for non-associative or non-commutative binary predicate.

Note: Complexity: $O(\text{last} - \text{first})$ applications of the operator+().

Note: The type of the initial value (and the result type) T is determined from the `value_type` of the used `FwdIterB`.

Note: `GENERALIZED_SUM(+, a1, ..., aN)` is defined as follows:

- a_1 when N is 1
 - $\text{op}(\text{GENERALIZED_SUM}(+, b_1, \dots, b_K), \text{GENERALIZED_SUM}(+, b_M, \dots, b_N))$, where:
 - $b_1, \dots, b_N$ may be any permutation of $a_1, \dots, a_N$ and
 - $1 < K+1 = M \leq N$.
-

Template Parameters

Rng – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

Parameters

rng – Refers to the sequence of elements the algorithm will be applied to.

Returns

The *reduce* algorithm returns T (where T is the `value_type` of `FwdIterB`). The *reduce* algorithm returns the result of the generalized sum (applying `operator+()`) over the elements given by the input range `[first, last)`.

hpx::ranges::uninitialized_value_construct

Defined in header `hpx/algorithm.hpp`⁷⁴⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::uninitialized_value_construct_n`

```
template<typename FwdIter, typename Sent>
```

```
FwdIter hpx::ranges::uninitialized_value_construct(FwdIter first, Sent last)
```

Constructs objects of type `typename iterator_traits<ForwardIt>::value_type` in the uninitialized storage designated by the range by value-initialization. If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized_value_construct* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

⁷⁴⁵ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for FwdIter.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.

Returns

The *uninitialized_value_construct* algorithm returns a returns *FwdIter*. The *uninitialized_value_construct* algorithm returns the output iterator to the element in the range, one past the last element constructed.

```
template<typename ExPolicy, typename FwdIter, typename Sent>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::ranges::uninitialized_value_construct(ExPolicy
                                                                 &&pol-
                                                                 icy,
                                                                 FwdIter
                                                                 first,
                                                                 Sent
                                                                 last)
```

Constructs objects of type `typename iterator_traits<ForwardIt>::value_type` in the uninitialized storage designated by the range by value-initialization. If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized_value_construct* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized_value_construct* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.

- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `FwdIter`.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.

Returns

The `uninitialized_value_construct` algorithm returns a `hpx::future<FwdIter>` if the execution policy is of type `sequenced_task_policy` or `parallel_task_policy` and returns `FwdIter` otherwise. The `uninitialized_value_construct` algorithm returns the iterator to the element in the source range, one past the last element constructed.

```
template<typename Rng>
hpx::traits::range_traits<Rng>::iterator_type hpx::ranges::uninitialized_value_construct(Rng &&rng)
```

Constructs objects of type `typename iterator_traits<ForwardIt>::value_type` in the uninitialized storage designated by the range by value-initialization. If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel `uninitialized_value_construct` algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

Rng – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

Parameters

rng – Refers to the range to which will be value constructed.

Returns

The `uninitialized_value_construct` algorithm returns a `hpx::traits::range_traits<Rng>::iterator_type`. The `uninitialized_value_construct` algorithm returns the output iterator to the element in the range, one past the last element constructed.

```
template<typename ExPolicy, typename Rng>
parallel::util::detail::algorithm_result<ExPolicy, typename hpx::traits::range_traits<Rng>::iterator_type>::type hpx::ranges::unini
```

Constructs objects of type `typename iterator_traits<ForwardIt>::value_type` in the uninitialized storage designated by the range by value-initialization. If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized_value_construct* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized_value_construct* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the range to which the value will be value constructed

Returns

The *uninitialized_value_construct* algorithm returns a `hpx::future<typename hpx::traits::range_traits<Rng>::iterator_type>`, if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *typename hpx::traits::range_traits<Rng>::iterator_type* otherwise. The *uninitialized_value_construct* algorithm returns the output iterator to the element in the range, one past the last element constructed.

hpx::ranges::uninitialized_value_construct_n

Defined in header `hpx/algorithm.hpp`⁷⁴⁶.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::uninitialized_value_construct`

template<typename **FwdIter**, typename **Size**>

FwdIter **hpx::ranges::uninitialized_value_construct_n**(*FwdIter* first, *Size* count)

Constructs objects of type `typename iterator_traits<ForwardIt>::value_type` in the uninitialized storage designated by the range `[first, first + count)` by value-initialization. If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized_value_construct_n* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *count* assignments, if *count* > 0, no assignments otherwise.

⁷⁴⁶ <http://github.com/TheHPXProject/hpx/blob/d402cead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply f to.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.

Returns

The *uninitialized_value_construct_n* algorithm returns a *FwdIter*. The *uninitialized_value_construct_n* algorithm returns the iterator to the element in the source range, one past the last element constructed.

```
template<typename ExPolicy, typename FwdIter, typename Size>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> hpx::ranges::uninitialized_value_construct_n(ExPolicy
&&pol-
icy,
FwdIter
first,
Size
count)
```

Constructs objects of type `typename iterator_traits<ForwardIt>::value_type` in the uninitialized storage designated by the range `[first, first + count)` by value-initialization. If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized_value_construct_n* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized_value_construct_n* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *count* assignments, if *count* > 0, no assignments otherwise.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply f to.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.

- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.

Returns

The *uninitialized_value_construct_n* algorithm returns a `hpx::future<FwdIter>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *uninitialized_value_construct_n* algorithm returns the iterator to the element in the source range, one past the last element constructed.

hpx::ranges::uninitialized_copy

Defined in header `hpx/algorithm.hpp`⁷⁴⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::uninitialized_copy_n`

```
template<typename InIter, typename Sent1, typename FwdIter, typename Sent2>
hpx::parallel::util::in_out_result<InIter, FwdIter> hpx::ranges::uninitialized_copy(InIter first1, Sent1 last1,
                                                                                   FwdIter first2, Sent2
                                                                                   last2)
```

Copies the elements in the range, defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the copy operation, the function has no effects.

The assignments in the parallel *uninitialized_copy* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent1** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent2** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter2.

Parameters

- **first1** – Refers to the beginning of the sequence of elements that will be copied from
- **last1** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied
- **first2** – Refers to the beginning of the destination range.
- **last2** – Refers to sentinel value denoting the end of the second range the algorithm will be applied to.

⁷⁴⁷ <http://github.com/TheHPXProject/hpx/blob/d402eoad7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Returns

The *uninitialized_copy* algorithm returns an *in_out_result<InIter, FwdIter>*. The *uninitialized_copy* algorithm returns an input iterator to one past the last element copied from and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Sent1, typename FwdIter2, typename Sent2>
parallel::util::detail::algorithm_result<ExPolicy, parallel::util::in_out_result<FwdIter1, FwdIter2>>::type hpx::ranges::uninitial
```

Copies the elements in the range, defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the copy operation, the function has no effects.

The assignments in the parallel *uninitialized_copy* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized_copy* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent1** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter*.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent2** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter2*.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements that will be copied from
- **last1** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **first2** – Refers to the beginning of the destination range.

- **last2** – Refers to sentinel value denoting the end of the second range the algorithm will be applied to.

Returns

The *uninitialized_copy* algorithm returns a `hpx::future<in_out_result<InIter, FwdIter>>`, if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *in_out_result<InIter, FwdIter>* otherwise. The *uninitialized_copy* algorithm returns an input iterator to one past the last element copied from and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng1, typename Rng2>
hpx::parallel::util::in_out_result<typename hpx::traits::range_traits<Rng1>::iterator_type, typename hpx::traits::range_traits<Rng2>::it
```

Copies the elements in the range, defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the copy operation, the function has no effects.

The assignments in the parallel *uninitialized_copy* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the destination range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.

Parameters

- **rng1** – Refers to the range from which the elements will be copied from
- **rng2** – Refers to the range to which the elements will be copied to

Returns

The *uninitialized_copy* algorithm returns an *in_out_result<typename hpx::traits::range_traits<Rng1>::iterator_type, typename hpx::traits::range_traits<Rng2>::iterator_type>*. The *uninitialized_copy* algorithm returns an input iterator to one past the last element copied from and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng1, typename Rng2>
parallel::util::detail::algorithm_result<ExPolicy, hpx::parallel::util::in_out_result<typename hpx::traits::range_traits<Rng1>::iterator_t
```

Copies the elements in the range, defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the copy operation, the function has no effects.

The assignments in the parallel *uninitialized_copy* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized_copy* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the destination range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the range from which the elements will be copied from
- **rng2** – Refers to the range to which the elements will be copied to

Returns

The *uninitialized_copy* algorithm returns a `hpx::future<in_out_result<InIter, FwdIter>>`, if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `in_out_result< typename hpx::traits::range_traits<Rng1>::iterator_type , typename hpx::traits::range_traits<Rng2>::iterator_type>` otherwise. The *uninitialized_copy* algorithm returns the input iterator to one past the last element copied from and the output iterator to the element in the destination range, one past the last element copied.

hpx::ranges::uninitialized_copy_n

Defined in header `hpx/algorithm.hpp`⁷⁴⁸.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::uninitialized_copy`

```
template<typename InIter, typename Size, typename FwdIter, typename Sent2>
hpx::parallel::util::in_out_result<InIter, FwdIter> hpx::ranges::uninitialized_copy_n(InIter first1, Size
                                                    count, FwdIter first2,
                                                    Sent2 last2)
```

Copies the elements in the range `[first, first + count)`, starting from `first` and proceeding to `first + count - 1.`, to another range beginning at `dest`. If an exception is thrown during the copy operation, the function has no effects.

⁷⁴⁸ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

The assignments in the parallel *uninitialized_copy_n* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent2** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.

Parameters

- **first1** – Refers to the beginning of the sequence of elements that will be copied from
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **first2** – Refers to the beginning of the destination range.
- **last2** – Refers to sentinel value denoting the end of the second range the algorithm will be applied to.

Returns

The *uninitialized_copy_n* algorithm returns *in_out_result<InIter, FwdIter>*. The *uninitialized_copy_n* algorithm returns the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Size, typename FwdIter2, typename Sent2>
parallel::util::detail::algorithm_result<ExPolicy, parallel::util::in_out_result<FwdIter1, FwdIter2>>::type hpx::ranges::uninitial
```

Copies the elements in the range [first, first + count), starting from first and proceeding to first + count - 1., to another range beginning at dest. If an exception is thrown during the copy operation, the function has no effects.

The assignments in the parallel *uninitialized_copy_n* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized_copy_n* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent2** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter2`.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements that will be copied from
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **first2** – Refers to the beginning of the destination range.
- **last2** – Refers to sentinel value denoting the end of the second range the algorithm will be applied to.

Returns

The `uninitialized_copy_n` algorithm returns a `hpx::future<in_out_result<FwdIter1, FwdIter2>>` if the execution policy is of type `sequenced_task_policy` or `parallel_task_policy` and returns `FwdIter2` otherwise. The `uninitialized_copy_n` algorithm returns the output iterator to the element in the destination range, one past the last element copied.

hpx::ranges::experimental::for_loop

Defined in header `hpx/algorithm.hpp`⁷⁴⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::experimental::for_loop_strided`
- `hpx::ranges::experimental::for_loop_n`
- `hpx::ranges::experimental::for_loop_n_strided`

template<typename **ExPolicy**, typename **Iter**, typename **Sent**, typename ...**Args**>

⁷⁴⁹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

`hpx::parallel::util::detail::algorithm_result<ExPolicy>::type hpx::ranges::experimental::for_loop(ExPolicy
&&pol-
icy, Iter
first,
Sent last,
Args&&...
args)`

The `for_loop` implements loop functionality over a range specified by iterator bounds. These algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator.

Requires: *Iter* shall meet the requirements of a forward iterator type. The *args* parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function, *f*. *f* shall meet the requirements of Move-Constructible.

Effects: Applies *f* to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the *args* parameter pack. The length of the input sequence is `last - first`.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

Remarks: If *f* returns a result, the result is ignored.

Note: As described in the C++ standard, arithmetic on non-random-access iterators is performed using `advance` and `distance`.

Note: The order of the elements of the input sequence is important for determining ordinal position of an application of *f*, even though the applications themselves may be unordered.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Iter** – The type of the iteration variable (forward iterator).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *Iter*.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last) should expose a signature equivalent to:

```
<ignored> pred(Iter const& a, ...);
```

The signature does not need to have `const&`. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

Returns

The `for_loop` algorithm returns a `hpx::future<void>` if the execution policy is of type `hpx::execution::sequenced_task_policy` or `hpx::execution::parallel_task_policy` and returns `void` otherwise.

```
template<typename Iter, typename Sent, typename ...Args>
void hpx::ranges::experimental::for_loop(Iter first, Sent last, Args&&... args)
```

The `for_loop` implements loop functionality over a range specified by iterator bounds. These algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator.

The execution of `for_loop` without specifying an execution policy is equivalent to specifying `hpx::execution::seq` as the execution policy.

Requires: *Iter* shall meet the requirements of an input iterator type. The *args* parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function, *f*. *f* shall meet the requirements of Move-Constructible.

Effects: Applies *f* to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the *args* parameter pack. The length of the input sequence is `last - first`.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

Remarks: If *f* returns a result, the result is ignored.

Note: As described in the C++ standard, arithmetic on non-random-access iterators is performed using advance and distance.

Note: The order of the elements of the input sequence is important for determining ordinal position of an application of f , even though the applications themselves may be unordered.

Template Parameters

- **Iter** – The type of the iteration variable (input iterator).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for Iter.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last) should expose a signature equivalent to:

```
<ignored> pred(Iter const& a, ...);
```

The signature does not need to have const&. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

```
template<typename ExPolicy, typename R, typename ...Args>
hpx::parallel::util::detail::algorithm_result<ExPolicy>::type hpx::ranges::experimental::for_loop(ExPolicy
                                                                                               &&pol-
                                                                                               icy, R
                                                                                               &&rng,
                                                                                               Args&&...
                                                                                               args)
```

The `for_loop` implements loop functionality over a range specified by a range. These algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator.

Requires: `Rng::iterator` shall meet the requirements of a forward iterator type. The `args` parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function, f . f shall meet the requirements of MoveConstructible.

Effects: Applies f to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the `args` parameter pack. The length of the input sequence is `last - first`.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

Remarks: If *f* returns a result, the result is ignored.

Note: As described in the C++ standard, arithmetic on non-random-access iterators is performed using advance and distance.

Note: The order of the elements of the input sequence is important for determining ordinal position of an application of *f*, even though the applications themselves may be unordered.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **R** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last) should expose a signature equivalent to:

```
<ignored> pred(Rng::iterator const& a, ...);
```

The signature does not need to have const&. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

Returns

The *for_loop* algorithm returns a *hpx::future<void>* if the execution policy is of type *hpx::execution::sequenced_task_policy* or *hpx::execution::parallel_task_policy* and returns *void* otherwise.

```
template<typename Rng, typename ...Args>
```

```
void hpx::ranges::experimental::for_loop(Rng &&rng, Args&&... args)
```

The `for_loop` implements loop functionality over a range specified by a range. These algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator.

The execution of `for_loop` without specifying an execution policy is equivalent to specifying `hpx::execution::seq` as the execution policy.

Requires: `Rng::iterator` shall meet the requirements of an input iterator type. The `args` parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function, *f*. *f* shall meet the requirements of `MoveConstructible`.

Effects: Applies *f* to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the `args` parameter pack. The length of the input sequence is `last - first`.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the `args` parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

Remarks: If *f* returns a result, the result is ignored.

Note: As described in the C++ standard, arithmetic on non-random-access iterators is performed using `advance` and `distance`.

Note: The order of the elements of the input sequence is important for determining ordinal position of an application of *f*, even though the applications themselves may be unordered.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by `[first, last)` should expose a signature equivalent to:

```
<ignored> pred(Rng::iterator const& a, ...);
```

The signature does not need to have `const&`. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

`hpx::ranges::experimental::for_loop_strided`

Defined in header `hpx/algorithm.hpp`⁷⁵⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::experimental::for_loop`
- `hpx::ranges::experimental::for_loop_n`
- `hpx::ranges::experimental::for_loop_n_strided`

```
template<typename ExPolicy, typename Iter, typename Sent, typename S, typename ...Args>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy> hpx::ranges::experimental::for_loop_strided(ExPolicy
                                                         &&pol-
                                                         icy,
                                                         Iter
                                                         first,
                                                         Sent
                                                         last,
                                                         S
                                                         stride,
                                                         Args&&...
                                                         args)
```

The `for_loop_strided` implements loop functionality over a range specified by iterator bounds. These algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator.

Requires: *Iter* shall meet the requirements of a forward iterator type. The *args* parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function, *f*. *f* shall meet the requirements of Move-Constructible.

Effects: Applies *f* to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the *args* parameter pack. The length of the input sequence is `last - first`.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

⁷⁵⁰ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Complexity: Applies f exactly once for each element of the input sequence.

Remarks: If f returns a result, the result is ignored.

Note: As described in the C++ standard, arithmetic on non-random-access iterators is performed using advance and distance.

Note: The order of the elements of the input sequence is important for determining ordinal position of an application of f , even though the applications themselves may be unordered.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Iter** – The type of the iteration variable (forward iterator).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for Iter.
- **S** – The type of the stride variable. This should be an integral type.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **stride** – Refers to the stride of the iteration steps. This shall have non-zero value and shall be negative only if Iter meets the requirements a bidirectional iterator.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last) should expose a signature equivalent to:

```
<ignored> pred(Iter const& a, ...);
```

The signature does not need to have const&. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

Returns

The *for_loop_strided* algorithm returns a `hpx::future<void>` if the execution policy is of type `hpx::execution::sequenced_task_policy` or `hpx::execution::parallel_task_policy` and returns `void` otherwise.

```
template<typename Iter, typename Sent, typename S, typename ...Args>
```


void `hpx::ranges::experimental::for_loop_strided`(*Iter* first, *Sent* last, *S* stride, *Args*&&... args)

The `for_loop_strided` implements loop functionality over a range specified by iterator bounds. These algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator.

The execution of `for_loop_strided` without specifying an execution policy is equivalent to specifying `hpx::execution::seq` as the execution policy.

Requires: *Iter* shall meet the requirements of an input iterator type. The *args* parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function, *f*. *f* shall meet the requirements of Move-Constructible.

Effects: Applies *f* to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the *args* parameter pack. The length of the input sequence is `last - first`.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

Remarks: If *f* returns a result, the result is ignored.

Note: As described in the C++ standard, arithmetic on non-random-access iterators is performed using `advance` and `distance`.

Note: The order of the elements of the input sequence is important for determining ordinal position of an application of *f*, even though the applications themselves may be unordered.

Template Parameters

- **Iter** – The type of the iteration variable (input iterator).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *Iter*.
- **S** – The type of the stride variable. This should be an integral type.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.

- **stride** – Refers to the stride of the iteration steps. This shall have non-zero value and shall be negative only if *Iter* meets the requirements a bidirectional iterator.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last) should expose a signature equivalent to:

`<ignored> pred(Iter const& a, ...);`

The signature does not need to have `const&`. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

```
template<typename ExPolicy, typename Rng, typename S, typename ...Args>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy> hpx::ranges::experimental::for_loop_strided(ExPolicy
                                                         &&pol-
                                                         icy,
                                                         Rng
                                                         &&rng,
                                                         S
                                                         stride,
                                                         Args&&...
                                                         args)
```

The `for_loop_strided` implements loop functionality over a range specified by a range. These algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator.

Requires: *Rng::iterator* shall meet the requirements of a forward iterator type. The *args* parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function, *f*. *f* shall meet the requirements of `MoveConstructible`.

Effects: Applies *f* to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the *args* parameter pack. The length of the input sequence is `last - first`.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

Remarks: If *f* returns a result, the result is ignored.

Note: As described in the C++ standard, arithmetic on non-random-access iterators is performed using `advance` and `distance`.

Note: The order of the elements of the input sequence is important for determining ordinal position of an application of f , even though the applications themselves may be unordered.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **S** – The type of the stride variable. This should be an integral type.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **stride** – Refers to the stride of the iteration steps. This shall have non-zero value and shall be negative only if `Rng::iterator` meets the requirements a bidirectional iterator.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last) should expose a signature equivalent to:

```
<ignored> pred(Rng::iterator const& a, ...);
```

The signature does not need to have `const&`. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

Returns

The `for_loop_strided` algorithm returns a `hpx::future<void>` if the execution policy is of type `hpx::execution::sequenced_task_policy` or `hpx::execution::parallel_task_policy` and returns `void` otherwise.

```
template<typename Rng, typename S, typename ...Args>
```

```
void hpx::ranges::experimental::for_loop_strided(Rng &&rng, S stride, Args&&... args)
```

The `for_loop_strided` implements loop functionality over a range specified by a range. These algorithms resemble `for_each` from the Parallelism TS, but leave to the programmer when and if to dereference the iterator.

The execution of `for_loop_strided` without specifying an execution policy is equivalent to specifying `hpx::execution::seq` as the execution policy.

Requires: `Rng::iterator` shall meet the requirements of an input iterator type. The `args` parameter pack shall have at least one element, comprising objects returned by invocations of *reduction* and/or *induction* function templates followed by exactly one element invocable element-access function, f . f shall meet the requirements of `MoveConstructible`.

Effects: Applies f to each element in the input sequence, with additional arguments corresponding to the reductions and inductions in the `args` parameter pack. The length of the input sequence is `last - first`.

The first element in the input sequence is specified by *first*. Each subsequent element is generated by incrementing the previous element.

Along with an element from the input sequence, for each member of the *args* parameter pack excluding *f*, an additional argument is passed to each application of *f* as follows:

If the pack member is an object returned by a call to a reduction function listed in section, then the additional argument is a reference to a view of that reduction object. If the pack member is an object returned by a call to induction, then the additional argument is the induction value for that induction object corresponding to the position of the application of *f* in the input sequence.

Complexity: Applies *f* exactly once for each element of the input sequence.

Remarks: If *f* returns a result, the result is ignored.

Note: As described in the C++ standard, arithmetic on non-random-access iterators is performed using advance and distance.

Note: The order of the elements of the input sequence is important for determining ordinal position of an application of *f*, even though the applications themselves may be unordered.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **S** – The type of the stride variable. This should be an integral type.
- **Args** – A parameter pack, it's last element is a function object to be invoked for each iteration, the others have to be either conforming to the induction or reduction concept.

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **stride** – Refers to the stride of the iteration steps. This shall have non-zero value and shall be negative only if `Rng::iterator` meets the requirements a bidirectional iterator.
- **args** – The last element of this parameter pack is the function (object) to invoke, while the remaining elements of the parameter pack are instances of either induction or reduction objects. The function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last) should expose a signature equivalent to:

```
<ignored> pred(Rng::iterator const& a, ...);
```

The signature does not need to have `const&`. It will receive the current value of the iteration variable and one argument for each of the induction or reduction objects passed to the algorithms, representing their current values.

hpx::ranges::experimental::for_loop_n

Defined in header [hpx/algorithm.hpp](#)⁷⁵¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- [hpx::ranges::experimental::for_loop](#)
- [hpx::ranges::experimental::for_loop_strided](#)
- [hpx::ranges::experimental::for_loop_n_strided](#)

Warning: doxygenfunction: Cannot find function “hpx::ranges::experimental::for_loop_n” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

hpx::ranges::experimental::for_loop_n_strided

Defined in header [hpx/algorithm.hpp](#)⁷⁵².

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- [hpx::ranges::experimental::for_loop](#)
- [hpx::ranges::experimental::for_loop_strided](#)
- [hpx::ranges::experimental::for_loop_n](#)

Warning: doxygenfunction: Cannot find function “hpx::ranges::experimental::for_loop_n_strided” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

hpx::ranges::uninitialized_move

Defined in header [hpx/algorithm.hpp](#)⁷⁵³.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- [hpx::ranges::uninitialized_move_n](#)

template<typename **InIter**, typename **Sent1**, typename **FwdIter**, typename **Sent2**>

[hpx::parallel::util::in_out_result<InIter, FwdIter>](#) **hpx::ranges::uninitialized_move**(*InIter* first1, *Sent1* last1, *FwdIter* first2, *Sent2* last2)

Moves the elements in the range, defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the initialization, some objects in [first, last) are left in a valid but unspecified state.

⁷⁵¹ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

⁷⁵² <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

⁷⁵³ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

The assignments in the parallel *uninitialized_move* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent1** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent2** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter2.

Parameters

- **first1** – Refers to the beginning of the sequence of elements that will be moved from
- **last1** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied
- **first2** – Refers to the beginning of the destination range.
- **last2** – Refers to sentinel value denoting the end of the second range the algorithm will be applied to.

Returns

The *uninitialized_move* algorithm returns an *in_out_result<InIter, FwdIter>*. The *uninitialized_move* algorithm returns an input iterator to one past the last element moved from and the output iterator to the element in the destination range, one past the last element moved.

```
template<typename ExPolicy, typename FwdIter1, typename Sent1, typename FwdIter2, typename Sent2>
parallel::util::detail::algorithm_result<ExPolicy, parallel::util::in_out_result<FwdIter1, FwdIter2>>::type hpx::ranges::uninitial
```

Moves the elements in the range, defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the initialization, some objects in [first, last) are left in a valid but unspecified state.

The assignments in the parallel *uninitialized_move* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized_move* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent1** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter`.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent2** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter2`.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements that will be moved from
- **last1** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **first2** – Refers to the beginning of the destination range.
- **last2** – Refers to sentinel value denoting the end of the second range the algorithm will be applied to.

Returns

The *uninitialized_move* algorithm returns a `hpx::future<in_out_result<InIter, FwdIter>>`, if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *in_out_result<InIter, FwdIter>* otherwise. The *uninitialized_move* algorithm returns an input iterator to one past the last element moved from and the output iterator to the element in the destination range, one past the last element moved.

```
template<typename Rng1, typename Rng2>
hpx::parallel::util::in_out_result<typename hpx::traits::range_traits<Rng1>::iterator_type, typename hpx::traits::range_traits<Rng2>::it
```

Moves the elements in the range, defined by `[first, last)`, to an uninitialized memory area beginning at *dest*. If an exception is thrown during the initialization, some objects in `[first, last)` are left in a valid but unspecified state.

The assignments in the parallel *uninitialized_move* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the destination range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.

Parameters

- **rng1** – Refers to the range from which the elements will be moved from
- **rng2** – Refers to the range to which the elements will be moved to

Returns

The *uninitialized_move* algorithm returns an *in_out_result*<typename `hpx::traits::range_traits<Rng1>::iterator_type`, typename `hpx::traits::range_traits<Rng2>::iterator_type`>. The *uninitialized_move* algorithm returns an input iterator to one past the last element moved from and the output iterator to the element in the destination range, one past the last element moved.

```
template<typename ExPolicy, typename Rng1, typename Rng2>  
parallel::util::detail::algorithm_result<ExPolicy, hpx::parallel::util::in_out_result<typename hpx::traits::range_traits<Rng1>::iterator_t
```

Moves the elements in the range, defined by [first, last), to an uninitialized memory area beginning at *dest*. If an exception is thrown during the initialization, some objects in [first, last) are left in a valid but unspecified state.

The assignments in the parallel *uninitialized_move* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized_move* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the destination range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the range from which the elements will be moved from
- **rng2** – Refers to the range to which the elements will be moved to

Returns

The *uninitialized_move* algorithm returns a `hpx::future<in_out_result<InIter, FwdIter>>`, if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `in_out_result< typename hpx::traits::range_traits<Rng1>::iterator_type , typename hpx::traits::range_traits<Rng2>::iterator_type>` otherwise. The *uninitialized_move* algorithm returns the input iterator to one past the last element moved from and the output iterator to the element in the destination range, one past the last element moved.

hpx::ranges::uninitialized_move_n

Defined in header `hpx/algorithm.hpp`⁷⁵⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::uninitialized_move`

```
template<typename InIter, typename Size, typename FwdIter, typename Sent2>
hpx::parallel::util::in_out_result<InIter, FwdIter> hpx::ranges::uninitialized_move_n(InIter first1, Size
count, FwdIter first2,
Sent2 last2)
```

Moves the elements in the range `[first, first + count)`, starting from `first` and proceeding to `first + count - 1`, to another range beginning at `dest`. If an exception is thrown during the initialization, some objects in `[first, first + count)` are left in a valid but unspecified state.

The assignments in the parallel *uninitialized_move_n* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *count* movements, if *count* > 0, no move operations otherwise.

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent2** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `FwdIter`.

Parameters

- **first1** – Refers to the beginning of the sequence of elements that will be moved from
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **first2** – Refers to the beginning of the destination range.
- **last2** – Refers to sentinel value denoting the end of the second range the algorithm will be applied to.

⁷⁵⁴ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Returns

The *uninitialized_move_n* algorithm returns *in_out_result<InIter, FwdIter>*. The *uninitialized_move_n* algorithm returns the output iterator to the element in the destination range, one past the last element moved.

```
template<typename ExPolicy, typename FwdIter1, typename Size, typename FwdIter2, typename Sent2>
parallel::util::detail::algorithm_result<ExPolicy, parallel::util::in_out_result<FwdIter1, FwdIter2>>::type hpx::ranges::uninitial
```

Moves the elements in the range [first, first + count), starting from first and proceeding to first + count - 1., to another range beginning at dest. If an exception is thrown during the initialization, some objects in [first, first + count) are left in a valid but unspecified state.

The assignments in the parallel *uninitialized_move_n* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized_move_n* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *count* movements, if count > 0, no move operations otherwise.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent2** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter2.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements that will be moved from
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.
- **first2** – Refers to the beginning of the destination range.

- **last2** – Refers to sentinel value denoting the end of the second range the algorithm will be applied to.

Returns

The *uninitialized_move_n* algorithm returns a `hpx::future<in_out_result<FwdIter1, FwdIter2>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter2* otherwise. The *uninitialized_move_n* algorithm returns the output iterator to the element in the destination range, one past the last element moved.

hpx::ranges::merge

Defined in header `hpx/algorithm.hpp`⁷⁵⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::ranges::inplace_merge*

```
template<typename ExPolicy, typename Rng1, typename Rng2, typename Iter3, typename Comp =
hpx::ranges::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, hpx::ranges::merge_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<
```

Merges two sorted ranges `[first1, last1)` and `[first2, last2)` into one sorted range beginning at *dest*. The order of equivalent elements in the each of original two ranges is preserved. For equivalent elements in the original two ranges, the elements from the first range precede the elements from the second range. The destination range cannot overlap with either of the input ranges.

The assignments in the parallel *merge* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *merge* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

⁷⁵⁵ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Note: Complexity: Performs $O(\text{std::distance}(\text{first1}, \text{last1}) + \text{std::distance}(\text{first2}, \text{last2}))$ applications of the comparison *comp* and the each projection.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng1** – The type of the first source range used (deduced). The iterators extracted from this range type must meet the requirements of a random access iterator.
- **Rng2** – The type of the second source range used (deduced). The iterators extracted from this range type must meet the requirements of a random access iterator.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *merge* requires *Comp* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function to be used for elements of the first range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function to be used for elements of the second range. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the first range of elements the algorithm will be applied to.
- **rng2** – Refers to the second range of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **comp** – *comp* is a callable object which returns true if the first argument is less than the second, and false otherwise. The signature of this comparison should be equivalent to:

```
bool comp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *Iter1* and *Iter2* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual comparison *comp* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual comparison *comp* is invoked.

Returns

The *merge* algorithm returns a `hpx::future<merge_result<Iter1, Iter2, Iter3>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *merge_result<Iter1, Iter2, Iter3>* otherwise. The *merge* algorithm returns the tuple of the source iterator *last1*, the source iterator *last2*, the destination iterator to the end of the *dest* range.

```

template<typename ExPolicy, typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename
Iter3, typename Comp = hpx::ranges::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
hpx::parallel::util::detail::algorithm_result<ExPolicy, hpx::ranges::merge_result<Iter1, Iter2, Iter3>>::type hpx::ranges::merge(Ex
&
icy
Iter
fir
Se
las
Iter
fir
Se
las
Iter
de
Co
&
=
Co
Pr
&
=
Pr
Pr
&
=
Pr

```

Merges two sorted ranges `[first1, last1)` and `[first2, last2)` into one sorted range beginning at *dest*. The order of equivalent elements in the each of original two ranges is preserved. For equivalent elements in the original two ranges, the elements from the first range precede the elements from the second range. The destination range cannot overlap with either of the input ranges.

The assignments in the parallel *merge* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *merge* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs $O(\text{std::distance}(\text{first1}, \text{last1}) + \text{std::distance}(\text{first2}, \text{last2}))$ applications of the comparison *comp* and the each projection.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of a random access iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of a sentinel for *Iter1*.

- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an random access iterator.
- **Sent2** – The type of the end source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an sentinel for Iter2.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an random access iterator.
- **Comp** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *merge* requires *Comp* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function to be used for elements of the first range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function to be used for elements of the second range. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **comp** – *comp* is a callable object which returns true if the first argument is less than the second, and false otherwise. The signature of this comparison should be equivalent to:

```
bool comp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *Iter1* and *Iter2* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual comparison *comp* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual comparison *comp* is invoked.

Returns

The *merge* algorithm returns a `hpx::future<merge_result<Iter1, Iter2, Iter3>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *merge_result<Iter1, Iter2, Iter3>* otherwise. The *merge* algorithm returns the tuple of the source iterator *last1*, the source iterator *last2*, the destination iterator to the end of the *dest* range.

```
template<typename Rng1, typename Rng2, typename Iter3, typename Comp = hpx::ranges::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
```

```

hpx::ranges::merge_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng2>, Iter3> hpx::ranges::merge(Rng1
&&rng1,
Rng2
&&rng2,
Iter3
dest,
Comp
&&comp
=
Comp(),
Proj1
&&proj1
=
Proj1(),
Proj2
&&proj2
=
Proj2())

```

Merges two sorted ranges [first1, last1) and [first2, last2) into one sorted range beginning at *dest*. The order of equivalent elements in the each of original two ranges is preserved. For equivalent elements in the original two ranges, the elements from the first range precede the elements from the second range. The destination range cannot overlap with either of the input ranges.

Note: Complexity: Performs $O(\text{std::distance}(\text{first1}, \text{last1}) + \text{std::distance}(\text{first2}, \text{last2}))$ applications of the comparison *comp* and the each projection.

Template Parameters

- **Rng1** – The type of the first source range used (deduced). The iterators extracted from this range type must meet the requirements of a random access iterator.
- **Rng2** – The type of the second source range used (deduced). The iterators extracted from this range type must meet the requirements of a random access iterator.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *merge* requires *Comp* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function to be used for elements of the first range. This defaults to `hpx::identity`
- **Proj2** – The type of an optional projection function to be used for elements of the second range. This defaults to `hpx::identity`

Parameters

- **rng1** – Refers to the first range of elements the algorithm will be applied to.
- **rng2** – Refers to the second range of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **comp** – *comp* is a callable object which returns true if the first argument is less than the second, and false otherwise. The signature of this comparison should be equivalent to:

```
bool comp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *Iter1* and *Iter2* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual comparison *comp* is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual comparison *comp* is invoked.

Returns

The *merge* algorithm returns *merge_result*<*Iter1*, *Iter2*, *Iter3*>. The *merge* algorithm returns the tuple of the source iterator *last1*, the source iterator *last2*, the destination iterator to the end of the *dest* range.

```
template<typename Iter1, typename Sent1, typename Iter2, typename Sent2, typename Iter3, typename Comp
= hpx::ranges::less, typename Proj1 = hpx::identity, typename Proj2 = hpx::identity>
hpx::ranges::merge_result<Iter1, Iter2, Iter3> hpx::ranges::merge(Iter1 first1, Sent1 last1, Iter2 first2, Sent2
                                                                    last2, Iter3 dest, Comp &&comp = Comp(),
                                                                    Proj1 &&proj1 = Proj1(), Proj2 &&proj2 =
                                                                    Proj2())
```

Merges two sorted ranges [*first1*, *last1*) and [*first2*, *last2*) into one sorted range beginning at *dest*. The order of equivalent elements in the each of original two ranges is preserved. For equivalent elements in the original two ranges, the elements from the first range precede the elements from the second range. The destination range cannot overlap with either of the input ranges.

Note: Complexity: Performs $O(\text{std::distance}(\text{first1}, \text{last1}) + \text{std::distance}(\text{first2}, \text{last2}))$ applications of the comparison *comp* and the each projection.

Template Parameters

- **Iter1** – The type of the source iterators used (deduced) representing the first sequence. This iterator type must meet the requirements of an random access iterator.
- **Sent1** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of an sentinel for *Iter1*.
- **Iter2** – The type of the source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an random access iterator.
- **Sent2** – The type of the end source iterators used (deduced) representing the second sequence. This iterator type must meet the requirements of an sentinel for *Iter2*.
- **Iter3** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an random access iterator.
- **Comp** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *merge* requires *Comp* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj1** – The type of an optional projection function to be used for elements of the first range. This defaults to *hpx::identity*

- **Proj2** – The type of an optional projection function to be used for elements of the second range. This defaults to `hpx::identity`

Parameters

- **first1** – Refers to the beginning of the sequence of elements of the first range the algorithm will be applied to.
- **last1** – Refers to the end of the sequence of elements of the first range the algorithm will be applied to.
- **first2** – Refers to the beginning of the sequence of elements of the second range the algorithm will be applied to.
- **last2** – Refers to the end of the sequence of elements of the second range the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **comp** – `comp` is a callable object which returns true if the first argument is less than the second, and false otherwise. The signature of this comparison should be equivalent to:

```
bool comp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *Iter1* and *Iter2* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

- **proj1** – Specifies the function (or function object) which will be invoked for each of the elements of the first range as a projection operation before the actual comparison `comp` is invoked.
- **proj2** – Specifies the function (or function object) which will be invoked for each of the elements of the second range as a projection operation before the actual comparison `comp` is invoked.

Returns

The *merge* algorithm returns `merge_result<Iter1, Iter2, Iter3>`. The *merge* algorithm returns the tuple of the source iterator *last1*, the source iterator *last2*, the destination iterator to the end of the *dest* range.

hpx::ranges::inplace_merge

Defined in header `hpx/algorithm.hpp`⁷⁵⁶.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::merge`

```
template<typename ExPolicy, typename Rng, typename Iter, typename Comp = hpx::ranges::less, typename Proj = hpx::identity>
```

⁷⁵⁶ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

```
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, Iter> hpx::ranges::inplace_merge(ExPolicy
                                                                    &&policy, Rng
                                                                    &&rng, Iter
                                                                    middle, Comp
                                                                    &&comp =
                                                                    Comp(), Proj
                                                                    &&proj = Proj())
```

Merges two consecutive sorted ranges [first, middle) and [middle, last) into one sorted range [first, last). The order of equivalent elements in the each of original two ranges is preserved. For equivalent elements in the original two ranges, the elements from the first range precede the elements from the second range.

The assignments in the parallel *inplace_merge* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *inplace_merge* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs $O(\text{std::distance}(\text{first}, \text{last}))$ applications of the comparison *comp* and the each projection.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a random access iterator.
- **Iter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Comp** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *inplace_merge* requires *Comp* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the range of elements the algorithm will be applied to.
- **middle** – Refers to the end of the first sorted range and the beginning of the second sorted range the algorithm will be applied to.
- **comp** – *comp* is a callable object which returns true if the first argument is less than the second, and false otherwise. The signature of this comparison should be equivalent to:

```
bool comp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *Iter* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *inplace_merge* algorithm returns a `hpx::future<Iter>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *Iter* otherwise. The *inplace_merge* algorithm returns the source iterator *last*

```
template<typename ExPolicy, typename Iter, typename Sent, typename Comp = hpx::ranges::less, typename
Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, Iter> hpx::ranges::inplace_merge(ExPolicy
&&policy, Iter
first, Iter middle,
Sent last, Comp
&&comp =
Comp(), Proj
&&proj = Proj())
```

Merges two consecutive sorted ranges [first, middle) and [middle, last) into one sorted range [first, last). The order of equivalent elements in the each of original two ranges is preserved. For equivalent elements in the original two ranges, the elements from the first range precede the elements from the second range.

The assignments in the parallel *inplace_merge* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *inplace_merge* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs $O(\text{std::distance}(\text{first}, \text{last}))$ applications of the comparison *comp* and the each projection.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of a sentinel for *Iter*1.
- **Comp** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *inplace_merge* requires *Comp* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the first sorted range the algorithm will be applied to.
- **middle** – Refers to the end of the first sorted range and the beginning of the second sorted range the algorithm will be applied to.

- **last** – Refers to the end of the second sorted range the algorithm will be applied to.
- **comp** – *comp* is a callable object which returns true if the first argument is less than the second, and false otherwise. The signature of this comparison should be equivalent to:

```
bool comp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *Iter* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *inplace_merge* algorithm returns a `hpx::future<Iter>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *Iter* otherwise. The *inplace_merge* algorithm returns the source iterator *last*

```
template<typename Rng, typename Iter, typename Comp = hpx::ranges::less, typename Proj = hpx::identity>  
Iter hpx::ranges::inplace_merge(Rng &&rng, Iter middle, Comp &&comp = Comp(), Proj &&proj = Proj())
```

Merges two consecutive sorted ranges [first, middle) and [middle, last) into one sorted range [first, last). The order of equivalent elements in the each of original two ranges is preserved. For equivalent elements in the original two ranges, the elements from the first range precede the elements from the second range.

Note: Complexity: Performs $O(\text{std::distance}(\text{first}, \text{last}))$ applications of the comparison *comp* and the each projection.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an random access iterator.
- **Iter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an random access iterator.
- **Comp** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *inplace_merge* requires *Comp* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **rng** – Refers to the range of elements the algorithm will be applied to.
- **middle** – Refers to the end of the first sorted range and the beginning of the second sorted range the algorithm will be applied to.
- **comp** – *comp* is a callable object which returns true if the first argument is less than the second, and false otherwise. The signature of this comparison should be equivalent to:

```
bool comp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *Iter* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *inplace_merge* algorithm returns *Iter*. The *inplace_merge* algorithm returns the source iterator *last*

```
template<typename Iter, typename Sent, typename Comp = hpx::ranges::less, typename Proj = hpx::identity>
Iter hpx::ranges::inplace_merge(Iter first, Iter middle, Sent last, Comp &&comp = Comp(), Proj &&proj =
    Proj())
```

Merges two consecutive sorted ranges [first, middle) and [middle, last) into one sorted range [first, last). The order of equivalent elements in the each of original two ranges is preserved. For equivalent elements in the original two ranges, the elements from the first range precede the elements from the second range.

Note: Complexity: Performs $O(\text{std::distance}(\text{first}, \text{last}))$ applications of the comparison *comp* and the each projection.

Template Parameters

- **Iter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Sent** – The type of the end source iterators used (deduced). This iterator type must meet the requirements of a sentinel for *Iter*.
- **Comp** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *inplace_merge* requires *Comp* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **first** – Refers to the beginning of the first sorted range the algorithm will be applied to.
- **middle** – Refers to the end of the first sorted range and the beginning of the second sorted range the algorithm will be applied to.
- **last** – Refers to the end of the second sorted range the algorithm will be applied to.
- **comp** – *comp* is a callable object which returns true if the first argument is less than the second, and false otherwise. The signature of this comparison should be equivalent to:

```
bool comp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *Iter* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *inplace_merge* algorithm *Iter*. The *inplace_merge* algorithm returns the source iterator *last*

hpx::ranges::replace

Defined in header [hpx/algorithm.hpp](#)⁷⁵⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::replace_if`
- `hpx::ranges::replace_copy`
- `hpx::ranges::replace_copy_if`

```
template<typename Iter, typename Sent, typename Proj = hpx::identity, typename T1 = typename  
hpx::parallel::traits::projected<Iter, Proj>::value_type, typename T2 = T1>  
Iter hpx::ranges::replace(Iter first, Sent sent, T1 const &old_value, T2 const &new_value, Proj &&proj =  
Proj())
```

Replaces all elements satisfying specific criteria with *new_value* in the range [first, last).

Effects: Substitutes elements referred by the iterator *it* in the range [first,last) with *new_value*, when the following corresponding conditions hold: `INVOKE(proj, *i) == old_value`

The assignments in the parallel *replace* algorithm execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **Iter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *Iter*.
- **T1** – The type of the old value to replace (deduced).
- **T2** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **old_value** – Refers to the old value of the elements to replace.
- **new_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *replace* algorithm returns an *Iter*.

⁷⁵⁷ <http://github.com/TheHPXProject/hpx/blob/d402cead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

```
template<typename Rng, typename Proj = hpx::identity, typename T1 = typename
hpx::parallel::traits::projected<std::ranges::iterator_t<Rng>, Proj>::value_type, typename T2 = T1>
std::ranges::iterator_t<Rng> hpx::ranges::replace(Rng &&rng, T1 const &old_value, T2 const &new_value,
                                                Proj &&proj = Proj())
```

Replaces all elements satisfying specific criteria with *new_value* in the range *rng*.

Effects: Substitutes elements referred by the iterator it in the range *rng* with *new_value*, when the following corresponding conditions hold: `INVOKE(proj, *i) == old_value`

The assignments in the parallel *replace* algorithm execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *util::end(rng) - util::begin(rng)* assignments.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a forward iterator.
- **T1** – The type of the old value to replace (deduced).
- **T2** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **old_value** – Refers to the old value of the elements to replace.
- **new_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *replace* algorithm returns an *hpx::traits::range_iterator<Rng>::type*.

```
template<typename ExPolicy, typename Iter, typename Sent, typename Proj = hpx::identity, typename T1 =
typename hpx::parallel::traits::projected<Iter, Proj>::value_type, typename T2 = T1>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, Iter> hpx::ranges::replace(ExPolicy &&policy, Iter
                                                                                   first, Sent sent, T1 const
                                                                                   &old_value, T2 const
                                                                                   &new_value, Proj &&proj
                                                                                   = Proj())
```

Replaces all elements satisfying specific criteria with *new_value* in the range [first, last).

Effects: Substitutes elements referred by the iterator it in the range [first,last) with *new_value*, when the following corresponding conditions hold: `INVOKE(proj, *i) == old_value`

The assignments in the parallel *replace* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *replace* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for Iter.
- **T1** – The type of the old value to replace (deduced).
- **T2** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **old_value** – Refers to the old value of the elements to replace.
- **new_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *replace* algorithm returns a *hpx::future<Iter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *Iter* otherwise.

```
template<typename ExPolicy, typename Rng, typename Proj = hpx::identity, typename T1 = typename  
hpx::parallel::traits::projected<std::ranges::iterator_t<Rng>, Proj>::value_type, typename T2 = T1>  
parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng>> hpx::ranges::replace(ExPolicy  
                                                    &&pol-  
                                                    icy, Rng  
                                                    &&rng,  
                                                    T1 const  
                                                    &old_value,  
                                                    T2 const  
                                                    &new_value,  
                                                    Proj  
                                                    &&proj =  
                                                    Proj())
```

Replaces all elements satisfying specific criteria with *new_value* in the range *rng*.

Effects: Substitutes elements referred by the iterator *it* in the range *rng* with *new_value*, when the following corresponding conditions hold: `INVOKE(proj, *i) == old_value`

The assignments in the parallel *replace* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *util::end(rng) - util::begin(rng)* assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a forward iterator.
- **T1** – The type of the old value to replace (deduced).
- **T2** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **old_value** – Refers to the old value of the elements to replace.
- **new_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked. The assignments in the parallel *replace* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

Returns

The *replace* algorithm returns an `hpx::future<hpx::traits::range_iterator<Rng>::type>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `hpx::traits::range_iterator<Rng>::type` otherwise.

`hpx::ranges::replace_if`

Defined in header `hpx/algorithm.hpp`⁷⁵⁸.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::replace`
- `hpx::ranges::replace_copy`

⁷⁵⁸ <http://github.com/TheHPXProject/hpx/blob/d402eed7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- `hpx::ranges::replace_copy_if`

```
template<typename Iter, typename Sent, typename Pred, typename Proj = hpx::identity, typename T = typename  
hpx::parallel::traits::projected<Iter, Proj>::value_type>
```

```
Iter hpx::ranges::replace_if(Iter first, Sent sent, Pred &&pred, T const &new_value, Proj &&proj = Proj())
```

Replaces all elements satisfying specific criteria (for which predicate *f* returns true) with *new_value* in the range [first, sent).

Effects: Substitutes elements referred by the iterator *it* in the range [first, sent) with *new_value*, when the following corresponding conditions hold: `INVOKE(f, INVOKE(proj, *it)) != false`

The assignments in the parallel *replace_if* algorithm execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *sent - first* applications of the predicate.

Template Parameters

- **Iter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *Iter*.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*. (deduced).
- **T** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the elements which need to be replaced. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *Iter* can be dereferenced and then implicitly converted to *Type*.

- **new_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *replace_if* algorithm returns *Iter*. It returns *last*.

```
template<typename Rng, typename Pred, typename Proj = hpx::identity, typename T = typename
hpx::parallel::traits::projected<std::ranges::iterator_t<Rng>, Proj>::value_type>
std::ranges::iterator_t<Rng> hpx::ranges::replace_if(Rng &&rng, Pred &&pred, T const &new_value, Proj
&&proj = Proj())
```

Replaces all elements satisfying specific criteria (for which predicate *pred* returns true) with *new_value* in the range *rng*.

Effects: Substitutes elements referred by the iterator it in the range *rng* with *new_value*, when the following corresponding conditions hold: `INVOKE(f, INVOKE(proj, *it)) != false`

Note: Complexity: Performs exactly `util::end(rng) - util::begin(rng)` applications of the predicate.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*. (deduced).
- **T** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by *rng*. This is an unary predicate which returns *true* for the elements which need to be replaced. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **new_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *replace_if* algorithm returns an *hpx::traits::range_iterator<Rng>::type*. It returns *last*.

```
template<typename ExPolicy, typename Iter, typename Sent, typename Pred, typename Proj = hpx::identity,
typename T = typename hpx::parallel::traits::projected<Iter, Proj>::value_type>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, Iter> hpx::ranges::replace_if(ExPolicy &&policy,
Iter first, Sent sent,
Pred &&pred, T const
&new_value, Proj
&&proj = Proj())
```

Replaces all elements satisfying specific criteria (for which predicate *pred* returns true) with *new_value* in the range *rng*.

Effects: Substitutes elements referred by the iterator *it* in the range *rng* with *new_value*, when the following corresponding conditions hold: `INVOKE(f, INVOKE(proj, *it)) != false`

The assignments in the parallel *replace_if* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *replace_if* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly `util::end(rng) - util::begin(rng)` applications of the predicate.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Iter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *Iter*.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. (deduced).
- **T** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [*first*, *last*). This is a unary predicate which returns *true* for the elements which need to be replaced. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **new_value** – Refers to the new value to use as the replacement.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *replace_if* algorithm returns a `hpx::future<Iter>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy*. It returns *last*.

```
template<typename ExPolicy, typename Rng, typename Pred, typename Proj = hpx::identity, typename T =
typename hpx::parallel::traits::projected<std::ranges::iterator_t<Rng>, Proj>::value_type>
parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng>> hpx::ranges::replace_if(ExPolicy
&&pol-
icy,
Rng
&&rng,
Pred
&&pred,
T
const
&new_value,
Proj
&&proj
=
Proj())
```

Replaces all elements satisfying specific criteria (for which predicate *pred* returns true) with *new_value* in the range *rng*.

Effects: Substitutes elements referred by the iterator *it* in the range *rng* with *new_value*, when the following corresponding conditions hold: `INVOKE(f, INVOKE(proj, *it)) != false`

The assignments in the parallel *replace* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *replace* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly `util::end(rng) - util::begin(rng)` applications of the predicate.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *F* to meet the requirements of *CopyConstructible*. (deduced).
- **T** – The type of the new values to replace (deduced).

- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by *rng*. This is an unary predicate which returns *true* for the elements which need to be replaced. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **new_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *replace_if* algorithm returns a `hpx::future<std::ranges::iterator_t<Rng>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy*. It returns *last*.

hpx::ranges::replace_copy

Defined in header `hpx/algorithm.hpp`⁷⁵⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::ranges::replace*
- *hpx::ranges::replace_if*
- *hpx::ranges::replace_copy_if*

```
template<typename InIter, typename Sent, typename OutIter, typename Proj = hpx::identity, typename T1 =
typename hpx::parallel::traits::projected<InIter, Proj>::value_type, typename T2 = T1>
replace_copy_result<InIter, OutIter> hpx::ranges::replace_copy(InIter first, Sent sent, OutIter dest, T1 const
&old_value, T2 const &new_value, Proj
&&proj = Proj())
```

Copies the all elements from the range `[first, sent)` to another range beginning at *dest* replacing all elements satisfying a specific criteria with *new_value*.

Effects: Assigns to every iterator *it* in the range `[result, result + (sent - first))` either *new_value* or `*(first + (it - result))` depending on whether the following corresponding condition holds: `INVOKE(proj, *(first + (i - result))) == old_value`

The assignments in the parallel *replace_copy* algorithm execute in sequential order in the calling thread.

⁷⁵⁹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e77/libs/full/include/include/hpx/algorithm.hpp>

Note: Complexity: Performs exactly *sent - first* applications of the predicate.

Template Parameters

- **InIter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for Iter.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **T1** – The type of the old value to replace (deduced).
- **T2** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **old_value** – Refers to the old value of the elements to replace.
- **new_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *replace_copy* algorithm returns an *in_out_result<InIter, OutIter>*. The *copy* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng, typename OutIter, typename Proj = hpx::identity, typename T1 = typename
hpx::parallel::traits::projected<std::ranges::iterator_t<Rng>, Proj>::value_type, typename T2 = T1>
replace_copy_result<std::ranges::iterator_t<Rng>, OutIter> hpx::ranges::replace_copy(Rng &&rng, OutIter
                                                                    dest, T1 const
                                                                    &old_value, T2 const
                                                                    &new_value, Proj
                                                                    &&proj = Proj())
```

Copies the all elements from the range *rbg* to another range beginning at *dest* replacing all elements satisfying a specific criteria with *new_value*.

Effects: Assigns to every iterator *it* in the range $[result, result + (util::end(rng) - util::begin(rng)))$ either *new_value* or $*(first + (it - result))$ depending on whether the following corresponding condition holds: `INVOKE(proj, *(first + (i - result))) == old_value`

The assignments in the parallel *replace_copy* algorithm execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *util::end(rng) - util::begin(rng)* applications of the predicate.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **T1** – The type of the old value to replace (deduced).
- **T2** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **old_value** – Refers to the old value of the elements to replace.
- **new_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *replace_copy* algorithm returns an *in_out_result*<*std::ranges::iterator_t*<*Rng*>, *OutIter*>. The *copy* algorithm returns the pair of the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Sent, typename FwdIter2, typename Proj =  
hpx::identity, typename T1 = typename hpx::parallel::traits::projected<FwdIter1, Proj>::value_type, typename T2 =  
T1>
```

```
parallel::util::detail::algorithm_result<ExPolicy, replace_copy_result<FwdIter1, FwdIter2>>::type hpx::ranges::replace_copy(E
```

Copies the all elements from the range [first, sent) to another range beginning at *dest* replacing all elements satisfying a specific criteria with *new_value*.

Effects: Assigns to every iterator it in the range [result, result + (sent - first)) either *new_value* or **(first + (it - result))* depending on whether the following corresponding condition holds: *INVOKE(proj, *(first + (i - result))) == old_value*

The assignments in the parallel *replace_copy* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *replace_copy* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *sent - first* applications of the predicate.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for Iter.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **T1** – The type of the old value to replace (deduced).
- **T2** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **old_value** – Refers to the old value of the elements to replace.
- **new_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *replace_copy* algorithm returns a `hpx::future<in_out_result<FwdIter1, FwdIter2>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `in_out_result<FwdIter1, FwdIter2>` otherwise. The *copy* algorithm returns the pair of the forward iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng, typename FwdIter, typename Proj = hpx::identity, typename T1 =
typename hpx::parallel::traits::projected<std::ranges::iterator_t<Rng>, Proj>::value_type, typename T2 = T1>
```

`parallel::util::detail::algorithm_result<ExPolicy, replace_copy_result<std::ranges::iterator_t<Rng>, FwdIter>>::type hpx::ranges:: :`

Copies the all elements from the range `rbg` to another range beginning at `dest` replacing all elements satisfying a specific criteria with `new_value`.

Effects: Assigns to every iterator `it` in the range `[result, result + (util::end(rng) - util::begin(rng))]` either `new_value` or `*(first + (it - result))` depending on whether the following corresponding condition holds: `INVOKE(proj, *(first + (i - result))) == old_value`

The assignments in the parallel `replace_copy` algorithm invoked with an execution policy object of type `sequenced_policy` execute in sequential order in the calling thread.

The assignments in the parallel `replace_copy` algorithm invoked with an execution policy object of type `parallel_policy` or `parallel_task_policy` are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly `util::end(rng) - util::begin(rng)` applications of the predicate.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **T1** – The type of the old value to replace (deduced).
- **T2** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.

- **dest** – Refers to the beginning of the destination range.
- **old_value** – Refers to the old value of the elements to replace.
- **new_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *replace_copy* algorithm returns a `hpx::future<in_out_result< std::ranges::iterator_t<Rng>, FwdIter>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `in_out_result< std::ranges::iterator_t<Rng>, FwdIter>>` The *copy* algorithm returns the pair of the input iterator *last* and the forward iterator to the element in the destination range, one past the last element copied.

hpx::ranges::replace_copy_if

Defined in header `hpx/algorithm.hpp`⁷⁶⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::replace`
- `hpx::ranges::replace_if`
- `hpx::ranges::replace_copy`

```
template<typename InIter, typename Sent, typename OutIter, typename Pred, typename T = typename
std::iterator_traits<OutIter>::value_type, typename Proj = hpx::identity>
replace_copy_if_result<InIter, OutIter> hpx::ranges::replace_copy_if(InIter first, Sent sent, OutIter dest,
                                                                    Pred &&pred, T const &new_value,
                                                                    Proj &&proj = Proj())
```

Copies the all elements from the range `[first, sent)` to another range beginning at *dest* replacing all elements satisfying a specific criteria with *new_value*.

Effects: Assigns to every iterator *it* in the range `[result, result + (sent - first))` either *new_value* or `*(first + (it - result))` depending on whether the following corresponding condition holds: `INVOKE(f, INVOKE(proj, *(first + (i - result)))) != false`

The assignments in the parallel *replace_copy_if* algorithm execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *sent - first* applications of the predicate.

Template Parameters

- **InIter** – The type of the source iterator used (deduced). The iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for *InIter*.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.

⁷⁶⁰ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. (deduced).
- **T** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the elements which need to be replaced. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **new_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *replace_copy_if* algorithm returns a *in_out_result<InIter, OutIter>*. The *replace_copy_if* algorithm returns the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename Rng, typename OutIter, typename Pred, typename T = typename
std::iterator_traits<OutIter>::value_type, typename Proj = hpx::identity>
replace_copy_if_result<std::ranges::iterator_t<Rng>, OutIter> hpx::ranges::replace_copy_if(Rng &&rng,
                                                                                          OutIter dest,
                                                                                          Pred &&pred,
                                                                                          T const
                                                                                          &new_value,
                                                                                          Proj &&proj =
                                                                                          Proj())
```

Copies the all elements from the range *rng* to another range beginning at *dest* replacing all elements satisfying a specific criteria with *new_value*.

Effects: Assigns to every iterator *it* in the range `[result, result + (util::end(rng) - util::begin(rng)))` either *new_value* or `*(first + (it - result))` depending on whether the following corresponding condition holds: `INVOKE(f, INVOKE(proj, *(first + (i - result)))) != false`

The assignments in the parallel *replace_copy_if* algorithm execute in sequential order in the calling thread.

Note: Complexity: Performs exactly `util::end(rng) - util::begin(rng)` applications of the predicate.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **OutIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an output iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. (deduced).
- **T** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the elements which need to be replaced. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **new_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *replace_copy_if* algorithm returns an *in_out_result<std::ranges::iterator_t<Rng>, OutIter>*. The *replace_copy_if* algorithm returns the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename FwdIter1, typename Sent, typename FwdIter2, typename Pred,
        typename T = typename std::iterator_traits<FwdIter2>::value_type, typename Proj = hpx::identity>
```

parallel::util::detail::algorithm_result<ExPolicy, replace_copy_if_result<FwdIter1, FwdIter2>>::type hpx::ranges::replace_copy

Copies the all elements from the range [first, sent) to another range beginning at *dest* replacing all elements satisfying a specific criteria with *new_value*.

Effects: Assigns to every iterator it in the range [result, result + (sent - first)) either *new_value* or $*(first + (it - result))$ depending on whether the following corresponding condition holds: `INVOKE(f, INVOKE(proj, *(first + (i - result)))) != false`

The assignments in the parallel *replace_copy_if* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *replace_copy_if* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *sent - first* applications of the predicate.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the source iterator used (deduced). The iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the end iterators used (deduced). This sentinel type must be a sentinel for InIter.
- **FwdIter2** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. (deduced).
- **T** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **sent** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the elements which need to be replaced. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **new_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *replace_copy_if* algorithm returns an `hpx::future<FwdIter1, FwdIter2>`. The *replace_copy_if* algorithm returns the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

```
template<typename ExPolicy, typename Rng, typename FwdIter, typename Pred, typename T = typename
std::iterator_traits<FwdIter>::value_type, typename Proj = hpx::identity>
parallel::util::detail::algorithm_result<ExPolicy, replace_copy_if_result<std::ranges::iterator_t<Rng>, FwdIter>>::type hpx::ranges::
```

Copies the all elements from the range *rng* to another range beginning at *dest* replacing all elements satisfying a specific criteria with *new_value*.

Effects: Assigns to every iterator *it* in the range `[result, result + (util::end(rng) - util::begin(rng))]` either *new_value* or `*(first + (it - result))` depending on whether the following corresponding condition holds: `INVOKE(f, INVOKE(proj, *(first + (i - result)))) != false`

The assignments in the parallel *replace_copy_if* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *replace_copy_if* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *util::end(rng) - util::begin(rng)* applications of the predicate.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **FwdIter** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *equal* requires *Pred* to meet the requirements of *CopyConstructible*. (deduced).
- **T** – The type of the new values to replace (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is an unary predicate which returns *true* for the elements which need to be replaced. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have *const&*, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **new_value** – Refers to the new value to use as the replacement.
- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *replace_copy_if* algorithm returns an `hpx::future<in_out_result<std::ranges::iterator_t<Rng>, OutIter>>`. The *replace_copy_if* algorithm returns the input iterator *last* and the output iterator to the element in the destination range, one past the last element copied.

hpx::ranges::uninitialized_default_construct

Defined in header `hpx/algorithm.hpp`⁷⁶¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::uninitialized_default_construct_n`

template<typename **FwdIter**, typename **Sent**>

FwdIter **hpx::ranges::uninitialized_default_construct**(*FwdIter* first, *Sent* last)

Constructs objects of type `typename iterator_traits<ForwardIt>::value_type` in the uninitialized storage designated by the range by default-initialization. If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized_default_construct* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.

Returns

The *uninitialized_default_construct* algorithm returns a returns *FwdIter*. The *uninitialized_default_construct* algorithm returns the output iterator to the element in the range, one past the last element constructed.

template<typename **ExPolicy**, typename **FwdIter**, typename **Sent**>

hpx::parallel::util::detail::algorithm_result_t<ExPolicy, FwdIter> **hpx::ranges::uninitialized_default_construct**(*ExPolicy* &&policy, *FwdIter* first, *Sent* last)

Constructs objects of type `typename iterator_traits<ForwardIt>::value_type` in the uninitialized storage designated by the range by default-initialization. If an exception is thrown during the initialization, the function has no effects.

⁷⁶¹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

The assignments in the parallel *uninitialized_default_construct* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized_default_construct* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for FwdIter.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.

Returns

The *uninitialized_default_construct* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *uninitialized_default_construct* algorithm returns the iterator to the element in the source range, one past the last element constructed.

```
template<typename Rng>
hpx::traits::range_traits<Rng>::iterator_type hpx::ranges::uninitialized_default_construct(Rng
                                                                                               &&rng)
```

Constructs objects of type *typename iterator_traits<ForwardIt>::value_type* in the uninitialized storage designated by the range by default-initialization. If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel *uninitialized_default_construct* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

Rng – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

Parameters

rng – Refers to the range to which will be default constructed.

Returns

The `uninitialized_default_construct` algorithm returns a `hpx::traits::range_traits<Rng>::iterator_type`. The `uninitialized_default_construct` algorithm returns the output iterator to the element in the range, one past the last element constructed.

```
template<typename ExPolicy, typename Rng>
parallel::util::detail::algorithm_result<ExPolicy, typename hpx::traits::range_traits<Rng>::iterator_type>::type hpx::ranges::unini
```

Constructs objects of type `typename iterator_traits<ForwardIt>::value_type` in the uninitialized storage designated by the range by default-initialization. If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel `uninitialized_default_construct` algorithm invoked with an execution policy object of type `sequenced_policy` execute in sequential order in the calling thread.

The assignments in the parallel `uninitialized_default_construct` algorithm invoked with an execution policy object of type `parallel_policy` or `parallel_task_policy` are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *last - first* assignments.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the range to which the value will be default constructed

Returns

The `uninitialized_default_construct` algorithm returns a `hpx::future<typename hpx::traits::range_traits<Rng>::iterator_type>`, if the execution policy is of type `sequenced_task_policy` or `parallel_task_policy` and returns `typename hpx::traits::range_traits<Rng>::iterator_type` otherwise. The `uninitialized_default_construct` algorithm returns the output iterator to the element in the range, one past the last element constructed.

hpx::ranges::uninitialized_default_construct_n

Defined in header `hpx/algorithm.hpp`⁷⁶².

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::uninitialized_default_construct`

template<typename **FwdIter**, typename **Size**>

FwdIter **hpx::ranges::uninitialized_default_construct_n**(*FwdIter* first, *Size* count)

Constructs objects of type `typename iterator_traits<ForwardIt>::value_type` in the uninitialized storage designated by the range `[first, first + count)` by default-initialization. If an exception is thrown during the initialization, the function has no effects.

The assignments in the parallel `uninitialized_default_construct_n` algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: Performs exactly *count* assignments, if *count* > 0, no assignments otherwise.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.

Returns

The `uninitialized_default_construct_n` algorithm returns a `returns FwdIter`. The `uninitialized_default_construct_n` algorithm returns the iterator to the element in the source range, one past the last element constructed.

template<typename **ExPolicy**, typename **FwdIter**, typename **Size**>

parallel::util::detail::algorithm_result<ExPolicy, FwdIter>::type **hpx::ranges::uninitialized_default_construct_n**(*ExPolicy* &&policy, *FwdIter* first, *Size* count)

Constructs objects of type `typename iterator_traits<ForwardIt>::value_type` in the uninitialized storage designated by the range `[first, first + count)` by default-initialization. If an exception is thrown during the initialization, the function has no effects.

⁷⁶² <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

The assignments in the parallel *uninitialized_default_construct_n* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *uninitialized_default_construct_n* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs exactly *count* assignments, if *count* > 0, no assignments otherwise.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Size** – The type of the argument specifying the number of elements to apply *f* to.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **count** – Refers to the number of elements starting at *first* the algorithm will be applied to.

Returns

The *uninitialized_default_construct_n* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *FwdIter* otherwise. The *uninitialized_default_construct_n* algorithm returns the iterator to the element in the source range, one past the last element constructed.

hpx::ranges::find

Defined in header [hpx/algorithm.hpp](http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp)⁷⁶³.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::ranges::find_if*
- *hpx::ranges::find_if_not*
- *hpx::ranges::find_end*
- *hpx::ranges::find_first_of*
- *hpx::ranges::find_last*
- *hpx::ranges::find_last_if*
- *hpx::ranges::find_last_if_not*

⁷⁶³ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

Warning: doxygenfunction: Cannot find function “hpx::ranges::find” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

hpx::ranges::find_if

Defined in header [hpx/algorithm.hpp](#)⁷⁶⁴.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- [hpx::ranges::find](#)
- [hpx::ranges::find_if_not](#)
- [hpx::ranges::find_end](#)
- [hpx::ranges::find_first_of](#)
- [hpx::ranges::find_last](#)
- [hpx::ranges::find_last_if](#)
- [hpx::ranges::find_last_if_not](#)

Warning: doxygenfunction: Cannot find function “hpx::ranges::find_if” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

hpx::ranges::find_if_not

Defined in header [hpx/algorithm.hpp](#)⁷⁶⁵.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- [hpx::ranges::find](#)
- [hpx::ranges::find_if](#)
- [hpx::ranges::find_end](#)
- [hpx::ranges::find_first_of](#)
- [hpx::ranges::find_last](#)
- [hpx::ranges::find_last_if](#)
- [hpx::ranges::find_last_if_not](#)

Warning: doxygenfunction: Cannot find function “hpx::ranges::find_if_not” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

⁷⁶⁴ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

⁷⁶⁵ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

hpx::ranges::find_end

Defined in header [hpx/algorithm.hpp](#)⁷⁶⁶.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::ranges::find`
- `hpx::ranges::find_if`
- `hpx::ranges::find_if_not`
- `hpx::ranges::find_first_of`
- `hpx::ranges::find_last`
- `hpx::ranges::find_last_if`
- `hpx::ranges::find_last_if_not`

Warning: doxygenfunction: Cannot find function “hpx::ranges::find_end” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

hpx::ranges::find_first_of

Defined in header [hpx/algorithm.hpp](#)⁷⁶⁷.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::ranges::find`
- `hpx::ranges::find_if`
- `hpx::ranges::find_if_not`
- `hpx::ranges::find_end`
- `hpx::ranges::find_last`
- `hpx::ranges::find_last_if`
- `hpx::ranges::find_last_if_not`

Warning: doxygenfunction: Cannot find function “hpx::ranges::find_first_of” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

⁷⁶⁶ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

⁷⁶⁷ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

hpx::ranges::find_last

Defined in header [hpx/algorithm.hpp](#)⁷⁶⁸.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::ranges::find*
- *hpx::ranges::find_if*
- *hpx::ranges::find_if_not*
- *hpx::ranges::find_end*
- *hpx::ranges::find_first_of*
- *hpx::ranges::find_last_if*
- *hpx::ranges::find_last_if_not*

Warning: doxygenfunction: Cannot find function “hpx::ranges::find_last” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

hpx::ranges::find_last_if

Defined in header [hpx/algorithm.hpp](#)⁷⁶⁹.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::ranges::find*
- *hpx::ranges::find_if*
- *hpx::ranges::find_if_not*
- *hpx::ranges::find_end*
- *hpx::ranges::find_first_of*
- *hpx::ranges::find_last*
- *hpx::ranges::find_last_if_not*

Warning: doxygenfunction: Cannot find function “hpx::ranges::find_last_if” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

⁷⁶⁸ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

⁷⁶⁹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

hpx::ranges::find_last_if_not

Defined in header [hpx/algorithm.hpp](#)⁷⁷⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::find`
- `hpx::ranges::find_if`
- `hpx::ranges::find_if_not`
- `hpx::ranges::find_end`
- `hpx::ranges::find_first_of`
- `hpx::ranges::find_last`
- `hpx::ranges::find_last_if`

Warning: doxygenfunction: Cannot find function “hpx::ranges::find_last_if_not” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

hpx::ranges::unique

Defined in header [hpx/algorithm.hpp](#)⁷⁷¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::unique_copy`

```
template<typename FwdIter, typename Sent, typename Pred = ranges::equal_to, typename Proj = hpx::identity>
subrange_t<FwdIter, Sent> hpx::ranges::unique(FwdIter first, Sent last, Pred &&pred = Pred(), Proj &&proj = Proj())
```

Eliminates all but the first element from every consecutive group of equivalent elements from the range [first, last) and returns a past-the-end iterator for the new logical end of the range.

The assignments in the parallel *unique* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: Performs not more than *last - first* assignments, exactly *last - first - 1* applications of the predicate *pred* and no more than twice as many applications of the projection *proj*.

Template Parameters

- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for **FwdIter**.

⁷⁷⁰ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

⁷⁷¹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *unique* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *unique* algorithm returns *subrange_t<FwdIter, Sent>*. The *unique* algorithm returns an object {ret, last}, where ret is a past-the-end iterator for a new subrange.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename Pred = ranges::equal_to, typename Proj = hpx::identity>
```

```
parallel::util::detail::algorithm_result<ExPolicy, subrange_t<FwdIter, Sent>>::type hpx::ranges::unique(ExPolicy
&&pol-
icy,
FwdIter
first,
Sent
last,
Pred
&&pred
=
Pred(),
Proj
&&proj
=
Proj())
```

Eliminates all but the first element from every consecutive group of equivalent elements from the range [first, last) and returns a past-the-end iterator for the new logical end of the range.

The assignments in the parallel *unique* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *unique* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs not more than *last - first* assignments, exactly *last - first - 1* applications of the predicate *pred* and no more than twice as many applications of the projection *proj*.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for FwdIter.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *unique* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The types *Type1* and *Type2* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to both *Type1* and *Type2*

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *unique* algorithm returns *subrange_t<FwdIter, Sent>*. The *unique* algorithm returns an object {ret, last}, where ret is a past-the-end iterator for a new subrange.

```
template<typename Rng, typename Pred = ranges::equal_to, typename Proj = hpx::identity>
subrange_t<std::ranges::iterator_t<Rng>, std::ranges::iterator_t<Rng>> hpx::ranges::unique(Rng &&rng, Pred
&&pred = Pred(),
Proj &&proj =
Proj())
```

Eliminates all but the first element from every consecutive group of equivalent elements from the range *rng* and returns a past-the-end iterator for the new logical end of the range.

The assignments in the parallel *unique* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: Performs not more than N assignments, exactly $N - 1$ applications of the predicate *pred* and no more than twice as many applications of the projection *proj*, where $N = \text{std::distance}(\text{begin}(\text{rng}), \text{end}(\text{rng}))$.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *unique* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *unique* algorithm returns `subrange_t<std::ranges::iterator_t<Rng>, std::ranges::iterator_t<Rng>>`. The *unique* algorithm returns an object {ret, last}, where ret is a past-the-end iterator for a new subrange.

```
template<typename ExPolicy, typename Rng, typename Pred = ranges::equal_to, typename Proj = hpx::identity>
parallel::util::detail::algorithm_result<ExPolicy, subrange_t<std::ranges::iterator_t<Rng>, std::ranges::iterator_t<Rng>>> hpx::range
```

Eliminates all but the first element from every consecutive group of equivalent elements from the range *rng* and returns a past-the-end iterator for the new logical end of the range.

The assignments in the parallel *unique* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *unique* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs not more than N assignments, exactly N - 1 applications of the predicate *pred* and no more than twice as many applications of the projection *proj*, where $N = \text{std::distance}(\text{begin}(\text{rng}), \text{end}(\text{rng}))$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *unique* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *unique* algorithm returns a `hpx::future <subrange_t<std::ranges::iterator_t<Rng>, std::ranges::iterator_t<Rng>>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `subrange_t<std::ranges::iterator_t<Rng>, std::ranges::iterator_t<Rng>>` otherwise. The *unique* algorithm returns an object {ret, last}, where ret is a past-the-end iterator for a new subrange.

hpx::ranges::unique_copy

Defined in header [hpx/algorithm.hpp](#)⁷⁷².

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::ranges::unique*

```
template<typename InIter, typename Sent, typename O, typename Pred = ranges::equal_to, typename Proj =  
hpx::identity>  
unique_copy_result<InIter, O> hpx::ranges::unique_copy(InIter first, Sent last, O dest, Pred &&pred = Pred(),  
Proj &&proj = Proj())
```

Copies the elements from the range [first, last), to another range beginning at *dest* in such a way that there are no consecutive equal elements. Only the first element of each group of equal elements is copied.

The assignments in the parallel *unique_copy* algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

Note: Complexity: Performs not more than *last - first* assignments, exactly *last - first - 1* applications of the predicate *pred* and no more than twice as many applications of the projection *proj*

Template Parameters

- **InIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an input iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for InIter.
- **O** – The type of the iterator representing the destination range (deduced).
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *unique_copy* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a, const Type &b);
```

⁷⁷² <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *InIter1* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *unique_copy* algorithm returns a `unique_copy_result<InIter, O>`. The *unique_copy* algorithm returns an `in_out_result` with the source iterator to one past the last element and out containing the destination iterator to the end of the *dest* range.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename O, typename Pred =
ranges::equal_to, typename Proj = hpx::identity>
parallel::util::detail::algorithm_result<ExPolicy, unique_copy_result<FwdIter, O>>::type hpx::ranges::unique_copy(ExPolicy
&&pol-
icy,
FwdIter
first,
Sent
last,
O
dest,
Pred
&&pred
=
Pred(),
Proj
&&proj
=
Proj())
```

Copies the elements from the range `[first, last)`, to another range beginning at *dest* in such a way that there are no consecutive equal elements. Only the first element of each group of equal elements is copied.

The assignments in the parallel *unique_copy* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *unique_copy* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs not more than *last - first* assignments, exactly *last - first - 1* applications of the predicate *pred* and no more than twice as many applications of the projection *proj*

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a forward iterator.

- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `FwdIter1`.
- **O** – The type of the iterator representing the destination range (deduced).
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of `unique_copy` requires `Pred` to meet the requirements of `CopyConstructible`. This defaults to `std::equal_to<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). This is a binary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The `unique_copy` algorithm returns a `hpx::future< unique_copy_result<FwdIter, O>>` if the execution policy is of type `sequenced_task_policy` or `parallel_task_policy` and returns `unique_copy_result<FwdIter, O>` otherwise. The `unique_copy` algorithm returns an `in_out_result` with the source iterator to one past the last element and out containing the destination iterator to the end of the `dest` range.

```
template<typename Rng, typename O, typename Pred = ranges::equal_to, typename Proj = hpx::identity>
unique_copy_result<std::ranges::iterator_t<Rng>, O> hpx::ranges::unique_copy(Rng &&rng, O dest, Pred
&&pred = Pred(), Proj
&&proj = Proj())
```

Copies the elements from the range *rng*, to another range beginning at *dest* in such a way that there are no consecutive equal elements. Only the first element of each group of equal elements is copied.

The assignments in the parallel `unique_copy` algorithm invoked without an execution policy object will execute in sequential order in the calling thread.

Note: Complexity: Performs not more than *N* assignments, exactly *N* - 1 applications of the predicate *pred*, where *N* = `std::distance(begin(rng), end(rng))`.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **O** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *unique_copy* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by the range *rng*. This is an binary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *unique_copy* algorithm returns *unique_copy_result*< `std::ranges::iterator_t<Rng>`, *O*>. The *unique_copy* algorithm returns the pair of the source iterator to *last*, and the destination iterator to the end of the *dest* range.

```
template<typename ExPolicy, typename Rng, typename O, typename Pred = ranges::equal_to, typename Proj =
hpx::identity>
parallel::util::detail::algorithm_result<ExPolicy, unique_copy_result<std::ranges::iterator_t<Rng>, O>> hpx::ranges::unique_cop
```

Copies the elements from the range *rng*, to another range beginning at *dest* in such a way that there are no consecutive equal elements. Only the first element of each group of equal elements is copied.

The assignments in the parallel *unique_copy* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *unique_copy* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Performs not more than N assignments, exactly $N - 1$ applications of the predicate *pred*, where $N = \text{std::distance}(\text{begin}(\text{rng}), \text{end}(\text{rng}))$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a forward iterator.
- **O** – The type of the iterator representing the destination range (deduced). This iterator type must meet the requirements of a forward iterator.
- **Pred** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *unique_copy* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::equal_to<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **dest** – Refers to the beginning of the destination range.
- **pred** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by the range *rng*. This is a binary predicate which returns *true* for the required elements. The signature of this predicate should be equivalent to:

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter1* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *unique_copy* algorithm returns a `hpx::future<unique_copy_result<std::ranges::iterator_t<Rng>, O>>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *unique_copy_result<std::ranges::iterator_t<Rng>, O>* otherwise. The *unique_copy* algorithm returns the pair of the source iterator to *last*, and the destination iterator to the end of the *dest* range.

hpx::ranges::all_of

Defined in header `hpx/algorithm.hpp`⁷⁷³.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::any_of`
- `hpx::ranges::none_of`

```
template<typename ExPolicy, typename Rng, typename F, typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx::ranges::all_of(ExPolicy &&policy, Rng
&&rng, F &&f, Proj
&&proj = Proj())
```

Checks if unary predicate *f* returns true for all elements in the range *rng*.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most `std::distance(begin(rng), end(rng))` applications of the predicate *f*

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *none_of* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

⁷⁷³ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *all_of* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *all_of* algorithm returns true if the unary predicate *f* returns true for all elements in the range, false otherwise. It returns true if the range is empty.

```
template<typename ExPolicy, typename Iter, typename Sent, typename F, typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx::ranges::all_of(ExPolicy &&policy, Iter
                                                                    first, Sent last, F &&f, Proj
                                                                    &&proj = Proj())
```

Checks if unary predicate *f* returns true for all elements in the range *rng*.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most `std::distance(begin(rng), end(rng))` applications of the predicate *f*

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter`.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *none_of* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *all_of* algorithm returns a `hpx::future<bool>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *all_of* algorithm returns true if the unary predicate *f* returns true for all elements in the range, false otherwise. It returns true if the range is empty.

```
template<typename Rng, typename F, typename Proj = hpx::identity>
bool hpx::ranges::all_of(Rng &&rng, F &&f, Proj &&proj = Proj())
```

Checks if unary predicate *f* returns true for all elements in the range *rng*.

Note: Complexity: At most `std::distance(begin(rng), end(rng))` applications of the predicate *f*

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *none_of* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *all_of* algorithm returns true if the unary predicate *f* returns true for all elements in the range, false otherwise. It returns true if the range is empty.

```
template<typename Iter, typename Sent, typename F, typename Proj = hpx::identity>
bool hpx::ranges::all_of(Iter first, Sent last, F &&f, Proj &&proj = Proj())
```

Checks if unary predicate *f* returns true for all elements in the range *rng*.

Note: Complexity: At most `std::distance(begin(rng), end(rng))` applications of the predicate *f*

Template Parameters

- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter`.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *none_of* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *all_of* algorithm returns true if the unary predicate *f* returns true for all elements in the range, false otherwise. It returns true if the range is empty.

hpx::ranges::any_of

Defined in header `hpx/algorithm.hpp`⁷⁷⁴.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::ranges::all_of*
- *hpx::ranges::none_of*

```
template<typename ExPolicy, typename Rng, typename F, typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx::ranges::any_of(ExPolicy &&policy, Rng
&&rng, F &&f, Proj
&&proj = Proj())
```

Checks if unary predicate *f* returns true for at least one element in the range *rng*.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

⁷⁷⁴ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e77/libs/full/include/include/hpx/algorithm.hpp>

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most `std::distance(begin(rng), end(rng))` applications of the predicate *f*

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *none_of* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *any_of* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *any_of* algorithm returns true if the unary predicate *f* returns true for at least one element in the range, false otherwise. It returns false if the range is empty.

```
template<typename ExPolicy, typename Iter, typename Sent, typename F, typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx::ranges::any_of(ExPolicy &&policy, Iter
                                                                    first, Sent last, F &&f, Proj
                                                                    &&proj = Proj())
```

Checks if unary predicate *f* returns true for at least one element in the range *rng*.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most `std::distance(begin(rng), end(rng))` applications of the predicate *f*

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter`.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *none_of* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *any_of* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *any_of* algorithm returns true if the unary predicate *f* returns true for at least one element in the range, false otherwise. It returns false if the range is empty.

```
template<typename Rng, typename F, typename Proj = hpx::identity>  
bool hpx::ranges::any_of(Rng &&rng, F &&f, Proj &&proj = Proj())
```

Checks if unary predicate *f* returns true for at least one element in the range *rng*.

Note: Complexity: At most `std::distance(begin(rng), end(rng))` applications of the predicate *f*

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *none_of* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *any_of* algorithm returns true if the unary predicate *f* returns true for at least one element in the range, false otherwise. It returns false if the range is empty.

```
template<typename Iter, typename Sent, typename F, typename Proj = hpx::identity>
bool hpx::ranges::any_of(Iter first, Sent last, F &&f, Proj &&proj = Proj())
```

Checks if unary predicate *f* returns true for at least one element in the range *rng*.

Note: Complexity: At most `std::distance(begin(rng), end(rng))` applications of the predicate *f*

Template Parameters

- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter*.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *none_of* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *any_of* algorithm returns true if the unary predicate *f* returns true for at least one element in the range, false otherwise. It returns false if the range is empty.

hpx::ranges::none_of

Defined in header `hpx/algorithm.hpp`⁷⁷⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::all_of`
- `hpx::ranges::any_of`

```
template<typename ExPolicy, typename Rng, typename F, typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx::ranges::none_of(ExPolicy &&policy, Rng
&&rng, F &&f, Proj
&&proj = Proj())
```

Checks if unary predicate *f* returns true for no elements in the range *rng*.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most `std::distance(begin(rng), end(rng))` applications of the predicate *f*

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *none_of* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

⁷⁷⁵ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *none_of* algorithm returns a `hpx::future<bool>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *none_of* algorithm returns true if the unary predicate *f* returns true for no elements in the range, false otherwise. It returns true if the range is empty.

```
template<typename ExPolicy, typename Iter, typename Sent, typename F, typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx::ranges::none_of(ExPolicy &&policy, Iter
                                                                    first, Sent last, F &&f,
                                                                    Proj &&proj = Proj())
```

Checks if unary predicate *f* returns true for no elements in the range [first, last).

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: At most *last - first* applications of the predicate *f*

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter*.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *none_of* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.

- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

`bool pred(const Type &a);`

The signature does not need to have const&, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *none_of* algorithm returns a *hpx::future<bool>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *bool* otherwise. The *none_of* algorithm returns true if the unary predicate *f* returns true for no elements in the range, false otherwise. It returns true if the range is empty.

```
template<typename Rng, typename F, typename Proj = hpx::identity>
bool hpx::ranges::none_of(Rng &&rng, F &&f, Proj &&proj = Proj())
    Checks if unary predicate f returns true for no elements in the range rng.
```

Note: Complexity: At most `std::distance(begin(rng), end(rng))` applications of the predicate *f*

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *none_of* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

`bool pred(const Type &a);`

The signature does not need to have const&, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *none_of* algorithm returns true if the unary predicate *f* returns true for no elements in the range, false otherwise. It returns true if the range is empty.

```
template<typename Iter, typename Sent, typename F, typename Proj = hpx::identity>
bool hpx::ranges::none_of(Iter first, Sent last, F &&f, Proj &&proj = Proj())
```

Checks if unary predicate *f* returns true for no elements in the range [first, last).

Note: Complexity: At most *last - first* applications of the predicate *f*

Template Parameters

- **Iter** – The type of the source iterators used for the range (deduced).
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `InIter`.
- **F** – The type of the function/function object to use (deduced). Unlike its sequential form, the parallel overload of *none_of* requires *F* to meet the requirements of *CopyConstructible*.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **first** – Refers to the beginning of the sequence of elements of the range the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of the range the algorithm will be applied to.
- **f** – Specifies the function (or function object) which will be invoked for each of the elements in the sequence specified by [first, last). The signature of this predicate should be equivalent to:

```
bool pred(const Type &a);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type *Type* must be such that an object of type *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *none_of* algorithm returns true if the unary predicate *f* returns true for no elements in the range, false otherwise. It returns true if the range is empty.

hpx::ranges::is_sorted

Defined in header [hpx/algorithm.hpp](#)⁷⁷⁶.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::ranges::is_sorted_until*

```
template<typename FwdIter, typename Sent, typename Pred = hpx::parallel::detail::less, typename Proj = hpx::identity>
```

```
bool hpx::ranges::is_sorted(FwdIter first, Sent last, Pred &&pred = Pred(), Proj &&proj = Proj())
```

Determines if the range [first, last) is sorted. Uses pred to compare elements.

The comparison operations in the parallel *is_sorted* algorithm executes in sequential order in the calling thread.

Note: Complexity: at most (N+S-1) comparisons where *N* = distance(first, last). *S* = number of partitions

Template Parameters

- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for FwdIter.
- **Pred** – The type of an optional function/function object to use.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.
- **pred** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second argument. The signature of the function should be equivalent to

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have const &, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to Type.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *is_sorted* algorithm returns a *bool*. The *is_sorted* algorithm returns true if each element in the sequence [first, last) satisfies the predicate passed. If the range [first, last) contains less than two elements, the function always returns true.

⁷⁷⁶ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename Pred = hpx::parallel::detail::less,
typename Proj = hpx::identity>
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx::ranges::is_sorted(ExPolicy &&policy,
                                                                                      FwdIter first, Sent last,
                                                                                      Pred &&pred = Pred(),
                                                                                      Proj &&proj = Proj())
```

Determines if the range [first, last) is sorted. Uses pred to compare elements.

The comparison operations in the parallel *is_sorted* algorithm invoked with an execution policy object of type *sequenced_policy* executes in sequential order in the calling thread.

The comparison operations in the parallel *is_sorted* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: at most (N+S-1) comparisons where *N* = distance(first, last). *S* = number of partitions

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *is_sorted* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.
- **pred** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second argument. The signature of the function should be equivalent to

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *is_sorted* algorithm returns a *hpx::future<bool>* if the execution policy is of type *task_execution_policy* and returns *bool* otherwise. The *is_sorted* algorithm returns a *bool* if each element in the sequence [first, last) satisfies the predicate passed. If the range [first, last) contains less than two elements, the function always returns true.

```
template<typename Rng, typename Pred = hpx::parallel::detail::less, typename Proj = hpx::identity>  
bool hpx::ranges::is_sorted(Rng &&rng, Pred &&pred = Pred(), Proj &&proj = Proj())
```

Determines if the range *rng* is sorted. Uses *pred* to compare elements.

The comparison operations in the parallel *is_sorted* algorithm executes in sequential order in the calling thread.

Note: Complexity: at most $(N+S-1)$ comparisons where $N = \text{size}(\text{rng})$. $S = \text{number of partitions}$

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second argument. The signature of the function should be equivalent to

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have *const &*, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *is_sorted* algorithm returns a *bool*. The *is_sorted* algorithm returns true if each element in the *rng* satisfies the predicate passed. If the range *rng* contains less than two elements, the function always returns true.

```
template<typename ExPolicy, typename Rng, typename Pred = hpx::parallel::detail::less, typename Proj =  
hpx::identity>  
hpx::parallel::util::detail::algorithm_result_t<ExPolicy, bool> hpx::ranges::is_sorted(ExPolicy &&policy,  
                                           Rng &&rng, Pred  
                                           &&pred = Pred(), Proj  
                                           &&proj = Proj())
```

Determines if the range *rng* is sorted. Uses *pred* to compare elements.

The comparison operations in the parallel *is_sorted* algorithm invoked with an execution policy object of type *sequenced_policy* executes in sequential order in the calling thread.

The comparison operations in the parallel *is_sorted* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: at most $(N+S-1)$ comparisons where $N = \text{size}(\text{rng})$. S = number of partitions

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *is_sorted* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second argument. The signature of the function should be equivalent to

<pre>bool pred(const Type &a, const Type &b);</pre>

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *is_sorted* algorithm returns a *hpx::future<bool>* if the execution policy is of type *task_execution_policy* and returns *bool* otherwise. The *is_sorted* algorithm returns a *bool* if each element in the range *rng* satisfies the predicate passed. If the range *rng* contains less than two elements, the function always returns true.

hpx::ranges::is_sorted_until

Defined in header `hpx/algorithm.hpp`⁷⁷⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::ranges::is_sorted*

```
template<typename FwdIter, typename Sent, typename Pred = hpx::parallel::detail::less, typename Proj = hpx::identity>
```

⁷⁷⁷ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

FwdIter `hpx::ranges::is_sorted_until`(*FwdIter* first, *Sent* last, *Pred* &&pred = *Pred*(), *Proj* &&proj = *Proj*())

Returns the first element in the range [first, last) that is not sorted. Uses a predicate to compare elements or the less than operator.

The comparison operations in the parallel *is_sorted_until* algorithm execute in sequential order in the calling thread.

Note: Complexity: at most $(N+S-1)$ comparisons where N = distance(first, last). S = number of partitions

Template Parameters

- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *FwdIter*.
- **Pred** – The type of an optional function/function object to use.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.
- **pred** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second argument. The signature of the function should be equivalent to

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *is_sorted_until* algorithm returns a *FwdIter*. The *is_sorted_until* algorithm returns the first unsorted element. If the sequence has less than two elements or the sequence is sorted, last is returned.

```
template<typename ExPolicy, typename FwdIter, typename Sent, typename Pred = hpx::parallel::detail::less,  
typename Proj = hpx::identity>
```

```

hpx::parallel::util::detail::algorithm_result<ExPolicy, FwdIter>::type hpx::ranges::is_sorted_until(ExPolicy
                                                                    &&pol-
                                                                    icy,
                                                                    FwdIter
                                                                    first,
                                                                    Sent last,
                                                                    Pred
                                                                    &&pred
                                                                    =
                                                                    Pred(),
                                                                    Proj
                                                                    &&proj
                                                                    =
                                                                    Proj())

```

Returns the first element in the range [first, last) that is not sorted. Uses a predicate to compare elements or the less than operator.

The comparison operations in the parallel *is_sorted_until* algorithm invoked with an execution policy object of type *sequenced_policy* executes in sequential order in the calling thread.

The comparison operations in the parallel *is_sorted_until* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: at most (N+S-1) comparisons where *N* = distance(first, last). *S* = number of partitions

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter** – The type of the source iterators used for the This iterator type must meet the requirements of a forward iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for FwdIter.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *is_sorted_until* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements of that the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements of that the algorithm will be applied to.
- **pred** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second argument. The signature of the function should be equivalent to

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *is_sorted_until* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *task_execution_policy* and returns *FwdIter* otherwise. The *is_sorted_until* algorithm returns the first unsorted element. If the sequence has less than two elements or the sequence is sorted, last is returned.

```
template<typename Rng, typename Pred = hpx::parallel::detail::less, typename Proj = hpx::identity>
std::ranges::iterator_t<Rng> hpx::ranges::is_sorted_until(Rng &&rng, Pred &&pred = Pred(), Proj
&&proj = Proj())
```

Returns the first element in the range *rng* that is not sorted. Uses a predicate to compare elements or the less than operator.

Note: Complexity: at most $(N+S-1)$ comparisons where $N = \text{size}(\text{rng})$. $S = \text{number of partitions}$

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of *is_sorted_until* requires *Pred* to meet the requirements of *CopyConstructible*. This defaults to `std::less<>`
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second argument. The signature of the function should be equivalent to

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type *Type* must be such that objects of types *FwdIter* can be dereferenced and then implicitly converted to *Type*.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *is_sorted_until* returns *FwdIter*. The *is_sorted_until* algorithm returns the first unsorted element. If the sequence has less than two elements or the sequence is sorted, last is returned.

```
template<typename ExPolicy, typename Rng, typename Pred = hpx::parallel::detail::less, typename Proj =
hpx::identity>
```

```

hpx::parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng>> hpx::ranges::is_sorted_until(ExPolicy
&&pol-
icy,
Rng
&&rng,
Pred
&&pred
=
Pred(),
Proj
&&proj
=
Proj())

```

Returns the first element in the range `rng` that is not sorted. Uses a predicate to compare elements or the less than operator.

The comparison operations in the parallel `is_sorted_until` algorithm invoked with an execution policy object of type `sequenced_policy` executes in sequential order in the calling thread.

The comparison operations in the parallel `is_sorted_until` algorithm invoked with an execution policy object of type `parallel_policy` or `parallel_task_policy` are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: at most $(N+S-1)$ comparisons where $N = \text{size}(\text{rng})$. S = number of partitions

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an forward iterator.
- **Pred** – The type of an optional function/function object to use. Unlike its sequential form, the parallel overload of `is_sorted_until` requires `Pred` to meet the requirements of `CopyConstructible`. This defaults to `std::less<>`
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **pred** – Refers to the binary predicate which returns true if the first argument should be treated as less than the second argument. The signature of the function should be equivalent to

```
bool pred(const Type &a, const Type &b);
```

The signature does not need to have `const &`, but the function must not modify the objects passed to it. The type `Type` must be such that objects of types `FwdIter` can be dereferenced and then implicitly converted to `Type`.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked.

Returns

The *is_sorted_until* algorithm returns a *hpx::future<FwdIter>* if the execution policy is of type *task_execution_policy* and returns *FwdIter* otherwise. The *is_sorted_until* algorithm returns the first unsorted element. If the sequence has less than two elements or the sequence is sorted, last is returned.

hpx::ranges::nth_element

Defined in header [hpx/algorithm.hpp](#)⁷⁷⁸.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename RandomIt, typename Sent, typename Pred = hpx::parallel::detail::less, typename Proj = hpx::identity>
```

```
RandomIt hpx::ranges::nth_element(RandomIt first, RandomIt nth, Sent last, Pred &&pred = Pred(), Proj &&proj = Proj())
```

nth_element is a partial sorting algorithm that rearranges elements in $[first, last)$ such that the element pointed at by *nth* is changed to whatever element would occur in that position if $[first, last)$ were sorted and all of the elements before this new *nth* element are less than or equal to the elements after the new *nth* element.

The comparison operations in the parallel *nth_element* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: Linear in $\text{std::distance}(first, last)$ on average. $O(N)$ applications of the predicate, and $O(N \log N)$ swaps, where $N = last - first$.

Template Parameters

- **RandomIt** – The type of the source begin, *nth*, and end iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *RandomIt*.
- **Pred** – Comparison function object which returns true if the first argument is less than the second.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **nth** – Refers to the iterator defining the sort partition point
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **pred** – Specifies the comparison function object which returns true if the first argument is less than (i.e. is ordered before) the second. The signature of this comparison function should be equivalent to:

⁷⁷⁸ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

```
bool cmp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type must be such that an object of type *randomIt* can be dereferenced and then implicitly converted to `Type`. This defaults to `std::less<>`.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked. This defaults to `hpx::identity`.

Returns

The *nth_element* algorithm returns *RandomIt*. The *nth_element* algorithm returns an iterator equal to last.

```
template<typename ExPolicy, typename RandomIt, typename Sent, typename Pred = hpx::parallel::detail::less,
typename Proj = hpx::identity>
parallel::util::detail::algorithm_result_t<ExPolicy, RandomIt> hpx::ranges::nth_element(ExPolicy &&policy,
                                                                                      RandomIt first,
                                                                                      RandomIt nth, Sent
                                                                                      last, Pred &&pred =
                                                                                      Pred(), Proj &&proj
                                                                                      = Proj())
```

nth_element is a partial sorting algorithm that rearranges elements in `[first, last)` such that the element pointed at by *nth* is changed to whatever element would occur in that position if `[first, last)` were sorted and all of the elements before this new *nth* element are less than or equal to the elements after the new *nth* element.

The comparison operations in the parallel *nth_element* invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *nth_element* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Linear in `std::distance(first, last)` on average. $O(N)$ applications of the predicate, and $O(N \log N)$ swaps, where $N = \text{last} - \text{first}$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **RandomIt** – The type of the source begin, *nth*, and end iterators used (deduced). This iterator type must meet the requirements of a random access iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *RandomIt*.
- **Pred** – Comparison function object which returns true if the first argument is less than the second.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **nth** – Refers to the iterator defining the sort partition point
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **pred** – Specifies the comparison function object which returns true if the first argument is less than (i.e. is ordered before) the second. The signature of this comparison function should be equivalent to:

```
bool cmp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type must be such that an object of type *randomIt* can be dereferenced and then implicitly converted to *Type*. This defaults to `std::less<>`.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked. This defaults to `hpx::identity`.

Returns

The *partition* algorithm returns a `hpx::future<RandomIt>` if the execution policy is of type *parallel_task_policy* and returns *RandomIt* otherwise. The *nth_element* algorithm returns an iterator equal to last.

```
template<typename Rng, typename Pred = hpx::parallel::detail::less, typename Proj = hpx::identity>
std::ranges::iterator_t<Rng> hpx::ranges::nth_element(Rng &&rng, std::ranges::iterator_t<Rng> nth, Pred
&&pred = Pred(), Proj &&proj = Proj())
```

nth_element is a partial sorting algorithm that rearranges elements in `[first, last)` such that the element pointed at by *nth* is changed to whatever element would occur in that position if `[first, last)` were sorted and all of the elements before this new *nth* element are less than or equal to the elements after the new *nth* element.

The comparison operations in the parallel *nth_element* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: Linear in `std::distance(first, last)` on average. $O(N)$ applications of the predicate, and $O(N \log N)$ swaps, where $N = \text{last} - \text{first}$.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a random access iterator.
- **Pred** – Comparison function object which returns true if the first argument is less than the second.
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **nth** – Refers to the iterator defining the sort partition point

- **pred** – Specifies the comparison function object which returns true if the first argument is less than (i.e. is ordered before) the second. The signature of this comparison function should be equivalent to:

```
bool cmp(const Type1 &a, const Type2 &b);
```

The signature does not need to have const&, but the function must not modify the objects passed to it. The type must be such that an object of type *randomIt* can be dereferenced and then implicitly converted to Type. This defaults to `std::less<>`.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked. This defaults to `hpx::identity`.

Returns

The *nth_element* algorithm returns `std::ranges::iterator_t<Rng>`. The *nth_element* algorithm returns an iterator equal to last.

```
template<typename ExPolicy, typename Rng, typename Pred = hpx::parallel::detail::less, typename Proj =
hpx::identity>
parallel::util::detail::algorithm_result_t<ExPolicy, std::ranges::iterator_t<Rng>> hpx::ranges::nth_element(ExPolicy
&&pol-
icy,
Rng
&&rng,
std::ranges::iterator_t
nth,
Pred
&&pred
=
Pred(),
Proj
&&proj
=
Proj())
```

nth_element is a partial sorting algorithm that rearranges elements in `[first, last)` such that the element pointed at by *nth* is changed to whatever element would occur in that position if `[first, last)` were sorted and all of the elements before this new *nth* element are less than or equal to the elements after the new *nth* element.

The comparison operations in the parallel *nth_element* invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The assignments in the parallel *nth_element* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Linear in `std::distance(first, last)` on average. $O(N)$ applications of the predicate, and $O(N \log N)$ swaps, where $N = \text{last} - \text{first}$.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes

the assignments.

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of a random access iterator.
- **Pred** – Comparison function object which returns true if the first argument is less than the second.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **nth** – Refers to the iterator defining the sort partition point
- **pred** – Specifies the comparison function object which returns true if the first argument is less than (i.e. is ordered before) the second. The signature of this comparison function should be equivalent to:

```
bool cmp(const Type1 &a, const Type2 &b);
```

The signature does not need to have `const&`, but the function must not modify the objects passed to it. The type must be such that an object of type *randomIt* can be dereferenced and then implicitly converted to *Type*. This defaults to `std::less<>`.

- **proj** – Specifies the function (or function object) which will be invoked for each of the elements as a projection operation before the actual predicate *is* invoked. This defaults to *hpx::identity*.

Returns

The *partition* algorithm returns a `hpx::future<std::ranges::iterator_t<Rng>>` if the execution policy is of type *parallel_task_policy* and returns `std::ranges::iterator_t<Rng>` otherwise. The *nth_element* algorithm returns an iterator equal to last.

hpx::ranges::transform_reduce

Defined in header `hpx/algorithm.hpp`⁷⁷⁹.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Warning: doxygenfunction: Cannot find function “hpx::ranges::transform_reduce” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

⁷⁷⁹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

hpx::ranges::partial_sort

Defined in header `hpx/algorithm.hpp`⁷⁸⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename RandomIt, typename Sent, typename Comp = ranges::less, typename Proj = hpx::identity>
RandomIt hpx::ranges::partial_sort(RandomIt first, RandomIt middle, Sent last, Comp &&comp = Comp(),
                                   Proj &&proj = Proj())
```

Places the first middle - first elements from the range [first, last) as sorted with respect to comp into the range [first, middle). The rest of the elements in the range [middle, last) are placed in an unspecified order.

The assignments in the parallel *partial_sort* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: Approximately (last - first) * log(middle - first) comparisons.

Template Parameters

- **RandomIt** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for RandomIt.
- **Comp** – The type of the function/function object to use (deduced). Comp defaults to detail::less.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the middle of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **comp** – comp is a callable object. The return value of the INVOKE operation applied to an object of type Comp, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that comp will not apply any non-constant function through the dereferenced iterator. Comp defaults to detail::less.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

Returns

The *partial_sort* algorithm returns *RandomIt*. The algorithm returns an iterator pointing to the first element after the last element in the input sequence.

```
template<typename ExPolicy, typename RandomIt, typename Sent, typename Comp = ranges::less, typename Proj = hpx::identity>
```

⁷⁸⁰ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

```
parallel::util::detail::algorithm_result<ExPolicy, RandomIt>::type hpx::ranges::partial_sort(ExPolicy  
                                                                                      &&policy,  
                                                                                      RandomIt first,  
                                                                                      RandomIt  
                                                                                      middle, Sent  
                                                                                      last, Comp  
                                                                                      &&comp =  
                                                                                      Comp(), Proj  
                                                                                      &&proj =  
                                                                                      Proj())
```

Places the first middle - first elements from the range [first, last) as sorted with respect to comp into the range [first, middle). The rest of the elements in the range [middle, last) are placed in an unspecified order.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Approximately (last - first) * log(middle - first) comparisons.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **RandomIt** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an random iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for RandomIt.
- **Comp** – The type of the function/function object to use (deduced). Comp defaults to detail::less.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the middle of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **comp** – comp is a callable object. The return value of the INVOKE operation applied to an object of type Comp, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that comp will not apply any non-constant function through the dereferenced iterator. Comp defaults to detail::less.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

Returns

The *partial_sort* algorithm returns a *hpx::future<RandomIt>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *RandomIt* otherwise. The algorithm returns an iterator pointing to the first element after the last element in the input sequence.

```
template<typename Rng, typename Comp = ranges::less, typename Proj = hpx::identity>
std::ranges::iterator_t<Rng> hpx::ranges::partial_sort(Rng &&rng, std::ranges::iterator_t<Rng> middle,
                                                    Comp &&comp = Comp(), Proj &&proj = Proj())
```

Places the first middle - first elements from the range [first, last) as sorted with respect to comp into the range [first, middle). The rest of the elements in the range [middle, last) are placed in an unspecified order.

The assignments in the parallel *partial_sort* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: Approximately (last - first) * log(middle - first) comparisons.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Comp** – The type of the function/function object to use (deduced). Comp defaults to detail::less.
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the middle of the sequence of elements the algorithm will be applied to.
- **comp** – comp is a callable object. The return value of the INVOKE operation applied to an object of type Comp, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that comp will not apply any non-constant function through the dereferenced iterator. Comp defaults to detail::less.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

Returns

The *partial_sort* algorithm returns *std::ranges::iterator_t<Rng>*. It returns *last*.

```
template<typename ExPolicy, typename Rng, typename Comp = ranges::less, typename Proj = hpx::identity>
```

```
parallel::util::detail::algorithm_result_t<ExPolicy, std::ranges::iterator_t<Rng>> hpx::ranges::partial_sort(ExPolicy
                                                                    &&pol-
                                                                    icy,
                                                                    Rng
                                                                    &&rng,
                                                                    std::ranges::iterator,
                                                                    mid-
                                                                    dle,
                                                                    Comp
                                                                    &&comp
                                                                    =
                                                                    Comp(),
                                                                    Proj
                                                                    &&proj
                                                                    =
                                                                    Proj())
```

Sorts the elements in the range `[first, last)` in ascending order. The relative order of equal elements is preserved. The function uses the given comparison function object `comp` (defaults to using `operator<()`).

A sequence is sorted with respect to a comparator `comp` and a projection `proj` if for every iterator `i` pointing to the sequence and every non-negative integer `n` such that `i + n` is a valid iterator pointing to an element of the sequence, and `INVOKE(comp, INVOKE(proj, *(i + n)), INVOKE(proj, *i)) == false`.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: $O(N \log(N))$, where $N = \text{std::distance}(\text{first}, \text{last})$ comparisons.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Comp** – The type of the function/function object to use (deduced). `Comp` defaults to `detail::less`;
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **middle** – Refers to the middle of the sequence of elements the algorithm will be applied to.
- **comp** – `comp` is a callable object. The return value of the `INVOKE` operation applied to an object of type `Comp`, when contextually converted to `bool`, yields `true` if the first argument

of the call is less than the second, and false otherwise. It is assumed that *comp* will not apply any non-constant function through the dereferenced iterator. *Comp* defaults to `detail::less`.

- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

Returns

The *partial_sort* algorithm returns a `hpx::future<std::ranges::iterator_t<Rng>` if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns `std::ranges::iterator_t<Rng>` otherwise. It returns *last*.

hpx::ranges::fold_left

Defined in header `hpx/algorithm.hpp`⁷⁸¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::fold_left_first`
- `hpx::ranges::fold_right`
- `hpx::ranges::fold_right_last`
- `hpx::ranges::fold_left_with_iter`
- `hpx::ranges::fold_left_first_with_iter`

```
template<typename InIter, typename Sent, typename T, typename F>
auto hpx::ranges::fold_left(InIter first, Sent last, T init, F f)
```

Left-folds a range of elements using a binary operator.

Note: Complexity: Exactly N applications of the binary operator, where $N = \text{std::distance}(\text{first}, \text{last})$.

Template Parameters

- **InIter** – The type of the source begin iterator (deduced). This iterator must meet the requirements of an input iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for *InIter*.
- **T** – The type of the initial value.
- **F** – The type of the binary function object.

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to the end of the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the fold operation.
- **f** – Binary function object that will be applied.

Returns

The result of left-folding the range.

⁷⁸¹ <http://github.com/TheHPXProject/hpx/blob/d402cead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

```
template<typename Rng, typename T, typename F>  
auto hpx::ranges::fold_left(Rng &&rng, T init, F f)
```

Left-folds a range of elements using a binary operator.

Note: Complexity: Exactly N applications of the binary operator, where N = std::distance(begin(rng), end(rng)).

Template Parameters

- **Rng** – The type of the source range (deduced).
- **T** – The type of the initial value.
- **F** – The type of the binary function object.

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **init** – The initial value for the fold operation.
- **f** – Binary function object that will be applied.

Returns

The result of left-folding the range.

hpx::ranges::fold_left_first

Defined in header [hpx/algorithm.hpp](#)⁷⁸².

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::ranges::fold_left*
- *hpx::ranges::fold_right*
- *hpx::ranges::fold_right_last*
- *hpx::ranges::fold_left_with_iter*
- *hpx::ranges::fold_left_first_with_iter*

```
template<typename InIter, typename Sent, typename F>  
auto hpx::ranges::fold_left_first(InIter first, Sent last, F f)
```

Left-folds a range using the first element as the initial value.

Note: Complexity: Exactly N-1 applications of the binary operator, where N = std::distance(first, last).

Template Parameters

- **InIter** – The type of the source begin iterator (deduced).
- **Sent** – The type of the source sentinel (deduced).
- **F** – The type of the binary function object.

Parameters

⁷⁸² <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **first** – Refers to the beginning of the sequence.
- **last** – Refers to the end of the sequence.
- **f** – Binary function object that will be applied.

Returns

An optional containing the fold result, or `std::nullopt` if the range is empty.

```
template<typename Rng, typename F>
auto hpx::ranges::fold_left_first(Rng &&rng, F f)
```

Left-folds a range using the first element as the initial value.

Template Parameters

- **Rng** – The type of the source range (deduced).
- **F** – The type of the binary function object.

Parameters

- **rng** – Refers to the sequence of elements.
- **f** – Binary function object that will be applied.

Returns

An optional containing the fold result, or `std::nullopt` if the range is empty.

hpx::ranges::fold_right

Defined in header [hpx/algorithm.hpp](#)⁷⁸³.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::ranges::fold_left`
- `hpx::ranges::fold_left_first`
- `hpx::ranges::fold_right_last`
- `hpx::ranges::fold_left_with_iter`
- `hpx::ranges::fold_left_first_with_iter`

```
template<typename BidIter, typename Sent, typename T, typename F>
auto hpx::ranges::fold_right(BidIter first, Sent last, T init, F f)
```

Right-folds a range of elements using a binary operator.

Note: Complexity: Exactly N applications of the binary operator.

Template Parameters

- **BidIter** – The type of the source begin iterator (deduced). This iterator must meet the requirements of a bidirectional iterator.
- **Sent** – The type of the source sentinel (deduced).
- **T** – The type of the initial value.

⁷⁸³ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **F** – The type of the binary function object.

Parameters

- **first** – Refers to the beginning of the sequence.
- **last** – Refers to the end of the sequence.
- **init** – The initial value for the fold operation.
- **f** – Binary function object that will be applied.

Returns

The result of right-folding the range.

```
template<typename Rng, typename T, typename F>  
auto hpx::ranges::fold_right(Rng &&rng, T init, F f)
```

Right-folds a range of elements using a binary operator.

Template Parameters

- **Rng** – The type of the source range (deduced).
- **T** – The type of the initial value.
- **F** – The type of the binary function object.

Parameters

- **rng** – Refers to the sequence of elements.
- **init** – The initial value for the fold operation.
- **f** – Binary function object that will be applied.

Returns

The result of right-folding the range.

`hpx::ranges::fold_right_last`

Defined in header `hpx/algorithm.hpp`⁷⁸⁴.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::ranges::fold_left`
- `hpx::ranges::fold_left_first`
- `hpx::ranges::fold_right`
- `hpx::ranges::fold_left_with_iter`
- `hpx::ranges::fold_left_first_with_iter`

```
template<typename BidIter, typename Sent, typename F>  
auto hpx::ranges::fold_right_last(BidIter first, Sent last, F f)
```

Right-folds a range using the last element as the initial value.

Template Parameters

- **BidIter** – The type of the source begin iterator (deduced).

⁷⁸⁴ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **Sent** – The type of the source sentinel (deduced).
- **F** – The type of the binary function object.

Parameters

- **first** – Refers to the beginning of the sequence.
- **last** – Refers to the end of the sequence.
- **f** – Binary function object that will be applied.

Returns

An optional containing the fold result, or `std::nullopt` if the range is empty.

```
template<typename Rng, typename F>
auto hpx::ranges::fold_right_last(Rng &&rng, F f)
```

Right-folds a range using the last element as the initial value.

Template Parameters

- **Rng** – The type of the source range (deduced).
- **F** – The type of the binary function object.

Parameters

- **rng** – Refers to the sequence of elements.
- **f** – Binary function object that will be applied.

Returns

An optional containing the fold result, or `std::nullopt` if the range is empty.

hpx::ranges::fold_left_with_iter

Defined in header [hpx/algorithm.hpp](#)⁷⁸⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ranges::fold_left`
- `hpx::ranges::fold_left_first`
- `hpx::ranges::fold_right`
- `hpx::ranges::fold_right_last`
- `hpx::ranges::fold_left_first_with_iter`

```
template<typename InIter, typename Sent, typename T, typename F>
auto hpx::ranges::fold_left_with_iter(InIter first, Sent last, T init, F f) ->
    fold_left_with_iter_result<InIter,
    std::decay_t<hpx::util::invoke_result_t<F&, T,
    hpx::traits::iter_reference_t<InIter>>>>
```

Left-folds a range and returns both the result and the end iterator.

Template Parameters

- **InIter** – The type of the source begin iterator (deduced).

⁷⁸⁵ <http://github.com/TheHPXProject/hpx/blob/d402cead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **Sent** – The type of the source sentinel (deduced).
- **T** – The type of the initial value.
- **F** – The type of the binary function object.

Parameters

- **first** – Refers to the beginning of the sequence.
- **last** – Refers to the end of the sequence.
- **init** – The initial value for the fold operation.
- **f** – Binary function object that will be applied.

Returns

A `fold_left_with_iter_result` containing the end iterator and the fold result.

```
template<typename Rng, typename T, typename F>
auto hpx::ranges::fold_left_with_iter(Rng &&rng, T init, F f) ->
    fold_left_with_iter_result<std::ranges::iterator_t<Rng>,
    std::decay_t<hpx::util::invoke_result_t<F&, T,
    hpx::traits::range_reference_t<Rng>>>>
```

Left-folds a range and returns both the result and the end iterator.

Template Parameters

- **Rng** – The type of the source range (deduced).
- **T** – The type of the initial value.
- **F** – The type of the binary function object.

Parameters

- **rng** – Refers to the sequence of elements.
- **init** – The initial value for the fold operation.
- **f** – Binary function object that will be applied.

Returns

A `fold_left_with_iter_result` containing the end iterator and the fold result.

`hpx::ranges::fold_left_first_with_iter`

Defined in header `hpx/algorithm.hpp`⁷⁸⁶.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::ranges::fold_left`
- `hpx::ranges::fold_left_first`
- `hpx::ranges::fold_right`
- `hpx::ranges::fold_right_last`
- `hpx::ranges::fold_left_with_iter`

```
template<typename InIter, typename Sent, typename F>
```

⁷⁸⁶ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

```
auto hpx::ranges::fold_left_first_with_iter(InIter first, Sent last, F f) ->
    fold_left_first_with_iter_result<InIter,
    hpx::optional<std::decay_t<hpx::util::invoke_result_t<F&,
    hpx::traits::iter_reference_t<InIter>,
    hpx::traits::iter_reference_t<InIter>>>>>
```

Left-folds a range using the first element and returns both an optional result and the end iterator.

Template Parameters

- **InIter** – The type of the source begin iterator (deduced).
- **Sent** – The type of the source sentinel (deduced).
- **F** – The type of the binary function object.

Parameters

- **first** – Refers to the beginning of the sequence.
- **last** – Refers to the end of the sequence.
- **f** – Binary function object that will be applied.

Returns

A `fold_left_first_with_iter_result` containing the end iterator and an optional with the fold result.

```
template<typename Rng, typename F>
auto hpx::ranges::fold_left_first_with_iter(Rng &&rng, F f) ->
    fold_left_first_with_iter_result<std::ranges::iterator_t<Rng>,
    hpx::optional<std::decay_t<hpx::util::invoke_result_t<F&,
    hpx::traits::range_reference_t<Rng>,
    hpx::traits::range_reference_t<Rng>>>>>
```

Left-folds a range using the first element and returns both an optional result and the end iterator.

Template Parameters

- **Rng** – The type of the source range (deduced).
- **F** – The type of the binary function object.

Parameters

- **rng** – Refers to the sequence of elements.
- **f** – Binary function object that will be applied.

Returns

A `fold_left_first_with_iter_result` containing the end iterator and an optional with the fold result.

hpx::ranges::swap_ranges

Defined in header `hpx/algorithm.hpp`⁷⁸⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename InIter1, typename Sent1, typename InIter2, typename Sent2>
swap_ranges_result<InIter1, InIter2> hpx::ranges::swap_ranges(InIter1 first1, Sent1 last1, InIter2 first2, Sent2
    last2)
```

⁷⁸⁷ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e77/libs/full/include/include/hpx/algorithm.hpp>

Exchanges elements between range [first1, last1) and another range starting at *first2*.

The swap operations in the parallel *swap_ranges* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: Linear in the distance between *first1* and *last1*

Template Parameters

- **InIter1** – The type of the first range of iterators to swap (deduced).
- **Sent1** – The type of the first sentinel (deduced). This sentinel type must be a sentinel for InIter1.
- **InIter2** – The type of the second range of iterators to swap (deduced).
- **Sent2** – The type of the second sentinel (deduced). This sentinel type must be a sentinel for InIter2.

Parameters

- **first1** – Refers to the beginning of the sequence of elements for the first range.
- **last1** – Refers to sentinel value denoting the end of the sequence of elements for the first range.
- **first2** – Refers to the beginning of the sequence of elements for the second range.
- **last2** – Refers to sentinel value denoting the end of the sequence of elements for the second range.

Returns

The *swap_ranges* algorithm returns *swap_ranges_result<InIter1, InIter2>*. The *swap_ranges* algorithm returns in_in_result with the first element as the iterator to the element past the last element exchanged in range beginning with *first1* and the second element as the iterator to the element past the last element exchanged in the range beginning with *first2*.

```
template<typename ExPolicy, typename FwdIter1, typename Sent1, typename FwdIter2, typename Sent2>
parallel::util::detail::algorithm_result<ExPolicy, swap_ranges_result<FwdIter1, FwdIter2>>::type hpx::ranges::swap_ranges(Ex
```

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Exchanges elements between range [first1, last1) and another range starting at *first2*.

The swap operations in the parallel *swap_ranges* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The swap operations in the parallel *swap_ranges* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Linear in the distance between *first1* and *last1*

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **FwdIter1** – The type of the first range of iterators to swap (deduced).
- **Sent1** – The type of the first sentinel (deduced). This sentinel type must be a sentinel for FwdIter1.
- **FwdIter2** – The type of the second range of iterators to swap (deduced).
- **Sent2** – The type of the second sentinel (deduced). This sentinel type must be a sentinel for FwdIter2.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first1** – Refers to the beginning of the sequence of elements for the first range.
- **last1** – Refers to sentinel value denoting the end of the sequence of elements for the first range.
- **first2** – Refers to the beginning of the sequence of elements for the second range.
- **last2** – Refers to sentinel value denoting the end of the sequence of elements for the second range.

Returns

The *swap_ranges* algorithm returns a `hpx::future<swap_ranges_result<FwdIter1, FwdIter2>>` if the execution policy is of type *parallel_task_policy* and returns *FwdIter2* otherwise. The *swap_ranges* algorithm returns `in_in_result` with the first element as the iterator to the element past the last element exchanged in range beginning with *first1* and the second element as the iterator to the element past the last element exchanged in the range beginning with *first2*.

```
template<typename Rng1, typename Rng2>
swap_ranges_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng2>> hpx::ranges::swap_ranges(Rng1
                                                                                                     &&rng1,
                                                                                                     Rng2
                                                                                                     &&rng2)
```

Exchanges elements between range [first1, last1) and another range starting at *first2*.

The swap operations in the parallel *swap_ranges* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: Linear in the distance between *first1* and *last1*

Template Parameters

- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the destination range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

Parameters

- **rng1** – Refers to the sequence of elements of the first range.
- **rng2** – Refers to the sequence of elements of the second range.

Returns

The *swap_ranges* algorithm returns *swap_ranges_result*< std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng1>>. The *swap_ranges* algorithm returns in *in_result* with the first element as the iterator to the element past the last element exchanged in range beginning with *first1* and the second element as the iterator to the element past the last element exchanged in the range beginning with *first2*.

```
template<typename ExPolicy, typename Rng1, typename Rng2>
parallel::util::detail::algorithm_result<ExPolicy, swap_ranges_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng2>>> h
```

Exchanges elements between range [first1, last1) and another range starting at *first2*.

The swap operations in the parallel *swap_ranges* algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The swap operations in the parallel *swap_ranges* algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: Linear in the distance between *first1* and *last1*

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **Rng1** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Rng2** – The type of the destination range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng1** – Refers to the sequence of elements of the first range.
- **rng2** – Refers to the sequence of elements of the second range.

Returns

The *swap_ranges* algorithm returns a `hpx::future<swap_ranges_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng1>>>` if the execution policy is of type *parallel_task_policy* and returns *swap_ranges_result<std::ranges::iterator_t<Rng1>, std::ranges::iterator_t<Rng1>>*. otherwise. The *swap_ranges* algorithm returns *in_in_result* with the first element as the iterator to the element past the last element exchanged in range beginning with *first1* and the second element as the iterator to the element past the last element exchanged in the range beginning with *first2*.

hpx::ranges::sort

Defined in header `hpx/algorithm.hpp`⁷⁸⁸.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename RandomIt, typename Sent, typename Comp = ranges::less, typename Proj = hpx::identity>
RandomIt hpx::ranges::sort(RandomIt first, Sent last, Comp &&comp = Comp(), Proj &&proj = Proj())
```

Sorts the elements in the range `[first, last)` in ascending order. The order of equal elements is not guaranteed to be preserved. The function uses the given comparison function object *comp* (defaults to using `operator<()`).

A sequence is sorted with respect to a comparator *comp* and a projection *proj* if for every iterator *i* pointing to the sequence and every non-negative integer *n* such that *i + n* is a valid iterator pointing to an element of the sequence, and `INVOKE(comp, INVOKE(proj, *(i + n)), INVOKE(proj, *i)) == false`.

comp has to induce a strict weak ordering on the values.

The assignments in the parallel *sort* algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: $O(N \log(N))$, where $N = \text{detail::distance}(\text{first}, \text{last})$ comparisons.

Template Parameters

- **RandomIt** – The type of the source iterators used (deduced). This iterator type must meet the requirements of an random iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for `RandomIt`.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **comp** – *comp* is a callable object. The return value of the `INVOKE` operation applied to an object of type *Comp*, when contextually converted to `bool`, yields `true` if the first argument of the call is less than the second, and `false` otherwise. It is assumed that *comp* will not apply any non-constant function through the dereferenced iterator.

⁷⁸⁸ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/algorithm.hpp>

- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

Returns

The *sort* algorithm returns *RandomIt*. The algorithm returns an iterator pointing to the first element after the last element in the input sequence.

```
template<typename ExPolicy, typename RandomIt, typename Sent, typename Comp = ranges::less, typename  
Proj = hpx::identity>  
parallel::util::detail::algorithm_result<ExPolicy, RandomIt>::type hpx::ranges::sort(ExPolicy &&policy,  
RandomIt first, Sent last,  
Comp &&comp = Comp(),  
Proj &&proj = Proj())
```

Sorts the elements in the range [first, last) in ascending order. The order of equal elements is not guaranteed to be preserved. The function uses the given comparison function object *comp* (defaults to using operator<()).

A sequence is sorted with respect to a comparator *comp* and a projection *proj* if for every iterator *i* pointing to the sequence and every non-negative integer *n* such that *i* + *n* is a valid iterator pointing to an element of the sequence, and INVOKE(comp, INVOKE(proj, *(i + n)), INVOKE(proj, *i)) == false.

comp has to induce a strict weak ordering on the values.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: O(N log(N)), where N = detail::distance(first, last) comparisons.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it executes the assignments.
- **RandomIt** – The type of the source iterators used (deduced). This iterator type must meet the requirements of a random iterator.
- **Sent** – The type of the source sentinel (deduced). This sentinel type must be a sentinel for RandomIt.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **first** – Refers to the beginning of the sequence of elements the algorithm will be applied to.
- **last** – Refers to sentinel value denoting the end of the sequence of elements the algorithm will be applied.
- **comp** – *comp* is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to bool, yields true if the first argument

of the call is less than the second, and false otherwise. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator.

- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate `comp` is invoked.

Returns

The `sort` algorithm returns a `hpx::future<RandomIt>` if the execution policy is of type `sequenced_task_policy` or `parallel_task_policy` and returns `RandomIt` otherwise. The algorithm returns an iterator pointing to the first element after the last element in the input sequence.

```
template<typename Rng, typename Comp, typename Proj>
```

```
std::ranges::iterator_t<Rng> hpx::ranges::sort(Rng &&rng, Comp &&comp = Comp(), Proj &&proj = Proj())
```

Sorts the elements in the range `rng` in ascending order. The order of equal elements is not guaranteed to be preserved. The function uses the given comparison function object `comp` (defaults to using `operator<()`).

A sequence is sorted with respect to a comparator `comp` and a projection `proj` if for every iterator `i` pointing to the sequence and every non-negative integer `n` such that `i + n` is a valid iterator pointing to an element of the sequence, and `INVOKE(comp, INVOKE(proj, *(i + n)), INVOKE(proj, *i)) == false`.

`comp` has to induce a strict weak ordering on the values.

The assignments in the parallel `sort` algorithm invoked without an execution policy object execute in sequential order in the calling thread.

Note: Complexity: $O(N \log(N))$, where $N = \text{std::distance}(\text{begin}(\text{rng}), \text{end}(\text{rng}))$ comparisons.

Template Parameters

- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to `hpx::identity`

Parameters

- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **comp** – `comp` is a callable object. The return value of the `INVOKE` operation applied to an object of type `Comp`, when contextually converted to `bool`, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that `comp` will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate `comp` is invoked.

Returns

The `sort` algorithm returns `std::ranges::iterator_t<Rng>`. It returns `last`.

```
template<typename ExPolicy, typename Rng, typename Comp = ranges::less, typename Proj = hpx::identity>
```

```
parallel::util::detail::algorithm_result<ExPolicy, std::ranges::iterator_t<Rng>> hpx::ranges::sort(ExPolicy
                                                                 &&policy,
                                                                 Rng &&rng,
                                                                 Comp
                                                                 &&comp =
                                                                 Comp(), Proj
                                                                 &&proj =
                                                                 Proj())
```

Sorts the elements in the range *rng* in ascending order. The order of equal elements is not guaranteed to be preserved. The function uses the given comparison function object *comp* (defaults to using operator<()).

A sequence is sorted with respect to a comparator *comp* and a projection *proj* if for every iterator *i* pointing to the sequence and every non-negative integer *n* such that *i* + *n* is a valid iterator pointing to an element of the sequence, and INVOKE(*comp*, INVOKE(*proj*, *(*i* + *n*)), INVOKE(*proj*, **i*)) == false.

comp has to induce a strict weak ordering on the values.

The application of function objects in parallel algorithm invoked with an execution policy object of type *sequenced_policy* execute in sequential order in the calling thread.

The application of function objects in parallel algorithm invoked with an execution policy object of type *parallel_policy* or *parallel_task_policy* are permitted to execute in an unordered fashion in unspecified threads, and indeterminately sequenced within each thread.

Note: Complexity: $O(N \log(N))$, where $N = \text{std::distance}(\text{begin}(\text{rng}), \text{end}(\text{rng}))$ comparisons.

Template Parameters

- **ExPolicy** – The type of the execution policy to use (deduced). It describes the manner in which the execution of the algorithm may be parallelized and the manner in which it applies user-provided function objects.
- **Rng** – The type of the source range used (deduced). The iterators extracted from this range type must meet the requirements of an input iterator.
- **Comp** – The type of the function/function object to use (deduced).
- **Proj** – The type of an optional projection function. This defaults to *hpx::identity*

Parameters

- **policy** – The execution policy to use for the scheduling of the iterations.
- **rng** – Refers to the sequence of elements the algorithm will be applied to.
- **comp** – *comp* is a callable object. The return value of the INVOKE operation applied to an object of type *Comp*, when contextually converted to bool, yields true if the first argument of the call is less than the second, and false otherwise. It is assumed that *comp* will not apply any non-constant function through the dereferenced iterator.
- **proj** – Specifies the function (or function object) which will be invoked for each pair of elements as a projection operation before the actual predicate *comp* is invoked.

Returns

The *sort* algorithm returns a *hpx::future<std::ranges::iterator_t<Rng>* if the execution policy is of type *sequenced_task_policy* or *parallel_task_policy* and returns *std::ranges::iterator_t<Rng>* otherwise. It returns *last*.

asio

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/asio/asio_util.hpp

Defined in header hpx/asio/asio_util.hpp.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **util**

Functions

bool **get_endpoint**(std::string const &addr, std::uint16_t port, ::asio::ip::tcp::endpoint &ep, bool force_ipv4 = false)

std::string **get_endpoint_name**(::asio::ip::tcp::endpoint const &ep)

::asio::ip::tcp::endpoint **resolve_hostname**(std::string const &hostname, std::uint16_t port, ::asio::io_context &io_service, bool force_ipv4 = false)

detail::endpoint_iterator_type **connect_begin**(std::string const &address, std::uint16_t port, ::asio::io_context &io_service)

template<typename **Locality**>

detail::endpoint_iterator_type **connect_begin**(Locality const &loc, ::asio::io_context &io_service)

Returns an iterator which when dereferenced will give an endpoint suitable for a call to connect() related to this locality.

inline detail::endpoint_iterator_type **connect_end**()

detail::endpoint_iterator_type **accept_begin**(std::string const &address, std::uint16_t port, ::asio::io_context &io_service)

template<typename **Locality**>

detail::endpoint_iterator_type **accept_begin**(Locality const &loc, ::asio::io_context &io_service)

Returns an iterator which when dereferenced will give an endpoint suitable for a call to accept() related to this locality.

inline detail::endpoint_iterator_type **accept_end**()

assertion

See *Public API* for a list of names and headers that are part of the public *HPX* API.

HPX_CURRENT_SOURCE_LOCATION

Defined in header [hpx/source_location.hpp](#)⁷⁸⁹.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::source_location*

Warning: doxygenfunction: Cannot find function “HPX_CURRENT_SOURCE_LOCATION” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

hpx::source_location

Defined in header [hpx/source_location.hpp](#)⁷⁹⁰.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *HPX_CURRENT_SOURCE_LOCATION*

Warning: doxygenfunction: Cannot find function “hpx::source_location” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

HPX_ASSERT

Defined in header [hpx/assert.hpp](#).

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *HPX_ASSERT_MSG*

Warning: doxygenfunction: Cannot find function “HPX_ASSERT” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

⁷⁸⁹ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/source_location.hpp

⁷⁹⁰ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/source_location.hpp

HPX_ASSERT_MSG

Defined in header `hpx/assert.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *HPX_ASSERT*

Warning: doxygenfunction: Cannot find function “HPX_ASSERT_MSG” in doxygen xml output for project “hpx” from directory: `../hpx_autodoc`

hpx/assertion/api.hpp

Defined in header `hpx/assertion/api.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **assertion**

Typedefs

using **assertion_handler** = void (*)(hpx::source_location const &loc, char const *expr, std::string const &msg)

The signature for an assertion handler.

Functions

assertion_handler **set_assertion_handler**(*assertion_handler* handler)

Set the assertion handler to be used within a program. If the handler has been set already once, the call to this function will be ignored.

Note: This function is not thread safe

hpx/assertion/evaluate_assert.hpp

Defined in header `hpx/assertion/evaluate_assert.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **assertion**

async_base

See *Public API* for a list of names and headers that are part of the public HPX API.

hpx::launch

Defined in header `hpx/future.hpp`⁷⁹¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
struct launch : public hpx::detail::policy_holder<>  
    Launch policies for hpx::async etc.
```

hpx::post

Defined in header `hpx/future.hpp`⁷⁹².

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename F, typename ...Ts>  
bool hpx::post(F &&f, Ts&&... ts)
```

Runs the function *f* asynchronously (potentially in a separate thread which might be a part of a thread pool). This is done in a fire-and-forget manner, meaning there is no return value or way to synchronize with the function execution (it does not return an *hpx::future* that would hold the result of that function call).

hpx::post is particularly useful when synchronization mechanisms as heavyweight as futures are not desired, and instead, more lightweight mechanisms like latches or atomic variables are preferred. Essentially, the post function enables the launch of a new thread without the overhead of creating a future.

Note: *hpx::post* is similar to *hpx::async* but does not return a future. This is why there is no way of finding out the result/failure of the execution of this function.

```
template<typename Action, typename ...Ts>  
bool hpx::post(hpx::id_type const &id, Ts&&... vs)
```

```
template<typename Action, typename Client, typename Stub, typename Data, typename ...Ts>  
bool hpx::post(components::client_base<Client, Stub, Data> const &c, Ts&&... vs)
```

```
template<typename Action, typename DistPolicy, typename ...Ts>  
bool hpx::post(DistPolicy const &policy, Ts&&... vs)
```

```
template<typename Action, typename Continuation, typename ...Ts>  
bool hpx::post(Continuation &&c, hpx::id_type const &gid, Ts&&... vs)
```

```
template<typename Action, typename Continuation, typename Client, typename Stub, typename Data,  
typename ...Ts>  
bool hpx::post(Continuation &&cont, components::client_base<Client, Stub, Data> const &c, Ts&&... vs)
```

```
template<typename Action, typename Continuation, typename DistPolicy, typename ...Ts>
```

⁷⁹¹ <http://github.com/TheHPXProject/hpx/blob/d402eoad7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

⁷⁹² <http://github.com/TheHPXProject/hpx/blob/d402eoad7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

bool `hpx::post`(*Continuation* &&c, *DistPolicy* const &policy, *Ts*&&... vs)

```
template<typename Action, typename Target, typename ...Ts>
bool hpx::post(Action &&action, Target &&target, Ts&&... ts)
```

The distributed implementation of `hpx::post` can be used by giving an action instance as argument instead of a function, and also by providing another argument with the locality ID or the target ID.

Template Parameters

- **Action** – The type of action instance
- **Target** – The type of target where the action should be executed
- **Ts** – The type of any additional arguments

Parameters

- **action** – The action instance to be executed
- **target** – The target where the action should be executed
- **ts** – Additional arguments

Returns

true if the action was successfully posted, false otherwise.

hpx::sync

Defined in header `hpx/future.hpp`⁷⁹³.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename Action, typename F, typename ...Ts>
auto hpx::sync(F &&f, Ts&&... ts) -> decltype(detail::sync_action_dispatch<Action,
std::decay_t<F>>::call(HPX_FORWARD(F, f), HPX_FORWARD(Ts, ts)...))
```

The function template `sync` runs the function `f` synchronously and returns a `hpx::future` that will eventually hold the result of that function call.

```
template<typename Action, typename F, typename ...Ts>
auto hpx::sync(F &&f, Ts&&... ts) -> decltype(detail::sync_action_dispatch<Action,
std::decay_t<F>>::call(HPX_FORWARD(F, f), HPX_FORWARD(Ts, ts)...))
```

The function template `sync` runs the function `f` synchronously and returns a `hpx::future` that will eventually hold the result of that function call.

Warning: doxygenfunction: Unable to resolve function “hpx::sync” with arguments (Action&&, Target&&, Ts&&...) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:

```
- template<typename Action, typename F, typename ...Ts> auto sync(F &&f, Ts&&... ts) -
-> decltype(detail::sync_action_dispatch<Action, std::decay_t<F>>::call(HPX_
->FORWARD(F, f), HPX_FORWARD(Ts, ts)...))
- template<typename Action, typename Target, typename ...Ts> decltype(auto)
->sync(Action &&action, Target &&target, Ts&&... ts)
- template<typename F, typename ...Ts> decltype(auto) sync(F &&f, Ts&&... ts)
```

⁷⁹³ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/future.hpp>

hpx::dataflow

Defined in header `hpx/future.hpp`⁷⁹⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

Warning: doxygenfunction: Unable to resolve function “hpx::dataflow” with arguments (F&&, Ts&&...) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:

```

- template<typename Action, typename Target, typename ...Ts> decltype(auto)
  ↳dataflow(Action &&action, Target &&target, Ts&&... ts)
- template<typename F, typename ...Ts> decltype(auto) dataflow(F &&f, Ts&&... ts)

```

Warning: doxygenfunction: Unable to resolve function “hpx::dataflow” with arguments (Action&&, Target&&, Ts&&...) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:

```

- template<typename Action, typename Target, typename ...Ts> decltype(auto)
  ↳dataflow(Action &&action, Target &&target, Ts&&... ts)
- template<typename F, typename ...Ts> decltype(auto) dataflow(F &&f, Ts&&... ts)

```

hpx::async

Defined in header `hpx/future.hpp`⁷⁹⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

```

template<typename Action, typename F, typename ...Ts>
auto hpx::async(F &&f, Ts&&... ts) -> decltype(detail::async_action_dispatch<Action,
  std::decay_t<F>>::call(HPX_FORWARD(F, f), HPX_FORWARD(Ts, ts)...))

```

The function template *async* runs the function *f* asynchronously (potentially in a separate thread which might be a part of a thread pool) and returns a `hpx::future` that will eventually hold the result of that function call. If no policy is defined, *async* behaves as if it is called with policy being `hpx::launch::async` | `hpx::launch::deferred`. Otherwise, it calls a function *f* with arguments *ts* according to a specific launch policy.

- If the `async` flag is set (i.e. `(policy & hpx::launch::async) != 0`), then *async* executes the callable object *f* on a new thread of execution (with all thread-locals initialized) as if spawned by `hpx::thread(std::forward<F>(f), std::forward<Ts>(ts)...)`, except that if the function *f* returns a value or throws an exception, it is stored in the shared state accessible through the `hpx::future` that *async* returns to the caller.
- If the `deferred` flag is set (i.e. `(policy & hpx::launch::deferred) != 0`), then *async* converts *f* and *ts...* the same way as by `hpx::thread` constructor, but does not spawn a new thread of execution. Instead, lazy evaluation is performed: the first call to a non-timed wait function on the `hpx::future` that *async* returned to the caller will cause the copy of *f* to be invoked (as a rvalue) with the copies of *ts...* (also passed as rvalues) in the current thread (which does not have to be the thread that originally called `hpx::async`). The result or exception is placed in the shared state associated with the future and only then it is made ready. All further accesses to the same `hpx::future` will return the result immediately.
- If neither `hpx::launch::async` nor `hpx::launch::deferred`, nor any implementation-defined policy flag is set in policy, the behavior is undefined.

⁷⁹⁴ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

⁷⁹⁵ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

If more than one flag is set, it is implementation-defined which policy is selected. For the default (both the `hpx::launch::async` and `hpx::launch::deferred` flags are set in policy), standard recommends (but doesn't require) utilizing available concurrency, and deferring any additional tasks.

In any case, the call to `hpx::async` synchronizes-with (as defined in `std::memory_order`) the call to `f`, and the completion of `f` is sequenced-before making the shared state ready. If the `async` policy is chosen, the associated thread completion synchronizes-with the successful return from the first function that is waiting on the shared state, or with the return of the last function that releases the shared state, whichever comes first. If `std::decay<Function>::type` or each type in `std::decay<Ts>::type` is not constructible from its corresponding argument, the program is ill-formed.

Parameters

- **f** – Callable object to call
- **ts** – parameters to pass to `f`

Returns

hpx::future referring to the shared state created by this call to `hpx::async`.

```
template<typename Action, typename F, typename ...Ts>
auto hpx::async(F &&f, Ts&&... ts) -> decltype(detail::async_action_dispatch<Action,
std::decay_t<F>>::call(HPX_FORWARD(F, f), HPX_FORWARD(Ts, ts)...))
```

The function template `async` runs the function `f` asynchronously (potentially in a separate thread which might be a part of a thread pool) and returns a `hpx::future` that will eventually hold the result of that function call. If no policy is defined, `async` behaves as if it is called with policy being `hpx::launch::async` | `hpx::launch::deferred`. Otherwise, it calls a function `f` with arguments `ts` according to a specific launch policy.

- If the `async` flag is set (i.e. `(policy & hpx::launch::async) != 0`), then `async` executes the callable object `f` on a new thread of execution (with all thread-locals initialized) as if spawned by `hpx::thread(std::forward<F>(f), std::forward<Ts>(ts)...)`, except that if the function `f` returns a value or throws an exception, it is stored in the shared state accessible through the `hpx::future` that `async` returns to the caller.
- If the `deferred` flag is set (i.e. `(policy & hpx::launch::deferred) != 0`), then `async` converts `f` and `ts...` the same way as by `hpx::thread` constructor, but does not spawn a new thread of execution. Instead, lazy evaluation is performed: the first call to a non-timed wait function on the `hpx::future` that `async` returned to the caller will cause the copy of `f` to be invoked (as a rvalue) with the copies of `ts...` (also passed as rvalues) in the current thread (which does not have to be the thread that originally called `hpx::async`). The result or exception is placed in the shared state associated with the future and only then it is made ready. All further accesses to the same `hpx::future` will return the result immediately.
- If neither `hpx::launch::async` nor `hpx::launch::deferred`, nor any implementation-defined policy flag is set in policy, the behavior is undefined.

If more than one flag is set, it is implementation-defined which policy is selected. For the default (both the `hpx::launch::async` and `hpx::launch::deferred` flags are set in policy), standard recommends (but doesn't require) utilizing available concurrency, and deferring any additional tasks.

In any case, the call to `hpx::async` synchronizes-with (as defined in `std::memory_order`) the call to `f`, and the completion of `f` is sequenced-before making the shared state ready. If the `async` policy is chosen, the associated thread completion synchronizes-with the successful return from the first function that is waiting on the shared state, or with the return of the last function that releases the shared state, whichever comes first. If `std::decay<Function>::type` or each type in `std::decay<Ts>::type` is not constructible from its corresponding argument, the program is ill-formed.

Parameters

- **f** – Callable object to call
- **ts** – parameters to pass to `f`

Returns

hpx::future referring to the shared state created by this call to *hpx::async*.

Warning: doxygenfunction: Unable to resolve function “hpx::async” with arguments (Executor&&, hpx::sycl::experimental::sycl_executor::queue_function_ptr_t<Ts...>&&, Ts&&...) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:

```
- template<typename Action, typename F, typename ...Ts> auto async(F &&f, Ts&&... ts)
  ↳-> decltype(detail::async_action_dispatch<Action, std::decay_t<F>>::call(HPX_
  ↳FORWARD(F, f), HPX_FORWARD(Ts, ts)...))
- template<typename Action, typename Target, typename ...Ts> auto async(Action &&
  ↳action, Target &&target, Ts&&... ts)
- template<typename Executor, typename ...Ts> decltype(auto) async(Executor &&exec,
  ↳hpx::sycl::experimental::sycl_executor::queue_function_ptr_t<Ts...> &&f, Ts&&... ts)
- template<typename F, typename ...Ts> decltype(auto) async(F &&f, Ts&&... ts)
```

```
template<typename Action, typename Target, typename ...Ts>
auto hpx::async(Action &&action, Target &&target, Ts&&... ts)
```

The distributed implementation of *hpx::async* can be used by giving an action instance as argument instead of a function, and also by providing another argument with the locality ID or the target ID. The action executes asynchronously.

Template Parameters

- **Action** – The type of action instance
- **Target** – The type of target where the action should be executed
- **Ts** – The type of any additional arguments

Parameters

- **action** – The action instance to be executed
- **target** – The target where the action should be executed
- **ts** – Additional arguments

Returns

hpx::future referring to the shared state created by this call to *hpx::async*

async_combinators

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx::wait_some

Defined in header `hpx/future.hpp`⁷⁹⁶.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::wait_some_nothrow*
- *hpx::wait_some_n*

⁷⁹⁶ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e77/libs/full/include/include/hpx/future.hpp>

- `hpx::wait_some_n_nothrow`

```
template<typename InputIter>
```

```
void hpx::wait_some(std::size_t n, InputIter first, InputIter last)
```

The function `wait_some` is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns a new future object representing the same list of futures after `n` of them finished executing.

Note: The function `wait_some` returns after `n` futures have become ready. All input futures are still valid after `wait_some` returns.

Note: The function `wait_some` will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use `wait_some_nothrow` instead.

Parameters

- **`n`** – [in] The number of futures out of the arguments which have to become ready in order for the function to return.
- **`first`** – [in] The iterator pointing to the first element of a sequence of *future* or *shared_future* objects for which *when_all* should wait.
- **`last`** – [in] The iterator pointing to the last element of a sequence of *future* or *shared_future* objects for which *when_all* should wait.

```
template<typename R>
```

```
void hpx::wait_some(std::size_t n, std::vector<future<R>> &&futures)
```

The function `wait_some` is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns a new future object representing the same list of futures after `n` of them finished executing.

Note: The function `wait_some` returns after `n` futures have become ready. All input futures are still valid after `wait_some` returns.

Note: The function `wait_some` will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use `wait_some_nothrow` instead.

Parameters

- **`n`** – [in] The number of futures out of the arguments which have to become ready in order for the returned future to get ready.
- **`futures`** – [in] A vector holding an arbitrary amount of *future* or *shared_future* objects for which `wait_some` should wait.

```
template<typename R, std::size_t N>
```

```
void hpx::wait_some(std::size_t n, std::array<future<R>, N> &&futures)
```

The function `wait_some` is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns a new future object representing the same list of futures after `n` of them finished executing.

Note: The function `wait_some` returns after n futures have become ready. All input futures are still valid after `wait_some` returns.

Note: The function `wait_some` will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use `wait_some_nothrow` instead.

Parameters

- **n** – [in] The number of futures out of the arguments which have to become ready in order for the returned future to get ready.
- **futures** – [in] An array holding an arbitrary amount of `future` or `shared_future` objects for which `wait_some` should wait.

```
template<typename ...T>
void hpx::wait_some(std::size_t n, T&&... futures)
```

The function `wait_some` is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns a new future object representing the same list of futures after n of them finished executing.

Note: The function `wait_all` returns after n futures have become ready. All input futures are still valid after `wait_all` returns.

Note: The function `wait_some` will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use `wait_some_nothrow` instead.

Parameters

- **n** – [in] The number of futures out of the arguments which have to become ready in order for the returned future to get ready.
- **futures** – [in] An arbitrary number of `future` or `shared_future` objects, possibly holding different types for which `wait_some` should wait.

`hpx::wait_some_nothrow`

Defined in header `hpx/future.hpp`⁷⁹⁷.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::wait_some`
- `hpx::wait_some_n`
- `hpx::wait_some_n_nothrow`

⁷⁹⁷ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

Warning: doxygenfunction: Cannot find function “hpx::wait_some_nothrow” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

hpx::wait_some_n

Defined in header [hpx/future.hpp](#)⁷⁹⁸.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- [hpx::wait_some](#)
- [hpx::wait_some_nothrow](#)
- [hpx::wait_some_n_nothrow](#)

```
template<typename InputIter>
```

```
void hpx::wait_some_n(std::size_t n, InputIter first, std::size_t count)
```

The function `wait_some_n` is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns a new future object representing the same list of futures after `n` of them finished executing.

Note: The function `wait_some_n` returns after `n` futures have become ready. All input futures are still valid after `wait_some_n` returns.

Note: The function `wait_some_n` will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use `wait_some_n_nothrow` instead.

Parameters

- **n** – [in] The number of futures out of the arguments which have to become ready in order for the returned future to get ready.
- **first** – [in] The iterator pointing to the first element of a sequence of *future* or *shared_future* objects for which *when_all* should wait.
- **count** – [in] The number of elements in the sequence starting at *first*.

hpx::wait_some_n_nothrow

Defined in header [hpx/future.hpp](#)⁷⁹⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- [hpx::wait_some](#)
- [hpx::wait_some_nothrow](#)
- [hpx::wait_some_n](#)

⁷⁹⁸ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

⁷⁹⁹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

Warning: doxygenfunction: Cannot find function “hpx::wait_some_n_nothrow” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

hpx::wait_any

Defined in header `hpx/future.hpp`⁸⁰⁰.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::wait_any_nothrow`
- `hpx::wait_any_n`
- `hpx::wait_any_n_nothrow`

template<typename **InputIter**>

void **hpx::wait_any**(*InputIter* first, *InputIter* last)

The function *wait_any* is a non-deterministic choice operator. It OR-composes all future objects given and returns after one future of that list finishes execution.

Note: The function *wait_any* returns after at least one future has become ready. All input futures are still valid after *wait_any* returns.

Note: The function *wait_any* will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use *wait_any_nothrow* instead.

Parameters

- **first** – [in] The iterator pointing to the first element of a sequence of *future* or *shared_future* objects for which *wait_any* should wait.
- **last** – [in] The iterator pointing to the last element of a sequence of *future* or *shared_future* objects for which *wait_any* should wait.

template<typename **R**>

void **hpx::wait_any**(*std::vector<future<R>>* &futures)

The function *wait_any* is a non-deterministic choice operator. It OR-composes all future objects given and returns after one future of that list finishes execution.

Note: The function *wait_any* returns after at least one future has become ready. All input futures are still valid after *wait_any* returns.

Note: The function *wait_any* will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use *wait_any_nothrow* instead.

⁸⁰⁰ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

Parameters

futures – [in] A vector holding an arbitrary amount of *future* or *shared_future* objects for which *wait_any* should wait.

```
template<typename R, std::size_t N>
void hpx::wait_any(std::array<future<R>, N> &futures)
```

The function *wait_any* is a non-deterministic choice operator. It OR-composes all future objects given and returns after one future of that list finishes execution.

Note: The function *wait_any* returns after at least one future has become ready. All input futures are still valid after *wait_any* returns.

Note: The function *wait_any* will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use *wait_any_nothrow* instead.

Parameters

futures – [in] Amn array holding an arbitrary amount of *future* or *shared_future* objects for which *wait_any* should wait.

```
template<typename ...T>
void hpx::wait_any(T&&... futures)
```

The function *wait_any* is a non-deterministic choice operator. It OR-composes all future objects given and returns after one future of that list finishes execution.

Note: The function *wait_any* returns after at least one future has become ready. All input futures are still valid after *wait_any* returns.

Note: The function *wait_any* will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use *wait_any_nothrow* instead.

Parameters

futures – [in] An arbitrary number of *future* or *shared_future* objects, possibly holding different types for which *wait_any* should wait.

hpx::wait_any_nothrow

Defined in header [hpx/future.hpp](#)⁸⁰¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::wait_any*
- *hpx::wait_any_n*
- *hpx::wait_any_n_nothrow*

⁸⁰¹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

Warning: doxygenfunction: Cannot find function “hpx::wait_any_nothrow” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

hpx::wait_any_n

Defined in header [hpx/future.hpp](#)⁸⁰².

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::wait_any`
- `hpx::wait_any_nothrow`
- `hpx::wait_any_n_nothrow`

template<typename **InputIter**>

void **hpx::wait_any_n**(*InputIter* first, *std::size_t* count)

The function `wait_any_n` is a non-deterministic choice operator. It OR-composes all future objects given and returns after one future of that list finishes execution.

Note: The function `wait_any_n` returns after at least one future has become ready. All input futures are still valid after `wait_any_n` returns.

Note: The function `wait_any_n` will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use `wait_any_n_nothrow` instead.

Parameters

- **first** – [in] The iterator pointing to the first element of a sequence of *future* or *shared_future* objects for which `wait_any_n` should wait.
- **count** – [in] The number of elements in the sequence starting at *first*.

hpx::wait_any_n_nothrow

Defined in header [hpx/future.hpp](#)⁸⁰³.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::wait_any`
- `hpx::wait_any_nothrow`
- `hpx::wait_any_n`

⁸⁰² <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

⁸⁰³ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

Warning: doxygenfunction: Cannot find function “hpx::wait_any_n_nothrow” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

hpx::wait_each

Defined in header `hpx/future.hpp`⁸⁰⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::wait_each_nothrow`
- `hpx::wait_each_n`
- `hpx::wait_each_n_nothrow`

```
template<typename F, typename Future>
void hpx::wait_each(F &&f, std::vector<Future> &&futures)
```

The function `wait_each` is an operator allowing to join on the results of all given futures. It AND-composes all future objects given and returns after they finished executing. Additionally, the supplied function is called for each of the passed futures as soon as the future has become ready. `wait_each` returns after all futures have been become ready.

Note: This function consumes the futures as they are passed on to the supplied function. The callback should take one or two parameters, namely either a *future* to be processed or a type that `std::size_t` is implicitly convertible to as the first parameter and the *future* as the second parameter. The first parameter will correspond to the index of the current *future* in the collection.

Parameters

- **f** – The function which will be called for each of the input futures once the future has become ready.
- **futures** – A vector holding an arbitrary amount of *future* or *shared_future* objects for which `wait_each` should wait.

```
template<typename F, typename Iterator>
void hpx::wait_each(F &&f, Iterator begin, Iterator end)
```

The function `wait_each` is an operator allowing to join on the results of all given futures. It AND-composes all future objects given and returns after they finished executing. Additionally, the supplied function is called for each of the passed futures as soon as the future has become ready. `wait_each` returns after all futures have been become ready.

Note: This function consumes the futures as they are passed on to the supplied function. The callback should take one or two parameters, namely either a *future* to be processed or a type that `std::size_t` is implicitly convertible to as the first parameter and the *future* as the second parameter. The first parameter will correspond to the index of the current *future* in the collection.

Parameters

⁸⁰⁴ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e77/libs/full/include/include/hpx/future.hpp>

- **f** – The function which will be called for each of the input futures once the future has become ready.
- **begin** – The iterator pointing to the first element of a sequence of *future* or *shared_future* objects for which *wait_each* should wait.
- **end** – The iterator pointing to the last element of a sequence of *future* or *shared_future* objects for which *wait_each* should wait.

```
template<typename F, typename ...T>  
void hpx::wait_each(F &&f, T&&... futures)
```

The function *wait_each* is an operator allowing to join on the results of all given futures. It AND-composes all future objects given and returns after they finished executing. Additionally, the supplied function is called for each of the passed futures as soon as the future has become ready. *wait_each* returns after all futures have been become ready.

Note: This function consumes the futures as they are passed on to the supplied function. The callback should take one or two parameters, namely either a *future* to be processed or a type that *std::size_t* is implicitly convertible to as the first parameter and the *future* as the second parameter. The first parameter will correspond to the index of the current *future* in the collection.

Parameters

- **f** – The function which will be called for each of the input futures once the future has become ready.
- **futures** – An arbitrary number of *future* or *shared_future* objects, possibly holding different types for which *wait_each* should wait.

hpx::wait_each_nothrow

Defined in header [hpx/future.hpp](#)⁸⁰⁵.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::wait_each*
- *hpx::wait_each_n*
- *hpx::wait_each_n_nothrow*

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⁸⁰⁵ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

hpx::wait_each_n

Defined in header `hpx/future.hpp`⁸⁰⁶.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::wait_each`
- `hpx::wait_each_nothrow`
- `hpx::wait_each_n_nothrow`

template<typename **F**, typename **Iterator**>

void **hpx::wait_each_n**(*F* &&*f*, *Iterator* begin, *std::size_t* count)

The function `wait_each` is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns after they finished executing. Additionally, the supplied function is called for each of the passed futures as soon as the future has become ready.

Note: This function consumes the futures as they are passed on to the supplied function. The callback should take one or two parameters, namely either a *future* to be processed or a type that *std::size_t* is implicitly convertible to as the first parameter and the *future* as the second parameter. The first parameter will correspond to the index of the current *future* in the collection.

Parameters

- **f** – The function which will be called for each of the input futures once the future has become ready.
- **begin** – The iterator pointing to the first element of a sequence of *future* or *shared_future* objects for which `wait_each_n` should wait.
- **count** – The number of elements in the sequence starting at *first*.

hpx::wait_each_n_nothrow

Defined in header `hpx/future.hpp`⁸⁰⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::wait_each`
- `hpx::wait_each_nothrow`
- `hpx::wait_each_n`

Warning: doxygenfunction: Cannot find function “hpx::wait_each_n_nothrow” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

⁸⁰⁶ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

⁸⁰⁷ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

hpx::when_any

Defined in header `hpx/future.hpp`⁸⁰⁸.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename InputIter, typename Container = vector<future<typename  
std::iterator_traits<InputIter>::value_type>>>  
future<when_any_result<Container>>> hpx::when_any(InputIter first, InputIter last)
```

function *when_any* creates a future object that becomes when at least one element in a set of *future* and *shared_future* objects becomes ready. It is a non-deterministic choice operator. It OR-composes all given future objects and returns a new future object representing the same list of futures after one future of that list finishes execution.

Parameters

- **first** – [in] The iterator pointing to the first element of a sequence of *future* or *shared_future* objects for which *when_any* should wait.
- **last** – [in] The iterator pointing to the last element of a sequence of *future* or *shared_future* objects for which *when_any* should wait.

Returns

Returns a *when_any_result* holding the same list of futures as has been passed to *when_any* and an index pointing to a ready future.

- `future<when_any_result<Container<future<R>>>>>`: If the input cardinality is unknown at compile time and the futures are all have the same type. The order of the futures in the output container will be the same as given by the input iterator.

```
template<typename Range>  
future<when_any_result<Range>>> hpx::when_any(Range &values)
```

function *when_any* creates a future object that becomes when at least one element in a set of *future* and *shared_future* objects becomes ready. It is a non-deterministic choice operator. It OR-composes all given future objects and returns a new future object representing the same list of futures after one future of that list finishes execution.

Parameters

values – [in] A range holding an arbitrary amount of *futures* or *shared_future* objects for which *when_any* should wait.

Returns

Returns a *when_any_result* holding the same list of futures as has been passed to *when_any* and an index pointing to a ready future.

- `future<when_any_result<Container<future<R>>>>>`: If the input cardinality is unknown at compile time and the futures are all have the same type. The order of the futures in the output container will be the same as given by the input iterator.

```
template<typename ...T>  
future<when_any_result<tuple<future<T>...>>> hpx::when_any(T&&... futures)
```

function *when_any* creates a future object that becomes when at least one element in a set of *future* and *shared_future* objects becomes ready. It is a non-deterministic choice operator. It OR-composes all given future objects and returns a new future object representing the same list of futures after one future of that list finishes execution.

⁸⁰⁸ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

Parameters

futures – [in] An arbitrary number of *future* or *shared_future* objects, possibly holding different types for which *when_any* should wait.

Returns

Returns a *when_any_result* holding the same list of futures as has been passed to *when_any* and an index pointing to a ready future.

- `future<when_any_result<tuple<future<T0>, future<T1>...>>>`: If inputs are fixed in number and are of heterogeneous types. The inputs can be any arbitrary number of future objects.
- `future<when_any_result<tuple<>>>` if *when_any* is called with zero arguments. The returned future will be initially ready.

hpx::when_all

Defined in header `hpx/future.hpp`⁸⁰⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename InputIter, typename Container = vector<future<typename
std::iterator_traits<InputIter>::value_type>>>
hpx::future<Container> hpx::when_all(InputIter first, InputIter last)
```

function *when_all* creates a future object that becomes ready when all elements in a set of *future* and *shared_future* objects become ready. It is an operator allowing to join on the result of all given futures. It AND-composes all given future objects and returns a new future object representing the same list of futures after they finished executing.

Note: Calling this version of *when_all* where `first == last`, returns a future with an empty container that is immediately ready. Each future and *shared_future* is waited upon and then copied into the collection of the output (returned) future, maintaining the order of the futures in the input collection. The future returned by *when_all* will not throw an exception, but the futures held in the output collection may.

Parameters

- **first** – [in] The iterator pointing to the first element of a sequence of *future* or *shared_future* objects for which *when_all* should wait.
- **last** – [in] The iterator pointing to the last element of a sequence of *future* or *shared_future* objects for which *when_all* should wait.

Returns

Returns a future holding the same list of futures as has been passed to *when_all*.

- `future<Container<future<R>>>`: If the input cardinality is unknown at compile time and the futures are all have the same type. The order of the futures in the output container will be the same as given by the input iterator.

```
template<typename Range>
hpx::future<Range> hpx::when_all(Range &&values)
```

function *when_all* creates a future object that becomes ready when all elements in a set of *future* and *shared_future* objects become ready. It is an operator allowing to join on the result of all given futures. It AND-composes all given future objects and returns a new future object representing the same list of futures after they finished executing.

⁸⁰⁹ <http://github.com/TheHPXProject/hpx/blob/d402cead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

Note: Calling this version of *when_all* where the input container is empty, returns a future with an empty container that is immediately ready. Each future and *shared_future* is waited upon and then copied into the collection of the output (returned) future, maintaining the order of the futures in the input collection. The future returned by *when_all* will not throw an exception, but the futures held in the output collection may.

Parameters

values – [in] A range holding an arbitrary amount of *future* or *shared_future* objects for which *when_all* should wait.

Returns

Returns a future holding the same list of futures as has been passed to *when_all*.

- `future<Container<future<R>>>`: If the input cardinality is unknown at compile time and the futures are all have the same type.

```
template<typename ...T>
```

```
hpx::future<hpx::tuple<hpx::future<T>...>> hpx::when_all(T&&... futures)
```

function *when_all* creates a future object that becomes ready when all elements in a set of *future* and *shared_future* objects become ready. It is an operator allowing to join on the result of all given futures. It AND-composes all given future objects and returns a new future object representing the same list of futures after they finished executing.

Note: Each future and *shared_future* is waited upon and then copied into the collection of the output (returned) future, maintaining the order of the futures in the input collection. The future returned by *when_all* will not throw an exception, but the futures held in the output collection may.

Parameters

futures – [in] An arbitrary number of *future* or *shared_future* objects, possibly holding different types for which *when_all* should wait.

Returns

Returns a future holding the same list of futures as has been passed to *when_all*.

- `future<tuple<future<T0>, future<T1>, future<T2>...>>`: If inputs are fixed in number and are of heterogeneous types. The inputs can be any arbitrary number of future objects.
- `future<tuple<>>` if *when_all* is called with zero arguments. The returned future will be initially ready.

hpx/async_combinators/split_future.hpp

Defined in header `hpx/async_combinators/split_future.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Top level HPX namespace.

Functions

```
template<typename ...Ts>
inline tuple<future<Ts>...> split_future(future<tuple<Ts...>> &&f)
```

The function *split_future* is an operator allowing to split a given future of a sequence of values (any tuple, `std::pair`, or `std::array`) into an equivalent container of futures where each future represents one of the values from the original future. In some sense this function provides the inverse operation of *when_all*.

Note: The following cases are special:

```
tuple<future<void> > split_future(future<tuple<> > && f);
array<future<void>, 1> split_future(future<array<T, 0> > && f);
```

here the returned futures are directly representing the futures which were passed to the function.

Parameters

f – [in] A future holding an arbitrary sequence of values stored in a tuple-like container. This facility supports *hpx::tuple<>*, *std::pair<T1, T2>*, and *std::array<T, N>*

Returns

Returns an equivalent container (same container type as passed as the argument) of futures, where each future refers to the corresponding value in the input parameter. All the returned futures become ready once the input future has become ready. If the input future is exceptional, all output futures will be exceptional as well.

```
template<typename T>
inline std::vector<future<T>> split_future(future<std::vector<T>> &&f, std::size_t size)
```

The function *split_future* is an operator allowing to split a given future of a sequence of values (any `std::vector`) into a `std::vector` of futures where each future represents one of the values from the original `std::vector`. In some sense this function provides the inverse operation of *when_all*.

Parameters

- **f** – [in] A future holding an arbitrary sequence of values stored in a `std::vector`.
- **size** – [in] The number of elements the vector will hold once the input future has become ready

Returns

Returns a `std::vector` of futures, where each future refers to the corresponding value in the input parameter. All the returned futures become ready once the input future has become ready. If the input future is exceptional, all output futures will be exceptional as well.

hpx::when_each

Defined in header `hpx/future.hpp`⁸¹⁰.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<typename F, typename Future>
```

⁸¹⁰ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/future.hpp>

```
future<void> hpx::when_each(F &&f, std::vector<Future> &&futures)
```

The function *when_each* is an operator allowing to join on the results of all given futures. It AND-composes all future objects given and returns a new future object representing the event of all those futures having finished executing. It also calls the supplied callback for each of the futures which becomes ready.

Note: This function consumes the futures as they are passed on to the supplied function. The callback should take one or two parameters, namely either a *future* to be processed or a type that *std::size_t* is implicitly convertible to as the first parameter and the *future* as the second parameter. The first parameter will correspond to the index of the current *future* in the collection.

Parameters

- **f** – The function which will be called for each of the input futures once the future has become ready.
- **futures** – A vector holding an arbitrary amount of *future* or *shared_future* objects for which *wait_each* should wait.

Returns

Returns a future representing the event of all input futures being ready.

```
template<typename F, typename Iterator>  
future<Iterator> hpx::when_each(F &&f, Iterator begin, Iterator end)
```

The function *when_each* is an operator allowing to join on the results of all given futures. It AND-composes all future objects given and returns a new future object representing the event of all those futures having finished executing. It also calls the supplied callback for each of the futures which becomes ready.

Note: This function consumes the futures as they are passed on to the supplied function. The callback should take one or two parameters, namely either a *future* to be processed or a type that *std::size_t* is implicitly convertible to as the first parameter and the *future* as the second parameter. The first parameter will correspond to the index of the current *future* in the collection.

Parameters

- **f** – The function which will be called for each of the input futures once the future has become ready.
- **begin** – The iterator pointing to the first element of a sequence of *future* or *shared_future* objects for which *wait_each* should wait.
- **end** – The iterator pointing to the last element of a sequence of *future* or *shared_future* objects for which *wait_each* should wait.

Returns

Returns a future representing the event of all input futures being ready.

```
template<typename F, typename ...Ts>  
future<void> hpx::when_each(F &&f, Ts&&... futures)
```

The function *when_each* is an operator allowing to join on the results of all given futures. It AND-composes all future objects given and returns a new future object representing the event of all those futures having finished executing. It also calls the supplied callback for each of the futures which becomes ready.

Note: This function consumes the futures as they are passed on to the supplied function. The callback should take one or two parameters, namely either a *future* to be processed or a type that *std::size_t* is implicitly convertible to as the first parameter and the *future* as the second parameter. The first parameter will correspond to the index of the current *future* in the collection.

Parameters

- **f** – The function which will be called for each of the input futures once the future has become ready.
- **futures** – An arbitrary number of *future* or *shared_future* objects, possibly holding different types for which *wait_each* should wait.

Returns

Returns a future representing the event of all input futures being ready.

hpx::when_some

Defined in header [hpx/future.hpp](#)⁸¹¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename InputIter, typename Container = vector<future<typename
std::iterator_traits<InputIter>::value_type>>>
future<when_some_result<Container>> hpx::when_some(std::size_t n, Iterator first, Iterator last)
```

The function *when_some* is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns a new future object representing the same list of futures after *n* of them finished executing.

Note: The future returned by the function *when_some* becomes ready when at least *n* argument futures have become ready.

Note: Calling this version of *when_some* where *first == last*, returns a future with an empty container that is immediately ready. Each future and *shared_future* is waited upon and then copied into the collection of the output (returned) future, maintaining the order of the futures in the input collection. The future returned by *when_some* will not throw an exception, but the futures held in the output collection may.

Parameters

- **n** – [in] The number of futures out of the arguments which have to become ready in order for the returned future to get ready.
- **first** – [in] The iterator pointing to the first element of a sequence of *future* or *shared_future* objects for which *when_all* should wait.
- **last** – [in] The iterator pointing to the last element of a sequence of *future* or *shared_future* objects for which *when_all* should wait.

Returns

Returns a *when_some_result* holding the same list of futures as has been passed to *when_some* and indices pointing to ready futures.

⁸¹¹ <http://github.com/TheHPXProject/hpx/blob/d402e6ad7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

- `future<when_some_result<Container<future<R>>>>`: If the input cardinality is unknown at compile time and the futures are all of the same type. The order of the futures in the output container will be the same as given by the input iterator.

```
template<typename Range>
```

```
future<when_some_result<Range>> hpx::when_some(std::size_t n, Range &&futures)
```

The function *when_some* is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns a new future object representing the same list of futures after *n* of them finished executing.

Note: The future returned by the function *when_some* becomes ready when at least *n* argument futures have become ready.

Note: Each future and *shared_future* is waited upon and then copied into the collection of the output (returned) future, maintaining the order of the futures in the input collection. The future returned by *when_some* will not throw an exception, but the futures held in the output collection may.

Parameters

- **n** – [in] The number of futures out of the arguments which have to become ready in order for the returned future to get ready.
- **futures** – [in] A container holding an arbitrary amount of *future* or *shared_future* objects for which *when_some* should wait.

Returns

Returns a *when_some_result* holding the same list of futures as has been passed to *when_some* and indices pointing to ready futures.

- `future<when_some_result<Container<future<R>>>>`: If the input cardinality is unknown at compile time and the futures are all of the same type. The order of the futures in the output container will be the same as given by the input iterator.

```
template<typename ...Ts>
```

```
future<when_some_result<tuple<future<T>...>>> hpx::when_some(std::size_t n, Ts&&... futures)
```

The function *when_some* is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns a new future object representing the same list of futures after *n* of them finished executing.

Note: The future returned by the function *when_some* becomes ready when at least *n* argument futures have become ready.

Note: Each future and *shared_future* is waited upon and then copied into the collection of the output (returned) future, maintaining the order of the futures in the input collection. The future returned by *when_some* will not throw an exception, but the futures held in the output collection may.

Parameters

- **n** – [in] The number of futures out of the arguments which have to become ready in order for the returned future to get ready.

- **futures** – [in] An arbitrary number of *future* or *shared_future* objects, possibly holding different types for which *when_some* should wait.

Returns

Returns a *when_some_result* holding the same list of futures as has been passed to *when_some* and an index pointing to a ready future..

- `future<when_some_result<tuple<future<T0>, future<T1>...>>>`: If inputs are fixed in number and are of heterogeneous types. The inputs can be any arbitrary number of future objects.
- `future<when_some_result<tuple<>>>` if *when_some* is called with zero arguments. The returned future will be initially ready.

hpx::wait_all

Defined in header `hpx/future.hpp`⁸¹².

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::wait_all_nothrow`
- `hpx::wait_all_n`
- `hpx::wait_all_n_nothrow`

```
template<typename InputIter>
```

```
void hpx::wait_all(InputIter first, InputIter last)
```

The function *wait_all* is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns after they finished executing.

Note: The function *wait_all* returns after all futures have become ready. All input futures are still valid after *wait_all* returns.

Note: The function *wait_all* will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use *wait_all_nothrow* instead.

Note: The caller is responsible for keeping the *futures* container alive, unmoved, and unmodified until *wait_all* returns and all inspection of the futures is complete.

Parameters

- **first** – The iterator pointing to the first element of a sequence of *future* or *shared_future* objects for which *wait_all* should wait.
- **last** – The iterator pointing to the element after the last one of a sequence of *future* or *shared_future* objects for which *wait_all* should wait.

```
template<typename R>
```

⁸¹² <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

```
void hpx::wait_all(std::vector<future<R>> &&futures)
```

The function *wait_all* is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns after they finished executing.

Note: The function *wait_all* returns after all futures have become ready. All input futures are still valid after *wait_all* returns.

Note: The function *wait_all* will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use *wait_all_nothrow* instead.

Note: The caller is responsible for keeping the *futures* container alive, unmoved, and unmodified until *wait_all* returns and all inspection of the futures is complete.

Parameters

futures – A vector or array holding an arbitrary amount of *future* or *shared_future* objects for which *wait_all* should wait.

```
template<typename R, std::size_t N>
```

```
void hpx::wait_all(std::array<future<R>, N> &&futures)
```

The function *wait_all* is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns after they finished executing.

Note: The function *wait_all* returns after all futures have become ready. All input futures are still valid after *wait_all* returns.

Note: The function *wait_all* will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use *wait_all_nothrow* instead.

Note: The caller is responsible for keeping the *futures* container alive, unmoved, and unmodified until *wait_all* returns and all inspection of the futures is complete.

Parameters

futures – A vector or array holding an arbitrary amount of *future* or *shared_future* objects for which *wait_all* should wait.

```
template<typename T>
```

```
void hpx::wait_all(hpx::future<T> const &f)
```

The function *wait_all* is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns after they finished executing.

Note: The function *wait_all* returns after the future has become ready. The input future is still valid after *wait_all* returns.

Note: The function `wait_all` will rethrow any exceptions captured by the future while becoming ready. If this behavior is undesirable, use `wait_all_nothrow` instead.

Parameters

f – A *future* or *shared_future* for which `wait_all` should wait.

```
template<typename ...T>
void hpx::wait_all(T&&... futures)
```

The function `wait_all` is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns after they finished executing.

Note: The function `wait_all` returns after all futures have become ready. All input futures are still valid after `wait_all` returns.

Note: The function `wait_all` will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use `wait_all_nothrow` instead.

Parameters

futures – An arbitrary number of *future* or *shared_future* objects, possibly holding different types for which `wait_all` should wait.

hpx::wait_all_nothrow

Defined in header `hpx/future.hpp`⁸¹³.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::wait_all`
- `hpx::wait_all_n`
- `hpx::wait_all_n_nothrow`

```
template<typename InputIter>
bool hpx::wait_all_nothrow(InputIter first, InputIter last)
```

The function `wait_all_nothrow` is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns after they finished executing, similar to `wait_all`. It differs from `wait_all` in that it will not rethrow any exceptions captured by the given futures while they became ready.

Note: The function `wait_all_nothrow` returns after all futures have become ready. All input futures are still valid after `wait_all_nothrow` returns.

Note: Unlike `wait_all`, this function will not rethrow any exceptions captured by the futures while becoming ready. Any such exceptions are not rethrown by this call; the caller can use the returned *bool* to detect their

⁸¹³ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/future.hpp>

presence.

Note: The caller is responsible for keeping the *futures* container alive, unmodified, and unmodified until *wait_all* returns and all inspection of the futures is complete.

Parameters

- **first** – The iterator pointing to the first element of a sequence of *future* or *shared_future* objects for which *wait_all_nothrow* should wait.
- **last** – The iterator pointing to the element after the last one of a sequence of *future* or *shared_future* objects for which *wait_all_nothrow* should wait.

Returns

Returns *true* if any of the given futures held an exception once it became ready, and *false* otherwise.

```
template<typename R>
bool hpx::wait_all_nothrow(std::vector<future<R>> &&futures)
```

The function *wait_all_nothrow* is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns after they finished executing, similar to *wait_all*. It differs from *wait_all* in that it will not rethrow any exceptions captured by the given futures while they became ready.

Note: The function *wait_all_nothrow* returns after all futures have become ready. All input futures are still valid after *wait_all_nothrow* returns.

Note: Unlike *wait_all*, this function will not rethrow any exceptions captured by the futures while becoming ready. Any such exceptions are not rethrown by this call; the caller can use the returned *bool* to detect their presence.

Parameters

futures – A vector or array holding an arbitrary amount of *future* or *shared_future* objects for which *wait_all_nothrow* should wait.

Returns

Returns *true* if any of the given futures held an exception once it became ready, and *false* otherwise.

```
template<typename R, std::size_t N>
bool hpx::wait_all_nothrow(std::array<future<R>, N> &&futures)
```

The function *wait_all_nothrow* is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns after they finished executing, similar to *wait_all*. It differs from *wait_all* in that it will not rethrow any exceptions captured by the given futures while they became ready.

Note: The function *wait_all_nothrow* returns after all futures have become ready. All input futures are still valid after *wait_all_nothrow* returns.

Note: Unlike *wait_all*, this function will not rethrow any exceptions captured by the futures while becoming ready. Any such exceptions are not rethrown by this call; the caller can use the returned *bool* to detect their presence.

Parameters

futures – A vector or array holding an arbitrary amount of *future* or *shared_future* objects for which *wait_all_nothrow* should wait.

Returns

Returns *true* if any of the given futures held an exception once it became ready, and *false* otherwise.

```
template<typename T>
bool hpx::wait_all_nothrow(hpx::future<T> const &f)
```

The function *wait_all_nothrow* is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns after they finished executing, similar to *wait_all*. It differs from *wait_all* in that it will not rethrow any exceptions captured by the given future while it became ready.

Note: The function *wait_all_nothrow* returns after the future has become ready. The input future is still valid after *wait_all_nothrow* returns.

Note: Unlike *wait_all*, this function will not rethrow any exception captured by the future while becoming ready. Any such exception is silently discarded; the caller can use the returned *bool* to detect its presence.

Parameters

f – A *future* or *shared_future* for which *wait_all_nothrow* should wait.

Returns

Returns *true* if the given future held an exception once it became ready, and *false* otherwise.

```
template<typename ...T>
bool hpx::wait_all_nothrow(T&&... futures)
```

The function *wait_all_nothrow* is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns after they finished executing, similar to *wait_all*. It differs from *wait_all* in that it will not rethrow any exceptions captured by the given futures while they became ready.

Note: The function *wait_all_nothrow* returns after all futures have become ready. All input futures are still valid after *wait_all_nothrow* returns.

Note: Unlike *wait_all*, this function will not rethrow any exceptions captured by the futures while becoming ready. Any such exceptions are not rethrown by this call; the caller can use the returned *bool* to detect their presence.

Parameters

futures – An arbitrary number of *future* or *shared_future* objects, possibly holding different types for which *wait_all_nothrow* should wait.

Returns

Returns *true* if any of the given futures held an exception once it became ready, and *false* otherwise.

hpx::wait_all_n

Defined in header [hpx/future.hpp](#)⁸¹⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::wait_all*
- *hpx::wait_all_nothrow*
- *hpx::wait_all_n_nothrow*

template<typename **InputIter**>

void **hpx::wait_all_n**(*InputIter* begin, *std::size_t* count)

The function *wait_all_n* is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns after they finished executing.

Note: The function *wait_all_n* returns after all futures have become ready. All input futures are still valid after *wait_all_n* returns.

Note: The function *wait_all_n* will rethrow any exceptions captured by the futures while becoming ready. If this behavior is undesirable, use *wait_all_n_nothrow* instead.

Note: The caller is responsible for keeping the *futures* container alive, unmoved, and unmodified until *wait_all* returns and all inspection of the futures is complete.

Parameters

- **begin** – The iterator pointing to the first element of a sequence of *future* or *shared_future* objects for which *wait_all_n* should wait.
- **count** – The number of elements in the sequence starting at *first*.

Returns

The function *wait_all_n* will return an iterator referring to the first element in the input sequence after the last processed element.

⁸¹⁴ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

hpx::wait_all_n_nothrow

Defined in header `hpx/future.hpp`⁸¹⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::wait_all`
- `hpx::wait_all_nothrow`
- `hpx::wait_all_n`

template<typename **InputIter**>

`std::pair<InputIter, bool> hpx::wait_all_n_nothrow(InputIter begin, std::size_t count)`

The function `wait_all_n_nothrow` is an operator allowing to join on the result of all given futures. It AND-composes all future objects given and returns after they finished executing, similar to `wait_all_n`. It differs from `wait_all_n` in that it will not rethrow any exceptions captured by the given futures while they became ready.

Note: The function `wait_all_n_nothrow` returns after all futures have become ready. All input futures are still valid after `wait_all_n_nothrow` returns.

Note: Unlike `wait_all_n`, this function will not rethrow any exceptions captured by the futures while becoming ready. Any such exceptions are not rethrown by this call; the caller can use the returned `bool` to detect their presence.

Note: The caller is responsible for keeping the *futures* container alive, unmoved, and unmodified until `wait_all` returns and all inspection of the futures is complete.

Parameters

- **begin** – The iterator pointing to the first element of a sequence of *future* or *shared_future* objects for which `wait_all_n_nothrow` should wait.
- **count** – The number of elements in the sequence starting at *first*.

Returns

The function `wait_all_n_nothrow` returns a pair consisting of an iterator referring to the first element in the input sequence after the last processed element, and a `bool` that is `true` if any of the given futures held an exception once it became ready, and `false` otherwise.

⁸¹⁵ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

hpx::wait_all_for

Defined in header `hpx/future.hpp`⁸¹⁶.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::wait_all_for_nothrow`
- `hpx::wait_all_for_n`
- `hpx::wait_all_for_n_nothrow`

```
template<typename InputIter>
hpx::future_status hpx::wait_all_for(hpx::chrono::steady_duration const &timeout, InputIter first, InputIter
                                     last)
```

The function `wait_all_for` is an operator allowing to join on the result of all given futures, similar to `wait_all`. It differs from `wait_all` in that it will not suspend indefinitely if one or more of the given futures never become ready; instead it returns as soon as either all futures have become ready, or the given timeout has elapsed, whichever happens first.

Note: The function `wait_all_for` returns after all futures have become ready, or after the given timeout has expired, whichever comes first. All input futures are still valid after `wait_all_for` returns, independently of the returned status, and can be inspected (e.g. using `is_ready()`) to determine which of the given futures have become ready.

Note: The function `wait_all_for` will rethrow any exceptions captured by the futures while becoming ready, as long as all the given futures became ready before the timeout elapsed. If this behavior is undesirable, use `wait_all_for_nothrow` instead.

Parameters

- **timeout** – The maximum duration to wait for all the given futures to become ready.
- **first** – The iterator pointing to the first element of a sequence of `future` or `shared_future` objects for which `wait_all_for` should wait.
- **last** – The iterator pointing to the element after the last one of a sequence of `future` or `shared_future` objects for which `wait_all_for` should wait.

Returns

Returns `hpx::future_status::ready` if all the given futures have become ready before `timeout` has elapsed, and `hpx::future_status::timeout` otherwise.

```
template<typename R>
hpx::future_status hpx::wait_all_for(hpx::chrono::steady_duration const &timeout, std::vector<future<R>>
                                     const &futures)
```

The function `wait_all_for` is an operator allowing to join on the result of all given futures, similar to `wait_all`. It differs from `wait_all` in that it will not suspend indefinitely if one or more of the given futures never become ready; instead it returns as soon as either all futures have become ready, or the given timeout has elapsed, whichever happens first.

⁸¹⁶ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

Note: The function `wait_all_for` returns after all futures have become ready, or after the given timeout has expired, whichever comes first. All input futures are still valid after `wait_all_for` returns, independently of the returned status, and can be inspected (e.g. using `is_ready()`) to determine which of the given futures have become ready.

Note: The function `wait_all_for` will rethrow any exceptions captured by the futures while becoming ready, as long as all the given futures became ready before the timeout elapsed. If this behavior is undesirable, use `wait_all_for_nothrow` instead.

Note: The function `wait_all_for` returns after all futures have become ready, or after the given timeout has expired, whichever comes first. All input futures are still valid after `wait_all_for` returns, independently of the returned status, and can be inspected (e.g. using `is_ready()`) to determine which of the given futures have become ready.

Note: The caller is responsible for keeping the *futures* container itself (and, for the iterator-range overload, the underlying sequence) alive, unmoved, and unmodified until it is done inspecting the futures' state. If the timeout expires before all futures become ready, `wait_all_for` may leave asynchronous continuations attached to the not-yet-ready futures; destroying, moving, or reallocating the container before those futures settle results in undefined behavior.

Parameters

- **timeout** – The maximum duration to wait for all the given futures to become ready.
- **futures** – A vector or array holding an arbitrary amount of *future* or *shared_future* objects for which `wait_all_for` should wait.

Returns

Returns `hpx::future_status::ready` if all the given futures have become ready before *timeout* has elapsed, and `hpx::future_status::timeout` otherwise.

```
template<typename R, std::size_t N>
hpx::future_status hpx::wait_all_for(hpx::chrono::steady_duration const &timeout, std::array<future<R>, N>
                                     const &futures)
```

The function `wait_all_for` is an operator allowing to join on the result of all given futures, similar to `wait_all`. It differs from `wait_all` in that it will not suspend indefinitely if one or more of the given futures never become ready; instead it returns as soon as either all futures have become ready, or the given timeout has elapsed, whichever happens first.

Note: The function `wait_all_for` returns after all futures have become ready, or after the given timeout has expired, whichever comes first. All input futures are still valid after `wait_all_for` returns, independently of the returned status, and can be inspected (e.g. using `is_ready()`) to determine which of the given futures have become ready.

Note: The function `wait_all_for` will rethrow any exceptions captured by the futures while becoming ready, as long as all the given futures became ready before the timeout elapsed. If this behavior is undesirable, use

wait_all_for_nothrow instead.

Note: The function *wait_all_for* returns after all futures have become ready, or after the given timeout has expired, whichever comes first. All input futures are still valid after *wait_all_for* returns, independently of the returned status, and can be inspected (e.g. using *is_ready()*) to determine which of the given futures have become ready.

Note: The caller is responsible for keeping the *futures* container itself (and, for the iterator-range overload, the underlying sequence) alive, unmove, and unmodified until it is done inspecting the futures' state. If the timeout expires before all futures become ready, *wait_all_for* may leave asynchronous continuations attached to the not-yet-ready futures; destroying, moving, or reallocating the container before those futures settle results in undefined behavior.

Parameters

- **timeout** – The maximum duration to wait for all the given futures to become ready.
- **futures** – A vector or array holding an arbitrary amount of *future* or *shared_future* objects for which *wait_all_for* should wait.

Returns

Returns *hpx::future_status::ready* if all the given futures have become ready before *timeout* has elapsed, and *hpx::future_status::timeout* otherwise.

template<typename T>

hpx::future_status **hpx::wait_all_for**(*hpx::chrono::steady_duration* const &timeout, *hpx::future*<T> const &f)

The function *wait_all_for* is an operator allowing to join on the result of the given future, similar to *wait_all*. It differs from *wait_all* in that it will not suspend indefinitely if the given future never becomes ready; instead it returns as soon as either the future has become ready, or the given timeout has elapsed, whichever happens first.

Note: The function *wait_all_for* returns after the future has become ready, or after the given timeout has expired, whichever comes first. The input future is still valid after *wait_all_for* returns, independently of the returned status.

Note: The function *wait_all_for* will rethrow any exceptions captured by the future while becoming ready, as long as the given future became ready before the timeout elapsed. If this behavior is undesirable, use *wait_all_for_nothrow* instead.

Parameters

- **timeout** – The maximum duration to wait for the given future to become ready.
- **f** – A *future* or *shared_future* for which *wait_all_for* should wait.

Returns

Returns *hpx::future_status::ready* if the given future has become ready before *timeout* has elapsed, and *hpx::future_status::timeout* otherwise.

template<typename ...T>

`hpx::future_status hpx::wait_all_for(hpx::chrono::steady_duration const &timeout, T const&... futures)`

The function `wait_all_for` is an operator allowing to join on the result of all given futures, similar to `wait_all`. It differs from `wait_all` in that it will not suspend indefinitely if one or more of the given futures never become ready; instead it returns as soon as either all futures have become ready, or the given timeout has elapsed, whichever happens first.

Note: The function `wait_all_for` returns after all futures have become ready, or after the given timeout has expired, whichever comes first. All input futures are still valid after `wait_all_for` returns, independently of the returned status, and can be inspected (e.g. using `is_ready()`) to determine which of the given futures have become ready.

Note: The function `wait_all_for` will rethrow any exceptions captured by the futures while becoming ready, as long as all the given futures became ready before the timeout elapsed. If this behavior is undesirable, use `wait_all_for_nothrow` instead.

Parameters

- **timeout** – The maximum duration to wait for all the given futures to become ready.
- **futures** – An arbitrary number of `future` or `shared_future` objects, possibly holding different types for which `wait_all_for` should wait.

Returns

Returns `hpx::future_status::ready` if all the given futures have become ready before `timeout` has elapsed, and `hpx::future_status::timeout` otherwise.

`hpx::wait_all_for_nothrow`

Defined in header `hpx/future.hpp`⁸¹⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::wait_all_for`
- `hpx::wait_all_for_n`
- `hpx::wait_all_for_n_nothrow`

template<typename **InputIter**>

`hpx::wait_all_for_nothrow_result hpx::wait_all_for_nothrow(hpx::chrono::steady_duration const &timeout, InputIter first, InputIter last)`

The function `wait_all_for_nothrow` is an operator allowing to join on the result of all given futures, similar to `wait_all_for`. It differs from `wait_all_for` in that it will not rethrow any exceptions captured by the given futures while they became ready; instead it returns as soon as either all futures have become ready, or the given timeout has elapsed, whichever happens first.

Note: The function `wait_all_for_nothrow` returns after all futures have become ready, or after the given timeout has expired, whichever comes first. All input futures are still valid after `wait_all_for_nothrow` returns, indepen-

⁸¹⁷ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/future.hpp>

dently of the returned status, and can be inspected (e.g. using *is_ready()*) to determine which of the given futures have become ready.

Note: Unlike *wait_all_for_nothrow*, this function will not rethrow any exceptions captured by the futures while becoming ready. Any such exceptions are not rethrown by this call.

Parameters

- **timeout** – The maximum duration to wait for all the given futures to become ready.
- **first** – The iterator pointing to the first element of a sequence of *future* or *shared_future* objects for which *wait_all_for_nothrow* should wait.
- **last** – The iterator pointing to the element after the last one of a sequence of *future* or *shared_future* objects for which *wait_all_for_nothrow* should wait.

Returns

Returns a *wait_all_for_nothrow_result* object whose *status* member is *hpx::future_status::ready* if all the given futures have become ready before *timeout* has elapsed, and *hpx::future_status::timeout* otherwise; and whose *has_exceptional_results* member indicates whether any of the futures that became ready had captured an exception.

```
template<typename R>
```

```
hpx::wait_all_for_nothrow_result hpx::wait_all_for_nothrow(hpx::chrono::steady_duration const &timeout,  
                                                         std::vector<future<R>> const &futures)
```

The function *wait_all_for_nothrow* is an operator allowing to join on the result of all given futures, similar to *wait_all_for*. It differs from *wait_all_for* in that it will not rethrow any exceptions captured by the given futures while they became ready; instead it returns as soon as either all futures have become ready, or the given timeout has elapsed, whichever happens first.

Note: The function *wait_all_for_nothrow* returns after all futures have become ready, or after the given timeout has expired, whichever comes first. All input futures are still valid after *wait_all_for_nothrow* returns, independently of the returned status, and can be inspected (e.g. using *is_ready()*) to determine which of the given futures have become ready.

Note: Unlike *wait_all_for*, this function will not rethrow any exceptions captured by the futures while becoming ready. Any such exceptions are not rethrown by this call.

Note: The function *wait_all_for_nothrow* returns after all futures have become ready, or after the given timeout has expired, whichever comes first. All input futures are still valid after *wait_all_for* returns, independently of the returned status, and can be inspected (e.g. using *is_ready()*) to determine which of the given futures have become ready.

Note: The caller is responsible for keeping the *futures* container itself (and, for the iterator-range overload, the underlying sequence) alive, unmoved, and unmodified until it is done inspecting the futures' state. If the timeout expires before all futures become ready, *wait_all_for* may leave asynchronous continuations attached to

the not-yet-ready futures; destroying, moving, or reallocating the container before those futures settle results in undefined behavior.

Parameters

- **timeout** – The maximum duration to wait for all the given futures to become ready.
- **futures** – A vector or array holding an arbitrary amount of *future* or *shared_future* objects for which *wait_all_for_nothrow* should wait.

Returns

Returns a *wait_all_for_nothrow_result* object whose *status* member is *hpx::future_status::ready* if all the given futures have become ready before *timeout* has elapsed, and *hpx::future_status::timeout* otherwise; and whose *has_exceptional_results* member indicates whether any of the futures that became ready had captured an exception.

```
template<typename R, std::size_t N>
hpx::wait_all_for_nothrow_result hpx::wait_all_for_nothrow(hpx::chrono::steady_duration const &timeout,
    std::array<future<R>, N> const &futures)
```

The function *wait_all_for_nothrow* is an operator allowing to join on the result of all given futures, similar to *wait_all_for*. It differs from *wait_all_for* in that it will not rethrow any exceptions captured by the given futures while they became ready; instead it returns as soon as either all futures have become ready, or the given timeout has elapsed, whichever happens first.

Note: The function *wait_all_for_nothrow* returns after all futures have become ready, or after the given timeout has expired, whichever comes first. All input futures are still valid after *wait_all_for_nothrow* returns, independently of the returned status, and can be inspected (e.g. using *is_ready()*) to determine which of the given futures have become ready.

Note: Unlike *wait_all_for*, this function will not rethrow any exceptions captured by the futures while becoming ready. Any such exceptions are not rethrown by this call.

Note: The function *wait_all_for_nothrow* returns after all futures have become ready, or after the given timeout has expired, whichever comes first. All input futures are still valid after *wait_all_for* returns, independently of the returned status, and can be inspected (e.g. using *is_ready()*) to determine which of the given futures have become ready.

Note: The caller is responsible for keeping the *futures* container itself (and, for the iterator-range overload, the underlying sequence) alive, unmoved, and unmodified until it is done inspecting the futures' state. If the timeout expires before all futures become ready, *wait_all_for* may leave asynchronous continuations attached to the not-yet-ready futures; destroying, moving, or reallocating the container before those futures settle results in undefined behavior.

Parameters

- **timeout** – The maximum duration to wait for all the given futures to become ready.
- **futures** – A vector or array holding an arbitrary amount of *future* or *shared_future* objects for which *wait_all_for_nothrow* should wait.

Returns

Returns a *wait_all_for_nothrow_result* object whose *status* member is *hpx::future_status::ready* if all the given futures have become ready before *timeout* has elapsed, and *hpx::future_status::timeout* otherwise; and whose *has_exceptional_results* member indicates whether any of the futures that became ready had captured an exception.

```
template<typename T>
```

```
hpx::wait_all_for_nothrow_result hpx::wait_all_for_nothrow(hpx::chrono::steady_duration const &timeout,  
                                                         hpx::future<T> const &f)
```

The function *wait_all_for_nothrow* is an operator allowing to join on the result of the given future, similar to *wait_all_for*. It differs from *wait_all_for* in that it will not rethrow any exceptions captured by the given future while it became ready; instead it returns as soon as either the future has become ready, or the given timeout has elapsed, whichever happens first.

Note: The function *wait_all_for_nothrow* returns after the future has become ready, or after the given timeout has expired, whichever comes first. The input future is still valid after *wait_all_for_nothrow* returns, independently of the returned status.

Note: Unlike *wait_all_for*, this function will not rethrow any exceptions captured by the future while becoming ready. Any such exception is silently discarded.

Parameters

- **timeout** – The maximum duration to wait for the given future to become ready.
- **f** – A *future* or *shared_future* for which *wait_all_for_nothrow* should wait.

Returns

Returns a *wait_all_for_nothrow_result* object whose *status* member is *hpx::future_status::ready* if all the given futures have become ready before *timeout* has elapsed, and *hpx::future_status::timeout* otherwise; and whose *has_exceptional_results* member indicates whether any of the futures that became ready had captured an exception.

```
template<typename ...T>
```

```
hpx::wait_all_for_nothrow_result hpx::wait_all_for_nothrow(hpx::chrono::steady_duration const &timeout, T  
                                                         const&... futures)
```

The function *wait_all_for_nothrow* is an operator allowing to join on the result of all given futures, similar to *wait_all_for*. It differs from *wait_all_for* in that it will not rethrow any exceptions captured by the given futures while they became ready; instead it returns as soon as either all futures have become ready, or the given timeout has elapsed, whichever happens first.

Note: The function *wait_all_for_nothrow* returns after all futures have become ready, or after the given timeout has expired, whichever comes first. All input futures are still valid after *wait_all_for_nothrow* returns, independently of the returned status, and can be inspected (e.g. using *is_ready()*) to determine which of the given futures have become ready.

Note: Unlike *wait_all_for*, this function will not rethrow any exceptions captured by the futures while becoming ready. Any such exceptions are not rethrown by this call.

Parameters

- **timeout** – The maximum duration to wait for all the given futures to become ready.
- **futures** – An arbitrary number of *future* or *shared_future* objects, possibly holding different types for which *wait_all_for_nothrow* should wait.

Returns

Returns a *wait_all_for_nothrow_result* object whose *status* member is *hpx::future_status::ready* if all the given futures have become ready before *timeout* has elapsed, and *hpx::future_status::timeout* otherwise; and whose *has_exceptional_results* member indicates whether any of the futures that became ready had captured an exception.

hpx::wait_all_for_n

Defined in header [hpx/future.hpp](#)⁸¹⁸.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::wait_all_for*
- *hpx::wait_all_for_nothrow*
- *hpx::wait_all_for_n_nothrow*

```
template<typename InputIter>
hpx::future_status hpx::wait_all_for_n(hpx::chrono::steady_duration const &timeout, InputIter begin,
                                       std::size_t count)
```

The function *wait_all_for_n* is an operator allowing to join on the result of all given futures, similar to *wait_all_n*. It differs from *wait_all_n* in that it will not suspend indefinitely if one or more of the given futures never become ready; instead it returns as soon as either all futures have become ready, or the given timeout has elapsed, whichever happens first.

Note: The function *wait_all_for_n* returns after all futures have become ready, or after the given timeout has expired, whichever comes first. All input futures are still valid after *wait_all_for_n* returns, independently of the returned status, and can be inspected (e.g. using *is_ready()*) to determine which of the given futures have become ready.

Note: The function *wait_all_for_n* will rethrow any exceptions captured by the futures while becoming ready, as long as all the given futures became ready before the timeout elapsed. If this behavior is undesirable, use *wait_all_for_n_nothrow* instead.

Parameters

- **timeout** – The maximum duration to wait for all the given futures to become ready.
- **begin** – The iterator pointing to the first element of a sequence of *future* or *shared_future* objects for which *wait_all_for_n* should wait.
- **count** – The number of elements in the sequence starting at *first*.

⁸¹⁸ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

Returns

Returns `hpx::future_status::ready` if all the given futures have become ready before *timeout* has elapsed, and `hpx::future_status::timeout` otherwise.

`hpx::wait_all_for_n_nothrow`

Defined in header `hpx/future.hpp`⁸¹⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::wait_all_for`
- `hpx::wait_all_for_nothrow`
- `hpx::wait_all_for_n`

template<typename **InputIter**>

`hpx::wait_all_for_n_nothrow_result hpx::wait_all_for_n_nothrow(hpx::chrono::steady_duration const &timeout, InputIter begin, std::size_t count)`

The function `wait_all_for_n_nothrow` is an operator allowing to join on the result of all given futures, similar to `wait_all_n`. It differs from `wait_all_for_n` in that it will not rethrow any exceptions captured by the given futures while they became ready; instead it returns as soon as either all futures have become ready, or the given timeout has elapsed, whichever happens first.

Note: The function `wait_all_for_n_nothrow` returns after all futures have become ready, or after the given timeout has expired, whichever comes first. All input futures are still valid after `wait_all_for_n_nothrow` returns, independently of the returned status, and can be inspected (e.g. using `is_ready()`) to determine which of the given futures have become ready.

Note: Unlike `wait_all_for_n`, this function will not rethrow any exceptions captured by the futures while becoming ready. Any such exceptions are not rethrown by this call.

Parameters

- **timeout** – The maximum duration to wait for all the given futures to become ready.
- **begin** – The iterator pointing to the first element of a sequence of *future* or *shared_future* objects for which `wait_all_for_n_nothrow` should wait.
- **count** – The number of elements in the sequence starting at *first*.

Returns

Returns a `wait_all_for_n_nothrow_result` object whose *status* member is `hpx::future_status::ready` if all the given futures have become ready before *timeout* has elapsed, and `hpx::future_status::timeout` otherwise; and whose *has_exceptional_results* member indicates whether any of the futures that became ready had captured an exception.

⁸¹⁹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

async_cuda

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/async_cuda/cublas_executor.hpp

Defined in header `hpx/async_cuda/cublas_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/async_cuda/cuda_executor.hpp

Defined in header `hpx/async_cuda/cuda_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **cuda**

namespace **experimental**

struct **cuda_executor** : public *hpx::cuda::experimental::cuda_executor_base*

Public Functions

inline explicit **cuda_executor**(*std::size_t* const device, bool const event_mode = true)

inline ~**cuda_executor**()

template<typename **F**, typename ...**Ts**>

inline void **post**(*F* &&*f*, *Ts*&&... *ts*) const

template<typename **F**, typename ...**Ts**>

inline decltype(auto) **async_execute**(*F* &&*f*, *Ts*&&... *ts*) const

Protected Functions

template<typename **R**, typename ...**Params**, typename ...**Args**>

inline void **post_impl**(*R* (**cuda_function*)(*Params*...), *Args*&&... *args*) const

template<typename **R**, typename ...**Params**, typename ...**Args**>

inline *hpx::future*<void> **async_impl**(*R* (**cuda_kernel*)(*Params*...), *Args*&&... *args*) const

struct **cuda_executor_base**

Subclassed by *hpx::cuda::experimental::cuda_executor*

Public Types

```
using future_type = hpx::future<void>
```

Public Functions

```
inline cuda_executor_base(std::size_t device, bool const event_mode)
```

```
inline future_type get_future() const
```

Protected Attributes

```
int device_
```

```
bool event_mode_
```

```
cudaStream_t stream_
```

```
std::shared_ptr<hpx::cuda::experimental::target> target_
```

```
namespace execution
```

```
namespace experimental
```

async_mpi

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/async_mpi/mpi_executor.hpp

Defined in header `hpx/async_mpi/mpi_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
namespace mpi
```

```
namespace experimental
```

```
struct executor
```

Public Types

```
using execution_category = hpx::execution::parallel_execution_tag
```

```
using executor_parameters_type = hpx::execution::experimental::default_parameters
```

Public Functions

```
inline explicit constexpr executor(MPI_Comm communicator = MPI_COMM_WORLD)
```

```
template<typename F, typename ...Ts>
```

```
inline decltype(auto) async_execute(F &&f, Ts&&... ts) const
```

```
inline std::size_t in_flight_estimate() const
```

Private Members

```
MPI_Comm communicator_
```

hpx/async_mpi/transform_mpi.hpp

Defined in header `hpx/async_mpi/transform_mpi.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace mpi
```

```
        namespace experimental
```

Variables

```
constexpr struct hpx::mpi::experimental::transform_mpi_t transform_mpi
```

```
struct transform_mpi_t
```

Public Functions

```
template<typename ...Args>
inline constexpr decltype(auto) operator()(Args&&... args) const

template<typename Sender, typename F>
inline constexpr auto operator()(Sender &&s, F &&f) const

template<typename F>
inline constexpr auto operator()(F &&f) const
```

async_sycl

See *Public API* for a list of names and headers that are part of the public HPX API.

hpx/async_sycl/sycl_executor.hpp

Defined in header `hpx/async_sycl/sycl_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

Functions

```
template<typename Executor, typename ...Ts>
decltype(auto) async(Executor &&exec, hpx::sycl::experimental::sycl_executor::queue_function_ptr_t<Ts...>
    &&f, Ts&&... ts)
```

hpx::async overload for launching sycl queue member functions with an sycl executor

```
template<typename Executor, typename ...Ts>
bool apply(Executor &&exec, hpx::sycl::experimental::sycl_executor::queue_function_ptr_t<Ts...> &&f,
    Ts&&... ts)
```

hpx::apply overload for launching sycl queue member functions with an sycl executor

namespace **execution**

namespace **experimental**

namespace **sycl**

namespace **experimental**

struct **sycl_executor**

Public Types

using **future_type** = *hpx::future*<void>

template<typename ...**Params**>

using **queue_function_ptr_t** = ::sycl::event
 (::sycl::queue::*)(*std::conditional_t*<*std::is_trivial_v*<*std::remove_reference_t*<*Params*>>, *std::decay_t*<*Params*>, *Params*>...)

Default Implementation without the extra intel code_location parameter. Removes the reference for trivial types to make the function matching easier (see *sycl_stream.cpp* test)

Public Functions

template<typename **sycl_device_selector**>

inline explicit **sycl_executor**(*sycl_device_selector* const &selector)

Create a SYCL executor (based on a sycl queue)

~sycl_executor() = default

inline *future_type* **get_future**()

Get future for this command_queue (NOTE will be more efficient if an event is provided — otherwise a dummy kernel must be submitted to get an event)

inline *future_type* **get_future**(::sycl::event event)

Get future for that becomes ready when the given event completes.

template<typename ...**Params**>

inline void **post**(*queue_function_ptr_t*<*Params*...> &&queue_member_function, *Params*&&... args)

Invoke queue member function given queue and parameters — do not use event to return a *hpx::future* (One way)

template<typename ...**Params**>

inline *hpx::future*<void> **async_execute**(*queue_function_ptr_t*<*Params*...> &&queue_member_function, *Params*&&... args)

Invoke queue member function given queue and parameters — *hpx::future* tied to the sycl event / (two way)

inline ::sycl::device **get_device**() const

Return the device used by the underlying SYCL queue.

inline ::sycl::context **get_context**() const

Return the context used by the underlying SYCL queue.

inline ::sycl::queue &**get_queue**()

Return the underlying queue for direct access.

Protected Attributes

`::sycl::queue command_queue`

cache

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/cache/local_cache.hpp

Defined in header `hpx/cache/local_cache.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **util**

namespace **cache**

```
template<typename Key, typename Entry, typename UpdatePolicy = std::less<Entry>, typename
InsertPolicy = policies::always<Entry>, typename CacheStorage = std::map<Key, Entry>,
typename Statistics = statistics::no_statistics>
class local_cache
```

#include <hpx/cache/local_cache.hpp> The `local_cache` implements the basic functionality needed for a local (non-distributed) cache.

Template Parameters

- **Key** – The type of the keys to use to identify the entries stored in the cache
- **Entry** – The type of the items to be held in the cache, must model the `CacheEntry` concept
- **UpdatePolicy** – A (optional) type specifying a (binary) function object used to sort the cache entries based on their ‘age’. The ‘oldest’ entries (according to this sorting criteria) will be discarded first if the maximum capacity of the cache is reached. The default is `std::less<Entry>`. The function object will be invoked using 2 entry instances of the type `Entry`. This type must model the `UpdatePolicy` model.
- **InsertPolicy** – A (optional) type specifying a (unary) function object used to allow global decisions whether a particular entry should be added to the cache or not. The default is `policies::always`, imposing no global insert related criteria on the cache. The function object will be invoked using the entry instance to be inserted into the cache. This type must model the `InsertPolicy` model.
- **CacheStorage** – A (optional) container type used to store the cache items. The container must be an associative and STL compatible container. The default is a `std::map<Key, Entry>`.
- **Statistics** – A (optional) type allowing to collect some basic statistics about the operation of the cache instance. The type must conform to the `CacheStatistics` concept. The default value is the type `statistics::no_statistics` which does not collect any numbers, but provides empty stubs allowing the code to compile.

Public Types

```
using key_type = Key

using entry_type = Entry

using update_policy_type = UpdatePolicy

using insert_policy_type = InsertPolicy

using storage_type = CacheStorage

using statistics_type = Statistics

using value_type = typename entry_type::value_type

using size_type = typename storage_type::size_type

using storage_value_type = typename storage_type::value_type
```

Public Functions

```
inline explicit local_cache(size_type max_size = 0, update_policy_type const &up =
    update_policy_type(), insert_policy_type const &ip =
    insert_policy_type())
```

Construct an instance of a *local_cache*.

Parameters

- **max_size** – [in] The maximal size this cache is allowed to reach any time. The default is zero (no size limitation). The unit of this value is usually determined by the unit of the values returned by the entry's *get_size* function.
- **up** – [in] An instance of the *UpdatePolicy* to use for this cache. The default is to use a default constructed instance of the type as defined by the *UpdatePolicy* template parameter.
- **ip** – [in] An instance of the *InsertPolicy* to use for this cache. The default is to use a default constructed instance of the type as defined by the *InsertPolicy* template parameter.

```
local_cache(local_cache const &other) = default
```

```
local_cache(local_cache &&other) = default
```

```
local_cache &operator=(local_cache const &other) = default
```

```
local_cache &operator=(local_cache &&other) = default
```

```
~local_cache() = default
```

inline constexpr *size_type* **size**() const noexcept

Return current size of the cache.

Returns

The current size of this cache instance.

inline constexpr *size_type* **capacity**() const noexcept

Access the maximum size the cache is allowed to grow to.

Note: The unit of this value is usually determined by the unit of the return values of the entry's function *entry::get_size*.

Returns

The maximum size this cache instance is currently allowed to reach. If this number is zero the cache has no limitation with regard to a maximum size.

inline bool **reserve**(*size_type* max_size)

Change the maximum size this cache can grow to.

Parameters

max_size – [in] The new maximum size this cache will be allowed to grow to.

Returns

This function returns *true* if successful. It returns *false* if the new *max_size* is smaller than the current limit and the cache could not be shrunk to the new maximum size.

inline bool **holds_key**(*key_type* const &k) const

Check whether the cache currently holds an entry identified by the given key.

Note: This function does not call the entry's function *entry::touch*. It just checks if the cache contains an entry corresponding to the given key.

Parameters

k – [in] The key for the entry which should be looked up in the cache.

Returns

This function returns *true* if the cache holds the referenced entry, otherwise it returns *false*.

inline bool **get_entry**(*key_type* const &k, *key_type* &realkey, *entry_type* &val)

Get a specific entry identified by the given key.

Note: The function will call the entry's *entry::touch* function if the value corresponding to the provided key is found in the cache.

Parameters

- **k** – [in] The key for the entry which should be retrieved from the cache.
- **realkey[out]** – Return the full real key found in the cache
- **val** – [out] If the entry indexed by the key is found in the cache this value on successful return will be a copy of the corresponding entry.

Returns

This function returns *true* if the cache holds the referenced entry, otherwise it returns *false*.

inline bool **get_entry**(*key_type* const &k, *entry_type* &val)

Get a specific entry identified by the given key.

Note: The function will call the entry's `entry::touch` function if the value corresponding to the provided key is found in the cache.

Parameters

- **k** – [in] The key for the entry which should be retrieved from the cache.
- **val** – [out] If the entry indexed by the key is found in the cache this value on successful return will be a copy of the corresponding entry.

Returns

This function returns *true* if the cache holds the referenced entry, otherwise it returns *false*.

```
inline bool get_entry(key_type const &k, value_type &val)
```

Get a specific entry identified by the given key.

Note: The function will call the entry's `entry::touch` function if the value corresponding to the provided is found in the cache.

Parameters

- **k** – [in] The key for the entry which should be retrieved from the cache
- **val** – [out] If the entry indexed by the key is found in the cache this value on successful return will be a copy of the corresponding value.

Returns

This function returns *true* if the cache holds the referenced entry, otherwise it returns *false*.

```
inline bool insert(key_type const &k, value_type const &val)
```

Insert a new element into this cache.

Note: This function invokes both, the insert policy as provided to the constructor and the function `entry::insert` of the newly constructed entry instance. If either of these functions returns false the key/value pair doesn't get inserted into the cache and the `insert` function will return *false*. Other reasons for this function to fail (return *false*) are a) the key/value pair is already held in the cache or b) inserting the new value into the cache maxed out its capacity and it was not possible to free any of the existing entries.

Parameters

- **k** – [in] The key for the entry which should be added to the cache.
- **val** – [in] The value which should be added to the cache.

Returns

This function returns *true* if the entry has been successfully added to the cache, otherwise it returns *false*.

```
inline bool insert(key_type const &k, value_type &&val)
```

```
template<typename Entry_, std::enable_if_t<std::is_convertible_v<std::decay_t<Entry_>, entry_type>, int> = 0>
```

```
inline bool insert(key_type const &k, Entry_ &&e)
```

Insert a new entry into this cache.

Note: This function invokes both, the insert policy as provided to the constructor and the function `entry::insert` of the provided entry instance. If either of these functions returns false

the key/value pair doesn't get inserted into the cache and the *insert* function will return *false*. Other reasons for this function to fail (return *false*) are a) the key/value pair is already held in the cache or b) inserting the new value into the cache maxed out its capacity and it was not possible to free any of the existing entries.

Parameters

- **k** – [in] The key for the entry which should be added to the cache.
- **e** – [in] The entry which should be added to the cache.

Returns

This function returns *true* if the entry has been successfully added to the cache, otherwise it returns *false*.

```
template<typename Value, std::enable_if_t<std::is_convertible_v<std::decay_t<Value>, value_type>, int> = 0>
```

```
inline bool update(key_type const &k, Value &&val)
```

Update an existing element in this cache.

Note: The function will call the entry's *entry::touch* function if the indexed value is found in the cache.

Note: The difference to the other overload of the *insert* function is that this overload replaces the cached value only, while the other overload replaces the whole cache entry, updating the cache entry properties.

Parameters

- **k** – [in] The key for the value which should be updated in the cache.
- **val** – [in] The value which should be used as a replacement for the existing value in the cache. Any existing cache entry is not changed except for its value.

Returns

This function returns *true* if the entry has been successfully updated, otherwise it returns *false*. If the entry currently is not held by the cache it is added and the return value reflects the outcome of the corresponding insert operation.

```
template<typename F, typename Value, typename =  
std::enable_if_t<std::is_convertible_v<std::decay_t<Value>, value_type>>>
```

```
inline bool update_if(key_type const &k, Value &&val, F &&f)
```

Update an existing element in this cache.

Note: The function will call the entry's *entry::touch* function if the indexed value is found in the cache.

Note: The difference to the other overload of the *insert* function is that this overload replaces the cached value only, while the other overload replaces the whole cache entry, updating the cache entry properties.

Parameters

- **k** – [in] The key for the value which should be updated in the cache.
- **val** – [in] The value which should be used as a replacement for the existing value in the cache. Any existing cache entry is not changed except for its value.

- **f** – [in] A callable taking two arguments, *k* and the key found in the cache (in that order). If *f* returns true, then the update will continue. If *f* returns false, then the update will not succeed.

Returns

This function returns *true* if the entry has been successfully updated, otherwise it returns *false*. If the entry currently is not held by the cache it is added and the return value reflects the outcome of the corresponding insert operation.

```
template<typename Entry_, std::enable_if_t<std::is_convertible_v<std::decay_t<Entry_>,
entry_type>, int> = 0>
inline bool update(key_type const &k, Entry_ &&e)
```

Update an existing entry in this cache.

Note: The function will call the entry's *entry::touch* function if the indexed value is found in the cache.

Note: The difference to the other overload of the *insert* function is that this overload replaces the whole cache entry, while the other overload replaces the cached value only, leaving the cache entry properties untouched.

Parameters

- **k** – [in] The key for the entry which should be updated in the cache.
- **e** – [in] The entry which should be used as a replacement for the existing entry in the cache. Any existing entry is first removed and then this entry is added.

Returns

This function returns *true* if the entry has been successfully updated, otherwise it returns *false*. If the entry currently is not held by the cache it is added and the return value reflects the outcome of the corresponding insert operation.

```
template<typename Func = policies::always<storage_value_type>>
inline size_type erase(Func &&ep = Func())
```

Remove stored entries from the cache for which the supplied function object returns true.

Parameters

ep – [in] This parameter has to be a (unary) function object. It is invoked for each of the entries currently held in the cache. An entry is considered for removal from the cache whenever the value returned from this invocation is *true*. Even then the entry might not be removed from the cache as its *entry::remove* function might return false.

Returns

This function returns the overall size of the removed entries (which is the sum of the values returned by the *entry::get_size* functions of the removed entries).

```
inline size_type erase()
```

Remove all stored entries from the cache.

Note: All entries are considered for removal, but in the end an entry might not be removed from the cache as its *entry::remove* function might return false. This function is very useful for instance in conjunction with an entry's *entry::remove* function enforcing additional criteria like entry expiration, etc.

Returns

This function returns the overall size of the removed entries (which is the sum of the

values returned by the *entry::get_size* functions of the removed entries).

inline void **clear**()

Clear the cache.

Unconditionally removes all stored entries from the cache.

inline constexpr *statistics_type* const &**get_statistics**() const noexcept

Allow to access the embedded statistics instance.

Returns

This function returns a reference to the statistics instance embedded inside this cache

inline *statistics_type* &**get_statistics**() noexcept

Protected Functions

inline bool **free_space**(long num_free)

Private Types

using **iterator** = typename *storage_type*::iterator

using **const_iterator** = typename *storage_type*::const_iterator

using **heap_type** = std::deque<iterator>

using **heap_iterator** = typename *heap_type*::iterator

using **adapted_update_policy_type** = adapt<UpdatePolicy, iterator>

using **update_on_exit** = typename *statistics_type*::update_on_exit

Private Members

size_type **max_size_**

size_type **current_size_**

storage_type **store_**

heap_type **entry_heap_**

adapted_update_policy_type **update_policy_**

insert_policy_type **insert_policy_**


```

statistics_type statistics_

template<typename Func, typename Iterator>

struct adapt

```

Public Functions

```

inline explicit adapt(Func const &f)

inline explicit adapt(Func &&f) noexcept

inline bool operator()(Iterator const &lhs, Iterator const &rhs) const

```

Public Members

```

Func f_

```

hpx/cache/lru_cache.hpp

Defined in header `hpx/cache/lru_cache.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace util
```

```
        namespace cache
```

```
            template<typename Key, typename Entry, typename Statistics = statistics::no_statistics>
```

```
            class lru_cache
```

#include <hpx/cache/lru_cache.hpp> The `lru_cache` implements the basic functionality needed for a local (non-distributed) LRU cache.

Template Parameters

- **Key** – The type of the keys to use to identify the entries stored in the cache
- **Entry** – The type of the items to be held in the cache.
- **Statistics** – A (optional) type allowing to collect some basic statistics about the operation of the cache instance. The type must conform to the `CacheStatistics` concept. The default value is the type `statistics::no_statistics` which does not collect any numbers, but provides empty stubs allowing the code to compile.

Public Types

```
using key_type = Key

using entry_type = Entry

using statistics_type = Statistics

using entry_pair = std::pair<key_type, entry_type>

using storage_type = std::list<entry_pair>

using map_type = std::map<Key, typename storage_type::iterator>

using size_type = std::size_t
```

Public Functions

```
inline explicit lru_cache(size_type max_size = 0)
```

Construct an instance of a *lru_cache*.

Parameters

max_size – [in] The maximal size this cache is allowed to reach any time. The default is zero (no size limitation). The unit of this value is usually determined by the unit of the values returned by the entry's *get_size* function.

```
lru_cache(lru_cache const &other) = default
```

```
lru_cache(lru_cache &&other) = default
```

```
lru_cache &operator=(lru_cache const &other) = default
```

```
lru_cache &operator=(lru_cache &&other) = default
```

```
~lru_cache() = default
```

```
inline constexpr size_type size() const noexcept
```

Return current size of the cache.

Returns

The current size of this cache instance.

```
inline constexpr size_type capacity() const noexcept
```

Access the maximum size the cache is allowed to grow to.

Note: The unit of this value is usually determined by the unit of the return values of the entry's function *entry::get_size*.

Returns

The maximum size this cache instance is currently allowed to reach. If this number is zero the cache has no limitation with regard to a maximum size.

```
inline void reserve(size_type max_size)
```

Change the maximum size this cache can grow to.

Parameters

max_size – [in] The new maximum size this cache will be allowed to grow to.

```
inline bool holds_key(key_type const &key) const
```

Check whether the cache currently holds an entry identified by the given key.

Note: This function does not call the entry’s function *entry::touch*. It just checks if the cache contains an entry corresponding to the given key.

Parameters

key – [in] The key for the entry which should be looked up in the cache.

Returns

This function returns *true* if the cache holds the referenced entry, otherwise it returns *false*.

```
inline bool get_entry(key_type const &key, key_type &realkey, entry_type &entry)
```

Get a specific entry identified by the given key.

Note: The function will “touch” the entry and mark it as recently used if the key was found in the cache.

Parameters

- **key** – [in] The key for the entry which should be retrieved from the cache.
- **realkey[out]** – Return the full real key found in the cache
- **entry** – [out] If the entry indexed by the key is found in the cache this value on successful return will be a copy of the corresponding entry.

Returns

This function returns *true* if the cache holds the referenced entry, otherwise it returns *false*.

```
inline bool get_entry(key_type const &key, entry_type const &entry)
```

Get a specific entry identified by the given key.

Note: The function will “touch” the entry and mark it as recently used if the key was found in the cache.

Parameters

- **key** – [in] The key for the entry which should be retrieved from the cache.
- **entry** – [out] If the entry indexed by the key is found in the cache this value on successful return will be a copy of the corresponding entry.

Returns

This function returns *true* if the cache holds the referenced entry, otherwise it returns *false*.

```
template<typename Entry_, typename =  
std::enable_if_t<std::is_convertible_v<std::decay_t<Entry_>, entry_type>>>  
inline bool insert(key_type const &key, Entry_ &&entry)
```

Insert a new entry into this cache.

Note: This function assumes that the entry is not in the cache already. Inserting an already existing entry is considered undefined behavior

Parameters

- **key** – [in] The key for the entry which should be added to the cache.
- **entry** – [in] The entry which should be added to the cache.

```
template<typename Entry_, typename =  
std::enable_if_t<std::is_convertible_v<std::decay_t<Entry_>, entry_type>>>  
inline void update(key_type const &key, Entry_ &&entry)
```

Update an existing element in this cache.

Note: The function will “touch” the entry and mark it as recently used if the key was found in the cache.

Note: The difference to the other overload of the *insert* function is that this overload replaces the cached value only, while the other overload replaces the whole cache entry, updating the cache entry properties.

Parameters

- **key** – [in] The key for the value which should be updated in the cache.
- **entry** – [in] The entry which should be used as a replacement for the existing value in the cache. Any existing cache entry is not changed except for its value.

```
template<typename F, typename Entry_,  
std::enable_if_t<std::is_convertible_v<std::decay_t<Entry_>, entry_type>, int> = 0>  
inline bool update_if(key_type const &key, Entry_ &&entry, F &&f)
```

Update an existing element in this cache.

Note: The function will “touch” the entry and mark it as recently used if the key was found in the cache.

Note: The difference to the other overload of the *insert* function is that this overload replaces the cached value only, while the other overload replaces the whole cache entry, updating the cache entry properties.

Parameters

- **key** – [in] The key for the value which should be updated in the cache.
- **entry** – [in] The value which should be used as a replacement for the existing value in the cache. Any existing cache entry is not changed except for its value.
- **f** – [in] A callable taking two arguments, *k* and the key found in the cache (in that order). If *f* returns true, then the update will continue. If *f* returns false, then the update will not succeed.

Returns

This function returns *true* if the entry has been successfully updated, otherwise it returns *false*. If the entry currently is not held by the cache it is added and the return value reflects the outcome of the corresponding insert operation.

```
template<typename Func>
inline size_type erase(Func const &ep)
```

Remove stored entries from the cache for which the supplied function object returns true.

Parameters

ep – [in] This parameter has to be a (unary) function object. It is invoked for each of the entries currently held in the cache. An entry is considered for removal from the cache whenever the value returned from this invocation is *true*.

Returns

This function returns the overall size of the removed entries (which is the sum of the values returned by the *entry::get_size* functions of the removed entries).

```
inline size_type erase()
```

Remove all stored entries from the cache.

Returns

This function returns the overall size of the removed entries (which is the sum of the values returned by the *entry::get_size* functions of the removed entries).

```
inline size_type clear()
```

Clear the cache.

Unconditionally removes all stored entries from the cache.

```
inline constexpr statistics_type const &get_statistics() const noexcept
```

Allow to access the embedded statistics instance.

Returns

This function returns a reference to the statistics instance embedded inside this cache

```
inline statistics_type &get_statistics() noexcept
```

Private Types

```
using update_on_exit = typename statistics_type::update_on_exit
```

Private Functions

```
template<typename Entry_, typename =
std::enable_if_t<std::is_convertible_v<std::decay_t<Entry_>, entry_type>>>
inline void insert_nonexist(key_type const &key, Entry_ &&entry)
```

```
inline void touch(typename storage_type::iterator it)
```

```
inline void evict()
```

Private Members

```
size_type max_size_
```

```
size_type current_size_ = 0
```

```
storage_type storage_
```

map_type **map_**

statistics_type **statistics_**

hpx/cache/statistics/local_statistics.hpp

Defined in header `hpx/cache/statistics/local_statistics.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **util**

namespace **cache**

namespace **statistics**

class **local_statistics** : public *hpx::util::cache::statistics::no_statistics*

Public Functions

local_statistics() = default

inline constexpr *std::size_t* **hits**() const noexcept

inline constexpr *std::size_t* **misses**() const noexcept

inline constexpr *std::size_t* **insertions**() const noexcept

inline constexpr *std::size_t* **evictions**() const noexcept

inline *std::size_t* **hits**(bool const reset) noexcept

inline *std::size_t* **misses**(bool const reset) noexcept

inline *std::size_t* **insertions**(bool const reset) noexcept

inline *std::size_t* **evictions**(bool const reset) noexcept

inline void **got_hit**() noexcept

The function *got_hit* will be called by a cache instance whenever an entry got touched.

inline void **got_miss**() noexcept

The function *got_miss* will be called by a cache instance whenever a requested entry has not been found in the cache.

inline void **got_insertion**() noexcept

The function *got_insertion* will be called by a cache instance whenever a new entry has been inserted.

```
inline void got_eviction() noexcept
```

The function *got_eviction* will be called by a cache instance whenever an entry has been removed from the cache because a new inserted entry let the cache grow beyond its capacity.

```
inline void clear() noexcept
```

Reset all statistics.

Private Members

```
std::size_t hits_ = 0
```

```
std::size_t misses_ = 0
```

```
std::size_t insertions_ = 0
```

```
std::size_t evictions_ = 0
```

Private Static Functions

```
static inline std::size_t get_and_reset(std::size_t &value, bool const reset) noexcept
```

hpx/cache/statistics/no_statistics.hpp

Defined in header `hpx/cache/statistics/no_statistics.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace util
```

```
        namespace cache
```

```
            namespace statistics
```

Enums

```
enum class method
```

Values:

```
    enumerator get_entry
```

```
    enumerator insert_entry
```

enumerator **update_entry**

enumerator **erase_entry**

class **no_statistics**

Subclassed by *hpx::util::cache::statistics::local_statistics*

Public Static Functions

static inline constexpr void **got_hit**() noexcept

The function *got_hit* will be called by a cache instance whenever an entry got touched.

static inline constexpr void **got_miss**() noexcept

The function *got_miss* will be called by a cache instance whenever a requested entry has not been found in the cache.

static inline constexpr void **got_insertion**() noexcept

The function *got_insertion* will be called by a cache instance whenever a new entry has been inserted.

static inline constexpr void **got_eviction**() noexcept

The function *got_eviction* will be called by a cache instance whenever an entry has been removed from the cache because a new inserted entry let the cache grow beyond its capacity.

static inline constexpr void **clear**() noexcept

Reset all statistics.

static inline constexpr std::int64_t **get_get_entry_count**(bool) noexcept

The function *get_get_entry_count* returns the number of invocations of the *get_entry()* API function of the cache.

static inline constexpr std::int64_t **get_insert_entry_count**(bool) noexcept

The function *get_insert_entry_count* returns the number of invocations of the *insert_entry()* API function of the cache.

static inline constexpr std::int64_t **get_update_entry_count**(bool) noexcept

The function *get_update_entry_count* returns the number of invocations of the *update_entry()* API function of the cache.

static inline constexpr std::int64_t **get_erase_entry_count**(bool) noexcept

The function *get_erase_entry_count* returns the number of invocations of the *erase()* API function of the cache.

static inline constexpr std::int64_t **get_get_entry_time**(bool) noexcept

The function *get_get_entry_time* returns the overall time spent executing of the *get_entry()* API function of the cache.

static inline constexpr std::int64_t **get_insert_entry_time**(bool) noexcept

The function *get_insert_entry_time* returns the overall time spent executing of the *insert_entry()* API function of the cache.

static inline constexpr std::int64_t **get_update_entry_time**(bool) noexcept

The function *get_update_entry_time* returns the overall time spent executing of the *update_entry()* API function of the cache.


```
static inline constexpr std::int64_t get_erase_entry_time(bool) noexcept
```

The function *get_erase_entry_time* returns the overall time spent executing of the *erase()* API function of the cache.

```
struct update_on_exit
```

#include <no_statistics.hpp> Helper class to update timings and counts on function exit.

Public Functions

```
inline constexpr update_on_exit(no_statistics const&, method) noexcept
```

```
~update_on_exit() = default
```

```
update_on_exit(update_on_exit const&) = delete
```

```
update_on_exit(update_on_exit&&) = delete
```

```
update_on_exit &operator=(update_on_exit const&) = delete
```

```
update_on_exit &operator=(update_on_exit&&) = delete
```

hpx/cache/entries/lru_entry.hpp

Defined in header *hpx/cache/entries/lru_entry.hpp*.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
namespace util
```

```
namespace cache
```

```
namespace entries
```

```
template<typename Value>
```

```
class lru_entry : public hpx::util::cache::entries::entry<Value>
```

#include <hpx/cache/entries/lru_entry.hpp> The *lru_entry* type can be used to store arbitrary values in a cache. Using this type as the cache's entry type makes sure that the least recently used entries are discarded from the cache first.

Note: The *lru_entry* conforms to the *CacheEntry* concept.

Note: This type can be used to model a 'most recently used' cache policy if it is used with a *std::greater* as the caches' *UpdatePolicy* (instead of the default *std::less*).

Template Parameters

Value – The data type to be stored in a cache. It has to be default constructible, copy constructible and *less_than_comparable*.

Public Functions

inline **lru_entry**()

Any cache entry has to be default constructible.

inline explicit **lru_entry**(*Value* const &val)
noexcept(*std::is_nothrow_constructible_v<base_type, Value const&>*)

Construct a new instance of a cache entry holding the given value.

inline explicit **lru_entry**(*Value* &&val) noexcept

Construct a new instance of a cache entry holding the given value.

inline bool **touch**()

The function *touch* is called by a cache holding this instance whenever it has been requested (touched).

In the case of the LRU entry we store the time of the last access which will be used to compare the age of an entry during the invocation of the *operator<()*.

Returns

This function should return true if the cache needs to update its internal heap. Usually this is needed if the entry has been changed by *touch()* in a way influencing the sort order as mandated by the cache's UpdatePolicy

inline constexpr *time_point* const &**get_access_time**() const noexcept

Returns the last access time of the entry.

Private Types

using **base_type** = *entry<Value>*

using **time_point** = *std::chrono::steady_clock::time_point*

Private Members

time_point **access_time_**

Friends

inline friend auto **operator**<(*lru_entry* const &lhs, *lru_entry* const &rhs)

Compare the 'age' of two entries. An entry is 'older' than another entry if it has been accessed less recently (LRU).

inline friend auto **operator**>(*lru_entry* const &lhs, *lru_entry* const &rhs)

inline friend auto **operator**<= (*lru_entry* const &lhs, *lru_entry* const &rhs)

inline friend auto **operator**>= (*lru_entry* const &lhs, *lru_entry* const &rhs)

inline friend auto **operator**== (*lru_entry* const &lhs, *lru_entry* const &rhs)

inline friend auto **operator**!= (*lru_entry* const &lhs, *lru_entry* const &rhs)

hpx/cache/entries/size_entry.hpp

Defined in header `hpx/cache/entries/size_entry.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **util**

namespace **cache**

namespace **entries**

template<typename **Value**>

class **size_entry** : public *hpx::util::cache::entries::entry<Value>*

#include <hpx/cache/entries/size_entry.hpp> The `size_entry` type can be used to store values in a cache which have a size associated (such as files, etc.). Using this type as the cache's entry type makes sure that the entries with the biggest size are discarded from the cache first.

Note: The `size_entry` conforms to the `CacheEntry` concept.

Note: This type can be used to model a 'discard smallest first' cache policy if it is used with a `std::greater` as the caches' `UpdatePolicy` (instead of the default `std::less`).

Template Parameters

Value – The data type to be stored in a cache. It has to be default constructible, copy constructible and `less_than_comparable`.

Public Functions

size_entry() = default

Any cache entry has to be default constructible.

inline explicit **size_entry**(*Value* const &val, *std::size_t* size = 0)
noexcept(*std::is_nothrow_constructible_v<base_type, Value*
const&>)

Construct a new instance of a cache entry holding the given value.

inline explicit **size_entry**(*Value* &&val, *std::size_t* size = 0) noexcept

Construct a new instance of a cache entry holding the given value.

inline constexpr *std::size_t* **get_size**() const noexcept

Return the 'size' of this entry.

Private Types

```
using base_type = entry<Value>
```

Private Members

```
std::size_t size_ = 0
```

Friends

```
inline friend auto operator<(size_entry const &lhs, size_entry const &rhs) noexcept
```

Compare the ‘age’ of two entries. An entry is ‘older’ than another entry if it has a bigger size.

```
inline friend auto operator>(size_entry const &lhs, size_entry const &rhs) noexcept
```

```
inline friend auto operator<=(size_entry const &lhs, size_entry const &rhs) noexcept
```

```
inline friend auto operator>=(size_entry const &lhs, size_entry const &rhs) noexcept
```

```
inline friend auto operator==(size_entry const &lhs, size_entry const &rhs) noexcept
```

```
inline friend auto operator!=(size_entry const &lhs, size_entry const &rhs) noexcept
```

hpx/cache/entries/entry.hpp

Defined in header hpx/cache/entries/entry.hpp.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
namespace hpx
```

```
    namespace util
```

```
        namespace cache
```

```
            namespace entries
```

```
                template<typename Value>
```

```
                class entry
```

```
                    #include <hpx/cache/entries/entry.hpp>
```

Template Parameters

Value – The data type to be stored in a cache. It has to be default constructible, copy constructible and less_than_comparable.

Subclassed by `hpx::util::cache::entries::fifo_entry< Value >`,
`hpx::util::cache::entries::lfu_entry< Value >`, `hpx::util::cache::entries::lru_entry< Value >`,
`hpx::util::cache::entries::size_entry< Value >`

Public Types

using **value_type** = *Value*

Public Functions

entry() = default

Any cache entry has to be default constructible.

inline explicit **entry**(*value_type* const &val)
noexcept(*std::is_nothrow_copy_constructible_v*<*value_type*>)

Construct a new instance of a cache entry holding the given value.

inline explicit **entry**(*value_type* &&val) noexcept

Construct a new instance of a cache entry holding the given value.

inline *value_type* &**get**() noexcept

Get a reference to the stored data value.

Note: This function is part of the CacheEntry concept

inline constexpr *value_type* const &**get**() const noexcept

Public Static Functions

static inline constexpr bool **touch**() noexcept

The function *touch* is called by a cache holding this instance whenever it has been requested (touched).

Note: It is possible to change the entry in a way influencing the sort criteria mandated by the UpdatePolicy. In this case the function should return *true* to indicate this to the cache, forcing to reorder the cache entries.

Note: This function is part of the CacheEntry concept

Returns

This function should return *true* if the cache needs to update its internal heap. Usually this is needed if the entry has been changed by *touch()* in a way influencing the sort order as mandated by the cache's UpdatePolicy

static inline constexpr bool **insert**() noexcept

The function *insert* is called by a cache whenever it is about to be inserted into the cache.

Note: This function is part of the CacheEntry concept

Returns

This function should return *true* if the entry should be added to the cache, otherwise it should return *false*.

static inline constexpr bool **remove**() noexcept

The function *remove* is called by a cache holding this instance whenever it is about to be removed from the cache.

Note: This function is part of the CacheEntry concept

Returns

The return value can be used to avoid removing this instance from the cache. If the value is *true* it is ok to remove the entry, otherwise it will stay in the cache.

static inline constexpr std::size_t **get_size**() noexcept

Return the 'size' of this entry. By default, the size of each entry is just one (1), which is sensible if the cache has a limit (capacity) measured in number of entries.

Private Members

value_type **value_**

Friends

friend auto operator (entry const &, entry const &)=default

Forwarding operator<=> allowing to compare entries instead of the values.

hpx/cache/entries/fifo_entry.hpp

Defined in header hpx/cache/entries/fifo_entry.hpp.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **util**

namespace **cache**

namespace **entries**

template<typename **Value**>

class **fifo_entry** : public hpx::util::cache::entries::entry<Value>

#include <hpx/cache/entries/fifo_entry.hpp> The *fifo_entry* type can be used to store arbitrary values in a cache. Using this type as the cache's entry type makes sure that the least recently inserted entries are discarded from the cache first.

Note: The *fifo_entry* conforms to the CacheEntry concept.

Note: This type can be used to model a ‘last in first out’ cache policy if it is used with a `std::greater` as the caches’ `UpdatePolicy` (instead of the default `std::less`).

Template Parameters

Value – The data type to be stored in a cache. It has to be default constructible, copy constructible and `less_than_comparable`.

Public Functions

fifo_entry() = default

Any cache entry has to be default constructible.

```
inline explicit fifo_entry(Value const &val)
    noexcept(std::is_nothrow_constructible_v<base_type, Value
    const&>)
```

Construct a new instance of a cache entry holding the given value.

```
inline explicit fifo_entry(Value &&val) noexcept
```

Construct a new instance of a cache entry holding the given value.

```
inline constexpr bool insert()
```

The function *insert* is called by a cache whenever it is about to be inserted into the cache.

Note: This function is part of the `CacheEntry` concept

Returns

This function should return *true* if the entry should be added to the cache, otherwise it should return *false*.

```
inline constexpr time_point const &get_creation_time() const noexcept
```

Private Types

```
using base_type = entry<Value>
```

```
using time_point = std::chrono::steady_clock::time_point
```

Private Members

```
time_point insertion_time_
```

Friends

```
inline friend auto operator<(fifo_entry const &lhs, fifo_entry const &rhs)
```

Compare the ‘age’ of two entries. An entry is ‘older’ than another entry if it has been created earlier (FIFO).

```
inline friend auto operator>(fifo_entry const &lhs, fifo_entry const &rhs)
```

```
inline friend auto operator<= (fifo_entry const &lhs, fifo_entry const &rhs)
```

```
inline friend auto operator>= (fifo_entry const &lhs, fifo_entry const &rhs)
```

```
inline friend auto operator== (fifo_entry const &lhs, fifo_entry const &rhs)
```

```
inline friend auto operator!= (fifo_entry const &lhs, fifo_entry const &rhs)
```

hpx/cache/entries/lfu_entry.hpp

Defined in header `hpx/cache/entries/lfu_entry.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **util**

namespace **cache**

namespace **entries**

```
template<typename Value>
```

```
class lfu_entry : public hpx::util::cache::entries::entry<Value>
```

#include <hpx/cache/entries/lfu_entry.hpp> The `lfu_entry` type can be used to store arbitrary values in a cache. Using this type as the cache’s entry type makes sure that the least frequently used entries are discarded from the cache first.

Note: The `lfu_entry` conforms to the `CacheEntry` concept.

Note: This type can be used to model a ‘most frequently used’ cache policy if it is used with a `std::greater` as the caches’ `UpdatePolicy` (instead of the default `std::less`).

Template Parameters

Value – The data type to be stored in a cache. It has to be default constructible, copy constructible and `less_than_comparable`.

Public Functions

lfu_entry() = default

Any cache entry has to be default constructible.

```
inline explicit lfu_entry(Value const &val)
    noexcept(std::is_nothrow_constructible_v<base_type, Value
    const&>)
```

Construct a new instance of a cache entry holding the given value.

```
inline explicit lfu_entry(Value &&val) noexcept
```

Construct a new instance of a cache entry holding the given value.

```
inline bool touch() noexcept
```

The function *touch* is called by a cache holding this instance whenever it has been requested (touched).

In the case of the LFU entry we store the reference count tracking the number of times this entry has been requested. This which will be used to compare the age of an entry during the invocation of the *operator<()*.

Returns

This function should return true if the cache needs to update its internal heap. Usually this is needed if the entry has been changed by *touch()* in a way influencing the sort order as mandated by the cache's UpdatePolicy

```
inline constexpr unsigned long const &get_access_count() const noexcept
```

Private Types

```
using base_type = entry<Value>
```

Private Members

```
unsigned long ref_count_ = 0
```

Friends

```
inline friend auto operator<(lfu_entry const &lhs, lfu_entry const &rhs)
```

Compare the 'age' of two entries. An entry is 'older' than another entry if it has been accessed less frequently (LFU).

```
inline friend auto operator>(lfu_entry const &lhs, lfu_entry const &rhs)
```

```
inline friend auto operator<= (lfu_entry const &lhs, lfu_entry const &rhs)
```

```
inline friend auto operator>= (lfu_entry const &lhs, lfu_entry const &rhs)
```

```
inline friend auto operator== (lfu_entry const &lhs, lfu_entry const &rhs)
```

```
inline friend auto operator!= (lfu_entry const &lhs, lfu_entry const &rhs)
```

checkpoint_base

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/checkpoint_base/checkpoint_data.hpp

Defined in header `hpx/checkpoint_base/checkpoint_data.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **util**

Functions

```
template<typename Container, typename ...Ts>
void save_checkpoint_data(Container &data, Ts&&... ts)
    save_checkpoint_data
```

`save_checkpoint_data` takes any number of objects which a user may wish to store in the given container.

Template Parameters

- **Container** – Container used to store the check-pointed data.
- **Ts** – Types of variables to checkpoint

Parameters

- **data** – Container instance used to store the checkpoint data
- **ts** – Variable instances to be inserted into the checkpoint.

```
template<typename ...Ts>
std::size_t prepare_checkpoint_data(Ts const&... ts)
    prepare_checkpoint_data
```

`prepare_checkpoint_data` takes any number of objects which a user may wish to store in a subsequent `save_checkpoint_data` operation. The function will return the number of bytes necessary to store the data that will be produced.

Template Parameters

- **Ts** – Types of variables to checkpoint

Parameters

- **ts** – Variable instances to be inserted into the checkpoint.

```
template<typename Container, typename ...Ts>
void restore_checkpoint_data(Container const &cont, Ts&... ts)
    restore_checkpoint_data
```

`restore_checkpoint_data` takes any number of objects which a user may wish to restore from the given container. The sequence of objects has to correspond to the sequence of objects for the corresponding call to `save_checkpoint_data` that had used the given container instance.

Template Parameters

- **Container** – Container used to restore the check-pointed data.
- **Ts** – Types of variables to restore

Parameters

- **cont** – Container instance used to restore the checkpoint data
- **ts** – Variable instances to be restored from the container

```
struct checkpointing_tag

template<>

struct extra_data_helper<checkpointing_tag>
```

Public Static Functions

```
static extra_data_id_type id() noexcept

static inline constexpr void reset(checkpointing_tag*) noexcept
```

compute_local

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/compute_local/vector.hpp

Defined in header `hpx/compute_local/vector.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace compute
```

Functions

```
template<typename T, typename Allocator>
void swap(vector<T, Allocator> &x, vector<T, Allocator> &y) noexcept
    Effects: x.swap(y);.

template<typename T, typename Allocator = std::allocator<T>>
class vector
```

Public Types

using **value_type** = *T*

Member types (FIXME: add reference to std.

using **allocator_type** = *Allocator*

using **access_target** = typename *alloc_traits::access_target*

using **size_type** = *std::size_t*

using **difference_type** = *std::ptrdiff_t*

using **reference** = typename *alloc_traits::reference*

using **const_reference** = typename *alloc_traits::const_reference*

using **pointer** = typename *alloc_traits::pointer*

using **const_pointer** = typename *alloc_traits::const_pointer*

using **iterator** = detail::iterator<*T*, *Allocator*>

using **const_iterator** = detail::iterator<*T* const, *Allocator*>

using **reverse_iterator** = *std::reverse_iterator<iterator>*

using **const_reverse_iterator** = *std::reverse_iterator<const_iterator>*

Public Functions

inline explicit **vector**(*Allocator* const &alloc = *Allocator*())

inline **vector**(*size_type* count, *T* const &value, *Allocator* const &alloc = *Allocator*())

inline explicit **vector**(*size_type* count, *Allocator* const &alloc = *Allocator*())

template<typename **InIter**>

inline **vector**(*InIter* first, *InIter* last, *Allocator* const &alloc = *Allocator*())

inline **vector**(*vector* const &other)

inline **vector**(*vector* const &other, *Allocator* const &alloc)

inline **vector**(*vector* &&other) noexcept

inline **vector**(*vector* &&other, *Allocator* const &alloc)

```

inline vector(std::initializer_list<T> init, Allocator const &alloc)

inline ~vector()

inline vector &operator=(vector const &other)

inline vector &operator=(vector &&other) noexcept

inline void assign(size_type count, T const &value)

template<typename InIter>
inline void assign(InIter first, InIter last)

inline void assign(std::initializer_list<T> ilist)

inline allocator_type get_allocator() const noexcept
    Returns the allocator associated with the container.

inline reference at(size_type pos)

inline const_reference at(size_type pos) const

inline reference operator[](size_type pos)

inline const_reference operator[](size_type pos) const

inline reference front()
    Returns a reference to the first element in the container. Calling front on an empty container is
    undefined.

inline const_reference front() const
    Returns a reference to the first element in the container. Calling front on an empty container is
    undefined.

inline reference back()
    Returns a reference to the last element in the container. Calling back on an empty container is
    undefined.

inline const_reference back() const
    Returns a reference to the last element in the container. Calling back on an empty container is
    undefined.

inline pointer data() noexcept
    Returns pointer to the underlying array serving as element storage. The pointer is such that range
    [data(); data() + size()) is always a valid range, even if the container is empty (data() is not
    dereference-able in that case).

inline const_pointer data() const noexcept
    Returns pointer to the underlying array serving as element storage. The pointer is such that range
    [data(); data() + size()) is always a valid range, even if the container is empty (data() is not
    dereference-able in that case).

inline T *device_data() const noexcept
    Returns a raw pointer corresponding to the address of the data allocated on the device.

inline std::size_t size() const noexcept

inline std::size_t capacity() const noexcept

```

inline bool **empty**() const noexcept

Returns: `size() == 0`.

inline void **resize**(*size_type*)

Effects: If `size <= size()`, equivalent to calling `pop_back()` `size()`

- `size` times. If `size() < size`, appends `size - size()` default-inserted elements to the sequence.

Requires: T shall be MoveInsertable and DefaultInsertable into *this.

Remarks: If an exception is thrown other than by the move constructor of a non-CopyInsertable T there are no effects.

inline void **resize**(*size_type*, T const&)

Effects: If `size <= size()`, equivalent to calling `pop_back()` `size() - size` times. If `size() < size`, appends `size - size()` copies of `val` to the sequence.

Requires: T shall be CopyInsertable into *this.

Remarks: If an exception is thrown there are no effects.

inline *iterator* **begin**() noexcept

inline *iterator* **end**() noexcept

inline *const_iterator* **cbegin**() const noexcept

inline *const_iterator* **cend**() const noexcept

inline *const_iterator* **begin**() const noexcept

inline *const_iterator* **end**() const noexcept

inline *reverse_iterator* **rbegin**() noexcept

inline *reverse_iterator* **rend**() noexcept

inline *const_reverse_iterator* **crbegin**() const noexcept

inline *const_reverse_iterator* **crend**() const noexcept

inline *const_reverse_iterator* **rbegin**() const noexcept

inline *const_reverse_iterator* **rend**() const noexcept

inline void **swap**(*vector* &other) noexcept

Effects: Exchanges the contents and capacity() of *this with that of x.

Complexity: Constant time.

inline void **clear**() noexcept

Effects: Erases all elements in the range `[begin(),end())`. Destroys all elements in 'a'. Invalidates all references, pointers, and iterators referring to the elements of a and may invalidate the past-the-end iterator.

Post: `a.empty()` returns true.

Complexity: Linear.

Private Types

```
using alloc_traits = traits::allocator_traits<Allocator>
```

Private Members

```
size_type size_
```

```
size_type capacity_
```

```
allocator_type alloc_
```

```
pointer data_
```

```
namespace traits
```

hpx/compute_local/host/block_fork_join_executor.hpp

Defined in header `hpx/compute_local/host/block_fork_join_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
namespace execution
```

```
namespace experimental
```

```
class block_fork_join_executor
```

#include <block_fork_join_executor.hpp> An executor with fork-join (blocking) semantics.

The *block_fork_join_executor* creates on construction a set of worker threads that are kept alive for the duration of the executor. Copying the executor has reference semantics, i.e. copies of a *fork_join_executor* hold a reference to the worker threads of the original instance. Scheduling work through the executor concurrently from different threads is undefined behaviour.

The executor keeps a set of worker threads alive for the lifetime of the executor, meaning other work will not be executed while the executor is busy or waiting for work. The executor has a customizable delay after which it will yield to other work. Since starting and resuming the worker threads is a slow operation the executor should be reused whenever possible for multiple adjacent parallel algorithms or invocations of `bulk_(a)sync_execute`.

This behaviour is similar to the plain *fork_join_executor* except that the *block_fork_join_executor* creates a hierarchy of *fork_join_executors*, one for each target used to initialize it.

Public Functions

```
inline explicit block_fork_join_executor(threads::thread_priority const priority =  
                                         threads::thread_priority::bound,  
                                         threads::thread_stacksize const stacksize =  
                                         threads::thread_stacksize::small,  
                                         fork_join_executor::loop_schedule const  
                                         schedule =  
                                         fork_join_executor::loop_schedule::static,  
                                         std::chrono::nanoseconds const yield_delay =  
                                         std::chrono::milliseconds(1))
```

Construct a *block_fork_join_executor*.

Note: This constructor will create one *fork_join_executor* for each numa domain

Parameters

- **priority** – The priority of the worker threads.
- **stacksize** – The stacksize of the worker threads. Must not be *nostack*.
- **schedule** – The loop schedule of the parallel regions.
- **yield_delay** – The time after which the executor yields to other work if it has not received any new work for execution.

```
inline explicit block_fork_join_executor(std::vector<compute::host::target> const  
                                         &targets, threads::thread_priority const priority =  
                                         threads::thread_priority::bound,  
                                         threads::thread_stacksize const stacksize =  
                                         threads::thread_stacksize::small,  
                                         fork_join_executor::loop_schedule const  
                                         schedule =  
                                         fork_join_executor::loop_schedule::static,  
                                         std::chrono::nanoseconds const yield_delay =  
                                         std::chrono::milliseconds(1))
```

Construct a *block_fork_join_executor*.

Note: This constructor will create one *fork_join_executor* for each given target

Parameters

- **targets** – The list of targets to use for thread placement
- **priority** – The priority of the worker threads.
- **stacksize** – The stacksize of the worker threads. Must not be *nostack*.
- **schedule** – The loop schedule of the parallel regions.
- **yield_delay** – The time after which the executor yields to other work if it has not received any new work for execution.

Private Members

fork_join_executor **exec_**

`std::vector<fork_join_executor>` **block_execs_**

Private Static Functions

static inline `hpx::threads::mask_type` **cores_for_targets**(`std::vector<compute::host::target>`
const &targets)

hpx/compute_local/host/block_executor.hpp

Defined in header `hpx/compute_local/host/block_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

template<typename **Executor**>

struct **executor_execution_category**<compute::host::block_executor<Executor>>

Public Types

using **type** = `hpx::execution::parallel_execution_tag`

template<typename **Executor**>

struct **is_never_blocking_one_way_executor**<compute::host::block_executor<Executor>> : public
`is_never_blocking_one_way_executor<Executor>`

template<typename **Executor**>

struct **is_one_way_executor**<compute::host::block_executor<Executor>> : public
`is_one_way_executor<Executor>`

template<typename **Executor**>

struct **is_two_way_executor**<compute::host::block_executor<Executor>> : public
`is_two_way_executor<Executor>`

template<typename **Executor**>

struct **is_bulk_one_way_executor**<compute::host::block_executor<Executor>> : public
`is_bulk_one_way_executor<Executor>`

template<typename **Executor**>

struct **is_bulk_two_way_executor**<compute::host::block_executor<Executor>> : public
`is_bulk_two_way_executor<Executor>`

namespace **hpx**

```
namespace host
```

```
template<typename Executor = hpx::execution::experimental::restricted_thread_pool_executor>
```

struct **block_executor**

#include <block_executor.hpp> The block executor can be used to build NUMA aware programs. It will distribute work evenly across the passed targets

Template Parameters

Executor – The underlying executor to use

Public Types

```
using executor_parameters_type = hpx::execution::experimental::default_parameters
```

Public Functions

```
inline explicit block_executor(std::vector<host::target> const &targets,
                                threads::thread_priority priority =
                                threads::thread_priority::high, threads::thread_stacksize
                                stacksize = threads::thread_stacksize::default_,
                                threads::thread_schedule_hint schedulehint = { })
```

```
inline explicit block_executor(std::vector<host::target> &&targets)
```

```
inline block_executor(block_executor const &other)
```

```
inline block_executor(block_executor &&other) noexcept
```

```
inline block_executor &operator=(block_executor const &other)
```

```
inline block_executor &operator=(block_executor &&other) noexcept
```

```
template<typename F, typename ...Ts>
inline decltype(auto) post(F &&f, Ts&&... ts) const
```

```
template<typename F, typename ...Ts>
inline decltype(auto) async_execute(F &&f, Ts&&... ts) const
```

```
template<typename F, typename ...Ts>
inline decltype(auto) sync_execute(F &&f, Ts&&... ts) const
```

```
template<typename F, typename Shape, typename ...Ts>
inline decltype(auto) bulk_async_execute_impl(F &&f, Shape const &shape, Ts&&... ts)
                                             const
```

```
template<typename F, typename Shape, typename ...Ts>
inline decltype(auto) bulk_async_execute(F &&f, Shape const &shape, Ts&&... ts) const
```

```
template<typename F, typename Shape, typename ...Ts>
```

```

inline decltype(auto) bulk_sync_execute_impl(F &&f, Shape const &shape, Ts&&... ts)
    const

template<typename F, typename Shape, typename ...Ts>
inline decltype(auto) bulk_sync_execute(F &&f, Shape const &shape, Ts&&... ts) const

inline std::vector<host::target> const &targets() const noexcept

```

Private Functions

```

inline auto get_next_executor() const

inline void init_executors()

```

Private Members

```

std::vector<host::target> targets_

mutable std::atomic<std::size_t> current_

std::vector<Executor> executors_

threads::thread_priority priority_ = threads::thread_priority::high

threads::thread_stacksize stacksize_ = threads::thread_stacksize::default_

threads::thread_schedule_hint schedulehint_ = {}

```

namespace **execution**

namespace **experimental**

```

template<typename Executor> block_executor< Executor > >

```

Public Types

```

using type = hpx::execution::parallel_execution_tag

template<typename Executor> block_executor< Executor > > : public is_bulk_one_way_executor<...

template<typename Executor> block_executor< Executor > > : public is_bulk_two_way_executor<...

template<typename Executor> block_executor< Executor > > : public is_never_blocking_one_way<...

```

```
template<typename Executor> block_executor< Executor > > : public is_one_way_executor< Exec  
template<typename Executor> block_executor< Executor > > : public is_two_way_executor< Exec
```

config

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/config/endian.hpp

Defined in header `hpx/config/endian.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

contracts

See *Public API* for a list of names and headers that are part of the public *HPX* API.

HPX_PRE

Defined in header `hpx/contracts.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *HPX_POST*
- *HPX_CONTRACT_ASSERT*

Warning: doxygenfunction: Cannot find function “HPX_PRE” in doxygen xml output for project “hpx” from directory: `../hpx_autodoc`

HPX_POST

Defined in header `hpx/contracts.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *HPX_PRE*
- *HPX_CONTRACT_ASSERT*

Warning: doxygenfunction: Cannot find function “HPX_POST” in doxygen xml output for project “hpx” from directory: `../hpx_autodoc`

HPX_CONTRACT_ASSERT

Defined in header `hpx/contracts.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *HPX_PRE*
- *HPX_POST*

Warning: doxygenfunction: Cannot find function “HPX_CONTRACT_ASSERT” in doxygen xml output for project “hpx” from directory: `../hpx_autodoc`

coroutines

See *Public API* for a list of names and headers that are part of the public *HPX* API.

`hpx/coroutines/thread_enums.hpp`

Defined in header `hpx/coroutines/thread_enums.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **threads**

Enums

enum class **thread_schedule_state** : `std::int8_t`

The *thread_schedule_state* enumerator encodes the current state of a *thread* instance

Task Lifecycle State Transitions:

```
stateDiagram-v2
    %% Creation Phase
    [*] --> Staged : task_staged
    [*] --> Pending : task_created
    [*] --> Suspended : task_created
    Staged --> Pending : task_created

    %% Execution Loop
    Pending --> Active : task_executing
    Active --> Pending : task_yielded

    %% Sleep / Wake Cycle
    Active --> Suspended : task_suspended
    Suspended --> Pending : task_resumed
```

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```
%% Termination Paths
Active --> Terminated : task_completed
Active --> Deleted : task_completed (Inline fast-path)
Pending --> Terminated : task_completed (Aborted)
Suspended --> Terminated : task_completed (Aborted)

%% Cleanup (Destruction / Recycling)
Terminated --> Deleted : task_deleted
Deleted --> [*]
```

Values:

enumerator **unknown**

enumerator **active**

thread is currently active (running, has resources)

enumerator **pending**

thread is pending (ready to run, but no hardware resource available)

enumerator **suspended**

thread has been suspended (waiting for synchronization event, but still known and under control of the thread-manager)

enumerator **depleted**

thread has been depleted (deeply suspended, it is not known to the thread-manager)

enumerator **terminated**

thread has been stopped and may be garbage collected

enumerator **staged**

this is not a real thread state, but allows to reference staged task descriptions, which eventually will be converted into thread objects

enumerator **pending_do_not_schedule**

this is not a real thread state, but allows to create a thread in pending state without scheduling it (internal, do not use)

enumerator **pending_boost**

this is not a real thread state, but allows to suspend a thread in pending state without high priority rescheduling

enumerator **deleted**

thread has been stopped and was deleted

enum class **thread_priority** : *std::int8_t*

This enumeration lists all possible thread-priorities for HPX threads.

Values:

enumerator **unknown**

enumerator **default_**

Will assign the priority of the task to the default (normal) priority.

enumerator **low**

Task goes onto a special low priority queue and will not be executed until all high/normal priority tasks are done, even if they are added after the low priority task.

enumerator **normal**

Task will be executed when it is taken from the normal priority queue, this is usually a first in-first-out ordering of tasks (depending on scheduler choice). This is the default priority.

enumerator **high_recursive**

The task is a high priority task and any child tasks spawned by this task will be made high priority as well - unless they are specifically flagged as non default priority.

enumerator **boost**

Same as *thread_priority_high* except that the thread will fall back to *thread_priority_normal* if resumed after being suspended.

enumerator **high**

Task goes onto a special high priority queue and will be executed before normal/low priority tasks are taken (some schedulers modify the behavior slightly and the documentation for those should be consulted).

enumerator **bound**

Task goes onto a special high priority queue and will never be stolen by another thread after initial assignment. This should be used for thread placement tasks such as OpenMP type for loops.

enumerator **initially_bound**

Task initially goes onto a special high priority queue and will never be stolen by another thread after initial assignment. If the thread is rescheduled later, it will use normal priority.

enum class **thread_restart_state** : *std::int8_t*

The *thread_restart_state* enumerator encodes the reason why a thread is being restarted

Values:

enumerator **unknown**

enumerator **signaled**

The thread has been signaled.

enumerator **timeout**

The thread has been reactivated after a timeout.

enumerator **terminate**

The thread needs to be terminated.

enumerator **abort**

The thread needs to be aborted.

enum class **thread_stacksize** : *std::int8_t*

A *thread_stacksize* references any of the possible stack-sizes for HPX threads.

Values:

enumerator **unknown**

enumerator **small_**

use small stack size (the underscore is to work around 'small' being defined to char on Windows)

enumerator **medium**

use medium sized stack size

enumerator **large**

use large stack size

enumerator **huge**

use very large stack size

enumerator **nostack**

this thread does not suspend (does not need a stack)

enumerator **current**

use size of current thread's stack

enumerator **default_**

use default stack size

enumerator **minimal**

use minimally stack size

enumerator **maximal**

use maximally stack size

enum class **thread_schedule_hint_mode** : *std::int8_t*

The type of hint given when creating new tasks.

Values:

enumerator **none**

A hint that leaves the choice of scheduling entirely up to the scheduler.

enumerator **thread**

A hint that tells the scheduler to prefer scheduling a task on the local thread number associated with this hint. Local thread numbers are indexed from zero. It is up to the scheduler to decide how to interpret thread numbers that are larger than the number of threads available to the scheduler. Typically, thread numbers will wrap around when too large.

enumerator **numa**

A hint that tells the scheduler to prefer scheduling a task on the NUMA domain associated with this hint. NUMA domains are indexed from zero. It is up to the scheduler to decide how to interpret NUMA domain indices that are larger than the number of available NUMA domains to the scheduler. Typically, indices will wrap around when too large.

enum class **thread_placement_hint** : *std::int8_t*

The type of hint given to the scheduler related to thread placement

The type of hint given to the scheduler related running a thread as a child directly in the context of the parent thread

Values:

enumerator **none**

No hint is specified. The implementation is free to chose what placement methods to use.

enumerator **depth_first**

A hint that tells the scheduler to prefer spreading thread placement on a depth-first basis (i.e. consecutively scheduled threads are placed on the same core).

enumerator **breadth_first**

A hint that tells the scheduler to prefer spreading thread placement on a breadth-first basis (i.e. consecutively scheduled threads are placed on the neighboring cores).

enumerator **depth_first_reverse**

A hint that tells the scheduler to prefer spreading thread placement on a depth-first basis (i.e. consecutively scheduled threads are placed on the same core). Threads are being scheduled in reverse order.

enumerator **breadth_first_reverse**

A hint that tells the scheduler to prefer spreading thread placement on a breadth-first basis (i.e. consecutively scheduled threads are placed on the neighboring cores). Threads are being scheduled in reverse order.

enum class **thread_sharing_hint** : *std::int8_t*

The type of hint given to the scheduler related to whether it is ok to share the invoked function object between threads

Values:

enumerator **none**

No hint is specified. The implementation is free to chose what sharing methods to use.

enumerator **do_not_share_function**

A hint that tells the scheduler to avoid sharing the given function (object) between threads.

enumerator **do_not_combine_tasks**

A hint that tells the scheduler to avoid combining tasks on the same thread. This is important for tasks that may synchronize between each other, which could lead to deadlocks if those tasks are combined running by the same thread.

enum class **thread_execution_hint** : *std::int8_t*

Values:

enumerator **none**

No hint is specified. Always run the thread in its own execution environment.

enumerator **run_as_child**

Attempt to run the thread in the execution context of the parent thread.

Functions

std::ostream &**operator<<**(*std::ostream* &os, *thread_schedule_state* t)

char const ***get_thread_state_name**(*thread_schedule_state* state) noexcept

Returns the name of the given state.

Get the readable string representing the name of the given *thread_state* constant.

Parameters

state – this represents the thread state.

std::ostream &**operator<<**(*std::ostream* &os, *thread_priority* t)

char const ***get_thread_priority_name**(*thread_priority* priority) noexcept

Return the thread priority name.

Get the readable string representing the name of the given *thread_priority* constant.

Parameters

priority – this represents the thread priority.

std::ostream &**operator<<**(*std::ostream* &os, *thread_restart_state* t)

char const ***get_thread_state_ex_name**(*thread_restart_state* state) noexcept

Get the readable string representing the name of the given *thread_restart_state* constant.

char const ***get_thread_state_name**(*thread_state* state) noexcept

Get the readable string representing the name of the given *thread_state* constant.

std::ostream &**operator<<**(*std::ostream* &os, *thread_stacksize* t)

char const ***get_stack_size_enum_name**(*thread_stacksize* size) noexcept

Returns the stack size name.

Get the readable string representing the given stack size constant.

Parameters

size – this represents the stack size

constexpr bool **do_not_share_function**(*thread_sharing_hint* hint) noexcept

constexpr bool **do_not_combine_tasks**(*thread_sharing_hint* hint) noexcept

constexpr *thread_sharing_hint* **operator|**(*thread_sharing_hint* lhs, *thread_sharing_hint* rhs) noexcept

constexpr bool **run_as_child**(*thread_execution_hint* hint) noexcept

Variables

constexpr *thread_execution_hint* **default_runs_as_child_hint** = *thread_execution_hint::none*

Default value to use for runs-as-child mode (if *true*, then futures will attempt to execute associated threads directly if they have not started running).

struct **thread_schedule_hint**

#include <thread_enums.hpp> A hint given to a scheduler to guide where a task should be scheduled.

A scheduler is free to ignore the hint, or modify the hint to suit the resources available to the scheduler.

Public Functions

inline constexpr **thread_schedule_hint**() noexcept

Construct a default hint with mode *thread_schedule_hint_mode::none*.

inline explicit constexpr **thread_schedule_hint**(*std::int16_t* const thread_hint,
 thread_placement_hint placement =
 thread_placement_hint::none,
 thread_execution_hint runs_as_child =
 default_runs_as_child_hint,
 thread_sharing_hint sharing =
 thread_sharing_hint::none) noexcept

Construct a hint with mode *thread_schedule_hint_mode::thread* and the given hint as the local thread number.

inline constexpr **thread_schedule_hint**(*thread_schedule_hint_mode* const mode, *std::int16_t*
 const hint, *thread_placement_hint* placement =
 thread_placement_hint::none, *thread_execution_hint*
 runs_as_child = *default_runs_as_child_hint*,
 thread_sharing_hint sharing =
 thread_sharing_hint::none) noexcept

Construct a hint with the given mode and hint. The numerical hint is unused when the mode is *thread_schedule_hint_mode::none*.

inline explicit constexpr **thread_schedule_hint**(*thread_placement_hint* placement) noexcept

Construct a hint with the given placement hint.

inline explicit constexpr **thread_schedule_hint**(*thread_execution_hint* runs_as_child) noexcept

Construct a hint with the given execution hint.

inline explicit constexpr **thread_schedule_hint**(*thread_sharing_hint* sharing) noexcept

Construct a hint with the given sharing hint.

inline constexpr *thread_placement_hint* **placement_mode**() const noexcept

inline constexpr void **placement_mode**(*thread_placement_hint* bits) noexcept

inline constexpr *thread_sharing_hint* **sharing_mode**() const noexcept

inline constexpr void **sharing_mode**(*thread_sharing_hint* bits) noexcept

inline constexpr *thread_execution_hint* **runs_as_child_mode**() const noexcept

inline constexpr void **runs_as_child_mode**(*thread_execution_hint* bits) noexcept

inline constexpr void **schedule_hint**(std::int16_t core) noexcept

Public Members

std::int16_t **hint** = -1

The hint associated with the mode. The interpretation of this hint depends on the given mode.

thread_schedule_hint_mode **mode** = *thread_schedule_hint_mode::none*

The mode of the scheduling hint.

std::uint8_t **placement_mode_bits**

The mode of the desired thread placement.

std::uint8_t **sharing_mode_bits**

The mode of the desired sharing hint.

std::uint8_t **runs_as_child_mode_bits**

The thread will run as a child directly in the context of the current thread

hpx/coroutines/thread_id_type.hpp

Defined in header `hpx/coroutines/thread_id_type.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

template<>

struct **hash**<::hpx::threads::thread_id>

Public Functions

```
inline std::size_t operator()(::hpx::threads::thread_id const &v) const noexcept
template<>
struct hash<::hpx::threads::thread_id_ref>
```

Public Functions

```
inline std::size_t operator()(::hpx::threads::thread_id_ref const &v) const noexcept

namespace hpx

    namespace threads
```

Enums

```
enum class thread_id_addrf : std::uint8_t
    Values:

    enumerator yes

    enumerator no
```

Variables

```
constexpr const thread_id invalid_thread_id

struct keep_alive_thread_id
```

Public Functions

```
inline explicit keep_alive_thread_id(hpx::threads::thread_id id, bool const add_ref) noexcept
keep_alive_thread_id(keep_alive_thread_id const &other) = delete
keep_alive_thread_id(keep_alive_thread_id &&other) = default
keep_alive_thread_id &operator=(keep_alive_thread_id const &other) = delete
inline keep_alive_thread_id &operator=(keep_alive_thread_id &&other) noexcept
inline ~keep_alive_thread_id()
inline void detach_if_needed() noexcept
```

Public Members

bool **add_ref**

hpx::threads::thread_id_ref **id**

struct **thread_id**

Public Functions

constexpr **thread_id**() noexcept = default

thread_id(*thread_id* const&) noexcept = default

thread_id &**operator**=(*thread_id* const&) noexcept = default

~thread_id() = default

inline constexpr **thread_id**(*thread_id* &&rhs) noexcept

inline constexpr *thread_id* &**operator**=(*thread_id* &&rhs) noexcept

inline explicit constexpr **thread_id**(*thread_id_repr* const thrd) noexcept

inline constexpr *thread_id* &**operator**=(*thread_id_repr* const rhs) noexcept

inline explicit constexpr **operator bool**() const noexcept

inline constexpr *thread_id_repr* **get**() const noexcept

inline constexpr void **reset**() noexcept

Private Types

using **thread_id_repr** = void*

Private Members

thread_id_repr **thrd_** = nullptr

Friends

inline friend constexpr friend bool **operator==** (std::nullptr_t,
thread_id const &rhs) noexcept

inline friend constexpr friend bool **operator!=** (std::nullptr_t,
thread_id const &rhs) noexcept

```
inline friend constexpr friend bool operator== (thread_id const &lhs,
std::nullptr_t) noexcept
```

```
inline friend constexpr friend bool operator!= (thread_id const &lhs,
std::nullptr_t) noexcept
```

```
inline friend constexpr friend bool operator== (thread_id const &lhs,
thread_id const &rhs) noexcept
```

```
inline friend constexpr friend bool operator!= (thread_id const &lhs,
thread_id const &rhs) noexcept
```

```
inline friend constexpr friend bool operator< (thread_id const &lhs,
thread_id const &rhs) noexcept
```

```
inline friend constexpr friend bool operator> (thread_id const &lhs,
thread_id const &rhs) noexcept
```

```
inline friend constexpr friend bool operator<= (thread_id const &lhs,
thread_id const &rhs) noexcept
```

```
inline friend constexpr friend bool operator>= (thread_id const &lhs,
thread_id const &rhs) noexcept
```

```
friend std::ostream &operator<<(std::ostream &os, thread_id const &id)
```

```
friend void format_value(std::ostream &os, std::string_view spec, thread_id const &id)
```

```
struct thread_id_ref
```

Public Types

```
using thread_repr = detail::thread_data_reference_counting
```

Public Functions

```
thread_id_ref() noexcept = default
```

```
thread_id_ref(thread_id_ref const&) = default
```

```
thread_id_ref &operator=(thread_id_ref const&) = default
```

```
thread_id_ref(thread_id_ref &&rhs) noexcept = default
```

```
thread_id_ref &operator=(thread_id_ref &&rhs) noexcept = default
```

```
~thread_id_ref() = default
```

```
inline explicit thread_id_ref(thread_id_repr const &thrd) noexcept
inline explicit thread_id_ref(thread_id_repr &&thrd) noexcept
inline thread_id_ref &operator=(thread_id_repr const &rhs) noexcept
inline thread_id_ref &operator=(thread_id_repr &&rhs) noexcept
inline explicit thread_id_ref(thread_repr *thrd, thread_id_addr const &addr =
                               thread_id_addr::yes) noexcept

inline thread_id_ref &operator=(thread_repr *rhs) noexcept
inline thread_id_ref(thread_id const &noref)
inline thread_id_ref(thread_id &&noref, bool const add_ref = true) noexcept
inline thread_id_ref &operator=(thread_id const &noref)
inline thread_id_ref &operator=(thread_id &&noref) noexcept
inline explicit constexpr operator bool() const noexcept
inline constexpr thread_id noref() const noexcept
inline constexpr thread_id_repr &get() & noexcept
inline thread_id_repr &&get() && noexcept
inline constexpr thread_id_repr const &get() const & noexcept
inline void reset() noexcept
inline void reset(thread_repr *thrd, bool const add_ref = true) noexcept
inline constexpr thread_repr *detach() noexcept
```

Private Types

```
using thread_id_repr = hpx::intrusive_ptr<detail::thread_data_reference_counting>
```

Private Members

```
thread_id_repr thrd_
```

Friends

```
inline friend constexpr friend bool operator== (std::nullptr_t,
thread_id_ref const &rhs) noexcept
```

```
inline friend constexpr friend bool operator!= (std::nullptr_t,
thread_id_ref const &rhs) noexcept
```



```
inline friend constexpr friend bool operator== (thread_id_ref const &lhs,
std::nullptr_t) noexcept
```

```
inline friend constexpr friend bool operator!= (thread_id_ref const &lhs,
std::nullptr_t) noexcept
```

```
inline friend constexpr friend bool operator== (thread_id_ref const &lhs,
thread_id_ref const &rhs) noexcept
```

```
inline friend constexpr friend bool operator!= (thread_id_ref const &lhs,
thread_id_ref const &rhs) noexcept
```

```
inline friend constexpr friend bool operator< (thread_id_ref const &lhs,
thread_id_ref const &rhs) noexcept
```

```
inline friend constexpr friend bool operator> (thread_id_ref const &lhs,
thread_id_ref const &rhs) noexcept
```

```
inline friend constexpr friend bool operator<= (thread_id_ref const &lhs,
thread_id_ref const &rhs) noexcept
```

```
inline friend constexpr friend bool operator>= (thread_id_ref const &lhs,
thread_id_ref const &rhs) noexcept
```

```
friend std::ostream &operator<<(std::ostream &os, thread_id_ref const &id)
```

```
friend void format_value(std::ostream &os, std::string_view spec, thread_id_ref const &id)
```

namespace **std**

```
template<> thread_id >
```

Public Functions

```
inline std::size_t operator() (::hpx::threads::thread_id const &v) const noexcept
```

```
template<> thread_id_ref >
```

Public Functions

```
inline std::size_t operator() (::hpx::threads::thread_id_ref const &v) const noexcept
```

datastructures

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx::any_nonser. hpx::bad_any_cast

Defined in header [hpx/any.hpp](#)⁸²⁰.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- [hpx::unique_any_nonser](#)
- [hpx::any_cast](#)
- [hpx::make_any_nonser](#)
- [hpx::make_unique_any_nonser](#)

Warning: doxygenfunction: Cannot find function “hpx::any_nonser. hpx::bad_any_cast” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

hpx::unique_any_nonser

Defined in header [hpx/any.hpp](#)⁸²¹.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- [hpx::any_nonser. hpx::bad_any_cast](#)
- [hpx::any_cast](#)
- [hpx::make_any_nonser](#)
- [hpx::make_unique_any_nonser](#)

using `hpx::unique_any_nonser = util::basic_any<void, void, void, std::false_type>`

hpx::any_cast

Defined in header [hpx/any.hpp](#)⁸²².

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- [hpx::any_nonser. hpx::bad_any_cast](#)
- [hpx::unique_any_nonser](#)
- [hpx::make_any_nonser](#)

⁸²⁰ http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/any.hpp

⁸²¹ http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/any.hpp

⁸²² http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/any.hpp

- *hpx::make_unique_any_nonser*

```
template<typename T, typename IArch, typename OArch, typename Char, typename Copyable>
T *hpx::any_cast(util::basic_any<IArch, OArch, Char, Copyable> *operand) noexcept
```

Performs type-safe access to the contained object.

Parameters

operand – target any object

Returns

If operand is not a null pointer, and the *typeid* of the requested *T* matches that of the contents of *operand*, a pointer to the value contained by *operand*, otherwise a null pointer.

```
template<typename T, typename IArch, typename OArch, typename Char, typename Copyable>
T const *hpx::any_cast(util::basic_any<IArch, OArch, Char, Copyable> const *operand) noexcept
```

Performs type-safe access to the contained object.

Parameters

operand – target any object

Returns

If operand is not a null pointer, and the *typeid* of the requested *T* matches that of the contents of *operand*, a pointer to the value contained by *operand*, otherwise a null pointer.

```
template<typename T, typename IArch, typename OArch, typename Char, typename Copyable>
T hpx::any_cast(util::basic_any<IArch, OArch, Char, Copyable> &operand)
```

Performs type-safe access to the contained object. Let *U* be *std::remove_cv_t<std::remove_reference_t<T>>*. The program is ill-formed if *std::is_constructible_v<T, U>* is false.

Parameters

operand – target any object

Returns

*static_cast<T>(*std::any_cast<U>(&operand))*

```
template<typename T, typename IArch, typename OArch, typename Char, typename Copyable>
T const &hpx::any_cast(util::basic_any<IArch, OArch, Char, Copyable> const &operand)
```

Performs type-safe access to the contained object. Let *U* be *std::remove_cv_t<std::remove_reference_t<T>>*. The program is ill-formed if *std::is_constructible_v<T, const U>* is false.

Parameters

operand – target any object

Returns

*static_cast<T>(*std::any_cast<U>(&operand))*

hpx::make_any_nonser

Defined in header [hpx/any.hpp](#)⁸²³.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::any_nonser*, *hpx::bad_any_cast*
- *hpx::unique_any_nonser*
- *hpx::any_cast*

⁸²³ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/any.hpp

- `hpx::make_unique_any_nonser`

```
template<typename T, typename ...Ts>
```

```
util::basic_any<void, void, void, std::true_type> hpx::make_any_nonser(Ts&&... ts)
```

```
template<typename T, typename U, typename ...Ts>
```

```
util::basic_any<void, void, void, std::true_type> hpx::make_any_nonser(std::initializer_list<U> il, Ts&&... ts)
```

```
template<typename T>
```

```
util::basic_any<void, void, void, std::true_type> hpx::make_any_nonser(T &&t)
```

hpx::make_unique_any_nonser

Defined in header [hpx/any.hpp](#)⁸²⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::any_nonser`, `hpx::bad_any_cast`
- `hpx::unique_any_nonser`
- `hpx::any_cast`
- `hpx::make_any_nonser`

```
template<typename T, typename ...Ts>
```

```
util::basic_any<void, void, void, std::false_type> hpx::make_unique_any_nonser(Ts&&... ts)
```

```
template<typename T, typename U, typename ...Ts>
```

```
util::basic_any<void, void, void, std::false_type> hpx::make_unique_any_nonser(std::initializer_list<U> il,  
Ts&&... ts)
```

```
template<typename T>
```

```
util::basic_any<void, void, void, std::false_type> hpx::make_unique_any_nonser(T &&t)
```

hpx::ignore

Defined in header [hpx/tuple.hpp](#)⁸²⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::tuple`
- `hpx::tuple_size`
- `hpx::tuple_element`
- `hpx::make_tuple`
- `hpx::tie`
- `hpx::forward_as_tuple`
- `hpx::tuple_cat`

⁸²⁴ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/any.hpp

⁸²⁵ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/tuple.hpp

- `hpx::get`

constexpr `hpx::detail::ignore_type hpx::ignore = {}`

An object of unspecified type such that any value can be assigned to it with no effect. Intended for use with `hpx::tie` when unpacking a `hpx::tuple`, as a placeholder for the arguments that are not used. While the behavior of `hpx::ignore` outside of `hpx::tie` is not formally specified, some code guides recommend using `hpx::ignore` to avoid warnings from unused return values of `[[nodiscard]]` functions.

`hpx::tuple`

Defined in header `hpx/tuple.hpp`⁸²⁶.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ignore`
- `hpx::tuple_size`
- `hpx::tuple_element`
- `hpx::make_tuple`
- `hpx::tie`
- `hpx::forward_as_tuple`
- `hpx::tuple_cat`
- `hpx::get`

template<typename ...**Ts**>

class **tuple**

Class template `hpx::tuple` is a fixed-size collection of heterogeneous values. It is a generalization of `hpx::pair`. If `std::is_trivially_destructible<Ti>::value` is true for every `Ti` in `Ts`, the destructor of tuple is trivial.

Param Ts...

the types of the elements that the tuple stores.

`hpx::tuple_size`

Defined in header `hpx/tuple.hpp`⁸²⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::ignore`
- `hpx::tuple`
- `hpx::tuple_element`
- `hpx::make_tuple`
- `hpx::tie`

⁸²⁶ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/tuple.hpp

⁸²⁷ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/tuple.hpp

- `hpx::forward_as_tuple`
- `hpx::tuple_cat`
- `hpx::get`

template<typename **T**>

struct **tuple_size**

Provides access to the number of elements in a tuple-like type as a compile-time constant expression.

The primary template is not defined. An explicit (full) or partial specialization is required to make a type tuple-like.

`hpx::tuple_element`

Defined in header [hpx/tuple.hpp](#)⁸²⁸.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::ignore`
- `hpx::tuple`
- `hpx::tuple_size`
- `hpx::make_tuple`
- `hpx::tie`
- `hpx::forward_as_tuple`
- `hpx::tuple_cat`
- `hpx::get`

template<std::size_t **I**, typename **T**, typename **Enable** = void>

struct **tuple_element**

Provides compile-time indexed access to the types of the elements of a tuple-like type.

The primary template is not defined. An explicit (full) or partial specialization is required to make a type tuple-like.

`hpx::make_tuple`

Defined in header [hpx/tuple.hpp](#)⁸²⁹.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::ignore`
- `hpx::tuple`
- `hpx::tuple_size`
- `hpx::tuple_element`

⁸²⁸ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/tuple.hpp

⁸²⁹ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/tuple.hpp

- `hpx::tie`
- `hpx::forward_as_tuple`
- `hpx::tuple_cat`
- `hpx::get`

```
template<typename ...Ts>
constexpr tuple<util::decay_unwrap_t<Ts>...> hpx::make_tuple(Ts&&... ts)
```

Provides compile-time indexed access to the types of the elements of the tuple.

`hpx::tie`

Defined in header `hpx/tuple.hpp`⁸³⁰.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::ignore`
- `hpx::tuple`
- `hpx::tuple_size`
- `hpx::tuple_element`
- `hpx::make_tuple`
- `hpx::forward_as_tuple`
- `hpx::tuple_cat`
- `hpx::get`

```
template<typename ...Ts>
constexpr tuple<Ts&...> hpx::tie(Ts&... ts)
```

Creates a tuple of lvalue references to its arguments or instances of `hpx::ignore`.

Parameters

ts – zero or more lvalue arguments to construct the tuple from.

Returns

`hpx::tuple` object containing lvalue references.

`hpx::forward_as_tuple`

Defined in header `hpx/tuple.hpp`⁸³¹.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::ignore`
- `hpx::tuple`
- `hpx::tuple_size`
- `hpx::tuple_element`

⁸³⁰ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/tuple.hpp

⁸³¹ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/tuple.hpp

- *hpx::make_tuple*
- *hpx::tie*
- *hpx::tuple_cat*
- *hpx::get*

```
template<typename ...Ts>  
constexpr tuple<Ts&&...> hpx::forward_as_tuple(Ts&&... ts)
```

Constructs a tuple of references to the arguments in *args* suitable for forwarding as an argument to a function. The tuple has rvalue reference data members when rvalues are used as arguments, and otherwise has lvalue reference data members.

Parameters

ts – zero or more arguments to construct the tuple from

Returns

hpx::tuple object created as if by

```
hpx::tuple<Ts&&...>(HPX_FORWARD(Ts, ts)...) 
```

hpx::tuple_cat

Defined in header [hpx/tuple.hpp](#)⁸³².

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::ignore*
- *hpx::tuple*
- *hpx::tuple_size*
- *hpx::tuple_element*
- *hpx::make_tuple*
- *hpx::tie*
- *hpx::forward_as_tuple*
- *hpx::get*

```
template<typename ...Tuples>  
constexpr auto hpx::tuple_cat(Tuples&&... tuples)
```

Constructs a tuple that is a concatenation of all tuples in *tuples*. The behavior is undefined if any type in `std::decay_t<Tuples>...` is not a specialization of *hpx::tuple*. However, an implementation may choose to support types (such as `std::array` and `std::pair`) that follow the tuple-like protocol.

Parameters

tuples – zero or more tuples to concatenate

Returns

hpx::tuple object composed of all elements of all argument tuples constructed from *hpx::get<Is>(HPX_FORWARD(UTuple,t))* for each individual element.

⁸³² http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/tuple.hpp

hpx::get

Defined in header [hpx/tuple.hpp](#)⁸³³.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::ignore*
- *hpx::tuple*
- *hpx::tuple_size*
- *hpx::tuple_element*
- *hpx::make_tuple*
- *hpx::tie*
- *hpx::forward_as_tuple*
- *hpx::tuple_cat*

template<std::size_t I>

util::at_index<I, Ts...>::type &hpx::get() noexcept

Extracts the Ith element from the tuple. I must be an integer value in [0, sizeof...(Ts)).

template<std::size_t I>

util::at_index<I, Ts...>::type const &hpx::get() const noexcept

Extracts the Ith element from the tuple. I must be an integer value in [0, sizeof...(Ts)).

hpx::any

Defined in header [hpx/any.hpp](#)⁸³⁴.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::make_any*

using hpx::any = util::basic_any<serialization::input_archive, serialization::output_archive, char, std::true_type>

hpx::make_any

Defined in header [hpx/any.hpp](#)⁸³⁵.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::any*

template<typename T, typename Char>

⁸³³ http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/tuple.hpp

⁸³⁴ http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/any.hpp

⁸³⁵ http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/any.hpp

`util::basic_any<serialization::input_archive, serialization::output_archive, Char> hpx::make_any(T &&t)`

Constructs an any object containing an object of type *T*, passing the provided arguments to *T*'s constructor. Equivalent to:

```
return std::any(std::in_place_type<T>, std::forward<Args>(args)...);
```

debugging

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/debugging/print.hpp

Defined in header `hpx/debugging/print.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

errors

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx::error_code

Defined in header `hpx/system_error.hpp`⁸³⁶.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

class **error_code** : public *error_code*

A *hpx::error_code* represents an arbitrary error condition.

The class *hpx::error_code* describes an object used to hold error code values, such as those originating from the operating system or other low-level application program interfaces.

Note: Class *hpx::error_code* is an adjunct to error reporting by exception

hpx/errors/exception_list.hpp

Defined in header `hpx/errors/exception_list.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

⁸³⁶ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/system_error.hpp

class **exception_list** : public *hpx::exception*

#include <exception_list.hpp> The class *exception_list* is a container of *exception_ptr* objects parallel algorithms may use to communicate uncaught exceptions encountered during parallel execution to the caller of the algorithm

The type *exception_list::const_iterator* fulfills the requirements of a forward iterator.

Public Types

using **iterator** = *exception_list_type::const_iterator*
 bidirectional iterator

Public Functions

inline *std::size_t* **size()** const noexcept

The number of *exception_ptr* objects contained within the *exception_list*.

Note: Complexity: Constant time.

inline *exception_list_type::const_iterator* **begin()** const noexcept

An iterator referring to the first *exception_ptr* object contained within the *exception_list*.

inline *exception_list_type::const_iterator* **end()** const noexcept

An iterator which is the past-the-end value for the *exception_list*.

HPX_THROW_EXCEPTION

Defined in header *hpx/exception.hpp*⁸³⁷.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *HPX_THROW_BAD_ALLOC*
- *HPX_THROWS_IF*

Warning: doxygenfunction: Cannot find function “HPX_THROW_EXCEPTION” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

⁸³⁷ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/exception.hpp

HPX_THROW_BAD_ALLOC

Defined in header [hpx/exception.hpp](#)⁸³⁸.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *HPX_THROW_EXCEPTION*
- *HPX_THROWS_IF*

Warning: doxygenfunction: Cannot find function “HPX_THROW_BAD_ALLOC” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

HPX_THROWS_IF

Defined in header [hpx/exception.hpp](#)⁸³⁹.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *HPX_THROW_EXCEPTION*
- *HPX_THROW_BAD_ALLOC*

Warning: doxygenfunction: Cannot find function “HPX_THROWS_IF” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

[hpx/errors/exception_fwd.hpp](#)

Defined in header [hpx/errors/exception_fwd.hpp](#).

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Enums

enum class **throwmode** : *std::uint8_t*

Encode error category for new *error_code*.

Values:

enumerator **plain**

⁸³⁸ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/exception.hpp

⁸³⁹ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/exception.hpp

enumerator **rethrow**

enumerator **lightweight**

Functions

constexpr bool **operator&**(*throwmode* lhs, *throwmode* rhs) noexcept

Variables

error_code **throws**

Predefined *error_code* object used as “throw on error” tag.

The predefined *hpx::error_code* object *hpx::throws* is supplied for use as a “throw on error” tag.

Functions that specify an argument in the form ‘*error_code*& ec=throws’ (with appropriate namespace qualifiers), have the following error handling semantics:

If &ec != &throws and an error occurred: ec.value() returns the implementation specific error number for the particular error that occurred and ec.category() returns the error_category for ec.value().

If &ec != &throws and an error did not occur, ec.clear().

If an error occurs and &ec == &throws, the function throws an exception of type *hpx::exception* or of a type derived from it. The exception’s *get_errcode()* member function returns a reference to a *hpx::error_code* object with the behavior as specified above.

hpx/errors/macros.hpp

Defined in header *hpx/errors/macros.hpp*.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Defines

HPX_THROW_EXCEPTION(errcode, f, ...)

Throw a *hpx::exception* initialized from the given parameters.

The macro *HPX_THROW_EXCEPTION* can be used to throw a *hpx::exception*. The purpose of this macro is to prepend the source file name and line number of the position where the exception is thrown to the error message. Moreover, this associates additional diagnostic information with the exception, such as file name and line number, locality id and thread id, and stack backtrace from the point where the exception was thrown.

The parameter *errcode* holds the *hpx::error* code the new exception should encapsulate. The parameter *f* is expected to hold the name of the function exception is thrown from and the parameter *msg* holds the error message the new exception should encapsulate.

```
void raise_exception()
{
    // Throw a hpx::exception initialized from the given parameters.
    // Additionally associate with this exception some detailed
    // diagnostic information about the throw-site.
    HPX_THROW_EXCEPTION(hpx::error::no_success, "raise_exception",
        "simulated error");
}
```

Example:**HPX_THROW_BAD_ALLOC(f)**

Throw a `hpx::bad_alloc_exception` initialized from the given parameters.

The macro `HPX_THROW_BAD_ALLOC` can be used to throw a `hpx::exception`. The purpose of this macro is to prepend the source file name and line number of the position where the exception is thrown to the error message. Moreover, this associates additional diagnostic information with the exception, such as file name and line number, locality id and thread id, and stack backtrace from the point where the exception was thrown.

The parameter `errcode` holds the `hpx::error` code the new exception should encapsulate. The parameter `f` is expected to hold the name of the function exception is thrown from and the parameter `msg` holds the error message the new exception should encapsulate.

```
void raise_exception()
{
    // Throw a hpx::exception initialized from the given parameters.
    // Additionally associate with this exception some detailed
    // diagnostic information about the throw-site.
    HPX_THROW_BAD_ALLOC("raise_exception", "simulated error");
}
```

Example:**HPX_THROWS_IF(ec, errcode, f, ...)**

Either throw a `hpx::exception` or initialize `hpx::error_code` from the given parameters.

The macro `HPX_THROWS_IF` can be used to either throw a `hpx::exception` or to initialize a `hpx::error_code` from the given parameters. If `&ec == &hpx::throws`, the semantics of this macro are equivalent to `HPX_THROW_EXCEPTION`. If `&ec != &hpx::throws`, the `hpx::error_code` instance `ec` is initialized instead.

The parameter `errcode` holds the `hpx::error` code from which the new exception should be initialized. The parameter `f` is expected to hold the name of the function exception is thrown from and the parameter `msg` holds the error message the new exception should encapsulate.

HPX_THROWS_BAD_ALLOC_IF(ec, f)

Either throw a `hpx::bad_alloc_exception` or `hpx::error_code` to `out_of_memory`.

The macro `HPX_THROWS_BAD_ALLOC_IF` can be used to either throw a `hpx::bad_alloc_exception` or to initialize a `hpx::error_code` to `hpx::error::out_of_memory`. If `&ec == &hpx::throws`, the semantics of this macro are equivalent to `HPX_THROW_BAD_ALLOC`. If `&ec != &hpx::throws`, the `hpx::error_code` instance `ec` is initialized instead.

HPX_DEFINE_ERROR_INFO(NAME, TYPE)

hpx::exception

Defined in header [hpx/exception.hpp](#)⁸⁴⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

class **exception** : public system_error

A *hpx::exception* is the main exception type used by HPX to report errors.

The *hpx::exception* type is the main exception type used by HPX to report errors. Any exceptions thrown by functions in the HPX library are either of this type or of a type derived from it. This implies that it is always safe to use this type only in catch statements guarding HPX library calls.

Subclassed by *hpx::bad_alloc_exception*, *hpx::cuda::experimental::cuda_exception*, *hpx::exception_list*, *hpx::experimental::task_canceled_exception*, *hpx::mpi::experimental::mpi_exception*

hpx/errors/error.hpp

Defined in header [hpx/errors/error.hpp](#).

See *Public API* for a list of names and headers that are part of the public HPX API.

Defines

HPX_ERROR_UNSCOPED_ENUM_DEPRECATION_MSG

namespace **hpx**

Enums

enum class **error** : std::int16_t

Possible error conditions.

This enumeration lists all possible error conditions which can be reported from any of the API functions.

Values:

enumerator **success**

The operation was successful.

enumerator **no_success**

The operation did failed, but not in an unexpected manner.

enumerator **not_implemented**

The operation is not implemented.

⁸⁴⁰ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/exception.hpp

enumerator **out_of_memory**

The operation caused an out of memory condition.

enumerator **bad_action_code**

enumerator **bad_component_type**

The specified component type is not known or otherwise invalid.

enumerator **network_error**

A generic network error occurred.

enumerator **version_too_new**

The version of the network representation for this object is too new.

enumerator **version_too_old**

The version of the network representation for this object is too old.

enumerator **version_unknown**

The version of the network representation for this object is unknown.

enumerator **unknown_component_address**

enumerator **duplicate_component_address**

The given global id has already been registered.

enumerator **invalid_status**

The operation was executed in an invalid status.

enumerator **bad_parameter**

One of the supplied parameters is invalid.

enumerator **internal_server_error**

enumerator **service_unavailable**

enumerator **bad_request**

enumerator **repeated_request**

enumerator **lock_error**

enumerator **duplicate_console**

There is more than one console locality.

enumerator **no_registered_console**

There is no registered console locality available.

enumerator **startup_timed_out**

enumerator **uninitialized_value**

enumerator **bad_response_type**

enumerator **deadlock**

enumerator **assertion_failure**

enumerator **null_thread_id**

Attempt to invoke an API function from a non-HPX thread.

enumerator **invalid_data**

enumerator **yield_aborted**

The yield operation was aborted.

enumerator **dynamic_link_failure**

enumerator **commandline_option_error**

One of the options given on the command line is erroneous.

enumerator **serialization_error**

There was an error during serialization of this object.

enumerator **unhandled_exception**

An unhandled exception has been caught.

enumerator **kernel_error**

The OS kernel reported an error.

enumerator **broken_task**

The task associated with this future object is not available anymore.

enumerator **task_moved**

The task associated with this future object has been moved.

enumerator **task_already_started**

The task associated with this future object has already been started.

enumerator **future_already_retrieved**

The future object has already been retrieved.

enumerator **promise_already_satisfied**

The value for this future object has already been set.

enumerator **future_does_not_support_cancellation**

The future object does not support cancellation.

enumerator **future_can_not_be_cancelled**

The future can't be canceled at this time.

enumerator **no_state**

The future object has no valid shared state.

enumerator **broken_promise**

The promise has been deleted.

enumerator **thread_resource_error**

enumerator **future_cancelled**

enumerator **thread_cancelled**

enumerator **thread_not_interruptable**

enumerator **duplicate_component_id**

The component type has already been registered.

enumerator **unknown_error**

An unknown error occurred.

enumerator **bad_plugin_type**

The specified plugin type is not known or otherwise invalid.

enumerator **filesystem_error**

The specified file does not exist or other filesystem related error.

enumerator **bad_function_call**

equivalent of `std::bad_function_call`

enumerator **task_canceled_exception**

`parallel::task_canceled_exception`

enumerator **task_block_not_active**

task_region is not active

enumerator **out_of_range**

Equivalent to `std::out_of_range`.

enumerator **length_error**

Equivalent to `std::length_error`.

enumerator **migration_needs_retry**

migration failed because of global race, retry

enumerator **stale_state**

The queried state may be stale, e.g. because no state has been recorded yet or a preceding notification has not yet been delivered

enumerator **target_fenced**

A dispatch was deliberately and correctly refused because the target had already latched a terminal lifecycle event; this is a final rejection, not a staleness signal, and should not be retried

Functions

constexpr bool **operator==(int lhs, error rhs)** noexcept

constexpr bool **operator==(error lhs, int rhs)** noexcept

constexpr bool **operator!=(int lhs, error rhs)** noexcept

constexpr bool **operator!=(error lhs, int rhs)** noexcept

constexpr bool **operator<(int lhs, error rhs)** noexcept

constexpr bool **operator>=(int lhs, error rhs)** noexcept

constexpr int **operator&(error lhs, error rhs)** noexcept

constexpr int **operator&(int lhs, error rhs)** noexcept

constexpr int **operator|=(int &lhs, error rhs)** noexcept

char const ***get_error_name(error e)** noexcept

Variables

constexpr error **success** = *error::success*

constexpr error **no_success** = *error::no_success*

```
constexpr error not_implemented = error::not_implemented

constexpr error out_of_memory = error::out_of_memory

constexpr error bad_action_code = error::bad_action_code

constexpr error bad_component_type = error::bad_component_type

constexpr error network_error = error::network_error

constexpr error version_too_new = error::version_too_new

constexpr error version_too_old = error::version_too_old

constexpr error version_unknown = error::version_unknown

constexpr error unknown_component_address = error::unknown_component_address

constexpr error duplicate_component_address = error::duplicate_component_address

constexpr error invalid_status = error::invalid_status

constexpr error bad_parameter = error::bad_parameter

constexpr error internal_server_error = error::internal_server_error

constexpr error service_unavailable = error::service_unavailable

constexpr error bad_request = error::bad_request

constexpr error repeated_request = error::repeated_request

constexpr error lock_error = error::lock_error

constexpr error duplicate_console = error::duplicate_console

constexpr error no_registered_console = error::no_registered_console

constexpr error startup_timed_out = error::startup_timed_out

constexpr error uninitialized_value = error::uninitialized_value
```

```
constexpr error bad_response_type = error::bad_response_type

constexpr error deadlock = error::deadlock

constexpr error assertion_failure = error::assertion_failure

constexpr error null_thread_id = error::null_thread_id

constexpr error invalid_data = error::invalid_data

constexpr error yield_aborted = error::yield_aborted

constexpr error dynamic_link_failure = error::dynamic_link_failure

constexpr error commandline_option_error = error::commandline_option_error

constexpr error serialization_error = error::serialization_error

constexpr error unhandled_exception = error::unhandled_exception

constexpr error kernel_error = error::kernel_error

constexpr error broken_task = error::broken_task

constexpr error task_moved = error::task_moved

constexpr error task_already_started = error::task_already_started

constexpr error future_already_retrieved = error::future_already_retrieved

constexpr error promise_already_satisfied = error::promise_already_satisfied

constexpr error future_does_not_support_cancellation =
error::future_does_not_support_cancellation

constexpr error future_can_not_be_cancelled = error::future_can_not_be_cancelled

constexpr error no_state = error::no_state

constexpr error broken_promise = error::broken_promise

constexpr error thread_resource_error = error::thread_resource_error
```

```
constexpr error future_cancelled = error::future_cancelled

constexpr error thread_cancelled = error::thread_cancelled

constexpr error thread_not_interruptable = error::thread_not_interruptable

constexpr error duplicate_component_id = error::duplicate_component_id

constexpr error unknown_error = error::unknown_error

constexpr error bad_plugin_type = error::bad_plugin_type

constexpr error filesystem_error = error::filesystem_error

constexpr error bad_function_call = error::bad_function_call

constexpr error task_canceled_exception = error::task_canceled_exception

constexpr error task_block_not_active = error::task_block_not_active

constexpr error out_of_range = error::out_of_range

constexpr error length_error = error::length_error

constexpr error migration_needs_retry = error::migration_needs_retry
```

hpx/errors/define_error_info.hpp

Defined in header `hpx/errors/define_error_info.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

execution

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/execution/traits/is_execution_policy.hpp

Defined in header `hpx/execution/traits/is_execution_policy.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Variables

```
template<typename T>
```

```
constexpr bool is_execution_policy_v = is_execution_policy<T>::value
```

```
template<typename T> concept execution_policy    = is_execution_policy<T>::value
```

```
template<typename T>
```

```
constexpr bool is_parallel_execution_policy_v = is_parallel_execution_policy<T>::value
```

```
template<typename T>
```

```
constexpr bool is_sequenced_execution_policy_v = is_sequenced_execution_policy<T>::value
```

```
template<typename T>
```

```
constexpr bool is_async_execution_policy_v = is_async_execution_policy<T>::value
```

```
template<typename T>
```

```
struct is_async_execution_policy : public hpx::detail::is_async_execution_policy<std::decay_t<T>>
```

```
    #include <is_execution_policy.hpp> Extension: Detect whether given execution policy makes algorithms asynchronous
```

- i. The type *is_async_execution_policy* can be used to detect asynchronous execution policies for the purpose of excluding function signatures from otherwise ambiguous overload resolution participation.
- ii. If T is the type of the standard or implementation-defined execution policy, *is_async_execution_policy*<T> shall be publicly derived from *integral_constant*<bool, true>, otherwise from *integral_constant*<bool, false>.
- iii. The behavior of a program that adds specializations for *is_async_execution_policy* is undefined.

```
template<typename T>
```

```
struct is_execution_policy : public hpx::detail::is_execution_policy<std::decay_t<T>>
```

```
    #include <is_execution_policy.hpp>
```

- i. The type *is_execution_policy* can be used to detect execution policies for the purpose of excluding function signatures from otherwise ambiguous overload resolution participation.
- ii. If T is the type of the standard or implementation-defined execution policy, *is_execution_policy*<T> shall be publicly derived from *integral_constant*<bool, true>, otherwise from *integral_constant*<bool, false>.
- iii. The behavior of a program that adds specializations for *is_execution_policy* is undefined.

```
template<typename T>
struct is_parallel_execution_policy : public
hpx::detail::is_parallel_execution_policy<std::decay_t<T>>
    #include <is_execution_policy.hpp> Extension: Detect whether given execution policy enables paralleliza-
    tion
```

- i. The type *is_parallel_execution_policy* can be used to detect parallel execution policies for the purpose of excluding function signatures from otherwise ambiguous overload resolution participation.
- ii. If *T* is the type of the standard or implementation-defined execution policy, *is_parallel_execution_policy*<*T*> shall be publicly derived from *integral_constant*<bool, true>, otherwise from *integral_constant*<bool, false>.
- iii. The behavior of a program that adds specializations for *is_parallel_execution_policy* is undefined.

```
template<typename T>
struct is_sequenced_execution_policy : public
hpx::detail::is_sequenced_execution_policy<std::decay_t<T>>
    #include <is_execution_policy.hpp> Extension: Detect whether given execution policy does not enable
    parallelization
```

- i. The type *is_sequenced_execution_policy* can be used to detect non-parallel execution policies for the purpose of excluding function signatures from otherwise ambiguous overload resolution participation.
- ii. If *T* is the type of the standard or implementation-defined execution policy, *is_sequenced_execution_policy*<*T*> shall be publicly derived from *integral_constant*<bool, true>, otherwise from *integral_constant*<bool, false>.
- iii. The behavior of a program that adds specializations for *is_sequenced_execution_policy* is undefined.

hpx/execution/executors/default_parameters.hpp

Defined in header `hpx/execution/executors/default_parameters.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **execution**

namespace **experimental**

struct **default_parameters**

#include <default_parameters.hpp> Loop iterations are divided into pieces of size *chunk_size* and then assigned to threads. If *chunk_size* is not specified, the iterations are evenly (if possible) divided contiguously among the threads.

Note: This executor parameters type is equivalent to OpenMP's *STATIC* scheduling directive.

Public Functions

default_parameters() = default

Construct a default_parameters executor parameters object

Note: By default, the number of loop iterations is determined from the number of available cores and the overall number of loop iterations to schedule.

hpx/execution/executors/execution_parameters.hpp

Defined in header hpx/execution/executors/execution_parameters.hpp.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **execution**

namespace **experimental**

Functions

```
template<typename ...Params>
constexpr executor_parameters_join<Params...>::type join_executor_parameters(Params&&...
                                                                    params)
```

```
template<typename Param>
constexpr Param &&join_executor_parameters(Param &&param) noexcept
```

```
template<typename ...Params>
struct executor_parameters_join
```

Public Types

```
using type = detail::executor_parameters<std::decay_t<Params>...>
```

```
template<typename Param>
struct executor_parameters_join<Param>
```

Public Types

using **type** = *Param*

hpx::execution::experimental::guided_chunk_size

Defined in header [hpx/execution.hpp](#)⁸⁴¹.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

struct **guided_chunk_size**

Iterations are dynamically assigned to threads in blocks as threads request those until no blocks remain to be assigned. Similar to `dynamic_chunk_size` except that the block size decreases each time a number of loop iterations is given to a thread. The size of the initial block is proportional to $\text{number_of_iterations} / \text{number_of_cores}$. Subsequent blocks are proportional to $\text{number_of_iterations_remaining} / \text{number_of_cores}$. The optional chunk size parameter defines the minimum block size. The default chunk size is 1.

Note: This executor parameters type is equivalent to OpenMP's GUIDED scheduling directive.

hpx::execution::experimental::persistent_auto_chunk_size

Defined in header [hpx/execution.hpp](#)⁸⁴².

See *Public API* for a list of names and headers that are part of the public *HPX* API.

struct **persistent_auto_chunk_size**

Loop iterations are divided into pieces and then assigned to threads. The number of loop iterations combined is determined based on measurements of how long the execution of 1% of the overall number of iterations takes. This executor parameters type makes sure that as many loop iterations are combined as necessary to run for the amount of time specified.

hpx/execution/executors/polymorphic_executor.hpp

Defined in header `hpx/execution/executors/polymorphic_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **execution**

namespace **experimental**

⁸⁴¹ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/execution.

hpp

⁸⁴² http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/execution.

hpp

namespace **parallel**

namespace **execution**

template<typename **Sig**>

class **polymorphic_executor**

template<typename **R**, typename ...**Ts**>

class **polymorphic_executor**<*R(Ts...)*> : private

hpx::parallel::execution::detail::polymorphic_executor_base

Public Types

template<typename>

using **future_type** = *hpx::future<R>*

Public Functions

inline constexpr **polymorphic_executor**() noexcept

inline **polymorphic_executor**(*polymorphic_executor* const &other)

inline **polymorphic_executor**(*polymorphic_executor* &&other) noexcept

inline *polymorphic_executor* &**operator**=(*polymorphic_executor* const &other)

inline *polymorphic_executor* &**operator**=(*polymorphic_executor* &&other) noexcept

template<typename **Exec**, typename **PE** = *std::decay_t<Exec>*, typename **Enable** =
std::enable_if_t<!std::is_same_v<PE, polymorphic_executor>>>

inline **polymorphic_executor**(*Exec* &&exec)

template<typename **Exec**, typename **PE** = *std::decay_t<Exec>*, typename **Enable** =
std::enable_if_t<!std::is_same_v<PE, polymorphic_executor>>>

inline *polymorphic_executor* &**operator**=(*Exec* &&exec)

inline void **reset**() noexcept

template<typename **F**>

inline void **post**(*F* &&f, *Ts... ts*) const

template<typename **F**>

inline *R* **sync_execute**(*F* &&f, *Ts... ts*) const

template<typename **F**>

inline *hpx::future<R>* **async_execute**(*F* &&f, *Ts... ts*) const

template<typename **F**, typename **Future**>

inline *hpx::future<R>* **then_execute**(*F* &&f, *Future* &&predecessor, *Ts&&... ts*) const

template<typename **F**, typename **Shape**>

```
inline std::vector<R> bulk_sync_execute(F &&f, Shape const &s, Ts&&... ts) const

template<typename F, typename Shape>
inline hpx::future<std::vector<R>> bulk_async_execute(F &&f, Shape const &s, Ts&&...
                                                    ts) const

template<typename F, typename Shape>
inline hpx::future<std::vector<R>> bulk_then_execute(F &&f, Shape const &s,
                                                    hpx::shared_future<void> const
                                                    &predecessor, Ts&&... ts) const
```

Private Types

```
using base_type = detail::polymorphic_executor_base

using vtable = detail::polymorphic_executor_vtable<R(Ts...)>
```

Private Functions

```
inline void assign(std::nullptr_t) noexcept

template<typename Exec>
inline void assign(Exec &&exec)
```

Private Static Functions

```
static inline constexpr vtable const *get_empty_vtable() noexcept

template<typename T>
static inline constexpr vtable const *get_vtable() noexcept
```

hpx::execution::experimental::collect_chunking_parameters

Defined in header [hpx/execution.hpp](#)⁸⁴³.

See *Public API* for a list of names and headers that are part of the public HPX API.

struct **collect_chunking_parameters**

Collect various parameters used for running a parallel algorithm.

⁸⁴³ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/execution.hpp

hpx/execution/executors/execution_information.hpp

Defined in header `hpx/execution/executors/execution_information.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **execution**

namespace **experimental**

Variables

constexpr struct `hpx::execution::experimental::has_pending_closures_t` **has_pending_closures**

constexpr struct `hpx::execution::experimental::get_pu_mask_t` **get_pu_mask**

constexpr struct `hpx::execution::experimental::set_scheduler_mode_t` **set_scheduler_mode**

struct **get_pu_mask_t**

#include <execution_information.hpp> Retrieve the bitmask describing the processing units the given thread is allowed to run on

All `threads::executors` invoke `sched.get_pu_mask()`.

Note: If the executor does not support this operation, this call will always invoke `hpx::threads::get_pu_mask()`

Param exec

[in] The executor object to use for querying the number of pending tasks.

Param topo

[in] The topology object to use to extract the requested information.

Param threan_num

[in] The sequence number of the thread to retrieve information for.

Public Functions

template<typename **Executor**>

inline decltype(auto) **operator()** (*Executor* &&exec, *threads::topology* &topo, *std::size_t* thread_num) const

template<typename **Executor**>

inline decltype(auto) **operator()** (*Executor*&&, *threads::topology* &topo, *std::size_t* thread_num) const

struct **has_pending_closures_t**

#include <execution_information.hpp> Retrieve whether this executor has operations pending or not.

Note: If the executor does not expose this information, this call will always return *false*

Param exec

[in] The executor object to use to extract the requested information for.

Public Functions

template<typename **Executor**>
inline decltype(auto) **operator()** (*Executor* &&exec) const

template<typename **Executor**>
inline decltype(auto) **operator()** (*Executor*&&) const

struct **set_scheduler_mode_t**

#include <execution_information.hpp> Set various modes of operation on the scheduler underneath the given executor.

Note: This calls `exec.set_scheduler_mode(mode)` if it exists; otherwise it does nothing.

Param exec

[in] The executor object to use.

Param mode

[in] The new mode for the scheduler to pick up

Public Functions

template<typename **Executor**, typename **Mode**>
inline void **operator()** (*Executor* &&exec, *Mode* const &mode) const

template<typename **Executor**, typename **Mode**>
inline void **operator()** (*Executor*&&, *Mode* const&) const

hpx::execution::experimental::auto_chunk_size

Defined in header `hpx/execution.hpp`⁸⁴⁴.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

struct **auto_chunk_size**

Loop iterations are divided into pieces and then assigned to threads. The number of loop iterations combined is determined based on measurements of how long the execution of 1% of the overall number of iterations takes.

⁸⁴⁴ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/execution.hpp

This executor parameters type makes sure that as many loop iterations are combined as necessary to run for the amount of time specified.

hpx/execution/executors/adaptive_static_chunk_size.hpp

Defined in header `hpx/execution/executors/adaptive_static_chunk_size.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **execution**

namespace **experimental**

struct **adaptive_static_chunk_size**

#include <adaptive_static_chunk_size.hpp> Loop iterations are divided into pieces of size *chunk_size* and then assigned to threads. If *chunk_size* is not specified, the iterations are evenly (if possible) divided contiguously among the threads.

Note: This executor parameters type is equivalent to OpenMP's STATIC scheduling directive.

Public Functions

adaptive_static_chunk_size() = default

Construct a `adaptive_static_chunk_size` executor parameters object

Note: By default the number of loop iterations is determined from the number of available cores and the overall number of loop iterations to schedule.

inline explicit constexpr **adaptive_static_chunk_size**(*std::size_t* const *chunk_size*)
noexcept

Construct a `adaptive_static_chunk_size` executor parameters object

Parameters

chunk_size – [in] The optional chunk size to use as the number of loop iterations to run on a single thread.

hpx::execution::experimental::dynamic_chunk_size

Defined in header `hpx/execution.hpp`⁸⁴⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

⁸⁴⁵ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/execution.hpp

struct **dynamic_chunk_size**

Loop iterations are divided into pieces of size *chunk_size* and then dynamically scheduled among the threads; when a thread finishes one chunk, it is dynamically assigned another. If *chunk_size* is not specified, the default chunk size is 1.

Note: This executor parameters type is equivalent to OpenMP's DYNAMIC scheduling directive.

hpx/execution/executors/rebind_executor.hpp

Defined in header `hpx/execution/executors/rebind_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **execution**

namespace **experimental**

Typedefs

```
template<typename ExPolicy, typename Executor, typename Parameters>
using rebind_executor_t = typename rebind_executor<ExPolicy, Executor, Parameters>::type
```

Variables

```
constexpr struct hpx::execution::experimental::create_rebound_policy_t create_rebound_policy
```

```
struct create_rebound_policy_t
```

Public Functions

```
template<typename ExPolicy, typename Executor, typename Parameters>
inline constexpr decltype(auto) operator() (ExPolicy&&, Executor &&exec, Parameters
&&parameters) const
```

```
template<typename ExPolicy, typename Executor>
inline constexpr decltype(auto) operator() (ExPolicy &&policy, Executor &&exec) const
```

```
template<typename ExPolicy, typename Parameters>
inline constexpr decltype(auto) operator() (ExPolicy &&policy, Parameters &&parameters)
const
```

```
template<typename ExPolicy, typename Executor, typename Parameters>
```


struct **rebind_executor**

#include <rebind_executor.hpp> Rebind the type of executor used by an execution policy. The execution category of Executor shall not be weaker than that of ExecutionPolicy.

Public Types

using **type** = typename policy_type::template rebind<executor_type, parameters_type>::type
The type of the rebound execution policy.

hpx/execution/executors/execution_parameters_fwd.hpp

Defined in header hpx/execution/executors/execution_parameters_fwd.hpp.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **execution**

namespace **experimental**

Variables

constexpr struct *hpx::execution::experimental::null_parameters_t* **null_parameters**

constexpr struct *hpx::execution::experimental::get_chunk_size_t* **get_chunk_size**

constexpr struct *hpx::execution::experimental::measure_iteration_t* **measure_iteration**

constexpr struct *hpx::execution::experimental::maximal_number_of_chunks_t*
maximal_number_of_chunks

constexpr struct *hpx::execution::experimental::reset_thread_distribution_t*
reset_thread_distribution

constexpr struct *hpx::execution::experimental::processing_units_count_t*
processing_units_count

constexpr struct *hpx::execution::experimental::with_processing_units_count_t*
with_processing_units_count

constexpr struct *hpx::execution::experimental::mark_begin_execution_t* **mark_begin_execution**

```
constexpr struct hpx::execution::experimental::mark_end_of_scheduling_t  
mark_end_of_scheduling
```

```
constexpr struct hpx::execution::experimental::mark_end_execution_t mark_end_execution
```

```
constexpr struct hpx::execution::experimental::mark_partition_t mark_partition
```

```
constexpr struct hpx::execution::experimental::collect_execution_parameters_t  
collect_execution_parameters
```

```
struct collect_execution_parameters_t
```

```
#include <execution_parameters_fwd.hpp> Collect various parameters of the chunking for this  
parallel algorithm execution
```

Note: This calls `params.mark_begin_execution(exec)` if it exists; otherwise it does nothing.

Param params

[in] The executor parameters object to use as a fallback if the executor does not expose

Param exec

[in] The executor object which will be used for scheduling of the loop iterations.

Param num_elements

[in] The overall number of elements for the algorithm.

Param num_cores

[in] The overall number of cores to utilize for the algorithm.

Param num_chunks

[in] The overall number of chunks for the algorithm.

Param chunk_size

[in] The size of the chunks created for the algorithm.

Public Functions

```
template<typename Parameters, typename Executor>  
inline decltype(auto) operator()(Parameters &&params, Executor &&exec, std::size_t  
                                num_elements, std::size_t num_cores, std::size_t  
                                num_chunks, std::size_t chunk_size) const
```

```
struct get_chunk_size_t
```

```
#include <execution_parameters_fwd.hpp> Return the number of invocations of the given func-  
tion f which should be combined into a single task
```

Param params

[in] The executor parameters object to use for determining the chunk size for the given
number of tasks *num_tasks*.

Param exec

[in] The executor object which will be used for scheduling of the loop iterations.

Param iteration_duration

[in] The time one of the tasks require to be executed.

Param cores

[in] The number of cores the number of chunks should be determined for.

Param num_tasks

[in] The number of tasks the chunk size should be determined for

Return

The size of the chunks (number of iterations per chunk) that should be used for parallel execution.

Public Functions

```
template<typename Parameters, typename Executor>
inline decltype(auto) operator()(Parameters &&params, Executor &&exec,
                                   hpx::chrono::steady_duration const &iteration_duration,
                                   std::size_t cores, std::size_t num_tasks) const

template<typename Parameters, typename Executor>
inline decltype(auto) operator()(Parameters &&params, Executor &&exec, std::size_t cores,
                                   std::size_t num_tasks) const
```

```
template<>
```

```
struct is_scheduling_property<with_processing_units_count_t> : public true_type
```

```
struct mark_begin_execution_t
```

```
#include <execution_parameters_fwd.hpp> Mark the begin of a parallel algorithm execution
```

Note: This calls `params.mark_begin_execution(exec)` if it exists; otherwise it does nothing.

Param params

[in] The executor parameters object to use as a fallback if the executor does not expose

Param exec

[in] The executor object which will be used for scheduling of the loop iterations.

Public Functions

```
template<typename Parameters, typename Executor>
inline decltype(auto) operator()(Parameters &&params, Executor &&exec) const
```

```
struct mark_end_execution_t
```

```
#include <execution_parameters_fwd.hpp> Mark the end of a parallel algorithm execution
```

Note: This calls `params.mark_end_execution(exec)` if it exists; otherwise it does nothing.

Param params

[in] The executor parameters object to use as a fallback if the executor does not expose

Param exec

[in] The executor object which will be used for scheduling of the loop iterations.

Public Functions

```
template<typename Parameters, typename Executor>  
inline decltype(auto) operator() (Parameters &&params, Executor &&exec) const
```

```
struct mark_end_of_scheduling_t
```

```
#include <execution_parameters_fwd.hpp> Mark the end of scheduling tasks during parallel algorithm execution
```

Note: This calls `params.mark_begin_execution(exec)` if it exists; otherwise it does nothing.

Param params

[in] The executor parameters object to use as a fallback if the executor does not expose

Param exec

[in] The executor object which will be used for scheduling of the loop iterations.

Public Functions

```
template<typename Parameters, typename Executor>  
inline decltype(auto) operator() (Parameters &&params, Executor &&exec) const
```

```
struct mark_partition_t
```

```
#include <execution_parameters_fwd.hpp> Mark custom point in parallel algorithm execution
```

Note: This calls `params.mark_partition(exec, partition, args...)` if it exists; otherwise it does nothing.

Param params

[in] The executor parameters object to use as a fallback if the executor does not expose

Param exec

[in] The executor object which will be used for scheduling of the loop iterations.

Public Functions

```
template<typename Parameters, typename Executor, typename ...Args>  
inline decltype(auto) operator() (Parameters &&params, Executor &&exec, std::size_t  
partition, Args&&... args) const
```

```
struct maximal_number_of_chunks_t
```

```
#include <execution_parameters_fwd.hpp> Return the largest reasonable number of chunks to create for a single algorithm invocation.
```

Param params

[in] The executor parameters object to use for determining the number of chunks for the given number of *cores*.

Param exec

[in] The executor object which will be used for scheduling of the loop iterations.

Param cores

[in] The number of cores the number of chunks should be determined for.

Param num_tasks

[in] The number of tasks the chunk size should be determined for

Public Functions

```
template<typename Parameters, typename Executor>
inline decltype(auto) operator() (Parameters &&params, Executor &&exec, std::size_t cores,
                                   std::size_t num_tasks) const
```

```
struct measure_iteration_t
```

#include <execution_parameters_fwd.hpp> Return the measured execution time for one iteration based on running the given function.

Note: The parameter *f* is expected to be a nullary function returning a `std::size_t` representing the number of iteration the function has already executed (i.e. which don't have to be scheduled anymore).

Param params

[in] The executor parameters object to use for determining the chunk size for the given number of tasks *num_tasks*.

Param exec

[in] The executor object which will be used for scheduling of the loop iterations.

Param f

[in] The function which will be optionally scheduled using the given executor.

Param num_tasks

[in] The number of tasks the chunk size should be determined for

Return

The execution time for one of the tasks.

Public Functions

```
template<typename Parameters, typename Executor, typename F>
inline decltype(auto) operator() (Parameters &&params, Executor &&exec, F &&f,
                                   std::size_t num_tasks) const
```

```
struct null_parameters_t
```

```
struct processing_units_count_t
```

#include <execution_parameters_fwd.hpp> Retrieve the number of (kernel-)threads used by the associated executor.

Note: This calls `params.processing_units_count(Executor&&)` if it exists; otherwise it forwards the request to the executor parameters object.

Param params

[in] The executor parameters object to use as a fallback if the executor does not expose

Param exec

[in] The executor object which will be used for scheduling of the loop iterations.

Param iteration_duration

[in] The time one of the tasks require to be executed.

Param num_tasks

[in] The number of tasks the number of cores should be determined for

Return

The number of cores to use

Public Functions

```
template<typename Parameters, typename Executor>
inline decltype(auto) operator() (Parameters &&params, Executor &&exec,
                                   hpx::chrono::steady_duration const &iteration_duration,
                                   std::size_t num_tasks) const
```

```
template<typename Parameters, typename Executor>
inline decltype(auto) operator() (Parameters &&params, Executor &&exec, std::size_t
                                   num_tasks) const
```

```
template<typename Executor>
inline decltype(auto) operator() (Executor &&exec, hpx::chrono::steady_duration const
                                   &iteration_duration, std::size_t num_tasks) const
```

```
template<typename Executor>
inline decltype(auto) operator() (Executor &&exec, std::size_t num_tasks) const
```

```
template<typename Parameters, typename Executor>
inline decltype(auto) operator() (Parameters &&params, Executor &&exec) const
```

```
template<typename Executor>
inline decltype(auto) operator() (Executor &&exec) const
```

```
template<typename Parameters, typename Executor>
inline decltype(auto) operator() (Parameters &&params, Executor &&exec,
                                   hpx::chrono::steady_duration const &iteration_duration,
                                   std::size_t num_tasks) const
```

```
template<typename Parameters, typename Executor>
inline decltype(auto) operator() (Parameters &&params, Executor &&exec, std::size_t
                                   num_tasks = 0) const
```

```
template<typename Executor>
inline decltype(auto) operator() (Executor &&exec, hpx::chrono::steady_duration const
                                   &iteration_duration, std::size_t num_tasks) const
```

```
template<typename Executor>
inline decltype(auto) operator() (Executor &&exec, std::size_t num_tasks = 0) const
```

```
template<typename ExPolicy>
inline decltype(auto) operator() (ExPolicy &&policy) const
```

```
struct reset_thread_distribution_t
```

```
#include <execution_parameters_fwd.hpp> Reset the internal round robin thread distribution
scheme for the given executor.
```

Note: This calls `params.reset_thread_distribution(exec)` if it exists; otherwise it does nothing.

Param params

[in] The executor parameters object to use for resetting the thread distribution scheme.

Param exec

[in] The executor object to use.

Public Functions

```
template<typename Parameters, typename Executor>
inline decltype(auto) operator() (Parameters &&params, Executor &&exec) const
```

```
struct with_processing_units_count_t
```

#include <execution_parameters_fwd.hpp> Generate a policy that supports setting the number of cores for execution.

Public Functions

```
template<typename Executor>
inline decltype(auto) operator() (Executor &&exec, std::size_t num_cores) const
```

```
template<typename Policy, typename Property>
inline decltype(auto) operator() (Policy &&policy, Property &&property) const
```

hpx::execution::experimental::static_chunk_size

Defined in header `hpx/execution.hpp`⁸⁴⁶.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
struct static_chunk_size
```

Loop iterations are divided into pieces of size *chunk_size* and then assigned to threads. If *chunk_size* is not specified, the iterations are evenly (if possible) divided contiguously among the threads.

Note: This executor parameters type is equivalent to OpenMP's *STATIC* scheduling directive.

⁸⁴⁶ http://github.com/TheHPXProject/hpx/blob/d402eed7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/execution.hpp

hpx/execution/executors/execution.hpp

Defined in header `hpx/execution/executors/execution.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **execution**

hpx::execution::experimental::num_cores

Defined in header `hpx/execution.hpp`⁸⁴⁷.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

struct **num_cores**

Control number of cores in executors which need a functionality for setting the number of cores to be used by an algorithm directly

hpx::execution::experimental::max_num_chunks

Defined in header `hpx/execution.hpp`⁸⁴⁸.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

struct **max_num_chunks**

Loop iterations are divided into not more than *num_chunks* partitions that are assigned to threads. If *num_chunks* is not specified, the number of chunks is determined based on the number of available cores.

hpx/execution/algorithms/sender_future.hpp

Defined in header `hpx/execution/algorithms/sender_future.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Converts any P2300 `std::exec` sender into an `hpx::future<T>`.

Usage:

```
// Explicit T:
auto sender = std::exec::just(42) | std::exec::then([](int x){ return x; });
hpx::future<int> f =
    hpx::execution::experimental::as_future<int>(std::move(sender));
int val = f.get();    // == 42
```

⁸⁴⁷ http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/execution.hpp

⁸⁴⁸ http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/execution.hpp

namespace **hpx**

namespace **execution**

namespace **experimental**

Variables

```
template<typename T>
constexpr as_future_t<T> as_future = {}

template<typename T>
struct as_future_t
```

Public Functions

```
template<typename Sender, typename =
    std::enable_if_t<hpx::execution::experimental::is_sender_v<std::decay_t<Sender>>>>
inline constexpr hpx::future<T> operator()(Sender &&sender) const
```

hpx/execution/algorithms/future_sender.hpp

Defined in header `hpx/execution/algorithms/future_sender.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Implementation details for adapting HPX futures as P2300 senders.

Usage:

```
hpx::future<int> f = hpx::async([]{ return 42; });
auto result = hpx::execution::experimental::as_sender(std::move(f))
    | hpx::execution::experimental::then([](int x){ return x * 2; })
    | hpx::execution::experimental::sync_wait();
```

namespace **hpx**

namespace **execution**

namespace **experimental**

execution_base

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/execution_base/execution.hpp

Defined in header `hpx/execution_base/execution.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **execution**

struct **parallel_execution_tag**

#include <execution.hpp> Function invocations executed by a group of parallel execution agents execute in unordered fashion. Any such invocations executing in the same thread are indeterminately sequenced with respect to each other.

Note: `parallel_execution_tag` is weaker than `sequenced_execution_tag`.

struct **sequenced_execution_tag**

#include <execution.hpp> Function invocations executed by a group of sequential execution agents execute in sequential order.

struct **unsequenced_execution_tag**

#include <execution.hpp> Function invocations executed by a group of vector execution agents are permitted to execute in unordered fashion when executed in different threads, and un-sequenced with respect to one another when executed in the same thread.

Note: `unsequenced_execution_tag` is weaker than `parallel_execution_tag`.

namespace **parallel**

namespace **execution**

Variables

constexpr struct `hpx::parallel::execution::sync_execute_t` **sync_execute**

constexpr struct `hpx::parallel::execution::async_execute_t` **async_execute**

constexpr struct `hpx::parallel::execution::then_execute_t` **then_execute**

```
constexpr struct hpx::parallel::execution::post_t post
```

```
constexpr struct hpx::parallel::execution::bulk_sync_execute_t bulk_sync_execute
```

```
constexpr struct hpx::parallel::execution::bulk_async_execute_t bulk_async_execute
```

```
constexpr struct hpx::parallel::execution::bulk_then_execute_t bulk_then_execute
```

```
constexpr struct hpx::parallel::execution::async_invoke_t async_invoke
```

```
constexpr struct hpx::parallel::execution::sync_invoke_t sync_invoke
```

```
struct async_execute_t
```

#include <execution.hpp> Customization point for asynchronous execution agent creation.

This asynchronously creates a single function invocation *f*() using the associated executor.

Note: Executors have to implement only `async_execute()`. All other functions will be emulated by this or other customization points in terms of this single basic primitive. However, some executors will naturally specialize all operations for maximum efficiency.

Note: This is valid for one way executors (calls `make_ready_future(exec.sync_execute(f, ts...))` if it exists) and for two-way executors (calls `exec.async_execute(f, ts...)` if it exists).

Param *exec*

[in] The executor object to use for scheduling of the function *f*.

Param *f*

[in] The function which will be scheduled using the given executor.

Param *ts*

[in] Additional arguments to use to invoke *f*.

Return

f(*ts...*)'s result through a future

Public Functions

```
template<typename Executor, typename F, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, F &&f, Ts&&... ts) const
```

```
template<executor_any Executor, typename F, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, F &&f, Ts&&... ts) const
```

```
struct async_invoke_t
```

#include <execution.hpp> Asynchronously invoke the given set of nullary functions, each on its own execution agent

This creates a group of function invocations whose ordering is given by the `execution_category` associated with the executor.

All exceptions thrown by invocations of the functions are reported in a manner consistent with parallel algorithm execution through the returned future.

Note: This calls `exec.async_invoke(fs...)` if it exists; otherwise it executes `async_execute(fs)` for each `fs`.

Param exec

[in] The executor object to use for scheduling of the functions *fs*.

Param fs

[in] The functions which will be scheduled using the given executor.

Return

The return type of `executor_type::async_invoke` if defined by `executor_type`. Otherwise, a `future<void>` representing finishing the execution of all functions *fs*.

Public Functions

```
template<typename Executor, typename F, typename ...Fs>
inline decltype(auto) operator() (Executor &&exec, F &&f, Fs&&... fs) const
```

```
template<executor_any Executor, typename F, typename ...Fs>
inline decltype(auto) operator() (Executor &&exec, F &&f, Fs&&... fs) const
```

```
struct bulk_async_execute_t
```

```
#include <execution.hpp> Bulk form of asynchronous execution agent creation.
```

This asynchronously creates a group of function invocations `f(i)` whose ordering is given by the `execution_category` associated with the executor.

Here *i* takes on all values in the index space implied by `shape`. All exceptions thrown by invocations of `f(i)` are reported in a manner consistent with parallel algorithm execution through the returned future.

Note: This is deliberately different from the `bulk_async_execute` customization points specified in P0443. The `bulk_async_execute` customization point defined here is more generic and is used as the workhorse for implementing the specified APIs.

Note: This calls `exec.bulk_async_execute(f, shape, ts...)` if it exists; otherwise it executes `async_execute(f, shape, ts...)` as often as needed.

Param exec

[in] The executor object to use for scheduling of the function *f*.

Param f

[in] The function which will be scheduled using the given executor.

Param shape

[in] The shape objects which defines the iteration boundaries for the arguments to be passed to *f*.

Param ts

[in] Additional arguments to use to invoke *f*.

Return

The return type of `executor_type::bulk_async_execute` if defined by `executor_type`. Otherwise, a vector of futures holding the returned values of each invocation of `f`.

Public Functions

```
template<typename Executor, typename F, typename Shape, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, F &&f, Shape const &shape, Ts&&... ts)
    const

template<typename Executor, typename F, typename Shape, typename ...Ts>
inline auto operator() (Executor &&exec, F &&f, Shape const &shape, Ts&&... ts) const ->
    de-
    cltype(std::forward<Executor>(exec).bulk_async_execute(std::forward<F>(f),
    shape, std::forward<Ts>(ts)...))

template<executor_any Executor, typename F, typename Shape, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, F &&f, Shape const &shape, Ts&&... ts)
    const

template<executor_any Executor, typename F, typename Shape, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, F &&f, Shape const &shape, Ts&&... ts)
    const
```

struct **bulk_sync_execute_t**

#include <execution.hpp> Bulk form of synchronous execution agent creation.

This synchronously creates a group of function invocations `f(i)` whose ordering is given by the `execution_category` associated with the executor. The function synchronizes the execution of all scheduled functions with the caller.

Here *i* takes on all values in the index space implied by `shape`. All exceptions thrown by invocations of `f(i)` are reported in a manner consistent with parallel algorithm execution through the returned future.

Note: This is deliberately different from the `bulk_sync_execute` customization points specified in P0443. The `bulk_sync_execute` customization point defined here is more generic and is used as the workhorse for implementing the specified APIs.

Note: This calls `exec.bulk_sync_execute(f, shape, ts...)` if it exists; otherwise it executes `sync_execute(f, shape, ts...)` as often as needed.

Param exec

[in] The executor object to use for scheduling of the function `f`.

Param f

[in] The function which will be scheduled using the given executor.

Param shape

[in] The shape objects which defines the iteration boundaries for the arguments to be passed to `f`.

Param ts

[in] Additional arguments to use to invoke *f*.

Return

The return type of *executor_type::bulk_sync_execute* if defined by *executor_type*. Otherwise, a vector holding the returned values of each invocation of *f* except when *f* returns void, which case void is returned.

Public Functions

```
template<typename Executor, typename F, typename Shape, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, F &&f, Shape const &shape, Ts&&... ts)
    const
```

```
template<executor_any Executor, typename F, typename Shape, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, F &&f, Shape const &shape, Ts&&... ts)
    const
```

```
template<executor_any Executor, typename F, typename Shape, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, F &&f, Shape const &shape, Ts&&... ts)
    const
```

struct **bulk_then_execute_t**

#include <execution.hpp> Bulk form of execution agent creation depending on a given future.

This creates a group of function invocations *f*(*i*) whose ordering is given by the *execution_category* associated with the executor.

Here *i* takes on all values in the index space implied by *shape*. All exceptions thrown by invocations of *f*(*i*) are reported in a manner consistent with parallel algorithm execution through the returned future.

Note: This is deliberately different from the *then_sync_execute* customization points specified in P0443. The *bulk_then_execute* customization point defined here is more generic and is used as the workhorse for implementing the specified APIs.

Note: This calls *exec.bulk_then_execute*(*f*, *shape*, *pred*, *ts*...) if it exists; otherwise it executes *sync_execute*(*f*, *shape*, *pred.share*(), *ts*...) (if this executor is also an *OneWayExecutor*), or *async_execute*(*f*, *shape*, *pred.share*(), *ts*...) (if this executor is also a *TwoWayExecutor*) - as often as needed.

Param exec

[in] The executor object to use for scheduling of the function *f*.

Param f

[in] The function which will be scheduled using the given executor.

Param shape

[in] The shape objects which defines the iteration boundaries for the arguments to be passed to *f*.

Param predecessor

[in] The future object the execution of the given function depends on.

Params

[in] Additional arguments to use to invoke *f*.

Return

The return type of *executor_type::bulk_then_execute* if defined by *executor_type*. Otherwise, a vector holding the returned values of each invocation of *f*.

Public Functions

```
template<typename Executor, typename F, typename Shape, typename Future, typename
...Ts>
```

```
inline decltype(auto) operator() (Executor &&exec, F &&f, Shape const &shape, Future
&&predecessor, Ts&&... ts) const
```

```
template<executor_any Executor, typename F, typename Shape, typename Future, typename
...Ts>
```

```
inline decltype(auto) operator() (Executor &&exec, F &&f, Shape const &shape, Future
&&predecessor, Ts&&... ts) const
```

```
template<executor_any Executor, typename F, typename Shape, typename Future, typename
...Ts>
```

```
inline decltype(auto) operator() (Executor &&exec, F &&f, Shape const &shape, Future
&&predecessor, Ts&&... ts) const
```

struct post_t

#include <execution.hpp> Customization point for asynchronous fire & forget execution agent creation.

This asynchronously (fire & forget) creates a single function invocation *f()* using the associated executor.

Note: This is valid for two-way executors (calls *exec.post(f, ts...)*, if available, otherwise it calls *exec.async_execute(f, ts...)* while discarding the returned future), and for non-blocking two-way executors (calls *exec.post(f, ts...)* if it exists).

Param exec

[in] The executor object to use for scheduling of the function *f*.

Param f

[in] The function which will be scheduled using the given executor.

Params

[in] Additional arguments to use to invoke *f*.

Public Functions

```
template<typename Executor, typename F, typename ...Ts>
```

```
inline decltype(auto) operator() (Executor &&exec, F &&f, Ts&&... ts) const
```

```
template<executor_any Executor, typename F, typename ...Ts>
```

```
inline decltype(auto) operator() (Executor &&exec, F &&f, Ts&&... ts) const
```

struct **sync_execute_t**

#include <execution.hpp> Customization point for synchronous execution agent creation.

This synchronously creates a single function invocation *f()* using the associated executor. The execution of the supplied function synchronizes with the caller

Note: It will call `exec.sync_execute(f, ts...)` if the executor exposes a corresponding member function. For two-way executors it will invoke `async_execute` and wait for the task's completion before returning.

Param exec

[in] The executor object to use for scheduling of the function *f*.

Param f

[in] The function which will be scheduled using the given executor.

Param ts

[in] Additional arguments to use to invoke *f*.

Return

f(ts...)'s result

Public Functions

```
template<typename Executor, typename F, typename ...Ts>
inline decltype(auto) operator()(Executor &&exec, F &&f, Ts&&... ts) const
```

```
template<executor_any Executor, typename F, typename ...Ts>
inline decltype(auto) operator()(Executor &&exec, F &&f, Ts&&... ts) const
```

struct **sync_invoke_t**

#include <execution.hpp> Synchronously invoke the given set of nullary functions, each on its own execution agent

This creates a group of function invocations whose ordering is given by the `execution_category` associated with the executor.

All exceptions thrown by invocations of the functions are reported in a manner consistent with parallel algorithm execution through the returned future.

Note: This calls `exec.sync_invoke(fs...)` if it exists; otherwise it executes `sync_execute(fs)` for each *fs*.

Param exec

[in] The executor object to use for scheduling of the functions *fs*.

Param fs

[in] The functions which will be scheduled using the given executor.

Return

The return type of `executor_type::async_invoke` if defined by `executor_type`.

Public Functions

```
template<typename Executor, typename F, typename ...Fs>
inline decltype(auto) operator() (Executor &&exec, F &&f, Fs&&... fs) const
```

```
template<executor_any Executor, typename F, typename ...Fs>
inline decltype(auto) operator() (Executor &&exec, F &&f, Fs&&... fs) const
```

```
struct then_execute_t
```

#include <execution.hpp> Customization point for execution agent creation depending on a given future.

This creates a single function invocation *f()* using the associated executor after the given future object has become ready.

Note: This is valid for two-way executors (calls *exec.then_execute(f, predecessor, ts...)* if it exists) and for one way executors (calls *predecessor.then(bind(f, ts...))*).

Param *exec*

[in] The executor object to use for scheduling of the function *f*.

Param *f*

[in] The function which will be scheduled using the given executor.

Param *predecessor*

[in] The future object the execution of the given function depends on.

Param *ts*

[in] Additional arguments to use to invoke *f*.

Return

f(ts...)'s result through a future

Public Functions

```
template<typename Executor, typename F, typename Future, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, F &&f, Future &&predecessor, Ts&&...
ts) const
```

```
template<executor_any Executor, typename F, typename Future, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, F &&f, Future &&predecessor, Ts&&...
ts) const
```

hpx/execution_base/traits/is_executor_parameters.hpp

Defined in header *hpx/execution_base/traits/is_executor_parameters.hpp*.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<typename Executor>
```

```
struct extract_executor_parameters<Executor, std::void_t<typename Executor::executor_parameters_type>>
```

Public Types

```
using type = typename Executor::executor_parameters_type

template<typename Parameters>
struct extract_has_variable_chunk_size<Parameters, std::void_t<typename
Parameters::has_variable_chunk_size>> : public true_type

template<typename Parameters>
struct extract_has_variable_chunk_size<::std::reference_wrapper<Parameters>> : public
hpx::execution::experimental::extract_has_variable_chunk_size<Parameters>

template<typename Parameters>
struct extract_invokes_testing_function<::std::reference_wrapper<Parameters>> : public
hpx::execution::experimental::extract_invokes_testing_function<Parameters>

namespace hpx
```

Variables

```
template<typename Parameters> concept executor_parameters = hpx::traits::is_executor_parameters_v<Parameters>

namespace execution

    namespace experimental
```

Typedefs

```
template<typename Executor>
using extract_executor_parameters_t = typename
extract_executor_parameters<Executor>::type
```

Variables

```
template<typename Parameters>
constexpr bool extract_has_variable_chunk_size_v =
extract_has_variable_chunk_size<Parameters>::value

template<typename Parameters>
constexpr bool extract_invokes_testing_function_v =
extract_invokes_testing_function<Parameters>::value

template<typename T>
```

```
constexpr bool is_executor_parameters_v = is_executor_parameters<T>::value

template<typename Executor, typename Enable = void>

struct extract_executor_parameters
```

Public Types

```
using type = sequential_executor_parameters

template<typename Executor> executor_parameters_type > >
```

Public Types

```
using type = typename Executor::executor_parameters_type

template<typename Parameters, typename Enable = void>

struct extract_has_variable_chunk_size : public false_type

template<typename Parameters> has_variable_chunk_size > > : public true_type

template<typename Parameters> reference_wrapper< Parameters > > : public hpx::execution::ex

template<typename Parameters, typename Enable = void>

struct extract_invokes_testing_function : public false_type

template<typename Parameters> reference_wrapper< Parameters > > : public hpx::execution::ex

template<typename T>

struct is_executor_parameters : public detail::is_executor_parameters<std::decay_t<T>>

struct sequential_executor_parameters
```

namespace **traits**

Variables

```
template<typename T>

constexpr bool is_executor_parameters_v = is_executor_parameters<T>::value

template<typename Parameters, typename Enable>

struct is_executor_parameters
```

executors

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/executors/thread_pool_scheduler.hpp

Defined in header `hpx/executors/thread_pool_scheduler.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **execution**

namespace **experimental**

Typedefs

using **thread_pool_scheduler** = *thread_pool_policy_scheduler*<*hpx::launch*>

Variables

```
template<typename Sender> concept bulk_chunked_or_unchunked_sender = sender_invokes_algorithm_v<Sender,
hpx::execution::experimental::bulk_t> || sender_invokes_algorithm_v<Sender,
hpx::execution::experimental::bulk_chunked_t> || sender_invokes_algorithm_v<Sender,
hpx::execution::experimental::bulk_unchunked_t>
```

```
template<typename Policy>
```

```
constexpr bool is_sequenced_policy_v = false
```

```
template<>
```

```
constexpr bool is_sequenced_policy_v<sequenced_policy> = true
```

```
template<>
```

```
constexpr bool is_sequenced_policy_v<unsequenced_policy> = true
```

```
template<typename Policy>
```

```
constexpr bool is_unsequenced_bulk_policy_v = false
```

```
template<>
```

```
constexpr bool is_unsequenced_bulk_policy_v<unsequenced_policy> = true
```

```
template<>
```

```
constexpr bool is_unsequenced_bulk_policy_v<parallel_unsequenced_policy> = true
```

```
template<typename Policy>
```

```
struct thread_pool_domain : public sync_wait_domain
```

Public Functions

```
template<bulk_chunked_or_unchunked_sender Sender, typename Env>
inline constexpr thread_pool_policy_scheduler<Policy> auto transform_sender (hpx::execution::experimental::
    Sender
    &&sndr,
    Env const
    &env) const
    noexcept
```

```
template<typename Sender,
typename Env> inline requires constexpr sender_invokes_algorithm_v< Sender,
hpx::execution::experimental::continues_on_t > auto transform_sender (hpx::execution::ex
Sender &&sndr, Env const &) const noexcept
```

```
template<typename Policy>
```

```
struct thread_pool_policy_scheduler
```

Public Types

```
using policy_type = Policy
```

```
using execution_category = std::conditional_t<std::is_same_v<Policy,
launch::sync_policy>, sequenced_execution_tag, parallel_execution_tag>
```

Public Functions

```
inline explicit constexpr thread_pool_policy_scheduler (Policy l = experimen-
    tal::detail::get_default_scheduler_policy<Policy>::c
```

```
inline explicit thread_pool_policy_scheduler (hpx::threads::thread_pool_base *pool,
    Policy l = experimen-
    tal::detail::get_default_scheduler_policy<Policy>::call()
    noexcept
```

hpx/executors/execution_policy_annotation.hpp

Defined in header `hpx/executors/execution_policy_annotation.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace execution
```

```
namespace experimental
```

hpx/executors/executor_scheduler.hpp

Defined in header `hpx/executors/executor_scheduler.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
namespace execution
```

```
namespace experimental
```

```
template<typename Executor>
```

```
struct executor_scheduler
```

Public Types

```
using executor_type = std::decay_t<Executor>
```

Public Functions

```
constexpr executor_scheduler() = default
```

```
constexpr executor_scheduler(executor_scheduler const&) = default
```

```
constexpr executor_scheduler &operator=(executor_scheduler const&) = default
```

```
inline constexpr executor_scheduler(executor_scheduler &&other) noexcept
```

```
inline constexpr executor_scheduler &operator=(executor_scheduler &&other) noexcept
```

```
template<typename Exec>
```

```
inline explicit constexpr executor_scheduler(Exec &&exec) noexcept(  
    cept(std::is_nothrow_constructible_v<executor_type,  
    Exec>)
```

```
inline constexpr bool operator==(executor_scheduler const &rhs) const noexcept
```

```
inline constexpr bool operator!=(executor_scheduler const &rhs) const noexcept
```

```
inline constexpr detail::executor_sender<Executor> schedule() const & noexcept(  
    cept(std::is_nothrow_copy_constructible_v<executor_type,
```

```
inline constexpr detail::executor_sender<Executor> schedule() && noexcept(  
    cept(std::is_nothrow_move_constructible_v<executor_type,
```

```
template<typename Sender, typename Shape, typename F>
```

```
auto bulk(Sender &&sender, Shape const &shape, F &&f) const
```

Public Members

HPX_NO_UNIQUE_ADDRESS **executor_type** **exec_**

hpx/executors/std_execution_policy.hpp

Defined in header `hpx/executors/std_execution_policy.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/executors/thread_pool_continues_on_sender.hpp

Defined in header `hpx/executors/thread_pool_continues_on_sender.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Domain-optimized `continues_on` sender for HPX thread pool schedulers.

When `stdexec::continues_on` is applied with an HPX `thread_pool_scheduler`, the generic `stdexec` implementation creates two operation states: one for the source sender and one for the scheduler hop (`schedule_from`). This double-state overhead adds ~1 us per task in microbenchmarks.

This header provides a custom sender that replaces the generic path by:

1. Connecting the source sender to a lightweight receiver that captures the downstream receiver and the scheduler.
2. On `set_value`: if we are already on the same HPX thread pool, forwarding inline; otherwise, posting a single HPX task to forward the values.

This eliminates the intermediate `__storage_for_t`, the second operation state, and the `run_loop` overhead.

namespace **hpx**

namespace **execution**

namespace **experimental**

hpx/executors/execution_policy_scheduling_property.hpp

Defined in header `hpx/executors/execution_policy_scheduling_property.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **execution**

namespace **experimental**

hpx/executors/sequenced_executor.hpp

Defined in header `hpx/executors/sequenced_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **execution**

struct **sequenced_executor**

#include <sequenced_executor.hpp> A *sequential_executor* creates groups of sequential execution agents which execute in the calling thread. The sequential order is given by the lexicographical order of indices in the index space.

Private Static Functions

```
template<typename F, typename ...Ts>
static inline decltype(auto) sync_execute_impl(F &&f, Ts&&... ts)
```

```
template<typename F, typename ...Ts>
static inline decltype(auto) async_execute_impl(hpx::threads::thread_description const &desc, F
&&f, Ts&&... ts)
```

```
template<typename Archive>
static inline constexpr void serialize(Archive&, unsigned int const) noexcept
```

Friends

```
friend class hpx::serialization::access
```

namespace **experimental**

hpx/executors/scheduler_executor.hpp

Defined in header `hpx/executors/scheduler_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **execution**

namespace **experimental**

Functions

```
template<typename BaseScheduler>
explicit scheduler_executor(BaseScheduler &&sched) ->
    scheduler_executor<std::decay_t<BaseScheduler>>

template<typename BaseScheduler>

struct scheduler_executor
```

hpx/executors/parallel_executor.hpp

Defined in header `hpx/executors/parallel_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **execution**

Typedefs

```
using parallel_executor = parallel_policy_executor<hpx::launch>

using parallel_executor_spawn_hierarchically = parallel_policy_executor<hpx::launch, true>
```

Functions

```
template<typename Policy>
constexpr parallel_policy_executor<Policy, true> to_hierarchical_spawning(parallel_policy_executor<Policy>
    &exec) noexcept

template<typename Policy>
constexpr parallel_policy_executor<Policy, true> to_hierarchical_spawning(parallel_policy_executor<Policy>
    &&exec) noexcept

template<typename Policy>
constexpr parallel_policy_executor<Policy, true> to_hierarchical_spawning(parallel_policy_executor<Policy>
    const &exec)
    noexcept

template<typename Executor>
constexpr Executor to_hierarchical_spawning(Executor &&exec) noexcept

template<typename Policy>
constexpr parallel_policy_executor<Policy> to_non_hierarchical_spawning(parallel_policy_executor<Policy,
    true> &exec)
    noexcept

template<typename Policy>
```

```
constexpr parallel_policy_executor<Policy> to_non_hierarchical_spawning(parallel_policy_executor<Policy,
                                                                    true> &&exec)
                                                                    noexcept

template<typename Policy>
constexpr parallel_policy_executor<Policy> to_non_hierarchical_spawning(parallel_policy_executor<Policy,
                                                                    true> const &exec)
                                                                    noexcept

template<typename Executor>
constexpr Executor to_non_hierarchical_spawning(Executor &&exec) noexcept

template<typename Policy, bool HierarchicalSpawning = false>
struct parallel_policy_executor : public hpx::execution::parallel_policy_executor_base<Policy>
```

Public Types

```
using base_type = parallel_policy_executor_base<Policy>
```

Public Functions

```
inline explicit constexpr parallel_policy_executor(threads::thread_priority priority,
                                                    threads::thread_stacksize stacksize =
                                                    threads::thread_stacksize::default_,
                                                    threads::thread_schedule_hint
                                                    schedulehint = { }, Policy l = parallel::execution::detail::get_default_policy<Policy>::call())
                                                    noexcept

inline explicit constexpr parallel_policy_executor(threads::thread_stacksize stacksize,
                                                    threads::thread_schedule_hint
                                                    schedulehint = { }, Policy l = parallel::execution::detail::get_default_policy<Policy>::call())
                                                    noexcept

inline explicit constexpr parallel_policy_executor(threads::thread_schedule_hint
                                                    schedulehint, Policy l = parallel::execution::detail::get_default_policy<Policy>::call())
                                                    noexcept

inline explicit constexpr parallel_policy_executor(Policy l) noexcept

constexpr parallel_policy_executor() noexcept = default

inline explicit constexpr parallel_policy_executor(threads::thread_pool_base *pool, Policy l)
                                                    noexcept
```

```

inline explicit constexpr parallel_policy_executor(threads::thread_pool_base *pool,
                                                    threads::thread_priority priority =
                                                    threads::thread_priority::default_,
                                                    threads::thread_stacksize stacksize =
                                                    threads::thread_stacksize::default_,
                                                    threads::thread_schedule_hint
                                                    schedulehint = { }, Policy l = parallel::execution::detail::get_default_policy<Policy>::call())
                                                    noexcept

parallel_policy_executor(parallel_policy_executor const&) = default
parallel_policy_executor(parallel_policy_executor&&) = default

parallel_policy_executor &operator=(parallel_policy_executor const&) = default
parallel_policy_executor &operator=(parallel_policy_executor&&) = default

inline auto query(experimental::with_processing_units_count_t, std::size_t num_cores) const

template<typename Parameters>
inline std::size_t query(experimental::processing_units_count_t, Parameters &&params,
                        hpx::chrono::steady_duration const &iter_dur = hpx::chrono::null_duration,
                        std::size_t num_tasks = 0) const

inline auto query(experimental::with_first_core_t, std::size_t first_core) const noexcept

inline constexpr std::size_t query(experimental::get_first_core_t) const noexcept

inline auto query(experimental::get_processing_units_mask_t) const

inline auto query(experimental::get_cores_mask_t) const

template<typename Tag, typename Property>
inline auto query(Tag tag, Property &&prop) const

template<typename Tag>
inline auto query(Tag tag) const

template<typename F, typename ...Ts>
inline decltype(auto) sync_execute(F &&f, Ts&&... ts) const

template<typename F, typename ...Ts>
inline decltype(auto) async_execute(F &&f, Ts&&... ts) const

template<typename F, typename Future, typename ...Ts>
inline decltype(auto) then_execute(F &&f, Future &&predecessor, Ts&&... ts) const

template<typename F, typename ...Ts>
inline void post(F &&f, Ts&&... ts) const

template<typename F, typename S, typename Future, typename ...Ts>
inline decltype(auto) bulk_then_execute(F &&f, S const &shape, Future &&predecessor,
                                         Ts&&... ts) const

template<typename F, typename S, typename ...Ts>
inline decltype(auto) bulk_sync_execute(F &&f, S const &shape, Ts&&... ts) const

template<typename F, typename S, typename ...Ts>

```

```
inline decltype(auto) bulk_async_execute(F &&f, S const &shape, Ts&&... ts) const
```

```
inline decltype(auto) to_non_par() const
```

```
template<typename Policy, bool HierarchicalSpawning>
```

```
constexpr hpx::execution::experimental::executor_scheduler<parallel_policy_executor<Policy, HierarchicalSpawning>
```

```
template<typename Policy>
```

```
struct parallel_policy_executor<Policy, true> : public
```

```
hpx::execution::parallel_policy_executor_base<Policy>
```

Public Types

```
using base_type = parallel_policy_executor_base<Policy>
```

Public Functions

```
inline explicit constexpr parallel_policy_executor(threads::thread_priority priority,  
                                                    threads::thread_stacksize stacksize =  
                                                    threads::thread_stacksize::default_,  
                                                    threads::thread_schedule_hint  
                                                    schedulehint = { }, Policy l = parallel::execution::detail::get_default_policy<Policy>::call(),  
                                                    std::size_t const hierarchical_threshold =  
                                                    hierarchical_threshold_default_) noexcept
```

```
inline explicit constexpr parallel_policy_executor(threads::thread_stacksize stacksize,  
                                                    threads::thread_schedule_hint  
                                                    schedulehint = { }, Policy l = parallel::execution::detail::get_default_policy<Policy>::call())  
                                                    noexcept
```

```
inline explicit constexpr parallel_policy_executor(threads::thread_schedule_hint  
                                                    schedulehint, Policy l = parallel::execution::detail::get_default_policy<Policy>::call())  
                                                    noexcept
```

```
inline explicit constexpr parallel_policy_executor(Policy l) noexcept
```

```
constexpr parallel_policy_executor() noexcept = default
```

```
inline explicit constexpr parallel_policy_executor(threads::thread_pool_base *pool, Policy l,  
                                                    std::size_t const hierarchical_threshold =  
                                                    hierarchical_threshold_default_) noexcept
```

```

inline explicit constexpr parallel_policy_executor(threads::thread_pool_base *pool,
                                                    threads::thread_priority priority =
                                                    threads::thread_priority::default_,
                                                    threads::thread_stacksize stacksize =
                                                    threads::thread_stacksize::default_,
                                                    threads::thread_schedule_hint
                                                    schedulehint = { }, Policy l = parallel::execution::detail::get_default_policy<Policy>::call(),
                                                    std::size_t const hierarchical_threshold =
                                                    hierarchical_threshold_default_) noexcept

inline constexpr void set_hierarchical_threshold(std::size_t const threshold) noexcept

parallel_policy_executor(parallel_policy_executor const&) = default

parallel_policy_executor(parallel_policy_executor&&) = default

parallel_policy_executor &operator=(parallel_policy_executor const&) = default

parallel_policy_executor &operator=(parallel_policy_executor&&) = default

~parallel_policy_executor() = default

inline auto query(experimental::with_processing_units_count_t, std::size_t num_cores) const

template<typename Parameters>
inline std::size_t query(experimental::processing_units_count_t, Parameters &&params,
                        hpx::chrono::steady_duration const &iter_dur = hpx::chrono::null_duration,
                        std::size_t num_tasks = 0) const

inline auto query(experimental::with_first_core_t, std::size_t first_core) const noexcept

inline constexpr std::size_t query(experimental::get_first_core_t) const noexcept

inline auto query(experimental::get_processing_units_mask_t) const

inline auto query(experimental::get_cores_mask_t) const

template<typename Tag, typename Property>
inline auto query(Tag tag, Property &&prop) const

template<typename Tag>
inline auto query(Tag tag) const

template<typename F, typename ...Ts>
inline decltype(auto) sync_execute(F &&f, Ts&&... ts) const

template<typename F, typename ...Ts>
inline decltype(auto) async_execute(F &&f, Ts&&... ts) const

template<typename F, typename Future, typename ...Ts>
inline decltype(auto) then_execute(F &&f, Future &&predecessor, Ts&&... ts) const

template<typename F, typename ...Ts>
inline void post(F &&f, Ts&&... ts) const

template<typename F, typename S, typename Future, typename ...Ts>

```

```
inline decltype(auto) bulk_then_execute(F &&f, S const &shape, Future &&predecessor,
                                         Ts&&... ts) const
```

```
template<typename F, typename S, typename ...Ts>
inline decltype(auto) bulk_sync_execute(F &&f, S const &shape, Ts&&... ts) const
```

```
template<typename F, typename S, typename ...Ts>
inline decltype(auto) bulk_async_execute(F &&f, S const &shape, Ts&&... ts) const
```

```
inline decltype(auto) to_non_par() const
```

```
template<typename Policy>
```

```
struct parallel_policy_executor_base
```

#include <parallel_executor.hpp> A *parallel_executor* creates groups of parallel execution agents which execute in threads implicitly created by the executor. This executor prefers continuing with the creating thread first before executing newly created threads.

This executor conforms to the concepts of a *TwoWayExecutor*, and a *BulkTwoWayExecutor*

Subclassed by *hpx::execution::parallel_policy_executor< Policy, HierarchicalSpawning >*, *hpx::execution::parallel_policy_executor< Policy, true >*, *hpx::execution::parallel_policy_executor< Policy >*

Public Types

```
using execution_category = std::conditional_t<std::is_same_v<Policy, launch::sync_policy>,
sequenced_execution_tag, parallel_execution_tag>
```

Associate the *parallel_execution_tag* executor tag type as a default with this executor, except if the given launch policy is sync.

```
using executor_parameters_type = experimental::default_parameters
```

Associate the *default_parameters* executor parameters type as a default with this executor.

Public Functions

```
inline parallel_policy_executor_base(parallel_policy_executor_base const &rhs) noexcept
```

```
inline parallel_policy_executor_base &operator=(parallel_policy_executor_base const &rhs)
noexcept
```

```
constexpr ~parallel_policy_executor_base() = default
```

```
template<typename Parameters>
inline std::size_t processing_units_count(Parameters&&, hpx::chrono::steady_duration const&
= hpx::chrono::null_duration, std::size_t = 0) const
```

Public Static Functions

```
template<typename Self>
static inline Self with_num_cores(Self self, std::size_t num_cores)

template<typename Self>
static inline Self with_first_core(Self self, std::size_t first_core) noexcept
```

Protected Functions

```
inline explicit constexpr parallel_policy_executor_base(threads::thread_priority priority,
                                                         threads::thread_stacksize stacksize =
                                                         threads::thread_stacksize::default_,
                                                         threads::thread_schedule_hint
                                                         schedulehint = {}, Policy l = paral-
                                                         lel::execution::detail::get_default_policy<Policy>::call()
                                                         noexcept
```

Create a new parallel executor.

```
inline explicit constexpr parallel_policy_executor_base(threads::thread_stacksize stacksize,
                                                         threads::thread_schedule_hint
                                                         schedulehint = {}, Policy l = paral-
                                                         lel::execution::detail::get_default_policy<Policy>::call()
                                                         noexcept
```

```
inline explicit constexpr parallel_policy_executor_base(threads::thread_schedule_hint
                                                         schedulehint, Policy l = paral-
                                                         lel::execution::detail::get_default_policy<Policy>::call()
                                                         noexcept
```

```
inline explicit constexpr parallel_policy_executor_base(Policy l) noexcept
```

```
inline constexpr parallel_policy_executor_base() noexcept
```

```
inline explicit constexpr parallel_policy_executor_base(threads::thread_pool_base *pool,
                                                         Policy l) noexcept
```

```
inline explicit constexpr parallel_policy_executor_base(threads::thread_pool_base *pool,
                                                         threads::thread_priority priority =
                                                         threads::thread_priority::default_,
                                                         threads::thread_stacksize stacksize =
                                                         threads::thread_stacksize::default_,
                                                         threads::thread_schedule_hint
                                                         schedulehint = {}, Policy l = paral-
                                                         lel::execution::detail::get_default_policy<Policy>::call()
```

```
namespace experimental
```

```
template<typename Executor>
struct executor_scheduler
```

```
namespace parallel
```

```
namespace execution
```

hpx/executors/execution_policy_mappings.hpp

Defined in header `hpx/executors/execution_policy_mappings.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
namespace execution
```

```
namespace experimental
```

Variables

```
template<typename Tag>
```

```
constexpr bool is_execution_policy_mapping_v = is_execution_policy_mapping<Tag>::value
```

```
constexpr struct hpx::execution::experimental::to_non_par_t to_non_par
```

```
constexpr struct hpx::execution::experimental::to_par_t to_par
```

```
constexpr struct hpx::execution::experimental::to_non_task_t to_non_task
```

```
constexpr struct hpx::execution::experimental::to_task_t to_task
```

```
constexpr struct hpx::execution::experimental::to_non_unseq_t to_non_unseq
```

```
constexpr struct hpx::execution::experimental::to_unseq_t to_unseq
```

```
template<typename Tag>
```

```
struct is_execution_policy_mapping : public false_type
```

```
template<>
```

```
struct is_execution_policy_mapping<to_non_par_t> : public true_type
```

```
template<>
```

```
struct is_execution_policy_mapping<to_non_task_t> : public true_type
```

```
template<>
```

```
struct is_execution_policy_mapping<to_non_unseq_t> : public true_type
```

```
template<>
```



```

struct is_execution_policy_mapping<to_par_t> : public true_type
template<>
struct is_execution_policy_mapping<to_task_t> : public true_type
template<>
struct is_execution_policy_mapping<to_unseq_t> : public true_type

struct to_non_par_t

```

Public Functions

```

inline constexpr decltype(auto) operator() (Target const &target) const
template<execution_policy ExPolicy>
inline constexpr decltype(auto) operator() (ExPolicy &&policy) const noexcept

```

Public Members

```

template<typename Target> requires = { t.to_non_par()

```

```

struct to_non_task_t
    Subclassed by hpx::execution::non_task_policy_tag

```

Public Functions

```

inline constexpr decltype(auto) operator() (Target const &target) const
template<execution_policy ExPolicy>
inline constexpr decltype(auto) operator() (ExPolicy &&policy) const noexcept

```

Public Members

```

template<typename Target> requires = { t.to_non_task()

```

```

struct to_non_unseq_t

```

Public Functions

```
inline constexpr decltype(auto) operator()(Target const &target) const
```

```
template<execution_policy ExPolicy>
```

```
inline constexpr decltype(auto) operator()(ExPolicy &&policy) const noexcept
```

Public Members

```
template<typename Target> requires = { t.to_non_unseq()
```

```
struct to_par_t
```

Public Functions

```
inline constexpr decltype(auto) operator()(Target const &target) const
```

```
template<execution_policy ExPolicy>
```

```
inline constexpr decltype(auto) operator()(ExPolicy &&policy) const noexcept
```

Public Members

```
template<typename Target> requires = { t.to_par()
```

```
struct to_task_t
```

Subclassed by `hpx::execution::task_policy_tag`

Public Functions

```
inline constexpr decltype(auto) operator()(Target const &target) const
```

```
template<execution_policy ExPolicy>
```

```
inline constexpr decltype(auto) operator()(ExPolicy &&policy) const noexcept
```

Public Members

```
template<typename Target> requires = { t.to_task()
```

```
struct to_unseq_t
```

Public Functions

```
inline constexpr decltype(auto) operator()(Target const &target) const

template<execution_policy ExPolicy>
inline constexpr decltype(auto) operator()(ExPolicy &&policy) const noexcept
```

Public Members

```
template<typename Target> requires = { t.to_unseq()
```

hpx/executors/execution_policy_parameters.hpp

Defined in header `hpx/executors/execution_policy_parameters.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/executors/restricted_thread_pool_executor.hpp

Defined in header `hpx/executors/restricted_thread_pool_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace execution
```

```
        namespace experimental
```

Typedefs

```
using restricted_thread_pool_executor = restricted_policy_executor<hpx::launch>

template<typename Policy>

class restricted_policy_executor
```

Public Types

```
using execution_category = typename embedded_executor::execution_category

    Associate the parallel_execution_tag executor tag type as a default with this executor.

using executor_parameters_type = typename
embedded_executor::executor_parameters_type
```

Public Functions

```
inline explicit restricted_policy_executor(std::size_t first_thread = 0, std::size_t
                                           num_threads = 1, threads::thread_priority
                                           priority = threads::thread_priority::default_,
                                           threads::thread_stacksize stacksize =
                                           threads::thread_stacksize::default_,
                                           threads::thread_schedule_hint schedulehint =
                                           { }, [[maybe_unused]] std::size_t
                                           hierarchical_threshold =
                                           hierarchical_threshold_default_)
```

Create a new parallel executor.

```
inline restricted_policy_executor(restricted_policy_executor const &other) noexcept
    cept(std::is_nothrow_copy_constructible_v<embedded_executor>)

inline restricted_policy_executor(restricted_policy_executor &&other) noexcept

inline restricted_policy_executor &operator=(restricted_policy_executor const &rhs) noexcept
    cept(std::is_nothrow_copy_assignable_v<embedded_executor>)

inline restricted_policy_executor &operator=(restricted_policy_executor &&rhs) noexcept

inline constexpr hpx::execution::experimental::executor_scheduler<restricted_policy_executor> query(hpx::exec
                                                                                               const
                                                                                               noexcept
                                                                                               cept(std::i
```

Private Types

```
using embedded_executor = hpx::execution::parallel_policy_executor<Policy>
```

Private Members

```
std::uint16_t first_thread_
```

```
mutable std::atomic<std::size_t> os_thread_
```

```
embedded_executor exec_
```

Private Static Attributes

```
static constexpr std::size_t hierarchical_threshold_default_ = 6
```

hpx/executors/exception_list.hpp

Defined in header `hpx/executors/exception_list.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace parallel
```

hpx::execution::seq

Defined in header `hpx/execution.hpp`⁸⁴⁹.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::execution::par`
- `hpx::execution::par_unseq`
- `hpx::execution::task`
- `hpx::execution::sequenced_policy`
- `hpx::execution::parallel_policy`
- `hpx::execution::parallel_unsequenced_policy`
- `hpx::execution::sequenced_task_policy`
- `hpx::execution::parallel_task_policy`

```
constexpr sequenced_policy hpx::execution::seq = { }
```

Default sequential execution policy object.

hpx::execution::par

Defined in header `hpx/execution.hpp`⁸⁵⁰.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::execution::seq`
- `hpx::execution::par_unseq`
- `hpx::execution::task`
- `hpx::execution::sequenced_policy`
- `hpx::execution::parallel_policy`
- `hpx::execution::parallel_unsequenced_policy`

⁸⁴⁹ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/execution.hpp

⁸⁵⁰ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/execution.hpp

- `hpx::execution::sequenced_task_policy`
- `hpx::execution::parallel_task_policy`

constexpr `parallel_policy` `hpx::execution::par` = {}

Default parallel execution policy object.

`hpx::execution::par_unseq`

Defined in header `hpx/execution.hpp`⁸⁵¹.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::execution::seq`
- `hpx::execution::par`
- `hpx::execution::task`
- `hpx::execution::sequenced_policy`
- `hpx::execution::parallel_policy`
- `hpx::execution::parallel_unsequenced_policy`
- `hpx::execution::sequenced_task_policy`
- `hpx::execution::parallel_task_policy`

constexpr `parallel_unsequenced_policy` `hpx::execution::par_unseq` = {}

Default vector execution policy object.

`hpx::execution::task`

Defined in header `hpx/execution.hpp`⁸⁵².

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::execution::seq`
- `hpx::execution::par`
- `hpx::execution::par_unseq`
- `hpx::execution::sequenced_policy`
- `hpx::execution::parallel_policy`
- `hpx::execution::parallel_unsequenced_policy`
- `hpx::execution::sequenced_task_policy`
- `hpx::execution::parallel_task_policy`

⁸⁵¹ [http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/execution.](http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/execution.hpp)

⁸⁵² [http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/execution.](http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/execution.hpp)

```
constexpr task_policy_tag hpx::execution::task = {}
```

hpx::execution::sequenced_policy

Defined in header [hpx/execution.hpp](#)⁸⁵³.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::execution::seq`
- `hpx::execution::par`
- `hpx::execution::par_unseq`
- `hpx::execution::task`
- `hpx::execution::parallel_policy`
- `hpx::execution::parallel_unsequenced_policy`
- `hpx::execution::sequenced_task_policy`
- `hpx::execution::parallel_task_policy`

```
using hpx::execution::sequenced_policy = detail::sequenced_policy_shim<sequenced_executor,  
hpx::traits::executor_parameters_type_t<sequenced_executor>>
```

The class `sequenced_policy` is an execution policy type used as a unique type to disambiguate parallel algorithm overloading and require that a parallel algorithm's execution may not be parallelized.

hpx::execution::parallel_policy

Defined in header [hpx/execution.hpp](#)⁸⁵⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::execution::seq`
- `hpx::execution::par`
- `hpx::execution::par_unseq`
- `hpx::execution::task`
- `hpx::execution::sequenced_policy`
- `hpx::execution::parallel_unsequenced_policy`
- `hpx::execution::sequenced_task_policy`
- `hpx::execution::parallel_task_policy`

⁸⁵³ http://github.com/TheHPXProject/hpx/blob/d402eada7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/execution.hpp

⁸⁵⁴ http://github.com/TheHPXProject/hpx/blob/d402eada7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/execution.hpp

```
using hpx::execution::parallel_policy = detail::parallel_policy_shim<parallel_executor,  
hpx::traits::executor_parameters_type_t<parallel_executor>>
```

The class `parallel_policy` is an execution policy type used as a unique type to disambiguate parallel algorithm overloading and indicate that a parallel algorithm's execution may be parallelized.

`hpx::execution::parallel_unsequenced_policy`

Defined in header `hpx/executation.hpp`⁸⁵⁵.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::execution::seq`
- `hpx::execution::par`
- `hpx::execution::par_unseq`
- `hpx::execution::task`
- `hpx::execution::sequenced_policy`
- `hpx::execution::parallel_policy`
- `hpx::execution::sequenced_task_policy`
- `hpx::execution::parallel_task_policy`

```
using hpx::execution::parallel_unsequenced_policy =  
detail::parallel_unsequenced_policy_shim<parallel_executor,  
hpx::traits::executor_parameters_type_t<parallel_executor>>
```

The class `parallel_unsequenced_policy` is an execution policy type used as a unique type to disambiguate parallel algorithm overloading and indicate that a parallel algorithm's execution may be parallelized and vectorized.

`hpx::execution::sequenced_task_policy`

Defined in header `hpx/executation.hpp`⁸⁵⁶.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::execution::seq`
- `hpx::execution::par`
- `hpx::execution::par_unseq`
- `hpx::execution::task`
- `hpx::execution::sequenced_policy`
- `hpx::execution::parallel_policy`
- `hpx::execution::parallel_unsequenced_policy`

⁸⁵⁵ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/executation.hpp

⁸⁵⁶ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/executation.hpp

- `hpx::execution::parallel_task_policy`

using `hpx::execution::sequenced_task_policy` = detail::sequenced_task_policy_shim<sequenced_executor, hpx::traits::executor_parameters_type_t<sequenced_executor>>

Extension: The class `sequenced_task_policy` is an execution policy type used as a unique type to disambiguate parallel algorithm overloading and indicate that a parallel algorithm's execution may not be parallelized (has to run sequentially).

The algorithm returns a future representing the result of the corresponding algorithm when invoked with the `sequenced_policy`.

`hpx::execution::parallel_task_policy`

Defined in header `hpx/execution.hpp`⁸⁵⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::execution::seq`
- `hpx::execution::par`
- `hpx::execution::par_unseq`
- `hpx::execution::task`
- `hpx::execution::sequenced_policy`
- `hpx::execution::parallel_policy`
- `hpx::execution::parallel_unsequenced_policy`
- `hpx::execution::sequenced_task_policy`

using `hpx::execution::parallel_task_policy` = detail::parallel_task_policy_shim<parallel_executor, hpx::traits::executor_parameters_type_t<parallel_executor>>

Extension: The class `parallel_task_policy` is an execution policy type used as a unique type to disambiguate parallel algorithm overloading and indicate that a parallel algorithm's execution may be parallelized.

The algorithm returns a future representing the result of the corresponding algorithm when invoked with the `parallel_policy`.

`hpx/executors/current_executor.hpp`

Defined in header `hpx/executors/current_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **parallel**

namespace **execution**

⁸⁵⁷ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/execution.hpp

Typedefs

```
typedef hpx::execution::parallel_executor instead
```

```
namespace this_thread
```

Functions

```
hpx::execution::parallel_executor get_executor(error_code &ec = throws)
```

Returns a reference to the executor that was used to create the current thread.

Throws

If – &ec != &throws, never throws, but will set *ec* to an appropriate value when an error occurs. Otherwise, this function will throw an *hpx::exception* with an error code of *hpx::error::yield_aborted* if it is signaled with *wait_aborted*. If called outside a HPX-thread, this function will throw a *hpx::exception* with an error code of *hpx::error::null_thread_id*. If this function is called while the thread-manager is not running, it will throw an *hpx::exception* with an error code of *hpx::error::invalid_status*.

```
namespace threads
```

Functions

```
hpx::execution::parallel_executor get_executor(thread_id_type const &id, error_code &ec = throws)
```

Returns a reference to the executor that was used to create the given thread.

Throws

If – &ec != &throws, never throws, but will set *ec* to an appropriate value when an error occurs. Otherwise, this function will throw an *hpx::exception* with an error code of *hpx::error::yield_aborted* if it is signaled with *wait_aborted*. If called outside a HPX-thread, this function will throw a *hpx::exception* with an error code of *hpx::error::null_thread_id*. If this function is called while the thread-manager is not running, it will throw an *hpx::exception* with an error code of *hpx::error::invalid_status*.

`hpx/executors/annotating_executor.hpp`

Defined in header `hpx/executors/annotating_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
namespace execution
```

```
namespace experimental
```

```
template<executor_any BaseExecutor>
```

```
struct annotating_executor
```

#include <*annotating_executor.hpp*> An *annotating_executor* wraps any other executor and adds the capability to add annotations to the launched threads.

Public Functions

```
template<executor_any Executor>
inline explicit constexpr annotating_executor(Executor &&exec, char const *annotation =
                                              nullptr)
```

```
template<executor_any Executor>
inline explicit annotating_executor(Executor &&exec, std::string annotation)
```

hpx/executors/service_executors.hpp

Defined in header `hpx/executors/service_executors.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace parallel
```

```
        namespace execution
```

hpx/executors/fork_join_executor.hpp

Defined in header `hpx/executors/fork_join_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/executors/explicit_scheduler_executor.hpp

Defined in header `hpx/executors/explicit_scheduler_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace execution
```

```
        namespace experimental
```

Functions

```
template<typename BaseScheduler>
explicit explicit_scheduler_executor(BaseScheduler &&sched) -> explicit_scheduler_executor<std::decay_t<BaseScheduler>>
```

```
template<typename BaseScheduler>
struct explicit_scheduler_executor
```

hpx/executors/parallel_executor_aggregated.hpp

Defined in header `hpx/executors/parallel_executor_aggregated.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace parallel
```

```
        namespace execution
```

hpx/executors/datapar/execution_policy_mappings.hpp

Defined in header `hpx/executors/datapar/execution_policy_mappings.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/executors/datapar/execution_policy.hpp

Defined in header `hpx/executors/datapar/execution_policy.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

filesystem

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/filesystem/api.hpp

Defined in header `hpx/filesystem/api.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

This file provides a compatibility layer using Boost.Filesystem for the C++17 filesystem library. It is *not* intended to be a complete compatibility layer. It only contains functions required by the HPX codebase. It also provides some functions only available in Boost.Filesystem when using C++17 filesystem.

```
namespace hpx
```

```
    namespace filesystem
```

Functions

inline path **initial_path**()

inline *std::string* **to_string**(path const &p)

inline *std::string* **basename**(path const &p)

inline path **canonical**(path const &p, path const &base)

inline path **canonical**(path const &p, path const &base, *std::error_code* &ec)

namespace **filesystem**

functional

See *Public API* for a list of names and headers that are part of the public HPX API.

hpx::function_ref

Defined in header `hpx/function.hpp`⁸⁵⁸.

See *Public API* for a list of names and headers that are part of the public HPX API.

template<typename **Sig**>

class **function_ref**

function_ref class is a vocabulary type with reference semantics for passing entities to call.

An example use case that benefits from higher-order functions is `retry(n, f)` which attempts to call `f` up to `n` times synchronously until success. This example might model the real-world scenario of repeatedly querying a flaky web service.

```
using payload = std::optional< /* ... */ >;
// Repeatedly invokes `action` up to `times` repetitions.
// Immediately returns if `action` returns a valid `payload`.
// Returns `std::nullopt` otherwise.
payload retry(size_t times, /* ????? */ action);
```

The passed-in action should be a callable entity that takes no arguments and returns a payload. This can be done with function pointers, `hpx::function` or a template but it is much simpler with *function_ref* as seen below:

```
payload retry(size_t times, function_ref<payload()> action);
```

⁸⁵⁸ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/function.hpp

hpx::mem_fn

Defined in header [hpx/functional.hpp](#)⁸⁵⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

Warning: doxygenfunction: Unable to resolve function “hpx::mem_fn” with arguments “(M C::*)”. Candidate function could not be parsed. Parsing error is Invalid C++ declaration: Expected identifier in nested name. [error at 4] (C::*)(Ps...)> mem_fn(R (C::* pm)(Ps...)) noexcept —^

Warning: doxygenfunction: Unable to resolve function “hpx::mem_fn” with arguments “(R (C::*)(Ps...))”. Candidate function could not be parsed. Parsing error is Invalid C++ declaration: Expected identifier in nested name. [error at 4] (C::*)(Ps...)> mem_fn(R (C::* pm)(Ps...)) noexcept —^

Warning: doxygenfunction: Unable to resolve function “hpx::mem_fn” with arguments “(R (C::*)(Ps...) const)”. Candidate function could not be parsed. Parsing error is Invalid C++ declaration: Expected identifier in nested name. [error at 4] (C::*)(Ps...)> mem_fn(R (C::* pm)(Ps...)) noexcept —^

hpx::invoke_fused

Defined in header [hpx/functional.hpp](#)⁸⁶⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- [hpx::invoke_fused_r](#)

template<typename **F**, typename **Tuple**>

```
constexpr detail::invoke_fused_result_t<F, Tuple> hpx::invoke_fused(F &&f, Tuple &&t) noexcept
    noexcept(detail::invoke_fused_impl(detail::fused_index,
    HPX_FORWARD(F, f),
    HPX_FORWARD(Tuple, t))))
```

Invokes the given callable object *f* with the content of the sequenced type *t* (tuples, pairs).

Note: This function is similar to `std::apply` (C++17). The difference between `hpx::invoke` and `hpx::invoke_fused` is that the later unpacks the tuples while the former cannot. Turning a tuple into a parameter pack is not a trivial operation which makes `hpx::invoke_fused` rather useful.

Parameters

- **f** – Must be a callable object. If *f* is a member function pointer, the first argument in the sequenced type will be treated as the callee (this object).
- **t** – A type whose contents are accessible through a call to `hpx::get`.

⁸⁵⁹ http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/functional.hpp

⁸⁶⁰ http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/functional.hpp

Throws

`std::exception` – like objects thrown by call to object `f` with the arguments contained in the sequenceable type `t`.

Returns

The result of the callable object when it's called with the content of the given sequenced type.

`hpx::invoke_fused_r`

Defined in header `hpx/functional.hpp`⁸⁶¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::invoke_fused`

```
template<typename R, typename F, typename Tuple>
constexpr R hpx::invoke_fused_r(F &&f, Tuple &&t) noexcept(
    noexcept(detail::invoke_fused_impl(detail::fused_index_pack_t<Tuple>{ },
    HPX_FORWARD(F, f), HPX_FORWARD(Tuple, t))))
```

Invokes the given callable object `f` with the content of the sequenced type `t` (tuples, pairs).

Note: This function is similar to `std::apply` (C++17). The difference between `hpx::invoke` and `hpx::invoke_fused` is that the later unpacks the tuples while the former cannot. Turning a tuple into a parameter pack is not a trivial operation which makes `hpx::invoke_fused` rather useful.

Note: The difference between `hpx::invoke_fused` and `hpx::invoke_fused_r` is that the later allows to specify the return type as well.

Parameters

- **`f`** – Must be a callable object. If `f` is a member function pointer, the first argument in the sequenced type will be treated as the callee (this object).
- **`t`** – A type whose contents are accessible through a call to `hpx::get`.

Throws

`std::exception` – like objects thrown by call to object `f` with the arguments contained in the sequenceable type `t`.

Template Parameters

`R` – The result type of the function when it's called with the content of the given sequenced type.

Returns

The result of the callable object when it's called with the content of the given sequenced type.

⁸⁶¹ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/functional.hpp

hpx::bind_front

Defined in header [hpx/functional.hpp](#)⁸⁶².

See *Public API* for a list of names and headers that are part of the public HPX API.

Warning: doxygenfunction: Unable to resolve function “hpx::bind_front” with arguments (F&&, Ts&&...) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:

```
- template<typename F, typename ...Ts> constexpr detail::bound_front<std::decay_t<F>,   
→util::make_index_pack_t<sizeof...(Ts)>, util::decay_unwrap_t<Ts>...> bind_front(F &&  
→f, Ts&&... vs)  
- template<typename F> constexpr std::decay_t<F> bind_front(F &&f)
```

```
template<typename F>  
constexpr std::decay_t<F> hpx::bind_front(F &&f)
```

hpx::move_only_function

Defined in header [hpx/functional.hpp](#)⁸⁶³.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename Sig, bool Serializable = false>
```

```
class move_only_function
```

Class template *hpx::move_only_function* is a general-purpose polymorphic function wrapper. *hpx::move_only_function* objects can store and invoke any constructible (not required to be move constructible) Callable target [§8212](#); functions, lambda expressions, bind expressions, or other function objects, as well as pointers to member functions and pointers to member objects.

The stored callable object is called the target of *hpx::move_only_function*. If an *hpx::move_only_function* contains no target, it is called empty. Unlike *hpx::function*, invoking an empty *hpx::move_only_function* results in undefined behavior.

hpx::move_only_functions supports every possible combination of cv-qualifiers, ref-qualifiers, and noexcept-specifiers not including volatile provided in its template parameter. These qualifiers and specifier (if any) are added to its operator(). *hpx::move_only_function* satisfies the requirements of MoveConstructible and MoveAssignable, but does not satisfy CopyConstructible or CopyAssignable.

⁸⁶² http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/functional.

hpp

⁸⁶³ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/functional.

hpp

hpx::bind_back

Defined in header [hpx/functional.hpp](#)⁸⁶⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

Warning: doxygenfunction: Unable to resolve function “hpx::bind_back” with arguments (F&&, Ts&&...) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:

```
- template<typename F, typename ...Ts> constexpr hpx::detail::bound_back<std::decay_t
  ↳<F>, util::make_index_pack_t<sizeof...(Ts)>, util::decay_unwrap_t<Ts>...> bind_
  ↳back(F &&f, Ts&&... vs)
- template<typename F> constexpr std::decay_t<F> bind_back(F &&f)
```

```
template<typename F>
constexpr std::decay_t<F> hpx::bind_back(F &&f)
```

hpx::bind

Defined in header [hpx/functional.hpp](#)⁸⁶⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- [hpx::placeholders::_1](#)
- [hpx::placeholders::_2](#)
- ...
- [hpx::placeholders::_9](#)

Warning: doxygenfunction: Unable to resolve function “hpx::bind” with arguments (F&&, Ts&&...) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:

```
- template<typename Action, typename ...Ts> detail::bound_action<std::decay_t<Action>,
  ↳ util::make_index_pack_t<sizeof...(Ts)>, hpx::util::decay_unwrap_t<Ts>...> bind(Ts&&
  ↳... vs)
- template<typename Component, typename Signature, typename Derived, typename ...Ts>
  ↳ detail::bound_action<Derived, util::make_index_pack_t<sizeof...(Ts)>, std::decay_t
  ↳<Ts>...> bind(hpx::actions::basic_action<Component, Signature, Derived> action, Ts&&
  ↳... vs)
- template<typename F, typename ...Ts> constexpr detail::bound<std::decay_t<F>,
  ↳ util::make_index_pack_t<sizeof...(Ts)>, util::decay_unwrap_t<Ts>...> bind(F &&f, Ts&
  ↳&... vs)
```

⁸⁶⁴ http://github.com/TheHPXProject/hpx/blob/d402eada7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/functional.hpp

⁸⁶⁵ http://github.com/TheHPXProject/hpx/blob/d402eada7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/functional.hpp

Warning: doxygenfunction: Unable to resolve function “hpx::bind” with arguments (Ts&&...) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:

```
- template<typename Action, typename ...Ts> detail::bound_action<std::decay_t<Action>,
  ↳ util::make_index_pack_t<sizeof...(Ts)>, hpx::util::decay_unwrap_t<Ts>...> bind(Ts&&
  ↳ ... vs)
- template<typename Component, typename Signature, typename Derived, typename ...Ts>↳
  ↳ detail::bound_action<Derived, util::make_index_pack_t<sizeof...(Ts)>, std::decay_t
  ↳ <Ts>...> bind(hpx::actions::basic_action<Component, Signature, Derived> action, Ts&&
  ↳ ... vs)
- template<typename F, typename ...Ts> constexpr detail::bound<std::decay_t<F>,↳
  ↳ util::make_index_pack_t<sizeof...(Ts)>, util::decay_unwrap_t<Ts>...> bind(F &&f, Ts&
  ↳ &... vs)
```

Warning: doxygenfunction: Unable to resolve function “hpx::bind” with arguments (hpx::actions::basic_action<Component, Signature, Derived>, Ts&&...) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:

```
- template<typename Action, typename ...Ts> detail::bound_action<std::decay_t<Action>,
  ↳ util::make_index_pack_t<sizeof...(Ts)>, hpx::util::decay_unwrap_t<Ts>...> bind(Ts&&
  ↳ ... vs)
- template<typename Component, typename Signature, typename Derived, typename ...Ts>↳
  ↳ detail::bound_action<Derived, util::make_index_pack_t<sizeof...(Ts)>, std::decay_t
  ↳ <Ts>...> bind(hpx::actions::basic_action<Component, Signature, Derived> action, Ts&&
  ↳ ... vs)
- template<typename F, typename ...Ts> constexpr detail::bound<std::decay_t<F>,↳
  ↳ util::make_index_pack_t<sizeof...(Ts)>, util::decay_unwrap_t<Ts>...> bind(F &&f, Ts&
  ↳ &... vs)
```

hpx::placeholders::_1

Defined in header `hpx/functional.hpp`⁸⁶⁶.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::bind`
- `hpx::placeholders::_2`
- ...
- `hpx::placeholders::_9`

constexpr detail::placeholder<1> `hpx::placeholders::_1` = {}

⁸⁶⁶ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbdc0c8044e7/libs/core/include_local/include/hpx/functional.hpp

hpx::placeholders::_2

Defined in header [hpx/functional.hpp](#)⁸⁶⁷.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- [hpx::bind](#)
- [hpx::placeholders::_1](#)
- ...
- [hpx::placeholders::_9](#)

```
constexpr detail::placeholder<2> hpx::placeholders::_2 = {}
```

—

—

Defined in header [hpx/functional.hpp](#)⁸⁶⁸.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- [hpx::bind](#)
- [hpx::placeholders::_1](#)
- [hpx::placeholders::_2](#)
- [hpx::placeholders::_9](#)

Warning: doxygenfunction: Cannot find function “...” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

hpx::placeholders::_9

Defined in header [hpx/functional.hpp](#)⁸⁶⁹.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- [hpx::bind](#)
- [hpx::placeholders::_1](#)
- [hpx::placeholders::_2](#)
- ...

⁸⁶⁷ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/functional.hpp

⁸⁶⁸ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/functional.hpp

⁸⁶⁹ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/functional.hpp

```
constexpr detail::placeholder<9> hpx::placeholders::_9 = {}
```

hpx::reference_wrapper

Defined in header [hpx/functional.hpp](#)⁸⁷⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- [hpx::ref](#)
- [hpx::cref](#)

```
template<typename T, typename Enable = void>
```

```
struct reference_wrapper : public std::reference_wrapper<T>
```

hpx::ref

Defined in header [hpx/functional.hpp](#)⁸⁷¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- [hpx::reference_wrapper](#)
- [hpx::cref](#)

```
template<typename T>
```

```
constexpr reference_wrapper<T> hpx::ref(T &val) noexcept
```

```
template<typename T>
```

```
void hpx::ref(T const&&) = delete
```

```
template<typename T>
```

```
constexpr reference_wrapper<T> hpx::ref(reference_wrapper<T> val) noexcept
```

```
template<typename T>
```

```
reference_wrapper<T> hpx::ref(T &&val) noexcept
```

hpx::cref

Defined in header [hpx/functional.hpp](#)⁸⁷².

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- [hpx::reference_wrapper](#)

⁸⁷⁰ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/functional.hpp

⁸⁷¹ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/functional.hpp

⁸⁷² http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/functional.hpp

hpp
hpp

- `hpx::ref`

```
template<typename T>
constexpr reference_wrapper<T const> hpx::cref(T const &val) noexcept
```

```
template<typename T>
void hpx::cref(T const&&) = delete
```

```
template<typename T>
constexpr reference_wrapper<T const> hpx::cref(reference_wrapper<T> val) noexcept
```

hpx::function

Defined in header `hpx/functional.hpp`⁸⁷³.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<typename Sig, bool Serializable = false>
```

```
class function
```

Class template `hpx::function` is a general-purpose polymorphic function wrapper. Instances of `hpx::function` can store, copy, and invoke any CopyConstructible Callable target⁸⁷³; functions, lambda expressions, bind expressions, or other function objects, as well as pointers to member functions and pointers to data members. The stored callable object is called the target of `hpx::function`. If an `hpx::function` contains no target, it is called empty. Invoking the target of an empty `hpx::function` results in `hpx::error::bad_function_call` exception being thrown. `hpx::function` satisfies the requirements of CopyConstructible and CopyAssignable.

hpx::is_bind_expression

Defined in header `hpx/functional.hpp`⁸⁷⁴.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<typename T>
```

```
struct is_bind_expression : public std::is_bind_expression<T>
```

If T is the type produced by a call to `hpx::bind`, this template is derived from `std::true_type`. For any other type, this template is derived from `std::false_type`.

This template may be specialized for a user-defined type T to implement *UnaryTypeTrait* with base characteristic of `std::true_type` to indicate that T should be treated by `hpx::bind` as if it were the type of a bind subexpression: when a bind-generated function object is invoked, a bound argument of this type will be invoked as a function object and will be given all the unbound arguments passed to the bind-generated object.

Subclassed by `hpx::is_bind_expression< T const >`

⁸⁷³ [http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/functional.](http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/functional.hpp)

⁸⁷⁴ [http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/functional.](http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/functional.hpp)
hpp

hpx::is_placeholder

Defined in header [hpx/functional.hpp](#)⁸⁷⁵.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<typename T>
```

```
struct is_placeholder
```

If *T* is a standard, Boost, or HPX placeholder (*_1*, *_2*, *_3*, ...) then this template is derived from `std::integral_constant<int,1>`, `std::integral_constant<int,2>`, `std::integral_constant<int,3>`, respectively. Otherwise, it is derived from `std::integral_constant<int,0>`.

The template may be specialized for any user-defined *T* type: the specialization must satisfy *UnaryTypeTrait* with base characteristic of `std::integral_constant<int,N>` with *N*>0 to indicate that *T* should be treated as *N*'th placeholder type. `hpx::bind` uses `hpx::is_placeholder` to detect placeholders for unbound arguments.

hpx::experimental::scope_success

Defined in header `hpx/experimental/scope.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<typename F>
```

```
auto hpx::experimental::scope_success(F &&f)
```

The class template `scope_success` is a general-purpose scope guard intended to call its exit function when a scope is normally exited.

Template Parameters

F – type of stored exit function

Parameters

f – stored exit function

hpx::experimental::scope_exit

Defined in header `hpx/experimental/scope.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<typename F>
```

```
auto hpx::experimental::scope_exit(F &&f)
```

The class template `scope_exit` is a general-purpose scope guard intended to call its exit function when a scope is exited.

Template Parameters

F – type of stored exit function

Parameters

f – stored exit function

⁸⁷⁵ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/functional.hpp

hpx::experimental::scope_fail

Defined in header `hpx/experimental/scope.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<typename F>
auto hpx::experimental::scope_fail(F &&f)
```

The class template `scope_fail` is a general-purpose scope guard intended to call its exit function when a scope is exited via an exception.

Template Parameters

F – type of stored exit function

Parameters

f – stored exit function

futures

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx::promise

Defined in header `hpx/future.hpp`⁸⁷⁶.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<typename R>
```

```
class promise
```

The class template `hpx::promise` provides a facility to store a value or an exception that is later acquired asynchronously via a `hpx::future` object created by the `hpx::promise` object. Note that the `hpx::promise` object is meant to be used only once. Each promise is associated with a shared state, which contains some state information and a result which may be not yet evaluated, evaluated to a value (possibly void) or evaluated to an exception. A promise may do three things with the shared state:

- **make ready:** the promise stores the result or the exception in the shared state. Marks the state ready and unblocks any thread waiting on a future associated with the shared state.
- **release:** the promise gives up its reference to the shared state. If this was the last such reference, the shared state is destroyed. Unless this was a shared state created by `hpx::async` which is not yet ready, this operation does not block.
- **abandon:** the promise stores the exception of type `hpx::future_error` with error code `hpx::error::broken_promise`, makes the shared state ready, and then releases it. The promise is the “push” end of the promise-future communication channel: the operation that stores a value in the shared state synchronizes-with (as defined in `hpx::memory_order`) the successful return from any function that is waiting on the shared state (such as `hpx::future::get`). Concurrent access to the same shared state may conflict otherwise: for example multiple callers of `hpx::shared_future::get` must either all be read-only or provide external synchronization.

⁸⁷⁶ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

hpx/futures/future_fwd.hpp

Defined in header `hpx/futures/future_fwd.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Top level HPX namespace.

template<typename **R**>

class **future**

#include <future_fwd.hpp> The class template `hpx::future` provides a mechanism to access the result of asynchronous operations:

- An asynchronous operation (created via `hpx::async`, `hpx::packaged_task`, or `hpx::promise`) can provide a `hpx::future` object to the creator of that asynchronous operation.
- The creator of the asynchronous operation can then use a variety of methods to query, wait for, or extract a value from the `hpx::future`. These methods may block if the asynchronous operation has not yet provided a value.
- When the asynchronous operation is ready to send a result to the creator, it can do so by modifying shared state (e.g. `hpx::promise::set_value`) that is linked to the creator's `hpx::future`. Note that `hpx::future` references shared state that is not shared with any other asynchronous return objects (as opposed to `hpx::shared_future`).

template<typename **R**>

class **promise** : public `hpx::detail::promise_base<R>`

#include <promise.hpp>

template<typename **R**>

class **shared_future**

#include <future_fwd.hpp> The class template `hpx::shared_future` provides a mechanism to access the result of asynchronous operations, similar to `hpx::future`, except that multiple threads are allowed to wait for the same shared state. Unlike `hpx::future`, which is only moveable (so only one instance can refer to any particular asynchronous result), `hpx::shared_future` is copyable and multiple shared future objects may refer to the same shared state. Access to the same shared state from multiple threads is safe if each thread does it through its own copy of a `shared_future` object.

namespace **lcos**

namespace **lcos**

hpx::packaged_task

Defined in header `hpx/future.hpp`⁸⁷⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

template<typename **Sig**>

class **packaged_task**

The class template `hpx::packaged_task` wraps any Callable⁸ target (function, lambda expression, bind expression, or another function object) so that it can be invoked asynchronously. Its return value or exception thrown is stored in a shared state which can be accessed through `hpx::future` objects. Just like `hpx::function`, `hpx::packaged_task` is a polymorphic, allocator-aware container: the stored callable target may be allocated on heap or with a provided allocator.

hpx::future

Defined in header `hpx/future.hpp`⁸⁷⁸.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::shared_future`
- `hpx::make_future`
- `hpx::make_shared_future`
- `hpx::make_ready_future`
- `hpx::make_ready_future_alloc`
- `hpx::make_ready_future_at`
- `hpx::make_ready_future_after`
- `hpx::make_exceptional_future`

template<typename **R**>

class **future** : public `hpx::lcos::detail::future_base<future<R>, R>`

The class template `hpx::future` provides a mechanism to access the result of asynchronous operations:

- An asynchronous operation (created via `hpx::async`, `hpx::packaged_task`, or `hpx::promise`) can provide a `hpx::future` object to the creator of that asynchronous operation.
- The creator of the asynchronous operation can then use a variety of methods to query, wait for, or extract a value from the `hpx::future`. These methods may block if the asynchronous operation has not yet provided a value.
- When the asynchronous operation is ready to send a result to the creator, it can do so by modifying shared state (e.g. `hpx::promise::set_value`) that is linked to the creator's `hpx::future`. Note that `hpx::future` references shared state that is not shared with any other asynchronous return objects (as opposed to `hpx::shared_future`).

⁸⁷⁷ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

⁸⁷⁸ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

hpx::shared_future

Defined in header [hpx/future.hpp](#)⁸⁷⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::future`
- `hpx::make_future`
- `hpx::make_shared_future`
- `hpx::make_ready_future`
- `hpx::make_ready_future_alloc`
- `hpx::make_ready_future_at`
- `hpx::make_ready_future_after`
- `hpx::make_exceptional_future`

template<typename **R**>

class **shared_future** : public `hpx::lcos::detail::future_base<shared_future<R>, R>`

The class template `hpx::shared_future` provides a mechanism to access the result of asynchronous operations, similar to `hpx::future`, except that multiple threads are allowed to wait for the same shared state. Unlike `hpx::future`, which is only moveable (so only one instance can refer to any particular asynchronous result), `hpx::shared_future` is copyable and multiple shared future objects may refer to the same shared state. Access to the same shared state from multiple threads is safe if each thread does it through its own copy of a `shared_future` object.

hpx::make_future

Defined in header [hpx/future.hpp](#)⁸⁸⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::future`
- `hpx::shared_future`
- `hpx::make_shared_future`
- `hpx::make_ready_future`
- `hpx::make_ready_future_alloc`
- `hpx::make_ready_future_at`
- `hpx::make_ready_future_after`
- `hpx::make_exceptional_future`

template<typename **R**, typename **U**>

⁸⁷⁹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

⁸⁸⁰ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

hpx::future<R> *hpx::make_future(hpx::future<U> &&f)*

Converts any future of type U to any other future of type R based on an existing conversion path from U to R.

template<typename **R**, typename **U**, typename **Conv**>

hpx::future<R> *hpx::make_future(hpx::future<U> &&f, Conv &&conv)*

Converts any future of type U to any other future of type R based on a given conversion function: R conv(U).

template<typename **R**, typename **U**>

hpx::future<R> *hpx::make_future(hpx::shared_future<U> f)*

Converts any *shared_future* of type U to any other future of type R based on an existing conversion path from U to R.

template<typename **R**, typename **U**, typename **Conv**>

hpx::future<R> *hpx::make_future(hpx::shared_future<U> f, Conv &&conv)*

Converts any future of type U to any other future of type R based on an existing conversion path from U to R.

hpx::make_shared_future

Defined in header [hpx/future.hpp](#)⁸⁸¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::future*
- *hpx::shared_future*
- *hpx::make_future*
- *hpx::make_ready_future*
- *hpx::make_ready_future_alloc*
- *hpx::make_ready_future_at*
- *hpx::make_ready_future_after*
- *hpx::make_exceptional_future*

template<typename **R**>

hpx::shared_future<R> *hpx::make_shared_future(hpx::future<R> &&f) noexcept*

Converts any future or *shared_future* of type T to a corresponding *shared_future* of type T.

template<typename **R**>

hpx::shared_future<R> &*hpx::make_shared_future(hpx::shared_future<R> &f) noexcept*

Converts any future or *shared_future* of type T to a corresponding *shared_future* of type T.

template<typename **R**>

hpx::shared_future<R> &&*hpx::make_shared_future(hpx::shared_future<R> &&f) noexcept*

Converts any future or *shared_future* of type T to a corresponding *shared_future* of type T.

template<typename **R**>

hpx::shared_future<R> const &*hpx::make_shared_future(hpx::shared_future<R> const &f) noexcept*

Converts any future or *shared_future* of type T to a corresponding *shared_future* of type T.

⁸⁸¹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/future.hpp>

hpx::make_ready_future

Defined in header [hpx/future.hpp](#)⁸⁸².

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::future`
- `hpx::shared_future`
- `hpx::make_future`
- `hpx::make_shared_future`
- `hpx::make_ready_future_alloc`
- `hpx::make_ready_future_at`
- `hpx::make_ready_future_after`
- `hpx::make_exceptional_future`

```
template<typename T, typename ...Ts>  
std::enable_if_t<std::is_constructible_v<T, Ts&&...> || std::is_void_v<T>, future<T>> hpx::make_ready_future(Ts&&...  
ts)
```

The function creates a shared state that is immediately ready and returns a future associated with that shared state. For the returned future, `valid() == true` and `is_ready() == true`.

```
template<int DeductionGuard = 0, typename T>  
future<hpx::util::decay_unwrap_t<T>> hpx::make_ready_future(T &&init)
```

The function creates a shared state that is immediately ready and returns a future associated with that shared state. For the returned future, `valid() == true` and `is_ready() == true`.

```
future<void> hpx::make_ready_future()
```

The function creates a shared state that is immediately ready and returns a future associated with that shared state. For the returned future, `valid() == true` and `is_ready() == true`.

hpx::make_ready_future_alloc

Defined in header [hpx/future.hpp](#)⁸⁸³.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::future`
- `hpx::shared_future`
- `hpx::make_future`
- `hpx::make_shared_future`
- `hpx::make_ready_future`
- `hpx::make_ready_future_at`
- `hpx::make_ready_future_after`

⁸⁸² <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

⁸⁸³ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

- `hpx::make_exceptional_future`

```
template<typename T, typename Allocator, typename ...Ts>
std::enable_if_t<std::is_constructible_v<T, Ts&&...> || std::is_void_v<T>, future<T>> hpx::make_ready_future_alloc(Allocator const
&a,
Ts&&...
ts)
```

Creates a pre-initialized future object with allocator (extension)

```
template<int DeductionGuard = 0, typename Allocator, typename T>
future<hpx::util::decay_unwrap_t<T>> hpx::make_ready_future_alloc(Allocator const &a, T &&init)
```

```
template<typename Allocator>
inline future<void> hpx::make_ready_future_alloc(Allocator const &a)
```

hpx::make_ready_future_at

Defined in header `hpx/future.hpp`⁸⁸⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::future`
- `hpx::shared_future`
- `hpx::make_future`
- `hpx::make_shared_future`
- `hpx::make_ready_future`
- `hpx::make_ready_future_alloc`
- `hpx::make_ready_future_after`
- `hpx::make_exceptional_future`

```
template<int DeductionGuard = 0, typename T>
future<hpx::util::decay_unwrap_t<T>> hpx::make_ready_future_at(hpx::chrono::steady_time_point const
&abs_time, T &&init)
```

Creates a pre-initialized future object which gets ready at a given point in time (extension)

```
inline future<void> hpx::make_ready_future_at(hpx::chrono::steady_time_point const &abs_time)
```

Creates a pre-initialized future object which gets ready at a given point in time (extension)

⁸⁸⁴ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

hpx::make_ready_future_after

Defined in header [hpx/future.hpp](#)⁸⁸⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::future`
- `hpx::shared_future`
- `hpx::make_future`
- `hpx::make_shared_future`
- `hpx::make_ready_future`
- `hpx::make_ready_future_alloc`
- `hpx::make_ready_future_at`
- `hpx::make_exceptional_future`

```
template<int DeductionGuard = 0, typename T>
future<hpx::util::decay_unwrap_t<T>> hpx::make_ready_future_after(hpx::chrono::steady_duration const
                                                                    &rel_time, T &&init)
```

Creates a pre-initialized future object which gets ready after a given point in time (extension)

```
inline future<void> hpx::make_ready_future_after(hpx::chrono::steady_duration const &rel_time)
```

Creates a pre-initialized future object which gets ready after a given point in time (extension)

hpx::make_exceptional_future

Defined in header [hpx/future.hpp](#)⁸⁸⁶.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::future`
- `hpx::shared_future`
- `hpx::make_future`
- `hpx::make_shared_future`
- `hpx::make_ready_future`
- `hpx::make_ready_future_alloc`
- `hpx::make_ready_future_at`
- `hpx::make_ready_future_after`

```
template<typename T>
future<T> hpx::make_exceptional_future(std::exception_ptr const &e)
```

Creates a pre-initialized future object which holds the given error (extension)

```
template<typename T, typename E>
```

⁸⁸⁵ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

⁸⁸⁶ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/future.hpp>

future<*T*> **hpx::make_exceptional_future**(*E* *e*)

Creates a pre-initialized future object which holds the given error (extension)

io_service

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/io_service/io_service_pool.hpp

Defined in header `hpx/io_service/io_service_pool.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **util**

class **io_service_pool**

#include <io_service_pool.hpp> A pool of `io_service` objects.

Public Functions

explicit **io_service_pool**(*std::size_t* pool_size = 2, *threads::policies::callback_notifier* const ¬ifier = *threads::policies::callback_notifier*(), char const *pool_name = "", char const *name_postfix = "")

Construct the `io_service` pool.

Parameters

- **pool_size** – [in] The number of threads to run to serve incoming requests
- **notifier** – [in]
- **pool_name** – [in]
- **name_postfix** – [in]

explicit **io_service_pool**(*threads::policies::callback_notifier* const ¬ifier, char const *pool_name = "", char const *name_postfix = "")

Construct the `io_service` pool.

Parameters

- **notifier** – [in]
- **pool_name** – [in]
- **name_postfix** – [in]

io_service_pool(*io_service_pool* const&) = delete

io_service_pool(*io_service_pool*&&) = delete

io_service_pool &**operator**=(*io_service_pool* const&) = delete

io_service_pool &**operator**=(*io_service_pool*&&) = delete

~io_service_pool()

bool **run**(bool join_threads = true, *barrier* *startup = nullptr) const

Run all io_service objects in the pool. If join_threads is true this will also wait for all threads to complete

bool **run**(std::size_t num_threads, bool join_threads = true, *barrier* *startup = nullptr) const

Run all io_service objects in the pool. If join_threads is true this will also wait for all threads to complete

void **stop**() const

Stop all io_service objects in the pool.

void **join**() const

Join all io_service threads in the pool.

void **clear**() const

Clear all internal data structures.

void **wait**() const

Wait for all work to be done.

bool **stopped**() const

::asio::io_context &**get_io_service**(int index = -1) const

Get an io_service to use.

std::thread &**get_os_thread_handle**(std::size_t thread_num) const

access underlying thread handle

std::size_t **size**() const noexcept

Get number of threads associated with this I/O service.

void **thread_run**(std::size_t index, *barrier* *startup = nullptr) const

Activate the thread *index* for this thread pool.

char const ***get_name**() const noexcept

Return name of this pool.

void **init**(std::size_t pool_size) const

Private Members

detail::io_service_pool_base ***pool_**

lcos_local

See *Public API* for a list of names and headers that are part of the public HPX API.

hpx::lcos::local::and_gate

Defined in header `hpx/modules/lcos_local.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
struct and_gate : public hpx::lcos::local::base_and_gate<hpx::no_mutex>
```

An AND-gate synchronization primitive that fires once all of its expected inputs have been received.

Note: This type is not thread-safe. It has to be protected from concurrent access by different threads by the code using instances of this type.

hpx/lcos_local/trigger.hpp

Defined in header `hpx/lcos_local/trigger.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace lcos
```

```
        namespace local
```

```
            template<typename Mutex = hpx::spinlock>
```

```
            struct base_trigger
```

Public Functions

```
            inline base_trigger() noexcept
```

```
            inline base_trigger(base_trigger &&rhs) noexcept
```

```
            inline base_trigger &operator=(base_trigger &&rhs) noexcept
```

```
            inline hpx::future<void> get_future(std::size_t *generation_value = nullptr, error_code &ec  
                                              = hpx::throws)
```

get a future allowing to wait for the trigger to fire

```
            inline bool set(error_code &ec = throws)
```

Trigger this object.

```
            inline void synchronize(std::size_t generation_value, char const *function_name =  
                                     "trigger::synchronize", error_code &ec = throws)
```

Wait for the generational counter to reach the requested stage.

```
            inline std::size_t next_generation()
```

```
            inline std::size_t generation() const
```

Protected Types

```
using mutex_type = Mutex
```

Protected Functions

```
inline bool trigger_conditions(error_code &ec = throws)
```

```
template<typename Lock>
```

```
inline void synchronize(std::size_t generation_value, Lock &l, char const *function_name =  
    "trigger::synchronize", error_code &ec = throws)
```

Private Types

```
using condition_list_type = hpx::detail::intrusive_list<condition_list_entry>
```

Private Functions

```
inline bool test_condition(std::size_t const generation_value) const noexcept
```

Private Members

```
mutable mutex_type mtx_
```

```
hpx::promise<void> promise_
```

```
std::size_t generation_
```

```
condition_list_type conditions_
```

```
struct condition_list_entry : public conditional_trigger
```

Public Functions

```
condition_list_entry() = default
```

Public Members

condition_list_entry ***prev** = nullptr

condition_list_entry ***next** = nullptr

struct **manage_condition**

Public Functions

inline **manage_condition**(*base_trigger* &gate, *condition_list_entry* &cond) noexcept

inline ~**manage_condition**()

template<typename **Condition**>

inline *hpx::future*<void> **get_future**(*Condition* &&func, *error_code* &ec = *hpx::throws*)

Public Members

base_trigger &**this_**

condition_list_entry &**e_**

struct **trigger** : public *hpx::lcos::local::base_trigger*<*hpx::no_mutex*>

Public Functions

trigger() = default

inline **trigger**(*trigger* &&rhs) noexcept

inline *trigger* &**operator=**(*trigger* &&rhs) noexcept

template<typename **Lock**>

inline void **synchronize**(*std::size_t* generation_value, *Lock* &l, char const *function_name =
"trigger::synchronize", *error_code* &ec = *throws*)

Private Types

using **base_type** = *base_trigger*<*hpx::no_mutex*>

pack_traversal

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/pack_traversal/pack_traversal.hpp

Defined in header `hpx/pack_traversal/pack_traversal.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **util**

Functions

```
template<typename Mapper, typename...  
T> < unspecified > map_pack (Mapper &&mapper, T &&... pack)
```

Maps the pack with the given mapper.

This function tries to visit all plain elements which may be wrapped in:

- homogeneous containers (`std::vector`, `std::list`)
- heterogeneous containers (`hpx::tuple`, `std::pair`, `std::array`) and re-assembles the pack with the result of the mapper. Mapping from one type to a different one is supported.

Elements that aren't accepted by the mapper are routed through and preserved through the hierarchy.

```
// Maps all integers to floats  
map_pack([](int value) {  
    return float(value);  
},  
1, hpx::make_tuple(2, std::vector<int>{3, 4}), 5);
```

Throws

`std::exception` – like objects which are thrown by an invocation to the mapper.

Parameters

- **mapper** – A callable object, which accept an arbitrary type and maps it to another type or the same one.
- **pack** – An arbitrary variadic pack which may contain any type.

Returns

The mapped element or in case the pack contains multiple elements, the pack is wrapped into a `hpx::tuple`.

hpx/pack_traversal/pack_traversal_async.hpp

Defined in header `hpx/pack_traversal/pack_traversal_async.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **util**

Functions

```
template<typename Visitor, typename ...T>
auto traverse_pack_async(Visitor &&visitor, T&&... pack) -> de-
    cltype(detail::apply_pack_transform_async(std::forward<Visitor>(visitor),
    std::forward<T>(pack)...))
```

Traverses the pack with the given visitor in an asynchronous way.

This function works in the same way as `traverse_pack`, however, we are able to suspend and continue the traversal at later time. Thus, we require a visitor callable object which provides three `operator()` overloads as depicted by the code sample below:

```
struct my_async_visitor {
    // The synchronous overload is called for each object. It may
    // return false to suspend the current control. In that case the
    // overload below is called.
    template <typename T>
    bool operator()(async_traverse_visit_tag, T&& element)
    {
        return true;
    }

    // The asynchronous overload is called when the synchronous
    // overload returned false. In addition to the current visited
    // element, the overload is called with a continuation callable
    // object which resumes the traversal when called later.
    template <typename T, typename N>
    void operator()(async_traverse_detach_tag, T&& element, N&& next)
    {
    }

    // The overload is called when the traversal was finished. The
    // whole pack which was traversed asynchronously is passed.
    template <typename T>
    void operator()(async_traverse_complete_tag, T&& pack)
    {
    }
};
```

See `traverse_pack` for a detailed description about the traversal behavior and capabilities.

Parameters

- **visitor** – A visitor object which provides the three `operator()` overloads that were described above. Additionally, the visitor must be compatible for referencing it from a `hpx::intrusive_ptr`. The visitor should have a virtual destructor!
- **pack** – The arbitrary parameter pack which is traversed asynchronously. Nested objects inside containers and tuple like types are traversed recursively.

Returns

A `hpx::intrusive_ptr` that references an instance of the given visitor object.

```
template<typename Allocator, typename Visitor, typename ...T>
auto traverse_pack_async_allocator(Allocator const &alloc, Visitor &&visitor, T&&... pack) ->
    de-
    cltype(detail::apply_pack_transform_async_allocator(alloc,
        std::forward<Visitor>(visitor), std::forward<T>(pack)...))
```

Traverses the pack with the given visitor in an asynchronous way.

This function works in the same way as `traverse_pack`, however, we are able to suspend and continue the traversal at later time. Thus, we require a visitor callable object which provides three `operator()` overloads as depicted by the code sample below:

```
struct my_async_visitor {
    // The synchronous overload is called for each object. It may
    // return false to suspend the current control. In that case the
    // overload below is called.
    template <typename T>
    bool operator()(async_traverse_visit_tag, T&& element)
    {
        return true;
    }

    // The asynchronous overload is called when the synchronous
    // overload returned false. In addition to the current visited
    // element, this overload receives a continuation callable object
    // which resumes the traversal when called later. The continuation
    // may be stored and called later or dropped completely to abort
    // the traversal early.
    template <typename T, typename N>
    void operator()(async_traverse_detach_tag, T&& element, N&& next)
    {
    }

    // This overload is called when the traversal is finished. The
    // complete pack that was traversed asynchronously is passed.
    template <typename T>
    void operator()(async_traverse_complete_tag, T&& pack)
    {
    }
};
```

See `traverse_pack` for a detailed description about the traversal behavior and capabilities.

Parameters

- **visitor** – A visitor object which provides the three `operator()` overloads that were described above. Additionally, the visitor must be compatible for referencing it from a `hpx::intrusive_ptr`. The visitor should have a virtual destructor!

- **pack** – The arbitrary parameter pack which is traversed asynchronously. Nested objects inside containers and tuple like types are traversed recursively.
- **alloc** – Allocator instance to use to create the traversal frame.

Returns

A `hpx::intrusive_ptr` that references an instance of the given visitor object.

hpx::functional::unwrap

Defined in header [hpx/unwrap.hpp](#)⁸⁸⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::functional::unwrap_n`
- `hpx::functional::unwrap_all`
- `hpx::unwrap`
- `hpx::unwrap_n`
- `hpx::unwrap_all`
- `hpx::unwrapping`
- `hpx::unwrapping_n`
- `hpx::unwrapping_all`

struct unwrap

A helper function object for functionally invoking `hpx::unwrap`. For more information please refer to its documentation.

hpx::functional::unwrap_n

Defined in header [hpx/unwrap.hpp](#)⁸⁸⁸.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::functional::unwrap`
- `hpx::functional::unwrap_all`
- `hpx::unwrap`
- `hpx::unwrap_n`
- `hpx::unwrap_all`
- `hpx::unwrapping`
- `hpx::unwrapping_n`
- `hpx::unwrapping_all`

template<std::size_t **Depth**>

⁸⁸⁷ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/unwrap.hpp

⁸⁸⁸ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/unwrap.hpp

struct **unwrap_n**

A helper function object for functionally invoking `hpx::unwrap_n`. For more information please refer to its documentation.

hpx::functional::unwrap_all

Defined in header [hpx/unwrap.hpp](#)⁸⁸⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::functional::unwrap`
- `hpx::functional::unwrap_n`
- `hpx::unwrap`
- `hpx::unwrap_n`
- `hpx::unwrap_all`
- `hpx::unwrapping`
- `hpx::unwrapping_n`
- `hpx::unwrapping_all`

struct **unwrap_all**

A helper function object for functionally invoking `hpx::unwrap_all`. For more information please refer to its documentation.

hpx::unwrap

Defined in header [hpx/unwrap.hpp](#)⁸⁹⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::functional::unwrap`
- `hpx::functional::unwrap_n`
- `hpx::functional::unwrap_all`
- `hpx::unwrap_n`
- `hpx::unwrap_all`
- `hpx::unwrapping`
- `hpx::unwrapping_n`
- `hpx::unwrapping_all`

template<typename ...**Args**>

⁸⁸⁹ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/unwrap.hpp

⁸⁹⁰ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/unwrap.hpp


```
auto hpx::unwrap(Args&&... args) -> decltype(util::detail::unwrap_depth_impl<1U>(HPX_FORWARD(Args,
args)...))
```

A helper function for retrieving the actual result of any *hpx::future* like type which is wrapped in an arbitrary way.

Unwraps the given pack of arguments, so that any *hpx::future* object is replaced by its future result type in the argument pack:

- *hpx::future<int>* -> *int*
- *hpx::future<std::vector<float>>* -> *std::vector<float>*
- *std::vector<future<float>>* -> *std::vector<float>*

The function is capable of unwrapping *hpx::future* like objects that are wrapped inside any container or tuple like type, see *hpx::util::map_pack()* for a detailed description about which surrounding types are supported. Non *hpx::future* like types are permitted as arguments and passed through.

```
hpx::unwrap(hpx::make_ready_future(0));

// Multiple arguments hpx::tuple<int, int> i2 =
    hpx::unwrap(hpx::make_ready_future(1),
                hpx::make_ready_future(2));
```

Note: This function unwraps the given arguments until the first traversed nested *hpx::future* which corresponds to an unwrapping depth of one. See *hpx::unwrap_n()* for a function which unwraps the given arguments to a particular depth or *hpx::unwrap_all()* that unwraps all future like objects recursively which are contained in the arguments.

Parameters

args – the arguments that are unwrapped which may contain any arbitrary future or non-future type.

Throws

std::exception – like objects in case any of the given wrapped *hpx::future* objects were resolved through an exception. See *hpx::future::get()* for details.

Returns

Depending on the count of arguments this function returns a *hpx::tuple* containing the unwrapped arguments if multiple arguments are given. In case the function is called with a single argument, the argument is unwrapped and returned.

hpx::unwrap_n

Defined in header [hpx/unwrap.hpp](#)⁸⁹¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::functional::unwrap*
- *hpx::functional::unwrap_n*
- *hpx::functional::unwrap_all*

⁸⁹¹ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/unwrap.hpp

- `hpx::unwrap`
- `hpx::unwrap_all`
- `hpx::unwrapping`
- `hpx::unwrapping_n`
- `hpx::unwrapping_all`

```
template<std::size_t Depth, typename ...Args>  
auto hpx::unwrap_n(Args&&... args) ->  
    decltype(util::detail::unwrap_depth_impl<Depth>(HPX_FORWARD(Args, args)...))
```

An alternative version of `hpx::unwrap()`, which unwraps the given arguments to a certain depth of `hpx::future` like objects.

See `unwrap` for a detailed description.

Template Parameters

Depth – The count of `hpx::future` like objects which are unwrapped maximally.

`hpx::unwrap_all`

Defined in header `hpx/unwrap.hpp`⁸⁹².

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::functional::unwrap`
- `hpx::functional::unwrap_n`
- `hpx::functional::unwrap_all`
- `hpx::unwrap`
- `hpx::unwrap_n`
- `hpx::unwrapping`
- `hpx::unwrapping_n`
- `hpx::unwrapping_all`

```
template<typename ...Args>  
auto hpx::unwrap_all(Args&&... args) ->  
    decltype(util::detail::unwrap_depth_impl<0U>(HPX_FORWARD(Args, args)...))
```

An alternative version of `hpx::unwrap()`, which unwraps the given arguments recursively so that all contained `hpx::future` like objects are replaced by their actual value.

See `hpx::unwrap()` for a detailed description.

⁸⁹² http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/unwrap.hpp

hpx::unwrapping

Defined in header [hpx/unwrap.hpp](#)⁸⁹³.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::functional::unwrap`
- `hpx::functional::unwrap_n`
- `hpx::functional::unwrap_all`
- `hpx::unwrap`
- `hpx::unwrap_n`
- `hpx::unwrap_all`
- `hpx::unwrapping_n`
- `hpx::unwrapping_all`

template<typename T>

```
auto hpx::unwrapping(T &&callable) ->
    decltype(util::detail::functional_unwrap_depth_impl<1U>(HPX_FORWARD(T,
        callable)))
```

Returns a callable object which unwraps its arguments upon invocation using the `hpx::unwrap()` function and then passes the result to the given callable object.

```
    return left + right;
});

int i1 = callable(hpx::make_ready_future(1),
                  hpx::make_ready_future(2));
```

See `hpx::unwrap()` for a detailed description.

Parameters

callable – the callable object which is called with the result of the corresponding unwrap function.

hpx::unwrapping_n

Defined in header [hpx/unwrap.hpp](#)⁸⁹⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::functional::unwrap`
- `hpx::functional::unwrap_n`
- `hpx::functional::unwrap_all`
- `hpx::unwrap`
- `hpx::unwrap_n`

⁸⁹³ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/unwrap.hpp

⁸⁹⁴ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/unwrap.hpp

- `hpx::unwrap_all`
- `hpx::unwrapping`
- `hpx::unwrapping_all`

```
template<std::size_t Depth, typename T>
auto hpx::unwrapping_n(T &&callable) ->
    decltype(util::detail::functional_unwrap_depth_impl<Depth>(HPX_FORWARD(T,
        callable)))
```

Returns a callable object which unwraps its arguments upon invocation using the `hpx::unwrap_n()` function and then passes the result to the given callable object.

See `hpx::unwrapping()` for a detailed description.

`hpx::unwrapping_all`

Defined in header `hpx/unwrap.hpp`⁸⁹⁵.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::functional::unwrap`
- `hpx::functional::unwrap_n`
- `hpx::functional::unwrap_all`
- `hpx::unwrap`
- `hpx::unwrap_n`
- `hpx::unwrap_all`
- `hpx::unwrapping`
- `hpx::unwrapping_n`

```
template<typename T>
auto hpx::unwrapping_all(T &&callable) ->
    decltype(util::detail::functional_unwrap_depth_impl<0U>(HPX_FORWARD(T,
        callable)))
```

Returns a callable object which unwraps its arguments upon invocation using the `hpx::unwrap_all()` function and then passes the result to the given callable object.

See `hpx::unwrapping()` for a detailed description.

preprocessor

See *Public API* for a list of names and headers that are part of the public *HPX* API.

⁸⁹⁵ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/unwrap.hpp

hpx/preprocessor/strip_parens.hpp

Defined in header `hpx/preprocessor/strip_parens.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Defines

HPX_PP_STRIP_PARENS(X)

For any symbol *X*, this macro returns the same symbol from which potential outer parens have been removed. If no outer parens are found, this macros evaluates to *X* itself without error.

The original implementation of this macro is from Steven Watanbe as shown in <http://boost.2283326.n4.nabble.com/preprocessor-removing-parentheses-td2591973.html#a2591976>

```

HPX_PP_STRIP_PARENS(no_parens)
HPX_PP_STRIP_PARENS((with_parens))

```

Example Usage:

This produces the following output

```

no_parens
with_parens

```

Parameters

- **X** – Symbol to strip parens from

hpx/preprocessor/nargs.hpp

Defined in header `hpx/preprocessor/nargs.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Defines

HPX_PP_NARGS(...)

Expands to the number of arguments passed in

Example Usage:

```

HPX_PP_NARGS(hpx, pp, nargs)
HPX_PP_NARGS(hpx, pp)
HPX_PP_NARGS(hpx)

```

Expands to:

```
3
2
1
```

Parameters

- ... – The variadic number of arguments

hpx/preprocessor/stringize.hpp

Defined in header `hpx/preprocessor/stringize.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Defines

HPX_PP_STRINGIZE(X)

The `HPX_PP_STRINGIZE` macro stringizes its argument after it has been expanded.

The passed argument `X` will expand to `"X"`. Note that the stringizing operator (`#`) prevents arguments from expanding. This macro circumvents this shortcoming.

Parameters

- `X` – The text to be converted to a string literal

hpx/preprocessor/cat.hpp

Defined in header `hpx/preprocessor/cat.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Defines

HPX_PP_CAT(A, B)

Concatenates the tokens `A` and `B` into a single token. Evaluates to `AB`

Parameters

- `A` – First token
- `B` – Second token

hpx/preprocessor/expand.hpp

Defined in header `hpx/preprocessor/expand.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Defines

HPX_PP_EXPAND(X)

The `HPX_PP_EXPAND` macro performs a double macro-expansion on its argument.

This macro can be used to produce a delayed preprocessor expansion.

Example:

```
#define MACRO(a, b, c) (a)(b)(c)
#define ARGS() (1, 2, 3)

HPX_PP_EXPAND(MACRO ARGS()) // expands to (1)(2)(3)
```

Parameters

- **X** – Token to be expanded twice

resiliency

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/resiliency/replay_executor.hpp

Defined in header `hpx/resiliency/replay_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<typename BaseExecutor, typename Validator>
```

```
struct is_two_way_executor<hpx::resiliency::experimental::replay_executor<BaseExecutor, Validator>> : public  
true_type
```

```
template<typename BaseExecutor, typename Validator>
```

```
struct is_bulk_two_way_executor<hpx::resiliency::experimental::replay_executor<BaseExecutor, Validator>> :  
public true_type
```

```
namespace hpx
```

```
namespace execution
```

```
namespace experimental
```

```
template<typename BaseExecutor,  
typename Validator> replay_executor< BaseExecutor,  
Validator > > : public true_type
```

```
template<typename BaseExecutor,  
typename Validator> replay_executor< BaseExecutor,  
Validator > > : public true_type
```

namespace **resiliency**

namespace **experimental**

Functions

```
template<executor_any BaseExecutor, typename Validate>  
replay_executor<BaseExecutor, std::decay_t<Validate>> make_replay_executor(BaseExecutor  
                                                                    &exec,  
                                                                    std::size_t n,  
                                                                    Validate  
                                                                    &&validate)
```

```
template<executor_any BaseExecutor>  
replay_executor<BaseExecutor, detail::replay_validator> make_replay_executor(BaseExecutor  
                                                                    &exec,  
                                                                    std::size_t n)
```

```
template<typename BaseExecutor, typename Validate>
```

```
class replay_executor
```

Public Types

```
using execution_category = hpx::traits::executor_execution_category_t<BaseExecutor>
```

```
using executor_parameters_type = hpx::traits::executor_parameters_type_t<BaseExecutor>
```

```
template<typename Result>
```

```
using future_type = hpx::traits::executor_future_t<BaseExecutor, Result>
```


Public Functions

```

template<typename BaseExecutor_, typename F>
inline explicit replay_executor(BaseExecutor_ &&exec, std::size_t n, F &&f)

inline bool operator==(replay_executor const &rhs) const noexcept

inline bool operator!=(replay_executor const &rhs) const noexcept

inline constexpr replay_executor const &context() const noexcept

template<typename F, typename ...Ts>
inline decltype(auto) async_execute(F &&f, Ts&&... ts) const

template<typename F, typename S, typename ...Ts>
inline decltype(auto) bulk_async_execute(F &&f, S const &shape, Ts&&... ts) const

inline BaseExecutor const &get_executor() const

inline std::size_t get_replay_count() const

inline Validate const &get_validator() const

template<typename Tag, typename ...Args>
inline auto query(Tag tag, Args&&... args) const -> decltype(replay_executor<BaseExecutor,
    Validate>(std::declval<Tag>()(std::declval<BaseExecutor>()),
    std::forward<Args>(args)...), std::declval<std::size_t>(),
    std::declval<Validate>()))

```

Public Static Attributes

```

static constexpr int num_spread = 4

static constexpr int num_tasks = 128

```

Private Members

```

BaseExecutor exec_

std::size_t replay_count_

Validate validator_

```

hpx/resiliency/async_replicate_executor.hpp

Defined in header `hpx/resiliency/async_replicate_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace resiliency
```

```
        namespace experimental
```

Functions

```
template<typename Executor, typename Vote, typename Pred, typename F, typename ...Ts>  
decltype(auto) hpx_invoke(async_replicate_vote_validate_t, Executor &&exec, std::size_t n, Vote  
    &&vote, Pred &&pred, F &&f, Ts&&... ts)
```

ADL hook for `async_replicate_vote_validate` CPO with executor.

Asynchronously launches *f* exactly *n* times on *exec*, validates results with *pred*, then runs valid results through *vote*.

Parameters

- **tag** – CPO tag (*async_replicate_vote_validate_t*).
- **exec** – Executor to run *f* on.
- **n** – Number of replicated invocations.
- **vote** – Voting function selecting among valid results.
- **pred** – Predicate validating each invocation result.
- **f** – Callable to invoke asynchronously.
- **ts** – Arguments forwarded to *f*.

Returns

future with the voted-upon result.

```
template<typename Executor, typename Vote, typename F, typename ...Ts>  
decltype(auto) hpx_invoke(async_replicate_vote_t, Executor &&exec, std::size_t n, Vote &&vote,  
    F &&f, Ts&&... ts)
```

ADL hook for `async_replicate_vote` CPO with executor.

Asynchronously launches *f* exactly *n* times on *exec*, runs valid results through *vote*. Validation uses the default exception check.

Parameters

- **tag** – CPO tag (*async_replicate_vote_t*).
- **exec** – Executor to run *f* on.
- **n** – Number of replicated invocations.
- **vote** – Voting function selecting among valid results.
- **f** – Callable to invoke asynchronously.
- **ts** – Arguments forwarded to *f*.

Returns

future with the voted-upon result.

```
template<typename Executor, typename Pred, typename F, typename ...Ts>  
decltype(auto) hpx_invoke(async_replicate_validate_t, Executor &&exec, std::size_t n, Pred  
    &&pred, F &&f, Ts&&... ts)
```

ADL hook for `async_replicate_validate` CPO with executor.

Asynchronously launches *f* exactly *n* times on *exec*, validates results with *pred*. Returns the first valid result.

Parameters

- **tag** – CPO tag (*async_replicate_validate_t*).
- **exec** – Executor to run *f* on.
- **n** – Number of replicated invocations.
- **pred** – Predicate validating each invocation result.
- **f** – Callable to invoke asynchronously.
- **ts** – Arguments forwarded to *f*.

Returns

future with the first valid result.

```
template<typename Executor, typename F, typename ...Ts>
decltype(auto) hpx_invoke(async_replicate_t, Executor &&exec, std::size_t n, F &&f, Ts&&... ts)
```

ADL hook for *async_replicate* CPO with executor.

Asynchronously launches *f* exactly *n* times on *exec*. Validates by checking for exceptions. Returns the first valid result.

Parameters

- **tag** – CPO tag (*async_replicate_t*).
- **exec** – Executor to run *f* on.
- **n** – Number of replicated invocations.
- **f** – Callable to invoke asynchronously.
- **ts** – Arguments forwarded to *f*.

Returns

future with the first non-throwing result.

hpx/resiliency/async_replay_executor.hpp

Defined in header `hpx/resiliency/async_replay_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **resiliency**

namespace **experimental**

Functions

```
template<typename Executor, typename Pred, typename F, typename ...Ts>
decltype(auto) hpx_invoke(async_replay_validate_t, Executor &&exec, std::size_t n, Pred
&&pred, F &&f, Ts&&... ts)
```

ADL hook for *async_replay_validate* CPO with executor.

Asynchronously launches *f* on *exec*, validating results with *pred*. Repeats on error up to *n* times (aborts on *abort_replay_exception*).

Parameters

- **tag** – CPO tag (*async_replay_validate_t*).
- **exec** – Executor to run *f* on.
- **n** – Maximum number of retry attempts.

- **pred** – Predicate validating each invocation result.
- **f** – Callable to invoke asynchronously.
- **ts** – Arguments forwarded to *f*.

Returns

future with the first valid result of *f*.

```
template<typename Executor, typename F, typename ...Ts>  
decltype(auto) hpx_invoke(async_replay_t, Executor &&exec, std::size_t n, F &&f, Ts&&... ts)
```

ADL hook for *async_replay* CPO with executor.

Asynchronously launches *f* on *exec*, repeating on error up to *n* times (aborts on *abort_replay_exception*).

Parameters

- **tag** – CPO tag (*async_replay_t*).
- **exec** – Executor to run *f* on.
- **n** – Maximum number of retry attempts.
- **f** – Callable to invoke asynchronously.
- **ts** – Arguments forwarded to *f*.

Returns

future with the first successful result of *f*.

hpx/resiliency/async_replay.hpp

Defined in header `hpx/resiliency/async_replay.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **resiliency**

namespace **experimental**

Functions

```
template<typename Pred, typename F, typename ...Ts>  
hpx::future<hpx::util::detail::invoke_deferred_result_t<F, Ts...>> hpx_invoke(async_replay_validate_t,  
                                                                    std::size_t n, Pred  
                                                                    &&pred, F &&f,  
                                                                    Ts&&... ts)
```

ADL hook for *async_replay_validate* CPO.

Asynchronously launches *f*, validating results with *pred*. Repeats on error up to *n* times (aborts on *abort_replay_exception*).

Parameters

- **tag** – CPO tag (*async_replay_validate_t*).
- **n** – Maximum number of retry attempts.
- **pred** – Predicate validating each invocation result.
- **f** – Callable to invoke asynchronously.
- **ts** – Arguments forwarded to *f*.

Returns

future with the first valid result of *f*.

```
template<typename F, typename ...Ts>
hpx::future<hpx::util::detail::invoke_deferred_result_t<F, Ts...>> hpx_invoke(async_replay_t,
                                                                    std::size_t n, F
                                                                    &&f, Ts&&... ts)
```

ADL hook for `async_replay` CPO.

Asynchronously launches *f*, repeating on error up to *n* times (aborts on `abort_replay_exception`).

Parameters

- **tag** – CPO tag (*async_replay_t*).
- **n** – Maximum number of retry attempts.
- **f** – Callable to invoke asynchronously.
- **ts** – Arguments forwarded to *f*.

Returns

future with the first successful result of *f*.

hpx/resiliency/replicate_executor.hpp

Defined in header `hpx/resiliency/replicate_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<typename BaseExecutor, typename Voter, typename Validator>
```

```
struct is_two_way_executor<hpx::resiliency::experimental::replicate_executor<BaseExecutor, Voter,
Validator>> : public true_type
```

```
template<typename BaseExecutor, typename Voter, typename Validator>
```

```
struct is_bulk_two_way_executor<hpx::resiliency::experimental::replicate_executor<BaseExecutor, Voter,
Validator>> : public true_type
```

```
namespace hpx
```

```
    namespace execution
```

```
        namespace experimental
```

```
            template<typename BaseExecutor, typename Voter,
                    typename Validator> replicate_executor< BaseExecutor, Voter,
                    Validator > > : public true_type
```

```
            template<typename BaseExecutor, typename Voter,
                    typename Validator> replicate_executor< BaseExecutor, Voter,
                    Validator > > : public true_type
```

```
        namespace resiliency
```

```
            namespace experimental
```

Functions

```
template<executor_any BaseExecutor, typename Voter, typename Validate>
replicate_executor<BaseExecutor, std::decay_t<Voter>, std::decay_t<Validate>> make_replicate_executor(BaseExecutor
&exe
std::s
n,
Voter
&&v
Val-
i-
date
&&v
i-
date)
```

```
template<executor_any BaseExecutor, typename Validate>
replicate_executor<BaseExecutor, detail::replicate_voter, std::decay_t<Validate>> make_replicate_executor(BaseExecutor
&ex
std::
n,
Val-
i-
date
&&
i-
date
```

```
template<executor_any BaseExecutor>
replicate_executor<BaseExecutor, detail::replicate_voter, detail::replicate_validator> make_replicate_executor(BaseExecutor
&
st
n,
```

```
template<typename BaseExecutor, typename Vote, typename Validate>
class replicate_executor
```

Public Types

```
using execution_category = hpx::traits::executor_execution_category_t<BaseExecutor>
```

```
using executor_parameters_type = hpx::traits::executor_parameters_type_t<BaseExecutor>
```

```
template<typename Result>
```

```
using future_type = hpx::traits::executor_future_t<BaseExecutor, Result>
```

Public Functions

```

template<typename BaseExecutor_, typename V, typename F>
inline explicit replicate_executor(BaseExecutor_ &&exec, std::size_t n, V &&v, F &&f)

inline bool operator==(replicate_executor const &rhs) const noexcept

inline bool operator!=(replicate_executor const &rhs) const noexcept

inline constexpr replicate_executor const &context() const noexcept

template<typename F, typename ...Ts>
inline decltype(auto) async_execute(F &&f, Ts&&... ts) const

template<typename F, typename S, typename ...Ts>
inline decltype(auto) bulk_async_execute(F &&f, S const &shape, Ts&&... ts) const

inline BaseExecutor const &get_executor() const

inline std::size_t get_replicate_count() const

inline Vote const &get_voter() const

inline Validate const &get_validator() const

template<typename Tag, typename ...Args>
inline auto query(Tag tag, Args&&... args) const ->
    decltype(replicate_executor<BaseExecutor, Vote,
        Validate>(std::declval<Tag>()(std::declval<BaseExecutor>()),
            std::forward<Args>(args)...), std::declval<std::size_t>(),
            std::declval<Vote>(), std::declval<Validate>()))

```

Public Static Attributes

```

static constexpr int num_spread = 4

static constexpr int num_tasks = 128

```

Private Members

```

BaseExecutor exec_

std::size_t replicate_count_

Vote voter_

Validate validator_

```

hpx/resiliency/resiliency_cpos.hpp

Defined in header `hpx/resiliency/resiliency_cpos.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace resiliency
```

```
        namespace experimental
```

Variables

```
hpx::resiliency::experimental::async_replay_validate_t async_replay_validate
```

```
hpx::resiliency::experimental::async_replay_t async_replay
```

```
hpx::resiliency::experimental::dataflow_replay_validate_t dataflow_replay_validate
```

```
hpx::resiliency::experimental::dataflow_replay_t dataflow_replay
```

```
hpx::resiliency::experimental::async_replicate_vote_validate_t  
async_replicate_vote_validate
```

```
hpx::resiliency::experimental::async_replicate_vote_t async_replicate_vote
```

```
hpx::resiliency::experimental::async_replicate_validate_t async_replicate_validate
```

```
hpx::resiliency::experimental::async_replicate_t async_replicate
```

```
hpx::resiliency::experimental::dataflow_replicate_vote_validate_t  
dataflow_replicate_vote_validate
```

```
hpx::resiliency::experimental::dataflow_replicate_vote_t dataflow_replicate_vote
```

```
hpx::resiliency::experimental::dataflow_replicate_validate_t dataflow_replicate_validate
```

```
hpx::resiliency::experimental::dataflow_replicate_t dataflow_replicate
```

```
struct async_replay_t : public
```

```
hpx::resiliency::experimental::detail::dispatch_cpo<async_replay_t>
```

```
    #include <resiliency_cpos.hpp> Customization point for asynchronously launching given func-  
    tion f repeatedly. Repeat launching on error exactly n times (except if abort_replay_exception is  
    thrown).
```



```
struct async_replay_validate_t : public
hpx::resiliency::experimental::detail::dispatch_cpo<async_replay_validate_t>
    #include <resiliency_cpos.hpp> Customization point for asynchronously launching the given
    function f. repeatedly. Verify the result of those invocations using the given predicate pred. Repeat
    launching on error exactly n times (except if abort_replay_exception is thrown).
```

```
struct async_replicate_t : public
hpx::resiliency::experimental::detail::dispatch_cpo<async_replicate_t>
    #include <resiliency_cpos.hpp> Customization point for asynchronously launching the given
    function f exactly n times concurrently. Verify the result of those invocations by checking for
    exception. Return the first valid result.
```

```
struct async_replicate_validate_t : public
hpx::resiliency::experimental::detail::dispatch_cpo<async_replicate_validate_t>
    #include <resiliency_cpos.hpp> Customization point for asynchronously launching the given
    function f exactly n times concurrently. Verify the result of those invocations using the given
    predicate pred. Return the first valid result.
```

```
struct async_replicate_vote_t : public
hpx::resiliency::experimental::detail::dispatch_cpo<async_replicate_vote_t>
    #include <resiliency_cpos.hpp> Customization point for asynchronously launching the given
    function f exactly n times concurrently. Verify the result of those invocations using the given
    predicate pred. Run all the valid results against a user provided voting function. Return the valid
    output.
```

```
struct async_replicate_vote_validate_t : public
hpx::resiliency::experimental::detail::dispatch_cpo<async_replicate_vote_validate_t>
    #include <resiliency_cpos.hpp> Customization point for asynchronously launching the given
    function f exactly n times concurrently. Verify the result of those invocations using the given
    predicate pred. Run all the valid results against a user provided voting function. Return the valid
    output.
```

```
struct dataflow_replay_t : public
hpx::resiliency::experimental::detail::tag_deferred<dataflow_replay_t, async_replay_t>
    #include <resiliency_cpos.hpp> Customization point for asynchronously launching the given
    function f. repeatedly. Repeat launching on error exactly n times.

    Delay the invocation of f if any of the arguments to f are futures.
```

```
struct dataflow_replay_validate_t : public
hpx::resiliency::experimental::detail::tag_deferred<dataflow_replay_validate_t,
async_replay_validate_t>
    #include <resiliency_cpos.hpp> Customization point for asynchronously launching the given
    function f. repeatedly. Verify the result of those invocations using the given predicate pred. Repeat
    launching on error exactly n times.

    Delay the invocation of f if any of the arguments to f are futures.
```

```
struct dataflow_replicate_t : public
hpx::resiliency::experimental::detail::tag_deferred<dataflow_replicate_t, async_replicate_t>
    #include <resiliency_cpos.hpp> Customization point for asynchronously launching the given
    function f exactly n times concurrently. Return the first valid result.
```

Delay the invocation of f if any of the arguments to f are futures.

```
struct dataflow_replicate_validate_t : public
hpx::resiliency::experimental::detail::tag_deferred<dataflow_replicate_validate_t,
async_replicate_validate_t>
#include <resiliency_cpos.hpp> Customization point for asynchronously launching the given
function  $f$  exactly  $n$  times concurrently. Verify the result of those invocations using the given
predicate  $pred$ . Return the first valid result.
```

Delay the invocation of f if any of the arguments to f are futures.

```
struct dataflow_replicate_vote_t : public
hpx::resiliency::experimental::detail::tag_deferred<dataflow_replicate_vote_t,
async_replicate_vote_t>
#include <resiliency_cpos.hpp> Customization point for asynchronously launching the given
function  $f$  exactly  $n$  times concurrently. Run all the valid results against a user provided voting
function. Return the valid output.
```

Delay the invocation of f if any of the arguments to f are futures.

```
struct dataflow_replicate_vote_validate_t : public
hpx::resiliency::experimental::detail::tag_deferred<dataflow_replicate_vote_validate_t,
async_replicate_vote_validate_t>
#include <resiliency_cpos.hpp> Customization point for asynchronously launching the given
function  $f$  exactly  $n$  times concurrently. Run all the valid results against a user provided voting
function. Return the valid output.
```

Delay the invocation of f if any of the arguments to f are futures.

hpx/resiliency/async_replicate.hpp

Defined in header `hpx/resiliency/async_replicate.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **resiliency**

namespace **experimental**

Functions

```
template<typename Vote, typename Pred, typename F, typename ...Ts>
hpx::future<hpx::util::detail::invoke_deferred_result_t<F, Ts...>> hpx_invoke(async_replicate_vote_validate_t,
std::size_t n, Vote
&&vote, Pred
&&pred, F &&f,
Ts&&... ts)
```

ADL hook for `async_replicate_vote_validate` CPO.

Asynchronously launches *f* exactly *n* times, validates results with *pred*, then runs valid results through *vote*.

Parameters

- **tag** – CPO tag (*async_replicate_vote_validate_t*).
- **n** – Number of replicated invocations.
- **vote** – Voting function selecting among valid results.
- **pred** – Predicate validating each invocation result.
- **f** – Callable to invoke asynchronously.
- **ts** – Arguments forwarded to *f*.

Returns

future with the voted-upon result.

```
template<typename Vote, typename F, typename ...Ts>
hpx::future<hpx::util::detail::invoke_deferred_result_t<F, Ts...>> hpx_invoke(async_replicate_vote_t,
                                                                    std::size_t n, Vote
                                                                    &&vote, F &&f,
                                                                    Ts&&... ts)
```

ADL hook for *async_replicate_vote* CPO.

Asynchronously launches *f* exactly *n* times, runs valid results through *vote*. Validation uses the default exception check.

Parameters

- **tag** – CPO tag (*async_replicate_vote_t*).
- **n** – Number of replicated invocations.
- **vote** – Voting function selecting among valid results.
- **f** – Callable to invoke asynchronously.
- **ts** – Arguments forwarded to *f*.

Returns

future with the voted-upon result.

```
template<typename Pred, typename F, typename ...Ts>
hpx::future<hpx::util::detail::invoke_deferred_result_t<F, Ts...>> hpx_invoke(async_replicate_validate_t,
                                                                    std::size_t n, Pred
                                                                    &&pred, F &&f,
                                                                    Ts&&... ts)
```

ADL hook for *async_replicate_validate* CPO.

Asynchronously launches *f* exactly *n* times, validates results with *pred*. Returns the first valid result.

Parameters

- **tag** – CPO tag (*async_replicate_validate_t*).
- **n** – Number of replicated invocations.
- **pred** – Predicate validating each invocation result.
- **f** – Callable to invoke asynchronously.
- **ts** – Arguments forwarded to *f*.

Returns

future with the first valid result.

```
template<typename F, typename ...Ts>
hpx::future<hpx::util::detail::invoke_deferred_result_t<F, Ts...>> hpx_invoke(async_replicate_t,
                                                                    std::size_t n, F
                                                                    &&f, Ts&&... ts)
```

ADL hook for *async_replicate* CPO.

Asynchronously launches *f* exactly *n* times. Validates by checking for exceptions. Returns the first valid result.

Parameters

- **tag** – CPO tag (*async_replicate_t*).
- **n** – Number of replicated invocations.
- **f** – Callable to invoke asynchronously.
- **ts** – Arguments forwarded to *f*.

Returns

future with the first non-throwing result.

runtime_configuration

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx::components::component_commandline_base

Defined in header `hpx/components.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

struct **component_commandline_base**

The `component_commandline_base` has to be used as a base class for all component command-line line handling registries.

Subclassed by `hpx::components::component_commandline`

hpx/runtime_configuration/component_registry_base.hpp

Defined in header `hpx/runtime_configuration/component_registry_base.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **components**

struct **component_registry_base**

#include <component_registry_base.hpp> The `component_registry_base` has to be used as a base class for all component registries.

Public Functions

virtual **~component_registry_base**() = default

virtual bool **get_component_info**(*std::vector<std::string>* &fillini, *std::string* const &filepath, bool is_static = false) = 0

Return the ini-information for all contained components.

Parameters

- **fillini** – [in, out] The module is expected to fill this vector with the ini-information (one line per vector element) for all components implemented in this module.
- **filepath** – [in]
- **is_static** – [in]

Returns

Returns *true* if the parameter *fillini* has been successfully initialized with the registry data of all implemented in this module.

virtual void **register_component_type**() = 0

Register the component type represented by this component.

HPX_REGISTER_COMPONENT_MODULE

Defined in header `hpx/components.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

Warning: doxygenfunction: Cannot find function “HPX_REGISTER_COMPONENT_MODULE” in doxygen xml output for project “hpx” from directory: ../hpx_autodoc

hpx/runtime_configuration/plugin_registry_base.hpp

Defined in header `hpx/runtime_configuration/plugin_registry_base.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **plugins**

struct **plugin_registry_base**

#include <plugin_registry_base.hpp> The `plugin_registry_base` has to be used as a base class for all plugin registries.

Public Functions

virtual **~plugin_registry_base**() = default

virtual bool **get_plugin_info**(*std::vector<std::string>* &fillini) = 0

Return the configuration information for any plugin implemented by this module

Parameters

fillini – [in, out] The module is expected to fill this vector with the ini-information (one line per vector element) for all plugins implemented in this module.

Returns

Returns *true* if the parameter *fillini* has been successfully initialized with the registry data of all implemented in this module.

inline virtual void **init**(int*, char***, *util::runtime_configuration*&)

hpx::runtime_mode

Defined in header [hpx/init.hpp](#)⁸⁹⁶.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

enum class **hpx::runtime_mode** : *std::int8_t*

A HPX runtime can be executed in two different modes: console mode and worker mode.

Values:

enumerator **invalid**

enumerator **console**

The runtime is the console locality.

enumerator **worker**

The runtime is a worker locality.

enumerator **connect**

The runtime is a worker locality connecting late

enumerator **local**

The runtime is fully local.

enumerator **default_**

The runtime mode will be determined based on the command line arguments

enumerator **last**

runtime_local

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx::get_initial_num_localities

Defined in header [hpx/runtime.hpp](#)⁸⁹⁷.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::get_num_localities*

⁸⁹⁶ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/init_runtime/include/hpx/init.hpp

⁸⁹⁷ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

`std::uint32_t hpx::get_initial_num_localities()`

Return the number of localities which were registered at startup for the running application.

The function `get_initial_num_localities` returns the number of localities which were connected to the console at application startup.

See also:

`hpx::find_all_localities`, `hpx::get_num_localities`

Note: As long as `ec` is not pre-initialized to `hpx::throws` this function doesn't throw but returns the result code using the parameter `ec`. Otherwise, it throws an instance of `hpx::exception`.

hpx::get_num_localities

Defined in header `hpx/runtime.hpp`⁸⁹⁸.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::get_initial_num_localities`

`hpx::future<std::uint32_t> hpx::get_num_localities()`

Asynchronously return the number of localities which are currently registered for the running application.

The function `get_num_localities` asynchronously returns the number of localities currently connected to the console. The returned future represents the actual result.

See also:

`hpx::find_all_localities`, `hpx::get_num_localities`

Note: This function will return meaningful results only if called from an HPX-thread. It will return 0 otherwise.

inline `std::uint32_t hpx::get_num_localities(launch::sync_policy, error_code &ec = throws)`

Return the number of localities which are currently registered for the running application.

The function `get_num_localities` returns the number of localities currently connected to the console.

See also:

`hpx::find_all_localities`, `hpx::get_num_localities`

Note: This function will return meaningful results only if called from an HPX-thread. It will return 0 otherwise.

Note: As long as `ec` is not pre-initialized to `hpx::throws` this function doesn't throw but returns the result code using the parameter `ec`. Otherwise, it throws an instance of `hpx::exception`.

⁸⁹⁸ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

Parameters

ec – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

`std::uint32_t hpx::get_num_localities(hpx::launch::sync_policy, error_code &ec)`

Return the number of localities which are currently registered for the running application.

`hpx::future<std::uint32_t> hpx::get_num_localities(components::component_type t)`

Asynchronously return the number of localities which are currently registered for the running application.

The function *get_num_localities* asynchronously returns the number of localities currently connected to the console which support the creation of the given component type. The returned future represents the actual result.

See also:

`hpx::find_all_localities`, `hpx::get_num_localities`

Note: This function will return meaningful results only if called from an HPX-thread. It will return 0 otherwise.

Parameters

t – The component type for which the number of connected localities should be retrieved.

`std::uint32_t hpx::get_num_localities(launch::sync_policy, components::component_type t, error_code &ec = throws)`

Synchronously return the number of localities which are currently registered for the running application.

The function *get_num_localities* returns the number of localities currently connected to the console which support the creation of the given component type. The returned future represents the actual result.

See also:

`hpx::find_all_localities`, `hpx::get_num_localities`

Note: This function will return meaningful results only if called from an HPX-thread. It will return 0 otherwise.

Parameters

- **t** – The component type for which the number of connected localities should be retrieved.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

`std::uint32_t hpx::get_num_localities(hpx::launch::sync_policy, components::component_type type, error_code &ec)`

hpx/runtime_local/runtime_local.hpp

Defined in header `hpx/runtime_local/runtime_local.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Functions

void **set_error_handlers**(*hpx::util::runtime_configuration* const &cfg)

class **runtime**

Public Types

using **notification_policy_type** = *threads::policies::callback_notifier*

Generate a new notification policy instance for the given thread name prefix

using **hpx_main_function_type** = int()

The *hpx_main_function_type* is the default function type usable as the main HPX thread function.

using **hpx_errorsink_function_type** = void(*std::uint32_t*, *std::string* const&)

Public Functions

virtual *notification_policy_type* **get_notification_policy**(char const *prefix,
runtime_local::os_thread_type type)

state **get_state**() const

void **set_state**(state s)

explicit **runtime**(*hpx::util::runtime_configuration* rtcfg, bool initialize)

Construct a new HPX runtime instance.

virtual **~runtime**()

The destructor makes sure all HPX runtime services are properly shut down before exiting.

void **on_exit**(*hpx::function*<void()> const &f)

Manage list of functions to call on exit.

void **starting**()

Manage runtime 'stopped' state.

void **stopping**()

Call all registered on_exit functions.

bool **stopped**() const

This accessor returns whether the runtime instance has been stopped.

`hpx::util::runtime_configuration &get_config()`

access configuration information

`hpx::util::runtime_configuration const &get_config() const`

`std::size_t get_instance_number() const`

`util::thread_mapper &get_thread_mapper() const`

Return a reference to the internal PAPI thread manager.

`threads::topology const &get_topology() const`

virtual int **run**(`hpx::function<hpx_main_function_type> const &func`)

Run the HPX runtime system, use the given function for the main *thread* and block waiting for all threads to finish.

Note: The parameter *func* is optional. If no function is supplied, the runtime system will simply wait for the shutdown action without explicitly executing any main thread.

Parameters

func – [in] This is the main function of an HPX application. It will be scheduled for execution by the thread manager as soon as the runtime has been initialized. This function is expected to expose an interface as defined by the typedef *hpx_main_function_type*. This parameter is optional and defaults to none main thread function, in which case all threads have to be scheduled explicitly.

Returns

This function will return the value as returned as the result of the invocation of the function object given by the parameter **func**.

virtual int **run**()

Run the HPX runtime system, initially use the given number of (OS) threads in the thread-manager and block waiting for all threads to finish.

Returns

This function will always return 0 (zero).

virtual void **rethrow_exception**()

Rethrow any stored exception (to be called after *stop()*)

virtual int **start**(`hpx::function<hpx_main_function_type> const &func, bool blocking = false`)

Start the runtime system.

Parameters

- **func** – [in] This is the main function of an HPX application. It will be scheduled for execution by the thread manager as soon as the runtime has been initialized. This function is expected to expose an interface as defined by the typedef *hpx_main_function_type*.
- **blocking** – [in] This allows to control whether this call blocks until the runtime system has been stopped. If this parameter is *true* the function `runtime::start` will call `runtime::wait` internally.

Returns

If a blocking is a true, this function will return the value as returned as the result of the invocation of the function object given by the parameter **func**. Otherwise, it will return zero.

virtual int **start**(bool blocking = false)

Start the runtime system.

Parameters

blocking – [in] This allows to control whether this call blocks until the runtime system has been stopped. If this parameter is *true* the function `runtime::start` will call `runtime::wait` internally .

Returns

If a blocking is a true, this function will return the value as returned as the result of the invocation of the function object given by the parameter `func`. Otherwise, it will return zero.

virtual int **wait**()

Wait for the shutdown action to be executed.

Returns

This function will return the value as returned as the result of the invocation of the function object given by the parameter `func`.

virtual void **stop**(bool blocking = true)

Initiate termination of the runtime system.

Parameters

blocking – [in] This allows to control whether this call blocks until the runtime system has been fully stopped. If this parameter is *false* then this call will initiate the stop action but will return immediately. Use a second call to stop with this parameter set to *true* to wait for all internal work to be completed.

virtual int **suspend**()

Suspend the runtime system.

virtual int **resume**()

Resume the runtime system.

virtual int **finalize**(double)

virtual bool **is_networking_enabled**()

Return true if networking is enabled.

virtual *hpx::threads::threadmanager* &**get_thread_manager**()

Allow access to the thread manager instance used by the HPX runtime.

virtual *std::string* **here**() const

Returns a string of the locality endpoints (usable in debug output)

virtual bool **report_error**(*std::size_t* num_thread, *std::exception_ptr* const &*e*, bool terminate_all = true)

Report a non-recoverable error to the runtime system.

Parameters

- **num_thread** – [in] The number of the operating system thread the error has been detected in.
- **e** – [in] This is an instance encapsulating an exception which lead to this function call.
- **terminate_all** – [in] signal whether all localities should be terminated

virtual bool **report_error**(*std::exception_ptr* const &*e*, bool terminate_all = true)

Report a non-recoverable error to the runtime system.

Note: This function will retrieve the number of the current shepherd thread and forward to the `report_error` function above.

Parameters

- **e** – [in] This is an instance encapsulating an exception which lead to this function call.
- **terminate_all** – [in] signal whether all localities should be terminated

virtual void **add_pre_startup_function**(*startup_function_type* f)

Add a function to be executed inside a HPX thread before `hpx_main` but guaranteed to be executed before any startup function registered with `add_startup_function`.

Note: The difference to a startup function is that all pre-startup functions will be (system-wide) executed before any startup function.

Parameters

f – The function ‘f’ will be called from inside a HPX thread before `hpx_main` is executed. This is very useful to set up the runtime environment of the application (install performance counters, etc.)

virtual void **add_startup_function**(*startup_function_type* f)

Add a function to be executed inside a HPX thread before `hpx_main`

Parameters

f – The function ‘f’ will be called from inside a HPX thread before `hpx_main` is executed. This is very useful to set up the runtime environment of the application (install performance counters, etc.)

virtual void **add_pre_shutdown_function**(*shutdown_function_type* f)

Add a function to be executed inside a HPX thread during `hpx::finalize`, but guaranteed before any of the shutdown functions is executed.

Note: The difference to a shutdown function is that all pre-shutdown functions will be (system-wide) executed before any shutdown function.

Parameters

f – The function ‘f’ will be called from inside a HPX thread while `hpx::finalize` is executed. This is very useful to tear down the runtime environment of the application (uninstall performance counters, etc.)

virtual void **add_shutdown_function**(*shutdown_function_type* f)

Add a function to be executed inside a HPX thread during `hpx::finalize`

Parameters

f – The function ‘f’ will be called from inside a HPX thread while `hpx::finalize` is executed. This is very useful to tear down the runtime environment of the application (uninstall performance counters, etc.)

virtual `hpx::util::io_service_pool` ***get_thread_pool**(char const *name)

Access one of the internal thread pools (`io_service` instances) HPX is using to perform specific tasks. The three possible values for the argument `name` are “`main_pool`”, “`io_pool`”, “`parcel_pool`”, and “`timer_pool`”. For any other argument value the function will return zero.

virtual bool **register_thread**(char const *name, `std::size_t` num = 0, bool service_thread = true, `error_code` &ec = throws)

Register an external OS-thread with HPX.

This function should be called from any OS-thread which is external to HPX (not created by HPX), but which needs to access HPX functionality, such as setting a value on a promise or similar.

‘main’, ‘io’, ‘timer’, ‘parcel’, ‘worker’

Note: The function will compose a thread name of the form ‘<name>-thread#<num>’ which is used to register the thread. It is the user’s responsibility to ensure that each (composed) thread name is unique. HPX internally uses the following names for the threads it creates, do not reuse those:

Note: This function should be called for each thread exactly once. It will fail if it is called more than once.

Parameters

- **name** – [in] The name to use for thread registration.
- **num** – [in] The sequence number to use for thread registration. The default for this parameter is zero.
- **service_thread** – [in] The thread should be registered as a service thread. The default for this parameter is ‘true’. Any service threads will be pinned to cores not currently used by any of the HPX worker threads.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

This function will return whether the requested operation succeeded or not.

virtual bool **unregister_thread()**

Unregister an external OS-thread with HPX.

This function will unregister any external OS-thread from HPX.

Note: This function should be called for each thread exactly once. It will fail if it is called more than once. It will fail as well if the thread has not been registered before (see *register_thread*).

Returns

This function will return whether the requested operation succeeded or not.

virtual runtime_local::os_thread_data **get_os_thread_data**(std::string const &label) const

Access data for a given OS thread that was previously registered by *register_thread*.

virtual bool **enumerate_os_threads**(hpx::function<bool(runtime_local::os_thread_data const&)> const &f) const

Enumerate all OS threads that have registered with the runtime.

notification_policy_type::on_startstop_type **on_start_func**() const

notification_policy_type::on_startstop_type **on_stop_func**() const

notification_policy_type::on_error_type **on_error_func**() const

notification_policy_type::on_startstop_type **on_start_func**(notification_policy_type::on_startstop_type&&)

notification_policy_type::on_startstop_type **on_stop_func**(notification_policy_type::on_startstop_type&&)

notification_policy_type::on_error_type **on_error_func**(notification_policy_type::on_error_type&&)

virtual std::uint32_t **get_locality_id**(error_code &ec) const

```
virtual std::size_t get_num_worker_threads() const  
virtual std::uint32_t get_num_localities(hpx::launch::sync_policy, error_code &ec) const  
virtual std::uint32_t get_initial_num_localities() const  
virtual hpx::future<std::uint32_t> get_num_localities() const  
virtual std::string get_locality_name() const  
virtual std::uint32_t assign_cores(std::string const&, std::uint32_t)  
virtual std::uint32_t assign_cores()  
inline hpx::program_options::options_description const &get_app_options() const  
inline void set_app_options(hpx::program_options::options_description const &app_options)
```

Public Static Functions

```
static std::uint64_t get_system_uptime()  
    Return the system uptime measure on the thread executing this call.
```

Protected Types

```
using on_exit_type = std::vector<hpx::function<void()>>
```

Protected Functions

```
explicit runtime(hpx::util::runtime_configuration rtcfg)  
void set_notification_policies(notification_policy_type &&notifier,  
                                threads::detail::network_background_callback_type const  
                                &network_background_callback)  
void init()  
    Common initialization for different constructors.  
void init_global_data()  
threads::thread_result_type run_helper(hpx::function<runtime::hpx_main_function_type> const  
                                         &func, int &result, bool call_startup_functions, void  
                                         (*handle_print_bind)(std::size_t) = nullptr)  
void wait_helper(std::mutex &mtx, std::condition_variable &cond, bool &running)
```

Protected Attributes

on_exit_type **on_exit_functions_**

mutable *std::mutex* **mtx_**

hpx::util::runtime_configuration **rtcfg_**

long **instance_number_**

std::unique_ptr<util::thread_mapper> **thread_support_**

threads::topology &**topology_**

std::atomic<state> **state_**

notification_policy_type::on_startstop_type **on_start_func_**

notification_policy_type::on_startstop_type **on_stop_func_**

notification_policy_type::on_error_type **on_error_func_**

int **result_**

std::exception_ptr **exception_**

notification_policy_type **main_pool_notifier_**

std::unique_ptr<util::io_service_pool> **main_pool_**

notification_policy_type **notifier_**

std::unique_ptr<hpx::threads::threadmanager> **thread_manager_**

Protected Static Functions

static void **deinit_global_data()**

Protected Static Attributes

`static std::atomic<int> instance_number_counter_`

Private Functions

void **stop_helper**(bool blocking, `std::condition_variable &cond`, `std::mutex &mtx`) const

Helper function to stop the runtime.

Parameters

- **blocking** – [in] This allows to control whether this call blocks until the runtime system has been fully stopped. If this parameter is *false* then this call will initiate the stop action but will return immediately. Use a second call to stop with this parameter set to *true* to wait for all internal work to be completed.
- **cond** –
- **mtx** –

void **deinit_tss_helper**(char const *context, `std::size_t num`) const

void **init_tss_ex**(char const *context, `runtime_local::os_thread_type type`, `std::size_t local_thread_num`, `std::size_t global_thread_num`, char const *pool_name, char const *postfix, bool service_thread, `error_code &ec`) const

void **init_tss_helper**(char const *context, `runtime_local::os_thread_type type`, `std::size_t local_thread_num`, `std::size_t global_thread_num`, char const *pool_name, char const *postfix, bool service_thread) const

void **notify_finalize**()

void **wait_finalize**()

void **call_startup_functions**(bool pre_startup)

Private Members

`std::list<startup_function_type> pre_startup_functions_`

`std::list<startup_function_type> startup_functions_`

`std::list<shutdown_function_type> pre_shutdown_functions_`

`std::list<shutdown_function_type> shutdown_functions_`

bool **stop_called_**

bool **stop_done_**

`std::condition_variable wait_condition_`

`hpx::program_options::options_description app_options_`

namespace **threads**

Functions

char const ***get_stack_size_name**(std::ptrdiff_t size)

Returns the stack size name.

Get the readable string representing the given stack size constant.

Parameters

size – this represents the stack size

std::ptrdiff_t **get_default_stack_size**()

Returns the default stack size.

Get the default stack size in bytes.

std::ptrdiff_t **get_stack_size**(thread_stacksize size)

Returns the stack size corresponding to the given stack size enumeration.

Get the stack size corresponding to the given stack size enumeration.

Parameters

size – this represents the stack size

namespace **util**

Functions

bool **retrieve_commandline_arguments**(hpx::program_options::options_description const &app_options, hpx::program_options::variables_map &vm)

bool **retrieve_commandline_arguments**(std::string const &app_name, hpx::program_options::variables_map &vm)

hpx::shutdown_function_type

Defined in header `hpx/runtime.hpp`⁸⁹⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::register_pre_shutdown_function`
- `hpx::register_shutdown_function`

using `hpx::shutdown_function_type` = `hpx::move_only_function<void()>`

The type of the function which is registered to be executed as a shutdown or pre-shutdown function.

⁸⁹⁹ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

hpx::register_pre_shutdown_function

Defined in header [hpx/runtime.hpp](#)⁹⁰⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::shutdown_function_type`
- `hpx::register_shutdown_function`

void **hpx::register_pre_shutdown_function**(*shutdown_function_type* f)

Add a function to be executed by a HPX thread during `hpx::finalize()` but guaranteed before any shutdown function is executed (system-wide)

Any of the functions registered with `register_pre_shutdown_function` are guaranteed to be executed by an HPX thread during the execution of `hpx::finalize()` before any of the registered shutdown functions are executed (see: `hpx::register_shutdown_function()`).

See also:

`hpx::register_shutdown_function()`

Note: If this function is called while the pre-shutdown functions are being executed, or after that point, it will raise an `invalid_status` exception.

Parameters

f – [in] The function to be registered to run by an HPX thread as a pre-shutdown function.

hpx::register_shutdown_function

Defined in header [hpx/runtime.hpp](#)⁹⁰¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::shutdown_function_type`
- `hpx::register_pre_shutdown_function`

void **hpx::register_shutdown_function**(*shutdown_function_type* f)

Add a function to be executed by a HPX thread during `hpx::finalize()` but guaranteed after any pre-shutdown function is executed (system-wide)

Any of the functions registered with `register_shutdown_function` are guaranteed to be executed by an HPX thread during the execution of `hpx::finalize()` after any of the registered pre-shutdown functions are executed (see: `hpx::register_pre_shutdown_function()`).

See also:

`hpx::register_pre_shutdown_function()`

⁹⁰⁰ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

⁹⁰¹ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

Note: If this function is called while the shutdown functions are being executed, or after that point, it will raise an `invalid_status` exception.

Parameters

f – [in] The function to be registered to run by an HPX thread as a shutdown function.

hpx/runtime_local/termination_detection.hpp

Defined in header `hpx/runtime_local/termination_detection.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **local**

Functions

void **termination_detection()**

Wait for all local HPX threads to complete execution.

This function blocks the calling thread until all HPX worker threads on the current locality have completed their work and become idle. It performs local termination detection by waiting for the thread manager to drain all pending tasks.

Example:

```
#include <hpx/hpx_init.hpp>
#include <hpx/runtime_local/termination_detection.hpp>

int hpx_main()
{
    // Launch some asynchronous work
    for (int i = 0; i < 100; ++i)
    {
        hpx::post([]{ /* do work */ });
    }

    // Wait for all local threads to complete
    hpx::local::termination_detection();

    return hpx::finalize();
}
```

Note: This function should be called from an HPX thread or after the HPX runtime has been initialized.

Throws

hpx::exception – if called when the runtime is not in a valid state (e.g., before initialization or after shutdown).

bool **termination_detection**(*hpx::chrono::steady_duration* const &timeout)

Wait for all local HPX threads to complete with a timeout.

This function blocks the calling thread until either all HPX worker threads have completed their work, or the specified timeout duration has elapsed.

See also:

termination_detection()

Example:

```
#include <hpx/hpx_init.hpp>
#include <hpx/runtime_local/termination_detection.hpp>
#include <chrono>

int hpx_main()
{
    // Launch some work
    hpx::post([]{ /* long running task */ });

    // Wait up to 5 seconds
    if (!hpx::local::termination_detection(std::chrono::seconds(5)))
    {
        std::cerr << "Warning: Work did not complete within timeout\n";
        // Implement fallback strategy
    }

    return hpx::finalize();
}
```

Parameters

timeout – Maximum duration to wait for termination. The function will return false if this duration elapses before all threads complete.

Throws

hpx::exception – if called when the runtime is not in a valid state.

Returns

true if all threads completed before the timeout, false if the timeout elapsed.

bool **termination_detection**(*hpx::chrono::steady_time_point* const &deadline)

Wait for all local HPX threads to complete until a deadline.

This function blocks the calling thread until either all HPX worker threads have completed their work, or the specified deadline time point is reached.

See also:

termination_detection(std::chrono::duration<double>)

Example:

```

#include <hpx/hpx_init.hpp>
#include <hpx/runtime_local/termination_detection.hpp>
#include <chrono>

int hpx_main()
{
    auto deadline = std::chrono::steady_clock::now() +
        std::chrono::seconds(10);

    hpx::post([]{ /* work */ });

    if (!hpx::local::termination_detection(deadline))
    {
        std::cerr << "Deadline reached\n";
    }

    return hpx::finalize();
}

```

Parameters

deadline – Absolute time point when waiting should stop. The function will return false if this time is reached before all threads complete.

Throws

hpx::exception – if called when the runtime is not in a valid state.

Returns

true if all threads completed before the deadline, false if the deadline was reached.

bool **termination_detection**(*hpx::stop_token* stop_token, *hpx::chrono::steady_duration* const &timeout = *hpx::chrono::steady_duration((hpx::chrono::steady_clock::duration::max)())*)

Wait for all local HPX threads with cancellation support.

This function blocks the calling thread until all HPX worker threads have completed, the timeout elapses, or a stop is requested via the stop_token. This enables cooperative cancellation of the wait operation.

See also:

termination_detection(std::chrono::duration<double>)

Example:

```

#include <hpx/hpx_init.hpp>
#include <hpx/runtime_local/termination_detection.hpp>
#include <hpx/stop_token.hpp>
#include <chrono>

int hpx_main()
{
    hpx::stop_source stop_src;

    // In shutdown handler
    hpx::thread shutdown_thread([&stop_src]() {

```

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```

        hpx::this_thread::sleep_for(std::chrono::seconds(30));
        stop_src.request_stop();
    });

    hpx::post([]{ /* work */ });

    // Wait with cancellation support
    bool completed = hpx::local::termination_detection(
        stop_src.get_token(),
        std::chrono::minutes(1)
    );

    if (!completed)
    {
        std::cerr << "Cancelled or timed out\n";
    }

    shutdown_thread.join();
    return hpx::finalize();
}

```

Parameters

- **stop_token** – Token that can be used to request cancellation of the wait operation.
- **timeout** – Maximum duration to wait. Defaults to maximum duration (effectively infinite).

Throws

hpx::exception – if called when the runtime is not in a valid state.

Returns

true if all threads completed, false if timeout elapsed or stop was requested.

hpx/runtime_local/get_os_thread_count.hpp

Defined in header `hpx/runtime_local/get_os_thread_count.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Functions

`std::size_t` **get_os_thread_count()**

Return the number of OS-threads running in the runtime instance the current HPX-thread is associated with.

`std::size_t` **get_os_thread_count**(*threads::executor* const &exec)

Return the number of worker OS- threads used by the given executor to execute HPX threads.

This function returns the number of cores used to execute HPX threads for the given executor. If the function is called while no HPX runtime system is active, it will return zero. If the executor is not valid, this function will fall back to retrieving the number of OS threads used by HPX.

Parameters**exec** – [in] The executor to be used.namespace **threads****hpx::get_locality_name**Defined in header [hpx/runtime.hpp](#)⁹⁰².See *Public API* for a list of names and headers that are part of the public *HPX* API.*std::string* **hpx::get_locality_name()**

Return the name of the locality this function is called on.

This function returns the name for the locality on which this function is called.

See also:*future<std::string>* **get_locality_name(hpx::id_type const& id)****Returns**

This function returns the name for the locality on which the function is called. The name is retrieved from the underlying networking layer and may be different for different parcelports.

future<std::string> **hpx::get_locality_name(hpx::id_type const &id)**

Return the name of the referenced locality.

This function returns a future referring to the name for the locality of the given id.

See also:*std::string* **get_locality_name()****Parameters****id** – [in] The global id of the locality for which the name should be retrieved**Returns**

This function returns the name for the locality of the given id. The name is retrieved from the underlying networking layer and may be different for different parcel ports.

hpx::get_locality_idDefined in header [hpx/runtime.hpp](#)⁹⁰³.See *Public API* for a list of names and headers that are part of the public *HPX* API.⁹⁰² <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>⁹⁰³ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

`std::uint32_t hpx::get_locality_id(error_code &ec = throws)`

Return the number of the locality this function is being called from.

This function returns the id of the current locality.

Note: The returned value is zero based and its maximum value is smaller than the overall number of localities the current application is running on (as returned by `get_num_localities()`).

Note: As long as *ec* is not pre-initialized to `hpx::throws` this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of `hpx::exception`.

Note: This function needs to be executed on a HPX-thread. It will fail otherwise (it will return -1).

Parameters

ec – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

`hpx::startup_function_type`

Defined in header `hpx/runtime.hpp`⁹⁰⁴.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::register_pre_startup_function`
- `hpx::register_startup_function`

using `hpx::startup_function_type = hpx::move_only_function<void()>`

The type of the function which is registered to be executed as a startup or pre-startup function.

`hpx::register_pre_startup_function`

Defined in header `hpx/runtime.hpp`⁹⁰⁵.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::startup_function_type`
- `hpx::register_startup_function`

void `hpx::register_pre_startup_function(startup_function_type f)`

Add a function to be executed by a HPX thread before `hpx_main` but guaranteed before any startup function is executed (system-wide).

Any of the functions registered with `register_pre_startup_function` are guaranteed to be executed by an HPX thread before any of the registered startup functions are executed (see `hpx::register_startup_function()`).

⁹⁰⁴ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

⁹⁰⁵ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

This function is one of the few API functions which can be called before the runtime system has been fully initialized. It will automatically stage the provided startup function to the runtime system during its initialization (if necessary).

See also:

`hpx::register_startup_function()`

Note: If this function is called while the pre-startup functions are being executed or after that point, it will raise an `invalid_status` exception.

Parameters

f – [in] The function to be registered to run by an HPX thread as a pre-startup function.

hpx::register_startup_function

Defined in header [hpx/runtime.hpp](#)⁹⁰⁶.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::startup_function_type`
- `hpx::register_pre_startup_function`

void **hpx::register_startup_function**(*startup_function_type* f)

Add a function to be executed by a HPX thread before `hpx_main` but guaranteed after any pre-startup function is executed (system-wide).

Any of the functions registered with *register_startup_function* are guaranteed to be executed by an HPX thread after any of the registered pre-startup functions are executed (see: `hpx::register_pre_startup_function()`), but before *hpx_main* is being called.

This function is one of the few API functions which can be called before the runtime system has been fully initialized. It will automatically stage the provided startup function to the runtime system during its initialization (if necessary).

See also:

`hpx::register_pre_startup_function()`

Note: If this function is called while the startup functions are being executed or after that point, it will raise an `invalid_status` exception.

Parameters

f – [in] The function to be registered to run by an HPX thread as a startup function.

⁹⁰⁶ <http://github.com/TheHPXProject/hpx/blob/d402cead7405eedff56d454c9380fbdc0c8044e7/libs/full/include/include/hpx/runtime.hpp>

hpx/runtime_local/component_startup_shutdown_base.hpp

Defined in header `hpx/runtime_local/component_startup_shutdown_base.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **components**

struct **component_startup_shutdown_base**

#include <component_startup_shutdown_base.hpp> The `component_startup_shutdown_base` has to be used as a base class for all component startup/shutdown registries.

Public Functions

virtual `~component_startup_shutdown_base()` = default

virtual bool **get_startup_function**(*startup_function_type* &startup, bool &pre_startup) = 0

Return any startup function for this component.

Parameters

- **startup** – [in, out] The module is expected to fill this function object with a reference to a startup function. This function will be executed by the runtime system during system startup.
- **pre_startup** – [in, out] Will be set to true if the returned startup function is executed during the first round of calls.

Returns

Returns *true* if the parameter *startup* has been successfully initialized with the startup function.

virtual bool **get_shutdown_function**(*shutdown_function_type* &shutdown, bool &pre_shutdown) = 0

Return any startup function for this component.

Parameters

- **shutdown** – [in, out] The module is expected to fill this function object with a reference to a startup function. This function will be executed by the runtime system during system startup.
- **pre_shutdown** – [in, out] Will be set to true if the returned shutdown function is executed during the first round of calls.

Returns

Returns *true* if the parameter *shutdown* has been successfully initialized with the shutdown function.

hpx/runtime_local/report_error.hpp

Defined in header `hpx/runtime_local/report_error.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Functions

void **report_error**(*std::size_t* num_thread, *std::exception_ptr* const &e)

The function `report_error` reports the given exception to the console.

void **report_error**(*std::exception_ptr* const &e)

The function `report_error` reports the given exception to the console.

hpx/runtime_local/service_executors.hpp

Defined in header `hpx/runtime_local/service_executors.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **execution**

namespace **experimental**

Enums

enum class **service_executor_type** : *std::uint8_t*

Values:

enumerator **io_thread_pool**

Selects creating a service executor using the I/O pool of threads

enumerator **parcel_thread_pool**

Selects creating a service executor using the parcel pool of threads

enumerator **timer_thread_pool**

Selects creating a service executor using the timer pool of threads

enumerator **main_thread**

Selects creating a service executor using the main thread

struct **io_pool_executor** : public *hpx::execution::experimental::service_executor*

Public Functions

io_pool_executor()

struct **main_pool_executor** : public *hpx::execution::experimental::service_executor*

Public Functions

main_pool_executor()

struct **parcel_pool_executor** : public *hpx::execution::experimental::service_executor*

Public Functions

explicit **parcel_pool_executor**(char const *name_suffix = "-tcp")

struct **service_executor** : public *service_executor*

Subclassed by *hpx::execution::experimental::io_pool_executor*,
hpx::execution::experimental::main_pool_executor, *hpx::execution::experimental::parcel_pool_executor*,
hpx::execution::experimental::timer_pool_executor

Public Functions

explicit **service_executor**(*service_executor_type* t, char const *name_suffix = "")

struct **timer_pool_executor** : public *hpx::execution::experimental::service_executor*

Public Functions

timer_pool_executor()

hpx/runtime_local/thread_hooks.hpp

Defined in header `hpx/runtime_local/thread_hooks.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Functions

threads::policies::callback_notifier::on_startstop_type **get_thread_on_start_func()**

Retrieve the currently installed start handler function. This is a function that will be called by HPX for each newly created thread that is made known to the runtime. HPX stores exactly one such function reference, thus the caller needs to make sure any newly registered start function chains into the previous one (see *register_thread_on_start_func*).

Note: This function can be called before the HPX runtime is initialized.

Returns

The currently installed error handler function.

threads::policies::callback_notifier::on_startstop_type **get_thread_on_stop_func()**

Retrieve the currently installed stop handler function. This is a function that will be called by HPX for each newly created thread that is made known to the runtime. HPX stores exactly one such function reference, thus the caller needs to make sure any newly registered stop function chains into the previous one (see *register_thread_on_stop_func*).

Note: This function can be called before the HPX runtime is initialized.

Returns

The currently installed error handler function.

threads::policies::callback_notifier::on_error_type **get_thread_on_error_func()**

Retrieve the currently installed error handler function. This is a function that will be called by HPX for each newly created thread that is made known to the runtime. HPX stores exactly one such function reference, thus the caller needs to make sure any newly registered error function chains into the previous one (see *register_thread_on_error_func*).

Note: This function can be called before the HPX runtime is initialized.

Returns

The currently installed error handler function.

threads::policies::callback_notifier::on_startstop_type **register_thread_on_start_func**(*threads::policies::callback_notifier::*
&&f)

Set the currently installed start handler function. This is a function that will be called by HPX for each newly created thread that is made known to the runtime. HPX stores exactly one such function reference, thus the caller needs to make sure any newly registered start function chains into the previous one (see *get_thread_on_start_func*).

Note: This function can be called before the HPX runtime is initialized.

Parameters

f – The function to install as the new start handler.

Returns

The previously registered function of this category. It is the user's responsibility to call that function if the callback is invoked by HPX.

threads::policies::callback_notifier::on_startstop_type **register_thread_on_stop_func**(*threads::policies::callback_notifier::on_startstop_type* &&f)

Set the currently installed stop handler function. This is a function that will be called by HPX for each newly created thread that is made known to the runtime. HPX stores exactly one such function reference, thus the caller needs to make sure any newly registered stop function chains into the previous one (see *get_thread_on_stop_func*).

Note: This function can be called before the HPX runtime is initialized.

Parameters

f – The function to install as the new stop handler.

Returns

The previously registered function of this category. It is the user's responsibility to call that function if the callback is invoked by HPX.

threads::policies::callback_notifier::on_error_type **register_thread_on_error_func**(*threads::policies::callback_notifier::on_error_type* &&f)

Set the currently installed error handler function. This is a function that will be called by HPX for each newly created thread that is made known to the runtime. HPX stores exactly one such function reference, thus the caller needs to make sure any newly registered error function chains into the previous one (see *get_thread_on_error_func*).

Note: This function can be called before the HPX runtime is initialized.

Parameters

f – The function to install as the new error handler.

Returns

The previously registered function of this category. It is the user's responsibility to call that function if the callback is invoked by HPX.

hpx/runtime_local/custom_exception_info.hpp

Defined in header `hpx/runtime_local/custom_exception_info.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Functions

`std::string diagnostic_information(exception_info const &xi)`

Extract the diagnostic information embedded in the given exception and return a string holding a formatted message.

The function `hpx::diagnostic_information` can be used to extract all diagnostic information stored in the given exception instance as a formatted string. This simplifies debug output as it composes the diagnostics into one, easy to use function call. This includes the name of the source file and line number, the sequence number of the OS-thread and the HPX-thread id, the locality id and the stack backtrace of the point where the original exception was thrown.

See also:

`hpx::get_error_locality_id()`, `hpx::get_error_host_name()`, `hpx::get_error_process_id()`,
`hpx::get_error_function_name()`, `hpx::get_error_file_name()`, `hpx::get_error_line_number()`,
`hpx::get_error_os_thread()`, `hpx::get_error_thread_id()`, `hpx::get_error_thread_description()`,
`hpx::get_error()`, `hpx::get_error_backtrace()`, `hpx::get_error_env()`, `hpx::get_error_what()`,
`hpx::get_error_config()`, `hpx::get_error_state()`

Parameters

xi – The parameter `e` will be inspected for all diagnostic information elements which have been stored at the point where the exception was thrown. This parameter can be one of the following types: `hpx::exception_info`, `hpx::error_code`, `std::exception`, or `std::exception_ptr`.

Throws

`std::bad_alloc` – (if any of the required allocation operations fail)

Returns

The formatted string holding all the available diagnostic information stored in the given exception instance.

`std::string default_diagnostic_information(std::exception_ptr const &e)`

`std::uint32_t get_error_locality_id(hpx::exception_info const &xi) noexcept`

Return the locality id where the exception was thrown.

The function `hpx::get_error_locality_id` can be used to extract the diagnostic information element representing the locality id as stored in the given exception instance.

See also:

`hpx::diagnostic_information()`, `hpx::get_error_host_name()`, `hpx::get_error_process_id()`,
`hpx::get_error_function_name()`, `hpx::get_error_file_name()`, `hpx::get_error_line_number()`,
`hpx::get_error_os_thread()`, `hpx::get_error_thread_id()`, `hpx::get_error_thread_description()`,
`hpx::get_error()`, `hpx::get_error_backtrace()`, `hpx::get_error_env()`, `hpx::get_error_what()`,
`hpx::get_error_config()`, `hpx::get_error_state()`

Parameters

xi – The parameter `e` will be inspected for the requested diagnostic information elements which have been stored at the point where the exception was thrown. This parameter can be one of the following types: `hpx::exception_info`, `hpx::error_code`, `std::exception`, or `std::exception_ptr`.

Throws

nothing –

Returns

The locality id of the locality where the exception was thrown. If the exception instance does not hold this information, the function will return *hpx::naming::invalid_locality_id*.

std::string **get_error_host_name**(*hpx::exception_info* const &xi)

Return the hostname of the locality where the exception was thrown.

The function *hpx::get_error_host_name* can be used to extract the diagnostic information element representing the host name as stored in the given exception instance.

See also:

hpx::diagnostic_information() *hpx::get_error_process_id()*, *hpx::get_error_function_name()*,
hpx::get_error_file_name(), *hpx::get_error_line_number()*, *hpx::get_error_os_thread()*,
hpx::get_error_thread_id(), *hpx::get_error_thread_description()*, *hpx::get_error()*
hpx::get_error_backtrace(), *hpx::get_error_env()*, *hpx::get_error_what()*, *hpx::get_error_config()*,
hpx::get_error_state()

Parameters

xi – The parameter *e* will be inspected for the requested diagnostic information elements which have been stored at the point where the exception was thrown. This parameter can be one of the following types: *hpx::exception_info*, *hpx::error_code*, *std::exception*, or *std::exception_ptr*.

Throws

std::bad_alloc – (if one of the required allocations fails)

Returns

The hostname of the locality where the exception was thrown. If the exception instance does not hold this information, the function will return an empty string.

std::int64_t **get_error_process_id**(*hpx::exception_info* const &xi) noexcept

Return the (operating system) process id of the locality where the exception was thrown.

The function *hpx::get_error_process_id* can be used to extract the diagnostic information element representing the process id as stored in the given exception instance.

See also:

hpx::diagnostic_information(), *hpx::get_error_host_name()*, *hpx::get_error_function_name()*,
hpx::get_error_file_name(), *hpx::get_error_line_number()*, *hpx::get_error_os_thread()*,
hpx::get_error_thread_id(), *hpx::get_error_thread_description()*, *hpx::get_error()*,
hpx::get_error_backtrace(), *hpx::get_error_env()*, *hpx::get_error_what()*, *hpx::get_error_config()*,
hpx::get_error_state()

Parameters

xi – The parameter *e* will be inspected for the requested diagnostic information elements which have been stored at the point where the exception was thrown. This parameter can be one of the following types: *hpx::exception_info*, *hpx::error_code*, *std::exception*, or *std::exception_ptr*.

Throws

nothing –

Returns

The process id of the OS-process which threw the exception. If the exception instance does not hold this information, the function will return 0.

`std::string get_error_env(hpx::exception_info const &xi)`

Return the environment of the OS-process at the point the exception was thrown.

The function `hpx::get_error_env` can be used to extract the diagnostic information element representing the environment of the OS-process collected at the point the exception was thrown.

See also:

`hpx::diagnostic_information()`, `hpx::get_error_host_name()`, `hpx::get_error_process_id()`,
`hpx::get_error_function_name()`, `hpx::get_error_file_name()`, `hpx::get_error_line_number()`,
`hpx::get_error_os_thread()`, `hpx::get_error_thread_id()`, `hpx::get_error_thread_description()`,
`hpx::get_error()`, `hpx::get_error_backtrace()`, `hpx::get_error_what()`, `hpx::get_error_config()`,
`hpx::get_error_state()`

Parameters

xi – The parameter `e` will be inspected for the requested diagnostic information elements which have been stored at the point where the exception was thrown. This parameter can be one of the following types: `hpx::exception_info`, `hpx::error_code`, `std::exception`, or `std::exception_ptr`.

Throws

`std::bad_alloc` – (if one of the required allocations fails)

Returns

The environment from the point the exception was thrown. If the exception instance does not hold this information, the function will return an empty string.

`std::string get_error_backtrace(hpx::exception_info const &xi)`

Return the stack backtrace from the point the exception was thrown.

The function `hpx::get_error_backtrace` can be used to extract the diagnostic information element representing the stack backtrace collected at the point the exception was thrown.

See also:

`hpx::diagnostic_information()`, `hpx::get_error_host_name()`, `hpx::get_error_process_id()`,
`hpx::get_error_function_name()`, `hpx::get_error_file_name()`, `hpx::get_error_line_number()`,
`hpx::get_error_os_thread()`, `hpx::get_error_thread_id()`, `hpx::get_error_thread_description()`,
`hpx::get_error()`, `hpx::get_error_env()`, `hpx::get_error_what()`, `hpx::get_error_config()`,
`hpx::get_error_state()`

Parameters

xi – The parameter `e` will be inspected for the requested diagnostic information elements which have been stored at the point where the exception was thrown. This parameter can be one of the following types: `hpx::exception_info`, `hpx::error_code`, `std::exception`, or `std::exception_ptr`.

Throws

`std::bad_alloc` – (if one of the required allocations fails)

Returns

The stack back trace from the point the exception was thrown. If the exception instance does not hold this information, the function will return an empty string.

`std::size_t` **get_error_os_thread**(*hpx::exception_info* const &xi) noexcept

Return the sequence number of the OS-thread used to execute HPX-threads from which the exception was thrown.

The function *hpx::get_error_os_thread* can be used to extract the diagnostic information element representing the sequence number of the OS-thread as stored in the given exception instance.

See also:

hpx::diagnostic_information(), *hpx::get_error_host_name()*, *hpx::get_error_process_id()*,
hpx::get_error_function_name(), *hpx::get_error_file_name()*, *hpx::get_error_line_number()*,
hpx::get_error_thread_id(), *hpx::get_error_thread_description()*, *hpx::get_error()*,
hpx::get_error_backtrace(), *hpx::get_error_env()*, *hpx::get_error_what()*, *hpx::get_error_config()*,
hpx::get_error_state()

Parameters

xi – The parameter *e* will be inspected for the requested diagnostic information elements which have been stored at the point where the exception was thrown. This parameter can be one of the following types: *hpx::exception_info*, *hpx::error_code*, *std::exception*, or *std::exception_ptr*.

Throws

nothing –

Returns

The sequence number of the OS-thread used to execute the HPX-thread from which the exception was thrown. If the exception instance does not hold this information, the function will return `std::size(-1)`.

`std::size_t` **get_error_thread_id**(*hpx::exception_info* const &xi) noexcept

Return the unique thread id of the HPX-thread from which the exception was thrown.

The function *hpx::get_error_thread_id* can be used to extract the diagnostic information element representing the HPX-thread id as stored in the given exception instance.

See also:

hpx::diagnostic_information(), *hpx::get_error_host_name()*, *hpx::get_error_process_id()*,
hpx::get_error_function_name(), *hpx::get_error_file_name()*, *hpx::get_error_line_number()*,
hpx::get_error_os_thread(), *hpx::get_error_thread_description()*, *hpx::get_error()*,
hpx::get_error_backtrace(), *hpx::get_error_env()*, *hpx::get_error_what()*, *hpx::get_error_config()*,
hpx::get_error_state()

Parameters

xi – The parameter *e* will be inspected for the requested diagnostic information elements which have been stored at the point where the exception was thrown. This parameter can be one of the following types: *hpx::exception_info*, *hpx::error_code*, *std::exception*, or *std::exception_ptr*.

Throws

nothing –

Returns

The unique thread id of the HPX-thread from which the exception was thrown. If the exception instance does not hold this information, the function will return `std::size_t(0)`.

`std::string` **get_error_thread_description**(*hpx::exception_info* const &xi)

Return any additionally available thread description of the HPX-thread from which the exception was thrown.

The function *hpx::get_error_thread_description* can be used to extract the diagnostic information element representing the additional thread description as stored in the given exception instance.

See also:

hpx::diagnostic_information(), *hpx::get_error_host_name()*, *hpx::get_error_process_id()*,
hpx::get_error_function_name(), *hpx::get_error_file_name()*, *hpx::get_error_line_number()*,
hpx::get_error_os_thread(), *hpx::get_error_thread_id()*, *hpx::get_error_backtrace()*,
hpx::get_error_env(), *hpx::get_error()*, *hpx::get_error_state()*, *hpx::get_error_what()*,
hpx::get_error_config()

Parameters

xi – The parameter *e* will be inspected for the requested diagnostic information elements which have been stored at the point where the exception was thrown. This parameter can be one of the following types: *hpx::exception_info*, *hpx::error_code*, *std::exception*, or *std::exception_ptr*.

Throws

`std::bad_alloc` – (if one of the required allocations fails)

Returns

Any additionally available thread description of the HPX-thread from which the exception was thrown. If the exception instance does not hold this information, the function will return an empty string.

`std::string` **get_error_config**(*hpx::exception_info* const &xi)

Return the HPX configuration information point from which the exception was thrown.

The function *hpx::get_error_config* can be used to extract the HPX configuration information element representing the full HPX configuration information as stored in the given exception instance.

See also:

hpx::diagnostic_information(), *hpx::get_error_host_name()*, *hpx::get_error_process_id()*,
hpx::get_error_function_name(), *hpx::get_error_file_name()*, *hpx::get_error_line_number()*,
hpx::get_error_os_thread(), *hpx::get_error_thread_id()*, *hpx::get_error_backtrace()*,
hpx::get_error_env(), *hpx::get_error()*, *hpx::get_error_state()*, *hpx::get_error_what()*,
hpx::get_error_thread_description()

Parameters

xi – The parameter *e* will be inspected for the requested diagnostic information elements which have been stored at the point where the exception was thrown. This parameter can be one of the following types: *hpx::exception_info*, *hpx::error_code*, *std::exception*, or *std::exception_ptr*.

Throws

`std::bad_alloc` – (if one of the required allocations fails)

Returns

Any additionally available HPX configuration information the point from which the exception was thrown. If the exception instance does not hold this information, the function will return an empty string.

`std::string` **get_error_state**(`hpx::exception_info` const &xi)

Return the HPX runtime state information at which the exception was thrown.

The function `hpx::get_error_state` can be used to extract the HPX runtime state information element representing the state the runtime system is currently in as stored in the given exception instance.

See also:

`hpx::diagnostic_information()`, `hpx::get_error_host_name()`, `hpx::get_error_process_id()`,
`hpx::get_error_function_name()`, `hpx::get_error_file_name()`, `hpx::get_error_line_number()`,
`hpx::get_error_os_thread()`, `hpx::get_error_thread_id()`, `hpx::get_error_backtrace()`,
`hpx::get_error_env()`, `hpx::get_error()`, `hpx::get_error_what()`, `hpx::get_error_thread_description()`

Parameters

xi – The parameter `e` will be inspected for the requested diagnostic information elements which have been stored at the point where the exception was thrown. This parameter can be one of the following types: `hpx::exception_info`, `hpx::error_code`, `std::exception`, or `std::exception_ptr`.

Throws

`std::bad_alloc` – (if one of the required allocations fails)

Returns

The point runtime state at the point at which the exception was thrown. If the exception instance does not hold this information, the function will return an empty string.

hpx::register_thread

Defined in header `hpx/runtime.hpp`⁹⁰⁷.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::unregister_thread`
- `hpx::get_os_thread_data`
- `hpx::enumerate_os_threads`
- `hpx::get_runtime_instance_number`
- `hpx::register_on_exit`
- `hpx::is_starting`
- `hpx::tolerate_node_faults`
- `hpx::is_running`

⁹⁰⁷ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

- `hpx::is_stopped`
- `hpx::is_stopped_or_shutting_down`
- `hpx::get_num_worker_threads`
- `hpx::get_system_uptime`

bool `hpx::register_thread`(*runtime* *rt, char const *name, *error_code* &ec = *throws*)

Register the current kernel thread with HPX, this should be done once for each external OS-thread intended to invoke HPX functionality. Calling this function more than once will return false.

`hpx::unregister_thread`

Defined in header `hpx/runtime.hpp`⁹⁰⁸.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::register_thread`
- `hpx::get_os_thread_data`
- `hpx::enumerate_os_threads`
- `hpx::get_runtime_instance_number`
- `hpx::register_on_exit`
- `hpx::is_starting`
- `hpx::tolerate_node_faults`
- `hpx::is_running`
- `hpx::is_stopped`
- `hpx::is_stopped_or_shutting_down`
- `hpx::get_num_worker_threads`
- `hpx::get_system_uptime`

void `hpx::unregister_thread`(*runtime* *rt)

Unregister the thread from HPX, this should be done once in the end before the external thread exists.

`hpx::get_os_thread_data`

Defined in header `hpx/runtime.hpp`⁹⁰⁹.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::register_thread`
- `hpx::unregister_thread`
- `hpx::enumerate_os_threads`
- `hpx::get_runtime_instance_number`

⁹⁰⁸ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

⁹⁰⁹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

- `hpx::register_on_exit`
- `hpx::is_starting`
- `hpx::tolerate_node_faults`
- `hpx::is_running`
- `hpx::is_stopped`
- `hpx::is_stopped_or_shutting_down`
- `hpx::get_num_worker_threads`
- `hpx::get_system_uptime`

`runtime_local::os_thread_data hpx::get_os_thread_data(std::string const &label)`

Access data for a given OS thread that was previously registered by `register_thread`. This function must be called from a thread that was previously registered with the runtime.

`hpx::enumerate_os_threads`

Defined in header `hpx/runtime.hpp`⁹¹⁰.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::register_thread`
- `hpx::unregister_thread`
- `hpx::get_os_thread_data`
- `hpx::get_runtime_instance_number`
- `hpx::register_on_exit`
- `hpx::is_starting`
- `hpx::tolerate_node_faults`
- `hpx::is_running`
- `hpx::is_stopped`
- `hpx::is_stopped_or_shutting_down`
- `hpx::get_num_worker_threads`
- `hpx::get_system_uptime`

`bool hpx::enumerate_os_threads(hpx::function<bool(os_thread_data const&)> const &f)`

Enumerate all OS threads that have registered with the runtime.

⁹¹⁰ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

hpx::get_runtime_instance_number

Defined in header [hpx/runtime.hpp](#)⁹¹¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::register_thread`
- `hpx::unregister_thread`
- `hpx::get_os_thread_data`
- `hpx::enumerate_os_threads`
- `hpx::register_on_exit`
- `hpx::is_starting`
- `hpx::tolerate_node_faults`
- `hpx::is_running`
- `hpx::is_stopped`
- `hpx::is_stopped_or_shutting_down`
- `hpx::get_num_worker_threads`
- `hpx::get_system_uptime`

`std::size_t hpx::get_runtime_instance_number()`

Return the runtime instance number associated with the runtime instance the current thread is running in.

hpx::register_on_exit

Defined in header [hpx/runtime.hpp](#)⁹¹².

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::register_thread`
- `hpx::unregister_thread`
- `hpx::get_os_thread_data`
- `hpx::enumerate_os_threads`
- `hpx::get_runtime_instance_number`
- `hpx::is_starting`
- `hpx::tolerate_node_faults`
- `hpx::is_running`
- `hpx::is_stopped`
- `hpx::is_stopped_or_shutting_down`
- `hpx::get_num_worker_threads`

⁹¹¹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

⁹¹² <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

- *hpx::get_system_uptime*

bool **hpx::register_on_exit**(*hpx::function*<void()> const&)

Register a function to be called during system shutdown.

hpx::is_starting

Defined in header [hpx/runtime.hpp](#)⁹¹³.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::register_thread*
- *hpx::unregister_thread*
- *hpx::get_os_thread_data*
- *hpx::enumerate_os_threads*
- *hpx::get_runtime_instance_number*
- *hpx::register_on_exit*
- *hpx::tolerate_node_faults*
- *hpx::is_running*
- *hpx::is_stopped*
- *hpx::is_stopped_or_shutting_down*
- *hpx::get_num_worker_threads*
- *hpx::get_system_uptime*

bool **hpx::is_starting**()

Test whether the runtime system is currently being started.

This function returns whether the runtime system is currently being started or not, e.g. whether the current state of the runtime system is *hpx::state::startup*

Note: This function needs to be executed on a HPX-thread. It will return false otherwise.

hpx::tolerate_node_faults

Defined in header [hpx/runtime.hpp](#)⁹¹⁴.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::register_thread*
- *hpx::unregister_thread*
- *hpx::get_os_thread_data*

⁹¹³ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

⁹¹⁴ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

- `hpx::enumerate_os_threads`
- `hpx::get_runtime_instance_number`
- `hpx::register_on_exit`
- `hpx::is_starting`
- `hpx::is_running`
- `hpx::is_stopped`
- `hpx::is_stopped_or_shutting_down`
- `hpx::get_num_worker_threads`
- `hpx::get_system_uptime`

bool `hpx::tolerate_node_faults()`

Test if HPX runs in fault-tolerant mode.

This function returns whether the runtime system is running in fault-tolerant mode

`hpx::is_running`

Defined in header `hpx/runtime.hpp`⁹¹⁵.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::register_thread`
- `hpx::unregister_thread`
- `hpx::get_os_thread_data`
- `hpx::enumerate_os_threads`
- `hpx::get_runtime_instance_number`
- `hpx::register_on_exit`
- `hpx::is_starting`
- `hpx::tolerate_node_faults`
- `hpx::is_stopped`
- `hpx::is_stopped_or_shutting_down`
- `hpx::get_num_worker_threads`
- `hpx::get_system_uptime`

bool `hpx::is_running()`

Test whether the runtime system is currently running.

This function returns whether the runtime system is currently running or not, e.g. whether the current state of the runtime system is `hpx::state::running`

Note: This function needs to be executed on a HPX-thread. It will return false otherwise.

⁹¹⁵ <http://github.com/TheHPXProject/hpx/blob/d402eed7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

hpx::is_stopped

Defined in header [hpx/runtime.hpp](#)⁹¹⁶.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::register_thread*
- *hpx::unregister_thread*
- *hpx::get_os_thread_data*
- *hpx::enumerate_os_threads*
- *hpx::get_runtime_instance_number*
- *hpx::register_on_exit*
- *hpx::is_starting*
- *hpx::tolerate_node_faults*
- *hpx::is_running*
- *hpx::is_stopped_or_shutting_down*
- *hpx::get_num_worker_threads*
- *hpx::get_system_uptime*

bool **hpx::is_stopped()**

Test whether the runtime system is currently stopped.

This function returns whether the runtime system is currently stopped or not, e.g. whether the current state of the runtime system is *hpx::state::stopped*

Note: This function needs to be executed on a HPX-thread. It will return false otherwise.

hpx::is_stopped_or_shutting_down

Defined in header [hpx/runtime.hpp](#)⁹¹⁷.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::register_thread*
- *hpx::unregister_thread*
- *hpx::get_os_thread_data*
- *hpx::enumerate_os_threads*
- *hpx::get_runtime_instance_number*
- *hpx::register_on_exit*
- *hpx::is_starting*

⁹¹⁶ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

⁹¹⁷ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

- `hpx::tolerate_node_faults`
- `hpx::is_running`
- `hpx::is_stopped`
- `hpx::get_num_worker_threads`
- `hpx::get_system_uptime`

bool `hpx::is_stopped_or_shutting_down()`

Test whether the runtime system is currently being shut down.

This function returns whether the runtime system is currently being shut down or not, e.g. whether the current state of the runtime system is `hpx::state::stopped` or `hpx::state::shutdown`

Note: This function needs to be executed on a HPX-thread. It will return false otherwise.

`hpx::get_num_worker_threads`

Defined in header `hpx/runtime.hpp`⁹¹⁸.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::register_thread`
- `hpx::unregister_thread`
- `hpx::get_os_thread_data`
- `hpx::enumerate_os_threads`
- `hpx::get_runtime_instance_number`
- `hpx::register_on_exit`
- `hpx::is_starting`
- `hpx::tolerate_node_faults`
- `hpx::is_running`
- `hpx::is_stopped`
- `hpx::is_stopped_or_shutting_down`
- `hpx::get_system_uptime`

`std::size_t hpx::get_num_worker_threads()`

Return the number of worker OS- threads used to execute HPX threads.

This function returns the number of OS-threads used to execute HPX threads. If the function is called while no HPX runtime system is active, it will return zero.

⁹¹⁸ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

hpx::get_system_uptime

Defined in header `hpx/runtime.hpp`⁹¹⁹.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::register_thread`
- `hpx::unregister_thread`
- `hpx::get_os_thread_data`
- `hpx::enumerate_os_threads`
- `hpx::get_runtime_instance_number`
- `hpx::register_on_exit`
- `hpx::is_starting`
- `hpx::tolerate_node_faults`
- `hpx::is_running`
- `hpx::is_stopped`
- `hpx::is_stopped_or_shutting_down`
- `hpx::get_num_worker_threads`

`std::uint64_t hpx::get_system_uptime()`

Return the system uptime measure on the thread executing this call.

This function returns the system uptime measured in nanoseconds for the thread executing this call. If the function is called while no HPX runtime system is active, it will return zero.

hpx/runtime_local/thread_pool_helpers.hpp

Defined in header `hpx/runtime_local/thread_pool_helpers.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **resource**

⁹¹⁹ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

Functions

`std::size_t get_num_thread_pools()`

Return the number of thread pools currently managed by the *resource_partitioner*

`std::size_t get_num_threads()`

Return the number of threads in all thread pools currently managed by the *resource_partitioner*

`std::size_t get_num_threads(std::string const &pool_name)`

Return the number of threads in the given thread pool currently managed by the *resource_partitioner*

`std::size_t get_num_threads(std::size_t pool_index)`

Return the number of threads in the given thread pool currently managed by the *resource_partitioner*

`std::size_t get_pool_index(std::string const &pool_name)`

Return the internal index of the pool given its name.

`std::string const &get_pool_name(std::size_t pool_index)`

Return the name of the pool given its internal index.

`threads::thread_pool_base &get_thread_pool(std::string const &pool_name)`

Return the name of the pool given its name.

`threads::thread_pool_base &get_thread_pool(std::size_t pool_index)`

Return the thread pool given its internal index.

`bool pool_exists(std::string const &pool_name)`

Return true if the pool with the given name exists.

`bool pool_exists(std::size_t pool_index)`

Return true if the pool with the given index exists.

namespace **threads**

Functions

`std::int64_t get_thread_count(thread_schedule_state state = thread_schedule_state::unknown)`

The function *get_thread_count* returns the number of currently known threads.

Note: If `state == unknown` this function will not only return the number of currently existing threads, but will add the number of registered task descriptions (which have not been converted into threads yet).

Parameters

state – [in] This specifies the thread-state for which the number of threads should be retrieved.

`std::int64_t get_thread_count(thread_priority priority, thread_schedule_state state = thread_schedule_state::unknown)`

The function *get_thread_count* returns the number of currently known threads.

Note: If `state == unknown` this function will not only return the number of currently existing threads, but will add the number of registered task descriptions (which have not been converted into threads yet).

Parameters

- **priority** – [in] This specifies the thread-priority for which the number of threads should be retrieved.
- **state** – [in] This specifies the thread-state for which the number of threads should be retrieved.

`std::int64_t` **get_idle_core_count()**

The function `get_idle_core_count` returns the number of currently idling threads (cores).

`mask_type` **get_idle_core_mask()**

The function `get_idle_core_mask` returns a bit-mask representing the currently idling threads (cores).

`bool` **enumerate_threads**(*hpx::function<bool(thread_id_type)> const &f, thread_schedule_state state = thread_schedule_state::unknown*)

The function `enumerate_threads` will invoke the given function *f* for each thread with a matching thread state.

Parameters

- **f** – [in] The function which should be called for each matching thread. Returning ‘false’ from this function will stop the enumeration process.
- **state** – [in] This specifies the thread-state for which the threads should be enumerated.

hpx/runtime_local/get_worker_thread_num.hpp

Defined in header `hpx/runtime_local/get_worker_thread_num.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx::get_thread_name

Defined in header `hpx/runtime.hpp`⁹²⁰.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

`std::string` *hpx::***get_thread_name()**

Return the name of the calling thread.

This function returns the name of the calling thread. This name uniquely identifies the thread in the context of HPX. If the function is called while no HPX runtime system is active, the result will be “<unknown>”.

⁹²⁰ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

serialization

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/serialization/base_object.hpp

Defined in header `hpx/serialization/base_object.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<typename Derived, typename Base>
```

```
struct base_object_type<Derived, Base, std::enable_if_t<hpx::traits::is_intrusive_polymorphic_v<Derived>>>
```

Public Functions

```
inline explicit constexpr base_object_type(Derived &d) noexcept
```

```
template<typename Archive>
inline void save(Archive &ar, unsigned) const
```

```
template<typename Archive>
inline void load(Archive &ar, unsigned)
```

```
HPX_SERIALIZATION_SPLIT_MEMBER()
```

Public Members

Derived &**d_**

```
namespace hpx
```

```
namespace serialization
```

Functions

```
template<typename Base, typename Derived>
constexpr base_object_type<Derived, Base> base_object(Derived &d) noexcept
```

```
template<typename D, typename B>
output_archive &operator<<(output_archive &ar, base_object_type<D, B> t)
```

```
template<typename D, typename B>
input_archive &operator>>(input_archive &ar, base_object_type<D, B> t)
```

```
template<typename D, typename B>
output_archive &operator&(output_archive &ar, base_object_type<D, B> t)
```

```
template<typename D, typename B>
input_archive &operator&(input_archive &ar, base_object_type<D, B> t)
```

```
template<typename Derived, typename Base, typename Enable = void>
```

struct **base_object_type**

Public Functions

inline explicit constexpr **base_object_type**(*Derived* &d) noexcept

template<typename **Archive**>
inline void **serialize**(*Archive* &ar, unsigned)

Public Members

Derived &d_

template<typename **Derived**,
typename **Base**> **is_intrusive_polymorphic_v**< **Derived** > > >

Public Functions

inline explicit constexpr **base_object_type**(*Derived* &d) noexcept

template<typename **Archive**>
inline void **save**(*Archive* &ar, unsigned) const

template<typename **Archive**>
inline void **load**(*Archive* &ar, unsigned)

HPX_SERIALIZATION_SPLIT_MEMBER()

Public Members

Derived &d_

synchronization

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/synchronization/sliding_semaphore.hpp

Defined in header `hpx/synchronization/sliding_semaphore.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

Typedefs

```
using sliding_semaphore = sliding_semaphore_var<>
```

```
template<typename Mutex = hpx::spinlock>
```

```
class sliding_semaphore_var
```

#include <sliding_semaphore.hpp> A semaphore is a protected variable (an entity storing a value) or abstract data type (an entity grouping several variables that may or may not be numerical) which constitutes the classic method for restricting access to shared resources, such as shared memory, in a multiprogramming environment. Semaphores exist in many variants, though usually the term refers to a counting semaphore, since a binary semaphore is better known as a mutex. A counting semaphore is a counter for a set of available resources, rather than a locked/unlocked flag of a single resource. It was invented by Edsger Dijkstra. Semaphores are the classic solution to preventing race conditions in the dining philosophers problem, although they do not prevent resource deadlocks.

Sliding semaphores can be used for synchronizing multiple threads as well: one thread waiting for several other threads to touch (signal) the semaphore, or several threads waiting for one other thread to touch this semaphore. The difference to a counting semaphore is that a sliding semaphore will not limit the number of threads which are allowed to proceed, but will make sure that the difference between the (arbitrary) number passed to set and wait does not exceed a given threshold.

Public Functions

```
sliding_semaphore_var(sliding_semaphore_var const&) = delete
```

```
sliding_semaphore_var &operator=(sliding_semaphore_var const&) = delete
```

```
sliding_semaphore_var(sliding_semaphore_var&&) = delete
```

```
sliding_semaphore_var &operator=(sliding_semaphore_var&&) = delete
```

```
inline explicit sliding_semaphore_var(std::int64_t max_difference, std::int64_t lower_limit = 0) noexcept
```

Construct a new sliding semaphore.

Parameters

- **max_difference** – [in] The max difference between the upper limit (as set by *wait()*) and the lower limit (as set by *signal()*) which is allowed without suspending any thread calling *wait()*.
- **lower_limit** – [in] The initial lower limit.

```
inline void set_max_difference(std::int64_t max_difference, std::int64_t lower_limit = 0) noexcept
```

Set/Change the difference that will cause the semaphore to trigger

Parameters

- **max_difference** – [in] The max difference between the upper limit (as set by *wait()*) and the lower limit (as set by *signal()*) which is allowed without suspending any thread calling *wait()*.
- **lower_limit** – [in] The initial lower limit.

```
inline void wait(std::int64_t upper_limit)
```

Wait for the semaphore to be signaled.

Parameters

- **upper_limit** – [in] The new upper limit. The calling thread will be suspended if the difference between this value and the largest *lower_limit* which was set by *signal()* is larger than the *max_difference*.

```
inline bool try_wait(std::int64_t upper_limit = 1)
```

Try to wait for the semaphore to be signaled.

Parameters

upper_limit – [in] The new upper limit. The calling thread will be suspended if the difference between this value and the largest lower_limit which was set by *signal()* is larger than the max_difference.

Returns

The function returns true if the calling thread would not block if it was calling *wait()*.

```
inline void signal(std::int64_t lower_limit)
```

Signal the semaphore.

Parameters

lower_limit – [in] The new lower limit. This will update the current lower limit of this semaphore. It will also re-schedule all suspended threads for which their associated upper limit is not larger than the lower limit plus the max_difference.

```
inline std::int64_t signal_all()
```

Private Types

```
using mutex_type = Mutex
```

```
using data_type = lcos::local::detail::sliding_semaphore_data<mutex_type>
```

Private Members

```
hpx::intrusive_ptr<data_type> data_
```

hpx::nostopstate

Defined in header [hpx/stop_token.hpp](#)⁹²¹.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::stop_callback*
- *hpx::stop_source*
- *hpx::stop_token*
- *hpx::nostopstate_t*

```
constexpr nostopstate_t hpx::nostopstate = { }
```

This is a constant object instance of *hpx::nostopstate_t* for use in constructing an empty *hpx::stop_source*, as a placeholder value in the non-default constructor.

⁹²¹ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/stop_token.hpp

hpx::stop_callback

Defined in header `hpx/stop_token.hpp`⁹²².

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::nostopstate`
- `hpx::stop_source`
- `hpx::stop_token`
- `hpx::nostopstate_t`

template<typename **Callback**>

class **stop_callback**

hpx::stop_source

Defined in header `hpx/stop_token.hpp`⁹²³.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::nostopstate`
- `hpx::stop_callback`
- `hpx::stop_token`
- `hpx::nostopstate_t`

class **stop_source**

The `stop_source` class provides the means to issue a stop request, such as for `hpx::jthread` cancellation. A stop request made for one `stop_source` object is visible to all *stop_sources* and *hpx::stop_tokens* of the same associated stop-state; any `hpx::stop_callback(s)` registered for associated `hpx::stop_token(s)` will be invoked, and any *hpx::condition_variable_any* objects waiting on associated `hpx::stop_token(s)` will be awoken. Once a stop is requested, it cannot be withdrawn. Additional stop requests have no effect.

Note: For the purposes of `hpx::jthread` cancellation the `stop_source` object should be retrieved from the *hpx::jthread* object using `get_stop_source()`; or stop should be requested directly from the `hpx::jthread` object using `request_stop()`. This will then use the same associated stop-state as that passed into the `hpx::jthread`'s invoked function argument (i.e., the function being executed on its thread). For other uses, however, a `stop_source` can be constructed separately using the default constructor, which creates new stop-state.

⁹²² http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/stop_token.hpp

⁹²³ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/stop_token.hpp

hpx::stop_token

Defined in header [hpx/stop_token.hpp](#)⁹²⁴.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::nostopstate*
- *hpx::stop_callback*
- *hpx::stop_source*
- *hpx::nostopstate_t*

class **stop_token**

The `stop_token` class provides the means to check if a stop request has been made or can be made, for its associated `hpx::stop_source` object. It is essentially a thread-safe “view” of the associated stop-state. The *stop_token* can also be passed to the constructor of `hpx::stop_callback`, such that the callback will be invoked if the `stop_token`’s associated `hpx::stop_source` is requested to stop. And `stop_token` can be passed to the interruptible waiting functions of `hpx::condition_variable_any`, to interrupt the condition variable’s wait if stop is requested.

Note: A `stop_token` object is not generally constructed independently, but rather retrieved from a `hpx::jthread` or `hpx::stop_source`. This makes it share the same associated stop-state as the `hpx::jthread` or `hpx::stop_source`.

hpx::nostopstate_t

Defined in header [hpx/stop_token.hpp](#)⁹²⁵.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::nostopstate*
- *hpx::stop_callback*
- *hpx::stop_source*
- *hpx::stop_token*

struct **nostopstate_t**

Unit type intended for use as a placeholder in *hpx::stop_source* non-default constructor, that makes the constructed *hpx::stop_source* empty with no associated stop-state.

⁹²⁴ [http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/stop_token.](http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/stop_token.hpp)

⁹²⁵ [http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/stop_token.](http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/stop_token.hpp)
hpp

hpx::binary_semaphore

Defined in header [hpx/semaphore.hpp](#)⁹²⁶.

See *Public API* for a list of names and headers that are part of the public HPX API.

class **binary_semaphore**

A binary semaphore is a semaphore object that has only two states. `binary_semaphore` is an alias for specialization of `hpx::counting_semaphore` with *LeastMaxValue* being 1. HPX's implementation of `binary_semaphore` is more efficient than the default implementation of a counting semaphore with a unit resource count (`hpx::counting_semaphore`).

hpx::recursive_mutex

Defined in header [hpx/mutex.hpp](#)⁹²⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
using hpx::recursive_mutex = detail::recursive_mutex_impl<>
```

hpx::shared_mutex

Defined in header [hpx/shared_mutex.hpp](#)⁹²⁸.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
using hpx::shared_mutex = detail::shared_mutex<>
```

The *shared_mutex* class is a synchronization primitive that can be used to protect shared data from being simultaneously accessed by multiple threads. In contrast to other mutex types which facilitate exclusive access, a *shared_mutex* has two levels of access:

- *shared* - several threads can share ownership of the same mutex.
- *exclusive* - only one thread can own the mutex.

If one thread has acquired the exclusive lock (through *lock*, *try_lock*), no other threads can acquire the lock (including the shared). If one thread has acquired the shared lock (through *lock_shared*, *try_lock_shared*), no other thread can acquire the exclusive lock, but can acquire the shared lock. Only when the exclusive lock has not been acquired by any thread, the shared lock can be acquired by multiple threads. Within one thread, only one lock (shared or exclusive) can be acquired at the same time. Shared mutexes are especially useful when shared data can be safely read by any number of threads simultaneously, but a thread may only write the same data when no other thread is reading or writing at the same time. The *shared_mutex* class satisfies all requirements of *SharedMutex* and *StandardLayoutType*.

⁹²⁶ http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/semaphore.hpp

⁹²⁷ http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/mutex.hpp

⁹²⁸ http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/shared_mutex.hpp

hpx::counting_semaphore

Defined in header [hpx/semaphore.hpp](#)⁹²⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<std::ptrdiff_t LeastMaxValue = PTRDIFF_MAX>
```

```
class counting_semaphore
```

A semaphore is a protected variable (an entity storing a value) or abstract data type (an entity grouping several variables that may or may not be numerical) which constitutes the classic method for restricting access to shared resources, such as shared memory, in a multiprogramming environment. Semaphores exist in many variants, though usually the term refers to a counting semaphore, since a binary semaphore is better known as a mutex. A counting semaphore is a counter for a set of available resources, rather than a locked/unlocked flag of a single resource. It was invented by Edsger Dijkstra. Semaphores are the classic solution to preventing race conditions in the dining philosophers problem, although they do not prevent resource deadlocks.

Counting semaphores can be used for synchronizing multiple threads as well: one thread waiting for several other threads to touch (signal) the semaphore, or several threads waiting for one other thread to touch this semaphore. Unlike `hpx::mutex` a `counting_semaphore` is not tied to threads of execution^{—}; acquiring a semaphore can occur on a different thread than releasing the semaphore, for example. All operations on `counting_semaphore` can be performed concurrently and without any relation to specific threads of execution, with the exception of the destructor which cannot be performed concurrently but can be performed on a different thread.

Semaphores are lightweight synchronization primitives used to constrain concurrent access to a shared resource. They are widely used to implement other synchronization primitives and, whenever both are applicable, can be more efficient than condition variables.

A counting semaphore is a semaphore object that models a non-negative resource count.

Class template `counting_semaphore` maintains an internal counter that is initialized when the semaphore is created. The counter is decremented when a thread acquires the semaphore, and is incremented when a thread releases the semaphore. If a thread tries to acquire the semaphore when the counter is zero, the thread will block until another thread increments the counter by releasing the semaphore.

Specializations of `hpx::counting_semaphore` are not `<a`, `<a`, `<a`, `<a`, or `<a`.

Note: `counting_semaphore`'s `try_acquire()` can spuriously fail.

Template Parameters

LeastMaxValue – `counting_semaphore` allows more than one concurrent access to the same resource, for at least *LeastMaxValue* concurrent accessors. As its name indicates, the *LeastMaxValue* is the minimum max value, not the actual max value. Thus `max()` can yield a number larger than *LeastMaxValue*.

⁹²⁹ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/semaphore.hpp

hpx::once_flag

Defined in header [hpx/mutex.hpp](#)⁹³⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- [hpx::call_once](#)

struct once_flag

The class `hpx::once_flag` is a helper structure for `hpx::call_once`. An object of type `hpx::once_flag` that is passed to multiple calls to `hpx::call_once` allows those calls to coordinate with each other such that only one of the calls will actually run to completion. `hpx::once_flag` is neither copyable nor movable.

hpx::call_once

Defined in header [hpx/mutex.hpp](#)⁹³¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- [hpx::once_flag](#)

template<typename **F**, typename ...**Args**>

void `hpx::call_once`(`once_flag` &flag, *F* &&*f*, *Args*&&... args)

Executes the Callable object *f* exactly once, even if called concurrently, from several threads.

In detail:

- If, by the time `call_once` is called, flag indicates that *f* was already called, `call_once` returns right away (such a call to `call_once` is known as passive).
- Otherwise, `call_once` invokes `std::forward<Callable>(f)` with the arguments `std::forward<Args>(args)...` (as if by `hpx::invoke`). Unlike the `hpx::thread` constructor or `hpx::async`, the arguments are not moved or copied because they don't need to be transferred to another thread of execution. (such a call to `call_once` is known as active).
 - If that invocation throws an exception, it is propagated to the caller of `call_once`, and the flag is not flipped so that another call will be attempted (such a call to `call_once` is known as exceptional).
 - If that invocation returns normally (such a call to `call_once` is known as returning), the flag is flipped, and all other calls to `call_once` with the same flag are guaranteed to be passive. All active calls on the same flag form a single total order consisting of zero or more exceptional calls, followed by one returning call. The end of each active call synchronizes-with the next active call in that order. The return from the returning call synchronizes-with the returns from all passive calls on the same flag: this means that all concurrent calls to `call_once` are guaranteed to observe any side-effects made by the active call, with no additional synchronization.

Note: If concurrent calls to `call_once` pass different functions *f*, it is unspecified which *f* will be called. The selected function runs in the same thread as the `call_once` invocation it was passed to. Initialization of function-local statics is guaranteed to occur only once even when called from multiple threads, and may be more efficient than the equivalent code using `hpx::call_once`. The POSIX equivalent of this function is `pthread_once`.

⁹³⁰ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/mutex.hpp

⁹³¹ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/mutex.hpp

Parameters

- **flag** – an object, for which exactly one function gets executed
- **f** – Callable object to invoke
- **args** – arguments to pass to the function

Throws

`std::system_error` – if any condition prevents calls to *call_once* from executing as specified or any exception thrown by *f*

hpx::mutex

Defined in header [hpx/mutex.hpp](#)⁹³².

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::timed_mutex*

class mutex

mutex class is a synchronization primitive that can be used to protect shared data from being simultaneously accessed by multiple threads. *mutex* offers exclusive, non-recursive ownership semantics:

- A calling thread owns a mutex from the time that it successfully calls either *lock* or *try_lock* until it calls *unlock*.
- When a thread owns a *mutex*, all other threads will block (for calls to *lock*) or receive a *false* return value (for *try_lock*) if they attempt to claim ownership of the *mutex*.
- A calling thread must not own the *mutex* prior to calling *lock* or *try_lock*.

The behavior of a program is undefined if a *mutex* is destroyed while still owned by any threads, or a thread terminates while owning a *mutex*. The mutex class satisfies all requirements of <a and <a.

`hpx::mutex` is neither copyable nor movable.

Subclassed by *hpx::timed_mutex*

hpx::timed_mutex

Defined in header [hpx/mutex.hpp](#)⁹³³.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::mutex*

class timed_mutex : private hpx::mutex

The *timed_mutex* class is a synchronization primitive that can be used to protect shared data from being simultaneously accessed by multiple threads. In a manner similar to *mutex*, *timed_mutex* offers exclusive, non-recursive ownership semantics. In addition, *timed_mutex* provides the ability to attempt to claim ownership of

⁹³² http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/mutex.hpp

⁹³³ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/mutex.hpp

a `timed_mutex` with a timeout via the member functions `try_lock_for()` and `try_lock_until()`. The `timed_mutex` class satisfies all requirements of `<a` and `<a`.

`hpx::timed_mutex` is neither copyable nor movable.

`hpx::latch`

Defined in header [hpx/latch.hpp](#)⁹³⁴.

See *Public API* for a list of names and headers that are part of the public HPX API.

class `latch`

Latches are a thread coordination mechanism that allow one or more threads to block until an operation is completed. An individual latch is a single-use object; once the operation has been completed, the latch cannot be reused.

Subclassed by `hpx::lcos::local::latch`

`hpx::no_mutex`

Defined in header [hpx/mutex.hpp](#)⁹³⁵.

See *Public API* for a list of names and headers that are part of the public HPX API.

struct `no_mutex`

`no_mutex` class can be used in cases where the shared data between multiple threads can be accessed simultaneously without causing inconsistencies.

`hpx::barrier`

Defined in header [hpx/barrier.hpp](#)⁹³⁶.

See *Public API* for a list of names and headers that are part of the public HPX API.

template<typename **OnCompletion** = detail::empty_oncompletion>

class `barrier`

A barrier is a thread coordination mechanism whose lifetime consists of a sequence of barrier phases, where each phase allows at most an expected number of threads to block until the expected number of threads arrive at the barrier. [Note: A barrier is useful for managing repeated tasks that are handled by multiple threads. - end note] Each barrier phase consists of the following steps:

- The expected count is decremented by each call to `arrive` or `arrive_and_drop`.
- When the expected count reaches zero, the phase completion step is run. For the specialization with the default value of the `CompletionFunction` template parameter, the completion step is run as part of the call to `arrive` or `arrive_and_drop` that caused the expected count to reach zero. For other specializations, the completion step is run on one of the threads that arrived at the barrier during the phase.

⁹³⁴ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/latch.hpp>

⁹³⁵ http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/mutex.hpp

⁹³⁶ <http://github.com/TheHPXProject/hpx/blob/d402ead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/barrier.hpp>

- When the completion step finishes, the expected count is reset to what was specified by the expected argument to the constructor, possibly adjusted by calls to `arrive_and_drop`, and the next phase starts.

Each phase defines a phase synchronization point. Threads that arrive at the barrier during the phase can block on the phase synchronization point by calling `wait`, and will remain blocked until the phase completion step is run. The phase completion step that is executed at the end of each phase has the following effects:

- Invokes the completion function, equivalent to `completion()`.
- Unblocks all threads that are blocked on the phase synchronization point.

The end of the completion step strongly happens before the returns from all calls that were unblocked by the completion step. For specializations that do not have the default value of the `CompletionFunction` template parameter, the behavior is undefined if any of the barrier object's member functions other than `wait` are called while the completion step is in progress.

Concurrent invocations of the member functions of `barrier`, other than its destructor, do not introduce data races. The member functions `arrive` and `arrive_and_drop` execute atomically.

`CompletionFunction` shall meet the `Cpp17MoveConstructible` (Table 28) and `Cpp17Destructible` (Table 32) requirements. `std::is_nothrow_invocable_v<CompletionFunction&>` shall be true.

The default value of the `CompletionFunction` template parameter is an unspecified type, such that, in addition to satisfying the requirements of `CompletionFunction`, it meets the `Cpp17DefaultConstructible` requirements (Table 27) and `completion()` has no effects.

`barrier::arrival_token` is an unspecified type, such that it meets the `Cpp17MoveConstructible` (Table 28), `Cpp17MoveAssignable` (Table 30), and `Cpp17Destructible` (Table 32) requirements.

hpx/synchronization/event.hpp

Defined in header `hpx/synchronization/event.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace lcos
```

```
        namespace local
```

```
            class event
```

```
                #include <event.hpp> Event semaphores can be used for synchronizing multiple threads that need to wait for an event to occur. When the event occurs, all threads waiting for the event are woken up.
```

Public Functions

```
inline event() noexcept
    Construct a new event semaphore.

inline bool occurred() const noexcept
    Check if the event has occurred.

inline void wait()
    Wait for the event to occur.

inline void set()
    Release all threads waiting on this semaphore.

inline void reset() noexcept
    Reset the event.
```

Private Types

```
using mutex_type = hpx::spinlock
```

Private Functions

```
inline void wait_locked(std::unique_lock<mutex_type> &l)

inline void set_locked(std::unique_lock<mutex_type> l)
```

Private Members

```
mutex_type mtx_
    This mutex protects the queue.

local::detail::condition_variable cond_

std::atomic<bool> event_
```

hpx::spinlock

Defined in header [hpx/mutex.hpp](#)⁹³⁷.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
using hpx::spinlock = detail::spinlock<true>
```

spinlock is a type of lock that causes a thread attempting to obtain it to check for its availability while waiting in a loop continuously.

⁹³⁷ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/mutex.hpp

hpx::condition_variable

Defined in header `hpx/condition_variable.hpp`⁹³⁸.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::condition_variable_any`
- `hpx::cv_status`

class `condition_variable`

The `condition_variable` class is a synchronization primitive that can be used to block a thread, or multiple threads at the same time, until another thread both modifies a shared variable (the condition), and notifies the `condition_variable`.

The thread that intends to modify the shared variable has to

- a. acquire a `hpx::mutex` (typically via `std::lock_guard`)
- b. perform the modification while the lock is held
- c. execute `notify_one` or `notify_all` on the `condition_variable` (the lock does not need to be held for notification)

Even if the shared variable is atomic, it must be modified under the mutex in order to correctly publish the modification to the waiting thread. Any thread that intends to wait on `condition_variable` has to

- a. acquire a `std::unique_lock<hpx::mutex>`, on the same mutex as used to protect the shared variable
- b. either
 - i. check the condition, in case it was already updated and notified
 - ii. execute `wait`, `await_for`, or `wait_until`. The wait operations atomically release the mutex and suspend the execution of the thread.
 - iii. When the condition variable is notified, a timeout expires, or a spurious wakeup occurs, the thread is awakened, and the mutex is atomically reacquired. The thread should then check the condition and resume waiting if the wake-up was spurious. or
- i. use the predicated overload of `wait`, `wait_for`, and `wait_until`, which takes care of the three steps above.

`hpx::condition_variable` works only with `std::unique_lock<hpx::mutex>`. This restriction allows for maximal efficiency on some platforms. `hpx::condition_variable_any` provides a condition variable that works with any `<a` object, such as `std::shared_lock`.

Condition variables permit concurrent invocation of the `wait`, `wait_for`, `wait_until`, `notify_one` and `notify_all` member functions.

The class `hpx::condition_variable` is a `<a`. It is not `<a`, `<a`, `<a`, or `<a`.

⁹³⁸ http://github.com/TheHPXProject/hpx/blob/d402eed7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/condition_variable.hpp

hpx::condition_variable_any

Defined in header [hpx/condition_variable.hpp](#)⁹³⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::condition_variable*
- *hpx::cv_status*

class **condition_variable_any**

The `condition_variable_any` class is a generalization of `hpx::condition_variable`. Whereas `hpx::condition_variable` works only on `std::unique_lock<std::mutex>`, a `condition_variable_any` can operate on any lock that meets the `<a` requirements.

See `hpx::condition_variable` for the description of the semantics of condition variables. It is not `<a`, `<a`, `<a`, or `<a`.

hpx::cv_status

Defined in header [hpx/condition_variable.hpp](#)⁹⁴⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::condition_variable*
- *hpx::condition_variable_any*

enum class **hpx::cv_status**

The scoped enumeration `hpx::cv_status` describes whether a timed wait returned because of timeout or not. `hpx::cv_status` is used by the `wait_for` and `wait_until` member functions of `hpx::condition_variable` and `hpx::condition_variable_any`.

Values:

enumerator **no_timeout**

The condition variable was awakened with *notify_all*, *notify_one*, or spuriously

enumerator **timeout**

the condition variable was awakened by timeout expiration

enumerator **error**

there was an error

⁹³⁹ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/condition_variable.hpp

⁹⁴⁰ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/condition_variable.hpp

tag_invoke

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx::invoke

Defined in header `hpx/functional.hpp`⁹⁴¹.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
template<typename F, typename ...Ts>
constexpr util::invoke_result_t<F, Ts&&...> hpx::invoke(F &&f, Ts&&... vs) noexcept(
    noexcept(HPX_INVOKE(HPX_FORWARD(F, f),
        HPX_FORWARD(Ts, vs)...)))
```

Invokes the given callable object *f* with the content of the argument pack *vs*

Note: This function is similar to `std::invoke` (C++17)

Parameters

- **f** – Requires to be a callable object. If *f* is a member function pointer, the first argument in the pack will be treated as the callee (this object).
- **vs** – An arbitrary pack of arguments

Throws

`std::exception` – like objects thrown by call to object *f* with the argument types *vs*.

Returns

The result of the callable object when it's called with the given argument types.

hpx::is_invocable

Defined in header `hpx/type_traits.hpp`⁹⁴².

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::is_invocable_r`

```
template<typename F, typename ...Ts>
```

```
struct is_invocable : public hpx::detail::is_invocable_impl<F&&(Ts&&...)>
```

Determines whether *F* can be invoked with the arguments *Ts*... Formally, determines whether

`INVOKE(std::declval<F>(), std::declval<Ts>()...)`

is well-formed when treated as an unevaluated operand, where *INVOKE* is the operation defined in *Callable*.

F, *R* and all types in the parameter pack *Ts* shall each be a complete type, (possibly cv-qualified) void, or an array of unknown bound. Otherwise, the behavior is undefined. If an instantiation of a template above depends,

⁹⁴¹ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/functional.hpp

⁹⁴² http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/type_traits.hpp

directly or indirectly, on an incomplete type, and that instantiation could yield a different result if that type were hypothetically completed, the behavior is undefined.

hpx::is_invocable_r

Defined in header `hpx/type_traits.hpp`⁹⁴³.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::is_invocable`

template<typename **R**, typename **F**, typename ...**Ts**>

struct **is_invocable_r** : public `hpx::detail::is_invocable_r_impl<F&&(Ts&&...), R>`

Determines whether *F* can be invoked with the arguments *Ts...* to yield a result that is convertible to *R* and the implicit conversion does not bind a reference to a temporary object (since C++23). If *R* is *cv* void, the result can be any type. Formally, determines whether

```
INVOKE<R>(std::declval<F>(), std::declval<Ts>()...)
```

is well-formed when treated as an unevaluated operand, where *INVOKE* is the operation defined in *Callable*. Determines whether *F* can be invoked with the arguments *Ts...*. Formally, determines whether

```
INVOKE(std::declval<F>(), std::declval<Ts>()...)
```

is well-formed when treated as an unevaluated operand, where *INVOKE* is the operation defined in *Callable*.

F, *R* and all types in the parameter pack *Ts* shall each be a complete type, (possibly cv-qualified) void, or an array of unknown bound. Otherwise, the behavior is undefined. If an instantiation of a template above depends, directly or indirectly, on an incomplete type, and that instantiation could yield a different result if that type were hypothetically completed, the behavior is undefined.

thread_pool_util

See *Public API* for a list of names and headers that are part of the public HPX API.

hpx/thread_pool_util/thread_pool_suspension_helpers.hpp

Defined in header `hpx/thread_pool_util/thread_pool_suspension_helpers.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **threads**

⁹⁴³ http://github.com/TheHPXProject/hpx/blob/d402cead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/type_traits.hpp

Functions

hpx::future<void> **resume_processing_unit**(*thread_pool_base* &pool, *std::size_t* virt_core)

Resumes the given processing unit. When the processing unit has been resumed the returned future will be ready.

Note: Can only be called from an HPX thread. Use `resume_processing_unit_cb` or to resume the processing unit from outside HPX. Requires that the pool has `threads::policies::enable_elasticity` set.

Parameters

- **pool** – [in] The thread pool to resume a processing unit on.
- **virt_core** – [in] The processing unit on the pool to be resumed. The processing units are indexed starting from 0.

Returns

A `future<void>` which is ready when the given processing unit has been resumed.

void **resume_processing_unit_cb**(*thread_pool_base* &pool, *hpx::function<void()>* callback, *std::size_t* virt_core, *error_code* &ec = *throws*)

Resumes the given processing unit. Takes a callback as a parameter which will be called when the processing unit has been resumed.

Note: Requires that the pool has `threads::policies::enable_elasticity` set.

Parameters

- **pool** – [in] The thread pool to resume a processing unit on.
- **callback** – [in] Callback which is called when the processing unit has been suspended.
- **virt_core** – [in] The processing unit to resume.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

hpx::future<void> **suspend_processing_unit**(*thread_pool_base* &pool, *std::size_t* virt_core)

Suspends the given processing unit. When the processing unit has been suspended the returned future will be ready.

Note: Can only be called from an HPX thread. Use `suspend_processing_unit_cb` or to suspend the processing unit from outside HPX. Requires that the pool has `threads::policies::enable_elasticity` set.

Parameters

- **pool** – [in] The thread pool to suspend a processing unit from.
- **virt_core** – [in] The processing unit on the pool to be suspended. The processing units are indexed starting from 0.

Throws

hpx::exception – if called from outside the HPX runtime.

Returns

A `future<void>` which is ready when the given processing unit has been suspended.

void **suspend_processing_unit_cb**(*hpx::function<void()>* callback, *thread_pool_base* &pool, *std::size_t* virt_core, *error_code* &ec = *throws*)

Suspends the given processing unit. Takes a callback as a parameter which will be called when the processing unit has been suspended.

Note: Requires that the pool has `threads::policies::enable_elasticity` set.

Parameters

- **pool** – [in] The thread pool to suspend a processing unit from.
- **callback** – [in] Callback which is called when the processing unit has been suspended.
- **virt_core** – [in] The processing unit to suspend.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

`hpx::future<void> resume_pool(thread_pool_base &pool)`

Resumes the thread pool. When the all OS threads on the thread pool have been resumed the returned future will be ready.

Note: Can only be called from an HPX thread. Use `resume_cb` or `resume_direct` to suspend the pool from outside HPX.

Parameters

pool – [in] The thread pool to resume.

Throws

`hpx::exception` – if called from outside the HPX runtime.

Returns

A `future<void>` which is ready when the thread pool has been resumed.

`void resume_pool_cb(thread_pool_base &pool, hpx::function<void()> callback, error_code &ec = throws)`

Resumes the thread pool. Takes a callback as a parameter which will be called when all OS threads on the thread pool have been resumed.

Parameters

- **pool** – [in] The thread pool to resume.
- **callback** – [in] called when the thread pool has been resumed.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

`hpx::future<void> suspend_pool(thread_pool_base &pool)`

Suspends the thread pool. When the all OS threads on the thread pool have been suspended the returned future will be ready.

Note: Can only be called from an HPX thread. Use `suspend_cb` or `suspend_direct` to suspend the pool from outside HPX. A thread pool cannot be suspended from an HPX thread running on the pool itself.

Parameters

pool – [in] The thread pool to suspend.

Throws

`hpx::exception` – if called from outside the HPX runtime.

Returns

A `future<void>` which is ready when the thread pool has been suspended.

`void suspend_pool_cb(thread_pool_base &pool, hpx::function<void()> callback, error_code &ec = throws)`

Suspends the thread pool. Takes a callback as a parameter which will be called when all OS threads on the thread pool have been suspended.

Note: A thread pool cannot be suspended from an HPX thread running on the pool itself.

Parameters

- **pool** – [in] The thread pool to suspend.
- **callback** – [in] called when the thread pool has been suspended.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Throws

hpx::exception – if called from an HPX thread which is running on the pool itself.

thread_support

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx::unlock_guard

Defined in header `hpx/mutex.hpp`⁹⁴⁴.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

template<typename **Lock**>

class **unlock_guard**

The class *unlock_guard* is a mutex wrapper that provides a convenient mechanism for releasing a mutex for the duration of a scoped block.

unlock_guard performs the opposite functionality of *lock_guard*.

When a *lock_guard* object is created, it attempts to take ownership of the mutex it is given. When control leaves the scope in which the *lock_guard* object was created, the *lock_guard* is destructed and the mutex is released. Accordingly, when an *unlock_guard* object is created, it attempts to release the ownership of the mutex it is given. So, when control leaves the scope in which the *unlock_guard* object was created, the *unlock_guard* is destructed and the mutex is owned again. In this way, the mutex is unlocked in the constructor and locked in the destructor, so that one can have an unlocked section within a locked one.

Note: *unlock_guard* is non-copyable and non-movable.

Template Parameters

Lock – The type of the lockable object managed by the *unlock_guard*.

⁹⁴⁴ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/mutex.hpp

hpx/thread_support/spinlock.hpp

Defined in header `hpx/thread_support/spinlock.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **util**

threading

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx::thread

Defined in header `hpx/thread.hpp`⁹⁴⁵.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::this_thread::yield`
- `hpx::this_thread::get_id`
- `hpx::this_thread::sleep_for`
- `hpx::this_thread::sleep_until`

class **thread**

The class `thread` represents a single thread of execution. Threads allow multiple functions to execute concurrently. Threads begin execution immediately upon construction of the associated thread object (pending any OS scheduling delays), starting at the top-level function provided as a constructor argument. The return value of the top-level function is ignored and if it terminates by throwing an exception, `hpx::terminate` is called. The top-level function may communicate its return value or an exception to the caller via `hpx::promise` or by modifying shared variables (which may require synchronization, see `hpx::mutex` and `hpx::atomic`) `hpx::thread` objects may also be in the state that does not represent any thread (after default construction, move from, detach, or join), and a thread of execution may not be associated with any thread objects (after detach). No two `hpx::thread` objects may represent the same thread of execution; `hpx::thread` is not *CopyConstructible* or *CopyAssignable*, although it is *MoveConstructible* and *MoveAssignable*.

hpx::this_thread::yield

Defined in header `hpx/thread.hpp`⁹⁴⁶.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::thread`
- `hpx::this_thread::get_id`

⁹⁴⁵ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/thread.hpp

⁹⁴⁶ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/thread.hpp

- `hpx::this_thread::sleep_for`
- `hpx::this_thread::sleep_until`

void `hpx::this_thread::yield()` noexcept

Provides a hint to the implementation to reschedule the execution of threads, allowing other threads to run.

Note: The exact behavior of this function depends on the implementation, in particular on the mechanics of the OS scheduler in use and the state of the system. For example, a first-in-first-out realtime scheduler (SCHED_FIFO in Linux) would suspend the current thread and put it on the back of the queue of the same-priority threads that are ready to run (and if there are no other threads at the same priority, yield has no effect).

`hpx::this_thread::get_id`

Defined in header `hpx/thread.hpp`⁹⁴⁷.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::thread`
- `hpx::this_thread::yield`
- `hpx::this_thread::sleep_for`
- `hpx::this_thread::sleep_until`

`thread::id` `hpx::this_thread::get_id()` noexcept

Returns the id of the current thread.

`hpx::this_thread::sleep_for`

Defined in header `hpx/thread.hpp`⁹⁴⁸.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::thread`
- `hpx::this_thread::yield`
- `hpx::this_thread::get_id`
- `hpx::this_thread::sleep_until`

inline void `hpx::this_thread::sleep_for(hpx::chrono::steady_duration const &rel_time)`

Blocks the execution of the current thread for at least the specified *rel_time*. This function may block for longer than *rel_time* due to scheduling or resource contention delays.

It is recommended to use a steady clock to measure the duration. If an implementation uses a system clock instead, the wait time may also be sensitive to clock adjustments.

Parameters

rel_time – time duration to sleep

⁹⁴⁷ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/thread.hpp

⁹⁴⁸ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/thread.hpp

hpx::this_thread::sleep_until

Defined in header `hpx/thread.hpp`⁹⁴⁹.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::thread`
- `hpx::this_thread::yield`
- `hpx::this_thread::get_id`
- `hpx::this_thread::sleep_for`

void `hpx::this_thread::sleep_until`(`hpx::chrono::steady_time_point` const &`abs_time`)

Blocks the execution of the current thread until specified *abs_time* has been reached.

It is recommended to use the clock tied to *abs_time*, in which case adjustments of the clock may be taken into account. Thus, the duration of the block might be more or less than `abs_time-Clock::now()` at the time of the call, depending on the direction of the adjustment and whether it is honored by the implementation. The function also may block until after *abs_time* has been reached due to process scheduling or resource contention delays.

Parameters

abs_time – absolute time to block until

hpx::jthread

Defined in header `hpx/jthread.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

class jthread

The class *jthread* represents a single thread of execution. It has the same general behavior as `hpx::thread`, except that *jthread* automatically rejoins on destruction, and can be cancelled/stopped in certain situations. Threads begin execution immediately upon construction of the associated thread object (pending any OS scheduling delays), starting at the top-level function provided as a constructor argument. The return value of the top-level function is ignored and if it terminates by throwing an exception, `hpx::terminate` is called. The top-level function may communicate its return value or an exception to the caller via `hpx::promise` or by modifying shared variables (which may require synchronization, see `hpx::mutex` and `hpx::atomic`). Unlike `hpx::thread`, the *jthread* logically holds an internal private member of type `hpx::stop_source`, which maintains a shared stop-state. The *jthread* constructor accepts a function that takes a `hpx::stop_token` as its first argument, which will be passed in by the *jthread* from its internal `stop_source`. This allows the function to check if stop has been requested during its execution, and return if it has. `hpx::jthread` objects may also be in the state that does not represent any thread (after default construction, move from, detach, or join), and a thread of execution may be not associated with any *jthread* objects (after detach). No two `hpx::jthread` objects may represent the same thread of execution; `hpx::jthread` is not *CopyConstructible* or *CopyAssignable*, although it is *MoveConstructible* and *MoveAssignable*.

⁹⁴⁹ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbdc0c8044e7/libs/core/include_local/include/hpx/thread.hpp

threading_base

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/threading_base/thread_description.hpp

Defined in header `hpx/threading_base/thread_description.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **threads**

Functions

```
std::ostream &operator<<(std::ostream&, thread_description const&)
```

```
std::string as_string(thread_description const &desc)
```

```
threads::thread_description get_thread_description(thread_id_type const &id, error_code &ec =  
                                                    throws)
```

The function `get_thread_description` is part of the thread related API allows to query the description of one of the threads known to the thread-manager.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise it throws an instance of *hpx::exception*.

Parameters

- **id** – [in] The thread id of the thread being queried.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

This function returns the description of the thread referenced by the *id* parameter. If the thread is not known to the thread-manager the return value will be the string “<unknown>”.

```
threads::thread_description set_thread_description(thread_id_type const &id,  
                                                  threads::thread_description const &desc =  
                                                  threads::thread_description(), error_code &ec =  
                                                  throws)
```

```
threads::thread_description get_thread_lco_description(thread_id_type const &id, error_code  
                                                       &ec = throws)
```

```
threads::thread_description set_thread_lco_description(thread_id_type const &id,  
                                                       threads::thread_description const &desc  
                                                       = threads::thread_description(),  
                                                       error_code &ec = throws)
```

```
struct thread_description
```

Public Types

enum class **data_type** : *std::uint8_t*

Values:

enumerator **description**

enumerator **address**

Public Functions

thread_description() noexcept = default

inline constexpr **thread_description**(char const*) noexcept

inline explicit constexpr **thread_description**(*std::string* const&) noexcept

template<typename **F**, typename = *std::enable_if_t<!std::is_same_v<F, thread_description> && !traits::is_action_v<F>>>*

inline explicit constexpr **thread_description**(*F* const&, char const* = nullptr) noexcept

template<typename **Action**, typename = *std::enable_if_t<traits::is_action_v<Action>>>*

inline explicit constexpr **thread_description**(*Action*, char const* = nullptr) noexcept

inline explicit constexpr **operator bool**() const noexcept

Public Static Functions

static inline constexpr *data_type* **kind**() noexcept

static inline constexpr char const* **get_description**() noexcept

static inline constexpr *std::size_t* **get_address**() noexcept

static inline constexpr bool **valid**() noexcept

Private Functions

void **init_from_alternative_name**(char const* altname)

namespace **util**

Typedefs

using **instead** = *hpx::threads::thread_description*

hpx/threading_base/print.hpp

Defined in header `hpx/threading_base/print.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/threading_base/threading_base_fwd.hpp

Defined in header `hpx/threading_base/threading_base_fwd.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **threads**

Functions

thread_data ***get_self_id_data**() noexcept

The function *get_self_id_data* returns the data of the HPX thread id associated with the current thread (or nullptr if the current thread is not a HPX thread).

thread_self &**get_self**()

The function *get_self* returns a reference to the (OS thread specific) self reference to the current HPX thread.

thread_self ***get_self_ptr**() noexcept

The function *get_self_ptr* returns a pointer to the (OS thread specific) self reference to the current HPX thread.

thread_self_impl_type ***get_ctx_ptr**()

The function *get_ctx_ptr* returns a pointer to the internal data associated with each coroutine.

thread_self ***get_self_ptr_checked**(*error_code* &ec = *throws*)

The function *get_self_ptr_checked* returns a pointer to the (OS thread specific) self reference to the current HPX thread.

thread_id_type **get_self_id**() noexcept

The function *get_self_id* returns the HPX thread id of the current thread (or zero if the current thread is not a HPX thread).

HPX_DEPRECATED_V(2, 0, "hpx::threads::get_outer_self_id is deprecated, use " "hpx::threads::get_self_id instead") inline thread_id_type get_outer_self_id() noexcept

The function *get_outer_self_id* returns the HPX thread id of the current outer thread (or zero if the current thread is not a HPX thread). This now always returns the same as *get_self_id*, even for directly executed threads.

thread_id_type **get_parent_id()** noexcept

The function *get_parent_id* returns the HPX thread id of the current thread's parent (or zero if the current thread is not a HPX thread).

Note: This function will return a meaningful value only if the code was compiled with HPX_HAVE_THREAD_PARENT_REFERENCE being defined.

std::size_t **get_parent_phase()** noexcept

The function *get_parent_phase* returns the HPX phase of the current thread's parent (or zero if the current thread is not a HPX thread).

Note: This function will return a meaningful value only if the code was compiled with HPX_HAVE_THREAD_PARENT_REFERENCE being defined.

std::ptrdiff_t **get_self_stacksize()** noexcept

The function *get_self_stacksize* returns the stack size of the current thread (or zero if the current thread is not a HPX thread).

thread_stacksize **get_self_stacksize_enum()** noexcept

The function *get_self_stacksize_enum* returns the stack size of the /.

std::uint32_t **get_parent_locality_id()** noexcept

The function *get_parent_locality_id* returns the id of the locality of the current thread's parent (or zero if the current thread is not a HPX thread).

Note: This function will return a meaningful value only if the code was compiled with HPX_HAVE_THREAD_PARENT_REFERENCE being defined.

std::uint64_t **get_self_component_id()** noexcept

The function *get_self_component_id* returns the lva of the component the current thread is acting on

Note: This function will return a meaningful value only if the code was compiled with HPX_HAVE_THREAD_TARGET_ADDRESS being defined.

namespace **policies**

namespace **tracing**

hpx/threading_base/thread_pool_base.hpp

Defined in header `hpx/threading_base/thread_pool_base.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **threads**

Functions

`std::ostream &operator<<(std::ostream &os, thread_pool_base const &thread_pool)`

class **thread_pool_base**

#include <thread_pool_base.hpp> The base class used to manage a pool of OS threads.

Public Functions

virtual void **suspend_processing_unit_direct**(`std::size_t virt_core, error_code &ec = throws`)
= 0

Suspends the given processing unit. Blocks until the processing unit has been suspended.

Parameters

- **virt_core** – [in] The processing unit on the pool to be suspended. The processing units are indexed starting from 0.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

virtual void **resume_processing_unit_direct**(`std::size_t virt_core, error_code &ec = throws`)
= 0

Resumes the given processing unit. Blocks until the processing unit has been resumed.

Parameters

- **virt_core** – [in] The processing unit on the pool to be resumed. The processing units are indexed starting from 0.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

virtual void **resume_direct**(`error_code &ec = throws`) = 0

Resumes the thread pool. Blocks until all OS threads on the thread pool have been resumed.

Parameters

- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

virtual void **suspend_direct**(`error_code &ec = throws`) = 0

Suspends the thread pool. Blocks until all OS threads on the thread pool have been suspended.

Note: A thread pool cannot be suspended from an HPX thread running on the pool itself.

Parameters

- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to `hpx::throws` the function will throw on error instead.

Throws

hpx::exception – if called from an HPX thread which is running on the pool itself.

struct **thread_pool_init_parameters**

Public Functions

```
inline thread_pool_init_parameters(std::string const &name, std::size_t index,
                                   policies::scheduler_mode mode, std::size_t num_threads,
                                   std::size_t thread_offset,
                                   hpx::threads::policies::callback_notifier &notifier,
                                   hpx::threads::policies::detail::affinity_data const
                                   &affinity_data,
                                   hpx::threads::detail::network_background_callback_type
                                   const &network_background_callback =
                                   hpx::threads::detail::network_background_callback_type(),
                                   std::size_t max_background_threads =
                                   static_cast<std::size_t>(-1), std::size_t
                                   max_idle_loop_count =
                                   HPX_IDLE_LOOP_COUNT_MAX, std::size_t
                                   max_busy_loop_count =
                                   HPX_BUSY_LOOP_COUNT_MAX, std::size_t
                                   shutdown_check_count = 10)
```

Public Members

std::string const &**name_**

std::size_t **index_**

policies::scheduler_mode **mode_**

std::size_t **num_threads_**

std::size_t **thread_offset_**

hpx::threads::policies::callback_notifier &**notifier_**

hpx::threads::policies::detail::affinity_data const &**affinity_data_**

hpx::threads::detail::network_background_callback_type const
&**network_background_callback_**

std::size_t **max_background_threads_**

std::size_t **max_idle_loop_count_**

```
std::size_t max_busy_loop_count_
```

```
std::size_t shutdown_check_count_
```

hpx::annotated_function

Defined in header [hpx/functional.hpp](#)⁹⁵⁰.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
template<typename F>
constexpr F &&hpx::annotated_function(F &&f, char const* = nullptr) noexcept
```

Returns a function annotated with the given annotation.

Annotating includes setting the thread description per thread id.

Parameters

f – Function to annotate

```
template<typename F>
constexpr F &&hpx::annotated_function(F &&f, std::string const&) noexcept
```

hpx/threading_base/thread_helpers.hpp

Defined in header `hpx/threading_base/thread_helpers.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

```
namespace hpx
```

```
namespace this_thread
```

Functions

```
threads::thread_restart_state suspend(threads::thread_schedule_state state, threads::thread_id_type
                                     nextid, threads::thread_description const &description =
                                     threads::thread_description("this_thread::suspend"),
                                     hpx::error_code &ec = hpx::throws)
```

The function *suspend* will return control to the thread manager (suspends the current thread). It sets the new state of this thread to the thread state passed as the parameter.

Note: Must be called from within a HPX-thread.

Throws

If `&ec != &throws`, never throws, but will set *ec* to an appropriate value when an error occurs. Otherwise, this function will throw an *hpx::exception* with an error code of *hpx::error::yield_aborted* if it is signaled with *wait_aborted*. If called outside a HPX-thread, this function will throw an *hpx::exception* with an error code of

⁹⁵⁰ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/functional.hpp

hpx::error::null_thread_id. If this function is called while the thread-manager is not running, it will throw an *hpx::exception* with an error code of *hpx::error::invalid_status*.

```
inline threads::thread_restart_state suspend(threads::thread_schedule_state state =
                                             threads::thread_schedule_state::pending,
                                             threads::thread_description const &description =
                                             threads::thread_description("this_thread::suspend"),
                                             hpx::error_code &ec = hpx::throws)
```

The function *suspend* will return control to the thread manager (suspends the current thread). It sets the new state of this thread to the thread state passed as the parameter.

Note: Must be called from within a HPX-thread.

Throws

If – &ec != &throws, never throws, but will set *ec* to an appropriate value when an error occurs. Otherwise, this function will throw an *hpx::exception* with an error code of *hpx::error::yield_aborted* if it is signaled with *wait_aborted*. If called outside a HPX-thread, this function will throw an *hpx::exception* with an error code of *hpx::error::null_thread_id*. If this function is called while the thread-manager is not running, it will throw an *hpx::exception* with an error code of *hpx::error::invalid_status*.

```
threads::thread_restart_state suspend(hpx::chrono::steady_time_point const &abs_time,
                                       threads::thread_id_type id, threads::thread_description const
                                       &description =
                                       threads::thread_description("this_thread::suspend"),
                                       hpx::error_code &ec = hpx::throws)
```

The function *suspend* will return control to the thread manager (suspends the current thread). It sets the new state of this thread to *suspended* and schedules a wakeup for this threads at the given time.

Note: Must be called from within a HPX-thread.

Throws

If – &ec != &throws, never throws, but will set *ec* to an appropriate value when an error occurs. Otherwise, this function will throw an *hpx::exception* with an error code of *hpx::error::yield_aborted* if it is signaled with *wait_aborted*. If called outside a HPX-thread, this function will throw an *hpx::exception* with an error code of *hpx::error::null_thread_id*. If this function is called while the thread-manager is not running, it will throw an *hpx::exception* with an error code of *hpx::error::invalid_status*.

```
inline threads::thread_restart_state suspend(hpx::chrono::steady_time_point const &abs_time,
                                             threads::thread_description const &description =
                                             threads::thread_description("this_thread::suspend"),
                                             hpx::error_code &ec = hpx::throws)
```

The function *suspend* will return control to the thread manager (suspends the current thread). It sets the new state of this thread to *suspended* and schedules a wakeup for this threads at the given time.

Note: Must be called from within a HPX-thread.

Throws

If – &ec != &throws, never throws, but will set *ec* to an appropriate value when an error occurs. Otherwise, this function will throw an *hpx::exception* with an error

code of *hpx::error::yield_aborted* if it is signaled with *wait_aborted*. If called outside a HPX-thread, this function will throw an *hpx::exception* with an error code of *hpx::error::null_thread_id*. If this function is called while the thread-manager is not running, it will throw an *hpx::exception* with an error code of *hpx::error::invalid_status*.

```
inline threads::thread_restart_state suspend(hpx::chrono::steady_duration const &rel_time,  
                                             threads::thread_description const &description =  
                                             threads::thread_description("this_thread::suspend"),  
                                             hpx::error_code &ec = hpx::throws)
```

The function *suspend* will return control to the thread manager (suspends the current thread). It sets the new state of this thread to *suspended* and schedules a wakeup for this threads after the given duration.

Note: Must be called from within a HPX-thread.

Throws

If – &ec != &throws, never throws, but will set *ec* to an appropriate value when an error occurs. Otherwise, this function will throw an *hpx::exception* with an error code of *hpx::error::yield_aborted* if it is signaled with *wait_aborted*. If called outside a HPX-thread, this function will throw an *hpx::exception* with an error code of *hpx::error::null_thread_id*. If this function is called while the thread-manager is not running, it will throw an *hpx::exception* with an error code of *hpx::error::invalid_status*.

```
inline threads::thread_restart_state suspend(hpx::chrono::steady_duration const &rel_time,  
                                             threads::thread_id_type const &id,  
                                             threads::thread_description const &description =  
                                             threads::thread_description("this_thread::suspend"),  
                                             hpx::error_code &ec = hpx::throws)
```

The function *suspend* will return control to the thread manager (suspends the current thread). It sets the new state of this thread to *suspended* and schedules a wakeup for this threads after the given duration.

Note: Must be called from within a HPX-thread.

Throws

If – &ec != &throws, never throws, but will set *ec* to an appropriate value when an error occurs. Otherwise, this function will throw an *hpx::exception* with an error code of *hpx::error::yield_aborted* if it is signaled with *wait_aborted*. If called outside a HPX-thread, this function will throw an *hpx::exception* with an error code of *hpx::error::null_thread_id*. If this function is called while the thread-manager is not running, it will throw an *hpx::exception* with an error code of *hpx::error::invalid_status*.

```
inline threads::thread_restart_state suspend(std::uint64_t ms, threads::thread_description const  
                                             &description =  
                                             threads::thread_description("this_thread::suspend"),  
                                             hpx::error_code &ec = hpx::throws)
```

The function *suspend* will return control to the thread manager (suspends the current thread). It sets the new state of this thread to *suspended* and schedules a wakeup for this threads after the given time (specified in milliseconds).

Note: Must be called from within a HPX-thread.

Throws

If – &ec != &throws, never throws, but will set *ec* to an appropriate value when

an error occurs. Otherwise, this function will throw an *hpx::exception* with an error code of *hpx::error::yield_aborted* if it is signaled with *wait_aborted*. If called outside a HPX-thread, this function will throw an *hpx::exception* with an error code of *hpx::error::null_thread_id*. If this function is called while the thread-manager is not running, it will throw an *hpx::exception* with an error code of *hpx::error::invalid_status*.

threads::thread_pool_base ***get_pool**(*hpx::error_code* &ec = *hpx::throws*)

Returns a pointer to the pool that was used to run the current thread

Throws

If – &ec != &throws, never throws, but will set *ec* to an appropriate value when an error occurs. Otherwise, this function will throw an *hpx::exception* with an error code of *hpx::error::yield_aborted* if it is signaled with *wait_aborted*. If called outside a HPX-thread, this function will throw an *hpx::exception* with an error code of *hpx::error::null_thread_id*. If this function is called while the thread-manager is not running, it will throw an *hpx::exception* with an error code of *hpx::error::invalid_status*.

namespace **threads**

Functions

thread_state **set_thread_state**(*thread_id_type* const &id, *thread_schedule_state* state = *thread_schedule_state::pending*, *thread_restart_state* stateex = *thread_restart_state::signaled*, *thread_priority* priority = *thread_priority::normal*, bool retry_on_active = true, *hpx::error_code* &ec = *hpx::throws*)

Set the thread state of the *thread* referenced by the *thread_id* *id*.

Note: If the thread referenced by the parameter *id* is in *thread_state::active* state this function schedules a new thread which will set the state of the thread as soon as it is not active anymore. The function returns *thread_state::active* in this case.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

Parameters

- **id** – [in] The thread id of the thread the state should be modified for.
- **state** – [in] The new state to be set for the thread referenced by the *id* parameter.
- **stateex** – [in] The new extended state to be set for the thread referenced by the *id* parameter.
- **priority** – [in]
- **retry_on_active** – [in]
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

This function returns the previous state of the thread referenced by the *id* parameter. It will return one of the values as defined by the *thread_state* enumeration. If the thread is not known to the thread-manager the return value will be *thread_state::unknown*.

```
thread_id_ref_type set_thread_state(thread_id_type const &id, hpx::chrono::steady_time_point
                                     const &abs_time, std::atomic<bool> *started,
                                     thread_schedule_state state = thread_schedule_state::pending,
                                     thread_restart_state stateex = thread_restart_state::timeout,
                                     thread_priority priority = thread_priority::normal, bool
                                     retry_on_active = true, hpx::error_code &ec = hpx::throws)
```

Set the thread state of the *thread* referenced by the thread_id *id*.

Set a timer to set the state of the given *thread* to the given new value after it expired (at the given time)

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

Parameters

- **id** – [in] The thread id of the thread the state should be modified for.
- **abs_time** – [in] Absolute point in time for the new thread to be run
- **started** – [in,out] A helper variable allowing to track the state of the timer helper thread
- **state** – [in] The new state to be set for the thread referenced by the *id* parameter.
- **stateex** – [in] The new extended state to be set for the thread referenced by the *id* parameter.
- **priority** – [in]
- **retry_on_active** – [in]
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

```
inline thread_id_ref_type set_thread_state(thread_id_type const &id,
                                             hpx::chrono::steady_time_point const &abs_time,
                                             thread_schedule_state state =
                                             thread_schedule_state::pending, thread_restart_state
                                             stateex = thread_restart_state::timeout, thread_priority
                                             priority = thread_priority::normal, bool retry_on_active
                                             = true, hpx::error_code& = throws)
```

```
inline thread_id_ref_type set_thread_state(thread_id_type const &id, hpx::chrono::steady_duration
                                             const &rel_time, thread_schedule_state state =
                                             thread_schedule_state::pending, thread_restart_state
                                             stateex = thread_restart_state::timeout, thread_priority
                                             priority = thread_priority::normal, bool retry_on_active
                                             = true, hpx::error_code &ec = hpx::throws)
```

Set the thread state of the *thread* referenced by the thread_id *id*.

Set a timer to set the state of the given *thread* to the given new value after it expired (after the given duration)

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

Parameters

- **id** – [in] The thread id of the thread the state should be modified for.
- **rel_time** – [in] Time duration after which the new thread should be run
- **state** – [in] The new state to be set for the thread referenced by the *id* parameter.

- **stateex** – [in] The new extended state to be set for the thread referenced by the *id* parameter.
- **priority** – [in]
- **retry_on_active** – [in]
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

thread_state **get_thread_state**(thread_id_type const &id, hpx::error_code &ec = hpx::throws) noexcept

The function `get_thread_backtrace` is part of the thread related API allows to query the currently stored thread back trace (which is captured during thread suspension).

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*. The function `get_thread_state` is part of the thread related API. It queries the state of one of the threads known to the thread-manager.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

Parameters

- **id** – [in] The thread id of the thread being queried.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.
- **id** – [in] The thread id of the thread the state should be modified for.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

This function returns the currently captured stack back trace of the thread referenced by the *id* parameter. If the thread is not known to the thread-manager the return value will be the zero.

Returns

This function returns the thread state of the thread referenced by the *id* parameter. If the thread is not known to the thread-manager the return value will be *terminated*.

std::size_t **get_thread_phase**(thread_id_type const &id, hpx::error_code &ec = hpx::throws) noexcept

The function `get_thread_phase` is part of the thread related API. It queries the phase of one of the threads known to the thread-manager.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

Parameters

- **id** – [in] The thread id of the thread the phase should be modified for.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

This function returns the thread phase of the thread referenced by the *id* parameter. If the thread is not known to the thread-manager the return value will be ~0.

```
bool get_thread_interruption_enabled(thread_id_type const &id, hpx::error_code &ec =  
                                     hpx::throws)
```

Returns whether the given thread can be interrupted at this point.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

Parameters

- **id** – [in] The thread id of the thread which should be queried.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

This function returns *true* if the given thread can be interrupted at this point in time. It will return *false* otherwise.

```
bool set_thread_interruption_enabled(thread_id_type const &id, bool enable, hpx::error_code  
                                     &ec = hpx::throws)
```

Set whether the given thread can be interrupted at this point.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

Parameters

- **id** – [in] The thread id of the thread which should receive the new value.
- **enable** – [in] This value will determine the new interruption enabled status for the given thread.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

This function returns the previous value of whether the given thread could have been interrupted.

```
bool get_thread_interruption_requested(thread_id_type const &id, hpx::error_code &ec =  
                                     hpx::throws)
```

Returns whether the given thread has been flagged for interruption.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

Parameters

- **id** – [in] The thread id of the thread which should be queried.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Returns

This function returns *true* if the given thread was flagged for interruption. It will return *false* otherwise.

```
void interrupt_thread(thread_id_type const &id, bool flag, hpx::error_code &ec = hpx::throws)
```

Flag the given thread for interruption.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

Parameters

- **id** – [in] The thread id of the thread which should be interrupted.
- **flag** – [in] The flag encodes whether the thread should be interrupted (if it is *true*), or 'uninterrupted' (if it is *false*).
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

inline void **interrupt_thread**(thread_id_type const &id, *hpx::error_code* &ec = *hpx::throws*)

void **interruption_point**(thread_id_type const &id, *hpx::error_code* &ec = *hpx::throws*)

Interrupt the current thread at this point if it was canceled. This will throw a *thread_interrupted* exception, which will cancel the thread.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

Parameters

- **id** – [in] The thread id of the thread which should be interrupted.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

threads::thread_priority **get_thread_priority**(thread_id_type const &id, *hpx::error_code* &ec = *hpx::throws*) noexcept

Return priority of the given thread

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

Parameters

- **id** – [in] The thread id of the thread whose priority is queried.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

std::ptrdiff_t **get_stack_size**(thread_id_type const &id, *hpx::error_code* &ec = *hpx::throws*) noexcept

Return stack size of the given thread

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

Parameters

- **id** – [in] The thread id of the thread whose priority is queried.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

*threads::thread_pool_base ****get_pool**(thread_id_type const &id, *hpx::error_code* &ec = *hpx::throws*)

Returns a pointer to the pool that was used to run the current thread

Throws

If – &ec != &throws, never throws, but will set *ec* to an appropriate value when

an error occurs. Otherwise, this function will throw an *hpx::exception* with an error code of *hpx::error::yield_aborted* if it is signaled with *wait_aborted*. If called outside a HPX-thread, this function will throw an *hpx::exception* with an error code of *hpx::error::null_thread_id*. If this function is called while the thread-manager is not running, it will throw an *hpx::exception* with an error code of *hpx::error::invalid_status*.

hpx::scoped_annotation

Defined in header [hpx/functional.hpp](#)⁹⁵¹.

See *Public API* for a list of names and headers that are part of the public HPX API.

struct **scoped_annotation**

scoped_annotation associates a **name** with a section of code (scope). It can be used to visualize code execution in profiling tools like *Intel VTune*, *Apex Profiler*, etc. That allows analyzing performance to figure out which part(s) of code is (are) responsible for performance degradation, etc.

hpx::get_worker_thread_num

Defined in header [hpx/runtime.hpp](#)⁹⁵².

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- *hpx::get_local_worker_thread_num*
- *hpx::get_thread_pool_num*

std::size_t **hpx::get_worker_thread_num()** noexcept

Return the number of the current OS-thread running in the runtime instance the current HPX-thread is executed with.

This function returns the zero based index of the OS-thread which executes the current HPX-thread.

Note: The returned value is zero based and its maximum value is smaller than the overall number of OS-threads executed (as returned by *get_os_thread_count()*).

Note: This function needs to be executed on a HPX-thread. It will fail otherwise (it will return -1).

std::size_t **hpx::get_worker_thread_num(error_code &ec)** noexcept

Return the number of the current OS-thread running in the runtime instance the current HPX-thread is executed with.

This function returns the zero based index of the OS-thread which executes the current HPX-thread.

Note: The returned value is zero based and its maximum value is smaller than the overall number of OS-threads executed (as returned by *get_os_thread_count()*). It will return -1 if the current thread is not a known thread or

⁹⁵¹ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/functional.hpp

⁹⁵² <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

if the runtime is not in running state.

Note: This function needs to be executed on a HPX-thread. It will fail otherwise (it will return -1).

Parameters

ec – [in,out] this represents the error status on exit (obsolete, ignored).

hpx::get_local_worker_thread_num

Defined in header [hpx/runtime.hpp](#)⁹⁵³.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::get_worker_thread_num`
- `hpx::get_thread_pool_num`

`std::size_t hpx::get_local_worker_thread_num()` noexcept

Return the number of the current OS-thread running in the current thread pool the current HPX-thread is executed with.

This function returns the zero based index of the OS-thread on the current thread pool which executes the current HPX-thread.

Note: The returned value is zero based and its maximum value is smaller than the number of OS-threads executed on the current thread pool. It will return -1 if the current thread is not a known thread or if the runtime is not in running state.

Note: This function needs to be executed on a HPX-thread. It will fail otherwise (it will return -1).

`std::size_t hpx::get_local_worker_thread_num(error_code &ec)` noexcept

Return the number of the current OS-thread running in the current thread pool the current HPX-thread is executed with.

This function returns the zero based index of the OS-thread on the current thread pool which executes the current HPX-thread.

Note: The returned value is zero based and its maximum value is smaller than the number of OS-threads executed on the current thread pool. It will return -1 if the current thread is not a known thread or if the runtime is not in running state.

Note: This function needs to be executed on a HPX-thread. It will fail otherwise (it will return -1).

Parameters

ec – [in,out] this represents the error status on exit (obsolete, ignored).

⁹⁵³ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

hpx::get_thread_pool_num

Defined in header [hpx/runtime.hpp](#)⁹⁵⁴.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::get_worker_thread_num`
- `hpx::get_local_worker_thread_num`

`std::size_t hpx::get_thread_pool_num()` noexcept

Return the number of the current thread pool the current HPX-thread is executed with.

This function returns the zero based index of the thread pool which executes the current HPX-thread.

Note: The returned value is zero based and its maximum value is smaller than the number of thread pools started by the runtime. It will return -1 if the current thread pool is not a known thread pool or if the runtime is not in running state.

Note: This function needs to be executed on a HPX-thread. It will fail otherwise (it will return -1).

`std::size_t hpx::get_thread_pool_num(error_code &ec)` noexcept

Return the number of the current thread pool the current HPX-thread is executed with.

This function returns the zero based index of the thread pool which executes the current HPX-thread.

Note: The returned value is zero based and its maximum value is smaller than the number of thread pools started by the runtime. It will return -1 if the current thread pool is not a known thread pool or if the runtime is not in running state.

Note: This function needs to be executed on a HPX-thread. It will fail otherwise (it will return -1).

Parameters

ec – [in,out] this represents the error status on exit (obsolete, ignored).

hpx/threading_base/thread_data.hpp

Defined in header `hpx/threading_base/thread_data.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **threads**

⁹⁵⁴ <http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/full/include/include/hpx/runtime.hpp>

Functions

```
constexpr thread_data *get_thread_id_data(thread_id_ref_type const &tid) noexcept
constexpr thread_data *get_thread_id_data(thread_id_type const &tid) noexcept
constexpr tracing::region_init_data get_region_init_data(thread_data const*) noexcept
constexpr tracing::fiber_region_init_data get_fiber_region_init_data(thread_data const*)
                                                                    noexcept
```

class **thread_data** : public *thread_data_reference_counting*

#include <thread_data.hpp> A *thread* is the representation of a HPX thread. It's a first class object in HPX. In our implementation this is a user level thread running on top of one of the OS threads spawned by the *thread-manager*.

A *thread* encapsulates:

- A thread status word (see the functions *thread::get_state* and *thread::set_state*)
- A function to execute (the thread function)
- A frame (in this implementation this is a block of memory used as the threads stack)
- A block of registers (not implemented yet)

Generally, *threads* are not created or executed directly. All functionality related to the management of *threads* is implemented by the *thread-manager*.

Public Types

```
using mutex_type = util::detail::spinlock_no_backoff
```

Public Functions

```
thread_data(thread_data const&) = delete
```

```
thread_data(thread_data&&) = delete
```

```
thread_data &operator=(thread_data const&) = delete
```

```
thread_data &operator=(thread_data&&) = delete
```

```
inline thread_state get_state(std::memory_order const order = std::memory_order_acquire) const
                                                                    noexcept
```

The *get_state* function queries the state of this thread instance.

Note: This function will seldom be used directly. Most of the time the state of a thread will be retrieved by using the function *threadmanager::get_state*.

Returns

This function returns the current state of this thread. It will return one of the values as defined by the *thread_state* enumeration.

```
inline thread_state set_state(thread_schedule_state const state, thread_restart_state state_ex =  
    thread_restart_state::unknown, std::memory_order const load_order  
    = std::memory_order_acquire, std::memory_order const  
    exchange_order = std::memory_order_acq_rel) const noexcept
```

The `set_state` function changes the state of this thread instance.

Note: This function will seldom be used directly. Most of the time the state of a thread will have to be changed using the thread-manager. Moreover, changing the thread state using this function does not change its scheduling status. It only sets the thread's status word. To change the thread's scheduling status `threadmanager::set_state` should be used.

Parameters

- **state** – [in] The new state to be set for the thread.
- **state_ex** – [in]
- **load_order** – [in]
- **exchange_order** – [in]

```
inline bool set_state_tagged(thread_schedule_state const newstate, thread_state const  
    &prev_state, thread_state &new_tagged_state, std::memory_order  
    const exchange_order = std::memory_order_acq_rel) const  
    noexcept
```

```
inline bool restore_state(thread_state const new_state, thread_state const old_state,  
    std::memory_order const load_order = std::memory_order_relaxed,  
    std::memory_order const load_exchange =  
    std::memory_order_acq_rel) const noexcept
```

The `restore_state` function changes the state of this thread instance depending on its current state. It will change the state atomically only if the current state is still the same as passed as the second parameter. Otherwise, it won't touch the thread state of this instance.

Note: This function will seldom be used directly. Most of the time the state of a thread will have to be changed using the threadmanager. Moreover, changing the thread state using this function does not change its scheduling status. It only sets the thread's status word. To change the thread's scheduling status `threadmanager::set_state` should be used.

Parameters

- **new_state** – [in] The new state to be set for the thread.
- **old_state** – [in] The old state of the thread which still has to be the current state.
- **load_order** – [in]
- **load_exchange** – [in]

Returns

This function returns `true` if the state has been changed successfully

```
inline bool restore_state(thread_schedule_state const new_state, thread_restart_state const  
    state_ex, thread_state old_state, std::memory_order const  
    load_exchange = std::memory_order_acq_rel) const noexcept
```

```
char const *get_fiber_name() const noexcept
```

```
inline constexpr thread_priority get_priority() const noexcept
```

```
inline void set_priority(thread_priority const priority) noexcept
```



```

inline bool is_background() const noexcept

inline void set_is_background() noexcept

inline bool interruption_requested() const noexcept

inline bool interruption_enabled() const noexcept

inline bool set_interruption_enabled(bool const enable) noexcept

inline void interrupt(bool const flag = true)

bool interruption_point(bool throw_on_interrupt = true)

bool add_thread_exit_callback(function<void()> const &f)

void run_thread_exit_callbacks()

void free_thread_exit_callbacks()

inline bool runs_as_child(std::memory_order const mo = std::memory_order_acquire) const
    noexcept

inline bool is_stackless() const noexcept

void destroy_thread() override

inline constexpr policies::scheduler_base *get_scheduler_base() const noexcept

inline constexpr std::uint16_t get_last_worker_thread_num() const noexcept

inline void set_last_worker_thread_num(std::uint16_t const last_worker_thread_num)
    noexcept

inline constexpr std::ptrdiff_t get_stack_size() const noexcept

inline thread_stacksize get_stack_size_enum() const noexcept

template<typename ThreadQueue>
inline constexpr ThreadQueue &get_queue() noexcept

inline coroutine_type::result_type operator() (hpx::execution_base::this_thread::detail::agent_storage
    *agent_storage)

```

Execute the thread function.

Returns

This function returns the thread state the thread should be scheduled from this point on.
The thread manager will use the returned value to set the thread's scheduling status.

```

inline coroutine_type::result_type invoke_directly()

```

Directly execute the thread function (inline)

Returns

This function returns the thread state the thread should be scheduled from this point on.
The thread manager will use the returned value to set the thread's scheduling status.

```

inline virtual thread_id_type get_thread_id() const

```

```

inline virtual std::size_t get_thread_phase() const noexcept

```

```

virtual std::size_t get_thread_data() const = 0

```

```
virtual std::size_t set_thread_data(std::size_t data) = 0

virtual void init() = 0

virtual void rebind(thread_init_data &init_data) = 0

inline hpx::tracing::task_timer_data get_timer_data() const noexcept

inline void set_timer_data(hpx::tracing::task_timer_data data) noexcept

thread_data(thread_init_data &init_data, void *queue, std::ptrdiff_t stacksize, bool is_stackless =
    false, thread_id_addrf addrf = thread_id_addrf::yes)

~thread_data() override

virtual void destroy() noexcept = 0
```

Public Static Functions

```
static inline constexpr std::uint64_t get_component_id() noexcept
    Return the id of the component this thread is running in.

static inline constexpr threads::thread_description get_description() noexcept

static inline constexpr threads::thread_description set_description(threads::thread_description)
    noexcept

static inline constexpr threads::thread_description get_lco_description() noexcept

static inline constexpr threads::thread_description set_lco_description(threads::thread_description)
    noexcept

static char const *get_safe_description(threads::thread_description const &description, char
    const *fallback = "<unknown>") noexcept

static inline constexpr std::uint32_t get_parent_locality_id() noexcept
    Return the locality of the parent thread.

static inline constexpr thread_id_type get_parent_thread_id() noexcept
    Return the thread id of the parent thread.

static inline constexpr std::size_t get_parent_thread_phase() noexcept
    Return the phase of the parent thread.

static inline constexpr util::backtrace const *get_backtrace() noexcept

static inline constexpr util::backtrace const *set_backtrace(util::backtrace const*) noexcept
```

Protected Functions

```
inline thread_restart_state set_state_ex(thread_restart_state const new_state,
                                         std::memory_order const load_order =
                                         std::memory_order_acquire, std::memory_order const
                                         load_exchange = std::memory_order_acq_rel) const
                                         noexcept
```

The `set_state` function changes the extended state of this thread instance.

Note: This function will seldom be used directly. Most of the time the state of a thread will have to be changed using the threadmanager.

Parameters

- **new_state** – [in] The new extended state to be set for the thread.
- **load_order** – [in]
- **load_exchange** – [in]

```
void rebind_base(thread_init_data &init_data)
```

Private Types

```
using state = detail::thread_data_state
```

Private Members

```
mutable mutex_type mtx_
```

```
std::atomic<bool> runs_as_child_
```

```
state state_
```

```
thread_stacksize stacksize_enum_
```

```
std::int32_t stacksize_
```

```
std::uint16_t last_worker_thread_num_
```

```
thread_priority priority_
```

```
HPX_NO_UNIQUE_ADDRESS hpx::tracing::task_timer_data timer_data_
```

```
mutable std::atomic<thread_state> current_state_
```

```
hpx::detail::forward_list<hpx::function<void()>> exit_funcs_
```

```
policies::scheduler_base *scheduler_base_  
  
void *queue_
```

hpx/threading_base/register_thread.hpp

Defined in header `hpx/threading_base/register_thread.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **threads**

Functions

```
template<typename F>  
thread_function_type make_thread_function(F &&f)  
  
template<typename F, typename ...Ts>  
thread_function_type make_thread_function_nullary(F &&f, Ts&&... ts)  
  
void register_thread(threads::thread_init_data &data, threads::thread_pool_base *pool,  
                    threads::thread_id_ref_type &id, error_code &ec = hpx::throws)
```

Create a new *thread* using the given data.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

Parameters

- **data** – [in] The data to use for creating the thread.
- **pool** – [in] The thread pool to use for launching the work.
- **id** – [out] The id of the newly created thread (if applicable)
- **ec** – [in,out] This represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Throws

invalid_status – if the runtime system has not been started yet.

Returns

This function will return the internal id of the newly created HPX-thread.

```
threads::thread_id_ref_type register_thread(threads::thread_init_data &data,  
                                           threads::thread_pool_base *pool, error_code &ec =  
                                           hpx::throws)
```

```
void register_thread(threads::thread_init_data &data, threads::thread_id_ref_type &id, error_code  
                   &ec = throws)
```

Create a new *thread* using the given data on the same thread pool as the calling thread, or on the default thread pool if not on an HPX thread.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

Parameters

- **data** – [in] The data to use for creating the thread.
- **id** – [out] The id of the newly created thread (if applicable)
- **ec** – [in,out] This represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Throws

invalid_status – if the runtime system has not been started yet.

Returns

This function will return the internal id of the newly created HPX-thread.

```
threads::thread_id_ref_type register_thread(threads::thread_init_data &data, error_code &ec =
                                           throws)
```

```
thread_id_ref_type register_work(threads::thread_init_data &data, threads::thread_pool_base *pool,
                                error_code &ec = hpx::throws)
```

Create a new work item using the given data.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

Parameters

- **data** – [in] The data to use for creating the thread.
- **pool** – [in] The thread pool to use for launching the work.
- **ec** – [in,out] This represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Throws

invalid_status – if the runtime system has not been started yet.

```
thread_id_ref_type register_work(threads::thread_init_data &data, error_code &ec = throws)
```

Create a new work item using the given data on the same thread pool as the calling thread, or on the default thread pool if not on an HPX thread.

Note: As long as *ec* is not pre-initialized to *hpx::throws* this function doesn't throw but returns the result code using the parameter *ec*. Otherwise, it throws an instance of *hpx::exception*.

Parameters

- **data** – [in] The data to use for creating the thread.
- **ec** – [in,out] This represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

Throws

invalid_status – if the runtime system has not been started yet.

threadmanager

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/threadmanager/threadmanager.hpp

Defined in header `hpx/threadmanager/threadmanager.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **threads**

class **threadmanager**

#include <threadmanager.hpp> The *thread-manager* class is the central instance of management for all (non-depleted) threads

Public Types

using **notification_policy_type** = *threads::policies::callback_notifier*

using **pool_type** = *std::unique_ptr<thread_pool_base>*

using **pool_vector** = *std::vector<pool_type>*

Public Functions

threadmanager(*hpx::util::runtime_configuration &rtcfg_, notification_policy_type ¬ifier, detail::network_background_callback_type network_background_callback = detail::network_background_callback_type()*)

threadmanager(*threadmanager const&*) = delete

threadmanager(*threadmanager&&*) = delete

threadmanager &operator=(*threadmanager const&*) = delete

threadmanager &operator=(*threadmanager&&*) = delete

~threadmanager()

void **init**() const

void **create_pools**()

void **print_pools**(*std::ostream&*) const

FIXME move to private and add `—hpx:printpools` cmd line option.

thread_pool_base &default_pool() const

```
thread_pool_base &get_pool(std::string const &pool_name) const
```

```
thread_pool_base &get_pool(pool_id_type const &pool_id) const
```

```
thread_pool_base &get_pool(std::size_t thread_index) const
```

```
bool pool_exists(std::string const &pool_name) const
```

```
bool pool_exists(std::size_t pool_index) const
```

```
thread_id_ref_type register_work(thread_init_data &data, error_code &ec = throws) const
```

The function *register_work* adds a new work item to the thread manager. It doesn't immediately create a new *thread*, it just adds the task parameters (function, initial state and description) to the internal management data structures. The thread itself will be created when the number of existing threads drops below the number of threads specified by the constructors *max_count* parameter.

Parameters

- **data** – [in] The value of this parameter allows to specify a description of the thread to create. This information is used for logging purposes mainly, but might be useful for debugging as well. This parameter is optional and defaults to an empty string.
- **ec** –

```
void register_thread(thread_init_data &data, thread_id_ref_type &id, error_code &ec = throws) const
```

The function *register_thread* adds a new work item to the thread manager. It creates a new *thread*, adds it to the internal management data structures, and schedules the new thread, if appropriate.

Parameters

- **data** – [in] The value of this parameter allows to specify a description of the thread to create. This information is used for logging purposes mainly, but might be useful for debugging as well. This parameter is optional and defaults to an empty string.
- **id** – [out] This parameter will hold the id of the created thread. This id is guaranteed to be validly initialized before the thread function is executed.
- **ec** –

```
bool run() const
```

Run the thread manager's work queue. This function instantiates the specified number of OS threads in each pool. All OS threads are started to execute the function *tfunc*.

Returns

The function returns *true* if the thread manager has been started successfully, otherwise it returns *false*.

```
void stop(bool blocking = true) const
```

Forcefully stop the thread-manager.

Parameters

blocking –

```
bool is_busy() const
```

```
bool is_idle() const
```

```
void wait() const
```

```
bool wait_for(hpx::chrono::steady_duration const &rel_time) const
```

```
template<typename StopToken>
```

```
inline bool wait_for(StopToken &&stop_token, hpx::chrono::steady_duration const &rel_time) const
```

void **suspend**() const

void **resume**() const

hpx::state **status**() const

Return whether the thread manager is still running This returns the “minimal state”, i.e. the state of the least advanced thread pool

std::int64_t **get_thread_count**(*thread_schedule_state* state = *thread_schedule_state::unknown*,
 thread_priority priority = *thread_priority::default_*, *std::size_t*
 num_thread = static_cast<*std::size_t*>(-1), bool reset = false)
const

return the number of HPX-threads with the given state

Note: This function locks the internal OS lock in the thread manager

std::int64_t **get_idle_core_count**() const

mask_type **get_idle_core_mask**() const

std::int64_t **get_background_thread_count**() const

bool **enumerate_threads**(*hpx::function*<bool(*thread_id_type*)> const &f, *thread_schedule_state*
 state = *thread_schedule_state::unknown*) const

void **abort_all_suspended_threads**() const

bool **cleanup_terminated**(bool delete_all) const

std::size_t **get_os_thread_count**() const

Return the number of OS threads running in this thread-manager.

This function will return correct results only if the thread-manager is running.

std::thread &**get_os_thread_handle**(*std::size_t* num_thread) const

void **report_error**(*std::size_t* num_thread, *std::exception_ptr* const &e) const

API functions forwarding to notification policy.

This notifies the thread manager that the passed exception has been raised. The exception will be routed through the notifier and the scheduler (which will result in it being passed to the runtime object, which in turn will report it to the console, etc.).

mask_type **get_used_processing_units**() const

Returns the mask identifying all processing units used by this thread manager.

hwloc_bitmap_ptr **get_pool_numa_bitmap**(*std::string* const &pool_name) const

void **set_scheduler_mode**(*threads::policies::scheduler_mode* mode,
 hpx::threads::mask_cref_type pu_mask) const noexcept

void **add_scheduler_mode**(*threads::policies::scheduler_mode* mode,
 hpx::threads::mask_cref_type pu_mask) const noexcept

void **add_remove_scheduler_mode**(*threads::policies::scheduler_mode* to_add_mode,
 threads::policies::scheduler_mode to_remove_mode,
 hpx::threads::mask_cref_type pu_mask) const noexcept


```

void remove_scheduler_mode(threads::policies::scheduler_mode mode,
                           hpx::threads::mask_cref_type pu_mask) const noexcept

void reset_thread_distribution() const noexcept

std::int64_t get_queue_length(bool reset) const

std::int64_t get_cumulative_duration(bool reset) const

std::int64_t get_thread_count_unknown(bool reset) const

std::int64_t get_thread_count_active(bool reset) const

std::int64_t get_thread_count_pending(bool reset) const

std::int64_t get_thread_count_suspended(bool reset) const

std::int64_t get_thread_count_terminated(bool reset) const

std::int64_t get_thread_count_staged(bool reset) const

```

Public Static Functions

```

static void init_tss(std::size_t global_thread_num)

static void deinit_tss()

```

Private Types

```

using mutex_type = std::mutex

```

Private Functions

```

policies::thread_queue_init_parameters get_init_parameters() const

void create_scheduler_user_defined(hpx::resource::scheduler_function const&,
                                   thread_pool_init_parameters const&,
                                   policies::thread_queue_init_parameters const&)

void create_scheduler_local(thread_pool_init_parameters const&,
                             policies::thread_queue_init_parameters const&, std::size_t)

void create_scheduler_local_priority_fifo(thread_pool_init_parameters const&,
                                           policies::thread_queue_init_parameters
                                           const&, std::size_t)

void create_scheduler_local_priority_fifo_double(thread_pool_init_parameters const&,
                                                  poli-
                                                  cies::thread_queue_init_parameters
                                                  const&, std::size_t)

```

```
void create_scheduler_local_priority_lifo(thread_pool_init_parameters const&,
                                           policies::thread_queue_init_parameters
                                           const&, std::size_t)

void create_scheduler_static(thread_pool_init_parameters const&,
                             policies::thread_queue_init_parameters const&, std::size_t)

void create_scheduler_static_priority(thread_pool_init_parameters const&,
                                       policies::thread_queue_init_parameters const&,
                                       std::size_t)

void create_scheduler_abp_priority_fifo(thread_pool_init_parameters const&,
                                         policies::thread_queue_init_parameters const&,
                                         std::size_t)

void create_scheduler_abp_priority_lifo(thread_pool_init_parameters const&,
                                         policies::thread_queue_init_parameters const&,
                                         std::size_t)

void create_scheduler_shared_priority(thread_pool_init_parameters const&,
                                       policies::thread_queue_init_parameters const&,
                                       std::size_t)

void create_scheduler_local_workrequesting_fifo(thread_pool_init_parameters const&,
                                                 policies::thread_queue_init_parameters
                                                 const&, std::size_t)

void create_scheduler_local_workrequesting_lifo(thread_pool_init_parameters const&,
                                                 policies::thread_queue_init_parameters
                                                 const&, std::size_t)

void create_scheduler_local_workrequesting_mc(thread_pool_init_parameters const&,
                                              policies::thread_queue_init_parameters
                                              const&, std::size_t)
```

Private Members

```
mutable mutex_type mtx_

hpx::util::runtime_configuration &rtcfg_

std::vector<pool_id_type> threads_lookup_

pool_vector pools_

notification_policy_type &notifier_

detail::network_background_callback_type network_background_callback_
```

thrust

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/thrust/algorithms.hpp

Defined in header `hpx/thrust/algorithms.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **thrust**

Functions

```
template<typename Tag, typename ThrustPolicy, typename ...Args>
decltype(auto) hpx_invoke(Tag tag, ThrustPolicy &&policy, Args&&... args)
```

ADL hook dispatching HPX algorithm CPOs to Thrust backends.

When a Thrust execution policy is passed to an HPX algorithm CPO, this hook maps the call to the corresponding Thrust implementation. Async policies wrap the result in a future.

Template Parameters

- **Tag** – The algorithm CPO tag type.
- **ThrustPolicy** – A Thrust execution policy satisfying `is_thrust_execution_policy`.
- **Args** – Arguments forwarded to the Thrust algorithm.

Parameters

- **tag** – The algorithm CPO tag instance.
- **policy** – Thrust execution policy controlling dispatch.
- **args** – Arguments forwarded to the mapped algorithm.

Returns

The algorithm result, or a future thereof for async policies.

timed_execution

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/timed_execution/timed_execution_fwd.hpp

Defined in header `hpx/timed_execution/timed_execution_fwd.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **parallel**

namespace **execution**

Variables

constexpr struct *hpx::parallel::execution::post_at_t* **post_at**

constexpr struct *hpx::parallel::execution::post_after_t* **post_after**

constexpr struct *hpx::parallel::execution::async_execute_at_t* **async_execute_at**

constexpr struct *hpx::parallel::execution::async_execute_after_t* **async_execute_after**

constexpr struct *hpx::parallel::execution::sync_execute_at_t* **sync_execute_at**

constexpr struct *hpx::parallel::execution::sync_execute_after_t* **sync_execute_after**

struct **async_execute_after_t**

#include <timed_execution_fwd.hpp> Customization point of asynchronous execution agent creation supporting timed execution.

This asynchronously creates a single function invocation *f*() using the associated executor at the given point in time.

Note: This calls `exec.async_execute_after(rel_time, f, ts...)`, if available, otherwise it emulates timed scheduling by delaying calling `execution::async_execute()` on the underlying non-time-scheduled execution agent.

Param **exec**

[in] The executor object to use for scheduling of the function *f*.

Param **rel_time**

[in] The duration of time after which the given function should be scheduled to run.

Param **f**

[in] The function which will be scheduled using the given executor.

Param **ts...**

[in] Additional arguments to use to invoke *f*.

Return

f(*ts...*)'s result through a future

Public Functions

```
template<executor_any Executor, typename F, typename ...Ts>
inline decltype(auto) operator()(Executor &&exec, hpx::chrono::steady_duration const
    &rel_time, F &&f, Ts&&... ts) const
```

```
template<executor_any Executor, typename F, typename ...Ts>
inline decltype(auto) operator()(Executor &&exec, hpx::chrono::steady_duration const
    &rel_time, F &&f, Ts&&... ts) const
```

struct **async_execute_at_t**

#include <timed_execution_fwd.hpp> Customization point of asynchronous execution agent creation supporting timed execution.

This asynchronously creates a single function invocation *f()* using the associated executor at the given point in time.

Note: This calls `exec.async_execute_at(abs_time, f, ts...)`, if available, otherwise it emulates timed scheduling by delaying calling `execution::async_execute()` on the underlying non-time-scheduled execution agent.

Param exec

[in] The executor object to use for scheduling of the function *f*.

Param abs_time

[in] The point in time the given function should be scheduled at to run.

Param f

[in] The function which will be scheduled using the given executor.

Param ts...

[in] Additional arguments to use to invoke *f*.

Return

f(ts...)'s result through a future

Public Functions

```
template<executor_any Executor, typename F, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, hpx::chrono::steady_time_point const
                                   &abs_time, F &&f, Ts&&... ts) const
```

```
template<executor_any Executor, typename F, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, hpx::chrono::steady_time_point const
                                   &abs_time, F &&f, Ts&&... ts) const
```

struct **post_after_t**

#include <timed_execution_fwd.hpp> Customization point of asynchronous fire & forget execution agent creation supporting timed execution.

This asynchronously (fire & forget) creates a single function invocation *f()* using the associated executor at the given point in time.

Note: This calls `exec.post_after(rel_time, f, ts...)`, if available, otherwise it emulates timed scheduling by delaying calling `execution::post()` on the underlying non-time-scheduled execution agent.

Param exec

[in] The executor object to use for scheduling of the function *f*.

Param rel_time

[in] The duration of time after which the given function should be scheduled to run.

Param f

[in] The function which will be scheduled using the given executor.

Param ts...

[in] Additional arguments to use to invoke *f*.

Public Functions

```
template<executor_any Executor, typename F, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, hpx::chrono::steady_duration const
                                   &rel_time, F &&f, Ts&&... ts) const
```

```
template<executor_any Executor, typename F, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, hpx::chrono::steady_duration const
                                   &rel_time, F &&f, Ts&&... ts) const
```

struct **post_at_t**

#include <timed_execution_fwd.hpp> Customization point of asynchronous fire & forget execution agent creation supporting timed execution.

This asynchronously (fire & forget) creates a single function invocation *f*() using the associated executor at the given point in time.

Note: This calls `exec.post_at(abs_time, f, ts...)`, if available, otherwise it emulates timed scheduling by delaying calling `execution::post()` on the underlying non-time-scheduled execution agent.

Param *exec*

[in] The executor object to use for scheduling of the function *f*.

Param *abs_time*

[in] The point in time the given function should be scheduled at to run.

Param *f*

[in] The function which will be scheduled using the given executor.

Param *ts...*

[in] Additional arguments to use to invoke *f*.

Public Functions

```
template<executor_any Executor, typename F, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, hpx::chrono::steady_time_point const
                                   &abs_time, F &&f, Ts&&... ts) const
```

```
template<executor_any Executor, typename F, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, hpx::chrono::steady_time_point const
                                   &abs_time, F &&f, Ts&&... ts) const
```

struct **sync_execute_after_t**

#include <timed_execution_fwd.hpp> Customization point of synchronous execution agent creation supporting timed execution.

This synchronously creates a single function invocation *f*() using the associated executor at the given point in time.

Note: This calls `exec.sync_execute_after(rel_time, f, ts...)`, if available, otherwise it emulates timed scheduling by delaying calling `execution::sync_execute()` on the underlying non-time-scheduled execution agent.

Param exec

[in] The executor object to use for scheduling of the function *f*.

Param rel_time

[in] The duration of time after which the given function should be scheduled to run.

Param f

[in] The function which will be scheduled using the given executor.

Param ts...

[in] Additional arguments to use to invoke *f*.

Return

f(*ts...*)'s result

Public Functions

```
template<executor_any Executor, typename F, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, hpx::chrono::steady_duration const
                                   &rel_time, F &&f, Ts&&... ts) const
```

```
template<executor_any Executor, typename F, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, hpx::chrono::steady_duration const
                                   &rel_time, F &&f, Ts&&... ts) const
```

struct **sync_execute_at_t**

#include <timed_execution_fwd.hpp> Customization point of synchronous execution agent creation supporting timed execution.

This synchronously creates a single function invocation *f*() using the associated executor at the given point in time.

Note: This calls `exec.sync_execute_at(abs_time, f, ts...)`, if available, otherwise it emulates timed scheduling by delaying calling `execution::sync_execute()` on the underlying non-time-scheduled execution agent.

Param exec

[in] The executor object to use for scheduling of the function *f*.

Param abs_time

[in] The point in time the given function should be scheduled at to run.

Param f

[in] The function which will be scheduled using the given executor.

Param ts...

[in] Additional arguments to use to invoke *f*.

Return

f(*ts...*)'s result

Public Functions

```
template<executor_any Executor, typename F, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, hpx::chrono::steady_time_point const
                                     &abs_time, F &&f, Ts&&... ts) const
```

```
template<executor_any Executor, typename F, typename ...Ts>
inline decltype(auto) operator() (Executor &&exec, hpx::chrono::steady_time_point const
                                     &abs_time, F &&f, Ts&&... ts) const
```

```
template<executor_any BaseExecutor>
```

```
struct timed_executor
```

`hpx/timed_execution/timed_execution.hpp`

Defined in header `hpx/timed_execution/timed_execution.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace parallel
```

```
        namespace execution
```

`hpx/timed_execution/timed_executors.hpp`

Defined in header `hpx/timed_execution/timed_executors.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace execution
```

```
        namespace experimental
```

```
            namespace parallel
```

```
                namespace execution
```


Typedefs

```
using sequenced_timed_executor = timed_executor<hpx::execution::sequenced_executor>

using parallel_timed_executor = timed_executor<hpx::execution::parallel_executor>

template<executor_any BaseExecutor>

struct timed_executor
```

hpx/timed_execution/traits/is_timed_executor.hpp

Defined in header `hpx/timed_execution/traits/is_timed_executor.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

Variables

```
template<typename Executor> concept timed_executor    = hpx::traits::is_timed_executor_v<Executor>
```

```
namespace parallel
```

```
    namespace execution
```

Typedefs

```
template<typename T>

using is_timed_executor_t = typename is_timed_executor<T>::type
```

Variables

```
template<typename T>

constexpr bool is_timed_executor_v = is_timed_executor<T>::value

template<typename T>

struct is_timed_executor : public detail::is_timed_executor<std::decay_t<T>>
```

```
namespace traits
```

Variables

```
template<typename T>
constexpr bool is_timed_executor_v = is_timed_executor<T>::value

template<typename Executor, typename Enable = void>
struct is_timed_executor : public hpx::parallel::execution::is_timed_executor<Executor>
```

timing

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx::chrono::high_resolution_clock

Defined in header [hpx/chrono.hpp](#)⁹⁵⁵.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

struct **high_resolution_clock**

Class *hpx::chrono::high_resolution_clock* represents the clock with the smallest tick period provided by the implementation. It may be an alias of *std::chrono::system_clock* or *std::chrono::steady_clock*, or a third, independent clock. *hpx::chrono::high_resolution_clock* meets the requirements of *Trivial-Clock*.

hpx::chrono::high_resolution_timer

Defined in header [hpx/chrono.hpp](#)⁹⁵⁶.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

class **high_resolution_timer**

high_resolution_timer is a timer object which measures the elapsed time

topology

See *Public API* for a list of names and headers that are part of the public *HPX* API.

⁹⁵⁵ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/chrono.hpp

⁹⁵⁶ http://github.com/TheHPXProject/hpx/blob/d402eead7405eedff56d454c9380fbd0c8044e7/libs/core/include_local/include/hpx/chrono.hpp

hpx/topology/cpu_mask.hpp

Defined in header hpx/topology/cpu_mask.hpp.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace threads
```

hpx/topology/topology.hpp

Defined in header hpx/topology/topology.hpp.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace threads
```

Typedefs

```
using hwloc_bitmap_ptr = std::shared_ptr<hpx_hwloc_bitmap_wrapper>
```

Enums

```
enum hpx_hwloc_membind_policy
```

Please see hwloc documentation for the corresponding enums HWLOC_MEMBIND_XXX.

Values:

```
    enumerator membind_default
```

```
    enumerator membind_firsttouch
```

```
    enumerator membind_bind
```

```
    enumerator membind_interleave
```

```
    enumerator membind_replicate
```

```
    enumerator membind_nexttouch
```

```
    enumerator membind_mixed
```

enumerator **membind_user**

Functions

topology &**create_topology**()

inline *topology* const &**get_topology**()

Get the global topology instance.

inline *std::size_t* **get_memory_page_size**()

struct **hpx_hwloc_bitmap_wrapper**

Public Functions

HPX_NON_COPYABLE(*hpx_hwloc_bitmap_wrapper*)

inline **hpx_hwloc_bitmap_wrapper**() noexcept

inline explicit **hpx_hwloc_bitmap_wrapper**(void *bmp) noexcept

inline ~**hpx_hwloc_bitmap_wrapper**()

inline void **reset**(hwloc_bitmap_t bmp) noexcept

inline explicit constexpr **operator bool**() const noexcept

inline hwloc_bitmap_t **get_bmp**() const noexcept

Private Members

hwloc_bitmap_t **bmp_**

Friends

friend *std::ostream* &**operator<<**(*std::ostream* &os, *hpx_hwloc_bitmap_wrapper* const *bmp)

struct **topology**

Public Functions

topology()

topology(*topology* const&) = delete

topology(*topology*&&) = delete

topology &**operator=**(*topology* const&) = delete

topology &**operator**=(*topology*&&) = delete

~topology()

inline *std::size_t* **get_socket_number**(*std::size_t* num_thread, [[maybe_unused]] *error_code* &ec = *throws*) const noexcept

Return the Socket number of the processing unit the given thread is running on.

Parameters

- **num_thread** – [in]
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

inline *std::size_t* **get_numa_node_number**(*std::size_t* num_thread, [[maybe_unused]] *error_code* &ec = *throws*) const noexcept

Return the NUMA node number of the processing unit the given thread is running on.

Parameters

- **num_thread** – [in]
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

inline *std::size_t* **get_numa_distance**(*std::size_t* node_a, *std::size_t* node_b) const noexcept

Return the hardware latency distance between two NUMA nodes. Values are sourced from hwloc's distance matrix when available; otherwise a standard default (10 = local, 20 = remote) is returned.

Parameters

- **node_a** – [in] logical index of the first NUMA node
- **node_b** – [in] logical index of the second NUMA node

mask_cref_type **get_machine_affinity_mask**(*error_code* &ec = *throws*) const noexcept

Return a bit mask where each set bit corresponds to a processing unit available to the application.

Parameters

- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

mask_type **get_service_affinity_mask**(mask_cref_type used_processing_units, *error_code* &ec = *throws*) const

Return a bit mask where each set bit corresponds to a processing unit available to the service threads in the application.

Parameters

- **used_processing_units** – [in] This is the mask of processing units which are not available for service threads.
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

mask_cref_type **get_socket_affinity_mask**(*std::size_t* num_thread, *error_code* &ec = *throws*) const

Return a bit mask where each set bit corresponds to a processing unit available to the given thread inside the socket it is running on.

Parameters

- **num_thread** – [in]
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

mask_cref_type **get_numa_node_affinity_mask**(*std::size_t* num_thread, *error_code* &ec = *throws*) const

Return a bit mask where each set bit corresponds to a processing unit available to the given thread inside the NUMA domain it is running on.

Parameters

- **num_thread** – [in]
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

mask_cref_type **get_core_affinity_mask**(*std::size_t* num_thread, *error_code* &ec = *throws*)
const

Return a bit mask where each set bit corresponds to a processing unit available to the given thread inside the core it is running on.

Parameters

- **num_thread** – [in]
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

mask_cref_type **get_thread_affinity_mask**(*std::size_t* num_thread, *error_code* &ec = *throws*)
const

Return a bit mask where each set bit corresponds to a processing unit available to the given thread.

Parameters

- **num_thread** – [in]
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

void **set_thread_affinity_mask**(mask_cref_type mask, *error_code* &ec = *throws*) const

Use the given bit mask to set the affinity of the given thread. Each set bit corresponds to a processing unit the thread will be allowed to run on.

Note: Use this function on systems where the affinity must be set from inside the thread itself.

Parameters

- **mask** – [in]
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

mask_type **get_thread_affinity_mask_from_lva**(void const *lva, *error_code* &ec = *throws*)
const

Return a bit mask where each set bit corresponds to a processing unit co-located with the memory the given address is currently allocated on.

Parameters

- **lva** – [in]
- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

void **print_affinity_mask**(*std::ostream* &os, *std::size_t* num_thread, mask_cref_type m, *std::string* const &pool_name) const

Prints the given mask *m* to *os* in a human-readable form.

bool **reduce_thread_priority**(*error_code* &ec = *throws*) const

Reduce thread priority of the current thread.

Parameters

- **ec** – [in,out] this represents the error status on exit, if this is pre-initialized to *hpx::throws* the function will throw on error instead.

std::size_t **get_number_of_sockets**() const

Return the number of available NUMA domains.

```

std::size_t get_number_of_numa_nodes() const
    Return the number of available NUMA domains.

std::size_t get_number_of_cores() const
    Return the number of available cores.

std::size_t get_number_of_pus() const noexcept
    Return the number of available hardware processing units.

std::size_t get_number_of_numa_node_cores(std::size_t numa) const
    Return number of cores in given numa domain.

std::size_t get_number_of_numa_node_pus(std::size_t numa) const
    Return number of processing units in a given numa domain.

std::size_t get_number_of_socket_pus(std::size_t socket) const
    Return number of processing units in a given socket.

std::size_t get_number_of_core_pus(std::size_t core) const
    Return number of processing units in given core.

std::size_t get_number_of_socket_cores(std::size_t socket) const
    Return number of cores units in given socket.

inline std::size_t get_core_number(std::size_t num_thread, error_code& = throws) const

std::size_t get_pu_number(std::size_t num_core, std::size_t num_pu, error_code &ec = throws)
    const

std::size_t get_cache_size(mask_cref_type mask, int level) const
    Return the size of the cache associated with the given mask.

mask_type get_cpubind_mask(error_code &ec = throws) const

mask_type get_cpubind_mask(std::thread &handle, error_code &ec = throws) const

hwloc_bitmap_ptr cpuset_to_nodeset(mask_cref_type mask) const
    convert a cpu mask into a numa node mask in hwloc bitmap form

void write_to_log() const

void *allocate(std::size_t len) const
    This is equivalent to malloc(), except that it tries to allocate page-aligned memory from the OS.

void *allocate_membind(std::size_t len, hwloc_bitmap_ptr const &bitmap,
    hpx_hwloc_membind_policy policy, int flags) const
    allocate memory with binding to a numa node set as specified by the policy and flags (see hwloc
    docs)

threads::mask_type get_area_membind_nodeset(void const *addr, std::size_t len) const

bool set_area_membind_nodeset(void const *addr, std::size_t len, void *nodeset) const

int get_numa_domain(void const *addr) const

void deallocate(void *addr, std::size_t len) const noexcept
    Free memory that was previously allocated by allocate.

```

```
void print_hwloc(std::ostream&) const

mask_type init_socket_affinity_mask_from_socket(std::size_t num_socket) const

mask_type init_numa_node_affinity_mask_from_numa_node(std::size_t num_numa_node)
                                                    const

mask_type init_core_affinity_mask_from_core(std::size_t num_core, mask_cref_type
                                           default_mask = empty_mask) const

mask_type init_thread_affinity_mask(std::size_t num_thread) const

mask_type init_thread_affinity_mask(std::size_t num_core, std::size_t num_pu) const

hwloc_bitmap_t mask_to_bitmap(mask_cref_type mask, hwloc_obj_type_t htype, unsigned
                              *count = nullptr) const

mask_type bitmap_to_mask(hwloc_bitmap_t bitmap, hwloc_obj_type_t htype) const
```

Public Static Functions

```
static void print_vector(std::ostream &os, std::vector<std::size_t> const &v)

static void print_mask_vector(std::ostream &os, std::vector<mask_type> const &v)
```

Private Types

```
using mutex_type = hpx::util::spinlock
```

Private Functions

```
std::size_t init_node_number(std::size_t num_thread, hwloc_obj_type_t type) const

inline std::size_t init_socket_number(std::size_t num_thread) const

std::size_t init_numa_node_number(std::size_t num_thread) const

inline std::size_t init_core_number(std::size_t num_thread) const

void extract_node_mask(hwloc_obj_t parent, mask_type &mask) const

std::size_t get_number_of_core_pus_locked(std::size_t core) const

std::size_t extract_node_count(hwloc_obj_t parent, hwloc_obj_type_t type, std::size_t count)
                               const

std::size_t extract_node_count_locked(hwloc_obj_t parent, hwloc_obj_type_t type, std::size_t
                                       count) const

mask_type init_machine_affinity_mask() const

inline mask_type init_socket_affinity_mask(std::size_t num_thread) const

inline mask_type init_numa_node_affinity_mask(std::size_t num_thread) const
```



```

inline mask_type init_core_affinity_mask(std::size_t num_thread) const

void init_num_of_pus()

hwloc_obj_t get_pu_obj(std::size_t num_pu) const

void init_numa_node_distances()

```

Private Members

```

hwloc_topology_t topo = nullptr

std::size_t num_of_pus_ = 0

bool use_pus_as_cores_ = false

mutable mutex_type topo_mtx

std::vector<std::size_t> socket_numbers_

std::vector<std::size_t> numa_node_numbers_

std::vector<std::size_t> core_numbers_

mask_type machine_affinity_mask_ = mask_type()

std::vector<mask_type> socket_affinity_masks_

std::vector<mask_type> numa_node_affinity_masks_

std::vector<mask_type> core_affinity_masks_

std::vector<mask_type> thread_affinity_masks_

std::vector<std::vector<std::size_t>> numa_node_distances_

```

Private Static Attributes

```

static mask_type empty_mask

static std::size_t memory_page_size_

static constexpr std::size_t pu_offset = 0

```

```
static constexpr std::size_t core_offset = 0
```

Friends

```
friend std::size_t get_memory_page_size()
```

tracing

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/tracing/tracing.hpp

Defined in header `hpx/tracing/tracing.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
namespace threads
```

```
namespace tracing
```

Functions

```
template<typename ThreadData>
constexpr void task_created([[maybe_unused]] ThreadData *thrdptr, [[maybe_unused]] void const
                           *parent_task_id = nullptr) noexcept
```

```
template<typename ThreadData>
constexpr void task_executing([[maybe_unused]] ThreadData *thrdptr) noexcept
```

```
template<typename ThreadId>
constexpr void task_yielded([[maybe_unused]] ThreadId const &id) noexcept
```

```
template<typename ThreadId, typename String>
constexpr void task_suspended([[maybe_unused]] ThreadId const &id, [[maybe_unused]] String const
                              &reason) noexcept
```

```
template<typename ThreadId, typename StateX>
constexpr void task_resumed([[maybe_unused]] ThreadId const &id, [[maybe_unused]] StateX statex)
                           noexcept
```

```
template<typename ThreadData>
constexpr void task_completed([[maybe_unused]] ThreadData *thrdptr) noexcept
```

```
struct lock_context
```

```
    #include <tracing.hpp> Context for annotating lock acquisitions and releases.
```

This struct is used to provide tracing annotations for synchronization primitives such as spinlocks and mutexes. It registers the creation, acquisition, and release of locks to the active tracing backend.

Public Functions

explicit **lock_context**(char const *name = nullptr, void const *addr = nullptr) noexcept

Constructs a lock context.

Parameters

- **name** – The name or description of the lock.
- **addr** – The memory address of the lock, used to track it uniquely.

explicit **lock_context**(char const *prefix, char const *suffix, void const *addr = nullptr) noexcept

Constructs a lock context with a dynamic name.

Parameters

- **prefix** – The prefix for the lock name.
- **suffix** – The suffix for the lock name.
- **addr** – The memory address of the lock, used to track it uniquely.

bool **before_lock**() const noexcept

Called immediately before attempting to acquire the lock.

Returns

true if the prepare event was recorded and `after_lock` should be called.

void **after_lock**() const noexcept

Called immediately after the lock is successfully acquired.

void **after_try_lock**(bool success) const noexcept

Called after a try-lock operation completes.

Parameters

success – true if the lock was acquired, false if it failed.

void **before_unlock**() const noexcept

Called immediately before releasing the lock.

void **after_unlock**() const noexcept

Called immediately after releasing the lock.

namespace **util**

namespace **external_timer**

hpx/tracing/backends/apex.hpp

Defined in header `hpx/tracing/backends/apex.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **tracing**

Typedefs

bool(*)() enable_parent_task_handler_type

typedef char const ***annotation_handle**

Functions

inline constexpr *annotation_handle* **create_annotation_handle**(char const *name) noexcept

inline constexpr void **set_thread_name**(char const*) noexcept

constexpr char const ***rename_region**(char const*) noexcept

constexpr void **task_staged**(char const*, void const* = nullptr) noexcept

constexpr void **task_created**(char const*, void const*, void const* = nullptr) noexcept

constexpr void **task_executing**(void const*, char const*, *std::size_t*) noexcept

constexpr void **task_yielded**(void const*, char const*) noexcept

constexpr void **task_suspended**(void const*, char const*, char const* = nullptr) noexcept

constexpr void **task_resumed**(void const*, char const*, char const* = nullptr) noexcept

constexpr void **task_completed**(void const*, char const*) noexcept

constexpr void **task_deleted**(void const*) noexcept

constexpr void **future_fulfilled**(void const*, char const* = nullptr) noexcept

Producer-side signal: future state fulfilled (no-op stub for APEX).

Producer-side signal: future state fulfilled (no-op stub).

Producer-side signal: a future shared state was set to a value.

Producer-side signal: future state fulfilled (no-op stub for ITTNotify).

Parameters

- **future_id** – Pointer to the *future_data* shared state. Used as a stable identifier to correlate producer and consumer events in the Tracy message log.
- **desc** – Optional description of the producing context (e.g. the name of the promise or task).

constexpr void **future_exception_set**(void const*, char const* = nullptr) noexcept

Producer-side signal: future exception set (no-op stub for APEX).

Producer-side signal: future exception set (no-op stub).

Producer-side signal: a future shared state was set with an exception.

Producer-side signal: future exception set (no-op stub for ITTNotify).

Parameters

- **future_id** – Pointer to the *future_data* shared state.
- **desc** – Optional description of the producing context.

constexpr void **continuation_run**(void const* = nullptr) noexcept

Consumer-side signal: continuation started (no-op stub for APEX).

Consumer-side signal: continuation started (no-op stub).

Consumer-side signal: a continuation has started executing, correlated with the source future state.

Consumer-side signal: continuation started (no-op stub for ITTNotify).

constexpr void **continuation_finished**(void const* = nullptr) noexcept

Consumer-side signal: continuation finished (no-op stub for APEX).

Consumer-side signal: continuation finished (no-op stub).

Consumer-side signal: a continuation has finished executing.

Consumer-side signal: continuation finished (no-op stub for ITTNotify).

constexpr void **handle_on_completed_fired**(void const* = nullptr) noexcept

Consumer-side signal: handle_on_completed fired (no-op stub for APEX).

Consumer-side signal: handle_on_completed fired (no-op stub).

Consumer-side signal: handle_on_completed fired, dispatching registered continuations.

Consumer-side signal: handle_on_completed fired (no-op stub for ITTNotify).

constexpr void **frame_mark**(char const* = nullptr) noexcept

Frame boundary marker (no-op stub for APEX).

Frame boundary marker (no-op stub).

Emit a frame/stage boundary marker in Tracy timeline.

Frame boundary marker (no-op stub for ITTNotify).

Parameters

name – Optional name for the frame/stage.

struct **fiber_region**

Public Functions

inline explicit constexpr **fiber_region**(*fiber_region_init_data* const&, *std::size_t*) noexcept

struct **fiber_region_init_data**

struct **fiber_suspend_region**

Public Functions

inline explicit constexpr **fiber_suspend_region**(char const*) noexcept

struct **lock_context**
#include <tracing.hpp>

struct **loop_context**

Public Functions

inline explicit constexpr **loop_context**() noexcept

~loop_context() = default

loop_context(*loop_context* const&) = delete

loop_context &**operator**=(*loop_context* const&) = delete

struct **mark_event**

Public Functions

inline explicit constexpr **mark_event**(char const*) noexcept

struct **region**

Public Functions

inline explicit constexpr **region**(*loop_context*&, *region_init_data* const&, *std::size_t*) noexcept

struct **region_init_data**

struct **scoped_task_timer**

Public Functions

explicit **scoped_task_timer**(*task_timer_data* data) noexcept

~scoped_task_timer()

scoped_task_timer(*scoped_task_timer* const&) = delete

scoped_task_timer &**operator**=(*scoped_task_timer* const&) = delete

void **stop**() noexcept

```
void yield() noexcept

template<typename T, typename State>
inline void handle_post_execution(T *thrdptr, State s) noexcept
```

Private Members

```
bool stopped_
```

```
task_timer_data data_
```

```
struct task_timer_data
```

Public Functions

```
task_timer_data() noexcept = default

inline explicit task_timer_data(std::shared_ptr<util::external_timer::task_wrapper> data)
                                noexcept

inline operator std::shared_ptr<util::external_timer::task_wrapper>() const
                                                    noexcept

inline constexpr bool valid() const noexcept
```

Public Members

```
std::shared_ptr<util::external_timer::task_wrapper> data
```

hpx/tracing/backends/ittnotify.hpp

Defined in header `hpx/tracing/backends/ittnotify.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace tracing
```

Functions

```
task_timer_data create_task_timer(threads::thread_description const &description, std::uint32_t
                                     parent_locality_id, threads::thread_id const &parent_task)

void update_task_timer(task_timer_data &timer, char const *new_name)

void tracing_init(char const *name, int argc, char **argv, std::uint32_t rank = 0, std::uint32_t size =
                    1)

void tracing_finalize()

void register_thread(char const *name)

void send_parcel(std::uint64_t tag, std::uint64_t size, std::uint64_t target_locality_id) noexcept

void recv_parcel(std::uint64_t tag, std::uint64_t size, std::uint64_t source_locality_id, std::uint64_t
                  source_thread_id) noexcept

void set_enable_parent_task_handler(enable_parent_task_handler_type f)

struct fiber_region

struct fiber_region_init_data

struct fiber_suspend_region

struct lock_context
    #include <tracing.hpp>

struct loop_context

struct mark_event

struct region

struct region_init_data

struct scoped_task_timer

struct task_timer_data
```


hpx/tracing/backends/tracy.hpp

Defined in header `hpx/tracing/backends/tracy.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx

    namespace tracing

        struct fiber_region

        struct fiber_region_init_data

        struct fiber_suspend_region

        struct lock_context
            #include <tracing.hpp>

        struct loop_context

        struct mark_event

        struct region

        struct region_init_data

        struct scoped_task_timer

        struct task_timer_data
```

hpx/tracing/backends/empty.hpp

Defined in header `hpx/tracing/backends/empty.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx

    namespace tracing
```

Functions

void **create_counter**(*std::string* const &full_name, *std::string* const &short_name) noexcept

void **sample_counter**(*std::string* const &name, *std::string* const &short_name, double value) noexcept

struct **annotation_handle**

struct **fiber_region**

struct **fiber_region_init_data**

struct **fiber_suspend_region**

struct **lock_context**
 #include <tracing.hpp>

struct **loop_context**

struct **mark_event**

struct **region**

struct **region_init_data**

struct **scoped_task_timer**

struct **task_timer_data**

tracy

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/tracy/tracy.hpp

Defined in header `hpx/tracy/tracy.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/tracy/tracy_tls.hpp

Defined in header `hpx/tracy/tracy_tls.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

util

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/util/insert_checked.hpp

Defined in header `hpx/util/insert_checked.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **util**

Functions

```
template<typename Iterator>
```

```
constexpr bool insert_checked(std::pair<Iterator, bool> const &r) noexcept
```

Helper function for writing predicates that test whether an `std::map` insertion succeeded. This inline template function negates the need to explicitly write the sometimes lengthy `std::pair<Iterator, bool>` type.

Parameters

r – [in] The return value of a `std::map` insert operation.

Returns

This function returns **r.second**.

```
template<typename Iterator>
```

```
bool insert_checked(std::pair<Iterator, bool> const &r, Iterator &it)
```

Helper function for writing predicates that test whether an `std::map` insertion succeeded. This inline template function negates the need to explicitly write the sometimes lengthy `std::pair<Iterator, bool>` type.

Parameters

- **r** – [in] The return value of a `std::map` insert operation.
- **r** – [out] A reference to an `Iterator`, which is set to **r.first**.
- **it** – [out] on exit, will hold the iterator referring to the inserted element

Returns

This function returns **r.second**.

hpx/util/sed_transform.hpp

Defined in header `hpx/util/sed_transform.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **util**

Functions

bool **parse_sed_expression**(*std::string* const &input, *std::string* &search, *std::string* &replace)

Parse a sed command.

Note: Currently, only supports search and replace syntax (s/search/replace/)

Parameters

- **input** – [in] The content to parse.
- **search** – [out] If the parsing is successful, this string is set to the search expression.
- **replace** – [out] If the parsing is successful, this string is set to the replace expression.

Returns

true if the parsing was successful, *false* otherwise.

struct **sed_transform**

#include <sed_transform.hpp> An unary function object which applies a sed command to its subject and returns the resulting string.

Note: Currently, only supports search and replace syntax (s/search/replace/)

Public Functions

sed_transform(*std::string* const &search, *std::string* replace)

explicit **sed_transform**(*std::string* const &expression)

std::string **operator()**(*std::string* const &input) const

inline explicit **operator bool**() const noexcept

inline bool **operator!**() const noexcept

Private Members

`std::shared_ptr<command> command_`

component_storage

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx/components/component_storage/migrate_to_storage.hpp

Defined in header `hpx/components/component_storage/migrate_to_storage.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **components**

Functions

```
template<typename Component>
future<hpx::id_type> migrate_to_storage(hpx::id_type const &to_migrate, hpx::id_type const
                                     &target_storage)
```

Migrate the component with the given id to the specified target storage

The function `migrate_to_storage<Component>` will migrate the component referenced by `to_migrate` to the storage facility specified with `target_storage`. It returns a future referring to the migrated component instance.

Parameters

- **to_migrate** – [in] The global id of the component to migrate.
- **target_storage** – [in] The id of the storage facility to migrate this object to.

Template Parameters

The – only template argument specifies the component type of the component to migrate to the given storage facility.

Returns

A future representing the global id of the migrated component instance. This should be the same as `migrate_to`.

```
template<typename Derived, typename Stub, typename Data>
inline Derived migrate_to_storage(client_base<Derived, Stub, Data> const &to_migrate,
                                hpx::components::component_storage const &target_storage)
```

Migrate the given component to the specified target storage

The function `migrate_to_storage` will migrate the component referenced by `to_migrate` to the storage facility specified with `target_storage`. It returns a future referring to the migrated component instance.

Parameters

- **to_migrate** – [in] The client side representation of the component to migrate.
- **target_storage** – [in] The id of the storage facility to migrate this object to.

Returns

A client side representation of representing of the migrated component instance. This should be the same as `migrate_to`.

hpx/components/component_storage/migrate_from_storage.hpp

Defined in header `hpx/components/component_storage/migrate_from_storage.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
namespace hpx
```

```
    namespace components
```

Functions

```
template<typename Component>
future<hpx::id_type> migrate_from_storage(hpx::id_type const &to_resurrect, hpx::id_type const
                                         &target = hpx::invalid_id)
```

Migrate the component with the given id from the specified target storage (resurrect the object)

The function `migrate_from_storage<Component>` will migrate the component referenced by `to_resurrect` from the storage facility specified where the object is currently stored on. It returns a future referring to the migrated component instance. The component instance is resurrected on the locality specified by `target_locality`.

Parameters

- **to_resurrect** – [in] The global id of the component to migrate.
- **target** – [in] The optional locality to resurrect the object on. By default the object is resurrected on the locality it was located on last.

Template Parameters

The – only template argument specifies the component type of the component to migrate from the given storage facility.

Returns

A future representing the global id of the migrated component instance. This should be the same as `to_resurrect`.

supervision_dispatch

See *Public API* for a list of names and headers that are part of the public *HPX* API.

hpx::supervision::discover_peers

Defined in header `hpx/supervision_dispatch.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::supervision::fan_out_join`
- `hpx::supervision::discover_and_join`

```
std::vector<discovered_peer> hpx::supervision::discover_peers(chrono::steady_duration const &timeout =
                                                                default_discovery_timeout)
```

Performs a one-time discovery pull for `supervision_dispatch` peers: concurrently resolves the pinned sentinel and registry names (see `register_name()`) of every remote locality (`hpx::find_remote_localities()`), which already excludes this locality), bounded by a single wait for `timeout` across all candidates.

Because constructing a sentinel/registry client from a symbolic name (see `client_base`'s symbolic-name constructor) waits/resolves once the symbol is bound rather than failing fast on one that is not (yet) registered, localities that never called `register_name()` (e.g. because they have not run `init_supervision()` yet) would otherwise hang this call indefinitely. Instead, once *timeout* elapses, only the candidates whose sentinel *and* registry clients had both resolved by then (checked via the non-blocking `is_ready()`) are returned; every other candidate is simply excluded as not currently participating in supervision - *discover_peers()* itself never fails or throws on their account.

Parameters

timeout – The maximum duration to wait, across all candidates combined, for their sentinel and registry names to resolve.

Returns

The peers whose sentinel and registry clients both resolved within *timeout*, paired with the numeric locality id they are pinned to.

hpx::supervision::fan_out_join

Defined in header `hpx/supervision_dispatch.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::supervision::discover_peers`
- `hpx::supervision::discover_and_join`

```
std::vector<shadow_id> hpx::supervision::fan_out_join(registry const &local_registry,
                                                    std::vector<discovered_peer> const &peers,
                                                    chrono::steady_duration const &timeout =
                                                    default_discovery_timeout)
```

Reactively fans out `join()` calls from *local_registry* to every peer in *peers* (typically the result of a prior *discover_peers()* call): for each entry, this calls `local_registry.join(peer.sentinel_client, peer.locality)`.

This reuses `registry::join()`'s existing reservation/idempotency machinery unchanged (see `server::registry::reserve_ownership()`) rather than duplicating any of it, so calling *fan_out_join()* more than once with (partially) overlapping peer lists - or racing with an inbound `join()` call from one of those same peers, e.g. because that peer is concurrently running its own *fan_out_join()* against this locality - is safe and never creates more than one shadow per peer sentinel.

fan_out_join() itself is reactive only: it performs exactly one round of `join()` calls for the given *peers* and starts no background/polling activity of its own.

Parameters

- **local_registry** – The registry to join every peer in *peers* to.
- **peers** – The peers to join, typically returned by a prior *discover_peers()* call.
- **timeout** – The maximum duration to wait, across all peers combined, for their `join()` calls to settle. Peers whose `join()` has not settled (or failed) within this bound are silently dropped from the result, rather than hanging or throwing.

Returns

The shadow target IDs for peers whose `join()` calls settled successfully within *timeout*, in the same relative order as *peers*.

hpx::supervision::discover_and_join

Defined in header `hpx/supervision_dispatch.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::supervision::discover_peers`
- `hpx::supervision::fan_out_join`

```
std::vector<shadow_id> hpx::supervision::discover_and_join(registry const &local_registry,
                                                         chrono::steady_duration const &timeout =
                                                         default_discovery_timeout)
```

Composes a single reactive discovery-and-join pass for `supervision_dispatch` peers: performs exactly one `discover_peers()` pull to resolve the pinned sentinel and registry names of every remote locality (`hpx::find_remote_localities()` already excludes this locality, so no separate self-exclusion is needed here), then fans the resulting peer list out to `local_registry` via a single `fan_out_join()` pass.

`discover_and_join()` introduces no state, polling thread, timer, or repeated broadcast of its own - it is a pure composition of `discover_peers()` and `fan_out_join()`, each of which is itself bounded to a single round of resolution/join calls. Because `fan_out_join()` reuses `registry::join()`'s existing reservation/idempotency machinery unchanged (see `server::registry::reserve_ownership()`), calling `discover_and_join()` more than once - whether to pick up newly started peers or purely by accident - is safe and never creates more than one shadow per peer sentinel; repeated invocations simply return the (stable) shadow ids already owned for peers that were previously joined.

Parameters

- **local_registry** – The registry to join every discovered peer to.
- **timeout** – The maximum duration to wait, across all discovery candidates combined, for their sentinel and registry names to resolve (see `discover_peers()`).

Returns

The shadow target id that `local_registry` created (or already had) for each peer discovered within `timeout`, in the same order as returned by `discover_peers()`.

hpx::supervision::sentinel

Defined in header `hpx/supervision_dispatch.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

```
class sentinel : public hpx::components::client_base<sentinel, server::sentinel>
```

A sentinel is a lightweight, self-supervising handle: constructing it creates a component on the given target locality, and calling `start()` publishes the `started` lifecycle event for that component to the supervision manager running on the same locality. No registry lookup or discovery step is required.

hpx::supervision::dispatch_work

Defined in header `hpx/supervision_dispatch.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::supervision::invoke_fenced_action`

Warning: doxygenfunction: Unable to resolve function “hpx::supervision::dispatch_work” with arguments (shadow_id const&, hpx::id_type const&, std::uint64_t const, Ts&&...) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:

```
- template<typename Action, typename ...Ts> decltype(auto) dispatch_work(Action,
↳joined_peer const &peer, std::uint64_t const epoch, Ts&&... ts)
- template<typename Action, typename ...Ts> decltype(auto) dispatch_work(Action,
↳shadow_id const &shadow, hpx::id_type const &target, std::uint64_t const epoch, Ts&&
↳... ts)
- template<typename Action, typename ...Ts> decltype(auto) dispatch_work(joined_peer
↳const &peer, std::uint64_t const epoch, Ts&&... ts)
- template<typename Action, typename ...Ts> decltype(auto) dispatch_work(shadow_id
↳const &shadow, hpx::id_type const &target, std::uint64_t const epoch, Ts&&... ts)
```

Warning: doxygenfunction: Unable to resolve function “hpx::supervision::dispatch_work” with arguments (Action, shadow_id const&, hpx::id_type const&, std::uint64_t const, Ts&&...) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:

```
- template<typename Action, typename ...Ts> decltype(auto) dispatch_work(Action,
↳joined_peer const &peer, std::uint64_t const epoch, Ts&&... ts)
- template<typename Action, typename ...Ts> decltype(auto) dispatch_work(Action,
↳shadow_id const &shadow, hpx::id_type const &target, std::uint64_t const epoch, Ts&&
↳... ts)
- template<typename Action, typename ...Ts> decltype(auto) dispatch_work(joined_peer
↳const &peer, std::uint64_t const epoch, Ts&&... ts)
- template<typename Action, typename ...Ts> decltype(auto) dispatch_work(shadow_id
↳const &shadow, hpx::id_type const &target, std::uint64_t const epoch, Ts&&... ts)
```

Warning: doxygenfunction: Unable to resolve function “hpx::supervision::dispatch_work” with arguments (joined_peer const&, std::uint64_t const, Ts&&...) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:

```
- template<typename Action, typename ...Ts> decltype(auto) dispatch_work(Action,
↳joined_peer const &peer, std::uint64_t const epoch, Ts&&... ts)
- template<typename Action, typename ...Ts> decltype(auto) dispatch_work(Action,
↳shadow_id const &shadow, hpx::id_type const &target, std::uint64_t const epoch, Ts&&
↳... ts)
- template<typename Action, typename ...Ts> decltype(auto) dispatch_work(joined_peer
↳const &peer, std::uint64_t const epoch, Ts&&... ts)
- template<typename Action, typename ...Ts> decltype(auto) dispatch_work(shadow_id
↳const &shadow, hpx::id_type const &target, std::uint64_t const epoch, Ts&&... ts)
```

Warning: doxygenfunction: Unable to resolve function “hpx::supervision::dispatch_work” with arguments (Action, joined_peer const&, std::uint64_t const, Ts&&...) in doxygen xml output for project “hpx” from directory: ../hpx_autodoc. Potential matches:

```
- template<typename Action, typename ...Ts> decltype(auto) dispatch_work(Action,
→joined_peer const &peer, std::uint64_t const epoch, Ts&&... ts)
- template<typename Action, typename ...Ts> decltype(auto) dispatch_work(Action,
→shadow_id const &shadow, hpx::id_type const &target, std::uint64_t const epoch, Ts&&
→... ts)
- template<typename Action, typename ...Ts> decltype(auto) dispatch_work(joined_peer,
→const &peer, std::uint64_t const epoch, Ts&&... ts)
- template<typename Action, typename ...Ts> decltype(auto) dispatch_work(shadow_id,
→const &shadow, hpx::id_type const &target, std::uint64_t const epoch, Ts&&... ts)
```

hpx::supervision::invoke_fenced_action

Defined in header `hpx/supervision_dispatch.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::supervision::dispatch_work`

```
template<typename Action, typename ShadowId, typename IdType, typename Epoch, typename ...Ts>
decltype(auto) hpx::supervision::invoke_fenced_action(Action act, ShadowId shadow, IdType target,
Epoch const epoch, Ts... ts)
```

Remote dispatch point that performs the authoritative admission re-check and invokes the wrapped action.

This is the function body that actually runs as the action on the *destination* locality (the target’s own locality, reached via `hpx::colocated` in `dispatch_work()`). It is invoked by `fenced_action` once the action has been dispatched.

The admission check performed here runs on the same locality/thread that owns the target’s *supervision_manager* state, so there is no suspension point between this check and the actual invocation below. This is what closes the client-side/dispatch race: even if the client’s own (non-authoritative) `check_admission()` in `dispatch_work()` returned `admitted`, the target may have latched a terminal event for this epoch in the meantime, and this re-check will catch it.

Template Parameters

- **Action** – The wrapped HPX action type to invoke on admission.
- **ShadowId** – Type of the shadow identifier.
- **IdType** – Type of the target identifier forwarded to `act` (this type is always `hpx::id_type`).
- **Epoch** – Type of the fencing epoch value (this type is always `std::uint64_t`).
- **Ts** – Types of the additional arguments forwarded to `act`.

Parameters

- **act** – The action instance to invoke once admission succeeds.
- **shadow** – Identifier of the shadow id representing the target component instance.

- **target** – Identifier of the component instance being dispatched to; forwarded to `act` (the epoch is fencing metadata only and is not part of `act`'s argument list).
- **epoch** – The fencing epoch to check admission for.
- **ts** – Additional arguments forwarded to `act`.

Throws

`hpx::exception` – with `hpx::error::target_fenced` if the target has latched a terminal event for epoch since the client's own admission check, in which case the wrapped action is not invoked.

Returns

The result of invoking `act`, obtained via `hpx::sync` (run directly/locally, without an extra parcel hop, since this function already executes on the target's locality).

hpx::supervision::registry

Defined in header `hpx/supervision_dispatch.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX API*.

class **registry** : public `hpx::components::client_base<registry, server::registry>`

A registry is a lightweight, self-supervising handle for pairing with peer sentinels. It mirrors a peer's lifecycle state locally via `join()`, and can itself be discovered by name in AGAS like other `supervision_dispatch` components.

hpx::supervision::testing::local_snapshot_peers

Defined in header `hpx/supervision_dispatch.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX API*.

See also:

- `hpx::supervision::testing::set_failure_detection_poll_timeout_for_testing`
- `hpx::supervision::testing::last_join_shadow`
- `hpx::supervision::testing::suspend_heartbeat_for_testing`
- `hpx::supervision::testing::failure_detection_sweep_in_flight_for_testing`

`std::vector<server::peer_snapshot> hpx::supervision::testing::local_snapshot_peers()`

Returns a snapshot of this locality's currently-joined, ready peers, as tracked by the local registry. Not part of the public dispatch API - exists so tests can inspect registry state directly.

hpx::supervision::testing::set_failure_detection_poll_timeout_for_testing

Defined in header `hpx/supervision_dispatch.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX API*.

See also:

- `hpx::supervision::testing::local_snapshot_peers`
- `hpx::supervision::testing::last_join_shadow`
- `hpx::supervision::testing::suspend_heartbeat_for_testing`

- `hpx::supervision::testing::failure_detection_sweep_in_flight_for_testing`

void `hpx::supervision::testing::set_failure_detection_poll_timeout_for_testing`(`hpx::chrono::steady_duration`
const
&timeout)

Overrides the timeout `failure_detection_loop()` passes to `await_terminal()` on each sweep. Must be called before `init()` starts the loop to take effect for that lifecycle; has no effect on a sweep already in flight. Not part of the public dispatch API - exists only to make failure-detection tests deterministic and fast instead of bound by the real 60s `default_discovery_timeout`.

Parameters

timeout – The new poll timeout.

`hpx::supervision::testing::last_join_shadow`

Defined in header `hpx/supervision_dispatch.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::supervision::testing::local_snapshot_peers`
- `hpx::supervision::testing::set_failure_detection_poll_timeout_for_testing`
- `hpx::supervision::testing::suspend_heartbeat_for_testing`
- `hpx::supervision::testing::failure_detection_sweep_in_flight_for_testing`

`shadow_id` `hpx::supervision::testing::last_join_shadow()`

Returns the shadow target most recently minted by `join()`, regardless of whether the `register_observers()` call that followed it went on to succeed or fail. Lets tests verify that a failed `join()` does not leak the shadow's locally tracked supervision state (see `registry::register_observers()`'s catch block).

`hpx::supervision::testing::suspend_heartbeat_for_testing`

Defined in header `hpx/supervision_dispatch.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::supervision::testing::local_snapshot_peers`
- `hpx::supervision::testing::set_failure_detection_poll_timeout_for_testing`
- `hpx::supervision::testing::last_join_shadow`
- `hpx::supervision::testing::failure_detection_sweep_in_flight_for_testing`

void `hpx::supervision::testing::suspend_heartbeat_for_testing()`

Stops this locality's own `heartbeat_loop()` without finalizing the dispatcher, simulating a hard crash (all lifecycle activity ceases, including the periodic `event::running` pulse) while staying joined in peers' registries. Idempotent; no-op if `init()` was never called or the loop is already stopped. Not part of the public dispatch API - exists only so failure-detection tests can deterministically simulate silence instead of relying on the absence of explicit calls, which no longer implies silence now that heartbeats are mirrored onto observers' shadows.

hpx::supervision::testing::failure_detection_sweep_in_flight_for_testing

Defined in header `hpx/supervision_dispatch.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::supervision::testing::local_snapshot_peers`
- `hpx::supervision::testing::set_failure_detection_poll_timeout_for_testing`
- `hpx::supervision::testing::last_join_shadow`
- `hpx::supervision::testing::suspend_heartbeat_for_testing`

bool `hpx::supervision::testing::failure_detection_sweep_in_flight_for_testing()`

Reports whether `failure_detection_loop()` currently has one or more `await_terminal()` calls outstanding against unresponsive peers (i.e. a sweep is mid-flight). Returns false before `init()` is called, between sweeps, and after `finalize()`. Not part of the public dispatch API - exists only so tests can deterministically synchronize on a real in-flight sweep instead of sleeping and hoping the loop got there first.

hpx::supervision::shadow_id

Defined in header `hpx/supervision_dispatch.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::supervision::invalid_shadow_id`
- `hpx::supervision::joined_peer`

class `shadow_id`

A distinct, non-dereferenceable identifier for a peer's local supervision "shadow" state.

A `shadow_id` wraps the opaque, unmanaged `hpx::id_type` minted by `make_shadow_target()` to key a peer's fencing/lifecycle state inside this locality's own `supervision_manager`. It is intentionally *not* implicitly convertible to or from `hpx::id_type`: a shadow id must never be resolved by AGAS, passed to `hpx::colocated()`, or used as the destination of an action dispatch (e.g. `hpx::sync(act, id, ...)`) - doing so is undefined behavior at the id level and will throw at invocation time for plain actions, since a shadow id does not satisfy `naming::is_locality()`.

Functions that operate on generic, id-type-agnostic supervision primitives (`query_state()`, `publish_event()`, `check_admission()`, `await_terminal()`) continue to take plain `hpx::id_type`, since those primitives are also used with real, remotely-addressable ids. Call `get()` to unwrap a `shadow_id` only at the boundary where it is passed into one of those primitives as an opaque lookup key.

See also:

`make_shadow_target()`

See also:

`dispatch_work()`

hpx::supervision::invalid_shadow_id

Defined in header `hpx/supervision_dispatch.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::supervision::shadow_id`
- `hpx::supervision::joined_peer`

`const shadow_id hpx::supervision::invalid_shadow_id = shadow_id()`

A well-known, invalid `shadow_id` sentinel value.

Equivalent to a default-constructed `shadow_id`. Use this to signal “no

shadow” (e.g. when a peer has not yet joined or has been fully evicted) without relying on default construction at each use site.

hpx::supervision::joined_peer

Defined in header `hpx/supervision_dispatch.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- `hpx::supervision::shadow_id`
- `hpx::supervision::invalid_shadow_id`

struct **joined_peer**

Pairs the two identifiers a caller obtains from a successful `registry::join()` and needs to perform a fenced dispatch.

`dispatch_work()` requires two distinct identifiers that must never be confused with one another:

- a `shadow_id` used purely as an opaque, local fencing/admission lookup key (never dereferenced, resolved, or dispatched to), and
- a real, colocatable `hpx::id_type` identifying the locality the wrapped action must actually execute on.

`joined_peer` bundles both values together so callers can carry a single object returned from `registry::join()` instead of two loose parameters, without introducing any dependency on the registry itself into `dispatch_work.hpp` - the struct is a plain value type composed solely from `shadow_id` and `hpx::id_type`.

See also:

`shadow_id`

See also:

`dispatch_work()`

See also:

`registry::join()`

hpx::supervision::init

Defined in header `hpx/supervision_dispatch.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

See also:

- `hpx::supervision::finalize`
- `hpx::supervision::is_initialized`

`hpx::shared_future<void> hpx::supervision::init(hpx::chrono::steady_duration const &discovery_timeout = default_discovery_timeout)`

Performs one-shot, idempotent initialization of the supervision-dispatch runtime for this locality. On the first successful call, this function atomically transitions the internal lifecycle state from *uninitialized* to *initializing*, after incrementing the internal epoch, then owns the entire startup sequence: it creates a local *sentinel* and *registry* for `hpx::find_here()`, publishes *event::started* on the sentinel *before* either component's symbol name is registered (so that no peer can discover and join a not-yet-started sentinel), registers both symbol names, and finally performs a single *discover_and_join()* pass to join every currently reachable peer registry. On success the state transitions to *active*; if any step throws, all partial state is rolled back and the lifecycle reverts to *uninitialized* so that a subsequent call can retry from scratch.

Calling `init()` while already *active* is a no-op that resolves immediately without repeating any side effect. Calling it while another call is still *initializing* attaches the caller to the in-flight operation's outcome rather than racing it, so concurrent callers never create more than one sentinel/registry pair. Calling it while *finalizing* is in progress is rejected; callers should wait for *finalize()* to complete before re-initializing.

Note: This function has no effect on any state that has not yet been created by a successful call — in particular, no symbol name is ever registered and no event is ever published unless this function (or a concurrent call it is attached to) actually runs the initialization sequence.

Parameters

discovery_timeout – The maximum duration to wait, across all discovery candidates combined, for peer sentinel/registry symbol names to resolve during the *discover_and_join()* step (see *discover_peers()*).

Returns

A shared future that becomes ready (with a void value) once the supervision runtime has reached the *active* state, or that becomes exceptional if initialization could not complete (e.g. due to a concurrent *finalize* still in progress).

`void hpx::supervision::init(hpx::launch::sync_policy, hpx::chrono::steady_duration const &discovery_timeout = default_discovery_timeout)`

Performs one-shot, idempotent initialization of the supervision-dispatch runtime for this locality, blocking the calling thread until the operation completes. This is the synchronous counterpart of `init(hpx::chrono::steady_duration const&)`: it carries out the exact same lifecycle transition and startup sequence (sentinel/registry creation, *event::started* publication, symbol name registration, and a single *discover_and_join()* pass), but waits for the resulting future to become ready (or exceptional) before returning, rather than handing the future back to the caller.

Note: As with the asynchronous overload, no symbol name is registered and no event is published unless this call (or a concurrent call it attaches to) actually performs the initialization sequence.

Parameters

- **launch::sync_policy** – Tag selecting the blocking overload of `init()`.
- **discovery_timeout** – The maximum duration to wait, across all discovery candidates combined, for peer sentinel/registry symbol names to resolve during the *discover_and_join()* step (see *discover_peers()*).

Throws

hpx::exception – if initialization could not complete (e.g. due to a concurrent *finalize()* still in progress). Calling this while already *active* is a no-op that returns immediately.

hpx::supervision::finalize

Defined in header `hpx/supervision_dispatch.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::supervision::init*
- *hpx::supervision::is_initialized*

void **hpx::supervision::finalize()**

Performs one-shot, idempotent teardown of the supervision-dispatch runtime previously started by *init()*. On the first call observing the *active* state, this function atomically transitions the lifecycle to *finalizing*, publishes *event::completed* for the local sentinel at the new epoch, unregisters both the sentinel and registry symbol names, and then releases ownership of both components (relying on *client_base*'s default lifetime management to destroy them once no other references remain) before resetting the lifecycle back to *uninitialized*, allowing a subsequent *init()* call to re-arm the runtime from scratch.

Calling *finalize()* when the runtime is *uninitialized*, still *initializing*, or already *finalizing* is a documented no-op: it resolves immediately without publishing any event, unregistering any symbol name, or otherwise touching component state.

Note: Because *finalize()* is a no-op unless the runtime is currently *active*, it is always safe to call speculatively, e.g. during shutdown, without first checking *is_initialized()*.

hpx::supervision::is_initialized

Defined in header `hpx/supervision_dispatch.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

See also:

- *hpx::supervision::init*
- *hpx::supervision::finalize*

bool **hpx::supervision::is_initialized()** noexcept

Returns whether the supervision-dispatch runtime for this locality is currently *active*, i.e. has completed *init()* and has not yet been torn down by *finalize()*.

Note: This function never blocks and performs a single atomic load; it is safe to call from any thread, including while *init()* or *finalize()* is concurrently in flight.

Returns

true if the lifecycle state is *active*, *false* if it is *uninitialized*, *initializing*, or *finalizing*.

hpx/supervision_dispatch/server/sentinel.hpp

Defined in header `hpx/supervision_dispatch/server/sentinel.hpp`.

See *Public API* for a list of names and headers that are part of the public HPX API.

namespace **hpx**

namespace **supervision**

Fenced action dispatch for the supervision subsystem.

This header provides the machinery to dispatch an HPX action to a target component while enforcing supervision “fencing” semantics: once a target has latched a terminal event for a given epoch, no further actions for that epoch may execute against it. Dispatch performs a cheap, non-authoritative admission check on the caller’s locality and an authoritative re-check on the target’s own locality (via `hpx::colocated`), immediately before invoking the wrapped action, closing the race between admission and invocation.

Types and functions implementing HPX’s supervision and fenced dispatch facilities.

namespace **server**

class **sentinel** : public `hpx::components::component_base<sentinel>`

Public Functions

sentinel()

`hpx::supervision::publish_result start(std::uint64_t epoch) const`

Publish event::started for this sentinel at the given epoch.

Parameters

epoch – The epoch associated with the event.

Returns

The result of publishing the event.

struct **start_action** : public `hpx::actions::make_action_t<decltype(&sentinel::start), &sentinel::start, start_action>`

hpx/supervision_dispatch/server/registry.hpp

Defined in header `hpx/supervision_dispatch/server/registry.hpp`.

See *Public API* for a list of names and headers that are part of the public *HPX* API.

namespace **hpx**

namespace **supervision**

Fenced action dispatch for the supervision subsystem.

This header provides the machinery to dispatch an HPX action to a target component while enforcing supervision “fencing” semantics: once a target has latched a terminal event for a given epoch, no further actions for that epoch may execute against it. Dispatch performs a cheap, non-authoritative admission check on the caller’s locality and an authoritative re-check on the target’s own locality (via `hpx::colocated`), immediately before invoking the wrapped action, closing the race between admission and invocation.

Types and functions implementing HPX’s supervision and fenced dispatch facilities.

namespace **server**

struct **peer_snapshot**

#include <registry.hpp> Plain-data view of a single joined peer, safe to hand out to callers outside the registry (unlike `peer_entry`, does not expose the ready/evict_pending bookkeeping used internally to coordinate concurrent `join()`/`evict_peer()` calls).

Public Members

hpx::id_type **peer_sentinel**

The peer sentinel this entry was joined against.

hpx::id_type **peer_locality**

The locality that owns `peer_sentinel`, as recorded by `join()`.

shadow_id **shadow**

The locally tracked shadow mirroring `peer_sentinel`’s supervision state.

std::uint64_t **join_epoch**

The epoch at which `peer_sentinel` was joined.

class **registry** : public *hpx::components::component_base<registry>*

Public Functions

registry()

joined_peer **join**(*hpx::id_type* const &peer_sentinel, *hpx::id_type* const &peer_locality)

Joins a peer sentinel and creates or reuses its shadow.

Parameters

- **peer_sentinel** – The peer sentinel to observe.
- **peer_locality** – The locality that owns **peer_sentinel**.

std::vector<peer_snapshot> **snapshot_peers**() const

Returns a point-in-time snapshot of all fully joined, non-evicting peers.

Intended for a future failure-detection poller (or other local consumers) that needs a stable, lock-safe list of peers to iterate over without reaching into **peers_** directly. Entries mid-join (**ready** == **false**) or already scheduled for eviction (**evict_pending** == **true**) are excluded, so a poller never observes a half-registered or torn-down peer.

Returns

A snapshot of every peer entry with **ready** == **true** and **evict_pending** == **false** at the time of the call.

void **leave**(*hpx::id_type* const &peer_sentinel, *hpx::id_type* const &peer_locality, *shadow_id* const &shadow)

Protected Functions

std::pair<shadow_id, bool> **reserve_ownership**(*hpx::id_type* const &peer_sentinel)

std::pair<hpx::id_type, hpx::id_type> **register_observers**(*hpx::id_type* const &peer_sentinel, *hpx::id_type* const &peer_locality, *shadow_id* const &shadow)

void **evict_peer**(*hpx::id_type* const &peer_sentinel, *hpx::id_type* const &peer_locality, *shadow_id* const &shadow, *hpx::id_type* keep_alive)

Protected Static Functions

static void **cleanup_peer**(*hpx::id_type* const &peer_locality, *hpx::id_type* const &lifecycle_observer, *hpx::id_type* const &activity_observer, *shadow_id* const &shadow)

Unregisters a peer's observers and removes its shadow state.

Performs the actual peer teardown deferred by **evict_peer**(): unregisters **lifecycle_observer** and **activity_observer** on **peer_locality** and removes the shadow's locally tracked supervision state. Kept separate from **evict_peer**() because **evict_peer**() only posts this work as a task (via **hpx::post**()) rather than performing it inline.

Parameters

- **peer_locality** – The locality that owns the observers.
- **lifecycle_observer** – The lifecycle observer id to unregister.
- **activity_observer** – The activity observer id to unregister.
- **shadow** – The peer's shadow whose local state is removed.

Private Members

mutable *hpx::spinlock* **mtx_**

hpx::lcos::local::detail::condition_variable **cond_**

std::map<hpx::id_type, peer_entry> **peers_**

struct **peer_entry**

Internal bookkeeping for a single joined (or joining) peer.

Not exposed outside the registry; see *peer_snapshot* for the plain-data subset safe to hand to external callers.

Public Members

hpx::id_type **peer_locality**

The locality that owns this peer's sentinel, recorded by *join()* when the entry is marked ready. Surfaced read-only via *peer_snapshot::peer_locality*.

shadow_id **shadow**

The locally tracked shadow mirroring this peer's supervision state (the same shadow returned by *join()* and referenced by *evict_peer()/cleanup_peer()*).

hpx::id_type **lifecycle_observer**

The observer id registered on *peer_locality* to receive this peer's terminal lifecycle notification (drives *evict_peer()* via *hpx::post()*).

hpx::id_type **activity_observer**

The observer id registered on *peer_locality* to receive this peer's activity notifications (used to keep the shadow's local state current between terminal events).

std::uint64_t **join_epoch**

The epoch at which this peer was joined, recorded by *join()*.

bool **ready** = false

False while a *join()* call has reserved ownership of this peer sentinel (i.e. is in the process of performing the remote observer registrations) but has not yet filled in the rest of the entry. Concurrent *join()* calls for the same peer wait on *cond_* instead of racing to register duplicate observers.

bool **evict_pending** = false

True once a terminal notification has been observed for this peer but eviction has not yet completed (deferred via *hpx::post()* to *evict_peer()*). Ensures a terminal notification racing ahead of *join()* completion is deferred instead of dropped, and lets *snapshot_peers()* exclude a peer that is already tearing down.

2.10 Contributing to *HPX*

HPX development happens on Github. The following sections are a collection of useful information related to *HPX* development.

2.10.1 Contributing to *HPX*

The main source of information to understand the process of how to contribute to *HPX* can be found in [this document](#)⁹⁵⁷. This is a living document that is constantly updated with relevant information.

2.10.2 *HPX* governance model

The *HPX* project is a meritocratic, consensus-based community project. Anyone with an interest in the project can join the community, contribute to the project design and participate in the decision making process. [This document](#)⁹⁵⁸ describes how that participation takes place and how to set about earning merit within the project community.

2.10.3 Release procedure for *HPX*

Below is a step by step procedure for making an *HPX* release. We aim to produce two releases per year: one in March-April, and one in September-October.

This is a living document and may not be totally current or accurate. It is an attempt to capture current practices in making an *HPX* release. Please update it as appropriate.

One way to use this procedure is to print a copy and check off the lines as they are completed to avoid confusion.

1. Notify developers that a release is imminent.
2. For minor and major releases: create and check out a new branch at an appropriate point on `master` with the name `release-major.minor.X`. `major` and `minor` should be the major and minor versions of the release. For patch releases: check out the corresponding `release-major.minor.X` branch.
3. Write release notes in `docs/sphinx/releases/whats_new_${VERSION}.rst`. Keep adding merged PRs and closed issues to this until just before the release is made. Use `tools/generate_pr_issue_list.sh` to generate the lists. Add the new release notes to the table of contents in `docs/sphinx/releases.rst`.
4. Build the docs, and proof-read them. Update any documentation that may have changed, and correct any typos. Pay special attention to:
 - `$HPX_SOURCE/README.rst`
 - Update grant information
 - `docs/sphinx/releases/whats_new_${VERSION}.rst`
 - `docs/sphinx/about_hpx/people.rst`
 - Update collaborators
 - Update grant information
5. This step does not apply to patch releases. For APEX:

⁹⁵⁷ <https://github.com/TheHPXProject/hpx/blob/master/.github/CONTRIBUTING.md>

⁹⁵⁸ <http://hpx.stellar-group.org/documents/governance/>

- Change the release branch to be the most current release tag available in the APEX `git_external` section in the main `CMakeLists.txt`. Please contact the maintainers of the respective packages to generate a new release to synchronize with the *HPX* release ([APEX⁹⁵⁹](#)).

6. Make sure `HPX_VERSION_MAJOR/MINOR/SUBMINOR` in `CMakeLists.txt` contain the correct values. Change them if needed.
7. Change version references in `CITATION.cff`. There are two occurrences. Change year in the copyright file under `/libs/core/version/src/version.cpp`.
8. This step does not apply to patch releases. Remove features which have been deprecated for at least 2 releases. This involves removing build options which enable those features from the main `CMakeLists.txt` and also deleting all related code and tests from the main source tree.

The general deprecation policy involves a three-step process we have to go through in order to introduce a breaking change:

- a. First release cycle: add a build option that allows for explicitly disabling any old (now deprecated) code.
- b. Second release cycle: turn this build option OFF by default.
- c. Third release cycle: completely remove the old code.

The main `CMakeLists.txt` contains a comment indicating for which version the breaking change was introduced first. In the case of deprecated features which don't have a replacement yet, we keep them around in case (like `Vc` for example).

9. Update the minimum required versions if necessary (compilers, dependencies, etc.) in `prerequisites.rst`. Update latest tested versions.
10. Verify that the Jenkins setups for the release branch on Rostam are running and do not display any errors.
11. Repeat the following steps until satisfied with the release.
 1. Change `HPX_VERSION_TAG` in `CMakeLists.txt` to `-rcN`, where `N` is the current iteration of this step. Start with `-rc1`.
 2. Create a pre-release on GitHub using the script `tools/roll_release.sh`. This script automatically tag with the corresponding release number. The script requires that you have the STE||AR Group signing key.
 3. This step is not necessary for patch releases. Notify hpx-users@stellar-group.org of the availability of the release candidate. Ask users to test the candidate by checking out the release candidate tag.
 4. Allow at least a week for testing of the release candidate.
 - Use `git merge` when possible, and fall back to `git cherry-pick` when needed. For patch releases `git cherry-pick` is most likely your only choice if there have been significant unrelated changes on master since the previous release.
 - Go back to the first step when enough patches have been added.
 - If there are no more patches, continue to make the final release.
12. Update any occurrences of the latest stable release to refer to the version about to be released. For example, `quickstart.rst` contains instructions to check out the latest stable tag. Make sure that refers to the new version.
13. Add a new entry to the RPM changelog (`cmake/packaging/rpm/Changelog.txt`) with the new version number and a link to the corresponding changelog.
14. Change `HPX_VERSION_TAG` in `CMakeLists.txt` to an empty string.
15. Add the release date to the caption of the current “What’s New” section in the docs, and change the value of `HPX_VERSION_DATE` in `CMakeLists.txt`.

⁹⁵⁹ <http://github.com/UO-OACISS/xpress-apex>

16. Create a release on GitHub using the script `tools/roll_release.sh`. This script automatically tag the with the corresponding release number. The script requires that you have the STE||AR Group signing key.
17. Update the websites (hpx.stellar-group.org⁹⁶⁰ and stellar-group.org <<https://stellar-group.org>>). You can login on wordpress through *this page* <<https://hpx.stellar-group.org/wp-login.php>>. You can update the pages with the following:
 - Update links on the downloads page. Link to the release on GitHub.
 - Documentation links on the docs page (link to generated documentation on GitHub Pages). Follow the style of previous releases.
 - A new blog post announcing the release, which links to downloads and the “What’s New” section in the documentation (see previous releases for examples).
18. Merge release branch into master.
19. Post-release cleanup. Create a new pull request against master with the following changes:
 1. Modify the release procedure if necessary.
 2. Change `HPX_VERSION_TAG` in `CMakeLists.txt` back to `-trunk`.
 3. Increment `HPX_VERSION_MINOR` in `CMakeLists.txt`.
20. Update Vcpkg (<https://github.com/Microsoft/vcpkg>) to pull from latest release.
 - Update version number in `CONTROL`
 - Update tag and SHA512 to that of the new release
21. Update spack (<https://github.com/spack/spack>) with the latest HPX package.
 - Update version number in `hpx/package.py` and SHA256 to that of the new release
22. Announce the release on hpx-users@stellar-group.org, stellar@cct.lsu.edu, allcct@cct.lsu.edu, faculty@csc.lsu.edu, faculty@ece.lsu.edu, the *HPX* Slack channel, the STE||AR-GROUP Discord channel, our list of external collaborators, isocpp.org, reddit.com, [HPC Wire](https://hpcwire.com), [Inside HPC](https://insidehpc.com), [Heise Online](https://heiseonline.com), and a CCT press release.
23. Beer and pizza.

2.10.4 Testing *HPX*

To ensure correctness of *HPX*, we ship a large variety of unit and regression tests. The tests are driven by the `CTest`⁹⁶¹ tool and are executed automatically on each commit to the *HPX* [Github](https://github.com)⁹⁶² repository. In addition, it is encouraged to run the test suite manually to ensure proper operation on your target system. If a test fails for your platform, we highly recommend submitting an issue on our *HPX* [Issues](https://github.com/TheHPXProject/hpx/issues)⁹⁶³ tracker with detailed information about the target system.

⁹⁶⁰ <https://hpx.stellar-group.org>

⁹⁶¹ <https://gitlab.kitware.com/cmake/community/wikis/doc/ctest/Testing-With-CTest>

⁹⁶² <https://github.com/TheHPXProject/hpx/>

⁹⁶³ <https://github.com/TheHPXProject/hpx/issues>

Running tests manually

Running the tests manually is as easy as typing `make tests` && `make test`. This will build all tests and run them once the tests are built successfully. After the tests have been built, you can invoke separate tests with the help of the `ctest` command. You can list all available test targets using `make help | grep tests`. Please see the [CTest Documentation](#)⁹⁶⁴ for further details.

Running performance tests

We run performance tests on Piz Daint for each pull request using Jenkins. To run those performance tests locally or on Piz Daint, a script is provided under `tools/perftests_ci/local_run.sh` (to be run in the build directory specifying the *HPX* source directory as the argument to the script, default is `$HOME/projects/hpx_perftests_ci`).

Adding new performance tests

To add a new performance test, you need to wrap the portion of code to benchmark with `hpx::util::perftests_report`, passing the test name, the executor name and the function to time (can be a lambda). This facility is used to output the time results in a json format (format needed to compare the results and plot them). To effectively print them at the end of your test, call `hpx::util::perftests_print_times`. To see an example of use, see `future_overhead_report.cpp`. Finally, you can add the test to the CI report editing the `hpx_targets` variable for the executable name and the `hpx_test_options` variable for the corresponding options to use for the run in the performance test script `.jenkins/cscs-perftests/launch_perftests.sh`. And then run the `tools/perftests_ci/local_run.sh` script to get a reference json run (use the name of the test) to be added in the `tools/perftests_ci/perftest/references/daint_default` directory.

Issue tracker

If you stumble over a bug or missing feature in *HPX*, please submit an issue to our [HPX Issues](#)⁹⁶⁵ page. For more information on how to submit support requests or other means of getting in contact with the developers, please see the [Support Website](#)⁹⁶⁶ page.

Continuous testing

In addition to manual testing, we run automated tests on various platforms. We also run tests on all pull requests using both [\[github\]](#). You can see the dashboards [CDash HPX dashboard](#)⁹⁶⁷.

2.10.5 Using docker for development

Although it can often be useful to set up a local development environment with system-provided or self-built dependencies, [Docker](#)⁹⁶⁸ provides a convenient alternative to quickly get all the dependencies needed to start development of *HPX*. Our testing setup on [\[github\]](#) uses a docker image to run all tests.

⁹⁶⁴ <https://www.cmake.org/cmake/help/latest/manual/ctest.1.html>

⁹⁶⁵ <https://github.com/TheHPXProject/hpx/issues>

⁹⁶⁶ <https://stellar.cct.lsu.edu/support/>

⁹⁶⁷ <https://cdash.rostam.cct.lsu.edu/index.php?project=HPX>

⁹⁶⁸ <https://www.docker.com>

To get started you need to install [Docker](#)⁹⁶⁹ using whatever means is most convenient on your system. Once you have [Docker](#)⁹⁷⁰ installed, you can pull or directly run the docker image. The image is based on Debian and Clang, and can be found on [Docker Hub](#)⁹⁷¹. To start a container using the *HPX* build environment, run:

```
$ docker run --interactive --tty stellargroup/build_env:latest bash
```

You are now in an environment where all the *HPX* build and runtime dependencies are present. You can install additional packages according to your own needs. Please see the [Docker Documentation](#)⁹⁷² for more information on using [Docker](#)⁹⁷³.

Warning: All changes made within the container are lost when the container is closed. If you want files to persist (e.g., the *HPX* source tree) after closing the container, you can bind directories from the host system into the container (see [Docker Documentation \(Bind mounts\)](#)⁹⁷⁴).

2.10.6 Documentation

This documentation is built using [Sphinx](#)⁹⁷⁵, and an automatically generated API reference using [Doxygen](#)⁹⁷⁶ and [Breathe](#)⁹⁷⁷.

We always welcome suggestions on how to improve our documentation, as well as pull requests with corrections and additions.

Prerequisites

To build the *HPX* documentation, you need recent versions of the following packages:

- python3
- sphinx 4.5.0 (Python package)
- sphinx-book-theme (Python package)
- breathe 4.33.1 (Python package)
- doxygen
- sphinxcontrib-bibtex
- sphinx-copybutton

If the [Python](#)⁹⁷⁸ dependencies are not available through your system package manager, you can install them using the Python package manager `pip`:

```
pip install --user "sphinx<5" sphinx-book-theme breathe sphinxcontrib-bibtex sphinx-
↪ copybutton
```

You may need to set the following CMake variables to make sure CMake can find the required dependencies.

⁹⁶⁹ <https://www.docker.com>

⁹⁷⁰ <https://www.docker.com>

⁹⁷¹ https://hub.docker.com/r/stellargroup/build_env/

⁹⁷² <https://docs.docker.com/>

⁹⁷³ <https://www.docker.com>

⁹⁷⁴ <https://docs.docker.com/storage/bind-mounts/>

⁹⁷⁵ <http://www.sphinx-doc.org>

⁹⁷⁶ <https://www.doxygen.org>

⁹⁷⁷ <https://breathe.readthedocs.io/en/latest>

⁹⁷⁸ <https://www.python.org>

Doxygen_ROOT:PATH

Specifies where to look for the installation of the [Doxygen](https://www.doxygen.org)⁹⁷⁹ tool.

Sphinx_ROOT:PATH

Specifies where to look for the installation of the [Sphinx](http://www.sphinx-doc.org)⁹⁸⁰ tool.

Breathe_APIDOC_ROOT:PATH

Specifies where to look for the installation of the [Breathe](https://breathe.readthedocs.io/en/latest)⁹⁸¹ tool.

Building documentation

Enable building of the documentation by setting `HPX_WITH_DOCUMENTATION=ON` during [CMake](https://www.cmake.org)⁹⁸² configuration. To build the documentation, build the `docs` target using your build tool. The default output format is HTML documentation. You can choose alternative output formats (single-page HTML, PDF, and man) with the `HPX_WITH_DOCUMENTATION_OUTPUT_FORMATS` CMake option.

Note: If you add new source files to the Sphinx documentation, you have to run CMake again to have the files included in the build.

Style guide

The documentation is written using reStructuredText. These are the conventions used for formatting the documentation:

- Use, at most, 80 characters per line.
- Top-level headings use over- and underlines with `=`.
- Sub-headings use only underlines with characters in decreasing level of importance: `=`, `-` and `..`
- Use sentence case in headings.
- Refer to common terminology using `:term:`Component``.
- Indent content of directives (`.. directive::`) by three spaces.
- For C++ code samples at the end of paragraphs, use `::` and indent the code sample by 4 spaces.
 - For other languages (or if you don't want a colon at the end of the paragraph), use `.. code-block:: language` and indent by three spaces as with other directives.
- Use `.. list-table::` to wrap tables with a lot of text in cells.

⁹⁷⁹ <https://www.doxygen.org>

⁹⁸⁰ <http://www.sphinx-doc.org>

⁹⁸¹ <https://breathe.readthedocs.io/en/latest>

⁹⁸² <https://www.cmake.org>

API documentation

The source code is documented using Doxygen. If you add new API documentation either to existing or new source files, make sure that you add the documented source files to the `doxygen_dependencies` variable in `docs/CMakeLists.txt`.

Adding new API documentation

When you add new public APIs to *HPX*, follow these guidelines to ensure the documentation is correctly generated and easy to use.

Documenting headers

At the top of a new header file, include the standard Doxygen documentation block. For example, if the new feature is called *my_new_feature*, you should:

- List the name of the new file under `\file`.
- Specify the page name with `\page` (e.g., `hpx:my_new_feature`).
- Reference the existing header that will include your new file under `\headerfile`.

```
/// \file my_new_feature.hpp
/// \page hpx:my_new_feature
/// \headerfile hpx/existing_header_of_new_feature.hpp
```

Adding to the public API reference

For the documentation to appear in the generated API reference, add your new API to `docs/sphinx/api/public_api.rst` under the appropriate table (usually “Functions”) for the relevant header (for example, `hpx/existing_header_of_new_feature.hpp`).

Including examples

Do **not** inline examples in the API reference. If you want to provide an example of how to use the new API, there are two recommended options:

1. **Documentation page option** Edit an existing relevant manual page to explain the new functionality and include a code example. Use this option if you want to provide additional context about how the API works.
2. **Standalone examples option** Create a separate example under `docs/sphinx/examples/`. Examples in this directory are typically more comprehensive and executable, making them suitable for richer or more complex scenarios.

2.10.7 Module structure

This section explains the structure of an *HPX* module.

The tool `create_library_skeleton.py`⁹⁸³ can be used to generate a basic skeleton. To create a library skeleton, run the tool in the `libs` subdirectory with the module name as an argument:

```
$ ./create_library_skeleton <lib_name>
```

This creates a skeleton with the necessary files for an *HPX* module. It will not create any actual source files. The structure of this skeleton is as follows:

- `<lib_name>/`
 - `README.rst`
 - `CMakeLists.txt`
 - `cmake`
 - `docs/`
 - * `index.rst`
 - `examples/`
 - * `CMakeLists.txt`
 - `include/`
 - * `hpx/`
 - `<lib_name>`
 - `src/`
 - * `CMakeLists.txt`
 - `tests/`
 - * `CMakeLists.txt`
 - * `unit/`
 - `CMakeLists.txt`
 - * `regressions/`
 - `CMakeLists.txt`
 - * `performance/`
 - `CMakeLists.txt`

A `README.rst` should be always included which explains the basic purpose of the library and a link to the generated documentation.

A main `CMakeLists.txt` is created in the root directory of the module. By default it contains a call to `add_hpx_module` which takes care of most of the boilerplate required for a module. You only need to fill in the source and header files in most cases.

`add_hpx_module` requires a module name. Optional flags are:

Optional single-value arguments are:

- `INSTALL_BINARIES`: Install the resulting library.

⁹⁸³ https://github.com/TheHPXProject/hpx/blob/master/libs/create_module_skeleton.py

Optional multi-value arguments-are:

- **SOURCES**: List of source files.
- **HEADERS**: List of header files.
- **COMPAT_HEADERS**: List of compatibility header files.
- **DEPENDENCIES**: Libraries that this module depends on, such as other modules.
- **CMAKE_SUBDIRS**: List of subdirectories to add to the module.

The `include` directory should contain only headers that other libraries need. For each of those headers, an automatic header test to check for self containment will be generated. Private headers should be placed under the `src` directory. This allows for clear separation. The `cmake` subdirectory may include additional [CMake](https://www.cmake.org)⁹⁸⁴ scripts needed to generate the respective build configurations.

Compatibility headers (forwarding headers for headers whose location is changed when creating a module, if moving them from the main library) should be placed in an `include_compatibility` directory. This directory is not created by default.

Documentation is placed in the `docs` folder. A empty skeleton for the index is created, which is picked up by the main build system and will be part of the generated documentation. Each header inside the `include` directory will automatically be processed by Doxygen and included into the documentation.

Tests are placed in suitable subdirectories of `tests`.

When in doubt, consult existing modules for examples on how to structure the module.

Finding circular dependencies

Our CI will perform a check to see if there are circular dependencies between modules. In cases where it's not clear what is causing the circular dependency, running the `cpp-dependencies`⁹⁸⁵ tool manually can be helpful. It can give you detailed information on exactly which files are causing the circular dependency. If you do not have the `cpp-dependencies` tool already installed, one way of obtaining it is by using our docker image. This way you will have exactly the same environment as on the CI. See *Using docker for development* for details on how to use the docker image.

To produce the graph produced by CI run the following command (`HPX_SOURCE` is assumed to hold the path to the *HPX* source directory):

```
$ cpp-dependencies --dir $HPX_SOURCE/libs --graph-cycles circular_dependencies.dot
```

This will produce a `dot` file in the current directory. You can inspect this manually with a text editor. You can also convert this to an image if you have `graphviz` installed:

```
$ dot circular_dependencies.dot -Tsvg -o circular_dependencies.svg
```

This produces an `svg` file in the current directory which shows the circular dependencies. Note that if there are no cycles the image will be empty.

You can use `cpp-dependencies` to print the include paths between two modules.

```
$ cpp-dependencies --dir $HPX_SOURCE/libs --shortest <from> <to>
```

prints all possible paths from the module `<from>` to the module `<to>`. For example, as most modules depend on `config`, the following should give you a long list of paths from `algorithms` to `config`:

⁹⁸⁴ <https://www.cmake.org>

⁹⁸⁵ <https://github.com/tomtom-international/cpp-dependencies>

```
$ cpp-dependencies --dir $HPX_SOURCE/libs --shortest algorithms config
```

The following should report that it can't find a path between the two modules:

```
$ cpp-dependencies --dir $HPX_SOURCE/libs --shortest config algorithms
```

2.11 Releases

2.11.1 List of supported releases

HPX V2.0.0 (TBD)

General changes

Breaking changes

Closed issues

Closed pull requests

HPX V1.11.0 (Jun 30, 2025)

This release is the last version of HPX that supports C++17. Future versions of HPX will require compilation using C++20 or above.

General changes

- Added synchronous versions of all collective operations. Added global predefined communicator objects that are accessible through new APIs: `hpx::collectives::get_world_communicator()` refers to all localities and `hpx::collectives::get_local_communicator()` refers to all threads on the calling locality. We unified the interfaces of the different communicator objects.
- Added `hpx::experimental::run_on_all` allowing to run a given function (possibly concurrently) using a given execution policy.
- Added a helper object `hpx::runtime_manager` that simplifies the initialization of HPX without needing to modify the main function of the application.
- We improved the compatibility with various accelerator frameworks (SYCL, OneAPI).
- Applied build system changes that allow building HPX without any prerequisites. This requires to pass `-DHPX_WITH_FETCH_HWLOC=On` and `-DHPX_WITH_FETCH_BOOST=On` to the [CMake](https://www.cmake.org)⁹⁸⁶ configuration.
- We have performed a lot of code cleanup and refactoring to improve the overall code quality and decrease compile times.
- Added the `hpx::contains` and `hpx::contains_subrange` parallel algorithms.
- Adapted many of HPX' parallel algorithms to be usable with senders/receivers.

⁹⁸⁶ <https://www.cmake.org>

Breaking changes

- We have moved most of the APIs that were defined in the namespace `hpx::parallel::execution` to the namespace `hpx::execution::experimental`. It was not possible to add compatibility facilities that will allow to continue using the old APIs, applications will have to be changed in order to continue functioning correctly.
- The CMake configuration parameter `HPX_WITH_RUN_MAIN_EVERYWHERE` is now deprecated and will be removed in the future. Use the preprocessor macro `HPX_HAVE_RUN_MAIN_EVERYWHERE` on a target-by-target case instead.
- Removed the dysfunctional libfabric parcelport.
- Removed features that were long deprecated (starting V1.8): - `hpx::flush`, `hpx::endl`, `hpx::async_flush`, `hpx::async_endl` - Various enumerator types are now only available as `class enum` requiring explicit scoped use of the enumerator values - Various non-conforming overloads of parallel algorithms - `hpx::for_loop` and friends (now only available as

`hpx::experimental::for_loop`)

- `hpx::parallel::induction` is now only available as `hpx::experimental::induction`
- `hpx::parallel::reduction` and friends are now only available as `hpx::experimental::reduction`
- `hpx::assertion::source_location` is now only available as `hpx::source_location`
- `hpx::lcos::split_future` is now only available as `hpx::split_future`
- `hpx::lcos::wait` and friends have been removed altogether
- `hpx::lcos::wait_any` and friends are now only available as `hpx::wait_any`
- `hpx::lcos::wait_some` and friends are now only available as `hpx::wait_some`
- `hpx::lcos::wait_each` and friends are now only available as `hpx::wait_each`
- `hpx::lcos::wait_all` and friends are now only available as `hpx::wait_all`
- `hpx::lcos::when_all` and friends are now only available as `hpx::when_all`
- `hpx::lcos::when_any` and friends are now only available as `hpx::when_any`
- `hpx::lcos::when_each` and friends are now only available as `hpx::when_each`
- `hpx::lcos::when_some` and friends are now only available as `hpx::when_some`
- `hpx::util::optional` and related facilities are now only available as `hpx::optional`
- `hpx::util::bind` and related facilities are now only available as `hpx::bind`
- `hpx::util::function` and friends are now only available as `hpx::function`
- `hpx::traits::is_bound_action` and related facilities are now only available as `hpx::is_bound_action`
- `hpx::traits::is_bind_expression` and related facilities are now only available as `hpx::is_bind_expression`
- `hpx::traits::is_placeholder` and related facilities are now only available as `hpx::is_placeholder`
- `hpx::lcos::future` and related facilities are now only available as `hpx::future`
- `hpx::memory::intrusive_ptr` is now only available as `hpx::intrusive_ptr`
- `hpx::lcos::local::barrier` is now only available as `hpx::barrier`
- `hpx::lcos::barrier` is now only available as `hpx::distributed::barrier`

- `hpx::lcos::local::cpp20_binary_semaphore` is now only available as `hpx::detail::binary_semaphore`
- `hpx::lcos::local::condition_variable` and friends are now only available as `hpx::condition_variable`
- `hpx::lcos::local::counting_semaphore` and friends are now only available as `hpx::counting_semaphore`
- `hpx::lcos::local::cpp20_latch` and is now only available as `hpx::latch`
- `hpx::lcos::latch` and is now only available as `hpx::distributed::latch`
- `hpx::lcos::local::upgrade_lock` and friends are now only available as `hpx::upgrade_lock`
- `hpx::lcos::local::mutex` and friends are now only available as `hpx::mutex`
- `hpx::lcos::local::spinlock` and friends are now only available as `hpx::spinlock`
- `hpx::lcos::local::call_once` and friends are now only available as `hpx::call_once`
- `hpx::util::annotated_function` and is now only available as `hpx::annotated_function`
- `hpx::components::abstract_simple_component_base` and is now only available as `hpx::components::abstract_component_base`
- `hpx::naming::id_type` and is now only available as `hpx::id_type`

Closed issues

- [Issue #6699⁹⁸⁷](#) - Catch lower-level runtime error
- [Issue #6696⁹⁸⁸](#) - HPX master breaks with Kokkos
- [Issue #6691⁹⁸⁹](#) - `minimum_category` doesn't work with custom iterator categories
- [Issue #6681⁹⁹⁰](#) - build break - missing `';`
- [Issue #6658⁹⁹¹](#) - CMake error upon building HPX manually
- [Issue #6648⁹⁹²](#) - Asio V1.34 deprecates `io_context::work`
- [Issue #6640⁹⁹³](#) - `iterator_facade` doesn't work with custom iterator categories
- [Issue #6636⁹⁹⁴](#) - problem with `hpx::collectives::exclusive_scan`
- [Issue #6623⁹⁹⁵](#) - HPX serialization error with `std::vector<std::vector<std::vector<float>>>`
- [Issue #6616⁹⁹⁶](#) - Add flux support to HPX to run on El Cap
- [Issue #6615⁹⁹⁷](#) - Too many fails test after installed hpx
- [Issue #6605⁹⁹⁸](#) - Partitionend vector copy constructor is broken

⁹⁸⁷ <https://github.com/TheHPXProject/hpx/issues/6699>

⁹⁸⁸ <https://github.com/TheHPXProject/hpx/issues/6696>

⁹⁸⁹ <https://github.com/TheHPXProject/hpx/issues/6691>

⁹⁹⁰ <https://github.com/TheHPXProject/hpx/issues/6681>

⁹⁹¹ <https://github.com/TheHPXProject/hpx/issues/6658>

⁹⁹² <https://github.com/TheHPXProject/hpx/issues/6648>

⁹⁹³ <https://github.com/TheHPXProject/hpx/issues/6640>

⁹⁹⁴ <https://github.com/TheHPXProject/hpx/issues/6636>

⁹⁹⁵ <https://github.com/TheHPXProject/hpx/issues/6623>

⁹⁹⁶ <https://github.com/TheHPXProject/hpx/issues/6616>

⁹⁹⁷ <https://github.com/TheHPXProject/hpx/issues/6615>

⁹⁹⁸ <https://github.com/TheHPXProject/hpx/issues/6605>

- [Issue #6586⁹⁹⁹](#) - Bullet points in quick start/installing HPX section in documentation incorrectly rendered
- [Issue #6563¹⁰⁰⁰](#) - Compilation issues on Grace Hopper
- [Issue #6544¹⁰⁰¹](#) - Errors in Public Distributed Api for all_to_all and gather_there
- [Issue #6519¹⁰⁰²](#) - Option `-hpx:queuing=local-priority-lifo` is not configured
- [Issue #6501¹⁰⁰³](#) - HPX 1.10 Failed Linking CXX executable for arm64-osx
- [Issue #5728¹⁰⁰⁴](#) - Add optional `fetch_content` support for needed Boost libraries

Closed pull requests

- [PR #6716¹⁰⁰⁵](#) - Fixing some of the reported linker warnings
- [PR #6705¹⁰⁰⁶](#) - Adding gcc/15 to jenkins
- [PR #6701¹⁰⁰⁷](#) - Attempting to fix shutdown hang on `exception_info`
- [PR #6698¹⁰⁰⁸](#) - Making sure `.hpp.in` files are not being installed
- [PR #6697¹⁰⁰⁹](#) - Minor docs fix
- [PR #6695¹⁰¹⁰](#) - Adding missing `‘;’`
- [PR #6693¹⁰¹¹](#) - Adding llvm/19 and 20 and cmake/4 Jenkins
- [PR #6692¹⁰¹²](#) - Better implementation of `minimal_category`
- [PR #6690¹⁰¹³](#) - Fixing bad `#include` in example
- [PR #6689¹⁰¹⁴](#) - Fix unreachable code warning in `wait_all`
- [PR #6687¹⁰¹⁵](#) - lci pp: change default `ndevices=2` and `progress_type=worker`; improve document
- [PR #6686¹⁰¹⁶](#) - lci pp: upgrade LCI autofetch target to 1.7.9
- [PR #6685¹⁰¹⁷](#) - Improve `run_on_all` implementation and tests
- [PR #6683¹⁰¹⁸](#) - Fix bad element comparison for `reduce_by_key`
- [PR #6682¹⁰¹⁹](#) - Add C++23 `std::generator` equivalence test and fix missing semicolon
- [PR #6680¹⁰²⁰](#) - Add oneapi device init workaround

⁹⁹⁹ <https://github.com/TheHPXProject/hpx/issues/6586>

¹⁰⁰⁰ <https://github.com/TheHPXProject/hpx/issues/6563>

¹⁰⁰¹ <https://github.com/TheHPXProject/hpx/issues/6544>

¹⁰⁰² <https://github.com/TheHPXProject/hpx/issues/6519>

¹⁰⁰³ <https://github.com/TheHPXProject/hpx/issues/6501>

¹⁰⁰⁴ <https://github.com/TheHPXProject/hpx/issues/5728>

¹⁰⁰⁵ <https://github.com/TheHPXProject/hpx/pull/6716>

¹⁰⁰⁶ <https://github.com/TheHPXProject/hpx/pull/6705>

¹⁰⁰⁷ <https://github.com/TheHPXProject/hpx/pull/6701>

¹⁰⁰⁸ <https://github.com/TheHPXProject/hpx/pull/6698>

¹⁰⁰⁹ <https://github.com/TheHPXProject/hpx/pull/6697>

¹⁰¹⁰ <https://github.com/TheHPXProject/hpx/pull/6695>

¹⁰¹¹ <https://github.com/TheHPXProject/hpx/pull/6693>

¹⁰¹² <https://github.com/TheHPXProject/hpx/pull/6692>

¹⁰¹³ <https://github.com/TheHPXProject/hpx/pull/6690>

¹⁰¹⁴ <https://github.com/TheHPXProject/hpx/pull/6689>

¹⁰¹⁵ <https://github.com/TheHPXProject/hpx/pull/6687>

¹⁰¹⁶ <https://github.com/TheHPXProject/hpx/pull/6686>

¹⁰¹⁷ <https://github.com/TheHPXProject/hpx/pull/6685>

¹⁰¹⁸ <https://github.com/TheHPXProject/hpx/pull/6683>

¹⁰¹⁹ <https://github.com/TheHPXProject/hpx/pull/6682>

¹⁰²⁰ <https://github.com/TheHPXProject/hpx/pull/6680>

- PR #6679¹⁰²¹ - Fix sycl deprecations
- PR #6678¹⁰²² - Fix oneapi overloads
- PR #6677¹⁰²³ - Offer a runtime manager object
- PR #6676¹⁰²⁴ - Mention the HPX book
- PR #6675¹⁰²⁵ - Bump required version of JSON library
- PR #6674¹⁰²⁶ - Issue 6631
- PR #6673¹⁰²⁷ - Fix: FindTBB.cmake cannot find correct TBB library. #6504
- PR #6672¹⁰²⁸ - Update modules.rst
- PR #6670¹⁰²⁹ - Add base template template param to execution_policy
- PR #6669¹⁰³⁰ - Add execution policy support to run_on_all
- PR #6667¹⁰³¹ - Making sure bound threads are rescheduled on their original core
- PR #6666¹⁰³² - Improve documentation for reduction operations
- PR #6664¹⁰³³ - Fix CMake template when fetching Boost
- PR #6663¹⁰³⁴ - More run_on_all overloads
- PR #6662¹⁰³⁵ - Fix “unary minus operator applied to unsigned type” warning
- PR #6661¹⁰³⁶ - Adding simple experimental::run_on_all
- PR #6659¹⁰³⁷ - fix(reduce): Initialize accumulator with init instead of first element
- PR #6656¹⁰³⁸ - Add missing channel_communicator::get_info
- PR #6652¹⁰³⁹ - Adding channel-based ping-pong example
- PR #6650¹⁰⁴⁰ - Adding constructor overloads to partitioned_vector
- PR #6649¹⁰⁴¹ - Remove the use of deprecated asio::io_context::work
- PR #6645¹⁰⁴² - Fixing collectives::exclusive_scan
- PR #6644¹⁰⁴³ - Update result_type in set_union.hpp

¹⁰²¹ <https://github.com/TheHPXProject/hpx/pull/6679>

¹⁰²² <https://github.com/TheHPXProject/hpx/pull/6678>

¹⁰²³ <https://github.com/TheHPXProject/hpx/pull/6677>

¹⁰²⁴ <https://github.com/TheHPXProject/hpx/pull/6676>

¹⁰²⁵ <https://github.com/TheHPXProject/hpx/pull/6675>

¹⁰²⁶ <https://github.com/TheHPXProject/hpx/pull/6674>

¹⁰²⁷ <https://github.com/TheHPXProject/hpx/pull/6673>

¹⁰²⁸ <https://github.com/TheHPXProject/hpx/pull/6672>

¹⁰²⁹ <https://github.com/TheHPXProject/hpx/pull/6670>

¹⁰³⁰ <https://github.com/TheHPXProject/hpx/pull/6669>

¹⁰³¹ <https://github.com/TheHPXProject/hpx/pull/6667>

¹⁰³² <https://github.com/TheHPXProject/hpx/pull/6666>

¹⁰³³ <https://github.com/TheHPXProject/hpx/pull/6664>

¹⁰³⁴ <https://github.com/TheHPXProject/hpx/pull/6663>

¹⁰³⁵ <https://github.com/TheHPXProject/hpx/pull/6662>

¹⁰³⁶ <https://github.com/TheHPXProject/hpx/pull/6661>

¹⁰³⁷ <https://github.com/TheHPXProject/hpx/pull/6659>

¹⁰³⁸ <https://github.com/TheHPXProject/hpx/pull/6656>

¹⁰³⁹ <https://github.com/TheHPXProject/hpx/pull/6652>

¹⁰⁴⁰ <https://github.com/TheHPXProject/hpx/pull/6650>

¹⁰⁴¹ <https://github.com/TheHPXProject/hpx/pull/6649>

¹⁰⁴² <https://github.com/TheHPXProject/hpx/pull/6645>

¹⁰⁴³ <https://github.com/TheHPXProject/hpx/pull/6644>

- PR #6643¹⁰⁴⁴ - Update result_type in set_union.hpp
- PR #6642¹⁰⁴⁵ - Allowing to use custom iterator tags with iterator_facade
- PR #6641¹⁰⁴⁶ - Allowing for zip-iterator to refer to sequences of different length
- PR #6639¹⁰⁴⁷ - docs: Fix spelling in example dictionary
- PR #6638¹⁰⁴⁸ - Update set_union.hpp
- PR #6637¹⁰⁴⁹ - lci/mpi pp: fix the case when non-zero-copy data is larger than INT_MAX
- PR #6635¹⁰⁵⁰ - Adding simplified reduction overload
- PR #6634¹⁰⁵¹ - Fixed issue 6634: Unqualified calls to insertion_sort
- PR #6633¹⁰⁵² - Increase timeouts for CircleCI tests
- PR #6630¹⁰⁵³ - Fix CPUId test
- PR #6628¹⁰⁵⁴ - Link aclocal with aclocal-1.16 as hwloc asks for it
- PR #6626¹⁰⁵⁵ - Fixing MPI parcel port issue exposed by #6623
- PR #6622¹⁰⁵⁶ - Newbranch:HPX-Based Task Scheduler with CUDA-Quantum Integration & Benchmarking
- PR #6621¹⁰⁵⁷ - HPX-Based Task Scheduler with CUDA-Quantum Integration & Benchmarking
- PR #6620¹⁰⁵⁸ - new test: very big tchunk
- PR #6619¹⁰⁵⁹ - mpi pp: fix transmission chunk send
- PR #6617¹⁰⁶⁰ - Adding support for the Flux job scheduling environment
- PR #6614¹⁰⁶¹ - Fix fallback to module mode for CMake finding Boost
- PR #6613¹⁰⁶² - Fix partitioned_vector_handle_values test
- PR #6612¹⁰⁶³ - Fixing naming convention for pp constant
- PR #6611¹⁰⁶⁴ - Fix Hwloc fetch content
- PR #6610¹⁰⁶⁵ - Add docs for synchronous collective operations
- PR #6609¹⁰⁶⁶ - Update perfest CI reference measurements

¹⁰⁴⁴ <https://github.com/TheHPXProject/hpx/pull/6643>

¹⁰⁴⁵ <https://github.com/TheHPXProject/hpx/pull/6642>

¹⁰⁴⁶ <https://github.com/TheHPXProject/hpx/pull/6641>

¹⁰⁴⁷ <https://github.com/TheHPXProject/hpx/pull/6639>

¹⁰⁴⁸ <https://github.com/TheHPXProject/hpx/pull/6638>

¹⁰⁴⁹ <https://github.com/TheHPXProject/hpx/pull/6637>

¹⁰⁵⁰ <https://github.com/TheHPXProject/hpx/pull/6635>

¹⁰⁵¹ <https://github.com/TheHPXProject/hpx/pull/6634>

¹⁰⁵² <https://github.com/TheHPXProject/hpx/pull/6633>

¹⁰⁵³ <https://github.com/TheHPXProject/hpx/pull/6630>

¹⁰⁵⁴ <https://github.com/TheHPXProject/hpx/pull/6628>

¹⁰⁵⁵ <https://github.com/TheHPXProject/hpx/pull/6626>

¹⁰⁵⁶ <https://github.com/TheHPXProject/hpx/pull/6622>

¹⁰⁵⁷ <https://github.com/TheHPXProject/hpx/pull/6621>

¹⁰⁵⁸ <https://github.com/TheHPXProject/hpx/pull/6620>

¹⁰⁵⁹ <https://github.com/TheHPXProject/hpx/pull/6619>

¹⁰⁶⁰ <https://github.com/TheHPXProject/hpx/pull/6617>

¹⁰⁶¹ <https://github.com/TheHPXProject/hpx/pull/6614>

¹⁰⁶² <https://github.com/TheHPXProject/hpx/pull/6613>

¹⁰⁶³ <https://github.com/TheHPXProject/hpx/pull/6612>

¹⁰⁶⁴ <https://github.com/TheHPXProject/hpx/pull/6611>

¹⁰⁶⁵ <https://github.com/TheHPXProject/hpx/pull/6610>

¹⁰⁶⁶ <https://github.com/TheHPXProject/hpx/pull/6609>

- PR #6608¹⁰⁶⁷ - Partially support data parallel for_loop
- PR #6607¹⁰⁶⁸ - Cleaning up copy_component facility
- PR #6606¹⁰⁶⁹ - Making sure copy_component creates a new gid
- PR #6600¹⁰⁷⁰ - Fixing sync collectives
- PR #6599¹⁰⁷¹ - Make HPX_HAVE_RUN_MAIN_EVERYWHERE application specific
- PR #6598¹⁰⁷² - Adding synchronous collective operations
- PR #6596¹⁰⁷³ - Minor fixes and optimizations
- PR #6595¹⁰⁷⁴ - Rfa parallel
- PR #6594¹⁰⁷⁵ - Move get_stack_ptr to source
- PR #6593¹⁰⁷⁶ - Fix outdated documentation and missing flags
- PR #6592¹⁰⁷⁷ - HPX_HAVE_THREADS_GET_STACK_POINTER to match builtin_frame_address feature test
- PR #6591¹⁰⁷⁸ - Feature test for __builtin_frame_address
- PR #6590¹⁰⁷⁹ - Add device guard for noexcept
- PR #6587¹⁰⁸⁰ - Fix bullet points in Quickstart
- PR #6585¹⁰⁸¹ - Fixed escape characters format to handle warning due to misinterpretation of syntax
- PR #6583¹⁰⁸² - Execute feature test for at_quick_exit
- PR #6582¹⁰⁸³ - Accommodate for CircleCI reduce available number of cores to two
- PR #6581¹⁰⁸⁴ - Attempting to work around a Boost.Spirit problem
- PR #6580¹⁰⁸⁵ - mpi pp: fix messages larger than INT_MAX
- PR #6578¹⁰⁸⁶ - Remove leftovers from libfabric parcelport
- PR #6577¹⁰⁸⁷ - Download Boost from their own archives, not from Sourceforge
- PR #6576¹⁰⁸⁸ - Fix CMake warning issued since CMake V3.30
- PR #6575¹⁰⁸⁹ - Replace previously downloaded CDash conv.xml with local version

¹⁰⁶⁷ <https://github.com/TheHPXProject/hpx/pull/6608>

¹⁰⁶⁸ <https://github.com/TheHPXProject/hpx/pull/6607>

¹⁰⁶⁹ <https://github.com/TheHPXProject/hpx/pull/6606>

¹⁰⁷⁰ <https://github.com/TheHPXProject/hpx/pull/6600>

¹⁰⁷¹ <https://github.com/TheHPXProject/hpx/pull/6599>

¹⁰⁷² <https://github.com/TheHPXProject/hpx/pull/6598>

¹⁰⁷³ <https://github.com/TheHPXProject/hpx/pull/6596>

¹⁰⁷⁴ <https://github.com/TheHPXProject/hpx/pull/6595>

¹⁰⁷⁵ <https://github.com/TheHPXProject/hpx/pull/6594>

¹⁰⁷⁶ <https://github.com/TheHPXProject/hpx/pull/6593>

¹⁰⁷⁷ <https://github.com/TheHPXProject/hpx/pull/6592>

¹⁰⁷⁸ <https://github.com/TheHPXProject/hpx/pull/6591>

¹⁰⁷⁹ <https://github.com/TheHPXProject/hpx/pull/6590>

¹⁰⁸⁰ <https://github.com/TheHPXProject/hpx/pull/6587>

¹⁰⁸¹ <https://github.com/TheHPXProject/hpx/pull/6585>

¹⁰⁸² <https://github.com/TheHPXProject/hpx/pull/6583>

¹⁰⁸³ <https://github.com/TheHPXProject/hpx/pull/6582>

¹⁰⁸⁴ <https://github.com/TheHPXProject/hpx/pull/6581>

¹⁰⁸⁵ <https://github.com/TheHPXProject/hpx/pull/6580>

¹⁰⁸⁶ <https://github.com/TheHPXProject/hpx/pull/6578>

¹⁰⁸⁷ <https://github.com/TheHPXProject/hpx/pull/6577>

¹⁰⁸⁸ <https://github.com/TheHPXProject/hpx/pull/6576>

¹⁰⁸⁹ <https://github.com/TheHPXProject/hpx/pull/6575>

- PR #6570¹⁰⁹⁰ - Update exception_list.hpp
- PR #6569¹⁰⁹¹ - Update exception_list.hpp
- PR #6567¹⁰⁹² - Fix vectorization error on copy algorithm
- PR #6566¹⁰⁹³ - lci pp: fix messages larger than INT_MAX
- PR #6565¹⁰⁹⁴ - Moving most of APIs from hpx::parallel::execution to hpx::execution::experimental
- PR #6564¹⁰⁹⁵ - Remove superfluous HPX_MOVE()
- PR #6562¹⁰⁹⁶ - Fix doc return type of broadcast_to
- PR #6560¹⁰⁹⁷ - Fixes for bit_cast on 32bit systems
- PR #6559¹⁰⁹⁸ - Making sure that all parcellport counters are unavailable if no networking is needed or configured
- PR #6558¹⁰⁹⁹ - Remove CSCS CI's
- PR #6556¹¹⁰⁰ - Set copyright year in generated files
- PR #6553¹¹⁰¹ - Fix omp vectorization pragma errors
- PR #6551¹¹⁰² - Update building_hpx.rst
- PR #6550¹¹⁰³ - Partitioned vector updates
- PR #6549¹¹⁰⁴ - Fix CMake conditionals checking ENV variables
- PR #6548¹¹⁰⁵ - Update CONTRIBUTING.md
- PR #6546¹¹⁰⁶ - Fix incorrect signature of distributed API functions
- PR #6543¹¹⁰⁷ - Throwing an exception derived from std::bad_alloc on OOM conditions
- PR #6539¹¹⁰⁸ - Use thread-safe cache in thread_local_caching_allocator
- PR #6537¹¹⁰⁹ - Update README.rst
- PR #6531¹¹¹⁰ - More fixes for the Boost package
- PR #6527¹¹¹¹ - Improve the LCI parcellport documentation
- PR #6525¹¹¹² - Addressing cmake warnings issued starting V3.30

¹⁰⁹⁰ <https://github.com/TheHPXProject/hpx/pull/6570>

¹⁰⁹¹ <https://github.com/TheHPXProject/hpx/pull/6569>

¹⁰⁹² <https://github.com/TheHPXProject/hpx/pull/6567>

¹⁰⁹³ <https://github.com/TheHPXProject/hpx/pull/6566>

¹⁰⁹⁴ <https://github.com/TheHPXProject/hpx/pull/6565>

¹⁰⁹⁵ <https://github.com/TheHPXProject/hpx/pull/6564>

¹⁰⁹⁶ <https://github.com/TheHPXProject/hpx/pull/6562>

¹⁰⁹⁷ <https://github.com/TheHPXProject/hpx/pull/6560>

¹⁰⁹⁸ <https://github.com/TheHPXProject/hpx/pull/6559>

¹⁰⁹⁹ <https://github.com/TheHPXProject/hpx/pull/6558>

¹¹⁰⁰ <https://github.com/TheHPXProject/hpx/pull/6556>

¹¹⁰¹ <https://github.com/TheHPXProject/hpx/pull/6553>

¹¹⁰² <https://github.com/TheHPXProject/hpx/pull/6551>

¹¹⁰³ <https://github.com/TheHPXProject/hpx/pull/6550>

¹¹⁰⁴ <https://github.com/TheHPXProject/hpx/pull/6549>

¹¹⁰⁵ <https://github.com/TheHPXProject/hpx/pull/6548>

¹¹⁰⁶ <https://github.com/TheHPXProject/hpx/pull/6546>

¹¹⁰⁷ <https://github.com/TheHPXProject/hpx/pull/6543>

¹¹⁰⁸ <https://github.com/TheHPXProject/hpx/pull/6539>

¹¹⁰⁹ <https://github.com/TheHPXProject/hpx/pull/6537>

¹¹¹⁰ <https://github.com/TheHPXProject/hpx/pull/6531>

¹¹¹¹ <https://github.com/TheHPXProject/hpx/pull/6527>

¹¹¹² <https://github.com/TheHPXProject/hpx/pull/6525>

- PR #6522¹¹¹³ - Fixing distance test
- PR #6520¹¹¹⁴ - Adding optional handshakes to acknowledge the received data
- PR #6518¹¹¹⁵ - Make sure that `-hpx:ini` log settings take effect
- PR #6512¹¹¹⁶ - Minor cleanup of `future_data`
- PR #6510¹¹¹⁷ - Include Boost as CMake subproject
- PR #6509¹¹¹⁸ - Add components documentation
- PR #6508¹¹¹⁹ - Fix typo: `s/unititialized/uninitialized/`
- PR #6507¹¹²⁰ - Update LSU Jenkins libraries to match Rostam 3.0 with RHEL9
- PR #6503¹¹²¹ - Fix 2 tests on FreeBSD by initializing `freebsd_envron`
- PR #6499¹¹²² - Fix crash in `get_executable_filename` on FreeBSD
- PR #6498¹¹²³ - Avoid rewriting `defines.hpp`
- PR #6497¹¹²⁴ - Contains and `contains_subrange` parallel algorithm implementation GSOC 2024
- PR #6496¹¹²⁵ - Prevent usage of CMake `try_run` on crosscompiling
- PR #6494¹¹²⁶ - Add unit test cases and fixes for the S/R versions of the parallel algorithms
- PR #6487¹¹²⁷ - Fixing security vulnerabilities reported by MSVC security checks
- PR #6486¹¹²⁸ - Create `codeql.yml`
- PR #6474¹¹²⁹ - Remove remnants of `libfabric` `parcelport`
- PR #6473¹¹³⁰ - Add documentation for distributed implementations of `post`, `async`, `sync` and `dataflow`
- PR #6471¹¹³¹ - Add `distance.cpp` test in CMake
- PR #6468¹¹³² - Small vector relocation
- PR #6448¹¹³³ - Standardising Benchmarks, with support for `nanobench` as an option for its backend
- PR #6365¹¹³⁴ - Release V1.10.0
- PR #6089¹¹³⁵ - Implementing `p2079`

¹¹¹³ <https://github.com/TheHPXProject/hpx/pull/6522>

¹¹¹⁴ <https://github.com/TheHPXProject/hpx/pull/6520>

¹¹¹⁵ <https://github.com/TheHPXProject/hpx/pull/6518>

¹¹¹⁶ <https://github.com/TheHPXProject/hpx/pull/6512>

¹¹¹⁷ <https://github.com/TheHPXProject/hpx/pull/6510>

¹¹¹⁸ <https://github.com/TheHPXProject/hpx/pull/6509>

¹¹¹⁹ <https://github.com/TheHPXProject/hpx/pull/6508>

¹¹²⁰ <https://github.com/TheHPXProject/hpx/pull/6507>

¹¹²¹ <https://github.com/TheHPXProject/hpx/pull/6503>

¹¹²² <https://github.com/TheHPXProject/hpx/pull/6499>

¹¹²³ <https://github.com/TheHPXProject/hpx/pull/6498>

¹¹²⁴ <https://github.com/TheHPXProject/hpx/pull/6497>

¹¹²⁵ <https://github.com/TheHPXProject/hpx/pull/6496>

¹¹²⁶ <https://github.com/TheHPXProject/hpx/pull/6494>

¹¹²⁷ <https://github.com/TheHPXProject/hpx/pull/6487>

¹¹²⁸ <https://github.com/TheHPXProject/hpx/pull/6486>

¹¹²⁹ <https://github.com/TheHPXProject/hpx/pull/6474>

¹¹³⁰ <https://github.com/TheHPXProject/hpx/pull/6473>

¹¹³¹ <https://github.com/TheHPXProject/hpx/pull/6471>

¹¹³² <https://github.com/TheHPXProject/hpx/pull/6468>

¹¹³³ <https://github.com/TheHPXProject/hpx/pull/6448>

¹¹³⁴ <https://github.com/TheHPXProject/hpx/pull/6365>

¹¹³⁵ <https://github.com/TheHPXProject/hpx/pull/6089>

2.11.2 List of older releases

HPX V1.10.0 (May 29, 2024)

General changes

- The HPX documentation has seen a major overhaul for this release. We finished documenting the public local HPX API, we have added migration guides from widely used parallelization platforms to HPX (OpenMP, TBB, and MPI).
- We have added facilities enabling optimizations for trivially-relocatable types (see [P1144](#)¹¹³⁶ for more details).
- We have added (and use) the `scope_xxx` helper facilities as specified by the C++ library fundamentals TS v3 (see: [N4948](#)¹¹³⁷).
- We have added configuration options that allow to build HPX without pre-installing any prerequisites. Use `HPX_WITH_FETCH_HWLOC=On` to have [Portable Hardware Locality \(HWLOC\)](#)¹¹³⁸ installed for you. Similarly, setting `HPX_WITH_FETCH_BOOST=On` during configuration time will install the necessary [Boost](#)¹¹³⁹ libraries (currently V1.84.0).
- We have performed a lot of code cleanup and refactoring to improve the overall code quality and decrease compile times.
- The collective operations APIs have seen an unification, we have fixed issues and performance problems for the collectives.
- The HPX executors have seen a streamlining and some consistency changes. We have applied many performance improvements to the executor implementations that directly positively impact the performance of our parallel algorithms.
- We have added a new parcelport allowing to use Gasnet as a communication platform.
- We have added optimizations to various parcelports improving overall communication performance. This includes - amongst other things - send immediate optimizations and receiver-side zero-copy optimizations.
- Futures will now execute the associated task eagerly and inline on any wait operation if the task has not started running yet. This feature can be enabled using the `HPX_COROUTINES_WITH_THREAD_SCHEDULE_HINT_RUNS_AS_CHILD=On` configuration setting (which is *Off* by default).
- We have enabled using json files to supply configuration information through the command line. This feature can be enabled with the configuration option `HPX_COMMAND_LINE_HANDLING_WITH_JSON_CONFIGURATION_FILES=On`. This functionality depends on the external [JSON library](#)¹¹⁴⁰, which can be built at configuration time by supplying `HPX_WITH_FETCH_JSON=On` to [CMake](#)¹¹⁴¹.
- We have applied many fixes to our CUDA, ROCm, and SYCL build environments.

¹¹³⁶ <https://wg21.link/p1144>

¹¹³⁷ <http://wg21.link/n4948>

¹¹³⁸ <https://www.open-mpi.org/projects/hwloc/>

¹¹³⁹ <https://www.boost.org/>

¹¹⁴⁰ <https://github.com/nlohmann/json>

¹¹⁴¹ <https://www.cmake.org>

Breaking changes

- The `CMake`¹¹⁴² configuration keys `SOMELIB_ROOT` (e.g., `BOOST_ROOT`) have been renamed to `Somelib_ROOT` (e.g., `Boost_ROOT`) to avoid warnings when using newer versions of `CMake`¹¹⁴³. Please update your scripts accordingly. For now, the old variable names are re-assigned to the new names and unset in the `CMake`¹¹⁴⁴ cache.

Closed issues

- [Issue #6466](#)¹¹⁴⁵ - No access limitations to Wiki
- [Issue #6461](#)¹¹⁴⁶ - `handle_received_parcel`s may never return
- [Issue #6459](#)¹¹⁴⁷ - Building HPX
- [Issue #6451](#)¹¹⁴⁸ - HPX hangs at the very end
- [Issue #6446](#)¹¹⁴⁹ - Issue on page `/manual/getting_hpx.html`
- [Issue #6443](#)¹¹⁵⁰ - PR #6435 (`parcel_layer_tweaks`) broke Octo-Tiger
- [Issue #6440](#)¹¹⁵¹ - HPX does not compile with MSVC of Visual Studio 2022 17.9+
- [Issue #6437](#)¹¹⁵² - HPX 1.9.1 does not compile on Fedora with `#pragma message: [Parallel STL message]: "Vectorized algorithm unimplemented, redirected to serial"`
- [Issue #6419](#)¹¹⁵³ - Enhancement of the macro functionalities within `hpx`
- [Issue #6417](#)¹¹⁵⁴ - The current HPX master branch is still not compatible with Kokkos 4.0.1
- [Issue #6414](#)¹¹⁵⁵ - Current HPX master causes segfaults within Octo-Tiger
- [Issue #6412](#)¹¹⁵⁶ - Clangd (Language Server) throws error for `__integer_pack` at `pack.hpp`
- [Issue #6407](#)¹¹⁵⁷ - Cannot build Kokkos 4.0.01 with current HPX master
- [Issue #6405](#)¹¹⁵⁸ - Spack Build Error with ROCm 5.7.0
- [Issue #6398](#)¹¹⁵⁹ - HPX sets affinity wrong with multiple processes per node and LCI `parcelport` enabled
- [Issue #6392](#)¹¹⁶⁰ - [Feature] Install dependencies using `CMake`
- [Issue #6388](#)¹¹⁶¹ - HPX error: "Host not found" when running on Expanse with 128 nodes

¹¹⁴² <https://www.cmake.org>

¹¹⁴³ <https://www.cmake.org>

¹¹⁴⁴ <https://www.cmake.org>

¹¹⁴⁵ <https://github.com/TheHPXProject/hpx/issues/6466>

¹¹⁴⁶ <https://github.com/TheHPXProject/hpx/issues/6461>

¹¹⁴⁷ <https://github.com/TheHPXProject/hpx/issues/6459>

¹¹⁴⁸ <https://github.com/TheHPXProject/hpx/issues/6451>

¹¹⁴⁹ <https://github.com/TheHPXProject/hpx/issues/6446>

¹¹⁵⁰ <https://github.com/TheHPXProject/hpx/issues/6443>

¹¹⁵¹ <https://github.com/TheHPXProject/hpx/issues/6440>

¹¹⁵² <https://github.com/TheHPXProject/hpx/issues/6437>

¹¹⁵³ <https://github.com/TheHPXProject/hpx/issues/6419>

¹¹⁵⁴ <https://github.com/TheHPXProject/hpx/issues/6417>

¹¹⁵⁵ <https://github.com/TheHPXProject/hpx/issues/6414>

¹¹⁵⁶ <https://github.com/TheHPXProject/hpx/issues/6412>

¹¹⁵⁷ <https://github.com/TheHPXProject/hpx/issues/6407>

¹¹⁵⁸ <https://github.com/TheHPXProject/hpx/issues/6405>

¹¹⁵⁹ <https://github.com/TheHPXProject/hpx/issues/6398>

¹¹⁶⁰ <https://github.com/TheHPXProject/hpx/issues/6392>

¹¹⁶¹ <https://github.com/TheHPXProject/hpx/issues/6388>

- Issue #6366¹¹⁶² - serialize_buffer allocator support needs adjustments
- Issue #6361¹¹⁶³ - HPX 1.9.1 does not compile on Fedora 40
- Issue #6355¹¹⁶⁴ - Single page documentation is broken
- Issue #6334¹¹⁶⁵ - Segmentation fault after adding a padding in one_size_heap_list
- Issue #6329¹¹⁶⁶ - Log hpx threads on forced shutdown
- Issue #6316¹¹⁶⁷ - Build breaks on FreeBSD
- Issue #6299¹¹⁶⁸ - HPX does not use distributed localities on Fugaku
- Issue #6298¹¹⁶⁹ - Update config for coroutines on ARM
- Issue #6291¹¹⁷⁰ - Zero-copy receive optimization disabled the invocation of direct actions
- Issue #6261¹¹⁷¹ - Add optional reading of json files for command line options
- Issue #6087¹¹⁷² - Support for vcpkg on Linux is broken
- Issue #5921¹¹⁷³ - hpx::info claims that async_mpi was not built, while cmake assures its existence
- Issue #5893¹¹⁷⁴ - Tests fail on FreeBSD: Executable copyn_test does not exist
- Issue #5833¹¹⁷⁵ - barrier lockup
- Issue #5799¹¹⁷⁶ - Investigate CUDA compilation problems
- Issue #5340¹¹⁷⁷ - Examples do not run on Mac OSX using the M1 chip

Closed pull requests

- PR #6493¹¹⁷⁸ - Fix distributed latch documentation
- PR #6492¹¹⁷⁹ - Fix kokkos hpx nvcc compilation
- PR #6491¹¹⁸⁰ - More fixes to handling bool arguments for collective operations
- PR #6490¹¹⁸¹ - Remove the default max cpu count
- PR #6489¹¹⁸² - Ensure TCP parcelport is deactivated if not needed
- PR #6488¹¹⁸³ - Fixing handling of bool value type for collective operations

¹¹⁶² <https://github.com/TheHPXProject/hpx/issues/6366>

¹¹⁶³ <https://github.com/TheHPXProject/hpx/issues/6361>

¹¹⁶⁴ <https://github.com/TheHPXProject/hpx/issues/6355>

¹¹⁶⁵ <https://github.com/TheHPXProject/hpx/issues/6334>

¹¹⁶⁶ <https://github.com/TheHPXProject/hpx/issues/6329>

¹¹⁶⁷ <https://github.com/TheHPXProject/hpx/issues/6316>

¹¹⁶⁸ <https://github.com/TheHPXProject/hpx/issues/6299>

¹¹⁶⁹ <https://github.com/TheHPXProject/hpx/issues/6298>

¹¹⁷⁰ <https://github.com/TheHPXProject/hpx/issues/6291>

¹¹⁷¹ <https://github.com/TheHPXProject/hpx/issues/6261>

¹¹⁷² <https://github.com/TheHPXProject/hpx/issues/6087>

¹¹⁷³ <https://github.com/TheHPXProject/hpx/issues/5921>

¹¹⁷⁴ <https://github.com/TheHPXProject/hpx/issues/5893>

¹¹⁷⁵ <https://github.com/TheHPXProject/hpx/issues/5833>

¹¹⁷⁶ <https://github.com/TheHPXProject/hpx/issues/5799>

¹¹⁷⁷ <https://github.com/TheHPXProject/hpx/issues/5340>

¹¹⁷⁸ <https://github.com/TheHPXProject/hpx/pull/6493>

¹¹⁷⁹ <https://github.com/TheHPXProject/hpx/pull/6492>

¹¹⁸⁰ <https://github.com/TheHPXProject/hpx/pull/6491>

¹¹⁸¹ <https://github.com/TheHPXProject/hpx/pull/6490>

¹¹⁸² <https://github.com/TheHPXProject/hpx/pull/6489>

¹¹⁸³ <https://github.com/TheHPXProject/hpx/pull/6488>

- PR #6485¹¹⁸⁴ - Destructive interference size
- PR #6484¹¹⁸⁵ - Improve performance counter error handling
- PR #6482¹¹⁸⁶ - Generalize the notion of bitwise serialization
- PR #6481¹¹⁸⁷ - Fixing use of HPX_WITH_CXX_STANDARD
- PR #6480¹¹⁸⁸ - Remove equal_to from hpx::any
- PR #6479¹¹⁸⁹ - Remove optimizations for certain built-in compiler intrinsics
- PR #6478¹¹⁹⁰ - Fixing issues on MacOS
- PR #6477¹¹⁹¹ - lci pp: lci's github repo name changed from LC to lci
- PR #6476¹¹⁹² - Fixing binary filter test target names
- PR #6475¹¹⁹³ - Fix mac os github actions
- PR #6472¹¹⁹⁴ - Troubleshoot CI hangs
- PR #6469¹¹⁹⁵ - improve(lci pp): more options to control the LCI parcelport
- PR #6467¹¹⁹⁶ - Bump jwlawson/actions-setup-cmake from 1.14 to 2.0
- PR #6464¹¹⁹⁷ - Update docs of "Writing distributed applications" page
- PR #6463¹¹⁹⁸ - Revert "Always return outermost thread id"
- PR #6458¹¹⁹⁹ - Reduce test workload to fix CI/CD time-out
- PR #6457¹²⁰⁰ - replace boost::array with std::array and update file name
- PR #6456¹²⁰¹ - Move APEX CI to rostam
- PR #6455¹²⁰² - Fixing compilation if HPX_HAVE_THREAD_QUEUE_WAITTIME is defined
- PR #6454¹²⁰³ - Update perftests reference measurements
- PR #6453¹²⁰⁴ - Update supported platforms of Manual/Prerequisites page
- PR #6452¹²⁰⁵ - Fix nvcc crashes in transform_stream.cu and synchronize.cu
- PR #6450¹²⁰⁶ - Fix git tag name in Getting HPX page

¹¹⁸⁴ <https://github.com/TheHPXProject/hpx/pull/6485>

¹¹⁸⁵ <https://github.com/TheHPXProject/hpx/pull/6484>

¹¹⁸⁶ <https://github.com/TheHPXProject/hpx/pull/6482>

¹¹⁸⁷ <https://github.com/TheHPXProject/hpx/pull/6481>

¹¹⁸⁸ <https://github.com/TheHPXProject/hpx/pull/6480>

¹¹⁸⁹ <https://github.com/TheHPXProject/hpx/pull/6479>

¹¹⁹⁰ <https://github.com/TheHPXProject/hpx/pull/6478>

¹¹⁹¹ <https://github.com/TheHPXProject/hpx/pull/6477>

¹¹⁹² <https://github.com/TheHPXProject/hpx/pull/6476>

¹¹⁹³ <https://github.com/TheHPXProject/hpx/pull/6475>

¹¹⁹⁴ <https://github.com/TheHPXProject/hpx/pull/6472>

¹¹⁹⁵ <https://github.com/TheHPXProject/hpx/pull/6469>

¹¹⁹⁶ <https://github.com/TheHPXProject/hpx/pull/6467>

¹¹⁹⁷ <https://github.com/TheHPXProject/hpx/pull/6464>

¹¹⁹⁸ <https://github.com/TheHPXProject/hpx/pull/6463>

¹¹⁹⁹ <https://github.com/TheHPXProject/hpx/pull/6458>

¹²⁰⁰ <https://github.com/TheHPXProject/hpx/pull/6457>

¹²⁰¹ <https://github.com/TheHPXProject/hpx/pull/6456>

¹²⁰² <https://github.com/TheHPXProject/hpx/pull/6455>

¹²⁰³ <https://github.com/TheHPXProject/hpx/pull/6454>

¹²⁰⁴ <https://github.com/TheHPXProject/hpx/pull/6453>

¹²⁰⁵ <https://github.com/TheHPXProject/hpx/pull/6452>

¹²⁰⁶ <https://github.com/TheHPXProject/hpx/pull/6450>

- PR #6449¹²⁰⁷ - LCI parcelport: add yield to potentially infinite retry loop
- PR #6447¹²⁰⁸ - Use compressed ptr in schedulers when 128 atomics are not lockfree
- PR #6445¹²⁰⁹ - Fix agas addressing cache
- PR #6444¹²¹⁰ - Update CTestConfig.cmake
- PR #6442¹²¹¹ - Update CMakeLists.txt
- PR #6441¹²¹² - Minor documentation fixes
- PR #6439¹²¹³ - Optimizing use of certain #includes
- PR #6438¹²¹⁴ - Bump jwlawson/actions-setup-cmake from 1.14 to 2.0
- PR #6436¹²¹⁵ - Update docs
- PR #6435¹²¹⁶ - Parcel layer tweaks
- PR #6434¹²¹⁷ - improve termination detection: removing lock from critical path
- PR #6433¹²¹⁸ - Use shared mutex for resolve_locality procedure
- PR #6432¹²¹⁹ - Module cleanup up to level 30
- PR #6429¹²²⁰ - Making sure HPX_WITH_ASYNC_MPI is reported properly
- PR #6427¹²²¹ - Modifying CMakeLists to copy libhwloc-15.dll to the binary folder in Windows, independently
- PR #6425¹²²² - Fix macOS failing test
- PR #6424¹²²³ - Adding option for downloading Boost using CMake FetchContent
- PR #6423¹²²⁴ - Move adjacent_difference to numeric header file
- PR #6422¹²²⁵ - Adding steal-half functionalities to work-requesting scheduler
- PR #6421¹²²⁶ - Bump actions/checkout from 2 to 4
- PR #6418¹²²⁷ - Working around nvcc problems to use CTAD
- PR #6416¹²²⁸ - Change run_as_os_thread deprecation forwarding due to hipcc compilation issue
- PR #6415¹²²⁹ - Attempting to avoid segfault in OctoTiger during initialization

¹²⁰⁷ <https://github.com/TheHPXProject/hpx/pull/6449>

¹²⁰⁸ <https://github.com/TheHPXProject/hpx/pull/6447>

¹²⁰⁹ <https://github.com/TheHPXProject/hpx/pull/6445>

¹²¹⁰ <https://github.com/TheHPXProject/hpx/pull/6444>

¹²¹¹ <https://github.com/TheHPXProject/hpx/pull/6442>

¹²¹² <https://github.com/TheHPXProject/hpx/pull/6441>

¹²¹³ <https://github.com/TheHPXProject/hpx/pull/6439>

¹²¹⁴ <https://github.com/TheHPXProject/hpx/pull/6438>

¹²¹⁵ <https://github.com/TheHPXProject/hpx/pull/6436>

¹²¹⁶ <https://github.com/TheHPXProject/hpx/pull/6435>

¹²¹⁷ <https://github.com/TheHPXProject/hpx/pull/6434>

¹²¹⁸ <https://github.com/TheHPXProject/hpx/pull/6433>

¹²¹⁹ <https://github.com/TheHPXProject/hpx/pull/6432>

¹²²⁰ <https://github.com/TheHPXProject/hpx/pull/6429>

¹²²¹ <https://github.com/TheHPXProject/hpx/pull/6427>

¹²²² <https://github.com/TheHPXProject/hpx/pull/6425>

¹²²³ <https://github.com/TheHPXProject/hpx/pull/6424>

¹²²⁴ <https://github.com/TheHPXProject/hpx/pull/6423>

¹²²⁵ <https://github.com/TheHPXProject/hpx/pull/6422>

¹²²⁶ <https://github.com/TheHPXProject/hpx/pull/6421>

¹²²⁷ <https://github.com/TheHPXProject/hpx/pull/6418>

¹²²⁸ <https://github.com/TheHPXProject/hpx/pull/6416>

¹²²⁹ <https://github.com/TheHPXProject/hpx/pull/6415>

- PR #6413¹²³⁰ - Always return outermost thread id
- PR #6411¹²³¹ - Minor refactoring and fixes to the LCI parcellport and pingpong_performance2 benchmark
- PR #6410¹²³² - Adding scope_xxx from library fundamentals TS v3
- PR #6409¹²³³ - Working around CUDA issue
- PR #6408¹²³⁴ - Tightening up collective operation semantics
- PR #6406¹²³⁵ - Working around ROCm compiler issue
- PR #6404¹²³⁶ - Allow to disable use of [[no_unique_address]] attribute
- PR #6403¹²³⁷ - Fixing copyright year
- PR #6402¹²³⁸ - fix(lci pp): fix deadlocks with too many failed sends
- PR #6401¹²³⁹ - fix(lci pp): fix the null_thread_id bug in the LCI parcellport
- PR #6400¹²⁴⁰ - Fix the affinity setting bug when using LCI pp and multiple localities per node
- PR #6397¹²⁴¹ - Change API header titles and info
- PR #6396¹²⁴² - Making is_bitwise_serializable SFINAE-friendly
- PR #6395¹²⁴³ - Adapt amount of collective testing
- PR #6394¹²⁴⁴ - Adding option for installing Hwloc using CMake FetchContent
- PR #6393¹²⁴⁵ - Optionally disable caching allocator
- PR #6391¹²⁴⁶ - Cleaning up collective operations
- PR #6390¹²⁴⁷ - Making function local constexpr variables non-static
- PR #6389¹²⁴⁸ - Disable resolving hostnames if TCP is disabled
- PR #6387¹²⁴⁹ - Need to break out of the loop when searching the suffixes.
- PR #6384¹²⁵⁰ - Fixing allocation/deallocation mismatch in serialize_buffer
- PR #6383¹²⁵¹ - Enable fork_join_executor to handle return values from scheduled functions
- PR #6381¹²⁵² - Consistently treat conflicting parameters provided by executors and parameter objects

¹²³⁰ <https://github.com/TheHPXProject/hpx/pull/6413>

¹²³¹ <https://github.com/TheHPXProject/hpx/pull/6411>

¹²³² <https://github.com/TheHPXProject/hpx/pull/6410>

¹²³³ <https://github.com/TheHPXProject/hpx/pull/6409>

¹²³⁴ <https://github.com/TheHPXProject/hpx/pull/6408>

¹²³⁵ <https://github.com/TheHPXProject/hpx/pull/6406>

¹²³⁶ <https://github.com/TheHPXProject/hpx/pull/6404>

¹²³⁷ <https://github.com/TheHPXProject/hpx/pull/6403>

¹²³⁸ <https://github.com/TheHPXProject/hpx/pull/6402>

¹²³⁹ <https://github.com/TheHPXProject/hpx/pull/6401>

¹²⁴⁰ <https://github.com/TheHPXProject/hpx/pull/6400>

¹²⁴¹ <https://github.com/TheHPXProject/hpx/pull/6397>

¹²⁴² <https://github.com/TheHPXProject/hpx/pull/6396>

¹²⁴³ <https://github.com/TheHPXProject/hpx/pull/6395>

¹²⁴⁴ <https://github.com/TheHPXProject/hpx/pull/6394>

¹²⁴⁵ <https://github.com/TheHPXProject/hpx/pull/6393>

¹²⁴⁶ <https://github.com/TheHPXProject/hpx/pull/6391>

¹²⁴⁷ <https://github.com/TheHPXProject/hpx/pull/6390>

¹²⁴⁸ <https://github.com/TheHPXProject/hpx/pull/6389>

¹²⁴⁹ <https://github.com/TheHPXProject/hpx/pull/6387>

¹²⁵⁰ <https://github.com/TheHPXProject/hpx/pull/6384>

¹²⁵¹ <https://github.com/TheHPXProject/hpx/pull/6383>

¹²⁵² <https://github.com/TheHPXProject/hpx/pull/6381>

- PR #6380¹²⁵³ - Fixing setting an annotation for an execution policy
- PR #6378¹²⁵⁴ - Allowing to disable signal handlers
- PR #6377¹²⁵⁵ - Fix gasnet-related test failures
- PR #6375¹²⁵⁶ - Update LSU Jenkins with 2023-10 libraries
- PR #6374¹²⁵⁷ - Investigate builder gasnet failure
- PR #6373¹²⁵⁸ - Fixing communicator API, adding docs
- PR #6372¹²⁵⁹ - Fix resource partitioner tests for small thread count
- PR #6371¹²⁶⁰ - Fix jacobi omp examples.
- PR #6370¹²⁶¹ - improve one_size_heap_list: use rwlock to speedup the allocation/free
- PR #6369¹²⁶² - working issue with MPI_CC / CC conflict in automake
- PR #6368¹²⁶³ - Making sure serialize_buffer properly destroys buffer, if needed.
- PR #6367¹²⁶⁴ - Fix parallel relocation test
- PR #6364¹²⁶⁵ - Relocation variants
- PR #6363¹²⁶⁶ - Update the lci parcelport to use LCI v1.7.6
- PR #6362¹²⁶⁷ - Fixing compilation problems on 32 Linux systems
- PR #6360¹²⁶⁸ - Fix broken links in docs: PDF, Single HTML page, Dependency report
- PR #6359¹²⁶⁹ - Fix header file links in Public API page
- PR #6358¹²⁷⁰ - Fix CMake find_library for HWLOC
- PR #6357¹²⁷¹ - Replace Custom Benchmarking Code with Nanobench
- PR #6356¹²⁷² - Fixed matrix multiplication example output
- PR #6354¹²⁷³ - Fix broken links for header files in Public API page
- PR #6353¹²⁷⁴ - Enable using std::reference_wrapper with executor parameters
- PR #6352¹²⁷⁵ - Add Public distributed API documentation

¹²⁵³ <https://github.com/TheHPXProject/hpx/pull/6380>

¹²⁵⁴ <https://github.com/TheHPXProject/hpx/pull/6378>

¹²⁵⁵ <https://github.com/TheHPXProject/hpx/pull/6377>

¹²⁵⁶ <https://github.com/TheHPXProject/hpx/pull/6375>

¹²⁵⁷ <https://github.com/TheHPXProject/hpx/pull/6374>

¹²⁵⁸ <https://github.com/TheHPXProject/hpx/pull/6373>

¹²⁵⁹ <https://github.com/TheHPXProject/hpx/pull/6372>

¹²⁶⁰ <https://github.com/TheHPXProject/hpx/pull/6371>

¹²⁶¹ <https://github.com/TheHPXProject/hpx/pull/6370>

¹²⁶² <https://github.com/TheHPXProject/hpx/pull/6369>

¹²⁶³ <https://github.com/TheHPXProject/hpx/pull/6368>

¹²⁶⁴ <https://github.com/TheHPXProject/hpx/pull/6367>

¹²⁶⁵ <https://github.com/TheHPXProject/hpx/pull/6364>

¹²⁶⁶ <https://github.com/TheHPXProject/hpx/pull/6363>

¹²⁶⁷ <https://github.com/TheHPXProject/hpx/pull/6362>

¹²⁶⁸ <https://github.com/TheHPXProject/hpx/pull/6360>

¹²⁶⁹ <https://github.com/TheHPXProject/hpx/pull/6359>

¹²⁷⁰ <https://github.com/TheHPXProject/hpx/pull/6358>

¹²⁷¹ <https://github.com/TheHPXProject/hpx/pull/6357>

¹²⁷² <https://github.com/TheHPXProject/hpx/pull/6356>

¹²⁷³ <https://github.com/TheHPXProject/hpx/pull/6354>

¹²⁷⁴ <https://github.com/TheHPXProject/hpx/pull/6353>

¹²⁷⁵ <https://github.com/TheHPXProject/hpx/pull/6352>

- PR #6350¹²⁷⁶ - Make coverage work with Jenkins Github Branch Source plugin
- PR #6349¹²⁷⁷ - Moving hpx::threads::run_as_xxx to namespace hpx
- PR #6348¹²⁷⁸ - Adding `--exclusive` to launching tests on rostam
- PR #6346¹²⁷⁹ - changed chat link to discord
- PR #6344¹²⁸⁰ - uninitialized_relocate w/ type_support primitive
- PR #6343¹²⁸¹ - Bump actions/checkout from 3 to 4
- PR #6342¹²⁸² - Fix HPX-APEX cmake integration
- PR #6341¹²⁸³ - Fix shared_future_continuation_order regression test
- PR #6340¹²⁸⁴ - Log alive hpx threads on exit
- PR #6339¹²⁸⁵ - Add coverage testing on Jenkins
- PR #6338¹²⁸⁶ - Fixing HPX_CURRENT_SOURCE_LOCATION when std::source_location exists
- PR #6337¹²⁸⁷ - Remove aurianer, biddisco, and msimberg from codeowners
- PR #6336¹²⁸⁸ - More cleaning up for module levels 19-20
- PR #6335¹²⁸⁹ - Finalize the MPI docs of the Migration Guide
- PR #6332¹²⁹⁰ - More fixes for CMake V3.27
- PR #6330¹²⁹¹ - Adding basic logging to collective operations
- PR #6328¹²⁹² - Cleanup previous patch adapting to CMake V3.27
- PR #6327¹²⁹³ - Modernize modules in level 17 and 18
- PR #6324¹²⁹⁴ - P1144 Relocation primitives
- PR #6321¹²⁹⁵ - Ensure hpx_main is a proper thread_function
- PR #6320¹²⁹⁶ - Fixing cyclic dependencies in naming and agas modules
- PR #6319¹²⁹⁷ - Generate git tag if needed but it is not available
- PR #6317¹²⁹⁸ - Fixing linker problem on FreeBSD

¹²⁷⁶ <https://github.com/TheHPXProject/hpx/pull/6350>

¹²⁷⁷ <https://github.com/TheHPXProject/hpx/pull/6349>

¹²⁷⁸ <https://github.com/TheHPXProject/hpx/pull/6348>

¹²⁷⁹ <https://github.com/TheHPXProject/hpx/pull/6346>

¹²⁸⁰ <https://github.com/TheHPXProject/hpx/pull/6344>

¹²⁸¹ <https://github.com/TheHPXProject/hpx/pull/6343>

¹²⁸² <https://github.com/TheHPXProject/hpx/pull/6342>

¹²⁸³ <https://github.com/TheHPXProject/hpx/pull/6341>

¹²⁸⁴ <https://github.com/TheHPXProject/hpx/pull/6340>

¹²⁸⁵ <https://github.com/TheHPXProject/hpx/pull/6339>

¹²⁸⁶ <https://github.com/TheHPXProject/hpx/pull/6338>

¹²⁸⁷ <https://github.com/TheHPXProject/hpx/pull/6337>

¹²⁸⁸ <https://github.com/TheHPXProject/hpx/pull/6336>

¹²⁸⁹ <https://github.com/TheHPXProject/hpx/pull/6335>

¹²⁹⁰ <https://github.com/TheHPXProject/hpx/pull/6332>

¹²⁹¹ <https://github.com/TheHPXProject/hpx/pull/6330>

¹²⁹² <https://github.com/TheHPXProject/hpx/pull/6328>

¹²⁹³ <https://github.com/TheHPXProject/hpx/pull/6327>

¹²⁹⁴ <https://github.com/TheHPXProject/hpx/pull/6324>

¹²⁹⁵ <https://github.com/TheHPXProject/hpx/pull/6321>

¹²⁹⁶ <https://github.com/TheHPXProject/hpx/pull/6320>

¹²⁹⁷ <https://github.com/TheHPXProject/hpx/pull/6319>

¹²⁹⁸ <https://github.com/TheHPXProject/hpx/pull/6317>

- PR #6315¹²⁹⁹ - acknowledge triv-rel and nothrow-rel types
- PR #6314¹³⁰⁰ - Relocation algorithms Clean
- PR #6313¹³⁰¹ - Trivial relocation of c-v-ref-array types
- PR #6312¹³⁰² - Fixing warning/error
- PR #6311¹³⁰³ - Adding executor parallel invoke CPOs
- PR #6310¹³⁰⁴ - Define HPX_COMPUTE_CODE in builds with SYCL
- PR #6309¹³⁰⁵ - Making sure changed number of cores is propagated to executor
- PR #6308¹³⁰⁶ - openshmem-parcelport initial import
- PR #6306¹³⁰⁷ - The hpccxx script was broken such that it could only compile for _release
- PR #6305¹³⁰⁸ - Adapting build system for CMake V3.27
- PR #6304¹³⁰⁹ - Fixing an integral type mismatch warning
- PR #6303¹³¹⁰ - omp for default vectorization
- PR #6301¹³¹¹ - Add MPI migration guide
- PR #6294¹³¹² - Add internal reference counting to semaphores
- PR #6286¹³¹³ - Simd helpers
- PR #6280¹³¹⁴ - Add TBB to HPX documentation in Migration Guide
- PR #6276¹³¹⁵ - Add dependabot.yml
- PR #6275¹³¹⁶ - Revert “Move dependabot.yml into correct directory”
- PR #6272¹³¹⁷ - set thread name for linux
- PR #6271¹³¹⁸ - Uninitialised algorithms, move using std::memcpy
- PR #6270¹³¹⁹ - Bump jwlawson/actions-setup-cmake from 1.9 to 1.14
- PR #6269¹³²⁰ - Bump actions/checkout from 2 to 3
- PR #6268¹³²¹ - Move dependabot.yml into correct directory

¹²⁹⁹ <https://github.com/TheHPXProject/hpx/pull/6315>

¹³⁰⁰ <https://github.com/TheHPXProject/hpx/pull/6314>

¹³⁰¹ <https://github.com/TheHPXProject/hpx/pull/6313>

¹³⁰² <https://github.com/TheHPXProject/hpx/pull/6312>

¹³⁰³ <https://github.com/TheHPXProject/hpx/pull/6311>

¹³⁰⁴ <https://github.com/TheHPXProject/hpx/pull/6310>

¹³⁰⁵ <https://github.com/TheHPXProject/hpx/pull/6309>

¹³⁰⁶ <https://github.com/TheHPXProject/hpx/pull/6308>

¹³⁰⁷ <https://github.com/TheHPXProject/hpx/pull/6306>

¹³⁰⁸ <https://github.com/TheHPXProject/hpx/pull/6305>

¹³⁰⁹ <https://github.com/TheHPXProject/hpx/pull/6304>

¹³¹⁰ <https://github.com/TheHPXProject/hpx/pull/6303>

¹³¹¹ <https://github.com/TheHPXProject/hpx/pull/6301>

¹³¹² <https://github.com/TheHPXProject/hpx/pull/6294>

¹³¹³ <https://github.com/TheHPXProject/hpx/pull/6286>

¹³¹⁴ <https://github.com/TheHPXProject/hpx/pull/6280>

¹³¹⁵ <https://github.com/TheHPXProject/hpx/pull/6276>

¹³¹⁶ <https://github.com/TheHPXProject/hpx/pull/6275>

¹³¹⁷ <https://github.com/TheHPXProject/hpx/pull/6272>

¹³¹⁸ <https://github.com/TheHPXProject/hpx/pull/6271>

¹³¹⁹ <https://github.com/TheHPXProject/hpx/pull/6270>

¹³²⁰ <https://github.com/TheHPXProject/hpx/pull/6269>

¹³²¹ <https://github.com/TheHPXProject/hpx/pull/6268>

- PR #6265¹³²² - Create dependabot.yml
- PR #6264¹³²³ - `hpx::is_trivially_relocatable` trait implementation
- PR #6263¹³²⁴ - Adding support for reading json configuration files for command line options
- PR #6249¹³²⁵ - Implement the send immediate optimization for the MPI parcelport.
- PR #6237¹³²⁶ - Improve compilation performance
- PR #6234¹³²⁷ - Adding release notes page for next release
- PR #6233¹³²⁸ - Moving `is_relocatable` to namespace `hpx`
- PR #6230¹³²⁹ - gasnet based parcelport
- PR #6226¹³³⁰ - Re-enable dependency on segmented algorithms on CircleCI
- PR #6220¹³³¹ - Add execution on
- PR #6212¹³³² - Initial trait definition for *relocatable*
- PR #6199¹³³³ - added support for `unseq`, `par_unseq` for `hpx::make_heap` algorithm
- PR #6173¹³³⁴ - C++ modules
- PR #6122¹³³⁵ - Add Module support
- PR #6099¹³³⁶ - Futures attempt to execute threads directly if those have not started executing
- PR #6050¹³³⁷ - Investigating `partitioned_vector` problems
- PR #5988¹³³⁸ - Adding CI configuration for DGX-A100 at LSU
- PR #5910¹³³⁹ - Improve MPI initialization
- PR #5845¹³⁴⁰ - Adding local work requesting scheduler that is based on message passing internally

¹³²² <https://github.com/TheHPXProject/hpx/pull/6265>

¹³²³ <https://github.com/TheHPXProject/hpx/pull/6264>

¹³²⁴ <https://github.com/TheHPXProject/hpx/pull/6263>

¹³²⁵ <https://github.com/TheHPXProject/hpx/pull/6249>

¹³²⁶ <https://github.com/TheHPXProject/hpx/pull/6237>

¹³²⁷ <https://github.com/TheHPXProject/hpx/pull/6234>

¹³²⁸ <https://github.com/TheHPXProject/hpx/pull/6233>

¹³²⁹ <https://github.com/TheHPXProject/hpx/pull/6230>

¹³³⁰ <https://github.com/TheHPXProject/hpx/pull/6226>

¹³³¹ <https://github.com/TheHPXProject/hpx/pull/6220>

¹³³² <https://github.com/TheHPXProject/hpx/pull/6212>

¹³³³ <https://github.com/TheHPXProject/hpx/pull/6199>

¹³³⁴ <https://github.com/TheHPXProject/hpx/pull/6173>

¹³³⁵ <https://github.com/TheHPXProject/hpx/pull/6122>

¹³³⁶ <https://github.com/TheHPXProject/hpx/pull/6099>

¹³³⁷ <https://github.com/TheHPXProject/hpx/pull/6050>

¹³³⁸ <https://github.com/TheHPXProject/hpx/pull/5988>

¹³³⁹ <https://github.com/TheHPXProject/hpx/pull/5910>

¹³⁴⁰ <https://github.com/TheHPXProject/hpx/pull/5845>

HPX V1.9.1 (August 4, 2023)

General changes

This point release fixes a couple of problems reported for the V1.9.0 release. Most importantly, we fixed various occasional hanging during startup and shutdown in distributed scenarios. We also added support for zero-copy serialization on the receiving side to the TCP, MPI, and LCI parcelports. Last but not least, we have added support for Visual Studio 2019 and gcc using MINGW on Windows, and also support for gcc V13 and clang V15.

HPX headers are now made consistently named the same as their standard library counterparts, e.g. `#include <thread>` now corresponds to `#include <hpx/thread.hpp>`. This significantly simplifies porting existing standards conforming codes to HPX.

A lot of work has been done to improve and optimize our network communication layers. Primary focus of this work was on the LCI parcelport, but we have also cleaned up and improved the MPI parcelport.

Additionally, we have continued working on our documentation. The main focus here was on completing the API documentation of the most important API functions. We have started adding migration guides for people interested in moving their codes away from other, commonplace parallelization frameworks like OpenMP.

Breaking changes

None

Closed issues

- [Issue #6155](https://github.com/TheHPXProject/hpx/issues/6155)¹³⁴¹ - `hpxcxx` and `hpxrun.py` do not work if `HPX_WITH_TESTS=OFF`
- [Issue #6164](https://github.com/TheHPXProject/hpx/issues/6164)¹³⁴² - `HPX_WITH_DATAPAR_BACKEND=EVE` causes compile errors with C++17
- [Issue #6175](https://github.com/TheHPXProject/hpx/issues/6175)¹³⁴³ - Make sure all our parallel algorithms accept the predicates by value
- [Issue #6194](https://github.com/TheHPXProject/hpx/issues/6194)¹³⁴⁴ - `tests.regressions.threads.threads_all_1422` failed at Perlmutter
- [Issue #6198](https://github.com/TheHPXProject/hpx/issues/6198)¹³⁴⁵ - `set_intersection/set_difference` fails when run with `execution::par`
- [Issue #6214](https://github.com/TheHPXProject/hpx/issues/6214)¹³⁴⁶ - Broken Links to the Documentation page in `readme.rst`
- [Issue #6217](https://github.com/TheHPXProject/hpx/issues/6217)¹³⁴⁷ - `hpx::make_heap` does not terminate when `exPolicy` is `par` (or `par_unseq`) and size of vector is 4
- [Issue #6246](https://github.com/TheHPXProject/hpx/issues/6246)¹³⁴⁸ - HPX fails to compile under `cxx 20` (fresh system)
- [Issue #6247](https://github.com/TheHPXProject/hpx/issues/6247)¹³⁴⁹ - HPX 1.9.0 does not compile with GCC on Windows
- [Issue #6282](https://github.com/TheHPXProject/hpx/issues/6282)¹³⁵⁰ - The “attach-debugger” option is broken on the current master branch.

¹³⁴¹ <https://github.com/TheHPXProject/hpx/issues/6155>

¹³⁴² <https://github.com/TheHPXProject/hpx/issues/6164>

¹³⁴³ <https://github.com/TheHPXProject/hpx/issues/6175>

¹³⁴⁴ <https://github.com/TheHPXProject/hpx/issues/6194>

¹³⁴⁵ <https://github.com/TheHPXProject/hpx/issues/6198>

¹³⁴⁶ <https://github.com/TheHPXProject/hpx/issues/6214>

¹³⁴⁷ <https://github.com/TheHPXProject/hpx/issues/6217>

¹³⁴⁸ <https://github.com/TheHPXProject/hpx/issues/6246>

¹³⁴⁹ <https://github.com/TheHPXProject/hpx/issues/6247>

¹³⁵⁰ <https://github.com/TheHPXProject/hpx/issues/6282>

Closed pull requests

- PR #6219¹³⁵¹ - Cleaning up #includes in hpx/ folder
- PR #6223¹³⁵² - Move documentation from README.rst to index.rst files under libs directory
- PR #6229¹³⁵³ - Adding zero-copy support on the receiving end of the TCP and MPI parcel ports
- PR #6231¹³⁵⁴ - Remove deprecated email from release procedure
- PR #6235¹³⁵⁵ - Modernize more modules (levels 12-16)
- PR #6236¹³⁵⁶ - Attempt to resolve occasional shutdown hangs in distributed operation
- PR #6239¹³⁵⁷ - Fix Optimizing HPX applications page of Manual
- PR #6241¹³⁵⁸ - LCI parcelport: Refactor, add more variants, zero copy receives.
- PR #6242¹³⁵⁹ - updated deprecated headers
- PR #6243¹³⁶⁰ - Adding github action builders using VS2019
- PR #6248¹³⁶¹ - Fix CUDA/HIP Jenkins pipelines
- PR #6250¹³⁶² - Resolve gcc problems on Windows
- PR #6251¹³⁶³ - Attempting to fix problems in barrier causing hangs
- PR #6253¹³⁶⁴ - Modernize set_thread_name on Windows
- PR #6256¹³⁶⁵ - Fix nvcc/gcc-10 (Octo-Tiger) compilation issue
- PR #6257¹³⁶⁶ - Cmake Tests: Delete operator check for size_t arg
- PR #6258¹³⁶⁷ - Rewriting wait_some to circumvent data races causing hangs
- PR #6260¹³⁶⁸ - Add migration guide to manual
- PR #6262¹³⁶⁹ - Fixing wrong command line options in local command line handling
- PR #6266¹³⁷⁰ - Attempt to resolve occasional hang in run_loop
- PR #6267¹³⁷¹ - Attempting to fix migration tests
- PR #6278¹³⁷² - Making sure the future's shared state doesn't go out of scope prematurely

¹³⁵¹ <https://github.com/TheHPXProject/hpx/pull/6219>

¹³⁵² <https://github.com/TheHPXProject/hpx/pull/6223>

¹³⁵³ <https://github.com/TheHPXProject/hpx/pull/6229>

¹³⁵⁴ <https://github.com/TheHPXProject/hpx/pull/6231>

¹³⁵⁵ <https://github.com/TheHPXProject/hpx/pull/6235>

¹³⁵⁶ <https://github.com/TheHPXProject/hpx/pull/6236>

¹³⁵⁷ <https://github.com/TheHPXProject/hpx/pull/6239>

¹³⁵⁸ <https://github.com/TheHPXProject/hpx/pull/6241>

¹³⁵⁹ <https://github.com/TheHPXProject/hpx/pull/6242>

¹³⁶⁰ <https://github.com/TheHPXProject/hpx/pull/6243>

¹³⁶¹ <https://github.com/TheHPXProject/hpx/pull/6248>

¹³⁶² <https://github.com/TheHPXProject/hpx/pull/6250>

¹³⁶³ <https://github.com/TheHPXProject/hpx/pull/6251>

¹³⁶⁴ <https://github.com/TheHPXProject/hpx/pull/6253>

¹³⁶⁵ <https://github.com/TheHPXProject/hpx/pull/6256>

¹³⁶⁶ <https://github.com/TheHPXProject/hpx/pull/6257>

¹³⁶⁷ <https://github.com/TheHPXProject/hpx/pull/6258>

¹³⁶⁸ <https://github.com/TheHPXProject/hpx/pull/6260>

¹³⁶⁹ <https://github.com/TheHPXProject/hpx/pull/6262>

¹³⁷⁰ <https://github.com/TheHPXProject/hpx/pull/6266>

¹³⁷¹ <https://github.com/TheHPXProject/hpx/pull/6267>

¹³⁷² <https://github.com/TheHPXProject/hpx/pull/6278>

- PR #6279¹³⁷³ - Re-expose error names
- PR #6281¹³⁷⁴ - Creating directory for file copy
- PR #6283¹³⁷⁵ - Consistently #include unistd.h for _POSIX_VERSION

HPX V1.9.0 (May 2, 2023)

General changes

- Added RISC-V 64bit support. HPX is now compatible with RISC-V architectures which have revolutionized the HPC world.
- LCI parcelport has been optimized to transfer parcels with fewer messages and use the HPX resource partitioner for its progress thread allocation. It should generally provide better performance than before. It also removes its dependency on the MPI library.
- HPX dependency on Boost was further relaxed by replacing headers from Boost.Range, Boost.Tokenizer and Boost.Lockfree.
- Improvements took place on our parallel algorithms implementation.
- Our Senders/Receivers (P2300) integration was extended:
 - Coroutines were integrated with senders/receivers.

`get_completion_signatures` now works with awaitable senders. - `with_awaitable_senders` allows the passed senders to retrieve the value i.e. senders are transparently awaitable from within a coroutine. - `when_all_vector` was added.
- `sync_wait` and `sync_wait_with_variant` sender consumers were added. The user can now initiate the execution of their asynchronous pipeline by blocking the current thread that executes the `main()` function until the result is retrieved.
- The combinators for futures (a.k.a. `async_combinators`) `when_*`, `wait_*`, `wait_*_nothrow` were turned into CPOs allowing for end-user customization. For more information on the `async_combinators` refer to the documentation, https://hpx-docs.stellar-group.org/latest/html/libs/core/async_combinators/docs/index.html?highlight=combinators.
- The new datapar backend SVE allows `simd` and `par_simd` execution policies to exploit dataparallelism in the processors that have SVE vector registers like A64FX and Neoverse V1.
- The documentation for parallel algorithms, container algorithms was further improved. The Public API page was vastly enriched.
- Copy button shortkey was added at the top-right of code-blocks.
- Pragma directive that reports warnings as errors on MSVC was fixed.
- Command line argument `--hpx:loopback_network` was added to facilitate debugging with networks.
- We added an HPX-SYCL integration, allowing users to obtain HPX futures for SYCL events. This effectively enables the integration of arbitrary asynchronous SYCL operations into the HPX task graph. Bolted on top of this integration, we further added an HPX-SYCL executor for ease of use.

¹³⁷³ <https://github.com/TheHPXProject/hpx/pull/6279>

¹³⁷⁴ <https://github.com/TheHPXProject/hpx/pull/6281>

¹³⁷⁵ <https://github.com/TheHPXProject/hpx/pull/6283>

Breaking changes

- Stopped supporting Clang V8, the minimal version supported is now Clang V10.
- Stopped supporting gcc V8, the minimal version supported is now gcc V9.
- Stopped supporting Visual Studio 2015, the minimal version supported is now Visual Studio 2019.
- `tag_policy_tag` et.al. were re-added after HPX V1.8.1 deprecation.
- `get_chunk_size` and `processing_units_count` API is now expecting the time for one iteration as an argument.
- The list of all the namespace changes can be found here: *HPX V1.9.0 Namespace changes*.

Closed issues

- Issue #6203¹³⁷⁶ - Compilation error with `-mcpu=a64fx` on Oookami
- Issue #6196¹³⁷⁷ - Incorrect log destination
- Issue #6191¹³⁷⁸ - installing HPX
- Issue #6184¹³⁷⁹ - Wrong `processing_units_count` of `restricted_thread_pool_executor`
- Issue #6171¹³⁸⁰ - Release Tag Name Request
- Issue #6162¹³⁸¹ - Current master does not compile on ROSTAM
- Issue #6156¹³⁸² - `hpxcxx` does not work if `HPX_WITH_PKGCONFIG=OFF`
- Issue #6108¹³⁸³ - `cxx17_aligned_new.cpp` on `msvc` fails due to wrong pragma directive
- Issue #6045¹³⁸⁴ - Can't call nullary callables wrapped with `hpx::unwrapping`
- Issue #6013¹³⁸⁵ - Unable to build subprojects `hpx_collectives/hpx_compute` with MSVC
- Issue #6008¹³⁸⁶ - Missing `constexpr` default constructor for `hpx::mutex`
- Issue #5999¹³⁸⁷ - Add HPX Conda package to conda-forge
- Issue #5998¹³⁸⁸ - Serializing multiple arguments when applying distributed action results in segfault
- Issue #5958¹³⁸⁹ - HPX 1.8.0 and Blaze issues
- Issue #5908¹³⁹⁰ - Windows: duplicated symbols in static builds
- Issue #5802¹³⁹¹ - Lost status `is_ready` from future

¹³⁷⁶ <https://github.com/TheHPXProject/hpx/issues/6203>

¹³⁷⁷ <https://github.com/TheHPXProject/hpx/issues/6196>

¹³⁷⁸ <https://github.com/TheHPXProject/hpx/issues/6191>

¹³⁷⁹ <https://github.com/TheHPXProject/hpx/issues/6184>

¹³⁸⁰ <https://github.com/TheHPXProject/hpx/issues/6171>

¹³⁸¹ <https://github.com/TheHPXProject/hpx/issues/6162>

¹³⁸² <https://github.com/TheHPXProject/hpx/issues/6156>

¹³⁸³ <https://github.com/TheHPXProject/hpx/issues/6108>

¹³⁸⁴ <https://github.com/TheHPXProject/hpx/issues/6045>

¹³⁸⁵ <https://github.com/TheHPXProject/hpx/issues/6013>

¹³⁸⁶ <https://github.com/TheHPXProject/hpx/issues/6008>

¹³⁸⁷ <https://github.com/TheHPXProject/hpx/issues/5999>

¹³⁸⁸ <https://github.com/TheHPXProject/hpx/issues/5998>

¹³⁸⁹ <https://github.com/TheHPXProject/hpx/issues/5958>

¹³⁹⁰ <https://github.com/TheHPXProject/hpx/issues/5908>

¹³⁹¹ <https://github.com/TheHPXProject/hpx/issues/5802>

- Issue #5767¹³⁹² - Performance drop on Piz Daint
- Issue #5752¹³⁹³ - Implement stride_view from P1899 (experimental)
- Issue #5744¹³⁹⁴ - HPX_WITH_FETCH_ASIO not working on Ookami
- Issue #5561¹³⁹⁵ - Possible race condition in helper thread / hpx::cout

Closed pull requests

- PR #6228¹³⁹⁶ - Fixing algorithms for zero length sequences when run with s/r scheduler
- PR #6227¹³⁹⁷ - Reliably disable background work when no networking is enabled
- PR #6225¹³⁹⁸ - Make heap fails in par for small sized heaps #6217
- PR #6222¹³⁹⁹ - Add documentation for *hpx::post*
- PR #6221¹⁴⁰⁰ - Fix segmented algorithms tests
- PR #6218¹⁴⁰¹ - Creating INSTALL component 'runtime' to enable installing binaries only
- PR #6216¹⁴⁰² - added tests for set_difference, updated set_operation.hpp to fix #6198
- PR #6213¹⁴⁰³ - Modernize and streamline MPI parcellport
- PR #6211¹⁴⁰⁴ - Modernize modules of level 11, 12, and 13
- PR #6210¹⁴⁰⁵ - Fixing MPI parcellport initialization if MPI is initialized outside of HPX
- PR #6209¹⁴⁰⁶ - Prevent thread stealing during scheduler shutdown
- PR #6208¹⁴⁰⁷ - Fix the compilation warning in the MPI parcellport with gcc 11.2
- PR #6207¹⁴⁰⁸ - Automatically enable Boost.Context when compiling for arm64.
- PR #6206¹⁴⁰⁹ - Update CMakeLists.txt
- PR #6205¹⁴¹⁰ - Do not generate hpxcxx if support for pkgconfig was disabled
- PR #6204¹⁴¹¹ - Use **LRT_** instead of **LAPP_** logging in barrier implementation
- PR #6202¹⁴¹² - Fixing Fedora build errors on Power systems
- PR #6201¹⁴¹³ - Update the LCI parcellport documents

¹³⁹² <https://github.com/TheHPXProject/hpx/issues/5767>

¹³⁹³ <https://github.com/TheHPXProject/hpx/issues/5752>

¹³⁹⁴ <https://github.com/TheHPXProject/hpx/issues/5744>

¹³⁹⁵ <https://github.com/TheHPXProject/hpx/issues/5561>

¹³⁹⁶ <https://github.com/TheHPXProject/hpx/pull/6228>

¹³⁹⁷ <https://github.com/TheHPXProject/hpx/pull/6227>

¹³⁹⁸ <https://github.com/TheHPXProject/hpx/pull/6225>

¹³⁹⁹ <https://github.com/TheHPXProject/hpx/pull/6222>

¹⁴⁰⁰ <https://github.com/TheHPXProject/hpx/pull/6221>

¹⁴⁰¹ <https://github.com/TheHPXProject/hpx/pull/6218>

¹⁴⁰² <https://github.com/TheHPXProject/hpx/pull/6216>

¹⁴⁰³ <https://github.com/TheHPXProject/hpx/pull/6213>

¹⁴⁰⁴ <https://github.com/TheHPXProject/hpx/pull/6211>

¹⁴⁰⁵ <https://github.com/TheHPXProject/hpx/pull/6210>

¹⁴⁰⁶ <https://github.com/TheHPXProject/hpx/pull/6209>

¹⁴⁰⁷ <https://github.com/TheHPXProject/hpx/pull/6208>

¹⁴⁰⁸ <https://github.com/TheHPXProject/hpx/pull/6207>

¹⁴⁰⁹ <https://github.com/TheHPXProject/hpx/pull/6206>

¹⁴¹⁰ <https://github.com/TheHPXProject/hpx/pull/6205>

¹⁴¹¹ <https://github.com/TheHPXProject/hpx/pull/6204>

¹⁴¹² <https://github.com/TheHPXProject/hpx/pull/6202>

¹⁴¹³ <https://github.com/TheHPXProject/hpx/pull/6201>

- PR #6200¹⁴¹⁴ - Par link jobs
- PR #6197¹⁴¹⁵ - LCI parcelport: add doc, upgrade to v1.7.4, refactor cmake autofetch.
- PR #6195¹⁴¹⁶ - Change the default tag of autofetch LCI to v1.7.3.
- PR #6192¹⁴¹⁷ - Fix page *Writing single-node applications*
- PR #6189¹⁴¹⁸ - Making sure restricted_thread_pool_executor properly reports used number of cores
- PR #6187¹⁴¹⁹ - Enable using for_loop with range generators
- PR #6186¹⁴²⁰ - thread_support/CMakeLists: Fix build issue
- PR #6185¹⁴²¹ - Fix EVE datapar with cxx_standard less than 20
- PR #6183¹⁴²² - Update CI integration for EVE
- PR #6182¹⁴²³ - Fixing performance regressions
- PR #6181¹⁴²⁴ - LCI parcelport: backlog queue, aggregation, separate devices, and more
- PR #6180¹⁴²⁵ - Fixing use of for_loop with rebound execution policy (using *.with()*)
- PR #6179¹⁴²⁶ - Taking predicates for algorithms by value
- PR #6178¹⁴²⁷ - Changes needed to make chapel_hpx examples work
- PR #6176¹⁴²⁸ - Fixing warnings that were generated by PVS Studio
- PR #6174¹⁴²⁹ - Replace boost::integer::gcd with std::gcd
- PR #6172¹⁴³⁰ - [Docs] Fix example of how to run single/specific test(s)
- PR #6170¹⁴³¹ - Adding missing fallback for processing_units_count customization point
- PR #6169¹⁴³² - LCI parcelport: bypass the parcel queue and connection cache.
- PR #6167¹⁴³³ - Add create_local_communicator API function
- PR #6166¹⁴³⁴ - Add missing header for std::intmax_t
- PR #6165¹⁴³⁵ - Attempt to work around MSVC problem
- PR #6161¹⁴³⁶ - Update EVE integration

¹⁴¹⁴ <https://github.com/TheHPXProject/hpx/pull/6200>

¹⁴¹⁵ <https://github.com/TheHPXProject/hpx/pull/6197>

¹⁴¹⁶ <https://github.com/TheHPXProject/hpx/pull/6195>

¹⁴¹⁷ <https://github.com/TheHPXProject/hpx/pull/6192>

¹⁴¹⁸ <https://github.com/TheHPXProject/hpx/pull/6189>

¹⁴¹⁹ <https://github.com/TheHPXProject/hpx/pull/6187>

¹⁴²⁰ <https://github.com/TheHPXProject/hpx/pull/6186>

¹⁴²¹ <https://github.com/TheHPXProject/hpx/pull/6185>

¹⁴²² <https://github.com/TheHPXProject/hpx/pull/6183>

¹⁴²³ <https://github.com/TheHPXProject/hpx/pull/6182>

¹⁴²⁴ <https://github.com/TheHPXProject/hpx/pull/6181>

¹⁴²⁵ <https://github.com/TheHPXProject/hpx/pull/6180>

¹⁴²⁶ <https://github.com/TheHPXProject/hpx/pull/6179>

¹⁴²⁷ <https://github.com/TheHPXProject/hpx/pull/6178>

¹⁴²⁸ <https://github.com/TheHPXProject/hpx/pull/6176>

¹⁴²⁹ <https://github.com/TheHPXProject/hpx/pull/6174>

¹⁴³⁰ <https://github.com/TheHPXProject/hpx/pull/6172>

¹⁴³¹ <https://github.com/TheHPXProject/hpx/pull/6170>

¹⁴³² <https://github.com/TheHPXProject/hpx/pull/6169>

¹⁴³³ <https://github.com/TheHPXProject/hpx/pull/6167>

¹⁴³⁴ <https://github.com/TheHPXProject/hpx/pull/6166>

¹⁴³⁵ <https://github.com/TheHPXProject/hpx/pull/6165>

¹⁴³⁶ <https://github.com/TheHPXProject/hpx/pull/6161>

- PR #6160¹⁴³⁷ - More cleanup for module levels 0 to 10
- PR #6159¹⁴³⁸ - Fix minor spelling mistake in generate_issue_pr_list.sh
- PR #6158¹⁴³⁹ - Update documentation in *writing single-node applications* page
- PR #6157¹⁴⁴⁰ - Improve index_queue_spawning
- PR #6154¹⁴⁴¹ - Avoid performing late command line handling twice in distributed runtime
- PR #6152¹⁴⁴² - The -rd and -mr options didn't work, and they should have been -rd and -mr
- PR #6151¹⁴⁴³ - Refactoring the Manual page in documentation
- PR #6148¹⁴⁴⁴ - Investigate the failure of the LCI parcelport.
- PR #6147¹⁴⁴⁵ - Make posix co-routine stacks non-executable
- PR #6146¹⁴⁴⁶ - Avoid ambiguities wrt tag_invoke
- PR #6144¹⁴⁴⁷ - General improvements to scheduling and related fixes
- PR #6143¹⁴⁴⁸ - Add list of new namespaces for new release
- PR #6140¹⁴⁴⁹ - Fixing background scheduler to properly exit in the end
- PR #6139¹⁴⁵⁰ - [P2300] execution: Cleanup coroutines integration and improve ADL isolation
- PR #6137¹⁴⁵¹ - Adding example of a simple master/slave distributed application
- PR #6136¹⁴⁵² - Deprecate *execution::experimental::task_group* in favor of *experimental::task_group*
- PR #6135¹⁴⁵³ - Fixing warnings reported by MSVC analysis
- PR #6134¹⁴⁵⁴ - Adding notification function for parcelports to be called after early parcel handling
- PR #6132¹⁴⁵⁵ - Fixing to_non_par() for parallel simd policies
- PR #6131¹⁴⁵⁶ - modernize modules from level 25
- PR #6130¹⁴⁵⁷ - Remove the mutex lock in the critical path of get_partitioner.
- PR #6129¹⁴⁵⁸ - Modernize module from levels 22, 23
- PR #6127¹⁴⁵⁹ - Working around gccV9 problem that prevent us from storing enum classes in bit fields

¹⁴³⁷ <https://github.com/TheHPXProject/hpx/pull/6160>

¹⁴³⁸ <https://github.com/TheHPXProject/hpx/pull/6159>

¹⁴³⁹ <https://github.com/TheHPXProject/hpx/pull/6158>

¹⁴⁴⁰ <https://github.com/TheHPXProject/hpx/pull/6157>

¹⁴⁴¹ <https://github.com/TheHPXProject/hpx/pull/6154>

¹⁴⁴² <https://github.com/TheHPXProject/hpx/pull/6152>

¹⁴⁴³ <https://github.com/TheHPXProject/hpx/pull/6151>

¹⁴⁴⁴ <https://github.com/TheHPXProject/hpx/pull/6148>

¹⁴⁴⁵ <https://github.com/TheHPXProject/hpx/pull/6147>

¹⁴⁴⁶ <https://github.com/TheHPXProject/hpx/pull/6146>

¹⁴⁴⁷ <https://github.com/TheHPXProject/hpx/pull/6144>

¹⁴⁴⁸ <https://github.com/TheHPXProject/hpx/pull/6143>

¹⁴⁴⁹ <https://github.com/TheHPXProject/hpx/pull/6140>

¹⁴⁵⁰ <https://github.com/TheHPXProject/hpx/pull/6139>

¹⁴⁵¹ <https://github.com/TheHPXProject/hpx/pull/6137>

¹⁴⁵² <https://github.com/TheHPXProject/hpx/pull/6136>

¹⁴⁵³ <https://github.com/TheHPXProject/hpx/pull/6135>

¹⁴⁵⁴ <https://github.com/TheHPXProject/hpx/pull/6134>

¹⁴⁵⁵ <https://github.com/TheHPXProject/hpx/pull/6132>

¹⁴⁵⁶ <https://github.com/TheHPXProject/hpx/pull/6131>

¹⁴⁵⁷ <https://github.com/TheHPXProject/hpx/pull/6130>

¹⁴⁵⁸ <https://github.com/TheHPXProject/hpx/pull/6129>

¹⁴⁵⁹ <https://github.com/TheHPXProject/hpx/pull/6127>

- PR #6126¹⁴⁶⁰ - Deprecate `hpx::parallel::task_block` in favor of `hpx::experimental::task_block`
- PR #6125¹⁴⁶¹ - Making sure `sync_wait` compiles when used with an lvalue sender involving bulk
- PR #6124¹⁴⁶² - Fixing use of `any_sender` in combination with `when_all`
- PR #6123¹⁴⁶³ - Fixed issues found by PVS-Studio
- PR #6121¹⁴⁶⁴ - Modernize modules of level 21, 22
- PR #6120¹⁴⁶⁵ - Use `index_queue` for parallel executors `bulk_async_execute`
- PR #6119¹⁴⁶⁶ - Update `CMakeLists.txt`
- PR #6118¹⁴⁶⁷ - Modernize modules from level 17, 18, 19, and 20
- PR #6117¹⁴⁶⁸ - Initialize `buffer_allocate_time_` to 0
- PR #6116¹⁴⁶⁹ - Add new command line argument `-hpx:loopback_network`
- PR #6115¹⁴⁷⁰ - Modernize modules of levels 14, 15, and 16
- PR #6114¹⁴⁷¹ - Enhance the formatting of the documentation
- PR #6113¹⁴⁷² - Modernize modules in module level 11, 12, and 13
- PR #6112¹⁴⁷³ - Modernize modules from levels 9 and 10
- PR #6111¹⁴⁷⁴ - Modernize all modules from module level 8
- PR #6110¹⁴⁷⁵ - Use pragma error directive to report warnings as errors on msvc
- PR #6109¹⁴⁷⁶ - Modernize serialization module
- PR #6107¹⁴⁷⁷ - Modernize error module
- PR #6106¹⁴⁷⁸ - Modernizing modules of levels 0 to 5
- PR #6105¹⁴⁷⁹ - Optimizations on LCI parcelport: merge small messages; remove sender mutex lock.
- PR #6104¹⁴⁸⁰ - Adding parameters API: `measure_iteration`
- PR #6103¹⁴⁸¹ - Document `task_group` and include in Public API
- PR #6102¹⁴⁸² - Prevent warnings generated by clang-cl

¹⁴⁶⁰ <https://github.com/TheHPXProject/hpx/pull/6126>

¹⁴⁶¹ <https://github.com/TheHPXProject/hpx/pull/6125>

¹⁴⁶² <https://github.com/TheHPXProject/hpx/pull/6124>

¹⁴⁶³ <https://github.com/TheHPXProject/hpx/pull/6123>

¹⁴⁶⁴ <https://github.com/TheHPXProject/hpx/pull/6121>

¹⁴⁶⁵ <https://github.com/TheHPXProject/hpx/pull/6120>

¹⁴⁶⁶ <https://github.com/TheHPXProject/hpx/pull/6119>

¹⁴⁶⁷ <https://github.com/TheHPXProject/hpx/pull/6118>

¹⁴⁶⁸ <https://github.com/TheHPXProject/hpx/pull/6117>

¹⁴⁶⁹ <https://github.com/TheHPXProject/hpx/pull/6116>

¹⁴⁷⁰ <https://github.com/TheHPXProject/hpx/pull/6115>

¹⁴⁷¹ <https://github.com/TheHPXProject/hpx/pull/6114>

¹⁴⁷² <https://github.com/TheHPXProject/hpx/pull/6113>

¹⁴⁷³ <https://github.com/TheHPXProject/hpx/pull/6112>

¹⁴⁷⁴ <https://github.com/TheHPXProject/hpx/pull/6111>

¹⁴⁷⁵ <https://github.com/TheHPXProject/hpx/pull/6110>

¹⁴⁷⁶ <https://github.com/TheHPXProject/hpx/pull/6109>

¹⁴⁷⁷ <https://github.com/TheHPXProject/hpx/pull/6107>

¹⁴⁷⁸ <https://github.com/TheHPXProject/hpx/pull/6106>

¹⁴⁷⁹ <https://github.com/TheHPXProject/hpx/pull/6105>

¹⁴⁸⁰ <https://github.com/TheHPXProject/hpx/pull/6104>

¹⁴⁸¹ <https://github.com/TheHPXProject/hpx/pull/6103>

¹⁴⁸² <https://github.com/TheHPXProject/hpx/pull/6102>

- PR #6101¹⁴⁸³ - Using more fold expressions
- PR #6100¹⁴⁸⁴ - Deprecate `hpx::parallel::reduce_by_key` in favor of `hpx::experimental::reduce_by_key`
- PR #6098¹⁴⁸⁵ - Forking Boost.Lockfree
- PR #6096¹⁴⁸⁶ - Forking Boost.Tokenizer
- PR #6095¹⁴⁸⁷ - Replacing facilities from Boost.Range
- PR #6094¹⁴⁸⁸ - Removing object_semaphore
- PR #6093¹⁴⁸⁹ - Replace `boost::string_ref` with `std::string_view`
- PR #6092¹⁴⁹⁰ - Use C++17 `static_assert` where possible
- PR #6091¹⁴⁹¹ - Replace artificial sequencing with fold expressions
- PR #6090¹⁴⁹² - Fixing use of `get_chunk_size` customization point
- PR #6088¹⁴⁹³ - Add/fix Public API documentation
- PR #6086¹⁴⁹⁴ - Deprecate `hpx::util::unlock_guard` in favor of `hpx::unlock_guard`
- PR #6085¹⁴⁹⁵ - Add experimental sycl integration/executor
- PR #6084¹⁴⁹⁶ - Renaming `hpx::apply` and friends to `hpx::post`
- PR #6083¹⁴⁹⁷ - Using `if constexpr` instead of tag-dispatching, where possible
- PR #6082¹⁴⁹⁸ - Replace `util::always_void_t` with `std::void_t`
- PR #6081¹⁴⁹⁹ - Update github actions to avoid warnings
- PR #6080¹⁵⁰⁰ - Disable some tests that fail on LCI
- PR #6079¹⁵⁰¹ - Adding more natvis files, correct existing
- PR #6078¹⁵⁰² - Changing target name of `memory_counters` component
- PR #6077¹⁵⁰³ - Making default constructor of `hpx::mutex` `constexpr`
- PR #6076¹⁵⁰⁴ - Cleaning up functionality that was deprecated in V1.7
- PR #6075¹⁵⁰⁵ - Remove conditional code for gcc V7 and below

¹⁴⁸³ <https://github.com/TheHPXProject/hpx/pull/6101>

¹⁴⁸⁴ <https://github.com/TheHPXProject/hpx/pull/6100>

¹⁴⁸⁵ <https://github.com/TheHPXProject/hpx/pull/6098>

¹⁴⁸⁶ <https://github.com/TheHPXProject/hpx/pull/6096>

¹⁴⁸⁷ <https://github.com/TheHPXProject/hpx/pull/6095>

¹⁴⁸⁸ <https://github.com/TheHPXProject/hpx/pull/6094>

¹⁴⁸⁹ <https://github.com/TheHPXProject/hpx/pull/6093>

¹⁴⁹⁰ <https://github.com/TheHPXProject/hpx/pull/6092>

¹⁴⁹¹ <https://github.com/TheHPXProject/hpx/pull/6091>

¹⁴⁹² <https://github.com/TheHPXProject/hpx/pull/6090>

¹⁴⁹³ <https://github.com/TheHPXProject/hpx/pull/6088>

¹⁴⁹⁴ <https://github.com/TheHPXProject/hpx/pull/6086>

¹⁴⁹⁵ <https://github.com/TheHPXProject/hpx/pull/6085>

¹⁴⁹⁶ <https://github.com/TheHPXProject/hpx/pull/6084>

¹⁴⁹⁷ <https://github.com/TheHPXProject/hpx/pull/6083>

¹⁴⁹⁸ <https://github.com/TheHPXProject/hpx/pull/6082>

¹⁴⁹⁹ <https://github.com/TheHPXProject/hpx/pull/6081>

¹⁵⁰⁰ <https://github.com/TheHPXProject/hpx/pull/6080>

¹⁵⁰¹ <https://github.com/TheHPXProject/hpx/pull/6079>

¹⁵⁰² <https://github.com/TheHPXProject/hpx/pull/6078>

¹⁵⁰³ <https://github.com/TheHPXProject/hpx/pull/6077>

¹⁵⁰⁴ <https://github.com/TheHPXProject/hpx/pull/6076>

¹⁵⁰⁵ <https://github.com/TheHPXProject/hpx/pull/6075>

- PR #6074¹⁵⁰⁶ - Fixing compilation issues on gcc V8
- PR #6073¹⁵⁰⁷ - Fixing PAPI counter component compilation
- PR #6072¹⁵⁰⁸ - Adding `ex::when_all_vector`
- PR #6071¹⁵⁰⁹ - Making `get_forward_progress_guarantee_t` specializations `constexpr`
- PR #6070¹⁵¹⁰ - Implement P2690 for our algorithms
- PR #6069¹⁵¹¹ - Do not check for cancellation during each iteration but only once per partition
- PR #6068¹⁵¹² - Prevent using `task` and `non_task` as a CPO
- PR #6067¹⁵¹³ - Deprecated `hpx::util::mem_fn` in favor of `hpx::mem_fn`
- PR #6066¹⁵¹⁴ - Create `codeql.yml`
- PR #6064¹⁵¹⁵ - Adapting `adjacent_difference` for S/R execution
- PR #6063¹⁵¹⁶ - Modernize `iterator_support` module
- PR #6062¹⁵¹⁷ - Make sure wrapping executor does not go out of scope prematurely
- PR #6061¹⁵¹⁸ - Minor fix in `small_vector` (from upstream)
- PR #6060¹⁵¹⁹ - Allow to disable registering signal handlers
- PR #6059¹⁵²⁰ - [P2300] Fix: `declval` cannot be ODR used
- PR #6058¹⁵²¹ - Avoid ambiguity for `hpx::get` used with `std::variant`
- PR #6057¹⁵²² - Create a dedicated thread pool to run `LCI_progress`.
- PR #6056¹⁵²³ - Fix coroutine test for clang
- PR #6055¹⁵²⁴ - Patches needed to be able to build HPX 1.8.1 on various platforms
- PR #6054¹⁵²⁵ - Use MSVC specific attribute `[[msvc::no_unique_address]]`
- PR #6052¹⁵²⁶ - Deprecated `hpx::util::invoke_fused` in favor of `hpx::invoke_fused`
- PR #6051¹⁵²⁷ - Add non-contiguous index queue and use it in `thread_pool_bulk_scheduler`
- PR #6049¹⁵²⁸ - Crosscompile arm sve

¹⁵⁰⁶ <https://github.com/TheHPXProject/hpx/pull/6074>

¹⁵⁰⁷ <https://github.com/TheHPXProject/hpx/pull/6073>

¹⁵⁰⁸ <https://github.com/TheHPXProject/hpx/pull/6072>

¹⁵⁰⁹ <https://github.com/TheHPXProject/hpx/pull/6071>

¹⁵¹⁰ <https://github.com/TheHPXProject/hpx/pull/6070>

¹⁵¹¹ <https://github.com/TheHPXProject/hpx/pull/6069>

¹⁵¹² <https://github.com/TheHPXProject/hpx/pull/6068>

¹⁵¹³ <https://github.com/TheHPXProject/hpx/pull/6067>

¹⁵¹⁴ <https://github.com/TheHPXProject/hpx/pull/6066>

¹⁵¹⁵ <https://github.com/TheHPXProject/hpx/pull/6064>

¹⁵¹⁶ <https://github.com/TheHPXProject/hpx/pull/6063>

¹⁵¹⁷ <https://github.com/TheHPXProject/hpx/pull/6062>

¹⁵¹⁸ <https://github.com/TheHPXProject/hpx/pull/6061>

¹⁵¹⁹ <https://github.com/TheHPXProject/hpx/pull/6060>

¹⁵²⁰ <https://github.com/TheHPXProject/hpx/pull/6059>

¹⁵²¹ <https://github.com/TheHPXProject/hpx/pull/6058>

¹⁵²² <https://github.com/TheHPXProject/hpx/pull/6057>

¹⁵²³ <https://github.com/TheHPXProject/hpx/pull/6056>

¹⁵²⁴ <https://github.com/TheHPXProject/hpx/pull/6055>

¹⁵²⁵ <https://github.com/TheHPXProject/hpx/pull/6054>

¹⁵²⁶ <https://github.com/TheHPXProject/hpx/pull/6052>

¹⁵²⁷ <https://github.com/TheHPXProject/hpx/pull/6051>

¹⁵²⁸ <https://github.com/TheHPXProject/hpx/pull/6049>

- PR #6048¹⁵²⁹ - Deprecated `hpx::util::invoke` in favor of `hpx::invoke`
- PR #6047¹⁵³⁰ - Separating `binary_semaphore` into its own file
- PR #6046¹⁵³¹ - Support using unwrapping with nullary function objects
- PR #6044¹⁵³² - Generalize the use of `then()` and dataflow
- PR #6043¹⁵³³ - Clean up `scan_partitioner`
- PR #6042¹⁵³⁴ - Modernize dataflow API
- PR #6041¹⁵³⁵ - docs: document semaphores
- PR #6040¹⁵³⁶ - Add/Fix documentation of Public API page
- PR #6039¹⁵³⁷ - remove MPI dependency when only using LCI parcelport
- PR #6038¹⁵³⁸ - Clean up command line handling
- PR #6037¹⁵³⁹ - Avoid performing parcel related background work if networking is disabled
- PR #6036¹⁵⁴⁰ - Support new datapar backend : SVE
- PR #6035¹⁵⁴¹ - Simplify datapar replace copy if
- PR #6034¹⁵⁴² - Add/Fix documentation of Public API
- PR #6033¹⁵⁴³ - Support for data-parallelism for `replace`, `replace_if`, `replace_copy`, `replace_copy_if` algorithms
- PR #6032¹⁵⁴⁴ - Add documentation in public API
- PR #6031¹⁵⁴⁵ - Expose available cache sizes from topology object
- PR #6030¹⁵⁴⁶ - Adding parcelport initialization hook for resource partitioner operation
- PR #6029¹⁵⁴⁷ - Simplify startup code
- PR #6027¹⁵⁴⁸ - Add/Fix documentation in Public API page
- PR #6026¹⁵⁴⁹ - add option `hpx:force_ipv4` to force resolving hostnames to ipv4 addresses
- PR #6025¹⁵⁵⁰ - build(docs): remove leftover sections
- PR #6023¹⁵⁵¹ - Minor fixes on “How to build on Windows”

¹⁵²⁹ <https://github.com/TheHPXProject/hpx/pull/6048>

¹⁵³⁰ <https://github.com/TheHPXProject/hpx/pull/6047>

¹⁵³¹ <https://github.com/TheHPXProject/hpx/pull/6046>

¹⁵³² <https://github.com/TheHPXProject/hpx/pull/6044>

¹⁵³³ <https://github.com/TheHPXProject/hpx/pull/6043>

¹⁵³⁴ <https://github.com/TheHPXProject/hpx/pull/6042>

¹⁵³⁵ <https://github.com/TheHPXProject/hpx/pull/6041>

¹⁵³⁶ <https://github.com/TheHPXProject/hpx/pull/6040>

¹⁵³⁷ <https://github.com/TheHPXProject/hpx/pull/6039>

¹⁵³⁸ <https://github.com/TheHPXProject/hpx/pull/6038>

¹⁵³⁹ <https://github.com/TheHPXProject/hpx/pull/6037>

¹⁵⁴⁰ <https://github.com/TheHPXProject/hpx/pull/6036>

¹⁵⁴¹ <https://github.com/TheHPXProject/hpx/pull/6035>

¹⁵⁴² <https://github.com/TheHPXProject/hpx/pull/6034>

¹⁵⁴³ <https://github.com/TheHPXProject/hpx/pull/6033>

¹⁵⁴⁴ <https://github.com/TheHPXProject/hpx/pull/6032>

¹⁵⁴⁵ <https://github.com/TheHPXProject/hpx/pull/6031>

¹⁵⁴⁶ <https://github.com/TheHPXProject/hpx/pull/6030>

¹⁵⁴⁷ <https://github.com/TheHPXProject/hpx/pull/6029>

¹⁵⁴⁸ <https://github.com/TheHPXProject/hpx/pull/6027>

¹⁵⁴⁹ <https://github.com/TheHPXProject/hpx/pull/6026>

¹⁵⁵⁰ <https://github.com/TheHPXProject/hpx/pull/6025>

¹⁵⁵¹ <https://github.com/TheHPXProject/hpx/pull/6023>

- PR #6022¹⁵⁵² - build(doxy): don't extract private members
- PR #6021¹⁵⁵³ - Adding pu_mask to thread_pool_bulk_scheduler
- PR #6020¹⁵⁵⁴ - docs: add cppref NamedRequirements support
- PR #6018¹⁵⁵⁵ - Unseq adaptation for for_each, transform, reduce, transform_reduce, etc.
- PR #6017¹⁵⁵⁶ - loop and transform_loop unseq adaptation
- PR #6016¹⁵⁵⁷ - Config and structural updates to support unseq implementation
- PR #6015¹⁵⁵⁸ - Integrating sync_wait & sync_wait_with_variant
- PR #6012¹⁵⁵⁹ - docs: add missing links to public api
- PR #6009¹⁵⁶⁰ - Fixing sender&receiver integration with for_each and for_loop
- PR #6007¹⁵⁶¹ - docs: add docs for mutex.hpp
- PR #6006¹⁵⁶² - Relax future::is_ready where possible
- PR #6005¹⁵⁶³ - reshuffle header tests to different instances
- PR #6004¹⁵⁶⁴ - Add documentation Public API
- PR #6003¹⁵⁶⁵ - Always exporting get_component_name implementations
- PR #6002¹⁵⁶⁶ - Making sure that default constructible arguments are properly constructed during deserialization
- PR #5996¹⁵⁶⁷ - Add back explicit template parameters to lock_guards for nvcc
- PR #5994¹⁵⁶⁸ - Fix CTRL+C on windows
- PR #5993¹⁵⁶⁹ - Using EVE requires C++20
- PR #5992¹⁵⁷⁰ - This properly terminates an application on Ctrl-C on Windows
- PR #5991¹⁵⁷¹ - Support IPV6 on command line for explicit network initialization
- PR #5990¹⁵⁷² - P2300 enhancements
- PR #5989¹⁵⁷³ - Fix missing documentation in Public API page
- PR #5987¹⁵⁷⁴ - Attempting to fix timed executor API

¹⁵⁵² <https://github.com/TheHPXProject/hpx/pull/6022>

¹⁵⁵³ <https://github.com/TheHPXProject/hpx/pull/6021>

¹⁵⁵⁴ <https://github.com/TheHPXProject/hpx/pull/6020>

¹⁵⁵⁵ <https://github.com/TheHPXProject/hpx/pull/6018>

¹⁵⁵⁶ <https://github.com/TheHPXProject/hpx/pull/6017>

¹⁵⁵⁷ <https://github.com/TheHPXProject/hpx/pull/6016>

¹⁵⁵⁸ <https://github.com/TheHPXProject/hpx/pull/6015>

¹⁵⁵⁹ <https://github.com/TheHPXProject/hpx/pull/6012>

¹⁵⁶⁰ <https://github.com/TheHPXProject/hpx/pull/6009>

¹⁵⁶¹ <https://github.com/TheHPXProject/hpx/pull/6007>

¹⁵⁶² <https://github.com/TheHPXProject/hpx/pull/6006>

¹⁵⁶³ <https://github.com/TheHPXProject/hpx/pull/6005>

¹⁵⁶⁴ <https://github.com/TheHPXProject/hpx/pull/6004>

¹⁵⁶⁵ <https://github.com/TheHPXProject/hpx/pull/6003>

¹⁵⁶⁶ <https://github.com/TheHPXProject/hpx/pull/6002>

¹⁵⁶⁷ <https://github.com/TheHPXProject/hpx/pull/5996>

¹⁵⁶⁸ <https://github.com/TheHPXProject/hpx/pull/5994>

¹⁵⁶⁹ <https://github.com/TheHPXProject/hpx/pull/5993>

¹⁵⁷⁰ <https://github.com/TheHPXProject/hpx/pull/5992>

¹⁵⁷¹ <https://github.com/TheHPXProject/hpx/pull/5991>

¹⁵⁷² <https://github.com/TheHPXProject/hpx/pull/5990>

¹⁵⁷³ <https://github.com/TheHPXProject/hpx/pull/5989>

¹⁵⁷⁴ <https://github.com/TheHPXProject/hpx/pull/5987>

- PR #5986¹⁵⁷⁵ - Fix warnings when building docs
- PR #5985¹⁵⁷⁶ - Re-add deprecated tag_policy_tag et.al. types that were removed in V1.8.1
- PR #5981¹⁵⁷⁷ - docs: add docs for condition_variable.hpp
- PR #5980¹⁵⁷⁸ - More work on execution::read
- PR #5979¹⁵⁷⁹ - Remove support for clang-v8 and clang-v9, switch LSU clang-v13 to C++17
- PR #5977¹⁵⁸⁰ - fix: Compilation errors for -std=c++17 builders
- PR #5975¹⁵⁸¹ - docs: fix & improve parallel algorithms documentation 5
- PR #5974¹⁵⁸² - [P2300] Adapt get completion signatures for awaitable senders
- PR #5973¹⁵⁸³ - defaults boost.context on riscv64
- PR #5972¹⁵⁸⁴ - Fix documentation for container algorithms
- PR #5971¹⁵⁸⁵ - added logic to detect riscv compiler configured for 64 bit target
- PR #5968¹⁵⁸⁶ - adds risc-v 64 bit support
- PR #5967¹⁵⁸⁷ - Adding missing pieces to sync_wait, adding run_loop
- PR #5966¹⁵⁸⁸ - docs: fix & improve parallel algorithms documentation 4
- PR #5965¹⁵⁸⁹ - Fixing inspect problems, adding missing header file
- PR #5962¹⁵⁹⁰ - Changes in html page of documentation
- PR #5961¹⁵⁹¹ - Prevent stalling during shutdown when running hello_world_distributed
- PR #5955¹⁵⁹² - Fix documentation for container algorithms
- PR #5952¹⁵⁹³ - docs: fix & improve parallel algorithms documentation 3
- PR #5950¹⁵⁹⁴ - Change executors to directly implement the executor CPOs
- PR #5949¹⁵⁹⁵ - Converting async combinators into CPOs
- PR #5948¹⁵⁹⁶ - Adding support for pure sender/receiver based executors to parallel algorithms
- PR #5945¹⁵⁹⁷ - [P2300] Added fundamental coroutine_traits for S/R

¹⁵⁷⁵ <https://github.com/TheHPXProject/hpx/pull/5986>

¹⁵⁷⁶ <https://github.com/TheHPXProject/hpx/pull/5985>

¹⁵⁷⁷ <https://github.com/TheHPXProject/hpx/pull/5981>

¹⁵⁷⁸ <https://github.com/TheHPXProject/hpx/pull/5980>

¹⁵⁷⁹ <https://github.com/TheHPXProject/hpx/pull/5979>

¹⁵⁸⁰ <https://github.com/TheHPXProject/hpx/pull/5977>

¹⁵⁸¹ <https://github.com/TheHPXProject/hpx/pull/5975>

¹⁵⁸² <https://github.com/TheHPXProject/hpx/pull/5974>

¹⁵⁸³ <https://github.com/TheHPXProject/hpx/pull/5973>

¹⁵⁸⁴ <https://github.com/TheHPXProject/hpx/pull/5972>

¹⁵⁸⁵ <https://github.com/TheHPXProject/hpx/pull/5971>

¹⁵⁸⁶ <https://github.com/TheHPXProject/hpx/pull/5968>

¹⁵⁸⁷ <https://github.com/TheHPXProject/hpx/pull/5967>

¹⁵⁸⁸ <https://github.com/TheHPXProject/hpx/pull/5966>

¹⁵⁸⁹ <https://github.com/TheHPXProject/hpx/pull/5965>

¹⁵⁹⁰ <https://github.com/TheHPXProject/hpx/pull/5962>

¹⁵⁹¹ <https://github.com/TheHPXProject/hpx/pull/5961>

¹⁵⁹² <https://github.com/TheHPXProject/hpx/pull/5955>

¹⁵⁹³ <https://github.com/TheHPXProject/hpx/pull/5952>

¹⁵⁹⁴ <https://github.com/TheHPXProject/hpx/pull/5950>

¹⁵⁹⁵ <https://github.com/TheHPXProject/hpx/pull/5949>

¹⁵⁹⁶ <https://github.com/TheHPXProject/hpx/pull/5948>

¹⁵⁹⁷ <https://github.com/TheHPXProject/hpx/pull/5945>

- PR #5883¹⁵⁹⁸ - Optimization on LCI parcelport: uses LCI_putva
- PR #5872¹⁵⁹⁹ - Block fork join executor
- PR #5855¹⁶⁰⁰ - Adding performance test Jenkins builder at LSU

HPX V1.8.1 (Aug 5, 2022)

This is a bugfix release with a few minor additions and resolved problems.

General changes

This patch release adds a number of small new features and fixes a handful of problems discovered since the last release, in particular:

- A lot of work has been done to improve vectorization support for our parallel algorithms. HPX now supports using EVE - the Expressive Vector Engine as a vectorization backend.
- Added a simple average power consumption performance counter.
- Added performance counters related to the use of zero-copy chunks in the networking layer.
- More work was done towards full compatibility with the sender/receivers proposal P2300.
- Fixing `sync_wait` to decay the result types
- Fixed collective operations to properly avoid overalapping consecutive operations on the same communicator.
- Simplified the implementation of our execution policies and added mapping functions between those.
- Fixed performance issues with our implementation of *small_vector*.
- Serialization now works with buffers of unsigned characters.
- Fixing dangling reference in serialization of non-default constructible types
- Fixed static linking on Windows.
- Fixed support for M1/MacOS based architectures.
- Fixed support for gentoo/musl.
- Fixed `hpx::counting_semaphore_var`.
- Properly check start and end bounds for `hpx::for_loop`
- A lot of changes and fixes to the documentation (see <https://hpx-docs.stellar-group.org>).

¹⁵⁹⁸ <https://github.com/TheHPXProject/hpx/pull/5883>

¹⁵⁹⁹ <https://github.com/TheHPXProject/hpx/pull/5872>

¹⁶⁰⁰ <https://github.com/TheHPXProject/hpx/pull/5855>

Breaking changes

- No breaking changes have been introduced.

Closed issues

- [Issue #5964](#)¹⁶⁰¹ - component with multiple inheritance
- [Issue #5946](#)¹⁶⁰² - dll_dlopen.hpp: error: RTLD_DI_ORIGIN was not declared in this scope with musl libc
- [Issue #5925](#)¹⁶⁰³ - Simplify implementation of execution policies
- [Issue #5924](#)¹⁶⁰⁴ - {what}: mmap() failed to allocate thread stack: HPX(unhandled_exception)
- [Issue #5912](#)¹⁶⁰⁵ - collectives all gather hangs if rank 0 is not involved
- [Issue #5902](#)¹⁶⁰⁶ - MPI parcelport issue on Fugaku
- [Issue #5900](#)¹⁶⁰⁷ - Unable to build hello_world_distributed.cpp.
- [Issue #5892](#)¹⁶⁰⁸ - Problems with HPX serialization as a standalone feature. Testcase provided.
- [Issue #5886](#)¹⁶⁰⁹ - Segfault when serializing non default constructible class with stl containers data members
- [Issue #5832](#)¹⁶¹⁰ - Distributed execution crash
- [Issue #5768](#)¹⁶¹¹ - HPX hangs on Perlmutter
- [Issue #5735](#)¹⁶¹² - hpx::for_loop executes without checking start and end bounds
- [Issue #5700](#)¹⁶¹³ - HPX(serialization_error)

Closed pull requests

- [PR #5970](#)¹⁶¹⁴ - Fixing component multiple inheritance
- [PR #5969](#)¹⁶¹⁵ - Fixing sync_wait to avoid dangling references
- [PR #5963](#)¹⁶¹⁶ - Fixing sync_wait to decay the result types
- [PR #5960](#)¹⁶¹⁷ - docs: added name to documentation contributors list
- [PR #5959](#)¹⁶¹⁸ - Fixing sync_wait to decay the result types

¹⁶⁰¹ <https://github.com/TheHPXProject/hpx/issues/5964>

¹⁶⁰² <https://github.com/TheHPXProject/hpx/issues/5946>

¹⁶⁰³ <https://github.com/TheHPXProject/hpx/issues/5925>

¹⁶⁰⁴ <https://github.com/TheHPXProject/hpx/issues/5924>

¹⁶⁰⁵ <https://github.com/TheHPXProject/hpx/issues/5912>

¹⁶⁰⁶ <https://github.com/TheHPXProject/hpx/issues/5902>

¹⁶⁰⁷ <https://github.com/TheHPXProject/hpx/issues/5900>

¹⁶⁰⁸ <https://github.com/TheHPXProject/hpx/issues/5892>

¹⁶⁰⁹ <https://github.com/TheHPXProject/hpx/issues/5886>

¹⁶¹⁰ <https://github.com/TheHPXProject/hpx/issues/5832>

¹⁶¹¹ <https://github.com/TheHPXProject/hpx/issues/5768>

¹⁶¹² <https://github.com/TheHPXProject/hpx/issues/5735>

¹⁶¹³ <https://github.com/TheHPXProject/hpx/issues/5700>

¹⁶¹⁴ <https://github.com/TheHPXProject/hpx/pull/5970>

¹⁶¹⁵ <https://github.com/TheHPXProject/hpx/pull/5969>

¹⁶¹⁶ <https://github.com/TheHPXProject/hpx/pull/5963>

¹⁶¹⁷ <https://github.com/TheHPXProject/hpx/pull/5960>

¹⁶¹⁸ <https://github.com/TheHPXProject/hpx/pull/5959>

- PR #5954¹⁶¹⁹ - refactor: rename itr to correct type (*reduce*)
- PR #5954¹⁶²⁰ - refactor: rename itr to correct type (*reduce*)
- PR #5953¹⁶²¹ - Fixed property handling in hierarchical_spawning
- PR #5951¹⁶²² - Fixing static linking (for Windows)
- PR #5947¹⁶²³ - Fix building on musl.
- PR #5944¹⁶²⁴ - added adaptive_static_chunk_size
- PR #5943¹⁶²⁵ - Fix sync_wait
- PR #5942¹⁶²⁶ - Fix doc warnings
- PR #5941¹⁶²⁷ - Fix sync_wait
- PR #5940¹⁶²⁸ - Protect collective operations against std::vector<bool> idiosyncrasies
- PR #5939¹⁶²⁹ - docs: fix & improve parallel algorithms documentation 2
- PR #5938¹⁶³⁰ - Properly implement generation support for collective operations
- PR #5937¹⁶³¹ - Remove leftover files from PMR based small_vector
- PR #5936¹⁶³² - Adding mapping functions between execution policies
- PR #5935¹⁶³³ - Fixing serialization to work with buffers of unsigned chars
- PR #5934¹⁶³⁴ - Attempting to fix datapar issues on CircleCI
- PR #5933¹⁶³⁵ - Fix documentation for ranges algorithms
- PR #5932¹⁶³⁶ - Remove mimalloc version constraint
- PR #5931¹⁶³⁷ - docs: fix & improve parallel algorithms documentation
- PR #5930¹⁶³⁸ - Add boost to hip builder
- PR #5929¹⁶³⁹ - Apply fixes to M1/MacOS related stack allocation to all relevant spots
- PR #5928¹⁶⁴⁰ - updated context_generic_context to accommodate arm64_arch_8/Apple architecture
- PR #5927¹⁶⁴¹ - Public derivation for counting_semaphore_var

¹⁶¹⁹ <https://github.com/TheHPXProject/hpx/pull/5954>

¹⁶²⁰ <https://github.com/TheHPXProject/hpx/pull/5954>

¹⁶²¹ <https://github.com/TheHPXProject/hpx/pull/5953>

¹⁶²² <https://github.com/TheHPXProject/hpx/pull/5951>

¹⁶²³ <https://github.com/TheHPXProject/hpx/pull/5947>

¹⁶²⁴ <https://github.com/TheHPXProject/hpx/pull/5944>

¹⁶²⁵ <https://github.com/TheHPXProject/hpx/pull/5943>

¹⁶²⁶ <https://github.com/TheHPXProject/hpx/pull/5942>

¹⁶²⁷ <https://github.com/TheHPXProject/hpx/pull/5941>

¹⁶²⁸ <https://github.com/TheHPXProject/hpx/pull/5940>

¹⁶²⁹ <https://github.com/TheHPXProject/hpx/pull/5939>

¹⁶³⁰ <https://github.com/TheHPXProject/hpx/pull/5938>

¹⁶³¹ <https://github.com/TheHPXProject/hpx/pull/5937>

¹⁶³² <https://github.com/TheHPXProject/hpx/pull/5936>

¹⁶³³ <https://github.com/TheHPXProject/hpx/pull/5935>

¹⁶³⁴ <https://github.com/TheHPXProject/hpx/pull/5934>

¹⁶³⁵ <https://github.com/TheHPXProject/hpx/pull/5933>

¹⁶³⁶ <https://github.com/TheHPXProject/hpx/pull/5932>

¹⁶³⁷ <https://github.com/TheHPXProject/hpx/pull/5931>

¹⁶³⁸ <https://github.com/TheHPXProject/hpx/pull/5930>

¹⁶³⁹ <https://github.com/TheHPXProject/hpx/pull/5929>

¹⁶⁴⁰ <https://github.com/TheHPXProject/hpx/pull/5928>

¹⁶⁴¹ <https://github.com/TheHPXProject/hpx/pull/5927>

- PR #5926¹⁶⁴² - Fix doxygen warnings when building documentation
- PR #5923¹⁶⁴³ - Fixing git checkout to reflect latest version tag
- PR #5922¹⁶⁴⁴ - A couple of unrelated changes in support of implementing P1673
- PR #5920¹⁶⁴⁵ - [P2300] enhancements: receiver_of, sender_of improvements
- PR #5917¹⁶⁴⁶ - Fixing various ‘held lock while suspending’ problems
- PR #5916¹⁶⁴⁷ - Fix minor doxygen parsing typo
- PR #5915¹⁶⁴⁸ - docs: fix broken api algo links
- PR #5914¹⁶⁴⁹ - Remove CSS rules - update sphinx version
- PR #5911¹⁶⁵⁰ - Removed references to hpx::vector in comments
- PR #5909¹⁶⁵¹ - Remove stuff which is defined in the header
- PR #5906¹⁶⁵² - Use BUILD_SHARED_LIBS correctly
- PR #5905¹⁶⁵³ - Fix incorrect usage of generator expressions
- PR #5904¹⁶⁵⁴ - Delete FindBZip2.cmake
- PR #5901¹⁶⁵⁵ - Fix #5900
- PR #5899¹⁶⁵⁶ - Replace PMR based version of small_vector
- PR #5897¹⁶⁵⁷ - Add missing “”
- PR #5896¹⁶⁵⁸ - Docs: Add serialization tutorial.
- PR #5895¹⁶⁵⁹ - Update to V1.9.0 on master
- PR #5894¹⁶⁶⁰ - Fix executor_with_thread_hooks example
- PR #5890¹⁶⁶¹ - Adding simple average power consumption performance counter
- PR #5889¹⁶⁶² - Par unseq/unseq adding
- PR #5888¹⁶⁶³ - Support for data-parallelism for reduce, transform reduce, transform_binary_reduce algorithms
- PR #5887¹⁶⁶⁴ - Fixing dangling reference in serialization of non-default constructible types

¹⁶⁴² <https://github.com/TheHPXProject/hpx/pull/5926>

¹⁶⁴³ <https://github.com/TheHPXProject/hpx/pull/5923>

¹⁶⁴⁴ <https://github.com/TheHPXProject/hpx/pull/5922>

¹⁶⁴⁵ <https://github.com/TheHPXProject/hpx/pull/5920>

¹⁶⁴⁶ <https://github.com/TheHPXProject/hpx/pull/5917>

¹⁶⁴⁷ <https://github.com/TheHPXProject/hpx/pull/5916>

¹⁶⁴⁸ <https://github.com/TheHPXProject/hpx/pull/5915>

¹⁶⁴⁹ <https://github.com/TheHPXProject/hpx/pull/5914>

¹⁶⁵⁰ <https://github.com/TheHPXProject/hpx/pull/5911>

¹⁶⁵¹ <https://github.com/TheHPXProject/hpx/pull/5909>

¹⁶⁵² <https://github.com/TheHPXProject/hpx/pull/5906>

¹⁶⁵³ <https://github.com/TheHPXProject/hpx/pull/5905>

¹⁶⁵⁴ <https://github.com/TheHPXProject/hpx/pull/5904>

¹⁶⁵⁵ <https://github.com/TheHPXProject/hpx/pull/5901>

¹⁶⁵⁶ <https://github.com/TheHPXProject/hpx/pull/5899>

¹⁶⁵⁷ <https://github.com/TheHPXProject/hpx/pull/5897>

¹⁶⁵⁸ <https://github.com/TheHPXProject/hpx/pull/5896>

¹⁶⁵⁹ <https://github.com/TheHPXProject/hpx/pull/5895>

¹⁶⁶⁰ <https://github.com/TheHPXProject/hpx/pull/5894>

¹⁶⁶¹ <https://github.com/TheHPXProject/hpx/pull/5890>

¹⁶⁶² <https://github.com/TheHPXProject/hpx/pull/5889>

¹⁶⁶³ <https://github.com/TheHPXProject/hpx/pull/5888>

¹⁶⁶⁴ <https://github.com/TheHPXProject/hpx/pull/5887>

- [PR #5879](#)¹⁶⁶⁵ - New performance counters related to zero-copy chunks.

HPX V1.8.0 (May 18, 2022)

With HPX parallel algorithms been fully adapted to C++20 the new release achieves full conformance with C++20 concurrency and parallelism facilities. HPX now supports all of the algorithms as specified by C++20. We have added support for vectorization to more of our algorithms. Much work has been done towards implementing P2300 (“std::execution”) and implementing the underlying senders/receivers facilities. Finally, The new release comes with a brand new documentation interface!

General changes

- The new documentation can now be found on our webpage: <https://hpx-docs.stellar-group.org>. This includes a completely new and user-friendly interface environment along with restructuring of certain components. The content in the “Quick start”, “Manual” and “Examples” was improved, while the “Build system” page was adapted to include necessary information for newcomers.
- With the vectorization support available in modern hardware architectures HPX now provides new data-parallel vector execution policies `hpx::execution::simd` and `hpx::execution::par_simd` that enable significant speed-up of our parallel algorithm implementations. The following algorithms now support SIMD execution:
 - `copy`, `copy_n`
 - `generate`
 - `adjacent_difference`, `adjacent_find`
 - `all_of`, `any_of`, `none_of`
 - `equal`, `mismatch`,
 - `inner_product`
 - `count`, `count_if`
 - `fill`, `fill_n`
 - `find`, `find_end`, `find_first_of`, `find_if`, `find_if_not`
 - `for_each`, `for_each_n`
 - `generate`, `generate_n`.
- Based on top of P2300 the HPX parallel algorithms now support the pipeline syntax towards an effort to unify their usage along with senders/receivers. The HPX parallel algorithms can now bind with senders/receivers using the pipeline operator.
- Several changes took place on the executors provided by HPX:
- The executors now support the `num_cores` options in order for the user to be able to specify the desired number of cores to be used in the corresponding execution.
- The `scheduler` executor was implemented on top of senders/receivers and can be used with all HPX facilities that schedule new work, such as parallel algorithms, `hpx::async`, `hpx::dataflow`, etc.
- The performance of `fork_join_executor` was improved.
- The following algorithms have been added/adapted to be C++20 conformant:
 - `min_element`

¹⁶⁶⁵ <https://github.com/TheHPXProject/hpx/pull/5879>

- max_element
 - minmax_element
 - starts_with
 - ends_with
 - swap_ranges
 - unique
 - unique_copy
 - rotate
 - rotate_copy
 - sort
 - shift_left
 - shift_right
 - stable_sort
 - partition
 - partition_copy
 - stable_partition
 - adjacent_difference
 - nth_element
 - partial_sort
 - partial_sort_copy.
- HPX_FORWARD/HPX_MOVE macros were introduced that replaced the `std::move` and `std::forward` facilities that in the library code.
 - Hangs on distributed barrier were fixed.
 - The performance of `scan_partitioner` was improved.
 - Support was added for `thread_priority` to the `parallel_execution_policy`
 - Regarding senders/receivers and the P2300 proposal various actions took place. `stop_token` was adapted to the recent proposal version (`in_place_stop_token` was introduced). Also hint, annotation, priority and stacksize properties were added to the scheduler executor. Stop support was added to `when_all`. Support for completion signatures was added. The following schedulers and algorithms were added:
 - get_completion_scheduler
 - any_sender and unique_any_sender
 - split sender
 - transform_mpi sender
 - transfer sender
 - let_error, let_stopped
 - get_env and related environment queries
 - schedule, set_value, set_error, set_done, start and connect are now proper customization points as defined in P2300.

- Several namespaces were altered towards conformance with C++20. Compatibility layers have been added and the old versions will be removed in next releases. The namespace changes are the following:
 - `hpx::parallel::induction/reduction` were moved into namespace `hpx::experimental`
 - `for_loop` and friends were moved into namespace `hpx::experimental`.
 - `hpx::util::optional` and friends were moved into namespace `hpx`.
 - `hpx::lcos::barrier` has been moved into the `hpx::distributed` namespace and `hpx::lcos::local::cpp20_barrier` has been renamed to `barrier` and moved into the `hpx` namespace.
 - `hpx::lcos::latch` has been moved into the `hpx::distributed` namespace and `lcos::local::latch` has been moved into the `hpx` namespace. The `count_down_and_wait()` functionality of `latch` has been renamed to `arrive_and_wait()`.
 - `hpx::util::unique_function_nonsr` has been renamed to `hpx::move_only_function`.
 - `hpx::util::unique_function` has been renamed to `hpx::distributed::move_only_function`.
 - `hpx::util::function` has been renamed to `hpx::distributed::function`.
 - `hpx::util::function_nonsr` has been renamed to `hpx::function`.
 - `hpx::util::function_ref` have been moved to namespace `hpx`.
 - `hpx::lcos::split_future` changed namespace and is now used as `hpx::split_future`.
 - `hpx::lcos::local::counting_semaphore` has been deprecated and `hpx::lcos::local::cpp20_counting_semaphore` has been renamed to `hpx::counting_semaphore`.
 - `hpx::lcos::local::cpp20_binary_semaphore` has been renamed to `hpx::binary_semaphore`.
 - `hpx::lcos::local::sliding_semaphore` has been renamed to `hpx::sliding_semaphore` and
 - `hpx::lcos::local::sliding_semaphore_var` has been renamed to `hpx::sliding_semaphore_var`.
 - `hpx::lcos::local::spinlock` has been renamed to `hpx::spinlock`.
 - `hpx::lcos::local::mutex` has been renamed to `hpx::mutex`.
 - `hpx::lcos::local::timed_mutex` has been renamed to `hpx::timed_mutex`.
 - `hpx::lcos::local::no_mutex` has been renamed to `hpx::no_mutex`.
 - `hpx::lcos::local::recursive_mutex` has been renamed to `hpx::recursive_mutex`.
 - `hpx::lcos::local::shared_mutex` has been renamed to `hpx::shared_mutex`.
 - `hpx::lcos::local::upgrade_lock` has been renamed to `hpx::upgrade_lock`.
 - `hpx::lcos::local::upgrade_to_unique_lock` has been renamed to `hpx::upgrade_to_unique_lock`.
 - `hpx::lcos::local::condition_variable` has been renamed to `hpx::condition_variable`. `hpx::lcos::local::condition_variable_var` has been renamed to `hpx::condition_variable_var`.
 - `hpx::lcos::local::once_flag` has been renamed to `hpx::once_flag`, and `hpx::lcos::local::call_once` has been renamed to `hpx::call_once`.
- The new LCI (Lightweight Communication Interface) `parcelport` was added that supports irregular and asynchronous applications like graph analysis, sparse linear algebra, modern parallel architectures etc. Major features include:

- Support for advanced communication primitives like two sided send/recv and one sided remote put.
- Better multi-threaded performance.
- Explicit user control of communication resource.
- Flexible signaling mechanisms (synchronizer, completion queue, active message handler).
- The following CMake flags were added, mostly to support using HPX as a backend for SHAD (<https://github.com/pnnl/SHAD>). Please note that these options enable questionable functionalities, partially they even enable undefined behavior. Please only use any of them if you know what you're doing:
 - HPX_SERIALIZATION_WITH_ALLOW_RAW_POINTER_SERIALIZATION
 - HPX_SERIALIZATION_WITH_ALL_TYPES_ARE_BITWISE_SERIALIZABLE
 - HPX_SERIALIZATION_WITH_ALLOW_CONST_TUPLE_MEMBERS

Breaking changes

- Minimum required C++ standard library is C++17.
- Support for GCC 7 and Clang 8.0.0 and below has been removed.
- CUDA version required updated to 11.4.
- CMake version required updated to 3.18.
- The default version of Asio used was updated to 1.20.0.
- The default version of APEX used was updated to 2.5.1.
- APEX version was updated to 2.5.1.
- `tagged_pair` and `tagged_tuple` were removed.
- `tag_dispatch` was renamed to `tag_invoke`.
- `hpx.max_backgroud_threads` was renamed to `hpx.parcel.max_background_threads`.
- The following CMake flags were removed after being deprecated for at least two releases:
 - HPX_SCHEDULER_MAX_TERMINATED_THREADS
 - HPX_WITH_GOOGLE_PERFTOOLS
 - HPX_WITH_INIT_START_OVERLOADS_COMPATIBILITY
 - HPX_HAVE_{COROUTINE,PLUGIN}_GCC_HIDDEN_VISIBILITY
 - HPX_TOP_LEVEL
 - HPX_WITH_COMPUTE_CUDA
 - HPX_WITH_ASYNC_CUDA
- `annotate_function` was renamed to `scoped_annotation`.
- `execution::transform` was renamed to `execution::then`.
- `execution::detach` was renamed to `execution::start_detached`.
- `execution::on_sender` was renamed to `execution::schedule_on`.
- `execution::just_on` was renamed to `execution::just_transfer`.
- `execution::set_done` was renamed to `execution::set_stopped`.

Closed issues

- [Issue #5871¹⁶⁶⁶](#) - `distributed::channel.regisiter_as` terminates the active task.
- [Issue #5856¹⁶⁶⁷](#) - Performance counters do not compile
- [Issue #5828¹⁶⁶⁸](#) - `hpx::distributed:barrier` errors
- [Issue #5812¹⁶⁶⁹](#) - OctoTiger does not compile with HPX master and CUDA 11.5
- [Issue #5784¹⁶⁷⁰](#) - HPX failing with `co_await` and `hpx::when_all(futures)`
- [Issue #5774¹⁶⁷¹](#) - CMake can't find `HPXCacheVariables.cmake`
- [Issue #5764¹⁶⁷²](#) - Fix HIP problem
- [Issue #5724¹⁶⁷³](#) - Missing binary filter compression header
- [Issue #5721¹⁶⁷⁴](#) - Cleanup after repository split
- [Issue #5701¹⁶⁷⁵](#) - It seems that the `tcp` `parcelport` is running, and the `MPI` `parcelport` is ignored
- [Issue #5692¹⁶⁷⁶](#) - Kokkos compilation fails when using both HPX and CUDA execution spaces with gcc 9.3.0
- [Issue #5686¹⁶⁷⁷](#) - Rename `annotate_function`
- [Issue #5668¹⁶⁷⁸](#) - HPX does not detect the C++ 20 standard using gcc 11.2
- [Issue #5666¹⁶⁷⁹](#) - Compilation error using boost 1.76 and gcc 11.2.1
- [Issue #5653¹⁶⁸⁰](#) - Implement P2248 for our algorithms
- [Issue #5647¹⁶⁸¹](#) - [User input needed] Remove (CUDA) compute functionality?
- [Issue #5590¹⁶⁸²](#) - `hello_world_distributed` fails on startup with HPX stable, MPICH 3.3.2, on Deep Bayou
- [Issue #5570¹⁶⁸³](#) - Rename `tag_dispatch` to `tag_invoke`
- [Issue #5566¹⁶⁸⁴](#) - can't build simple example: "Cannot use the dummy implementation of `future_then_dispatch`"
- [Issue #5565¹⁶⁸⁵](#) - build failure: `hpx::string_util::trim()`
- [Issue #5553¹⁶⁸⁶](#) - Github action to validate the `cff` file refs #5471
- [Issue #5504¹⁶⁸⁷](#) - CMake does not work for HPX 1.7.0 on Piz Daint

¹⁶⁶⁶ <https://github.com/TheHPXProject/hpx/issues/5871>

¹⁶⁶⁷ <https://github.com/TheHPXProject/hpx/issues/5856>

¹⁶⁶⁸ <https://github.com/TheHPXProject/hpx/issues/5828>

¹⁶⁶⁹ <https://github.com/TheHPXProject/hpx/issues/5812>

¹⁶⁷⁰ <https://github.com/TheHPXProject/hpx/issues/5784>

¹⁶⁷¹ <https://github.com/TheHPXProject/hpx/issues/5774>

¹⁶⁷² <https://github.com/TheHPXProject/hpx/issues/5764>

¹⁶⁷³ <https://github.com/TheHPXProject/hpx/issues/5724>

¹⁶⁷⁴ <https://github.com/TheHPXProject/hpx/issues/5721>

¹⁶⁷⁵ <https://github.com/TheHPXProject/hpx/issues/5701>

¹⁶⁷⁶ <https://github.com/TheHPXProject/hpx/issues/5692>

¹⁶⁷⁷ <https://github.com/TheHPXProject/hpx/issues/5686>

¹⁶⁷⁸ <https://github.com/TheHPXProject/hpx/issues/5668>

¹⁶⁷⁹ <https://github.com/TheHPXProject/hpx/issues/5666>

¹⁶⁸⁰ <https://github.com/TheHPXProject/hpx/issues/5653>

¹⁶⁸¹ <https://github.com/TheHPXProject/hpx/issues/5647>

¹⁶⁸² <https://github.com/TheHPXProject/hpx/issues/5590>

¹⁶⁸³ <https://github.com/TheHPXProject/hpx/issues/5570>

¹⁶⁸⁴ <https://github.com/TheHPXProject/hpx/issues/5566>

¹⁶⁸⁵ <https://github.com/TheHPXProject/hpx/issues/5565>

¹⁶⁸⁶ <https://github.com/TheHPXProject/hpx/issues/5553>

¹⁶⁸⁷ <https://github.com/TheHPXProject/hpx/issues/5504>

- [Issue #5503](#)¹⁶⁸⁸ - Use contiguous index queue in bulk execution to reduce number of spawned tasks
- [Issue #5502](#)¹⁶⁸⁹ - C++20 `std::coroutine` cmake detection
- [Issue #5478](#)¹⁶⁹⁰ - `hpx.dll` built with `vcpkg` got functions pointing to the same location
- [Issue #5472](#)¹⁶⁹¹ - Compilation error with `cuda/11.3`
- [Issue #5469](#)¹⁶⁹² - Compiler warning about `HPX_NODISCARD` when building with APEX
- [Issue #5463](#)¹⁶⁹³ - Address minor comments of the C++17 PR bump
- [Issue #5456](#)¹⁶⁹⁴ - Use `std::ranges::iter_swap` where available
- [Issue #5404](#)¹⁶⁹⁵ - Build fails with error “Cannot open include file `asio/io_context.hpp`”
- [Issue #5381](#)¹⁶⁹⁶ - Add `starts_with` and `ends_with` algorithms
- [Issue #5344](#)¹⁶⁹⁷ - Further simplify `tag_invoke` helpers
- [Issue #5269](#)¹⁶⁹⁸ - Allow setting a label on executors/policies
- [Issue #5219](#)¹⁶⁹⁹ - (Re-)Implement executor API on top of sender/receiver infrastructure
- [Issue #5216](#)¹⁷⁰⁰ - Performance counter module not loading
- [Issue #5162](#)¹⁷⁰¹ - Require C++17 support
- [Issue #5156](#)¹⁷⁰² - Disentangle segmented algorithms
- [Issue #5118](#)¹⁷⁰³ - Lock held while suspending
- [Issue #5111](#)¹⁷⁰⁴ - Tests fail to build with `binary_filter` plugins enabled
- [Issue #5110](#)¹⁷⁰⁵ - Tests don't get built
- [Issue #5105](#)¹⁷⁰⁶ - PAPI performance counters not available
- [Issue #5002](#)¹⁷⁰⁷ - `hpx::lcos::barrier()` results in deadlock
- [Issue #4992](#)¹⁷⁰⁸ - Clang-format the rest of the files
- [Issue #4987](#)¹⁷⁰⁹ - Use `std::function` in public APIs
- [Issue #4871](#)¹⁷¹⁰ - HEP: conformance to C++20

¹⁶⁸⁸ <https://github.com/TheHPXProject/hpx/issues/5503>

¹⁶⁸⁹ <https://github.com/TheHPXProject/hpx/issues/5502>

¹⁶⁹⁰ <https://github.com/TheHPXProject/hpx/issues/5478>

¹⁶⁹¹ <https://github.com/TheHPXProject/hpx/issues/5472>

¹⁶⁹² <https://github.com/TheHPXProject/hpx/issues/5469>

¹⁶⁹³ <https://github.com/TheHPXProject/hpx/issues/5463>

¹⁶⁹⁴ <https://github.com/TheHPXProject/hpx/issues/5456>

¹⁶⁹⁵ <https://github.com/TheHPXProject/hpx/issues/5404>

¹⁶⁹⁶ <https://github.com/TheHPXProject/hpx/issues/5381>

¹⁶⁹⁷ <https://github.com/TheHPXProject/hpx/issues/5344>

¹⁶⁹⁸ <https://github.com/TheHPXProject/hpx/issues/5269>

¹⁶⁹⁹ <https://github.com/TheHPXProject/hpx/issues/5219>

¹⁷⁰⁰ <https://github.com/TheHPXProject/hpx/issues/5216>

¹⁷⁰¹ <https://github.com/TheHPXProject/hpx/issues/5162>

¹⁷⁰² <https://github.com/TheHPXProject/hpx/issues/5156>

¹⁷⁰³ <https://github.com/TheHPXProject/hpx/issues/5118>

¹⁷⁰⁴ <https://github.com/TheHPXProject/hpx/issues/5111>

¹⁷⁰⁵ <https://github.com/TheHPXProject/hpx/issues/5110>

¹⁷⁰⁶ <https://github.com/TheHPXProject/hpx/issues/5105>

¹⁷⁰⁷ <https://github.com/TheHPXProject/hpx/issues/5002>

¹⁷⁰⁸ <https://github.com/TheHPXProject/hpx/issues/4992>

¹⁷⁰⁹ <https://github.com/TheHPXProject/hpx/issues/4987>

¹⁷¹⁰ <https://github.com/TheHPXProject/hpx/issues/4871>

- [Issue #4822¹⁷¹¹](#) - Adapt parallel algorithms to C++20
- [Issue #4736¹⁷¹²](#) - Deprecate `hpx::flush` and `hpx::endl`
- [Issue #4558¹⁷¹³](#) - Prevent work-stealing from stalling
- [Issue #4495¹⁷¹⁴](#) - Add anchor links to table rows in documentation
- [Issue #4469¹⁷¹⁵](#) - New thread state: *pending_low*
- [Issue #4321¹⁷¹⁶](#) - After the modularization the `libfabric` `parcelpoint` does not compile
- [Issue #4308¹⁷¹⁷](#) - Using APEX on multinode jobs when `HPX_WITH_NETWORKING = OFF`
- [Issue #3995¹⁷¹⁸](#) - Use C++20 `std::source_location` where available, adapt ours to conform
- [Issue #3861¹⁷¹⁹](#) - Selected processor does not support ‘yield’ in ARM mode
- [Issue #3706¹⁷²⁰](#) - Add `shift_left` and `shift_right` algorithms
- [Issue #3646¹⁷²¹](#) - Parallel algorithms should accept iterator/sentinel pairs
- [Issue #3636¹⁷²²](#) - HPX Modularization
- [Issue #3546¹⁷²³](#) - Modularization of HPX
- [Issue #3474¹⁷²⁴](#) - Modernize CMake used in HPX
- [Issue #1836¹⁷²⁵](#) - `hpx::parallel` does not have a sort implementation
- [Issue #1668¹⁷²⁶](#) - Adapt all parallel algorithms to Ranges TS
- [Issue #1141¹⁷²⁷](#) - Implement N4409 on top of HPX

Closed pull requests

- [PR #5885¹⁷²⁸](#) - Testing newer ASIO version
- [PR #5884¹⁷²⁹](#) - Fix miscellaneous doc sections
- [PR #5882¹⁷³⁰](#) - Fixing OctoTiger incompatibility introduced recently
- [PR #5881¹⁷³¹](#) - Fixing recent patch that disables `ATOMIC_FLAG_INIT` for C++20 and up
- [PR #5880¹⁷³²](#) - refactor: convert *counter_status* enum to enum class

¹⁷¹¹ <https://github.com/TheHPXProject/hpx/issues/4822>

¹⁷¹² <https://github.com/TheHPXProject/hpx/issues/4736>

¹⁷¹³ <https://github.com/TheHPXProject/hpx/issues/4558>

¹⁷¹⁴ <https://github.com/TheHPXProject/hpx/issues/4495>

¹⁷¹⁵ <https://github.com/TheHPXProject/hpx/issues/4469>

¹⁷¹⁶ <https://github.com/TheHPXProject/hpx/issues/4321>

¹⁷¹⁷ <https://github.com/TheHPXProject/hpx/issues/4308>

¹⁷¹⁸ <https://github.com/TheHPXProject/hpx/issues/3995>

¹⁷¹⁹ <https://github.com/TheHPXProject/hpx/issues/3861>

¹⁷²⁰ <https://github.com/TheHPXProject/hpx/issues/3706>

¹⁷²¹ <https://github.com/TheHPXProject/hpx/issues/3646>

¹⁷²² <https://github.com/TheHPXProject/hpx/issues/3636>

¹⁷²³ <https://github.com/TheHPXProject/hpx/issues/3546>

¹⁷²⁴ <https://github.com/TheHPXProject/hpx/issues/3474>

¹⁷²⁵ <https://github.com/TheHPXProject/hpx/issues/1836>

¹⁷²⁶ <https://github.com/TheHPXProject/hpx/issues/1668>

¹⁷²⁷ <https://github.com/TheHPXProject/hpx/issues/1141>

¹⁷²⁸ <https://github.com/TheHPXProject/hpx/pull/5885>

¹⁷²⁹ <https://github.com/TheHPXProject/hpx/pull/5884>

¹⁷³⁰ <https://github.com/TheHPXProject/hpx/pull/5882>

¹⁷³¹ <https://github.com/TheHPXProject/hpx/pull/5881>

¹⁷³² <https://github.com/TheHPXProject/hpx/pull/5880>

- [PR #5878](#)¹⁷³³ - Docs: Replaced non-existent create_reducer function with create_communicator
- [PR #5877](#)¹⁷³⁴ - Doc updates hpx runtime and resources
- [PR #5876](#)¹⁷³⁵ - Updates to documentation; grammar edits.
- [PR #5875](#)¹⁷³⁶ - Doc updates starting the hpx runtime
- [PR #5874](#)¹⁷³⁷ - Doc updates launching configuring
- [PR #5873](#)¹⁷³⁸ - Prevent certain generated files from being deleted on reconfigure
- [PR #5870](#)¹⁷³⁹ - Adding support for the PJM batch environment
- [PR #5867](#)¹⁷⁴⁰ - Update CMakeLists.txt
- [PR #5866](#)¹⁷⁴¹ - add cmake option HPX_WITH_PARCELPORNT_COUNTERS
- [PR #5864](#)¹⁷⁴² - ATOMIC_INIT_FLAG is deprecated starting C++20
- [PR #5863](#)¹⁷⁴³ - Adding llvm 14.0.0 with boost 1.79.0 to Jenkins
- [PR #5861](#)¹⁷⁴⁴ - Let install step proceed on CircleCI even if the segmented algorithms fail
- [PR #5860](#)¹⁷⁴⁵ - Updating APEX tag
- [PR #5859](#)¹⁷⁴⁶ - Splitting documentation generation steps on CircleCI
- [PR #5854](#)¹⁷⁴⁷ - Fixing left-overs from changing counter_type to enum class
- [PR #5853](#)¹⁷⁴⁸ - Adding HPX dependency tool (adapted from Boostdep tool)
- [PR #5852](#)¹⁷⁴⁹ - Optimize LCI parcellport
- [PR #5851](#)¹⁷⁵⁰ - Forking dynamic_bitset from Boost
- [PR #5850](#)¹⁷⁵¹ - Convert perf_counters::counter_type enum to enum class.
- [PR #5849](#)¹⁷⁵² - Update LCI parcellport to LCI v1.7.1
- [PR #5848](#)¹⁷⁵³ - Fedora related fixes
- [PR #5847](#)¹⁷⁵⁴ - Fix API, troubleshooting & people
- [PR #5844](#)¹⁷⁵⁵ - Attempting to fix timeouts of segmented iterator tests

¹⁷³³ <https://github.com/TheHPXProject/hpx/pull/5878>

¹⁷³⁴ <https://github.com/TheHPXProject/hpx/pull/5877>

¹⁷³⁵ <https://github.com/TheHPXProject/hpx/pull/5876>

¹⁷³⁶ <https://github.com/TheHPXProject/hpx/pull/5875>

¹⁷³⁷ <https://github.com/TheHPXProject/hpx/pull/5874>

¹⁷³⁸ <https://github.com/TheHPXProject/hpx/pull/5873>

¹⁷³⁹ <https://github.com/TheHPXProject/hpx/pull/5870>

¹⁷⁴⁰ <https://github.com/TheHPXProject/hpx/pull/5867>

¹⁷⁴¹ <https://github.com/TheHPXProject/hpx/pull/5866>

¹⁷⁴² <https://github.com/TheHPXProject/hpx/pull/5864>

¹⁷⁴³ <https://github.com/TheHPXProject/hpx/pull/5863>

¹⁷⁴⁴ <https://github.com/TheHPXProject/hpx/pull/5861>

¹⁷⁴⁵ <https://github.com/TheHPXProject/hpx/pull/5860>

¹⁷⁴⁶ <https://github.com/TheHPXProject/hpx/pull/5859>

¹⁷⁴⁷ <https://github.com/TheHPXProject/hpx/pull/5854>

¹⁷⁴⁸ <https://github.com/TheHPXProject/hpx/pull/5853>

¹⁷⁴⁹ <https://github.com/TheHPXProject/hpx/pull/5852>

¹⁷⁵⁰ <https://github.com/TheHPXProject/hpx/pull/5851>

¹⁷⁵¹ <https://github.com/TheHPXProject/hpx/pull/5850>

¹⁷⁵² <https://github.com/TheHPXProject/hpx/pull/5849>

¹⁷⁵³ <https://github.com/TheHPXProject/hpx/pull/5848>

¹⁷⁵⁴ <https://github.com/TheHPXProject/hpx/pull/5847>

¹⁷⁵⁵ <https://github.com/TheHPXProject/hpx/pull/5844>

- PR #5842¹⁷⁵⁶ - change the default value of HPX_WITH_LCI_TAG to v1.7
- PR #5841¹⁷⁵⁷ - Move the split_future facilities into the namespace hpx
- PR #5840¹⁷⁵⁸ - wait_XXX_nothrow functions return whether one of the futures is exceptional
- PR #5839¹⁷⁵⁹ - Moving a list of synchronization primitives into namespace hpx
- PR #5837¹⁷⁶⁰ - Moving latch types to hpx and hpx::distributed namespaces
- PR #5835¹⁷⁶¹ - Add missing compatibility layer for id_type::management_type values
- PR #5834¹⁷⁶² - API docs changes
- PR #5831¹⁷⁶³ - Further improvement actions to rotate
- PR #5830¹⁷⁶⁴ - Exposing zero-copy serialization threshold through configuration option
- PR #5829¹⁷⁶⁵ - Attempting to fix failing barrier test
- PR #5827¹⁷⁶⁶ - Add back explicit template parameter to *ignore_while_checking* to compile with nvcc
- PR #5826¹⁷⁶⁷ - Reduce number of allocations while calling *async_bulk_execute*
- PR #5825¹⁷⁶⁸ - Steal from neighboring NUMA domain only
- PR #5823¹⁷⁶⁹ - Remove obsolete directories and adjust build system
- PR #5822¹⁷⁷⁰ - Clang-format remaining files
- PR #5821¹⁷⁷¹ - Enable permissive- flag on Windows GitHub actions builders
- PR #5820¹⁷⁷² - Convert throwmode enum to enum class
- PR #5819¹⁷⁷³ - Marking customization points for intrusive_ptr as noexcept
- PR #5818¹⁷⁷⁴ - Unconditionally use C++17 attributes
- PR #5817¹⁷⁷⁵ - Modernize naming modules
- PR #5816¹⁷⁷⁶ - Modernize cache module
- PR #5815¹⁷⁷⁷ - Reapply flyby changes from #5467
- PR #5814¹⁷⁷⁸ - Avoid test timeouts by reducing test sizes

¹⁷⁵⁶ <https://github.com/TheHPXProject/hpx/pull/5842>

¹⁷⁵⁷ <https://github.com/TheHPXProject/hpx/pull/5841>

¹⁷⁵⁸ <https://github.com/TheHPXProject/hpx/pull/5840>

¹⁷⁵⁹ <https://github.com/TheHPXProject/hpx/pull/5839>

¹⁷⁶⁰ <https://github.com/TheHPXProject/hpx/pull/5837>

¹⁷⁶¹ <https://github.com/TheHPXProject/hpx/pull/5835>

¹⁷⁶² <https://github.com/TheHPXProject/hpx/pull/5834>

¹⁷⁶³ <https://github.com/TheHPXProject/hpx/pull/5831>

¹⁷⁶⁴ <https://github.com/TheHPXProject/hpx/pull/5830>

¹⁷⁶⁵ <https://github.com/TheHPXProject/hpx/pull/5829>

¹⁷⁶⁶ <https://github.com/TheHPXProject/hpx/pull/5827>

¹⁷⁶⁷ <https://github.com/TheHPXProject/hpx/pull/5826>

¹⁷⁶⁸ <https://github.com/TheHPXProject/hpx/pull/5825>

¹⁷⁶⁹ <https://github.com/TheHPXProject/hpx/pull/5823>

¹⁷⁷⁰ <https://github.com/TheHPXProject/hpx/pull/5822>

¹⁷⁷¹ <https://github.com/TheHPXProject/hpx/pull/5821>

¹⁷⁷² <https://github.com/TheHPXProject/hpx/pull/5820>

¹⁷⁷³ <https://github.com/TheHPXProject/hpx/pull/5819>

¹⁷⁷⁴ <https://github.com/TheHPXProject/hpx/pull/5818>

¹⁷⁷⁵ <https://github.com/TheHPXProject/hpx/pull/5817>

¹⁷⁷⁶ <https://github.com/TheHPXProject/hpx/pull/5816>

¹⁷⁷⁷ <https://github.com/TheHPXProject/hpx/pull/5815>

¹⁷⁷⁸ <https://github.com/TheHPXProject/hpx/pull/5814>

- PR #5813¹⁷⁷⁹ - The CUDA problem is not fixed in V11.5 yet...
- PR #5811¹⁷⁸⁰ - Make sure reduction value is properly moved, when possible
- PR #5810¹⁷⁸¹ - Improve error reporting during device initialization in HIP environments
- PR #5809¹⁷⁸² - Converting scheduler enums into enum class
- PR #5808¹⁷⁸³ - Deprecate `hpx::flush` and friends
- PR #5807¹⁷⁸⁴ - Use C++20 `std::source_location`, if available
- PR #5806¹⁷⁸⁵ - Moving `promise` and `packaged_task` to new namespaces
- PR #5805¹⁷⁸⁶ - Attempting to fix a test failure when using the LCI `parcelpor`
- PR #5803¹⁷⁸⁷ - Attempt to fix CUDA related OctoTiger problems
- PR #5800¹⁷⁸⁸ - Add option to restrict MPI background work to subset of cores
- PR #5798¹⁷⁸⁹ - Adding MPI as a dependency to APEX
- PR #5797¹⁷⁹⁰ - Extend Sphinx role to support arbitrary text to display on a link
- PR #5796¹⁷⁹¹ - Disable CUDA tests that cause NVCC to silently fail without error messages
- PR #5795¹⁷⁹² - Avoid writing path and directories into `HPXCacheVariables.cmake`
- PR #5793¹⁷⁹³ - Remove features that are deprecated since V1.6
- PR #5792¹⁷⁹⁴ - Making sure `num_cores` is properly handled by `parallel_executor`
- PR #5791¹⁷⁹⁵ - Moving `bind`, `bind_front`, `bind_back` to namespace `hpx`
- PR #5790¹⁷⁹⁶ - Moving serializable function/`move_only_function` into namespace `hpx::distributed`
- PR #5787¹⁷⁹⁷ - Remove unneeded (and commented) tests
- PR #5786¹⁷⁹⁸ - Attempting to fix hangs in distributed barrier
- PR #5785¹⁷⁹⁹ - add `cmake` code to detect `arm64` on `macOS`
- PR #5783¹⁸⁰⁰ - Moving `function` and `function_ref` into namespace `hpx`
- PR #5781¹⁸⁰¹ - Updating used version of Visual Studio

¹⁷⁷⁹ <https://github.com/TheHPXProject/hpx/pull/5813>

¹⁷⁸⁰ <https://github.com/TheHPXProject/hpx/pull/5811>

¹⁷⁸¹ <https://github.com/TheHPXProject/hpx/pull/5810>

¹⁷⁸² <https://github.com/TheHPXProject/hpx/pull/5809>

¹⁷⁸³ <https://github.com/TheHPXProject/hpx/pull/5808>

¹⁷⁸⁴ <https://github.com/TheHPXProject/hpx/pull/5807>

¹⁷⁸⁵ <https://github.com/TheHPXProject/hpx/pull/5806>

¹⁷⁸⁶ <https://github.com/TheHPXProject/hpx/pull/5805>

¹⁷⁸⁷ <https://github.com/TheHPXProject/hpx/pull/5803>

¹⁷⁸⁸ <https://github.com/TheHPXProject/hpx/pull/5800>

¹⁷⁸⁹ <https://github.com/TheHPXProject/hpx/pull/5798>

¹⁷⁹⁰ <https://github.com/TheHPXProject/hpx/pull/5797>

¹⁷⁹¹ <https://github.com/TheHPXProject/hpx/pull/5796>

¹⁷⁹² <https://github.com/TheHPXProject/hpx/pull/5795>

¹⁷⁹³ <https://github.com/TheHPXProject/hpx/pull/5793>

¹⁷⁹⁴ <https://github.com/TheHPXProject/hpx/pull/5792>

¹⁷⁹⁵ <https://github.com/TheHPXProject/hpx/pull/5791>

¹⁷⁹⁶ <https://github.com/TheHPXProject/hpx/pull/5790>

¹⁷⁹⁷ <https://github.com/TheHPXProject/hpx/pull/5787>

¹⁷⁹⁸ <https://github.com/TheHPXProject/hpx/pull/5786>

¹⁷⁹⁹ <https://github.com/TheHPXProject/hpx/pull/5785>

¹⁸⁰⁰ <https://github.com/TheHPXProject/hpx/pull/5783>

¹⁸⁰¹ <https://github.com/TheHPXProject/hpx/pull/5781>

- PR #5780¹⁸⁰² - Update Piz Daint Jenkins configurations from gcc/clang 7 to 8
- PR #5778¹⁸⁰³ - Updated for_loop.hpp
- PR #5777¹⁸⁰⁴ - Update reference for foreach benchmark
- PR #5775¹⁸⁰⁵ - Move optional into namespace hpx
- PR #5773¹⁸⁰⁶ - Moving barrier to consolidated namespaces
- PR #5772¹⁸⁰⁷ - Adding missing docs for ranges::find_if and find_if_not algorithms
- PR #5771¹⁸⁰⁸ - Moving for_loop into namespace hpx::experimental
- PR #5770¹⁸⁰⁹ - Fixing HIP issues
- PR #5769¹⁸¹⁰ - Slight improvement of small_vector performance
- PR #5766¹⁸¹¹ - Fixing a integral conversion warning
- PR #5765¹⁸¹² - Adding a sphinx role allowing to link to a file directly in github
- PR #5763¹⁸¹³ - add num_cores facility
- PR #5762¹⁸¹⁴ - Fix Public API main page
- PR #5761¹⁸¹⁵ - Add missing inline to mpi_exception.hpp error_message function
- PR #5760¹⁸¹⁶ - Update cdash build url
- PR #5759¹⁸¹⁷ - Switch to use generic rostam SLURM partitions
- PR #5758¹⁸¹⁸ - Adding support for P2300 completion signatures
- PR #5757¹⁸¹⁹ - Fix missing links in Public API
- PR #5756¹⁸²⁰ - Add stop support to when_all
- PR #5755¹⁸²¹ - Support for data-parallelism for mismatch algorithm
- PR #5754¹⁸²² - Support for data-parallelism for equal algorithm
- PR #5751¹⁸²³ - Propagate MPI dependencies to command line handling
- PR #5750¹⁸²⁴ - Make sure required MPI initialization flags are properly applied and supported

¹⁸⁰² <https://github.com/TheHPXProject/hpx/pull/5780>

¹⁸⁰³ <https://github.com/TheHPXProject/hpx/pull/5778>

¹⁸⁰⁴ <https://github.com/TheHPXProject/hpx/pull/5777>

¹⁸⁰⁵ <https://github.com/TheHPXProject/hpx/pull/5775>

¹⁸⁰⁶ <https://github.com/TheHPXProject/hpx/pull/5773>

¹⁸⁰⁷ <https://github.com/TheHPXProject/hpx/pull/5772>

¹⁸⁰⁸ <https://github.com/TheHPXProject/hpx/pull/5771>

¹⁸⁰⁹ <https://github.com/TheHPXProject/hpx/pull/5770>

¹⁸¹⁰ <https://github.com/TheHPXProject/hpx/pull/5769>

¹⁸¹¹ <https://github.com/TheHPXProject/hpx/pull/5766>

¹⁸¹² <https://github.com/TheHPXProject/hpx/pull/5765>

¹⁸¹³ <https://github.com/TheHPXProject/hpx/pull/5763>

¹⁸¹⁴ <https://github.com/TheHPXProject/hpx/pull/5762>

¹⁸¹⁵ <https://github.com/TheHPXProject/hpx/pull/5761>

¹⁸¹⁶ <https://github.com/TheHPXProject/hpx/pull/5760>

¹⁸¹⁷ <https://github.com/TheHPXProject/hpx/pull/5759>

¹⁸¹⁸ <https://github.com/TheHPXProject/hpx/pull/5758>

¹⁸¹⁹ <https://github.com/TheHPXProject/hpx/pull/5757>

¹⁸²⁰ <https://github.com/TheHPXProject/hpx/pull/5756>

¹⁸²¹ <https://github.com/TheHPXProject/hpx/pull/5755>

¹⁸²² <https://github.com/TheHPXProject/hpx/pull/5754>

¹⁸²³ <https://github.com/TheHPXProject/hpx/pull/5751>

¹⁸²⁴ <https://github.com/TheHPXProject/hpx/pull/5750>

- PR #5749¹⁸²⁵ - P2300 stop token
- PR #5748¹⁸²⁶ - Adding environmental query CPOs
- PR #5747¹⁸²⁷ - Renaming set_done to set_stopped (as per P2300)
- PR #5745¹⁸²⁸ - Modernize serialization module
- PR #5743¹⁸²⁹ - Add check for MPICH and set the correct env to support multi-threaded
- PR #5742¹⁸³⁰ - Remove obsolete files related to cpuid, etc.
- PR #5741¹⁸³¹ - Support for data-parallelism for adjacent find
- PR #5740¹⁸³² - Support for data-parallelism for find algorithms
- PR #5739¹⁸³³ - Enable the option to attach a debugger on a segmentation fault (linux)
- PR #5738¹⁸³⁴ - Fixing spell-checking errors
- PR #5737¹⁸³⁵ - Attempt to fix migrate_component issue
- PR #5736¹⁸³⁶ - Set commit status from Jenkins also for special branches
- PR #5734¹⁸³⁷ - Revert #5586
- PR #5732¹⁸³⁸ - Attempt to improve build-id reporting to cdash
- PR #5731¹⁸³⁹ - Randomly delay execution of bash scripts launched by Jenkins
- PR #5729¹⁸⁴⁰ - Workaround for CMake/Ninja generator OOM problem
- PR #5727¹⁸⁴¹ - Moving compression plugins to components directory
- PR #5726¹⁸⁴² - Moving/consolidating parcel coalescing plugin sources
- PR #5725¹⁸⁴³ - Making sure headers for serialization filters are being installed
- PR #5723¹⁸⁴⁴ - Moving more tests to modules
- PR #5722¹⁸⁴⁵ - Removing superfluous semicolons
- PR #5720¹⁸⁴⁶ - Moving parcelports into modules
- PR #5719¹⁸⁴⁷ - Moving more files to parcelset module

¹⁸²⁵ <https://github.com/TheHPXProject/hpx/pull/5749>

¹⁸²⁶ <https://github.com/TheHPXProject/hpx/pull/5748>

¹⁸²⁷ <https://github.com/TheHPXProject/hpx/pull/5747>

¹⁸²⁸ <https://github.com/TheHPXProject/hpx/pull/5745>

¹⁸²⁹ <https://github.com/TheHPXProject/hpx/pull/5743>

¹⁸³⁰ <https://github.com/TheHPXProject/hpx/pull/5742>

¹⁸³¹ <https://github.com/TheHPXProject/hpx/pull/5741>

¹⁸³² <https://github.com/TheHPXProject/hpx/pull/5740>

¹⁸³³ <https://github.com/TheHPXProject/hpx/pull/5739>

¹⁸³⁴ <https://github.com/TheHPXProject/hpx/pull/5738>

¹⁸³⁵ <https://github.com/TheHPXProject/hpx/pull/5737>

¹⁸³⁶ <https://github.com/TheHPXProject/hpx/pull/5736>

¹⁸³⁷ <https://github.com/TheHPXProject/hpx/pull/5734>

¹⁸³⁸ <https://github.com/TheHPXProject/hpx/pull/5732>

¹⁸³⁹ <https://github.com/TheHPXProject/hpx/pull/5731>

¹⁸⁴⁰ <https://github.com/TheHPXProject/hpx/pull/5729>

¹⁸⁴¹ <https://github.com/TheHPXProject/hpx/pull/5727>

¹⁸⁴² <https://github.com/TheHPXProject/hpx/pull/5726>

¹⁸⁴³ <https://github.com/TheHPXProject/hpx/pull/5725>

¹⁸⁴⁴ <https://github.com/TheHPXProject/hpx/pull/5723>

¹⁸⁴⁵ <https://github.com/TheHPXProject/hpx/pull/5722>

¹⁸⁴⁶ <https://github.com/TheHPXProject/hpx/pull/5720>

¹⁸⁴⁷ <https://github.com/TheHPXProject/hpx/pull/5719>

- PR #5718¹⁸⁴⁸ - build: refactor sphinx config file
- PR #5717¹⁸⁴⁹ - Creating parcelset modules
- PR #5716¹⁸⁵⁰ - Avoid duplicate definition error
- PR #5715¹⁸⁵¹ - The new LCI parcelport for HPX
- PR #5714¹⁸⁵² - Refine propagation of **HPX_WITH**... options
- PR #5713¹⁸⁵³ - Significantly reduce CI jobs run on Piz Daint
- PR #5712¹⁸⁵⁴ - Updating jenkins configuration for Rostam2.2
- PR #5711¹⁸⁵⁵ - Refactor manual sections
- PR #5710¹⁸⁵⁶ - Making task_group serializable
- PR #5709¹⁸⁵⁷ - Update the MPI cmake setup
- PR #5707¹⁸⁵⁸ - Better diagnose parcel bootstrap problems
- PR #5704¹⁸⁵⁹ - Test with hwloc 2.7.0 with GCC 11
- PR #5703¹⁸⁶⁰ - Fix *counting_iterator* container tests
- PR #5702¹⁸⁶¹ - Attempting to fix CircleCI timeouts
- PR #5699¹⁸⁶² - Update CI to use Boost 1.78.0
- PR #5697¹⁸⁶³ - Adding fork_join_executor to foreach_benchmark
- PR #5696¹⁸⁶⁴ - Modernize when_all and friends (when_any, when_some, when_each)
- PR #5693¹⁸⁶⁵ - Fix test errors with *_GLIBCXX_DEBUG* defined
- PR #5691¹⁸⁶⁶ - Rename *annotate_function* to *scoped_annotation*
- PR #5690¹⁸⁶⁷ - Replace tag_dispatch with tag_invoke in minmax segmented
- PR #5688¹⁸⁶⁸ - Remove more deprecated macros
- PR #5687¹⁸⁶⁹ - Add most important CMake options
- PR #5685¹⁸⁷⁰ - Fix future API

¹⁸⁴⁸ <https://github.com/TheHPXProject/hpx/pull/5718>

¹⁸⁴⁹ <https://github.com/TheHPXProject/hpx/pull/5717>

¹⁸⁵⁰ <https://github.com/TheHPXProject/hpx/pull/5716>

¹⁸⁵¹ <https://github.com/TheHPXProject/hpx/pull/5715>

¹⁸⁵² <https://github.com/TheHPXProject/hpx/pull/5714>

¹⁸⁵³ <https://github.com/TheHPXProject/hpx/pull/5713>

¹⁸⁵⁴ <https://github.com/TheHPXProject/hpx/pull/5712>

¹⁸⁵⁵ <https://github.com/TheHPXProject/hpx/pull/5711>

¹⁸⁵⁶ <https://github.com/TheHPXProject/hpx/pull/5710>

¹⁸⁵⁷ <https://github.com/TheHPXProject/hpx/pull/5709>

¹⁸⁵⁸ <https://github.com/TheHPXProject/hpx/pull/5707>

¹⁸⁵⁹ <https://github.com/TheHPXProject/hpx/pull/5704>

¹⁸⁶⁰ <https://github.com/TheHPXProject/hpx/pull/5703>

¹⁸⁶¹ <https://github.com/TheHPXProject/hpx/pull/5702>

¹⁸⁶² <https://github.com/TheHPXProject/hpx/pull/5699>

¹⁸⁶³ <https://github.com/TheHPXProject/hpx/pull/5697>

¹⁸⁶⁴ <https://github.com/TheHPXProject/hpx/pull/5696>

¹⁸⁶⁵ <https://github.com/TheHPXProject/hpx/pull/5693>

¹⁸⁶⁶ <https://github.com/TheHPXProject/hpx/pull/5691>

¹⁸⁶⁷ <https://github.com/TheHPXProject/hpx/pull/5690>

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¹⁸⁶⁹ <https://github.com/TheHPXProject/hpx/pull/5687>

¹⁸⁷⁰ <https://github.com/TheHPXProject/hpx/pull/5685>

- PR #5684¹⁸⁷¹ - Move lock registration to separate module and remove global lock registration
- PR #5683¹⁸⁷² - Make `hpx::wait_all` etc. throw exceptions when waited futures hold exceptions and deprecate `hpx::lcos::wait_all[_n]` in favor of `hpx::wait_all[_n]`
- PR #5682¹⁸⁷³ - Fix macOS test exceptions
- PR #5681¹⁸⁷⁴ - docs: add links to hpx recepies
- PR #5680¹⁸⁷⁵ - Embed base execution policies to datapar execution policies
- PR #5679¹⁸⁷⁶ - Fix *fork_join_executor* with dynamic schedule
- PR #5678¹⁸⁷⁷ - Fix compilation of service executors with nvcc
- PR #5677¹⁸⁷⁸ - Remove `compute_cuda` module
- PR #5676¹⁸⁷⁹ - Don't require up-to-date approvals for bors
- PR #5675¹⁸⁸⁰ - Add default template type parameters for algorithms
- PR #5674¹⁸⁸¹ - Allow using *any_sender* in global variables
- PR #5671¹⁸⁸² - Making sure `task_group` can be reused
- PR #5670¹⁸⁸³ - Relax constraints on *execution::when_all*
- PR #5669¹⁸⁸⁴ - Use `HPX_WITH_CXX_STANDARD` for controlling C++ version
- PR #5667¹⁸⁸⁵ - Attempt to fix compilation issues with Boost V1.76
- PR #5664¹⁸⁸⁶ - Change logging errors to warnings in schedulers
- PR #5663¹⁸⁸⁷ - Use dynamic bitsets by default for CPU masks
- PR #5662¹⁸⁸⁸ - Disambiguate namespace for MSVC
- PR #5660¹⁸⁸⁹ - Replacing remaining `std::forward` and `std::move` with `HPX_FORWARD` and `HPX_MOVE`
- PR #5659¹⁸⁹⁰ - Modernize `hpx::future` and related facilities
- PR #5658¹⁸⁹¹ - Replace `HPX_INLINE_CONSTEXPR_VARIABLE` with `inline constexpr`
- PR #5657¹⁸⁹² - Remove `tagged`, `tagged_pair` and `tagged_tuple`, remove tuple/pair specializations
- PR #5656¹⁸⁹³ - Rename `on_execution::schedule_from`, rename `just_on` to `just_transfer`, and add `transfer`

¹⁸⁷¹ <https://github.com/TheHPXProject/hpx/pull/5684>

¹⁸⁷² <https://github.com/TheHPXProject/hpx/pull/5683>

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¹⁸⁹³ <https://github.com/TheHPXProject/hpx/pull/5656>

- [PR #5655](#)¹⁸⁹⁴ - Avoid for module lists to grow indefinitely in cmake cache
- [PR #5649](#)¹⁸⁹⁵ - build: replace usage of Python's reserved words and functions as variable names
- [PR #5648](#)¹⁸⁹⁶ - Modernize action modules and related code
- [PR #5646](#)¹⁸⁹⁷ - Fix ends_with test
- [PR #5645](#)¹⁸⁹⁸ - Add matrix multiplication example
- [PR #5644](#)¹⁸⁹⁹ - Rename execution::transform to execution::then and execution::detach to execution::start_detached
- [PR #5643](#)¹⁹⁰⁰ - Update performance test references
- [PR #5642](#)¹⁹⁰¹ - Adapting adjacent_difference to work with proxy iterators
- [PR #5641](#)¹⁹⁰² - Factorize perftests scripts
- [PR #5640](#)¹⁹⁰³ - Fixed links to sources in Sphinx documentation
- [PR #5639](#)¹⁹⁰⁴ - Fix generate datapar tests for Vc
- [PR #5638](#)¹⁹⁰⁵ - Simd all any none
- [PR #5637](#)¹⁹⁰⁶ - Use bors for merging pull requests
- [PR #5636](#)¹⁹⁰⁷ - Fix leftover std::holds_alternative usage
- [PR #5635](#)¹⁹⁰⁸ - Update container image tag in GitHub actions HIP configuration
- [PR #5633](#)¹⁹⁰⁹ - Moving packaged_task to module futures
- [PR #5632](#)¹⁹¹⁰ - Tell Asio to use std::aligned_new only if available
- [PR #5631](#)¹⁹¹¹ - Adding tag parameter to channel communicator get/set
- [PR #5630](#)¹⁹¹² - Add partial_sort_copy and adapt partial sort to c++ 20
- [PR #5629](#)¹⁹¹³ - Set HPX_WITH_FETCH_ASIO to OFF as available in the docker image
- [PR #5628](#)¹⁹¹⁴ - Add Clang 13 CI configuration
- [PR #5627](#)¹⁹¹⁵ - Replace alternative keyword
- [PR #5626](#)¹⁹¹⁶ - docs: add support for BibTeX references in Sphinx docs

¹⁸⁹⁴ <https://github.com/TheHPXProject/hpx/pull/5655>

¹⁸⁹⁵ <https://github.com/TheHPXProject/hpx/pull/5649>

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¹⁹¹⁵ <https://github.com/TheHPXProject/hpx/pull/5627>

¹⁹¹⁶ <https://github.com/TheHPXProject/hpx/pull/5626>

- PR #5624¹⁹¹⁷ - Fix pkgconfig replacements involving CMAKE_INSTALL_PREFIX
- PR #5623¹⁹¹⁸ - build: remove unused import from conf.py.in
- PR #5622¹⁹¹⁹ - Remove HPX_WITH_VCPKG CMake option
- PR #5621¹⁹²⁰ - Replacing boost::container::small_vector
- PR #5620¹⁹²¹ - Update Asio tag from 1.18.2 to 1.20.0
- PR #5619¹⁹²² - Fix block_os_threads_1036 test
- PR #5618¹⁹²³ - Make sure condition variables are notified under a lock in the thread_pool_scheduler test
- PR #5617¹⁹²⁴ - Use advance_and_get_distance where required
- PR #5616¹⁹²⁵ - Remove separately building segmented algorithms on CircleCI
- PR #5613¹⁹²⁶ - Fix Vc datapar adjacent_difference
- PR #5609¹⁹²⁷ - docs: add anchor links to performance counter tables
- PR #5608¹⁹²⁸ - Fix header test error by adding missing numeric
- PR #5607¹⁹²⁹ - Fix simd adj diff
- PR #5605¹⁹³⁰ - Fix usage of HPX_INVOKE macro
- PR #5604¹⁹³¹ - Make use of shell-session to allow non-copyable \$
- PR #5603¹⁹³² - Suppress some MSVC warnings in C++20 mode
- PR #5602¹⁹³³ - Test HPX_DATASTRUCTURES_WITH_ADAPT_STD_TUPLE=OFF to one CI configuration
- PR #5601¹⁹³⁴ - Test case for any_sender should use hpx::tuple
- PR #5600¹⁹³⁵ - Rename tag_dispatch back to tag_invoke
- PR #5599¹⁹³⁶ - Change theme, fix Quickstart & Examples
- PR #5596¹⁹³⁷ - Use precompiled headers in tests
- PR #5595¹⁹³⁸ - Drop semicolons for macro calls
- PR #5594¹⁹³⁹ - Adapt datapar generate

¹⁹¹⁷ <https://github.com/TheHPXProject/hpx/pull/5624>

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¹⁹³⁹ <https://github.com/TheHPXProject/hpx/pull/5594>

- PR #5593¹⁹⁴⁰ - Update any_sender to use tag_dispatch for execution customizations
- PR #5592¹⁹⁴¹ - Add nth_element
- PR #5591¹⁹⁴² - Remove unnecessary checks for C++17 for tests
- PR #5589¹⁹⁴³ - Add HPX_FORWARD/HPX_MOVE macros
- PR #5588¹⁹⁴⁴ - Fixing the output formatting for id_types
- PR #5586¹⁹⁴⁵ - Remove local functionality
- PR #5585¹⁹⁴⁶ - Delete GitExternal.cmake
- PR #5584¹⁹⁴⁷ - Serialization of hpx::tuple must use hpx::get
- PR #5583¹⁹⁴⁸ - fix coroutine_traits allocate calls, add unhandled_exception() implementation.
- PR #5582¹⁹⁴⁹ - Make more examples work with local runtime
- PR #5581¹⁹⁵⁰ - Add support for several performance tests in CI
- PR #5580¹⁹⁵¹ - Adapt simd adj diff
- PR #5579¹⁹⁵² - Split absolute paths for generated pkg-config files into -L/-I parts
- PR #5577¹⁹⁵³ - fix unit fill test for datapar with Vc
- PR #5576¹⁹⁵⁴ - Update forgotten “Full” names
- PR #5575¹⁹⁵⁵ - Change scan partitioner implementation
- PR #5574¹⁹⁵⁶ - Remove a few deprecated and unused CMake options
- PR #5572¹⁹⁵⁷ - Remove more guards for the distributed runtime
- PR #5571¹⁹⁵⁸ - Add workaround for libstc++ in string_util trim
- PR #5569¹⁹⁵⁹ - Use no_unique_address in sender adaptors
- PR #5568¹⁹⁶⁰ - Change try catch block to try_catch_exception_ptr
- PR #5567¹⁹⁶¹ - Make default_agent::yield actually yield
- PR #5564¹⁹⁶² - Adjacent

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¹⁹⁶² <https://github.com/TheHPXProject/hpx/pull/5564>

- PR #5562¹⁹⁶³ - More changes to overcome build problems on Windows after recent module rearrangements
- PR #5560¹⁹⁶⁴ - Update tests and examples
- PR #5559¹⁹⁶⁵ - Fixing cmake folder names after module restructuring
- PR #5558¹⁹⁶⁶ - Fixing wrong module dependencies
- PR #5557¹⁹⁶⁷ - Adding an example for the new channel_communicator API
- PR #5556¹⁹⁶⁸ - Remove leftover thread pool os executor tests
- PR #5555¹⁹⁶⁹ - Add option enabling serializing raw pointers
- PR #5554¹⁹⁷⁰ - Make sure command line aliasing is properly handled
- PR #5552¹⁹⁷¹ - Modernizing some of the async facilities
- PR #5551¹⁹⁷² - Fixing for local executions of actions to properly set task names
- PR #5550¹⁹⁷³ - Update CUDA module in clang-cuda configuration
- PR #5549¹⁹⁷⁴ - Fixing agent_ref::yield_k to actually call yield_k
- PR #5548¹⁹⁷⁵ - Making get_action_name() noexcept
- PR #5547¹⁹⁷⁶ - Fixing communication set
- PR #5546¹⁹⁷⁷ - Fixing shutdown problems caused by missing ref-counting
- PR #5545¹⁹⁷⁸ - Remove wrong move in thread_pool_scheduler_bulk.hpp
- PR #5543¹⁹⁷⁹ - Extend launch policy to carry stack size and scheduling hint in addition to priority
- PR #5542¹⁹⁸⁰ - Simplify execution CPOs
- PR #5540¹⁹⁸¹ - Adapt partition, partition_copy and stable_partition to C++ 20
- PR #5539¹⁹⁸² - Adapt mismatch to support sentinels
- PR #5538¹⁹⁸³ - Document specific sphinx version required for the documentation
- PR #5537¹⁹⁸⁴ - Test release and debug builds on Piz Daint
- PR #5536¹⁹⁸⁵ - This fixes referencing stale iterators during the execution of binary mismatch

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¹⁹⁸⁵ <https://github.com/TheHPXProject/hpx/pull/5536>

- PR #5535¹⁹⁸⁶ - Rename `simdpar` to `par_simd`
- PR #5534¹⁹⁸⁷ - Fix Quick start & Manual Docs
- PR #5533¹⁹⁸⁸ - Fix `annotate_function` for `std::string`
- PR #5532¹⁹⁸⁹ - Update two remaining apex links from khuck to UO-OACISS
- PR #5531¹⁹⁹⁰ - Use `contiguous_index_queue` in `thread_pool_scheduler`
- PR #5530¹⁹⁹¹ - Eagerly initialize a configurable number of threads on scheduler/thread queue init
- PR #5529¹⁹⁹² - Update benchmarks and add support for `scheduler_executor`
- PR #5528¹⁹⁹³ - Add missing properties to executors/schedulers
- PR #5527¹⁹⁹⁴ - Set local thread/pool number in `local/static_queue_scheduler`
- PR #5526¹⁹⁹⁵ - Update Rostam HIP configuration to use 4.3.0
- PR #5525¹⁹⁹⁶ - Fix Building HPX in Quick start
- PR #5524¹⁹⁹⁷ - Upload image on cdash
- PR #5523¹⁹⁹⁸ - Modernize facilities related to `hpx::sync`
- PR #5522¹⁹⁹⁹ - Add sender overloads for remaining algorithms
- PR #5521²⁰⁰⁰ - Minor changes that improve performance
- PR #5520²⁰⁰¹ - Update reference as perftests failing regularly
- PR #5519²⁰⁰² - Add `transform_mpi` sender adapter
- PR #5518²⁰⁰³ - Add sender overloads to `rotate/rotate_copy`
- PR #5517²⁰⁰⁴ - Fix coroutine integration
- PR #5515²⁰⁰⁵ - Avoid deadlock in `ignore_while_locked_1485` test
- PR #5514²⁰⁰⁶ - Add split sender adapter
- PR #5512²⁰⁰⁷ - Update Rostam HIP configuration
- PR #5511²⁰⁰⁸ - Fix Asio target name for precompiled headers

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¹⁹⁹¹ <https://github.com/TheHPXProject/hpx/pull/5530>

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¹⁹⁹⁶ <https://github.com/TheHPXProject/hpx/pull/5525>

¹⁹⁹⁷ <https://github.com/TheHPXProject/hpx/pull/5524>

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²⁰⁰⁵ <https://github.com/TheHPXProject/hpx/pull/5515>

²⁰⁰⁶ <https://github.com/TheHPXProject/hpx/pull/5514>

²⁰⁰⁷ <https://github.com/TheHPXProject/hpx/pull/5512>

²⁰⁰⁸ <https://github.com/TheHPXProject/hpx/pull/5511>

- PR #5510²⁰⁰⁹ - Add any_sender and unique_any_sender
- PR #5509²⁰¹⁰ - Test with Boost 1.77 on gcc/clang-newest configurations
- PR #5508²⁰¹¹ - Minor release changes from 1.7.1
- PR #5507²⁰¹² - Add missing commits from scheduler_executor PR
- PR #5506²⁰¹³ - Fix condition for checking if we should use our own variant
- PR #5501²⁰¹⁴ - Attempt to fix thread_pool_scheduler test
- PR #5493²⁰¹⁵ - Update Jenkins GitHub token to use StellarBot GitHub account
- PR #5490²⁰¹⁶ - Fix clang-format error on master
- PR #5487²⁰¹⁷ - Add get_completion_scheduler CPO and customize bulk for thread_pool_scheduler
- PR #5484²⁰¹⁸ - Add missing header to jacobi_component/server/solver.hpp
- PR #5481²⁰¹⁹ - Changing the APEX repository to the new location
- PR #5479²⁰²⁰ - Fix version check for CUDA noexcept/result_of bug
- PR #5477²⁰²¹ - Require cxx17 minor comments
- PR #5476²⁰²² - Fix cmake format error
- PR #5475²⁰²³ - Require CMake 3.18 as it is already a requirement for CUDA
- PR #5474²⁰²⁴ - Make the cuda parameters of try_compile optional
- PR #5473²⁰²⁵ - Update cuda arch and change cuda version
- PR #5471²⁰²⁶ - Add corrected citation.cff
- PR #5470²⁰²⁷ - Adapt stable_sort to C++ 20
- PR #5468²⁰²⁸ - Experimentation to make the perftest report public
- PR #5466²⁰²⁹ - Add shift_left and shift_right algorithms
- PR #5465²⁰³⁰ - Adapt datapar fill
- PR #5464²⁰³¹ - Moving tag_dispatch to separate module

²⁰⁰⁹ <https://github.com/TheHPXProject/hpx/pull/5510>

²⁰¹⁰ <https://github.com/TheHPXProject/hpx/pull/5509>

²⁰¹¹ <https://github.com/TheHPXProject/hpx/pull/5508>

²⁰¹² <https://github.com/TheHPXProject/hpx/pull/5507>

²⁰¹³ <https://github.com/TheHPXProject/hpx/pull/5506>

²⁰¹⁴ <https://github.com/TheHPXProject/hpx/pull/5501>

²⁰¹⁵ <https://github.com/TheHPXProject/hpx/pull/5493>

²⁰¹⁶ <https://github.com/TheHPXProject/hpx/pull/5490>

²⁰¹⁷ <https://github.com/TheHPXProject/hpx/pull/5487>

²⁰¹⁸ <https://github.com/TheHPXProject/hpx/pull/5484>

²⁰¹⁹ <https://github.com/TheHPXProject/hpx/pull/5481>

²⁰²⁰ <https://github.com/TheHPXProject/hpx/pull/5479>

²⁰²¹ <https://github.com/TheHPXProject/hpx/pull/5477>

²⁰²² <https://github.com/TheHPXProject/hpx/pull/5476>

²⁰²³ <https://github.com/TheHPXProject/hpx/pull/5475>

²⁰²⁴ <https://github.com/TheHPXProject/hpx/pull/5474>

²⁰²⁵ <https://github.com/TheHPXProject/hpx/pull/5473>

²⁰²⁶ <https://github.com/TheHPXProject/hpx/pull/5471>

²⁰²⁷ <https://github.com/TheHPXProject/hpx/pull/5470>

²⁰²⁸ <https://github.com/TheHPXProject/hpx/pull/5468>

²⁰²⁹ <https://github.com/TheHPXProject/hpx/pull/5466>

²⁰³⁰ <https://github.com/TheHPXProject/hpx/pull/5465>

²⁰³¹ <https://github.com/TheHPXProject/hpx/pull/5464>

- PR #5461²⁰³² - Rename HPX_WITH_CUDA_COMPUTE with HPX_WITH_COMPUTE_CUDA
- PR #5460²⁰³³ - Adapt sort to C++ 20
- PR #5459²⁰³⁴ - Adapt rotate/rotate_copy to C++20
- PR #5458²⁰³⁵ - Adapt unique and unique_copy to C++ 20
- PR #5455²⁰³⁶ - Remove and clean up fallback sender implementations
- PR #5454²⁰³⁷ - Make performance plot show even if similar performance
- PR #5453²⁰³⁸ - Post 1.7.0 version bump
- PR #5452²⁰³⁹ - Fix find_end parallel overload
- PR #5450²⁰⁴⁰ - Change the print-bind output to be more precise
- PR #5449²⁰⁴¹ - Adapt swap_ranges to C++ 20
- PR #5446²⁰⁴² - Use more verbose names in sender algorithms
- PR #5443²⁰⁴³ - Properly support ASAN with MSVC
- PR #5441²⁰⁴⁴ - Adding reference counting to thread_data
- PR #5429²⁰⁴⁵ - Scheduler executor
- PR #5428²⁰⁴⁶ - Adapt datapar copy
- PR #5421²⁰⁴⁷ - Update CI base image to use clang-format 11
- PR #5410²⁰⁴⁸ - Add ranges starts_with and ends_with algorithms
- PR #5383²⁰⁴⁹ - Tentatively remove runtime_registration_wrapper from cuda futures
- PR #5377²⁰⁵⁰ - Fewer Asio includes and more precompiled headers
- PR #5329²⁰⁵¹ - Sender overloads for parallel algorithms
- PR #5313²⁰⁵² - Rearrange modules between libraries
- PR #5283²⁰⁵³ - Require minimum C++17 and change CUDA handling
- PR #5241²⁰⁵⁴ - Adapt min_element, max_element and minmax_element to C++20

²⁰³² <https://github.com/TheHPXProject/hpx/pull/5461>

²⁰³³ <https://github.com/TheHPXProject/hpx/pull/5460>

²⁰³⁴ <https://github.com/TheHPXProject/hpx/pull/5459>

²⁰³⁵ <https://github.com/TheHPXProject/hpx/pull/5458>

²⁰³⁶ <https://github.com/TheHPXProject/hpx/pull/5455>

²⁰³⁷ <https://github.com/TheHPXProject/hpx/pull/5454>

²⁰³⁸ <https://github.com/TheHPXProject/hpx/pull/5453>

²⁰³⁹ <https://github.com/TheHPXProject/hpx/pull/5452>

²⁰⁴⁰ <https://github.com/TheHPXProject/hpx/pull/5450>

²⁰⁴¹ <https://github.com/TheHPXProject/hpx/pull/5449>

²⁰⁴² <https://github.com/TheHPXProject/hpx/pull/5446>

²⁰⁴³ <https://github.com/TheHPXProject/hpx/pull/5443>

²⁰⁴⁴ <https://github.com/TheHPXProject/hpx/pull/5441>

²⁰⁴⁵ <https://github.com/TheHPXProject/hpx/pull/5429>

²⁰⁴⁶ <https://github.com/TheHPXProject/hpx/pull/5428>

²⁰⁴⁷ <https://github.com/TheHPXProject/hpx/pull/5421>

²⁰⁴⁸ <https://github.com/TheHPXProject/hpx/pull/5410>

²⁰⁴⁹ <https://github.com/TheHPXProject/hpx/pull/5383>

²⁰⁵⁰ <https://github.com/TheHPXProject/hpx/pull/5377>

²⁰⁵¹ <https://github.com/TheHPXProject/hpx/pull/5329>

²⁰⁵² <https://github.com/TheHPXProject/hpx/pull/5313>

²⁰⁵³ <https://github.com/TheHPXProject/hpx/pull/5283>

²⁰⁵⁴ <https://github.com/TheHPXProject/hpx/pull/5241>

HPX V1.7.1 (Aug 12, 2021)

This is a bugfix release with a few minor fixes.

General changes

- Added a CMake option to assume that all types are bitwise serializable by default: `HPX_SERIALIZATION_WITH_ALL_TYPES_ARE_BITWISE_SERIALIZABLE`. The default value `OFF` corresponds to the old behaviour.
- Added a version check for Asio. The minimum Asio version supported by *HPX* is 1.12.0.
- Fixed a bug affecting usage of actions, where the internals of *HPX* relied on function addresses being unique. This was fixed by relying on variable addresses being unique instead.
- Made `hpx::util::bind` more strict in checking the validity of placeholders.
- Small performance improvement to spinlocks.
- Adapted the following parallel algorithms to C++20: `inclusive_scan`, `exclusive_scan`, `transform_inclusive_scan`, `transform_exclusive_scan`.

Breaking changes

- The experimental `hpx::execution::simdpair` execution policy (introduced in 1.7.0) was renamed to `hpx::execution::par_simd` for consistency with the other parallel policies.

Closed issues

- [Issue #5494](#)²⁰⁵⁵ - Rename *simdpair* execution policy to *par_simd*
- [Issue #5488](#)²⁰⁵⁶ - `hpx::util::bind` doesn't bounds-check placeholders
- [Issue #5486](#)²⁰⁵⁷ - Possible V1.7.1 release

Closed pull requests

- [PR #5500](#)²⁰⁵⁸ - Minor bug fix in transform exclusive and inclusive scan tests
- [PR #5499](#)²⁰⁵⁹ - Rename *simdpair* to *par_simd*
- [PR #5489](#)²⁰⁶⁰ - Adding bound-checking for bind placeholders
- [PR #5485](#)²⁰⁶¹ - Add Asio version check
- [PR #5482](#)²⁰⁶² - Change extra archive data to rely on uniqueness of a variable address, not a function address
- [PR #5448](#)²⁰⁶³ - More fixes to enable for all types to be assumed to be bitwise copyable

²⁰⁵⁵ <https://github.com/TheHPXProject/hpx/issues/5494>

²⁰⁵⁶ <https://github.com/TheHPXProject/hpx/issues/5488>

²⁰⁵⁷ <https://github.com/TheHPXProject/hpx/issues/5486>

²⁰⁵⁸ <https://github.com/TheHPXProject/hpx/pull/5500>

²⁰⁵⁹ <https://github.com/TheHPXProject/hpx/pull/5499>

²⁰⁶⁰ <https://github.com/TheHPXProject/hpx/pull/5489>

²⁰⁶¹ <https://github.com/TheHPXProject/hpx/pull/5485>

²⁰⁶² <https://github.com/TheHPXProject/hpx/pull/5482>

²⁰⁶³ <https://github.com/TheHPXProject/hpx/pull/5448>

- PR #5445²⁰⁶⁴ - Improve performance of Spinlocks
- PR #5444²⁰⁶⁵ - Adapt transform_inclusive_scan to C++ 20
- PR #5440²⁰⁶⁶ - Adapt transform_exclusive_scan to C++ 20
- PR #5439²⁰⁶⁷ - Adapt inclusive_scan to C++ 20
- PR #5436²⁰⁶⁸ - Adapt exclusive_scan to C++20

HPX V1.7.0 (Jul 14, 2021)

This release is again focused on C++20 conformance of algorithms. Additionally, many new experimental sender-based algorithms have been added based on the latest proposals.

General changes

- The following algorithms have been adapted to be C++20 conformant:
 - `remove`,
 - `remove_if`,
 - `remove_copy`,
 - `remove_copy_if`,
 - `replace`,
 - `replace_if`,
 - `reverse`, and
 - `lexicographical_compare`.
- When the compiler and standard library support the standard execution policies `std::execution::seq`, `std::execution::par`, and `std::execution::par_unseq` they can now be used in all *HPX* parallel algorithms with equivalent behaviour to the non-task policies `hpx::execution::seq`, `hpx::execution::par`, and `hpx::execution::par_unseq`.
- Vc support has been fixed, after being broken in 1.6.0. In addition, *HPX* now experimentally supports GCC's SIMD implementation, when available. The implementation can be used through the `hpx::execution::simd` and `hpx::execution::simdpair` execution policies.
- The customization points `sync_execute`, `async_execute`, `then_execute`, `post`, `bulk_sync_execute`, `bulk_async_execute`, and `bulk_then_execute` are now implemented using `tag_dispatch` (previously `tag_invoke`). Executors can still be implemented by providing the aforementioned functions as member functions of an executor.
- New functionality, enhancements, and fixes based on P0443r14 (executors proposal) and P1897 (sender-based algorithms) have been added to the `hpx::execution::experimental` namespace. These can be accessed through the `hpx/execution.hpp` and `hpx/local/execution.hpp` headers. In particular, the following sender-based algorithms have been added:
 - `detach`,

²⁰⁶⁴ <https://github.com/TheHPXProject/hpx/pull/5445>

²⁰⁶⁵ <https://github.com/TheHPXProject/hpx/pull/5444>

²⁰⁶⁶ <https://github.com/TheHPXProject/hpx/pull/5440>

²⁰⁶⁷ <https://github.com/TheHPXProject/hpx/pull/5439>

²⁰⁶⁸ <https://github.com/TheHPXProject/hpx/pull/5436>

- `ensure_started`,
- `just`,
- `just_on`,
- `let_error`,
- `let_value`,
- `on`,
- `transform`, and
- `when_all`.

Additionally, futures now implement the sender concept. `make_future` can be used to turn a sender into a future. All functionality is experimental and can change without notice.

- All `hpx::init` and `hpx::start` overloads now take `std::functions` instead of `hpx::util::function_nonser`. No changes should be required in user code to accommodate this change.
- `hpx::util::unwrapping` and other related unwrapping functionality has been moved up into the `hpx` namespace. Names in `hpx::util` are still usable with a deprecation warning. This functionality can now be accessed through the `hpx/unwrap.hpp` and `hpx/local/unwrap.hpp` headers.
- The default tag for APEX has been update from 2.3.1 to 2.4.0. In particular, this fixes a bug which could lead to hangs in distributed runs.
- The dependency on Boost.Asio has been replaced with the standalone Asio available at <https://github.com/chriskohlhoff/asio>. By default, a system-installed Asio will be used. `ASIO_ROOT` can be given as a hint to tell CMake where to find Asio. Alternatively, Asio can be fetched automatically using CMake's `fetchcontent` by setting `HPX_WITH_FETCH_ASIO=ON`. In general, dependencies on Boost have again been reduced.
- Modularization of the library has continued. In this release almost all functionality has been moved into modules. These changes do not generally affect user code. Warnings are still issued for headers that have moved.
- `hipBLAS` is now optional when compiling with `hipcc`. A warning instead of an error will be printed if `hipBLAS` is not found during configuration.
- Previously `HPX_COMPUTE_HOST_CODE` was defined in host code only if HPX was configured with CUDA or HIP. In this release `HPX_COMPUTE_HOST_CODE` is always defined in host code.
- An experimental `HPX_WITH_PRECOMPILED_HEADERS` CMake option has been added to use precompiled headers when building *HPX*. This option should not be used on Windows.
- Numerous bug fixes.

Breaking changes

- The minimum required CMake version is now 3.17.
- The minimum required Boost version is now 1.71.0.
- The customization mechanism used to implement and extend sender functionality and algorithms has been re-named from `tag_invoke` to `tag_dispatch`. All customization of sender functionality should be done by overloading `tag_dispatch`.
- The following compatibility options have been removed, along with their compatibility implementations:
 - `HPX_PROGRAM_OPTIONS_WITH_BOOST_PROGRAM_OPTIONS_COMPATIBILITY`
 - `HPX_WITH_ACTION_BASE_COMPATIBILITY`
 - `HPX_WITH_EMBEDDED_THREAD_POOLS_COMPATIBILITY`
 - `HPX_WITH_POOL_EXECUTOR_COMPATIBILITY`
 - `HPX_WITH_PROMISE_ALIAS_COMPATIBILITY`

HPX_WITH_REGISTER_THREAD_COMPATIBILITY - HPX_WITH_REGISTER_THREAD_OVERLOADS_COMPATIBILITY
- HPX_WITH_THREAD_AWARE_TIMER_COMPATIBILITY - HPX_WITH_THREAD_EXECUTORS_COMPATIBILITY -
HPX_WITH_THREAD_POOL_OS_EXECUTOR_COMPATIBILITY

- The HPX_WITH_THREAD_SCHEDULERS CMake option has been removed. All schedulers are now enabled when possible.
- HPX_WITH_INIT_START_OVERLOADS_COMPATIBILITY has been turned off by default.

Closed issues

- [Issue #5423](#)²⁰⁶⁹ - Fix lvalue-ref qualified connect for when_all-sender
- [Issue #5412](#)²⁰⁷⁰ - Link error
- [Issue #5397](#)²⁰⁷¹ - Performance regression in thread annotations
- [Issue #5395](#)²⁰⁷² - HPX 1.7.0-rc1 fails to build icw APEX + OTF2
- [Issue #5385](#)²⁰⁷³ - HPX 1.7 crashes on Piz Daint > 64 nodes
- [Issue #5380](#)²⁰⁷⁴ - CMake should search for asio package installed on the system
- [Issue #5378](#)²⁰⁷⁵ - HPX 1.7.0 stopped building on Fedora
- [Issue #5369](#)²⁰⁷⁶ - HPX 1.6 and master hangs on Summit for > 64 nodes
- [Issue #5358](#)²⁰⁷⁷ - HPX init fails for single-core environments
- [Issue #5345](#)²⁰⁷⁸ - Rename P2220 property CPOs?
- [Issue #5333](#)²⁰⁷⁹ - HPX does not compile on the new Mac OSX using the M1 chip
- [Issue #5317](#)²⁰⁸⁰ - Consider making hipblas optional
- [Issue #5306](#)²⁰⁸¹ - asio fails to build with CUDA 10.0
- [Issue #5294](#)²⁰⁸² - `execution::on` should be based on `execution::schedule`
- [Issue #5275](#)²⁰⁸³ - HPX V1.6.0 fails on Fedora release
- [Issue #5270](#)²⁰⁸⁴ - HPX-1.6.0 fails to build on Windows 10
- [Issue #5257](#)²⁰⁸⁵ - Allow triggering the output of OS thread affinity from configuration settings
- [Issue #5246](#)²⁰⁸⁶ - HPX fails to build on ppc64le

²⁰⁶⁹ <https://github.com/TheHPXProject/hpx/issues/5423>

²⁰⁷⁰ <https://github.com/TheHPXProject/hpx/issues/5412>

²⁰⁷¹ <https://github.com/TheHPXProject/hpx/issues/5397>

²⁰⁷² <https://github.com/TheHPXProject/hpx/issues/5395>

²⁰⁷³ <https://github.com/TheHPXProject/hpx/issues/5385>

²⁰⁷⁴ <https://github.com/TheHPXProject/hpx/issues/5380>

²⁰⁷⁵ <https://github.com/TheHPXProject/hpx/issues/5378>

²⁰⁷⁶ <https://github.com/TheHPXProject/hpx/issues/5369>

²⁰⁷⁷ <https://github.com/TheHPXProject/hpx/issues/5358>

²⁰⁷⁸ <https://github.com/TheHPXProject/hpx/issues/5345>

²⁰⁷⁹ <https://github.com/TheHPXProject/hpx/issues/5333>

²⁰⁸⁰ <https://github.com/TheHPXProject/hpx/issues/5317>

²⁰⁸¹ <https://github.com/TheHPXProject/hpx/issues/5306>

²⁰⁸² <https://github.com/TheHPXProject/hpx/issues/5294>

²⁰⁸³ <https://github.com/TheHPXProject/hpx/issues/5275>

²⁰⁸⁴ <https://github.com/TheHPXProject/hpx/issues/5270>

²⁰⁸⁵ <https://github.com/TheHPXProject/hpx/issues/5257>

²⁰⁸⁶ <https://github.com/TheHPXProject/hpx/issues/5246>

- [Issue #5232²⁰⁸⁷](#) - Annotation using `hpx::util::annotated_function` not working
- [Issue #5222²⁰⁸⁸](#) - Build and link errors with `itnotify` enabled
- [Issue #5204²⁰⁸⁹](#) - Move algorithms to `tag_fallback_dispatch`
- [Issue #5163²⁰⁹⁰](#) - Remove module-specific compatibility and deprecation options
- [Issue #5161²⁰⁹¹](#) - Bump required CMake version to 3.17
- [Issue #5143²⁰⁹²](#) - Searching for HPX-Application to generate work on multiple Nodes

Closed pull requests

- [PR #5438²⁰⁹³](#) - Delete `datapar/foreach_tests.hpp`
- [PR #5437²⁰⁹⁴](#) - Add back explicit `-pthread` flags when available
- [PR #5435²⁰⁹⁵](#) - This adds support for systems that assume all types are bitwise serializable by default
- [PR #5434²⁰⁹⁶](#) - Update CUDA polling logging to be more verbose
- [PR #5433²⁰⁹⁷](#) - Fix `when_all_sender` connect for references
- [PR #5432²⁰⁹⁸](#) - Add deprecation warnings for v1.8
- [PR #5431²⁰⁹⁹](#) - Rename the new P0443/P2300 executor to `thread_pool_scheduler`
- [PR #5430²¹⁰⁰](#) - Revert “Adding the missing defined for `HPX_HAVE_DEPRECATED_WARNINGS`”
- [PR #5427²¹⁰¹](#) - Removing unneeded `typedef`
- [PR #5426²¹⁰²](#) - Adding more concept checks for sender/receiver algorithms
- [PR #5425²¹⁰³](#) - Adding the missing defined for `HPX_HAVE_DEPRECATED_WARNINGS`
- [PR #5424²¹⁰⁴](#) - Disable Vc in final docker image created in CI
- [PR #5422²¹⁰⁵](#) - Adding `execution::experimental::bulk` algorithm
- [PR #5420²¹⁰⁶](#) - Update logic to find threading library
- [PR #5418²¹⁰⁷](#) - Reduce max size and number of files in `ccache` cache

²⁰⁸⁷ <https://github.com/TheHPXProject/hpx/issues/5232>

²⁰⁸⁸ <https://github.com/TheHPXProject/hpx/issues/5222>

²⁰⁸⁹ <https://github.com/TheHPXProject/hpx/issues/5204>

²⁰⁹⁰ <https://github.com/TheHPXProject/hpx/issues/5163>

²⁰⁹¹ <https://github.com/TheHPXProject/hpx/issues/5161>

²⁰⁹² <https://github.com/TheHPXProject/hpx/issues/5143>

²⁰⁹³ <https://github.com/TheHPXProject/hpx/pull/5438>

²⁰⁹⁴ <https://github.com/TheHPXProject/hpx/pull/5437>

²⁰⁹⁵ <https://github.com/TheHPXProject/hpx/pull/5435>

²⁰⁹⁶ <https://github.com/TheHPXProject/hpx/pull/5434>

²⁰⁹⁷ <https://github.com/TheHPXProject/hpx/pull/5433>

²⁰⁹⁸ <https://github.com/TheHPXProject/hpx/pull/5432>

²⁰⁹⁹ <https://github.com/TheHPXProject/hpx/pull/5431>

²¹⁰⁰ <https://github.com/TheHPXProject/hpx/pull/5430>

²¹⁰¹ <https://github.com/TheHPXProject/hpx/pull/5427>

²¹⁰² <https://github.com/TheHPXProject/hpx/pull/5426>

²¹⁰³ <https://github.com/TheHPXProject/hpx/pull/5425>

²¹⁰⁴ <https://github.com/TheHPXProject/hpx/pull/5424>

²¹⁰⁵ <https://github.com/TheHPXProject/hpx/pull/5422>

²¹⁰⁶ <https://github.com/TheHPXProject/hpx/pull/5420>

²¹⁰⁷ <https://github.com/TheHPXProject/hpx/pull/5418>

- [PR #5417](#)²¹⁰⁸ - Final release notes for 1.7.0
- [PR #5416](#)²¹⁰⁹ - Adapt `uninitialized_value_construct` and `uninitialized_value_construct_n` to C++ 20
- [PR #5415](#)²¹¹⁰ - Adapt `uninitialized_default_construct` and `uninitialized_default_construct_n` to C++ 20
- [PR #5414](#)²¹¹¹ - Improve integration of futures and senders
- [PR #5413](#)²¹¹² - Fixing sender/receiver code base to compile with MSVC
- [PR #5407](#)²¹¹³ - Handle exceptions thrown during initialization of parcel handler
- [PR #5406](#)²¹¹⁴ - Simplify dispatching to annotation handlers
- [PR #5405](#)²¹¹⁵ - Fetch Asio automatically in perf tests CI
- [PR #5403](#)²¹¹⁶ - Create generic executor that adds annotations to any other executor
- [PR #5402](#)²¹¹⁷ - Adapt `uninitialized_fill` and `uninitialized_fill_n` to C++ 20
- [PR #5401](#)²¹¹⁸ - Modernize a variety of facilities related to parallel algorithms
- [PR #5400](#)²¹¹⁹ - Fix sliding semaphore test
- [PR #5399](#)²¹²⁰ - Rename leftover `tag_fallback_invoke` to `tag_fallback_dispatch`
- [PR #5398](#)²¹²¹ - Improve logging in AGAS symbol namespace
- [PR #5396](#)²¹²² - Introduce compatibility layer for collective operations
- [PR #5394](#)²¹²³ - Enable OTF2 in APEX CI configuration
- [PR #5393](#)²¹²⁴ - Update APEX tag
- [PR #5392](#)²¹²⁵ - Fixing wrong usage of `std::forward`
- [PR #5391](#)²¹²⁶ - Fix forwarding in `transform_receiver` constructor
- [PR #5390](#)²¹²⁷ - Make sure shared priority scheduler steals tasks on the current NUMA domain when (core) stealing is enabled
- [PR #5389](#)²¹²⁸ - Adapt `uninitialized_move` and `uninitialized_move_n` to C++ 20
- [PR #5388](#)²¹²⁹ - Fixing `gather_there` for used with lvalue reference argument

²¹⁰⁸ <https://github.com/TheHPXProject/hpx/pull/5417>

²¹⁰⁹ <https://github.com/TheHPXProject/hpx/pull/5416>

²¹¹⁰ <https://github.com/TheHPXProject/hpx/pull/5415>

²¹¹¹ <https://github.com/TheHPXProject/hpx/pull/5414>

²¹¹² <https://github.com/TheHPXProject/hpx/pull/5413>

²¹¹³ <https://github.com/TheHPXProject/hpx/pull/5407>

²¹¹⁴ <https://github.com/TheHPXProject/hpx/pull/5406>

²¹¹⁵ <https://github.com/TheHPXProject/hpx/pull/5405>

²¹¹⁶ <https://github.com/TheHPXProject/hpx/pull/5403>

²¹¹⁷ <https://github.com/TheHPXProject/hpx/pull/5402>

²¹¹⁸ <https://github.com/TheHPXProject/hpx/pull/5401>

²¹¹⁹ <https://github.com/TheHPXProject/hpx/pull/5400>

²¹²⁰ <https://github.com/TheHPXProject/hpx/pull/5399>

²¹²¹ <https://github.com/TheHPXProject/hpx/pull/5398>

²¹²² <https://github.com/TheHPXProject/hpx/pull/5396>

²¹²³ <https://github.com/TheHPXProject/hpx/pull/5394>

²¹²⁴ <https://github.com/TheHPXProject/hpx/pull/5393>

²¹²⁵ <https://github.com/TheHPXProject/hpx/pull/5392>

²¹²⁶ <https://github.com/TheHPXProject/hpx/pull/5391>

²¹²⁷ <https://github.com/TheHPXProject/hpx/pull/5390>

²¹²⁸ <https://github.com/TheHPXProject/hpx/pull/5389>

²¹²⁹ <https://github.com/TheHPXProject/hpx/pull/5388>

- PR #5387²¹³⁰ - Extend thread state logging and change default stealing parameters
- PR #5386²¹³¹ - Attempt to fix the startup hang with nodes > 32
- PR #5384²¹³² - Remove HPX 1.5.0 deprecations
- PR #5382²¹³³ - Prefer installed Asio before considering FetchContent
- PR #5379²¹³⁴ - Allow using pre-downloaded (not installed) versions of Asio and/or Apex
- PR #5376²¹³⁵ - Remove unnecessary explicit listing of library modules.rst files in CMakeLists.txt
- PR #5375²¹³⁶ - Slight performance improvement for `hpx::copy` and `hpx::move` et.al.
- PR #5374²¹³⁷ - Remove unnecessary moves from future sender implementations
- PR #5373²¹³⁸ - More changes to clang-cuda Jenkins configuration
- PR #5372²¹³⁹ - Slight improvements to min/max/minmax_element algorithms
- PR #5371²¹⁴⁰ - Adapt uninitialized_copy and uninitialized_copy_n to C++ 20
- PR #5370²¹⁴¹ - Decay types in just_sender value_types to match stored types
- PR #5367²¹⁴² - Disable pkgconfig by default again on macOS
- PR #5365²¹⁴³ - Use ccache for Jenkins builds on Piz Daint
- PR #5363²¹⁴⁴ - Update cudatoolkit module name in clang-cuda Jenkins configuration
- PR #5362²¹⁴⁵ - Adding channel_communicator
- PR #5361²¹⁴⁶ - Fix compilation with MPI enabled
- PR #5360²¹⁴⁷ - Update APEX and asio tags
- PR #5359²¹⁴⁸ - Fix check for pu-step in single-core case
- PR #5357²¹⁴⁹ - Making sure collective operations can be reused by preallocating communicator
- PR #5356²¹⁵⁰ - Update API documentation
- PR #5355²¹⁵¹ - Make the sequenced_executor processing_units_count member function const
- PR #5354²¹⁵² - Making sure default_stack_size is defined whenever declared

²¹³⁰ <https://github.com/TheHPXProject/hpx/pull/5387>

²¹³¹ <https://github.com/TheHPXProject/hpx/pull/5386>

²¹³² <https://github.com/TheHPXProject/hpx/pull/5384>

²¹³³ <https://github.com/TheHPXProject/hpx/pull/5382>

²¹³⁴ <https://github.com/TheHPXProject/hpx/pull/5379>

²¹³⁵ <https://github.com/TheHPXProject/hpx/pull/5376>

²¹³⁶ <https://github.com/TheHPXProject/hpx/pull/5375>

²¹³⁷ <https://github.com/TheHPXProject/hpx/pull/5374>

²¹³⁸ <https://github.com/TheHPXProject/hpx/pull/5373>

²¹³⁹ <https://github.com/TheHPXProject/hpx/pull/5372>

²¹⁴⁰ <https://github.com/TheHPXProject/hpx/pull/5371>

²¹⁴¹ <https://github.com/TheHPXProject/hpx/pull/5370>

²¹⁴² <https://github.com/TheHPXProject/hpx/pull/5367>

²¹⁴³ <https://github.com/TheHPXProject/hpx/pull/5365>

²¹⁴⁴ <https://github.com/TheHPXProject/hpx/pull/5363>

²¹⁴⁵ <https://github.com/TheHPXProject/hpx/pull/5362>

²¹⁴⁶ <https://github.com/TheHPXProject/hpx/pull/5361>

²¹⁴⁷ <https://github.com/TheHPXProject/hpx/pull/5360>

²¹⁴⁸ <https://github.com/TheHPXProject/hpx/pull/5359>

²¹⁴⁹ <https://github.com/TheHPXProject/hpx/pull/5357>

²¹⁵⁰ <https://github.com/TheHPXProject/hpx/pull/5356>

²¹⁵¹ <https://github.com/TheHPXProject/hpx/pull/5355>

²¹⁵² <https://github.com/TheHPXProject/hpx/pull/5354>

- PR #5353²¹⁵³ - Add CUDA timestamp support to HPX Hardware Clock
- PR #5352²¹⁵⁴ - Adding missing includes
- PR #5351²¹⁵⁵ - Adding `enable_logging/disable_logging` API functions
- PR #5350²¹⁵⁶ - Adapt `lexicographical_compare` to C++20
- PR #5349²¹⁵⁷ - Update minimum boost version needed on the docs
- PR #5348²¹⁵⁸ - Rename `tag_invoke` and related facilities to `tag_dispatch`
- PR #5347²¹⁵⁹ - Remove `make_` prefix for executor properties
- PR #5346²¹⁶⁰ - Remove and disable compatibility options for 1.7.0
- PR #5343²¹⁶¹ - Fix `timed_executor` static cast conversion
- PR #5342²¹⁶² - Refactor CUDA event polling
- PR #5341²¹⁶³ - Adding `make_with_annotation` and `get_annotation` properties
- PR #5339²¹⁶⁴ - Making sure `hpx::util::hardware::timestamp()` is always defined
- PR #5338²¹⁶⁵ - Fixing `timed_executor` specializations of customization points
- PR #5335²¹⁶⁶ - Make `partial_algorithm` work with any number of arguments
- PR #5334²¹⁶⁷ - Follow up `iter_sent` include on #5225
- PR #5332²¹⁶⁸ - Simplify `tag_invoke` and friends
- PR #5331²¹⁶⁹ - More work on cleaning up executor CPOs
- PR #5330²¹⁷⁰ - Add option to disable `pkgconfig` generation
- PR #5328²¹⁷¹ - Adapt data parallel support using `std-simd`
- PR #5327²¹⁷² - Fix missing `ifdef HPX_SMT_PAUSE`
- PR #5326²¹⁷³ - Adding `resize()` to `serialize_buffer` allowing to shrink its size
- PR #5324²¹⁷⁴ - Add get member functions to `async_rw_mutex` proxy objects for explicitly getting the wrapped value
- PR #5323²¹⁷⁵ - Add `keep_future` algorithm

²¹⁵³ <https://github.com/TheHPXProject/hpx/pull/5353>

²¹⁵⁴ <https://github.com/TheHPXProject/hpx/pull/5352>

²¹⁵⁵ <https://github.com/TheHPXProject/hpx/pull/5351>

²¹⁵⁶ <https://github.com/TheHPXProject/hpx/pull/5350>

²¹⁵⁷ <https://github.com/TheHPXProject/hpx/pull/5349>

²¹⁵⁸ <https://github.com/TheHPXProject/hpx/pull/5348>

²¹⁵⁹ <https://github.com/TheHPXProject/hpx/pull/5347>

²¹⁶⁰ <https://github.com/TheHPXProject/hpx/pull/5346>

²¹⁶¹ <https://github.com/TheHPXProject/hpx/pull/5343>

²¹⁶² <https://github.com/TheHPXProject/hpx/pull/5342>

²¹⁶³ <https://github.com/TheHPXProject/hpx/pull/5341>

²¹⁶⁴ <https://github.com/TheHPXProject/hpx/pull/5339>

²¹⁶⁵ <https://github.com/TheHPXProject/hpx/pull/5338>

²¹⁶⁶ <https://github.com/TheHPXProject/hpx/pull/5335>

²¹⁶⁷ <https://github.com/TheHPXProject/hpx/pull/5334>

²¹⁶⁸ <https://github.com/TheHPXProject/hpx/pull/5332>

²¹⁶⁹ <https://github.com/TheHPXProject/hpx/pull/5331>

²¹⁷⁰ <https://github.com/TheHPXProject/hpx/pull/5330>

²¹⁷¹ <https://github.com/TheHPXProject/hpx/pull/5328>

²¹⁷² <https://github.com/TheHPXProject/hpx/pull/5327>

²¹⁷³ <https://github.com/TheHPXProject/hpx/pull/5326>

²¹⁷⁴ <https://github.com/TheHPXProject/hpx/pull/5324>

²¹⁷⁵ <https://github.com/TheHPXProject/hpx/pull/5323>

- PR #5322²¹⁷⁶ - Replace executor customization point implementations with `tag_invoke`
- PR #5321²¹⁷⁷ - Seperate segmented algorithms for `reduce`
- PR #5320²¹⁷⁸ - Fix `is_sender` trait and other small fixes to p0443 traits
- PR #5319²¹⁷⁹ - gcc 11.1 c++20 build fixes
- PR #5318²¹⁸⁰ - Make hipblas dependency optional as not always available
- PR #5316²¹⁸¹ - Attempt to fix checking for `libatomic`
- PR #5315²¹⁸² - Add `explicit` keyword to fixture constructor
- PR #5314²¹⁸³ - Fix a race condition in `async_mpi` affecting limiting executor
- PR #5312²¹⁸⁴ - Use local runtime and local headers in local-only modules and tests
- PR #5311²¹⁸⁵ - Add GCC 11 builder to jenkins
- PR #5310²¹⁸⁶ - Adding `hpx::execution::experimental::task_group`
- PR #5309²¹⁸⁷ - Seperate `datapar`
- PR #5308²¹⁸⁸ - Seperate segmented algorithms for `find`, `find_if`, `find_if_not`
- PR #5307²¹⁸⁹ - Seperate segmented algorithms for `fill` and `generate`
- PR #5304²¹⁹⁰ - Fix compilation of sender CPOs with `nvcc`
- PR #5300²¹⁹¹ - Remove `PRIVATE` flag that was propagated into the `LANGUAGES`
- PR #5298²¹⁹² - Seperate `datapar`
- PR #5297²¹⁹³ - Specify exact `cmake` and `ninja` versions when loading them in jenkins jobs
- PR #5295²¹⁹⁴ - Update `clang-newest` configuration to use `clang 12` and `Boost 1.76.0`
- PR #5293²¹⁹⁵ - Fix `Clang 11` `cuda_future` test bug
- PR #5292²¹⁹⁶ - Add `async_rw_mutex` based on senders
- PR #5291²¹⁹⁷ - “Fix” termination detection
- PR #5290²¹⁹⁸ - Fixed source file line statements in examples documentation

²¹⁷⁶ <https://github.com/TheHPXProject/hpx/pull/5322>

²¹⁷⁷ <https://github.com/TheHPXProject/hpx/pull/5321>

²¹⁷⁸ <https://github.com/TheHPXProject/hpx/pull/5320>

²¹⁷⁹ <https://github.com/TheHPXProject/hpx/pull/5319>

²¹⁸⁰ <https://github.com/TheHPXProject/hpx/pull/5318>

²¹⁸¹ <https://github.com/TheHPXProject/hpx/pull/5316>

²¹⁸² <https://github.com/TheHPXProject/hpx/pull/5315>

²¹⁸³ <https://github.com/TheHPXProject/hpx/pull/5314>

²¹⁸⁴ <https://github.com/TheHPXProject/hpx/pull/5312>

²¹⁸⁵ <https://github.com/TheHPXProject/hpx/pull/5311>

²¹⁸⁶ <https://github.com/TheHPXProject/hpx/pull/5310>

²¹⁸⁷ <https://github.com/TheHPXProject/hpx/pull/5309>

²¹⁸⁸ <https://github.com/TheHPXProject/hpx/pull/5308>

²¹⁸⁹ <https://github.com/TheHPXProject/hpx/pull/5307>

²¹⁹⁰ <https://github.com/TheHPXProject/hpx/pull/5304>

²¹⁹¹ <https://github.com/TheHPXProject/hpx/pull/5300>

²¹⁹² <https://github.com/TheHPXProject/hpx/pull/5298>

²¹⁹³ <https://github.com/TheHPXProject/hpx/pull/5297>

²¹⁹⁴ <https://github.com/TheHPXProject/hpx/pull/5295>

²¹⁹⁵ <https://github.com/TheHPXProject/hpx/pull/5293>

²¹⁹⁶ <https://github.com/TheHPXProject/hpx/pull/5292>

²¹⁹⁷ <https://github.com/TheHPXProject/hpx/pull/5291>

²¹⁹⁸ <https://github.com/TheHPXProject/hpx/pull/5290>

- PR #5289²¹⁹⁹ - Allow splitting of futures holding `std::tuple`
- PR #5288²²⁰⁰ - Move algorithms to `tag_fallback_invoke`
- PR #5287²²⁰¹ - Move algorithms to `tag_fallback_invoke`
- PR #5285²²⁰² - Fix clang-format failure on master
- PR #5284²²⁰³ - Replacing `util::function_nonser` on `std::function` in `hpx_init`
- PR #5282²²⁰⁴ - Update Boost for daint 20.11 after update
- PR #5281²²⁰⁵ - Fix Segmentation fault on `foreach_datapar_zipiter`
- PR #5280²²⁰⁶ - Avoid modulo by zero in `counting_iterator` test
- PR #5279²²⁰⁷ - Fix more GCC 10 deprecation warnings
- PR #5277²²⁰⁸ - Small fixes and improvements to CUDA/MPI polling
- PR #5276²²⁰⁹ - Fix typo in docs
- PR #5274²²¹⁰ - More P1897 algorithms
- PR #5273²²¹¹ - Retry CDash submissions on failure
- PR #5272²²¹² - Fix bogus deprecation warnings with GCC 10
- PR #5271²²¹³ - Correcting target ids for `symbol_namespace::iterate`
- PR #5268²²¹⁴ - Adding generic `require`, `require_concept`, and `query` properties
- PR #5267²²¹⁵ - Support annotations in `hpx::transform_reduce`
- PR #5266²²¹⁶ - Making late command line options available for local runtime
- PR #5265²²¹⁷ - Leverage `no_unique_address` for `member_pack`
- PR #5264²²¹⁸ - Adopt format in more places
- PR #5262²²¹⁹ - Install HPX in Rostam Jenkins jobs
- PR #5261²²²⁰ - Limit Rostam Jenkins jobs to marvin partition temporarily
- PR #5260²²²¹ - Separate segmented algorithms for `transform_reduce`

²¹⁹⁹ <https://github.com/TheHPXProject/hpx/pull/5289>

²²⁰⁰ <https://github.com/TheHPXProject/hpx/pull/5288>

²²⁰¹ <https://github.com/TheHPXProject/hpx/pull/5287>

²²⁰² <https://github.com/TheHPXProject/hpx/pull/5285>

²²⁰³ <https://github.com/TheHPXProject/hpx/pull/5284>

²²⁰⁴ <https://github.com/TheHPXProject/hpx/pull/5282>

²²⁰⁵ <https://github.com/TheHPXProject/hpx/pull/5281>

²²⁰⁶ <https://github.com/TheHPXProject/hpx/pull/5280>

²²⁰⁷ <https://github.com/TheHPXProject/hpx/pull/5279>

²²⁰⁸ <https://github.com/TheHPXProject/hpx/pull/5277>

²²⁰⁹ <https://github.com/TheHPXProject/hpx/pull/5276>

²²¹⁰ <https://github.com/TheHPXProject/hpx/pull/5274>

²²¹¹ <https://github.com/TheHPXProject/hpx/pull/5273>

²²¹² <https://github.com/TheHPXProject/hpx/pull/5272>

²²¹³ <https://github.com/TheHPXProject/hpx/pull/5271>

²²¹⁴ <https://github.com/TheHPXProject/hpx/pull/5268>

²²¹⁵ <https://github.com/TheHPXProject/hpx/pull/5267>

²²¹⁶ <https://github.com/TheHPXProject/hpx/pull/5266>

²²¹⁷ <https://github.com/TheHPXProject/hpx/pull/5265>

²²¹⁸ <https://github.com/TheHPXProject/hpx/pull/5264>

²²¹⁹ <https://github.com/TheHPXProject/hpx/pull/5262>

²²²⁰ <https://github.com/TheHPXProject/hpx/pull/5261>

²²²¹ <https://github.com/TheHPXProject/hpx/pull/5260>

- PR #5259²²²² - Making sure late command line options are recognized as configuration options
- PR #5258²²²³ - Allow for HPX algorithms being invoked with std execution policies
- PR #5256²²²⁴ - Separate segmented algorithms for transform
- PR #5255²²²⁵ - Future/sender adapters
- PR #5254²²²⁶ - Fixing datapar
- PR #5253²²²⁷ - Add utility to format ranges
- PR #5252²²²⁸ - Remove uses of Boost.Bimap
- PR #5251²²²⁹ - Banish <iostream> from library headers
- PR #5250²²³⁰ - Try fixing vc circle ci
- PR #5249²²³¹ - Adding missing header
- PR #5248²²³² - Use old Piz Daint modules after upgrade
- PR #5247²²³³ - Significantly speedup simple for_each, for_loop, and transform
- PR #5245²²³⁴ - P1897 operator| overloads
- PR #5244²²³⁵ - P1897 when_all
- PR #5243²²³⁶ - Make sure HPX_DEBUG is set based on HPX's build type, not consuming project's build type
- PR #5242²²³⁷ - Moving last files unrelated to parcel layer to modules
- PR #5240²²³⁸ - change namespace for transform_loop.hpp
- PR #5238²²³⁹ - Make sure annotations are used in the binary transform
- PR #5237²²⁴⁰ - Add P1897 just, just_on, and on algorithms
- PR #5236²²⁴¹ - Add an example demonstrating the use of the invoke_function_action facility
- PR #5235²²⁴² - Attempting to fix datapar compilation issues
- PR #5234²²⁴³ - Fix small typo in --hpx:local option description
- PR #5233²²⁴⁴ - Only find Boost.Iostreams if required for plugins

²²²² <https://github.com/TheHPXProject/hpx/pull/5259>

²²²³ <https://github.com/TheHPXProject/hpx/pull/5258>

²²²⁴ <https://github.com/TheHPXProject/hpx/pull/5256>

²²²⁵ <https://github.com/TheHPXProject/hpx/pull/5255>

²²²⁶ <https://github.com/TheHPXProject/hpx/pull/5254>

²²²⁷ <https://github.com/TheHPXProject/hpx/pull/5253>

²²²⁸ <https://github.com/TheHPXProject/hpx/pull/5252>

²²²⁹ <https://github.com/TheHPXProject/hpx/pull/5251>

²²³⁰ <https://github.com/TheHPXProject/hpx/pull/5250>

²²³¹ <https://github.com/TheHPXProject/hpx/pull/5249>

²²³² <https://github.com/TheHPXProject/hpx/pull/5248>

²²³³ <https://github.com/TheHPXProject/hpx/pull/5247>

²²³⁴ <https://github.com/TheHPXProject/hpx/pull/5245>

²²³⁵ <https://github.com/TheHPXProject/hpx/pull/5244>

²²³⁶ <https://github.com/TheHPXProject/hpx/pull/5243>

²²³⁷ <https://github.com/TheHPXProject/hpx/pull/5242>

²²³⁸ <https://github.com/TheHPXProject/hpx/pull/5240>

²²³⁹ <https://github.com/TheHPXProject/hpx/pull/5238>

²²⁴⁰ <https://github.com/TheHPXProject/hpx/pull/5237>

²²⁴¹ <https://github.com/TheHPXProject/hpx/pull/5236>

²²⁴² <https://github.com/TheHPXProject/hpx/pull/5235>

²²⁴³ <https://github.com/TheHPXProject/hpx/pull/5234>

²²⁴⁴ <https://github.com/TheHPXProject/hpx/pull/5233>

- PR #5231²²⁴⁵ - Sort printed config options
- PR #5230²²⁴⁶ - Fix C++20 replace algo adaptation misses
- PR #5229²²⁴⁷ - Remove leftover Boost include from `sync_wait.hpp`
- PR #5228²²⁴⁸ - Print module name only if it has custom configuration settings
- PR #5227²²⁴⁹ - Update `.codespell_whitelist`
- PR #5226²²⁵⁰ - Use new docker image in all CircleCI steps
- PR #5225²²⁵¹ - Adapt reverse to C++20
- PR #5224²²⁵² - Separate segmented algorithms for `none_of`, `any_of` and `all_of`
- PR #5223²²⁵³ - Fixing build system for `itnotify`
- PR #5221²²⁵⁴ - Moving LCO related files to modules
- PR #5220²²⁵⁵ - Separate segmented algorithms for `count` and `count_if`
- PR #5218²²⁵⁶ - Separate segmented algorithms for `adjacent_find`
- PR #5217²²⁵⁷ - Add a HIP github action
- PR #5215²²⁵⁸ - Update ROCm to 4.0.1 on Rostam
- PR #5214²²⁵⁹ - Fix clang-format error in `sender.hpp`
- PR #5213²²⁶⁰ - Removing ESSENTIAL option to the doc example
- PR #5212²²⁶¹ - Separate segmented algorithms for `for_each_n`
- PR #5211²²⁶² - Minor adapted algos fixes
- PR #5210²²⁶³ - Fixing `is_invocable` deprecation warnings
- PR #5209²²⁶⁴ - Moving more files into modules (actions, components, `init_runtime`, etc.)
- PR #5208²²⁶⁵ - Add examples and explanation on when `tag_fallback/priority` are useful
- PR #5207²²⁶⁶ - Always define `HPX_COMPUTE_HOST_CODE` for host code
- PR #5206²²⁶⁷ - Add formatting exceptions for `libhpx` to `create_module_skeleton.py`

²²⁴⁵ <https://github.com/TheHPXProject/hpx/pull/5231>

²²⁴⁶ <https://github.com/TheHPXProject/hpx/pull/5230>

²²⁴⁷ <https://github.com/TheHPXProject/hpx/pull/5229>

²²⁴⁸ <https://github.com/TheHPXProject/hpx/pull/5228>

²²⁴⁹ <https://github.com/TheHPXProject/hpx/pull/5227>

²²⁵⁰ <https://github.com/TheHPXProject/hpx/pull/5226>

²²⁵¹ <https://github.com/TheHPXProject/hpx/pull/5225>

²²⁵² <https://github.com/TheHPXProject/hpx/pull/5224>

²²⁵³ <https://github.com/TheHPXProject/hpx/pull/5223>

²²⁵⁴ <https://github.com/TheHPXProject/hpx/pull/5221>

²²⁵⁵ <https://github.com/TheHPXProject/hpx/pull/5220>

²²⁵⁶ <https://github.com/TheHPXProject/hpx/pull/5218>

²²⁵⁷ <https://github.com/TheHPXProject/hpx/pull/5217>

²²⁵⁸ <https://github.com/TheHPXProject/hpx/pull/5215>

²²⁵⁹ <https://github.com/TheHPXProject/hpx/pull/5214>

²²⁶⁰ <https://github.com/TheHPXProject/hpx/pull/5213>

²²⁶¹ <https://github.com/TheHPXProject/hpx/pull/5212>

²²⁶² <https://github.com/TheHPXProject/hpx/pull/5211>

²²⁶³ <https://github.com/TheHPXProject/hpx/pull/5210>

²²⁶⁴ <https://github.com/TheHPXProject/hpx/pull/5209>

²²⁶⁵ <https://github.com/TheHPXProject/hpx/pull/5208>

²²⁶⁶ <https://github.com/TheHPXProject/hpx/pull/5207>

²²⁶⁷ <https://github.com/TheHPXProject/hpx/pull/5206>

- PR #5205²²⁶⁸ - Moving all distribution policies into modules
- PR #5203²²⁶⁹ - Move copy algorithms to tag_fallback_invoke
- PR #5202²²⁷⁰ - Make HPX_WITH_PSEUDO_DEPENDENCIES a cache variable
- PR #5201²²⁷¹ - Replaced tag_invoke with tag_fallback_invoke for adjacent_find algorithm
- PR #5200²²⁷² - Moving files to (distributed) runtime module
- PR #5199²²⁷³ - Update ICC module name on Piz Daint Jenkins configuration
- PR #5198²²⁷⁴ - Add doxygen documentation for thread_schedule_hint
- PR #5197²²⁷⁵ - Attempt to fix compilation of context implementations with unity build enabled
- PR #5196²²⁷⁶ - Re-enable component tests
- PR #5195²²⁷⁷ - Moving files related to colocation logic
- PR #5194²²⁷⁸ - Another attempt at fixing the Fedora 35 problem
- PR #5193²²⁷⁹ - Components module
- PR #5192²²⁸⁰ - Adapt replace(_if) to C++20
- PR #5190²²⁸¹ - Set compatibility headers by default to on
- PR #5188²²⁸² - Bump Boost minimum version to 1.71.0
- PR #5187²²⁸³ - Force CMake to set the -std=c++XX flag
- PR #5186²²⁸⁴ - Remove message to print .cu extension whenever .cu files are encountered
- PR #5185²²⁸⁵ - Remove some minor unnecessary CMake options
- PR #5184²²⁸⁶ - Remove some leftover HPX_WITH_*_SCHEDULER uses
- PR #5183²²⁸⁷ - Remove dependency on boost/iterators/iterator_categories.hpp
- PR #5182²²⁸⁸ - Fixing Fedora 35 for Power architectures
- PR #5181²²⁸⁹ - Bump version number and tag post 1.6.0 release
- PR #5180²²⁹⁰ - Fix https_v2 tests linking

²²⁶⁸ <https://github.com/TheHPXProject/hpx/pull/5205>

²²⁶⁹ <https://github.com/TheHPXProject/hpx/pull/5203>

²²⁷⁰ <https://github.com/TheHPXProject/hpx/pull/5202>

²²⁷¹ <https://github.com/TheHPXProject/hpx/pull/5201>

²²⁷² <https://github.com/TheHPXProject/hpx/pull/5200>

²²⁷³ <https://github.com/TheHPXProject/hpx/pull/5199>

²²⁷⁴ <https://github.com/TheHPXProject/hpx/pull/5198>

²²⁷⁵ <https://github.com/TheHPXProject/hpx/pull/5197>

²²⁷⁶ <https://github.com/TheHPXProject/hpx/pull/5196>

²²⁷⁷ <https://github.com/TheHPXProject/hpx/pull/5195>

²²⁷⁸ <https://github.com/TheHPXProject/hpx/pull/5194>

²²⁷⁹ <https://github.com/TheHPXProject/hpx/pull/5193>

²²⁸⁰ <https://github.com/TheHPXProject/hpx/pull/5192>

²²⁸¹ <https://github.com/TheHPXProject/hpx/pull/5190>

²²⁸² <https://github.com/TheHPXProject/hpx/pull/5188>

²²⁸³ <https://github.com/TheHPXProject/hpx/pull/5187>

²²⁸⁴ <https://github.com/TheHPXProject/hpx/pull/5186>

²²⁸⁵ <https://github.com/TheHPXProject/hpx/pull/5185>

²²⁸⁶ <https://github.com/TheHPXProject/hpx/pull/5184>

²²⁸⁷ <https://github.com/TheHPXProject/hpx/pull/5183>

²²⁸⁸ <https://github.com/TheHPXProject/hpx/pull/5182>

²²⁸⁹ <https://github.com/TheHPXProject/hpx/pull/5181>

²²⁹⁰ <https://github.com/TheHPXProject/hpx/pull/5180>

- [PR #5179](#)²²⁹¹ - Make sure `--hpx:local` command line option is respected with networking is off but distributed runtime is on
- [PR #5177](#)²²⁹² - Remove module cmake options
- [PR #5176](#)²²⁹³ - Starting to separate segmented algorithms: `for_each`
- [PR #5174](#)²²⁹⁴ - Don't run segmented algorithms twice on CircleCI
- [PR #5173](#)²²⁹⁵ - Fetching APEX using cmake `FetchContent`
- [PR #5172](#)²²⁹⁶ - Add separate local-only entry point
- [PR #5171](#)²²⁹⁷ - Remove `HPX_WITH_THREAD_SCHEDULERS` CMake option
- [PR #5170](#)²²⁹⁸ - Add `HPX_WITH_PRECOMPILED_HEADERS` option
- [PR #5166](#)²²⁹⁹ - Moving some action tests to modules
- [PR #5165](#)²³⁰⁰ - Require cmake 3.17
- [PR #5164](#)²³⁰¹ - Move `thread_pool_suspension_helper` files to small utility module
- [PR #5160](#)²³⁰² - Adding checks ensuring modules are not cross-referenced from other module categories
- [PR #5158](#)²³⁰³ - Replace `boost::asio` with standalone `asio`
- [PR #5155](#)²³⁰⁴ - Allow logging when distributed runtime is off
- [PR #5153](#)²³⁰⁵ - Components module
- [PR #5152](#)²³⁰⁶ - Move more files to performance counter module
- [PR #5150](#)²³⁰⁷ - Adapt `remove_copy(_if)` to C++20
- [PR #5144](#)²³⁰⁸ - AGAS module
- [PR #5125](#)²³⁰⁹ - Adapt `remove` and `remove_if` to C++20
- [PR #5117](#)²³¹⁰ - Attempt to fix segfaults assumed to be caused by `future_data` instances going out of scope.
- [PR #5099](#)²³¹¹ - Allow mixing debug and release builds
- [PR #5092](#)²³¹² - Replace `spirit.qi` with `x3`
- [PR #5053](#)²³¹³ - Add P0443r14 executor and a few P1897 algorithms

²²⁹¹ <https://github.com/TheHPXProject/hpx/pull/5179>

²²⁹² <https://github.com/TheHPXProject/hpx/pull/5177>

²²⁹³ <https://github.com/TheHPXProject/hpx/pull/5176>

²²⁹⁴ <https://github.com/TheHPXProject/hpx/pull/5174>

²²⁹⁵ <https://github.com/TheHPXProject/hpx/pull/5173>

²²⁹⁶ <https://github.com/TheHPXProject/hpx/pull/5172>

²²⁹⁷ <https://github.com/TheHPXProject/hpx/pull/5171>

²²⁹⁸ <https://github.com/TheHPXProject/hpx/pull/5170>

²²⁹⁹ <https://github.com/TheHPXProject/hpx/pull/5166>

²³⁰⁰ <https://github.com/TheHPXProject/hpx/pull/5165>

²³⁰¹ <https://github.com/TheHPXProject/hpx/pull/5164>

²³⁰² <https://github.com/TheHPXProject/hpx/pull/5160>

²³⁰³ <https://github.com/TheHPXProject/hpx/pull/5158>

²³⁰⁴ <https://github.com/TheHPXProject/hpx/pull/5155>

²³⁰⁵ <https://github.com/TheHPXProject/hpx/pull/5153>

²³⁰⁶ <https://github.com/TheHPXProject/hpx/pull/5152>

²³⁰⁷ <https://github.com/TheHPXProject/hpx/pull/5150>

²³⁰⁸ <https://github.com/TheHPXProject/hpx/pull/5144>

²³⁰⁹ <https://github.com/TheHPXProject/hpx/pull/5125>

²³¹⁰ <https://github.com/TheHPXProject/hpx/pull/5117>

²³¹¹ <https://github.com/TheHPXProject/hpx/pull/5099>

²³¹² <https://github.com/TheHPXProject/hpx/pull/5092>

²³¹³ <https://github.com/TheHPXProject/hpx/pull/5053>

- [PR #5044](#)²³¹⁴ - Add performance test in jenkins and reports

HPX V1.6.0 (Feb 17, 2021)

General changes

This release continues the focus on C++20 conformance with multiple new algorithms adapted to be C++20 conformant and becoming customization point objects (CPOs). We have also added experimental support for HIP, allowing previous CUDA features to now be compiled with `hipcc` and run on AMD GPUs.

- The following algorithms have been adapted to be C++20 conformant: `adjacent_find`, `includes`, `inplace_merge`, `is_heap`, `is_heap_until`, `is_partitioned`, `is_sorted`, `is_sorted_until`, `merge`, `set_difference`, `set_intersection`, `set_symmetric_difference`, `set_union`.
- Experimental HIP support can be enabled by compiling *HPX* with `hipcc`. All CUDA functionality in *HPX* can now be used with HIP. The HIP functionality is for the time being exposed through the same API as the CUDA functionality, i.e. no changes are required in user code. The CUDA, and now HIP, functionality is in the `hpx::cuda` namespace.
- We have added `partial_sort` based on Francisco Tapia's implementation.
- `hpx::init` and `hpx::start` gained new overloads taking an `hpx::init_params` struct in 1.5.0. All overloads not taking an `hpx::init_params` are now deprecated.
- We have added an experimental `fork_join_executor`. This executor can be used for OpenMP-style fork-join parallelism, where the latency of a parallel region is important for performance.
- The `parallel_executor` now uses a hierarchical spawning scheme for bulk execution, which improves data locality and performance.
- `hpx::dataflow` can now be used with executors that inject additional parameters into the call of the user-provided function.
- We have added experimental support for properties as proposed in [P2220](#)²³¹⁵. Currently the only supported property is the scheduling hint on `parallel_executor`.
- `hpx::util::annotated_function` can now be passed a dynamically generated `std::string`.
- In moving functionality to new namespaces, old names have been deprecated. A deprecation warning will be issued if you are using deprecated functionality, with instructions on how to correct or ignore the warning.
- We have removed all support for C and Fortran from our build system.
- We have further reduced the use of Boost types within *HPX* (`boost::system::error_code` and `boost::detail::spinlock`).
- We have enabled more warnings in our CI builds (unused variables and unused typedefs).

²³¹⁴ <https://github.com/TheHPXProject/hpx/pull/5044>

²³¹⁵ <https://wg21.link/p2220>

Breaking changes

- hpxMP support has been completely removed.
- The verbs parcelport has been removed.
- The following compatibility options have been disabled by default: `HPX_WITH_ACTION_BASE_COMPATIBILITY`, `HPX_WITH_REGISTER_THREAD_COMPATIBILITY`, `HPX_WITH_PROMISE_ALIAS_COMPATIBILITY`, `HPX_WITH_UNSCOPED_ENUM_COMPATIBILITY`, `HPX_PROGRAM_OPTIONS_WITH_BOOST_PROGRAM_OPTIONS_COMPATIBILITY`, `HPX_WITH_EMBEDDED_THREAD_POOLS_COMPATIBILITY`, `HPX_WITH_THREAD_POOL_OS_EXECUTOR_COMPATIBILITY`, `HPX_WITH_THREAD_EXECUTORS_COMPATIBILITY`, `HPX_THREAD_AWARE_TIMER_COMPATIBILITY`, `HPX_WITH_POOL_EXECUTOR_COMPATIBILITY`. Unless noted here, the above functionalities do not come with replacements. Unscoped enumerations have been replaced by scoped enumerations. Previously deprecated unscoped enumerations are disabled by `HPX_WITH_UNSCOPED_ENUM_COMPATIBILITY`. Newly deprecated unscoped enumerations have been given deprecation warnings and replaced by scoped enumerations. `hpx::promise` has been replaced with `hpx::distributed::promise`. `hpx::program_options` is a drop-in replacement for `boost::program_options`. `hpx::execution::parallel_executor` now has constructors which take a thread pool, covering the use case of `hpx::threads::executors::pool_executor`. A pool can be supplied with `hpx::resource::get_thread_pool`.

Closed issues

- [Issue #5148²³¹⁶](#) - `runtime_support.hpp` does not work with newer clang
- [Issue #5147²³¹⁷](#) - Wrong results with parallel reduce
- [Issue #5129²³¹⁸](#) - Missing specialization for `std::hash<hpx::thread::id>`
- [Issue #5126²³¹⁹](#) - Use `std::string` for task annotations
- [Issue #5115²³²⁰](#) - Don't expect hwloc to always report Cores
- [Issue #5113²³²¹](#) - Handle threadmanager exceptions during startup
- [Issue #5112²³²²](#) - libatomic problems causing unexpected fails
- [Issue #5089²³²³](#) - Remove non-BSL files
- [Issue #5088²³²⁴](#) - Unwrapping problem
- [Issue #5087²³²⁵](#) - Remove hpxMP support
- [Issue #5077²³²⁶](#) - PAPI counters are not accessible when HPX is installed
- [Issue #5075²³²⁷](#) - Make the structs in all `iter_sent.hpp` lower case
- [Issue #5067²³²⁸](#) - Bug `string_util/split.hpp`

²³¹⁶ <https://github.com/TheHPXProject/hpx/issues/5148>

²³¹⁷ <https://github.com/TheHPXProject/hpx/issues/5147>

²³¹⁸ <https://github.com/TheHPXProject/hpx/issues/5129>

²³¹⁹ <https://github.com/TheHPXProject/hpx/issues/5126>

²³²⁰ <https://github.com/TheHPXProject/hpx/issues/5115>

²³²¹ <https://github.com/TheHPXProject/hpx/issues/5113>

²³²² <https://github.com/TheHPXProject/hpx/issues/5112>

²³²³ <https://github.com/TheHPXProject/hpx/issues/5089>

²³²⁴ <https://github.com/TheHPXProject/hpx/issues/5088>

²³²⁵ <https://github.com/TheHPXProject/hpx/issues/5087>

²³²⁶ <https://github.com/TheHPXProject/hpx/issues/5077>

²³²⁷ <https://github.com/TheHPXProject/hpx/issues/5075>

²³²⁸ <https://github.com/TheHPXProject/hpx/issues/5067>

- [Issue #5049](#)²³²⁹ - Change back the hipcc jenkins config to the fury partition on rostam
- [Issue #5038](#)²³³⁰ - Not all examples link in the latest HPX master
- [Issue #5035](#)²³³¹ - Build with HPX_WITH_EXAMPLES fails
- [Issue #5019](#)²³³² - Broken help string for hpx
- [Issue #5016](#)²³³³ - `hpx::parallel::fill` fails compiling
- [Issue #5014](#)²³³⁴ - Rename all `.cc` to `.cpp` and `.hh` to `.hpp`
- [Issue #4988](#)²³³⁵ - MPI is not finalized if running with only one locality
- [Issue #4978](#)²³³⁶ - Change feature test macros to expand to zero/one
- [Issue #4949](#)²³³⁷ - Crash when not enabling TCP parcelport
- [Issue #4933](#)²³³⁸ - Improve test coverage for unused variable warnings etc.
- [Issue #4878](#)²³³⁹ - HPX mpi async might call `MPI_FINALIZE` before app calls it
- [Issue #4127](#)²³⁴⁰ - Local runtime entry-points

Closed pull requests

- [PR #5178](#)²³⁴¹ - Fix parallel `remove/remove_copy/transform` namespace references in docs
- [PR #5169](#)²³⁴² - Attempt to get Piz Daint jenkins setup running after maintenance
- [PR #5168](#)²³⁴³ - Remove include of itself
- [PR #5167](#)²³⁴⁴ - Fixing deprecation warnings that slipped through the net
- [PR #5159](#)²³⁴⁵ - Update APEX tag to 2.3.1
- [PR #5154](#)²³⁴⁶ - Splitting unit tests on circleci to avoid timeouts
- [PR #5151](#)²³⁴⁷ - Use C++20 on clang-newest Jenkins CI configuration
- [PR #5149](#)²³⁴⁸ - Rename 'module' symbols to avoid keyword conflict
- [PR #5145](#)²³⁴⁹ - Adjust handling of CUDA/HIP options in CMake
- [PR #5142](#)²³⁵⁰ - Store annotated_function annotations as `std::strings`

²³²⁹ <https://github.com/TheHPXProject/hpx/issues/5049>

²³³⁰ <https://github.com/TheHPXProject/hpx/issues/5038>

²³³¹ <https://github.com/TheHPXProject/hpx/issues/5035>

²³³² <https://github.com/TheHPXProject/hpx/issues/5019>

²³³³ <https://github.com/TheHPXProject/hpx/issues/5016>

²³³⁴ <https://github.com/TheHPXProject/hpx/issues/5014>

²³³⁵ <https://github.com/TheHPXProject/hpx/issues/4988>

²³³⁶ <https://github.com/TheHPXProject/hpx/issues/4978>

²³³⁷ <https://github.com/TheHPXProject/hpx/issues/4949>

²³³⁸ <https://github.com/TheHPXProject/hpx/issues/4933>

²³³⁹ <https://github.com/TheHPXProject/hpx/issues/4878>

²³⁴⁰ <https://github.com/TheHPXProject/hpx/issues/4127>

²³⁴¹ <https://github.com/TheHPXProject/hpx/pull/5178>

²³⁴² <https://github.com/TheHPXProject/hpx/pull/5169>

²³⁴³ <https://github.com/TheHPXProject/hpx/pull/5168>

²³⁴⁴ <https://github.com/TheHPXProject/hpx/pull/5167>

²³⁴⁵ <https://github.com/TheHPXProject/hpx/pull/5159>

²³⁴⁶ <https://github.com/TheHPXProject/hpx/pull/5154>

²³⁴⁷ <https://github.com/TheHPXProject/hpx/pull/5151>

²³⁴⁸ <https://github.com/TheHPXProject/hpx/pull/5149>

²³⁴⁹ <https://github.com/TheHPXProject/hpx/pull/5145>

²³⁵⁰ <https://github.com/TheHPXProject/hpx/pull/5142>

- PR #5140²³⁵¹ - Scheduler mode
- PR #5139²³⁵² - Fix path problem in pre-commit hook, add summary commit line
- PR #5138²³⁵³ - Add program options variable map to resource partitioner init
- PR #5137²³⁵⁴ - Remove the use of `boost::throw_exception`
- PR #5136²³⁵⁵ - Make sure codespell checks run on CircleCI
- PR #5132²³⁵⁶ - Fixing spelling errors
- PR #5131²³⁵⁷ - Mark `counting_iterator` member functions as `HPX_HOST_DEVICE`
- PR #5130²³⁵⁸ - Adding specialization for `std::hash<hpx::thread::id>`
- PR #5128²³⁵⁹ - Fixing environment handling for FreeBSD
- PR #5127²³⁶⁰ - Fix typo in fibonacci documentation
- PR #5123²³⁶¹ - Reduce vector sizes in partial sort benchmarks when running in debug mode
- PR #5122²³⁶² - Making sure exceptions during runtime initialization are correctly reported
- PR #5121²³⁶³ - Working around hwloc limitation on certain platforms
- PR #5120²³⁶⁴ - Fixing compatibility warnings in `hpx::transform` implementation
- PR #5119²³⁶⁵ - Use `sequential_find` and friends from separate detail header
- PR #5116²³⁶⁶ - Fix compilation with timer pool off
- PR #5114²³⁶⁷ - Fix 5112 - make sure `libatomic` is used when needed
- PR #5109²³⁶⁸ - Remove default runtime mode argument from `init` overload, again
- PR #5108²³⁶⁹ - Refactor `iter_sent.hpp` to make structs lowercase
- PR #5107²³⁷⁰ - Relax `dataflow` internals
- PR #5106²³⁷¹ - Change initialization of property CPOs to satisfy older `nvcc` versions
- PR #5104²³⁷² - Fix regeneration of two files that trigger unnecessary rebuilds
- PR #5103²³⁷³ - Remove default runtime mode argument from `start/init` overloads

²³⁵¹ <https://github.com/TheHPXProject/hpx/pull/5140>

²³⁵² <https://github.com/TheHPXProject/hpx/pull/5139>

²³⁵³ <https://github.com/TheHPXProject/hpx/pull/5138>

²³⁵⁴ <https://github.com/TheHPXProject/hpx/pull/5137>

²³⁵⁵ <https://github.com/TheHPXProject/hpx/pull/5136>

²³⁵⁶ <https://github.com/TheHPXProject/hpx/pull/5132>

²³⁵⁷ <https://github.com/TheHPXProject/hpx/pull/5131>

²³⁵⁸ <https://github.com/TheHPXProject/hpx/pull/5130>

²³⁵⁹ <https://github.com/TheHPXProject/hpx/pull/5128>

²³⁶⁰ <https://github.com/TheHPXProject/hpx/pull/5127>

²³⁶¹ <https://github.com/TheHPXProject/hpx/pull/5123>

²³⁶² <https://github.com/TheHPXProject/hpx/pull/5122>

²³⁶³ <https://github.com/TheHPXProject/hpx/pull/5121>

²³⁶⁴ <https://github.com/TheHPXProject/hpx/pull/5120>

²³⁶⁵ <https://github.com/TheHPXProject/hpx/pull/5119>

²³⁶⁶ <https://github.com/TheHPXProject/hpx/pull/5116>

²³⁶⁷ <https://github.com/TheHPXProject/hpx/pull/5114>

²³⁶⁸ <https://github.com/TheHPXProject/hpx/pull/5109>

²³⁶⁹ <https://github.com/TheHPXProject/hpx/pull/5108>

²³⁷⁰ <https://github.com/TheHPXProject/hpx/pull/5107>

²³⁷¹ <https://github.com/TheHPXProject/hpx/pull/5106>

²³⁷² <https://github.com/TheHPXProject/hpx/pull/5104>

²³⁷³ <https://github.com/TheHPXProject/hpx/pull/5103>

- PR #5102²³⁷⁴ - Untie deprecated thread enums from the CMake option
- PR #5101²³⁷⁵ - Update APEX tag for 1.6.0
- PR #5100²³⁷⁶ - Bump minimum required Boost version to 1.66 and update CI configurations
- PR #5098²³⁷⁷ - Minor fixes to public API listing
- PR #5097²³⁷⁸ - Remove hpxMP support
- PR #5096²³⁷⁹ - Remove fractals examples
- PR #5095²³⁸⁰ - Use all AMD nodes again on rostam
- PR #5094²³⁸¹ - Attempt to remove macOS workaround for GH actions environment
- PR #5093²³⁸² - Remove verbs parcelport
- PR #5091²³⁸³ - Avoid moving from lvalues
- PR #5090²³⁸⁴ - Adopt C++20 `std::endian`
- PR #5085²³⁸⁵ - Update daint CI to use Boost 1.75.0
- PR #5084²³⁸⁶ - Disable compatibility options for 1.6.0 release
- PR #5083²³⁸⁷ - Remove duplicated call to the `limiting_executor` in `future_overhead` test
- PR #5079²³⁸⁸ - Add checks to make sure that MPI/CUDA polling is enabled/not disabled too early
- PR #5078²³⁸⁹ - Add install lib directory to list of component search paths
- PR #5076²³⁹⁰ - Fix a typo in the jenkins `clang-newest` cmake config
- PR #5074²³⁹¹ - Fixing warnings generated by MSVC
- PR #5073²³⁹² - Allow using noncopyable types with unwrapping
- PR #5072²³⁹³ - Fix `is_convertible` args in `result_types`
- PR #5071²³⁹⁴ - Fix unused parameters
- PR #5070²³⁹⁵ - Fix unused variables warnings in `hipcc`
- PR #5069²³⁹⁶ - Add support for sentinels to `adjacent_find`

²³⁷⁴ <https://github.com/TheHPXProject/hpx/pull/5102>

²³⁷⁵ <https://github.com/TheHPXProject/hpx/pull/5101>

²³⁷⁶ <https://github.com/TheHPXProject/hpx/pull/5100>

²³⁷⁷ <https://github.com/TheHPXProject/hpx/pull/5098>

²³⁷⁸ <https://github.com/TheHPXProject/hpx/pull/5097>

²³⁷⁹ <https://github.com/TheHPXProject/hpx/pull/5096>

²³⁸⁰ <https://github.com/TheHPXProject/hpx/pull/5095>

²³⁸¹ <https://github.com/TheHPXProject/hpx/pull/5094>

²³⁸² <https://github.com/TheHPXProject/hpx/pull/5093>

²³⁸³ <https://github.com/TheHPXProject/hpx/pull/5091>

²³⁸⁴ <https://github.com/TheHPXProject/hpx/pull/5090>

²³⁸⁵ <https://github.com/TheHPXProject/hpx/pull/5085>

²³⁸⁶ <https://github.com/TheHPXProject/hpx/pull/5084>

²³⁸⁷ <https://github.com/TheHPXProject/hpx/pull/5083>

²³⁸⁸ <https://github.com/TheHPXProject/hpx/pull/5079>

²³⁸⁹ <https://github.com/TheHPXProject/hpx/pull/5078>

²³⁹⁰ <https://github.com/TheHPXProject/hpx/pull/5076>

²³⁹¹ <https://github.com/TheHPXProject/hpx/pull/5074>

²³⁹² <https://github.com/TheHPXProject/hpx/pull/5073>

²³⁹³ <https://github.com/TheHPXProject/hpx/pull/5072>

²³⁹⁴ <https://github.com/TheHPXProject/hpx/pull/5071>

²³⁹⁵ <https://github.com/TheHPXProject/hpx/pull/5070>

²³⁹⁶ <https://github.com/TheHPXProject/hpx/pull/5069>

- PR #5068²³⁹⁷ - Fix string split function
- PR #5066²³⁹⁸ - Adapt search to C++20 and Range TS
- PR #5065²³⁹⁹ - Fix `hpx::range::adjacent_find` doxygen function signatures
- PR #5064²⁴⁰⁰ - Refactor runtime configuration, command line handling, and resource partitioner
- PR #5063²⁴⁰¹ - Limit the device code guards to the distributed parts of the `future_overhead` bench
- PR #5061²⁴⁰² - Remove hipcc guards in examples and tests
- PR #5060²⁴⁰³ - Fix deprecation warnings generated by msvc
- PR #5059²⁴⁰⁴ - Add warning about suspending/resuming the runtime in multi-locality scenarios
- PR #5057²⁴⁰⁵ - Fix unused variable warnings
- PR #5056²⁴⁰⁶ - Fix `hpx::util::get`
- PR #5055²⁴⁰⁷ - Remove hipcc guards
- PR #5054²⁴⁰⁸ - Fix typo
- PR #5051²⁴⁰⁹ - Adapt transform to C++20
- PR #5050²⁴¹⁰ - Replace old init overloads in tests and examples
- PR #5048²⁴¹¹ - Limit jenkins hipcc to the reno node
- PR #5047²⁴¹² - Limit cuda jenkins run to nodes with exclusively Nvidia GPUs
- PR #5046²⁴¹³ - Convert thread and future enums to class enums
- PR #5043²⁴¹⁴ - Improve `hpxrun.py` for Phylanx
- PR #5042²⁴¹⁵ - Add missing header to partial sort test
- PR #5041²⁴¹⁶ - Adding Francisco Tapia's implementation of `partial_sort`
- PR #5040²⁴¹⁷ - Remove generated headers left behind from a previous configuration
- PR #5039²⁴¹⁸ - Fix GCC 10 release builds
- PR #5037²⁴¹⁹ - Add `is_invocable` typedefs to top-level `hpx` namespace and public API list

²³⁹⁷ <https://github.com/TheHPXProject/hpx/pull/5068>

²³⁹⁸ <https://github.com/TheHPXProject/hpx/pull/5066>

²³⁹⁹ <https://github.com/TheHPXProject/hpx/pull/5065>

²⁴⁰⁰ <https://github.com/TheHPXProject/hpx/pull/5064>

²⁴⁰¹ <https://github.com/TheHPXProject/hpx/pull/5063>

²⁴⁰² <https://github.com/TheHPXProject/hpx/pull/5061>

²⁴⁰³ <https://github.com/TheHPXProject/hpx/pull/5060>

²⁴⁰⁴ <https://github.com/TheHPXProject/hpx/pull/5059>

²⁴⁰⁵ <https://github.com/TheHPXProject/hpx/pull/5057>

²⁴⁰⁶ <https://github.com/TheHPXProject/hpx/pull/5056>

²⁴⁰⁷ <https://github.com/TheHPXProject/hpx/pull/5055>

²⁴⁰⁸ <https://github.com/TheHPXProject/hpx/pull/5054>

²⁴⁰⁹ <https://github.com/TheHPXProject/hpx/pull/5051>

²⁴¹⁰ <https://github.com/TheHPXProject/hpx/pull/5050>

²⁴¹¹ <https://github.com/TheHPXProject/hpx/pull/5048>

²⁴¹² <https://github.com/TheHPXProject/hpx/pull/5047>

²⁴¹³ <https://github.com/TheHPXProject/hpx/pull/5046>

²⁴¹⁴ <https://github.com/TheHPXProject/hpx/pull/5043>

²⁴¹⁵ <https://github.com/TheHPXProject/hpx/pull/5042>

²⁴¹⁶ <https://github.com/TheHPXProject/hpx/pull/5041>

²⁴¹⁷ <https://github.com/TheHPXProject/hpx/pull/5040>

²⁴¹⁸ <https://github.com/TheHPXProject/hpx/pull/5039>

²⁴¹⁹ <https://github.com/TheHPXProject/hpx/pull/5037>

- PR #5036²⁴²⁰ - Deprecate `hpx::util::decay` in favor of `std::decay`
- PR #5034²⁴²¹ - Use versioned container image on CircleCI
- PR #5033²⁴²² - Implement P2220 properties module
- PR #5032²⁴²³ - Do codespell comparison only on files changed from common ancestor
- PR #5031²⁴²⁴ - Moving traits files to `actions_base`
- PR #5030²⁴²⁵ - Add codespell version print in circleci
- PR #5029²⁴²⁶ - Work around problems in GitHub actions macOS builder
- PR #5028²⁴²⁷ - Moving move files to `naming` and `naming_base`
- PR #5027²⁴²⁸ - Lessen constraints on certain algorithm arguments
- PR #5025²⁴²⁹ - Adapt `is_sorted` and `is_sorted_until` to C++20
- PR #5024²⁴³⁰ - Moving `naming_base` to full modules
- PR #5022²⁴³¹ - Remove C language from `CMakeLists.txt`
- PR #5021²⁴³² - Warn about unused arguments given to `add_hpx_module`
- PR #5020²⁴³³ - Fixing help string
- PR #5018²⁴³⁴ - Update CSCS jenkins configuration to clang 11
- PR #5017²⁴³⁵ - Fixing broken backwards compatibility for `hpx::parallel::fill`
- PR #5015²⁴³⁶ - Detect if generated global header conflicts with explicitly listed module headers
- PR #5012²⁴³⁷ - Properly reset pointer tracking data in `output_archive`
- PR #5011²⁴³⁸ - Inspect command line tweaks
- PR #5010²⁴³⁹ - Creating AGAS module
- PR #5009²⁴⁴⁰ - Replace `boost::system::error_code` with `std::error_code`
- PR #5008²⁴⁴¹ - Replace uses of `boost::detail::spinlock`
- PR #5007²⁴⁴² - Bump minimal Boost version to 1.65.0

²⁴²⁰ <https://github.com/TheHPXProject/hpx/pull/5036>

²⁴²¹ <https://github.com/TheHPXProject/hpx/pull/5034>

²⁴²² <https://github.com/TheHPXProject/hpx/pull/5033>

²⁴²³ <https://github.com/TheHPXProject/hpx/pull/5032>

²⁴²⁴ <https://github.com/TheHPXProject/hpx/pull/5031>

²⁴²⁵ <https://github.com/TheHPXProject/hpx/pull/5030>

²⁴²⁶ <https://github.com/TheHPXProject/hpx/pull/5029>

²⁴²⁷ <https://github.com/TheHPXProject/hpx/pull/5028>

²⁴²⁸ <https://github.com/TheHPXProject/hpx/pull/5027>

²⁴²⁹ <https://github.com/TheHPXProject/hpx/pull/5025>

²⁴³⁰ <https://github.com/TheHPXProject/hpx/pull/5024>

²⁴³¹ <https://github.com/TheHPXProject/hpx/pull/5022>

²⁴³² <https://github.com/TheHPXProject/hpx/pull/5021>

²⁴³³ <https://github.com/TheHPXProject/hpx/pull/5020>

²⁴³⁴ <https://github.com/TheHPXProject/hpx/pull/5018>

²⁴³⁵ <https://github.com/TheHPXProject/hpx/pull/5017>

²⁴³⁶ <https://github.com/TheHPXProject/hpx/pull/5015>

²⁴³⁷ <https://github.com/TheHPXProject/hpx/pull/5012>

²⁴³⁸ <https://github.com/TheHPXProject/hpx/pull/5011>

²⁴³⁹ <https://github.com/TheHPXProject/hpx/pull/5010>

²⁴⁴⁰ <https://github.com/TheHPXProject/hpx/pull/5009>

²⁴⁴¹ <https://github.com/TheHPXProject/hpx/pull/5008>

²⁴⁴² <https://github.com/TheHPXProject/hpx/pull/5007>

- [PR #5006²⁴⁴³](#) - Adapt `is_partitioned` to C++20
- [PR #5005²⁴⁴⁴](#) - Making sure `reduce_by_key` compiles again
- [PR #5004²⁴⁴⁵](#) - Fixing template specializations that make extra archive data types unique across module boundaries
- [PR #5003²⁴⁴⁶](#) - Relax `dataflow` argument constraints
- [PR #5001²⁴⁴⁷](#) - Add `<random>` inspect check
- [PR #4999²⁴⁴⁸](#) - Attempt to fix MacOS Github action error
- [PR #4997²⁴⁴⁹](#) - Fix unused variable and typedef warnings
- [PR #4996²⁴⁵⁰](#) - Adapt `adjacent_find` to C++20
- [PR #4995²⁴⁵¹](#) - Test all schedulers in `cross_pool_injection` test except `shared_priority_queue_scheduler`
- [PR #4993²⁴⁵²](#) - Fix deprecation warnings
- [PR #4991²⁴⁵³](#) - Avoid unnecessarily including entire modules
- [PR #4990²⁴⁵⁴](#) - Fixing some warnings from HPX complaining about use of obsolete types
- [PR #4989²⁴⁵⁵](#) - add a `*destroy*` trait for `ParcelPort` plugins
- [PR #4986²⁴⁵⁶](#) - Remove serialization to functional module dependency
- [PR #4985²⁴⁵⁷](#) - Compatibility header generation
- [PR #4980²⁴⁵⁸](#) - Add ranges overloads to `for_loop` (and variants)
- [PR #4979²⁴⁵⁹](#) - Actually enable unity builds on Jenkins
- [PR #4977²⁴⁶⁰](#) - Cleaning up `debug::print` functionalities
- [PR #4976²⁴⁶¹](#) - Remove indirection layer in `at_index_impl`
- [PR #4975²⁴⁶²](#) - Remove indirection layer in `at_index_impl`
- [PR #4973²⁴⁶³](#) - Avoid warnings/errors for older gcc complaining about multi-line comments
- [PR #4970²⁴⁶⁴](#) - Making set algorithms conform to C++20

²⁴⁴³ <https://github.com/TheHPXProject/hpx/pull/5006>

²⁴⁴⁴ <https://github.com/TheHPXProject/hpx/pull/5005>

²⁴⁴⁵ <https://github.com/TheHPXProject/hpx/pull/5004>

²⁴⁴⁶ <https://github.com/TheHPXProject/hpx/pull/5003>

²⁴⁴⁷ <https://github.com/TheHPXProject/hpx/pull/5001>

²⁴⁴⁸ <https://github.com/TheHPXProject/hpx/pull/4999>

²⁴⁴⁹ <https://github.com/TheHPXProject/hpx/pull/4997>

²⁴⁵⁰ <https://github.com/TheHPXProject/hpx/pull/4996>

²⁴⁵¹ <https://github.com/TheHPXProject/hpx/pull/4995>

²⁴⁵² <https://github.com/TheHPXProject/hpx/pull/4993>

²⁴⁵³ <https://github.com/TheHPXProject/hpx/pull/4991>

²⁴⁵⁴ <https://github.com/TheHPXProject/hpx/pull/4990>

²⁴⁵⁵ <https://github.com/TheHPXProject/hpx/pull/4989>

²⁴⁵⁶ <https://github.com/TheHPXProject/hpx/pull/4986>

²⁴⁵⁷ <https://github.com/TheHPXProject/hpx/pull/4985>

²⁴⁵⁸ <https://github.com/TheHPXProject/hpx/pull/4980>

²⁴⁵⁹ <https://github.com/TheHPXProject/hpx/pull/4979>

²⁴⁶⁰ <https://github.com/TheHPXProject/hpx/pull/4977>

²⁴⁶¹ <https://github.com/TheHPXProject/hpx/pull/4976>

²⁴⁶² <https://github.com/TheHPXProject/hpx/pull/4975>

²⁴⁶³ <https://github.com/TheHPXProject/hpx/pull/4973>

²⁴⁶⁴ <https://github.com/TheHPXProject/hpx/pull/4970>

- PR #4969²⁴⁶⁵ - Moving `is_execution_policy` and friends into namespace `hpx`
- PR #4968²⁴⁶⁶ - Enable deprecation warnings for 1.6.0 and move any functionality to `hpx` namespace
- PR #4967²⁴⁶⁷ - Define deprecation macros conditionally
- PR #4966²⁴⁶⁸ - Add `clang-format` and `cmake-format` version prints
- PR #4965²⁴⁶⁹ - Making `is_heap` and `is_heap_until` conforming to C++20
- PR #4964²⁴⁷⁰ - Adding parallel `make_heap`
- PR #4962²⁴⁷¹ - Fix external timer function pointer exports
- PR #4960²⁴⁷² - Fixing folder names for module tests and examples
- PR #4959²⁴⁷³ - Adding communications set
- PR #4958²⁴⁷⁴ - Deprecate tuple and timing functionality in `hpx::util`
- PR #4957²⁴⁷⁵ - Fixing unity build option for parcelports
- PR #4953²⁴⁷⁶ - Fixing MSVC problems after recent restructurings
- PR #4952²⁴⁷⁷ - Make `parallel_executor` use `thread_pool_executor` spawning mechanism
- PR #4948²⁴⁷⁸ - Clean up old artifacts better and more aggressively on Jenkins
- PR #4947²⁴⁷⁹ - Add HIP support for AMD GPUs
- PR #4945²⁴⁸⁰ - Enable `HPX_WITH_UNITY_BUILD` option on one of the Jenkins configurations
- PR #4943²⁴⁸¹ - Move public `hpx::parallel::execution` functionality to `hpx::execution`
- PR #4938²⁴⁸² - Post release cleanup
- PR #4858²⁴⁸³ - Extending resilience APIs to support distributed invocations
- PR #4744²⁴⁸⁴ - Fork-join executor
- PR #4665²⁴⁸⁵ - Implementing sender, receiver, and `operation_state` concepts in terms of P0443r13
- PR #4649²⁴⁸⁶ - Split `libhpx` into multiple libraries
- PR #4642²⁴⁸⁷ - Implementing `operation_state` concept in terms of P0443r13

²⁴⁶⁵ <https://github.com/TheHPXProject/hpx/pull/4969>

²⁴⁶⁶ <https://github.com/TheHPXProject/hpx/pull/4968>

²⁴⁶⁷ <https://github.com/TheHPXProject/hpx/pull/4967>

²⁴⁶⁸ <https://github.com/TheHPXProject/hpx/pull/4966>

²⁴⁶⁹ <https://github.com/TheHPXProject/hpx/pull/4965>

²⁴⁷⁰ <https://github.com/TheHPXProject/hpx/pull/4964>

²⁴⁷¹ <https://github.com/TheHPXProject/hpx/pull/4962>

²⁴⁷² <https://github.com/TheHPXProject/hpx/pull/4960>

²⁴⁷³ <https://github.com/TheHPXProject/hpx/pull/4959>

²⁴⁷⁴ <https://github.com/TheHPXProject/hpx/pull/4958>

²⁴⁷⁵ <https://github.com/TheHPXProject/hpx/pull/4957>

²⁴⁷⁶ <https://github.com/TheHPXProject/hpx/pull/4953>

²⁴⁷⁷ <https://github.com/TheHPXProject/hpx/pull/4952>

²⁴⁷⁸ <https://github.com/TheHPXProject/hpx/pull/4948>

²⁴⁷⁹ <https://github.com/TheHPXProject/hpx/pull/4947>

²⁴⁸⁰ <https://github.com/TheHPXProject/hpx/pull/4945>

²⁴⁸¹ <https://github.com/TheHPXProject/hpx/pull/4943>

²⁴⁸² <https://github.com/TheHPXProject/hpx/pull/4938>

²⁴⁸³ <https://github.com/TheHPXProject/hpx/pull/4858>

²⁴⁸⁴ <https://github.com/TheHPXProject/hpx/pull/4744>

²⁴⁸⁵ <https://github.com/TheHPXProject/hpx/pull/4665>

²⁴⁸⁶ <https://github.com/TheHPXProject/hpx/pull/4649>

²⁴⁸⁷ <https://github.com/TheHPXProject/hpx/pull/4642>

- [PR #4640²⁴⁸⁸](#) - Implementing receiver concept in terms of P0443r13
- [PR #4622²⁴⁸⁹](#) - Sanitizer fixes

HPX V1.5.1 (Sep 30, 2020)

General changes

This is a patch release. It contains the following changes:

- Remove restriction on suspending runtime with multiple localities, users are now responsible for synchronizing work between localities before suspending.
- Fixes several compilation problems and warnings.
- Adds notes in the documentation explaining how to cite HPX.

Closed issues

- [Issue #4971²⁴⁹⁰](#) - Parallel sort fails to compile with C++20
- [Issue #4950²⁴⁹¹](#) - Build with `HPX_WITH_PARCELPORT_ACTION_COUNTERS_ON` fails
- [Issue #4940²⁴⁹²](#) - Codespell report for “HPX” (on fossies.org)
- [Issue #4937²⁴⁹³](#) - Allow suspension of runtime for multiple localities

Closed pull requests

- [PR #4982²⁴⁹⁴](#) - Add page about citing HPX to documentation
- [PR #4981²⁴⁹⁵](#) - Adding the missing include
- [PR #4974²⁴⁹⁶](#) - Remove leftover format export hack
- [PR #4972²⁴⁹⁷](#) - Removing use of `get_temporary_buffer` and `return_temporary_buffer`
- [PR #4963²⁴⁹⁸](#) - Renaming files to avoid warnings from the vs build system
- [PR #4951²⁴⁹⁹](#) - Fixing build if `HPX_WITH_PARCELPORT_ACTION_COUNTERS=On`
- [PR #4946²⁵⁰⁰](#) - Allow suspension on multiple localities
- [PR #4944²⁵⁰¹](#) - Fix typos reported by fossies codespell report
- [PR #4941²⁵⁰²](#) - Adding some explanation to README about how to cite HPX

²⁴⁸⁸ <https://github.com/TheHPXProject/hpx/pull/4640>
²⁴⁸⁹ <https://github.com/TheHPXProject/hpx/pull/4622>
²⁴⁹⁰ <https://github.com/TheHPXProject/hpx/issues/4971>
²⁴⁹¹ <https://github.com/TheHPXProject/hpx/issues/4950>
²⁴⁹² <https://github.com/TheHPXProject/hpx/issues/4940>
²⁴⁹³ <https://github.com/TheHPXProject/hpx/issues/4937>
²⁴⁹⁴ <https://github.com/TheHPXProject/hpx/pull/4982>
²⁴⁹⁵ <https://github.com/TheHPXProject/hpx/pull/4981>
²⁴⁹⁶ <https://github.com/TheHPXProject/hpx/pull/4974>
²⁴⁹⁷ <https://github.com/TheHPXProject/hpx/pull/4972>
²⁴⁹⁸ <https://github.com/TheHPXProject/hpx/pull/4963>
²⁴⁹⁹ <https://github.com/TheHPXProject/hpx/pull/4951>
²⁵⁰⁰ <https://github.com/TheHPXProject/hpx/pull/4946>
²⁵⁰¹ <https://github.com/TheHPXProject/hpx/pull/4944>
²⁵⁰² <https://github.com/TheHPXProject/hpx/pull/4941>

- [PR #4939²⁵⁰³](#) - Small changes

HPX V1.5.0 (Sep 02, 2020)

General changes

The main focus of this release is on APIs and C++20 conformance. We have added many new C++20 features and adapted multiple algorithms to be fully C++20 conformant. As part of the modularization we have begun specifying the public API of HPX in terms of headers and functionality, and aligning it more closely to the C++ standard. All non-distributed modules are now in place, along with an experimental option to completely disable distributed features in HPX. We have also added experimental asynchronous MPI and CUDA executors. Lastly this release introduces CMake targets for depending projects, performance improvements, and many bug fixes.

- We have added the C++20 features `hpx::jthread` and `hpx::stop_token`. `hpx::condition_variable_any` now exposes new functions supporting `hpx::stop_token`.
- We have added `hpx::stable_sort` based on Francisco Tapia's implementation.
- We have adapted existing synchronization primitives to be fully conformant C++20: `hpx::barrier`, `hpx::latch`, `hpx::counting_semaphore`, and `hpx::binary_semaphore`.
- We have started using customization point objects (CPOs) to make the corresponding algorithms fully conformant to C++20 as well as to make algorithm extension easier for the user. `all_of/any_of/none_of`, `copy`, `count`, `destroy`, `equal`, `fill`, `find`, `for_each`, `generate`, `mismatch`, `move`, `reduce`, `transform_reduce` are using those CPOs (all in namespace `hpx`). We also have adapted their corresponding `hpx::ranges` versions to be conforming to C++20 in this release.
- We have adapted support for `co_await` to C++20, in addition to `hpx::future` it now also supports `hpx::shared_future`. We have also added allocator support for futures returned by `co_return`. It is no longer in the `experimental` namespace.
- We added serialization support for `std::variant` and `std::tuple`.
- `result_of` and `is_callable` are now deprecated and replaced by `invoke_result` and `is_invocable` to conform to C++20.
- We continued with the modularization, making it easier for us to add the new experimental `HPX_WITH_DISTRIBUTED_RUNTIME` CMake option (see below) . A significant amount of headers have been deprecated. We adapted the namespaces and headers we could to be closer to the standard ones (*Public API*). Depending code should still compile, however warnings are now generated instructing to change the include statements accordingly.
- It is now possible to have a basic CUDA support including a helper function to get a future from a CUDA stream and target handling. They are available under the `hpx::cuda::experimental` namespace and they can be enabled with the `-DHPX_WITH_ASYNC_CUDA=ON` CMake option.
- We added a new `hpx::mpi::experimental` namespace for getting futures from an asynchronous MPI call and a new minimal MPI executor `hpx::mpi::experimental::executor`. These can be enabled with the `-DHPX_WITH_ASYNC_MPI=On` CMake option.
- A polymorphic executor has been implemented to reduce compile times as a function accepting executors can potentially be instantiated only once instead of multiple times with different executors. It accepts the function signature as a template argument. It needs to be constructed from any other executor. Please note, that the function signatures that can be scheduled using `then_execute`, `bulk_sync_execute`, `bulk_async_execute` and `bulk_then_execute` are slightly different (See the comment in [PR #4514²⁵⁰⁴](#) for more details).
- The underlying executor of `block_executor` has been updated to a newer one.

²⁵⁰³ <https://github.com/TheHPXProject/hpx/pull/4939>

²⁵⁰⁴ <https://github.com/TheHPXProject/hpx/pull/4514>

- We have added a parameter to `auto_chunk_size` to control the amount of iterations to measure.
- All executor parameter hooks can now be exposed through the executor itself. This will allow to deprecate the `.with()` functionality on execution policies in the future. This is also a first step towards simplifying our executor APIs in preparation for the upcoming C++23 executors (senders/receivers).
- We have moved all of the existing APIs related to resiliency into the namespace `hpx::resiliency::experimental`. Please note this is a breaking change without backwards-compatibility option. We have converted all of those APIs to be based on customization point objects. Two new executors have been added to enable easy integration of the existing resiliency features with other facilities (like the parallel algorithms): `replay_executor` and `replicate_executor`.
- We have added performance counters type information (aggregating, monotonically increasing, average count, average timer, etc.).
- HPX threads are now re-scheduled on the same worker thread they were suspended on to avoid cache misses from moving from one thread to the other. This behavior doesn't prevent the thread from being stolen, however.
- We have added a new configuration option `hpx.exception_verbosity` to allow to control the level of verbosity of the exceptions (3 levels available).
- `broadcast_to`, `broadcast_from`, `scatter_to` and `scatter_from` have been added to the collectives, modernization of `gather_here` and `gather_there` with futures taken by rvalue references. See the breaking change on `all_to_all` in the next section. None of the collectives need supporting macros anymore (e.g. specifying the data types used for a collective operation using `HPX_REGISTER_ALLGATHER` and similar is not needed anymore).
- New API functions have been added: a) to get the number of cores which are idle (`hpx::get_idle_core_count`) and b) returning a bitmask representing the currently idle cores (`hpx::get_idle_core_mask`).
- We have added an experimental option to only enable the local runtime, you can disable the distributed runtime with `HPX_WITH_DISTRIBUTED_RUNTIME=OFF`. You can also enable the local runtime by using the `--hpx:local` runtime option.
- We fixed task annotations for actions.
- The alias `hpx::promise` to `hpx::lcos::promise` is now deprecated. You can use `hpx::lcos::promise` directly instead. `hpx::promise` will refer to the local-only promise in the future.
- We have added a `prepare_checkpoint` API function that calculates the amount of necessary buffer space for a particular set of arguments checkpointed.
- We have added `hpx::upgrade_lock` and `hpx::upgrade_to_unique_lock`, which make `hpx::shared_mutex` (and similar) usable in more flexible ways.
- We have changed the CMake targets exposed to the user, it now includes `HPX:hpx`, `HPX:wrap_main` (int main as the first *HPX* thread of the application, see *Starting the HPX runtime*), `HPX:plugin`, `HPX:component`. The CMake variables `HPX_INCLUDE_DIRS` and `HPX_LIBRARIES` are deprecated and will be removed in a future release, you should now link directly to the `HPX:hpx` CMake target.
- A new example is demonstrating how to create and use a wrapping executor (`quickstart/executor_with_thread_hooks.cpp`)
- A new example is demonstrating how to disable thread stealing during the execution of parallel algorithms (`quickstart/disable_thread_stealing_executor.cpp`)
- We now require for our CMake build system configuration files to be formatted using `cmake-format`.
- We have removed more dependencies on various Boost libraries.
- We have added an experimental option enabling unity builds of HPX using the `-DHPX_WITH_UNITY_BUILD=On` CMake option.

- Many bug fixes.

Breaking changes

- *HPX* now requires a C++14 capable compiler. We have set the *HPX* C++ standard automatically to C++14 and if it needs to be set explicitly, it should be specified through the `CMAKE_CXX_STANDARD` setting as mandated by CMake. The `HPX_WITH_CXX*` variables are now deprecated and will be removed in the future.
- Building and using *HPX* is now supported only when using CMake V3.13 or later, Boost V1.64 or newer, and when compiling with clang V5, gcc V7, or VS2019, or later. Other compilers might still work but have not been tested thoroughly.
- We have added a `hpx::init_params` struct to pass parameters for *HPX* initialization e.g. the resource partitioner callback to initialize thread pools (*Using the resource partitioner*).
- The `all_to_all` algorithm is renamed to `all_gather`, and the new `all_to_all` algorithm is not compatible with the old one.
- We have moved all of the existing APIs related to resiliency into the namespace `hpx::resiliency::experimental`.

Closed issues

- [Issue #4918](#)²⁵⁰⁵ - Rename distributed_executors module
- [Issue #4900](#)²⁵⁰⁶ - Adding JOSS status badge to README
- [Issue #4897](#)²⁵⁰⁷ - Compiler warning, deprecated header used by *HPX* itself
- [Issue #4886](#)²⁵⁰⁸ - A future bound to an action executing on a different locality doesn't capture exception state
- [Issue #4880](#)²⁵⁰⁹ - Undefined reference to main build error when `HPX_WITH_DYNAMIC_HP_X_MAIN=OFF`
- [Issue #4877](#)²⁵¹⁰ - `hpx_main` might not able to start *hpx* runtime properly
- [Issue #4850](#)²⁵¹¹ - Issues creating templated component
- [Issue #4829](#)²⁵¹² - Spack package & `HPX_WITH_GENERIC_CONTEXT_COROUTINES`
- [Issue #4820](#)²⁵¹³ - PAPI counters don't work
- [Issue #4818](#)²⁵¹⁴ - *HPX* can't be used with IO pool turned off
- [Issue #4816](#)²⁵¹⁵ - Build of *HPX* fails when `find_package(Boost)` is called before `FetchContent_MakeAvailable(hpx)`
- [Issue #4813](#)²⁵¹⁶ - *HPX* MPI Future failed
- [Issue #4811](#)²⁵¹⁷ - Remove `HPX::hpx_no_wrap_main` target before 1.5.0 release

²⁵⁰⁵ <https://github.com/TheHPXProject/hpx/issues/4918>

²⁵⁰⁶ <https://github.com/TheHPXProject/hpx/issues/4900>

²⁵⁰⁷ <https://github.com/TheHPXProject/hpx/issues/4897>

²⁵⁰⁸ <https://github.com/TheHPXProject/hpx/issues/4886>

²⁵⁰⁹ <https://github.com/TheHPXProject/hpx/issues/4880>

²⁵¹⁰ <https://github.com/TheHPXProject/hpx/issues/4877>

²⁵¹¹ <https://github.com/TheHPXProject/hpx/issues/4850>

²⁵¹² <https://github.com/TheHPXProject/hpx/issues/4829>

²⁵¹³ <https://github.com/TheHPXProject/hpx/issues/4820>

²⁵¹⁴ <https://github.com/TheHPXProject/hpx/issues/4818>

²⁵¹⁵ <https://github.com/TheHPXProject/hpx/issues/4816>

²⁵¹⁶ <https://github.com/TheHPXProject/hpx/issues/4813>

²⁵¹⁷ <https://github.com/TheHPXProject/hpx/issues/4811>

- [Issue #4810](#)²⁵¹⁸ - In `hpx::for_each::invoke_projected` the `hpx::util::decay` is misguided
- [Issue #4787](#)²⁵¹⁹ - `transform_inclusive_scan` gives incorrect results for non-commutative operator
- [Issue #4786](#)²⁵²⁰ - `transform_inclusive_scan` tries to implicitly convert between types, instead of using the provided `conv` function
- [Issue #4779](#)²⁵²¹ - HPX build error with GCC 10.1
- [Issue #4766](#)²⁵²² - Move HPX.Compute functionality to experimental namespace
- [Issue #4763](#)²⁵²³ - License file name
- [Issue #4758](#)²⁵²⁴ - CMake profiling results
- [Issue #4755](#)²⁵²⁵ - Building HPX with support for PAPI fails
- [Issue #4754](#)²⁵²⁶ - CMake cache creation breaks when using HPX with mimalloc
- [Issue #4752](#)²⁵²⁷ - HPX MPI Future build failed
- [Issue #4746](#)²⁵²⁸ - Memory leak when using dataflow icw components
- [Issue #4731](#)²⁵²⁹ - Bug in stencil example, calculation of locality IDs
- [Issue #4723](#)²⁵³⁰ - Build fail with NETWORKING OFF
- [Issue #4720](#)²⁵³¹ - Add compatibility headers for modules that had their module headers implicitly generated in 1.4.1
- [Issue #4719](#)²⁵³² - Undeprecate some module headers
- [Issue #4712](#)²⁵³³ - Rename HPX_MPI_WITH_FUTURES option
- [Issue #4709](#)²⁵³⁴ - Make deprecation warnings overridable in dependent projects
- [Issue #4691](#)²⁵³⁵ - Suggestion to fix and enhance the `thread_mapper` API
- [Issue #4686](#)²⁵³⁶ - Fix tutorials examples
- [Issue #4685](#)²⁵³⁷ - HPX distributed map fails to compile
- [Issue #4680](#)²⁵³⁸ - Build error with `HPX_WITH_DYNAMIC_HPX_MAIN=OFF`
- [Issue #4679](#)²⁵³⁹ - Build error for hpx w/ Apex on Summit

²⁵¹⁸ <https://github.com/TheHPXProject/hpx/issues/4810>

²⁵¹⁹ <https://github.com/TheHPXProject/hpx/issues/4787>

²⁵²⁰ <https://github.com/TheHPXProject/hpx/issues/4786>

²⁵²¹ <https://github.com/TheHPXProject/hpx/issues/4779>

²⁵²² <https://github.com/TheHPXProject/hpx/issues/4766>

²⁵²³ <https://github.com/TheHPXProject/hpx/issues/4763>

²⁵²⁴ <https://github.com/TheHPXProject/hpx/issues/4758>

²⁵²⁵ <https://github.com/TheHPXProject/hpx/issues/4755>

²⁵²⁶ <https://github.com/TheHPXProject/hpx/issues/4754>

²⁵²⁷ <https://github.com/TheHPXProject/hpx/issues/4752>

²⁵²⁸ <https://github.com/TheHPXProject/hpx/issues/4746>

²⁵²⁹ <https://github.com/TheHPXProject/hpx/issues/4731>

²⁵³⁰ <https://github.com/TheHPXProject/hpx/issues/4723>

²⁵³¹ <https://github.com/TheHPXProject/hpx/issues/4720>

²⁵³² <https://github.com/TheHPXProject/hpx/issues/4719>

²⁵³³ <https://github.com/TheHPXProject/hpx/issues/4712>

²⁵³⁴ <https://github.com/TheHPXProject/hpx/issues/4709>

²⁵³⁵ <https://github.com/TheHPXProject/hpx/issues/4691>

²⁵³⁶ <https://github.com/TheHPXProject/hpx/issues/4686>

²⁵³⁷ <https://github.com/TheHPXProject/hpx/issues/4685>

²⁵³⁸ <https://github.com/TheHPXProject/hpx/issues/4680>

²⁵³⁹ <https://github.com/TheHPXProject/hpx/issues/4679>

- Issue #4675²⁵⁴⁰ - build error with HPX_WITH_NETWORKING=OFF
- Issue #4674²⁵⁴¹ - Error running Quickstart tests on OS X
- Issue #4662²⁵⁴² - MPI initialization broken when networking off
- Issue #4652²⁵⁴³ - How to fix distributed action annotation
- Issue #4650²⁵⁴⁴ - thread descriptions are broken... again
- Issue #4648²⁵⁴⁵ - Thread stacksize not properly set
- Issue #4647²⁵⁴⁶ - Rename generated collective headers in modules
- Issue #4639²⁵⁴⁷ - Update deprecation warnings in compatibility headers to point to collective headers
- Issue #4628²⁵⁴⁸ - mpi parcellport totally broken
- Issue #4619²⁵⁴⁹ - Fully document hpx_wrap behaviour and targets
- Issue #4612²⁵⁵⁰ - Compilation issue with HPX 1.4.1 and 1.4.0
- Issue #4594²⁵⁵¹ - Rename modules
- Issue #4578²⁵⁵² - Default value for HPX_WITH_THREAD_BACKTRACE_DEPTH
- Issue #4572²⁵⁵³ - Thread manager should be given a runtime_configuration
- Issue #4571²⁵⁵⁴ - Add high-level documentation to new modules
- Issue #4569²⁵⁵⁵ - Annoying warning when compiling - pls suppress or fix it.
- Issue #4555²⁵⁵⁶ - HPX_HAVE_THREAD_BACKTRACE_ON_SUSPENSION compilation error
- Issue #4543²⁵⁵⁷ - Segfaults in Release builds using *sleep_for*
- Issue #4539²⁵⁵⁸ - Compilation Error when HPX_MPI_WITH_FUTURES=ON
- Issue #4537²⁵⁵⁹ - Linking issue with libhpx_initd.a
- Issue #4535²⁵⁶⁰ - API for checking if pool with a given name exists
- Issue #4523²⁵⁶¹ - Build of PR #4311 (git tag 9955e8e) fails
- Issue #4519²⁵⁶² - Documentation problem

²⁵⁴⁰ <https://github.com/TheHPXProject/hpx/issues/4675>

²⁵⁴¹ <https://github.com/TheHPXProject/hpx/issues/4674>

²⁵⁴² <https://github.com/TheHPXProject/hpx/issues/4662>

²⁵⁴³ <https://github.com/TheHPXProject/hpx/issues/4652>

²⁵⁴⁴ <https://github.com/TheHPXProject/hpx/issues/4650>

²⁵⁴⁵ <https://github.com/TheHPXProject/hpx/issues/4648>

²⁵⁴⁶ <https://github.com/TheHPXProject/hpx/issues/4647>

²⁵⁴⁷ <https://github.com/TheHPXProject/hpx/issues/4639>

²⁵⁴⁸ <https://github.com/TheHPXProject/hpx/issues/4628>

²⁵⁴⁹ <https://github.com/TheHPXProject/hpx/issues/4619>

²⁵⁵⁰ <https://github.com/TheHPXProject/hpx/issues/4612>

²⁵⁵¹ <https://github.com/TheHPXProject/hpx/issues/4594>

²⁵⁵² <https://github.com/TheHPXProject/hpx/issues/4578>

²⁵⁵³ <https://github.com/TheHPXProject/hpx/issues/4572>

²⁵⁵⁴ <https://github.com/TheHPXProject/hpx/issues/4571>

²⁵⁵⁵ <https://github.com/TheHPXProject/hpx/issues/4569>

²⁵⁵⁶ <https://github.com/TheHPXProject/hpx/issues/4555>

²⁵⁵⁷ <https://github.com/TheHPXProject/hpx/issues/4543>

²⁵⁵⁸ <https://github.com/TheHPXProject/hpx/issues/4539>

²⁵⁵⁹ <https://github.com/TheHPXProject/hpx/issues/4537>

²⁵⁶⁰ <https://github.com/TheHPXProject/hpx/issues/4535>

²⁵⁶¹ <https://github.com/TheHPXProject/hpx/issues/4523>

²⁵⁶² <https://github.com/TheHPXProject/hpx/issues/4519>

- [Issue #4513](#)²⁵⁶³ - HPXConfig.cmake contains ill-formed paths when library paths use backslashes
- [Issue #4507](#)²⁵⁶⁴ - User-polling introduced by MPI futures module should be more generally usable
- [Issue #4506](#)²⁵⁶⁵ - Make sure force_linking.hpp is not included in main module header
- [Issue #4501](#)²⁵⁶⁶ - Fix compilation of PAPI tests
- [Issue #4497](#)²⁵⁶⁷ - Add modules CI checks
- [Issue #4489](#)²⁵⁶⁸ - Polymorphic executor
- [Issue #4476](#)²⁵⁶⁹ - Use CMake targets defined by FindBoost
- [Issue #4473](#)²⁵⁷⁰ - Add vcpkg installation instructions
- [Issue #4470](#)²⁵⁷¹ - Adapt hpx::future to C++20 co_await
- [Issue #4468](#)²⁵⁷² - Compile error on Raspberry Pi 4
- [Issue #4466](#)²⁵⁷³ - Compile error on Windows, current stable:
- [Issue #4453](#)²⁵⁷⁴ - Installing HPX on fedora with dnf is not adding cmake files
- [Issue #4448](#)²⁵⁷⁵ - New std::variant serialization broken
- [Issue #4438](#)²⁵⁷⁶ - Add performance counter flag is monotonically increasing
- [Issue #4436](#)²⁵⁷⁷ - Build problem: same code build and works with 1.4.0 but it doesn't with 1.4.1
- [Issue #4429](#)²⁵⁷⁸ - Function descriptions not supported in distributed
- [Issue #4423](#)²⁵⁷⁹ - `-hpx:ini=hpx.lock_detection=0` has no effect
- [Issue #4422](#)²⁵⁸⁰ - Add performance counter metadata
- [Issue #4419](#)²⁵⁸¹ - Weird behavior for `-hpx:print-counter-interval` with large numbers
- [Issue #4401](#)²⁵⁸² - Create module repository
- [Issue #4400](#)²⁵⁸³ - Command line options conflict related to performance counters
- [Issue #4349](#)²⁵⁸⁴ - `-hpx:use-process-mask` option throw an exception on OS X
- [Issue #4345](#)²⁵⁸⁵ - Move gh-pages branch out of hpx repo

²⁵⁶³ <https://github.com/TheHPXProject/hpx/issues/4513>

²⁵⁶⁴ <https://github.com/TheHPXProject/hpx/issues/4507>

²⁵⁶⁵ <https://github.com/TheHPXProject/hpx/issues/4506>

²⁵⁶⁶ <https://github.com/TheHPXProject/hpx/issues/4501>

²⁵⁶⁷ <https://github.com/TheHPXProject/hpx/issues/4497>

²⁵⁶⁸ <https://github.com/TheHPXProject/hpx/issues/4489>

²⁵⁶⁹ <https://github.com/TheHPXProject/hpx/issues/4476>

²⁵⁷⁰ <https://github.com/TheHPXProject/hpx/issues/4473>

²⁵⁷¹ <https://github.com/TheHPXProject/hpx/issues/4470>

²⁵⁷² <https://github.com/TheHPXProject/hpx/issues/4468>

²⁵⁷³ <https://github.com/TheHPXProject/hpx/issues/4466>

²⁵⁷⁴ <https://github.com/TheHPXProject/hpx/issues/4453>

²⁵⁷⁵ <https://github.com/TheHPXProject/hpx/issues/4448>

²⁵⁷⁶ <https://github.com/TheHPXProject/hpx/issues/4438>

²⁵⁷⁷ <https://github.com/TheHPXProject/hpx/issues/4436>

²⁵⁷⁸ <https://github.com/TheHPXProject/hpx/issues/4429>

²⁵⁷⁹ <https://github.com/TheHPXProject/hpx/issues/4423>

²⁵⁸⁰ <https://github.com/TheHPXProject/hpx/issues/4422>

²⁵⁸¹ <https://github.com/TheHPXProject/hpx/issues/4419>

²⁵⁸² <https://github.com/TheHPXProject/hpx/issues/4401>

²⁵⁸³ <https://github.com/TheHPXProject/hpx/issues/4400>

²⁵⁸⁴ <https://github.com/TheHPXProject/hpx/issues/4349>

²⁵⁸⁵ <https://github.com/TheHPXProject/hpx/issues/4345>

- [Issue #4323²⁵⁸⁶](#) - Const-correctness error in assignment operator of `compute::vector`
- [Issue #4318²⁵⁸⁷](#) - ASIO breaks with C++2a concepts
- [Issue #4317²⁵⁸⁸](#) - Application runs even if `-hpx:help` is specified
- [Issue #4063²⁵⁸⁹](#) - Document `hpxcxx` compiler wrapper
- [Issue #3983²⁵⁹⁰](#) - Implement the C++20 Synchronization Library
- [Issue #3696²⁵⁹¹](#) - C++11 `constexpr` support is now required
- [Issue #3623²⁵⁹²](#) - Modular HPX branch and an alternative project layout
- [Issue #2836²⁵⁹³](#) - The worst-case time complexity of `parallel::sort` seems to be $O(N^2)$.

Closed pull requests

- [PR #4936²⁵⁹⁴](#) - Minor documentation fixes part 2
- [PR #4935²⁵⁹⁵](#) - Add copyright and license to joss paper file
- [PR #4934²⁵⁹⁶](#) - Adding Semicolon in Documentation
- [PR #4932²⁵⁹⁷](#) - Fixing compiler warnings
- [PR #4931²⁵⁹⁸](#) - Small documentation formatting fixes
- [PR #4930²⁵⁹⁹](#) - Documentation Distributed HPX applications `localvv` with `local_vv`
- [PR #4929²⁶⁰⁰](#) - Add final version of the JOSS paper
- [PR #4928²⁶⁰¹](#) - Add `HPX_NODISCARD` to `enable_user_polling` structs
- [PR #4926²⁶⁰²](#) - Rename `distributed_executors` module to `executors_distributed`
- [PR #4925²⁶⁰³](#) - Making `transform_reduce` conforming to C++20
- [PR #4923²⁶⁰⁴](#) - Don't acquire lock if not needed
- [PR #4921²⁶⁰⁵](#) - Update the release notes for the release candidate 3
- [PR #4920²⁶⁰⁶](#) - Disable `libcdis` release
- [PR #4919²⁶⁰⁷](#) - Make `cuda` event pool dynamic instead of fixed size

²⁵⁸⁶ <https://github.com/TheHPXProject/hpx/issues/4323>

²⁵⁸⁷ <https://github.com/TheHPXProject/hpx/issues/4318>

²⁵⁸⁸ <https://github.com/TheHPXProject/hpx/issues/4317>

²⁵⁸⁹ <https://github.com/TheHPXProject/hpx/issues/4063>

²⁵⁹⁰ <https://github.com/TheHPXProject/hpx/issues/3983>

²⁵⁹¹ <https://github.com/TheHPXProject/hpx/issues/3696>

²⁵⁹² <https://github.com/TheHPXProject/hpx/issues/3623>

²⁵⁹³ <https://github.com/TheHPXProject/hpx/issues/2836>

²⁵⁹⁴ <https://github.com/TheHPXProject/hpx/pull/4936>

²⁵⁹⁵ <https://github.com/TheHPXProject/hpx/pull/4935>

²⁵⁹⁶ <https://github.com/TheHPXProject/hpx/pull/4934>

²⁵⁹⁷ <https://github.com/TheHPXProject/hpx/pull/4932>

²⁵⁹⁸ <https://github.com/TheHPXProject/hpx/pull/4931>

²⁵⁹⁹ <https://github.com/TheHPXProject/hpx/pull/4930>

²⁶⁰⁰ <https://github.com/TheHPXProject/hpx/pull/4929>

²⁶⁰¹ <https://github.com/TheHPXProject/hpx/pull/4928>

²⁶⁰² <https://github.com/TheHPXProject/hpx/pull/4926>

²⁶⁰³ <https://github.com/TheHPXProject/hpx/pull/4925>

²⁶⁰⁴ <https://github.com/TheHPXProject/hpx/pull/4923>

²⁶⁰⁵ <https://github.com/TheHPXProject/hpx/pull/4921>

²⁶⁰⁶ <https://github.com/TheHPXProject/hpx/pull/4920>

²⁶⁰⁷ <https://github.com/TheHPXProject/hpx/pull/4919>

- PR #4917²⁶⁰⁸ - Move chrono functionality to hpx::chrono namespace
- PR #4916²⁶⁰⁹ - HPX_HAVE_DEPRECATED_WARNINGS needs to be set even when disabled
- PR #4915²⁶¹⁰ - Moving more action related files to actions modules
- PR #4914²⁶¹¹ - Add alias targets with namespaces used for exporting
- PR #4912²⁶¹² - Aggregate initialize CPOs
- PR #4910²⁶¹³ - Explicitly specify hwloc root on Jenkins CSCS builds
- PR #4908²⁶¹⁴ - Fix algorithms documentation
- PR #4907²⁶¹⁵ - Remove HPX::hpx_no_wrap_main target
- PR #4906²⁶¹⁶ - Fixing unused variable warning
- PR #4905²⁶¹⁷ - Adding specializations for simple for_loops
- PR #4904²⁶¹⁸ - Update boost to 1.74.0 for the newest jenkins configs
- PR #4903²⁶¹⁹ - Hide GITHUB_TOKEN environment variables from environment variable output
- PR #4902²⁶²⁰ - Cancel previous pull requests builds before starting a new one with Jenkins
- PR #4901²⁶²¹ - Update public API list with updated algorithms
- PR #4899²⁶²² - Suggested changes for HPX V1.5 release notes
- PR #4898²⁶²³ - Minor tweak to hpx::equal implementation
- PR #4896²⁶²⁴ - Making generate() and generate_n conforming to C++20
- PR #4895²⁶²⁵ - Update apex tag
- PR #4894²⁶²⁶ - Fix exception handling for tasks
- PR #4893²⁶²⁷ - Remove last use of std::result_of, removed in C++20
- PR #4892²⁶²⁸ - Adding replay_executor and replicate_executor
- PR #4889²⁶²⁹ - Restore old behaviour of not requiring linking to hpx_wrap when HPX_WITH_DYNAMIC_HP_X_MAIN=OFF
- PR #4887²⁶³⁰ - Making sure remotely thrown (non-hpx) exceptions are properly marshaled back to invocation

²⁶⁰⁸ <https://github.com/TheHPXProject/hpx/pull/4917>

²⁶⁰⁹ <https://github.com/TheHPXProject/hpx/pull/4916>

²⁶¹⁰ <https://github.com/TheHPXProject/hpx/pull/4915>

²⁶¹¹ <https://github.com/TheHPXProject/hpx/pull/4914>

²⁶¹² <https://github.com/TheHPXProject/hpx/pull/4912>

²⁶¹³ <https://github.com/TheHPXProject/hpx/pull/4910>

²⁶¹⁴ <https://github.com/TheHPXProject/hpx/pull/4908>

²⁶¹⁵ <https://github.com/TheHPXProject/hpx/pull/4907>

²⁶¹⁶ <https://github.com/TheHPXProject/hpx/pull/4906>

²⁶¹⁷ <https://github.com/TheHPXProject/hpx/pull/4905>

²⁶¹⁸ <https://github.com/TheHPXProject/hpx/pull/4904>

²⁶¹⁹ <https://github.com/TheHPXProject/hpx/pull/4903>

²⁶²⁰ <https://github.com/TheHPXProject/hpx/pull/4902>

²⁶²¹ <https://github.com/TheHPXProject/hpx/pull/4901>

²⁶²² <https://github.com/TheHPXProject/hpx/pull/4899>

²⁶²³ <https://github.com/TheHPXProject/hpx/pull/4898>

²⁶²⁴ <https://github.com/TheHPXProject/hpx/pull/4896>

²⁶²⁵ <https://github.com/TheHPXProject/hpx/pull/4895>

²⁶²⁶ <https://github.com/TheHPXProject/hpx/pull/4894>

²⁶²⁷ <https://github.com/TheHPXProject/hpx/pull/4893>

²⁶²⁸ <https://github.com/TheHPXProject/hpx/pull/4892>

²⁶²⁹ <https://github.com/TheHPXProject/hpx/pull/4889>

²⁶³⁰ <https://github.com/TheHPXProject/hpx/pull/4887>

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- [PR #4885](#)²⁶³¹ - Adapting `hpx::find` and `friends` to C++20
- [PR #4884](#)²⁶³² - Adapting `mismatch` to C++20
- [PR #4883](#)²⁶³³ - Adapting `hpx::equal` to be conforming to C++20
- [PR #4882](#)²⁶³⁴ - Fixing exception handling for `hpx::copy` and adding missing tests
- [PR #4881](#)²⁶³⁵ - Adds different runtime exception when registering thread with the HPX runtime
- [PR #4876](#)²⁶³⁶ - Adding example demonstrating how to disable thread stealing during the execution of parallel algorithms
- [PR #4874](#)²⁶³⁷ - Adding non-policy tests to `all_of`, `any_of`, and `none_of`
- [PR #4873](#)²⁶³⁸ - Set CUDA compute capability on rostam Jenkins builds
- [PR #4872](#)²⁶³⁹ - Force partitioned vector scan tests to run serially
- [PR #4870](#)²⁶⁴⁰ - Making move conforming with C++20
- [PR #4869](#)²⁶⁴¹ - Making `destroy` and `destroy_n` conforming to C++20
- [PR #4868](#)²⁶⁴² - Fix miscellaneous header problems
- [PR #4867](#)²⁶⁴³ - Add CPOs for `for_each`
- [PR #4865](#)²⁶⁴⁴ - Adapting `count` and `count_if` to be conforming to C++20
- [PR #4864](#)²⁶⁴⁵ - Release notes 1.5.0
- [PR #4863](#)²⁶⁴⁶ - adding `libcds-hpx` tag to prepare for `hpx1.5` release
- [PR #4862](#)²⁶⁴⁷ - Adding version specific deprecation options
- [PR #4861](#)²⁶⁴⁸ - Limiting executor improvements
- [PR #4860](#)²⁶⁴⁹ - Making `fill` and `fill_n` compatible with C++20
- [PR #4859](#)²⁶⁵⁰ - Adapting `all_of`, `any_of`, and `none_of` to C++20
- [PR #4857](#)²⁶⁵¹ - Improve `libCDS` integration
- [PR #4856](#)²⁶⁵² - Correct typos in the documentation of the `hpx` performance counters

²⁶³¹ <https://github.com/TheHPXProject/hpx/pull/4885>

²⁶³² <https://github.com/TheHPXProject/hpx/pull/4884>

²⁶³³ <https://github.com/TheHPXProject/hpx/pull/4883>

²⁶³⁴ <https://github.com/TheHPXProject/hpx/pull/4882>

²⁶³⁵ <https://github.com/TheHPXProject/hpx/pull/4881>

²⁶³⁶ <https://github.com/TheHPXProject/hpx/pull/4876>

²⁶³⁷ <https://github.com/TheHPXProject/hpx/pull/4874>

²⁶³⁸ <https://github.com/TheHPXProject/hpx/pull/4873>

²⁶³⁹ <https://github.com/TheHPXProject/hpx/pull/4872>

²⁶⁴⁰ <https://github.com/TheHPXProject/hpx/pull/4870>

²⁶⁴¹ <https://github.com/TheHPXProject/hpx/pull/4869>

²⁶⁴² <https://github.com/TheHPXProject/hpx/pull/4868>

²⁶⁴³ <https://github.com/TheHPXProject/hpx/pull/4867>

²⁶⁴⁴ <https://github.com/TheHPXProject/hpx/pull/4865>

²⁶⁴⁵ <https://github.com/TheHPXProject/hpx/pull/4864>

²⁶⁴⁶ <https://github.com/TheHPXProject/hpx/pull/4863>

²⁶⁴⁷ <https://github.com/TheHPXProject/hpx/pull/4862>

²⁶⁴⁸ <https://github.com/TheHPXProject/hpx/pull/4861>

²⁶⁴⁹ <https://github.com/TheHPXProject/hpx/pull/4860>

²⁶⁵⁰ <https://github.com/TheHPXProject/hpx/pull/4859>

²⁶⁵¹ <https://github.com/TheHPXProject/hpx/pull/4857>

²⁶⁵² <https://github.com/TheHPXProject/hpx/pull/4856>

- PR #4854²⁶⁵³ - Removing obsolete code
- PR #4853²⁶⁵⁴ - Adding test that derives component from two other components
- PR #4852²⁶⁵⁵ - Fix mpi_ring test in distributed mode by ensuring all ranks run hpx_main
- PR #4851²⁶⁵⁶ - Converting resiliency APIs to tag_invoke based CPOs
- PR #4849²⁶⁵⁷ - Enable use of future_overhead test when DISTRIBUTED_RUNTIME is OFF
- PR #4847²⁶⁵⁸ - Fixing 'error prone' constructs as reported by Codacy
- PR #4846²⁶⁵⁹ - Disable Boost.Asio concepts support
- PR #4845²⁶⁶⁰ - Fix PAPI counters
- PR #4843²⁶⁶¹ - Remove dependency on various Boost headers
- PR #4841²⁶⁶² - Rearrange public API headers
- PR #4840²⁶⁶³ - Fixing TSS problems during thread termination
- PR #4839²⁶⁶⁴ - Fix async_cuda build problems when distributed runtime is disabled
- PR #4837²⁶⁶⁵ - Restore compatibility for old (now deprecated) copy algorithms
- PR #4836²⁶⁶⁶ - Adding CPOs for hpx::reduce
- PR #4835²⁶⁶⁷ - Remove *using util::result_of* from namespace hpx
- PR #4834²⁶⁶⁸ - Fixing the calculation of the number of idle cores and the corresponding idle masks
- PR #4833²⁶⁶⁹ - Allow thread function destructors to yield
- PR #4832²⁶⁷⁰ - Fixing assertion in split_gids and memory leaks in 1d_stencil_7
- PR #4831²⁶⁷¹ - Making sure MPI_CXX_COMPILE_FLAGS is interpreted as a sequence of options
- PR #4830²⁶⁷² - Update documentation on using HPX::wrap_main
- PR #4827²⁶⁷³ - Update clang-newest configuration to use clang 10
- PR #4826²⁶⁷⁴ - Add Jenkins configuration for rostam
- PR #4825²⁶⁷⁵ - Move all CUDA functionality to hpx::cuda::experimental namespace

²⁶⁵³ <https://github.com/TheHPXProject/hpx/pull/4854>

²⁶⁵⁴ <https://github.com/TheHPXProject/hpx/pull/4853>

²⁶⁵⁵ <https://github.com/TheHPXProject/hpx/pull/4852>

²⁶⁵⁶ <https://github.com/TheHPXProject/hpx/pull/4851>

²⁶⁵⁷ <https://github.com/TheHPXProject/hpx/pull/4849>

²⁶⁵⁸ <https://github.com/TheHPXProject/hpx/pull/4847>

²⁶⁵⁹ <https://github.com/TheHPXProject/hpx/pull/4846>

²⁶⁶⁰ <https://github.com/TheHPXProject/hpx/pull/4845>

²⁶⁶¹ <https://github.com/TheHPXProject/hpx/pull/4843>

²⁶⁶² <https://github.com/TheHPXProject/hpx/pull/4841>

²⁶⁶³ <https://github.com/TheHPXProject/hpx/pull/4840>

²⁶⁶⁴ <https://github.com/TheHPXProject/hpx/pull/4839>

²⁶⁶⁵ <https://github.com/TheHPXProject/hpx/pull/4837>

²⁶⁶⁶ <https://github.com/TheHPXProject/hpx/pull/4836>

²⁶⁶⁷ <https://github.com/TheHPXProject/hpx/pull/4835>

²⁶⁶⁸ <https://github.com/TheHPXProject/hpx/pull/4834>

²⁶⁶⁹ <https://github.com/TheHPXProject/hpx/pull/4833>

²⁶⁷⁰ <https://github.com/TheHPXProject/hpx/pull/4832>

²⁶⁷¹ <https://github.com/TheHPXProject/hpx/pull/4831>

²⁶⁷² <https://github.com/TheHPXProject/hpx/pull/4830>

²⁶⁷³ <https://github.com/TheHPXProject/hpx/pull/4827>

²⁶⁷⁴ <https://github.com/TheHPXProject/hpx/pull/4826>

²⁶⁷⁵ <https://github.com/TheHPXProject/hpx/pull/4825>

- PR #4824²⁶⁷⁶ - Add support for building master/release branches to Jenkins configuration
- PR #4821²⁶⁷⁷ - Implement customization point for `hpx::copy` and `hpx::ranges::copy`
- PR #4819²⁶⁷⁸ - Allow finding Boost components before finding HPX
- PR #4817²⁶⁷⁹ - Adding range version of stable sort
- PR #4815²⁶⁸⁰ - Fix a wrong `#ifdef` for IO/TIMER pools causing build errors
- PR #4814²⁶⁸¹ - Replace `hpx::function_nonsr` with `std::function` in error module
- PR #4809²⁶⁸² - Foreach adapt
- PR #4808²⁶⁸³ - Make internal algorithms functions const
- PR #4807²⁶⁸⁴ - Add Jenkins configuration for running on Piz Daint
- PR #4806²⁶⁸⁵ - Update documentation links to new domain name
- PR #4805²⁶⁸⁶ - Applying changes that resolve time complexity issues in sort
- PR #4803²⁶⁸⁷ - Adding implementation of `stable_sort`
- PR #4802²⁶⁸⁸ - Fix datapar header paths
- PR #4801²⁶⁸⁹ - Replace `boost::shared_array<T>` with `std::shared_ptr<T[]>` if supported
- PR #4799²⁶⁹⁰ - Fixing `#include` paths in compatibility headers
- PR #4798²⁶⁹¹ - Include the main module header (fixes partially #4488)
- PR #4797²⁶⁹² - Change cmake targets
- PR #4794²⁶⁹³ - Removing 128bit integer emulation
- PR #4793²⁶⁹⁴ - Make sure global variable is handled properly
- PR #4792²⁶⁹⁵ - Replace `enable_if` with **HPX_CONCEPT_REQUIRES_** and add `is_sentinel_for` constraint
- PR #4790²⁶⁹⁶ - Move deprecation warnings from base template to template specializations for `result_of` etc. structs
- PR #4789²⁶⁹⁷ - Fix hangs during assertion handling and distributed runtime construction
- PR #4788²⁶⁹⁸ - Fixing inclusive transform scan algorithm to properly handle initial value

²⁶⁷⁶ <https://github.com/TheHPXProject/hpx/pull/4824>

²⁶⁷⁷ <https://github.com/TheHPXProject/hpx/pull/4821>

²⁶⁷⁸ <https://github.com/TheHPXProject/hpx/pull/4819>

²⁶⁷⁹ <https://github.com/TheHPXProject/hpx/pull/4817>

²⁶⁸⁰ <https://github.com/TheHPXProject/hpx/pull/4815>

²⁶⁸¹ <https://github.com/TheHPXProject/hpx/pull/4814>

²⁶⁸² <https://github.com/TheHPXProject/hpx/pull/4809>

²⁶⁸³ <https://github.com/TheHPXProject/hpx/pull/4808>

²⁶⁸⁴ <https://github.com/TheHPXProject/hpx/pull/4807>

²⁶⁸⁵ <https://github.com/TheHPXProject/hpx/pull/4806>

²⁶⁸⁶ <https://github.com/TheHPXProject/hpx/pull/4805>

²⁶⁸⁷ <https://github.com/TheHPXProject/hpx/pull/4803>

²⁶⁸⁸ <https://github.com/TheHPXProject/hpx/pull/4802>

²⁶⁸⁹ <https://github.com/TheHPXProject/hpx/pull/4801>

²⁶⁹⁰ <https://github.com/TheHPXProject/hpx/pull/4799>

²⁶⁹¹ <https://github.com/TheHPXProject/hpx/pull/4798>

²⁶⁹² <https://github.com/TheHPXProject/hpx/pull/4797>

²⁶⁹³ <https://github.com/TheHPXProject/hpx/pull/4794>

²⁶⁹⁴ <https://github.com/TheHPXProject/hpx/pull/4793>

²⁶⁹⁵ <https://github.com/TheHPXProject/hpx/pull/4792>

²⁶⁹⁶ <https://github.com/TheHPXProject/hpx/pull/4790>

²⁶⁹⁷ <https://github.com/TheHPXProject/hpx/pull/4789>

²⁶⁹⁸ <https://github.com/TheHPXProject/hpx/pull/4788>

- PR #4785²⁶⁹⁹ - Fixing barrier test
- PR #4784²⁷⁰⁰ - Fixing deleter argument bindings in `serialize_buffer`
- PR #4783²⁷⁰¹ - Add coveralls badge
- PR #4782²⁷⁰² - Make header tests parallel again
- PR #4780²⁷⁰³ - Remove outdated comment about `hpx::stop` in documentation
- PR #4776²⁷⁰⁴ - debug print improvements
- PR #4775²⁷⁰⁵ - Checkpoint cleanup
- PR #4771²⁷⁰⁶ - Fix compilation with `HPX_WITH_NETWORKING=OFF`
- PR #4767²⁷⁰⁷ - Remove all force linking leftovers
- PR #4765²⁷⁰⁸ - Fix 1d stencil index calculation
- PR #4764²⁷⁰⁹ - Force some tests to run serially
- PR #4762²⁷¹⁰ - Update pointees in compatibility headers
- PR #4761²⁷¹¹ - Fix running and building of execution module tests on CircleCI
- PR #4760²⁷¹² - Storing `hpx_options` in global property to speed up summary report
- PR #4759²⁷¹³ - Reduce memory requirements for our main shared state
- PR #4757²⁷¹⁴ - Fix `mimalloc` linking on Windows
- PR #4756²⁷¹⁵ - Fix compilation issues
- PR #4753²⁷¹⁶ - Re-adding API functions that were lost during merges
- PR #4751²⁷¹⁷ - Revert “Create coverage reports and upload them to `codecov.io`”
- PR #4750²⁷¹⁸ - Fixing possible race condition during termination detection
- PR #4749²⁷¹⁹ - Deprecate `result_of` and friends
- PR #4748²⁷²⁰ - Create coverage reports and upload them to `codecov.io`
- PR #4747²⁷²¹ - Changing `#include` for MPI `parcelpport`

²⁶⁹⁹ <https://github.com/TheHPXProject/hpx/pull/4785>

²⁷⁰⁰ <https://github.com/TheHPXProject/hpx/pull/4784>

²⁷⁰¹ <https://github.com/TheHPXProject/hpx/pull/4783>

²⁷⁰² <https://github.com/TheHPXProject/hpx/pull/4782>

²⁷⁰³ <https://github.com/TheHPXProject/hpx/pull/4780>

²⁷⁰⁴ <https://github.com/TheHPXProject/hpx/pull/4776>

²⁷⁰⁵ <https://github.com/TheHPXProject/hpx/pull/4775>

²⁷⁰⁶ <https://github.com/TheHPXProject/hpx/pull/4771>

²⁷⁰⁷ <https://github.com/TheHPXProject/hpx/pull/4767>

²⁷⁰⁸ <https://github.com/TheHPXProject/hpx/pull/4765>

²⁷⁰⁹ <https://github.com/TheHPXProject/hpx/pull/4764>

²⁷¹⁰ <https://github.com/TheHPXProject/hpx/pull/4762>

²⁷¹¹ <https://github.com/TheHPXProject/hpx/pull/4761>

²⁷¹² <https://github.com/TheHPXProject/hpx/pull/4760>

²⁷¹³ <https://github.com/TheHPXProject/hpx/pull/4759>

²⁷¹⁴ <https://github.com/TheHPXProject/hpx/pull/4757>

²⁷¹⁵ <https://github.com/TheHPXProject/hpx/pull/4756>

²⁷¹⁶ <https://github.com/TheHPXProject/hpx/pull/4753>

²⁷¹⁷ <https://github.com/TheHPXProject/hpx/pull/4751>

²⁷¹⁸ <https://github.com/TheHPXProject/hpx/pull/4750>

²⁷¹⁹ <https://github.com/TheHPXProject/hpx/pull/4749>

²⁷²⁰ <https://github.com/TheHPXProject/hpx/pull/4748>

²⁷²¹ <https://github.com/TheHPXProject/hpx/pull/4747>

- PR #4745²⁷²² - Add *is_sentinel_for* trait implementation and test
- PR #4743²⁷²³ - Fix *init_globally* example after runtime mode changes
- PR #4742²⁷²⁴ - Update SUPPORT.md
- PR #4741²⁷²⁵ - Fixing a warning generated for unity builds with msvc
- PR #4740²⁷²⁶ - Rename *local_lcos* and *basic_execution* modules
- PR #4739²⁷²⁷ - Undeprecate a couple of *hpx/modulename.hpp* headers
- PR #4738²⁷²⁸ - Conditionally test schedulers in *thread_stacksize_current* test
- PR #4734²⁷²⁹ - Fixing a bunch of codacy warnings
- PR #4733²⁷³⁰ - Add experimental unity build option to CMake configuration
- PR #4730²⁷³¹ - Fixing compilation problems with *unordered_map*
- PR #4729²⁷³² - Fix APEX build
- PR #4727²⁷³³ - Fix missing runtime includes for distributed runtime
- PR #4726²⁷³⁴ - Add more API headers
- PR #4725²⁷³⁵ - Add more compatibility headers for deprecated module headers
- PR #4724²⁷³⁶ - Fix 4723
- PR #4721²⁷³⁷ - Attempt to fixing migration tests
- PR #4717²⁷³⁸ - Make the compatibility headers macro conditional
- PR #4716²⁷³⁹ - Add *hpx/runtime.hpp* and *hpx/distributed/runtime.hpp* API headers
- PR #4714²⁷⁴⁰ - Add *hpx/future.hpp* header
- PR #4713²⁷⁴¹ - Remove *hpx/runtime/threads_fwd.hpp* and *hpx/util_fwd.hpp*
- PR #4711²⁷⁴² - Make module deprecation warnings overridable
- PR #4710²⁷⁴³ - Add compatibility headers and other fixes after module header renaming
- PR #4708²⁷⁴⁴ - Add termination handler for parallel algorithms

²⁷²² <https://github.com/TheHPXProject/hpx/pull/4745>

²⁷²³ <https://github.com/TheHPXProject/hpx/pull/4743>

²⁷²⁴ <https://github.com/TheHPXProject/hpx/pull/4742>

²⁷²⁵ <https://github.com/TheHPXProject/hpx/pull/4741>

²⁷²⁶ <https://github.com/TheHPXProject/hpx/pull/4740>

²⁷²⁷ <https://github.com/TheHPXProject/hpx/pull/4739>

²⁷²⁸ <https://github.com/TheHPXProject/hpx/pull/4738>

²⁷²⁹ <https://github.com/TheHPXProject/hpx/pull/4734>

²⁷³⁰ <https://github.com/TheHPXProject/hpx/pull/4733>

²⁷³¹ <https://github.com/TheHPXProject/hpx/pull/4730>

²⁷³² <https://github.com/TheHPXProject/hpx/pull/4729>

²⁷³³ <https://github.com/TheHPXProject/hpx/pull/4727>

²⁷³⁴ <https://github.com/TheHPXProject/hpx/pull/4726>

²⁷³⁵ <https://github.com/TheHPXProject/hpx/pull/4725>

²⁷³⁶ <https://github.com/TheHPXProject/hpx/pull/4724>

²⁷³⁷ <https://github.com/TheHPXProject/hpx/pull/4721>

²⁷³⁸ <https://github.com/TheHPXProject/hpx/pull/4717>

²⁷³⁹ <https://github.com/TheHPXProject/hpx/pull/4716>

²⁷⁴⁰ <https://github.com/TheHPXProject/hpx/pull/4714>

²⁷⁴¹ <https://github.com/TheHPXProject/hpx/pull/4713>

²⁷⁴² <https://github.com/TheHPXProject/hpx/pull/4711>

²⁷⁴³ <https://github.com/TheHPXProject/hpx/pull/4710>

²⁷⁴⁴ <https://github.com/TheHPXProject/hpx/pull/4708>

- PR #4707²⁷⁴⁵ - Use `hpx::function_nonser` instead of `std::function` internally
- PR #4706²⁷⁴⁶ - Move header file to module
- PR #4705²⁷⁴⁷ - Fix incorrect behaviour of `cmake-format` check
- PR #4704²⁷⁴⁸ - Fix resource tests
- PR #4701²⁷⁴⁹ - Fix missing includes for `future::then` specializations
- PR #4700²⁷⁵⁰ - Removing obsolete memory component
- PR #4699²⁷⁵¹ - Add short descriptions to modules missing documentation
- PR #4696²⁷⁵² - Rename generated modules headers
- PR #4693²⁷⁵³ - Overhauling `thread_mapper` for public consumption
- PR #4688²⁷⁵⁴ - Fix thread stack size handling
- PR #4687²⁷⁵⁵ - Adding `all_gather` and fixing `all_to_all`
- PR #4684²⁷⁵⁶ - Miscellaneous compilation fixes
- PR #4683²⁷⁵⁷ - Fix `HPX_WITH_DYNAMIC_HPX_MAIN=OFF`
- PR #4682²⁷⁵⁸ - Fix compilation of `pack_traversal_rebind_container.hpp`
- PR #4681²⁷⁵⁹ - Add missing `hpx/execution.hpp` includes for `future::then`
- PR #4678²⁷⁶⁰ - Typeless communicator
- PR #4677²⁷⁶¹ - Forcing registry option to be accepted without checks.
- PR #4676²⁷⁶² - Adding `scatter_to/scatter_from` collective operations
- PR #4673²⁷⁶³ - Fix PAPI counters compilation
- PR #4671²⁷⁶⁴ - Deprecate `hpx::promise` alias to `hpx::lcos::promise`
- PR #4670²⁷⁶⁵ - Explicitly instantiate `get_exception`
- PR #4667²⁷⁶⁶ - Add `stopValue` in *Sentinel* struct instead of *Iterator*
- PR #4666²⁷⁶⁷ - Add release build on Windows to GitHub actions

²⁷⁴⁵ <https://github.com/TheHPXProject/hpx/pull/4707>

²⁷⁴⁶ <https://github.com/TheHPXProject/hpx/pull/4706>

²⁷⁴⁷ <https://github.com/TheHPXProject/hpx/pull/4705>

²⁷⁴⁸ <https://github.com/TheHPXProject/hpx/pull/4704>

²⁷⁴⁹ <https://github.com/TheHPXProject/hpx/pull/4701>

²⁷⁵⁰ <https://github.com/TheHPXProject/hpx/pull/4700>

²⁷⁵¹ <https://github.com/TheHPXProject/hpx/pull/4699>

²⁷⁵² <https://github.com/TheHPXProject/hpx/pull/4696>

²⁷⁵³ <https://github.com/TheHPXProject/hpx/pull/4693>

²⁷⁵⁴ <https://github.com/TheHPXProject/hpx/pull/4688>

²⁷⁵⁵ <https://github.com/TheHPXProject/hpx/pull/4687>

²⁷⁵⁶ <https://github.com/TheHPXProject/hpx/pull/4684>

²⁷⁵⁷ <https://github.com/TheHPXProject/hpx/pull/4683>

²⁷⁵⁸ <https://github.com/TheHPXProject/hpx/pull/4682>

²⁷⁵⁹ <https://github.com/TheHPXProject/hpx/pull/4681>

²⁷⁶⁰ <https://github.com/TheHPXProject/hpx/pull/4678>

²⁷⁶¹ <https://github.com/TheHPXProject/hpx/pull/4677>

²⁷⁶² <https://github.com/TheHPXProject/hpx/pull/4676>

²⁷⁶³ <https://github.com/TheHPXProject/hpx/pull/4673>

²⁷⁶⁴ <https://github.com/TheHPXProject/hpx/pull/4671>

²⁷⁶⁵ <https://github.com/TheHPXProject/hpx/pull/4670>

²⁷⁶⁶ <https://github.com/TheHPXProject/hpx/pull/4667>

²⁷⁶⁷ <https://github.com/TheHPXProject/hpx/pull/4666>

- PR #4664²⁷⁶⁸ - Creating itt_notify module.
- PR #4663²⁷⁶⁹ - Mpi fixes
- PR #4659²⁷⁷⁰ - Making sure declarations match definitions in register_locks implementation
- PR #4655²⁷⁷¹ - Fixing task annotations for actions
- PR #4653²⁷⁷² - Making sure APEX is linked into every application, if needed
- PR #4651²⁷⁷³ - Update get_function_annotation.hpp
- PR #4646²⁷⁷⁴ - Runtime type
- PR #4645²⁷⁷⁵ - Add a few more API headers
- PR #4644²⁷⁷⁶ - Fixing support for mpirun (and similar)
- PR #4643²⁷⁷⁷ - Fixing the fix for get_idle_core_count() API
- PR #4638²⁷⁷⁸ - Remove HPX_API_EXPORT missed in previous cleanup
- PR #4636²⁷⁷⁹ - Adding C++20 barrier
- PR #4635²⁷⁸⁰ - Adding C++20 latch API
- PR #4634²⁷⁸¹ - Adding C++20 counting semaphore API
- PR #4633²⁷⁸² - Unify execution parameters customization points
- PR #4632²⁷⁸³ - Adding missing bulk_sync_execute wrapper to example executor
- PR #4631²⁷⁸⁴ - Updates to documentation; grammar edits.
- PR #4630²⁷⁸⁵ - Updates to documentation; moved hyperlink
- PR #4624²⁷⁸⁶ - Export set_self_ptr in thread_data.hpp instead of with forward declarations where used
- PR #4623²⁷⁸⁷ - Clean up export macros
- PR #4621²⁷⁸⁸ - Trigger an error for older boost versions on power architectures
- PR #4617²⁷⁸⁹ - Ignore user-set compatibility header options if the module does not have compatibility headers
- PR #4616²⁷⁹⁰ - Fix cmake-format warning

²⁷⁶⁸ <https://github.com/TheHPXProject/hpx/pull/4664>

²⁷⁶⁹ <https://github.com/TheHPXProject/hpx/pull/4663>

²⁷⁷⁰ <https://github.com/TheHPXProject/hpx/pull/4659>

²⁷⁷¹ <https://github.com/TheHPXProject/hpx/pull/4655>

²⁷⁷² <https://github.com/TheHPXProject/hpx/pull/4653>

²⁷⁷³ <https://github.com/TheHPXProject/hpx/pull/4651>

²⁷⁷⁴ <https://github.com/TheHPXProject/hpx/pull/4646>

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²⁷⁸¹ <https://github.com/TheHPXProject/hpx/pull/4634>

²⁷⁸² <https://github.com/TheHPXProject/hpx/pull/4633>

²⁷⁸³ <https://github.com/TheHPXProject/hpx/pull/4632>

²⁷⁸⁴ <https://github.com/TheHPXProject/hpx/pull/4631>

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²⁷⁸⁷ <https://github.com/TheHPXProject/hpx/pull/4623>

²⁷⁸⁸ <https://github.com/TheHPXProject/hpx/pull/4621>

²⁷⁸⁹ <https://github.com/TheHPXProject/hpx/pull/4617>

²⁷⁹⁰ <https://github.com/TheHPXProject/hpx/pull/4616>

- [PR #4615](#)²⁷⁹¹ - Add handler for serializing custom exceptions
- [PR #4614](#)²⁷⁹² - Fix error message when HPX_IGNORE_CMAKE_BUILD_TYPE_COMPATIBILITY=OFF
- [PR #4613](#)²⁷⁹³ - Make partitioner constructor private
- [PR #4611](#)²⁷⁹⁴ - Making auto_chunk_size execute the given function using the given executor
- [PR #4610](#)²⁷⁹⁵ - Making sure the thread-local lock registration data is moving to the core the suspended HPX thread is resumed on
- [PR #4609](#)²⁷⁹⁶ - Adding an API function that exposes the number of idle cores
- [PR #4608](#)²⁷⁹⁷ - Fixing moodycamel namespace
- [PR #4607](#)²⁷⁹⁸ - Moving winsocket initialization to core library
- [PR #4606](#)²⁷⁹⁹ - Local runtime module etc.
- [PR #4604](#)²⁸⁰⁰ - Add config_registry module
- [PR #4603](#)²⁸⁰¹ - Deal with distributed modules in their respective CMakeLists.txt
- [PR #4602](#)²⁸⁰² - Small module fixes
- [PR #4598](#)²⁸⁰³ - Making sure current_executor and service_executor functions are linked into the core library
- [PR #4597](#)²⁸⁰⁴ - Adding broadcast_to/broadcast_from to collectives module
- [PR #4596](#)²⁸⁰⁵ - Fix performance regression in block_executor
- [PR #4595](#)²⁸⁰⁶ - Making sure main.cpp is built as a library if HPX_WITH_DYNAMIC_MAIN=OFF
- [PR #4592](#)²⁸⁰⁷ - Futures module
- [PR #4591](#)²⁸⁰⁸ - Adapting co_await support for C++20
- [PR #4590](#)²⁸⁰⁹ - Adding missing exception test for for_loop()
- [PR #4587](#)²⁸¹⁰ - Move traits headers to hpx/modulename/traits directory
- [PR #4586](#)²⁸¹¹ - Remove Travis CI config
- [PR #4585](#)²⁸¹² - Update macOS test blacklist
- [PR #4584](#)²⁸¹³ - Attempting to fix missing symbols in stack trace

²⁷⁹¹ <https://github.com/TheHPXProject/hpx/pull/4615>

²⁷⁹² <https://github.com/TheHPXProject/hpx/pull/4614>

²⁷⁹³ <https://github.com/TheHPXProject/hpx/pull/4613>

²⁷⁹⁴ <https://github.com/TheHPXProject/hpx/pull/4611>

²⁷⁹⁵ <https://github.com/TheHPXProject/hpx/pull/4610>

²⁷⁹⁶ <https://github.com/TheHPXProject/hpx/pull/4609>

²⁷⁹⁷ <https://github.com/TheHPXProject/hpx/pull/4608>

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²⁷⁹⁹ <https://github.com/TheHPXProject/hpx/pull/4606>

²⁸⁰⁰ <https://github.com/TheHPXProject/hpx/pull/4604>

²⁸⁰¹ <https://github.com/TheHPXProject/hpx/pull/4603>

²⁸⁰² <https://github.com/TheHPXProject/hpx/pull/4602>

²⁸⁰³ <https://github.com/TheHPXProject/hpx/pull/4598>

²⁸⁰⁴ <https://github.com/TheHPXProject/hpx/pull/4597>

²⁸⁰⁵ <https://github.com/TheHPXProject/hpx/pull/4596>

²⁸⁰⁶ <https://github.com/TheHPXProject/hpx/pull/4595>

²⁸⁰⁷ <https://github.com/TheHPXProject/hpx/pull/4592>

²⁸⁰⁸ <https://github.com/TheHPXProject/hpx/pull/4591>

²⁸⁰⁹ <https://github.com/TheHPXProject/hpx/pull/4590>

²⁸¹⁰ <https://github.com/TheHPXProject/hpx/pull/4587>

²⁸¹¹ <https://github.com/TheHPXProject/hpx/pull/4586>

²⁸¹² <https://github.com/TheHPXProject/hpx/pull/4585>

²⁸¹³ <https://github.com/TheHPXProject/hpx/pull/4584>

- PR #4583²⁸¹⁴ - Fixing bad static_cast
- PR #4582²⁸¹⁵ - Changing download url for Windows prerequisites to circumvent bandwidth limitations
- PR #4581²⁸¹⁶ - Adding missing using placeholder::_X
- PR #4579²⁸¹⁷ - Move get_stack_size_name and related functions
- PR #4575²⁸¹⁸ - Excluding unconditional definition of class backtrace from global header
- PR #4574²⁸¹⁹ - Changing return type of hardware_concurrency() to unsigned int
- PR #4570²⁸²⁰ - Move tests to modules
- PR #4564²⁸²¹ - Reshuffle internal targets and add HPX::hpx_no_wrap_main target
- PR #4563²⁸²² - fix CMake option typo
- PR #4562²⁸²³ - Unregister lock earlier to avoid holding it while suspending
- PR #4561²⁸²⁴ - Adding test macros supporting custom output stream
- PR #4560²⁸²⁵ - Making sure hash_any::operator() is linked into core library
- PR #4559²⁸²⁶ - Fixing compilation if HPX_WITH_THREAD_BACKTRACE_ON_SUSPENSION=On
- PR #4557²⁸²⁷ - Improve spinlock implementation to perform better in high-contention situations
- PR #4553²⁸²⁸ - Fix a runtime_ptr problem at shutdown when apex is enabled
- PR #4552²⁸²⁹ - Add configuration option for making exceptions less noisy
- PR #4551²⁸³⁰ - Clean up thread creation parameters
- PR #4549²⁸³¹ - Test FetchContent build on GitHub actions
- PR #4548²⁸³² - Fix stack size
- PR #4545²⁸³³ - Fix header tests
- PR #4544²⁸³⁴ - Fix a typo in sanitizer build
- PR #4541²⁸³⁵ - Add API to check if a thread pool exists
- PR #4540²⁸³⁶ - Making sure MPI support is enabled if MPI futures are used but networking is disabled

²⁸¹⁴ <https://github.com/TheHPXProject/hpx/pull/4583>

²⁸¹⁵ <https://github.com/TheHPXProject/hpx/pull/4582>

²⁸¹⁶ <https://github.com/TheHPXProject/hpx/pull/4581>

²⁸¹⁷ <https://github.com/TheHPXProject/hpx/pull/4579>

²⁸¹⁸ <https://github.com/TheHPXProject/hpx/pull/4575>

²⁸¹⁹ <https://github.com/TheHPXProject/hpx/pull/4574>

²⁸²⁰ <https://github.com/TheHPXProject/hpx/pull/4570>

²⁸²¹ <https://github.com/TheHPXProject/hpx/pull/4564>

²⁸²² <https://github.com/TheHPXProject/hpx/pull/4563>

²⁸²³ <https://github.com/TheHPXProject/hpx/pull/4562>

²⁸²⁴ <https://github.com/TheHPXProject/hpx/pull/4561>

²⁸²⁵ <https://github.com/TheHPXProject/hpx/pull/4560>

²⁸²⁶ <https://github.com/TheHPXProject/hpx/pull/4559>

²⁸²⁷ <https://github.com/TheHPXProject/hpx/pull/4557>

²⁸²⁸ <https://github.com/TheHPXProject/hpx/pull/4553>

²⁸²⁹ <https://github.com/TheHPXProject/hpx/pull/4552>

²⁸³⁰ <https://github.com/TheHPXProject/hpx/pull/4551>

²⁸³¹ <https://github.com/TheHPXProject/hpx/pull/4549>

²⁸³² <https://github.com/TheHPXProject/hpx/pull/4548>

²⁸³³ <https://github.com/TheHPXProject/hpx/pull/4545>

²⁸³⁴ <https://github.com/TheHPXProject/hpx/pull/4544>

²⁸³⁵ <https://github.com/TheHPXProject/hpx/pull/4541>

²⁸³⁶ <https://github.com/TheHPXProject/hpx/pull/4540>

- PR #4538²⁸³⁷ - Move channel documentation examples to examples directory
- PR #4536²⁸³⁸ - Add generic allocator for execution policies
- PR #4534²⁸³⁹ - Enable compatibility headers for thread_executors module
- PR #4532²⁸⁴⁰ - Fixing broken url in README.rst
- PR #4531²⁸⁴¹ - Update scripts
- PR #4530²⁸⁴² - Make sure module API docs show up in correct order
- PR #4529²⁸⁴³ - Adding missing template code to module creation script
- PR #4528²⁸⁴⁴ - Make sure version module uses HPX's binary dir, not the parent's
- PR #4527²⁸⁴⁵ - Creating actions_base and actions module
- PR #4526²⁸⁴⁶ - Shared state for cv
- PR #4525²⁸⁴⁷ - Changing sub-name sequencing for experimental namespace
- PR #4524²⁸⁴⁸ - Add API guarantee notes to API reference documentation
- PR #4522²⁸⁴⁹ - Enable and fix deprecation warnings in execution module
- PR #4521²⁸⁵⁰ - Moves more miscellaneous files to modules
- PR #4520²⁸⁵¹ - Skip execution customization points when executor is known
- PR #4518²⁸⁵² - Module distributed lcos
- PR #4516²⁸⁵³ - Fix various builds
- PR #4515²⁸⁵⁴ - Replace backslashes by slashes in windows paths
- PR #4514²⁸⁵⁵ - Adding polymorphic_executor
- PR #4512²⁸⁵⁶ - Adding C++20 jthread and stop_token
- PR #4510²⁸⁵⁷ - Attempt to fix APEX linking in external packages again
- PR #4508²⁸⁵⁸ - Only test pull requests (not all branches) with GitHub actions
- PR #4505²⁸⁵⁹ - Fix duplicate linking in tests (ODR violations)

²⁸³⁷ <https://github.com/TheHPXProject/hpx/pull/4538>

²⁸³⁸ <https://github.com/TheHPXProject/hpx/pull/4536>

²⁸³⁹ <https://github.com/TheHPXProject/hpx/pull/4534>

²⁸⁴⁰ <https://github.com/TheHPXProject/hpx/pull/4532>

²⁸⁴¹ <https://github.com/TheHPXProject/hpx/pull/4531>

²⁸⁴² <https://github.com/TheHPXProject/hpx/pull/4530>

²⁸⁴³ <https://github.com/TheHPXProject/hpx/pull/4529>

²⁸⁴⁴ <https://github.com/TheHPXProject/hpx/pull/4528>

²⁸⁴⁵ <https://github.com/TheHPXProject/hpx/pull/4527>

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²⁸⁴⁷ <https://github.com/TheHPXProject/hpx/pull/4525>

²⁸⁴⁸ <https://github.com/TheHPXProject/hpx/pull/4524>

²⁸⁴⁹ <https://github.com/TheHPXProject/hpx/pull/4522>

²⁸⁵⁰ <https://github.com/TheHPXProject/hpx/pull/4521>

²⁸⁵¹ <https://github.com/TheHPXProject/hpx/pull/4520>

²⁸⁵² <https://github.com/TheHPXProject/hpx/pull/4518>

²⁸⁵³ <https://github.com/TheHPXProject/hpx/pull/4516>

²⁸⁵⁴ <https://github.com/TheHPXProject/hpx/pull/4515>

²⁸⁵⁵ <https://github.com/TheHPXProject/hpx/pull/4514>

²⁸⁵⁶ <https://github.com/TheHPXProject/hpx/pull/4512>

²⁸⁵⁷ <https://github.com/TheHPXProject/hpx/pull/4510>

²⁸⁵⁸ <https://github.com/TheHPXProject/hpx/pull/4508>

²⁸⁵⁹ <https://github.com/TheHPXProject/hpx/pull/4505>

- PR #4504²⁸⁶⁰ - Fix C++ standard handling
- PR #4503²⁸⁶¹ - Add CMakeLists file check
- PR #4500²⁸⁶² - Fix .clang-format version requirement comment
- PR #4499²⁸⁶³ - Attempting to fix hpx_init linking on macOS
- PR #4498²⁸⁶⁴ - Fix compatibility of *pool_executor*
- PR #4496²⁸⁶⁵ - Removing superfluous SPDX tags
- PR #4494²⁸⁶⁶ - Module executors
- PR #4493²⁸⁶⁷ - Pack traversal module
- PR #4492²⁸⁶⁸ - Update copyright year in documentation
- PR #4491²⁸⁶⁹ - Add missing current_executor header
- PR #4490²⁸⁷⁰ - Update GitHub actions configs
- PR #4487²⁸⁷¹ - Properly dispatch exceptions thrown from hpx_main to be rethrown from hpx::init/hpx::stop
- PR #4486²⁸⁷² - Fixing an initialization order problem
- PR #4485²⁸⁷³ - Move miscellaneous files to their rightful modules
- PR #4483²⁸⁷⁴ - Clean up imported CMake target naming
- PR #4481²⁸⁷⁵ - Add vcpkg installation instructions
- PR #4479²⁸⁷⁶ - Add hints to allow to specify MIMALLOC_ROOT
- PR #4478²⁸⁷⁷ - Async modules
- PR #4475²⁸⁷⁸ - Fix rp init changes
- PR #4474²⁸⁷⁹ - Use #pragma once in headers
- PR #4472²⁸⁸⁰ - Add more descriptive error message when using x86 coroutines on non-x86 platforms
- PR #4467²⁸⁸¹ - Add mimalloc find cmake script
- PR #4465²⁸⁸² - Add thread_executors module

²⁸⁶⁰ <https://github.com/TheHPXProject/hpx/pull/4504>

²⁸⁶¹ <https://github.com/TheHPXProject/hpx/pull/4503>

²⁸⁶² <https://github.com/TheHPXProject/hpx/pull/4500>

²⁸⁶³ <https://github.com/TheHPXProject/hpx/pull/4499>

²⁸⁶⁴ <https://github.com/TheHPXProject/hpx/pull/4498>

²⁸⁶⁵ <https://github.com/TheHPXProject/hpx/pull/4496>

²⁸⁶⁶ <https://github.com/TheHPXProject/hpx/pull/4494>

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²⁸⁶⁸ <https://github.com/TheHPXProject/hpx/pull/4492>

²⁸⁶⁹ <https://github.com/TheHPXProject/hpx/pull/4491>

²⁸⁷⁰ <https://github.com/TheHPXProject/hpx/pull/4490>

²⁸⁷¹ <https://github.com/TheHPXProject/hpx/pull/4487>

²⁸⁷² <https://github.com/TheHPXProject/hpx/pull/4486>

²⁸⁷³ <https://github.com/TheHPXProject/hpx/pull/4485>

²⁸⁷⁴ <https://github.com/TheHPXProject/hpx/pull/4483>

²⁸⁷⁵ <https://github.com/TheHPXProject/hpx/pull/4481>

²⁸⁷⁶ <https://github.com/TheHPXProject/hpx/pull/4479>

²⁸⁷⁷ <https://github.com/TheHPXProject/hpx/pull/4478>

²⁸⁷⁸ <https://github.com/TheHPXProject/hpx/pull/4475>

²⁸⁷⁹ <https://github.com/TheHPXProject/hpx/pull/4474>

²⁸⁸⁰ <https://github.com/TheHPXProject/hpx/pull/4472>

²⁸⁸¹ <https://github.com/TheHPXProject/hpx/pull/4467>

²⁸⁸² <https://github.com/TheHPXProject/hpx/pull/4465>

- [PR #4464](#)²⁸⁸³ - Include module
- [PR #4462](#)²⁸⁸⁴ - Merge hpx_init and hpx_wrap into one static library
- [PR #4461](#)²⁸⁸⁵ - Making thread_data test more realistic
- [PR #4460](#)²⁸⁸⁶ - Suppress MPI warnings in version.cpp
- [PR #4459](#)²⁸⁸⁷ - Make sure pkgconfig applications link with hpx_init
- [PR #4458](#)²⁸⁸⁸ - Added example demonstrating how to create and use a wrapping executor
- [PR #4457](#)²⁸⁸⁹ - Fixing execution of thread exit functions
- [PR #4456](#)²⁸⁹⁰ - Move backtrace files to debugging module
- [PR #4455](#)²⁸⁹¹ - Move deadlock_detection and maintain_queue_wait_times source files into schedulers module
- [PR #4450](#)²⁸⁹² - Fixing compilation with std::filesystem enabled
- [PR #4449](#)²⁸⁹³ - Fixing build system to actually build variant test
- [PR #4447](#)²⁸⁹⁴ - This fixes an obsolete #include
- [PR #4446](#)²⁸⁹⁵ - Resume tasks where they were suspended
- [PR #4444](#)²⁸⁹⁶ - Minor CUDA fixes
- [PR #4443](#)²⁸⁹⁷ - Add missing tests to CircleCI config
- [PR #4442](#)²⁸⁹⁸ - Adding a tag to all auto-generated files allowing for tools to visually distinguish those
- [PR #4441](#)²⁸⁹⁹ - Adding performance counter type information
- [PR #4440](#)²⁹⁰⁰ - Fixing MSVC build
- [PR #4439](#)²⁹⁰¹ - Link HPX::plugin and component privately in hpx_setup_target
- [PR #4437](#)²⁹⁰² - Adding a test that verifies the problem can be solved using a trait specialization
- [PR #4434](#)²⁹⁰³ - Clean up Boost dependencies and copy string algorithms to new module
- [PR #4433](#)²⁹⁰⁴ - Fixing compilation issues (!) if MPI parcelport is enabled
- [PR #4431](#)²⁹⁰⁵ - Ignore warnings about name mangling changing

²⁸⁸³ <https://github.com/TheHPXProject/hpx/pull/4464>

²⁸⁸⁴ <https://github.com/TheHPXProject/hpx/pull/4462>

²⁸⁸⁵ <https://github.com/TheHPXProject/hpx/pull/4461>

²⁸⁸⁶ <https://github.com/TheHPXProject/hpx/pull/4460>

²⁸⁸⁷ <https://github.com/TheHPXProject/hpx/pull/4459>

²⁸⁸⁸ <https://github.com/TheHPXProject/hpx/pull/4458>

²⁸⁸⁹ <https://github.com/TheHPXProject/hpx/pull/4457>

²⁸⁹⁰ <https://github.com/TheHPXProject/hpx/pull/4456>

²⁸⁹¹ <https://github.com/TheHPXProject/hpx/pull/4455>

²⁸⁹² <https://github.com/TheHPXProject/hpx/pull/4450>

²⁸⁹³ <https://github.com/TheHPXProject/hpx/pull/4449>

²⁸⁹⁴ <https://github.com/TheHPXProject/hpx/pull/4447>

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²⁸⁹⁶ <https://github.com/TheHPXProject/hpx/pull/4444>

²⁸⁹⁷ <https://github.com/TheHPXProject/hpx/pull/4443>

²⁸⁹⁸ <https://github.com/TheHPXProject/hpx/pull/4442>

²⁸⁹⁹ <https://github.com/TheHPXProject/hpx/pull/4441>

²⁹⁰⁰ <https://github.com/TheHPXProject/hpx/pull/4440>

²⁹⁰¹ <https://github.com/TheHPXProject/hpx/pull/4439>

²⁹⁰² <https://github.com/TheHPXProject/hpx/pull/4437>

²⁹⁰³ <https://github.com/TheHPXProject/hpx/pull/4434>

²⁹⁰⁴ <https://github.com/TheHPXProject/hpx/pull/4433>

²⁹⁰⁵ <https://github.com/TheHPXProject/hpx/pull/4431>

- PR #4430²⁹⁰⁶ - Add performance_counters module
- PR #4428²⁹⁰⁷ - Don't add compatibility headers to module API reference
- PR #4426²⁹⁰⁸ - Add currently failing tests on GitHub actions to blacklist
- PR #4425²⁹⁰⁹ - Clean up and correct minimum required versions
- PR #4424²⁹¹⁰ - Making sure hpx.lock_detection=0 works as advertized
- PR #4421²⁹¹¹ - Making sure interval time stops underlying timer thread on termination
- PR #4417²⁹¹² - Adding serialization support for std::variant (if available) and std::tuple
- PR #4415²⁹¹³ - Partially reverting changes applied by PR 4373
- PR #4414²⁹¹⁴ - Added documentation for the compiler-wrapper script hpxcxx.in in creating_hpx_projects.rst
- PR #4413²⁹¹⁵ - Merging from V1.4.1 release
- PR #4412²⁹¹⁶ - Making sure to issue a warning if a file specified using -hpx:options-file is not found
- PR #4411²⁹¹⁷ - Make test specific to HPX_WITH_SHARED_PRIORITY_SCHEDULER
- PR #4407²⁹¹⁸ - Adding minimal MPI executor
- PR #4405²⁹¹⁹ - Fix cross pool injection test, use default scheduler as fallback
- PR #4404²⁹²⁰ - Fix a race condition and clean-up usage of scheduler mode
- PR #4399²⁹²¹ - Add more threading modules
- PR #4398²⁹²² - Add CODEOWNERS file
- PR #4395²⁹²³ - Adding a parameter to auto_chunk_size allowing to control the amount of iterations to measure
- PR #4393²⁹²⁴ - Use appropriate cache-line size defaults for different platforms
- PR #4391²⁹²⁵ - Fixing use of allocator for C++20
- PR #4390²⁹²⁶ - Making -hpx:help behavior consistent
- PR #4388²⁹²⁷ - Change the resource partitioner initialization
- PR #4387²⁹²⁸ - Fix roll_release.sh

²⁹⁰⁶ <https://github.com/TheHPXProject/hpx/pull/4430>

²⁹⁰⁷ <https://github.com/TheHPXProject/hpx/pull/4428>

²⁹⁰⁸ <https://github.com/TheHPXProject/hpx/pull/4426>

²⁹⁰⁹ <https://github.com/TheHPXProject/hpx/pull/4425>

²⁹¹⁰ <https://github.com/TheHPXProject/hpx/pull/4424>

²⁹¹¹ <https://github.com/TheHPXProject/hpx/pull/4421>

²⁹¹² <https://github.com/TheHPXProject/hpx/pull/4417>

²⁹¹³ <https://github.com/TheHPXProject/hpx/pull/4415>

²⁹¹⁴ <https://github.com/TheHPXProject/hpx/pull/4414>

²⁹¹⁵ <https://github.com/TheHPXProject/hpx/pull/4413>

²⁹¹⁶ <https://github.com/TheHPXProject/hpx/pull/4412>

²⁹¹⁷ <https://github.com/TheHPXProject/hpx/pull/4411>

²⁹¹⁸ <https://github.com/TheHPXProject/hpx/pull/4407>

²⁹¹⁹ <https://github.com/TheHPXProject/hpx/pull/4405>

²⁹²⁰ <https://github.com/TheHPXProject/hpx/pull/4404>

²⁹²¹ <https://github.com/TheHPXProject/hpx/pull/4399>

²⁹²² <https://github.com/TheHPXProject/hpx/pull/4398>

²⁹²³ <https://github.com/TheHPXProject/hpx/pull/4395>

²⁹²⁴ <https://github.com/TheHPXProject/hpx/pull/4393>

²⁹²⁵ <https://github.com/TheHPXProject/hpx/pull/4391>

²⁹²⁶ <https://github.com/TheHPXProject/hpx/pull/4390>

²⁹²⁷ <https://github.com/TheHPXProject/hpx/pull/4388>

²⁹²⁸ <https://github.com/TheHPXProject/hpx/pull/4387>

- [PR #4386](#)²⁹²⁹ - Add warning messages for using thread binding options on macOS
- [PR #4385](#)²⁹³⁰ - Cuda futures
- [PR #4384](#)²⁹³¹ - Make enabling dynamic `hpx_main` on non-Linux systems a configuration error
- [PR #4383](#)²⁹³² - Use `configure_file` for `HPXCacheVariables.cmake`
- [PR #4382](#)²⁹³³ - Update spellchecking whitelist and fix more typos
- [PR #4380](#)²⁹³⁴ - Add a helper function to get a future from a cuda stream
- [PR #4379](#)²⁹³⁵ - Add Windows and macOS CI with GitHub actions
- [PR #4378](#)²⁹³⁶ - Change C++ standard handling
- [PR #4377](#)²⁹³⁷ - Remove Python scripts
- [PR #4374](#)²⁹³⁸ - Adding overload for `hpx::init/hpx::start` for use with resource partitioner
- [PR #4373](#)²⁹³⁹ - Adding test that verifies for 4369 to be fixed
- [PR #4372](#)²⁹⁴⁰ - Another attempt at fixing the integral mismatch and conversion warnings
- [PR #4370](#)²⁹⁴¹ - Doc updates quick start
- [PR #4368](#)²⁹⁴² - Add a whitelist of words for weird spelling suggestions
- [PR #4366](#)²⁹⁴³ - Suppress or fix clang-tidy-9 warnings
- [PR #4365](#)²⁹⁴⁴ - Removing more Boost dependencies
- [PR #4363](#)²⁹⁴⁵ - Update clang-format config file for version 9
- [PR #4362](#)²⁹⁴⁶ - Fix indices typo
- [PR #4361](#)²⁹⁴⁷ - Boost cleanup
- [PR #4360](#)²⁹⁴⁸ - Move plugins
- [PR #4358](#)²⁹⁴⁹ - Doc updates; generating documentation. Will likely need heavy editing.
- [PR #4356](#)²⁹⁵⁰ - Remove some minor unused and unnecessary Boost includes
- [PR #4355](#)²⁹⁵¹ - Fix spellcheck step in CircleCI config

²⁹²⁹ <https://github.com/TheHPXProject/hpx/pull/4386>

²⁹³⁰ <https://github.com/TheHPXProject/hpx/pull/4385>

²⁹³¹ <https://github.com/TheHPXProject/hpx/pull/4384>

²⁹³² <https://github.com/TheHPXProject/hpx/pull/4383>

²⁹³³ <https://github.com/TheHPXProject/hpx/pull/4382>

²⁹³⁴ <https://github.com/TheHPXProject/hpx/pull/4380>

²⁹³⁵ <https://github.com/TheHPXProject/hpx/pull/4379>

²⁹³⁶ <https://github.com/TheHPXProject/hpx/pull/4378>

²⁹³⁷ <https://github.com/TheHPXProject/hpx/pull/4377>

²⁹³⁸ <https://github.com/TheHPXProject/hpx/pull/4374>

²⁹³⁹ <https://github.com/TheHPXProject/hpx/pull/4373>

²⁹⁴⁰ <https://github.com/TheHPXProject/hpx/pull/4372>

²⁹⁴¹ <https://github.com/TheHPXProject/hpx/pull/4370>

²⁹⁴² <https://github.com/TheHPXProject/hpx/pull/4368>

²⁹⁴³ <https://github.com/TheHPXProject/hpx/pull/4366>

²⁹⁴⁴ <https://github.com/TheHPXProject/hpx/pull/4365>

²⁹⁴⁵ <https://github.com/TheHPXProject/hpx/pull/4363>

²⁹⁴⁶ <https://github.com/TheHPXProject/hpx/pull/4362>

²⁹⁴⁷ <https://github.com/TheHPXProject/hpx/pull/4361>

²⁹⁴⁸ <https://github.com/TheHPXProject/hpx/pull/4360>

²⁹⁴⁹ <https://github.com/TheHPXProject/hpx/pull/4358>

²⁹⁵⁰ <https://github.com/TheHPXProject/hpx/pull/4356>

²⁹⁵¹ <https://github.com/TheHPXProject/hpx/pull/4355>

- PR #4354²⁹⁵² - Lightweight utility to hold a pack as members
- PR #4352²⁹⁵³ - Minor fixes to the C++ standard detection for MSVC
- PR #4351²⁹⁵⁴ - Move generated documentation to hpx-docs repo
- PR #4347²⁹⁵⁵ - Add cmake policy - CMP0074
- PR #4346²⁹⁵⁶ - Remove file committed by mistake
- PR #4342²⁹⁵⁷ - Remove HCC and SYCL options from CMakeLists.txt
- PR #4341²⁹⁵⁸ - Fix launch process test with APEX enabled
- PR #4340²⁹⁵⁹ - Testing Cirrus CI
- PR #4339²⁹⁶⁰ - Post 1.4.0 updates
- PR #4338²⁹⁶¹ - Spelling corrections and CircleCI spell check
- PR #4333²⁹⁶² - Flatten bound callables
- PR #4332²⁹⁶³ - This is a collection of mostly minor (cleanup) fixes
- PR #4331²⁹⁶⁴ - This adds the missing tests for `async_colocated` and `async_continue_colocated`
- PR #4330²⁹⁶⁵ - Remove `HPX.Compute` host default_executor
- PR #4328²⁹⁶⁶ - Generate global header for `basic_execution` module
- PR #4327²⁹⁶⁷ - Use `INTERNAL_FLAGS` option for all examples and components
- PR #4326²⁹⁶⁸ - Usage of temporary allocator in assignment operator of `compute::vector`
- PR #4325²⁹⁶⁹ - Use `hpx::threads::get_cache_line_size` in `prefetching.hpp`
- PR #4324²⁹⁷⁰ - Enable compatibility headers option for execution module
- PR #4316²⁹⁷¹ - Add clang format indentppdirectives
- PR #4313²⁹⁷² - Introduce `index_pack` alias to pack of `size_t`
- PR #4312²⁹⁷³ - Fixing compatibility header for `pack.hpp`
- PR #4311²⁹⁷⁴ - Dataflow annotations for APEX

²⁹⁵² <https://github.com/TheHPXProject/hpx/pull/4354>

²⁹⁵³ <https://github.com/TheHPXProject/hpx/pull/4352>

²⁹⁵⁴ <https://github.com/TheHPXProject/hpx/pull/4351>

²⁹⁵⁵ <https://github.com/TheHPXProject/hpx/pull/4347>

²⁹⁵⁶ <https://github.com/TheHPXProject/hpx/pull/4346>

²⁹⁵⁷ <https://github.com/TheHPXProject/hpx/pull/4342>

²⁹⁵⁸ <https://github.com/TheHPXProject/hpx/pull/4341>

²⁹⁵⁹ <https://github.com/TheHPXProject/hpx/pull/4340>

²⁹⁶⁰ <https://github.com/TheHPXProject/hpx/pull/4339>

²⁹⁶¹ <https://github.com/TheHPXProject/hpx/pull/4338>

²⁹⁶² <https://github.com/TheHPXProject/hpx/pull/4333>

²⁹⁶³ <https://github.com/TheHPXProject/hpx/pull/4332>

²⁹⁶⁴ <https://github.com/TheHPXProject/hpx/pull/4331>

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²⁹⁶⁶ <https://github.com/TheHPXProject/hpx/pull/4328>

²⁹⁶⁷ <https://github.com/TheHPXProject/hpx/pull/4327>

²⁹⁶⁸ <https://github.com/TheHPXProject/hpx/pull/4326>

²⁹⁶⁹ <https://github.com/TheHPXProject/hpx/pull/4325>

²⁹⁷⁰ <https://github.com/TheHPXProject/hpx/pull/4324>

²⁹⁷¹ <https://github.com/TheHPXProject/hpx/pull/4316>

²⁹⁷² <https://github.com/TheHPXProject/hpx/pull/4313>

²⁹⁷³ <https://github.com/TheHPXProject/hpx/pull/4312>

²⁹⁷⁴ <https://github.com/TheHPXProject/hpx/pull/4311>

- PR #4309²⁹⁷⁵ - Update launching_and_configuring_hpx_applications.rst
- PR #4306²⁹⁷⁶ - Fix schedule hint not being taken from executor
- PR #4305²⁹⁷⁷ - Implementing *hpx::functional::tag_invoke*
- PR #4304²⁹⁷⁸ - Improve pack support utilities
- PR #4303²⁹⁷⁹ - Remove errors module dependency on datastructures
- PR #4301²⁹⁸⁰ - Clean up thread executors
- PR #4294²⁹⁸¹ - Logging revamp
- PR #4292²⁹⁸² - Remove SPDX tag from Boost License file to allow for github to recognize it
- PR #4291²⁹⁸³ - Add format support for std::tm
- PR #4290²⁹⁸⁴ - Simplify compatible tuples check
- PR #4288²⁹⁸⁵ - A lightweight take on boost::lexical_cast
- PR #4287²⁹⁸⁶ - Forking boost::lexical_cast as a new module
- PR #4277²⁹⁸⁷ - MPI_futures
- PR #4270²⁹⁸⁸ - Refactor future implementation
- PR #4265²⁹⁸⁹ - Threading module
- PR #4259²⁹⁹⁰ - Module naming base
- PR #4251²⁹⁹¹ - Local workrequesting scheduler
- PR #4250²⁹⁹² - Inline execution of scoped tasks, if possible
- PR #4247²⁹⁹³ - Add execution in module headers
- PR #4246²⁹⁹⁴ - Expose CMake targets officially
- PR #4239²⁹⁹⁵ - Doc updates miscellaneous (partially completed during Google Season of Docs)
- PR #4233²⁹⁹⁶ - Remove project() from modules + fix CMAKE_SOURCE_DIR issue
- PR #4231²⁹⁹⁷ - Module local lcos

²⁹⁷⁵ <https://github.com/TheHPXProject/hpx/pull/4309>

²⁹⁷⁶ <https://github.com/TheHPXProject/hpx/pull/4306>

²⁹⁷⁷ <https://github.com/TheHPXProject/hpx/pull/4305>

²⁹⁷⁸ <https://github.com/TheHPXProject/hpx/pull/4304>

²⁹⁷⁹ <https://github.com/TheHPXProject/hpx/pull/4303>

²⁹⁸⁰ <https://github.com/TheHPXProject/hpx/pull/4301>

²⁹⁸¹ <https://github.com/TheHPXProject/hpx/pull/4294>

²⁹⁸² <https://github.com/TheHPXProject/hpx/pull/4292>

²⁹⁸³ <https://github.com/TheHPXProject/hpx/pull/4291>

²⁹⁸⁴ <https://github.com/TheHPXProject/hpx/pull/4290>

²⁹⁸⁵ <https://github.com/TheHPXProject/hpx/pull/4288>

²⁹⁸⁶ <https://github.com/TheHPXProject/hpx/pull/4287>

²⁹⁸⁷ <https://github.com/TheHPXProject/hpx/pull/4277>

²⁹⁸⁸ <https://github.com/TheHPXProject/hpx/pull/4270>

²⁹⁸⁹ <https://github.com/TheHPXProject/hpx/pull/4265>

²⁹⁹⁰ <https://github.com/TheHPXProject/hpx/pull/4259>

²⁹⁹¹ <https://github.com/TheHPXProject/hpx/pull/4251>

²⁹⁹² <https://github.com/TheHPXProject/hpx/pull/4250>

²⁹⁹³ <https://github.com/TheHPXProject/hpx/pull/4247>

²⁹⁹⁴ <https://github.com/TheHPXProject/hpx/pull/4246>

²⁹⁹⁵ <https://github.com/TheHPXProject/hpx/pull/4239>

²⁹⁹⁶ <https://github.com/TheHPXProject/hpx/pull/4233>

²⁹⁹⁷ <https://github.com/TheHPXProject/hpx/pull/4231>

- PR #4207²⁹⁹⁸ - Command line handling module
- PR #4206²⁹⁹⁹ - Runtime configuration module
- PR #4141³⁰⁰⁰ - Doc updates examples local to remote (partially completed during Google Season of Docs)
- PR #4091³⁰⁰¹ - Split runtime into local and distributed parts
- PR #4017³⁰⁰² - Require C++14

HPX V1.4.1 (Feb 12, 2020)

General changes

This is a bugfix release. It contains the following changes:

- Fix compilation issues on Windows, macOS, FreeBSD, and with gcc 10
- Install missing pdb files on Windows
- Allow running tests using an installed version of *HPX*
- Skip MPI finalization if HPX has not initialized MPI
- Give a hard error when attempting to use IO counters on Windows

Closed issues

- Issue #4320³⁰⁰³ - HPX 1.4.0 does not compile with gcc 10
- Issue #4336³⁰⁰⁴ - Building HPX 1.4.0 with IO Counters breaks (Windows)
- Issue #4334³⁰⁰⁵ - HPX Debug and RelWithDebinfo builds on Windows not installing .pdb files
- Issue #4322³⁰⁰⁶ - Undefine VT1 and VT2 after boost includes
- Issue #4314³⁰⁰⁷ - Compile error on 1.4.0
- Issue #4307³⁰⁰⁸ - ld: error: duplicate symbol: freebsd_envron

²⁹⁹⁸ <https://github.com/TheHPXProject/hpx/pull/4207>

²⁹⁹⁹ <https://github.com/TheHPXProject/hpx/pull/4206>

³⁰⁰⁰ <https://github.com/TheHPXProject/hpx/pull/4141>

³⁰⁰¹ <https://github.com/TheHPXProject/hpx/pull/4091>

³⁰⁰² <https://github.com/TheHPXProject/hpx/pull/4017>

³⁰⁰³ <https://github.com/TheHPXProject/hpx/issues/4320>

³⁰⁰⁴ <https://github.com/TheHPXProject/hpx/issues/4336>

³⁰⁰⁵ <https://github.com/TheHPXProject/hpx/issues/4334>

³⁰⁰⁶ <https://github.com/TheHPXProject/hpx/issues/4322>

³⁰⁰⁷ <https://github.com/TheHPXProject/hpx/issues/4314>

³⁰⁰⁸ <https://github.com/TheHPXProject/hpx/issues/4307>

Closed pull requests

- [PR #4376³⁰⁰⁹](#) - Attempt to fix some test build errors on Windows
- [PR #4357³⁰¹⁰](#) - Adding missing `#includes` to fix gcc V10 linker problems
- [PR #4353³⁰¹¹](#) - Skip `MPI_Finalize` if `MPI_Init` is not called from HPX
- [PR #4343³⁰¹²](#) - Give a hard error if IO counters are enabled on non-Linux systems
- [PR #4337³⁰¹³](#) - Installing `pdb` files on Windows
- [PR #4335³⁰¹⁴](#) - Adding capability to `buildsystem` to use an installed version of HPX
- [PR #4315³⁰¹⁵](#) - Forcing exported symbols from `composable_guard` to be linked into core library
- [PR #4310³⁰¹⁶](#) - Remove environment handling from `exception.cpp`

HPX V1.4.0 (January 15, 2020)

General changes

- We have added the collectives `all_to_all` and `all_reduce`.
- We have added APIs for resiliency, which allows replication and replay for failed tasks. See the *documentation* for more details.
- Components can now be checkpointed.
- Performance improvements to schedulers and coroutines. A significant change is the addition of stackless coroutines. These are to be used for tasks that do not need to be suspended and can reduce overheads noticeably in applications with short tasks. A stackless coroutine can be created with the new `stack_size_thread_stacksize_nostack`.
- We have added an implementation of `unique_any`, which is a non-copyable version of `any`.
- The `shared_priority_queue_scheduler` has been improved. It now has lower overheads than the default scheduler in many situations. Unlike the default scheduler it fully supports NUMA scheduling hints. Enable it with the command line option `--hpx:queuing=shared-priority`. This scheduler should still be considered experimental, but its use is encouraged in real applications to help us make it production ready.
- We have added the performance counters `background-receive-duration` and `background-receive-overhead` for inspecting the time and overhead spent on receiving parcels in the background.
- Compilation time has been further improved when `HPX_WITH_NETWORKING=OFF`.
- We no longer require compiled Boost dependencies in certain configurations. This requires at least Boost 1.70, compiling on x86 with GCC 9, clang (libc++) 9, or VS2019 in C++17 mode. The dependency on `Boost.Filesystem` can explicitly be turned on with `HPX_FILESYSTEM_WITH_BOOST_FILESYSTEM_COMPATIBILITY=ON` (it is off by default if the standard library supports `std::filesystem`). `Boost.ProgramOptions` has been copied into the HPX repository. We have a compatibility layer for users who must explicitly use

³⁰⁰⁹ <https://github.com/TheHPXProject/hpx/pull/4376>

³⁰¹⁰ <https://github.com/TheHPXProject/hpx/pull/4357>

³⁰¹¹ <https://github.com/TheHPXProject/hpx/pull/4353>

³⁰¹² <https://github.com/TheHPXProject/hpx/pull/4343>

³⁰¹³ <https://github.com/TheHPXProject/hpx/pull/4337>

³⁰¹⁴ <https://github.com/TheHPXProject/hpx/pull/4335>

³⁰¹⁵ <https://github.com/TheHPXProject/hpx/pull/4315>

³⁰¹⁶ <https://github.com/TheHPXProject/hpx/pull/4310>

Boost.ProgramOptions instead of the ProgramOptions provided by HPX. To remove the dependency HPX_PROGRAM_OPTIONS_WITH_BOOST_PROGRAM_OPTIONS_COMPATIBILITY must be explicitly set to OFF. This option will be removed in a future release. We have also removed several other header-only dependencies on Boost.

- It is now possible to use the process affinity mask set by tools like numactl and various batch environments with the command line option `--hpx:use-process-mask`. Enabling this option implies `--hpx:ignore-batch-env`.
- It is now possible to create standalone thread pools without starting the runtime. See the `standalone_thread_pool_executor.cpp` test in the `execution` module for an example.
- Tasks annotated with `hpx::util::annotated_function` now have their correct name when using APEX to generate OTF2 files.
- Cloning of APEX was defective in previous releases (it required manual intervention to check out the correct tag or branch). This has been fixed.
- The option `HPX_WITH_MORE_THAN_64_THREADS` is now ignored and will be removed in a future release. The value is instead derived directly from `HPX_WITH_MAX_CPU_COUNT` option.
- We have deprecated compiling in C++11 mode. The next release will require a C++14 capable compiler.
- We have deprecated support for the Vc library. This option will be replaced with SIMD support from the standard library in a future release.
- We have significantly refactored our CMake setup. This is intended to be a non-breaking change and will allow for using HPX through CMake targets in the future.
- We have continued modularizing the HPX library. In the process we have rearranged many header files into module-specific directories. All moved headers have compatibility headers which forward from the old location to the new location, together with a deprecation warning. The compatibility headers will eventually be removed.
- We now enforce formatting with `clang-format` on the majority of our source files.
- We have added SPDX license tags to all files.
- Many bugfixes.

Breaking changes

- The `HPX_WITH_THREAD_COMPATIBILITY` option and the associated compatibility layer has been removed.
- The `HPX_WITH_INCLUSIVE_SCAN_COMPATIBILITY` option and the associated compatibility layer has been removed.
- The `HPX_WITH_UNWRAPPED_COMPATIBILITY` option and the associated compatibility layer has been removed.

Closed issues

- [Issue #4282³⁰¹⁷](https://github.com/TheHPXProject/hpx/issues/4282) - Build Issues with Release on Windows
- [Issue #4278³⁰¹⁸](https://github.com/TheHPXProject/hpx/issues/4278) - Build Issues with CMake 3.14.4
- [Issue #4273³⁰¹⁹](https://github.com/TheHPXProject/hpx/issues/4273) - Clients of HPX 1.4.0-rc2 with APEX are not linked to libhpx-apex
- [Issue #4269³⁰²⁰](https://github.com/TheHPXProject/hpx/issues/4269) - Building HPX 1.4.0-rc2 with support for APEX fails

³⁰¹⁷ <https://github.com/TheHPXProject/hpx/issues/4282>

³⁰¹⁸ <https://github.com/TheHPXProject/hpx/issues/4278>

³⁰¹⁹ <https://github.com/TheHPXProject/hpx/issues/4273>

³⁰²⁰ <https://github.com/TheHPXProject/hpx/issues/4269>

- Issue #4263³⁰²¹ - Compilation fail on latest master
- Issue #4232³⁰²² - Configure of HPX project using CMake FetchContent fails
- Issue #4223³⁰²³ - “Re-using the main() function as the main HPX entry point” doesn’t work
- Issue #4220³⁰²⁴ - HPX won’t compile - error building resource_partitioner
- Issue #4215³⁰²⁵ - HPX 1.4.0rc1 does not link on s390x
- Issue #4204³⁰²⁶ - Trouble compiling HPX with Intel compiler
- Issue #4199³⁰²⁷ - Refactor APEX to eliminate circular dependency
- Issue #4187³⁰²⁸ - HPX can’t build on OSX
- Issue #4185³⁰²⁹ - Simple debug output for development
- Issue #4182³⁰³⁰ - @HPX_CONF_PREFIX@ is the empty string
- Issue #4169³⁰³¹ - HPX won’t build with APEX
- Issue #4163³⁰³² - Add back HPX_LIBRARIES and HPX_INCLUDE_DIRS
- Issue #4161³⁰³³ - It should be possible to call find_package(HPX) multiple times
- Issue #4155³⁰³⁴ - get_self_id() for stackless threads returns invalid_thread_id
- Issue #4151³⁰³⁵ - build error with MPI code
- Issue #4150³⁰³⁶ - hpx won’t build on POWER9 with clang 8
- Issue #4148³⁰³⁷ - cacheline_data delivers poor performance with C++17 compared to C++14
- Issue #4144³⁰³⁸ - target general in HPX_LIBRARIES does not exist
- Issue #4134³⁰³⁹ - CMake Error when -DHPX_WITH_HPXMP=ON
- Issue #4132³⁰⁴⁰ - parallel fill leaves elements unfilled
- Issue #4123³⁰⁴¹ - PAPI performance counters are inaccessible
- Issue #4118³⁰⁴² - static_chunk_size is not obeyed in scan algorithms
- Issue #4115³⁰⁴³ - dependency chaining error with APEX

³⁰²¹ <https://github.com/TheHPXProject/hpx/issues/4263>

³⁰²² <https://github.com/TheHPXProject/hpx/issues/4232>

³⁰²³ <https://github.com/TheHPXProject/hpx/issues/4223>

³⁰²⁴ <https://github.com/TheHPXProject/hpx/issues/4220>

³⁰²⁵ <https://github.com/TheHPXProject/hpx/issues/4215>

³⁰²⁶ <https://github.com/TheHPXProject/hpx/issues/4204>

³⁰²⁷ <https://github.com/TheHPXProject/hpx/issues/4199>

³⁰²⁸ <https://github.com/TheHPXProject/hpx/issues/4187>

³⁰²⁹ <https://github.com/TheHPXProject/hpx/issues/4185>

³⁰³⁰ <https://github.com/TheHPXProject/hpx/issues/4182>

³⁰³¹ <https://github.com/TheHPXProject/hpx/issues/4169>

³⁰³² <https://github.com/TheHPXProject/hpx/issues/4163>

³⁰³³ <https://github.com/TheHPXProject/hpx/issues/4161>

³⁰³⁴ <https://github.com/TheHPXProject/hpx/issues/4155>

³⁰³⁵ <https://github.com/TheHPXProject/hpx/issues/4151>

³⁰³⁶ <https://github.com/TheHPXProject/hpx/issues/4150>

³⁰³⁷ <https://github.com/TheHPXProject/hpx/issues/4148>

³⁰³⁸ <https://github.com/TheHPXProject/hpx/issues/4144>

³⁰³⁹ <https://github.com/TheHPXProject/hpx/issues/4134>

³⁰⁴⁰ <https://github.com/TheHPXProject/hpx/issues/4132>

³⁰⁴¹ <https://github.com/TheHPXProject/hpx/issues/4123>

³⁰⁴² <https://github.com/TheHPXProject/hpx/issues/4118>

³⁰⁴³ <https://github.com/TheHPXProject/hpx/issues/4115>

- [Issue #4107](#)³⁰⁴⁴ - Initializing runtime without entry point function and command line arguments
- [Issue #4105](#)³⁰⁴⁵ - Bug in `hpx:bind=numa-balanced`
- [Issue #4101](#)³⁰⁴⁶ - Bound tasks
- [Issue #4100](#)³⁰⁴⁷ - Add SPDX identifier to all files
- [Issue #4085](#)³⁰⁴⁸ - `hpx_topology` library should depend on `hwloc`
- [Issue #4067](#)³⁰⁴⁹ - HPX fails to build on macOS
- [Issue #4056](#)³⁰⁵⁰ - Building without thread manager idle backoff fails
- [Issue #4052](#)³⁰⁵¹ - Enforce `clang-format` style for modules
- [Issue #4032](#)³⁰⁵² - Simple hello world fails to launch correctly
- [Issue #4030](#)³⁰⁵³ - Allow threads to skip context switching
- [Issue #4029](#)³⁰⁵⁴ - Add support for `mimalloc`
- [Issue #4005](#)³⁰⁵⁵ - Can't link HPX when APEX enabled
- [Issue #4002](#)³⁰⁵⁶ - Missing header for algorithm module
- [Issue #3989](#)³⁰⁵⁷ - conversion from `long` to `unsigned int` requires a narrowing conversion on MSVC
- [Issue #3958](#)³⁰⁵⁸ - `/statistics/average@` perf counter can't be created
- [Issue #3953](#)³⁰⁵⁹ - CMake errors from `HPX_AddPseudoDependencies`
- [Issue #3941](#)³⁰⁶⁰ - CMake error for APEX install target
- [Issue #3940](#)³⁰⁶¹ - Convert pseudo-doxygen function documentation into actual doxygen documentation
- [Issue #3935](#)³⁰⁶² - HPX compiler match too strict?
- [Issue #3929](#)³⁰⁶³ - Buildbot failures on latest HPX stable
- [Issue #3912](#)³⁰⁶⁴ - I recommend publishing a version that does not depend on the boost library
- [Issue #3890](#)³⁰⁶⁵ - `hpx.ini` not working
- [Issue #3883](#)³⁰⁶⁶ - cuda compilation fails because of `-faligned-new`

³⁰⁴⁴ <https://github.com/TheHPXProject/hpx/issues/4107>

³⁰⁴⁵ <https://github.com/TheHPXProject/hpx/issues/4105>

³⁰⁴⁶ <https://github.com/TheHPXProject/hpx/issues/4101>

³⁰⁴⁷ <https://github.com/TheHPXProject/hpx/issues/4100>

³⁰⁴⁸ <https://github.com/TheHPXProject/hpx/issues/4085>

³⁰⁴⁹ <https://github.com/TheHPXProject/hpx/issues/4067>

³⁰⁵⁰ <https://github.com/TheHPXProject/hpx/issues/4056>

³⁰⁵¹ <https://github.com/TheHPXProject/hpx/issues/4052>

³⁰⁵² <https://github.com/TheHPXProject/hpx/issues/4032>

³⁰⁵³ <https://github.com/TheHPXProject/hpx/issues/4030>

³⁰⁵⁴ <https://github.com/TheHPXProject/hpx/issues/4029>

³⁰⁵⁵ <https://github.com/TheHPXProject/hpx/issues/4005>

³⁰⁵⁶ <https://github.com/TheHPXProject/hpx/issues/4002>

³⁰⁵⁷ <https://github.com/TheHPXProject/hpx/issues/3989>

³⁰⁵⁸ <https://github.com/TheHPXProject/hpx/issues/3958>

³⁰⁵⁹ <https://github.com/TheHPXProject/hpx/issues/3953>

³⁰⁶⁰ <https://github.com/TheHPXProject/hpx/issues/3941>

³⁰⁶¹ <https://github.com/TheHPXProject/hpx/issues/3940>

³⁰⁶² <https://github.com/TheHPXProject/hpx/issues/3935>

³⁰⁶³ <https://github.com/TheHPXProject/hpx/issues/3929>

³⁰⁶⁴ <https://github.com/TheHPXProject/hpx/issues/3912>

³⁰⁶⁵ <https://github.com/TheHPXProject/hpx/issues/3890>

³⁰⁶⁶ <https://github.com/TheHPXProject/hpx/issues/3883>

- Issue #3879³⁰⁶⁷ - HPX fails to configure with `-DHPX_WITH_TESTS=OFF`
- Issue #3871³⁰⁶⁸ - `dataflow` does not support void allocators
- Issue #3867³⁰⁶⁹ - Latest HTML docs placed in wrong directory on GitHub pages
- Issue #3866³⁰⁷⁰ - Make sure all tests use `HPX_TEST*` macros and not `HPX_ASSERT`
- Issue #3857³⁰⁷¹ - CMake all-keyword or all-plain for `target_link_libraries`
- Issue #3856³⁰⁷² - `hpx_setup_target` adds rogue flags
- Issue #3850³⁰⁷³ - HPX fails to build on POWER8 with Clang7
- Issue #3848³⁰⁷⁴ - Remove `lva` member from `thread_init_data`
- Issue #3838³⁰⁷⁵ - `hpx::parallel::count/count_if` failing tests
- Issue #3651³⁰⁷⁶ - `hpx::parallel::transform_reduce` with non const reference as lambda parameter
- Issue #3560³⁰⁷⁷ - Apex integration with HPX not working properly
- Issue #3322³⁰⁷⁸ - No warning when mixing debug/release builds

Closed pull requests

- PR #4300³⁰⁷⁹ - Checks for `MPI_Init` being called twice
- PR #4299³⁰⁸⁰ - Small CMake fixes
- PR #4298³⁰⁸¹ - Remove extra call to annotate function that messes up traces
- PR #4296³⁰⁸² - Fixing collectives locking problem
- PR #4295³⁰⁸³ - Do not check `LICENSE_1_0.txt` for inspect violations
- PR #4293³⁰⁸⁴ - Applying two small changes fixing carious MSVC/Windows problems
- PR #4285³⁰⁸⁵ - Delete `apex.hpp`
- PR #4276³⁰⁸⁶ - Disable doxygen generation for `hpx/debugging/print.hpp` file
- PR #4275³⁰⁸⁷ - Make sure APEX is linked to even when not explicitly referenced
- PR #4272³⁰⁸⁸ - Fix pushing of documentation

³⁰⁶⁷ <https://github.com/TheHPXProject/hpx/issues/3879>

³⁰⁶⁸ <https://github.com/TheHPXProject/hpx/issues/3871>

³⁰⁶⁹ <https://github.com/TheHPXProject/hpx/issues/3867>

³⁰⁷⁰ <https://github.com/TheHPXProject/hpx/issues/3866>

³⁰⁷¹ <https://github.com/TheHPXProject/hpx/issues/3857>

³⁰⁷² <https://github.com/TheHPXProject/hpx/issues/3856>

³⁰⁷³ <https://github.com/TheHPXProject/hpx/issues/3850>

³⁰⁷⁴ <https://github.com/TheHPXProject/hpx/issues/3848>

³⁰⁷⁵ <https://github.com/TheHPXProject/hpx/issues/3838>

³⁰⁷⁶ <https://github.com/TheHPXProject/hpx/issues/3651>

³⁰⁷⁷ <https://github.com/TheHPXProject/hpx/issues/3560>

³⁰⁷⁸ <https://github.com/TheHPXProject/hpx/issues/3322>

³⁰⁷⁹ <https://github.com/TheHPXProject/hpx/pull/4300>

³⁰⁸⁰ <https://github.com/TheHPXProject/hpx/pull/4299>

³⁰⁸¹ <https://github.com/TheHPXProject/hpx/pull/4298>

³⁰⁸² <https://github.com/TheHPXProject/hpx/pull/4296>

³⁰⁸³ <https://github.com/TheHPXProject/hpx/pull/4295>

³⁰⁸⁴ <https://github.com/TheHPXProject/hpx/pull/4293>

³⁰⁸⁵ <https://github.com/TheHPXProject/hpx/pull/4285>

³⁰⁸⁶ <https://github.com/TheHPXProject/hpx/pull/4276>

³⁰⁸⁷ <https://github.com/TheHPXProject/hpx/pull/4275>

³⁰⁸⁸ <https://github.com/TheHPXProject/hpx/pull/4272>

- PR #4271³⁰⁸⁹ - Updating APEX tag, don't create new task_wrapper on operator= of hpx_thread object
- PR #4268³⁰⁹⁰ - Testing for noexcept function specializations in C++11/14 mode
- PR #4267³⁰⁹¹ - Fixing MSVC warning
- PR #4266³⁰⁹² - Make sure macOS Travis CI fails if build step fails
- PR #4264³⁰⁹³ - Clean up compatibility header options
- PR #4262³⁰⁹⁴ - Cleanup modules CMakeLists.txt
- PR #4261³⁰⁹⁵ - Fixing HPX/APEX linking and dependencies for external projects like Phylanx
- PR #4260³⁰⁹⁶ - Fix docs compilation problems
- PR #4258³⁰⁹⁷ - Couple of minor changes
- PR #4257³⁰⁹⁸ - Fix apex annotation for async dispatch
- PR #4256³⁰⁹⁹ - Remove lambdas from assert expressions
- PR #4255³¹⁰⁰ - Ignoring lock in all_to_all and all_reduce
- PR #4254³¹⁰¹ - Adding action specializations for noexcept functions
- PR #4253³¹⁰² - Move partlit.hpp to affinity module
- PR #4252³¹⁰³ - Make mismatching build types a hard error in CMake
- PR #4249³¹⁰⁴ - Scheduler improvement
- PR #4248³¹⁰⁵ - update hpxmp tag to v0.3.0
- PR #4245³¹⁰⁶ - Adding high performance channels
- PR #4244³¹⁰⁷ - Ignore lock in ignore_while_locked_1485 test
- PR #4243³¹⁰⁸ - Fix PAPI command line option documentation
- PR #4242³¹⁰⁹ - Ignore lock in target_distribution_policy
- PR #4241³¹¹⁰ - Fix start_stop_callbacks test
- PR #4240³¹¹¹ - Mostly fix clang CUDA compilation

³⁰⁸⁹ <https://github.com/TheHPXProject/hpx/pull/4271>

³⁰⁹⁰ <https://github.com/TheHPXProject/hpx/pull/4268>

³⁰⁹¹ <https://github.com/TheHPXProject/hpx/pull/4267>

³⁰⁹² <https://github.com/TheHPXProject/hpx/pull/4266>

³⁰⁹³ <https://github.com/TheHPXProject/hpx/pull/4264>

³⁰⁹⁴ <https://github.com/TheHPXProject/hpx/pull/4262>

³⁰⁹⁵ <https://github.com/TheHPXProject/hpx/pull/4261>

³⁰⁹⁶ <https://github.com/TheHPXProject/hpx/pull/4260>

³⁰⁹⁷ <https://github.com/TheHPXProject/hpx/pull/4258>

³⁰⁹⁸ <https://github.com/TheHPXProject/hpx/pull/4257>

³⁰⁹⁹ <https://github.com/TheHPXProject/hpx/pull/4256>

³¹⁰⁰ <https://github.com/TheHPXProject/hpx/pull/4255>

³¹⁰¹ <https://github.com/TheHPXProject/hpx/pull/4254>

³¹⁰² <https://github.com/TheHPXProject/hpx/pull/4253>

³¹⁰³ <https://github.com/TheHPXProject/hpx/pull/4252>

³¹⁰⁴ <https://github.com/TheHPXProject/hpx/pull/4249>

³¹⁰⁵ <https://github.com/TheHPXProject/hpx/pull/4248>

³¹⁰⁶ <https://github.com/TheHPXProject/hpx/pull/4245>

³¹⁰⁷ <https://github.com/TheHPXProject/hpx/pull/4244>

³¹⁰⁸ <https://github.com/TheHPXProject/hpx/pull/4243>

³¹⁰⁹ <https://github.com/TheHPXProject/hpx/pull/4242>

³¹¹⁰ <https://github.com/TheHPXProject/hpx/pull/4241>

³¹¹¹ <https://github.com/TheHPXProject/hpx/pull/4240>

- [PR #4238](#)³¹¹² - Google Season of Docs updates to documentation; grammar edits.
- [PR #4237](#)³¹¹³ - fixing annotated task to use the name, not the desc
- [PR #4236](#)³¹¹⁴ - Move module print summary to modules
- [PR #4235](#)³¹¹⁵ - Don't use alignas in `cache_{aligned,line}_data`
- [PR #4234](#)³¹¹⁶ - Add basic overview sentence to all modules
- [PR #4230](#)³¹¹⁷ - Add OS X builds to Travis CI
- [PR #4229](#)³¹¹⁸ - Remove leftover queue compatibility checks
- [PR #4226](#)³¹¹⁹ - Fixing APEX shutdown by explicitly shutting down throttling
- [PR #4225](#)³¹²⁰ - Allow `CMAKE_INSTALL_PREFIX` to be a relative path
- [PR #4224](#)³¹²¹ - Deprecate verbs `parcelport`
- [PR #4222](#)³¹²² - Update `register_{thread,work}` namespaces
- [PR #4221](#)³¹²³ - Changing `HPX_GCC_VERSION` check from `70000` to `70300`
- [PR #4218](#)³¹²⁴ - Google Season of Docs updates to documentation; grammar edits.
- [PR #4217](#)³¹²⁵ - Google Season of Docs updates to documentation; grammar edits.
- [PR #4216](#)³¹²⁶ - Fixing gcc warning on 32bit platforms (integer truncation)
- [PR #4214](#)³¹²⁷ - Apex callback refactoring
- [PR #4213](#)³¹²⁸ - Clean up allocator checks for dependent projects
- [PR #4212](#)³¹²⁹ - Google Season of Docs updates to documentation; grammar edits.
- [PR #4211](#)³¹³⁰ - Google Season of Docs updates to documentation; contributing to hpx
- [PR #4210](#)³¹³¹ - Attempting to fix Intel compilation
- [PR #4209](#)³¹³² - Fix CUDA 10 build
- [PR #4205](#)³¹³³ - Making sure that differences in `CMAKE_BUILD_TYPE` are not reported on multi-configuration cmake generators
- [PR #4203](#)³¹³⁴ - Deprecate `Vc`

³¹¹² <https://github.com/TheHPXProject/hpx/pull/4238>

³¹¹³ <https://github.com/TheHPXProject/hpx/pull/4237>

³¹¹⁴ <https://github.com/TheHPXProject/hpx/pull/4236>

³¹¹⁵ <https://github.com/TheHPXProject/hpx/pull/4235>

³¹¹⁶ <https://github.com/TheHPXProject/hpx/pull/4234>

³¹¹⁷ <https://github.com/TheHPXProject/hpx/pull/4230>

³¹¹⁸ <https://github.com/TheHPXProject/hpx/pull/4229>

³¹¹⁹ <https://github.com/TheHPXProject/hpx/pull/4226>

³¹²⁰ <https://github.com/TheHPXProject/hpx/pull/4225>

³¹²¹ <https://github.com/TheHPXProject/hpx/pull/4224>

³¹²² <https://github.com/TheHPXProject/hpx/pull/4222>

³¹²³ <https://github.com/TheHPXProject/hpx/pull/4221>

³¹²⁴ <https://github.com/TheHPXProject/hpx/pull/4218>

³¹²⁵ <https://github.com/TheHPXProject/hpx/pull/4217>

³¹²⁶ <https://github.com/TheHPXProject/hpx/pull/4216>

³¹²⁷ <https://github.com/TheHPXProject/hpx/pull/4214>

³¹²⁸ <https://github.com/TheHPXProject/hpx/pull/4213>

³¹²⁹ <https://github.com/TheHPXProject/hpx/pull/4212>

³¹³⁰ <https://github.com/TheHPXProject/hpx/pull/4211>

³¹³¹ <https://github.com/TheHPXProject/hpx/pull/4210>

³¹³² <https://github.com/TheHPXProject/hpx/pull/4209>

³¹³³ <https://github.com/TheHPXProject/hpx/pull/4205>

³¹³⁴ <https://github.com/TheHPXProject/hpx/pull/4203>

- PR #4202³¹³⁵ - Fix CUDA configuration
- PR #4200³¹³⁶ - Making sure `hpx_wrap` is not passed on to linker on non-Linux systems
- PR #4198³¹³⁷ - Fix `execution_agent.cpp` compilation with GCC 5
- PR #4197³¹³⁸ - Remove deprecated options for 1.4.0 release
- PR #4196³¹³⁹ - minor fixes for building on OSX Darwin
- PR #4195³¹⁴⁰ - Use full clone on CircleCI for pushing stable tag
- PR #4193³¹⁴¹ - Add scheduling hints to `hello_world_distributed`
- PR #4192³¹⁴² - Set up CUDA in `HPXConfig.cmake`
- PR #4191³¹⁴³ - Export allocators root variables
- PR #4190³¹⁴⁴ - Don't use `constexpr` in `thread_data` with GCC <= 6
- PR #4189³¹⁴⁵ - Only use `quick_exit` if available
- PR #4188³¹⁴⁶ - Google Season of Docs updates to documentation; writing single node hpx applications
- PR #4186³¹⁴⁷ - correct `vc` to `cuda` in `cuda` `cmake`
- PR #4184³¹⁴⁸ - Resetting some cached variables to make sure those are re-filled
- PR #4183³¹⁴⁹ - Fix `hpxcxx` configuration
- PR #4181³¹⁵⁰ - Rename base libraries var
- PR #4180³¹⁵¹ - Move header left behind earlier to plugin module
- PR #4179³¹⁵² - Moving `zip_iterator` and `transform_iterator` to `iterator_support` module
- PR #4178³¹⁵³ - Move checkpointing support to its own module
- PR #4177³¹⁵⁴ - Small `const` fix to `basic_execution` module
- PR #4176³¹⁵⁵ - Add back `HPX_LIBRARIES` and friends to `HPXConfig.cmake`
- PR #4175³¹⁵⁶ - Make `Vc` public and add it to `HPXConfig.cmake`
- PR #4173³¹⁵⁷ - Wait for runtime to be running before returning from `hpx::start`

³¹³⁵ <https://github.com/TheHPXProject/hpx/pull/4202>

³¹³⁶ <https://github.com/TheHPXProject/hpx/pull/4200>

³¹³⁷ <https://github.com/TheHPXProject/hpx/pull/4198>

³¹³⁸ <https://github.com/TheHPXProject/hpx/pull/4197>

³¹³⁹ <https://github.com/TheHPXProject/hpx/pull/4196>

³¹⁴⁰ <https://github.com/TheHPXProject/hpx/pull/4195>

³¹⁴¹ <https://github.com/TheHPXProject/hpx/pull/4193>

³¹⁴² <https://github.com/TheHPXProject/hpx/pull/4192>

³¹⁴³ <https://github.com/TheHPXProject/hpx/pull/4191>

³¹⁴⁴ <https://github.com/TheHPXProject/hpx/pull/4190>

³¹⁴⁵ <https://github.com/TheHPXProject/hpx/pull/4189>

³¹⁴⁶ <https://github.com/TheHPXProject/hpx/pull/4188>

³¹⁴⁷ <https://github.com/TheHPXProject/hpx/pull/4186>

³¹⁴⁸ <https://github.com/TheHPXProject/hpx/pull/4184>

³¹⁴⁹ <https://github.com/TheHPXProject/hpx/pull/4183>

³¹⁵⁰ <https://github.com/TheHPXProject/hpx/pull/4181>

³¹⁵¹ <https://github.com/TheHPXProject/hpx/pull/4180>

³¹⁵² <https://github.com/TheHPXProject/hpx/pull/4179>

³¹⁵³ <https://github.com/TheHPXProject/hpx/pull/4178>

³¹⁵⁴ <https://github.com/TheHPXProject/hpx/pull/4177>

³¹⁵⁵ <https://github.com/TheHPXProject/hpx/pull/4176>

³¹⁵⁶ <https://github.com/TheHPXProject/hpx/pull/4175>

³¹⁵⁷ <https://github.com/TheHPXProject/hpx/pull/4173>

- PR #4172³¹⁵⁸ - More protection against shutdown problems in error handling scenarios.
- PR #4171³¹⁵⁹ - Ignore lock in `condition_variable::wait`
- PR #4170³¹⁶⁰ - Adding APEX dependency to MPI parcellport
- PR #4168³¹⁶¹ - Adding utility include
- PR #4167³¹⁶² - Add a condition to setup the external libraries
- PR #4166³¹⁶³ - Add an `INTERNAL_FLAGS` option to link to `hpx_internal_flags`
- PR #4165³¹⁶⁴ - Forward `HPX_*` cmake cache variables to external projects
- PR #4164³¹⁶⁵ - Affinity and batch environment modules
- PR #4162³¹⁶⁶ - Handle quick `exit`
- PR #4160³¹⁶⁷ - Using `target_link_libraries` for cmake versions ≥ 3.12
- PR #4159³¹⁶⁸ - Make sure `HPX_WITH_NATIVE_TLS` is forwarded to dependent projects
- PR #4158³¹⁶⁹ - Adding allocator imported target as a dependency of allocator module
- PR #4157³¹⁷⁰ - Add `hpx_memory` as a dependency of parcellport plugins
- PR #4156³¹⁷¹ - Stackless coroutines now can refer to themselves (through `get_self()` and friends)
- PR #4154³¹⁷² - Added CMake policy CMP0060 for HPX applications.
- PR #4153³¹⁷³ - add header `iomanip` to tests and tool
- PR #4152³¹⁷⁴ - Casting MPI tag value
- PR #4149³¹⁷⁵ - Add back private `m_desc` member variable in `program_options` module
- PR #4147³¹⁷⁶ - Resource partitioner and threadmanager modules
- PR #4146³¹⁷⁷ - Google Season of Docs updates to documentation; creating hpx projects
- PR #4145³¹⁷⁸ - Adding basic support for stackless threads
- PR #4143³¹⁷⁹ - Exclude `test_client_1950` from all target
- PR #4142³¹⁸⁰ - Add a new `thread_pool_executor`

³¹⁵⁸ <https://github.com/TheHPXProject/hpx/pull/4172>

³¹⁵⁹ <https://github.com/TheHPXProject/hpx/pull/4171>

³¹⁶⁰ <https://github.com/TheHPXProject/hpx/pull/4170>

³¹⁶¹ <https://github.com/TheHPXProject/hpx/pull/4168>

³¹⁶² <https://github.com/TheHPXProject/hpx/pull/4167>

³¹⁶³ <https://github.com/TheHPXProject/hpx/pull/4166>

³¹⁶⁴ <https://github.com/TheHPXProject/hpx/pull/4165>

³¹⁶⁵ <https://github.com/TheHPXProject/hpx/pull/4164>

³¹⁶⁶ <https://github.com/TheHPXProject/hpx/pull/4162>

³¹⁶⁷ <https://github.com/TheHPXProject/hpx/pull/4160>

³¹⁶⁸ <https://github.com/TheHPXProject/hpx/pull/4159>

³¹⁶⁹ <https://github.com/TheHPXProject/hpx/pull/4158>

³¹⁷⁰ <https://github.com/TheHPXProject/hpx/pull/4157>

³¹⁷¹ <https://github.com/TheHPXProject/hpx/pull/4156>

³¹⁷² <https://github.com/TheHPXProject/hpx/pull/4154>

³¹⁷³ <https://github.com/TheHPXProject/hpx/pull/4153>

³¹⁷⁴ <https://github.com/TheHPXProject/hpx/pull/4152>

³¹⁷⁵ <https://github.com/TheHPXProject/hpx/pull/4149>

³¹⁷⁶ <https://github.com/TheHPXProject/hpx/pull/4147>

³¹⁷⁷ <https://github.com/TheHPXProject/hpx/pull/4146>

³¹⁷⁸ <https://github.com/TheHPXProject/hpx/pull/4145>

³¹⁷⁹ <https://github.com/TheHPXProject/hpx/pull/4143>

³¹⁸⁰ <https://github.com/TheHPXProject/hpx/pull/4142>

- PR #4140³¹⁸¹ - Google Season of Docs updates to documentation; why hpx
- PR #4139³¹⁸² - Remove runtime includes from coroutines module
- PR #4138³¹⁸³ - Forking `boost::intrusive_ptr` and adding it as `hpx::intrusive_ptr`
- PR #4137³¹⁸⁴ - Fixing TSS destruction
- PR #4136³¹⁸⁵ - HPX.Compute modules
- PR #4133³¹⁸⁶ - Fix `block_executor`
- PR #4131³¹⁸⁷ - Applying fixes based on reports from PVS Studio
- PR #4130³¹⁸⁸ - Adding missing header to build system
- PR #4129³¹⁸⁹ - Fixing compilation if `HPX_WITH_DATAPAR_VC` is enabled
- PR #4128³¹⁹⁰ - Renaming `moveonly_any` to `unique_any`
- PR #4126³¹⁹¹ - Attempt to fix `basic_any` constructor for gcc 7
- PR #4125³¹⁹² - Changing `extra_archive_data` implementation
- PR #4124³¹⁹³ - Don't link to Boost.System unless required
- PR #4122³¹⁹⁴ - Add kernel launch helper utility (+saxpy demo) and merge in octotiger changes
- PR #4121³¹⁹⁵ - Fixing migration test if networking is disabled.
- PR #4120³¹⁹⁶ - Google Season of Docs updates to documentation; hpx build system v1
- PR #4119³¹⁹⁷ - Making sure `chunk_size` and `max_chunk` are actually applied to parallel algorithms if specified
- PR #4117³¹⁹⁸ - Make CircleCI formatting check store diff
- PR #4116³¹⁹⁹ - Fix automatically setting C++ standard
- PR #4114³²⁰⁰ - Module serialization
- PR #4113³²⁰¹ - Module datastructures
- PR #4111³²⁰² - Fixing performance regression introduced earlier
- PR #4110³²⁰³ - Adding missing SPDX tags

³¹⁸¹ <https://github.com/TheHPXProject/hpx/pull/4140>

³¹⁸² <https://github.com/TheHPXProject/hpx/pull/4139>

³¹⁸³ <https://github.com/TheHPXProject/hpx/pull/4138>

³¹⁸⁴ <https://github.com/TheHPXProject/hpx/pull/4137>

³¹⁸⁵ <https://github.com/TheHPXProject/hpx/pull/4136>

³¹⁸⁶ <https://github.com/TheHPXProject/hpx/pull/4133>

³¹⁸⁷ <https://github.com/TheHPXProject/hpx/pull/4131>

³¹⁸⁸ <https://github.com/TheHPXProject/hpx/pull/4130>

³¹⁸⁹ <https://github.com/TheHPXProject/hpx/pull/4129>

³¹⁹⁰ <https://github.com/TheHPXProject/hpx/pull/4128>

³¹⁹¹ <https://github.com/TheHPXProject/hpx/pull/4126>

³¹⁹² <https://github.com/TheHPXProject/hpx/pull/4125>

³¹⁹³ <https://github.com/TheHPXProject/hpx/pull/4124>

³¹⁹⁴ <https://github.com/TheHPXProject/hpx/pull/4122>

³¹⁹⁵ <https://github.com/TheHPXProject/hpx/pull/4121>

³¹⁹⁶ <https://github.com/TheHPXProject/hpx/pull/4120>

³¹⁹⁷ <https://github.com/TheHPXProject/hpx/pull/4119>

³¹⁹⁸ <https://github.com/TheHPXProject/hpx/pull/4117>

³¹⁹⁹ <https://github.com/TheHPXProject/hpx/pull/4116>

³²⁰⁰ <https://github.com/TheHPXProject/hpx/pull/4114>

³²⁰¹ <https://github.com/TheHPXProject/hpx/pull/4113>

³²⁰² <https://github.com/TheHPXProject/hpx/pull/4111>

³²⁰³ <https://github.com/TheHPXProject/hpx/pull/4110>

- PR #4109³²⁰⁴ - Overload for start without entry point/argv.
- PR #4108³²⁰⁵ - Making sure C++ standard is properly detected and propagated
- PR #4106³²⁰⁶ - use `std::round` for guaranteed rounding without errors
- PR #4104³²⁰⁷ - Extend `scheduler_mode` with new `work_stealing` and task assignment modes
- PR #4103³²⁰⁸ - Add this to lambda capture list
- PR #4102³²⁰⁹ - Add spdx license and check
- PR #4099³²¹⁰ - Module coroutines
- PR #4098³²¹¹ - Fix append module path in module CMakeLists template
- PR #4097³²¹² - Function tests
- PR #4096³²¹³ - Removing return of `thread_result_type` from functions not needing them
- PR #4095³²¹⁴ - Stop-gap measure until cmake overhaul is in place
- PR #4094³²¹⁵ - Deprecate `HPX_WITH_MORE_THAN_64_THREADS`
- PR #4093³²¹⁶ - Fix initialization of `global_num_tasks` in `parallel_executor`
- PR #4092³²¹⁷ - Add support for mi-malloc
- PR #4090³²¹⁸ - Execution context
- PR #4089³²¹⁹ - Make counters in coroutines optional
- PR #4087³²²⁰ - Making `hpx::util::any` compatible with C++17
- PR #4084³²²¹ - Making sure destination array for `std::transform` is properly resized
- PR #4083³²²² - Adapting `thread_queue_mc` to behave even if no 128bit atomics are available
- PR #4082³²²³ - Fix compilation on GCC 5
- PR #4081³²²⁴ - Adding option allowing to force using Boost.FileSystem
- PR #4080³²²⁵ - Updating module dependencies
- PR #4079³²²⁶ - Add missing tests for `iterator_support` module

³²⁰⁴ <https://github.com/TheHPXProject/hpx/pull/4109>

³²⁰⁵ <https://github.com/TheHPXProject/hpx/pull/4108>

³²⁰⁶ <https://github.com/TheHPXProject/hpx/pull/4106>

³²⁰⁷ <https://github.com/TheHPXProject/hpx/pull/4104>

³²⁰⁸ <https://github.com/TheHPXProject/hpx/pull/4103>

³²⁰⁹ <https://github.com/TheHPXProject/hpx/pull/4102>

³²¹⁰ <https://github.com/TheHPXProject/hpx/pull/4099>

³²¹¹ <https://github.com/TheHPXProject/hpx/pull/4098>

³²¹² <https://github.com/TheHPXProject/hpx/pull/4097>

³²¹³ <https://github.com/TheHPXProject/hpx/pull/4096>

³²¹⁴ <https://github.com/TheHPXProject/hpx/pull/4095>

³²¹⁵ <https://github.com/TheHPXProject/hpx/pull/4094>

³²¹⁶ <https://github.com/TheHPXProject/hpx/pull/4093>

³²¹⁷ <https://github.com/TheHPXProject/hpx/pull/4092>

³²¹⁸ <https://github.com/TheHPXProject/hpx/pull/4090>

³²¹⁹ <https://github.com/TheHPXProject/hpx/pull/4089>

³²²⁰ <https://github.com/TheHPXProject/hpx/pull/4087>

³²²¹ <https://github.com/TheHPXProject/hpx/pull/4084>

³²²² <https://github.com/TheHPXProject/hpx/pull/4083>

³²²³ <https://github.com/TheHPXProject/hpx/pull/4082>

³²²⁴ <https://github.com/TheHPXProject/hpx/pull/4081>

³²²⁵ <https://github.com/TheHPXProject/hpx/pull/4080>

³²²⁶ <https://github.com/TheHPXProject/hpx/pull/4079>

- PR #4078³²²⁷ - Disable parcel-layer if networking is disabled
- PR #4077³²²⁸ - Add missing include that causes build fails
- PR #4076³²²⁹ - Enable compatibility headers for functional module
- PR #4075³²³⁰ - Coroutines module
- PR #4073³²³¹ - Use `configure_file` for generated files in modules
- PR #4071³²³² - Fixing MPI detection for PMIx
- PR #4070³²³³ - Fix macOS builds
- PR #4069³²³⁴ - Moving more facilities to the collectives module
- PR #4068³²³⁵ - Adding main HPX `#include` directory to modules
- PR #4066³²³⁶ - Switching the use of `message(STATUS "...")` to `hpx_info`
- PR #4065³²³⁷ - Move Boost.Filesystem handling to filesystem module
- PR #4064³²³⁸ - Fix program_options test with older boost versions
- PR #4062³²³⁹ - The `cpu_features` tool fails to compile on anything but x86 architectures
- PR #4061³²⁴⁰ - Add `clang-format` checking step for modules
- PR #4060³²⁴¹ - Making sure `HPX_IDLE_BACKOFF_TIME_MAX` is always defined (even if its unused)
- PR #4059³²⁴² - Renaming module `hpx_parallel_executors` into `hpx_execution`
- PR #4058³²⁴³ - Do not build networking tests when networking disabled
- PR #4057³²⁴⁴ - Printing configuration summary for modules as well
- PR #4055³²⁴⁵ - Google Season of Docs updates to documentation; hpx build systems
- PR #4054³²⁴⁶ - Add troubleshooting section to manual
- PR #4051³²⁴⁷ - Add more variations to `future_overhead` test
- PR #4050³²⁴⁸ - Creating plugin module
- PR #4049³²⁴⁹ - Move missing modules tests

³²²⁷ <https://github.com/TheHPXProject/hpx/pull/4078>

³²²⁸ <https://github.com/TheHPXProject/hpx/pull/4077>

³²²⁹ <https://github.com/TheHPXProject/hpx/pull/4076>

³²³⁰ <https://github.com/TheHPXProject/hpx/pull/4075>

³²³¹ <https://github.com/TheHPXProject/hpx/pull/4073>

³²³² <https://github.com/TheHPXProject/hpx/pull/4071>

³²³³ <https://github.com/TheHPXProject/hpx/pull/4070>

³²³⁴ <https://github.com/TheHPXProject/hpx/pull/4069>

³²³⁵ <https://github.com/TheHPXProject/hpx/pull/4068>

³²³⁶ <https://github.com/TheHPXProject/hpx/pull/4066>

³²³⁷ <https://github.com/TheHPXProject/hpx/pull/4065>

³²³⁸ <https://github.com/TheHPXProject/hpx/pull/4064>

³²³⁹ <https://github.com/TheHPXProject/hpx/pull/4062>

³²⁴⁰ <https://github.com/TheHPXProject/hpx/pull/4061>

³²⁴¹ <https://github.com/TheHPXProject/hpx/pull/4060>

³²⁴² <https://github.com/TheHPXProject/hpx/pull/4059>

³²⁴³ <https://github.com/TheHPXProject/hpx/pull/4058>

³²⁴⁴ <https://github.com/TheHPXProject/hpx/pull/4057>

³²⁴⁵ <https://github.com/TheHPXProject/hpx/pull/4055>

³²⁴⁶ <https://github.com/TheHPXProject/hpx/pull/4054>

³²⁴⁷ <https://github.com/TheHPXProject/hpx/pull/4051>

³²⁴⁸ <https://github.com/TheHPXProject/hpx/pull/4050>

³²⁴⁹ <https://github.com/TheHPXProject/hpx/pull/4049>

- PR #4047³²⁵⁰ - Add boost/filesystem headers to inspect deprecated headers
- PR #4045³²⁵¹ - Module functional
- PR #4043³²⁵² - Fix preconditions and error messages for suspension functions
- PR #4041³²⁵³ - Pass HPX_STANDARD on to dependent projects via HPXConfig.cmake
- PR #4040³²⁵⁴ - Program options module
- PR #4039³²⁵⁵ - Moving non-serializable any (any_nonser) to datastructures module
- PR #4038³²⁵⁶ - Adding MPark's variant (V1.4.0) to HPX
- PR #4037³²⁵⁷ - Adding resiliency module
- PR #4036³²⁵⁸ - Add C++17 filesystem compatibility header
- PR #4035³²⁵⁹ - Fixing support for mpirun
- PR #4028³²⁶⁰ - CMake to target based directives
- PR #4027³²⁶¹ - Remove GitLab CI configuration
- PR #4026³²⁶² - Threading refactoring
- PR #4025³²⁶³ - Refactoring thread queue configuration options
- PR #4024³²⁶⁴ - Fix padding calculation in cache_aligned_data.hpp
- PR #4023³²⁶⁵ - Fixing Codacy issues
- PR #4022³²⁶⁶ - Make sure process mask option is passed to affinity_data
- PR #4021³²⁶⁷ - Warn about compiling in C++11 mode
- PR #4020³²⁶⁸ - Module concurrency
- PR #4019³²⁶⁹ - Module topology
- PR #4018³²⁷⁰ - Update deprecated header in thread_queue_mc.hpp
- PR #4015³²⁷¹ - Avoid overwriting artifacts
- PR #4014³²⁷² - Future overheads

³²⁵⁰ <https://github.com/TheHPXProject/hpx/pull/4047>

³²⁵¹ <https://github.com/TheHPXProject/hpx/pull/4045>

³²⁵² <https://github.com/TheHPXProject/hpx/pull/4043>

³²⁵³ <https://github.com/TheHPXProject/hpx/pull/4041>

³²⁵⁴ <https://github.com/TheHPXProject/hpx/pull/4040>

³²⁵⁵ <https://github.com/TheHPXProject/hpx/pull/4039>

³²⁵⁶ <https://github.com/TheHPXProject/hpx/pull/4038>

³²⁵⁷ <https://github.com/TheHPXProject/hpx/pull/4037>

³²⁵⁸ <https://github.com/TheHPXProject/hpx/pull/4036>

³²⁵⁹ <https://github.com/TheHPXProject/hpx/pull/4035>

³²⁶⁰ <https://github.com/TheHPXProject/hpx/pull/4028>

³²⁶¹ <https://github.com/TheHPXProject/hpx/pull/4027>

³²⁶² <https://github.com/TheHPXProject/hpx/pull/4026>

³²⁶³ <https://github.com/TheHPXProject/hpx/pull/4025>

³²⁶⁴ <https://github.com/TheHPXProject/hpx/pull/4024>

³²⁶⁵ <https://github.com/TheHPXProject/hpx/pull/4023>

³²⁶⁶ <https://github.com/TheHPXProject/hpx/pull/4022>

³²⁶⁷ <https://github.com/TheHPXProject/hpx/pull/4021>

³²⁶⁸ <https://github.com/TheHPXProject/hpx/pull/4020>

³²⁶⁹ <https://github.com/TheHPXProject/hpx/pull/4019>

³²⁷⁰ <https://github.com/TheHPXProject/hpx/pull/4018>

³²⁷¹ <https://github.com/TheHPXProject/hpx/pull/4015>

³²⁷² <https://github.com/TheHPXProject/hpx/pull/4014>

- PR #4013³²⁷³ - Update URL to test output conversion script
- PR #4012³²⁷⁴ - Fix CUDA compilation
- PR #4011³²⁷⁵ - Fixing cyclic dependencies between modules
- PR #4010³²⁷⁶ - Ignore stable tag on CircleCI
- PR #4009³²⁷⁷ - Check circular dependencies in a circle ci step
- PR #4008³²⁷⁸ - Extend cache aligned data to handle tuple-like data
- PR #4007³²⁷⁹ - Fixing migration for components that have actions returning a client
- PR #4006³²⁸⁰ - Move is_value_proxy.hpp to algorithms module
- PR #4004³²⁸¹ - Shorten CTest timeout on CircleCI
- PR #4003³²⁸² - Refactoring to remove (internal) dependencies
- PR #4001³²⁸³ - Exclude tests from all target
- PR #4000³²⁸⁴ - Module errors
- PR #3999³²⁸⁵ - Enable support for compatibility headers for logging module
- PR #3998³²⁸⁶ - Add process thread binding option
- PR #3997³²⁸⁷ - Export handle_assert function
- PR #3996³²⁸⁸ - Attempt to solve issue where -latomic does not support 128bit atomics
- PR #3993³²⁸⁹ - Make sure __LINE__ is an unsigned
- PR #3991³²⁹⁰ - Fix dependencies and flags for header tests
- PR #3990³²⁹¹ - Documentation tags fixes
- PR #3988³²⁹² - Adding missing solution folder for format module test
- PR #3987³²⁹³ - Move runtime-dependent functions out of command line handling
- PR #3986³²⁹⁴ - Fix CMake configuration with PAPI on
- PR #3985³²⁹⁵ - Module timing

³²⁷³ <https://github.com/TheHPXProject/hpx/pull/4013>

³²⁷⁴ <https://github.com/TheHPXProject/hpx/pull/4012>

³²⁷⁵ <https://github.com/TheHPXProject/hpx/pull/4011>

³²⁷⁶ <https://github.com/TheHPXProject/hpx/pull/4010>

³²⁷⁷ <https://github.com/TheHPXProject/hpx/pull/4009>

³²⁷⁸ <https://github.com/TheHPXProject/hpx/pull/4008>

³²⁷⁹ <https://github.com/TheHPXProject/hpx/pull/4007>

³²⁸⁰ <https://github.com/TheHPXProject/hpx/pull/4006>

³²⁸¹ <https://github.com/TheHPXProject/hpx/pull/4004>

³²⁸² <https://github.com/TheHPXProject/hpx/pull/4003>

³²⁸³ <https://github.com/TheHPXProject/hpx/pull/4001>

³²⁸⁴ <https://github.com/TheHPXProject/hpx/pull/4000>

³²⁸⁵ <https://github.com/TheHPXProject/hpx/pull/3999>

³²⁸⁶ <https://github.com/TheHPXProject/hpx/pull/3998>

³²⁸⁷ <https://github.com/TheHPXProject/hpx/pull/3997>

³²⁸⁸ <https://github.com/TheHPXProject/hpx/pull/3996>

³²⁸⁹ <https://github.com/TheHPXProject/hpx/pull/3993>

³²⁹⁰ <https://github.com/TheHPXProject/hpx/pull/3991>

³²⁹¹ <https://github.com/TheHPXProject/hpx/pull/3990>

³²⁹² <https://github.com/TheHPXProject/hpx/pull/3988>

³²⁹³ <https://github.com/TheHPXProject/hpx/pull/3987>

³²⁹⁴ <https://github.com/TheHPXProject/hpx/pull/3986>

³²⁹⁵ <https://github.com/TheHPXProject/hpx/pull/3985>

- PR #3984³²⁹⁶ - Fix default behaviour of paths in `add_hpx_component`
- PR #3982³²⁹⁷ - Parallel executors module
- PR #3981³²⁹⁸ - Segmented algorithms module
- PR #3980³²⁹⁹ - Module logging
- PR #3979³³⁰⁰ - Module util
- PR #3978³³⁰¹ - Fix `clang-tidy` step on CircleCI
- PR #3977³³⁰² - Fixing solution folders for moved components
- PR #3976³³⁰³ - Module format
- PR #3975³³⁰⁴ - Enable deprecation warnings on CircleCI
- PR #3974³³⁰⁵ - Fix typos in documentation
- PR #3973³³⁰⁶ - Fix compilation with GCC 9
- PR #3972³³⁰⁷ - Add condition to clone apex + use of new cmake var `APEX_ROOT`
- PR #3971³³⁰⁸ - Add testing module
- PR #3968³³⁰⁹ - Remove unneeded file in hardware module
- PR #3967³³¹⁰ - Remove leftover PIC settings from main `CMakeLists.txt`
- PR #3966³³¹¹ - Add missing export option in `add_hpx_module`
- PR #3965³³¹² - Change `current_function_helper` back to non-constexpr
- PR #3964³³¹³ - Fixing merge problems
- PR #3962³³¹⁴ - Add a trait for `std::array` for unwrapping
- PR #3961³³¹⁵ - Making `hpx::util::tuple<Ts...>` and `std::tuple<Ts...>` convertible
- PR #3960³³¹⁶ - fix compilation with CUDA 10 and GCC 6
- PR #3959³³¹⁷ - Fix C++11 incompatibility
- PR #3957³³¹⁸ - Algorithms module

³²⁹⁶ <https://github.com/TheHPXProject/hpx/pull/3984>

³²⁹⁷ <https://github.com/TheHPXProject/hpx/pull/3982>

³²⁹⁸ <https://github.com/TheHPXProject/hpx/pull/3981>

³²⁹⁹ <https://github.com/TheHPXProject/hpx/pull/3980>

³³⁰⁰ <https://github.com/TheHPXProject/hpx/pull/3979>

³³⁰¹ <https://github.com/TheHPXProject/hpx/pull/3978>

³³⁰² <https://github.com/TheHPXProject/hpx/pull/3977>

³³⁰³ <https://github.com/TheHPXProject/hpx/pull/3976>

³³⁰⁴ <https://github.com/TheHPXProject/hpx/pull/3975>

³³⁰⁵ <https://github.com/TheHPXProject/hpx/pull/3974>

³³⁰⁶ <https://github.com/TheHPXProject/hpx/pull/3973>

³³⁰⁷ <https://github.com/TheHPXProject/hpx/pull/3972>

³³⁰⁸ <https://github.com/TheHPXProject/hpx/pull/3971>

³³⁰⁹ <https://github.com/TheHPXProject/hpx/pull/3968>

³³¹⁰ <https://github.com/TheHPXProject/hpx/pull/3967>

³³¹¹ <https://github.com/TheHPXProject/hpx/pull/3966>

³³¹² <https://github.com/TheHPXProject/hpx/pull/3965>

³³¹³ <https://github.com/TheHPXProject/hpx/pull/3964>

³³¹⁴ <https://github.com/TheHPXProject/hpx/pull/3962>

³³¹⁵ <https://github.com/TheHPXProject/hpx/pull/3961>

³³¹⁶ <https://github.com/TheHPXProject/hpx/pull/3960>

³³¹⁷ <https://github.com/TheHPXProject/hpx/pull/3959>

³³¹⁸ <https://github.com/TheHPXProject/hpx/pull/3957>

- PR #3956³³¹⁹ - [HPX_AddModule] Fix lower name var to upper
- PR #3955³³²⁰ - Fix CMake configuration with examples off and tests on
- PR #3954³³²¹ - Move components to separate subdirectory in root of repository
- PR #3952³³²² - Update `papi.cpp`
- PR #3951³³²³ - Exclude modules header tests from all target
- PR #3950³³²⁴ - Adding `all_reduce` facility to collectives module
- PR #3949³³²⁵ - This adds a configuration file that will cause for stale issues to be automatically closed
- PR #3948³³²⁶ - Fixing ALPS environment
- PR #3947³³²⁷ - Add major compiler version check for building hpx as a binary package
- PR #3946³³²⁸ - [Modules] Move the location of the generated headers
- PR #3945³³²⁹ - Simplify tests and examples cmake
- PR #3943³³³⁰ - Remove example module
- PR #3942³³³¹ - Add `NOEXPORT` option to `add_hpx_{component,library}`
- PR #3938³³³² - Use https for CDash submissions
- PR #3937³³³³ - Add `HPX_WITH_BUILD_BINARY_PACKAGE` to the compiler check (refs #3935)
- PR #3936³³³⁴ - Fixing installation of binaries on windows
- PR #3934³³³⁵ - Add set function for `sliding_semaphore_max_difference`
- PR #3933³³³⁶ - Remove `cuDart` from compile/link flags as it breaks downstream projects
- PR #3932³³³⁷ - Fixing 3929
- PR #3931³³³⁸ - Adding `all_to_all`
- PR #3930³³³⁹ - Add test demonstrating the use of broadcast with component actions
- PR #3928³³⁴⁰ - fixed number of tasks and number of threads for heterogeneous slurm environments
- PR #3927³³⁴¹ - Moving Cache module's tests into separate solution folder

³³¹⁹ <https://github.com/TheHPXProject/hpx/pull/3956>

³³²⁰ <https://github.com/TheHPXProject/hpx/pull/3955>

³³²¹ <https://github.com/TheHPXProject/hpx/pull/3954>

³³²² <https://github.com/TheHPXProject/hpx/pull/3952>

³³²³ <https://github.com/TheHPXProject/hpx/pull/3951>

³³²⁴ <https://github.com/TheHPXProject/hpx/pull/3950>

³³²⁵ <https://github.com/TheHPXProject/hpx/pull/3949>

³³²⁶ <https://github.com/TheHPXProject/hpx/pull/3948>

³³²⁷ <https://github.com/TheHPXProject/hpx/pull/3947>

³³²⁸ <https://github.com/TheHPXProject/hpx/pull/3946>

³³²⁹ <https://github.com/TheHPXProject/hpx/pull/3945>

³³³⁰ <https://github.com/TheHPXProject/hpx/pull/3943>

³³³¹ <https://github.com/TheHPXProject/hpx/pull/3942>

³³³² <https://github.com/TheHPXProject/hpx/pull/3938>

³³³³ <https://github.com/TheHPXProject/hpx/pull/3937>

³³³⁴ <https://github.com/TheHPXProject/hpx/pull/3936>

³³³⁵ <https://github.com/TheHPXProject/hpx/pull/3934>

³³³⁶ <https://github.com/TheHPXProject/hpx/pull/3933>

³³³⁷ <https://github.com/TheHPXProject/hpx/pull/3932>

³³³⁸ <https://github.com/TheHPXProject/hpx/pull/3931>

³³³⁹ <https://github.com/TheHPXProject/hpx/pull/3930>

³³⁴⁰ <https://github.com/TheHPXProject/hpx/pull/3928>

³³⁴¹ <https://github.com/TheHPXProject/hpx/pull/3927>

- PR #3926³³⁴² - Move unit tests to cache module
- PR #3925³³⁴³ - Move version check to config module
- PR #3924³³⁴⁴ - Add schedule hint executor parameters
- PR #3923³³⁴⁵ - Allow aligning objects bigger than the cache line size
- PR #3922³³⁴⁶ - Add Windows builds with Travis CI
- PR #3921³³⁴⁷ - Add ccls cache directory to gitignore
- PR #3920³³⁴⁸ - Fix `git_external` fetching of tags
- PR #3905³³⁴⁹ - Correct rostambod url. Fix typo in doc
- PR #3904³³⁵⁰ - Fix bug in `context_base.hpp`
- PR #3903³³⁵¹ - Adding new performance counters
- PR #3902³³⁵² - Add `add_hpx_module` function
- PR #3901³³⁵³ - Factoring out container remapping into a separate trait
- PR #3900³³⁵⁴ - Making sure errors during command line processing are properly reported and will not cause assertions
- PR #3899³³⁵⁵ - Remove old compatibility bases from `make_action`
- PR #3898³³⁵⁶ - Make parameter size be of type `size_t`
- PR #3897³³⁵⁷ - Making sure all tests are disabled if `HPX_WITH_TESTS=OFF`
- PR #3895³³⁵⁸ - Add documentation for `annotated_function`
- PR #3894³³⁵⁹ - Working around VS2019 problem with `make_action`
- PR #3892³³⁶⁰ - Avoid MSVC compatibility warning in internal allocator
- PR #3891³³⁶¹ - Removal of the default intel config include
- PR #3888³³⁶² - Fix `async_customization` dataflow example and Clarify what's being tested
- PR #3887³³⁶³ - Add Doxygen documentation
- PR #3882³³⁶⁴ - Minor docs fixes

³³⁴² <https://github.com/TheHPXProject/hpx/pull/3926>

³³⁴³ <https://github.com/TheHPXProject/hpx/pull/3925>

³³⁴⁴ <https://github.com/TheHPXProject/hpx/pull/3924>

³³⁴⁵ <https://github.com/TheHPXProject/hpx/pull/3923>

³³⁴⁶ <https://github.com/TheHPXProject/hpx/pull/3922>

³³⁴⁷ <https://github.com/TheHPXProject/hpx/pull/3921>

³³⁴⁸ <https://github.com/TheHPXProject/hpx/pull/3920>

³³⁴⁹ <https://github.com/TheHPXProject/hpx/pull/3905>

³³⁵⁰ <https://github.com/TheHPXProject/hpx/pull/3904>

³³⁵¹ <https://github.com/TheHPXProject/hpx/pull/3903>

³³⁵² <https://github.com/TheHPXProject/hpx/pull/3902>

³³⁵³ <https://github.com/TheHPXProject/hpx/pull/3901>

³³⁵⁴ <https://github.com/TheHPXProject/hpx/pull/3900>

³³⁵⁵ <https://github.com/TheHPXProject/hpx/pull/3899>

³³⁵⁶ <https://github.com/TheHPXProject/hpx/pull/3898>

³³⁵⁷ <https://github.com/TheHPXProject/hpx/pull/3897>

³³⁵⁸ <https://github.com/TheHPXProject/hpx/pull/3895>

³³⁵⁹ <https://github.com/TheHPXProject/hpx/pull/3894>

³³⁶⁰ <https://github.com/TheHPXProject/hpx/pull/3892>

³³⁶¹ <https://github.com/TheHPXProject/hpx/pull/3891>

³³⁶² <https://github.com/TheHPXProject/hpx/pull/3888>

³³⁶³ <https://github.com/TheHPXProject/hpx/pull/3887>

³³⁶⁴ <https://github.com/TheHPXProject/hpx/pull/3882>

- PR #3880³³⁶⁵ - Updating APEX version tag
- PR #3878³³⁶⁶ - Making sure symbols are properly exported from modules (needed for Windows/MacOS)
- PR #3877³³⁶⁷ - Documentation
- PR #3876³³⁶⁸ - Module hardware
- PR #3875³³⁶⁹ - Converted typedefs in actions submodule to using directives
- PR #3874³³⁷⁰ - Allow one to suppress target keywords in `hpx_setup_target` for backwards compatibility
- PR #3873³³⁷¹ - Add scripts to create releases and generate lists of PRs and issues
- PR #3872³³⁷² - Fix latest HTML docs location
- PR #3870³³⁷³ - Module cache
- PR #3869³³⁷⁴ - Post 1.3.0 version bumps
- PR #3868³³⁷⁵ - Replace the macro `HPX_ASSERT` by `HPX_TEST` in tests
- PR #3845³³⁷⁶ - Assertion module
- PR #3839³³⁷⁷ - Make tuple serialization non-intrusive
- PR #3832³³⁷⁸ - Config module
- PR #3799³³⁷⁹ - Remove compat namespace and its contents
- PR #3701³³⁸⁰ - MoodyCamel lockfree
- PR #3496³³⁸¹ - Disabling MPI's (deprecated) C++ interface
- PR #3192³³⁸² - Move type info into `hpx::debug` namespace and add print helper functions
- PR #3159³³⁸³ - Support Checkpointing Components

³³⁶⁵ <https://github.com/TheHPXProject/hpx/pull/3880>

³³⁶⁶ <https://github.com/TheHPXProject/hpx/pull/3878>

³³⁶⁷ <https://github.com/TheHPXProject/hpx/pull/3877>

³³⁶⁸ <https://github.com/TheHPXProject/hpx/pull/3876>

³³⁶⁹ <https://github.com/TheHPXProject/hpx/pull/3875>

³³⁷⁰ <https://github.com/TheHPXProject/hpx/pull/3874>

³³⁷¹ <https://github.com/TheHPXProject/hpx/pull/3873>

³³⁷² <https://github.com/TheHPXProject/hpx/pull/3872>

³³⁷³ <https://github.com/TheHPXProject/hpx/pull/3870>

³³⁷⁴ <https://github.com/TheHPXProject/hpx/pull/3869>

³³⁷⁵ <https://github.com/TheHPXProject/hpx/pull/3868>

³³⁷⁶ <https://github.com/TheHPXProject/hpx/pull/3845>

³³⁷⁷ <https://github.com/TheHPXProject/hpx/pull/3839>

³³⁷⁸ <https://github.com/TheHPXProject/hpx/pull/3832>

³³⁷⁹ <https://github.com/TheHPXProject/hpx/pull/3799>

³³⁸⁰ <https://github.com/TheHPXProject/hpx/pull/3701>

³³⁸¹ <https://github.com/TheHPXProject/hpx/pull/3496>

³³⁸² <https://github.com/TheHPXProject/hpx/pull/3192>

³³⁸³ <https://github.com/TheHPXProject/hpx/pull/3159>

HPX V1.3.0 (May 23, 2019)

General changes

- Performance improvements: the schedulers have significantly reduced overheads from removing false sharing and the parallel executor has been updated to create fewer futures.
- HPX now defaults to not turning on networking when running on one locality. This means that you can run multiple instances on the same system without adding command line options.
- Multiple issues reported by Clang sanitizers have been fixed.
- We have added (back) single-page HTML documentation and PDF documentation.
- We have started modularizing the HPX library. This is useful both for developers and users. In the long term users will be able to consume only parts of the HPX libraries if they do not require all the functionality that HPX currently provides.
- We have added an implementation of `function_ref`.
- The `barrier` and `latch` classes have gained a few additional member functions.

Breaking changes

- Executable and library targets are now created without the `_exe` and `_lib` suffix respectively. For example, the target `1d_stencil_1_exe` is now simply called `1d_stencil_1`.
- We have removed the following deprecated functionality: `queue`, `scoped_unlock`, and support for input iterators in algorithms.
- We have turned off the compatibility layer for `unwrapped` by default. The functionality will be removed in the next release. The option can still be turned on using the [CMake](https://www.cmake.org)³³⁸⁴ option `HPX_WITH_UNWRAPPED_SUPPORT`. Likewise, `inclusive_scan` compatibility overloads have been turned off by default. They can still be turned on with `HPX_WITH_INCLUSIVE_SCAN_COMPATIBILITY`.
- The minimum compiler and dependency versions have been updated. We now support GCC from version 5 onwards, Clang from version 4 onwards, and Boost from version 1.61.0 onwards.
- The headers for preprocessor macros have moved as a result of the functionality being moved to a separate module. The old headers are deprecated and will be removed in a future version of HPX. You can turn off the warnings by setting `HPX_PREPROCESSOR_WITH_DEPRECATION_WARNINGS=OFF` or turn off the compatibility headers completely with `HPX_PREPROCESSOR_WITH_COMPATIBILITY_HEADERS=OFF`.

Closed issues

- [Issue #3863](https://github.com/TheHPXProject/hpx/issues/3863)³³⁸⁵ - shouldn't “-faligned-new” be a usage requirement?
- [Issue #3841](https://github.com/TheHPXProject/hpx/issues/3841)³³⁸⁶ - Build error with msvc 19 caused by SFINAE and C++17
- [Issue #3836](https://github.com/TheHPXProject/hpx/issues/3836)³³⁸⁷ - master branch does not build with idle rate counters enabled
- [Issue #3819](https://github.com/TheHPXProject/hpx/issues/3819)³³⁸⁸ - Add debug suffix to modules built in debug mode

³³⁸⁴ <https://www.cmake.org>

³³⁸⁵ <https://github.com/TheHPXProject/hpx/issues/3863>

³³⁸⁶ <https://github.com/TheHPXProject/hpx/issues/3841>

³³⁸⁷ <https://github.com/TheHPXProject/hpx/issues/3836>

³³⁸⁸ <https://github.com/TheHPXProject/hpx/issues/3819>

- Issue #3817³³⁸⁹ - HPX_INCLUDE_DIRS contains non-existent directory
- Issue #3810³³⁹⁰ - Source groups are not created for files in modules
- Issue #3805³³⁹¹ - HPX won't compile with -DHPX_WITH_APEX=TRUE
- Issue #3792³³⁹² - Barrier Hangs When Locality Zero not included
- Issue #3778³³⁹³ - Replace throw() with noexcept
- Issue #3763³³⁹⁴ - configurable sort limit per task
- Issue #3758³³⁹⁵ - dataflow doesn't convert future<future<T>> to future<T>
- Issue #3757³³⁹⁶ - When compiling undefined reference to hpx::hpx_check_version_1_2 HPX V1.2.1, Ubuntu 18.04.01 Server Edition
- Issue #3753³³⁹⁷ - --hpx:list-counters=full crashes
- Issue #3746³³⁹⁸ - Detection of MPI with pmix
- Issue #3744³³⁹⁹ - Separate spinlock from same cacheline as internal data for all LCOs
- Issue #3743³⁴⁰⁰ - hpxcxx's shebang doesn't specify the python version
- Issue #3738³⁴⁰¹ - Unable to debug parcellport on a single node
- Issue #3735³⁴⁰² - Latest master: Can't compile in MSVC
- Issue #3731³⁴⁰³ - util::bound seems broken on Clang with older libstdc++
- Issue #3724³⁴⁰⁴ - Allow to pre-set command line options through environment
- Issue #3723³⁴⁰⁵ - examples/resource_partitioner build issue on master branch / ubuntu 18
- Issue #3721³⁴⁰⁶ - faced a building error
- Issue #3720³⁴⁰⁷ - Hello World example fails to link
- Issue #3719³⁴⁰⁸ - pkg-config produces invalid output: -l-pthread
- Issue #3718³⁴⁰⁹ - Please make the python executable configurable through cmake
- Issue #3717³⁴¹⁰ - interested to contribute to the organisation
- Issue #3699³⁴¹¹ - Remove 'HPX runtime' executable

³³⁸⁹ <https://github.com/TheHPXProject/hpx/issues/3817>

³³⁹⁰ <https://github.com/TheHPXProject/hpx/issues/3810>

³³⁹¹ <https://github.com/TheHPXProject/hpx/issues/3805>

³³⁹² <https://github.com/TheHPXProject/hpx/issues/3792>

³³⁹³ <https://github.com/TheHPXProject/hpx/issues/3778>

³³⁹⁴ <https://github.com/TheHPXProject/hpx/issues/3763>

³³⁹⁵ <https://github.com/TheHPXProject/hpx/issues/3758>

³³⁹⁶ <https://github.com/TheHPXProject/hpx/issues/3757>

³³⁹⁷ <https://github.com/TheHPXProject/hpx/issues/3753>

³³⁹⁸ <https://github.com/TheHPXProject/hpx/issues/3746>

³³⁹⁹ <https://github.com/TheHPXProject/hpx/issues/3744>

³⁴⁰⁰ <https://github.com/TheHPXProject/hpx/issues/3743>

³⁴⁰¹ <https://github.com/TheHPXProject/hpx/issues/3738>

³⁴⁰² <https://github.com/TheHPXProject/hpx/issues/3735>

³⁴⁰³ <https://github.com/TheHPXProject/hpx/issues/3731>

³⁴⁰⁴ <https://github.com/TheHPXProject/hpx/issues/3724>

³⁴⁰⁵ <https://github.com/TheHPXProject/hpx/issues/3723>

³⁴⁰⁶ <https://github.com/TheHPXProject/hpx/issues/3721>

³⁴⁰⁷ <https://github.com/TheHPXProject/hpx/issues/3720>

³⁴⁰⁸ <https://github.com/TheHPXProject/hpx/issues/3719>

³⁴⁰⁹ <https://github.com/TheHPXProject/hpx/issues/3718>

³⁴¹⁰ <https://github.com/TheHPXProject/hpx/issues/3717>

³⁴¹¹ <https://github.com/TheHPXProject/hpx/issues/3699>

- Issue #3698³⁴¹² - Ignore all locks while handling asserts
- Issue #3689³⁴¹³ - Incorrect and inconsistent website structure <http://stellar.cct.lsu.edu/downloads/>.
- Issue #3681³⁴¹⁴ - Broken links on <http://stellar.cct.lsu.edu/2015/05/hpx-archives-now-on-gmane/>
- Issue #3676³⁴¹⁵ - HPX master built from source, cmake fails to link main.cpp example in docs
- Issue #3673³⁴¹⁶ - HPX build fails with `std::atomic` missing error
- Issue #3670³⁴¹⁷ - Generate PDF again from documentation (with Sphinx)
- Issue #3643³⁴¹⁸ - Warnings when compiling HPX 1.2.1 with gcc 9
- Issue #3641³⁴¹⁹ - Trouble with using ranges-v3 and `hpx::parallel::reduce`
- Issue #3639³⁴²⁰ - `util::unwrapping` does not work well with member functions
- Issue #3634³⁴²¹ - The build fails if `shared_future<>::then` is called with a thread executor
- Issue #3622³⁴²² - VTune Amplifier 2019 not working with `use_itt_notify=1`
- Issue #3616³⁴²³ - HPX Fails to Build with CUDA 10
- Issue #3612³⁴²⁴ - False sharing of scheduling counters
- Issue #3609³⁴²⁵ - `executor_parameters` timeout with gcc ≤ 7 and Debug mode
- Issue #3601³⁴²⁶ - Misleading error message on power pc for `rdtsc` and `rdtscp`
- Issue #3598³⁴²⁷ - Build of some examples fails when using Vc
- Issue #3594³⁴²⁸ - Error: The number of OS threads requested (20) does not match the number of threads to bind (12): HPX(bad_parameter)
- Issue #3592³⁴²⁹ - Undefined Reference Error
- Issue #3589³⁴³⁰ - include could not find load file: HPX_Utils.cmake
- Issue #3587³⁴³¹ - HPX won't compile on POWER8 with Clang 7
- Issue #3583³⁴³² - Fedora and openSUSE instructions missing on "Distribution Packages" page
- Issue #3578³⁴³³ - Build error when configuring with `HPX_HAVE_ALGORITHM_INPUT_ITERATOR_SUPPORT=ON`
- Issue #3575³⁴³⁴ - Merge openSUSE reproducible patch

³⁴¹² <https://github.com/TheHPXProject/hpx/issues/3698>

³⁴¹³ <https://github.com/TheHPXProject/hpx/issues/3689>

³⁴¹⁴ <https://github.com/TheHPXProject/hpx/issues/3681>

³⁴¹⁵ <https://github.com/TheHPXProject/hpx/issues/3676>

³⁴¹⁶ <https://github.com/TheHPXProject/hpx/issues/3673>

³⁴¹⁷ <https://github.com/TheHPXProject/hpx/issues/3670>

³⁴¹⁸ <https://github.com/TheHPXProject/hpx/issues/3643>

³⁴¹⁹ <https://github.com/TheHPXProject/hpx/issues/3641>

³⁴²⁰ <https://github.com/TheHPXProject/hpx/issues/3639>

³⁴²¹ <https://github.com/TheHPXProject/hpx/issues/3634>

³⁴²² <https://github.com/TheHPXProject/hpx/issues/3622>

³⁴²³ <https://github.com/TheHPXProject/hpx/issues/3616>

³⁴²⁴ <https://github.com/TheHPXProject/hpx/issues/3612>

³⁴²⁵ <https://github.com/TheHPXProject/hpx/issues/3609>

³⁴²⁶ <https://github.com/TheHPXProject/hpx/issues/3601>

³⁴²⁷ <https://github.com/TheHPXProject/hpx/issues/3598>

³⁴²⁸ <https://github.com/TheHPXProject/hpx/issues/3594>

³⁴²⁹ <https://github.com/TheHPXProject/hpx/issues/3592>

³⁴³⁰ <https://github.com/TheHPXProject/hpx/issues/3589>

³⁴³¹ <https://github.com/TheHPXProject/hpx/issues/3587>

³⁴³² <https://github.com/TheHPXProject/hpx/issues/3583>

³⁴³³ <https://github.com/TheHPXProject/hpx/issues/3578>

³⁴³⁴ <https://github.com/TheHPXProject/hpx/issues/3575>

- [Issue #3570](#)³⁴³⁵ - Update HPX to work with the latest VC version
- [Issue #3567](#)³⁴³⁶ - Build succeed and make failed for `hpx:cout`
- [Issue #3565](#)³⁴³⁷ - Polymorphic simple component destructor not getting called
- [Issue #3559](#)³⁴³⁸ - 1.2.0 is missing from download page
- [Issue #3554](#)³⁴³⁹ - Clang 6.0 warning of hiding overloaded virtual function
- [Issue #3510](#)³⁴⁴⁰ - Build on ppc64 fails
- [Issue #3482](#)³⁴⁴¹ - Improve error message when `HPX_WITH_MAX_CPU_COUNT` is too low for given system
- [Issue #3453](#)³⁴⁴² - Two HPX applications can't run at the same time.
- [Issue #3452](#)³⁴⁴³ - Scaling issue on the change to 2 NUMA domains
- [Issue #3442](#)³⁴⁴⁴ - HPX `set_difference`, `set_intersection` failure cases
- [Issue #3437](#)³⁴⁴⁵ - Ensure `parent_task` pointer when child task is created and child/parent are on same locality
- [Issue #3255](#)³⁴⁴⁶ - Suspension with lock for `--hpx:list-component-types`
- [Issue #3034](#)³⁴⁴⁷ - Use C++17 structured bindings for serialization
- [Issue #2999](#)³⁴⁴⁸ - Change thread scheduling use of `size_t` for thread indexing

Closed pull requests

- [PR #3865](#)³⁴⁴⁹ - adds `hpx_target_compile_option_if_available`
- [PR #3864](#)³⁴⁵⁰ - Helper functions that are useful in numa binding and testing of allocator
- [PR #3862](#)³⁴⁵¹ - Temporary fix to `local_dataflow_boost_small_vector` test
- [PR #3860](#)³⁴⁵² - Add cache line padding to intermediate results in for loop reduction
- [PR #3859](#)³⁴⁵³ - Remove `HPX_TLL_PUBLIC` and `HPX_TLL_PRIVATE` from CMake files
- [PR #3858](#)³⁴⁵⁴ - Add compile flags and definitions to modules
- [PR #3851](#)³⁴⁵⁵ - update `hpxmp` release tag to v0.2.0
- [PR #3849](#)³⁴⁵⁶ - Correct `BOOST_ROOT` variable name in quick start guide

³⁴³⁵ <https://github.com/TheHPXProject/hpx/issues/3570>

³⁴³⁶ <https://github.com/TheHPXProject/hpx/issues/3567>

³⁴³⁷ <https://github.com/TheHPXProject/hpx/issues/3565>

³⁴³⁸ <https://github.com/TheHPXProject/hpx/issues/3559>

³⁴³⁹ <https://github.com/TheHPXProject/hpx/issues/3554>

³⁴⁴⁰ <https://github.com/TheHPXProject/hpx/issues/3510>

³⁴⁴¹ <https://github.com/TheHPXProject/hpx/issues/3482>

³⁴⁴² <https://github.com/TheHPXProject/hpx/issues/3453>

³⁴⁴³ <https://github.com/TheHPXProject/hpx/issues/3452>

³⁴⁴⁴ <https://github.com/TheHPXProject/hpx/issues/3442>

³⁴⁴⁵ <https://github.com/TheHPXProject/hpx/issues/3437>

³⁴⁴⁶ <https://github.com/TheHPXProject/hpx/issues/3255>

³⁴⁴⁷ <https://github.com/TheHPXProject/hpx/issues/3034>

³⁴⁴⁸ <https://github.com/TheHPXProject/hpx/issues/2999>

³⁴⁴⁹ <https://github.com/TheHPXProject/hpx/pull/3865>

³⁴⁵⁰ <https://github.com/TheHPXProject/hpx/pull/3864>

³⁴⁵¹ <https://github.com/TheHPXProject/hpx/pull/3862>

³⁴⁵² <https://github.com/TheHPXProject/hpx/pull/3860>

³⁴⁵³ <https://github.com/TheHPXProject/hpx/pull/3859>

³⁴⁵⁴ <https://github.com/TheHPXProject/hpx/pull/3858>

³⁴⁵⁵ <https://github.com/TheHPXProject/hpx/pull/3851>

³⁴⁵⁶ <https://github.com/TheHPXProject/hpx/pull/3849>

- PR #3847³⁴⁵⁷ - Fix attach_debugger configuration option
- PR #3846³⁴⁵⁸ - Add tests for libs header tests
- PR #3844³⁴⁵⁹ - Fixing source_groups in preprocessor module to properly handle compatibility headers
- PR #3843³⁴⁶⁰ - This fixes the launch_process/launched_process pair of tests
- PR #3842³⁴⁶¹ - Fix macro call with ITTNOTIFY enabled
- PR #3840³⁴⁶² - Fixing SLURM environment parsing
- PR #3837³⁴⁶³ - Fixing misplaced #endif
- PR #3835³⁴⁶⁴ - make all latch members protected for consistency
- PR #3834³⁴⁶⁵ - Disable transpose_block_numa example on CircleCI
- PR #3833³⁴⁶⁶ - make latch **counter_** protected for deriving latch in hpxmp
- PR #3831³⁴⁶⁷ - Fix CircleCI config for modules
- PR #3830³⁴⁶⁸ - minor fix: option HPX_WITH_TEST was not working correctly
- PR #3828³⁴⁶⁹ - Avoid for binaries that depend on HPX to directly link against internal modules
- PR #3827³⁴⁷⁰ - Adding shortcut for `hpx::get_ptr<>(sync, id)` for a local, non-migratable objects
- PR #3826³⁴⁷¹ - Fix and update modules documentation
- PR #3825³⁴⁷² - Updating default APEX version to 2.1.3 with HPX
- PR #3823³⁴⁷³ - Fix pkgconfig libs handling
- PR #3822³⁴⁷⁴ - Change includes in hpx_wrap.cpp to more specific includes
- PR #3821³⁴⁷⁵ - Disable barrier_3792 test when networking is disabled
- PR #3820³⁴⁷⁶ - Assorted CMake fixes
- PR #3815³⁴⁷⁷ - Removing left-over debug output
- PR #3814³⁴⁷⁸ - Allow setting default scheduler mode via the configuration database
- PR #3813³⁴⁷⁹ - Make the deprecation warnings issued by the old pp headers optional

³⁴⁵⁷ <https://github.com/TheHPXProject/hpx/pull/3847>

³⁴⁵⁸ <https://github.com/TheHPXProject/hpx/pull/3846>

³⁴⁵⁹ <https://github.com/TheHPXProject/hpx/pull/3844>

³⁴⁶⁰ <https://github.com/TheHPXProject/hpx/pull/3843>

³⁴⁶¹ <https://github.com/TheHPXProject/hpx/pull/3842>

³⁴⁶² <https://github.com/TheHPXProject/hpx/pull/3840>

³⁴⁶³ <https://github.com/TheHPXProject/hpx/pull/3837>

³⁴⁶⁴ <https://github.com/TheHPXProject/hpx/pull/3835>

³⁴⁶⁵ <https://github.com/TheHPXProject/hpx/pull/3834>

³⁴⁶⁶ <https://github.com/TheHPXProject/hpx/pull/3833>

³⁴⁶⁷ <https://github.com/TheHPXProject/hpx/pull/3831>

³⁴⁶⁸ <https://github.com/TheHPXProject/hpx/pull/3830>

³⁴⁶⁹ <https://github.com/TheHPXProject/hpx/pull/3828>

³⁴⁷⁰ <https://github.com/TheHPXProject/hpx/pull/3827>

³⁴⁷¹ <https://github.com/TheHPXProject/hpx/pull/3826>

³⁴⁷² <https://github.com/TheHPXProject/hpx/pull/3825>

³⁴⁷³ <https://github.com/TheHPXProject/hpx/pull/3823>

³⁴⁷⁴ <https://github.com/TheHPXProject/hpx/pull/3822>

³⁴⁷⁵ <https://github.com/TheHPXProject/hpx/pull/3821>

³⁴⁷⁶ <https://github.com/TheHPXProject/hpx/pull/3820>

³⁴⁷⁷ <https://github.com/TheHPXProject/hpx/pull/3815>

³⁴⁷⁸ <https://github.com/TheHPXProject/hpx/pull/3814>

³⁴⁷⁹ <https://github.com/TheHPXProject/hpx/pull/3813>

- PR #3812³⁴⁸⁰ - Windows requires to handle symlinks to directories differently from those linking files
- PR #3811³⁴⁸¹ - Clean up PP module and library skeleton
- PR #3806³⁴⁸² - Moving include path configuration to before APEX
- PR #3804³⁴⁸³ - Fix latch
- PR #3803³⁴⁸⁴ - Update hpxcxx to look at lib64 and use python3
- PR #3802³⁴⁸⁵ - Numa binding allocator
- PR #3801³⁴⁸⁶ - Remove duplicated includes
- PR #3800³⁴⁸⁷ - Attempt to fix Posix context switching after lazy init changes
- PR #3798³⁴⁸⁸ - count and count_if accepts different iterator types
- PR #3797³⁴⁸⁹ - Adding a couple of `override` keywords to overloaded virtual functions
- PR #3796³⁴⁹⁰ - Re-enable testing all schedulers in `shutdown_suspended_test`
- PR #3795³⁴⁹¹ - Change `std::terminate` to `std::abort` in SIGSEGV handler
- PR #3794³⁴⁹² - Fixing #3792
- PR #3793³⁴⁹³ - Extending `migrate_polymorphic_component` unit test
- PR #3791³⁴⁹⁴ - Change `throw()` to `noexcept`
- PR #3790³⁴⁹⁵ - Remove deprecated options for 1.3.0 release
- PR #3789³⁴⁹⁶ - Remove Boost filesystem compatibility header
- PR #3788³⁴⁹⁷ - Disabled even more spots that should not execute if networking is disabled
- PR #3787³⁴⁹⁸ - Bump minimal boost supported version to 1.61.0
- PR #3786³⁴⁹⁹ - Bump minimum required versions for 1.3.0 release
- PR #3785³⁵⁰⁰ - Explicitly set number of jobs for all ninja invocations on CircleCI
- PR #3784³⁵⁰¹ - Fix leak and address sanitizer problems
- PR #3783³⁵⁰² - Disabled even more spots that should not execute is networking is disabled

³⁴⁸⁰ <https://github.com/TheHPXProject/hpx/pull/3812>

³⁴⁸¹ <https://github.com/TheHPXProject/hpx/pull/3811>

³⁴⁸² <https://github.com/TheHPXProject/hpx/pull/3806>

³⁴⁸³ <https://github.com/TheHPXProject/hpx/pull/3804>

³⁴⁸⁴ <https://github.com/TheHPXProject/hpx/pull/3803>

³⁴⁸⁵ <https://github.com/TheHPXProject/hpx/pull/3802>

³⁴⁸⁶ <https://github.com/TheHPXProject/hpx/pull/3801>

³⁴⁸⁷ <https://github.com/TheHPXProject/hpx/pull/3800>

³⁴⁸⁸ <https://github.com/TheHPXProject/hpx/pull/3798>

³⁴⁸⁹ <https://github.com/TheHPXProject/hpx/pull/3797>

³⁴⁹⁰ <https://github.com/TheHPXProject/hpx/pull/3796>

³⁴⁹¹ <https://github.com/TheHPXProject/hpx/pull/3795>

³⁴⁹² <https://github.com/TheHPXProject/hpx/pull/3794>

³⁴⁹³ <https://github.com/TheHPXProject/hpx/pull/3793>

³⁴⁹⁴ <https://github.com/TheHPXProject/hpx/pull/3791>

³⁴⁹⁵ <https://github.com/TheHPXProject/hpx/pull/3790>

³⁴⁹⁶ <https://github.com/TheHPXProject/hpx/pull/3789>

³⁴⁹⁷ <https://github.com/TheHPXProject/hpx/pull/3788>

³⁴⁹⁸ <https://github.com/TheHPXProject/hpx/pull/3787>

³⁴⁹⁹ <https://github.com/TheHPXProject/hpx/pull/3786>

³⁵⁰⁰ <https://github.com/TheHPXProject/hpx/pull/3785>

³⁵⁰¹ <https://github.com/TheHPXProject/hpx/pull/3784>

³⁵⁰² <https://github.com/TheHPXProject/hpx/pull/3783>

- PR #3782³⁵⁰³ - Cherry-picked tuple and thread_init_data fixes from #3701
- PR #3781³⁵⁰⁴ - Fix generic context coroutines after lazy stack allocation changes
- PR #3780³⁵⁰⁵ - Rename hello world examples
- PR #3776³⁵⁰⁶ - Sort algorithms now use the supplied chunker to determine the required minimal chunk size
- PR #3775³⁵⁰⁷ - Disable Boost auto-linking
- PR #3774³⁵⁰⁸ - Tag and push stable builds
- PR #3773³⁵⁰⁹ - Enable migration of polymorphic components
- PR #3771³⁵¹⁰ - Fix link to stackoverflow in documentation
- PR #3770³⁵¹¹ - Replacing constexpr if in brace-serialization code
- PR #3769³⁵¹² - Fix SIGSEGV handler
- PR #3768³⁵¹³ - Adding flags to scheduler allowing to control thread stealing and idle back-off
- PR #3767³⁵¹⁴ - Fix help formatting in hpxrun.py
- PR #3765³⁵¹⁵ - Fix a couple of bugs in the thread test
- PR #3764³⁵¹⁶ - Workaround for SFINAE regression in msvc14.2
- PR #3762³⁵¹⁷ - Prevent MSVC from prematurely instantiating things
- PR #3761³⁵¹⁸ - Update python scripts to work with python 3
- PR #3760³⁵¹⁹ - Fix callable vtable for GCC4.9
- PR #3759³⁵²⁰ - Rename PAGE_SIZE to PAGE_SIZE_ because AppleClang
- PR #3755³⁵²¹ - Making sure locks are not held during suspension
- PR #3754³⁵²² - Disable more code if networking is not available/not enabled
- PR #3752³⁵²³ - Move util::format implementation to source file
- PR #3751³⁵²⁴ - Fixing problems with lcos::barrier and iostreams
- PR #3750³⁵²⁵ - Change error message to take into account use_guard_page setting

³⁵⁰³ <https://github.com/TheHPXProject/hpx/pull/3782>

³⁵⁰⁴ <https://github.com/TheHPXProject/hpx/pull/3781>

³⁵⁰⁵ <https://github.com/TheHPXProject/hpx/pull/3780>

³⁵⁰⁶ <https://github.com/TheHPXProject/hpx/pull/3776>

³⁵⁰⁷ <https://github.com/TheHPXProject/hpx/pull/3775>

³⁵⁰⁸ <https://github.com/TheHPXProject/hpx/pull/3774>

³⁵⁰⁹ <https://github.com/TheHPXProject/hpx/pull/3773>

³⁵¹⁰ <https://github.com/TheHPXProject/hpx/pull/3771>

³⁵¹¹ <https://github.com/TheHPXProject/hpx/pull/3770>

³⁵¹² <https://github.com/TheHPXProject/hpx/pull/3769>

³⁵¹³ <https://github.com/TheHPXProject/hpx/pull/3768>

³⁵¹⁴ <https://github.com/TheHPXProject/hpx/pull/3767>

³⁵¹⁵ <https://github.com/TheHPXProject/hpx/pull/3765>

³⁵¹⁶ <https://github.com/TheHPXProject/hpx/pull/3764>

³⁵¹⁷ <https://github.com/TheHPXProject/hpx/pull/3762>

³⁵¹⁸ <https://github.com/TheHPXProject/hpx/pull/3761>

³⁵¹⁹ <https://github.com/TheHPXProject/hpx/pull/3760>

³⁵²⁰ <https://github.com/TheHPXProject/hpx/pull/3759>

³⁵²¹ <https://github.com/TheHPXProject/hpx/pull/3755>

³⁵²² <https://github.com/TheHPXProject/hpx/pull/3754>

³⁵²³ <https://github.com/TheHPXProject/hpx/pull/3752>

³⁵²⁴ <https://github.com/TheHPXProject/hpx/pull/3751>

³⁵²⁵ <https://github.com/TheHPXProject/hpx/pull/3750>

- PR #3749³⁵²⁶ - Fix lifetime problem in `run_as_hpx_thread`
- PR #3748³⁵²⁷ - Fixed unusable behavior of the clang code analyzer.
- PR #3747³⁵²⁸ - Added `PMIX_RANK` to the defaults of `HPX_WITH_PARCELPOR_MPI_ENV`.
- PR #3745³⁵²⁹ - Introduced `cache_aligned_data` and `cache_line_data` helper structure
- PR #3742³⁵³⁰ - Remove more unused functionality from `util/logging`
- PR #3740³⁵³¹ - Fix includes in partitioned vector tests
- PR #3739³⁵³² - More fixes to make sure that `std::flush` really flushes all output
- PR #3737³⁵³³ - Fix potential shutdown problems
- PR #3736³⁵³⁴ - Fix `guided_pool_executor` after dataflow changes caused compilation fail
- PR #3734³⁵³⁵ - Limiting executor
- PR #3732³⁵³⁶ - More constrained bound constructors
- PR #3730³⁵³⁷ - Attempt to fix deadlocks during component loading
- PR #3729³⁵³⁸ - Add latch member function `count_up` and `reset`, requested by `hpxMP`
- PR #3728³⁵³⁹ - Send even empty buffers on `hpx::endl` and `hpx::flush`
- PR #3727³⁵⁴⁰ - Adding example demonstrating how to customize the memory management for a component
- PR #3726³⁵⁴¹ - Adding support for passing command line options through the `HPX_COMMANDLINE_OPTIONS` environment variable
- PR #3722³⁵⁴² - Document known broken OpenMPI builds
- PR #3716³⁵⁴³ - Add barrier reset function, requested by `hpxMP` for reusing barrier
- PR #3715³⁵⁴⁴ - More work on functions and `vtables`
- PR #3714³⁵⁴⁵ - Generate single-page HTML, PDF, manpage from documentation
- PR #3713³⁵⁴⁶ - Updating default APEX version to 2.1.2
- PR #3712³⁵⁴⁷ - Update release procedure
- PR #3710³⁵⁴⁸ - Fix the C++11 build, after #3704

³⁵²⁶ <https://github.com/TheHPXProject/hpx/pull/3749>

³⁵²⁷ <https://github.com/TheHPXProject/hpx/pull/3748>

³⁵²⁸ <https://github.com/TheHPXProject/hpx/pull/3747>

³⁵²⁹ <https://github.com/TheHPXProject/hpx/pull/3745>

³⁵³⁰ <https://github.com/TheHPXProject/hpx/pull/3742>

³⁵³¹ <https://github.com/TheHPXProject/hpx/pull/3740>

³⁵³² <https://github.com/TheHPXProject/hpx/pull/3739>

³⁵³³ <https://github.com/TheHPXProject/hpx/pull/3737>

³⁵³⁴ <https://github.com/TheHPXProject/hpx/pull/3736>

³⁵³⁵ <https://github.com/TheHPXProject/hpx/pull/3734>

³⁵³⁶ <https://github.com/TheHPXProject/hpx/pull/3732>

³⁵³⁷ <https://github.com/TheHPXProject/hpx/pull/3730>

³⁵³⁸ <https://github.com/TheHPXProject/hpx/pull/3729>

³⁵³⁹ <https://github.com/TheHPXProject/hpx/pull/3728>

³⁵⁴⁰ <https://github.com/TheHPXProject/hpx/pull/3727>

³⁵⁴¹ <https://github.com/TheHPXProject/hpx/pull/3726>

³⁵⁴² <https://github.com/TheHPXProject/hpx/pull/3722>

³⁵⁴³ <https://github.com/TheHPXProject/hpx/pull/3716>

³⁵⁴⁴ <https://github.com/TheHPXProject/hpx/pull/3715>

³⁵⁴⁵ <https://github.com/TheHPXProject/hpx/pull/3714>

³⁵⁴⁶ <https://github.com/TheHPXProject/hpx/pull/3713>

³⁵⁴⁷ <https://github.com/TheHPXProject/hpx/pull/3712>

³⁵⁴⁸ <https://github.com/TheHPXProject/hpx/pull/3710>

- PR #3709³⁵⁴⁹ - Move some component_registry functionality to source file
- PR #3708³⁵⁵⁰ - Ignore all locks while handling assertions
- PR #3707³⁵⁵¹ - Remove obsolete hpx runtime executable
- PR #3705³⁵⁵² - Fix and simplify make_ready_future overload sets
- PR #3704³⁵⁵³ - Reduce use of binders
- PR #3703³⁵⁵⁴ - Ini
- PR #3702³⁵⁵⁵ - Fixing CUDA compiler errors
- PR #3700³⁵⁵⁶ - Added barrier::increment function to increase total number of thread
- PR #3697³⁵⁵⁷ - One more attempt to fix migration...
- PR #3694³⁵⁵⁸ - Fixing component migration
- PR #3693³⁵⁵⁹ - Print thread state when getting disallowed value in set_thread_state
- PR #3692³⁵⁶⁰ - Only disable constexpr with clang-cuda, not nvcc+gcc
- PR #3691³⁵⁶¹ - Link with libsupc++ if needed for thread_local
- PR #3690³⁵⁶² - Remove thousands separators in set_operations_3442 to comply with C++11
- PR #3688³⁵⁶³ - Decouple serialization from function vtables
- PR #3687³⁵⁶⁴ - Fix a couple of test failures
- PR #3686³⁵⁶⁵ - Make sure tests.unit.build are run after install on CircleCI
- PR #3685³⁵⁶⁶ - Revise quickstart CMakeLists.txt explanation
- PR #3684³⁵⁶⁷ - Provide concept emulation for Ranges-TS concepts
- PR #3683³⁵⁶⁸ - Ignore uninitialized chunks
- PR #3682³⁵⁶⁹ - Ignore uninitialized chunks. Check proper indices.
- PR #3680³⁵⁷⁰ - Ignore uninitialized chunks. Check proper range indices
- PR #3679³⁵⁷¹ - Simplify basic action implementations

³⁵⁴⁹ <https://github.com/TheHPXProject/hpx/pull/3709>

³⁵⁵⁰ <https://github.com/TheHPXProject/hpx/pull/3708>

³⁵⁵¹ <https://github.com/TheHPXProject/hpx/pull/3707>

³⁵⁵² <https://github.com/TheHPXProject/hpx/pull/3705>

³⁵⁵³ <https://github.com/TheHPXProject/hpx/pull/3704>

³⁵⁵⁴ <https://github.com/TheHPXProject/hpx/pull/3703>

³⁵⁵⁵ <https://github.com/TheHPXProject/hpx/pull/3702>

³⁵⁵⁶ <https://github.com/TheHPXProject/hpx/pull/3700>

³⁵⁵⁷ <https://github.com/TheHPXProject/hpx/pull/3697>

³⁵⁵⁸ <https://github.com/TheHPXProject/hpx/pull/3694>

³⁵⁵⁹ <https://github.com/TheHPXProject/hpx/pull/3693>

³⁵⁶⁰ <https://github.com/TheHPXProject/hpx/pull/3692>

³⁵⁶¹ <https://github.com/TheHPXProject/hpx/pull/3691>

³⁵⁶² <https://github.com/TheHPXProject/hpx/pull/3690>

³⁵⁶³ <https://github.com/TheHPXProject/hpx/pull/3688>

³⁵⁶⁴ <https://github.com/TheHPXProject/hpx/pull/3687>

³⁵⁶⁵ <https://github.com/TheHPXProject/hpx/pull/3686>

³⁵⁶⁶ <https://github.com/TheHPXProject/hpx/pull/3685>

³⁵⁶⁷ <https://github.com/TheHPXProject/hpx/pull/3684>

³⁵⁶⁸ <https://github.com/TheHPXProject/hpx/pull/3683>

³⁵⁶⁹ <https://github.com/TheHPXProject/hpx/pull/3682>

³⁵⁷⁰ <https://github.com/TheHPXProject/hpx/pull/3680>

³⁵⁷¹ <https://github.com/TheHPXProject/hpx/pull/3679>

- PR #3678³⁵⁷² - Making sure HPX_HAVE_LIBATOMIC is unset before checking
- PR #3677³⁵⁷³ - Fix generated full version number to be usable in expressions
- PR #3674³⁵⁷⁴ - Reduce functional utilities call depth
- PR #3672³⁵⁷⁵ - Change new build system to use existing macros related to pseudo dependencies
- PR #3669³⁵⁷⁶ - Remove indirection in `function_ref` when thread description is disabled
- PR #3668³⁵⁷⁷ - Unbreaking `async_*cb*` tests
- PR #3667³⁵⁷⁸ - Generate `version.hpp`
- PR #3665³⁵⁷⁹ - Enabling MPI parcelport for gitlab runners
- PR #3664³⁵⁸⁰ - making clang-tidy work properly again
- PR #3662³⁵⁸¹ - Attempt to fix exception handling
- PR #3661³⁵⁸² - Move `lcos::latch` to source file
- PR #3660³⁵⁸³ - Fix accidentally explicit `gid_type` default constructor
- PR #3659³⁵⁸⁴ - Parallel executor latch
- PR #3658³⁵⁸⁵ - Fixing `execution_parameters`
- PR #3657³⁵⁸⁶ - Avoid dangling references in `wait_all`
- PR #3656³⁵⁸⁷ - Avoiding lifetime problems with `sync_put_parcel`
- PR #3655³⁵⁸⁸ - Fixing `nullptr` dereference inside of function
- PR #3652³⁵⁸⁹ - Attempt to fix `thread_map_type` definition with C++11
- PR #3650³⁵⁹⁰ - Allowing for end iterator being different from begin iterator
- PR #3649³⁵⁹¹ - Added architecture identification to cmake to be able to detect timestamp support
- PR #3645³⁵⁹² - Enabling sanitizers on gitlab runner
- PR #3644³⁵⁹³ - Attempt to tackle timeouts during startup
- PR #3642³⁵⁹⁴ - Cleanup parallel partitioners

³⁵⁷² <https://github.com/TheHPXProject/hpx/pull/3678>

³⁵⁷³ <https://github.com/TheHPXProject/hpx/pull/3677>

³⁵⁷⁴ <https://github.com/TheHPXProject/hpx/pull/3674>

³⁵⁷⁵ <https://github.com/TheHPXProject/hpx/pull/3672>

³⁵⁷⁶ <https://github.com/TheHPXProject/hpx/pull/3669>

³⁵⁷⁷ <https://github.com/TheHPXProject/hpx/pull/3668>

³⁵⁷⁸ <https://github.com/TheHPXProject/hpx/pull/3667>

³⁵⁷⁹ <https://github.com/TheHPXProject/hpx/pull/3665>

³⁵⁸⁰ <https://github.com/TheHPXProject/hpx/pull/3664>

³⁵⁸¹ <https://github.com/TheHPXProject/hpx/pull/3662>

³⁵⁸² <https://github.com/TheHPXProject/hpx/pull/3661>

³⁵⁸³ <https://github.com/TheHPXProject/hpx/pull/3660>

³⁵⁸⁴ <https://github.com/TheHPXProject/hpx/pull/3659>

³⁵⁸⁵ <https://github.com/TheHPXProject/hpx/pull/3658>

³⁵⁸⁶ <https://github.com/TheHPXProject/hpx/pull/3657>

³⁵⁸⁷ <https://github.com/TheHPXProject/hpx/pull/3656>

³⁵⁸⁸ <https://github.com/TheHPXProject/hpx/pull/3655>

³⁵⁸⁹ <https://github.com/TheHPXProject/hpx/pull/3652>

³⁵⁹⁰ <https://github.com/TheHPXProject/hpx/pull/3650>

³⁵⁹¹ <https://github.com/TheHPXProject/hpx/pull/3649>

³⁵⁹² <https://github.com/TheHPXProject/hpx/pull/3645>

³⁵⁹³ <https://github.com/TheHPXProject/hpx/pull/3644>

³⁵⁹⁴ <https://github.com/TheHPXProject/hpx/pull/3642>

- PR #3640³⁵⁹⁵ - Dataflow now works with functions that return a reference
- PR #3637³⁵⁹⁶ - Merging the executor-enabled overloads of `shared_future<>::then`
- PR #3633³⁵⁹⁷ - Replace deprecated boost endian macros
- PR #3632³⁵⁹⁸ - Add instructions on getting HPX to documentation
- PR #3631³⁵⁹⁹ - Simplify parcel creation
- PR #3630³⁶⁰⁰ - Small additions and fixes to release procedure
- PR #3629³⁶⁰¹ - Modular pp
- PR #3627³⁶⁰² - Implement `util::function_ref`
- PR #3626³⁶⁰³ - Fix cancelable_action_client example
- PR #3625³⁶⁰⁴ - Added automatic serialization for simple structs (see #3034)
- PR #3624³⁶⁰⁵ - Updating the default order of priority for `thread_description`
- PR #3621³⁶⁰⁶ - Update copyright year and other small formatting fixes
- PR #3620³⁶⁰⁷ - Adding support for gitlab runner
- PR #3619³⁶⁰⁸ - Store debug logs and core dumps on CircleCI
- PR #3618³⁶⁰⁹ - Various optimizations
- PR #3617³⁶¹⁰ - Fix link to the gpg key (#2)
- PR #3615³⁶¹¹ - Fix unused variable warnings with networking off
- PR #3614³⁶¹² - Restructuring counter data in scheduler to reduce false sharing
- PR #3613³⁶¹³ - Adding support for gitlab runners
- PR #3610³⁶¹⁴ - Don't wait for `stop_condition` in main thread
- PR #3608³⁶¹⁵ - Add inline keyword to `invalid_thread_id` definition for nvcc
- PR #3607³⁶¹⁶ - Adding configuration key that allows one to explicitly add a directory to the component search path
- PR #3606³⁶¹⁷ - Add nvcc to exclude constexpress since is it not supported by nvcc

³⁵⁹⁵ <https://github.com/TheHPXProject/hpx/pull/3640>

³⁵⁹⁶ <https://github.com/TheHPXProject/hpx/pull/3637>

³⁵⁹⁷ <https://github.com/TheHPXProject/hpx/pull/3633>

³⁵⁹⁸ <https://github.com/TheHPXProject/hpx/pull/3632>

³⁵⁹⁹ <https://github.com/TheHPXProject/hpx/pull/3631>

³⁶⁰⁰ <https://github.com/TheHPXProject/hpx/pull/3630>

³⁶⁰¹ <https://github.com/TheHPXProject/hpx/pull/3629>

³⁶⁰² <https://github.com/TheHPXProject/hpx/pull/3627>

³⁶⁰³ <https://github.com/TheHPXProject/hpx/pull/3626>

³⁶⁰⁴ <https://github.com/TheHPXProject/hpx/pull/3625>

³⁶⁰⁵ <https://github.com/TheHPXProject/hpx/pull/3624>

³⁶⁰⁶ <https://github.com/TheHPXProject/hpx/pull/3621>

³⁶⁰⁷ <https://github.com/TheHPXProject/hpx/pull/3620>

³⁶⁰⁸ <https://github.com/TheHPXProject/hpx/pull/3619>

³⁶⁰⁹ <https://github.com/TheHPXProject/hpx/pull/3618>

³⁶¹⁰ <https://github.com/TheHPXProject/hpx/pull/3617>

³⁶¹¹ <https://github.com/TheHPXProject/hpx/pull/3615>

³⁶¹² <https://github.com/TheHPXProject/hpx/pull/3614>

³⁶¹³ <https://github.com/TheHPXProject/hpx/pull/3613>

³⁶¹⁴ <https://github.com/TheHPXProject/hpx/pull/3610>

³⁶¹⁵ <https://github.com/TheHPXProject/hpx/pull/3608>

³⁶¹⁶ <https://github.com/TheHPXProject/hpx/pull/3607>

³⁶¹⁷ <https://github.com/TheHPXProject/hpx/pull/3606>

- PR #3605³⁶¹⁸ - Add `inline` to definition of checkpoint stream operators to fix link error
- PR #3604³⁶¹⁹ - Use format for string formatting
- PR #3603³⁶²⁰ - Improve the error message for using to less `MAX_CPU_COUNT`
- PR #3602³⁶²¹ - Improve the error message for to small values of `MAX_CPU_COUNT`
- PR #3600³⁶²² - Parallel executor aggregated
- PR #3599³⁶²³ - Making sure networking is disabled for default one-locality-runs
- PR #3596³⁶²⁴ - Store thread exit functions in `forward_list` instead of `deque` to avoid allocations
- PR #3590³⁶²⁵ - Fix typo/mistake in thread queue `cleanup_terminated`
- PR #3588³⁶²⁶ - Fix formatting errors in `launching_and_configuring_hpx_applications.rst`
- PR #3586³⁶²⁷ - Make `bind` propagate value category
- PR #3585³⁶²⁸ - Extend Cmake for building `hpx` as distribution packages (refs #3575)
- PR #3584³⁶²⁹ - Untangle function storage from object pointer
- PR #3582³⁶³⁰ - Towards Modularized HPX
- PR #3580³⁶³¹ - Remove extra `||` in `merge.hpp`
- PR #3577³⁶³² - Partially revert “Remove vtable empty flag”
- PR #3576³⁶³³ - Make sure empty startup/shutdown functions are not being used
- PR #3574³⁶³⁴ - Make sure `DATAPAR` settings are conveyed to depending projects
- PR #3573³⁶³⁵ - Make sure HPX is usable with latest released version of Vc (V1.4.1)
- PR #3572³⁶³⁶ - Adding test ensuring ticket 3565 is fixed
- PR #3571³⁶³⁷ - Make empty `[unique_]function` vtable non-dependent
- PR #3566³⁶³⁸ - Fix compilation with dynamic bitset for CPU masks
- PR #3563³⁶³⁹ - Drop `util::[unique_]function` target_type
- PR #3562³⁶⁴⁰ - Removing the target suffixes

³⁶¹⁸ <https://github.com/TheHPXProject/hpx/pull/3605>

³⁶¹⁹ <https://github.com/TheHPXProject/hpx/pull/3604>

³⁶²⁰ <https://github.com/TheHPXProject/hpx/pull/3603>

³⁶²¹ <https://github.com/TheHPXProject/hpx/pull/3602>

³⁶²² <https://github.com/TheHPXProject/hpx/pull/3600>

³⁶²³ <https://github.com/TheHPXProject/hpx/pull/3599>

³⁶²⁴ <https://github.com/TheHPXProject/hpx/pull/3596>

³⁶²⁵ <https://github.com/TheHPXProject/hpx/pull/3590>

³⁶²⁶ <https://github.com/TheHPXProject/hpx/pull/3588>

³⁶²⁷ <https://github.com/TheHPXProject/hpx/pull/3586>

³⁶²⁸ <https://github.com/TheHPXProject/hpx/pull/3585>

³⁶²⁹ <https://github.com/TheHPXProject/hpx/pull/3584>

³⁶³⁰ <https://github.com/TheHPXProject/hpx/pull/3582>

³⁶³¹ <https://github.com/TheHPXProject/hpx/pull/3580>

³⁶³² <https://github.com/TheHPXProject/hpx/pull/3577>

³⁶³³ <https://github.com/TheHPXProject/hpx/pull/3576>

³⁶³⁴ <https://github.com/TheHPXProject/hpx/pull/3574>

³⁶³⁵ <https://github.com/TheHPXProject/hpx/pull/3573>

³⁶³⁶ <https://github.com/TheHPXProject/hpx/pull/3572>

³⁶³⁷ <https://github.com/TheHPXProject/hpx/pull/3571>

³⁶³⁸ <https://github.com/TheHPXProject/hpx/pull/3566>

³⁶³⁹ <https://github.com/TheHPXProject/hpx/pull/3563>

³⁶⁴⁰ <https://github.com/TheHPXProject/hpx/pull/3562>

- [PR #3561](#)³⁶⁴¹ - Replace executor traits return type deduction (keep non-SFINAE)
- [PR #3557](#)³⁶⁴² - Replace the last usages of `boost::atomic`
- [PR #3556](#)³⁶⁴³ - Replace `boost::scoped_array` with `std::unique_ptr`
- [PR #3552](#)³⁶⁴⁴ - (Re)move APEX readme
- [PR #3548](#)³⁶⁴⁵ - Replace `boost::scoped_ptr` with `std::unique_ptr`
- [PR #3547](#)³⁶⁴⁶ - Remove last use of `Boost.Signals2`
- [PR #3544](#)³⁶⁴⁷ - Post 1.2.0 version bumps
- [PR #3543](#)³⁶⁴⁸ - added Ubuntu dependency list to readme
- [PR #3531](#)³⁶⁴⁹ - Warnings, warnings...
- [PR #3527](#)³⁶⁵⁰ - Add CircleCI filter for building all tags
- [PR #3525](#)³⁶⁵¹ - Segmented algorithms
- [PR #3517](#)³⁶⁵² - Replace `boost::regex` with C++11 `<regex>`
- [PR #3514](#)³⁶⁵³ - Cleaning up the build system
- [PR #3505](#)³⁶⁵⁴ - Fixing type attribute warning for `transfer_action`
- [PR #3504](#)³⁶⁵⁵ - Add support for rpm packaging
- [PR #3499](#)³⁶⁵⁶ - Improving spinlock pools
- [PR #3498](#)³⁶⁵⁷ - Remove thread specific ptr
- [PR #3486](#)³⁶⁵⁸ - Fix comparison for `expect_connecting_localities` config entry
- [PR #3469](#)³⁶⁵⁹ - Enable (existing) code for extracting stack pointer on Power platform

³⁶⁴¹ <https://github.com/TheHPXProject/hpx/pull/3561>

³⁶⁴² <https://github.com/TheHPXProject/hpx/pull/3557>

³⁶⁴³ <https://github.com/TheHPXProject/hpx/pull/3556>

³⁶⁴⁴ <https://github.com/TheHPXProject/hpx/pull/3552>

³⁶⁴⁵ <https://github.com/TheHPXProject/hpx/pull/3548>

³⁶⁴⁶ <https://github.com/TheHPXProject/hpx/pull/3547>

³⁶⁴⁷ <https://github.com/TheHPXProject/hpx/pull/3544>

³⁶⁴⁸ <https://github.com/TheHPXProject/hpx/pull/3543>

³⁶⁴⁹ <https://github.com/TheHPXProject/hpx/pull/3531>

³⁶⁵⁰ <https://github.com/TheHPXProject/hpx/pull/3527>

³⁶⁵¹ <https://github.com/TheHPXProject/hpx/pull/3525>

³⁶⁵² <https://github.com/TheHPXProject/hpx/pull/3517>

³⁶⁵³ <https://github.com/TheHPXProject/hpx/pull/3514>

³⁶⁵⁴ <https://github.com/TheHPXProject/hpx/pull/3505>

³⁶⁵⁵ <https://github.com/TheHPXProject/hpx/pull/3504>

³⁶⁵⁶ <https://github.com/TheHPXProject/hpx/pull/3499>

³⁶⁵⁷ <https://github.com/TheHPXProject/hpx/pull/3498>

³⁶⁵⁸ <https://github.com/TheHPXProject/hpx/pull/3486>

³⁶⁵⁹ <https://github.com/TheHPXProject/hpx/pull/3469>

HPX V1.2.1 (Feb 19, 2019)

General changes

This is a bugfix release. It contains the following changes:

- Fix compilation on ARM, s390x and 32-bit architectures.
- Fix a critical bug in the `future` implementation.
- Fix several problems in the CMake configuration which affects external projects.
- Add support for Boost 1.69.0.

Closed issues

- Issue #3638³⁶⁶⁰ - Build HPX 1.2 with boost 1.69
- Issue #3635³⁶⁶¹ - Non-deterministic crashing on Stampede2
- Issue #3550³⁶⁶² - `1>e:000workhpxsrcthrow_exception.cpp(54): error C2440: '<function-style-cast>': cannot convert from 'boost::system::error_code' to 'hpx::exception'`
- Issue #3549³⁶⁶³ - HPX 1.2.0 does not build on i686, but release candidate did
- Issue #3511³⁶⁶⁴ - Build on s390x fails
- Issue #3509³⁶⁶⁵ - Build on armv7l fails

Closed pull requests

- PR #3695³⁶⁶⁶ - Don't install CMake templates and packaging files
- PR #3666³⁶⁶⁷ - Fixing yet another race in `future_data`
- PR #3663³⁶⁶⁸ - Fixing race between setting and getting the value inside `future_data`
- PR #3648³⁶⁶⁹ - Adding timestamp option for S390x platform
- PR #3647³⁶⁷⁰ - Blind attempt to fix warnings issued by gcc V9
- PR #3611³⁶⁷¹ - Include `GNUInstallDirs` earlier to have it available for subdirectories
- PR #3595³⁶⁷² - Use `GNUInstallDirs` lib path in `pkgconfig` config file
- PR #3593³⁶⁷³ - Add `include(GNUInstallDirs)` to `HPXMacros.cmake`

³⁶⁶⁰ <https://github.com/TheHPXProject/hpx/issues/3638>

³⁶⁶¹ <https://github.com/TheHPXProject/hpx/issues/3635>

³⁶⁶² <https://github.com/TheHPXProject/hpx/issues/3550>

³⁶⁶³ <https://github.com/TheHPXProject/hpx/issues/3549>

³⁶⁶⁴ <https://github.com/TheHPXProject/hpx/issues/3511>

³⁶⁶⁵ <https://github.com/TheHPXProject/hpx/issues/3509>

³⁶⁶⁶ <https://github.com/TheHPXProject/hpx/pull/3695>

³⁶⁶⁷ <https://github.com/TheHPXProject/hpx/pull/3666>

³⁶⁶⁸ <https://github.com/TheHPXProject/hpx/pull/3663>

³⁶⁶⁹ <https://github.com/TheHPXProject/hpx/pull/3648>

³⁶⁷⁰ <https://github.com/TheHPXProject/hpx/pull/3647>

³⁶⁷¹ <https://github.com/TheHPXProject/hpx/pull/3611>

³⁶⁷² <https://github.com/TheHPXProject/hpx/pull/3595>

³⁶⁷³ <https://github.com/TheHPXProject/hpx/pull/3593>

- [PR #3591](#)³⁶⁷⁴ - Fix compilation error on arm7 architecture. Compiles and runs on Fedora 29 on Pi 3.
- [PR #3558](#)³⁶⁷⁵ - Adding constructor *exception(boost::system::error_code const&)*
- [PR #3555](#)³⁶⁷⁶ - cmake: make install locations configurable
- [PR #3551](#)³⁶⁷⁷ - Fix uint64_t causing compilation fail on i686

HPX V1.2.0 (Nov 12, 2018)

General changes

Here are some of the main highlights and changes for this release:

- Thanks to the work of our Google Summer of Code student, Nikunj Gupta, we now have a new implementation of `hpx_main.hpp` on supported platforms (Linux, BSD and MacOS). This is intended to be a less fragile drop-in replacement for the old implementation relying on preprocessor macros. The new implementation does not require changes if you are using the [CMake](#)³⁶⁷⁸ or `pkg-config`. The old behaviour can be restored by setting `HPX_WITH_DYNAMIC_HPX_MAIN=OFF` during [CMake](#)³⁶⁷⁹ configuration. The implementation on Windows is unchanged.
- We have added functionality to allow passing scheduling hints to our schedulers. These will allow us to create executors that for example target a specific NUMA domain or allow for *HPX* threads to be pinned to a particular worker thread.
- We have significantly improved the performance of our futures implementation by making the shared state atomic.
- We have replaced Boostbook by Sphinx for our documentation. This means the documentation is easier to navigate with built-in search and table of contents. We have also added a quick start section and restructured the documentation to be easier to follow for new users.
- We have added a new option to the `--hpx:threads` command line option. It is now possible to use `cores` to tell *HPX* to only use one worker thread per core, unlike the existing option `all` which uses one worker thread per processing unit (processing unit can be a hyperthread if hyperthreads are available). The default value of `--hpx:threads` has also been changed to `cores` as this leads to better performance in most cases.
- All command line options can now be passed alongside configuration options when initializing *HPX*. This means that some options that were previously only available on the command line can now be set as configuration options.
- *HPXMP* is a portable, scalable, and flexible application programming interface using the OpenMP specification that supports multi-platform shared memory multiprocessing programming in C and C++. *HPXMP* can be enabled within *HPX* by setting `DHPX_WITH_HPXMP=ON` during [CMake](#)³⁶⁸⁰ configuration.
- Two new performance counters were added for measuring the time spent doing background work. `/threads/time/background-work-duration` returns the time spent doing background on a given thread or locality, while `/threads/time/background-overhead` returns the fraction of time spent doing background work with respect to the overall time spent running the scheduler. The new performance counters are disabled by default and can be turned on by setting `HPX_WITH_BACKGROUND_THREAD_COUNTERS=ON` during [CMake](#)³⁶⁸¹ configuration.

³⁶⁷⁴ <https://github.com/TheHPXProject/hpx/pull/3591>

³⁶⁷⁵ <https://github.com/TheHPXProject/hpx/pull/3558>

³⁶⁷⁶ <https://github.com/TheHPXProject/hpx/pull/3555>

³⁶⁷⁷ <https://github.com/TheHPXProject/hpx/pull/3551>

³⁶⁷⁸ <https://www.cmake.org>

³⁶⁷⁹ <https://www.cmake.org>

³⁶⁸⁰ <https://www.cmake.org>

³⁶⁸¹ <https://www.cmake.org>

- The idling behaviour of *HPX* has been tweaked to allow for faster idling. This is useful in interactive applications where the *HPX* worker threads may not have work all the time. This behaviour can be tweaked and turned off as before with `HPX_WITH_THREAD_MANAGER_IDLE_BACKOFF=OFF` during [CMake](#)³⁶⁸² configuration.
- It is now possible to register callback functions for *HPX* worker thread events. Callbacks can be registered for starting and stopping worker threads, and for when errors occur.

Breaking changes

- The implementation of `hpx_main.hpp` has changed. If you are using custom Makefiles you will need to make changes. Please see the documentation on *using Makefiles* for more details.
- The default value of `--hpx:threads` has changed from `all` to `cores`. The new option `cores` only starts one worker thread per core.
- We have dropped support for Boost 1.56 and 1.57. The minimal version of Boost we now test is 1.58.
- Our `boost::format`-based formatting implementation has been revised and replaced with a custom implementation. This changes the formatting syntax and requires changes if you are relying on `hpx::util::format` or `hpx::util::format_to`. The pull request for this change contains more information: [PR #3266](#)³⁶⁸³.
- The following deprecated options have now been completely removed:
`HPX_WITH_ASYNC_FUNCTION_COMPATIBILITY`, `HPX_WITH_LOCAL_DATAFLOW`,
`HPX_WITH_GENERIC_EXECUTION_POLICY`, `HPX_WITH_BOOST_CHRONO_COMPATIBILITY`,
`HPX_WITH_EXECUTOR_COMPATIBILITY`, `HPX_WITH_EXECUTION_POLICY_COMPATIBILITY`, and
`HPX_WITH_TRANSFORM_REDUCE_COMPATIBILITY`.

Closed issues

- [Issue #3538](#)³⁶⁸⁴ - numa handling incorrect for hwloc 2
- [Issue #3533](#)³⁶⁸⁵ - Cmake version 3.5.1 does not work (git ff26b35 2018-11-06)
- [Issue #3526](#)³⁶⁸⁶ - Failed building hpx-1.2.0-rc1 on Ubuntu16.04 x86-64 Virtualbox VM
- [Issue #3512](#)³⁶⁸⁷ - Build on aarch64 fails
- [Issue #3475](#)³⁶⁸⁸ - HPX fails to link if the MPI parcellport is enabled
- [Issue #3462](#)³⁶⁸⁹ - CMake configuration shows a minor and inconsequential failure to create a symlink
- [Issue #3461](#)³⁶⁹⁰ - Compilation Problems with the most recent Clang
- [Issue #3460](#)³⁶⁹¹ - Deadlock when `create_partitioner` fails (assertion fails) in debug mode
- [Issue #3455](#)³⁶⁹² - HPX build failing with HWLOC errors on POWER8 with hwloc 1.8
- [Issue #3438](#)³⁶⁹³ - HPX no longer builds on IBM POWER8

³⁶⁸² <https://www.cmake.org>

³⁶⁸³ <https://github.com/TheHPXProject/hpx/pull/3266>

³⁶⁸⁴ <https://github.com/TheHPXProject/hpx/issues/3538>

³⁶⁸⁵ <https://github.com/TheHPXProject/hpx/issues/3533>

³⁶⁸⁶ <https://github.com/TheHPXProject/hpx/issues/3526>

³⁶⁸⁷ <https://github.com/TheHPXProject/hpx/issues/3512>

³⁶⁸⁸ <https://github.com/TheHPXProject/hpx/issues/3475>

³⁶⁸⁹ <https://github.com/TheHPXProject/hpx/issues/3462>

³⁶⁹⁰ <https://github.com/TheHPXProject/hpx/issues/3461>

³⁶⁹¹ <https://github.com/TheHPXProject/hpx/issues/3460>

³⁶⁹² <https://github.com/TheHPXProject/hpx/issues/3455>

³⁶⁹³ <https://github.com/TheHPXProject/hpx/issues/3438>

- Issue #3426³⁶⁹⁴ - hpx build failed on MacOS
- Issue #3424³⁶⁹⁵ - CircleCI builds broken for forked repositories
- Issue #3422³⁶⁹⁶ - Benchmarks in tests.performance.local are not run nightly
- Issue #3408³⁶⁹⁷ - CMake Targets for HPX
- Issue #3399³⁶⁹⁸ - processing unit out of bounds
- Issue #3395³⁶⁹⁹ - Floating point bug in hpx/runtime/threads/policies/scheduler_base.hpp
- Issue #3378³⁷⁰⁰ - compile error with lcos::communicator
- Issue #3376³⁷⁰¹ - Failed to build HPX with APEX using clang
- Issue #3366³⁷⁰² - Adapted Safe_Object example fails for `-hpx:threads > 1`
- Issue #3360³⁷⁰³ - Segmentation fault when passing component id as parameter
- Issue #3358³⁷⁰⁴ - HPX runtime hangs after multiple (~thousands) start-stop sequences
- Issue #3352³⁷⁰⁵ - Support TCP provider in libfabric ParcelPort
- Issue #3342³⁷⁰⁶ - undefined reference to `__atomic_load_16`
- Issue #3339³⁷⁰⁷ - setting command line options/flags from init cfg is not obvious
- Issue #3325³⁷⁰⁸ - AGAS migrates components prematurely
- Issue #3321³⁷⁰⁹ - hpx bad_parameter handling is awful
- Issue #3318³⁷¹⁰ - Benchmarks fail to build with C++11
- Issue #3304³⁷¹¹ - `hpx::threads::run_as_hpx_thread` does not properly handle exceptions
- Issue #3300³⁷¹² - Setting pu step or offset results in no threads in default pool
- Issue #3297³⁷¹³ - Crash with APEX when running Phylanx lra_csv with > 1 thread
- Issue #3296³⁷¹⁴ - Building HPX with APEX configuration gives compiler warnings
- Issue #3290³⁷¹⁵ - make tests failing at hello_world_component
- Issue #3285³⁷¹⁶ - possible compilation error when “using namespace std;” is defined before including “hpx” headers files

³⁶⁹⁴ <https://github.com/TheHPXProject/hpx/issues/3426>

³⁶⁹⁵ <https://github.com/TheHPXProject/hpx/issues/3424>

³⁶⁹⁶ <https://github.com/TheHPXProject/hpx/issues/3422>

³⁶⁹⁷ <https://github.com/TheHPXProject/hpx/issues/3408>

³⁶⁹⁸ <https://github.com/TheHPXProject/hpx/issues/3399>

³⁶⁹⁹ <https://github.com/TheHPXProject/hpx/issues/3395>

³⁷⁰⁰ <https://github.com/TheHPXProject/hpx/issues/3378>

³⁷⁰¹ <https://github.com/TheHPXProject/hpx/issues/3376>

³⁷⁰² <https://github.com/TheHPXProject/hpx/issues/3366>

³⁷⁰³ <https://github.com/TheHPXProject/hpx/issues/3360>

³⁷⁰⁴ <https://github.com/TheHPXProject/hpx/issues/3358>

³⁷⁰⁵ <https://github.com/TheHPXProject/hpx/issues/3352>

³⁷⁰⁶ <https://github.com/TheHPXProject/hpx/issues/3342>

³⁷⁰⁷ <https://github.com/TheHPXProject/hpx/issues/3339>

³⁷⁰⁸ <https://github.com/TheHPXProject/hpx/issues/3325>

³⁷⁰⁹ <https://github.com/TheHPXProject/hpx/issues/3321>

³⁷¹⁰ <https://github.com/TheHPXProject/hpx/issues/3318>

³⁷¹¹ <https://github.com/TheHPXProject/hpx/issues/3304>

³⁷¹² <https://github.com/TheHPXProject/hpx/issues/3300>

³⁷¹³ <https://github.com/TheHPXProject/hpx/issues/3297>

³⁷¹⁴ <https://github.com/TheHPXProject/hpx/issues/3296>

³⁷¹⁵ <https://github.com/TheHPXProject/hpx/issues/3290>

³⁷¹⁶ <https://github.com/TheHPXProject/hpx/issues/3285>

- Issue #3280³⁷¹⁷ - HPX fails on OSX
- Issue #3272³⁷¹⁸ - CircleCI does not upload generated docker image any more
- Issue #3270³⁷¹⁹ - Error when compiling CUDA examples
- Issue #3267³⁷²⁰ - `tests.unit.host_.block_allocator` fails occasionally
- Issue #3264³⁷²¹ - Possible move to Sphinx for documentation
- Issue #3263³⁷²² - Documentation improvements
- Issue #3259³⁷²³ - `set_parcel_write_handler` test fails occasionally
- Issue #3258³⁷²⁴ - Links to source code in documentation are broken
- Issue #3247³⁷²⁵ - Rare `tests.unit.host_.block_allocator` test failure on 1.1.0-rc1
- Issue #3244³⁷²⁶ - Slowing down and speeding up an `interval_timer`
- Issue #3215³⁷²⁷ - Cannot build both tests and examples on MSVC with pseudo-dependencies enabled
- Issue #3195³⁷²⁸ - Unnecessary customization point route causing performance penalty
- Issue #3088³⁷²⁹ - A strange thing in `parallel::sort`.
- Issue #2650³⁷³⁰ - libfabric support for passive endpoints
- Issue #1205³⁷³¹ - TSS is broken

Closed pull requests

- PR #3542³⁷³² - Fix numa lookup from pu when using hwloc 2.x
- PR #3541³⁷³³ - Fixing the build system of the MPI parcelport
- PR #3540³⁷³⁴ - Updating HPX people section
- PR #3539³⁷³⁵ - Splitting test to avoid OOM on CircleCI
- PR #3537³⁷³⁶ - Fix guided exec
- PR #3536³⁷³⁷ - Updating grants which support the LSU team
- PR #3535³⁷³⁸ - Fix hiding of docker credentials

³⁷¹⁷ <https://github.com/TheHPXProject/hpx/issues/3280>

³⁷¹⁸ <https://github.com/TheHPXProject/hpx/issues/3272>

³⁷¹⁹ <https://github.com/TheHPXProject/hpx/issues/3270>

³⁷²⁰ <https://github.com/TheHPXProject/hpx/issues/3267>

³⁷²¹ <https://github.com/TheHPXProject/hpx/issues/3264>

³⁷²² <https://github.com/TheHPXProject/hpx/issues/3263>

³⁷²³ <https://github.com/TheHPXProject/hpx/issues/3259>

³⁷²⁴ <https://github.com/TheHPXProject/hpx/issues/3258>

³⁷²⁵ <https://github.com/TheHPXProject/hpx/issues/3247>

³⁷²⁶ <https://github.com/TheHPXProject/hpx/issues/3244>

³⁷²⁷ <https://github.com/TheHPXProject/hpx/issues/3215>

³⁷²⁸ <https://github.com/TheHPXProject/hpx/issues/3195>

³⁷²⁹ <https://github.com/TheHPXProject/hpx/issues/3088>

³⁷³⁰ <https://github.com/TheHPXProject/hpx/issues/2650>

³⁷³¹ <https://github.com/TheHPXProject/hpx/issues/1205>

³⁷³² <https://github.com/TheHPXProject/hpx/pull/3542>

³⁷³³ <https://github.com/TheHPXProject/hpx/pull/3541>

³⁷³⁴ <https://github.com/TheHPXProject/hpx/pull/3540>

³⁷³⁵ <https://github.com/TheHPXProject/hpx/pull/3539>

³⁷³⁶ <https://github.com/TheHPXProject/hpx/pull/3537>

³⁷³⁷ <https://github.com/TheHPXProject/hpx/pull/3536>

³⁷³⁸ <https://github.com/TheHPXProject/hpx/pull/3535>

- PR #3534³⁷³⁹ - Fixing #3533
- PR #3532³⁷⁴⁰ - fixing minor doc typo `-hpx:print-counter-at` arg
- PR #3530³⁷⁴¹ - Changing APEX default tag to v2.1.0
- PR #3529³⁷⁴² - Remove leftover security options and documentation
- PR #3528³⁷⁴³ - Fix hwloc version check
- PR #3524³⁷⁴⁴ - Do not build guided pool examples with older GCC compilers
- PR #3523³⁷⁴⁵ - Fix logging regression
- PR #3522³⁷⁴⁶ - Fix more warnings
- PR #3521³⁷⁴⁷ - Fixing argument handling in induction and reduction clauses for `parallel::for_loop`
- PR #3520³⁷⁴⁸ - Remove docs symlink and versioned docs folders
- PR #3519³⁷⁴⁹ - hpxMP release
- PR #3518³⁷⁵⁰ - Change all steps to use new docker image on CircleCI
- PR #3516³⁷⁵¹ - Drop usage of deprecated facilities removed in C++17
- PR #3515³⁷⁵² - Remove remaining uses of `Boost.TypeTraits`
- PR #3513³⁷⁵³ - Fixing a CMake problem when trying to use `libfabric`
- PR #3508³⁷⁵⁴ - Remove `memory_block` component
- PR #3507³⁷⁵⁵ - Propagating the MPI compile definitions to all relevant targets
- PR #3503³⁷⁵⁶ - Update documentation colors and logo
- PR #3502³⁷⁵⁷ - Fix bogus ``throws`` bindings in `scheduled_thread_pool_impl`
- PR #3501³⁷⁵⁸ - Split `parallel::remove_if` tests to avoid OOM on CircleCI
- PR #3500³⁷⁵⁹ - Support `NONAMEPREFIX` in `add_hpx_library()`
- PR #3497³⁷⁶⁰ - Note that cuda support requires cmake 3.9
- PR #3495³⁷⁶¹ - Fixing dataflow

³⁷³⁹ <https://github.com/TheHPXProject/hpx/pull/3534>

³⁷⁴⁰ <https://github.com/TheHPXProject/hpx/pull/3532>

³⁷⁴¹ <https://github.com/TheHPXProject/hpx/pull/3530>

³⁷⁴² <https://github.com/TheHPXProject/hpx/pull/3529>

³⁷⁴³ <https://github.com/TheHPXProject/hpx/pull/3528>

³⁷⁴⁴ <https://github.com/TheHPXProject/hpx/pull/3524>

³⁷⁴⁵ <https://github.com/TheHPXProject/hpx/pull/3523>

³⁷⁴⁶ <https://github.com/TheHPXProject/hpx/pull/3522>

³⁷⁴⁷ <https://github.com/TheHPXProject/hpx/pull/3521>

³⁷⁴⁸ <https://github.com/TheHPXProject/hpx/pull/3520>

³⁷⁴⁹ <https://github.com/TheHPXProject/hpx/pull/3519>

³⁷⁵⁰ <https://github.com/TheHPXProject/hpx/pull/3518>

³⁷⁵¹ <https://github.com/TheHPXProject/hpx/pull/3516>

³⁷⁵² <https://github.com/TheHPXProject/hpx/pull/3515>

³⁷⁵³ <https://github.com/TheHPXProject/hpx/pull/3513>

³⁷⁵⁴ <https://github.com/TheHPXProject/hpx/pull/3508>

³⁷⁵⁵ <https://github.com/TheHPXProject/hpx/pull/3507>

³⁷⁵⁶ <https://github.com/TheHPXProject/hpx/pull/3503>

³⁷⁵⁷ <https://github.com/TheHPXProject/hpx/pull/3502>

³⁷⁵⁸ <https://github.com/TheHPXProject/hpx/pull/3501>

³⁷⁵⁹ <https://github.com/TheHPXProject/hpx/pull/3500>

³⁷⁶⁰ <https://github.com/TheHPXProject/hpx/pull/3497>

³⁷⁶¹ <https://github.com/TheHPXProject/hpx/pull/3495>

- PR #3493³⁷⁶² - Remove deprecated options for 1.2.0 part 2
- PR #3492³⁷⁶³ - Add CUDA_LINK_LIBRARIES_KEYWORD to allow PRIVATE keyword in linkage t...
- PR #3491³⁷⁶⁴ - Changing Base docker image
- PR #3490³⁷⁶⁵ - Don't create tasks immediately with hpx::apply
- PR #3489³⁷⁶⁶ - Remove deprecated options for 1.2.0
- PR #3488³⁷⁶⁷ - Revert "Use BUILD_INTERFACE generator expression to fix cmake flag exports"
- PR #3487³⁷⁶⁸ - Revert "Fixing type attribute warning for transfer_action"
- PR #3485³⁷⁶⁹ - Use BUILD_INTERFACE generator expression to fix cmake flag exports
- PR #3483³⁷⁷⁰ - Fixing type attribute warning for transfer_action
- PR #3481³⁷⁷¹ - Remove unused variables
- PR #3480³⁷⁷² - Towards a more lightweight transfer action
- PR #3479³⁷⁷³ - Fix FLAGS - Use correct version of target_compile_options
- PR #3478³⁷⁷⁴ - Making sure the application's exit code is properly propagated back to the OS
- PR #3476³⁷⁷⁵ - Don't print docker credentials as part of the environment.
- PR #3473³⁷⁷⁶ - Fixing invalid cmake code if no jemalloc prefix was given
- PR #3472³⁷⁷⁷ - Attempting to work around recent clang test compilation failures
- PR #3471³⁷⁷⁸ - Enable jemalloc on windows
- PR #3470³⁷⁷⁹ - Updates readme
- PR #3468³⁷⁸⁰ - Avoid hang if there is an exception thrown during startup
- PR #3467³⁷⁸¹ - Add compiler specific fallback attributes if C++17 attribute is not available
- PR #3466³⁷⁸² - - bugfix : fix compilation with llvm-7.0
- PR #3465³⁷⁸³ - This patch adds various optimizations extracted from the thread_local_allocator work
- PR #3464³⁷⁸⁴ - Check for forked repos in CircleCI docker push step

³⁷⁶² <https://github.com/TheHPXProject/hpx/pull/3493>

³⁷⁶³ <https://github.com/TheHPXProject/hpx/pull/3492>

³⁷⁶⁴ <https://github.com/TheHPXProject/hpx/pull/3491>

³⁷⁶⁵ <https://github.com/TheHPXProject/hpx/pull/3490>

³⁷⁶⁶ <https://github.com/TheHPXProject/hpx/pull/3489>

³⁷⁶⁷ <https://github.com/TheHPXProject/hpx/pull/3488>

³⁷⁶⁸ <https://github.com/TheHPXProject/hpx/pull/3487>

³⁷⁶⁹ <https://github.com/TheHPXProject/hpx/pull/3485>

³⁷⁷⁰ <https://github.com/TheHPXProject/hpx/pull/3483>

³⁷⁷¹ <https://github.com/TheHPXProject/hpx/pull/3481>

³⁷⁷² <https://github.com/TheHPXProject/hpx/pull/3480>

³⁷⁷³ <https://github.com/TheHPXProject/hpx/pull/3479>

³⁷⁷⁴ <https://github.com/TheHPXProject/hpx/pull/3478>

³⁷⁷⁵ <https://github.com/TheHPXProject/hpx/pull/3476>

³⁷⁷⁶ <https://github.com/TheHPXProject/hpx/pull/3473>

³⁷⁷⁷ <https://github.com/TheHPXProject/hpx/pull/3472>

³⁷⁷⁸ <https://github.com/TheHPXProject/hpx/pull/3471>

³⁷⁷⁹ <https://github.com/TheHPXProject/hpx/pull/3470>

³⁷⁸⁰ <https://github.com/TheHPXProject/hpx/pull/3468>

³⁷⁸¹ <https://github.com/TheHPXProject/hpx/pull/3467>

³⁷⁸² <https://github.com/TheHPXProject/hpx/pull/3466>

³⁷⁸³ <https://github.com/TheHPXProject/hpx/pull/3465>

³⁷⁸⁴ <https://github.com/TheHPXProject/hpx/pull/3464>

- PR #3463³⁷⁸⁵ - - cmake : create the parent directory before symlinking
- PR #3459³⁷⁸⁶ - Remove unused/incomplete functionality from util/logging
- PR #3458³⁷⁸⁷ - Fix a problem with scope of CMAKE_CXX_FLAGS and hpx_add_compile_flag
- PR #3457³⁷⁸⁸ - Fixing more size_t -> int16_t (and similar) warnings
- PR #3456³⁷⁸⁹ - Add #ifdefs to topology.cpp to support old hwloc versions again
- PR #3454³⁷⁹⁰ - Fixing warnings related to silent conversion of size_t -> int16_t
- PR #3451³⁷⁹¹ - Add examples as unit tests
- PR #3450³⁷⁹² - Constexpr-fying bind and other functional facilities
- PR #3446³⁷⁹³ - Fix some thread suspension timeouts
- PR #3445³⁷⁹⁴ - Fix various warnings
- PR #3443³⁷⁹⁵ - Only enable service pool config options if pools are enabled
- PR #3441³⁷⁹⁶ - Fix missing closing brackets in documentation
- PR #3439³⁷⁹⁷ - Use correct MPI CXX libraries for MPI parcelport
- PR #3436³⁷⁹⁸ - Add projection function to find_* (and fix very bad bug)
- PR #3435³⁷⁹⁹ - Fixing 1205
- PR #3434³⁸⁰⁰ - Fix threads cores
- PR #3433³⁸⁰¹ - Add Heise Online to release announcement list
- PR #3432³⁸⁰² - Don't track task dependencies for distributed runs
- PR #3431³⁸⁰³ - Circle CI setting changes for hpxMP
- PR #3430³⁸⁰⁴ - Fix unused params warning
- PR #3429³⁸⁰⁵ - One thread per core
- PR #3428³⁸⁰⁶ - This suppresses a deprecation warning that is being issued by MSVC 19.15.26726
- PR #3427³⁸⁰⁷ - Fixes #3426

³⁷⁸⁵ <https://github.com/TheHPXProject/hpx/pull/3463>

³⁷⁸⁶ <https://github.com/TheHPXProject/hpx/pull/3459>

³⁷⁸⁷ <https://github.com/TheHPXProject/hpx/pull/3458>

³⁷⁸⁸ <https://github.com/TheHPXProject/hpx/pull/3457>

³⁷⁸⁹ <https://github.com/TheHPXProject/hpx/pull/3456>

³⁷⁹⁰ <https://github.com/TheHPXProject/hpx/pull/3454>

³⁷⁹¹ <https://github.com/TheHPXProject/hpx/pull/3451>

³⁷⁹² <https://github.com/TheHPXProject/hpx/pull/3450>

³⁷⁹³ <https://github.com/TheHPXProject/hpx/pull/3446>

³⁷⁹⁴ <https://github.com/TheHPXProject/hpx/pull/3445>

³⁷⁹⁵ <https://github.com/TheHPXProject/hpx/pull/3443>

³⁷⁹⁶ <https://github.com/TheHPXProject/hpx/pull/3441>

³⁷⁹⁷ <https://github.com/TheHPXProject/hpx/pull/3439>

³⁷⁹⁸ <https://github.com/TheHPXProject/hpx/pull/3436>

³⁷⁹⁹ <https://github.com/TheHPXProject/hpx/pull/3435>

³⁸⁰⁰ <https://github.com/TheHPXProject/hpx/pull/3434>

³⁸⁰¹ <https://github.com/TheHPXProject/hpx/pull/3433>

³⁸⁰² <https://github.com/TheHPXProject/hpx/pull/3432>

³⁸⁰³ <https://github.com/TheHPXProject/hpx/pull/3431>

³⁸⁰⁴ <https://github.com/TheHPXProject/hpx/pull/3430>

³⁸⁰⁵ <https://github.com/TheHPXProject/hpx/pull/3429>

³⁸⁰⁶ <https://github.com/TheHPXProject/hpx/pull/3428>

³⁸⁰⁷ <https://github.com/TheHPXProject/hpx/pull/3427>

- PR #3425³⁸⁰⁸ - Use source cache and workspace between job steps on CircleCI
- PR #3421³⁸⁰⁹ - Add CDash timing output to future overhead test (for graphs)
- PR #3420³⁸¹⁰ - Add guided_pool_executor
- PR #3419³⁸¹¹ - Fix typo in CircleCI config
- PR #3418³⁸¹² - Add sphinx documentation
- PR #3415³⁸¹³ - Scheduler NUMA hint and shared priority scheduler
- PR #3414³⁸¹⁴ - Adding step to synchronize the APEX release
- PR #3413³⁸¹⁵ - Fixing multiple defines of APEX_HAVE_HPX
- PR #3412³⁸¹⁶ - Fixes linking with libhpx_wrap error with BSD and Windows based systems
- PR #3410³⁸¹⁷ - Fix typo in CMakeLists.txt
- PR #3409³⁸¹⁸ - Fix brackets and indentation in existing_performance_counters.qbk
- PR #3407³⁸¹⁹ - Fix unused param and extra ; warnings emitted by gcc 8.x
- PR #3406³⁸²⁰ - Adding thread local allocator and use it for future shared states
- PR #3405³⁸²¹ - Adding DHPX_HAVE_THREAD_LOCAL_STORAGE=ON to builds
- PR #3404³⁸²² - fixing multiple definition of main() in linux
- PR #3402³⁸²³ - Allow debug option to be enabled only for Linux systems with dynamic main on
- PR #3401³⁸²⁴ - Fix cuda_future_helper.h when compiling with C++11
- PR #3400³⁸²⁵ - Fix floating point exception scheduler_base idle backoff
- PR #3398³⁸²⁶ - Atomic future state
- PR #3397³⁸²⁷ - Fixing code for older gcc versions
- PR #3396³⁸²⁸ - Allowing to register thread event functions (start/stop/error)
- PR #3394³⁸²⁹ - Fix small mistake in primary_namespace_server.cpp
- PR #3393³⁸³⁰ - Explicitly instantiate configured schedulers

³⁸⁰⁸ <https://github.com/TheHPXProject/hpx/pull/3425>

³⁸⁰⁹ <https://github.com/TheHPXProject/hpx/pull/3421>

³⁸¹⁰ <https://github.com/TheHPXProject/hpx/pull/3420>

³⁸¹¹ <https://github.com/TheHPXProject/hpx/pull/3419>

³⁸¹² <https://github.com/TheHPXProject/hpx/pull/3418>

³⁸¹³ <https://github.com/TheHPXProject/hpx/pull/3415>

³⁸¹⁴ <https://github.com/TheHPXProject/hpx/pull/3414>

³⁸¹⁵ <https://github.com/TheHPXProject/hpx/pull/3413>

³⁸¹⁶ <https://github.com/TheHPXProject/hpx/pull/3412>

³⁸¹⁷ <https://github.com/TheHPXProject/hpx/pull/3410>

³⁸¹⁸ <https://github.com/TheHPXProject/hpx/pull/3409>

³⁸¹⁹ <https://github.com/TheHPXProject/hpx/pull/3407>

³⁸²⁰ <https://github.com/TheHPXProject/hpx/pull/3406>

³⁸²¹ <https://github.com/TheHPXProject/hpx/pull/3405>

³⁸²² <https://github.com/TheHPXProject/hpx/pull/3404>

³⁸²³ <https://github.com/TheHPXProject/hpx/pull/3402>

³⁸²⁴ <https://github.com/TheHPXProject/hpx/pull/3401>

³⁸²⁵ <https://github.com/TheHPXProject/hpx/pull/3400>

³⁸²⁶ <https://github.com/TheHPXProject/hpx/pull/3398>

³⁸²⁷ <https://github.com/TheHPXProject/hpx/pull/3397>

³⁸²⁸ <https://github.com/TheHPXProject/hpx/pull/3396>

³⁸²⁹ <https://github.com/TheHPXProject/hpx/pull/3394>

³⁸³⁰ <https://github.com/TheHPXProject/hpx/pull/3393>

- PR #3392³⁸³¹ - Add performance counters background overhead and background work duration
- PR #3391³⁸³² - Adapt integration of HPXMP to latest build system changes
- PR #3390³⁸³³ - Make AGAS measurements optional
- PR #3389³⁸³⁴ - Fix deadlock during shutdown
- PR #3388³⁸³⁵ - Add several functionalities allowing to optimize synchronous action invocation
- PR #3387³⁸³⁶ - Add cmake option to opt out of fail-compile tests
- PR #3386³⁸³⁷ - Adding support for boost::container::small_vector to dataflow
- PR #3385³⁸³⁸ - Adds Debug option for hpx initializing from main
- PR #3384³⁸³⁹ - This hopefully fixes two tests that occasionally fail
- PR #3383³⁸⁴⁰ - Making sure thread local storage is enable for hpxMP
- PR #3382³⁸⁴¹ - Fix usage of HPX_CAPTURE together with default value capture [=]
- PR #3381³⁸⁴² - Replace undefined instantiations of uniform_int_distribution
- PR #3380³⁸⁴³ - Add missing semicolons to uses of HPX_COMPILER_FENCE
- PR #3379³⁸⁴⁴ - Fixing #3378
- PR #3377³⁸⁴⁵ - Adding build system support to integrate hpxmp into hpx at the user's machine
- PR #3375³⁸⁴⁶ - Replacing wrapper for __libc_start_main with main
- PR #3374³⁸⁴⁷ - Adds hpx_wrap to HPX_LINK_LIBRARIES which links only when specified.
- PR #3373³⁸⁴⁸ - Forcing cache settings in HPXConfig.cmake to guarantee updated values
- PR #3372³⁸⁴⁹ - Fix some more c++11 build problems
- PR #3371³⁸⁵⁰ - Adds HPX_LINKER_FLAGS to HPX applications without editing their source codes
- PR #3370³⁸⁵¹ - util::format: add type_specifier<> specializations for %!s(MISSING) and %!l(MISSING)s
- PR #3369³⁸⁵² - Adding configuration option to allow explicit disable of the new hpx_main feature on Linux
- PR #3368³⁸⁵³ - Updates doc with recent hpx_wrap implementation

³⁸³¹ <https://github.com/TheHPXProject/hpx/pull/3392>

³⁸³² <https://github.com/TheHPXProject/hpx/pull/3391>

³⁸³³ <https://github.com/TheHPXProject/hpx/pull/3390>

³⁸³⁴ <https://github.com/TheHPXProject/hpx/pull/3389>

³⁸³⁵ <https://github.com/TheHPXProject/hpx/pull/3388>

³⁸³⁶ <https://github.com/TheHPXProject/hpx/pull/3387>

³⁸³⁷ <https://github.com/TheHPXProject/hpx/pull/3386>

³⁸³⁸ <https://github.com/TheHPXProject/hpx/pull/3385>

³⁸³⁹ <https://github.com/TheHPXProject/hpx/pull/3384>

³⁸⁴⁰ <https://github.com/TheHPXProject/hpx/pull/3383>

³⁸⁴¹ <https://github.com/TheHPXProject/hpx/pull/3382>

³⁸⁴² <https://github.com/TheHPXProject/hpx/pull/3381>

³⁸⁴³ <https://github.com/TheHPXProject/hpx/pull/3380>

³⁸⁴⁴ <https://github.com/TheHPXProject/hpx/pull/3379>

³⁸⁴⁵ <https://github.com/TheHPXProject/hpx/pull/3377>

³⁸⁴⁶ <https://github.com/TheHPXProject/hpx/pull/3375>

³⁸⁴⁷ <https://github.com/TheHPXProject/hpx/pull/3374>

³⁸⁴⁸ <https://github.com/TheHPXProject/hpx/pull/3373>

³⁸⁴⁹ <https://github.com/TheHPXProject/hpx/pull/3372>

³⁸⁵⁰ <https://github.com/TheHPXProject/hpx/pull/3371>

³⁸⁵¹ <https://github.com/TheHPXProject/hpx/pull/3370>

³⁸⁵² <https://github.com/TheHPXProject/hpx/pull/3369>

³⁸⁵³ <https://github.com/TheHPXProject/hpx/pull/3368>

- PR #3367³⁸⁵⁴ - Adds Mac OS implementation to hpx_main.hpp
- PR #3365³⁸⁵⁵ - Fix order of hpx libs in HPX_CONF_LIBRARIES.
- PR #3363³⁸⁵⁶ - Apex fixing null wrapper
- PR #3361³⁸⁵⁷ - Making sure all parcels get destroyed on an HPX thread (TCP pp)
- PR #3359³⁸⁵⁸ - Feature/improve error for compiler
- PR #3357³⁸⁵⁹ - Static/dynamic executable implementation
- PR #3355³⁸⁶⁰ - Reverting changes introduced by #3283 as those make applications hang
- PR #3354³⁸⁶¹ - Add external dependencies to HPX_LIBRARY_DIR
- PR #3353³⁸⁶² - Fix libfabric tcp
- PR #3351³⁸⁶³ - Move obsolete header to tests directory.
- PR #3350³⁸⁶⁴ - Renaming two functions to avoid problem described in #3285
- PR #3349³⁸⁶⁵ - Make idle backoff exponential with maximum sleep time
- PR #3347³⁸⁶⁶ - Replace *simple_component** with *component** in the Documentation
- PR #3346³⁸⁶⁷ - Fix CMakeLists.txt example in quick start
- PR #3345³⁸⁶⁸ - Fix automatic setting of HPX_MORE_THAN_64_THREADS
- PR #3344³⁸⁶⁹ - Reduce amount of information printed for unknown command line options
- PR #3343³⁸⁷⁰ - Safeguard HPX against destruction in global contexts
- PR #3341³⁸⁷¹ - Allowing for all command line options to be used as configuration settings
- PR #3340³⁸⁷² - Always convert inspect results to JUnit XML
- PR #3336³⁸⁷³ - Only run docker push on master on CircleCI
- PR #3335³⁸⁷⁴ - Update description of hpx.os_threads config parameter.
- PR #3334³⁸⁷⁵ - Making sure early logging settings don't get mixed with others
- PR #3333³⁸⁷⁶ - Update CMake links and versions in documentation

³⁸⁵⁴ <https://github.com/TheHPXProject/hpx/pull/3367>

³⁸⁵⁵ <https://github.com/TheHPXProject/hpx/pull/3365>

³⁸⁵⁶ <https://github.com/TheHPXProject/hpx/pull/3363>

³⁸⁵⁷ <https://github.com/TheHPXProject/hpx/pull/3361>

³⁸⁵⁸ <https://github.com/TheHPXProject/hpx/pull/3359>

³⁸⁵⁹ <https://github.com/TheHPXProject/hpx/pull/3357>

³⁸⁶⁰ <https://github.com/TheHPXProject/hpx/pull/3355>

³⁸⁶¹ <https://github.com/TheHPXProject/hpx/pull/3354>

³⁸⁶² <https://github.com/TheHPXProject/hpx/pull/3353>

³⁸⁶³ <https://github.com/TheHPXProject/hpx/pull/3351>

³⁸⁶⁴ <https://github.com/TheHPXProject/hpx/pull/3350>

³⁸⁶⁵ <https://github.com/TheHPXProject/hpx/pull/3349>

³⁸⁶⁶ <https://github.com/TheHPXProject/hpx/pull/3347>

³⁸⁶⁷ <https://github.com/TheHPXProject/hpx/pull/3346>

³⁸⁶⁸ <https://github.com/TheHPXProject/hpx/pull/3345>

³⁸⁶⁹ <https://github.com/TheHPXProject/hpx/pull/3344>

³⁸⁷⁰ <https://github.com/TheHPXProject/hpx/pull/3343>

³⁸⁷¹ <https://github.com/TheHPXProject/hpx/pull/3341>

³⁸⁷² <https://github.com/TheHPXProject/hpx/pull/3340>

³⁸⁷³ <https://github.com/TheHPXProject/hpx/pull/3336>

³⁸⁷⁴ <https://github.com/TheHPXProject/hpx/pull/3335>

³⁸⁷⁵ <https://github.com/TheHPXProject/hpx/pull/3334>

³⁸⁷⁶ <https://github.com/TheHPXProject/hpx/pull/3333>

- PR #3332³⁸⁷⁷ - Add notes on target suffixes to CMake documentation
- PR #3331³⁸⁷⁸ - Add quickstart section to documentation
- PR #3330³⁸⁷⁹ - Rename resource_partitioner test to avoid conflicts with pseudodependencies
- PR #3328³⁸⁸⁰ - Making sure object is pinned while executing actions, even if action returns a future
- PR #3327³⁸⁸¹ - Add missing std::forward to tuple.hpp
- PR #3326³⁸⁸² - Make sure logging is up and running while modules are being discovered.
- PR #3324³⁸⁸³ - Replace C++14 overload of std::equal with C++11 code.
- PR #3323³⁸⁸⁴ - Fix a missing apex thread data (wrapper) initialization
- PR #3320³⁸⁸⁵ - Adding support for -std=c++2a (define *HPX_WITH_CXX2A=On*)
- PR #3319³⁸⁸⁶ - Replacing C++14 feature with equivalent C++11 code
- PR #3317³⁸⁸⁷ - Fix compilation with VS 15.7.1 and /std:c++latest
- PR #3316³⁸⁸⁸ - Fix includes for 1d_stencil_*_omp examples
- PR #3314³⁸⁸⁹ - Remove some unused parameter warnings
- PR #3313³⁸⁹⁰ - Fix pu-step and pu-offset command line options
- PR #3312³⁸⁹¹ - Add conversion of inspect reports to JUnit XML
- PR #3311³⁸⁹² - Fix escaping of closing braces in format specification syntax
- PR #3310³⁸⁹³ - Don't overwrite user settings with defaults in registration database
- PR #3309³⁸⁹⁴ - Fixing potential stack overflow for dataflow
- PR #3308³⁸⁹⁵ - This updates the .clang-format configuration file to utilize newer features
- PR #3306³⁸⁹⁶ - Marking migratable objects in their gid to allow not handling migration in AGAS
- PR #3305³⁸⁹⁷ - Add proper exception handling to run_as_hpx_thread
- PR #3303³⁸⁹⁸ - Changed std::rand to a better inbuilt PRNG Generator
- PR #3302³⁸⁹⁹ - All non-migratable (simple) components now encode their lva and component type in their gid

³⁸⁷⁷ <https://github.com/TheHPXProject/hpx/pull/3332>

³⁸⁷⁸ <https://github.com/TheHPXProject/hpx/pull/3331>

³⁸⁷⁹ <https://github.com/TheHPXProject/hpx/pull/3330>

³⁸⁸⁰ <https://github.com/TheHPXProject/hpx/pull/3328>

³⁸⁸¹ <https://github.com/TheHPXProject/hpx/pull/3327>

³⁸⁸² <https://github.com/TheHPXProject/hpx/pull/3326>

³⁸⁸³ <https://github.com/TheHPXProject/hpx/pull/3324>

³⁸⁸⁴ <https://github.com/TheHPXProject/hpx/pull/3323>

³⁸⁸⁵ <https://github.com/TheHPXProject/hpx/pull/3320>

³⁸⁸⁶ <https://github.com/TheHPXProject/hpx/pull/3319>

³⁸⁸⁷ <https://github.com/TheHPXProject/hpx/pull/3317>

³⁸⁸⁸ <https://github.com/TheHPXProject/hpx/pull/3316>

³⁸⁸⁹ <https://github.com/TheHPXProject/hpx/pull/3314>

³⁸⁹⁰ <https://github.com/TheHPXProject/hpx/pull/3313>

³⁸⁹¹ <https://github.com/TheHPXProject/hpx/pull/3312>

³⁸⁹² <https://github.com/TheHPXProject/hpx/pull/3311>

³⁸⁹³ <https://github.com/TheHPXProject/hpx/pull/3310>

³⁸⁹⁴ <https://github.com/TheHPXProject/hpx/pull/3309>

³⁸⁹⁵ <https://github.com/TheHPXProject/hpx/pull/3308>

³⁸⁹⁶ <https://github.com/TheHPXProject/hpx/pull/3306>

³⁸⁹⁷ <https://github.com/TheHPXProject/hpx/pull/3305>

³⁸⁹⁸ <https://github.com/TheHPXProject/hpx/pull/3303>

³⁸⁹⁹ <https://github.com/TheHPXProject/hpx/pull/3302>

- PR #3301³⁹⁰⁰ - Add nullptr_t overloads to resource partitioner
- PR #3298³⁹⁰¹ - Apex task wrapper memory bug
- PR #3295³⁹⁰² - Fix mistakes after merge of CircleCI config
- PR #3294³⁹⁰³ - Fix partitioned vector include in partitioned_vector_find tests
- PR #3293³⁹⁰⁴ - Adding emplace support to promise and make_ready_future
- PR #3292³⁹⁰⁵ - Add new cuda kernel synchronization with hpx::future demo
- PR #3291³⁹⁰⁶ - Fixes #3290
- PR #3289³⁹⁰⁷ - Fixing Docker image creation
- PR #3288³⁹⁰⁸ - Avoid allocating shared state for wait_all
- PR #3287³⁹⁰⁹ - Fixing /scheduler/utilization/instantaneous performance counter
- PR #3286³⁹¹⁰ - dataflow() and future::then() use sync policy where possible
- PR #3284³⁹¹¹ - Background thread can use relaxed atomics to manipulate thread state
- PR #3283³⁹¹² - Do not unwrap ready future
- PR #3282³⁹¹³ - Fix virtual method override warnings in static schedulers
- PR #3281³⁹¹⁴ - Disable set_area_membind_nodeset for OSX
- PR #3279³⁹¹⁵ - Add two variations to the future_overhead benchmark
- PR #3278³⁹¹⁶ - Fix circleci workspace
- PR #3277³⁹¹⁷ - Support external plugins
- PR #3276³⁹¹⁸ - Fix missing parenthesis in hello_compute.cu.
- PR #3274³⁹¹⁹ - Reinit counters synchronously in reinit_counters test
- PR #3273³⁹²⁰ - Splitting tests to avoid compiler OOM
- PR #3271³⁹²¹ - Remove leftover code from context_generic_context.hpp
- PR #3269³⁹²² - Fix bulk_construct with count = 0

³⁹⁰⁰ <https://github.com/TheHPXProject/hpx/pull/3301>

³⁹⁰¹ <https://github.com/TheHPXProject/hpx/pull/3298>

³⁹⁰² <https://github.com/TheHPXProject/hpx/pull/3295>

³⁹⁰³ <https://github.com/TheHPXProject/hpx/pull/3294>

³⁹⁰⁴ <https://github.com/TheHPXProject/hpx/pull/3293>

³⁹⁰⁵ <https://github.com/TheHPXProject/hpx/pull/3292>

³⁹⁰⁶ <https://github.com/TheHPXProject/hpx/pull/3291>

³⁹⁰⁷ <https://github.com/TheHPXProject/hpx/pull/3289>

³⁹⁰⁸ <https://github.com/TheHPXProject/hpx/pull/3288>

³⁹⁰⁹ <https://github.com/TheHPXProject/hpx/pull/3287>

³⁹¹⁰ <https://github.com/TheHPXProject/hpx/pull/3286>

³⁹¹¹ <https://github.com/TheHPXProject/hpx/pull/3284>

³⁹¹² <https://github.com/TheHPXProject/hpx/pull/3283>

³⁹¹³ <https://github.com/TheHPXProject/hpx/pull/3282>

³⁹¹⁴ <https://github.com/TheHPXProject/hpx/pull/3281>

³⁹¹⁵ <https://github.com/TheHPXProject/hpx/pull/3279>

³⁹¹⁶ <https://github.com/TheHPXProject/hpx/pull/3278>

³⁹¹⁷ <https://github.com/TheHPXProject/hpx/pull/3277>

³⁹¹⁸ <https://github.com/TheHPXProject/hpx/pull/3276>

³⁹¹⁹ <https://github.com/TheHPXProject/hpx/pull/3274>

³⁹²⁰ <https://github.com/TheHPXProject/hpx/pull/3273>

³⁹²¹ <https://github.com/TheHPXProject/hpx/pull/3271>

³⁹²² <https://github.com/TheHPXProject/hpx/pull/3269>

- [PR #3268](#)³⁹²³ - Replace `constexpr` with `HPX_CXX14_CONSTEXPR` and `HPX_CONSTEXPR`
- [PR #3266](#)³⁹²⁴ - Replace `boost::format` with custom `sprintf`-based implementation
- [PR #3265](#)³⁹²⁵ - Split parallel tests on CircleCI
- [PR #3262](#)³⁹²⁶ - Making sure documentation correctly links to source files
- [PR #3261](#)³⁹²⁷ - Apex refactoring fix rebind
- [PR #3260](#)³⁹²⁸ - Isolate performance counter parser into a separate TU
- [PR #3256](#)³⁹²⁹ - Post 1.1.0 version bumps
- [PR #3254](#)³⁹³⁰ - Adding trait for actions allowing to make runtime decision on whether to execute it directly
- [PR #3253](#)³⁹³¹ - Bump minimal supported Boost to 1.58.0
- [PR #3251](#)³⁹³² - Adds new feature: changing interval used in `interval_timer` (issue 3244)
- [PR #3239](#)³⁹³³ - Changing `std::rand()` to a better inbuilt PRNG generator.
- [PR #3234](#)³⁹³⁴ - Disable background thread when networking is off
- [PR #3232](#)³⁹³⁵ - Clean up suspension tests
- [PR #3230](#)³⁹³⁶ - Add optional scheduler mode parameter to `create_thread_pool` function
- [PR #3228](#)³⁹³⁷ - Allow suspension also on static schedulers
- [PR #3163](#)³⁹³⁸ - libfabric parcelport w/o `HPX_PARCELPORTR_LIBFABRIC_ENDPOINT_RDM`
- [PR #3036](#)³⁹³⁹ - Switching to CircleCI 2.0

HPX V1.1.0 (Mar 24, 2018)

General changes

Here are some of the main highlights and changes for this release (in no particular order):

- We have changed the way *HPX* manages the processing units on a node. We do not longer implicitly bind all available cores to a single thread pool. The user has now full control over what processing units are bound to what thread pool, each with a separate scheduler. It is now also possible to create your own scheduler implementation and control what processing units this scheduler should use. We added the `hpx::resource::partitioner` that manages all available processing units and assigns resources to the used thread pools. Thread pools can be

³⁹²³ <https://github.com/TheHPXProject/hpx/pull/3268>

³⁹²⁴ <https://github.com/TheHPXProject/hpx/pull/3266>

³⁹²⁵ <https://github.com/TheHPXProject/hpx/pull/3265>

³⁹²⁶ <https://github.com/TheHPXProject/hpx/pull/3262>

³⁹²⁷ <https://github.com/TheHPXProject/hpx/pull/3261>

³⁹²⁸ <https://github.com/TheHPXProject/hpx/pull/3260>

³⁹²⁹ <https://github.com/TheHPXProject/hpx/pull/3256>

³⁹³⁰ <https://github.com/TheHPXProject/hpx/pull/3254>

³⁹³¹ <https://github.com/TheHPXProject/hpx/pull/3253>

³⁹³² <https://github.com/TheHPXProject/hpx/pull/3251>

³⁹³³ <https://github.com/TheHPXProject/hpx/pull/3239>

³⁹³⁴ <https://github.com/TheHPXProject/hpx/pull/3234>

³⁹³⁵ <https://github.com/TheHPXProject/hpx/pull/3232>

³⁹³⁶ <https://github.com/TheHPXProject/hpx/pull/3230>

³⁹³⁷ <https://github.com/TheHPXProject/hpx/pull/3228>

³⁹³⁸ <https://github.com/TheHPXProject/hpx/pull/3163>

³⁹³⁹ <https://github.com/TheHPXProject/hpx/pull/3036>

now be suspended/resumed independently. This functionality helps in running *HPX* concurrently to code that is directly relying on [OpenMP](https://openmp.org/wp/)³⁹⁴⁰ and/or [MPI](https://en.wikipedia.org/wiki/Message_Passing_Interface)³⁹⁴¹.

- We have continued to implement various parallel algorithms. *HPX* now almost completely implements all of the parallel algorithms as specified by the [C++17 Standard](http://www.open-std.org/jtc1/sc22/wg21)³⁹⁴². We have also continued to implement these algorithms for the distributed use case (for segmented data structures, such as `hpx::partitioned_vector`).
- Added a compatibility layer for `std::thread`, `std::mutex`, and `std::condition_variable` allowing for the code to use those facilities where available and to fall back to the corresponding Boost facilities otherwise. The [CMake](https://www.cmake.org)³⁹⁴³ configuration option `-DHPX_WITH_THREAD_COMPATIBILITY=On` can be used to force using the Boost equivalents.
- The parameter sequence for the `hpx::parallel::transform_inclusive_scan` overload taking one iterator range has changed (again) to match the changes this algorithm has undergone while being moved to C++17. The old overloads can be still enabled at configure time by passing `-DHPX_WITH_TRANSFORM_REDUCE_COMPATIBILITY=On` to [CMake](https://www.cmake.org)³⁹⁴⁴.
- The parameter sequence for the `hpx::parallel::inclusive_scan` overload taking one iterator range has changed to match the changes this algorithm has undergone while being moved to C++17. The old overloads can be still enabled at configure time by passing `-DHPX_WITH_INCLUSIVE_SCAN_COMPATIBILITY=On` to [CMake](https://www.cmake.org).
- Added a helper facility `hpx::local_new` which is equivalent to `hpx::new_` except that it creates components locally only. As a consequence, the used component constructor may accept non-serializable argument types and/or non-const references or pointers.
- Removed the (broken) component type `hpx::lcos::queue<T>`. The old type is still available at configure time by passing `-DHPX_WITH_QUEUE_COMPATIBILITY=On` to [CMake](https://www.cmake.org).
- The parallel algorithms adopted for C++17 restrict the iterator categories usable with those to at least forward iterators. Our implementation of the parallel algorithms was supporting input iterators (and output iterators) as well by simply falling back to sequential execution. We have now made our implementations conforming by requiring at least forward iterators. In order to enable the old behavior use the compatibility option `-DHPX_WITH_ALGORITHM_INPUT_ITERATOR_SUPPORT=On` on the [CMake](https://www.cmake.org)³⁹⁴⁵ command line.
- We have added the functionalities allowing for LCOs being implemented using (simple) components. Before LCOs had to always be implemented using managed components.
- User defined components don't have to be default-constructible anymore. Return types from actions don't have to be default-constructible anymore either. Our serialization layer now in general supports non-default-constructible types.
- We have added a new launch policy `hpx::launch::lazy` that allows one to defer the decision on what launch policy to use to the point of execution. This policy is initialized with a function (object) that – when invoked – is expected to produce the desired launch policy.

³⁹⁴⁰ <https://openmp.org/wp/>

³⁹⁴¹ https://en.wikipedia.org/wiki/Message_Passing_Interface

³⁹⁴² <http://www.open-std.org/jtc1/sc22/wg21>

³⁹⁴³ <https://www.cmake.org>

³⁹⁴⁴ <https://www.cmake.org>

³⁹⁴⁵ <https://www.cmake.org>

Breaking changes

- We have dropped support for the gcc compiler version V4.8. The minimal gcc version we now test on is gcc V4.9. The minimally required version of [CMake](https://www.cmake.org)³⁹⁴⁶ is now V3.3.2.
- We have dropped support for the Visual Studio 2013 compiler version. The minimal Visual Studio version we now test on is Visual Studio 2015.5.
- We have dropped support for the Boost V1.51-V1.54. The minimal version of Boost we now test is Boost V1.55.
- We have dropped support for the `hpx::util::unwrapped` API. `hpx::util::unwrapped` will stay functional to some degree, until it finally gets removed in a later version of HPX. The functional usage of `hpx::util::unwrapped` should be changed to the new `hpx::util::unwrapping` function whereas the immediate usage should be replaced to `hpx::util::unwrap`.
- The performance counter names referring to properties as exposed by the threading subsystem have changes as those now additionally have to specify the thread-pool. See the corresponding documentation for more details.
- The overloads of `hpx::async` that invoke an action do not perform implicit unwrapping of the returned future anymore in case the invoked function does return a future in the first place. In this case `hpx::async` now returns a `hpx::future<future<T>>` making its behavior conforming to its local counterpart.
- We have replaced the use of `boost::exception_ptr` in our APIs with the equivalent `std::exception_ptr`. Please change your codes accordingly. No compatibility settings are provided.
- We have removed the compatibility settings for `HPX_WITH_COLOCATED_BACKWARDS_COMPATIBILITY` and `HPX_WITH_COMPONENT_GET_GID_COMPATIBILITY` as their life-cycle has reached its end.
- We have removed the experimental thread schedulers `hierarchy_scheduler`, `periodic_priority_scheduler` and `throttling_scheduler` in an effort to clean up and consolidate our thread schedulers.

Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this release.

- [PR #3250](#)³⁹⁴⁷ - Apex refactoring with guides
- [PR #3249](#)³⁹⁴⁸ - Updating People.qbk
- [PR #3246](#)³⁹⁴⁹ - Assorted fixes for CUDA
- [PR #3245](#)³⁹⁵⁰ - Apex refactoring with guides
- [PR #3242](#)³⁹⁵¹ - Modify task counting in `thread_queue.hpp`
- [PR #3240](#)³⁹⁵² - Fixed typos
- [PR #3238](#)³⁹⁵³ - Readding accidentally removed `std::abort`
- [PR #3237](#)³⁹⁵⁴ - Adding Pipeline example
- [PR #3236](#)³⁹⁵⁵ - Fixing `memory_block`

³⁹⁴⁶ <https://www.cmake.org>

³⁹⁴⁷ <https://github.com/TheHPXProject/hpx/pull/3250>

³⁹⁴⁸ <https://github.com/TheHPXProject/hpx/pull/3249>

³⁹⁴⁹ <https://github.com/TheHPXProject/hpx/pull/3246>

³⁹⁵⁰ <https://github.com/TheHPXProject/hpx/pull/3245>

³⁹⁵¹ <https://github.com/TheHPXProject/hpx/pull/3242>

³⁹⁵² <https://github.com/TheHPXProject/hpx/pull/3240>

³⁹⁵³ <https://github.com/TheHPXProject/hpx/pull/3238>

³⁹⁵⁴ <https://github.com/TheHPXProject/hpx/pull/3237>

³⁹⁵⁵ <https://github.com/TheHPXProject/hpx/pull/3236>

- PR #3233³⁹⁵⁶ - Make `schedule_thread` take suspended threads into account
- Issue #3226³⁹⁵⁷ - `memory_block` is breaking, signaling SIGSEGV on a thread on creation and freeing
- PR #3225³⁹⁵⁸ - Applying quick fix for `hwloc-2.0`
- Issue #3224³⁹⁵⁹ - HPX counters crashing the application
- PR #3223³⁹⁶⁰ - Fix returns when setting config entries
- Issue #3222³⁹⁶¹ - Errors linking `libhpx.so`
- Issue #3221³⁹⁶² - HPX on Mac OS X with `HWLoc 2.0.0` fails to run
- PR #3216³⁹⁶³ - Reorder a variadic array to satisfy VS 2017 15.6
- PR #3214³⁹⁶⁴ - Changed `prerequisites.qbk` to avoid confusion while building boost
- PR #3213³⁹⁶⁵ - Relax locks for thread suspension to avoid holding locks when yielding
- PR #3212³⁹⁶⁶ - Fix check in `sequenced_executor` test
- PR #3211³⁹⁶⁷ - Use `preinit_array` to set `argc/argv` in `init_globally` example
- PR #3210³⁹⁶⁸ - Adapted `parallel::{search | search_n}` for Ranges TS (see #1668)
- PR #3209³⁹⁶⁹ - Fix locking problems during shutdown
- Issue #3208³⁹⁷⁰ - `init_globally` throwing a run-time error
- PR #3206³⁹⁷¹ - Addition of new arithmetic performance counter “Count”
- PR #3205³⁹⁷² - Fixing return type calculation for `bulk_then_execute`
- PR #3204³⁹⁷³ - Changing `std::rand()` to a better inbuilt PRNG generator
- PR #3203³⁹⁷⁴ - Resolving problems during shutdown for VS2015
- PR #3202³⁹⁷⁵ - Making sure resource partitioner is not accessed if its not valid
- PR #3201³⁹⁷⁶ - Fixing `optional::swap`
- Issue #3200³⁹⁷⁷ - `hpx::util::optional` fails
- PR #3199³⁹⁷⁸ - Fix `sliding_semaphore` test

³⁹⁵⁶ <https://github.com/TheHPXProject/hpx/pull/3233>

³⁹⁵⁷ <https://github.com/TheHPXProject/hpx/issues/3226>

³⁹⁵⁸ <https://github.com/TheHPXProject/hpx/pull/3225>

³⁹⁵⁹ <https://github.com/TheHPXProject/hpx/issues/3224>

³⁹⁶⁰ <https://github.com/TheHPXProject/hpx/pull/3223>

³⁹⁶¹ <https://github.com/TheHPXProject/hpx/issues/3222>

³⁹⁶² <https://github.com/TheHPXProject/hpx/issues/3221>

³⁹⁶³ <https://github.com/TheHPXProject/hpx/pull/3216>

³⁹⁶⁴ <https://github.com/TheHPXProject/hpx/pull/3214>

³⁹⁶⁵ <https://github.com/TheHPXProject/hpx/pull/3213>

³⁹⁶⁶ <https://github.com/TheHPXProject/hpx/pull/3212>

³⁹⁶⁷ <https://github.com/TheHPXProject/hpx/pull/3211>

³⁹⁶⁸ <https://github.com/TheHPXProject/hpx/pull/3210>

³⁹⁶⁹ <https://github.com/TheHPXProject/hpx/pull/3209>

³⁹⁷⁰ <https://github.com/TheHPXProject/hpx/issues/3208>

³⁹⁷¹ <https://github.com/TheHPXProject/hpx/pull/3206>

³⁹⁷² <https://github.com/TheHPXProject/hpx/pull/3205>

³⁹⁷³ <https://github.com/TheHPXProject/hpx/pull/3204>

³⁹⁷⁴ <https://github.com/TheHPXProject/hpx/pull/3203>

³⁹⁷⁵ <https://github.com/TheHPXProject/hpx/pull/3202>

³⁹⁷⁶ <https://github.com/TheHPXProject/hpx/pull/3201>

³⁹⁷⁷ <https://github.com/TheHPXProject/hpx/issues/3200>

³⁹⁷⁸ <https://github.com/TheHPXProject/hpx/pull/3199>

- PR #3198³⁹⁷⁹ - Set `pre_main` status before launching `run_helper`
- PR #3197³⁹⁸⁰ - Update `README.rst`
- PR #3194³⁹⁸¹ - `parallel::fill` updated for Ranges TS
- PR #3193³⁹⁸² - Updating `Runtime.cpp` by adding correct description of Performance counters during register
- PR #3191³⁹⁸³ - Fix `sliding_semaphore_2338` test
- PR #3190³⁹⁸⁴ - Topology improvements
- PR #3189³⁹⁸⁵ - Deleting one include of median from BOOST library to `arithmetics_counter` file
- PR #3188³⁹⁸⁶ - Optionally disable printing of diagnostics during terminate
- PR #3187³⁹⁸⁷ - Suppressing cmake warning issued by `cmake > V3.11`
- PR #3185³⁹⁸⁸ - Remove unused `scoped_unlock`, `unlock_guard_try`
- PR #3184³⁹⁸⁹ - Fix nqueen example
- PR #3183³⁹⁹⁰ - Add runtime start/stop, resume/suspend and OpenMP benchmarks
- Issue #3182³⁹⁹¹ - `bulk_then_execute` has unexpected return type/does not compile
- Issue #3181³⁹⁹² - `hwloc 2.0` breaks topo class and cannot be used
- Issue #3180³⁹⁹³ - Schedulers that don't support suspend/resume are unusable
- PR #3179³⁹⁹⁴ - Various minor changes to support FLeCSI
- PR #3178³⁹⁹⁵ - Fix #3124
- PR #3177³⁹⁹⁶ - Removed `allgather`
- PR #3176³⁹⁹⁷ - Fixed Documentation for “`using_hpx_pkgconfig`”
- PR #3174³⁹⁹⁸ - Add `hpx::iostream::ostream` overload to `format_to`
- PR #3172³⁹⁹⁹ - Fix lifo queue backend
- PR #3171⁴⁰⁰⁰ - adding the missing `unset()` function to `cpu_mask()` for case of more than 64 threads
- PR #3170⁴⁰⁰¹ - Add cmake flag `-DHPX_WITH_FAULT_TOLERANCE=ON` (OFF by default)

³⁹⁷⁹ <https://github.com/TheHPXProject/hpx/pull/3198>

³⁹⁸⁰ <https://github.com/TheHPXProject/hpx/pull/3197>

³⁹⁸¹ <https://github.com/TheHPXProject/hpx/pull/3194>

³⁹⁸² <https://github.com/TheHPXProject/hpx/pull/3193>

³⁹⁸³ <https://github.com/TheHPXProject/hpx/pull/3191>

³⁹⁸⁴ <https://github.com/TheHPXProject/hpx/pull/3190>

³⁹⁸⁵ <https://github.com/TheHPXProject/hpx/pull/3189>

³⁹⁸⁶ <https://github.com/TheHPXProject/hpx/pull/3188>

³⁹⁸⁷ <https://github.com/TheHPXProject/hpx/pull/3187>

³⁹⁸⁸ <https://github.com/TheHPXProject/hpx/pull/3185>

³⁹⁸⁹ <https://github.com/TheHPXProject/hpx/pull/3184>

³⁹⁹⁰ <https://github.com/TheHPXProject/hpx/pull/3183>

³⁹⁹¹ <https://github.com/TheHPXProject/hpx/issues/3182>

³⁹⁹² <https://github.com/TheHPXProject/hpx/issues/3181>

³⁹⁹³ <https://github.com/TheHPXProject/hpx/issues/3180>

³⁹⁹⁴ <https://github.com/TheHPXProject/hpx/pull/3179>

³⁹⁹⁵ <https://github.com/TheHPXProject/hpx/pull/3178>

³⁹⁹⁶ <https://github.com/TheHPXProject/hpx/pull/3177>

³⁹⁹⁷ <https://github.com/TheHPXProject/hpx/pull/3176>

³⁹⁹⁸ <https://github.com/TheHPXProject/hpx/pull/3174>

³⁹⁹⁹ <https://github.com/TheHPXProject/hpx/pull/3172>

⁴⁰⁰⁰ <https://github.com/TheHPXProject/hpx/pull/3171>

⁴⁰⁰¹ <https://github.com/TheHPXProject/hpx/pull/3170>

- PR #3169⁴⁰⁰² - Adapted parallel::count/count_if for Ranges TS (see #1668)
- PR #3168⁴⁰⁰³ - Changing used namespace for seq execution policy
- Issue #3167⁴⁰⁰⁴ - Update GSoC projects
- Issue #3166⁴⁰⁰⁵ - Application (Octotiger) gets stuck on hpx::finalize when only using one thread
- Issue #3165⁴⁰⁰⁶ - Compilation of parallel algorithms with HPX_WITH_DATAPAR is broken
- PR #3164⁴⁰⁰⁷ - Fixing component migration
- PR #3162⁴⁰⁰⁸ - regex_from_pattern: escape regex special characters to avoid misinterpretation
- Issue #3161⁴⁰⁰⁹ - Building HPX with hwloc 2.0.0 fails
- PR #3160⁴⁰¹⁰ - Fixing the handling of quoted command line arguments.
- PR #3158⁴⁰¹¹ - Fixing a race with timed suspension (second attempt)
- PR #3157⁴⁰¹² - Revert “Fixing a race with timed suspension”
- PR #3156⁴⁰¹³ - Fixing serialization of classes with incompatible serialize signature
- PR #3154⁴⁰¹⁴ - More refactorings based on clang-tidy reports
- PR #3153⁴⁰¹⁵ - Fixing a race with timed suspension
- PR #3152⁴⁰¹⁶ - Documentation for runtime suspension
- PR #3151⁴⁰¹⁷ - Use small_vector only from boost version 1.59 onwards
- PR #3150⁴⁰¹⁸ - Avoiding more stack overflows
- PR #3148⁴⁰¹⁹ - Refactoring component_base and base_action/transfer_base_action
- PR #3147⁴⁰²⁰ - Move yield_while out of detail namespace and into own file
- PR #3145⁴⁰²¹ - Remove a leftover of the cxx11 std array cleanup
- PR #3144⁴⁰²² - Minor changes to how actions are executed
- PR #3143⁴⁰²³ - Fix stack overhead
- PR #3142⁴⁰²⁴ - Fix typo in config.hpp

⁴⁰⁰² <https://github.com/TheHPXProject/hpx/pull/3169>

⁴⁰⁰³ <https://github.com/TheHPXProject/hpx/pull/3168>

⁴⁰⁰⁴ <https://github.com/TheHPXProject/hpx/issues/3167>

⁴⁰⁰⁵ <https://github.com/TheHPXProject/hpx/issues/3166>

⁴⁰⁰⁶ <https://github.com/TheHPXProject/hpx/issues/3165>

⁴⁰⁰⁷ <https://github.com/TheHPXProject/hpx/pull/3164>

⁴⁰⁰⁸ <https://github.com/TheHPXProject/hpx/pull/3162>

⁴⁰⁰⁹ <https://github.com/TheHPXProject/hpx/issues/3161>

⁴⁰¹⁰ <https://github.com/TheHPXProject/hpx/pull/3160>

⁴⁰¹¹ <https://github.com/TheHPXProject/hpx/pull/3158>

⁴⁰¹² <https://github.com/TheHPXProject/hpx/pull/3157>

⁴⁰¹³ <https://github.com/TheHPXProject/hpx/pull/3156>

⁴⁰¹⁴ <https://github.com/TheHPXProject/hpx/pull/3154>

⁴⁰¹⁵ <https://github.com/TheHPXProject/hpx/pull/3153>

⁴⁰¹⁶ <https://github.com/TheHPXProject/hpx/pull/3152>

⁴⁰¹⁷ <https://github.com/TheHPXProject/hpx/pull/3151>

⁴⁰¹⁸ <https://github.com/TheHPXProject/hpx/pull/3150>

⁴⁰¹⁹ <https://github.com/TheHPXProject/hpx/pull/3148>

⁴⁰²⁰ <https://github.com/TheHPXProject/hpx/pull/3147>

⁴⁰²¹ <https://github.com/TheHPXProject/hpx/pull/3145>

⁴⁰²² <https://github.com/TheHPXProject/hpx/pull/3144>

⁴⁰²³ <https://github.com/TheHPXProject/hpx/pull/3143>

⁴⁰²⁴ <https://github.com/TheHPXProject/hpx/pull/3142>

- PR #3141⁴⁰²⁵ - Fixing small_vector compatibility with older boost version
- PR #3140⁴⁰²⁶ - is_heap_text fix
- Issue #3139⁴⁰²⁷ - Error in is_heap_tests.hpp
- PR #3138⁴⁰²⁸ - Partially reverting #3126
- PR #3137⁴⁰²⁹ - Suspend speedup
- PR #3136⁴⁰³⁰ - Revert “Fixing #2325”
- PR #3135⁴⁰³¹ - Improving destruction of threads
- Issue #3134⁴⁰³² - HPX_SERIALIZATION_SPLIT_FREE does not stop compiler from looking for serialize() method
- PR #3133⁴⁰³³ - Make hwloc compulsory
- PR #3132⁴⁰³⁴ - Update CXX14 constexpr feature test
- PR #3131⁴⁰³⁵ - Fixing #2325
- PR #3130⁴⁰³⁶ - Avoid completion handler allocation
- PR #3129⁴⁰³⁷ - Suspend runtime
- PR #3128⁴⁰³⁸ - Make docbook dtd and xsl path names consistent
- PR #3127⁴⁰³⁹ - Add hpx::start nullptr overloads
- PR #3126⁴⁰⁴⁰ - Cleaning up coroutine implementation
- PR #3125⁴⁰⁴¹ - Replacing nullptr with hpx::threads::invalid_thread_id
- Issue #3124⁴⁰⁴² - Add hello_world_component to CI builds
- PR #3123⁴⁰⁴³ - Add new constructor.
- PR #3122⁴⁰⁴⁴ - Fixing #3121
- Issue #3121⁴⁰⁴⁵ - HPX_SMT_PAUSE is broken on non-x86 platforms when __GNUC__ is defined
- PR #3120⁴⁰⁴⁶ - Don't use boost::intrusive_ptr for thread_id_type
- PR #3119⁴⁰⁴⁷ - Disable default executor compatibility with V1 executors

⁴⁰²⁵ <https://github.com/TheHPXProject/hpx/pull/3141>

⁴⁰²⁶ <https://github.com/TheHPXProject/hpx/pull/3140>

⁴⁰²⁷ <https://github.com/TheHPXProject/hpx/issues/3139>

⁴⁰²⁸ <https://github.com/TheHPXProject/hpx/pull/3138>

⁴⁰²⁹ <https://github.com/TheHPXProject/hpx/pull/3137>

⁴⁰³⁰ <https://github.com/TheHPXProject/hpx/pull/3136>

⁴⁰³¹ <https://github.com/TheHPXProject/hpx/pull/3135>

⁴⁰³² <https://github.com/TheHPXProject/hpx/issues/3134>

⁴⁰³³ <https://github.com/TheHPXProject/hpx/pull/3133>

⁴⁰³⁴ <https://github.com/TheHPXProject/hpx/pull/3132>

⁴⁰³⁵ <https://github.com/TheHPXProject/hpx/pull/3131>

⁴⁰³⁶ <https://github.com/TheHPXProject/hpx/pull/3130>

⁴⁰³⁷ <https://github.com/TheHPXProject/hpx/pull/3129>

⁴⁰³⁸ <https://github.com/TheHPXProject/hpx/pull/3128>

⁴⁰³⁹ <https://github.com/TheHPXProject/hpx/pull/3127>

⁴⁰⁴⁰ <https://github.com/TheHPXProject/hpx/pull/3126>

⁴⁰⁴¹ <https://github.com/TheHPXProject/hpx/pull/3125>

⁴⁰⁴² <https://github.com/TheHPXProject/hpx/issues/3124>

⁴⁰⁴³ <https://github.com/TheHPXProject/hpx/pull/3123>

⁴⁰⁴⁴ <https://github.com/TheHPXProject/hpx/pull/3122>

⁴⁰⁴⁵ <https://github.com/TheHPXProject/hpx/issues/3121>

⁴⁰⁴⁶ <https://github.com/TheHPXProject/hpx/pull/3120>

⁴⁰⁴⁷ <https://github.com/TheHPXProject/hpx/pull/3119>

- PR #3118⁴⁰⁴⁸ - Adding performance_counter::reinit to allow for dynamically changing counter sets
- PR #3117⁴⁰⁴⁹ - Replace uses of boost/experimental::optional with util::optional
- PR #3116⁴⁰⁵⁰ - Moving background thread APEX timer #2980
- PR #3115⁴⁰⁵¹ - Fixing race condition in channel test
- PR #3114⁴⁰⁵² - Avoid using util::function for thread function wrappers
- PR #3113⁴⁰⁵³ - cmake V3.10.2 has changed the variable names used for MPI
- PR #3112⁴⁰⁵⁴ - Minor fixes to exclusive_scan algorithm
- PR #3111⁴⁰⁵⁵ - Revert “fix detection of cxx11_std_atomic”
- PR #3110⁴⁰⁵⁶ - Suspend thread pool
- PR #3109⁴⁰⁵⁷ - Fixing thread scheduling when yielding a thread id
- PR #3108⁴⁰⁵⁸ - Revert “Suspend thread pool”
- PR #3107⁴⁰⁵⁹ - Remove UB from thread::id relational operators
- PR #3106⁴⁰⁶⁰ - Add cmake test for std::decay_t to fix cuda build
- PR #3105⁴⁰⁶¹ - Fixing refcount for async traversal frame
- PR #3104⁴⁰⁶² - Local execution of direct actions is now actually performed directly
- PR #3103⁴⁰⁶³ - Adding support for generic counter_raw_values performance counter type
- Issue #3102⁴⁰⁶⁴ - Introduce generic performance counter type returning an array of values
- PR #3101⁴⁰⁶⁵ - Revert “Adapting stack overhead limit for gcc 4.9”
- PR #3100⁴⁰⁶⁶ - Fix #3068 (condition_variable deadlock)
- PR #3099⁴⁰⁶⁷ - Fixing lock held during suspension in papi counter component
- PR #3098⁴⁰⁶⁸ - Unbreak broadcast_wait_for_2822 test
- PR #3097⁴⁰⁶⁹ - Adapting stack overhead limit for gcc 4.9
- PR #3096⁴⁰⁷⁰ - fix detection of cxx11_std_atomic

⁴⁰⁴⁸ <https://github.com/TheHPXProject/hpx/pull/3118>

⁴⁰⁴⁹ <https://github.com/TheHPXProject/hpx/pull/3117>

⁴⁰⁵⁰ <https://github.com/TheHPXProject/hpx/pull/3116>

⁴⁰⁵¹ <https://github.com/TheHPXProject/hpx/pull/3115>

⁴⁰⁵² <https://github.com/TheHPXProject/hpx/pull/3114>

⁴⁰⁵³ <https://github.com/TheHPXProject/hpx/pull/3113>

⁴⁰⁵⁴ <https://github.com/TheHPXProject/hpx/pull/3112>

⁴⁰⁵⁵ <https://github.com/TheHPXProject/hpx/pull/3111>

⁴⁰⁵⁶ <https://github.com/TheHPXProject/hpx/pull/3110>

⁴⁰⁵⁷ <https://github.com/TheHPXProject/hpx/pull/3109>

⁴⁰⁵⁸ <https://github.com/TheHPXProject/hpx/pull/3108>

⁴⁰⁵⁹ <https://github.com/TheHPXProject/hpx/pull/3107>

⁴⁰⁶⁰ <https://github.com/TheHPXProject/hpx/pull/3106>

⁴⁰⁶¹ <https://github.com/TheHPXProject/hpx/pull/3105>

⁴⁰⁶² <https://github.com/TheHPXProject/hpx/pull/3104>

⁴⁰⁶³ <https://github.com/TheHPXProject/hpx/pull/3103>

⁴⁰⁶⁴ <https://github.com/TheHPXProject/hpx/issues/3102>

⁴⁰⁶⁵ <https://github.com/TheHPXProject/hpx/pull/3101>

⁴⁰⁶⁶ <https://github.com/TheHPXProject/hpx/pull/3100>

⁴⁰⁶⁷ <https://github.com/TheHPXProject/hpx/pull/3099>

⁴⁰⁶⁸ <https://github.com/TheHPXProject/hpx/pull/3098>

⁴⁰⁶⁹ <https://github.com/TheHPXProject/hpx/pull/3097>

⁴⁰⁷⁰ <https://github.com/TheHPXProject/hpx/pull/3096>

- PR #3095⁴⁰⁷¹ - Add ciso646 header to get `_LIBCPP_VERSION` for testing inplace merge
- PR #3094⁴⁰⁷² - Relax atomic operations on performance counter values
- PR #3093⁴⁰⁷³ - Short-circuit `all_of/any_of/none_of` instantiations
- PR #3092⁴⁰⁷⁴ - Take advantage of C++14 lambda capture initialization syntax, where possible
- PR #3091⁴⁰⁷⁵ - Remove more references to Boost from logging code
- PR #3090⁴⁰⁷⁶ - Unify use of `yield/yield_k`
- PR #3089⁴⁰⁷⁷ - Fix a strange thing in `parallel::detail::handle_exception`. (Fix #2834.)
- Issue #3088⁴⁰⁷⁸ - A strange thing in `parallel::sort`.
- PR #3087⁴⁰⁷⁹ - Fixing assertion in `default_distribution_policy`
- PR #3086⁴⁰⁸⁰ - Implement `parallel::remove` and `parallel::remove_if`
- PR #3085⁴⁰⁸¹ - Addressing breaking changes in Boost V1.66
- PR #3084⁴⁰⁸² - Ignore build warnings round 2
- PR #3083⁴⁰⁸³ - Fix typo `HPX_WITH_MM_PREFECTH`
- PR #3081⁴⁰⁸⁴ - Pre-decay template arguments early
- PR #3080⁴⁰⁸⁵ - Suspend thread pool
- PR #3079⁴⁰⁸⁶ - Ignore build warnings
- PR #3078⁴⁰⁸⁷ - Don't test `inplace_merge` with `libc++`
- PR #3076⁴⁰⁸⁸ - Fixing 3075: Part 1
- PR #3074⁴⁰⁸⁹ - Fix more build warnings
- PR #3073⁴⁰⁹⁰ - Suspend thread cleanup
- PR #3072⁴⁰⁹¹ - Change existing `symbol_namespace::iterate` to return all data instead of invoking a callback
- PR #3071⁴⁰⁹² - Fixing `pack_traversal_async` test
- PR #3070⁴⁰⁹³ - Fix `dynamic_counters_loaded_1508` test by adding dependency to `memory_component`

⁴⁰⁷¹ <https://github.com/TheHPXProject/hpx/pull/3095>

⁴⁰⁷² <https://github.com/TheHPXProject/hpx/pull/3094>

⁴⁰⁷³ <https://github.com/TheHPXProject/hpx/pull/3093>

⁴⁰⁷⁴ <https://github.com/TheHPXProject/hpx/pull/3092>

⁴⁰⁷⁵ <https://github.com/TheHPXProject/hpx/pull/3091>

⁴⁰⁷⁶ <https://github.com/TheHPXProject/hpx/pull/3090>

⁴⁰⁷⁷ <https://github.com/TheHPXProject/hpx/pull/3089>

⁴⁰⁷⁸ <https://github.com/TheHPXProject/hpx/issues/3088>

⁴⁰⁷⁹ <https://github.com/TheHPXProject/hpx/pull/3087>

⁴⁰⁸⁰ <https://github.com/TheHPXProject/hpx/pull/3086>

⁴⁰⁸¹ <https://github.com/TheHPXProject/hpx/pull/3085>

⁴⁰⁸² <https://github.com/TheHPXProject/hpx/pull/3084>

⁴⁰⁸³ <https://github.com/TheHPXProject/hpx/pull/3083>

⁴⁰⁸⁴ <https://github.com/TheHPXProject/hpx/pull/3081>

⁴⁰⁸⁵ <https://github.com/TheHPXProject/hpx/pull/3080>

⁴⁰⁸⁶ <https://github.com/TheHPXProject/hpx/pull/3079>

⁴⁰⁸⁷ <https://github.com/TheHPXProject/hpx/pull/3078>

⁴⁰⁸⁸ <https://github.com/TheHPXProject/hpx/pull/3076>

⁴⁰⁸⁹ <https://github.com/TheHPXProject/hpx/pull/3074>

⁴⁰⁹⁰ <https://github.com/TheHPXProject/hpx/pull/3073>

⁴⁰⁹¹ <https://github.com/TheHPXProject/hpx/pull/3072>

⁴⁰⁹² <https://github.com/TheHPXProject/hpx/pull/3071>

⁴⁰⁹³ <https://github.com/TheHPXProject/hpx/pull/3070>

- PR #3069⁴⁰⁹⁴ - Fix scheduling loop exit
- Issue #3068⁴⁰⁹⁵ - `hpx::lcos::condition_variable` could be suspect to deadlocks
- PR #3067⁴⁰⁹⁶ - `#ifdef` out `random_shuffle` deprecated in later c++
- PR #3066⁴⁰⁹⁷ - Make coalescing test depend on coalescing library to ensure it gets built
- PR #3065⁴⁰⁹⁸ - Workaround for `minimal_timed_async_executor_test` compilation failures, attempts to copy a deferred call (in unevaluated context)
- PR #3064⁴⁰⁹⁹ - Fixing wrong condition in `wrapper_heap`
- PR #3062⁴¹⁰⁰ - Fix exception handling for `execution::seq`
- PR #3061⁴¹⁰¹ - Adapt MSVC C++ mode handling to VS15.5
- PR #3060⁴¹⁰² - Fix compiler problem in MSVC release mode
- PR #3059⁴¹⁰³ - Fixing #2931
- Issue #3058⁴¹⁰⁴ - `minimal_timed_async_executor_test_exe` fails to compile on master (d6f505c)
- PR #3057⁴¹⁰⁵ - Fix `stable_merge_2964` compilation problems
- PR #3056⁴¹⁰⁶ - Fix some build warnings caused by unused variables/unnecessary tests
- PR #3055⁴¹⁰⁷ - Update documentation for running tests
- Issue #3054⁴¹⁰⁸ - Assertion failure when using bulk `hpx::new_` in asynchronous mode
- PR #3052⁴¹⁰⁹ - Do not bind test running to `cmake` test build rule
- PR #3051⁴¹¹⁰ - Fix HPX-Qt interaction in Qt example.
- Issue #3048⁴¹¹¹ - `nqueen` example fails occasionally
- PR #3047⁴¹¹² - Fixing #3044
- PR #3046⁴¹¹³ - Add OS thread suspension
- PR #3042⁴¹¹⁴ - PyCicle - first attempt at a build tool for checking PR's
- PR #3041⁴¹¹⁵ - Fix a problem about asynchronous execution of `parallel::merge` and `parallel::partition`.
- PR #3040⁴¹¹⁶ - Fix a mistake about exception handling in asynchronous execution of `scan_partitioner`.

⁴⁰⁹⁴ <https://github.com/TheHPXProject/hpx/pull/3069>

⁴⁰⁹⁵ <https://github.com/TheHPXProject/hpx/issues/3068>

⁴⁰⁹⁶ <https://github.com/TheHPXProject/hpx/pull/3067>

⁴⁰⁹⁷ <https://github.com/TheHPXProject/hpx/pull/3066>

⁴⁰⁹⁸ <https://github.com/TheHPXProject/hpx/pull/3065>

⁴⁰⁹⁹ <https://github.com/TheHPXProject/hpx/pull/3064>

⁴¹⁰⁰ <https://github.com/TheHPXProject/hpx/pull/3062>

⁴¹⁰¹ <https://github.com/TheHPXProject/hpx/pull/3061>

⁴¹⁰² <https://github.com/TheHPXProject/hpx/pull/3060>

⁴¹⁰³ <https://github.com/TheHPXProject/hpx/pull/3059>

⁴¹⁰⁴ <https://github.com/TheHPXProject/hpx/issues/3058>

⁴¹⁰⁵ <https://github.com/TheHPXProject/hpx/pull/3057>

⁴¹⁰⁶ <https://github.com/TheHPXProject/hpx/pull/3056>

⁴¹⁰⁷ <https://github.com/TheHPXProject/hpx/pull/3055>

⁴¹⁰⁸ <https://github.com/TheHPXProject/hpx/issues/3054>

⁴¹⁰⁹ <https://github.com/TheHPXProject/hpx/pull/3052>

⁴¹¹⁰ <https://github.com/TheHPXProject/hpx/pull/3051>

⁴¹¹¹ <https://github.com/TheHPXProject/hpx/issues/3048>

⁴¹¹² <https://github.com/TheHPXProject/hpx/pull/3047>

⁴¹¹³ <https://github.com/TheHPXProject/hpx/pull/3046>

⁴¹¹⁴ <https://github.com/TheHPXProject/hpx/pull/3042>

⁴¹¹⁵ <https://github.com/TheHPXProject/hpx/pull/3041>

⁴¹¹⁶ <https://github.com/TheHPXProject/hpx/pull/3040>

- PR #3039⁴¹¹⁷ - Consistently use executors to schedule work
- PR #3038⁴¹¹⁸ - Fixing local direct function execution and lambda actions perfect forwarding
- PR #3035⁴¹¹⁹ - Make parallel unit test names match build target/folder names
- PR #3033⁴¹²⁰ - Fix setting of default build type
- Issue #3032⁴¹²¹ - Fix partitioner arg copy found in #2982
- Issue #3031⁴¹²² - Errors linking libhpx.so due to missing references (master branch, commit 6679a8882)
- PR #3030⁴¹²³ - Revert “implement executor then interface with && forwarding reference”
- PR #3029⁴¹²⁴ - Run CI inspect checks before building
- PR #3028⁴¹²⁵ - Added range version of parallel::move
- Issue #3027⁴¹²⁶ - Implement all scheduling APIs in terms of executors
- PR #3026⁴¹²⁷ - implement executor then interface with && forwarding reference
- PR #3025⁴¹²⁸ - Fix typo uninitialized to uninitialized
- PR #3024⁴¹²⁹ - Inspect fixes
- PR #3023⁴¹³⁰ - P0356 Simplified partial function application
- PR #3022⁴¹³¹ - Master fixes
- PR #3021⁴¹³² - Segfault fix
- PR #3020⁴¹³³ - Disable command-line aliasing for applications that use user_main
- PR #3019⁴¹³⁴ - Adding enable_elasticity option to pool configuration
- PR #3018⁴¹³⁵ - Fix stack overflow detection configuration in header files
- PR #3017⁴¹³⁶ - Speed up local action execution
- PR #3016⁴¹³⁷ - Unify stack-overflow detection options, remove reference to libsigsegv
- PR #3015⁴¹³⁸ - Speeding up accessing the resource partitioner and the topology info
- Issue #3014⁴¹³⁹ - HPX does not compile on POWER8 with gcc 5.4

⁴¹¹⁷ <https://github.com/TheHPXProject/hpx/pull/3039>

⁴¹¹⁸ <https://github.com/TheHPXProject/hpx/pull/3038>

⁴¹¹⁹ <https://github.com/TheHPXProject/hpx/pull/3035>

⁴¹²⁰ <https://github.com/TheHPXProject/hpx/pull/3033>

⁴¹²¹ <https://github.com/TheHPXProject/hpx/issues/3032>

⁴¹²² <https://github.com/TheHPXProject/hpx/issues/3031>

⁴¹²³ <https://github.com/TheHPXProject/hpx/pull/3030>

⁴¹²⁴ <https://github.com/TheHPXProject/hpx/pull/3029>

⁴¹²⁵ <https://github.com/TheHPXProject/hpx/pull/3028>

⁴¹²⁶ <https://github.com/TheHPXProject/hpx/issues/3027>

⁴¹²⁷ <https://github.com/TheHPXProject/hpx/pull/3026>

⁴¹²⁸ <https://github.com/TheHPXProject/hpx/pull/3025>

⁴¹²⁹ <https://github.com/TheHPXProject/hpx/pull/3024>

⁴¹³⁰ <https://github.com/TheHPXProject/hpx/pull/3023>

⁴¹³¹ <https://github.com/TheHPXProject/hpx/pull/3022>

⁴¹³² <https://github.com/TheHPXProject/hpx/pull/3021>

⁴¹³³ <https://github.com/TheHPXProject/hpx/pull/3020>

⁴¹³⁴ <https://github.com/TheHPXProject/hpx/pull/3019>

⁴¹³⁵ <https://github.com/TheHPXProject/hpx/pull/3018>

⁴¹³⁶ <https://github.com/TheHPXProject/hpx/pull/3017>

⁴¹³⁷ <https://github.com/TheHPXProject/hpx/pull/3016>

⁴¹³⁸ <https://github.com/TheHPXProject/hpx/pull/3015>

⁴¹³⁹ <https://github.com/TheHPXProject/hpx/issues/3014>

- [Issue #3013](#)⁴¹⁴⁰ - hello_world occasionally prints multiple lines from a single OS-thread
- [PR #3012](#)⁴¹⁴¹ - Silence warning about casting away qualifiers in itt_notify.hpp
- [PR #3011](#)⁴¹⁴² - Fix cpuset leak in hwloc_topology_info.cpp
- [PR #3010](#)⁴¹⁴³ - Remove useless decay_copy
- [PR #3009](#)⁴¹⁴⁴ - Fixing 2996
- [PR #3008](#)⁴¹⁴⁵ - Remove unused internal function
- [PR #3007](#)⁴¹⁴⁶ - Fixing wrapper_heap alignment problems
- [Issue #3006](#)⁴¹⁴⁷ - hwloc memory leak
- [PR #3004](#)⁴¹⁴⁸ - Silence C4251 (needs to have dll-interface) for future_data_void
- [Issue #3003](#)⁴¹⁴⁹ - Suspension of runtime
- [PR #3001](#)⁴¹⁵⁰ - Attempting to avoid data races in async_traversal while evaluating dataflow()
- [PR #3000](#)⁴¹⁵¹ - Adding hpx::util::optional as a first step to replace experimental::optional
- [PR #2998](#)⁴¹⁵² - Cleanup up and Fixing component creation and deletion
- [Issue #2996](#)⁴¹⁵³ - Build fails with HPX_WITH_HWLOC=OFF
- [PR #2995](#)⁴¹⁵⁴ - Push more future_data functionality to source file
- [PR #2994](#)⁴¹⁵⁵ - WIP: Fix throttle test
- [PR #2993](#)⁴¹⁵⁶ - Making sure -hpx:help does not throw for required (but missing) arguments
- [PR #2992](#)⁴¹⁵⁷ - Adding non-blocking (on destruction) service executors
- [Issue #2991](#)⁴¹⁵⁸ - run_as_os_thread locks up
- [Issue #2990](#)⁴¹⁵⁹ - -help will not work until all required options are provided
- [PR #2989](#)⁴¹⁶⁰ - Improve error messages caused by misuse of dataflow
- [PR #2988](#)⁴¹⁶¹ - Improve error messages caused by misuse of .then
- [Issue #2987](#)⁴¹⁶² - stack overflow detection producing false positives

⁴¹⁴⁰ <https://github.com/TheHPXProject/hpx/issues/3013>

⁴¹⁴¹ <https://github.com/TheHPXProject/hpx/pull/3012>

⁴¹⁴² <https://github.com/TheHPXProject/hpx/pull/3011>

⁴¹⁴³ <https://github.com/TheHPXProject/hpx/pull/3010>

⁴¹⁴⁴ <https://github.com/TheHPXProject/hpx/pull/3009>

⁴¹⁴⁵ <https://github.com/TheHPXProject/hpx/pull/3008>

⁴¹⁴⁶ <https://github.com/TheHPXProject/hpx/pull/3007>

⁴¹⁴⁷ <https://github.com/TheHPXProject/hpx/issues/3006>

⁴¹⁴⁸ <https://github.com/TheHPXProject/hpx/pull/3004>

⁴¹⁴⁹ <https://github.com/TheHPXProject/hpx/issues/3003>

⁴¹⁵⁰ <https://github.com/TheHPXProject/hpx/pull/3001>

⁴¹⁵¹ <https://github.com/TheHPXProject/hpx/pull/3000>

⁴¹⁵² <https://github.com/TheHPXProject/hpx/pull/2998>

⁴¹⁵³ <https://github.com/TheHPXProject/hpx/issues/2996>

⁴¹⁵⁴ <https://github.com/TheHPXProject/hpx/pull/2995>

⁴¹⁵⁵ <https://github.com/TheHPXProject/hpx/pull/2994>

⁴¹⁵⁶ <https://github.com/TheHPXProject/hpx/pull/2993>

⁴¹⁵⁷ <https://github.com/TheHPXProject/hpx/pull/2992>

⁴¹⁵⁸ <https://github.com/TheHPXProject/hpx/issues/2991>

⁴¹⁵⁹ <https://github.com/TheHPXProject/hpx/issues/2990>

⁴¹⁶⁰ <https://github.com/TheHPXProject/hpx/pull/2989>

⁴¹⁶¹ <https://github.com/TheHPXProject/hpx/pull/2988>

⁴¹⁶² <https://github.com/TheHPXProject/hpx/issues/2987>

- [PR #2986](#)⁴¹⁶³ - Deduplicate non-dependent thread_info logging types
- [PR #2985](#)⁴¹⁶⁴ - Adapted parallel::{all_of|any_of|none_of} for Ranges TS (see #1668)
- [PR #2984](#)⁴¹⁶⁵ - Refactor one_size_heap code to simplify code
- [PR #2983](#)⁴¹⁶⁶ - Fixing local_new_component
- [PR #2982](#)⁴¹⁶⁷ - Clang tidy
- [PR #2981](#)⁴¹⁶⁸ - Simplify allocator rebinding in pack traversal
- [PR #2979](#)⁴¹⁶⁹ - Fixing integer overflows
- [PR #2978](#)⁴¹⁷⁰ - Implement parallel::inplace_merge
- [Issue #2977](#)⁴¹⁷¹ - Make hwloc compulsory instead of optional
- [PR #2976](#)⁴¹⁷² - Making sure client_base instance that registered the component does not unregister it when being destructed
- [PR #2975](#)⁴¹⁷³ - Change version of pulled APEX to master
- [PR #2974](#)⁴¹⁷⁴ - Fix domain not being freed at the end of scheduling loop
- [PR #2973](#)⁴¹⁷⁵ - Fix small typos
- [PR #2972](#)⁴¹⁷⁶ - Adding uintstd.h header
- [PR #2971](#)⁴¹⁷⁷ - Fall back to creating local components using local_new
- [PR #2970](#)⁴¹⁷⁸ - Improve is_tuple_like trait
- [PR #2969](#)⁴¹⁷⁹ - Fix HPX_WITH_MORE_THAN_64_THREADS default value
- [PR #2968](#)⁴¹⁸⁰ - Cleaning up dataflow overload set
- [PR #2967](#)⁴¹⁸¹ - Make parallel::merge is stable. (Fix #2964.)
- [PR #2966](#)⁴¹⁸² - Fixing a couple of held locks during exception handling
- [PR #2965](#)⁴¹⁸³ - Adding missing #include
- [Issue #2964](#)⁴¹⁸⁴ - parallel merge is not stable
- [PR #2963](#)⁴¹⁸⁵ - Making sure any function object passed to dataflow is released after being invoked

⁴¹⁶³ <https://github.com/TheHPXProject/hpx/pull/2986>

⁴¹⁶⁴ <https://github.com/TheHPXProject/hpx/pull/2985>

⁴¹⁶⁵ <https://github.com/TheHPXProject/hpx/pull/2984>

⁴¹⁶⁶ <https://github.com/TheHPXProject/hpx/pull/2983>

⁴¹⁶⁷ <https://github.com/TheHPXProject/hpx/pull/2982>

⁴¹⁶⁸ <https://github.com/TheHPXProject/hpx/pull/2981>

⁴¹⁶⁹ <https://github.com/TheHPXProject/hpx/pull/2979>

⁴¹⁷⁰ <https://github.com/TheHPXProject/hpx/pull/2978>

⁴¹⁷¹ <https://github.com/TheHPXProject/hpx/issues/2977>

⁴¹⁷² <https://github.com/TheHPXProject/hpx/pull/2976>

⁴¹⁷³ <https://github.com/TheHPXProject/hpx/pull/2975>

⁴¹⁷⁴ <https://github.com/TheHPXProject/hpx/pull/2974>

⁴¹⁷⁵ <https://github.com/TheHPXProject/hpx/pull/2973>

⁴¹⁷⁶ <https://github.com/TheHPXProject/hpx/pull/2972>

⁴¹⁷⁷ <https://github.com/TheHPXProject/hpx/pull/2971>

⁴¹⁷⁸ <https://github.com/TheHPXProject/hpx/pull/2970>

⁴¹⁷⁹ <https://github.com/TheHPXProject/hpx/pull/2969>

⁴¹⁸⁰ <https://github.com/TheHPXProject/hpx/pull/2968>

⁴¹⁸¹ <https://github.com/TheHPXProject/hpx/pull/2967>

⁴¹⁸² <https://github.com/TheHPXProject/hpx/pull/2966>

⁴¹⁸³ <https://github.com/TheHPXProject/hpx/pull/2965>

⁴¹⁸⁴ <https://github.com/TheHPXProject/hpx/issues/2964>

⁴¹⁸⁵ <https://github.com/TheHPXProject/hpx/pull/2963>

- PR #2962⁴¹⁸⁶ - Partially reverting #2891
- PR #2961⁴¹⁸⁷ - Attempt to fix the gcc 4.9 problem with the async pack traversal
- Issue #2959⁴¹⁸⁸ - Program terminates during error handling
- Issue #2958⁴¹⁸⁹ - HPX_PLAIN_ACTION breaks due to missing include
- PR #2957⁴¹⁹⁰ - Fixing errors generated by mixing different attribute syntaxes
- Issue #2956⁴¹⁹¹ - Mixing attribute syntaxes leads to compiler errors
- Issue #2955⁴¹⁹² - Fix OS-Thread throttling
- PR #2953⁴¹⁹³ - Making sure any hpx.os_threads=N supplied through a -hpx::config file is taken into account
- PR #2952⁴¹⁹⁴ - Removing wrong call to cleanup_terminated_locked
- PR #2951⁴¹⁹⁵ - Revert “Make sure the function vttables are initialized before use”
- PR #2950⁴¹⁹⁶ - Fix a namespace compilation error when some schedulers are disabled
- Issue #2949⁴¹⁹⁷ - master branch giving lockups on shutdown
- Issue #2947⁴¹⁹⁸ - hpx.ini is not used correctly at initialization
- PR #2946⁴¹⁹⁹ - Adding explicit feature test for thread_local
- PR #2945⁴²⁰⁰ - Make sure the function vttables are initialized before use
- PR #2944⁴²⁰¹ - Attempting to solve affinity problems on CircleCI
- PR #2943⁴²⁰² - Changing channel actions to be direct
- PR #2942⁴²⁰³ - Adding split_future for std::vector
- PR #2941⁴²⁰⁴ - Add a feature test to test for CXX11 override
- Issue #2940⁴²⁰⁵ - Add split_future for future<vector<T>>
- PR #2939⁴²⁰⁶ - Making error reporting during problems with setting affinity masks more verbose
- PR #2938⁴²⁰⁷ - Fix this various executors
- PR #2937⁴²⁰⁸ - Fix some typos in documentation

⁴¹⁸⁶ <https://github.com/TheHPXProject/hpx/pull/2962>

⁴¹⁸⁷ <https://github.com/TheHPXProject/hpx/pull/2961>

⁴¹⁸⁸ <https://github.com/TheHPXProject/hpx/issues/2959>

⁴¹⁸⁹ <https://github.com/TheHPXProject/hpx/issues/2958>

⁴¹⁹⁰ <https://github.com/TheHPXProject/hpx/pull/2957>

⁴¹⁹¹ <https://github.com/TheHPXProject/hpx/issues/2956>

⁴¹⁹² <https://github.com/TheHPXProject/hpx/issues/2955>

⁴¹⁹³ <https://github.com/TheHPXProject/hpx/pull/2953>

⁴¹⁹⁴ <https://github.com/TheHPXProject/hpx/pull/2952>

⁴¹⁹⁵ <https://github.com/TheHPXProject/hpx/pull/2951>

⁴¹⁹⁶ <https://github.com/TheHPXProject/hpx/pull/2950>

⁴¹⁹⁷ <https://github.com/TheHPXProject/hpx/issues/2949>

⁴¹⁹⁸ <https://github.com/TheHPXProject/hpx/issues/2947>

⁴¹⁹⁹ <https://github.com/TheHPXProject/hpx/pull/2946>

⁴²⁰⁰ <https://github.com/TheHPXProject/hpx/pull/2945>

⁴²⁰¹ <https://github.com/TheHPXProject/hpx/pull/2944>

⁴²⁰² <https://github.com/TheHPXProject/hpx/pull/2943>

⁴²⁰³ <https://github.com/TheHPXProject/hpx/pull/2942>

⁴²⁰⁴ <https://github.com/TheHPXProject/hpx/pull/2941>

⁴²⁰⁵ <https://github.com/TheHPXProject/hpx/issues/2940>

⁴²⁰⁶ <https://github.com/TheHPXProject/hpx/pull/2939>

⁴²⁰⁷ <https://github.com/TheHPXProject/hpx/pull/2938>

⁴²⁰⁸ <https://github.com/TheHPXProject/hpx/pull/2937>

- PR #2934⁴²⁰⁹ - Remove the need for “complete” SFINAE checks
- PR #2933⁴²¹⁰ - Making sure parallel::for_loop is executed in parallel if requested
- PR #2932⁴²¹¹ - Classify chunk_size_iterator to input iterator tag. (Fix #2866)
- Issue #2931⁴²¹² - --hpx:help triggers unusual error with clang build
- PR #2930⁴²¹³ - Add #include files needed to set _POSIX_VERSION for debug check
- PR #2929⁴²¹⁴ - Fix a couple of deprecated c++ features
- PR #2928⁴²¹⁵ - Fixing execution parameters
- Issue #2927⁴²¹⁶ - CMake warning: ... cycle in constraint graph
- PR #2926⁴²¹⁷ - Default pool rename
- Issue #2925⁴²¹⁸ - Default pool cannot be renamed
- Issue #2924⁴²¹⁹ - hpx:attach-debugger=startup does not work any more
- PR #2923⁴²²⁰ - Alloc membind
- PR #2922⁴²²¹ - This fixes CircleCI errors when running with --hpx:bind=none
- PR #2921⁴²²² - Custom pool executor was missing priority and stacksize options
- PR #2920⁴²²³ - Adding test to trigger problem reported in #2916
- PR #2919⁴²²⁴ - Make sure the resource_partitioner is properly destructed on hpx::finalize
- Issue #2918⁴²²⁵ - hpx::init calls wrong (first) callback when called multiple times
- PR #2917⁴²²⁶ - Adding util::checkpoint
- Issue #2916⁴²²⁷ - Weird runtime failures when using a channel and chained continuations
- PR #2915⁴²²⁸ - Introduce executor parameters customization points
- Issue #2914⁴²²⁹ - Task assignment to current Pool has unintended consequences
- PR #2913⁴²³⁰ - Fix rp hang
- PR #2912⁴²³¹ - Update contributors

⁴²⁰⁹ <https://github.com/TheHPXProject/hpx/pull/2934>

⁴²¹⁰ <https://github.com/TheHPXProject/hpx/pull/2933>

⁴²¹¹ <https://github.com/TheHPXProject/hpx/pull/2932>

⁴²¹² <https://github.com/TheHPXProject/hpx/issues/2931>

⁴²¹³ <https://github.com/TheHPXProject/hpx/pull/2930>

⁴²¹⁴ <https://github.com/TheHPXProject/hpx/pull/2929>

⁴²¹⁵ <https://github.com/TheHPXProject/hpx/pull/2928>

⁴²¹⁶ <https://github.com/TheHPXProject/hpx/issues/2927>

⁴²¹⁷ <https://github.com/TheHPXProject/hpx/pull/2926>

⁴²¹⁸ <https://github.com/TheHPXProject/hpx/issues/2925>

⁴²¹⁹ <https://github.com/TheHPXProject/hpx/issues/2924>

⁴²²⁰ <https://github.com/TheHPXProject/hpx/pull/2923>

⁴²²¹ <https://github.com/TheHPXProject/hpx/pull/2922>

⁴²²² <https://github.com/TheHPXProject/hpx/pull/2921>

⁴²²³ <https://github.com/TheHPXProject/hpx/pull/2920>

⁴²²⁴ <https://github.com/TheHPXProject/hpx/pull/2919>

⁴²²⁵ <https://github.com/TheHPXProject/hpx/issues/2918>

⁴²²⁶ <https://github.com/TheHPXProject/hpx/pull/2917>

⁴²²⁷ <https://github.com/TheHPXProject/hpx/issues/2916>

⁴²²⁸ <https://github.com/TheHPXProject/hpx/pull/2915>

⁴²²⁹ <https://github.com/TheHPXProject/hpx/issues/2914>

⁴²³⁰ <https://github.com/TheHPXProject/hpx/pull/2913>

⁴²³¹ <https://github.com/TheHPXProject/hpx/pull/2912>

- PR #2911⁴²³² - Fixing CUDA problems
- PR #2910⁴²³³ - Improve error reporting for process component on POSIX systems
- PR #2909⁴²³⁴ - Fix typo in include path
- PR #2908⁴²³⁵ - Use proper container according to iterator tag in benchmarks of parallel algorithms
- PR #2907⁴²³⁶ - Optionally force-delete remaining channel items on close
- PR #2906⁴²³⁷ - Making sure generated performance counter names are correct
- Issue #2905⁴²³⁸ - collecting idle-rate performance counters on multiple localities produces an error
- Issue #2904⁴²³⁹ - build broken for Intel 17 compilers
- PR #2903⁴²⁴⁰ - Documentation Updates– Adding New People
- PR #2902⁴²⁴¹ - Fixing service_executor
- PR #2901⁴²⁴² - Fixing partitioned_vector creation
- PR #2900⁴²⁴³ - Add numa-balanced mode to hpx::bind, spread cores over numa domains
- Issue #2899⁴²⁴⁴ - hpx::bind does not have a mode that balances cores over numa domains
- PR #2898⁴²⁴⁵ - Adding missing #include and missing guard for optional code section
- PR #2897⁴²⁴⁶ - Removing dependency on Boost.ICL
- Issue #2896⁴²⁴⁷ - Debug build fails without -fpermissive with GCC 7.1 and Boost 1.65
- PR #2895⁴²⁴⁸ - Fixing SLURM environment parsing
- PR #2894⁴²⁴⁹ - Fix incorrect handling of compile definition with value 0
- Issue #2893⁴²⁵⁰ - Disabling schedulers causes build errors
- PR #2892⁴²⁵¹ - added list serializer
- PR #2891⁴²⁵² - Resource Partitioner Fixes
- Issue #2890⁴²⁵³ - Destroying a non-empty channel causes an assertion failure
- PR #2889⁴²⁵⁴ - Add check for libatomic

⁴²³² <https://github.com/TheHPXProject/hpx/pull/2911>

⁴²³³ <https://github.com/TheHPXProject/hpx/pull/2910>

⁴²³⁴ <https://github.com/TheHPXProject/hpx/pull/2909>

⁴²³⁵ <https://github.com/TheHPXProject/hpx/pull/2908>

⁴²³⁶ <https://github.com/TheHPXProject/hpx/pull/2907>

⁴²³⁷ <https://github.com/TheHPXProject/hpx/pull/2906>

⁴²³⁸ <https://github.com/TheHPXProject/hpx/issues/2905>

⁴²³⁹ <https://github.com/TheHPXProject/hpx/issues/2904>

⁴²⁴⁰ <https://github.com/TheHPXProject/hpx/pull/2903>

⁴²⁴¹ <https://github.com/TheHPXProject/hpx/pull/2902>

⁴²⁴² <https://github.com/TheHPXProject/hpx/pull/2901>

⁴²⁴³ <https://github.com/TheHPXProject/hpx/pull/2900>

⁴²⁴⁴ <https://github.com/TheHPXProject/hpx/issues/2899>

⁴²⁴⁵ <https://github.com/TheHPXProject/hpx/pull/2898>

⁴²⁴⁶ <https://github.com/TheHPXProject/hpx/pull/2897>

⁴²⁴⁷ <https://github.com/TheHPXProject/hpx/issues/2896>

⁴²⁴⁸ <https://github.com/TheHPXProject/hpx/pull/2895>

⁴²⁴⁹ <https://github.com/TheHPXProject/hpx/pull/2894>

⁴²⁵⁰ <https://github.com/TheHPXProject/hpx/issues/2893>

⁴²⁵¹ <https://github.com/TheHPXProject/hpx/pull/2892>

⁴²⁵² <https://github.com/TheHPXProject/hpx/pull/2891>

⁴²⁵³ <https://github.com/TheHPXProject/hpx/issues/2890>

⁴²⁵⁴ <https://github.com/TheHPXProject/hpx/pull/2889>

- PR #2888⁴²⁵⁵ - Fix compilation problems if HPX_WITH_ITT_NOTIFY=ON
- PR #2887⁴²⁵⁶ - Adapt broadcast() to non-unwrapping async<Action>
- PR #2886⁴²⁵⁷ - Replace Boost.Random with C++11 <random>
- Issue #2885⁴²⁵⁸ - regression in broadcast?
- Issue #2884⁴²⁵⁹ - linking -latomic is not portable
- PR #2883⁴²⁶⁰ - Explicitly set -pthread flag if available
- PR #2882⁴²⁶¹ - Wrap boost::format uses
- Issue #2881⁴²⁶² - hpx not compiling with HPX_WITH_ITTNOTIFY=On
- Issue #2880⁴²⁶³ - hpx::bind scatter/balanced give wrong pu masks
- PR #2878⁴²⁶⁴ - Fix incorrect pool usage masks setup in RP/thread manager
- PR #2877⁴²⁶⁵ - Require std::array by default
- PR #2875⁴²⁶⁶ - Deprecate use of BOOST_ASSERT
- PR #2874⁴²⁶⁷ - Changed serialization of boost.variant to use variadic templates
- Issue #2873⁴²⁶⁸ - building with parcelport_mpi fails on cori
- PR #2871⁴²⁶⁹ - Adding missing support for throttling scheduler
- PR #2870⁴²⁷⁰ - Disambiguate use of base_lco_with_value macros with channel
- Issue #2869⁴²⁷¹ - Difficulty compiling HPX_REGISTER_CHANNEL_DECLARATION(double)
- PR #2868⁴²⁷² - Removing unneeded assert
- PR #2867⁴²⁷³ - Implement parallel::unique
- Issue #2866⁴²⁷⁴ - The chunk_size_iterator violates multipass guarantee
- PR #2865⁴²⁷⁵ - Only use sched_getcpu on linux machines
- PR #2864⁴²⁷⁶ - Create redistribution archive for successful builds
- PR #2863⁴²⁷⁷ - Replace casts/assignments with hard-coded memcpy operations

⁴²⁵⁵ <https://github.com/TheHPXProject/hpx/pull/2888>

⁴²⁵⁶ <https://github.com/TheHPXProject/hpx/pull/2887>

⁴²⁵⁷ <https://github.com/TheHPXProject/hpx/pull/2886>

⁴²⁵⁸ <https://github.com/TheHPXProject/hpx/issues/2885>

⁴²⁵⁹ <https://github.com/TheHPXProject/hpx/issues/2884>

⁴²⁶⁰ <https://github.com/TheHPXProject/hpx/pull/2883>

⁴²⁶¹ <https://github.com/TheHPXProject/hpx/pull/2882>

⁴²⁶² <https://github.com/TheHPXProject/hpx/issues/2881>

⁴²⁶³ <https://github.com/TheHPXProject/hpx/issues/2880>

⁴²⁶⁴ <https://github.com/TheHPXProject/hpx/pull/2878>

⁴²⁶⁵ <https://github.com/TheHPXProject/hpx/pull/2877>

⁴²⁶⁶ <https://github.com/TheHPXProject/hpx/pull/2875>

⁴²⁶⁷ <https://github.com/TheHPXProject/hpx/pull/2874>

⁴²⁶⁸ <https://github.com/TheHPXProject/hpx/issues/2873>

⁴²⁶⁹ <https://github.com/TheHPXProject/hpx/pull/2871>

⁴²⁷⁰ <https://github.com/TheHPXProject/hpx/pull/2870>

⁴²⁷¹ <https://github.com/TheHPXProject/hpx/issues/2869>

⁴²⁷² <https://github.com/TheHPXProject/hpx/pull/2868>

⁴²⁷³ <https://github.com/TheHPXProject/hpx/pull/2867>

⁴²⁷⁴ <https://github.com/TheHPXProject/hpx/issues/2866>

⁴²⁷⁵ <https://github.com/TheHPXProject/hpx/pull/2865>

⁴²⁷⁶ <https://github.com/TheHPXProject/hpx/pull/2864>

⁴²⁷⁷ <https://github.com/TheHPXProject/hpx/pull/2863>

- Issue #2862⁴²⁷⁸ - sched_getcpu not available on MacOS
- PR #2861⁴²⁷⁹ - Fixing unmatched header defines and recursive inclusion of threadmanager
- Issue #2860⁴²⁸⁰ - Master program fails with assertion 'type == data_type_address' failed: HPX(assertion_failure)
- Issue #2852⁴²⁸¹ - Support for ARM64
- PR #2858⁴²⁸² - Fix misplaced #if #endif's that cause build failure without THREAD_CUMULATIVE_COUNTS
- PR #2857⁴²⁸³ - Fix some listing in documentation
- PR #2856⁴²⁸⁴ - Fixing component handling for lcos
- PR #2855⁴²⁸⁵ - Add documentation for coarrays
- PR #2854⁴²⁸⁶ - Support ARM64 in timestamps
- PR #2853⁴²⁸⁷ - Update Table 17. Non-modifying Parallel Algorithms in Documentation
- PR #2851⁴²⁸⁸ - Allowing for non-default-constructible component types
- PR #2850⁴²⁸⁹ - Enable returning future<R> from actions where R is not default-constructible
- PR #2849⁴²⁹⁰ - Unify serialization of non-default-constructable types
- Issue #2848⁴²⁹¹ - Components have to be default constructible
- Issue #2847⁴²⁹² - Returning a future<R> where R is not default-constructable broken
- Issue #2846⁴²⁹³ - Unify serialization of non-default-constructible types
- PR #2845⁴²⁹⁴ - Add Visual Studio 2015 to the tested toolchains in Appveyor
- Issue #2844⁴²⁹⁵ - Change the appveyor build to use the minimal required MSVC version
- Issue #2843⁴²⁹⁶ - multi node hello_world hangs
- PR #2842⁴²⁹⁷ - Correcting Spelling mistake in docs
- PR #2841⁴²⁹⁸ - Fix usage of std::aligned_storage
- PR #2840⁴²⁹⁹ - Remove constexpr from a void function
- Issue #2839⁴³⁰⁰ - memcpy buffer overflow: load_construct_data() and std::complex members

⁴²⁷⁸ <https://github.com/TheHPXProject/hpx/issues/2862>

⁴²⁷⁹ <https://github.com/TheHPXProject/hpx/pull/2861>

⁴²⁸⁰ <https://github.com/TheHPXProject/hpx/issues/2860>

⁴²⁸¹ <https://github.com/TheHPXProject/hpx/issues/2852>

⁴²⁸² <https://github.com/TheHPXProject/hpx/pull/2858>

⁴²⁸³ <https://github.com/TheHPXProject/hpx/pull/2857>

⁴²⁸⁴ <https://github.com/TheHPXProject/hpx/pull/2856>

⁴²⁸⁵ <https://github.com/TheHPXProject/hpx/pull/2855>

⁴²⁸⁶ <https://github.com/TheHPXProject/hpx/pull/2854>

⁴²⁸⁷ <https://github.com/TheHPXProject/hpx/pull/2853>

⁴²⁸⁸ <https://github.com/TheHPXProject/hpx/pull/2851>

⁴²⁸⁹ <https://github.com/TheHPXProject/hpx/pull/2850>

⁴²⁹⁰ <https://github.com/TheHPXProject/hpx/pull/2849>

⁴²⁹¹ <https://github.com/TheHPXProject/hpx/issues/2848>

⁴²⁹² <https://github.com/TheHPXProject/hpx/issues/2847>

⁴²⁹³ <https://github.com/TheHPXProject/hpx/issues/2846>

⁴²⁹⁴ <https://github.com/TheHPXProject/hpx/pull/2845>

⁴²⁹⁵ <https://github.com/TheHPXProject/hpx/issues/2844>

⁴²⁹⁶ <https://github.com/TheHPXProject/hpx/issues/2843>

⁴²⁹⁷ <https://github.com/TheHPXProject/hpx/pull/2842>

⁴²⁹⁸ <https://github.com/TheHPXProject/hpx/pull/2841>

⁴²⁹⁹ <https://github.com/TheHPXProject/hpx/pull/2840>

⁴³⁰⁰ <https://github.com/TheHPXProject/hpx/issues/2839>

- [Issue #2835](#)⁴³⁰¹ - `constexpr` functions with void return type break compilation with CUDA 8.0
- [Issue #2834](#)⁴³⁰² - One suspicion in `parallel::detail::handle_exception`
- [PR #2833](#)⁴³⁰³ - Implement `parallel::merge`
- [PR #2832](#)⁴³⁰⁴ - Fix a strange thing in `parallel::util::detail::handle_local_exceptions`. (Fix #2818)
- [PR #2830](#)⁴³⁰⁵ - Break the debugger when a test failed
- [Issue #2831](#)⁴³⁰⁶ - `parallel/executors/execution_fwd.hpp` causes compilation failure in C++11 mode.
- [PR #2829](#)⁴³⁰⁷ - Implement an API for asynchronous pack traversal
- [PR #2828](#)⁴³⁰⁸ - Split unit test builds on CircleCI to avoid timeouts
- [Issue #2827](#)⁴³⁰⁹ - failure to compile `hello_world` example with `-Werror`
- [PR #2824](#)⁴³¹⁰ - Making sure promises are marked as started when used as continuations
- [PR #2823](#)⁴³¹¹ - Add documentation for `partitioned_vector_view`
- [Issue #2822](#)⁴³¹² - Yet another issue with `wait_for` similar to #2796
- [PR #2821](#)⁴³¹³ - Fix bugs and improve that about `HPX_HAVE_CXX11_AUTO_RETURN_VALUE` of CMake
- [PR #2820](#)⁴³¹⁴ - Support C++11 in benchmark codes of `parallel::partition` and `parallel::partition_copy`
- [PR #2819](#)⁴³¹⁵ - Fix compile errors in unit test of container version of `parallel::partition`
- [Issue #2818](#)⁴³¹⁶ - A strange thing in `parallel::util::detail::handle_local_exceptions`
- [Issue #2815](#)⁴³¹⁷ - HPX fails to compile with `HPX_WITH_CUDA=ON` and the new CUDA 9.0 RC
- [Issue #2814](#)⁴³¹⁸ - Using ‘`gmakeN`’ after ‘`cmake`’ produces error in `src/CMakeFiles/hpx.dir/runtime/agas/addressing_service.cpp.o`
- [PR #2813](#)⁴³¹⁹ - Properly support `[[noreturn]]` attribute if available
- [Issue #2812](#)⁴³²⁰ - Compilation fails with `gcc 7.1.1`
- [PR #2811](#)⁴³²¹ - Adding `hpx::launch::lazy` and support for `async`, `dataflow`, and `future::then`
- [PR #2810](#)⁴³²² - Add option allowing to disable deprecation warning
- [PR #2809](#)⁴³²³ - Disable throttling scheduler if `HWLOC` is not found/used

⁴³⁰¹ <https://github.com/TheHPXProject/hpx/issues/2835>

⁴³⁰² <https://github.com/TheHPXProject/hpx/issues/2834>

⁴³⁰³ <https://github.com/TheHPXProject/hpx/pull/2833>

⁴³⁰⁴ <https://github.com/TheHPXProject/hpx/pull/2832>

⁴³⁰⁵ <https://github.com/TheHPXProject/hpx/pull/2830>

⁴³⁰⁶ <https://github.com/TheHPXProject/hpx/issues/2831>

⁴³⁰⁷ <https://github.com/TheHPXProject/hpx/pull/2829>

⁴³⁰⁸ <https://github.com/TheHPXProject/hpx/pull/2828>

⁴³⁰⁹ <https://github.com/TheHPXProject/hpx/issues/2827>

⁴³¹⁰ <https://github.com/TheHPXProject/hpx/pull/2824>

⁴³¹¹ <https://github.com/TheHPXProject/hpx/pull/2823>

⁴³¹² <https://github.com/TheHPXProject/hpx/issues/2822>

⁴³¹³ <https://github.com/TheHPXProject/hpx/pull/2821>

⁴³¹⁴ <https://github.com/TheHPXProject/hpx/pull/2820>

⁴³¹⁵ <https://github.com/TheHPXProject/hpx/pull/2819>

⁴³¹⁶ <https://github.com/TheHPXProject/hpx/issues/2818>

⁴³¹⁷ <https://github.com/TheHPXProject/hpx/issues/2815>

⁴³¹⁸ <https://github.com/TheHPXProject/hpx/issues/2814>

⁴³¹⁹ <https://github.com/TheHPXProject/hpx/pull/2813>

⁴³²⁰ <https://github.com/TheHPXProject/hpx/issues/2812>

⁴³²¹ <https://github.com/TheHPXProject/hpx/pull/2811>

⁴³²² <https://github.com/TheHPXProject/hpx/pull/2810>

⁴³²³ <https://github.com/TheHPXProject/hpx/pull/2809>

- PR #2808⁴³²⁴ - Fix compile errors on some environments of parallel::partition
- Issue #2807⁴³²⁵ - Difficulty building with HPX_WITH_HWLOC=Off
- PR #2806⁴³²⁶ - Partitioned vector
- PR #2805⁴³²⁷ - Serializing collections with non-default constructible data
- PR #2802⁴³²⁸ - Fix FreeBSD 11
- Issue #2801⁴³²⁹ - Rate limiting techniques in io_service
- Issue #2800⁴³³⁰ - New Launch Policy: async_if
- PR #2799⁴³³¹ - Fix a unit test failure on GCC in tuple_cat
- PR #2798⁴³³² - bump minimum required cmake to 3.0 in test
- PR #2797⁴³³³ - Making sure future::wait_for et.al. work properly for action results
- Issue #2796⁴³³⁴ - wait_for does always in “deferred” state for calls on remote localities
- Issue #2795⁴³³⁵ - Serialization of types without default constructor
- PR #2794⁴³³⁶ - Fixing test for partitioned_vector iteration
- PR #2792⁴³³⁷ - Implemented segmented find and its variations for partitioned vector
- PR #2791⁴³³⁸ - Circumvent scary warning about placement new
- PR #2790⁴³³⁹ - Fix OSX build
- PR #2789⁴³⁴⁰ - Resource partitioner
- PR #2788⁴³⁴¹ - Adapt parallel::is_heap and parallel::is_heap_until to Ranges TS
- PR #2787⁴³⁴² - Unwrap hotfixes
- PR #2786⁴³⁴³ - Update CMake Minimum Version to 3.3.2 (refs #2565)
- Issue #2785⁴³⁴⁴ - Issues with masks and cpuset
- PR #2784⁴³⁴⁵ - Error with reduce and transform reduce fixed
- PR #2783⁴³⁴⁶ - StackOverflow integration with libsigsegv

⁴³²⁴ <https://github.com/TheHPXProject/hpx/pull/2808>

⁴³²⁵ <https://github.com/TheHPXProject/hpx/issues/2807>

⁴³²⁶ <https://github.com/TheHPXProject/hpx/pull/2806>

⁴³²⁷ <https://github.com/TheHPXProject/hpx/pull/2805>

⁴³²⁸ <https://github.com/TheHPXProject/hpx/pull/2802>

⁴³²⁹ <https://github.com/TheHPXProject/hpx/issues/2801>

⁴³³⁰ <https://github.com/TheHPXProject/hpx/issues/2800>

⁴³³¹ <https://github.com/TheHPXProject/hpx/pull/2799>

⁴³³² <https://github.com/TheHPXProject/hpx/pull/2798>

⁴³³³ <https://github.com/TheHPXProject/hpx/pull/2797>

⁴³³⁴ <https://github.com/TheHPXProject/hpx/issues/2796>

⁴³³⁵ <https://github.com/TheHPXProject/hpx/issues/2795>

⁴³³⁶ <https://github.com/TheHPXProject/hpx/pull/2794>

⁴³³⁷ <https://github.com/TheHPXProject/hpx/pull/2792>

⁴³³⁸ <https://github.com/TheHPXProject/hpx/pull/2791>

⁴³³⁹ <https://github.com/TheHPXProject/hpx/pull/2790>

⁴³⁴⁰ <https://github.com/TheHPXProject/hpx/pull/2789>

⁴³⁴¹ <https://github.com/TheHPXProject/hpx/pull/2788>

⁴³⁴² <https://github.com/TheHPXProject/hpx/pull/2787>

⁴³⁴³ <https://github.com/TheHPXProject/hpx/pull/2786>

⁴³⁴⁴ <https://github.com/TheHPXProject/hpx/issues/2785>

⁴³⁴⁵ <https://github.com/TheHPXProject/hpx/pull/2784>

⁴³⁴⁶ <https://github.com/TheHPXProject/hpx/pull/2783>

- PR #2782⁴³⁴⁷ - Replace boost::atomic with std::atomic (where possible)
- PR #2781⁴³⁴⁸ - Check for and optionally use [[deprecated]] attribute
- PR #2780⁴³⁴⁹ - Adding empty (but non-trivial) destructor to circumvent warnings
- PR #2779⁴³⁵⁰ - Exception info tweaks
- PR #2778⁴³⁵¹ - Implement parallel::partition
- PR #2777⁴³⁵² - Improve error handling in gather_here/gather_there
- PR #2776⁴³⁵³ - Fix a bug in compiler version check
- PR #2775⁴³⁵⁴ - Fix compilation when HPX_WITH_LOGGING is OFF
- PR #2774⁴³⁵⁵ - Removing dependency on Boost.Date_Time
- PR #2773⁴³⁵⁶ - Add sync_images() method to spmd_block class
- PR #2772⁴³⁵⁷ - Adding documentation for PAPI counters
- PR #2771⁴³⁵⁸ - Removing boost preprocessor dependency
- PR #2770⁴³⁵⁹ - Adding test, fixing deadlock in config registry
- PR #2769⁴³⁶⁰ - Remove some other warnings and errors detected by clang 5.0
- Issue #2768⁴³⁶¹ - Is there iterator tag for HPX?
- PR #2767⁴³⁶² - Improvements to continuation annotation
- PR #2765⁴³⁶³ - gcc split stack support for HPX threads #620
- PR #2764⁴³⁶⁴ - Fix some uses of begin/end, remove unnecessary includes
- PR #2763⁴³⁶⁵ - Bump minimal Boost version to 1.55.0
- PR #2762⁴³⁶⁶ - hpx::partitioned_vector serializer
- PR #2761⁴³⁶⁷ - Adding configuration summary to cmake output and -hpx:info
- PR #2760⁴³⁶⁸ - Removing 1d_hydro example as it is broken
- PR #2758⁴³⁶⁹ - Remove various warnings detected by clang 5.0

⁴³⁴⁷ <https://github.com/TheHPXProject/hpx/pull/2782>
⁴³⁴⁸ <https://github.com/TheHPXProject/hpx/pull/2781>
⁴³⁴⁹ <https://github.com/TheHPXProject/hpx/pull/2780>
⁴³⁵⁰ <https://github.com/TheHPXProject/hpx/pull/2779>
⁴³⁵¹ <https://github.com/TheHPXProject/hpx/pull/2778>
⁴³⁵² <https://github.com/TheHPXProject/hpx/pull/2777>
⁴³⁵³ <https://github.com/TheHPXProject/hpx/pull/2776>
⁴³⁵⁴ <https://github.com/TheHPXProject/hpx/pull/2775>
⁴³⁵⁵ <https://github.com/TheHPXProject/hpx/pull/2774>
⁴³⁵⁶ <https://github.com/TheHPXProject/hpx/pull/2773>
⁴³⁵⁷ <https://github.com/TheHPXProject/hpx/pull/2772>
⁴³⁵⁸ <https://github.com/TheHPXProject/hpx/pull/2771>
⁴³⁵⁹ <https://github.com/TheHPXProject/hpx/pull/2770>
⁴³⁶⁰ <https://github.com/TheHPXProject/hpx/pull/2769>
⁴³⁶¹ <https://github.com/TheHPXProject/hpx/issues/2768>
⁴³⁶² <https://github.com/TheHPXProject/hpx/pull/2767>
⁴³⁶³ <https://github.com/TheHPXProject/hpx/pull/2765>
⁴³⁶⁴ <https://github.com/TheHPXProject/hpx/pull/2764>
⁴³⁶⁵ <https://github.com/TheHPXProject/hpx/pull/2763>
⁴³⁶⁶ <https://github.com/TheHPXProject/hpx/pull/2762>
⁴³⁶⁷ <https://github.com/TheHPXProject/hpx/pull/2761>
⁴³⁶⁸ <https://github.com/TheHPXProject/hpx/pull/2760>
⁴³⁶⁹ <https://github.com/TheHPXProject/hpx/pull/2758>

- [Issue #2757](#)⁴³⁷⁰ - In case of a “raw thread” is needed per core for implementing parallel algorithm, what is good practice in HPX?
- [PR #2756](#)⁴³⁷¹ - Allowing for LCOs to be simple components
- [PR #2755](#)⁴³⁷² - Removing make_index_pack_unrolled
- [PR #2754](#)⁴³⁷³ - Implement parallel::unique_copy
- [PR #2753](#)⁴³⁷⁴ - Fixing detection of [[fallthrough]] attribute
- [PR #2752](#)⁴³⁷⁵ - New thread priority names
- [PR #2751](#)⁴³⁷⁶ - Replace boost::exception with proposed exception_info
- [PR #2750](#)⁴³⁷⁷ - Replace boost::iterator_range
- [PR #2749](#)⁴³⁷⁸ - Fixing hdf5 examples
- [Issue #2748](#)⁴³⁷⁹ - HPX fails to build with enabled hdf5 examples
- [Issue #2747](#)⁴³⁸⁰ - Inherited task priorities break certain DAG optimizations
- [Issue #2746](#)⁴³⁸¹ - HPX segfaulting with valgrind
- [PR #2745](#)⁴³⁸² - Adding extended arithmetic performance counters
- [PR #2744](#)⁴³⁸³ - Adding ability to statistics counters to reset base counter
- [Issue #2743](#)⁴³⁸⁴ - Statistics counter does not support resetting
- [PR #2742](#)⁴³⁸⁵ - Making sure Vc V2 builds without additional HPX configuration flags
- [PR #2741](#)⁴³⁸⁶ - Deprecate unwrapped and implement unwrap and unwrapping
- [PR #2740](#)⁴³⁸⁷ - Coroutine stackoverflow detection for linux/posix; [Issue #2408](#)
- [PR #2739](#)⁴³⁸⁸ - Add files via upload
- [PR #2738](#)⁴³⁸⁹ - Appveyor support
- [PR #2737](#)⁴³⁹⁰ - Fixing 2735
- [Issue #2736](#)⁴³⁹¹ - 1d_hydro example doesn't work
- [Issue #2735](#)⁴³⁹² - partitioned_vector_subview test failing

⁴³⁷⁰ <https://github.com/TheHPXProject/hpx/issues/2757>

⁴³⁷¹ <https://github.com/TheHPXProject/hpx/pull/2756>

⁴³⁷² <https://github.com/TheHPXProject/hpx/pull/2755>

⁴³⁷³ <https://github.com/TheHPXProject/hpx/pull/2754>

⁴³⁷⁴ <https://github.com/TheHPXProject/hpx/pull/2753>

⁴³⁷⁵ <https://github.com/TheHPXProject/hpx/pull/2752>

⁴³⁷⁶ <https://github.com/TheHPXProject/hpx/pull/2751>

⁴³⁷⁷ <https://github.com/TheHPXProject/hpx/pull/2750>

⁴³⁷⁸ <https://github.com/TheHPXProject/hpx/pull/2749>

⁴³⁷⁹ <https://github.com/TheHPXProject/hpx/issues/2748>

⁴³⁸⁰ <https://github.com/TheHPXProject/hpx/issues/2747>

⁴³⁸¹ <https://github.com/TheHPXProject/hpx/issues/2746>

⁴³⁸² <https://github.com/TheHPXProject/hpx/pull/2745>

⁴³⁸³ <https://github.com/TheHPXProject/hpx/pull/2744>

⁴³⁸⁴ <https://github.com/TheHPXProject/hpx/issues/2743>

⁴³⁸⁵ <https://github.com/TheHPXProject/hpx/pull/2742>

⁴³⁸⁶ <https://github.com/TheHPXProject/hpx/pull/2741>

⁴³⁸⁷ <https://github.com/TheHPXProject/hpx/pull/2740>

⁴³⁸⁸ <https://github.com/TheHPXProject/hpx/pull/2739>

⁴³⁸⁹ <https://github.com/TheHPXProject/hpx/pull/2738>

⁴³⁹⁰ <https://github.com/TheHPXProject/hpx/pull/2737>

⁴³⁹¹ <https://github.com/TheHPXProject/hpx/issues/2736>

⁴³⁹² <https://github.com/TheHPXProject/hpx/issues/2735>

- PR #2734⁴³⁹³ - Add C++11 range utilities
- PR #2733⁴³⁹⁴ - Adapting iterator requirements for parallel algorithms
- PR #2732⁴³⁹⁵ - Integrate C++ Co-arrays
- PR #2731⁴³⁹⁶ - Adding on_migrated event handler to migratable component instances
- Issue #2729⁴³⁹⁷ - Add on_migrated() event handler to migratable components
- Issue #2728⁴³⁹⁸ - Why Projection is needed in parallel algorithms?
- PR #2727⁴³⁹⁹ - Cmake files for StackOverflow Detection
- PR #2726⁴⁴⁰⁰ - CMake for Stack Overflow Detection
- PR #2725⁴⁴⁰¹ - Implemented segmented algorithms for partitioned vector
- PR #2724⁴⁴⁰² - Fix examples in Action documentation
- PR #2723⁴⁴⁰³ - Enable lcos::channel<T>::register_as
- Issue #2722⁴⁴⁰⁴ - channel register_as() failing on compilation
- PR #2721⁴⁴⁰⁵ - Mind map
- PR #2720⁴⁴⁰⁶ - reorder forward declarations to get rid of C++14-only auto return types
- PR #2719⁴⁴⁰⁷ - Add documentation for partitioned_vector and add features in pack.hpp
- Issue #2718⁴⁴⁰⁸ - Some forward declarations in execution_fwd.hpp aren't C++11-compatible
- PR #2717⁴⁴⁰⁹ - Config support for fallthrough attribute
- PR #2716⁴⁴¹⁰ - Implement parallel::partition_copy
- PR #2715⁴⁴¹¹ - initial import of icu string serializer
- PR #2714⁴⁴¹² - initial import of valarray serializer
- PR #2713⁴⁴¹³ - Remove slashes before CMAKE_FILES_DIRECTORY variables
- PR #2712⁴⁴¹⁴ - Fixing wait for 1751
- PR #2711⁴⁴¹⁵ - Adjust code for minimal supported GCC having being bumped to 4.9

⁴³⁹³ <https://github.com/TheHPXProject/hpx/pull/2734>

⁴³⁹⁴ <https://github.com/TheHPXProject/hpx/pull/2733>

⁴³⁹⁵ <https://github.com/TheHPXProject/hpx/pull/2732>

⁴³⁹⁶ <https://github.com/TheHPXProject/hpx/pull/2731>

⁴³⁹⁷ <https://github.com/TheHPXProject/hpx/issues/2729>

⁴³⁹⁸ <https://github.com/TheHPXProject/hpx/issues/2728>

⁴³⁹⁹ <https://github.com/TheHPXProject/hpx/pull/2727>

⁴⁴⁰⁰ <https://github.com/TheHPXProject/hpx/pull/2726>

⁴⁴⁰¹ <https://github.com/TheHPXProject/hpx/pull/2725>

⁴⁴⁰² <https://github.com/TheHPXProject/hpx/pull/2724>

⁴⁴⁰³ <https://github.com/TheHPXProject/hpx/pull/2723>

⁴⁴⁰⁴ <https://github.com/TheHPXProject/hpx/issues/2722>

⁴⁴⁰⁵ <https://github.com/TheHPXProject/hpx/pull/2721>

⁴⁴⁰⁶ <https://github.com/TheHPXProject/hpx/pull/2720>

⁴⁴⁰⁷ <https://github.com/TheHPXProject/hpx/pull/2719>

⁴⁴⁰⁸ <https://github.com/TheHPXProject/hpx/issues/2718>

⁴⁴⁰⁹ <https://github.com/TheHPXProject/hpx/pull/2717>

⁴⁴¹⁰ <https://github.com/TheHPXProject/hpx/pull/2716>

⁴⁴¹¹ <https://github.com/TheHPXProject/hpx/pull/2715>

⁴⁴¹² <https://github.com/TheHPXProject/hpx/pull/2714>

⁴⁴¹³ <https://github.com/TheHPXProject/hpx/pull/2713>

⁴⁴¹⁴ <https://github.com/TheHPXProject/hpx/pull/2712>

⁴⁴¹⁵ <https://github.com/TheHPXProject/hpx/pull/2711>

- PR #2710⁴⁴¹⁶ - Adding code of conduct
- PR #2709⁴⁴¹⁷ - Fixing UB in destroy tests
- PR #2708⁴⁴¹⁸ - Add inline to prevent multiple definition issue
- Issue #2707⁴⁴¹⁹ - Multiple defined symbols for task_block.hpp in VS2015
- PR #2706⁴⁴²⁰ - Adding .clang-format file
- PR #2704⁴⁴²¹ - Add a synchronous mapping API
- Issue #2703⁴⁴²² - Request: Add the .clang-format file to the repository
- Issue #2702⁴⁴²³ - STELLAR-GROUP/Vc slower than VCv1 possibly due to wrong instructions generated
- Issue #2701⁴⁴²⁴ - Datapar with STELLAR-GROUP/Vc requires obscure flag
- Issue #2700⁴⁴²⁵ - Naming inconsistency in parallel algorithms
- Issue #2699⁴⁴²⁶ - Iterator requirements are different from standard in parallel copy_if.
- PR #2698⁴⁴²⁷ - Properly releasing parcelport write handlers
- Issue #2697⁴⁴²⁸ - Compile error in addressing_service.cpp
- Issue #2696⁴⁴²⁹ - Building and using HPX statically: undefined references from runtime_support_server.cpp
- Issue #2695⁴⁴³⁰ - Executor changes cause compilation failures
- PR #2694⁴⁴³¹ - Refining C++ language mode detection for MSVC
- PR #2693⁴⁴³² - P0443 r2
- PR #2692⁴⁴³³ - Partially reverting changes to parcel_await
- Issue #2689⁴⁴³⁴ - HPX build fails when HPX_WITH_CUDA is enabled
- PR #2688⁴⁴³⁵ - Make Cuda Clang builds pass
- PR #2687⁴⁴³⁶ - Add an is_tuple_like trait for sequenceable type detection
- PR #2686⁴⁴³⁷ - Allowing throttling scheduler to be used without idle backoff
- PR #2685⁴⁴³⁸ - Add support of std::array to hpx::util::tuple_size and tuple_element

⁴⁴¹⁶ <https://github.com/TheHPXProject/hpx/pull/2710>

⁴⁴¹⁷ <https://github.com/TheHPXProject/hpx/pull/2709>

⁴⁴¹⁸ <https://github.com/TheHPXProject/hpx/pull/2708>

⁴⁴¹⁹ <https://github.com/TheHPXProject/hpx/issues/2707>

⁴⁴²⁰ <https://github.com/TheHPXProject/hpx/pull/2706>

⁴⁴²¹ <https://github.com/TheHPXProject/hpx/pull/2704>

⁴⁴²² <https://github.com/TheHPXProject/hpx/issues/2703>

⁴⁴²³ <https://github.com/TheHPXProject/hpx/issues/2702>

⁴⁴²⁴ <https://github.com/TheHPXProject/hpx/issues/2701>

⁴⁴²⁵ <https://github.com/TheHPXProject/hpx/issues/2700>

⁴⁴²⁶ <https://github.com/TheHPXProject/hpx/issues/2699>

⁴⁴²⁷ <https://github.com/TheHPXProject/hpx/pull/2698>

⁴⁴²⁸ <https://github.com/TheHPXProject/hpx/issues/2697>

⁴⁴²⁹ <https://github.com/TheHPXProject/hpx/issues/2696>

⁴⁴³⁰ <https://github.com/TheHPXProject/hpx/issues/2695>

⁴⁴³¹ <https://github.com/TheHPXProject/hpx/pull/2694>

⁴⁴³² <https://github.com/TheHPXProject/hpx/pull/2693>

⁴⁴³³ <https://github.com/TheHPXProject/hpx/pull/2692>

⁴⁴³⁴ <https://github.com/TheHPXProject/hpx/issues/2689>

⁴⁴³⁵ <https://github.com/TheHPXProject/hpx/pull/2688>

⁴⁴³⁶ <https://github.com/TheHPXProject/hpx/pull/2687>

⁴⁴³⁷ <https://github.com/TheHPXProject/hpx/pull/2686>

⁴⁴³⁸ <https://github.com/TheHPXProject/hpx/pull/2685>

- [PR #2684](#)⁴⁴³⁹ - Adding new statistics performance counters
- [PR #2683](#)⁴⁴⁴⁰ - Replace `boost::exception_ptr` with `std::exception_ptr`
- [Issue #2682](#)⁴⁴⁴¹ - HPX does not compile with `HPX_WITH_THREAD_MANAGER_IDLE_BACKOFF=OFF`
- [PR #2681](#)⁴⁴⁴² - Attempt to fix problem in `managed_component_base`
- [PR #2680](#)⁴⁴⁴³ - Fix bad size during archive creation
- [Issue #2679](#)⁴⁴⁴⁴ - Mismatch between size of archive and container
- [Issue #2678](#)⁴⁴⁴⁵ - In parallel algorithm, other tasks are executed to the end even if an exception occurs in any task.
- [PR #2677](#)⁴⁴⁴⁶ - Adding include check for `std::addressof`
- [PR #2676](#)⁴⁴⁴⁷ - Adding `parallel::destroy` and `destroy_n`
- [PR #2675](#)⁴⁴⁴⁸ - Making sure statistics counters work as expected
- [PR #2674](#)⁴⁴⁴⁹ - Turning assertions into exceptions
- [PR #2673](#)⁴⁴⁵⁰ - Inhibit direct conversion from `future<future<T>>` -> `future<void>`
- [PR #2672](#)⁴⁴⁵¹ - C++17 invoke forms
- [PR #2671](#)⁴⁴⁵² - Adding `uninitialized_value_construct` and `uninitialized_value_construct_n`
- [PR #2670](#)⁴⁴⁵³ - Integrate `spmd` multidimensional views for `partitioned_vectors`
- [PR #2669](#)⁴⁴⁵⁴ - Adding `uninitialized_default_construct` and `uninitialized_default_construct_n`
- [PR #2668](#)⁴⁴⁵⁵ - Fixing documentation index
- [Issue #2667](#)⁴⁴⁵⁶ - Ambiguity of nested `hpx::future<void>`'s
- [Issue #2666](#)⁴⁴⁵⁷ - Statistics Performance counter is not working
- [PR #2664](#)⁴⁴⁵⁸ - Adding `uninitialized_move` and `uninitialized_move_n`
- [Issue #2663](#)⁴⁴⁵⁹ - Seg fault in `managed_component::get_base_gid`, possibly cause by `util::reinitializable_static`
- [Issue #2662](#)⁴⁴⁶⁰ - Crash in `managed_component::get_base_gid` due to problem with `util::reinitializable_static`
- [PR #2665](#)⁴⁴⁶¹ - Hide the `detail` namespace in `doxygen` per default

⁴⁴³⁹ <https://github.com/TheHPXProject/hpx/pull/2684>

⁴⁴⁴⁰ <https://github.com/TheHPXProject/hpx/pull/2683>

⁴⁴⁴¹ <https://github.com/TheHPXProject/hpx/issues/2682>

⁴⁴⁴² <https://github.com/TheHPXProject/hpx/pull/2681>

⁴⁴⁴³ <https://github.com/TheHPXProject/hpx/pull/2680>

⁴⁴⁴⁴ <https://github.com/TheHPXProject/hpx/issues/2679>

⁴⁴⁴⁵ <https://github.com/TheHPXProject/hpx/issues/2678>

⁴⁴⁴⁶ <https://github.com/TheHPXProject/hpx/pull/2677>

⁴⁴⁴⁷ <https://github.com/TheHPXProject/hpx/pull/2676>

⁴⁴⁴⁸ <https://github.com/TheHPXProject/hpx/pull/2675>

⁴⁴⁴⁹ <https://github.com/TheHPXProject/hpx/pull/2674>

⁴⁴⁵⁰ <https://github.com/TheHPXProject/hpx/pull/2673>

⁴⁴⁵¹ <https://github.com/TheHPXProject/hpx/pull/2672>

⁴⁴⁵² <https://github.com/TheHPXProject/hpx/pull/2671>

⁴⁴⁵³ <https://github.com/TheHPXProject/hpx/pull/2670>

⁴⁴⁵⁴ <https://github.com/TheHPXProject/hpx/pull/2669>

⁴⁴⁵⁵ <https://github.com/TheHPXProject/hpx/pull/2668>

⁴⁴⁵⁶ <https://github.com/TheHPXProject/hpx/issues/2667>

⁴⁴⁵⁷ <https://github.com/TheHPXProject/hpx/issues/2666>

⁴⁴⁵⁸ <https://github.com/TheHPXProject/hpx/pull/2664>

⁴⁴⁵⁹ <https://github.com/TheHPXProject/hpx/issues/2663>

⁴⁴⁶⁰ <https://github.com/TheHPXProject/hpx/issues/2662>

⁴⁴⁶¹ <https://github.com/TheHPXProject/hpx/pull/2665>

- PR #2660⁴⁴⁶² - Add documentation to `hpx::util::unwrapped` and `hpx::util::unwrapped2`
- PR #2659⁴⁴⁶³ - Improve integration with `vcpkg`
- PR #2658⁴⁴⁶⁴ - Unify `access_data` trait for use in both, serialization and de-serialization
- PR #2657⁴⁴⁶⁵ - Removing `hpx::lcos::queue<T>`
- PR #2656⁴⁴⁶⁶ - Reduce `MAX_TERMINATED_THREADS` default, improve memory use on manycore cpus
- PR #2655⁴⁴⁶⁷ - Maintenance for emulate-deleted macros
- PR #2654⁴⁴⁶⁸ - Implement parallel `is_heap` and `is_heap_until`
- PR #2653⁴⁴⁶⁹ - Drop support for VS2013
- PR #2652⁴⁴⁷⁰ - This patch makes sure that all parcels in a batch are properly handled
- PR #2649⁴⁴⁷¹ - Update docs (Table 18) - move `transform` to end
- Issue #2647⁴⁴⁷² - `hpx::parcels::detail::parcel_data::has_continuation_` is uninitialized
- Issue #2644⁴⁴⁷³ - Some `.vcxproj` in the `HPX.sln` fail to build
- Issue #2641⁴⁴⁷⁴ - `hpx::lcos::queue` should be deprecated
- PR #2640⁴⁴⁷⁵ - A new throttling policy with public APIs to suspend/resume
- PR #2639⁴⁴⁷⁶ - Fix a tiny typo in tutorial.
- Issue #2638⁴⁴⁷⁷ - Invalid return type 'void' of `constexpr` function
- PR #2636⁴⁴⁷⁸ - Add and use `HPX_MSVC_WARNING_PRAGMA` for `#pragma` warning
- PR #2633⁴⁴⁷⁹ - Distributed `define_spmd_block`
- PR #2632⁴⁴⁸⁰ - Making sure container serialization uses size-compatible types
- PR #2631⁴⁴⁸¹ - Add `lcos::local::one_element_channel`
- PR #2629⁴⁴⁸² - Move `unordered_map` out of `parcels` into `hpx/concurrent`
- PR #2628⁴⁴⁸³ - Making sure that shutdown does not hang
- PR #2627⁴⁴⁸⁴ - Fix serialization

⁴⁴⁶² <https://github.com/TheHPXProject/hpx/pull/2660>

⁴⁴⁶³ <https://github.com/TheHPXProject/hpx/pull/2659>

⁴⁴⁶⁴ <https://github.com/TheHPXProject/hpx/pull/2658>

⁴⁴⁶⁵ <https://github.com/TheHPXProject/hpx/pull/2657>

⁴⁴⁶⁶ <https://github.com/TheHPXProject/hpx/pull/2656>

⁴⁴⁶⁷ <https://github.com/TheHPXProject/hpx/pull/2655>

⁴⁴⁶⁸ <https://github.com/TheHPXProject/hpx/pull/2654>

⁴⁴⁶⁹ <https://github.com/TheHPXProject/hpx/pull/2653>

⁴⁴⁷⁰ <https://github.com/TheHPXProject/hpx/pull/2652>

⁴⁴⁷¹ <https://github.com/TheHPXProject/hpx/pull/2649>

⁴⁴⁷² <https://github.com/TheHPXProject/hpx/issues/2647>

⁴⁴⁷³ <https://github.com/TheHPXProject/hpx/issues/2644>

⁴⁴⁷⁴ <https://github.com/TheHPXProject/hpx/issues/2641>

⁴⁴⁷⁵ <https://github.com/TheHPXProject/hpx/pull/2640>

⁴⁴⁷⁶ <https://github.com/TheHPXProject/hpx/pull/2639>

⁴⁴⁷⁷ <https://github.com/TheHPXProject/hpx/issues/2638>

⁴⁴⁷⁸ <https://github.com/TheHPXProject/hpx/pull/2636>

⁴⁴⁷⁹ <https://github.com/TheHPXProject/hpx/pull/2633>

⁴⁴⁸⁰ <https://github.com/TheHPXProject/hpx/pull/2632>

⁴⁴⁸¹ <https://github.com/TheHPXProject/hpx/pull/2631>

⁴⁴⁸² <https://github.com/TheHPXProject/hpx/pull/2629>

⁴⁴⁸³ <https://github.com/TheHPXProject/hpx/pull/2628>

⁴⁴⁸⁴ <https://github.com/TheHPXProject/hpx/pull/2627>

- PR #2626⁴⁴⁸⁵ - Generate `cmake_variables.qbk` and `cmake_toolchains.qbk` outside of the source tree
- PR #2625⁴⁴⁸⁶ - Supporting `-std=c++17` flag
- PR #2624⁴⁴⁸⁷ - Fixing a small `cmake` typo
- PR #2622⁴⁴⁸⁸ - Update CMake minimum required version to 3.0.2 (closes #2621)
- Issue #2621⁴⁴⁸⁹ - Compiling `hpx` master fails with `/usr/bin/ld: final link failed: Bad value`
- PR #2620⁴⁴⁹⁰ - Remove warnings due to some captured variables
- PR #2619⁴⁴⁹¹ - LF multiple parcels
- PR #2618⁴⁴⁹² - Some fixes to `libfabric` that didn't get caught before the merge
- PR #2617⁴⁴⁹³ - Adding `hpx::local_new`
- PR #2616⁴⁴⁹⁴ - Documentation: Extract all entities in order to autolink functions correctly
- Issue #2615⁴⁴⁹⁵ - Documentation: Linking functions is broken
- PR #2614⁴⁴⁹⁶ - Adding serialization for `std::deque`
- PR #2613⁴⁴⁹⁷ - We need to link with `boost.thread` and `boost.chrono` if we use `boost.context`
- PR #2612⁴⁴⁹⁸ - Making sure `for_loop_n(par, ...)` is actually executed in parallel
- PR #2611⁴⁴⁹⁹ - Add documentation to `invoke_fused` and `friends NFC`
- PR #2610⁴⁵⁰⁰ - Added reduction templates using an identity value
- PR #2608⁴⁵⁰¹ - Fixing some unused vars in `inspect`
- PR #2607⁴⁵⁰² - Fixed build for `mingw`
- PR #2606⁴⁵⁰³ - Supporting generic context for `boost >= 1.61`
- PR #2605⁴⁵⁰⁴ - `Parcelport libfabric3`
- PR #2604⁴⁵⁰⁵ - Adding allocator support to `promise` and `friends`
- PR #2603⁴⁵⁰⁶ - Barrier hang
- PR #2602⁴⁵⁰⁷ - Changes to scheduler to steal from one high-priority queue

⁴⁴⁸⁵ <https://github.com/TheHPXProject/hpx/pull/2626>
⁴⁴⁸⁶ <https://github.com/TheHPXProject/hpx/pull/2625>
⁴⁴⁸⁷ <https://github.com/TheHPXProject/hpx/pull/2624>
⁴⁴⁸⁸ <https://github.com/TheHPXProject/hpx/pull/2622>
⁴⁴⁸⁹ <https://github.com/TheHPXProject/hpx/issues/2621>
⁴⁴⁹⁰ <https://github.com/TheHPXProject/hpx/pull/2620>
⁴⁴⁹¹ <https://github.com/TheHPXProject/hpx/pull/2619>
⁴⁴⁹² <https://github.com/TheHPXProject/hpx/pull/2618>
⁴⁴⁹³ <https://github.com/TheHPXProject/hpx/pull/2617>
⁴⁴⁹⁴ <https://github.com/TheHPXProject/hpx/pull/2616>
⁴⁴⁹⁵ <https://github.com/TheHPXProject/hpx/issues/2615>
⁴⁴⁹⁶ <https://github.com/TheHPXProject/hpx/pull/2614>
⁴⁴⁹⁷ <https://github.com/TheHPXProject/hpx/pull/2613>
⁴⁴⁹⁸ <https://github.com/TheHPXProject/hpx/pull/2612>
⁴⁴⁹⁹ <https://github.com/TheHPXProject/hpx/pull/2611>
⁴⁵⁰⁰ <https://github.com/TheHPXProject/hpx/pull/2610>
⁴⁵⁰¹ <https://github.com/TheHPXProject/hpx/pull/2608>
⁴⁵⁰² <https://github.com/TheHPXProject/hpx/pull/2607>
⁴⁵⁰³ <https://github.com/TheHPXProject/hpx/pull/2606>
⁴⁵⁰⁴ <https://github.com/TheHPXProject/hpx/pull/2605>
⁴⁵⁰⁵ <https://github.com/TheHPXProject/hpx/pull/2604>
⁴⁵⁰⁶ <https://github.com/TheHPXProject/hpx/pull/2603>
⁴⁵⁰⁷ <https://github.com/TheHPXProject/hpx/pull/2602>

- Issue #2601⁴⁵⁰⁸ - High priority tasks are not executed first
- PR #2600⁴⁵⁰⁹ - Compat fixes
- PR #2599⁴⁵¹⁰ - Compatibility layer for threading support
- PR #2598⁴⁵¹¹ - V1.1
- PR #2597⁴⁵¹² - Release V1.0
- PR #2592⁴⁵¹³ - First attempt to introduce `spmd_block` in `hpx`
- PR #2586⁴⁵¹⁴ - `local_segment` in `segmented_iterator_traits`
- Issue #2584⁴⁵¹⁵ - Add allocator support to `promise`, `packaged_task` and friends
- PR #2576⁴⁵¹⁶ - Add missing dependencies of `cuda` based tests
- PR #2575⁴⁵¹⁷ - Remove warnings due to some captured variables
- Issue #2574⁴⁵¹⁸ - MSVC 2015 Compiler crash when building HPX
- Issue #2568⁴⁵¹⁹ - Remove `throttle_scheduler` as it has been abandoned
- Issue #2566⁴⁵²⁰ - Add an inline versioning namespace before 1.0 release
- Issue #2565⁴⁵²¹ - Raise minimal `cmake` version requirement
- PR #2556⁴⁵²² - Fixing scan partitioner
- PR #2546⁴⁵²³ - Broadcast async
- Issue #2543⁴⁵²⁴ - `make install` fails due to a non-existing `.so` file
- PR #2495⁴⁵²⁵ - `wait_or_add_new` returning `thread_id_type`
- Issue #2480⁴⁵²⁶ - Unable to register new performance counter
- Issue #2471⁴⁵²⁷ - no type named `'fcontext_t'` in namespace
- Issue #2456⁴⁵²⁸ - Re-implement `hpx::util::unwrapped`
- Issue #2455⁴⁵²⁹ - Add more arithmetic performance counters
- PR #2454⁴⁵³⁰ - Fix a couple of warnings and compiler errors

⁴⁵⁰⁸ <https://github.com/TheHPXProject/hpx/issues/2601>

⁴⁵⁰⁹ <https://github.com/TheHPXProject/hpx/pull/2600>

⁴⁵¹⁰ <https://github.com/TheHPXProject/hpx/pull/2599>

⁴⁵¹¹ <https://github.com/TheHPXProject/hpx/pull/2598>

⁴⁵¹² <https://github.com/TheHPXProject/hpx/pull/2597>

⁴⁵¹³ <https://github.com/TheHPXProject/hpx/pull/2592>

⁴⁵¹⁴ <https://github.com/TheHPXProject/hpx/pull/2586>

⁴⁵¹⁵ <https://github.com/TheHPXProject/hpx/issues/2584>

⁴⁵¹⁶ <https://github.com/TheHPXProject/hpx/pull/2576>

⁴⁵¹⁷ <https://github.com/TheHPXProject/hpx/pull/2575>

⁴⁵¹⁸ <https://github.com/TheHPXProject/hpx/issues/2574>

⁴⁵¹⁹ <https://github.com/TheHPXProject/hpx/issues/2568>

⁴⁵²⁰ <https://github.com/TheHPXProject/hpx/issues/2566>

⁴⁵²¹ <https://github.com/TheHPXProject/hpx/issues/2565>

⁴⁵²² <https://github.com/TheHPXProject/hpx/pull/2556>

⁴⁵²³ <https://github.com/TheHPXProject/hpx/pull/2546>

⁴⁵²⁴ <https://github.com/TheHPXProject/hpx/issues/2543>

⁴⁵²⁵ <https://github.com/TheHPXProject/hpx/pull/2495>

⁴⁵²⁶ <https://github.com/TheHPXProject/hpx/issues/2480>

⁴⁵²⁷ <https://github.com/TheHPXProject/hpx/issues/2471>

⁴⁵²⁸ <https://github.com/TheHPXProject/hpx/issues/2456>

⁴⁵²⁹ <https://github.com/TheHPXProject/hpx/issues/2455>

⁴⁵³⁰ <https://github.com/TheHPXProject/hpx/pull/2454>

- PR #2453⁴⁵³¹ - Timed executor support
- PR #2447⁴⁵³² - Implementing new executor API (P0443)
- Issue #2439⁴⁵³³ - Implement executor proposal
- Issue #2408⁴⁵³⁴ - Stackoverflow detection for linux, e.g. based on libsigsegv
- PR #2377⁴⁵³⁵ - Add a customization point for `put_parcel` so we can override actions
- Issue #2368⁴⁵³⁶ - HPX_ASSERT problem
- Issue #2324⁴⁵³⁷ - Change default number of threads used to the maximum of the system
- Issue #2266⁴⁵³⁸ - `hpx_0.9.99` make tests fail
- PR #2195⁴⁵³⁹ - Support for code completion in VIM
- Issue #2137⁴⁵⁴⁰ - Hpx does not compile over osx
- Issue #2092⁴⁵⁴¹ - make tests should just build the tests
- Issue #2026⁴⁵⁴² - Build HPX with Apple's clang
- Issue #1932⁴⁵⁴³ - hpx with PBS fails on multiple localities
- PR #1914⁴⁵⁴⁴ - Parallel heap algorithm implementations WIP
- Issue #1598⁴⁵⁴⁵ - Disconnecting a locality results in segfault using heartbeat example
- Issue #1404⁴⁵⁴⁶ - unwrapped doesn't work with movable only types
- Issue #1400⁴⁵⁴⁷ - `hpx::util::unwrapped` doesn't work with non-future types
- Issue #1205⁴⁵⁴⁸ - TSS is broken
- Issue #1126⁴⁵⁴⁹ - `vector<future<T>>` does not work gracefully with dataflow, `when_all` and `unwrapped`
- Issue #1056⁴⁵⁵⁰ - Thread manager cleanup
- Issue #863⁴⁵⁵¹ - Futures should not require a default constructor
- Issue #856⁴⁵⁵² - Allow `runtime_mode_connect` to be used with security enabled
- Issue #726⁴⁵⁵³ - Valgrind

⁴⁵³¹ <https://github.com/TheHPXProject/hpx/pull/2453>

⁴⁵³² <https://github.com/TheHPXProject/hpx/pull/2447>

⁴⁵³³ <https://github.com/TheHPXProject/hpx/issues/2439>

⁴⁵³⁴ <https://github.com/TheHPXProject/hpx/issues/2408>

⁴⁵³⁵ <https://github.com/TheHPXProject/hpx/pull/2377>

⁴⁵³⁶ <https://github.com/TheHPXProject/hpx/issues/2368>

⁴⁵³⁷ <https://github.com/TheHPXProject/hpx/issues/2324>

⁴⁵³⁸ <https://github.com/TheHPXProject/hpx/issues/2266>

⁴⁵³⁹ <https://github.com/TheHPXProject/hpx/pull/2195>

⁴⁵⁴⁰ <https://github.com/TheHPXProject/hpx/issues/2137>

⁴⁵⁴¹ <https://github.com/TheHPXProject/hpx/issues/2092>

⁴⁵⁴² <https://github.com/TheHPXProject/hpx/issues/2026>

⁴⁵⁴³ <https://github.com/TheHPXProject/hpx/issues/1932>

⁴⁵⁴⁴ <https://github.com/TheHPXProject/hpx/pull/1914>

⁴⁵⁴⁵ <https://github.com/TheHPXProject/hpx/issues/1598>

⁴⁵⁴⁶ <https://github.com/TheHPXProject/hpx/issues/1404>

⁴⁵⁴⁷ <https://github.com/TheHPXProject/hpx/issues/1400>

⁴⁵⁴⁸ <https://github.com/TheHPXProject/hpx/issues/1205>

⁴⁵⁴⁹ <https://github.com/TheHPXProject/hpx/issues/1126>

⁴⁵⁵⁰ <https://github.com/TheHPXProject/hpx/issues/1056>

⁴⁵⁵¹ <https://github.com/TheHPXProject/hpx/issues/863>

⁴⁵⁵² <https://github.com/TheHPXProject/hpx/issues/856>

⁴⁵⁵³ <https://github.com/TheHPXProject/hpx/issues/726>

- [Issue #701](#)⁴⁵⁵⁴ - Add RCR performance counter component
- [Issue #528](#)⁴⁵⁵⁵ - Add support for known failures and warning count/comparisons to `hpx_run_tests.py`

HPX V1.0.0 (Apr 24, 2017)

General changes

Here are some of the main highlights and changes for this release (in no particular order):

- Added the facility `hpx::split_future` which allows one to convert a `future<tuple<Ts...>>` into a `tuple<future<Ts>...>`. This functionality is not available when compiling *HPX* with VS2012.
- Added a new type of performance counter which allows one to return a list of values for each invocation. We also added a first counter of this type which collects a histogram of the times between parcels being created.
- Added new LCOs: `hpx::lcos::channel` and `hpx::lcos::local::channel` which are very similar to the well known channel constructs used in the Go language.
- Added new performance counters reporting the amount of data handled by the networking layer on a action-by-action basis (please see [PR #2289](#)⁴⁵⁵⁶ for more details).
- Added a new facility `hpx::lcos::barrier`, replacing the equally named older one. The new facility has a slightly changed API and is much more efficient. Most notable, the new facility exposes a (global) function `hpx::lcos::barrier::synchronize()` which represents a global barrier across all localities.
- We have started to add support for vectorization to our parallel algorithm implementations. This support depends on using an external library, currently either Vc Library or `|boost_simd|`. Please see [Issue #2333](#)⁴⁵⁵⁷ for a list of currently supported algorithms. This is an experimental feature and its implementation and/or API might change in the future. Please see this [blog-post](#)⁴⁵⁵⁸ for more information.
- The parameter sequence for the `hpx::parallel::transform_reduce` overload taking one iterator range has changed to match the changes this algorithm has undergone while being moved to C++17. The old overload can be still enabled at configure time by specifying `-DHPX_WITH_TRANSFORM_REDUCE_COMPATIBILITY=On` to CMake.
- The algorithm `hpx::parallel::inner_product` has been renamed to `hpx::parallel::transform_reduce` to match the changes this algorithm has undergone while being moved to C++17. The old `inner_product` names can be still enabled at configure time by specifying `-DHPX_WITH_TRANSFORM_REDUCE_COMPATIBILITY=On` to CMake.
- Added versions of `hpx::get_ptr` taking client side representations for component instances as their parameter (instead of a global id).
- Added the helper utility `hpx::performance_counters::performance_counter_set` helping to encapsulate a set of performance counters to be managed concurrently.
- All execution policies and related classes have been renamed to be consistent with the naming changes applied for C++17. All policies now live in the namespace `hpx::parallel::execution`. The old names can be still enabled at configure time by specifying `-DHPX_WITH_EXECUTION_POLICY_COMPATIBILITY=On` to CMake.
- The thread scheduling subsystem has undergone a major refactoring which results in significant performance improvements. We have also improved the performance of creating `hpx::future` and of various facilities handling those.

⁴⁵⁵⁴ <https://github.com/TheHPXProject/hpx/issues/701>

⁴⁵⁵⁵ <https://github.com/TheHPXProject/hpx/issues/528>

⁴⁵⁵⁶ <https://github.com/TheHPXProject/hpx/pull/2289>

⁴⁵⁵⁷ <https://github.com/TheHPXProject/hpx/issues/2333>

⁴⁵⁵⁸ <http://stellar-group.org/2016/09/vectorized-cpp-parallel-algorithms-with-hpx/>

- We have consolidated all of the code in `HPX.Compute` related to the integration of CUDA. `hpx::partitioned_vector` has been enabled to be usable with `hpx::compute::vector` which allows one to place the partitions on one or more GPU devices.
- Added new performance counters exposing various internals of the thread scheduling subsystem, such as the current idle- and busy-loop counters and instantaneous scheduler utilization.
- Extended and improved the use of the ITTNotify hooks allowing to collect performance counter data and function annotation information from within the Intel Amplifier tool.

Breaking changes

- We have dropped support for the gcc compiler versions V4.6 and 4.7. The minimal gcc version we now test on is gcc V4.8.
- We have removed (default) support for `boost::chrono` in interfaces, uses of it have been replaced with `std::chrono`. This facility can be still enabled at configure time by specifying `-DHPX_WITH_BOOST_CHRONO_COMPATIBILITY=On` to CMake.
- The parameter sequence for the `hpx::parallel::transform_reduce` overload taking one iterator range has changed to match the changes this algorithm has undergone while being moved to C++17.
- The algorithm `hpx::parallel::inner_product` has been renamed to `hpx::parallel::transform_reduce` to match the changes this algorithm has undergone while being moved to C++17.
- the build options `HPX_WITH_COLOCATED_BACKWARDS_COMPATIBILITY` and `HPX_WITH_COMPONENT_GET_GID_COMPATIBILITY` are now disabled by default. Please change your code still depending on the deprecated interfaces.

Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this release.

- [PR #2596](#)⁴⁵⁵⁹ - Adding apex data
- [PR #2595](#)⁴⁵⁶⁰ - Remove obsolete file
- [Issue #2594](#)⁴⁵⁶¹ - FindOpenCL.cmake mismatch with the official cmake module
- [PR #2592](#)⁴⁵⁶² - First attempt to introduce `spmd_block` in `hpx`
- [Issue #2591](#)⁴⁵⁶³ - Feature request: continuation (then) which does not require the callable object to take a `future<R>` as parameter
- [PR #2588](#)⁴⁵⁶⁴ - Daint fixes
- [PR #2587](#)⁴⁵⁶⁵ - Fixing `transfer_(continuation)_action::schedule`
- [PR #2585](#)⁴⁵⁶⁶ - Work around MSVC having an ICE when compiling with `-Ob2`
- [PR #2583](#)⁴⁵⁶⁷ - changing `7zip` command to `7za` in `roll_release.sh`

⁴⁵⁵⁹ <https://github.com/TheHPXProject/hpx/pull/2596>

⁴⁵⁶⁰ <https://github.com/TheHPXProject/hpx/pull/2595>

⁴⁵⁶¹ <https://github.com/TheHPXProject/hpx/issues/2594>

⁴⁵⁶² <https://github.com/TheHPXProject/hpx/pull/2592>

⁴⁵⁶³ <https://github.com/TheHPXProject/hpx/issues/2591>

⁴⁵⁶⁴ <https://github.com/TheHPXProject/hpx/pull/2588>

⁴⁵⁶⁵ <https://github.com/TheHPXProject/hpx/pull/2587>

⁴⁵⁶⁶ <https://github.com/TheHPXProject/hpx/pull/2585>

⁴⁵⁶⁷ <https://github.com/TheHPXProject/hpx/pull/2583>

- PR #2582⁴⁵⁶⁸ - First attempt to introduce `spmd_block` in `hpx`
- PR #2581⁴⁵⁶⁹ - Enable annotated function for parallel algorithms
- PR #2580⁴⁵⁷⁰ - First attempt to introduce `spmd_block` in `hpx`
- PR #2579⁴⁵⁷¹ - Make thread NICE level setting an option
- PR #2578⁴⁵⁷² - Implementing `enqueue` instead of busy wait when no sender is available
- PR #2577⁴⁵⁷³ - Retrieve `-std=c++11` consistent `nvcc` flag
- PR #2576⁴⁵⁷⁴ - Add missing dependencies of `cuda` based tests
- PR #2575⁴⁵⁷⁵ - Remove warnings due to some captured variables
- PR #2573⁴⁵⁷⁶ - Attempt to resolve `resolve_locality`
- PR #2572⁴⁵⁷⁷ - Adding APEX hooks to background thread
- PR #2571⁴⁵⁷⁸ - Pick up `hpx.ignore_batch_env` from config map
- PR #2570⁴⁵⁷⁹ - Add cmdline options `-hpx:print-counters-locally`
- PR #2569⁴⁵⁸⁰ - Fix `computeapi` unit tests
- PR #2567⁴⁵⁸¹ - This adds another `barrier::synchronize` before registering performance counters
- PR #2564⁴⁵⁸² - Cray static toolchain support
- PR #2563⁴⁵⁸³ - Fixed unhandled exception during startup
- PR #2562⁴⁵⁸⁴ - Remove `partitioned_vector.cu` from build tree when `nvcc` is used
- Issue #2561⁴⁵⁸⁵ - octo-tiger crash with commit `6e921495ff6c26f125d62629cbaad0525f14f7ab`
- PR #2560⁴⁵⁸⁶ - Prevent `-Wundef` warnings on `Vc` version checks
- PR #2559⁴⁵⁸⁷ - Allowing `CUDA` callback to set the future directly from an OS thread
- PR #2558⁴⁵⁸⁸ - Remove warnings due to float precisions
- PR #2557⁴⁵⁸⁹ - Removing bogus handling of compile flags for `CUDA`
- PR #2556⁴⁵⁹⁰ - Fixing scan partitioner

⁴⁵⁶⁸ <https://github.com/TheHPXProject/hpx/pull/2582>
⁴⁵⁶⁹ <https://github.com/TheHPXProject/hpx/pull/2581>
⁴⁵⁷⁰ <https://github.com/TheHPXProject/hpx/pull/2580>
⁴⁵⁷¹ <https://github.com/TheHPXProject/hpx/pull/2579>
⁴⁵⁷² <https://github.com/TheHPXProject/hpx/pull/2578>
⁴⁵⁷³ <https://github.com/TheHPXProject/hpx/pull/2577>
⁴⁵⁷⁴ <https://github.com/TheHPXProject/hpx/pull/2576>
⁴⁵⁷⁵ <https://github.com/TheHPXProject/hpx/pull/2575>
⁴⁵⁷⁶ <https://github.com/TheHPXProject/hpx/pull/2573>
⁴⁵⁷⁷ <https://github.com/TheHPXProject/hpx/pull/2572>
⁴⁵⁷⁸ <https://github.com/TheHPXProject/hpx/pull/2571>
⁴⁵⁷⁹ <https://github.com/TheHPXProject/hpx/pull/2570>
⁴⁵⁸⁰ <https://github.com/TheHPXProject/hpx/pull/2569>
⁴⁵⁸¹ <https://github.com/TheHPXProject/hpx/pull/2567>
⁴⁵⁸² <https://github.com/TheHPXProject/hpx/pull/2564>
⁴⁵⁸³ <https://github.com/TheHPXProject/hpx/pull/2563>
⁴⁵⁸⁴ <https://github.com/TheHPXProject/hpx/pull/2562>
⁴⁵⁸⁵ <https://github.com/TheHPXProject/hpx/issues/2561>
⁴⁵⁸⁶ <https://github.com/TheHPXProject/hpx/pull/2560>
⁴⁵⁸⁷ <https://github.com/TheHPXProject/hpx/pull/2559>
⁴⁵⁸⁸ <https://github.com/TheHPXProject/hpx/pull/2558>
⁴⁵⁸⁹ <https://github.com/TheHPXProject/hpx/pull/2557>
⁴⁵⁹⁰ <https://github.com/TheHPXProject/hpx/pull/2556>

- PR #2554⁴⁵⁹¹ - Add more diagnostics to error thrown from `find_appropriate_destination`
- Issue #2555⁴⁵⁹² - No valid parcellport configured
- PR #2553⁴⁵⁹³ - Add `cmake cuda_arch` option
- PR #2552⁴⁵⁹⁴ - Remove incomplete datapar bindings to `libflatarray`
- PR #2551⁴⁵⁹⁵ - Rename `hwloc_topology` to `hwloc_topology_info`
- PR #2550⁴⁵⁹⁶ - Apex api updates
- PR #2549⁴⁵⁹⁷ - Pre-include `defines.hpp` to get the macro `HPX_HAVE_CUDA` value
- PR #2548⁴⁵⁹⁸ - Fixing issue with `disconnect`
- PR #2546⁴⁵⁹⁹ - Some fixes around `cuda clang partitioned_vector` example
- PR #2545⁴⁶⁰⁰ - Fix uses of the `Vc2` datapar flags; the value, not the type, should be passed to functions
- PR #2542⁴⁶⁰¹ - Make `HPX_WITH_MALLOC` easier to use
- PR #2541⁴⁶⁰² - avoid recompiles when enabling/disabling examples
- PR #2540⁴⁶⁰³ - Fixing usage of `target_link_libraries()`
- PR #2539⁴⁶⁰⁴ - fix `RPATH` behaviour
- Issue #2538⁴⁶⁰⁵ - `HPX_WITH_CUDA` corrupts compilation flags
- PR #2537⁴⁶⁰⁶ - Add output of a Bazel Skylark extension for paths and compile options
- PR #2536⁴⁶⁰⁷ - Add counter exposing total available memory to Windows as well
- PR #2535⁴⁶⁰⁸ - Remove obsolete support for security
- Issue #2534⁴⁶⁰⁹ - Remove command line option `--hpx:run-agas-server`
- PR #2533⁴⁶¹⁰ - Pre-cache locality endpoints during bootstrap
- PR #2532⁴⁶¹¹ - Fixing handling of GIDs during serialization preprocessing
- PR #2531⁴⁶¹² - Amend uses of the term “functor”
- PR #2529⁴⁶¹³ - added counter for reading available memory

⁴⁵⁹¹ <https://github.com/TheHPXProject/hpx/pull/2554>

⁴⁵⁹² <https://github.com/TheHPXProject/hpx/issues/2555>

⁴⁵⁹³ <https://github.com/TheHPXProject/hpx/pull/2553>

⁴⁵⁹⁴ <https://github.com/TheHPXProject/hpx/pull/2552>

⁴⁵⁹⁵ <https://github.com/TheHPXProject/hpx/pull/2551>

⁴⁵⁹⁶ <https://github.com/TheHPXProject/hpx/pull/2550>

⁴⁵⁹⁷ <https://github.com/TheHPXProject/hpx/pull/2549>

⁴⁵⁹⁸ <https://github.com/TheHPXProject/hpx/pull/2548>

⁴⁵⁹⁹ <https://github.com/TheHPXProject/hpx/pull/2546>

⁴⁶⁰⁰ <https://github.com/TheHPXProject/hpx/pull/2545>

⁴⁶⁰¹ <https://github.com/TheHPXProject/hpx/pull/2542>

⁴⁶⁰² <https://github.com/TheHPXProject/hpx/pull/2541>

⁴⁶⁰³ <https://github.com/TheHPXProject/hpx/pull/2540>

⁴⁶⁰⁴ <https://github.com/TheHPXProject/hpx/pull/2539>

⁴⁶⁰⁵ <https://github.com/TheHPXProject/hpx/issues/2538>

⁴⁶⁰⁶ <https://github.com/TheHPXProject/hpx/pull/2537>

⁴⁶⁰⁷ <https://github.com/TheHPXProject/hpx/pull/2536>

⁴⁶⁰⁸ <https://github.com/TheHPXProject/hpx/pull/2535>

⁴⁶⁰⁹ <https://github.com/TheHPXProject/hpx/issues/2534>

⁴⁶¹⁰ <https://github.com/TheHPXProject/hpx/pull/2533>

⁴⁶¹¹ <https://github.com/TheHPXProject/hpx/pull/2532>

⁴⁶¹² <https://github.com/TheHPXProject/hpx/pull/2531>

⁴⁶¹³ <https://github.com/TheHPXProject/hpx/pull/2529>

- PR #2527⁴⁶¹⁴ - Facilities to create actions from lambdas
- PR #2526⁴⁶¹⁵ - Updated docs: HPX_WITH_EXAMPLES
- PR #2525⁴⁶¹⁶ - Remove warnings related to unused captured variables
- Issue #2524⁴⁶¹⁷ - CMAKE failed because it is missing: TCMALLOC_LIBRARY TCMALLOC_INCLUDE_DIR
- PR #2523⁴⁶¹⁸ - Fixing compose_cb stack overflow
- PR #2522⁴⁶¹⁹ - Instead of unlocking, ignore the lock while creating the message handler
- PR #2521⁴⁶²⁰ - Create LPROGRESS_ logging macro to simplify progress tracking and timings
- PR #2520⁴⁶²¹ - Intel 17 support
- PR #2519⁴⁶²² - Fix components example
- PR #2518⁴⁶²³ - Fixing parcel scheduling
- Issue #2517⁴⁶²⁴ - Race condition during Parcel Coalescing Handler creation
- Issue #2516⁴⁶²⁵ - HPX locks up when using at least 256 localities
- Issue #2515⁴⁶²⁶ - error: Install cannot find “/lib/hpx/libparcel_coalescing.so.0.9.99” but I can see that file
- PR #2514⁴⁶²⁷ - Making sure that all continuations of a shared_future are invoked in order
- PR #2513⁴⁶²⁸ - Fixing locks held during suspension
- PR #2512⁴⁶²⁹ - MPI Parcelport improvements and fixes related to the background work changes
- PR #2511⁴⁶³⁰ - Fixing bit-wise (zero-copy) serialization
- Issue #2509⁴⁶³¹ - Linking errors in hwloc_topology
- PR #2508⁴⁶³² - Added documentation for debugging with core files
- PR #2506⁴⁶³³ - Fixing background work invocations
- PR #2505⁴⁶³⁴ - Fix tuple serialization
- Issue #2504⁴⁶³⁵ - Ensure continuations are called in the order they have been attached
- PR #2503⁴⁶³⁶ - Adding serialization support for Vc v2 (datapar)

⁴⁶¹⁴ <https://github.com/TheHPXProject/hpx/pull/2527>

⁴⁶¹⁵ <https://github.com/TheHPXProject/hpx/pull/2526>

⁴⁶¹⁶ <https://github.com/TheHPXProject/hpx/pull/2525>

⁴⁶¹⁷ <https://github.com/TheHPXProject/hpx/issues/2524>

⁴⁶¹⁸ <https://github.com/TheHPXProject/hpx/pull/2523>

⁴⁶¹⁹ <https://github.com/TheHPXProject/hpx/pull/2522>

⁴⁶²⁰ <https://github.com/TheHPXProject/hpx/pull/2521>

⁴⁶²¹ <https://github.com/TheHPXProject/hpx/pull/2520>

⁴⁶²² <https://github.com/TheHPXProject/hpx/pull/2519>

⁴⁶²³ <https://github.com/TheHPXProject/hpx/pull/2518>

⁴⁶²⁴ <https://github.com/TheHPXProject/hpx/issues/2517>

⁴⁶²⁵ <https://github.com/TheHPXProject/hpx/issues/2516>

⁴⁶²⁶ <https://github.com/TheHPXProject/hpx/issues/2515>

⁴⁶²⁷ <https://github.com/TheHPXProject/hpx/pull/2514>

⁴⁶²⁸ <https://github.com/TheHPXProject/hpx/pull/2513>

⁴⁶²⁹ <https://github.com/TheHPXProject/hpx/pull/2512>

⁴⁶³⁰ <https://github.com/TheHPXProject/hpx/pull/2511>

⁴⁶³¹ <https://github.com/TheHPXProject/hpx/issues/2509>

⁴⁶³² <https://github.com/TheHPXProject/hpx/pull/2508>

⁴⁶³³ <https://github.com/TheHPXProject/hpx/pull/2506>

⁴⁶³⁴ <https://github.com/TheHPXProject/hpx/pull/2505>

⁴⁶³⁵ <https://github.com/TheHPXProject/hpx/issues/2504>

⁴⁶³⁶ <https://github.com/TheHPXProject/hpx/pull/2503>

- PR #2502⁴⁶³⁷ - Resolve various, minor compiler warnings
- PR #2501⁴⁶³⁸ - Some other fixes around cuda examples
- Issue #2500⁴⁶³⁹ - nvcc / cuda clang issue due to a missing -DHPX_WITH_CUDA flag
- PR #2499⁴⁶⁴⁰ - Adding support for std::array to wait_all and friends
- PR #2498⁴⁶⁴¹ - Execute background work as HPX thread
- PR #2497⁴⁶⁴² - Fixing configuration options for spinlock-deadlock detection
- PR #2496⁴⁶⁴³ - Accounting for different compilers in CrayKNL toolchain file
- PR #2494⁴⁶⁴⁴ - Adding component base class which ties a component instance to a given executor
- PR #2493⁴⁶⁴⁵ - Enable controlling amount of pending threads which must be available to allow thread stealing
- PR #2492⁴⁶⁴⁶ - Adding new command line option -hpx:print-counter-reset
- PR #2491⁴⁶⁴⁷ - Resolve ambiguities when compiling with APEX
- PR #2490⁴⁶⁴⁸ - Resuming threads waiting on future with higher priority
- Issue #2489⁴⁶⁴⁹ - nvcc issue because -std=c++11 appears twice
- PR #2488⁴⁶⁵⁰ - Adding performance counters exposing the internal idle and busy-loop counters
- PR #2487⁴⁶⁵¹ - Allowing for plain suspend to reschedule thread right away
- PR #2486⁴⁶⁵² - Only flag HPX code for CUDA if HPX_WITH_CUDA is set
- PR #2485⁴⁶⁵³ - Making thread-queue parameters runtime-configurable
- PR #2484⁴⁶⁵⁴ - Added atomic counter for parcel-destinations
- PR #2483⁴⁶⁵⁵ - Added priority-queue lifo scheduler
- PR #2482⁴⁶⁵⁶ - Changing scheduler to steal only if more than a minimal number of tasks are available
- PR #2481⁴⁶⁵⁷ - Extending command line option -hpx:print-counter-destination to support value 'none'
- PR #2479⁴⁶⁵⁸ - Added option to disable signal handler
- PR #2478⁴⁶⁵⁹ - Making sure the sine performance counter module gets loaded only for the corresponding example

⁴⁶³⁷ <https://github.com/TheHPXProject/hpx/pull/2502>

⁴⁶³⁸ <https://github.com/TheHPXProject/hpx/pull/2501>

⁴⁶³⁹ <https://github.com/TheHPXProject/hpx/issues/2500>

⁴⁶⁴⁰ <https://github.com/TheHPXProject/hpx/pull/2499>

⁴⁶⁴¹ <https://github.com/TheHPXProject/hpx/pull/2498>

⁴⁶⁴² <https://github.com/TheHPXProject/hpx/pull/2497>

⁴⁶⁴³ <https://github.com/TheHPXProject/hpx/pull/2496>

⁴⁶⁴⁴ <https://github.com/TheHPXProject/hpx/pull/2494>

⁴⁶⁴⁵ <https://github.com/TheHPXProject/hpx/pull/2493>

⁴⁶⁴⁶ <https://github.com/TheHPXProject/hpx/pull/2492>

⁴⁶⁴⁷ <https://github.com/TheHPXProject/hpx/pull/2491>

⁴⁶⁴⁸ <https://github.com/TheHPXProject/hpx/pull/2490>

⁴⁶⁴⁹ <https://github.com/TheHPXProject/hpx/issues/2489>

⁴⁶⁵⁰ <https://github.com/TheHPXProject/hpx/pull/2488>

⁴⁶⁵¹ <https://github.com/TheHPXProject/hpx/pull/2487>

⁴⁶⁵² <https://github.com/TheHPXProject/hpx/pull/2486>

⁴⁶⁵³ <https://github.com/TheHPXProject/hpx/pull/2485>

⁴⁶⁵⁴ <https://github.com/TheHPXProject/hpx/pull/2484>

⁴⁶⁵⁵ <https://github.com/TheHPXProject/hpx/pull/2483>

⁴⁶⁵⁶ <https://github.com/TheHPXProject/hpx/pull/2482>

⁴⁶⁵⁷ <https://github.com/TheHPXProject/hpx/pull/2481>

⁴⁶⁵⁸ <https://github.com/TheHPXProject/hpx/pull/2479>

⁴⁶⁵⁹ <https://github.com/TheHPXProject/hpx/pull/2478>

- Issue #2477⁴⁶⁶⁰ - Breaking at a throw statement
- PR #2476⁴⁶⁶¹ - Annotated function
- PR #2475⁴⁶⁶² - Ensure that using `%osthread%` during logging will not throw for non-hpx threads
- PR #2474⁴⁶⁶³ - Remove now superficial `non_direct` actions from `base_lco` and friends
- PR #2473⁴⁶⁶⁴ - Refining support for `ITTNotify`
- PR #2472⁴⁶⁶⁵ - Some fixes around hpx compute
- Issue #2470⁴⁶⁶⁶ - redefinition of `boost::detail::spinlock`
- Issue #2469⁴⁶⁶⁷ - Dataflow performance issue
- PR #2468⁴⁶⁶⁸ - Perf docs update
- PR #2466⁴⁶⁶⁹ - Guarantee to execute remote direct actions on HPX-thread
- PR #2465⁴⁶⁷⁰ - Improve demo : Async copy and fixed device handling
- PR #2464⁴⁶⁷¹ - Adding performance counter exposing instantaneous scheduler utilization
- PR #2463⁴⁶⁷² - Downcast to `future<void>`
- PR #2462⁴⁶⁷³ - Fixed usage of `ITT-Notify` API with Intel Amplifier
- PR #2461⁴⁶⁷⁴ - Cublas demo
- PR #2460⁴⁶⁷⁵ - Fixing thread bindings
- PR #2459⁴⁶⁷⁶ - Make `-std=c++11` nvcc flag consistent for in-build and installed versions
- Issue #2457⁴⁶⁷⁷ - Segmentation fault when registering a partitioned vector
- PR #2452⁴⁶⁷⁸ - Properly releasing global barrier for unhandled exceptions
- PR #2451⁴⁶⁷⁹ - Fixing long shutdown times
- PR #2450⁴⁶⁸⁰ - Attempting to fix initialization errors on newer platforms (Boost V1.63)
- PR #2449⁴⁶⁸¹ - Replace `BOOST_COMPILER_FENCE` with an HPX version
- PR #2448⁴⁶⁸² - This fixes a possible race in the migration code

⁴⁶⁶⁰ <https://github.com/TheHPXProject/hpx/issues/2477>

⁴⁶⁶¹ <https://github.com/TheHPXProject/hpx/pull/2476>

⁴⁶⁶² <https://github.com/TheHPXProject/hpx/pull/2475>

⁴⁶⁶³ <https://github.com/TheHPXProject/hpx/pull/2474>

⁴⁶⁶⁴ <https://github.com/TheHPXProject/hpx/pull/2473>

⁴⁶⁶⁵ <https://github.com/TheHPXProject/hpx/pull/2472>

⁴⁶⁶⁶ <https://github.com/TheHPXProject/hpx/issues/2470>

⁴⁶⁶⁷ <https://github.com/TheHPXProject/hpx/issues/2469>

⁴⁶⁶⁸ <https://github.com/TheHPXProject/hpx/pull/2468>

⁴⁶⁶⁹ <https://github.com/TheHPXProject/hpx/pull/2466>

⁴⁶⁷⁰ <https://github.com/TheHPXProject/hpx/pull/2465>

⁴⁶⁷¹ <https://github.com/TheHPXProject/hpx/pull/2464>

⁴⁶⁷² <https://github.com/TheHPXProject/hpx/pull/2463>

⁴⁶⁷³ <https://github.com/TheHPXProject/hpx/pull/2462>

⁴⁶⁷⁴ <https://github.com/TheHPXProject/hpx/pull/2461>

⁴⁶⁷⁵ <https://github.com/TheHPXProject/hpx/pull/2460>

⁴⁶⁷⁶ <https://github.com/TheHPXProject/hpx/pull/2459>

⁴⁶⁷⁷ <https://github.com/TheHPXProject/hpx/issues/2457>

⁴⁶⁷⁸ <https://github.com/TheHPXProject/hpx/pull/2452>

⁴⁶⁷⁹ <https://github.com/TheHPXProject/hpx/pull/2451>

⁴⁶⁸⁰ <https://github.com/TheHPXProject/hpx/pull/2450>

⁴⁶⁸¹ <https://github.com/TheHPXProject/hpx/pull/2449>

⁴⁶⁸² <https://github.com/TheHPXProject/hpx/pull/2448>

- **PR #2445[?]** - Fixing dataflow et.al. for futures or future-ranges wrapped into ref()
- PR #2444⁴⁶⁸⁴ - Fix segfaults
- PR #2443⁴⁶⁸⁵ - Issue 2442
- Issue #2442⁴⁶⁸⁶ - Mismatch between #if/#endif and namespace scope brackets in this_thread_executors.hpp
- Issue #2441⁴⁶⁸⁷ - undeclared identifier BOOST_COMPILER_FENCE
- PR #2440⁴⁶⁸⁸ - Knl build
- PR #2438⁴⁶⁸⁹ - Datapar backend
- PR #2437⁴⁶⁹⁰ - Adapt algorithm parameter sequence changes from C++17
- PR #2436⁴⁶⁹¹ - Adapt execution policy name changes from C++17
- Issue #2435⁴⁶⁹² - Trunk broken, undefined reference to hpx::thread::interrupt(hpx::thread::id, bool)
- PR #2434⁴⁶⁹³ - More fixes to resource manager
- PR #2433⁴⁶⁹⁴ - Added versions of hpx::get_ptr taking client side representations
- PR #2432⁴⁶⁹⁵ - Warning fixes
- PR #2431⁴⁶⁹⁶ - Adding facility representing set of performance counters
- PR #2430⁴⁶⁹⁷ - Fix parallel_executor thread spawning
- PR #2429⁴⁶⁹⁸ - Fix attribute warning for gcc
- Issue #2427⁴⁶⁹⁹ - Seg fault running octo-tiger with latest HPX commit
- Issue #2426⁴⁷⁰⁰ - Bug in 9592f5c0bc29806fce0dbe73f35b6ca7e027edcb causes immediate crash in Octo-tiger
- PR #2425⁴⁷⁰¹ - Fix nvcc errors due to constexpr specifier
- Issue #2424⁴⁷⁰² - Async action on component present on hpx::find_here is executing synchronously
- PR #2423⁴⁷⁰³ - Fix nvcc errors due to constexpr specifier
- PR #2422⁴⁷⁰⁴ - Implementing hpx::this_thread thread data functions
- PR #2421⁴⁷⁰⁵ - Adding benchmark for wait_all

⁴⁶⁸³ <https://github.com/TheHPXProject/hpx/pull/2445>

⁴⁶⁸⁴ <https://github.com/TheHPXProject/hpx/pull/2444>

⁴⁶⁸⁵ <https://github.com/TheHPXProject/hpx/pull/2443>

⁴⁶⁸⁶ <https://github.com/TheHPXProject/hpx/issues/2442>

⁴⁶⁸⁷ <https://github.com/TheHPXProject/hpx/issues/2441>

⁴⁶⁸⁸ <https://github.com/TheHPXProject/hpx/pull/2440>

⁴⁶⁸⁹ <https://github.com/TheHPXProject/hpx/pull/2438>

⁴⁶⁹⁰ <https://github.com/TheHPXProject/hpx/pull/2437>

⁴⁶⁹¹ <https://github.com/TheHPXProject/hpx/pull/2436>

⁴⁶⁹² <https://github.com/TheHPXProject/hpx/issues/2435>

⁴⁶⁹³ <https://github.com/TheHPXProject/hpx/pull/2434>

⁴⁶⁹⁴ <https://github.com/TheHPXProject/hpx/pull/2433>

⁴⁶⁹⁵ <https://github.com/TheHPXProject/hpx/pull/2432>

⁴⁶⁹⁶ <https://github.com/TheHPXProject/hpx/pull/2431>

⁴⁶⁹⁷ <https://github.com/TheHPXProject/hpx/pull/2430>

⁴⁶⁹⁸ <https://github.com/TheHPXProject/hpx/pull/2429>

⁴⁶⁹⁹ <https://github.com/TheHPXProject/hpx/issues/2427>

⁴⁷⁰⁰ <https://github.com/TheHPXProject/hpx/issues/2426>

⁴⁷⁰¹ <https://github.com/TheHPXProject/hpx/pull/2425>

⁴⁷⁰² <https://github.com/TheHPXProject/hpx/issues/2424>

⁴⁷⁰³ <https://github.com/TheHPXProject/hpx/pull/2423>

⁴⁷⁰⁴ <https://github.com/TheHPXProject/hpx/pull/2422>

⁴⁷⁰⁵ <https://github.com/TheHPXProject/hpx/pull/2421>

- [Issue #2420](#)⁴⁷⁰⁶ - Returning object of a component client from another component action fails
- [PR #2419](#)⁴⁷⁰⁷ - Infiniband parcellport
- [Issue #2418](#)⁴⁷⁰⁸ - gcc + nvcc fails to compile code that uses `partitioned_vector`
- [PR #2417](#)⁴⁷⁰⁹ - Fixing context switching
- [PR #2416](#)⁴⁷¹⁰ - Adding fixes and workarounds to allow compilation with nvcc/msvc (VS2015up3)
- [PR #2415](#)⁴⁷¹¹ - Fix errors coming from hpx compute examples
- [PR #2414](#)⁴⁷¹² - Fixing msvc12
- [PR #2413](#)⁴⁷¹³ - Enable cuda/nvcc or cuda/clang when using `add_hpx_executable()`
- [PR #2412](#)⁴⁷¹⁴ - Fix issue in `HPX_SetupTarget.cmake` when cuda is used
- [PR #2411](#)⁴⁷¹⁵ - This fixes the core compilation issues with MSVC12
- [Issue #2410](#)⁴⁷¹⁶ - `undefined reference to opal_hwloc191_hwloc_.....`
- [PR #2409](#)⁴⁷¹⁷ - Fixing locking for channel and `receive_buffer`
- [PR #2407](#)⁴⁷¹⁸ - Solving #2402 and #2403
- [PR #2406](#)⁴⁷¹⁹ - Improve guards
- [PR #2405](#)⁴⁷²⁰ - Enable `parallel::for_each` for iterators returning proxy types
- [PR #2404](#)⁴⁷²¹ - Forward the explicitly given `result_type` in the hpx invoke
- [Issue #2403](#)⁴⁷²² - `datapar_execution + zip iterator`: lambda arguments aren't references
- [Issue #2402](#)⁴⁷²³ - `datapar` algorithm instantiated with wrong type #2402
- [PR #2401](#)⁴⁷²⁴ - Added support for imported libraries to `HPX_Libraries.cmake`
- [PR #2400](#)⁴⁷²⁵ - Use CMake policy `CMP0060`
- [Issue #2399](#)⁴⁷²⁶ - Error trying to push back vector of futures to vector
- [PR #2398](#)⁴⁷²⁷ - Allow `config` `#defines` to be written out to custom `config/defines.hpp`
- [Issue #2397](#)⁴⁷²⁸ - CMake generated `config` `defines` can cause tedious rebuilds category

⁴⁷⁰⁶ <https://github.com/TheHPXProject/hpx/issues/2420>

⁴⁷⁰⁷ <https://github.com/TheHPXProject/hpx/pull/2419>

⁴⁷⁰⁸ <https://github.com/TheHPXProject/hpx/issues/2418>

⁴⁷⁰⁹ <https://github.com/TheHPXProject/hpx/pull/2417>

⁴⁷¹⁰ <https://github.com/TheHPXProject/hpx/pull/2416>

⁴⁷¹¹ <https://github.com/TheHPXProject/hpx/pull/2415>

⁴⁷¹² <https://github.com/TheHPXProject/hpx/pull/2414>

⁴⁷¹³ <https://github.com/TheHPXProject/hpx/pull/2413>

⁴⁷¹⁴ <https://github.com/TheHPXProject/hpx/pull/2412>

⁴⁷¹⁵ <https://github.com/TheHPXProject/hpx/pull/2411>

⁴⁷¹⁶ <https://github.com/TheHPXProject/hpx/issues/2410>

⁴⁷¹⁷ <https://github.com/TheHPXProject/hpx/pull/2409>

⁴⁷¹⁸ <https://github.com/TheHPXProject/hpx/pull/2407>

⁴⁷¹⁹ <https://github.com/TheHPXProject/hpx/pull/2406>

⁴⁷²⁰ <https://github.com/TheHPXProject/hpx/pull/2405>

⁴⁷²¹ <https://github.com/TheHPXProject/hpx/pull/2404>

⁴⁷²² <https://github.com/TheHPXProject/hpx/issues/2403>

⁴⁷²³ <https://github.com/TheHPXProject/hpx/issues/2402>

⁴⁷²⁴ <https://github.com/TheHPXProject/hpx/pull/2401>

⁴⁷²⁵ <https://github.com/TheHPXProject/hpx/pull/2400>

⁴⁷²⁶ <https://github.com/TheHPXProject/hpx/issues/2399>

⁴⁷²⁷ <https://github.com/TheHPXProject/hpx/pull/2398>

⁴⁷²⁸ <https://github.com/TheHPXProject/hpx/issues/2397>

- [Issue #2396](#)⁴⁷²⁹ - BOOST_ROOT paths are not used at link time
- [PR #2395](#)⁴⁷³⁰ - Fix target_link_libraries() issue when HPX Cuda is enabled
- [Issue #2394](#)⁴⁷³¹ - Template compilation error using HPX_WITH_DATAPAR_LIBFLATARRAY
- [PR #2393](#)⁴⁷³² - Fixing lock registration for recursive mutex
- [PR #2392](#)⁴⁷³³ - Add keywords in target_link_libraries in hpx_setup_target
- [PR #2391](#)⁴⁷³⁴ - Clang goroutines
- [Issue #2390](#)⁴⁷³⁵ - Adapt execution policy name changes from C++17
- [PR #2389](#)⁴⁷³⁶ - Chunk allocator and pool are not used and are obsolete
- [PR #2388](#)⁴⁷³⁷ - Adding functionalities to datapar needed by octotiger
- [PR #2387](#)⁴⁷³⁸ - Fixing race condition for early parcels
- [Issue #2386](#)⁴⁷³⁹ - Lock registration broken for recursive_mutex
- [PR #2385](#)⁴⁷⁴⁰ - Datapar zip iterator
- [PR #2384](#)⁴⁷⁴¹ - Fixing race condition in for_loop_reduction
- [PR #2383](#)⁴⁷⁴² - Continuations
- [PR #2382](#)⁴⁷⁴³ - add LibFlatArray-based backend for datapar
- [PR #2381](#)⁴⁷⁴⁴ - remove unused typedef to get rid of compiler warnings
- [PR #2380](#)⁴⁷⁴⁵ - Tau cleanup
- [PR #2379](#)⁴⁷⁴⁶ - Can send immediate
- [PR #2378](#)⁴⁷⁴⁷ - Renaming copy_helper/copy_n_helper/move_helper/move_n_helper
- [Issue #2376](#)⁴⁷⁴⁸ - Boost trunk's spinlock initializer fails to compile
- [PR #2375](#)⁴⁷⁴⁹ - Add support for minimal thread local data
- [PR #2374](#)⁴⁷⁵⁰ - Adding API functions set_config_entry_callback
- [PR #2373](#)⁴⁷⁵¹ - Add a simple utility for debugging that gives suspended task backtraces

⁴⁷²⁹ <https://github.com/TheHPXProject/hpx/issues/2396>

⁴⁷³⁰ <https://github.com/TheHPXProject/hpx/pull/2395>

⁴⁷³¹ <https://github.com/TheHPXProject/hpx/issues/2394>

⁴⁷³² <https://github.com/TheHPXProject/hpx/pull/2393>

⁴⁷³³ <https://github.com/TheHPXProject/hpx/pull/2392>

⁴⁷³⁴ <https://github.com/TheHPXProject/hpx/pull/2391>

⁴⁷³⁵ <https://github.com/TheHPXProject/hpx/issues/2390>

⁴⁷³⁶ <https://github.com/TheHPXProject/hpx/pull/2389>

⁴⁷³⁷ <https://github.com/TheHPXProject/hpx/pull/2388>

⁴⁷³⁸ <https://github.com/TheHPXProject/hpx/pull/2387>

⁴⁷³⁹ <https://github.com/TheHPXProject/hpx/issues/2386>

⁴⁷⁴⁰ <https://github.com/TheHPXProject/hpx/pull/2385>

⁴⁷⁴¹ <https://github.com/TheHPXProject/hpx/pull/2384>

⁴⁷⁴² <https://github.com/TheHPXProject/hpx/pull/2383>

⁴⁷⁴³ <https://github.com/TheHPXProject/hpx/pull/2382>

⁴⁷⁴⁴ <https://github.com/TheHPXProject/hpx/pull/2381>

⁴⁷⁴⁵ <https://github.com/TheHPXProject/hpx/pull/2380>

⁴⁷⁴⁶ <https://github.com/TheHPXProject/hpx/pull/2379>

⁴⁷⁴⁷ <https://github.com/TheHPXProject/hpx/pull/2378>

⁴⁷⁴⁸ <https://github.com/TheHPXProject/hpx/issues/2376>

⁴⁷⁴⁹ <https://github.com/TheHPXProject/hpx/pull/2375>

⁴⁷⁵⁰ <https://github.com/TheHPXProject/hpx/pull/2374>

⁴⁷⁵¹ <https://github.com/TheHPXProject/hpx/pull/2373>

- PR #2372⁴⁷⁵² - Barrier Fixes
- Issue #2370⁴⁷⁵³ - Can't wait on a wrapped future
- PR #2369⁴⁷⁵⁴ - Fixing stable_partition
- PR #2367⁴⁷⁵⁵ - Fixing find_prefixes for Windows platforms
- PR #2366⁴⁷⁵⁶ - Testing for experimental/optional only in C++14 mode
- PR #2364⁴⁷⁵⁷ - Adding set_config_entry
- PR #2363⁴⁷⁵⁸ - Fix papi
- PR #2362⁴⁷⁵⁹ - Adding missing macros for new non-direct actions
- PR #2361⁴⁷⁶⁰ - Improve cmake output to help debug compiler incompatibility check
- PR #2360⁴⁷⁶¹ - Fixing race condition in condition_variable
- PR #2359⁴⁷⁶² - Fixing shutdown when parcels are still in flight
- Issue #2357⁴⁷⁶³ - failed to insert console_print_action into typename_to_id_t registry
- PR #2356⁴⁷⁶⁴ - Fixing return type of get_iterator_tuple
- PR #2355⁴⁷⁶⁵ - Fixing compilation against Boost 1.62
- PR #2354⁴⁷⁶⁶ - Adding serialization for mask_type if CPU_COUNT > 64
- PR #2353⁴⁷⁶⁷ - Adding hooks to tie in APEX into the parcel layer
- Issue #2352⁴⁷⁶⁸ - Compile errors when using intel 17 beta (for KNL) on edison
- PR #2351⁴⁷⁶⁹ - Fix function vtable get_function_address implementation
- Issue #2350⁴⁷⁷⁰ - Build failure - master branch (4de09f5) with Intel Compiler v17
- PR #2349⁴⁷⁷¹ - Enabling zero-copy serialization support for std::vector<>
- PR #2348⁴⁷⁷² - Adding test to verify #2334 is fixed
- PR #2347⁴⁷⁷³ - Bug fixes for hpx.compute and hpx::lcos::channel
- PR #2346⁴⁷⁷⁴ - Removing cmake "find" files that are in the APEX cmake Modules

⁴⁷⁵² <https://github.com/TheHPXProject/hpx/pull/2372>

⁴⁷⁵³ <https://github.com/TheHPXProject/hpx/issues/2370>

⁴⁷⁵⁴ <https://github.com/TheHPXProject/hpx/pull/2369>

⁴⁷⁵⁵ <https://github.com/TheHPXProject/hpx/pull/2367>

⁴⁷⁵⁶ <https://github.com/TheHPXProject/hpx/pull/2366>

⁴⁷⁵⁷ <https://github.com/TheHPXProject/hpx/pull/2364>

⁴⁷⁵⁸ <https://github.com/TheHPXProject/hpx/pull/2363>

⁴⁷⁵⁹ <https://github.com/TheHPXProject/hpx/pull/2362>

⁴⁷⁶⁰ <https://github.com/TheHPXProject/hpx/pull/2361>

⁴⁷⁶¹ <https://github.com/TheHPXProject/hpx/pull/2360>

⁴⁷⁶² <https://github.com/TheHPXProject/hpx/pull/2359>

⁴⁷⁶³ <https://github.com/TheHPXProject/hpx/issues/2357>

⁴⁷⁶⁴ <https://github.com/TheHPXProject/hpx/pull/2356>

⁴⁷⁶⁵ <https://github.com/TheHPXProject/hpx/pull/2355>

⁴⁷⁶⁶ <https://github.com/TheHPXProject/hpx/pull/2354>

⁴⁷⁶⁷ <https://github.com/TheHPXProject/hpx/pull/2353>

⁴⁷⁶⁸ <https://github.com/TheHPXProject/hpx/issues/2352>

⁴⁷⁶⁹ <https://github.com/TheHPXProject/hpx/pull/2351>

⁴⁷⁷⁰ <https://github.com/TheHPXProject/hpx/issues/2350>

⁴⁷⁷¹ <https://github.com/TheHPXProject/hpx/pull/2349>

⁴⁷⁷² <https://github.com/TheHPXProject/hpx/pull/2348>

⁴⁷⁷³ <https://github.com/TheHPXProject/hpx/pull/2347>

⁴⁷⁷⁴ <https://github.com/TheHPXProject/hpx/pull/2346>

- [PR #2345⁴⁷⁷⁵](#) - Implemented `parallel::stable_partition`
- [PR #2344⁴⁷⁷⁶](#) - Making `hpx::lcos::channel` usable with basename registration
- [PR #2343⁴⁷⁷⁷](#) - Fix a couple of examples that failed to compile after recent api changes
- [Issue #2342⁴⁷⁷⁸](#) - Enabling APEX causes link errors
- [PR #2341⁴⁷⁷⁹](#) - Removing cmake “find” files that are in the APEX cmake Modules
- [PR #2340⁴⁷⁸⁰](#) - Implemented all existing datapar algorithms using Boost.SIMD
- [PR #2339⁴⁷⁸¹](#) - Fixing 2338
- [PR #2338⁴⁷⁸²](#) - Possible race in sliding semaphore
- [PR #2337⁴⁷⁸³](#) - Adjust `osu_latency` test to measure `window_size` parcels in flight at once
- [PR #2336⁴⁷⁸⁴](#) - Allowing remote direct actions to be executed without spawning a task
- [PR #2335⁴⁷⁸⁵](#) - Making sure multiple components are properly initialized from arguments
- [Issue #2334⁴⁷⁸⁶](#) - Cannot construct component with large vector on a remote locality
- [PR #2332⁴⁷⁸⁷](#) - Fixing `hpx::lcos::local::barrier`
- [PR #2331⁴⁷⁸⁸](#) - Updating APEX support to include OTF2
- [PR #2330⁴⁷⁸⁹](#) - Support for data-parallelism for parallel algorithms
- [Issue #2329⁴⁷⁹⁰](#) - Coordinate settings in cmake
- [PR #2328⁴⁷⁹¹](#) - fix LibGeoDecomp builds with HPX + GCC 5.3.0 + CUDA 8RC
- [PR #2326⁴⁷⁹²](#) - Making `scan_partitioner` work (for now)
- [Issue #2323⁴⁷⁹³](#) - Constructing a vector of components only correctly initializes the first component
- [PR #2322⁴⁷⁹⁴](#) - Fix problems that bubbled up after merging #2278
- [PR #2321⁴⁷⁹⁵](#) - Scalable barrier
- [PR #2320⁴⁷⁹⁶](#) - Std flag fixes
- [Issue #2319⁴⁷⁹⁷](#) - -std=c++14 and -std=c++1y with Intel can't build recent Boost builds due to insufficient C++14 support; don't enable these flags by default for Intel

⁴⁷⁷⁵ <https://github.com/TheHPXProject/hpx/pull/2345>

⁴⁷⁷⁶ <https://github.com/TheHPXProject/hpx/pull/2344>

⁴⁷⁷⁷ <https://github.com/TheHPXProject/hpx/pull/2343>

⁴⁷⁷⁸ <https://github.com/TheHPXProject/hpx/issues/2342>

⁴⁷⁷⁹ <https://github.com/TheHPXProject/hpx/pull/2341>

⁴⁷⁸⁰ <https://github.com/TheHPXProject/hpx/pull/2340>

⁴⁷⁸¹ <https://github.com/TheHPXProject/hpx/pull/2339>

⁴⁷⁸² <https://github.com/TheHPXProject/hpx/pull/2338>

⁴⁷⁸³ <https://github.com/TheHPXProject/hpx/pull/2337>

⁴⁷⁸⁴ <https://github.com/TheHPXProject/hpx/pull/2336>

⁴⁷⁸⁵ <https://github.com/TheHPXProject/hpx/pull/2335>

⁴⁷⁸⁶ <https://github.com/TheHPXProject/hpx/issues/2334>

⁴⁷⁸⁷ <https://github.com/TheHPXProject/hpx/pull/2332>

⁴⁷⁸⁸ <https://github.com/TheHPXProject/hpx/pull/2331>

⁴⁷⁸⁹ <https://github.com/TheHPXProject/hpx/pull/2330>

⁴⁷⁹⁰ <https://github.com/TheHPXProject/hpx/issues/2329>

⁴⁷⁹¹ <https://github.com/TheHPXProject/hpx/pull/2328>

⁴⁷⁹² <https://github.com/TheHPXProject/hpx/pull/2326>

⁴⁷⁹³ <https://github.com/TheHPXProject/hpx/issues/2323>

⁴⁷⁹⁴ <https://github.com/TheHPXProject/hpx/pull/2322>

⁴⁷⁹⁵ <https://github.com/TheHPXProject/hpx/pull/2321>

⁴⁷⁹⁶ <https://github.com/TheHPXProject/hpx/pull/2320>

⁴⁷⁹⁷ <https://github.com/TheHPXProject/hpx/issues/2319>

- PR #2318⁴⁷⁹⁸ - Improve handling of `-hpx:bind=<bind-spec>`
- PR #2317⁴⁷⁹⁹ - Making sure command line warnings are printed once only
- PR #2316⁴⁸⁰⁰ - Fixing command line handling for default bind mode
- PR #2315⁴⁸⁰¹ - Set `id_retrieved` if `set_id` is present
- Issue #2314⁴⁸⁰² - Warning for requested/allocated thread discrepancy is printed twice
- Issue #2313⁴⁸⁰³ - `-hpx:print-bind` doesn't work with `-hpx:pu-step`
- Issue #2312⁴⁸⁰⁴ - `-hpx:bind` range specifier restrictions are overly restrictive
- Issue #2311⁴⁸⁰⁵ - `hpx_0.9.99` out of project build fails
- PR #2310⁴⁸⁰⁶ - Simplify function registration
- PR #2309⁴⁸⁰⁷ - Spelling and grammar revisions in documentation (and some code)
- PR #2306⁴⁸⁰⁸ - Correct minor typo in the documentation
- PR #2305⁴⁸⁰⁹ - Cleaning up and fixing parcel coalescing
- PR #2304⁴⁸¹⁰ - Inspect checks for stream related includes
- PR #2303⁴⁸¹¹ - Add functionality allowing to enumerate threads of given state
- PR #2301⁴⁸¹² - Algorithm overloads fix for VS2013
- PR #2300⁴⁸¹³ - Use `<cstdint>`, add inspect checks
- PR #2299⁴⁸¹⁴ - Replace `boost::[c]ref` with `std::[c]ref`, add inspect checks
- PR #2297⁴⁸¹⁵ - Fixing compilation with no `hw_loc`
- PR #2296⁴⁸¹⁶ - Hpx compute
- PR #2295⁴⁸¹⁷ - Making sure `for_loop(execution::par, 0, N, ...)` is actually executed in parallel
- PR #2294⁴⁸¹⁸ - Throwing exceptions if the runtime is not up and running
- PR #2293⁴⁸¹⁹ - Removing unused parcel port code
- PR #2292⁴⁸²⁰ - Refactor function vttables

⁴⁷⁹⁸ <https://github.com/TheHPXProject/hpx/pull/2318>
⁴⁷⁹⁹ <https://github.com/TheHPXProject/hpx/pull/2317>
⁴⁸⁰⁰ <https://github.com/TheHPXProject/hpx/pull/2316>
⁴⁸⁰¹ <https://github.com/TheHPXProject/hpx/pull/2315>
⁴⁸⁰² <https://github.com/TheHPXProject/hpx/issues/2314>
⁴⁸⁰³ <https://github.com/TheHPXProject/hpx/issues/2313>
⁴⁸⁰⁴ <https://github.com/TheHPXProject/hpx/issues/2312>
⁴⁸⁰⁵ <https://github.com/TheHPXProject/hpx/issues/2311>
⁴⁸⁰⁶ <https://github.com/TheHPXProject/hpx/pull/2310>
⁴⁸⁰⁷ <https://github.com/TheHPXProject/hpx/pull/2309>
⁴⁸⁰⁸ <https://github.com/TheHPXProject/hpx/pull/2306>
⁴⁸⁰⁹ <https://github.com/TheHPXProject/hpx/pull/2305>
⁴⁸¹⁰ <https://github.com/TheHPXProject/hpx/pull/2304>
⁴⁸¹¹ <https://github.com/TheHPXProject/hpx/pull/2303>
⁴⁸¹² <https://github.com/TheHPXProject/hpx/pull/2301>
⁴⁸¹³ <https://github.com/TheHPXProject/hpx/pull/2300>
⁴⁸¹⁴ <https://github.com/TheHPXProject/hpx/pull/2299>
⁴⁸¹⁵ <https://github.com/TheHPXProject/hpx/pull/2297>
⁴⁸¹⁶ <https://github.com/TheHPXProject/hpx/pull/2296>
⁴⁸¹⁷ <https://github.com/TheHPXProject/hpx/pull/2295>
⁴⁸¹⁸ <https://github.com/TheHPXProject/hpx/pull/2294>
⁴⁸¹⁹ <https://github.com/TheHPXProject/hpx/pull/2293>
⁴⁸²⁰ <https://github.com/TheHPXProject/hpx/pull/2292>

- PR #2291⁴⁸²¹ - Fixing 2286
- PR #2290⁴⁸²² - Simplify algorithm overloads
- PR #2289⁴⁸²³ - Adding performance counters reporting parcel related data on a per-action basis
- Issue #2288⁴⁸²⁴ - Remove dormant parcelports
- Issue #2286⁴⁸²⁵ - adjustments to parcel handling to support parcelports that do not need a connection cache
- PR #2285⁴⁸²⁶ - add CMake option to disable package export
- PR #2283⁴⁸²⁷ - Add more inspect checks for use of deprecated components
- Issue #2282⁴⁸²⁸ - Arithmetic exception in executor static chunker
- Issue #2281⁴⁸²⁹ - For loop doesn't parallelize
- PR #2280⁴⁸³⁰ - Fixing 2277: build failure with PAPI
- PR #2279⁴⁸³¹ - Child vs parent stealing
- Issue #2277⁴⁸³² - master branch build failure (53c5b4f) with papi
- PR #2276⁴⁸³³ - Compile time launch policies
- PR #2275⁴⁸³⁴ - Replace boost::chrono with std::chrono in interfaces
- PR #2274⁴⁸³⁵ - Replace most uses of Boost.Assign with initializer list
- PR #2273⁴⁸³⁶ - Fixed typos
- PR #2272⁴⁸³⁷ - Inspect checks
- PR #2270⁴⁸³⁸ - Adding test verifying -lhp.os_threads=all
- PR #2269⁴⁸³⁹ - Added inspect check for now obsolete boost type traits
- PR #2268⁴⁸⁴⁰ - Moving more code into source files
- Issue #2267⁴⁸⁴¹ - Add inspect support to deprecate Boost.TypeTraits
- PR #2265⁴⁸⁴² - Adding channel LCO
- PR #2264⁴⁸⁴³ - Make support for std::ref mandatory

⁴⁸²¹ <https://github.com/TheHPXProject/hpx/pull/2291>

⁴⁸²² <https://github.com/TheHPXProject/hpx/pull/2290>

⁴⁸²³ <https://github.com/TheHPXProject/hpx/pull/2289>

⁴⁸²⁴ <https://github.com/TheHPXProject/hpx/issues/2288>

⁴⁸²⁵ <https://github.com/TheHPXProject/hpx/issues/2286>

⁴⁸²⁶ <https://github.com/TheHPXProject/hpx/pull/2285>

⁴⁸²⁷ <https://github.com/TheHPXProject/hpx/pull/2283>

⁴⁸²⁸ <https://github.com/TheHPXProject/hpx/issues/2282>

⁴⁸²⁹ <https://github.com/TheHPXProject/hpx/issues/2281>

⁴⁸³⁰ <https://github.com/TheHPXProject/hpx/pull/2280>

⁴⁸³¹ <https://github.com/TheHPXProject/hpx/pull/2279>

⁴⁸³² <https://github.com/TheHPXProject/hpx/issues/2277>

⁴⁸³³ <https://github.com/TheHPXProject/hpx/pull/2276>

⁴⁸³⁴ <https://github.com/TheHPXProject/hpx/pull/2275>

⁴⁸³⁵ <https://github.com/TheHPXProject/hpx/pull/2274>

⁴⁸³⁶ <https://github.com/TheHPXProject/hpx/pull/2273>

⁴⁸³⁷ <https://github.com/TheHPXProject/hpx/pull/2272>

⁴⁸³⁸ <https://github.com/TheHPXProject/hpx/pull/2270>

⁴⁸³⁹ <https://github.com/TheHPXProject/hpx/pull/2269>

⁴⁸⁴⁰ <https://github.com/TheHPXProject/hpx/pull/2268>

⁴⁸⁴¹ <https://github.com/TheHPXProject/hpx/issues/2267>

⁴⁸⁴² <https://github.com/TheHPXProject/hpx/pull/2265>

⁴⁸⁴³ <https://github.com/TheHPXProject/hpx/pull/2264>

- PR #2263⁴⁸⁴⁴ - Constrain tuple_member forwarding constructor
- Issue #2262⁴⁸⁴⁵ - Test hpx.os_threads=all
- Issue #2261⁴⁸⁴⁶ - OS X: Error: no matching constructor for initialization of 'hpx::lcos::local::condition_variable_any'
- Issue #2260⁴⁸⁴⁷ - Make support for std::ref mandatory
- PR #2259⁴⁸⁴⁸ - Remove most of Boost.MPL, Boost.EnableIf and Boost.TypeTraits
- PR #2258⁴⁸⁴⁹ - Fixing #2256
- PR #2257⁴⁸⁵⁰ - Fixing launch process
- Issue #2256⁴⁸⁵¹ - Actions are not registered if not invoked
- PR #2255⁴⁸⁵² - Coalescing histogram
- PR #2254⁴⁸⁵³ - Silence explicit initialization in copy-constructor warnings
- PR #2253⁴⁸⁵⁴ - Drop support for GCC 4.6 and 4.7
- PR #2252⁴⁸⁵⁵ - Prepare V1.0
- PR #2251⁴⁸⁵⁶ - Convert to 0.9.99
- PR #2249⁴⁸⁵⁷ - Adding iterator_facade and iterator_adaptor
- Issue #2248⁴⁸⁵⁸ - Need a feature to yield to a new task immediately
- PR #2246⁴⁸⁵⁹ - Adding split_future
- PR #2245⁴⁸⁶⁰ - Add an example for handing over a component instance to a dynamically launched locality
- Issue #2243⁴⁸⁶¹ - Add example demonstrating AGAS symbolic name registration
- Issue #2242⁴⁸⁶² - pkgconfig test broken on CentOS 7 / Boost 1.61
- Issue #2241⁴⁸⁶³ - Compilation error for partitioned vector in hpx_compute branch
- PR #2240⁴⁸⁶⁴ - Fixing termination detection on one locality
- Issue #2239⁴⁸⁶⁵ - Create a new facility lcos::split_all
- Issue #2236⁴⁸⁶⁶ - hpx::cout vs. std::cout

⁴⁸⁴⁴ <https://github.com/TheHPXProject/hpx/pull/2263>

⁴⁸⁴⁵ <https://github.com/TheHPXProject/hpx/issues/2262>

⁴⁸⁴⁶ <https://github.com/TheHPXProject/hpx/issues/2261>

⁴⁸⁴⁷ <https://github.com/TheHPXProject/hpx/issues/2260>

⁴⁸⁴⁸ <https://github.com/TheHPXProject/hpx/pull/2259>

⁴⁸⁴⁹ <https://github.com/TheHPXProject/hpx/pull/2258>

⁴⁸⁵⁰ <https://github.com/TheHPXProject/hpx/pull/2257>

⁴⁸⁵¹ <https://github.com/TheHPXProject/hpx/issues/2256>

⁴⁸⁵² <https://github.com/TheHPXProject/hpx/pull/2255>

⁴⁸⁵³ <https://github.com/TheHPXProject/hpx/pull/2254>

⁴⁸⁵⁴ <https://github.com/TheHPXProject/hpx/pull/2253>

⁴⁸⁵⁵ <https://github.com/TheHPXProject/hpx/pull/2252>

⁴⁸⁵⁶ <https://github.com/TheHPXProject/hpx/pull/2251>

⁴⁸⁵⁷ <https://github.com/TheHPXProject/hpx/pull/2249>

⁴⁸⁵⁸ <https://github.com/TheHPXProject/hpx/issues/2248>

⁴⁸⁵⁹ <https://github.com/TheHPXProject/hpx/pull/2246>

⁴⁸⁶⁰ <https://github.com/TheHPXProject/hpx/pull/2245>

⁴⁸⁶¹ <https://github.com/TheHPXProject/hpx/issues/2243>

⁴⁸⁶² <https://github.com/TheHPXProject/hpx/issues/2242>

⁴⁸⁶³ <https://github.com/TheHPXProject/hpx/issues/2241>

⁴⁸⁶⁴ <https://github.com/TheHPXProject/hpx/pull/2240>

⁴⁸⁶⁵ <https://github.com/TheHPXProject/hpx/issues/2239>

⁴⁸⁶⁶ <https://github.com/TheHPXProject/hpx/issues/2236>

- PR #2232⁴⁸⁶⁷ - Implement local-only primary namespace service
- Issue #2147⁴⁸⁶⁸ - would like to know how much data is being routed by particular actions
- Issue #2109⁴⁸⁶⁹ - Warning while compiling hpx
- Issue #1973⁴⁸⁷⁰ - Setting INTERFACE_COMPILE_OPTIONS for hpx_init in CMake taints Fortran_FLAGS
- Issue #1864⁴⁸⁷¹ - run_guarded using bound function ignores reference
- Issue #1754⁴⁸⁷² - Running with TCP parcelport causes immediate crash or freeze
- Issue #1655⁴⁸⁷³ - Enable zip_iterator to be used with Boost traversal iterator categories
- Issue #1591⁴⁸⁷⁴ - Optimize AGAS for shared memory only operation
- Issue #1401⁴⁸⁷⁵ - Need an efficient infiniband parcelport
- Issue #1125⁴⁸⁷⁶ - Fix the IPC parcelport
- Issue #839⁴⁸⁷⁷ - Refactor ibverbs and shmем parcelport
- Issue #702⁴⁸⁷⁸ - Add instrumentation of parcel layer
- Issue #668⁴⁸⁷⁹ - Implement ispc task interface
- Issue #533⁴⁸⁸⁰ - Thread queue/dequeue internal parameters should be runtime configurable
- Issue #475⁴⁸⁸¹ - Create a means of combining performance counters into querysets

HPX V0.9.99 (Jul 15, 2016)

General changes

As the version number of this release hints, we consider this release to be a preview for the upcoming *HPX* V1.0. All of the functionalities we set out to implement for V1.0 are in place; all of the features we wanted to have exposed are ready. We are very happy with the stability and performance of *HPX* and we would like to present this release to the community in order for us to gather broad feedback before releasing V1.0. We still expect for some minor details to change, but on the whole this release represents what we would like to have in a V1.0.

Overall, since the last release we have had almost 1600 commits while closing almost 400 tickets. These numbers reflect the incredible development activity we have seen over the last couple of months. We would like to express a big ‘Thank you!’ to all contributors and those who helped to make this release happen.

The most notable addition in terms of new functionality available with this release is the full implementation of object migration (i.e. the ability to transparently move *HPX* components to a different compute node). Additionally, this release of *HPX* cleans up many minor issues and some API inconsistencies.

Here are some of the main highlights and changes for this release (in no particular order):

⁴⁸⁶⁷ <https://github.com/TheHPXProject/hpx/pull/2232>
⁴⁸⁶⁸ <https://github.com/TheHPXProject/hpx/issues/2147>
⁴⁸⁶⁹ <https://github.com/TheHPXProject/hpx/issues/2109>
⁴⁸⁷⁰ <https://github.com/TheHPXProject/hpx/issues/1973>
⁴⁸⁷¹ <https://github.com/TheHPXProject/hpx/issues/1864>
⁴⁸⁷² <https://github.com/TheHPXProject/hpx/issues/1754>
⁴⁸⁷³ <https://github.com/TheHPXProject/hpx/issues/1655>
⁴⁸⁷⁴ <https://github.com/TheHPXProject/hpx/issues/1591>
⁴⁸⁷⁵ <https://github.com/TheHPXProject/hpx/issues/1401>
⁴⁸⁷⁶ <https://github.com/TheHPXProject/hpx/issues/1125>
⁴⁸⁷⁷ <https://github.com/TheHPXProject/hpx/issues/839>
⁴⁸⁷⁸ <https://github.com/TheHPXProject/hpx/issues/702>
⁴⁸⁷⁹ <https://github.com/TheHPXProject/hpx/issues/668>
⁴⁸⁸⁰ <https://github.com/TheHPXProject/hpx/issues/533>
⁴⁸⁸¹ <https://github.com/TheHPXProject/hpx/issues/475>

- We have fixed a couple of issues in AGAS and the parcel layer which have caused hangs, segmentation faults at exit, and a slowdown of applications over time. Fixing those has significantly increased the overall stability and performance of distributed runs.
- We have started to add parallel algorithm overloads based on the C++ Extensions for Ranges (N4560⁴⁸⁸²) proposal. This also includes the addition of projections to the existing algorithms. Please see Issue #1668⁴⁸⁸³ for a list of algorithms which have been adapted to N4560⁴⁸⁸⁴.
- We have implemented index-based parallel for-loops based on a corresponding standardization proposal (P0075R1⁴⁸⁸⁵). Please see Issue #2016⁴⁸⁸⁶ for a list of available algorithms.
- We have added implementations for more parallel algorithms as proposed for the upcoming C++ 17 Standard. See Issue #1141⁴⁸⁸⁷ for an overview of which algorithms are available by now.
- We have started to implement a new prototypical functionality with *HPX.Compute* which uniformly exposes some of the higher level APIs to heterogeneous architectures (currently CUDA). This functionality is an early preview and should not be considered stable. It may change considerably in the future.
- We have pervasively added (optional) executor arguments to all API functions which schedule new work. Executors are now used throughout the code base as the main means of executing tasks.
- Added `hpx::make_future<R>(future<T> &&)` allowing to convert a future of any type T into a future of any other type R, either based on default conversion rules of the embedded types or using a given explicit conversion function.
- We finally finished the implementation of transparent migration of components to another locality. It is now possible to trigger a migration operation without ‘stopping the world’ for the object to migrate. *HPX* will make sure that no work is being performed on an object before it is migrated and that all subsequently scheduled work for the migrated object will be transparently forwarded to the new locality. Please note that the global id of the migrated object does not change, thus the application will not have to be changed in any way to support this new functionality. Please note that this feature is currently considered experimental. See Issue #559⁴⁸⁸⁸ and PR #1966⁴⁸⁸⁹ for more details.
- The `hpx::dataflow` facility is now usable with actions. Similarly to `hpx::async`, actions can be specified as an explicit template argument (`hpx::dataflow<Action>(target, ...)`) or as the first argument (`hpx::dataflow(Action(), target, ...)`). We have also enabled the use of distribution policies as the target for dataflow invocations. Please see Issue #1265⁴⁸⁹⁰ and PR #1912⁴⁸⁹¹ for more information.
- Adding overloads of `gather_here` and `gather_there` to accept the plain values of the data to gather (in addition to the existing overloads expecting futures).
- We have cleaned up and refactored large parts of the code base. This helped reducing compile and link times of *HPX* itself and also of applications depending on it. We have further decreased the dependency of *HPX* on the Boost libraries by replacing part of those with facilities available from the standard libraries.
- Wherever possible we have removed dependencies of our API on Boost by replacing those with the equivalent facility from the C++11 standard library.
- We have added new performance counters for parcel coalescing, file-IO, the AGAS cache, and overall scheduler time. Resetting performance counters has been overhauled and fixed.

⁴⁸⁸² <http://wg21.link/n4560>

⁴⁸⁸³ <https://github.com/TheHPXProject/hpx/issues/1668>

⁴⁸⁸⁴ <http://wg21.link/n4560>

⁴⁸⁸⁵ <http://wg21.link/p0075r1>

⁴⁸⁸⁶ <https://github.com/TheHPXProject/hpx/issues/2016>

⁴⁸⁸⁷ <https://github.com/TheHPXProject/hpx/issues/1141>

⁴⁸⁸⁸ <https://github.com/TheHPXProject/hpx/issues/559>

⁴⁸⁸⁹ <https://github.com/TheHPXProject/hpx/pull/1966>

⁴⁸⁹⁰ <https://github.com/TheHPXProject/hpx/issues/1265>

⁴⁸⁹¹ <https://github.com/TheHPXProject/hpx/pull/1912>

- We have introduced a generic client type `hpx::components::client<>` and added support for using it with `hpx::async`. This removes the necessity to implement specific client types for every component type without losing type safety. This deemphasizes the need for using the low level `hpx::id_type` for referencing (possibly remote) component instances. The plan is to deprecate the direct use of `hpx::id_type` in user code in the future.
- We have added a special iterator which supports automatic prefetching of one or more arrays for speeding up loop-like code (see `hpx::parallel::util::make_prefetcher_context()`).
- We have extended the interfaces exposed from executors (as proposed by [N4406](#)⁴⁸⁹²) to accept an arbitrary number of arguments.

Breaking changes

- In order to move the dataflow facility to namespace `hpx` we added a definition of `hpx::dataflow` which might create ambiguities in existing codes. The previous definition of this facility (`hpx::lcos::local::dataflow`) has been deprecated and is available only if the constant `-DHPX_WITH_LOCAL_DATAFLOW_COMPATIBILITY=On` to `CMake`⁴⁸⁹³ is defined at configuration time. Please explicitly qualify all uses of the dataflow facility if you enable this compatibility setting and encounter ambiguities.
- The adaptation of the C++ Extensions for Ranges ([N4560](#)⁴⁸⁹⁴) proposal imposes some breaking changes related to the return types of some of the parallel algorithms. Please see [Issue #1668](#)⁴⁸⁹⁵ for a list of algorithms which have already been adapted.
- The facility `hpx::lcos::make_future_void()` has been replaced by `hpx::make_future<void>()`.
- We have removed support for Intel V13 and gcc 4.4.x.
- We have removed (default) support for the generic `hpx::parallel::execution_policy` because it was removed from the Parallelism TS (`__cpp11_n4104__`) while it was being added to the upcoming C++17 Standard. This facility can be still enabled at configure time by specifying `-DHPX_WITH_GENERIC_EXECUTION_POLICY=On` to `CMake`.
- Uses of `boost::shared_ptr` and related facilities have been replaced with `std::shared_ptr` and friends. Uses of `boost::unique_lock`, `boost::lock_guard` etc. have also been replaced by the equivalent (and equally named) tools available from the C++11 standard library.
- Facilities that used to expect an explicit `boost::unique_lock` now take an `std::unique_lock`. Additionally, `condition_variable` no longer aliases `condition_variable_any`; its interface now only works with `std::unique_lock<local::mutex>`.
- Uses of `boost::function`, `boost::bind`, `boost::tuple` have been replaced by the corresponding facilities in *HPX* (`hpx::util::function`, `hpx::util::bind`, and `hpx::util::tuple`, respectively).

⁴⁸⁹² <http://wg21.link/n4406>

⁴⁸⁹³ <https://www.cmake.org>

⁴⁸⁹⁴ <http://wg21.link/n4560>

⁴⁸⁹⁵ <https://github.com/TheHPXProject/hpx/issues/1668>

Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this release.

- [PR #2250⁴⁸⁹⁶](#) - change default chunker of parallel executor to static one
- [PR #2247⁴⁸⁹⁷](#) - HPX on ppc64le
- [PR #2244⁴⁸⁹⁸](#) - Fixing MSVC problems
- [PR #2238⁴⁸⁹⁹](#) - Fixing small typos
- [PR #2237⁴⁹⁰⁰](#) - Fixing small typos
- [PR #2234⁴⁹⁰¹](#) - Fix broken add test macro when extra args are passed in
- [PR #2231⁴⁹⁰²](#) - Fixing possible race during future awaiting in serialization
- [PR #2230⁴⁹⁰³](#) - Fix stream nvcc
- [PR #2229⁴⁹⁰⁴](#) - Fixed run_as_hpx_thread
- [PR #2228⁴⁹⁰⁵](#) - On prefetching_test branch : adding prefetching_iterator and related tests used for prefetching containers within lambda functions
- [PR #2227⁴⁹⁰⁶](#) - Support for HPXCL's `opencil::event`
- [PR #2226⁴⁹⁰⁷](#) - Preparing for release of V0.9.99
- [PR #2225⁴⁹⁰⁸](#) - fix issue when compiling components with `hpxcxx`
- [PR #2224⁴⁹⁰⁹](#) - Compute alloc fix
- [PR #2223⁴⁹¹⁰](#) - Simplify promise
- [PR #2222⁴⁹¹¹](#) - Replace last uses of `boost::function` by `util::function_nonsr`
- [PR #2221⁴⁹¹²](#) - Fix config tests
- [PR #2220⁴⁹¹³](#) - Fixing gcc 4.6 compilation issues
- [PR #2219⁴⁹¹⁴](#) - nullptr support for `[unique_]function`
- [PR #2218⁴⁹¹⁵](#) - Introducing clang tidy
- [PR #2216⁴⁹¹⁶](#) - Replace NULL with nullptr

⁴⁸⁹⁶ <https://github.com/TheHPXProject/hpx/pull/2250>

⁴⁸⁹⁷ <https://github.com/TheHPXProject/hpx/pull/2247>

⁴⁸⁹⁸ <https://github.com/TheHPXProject/hpx/pull/2244>

⁴⁸⁹⁹ <https://github.com/TheHPXProject/hpx/pull/2238>

⁴⁹⁰⁰ <https://github.com/TheHPXProject/hpx/pull/2237>

⁴⁹⁰¹ <https://github.com/TheHPXProject/hpx/pull/2234>

⁴⁹⁰² <https://github.com/TheHPXProject/hpx/pull/2231>

⁴⁹⁰³ <https://github.com/TheHPXProject/hpx/pull/2230>

⁴⁹⁰⁴ <https://github.com/TheHPXProject/hpx/pull/2229>

⁴⁹⁰⁵ <https://github.com/TheHPXProject/hpx/pull/2228>

⁴⁹⁰⁶ <https://github.com/TheHPXProject/hpx/pull/2227>

⁴⁹⁰⁷ <https://github.com/TheHPXProject/hpx/pull/2226>

⁴⁹⁰⁸ <https://github.com/TheHPXProject/hpx/pull/2225>

⁴⁹⁰⁹ <https://github.com/TheHPXProject/hpx/pull/2224>

⁴⁹¹⁰ <https://github.com/TheHPXProject/hpx/pull/2223>

⁴⁹¹¹ <https://github.com/TheHPXProject/hpx/pull/2222>

⁴⁹¹² <https://github.com/TheHPXProject/hpx/pull/2221>

⁴⁹¹³ <https://github.com/TheHPXProject/hpx/pull/2220>

⁴⁹¹⁴ <https://github.com/TheHPXProject/hpx/pull/2219>

⁴⁹¹⁵ <https://github.com/TheHPXProject/hpx/pull/2218>

⁴⁹¹⁶ <https://github.com/TheHPXProject/hpx/pull/2216>

- Issue #2214⁴⁹¹⁷ - Let inspect flag use of NULL, suggest nullptr instead
- PR #2213⁴⁹¹⁸ - Require support for nullptr
- PR #2212⁴⁹¹⁹ - Properly find jemalloc through pkg-config
- PR #2211⁴⁹²⁰ - Disable a couple of warnings reported by Intel on Windows
- PR #2210⁴⁹²¹ - Fixed host::block_allocator::bulk_construct
- PR #2209⁴⁹²² - Started to clean up new sort algorithms, made things compile for sort_by_key
- PR #2208⁴⁹²³ - A couple of fixes that were exposed by a new sort algorithm
- PR #2207⁴⁹²⁴ - Adding missing includes in /hpx/include/serialization.hpp
- PR #2206⁴⁹²⁵ - Call package_action::get_future before package_action::apply
- PR #2205⁴⁹²⁶ - The indirect_packaged_task::operator() needs to be run on a HPX thread
- PR #2204⁴⁹²⁷ - Variadic executor parameters
- PR #2203⁴⁹²⁸ - Delay-initialize members of partitioned iterator
- PR #2202⁴⁹²⁹ - Added segmented fill for hpx::vector
- Issue #2201⁴⁹³⁰ - Null Thread id encountered on partitioned_vector
- PR #2200⁴⁹³¹ - Fix hangs
- PR #2199⁴⁹³² - Deprecating hpx/traits.hpp
- PR #2198⁴⁹³³ - Making explicit inclusion of external libraries into build
- PR #2197⁴⁹³⁴ - Fix typo in QT CMakeLists
- PR #2196⁴⁹³⁵ - Fixing a gcc warning about attributes being ignored
- PR #2194⁴⁹³⁶ - Fixing partitioned_vector_spmf_foreach example
- Issue #2193⁴⁹³⁷ - partitioned_vector_spmf_foreach seg faults
- PR #2192⁴⁹³⁸ - Support Boost.Thread v4
- PR #2191⁴⁹³⁹ - HPX.Compute prototype

⁴⁹¹⁷ <https://github.com/TheHPXProject/hpx/issues/2214>

⁴⁹¹⁸ <https://github.com/TheHPXProject/hpx/pull/2213>

⁴⁹¹⁹ <https://github.com/TheHPXProject/hpx/pull/2212>

⁴⁹²⁰ <https://github.com/TheHPXProject/hpx/pull/2211>

⁴⁹²¹ <https://github.com/TheHPXProject/hpx/pull/2210>

⁴⁹²² <https://github.com/TheHPXProject/hpx/pull/2209>

⁴⁹²³ <https://github.com/TheHPXProject/hpx/pull/2208>

⁴⁹²⁴ <https://github.com/TheHPXProject/hpx/pull/2207>

⁴⁹²⁵ <https://github.com/TheHPXProject/hpx/pull/2206>

⁴⁹²⁶ <https://github.com/TheHPXProject/hpx/pull/2205>

⁴⁹²⁷ <https://github.com/TheHPXProject/hpx/pull/2204>

⁴⁹²⁸ <https://github.com/TheHPXProject/hpx/pull/2203>

⁴⁹²⁹ <https://github.com/TheHPXProject/hpx/pull/2202>

⁴⁹³⁰ <https://github.com/TheHPXProject/hpx/issues/2201>

⁴⁹³¹ <https://github.com/TheHPXProject/hpx/pull/2200>

⁴⁹³² <https://github.com/TheHPXProject/hpx/pull/2199>

⁴⁹³³ <https://github.com/TheHPXProject/hpx/pull/2198>

⁴⁹³⁴ <https://github.com/TheHPXProject/hpx/pull/2197>

⁴⁹³⁵ <https://github.com/TheHPXProject/hpx/pull/2196>

⁴⁹³⁶ <https://github.com/TheHPXProject/hpx/pull/2194>

⁴⁹³⁷ <https://github.com/TheHPXProject/hpx/issues/2193>

⁴⁹³⁸ <https://github.com/TheHPXProject/hpx/pull/2192>

⁴⁹³⁹ <https://github.com/TheHPXProject/hpx/pull/2191>

- PR #2190⁴⁹⁴⁰ - Spawning operation on new thread if remaining stack space becomes too small
- PR #2189⁴⁹⁴¹ - Adding callback taking index and future to when_each
- PR #2188⁴⁹⁴² - Adding new example demonstrating receive_buffer
- PR #2187⁴⁹⁴³ - Mask 128-bit ints if CUDA is being used
- PR #2186⁴⁹⁴⁴ - Make startup & shutdown functions unique_function
- PR #2185⁴⁹⁴⁵ - Fixing logging output not to cause hang on shutdown
- PR #2184⁴⁹⁴⁶ - Allowing component clients as action return types
- Issue #2183⁴⁹⁴⁷ - Enabling logging output causes hang on shutdown
- Issue #2182⁴⁹⁴⁸ - 1d_stencil seg fault
- Issue #2181⁴⁹⁴⁹ - Setting small stack size does not change default
- PR #2180⁴⁹⁵⁰ - Changing default bind mode to balanced
- PR #2179⁴⁹⁵¹ - adding prefetching_iterator and related tests used for prefetching containers within lambda functions
- PR #2177⁴⁹⁵² - Fixing 2176
- Issue #2176⁴⁹⁵³ - Launch process test fails on OSX
- PR #2175⁴⁹⁵⁴ - Fix unbalanced config/warnings includes, add some new ones
- PR #2174⁴⁹⁵⁵ - Fix test categorization : regression not unit
- Issue #2172⁴⁹⁵⁶ - Different performance results
- Issue #2171⁴⁹⁵⁷ - “negative entry in reference count table” running octotiger on 32 nodes on queenbee
- Issue #2170⁴⁹⁵⁸ - Error while compiling on Mac + boost 1.60
- PR #2168⁴⁹⁵⁹ - Fixing problems with is_bitwise_serializable
- Issue #2167⁴⁹⁶⁰ - startup & shutdown function should accept unique_function
- Issue #2166⁴⁹⁶¹ - Simple receive_buffer example
- PR #2165⁴⁹⁶² - Fix wait all

⁴⁹⁴⁰ <https://github.com/TheHPXProject/hpx/pull/2190>

⁴⁹⁴¹ <https://github.com/TheHPXProject/hpx/pull/2189>

⁴⁹⁴² <https://github.com/TheHPXProject/hpx/pull/2188>

⁴⁹⁴³ <https://github.com/TheHPXProject/hpx/pull/2187>

⁴⁹⁴⁴ <https://github.com/TheHPXProject/hpx/pull/2186>

⁴⁹⁴⁵ <https://github.com/TheHPXProject/hpx/pull/2185>

⁴⁹⁴⁶ <https://github.com/TheHPXProject/hpx/pull/2184>

⁴⁹⁴⁷ <https://github.com/TheHPXProject/hpx/issues/2183>

⁴⁹⁴⁸ <https://github.com/TheHPXProject/hpx/issues/2182>

⁴⁹⁴⁹ <https://github.com/TheHPXProject/hpx/issues/2181>

⁴⁹⁵⁰ <https://github.com/TheHPXProject/hpx/pull/2180>

⁴⁹⁵¹ <https://github.com/TheHPXProject/hpx/pull/2179>

⁴⁹⁵² <https://github.com/TheHPXProject/hpx/pull/2177>

⁴⁹⁵³ <https://github.com/TheHPXProject/hpx/issues/2176>

⁴⁹⁵⁴ <https://github.com/TheHPXProject/hpx/pull/2175>

⁴⁹⁵⁵ <https://github.com/TheHPXProject/hpx/pull/2174>

⁴⁹⁵⁶ <https://github.com/TheHPXProject/hpx/issues/2172>

⁴⁹⁵⁷ <https://github.com/TheHPXProject/hpx/issues/2171>

⁴⁹⁵⁸ <https://github.com/TheHPXProject/hpx/issues/2170>

⁴⁹⁵⁹ <https://github.com/TheHPXProject/hpx/pull/2168>

⁴⁹⁶⁰ <https://github.com/TheHPXProject/hpx/issues/2167>

⁴⁹⁶¹ <https://github.com/TheHPXProject/hpx/issues/2166>

⁴⁹⁶² <https://github.com/TheHPXProject/hpx/pull/2165>

- PR #2164⁴⁹⁶³ - Fix wait all
- PR #2163⁴⁹⁶⁴ - Fix some typos in config tests
- PR #2162⁴⁹⁶⁵ - Improve #includes
- PR #2160⁴⁹⁶⁶ - Add inspect check for missing #include <list>
- PR #2159⁴⁹⁶⁷ - Add missing finalize call to stop test hanging
- PR #2158⁴⁹⁶⁸ - Algo fixes
- PR #2157⁴⁹⁶⁹ - Stack check
- Issue #2156⁴⁹⁷⁰ - OSX reports stack space incorrectly (generic context coroutines)
- Issue #2155⁴⁹⁷¹ - Race condition suspected in runtime
- PR #2154⁴⁹⁷² - Replace boost::detail::atomic_count with the new util::atomic_count
- PR #2153⁴⁹⁷³ - Fix stack overflow on OSX
- PR #2152⁴⁹⁷⁴ - Define is_bitwise_serializable as is_trivially_copyable when available
- PR #2151⁴⁹⁷⁵ - Adding missing <cstring> for std::mem* functions
- Issue #2150⁴⁹⁷⁶ - Unable to use component clients as action return types
- PR #2149⁴⁹⁷⁷ - std::memmove copies bytes, use bytes*sizeof(type) when copying larger types
- PR #2146⁴⁹⁷⁸ - Adding customization point for parallel copy/move
- PR #2145⁴⁹⁷⁹ - Applying changes to address warnings issued by latest version of PVS Studio
- Issue #2148⁴⁹⁸⁰ - hpx::parallel::copy is broken after trivially copyable changes
- PR #2144⁴⁹⁸¹ - Some minor tweaks to compute prototype
- PR #2143⁴⁹⁸² - Added Boost version support information over OSX platform
- PR #2142⁴⁹⁸³ - Fixing memory leak in example
- PR #2141⁴⁹⁸⁴ - Add missing specializations in execution policies
- PR #2139⁴⁹⁸⁵ - This PR fixes a few problems reported by Clang's Undefined Behavior sanitizer

⁴⁹⁶³ <https://github.com/TheHPXProject/hpx/pull/2164>

⁴⁹⁶⁴ <https://github.com/TheHPXProject/hpx/pull/2163>

⁴⁹⁶⁵ <https://github.com/TheHPXProject/hpx/pull/2162>

⁴⁹⁶⁶ <https://github.com/TheHPXProject/hpx/pull/2160>

⁴⁹⁶⁷ <https://github.com/TheHPXProject/hpx/pull/2159>

⁴⁹⁶⁸ <https://github.com/TheHPXProject/hpx/pull/2158>

⁴⁹⁶⁹ <https://github.com/TheHPXProject/hpx/pull/2157>

⁴⁹⁷⁰ <https://github.com/TheHPXProject/hpx/issues/2156>

⁴⁹⁷¹ <https://github.com/TheHPXProject/hpx/issues/2155>

⁴⁹⁷² <https://github.com/TheHPXProject/hpx/pull/2154>

⁴⁹⁷³ <https://github.com/TheHPXProject/hpx/pull/2153>

⁴⁹⁷⁴ <https://github.com/TheHPXProject/hpx/pull/2152>

⁴⁹⁷⁵ <https://github.com/TheHPXProject/hpx/pull/2151>

⁴⁹⁷⁶ <https://github.com/TheHPXProject/hpx/issues/2150>

⁴⁹⁷⁷ <https://github.com/TheHPXProject/hpx/pull/2149>

⁴⁹⁷⁸ <https://github.com/TheHPXProject/hpx/pull/2146>

⁴⁹⁷⁹ <https://github.com/TheHPXProject/hpx/pull/2145>

⁴⁹⁸⁰ <https://github.com/TheHPXProject/hpx/issues/2148>

⁴⁹⁸¹ <https://github.com/TheHPXProject/hpx/pull/2144>

⁴⁹⁸² <https://github.com/TheHPXProject/hpx/pull/2143>

⁴⁹⁸³ <https://github.com/TheHPXProject/hpx/pull/2142>

⁴⁹⁸⁴ <https://github.com/TheHPXProject/hpx/pull/2141>

⁴⁹⁸⁵ <https://github.com/TheHPXProject/hpx/pull/2139>

- PR #2138⁴⁹⁸⁶ - Revert “Adding fedora docs”
- PR #2136⁴⁹⁸⁷ - Removed double semicolon
- PR #2135⁴⁹⁸⁸ - Add deprecated #include check for hpx_fwd.hpp
- PR #2134⁴⁹⁸⁹ - Resolved memory leak in stencil_8
- PR #2133⁴⁹⁹⁰ - Replace uses of boost pointer containers
- PR #2132⁴⁹⁹¹ - Removing unused typedef
- PR #2131⁴⁹⁹² - Add several include checks for std facilities
- PR #2130⁴⁹⁹³ - Fixing parcel compression, adding test
- PR #2129⁴⁹⁹⁴ - Fix invalid attribute warnings
- Issue #2128⁴⁹⁹⁵ - hpx::init seems to segfault
- PR #2127⁴⁹⁹⁶ - Making executor_traits N-nary
- PR #2126⁴⁹⁹⁷ - GCC 4.6 fails to deduce the correct type in lambda
- PR #2125⁴⁹⁹⁸ - Making parcel coalescing test actually test something
- Issue #2124⁴⁹⁹⁹ - Make a testcase for parcel compression
- Issue #2123⁵⁰⁰⁰ - hpx/hpx/runtime/applier_fwd.hpp - Multiple defined types
- Issue #2122⁵⁰⁰¹ - Exception in primary_namespace::resolve_free_list
- Issue #2121⁵⁰⁰² - Possible memory leak in 1d_stencil_8
- PR #2120⁵⁰⁰³ - Fixing 2119
- Issue #2119⁵⁰⁰⁴ - reduce_by_key compilation problems
- Issue #2118⁵⁰⁰⁵ - Premature unwrapping of boost::ref'ed arguments
- PR #2117⁵⁰⁰⁶ - Added missing initializer on last constructor for thread_description
- PR #2116⁵⁰⁰⁷ - Use a lightweight bind implementation when no placeholders are given
- PR #2115⁵⁰⁰⁸ - Replace boost::shared_ptr with std::shared_ptr

⁴⁹⁸⁶ <https://github.com/TheHPXProject/hpx/pull/2138>
⁴⁹⁸⁷ <https://github.com/TheHPXProject/hpx/pull/2136>
⁴⁹⁸⁸ <https://github.com/TheHPXProject/hpx/pull/2135>
⁴⁹⁸⁹ <https://github.com/TheHPXProject/hpx/pull/2134>
⁴⁹⁹⁰ <https://github.com/TheHPXProject/hpx/pull/2133>
⁴⁹⁹¹ <https://github.com/TheHPXProject/hpx/pull/2132>
⁴⁹⁹² <https://github.com/TheHPXProject/hpx/pull/2131>
⁴⁹⁹³ <https://github.com/TheHPXProject/hpx/pull/2130>
⁴⁹⁹⁴ <https://github.com/TheHPXProject/hpx/pull/2129>
⁴⁹⁹⁵ <https://github.com/TheHPXProject/hpx/issues/2128>
⁴⁹⁹⁶ <https://github.com/TheHPXProject/hpx/pull/2127>
⁴⁹⁹⁷ <https://github.com/TheHPXProject/hpx/pull/2126>
⁴⁹⁹⁸ <https://github.com/TheHPXProject/hpx/pull/2125>
⁴⁹⁹⁹ <https://github.com/TheHPXProject/hpx/issues/2124>
⁵⁰⁰⁰ <https://github.com/TheHPXProject/hpx/issues/2123>
⁵⁰⁰¹ <https://github.com/TheHPXProject/hpx/issues/2122>
⁵⁰⁰² <https://github.com/TheHPXProject/hpx/issues/2121>
⁵⁰⁰³ <https://github.com/TheHPXProject/hpx/pull/2120>
⁵⁰⁰⁴ <https://github.com/TheHPXProject/hpx/issues/2119>
⁵⁰⁰⁵ <https://github.com/TheHPXProject/hpx/issues/2118>
⁵⁰⁰⁶ <https://github.com/TheHPXProject/hpx/pull/2117>
⁵⁰⁰⁷ <https://github.com/TheHPXProject/hpx/pull/2116>
⁵⁰⁰⁸ <https://github.com/TheHPXProject/hpx/pull/2115>

- PR #2114⁵⁰⁰⁹ - Adding hook functions for `executor_parameter_traits` supporting timers
- Issue #2113⁵⁰¹⁰ - Compilation error with gcc version 4.9.3 (MacPorts gcc49 4.9.3_0)
- PR #2112⁵⁰¹¹ - Replace uses of `safe_bool` with explicit operator `bool`
- Issue #2111⁵⁰¹² - Compilation error on QT example
- Issue #2110⁵⁰¹³ - Compilation error when passing non-future argument to unwrapped continuation in dataflow
- Issue #2109⁵⁰¹⁴ - Warning while compiling `hpx`
- Issue #2109⁵⁰¹⁵ - Stack trace of last bug causing issues with `octotiger`
- Issue #2108⁵⁰¹⁶ - Stack trace of last bug causing issues with `octotiger`
- PR #2107⁵⁰¹⁷ - Making sure that a missing `parcel_coalescing` module does not cause startup exceptions
- PR #2106⁵⁰¹⁸ - Stop using `hpx_fwd.hpp`
- Issue #2105⁵⁰¹⁹ - coalescing plugin handler is not optional any more
- Issue #2104⁵⁰²⁰ - Make `executor_traits` N-nary
- Issue #2103⁵⁰²¹ - Build error with `octotiger` and `hpx` commit e657426d
- PR #2102⁵⁰²² - Combining thread data storage
- PR #2101⁵⁰²³ - Added repartition version of 1d stencil that uses any performance counter
- PR #2100⁵⁰²⁴ - Drop obsolete TR1 `result_of` protocol
- PR #2099⁵⁰²⁵ - Replace uses of `boost::bind` with `util::bind`
- PR #2098⁵⁰²⁶ - Deprecated inspect checks
- PR #2097⁵⁰²⁷ - Reduce by key, extends #1141
- PR #2096⁵⁰²⁸ - Moving local cache from external to `hpx/util`
- PR #2095⁵⁰²⁹ - Bump minimum required Boost to 1.50.0
- PR #2094⁵⁰³⁰ - Add include checks for several Boost utilities
- Issue #2093⁵⁰³¹ - `./.../local_cache.hpp(89): error #303: explicit type is missing ("int" assumed)`

⁵⁰⁰⁹ <https://github.com/TheHPXProject/hpx/pull/2114>

⁵⁰¹⁰ <https://github.com/TheHPXProject/hpx/issues/2113>

⁵⁰¹¹ <https://github.com/TheHPXProject/hpx/pull/2112>

⁵⁰¹² <https://github.com/TheHPXProject/hpx/issues/2111>

⁵⁰¹³ <https://github.com/TheHPXProject/hpx/issues/2110>

⁵⁰¹⁴ <https://github.com/TheHPXProject/hpx/issues/2109>

⁵⁰¹⁵ <https://github.com/TheHPXProject/hpx/issues/2109>

⁵⁰¹⁶ <https://github.com/TheHPXProject/hpx/issues/2108>

⁵⁰¹⁷ <https://github.com/TheHPXProject/hpx/pull/2107>

⁵⁰¹⁸ <https://github.com/TheHPXProject/hpx/pull/2106>

⁵⁰¹⁹ <https://github.com/TheHPXProject/hpx/issues/2105>

⁵⁰²⁰ <https://github.com/TheHPXProject/hpx/issues/2104>

⁵⁰²¹ <https://github.com/TheHPXProject/hpx/issues/2103>

⁵⁰²² <https://github.com/TheHPXProject/hpx/pull/2102>

⁵⁰²³ <https://github.com/TheHPXProject/hpx/pull/2101>

⁵⁰²⁴ <https://github.com/TheHPXProject/hpx/pull/2100>

⁵⁰²⁵ <https://github.com/TheHPXProject/hpx/pull/2099>

⁵⁰²⁶ <https://github.com/TheHPXProject/hpx/pull/2098>

⁵⁰²⁷ <https://github.com/TheHPXProject/hpx/pull/2097>

⁵⁰²⁸ <https://github.com/TheHPXProject/hpx/pull/2096>

⁵⁰²⁹ <https://github.com/TheHPXProject/hpx/pull/2095>

⁵⁰³⁰ <https://github.com/TheHPXProject/hpx/pull/2094>

⁵⁰³¹ <https://github.com/TheHPXProject/hpx/issues/2093>

- PR #2091⁵⁰³² - Fix for Raspberry pi build
- PR #2090⁵⁰³³ - Fix storage size for util::function<>
- PR #2089⁵⁰³⁴ - Fix #2088
- Issue #2088⁵⁰³⁵ - More verbose output from cmake configuration
- PR #2087⁵⁰³⁶ - Making sure init_globally always executes hpx_main
- Issue #2086⁵⁰³⁷ - Race condition with recent HPX
- PR #2085⁵⁰³⁸ - Adding #include checker
- PR #2084⁵⁰³⁹ - Replace boost lock types with standard library ones
- PR #2083⁵⁰⁴⁰ - Simplify packaged task
- PR #2082⁵⁰⁴¹ - Updating APEX version for testing
- PR #2081⁵⁰⁴² - Cleanup exception headers
- PR #2080⁵⁰⁴³ - Make call_once variadic
- Issue #2079⁵⁰⁴⁴ - With GNU C++, line 85 of hpx/config/version.hpp causes link failure when linking application
- Issue #2078⁵⁰⁴⁵ - Simple test fails with _GLIBCXX_DEBUG defined
- PR #2077⁵⁰⁴⁶ - Instantiate board in nqueen client
- PR #2076⁵⁰⁴⁷ - Moving coalescing registration to TUs
- PR #2075⁵⁰⁴⁸ - Fixed some documentation typos
- PR #2074⁵⁰⁴⁹ - Adding flush-mode to message handler flush
- PR #2073⁵⁰⁵⁰ - Fixing performance regression introduced lately
- PR #2072⁵⁰⁵¹ - Refactor local::condition_variable
- PR #2071⁵⁰⁵² - Timer based on boost::asio::deadline_timer
- PR #2070⁵⁰⁵³ - Refactor tuple based functionality
- PR #2069⁵⁰⁵⁴ - Fixed typos

⁵⁰³² <https://github.com/TheHPXProject/hpx/pull/2091>

⁵⁰³³ <https://github.com/TheHPXProject/hpx/pull/2090>

⁵⁰³⁴ <https://github.com/TheHPXProject/hpx/pull/2089>

⁵⁰³⁵ <https://github.com/TheHPXProject/hpx/issues/2088>

⁵⁰³⁶ <https://github.com/TheHPXProject/hpx/pull/2087>

⁵⁰³⁷ <https://github.com/TheHPXProject/hpx/issues/2086>

⁵⁰³⁸ <https://github.com/TheHPXProject/hpx/pull/2085>

⁵⁰³⁹ <https://github.com/TheHPXProject/hpx/pull/2084>

⁵⁰⁴⁰ <https://github.com/TheHPXProject/hpx/pull/2083>

⁵⁰⁴¹ <https://github.com/TheHPXProject/hpx/pull/2082>

⁵⁰⁴² <https://github.com/TheHPXProject/hpx/pull/2081>

⁵⁰⁴³ <https://github.com/TheHPXProject/hpx/pull/2080>

⁵⁰⁴⁴ <https://github.com/TheHPXProject/hpx/issues/2079>

⁵⁰⁴⁵ <https://github.com/TheHPXProject/hpx/issues/2078>

⁵⁰⁴⁶ <https://github.com/TheHPXProject/hpx/pull/2077>

⁵⁰⁴⁷ <https://github.com/TheHPXProject/hpx/pull/2076>

⁵⁰⁴⁸ <https://github.com/TheHPXProject/hpx/pull/2075>

⁵⁰⁴⁹ <https://github.com/TheHPXProject/hpx/pull/2074>

⁵⁰⁵⁰ <https://github.com/TheHPXProject/hpx/pull/2073>

⁵⁰⁵¹ <https://github.com/TheHPXProject/hpx/pull/2072>

⁵⁰⁵² <https://github.com/TheHPXProject/hpx/pull/2071>

⁵⁰⁵³ <https://github.com/TheHPXProject/hpx/pull/2070>

⁵⁰⁵⁴ <https://github.com/TheHPXProject/hpx/pull/2069>

- Issue #2068⁵⁰⁵⁵ - Seg fault with octotiger
- PR #2067⁵⁰⁵⁶ - Algorithm cleanup
- PR #2066⁵⁰⁵⁷ - Split credit fixes
- PR #2065⁵⁰⁵⁸ - Rename HPX_MOVABLE_BUT_NOT_COPYABLE to HPX_MOVABLE_ONLY
- PR #2064⁵⁰⁵⁹ - Fixed some typos in docs
- PR #2063⁵⁰⁶⁰ - Adding example demonstrating template components
- Issue #2062⁵⁰⁶¹ - Support component templates
- PR #2061⁵⁰⁶² - Replace some uses of lexical_cast<string> with C++11 std::to_string
- PR #2060⁵⁰⁶³ - Replace uses of boost::noncopyable with HPX_NON_COPYABLE
- PR #2059⁵⁰⁶⁴ - Adding missing for_loop algorithms
- PR #2058⁵⁰⁶⁵ - Move several definitions to more appropriate headers
- PR #2057⁵⁰⁶⁶ - Simplify assert_owns_lock and ignore_while_checking
- PR #2056⁵⁰⁶⁷ - Replacing std::result_of with util::result_of
- PR #2055⁵⁰⁶⁸ - Fix process launching/connecting back
- PR #2054⁵⁰⁶⁹ - Add a forwarding coroutine header
- PR #2053⁵⁰⁷⁰ - Replace uses of boost::unordered_map with std::unordered_map
- PR #2052⁵⁰⁷¹ - Rewrite tuple unwrap
- PR #2050⁵⁰⁷² - Replace uses of BOOST_SCOPED_ENUM with C++11 scoped enums
- PR #2049⁵⁰⁷³ - Attempt to narrow down split_credit problem
- PR #2048⁵⁰⁷⁴ - Fixing gcc startup hangs
- PR #2047⁵⁰⁷⁵ - Fixing when_xxx and wait_xxx for MSVC12
- PR #2046⁵⁰⁷⁶ - adding persistent_auto_chunk_size and related tests for for_each
- PR #2045⁵⁰⁷⁷ - Fixing HPX_HAVE_THREAD_BACKTRACE_DEPTH build time configuration

⁵⁰⁵⁵ <https://github.com/TheHPXProject/hpx/issues/2068>

⁵⁰⁵⁶ <https://github.com/TheHPXProject/hpx/pull/2067>

⁵⁰⁵⁷ <https://github.com/TheHPXProject/hpx/pull/2066>

⁵⁰⁵⁸ <https://github.com/TheHPXProject/hpx/pull/2065>

⁵⁰⁵⁹ <https://github.com/TheHPXProject/hpx/pull/2064>

⁵⁰⁶⁰ <https://github.com/TheHPXProject/hpx/pull/2063>

⁵⁰⁶¹ <https://github.com/TheHPXProject/hpx/issues/2062>

⁵⁰⁶² <https://github.com/TheHPXProject/hpx/pull/2061>

⁵⁰⁶³ <https://github.com/TheHPXProject/hpx/pull/2060>

⁵⁰⁶⁴ <https://github.com/TheHPXProject/hpx/pull/2059>

⁵⁰⁶⁵ <https://github.com/TheHPXProject/hpx/pull/2058>

⁵⁰⁶⁶ <https://github.com/TheHPXProject/hpx/pull/2057>

⁵⁰⁶⁷ <https://github.com/TheHPXProject/hpx/pull/2056>

⁵⁰⁶⁸ <https://github.com/TheHPXProject/hpx/pull/2055>

⁵⁰⁶⁹ <https://github.com/TheHPXProject/hpx/pull/2054>

⁵⁰⁷⁰ <https://github.com/TheHPXProject/hpx/pull/2053>

⁵⁰⁷¹ <https://github.com/TheHPXProject/hpx/pull/2052>

⁵⁰⁷² <https://github.com/TheHPXProject/hpx/pull/2050>

⁵⁰⁷³ <https://github.com/TheHPXProject/hpx/pull/2049>

⁵⁰⁷⁴ <https://github.com/TheHPXProject/hpx/pull/2048>

⁵⁰⁷⁵ <https://github.com/TheHPXProject/hpx/pull/2047>

⁵⁰⁷⁶ <https://github.com/TheHPXProject/hpx/pull/2046>

⁵⁰⁷⁷ <https://github.com/TheHPXProject/hpx/pull/2045>

- PR #2044⁵⁰⁷⁸ - Adding missing service executor types
- PR #2043⁵⁰⁷⁹ - Removing ambiguous definitions for `is_future_range` and `future_range_traits`
- PR #2042⁵⁰⁸⁰ - Clarify that HPX builds can use (much) more than 2GB per process
- PR #2041⁵⁰⁸¹ - Changing `future_iterator_traits` to support pointers
- Issue #2040⁵⁰⁸² - Improve documentation memory usage warning?
- PR #2039⁵⁰⁸³ - Coroutine cleanup
- PR #2038⁵⁰⁸⁴ - Fix cmake policy CMP0042 warning MACOSX_RPATH
- PR #2037⁵⁰⁸⁵ - Avoid redundant specialization of `[unique_]function_nonser`
- PR #2036⁵⁰⁸⁶ - nvcc dies with an internal error upon pushing/popping warnings inside templates
- Issue #2035⁵⁰⁸⁷ - Use a less restrictive iterator definition in `hpx::lcos::detail::future_iterator_traits`
- PR #2034⁵⁰⁸⁸ - Fixing compilation error with thread queue wait time performance counter
- Issue #2033⁵⁰⁸⁹ - Compilation error when compiling with thread queue waittime performance counter
- Issue #2032⁵⁰⁹⁰ - Ambiguous template instantiation for `is_future_range` and `future_range_traits`.
- PR #2031⁵⁰⁹¹ - Don't restart timer on every incoming parcel
- PR #2030⁵⁰⁹² - Unify handling of execution policies in parallel algorithms
- PR #2029⁵⁰⁹³ - Make pkg-config .pc files use .dylib on OSX
- PR #2028⁵⁰⁹⁴ - Adding process component
- PR #2027⁵⁰⁹⁵ - Making check for compiler compatibility independent on compiler path
- PR #2025⁵⁰⁹⁶ - Fixing inspect tool
- PR #2024⁵⁰⁹⁷ - Intel13 removal
- PR #2023⁵⁰⁹⁸ - Fix errors related to older boost versions and parameter pack expansions in lambdas
- Issue #2022⁵⁰⁹⁹ - gmake fail: "No rule to make target /usr/lib46/libboost_context-mt.so"
- PR #2021⁵¹⁰⁰ - Added Sudoku example

⁵⁰⁷⁸ <https://github.com/TheHPXProject/hpx/pull/2044>

⁵⁰⁷⁹ <https://github.com/TheHPXProject/hpx/pull/2043>

⁵⁰⁸⁰ <https://github.com/TheHPXProject/hpx/pull/2042>

⁵⁰⁸¹ <https://github.com/TheHPXProject/hpx/pull/2041>

⁵⁰⁸² <https://github.com/TheHPXProject/hpx/issues/2040>

⁵⁰⁸³ <https://github.com/TheHPXProject/hpx/pull/2039>

⁵⁰⁸⁴ <https://github.com/TheHPXProject/hpx/pull/2038>

⁵⁰⁸⁵ <https://github.com/TheHPXProject/hpx/pull/2037>

⁵⁰⁸⁶ <https://github.com/TheHPXProject/hpx/pull/2036>

⁵⁰⁸⁷ <https://github.com/TheHPXProject/hpx/issues/2035>

⁵⁰⁸⁸ <https://github.com/TheHPXProject/hpx/pull/2034>

⁵⁰⁸⁹ <https://github.com/TheHPXProject/hpx/issues/2033>

⁵⁰⁹⁰ <https://github.com/TheHPXProject/hpx/issues/2032>

⁵⁰⁹¹ <https://github.com/TheHPXProject/hpx/pull/2031>

⁵⁰⁹² <https://github.com/TheHPXProject/hpx/pull/2030>

⁵⁰⁹³ <https://github.com/TheHPXProject/hpx/pull/2029>

⁵⁰⁹⁴ <https://github.com/TheHPXProject/hpx/pull/2028>

⁵⁰⁹⁵ <https://github.com/TheHPXProject/hpx/pull/2027>

⁵⁰⁹⁶ <https://github.com/TheHPXProject/hpx/pull/2025>

⁵⁰⁹⁷ <https://github.com/TheHPXProject/hpx/pull/2024>

⁵⁰⁹⁸ <https://github.com/TheHPXProject/hpx/pull/2023>

⁵⁰⁹⁹ <https://github.com/TheHPXProject/hpx/issues/2022>

⁵¹⁰⁰ <https://github.com/TheHPXProject/hpx/pull/2021>

- [Issue #2020⁵¹⁰¹](#) - Make errors related to `init_globally.cpp` example while building HPX out of the box
- [PR #2019⁵¹⁰²](#) - Fixed some compilation and cmake errors encountered in `nqueen` example
- [PR #2018⁵¹⁰³](#) - For loop algorithms
- [PR #2017⁵¹⁰⁴](#) - Non-recursive `at_index` implementation
- [Issue #2016⁵¹⁰⁵](#) - Add index-based for-loops
- [Issue #2015⁵¹⁰⁶](#) - Change default bind-mode to balanced
- [PR #2014⁵¹⁰⁷](#) - Fixed dataflow if invoked action returns a future
- [PR #2013⁵¹⁰⁸](#) - Fixing compilation issues with external example
- [PR #2012⁵¹⁰⁹](#) - Added Sierpinski Triangle example
- [Issue #2011⁵¹¹⁰](#) - Compilation error while running sample `hello_world_component` code
- [PR #2010⁵¹¹¹](#) - Segmented move implemented for `hpx::vector`
- [Issue #2009⁵¹¹²](#) - pkg-config order incorrect on 14.04 / GCC 4.8
- [Issue #2008⁵¹¹³](#) - Compilation error in dataflow of action returning a future
- [PR #2007⁵¹¹⁴](#) - Adding new performance counter exposing overall scheduler time
- [PR #2006⁵¹¹⁵](#) - Function includes
- [PR #2005⁵¹¹⁶](#) - Adding an example demonstrating how to initialize HPX from a global object
- [PR #2004⁵¹¹⁷](#) - Fixing 2000
- [PR #2003⁵¹¹⁸](#) - Adding generation parameter to `gather` to enable using it more than once
- [PR #2002⁵¹¹⁹](#) - Turn on position independent code to solve link problem with `hpx_init`
- [Issue #2001⁵¹²⁰](#) - Gathering more than once segfaults
- [Issue #2000⁵¹²¹](#) - Undefined reference to `hpx::assertion_failed`
- [Issue #1999⁵¹²²](#) - Seg fault in `hpx::lcos::base_lco_with_value<*>::set_value_nonvirt()` when running `octo-tiger`
- [PR #1998⁵¹²³](#) - Detect unknown command line options

⁵¹⁰¹ <https://github.com/TheHPXProject/hpx/issues/2020>

⁵¹⁰² <https://github.com/TheHPXProject/hpx/pull/2019>

⁵¹⁰³ <https://github.com/TheHPXProject/hpx/pull/2018>

⁵¹⁰⁴ <https://github.com/TheHPXProject/hpx/pull/2017>

⁵¹⁰⁵ <https://github.com/TheHPXProject/hpx/issues/2016>

⁵¹⁰⁶ <https://github.com/TheHPXProject/hpx/issues/2015>

⁵¹⁰⁷ <https://github.com/TheHPXProject/hpx/pull/2014>

⁵¹⁰⁸ <https://github.com/TheHPXProject/hpx/pull/2013>

⁵¹⁰⁹ <https://github.com/TheHPXProject/hpx/pull/2012>

⁵¹¹⁰ <https://github.com/TheHPXProject/hpx/issues/2011>

⁵¹¹¹ <https://github.com/TheHPXProject/hpx/pull/2010>

⁵¹¹² <https://github.com/TheHPXProject/hpx/issues/2009>

⁵¹¹³ <https://github.com/TheHPXProject/hpx/issues/2008>

⁵¹¹⁴ <https://github.com/TheHPXProject/hpx/pull/2007>

⁵¹¹⁵ <https://github.com/TheHPXProject/hpx/pull/2006>

⁵¹¹⁶ <https://github.com/TheHPXProject/hpx/pull/2005>

⁵¹¹⁷ <https://github.com/TheHPXProject/hpx/pull/2004>

⁵¹¹⁸ <https://github.com/TheHPXProject/hpx/pull/2003>

⁵¹¹⁹ <https://github.com/TheHPXProject/hpx/pull/2002>

⁵¹²⁰ <https://github.com/TheHPXProject/hpx/issues/2001>

⁵¹²¹ <https://github.com/TheHPXProject/hpx/issues/2000>

⁵¹²² <https://github.com/TheHPXProject/hpx/issues/1999>

⁵¹²³ <https://github.com/TheHPXProject/hpx/pull/1998>

- PR #1997⁵¹²⁴ - Extending thread description
- PR #1996⁵¹²⁵ - Adding natvis files to solution (MSVC only)
- Issue #1995⁵¹²⁶ - Command line handling does not produce error
- PR #1994⁵¹²⁷ - Possible missing include in test_utils.hpp
- PR #1993⁵¹²⁸ - Add missing LANGUAGES tag to a hpx_add_compile_flag_if_available() call in CMakeLists.txt
- PR #1992⁵¹²⁹ - Fixing shared_executor_test
- PR #1991⁵¹³⁰ - Making sure the winsock library is properly initialized
- PR #1990⁵¹³¹ - Fixing bind_test placeholder ambiguity coming from boost-1.60
- PR #1989⁵¹³² - Performance tuning
- PR #1987⁵¹³³ - Make configurable size of internal storage in util::function
- PR #1986⁵¹³⁴ - AGAS Refactoring+1753 Cache mods
- PR #1985⁵¹³⁵ - Adding missing task_block::run() overload taking an executor
- PR #1984⁵¹³⁶ - Adding an optimized LRU Cache implementation (for AGAS)
- PR #1983⁵¹³⁷ - Avoid invoking migration table look up for all objects
- PR #1981⁵¹³⁸ - Replacing uintptr_t (which is not defined everywhere) with std::size_t
- PR #1980⁵¹³⁹ - Optimizing LCO continuations
- PR #1979⁵¹⁴⁰ - Fixing Cori
- PR #1978⁵¹⁴¹ - Fix test check that got broken in hasty fix to memory overflow
- PR #1977⁵¹⁴² - Refactor action traits
- PR #1976⁵¹⁴³ - Fixes typo in README.rst
- PR #1975⁵¹⁴⁴ - Reduce size of benchmark timing arrays to fix test failures
- PR #1974⁵¹⁴⁵ - Add action to update data owned by the partitioned_vector component
- PR #1972⁵¹⁴⁶ - Adding partitioned_vector SPMD example

⁵¹²⁴ <https://github.com/TheHPXProject/hpx/pull/1997>
⁵¹²⁵ <https://github.com/TheHPXProject/hpx/pull/1996>
⁵¹²⁶ <https://github.com/TheHPXProject/hpx/issues/1995>
⁵¹²⁷ <https://github.com/TheHPXProject/hpx/pull/1994>
⁵¹²⁸ <https://github.com/TheHPXProject/hpx/pull/1993>
⁵¹²⁹ <https://github.com/TheHPXProject/hpx/pull/1992>
⁵¹³⁰ <https://github.com/TheHPXProject/hpx/pull/1991>
⁵¹³¹ <https://github.com/TheHPXProject/hpx/pull/1990>
⁵¹³² <https://github.com/TheHPXProject/hpx/pull/1989>
⁵¹³³ <https://github.com/TheHPXProject/hpx/pull/1987>
⁵¹³⁴ <https://github.com/TheHPXProject/hpx/pull/1986>
⁵¹³⁵ <https://github.com/TheHPXProject/hpx/pull/1985>
⁵¹³⁶ <https://github.com/TheHPXProject/hpx/pull/1984>
⁵¹³⁷ <https://github.com/TheHPXProject/hpx/pull/1983>
⁵¹³⁸ <https://github.com/TheHPXProject/hpx/pull/1981>
⁵¹³⁹ <https://github.com/TheHPXProject/hpx/pull/1980>
⁵¹⁴⁰ <https://github.com/TheHPXProject/hpx/pull/1979>
⁵¹⁴¹ <https://github.com/TheHPXProject/hpx/pull/1978>
⁵¹⁴² <https://github.com/TheHPXProject/hpx/pull/1977>
⁵¹⁴³ <https://github.com/TheHPXProject/hpx/pull/1976>
⁵¹⁴⁴ <https://github.com/TheHPXProject/hpx/pull/1975>
⁵¹⁴⁵ <https://github.com/TheHPXProject/hpx/pull/1974>
⁵¹⁴⁶ <https://github.com/TheHPXProject/hpx/pull/1972>

- PR #1971⁵¹⁴⁷ - Fixing 1965
- PR #1970⁵¹⁴⁸ - Papi fixes
- PR #1969⁵¹⁴⁹ - Fixing continuation recursions to not depend on fixed amount of recursions
- PR #1968⁵¹⁵⁰ - More segmented algorithms
- Issue #1967⁵¹⁵¹ - Simplify component implementations
- PR #1966⁵¹⁵² - Migrate components
- Issue #1964⁵¹⁵³ - fatal error: 'boost/lockfree/detail/branch_hints.hpp' file not found
- Issue #1962⁵¹⁵⁴ - parallel::copy_if has race condition when used on in place arrays
- PR #1963⁵¹⁵⁵ - Fixing Static Parcelport initialization
- PR #1961⁵¹⁵⁶ - Fix function target
- Issue #1960⁵¹⁵⁷ - Papi counters don't reset
- PR #1959⁵¹⁵⁸ - Fixing 1958
- Issue #1958⁵¹⁵⁹ - inclusive_scan gives incorrect results with non-commutative operator
- PR #1957⁵¹⁶⁰ - Fixing #1950
- PR #1956⁵¹⁶¹ - Sort by key example
- PR #1955⁵¹⁶² - Adding regression test for #1946: Hang in wait_all() in distributed run
- Issue #1954⁵¹⁶³ - HPX releases should not use -Werror
- PR #1953⁵¹⁶⁴ - Adding performance analysis for AGAS cache
- PR #1952⁵¹⁶⁵ - Adapting test for explicit variadics to fail for gcc 4.6
- PR #1951⁵¹⁶⁶ - Fixing memory leak
- Issue #1950⁵¹⁶⁷ - Simplify external builds
- PR #1949⁵¹⁶⁸ - Fixing yet another lock that is being held during suspension
- PR #1948⁵¹⁶⁹ - Fixed container algorithms for Intel

⁵¹⁴⁷ <https://github.com/TheHPXProject/hpx/pull/1971>

⁵¹⁴⁸ <https://github.com/TheHPXProject/hpx/pull/1970>

⁵¹⁴⁹ <https://github.com/TheHPXProject/hpx/pull/1969>

⁵¹⁵⁰ <https://github.com/TheHPXProject/hpx/pull/1968>

⁵¹⁵¹ <https://github.com/TheHPXProject/hpx/issues/1967>

⁵¹⁵² <https://github.com/TheHPXProject/hpx/pull/1966>

⁵¹⁵³ <https://github.com/TheHPXProject/hpx/issues/1964>

⁵¹⁵⁴ <https://github.com/TheHPXProject/hpx/issues/1962>

⁵¹⁵⁵ <https://github.com/TheHPXProject/hpx/pull/1963>

⁵¹⁵⁶ <https://github.com/TheHPXProject/hpx/pull/1961>

⁵¹⁵⁷ <https://github.com/TheHPXProject/hpx/issues/1960>

⁵¹⁵⁸ <https://github.com/TheHPXProject/hpx/pull/1959>

⁵¹⁵⁹ <https://github.com/TheHPXProject/hpx/issues/1958>

⁵¹⁶⁰ <https://github.com/TheHPXProject/hpx/pull/1957>

⁵¹⁶¹ <https://github.com/TheHPXProject/hpx/pull/1956>

⁵¹⁶² <https://github.com/TheHPXProject/hpx/pull/1955>

⁵¹⁶³ <https://github.com/TheHPXProject/hpx/issues/1954>

⁵¹⁶⁴ <https://github.com/TheHPXProject/hpx/pull/1953>

⁵¹⁶⁵ <https://github.com/TheHPXProject/hpx/pull/1952>

⁵¹⁶⁶ <https://github.com/TheHPXProject/hpx/pull/1951>

⁵¹⁶⁷ <https://github.com/TheHPXProject/hpx/issues/1950>

⁵¹⁶⁸ <https://github.com/TheHPXProject/hpx/pull/1949>

⁵¹⁶⁹ <https://github.com/TheHPXProject/hpx/pull/1948>

- PR #1947⁵¹⁷⁰ - Adding workaround for tagged_tuple
- Issue #1946⁵¹⁷¹ - Hang in wait_all() in distributed run
- PR #1945⁵¹⁷² - Fixed container algorithm tests
- Issue #1944⁵¹⁷³ - assertion 'p.destination_locality() == hpx::get_locality()' failed
- PR #1943⁵¹⁷⁴ - Fix a couple of compile errors with clang
- PR #1942⁵¹⁷⁵ - Making parcel coalescing functional
- Issue #1941⁵¹⁷⁶ - Re-enable parcel coalescing
- PR #1940⁵¹⁷⁷ - Touching up make_future
- PR #1939⁵¹⁷⁸ - Fixing problems in over-subscription management in the resource manager
- PR #1938⁵¹⁷⁹ - Removing use of unified Boost.Thread header
- PR #1937⁵¹⁸⁰ - Cleaning up the use of Boost.Accumulator headers
- PR #1936⁵¹⁸¹ - Making sure interval timer is started for aggregating performance counters
- PR #1935⁵¹⁸² - Tagged results
- PR #1934⁵¹⁸³ - Fix remote async with deferred launch policy
- Issue #1933⁵¹⁸⁴ - Floating point exception in statistics_counter<boost::accumulators::tag::mean>::get_counter_v
- PR #1932⁵¹⁸⁵ - Removing superfluous includes of boost/lockfree/detail/branch_hints.hpp
- PR #1931⁵¹⁸⁶ - fix compilation with clang 3.8.0
- Issue #1930⁵¹⁸⁷ - Missing online documentation for HPX 0.9.11
- PR #1929⁵¹⁸⁸ - LWG2485: get() should be overloaded for const tuple&&
- PR #1928⁵¹⁸⁹ - Revert "Using ninja for circle-ci builds"
- PR #1927⁵¹⁹⁰ - Using ninja for circle-ci builds
- PR #1926⁵¹⁹¹ - Fixing serialization of std::array
- Issue #1925⁵¹⁹² - Issues with static HPX libraries

⁵¹⁷⁰ <https://github.com/TheHPXProject/hpx/pull/1947>

⁵¹⁷¹ <https://github.com/TheHPXProject/hpx/issues/1946>

⁵¹⁷² <https://github.com/TheHPXProject/hpx/pull/1945>

⁵¹⁷³ <https://github.com/TheHPXProject/hpx/issues/1944>

⁵¹⁷⁴ <https://github.com/TheHPXProject/hpx/pull/1943>

⁵¹⁷⁵ <https://github.com/TheHPXProject/hpx/pull/1942>

⁵¹⁷⁶ <https://github.com/TheHPXProject/hpx/issues/1941>

⁵¹⁷⁷ <https://github.com/TheHPXProject/hpx/pull/1940>

⁵¹⁷⁸ <https://github.com/TheHPXProject/hpx/pull/1939>

⁵¹⁷⁹ <https://github.com/TheHPXProject/hpx/pull/1938>

⁵¹⁸⁰ <https://github.com/TheHPXProject/hpx/pull/1937>

⁵¹⁸¹ <https://github.com/TheHPXProject/hpx/pull/1936>

⁵¹⁸² <https://github.com/TheHPXProject/hpx/pull/1935>

⁵¹⁸³ <https://github.com/TheHPXProject/hpx/pull/1934>

⁵¹⁸⁴ <https://github.com/TheHPXProject/hpx/issues/1933>

⁵¹⁸⁵ <https://github.com/TheHPXProject/hpx/pull/1932>

⁵¹⁸⁶ <https://github.com/TheHPXProject/hpx/pull/1931>

⁵¹⁸⁷ <https://github.com/TheHPXProject/hpx/issues/1930>

⁵¹⁸⁸ <https://github.com/TheHPXProject/hpx/pull/1929>

⁵¹⁸⁹ <https://github.com/TheHPXProject/hpx/pull/1928>

⁵¹⁹⁰ <https://github.com/TheHPXProject/hpx/pull/1927>

⁵¹⁹¹ <https://github.com/TheHPXProject/hpx/pull/1926>

⁵¹⁹² <https://github.com/TheHPXProject/hpx/issues/1925>

- [Issue #1924](#)⁵¹⁹³ - Performance degrading over time
- [Issue #1923](#)⁵¹⁹⁴ - serialization of `std::array` appears broken in latest commit
- [PR #1922](#)⁵¹⁹⁵ - Container algorithms
- [PR #1921](#)⁵¹⁹⁶ - Tons of smaller quality improvements
- [Issue #1920](#)⁵¹⁹⁷ - Seg fault in `hpx::serialization::output_archive::add_gid` when running octotiger
- [Issue #1919](#)⁵¹⁹⁸ - Intel 15 compiler bug preventing HPX build
- [PR #1918](#)⁵¹⁹⁹ - Address sanitizer fixes
- [PR #1917](#)⁵²⁰⁰ - Fixing compilation problems of `parallel::sort` with Intel compilers
- [PR #1916](#)⁵²⁰¹ - Making sure code compiles if `HPX_WITH_HWLOC=Off`
- [Issue #1915](#)⁵²⁰² - `max_cores` undefined if `HPX_WITH_HWLOC=Off`
- [PR #1913](#)⁵²⁰³ - Add utility member functions for `partitioned_vector`
- [PR #1912](#)⁵²⁰⁴ - Adding support for invoking actions to dataflow
- [PR #1911](#)⁵²⁰⁵ - Adding first batch of container algorithms
- [PR #1910](#)⁵²⁰⁶ - Keep `cmake_module_path`
- [PR #1909](#)⁵²⁰⁷ - Fix `mpirun` with `pbs`
- [PR #1908](#)⁵²⁰⁸ - Changing `parallel::sort` to return the last iterator as proposed by N4560
- [PR #1907](#)⁵²⁰⁹ - Adding a minimum version for Open MPI
- [PR #1906](#)⁵²¹⁰ - Updates to the Release Procedure
- [PR #1905](#)⁵²¹¹ - Fixing #1903
- [PR #1904](#)⁵²¹² - Making sure `std` containers are cleared before serialization loads data
- [Issue #1903](#)⁵²¹³ - When running octotiger, I get: `assertion '(*new_gids_)[gid].size() == 1' failed: HPX(assertion_failure)`
- [Issue #1902](#)⁵²¹⁴ - Immediate crash when running `hpx/octotiger` with `_GLIBCXX_DEBUG` defined.
- [PR #1901](#)⁵²¹⁵ - Making non-serializable classes non-serializable

⁵¹⁹³ <https://github.com/TheHPXProject/hpx/issues/1924>

⁵¹⁹⁴ <https://github.com/TheHPXProject/hpx/issues/1923>

⁵¹⁹⁵ <https://github.com/TheHPXProject/hpx/pull/1922>

⁵¹⁹⁶ <https://github.com/TheHPXProject/hpx/pull/1921>

⁵¹⁹⁷ <https://github.com/TheHPXProject/hpx/issues/1920>

⁵¹⁹⁸ <https://github.com/TheHPXProject/hpx/issues/1919>

⁵¹⁹⁹ <https://github.com/TheHPXProject/hpx/pull/1918>

⁵²⁰⁰ <https://github.com/TheHPXProject/hpx/pull/1917>

⁵²⁰¹ <https://github.com/TheHPXProject/hpx/pull/1916>

⁵²⁰² <https://github.com/TheHPXProject/hpx/issues/1915>

⁵²⁰³ <https://github.com/TheHPXProject/hpx/pull/1913>

⁵²⁰⁴ <https://github.com/TheHPXProject/hpx/pull/1912>

⁵²⁰⁵ <https://github.com/TheHPXProject/hpx/pull/1911>

⁵²⁰⁶ <https://github.com/TheHPXProject/hpx/pull/1910>

⁵²⁰⁷ <https://github.com/TheHPXProject/hpx/pull/1909>

⁵²⁰⁸ <https://github.com/TheHPXProject/hpx/pull/1908>

⁵²⁰⁹ <https://github.com/TheHPXProject/hpx/pull/1907>

⁵²¹⁰ <https://github.com/TheHPXProject/hpx/pull/1906>

⁵²¹¹ <https://github.com/TheHPXProject/hpx/pull/1905>

⁵²¹² <https://github.com/TheHPXProject/hpx/pull/1904>

⁵²¹³ <https://github.com/TheHPXProject/hpx/issues/1903>

⁵²¹⁴ <https://github.com/TheHPXProject/hpx/issues/1902>

⁵²¹⁵ <https://github.com/TheHPXProject/hpx/pull/1901>

- [Issue #1900](#)⁵²¹⁶ - Two possible issues with std::list serialization
- [PR #1899](#)⁵²¹⁷ - Fixing a problem with credit splitting as revealed by #1898
- [Issue #1898](#)⁵²¹⁸ - Accessing component from locality where it was not created segfaults
- [PR #1897](#)⁵²¹⁹ - Changing parallel::sort to return the last iterator as proposed by N4560
- [Issue #1896](#)⁵²²⁰ - version 1.0?
- [Issue #1895](#)⁵²²¹ - Warning comment on numa_allocator is not very clear
- [PR #1894](#)⁵²²² - Add support for compilers that have thread_local
- [PR #1893](#)⁵²²³ - Fixing 1890
- [PR #1892](#)⁵²²⁴ - Adds typed future_type for executor_traits
- [PR #1891](#)⁵²²⁵ - Fix wording in certain parallel algorithm docs
- [Issue #1890](#)⁵²²⁶ - Invoking papi counters give segfault
- [PR #1889](#)⁵²²⁷ - Fixing problems as reported by clang-check
- [PR #1888](#)⁵²²⁸ - WIP parallel is_heap
- [PR #1887](#)⁵²²⁹ - Fixed resetting performance counters related to idle-rate, etc
- [Issue #1886](#)⁵²³⁰ - Run hpx with qsub does not work
- [PR #1885](#)⁵²³¹ - Warning cleaning pass
- [PR #1884](#)⁵²³² - Add missing parallel algorithm header
- [PR #1883](#)⁵²³³ - Add feature test for thread_local on Clang for TLS
- [PR #1882](#)⁵²³⁴ - Fix some redundant qualifiers
- [Issue #1881](#)⁵²³⁵ - Unable to compile Octotiger using HPX and Intel MPI on SuperMIC
- [Issue #1880](#)⁵²³⁶ - clang with libc++ on Linux needs TLS case
- [PR #1879](#)⁵²³⁷ - Doc fixes for #1868
- [PR #1878](#)⁵²³⁸ - Simplify functions

⁵²¹⁶ <https://github.com/TheHPXProject/hpx/issues/1900>

⁵²¹⁷ <https://github.com/TheHPXProject/hpx/pull/1899>

⁵²¹⁸ <https://github.com/TheHPXProject/hpx/issues/1898>

⁵²¹⁹ <https://github.com/TheHPXProject/hpx/pull/1897>

⁵²²⁰ <https://github.com/TheHPXProject/hpx/issues/1896>

⁵²²¹ <https://github.com/TheHPXProject/hpx/issues/1895>

⁵²²² <https://github.com/TheHPXProject/hpx/pull/1894>

⁵²²³ <https://github.com/TheHPXProject/hpx/pull/1893>

⁵²²⁴ <https://github.com/TheHPXProject/hpx/pull/1892>

⁵²²⁵ <https://github.com/TheHPXProject/hpx/pull/1891>

⁵²²⁶ <https://github.com/TheHPXProject/hpx/issues/1890>

⁵²²⁷ <https://github.com/TheHPXProject/hpx/pull/1889>

⁵²²⁸ <https://github.com/TheHPXProject/hpx/pull/1888>

⁵²²⁹ <https://github.com/TheHPXProject/hpx/pull/1887>

⁵²³⁰ <https://github.com/TheHPXProject/hpx/issues/1886>

⁵²³¹ <https://github.com/TheHPXProject/hpx/pull/1885>

⁵²³² <https://github.com/TheHPXProject/hpx/pull/1884>

⁵²³³ <https://github.com/TheHPXProject/hpx/pull/1883>

⁵²³⁴ <https://github.com/TheHPXProject/hpx/pull/1882>

⁵²³⁵ <https://github.com/TheHPXProject/hpx/issues/1881>

⁵²³⁶ <https://github.com/TheHPXProject/hpx/issues/1880>

⁵²³⁷ <https://github.com/TheHPXProject/hpx/pull/1879>

⁵²³⁸ <https://github.com/TheHPXProject/hpx/pull/1878>

- PR #1877⁵²³⁹ - Removing most usage of Boost.Config
- PR #1876⁵²⁴⁰ - Add missing parallel algorithms to algorithm.hpp
- PR #1875⁵²⁴¹ - Simplify callables
- PR #1874⁵²⁴² - Address long standing FIXME on using `std::unique_ptr` with incomplete types
- PR #1873⁵²⁴³ - Fixing 1871
- PR #1872⁵²⁴⁴ - Making sure PBS environment uses specified node list even if no PBS_NODEFILE env is available
- Issue #1871⁵²⁴⁵ - Fortran checks should be optional
- PR #1870⁵²⁴⁶ - Touch local::mutex
- PR #1869⁵²⁴⁷ - Documentation refactoring based off #1868
- PR #1867⁵²⁴⁸ - Embrace static_assert
- PR #1866⁵²⁴⁹ - Fix #1803 with documentation refactoring
- PR #1865⁵²⁵⁰ - Setting OUTPUT_NAME as target properties
- PR #1863⁵²⁵¹ - Use SYSTEM for boost includes
- PR #1862⁵²⁵² - Minor cleanups
- PR #1861⁵²⁵³ - Minor Corrections for Release
- PR #1860⁵²⁵⁴ - Fixing hpx gdb script
- Issue #1859⁵²⁵⁵ - reset_active_counters resets times and thread counts before some of the counters are evaluated
- PR #1858⁵²⁵⁶ - Release V0.9.11
- PR #1857⁵²⁵⁷ - removing diskperf example from 9.11 release
- PR #1856⁵²⁵⁸ - fix return in packaged_task_base::reset()
- Issue #1842⁵²⁵⁹ - Install error: file INSTALL cannot find libhpx_parcel_coalescing.so.0.9.11
- PR #1839⁵²⁶⁰ - Adding fedora docs
- PR #1824⁵²⁶¹ - Changing version on master to V0.9.12

⁵²³⁹ <https://github.com/TheHPXProject/hpx/pull/1877>
⁵²⁴⁰ <https://github.com/TheHPXProject/hpx/pull/1876>
⁵²⁴¹ <https://github.com/TheHPXProject/hpx/pull/1875>
⁵²⁴² <https://github.com/TheHPXProject/hpx/pull/1874>
⁵²⁴³ <https://github.com/TheHPXProject/hpx/pull/1873>
⁵²⁴⁴ <https://github.com/TheHPXProject/hpx/pull/1872>
⁵²⁴⁵ <https://github.com/TheHPXProject/hpx/issues/1871>
⁵²⁴⁶ <https://github.com/TheHPXProject/hpx/pull/1870>
⁵²⁴⁷ <https://github.com/TheHPXProject/hpx/pull/1869>
⁵²⁴⁸ <https://github.com/TheHPXProject/hpx/pull/1867>
⁵²⁴⁹ <https://github.com/TheHPXProject/hpx/pull/1866>
⁵²⁵⁰ <https://github.com/TheHPXProject/hpx/pull/1865>
⁵²⁵¹ <https://github.com/TheHPXProject/hpx/pull/1863>
⁵²⁵² <https://github.com/TheHPXProject/hpx/pull/1862>
⁵²⁵³ <https://github.com/TheHPXProject/hpx/pull/1861>
⁵²⁵⁴ <https://github.com/TheHPXProject/hpx/pull/1860>
⁵²⁵⁵ <https://github.com/TheHPXProject/hpx/issues/1859>
⁵²⁵⁶ <https://github.com/TheHPXProject/hpx/pull/1858>
⁵²⁵⁷ <https://github.com/TheHPXProject/hpx/pull/1857>
⁵²⁵⁸ <https://github.com/TheHPXProject/hpx/pull/1856>
⁵²⁵⁹ <https://github.com/TheHPXProject/hpx/issues/1842>
⁵²⁶⁰ <https://github.com/TheHPXProject/hpx/pull/1839>
⁵²⁶¹ <https://github.com/TheHPXProject/hpx/pull/1824>

- PR #1818⁵²⁶² - Fixing #1748
- Issue #1815⁵²⁶³ - seg fault in AGAS
- Issue #1803⁵²⁶⁴ - wait_all documentation
- Issue #1796⁵²⁶⁵ - Outdated documentation to be revised
- Issue #1759⁵²⁶⁶ - glibc munmap_chunk or free(): invalid pointer on SuperMIC
- Issue #1753⁵²⁶⁷ - HPX performance degrades with time since execution begins
- Issue #1748⁵²⁶⁸ - All public HPX headers need to be self contained
- PR #1719⁵²⁶⁹ - How to build HPX with Visual Studio
- Issue #1684⁵²⁷⁰ - Race condition when using -hpx:connect?
- PR #1658⁵²⁷¹ - Add serialization for std::set (as there is for std::vector and std::map)
- PR #1641⁵²⁷² - Generic client
- Issue #1632⁵²⁷³ - heartbeat example fails on separate nodes
- PR #1603⁵²⁷⁴ - Adds preferred namespace check to inspect tool
- Issue #1559⁵²⁷⁵ - Extend inspect tool
- Issue #1523⁵²⁷⁶ - Remote async with deferred launch policy never executes
- Issue #1472⁵²⁷⁷ - Serialization issues
- Issue #1457⁵²⁷⁸ - Implement N4392: C++ Latches and Barriers
- PR #1444⁵²⁷⁹ - Enabling usage of moveonly types for component construction
- Issue #1407⁵²⁸⁰ - The Intel 13 compiler has failing unit tests
- Issue #1405⁵²⁸¹ - Allow component constructors to take movable only types
- Issue #1265⁵²⁸² - Enable dataflow() to be usable with actions
- Issue #1236⁵²⁸³ - NUMA aware allocators
- Issue #802⁵²⁸⁴ - Fix Broken Examples

⁵²⁶² <https://github.com/TheHPXProject/hpx/pull/1818>

⁵²⁶³ <https://github.com/TheHPXProject/hpx/issues/1815>

⁵²⁶⁴ <https://github.com/TheHPXProject/hpx/issues/1803>

⁵²⁶⁵ <https://github.com/TheHPXProject/hpx/issues/1796>

⁵²⁶⁶ <https://github.com/TheHPXProject/hpx/issues/1759>

⁵²⁶⁷ <https://github.com/TheHPXProject/hpx/issues/1753>

⁵²⁶⁸ <https://github.com/TheHPXProject/hpx/issues/1748>

⁵²⁶⁹ <https://github.com/TheHPXProject/hpx/pull/1719>

⁵²⁷⁰ <https://github.com/TheHPXProject/hpx/issues/1684>

⁵²⁷¹ <https://github.com/TheHPXProject/hpx/pull/1658>

⁵²⁷² <https://github.com/TheHPXProject/hpx/pull/1641>

⁵²⁷³ <https://github.com/TheHPXProject/hpx/issues/1632>

⁵²⁷⁴ <https://github.com/TheHPXProject/hpx/pull/1603>

⁵²⁷⁵ <https://github.com/TheHPXProject/hpx/issues/1559>

⁵²⁷⁶ <https://github.com/TheHPXProject/hpx/issues/1523>

⁵²⁷⁷ <https://github.com/TheHPXProject/hpx/issues/1472>

⁵²⁷⁸ <https://github.com/TheHPXProject/hpx/issues/1457>

⁵²⁷⁹ <https://github.com/TheHPXProject/hpx/pull/1444>

⁵²⁸⁰ <https://github.com/TheHPXProject/hpx/issues/1407>

⁵²⁸¹ <https://github.com/TheHPXProject/hpx/issues/1405>

⁵²⁸² <https://github.com/TheHPXProject/hpx/issues/1265>

⁵²⁸³ <https://github.com/TheHPXProject/hpx/issues/1236>

⁵²⁸⁴ <https://github.com/TheHPXProject/hpx/issues/802>

- [Issue #559](#)⁵²⁸⁵ - Add `hpx::migrate` facility
- [Issue #449](#)⁵²⁸⁶ - Make actions with template arguments usable and add documentation
- [Issue #279](#)⁵²⁸⁷ - Refactor `addressing_service` into a base class and two derived classes
- [Issue #224](#)⁵²⁸⁸ - Changing thread state metadata is not thread safe
- [Issue #55](#)⁵²⁸⁹ - Uniform syntax for enums should be implemented

HPX V0.9.11 (Nov 11, 2015)

Our main focus for this release was the design and development of a coherent set of higher-level APIs exposing various types of parallelism to the application programmer. We introduced the concepts of an `executor`, which can be used to customize the `where` and `when` of execution of tasks in the context of parallelizing codes. We extended all APIs related to managing parallel tasks to support executors which gives the user the choice of either using one of the predefined executor types or to provide its own, possibly application specific, executor. We paid very close attention to align all of these changes with the existing C++ Standards documents or with the ongoing proposals for standardization.

This release is the first after our change to a new development policy. We switched all development to be strictly performed on branches only, all direct commits to our main branch (`master`) are prohibited. Any change has to go through a peer review before it will be merged to `master`. As a result the overall stability of our code base has significantly increased, the development process itself has been simplified. This change manifests itself in a large number of pull-requests which have been merged (please see below for a full list of closed issues and pull-requests). All in all for this release, we closed almost 100 issues and merged over 290 pull-requests. There have been over 1600 commits to the master branch since the last release.

General changes

- We are moving into the direction of unifying managed and simple components. As such, the classes `hpx::components::component` and `hpx::components::component_base` have been added which currently just forward to the currently existing simple component facilities. The examples have been converted to only use those two classes.
- Added integration with the [CircleCI](#)⁵²⁹⁰ hosted continuous integration service. This gives us constant and immediate feedback on the health of our master branch.
- The compiler configuration subsystem in the build system has been reimplemented. Instead of using `Boost.Config` we now use our own lightweight set of `cmake` scripts to determine the available language and library features supported by the used compiler.
- The API for creating instances of components has been consolidated. All component instances should be created using the `hpx::new_` only. It allows one to instantiate both, single component instances and multiple component instances. The placement of the created components can be controlled by special distribution policies. Please see the corresponding documentation outlining the use of `hpx::new_`.
- Introduced four new distribution policies which can be used with many API functions which traditionally expected to be used with a locality id. The new distribution policies are:
 - `hpx::components::default_distribution_policy` which tries to place multiple component instances as evenly as possible.

⁵²⁸⁵ <https://github.com/TheHPXProject/hpx/issues/559>

⁵²⁸⁶ <https://github.com/TheHPXProject/hpx/issues/449>

⁵²⁸⁷ <https://github.com/TheHPXProject/hpx/issues/279>

⁵²⁸⁸ <https://github.com/TheHPXProject/hpx/issues/224>

⁵²⁸⁹ <https://github.com/TheHPXProject/hpx/issues/55>

⁵²⁹⁰ <https://circleci.com/gh/TheHPXProject/hpx>

- `hpx::components::colocating_distribution_policy` which will refer to the locality where a given component instance is currently placed.
 - `hpx::components::binpacking_distribution_policy` which will place multiple component instances as evenly as possible based on any performance counter.
 - `hpx::components::target_distribution_policy` which allows one to represent a given locality in the context of a distribution policy.
- The new distribution policies can now be also used with `hpx::async`. This change also deprecates `hpx::async_colocated(id, ...)` which now is replaced by a distribution policy: `hpx::async(hpx::colocated(id), ...)`.
 - The `hpx::vector` and `hpx::unordered_map` data structures can now be used with the new distribution policies as well.
 - The parallel facility `hpx::parallel::task_region` has been renamed to `hpx::parallel::task_block` based on the changes in the corresponding standardization proposal N4411⁵²⁹¹.
 - Added extensions to the parallel facility `hpx::parallel::task_block` allowing to combine a `task_block` with an execution policy. This implies a minor breaking change as the `hpx::parallel::task_block` is now a template.
 - Added new LCOs: `hpx::lcos::latch` and `hpx::lcos::local::latch` which semantically conform to the proposed `std::latch` (see N4399⁵²⁹²).
 - Added performance counters exposing data related to data transferred by input/output (filesystem) operations (thanks to Maciej Brodowicz).
 - Added performance counters allowing to track the number of action invocations (local and remote invocations).
 - Added new command line options `-hpx:print-counter-at` and `-hpx:reset-counters`.
 - The `hpx::vector` component has been renamed to `hpx::partitioned_vector` to make it explicit that the underlying memory is not contiguous.
 - Introduced a completely new and uniform higher-level parallelism API which is based on executors. All existing parallelism APIs have been adapted to this. We have added a large number of different executor types, such as a numa-aware executor, a this-thread executor, etc.
 - Added support for the MingW toolchain on Windows (thanks to Eric Lemanissier).
 - HPX now includes support for APEX, (Autonomic Performance Environment for eXascale). APEX is an instrumentation and software adaptation library that provides an interface to TAU profiling / tracing as well as runtime adaptation of HPX applications through policy definitions. For more information and documentation, please see <https://github.com/UO-OACISS/xpress-apex>. To enable APEX at configuration time, specify `-DHPX_WITH_APEX=On`. To also include support for TAU profiling, specify `-DHPX_WITH_TAU=On` and specify the `-DTAU_ROOT`, `-DTAU_ARCH` and `-DTAU_OPTIONS` cmake parameters.
 - We have implemented many more of the *Using parallel algorithms*. Please see Issue #1141⁵²⁹³ for the list of all available parallel algorithms (thanks to Daniel Bourgeois and John Biddiscombe for contributing their work).

⁵²⁹¹ <http://wg21.link/n4411>

⁵²⁹² <http://wg21.link/n4399>

⁵²⁹³ <https://github.com/TheHPXProject/hpx/issues/1141>

Breaking changes

- We are moving into the direction of unifying managed and simple components. In order to stop exposing the old facilities, all examples have been converted to use the new classes. The breaking change in this release is that performance counters are now a `hpx::components::component_base` instead of `hpx::components::managed_component_base`.
- We removed the support for stackless threads. It turned out that there was no performance benefit when using stackless threads. As such, we decided to clean up our codebase. This feature was not documented.
- The CMake project name has changed from 'hpx' to 'HPX' for consistency and compatibility with naming conventions and other CMake projects. Generated config files go into `<prefix>/lib/cmake/HPX` and not `<prefix>/lib/cmake/hpx`.
- The macro `HPX_REGISTER_MINIMAL_COMPONENT_FACTORY` has been deprecated. Please use `HPX_REGISTER_COMPONENT` instead. The old macro will be removed in the next release.
- The obsolete `distributing_factory` and `binpacking_factory` components have been removed. The corresponding functionality is now provided by the `hpx::new_` API function in conjunction with the `hpx::default_layout` and `hpx::binpacking` distribution policies (`hpx::components::default_distribution_policy` and `hpx::components::binpacking_distribution_policy`).
- The API function `hpx::new_colocated` has been deprecated. Please use the consolidated API `hpx::new_` in conjunction with the new `hpx::colocated` distribution policy (`hpx::components::colocating_distribution_policy`) instead. The old API function will still be available for at least one release of HPX if the configuration variable `HPX_WITH_COLOCATED_BACKWARDS_COMPATIBILITY` is enabled.
- The API function `hpx::async_colocated` has been deprecated. Please use the consolidated API `hpx::async` in conjunction with the new `hpx::colocated` distribution policy (`hpx::components::colocating_distribution_policy`) instead. The old API function will still be available for at least one release of HPX if the configuration variable `HPX_WITH_COLOCATED_BACKWARDS_COMPATIBILITY` is enabled.
- The obsolete `remote_object` component has been removed.
- Replaced the use of Boost.Serialization with our own solution. While the new version is mostly compatible with Boost.Serialization, this change requires some minor code modifications in user code. For more information, please see the corresponding [announcement⁵²⁹⁴](http://thread.gmane.org/gmane.comp.lib.hpx.devel/196) on the `hpx-users@stellar.cct.lsu.edu` mailing list.
- The names used by cmake to influence various configuration options have been unified. The new naming scheme relies on all configuration constants to start with `HPX_WITH_...`, while the preprocessor constant which is used at build time starts with `HPX_HAVE_...`. For instance, the former cmake command line `-DHPX_MALLOC=...` now has to be specified as `-DHPX_WITH_MALLOC=...` and will cause the preprocessor constant `HPX_HAVE_MALLOC` to be defined. The actual name of the constant (i.e. `MALLOC`) has not changed. Please see the corresponding documentation for more details (*CMake options*).
- The `get_gid()` functions exposed by the component base classes `hpx::components::server::simple_component_base`, `hpx::components::server::managed_component_base`, and `hpx::components::server::fixed_component_base` have been replaced by two new functions: `get_unmanaged_id()` and `get_id()`. To enable the old function name for backwards compatibility, use the cmake configuration option `HPX_WITH_COMPONENT_GET_GID_COMPATIBILITY=On`.
- All functions which were named `get_gid()` but were returning `hpx::id_type` have been renamed to `get_id()`. To enable the old function names for backwards compatibility, use the cmake configuration option `HPX_WITH_COMPONENT_GET_GID_COMPATIBILITY=On`.

⁵²⁹⁴ <http://thread.gmane.org/gmane.comp.lib.hpx.devel/196>

Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this release.

- PR #1855⁵²⁹⁵ - Completely removing external/endian
- PR #1854⁵²⁹⁶ - Don't pollute CMAKE_CXX_FLAGS through find_package()
- PR #1853⁵²⁹⁷ - Updating CMake configuration to get correct version of TAU library
- PR #1852⁵²⁹⁸ - Fixing Performance Problems with MPI Parcelport
- PR #1851⁵²⁹⁹ - Fixing hpx_add_link_flag() and hpx_remove_link_flag()
- PR #1850⁵³⁰⁰ - Fixing 1836, adding parallel::sort
- PR #1849⁵³⁰¹ - Fixing configuration for use of more than 64 cores
- PR #1848⁵³⁰² - Change default APEX version for release
- PR #1847⁵³⁰³ - Fix client_base::then on release
- PR #1846⁵³⁰⁴ - Removing broken lcos::local::channel from release
- PR #1845⁵³⁰⁵ - Adding example demonstrating a possible safe-object implementation to release
- PR #1844⁵³⁰⁶ - Removing stubs from accumulator examples
- PR #1843⁵³⁰⁷ - Don't pollute CMAKE_CXX_FLAGS through find_package()
- PR #1841⁵³⁰⁸ - Fixing client_base<>::then
- PR #1840⁵³⁰⁹ - Adding example demonstrating a possible safe-object implementation
- PR #1838⁵³¹⁰ - Update version rc1
- PR #1837⁵³¹¹ - Removing broken lcos::local::channel
- PR #1835⁵³¹² - Adding explicit move constructor and assignment operator to hpx::lcos::promise
- PR #1834⁵³¹³ - Making hpx::lcos::promise move-only
- PR #1833⁵³¹⁴ - Adding fedora docs
- Issue #1832⁵³¹⁵ - hpx::lcos::promise<> must be move-only

⁵²⁹⁵ <https://github.com/TheHPXProject/hpx/pull/1855>

⁵²⁹⁶ <https://github.com/TheHPXProject/hpx/pull/1854>

⁵²⁹⁷ <https://github.com/TheHPXProject/hpx/pull/1853>

⁵²⁹⁸ <https://github.com/TheHPXProject/hpx/pull/1852>

⁵²⁹⁹ <https://github.com/TheHPXProject/hpx/pull/1851>

⁵³⁰⁰ <https://github.com/TheHPXProject/hpx/pull/1850>

⁵³⁰¹ <https://github.com/TheHPXProject/hpx/pull/1849>

⁵³⁰² <https://github.com/TheHPXProject/hpx/pull/1848>

⁵³⁰³ <https://github.com/TheHPXProject/hpx/pull/1847>

⁵³⁰⁴ <https://github.com/TheHPXProject/hpx/pull/1846>

⁵³⁰⁵ <https://github.com/TheHPXProject/hpx/pull/1845>

⁵³⁰⁶ <https://github.com/TheHPXProject/hpx/pull/1844>

⁵³⁰⁷ <https://github.com/TheHPXProject/hpx/pull/1843>

⁵³⁰⁸ <https://github.com/TheHPXProject/hpx/pull/1841>

⁵³⁰⁹ <https://github.com/TheHPXProject/hpx/pull/1840>

⁵³¹⁰ <https://github.com/TheHPXProject/hpx/pull/1838>

⁵³¹¹ <https://github.com/TheHPXProject/hpx/pull/1837>

⁵³¹² <https://github.com/TheHPXProject/hpx/pull/1835>

⁵³¹³ <https://github.com/TheHPXProject/hpx/pull/1834>

⁵³¹⁴ <https://github.com/TheHPXProject/hpx/pull/1833>

⁵³¹⁵ <https://github.com/TheHPXProject/hpx/issues/1832>

- [PR #1831](#)⁵³¹⁶ - Fixing resource manager gcc5.2
- [PR #1830](#)⁵³¹⁷ - Fix intel13
- [PR #1829](#)⁵³¹⁸ - Unbreaking thread test
- [PR #1828](#)⁵³¹⁹ - Fixing #1620
- [PR #1827](#)⁵³²⁰ - Fixing a memory management issue for the Parquet application
- [Issue #1826](#)⁵³²¹ - Memory management issue in `hpx::lcos::promise`
- [PR #1825](#)⁵³²² - Adding `hpx::components::component` and `hpx::components::component_base`
- [PR #1823](#)⁵³²³ - Adding git commit id to circleci build
- [PR #1822](#)⁵³²⁴ - applying fixes suggested by clang 3.7
- [PR #1821](#)⁵³²⁵ - Hyperlink fixes
- [PR #1820](#)⁵³²⁶ - added parallel multi-locality sanity test
- [PR #1819](#)⁵³²⁷ - Fixing #1667
- [Issue #1817](#)⁵³²⁸ - Hyperlinks generated by inspect tool are wrong
- [PR #1816](#)⁵³²⁹ - Support `hpxrx`
- [PR #1814](#)⁵³³⁰ - Fix `async` to dispatch to the correct locality in all cases
- [Issue #1813](#)⁵³³¹ - `async(launch::..., action(), ...)` always invokes locally
- [PR #1812](#)⁵³³² - fixed syntax error in `CMakeLists.txt`
- [PR #1811](#)⁵³³³ - Agas optimizations
- [PR #1810](#)⁵³³⁴ - drop superfluous typedefs
- [PR #1809](#)⁵³³⁵ - Allow HPX to be used as an optional package in 3rd party code
- [PR #1808](#)⁵³³⁶ - Fixing #1723
- [PR #1807](#)⁵³³⁷ - Making sure `resolve_localities` does not hang during normal operation
- [Issue #1806](#)⁵³³⁸ - Spinlock no longer movable and deletes operator `'='`, breaks MiniGhost

⁵³¹⁶ <https://github.com/TheHPXProject/hpx/pull/1831>
⁵³¹⁷ <https://github.com/TheHPXProject/hpx/pull/1830>
⁵³¹⁸ <https://github.com/TheHPXProject/hpx/pull/1829>
⁵³¹⁹ <https://github.com/TheHPXProject/hpx/pull/1828>
⁵³²⁰ <https://github.com/TheHPXProject/hpx/pull/1827>
⁵³²¹ <https://github.com/TheHPXProject/hpx/issues/1826>
⁵³²² <https://github.com/TheHPXProject/hpx/pull/1825>
⁵³²³ <https://github.com/TheHPXProject/hpx/pull/1823>
⁵³²⁴ <https://github.com/TheHPXProject/hpx/pull/1822>
⁵³²⁵ <https://github.com/TheHPXProject/hpx/pull/1821>
⁵³²⁶ <https://github.com/TheHPXProject/hpx/pull/1820>
⁵³²⁷ <https://github.com/TheHPXProject/hpx/pull/1819>
⁵³²⁸ <https://github.com/TheHPXProject/hpx/issues/1817>
⁵³²⁹ <https://github.com/TheHPXProject/hpx/pull/1816>
⁵³³⁰ <https://github.com/TheHPXProject/hpx/pull/1814>
⁵³³¹ <https://github.com/TheHPXProject/hpx/issues/1813>
⁵³³² <https://github.com/TheHPXProject/hpx/pull/1812>
⁵³³³ <https://github.com/TheHPXProject/hpx/pull/1811>
⁵³³⁴ <https://github.com/TheHPXProject/hpx/pull/1810>
⁵³³⁵ <https://github.com/TheHPXProject/hpx/pull/1809>
⁵³³⁶ <https://github.com/TheHPXProject/hpx/pull/1808>
⁵³³⁷ <https://github.com/TheHPXProject/hpx/pull/1807>
⁵³³⁸ <https://github.com/TheHPXProject/hpx/issues/1806>

- Issue #1804⁵³³⁹ - register_with_basename causes hangs
- PR #1801⁵³⁴⁰ - Enhanced the inspect tool to take user directly to the problem with hyperlinks
- Issue #1800⁵³⁴¹ - Problems compiling application on smic
- PR #1799⁵³⁴² - Fixing cv exceptions
- PR #1798⁵³⁴³ - Documentation refactoring & updating
- PR #1797⁵³⁴⁴ - Updating the activeharmony CMake module
- PR #1795⁵³⁴⁵ - Fixing cv
- PR #1794⁵³⁴⁶ - Fix connect with hpx::runtime_mode_connect
- PR #1793⁵³⁴⁷ - fix a wrong use of HPX_MAX_CPU_COUNT instead of HPX_HAVE_MAX_CPU_COUNT
- PR #1792⁵³⁴⁸ - Allow for default constructed parcel instances to be moved
- PR #1791⁵³⁴⁹ - Fix connect with hpx::runtime_mode_connect
- Issue #1790⁵³⁵⁰ - assertion action_.get() failed: HPX(assertion_failure) when running Octotiger with pull request 1786
- PR #1789⁵³⁵¹ - Fixing discover_counter_types API function
- Issue #1788⁵³⁵² - connect with hpx::runtime_mode_connect
- Issue #1787⁵³⁵³ - discover_counter_types not working
- PR #1786⁵³⁵⁴ - Changing addressing_service to use std::unordered_map instead of std::map
- PR #1785⁵³⁵⁵ - Fix is_iterator for container algorithms
- PR #1784⁵³⁵⁶ - Adding new command line options:
- PR #1783⁵³⁵⁷ - Minor changes for APEX support
- PR #1782⁵³⁵⁸ - Drop legacy forwarding action traits
- PR #1781⁵³⁵⁹ - Attempt to resolve the race between cv::wait_xxx and cv::notify_all
- PR #1780⁵³⁶⁰ - Removing serialize_sequence
- PR #1779⁵³⁶¹ - Fixed #1501: hwloc configuration options are wrong for MIC

⁵³³⁹ <https://github.com/TheHPXProject/hpx/issues/1804>

⁵³⁴⁰ <https://github.com/TheHPXProject/hpx/pull/1801>

⁵³⁴¹ <https://github.com/TheHPXProject/hpx/issues/1800>

⁵³⁴² <https://github.com/TheHPXProject/hpx/pull/1799>

⁵³⁴³ <https://github.com/TheHPXProject/hpx/pull/1798>

⁵³⁴⁴ <https://github.com/TheHPXProject/hpx/pull/1797>

⁵³⁴⁵ <https://github.com/TheHPXProject/hpx/pull/1795>

⁵³⁴⁶ <https://github.com/TheHPXProject/hpx/pull/1794>

⁵³⁴⁷ <https://github.com/TheHPXProject/hpx/pull/1793>

⁵³⁴⁸ <https://github.com/TheHPXProject/hpx/pull/1792>

⁵³⁴⁹ <https://github.com/TheHPXProject/hpx/pull/1791>

⁵³⁵⁰ <https://github.com/TheHPXProject/hpx/issues/1790>

⁵³⁵¹ <https://github.com/TheHPXProject/hpx/pull/1789>

⁵³⁵² <https://github.com/TheHPXProject/hpx/issues/1788>

⁵³⁵³ <https://github.com/TheHPXProject/hpx/issues/1787>

⁵³⁵⁴ <https://github.com/TheHPXProject/hpx/pull/1786>

⁵³⁵⁵ <https://github.com/TheHPXProject/hpx/pull/1785>

⁵³⁵⁶ <https://github.com/TheHPXProject/hpx/pull/1784>

⁵³⁵⁷ <https://github.com/TheHPXProject/hpx/pull/1783>

⁵³⁵⁸ <https://github.com/TheHPXProject/hpx/pull/1782>

⁵³⁵⁹ <https://github.com/TheHPXProject/hpx/pull/1781>

⁵³⁶⁰ <https://github.com/TheHPXProject/hpx/pull/1780>

⁵³⁶¹ <https://github.com/TheHPXProject/hpx/pull/1779>

- PR #1778⁵³⁶² - Removing ability to enable/disable parcel handling
- PR #1777⁵³⁶³ - Completely removing stackless threads
- PR #1776⁵³⁶⁴ - Cleaning up util/plugin
- PR #1775⁵³⁶⁵ - Agas fixes
- PR #1774⁵³⁶⁶ - Action invocation count
- PR #1773⁵³⁶⁷ - replaced MSVC variable with WIN32
- PR #1772⁵³⁶⁸ - Fixing Problems in MPI parcellport and future serialization.
- PR #1771⁵³⁶⁹ - Fixing intel 13 compiler errors related to variadic template template parameters for `lcos::when_` tests
- PR #1770⁵³⁷⁰ - Forwarding decay to `std::`
- PR #1769⁵³⁷¹ - Add more characters with special regex meaning to the existing patch
- PR #1768⁵³⁷² - Adding test for `receive_buffer`
- PR #1767⁵³⁷³ - Making sure that uptime counter throws exception on any attempt to be reset
- PR #1766⁵³⁷⁴ - Cleaning up code related to throttling scheduler
- PR #1765⁵³⁷⁵ - Restricting `thread_data` to creating only with `intrusive_pointers`
- PR #1764⁵³⁷⁶ - Fixing 1763
- Issue #1763⁵³⁷⁷ - UB in `thread_data::operator delete`
- PR #1762⁵³⁷⁸ - Making sure all serialization registries/factories are unique
- PR #1761⁵³⁷⁹ - Fixed #1751: `hpx::future::wait_for` fails a simple test
- PR #1758⁵³⁸⁰ - Fixing #1757
- Issue #1757⁵³⁸¹ - pinning not correct using `-hpx:bind`
- Issue #1756⁵³⁸² - compilation error with MinGW
- PR #1755⁵³⁸³ - Making output serialization const-correct
- Issue #1753⁵³⁸⁴ - HPX performance degrades with time since execution begins

⁵³⁶² <https://github.com/TheHPXProject/hpx/pull/1778>

⁵³⁶³ <https://github.com/TheHPXProject/hpx/pull/1777>

⁵³⁶⁴ <https://github.com/TheHPXProject/hpx/pull/1776>

⁵³⁶⁵ <https://github.com/TheHPXProject/hpx/pull/1775>

⁵³⁶⁶ <https://github.com/TheHPXProject/hpx/pull/1774>

⁵³⁶⁷ <https://github.com/TheHPXProject/hpx/pull/1773>

⁵³⁶⁸ <https://github.com/TheHPXProject/hpx/pull/1772>

⁵³⁶⁹ <https://github.com/TheHPXProject/hpx/pull/1771>

⁵³⁷⁰ <https://github.com/TheHPXProject/hpx/pull/1770>

⁵³⁷¹ <https://github.com/TheHPXProject/hpx/pull/1769>

⁵³⁷² <https://github.com/TheHPXProject/hpx/pull/1768>

⁵³⁷³ <https://github.com/TheHPXProject/hpx/pull/1767>

⁵³⁷⁴ <https://github.com/TheHPXProject/hpx/pull/1766>

⁵³⁷⁵ <https://github.com/TheHPXProject/hpx/pull/1765>

⁵³⁷⁶ <https://github.com/TheHPXProject/hpx/pull/1764>

⁵³⁷⁷ <https://github.com/TheHPXProject/hpx/issues/1763>

⁵³⁷⁸ <https://github.com/TheHPXProject/hpx/pull/1762>

⁵³⁷⁹ <https://github.com/TheHPXProject/hpx/pull/1761>

⁵³⁸⁰ <https://github.com/TheHPXProject/hpx/pull/1758>

⁵³⁸¹ <https://github.com/TheHPXProject/hpx/issues/1757>

⁵³⁸² <https://github.com/TheHPXProject/hpx/issues/1756>

⁵³⁸³ <https://github.com/TheHPXProject/hpx/pull/1755>

⁵³⁸⁴ <https://github.com/TheHPXProject/hpx/issues/1753>

- Issue #1752⁵³⁸⁵ - Error in AGAS
- Issue #1751⁵³⁸⁶ - `hpx::future::wait_for` fails a simple test
- PR #1750⁵³⁸⁷ - Removing `hpx_fwd.hpp` includes
- PR #1749⁵³⁸⁸ - Simplify `result_of` and friends
- PR #1747⁵³⁸⁹ - Removed superfluous code from `message_buffer.hpp`
- PR #1746⁵³⁹⁰ - Tuple dependencies
- Issue #1745⁵³⁹¹ - Broken when `_some` which takes iterators
- PR #1744⁵³⁹² - Refining archive interface
- PR #1743⁵³⁹³ - Fixing when `_all` when only a single future is passed
- PR #1742⁵³⁹⁴ - Config includes
- PR #1741⁵³⁹⁵ - Os executors
- Issue #1740⁵³⁹⁶ - `hpx::promise` has some problems
- PR #1739⁵³⁹⁷ - Parallel composition with generic containers
- Issue #1738⁵³⁹⁸ - After building program and successfully linking to a version of `hpx` `DHPX_DIR` seems to be ignored
- Issue #1737⁵³⁹⁹ - Uptime problems
- PR #1736⁵⁴⁰⁰ - added convenience `c-tor` and `begin()/end()` to `serialize_buffer`
- PR #1735⁵⁴⁰¹ - Config includes
- PR #1734⁵⁴⁰² - Fixed #1688: Add timer counters for `tfunc_total` and `exec_total`
- Issue #1733⁵⁴⁰³ - Add unit test for `hpx/lcos/local/receive_buffer.hpp`
- PR #1732⁵⁴⁰⁴ - Renaming `get_os_thread_count`
- PR #1731⁵⁴⁰⁵ - Basename registration
- Issue #1730⁵⁴⁰⁶ - Use after move of `thread_init_data`
- PR #1729⁵⁴⁰⁷ - Rewriting channel based on new gate component

⁵³⁸⁵ <https://github.com/TheHPXProject/hpx/issues/1752>

⁵³⁸⁶ <https://github.com/TheHPXProject/hpx/issues/1751>

⁵³⁸⁷ <https://github.com/TheHPXProject/hpx/pull/1750>

⁵³⁸⁸ <https://github.com/TheHPXProject/hpx/pull/1749>

⁵³⁸⁹ <https://github.com/TheHPXProject/hpx/pull/1747>

⁵³⁹⁰ <https://github.com/TheHPXProject/hpx/pull/1746>

⁵³⁹¹ <https://github.com/TheHPXProject/hpx/issues/1745>

⁵³⁹² <https://github.com/TheHPXProject/hpx/pull/1744>

⁵³⁹³ <https://github.com/TheHPXProject/hpx/pull/1743>

⁵³⁹⁴ <https://github.com/TheHPXProject/hpx/pull/1742>

⁵³⁹⁵ <https://github.com/TheHPXProject/hpx/pull/1741>

⁵³⁹⁶ <https://github.com/TheHPXProject/hpx/issues/1740>

⁵³⁹⁷ <https://github.com/TheHPXProject/hpx/pull/1739>

⁵³⁹⁸ <https://github.com/TheHPXProject/hpx/issues/1738>

⁵³⁹⁹ <https://github.com/TheHPXProject/hpx/issues/1737>

⁵⁴⁰⁰ <https://github.com/TheHPXProject/hpx/pull/1736>

⁵⁴⁰¹ <https://github.com/TheHPXProject/hpx/pull/1735>

⁵⁴⁰² <https://github.com/TheHPXProject/hpx/pull/1734>

⁵⁴⁰³ <https://github.com/TheHPXProject/hpx/issues/1733>

⁵⁴⁰⁴ <https://github.com/TheHPXProject/hpx/pull/1732>

⁵⁴⁰⁵ <https://github.com/TheHPXProject/hpx/pull/1731>

⁵⁴⁰⁶ <https://github.com/TheHPXProject/hpx/issues/1730>

⁵⁴⁰⁷ <https://github.com/TheHPXProject/hpx/pull/1729>

- PR #1728⁵⁴⁰⁸ - Fixing #1722
- PR #1727⁵⁴⁰⁹ - Fixing compile problems with `apply_collocated`
- PR #1726⁵⁴¹⁰ - Apex integration
- PR #1725⁵⁴¹¹ - fixed test timeouts
- PR #1724⁵⁴¹² - Renaming vector
- Issue #1723⁵⁴¹³ - Drop support for intel compilers and gcc 4.4. based standard libs
- Issue #1722⁵⁴¹⁴ - Add support for detecting non-ready futures before serialization
- PR #1721⁵⁴¹⁵ - Unifying parallel executors, initializing from launch policy
- PR #1720⁵⁴¹⁶ - dropped superfluous typedef
- Issue #1718⁵⁴¹⁷ - Windows 10 x64, VS 2015 - Unknown CMake command “`add_hpx_pseudo_target`”.
- PR #1717⁵⁴¹⁸ - Timed executor traits for thread-executors
- PR #1716⁵⁴¹⁹ - serialization of arrays didn’t work with non-pod types. fixed
- PR #1715⁵⁴²⁰ - List serialization
- PR #1714⁵⁴²¹ - changing misspellings
- PR #1713⁵⁴²² - Fixed distribution policy executors
- PR #1712⁵⁴²³ - Moving library detection to be executed after feature tests
- PR #1711⁵⁴²⁴ - Simplify parcel
- PR #1710⁵⁴²⁵ - Compile only tests
- PR #1709⁵⁴²⁶ - Implemented timed executors
- PR #1708⁵⁴²⁷ - Implement `parallel::executor_traits` for thread-executors
- PR #1707⁵⁴²⁸ - Various fixes to `threads::executors` to make custom schedulers work
- PR #1706⁵⁴²⁹ - Command line option `-hpx:cores` does not work as expected
- Issue #1705⁵⁴³⁰ - command line option `-hpx:cores` does not work as expected

⁵⁴⁰⁸ <https://github.com/TheHPXProject/hpx/pull/1728>
⁵⁴⁰⁹ <https://github.com/TheHPXProject/hpx/pull/1727>
⁵⁴¹⁰ <https://github.com/TheHPXProject/hpx/pull/1726>
⁵⁴¹¹ <https://github.com/TheHPXProject/hpx/pull/1725>
⁵⁴¹² <https://github.com/TheHPXProject/hpx/pull/1724>
⁵⁴¹³ <https://github.com/TheHPXProject/hpx/issues/1723>
⁵⁴¹⁴ <https://github.com/TheHPXProject/hpx/issues/1722>
⁵⁴¹⁵ <https://github.com/TheHPXProject/hpx/pull/1721>
⁵⁴¹⁶ <https://github.com/TheHPXProject/hpx/pull/1720>
⁵⁴¹⁷ <https://github.com/TheHPXProject/hpx/issues/1718>
⁵⁴¹⁸ <https://github.com/TheHPXProject/hpx/pull/1717>
⁵⁴¹⁹ <https://github.com/TheHPXProject/hpx/pull/1716>
⁵⁴²⁰ <https://github.com/TheHPXProject/hpx/pull/1715>
⁵⁴²¹ <https://github.com/TheHPXProject/hpx/pull/1714>
⁵⁴²² <https://github.com/TheHPXProject/hpx/pull/1713>
⁵⁴²³ <https://github.com/TheHPXProject/hpx/pull/1712>
⁵⁴²⁴ <https://github.com/TheHPXProject/hpx/pull/1711>
⁵⁴²⁵ <https://github.com/TheHPXProject/hpx/pull/1710>
⁵⁴²⁶ <https://github.com/TheHPXProject/hpx/pull/1709>
⁵⁴²⁷ <https://github.com/TheHPXProject/hpx/pull/1708>
⁵⁴²⁸ <https://github.com/TheHPXProject/hpx/pull/1707>
⁵⁴²⁹ <https://github.com/TheHPXProject/hpx/pull/1706>
⁵⁴³⁰ <https://github.com/TheHPXProject/hpx/issues/1705>

- PR #1704⁵⁴³¹ - vector deserialization is speeded up a little
- PR #1703⁵⁴³² - Fixing shared_mutes
- Issue #1702⁵⁴³³ - Shared_mutex does not compile with no_mutex cond_var
- PR #1701⁵⁴³⁴ - Add distribution_policy_executor
- PR #1700⁵⁴³⁵ - Executor parameters
- PR #1699⁵⁴³⁶ - Readers writer lock
- PR #1698⁵⁴³⁷ - Remove leftovers
- PR #1697⁵⁴³⁸ - Fixing held locks
- PR #1696⁵⁴³⁹ - Modified Scan Partitioner for Algorithms
- PR #1695⁵⁴⁴⁰ - This thread executors
- PR #1694⁵⁴⁴¹ - Fixed #1688: Add timer counters for tfunc_total and exec_total
- PR #1693⁵⁴⁴² - Fix #1691: is_executor template specification fails for inherited executors
- PR #1692⁵⁴⁴³ - Fixed #1662: Possible exception source in coalescing_message_handler
- Issue #1691⁵⁴⁴⁴ - is_executor template specification fails for inherited executors
- PR #1690⁵⁴⁴⁵ - added macro for non-intrusive serialization of classes without a default c-tor
- PR #1689⁵⁴⁴⁶ - Replace value_or_error with custom storage, unify future_data state
- Issue #1688⁵⁴⁴⁷ - Add timer counters for tfunc_total and exec_total
- PR #1687⁵⁴⁴⁸ - Fixed interval timer
- PR #1686⁵⁴⁴⁹ - Fixing cmake warnings about not existing pseudo target dependencies
- PR #1685⁵⁴⁵⁰ - Converting partitioners to use bulk async execute
- PR #1683⁵⁴⁵¹ - Adds a tool for inspect that checks for character limits
- PR #1682⁵⁴⁵² - Change project name to (uppercase) HPX
- PR #1681⁵⁴⁵³ - Counter shortnames

⁵⁴³¹ <https://github.com/TheHPXProject/hpx/pull/1704>

⁵⁴³² <https://github.com/TheHPXProject/hpx/pull/1703>

⁵⁴³³ <https://github.com/TheHPXProject/hpx/issues/1702>

⁵⁴³⁴ <https://github.com/TheHPXProject/hpx/pull/1701>

⁵⁴³⁵ <https://github.com/TheHPXProject/hpx/pull/1700>

⁵⁴³⁶ <https://github.com/TheHPXProject/hpx/pull/1699>

⁵⁴³⁷ <https://github.com/TheHPXProject/hpx/pull/1698>

⁵⁴³⁸ <https://github.com/TheHPXProject/hpx/pull/1697>

⁵⁴³⁹ <https://github.com/TheHPXProject/hpx/pull/1696>

⁵⁴⁴⁰ <https://github.com/TheHPXProject/hpx/pull/1695>

⁵⁴⁴¹ <https://github.com/TheHPXProject/hpx/pull/1694>

⁵⁴⁴² <https://github.com/TheHPXProject/hpx/pull/1693>

⁵⁴⁴³ <https://github.com/TheHPXProject/hpx/pull/1692>

⁵⁴⁴⁴ <https://github.com/TheHPXProject/hpx/issues/1691>

⁵⁴⁴⁵ <https://github.com/TheHPXProject/hpx/pull/1690>

⁵⁴⁴⁶ <https://github.com/TheHPXProject/hpx/pull/1689>

⁵⁴⁴⁷ <https://github.com/TheHPXProject/hpx/issues/1688>

⁵⁴⁴⁸ <https://github.com/TheHPXProject/hpx/pull/1687>

⁵⁴⁴⁹ <https://github.com/TheHPXProject/hpx/pull/1686>

⁵⁴⁵⁰ <https://github.com/TheHPXProject/hpx/pull/1685>

⁵⁴⁵¹ <https://github.com/TheHPXProject/hpx/pull/1683>

⁵⁴⁵² <https://github.com/TheHPXProject/hpx/pull/1682>

⁵⁴⁵³ <https://github.com/TheHPXProject/hpx/pull/1681>

- PR #1680⁵⁴⁵⁴ - Extended Non-intrusive Serialization to Ease Usage for Library Developers
- PR #1679⁵⁴⁵⁵ - Working on 1544: More executor changes
- PR #1678⁵⁴⁵⁶ - Transpose fixes
- PR #1677⁵⁴⁵⁷ - Improve Boost compatibility check
- PR #1676⁵⁴⁵⁸ - 1d stencil fix
- Issue #1675⁵⁴⁵⁹ - hpx project name is not HPX
- PR #1674⁵⁴⁶⁰ - Fixing the MPI parcellport
- PR #1673⁵⁴⁶¹ - added move semantics to map/vector deserialization
- PR #1672⁵⁴⁶² - Vs2015 await
- PR #1671⁵⁴⁶³ - Adapt transform for #1668
- PR #1670⁵⁴⁶⁴ - Started to work on #1668
- PR #1669⁵⁴⁶⁵ - Add this_thread_executors
- Issue #1667⁵⁴⁶⁶ - Apple build instructions in docs are out of date
- PR #1666⁵⁴⁶⁷ - Apex integration
- PR #1665⁵⁴⁶⁸ - Fixes an error with the whitespace check that showed the incorrect location of the error
- Issue #1664⁵⁴⁶⁹ - Inspect tool found incorrect endline whitespace
- PR #1663⁵⁴⁷⁰ - Improve use of locks
- Issue #1662⁵⁴⁷¹ - Possible exception source in coalescing_message_handler
- PR #1661⁵⁴⁷² - Added support for 128bit number serialization
- PR #1660⁵⁴⁷³ - Serialization 128bits
- PR #1659⁵⁴⁷⁴ - Implemented inner_product and adjacent_diff algos
- PR #1658⁵⁴⁷⁵ - Add serialization for std::set (as there is for std::vector and std::map)
- PR #1657⁵⁴⁷⁶ - Use of shared_ptr in io_service_pool changed to unique_ptr

⁵⁴⁵⁴ <https://github.com/TheHPXProject/hpx/pull/1680>
⁵⁴⁵⁵ <https://github.com/TheHPXProject/hpx/pull/1679>
⁵⁴⁵⁶ <https://github.com/TheHPXProject/hpx/pull/1678>
⁵⁴⁵⁷ <https://github.com/TheHPXProject/hpx/pull/1677>
⁵⁴⁵⁸ <https://github.com/TheHPXProject/hpx/pull/1676>
⁵⁴⁵⁹ <https://github.com/TheHPXProject/hpx/issues/1675>
⁵⁴⁶⁰ <https://github.com/TheHPXProject/hpx/pull/1674>
⁵⁴⁶¹ <https://github.com/TheHPXProject/hpx/pull/1673>
⁵⁴⁶² <https://github.com/TheHPXProject/hpx/pull/1672>
⁵⁴⁶³ <https://github.com/TheHPXProject/hpx/pull/1671>
⁵⁴⁶⁴ <https://github.com/TheHPXProject/hpx/pull/1670>
⁵⁴⁶⁵ <https://github.com/TheHPXProject/hpx/pull/1669>
⁵⁴⁶⁶ <https://github.com/TheHPXProject/hpx/issues/1667>
⁵⁴⁶⁷ <https://github.com/TheHPXProject/hpx/pull/1666>
⁵⁴⁶⁸ <https://github.com/TheHPXProject/hpx/pull/1665>
⁵⁴⁶⁹ <https://github.com/TheHPXProject/hpx/issues/1664>
⁵⁴⁷⁰ <https://github.com/TheHPXProject/hpx/pull/1663>
⁵⁴⁷¹ <https://github.com/TheHPXProject/hpx/issues/1662>
⁵⁴⁷² <https://github.com/TheHPXProject/hpx/pull/1661>
⁵⁴⁷³ <https://github.com/TheHPXProject/hpx/pull/1660>
⁵⁴⁷⁴ <https://github.com/TheHPXProject/hpx/pull/1659>
⁵⁴⁷⁵ <https://github.com/TheHPXProject/hpx/pull/1658>
⁵⁴⁷⁶ <https://github.com/TheHPXProject/hpx/pull/1657>

- Issue #1656⁵⁴⁷⁷ - 1d_stencil codes all have wrong factor
- PR #1654⁵⁴⁷⁸ - When using runtime_mode_connect, find the correct localhost public ip address
- PR #1653⁵⁴⁷⁹ - Fixing 1617
- PR #1652⁵⁴⁸⁰ - Remove traits::action_may_require_id_splitting
- PR #1651⁵⁴⁸¹ - Fixed performance counters related to AGAS cache timings
- PR #1650⁵⁴⁸² - Remove leftovers of traits::type_size
- PR #1649⁵⁴⁸³ - Shorten target names on Windows to shorten used path names
- PR #1648⁵⁴⁸⁴ - Fixing problems introduced by merging #1623 for older compilers
- PR #1647⁵⁴⁸⁵ - Simplify running automatic builds on Windows
- Issue #1646⁵⁴⁸⁶ - Cache insert and update performance counters are broken
- Issue #1644⁵⁴⁸⁷ - Remove leftovers of traits::type_size
- Issue #1643⁵⁴⁸⁸ - Remove traits::action_may_require_id_splitting
- PR #1642⁵⁴⁸⁹ - Adds spell checker to the inspect tool for qbk and doxygen comments
- PR #1640⁵⁴⁹⁰ - First step towards fixing 688
- PR #1639⁵⁴⁹¹ - Re-apply remaining changes from limit_dataflow_recursion branch
- PR #1638⁵⁴⁹² - This fixes possible deadlock in the test ignore_while_locked_1485
- PR #1637⁵⁴⁹³ - Fixing hpx::wait_all() invoked with two vector<future<T>>
- PR #1636⁵⁴⁹⁴ - Partially re-apply changes from limit_dataflow_recursion branch
- PR #1635⁵⁴⁹⁵ - Adding missing test for #1572
- PR #1634⁵⁴⁹⁶ - Revert “Limit recursion-depth in dataflow to a configurable constant”
- PR #1633⁵⁴⁹⁷ - Add command line option to ignore batch environment
- PR #1631⁵⁴⁹⁸ - hpx::lcos::queue exhibits strange behavior
- PR #1630⁵⁴⁹⁹ - Fixed endline_whitespace_check.cpp to detect lines with only whitespace

⁵⁴⁷⁷ <https://github.com/TheHPXProject/hpx/issues/1656>

⁵⁴⁷⁸ <https://github.com/TheHPXProject/hpx/pull/1654>

⁵⁴⁷⁹ <https://github.com/TheHPXProject/hpx/pull/1653>

⁵⁴⁸⁰ <https://github.com/TheHPXProject/hpx/pull/1652>

⁵⁴⁸¹ <https://github.com/TheHPXProject/hpx/pull/1651>

⁵⁴⁸² <https://github.com/TheHPXProject/hpx/pull/1650>

⁵⁴⁸³ <https://github.com/TheHPXProject/hpx/pull/1649>

⁵⁴⁸⁴ <https://github.com/TheHPXProject/hpx/pull/1648>

⁵⁴⁸⁵ <https://github.com/TheHPXProject/hpx/pull/1647>

⁵⁴⁸⁶ <https://github.com/TheHPXProject/hpx/issues/1646>

⁵⁴⁸⁷ <https://github.com/TheHPXProject/hpx/issues/1644>

⁵⁴⁸⁸ <https://github.com/TheHPXProject/hpx/issues/1643>

⁵⁴⁸⁹ <https://github.com/TheHPXProject/hpx/pull/1642>

⁵⁴⁹⁰ <https://github.com/TheHPXProject/hpx/pull/1640>

⁵⁴⁹¹ <https://github.com/TheHPXProject/hpx/pull/1639>

⁵⁴⁹² <https://github.com/TheHPXProject/hpx/pull/1638>

⁵⁴⁹³ <https://github.com/TheHPXProject/hpx/pull/1637>

⁵⁴⁹⁴ <https://github.com/TheHPXProject/hpx/pull/1636>

⁵⁴⁹⁵ <https://github.com/TheHPXProject/hpx/pull/1635>

⁵⁴⁹⁶ <https://github.com/TheHPXProject/hpx/pull/1634>

⁵⁴⁹⁷ <https://github.com/TheHPXProject/hpx/pull/1633>

⁵⁴⁹⁸ <https://github.com/TheHPXProject/hpx/pull/1631>

⁵⁴⁹⁹ <https://github.com/TheHPXProject/hpx/pull/1630>

- [Issue #1629⁵⁵⁰⁰](#) - Inspect trailing whitespace checker problem
- [PR #1628⁵⁵⁰¹](#) - Removed meaningless const qualifiers. Minor icpc fix.
- [PR #1627⁵⁵⁰²](#) - Fixing the queue LCO and add example demonstrating its use
- [PR #1626⁵⁵⁰³](#) - Deprecating `get_gid()`, add `get_id()` and `get_unmanaged_id()`
- [PR #1625⁵⁵⁰⁴](#) - Allowing to specify whether to send credits along with message
- [Issue #1624⁵⁵⁰⁵](#) - Lifetime issue
- [Issue #1623⁵⁵⁰⁶](#) - `hpx::wait_all()` invoked with two `vector<future<T>>` fails
- [PR #1622⁵⁵⁰⁷](#) - Executor partitioners
- [PR #1621⁵⁵⁰⁸](#) - Clean up coroutines implementation
- [Issue #1620⁵⁵⁰⁹](#) - Revert #1535
- [PR #1619⁵⁵¹⁰](#) - Fix result type calculation for `hpx::make_continuation`
- [PR #1618⁵⁵¹¹](#) - Fixing RDTSC on Xeon/Phi
- [Issue #1617⁵⁵¹²](#) - `hpx cmake` not working when run as a subproject
- [Issue #1616⁵⁵¹³](#) - `cmake` problem resulting in RDTSC not working correctly for Xeon Phi creates very strange results for duration counters
- [Issue #1615⁵⁵¹⁴](#) - `hpx::make_continuation` requires input and output to be the same
- [PR #1614⁵⁵¹⁵](#) - Fixed remove copy test
- [Issue #1613⁵⁵¹⁶](#) - Dataflow causes stack overflow
- [PR #1612⁵⁵¹⁷](#) - Modified foreach partitioner to use bulk execute
- [PR #1611⁵⁵¹⁸](#) - Limit recursion-depth in dataflow to a configurable constant
- [PR #1610⁵⁵¹⁹](#) - Increase timeout for CircleCI
- [PR #1609⁵⁵²⁰](#) - Refactoring thread manager, mainly extracting thread pool
- [PR #1608⁵⁵²¹](#) - Fixed running multiple localities without localities parameter
- [PR #1607⁵⁵²²](#) - More algorithm fixes to `adjacentfind`

⁵⁵⁰⁰ <https://github.com/TheHPXProject/hpx/issues/1629>

⁵⁵⁰¹ <https://github.com/TheHPXProject/hpx/pull/1628>

⁵⁵⁰² <https://github.com/TheHPXProject/hpx/pull/1627>

⁵⁵⁰³ <https://github.com/TheHPXProject/hpx/pull/1626>

⁵⁵⁰⁴ <https://github.com/TheHPXProject/hpx/pull/1625>

⁵⁵⁰⁵ <https://github.com/TheHPXProject/hpx/issues/1624>

⁵⁵⁰⁶ <https://github.com/TheHPXProject/hpx/issues/1623>

⁵⁵⁰⁷ <https://github.com/TheHPXProject/hpx/pull/1622>

⁵⁵⁰⁸ <https://github.com/TheHPXProject/hpx/pull/1621>

⁵⁵⁰⁹ <https://github.com/TheHPXProject/hpx/issues/1620>

⁵⁵¹⁰ <https://github.com/TheHPXProject/hpx/pull/1619>

⁵⁵¹¹ <https://github.com/TheHPXProject/hpx/pull/1618>

⁵⁵¹² <https://github.com/TheHPXProject/hpx/issues/1617>

⁵⁵¹³ <https://github.com/TheHPXProject/hpx/issues/1616>

⁵⁵¹⁴ <https://github.com/TheHPXProject/hpx/issues/1615>

⁵⁵¹⁵ <https://github.com/TheHPXProject/hpx/pull/1614>

⁵⁵¹⁶ <https://github.com/TheHPXProject/hpx/issues/1613>

⁵⁵¹⁷ <https://github.com/TheHPXProject/hpx/pull/1612>

⁵⁵¹⁸ <https://github.com/TheHPXProject/hpx/pull/1611>

⁵⁵¹⁹ <https://github.com/TheHPXProject/hpx/pull/1610>

⁵⁵²⁰ <https://github.com/TheHPXProject/hpx/pull/1609>

⁵⁵²¹ <https://github.com/TheHPXProject/hpx/pull/1608>

⁵⁵²² <https://github.com/TheHPXProject/hpx/pull/1607>

- [Issue #1606](#)⁵⁵²³ - Running without localities parameter binds to bogus port range
- [Issue #1605](#)⁵⁵²⁴ - Too many serializations
- [PR #1604](#)⁵⁵²⁵ - Changes the HPX image into a hyperlink
- [PR #1601](#)⁵⁵²⁶ - Fixing problems with remove_copy algorithm tests
- [PR #1600](#)⁵⁵²⁷ - Actions with ids cleanup
- [PR #1599](#)⁵⁵²⁸ - Duplicate binding of global ids should fail
- [PR #1598](#)⁵⁵²⁹ - Fixing array access
- [PR #1597](#)⁵⁵³⁰ - Improved the reliability of connecting/disconnecting localities
- [Issue #1596](#)⁵⁵³¹ - Duplicate id binding should fail
- [PR #1595](#)⁵⁵³² - Fixing more cmake config constants
- [PR #1594](#)⁵⁵³³ - Fixing preprocessor constant used to enable C++11 chrono
- [PR #1593](#)⁵⁵³⁴ - Adding operator|() for hpx::launch
- [Issue #1592](#)⁵⁵³⁵ - Error (typo) in the docs
- [Issue #1590](#)⁵⁵³⁶ - CMake fails when CMAKE_BINARY_DIR contains '+'
- [Issue #1589](#)⁵⁵³⁷ - Disconnecting a locality results in segfault using heartbeat example
- [PR #1588](#)⁵⁵³⁸ - Fix doc string for config option HPX_WITH_EXAMPLES
- [PR #1586](#)⁵⁵³⁹ - Fixing 1493
- [PR #1585](#)⁵⁵⁴⁰ - Additional Check for Inspect Tool to detect Endline Whitespace
- [Issue #1584](#)⁵⁵⁴¹ - Clean up coroutines implementation
- [PR #1583](#)⁵⁵⁴² - Adding a check for end line whitespace
- [PR #1582](#)⁵⁵⁴³ - Attempt to fix assert firing after scheduling loop was exited
- [PR #1581](#)⁵⁵⁴⁴ - Fixed adjacentfind_binary test
- [PR #1580](#)⁵⁵⁴⁵ - Prevent some of the internal cmake lists from growing indefinitely

⁵⁵²³ <https://github.com/TheHPXProject/hpx/issues/1606>

⁵⁵²⁴ <https://github.com/TheHPXProject/hpx/issues/1605>

⁵⁵²⁵ <https://github.com/TheHPXProject/hpx/pull/1604>

⁵⁵²⁶ <https://github.com/TheHPXProject/hpx/pull/1601>

⁵⁵²⁷ <https://github.com/TheHPXProject/hpx/pull/1600>

⁵⁵²⁸ <https://github.com/TheHPXProject/hpx/pull/1599>

⁵⁵²⁹ <https://github.com/TheHPXProject/hpx/pull/1598>

⁵⁵³⁰ <https://github.com/TheHPXProject/hpx/pull/1597>

⁵⁵³¹ <https://github.com/TheHPXProject/hpx/issues/1596>

⁵⁵³² <https://github.com/TheHPXProject/hpx/pull/1595>

⁵⁵³³ <https://github.com/TheHPXProject/hpx/pull/1594>

⁵⁵³⁴ <https://github.com/TheHPXProject/hpx/pull/1593>

⁵⁵³⁵ <https://github.com/TheHPXProject/hpx/issues/1592>

⁵⁵³⁶ <https://github.com/TheHPXProject/hpx/issues/1590>

⁵⁵³⁷ <https://github.com/TheHPXProject/hpx/issues/1589>

⁵⁵³⁸ <https://github.com/TheHPXProject/hpx/pull/1588>

⁵⁵³⁹ <https://github.com/TheHPXProject/hpx/pull/1586>

⁵⁵⁴⁰ <https://github.com/TheHPXProject/hpx/pull/1585>

⁵⁵⁴¹ <https://github.com/TheHPXProject/hpx/issues/1584>

⁵⁵⁴² <https://github.com/TheHPXProject/hpx/pull/1583>

⁵⁵⁴³ <https://github.com/TheHPXProject/hpx/pull/1582>

⁵⁵⁴⁴ <https://github.com/TheHPXProject/hpx/pull/1581>

⁵⁵⁴⁵ <https://github.com/TheHPXProject/hpx/pull/1580>

- [PR #1579](#)⁵⁵⁴⁶ - Removing type_size trait, replacing it with special archive type
- [Issue #1578](#)⁵⁵⁴⁷ - Remove demangle_helper
- [PR #1577](#)⁵⁵⁴⁸ - Get ptr problems
- [Issue #1576](#)⁵⁵⁴⁹ - Refactor async, dataflow, and future::then
- [PR #1575](#)⁵⁵⁵⁰ - Fixing tests for parallel rotate
- [PR #1574](#)⁵⁵⁵¹ - Cleaning up schedulers
- [PR #1573](#)⁵⁵⁵² - Fixing thread pool executor
- [PR #1572](#)⁵⁵⁵³ - Fixing number of configured localities
- [PR #1571](#)⁵⁵⁵⁴ - Reimplement decay
- [PR #1570](#)⁵⁵⁵⁵ - Refactoring async, apply, and dataflow APIs
- [PR #1569](#)⁵⁵⁵⁶ - Changed range for mach-o library lookup
- [PR #1568](#)⁵⁵⁵⁷ - Mark decltype support as required
- [PR #1567](#)⁵⁵⁵⁸ - Removed const from algorithms
- [Issue #1566](#)⁵⁵⁵⁹ - CMAKE Configuration Test Failures for clang 3.5 on debian
- [PR #1565](#)⁵⁵⁶⁰ - Dylib support
- [PR #1564](#)⁵⁵⁶¹ - Converted partitioners and some algorithms to use executors
- [PR #1563](#)⁵⁵⁶² - Fix several #includes for Boost.Preprocessor
- [PR #1562](#)⁵⁵⁶³ - Adding configuration option disabling/enabling all message handlers
- [PR #1561](#)⁵⁵⁶⁴ - Removed all occurrences of boost::move replacing it with std::move
- [Issue #1560](#)⁵⁵⁶⁵ - Leftover HPX_REGISTER_ACTION_DECLARATION_2
- [PR #1558](#)⁵⁵⁶⁶ - Revisit async/apply SFINAE conditions
- [PR #1557](#)⁵⁵⁶⁷ - Removing type_size trait, replacing it with special archive type
- [PR #1556](#)⁵⁵⁶⁸ - Executor algorithms

⁵⁵⁴⁶ <https://github.com/TheHPXProject/hpx/pull/1579>

⁵⁵⁴⁷ <https://github.com/TheHPXProject/hpx/issues/1578>

⁵⁵⁴⁸ <https://github.com/TheHPXProject/hpx/pull/1577>

⁵⁵⁴⁹ <https://github.com/TheHPXProject/hpx/issues/1576>

⁵⁵⁵⁰ <https://github.com/TheHPXProject/hpx/pull/1575>

⁵⁵⁵¹ <https://github.com/TheHPXProject/hpx/pull/1574>

⁵⁵⁵² <https://github.com/TheHPXProject/hpx/pull/1573>

⁵⁵⁵³ <https://github.com/TheHPXProject/hpx/pull/1572>

⁵⁵⁵⁴ <https://github.com/TheHPXProject/hpx/pull/1571>

⁵⁵⁵⁵ <https://github.com/TheHPXProject/hpx/pull/1570>

⁵⁵⁵⁶ <https://github.com/TheHPXProject/hpx/pull/1569>

⁵⁵⁵⁷ <https://github.com/TheHPXProject/hpx/pull/1568>

⁵⁵⁵⁸ <https://github.com/TheHPXProject/hpx/pull/1567>

⁵⁵⁵⁹ <https://github.com/TheHPXProject/hpx/issues/1566>

⁵⁵⁶⁰ <https://github.com/TheHPXProject/hpx/pull/1565>

⁵⁵⁶¹ <https://github.com/TheHPXProject/hpx/pull/1564>

⁵⁵⁶² <https://github.com/TheHPXProject/hpx/pull/1563>

⁵⁵⁶³ <https://github.com/TheHPXProject/hpx/pull/1562>

⁵⁵⁶⁴ <https://github.com/TheHPXProject/hpx/pull/1561>

⁵⁵⁶⁵ <https://github.com/TheHPXProject/hpx/issues/1560>

⁵⁵⁶⁶ <https://github.com/TheHPXProject/hpx/pull/1558>

⁵⁵⁶⁷ <https://github.com/TheHPXProject/hpx/pull/1557>

⁵⁵⁶⁸ <https://github.com/TheHPXProject/hpx/pull/1556>

- PR #1555⁵⁵⁶⁹ - Remove the necessity to specify archive flags on the receiving end
- PR #1554⁵⁵⁷⁰ - Removing obsolete Boost.Serialization macros
- PR #1553⁵⁵⁷¹ - Properly fix HPX_DEFINE_*_ACTION macros
- PR #1552⁵⁵⁷² - Fixed algorithms relying on copy_if implementation
- PR #1551⁵⁵⁷³ - Pxfs - Modifying FindOrangeFS.cmake based on OrangeFS 2.9.X
- Issue #1550⁵⁵⁷⁴ - Passing plain identifier inside HPX_DEFINE_PLAIN_ACTION_1
- PR #1549⁵⁵⁷⁵ - Fixing intel14/libstdc++4.4
- PR #1548⁵⁵⁷⁶ - Moving raw_ptr to detail namespace
- PR #1547⁵⁵⁷⁷ - Adding support for executors to future.then
- PR #1546⁵⁵⁷⁸ - Executor traits result types
- PR #1545⁵⁵⁷⁹ - Integrate executors with dataflow
- PR #1543⁵⁵⁸⁰ - Fix potential zero-copy for primarynamespace::bulk_service_async et.al.
- PR #1542⁵⁵⁸¹ - Merging HPX0.9.10 into pxfs branch
- PR #1541⁵⁵⁸² - Removed stale cmake tests, unused since the great cmake refactoring
- PR #1540⁵⁵⁸³ - Fix idle-rate on platforms without TSC
- PR #1539⁵⁵⁸⁴ - Reporting situation if zero-copy-serialization was performed by a parcel generated from a plain apply/async
- PR #1538⁵⁵⁸⁵ - Changed return type of bulk executors and added test
- Issue #1537⁵⁵⁸⁶ - Incorrect cpuid config tests
- PR #1536⁵⁵⁸⁷ - Changed return type of bulk executors and added test
- PR #1535⁵⁵⁸⁸ - Make sure promise::get_gid() can be called more than once
- PR #1534⁵⁵⁸⁹ - Fixed async_callback with bound callback
- PR #1533⁵⁵⁹⁰ - Updated the link in the documentation to a publically- accessible URL
- PR #1532⁵⁵⁹¹ - Make sure sync primitives are not copyable nor movable

⁵⁵⁶⁹ <https://github.com/TheHPXProject/hpx/pull/1555>
⁵⁵⁷⁰ <https://github.com/TheHPXProject/hpx/pull/1554>
⁵⁵⁷¹ <https://github.com/TheHPXProject/hpx/pull/1553>
⁵⁵⁷² <https://github.com/TheHPXProject/hpx/pull/1552>
⁵⁵⁷³ <https://github.com/TheHPXProject/hpx/pull/1551>
⁵⁵⁷⁴ <https://github.com/TheHPXProject/hpx/issues/1550>
⁵⁵⁷⁵ <https://github.com/TheHPXProject/hpx/pull/1549>
⁵⁵⁷⁶ <https://github.com/TheHPXProject/hpx/pull/1548>
⁵⁵⁷⁷ <https://github.com/TheHPXProject/hpx/pull/1547>
⁵⁵⁷⁸ <https://github.com/TheHPXProject/hpx/pull/1546>
⁵⁵⁷⁹ <https://github.com/TheHPXProject/hpx/pull/1545>
⁵⁵⁸⁰ <https://github.com/TheHPXProject/hpx/pull/1543>
⁵⁵⁸¹ <https://github.com/TheHPXProject/hpx/pull/1542>
⁵⁵⁸² <https://github.com/TheHPXProject/hpx/pull/1541>
⁵⁵⁸³ <https://github.com/TheHPXProject/hpx/pull/1540>
⁵⁵⁸⁴ <https://github.com/TheHPXProject/hpx/pull/1539>
⁵⁵⁸⁵ <https://github.com/TheHPXProject/hpx/pull/1538>
⁵⁵⁸⁶ <https://github.com/TheHPXProject/hpx/issues/1537>
⁵⁵⁸⁷ <https://github.com/TheHPXProject/hpx/pull/1536>
⁵⁵⁸⁸ <https://github.com/TheHPXProject/hpx/pull/1535>
⁵⁵⁸⁹ <https://github.com/TheHPXProject/hpx/pull/1534>
⁵⁵⁹⁰ <https://github.com/TheHPXProject/hpx/pull/1533>
⁵⁵⁹¹ <https://github.com/TheHPXProject/hpx/pull/1532>

- PR #1531⁵⁵⁹² - Fix unwrapped issue with future ranges of void type
- PR #1530⁵⁵⁹³ - Serialization complex
- Issue #1528⁵⁵⁹⁴ - Unwrapped issue with future<void>
- Issue #1527⁵⁵⁹⁵ - HPX does not build with Boost 1.58.0
- PR #1526⁵⁵⁹⁶ - Added support for boost.multi_array serialization
- PR #1525⁵⁵⁹⁷ - Properly handle deferred futures, fixes #1506
- PR #1524⁵⁵⁹⁸ - Making sure invalid action argument types generate clear error message
- Issue #1522⁵⁵⁹⁹ - Need serialization support for boost multi array
- Issue #1521⁵⁶⁰⁰ - Remote async and zero-copy serialization optimizations don't play well together
- PR #1520⁵⁶⁰¹ - Fixing UB while registering polymorphic classes for serialization
- PR #1519⁵⁶⁰² - Making detail::condition_variable safe to use
- PR #1518⁵⁶⁰³ - Fix when_some bug missing indices in its result
- Issue #1517⁵⁶⁰⁴ - Typo may affect CMake build system tests
- PR #1516⁵⁶⁰⁵ - Fixing Posix context
- PR #1515⁵⁶⁰⁶ - Fixing Posix context
- PR #1514⁵⁶⁰⁷ - Correct problems with loading dynamic components
- PR #1513⁵⁶⁰⁸ - Fixing intel glibc4 4
- Issue #1508⁵⁶⁰⁹ - memory and papi counters do not work
- Issue #1507⁵⁶¹⁰ - Unrecognized Command Line Option Error causing exit status 0
- Issue #1506⁵⁶¹¹ - Properly handle deferred futures
- PR #1505⁵⁶¹² - Adding #include - would not compile without this
- Issue #1502⁵⁶¹³ - boost::filesystem::exists throws unexpected exception
- Issue #1501⁵⁶¹⁴ - hwloc configuration options are wrong for MIC

⁵⁵⁹² <https://github.com/TheHPXProject/hpx/pull/1531>

⁵⁵⁹³ <https://github.com/TheHPXProject/hpx/pull/1530>

⁵⁵⁹⁴ <https://github.com/TheHPXProject/hpx/issues/1528>

⁵⁵⁹⁵ <https://github.com/TheHPXProject/hpx/issues/1527>

⁵⁵⁹⁶ <https://github.com/TheHPXProject/hpx/pull/1526>

⁵⁵⁹⁷ <https://github.com/TheHPXProject/hpx/pull/1525>

⁵⁵⁹⁸ <https://github.com/TheHPXProject/hpx/pull/1524>

⁵⁵⁹⁹ <https://github.com/TheHPXProject/hpx/issues/1522>

⁵⁶⁰⁰ <https://github.com/TheHPXProject/hpx/issues/1521>

⁵⁶⁰¹ <https://github.com/TheHPXProject/hpx/pull/1520>

⁵⁶⁰² <https://github.com/TheHPXProject/hpx/pull/1519>

⁵⁶⁰³ <https://github.com/TheHPXProject/hpx/pull/1518>

⁵⁶⁰⁴ <https://github.com/TheHPXProject/hpx/issues/1517>

⁵⁶⁰⁵ <https://github.com/TheHPXProject/hpx/pull/1516>

⁵⁶⁰⁶ <https://github.com/TheHPXProject/hpx/pull/1515>

⁵⁶⁰⁷ <https://github.com/TheHPXProject/hpx/pull/1514>

⁵⁶⁰⁸ <https://github.com/TheHPXProject/hpx/pull/1513>

⁵⁶⁰⁹ <https://github.com/TheHPXProject/hpx/issues/1508>

⁵⁶¹⁰ <https://github.com/TheHPXProject/hpx/issues/1507>

⁵⁶¹¹ <https://github.com/TheHPXProject/hpx/issues/1506>

⁵⁶¹² <https://github.com/TheHPXProject/hpx/pull/1505>

⁵⁶¹³ <https://github.com/TheHPXProject/hpx/issues/1502>

⁵⁶¹⁴ <https://github.com/TheHPXProject/hpx/issues/1501>

- PR #1504⁵⁶¹⁵ - Making sure `boost::filesystem::exists()` does not throw
- PR #1500⁵⁶¹⁶ - Exit application on `--hpx:version/-v` and `--hpx:info`
- PR #1498⁵⁶¹⁷ - Extended task block
- PR #1497⁵⁶¹⁸ - Unique ptr serialization
- PR #1496⁵⁶¹⁹ - Unique ptr serialization (closed)
- PR #1495⁵⁶²⁰ - Switching `circleci` build type to debug
- Issue #1494⁵⁶²¹ - `--hpx:version/-v` does not exit after printing version information
- Issue #1493⁵⁶²² - add an `hpx_` prefix to libraries and components to avoid name conflicts
- Issue #1492⁵⁶²³ - Define and ensure limitations for arguments to `async/apply`
- PR #1489⁵⁶²⁴ - Enable idle rate counter on demand
- PR #1488⁵⁶²⁵ - Made sure `detail::condition_variable` can be safely destroyed
- PR #1487⁵⁶²⁶ - Introduced default (main) template implementation for `ignore_while_checking`
- PR #1486⁵⁶²⁷ - Add HPX inspect tool
- Issue #1485⁵⁶²⁸ - `ignore_while_locked` doesn't support all Lockable types
- PR #1484⁵⁶²⁹ - Docker image generation
- PR #1483⁵⁶³⁰ - Move external endian library into HPX
- PR #1482⁵⁶³¹ - Actions with integer type ids
- Issue #1481⁵⁶³² - Sync primitives safe destruction
- Issue #1480⁵⁶³³ - Move external/boost/endian into `hpx/util`
- Issue #1478⁵⁶³⁴ - Boost inspect violations
- PR #1479⁵⁶³⁵ - Adds serialization for arrays; some further/minor fixes
- PR #1477⁵⁶³⁶ - Fixing problems with the Intel compiler using a GCC 4.4 std library
- PR #1476⁵⁶³⁷ - Adding `hpx::lcos::latch` and `hpx::lcos::local::latch`

⁵⁶¹⁵ <https://github.com/TheHPXProject/hpx/pull/1504>

⁵⁶¹⁶ <https://github.com/TheHPXProject/hpx/pull/1500>

⁵⁶¹⁷ <https://github.com/TheHPXProject/hpx/pull/1498>

⁵⁶¹⁸ <https://github.com/TheHPXProject/hpx/pull/1497>

⁵⁶¹⁹ <https://github.com/TheHPXProject/hpx/pull/1496>

⁵⁶²⁰ <https://github.com/TheHPXProject/hpx/pull/1495>

⁵⁶²¹ <https://github.com/TheHPXProject/hpx/issues/1494>

⁵⁶²² <https://github.com/TheHPXProject/hpx/issues/1493>

⁵⁶²³ <https://github.com/TheHPXProject/hpx/issues/1492>

⁵⁶²⁴ <https://github.com/TheHPXProject/hpx/pull/1489>

⁵⁶²⁵ <https://github.com/TheHPXProject/hpx/pull/1488>

⁵⁶²⁶ <https://github.com/TheHPXProject/hpx/pull/1487>

⁵⁶²⁷ <https://github.com/TheHPXProject/hpx/pull/1486>

⁵⁶²⁸ <https://github.com/TheHPXProject/hpx/issues/1485>

⁵⁶²⁹ <https://github.com/TheHPXProject/hpx/pull/1484>

⁵⁶³⁰ <https://github.com/TheHPXProject/hpx/pull/1483>

⁵⁶³¹ <https://github.com/TheHPXProject/hpx/pull/1482>

⁵⁶³² <https://github.com/TheHPXProject/hpx/issues/1481>

⁵⁶³³ <https://github.com/TheHPXProject/hpx/issues/1480>

⁵⁶³⁴ <https://github.com/TheHPXProject/hpx/issues/1478>

⁵⁶³⁵ <https://github.com/TheHPXProject/hpx/pull/1479>

⁵⁶³⁶ <https://github.com/TheHPXProject/hpx/pull/1477>

⁵⁶³⁷ <https://github.com/TheHPXProject/hpx/pull/1476>

- [Issue #1475](#)⁵⁶³⁸ - Boost inspect violations
- [PR #1473](#)⁵⁶³⁹ - Fixing action move tests
- [Issue #1471](#)⁵⁶⁴⁰ - Sync primitives should not be movable
- [PR #1470](#)⁵⁶⁴¹ - Removing `hpx::util::polymorphic_factory`
- [PR #1468](#)⁵⁶⁴² - Fixed container creation
- [Issue #1467](#)⁵⁶⁴³ - HPX application fail during finalization
- [Issue #1466](#)⁵⁶⁴⁴ - HPX doesn't pick up Torque's nodefile on SuperMIC
- [Issue #1464](#)⁵⁶⁴⁵ - HPX option for pre and post bootstrap performance counters
- [PR #1463](#)⁵⁶⁴⁶ - Replacing `async_colocated(id, ...)` with `async(colocated(id), ...)`
- [PR #1462](#)⁵⁶⁴⁷ - Consolidated `task_region` with N4411
- [PR #1461](#)⁵⁶⁴⁸ - Consolidate inconsistent CMake option names
- [Issue #1460](#)⁵⁶⁴⁹ - Which malloc is actually used? or at least which one is HPX built with
- [Issue #1459](#)⁵⁶⁵⁰ - Make cmake configure step fail explicitly if compiler version is not supported
- [Issue #1458](#)⁵⁶⁵¹ - Update `parallel::task_region` with N4411
- [PR #1456](#)⁵⁶⁵² - Consolidating `new_<>()`
- [Issue #1455](#)⁵⁶⁵³ - Replace `async_colocated(id, ...)` with `async(colocated(id), ...)`
- [PR #1454](#)⁵⁶⁵⁴ - Removed harmful `std::moves` from return statements
- [PR #1453](#)⁵⁶⁵⁵ - Use range-based for-loop instead of `Boost.Foreach`
- [PR #1452](#)⁵⁶⁵⁶ - C++ feature tests
- [PR #1451](#)⁵⁶⁵⁷ - When serializing, pass archive flags to `traits::get_type_size`
- [Issue #1450](#)⁵⁶⁵⁸ - `traits::get_type_size` needs archive flags to enable `zero_copy` optimizations
- [Issue #1449](#)⁵⁶⁵⁹ - "couldn't create performance counter" - AGAS
- [Issue #1448](#)⁵⁶⁶⁰ - Replace distributing factories with `new_<T[]>(...)`

⁵⁶³⁸ <https://github.com/TheHPXProject/hpx/issues/1475>

⁵⁶³⁹ <https://github.com/TheHPXProject/hpx/pull/1473>

⁵⁶⁴⁰ <https://github.com/TheHPXProject/hpx/issues/1471>

⁵⁶⁴¹ <https://github.com/TheHPXProject/hpx/pull/1470>

⁵⁶⁴² <https://github.com/TheHPXProject/hpx/pull/1468>

⁵⁶⁴³ <https://github.com/TheHPXProject/hpx/issues/1467>

⁵⁶⁴⁴ <https://github.com/TheHPXProject/hpx/issues/1466>

⁵⁶⁴⁵ <https://github.com/TheHPXProject/hpx/issues/1464>

⁵⁶⁴⁶ <https://github.com/TheHPXProject/hpx/pull/1463>

⁵⁶⁴⁷ <https://github.com/TheHPXProject/hpx/pull/1462>

⁵⁶⁴⁸ <https://github.com/TheHPXProject/hpx/pull/1461>

⁵⁶⁴⁹ <https://github.com/TheHPXProject/hpx/issues/1460>

⁵⁶⁵⁰ <https://github.com/TheHPXProject/hpx/issues/1459>

⁵⁶⁵¹ <https://github.com/TheHPXProject/hpx/issues/1458>

⁵⁶⁵² <https://github.com/TheHPXProject/hpx/pull/1456>

⁵⁶⁵³ <https://github.com/TheHPXProject/hpx/issues/1455>

⁵⁶⁵⁴ <https://github.com/TheHPXProject/hpx/pull/1454>

⁵⁶⁵⁵ <https://github.com/TheHPXProject/hpx/pull/1453>

⁵⁶⁵⁶ <https://github.com/TheHPXProject/hpx/pull/1452>

⁵⁶⁵⁷ <https://github.com/TheHPXProject/hpx/pull/1451>

⁵⁶⁵⁸ <https://github.com/TheHPXProject/hpx/issues/1450>

⁵⁶⁵⁹ <https://github.com/TheHPXProject/hpx/issues/1449>

⁵⁶⁶⁰ <https://github.com/TheHPXProject/hpx/issues/1448>

- PR #1447⁵⁶⁶¹ - Removing obsolete remote_object component
- PR #1446⁵⁶⁶² - Hpx serialization
- PR #1445⁵⁶⁶³ - Replacing travis with circleci
- PR #1443⁵⁶⁶⁴ - Always stripping HPX command line arguments before executing start function
- PR #1442⁵⁶⁶⁵ - Adding `-hpx:bind=none` to disable thread affinities
- Issue #1439⁵⁶⁶⁶ - Libraries get linked in multiple times, RPATH is not properly set
- PR #1438⁵⁶⁶⁷ - Removed superfluous typedefs
- Issue #1437⁵⁶⁶⁸ - `hpx::init()` should strip HPX-related flags from argv
- Issue #1436⁵⁶⁶⁹ - Add strong scaling option to https
- PR #1435⁵⁶⁷⁰ - Adding `async_cb`, `async_continue_cb`, and `async_colocated_cb`
- PR #1434⁵⁶⁷¹ - Added missing install rule, removed some dead CMake code
- PR #1433⁵⁶⁷² - Add GitExternal and SubProject cmake scripts from eyescale/cmake repo
- Issue #1432⁵⁶⁷³ - Add command line flag to disable thread pinning
- PR #1431⁵⁶⁷⁴ - Fix #1423
- Issue #1430⁵⁶⁷⁵ - Inconsistent CMake option names
- Issue #1429⁵⁶⁷⁶ - Configure setting `HPX_HAVE_PARCELPOR_MPI` is ignored
- PR #1428⁵⁶⁷⁷ - Fixes #1419 (closed)
- PR #1427⁵⁶⁷⁸ - Adding `stencil_iterator` and `transform_iterator`
- PR #1426⁵⁶⁷⁹ - Fixes #1419
- PR #1425⁵⁶⁸⁰ - During serialization memory allocation should honour allocator chunk size
- Issue #1424⁵⁶⁸¹ - chunk allocation during serialization does not use memory pool/allocator chunk size
- Issue #1423⁵⁶⁸² - Remove `HPX_STD_UNIQUE_PTR`
- Issue #1422⁵⁶⁸³ - `hpx:threads=all` allocates too many os threads

⁵⁶⁶¹ <https://github.com/TheHPXProject/hpx/pull/1447>

⁵⁶⁶² <https://github.com/TheHPXProject/hpx/pull/1446>

⁵⁶⁶³ <https://github.com/TheHPXProject/hpx/pull/1445>

⁵⁶⁶⁴ <https://github.com/TheHPXProject/hpx/pull/1443>

⁵⁶⁶⁵ <https://github.com/TheHPXProject/hpx/pull/1442>

⁵⁶⁶⁶ <https://github.com/TheHPXProject/hpx/issues/1439>

⁵⁶⁶⁷ <https://github.com/TheHPXProject/hpx/pull/1438>

⁵⁶⁶⁸ <https://github.com/TheHPXProject/hpx/issues/1437>

⁵⁶⁶⁹ <https://github.com/TheHPXProject/hpx/issues/1436>

⁵⁶⁷⁰ <https://github.com/TheHPXProject/hpx/pull/1435>

⁵⁶⁷¹ <https://github.com/TheHPXProject/hpx/pull/1434>

⁵⁶⁷² <https://github.com/TheHPXProject/hpx/pull/1433>

⁵⁶⁷³ <https://github.com/TheHPXProject/hpx/issues/1432>

⁵⁶⁷⁴ <https://github.com/TheHPXProject/hpx/pull/1431>

⁵⁶⁷⁵ <https://github.com/TheHPXProject/hpx/issues/1430>

⁵⁶⁷⁶ <https://github.com/TheHPXProject/hpx/issues/1429>

⁵⁶⁷⁷ <https://github.com/TheHPXProject/hpx/pull/1428>

⁵⁶⁷⁸ <https://github.com/TheHPXProject/hpx/pull/1427>

⁵⁶⁷⁹ <https://github.com/TheHPXProject/hpx/pull/1426>

⁵⁶⁸⁰ <https://github.com/TheHPXProject/hpx/pull/1425>

⁵⁶⁸¹ <https://github.com/TheHPXProject/hpx/issues/1424>

⁵⁶⁸² <https://github.com/TheHPXProject/hpx/issues/1423>

⁵⁶⁸³ <https://github.com/TheHPXProject/hpx/issues/1422>

- PR #1420⁵⁶⁸⁴ - added .travis.yml
- Issue #1419⁵⁶⁸⁵ - Unify enums: `hpx::runtime::state` and `hpx::state`
- PR #1416⁵⁶⁸⁶ - Adding travis builder
- Issue #1414⁵⁶⁸⁷ - Correct directory for `dispatch_gcc46.hpp` iteration
- Issue #1410⁵⁶⁸⁸ - Set operation algorithms
- Issue #1389⁵⁶⁸⁹ - Parallel algorithms relying on scan partitioner break for small number of elements
- Issue #1325⁵⁶⁹⁰ - Exceptions thrown during parcel handling are not handled correctly
- Issue #1315⁵⁶⁹¹ - Errors while running performance tests
- Issue #1309⁵⁶⁹² - `hpx::vector` partitions are not easily extendable by applications
- PR #1300⁵⁶⁹³ - Added serialization/de-serialization to `examples.tuplespace`
- Issue #1251⁵⁶⁹⁴ - `hpx::threads::get_thread_count` doesn't consider pending threads
- Issue #1008⁵⁶⁹⁵ - Decrease in application performance overtime; occasional spikes of major slowdown
- Issue #1001⁵⁶⁹⁶ - Zero copy serialization raises assert
- Issue #721⁵⁶⁹⁷ - Make HPX usable for Xeon Phi
- Issue #524⁵⁶⁹⁸ - Extend scheduler to support threads which can't be stolen

HPX V0.9.10 (Mar 24, 2015)

General changes

This is the 12th official release of *HPX*. It coincides with the 7th anniversary of the first commit to our source code repository. Since then, we have seen over 12300 commits amounting to more than 220000 lines of C++ code.

The major focus of this release was to improve the reliability of large scale runs. We believe to have achieved this goal as we now can reliably run *HPX* applications on up to ~24k cores. We have also shown that HPX can be used with success for symmetric runs (applications using both, host cores and Intel Xeon/Phi coprocessors). This is a huge step forward in terms of the usability of *HPX*. The main focus of this work involved isolating the causes of the segmentation faults at start up and shut down. Many of these issues were discovered to be the result of the suspension of threads which hold locks.

A very important improvement introduced with this release is the refactoring of the code representing our parcel-port implementation. Parcel- ports can now be implemented by 3rd parties as independent plugins which are dynamically loaded at runtime (static linking of parcel-ports is also supported). This refactoring also includes a massive improvement of the performance of our existing parcel-ports. We were able to significantly reduce the networking latencies

⁵⁶⁸⁴ <https://github.com/TheHPXProject/hpx/pull/1420>

⁵⁶⁸⁵ <https://github.com/TheHPXProject/hpx/issues/1419>

⁵⁶⁸⁶ <https://github.com/TheHPXProject/hpx/pull/1416>

⁵⁶⁸⁷ <https://github.com/TheHPXProject/hpx/issues/1414>

⁵⁶⁸⁸ <https://github.com/TheHPXProject/hpx/issues/1410>

⁵⁶⁸⁹ <https://github.com/TheHPXProject/hpx/issues/1389>

⁵⁶⁹⁰ <https://github.com/TheHPXProject/hpx/issues/1325>

⁵⁶⁹¹ <https://github.com/TheHPXProject/hpx/issues/1315>

⁵⁶⁹² <https://github.com/TheHPXProject/hpx/issues/1309>

⁵⁶⁹³ <https://github.com/TheHPXProject/hpx/pull/1300>

⁵⁶⁹⁴ <https://github.com/TheHPXProject/hpx/issues/1251>

⁵⁶⁹⁵ <https://github.com/TheHPXProject/hpx/issues/1008>

⁵⁶⁹⁶ <https://github.com/TheHPXProject/hpx/issues/1001>

⁵⁶⁹⁷ <https://github.com/TheHPXProject/hpx/issues/721>

⁵⁶⁹⁸ <https://github.com/TheHPXProject/hpx/issues/524>

and to improve the available networking bandwidth. Please note that in this release we disabled the `ibverbs` and `ipc` parcel ports as those have not been ported to the new plugin system yet (see [Issue #839](#)⁵⁶⁹⁹).

Another corner stone of this release is our work towards a complete implementation of `__cpp11_n4104__` (Working Draft, Technical Specification for C++ Extensions for Parallelism). This document defines a set of parallel algorithms to be added to the C++ standard library. We now have implemented about 75% of all specified parallel algorithms (see [\[link hpx.manual.parallel.parallel_algorithms Parallel Algorithms\]](#) for more details). We also implemented some extensions to `__cpp11_n4104__` allowing to invoke all of the algorithms asynchronously.

This release adds a first implementation of `hpx::vector` which is a distributed data structure closely aligned to the functionality of `std::vector`. The difference is that `hpx::vector` stores the data in partitions where the partitions can be distributed over different localities. We started to work on allowing to use the parallel algorithms with `hpx::vector`. At this point we have implemented only a few of the parallel algorithms to support distributed data structures (like `hpx::vector`) for testing purposes (see [Issue #1338](#)⁵⁷⁰⁰ for a documentation of our progress).

Breaking changes

With this release we put a lot of effort into changing the code base to be more compatible to C++11. These changes have caused the following issues for backward compatibility:

- **Move to Variadics-** All of the API now uses variadic templates. However, this change required to modify the argument sequence for some of the exiting API functions (`hpx::async_continue`, `hpx::apply_continue`, `hpx::when_each`, `hpx::wait_each`, synchronous invocation of actions).
- **Changes to Macros-** We also removed the macros `HPX_STD_FUNCTION` and `HPX_STD_TUPLE`. This shouldn't affect any user code as we replaced `HPX_STD_FUNCTION` with `hpx::util::function_nonser` which was the default expansion used for this macro. All `HPX` API functions which expect a `hpx::util::function_nonser` (or a `hpx::util::unique_function_nonser`) can now be transparently called with a compatible `std::function` instead. Similarly, `HPX_STD_TUPLE` was replaced by its default expansion as well: `hpx::util::tuple`.
- **Changes to `hpx::unique_future`-** `hpx::unique_future`, which was deprecated in the previous release for `hpx::future` is now completely removed from `HPX`. This completes the transition to a completely standards conforming implementation of `hpx::future`.
- **Changes to Supported Compilers.** Finally, in order to utilize more C++11 semantics, we have officially dropped support for GCC 4.4 and MSVC 2012. Please see our *Prerequisites* page for more details.

Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this release.

- [Issue #1402](#)⁵⁷⁰¹ - Internal `shared_future` serialization copies
- [Issue #1399](#)⁵⁷⁰² - Build takes unusually long time...
- [Issue #1398](#)⁵⁷⁰³ - Tests using the scan partitioner are broken on at least gcc 4.7 and intel compiler
- [Issue #1397](#)⁵⁷⁰⁴ - Completely remove `hpx::unique_future`
- [Issue #1396](#)⁵⁷⁰⁵ - Parallel scan algorithms with different initial values

⁵⁶⁹⁹ <https://github.com/TheHPXProject/hpx/issues/839>

⁵⁷⁰⁰ <https://github.com/TheHPXProject/hpx/issues/1338>

⁵⁷⁰¹ <https://github.com/TheHPXProject/hpx/issues/1402>

⁵⁷⁰² <https://github.com/TheHPXProject/hpx/issues/1399>

⁵⁷⁰³ <https://github.com/TheHPXProject/hpx/issues/1398>

⁵⁷⁰⁴ <https://github.com/TheHPXProject/hpx/issues/1397>

⁵⁷⁰⁵ <https://github.com/TheHPXProject/hpx/issues/1396>

- [Issue #1395⁵⁷⁰⁶](#) - Race Condition - 1d_stencil_8 - SuperMIC
- [Issue #1394⁵⁷⁰⁷](#) - “suspending thread while at least one lock is being held” - 1d_stencil_8 - SuperMIC
- [Issue #1393⁵⁷⁰⁸](#) - SEGFAULT in 1d_stencil_8 on SuperMIC
- [Issue #1392⁵⁷⁰⁹](#) - Fixing #1168
- [Issue #1391⁵⁷¹⁰](#) - Parallel Algorithms for scan partitioner for small number of elements
- [Issue #1387⁵⁷¹¹](#) - Failure with more than 4 localities
- [Issue #1386⁵⁷¹²](#) - Dispatching unhandled exceptions to outer user code
- [Issue #1385⁵⁷¹³](#) - Adding Copy algorithms, fixing `parallel::copy_if`
- [Issue #1384⁵⁷¹⁴](#) - Fixing 1325
- [Issue #1383⁵⁷¹⁵](#) - Fixed #504: Refactor Dataflow LCO to work with futures, this removes the dataflow component as it is obsolete
- [Issue #1382⁵⁷¹⁶](#) - `is_sorted`, `is_sorted_until` and `is_partitioned` algorithms
- [Issue #1381⁵⁷¹⁷](#) - fix for CMake versions prior to 3.1
- [Issue #1380⁵⁷¹⁸](#) - resolved warning in CMake 3.1 and newer
- [Issue #1379⁵⁷¹⁹](#) - Compilation error with papi
- [Issue #1378⁵⁷²⁰](#) - Towards safer migration
- [Issue #1377⁵⁷²¹](#) - HPXConfig.cmake should include TCMALLOC_LIBRARY and TCMALLOC_INCLUDE_DIR
- [Issue #1376⁵⁷²²](#) - Warning on uninitialized member
- [Issue #1375⁵⁷²³](#) - Fixing 1163
- [Issue #1374⁵⁷²⁴](#) - Fixing the MSVC 12 release builder
- [Issue #1373⁵⁷²⁵](#) - Modifying parallel search algorithm for zero length searches
- [Issue #1372⁵⁷²⁶](#) - Modifying parallel search algorithm for zero length searches
- [Issue #1371⁵⁷²⁷](#) - Avoid holding a lock during `agas::incrcf` while doing a credit split
- [Issue #1370⁵⁷²⁸](#) - `--hpx:bind` throws unexpected error

⁵⁷⁰⁶ <https://github.com/TheHPXProject/hpx/issues/1395>

⁵⁷⁰⁷ <https://github.com/TheHPXProject/hpx/issues/1394>

⁵⁷⁰⁸ <https://github.com/TheHPXProject/hpx/issues/1393>

⁵⁷⁰⁹ <https://github.com/TheHPXProject/hpx/issues/1392>

⁵⁷¹⁰ <https://github.com/TheHPXProject/hpx/issues/1391>

⁵⁷¹¹ <https://github.com/TheHPXProject/hpx/issues/1387>

⁵⁷¹² <https://github.com/TheHPXProject/hpx/issues/1386>

⁵⁷¹³ <https://github.com/TheHPXProject/hpx/issues/1385>

⁵⁷¹⁴ <https://github.com/TheHPXProject/hpx/issues/1384>

⁵⁷¹⁵ <https://github.com/TheHPXProject/hpx/issues/1383>

⁵⁷¹⁶ <https://github.com/TheHPXProject/hpx/issues/1382>

⁵⁷¹⁷ <https://github.com/TheHPXProject/hpx/issues/1381>

⁵⁷¹⁸ <https://github.com/TheHPXProject/hpx/issues/1380>

⁵⁷¹⁹ <https://github.com/TheHPXProject/hpx/issues/1379>

⁵⁷²⁰ <https://github.com/TheHPXProject/hpx/issues/1378>

⁵⁷²¹ <https://github.com/TheHPXProject/hpx/issues/1377>

⁵⁷²² <https://github.com/TheHPXProject/hpx/issues/1376>

⁵⁷²³ <https://github.com/TheHPXProject/hpx/issues/1375>

⁵⁷²⁴ <https://github.com/TheHPXProject/hpx/issues/1374>

⁵⁷²⁵ <https://github.com/TheHPXProject/hpx/issues/1373>

⁵⁷²⁶ <https://github.com/TheHPXProject/hpx/issues/1372>

⁵⁷²⁷ <https://github.com/TheHPXProject/hpx/issues/1371>

⁵⁷²⁸ <https://github.com/TheHPXProject/hpx/issues/1370>

- Issue #1369⁵⁷²⁹ - Getting rid of (void) in loops
- Issue #1368⁵⁷³⁰ - Variadic templates support for tuple
- Issue #1367⁵⁷³¹ - One last batch of variadic templates support
- Issue #1366⁵⁷³² - Fixing symbolic namespace hang
- Issue #1365⁵⁷³³ - More held locks
- Issue #1364⁵⁷³⁴ - Add counters 1363
- Issue #1363⁵⁷³⁵ - Add thread overhead counters
- Issue #1362⁵⁷³⁶ - Std config removal
- Issue #1361⁵⁷³⁷ - Parcelport plugins
- Issue #1360⁵⁷³⁸ - Detuplify transfer_action
- Issue #1359⁵⁷³⁹ - Removed obsolete checks
- Issue #1358⁵⁷⁴⁰ - Fixing 1352
- Issue #1357⁵⁷⁴¹ - Variadic templates support for runtime_support and components
- Issue #1356⁵⁷⁴² - fixed coordinate test for intel13
- Issue #1355⁵⁷⁴³ - fixed coordinate.hpp
- Issue #1354⁵⁷⁴⁴ - Lexicographical Compare completed
- Issue #1353⁵⁷⁴⁵ - HPX should set Boost_ADDITIONAL_VERSIONS flags
- Issue #1352⁵⁷⁴⁶ - Error: Cannot find action “ in type registry: HPX(bad_action_code)
- Issue #1351⁵⁷⁴⁷ - Variadic templates support for appliers
- Issue #1350⁵⁷⁴⁸ - Actions simplification
- Issue #1349⁵⁷⁴⁹ - Variadic when and wait functions
- Issue #1348⁵⁷⁵⁰ - Added hpx_init header to test files
- Issue #1347⁵⁷⁵¹ - Another batch of variadic templates support

⁵⁷²⁹ <https://github.com/TheHPXProject/hpx/issues/1369>

⁵⁷³⁰ <https://github.com/TheHPXProject/hpx/issues/1368>

⁵⁷³¹ <https://github.com/TheHPXProject/hpx/issues/1367>

⁵⁷³² <https://github.com/TheHPXProject/hpx/issues/1366>

⁵⁷³³ <https://github.com/TheHPXProject/hpx/issues/1365>

⁵⁷³⁴ <https://github.com/TheHPXProject/hpx/issues/1364>

⁵⁷³⁵ <https://github.com/TheHPXProject/hpx/issues/1363>

⁵⁷³⁶ <https://github.com/TheHPXProject/hpx/issues/1362>

⁵⁷³⁷ <https://github.com/TheHPXProject/hpx/issues/1361>

⁵⁷³⁸ <https://github.com/TheHPXProject/hpx/issues/1360>

⁵⁷³⁹ <https://github.com/TheHPXProject/hpx/issues/1359>

⁵⁷⁴⁰ <https://github.com/TheHPXProject/hpx/issues/1358>

⁵⁷⁴¹ <https://github.com/TheHPXProject/hpx/issues/1357>

⁵⁷⁴² <https://github.com/TheHPXProject/hpx/issues/1356>

⁵⁷⁴³ <https://github.com/TheHPXProject/hpx/issues/1355>

⁵⁷⁴⁴ <https://github.com/TheHPXProject/hpx/issues/1354>

⁵⁷⁴⁵ <https://github.com/TheHPXProject/hpx/issues/1353>

⁵⁷⁴⁶ <https://github.com/TheHPXProject/hpx/issues/1352>

⁵⁷⁴⁷ <https://github.com/TheHPXProject/hpx/issues/1351>

⁵⁷⁴⁸ <https://github.com/TheHPXProject/hpx/issues/1350>

⁵⁷⁴⁹ <https://github.com/TheHPXProject/hpx/issues/1349>

⁵⁷⁵⁰ <https://github.com/TheHPXProject/hpx/issues/1348>

⁵⁷⁵¹ <https://github.com/TheHPXProject/hpx/issues/1347>

- Issue #1346⁵⁷⁵² - Segmented copy
- Issue #1345⁵⁷⁵³ - Attempting to fix hangs during shutdown
- Issue #1344⁵⁷⁵⁴ - Std config removal
- Issue #1343⁵⁷⁵⁵ - Removing various distribution policies for `hpx::vector`
- Issue #1342⁵⁷⁵⁶ - Inclusive scan
- Issue #1341⁵⁷⁵⁷ - Exclusive scan
- Issue #1340⁵⁷⁵⁸ - Adding `parallel::count` for distributed data structures, adding tests
- Issue #1339⁵⁷⁵⁹ - Update argument order for `transform_reduce`
- Issue #1337⁵⁷⁶⁰ - Fix dataflow to handle properly ranges of futures
- Issue #1336⁵⁷⁶¹ - dataflow needs to hold onto futures passed to it
- Issue #1335⁵⁷⁶² - Fails to compile with `msvc14`
- Issue #1334⁵⁷⁶³ - Examples build problem
- Issue #1333⁵⁷⁶⁴ - Distributed transform reduce
- Issue #1332⁵⁷⁶⁵ - Variadic templates support for actions
- Issue #1331⁵⁷⁶⁶ - Some ambiguous calls of `map::erase` have been prevented by adding additional check in locality constructor.
- Issue #1330⁵⁷⁶⁷ - Defining Plain Actions does not work as described in the documentation
- Issue #1329⁵⁷⁶⁸ - Distributed vector cleanup
- Issue #1328⁵⁷⁶⁹ - Sync docs and comments with code in `hello_world` example
- Issue #1327⁵⁷⁷⁰ - Typos in docs
- Issue #1326⁵⁷⁷¹ - Documentation and code diverged in Fibonacci tutorial
- Issue #1325⁵⁷⁷² - Exceptions thrown during parcel handling are not handled correctly
- Issue #1324⁵⁷⁷³ - fixed bandwidth calculation
- Issue #1323⁵⁷⁷⁴ - `mmap()` failed to allocate thread stack due to insufficient resources

⁵⁷⁵² <https://github.com/TheHPXProject/hpx/issues/1346>

⁵⁷⁵³ <https://github.com/TheHPXProject/hpx/issues/1345>

⁵⁷⁵⁴ <https://github.com/TheHPXProject/hpx/issues/1344>

⁵⁷⁵⁵ <https://github.com/TheHPXProject/hpx/issues/1343>

⁵⁷⁵⁶ <https://github.com/TheHPXProject/hpx/issues/1342>

⁵⁷⁵⁷ <https://github.com/TheHPXProject/hpx/issues/1341>

⁵⁷⁵⁸ <https://github.com/TheHPXProject/hpx/issues/1340>

⁵⁷⁵⁹ <https://github.com/TheHPXProject/hpx/issues/1339>

⁵⁷⁶⁰ <https://github.com/TheHPXProject/hpx/issues/1337>

⁵⁷⁶¹ <https://github.com/TheHPXProject/hpx/issues/1336>

⁵⁷⁶² <https://github.com/TheHPXProject/hpx/issues/1335>

⁵⁷⁶³ <https://github.com/TheHPXProject/hpx/issues/1334>

⁵⁷⁶⁴ <https://github.com/TheHPXProject/hpx/issues/1333>

⁵⁷⁶⁵ <https://github.com/TheHPXProject/hpx/issues/1332>

⁵⁷⁶⁶ <https://github.com/TheHPXProject/hpx/issues/1331>

⁵⁷⁶⁷ <https://github.com/TheHPXProject/hpx/issues/1330>

⁵⁷⁶⁸ <https://github.com/TheHPXProject/hpx/issues/1329>

⁵⁷⁶⁹ <https://github.com/TheHPXProject/hpx/issues/1328>

⁵⁷⁷⁰ <https://github.com/TheHPXProject/hpx/issues/1327>

⁵⁷⁷¹ <https://github.com/TheHPXProject/hpx/issues/1326>

⁵⁷⁷² <https://github.com/TheHPXProject/hpx/issues/1325>

⁵⁷⁷³ <https://github.com/TheHPXProject/hpx/issues/1324>

⁵⁷⁷⁴ <https://github.com/TheHPXProject/hpx/issues/1323>

- Issue #1322⁵⁷⁷⁵ - HPX fails to build aa182cf
- Issue #1321⁵⁷⁷⁶ - Limiting size of outgoing messages while coalescing parcels
- Issue #1320⁵⁷⁷⁷ - passing a future with launch::deferred in remote function call causes hang
- Issue #1319⁵⁷⁷⁸ - An exception when tries to specify number high priority threads with abp-priority
- Issue #1318⁵⁷⁷⁹ - Unable to run program with abp-priority and numa-sensitivity enabled
- Issue #1317⁵⁷⁸⁰ - N4071 Search/Search_n finished, minor changes
- Issue #1316⁵⁷⁸¹ - Add config option to make -lhp.run_hpx_main!=1 the default
- Issue #1314⁵⁷⁸² - Variadic support for async and apply
- Issue #1313⁵⁷⁸³ - Adjust when_any/some to the latest proposed interfaces
- Issue #1312⁵⁷⁸⁴ - Fixing #857: hpx::naming::locality leaks parcelport specific information into the public interface
- Issue #1311⁵⁷⁸⁵ - Distributed get'er/set'er_values for distributed vector
- Issue #1310⁵⁷⁸⁶ - Crashing in hpx::parcelset::policies::mpi::connection_handler::handle_messages() on Super-MIC
- Issue #1308⁵⁷⁸⁷ - Unable to execute an application with -hpx:threads
- Issue #1307⁵⁷⁸⁸ - merge_graph linking issue
- Issue #1306⁵⁷⁸⁹ - First batch of variadic templates support
- Issue #1305⁵⁷⁹⁰ - Create a compiler wrapper
- Issue #1304⁵⁷⁹¹ - Provide a compiler wrapper for hpx
- Issue #1303⁵⁷⁹² - Drop support for GCC44
- Issue #1302⁵⁷⁹³ - Fixing #1297
- Issue #1301⁵⁷⁹⁴ - Compilation error when tried to use boost range iterators with wait_all
- Issue #1298⁵⁷⁹⁵ - Distributed vector
- Issue #1297⁵⁷⁹⁶ - Unable to invoke component actions recursively

⁵⁷⁷⁵ <https://github.com/TheHPXProject/hpx/issues/1322>

⁵⁷⁷⁶ <https://github.com/TheHPXProject/hpx/issues/1321>

⁵⁷⁷⁷ <https://github.com/TheHPXProject/hpx/issues/1320>

⁵⁷⁷⁸ <https://github.com/TheHPXProject/hpx/issues/1319>

⁵⁷⁷⁹ <https://github.com/TheHPXProject/hpx/issues/1318>

⁵⁷⁸⁰ <https://github.com/TheHPXProject/hpx/issues/1317>

⁵⁷⁸¹ <https://github.com/TheHPXProject/hpx/issues/1316>

⁵⁷⁸² <https://github.com/TheHPXProject/hpx/issues/1314>

⁵⁷⁸³ <https://github.com/TheHPXProject/hpx/issues/1313>

⁵⁷⁸⁴ <https://github.com/TheHPXProject/hpx/issues/1312>

⁵⁷⁸⁵ <https://github.com/TheHPXProject/hpx/issues/1311>

⁵⁷⁸⁶ <https://github.com/TheHPXProject/hpx/issues/1310>

⁵⁷⁸⁷ <https://github.com/TheHPXProject/hpx/issues/1308>

⁵⁷⁸⁸ <https://github.com/TheHPXProject/hpx/issues/1307>

⁵⁷⁸⁹ <https://github.com/TheHPXProject/hpx/issues/1306>

⁵⁷⁹⁰ <https://github.com/TheHPXProject/hpx/issues/1305>

⁵⁷⁹¹ <https://github.com/TheHPXProject/hpx/issues/1304>

⁵⁷⁹² <https://github.com/TheHPXProject/hpx/issues/1303>

⁵⁷⁹³ <https://github.com/TheHPXProject/hpx/issues/1302>

⁵⁷⁹⁴ <https://github.com/TheHPXProject/hpx/issues/1301>

⁵⁷⁹⁵ <https://github.com/TheHPXProject/hpx/issues/1298>

⁵⁷⁹⁶ <https://github.com/TheHPXProject/hpx/issues/1297>

- [Issue #1294⁵⁷⁹⁷](#) - HDF5 build error
- [Issue #1275⁵⁷⁹⁸](#) - The parcelport implementation is non-optimal
- [Issue #1267⁵⁷⁹⁹](#) - Added classes and unit tests for local_file, orangefs_file and pxfs_file
- [Issue #1264⁵⁸⁰⁰](#) - Error “assertion ‘!m_fun’ failed” randomly occurs when using TCP
- [Issue #1254⁵⁸⁰¹](#) - thread binding seems to not work properly
- [Issue #1220⁵⁸⁰²](#) - parallel::copy_if is broken
- [Issue #1217⁵⁸⁰³](#) - Find a better way of fixing the issue patched by #1216
- [Issue #1168⁵⁸⁰⁴](#) - Starting HPX on Cray machines using aprun isn’t working correctly
- [Issue #1085⁵⁸⁰⁵](#) - Replace startup and shutdown barriers with broadcasts
- [Issue #981⁵⁸⁰⁶](#) - With SLURM, -hpx:threads=8 should not be necessary
- [Issue #857⁵⁸⁰⁷](#) - hpx::naming::locality leaks parcelport specific information into the public interface
- [Issue #850⁵⁸⁰⁸](#) - “flush” not documented
- [Issue #763⁵⁸⁰⁹](#) - Create buildbot instance that uses std::bind as HPX_STD_BIND
- [Issue #680⁵⁸¹⁰](#) - Convert parcel ports into a plugin system
- [Issue #582⁵⁸¹¹](#) - Make exception thrown from HPX threads available from `hpx::init`
- [Issue #504⁵⁸¹²](#) - Refactor Dataflow LCO to work with futures
- [Issue #196⁵⁸¹³](#) - Don’t store copies of the locality network metadata in the gva table

HPX V0.9.9 (Oct 31, 2014, codename Spooky)

General changes

We have had over 1500 commits since the last release and we have closed over 200 tickets (bugs, feature requests, pull requests, etc.). These are by far the largest numbers of commits and resolved issues for any of the *HPX* releases so far. We are especially happy about the large number of people who contributed for the first time to *HPX*.

- We completed the transition from the older (non-conforming) implementation of `hpx::future` to the new and fully conforming version by removing the old code and by renaming the type `hpx::unique_future` to `hpx::future`. In order to maintain backwards compatibility with existing code which uses the type `hpx::unique_future` we support the configuration variable `HPX_UNIQUE_FUTURE_ALIAS`. If this variable is set to ON while running cmake it will additionally define a template alias for this type.

⁵⁷⁹⁷ <https://github.com/TheHPXProject/hpx/issues/1294>

⁵⁷⁹⁸ <https://github.com/TheHPXProject/hpx/issues/1275>

⁵⁷⁹⁹ <https://github.com/TheHPXProject/hpx/issues/1267>

⁵⁸⁰⁰ <https://github.com/TheHPXProject/hpx/issues/1264>

⁵⁸⁰¹ <https://github.com/TheHPXProject/hpx/issues/1254>

⁵⁸⁰² <https://github.com/TheHPXProject/hpx/issues/1220>

⁵⁸⁰³ <https://github.com/TheHPXProject/hpx/issues/1217>

⁵⁸⁰⁴ <https://github.com/TheHPXProject/hpx/issues/1168>

⁵⁸⁰⁵ <https://github.com/TheHPXProject/hpx/issues/1085>

⁵⁸⁰⁶ <https://github.com/TheHPXProject/hpx/issues/981>

⁵⁸⁰⁷ <https://github.com/TheHPXProject/hpx/issues/857>

⁵⁸⁰⁸ <https://github.com/TheHPXProject/hpx/issues/850>

⁵⁸⁰⁹ <https://github.com/TheHPXProject/hpx/issues/763>

⁵⁸¹⁰ <https://github.com/TheHPXProject/hpx/issues/680>

⁵⁸¹¹ <https://github.com/TheHPXProject/hpx/issues/582>

⁵⁸¹² <https://github.com/TheHPXProject/hpx/issues/504>

⁵⁸¹³ <https://github.com/TheHPXProject/hpx/issues/196>

- We rewrote and significantly changed our build system. Please have a look at the new (now generated) documentation here: *Building HPX*. Please revisit your build scripts to adapt to the changes. The most notable changes are:
 - HPX_NO_INSTALL is no longer necessary.
 - For external builds, you need to set HPX_DIR instead of HPX_ROOT as described here: *Using HPX with CMake-based projects*.
 - IDEs that support multiple configurations (Visual Studio and XCode) can now be used as intended. that means no build dir.
 - Building HPX statically (without dynamic libraries) is now supported (-DHPX_STATIC_LINKING=On).
 - Please note that many variables used to configure the build process have been renamed to unify the naming conventions (see the section *CMake options* for more information).
 - This also fixes a long list of issues, for more information see [Issue #1204](#)⁵⁸¹⁴.
- We started to implement various proposals to the C++ Standardization committee related to parallelism and concurrency, most notably [N4409](#)⁵⁸¹⁵ (Working Draft, Technical Specification for C++ Extensions for Parallelism), [N4411](#)⁵⁸¹⁶ (Task Region Rev. 3), and [N4313](#)⁵⁸¹⁷ (Working Draft, Technical Specification for C++ Extensions for Concurrency).
- We completely remodeled our automatic build system to run builds and unit tests on various systems and compilers. This allows us to find most bugs right as they were introduced and helps to maintain a high level of quality and compatibility. The newest build logs can be found at [HPX Buildbot Website](#)⁵⁸¹⁸.

Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this release.

- [Issue #1296](#)⁵⁸¹⁹ - Rename make_error_future to make_exceptional_future, adjust to N4123
- [Issue #1295](#)⁵⁸²⁰ - building issue
- [Issue #1293](#)⁵⁸²¹ - Transpose example
- [Issue #1292](#)⁵⁸²² - Wrong abs() function used in example
- [Issue #1291](#)⁵⁸²³ - non-synchronized shift operators have been removed
- [Issue #1290](#)⁵⁸²⁴ - RDTSCP is defined as true for Xeon Phi build
- [Issue #1289](#)⁵⁸²⁵ - Fixing 1288
- [Issue #1288](#)⁵⁸²⁶ - Add new performance counters
- [Issue #1287](#)⁵⁸²⁷ - Hierarchy scheduler broken performance counters

⁵⁸¹⁴ <https://github.com/TheHPXProject/hpx/issues/1204>

⁵⁸¹⁵ <http://wg21.link/n4409>

⁵⁸¹⁶ <http://wg21.link/n4411>

⁵⁸¹⁷ <http://wg21.link/n4313>

⁵⁸¹⁸ <http://rostam.cct.lsu.edu/>

⁵⁸¹⁹ <https://github.com/TheHPXProject/hpx/issues/1296>

⁵⁸²⁰ <https://github.com/TheHPXProject/hpx/issues/1295>

⁵⁸²¹ <https://github.com/TheHPXProject/hpx/issues/1293>

⁵⁸²² <https://github.com/TheHPXProject/hpx/issues/1292>

⁵⁸²³ <https://github.com/TheHPXProject/hpx/issues/1291>

⁵⁸²⁴ <https://github.com/TheHPXProject/hpx/issues/1290>

⁵⁸²⁵ <https://github.com/TheHPXProject/hpx/issues/1289>

⁵⁸²⁶ <https://github.com/TheHPXProject/hpx/issues/1288>

⁵⁸²⁷ <https://github.com/TheHPXProject/hpx/issues/1287>

- Issue #1286⁵⁸²⁸ - Algorithm cleanup
- Issue #1285⁵⁸²⁹ - Broken Links in Documentation
- Issue #1284⁵⁸³⁰ - Uninitialized copy
- Issue #1283⁵⁸³¹ - missing boost::scoped_ptr includes
- Issue #1282⁵⁸³² - Update documentation of build options for schedulers
- Issue #1281⁵⁸³³ - reset idle rate counter
- Issue #1280⁵⁸³⁴ - Bug when executing on Intel MIC
- Issue #1279⁵⁸³⁵ - Add improved when_all/wait_all
- Issue #1278⁵⁸³⁶ - Implement improved when_all/wait_all
- Issue #1277⁵⁸³⁷ - feature request: get access to argc argv and variables_map
- Issue #1276⁵⁸³⁸ - Remove merging map
- Issue #1274⁵⁸³⁹ - Weird (wrong) string code in papi.cpp
- Issue #1273⁵⁸⁴⁰ - Sequential task execution policy
- Issue #1272⁵⁸⁴¹ - Avoid CMake name clash for Boost.Thread library
- Issue #1271⁵⁸⁴² - Updates on HPX Test Units
- Issue #1270⁵⁸⁴³ - hpx/util/safe_lexical_cast.hpp is added
- Issue #1269⁵⁸⁴⁴ - Added default value for “LIB” cmake variable
- Issue #1268⁵⁸⁴⁵ - Memory Counters not working
- Issue #1266⁵⁸⁴⁶ - FindHPX.cmake is not installed
- Issue #1263⁵⁸⁴⁷ - apply_remote test takes too long
- Issue #1262⁵⁸⁴⁸ - Chrono cleanup
- Issue #1261⁵⁸⁴⁹ - Need make install for papi counters and this builds all the examples
- Issue #1260⁵⁸⁵⁰ - Documentation of Stencil example claims

⁵⁸²⁸ <https://github.com/TheHPXProject/hpx/issues/1286>

⁵⁸²⁹ <https://github.com/TheHPXProject/hpx/issues/1285>

⁵⁸³⁰ <https://github.com/TheHPXProject/hpx/issues/1284>

⁵⁸³¹ <https://github.com/TheHPXProject/hpx/issues/1283>

⁵⁸³² <https://github.com/TheHPXProject/hpx/issues/1282>

⁵⁸³³ <https://github.com/TheHPXProject/hpx/issues/1281>

⁵⁸³⁴ <https://github.com/TheHPXProject/hpx/issues/1280>

⁵⁸³⁵ <https://github.com/TheHPXProject/hpx/issues/1279>

⁵⁸³⁶ <https://github.com/TheHPXProject/hpx/issues/1278>

⁵⁸³⁷ <https://github.com/TheHPXProject/hpx/issues/1277>

⁵⁸³⁸ <https://github.com/TheHPXProject/hpx/issues/1276>

⁵⁸³⁹ <https://github.com/TheHPXProject/hpx/issues/1274>

⁵⁸⁴⁰ <https://github.com/TheHPXProject/hpx/issues/1273>

⁵⁸⁴¹ <https://github.com/TheHPXProject/hpx/issues/1272>

⁵⁸⁴² <https://github.com/TheHPXProject/hpx/issues/1271>

⁵⁸⁴³ <https://github.com/TheHPXProject/hpx/issues/1270>

⁵⁸⁴⁴ <https://github.com/TheHPXProject/hpx/issues/1269>

⁵⁸⁴⁵ <https://github.com/TheHPXProject/hpx/issues/1268>

⁵⁸⁴⁶ <https://github.com/TheHPXProject/hpx/issues/1266>

⁵⁸⁴⁷ <https://github.com/TheHPXProject/hpx/issues/1263>

⁵⁸⁴⁸ <https://github.com/TheHPXProject/hpx/issues/1262>

⁵⁸⁴⁹ <https://github.com/TheHPXProject/hpx/issues/1261>

⁵⁸⁵⁰ <https://github.com/TheHPXProject/hpx/issues/1260>

- Issue #1259⁵⁸⁵¹ - Avoid double-linking Boost on Windows
- Issue #1257⁵⁸⁵² - Adding additional parameter to create_thread
- Issue #1256⁵⁸⁵³ - added buildbot changes to release notes
- Issue #1255⁵⁸⁵⁴ - Cannot build MiniGhost
- Issue #1253⁵⁸⁵⁵ - hpx::thread defects
- Issue #1252⁵⁸⁵⁶ - HPX_PREFIX is too fragile
- Issue #1250⁵⁸⁵⁷ - switch_to_fiber_emulation does not work properly
- Issue #1249⁵⁸⁵⁸ - Documentation is generated under Release folder
- Issue #1248⁵⁸⁵⁹ - Fix usage of hpx_generic_coroutine_context and get tests passing on powerpc
- Issue #1247⁵⁸⁶⁰ - Dynamic linking error
- Issue #1246⁵⁸⁶¹ - Make cpuid.cpp C++11 compliant
- Issue #1245⁵⁸⁶² - HPX fails on startup (setting thread affinity mask)
- Issue #1244⁵⁸⁶³ - HPX_WITH_RDTSC configure test fails, but should succeed
- Issue #1243⁵⁸⁶⁴ - CTest dashboard info for CSCS CDash drop location
- Issue #1242⁵⁸⁶⁵ - Mac fixes
- Issue #1241⁵⁸⁶⁶ - Failure in Distributed with Boost 1.56
- Issue #1240⁵⁸⁶⁷ - fix a race condition in examples.diskperf
- Issue #1239⁵⁸⁶⁸ - fix wait_each in examples.diskperf
- Issue #1238⁵⁸⁶⁹ - Fixed #1237: hpx::util::portable_binary_iarchive failed
- Issue #1237⁵⁸⁷⁰ - hpx::util::portable_binary_iarchive faileds
- Issue #1235⁵⁸⁷¹ - Fixing clang warnings and errors
- Issue #1234⁵⁸⁷² - TCP runs fail: Transport endpoint is not connected
- Issue #1233⁵⁸⁷³ - Making sure the correct number of threads is registered with AGAS

⁵⁸⁵¹ <https://github.com/TheHPXProject/hpx/issues/1259>

⁵⁸⁵² <https://github.com/TheHPXProject/hpx/issues/1257>

⁵⁸⁵³ <https://github.com/TheHPXProject/hpx/issues/1256>

⁵⁸⁵⁴ <https://github.com/TheHPXProject/hpx/issues/1255>

⁵⁸⁵⁵ <https://github.com/TheHPXProject/hpx/issues/1253>

⁵⁸⁵⁶ <https://github.com/TheHPXProject/hpx/issues/1252>

⁵⁸⁵⁷ <https://github.com/TheHPXProject/hpx/issues/1250>

⁵⁸⁵⁸ <https://github.com/TheHPXProject/hpx/issues/1249>

⁵⁸⁵⁹ <https://github.com/TheHPXProject/hpx/issues/1248>

⁵⁸⁶⁰ <https://github.com/TheHPXProject/hpx/issues/1247>

⁵⁸⁶¹ <https://github.com/TheHPXProject/hpx/issues/1246>

⁵⁸⁶² <https://github.com/TheHPXProject/hpx/issues/1245>

⁵⁸⁶³ <https://github.com/TheHPXProject/hpx/issues/1244>

⁵⁸⁶⁴ <https://github.com/TheHPXProject/hpx/issues/1243>

⁵⁸⁶⁵ <https://github.com/TheHPXProject/hpx/issues/1242>

⁵⁸⁶⁶ <https://github.com/TheHPXProject/hpx/issues/1241>

⁵⁸⁶⁷ <https://github.com/TheHPXProject/hpx/issues/1240>

⁵⁸⁶⁸ <https://github.com/TheHPXProject/hpx/issues/1239>

⁵⁸⁶⁹ <https://github.com/TheHPXProject/hpx/issues/1238>

⁵⁸⁷⁰ <https://github.com/TheHPXProject/hpx/issues/1237>

⁵⁸⁷¹ <https://github.com/TheHPXProject/hpx/issues/1235>

⁵⁸⁷² <https://github.com/TheHPXProject/hpx/issues/1234>

⁵⁸⁷³ <https://github.com/TheHPXProject/hpx/issues/1233>

- Issue #1232⁵⁸⁷⁴ - Fixing race in wait_XXX
- Issue #1231⁵⁸⁷⁵ - Parallel minmax
- Issue #1230⁵⁸⁷⁶ - Distributed run of 1d_stencil_8 uses less threads than spec. & sometimes gives errors
- Issue #1229⁵⁸⁷⁷ - Unstable number of threads
- Issue #1228⁵⁸⁷⁸ - HPX link error (cmake / MPI)
- Issue #1226⁵⁸⁷⁹ - Warning about struct/class thread_counters
- Issue #1225⁵⁸⁸⁰ - Adding parallel::replace etc
- Issue #1224⁵⁸⁸¹ - Extending dataflow to pass through non-future arguments
- Issue #1223⁵⁸⁸² - Remaining find algorithms implemented, N4071
- Issue #1222⁵⁸⁸³ - Merging all the changes
- Issue #1221⁵⁸⁸⁴ - No error output when using mpirun with hpx
- Issue #1219⁵⁸⁸⁵ - Adding new AGAS cache performance counters
- Issue #1216⁵⁸⁸⁶ - Fixing using futures (clients) as arguments to actions
- Issue #1215⁵⁸⁸⁷ - Error compiling simple component
- Issue #1214⁵⁸⁸⁸ - Stencil docs
- Issue #1213⁵⁸⁸⁹ - Using more than a few dozen MPI processes on SuperMike results in a seg fault before getting to hpx_main
- Issue #1212⁵⁸⁹⁰ - Parallel rotate
- Issue #1211⁵⁸⁹¹ - Direct actions cause the future's shared_state to be leaked
- Issue #1210⁵⁸⁹² - Refactored local::promise to be standard conformant
- Issue #1209⁵⁸⁹³ - Improve command line handling
- Issue #1208⁵⁸⁹⁴ - Adding parallel::reverse and parallel::reverse_copy
- Issue #1207⁵⁸⁹⁵ - Add copy_backward and move_backward
- Issue #1206⁵⁸⁹⁶ - N4071 additional algorithms implemented

⁵⁸⁷⁴ <https://github.com/TheHPXProject/hpx/issues/1232>

⁵⁸⁷⁵ <https://github.com/TheHPXProject/hpx/issues/1231>

⁵⁸⁷⁶ <https://github.com/TheHPXProject/hpx/issues/1230>

⁵⁸⁷⁷ <https://github.com/TheHPXProject/hpx/issues/1229>

⁵⁸⁷⁸ <https://github.com/TheHPXProject/hpx/issues/1228>

⁵⁸⁷⁹ <https://github.com/TheHPXProject/hpx/issues/1226>

⁵⁸⁸⁰ <https://github.com/TheHPXProject/hpx/issues/1225>

⁵⁸⁸¹ <https://github.com/TheHPXProject/hpx/issues/1224>

⁵⁸⁸² <https://github.com/TheHPXProject/hpx/issues/1223>

⁵⁸⁸³ <https://github.com/TheHPXProject/hpx/issues/1222>

⁵⁸⁸⁴ <https://github.com/TheHPXProject/hpx/issues/1221>

⁵⁸⁸⁵ <https://github.com/TheHPXProject/hpx/issues/1219>

⁵⁸⁸⁶ <https://github.com/TheHPXProject/hpx/issues/1216>

⁵⁸⁸⁷ <https://github.com/TheHPXProject/hpx/issues/1215>

⁵⁸⁸⁸ <https://github.com/TheHPXProject/hpx/issues/1214>

⁵⁸⁸⁹ <https://github.com/TheHPXProject/hpx/issues/1213>

⁵⁸⁹⁰ <https://github.com/TheHPXProject/hpx/issues/1212>

⁵⁸⁹¹ <https://github.com/TheHPXProject/hpx/issues/1211>

⁵⁸⁹² <https://github.com/TheHPXProject/hpx/issues/1210>

⁵⁸⁹³ <https://github.com/TheHPXProject/hpx/issues/1209>

⁵⁸⁹⁴ <https://github.com/TheHPXProject/hpx/issues/1208>

⁵⁸⁹⁵ <https://github.com/TheHPXProject/hpx/issues/1207>

⁵⁸⁹⁶ <https://github.com/TheHPXProject/hpx/issues/1206>

- [Issue #1204⁵⁸⁹⁷](#) - Cmake simplification and various other minor changes
- [Issue #1203⁵⁸⁹⁸](#) - Implementing new launch policy for (local) `async: hpx::launch::fork`.
- [Issue #1202⁵⁸⁹⁹](#) - Failed assertion in `connection_cache.hpp`
- [Issue #1201⁵⁹⁰⁰](#) - `pkg-config` doesn't add `mpi` link directories
- [Issue #1200⁵⁹⁰¹](#) - Error when querying time performance counters
- [Issue #1199⁵⁹⁰²](#) - library path is now configurable (again)
- [Issue #1198⁵⁹⁰³](#) - Error when querying performance counters
- [Issue #1197⁵⁹⁰⁴](#) - tests fail with intel compiler
- [Issue #1196⁵⁹⁰⁵](#) - Silence several warnings
- [Issue #1195⁵⁹⁰⁶](#) - Rephrase initializers to work with VC++ 2012
- [Issue #1194⁵⁹⁰⁷](#) - Simplify parallel algorithms
- [Issue #1193⁵⁹⁰⁸](#) - Adding `parallel::equal`
- [Issue #1192⁵⁹⁰⁹](#) - `HPX(out_of_memory)` on including `<hpx/hpx.hpp>`
- [Issue #1191⁵⁹¹⁰](#) - Fixing #1189
- [Issue #1190⁵⁹¹¹](#) - Chrono cleanup
- [Issue #1189⁵⁹¹²](#) - Deadlock .. somewhere? (probably serialization)
- [Issue #1188⁵⁹¹³](#) - Removed `future::get_status()`
- [Issue #1186⁵⁹¹⁴](#) - Fixed `FindOpenCL` to find current AMD APP SDK
- [Issue #1184⁵⁹¹⁵](#) - Tweaking future unwrapping
- [Issue #1183⁵⁹¹⁶](#) - Extended `parallel::reduce`
- [Issue #1182⁵⁹¹⁷](#) - `future::unwrap` hangs for `launch::deferred`
- [Issue #1181⁵⁹¹⁸](#) - Adding `all_of`, `any_of`, and `none_of` and corresponding documentation
- [Issue #1180⁵⁹¹⁹](#) - `hpx::cout` defect

⁵⁸⁹⁷ <https://github.com/TheHPXProject/hpx/issues/1204>

⁵⁸⁹⁸ <https://github.com/TheHPXProject/hpx/issues/1203>

⁵⁸⁹⁹ <https://github.com/TheHPXProject/hpx/issues/1202>

⁵⁹⁰⁰ <https://github.com/TheHPXProject/hpx/issues/1201>

⁵⁹⁰¹ <https://github.com/TheHPXProject/hpx/issues/1200>

⁵⁹⁰² <https://github.com/TheHPXProject/hpx/issues/1199>

⁵⁹⁰³ <https://github.com/TheHPXProject/hpx/issues/1198>

⁵⁹⁰⁴ <https://github.com/TheHPXProject/hpx/issues/1197>

⁵⁹⁰⁵ <https://github.com/TheHPXProject/hpx/issues/1196>

⁵⁹⁰⁶ <https://github.com/TheHPXProject/hpx/issues/1195>

⁵⁹⁰⁷ <https://github.com/TheHPXProject/hpx/issues/1194>

⁵⁹⁰⁸ <https://github.com/TheHPXProject/hpx/issues/1193>

⁵⁹⁰⁹ <https://github.com/TheHPXProject/hpx/issues/1192>

⁵⁹¹⁰ <https://github.com/TheHPXProject/hpx/issues/1191>

⁵⁹¹¹ <https://github.com/TheHPXProject/hpx/issues/1190>

⁵⁹¹² <https://github.com/TheHPXProject/hpx/issues/1189>

⁵⁹¹³ <https://github.com/TheHPXProject/hpx/issues/1188>

⁵⁹¹⁴ <https://github.com/TheHPXProject/hpx/issues/1186>

⁵⁹¹⁵ <https://github.com/TheHPXProject/hpx/issues/1184>

⁵⁹¹⁶ <https://github.com/TheHPXProject/hpx/issues/1183>

⁵⁹¹⁷ <https://github.com/TheHPXProject/hpx/issues/1182>

⁵⁹¹⁸ <https://github.com/TheHPXProject/hpx/issues/1181>

⁵⁹¹⁹ <https://github.com/TheHPXProject/hpx/issues/1180>

- [Issue #1179](#)⁵⁹²⁰ - `hpx::async` does not work for member function pointers when called on types with self-defined unary operator*
- [Issue #1178](#)⁵⁹²¹ - Implemented variadic `hpx::util::zip_iterator`
- [Issue #1177](#)⁵⁹²² - MPI parcelport defect
- [Issue #1176](#)⁵⁹²³ - `HPX_DEFINE_COMPONENT_CONST_ACTION_TPL` does not have a 2-argument version
- [Issue #1175](#)⁵⁹²⁴ - Create `util::zip_iterator` working with `util::tuple<>`
- [Issue #1174](#)⁵⁹²⁵ - Error Building HPX on linux, `root_certificate_authority.cpp`
- [Issue #1173](#)⁵⁹²⁶ - `hpx::cout` output lost
- [Issue #1172](#)⁵⁹²⁷ - HPX build error with Clang 3.4.2
- [Issue #1171](#)⁵⁹²⁸ - `CMAKE_INSTALL_PREFIX` ignored
- [Issue #1170](#)⁵⁹²⁹ - Close `hpx_benchmarks` repository on Github
- [Issue #1169](#)⁵⁹³⁰ - Buildbot emails have syntax error in url
- [Issue #1167](#)⁵⁹³¹ - Merge partial implementation of standards proposal N3960
- [Issue #1166](#)⁵⁹³² - Fixed several compiler warnings
- [Issue #1165](#)⁵⁹³³ - `cmake` warns: “`tests.regressions.actions`” does not exist
- [Issue #1164](#)⁵⁹³⁴ - Want my own serialization of `hpx::future`
- [Issue #1162](#)⁵⁹³⁵ - Segfault in `hello_world` example
- [Issue #1161](#)⁵⁹³⁶ - Use `HPX_ASSERT` to aid the compiler
- [Issue #1160](#)⁵⁹³⁷ - Do not put `-DNDEBUG` into `hpx_application.pc`
- [Issue #1159](#)⁵⁹³⁸ - Support Clang 3.4.2
- [Issue #1158](#)⁵⁹³⁹ - Fixed #1157: Rename `when_n/wait_n`, add `when_xxx_n/wait_xxx_n`
- [Issue #1157](#)⁵⁹⁴⁰ - Rename `when_n/wait_n`, add `when_xxx_n/wait_xxx_n`
- [Issue #1156](#)⁵⁹⁴¹ - Force inlining fails
- [Issue #1155](#)⁵⁹⁴² - changed header of `printout` to be compatible with python csv module

⁵⁹²⁰ <https://github.com/TheHPXProject/hpx/issues/1179>

⁵⁹²¹ <https://github.com/TheHPXProject/hpx/issues/1178>

⁵⁹²² <https://github.com/TheHPXProject/hpx/issues/1177>

⁵⁹²³ <https://github.com/TheHPXProject/hpx/issues/1176>

⁵⁹²⁴ <https://github.com/TheHPXProject/hpx/issues/1175>

⁵⁹²⁵ <https://github.com/TheHPXProject/hpx/issues/1174>

⁵⁹²⁶ <https://github.com/TheHPXProject/hpx/issues/1173>

⁵⁹²⁷ <https://github.com/TheHPXProject/hpx/issues/1172>

⁵⁹²⁸ <https://github.com/TheHPXProject/hpx/issues/1171>

⁵⁹²⁹ <https://github.com/TheHPXProject/hpx/issues/1170>

⁵⁹³⁰ <https://github.com/TheHPXProject/hpx/issues/1169>

⁵⁹³¹ <https://github.com/TheHPXProject/hpx/issues/1167>

⁵⁹³² <https://github.com/TheHPXProject/hpx/issues/1166>

⁵⁹³³ <https://github.com/TheHPXProject/hpx/issues/1165>

⁵⁹³⁴ <https://github.com/TheHPXProject/hpx/issues/1164>

⁵⁹³⁵ <https://github.com/TheHPXProject/hpx/issues/1162>

⁵⁹³⁶ <https://github.com/TheHPXProject/hpx/issues/1161>

⁵⁹³⁷ <https://github.com/TheHPXProject/hpx/issues/1160>

⁵⁹³⁸ <https://github.com/TheHPXProject/hpx/issues/1159>

⁵⁹³⁹ <https://github.com/TheHPXProject/hpx/issues/1158>

⁵⁹⁴⁰ <https://github.com/TheHPXProject/hpx/issues/1157>

⁵⁹⁴¹ <https://github.com/TheHPXProject/hpx/issues/1156>

⁵⁹⁴² <https://github.com/TheHPXProject/hpx/issues/1155>

- Issue #1154⁵⁹⁴³ - Fixing iostreams
- Issue #1153⁵⁹⁴⁴ - Standard manipulators (like `std::endl`) do not work with `hpx::ostream`
- Issue #1152⁵⁹⁴⁵ - Functions revamp
- Issue #1151⁵⁹⁴⁶ - Suppressing cmake 3.0 policy warning for CMP0026
- Issue #1150⁵⁹⁴⁷ - Client Serialization error
- Issue #1149⁵⁹⁴⁸ - Segfault on Stampede
- Issue #1148⁵⁹⁴⁹ - Refactoring mini-ghost
- Issue #1147⁵⁹⁵⁰ - N3960 `copy_if` and `copy_n` implemented and tested
- Issue #1146⁵⁹⁵¹ - Stencil print
- Issue #1145⁵⁹⁵² - N3960 `hpx::parallel::copy` implemented and tested
- Issue #1144⁵⁹⁵³ - OpenMP examples `1d_stencil` do not build
- Issue #1143⁵⁹⁵⁴ - `1d_stencil` OpenMP examples do not build
- Issue #1142⁵⁹⁵⁵ - Cannot build HPX with gcc 4.6 on OS X
- Issue #1140⁵⁹⁵⁶ - Fix OpenMP lookup, enable usage of config tests in external CMake projects.
- Issue #1139⁵⁹⁵⁷ - `hpx/hpx/config/compiler_specific.hpp`
- Issue #1138⁵⁹⁵⁸ - clean up pkg-config files
- Issue #1137⁵⁹⁵⁹ - Improvements to create binary packages
- Issue #1136⁵⁹⁶⁰ - `HPX_GCC_VERSION` not defined on all compilers
- Issue #1135⁵⁹⁶¹ - Avoiding collision between `winsock2.h` and `windows.h`
- Issue #1134⁵⁹⁶² - Making sure, that `hpx::finalize` can be called from any locality
- Issue #1133⁵⁹⁶³ - `1d stencil` examples
- Issue #1131⁵⁹⁶⁴ - Refactor `unique_function` implementation
- Issue #1130⁵⁹⁶⁵ - Unique function

⁵⁹⁴³ <https://github.com/TheHPXProject/hpx/issues/1154>

⁵⁹⁴⁴ <https://github.com/TheHPXProject/hpx/issues/1153>

⁵⁹⁴⁵ <https://github.com/TheHPXProject/hpx/issues/1152>

⁵⁹⁴⁶ <https://github.com/TheHPXProject/hpx/issues/1151>

⁵⁹⁴⁷ <https://github.com/TheHPXProject/hpx/issues/1150>

⁵⁹⁴⁸ <https://github.com/TheHPXProject/hpx/issues/1149>

⁵⁹⁴⁹ <https://github.com/TheHPXProject/hpx/issues/1148>

⁵⁹⁵⁰ <https://github.com/TheHPXProject/hpx/issues/1147>

⁵⁹⁵¹ <https://github.com/TheHPXProject/hpx/issues/1146>

⁵⁹⁵² <https://github.com/TheHPXProject/hpx/issues/1145>

⁵⁹⁵³ <https://github.com/TheHPXProject/hpx/issues/1144>

⁵⁹⁵⁴ <https://github.com/TheHPXProject/hpx/issues/1143>

⁵⁹⁵⁵ <https://github.com/TheHPXProject/hpx/issues/1142>

⁵⁹⁵⁶ <https://github.com/TheHPXProject/hpx/issues/1140>

⁵⁹⁵⁷ <https://github.com/TheHPXProject/hpx/issues/1139>

⁵⁹⁵⁸ <https://github.com/TheHPXProject/hpx/issues/1138>

⁵⁹⁵⁹ <https://github.com/TheHPXProject/hpx/issues/1137>

⁵⁹⁶⁰ <https://github.com/TheHPXProject/hpx/issues/1136>

⁵⁹⁶¹ <https://github.com/TheHPXProject/hpx/issues/1135>

⁵⁹⁶² <https://github.com/TheHPXProject/hpx/issues/1134>

⁵⁹⁶³ <https://github.com/TheHPXProject/hpx/issues/1133>

⁵⁹⁶⁴ <https://github.com/TheHPXProject/hpx/issues/1131>

⁵⁹⁶⁵ <https://github.com/TheHPXProject/hpx/issues/1130>

- Issue #1129⁵⁹⁶⁶ - Some fixes to the Build system on OS X
- Issue #1128⁵⁹⁶⁷ - Action future args
- Issue #1127⁵⁹⁶⁸ - Executor causes segmentation fault
- Issue #1124⁵⁹⁶⁹ - Adding new API functions: `register_id_with_basename`, `unregister_id_with_basename`, `find_ids_from_basename`; adding test
- Issue #1123⁵⁹⁷⁰ - Reduce nesting of try-catch construct in `encode_parcel`s?
- Issue #1122⁵⁹⁷¹ - Client base fixes
- Issue #1121⁵⁹⁷² - Update `hpxrun.py.in`
- Issue #1120⁵⁹⁷³ - HTTPS2 tests compile errors on v110 (VS2012)
- Issue #1119⁵⁹⁷⁴ - Remove references to `boost::atomic` in accumulator example
- Issue #1118⁵⁹⁷⁵ - Only build test `thread_pool_executor_1114_test` if `HPX_SCHEDULER` is set
- Issue #1117⁵⁹⁷⁶ - `local_queue_executor` linker error on vc110
- Issue #1116⁵⁹⁷⁷ - Disabled performance counter should give runtime errors, not invalid data
- Issue #1115⁵⁹⁷⁸ - Compile error with Intel C++ 13.1
- Issue #1114⁵⁹⁷⁹ - Default constructed executor is not usable
- Issue #1113⁵⁹⁸⁰ - Fast compilation of logging causes ABI incompatibilities between different `NDEBUG` values
- Issue #1112⁵⁹⁸¹ - Using `thread_pool_executors` causes segfault
- Issue #1111⁵⁹⁸² - `hpx::threads::get_thread_data` always returns zero
- Issue #1110⁵⁹⁸³ - Remove unnecessary null pointer checks
- Issue #1109⁵⁹⁸⁴ - More tests adjustments
- Issue #1108⁵⁹⁸⁵ - Clarify build rules for “`libboost_atomic-mt.so`”?
- Issue #1107⁵⁹⁸⁶ - Remove unnecessary null pointer checks
- Issue #1106⁵⁹⁸⁷ - `network_storage` benchmark improvements, adding legends to plots and tidying layout
- Issue #1105⁵⁹⁸⁸ - Add more plot outputs and improve instructions doc

⁵⁹⁶⁶ <https://github.com/TheHPXProject/hpx/issues/1129>

⁵⁹⁶⁷ <https://github.com/TheHPXProject/hpx/issues/1128>

⁵⁹⁶⁸ <https://github.com/TheHPXProject/hpx/issues/1127>

⁵⁹⁶⁹ <https://github.com/TheHPXProject/hpx/issues/1124>

⁵⁹⁷⁰ <https://github.com/TheHPXProject/hpx/issues/1123>

⁵⁹⁷¹ <https://github.com/TheHPXProject/hpx/issues/1122>

⁵⁹⁷² <https://github.com/TheHPXProject/hpx/issues/1121>

⁵⁹⁷³ <https://github.com/TheHPXProject/hpx/issues/1120>

⁵⁹⁷⁴ <https://github.com/TheHPXProject/hpx/issues/1119>

⁵⁹⁷⁵ <https://github.com/TheHPXProject/hpx/issues/1118>

⁵⁹⁷⁶ <https://github.com/TheHPXProject/hpx/issues/1117>

⁵⁹⁷⁷ <https://github.com/TheHPXProject/hpx/issues/1116>

⁵⁹⁷⁸ <https://github.com/TheHPXProject/hpx/issues/1115>

⁵⁹⁷⁹ <https://github.com/TheHPXProject/hpx/issues/1114>

⁵⁹⁸⁰ <https://github.com/TheHPXProject/hpx/issues/1113>

⁵⁹⁸¹ <https://github.com/TheHPXProject/hpx/issues/1112>

⁵⁹⁸² <https://github.com/TheHPXProject/hpx/issues/1111>

⁵⁹⁸³ <https://github.com/TheHPXProject/hpx/issues/1110>

⁵⁹⁸⁴ <https://github.com/TheHPXProject/hpx/issues/1109>

⁵⁹⁸⁵ <https://github.com/TheHPXProject/hpx/issues/1108>

⁵⁹⁸⁶ <https://github.com/TheHPXProject/hpx/issues/1107>

⁵⁹⁸⁷ <https://github.com/TheHPXProject/hpx/issues/1106>

⁵⁹⁸⁸ <https://github.com/TheHPXProject/hpx/issues/1105>

- Issue #1104⁵⁹⁸⁹ - Complete quoting for parameters of some CMake commands
- Issue #1103⁵⁹⁹⁰ - Work on test/scripts
- Issue #1102⁵⁹⁹¹ - Changed minimum requirement of window install to 2012
- Issue #1101⁵⁹⁹² - Changed minimum requirement of window install to 2012
- Issue #1100⁵⁹⁹³ - Changed readme to no longer specify using MSVC 2010 compiler
- Issue #1099⁵⁹⁹⁴ - Error returning futures from component actions
- Issue #1098⁵⁹⁹⁵ - Improve storage test
- Issue #1097⁵⁹⁹⁶ - data_actions quickstart example calls missing function decorate_action of data_get_action
- Issue #1096⁵⁹⁹⁷ - MPI parcelport broken with new zero copy optimization
- Issue #1095⁵⁹⁹⁸ - Warning C4005: _WIN32_WINNT: Macro redefinition
- Issue #1094⁵⁹⁹⁹ - Syntax error for -DHPX_UNIQUE_FUTURE_ALIAS in master
- Issue #1093⁶⁰⁰⁰ - Syntax error for -DHPX_UNIQUE_FUTURE_ALIAS
- Issue #1092⁶⁰⁰¹ - Rename unique_future<> back to future<>
- Issue #1091⁶⁰⁰² - Inconsistent error message
- Issue #1090⁶⁰⁰³ - On windows 8.1 the examples crashed if using more than one os thread
- Issue #1089⁶⁰⁰⁴ - Components should be allowed to have their own executor
- Issue #1088⁶⁰⁰⁵ - Add possibility to select a network interface for the ibverbs parcelport
- Issue #1087⁶⁰⁰⁶ - ibverbs and ipc parcelport uses zero copy optimization
- Issue #1083⁶⁰⁰⁷ - Make shell examples copyable in docs
- Issue #1082⁶⁰⁰⁸ - Implement proper termination detection during shutdown
- Issue #1081⁶⁰⁰⁹ - Implement thread_specific_ptr for hpx::threads
- Issue #1072⁶⁰¹⁰ - make install not working properly
- Issue #1070⁶⁰¹¹ - Complete quoting for parameters of some CMake commands

⁵⁹⁸⁹ <https://github.com/TheHPXProject/hpx/issues/1104>

⁵⁹⁹⁰ <https://github.com/TheHPXProject/hpx/issues/1103>

⁵⁹⁹¹ <https://github.com/TheHPXProject/hpx/issues/1102>

⁵⁹⁹² <https://github.com/TheHPXProject/hpx/issues/1101>

⁵⁹⁹³ <https://github.com/TheHPXProject/hpx/issues/1100>

⁵⁹⁹⁴ <https://github.com/TheHPXProject/hpx/issues/1099>

⁵⁹⁹⁵ <https://github.com/TheHPXProject/hpx/issues/1098>

⁵⁹⁹⁶ <https://github.com/TheHPXProject/hpx/issues/1097>

⁵⁹⁹⁷ <https://github.com/TheHPXProject/hpx/issues/1096>

⁵⁹⁹⁸ <https://github.com/TheHPXProject/hpx/issues/1095>

⁵⁹⁹⁹ <https://github.com/TheHPXProject/hpx/issues/1094>

⁶⁰⁰⁰ <https://github.com/TheHPXProject/hpx/issues/1093>

⁶⁰⁰¹ <https://github.com/TheHPXProject/hpx/issues/1092>

⁶⁰⁰² <https://github.com/TheHPXProject/hpx/issues/1091>

⁶⁰⁰³ <https://github.com/TheHPXProject/hpx/issues/1090>

⁶⁰⁰⁴ <https://github.com/TheHPXProject/hpx/issues/1089>

⁶⁰⁰⁵ <https://github.com/TheHPXProject/hpx/issues/1088>

⁶⁰⁰⁶ <https://github.com/TheHPXProject/hpx/issues/1087>

⁶⁰⁰⁷ <https://github.com/TheHPXProject/hpx/issues/1083>

⁶⁰⁰⁸ <https://github.com/TheHPXProject/hpx/issues/1082>

⁶⁰⁰⁹ <https://github.com/TheHPXProject/hpx/issues/1081>

⁶⁰¹⁰ <https://github.com/TheHPXProject/hpx/issues/1072>

⁶⁰¹¹ <https://github.com/TheHPXProject/hpx/issues/1070>

- [Issue #1059](#)⁶⁰¹² - Fix more unused variable warnings
- [Issue #1051](#)⁶⁰¹³ - Implement when_each
- [Issue #973](#)⁶⁰¹⁴ - Would like option to report hwloc bindings
- [Issue #970](#)⁶⁰¹⁵ - Bad flags for Fortran compiler
- [Issue #941](#)⁶⁰¹⁶ - Create a proper user level context switching class for BG/Q
- [Issue #935](#)⁶⁰¹⁷ - Build error with gcc 4.6 and Boost 1.54.0 on hpx trunk and 0.9.6
- [Issue #934](#)⁶⁰¹⁸ - Want to build HPX without dynamic libraries
- [Issue #927](#)⁶⁰¹⁹ - Make hpx/lcos/reduce.hpp accept futures of id_type
- [Issue #926](#)⁶⁰²⁰ - All unit tests that are run with more than one thread with CTest/hpx_run_test should configure hpx.os_threads
- [Issue #925](#)⁶⁰²¹ - regression_dataflow_791 needs to be brought in line with HPX standards
- [Issue #899](#)⁶⁰²² - Fix race conditions in regression tests
- [Issue #879](#)⁶⁰²³ - Hung test leads to cascading test failure; make tests should support the MPI parcelport
- [Issue #865](#)⁶⁰²⁴ - future<T> and friends shall work for movable only Ts
- [Issue #847](#)⁶⁰²⁵ - Dynamic libraries are not installed on OS X
- [Issue #816](#)⁶⁰²⁶ - First Program tutorial pull request
- [Issue #799](#)⁶⁰²⁷ - Wrap lexical_cast to avoid exceptions
- [Issue #720](#)⁶⁰²⁸ - broken configuration when using cmake on Ubuntu
- [Issue #622](#)⁶⁰²⁹ - --hpx:hpx and --hpx:debug-hpx-log is nonsensical
- [Issue #525](#)⁶⁰³⁰ - Extend barrier LCO test to run in distributed
- [Issue #515](#)⁶⁰³¹ - Multi-destination version of hpx::apply is broken
- [Issue #509](#)⁶⁰³² - Push Boost.Atomic changes upstream
- [Issue #503](#)⁶⁰³³ - Running HPX applications on Windows should not require setting %PATH%
- [Issue #461](#)⁶⁰³⁴ - Add a compilation sanity test

⁶⁰¹² <https://github.com/TheHPXProject/hpx/issues/1059>

⁶⁰¹³ <https://github.com/TheHPXProject/hpx/issues/1051>

⁶⁰¹⁴ <https://github.com/TheHPXProject/hpx/issues/973>

⁶⁰¹⁵ <https://github.com/TheHPXProject/hpx/issues/970>

⁶⁰¹⁶ <https://github.com/TheHPXProject/hpx/issues/941>

⁶⁰¹⁷ <https://github.com/TheHPXProject/hpx/issues/935>

⁶⁰¹⁸ <https://github.com/TheHPXProject/hpx/issues/934>

⁶⁰¹⁹ <https://github.com/TheHPXProject/hpx/issues/927>

⁶⁰²⁰ <https://github.com/TheHPXProject/hpx/issues/926>

⁶⁰²¹ <https://github.com/TheHPXProject/hpx/issues/925>

⁶⁰²² <https://github.com/TheHPXProject/hpx/issues/899>

⁶⁰²³ <https://github.com/TheHPXProject/hpx/issues/879>

⁶⁰²⁴ <https://github.com/TheHPXProject/hpx/issues/865>

⁶⁰²⁵ <https://github.com/TheHPXProject/hpx/issues/847>

⁶⁰²⁶ <https://github.com/TheHPXProject/hpx/issues/816>

⁶⁰²⁷ <https://github.com/TheHPXProject/hpx/issues/799>

⁶⁰²⁸ <https://github.com/TheHPXProject/hpx/issues/720>

⁶⁰²⁹ <https://github.com/TheHPXProject/hpx/issues/622>

⁶⁰³⁰ <https://github.com/TheHPXProject/hpx/issues/525>

⁶⁰³¹ <https://github.com/TheHPXProject/hpx/issues/515>

⁶⁰³² <https://github.com/TheHPXProject/hpx/issues/509>

⁶⁰³³ <https://github.com/TheHPXProject/hpx/issues/503>

⁶⁰³⁴ <https://github.com/TheHPXProject/hpx/issues/461>

- Issue #456⁶⁰³⁵ - hpx_run_tests.py should log output from tests that timeout
- Issue #454⁶⁰³⁶ - Investigate threadmanager performance
- Issue #345⁶⁰³⁷ - Add more versatile environmental/cmake variable support to hpx_find_* CMake macros
- Issue #209⁶⁰³⁸ - Support multiple configurations in generated build files
- Issue #190⁶⁰³⁹ - hpx::cout should be a std::ostream
- Issue #189⁶⁰⁴⁰ - iostreams component should use startup/shutdown functions
- Issue #183⁶⁰⁴¹ - Use Boost.ICL for correctness in AGAS
- Issue #44⁶⁰⁴² - Implement real futures

HPX V0.9.8 (Mar 24, 2014)

We have had over 800 commits since the last release and we have closed over 65 tickets (bugs, feature requests, etc.).

With the changes below, *HPX* is once again leading the charge of a whole new era of computation. By intrinsically breaking down and synchronizing the work to be done, *HPX* insures that application developers will no longer have to fret about where a segment of code executes. That allows coders to focus their time and energy to understanding the data dependencies of their algorithms and thereby the core obstacles to an efficient code. Here are some of the advantages of using *HPX*:

- *HPX* is solidly rooted in a sophisticated theoretical execution model – ParalleX
- *HPX* exposes an API fully conforming to the C++11 and the draft C++14 standards, extended and applied to distributed computing. Everything programmers know about the concurrency primitives of the standard C++ library is still valid in the context of *HPX*.
- It provides a competitive, high performance implementation of modern, future-proof ideas which gives an smooth migration path from today's mainstream techniques
- There is no need for the programmer to worry about lower level parallelization paradigms like threads or message passing; no need to understand pthreads, MPI, OpenMP, or Windows threads, etc.
- There is no need to think about different types of parallelism such as tasks, pipelines, or fork-join, task or data parallelism.
- The same source of your program compiles and runs on Linux, BlueGene/Q, Mac OS X, Windows, and Android.
- The same code runs on shared memory multi-core systems and supercomputers, on handheld devices and Intel® Xeon Phi™ accelerators, or a heterogeneous mix of those.

⁶⁰³⁵ <https://github.com/TheHPXProject/hpx/issues/456>

⁶⁰³⁶ <https://github.com/TheHPXProject/hpx/issues/454>

⁶⁰³⁷ <https://github.com/TheHPXProject/hpx/issues/345>

⁶⁰³⁸ <https://github.com/TheHPXProject/hpx/issues/209>

⁶⁰³⁹ <https://github.com/TheHPXProject/hpx/issues/190>

⁶⁰⁴⁰ <https://github.com/TheHPXProject/hpx/issues/189>

⁶⁰⁴¹ <https://github.com/TheHPXProject/hpx/issues/183>

⁶⁰⁴² <https://github.com/TheHPXProject/hpx/issues/44>

General changes

- A major API breaking change for this release was introduced by implementing `hpx::future` and `hpx::shared_future` fully in conformance with the C++11 Standard⁶⁰⁴³. While `hpx::shared_future` is new and will not create any compatibility problems, we revised the interface and implementation of the existing `hpx::future`. For more details please see the [mailing list archive](#)⁶⁰⁴⁴. To avoid any incompatibilities for existing code we named the type which implements the `std::future` interface as `hpx::unique_future`. For the next release this will be renamed to `hpx::future`, making it full conforming to C++11 Standard⁶⁰⁴⁵.
- A large part of the code base of *HPX* has been refactored and partially re-implemented. The main changes were related to
 - The threading subsystem: these changes significantly reduce the amount of overheads caused by the schedulers, improve the modularity of the code base, and extend the variety of available scheduling algorithms.
 - The parcel subsystem: these changes improve the performance of the *HPX* networking layer, modularize the structure of the parcelports, and simplify the creation of new parcelports for other underlying networking libraries.
 - The API subsystem: these changes improved the conformance of the API to C++11 Standard, extend and unify the available API functionality, and decrease the overheads created by various elements of the API.
 - The robustness of the component loading subsystem has been improved significantly, allowing to more portably and more reliably register the components needed by an application as startup. This additionally speeds up general application initialization.
- We added new API functionality like `hpx::migrate` and `hpx::copy_component` which are the basic building blocks necessary for implementing higher level abstractions for system-wide load balancing, runtime-adaptive resource management, and object-oriented checkpointing and state-management.
- We removed the use of C++11 move emulation (using `Boost.Move`), replacing it with C++11 rvalue references. This is the first step towards using more and more native C++11 facilities which we plan to introduce in the future.
- We improved the reference counting scheme used by *HPX* which helps managing distributed objects and memory. This improves the overall stability of *HPX* and further simplifies writing real world applications.
- The minimal Boost version required to use *HPX* is now V1.49.0.
- This release coincides with the first release of HPXPI (V0.1.0), the first implementation of the [XPI specification](#)⁶⁰⁴⁶.

Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this release.

- [Issue #1086](#)⁶⁰⁴⁷ - Expose internal `boost::shared_array` to allow user management of array lifetime
- [Issue #1083](#)⁶⁰⁴⁸ - Make shell examples copyable in docs
- [Issue #1080](#)⁶⁰⁴⁹ - `/threads{locality#*/total}/count/cumulative` broken
- [Issue #1079](#)⁶⁰⁵⁰ - Build problems on OS X

⁶⁰⁴³ <http://www.open-std.org/jtc1/sc22/wg21>

⁶⁰⁴⁴ <http://mail.cct.lsu.edu/pipermail/hpx-users/2014-January/000141.html>

⁶⁰⁴⁵ <http://www.open-std.org/jtc1/sc22/wg21>

⁶⁰⁴⁶ <https://github.com/STELLAR-GROUP/hpxpi/blob/master/spec.pdf?raw=true>

⁶⁰⁴⁷ <https://github.com/TheHPXProject/hpx/issues/1086>

⁶⁰⁴⁸ <https://github.com/TheHPXProject/hpx/issues/1083>

⁶⁰⁴⁹ <https://github.com/TheHPXProject/hpx/issues/1080>

⁶⁰⁵⁰ <https://github.com/TheHPXProject/hpx/issues/1079>

- Issue #1078⁶⁰⁵¹ - Improve robustness of component loading
- Issue #1077⁶⁰⁵² - Fix a missing enum definition for ‘take’ mode
- Issue #1076⁶⁰⁵³ - Merge Jb master
- Issue #1075⁶⁰⁵⁴ - Unknown CMake command “add_hpx_pseudo_target”
- Issue #1074⁶⁰⁵⁵ - Implement `apply_continue_callback` and `apply_colocated_callback`
- Issue #1073⁶⁰⁵⁶ - The new `apply_colocated` and `async_colocated` functions lead to automatic registered functions
- Issue #1071⁶⁰⁵⁷ - Remove `deferred_packaged_task`
- Issue #1069⁶⁰⁵⁸ - `serialize_buffer` with allocator fails at destruction
- Issue #1068⁶⁰⁵⁹ - Coroutine include and forward declarations missing
- Issue #1067⁶⁰⁶⁰ - Add allocator support to `util::serialize_buffer`
- Issue #1066⁶⁰⁶¹ - Allow for `MPI_Init` being called before HPX launches
- Issue #1065⁶⁰⁶² - AGAS cache isn’t used/populated on worker localities
- Issue #1064⁶⁰⁶³ - Reorder includes to ensure `ws2` includes early
- Issue #1063⁶⁰⁶⁴ - Add `hpx::runtime::suspend` and `hpx::runtime::resume`
- Issue #1062⁶⁰⁶⁵ - Fix `async_continue` to properly handle return types
- Issue #1061⁶⁰⁶⁶ - Implement `async_colocated` and `apply_colocated`
- Issue #1060⁶⁰⁶⁷ - Implement minimal component migration
- Issue #1058⁶⁰⁶⁸ - Remove `HPX_UTIL_TUPLE` from code base
- Issue #1057⁶⁰⁶⁹ - Add performance counters for threading subsystem
- Issue #1055⁶⁰⁷⁰ - Thread allocation uses two memory pools
- Issue #1053⁶⁰⁷¹ - Work stealing flawed
- Issue #1052⁶⁰⁷² - Fix a number of warnings
- Issue #1049⁶⁰⁷³ - Fixes for TLS on OSX and more reliable test running

⁶⁰⁵¹ <https://github.com/TheHPXProject/hpx/issues/1078>

⁶⁰⁵² <https://github.com/TheHPXProject/hpx/issues/1077>

⁶⁰⁵³ <https://github.com/TheHPXProject/hpx/issues/1076>

⁶⁰⁵⁴ <https://github.com/TheHPXProject/hpx/issues/1075>

⁶⁰⁵⁵ <https://github.com/TheHPXProject/hpx/issues/1074>

⁶⁰⁵⁶ <https://github.com/TheHPXProject/hpx/issues/1073>

⁶⁰⁵⁷ <https://github.com/TheHPXProject/hpx/issues/1071>

⁶⁰⁵⁸ <https://github.com/TheHPXProject/hpx/issues/1069>

⁶⁰⁵⁹ <https://github.com/TheHPXProject/hpx/issues/1068>

⁶⁰⁶⁰ <https://github.com/TheHPXProject/hpx/issues/1067>

⁶⁰⁶¹ <https://github.com/TheHPXProject/hpx/issues/1066>

⁶⁰⁶² <https://github.com/TheHPXProject/hpx/issues/1065>

⁶⁰⁶³ <https://github.com/TheHPXProject/hpx/issues/1064>

⁶⁰⁶⁴ <https://github.com/TheHPXProject/hpx/issues/1063>

⁶⁰⁶⁵ <https://github.com/TheHPXProject/hpx/issues/1062>

⁶⁰⁶⁶ <https://github.com/TheHPXProject/hpx/issues/1061>

⁶⁰⁶⁷ <https://github.com/TheHPXProject/hpx/issues/1060>

⁶⁰⁶⁸ <https://github.com/TheHPXProject/hpx/issues/1058>

⁶⁰⁶⁹ <https://github.com/TheHPXProject/hpx/issues/1057>

⁶⁰⁷⁰ <https://github.com/TheHPXProject/hpx/issues/1055>

⁶⁰⁷¹ <https://github.com/TheHPXProject/hpx/issues/1053>

⁶⁰⁷² <https://github.com/TheHPXProject/hpx/issues/1052>

⁶⁰⁷³ <https://github.com/TheHPXProject/hpx/issues/1049>

- Issue #1048⁶⁰⁷⁴ - Fixing after 588 hang
- Issue #1047⁶⁰⁷⁵ - Use port '0' for networking when using one locality
- Issue #1046⁶⁰⁷⁶ - `composable_guard` test is broken when having more than one thread
- Issue #1045⁶⁰⁷⁷ - Security missing headers
- Issue #1044⁶⁰⁷⁸ - Native TLS on FreeBSD via `__thread`
- Issue #1043⁶⁰⁷⁹ - `async` et.al. compute the wrong result type
- Issue #1042⁶⁰⁸⁰ - `async` et.al. implicitly unwrap reference_wrappers
- Issue #1041⁶⁰⁸¹ - Remove redundant costly Kleene stars from regex searches
- Issue #1040⁶⁰⁸² - CMake script regex match patterns has unnecessary kleenes
- Issue #1039⁶⁰⁸³ - Remove use of `Boost.Move` and replace with `std::move` and real rvalue refs
- Issue #1038⁶⁰⁸⁴ - Bump minimal required Boost to 1.49.0
- Issue #1037⁶⁰⁸⁵ - Implicit unwrapping of futures in `async` broken
- Issue #1036⁶⁰⁸⁶ - Scheduler hangs when user code attempts to “block” OS-threads
- Issue #1035⁶⁰⁸⁷ - Idle-rate counter always reports 100% idle rate
- Issue #1034⁶⁰⁸⁸ - Symbolic name registration causes application hangs
- Issue #1033⁶⁰⁸⁹ - Application options read in from an options file generate an error message
- Issue #1032⁶⁰⁹⁰ - `hpx::id_type` local reference counting is wrong
- Issue #1031⁶⁰⁹¹ - Negative entry in reference count table
- Issue #1030⁶⁰⁹² - Implement `condition_variable`
- Issue #1029⁶⁰⁹³ - Deadlock in thread scheduling subsystem
- Issue #1028⁶⁰⁹⁴ - HPX-thread cumulative count performance counters report incorrect value
- Issue #1027⁶⁰⁹⁵ - Expose `hpx::thread_interrupted` error code as a separate exception type
- Issue #1026⁶⁰⁹⁶ - Exceptions thrown in asynchronous calls can be lost if the value of the future is never queried

⁶⁰⁷⁴ <https://github.com/TheHPXProject/hpx/issues/1048>

⁶⁰⁷⁵ <https://github.com/TheHPXProject/hpx/issues/1047>

⁶⁰⁷⁶ <https://github.com/TheHPXProject/hpx/issues/1046>

⁶⁰⁷⁷ <https://github.com/TheHPXProject/hpx/issues/1045>

⁶⁰⁷⁸ <https://github.com/TheHPXProject/hpx/issues/1044>

⁶⁰⁷⁹ <https://github.com/TheHPXProject/hpx/issues/1043>

⁶⁰⁸⁰ <https://github.com/TheHPXProject/hpx/issues/1042>

⁶⁰⁸¹ <https://github.com/TheHPXProject/hpx/issues/1041>

⁶⁰⁸² <https://github.com/TheHPXProject/hpx/issues/1040>

⁶⁰⁸³ <https://github.com/TheHPXProject/hpx/issues/1039>

⁶⁰⁸⁴ <https://github.com/TheHPXProject/hpx/issues/1038>

⁶⁰⁸⁵ <https://github.com/TheHPXProject/hpx/issues/1037>

⁶⁰⁸⁶ <https://github.com/TheHPXProject/hpx/issues/1036>

⁶⁰⁸⁷ <https://github.com/TheHPXProject/hpx/issues/1035>

⁶⁰⁸⁸ <https://github.com/TheHPXProject/hpx/issues/1034>

⁶⁰⁸⁹ <https://github.com/TheHPXProject/hpx/issues/1033>

⁶⁰⁹⁰ <https://github.com/TheHPXProject/hpx/issues/1032>

⁶⁰⁹¹ <https://github.com/TheHPXProject/hpx/issues/1031>

⁶⁰⁹² <https://github.com/TheHPXProject/hpx/issues/1030>

⁶⁰⁹³ <https://github.com/TheHPXProject/hpx/issues/1029>

⁶⁰⁹⁴ <https://github.com/TheHPXProject/hpx/issues/1028>

⁶⁰⁹⁵ <https://github.com/TheHPXProject/hpx/issues/1027>

⁶⁰⁹⁶ <https://github.com/TheHPXProject/hpx/issues/1026>

- Issue #1025⁶⁰⁹⁷ - `future::wait_for/wait_until` do not remove callback
- Issue #1024⁶⁰⁹⁸ - Remove dependence to boost assert and create hpx assert
- Issue #1023⁶⁰⁹⁹ - Segfaults with `tcmalloc`
- Issue #1022⁶¹⁰⁰ - prerequisites link in readme is broken
- Issue #1020⁶¹⁰¹ - HPX Deadlock on external synchronization
- Issue #1019⁶¹⁰² - Convert using `BOOST_ASSERT` to `HPX_ASSERT`
- Issue #1018⁶¹⁰³ - compiling bug with gcc 4.8.1
- Issue #1017⁶¹⁰⁴ - Possible crash in `io_pool` executor
- Issue #1016⁶¹⁰⁵ - Crash at startup
- Issue #1014⁶¹⁰⁶ - Implement Increment/Decrement Merging
- Issue #1013⁶¹⁰⁷ - Add more logging channels to enable greater control over logging granularity
- Issue #1012⁶¹⁰⁸ - `--hpx:debug-hpx-log` and `--hpx:debug-agas-log` lead to non-thread safe writes
- Issue #1011⁶¹⁰⁹ - After installation, running applications from the build/staging directory no longer works
- Issue #1010⁶¹¹⁰ - Mergeable decrement requests are not being merged
- Issue #1009⁶¹¹¹ - `--hpx:list-symbolic-names` crashes
- Issue #1007⁶¹¹² - Components are not properly destroyed
- Issue #1006⁶¹¹³ - Segfault/hang in `set_data`
- Issue #1003⁶¹¹⁴ - Performance counter naming issue
- Issue #982⁶¹¹⁵ - Race condition during startup
- Issue #912⁶¹¹⁶ - OS X: component type not found in map
- Issue #663⁶¹¹⁷ - Create a buildbot slave based on Clang 3.2/OSX
- Issue #636⁶¹¹⁸ - Expose `this_locality::apply<act>(p1, p2);` for local execution
- Issue #197⁶¹¹⁹ - Add `--console=address` option for PBS runs

⁶⁰⁹⁷ <https://github.com/TheHPXProject/hpx/issues/1025>

⁶⁰⁹⁸ <https://github.com/TheHPXProject/hpx/issues/1024>

⁶⁰⁹⁹ <https://github.com/TheHPXProject/hpx/issues/1023>

⁶¹⁰⁰ <https://github.com/TheHPXProject/hpx/issues/1022>

⁶¹⁰¹ <https://github.com/TheHPXProject/hpx/issues/1020>

⁶¹⁰² <https://github.com/TheHPXProject/hpx/issues/1019>

⁶¹⁰³ <https://github.com/TheHPXProject/hpx/issues/1018>

⁶¹⁰⁴ <https://github.com/TheHPXProject/hpx/issues/1017>

⁶¹⁰⁵ <https://github.com/TheHPXProject/hpx/issues/1016>

⁶¹⁰⁶ <https://github.com/TheHPXProject/hpx/issues/1014>

⁶¹⁰⁷ <https://github.com/TheHPXProject/hpx/issues/1013>

⁶¹⁰⁸ <https://github.com/TheHPXProject/hpx/issues/1012>

⁶¹⁰⁹ <https://github.com/TheHPXProject/hpx/issues/1011>

⁶¹¹⁰ <https://github.com/TheHPXProject/hpx/issues/1010>

⁶¹¹¹ <https://github.com/TheHPXProject/hpx/issues/1009>

⁶¹¹² <https://github.com/TheHPXProject/hpx/issues/1007>

⁶¹¹³ <https://github.com/TheHPXProject/hpx/issues/1006>

⁶¹¹⁴ <https://github.com/TheHPXProject/hpx/issues/1003>

⁶¹¹⁵ <https://github.com/TheHPXProject/hpx/issues/982>

⁶¹¹⁶ <https://github.com/TheHPXProject/hpx/issues/912>

⁶¹¹⁷ <https://github.com/TheHPXProject/hpx/issues/663>

⁶¹¹⁸ <https://github.com/TheHPXProject/hpx/issues/636>

⁶¹¹⁹ <https://github.com/TheHPXProject/hpx/issues/197>

- [Issue #175](#)⁶¹²⁰ - Asynchronous AGAS API

HPX V0.9.7 (Nov 13, 2013)

We have had over 1000 commits since the last release and we have closed over 180 tickets (bugs, feature requests, etc.).

General changes

- Ported HPX to BlueGene/Q
- Improved HPX support for Xeon/Phi accelerators
- Reimplemented `hpx::bind`, `hpx::tuple`, and `hpx::function` for better performance and better compliance with the C++11 Standard. Added `hpx::mem_fn`.
- Reworked `hpx::when_all` and `hpx::when_any` for better compliance with the ongoing C++ standardization effort, added heterogeneous version for those functions. Added `hpx::when_any_swapped`.
- Added `hpx::copy` as a precursor for a migrate functionality
- Added `hpx::get_ptr` allowing to directly access the memory underlying a given component
- Added the `hpx::lcos::broadcast`, `hpx::lcos::reduce`, and `hpx::lcos::fold` collective operations
- Added `hpx::get_locality_name` allowing to retrieve the name of any of the localities for the application.
- Added support for more flexible thread affinity control from the HPX command line, such as new modes for `--hpx:bind` (balanced, scattered, compact), improved default settings when running multiple localities on the same node.
- Added experimental executors for simpler thread pooling and scheduling. This API may change in the future as it will stay aligned with the ongoing C++ standardization efforts.
- Massively improved the performance of the HPX serialization code. Added partial support for zero copy serialization of array and bitwise-copyable types.
- General performance improvements of the code related to threads and futures.

Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this release.

- [Issue #1005](#)⁶¹²¹ - Allow one to disable array optimizations and zero copy optimizations for each parcelport
- [Issue #1004](#)⁶¹²² - Generate new HPX logo image for the docs
- [Issue #1002](#)⁶¹²³ - If MPI parcelport is not available, running HPX under mpirun should fail
- [Issue #1001](#)⁶¹²⁴ - Zero copy serialization raises assert
- [Issue #1000](#)⁶¹²⁵ - Can't connect to a HPX application running with the MPI parcelport from a non MPI parcelport locality
- [Issue #999](#)⁶¹²⁶ - Optimize `hpx::when_n`

⁶¹²⁰ <https://github.com/TheHPXProject/hpx/issues/175>

⁶¹²¹ <https://github.com/TheHPXProject/hpx/issues/1005>

⁶¹²² <https://github.com/TheHPXProject/hpx/issues/1004>

⁶¹²³ <https://github.com/TheHPXProject/hpx/issues/1002>

⁶¹²⁴ <https://github.com/TheHPXProject/hpx/issues/1001>

⁶¹²⁵ <https://github.com/TheHPXProject/hpx/issues/1000>

⁶¹²⁶ <https://github.com/TheHPXProject/hpx/issues/999>

- Issue #998⁶¹²⁷ - Fixed const-correctness
- Issue #997⁶¹²⁸ - Making `serialize_buffer::data()` type save
- Issue #996⁶¹²⁹ - Memory leak in `hpx::lcos::promise`
- Issue #995⁶¹³⁰ - Race while registering pre-shutdown functions
- Issue #994⁶¹³¹ - `thread_rescheduling` regression test does not compile
- Issue #992⁶¹³² - Correct comments and messages
- Issue #991⁶¹³³ - `setcap cap_sys_rawio=ep` for power profiling causes an HPX application to abort
- Issue #989⁶¹³⁴ - Jacobi hangs during execution
- Issue #988⁶¹³⁵ - `multiple_init` test is failing
- Issue #986⁶¹³⁶ - Can't call a function called "init" from "main" when using `<hpx/hpx_main.hpp>`
- Issue #984⁶¹³⁷ - Reference counting tests are failing
- Issue #983⁶¹³⁸ - `thread_suspension_executor` test fails
- Issue #980⁶¹³⁹ - Terminating HPX threads don't leave stack in virgin state
- Issue #979⁶¹⁴⁰ - Static scheduler not in documents
- Issue #978⁶¹⁴¹ - Preprocessing limits are broken
- Issue #977⁶¹⁴² - Make `tests.regressions.lcos.future_hang_on_get` shorter
- Issue #976⁶¹⁴³ - Wrong library order in `pkgconfig`
- Issue #975⁶¹⁴⁴ - Please reopen #963
- Issue #974⁶¹⁴⁵ - Option `pu-offset` ignored in `fixing_588` branch
- Issue #972⁶¹⁴⁶ - Cannot use MKL with HPX
- Issue #969⁶¹⁴⁷ - Non-existent INI files requested on the command line via `--hpx:config` do not cause warnings or errors.
- Issue #968⁶¹⁴⁸ - Cannot build examples in `fixing_588` branch
- Issue #967⁶¹⁴⁹ - Command line description of `--hpx:queuing` seems wrong

⁶¹²⁷ <https://github.com/TheHPXProject/hpx/issues/998>

⁶¹²⁸ <https://github.com/TheHPXProject/hpx/issues/997>

⁶¹²⁹ <https://github.com/TheHPXProject/hpx/issues/996>

⁶¹³⁰ <https://github.com/TheHPXProject/hpx/issues/995>

⁶¹³¹ <https://github.com/TheHPXProject/hpx/issues/994>

⁶¹³² <https://github.com/TheHPXProject/hpx/issues/992>

⁶¹³³ <https://github.com/TheHPXProject/hpx/issues/991>

⁶¹³⁴ <https://github.com/TheHPXProject/hpx/issues/989>

⁶¹³⁵ <https://github.com/TheHPXProject/hpx/issues/988>

⁶¹³⁶ <https://github.com/TheHPXProject/hpx/issues/986>

⁶¹³⁷ <https://github.com/TheHPXProject/hpx/issues/984>

⁶¹³⁸ <https://github.com/TheHPXProject/hpx/issues/983>

⁶¹³⁹ <https://github.com/TheHPXProject/hpx/issues/980>

⁶¹⁴⁰ <https://github.com/TheHPXProject/hpx/issues/979>

⁶¹⁴¹ <https://github.com/TheHPXProject/hpx/issues/978>

⁶¹⁴² <https://github.com/TheHPXProject/hpx/issues/977>

⁶¹⁴³ <https://github.com/TheHPXProject/hpx/issues/976>

⁶¹⁴⁴ <https://github.com/TheHPXProject/hpx/issues/975>

⁶¹⁴⁵ <https://github.com/TheHPXProject/hpx/issues/974>

⁶¹⁴⁶ <https://github.com/TheHPXProject/hpx/issues/972>

⁶¹⁴⁷ <https://github.com/TheHPXProject/hpx/issues/969>

⁶¹⁴⁸ <https://github.com/TheHPXProject/hpx/issues/968>

⁶¹⁴⁹ <https://github.com/TheHPXProject/hpx/issues/967>

- Issue #966⁶¹⁵⁰ - `--hpx:print-bind` physical core numbers are wrong
- Issue #965⁶¹⁵¹ - Deadlock when building in Release mode
- Issue #963⁶¹⁵² - Not all worker threads are working
- Issue #962⁶¹⁵³ - Problem with SLURM integration
- Issue #961⁶¹⁵⁴ - `--hpx:print-bind` outputs incorrect information
- Issue #960⁶¹⁵⁵ - Fix cut and paste error in documentation of `get_thread_priority`
- Issue #959⁶¹⁵⁶ - Change link to `boost.atomic` in documentation to point to `boost.org`
- Issue #958⁶¹⁵⁷ - Undefined reference to `intrusive_ptr_release`
- Issue #957⁶¹⁵⁸ - Make tuple standard compliant
- Issue #956⁶¹⁵⁹ - Segfault with `a3382fb`
- Issue #955⁶¹⁶⁰ - `--hpx:nodes` and `--hpx:nodefiles` do not work with foreign nodes
- Issue #954⁶¹⁶¹ - Make order of arguments for `hpx::async` and `hpx::broadcast` consistent
- Issue #953⁶¹⁶² - Cannot use MKL with HPX
- Issue #952⁶¹⁶³ - `register_[pre_]shutdown_function` never throw
- Issue #951⁶¹⁶⁴ - Assert when number of threads is greater than hardware concurrency
- Issue #948⁶¹⁶⁵ - `HPX_HAVE_GENERIC_CONTEXT_COROUTINES` conflicts with `HPX_HAVE_FIBER_BASED_COROUTINES`
- Issue #947⁶¹⁶⁶ - Need `MPI_THREAD_MULTIPLE` for backward compatibility
- Issue #946⁶¹⁶⁷ - HPX does not call `MPI_Finalize`
- Issue #945⁶¹⁶⁸ - Segfault with `hpx::lcos::broadcast`
- Issue #944⁶¹⁶⁹ - OS X: assertion `pu_offset_ < hardware_concurrency` failed
- Issue #943⁶¹⁷⁰ - `#include <hpx/hpx_main.hpp>` does not work
- Issue #942⁶¹⁷¹ - Make the BG/Q work with `-O3`
- Issue #940⁶¹⁷² - Use separator when concatenating locality name

⁶¹⁵⁰ <https://github.com/TheHPXProject/hpx/issues/966>

⁶¹⁵¹ <https://github.com/TheHPXProject/hpx/issues/965>

⁶¹⁵² <https://github.com/TheHPXProject/hpx/issues/963>

⁶¹⁵³ <https://github.com/TheHPXProject/hpx/issues/962>

⁶¹⁵⁴ <https://github.com/TheHPXProject/hpx/issues/961>

⁶¹⁵⁵ <https://github.com/TheHPXProject/hpx/issues/960>

⁶¹⁵⁶ <https://github.com/TheHPXProject/hpx/issues/959>

⁶¹⁵⁷ <https://github.com/TheHPXProject/hpx/issues/958>

⁶¹⁵⁸ <https://github.com/TheHPXProject/hpx/issues/957>

⁶¹⁵⁹ <https://github.com/TheHPXProject/hpx/issues/956>

⁶¹⁶⁰ <https://github.com/TheHPXProject/hpx/issues/955>

⁶¹⁶¹ <https://github.com/TheHPXProject/hpx/issues/954>

⁶¹⁶² <https://github.com/TheHPXProject/hpx/issues/953>

⁶¹⁶³ <https://github.com/TheHPXProject/hpx/issues/952>

⁶¹⁶⁴ <https://github.com/TheHPXProject/hpx/issues/951>

⁶¹⁶⁵ <https://github.com/TheHPXProject/hpx/issues/948>

⁶¹⁶⁶ <https://github.com/TheHPXProject/hpx/issues/947>

⁶¹⁶⁷ <https://github.com/TheHPXProject/hpx/issues/946>

⁶¹⁶⁸ <https://github.com/TheHPXProject/hpx/issues/945>

⁶¹⁶⁹ <https://github.com/TheHPXProject/hpx/issues/944>

⁶¹⁷⁰ <https://github.com/TheHPXProject/hpx/issues/943>

⁶¹⁷¹ <https://github.com/TheHPXProject/hpx/issues/942>

⁶¹⁷² <https://github.com/TheHPXProject/hpx/issues/940>

- [Issue #939](#)⁶¹⁷³ - Refactor MPI parcelport to use `MPI_Wait` instead of multiple `MPI_Test` calls
- [Issue #938](#)⁶¹⁷⁴ - Want to officially access `client_base::gid_`
- [Issue #937](#)⁶¹⁷⁵ - `client_base::gid_` should be private`
- [Issue #936](#)⁶¹⁷⁶ - Want doxygen-like source code index
- [Issue #935](#)⁶¹⁷⁷ - Build error with gcc 4.6 and Boost 1.54.0 on hpx trunk and 0.9.6
- [Issue #933](#)⁶¹⁷⁸ - Cannot build HPX with Boost 1.54.0
- [Issue #932](#)⁶¹⁷⁹ - Components are destructed too early
- [Issue #931](#)⁶¹⁸⁰ - Make HPX work on BG/Q
- [Issue #930](#)⁶¹⁸¹ - make git-docs is broken
- [Issue #929](#)⁶¹⁸² - Generating index in docs broken
- [Issue #928](#)⁶¹⁸³ - Optimize `hpx::util::static_` for C++11 compilers supporting magic statics
- [Issue #924](#)⁶¹⁸⁴ - Make `kill_process_tree` (in `process.py`) more robust on Mac OSX
- [Issue #923](#)⁶¹⁸⁵ - Correct BLAS and RNPL cmake tests
- [Issue #922](#)⁶¹⁸⁶ - Cannot link against BLAS
- [Issue #921](#)⁶¹⁸⁷ - Implement `hpx::mem_fn`
- [Issue #920](#)⁶¹⁸⁸ - Output locality with `--hpx:print-bind`
- [Issue #919](#)⁶¹⁸⁹ - Correct grammar; simplify boolean expressions
- [Issue #918](#)⁶¹⁹⁰ - Link to `hello_world.cpp` is broken
- [Issue #917](#)⁶¹⁹¹ - adapt cmake file to new boostbook version
- [Issue #916](#)⁶¹⁹² - fix problem building documentation with `xsltproc` $\geq$ 1.1.27
- [Issue #915](#)⁶¹⁹³ - Add another TBBMalloc library search path
- [Issue #914](#)⁶¹⁹⁴ - Build problem with Intel compiler on Stampede (TACC)
- [Issue #913](#)⁶¹⁹⁵ - fix error messages in fibonacci examples

⁶¹⁷³ <https://github.com/TheHPXProject/hpx/issues/939>

⁶¹⁷⁴ <https://github.com/TheHPXProject/hpx/issues/938>

⁶¹⁷⁵ <https://github.com/TheHPXProject/hpx/issues/937>

⁶¹⁷⁶ <https://github.com/TheHPXProject/hpx/issues/936>

⁶¹⁷⁷ <https://github.com/TheHPXProject/hpx/issues/935>

⁶¹⁷⁸ <https://github.com/TheHPXProject/hpx/issues/933>

⁶¹⁷⁹ <https://github.com/TheHPXProject/hpx/issues/932>

⁶¹⁸⁰ <https://github.com/TheHPXProject/hpx/issues/931>

⁶¹⁸¹ <https://github.com/TheHPXProject/hpx/issues/930>

⁶¹⁸² <https://github.com/TheHPXProject/hpx/issues/929>

⁶¹⁸³ <https://github.com/TheHPXProject/hpx/issues/928>

⁶¹⁸⁴ <https://github.com/TheHPXProject/hpx/issues/924>

⁶¹⁸⁵ <https://github.com/TheHPXProject/hpx/issues/923>

⁶¹⁸⁶ <https://github.com/TheHPXProject/hpx/issues/922>

⁶¹⁸⁷ <https://github.com/TheHPXProject/hpx/issues/921>

⁶¹⁸⁸ <https://github.com/TheHPXProject/hpx/issues/920>

⁶¹⁸⁹ <https://github.com/TheHPXProject/hpx/issues/919>

⁶¹⁹⁰ <https://github.com/TheHPXProject/hpx/issues/918>

⁶¹⁹¹ <https://github.com/TheHPXProject/hpx/issues/917>

⁶¹⁹² <https://github.com/TheHPXProject/hpx/issues/916>

⁶¹⁹³ <https://github.com/TheHPXProject/hpx/issues/915>

⁶¹⁹⁴ <https://github.com/TheHPXProject/hpx/issues/914>

⁶¹⁹⁵ <https://github.com/TheHPXProject/hpx/issues/913>

- [Issue #911](#)⁶¹⁹⁶ - Update OS X build instructions
- [Issue #910](#)⁶¹⁹⁷ - Want like to specify MPI_ROOT instead of compiler wrapper script
- [Issue #909](#)⁶¹⁹⁸ - Warning about void* arithmetic
- [Issue #908](#)⁶¹⁹⁹ - Buildbot for MIC is broken
- [Issue #906](#)⁶²⁰⁰ - Can't use --hpx:bind=balanced with multiple MPI processes
- [Issue #905](#)⁶²⁰¹ - --hpx:bind documentation should describe full grammar
- [Issue #904](#)⁶²⁰² - Add hpx::lcos::fold and hpx::lcos::inverse_fold collective operation
- [Issue #903](#)⁶²⁰³ - Add hpx::when_any_swapped()
- [Issue #902](#)⁶²⁰⁴ - Add hpx::lcos::reduce collective operation
- [Issue #901](#)⁶²⁰⁵ - Web documentation is not searchable
- [Issue #900](#)⁶²⁰⁶ - Web documentation for trunk has no index
- [Issue #898](#)⁶²⁰⁷ - Some tests fail with GCC 4.8.1 and MPI parcel port
- [Issue #897](#)⁶²⁰⁸ - HWLOC causes failures on Mac
- [Issue #896](#)⁶²⁰⁹ - pu-offset leads to startup error
- [Issue #895](#)⁶²¹⁰ - hpx::get_locality_name not defined
- [Issue #894](#)⁶²¹¹ - Race condition at shutdown
- [Issue #893](#)⁶²¹² - --hpx:print-bind switches std::cout to hexadecimal mode
- [Issue #892](#)⁶²¹³ - hwloc_topology_load can be expensive – don't call multiple times
- [Issue #891](#)⁶²¹⁴ - The documentation for get_locality_name is wrong
- [Issue #890](#)⁶²¹⁵ - --hpx:print-bind should not exit
- [Issue #889](#)⁶²¹⁶ - --hpx:debug-hpx-log=FILE does not work
- [Issue #888](#)⁶²¹⁷ - MPI parcelport does not exit cleanly for --hpx:print-bind
- [Issue #887](#)⁶²¹⁸ - Choose thread affinities more cleverly

⁶¹⁹⁶ <https://github.com/TheHPXProject/hpx/issues/911>

⁶¹⁹⁷ <https://github.com/TheHPXProject/hpx/issues/910>

⁶¹⁹⁸ <https://github.com/TheHPXProject/hpx/issues/909>

⁶¹⁹⁹ <https://github.com/TheHPXProject/hpx/issues/908>

⁶²⁰⁰ <https://github.com/TheHPXProject/hpx/issues/906>

⁶²⁰¹ <https://github.com/TheHPXProject/hpx/issues/905>

⁶²⁰² <https://github.com/TheHPXProject/hpx/issues/904>

⁶²⁰³ <https://github.com/TheHPXProject/hpx/issues/903>

⁶²⁰⁴ <https://github.com/TheHPXProject/hpx/issues/902>

⁶²⁰⁵ <https://github.com/TheHPXProject/hpx/issues/901>

⁶²⁰⁶ <https://github.com/TheHPXProject/hpx/issues/900>

⁶²⁰⁷ <https://github.com/TheHPXProject/hpx/issues/898>

⁶²⁰⁸ <https://github.com/TheHPXProject/hpx/issues/897>

⁶²⁰⁹ <https://github.com/TheHPXProject/hpx/issues/896>

⁶²¹⁰ <https://github.com/TheHPXProject/hpx/issues/895>

⁶²¹¹ <https://github.com/TheHPXProject/hpx/issues/894>

⁶²¹² <https://github.com/TheHPXProject/hpx/issues/893>

⁶²¹³ <https://github.com/TheHPXProject/hpx/issues/892>

⁶²¹⁴ <https://github.com/TheHPXProject/hpx/issues/891>

⁶²¹⁵ <https://github.com/TheHPXProject/hpx/issues/890>

⁶²¹⁶ <https://github.com/TheHPXProject/hpx/issues/889>

⁶²¹⁷ <https://github.com/TheHPXProject/hpx/issues/888>

⁶²¹⁸ <https://github.com/TheHPXProject/hpx/issues/887>

- Issue #886⁶²¹⁹ - Logging documentation is confusing
- Issue #885⁶²²⁰ - Two threads are slower than one
- Issue #884⁶²²¹ - `is_callable` failing with member pointers in C++11
- Issue #883⁶²²² - Need help with `is_callable_test`
- Issue #882⁶²²³ - `tests.regressions.lcos.future_hang_on_get` does not terminate
- Issue #881⁶²²⁴ - `tests.regressions/block_matrix/matrix.hh` won't compile with GCC 4.8.1
- Issue #880⁶²²⁵ - HPX does not work on OS X
- Issue #878⁶²²⁶ - `future::unwrap` triggers assertion
- Issue #877⁶²²⁷ - “make tests” has build errors on Ubuntu 12.10
- Issue #876⁶²²⁸ - `tcmalloc` is used by default, even if it is not present
- Issue #875⁶²²⁹ - `global_fixture` is defined in a header file
- Issue #874⁶²³⁰ - Some tests take very long
- Issue #873⁶²³¹ - Add block-matrix code as regression test
- Issue #872⁶²³² - HPX documentation does not say how to run tests with detailed output
- Issue #871⁶²³³ - All tests fail with “make test”
- Issue #870⁶²³⁴ - Please explicitly disable serialization in classes that don't support it
- Issue #868⁶²³⁵ - `boost_any` test failing
- Issue #867⁶²³⁶ - Reduce the number of copies of `hpx::function` arguments
- Issue #863⁶²³⁷ - Futures should not require a default constructor
- Issue #862⁶²³⁸ - `value_or_error` shall not default construct its result
- Issue #861⁶²³⁹ - `HPX_UNUSED` macro
- Issue #860⁶²⁴⁰ - Add functionality to copy construct a component
- Issue #859⁶²⁴¹ - `hpx::endl` should flush

⁶²¹⁹ <https://github.com/TheHPXProject/hpx/issues/886>

⁶²²⁰ <https://github.com/TheHPXProject/hpx/issues/885>

⁶²²¹ <https://github.com/TheHPXProject/hpx/issues/884>

⁶²²² <https://github.com/TheHPXProject/hpx/issues/883>

⁶²²³ <https://github.com/TheHPXProject/hpx/issues/882>

⁶²²⁴ <https://github.com/TheHPXProject/hpx/issues/881>

⁶²²⁵ <https://github.com/TheHPXProject/hpx/issues/880>

⁶²²⁶ <https://github.com/TheHPXProject/hpx/issues/878>

⁶²²⁷ <https://github.com/TheHPXProject/hpx/issues/877>

⁶²²⁸ <https://github.com/TheHPXProject/hpx/issues/876>

⁶²²⁹ <https://github.com/TheHPXProject/hpx/issues/875>

⁶²³⁰ <https://github.com/TheHPXProject/hpx/issues/874>

⁶²³¹ <https://github.com/TheHPXProject/hpx/issues/873>

⁶²³² <https://github.com/TheHPXProject/hpx/issues/872>

⁶²³³ <https://github.com/TheHPXProject/hpx/issues/871>

⁶²³⁴ <https://github.com/TheHPXProject/hpx/issues/870>

⁶²³⁵ <https://github.com/TheHPXProject/hpx/issues/868>

⁶²³⁶ <https://github.com/TheHPXProject/hpx/issues/867>

⁶²³⁷ <https://github.com/TheHPXProject/hpx/issues/863>

⁶²³⁸ <https://github.com/TheHPXProject/hpx/issues/862>

⁶²³⁹ <https://github.com/TheHPXProject/hpx/issues/861>

⁶²⁴⁰ <https://github.com/TheHPXProject/hpx/issues/860>

⁶²⁴¹ <https://github.com/TheHPXProject/hpx/issues/859>

- [Issue #858](#)⁶²⁴² - Create `hpx::get_ptr<>` allowing to access component implementation
- [Issue #855](#)⁶²⁴³ - Implement `hpx::INVOKE`
- [Issue #854](#)⁶²⁴⁴ - `hpx/hpx.hpp` does not include `hpx/include/iostreams.hpp`
- [Issue #853](#)⁶²⁴⁵ - Feature request: null future
- [Issue #852](#)⁶²⁴⁶ - Feature request: Locality names
- [Issue #851](#)⁶²⁴⁷ - `hpx::cout` output does not appear on screen
- [Issue #849](#)⁶²⁴⁸ - All tests fail on OS X after installing
- [Issue #848](#)⁶²⁴⁹ - Update OS X build instructions
- [Issue #846](#)⁶²⁵⁰ - Update `hpx_external_example`
- [Issue #845](#)⁶²⁵¹ - Issues with having both debug and release modules in the same directory
- [Issue #844](#)⁶²⁵² - Create configuration header
- [Issue #843](#)⁶²⁵³ - Tests should use CTest
- [Issue #842](#)⁶²⁵⁴ - Remove `buffer_pool` from MPI parcelport
- [Issue #841](#)⁶²⁵⁵ - Add possibility to broadcast an index with `hpx::lcos::broadcast`
- [Issue #838](#)⁶²⁵⁶ - Simplify `util::tuple`
- [Issue #837](#)⁶²⁵⁷ - Adopt `boost::tuple` tests for `util::tuple`
- [Issue #836](#)⁶²⁵⁸ - Adopt `boost::function` tests for `util::function`
- [Issue #835](#)⁶²⁵⁹ - Tuple interface missing pieces
- [Issue #833](#)⁶²⁶⁰ - Partially preprocessing files not working
- [Issue #832](#)⁶²⁶¹ - Native papi counters do not work with wild cards
- [Issue #831](#)⁶²⁶² - Arithmetics counter fails if only one parameter is given
- [Issue #830](#)⁶²⁶³ - Convert `hpx::util::function` to use new scheme for serializing its base pointer
- [Issue #829](#)⁶²⁶⁴ - Consistently use `decay<T>` instead of `remove_const< remove_reference<T>>`

⁶²⁴² <https://github.com/TheHPXProject/hpx/issues/858>

⁶²⁴³ <https://github.com/TheHPXProject/hpx/issues/855>

⁶²⁴⁴ <https://github.com/TheHPXProject/hpx/issues/854>

⁶²⁴⁵ <https://github.com/TheHPXProject/hpx/issues/853>

⁶²⁴⁶ <https://github.com/TheHPXProject/hpx/issues/852>

⁶²⁴⁷ <https://github.com/TheHPXProject/hpx/issues/851>

⁶²⁴⁸ <https://github.com/TheHPXProject/hpx/issues/849>

⁶²⁴⁹ <https://github.com/TheHPXProject/hpx/issues/848>

⁶²⁵⁰ <https://github.com/TheHPXProject/hpx/issues/846>

⁶²⁵¹ <https://github.com/TheHPXProject/hpx/issues/845>

⁶²⁵² <https://github.com/TheHPXProject/hpx/issues/844>

⁶²⁵³ <https://github.com/TheHPXProject/hpx/issues/843>

⁶²⁵⁴ <https://github.com/TheHPXProject/hpx/issues/842>

⁶²⁵⁵ <https://github.com/TheHPXProject/hpx/issues/841>

⁶²⁵⁶ <https://github.com/TheHPXProject/hpx/issues/838>

⁶²⁵⁷ <https://github.com/TheHPXProject/hpx/issues/837>

⁶²⁵⁸ <https://github.com/TheHPXProject/hpx/issues/836>

⁶²⁵⁹ <https://github.com/TheHPXProject/hpx/issues/835>

⁶²⁶⁰ <https://github.com/TheHPXProject/hpx/issues/833>

⁶²⁶¹ <https://github.com/TheHPXProject/hpx/issues/832>

⁶²⁶² <https://github.com/TheHPXProject/hpx/issues/831>

⁶²⁶³ <https://github.com/TheHPXProject/hpx/issues/830>

⁶²⁶⁴ <https://github.com/TheHPXProject/hpx/issues/829>

- [Issue #828](#)⁶²⁶⁵ - Update future implementation to N3721 and N3722
- [Issue #827](#)⁶²⁶⁶ - Enable MPI parcelport for bootstrapping whenever application was started using mpirun
- [Issue #826](#)⁶²⁶⁷ - Support command line option `--hpx:print-bind` even if `--hpx::bind` was not used
- [Issue #825](#)⁶²⁶⁸ - Memory counters give segfault when attempting to use thread wild cards or numbers only total works
- [Issue #824](#)⁶²⁶⁹ - Enable lambda functions to be used with `hpx::async/hpx::apply`
- [Issue #823](#)⁶²⁷⁰ - Using a hashing filter
- [Issue #822](#)⁶²⁷¹ - Silence unused variable warning
- [Issue #821](#)⁶²⁷² - Detect if a function object is callable with given arguments
- [Issue #820](#)⁶²⁷³ - Allow wildcards to be used for performance counter names
- [Issue #819](#)⁶²⁷⁴ - Make the AGAS symbolic name registry distributed
- [Issue #818](#)⁶²⁷⁵ - Add `future::then()` overload taking an executor
- [Issue #817](#)⁶²⁷⁶ - Fixed typo
- [Issue #815](#)⁶²⁷⁷ - Create an lco that is performing an efficient broadcast of actions
- [Issue #814](#)⁶²⁷⁸ - Papi counters cannot specify `thread#*` to get the counts for all threads
- [Issue #813](#)⁶²⁷⁹ - Scoped unlock
- [Issue #811](#)⁶²⁸⁰ - `simple_central_tuplespace_client` run error
- [Issue #810](#)⁶²⁸¹ - ostream error when `<<` any objects
- [Issue #809](#)⁶²⁸² - Optimize parcel serialization
- [Issue #808](#)⁶²⁸³ - HPX applications throw exception when executed from the build directory
- [Issue #807](#)⁶²⁸⁴ - Create performance counters exposing overall AGAS statistics
- [Issue #795](#)⁶²⁸⁵ - Create timed `make_ready_future`
- [Issue #794](#)⁶²⁸⁶ - Create heterogeneous `when_all/when_any/etc.`
- [Issue #721](#)⁶²⁸⁷ - Make HPX usable for Xeon Phi

⁶²⁶⁵ <https://github.com/TheHPXProject/hpx/issues/828>

⁶²⁶⁶ <https://github.com/TheHPXProject/hpx/issues/827>

⁶²⁶⁷ <https://github.com/TheHPXProject/hpx/issues/826>

⁶²⁶⁸ <https://github.com/TheHPXProject/hpx/issues/825>

⁶²⁶⁹ <https://github.com/TheHPXProject/hpx/issues/824>

⁶²⁷⁰ <https://github.com/TheHPXProject/hpx/issues/823>

⁶²⁷¹ <https://github.com/TheHPXProject/hpx/issues/822>

⁶²⁷² <https://github.com/TheHPXProject/hpx/issues/821>

⁶²⁷³ <https://github.com/TheHPXProject/hpx/issues/820>

⁶²⁷⁴ <https://github.com/TheHPXProject/hpx/issues/819>

⁶²⁷⁵ <https://github.com/TheHPXProject/hpx/issues/818>

⁶²⁷⁶ <https://github.com/TheHPXProject/hpx/issues/817>

⁶²⁷⁷ <https://github.com/TheHPXProject/hpx/issues/815>

⁶²⁷⁸ <https://github.com/TheHPXProject/hpx/issues/814>

⁶²⁷⁹ <https://github.com/TheHPXProject/hpx/issues/813>

⁶²⁸⁰ <https://github.com/TheHPXProject/hpx/issues/811>

⁶²⁸¹ <https://github.com/TheHPXProject/hpx/issues/810>

⁶²⁸² <https://github.com/TheHPXProject/hpx/issues/809>

⁶²⁸³ <https://github.com/TheHPXProject/hpx/issues/808>

⁶²⁸⁴ <https://github.com/TheHPXProject/hpx/issues/807>

⁶²⁸⁵ <https://github.com/TheHPXProject/hpx/issues/795>

⁶²⁸⁶ <https://github.com/TheHPXProject/hpx/issues/794>

⁶²⁸⁷ <https://github.com/TheHPXProject/hpx/issues/721>

- [Issue #694](#)⁶²⁸⁸ - CMake should complain if you attempt to build an example without its dependencies
- [Issue #692](#)⁶²⁸⁹ - SLURM support broken
- [Issue #683](#)⁶²⁹⁰ - python/hpx/process.py imports epoll on all platforms
- [Issue #619](#)⁶²⁹¹ - Automate the doc building process
- [Issue #600](#)⁶²⁹² - GTC performance broken
- [Issue #577](#)⁶²⁹³ - Allow for zero copy serialization/networking
- [Issue #551](#)⁶²⁹⁴ - Change executable names to have debug postfix in Debug builds
- [Issue #544](#)⁶²⁹⁵ - Write a custom .lib file on Windows pulling in hpx_init and hpx.dll, phase out hpx_init
- [Issue #534](#)⁶²⁹⁶ - `hpx::init` should take functions by `std::function` and should accept all forms of `hpx_main`
- [Issue #508](#)⁶²⁹⁷ - FindPackage fails to set FOO_LIBRARY_DIR
- [Issue #506](#)⁶²⁹⁸ - Add cmake support to generate ini files for external applications
- [Issue #470](#)⁶²⁹⁹ - Changing build-type after configure does not update boost library names
- [Issue #453](#)⁶³⁰⁰ - Document `hpx_run_tests.py`
- [Issue #445](#)⁶³⁰¹ - Significant performance mismatch between MPI and HPX in SMP for allgather example
- [Issue #443](#)⁶³⁰² - Make docs viewable from build directory
- [Issue #421](#)⁶³⁰³ - Support multiple HPX instances per node in a batch environment like PBS or SLURM
- [Issue #316](#)⁶³⁰⁴ - Add message size limitation
- [Issue #249](#)⁶³⁰⁵ - Clean up locking code in big boot barrier
- [Issue #136](#)⁶³⁰⁶ - Persistent CMake variables need to be marked as cache variables

⁶²⁸⁸ <https://github.com/TheHPXProject/hpx/issues/694>

⁶²⁸⁹ <https://github.com/TheHPXProject/hpx/issues/692>

⁶²⁹⁰ <https://github.com/TheHPXProject/hpx/issues/683>

⁶²⁹¹ <https://github.com/TheHPXProject/hpx/issues/619>

⁶²⁹² <https://github.com/TheHPXProject/hpx/issues/600>

⁶²⁹³ <https://github.com/TheHPXProject/hpx/issues/577>

⁶²⁹⁴ <https://github.com/TheHPXProject/hpx/issues/551>

⁶²⁹⁵ <https://github.com/TheHPXProject/hpx/issues/544>

⁶²⁹⁶ <https://github.com/TheHPXProject/hpx/issues/534>

⁶²⁹⁷ <https://github.com/TheHPXProject/hpx/issues/508>

⁶²⁹⁸ <https://github.com/TheHPXProject/hpx/issues/506>

⁶²⁹⁹ <https://github.com/TheHPXProject/hpx/issues/470>

⁶³⁰⁰ <https://github.com/TheHPXProject/hpx/issues/453>

⁶³⁰¹ <https://github.com/TheHPXProject/hpx/issues/445>

⁶³⁰² <https://github.com/TheHPXProject/hpx/issues/443>

⁶³⁰³ <https://github.com/TheHPXProject/hpx/issues/421>

⁶³⁰⁴ <https://github.com/TheHPXProject/hpx/issues/316>

⁶³⁰⁵ <https://github.com/TheHPXProject/hpx/issues/249>

⁶³⁰⁶ <https://github.com/TheHPXProject/hpx/issues/136>

HPX V0.9.6 (Jul 30, 2013)

We have had over 1200 commits since the last release and we have closed roughly 140 tickets (bugs, feature requests, etc.).

General changes

The major new features in this release are:

- We further consolidated the API exposed by *HPX*. We aligned our APIs as much as possible with the existing C++11 Standard⁶³⁰⁷ and related proposals to the C++ standardization committee (such as N3632⁶³⁰⁸ and N3857⁶³⁰⁹).
- We implemented a first version of a distributed AGAS service which essentially eliminates all explicit AGAS network traffic.
- We created a native *ibverbs* *parcelport* allowing to take advantage of the superior latency and bandwidth characteristics of *Infiniband* networks.
- We successfully ported *HPX* to the Xeon Phi platform.
- Support for the SLURM scheduling system was implemented.
- Major efforts have been dedicated to improving the performance counter framework, numerous new counters were implemented and new APIs were added.
- We added a modular *parcel* compression system allowing to improve bandwidth utilization (by reducing the overall size of the transferred data).
- We added a modular *parcel* coalescing system allowing to combine several *parcels* into larger messages. This reduces latencies introduced by the communication layer.
- Added an experimental *executors* API allowing to use different scheduling policies for different parts of the code. This API has been modelled after the Standards proposal N3562⁶³¹⁰. This API is bound to change in the future, though.
- Added minimal security support for localities which is enforced on the *parcelport* level. This support is preliminary and experimental and might change in the future.
- We created a *parcelport* using low level MPI functions. This is in support of legacy applications which are to be gradually ported and to support platforms where MPI is the only available portable networking layer.
- We added a preliminary and experimental implementation of a *tuple-space* object which exposes an interface similar to such systems described in the literature (see for instance *The Linda Coordination Language*⁶³¹¹).

⁶³⁰⁷ <http://www.open-std.org/jtc1/sc22/wg21>

⁶³⁰⁸ <http://wg21.link/n3632>

⁶³⁰⁹ <http://wg21.link/n3857>

⁶³¹⁰ <http://wg21.link/n3562>

⁶³¹¹ [https://en.wikipedia.org/wiki/Linda_\(coordination_language\)](https://en.wikipedia.org/wiki/Linda_(coordination_language))

Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this release. This is again a very long list of newly implemented features and fixed issues.

- [Issue #806](#)⁶³¹² - make (all) in examples folder does nothing
- [Issue #805](#)⁶³¹³ - Adding the introduction and fixing DOCBOOK dependencies for Windows use
- [Issue #804](#)⁶³¹⁴ - Add stackless (non-suspendable) thread type
- [Issue #803](#)⁶³¹⁵ - Create proper serialization support functions for util::tuple
- [Issue #800](#)⁶³¹⁶ - Add possibility to disable array optimizations during serialization
- [Issue #798](#)⁶³¹⁷ - HPX_LIMIT does not work for local dataflow
- [Issue #797](#)⁶³¹⁸ - Create a parcelport which uses MPI
- [Issue #796](#)⁶³¹⁹ - Problem with Large Numbers of Threads
- [Issue #793](#)⁶³²⁰ - Changing dataflow test case to hang consistently
- [Issue #792](#)⁶³²¹ - CMake Error
- [Issue #791](#)⁶³²² - Problems with local::dataflow
- [Issue #790](#)⁶³²³ - wait_for() doesn't compile
- [Issue #789](#)⁶³²⁴ - HPX with Intel compiler segfaults
- [Issue #788](#)⁶³²⁵ - Intel compiler support
- [Issue #787](#)⁶³²⁶ - Fixed SFINAE'd specializations
- [Issue #786](#)⁶³²⁷ - Memory issues during benchmarking.
- [Issue #785](#)⁶³²⁸ - Create an API allowing to register external threads with HPX
- [Issue #784](#)⁶³²⁹ - util::plugin is throwing an error when a symbol is not found
- [Issue #783](#)⁶³³⁰ - How does hpx::bind work?
- [Issue #782](#)⁶³³¹ - Added quotes around STRING REPLACE potentially empty arguments
- [Issue #781](#)⁶³³² - Make sure no exceptions propagate into the thread manager

⁶³¹² <https://github.com/TheHPXProject/hpx/issues/806>

⁶³¹³ <https://github.com/TheHPXProject/hpx/issues/805>

⁶³¹⁴ <https://github.com/TheHPXProject/hpx/issues/804>

⁶³¹⁵ <https://github.com/TheHPXProject/hpx/issues/803>

⁶³¹⁶ <https://github.com/TheHPXProject/hpx/issues/800>

⁶³¹⁷ <https://github.com/TheHPXProject/hpx/issues/798>

⁶³¹⁸ <https://github.com/TheHPXProject/hpx/issues/797>

⁶³¹⁹ <https://github.com/TheHPXProject/hpx/issues/796>

⁶³²⁰ <https://github.com/TheHPXProject/hpx/issues/793>

⁶³²¹ <https://github.com/TheHPXProject/hpx/issues/792>

⁶³²² <https://github.com/TheHPXProject/hpx/issues/791>

⁶³²³ <https://github.com/TheHPXProject/hpx/issues/790>

⁶³²⁴ <https://github.com/TheHPXProject/hpx/issues/789>

⁶³²⁵ <https://github.com/TheHPXProject/hpx/issues/788>

⁶³²⁶ <https://github.com/TheHPXProject/hpx/issues/787>

⁶³²⁷ <https://github.com/TheHPXProject/hpx/issues/786>

⁶³²⁸ <https://github.com/TheHPXProject/hpx/issues/785>

⁶³²⁹ <https://github.com/TheHPXProject/hpx/issues/784>

⁶³³⁰ <https://github.com/TheHPXProject/hpx/issues/783>

⁶³³¹ <https://github.com/TheHPXProject/hpx/issues/782>

⁶³³² <https://github.com/TheHPXProject/hpx/issues/781>

- Issue #780⁶³³³ - Allow arithmetics performance counters to expand its parameters
- Issue #779⁶³³⁴ - Test case for 778
- Issue #778⁶³³⁵ - Swapping futures segfaults
- Issue #777⁶³³⁶ - `hpx::lcos::details::when_XXX` don't restore completion handlers
- Issue #776⁶³³⁷ - Compiler chokes on dataflow overload with launch policy
- Issue #775⁶³³⁸ - Runtime error with local dataflow (copying futures?)
- Issue #774⁶³³⁹ - Using local dataflow without explicit namespace
- Issue #773⁶³⁴⁰ - Local dataflow with unwrap: functor operators need to be const
- Issue #772⁶³⁴¹ - Allow (remote) actions to return a future
- Issue #771⁶³⁴² - Setting `HPX_LIMIT` gives huge boost MPL errors
- Issue #770⁶³⁴³ - Add launch policy to (local) dataflow
- Issue #769⁶³⁴⁴ - Make compile time configuration information available
- Issue #768⁶³⁴⁵ - Const correctness problem in local dataflow
- Issue #767⁶³⁴⁶ - Add launch policies to async
- Issue #766⁶³⁴⁷ - Mark data structures for optimized (array based) serialization
- Issue #765⁶³⁴⁸ - Align `hpx::any` with N3508: Any Library Proposal (Revision 2)
- Issue #764⁶³⁴⁹ - Align `hpx::future` with newest N3558: A Standardized Representation of Asynchronous Operations
- Issue #762⁶³⁵⁰ - added a human readable output for the ping pong example
- Issue #761⁶³⁵¹ - Ambiguous typename when constructing derived component
- Issue #760⁶³⁵² - Simple components can not be derived
- Issue #759⁶³⁵³ - make install doesn't give a complete install
- Issue #758⁶³⁵⁴ - Stack overflow when using `locking_hook<>`
- Issue #757⁶³⁵⁵ - copy paste error; unsupported function overloading

⁶³³³ <https://github.com/TheHPXProject/hpx/issues/780>

⁶³³⁴ <https://github.com/TheHPXProject/hpx/issues/779>

⁶³³⁵ <https://github.com/TheHPXProject/hpx/issues/778>

⁶³³⁶ <https://github.com/TheHPXProject/hpx/issues/777>

⁶³³⁷ <https://github.com/TheHPXProject/hpx/issues/776>

⁶³³⁸ <https://github.com/TheHPXProject/hpx/issues/775>

⁶³³⁹ <https://github.com/TheHPXProject/hpx/issues/774>

⁶³⁴⁰ <https://github.com/TheHPXProject/hpx/issues/773>

⁶³⁴¹ <https://github.com/TheHPXProject/hpx/issues/772>

⁶³⁴² <https://github.com/TheHPXProject/hpx/issues/771>

⁶³⁴³ <https://github.com/TheHPXProject/hpx/issues/770>

⁶³⁴⁴ <https://github.com/TheHPXProject/hpx/issues/769>

⁶³⁴⁵ <https://github.com/TheHPXProject/hpx/issues/768>

⁶³⁴⁶ <https://github.com/TheHPXProject/hpx/issues/767>

⁶³⁴⁷ <https://github.com/TheHPXProject/hpx/issues/766>

⁶³⁴⁸ <https://github.com/TheHPXProject/hpx/issues/765>

⁶³⁴⁹ <https://github.com/TheHPXProject/hpx/issues/764>

⁶³⁵⁰ <https://github.com/TheHPXProject/hpx/issues/762>

⁶³⁵¹ <https://github.com/TheHPXProject/hpx/issues/761>

⁶³⁵² <https://github.com/TheHPXProject/hpx/issues/760>

⁶³⁵³ <https://github.com/TheHPXProject/hpx/issues/759>

⁶³⁵⁴ <https://github.com/TheHPXProject/hpx/issues/758>

⁶³⁵⁵ <https://github.com/TheHPXProject/hpx/issues/757>

- [Issue #756](#)⁶³⁵⁶ - GTCX runtime issue in Gordon
- [Issue #755](#)⁶³⁵⁷ - Papi counters don't work with reset and evaluate API's
- [Issue #753](#)⁶³⁵⁸ - cmake bugfix and improved component action docs
- [Issue #752](#)⁶³⁵⁹ - hpx simple component docs
- [Issue #750](#)⁶³⁶⁰ - Add `hpx::util::any`
- [Issue #749](#)⁶³⁶¹ - Thread phase counter is not reset
- [Issue #748](#)⁶³⁶² - Memory performance counter are not registered
- [Issue #747](#)⁶³⁶³ - Create performance counters exposing arithmetic operations
- [Issue #745](#)⁶³⁶⁴ - `apply_callback` needs to invoke callback when applied locally
- [Issue #744](#)⁶³⁶⁵ - CMake fixes
- [Issue #743](#)⁶³⁶⁶ - Problem Building github version of HPX
- [Issue #742](#)⁶³⁶⁷ - Remove `HPX_STD_BIND`
- [Issue #741](#)⁶³⁶⁸ - assertion '`px != 0`' failed: `HPX(assertion_failure)` for low numbers of OS threads
- [Issue #739](#)⁶³⁶⁹ - Performance counters do not count to the end of the program or evaluation
- [Issue #738](#)⁶³⁷⁰ - Dedicated AGAS server runs don't work; console ignores `-a` option.
- [Issue #737](#)⁶³⁷¹ - Missing bind overloads
- [Issue #736](#)⁶³⁷² - Performance counter wildcards do not always work
- [Issue #735](#)⁶³⁷³ - Create native `ibverbs` `parcelport` based on `rdma` operations
- [Issue #734](#)⁶³⁷⁴ - Threads stolen performance counter total is incorrect
- [Issue #733](#)⁶³⁷⁵ - Test benchmarks need to be checked and fixed
- [Issue #732](#)⁶³⁷⁶ - Build fails with Mac, using mac ports `clang-3.3` on latest git branch
- [Issue #731](#)⁶³⁷⁷ - Add global start/stop API for performance counters
- [Issue #730](#)⁶³⁷⁸ - Performance counter values are apparently incorrect

⁶³⁵⁶ <https://github.com/TheHPXProject/hpx/issues/756>

⁶³⁵⁷ <https://github.com/TheHPXProject/hpx/issues/755>

⁶³⁵⁸ <https://github.com/TheHPXProject/hpx/issues/753>

⁶³⁵⁹ <https://github.com/TheHPXProject/hpx/issues/752>

⁶³⁶⁰ <https://github.com/TheHPXProject/hpx/issues/750>

⁶³⁶¹ <https://github.com/TheHPXProject/hpx/issues/749>

⁶³⁶² <https://github.com/TheHPXProject/hpx/issues/748>

⁶³⁶³ <https://github.com/TheHPXProject/hpx/issues/747>

⁶³⁶⁴ <https://github.com/TheHPXProject/hpx/issues/745>

⁶³⁶⁵ <https://github.com/TheHPXProject/hpx/issues/744>

⁶³⁶⁶ <https://github.com/TheHPXProject/hpx/issues/743>

⁶³⁶⁷ <https://github.com/TheHPXProject/hpx/issues/742>

⁶³⁶⁸ <https://github.com/TheHPXProject/hpx/issues/741>

⁶³⁶⁹ <https://github.com/TheHPXProject/hpx/issues/739>

⁶³⁷⁰ <https://github.com/TheHPXProject/hpx/issues/738>

⁶³⁷¹ <https://github.com/TheHPXProject/hpx/issues/737>

⁶³⁷² <https://github.com/TheHPXProject/hpx/issues/736>

⁶³⁷³ <https://github.com/TheHPXProject/hpx/issues/735>

⁶³⁷⁴ <https://github.com/TheHPXProject/hpx/issues/734>

⁶³⁷⁵ <https://github.com/TheHPXProject/hpx/issues/733>

⁶³⁷⁶ <https://github.com/TheHPXProject/hpx/issues/732>

⁶³⁷⁷ <https://github.com/TheHPXProject/hpx/issues/731>

⁶³⁷⁸ <https://github.com/TheHPXProject/hpx/issues/730>

- Issue #729⁶³⁷⁹ - Unhandled switch
- Issue #728⁶³⁸⁰ - Serialization of `hpx::util::function` between two localities causes seg faults
- Issue #727⁶³⁸¹ - Memory counters on Mac OS X
- Issue #725⁶³⁸² - Restore original thread priority on resume
- Issue #724⁶³⁸³ - Performance benchmarks do not depend on main HPX libraries
- Issue #723⁶³⁸⁴ - `[teletype]-hpx:nodes=`cat $PBS_NODEFILE`` works; `-hpx:nodefile=$PBS_NODEFILE` does not.`[c++]`
- Issue #722⁶³⁸⁵ - Fix binding `const` member functions as actions
- Issue #719⁶³⁸⁶ - Create performance counter exposing compression ratio
- Issue #718⁶³⁸⁷ - Add possibility to compress parcel data
- Issue #717⁶³⁸⁸ - `strip_credit_from_gid` has misleading semantics
- Issue #716⁶³⁸⁹ - Non-option arguments to programs run using `pbsdsh` must be before `--hpx:nodes`, contrary to directions
- Issue #715⁶³⁹⁰ - Re-thrown exceptions should retain the original call site
- Issue #714⁶³⁹¹ - failed assertion in debug mode
- Issue #713⁶³⁹² - Add performance counters monitoring connection caches
- Issue #712⁶³⁹³ - Adjust parcel related performance counters to be connection type specific
- Issue #711⁶³⁹⁴ - configuration failure
- Issue #710⁶³⁹⁵ - Error “timed out while trying to find room in the connection cache” when trying to start multiple localities on a single computer
- Issue #709⁶³⁹⁶ - Add new thread state ‘staged’ referring to task descriptions
- Issue #708⁶³⁹⁷ - Detect/mitigate bad non-system installs of GCC on Redhat systems
- Issue #707⁶³⁹⁸ - Many examples do not link with Git HEAD version
- Issue #706⁶³⁹⁹ - `hpx::init` removes portions of non-option command line arguments before last = sign
- Issue #705⁶⁴⁰⁰ - Create rolling average and median aggregating performance counters

⁶³⁷⁹ <https://github.com/TheHPXProject/hpx/issues/729>

⁶³⁸⁰ <https://github.com/TheHPXProject/hpx/issues/728>

⁶³⁸¹ <https://github.com/TheHPXProject/hpx/issues/727>

⁶³⁸² <https://github.com/TheHPXProject/hpx/issues/725>

⁶³⁸³ <https://github.com/TheHPXProject/hpx/issues/724>

⁶³⁸⁴ <https://github.com/TheHPXProject/hpx/issues/723>

⁶³⁸⁵ <https://github.com/TheHPXProject/hpx/issues/722>

⁶³⁸⁶ <https://github.com/TheHPXProject/hpx/issues/719>

⁶³⁸⁷ <https://github.com/TheHPXProject/hpx/issues/718>

⁶³⁸⁸ <https://github.com/TheHPXProject/hpx/issues/717>

⁶³⁸⁹ <https://github.com/TheHPXProject/hpx/issues/716>

⁶³⁹⁰ <https://github.com/TheHPXProject/hpx/issues/715>

⁶³⁹¹ <https://github.com/TheHPXProject/hpx/issues/714>

⁶³⁹² <https://github.com/TheHPXProject/hpx/issues/713>

⁶³⁹³ <https://github.com/TheHPXProject/hpx/issues/712>

⁶³⁹⁴ <https://github.com/TheHPXProject/hpx/issues/711>

⁶³⁹⁵ <https://github.com/TheHPXProject/hpx/issues/710>

⁶³⁹⁶ <https://github.com/TheHPXProject/hpx/issues/709>

⁶³⁹⁷ <https://github.com/TheHPXProject/hpx/issues/708>

⁶³⁹⁸ <https://github.com/TheHPXProject/hpx/issues/707>

⁶³⁹⁹ <https://github.com/TheHPXProject/hpx/issues/706>

⁶⁴⁰⁰ <https://github.com/TheHPXProject/hpx/issues/705>

- [Issue #704](#)⁶⁴⁰¹ - Create performance counter to expose thread queue waiting time
- [Issue #703](#)⁶⁴⁰² - Add support to HPX build system to find librcrttool.a and related headers
- [Issue #699](#)⁶⁴⁰³ - Generalize instrumentation support
- [Issue #698](#)⁶⁴⁰⁴ - compilation failure with hwloc absent
- [Issue #697](#)⁶⁴⁰⁵ - Performance counter counts should be zero indexed
- [Issue #696](#)⁶⁴⁰⁶ - Distributed problem
- [Issue #695](#)⁶⁴⁰⁷ - Bad perf counter time printed
- [Issue #693](#)⁶⁴⁰⁸ - --help doesn't print component specific command line options
- [Issue #692](#)⁶⁴⁰⁹ - SLURM support broken
- [Issue #691](#)⁶⁴¹⁰ - exception while executing any application linked with hwloc
- [Issue #690](#)⁶⁴¹¹ - thread_id_test and thread_launcher_test failing
- [Issue #689](#)⁶⁴¹² - Make the buildbots use hwloc
- [Issue #687](#)⁶⁴¹³ - compilation error fix (hwloc_topology)
- [Issue #686](#)⁶⁴¹⁴ - Linker Error for Applications
- [Issue #684](#)⁶⁴¹⁵ - Pinning of service thread fails when number of worker threads equals the number of cores
- [Issue #682](#)⁶⁴¹⁶ - Add performance counters exposing number of stolen threads
- [Issue #681](#)⁶⁴¹⁷ - Add apply_continue for asynchronous chaining of actions
- [Issue #679](#)⁶⁴¹⁸ - Remove obsolete async_callback API functions
- [Issue #678](#)⁶⁴¹⁹ - Add new API for setting/triggering LCOs
- [Issue #677](#)⁶⁴²⁰ - Add async_continue for true continuation style actions
- [Issue #676](#)⁶⁴²¹ - Buildbot for gcc 4.4 broken
- [Issue #675](#)⁶⁴²² - Partial preprocessing broken
- [Issue #674](#)⁶⁴²³ - HPX segfaults when built with gcc 4.7

⁶⁴⁰¹ <https://github.com/TheHPXProject/hpx/issues/704>

⁶⁴⁰² <https://github.com/TheHPXProject/hpx/issues/703>

⁶⁴⁰³ <https://github.com/TheHPXProject/hpx/issues/699>

⁶⁴⁰⁴ <https://github.com/TheHPXProject/hpx/issues/698>

⁶⁴⁰⁵ <https://github.com/TheHPXProject/hpx/issues/697>

⁶⁴⁰⁶ <https://github.com/TheHPXProject/hpx/issues/696>

⁶⁴⁰⁷ <https://github.com/TheHPXProject/hpx/issues/695>

⁶⁴⁰⁸ <https://github.com/TheHPXProject/hpx/issues/693>

⁶⁴⁰⁹ <https://github.com/TheHPXProject/hpx/issues/692>

⁶⁴¹⁰ <https://github.com/TheHPXProject/hpx/issues/691>

⁶⁴¹¹ <https://github.com/TheHPXProject/hpx/issues/690>

⁶⁴¹² <https://github.com/TheHPXProject/hpx/issues/689>

⁶⁴¹³ <https://github.com/TheHPXProject/hpx/issues/687>

⁶⁴¹⁴ <https://github.com/TheHPXProject/hpx/issues/686>

⁶⁴¹⁵ <https://github.com/TheHPXProject/hpx/issues/684>

⁶⁴¹⁶ <https://github.com/TheHPXProject/hpx/issues/682>

⁶⁴¹⁷ <https://github.com/TheHPXProject/hpx/issues/681>

⁶⁴¹⁸ <https://github.com/TheHPXProject/hpx/issues/679>

⁶⁴¹⁹ <https://github.com/TheHPXProject/hpx/issues/678>

⁶⁴²⁰ <https://github.com/TheHPXProject/hpx/issues/677>

⁶⁴²¹ <https://github.com/TheHPXProject/hpx/issues/676>

⁶⁴²² <https://github.com/TheHPXProject/hpx/issues/675>

⁶⁴²³ <https://github.com/TheHPXProject/hpx/issues/674>

- Issue #673⁶⁴²⁴ - `use_guard_pages` has inconsistent preprocessor guards
- Issue #672⁶⁴²⁵ - External build breaks if library path has spaces
- Issue #671⁶⁴²⁶ - release tarballs are tarbombs
- Issue #670⁶⁴²⁷ - CMake won't find Boost headers in layout=versioned install
- Issue #669⁶⁴²⁸ - Links in docs to source files broken if not installed
- Issue #667⁶⁴²⁹ - Not reading ini file properly
- Issue #664⁶⁴³⁰ - Adapt new meanings of 'const' and 'mutable'
- Issue #661⁶⁴³¹ - Implement BTL Parcel port
- Issue #655⁶⁴³² - Make HPX work with the "decltype" result_of
- Issue #647⁶⁴³³ - documentation for specifying the number of high priority threads
--hpx:high-priority-threads
- Issue #643⁶⁴³⁴ - Error parsing host file
- Issue #642⁶⁴³⁵ - HWLoc issue with TAU
- Issue #639⁶⁴³⁶ - Logging potentially suspends a running thread
- Issue #634⁶⁴³⁷ - Improve error reporting from parcel layer
- Issue #627⁶⁴³⁸ - Add tests for async and apply overloads that accept regular C++ functions
- Issue #626⁶⁴³⁹ - `hpx/future.hpp` header
- Issue #601⁶⁴⁴⁰ - Intel support
- Issue #557⁶⁴⁴¹ - Remove action codes
- Issue #531⁶⁴⁴² - AGAS request and response classes should use switch statements
- Issue #529⁶⁴⁴³ - Investigate the state of hwloc support
- Issue #526⁶⁴⁴⁴ - Make HPX aware of hyper-threading
- Issue #518⁶⁴⁴⁵ - Create facilities allowing to use plain arrays as action arguments
- Issue #473⁶⁴⁴⁶ - hwloc thread binding is broken on CPUs with hyperthreading

⁶⁴²⁴ <https://github.com/TheHPXProject/hpx/issues/673>

⁶⁴²⁵ <https://github.com/TheHPXProject/hpx/issues/672>

⁶⁴²⁶ <https://github.com/TheHPXProject/hpx/issues/671>

⁶⁴²⁷ <https://github.com/TheHPXProject/hpx/issues/670>

⁶⁴²⁸ <https://github.com/TheHPXProject/hpx/issues/669>

⁶⁴²⁹ <https://github.com/TheHPXProject/hpx/issues/667>

⁶⁴³⁰ <https://github.com/TheHPXProject/hpx/issues/664>

⁶⁴³¹ <https://github.com/TheHPXProject/hpx/issues/661>

⁶⁴³² <https://github.com/TheHPXProject/hpx/issues/655>

⁶⁴³³ <https://github.com/TheHPXProject/hpx/issues/647>

⁶⁴³⁴ <https://github.com/TheHPXProject/hpx/issues/643>

⁶⁴³⁵ <https://github.com/TheHPXProject/hpx/issues/642>

⁶⁴³⁶ <https://github.com/TheHPXProject/hpx/issues/639>

⁶⁴³⁷ <https://github.com/TheHPXProject/hpx/issues/634>

⁶⁴³⁸ <https://github.com/TheHPXProject/hpx/issues/627>

⁶⁴³⁹ <https://github.com/TheHPXProject/hpx/issues/626>

⁶⁴⁴⁰ <https://github.com/TheHPXProject/hpx/issues/601>

⁶⁴⁴¹ <https://github.com/TheHPXProject/hpx/issues/557>

⁶⁴⁴² <https://github.com/TheHPXProject/hpx/issues/531>

⁶⁴⁴³ <https://github.com/TheHPXProject/hpx/issues/529>

⁶⁴⁴⁴ <https://github.com/TheHPXProject/hpx/issues/526>

⁶⁴⁴⁵ <https://github.com/TheHPXProject/hpx/issues/518>

⁶⁴⁴⁶ <https://github.com/TheHPXProject/hpx/issues/473>

- [Issue #383](#)⁶⁴⁴⁷ - Change result type detection for `hpx::util::bind` to use `result_of` protocol
- [Issue #341](#)⁶⁴⁴⁸ - Consolidate route code
- [Issue #219](#)⁶⁴⁴⁹ - Only copy arguments into actions once
- [Issue #177](#)⁶⁴⁵⁰ - Implement distributed AGAS
- [Issue #43](#)⁶⁴⁵¹ - Support for Darwin (Xcode + Clang)

HPX V0.9.5 (Jan 16, 2013)

We have had over 1000 commits since the last release and we have closed roughly 150 tickets (bugs, feature requests, etc.).

General changes

This release is continuing along the lines of code and API consolidation, and overall usability improvements. We dedicated much attention to performance and we were able to significantly improve the threading and networking subsystems.

We successfully ported *HPX* to the Android platform. *HPX* applications now not only can run on mobile devices, but we support heterogeneous applications running across architecture boundaries. At the Supercomputing Conference 2012 we demonstrated connecting Android tablets to simulations running on a Linux cluster. The Android tablet was used to query performance counters from the Linux simulation and to steer its parameters.

We successfully ported *HPX* to Mac OSX (using the Clang compiler). Thanks to Pyry Jähkölä for contributing the corresponding patches. Please see the section `macos_installation` for more details.

We made a special effort to make *HPX* usable in highly concurrent use cases. Many of the *HPX* API functions which possibly take longer than 100 microseconds to execute now can be invoked asynchronously. We added uniform support for composing futures which simplifies to write asynchronous code. *HPX* actions (function objects encapsulating possibly concurrent remote function invocations) are now well integrated with all other API facilities such like `hpx::bind`.

All of the API has been aligned as much as possible with established paradigms. *HPX* now mirrors many of the facilities as defined in the C++11 Standard, such as `hpx::thread`, `hpx::function`, `hpx::future`, etc.

A lot of work has been put into improving the documentation. Many of the API functions are documented now, concepts are explained in detail, and examples are better described than before. The new documentation index enables finding information with lesser effort.

This is the first release of *HPX* we perform after the move to [Github](#)⁶⁴⁵². This step has enabled a wider participation from the community and further encourages us in our decision to release *HPX* as a true open source library (*HPX* is licensed under the very liberal [Boost Software License](#)⁶⁴⁵³).

⁶⁴⁴⁷ <https://github.com/TheHPXProject/hpx/issues/383>

⁶⁴⁴⁸ <https://github.com/TheHPXProject/hpx/issues/341>

⁶⁴⁴⁹ <https://github.com/TheHPXProject/hpx/issues/219>

⁶⁴⁵⁰ <https://github.com/TheHPXProject/hpx/issues/177>

⁶⁴⁵¹ <https://github.com/TheHPXProject/hpx/issues/43>

⁶⁴⁵² <https://github.com/TheHPXProject/hpx/>

⁶⁴⁵³ https://www.boost.org/LICENSE_1_0.txt

Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this release. This is by far the longest list of newly implemented features and fixed issues for any of HPX' releases so far.

- [Issue #666](#)⁶⁴⁵⁴ - Segfault on calling `hpx::finalize` twice
- [Issue #665](#)⁶⁴⁵⁵ - Adding declaration `num_of_cores`
- [Issue #662](#)⁶⁴⁵⁶ - `pkgconfig` is building wrong
- [Issue #660](#)⁶⁴⁵⁷ - Need `uninterrupt` function
- [Issue #659](#)⁶⁴⁵⁸ - Move our logging library into a different namespace
- [Issue #658](#)⁶⁴⁵⁹ - Dynamic performance counter types are broken
- [Issue #657](#)⁶⁴⁶⁰ - HPX v0.9.5 (RC1) `hello_world` example segfaulting
- [Issue #656](#)⁶⁴⁶¹ - Define the affinity of `parcel-pool`, `io-pool`, and `timer-pool` threads
- [Issue #654](#)⁶⁴⁶² - Integrate the Boost `auto_index` tool with documentation
- [Issue #653](#)⁶⁴⁶³ - Make HPX build on OS X + Clang + `libc++`
- [Issue #651](#)⁶⁴⁶⁴ - Add fine-grained control for thread pinning
- [Issue #650](#)⁶⁴⁶⁵ - Command line no error message when using `-hpx:(anything)`
- [Issue #645](#)⁶⁴⁶⁶ - Command line aliases don't work in `[teletype]`@file`[c++]`
- [Issue #644](#)⁶⁴⁶⁷ - Terminated threads are not always properly cleaned up
- [Issue #640](#)⁶⁴⁶⁸ - `future_data<T>::set_on_completed_` used without locks
- [Issue #638](#)⁶⁴⁶⁹ - `hpx` build with intel compilers fails on linux
- [Issue #637](#)⁶⁴⁷⁰ - `-copy-dt-needed-entries` breaks with gold
- [Issue #635](#)⁶⁴⁷¹ - Boost V1.53 will add `Boost.Lockfree` and `Boost.Atomic`
- [Issue #633](#)⁶⁴⁷² - Re-add examples to final 0.9.5 release
- [Issue #632](#)⁶⁴⁷³ - Example `thread_aware_timer` is broken
- [Issue #631](#)⁶⁴⁷⁴ - FFT application throws error in `parcellayer`

⁶⁴⁵⁴ <https://github.com/TheHPXProject/hpx/issues/666>

⁶⁴⁵⁵ <https://github.com/TheHPXProject/hpx/issues/665>

⁶⁴⁵⁶ <https://github.com/TheHPXProject/hpx/issues/662>

⁶⁴⁵⁷ <https://github.com/TheHPXProject/hpx/issues/660>

⁶⁴⁵⁸ <https://github.com/TheHPXProject/hpx/issues/659>

⁶⁴⁵⁹ <https://github.com/TheHPXProject/hpx/issues/658>

⁶⁴⁶⁰ <https://github.com/TheHPXProject/hpx/issues/657>

⁶⁴⁶¹ <https://github.com/TheHPXProject/hpx/issues/656>

⁶⁴⁶² <https://github.com/TheHPXProject/hpx/issues/654>

⁶⁴⁶³ <https://github.com/TheHPXProject/hpx/issues/653>

⁶⁴⁶⁴ <https://github.com/TheHPXProject/hpx/issues/651>

⁶⁴⁶⁵ <https://github.com/TheHPXProject/hpx/issues/650>

⁶⁴⁶⁶ <https://github.com/TheHPXProject/hpx/issues/645>

⁶⁴⁶⁷ <https://github.com/TheHPXProject/hpx/issues/644>

⁶⁴⁶⁸ <https://github.com/TheHPXProject/hpx/issues/640>

⁶⁴⁶⁹ <https://github.com/TheHPXProject/hpx/issues/638>

⁶⁴⁷⁰ <https://github.com/TheHPXProject/hpx/issues/637>

⁶⁴⁷¹ <https://github.com/TheHPXProject/hpx/issues/635>

⁶⁴⁷² <https://github.com/TheHPXProject/hpx/issues/633>

⁶⁴⁷³ <https://github.com/TheHPXProject/hpx/issues/632>

⁶⁴⁷⁴ <https://github.com/TheHPXProject/hpx/issues/631>

- [Issue #630](#)⁶⁴⁷⁵ - Event synchronization example is broken
- [Issue #629](#)⁶⁴⁷⁶ - Waiting on futures hangs
- [Issue #628](#)⁶⁴⁷⁷ - Add an HPX_ALWAYS_ASSERT macro
- [Issue #625](#)⁶⁴⁷⁸ - Port coroutines context switch benchmark
- [Issue #621](#)⁶⁴⁷⁹ - New INI section for stack sizes
- [Issue #618](#)⁶⁴⁸⁰ - pkg_config support does not work with a HPX debug build
- [Issue #617](#)⁶⁴⁸¹ - hpx/external/logging/boost/logging/detail/cache_before_init.hpp:139:67: error: 'get_thread_id' was not declared in this scope
- [Issue #616](#)⁶⁴⁸² - Change wait_xxx not to use locking
- [Issue #615](#)⁶⁴⁸³ - Revert visibility 'fix' (fb0b6b8245dad1127b0c25ebafd9386b3945cca9)
- [Issue #614](#)⁶⁴⁸⁴ - Fix Dataflow linker error
- [Issue #613](#)⁶⁴⁸⁵ - find_here should throw an exception on failure
- [Issue #612](#)⁶⁴⁸⁶ - Thread phase doesn't show up in debug mode
- [Issue #611](#)⁶⁴⁸⁷ - Make stack guard pages configurable at runtime (initialization time)
- [Issue #610](#)⁶⁴⁸⁸ - Co-Locate Components
- [Issue #609](#)⁶⁴⁸⁹ - future_overhead
- [Issue #608](#)⁶⁴⁹⁰ - --hpx:list-counter-infos problem
- [Issue #607](#)⁶⁴⁹¹ - Update Boost.Context based backend for coroutines
- [Issue #606](#)⁶⁴⁹² - 1d_wave_equation is not working
- [Issue #605](#)⁶⁴⁹³ - Any C++ function that has serializable arguments and a serializable return type should be remotable
- [Issue #604](#)⁶⁴⁹⁴ - Connecting localities isn't working anymore
- [Issue #603](#)⁶⁴⁹⁵ - Do not verify any ini entries read from a file
- [Issue #602](#)⁶⁴⁹⁶ - Rename argument_size to type_size/ added implementation to get parcel size

⁶⁴⁷⁵ <https://github.com/TheHPXProject/hpx/issues/630>

⁶⁴⁷⁶ <https://github.com/TheHPXProject/hpx/issues/629>

⁶⁴⁷⁷ <https://github.com/TheHPXProject/hpx/issues/628>

⁶⁴⁷⁸ <https://github.com/TheHPXProject/hpx/issues/625>

⁶⁴⁷⁹ <https://github.com/TheHPXProject/hpx/issues/621>

⁶⁴⁸⁰ <https://github.com/TheHPXProject/hpx/issues/618>

⁶⁴⁸¹ <https://github.com/TheHPXProject/hpx/issues/617>

⁶⁴⁸² <https://github.com/TheHPXProject/hpx/issues/616>

⁶⁴⁸³ <https://github.com/TheHPXProject/hpx/issues/615>

⁶⁴⁸⁴ <https://github.com/TheHPXProject/hpx/issues/614>

⁶⁴⁸⁵ <https://github.com/TheHPXProject/hpx/issues/613>

⁶⁴⁸⁶ <https://github.com/TheHPXProject/hpx/issues/612>

⁶⁴⁸⁷ <https://github.com/TheHPXProject/hpx/issues/611>

⁶⁴⁸⁸ <https://github.com/TheHPXProject/hpx/issues/610>

⁶⁴⁸⁹ <https://github.com/TheHPXProject/hpx/issues/609>

⁶⁴⁹⁰ <https://github.com/TheHPXProject/hpx/issues/608>

⁶⁴⁹¹ <https://github.com/TheHPXProject/hpx/issues/607>

⁶⁴⁹² <https://github.com/TheHPXProject/hpx/issues/606>

⁶⁴⁹³ <https://github.com/TheHPXProject/hpx/issues/605>

⁶⁴⁹⁴ <https://github.com/TheHPXProject/hpx/issues/604>

⁶⁴⁹⁵ <https://github.com/TheHPXProject/hpx/issues/603>

⁶⁴⁹⁶ <https://github.com/TheHPXProject/hpx/issues/602>

- [Issue #599](#)⁶⁴⁹⁷ - Enable locality specific command line options
- [Issue #598](#)⁶⁴⁹⁸ - Need an API that accesses the performance counter reporting the system uptime
- [Issue #597](#)⁶⁴⁹⁹ - compiling on ranger
- [Issue #595](#)⁶⁵⁰⁰ - I need a place to store data in a thread self pointer
- [Issue #594](#)⁶⁵⁰¹ - 32/64 interoperability
- [Issue #593](#)⁶⁵⁰² - Warn if logging is disabled at compile time but requested at runtime
- [Issue #592](#)⁶⁵⁰³ - Add optional argument value to `--hpx:list-counters` and `--hpx:list-counter-infos`
- [Issue #591](#)⁶⁵⁰⁴ - Allow for wildcards in performance counter names specified with `--hpx:print-counter`
- [Issue #590](#)⁶⁵⁰⁵ - Local promise semantic differences
- [Issue #589](#)⁶⁵⁰⁶ - Create API to query performance counter names
- [Issue #587](#)⁶⁵⁰⁷ - Add `get_num_localities` and `get_num_threads` to AGAS API
- [Issue #586](#)⁶⁵⁰⁸ - Adjust local AGAS cache size based on number of localities
- [Issue #585](#)⁶⁵⁰⁹ - Error while using counters in HPX
- [Issue #584](#)⁶⁵¹⁰ - counting argument size of actions, initial pass.
- [Issue #581](#)⁶⁵¹¹ - Remove `RemoteResult` template parameter for `future<>`
- [Issue #580](#)⁶⁵¹² - Add possibility to hook into actions
- [Issue #578](#)⁶⁵¹³ - Use angle brackets in HPX error dumps
- [Issue #576](#)⁶⁵¹⁴ - Exception incorrectly thrown when `--help` is used
- [Issue #575](#)⁶⁵¹⁵ - `HPX(bad_component_type)` with gcc 4.7.2 and boost 1.51
- [Issue #574](#)⁶⁵¹⁶ - `--hpx:connect` command line parameter not working correctly
- [Issue #571](#)⁶⁵¹⁷ - `hpx::wait()` (callback version) should pass the future to the callback function
- [Issue #570](#)⁶⁵¹⁸ - `hpx::wait` should operate on `boost::arrays` and `std::lists`
- [Issue #569](#)⁶⁵¹⁹ - Add a logging sink for Android

⁶⁴⁹⁷ <https://github.com/TheHPXProject/hpx/issues/599>

⁶⁴⁹⁸ <https://github.com/TheHPXProject/hpx/issues/598>

⁶⁴⁹⁹ <https://github.com/TheHPXProject/hpx/issues/597>

⁶⁵⁰⁰ <https://github.com/TheHPXProject/hpx/issues/595>

⁶⁵⁰¹ <https://github.com/TheHPXProject/hpx/issues/594>

⁶⁵⁰² <https://github.com/TheHPXProject/hpx/issues/593>

⁶⁵⁰³ <https://github.com/TheHPXProject/hpx/issues/592>

⁶⁵⁰⁴ <https://github.com/TheHPXProject/hpx/issues/591>

⁶⁵⁰⁵ <https://github.com/TheHPXProject/hpx/issues/590>

⁶⁵⁰⁶ <https://github.com/TheHPXProject/hpx/issues/589>

⁶⁵⁰⁷ <https://github.com/TheHPXProject/hpx/issues/587>

⁶⁵⁰⁸ <https://github.com/TheHPXProject/hpx/issues/586>

⁶⁵⁰⁹ <https://github.com/TheHPXProject/hpx/issues/585>

⁶⁵¹⁰ <https://github.com/TheHPXProject/hpx/issues/584>

⁶⁵¹¹ <https://github.com/TheHPXProject/hpx/issues/581>

⁶⁵¹² <https://github.com/TheHPXProject/hpx/issues/580>

⁶⁵¹³ <https://github.com/TheHPXProject/hpx/issues/578>

⁶⁵¹⁴ <https://github.com/TheHPXProject/hpx/issues/576>

⁶⁵¹⁵ <https://github.com/TheHPXProject/hpx/issues/575>

⁶⁵¹⁶ <https://github.com/TheHPXProject/hpx/issues/574>

⁶⁵¹⁷ <https://github.com/TheHPXProject/hpx/issues/571>

⁶⁵¹⁸ <https://github.com/TheHPXProject/hpx/issues/570>

⁶⁵¹⁹ <https://github.com/TheHPXProject/hpx/issues/569>

- [Issue #568](#)⁶⁵²⁰ - 2-argument version of `HPX_DEFINE_COMPONENT_ACTION`
- [Issue #567](#)⁶⁵²¹ - Connecting to a running HPX application works only once
- [Issue #565](#)⁶⁵²² - HPX doesn't shutdown properly
- [Issue #564](#)⁶⁵²³ - Partial preprocessing of new component creation interface
- [Issue #563](#)⁶⁵²⁴ - Add `hpx::start/hpx::stop` to avoid blocking main thread
- [Issue #562](#)⁶⁵²⁵ - All command line arguments swallowed by `hpx`
- [Issue #561](#)⁶⁵²⁶ - `Boost.Tuple` is not move aware
- [Issue #558](#)⁶⁵²⁷ - `boost::shared_ptr<>` style semantics/syntax for client classes
- [Issue #556](#)⁶⁵²⁸ - Creation of partially preprocessed headers should be enabled for Boost newer than V1.50
- [Issue #555](#)⁶⁵²⁹ - `BOOST_FORCEINLINE` does not name a type
- [Issue #554](#)⁶⁵³⁰ - Possible race condition in thread `get_id()`
- [Issue #552](#)⁶⁵³¹ - Move enable `client_base`
- [Issue #550](#)⁶⁵³² - Add stack size category 'huge'
- [Issue #549](#)⁶⁵³³ - ShenEOS run seg-faults on single or distributed runs
- [Issue #545](#)⁶⁵³⁴ - `AUTOGLOB` broken for `add_hpx_component`
- [Issue #542](#)⁶⁵³⁵ - `FindHPX_HDF5` still searches multiple times
- [Issue #541](#)⁶⁵³⁶ - Quotes around application name in `hpx::init`
- [Issue #539](#)⁶⁵³⁷ - Race condition occurring with new lightweight threads
- [Issue #535](#)⁶⁵³⁸ - `hpx_run_tests.py` exits with no error code when tests are missing
- [Issue #530](#)⁶⁵³⁹ - Thread description(<unknown>) in logs
- [Issue #523](#)⁶⁵⁴⁰ - Make thread objects more lightweight
- [Issue #521](#)⁶⁵⁴¹ - `hpx::error_code` is not usable for lightweight error handling
- [Issue #520](#)⁶⁵⁴² - Add full user environment to HPX logs

⁶⁵²⁰ <https://github.com/TheHPXProject/hpx/issues/568>

⁶⁵²¹ <https://github.com/TheHPXProject/hpx/issues/567>

⁶⁵²² <https://github.com/TheHPXProject/hpx/issues/565>

⁶⁵²³ <https://github.com/TheHPXProject/hpx/issues/564>

⁶⁵²⁴ <https://github.com/TheHPXProject/hpx/issues/563>

⁶⁵²⁵ <https://github.com/TheHPXProject/hpx/issues/562>

⁶⁵²⁶ <https://github.com/TheHPXProject/hpx/issues/561>

⁶⁵²⁷ <https://github.com/TheHPXProject/hpx/issues/558>

⁶⁵²⁸ <https://github.com/TheHPXProject/hpx/issues/556>

⁶⁵²⁹ <https://github.com/TheHPXProject/hpx/issues/555>

⁶⁵³⁰ <https://github.com/TheHPXProject/hpx/issues/554>

⁶⁵³¹ <https://github.com/TheHPXProject/hpx/issues/552>

⁶⁵³² <https://github.com/TheHPXProject/hpx/issues/550>

⁶⁵³³ <https://github.com/TheHPXProject/hpx/issues/549>

⁶⁵³⁴ <https://github.com/TheHPXProject/hpx/issues/545>

⁶⁵³⁵ <https://github.com/TheHPXProject/hpx/issues/542>

⁶⁵³⁶ <https://github.com/TheHPXProject/hpx/issues/541>

⁶⁵³⁷ <https://github.com/TheHPXProject/hpx/issues/539>

⁶⁵³⁸ <https://github.com/TheHPXProject/hpx/issues/535>

⁶⁵³⁹ <https://github.com/TheHPXProject/hpx/issues/530>

⁶⁵⁴⁰ <https://github.com/TheHPXProject/hpx/issues/523>

⁶⁵⁴¹ <https://github.com/TheHPXProject/hpx/issues/521>

⁶⁵⁴² <https://github.com/TheHPXProject/hpx/issues/520>

- [Issue #519](#)⁶⁵⁴³ - Build succeeds, running fails
- [Issue #517](#)⁶⁵⁴⁴ - Add a guard page to linux coroutine stacks
- [Issue #516](#)⁶⁵⁴⁵ - `hpx::thread::detach` suspends while holding locks, leads to hang in debug
- [Issue #514](#)⁶⁵⁴⁶ - Preprocessed headers for `<hpx/apply.hpp>` don't compile
- [Issue #513](#)⁶⁵⁴⁷ - Buildbot configuration problem
- [Issue #512](#)⁶⁵⁴⁸ - Implement action based stack size customization
- [Issue #511](#)⁶⁵⁴⁹ - Move action priority into a separate type trait
- [Issue #510](#)⁶⁵⁵⁰ - trunk broken
- [Issue #507](#)⁶⁵⁵¹ - no matching function for call to `boost::scoped_ptr<hpx::threads::topology>::scoped_ptr(hpx::thre`
- [Issue #505](#)⁶⁵⁵² - `undefined_symbol` regression test currently failing
- [Issue #502](#)⁶⁵⁵³ - Adding OpenCL and OCLM support to HPX for Windows and Linux
- [Issue #501](#)⁶⁵⁵⁴ - `find_package(HPX)` sets cmake output variables
- [Issue #500](#)⁶⁵⁵⁵ - `wait_any/wait_all` are badly named
- [Issue #499](#)⁶⁵⁵⁶ - Add support for disabling pbs support in pbs runs
- [Issue #498](#)⁶⁵⁵⁷ - Error during no-cache runs
- [Issue #496](#)⁶⁵⁵⁸ - Add partial preprocessing support to cmake
- [Issue #495](#)⁶⁵⁵⁹ - Support HPX modules exporting startup/shutdown functions only
- [Issue #494](#)⁶⁵⁶⁰ - Allow modules to specify when to run startup/shutdown functions
- [Issue #493](#)⁶⁵⁶¹ - Avoid constructing a string in `make_success_code`
- [Issue #492](#)⁶⁵⁶² - Performance counter creation is no longer synchronized at startup
- [Issue #491](#)⁶⁵⁶³ - Performance counter creation is no longer synchronized at startup
- [Issue #490](#)⁶⁵⁶⁴ - Sheneos on `_completed_bulk` seg fault in distributed
- [Issue #489](#)⁶⁵⁶⁵ - compiling issue with `g++44`

⁶⁵⁴³ <https://github.com/TheHPXProject/hpx/issues/519>

⁶⁵⁴⁴ <https://github.com/TheHPXProject/hpx/issues/517>

⁶⁵⁴⁵ <https://github.com/TheHPXProject/hpx/issues/516>

⁶⁵⁴⁶ <https://github.com/TheHPXProject/hpx/issues/514>

⁶⁵⁴⁷ <https://github.com/TheHPXProject/hpx/issues/513>

⁶⁵⁴⁸ <https://github.com/TheHPXProject/hpx/issues/512>

⁶⁵⁴⁹ <https://github.com/TheHPXProject/hpx/issues/511>

⁶⁵⁵⁰ <https://github.com/TheHPXProject/hpx/issues/510>

⁶⁵⁵¹ <https://github.com/TheHPXProject/hpx/issues/507>

⁶⁵⁵² <https://github.com/TheHPXProject/hpx/issues/505>

⁶⁵⁵³ <https://github.com/TheHPXProject/hpx/issues/502>

⁶⁵⁵⁴ <https://github.com/TheHPXProject/hpx/issues/501>

⁶⁵⁵⁵ <https://github.com/TheHPXProject/hpx/issues/500>

⁶⁵⁵⁶ <https://github.com/TheHPXProject/hpx/issues/499>

⁶⁵⁵⁷ <https://github.com/TheHPXProject/hpx/issues/498>

⁶⁵⁵⁸ <https://github.com/TheHPXProject/hpx/issues/496>

⁶⁵⁵⁹ <https://github.com/TheHPXProject/hpx/issues/495>

⁶⁵⁶⁰ <https://github.com/TheHPXProject/hpx/issues/494>

⁶⁵⁶¹ <https://github.com/TheHPXProject/hpx/issues/493>

⁶⁵⁶² <https://github.com/TheHPXProject/hpx/issues/492>

⁶⁵⁶³ <https://github.com/TheHPXProject/hpx/issues/491>

⁶⁵⁶⁴ <https://github.com/TheHPXProject/hpx/issues/490>

⁶⁵⁶⁵ <https://github.com/TheHPXProject/hpx/issues/489>

- Issue #488⁶⁵⁶⁶ - Adding OpenCL and OCLM support to HPX for the MSVC platform
- Issue #487⁶⁵⁶⁷ - FindHPX.cmake problems
- Issue #485⁶⁵⁶⁸ - Change distributing_factory and binpacking_factory to use bulk creation
- Issue #484⁶⁵⁶⁹ - Change HPX_DONT_USE_PREPROCESSED_FILES to HPX_USE_PREPROCESSED_FILES
- Issue #483⁶⁵⁷⁰ - Memory counter for Windows
- Issue #479⁶⁵⁷¹ - strange errors appear when requesting performance counters on multiple nodes
- Issue #477⁶⁵⁷² - Create (global) timer for multi-threaded measurements
- Issue #472⁶⁵⁷³ - Add partial preprocessing using Wave
- Issue #471⁶⁵⁷⁴ - Segfault stack traces don't show up in release
- Issue #468⁶⁵⁷⁵ - External projects need to link with internal components
- Issue #462⁶⁵⁷⁶ - Startup/shutdown functions are called more than once
- Issue #458⁶⁵⁷⁷ - Consolidate hpx::util::high_resolution_timer and hpx::util::high_resolution_clock
- Issue #457⁶⁵⁷⁸ - index out of bounds in allgather_and_gate on 4 cores or more
- Issue #448⁶⁵⁷⁹ - Make HPX compile with clang
- Issue #447⁶⁵⁸⁰ - 'make tests' should execute tests on local installation
- Issue #446⁶⁵⁸¹ - Remove SVN-related code from the codebase
- Issue #444⁶⁵⁸² - race condition in smp
- Issue #441⁶⁵⁸³ - Patched Boost.Serialization headers should only be installed if needed
- Issue #439⁶⁵⁸⁴ - Components using HPX_REGISTER_STARTUP_MODULE fail to compile with MSVC
- Issue #436⁶⁵⁸⁵ - Verify that no locks are being held while threads are suspended
- Issue #435⁶⁵⁸⁶ - Installing HPX should not clobber existing Boost installation
- Issue #434⁶⁵⁸⁷ - Logging external component failed (Boost 1.50)
- Issue #433⁶⁵⁸⁸ - Runtime crash when building all examples

⁶⁵⁶⁶ <https://github.com/TheHPXProject/hpx/issues/488>

⁶⁵⁶⁷ <https://github.com/TheHPXProject/hpx/issues/487>

⁶⁵⁶⁸ <https://github.com/TheHPXProject/hpx/issues/485>

⁶⁵⁶⁹ <https://github.com/TheHPXProject/hpx/issues/484>

⁶⁵⁷⁰ <https://github.com/TheHPXProject/hpx/issues/483>

⁶⁵⁷¹ <https://github.com/TheHPXProject/hpx/issues/479>

⁶⁵⁷² <https://github.com/TheHPXProject/hpx/issues/477>

⁶⁵⁷³ <https://github.com/TheHPXProject/hpx/issues/472>

⁶⁵⁷⁴ <https://github.com/TheHPXProject/hpx/issues/471>

⁶⁵⁷⁵ <https://github.com/TheHPXProject/hpx/issues/468>

⁶⁵⁷⁶ <https://github.com/TheHPXProject/hpx/issues/462>

⁶⁵⁷⁷ <https://github.com/TheHPXProject/hpx/issues/458>

⁶⁵⁷⁸ <https://github.com/TheHPXProject/hpx/issues/457>

⁶⁵⁷⁹ <https://github.com/TheHPXProject/hpx/issues/448>

⁶⁵⁸⁰ <https://github.com/TheHPXProject/hpx/issues/447>

⁶⁵⁸¹ <https://github.com/TheHPXProject/hpx/issues/446>

⁶⁵⁸² <https://github.com/TheHPXProject/hpx/issues/444>

⁶⁵⁸³ <https://github.com/TheHPXProject/hpx/issues/441>

⁶⁵⁸⁴ <https://github.com/TheHPXProject/hpx/issues/439>

⁶⁵⁸⁵ <https://github.com/TheHPXProject/hpx/issues/436>

⁶⁵⁸⁶ <https://github.com/TheHPXProject/hpx/issues/435>

⁶⁵⁸⁷ <https://github.com/TheHPXProject/hpx/issues/434>

⁶⁵⁸⁸ <https://github.com/TheHPXProject/hpx/issues/433>

- Issue #432⁶⁵⁸⁹ - Dataflow hangs on 512 cores/64 nodes
- Issue #430⁶⁵⁹⁰ - Problem with distributing factory
- Issue #424⁶⁵⁹¹ - File paths referring to XSL-files need to be properly escaped
- Issue #417⁶⁵⁹² - Make dataflow LCOs work out of the box by using partial preprocessing
- Issue #413⁶⁵⁹³ - hpx_svnversion.py fails on Windows
- Issue #412⁶⁵⁹⁴ - Make hpx::error_code equivalent to hpx::exception
- Issue #398⁶⁵⁹⁵ - HPX clobbers out-of-tree application specific CMake variables (specifically CMAKE_BUILD_TYPE)
- Issue #394⁶⁵⁹⁶ - Remove code generating random port numbers for network
- Issue #378⁶⁵⁹⁷ - ShenEOS scaling issues
- Issue #354⁶⁵⁹⁸ - Create a coroutines wrapper for Boost.Context
- Issue #349⁶⁵⁹⁹ - Commandline option --localities=N/-lN should be necessary only on AGAS locality
- Issue #334⁶⁶⁰⁰ - Add auto_index support to cmake based documentation toolchain
- Issue #318⁶⁶⁰¹ - Network benchmarks
- Issue #317⁶⁶⁰² - Implement network performance counters
- Issue #310⁶⁶⁰³ - Duplicate logging entries
- Issue #230⁶⁶⁰⁴ - Add compile time option to disable thread debugging info
- Issue #171⁶⁶⁰⁵ - Add an INI option to turn off deadlock detection independently of logging
- Issue #170⁶⁶⁰⁶ - OSHL internal counters are incorrect
- Issue #103⁶⁶⁰⁷ - Better diagnostics for multiple component/action registrations under the same name
- Issue #48⁶⁶⁰⁸ - Support for Darwin (Xcode + Clang)
- Issue #21⁶⁶⁰⁹ - Build fails with GCC 4.6

⁶⁵⁸⁹ <https://github.com/TheHPXProject/hpx/issues/432>

⁶⁵⁹⁰ <https://github.com/TheHPXProject/hpx/issues/430>

⁶⁵⁹¹ <https://github.com/TheHPXProject/hpx/issues/424>

⁶⁵⁹² <https://github.com/TheHPXProject/hpx/issues/417>

⁶⁵⁹³ <https://github.com/TheHPXProject/hpx/issues/413>

⁶⁵⁹⁴ <https://github.com/TheHPXProject/hpx/issues/412>

⁶⁵⁹⁵ <https://github.com/TheHPXProject/hpx/issues/398>

⁶⁵⁹⁶ <https://github.com/TheHPXProject/hpx/issues/394>

⁶⁵⁹⁷ <https://github.com/TheHPXProject/hpx/issues/378>

⁶⁵⁹⁸ <https://github.com/TheHPXProject/hpx/issues/354>

⁶⁵⁹⁹ <https://github.com/TheHPXProject/hpx/issues/349>

⁶⁶⁰⁰ <https://github.com/TheHPXProject/hpx/issues/334>

⁶⁶⁰¹ <https://github.com/TheHPXProject/hpx/issues/318>

⁶⁶⁰² <https://github.com/TheHPXProject/hpx/issues/317>

⁶⁶⁰³ <https://github.com/TheHPXProject/hpx/issues/310>

⁶⁶⁰⁴ <https://github.com/TheHPXProject/hpx/issues/230>

⁶⁶⁰⁵ <https://github.com/TheHPXProject/hpx/issues/171>

⁶⁶⁰⁶ <https://github.com/TheHPXProject/hpx/issues/170>

⁶⁶⁰⁷ <https://github.com/TheHPXProject/hpx/issues/103>

⁶⁶⁰⁸ <https://github.com/TheHPXProject/hpx/issues/48>

⁶⁶⁰⁹ <https://github.com/TheHPXProject/hpx/issues/21>

HPX V0.9.0 (Jul 5, 2012)

We have had roughly 800 commits since the last release and we have closed approximately 80 tickets (bugs, feature requests, etc.).

General changes

- Significant improvements made to the usability of *HPX* in large-scale, distributed environments.
- Renamed `hpx::lcos::packaged_task` to `hpx::lcos::packaged_action` to reflect the semantic differences to a `packaged_task` as defined by the C++11 Standard⁶⁶¹⁰.
- *HPX* now exposes `hpx::thread` which is compliant to the C++11 `std::thread` type except that it (purely locally) represents an *HPX* thread. This new type does not expose any of the remote capabilities of the underlying *HPX*-thread implementation.
- The type `hpx::lcos::future` is now compliant to the C++11 `std::future<>` type. This type can be used to synchronize both, local and remote operations. In both cases the control flow will ‘return’ to the future in order to trigger any continuation.
- The types `hpx::lcos::local::promise` and `hpx::lcos::local::packaged_task` are now compliant to the C++11 `std::promise<>` and `std::packaged_task<>` types. These can be used to create a future representing local work only. Use the types `hpx::lcos::promise` and `hpx::lcos::packaged_action` to wrap any (possibly remote) action into a future.
- `hpx::thread` and `hpx::lcos::future` are now cancelable.
- Added support for sequential and logic composition of `hpx::lcos::futures`. The member function `hpx::lcos::future::when` permits futures to be sequentially composed. The helper functions `hpx::wait_all`, `hpx::wait_any`, and `hpx::wait_n` can be used to wait for more than one future at a time.
- *HPX* now exposes `hpx::apply` and `hpx::async` as the preferred way of creating (or invoking) any deferred work. These functions are usable with various types of functions, function objects, and actions and provide a uniform way to spawn deferred tasks.
- *HPX* now utilizes `hpx::util::bind` to (partially) bind local functions and function objects, and also actions. Remote bound actions can have placeholders as well.
- *HPX* continuations are now fully polymorphic. The class `hpx::actions::forwarding_continuation` is an example of how the user can write its own types of continuations. It can be used to execute any function as an continuation of a particular action.
- Reworked the action invocation API to be fully conformant to normal functions. Actions can now be invoked using `hpx::apply`, `hpx::async`, or using the `operator()` implemented on actions. Actions themselves can now be cheaply instantiated as they do not have any members anymore.
- Reworked the lazy action invocation API. Actions can now be directly bound using `hpx::util::bind` by passing an action instance as the first argument.
- A minimal *HPX* program now looks like this:

```
#include <hpx/hpx_init.hpp>

int hpx_main()
{
    return hpx::finalize();
}
```

(continues on next page)

⁶⁶¹⁰ <http://www.open-std.org/jtc1/sc22/wg21>

(continued from previous page)

```
int main()
{
    return hpx::init();
}
```

This removes the immediate dependency on the `Boost.Program_options`⁶⁶¹¹ library.

Note: This minimal version of an *HPX* program does not support any of the default command line arguments (such as `-help`, or command line options related to PBS). It is suggested to always pass `argc` and `argv` to *HPX* as shown in the example below.

- In order to support those, but still not to depend on `Boost.Program_options`⁶⁶¹², the minimal program can be written as:

```
#include <hpx/hpx_init.hpp>

// The arguments for hpx_main can be left off, which very similar to the
// behavior of ``main()`` as defined by C++.
int hpx_main(int argc, char* argv[])
{
    return hpx::finalize();
}

int main(int argc, char* argv[])
{
    return hpx::init(argc, argv);
}
```

- Added performance counters exposing the number of component instances which are alive on a given locality.
- Added performance counters exposing then number of messages sent and received, the number of parcels sent and received, the number of bytes sent and received, the overall time required to send and receive data, and the overall time required to serialize and deserialize the data.
- Added a new component: `hpx::components::binpacking_factory` which is equivalent to the existing `hpx::components::distributing_factory` component, except that it equalizes the overall population of the components to create. It exposes two factory methods, one based on the number of existing instances of the component type to create, and one based on an arbitrary performance counter which will be queried for all relevant localities.
- Added API functions allowing to access elements of the diagnostic information embedded in the given exception: `hpx::get_locality_id`, `hpx::get_host_name`, `hpx::get_process_id`, `hpx::get_function_name`, `hpx::get_file_name`, `hpx::get_line_number`, `hpx::get_os_thread`, `hpx::get_thread_id`, and `hpx::get_thread_description`.

⁶⁶¹¹ https://www.boost.org/doc/html/program_options.html

⁶⁶¹² https://www.boost.org/doc/html/program_options.html

Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this release:

- [Issue #71](#)⁶⁶¹³ - GIDs that are not serialized via `handle_gid<>` should raise an exception
- [Issue #105](#)⁶⁶¹⁴ - Allow for `hpx::util::functions` to be registered in the AGAS symbolic namespace
- [Issue #107](#)⁶⁶¹⁵ - Nasty threadmanger race condition (reproducible in `sheneos_test`)
- [Issue #108](#)⁶⁶¹⁶ - Add millisecond resolution to *HPX* logs on Linux
- [Issue #110](#)⁶⁶¹⁷ - Shutdown hang in distributed with release build
- [Issue #116](#)⁶⁶¹⁸ - Don't use TSS for the applier and runtime pointers
- [Issue #162](#)⁶⁶¹⁹ - Move local synchronous execution shortcut from `hpx::function` to the applier
- [Issue #172](#)⁶⁶²⁰ - Cache sources in CMake and check if they change manually
- [Issue #178](#)⁶⁶²¹ - Add an INI option to turn off ranged-based AGAS caching
- [Issue #187](#)⁶⁶²² - Support for disabling performance counter deployment
- [Issue #202](#)⁶⁶²³ - Support for sending performance counter data to a specific file
- [Issue #218](#)⁶⁶²⁴ - `boost.coroutines` allows different stack sizes, but stack pool is unaware of this
- [Issue #231](#)⁶⁶²⁵ - Implement movable `boost::bind`
- [Issue #232](#)⁶⁶²⁶ - Implement movable `boost::function`
- [Issue #236](#)⁶⁶²⁷ - Allow binding `hpx::util::function` to actions
- [Issue #239](#)⁶⁶²⁸ - Replace `hpx::function` with `hpx::util::function`
- [Issue #240](#)⁶⁶²⁹ - Can't specify `RemoteResult` with `lcos::async`
- [Issue #242](#)⁶⁶³⁰ - `REGISTER_TEMPLATE` support for plain actions
- [Issue #243](#)⁶⁶³¹ - `handle_gid<>` support for `hpx::util::function`
- [Issue #245](#)⁶⁶³² - `*_c_cache` code throws an exception if the queried GID is not in the local cache
- [Issue #246](#)⁶⁶³³ - Undefined references in `dataflow/adaptive1d` example

⁶⁶¹³ <https://github.com/TheHPXProject/hpx/issues/71>

⁶⁶¹⁴ <https://github.com/TheHPXProject/hpx/issues/105>

⁶⁶¹⁵ <https://github.com/TheHPXProject/hpx/issues/107>

⁶⁶¹⁶ <https://github.com/TheHPXProject/hpx/issues/108>

⁶⁶¹⁷ <https://github.com/TheHPXProject/hpx/issues/110>

⁶⁶¹⁸ <https://github.com/TheHPXProject/hpx/issues/116>

⁶⁶¹⁹ <https://github.com/TheHPXProject/hpx/issues/162>

⁶⁶²⁰ <https://github.com/TheHPXProject/hpx/issues/172>

⁶⁶²¹ <https://github.com/TheHPXProject/hpx/issues/178>

⁶⁶²² <https://github.com/TheHPXProject/hpx/issues/187>

⁶⁶²³ <https://github.com/TheHPXProject/hpx/issues/202>

⁶⁶²⁴ <https://github.com/TheHPXProject/hpx/issues/218>

⁶⁶²⁵ <https://github.com/TheHPXProject/hpx/issues/231>

⁶⁶²⁶ <https://github.com/TheHPXProject/hpx/issues/232>

⁶⁶²⁷ <https://github.com/TheHPXProject/hpx/issues/236>

⁶⁶²⁸ <https://github.com/TheHPXProject/hpx/issues/239>

⁶⁶²⁹ <https://github.com/TheHPXProject/hpx/issues/240>

⁶⁶³⁰ <https://github.com/TheHPXProject/hpx/issues/242>

⁶⁶³¹ <https://github.com/TheHPXProject/hpx/issues/243>

⁶⁶³² <https://github.com/TheHPXProject/hpx/issues/245>

⁶⁶³³ <https://github.com/TheHPXProject/hpx/issues/246>

- Issue #252⁶⁶³⁴ - Problems configuring sheneos with CMake
- Issue #254⁶⁶³⁵ - Lifetime of components doesn't end when client goes out of scope
- Issue #259⁶⁶³⁶ - CMake does not detect that MSVC10 has lambdas
- Issue #260⁶⁶³⁷ - io_service_pool segfault
- Issue #261⁶⁶³⁸ - Late parcel executed outside of pxtread
- Issue #263⁶⁶³⁹ - Cannot select allocator with CMake
- Issue #264⁶⁶⁴⁰ - Fix allocator select
- Issue #267⁶⁶⁴¹ - Runtime error for hello_world
- Issue #269⁶⁶⁴² - pthread_affinity_np test fails to compile
- Issue #270⁶⁶⁴³ - Compiler noise due to -Wcast-qual
- Issue #275⁶⁶⁴⁴ - Problem with configuration tests/include paths on Gentoo
- Issue #325⁶⁶⁴⁵ - Sheneos is 200-400 times slower than the fortran equivalent
- Issue #331⁶⁶⁴⁶ - `hpx::init` and `hpx_main()` should not depend on `program_options`
- Issue #333⁶⁶⁴⁷ - Add doxygen support to CMake for doc toolchain
- Issue #340⁶⁶⁴⁸ - Performance counters for parcels
- Issue #346⁶⁶⁴⁹ - Component loading error when running hello_world in distributed on MSVC2010
- Issue #362⁶⁶⁵⁰ - Missing initializer error
- Issue #363⁶⁶⁵¹ - Parcel port serialization error
- Issue #366⁶⁶⁵² - Parcel buffering leads to types incompatible exception
- Issue #368⁶⁶⁵³ - Scalable alternative to `rand()` needed for *HPX*
- Issue #369⁶⁶⁵⁴ - IB over IP is substantially slower than just using standard TCP/IP
- Issue #374⁶⁶⁵⁵ - `hpx::lcos::wait` should work with dataflows and arbitrary classes meeting the future interface
- Issue #375⁶⁶⁵⁶ - Conflicting/ambiguous overloads of `hpx::lcos::wait`

⁶⁶³⁴ <https://github.com/TheHPXProject/hpx/issues/252>

⁶⁶³⁵ <https://github.com/TheHPXProject/hpx/issues/254>

⁶⁶³⁶ <https://github.com/TheHPXProject/hpx/issues/259>

⁶⁶³⁷ <https://github.com/TheHPXProject/hpx/issues/260>

⁶⁶³⁸ <https://github.com/TheHPXProject/hpx/issues/261>

⁶⁶³⁹ <https://github.com/TheHPXProject/hpx/issues/263>

⁶⁶⁴⁰ <https://github.com/TheHPXProject/hpx/issues/264>

⁶⁶⁴¹ <https://github.com/TheHPXProject/hpx/issues/267>

⁶⁶⁴² <https://github.com/TheHPXProject/hpx/issues/269>

⁶⁶⁴³ <https://github.com/TheHPXProject/hpx/issues/270>

⁶⁶⁴⁴ <https://github.com/TheHPXProject/hpx/issues/275>

⁶⁶⁴⁵ <https://github.com/TheHPXProject/hpx/issues/325>

⁶⁶⁴⁶ <https://github.com/TheHPXProject/hpx/issues/331>

⁶⁶⁴⁷ <https://github.com/TheHPXProject/hpx/issues/333>

⁶⁶⁴⁸ <https://github.com/TheHPXProject/hpx/issues/340>

⁶⁶⁴⁹ <https://github.com/TheHPXProject/hpx/issues/346>

⁶⁶⁵⁰ <https://github.com/TheHPXProject/hpx/issues/362>

⁶⁶⁵¹ <https://github.com/TheHPXProject/hpx/issues/363>

⁶⁶⁵² <https://github.com/TheHPXProject/hpx/issues/366>

⁶⁶⁵³ <https://github.com/TheHPXProject/hpx/issues/368>

⁶⁶⁵⁴ <https://github.com/TheHPXProject/hpx/issues/369>

⁶⁶⁵⁵ <https://github.com/TheHPXProject/hpx/issues/374>

⁶⁶⁵⁶ <https://github.com/TheHPXProject/hpx/issues/375>

- [Issue #376](#)⁶⁶⁵⁷ - Find_HPX.cmake should set CMake variable HPX_FOUND for out of tree builds
- [Issue #377](#)⁶⁶⁵⁸ - ShenEOS interpolate bulk and interpolate_one_bulk are broken
- [Issue #379](#)⁶⁶⁵⁹ - Add support for distributed runs under SLURM
- [Issue #382](#)⁶⁶⁶⁰ - _Unwind_Word not declared in boost.backtrace
- [Issue #387](#)⁶⁶⁶¹ - Doxygen should look only at list of specified files
- [Issue #388](#)⁶⁶⁶² - Running make install on an out-of-tree application is broken
- [Issue #391](#)⁶⁶⁶³ - Out-of-tree application segfaults when running in qsub
- [Issue #392](#)⁶⁶⁶⁴ - Remove HPX_NO_INSTALL option from cmake build system
- [Issue #396](#)⁶⁶⁶⁵ - Pragma related warnings when compiling with older gcc versions
- [Issue #399](#)⁶⁶⁶⁶ - Out of tree component build problems
- [Issue #400](#)⁶⁶⁶⁷ - Out of source builds on Windows: linker should not receive compiler flags
- [Issue #401](#)⁶⁶⁶⁸ - Out of source builds on Windows: components need to be linked with hpx_serialization
- [Issue #404](#)⁶⁶⁶⁹ - gfortran fails to link automatically when fortran files are present
- [Issue #405](#)⁶⁶⁷⁰ - Inability to specify linking order for external libraries
- [Issue #406](#)⁶⁶⁷¹ - Adapt action limits such that dataflow applications work without additional defines
- [Issue #415](#)⁶⁶⁷² - locality_results is not a member of hpx::components::server
- [Issue #425](#)⁶⁶⁷³ - Breaking changes to traits::*result wrt std::vector<id_type>
- [Issue #426](#)⁶⁶⁷⁴ - AUTOGLOB needs to be updated to support fortran

HPX V0.8.1 (Apr 21, 2012)

This is a point release including important bug fixes for *HPX V0.8.0 (Mar 23, 2012)*.

⁶⁶⁵⁷ <https://github.com/TheHPXProject/hpx/issues/376>
⁶⁶⁵⁸ <https://github.com/TheHPXProject/hpx/issues/377>
⁶⁶⁵⁹ <https://github.com/TheHPXProject/hpx/issues/379>
⁶⁶⁶⁰ <https://github.com/TheHPXProject/hpx/issues/382>
⁶⁶⁶¹ <https://github.com/TheHPXProject/hpx/issues/387>
⁶⁶⁶² <https://github.com/TheHPXProject/hpx/issues/388>
⁶⁶⁶³ <https://github.com/TheHPXProject/hpx/issues/391>
⁶⁶⁶⁴ <https://github.com/TheHPXProject/hpx/issues/392>
⁶⁶⁶⁵ <https://github.com/TheHPXProject/hpx/issues/396>
⁶⁶⁶⁶ <https://github.com/TheHPXProject/hpx/issues/399>
⁶⁶⁶⁷ <https://github.com/TheHPXProject/hpx/issues/400>
⁶⁶⁶⁸ <https://github.com/TheHPXProject/hpx/issues/401>
⁶⁶⁶⁹ <https://github.com/TheHPXProject/hpx/issues/404>
⁶⁶⁷⁰ <https://github.com/TheHPXProject/hpx/issues/405>
⁶⁶⁷¹ <https://github.com/TheHPXProject/hpx/issues/406>
⁶⁶⁷² <https://github.com/TheHPXProject/hpx/issues/415>
⁶⁶⁷³ <https://github.com/TheHPXProject/hpx/issues/425>
⁶⁶⁷⁴ <https://github.com/TheHPXProject/hpx/issues/426>

General changes

- *HPX* does not need to be installed anymore to be functional.

Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this point release:

- [Issue #295⁶⁶⁷⁵](https://github.com/TheHPXProject/hpx/issues/295) - Don't require install path to be known at compile time.
- [Issue #371⁶⁶⁷⁶](https://github.com/TheHPXProject/hpx/issues/371) - Add `hpx::iostreams` to standard build.
- [Issue #384⁶⁶⁷⁷](https://github.com/TheHPXProject/hpx/issues/384) - Fix compilation with GCC 4.7.
- [Issue #390⁶⁶⁷⁸](https://github.com/TheHPXProject/hpx/issues/390) - Remove `keep_factory_alive` startup call from ShenEOS; add shutdown call to `H5close`.
- [Issue #393⁶⁶⁷⁹](https://github.com/TheHPXProject/hpx/issues/393) - Thread affinity control is broken.

Bug fixes (commits)

Here is a list of the important commits included in this point release:

- `r7642` - External: Fix backtrace memory violation.
- **`r7775` - Components: Fix symbol visibility bug with component startup providers.** This prevents one components providers from overriding another components.
- `r7778` - Components: Fix startup/shutdown provider shadowing issues.

HPX V0.8.0 (Mar 23, 2012)

We have had roughly 1000 commits since the last release and we have closed approximately 70 tickets (bugs, feature requests, etc.).

General changes

- Improved PBS support, allowing for arbitrary naming schemes of node-hostnames.
- Finished verification of the reference counting framework.
- Implemented decrement merging logic to optimize the distributed reference counting system.
- Restructured the LCO framework. Renamed `hpx::lcos::eager_future<>` and `hpx::lcos::lazy_future<>` into `hpx::lcos::packaged_task` and `hpx::lcos::deferred_packaged_task`. Split `hpx::lcos::promise` into `hpx::lcos::packaged_task` and `hpx::lcos::future`. Added 'local' futures (in namespace `hpx::lcos::local`).
- Improved the general performance of local and remote action invocations. This (under certain circumstances) drastically reduces the number of copies created for each of the parameters and return values.

⁶⁶⁷⁵ <https://github.com/TheHPXProject/hpx/issues/295>

⁶⁶⁷⁶ <https://github.com/TheHPXProject/hpx/issues/371>

⁶⁶⁷⁷ <https://github.com/TheHPXProject/hpx/issues/384>

⁶⁶⁷⁸ <https://github.com/TheHPXProject/hpx/issues/390>

⁶⁶⁷⁹ <https://github.com/TheHPXProject/hpx/issues/393>

- Reworked the performance counter framework. Performance counters are now created only when needed, which reduces the overall resource requirements. The new framework allows for much more flexible creation and management of performance counters. The new sine example application demonstrates some of the capabilities of the new infrastructure.
- Added a buildbot-based continuous build system which gives instant, automated feedback on each commit to SVN.
- Added more automated tests to verify proper functioning of *HPX*.
- Started to create documentation for *HPX* and its API.
- Added documentation toolchain to the build system.
- Added dataflow LCO.
- Changed default *HPX* command line options to have `hpx:` prefix. For instance, the former option `--threads` is now `--hpx:threads`. This has been done to make ambiguities with possible application specific command line options as unlikely as possible. See the section *HPX Command Line Options* for a full list of available options.
- Added the possibility to define command line aliases. The former short (one-letter) command line options have been predefined as aliases for backwards compatibility. See the section *HPX Command Line Options* for a detailed description of command line option aliasing.
- Network connections are now cached based on the connected host. The number of simultaneous connections to a particular host is now limited. Parcels are buffered and bundled if all connections are in use.
- Added more refined thread affinity control. This is based on the external library Portable Hardware Locality (HWLOC).
- Improved support for Windows builds with CMake.
- Added support for components to register their own command line options.
- Added the possibility to register custom startup/shutdown functions for any component. These functions are guaranteed to be executed by an *HPX* thread.
- Added two new experimental thread schedulers: `hierarchy_scheduler` and `periodic_priority_scheduler`. These can be activated by using the command line options `--hpx:queuing=hierarchy` or `--hpx:queuing=periodic`.

Example applications

- [Graph500 performance benchmark](http://www.graph500.org/)⁶⁶⁸⁰ (thanks to Matthew Anderson for contributing this application).
- [GTC \(Gyrokinetic Toroidal Code\)](http://www.nersc.gov/research-and-development/benchmarking-and-workload-characterization/nersc-6-benchmarks/gtc/)⁶⁶⁸¹: a skeleton for particle in cell type codes.
- Random Memory Access: an example demonstrating random memory accesses in a large array
- [ShenEOS example](http://stellarcollapse.org/equationofstate)⁶⁶⁸², demonstrating partitioning of large read-only data structures and exposing an interpolation API.
- Sine performance counter demo.
- Accumulator examples demonstrating how to write and use *HPX* components.
- Quickstart examples (like `hello_world`, `fibonacci`, `quicksort`, `factorial`, etc.) demonstrating simple *HPX* concepts which introduce some of the concepts in *HPX*.
- Load balancing and work stealing demos.

⁶⁶⁸⁰ <http://www.graph500.org/>

⁶⁶⁸¹ <http://www.nersc.gov/research-and-development/benchmarking-and-workload-characterization/nersc-6-benchmarks/gtc/>

⁶⁶⁸² <http://stellarcollapse.org/equationofstate>

API changes

- Moved all local LCOs into a separate namespace `hpx::lcos::local` (for instance, `hpx::lcos::local_mutex` is now `hpx::lcos::local::mutex`).
- Replaced `hpx::actions::function` with `hpx::util::function`. Cleaned up related code.
- Removed `hpx::traits::handle_gid` and moved handling of global reference counts into the corresponding serialization code.
- Changed terminology: `prefix` is now called `locality_id`, renamed the corresponding API functions (such as `hpx::get_prefix`, which is now called `hpx::get_locality_id`).
- Adding `hpx::find_remote_localities`, and `hpx::get_num_localities`.
- Changed performance counter naming scheme to make it more bash friendly. The new performance counter naming scheme is now

```
/object{parentname#parentindex/instance#index}/counter#parameters
```

- Added `hpx::get_worker_thread_num` replacing `hpx::threadmanager_base::get_thread_num`.
- Renamed `hpx::get_num_os_threads` to `hpx::get_os_threads_count`.
- Added `hpx::threads::get_thread_count`.
- Restructured the Futures sub-system, renaming types in accordance with the terminology used by the C++11 ISO standard.

Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this release:

- [Issue #31](#)⁶⁶⁸³ - Specialize `handle_gid<>` for examples and tests
- [Issue #72](#)⁶⁶⁸⁴ - Fix AGAS reference counting
- [Issue #104](#)⁶⁶⁸⁵ - heartbeat throws an exception when decrefing the performance counter it's watching
- [Issue #111](#)⁶⁶⁸⁶ - throttle causes an exception on the target application
- [Issue #142](#)⁶⁶⁸⁷ - One failed component loading causes an unrelated component to fail
- [Issue #165](#)⁶⁶⁸⁸ - Remote exception propagation bug in AGAS reference counting test
- [Issue #186](#)⁶⁶⁸⁹ - Test credit exhaustion/splitting (e.g. `prepare_gid` and symbol NS)
- [Issue #188](#)⁶⁶⁹⁰ - Implement remaining AGAS reference counting test cases
- [Issue #258](#)⁶⁶⁹¹ - No type checking of GIDs in stubs classes
- [Issue #271](#)⁶⁶⁹² - Seg fault/shared pointer assertion in distributed code

⁶⁶⁸³ <https://github.com/TheHPXProject/hpx/issues/31>

⁶⁶⁸⁴ <https://github.com/TheHPXProject/hpx/issues/72>

⁶⁶⁸⁵ <https://github.com/TheHPXProject/hpx/issues/104>

⁶⁶⁸⁶ <https://github.com/TheHPXProject/hpx/issues/111>

⁶⁶⁸⁷ <https://github.com/TheHPXProject/hpx/issues/142>

⁶⁶⁸⁸ <https://github.com/TheHPXProject/hpx/issues/165>

⁶⁶⁸⁹ <https://github.com/TheHPXProject/hpx/issues/186>

⁶⁶⁹⁰ <https://github.com/TheHPXProject/hpx/issues/188>

⁶⁶⁹¹ <https://github.com/TheHPXProject/hpx/issues/258>

⁶⁶⁹² <https://github.com/TheHPXProject/hpx/issues/271>

- [Issue #281](#)⁶⁶⁹³ - CMake options need descriptive text
- [Issue #283](#)⁶⁶⁹⁴ - AGAS caching broken (gva_cache needs to be rewritten with ICL)
- [Issue #285](#)⁶⁶⁹⁵ - HPX_INSTALL root directory not the same as CMAKE_INSTALL_PREFIX
- [Issue #286](#)⁶⁶⁹⁶ - New segfault in dataflow applications
- [Issue #289](#)⁶⁶⁹⁷ - Exceptions should only be logged if not handled
- [Issue #290](#)⁶⁶⁹⁸ - c++11 tests failure
- [Issue #293](#)⁶⁶⁹⁹ - Build target for component libraries
- [Issue #296](#)⁶⁷⁰⁰ - Compilation error with Boost V1.49rc1
- [Issue #298](#)⁶⁷⁰¹ - Illegal instructions on termination
- [Issue #299](#)⁶⁷⁰² - gravity aborts with multiple threads
- [Issue #301](#)⁶⁷⁰³ - Build error with Boost trunk
- [Issue #303](#)⁶⁷⁰⁴ - Logging assertion failure in distributed runs
- [Issue #304](#)⁶⁷⁰⁵ - Exception ‘what’ strings are lost when exceptions from decode_parcel are reported
- [Issue #306](#)⁶⁷⁰⁶ - Performance counter user interface issues
- [Issue #307](#)⁶⁷⁰⁷ - Logging exception in distributed runs
- [Issue #308](#)⁶⁷⁰⁸ - Logging deadlocks in distributed
- [Issue #309](#)⁶⁷⁰⁹ - Reference counting test failures and exceptions
- [Issue #311](#)⁶⁷¹⁰ - Merge AGAS remote_interface with the runtime_support object
- [Issue #314](#)⁶⁷¹¹ - Object tracking for id_types
- [Issue #315](#)⁶⁷¹² - Remove handle_gid and handle credit splitting in id_type serialization
- [Issue #320](#)⁶⁷¹³ - applier::get_locality_id() should return an error value (or throw an exception)
- [Issue #321](#)⁶⁷¹⁴ - Optimization for id_types which are never split should be restored
- [Issue #322](#)⁶⁷¹⁵ - Command line processing ignored with Boost 1.47.0

⁶⁶⁹³ <https://github.com/TheHPXProject/hpx/issues/281>

⁶⁶⁹⁴ <https://github.com/TheHPXProject/hpx/issues/283>

⁶⁶⁹⁵ <https://github.com/TheHPXProject/hpx/issues/285>

⁶⁶⁹⁶ <https://github.com/TheHPXProject/hpx/issues/286>

⁶⁶⁹⁷ <https://github.com/TheHPXProject/hpx/issues/289>

⁶⁶⁹⁸ <https://github.com/TheHPXProject/hpx/issues/290>

⁶⁶⁹⁹ <https://github.com/TheHPXProject/hpx/issues/293>

⁶⁷⁰⁰ <https://github.com/TheHPXProject/hpx/issues/296>

⁶⁷⁰¹ <https://github.com/TheHPXProject/hpx/issues/298>

⁶⁷⁰² <https://github.com/TheHPXProject/hpx/issues/299>

⁶⁷⁰³ <https://github.com/TheHPXProject/hpx/issues/301>

⁶⁷⁰⁴ <https://github.com/TheHPXProject/hpx/issues/303>

⁶⁷⁰⁵ <https://github.com/TheHPXProject/hpx/issues/304>

⁶⁷⁰⁶ <https://github.com/TheHPXProject/hpx/issues/306>

⁶⁷⁰⁷ <https://github.com/TheHPXProject/hpx/issues/307>

⁶⁷⁰⁸ <https://github.com/TheHPXProject/hpx/issues/308>

⁶⁷⁰⁹ <https://github.com/TheHPXProject/hpx/issues/309>

⁶⁷¹⁰ <https://github.com/TheHPXProject/hpx/issues/311>

⁶⁷¹¹ <https://github.com/TheHPXProject/hpx/issues/314>

⁶⁷¹² <https://github.com/TheHPXProject/hpx/issues/315>

⁶⁷¹³ <https://github.com/TheHPXProject/hpx/issues/320>

⁶⁷¹⁴ <https://github.com/TheHPXProject/hpx/issues/321>

⁶⁷¹⁵ <https://github.com/TheHPXProject/hpx/issues/322>

- Issue #323⁶⁷¹⁶ - Credit exhaustion causes object to stay alive
- Issue #324⁶⁷¹⁷ - Duplicate exception messages
- Issue #326⁶⁷¹⁸ - Integrate Quickbook with CMake
- Issue #329⁶⁷¹⁹ - --help and --version should still work
- Issue #330⁶⁷²⁰ - Create pkg-config files
- Issue #337⁶⁷²¹ - Improve usability of performance counter timestamps
- Issue #338⁶⁷²² - Non-std exceptions deriving from std::exceptions in tfunc may be sliced
- Issue #339⁶⁷²³ - Decrease the number of send_pending_parcel threads
- Issue #343⁶⁷²⁴ - Dynamically setting the stack size doesn't work
- Issue #351⁶⁷²⁵ - 'make install' does not update documents
- Issue #353⁶⁷²⁶ - Disable FIXMEs in the docs by default; add a doc developer CMake option to enable FIXMEs
- Issue #355⁶⁷²⁷ - 'make' doesn't do anything after correct configuration
- Issue #356⁶⁷²⁸ - Don't use `hpx::util::static_` in topology code
- Issue #359⁶⁷²⁹ - Infinite recursion in `hpx::tuple` serialization
- Issue #361⁶⁷³⁰ - Add compile time option to disable logging completely
- Issue #364⁶⁷³¹ - Installation seriously broken in r7443

HPX V0.7.0 (Dec 12, 2011)

We have had roughly 1000 commits since the last release and we have closed approximately 120 tickets (bugs, feature requests, etc.).

⁶⁷¹⁶ <https://github.com/TheHPXProject/hpx/issues/323>

⁶⁷¹⁷ <https://github.com/TheHPXProject/hpx/issues/324>

⁶⁷¹⁸ <https://github.com/TheHPXProject/hpx/issues/326>

⁶⁷¹⁹ <https://github.com/TheHPXProject/hpx/issues/329>

⁶⁷²⁰ <https://github.com/TheHPXProject/hpx/issues/330>

⁶⁷²¹ <https://github.com/TheHPXProject/hpx/issues/337>

⁶⁷²² <https://github.com/TheHPXProject/hpx/issues/338>

⁶⁷²³ <https://github.com/TheHPXProject/hpx/issues/339>

⁶⁷²⁴ <https://github.com/TheHPXProject/hpx/issues/343>

⁶⁷²⁵ <https://github.com/TheHPXProject/hpx/issues/351>

⁶⁷²⁶ <https://github.com/TheHPXProject/hpx/issues/353>

⁶⁷²⁷ <https://github.com/TheHPXProject/hpx/issues/355>

⁶⁷²⁸ <https://github.com/TheHPXProject/hpx/issues/356>

⁶⁷²⁹ <https://github.com/TheHPXProject/hpx/issues/359>

⁶⁷³⁰ <https://github.com/TheHPXProject/hpx/issues/361>

⁶⁷³¹ <https://github.com/TheHPXProject/hpx/issues/364>

General changes

- Completely removed code related to deprecated AGAS V1, started to work on AGAS V2.1.
- Started to clean up and streamline the exposed APIs (see ‘API changes’ below for more details).
- Revamped and unified performance counter framework, added a lot of new performance counter instances for monitoring of a diverse set of internal *HPX* parameters (queue lengths, access statistics, etc.).
- Improved general error handling and logging support.
- Fixed several race conditions, improved overall stability, decreased memory footprint, improved overall performance (major optimizations include native TLS support and ranged-based AGAS caching).
- Added support for running *HPX* applications with PBS.
- Many updates to the build system, added support for gcc 4.5.x and 4.6.x, added C++11 support.
- Many updates to default command line options.
- Added many tests, set up buildbot for continuous integration testing.
- Better shutdown handling of distributed applications.

Example applications

- quickstart/factorial and quickstart/fibonacci, future-recursive parallel algorithms.
- quickstart/hello_world, distributed hello world example.
- quickstart/rma, simple remote memory access example
- quickstart/quicksort, parallel quicksort implementation.
- gtc, gyrokinetic torodial code.
- bfs, breadth-first-search, example code for a graph application.
- sheneos, partitioning of large data sets.
- accumulator, simple component example.
- balancing/os_thread_num, balancing/px_thread_phase, examples demonstrating load balancing and work stealing.

API changes

- Added `hpx::find_all_localities`.
- Added `hpx::terminate` for non-graceful termination of applications.
- Added `hpx::lcos::async` functions for simpler asynchronous programming.
- Added new AGAS interface for handling of symbolic namespace (`hpx::agas::*`).
- Renamed `hpx::components::wait` to `hpx::lcos::wait`.
- Renamed `hpx::lcos::future_value` to `hpx::lcos::promise`.
- Renamed `hpx::lcos::recursive_mutex` to `hpx::lcos::local_recursive_mutex`, `hpx::lcos::mutex` to `hpx::lcos::local_mutex`
- Removed support for Boost versions older than V1.38, recommended Boost version is now V1.47 and newer.

- Removed `hpx::process` (this will be replaced by a real process implementation in the future).
- Removed non-functional LCO code (`hpx::lcos::dataflow`, `hpx::lcos::thunk`, `hpx::lcos::dataflow_variable`).
- Removed deprecated `hpx::naming::full_address`.

Bug fixes (closed tickets)

Here is a list of the important tickets we closed for this release:

- [Issue #28⁶⁷³²](#) - Integrate Windows/Linux CMake code for *HPX* core
- [Issue #32⁶⁷³³](#) - `hpx::cout()` should be `hpx::cout`
- [Issue #33⁶⁷³⁴](#) - AGAS V2 legacy client does not properly handle `error_code`
- [Issue #60⁶⁷³⁵](#) - AGAS: allow for `registerid` to optionally take ownership of the `gid`
- [Issue #62⁶⁷³⁶](#) - `adaptive1d` compilation failure in Fusion
- [Issue #64⁶⁷³⁷](#) - Parcel subsystem doesn't resolve domain names
- [Issue #83⁶⁷³⁸](#) - No error handling if no console is available
- [Issue #84⁶⁷³⁹](#) - No error handling if a hosted locality is treated as the bootstrap server
- [Issue #90⁶⁷⁴⁰](#) - Add general commandline option `-N`
- [Issue #91⁶⁷⁴¹](#) - Add possibility to read command line arguments from file
- [Issue #92⁶⁷⁴²](#) - Always log exceptions/errors to the log file
- [Issue #93⁶⁷⁴³](#) - Log the command line/program name
- [Issue #95⁶⁷⁴⁴](#) - Support for distributed launches
- [Issue #97⁶⁷⁴⁵](#) - Attempt to create a bad component type in AMR examples
- [Issue #100⁶⁷⁴⁶](#) - `factorial` and `factorial_get` examples trigger AGAS component type assertions
- [Issue #101⁶⁷⁴⁷](#) - Segfault when `hpx::process::here()` is called in `fibonacci2`
- [Issue #102⁶⁷⁴⁸](#) - `unknown_component_address` in `int_object_semaphore_client`
- [Issue #114⁶⁷⁴⁹](#) - `marduk` raises assertion with default parameters

⁶⁷³² <https://github.com/TheHPXProject/hpx/issues/28>

⁶⁷³³ <https://github.com/TheHPXProject/hpx/issues/32>

⁶⁷³⁴ <https://github.com/TheHPXProject/hpx/issues/33>

⁶⁷³⁵ <https://github.com/TheHPXProject/hpx/issues/60>

⁶⁷³⁶ <https://github.com/TheHPXProject/hpx/issues/62>

⁶⁷³⁷ <https://github.com/TheHPXProject/hpx/issues/64>

⁶⁷³⁸ <https://github.com/TheHPXProject/hpx/issues/83>

⁶⁷³⁹ <https://github.com/TheHPXProject/hpx/issues/84>

⁶⁷⁴⁰ <https://github.com/TheHPXProject/hpx/issues/90>

⁶⁷⁴¹ <https://github.com/TheHPXProject/hpx/issues/91>

⁶⁷⁴² <https://github.com/TheHPXProject/hpx/issues/92>

⁶⁷⁴³ <https://github.com/TheHPXProject/hpx/issues/93>

⁶⁷⁴⁴ <https://github.com/TheHPXProject/hpx/issues/95>

⁶⁷⁴⁵ <https://github.com/TheHPXProject/hpx/issues/97>

⁶⁷⁴⁶ <https://github.com/TheHPXProject/hpx/issues/100>

⁶⁷⁴⁷ <https://github.com/TheHPXProject/hpx/issues/101>

⁶⁷⁴⁸ <https://github.com/TheHPXProject/hpx/issues/102>

⁶⁷⁴⁹ <https://github.com/TheHPXProject/hpx/issues/114>

- [Issue #115](#)⁶⁷⁵⁰ - Logging messages for SMP runs (on the console) shouldn't be buffered
- [Issue #119](#)⁶⁷⁵¹ - marduk linking strategy breaks other applications
- [Issue #121](#)⁶⁷⁵² - pbsdsh problem
- [Issue #123](#)⁶⁷⁵³ - marduk, dataflow and adaptive1d fail to build
- [Issue #124](#)⁶⁷⁵⁴ - Lower default preprocessing arity
- [Issue #125](#)⁶⁷⁵⁵ - Move hpx::detail::diagnostic_information out of the detail namespace
- [Issue #126](#)⁶⁷⁵⁶ - Test definitions for AGAS reference counting
- [Issue #128](#)⁶⁷⁵⁷ - Add averaging performance counter
- [Issue #129](#)⁶⁷⁵⁸ - Error with endian.hpp while building adaptive1d
- [Issue #130](#)⁶⁷⁵⁹ - Bad initialization of performance counters
- [Issue #131](#)⁶⁷⁶⁰ - Add global startup/shutdown functions to component modules
- [Issue #132](#)⁶⁷⁶¹ - Avoid using auto_ptr
- [Issue #133](#)⁶⁷⁶² - On Windows hpx.dll doesn't get installed
- [Issue #134](#)⁶⁷⁶³ - HPX_LIBRARY does not reflect real library name (on Windows)
- [Issue #135](#)⁶⁷⁶⁴ - Add detection of unique_ptr to build system
- [Issue #137](#)⁶⁷⁶⁵ - Add command line option allowing to repeatedly evaluate performance counters
- [Issue #139](#)⁶⁷⁶⁶ - Logging is broken
- [Issue #140](#)⁶⁷⁶⁷ - CMake problem on windows
- [Issue #141](#)⁶⁷⁶⁸ - Move all non-component libraries into \$PREFIX/lib/hpx
- [Issue #143](#)⁶⁷⁶⁹ - adaptive1d throws an exception with the default command line options
- [Issue #146](#)⁶⁷⁷⁰ - Early exception handling is broken
- [Issue #147](#)⁶⁷⁷¹ - Sheneos doesn't link on Linux
- [Issue #149](#)⁶⁷⁷² - sheneos_test hangs

⁶⁷⁵⁰ <https://github.com/TheHPXProject/hpx/issues/115>

⁶⁷⁵¹ <https://github.com/TheHPXProject/hpx/issues/119>

⁶⁷⁵² <https://github.com/TheHPXProject/hpx/issues/121>

⁶⁷⁵³ <https://github.com/TheHPXProject/hpx/issues/123>

⁶⁷⁵⁴ <https://github.com/TheHPXProject/hpx/issues/124>

⁶⁷⁵⁵ <https://github.com/TheHPXProject/hpx/issues/125>

⁶⁷⁵⁶ <https://github.com/TheHPXProject/hpx/issues/126>

⁶⁷⁵⁷ <https://github.com/TheHPXProject/hpx/issues/128>

⁶⁷⁵⁸ <https://github.com/TheHPXProject/hpx/issues/129>

⁶⁷⁵⁹ <https://github.com/TheHPXProject/hpx/issues/130>

⁶⁷⁶⁰ <https://github.com/TheHPXProject/hpx/issues/131>

⁶⁷⁶¹ <https://github.com/TheHPXProject/hpx/issues/132>

⁶⁷⁶² <https://github.com/TheHPXProject/hpx/issues/133>

⁶⁷⁶³ <https://github.com/TheHPXProject/hpx/issues/134>

⁶⁷⁶⁴ <https://github.com/TheHPXProject/hpx/issues/135>

⁶⁷⁶⁵ <https://github.com/TheHPXProject/hpx/issues/137>

⁶⁷⁶⁶ <https://github.com/TheHPXProject/hpx/issues/139>

⁶⁷⁶⁷ <https://github.com/TheHPXProject/hpx/issues/140>

⁶⁷⁶⁸ <https://github.com/TheHPXProject/hpx/issues/141>

⁶⁷⁶⁹ <https://github.com/TheHPXProject/hpx/issues/143>

⁶⁷⁷⁰ <https://github.com/TheHPXProject/hpx/issues/146>

⁶⁷⁷¹ <https://github.com/TheHPXProject/hpx/issues/147>

⁶⁷⁷² <https://github.com/TheHPXProject/hpx/issues/149>

- Issue #154⁶⁷⁷³ - Compilation fails for r5661
- Issue #155⁶⁷⁷⁴ - Sine performance counters example chokes on chrono headers
- Issue #156⁶⁷⁷⁵ - Add build type to `--version`
- Issue #157⁶⁷⁷⁶ - Extend AGAS caching to store gid ranges
- Issue #158⁶⁷⁷⁷ - r5691 doesn't compile
- Issue #160⁶⁷⁷⁸ - Re-add AGAS function for resolving a locality to its prefix
- Issue #168⁶⁷⁷⁹ - Managed components should be able to access their own GID
- Issue #169⁶⁷⁸⁰ - Rewrite AGAS future pool
- Issue #179⁶⁷⁸¹ - Complete switch to request class for AGAS server interface
- Issue #182⁶⁷⁸² - Sine performance counter is loaded by other examples
- Issue #185⁶⁷⁸³ - Write tests for symbol namespace reference counting
- Issue #191⁶⁷⁸⁴ - Assignment of read-only variable in `point_geometry`
- Issue #200⁶⁷⁸⁵ - Seg faults when querying performance counters
- Issue #204⁶⁷⁸⁶ - `--ifnames` and suffix stripping needs to be more generic
- Issue #205⁶⁷⁸⁷ - `--list-*` and `--print-counter-*` options do not work together and produce no warning
- Issue #207⁶⁷⁸⁸ - Implement decrement entry merging
- Issue #208⁶⁷⁸⁹ - Replace the spinlocks in AGAS with `hpx::lcos::local_mutexes`
- Issue #210⁶⁷⁹⁰ - Add an `--ifprefix` option
- Issue #214⁶⁷⁹¹ - Performance test for PX-thread creation
- Issue #216⁶⁷⁹² - VS2010 compilation
- Issue #222⁶⁷⁹³ - r6045 `context_linux_x86.hpp`
- Issue #223⁶⁷⁹⁴ - fibonacci hangs when changing the state of an active thread
- Issue #225⁶⁷⁹⁵ - Active threads end up in the FEB wait queue

⁶⁷⁷³ <https://github.com/TheHPXProject/hpx/issues/154>

⁶⁷⁷⁴ <https://github.com/TheHPXProject/hpx/issues/155>

⁶⁷⁷⁵ <https://github.com/TheHPXProject/hpx/issues/156>

⁶⁷⁷⁶ <https://github.com/TheHPXProject/hpx/issues/157>

⁶⁷⁷⁷ <https://github.com/TheHPXProject/hpx/issues/158>

⁶⁷⁷⁸ <https://github.com/TheHPXProject/hpx/issues/160>

⁶⁷⁷⁹ <https://github.com/TheHPXProject/hpx/issues/168>

⁶⁷⁸⁰ <https://github.com/TheHPXProject/hpx/issues/169>

⁶⁷⁸¹ <https://github.com/TheHPXProject/hpx/issues/179>

⁶⁷⁸² <https://github.com/TheHPXProject/hpx/issues/182>

⁶⁷⁸³ <https://github.com/TheHPXProject/hpx/issues/185>

⁶⁷⁸⁴ <https://github.com/TheHPXProject/hpx/issues/191>

⁶⁷⁸⁵ <https://github.com/TheHPXProject/hpx/issues/200>

⁶⁷⁸⁶ <https://github.com/TheHPXProject/hpx/issues/204>

⁶⁷⁸⁷ <https://github.com/TheHPXProject/hpx/issues/205>

⁶⁷⁸⁸ <https://github.com/TheHPXProject/hpx/issues/207>

⁶⁷⁸⁹ <https://github.com/TheHPXProject/hpx/issues/208>

⁶⁷⁹⁰ <https://github.com/TheHPXProject/hpx/issues/210>

⁶⁷⁹¹ <https://github.com/TheHPXProject/hpx/issues/214>

⁶⁷⁹² <https://github.com/TheHPXProject/hpx/issues/216>

⁶⁷⁹³ <https://github.com/TheHPXProject/hpx/issues/222>

⁶⁷⁹⁴ <https://github.com/TheHPXProject/hpx/issues/223>

⁶⁷⁹⁵ <https://github.com/TheHPXProject/hpx/issues/225>

- [Issue #226](#)⁶⁷⁹⁶ - VS Build Error for Accumulator Client
- [Issue #228](#)⁶⁷⁹⁷ - Move all traits into namespace `hpx::traits`
- [Issue #229](#)⁶⁷⁹⁸ - Invalid initialization of reference in `thread_init_data`
- [Issue #235](#)⁶⁷⁹⁹ - Invalid GID in `iostreams`
- [Issue #238](#)⁶⁸⁰⁰ - Demangle type names for the default implementation of `get_action_name`
- [Issue #241](#)⁶⁸⁰¹ - C++11 support breaks GCC 4.5
- [Issue #247](#)⁶⁸⁰² - Reference to temporary with GCC 4.4
- [Issue #248](#)⁶⁸⁰³ - Seg fault at shutdown with GCC 4.4
- [Issue #253](#)⁶⁸⁰⁴ - Default component action registration kills compiler
- [Issue #272](#)⁶⁸⁰⁵ - G++ unrecognized command line option
- [Issue #273](#)⁶⁸⁰⁶ - quicksort example doesn't compile
- [Issue #277](#)⁶⁸⁰⁷ - Invalid CMake logic for Windows

2.11.3 Namespace changes

HPX V1.9.0 Namespace changes

The latest release includes amongst others changes in the namespaces so that *HPX* facilities correspond to the C++ Standard Library. The old namespaces are deprecated. Below is a comprehensive list of the namespace changes.

Table 2.187: Namespace changes in V1.9.0

Old namespace	New namespace
<code>hpx::util::mem_fn</code>	<code>hpx::mem_fn</code>
<code>hpx::util::invoke</code>	<code>hpx::invoke</code>
<code>hpx::util::invoke_r</code>	<code>hpx::invoke_r</code>
<code>hpx::util::invoke_fused</code>	<code>hpx::invoke_fused</code>
<code>hpx::util::invoke_fused_r</code>	<code>hpx::invoke_fused_r</code>
<code>hpx::util::unlock_guard</code>	<code>hpx::unlock_guard</code>
<code>hpx::parallel::v1::reduce_by_key</code>	<code>hpx::experimental::reduce_by_key</code>
<code>hpx::parallel::v1::sort_by_key</code>	<code>hpx::experimental::sort_by_key</code>
<code>hpx::parallel::task_canceled_exception</code>	<code>hpx::experimental::task_canceled_exception</code>
<code>hpx::parallel::task_block</code>	<code>hpx::experimental::task_block</code>
<code>hpx::parallel::define_task_block</code>	<code>hpx::experimental::define_task_block</code>
<code>hpx::parallel::define_task_block_restore_thread</code>	<code>hpx::experimental::define_task_block_restore_thread</code>
<code>hpx::execution::experimental::task_group</code>	<code>hpx::experimental::task_group</code>

⁶⁷⁹⁶ <https://github.com/TheHPXProject/hpx/issues/226>

⁶⁷⁹⁷ <https://github.com/TheHPXProject/hpx/issues/228>

⁶⁷⁹⁸ <https://github.com/TheHPXProject/hpx/issues/229>

⁶⁷⁹⁹ <https://github.com/TheHPXProject/hpx/issues/235>

⁶⁸⁰⁰ <https://github.com/TheHPXProject/hpx/issues/238>

⁶⁸⁰¹ <https://github.com/TheHPXProject/hpx/issues/241>

⁶⁸⁰² <https://github.com/TheHPXProject/hpx/issues/247>

⁶⁸⁰³ <https://github.com/TheHPXProject/hpx/issues/248>

⁶⁸⁰⁴ <https://github.com/TheHPXProject/hpx/issues/253>

⁶⁸⁰⁵ <https://github.com/TheHPXProject/hpx/issues/272>

⁶⁸⁰⁶ <https://github.com/TheHPXProject/hpx/issues/273>

⁶⁸⁰⁷ <https://github.com/TheHPXProject/hpx/issues/277>

2.12 Citing HPX

Please cite *HPX* whenever you use it for publications. Use our paper in The Journal of Open Source Software as the main citation for *HPX*: ⁶⁸⁰⁸. Use the Zenodo entry for referring to the latest version of *HPX*: ⁶⁸⁰⁹. Entries for citing specific versions of *HPX* can also be found at ⁶⁸¹⁰.

2.13 HPX users

A list of institutions and projects using *HPX* can be found on the *HPX Users*⁶⁸¹¹ page.

2.14 About HPX

2.14.1 History

The development of High Performance ParallelX (*HPX*) began in 2007. At that time, Hartmut Kaiser became interested in the work done by the ParallelX group at the [Center for Computation and Technology \(CCT\)](https://www.cct.lsu.edu)⁶⁸¹², a multi-disciplinary research institute at [Louisiana State University \(LSU\)](https://www.lsu.edu)⁶⁸¹³. The ParallelX group was working to develop a new and experimental execution model for future high performance computing architectures. This model was christened ParallelX. The first implementations of ParallelX were crude, and many of those designs had to be discarded entirely. However, over time the team learned quite a bit about how to design a parallel, distributed runtime system which implements the concepts of ParallelX.

From the very beginning, this endeavour has been a group effort. In addition to a handful of interested researchers, there have always been graduate and undergraduate students participating in the discussions, design, and implementation of *HPX*. In 2011 we decided to formalize our collective research efforts by creating the [STE||AR](https://stellar-group.org)⁶⁸¹⁴ group (Systems Technology, Emergent Parallelism, and Algorithm Research). Over time, the team grew to include researchers around the country and the world. In 2014, the [STE||AR](https://stellar-group.org)⁶⁸¹⁵ Group was reorganized to become the international community it is today. This consortium of researchers aims to develop stable, sustainable, and scalable tools which will enable application developers to exploit the parallelism latent in the machines of today and tomorrow. Our goal of the *HPX* project is to create a high quality, freely available, open source implementation of ParallelX concepts for conventional and future systems by building a modular and standards conforming runtime system for SMP and distributed application environments. The API exposed by *HPX* is conformant to the interfaces defined by the C++ ISO Standard and adheres to the programming guidelines used by the [Boost](https://www.boost.org/)⁶⁸¹⁶ collection of C++ libraries. We steer the development of *HPX* with real world applications and aim to provide a smooth migration path for domain scientists.

To learn more about [STE||AR](https://stellar-group.org)⁶⁸¹⁷ and ParallelX, see *People* and *Why HPX?*.

⁶⁸⁰⁸ <https://joss.theoj.org/papers/022e5917b95517dff20cd3742ab95eca>

⁶⁸⁰⁹ <https://doi.org/10.5281/zenodo.598202>

⁶⁸¹⁰ <https://doi.org/10.5281/zenodo.598202>

⁶⁸¹¹ <https://hpx.stellar-group.org/hpx-users/>

⁶⁸¹² <https://www.cct.lsu.edu>

⁶⁸¹³ <https://www.lsu.edu>

⁶⁸¹⁴ <https://stellar-group.org>

⁶⁸¹⁵ <https://stellar-group.org>

⁶⁸¹⁶ <https://www.boost.org/>

⁶⁸¹⁷ <https://stellar-group.org>

2.14.2 People

The [STE||AR](https://stellar-group.org)⁶⁸¹⁸ Group (pronounced as stellar) stands for “Systems Technology, Emergent Parallelism, and Algorithm Research”. We are an international group of faculty, researchers, and students working at various institutions around the world. The goal of the [STE||AR](https://stellar-group.org)⁶⁸¹⁹ Group is to promote the development of scalable parallel applications by providing a community for ideas, a framework for collaboration, and a platform for communicating these concepts to the broader community.

Our work is focused on building technologies for scalable parallel applications. *HPX*, our general purpose C++ runtime system for parallel and distributed applications, is no exception. We use *HPX* for a broad range of scientific applications, helping scientists and developers to write code which scales better and shows better performance compared to more conventional programming models such as MPI.

HPX is based on *ParalleX* which is a new (and still experimental) parallel execution model aiming to overcome the limitations imposed by the current hardware and the techniques we use to write applications today. Our group focuses on two types of applications - those requiring excellent strong scaling, allowing for a dramatic reduction of execution time for fixed workloads and those needing highest level of sustained performance through massive parallelism. These applications are presently unable (through conventional practices) to effectively exploit a relatively small number of cores in a multi-core system. By extension, these application will not be able to exploit high-end exascale computing systems which are likely to employ hundreds of millions of such cores by the end of this decade.

Critical bottlenecks to the effective use of new generation high performance computing (HPC) systems include:

- *Starvation*: due to lack of usable application parallelism and means of managing it,
- *Overhead*: reduction to permit strong scalability, improve efficiency, and enable dynamic resource management,
- *Latency*: from remote access across system or to local memories,
- *Contention*: due to multicore chip I/O pins, memory banks, and system interconnects.

The *ParalleX* model has been devised to address these challenges by enabling a new computing dynamic through the application of message-driven computation in a global address space context with lightweight synchronization. The work on *HPX* is centered around implementing the concepts as defined by the *ParalleX* model. *HPX* is currently targeted at conventional machines, such as classical Linux based Beowulf clusters and SMP nodes.

We fully understand that the success of *HPX* (and *ParalleX*) is very much the result of the work of many people. To see a list of who is contributing see our tables below.

⁶⁸¹⁸ <https://stellar-group.org>

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Table 2.188: Contributors

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⁶⁸³⁸ <https://www.cct.lsu.edu>

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⁶⁸⁴⁰ <https://www3.cs.fau.de>

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⁶⁸⁴⁷ <https://www.tudelft.nl/en/eemcs>

⁶⁸⁴⁸ <https://www.tudelft.nl/en/>

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⁶⁸⁴⁹ <https://www.cscs.ch>

⁶⁸⁵⁰ <https://www.cscs.ch>

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⁶⁸⁵⁹ <https://www.lsu.edu>

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⁶⁸⁶¹ <https://github.com/STELLAR-GROUP/hpxcl/>

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⁶⁸⁶³ <http://wg21.link/n4313>

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⁶⁸⁷⁸ <https://www.cscs.ch>

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⁶⁸⁷⁹ <https://www.nmsu.edu>

⁶⁸⁸⁰ <https://hpx.stellar-group.org/funding-acknowledgements/>

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⁶⁸⁸² <https://developers.google.com/season-of-docs>

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